Algebra

— *EYNTKA* Series —

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Contents

0.0.1 Reading the numbering A star marks *bonus material*: a chapter, section or subsection that belongs in the book but that nothing later depends on. Skipping it costs nothing downstream. The star sits on the number, as in §2.7$\star$; in divisions written before this convention it sits at the front of the title instead, as in **Uniform Spaces*. The two mean the same thing.

The title of the book says it all: the intention is that you will find "everything you need to know about undergraduate algebra" in these many pages. I chose the word "everything" purposefully knowing it is a contentious word, especially in the range of material that is considered at the undergraduate level. In some universities, only the first few chapters of this book are considered part of the undergraduate curriculum, while at others the entirety of the books content and more is available in the undergraduate program. Thus, parameters had to be set in choosing the material of this book while trying to remain true in using the word "everything" – they are the following:

1. By the end of the book, or any part of the book (ex. part on Groups, Rings, Modules, etc.), the reader should have had a deep enough understanding and intuition so as to be able to tackle graduate level mathematics. The reader should be comfortable with picking up almost any graduate level book in algebra, including topics like algebraic number theory, algebraic geometry, representation theory, lie theory, classical or geometric group theory, ring theory, homological algebra, and so forth.

2. By the end of the book, the reader should be comfortable to think in any (or at least most) major algebraic studies, to be able to more than just recite definitions and systematically follow proofs but to have a connection with the definition and the way of thinking that allowed the mathematician to come up with the proofs and ideas, which will hopefully encourage the reader to start thinking like a mathematician. The goal is that the reader has a deep pool of examples, mental structures, and intuitions that they can draw on to come up with their own ideas.

3. By the end of the book, the reader should get a sense of how interconnected mathematics is. There is over a thousand cross-references in the book interconnecting all parts of it together, and a mathematical language will be introduced early on that will allow the reader to study the connection between objects.

4. By the end of this book (or any section), the reader should be comfortable solving problems (or problems from that section). To achieve this, many exercises are added throughout the end of sections in the book. Doing as many as necessary to get comfortable with the material is very important; reading a math book and not doing the exercises is like reading a cook book without cooking – you won't get to taste the result. It is common to have a fake sense of understanding or a lack of intuition if exercises are skipped (I say this out of personal experience and that of many of the colleagues I've worked with).

Thus, in order to achieve these goals, the book is on the larger side of textbooks, and covers most of "core algebra" of undergraduate curriculums as well as most "optional" courses in algebra many undergraduate curriculum's offer. To insure the book stays within the scope of undergraduate knowledge, only material that I have personally covered or have learned about during my undergraduate years are included. Since algebra is my favourite topic, I've taken as many optional algebra courses and read as many books as possible on the subject, and so I wish to think that I've covered enough of a broad range of topics in great enough depth to use my personal experience to be a benchmark for the quantity of material[1]. There is also the question of what should be assumed to be the starting point. The solution I've done is the same as many authors solutions: have a preliminary chapter that assumes only an elementary level of knowledge of mathematics (for this book, knowledge of arithmetic's and factorization) and cover all necessary prerequisite material that will be used in the book. The preliminary chapter in this book is more extensive than most preliminary chapter to

[1] This also means that I chose to start considering some material as beyond the scope of undergraduate algebra, such as scheme theory and class field theory (dissuasion of omitted material at the end of this section)

insure the reader has no "knowledge gaps". In any case, I hope that the reader will find this textbook a useful accompaniment throughout their undergraduate studies in algebra.

Properties of the Book

The book covers a great multitude of topics and structures (there are at least 21 different structures in this book, including groups, rings, modules, sets, graphs, and so forth), which not only overlap in many important ways but also constantly interconnect, describe, and influence each other. Part of the goal of this book is to show how interconnected mathematics and algebra is, and so to accomplish this goal a light-version of category theory is introduced early on and gradually built upon throughout the book. It is the opinion of this author that shying away from all category theory (as some textbooks I've read have done) delays many interesting and important connections between so many parts of algebra, and gets rid of the opportunity to foreshadow some amazing results[2]. Naturally, the full generality of category theory is not needed. Similarly to how set theory is not introduced in terms of the Zermelo-Frenkel axioms but as naïve set theory, so too is category theory introduced in a simplified version, giving a sufficient amount of language and tools to study how mathematical objects interconnect while not overloading on the generalities and technicalities. Some key tools and vocabulary that will be introduced early on is that of universal properties and functors. Throughout the book, there are hundreds of instances of universal properties and functors being used, showing how pervasive of a tool they are in algebra and mathematics as a whole. Gaining an intuition by seeing these concepts naturally appear gradually but early is useful to gain deeper insights.

Each part and chapter of the book will start off with some motivation for the material. To accommodate many different learning styles, many forms of motivation are given, from historical context, to practical application, to important consequences, to how the material is connected to other areas of mathematics. The idea is to have the reader already start making connections while reading the main material. This is an approach that contrasts with Bourbaki which aims to present material minimally and coarsely, omitting any of the authors intuitions or insights with the goal of creating a standard for mathematics. This book is not intended to be an encyclopedia for senior mathematicians to double-check their notation understanding of some concepts, but to be a fun and interesting yet deep and insightful pedagogical tool to encourage the reader to build up their understanding and make as many connections as possible while challenging the readers ways of thinking by actively engaging in the interpretation of the material. Though the intuitions of this author will be present, it is my hope that the book will be written in such a way that the reader will develop their own intuition, solidifying in their own way what the material of this book has to offer.

The end of each chapter will contain a summary section. This summary serves both as a chance for readers to insure they have memorized the core ideas of the chapter, as well as a litmus test for readers coming from other EYTNKA books that reference a chapters in this book.

This book is an algebra textbook, and so avoids covering material outside of algebra. However, since mathematics is a very interconnected discipline, it is sometimes impossible to avoid talking about other domains of mathematics. As an example of the interconnectedness of mathematics, many definitions will have examples that lie in different domains of mathematics. Thus, it won't be uncommon that examples of definitions will be given that are not strictly algebraic, however there will always be an algebraic example accompanying a non-algebraic example, since the book should

[2]For example, Cayley's theorem, the existence of a homomorphism from any ring R to $\mathrm{End}_{\mathbf{Grp}}(R)$, or representation theory in general are all naturally connected via Yonneda's lemma. Another example would be the plethora of adjoints

have no prerequisites besides what is covered in the preliminary chapter. It will be necessary to cover some non-algebra material at the preliminary stage and late stage of the book. For the preliminary stage, it will be necessary to cover some basics of [naïve] set theory, number theory, graph theory, and logic. Elementary concepts not needed for algebra (such as supremum, limits, completeness of $\mathbb{R}$) are not covered, and concepts that play a tangential role (such as cardinality) will be left as optional. In the later stages of the books (starting near the end of part IV on page 1396), topology and geometry will start getting a more prominent role. Since these topics are so far into the book, it is perhaps not too wrong to think that the reader has a great interest in mathematics as a whole and has already covered some basics of topology, or perhaps even differential geometry. Looking through the curriculum of many universities, it seems that differential geometry is usually covered quite late, while topology is either taught as part of a 2nd or 3rd year analysis or as a separate course. Thus, the book will not rely on geometric thinking, however it will start to casually incorporate topological thinking and results to relevant theorems (which will be relatively few), starting in part V. A review of topology is given in appendix C. Finally, some intuition on the notion of the real numbers $\mathbb{R}$ and the Pythagorean theorem shall start appearing at chapter 13 for very few results.

A convention that is prominent in this book is that square brackets around words indicate that the word can be used to eliminate ambiguity. For example, we will define functions in the preliminary chapter. But in other domain of mathematics, the term function is used differently (ex. the definition of *function* in differential geometry would be defined as *real-valued function* in the preliminary chapter). Thus I will put a word in square parenthesis to show that the word can be added to the term to eliminate ambiguity. For example, though we will always say *function* throughout this book, where it might be ambiguous in general mathematical contexts and in the definition it will be written as *[set] function*.

Another convention will be that sections that have a starred in front of their title are consider as “optional”. This is material that does not fit in the appendix, which is material that fills in more advanced gaps for the curious reader (ex. properties of the Axiom of Choice, or topological interpretations of some results), while at the same time not being central to the main narratives of the chapter. They were included as side-quests for the interested readers, but the reader who isn’t interested can comfortably skip the section without worry of it being referenced later in the book. The exception to this referencing rule is the starred sections in the preliminary chapter; these shall only be referenced for a hand-full of proofs in the book.

The book shall also try to mix classical and modern motivation for the study of each given field. For example, Galois groups have a classical motivation in understanding the symmetries of roots of equation, while they also have a modern motivation in containing the structural information of primes or of being the right analogue of the fundamental group. The classical motivation shall often be the doorway that makes it easy to start having an interest in a certain theory, while the modern motivation explains why this theory persists in being interesting into present mathematical research. More modern motivations for the theories that will be presented shall become more prevalent in latter parts of the book and left as boxes for more advanced readers in the earlier parts of the book.

(mention how a lot of the footnotes are are for a 2nd reading).

Finally, as the reader goes through the book, they will gain skills and intuitions for solving many problems. Many of these skills will come from reading through all the proofs of the results, as well as the reader doing the exercises. Eventually, many of the results boxed as propositions, lemmas, and corollaries will stop being proved in detail and have part or all of it left as exercises (starting around 300 pages into the book). This pattern shall become more prominent as the reader progresses through the book and gains “mathematical maturity”. This pattern reflects the changing pedagogical need of the reader as they progress through the book; it shall become more fruitful for the reader to prove results on their own rather than having the detail laid out. In fact, at that stage of the book the reader may even find that it is easier to prove the results themselves, or feel it is cumbersome to read through a proof they can do themselves[3].

Exercises There are very few exercises in this textbook as compared a proper textbook. This will be remedied in the 1st edition.

Conventions

If any of the notation is unclear, the preliminary chapter will cover it in more detail.

- Notation $x = a, b$ is shorthand for $x \in \{a, b\}$. If S is a clearly enumerable set (ex. $S = \{1, 2, 3, 4, 5, 6, 7\}$, then we shall sometimes write $x = 1, .., 7$ to mean $x \in S$.
- (convention on fields notation (k vs K vs $\mathbb{F}$)

The numbering of definitions, lemmas, propositions, theorems, corollaries, examples, and box environments all share the same counter; the goal is to make it quicker to navigate to relevant results when flipping through the pages.

What is [Abstract] Algebra?[4]

Algebra as a field of study originated in the study of numbers and equations that can be formed through the elementary arithmetic operations: addition, subtraction, multiplication, and division $(+, -, \times, \mathrm{div})$. As mathematics has evolved, it was found that there exist many instances where we can find new forms of arithmetic on objects that are different from numbers with new operations. A simple example can be given by a square with corners distinctly labeled. We can rotate the square by $\pi/2$ radians (90 degrees). Doing four $\pi/2$ rotations shall result in the square with its original orientation. If we think of the symbol e as representing the original square in its original orientation, r the square rotated by $\pi/2$ radians, r^2 the square rotated by $2\pi/2$ radians, r^3 the square rotated by $3\pi/2$ radians, and r^4 the square rotate by $4\pi/2$ radians, we can imagine that $r^4 = e$. As a mathematician, we may consider this equation as a property of the "arithmetics of rotation" of a square. A similar example would be the hands of a clock: a clockhand rotates by $\pi/6$ radians every hour, and after 12 hours shall come back to its original position (as an equation, $r^{12} = e$), giving a form of "clock arithmetics". As mathematicians studied new systems, a huge quantity of new "arithmetics" where found. A list of some of the first new arithmetics systems that were found are:

1. The set of matrices can be added and multiplied
2. The set of networks (or direct graphs) can be combined to form larger graphs (in this case, the operation is the direct product)

[3] I, the author of this book, have had this experience the more mathematics I studied

[4] From a non-historical perspective

3. Functions can be composed (In this case, the operation is composition)
4. Sets can be unioned and intersected (in this case, the operation here is set-union and set-intersection)
5. vectors can be added and multiplied by a scalar (in this case, the operations are vector addition and scalar multiplication)
6. n-gons can be rotated by multiple of $2\pi/n$ radians to get back a square that is indistinguishable from the original square (in this case, the operation is rotation by $2\pi/n$)
7. Two paths $\alpha, \beta \subseteq \mathbb{R}^2$ whose endpoints match can be concatenated (in this case, the operation is path-concatenation)

This is a short list of thousands of new "arithmetics" that were found in a huge multitude of fields (from physics, to engineering, to linguistics, to geometry, and much more), many of which share commonalities in their properties, their "algebra". A new unified way of studying these systems had to put in place in order to systematically study these systems and better understand the rules manipulating these objects, the *algebra* of these systems.

The solution was to *abstract* a list of properties these systems share and study these properties. The abstraction was called an *algebraic structure*. Any algebraic structure consists of:

- a set of *objects* and
- a set of *operations* (i.e. functions) which must follow
- a collection of *axioms* (assumed rules they must follow)

All the examples that were given above are algebraic structures, for they may all be represented as a set along with one or multiple operations. Many of the operations in the above examples follow different axioms. Mathematicians have classified and given names to different types of algebraic objects depending on which set of axioms they follow. For example, the integers (with addition and multiplication) is an example of an algebraic structure called a *commutative ring*, while n-gons (with rotation by $2\pi/n$ radians) are an example of an algebraic structure called a *group*.

Abstract Algebra is the study of *family* or *types*[5] algebraic structures. As a consequence, in abstract algebra, we do not study a particular operation or "arithmetic system" such as the numbers with addition, rotations of a square, or any other given in the examples, but study operations that follow certain rules called *axioms*. This is the core idea behind abstract algebra: a new type algebraic structure is defined by either abstracting desirable properties from a collection of algebraic objects or creating axioms which are deemed representative of the studied structures. The advantage of this abstraction is we can study the property of the operations directly without any "noise" that may come from other properties of objects, giving us insights on all systems which can be represented an algebraic structure of a certain type. In this book, there shall be over 20 types algebraic structures that shall be studied, each at varying degrees of depth depending on their importance. Some of these elementary structures are, in the order presented in the book:

groups, group-actions, rings, fields, modules,
vector-spaces, [associative] algebras, chain complexes

[5]This is similar to the term types from *type theory* which holds a more precise meaning that the current casual use of the word type. The words "family" and "type" where both used here to accommodate type theory and set theory

Some other important objects that appear less often are (in no particular order):

monoids, groupoids, Lie algebras, inner product spaces, lattices, rngs, algebraic sets

Algebraic Equations

Given an algebraic structure, we may define *algebraic equations*: we have a collection of symbols called *variables* (three such common symbols in high school are x, y, z) which have restrictions placed on them. A prototypical example of such equations on numbers are *linear equations*: equations of the form $ax + b = 0$ for numbers a, b and variable x.

In abstract algebra, the operations no longer need to be addition and multiplication, and the objects no longer need to be numbers. If you know some calculus, then the following is an example of a more advanced equation[6]:

- *Differential Equations*: These are equations whose solutions are functions. The operations are addition of functions represented by $+$, multiplying by a number, and differentiating (we can treat differentiation as an operation that takes in one function and returns another). An example of a differential equation is:

  $$6f^{''} + 5cf' - f = 0$$

  where f is an at least twice differentiable function

More generally, an algebraic equation p is some composition of operations with elements of a set and variables. Often, we shall want to know if there exists an α (or multiple $\alpha_1, \alpha_2, ..., \alpha_n$) and β such that $p(\alpha) = \beta$ (or $p(\alpha_1, ..., \alpha_n) = \beta$). When looking for solutions, the set of objects of our algebraic structure matter. For example, if we restrict ourselves to working with the positive numbers, the equation $x + 5 = 0$ has no solutions, and if we allow for negative numbers then $x = -5$ is the solution. If we restrict ourselves to working with the whole numbers, then the equation $2x + 3 = 0$ has no solutions, and if we allow for rational numbers then $x = -3/2$ is the solution. As we see with the example with differential equations, the objects we are working with need not be numbers. Indeed, as long as we define the operation $+$ correctly in more general contexts, the solution to an equation $ax + b = 0$ need not be a number (it may be a function, a matrix, a network, or a shape)!

Though looking for solutions to equations may seem like a core goal of algebra, it turns out that equations themselves present many fascinating properties that are often much more valuable to study than looking for solutions to equations. In abstract algebra, we consider equations as holding "semantic information", often holding information about a "property" of an algebraic structure. For example, the equation $r^4 = e$ from earlier is an example of an algebraic equation that shows that rotating 4 times by $\pi/2$ radians gives us back the original square. The reader may start to imagine how this equation characterizes a square more than a pentagon or a triangle, and indeed this notion of an equation "representing" some core information become central to algebra[7]. Often, we shall be able to distinguish two particular instances of an algebraic structure by what equations the elements satisfy. More generally, algebraic equations may be hiding much more information (a famous example being the study of elliptic curves in algebraic number theory), and many mathematicians will invest a great deal of time uncovering these properties.

[6] The set of differential functions along with differentiation forms an algebraic structure known as a *lie algebra*

[7] To foreshadow, the construction of free-objects shall make this rigorous, and the Nullstellensatz shall be the bridge in thinking of equations as holding semantic information and interpreting them as functions

Goal of Abstract Algebra?

So what does an algebraist aim to achieve? In short:

1. *Solving and understanding equations*: Given an equation or system of equations, an algebraist wants to find solutions to the equation(s) (or determine that no solution exists). An algebraist will look for these solutions by "algebraic" means (for example, calculus or tools from ODE/PDE theory would be considered non-algebraic[8]). When the objects of study are not numbers, finding solutions may get quite complicated. As mentioned above, it shall also be the case that mathematicians look for intrinsic properties of algebraic equations themselves. How to think of this "information" which algebraic equations provide will be elucidated throughout the textbook.

2. *Finding properties of Algebraic Structures*: There is an incredible amount of different families of algebraic structures, each with varying properties. As can be surmised by the table of contents, a lot of energy will be dedicated to understanding the properties of algebraic structures. Some intriguing questions that shall be answered are:

   (a) When can the notion of "factorization" be defined, and when is it unique?

   (b) Given some algebraic structure, are there simple objects that can be used to construct all other objects (for example, there will be algebraic structures which will have the notion of "prime" like prime numbers)

   (c) when are objects "finite", "almost finite", or "infinite"?

   (d) Is the notion of "dimension" well-defined?

   (e) Can we define some reasonable notion of "size"?

   (f) Are some objects good at representing a problem in such a way that it elucidates some new property?

   these may seem like rather abstract questions, however this is a consequence of studying the abstraction of objects, which often implies that some "core" property of the object reveals itself in the abstraction

3. *Clasifying Algebraic Structures*: In the best case scenario, all possible types of algebraic objects of a given family can be found. For example, it is one of the great accomplishments of the 20th century is finding the building blocks of an algebraic structure known as *finite groups* (requiring over 10000 pages to write down the proof)[9]. More generally, algebraist tend to look for building blocks of algebraic structures and means of creating more complicated algebraic structures given the building blocks.

4. *Developping Algebraic Techniques*: Many algebraic structures allow "shift of paradigms" that reduce a seemingly very difficult problem into a simple solved problem. For example, many shapes (circles, tori, Möbus bands, klein bottles) can be distinguished by associating an algebraic structure to them which then allow mathematicians to find properties of these shapes given properties of the algebraic structures.

[8]The boundary of what is an "algebraic" method to solve equations becomes more murky in advanced algebraic topics, for example in the representation of Lie Algebras or Scheme Theory. This shall not be a problem for this textbook in over 99% of cases

[9]this shall be explored further in section 3.7

The general angle of attack in solving a problem "algebraically" is:

- to find a way of inducing an algebraic structure given the object the mathematician is working on. This is very commonly done in geometry where associating different algebraic structure to a shape give different properties of the shape, depending on the properties of the algebraic structure[10].
- to find some key properties, call them axioms, then study the consequences of these axioms. This shall be done repeatedly in this textbook, and most major algebraic structures have come about through this process[11]
- to break down your problem in pieces that we *know how to put it back together* through the use of several *functions*. The more mathematics that you encounter, the more often you shall see this strategy. For example, in Measure Theory we take advantage of the vector-space structure in the process of measuring diagonal planes in $\mathbb{R}^n$, and in Fourier analysis it is key that solutions to linear PDE's are closed under function-addition to form more complicated solutions. It is important to emphasize that we know how to put it back together using *functions* or else we would have something more akin to *topology*.

Advice for Beginners

For those experiencing abstract mathematics for the first time, perhaps with a background in calculus or some computational linear algebra, the first encounter with this sort of mathematical thinking may seem a bit obtuse. This section shall give some pointers to ease into abstraction.

As a first-time learner, it is tempting to try and memorize the definitions, lemmas, and theorems to be able to apply them on a test, similar to how a successful strategy for at least passing most calculus tests is to memorize the rules of differentiation and integration. This only works in minimal ways for abstract algebra, and mathematics more generally. In order to achieve mastery over mathematics, the reader should really form their own intuitive understanding of the topics, an internal world that naturally holds together all the concepts which the reader should know like the back of their hand. The reader should try to internalize a reason for why the definitions exist, for why the theorems and lemmas need to be proved in the first place, and how one would come up with the questions the theorems and lemmas are asking.

At the very beginning of the readers mathematical journey, this is only partly feasible. In my personal experience, it was very much akin to being a new-born who needs help understanding the language and the basics of how to "walk". However, at your own pace, you shall start "seeing" why a certain definition was chosen, shall start asking questions to yourself that on the next page will be a theorem the author (i.e. me) already answered, and shall start feeling that the concepts you have learned are "basic concepts" that anybody would know. To start picking up these skills, there are certain steps the reader should take:

1. Do as many exercises as the reader needs until they feel they understand the concepts. Some more advanced mathematics courses even go so far as to not present any theorems/lemmas and

[10] See a book in *algebraic topology* for more details

[11] Groups were an abstraction from bijective function, commutative rings from integers and modular arithmetic's, fields from solving polynomials, Dedekind Domain from rings of integers, and so forth

make the students deduce these results[12].

2. Before looking at the proof of any result, try to see if you can prove it yourself. For some results, this may be very challenging, however in these cases the reader shall find that even taking a moment to think of how to tackle the question creates enormous benefit in understanding the concept, even if the reader is totally off (which I must confess I myself have been multiple times). Doing this shall also give the added benefit of giving the reader more confidence in trying to attempt problems, and being able to prove a result before a proof is presented will certainly make the reader deeply feel like they understand what is going on. It may even be that you come up with a better proof that the one presented in the book! Even if the reader was completely off, having experienced exactly what *not* to think is itself insightful (though admittedly could be frustrating); it is inevitable that the reader shall not have the right idea for a concept in their mind, and the sooner the reader catches their misconception by being wrong, the better.

3. Try to tie together various concepts you have learned. This increases the number of ways you can recall and use your concepts, and greatly enriches ones appreciation of the topics. This book goes to great lengths to give some intriguing connections, however it is very fruitful to any reader to start seeking out these connections.

Once the readers masters these tools and ways of thinking, by the end of the book the material will truly feel elementary and introductory.

Omitted Material

No textbook can comprehensively cover all possible material as there are too many branching subjects with their own preliminaries. The following material was omitted for various reasons such as having relatively narrow application for a first-time algebraist, needing a whole other book to explore its basic properties and motivations, needing knowledge of fields outside of algebra to properly study their properties, and sometimes simply because the material (as of writing this book and the last few decades) has not had a strong branch of research within mathematics. Material falling under one of these categories include:

1. Schemes. They are the starting point of modern algebraic geometry, but the topics depth requires its own book to study (see [5]) for the continuation in algebraic geometry.

2. One of the culminating results of number theory in the 20th century is the proof of *Artin Reciprocity* which, *very* loosely speaking, connects the properties of primes to properties of some polynomials (more specifically, there is a connection between class groups of field extensions with galois groups of field extensions). A lot of modern algebraic research centers on generalizations of this result (ex. *Langland's Program*), however the amount of extra material required to cover would be too much and would require some use of complex and real analysis to cover properly.

3. Commutative algebra is a vast field that holds many very interesting and important results, more than any introductory textbook can cover. This textbook shall cover enough to introduce number theory and (classical) algebraic geometry, and prepare the reader for some further studies in these fields. See [37] or [49] for in-depth coverage.

[12] I have once jokingly seen an exercise in Lang's *Algebra* which was to pick up any textbook in algebra, and prove every result from beginning to end independently

4. Lie Algebras get a cursory overview due to their importance in representation theory, but a deeper study is greatly benefited when paired with Lie Groups.

5. Invariant Theory is an area of mathematics that has been greatly studied and has grown further in disciplines such as complexity theory[13], however we shall not have the time to cover the material in any level of depth.

6. The classification of finite groups is a project considered "complete" and so only a limited amount of effort is put in covering the mathematics behind it, enough to understand the magnitude of the project and the important principles leading to how the program was pushed forward.

7. Representation theory is a huge disciplines which shall only get enough coverage to elucidate the most important results (ex. Artin-Wedderburn) and its applications to many fields (finite groups, quivers and lie algebras, number theory).

8. An introduction to Category theory is covered in part XII, however it stops short of covering higher-order category theory (a tool useful in modern algebraic topology and algebraic geometry).

9. Cohomology theory is covered to the point where the core results are on full display and some "immediate" application is evident, but will not go into detail about many of the broader applications and theory (ex. Motivic Cohomology will be left for a textbook in algebraic geometry)

Some advanced algebraic material was chosen to be included due to its relevance in other textbooks in the EYTNKA series such as spectral sequences, quiver theory, infinite galois theory, and Hilbert polynomials.

Acknowledgments

This book started off as my personal notes for my first course in algebra, and then evolved as my algebra notes in general. I found it helpful to write the notes in a textbook format, as being able to explain why something is true proved to greatly increase my own understanding of the topic. Since the book was written while I was learning the topics, a lot of inspiration was drawn form multiple textbooks and followed their outlines closely. As the book expanded, it started to get its own character, however as of writing this acknowledgment there is still a lot of the book that is heavily influenced by other textbooks. It is thus in good conscience that I would like to give credit to these inspirations:

1. *Abstract Algebra* by Dummit and Foote [28] was the first textbook in algebra I used. A lot of the core material covered in this textbook stem from the material covered in this textbook. This textbook is an excellent introduction to the field of algebra with the abundance of details and examples, as well as some detailed proofs. The textbook is especially useful for its coverage of foundational material such as Groups and Rings. Some of the more computational aspects of the book (ex. Calculating Gröbner basis, finding RCF) is inspired by this textbook.

[13] The P vs. NP conjecture can be reformulated in the language of invariance theory

2. *Chapter 0* by Paolo Aluffi's [2] has been a huge influence on this textbook. I agree with the author that categorical thinking should not be shied away from and should be introduced as soon as possible, though as this textbook aims to have a more undergraduate starting point a lot of the categorical results are introduced more slowly in this textbook. A lot of the later material this book is also more strongly inspired by Chapter 0 (especially the chapters on Field/Galois Theory and Homological Algebra).

3. Richard E. Borcherds' youtube videos were often my key source for finding intuitions: https://www.youtube.com/@richarde.borcherds7998. His clarity and ability to communicate the core idea of a concept so succinctly has greatly helped my entire mathematical journey[14].

4. *Commutative Algebra* by Michael Atiyah [3] has had a strong influence on the commutative algebra part of the book. A large part of what was considered to be core results came from his book. A section that I was wavering on adding was the one on Hilbert polynomials as these do not show up again in the EYNTKA series until Hilbert Schemes and cohomology of schemes are introduced in [5], however I ultimately decided to keep the section on Hilbert Polynomials as there is no other natural place to introduce them in the EYNTKA series.

5. *Number Fields* by Marcus [36] and *Algebraic Number Theory* by Neukirch [44] were foundational to the chapter on number theory. Marcus's textbook offered a very detailed overview of the theory when only focusing on number fields, while Neukrich offered the abstraction I was looking for to ground these ideas in my mental framework.

6. *Algebraic Curves* by Fulton [54] *Basic Algebraic Geometry* by Shafarevich [46] were strong inspiration for the chapter on algebraic geometry. The initial sections more strongly draw inspiration form Dummit and Foote who go over the core algebraic-geometric concepts in great detail, while Shafarevich and Fulton covered the missing sections that would be necessary background for scheme theory and class field theory.

7. Milne's notes were often consulted for their lucidity and breath of subject. The section on Kummer Theory was heavily inspired by Milne's *Field and Galois Theory*, [34]

8. *Algebra* be Serge Lang [35] has often been consulted for some the more advanced material of this book, such as the section on infinite galois extensions, as well as ordered fields and derivations on fields.

[14]After struggling with a concept, I would often watch Richard Borcherds explanation and comment to my colleges that "As always, Richard Borcherds knocks it out of the park"

Preliminary Chapter

If naïve set theory is new to you, this section is going to go through a lot of terminology. The reader may want to skip to certain sections depending on their background:

1. Sets: sets, power-sets, unions and intersections. Logical quantifiers, basic propositions, vacuous truth. Induction. Relations, equivalence Relations, partial and total orders.
2. Functions: injective/surjective/bijective functions and their relation to equivalence relations. Image and restriction, composition of functions, associativity of composition, Cardinality
3. Number theory: GCD and LCM, divisibility, prime factorization, coprime numbers, the euclidean algorithm, Bézout's Identity, Modular Arithmetics, Fermat's Little Theorem
4. Graphs: basic properties
5. Category Theory: intuition, universal property, functors.

You can look over this section if you want a refresher, or skip to section 0.1.4 to start from equivalence relations, in particular if the reader is unaware of the relation between equivalence relations and functions, then this section will be important.

There are also a few starred sections that are not necessary for most of the book, and are added as they show up for a few proofs.

0.1 Sets

As in any field of study, we need to establish a common language for everyone in the field to communicate their ideas. The accepted language of mathematics today was invented by Georg Cantor in the 19th century and is called *set theory*. Before Cantor, there's wasn't one universal set of symbols and concepts mathematicians agreed upon to talk about basic concepts, with different groups of mathematician inventing their own symbology to talk about their ideas. Nowadays, every field of mathematics grounds their work in set theory [15] – algebra is no exception.

We'll be learning a small part of set theory which concerns itself with using the tools of set theory for talking about concepts. These tools are under the label of *naïve set theory*. We will be using these tools and question little why we are using these particular tools. The reason for the term "naïve" set theory is due to history: there were a couple of important paradoxes which showed some inconsistencies in set theory [16]. From the existence of these paradoxes, a generation of mathematician's tried to free set theory from these problems [17]. Their research became known as set theory, and the previous understanding of set theory came to be known as *naïve set theory*. Naïve set theory wasn't dismissed, instead mathematicians who were interested in making a solid foundation for mathematics tried figuring out exactly how we should be using naïve set theory without contradiction, while the rest of mathematicians used naïve set theory. When we use naïve set theory, we're essentially using the results of [the new] set theory without worrying about the details. It's like taking advantage of a building without knowing the architectural details of why a support was put here, or a door was needed there. We will be taking full advantage of this pre-built building for us. Some footnotes will be added for those who want to peek at the questions that lead mathematicians of the early 20th century to investigate the details of set theory. For those with a deeper interest in the foundations of mathematics, a class on logic is recommended, or the reader can checkout EYTNKA logic [21] for the logic book in the EYNTKA series.

As a consequence of using naïve set theory, we'll be defining certain concepts by relying on some [very] basic intuition. For example, we will assume the idea that we can conceptualize a collection of objects (for example, imagine a bunch of trees, or a bunch of numbers), or know the basics of numbers (ex. you know how to think of the symbols $1, 2, 3, \ldots$ as number, and you can add/subtract/multiply/divide). Definition's after the preliminary chapter will build on the definition's presented here, and so it's not a prerequisite for chapters after the preliminary chapter to rely heavily on intuition.

Basic Definitions: Sets

In mathematics, we work with collection of things, and with "mathematical sentences" about a collection of things, where a mathematical sentence must be either be true or false. We will start by learning how to describe the collection of things, and then move on to learning about mathematical sentences. We will make a distinction between a collection of things and the things inside the collection. We call a collection of things a *set*, and the things inside a set are *elements*. Another definition of a set is provided by the founder of set theory Georg Cantor

> *A set is a Many that allows itself to be thought of as a One*

In other words, we can think of any arbitrary collection of objects put together as a new single entity. The one restriction we give to sets is that all the elements in a set must be different, in other words,

[15] There is a movement to replace foundation of mathematics with *type theory* – for those interested see ref:HERE

[16] Famously, Russel's paradox showed a logical problem in how we can define sets. We'll actually cover Russel's Paradox in ref:HERE (Perhaps add it to the end of set theory section?) if you're curious to check it out after reading this section

[17] one major result of this research was the axiomatization of set theory (i.e. the development of the fundamental rules) by Zermelo and Frankel, their set of axioms (i.e. fundamental rules) being called ZF in honour of them. There are a couple of axioms that can be added or removed from ZF, but ZF generally forms the core of modern mathematics. If you keep doing mathematics, you will hear statements like "This statement is true in ZF"

we must have each element to be distinct. The reason for this is that set theory can be thought of pulling from universe of all possible objects of which there are no copies of objects[18].

When we write a set, we put curly braces ("{" and "}") around a collection of things (or really, symbols which could represent things [19]) which are separated by a comma.

Example 0.1.1: Example of sets

1. To write a set with the numbers $1, 2$ and 3, we write $\{1, 2, 3\}$. The numbers 1, 2, and 3 are the elements of $\{1, 2, 3\}$. Note that $\{1, 1, 2, 3\}$ would not be considered a set since the 1's repeat. The convention when a set has repeated elements is to have them "squished" into one element to create a set with only distinct elements. This would mean that we would consider the set $\{1, 1\}$ to be $\{1\}$ where we squished the two 1's into a single element. This convention also holds if the set has other elements too:

$$\{1, 1, 2, 3\} = \{1, 2, 3\}$$

   If we add an element to a set that's already there, we will also do this "squishing" [a]

2. As mentioned, sets contain can anything. The set

$$\{\text{ice-cream}, \text{cheesecake}, \text{chocolate}\}$$

   is a set that represents 3 desserts.

3. The set $\{1\}$ is a set with the element 1. We would not say that 1 and $\{1\}$ are the same thing, since $\{1\}$ is "a set containing the number one as its element" and 1 as "the number one". Using mathematical notation, $1 \neq \{1\}$

4. $\{\}$ is also a set, called the empty set. It pops up often enough that we have a special symbol for this set: $\emptyset$

5. Sets can contain sets, so $\{1, \{☺, \mathfrak{T}\}, \text{bat}\}$ is a set that contains the element 1, the element $\{☺, \mathfrak{T}\}$ (which is also a set), and the element bat.

6. Sets need not have an order, so for example

$$\{\text{ice-cream}, \text{cheesecake}, \text{chocolate}\} = \{\text{cheesecake}, \text{ice-cream}, \text{chocolate}\}$$

   This also means that a set with the words for days of the week is the same if we re-arrange the days.

[a] Formally, (explain what's actually happening here in terms of ZFC?)

Sometimes, we care about the order of elements. If that is the case, we will surround them in parenthesis, "(and)", and call these *tuples*, or if there are n elements then n-tuples where n represents the number of elements. For example: $(1, 2, 3)$ is a 3-tuple, and $(1, 2, 3) \neq (3, 2, 1)$ even

[18] Going deeper than this, asking questions like what are the possible things a set can contain, what are all possible sets, can there be sets that contain itself, all venture into set theory proper

[19] Some might be dubious on whether the symbols can represent "anything". For example, is it okay to say that x represents a statement that is both true and false? Is that "a thing". This type of question of what is a "well-defined" thing is subject to deeper explorations of set theory

though $\{1, 2, 3\} = \{3, 2, 1\}$ [20]. 2-tuples are sometimes called a *pair*. If it is clear from context, it is a common convention to drop the number in front of the word tuple, i.e., if the writer consistently uses 2-tuples they might choose to just call them tuples.

Elements

As mentioned earlier, If something is inside a set, we call that an *element* or *object* of the set. We usually (but not necessarily) denote elements with lower-case letters (ex. a, b)[21]. To write "a in A" in mathematical notation, we use the symbol $\in$. For example:

$$a \in A$$

is read as "a in A" (or "a is in A" to sound more grammatically correct). If we want to say that "a not in A", we add a slash across our symbol (/):

$$a \notin A$$

For example, $1 \in \{1, 2, 3\}$, but $4 \notin \{1, 2, 3\}$. If we have the empty set $\emptyset$ (i.e. $\{\}$), then it is always the case that nothing is inside it, i.e., $a \notin \emptyset$ for any a. Sets can also contain other sets, so $\{1\} \in \{\{1\}, 2, \{\text{cheesecake}, 6\}\}$. Note that $1 \notin \{\{1\}, 2, \{\text{cheesecake}, 6\}\}$ since the set $\{\{1\}, 2, \{\text{cheesecake}, 6\}\}$ doesn't contain "the element 1" but the "the set that contains the element 1".

0.1.2 Concept Check Is

$$\{1, 2, 3\} \in \{0, \{\{1, 2, 3\}\}, 4\}$$

Equality

The concept of equality was already used earlier based on our intuition. We will expand a bit further on how to think of equality in set theory, and we will distinguish two symbols for equality:

$$A = \{1, 2, 3\}$$
$$A := \{1, 2, 3\}$$

Starting with the first line, the symbol $=$ means *logically equal*. Being "logically equal" (which we will always abbreviate to saying equal[22]) means the two sets have the same elements (for example

[20]In set theory, every mathematical object is represented as a set. This is no exception in n-tuples. A 2-tuple (a, b) would be $\{a, \{a, b\}\}$. A similar pattern follows for n-tuples

[21]Here's another "beyond the scope of naive set theory" question: What are the possible elements that may exist? Question of this varieties lead to logicians like Butrard Russel to come up with *Russel's Paradox*. This paradox will actually be covered in the book near the end (see ref:HERE). It's written in such a way that you can skip to take a look at it if you're curious

[22]This is another big set-theoretic question: what does it really mean for something to be equal? Bertrand Russel and Alfred Whitehead in their book *Principia Mathematica* famously took 379 pages before they defined equality. In modern set theory, two sets are equal if they contain the same elements

$\{1,2\} = \{2,1\}$). If we want to say that the two are not equal (i.e. the symbols represent different things [23]), we do the same convention as with $\in$ and put a slash across the symbol: $\{1\} \neq \{1,2\}$.

Notice we were comparing sets, while "$=$" has been done between numbers:

$$\frac{1}{2} = \frac{2}{4} \quad \text{and} \quad \frac{1}{2} \neq \frac{1}{3} \tag{1}$$

In proper set theory, every object is actually a set. Thus, even comparing $\frac{1}{2}$ and $\frac{2}{4}$ will be a comparison with sets. We will construct $\mathbb{Q}$ and show how it's elements are sets in section 9.5. For this entire book, we will freely use the common intuition of equality between elements like done in the above equations (equation 1), and for equality between predicates (covered in section 0.1.1).

Example 0.1.3: Examples Of Equality

1. if we want to say two sets have the same elements, that they are each representing the same collection of objects, we write $A = B$
2. $\{1,2,3\} \neq \{2,3,4\}$. Even though both sets have the elements 2 and 3, they each contain a number the other doesn't have, and so they are not equal
3. Similarly, $\{1,2\} \neq \{1,2,3\}$ since $3 \notin \{1,2\}$

The symbol $:=$ means the same thing as $=$, that is, $:=$ is also logically equal. The difference between $:=$ and $=$ is that $:=$ is used to emphasize that the symbol on the left of $:=$ will now be used to represent the thing on the right of $:=$. This is most often used in definitions to make sure the reader understands that symbol will consistently be used to represent the same thing many times. Because of this, the symbol $:=$ is read as "define" – the statement $A := \{1,2,3\}$ can be read that "Define A to be the set $\{1,2,3\}$".

Set-Builder Notation

Many times, we want to have a set of elements that satisfy only certain properties. For example, from the set of all deserts, I only want the ones that contain ice-cream, or from the set of all numbers I want only the prime numbers, or from the set of all months, I want the months that end in "ber". The way we'll capture this idea with sets is by using *set builder-notation* to build a new set given an old one. If S is a set and P is some sentence describing a property (ex. being an even number, or being a prime number, or representing "this desert has ice-cream"), then we'd write

$$A = \{s \in S \mid P\}$$
$$\text{or}$$
$$A = \{s \in S \ : \ P\}$$

[23] This is another question beyond naïve set theory: Since all we're using are symbols that represent objects, where do the objects come from? One answer to this is some universal set with every possible thing as an element of it, but thinking about what exactly does it mean for a set to contain everything leads to interesting nuances (one of the problems of a set that contains everything is once again Russel's Paradox)

and is read "s in S such that [s satisfies the property P]". In the next section, we shall be more precise about what type of sentence can be used:

Example 0.1.4: Set Builder Notation

1. If

$$M = \{\text{Janurary, February, March, April, May, June,} \\ \text{July, August, September, October, November, December}\}$$

   and $D = \{m \in M \mid \text{m ends in "ber"}\}$, then

$$D = \{\text{September, October, November, December}\}$$

2. If $W = \{\text{hello, bonjour, cześć}\}$, and $V = \{w \in W \mid w \text{ is in polish}\}$, then

$$V = \{\text{cześć}\}$$

3. if $\mathbb{N} = \{0, 1, 2, 3, \cdots\}$ and $P = \{n \in \mathbb{N} \mid n \text{ is a prime number}\}$ then

$$P = \{2, 3, 5, 7, 11, \cdots\}$$

   Note that $P' = \{n \in \mathbb{N} \mid \text{the only numbers that divide } n \text{ are 1 and } n\}$ is the exact same set (i.e. $P = P'$). Different properties can produce the same sets.

4. if $\mathbb{N} = \{0, 1, 2, 3, \cdots\}$ represents the set of all non-negative numbers with no decimal places, the set

$$A = \{x \in \mathbb{N} \mid x = 2y,\ y \in \mathbb{N}\}$$

   would represent all even numbers [a], and can be read "x in the natural numbers such that x can be twice any natural number"

[a] This notation has it's limitations if you dig deeper in it. You can create some very interesting paradoxes (Russel's paradox being one of them). We will not construct such contradictory cases in the book

0.1.1 Propositions and Predicates

We will now explore how to formulate questions and mathematical statements using mathematical language. Again, a full exploration would be covered in a textbook in logic.

Intuitively, there is some understanding of the meaning of a sentence[24] . If I say "I think blue is pretty" then in your mind some information got passed on. If we make a sentence that has a truth value associated to it (the sentence can either be true or false) [25] , like "The number 2 is prime", then we call such a sentence a *proposition*. If we have a sentence that takes in a variable that will make it depend on its truth value, for example "The number x is a prime number", then this will be called a *predicate*. Propositions and predicates are at the heart of mathematical communication. We

[24] The nature of how we derive meaning from sentences is an interesting logical and philosophical debate that goes beyond the scope of this book. See Russel, Wittgenstein, ________, for more details

[25] Development in a field called *topoi theory* are trying to generalize our notion of proposition beyond true and false, an interesting developing field to look into

have already encountered predicates before when we used set-builder notations: the restriction *has to be a predicate.* For convenience, we assign a symbol to a proposition. For example, we can have $P =$ "2 is a prime number". Then P would be true. However, $Q =$ "2 is not a prime number" is a false. For predicates, we shall usually assign a symbol and a variables, for example:

$$R(x) = \text{“}x \text{ is a prime number”’}$$

where x may be any natural number, then this predicate truth-value (that is, whether it is true or false) depends on the value the variables x will take.

Example 0.1.5: Predicates

1. let $P(x) =$"x is a month of the year". Then

$$P(\text{September}) = \text{true} \qquad P(\text{Wednesday}) = \text{false} \qquad P(\text{Laughing}) = \text{false}$$

2. The variable x can also reused in other propositions. For example we can have $P(x) =$"x is a dog" and $Q(x) =$"x is an animal".
3. Propositions can be composed, for example consider $R(x) =$"$P(x)$, then $Q(x)$" where $P(x), Q(x)$ are the propositions from the above. When is this proposition true?
4. Proposition can also take in multiple variables, for example

$$P(x, y) = (x < y)$$

Predicates of the form

$$\text{if } P(x) \text{ then } Q(x)$$

are some of the most common, allowing for relations being made between different concepts. The symbol used to symbolize an "if, then" relation is $\Rightarrow$, so

$$\text{If } P(x) \text{ then } (Q) \qquad \text{becomes} \qquad P(x) \Rightarrow Q(x)$$

Which is usually read "$P(x)$ implies $Q(x)$". A common mistake is to assume that if $P(x) \Rightarrow Q(x)$, then $Q(x) \Rightarrow P(x)$. This need not be the case. For example, define the following two predicates

$$A(x) = (x|2) \Rightarrow (x|4) \qquad A'(x) = (x|4) \Rightarrow (x|2)$$

When is predicate $A(x)$ true? When is $A'(x)$ true? If it *is* true that $P(x) \Rightarrow Q(x)$ always implies $Q(x) \Rightarrow P(x)$ we use the symbol $P(x) \Leftrightarrow Q(x)$ and read $\Leftrightarrow$ as "if and only if" (sometimes abbreviated to iff).

Vacuous Truth

Let's say we had two proposition $P(x, y) = x \Rightarrow y$, and let's say that x was false, while y is true. What is the truth value of $P(x, y)$? For example, let's say that

$$x = \text{I'm walking in the rain}$$
$$y = \text{I am wet}$$

Then if you are *not* walking in the rain, but *are* wet, then what should be the value of $P(x, y)$? Perhaps more interesting is the following example:

1. Let's say there are no cell-phones in a room. Then is the statement "all phone are turned off" true?

2. A famous such example is the following: is the statement "The king of France is not wearing pants" true or false if France does not have a king?

There are many solutions to this problem. The mathematically accepted solution to this problem is something called *vacuous truth*, which gives the following resolution:

$$(\text{False} \Rightarrow \text{True}) \text{ is True, and } (\text{False} \Rightarrow \text{False}) \text{ is } \textit{also} \text{ True}$$

The idea being that any "nonsense" statement is always "true" in the most trivial way. This interpretation of truth is not a unique one, but it is one that is very convenient as shall be seen by the reader.

Logical Quantifiers

We will now explore how to create more complicated properties by adding to our tool-belt 6 "words" from which we can construct mathematical sentences:

Definition 0.1.6: Quantifiers And Connectives

We define the following 3 quantifiers to be used in propositions:

1. $\forall$: the for-all quantifier.

2. $\exists$: the *existence* quantifier

3. $\exists!$ the *unique existence quantifier*

and the following 4 connectives to be used for propositions

1. $\vee$: disjunction (or) connective. If P, Q are propositions, then $P \vee Q$ mean that if one of or both of P or Q is true, then the new poposition $P \vee Q$ is true.

2. $\wedge$: conjunction (and) connective. If P, Q are propositions, then $P \wedge Q$ mean that if both P and Q are true, then the new proposition $P \wedge Q$ is true. If one of them are false, then $P \wedge Q$ is false.

3. $\Rightarrow$: implication connective: $P \Rightarrow Q$ is false if P is true and Q is false. In other words, if P is true, then P implies Q means Q is true.

4. $\neg$ negation (not) connective. If P is a proposition, then if P is true then $\neg P$ is false, and similarly if P is false then $\neg P$ is true.

The keen reader may ask if there are any other quantifiers or connectives, and indeed there can be many more. However, it is in fact sufficient to define these connectives/quantifiers to completely describe any [first-order] mathematical sentence[26]. Using these symbols, can the reader translate some of the propositions earlier in the book using these symbols?

Recall that we have defined the symbol $\in$ earlier to specify an element belongs to a set. This is a special symbol which is neither a quantifier nor a connective. It is considered a special symbol that is part of the language of set theory.

The order of quantifiers is important. For example:

$$\forall x \in \mathbb{Z},\ \exists y \in \mathbb{Z},\ x < y$$

means that for every x, there exists a y that is greater than it, which is certainly true when working with numbers. However:

$$\exists x \in \mathbb{Z},\ \forall y \in \mathbb{Z},\ x < y$$

means that there exists an x that is smaller than all y, which is certainly not possible since we can always find a smaller number.

With these, we can define certain operations we can perform on sets:

Definition 0.1.7: Set Operations

1. Define a subset to be:
$$A \subseteq B \Leftrightarrow (x \in A \Rightarrow x \in B)$$
Sometimes the symbol $A \subset B$ is also used. Since it is possible that $A \subset B$ and $A = B$ (you can check this by the definition), the symbol $\subseteq$ will predominantly be used to emphasize this relation. If we want to make it clear that $A \neq B$, we write $A \subsetneq B$.

   For any set S, we always have $\emptyset \subseteq S$ and $S \subseteq S$. With this notation, we also have that $A \subseteq B$ and $B \subseteq A \Rightarrow A = B$
2. Define the intersection of two sets to be:
$$A \cap B := \{x \mid x \in A \text{ and } x \in B\}$$
3. Define the union of two sets to be:
$$A \cup B := \{x \mid x \in A \text{ or } x \in B\}$$
if $A \cap B = \emptyset$, we say that A and B are *disjoint*
4. Define difference between sets to be:
$$A \setminus B := \{x \mid x \in A \text{ and } x \notin B\}$$
sometimes also written as $A - B$. If $T \subseteq S$, then we say that $S \setminus T$ is the *compliment* of T, usually written T^c.

Another very important definition

[26] This would be explored in a course on logic

Definition 0.1.8: Product Of Sets

Let A and B be sets. Then

$$A \times B = \{(a, b) \mid a \in A,\ b \in B\}$$

is the set of all 2-tuples (called just tuples for short). This construction can be extended indefinitely:

$$X_1 \times X_2 \times \cdots = \{(x_1, x_2, ...) \mid x_1 \in X_1,\ x_2 \in X_2,\ ...\}$$

This notation is also abbreviated as:

$$\prod_i X_i$$

For example:

Example 0.1.9: Product Of Sets

$$\{1, 2\} \times \{a, b\} = \{(1, a), (2, a), (1, b), (2, b)\}$$

and

$$\{a, b\} \times \{1, 2\} = \{(a, 1), (b, 1), (a, 2), (b, 2)\}$$

(A word on arbitrary products? i.e. $\prod_{i \in I} X_i$)

Exercise 0.1.1

1. Let $\{E_n\}_{n=1}^{\infty}$ be a collection of sets indexed by the natural numbers. Define

$$\limsup\{E_n\}_{n=1}^{\infty} = \bigcap_{k=1}^{\infty} \bigcup_{n=k}^{\infty} E_n \qquad \liminf\{E_n\}_{n=1}^{\infty} = \bigcup_{k=1}^{\infty} \bigcap_{n=k}^{\infty} E_n$$

show that

$$\limsup\{E_n\}_{n=1}^{\infty} = \{x \mid x \in E_N \text{ for infinitely many } n\}$$
$$\liminf\{E_n\}_{n=1}^{\infty} = \{x \mid x \in E_N \text{ for all but finitely many } n\}$$

2. (Demorgans La)

Naming the Numbers and Common Sets

With the basics of set down, we can introduce some of the most common sets and their notation

Definition 0.1.10: Common Sets

1. As we've already seen, if a set has no elements, the symbol $\emptyset$ represents the empty-set
2. If a set has a single element, we call that set a *singleton*. For example, $\{1\}$ and $\{\pi\}$ are singletons.
3. If X is a set, $P(X)$ is the set of all subsets of X[a]. In set builder notation, it can be denoted $P(X) = \{S \mid S \subseteq X\}$. For example, if $X = \{a, b, c\}$, then

$$P(X) = \{\emptyset, \{a\}, \{b\}, \{c\}, \{a, b\}, \{a, c\}, \{b, c\}, \{a, b, c\}\}$$

[a]The existence of this set is an axiom in ZFC

Definition 0.1.11: Sets Representing Numbers

1. $\mathbb{N} := \{0, 1, 2, 3, \cdots\}$ is the set of all *natural numbers* (some call it the *counting numbers* or *whole numbers*). Natural numbers are all positive numbers which do not have any decimal places. Sometimes, people define $\mathbb{N}$ to not have 0. We will always have $\mathbb{N}$ contain the number 0 (for reasons which will be clear much later in the book [a]). If we want only the [strictly] positive number [b], we will write $\mathbb{N}_{>0}$.
2. $\mathbb{Z} := \{\cdots - 3, -2, -1, 0, 1, 2, 3, \cdots\}$ is the set of all *integers*. Integers are the natural numbers along with their negative counter-part. Sometimes, the natural numbers are called the non-negative integers.
3. $\mathbb{Q} := \left\{ \frac{p}{q} \;\middle|\; p, q \in \mathbb{Z},\ q \neq 0 \right\}$ is the set of all *rational numbers*. For example, $\frac{1}{2} \in \mathbb{Q}$. The symbol $\mathbb{Q}$ is used since numbers of the form $\frac{p}{q}$ are called ***quotients***. Notice also that this is one of the instances where we use the convention that if there is a repeated element, we "squish" to a single element. If we did not take this convention, this set wouldn't be defined, since there would be many repeated numbers (for example $\frac{1}{2} = \frac{2}{4} = \frac{4}{8}$ and so forth)
4. $\mathbb{R}$ represents any number with any decimal expansion (ex. $\pi, e, 1, \frac{1}{3}$), i.e., all the numbers on the number line. If you've learned about Complex numbers or Quaternions in high school, these numbers are not real numbers, and will be defined and studied later in this book. The set $\mathbb{R}$ is called the *real numbers*. We can abuse notation (meaning use the notation in a way bends the rules a little) and write $\mathbb{R}$ as

$$\mathbb{R} := \{\cdots D_3 D_2 D_1 . d_1 d_2 d_3 \cdots \mid D_i, d_i \in \mathbb{Z}\}$$

where $\cdots D_3 D_2 D_1 . d_1 d_2 d_3 \cdots$ represents some arbitrary string of numbers put together where there can only be finitely many D_i's, but there can be infinitely many d_i's. In other word, the number

$$0.123123123123\cdots$$

where 123 continues on forever on the right is in the set $\mathbb{R}$, but

$$\cdots 123123123$$

where 123 continues on forever on the left is not in the set $\mathbb{R}$.

[a]See section 7.1
[b]There is no complete agreement on whether 0 is positive or negative: 0 is either both positive and negative or neither positive nor negative. It usually comes down to the purposes of the context. If we allow 0 to be a positive number, we sometimes say that all numbers larger than 0 are *strictly positive*

Some find it intuitive to think about $\mathbb{R}$ has having "no gaps", which means we cannot perfectly split $\mathbb{R}$ into two using the less than and greater than operator ($<$ and $>$) without missing a number. For an example of a split, we can split $\mathbb{N}$ into $\mathbb{N}_{\frac{5}{2}^+} = \left\{x \in \mathbb{N} \;\middle|\; x > \frac{5}{2}\right\}$ and $\mathbb{N}_{\frac{5}{2}^-} = \left\{x \in \mathbb{N} \;\middle|\; x < \frac{5}{2}\right\}$, and we have successfully split $\mathbb{N}$ into two without missing a number (i.e. every natural number is either in $\mathbb{N}_{\frac{5}{2}^+}$ or $\mathbb{N}_{\frac{5}{2}^-}$). If we chose to split around 5, then neither the set $\mathbb{N}_{5^+}$ nor the set $\mathbb{N}_{5^-}$ would contain 5, meaning it does not split $\mathbb{N}$. For the Real numbers, there does not exist a perfect split. You might ask if $\mathbb{Q}$ has this same property, but we will prove that we can in fact always split it: We will show $\mathbb{Q}_{\sqrt{2}^+} = \left\{x \in \mathbb{Q} \;\middle|\; x > \sqrt{2}\right\}$ and $\mathbb{Q}_{\sqrt{2}^-} = \left\{x \in \mathbb{Q} \;\middle|\; x < \sqrt{2}\right\}$ splits $\mathbb{Q}$ (see ref:HERE).
The fact that $\mathbb{R}$ cannot be split into two using $>$ and $<$ is actually at the heart of many formal definitions of $\mathbb{R}$ which is covered in an Analysis I class. In this book, we will not need to think of $\mathbb{R}$ in this fashion, and so we'll stick with just using intuition for now. [27]

`https://math.stackexchange.com/questions/2834876/convergence-of-geometric-series-with-r1`

0.1.2 Induction

Here. The whole $(p(x) \Rightarrow p(x+1)) \Rightarrow$ here this as the formal definition? + some intuition?

0.1.3 *Pigeon-hole Principle

(Comes up in algeraic number theory, so should add it here)

0.1.4 Relations and Equivalence Relations

What does it mean two things to be equal? There are actually many ways, depending on what you want to be "equal". For example,

1. we may say that two people have the same birthday, even if they are not born in the same year.
2. more examples her

In this section, we shall explore how to capture this more general notion of equality.

Relations

We start of with the more general notion of *relation*. Many times when working with sets, some subsets can be thought of as having certain properties for example, we can split the set $\mathbb{Z}$ into even

[27]Google *Dedikend Cut construction* or *Completion of Rational Numbers* if you're curious to see a formal definition of $\mathbb{R}$

and odd numbers, or we have some numbers being bigger than others, or that some numbers are multiples of another. The following is this idea is formalized:

Definition 0.1.12: [Binary] Relation

Let X be a set. Then any collection of pairs (x, y), $x, y \in X$ is called a [binary] *relation.* If $(a, b) \in R$, we will say that *x is related to y.* If $(a, b) \in R$, we will write aRb.

More generally, we can increase the number of times we take the product (i.e. $n > 2$) (ex. $X \times X \times \cdots X$). This would be called an *n-relation.*

We shall almost always think of relations

Example 0.1.13: Relations

1. Take $\mathbb{Z}$ and define the set $< \subseteq \mathbb{Z} \times \mathbb{Z}$ to be

$$(x, y) \in < \text{ if and only if } x \text{ is strictly less than } y$$

   Then $(3, 4) \in <$, $(3, 3) \notin <$, and $(4, 3) \notin <$. If $(a, b) \in <$, we would usually write:

$$a < b$$

   and if $(a, b) \notin <$, we would write

$$a \not< b$$

   A similar relation is the $\leq$ relation, where we also allow for two numbers to be equal

2. Take $\mathbb{Z}$ and define the set R_{div} to be:

$$(x, y) \in R_{\text{div}} \text{ if and only if } x \text{ divides } y$$

   Then $(2, 4) \in R_{\text{div}}$. What other elements are in this set? This set is usually denoted $|$ so that if a divides b then:

$$a|b$$

3. for a non-mathematical example, if we let X be the set of all humans, then we may define a relation "is the ancestor of". Another similar but different relation is "is the parent of" where two humans are related is one is the parent of another.

A common relation to put on objects give a sense ordering:

Definition 0.1.14: Order

Let R be a relation on X. Then R is a *partial order* if it is:

1. **reflective**: For all $x \in X$, xRx (x always relates to itself)
2. **antisymmetric**: if xRy and yRx, then $x = y$[a]
3. **Transitive**: if xRy and yRz, then xRz.

Furthermore, a partial order is a *total order* (or simply an *order*) if it is furthermore:

4. **Connected**: For each $x, y \in X$, either xRy or yRx

[a] a relation is symmetric if xRy then yRx. Hence, antisymmetry means no elements can be symmetric with another elements besides itself

The canonical total order is $\leq$ on numbers, and the canonical partial order is $\subseteq$ on sets.

Example 0.1.15: Orders

1. Let $\mathbb{N}^2 = \mathbb{N} \times \mathbb{N}$. Give some examples on $\mathbb{N}^2$ of different ordering.
2. Let D be the set of words in a dictionary dictionary. Then D can be lexicographically ordered
3. All numbers introduced so far have an order defined: $(\mathbb{N}, \leq), (\mathbb{Z}, \leq), (\mathbb{Q}, \leq), (\mathbb{R}, \leq)$.

Equivalence Relations

The most common type of relation that we shall see is the equivalence relation which generalizes the notion of equality:

Definition 0.1.16: Equivalence Relation

Let S be a set and $x, y, z \in S$ be elements of S. An *equivalence relation* on a set S is relation $\sim$ satisfying

1. **(reflective)** $x \sim x$ (i.e. every element must be related to itself)
2. **(symmetric)** if $x \sim y$ then $y \sim x$
3. **(Transitive)** if $x \sim y$ and $y \sim z$ then $x \sim z$

Example 0.1.17: Equivalence Relations

1. Equality $(=)$ is the prototypical example of an equivalence relation. For example, If we have $a, b, c \in S$ with $aRb \Leftrightarrow a = b$, then
   (a) $a = a$
   (b) $a = b \Rightarrow b = a$

(c) If $a = b$ and $b = c$, then $a = c$.

In a ways, an equivalence relation is a generalization of equality.

2. Let X be any non-empty set. Then we may make all elements equivalent to each other. Similarly, we may make all elements only equivalent to themselves.

3. Let P be the set of all people. Check that "has the same birthday" is an equivalence relation

4. Two equivalence relations from high school that the reader may have encountered is with triangles. Check that being *similar to* and being *congruent to* are equivalence relations.

5. Define $\sim$ on $\mathbb{Z}$ to be $a \sim b$ if and only if $|a| = |b|$. Then $\sim$ is an equivalence relation.

6. Is $\leq$ an equivalence relation?

7. Let R be the empty relationship on any non-empty set. Is R an equivalence relation?

Given an equivalence relation $\sim$ on S, we may split S into subsets defined as:

$$(S/\sim) = \{T \subseteq S \mid x, y \in T \iff x \sim y\}$$

For example, if $S = \{1, 2, 3, 4, 5, 6, 7, 8, 9\}$, and:

$$1 \sim 2 \qquad 3 \sim 4 \sim 5 \sim 6 \qquad 7 \sim 8 \sim 9$$

Then:

$$S/\sim = \{\{1, 2\}, \{3, 4, 5, 6\}, \{7, 8, 9\}\}$$

In fact, we may define an equivalence relation on a set S by breaking it up into disjoint sets who union of elements would be S. This is called a partition:

Definition 0.1.18: Partition

Let S be a set. Then a *partition* of S is a collection of subsets of S such that

1. for each subset S_i, S_j, $S_i \cap S_J = \emptyset$

2. $\cup_i S_i = S$

each partition is called an *equivalence class*.

0.1.5 Functions

In mathematics, it is very desirable to relate one set to another set, or said differently how two sets may "interact". For example, it may be desirable that to each positive natural number $\mathbb{N}_{>0}$, we associate to it the number of prime factors it has, for example:

$$\begin{aligned} 1 &\mapsto 1 \\ 2 &\mapsto 1 \\ 3 &\mapsto 1 \\ 4 &\mapsto 2 \\ 5 &\mapsto 1 \\ 6 &\mapsto 2 \\ &\vdots \\ 16 &\mapsto 4 \\ &\vdots \end{aligned}$$

Another more abstract relation we may ask for is given a set of names, say

$$\{\text{Alfred}, \text{Socrates}, \text{Buddha}, \text{Oscar Petesron}\}$$

the function returns a set containing some information about that person:

$$\begin{aligned} \text{Alfred} &\mapsto \{\text{Batman's Butler}, \text{Friend of Bruce}\} \\ \text{Socrates} &\mapsto \{\text{Philosopher}, \text{lived in Athens}, \text{Never wrote his work down}\} \\ \text{Buddha} &\mapsto \{\text{Real name Siddhartha Gautama}, \text{Starter of a Religion}, \text{Born in Lumbini}\} \\ \text{Oscar Peterson} &\mapsto \{\text{Greatest Jazz Pianist}, \text{Canadian}, \text{``Maharaja of the keyboard''}, \text{Lived to 82}\} \end{aligned}$$

As a final example, we may imagine a every location of a map of the world being represented as an element in a set. Then we may relate a point on the map to the temperature at that location. Generalizing this pattern, given some set X, we may want to assign it an element from another set Y.

Definition 0.1.19: [set] Functions

Let X and Y be sets. Then a *function* is a subset $\Gamma(f) \subseteq \{(x,y) \mid x \in X,\ y \in Y\}$ such that

1. **Total** or **Serial** (left-total): For every $x \in X$, there exists an element $(x,y) \in \Gamma(f)$
2. **Functional** (right-unique): Every $x \in X$ is associated with one and only one $y \in Y$, that is, it cannot be that $(x,y),(x,y') \in \Gamma$, for $y \neq y'$ where $y, y' \in Y$

This is usually written as:

$$f : X \to Y \qquad f(x) = y \qquad \text{or} \qquad X \xrightarrow{f} Y$$

where f is called the *function* and $\Gamma(f)$ is called the graph of the function and is defined to be:

$$\Gamma(f) = \{(x,y) \mid x \in X,\ y \in Y,\ f(x) = y\}$$

If $f : X \to Y$, X is called the *domain* and Y is called the *codomain*. The collection of all functions from two sets X, Y is denoted:

$$\mathrm{Hom}_{\mathbf{Set}}(X,Y) \qquad \text{or} \qquad Y^X$$ [a]

[a] Notice this is a set operation. This particular notation shall be elaborated on with more category theory in section ref:HERE

Function are also sometime called maps or mappings. Sometimes, we want to define a function, but don't need to reference to it again (in computer science, this is called an *anonymous function*). In this case, we may use the following notation:

$$x \mapsto y$$

instead of $f(x) = y$. If $X = Y$ so that $f : X \to X$, then we may think of f as a relation which has the property of being functional and total. If more than one function with the same domain and codomain is defined (say f, g with domain X and codomain Y), then the declaration is often shorthanded to $f, g : X \to Y$.

Example 0.1.20: Functions

1. The first three examples given are examples of functions, since every element in the domain get's mapped to a unique element in the codomain.
2. Some other perhaps more familiar functions to the reader are
   (a) $x \mapsto e^x$, i.e. $f : \mathbb{R} \to \mathbb{R},\ f(x) = e^x$
   (b) $\log : \mathbb{R}_{>0} \to \mathbb{R}$, $x \mapsto \log(x)$
   (c) $p(x) : \mathbb{R} \to \mathbb{R}$, $p(x) = ax^2 + bx + c$, for $a, b, c \in \mathbb{R}$
3. Let X be an arbitrary non-empty set. Then there always exists at least one function from X to X, namely:
   $$\mathrm{id}_X : X \to X \qquad \mathrm{id}_X(x) = x$$
   that is, id_X maps every element to itself.

Definition 0.1.21: Image and Restriction of Function

1. Let $S \subseteq X$ and $f : X \to Y$. Then:

$$f(S) := \{f(x) \mid x \in S\}$$

this is called the *image* of S. If $S = X$, then $f(X)$ is called the *image* or *range* of the function f, and is commonly denoted $\operatorname{im}(f)$.

2. Let $S \subseteq X$. Then $f|_S$ is a function $f|_S : S \to Y$ such that

$$f|_S(x) = f(x)$$

that is, $f|_S$ is the *restriciton* of f by S.

As a function is just some "syntactic sugar" to more easily represent the set Γ_f, two functions f, g are equal if $\Gamma_f = \Gamma_g$. In terms of function notation this means that f, g are equal if for all elements x in the domain:

$$f(x) = g(x)$$

An advantage of the left-total and right-unique property is that it is easy to compose functions together, that is it is easy to chain together the output of one function to be the input of the next:

Proposition 0.1.22: Composition of Functions

Let $f : X \to Y$, $g : Y \to Z$, $h : Z \to W$ be function. Define $(g \circ f)$ to be $(g \circ f)(x) = g(f(x))$. Then $(g \circ f)$ is a function and is called *g compose f*. The process is called *function composition*.

***Proof*:**
Make sure that $g \circ f$ satisfies the properties of functions

Example 0.1.23: Function Composition

1. Let $f : \mathbb{R} \to \mathbb{R}$, $f(x) = x^2$ and $g : \mathbb{R} \to \mathbb{R}^2$ be $g(x) = (x, 0)$. Then:

$$(g \circ f)(x) = (x^2, 0)$$

A function may also be composed with itself, given the domain and codomain match:

$$(f \circ f)(x) = x^4$$

2. Let $f : X \to Y$ be any function, $\operatorname{id}_X, \operatorname{id}_Y$ the identity functions defined in example 0.1.20. Then:

$$f \circ \operatorname{id}_X = f \qquad \operatorname{id}_Y \circ f = f$$

This chaining together is in a way "state-independent": given three functions f, g, h, the function $(g \circ f)$ following by composing with h is equal to if we first just do f, then follow by taking the combined state $(h \circ g)$[28]:

Proposition 0.1.24: Associativity of Functions

Let $f : X \to Y$, $g : Y \to Z$, $h : Z \to W$ be function. Then:

$$h \circ (g \circ f) = (h \circ g) \circ f$$

Pictorially, function composition may be represented by the following diagrams:

$$\begin{tikzcd} A \arrow[r, "f"] \arrow[rr, bend left, "g\circ f"] & B \arrow[r, "g"] & C \end{tikzcd} \quad \text{or} \quad \begin{tikzcd} A \arrow[r, "f"] \arrow[dr, "g\circ f"'] & B \arrow[d, "g"] \\ & C \end{tikzcd} \quad \text{or} \quad \begin{tikzcd} & C \\ A \arrow[r, "f"] \arrow[ur, "g\circ f"] & B \arrow[u, "g"'] \end{tikzcd}$$

and associativity may be drawn as:

$$\begin{tikzcd} A \arrow[r, "f"] \arrow[rr, bend right, "g\circ f"'] & B \arrow[r, "g"] \arrow[rr, bend left, "h\circ g"] & C \arrow[r, "h"] & D \end{tikzcd}$$

Proof:
This is a long and tedious proof that the reader should attempt at least once in their life.

These diagrams have the following special property: start on one of sets in the diagram (say A) and travel down the arrows to reach another set (say to D in the last diagram). Now start again at A and travel down to D via a different path in the diagram. Then notice that the compositions of functions agree with each other in the two paths. For example, take the following path:

$$\begin{tikzcd} A \arrow[r, "f"] \arrow[rr, bend right, "g\circ f"'] & B \arrow[r, "g"] \arrow[rr, bend left, "h\circ g"] & C \arrow[r, "h"] & D \end{tikzcd} \quad \text{and} \quad \begin{tikzcd} A \arrow[r, "f"] \arrow[rr, bend right, "g\circ f"'] & B \arrow[r, "g"] \arrow[rr, bend left, "h\circ g"] & C \arrow[r, "h"] & D \end{tikzcd}$$

Then the 1st red line is the composition $(h \circ g) \circ f$ and the 2nd red line is the composition $h \circ (g \circ f)$. Then by associativity, these two functions are equal

$$(h \circ g) \circ f = h \circ (g \circ f)$$

a diagram which satisfies this property for all paths is called a *commutative diagram*. These shall show up everywhere in this book, and mathematics more generally. They shall be further explored in section 0.4.

Injective, Surjective, and Bijective

There are two special types of functions that deserve to be pointed out: *injective* functions and *surjeciton* functions. These shall each represent some partial "reverse" of a function. Starting off

[28]can you think of an example where this is not the case in your everyday life? This would greatly help in understanding associativity

with injectivity:

Definition 0.1.25: Injective

Let $f : X \to Y$ be a function. Then f is called *injective*, an *injection*, or *one-to-one* if the following condition holds: for all $a, b \in X$

$$f(a) = f(b) \;\Rightarrow\; a = b$$

To emphasize a function is injective, a hooked right arrow ($\hookrightarrow$) may be used like so:

$$f : X \hookrightarrow Y \qquad X \overset{f}{\hookrightarrow} Y \qquad X \hookrightarrow Y$$

Intuitively, being injective means we can embed the domain X directly into Y. As a consequence of this, we can always "recover" X in the following way

Proposition 0.1.26: Injective And Left Inverse

let f be an injective function. Then there exists a function $\ell f(X) \to X$ such that

$$\ell \circ f = \mathrm{id}$$

that is $(\ell \circ f)(x) = \mathrm{id}(x) = x$ is the identity function.

***Proof*:**
Define $\ell : Y \to X$, $\ell(y) = \ell(f(x)) = x$. Since f is injective, this is a well-defined function (if it were not injective, then $\ell(f(x))$ would be ambiguous). Then by definition $(\ell \circ f)(x) = \ell(f(x)) = x$, as we sought to show.

Notice that the inverse function ℓ inverted the function *after* applying f. What if there is some function $r : Y \to X$ such that f reverses it? The following shows when this is the case:

Definition 0.1.27: Surjective

Let $f : X \to Y$ be a function. Then f is *surjective*, a *surjection*, or *onto* if for all $y \in Y$, there exists an x such that $f(x) = y$, i.e. $f(X) = Y$. To emphasize a function is surjective, a double-headed right arrow ($\twoheadrightarrow$) may be used like so:

$$f : X \twoheadrightarrow Y \qquad X \overset{f}{\twoheadrightarrow} Y \qquad X \twoheadrightarrow Y$$

Intuitively, being surjective means that Y is "represented" by the elements of X, i.e., every element y has one or many elements of x associated to it, which is why another word for a function to be surjective is that it is "onto". As a consequence of this, we can pullback any element $y \in Y$ to some representative $x \in X$ such that $f(x) = y$. We can capture this intuition in the following way:

Proposition 0.1.28: Surjective And Right Inverse

Let $f : X \to Y$ be a surjective function. Then there exists a (and in fact more generally many) $r : Y \to X$ such that

$$r \circ f = \text{id}$$

that is $(f \circ r)(x) = \text{id}(x) = x$ is the identity function.

***Proof*:**
Define $r : Y \to X$ by mapping each $y \in Y$ to some element $x \in X$ such that $f(x) = y$ (such an element exists since f is surjective). Then by definition for any $y \in Y$, $(f \circ r)(y) = f(r(y)) = f(x) = y$ as we sought to show.

Another perhaps more intuitive way of seeing how surjections make every y "represented" by X is by seeing that every surjection defines an equivalence class:

Proposition 0.1.29: Surjective And Equivalence Class

Let $f : X \to Y$ be a surjective function. Then we can define an equivalence class $\sim_f$ on X by defining:

$$a \sim_f b \qquad \Leftrightarrow \qquad f(a) = f(b)$$

If $[x] \in X/\sim_f$ is an equivalence class, such that $f(x) = y$, then $[x]$ is called the *fiber* over y.

***Proof*:**

1. For each $a \in X$, $f(a) = f(a)$ by right-uniqueness, hence $a \sim_f a$
2. If $a \sim_f b$ for $a, b \in X$, then $f(a) = f(b)$. By symmetry of the equivalence relation $=$, $f(b) = f(a)$, and so $b \sim_f a$
3. Can the reader show that if $a \sim_f b$ and $b \sim_f c$, then $a \sim_f c$?

If f is not surjective, this equivalence relation may still be defined on the image of f. If f is injective, then each equivalence class has size 1. Many functions that are present in mathematics are both injective and surjective; to give a simple example:

$$f : \{1, 2, 3\} \to \{a, b, c\}, \qquad f(n) = \begin{cases} a & n = 1 \\ b & n = 2 \\ c & n = 3 \end{cases} \tag{2}$$

Sch functions are given a name:

Definition 0.1.30: Bijective

Let $f : X \to Y$ be a function. Then f is said to be *bijective* or *isomorphic in* **Set** if f is injective and surjective. If X and Y are bijective, then we may write:

$$X \cong_{\mathbf{Set}} Y \qquad X \cong Y \qquad X \xrightarrow{\sim} Y$$

The intuition of bijective functions can combine the intuition behind injective and surjective functions, however they may equivalently be seen as formally defining a notion of "size". As seen in equation 2, two sets that are bijective have the same size. The issue becomes more subtle when the domain/codomain are infinite:

Example 0.1.31: Size Of Infinite Sets

Define $2\mathbb{N} = \{2n \mid n \in \mathbb{N}\}$ and $f : \mathbb{N} \to 2\mathbb{N}$ by:

$$f(n) = 2n$$

Show that f is bijective.

This gives rise to the following definition:

Definition 0.1.32: Cardinality

Let X, Y be sets. Then X is said to have the same *cardinality* as Y if there exists a bijective function $f : X \to Y$. If X, Y have the same cardinality, they are said to have the same size.

Another important property of bijective functions is that they give rise to a *unique* inverse:

Proposition 0.1.33: Bijectivity And Unique Inverse

Let $f : X \to Y$ be a bijective function. Then there exists a unique function g such that

$$g \circ f = \mathrm{id}_X \qquad f \circ g = \mathrm{id}_Y$$

where $\mathrm{id}_X : X \to X$ and $\mathrm{id}_Y : Y \to Y$ are the identity maps on the respective sets. Often, the subscript on the identity maps are dropped and the following abuse of notation is carried out:

$$g \circ f = f \circ g = \mathrm{id}$$

The function g is often denoted f^{-1}

***Proof*:**

Since f is surjective, there exists an inverse function $g : Y \to X$ such that $f \circ g = \mathrm{id}_Y$. Since f is injective, there can only be one such function (there is only element in the fiber above each $y \in Y$). Furthermore, $\mathrm{im}(f) = Y$, hence g satisfies the condition $g \circ f = \mathrm{id}_X$, as we sought to show.

0.1.34 Remark: inverse notation abuse The symbol f^{-1} has one other of notations that the reader should be aware of, in particular this symbol may represent one of two things depending on the context. Given $f : X \to Y$:

1. if f is bijective, f^{-1} represents the unique inverse function
2. $f^{-1} : P(Y) \to P(X)$ is the function

$$f^{-1}(T) = S \in P(X) \qquad S = \{x \in X \mid f(x) = y\}$$

the images of f^{-1} are called the *pre-images* of f. This notation is particularly prevalent in topology where this the pre-image function takes center stage through continuous functions, however it will also have many applications in algebra.

Given any function $f : X \to Y$, there are certain natural functions that we may define that allows us to decompose f as a composition of injective and surjective functions with an intermediary bijective function

Proposition 0.1.35: Function Decomposition

Let $f : X \to Y$ be a function. Then there exists functions such that

$$\begin{tikzcd} X \arrow[r, two heads] \arrow[rrr, bend left, "f"] & X/\sim_f \arrow[r, "\sim"] & \mathrm{im}(f) \arrow[r, hook] & Y \end{tikzcd}$$

where the 1st function is surjective, the 2nd bijective, and the 3rd injective

***Proof*:**
Define $\pi_f : X \to X/\sim_f$ by

$$\pi_f(x) = [x]_f$$

that is, π_f maps x to it's equivalence class. Next, define $F : X/\sim_f \to \mathrm{im}(f)$ by

$$F([x]) = y$$

where $y \in Y$ is chosen so that $f(x) = y$. Finally, define $\iota : \mathrm{im}(f) \to Y$ by

$$\iota(y) = y$$

that is, ι maps y to itself (since $\mathrm{im}(f) \subseteq Y$). The reader should verify that these maps have the desired properties.

This proposition gives an intriguing relation between equivalence relations and functions, namely any function may be "represented" as an equivalence relation, and equivalently any equivalence relation my be "represented" by a function.

Using functions, we may define the notion of a multiset:

Definition 0.1.36: Multiset

Let X be a set. Then a *multiset* given by elements of X is a function:

$$m : X \to \mathbb{N}$$

which assigns each element $x \in X$ a multiplicity.

Example 0.1.37: Multiset

Let $X = \{a, b, c\}$ and:

$$m(a) = 3 \qquad m(b) = 1 \qquad m(c) = 0$$

Then m defines a multiset. Commonly, this set is written as:

$$\{a, a, a, b\}$$

As practice, define another multiset n and show how to define the notion of inclusion, union, intersection, and so forth.

As a closing remark, it is a common convention in the study of Analysis (one of the major studies of mathematics) that if we have a function $f : X \to Y$, we might abuse notation and use a subset of X as the actual domain. For example, $f : \mathbb{R} \to \mathbb{R}$, $f(x) = \frac{1}{x}$ is not a function as we've defined the concept, since $f(0) = \frac{1}{0}$ is not defined [29]. If we were to re-write this to be a function, we'd write $f : \mathbb{R} - \{0\} \to \mathbb{R}$. However, it is often implied in analysis that we will restrict the domain to the points for which f is defined.

Exercise 0.1.2

1. Let $f : X \to Y$ and take $A \subseteq X$, $B \subseteq Y$. Show that $f^{-1}(A) \subseteq B \Leftrightarrow A \subseteq f(B)$
2. Let $f : X \to Y$. Then is it true that $f(A \cup B) = f(A) \cup f(B)$? What about $f(A \cap B) = f(A) \cap f(B)$? What if there were more unions or intersections?
3. Let $f : X \to Y$, and define $f^{-1} : P(Y) \to P(X), f^{-1}(Y) = \{x \in X \mid f(x) \in Y\}$. Is $f^{-1}(A \cup B) = f^{-1}(A) \cup f^{-1}(B)$? What about $f^{-1}(A \cap B) = f^{-1}(A \cap B)$? What if there were more unions or intersections?

   (In a way, equivalence relations capture some idea of "projecting". If you have some set S an you want to put together all equivalent objects, where equivalence here can be thought as "similar enough so that we can define the concept of equality"
4. (partitioning property. Like creating the "new objects" where we ignore the "details" that is "consumed" by the equivalence relation)
5. (natural projection?? Give $\pi : G \to \{0\}$ a natural projection that always exists?)
6. (formalities behind natural numbers). Take:

$$S = \{\{\},\ \{\{\}\},\ \{\{\{\}\}\}, \ldots\}$$

[29] This would be covered rigorously in a calculus or analysis class

to be the set containing the empty set, the set containing the empty set, and so forth. For convenience, label these elements:

$$S = \{0, 1, 2, 3, ...\}$$

Define $+ : S \times S \to S$ to be the function that takes in the tuple (n, m) of n-nested empty sets and m nested empty and returns the $n+m$-nested empty sets. In other words $+(n, m) = n+m$. Let $S^* = \{1, 2, 3, ...\}$ (i.e. exclude $\{\} = 0$). Define $*(1, 1) = 1$ and $*(n, m) = nm$ (i.e. for the n-nested empty set, repeat the nesting m-times). Show these create the usual natural numbers.

0.1.6 Adding Structures to Sets

So far, we have been extensively working with sets and functions. Sets by themselves have no notion of size, order, arithmetic operations, geometry, among others possible properties. Many interesting subjects in mathematics deal with these additional concepts. For this reason, mathematicians often "augment", so to speak, sets to have some extra structure:

Example 0.1.38: Extra Structure

1. (Order Structure). A set $(A, \leq)$ with an order $\leq$ is called an *ordered set*. Depending on the type of order given (ex. partial oder, total order, well order etc.), the set would be called the appropraited order (ex. paritally ordere set, totally ordered set, well ordered set, etc.)

2. (Algebraic structure) The set $\mathbb{Z}$ on its own is only numbers (and so is $\mathbb{N}, \mathbb{Q}, \mathbb{R}$). If we add a function $+ : \mathbb{Z} \times \mathbb{Z} \to \mathbb{Z}$, where $+$ acts exactly like the familiar addition we have learnt earlier in life, then $(\mathbb{Z}, +)$ now also has addition! We can add more functions to sets, for example $(\mathbb{Z}, +, \cdot)$ where $\cdot : \mathbb{Z} \times \mathbb{Z} \to \mathbb{Z}$ is the usual multiplication. If we augment a set with function(s), we call this an *algebraic structure*. This whole book will be dedicated to the study of algebraic structures.

3. (Topological Structures). Another common augmentation of a set is something called a topology. Given a set A, a set $\mathcal{T}_A$ is a set of subsets of A that follows three rules:

   (a) $\emptyset \in \mathcal{T}_A$ and $A \in \mathcal{T}_A$

   (b) if $\{U_i\}_{i \in I} \subseteq \mathcal{T}_A$ is some arbitrary subset of $\mathcal{T}_A$ indexed by I, then $\bigcup_{i \in I} U_i \in \mathcal{T}_A$ (i.e. an arbitrary union of sets in $\mathcal{T}_A$ is in $\mathcal{T}_A$)

   (c) if $U_1, U_2, ..., U_n \in \mathcal{T}_A$ is a finite subset of $\mathcal{T}_A$, then $\bigcap_i U_i \in \mathcal{T}_A$ (i.e. a finite intersections of sets in $\mathcal{T}_A$ is in $\mathcal{T}_A$)

   Then $(A, \mathcal{T}_A)$ is called a *topology* or sometimes a *space*. If a set is augmented with a sets of sets that follow these rules, it's called a *topological structure*. Topology is the natural space in which much of Real and Complex Analysis, as well as [non-classical] geometry takes place. It is the "coarsest" structure we add to sets to give sets some notion of locality.

4. (Measure Structures). Similar to a topological structure, a measure structure is one where

   (a) $\emptyset \in \mathcal{T}_A$ and $A \in \mathcal{T}_A$

   (b) if $U_1, U_2, ..., U_n, ... \in \mathcal{T}_A$, then $\bigcup_i U_i \in \mathcal{T}_A$ (i.e. an countable union of sets in $\mathcal{T}_A$ is in $\mathcal{T}_A$)

(c) if $U_1, U_2, ..., U_n, ... \in \mathcal{T}_A$, then $\bigcap_i U_i \in \mathcal{T}_A$ (i.e. a countable intersections of sets in $\mathcal{T}_A$ is in $\mathcal{T}_A$)

This structure is central to generalizing the notion of "size" for general sets and is the basis for defining the integral[a].

5. (Differential structure) If you have done calculus, then you are familiar with the notion of being differentiable. There is a field of study in which we can call "sets" differentiable. Given a set S, there is a collection of functions called an *atlas* (say, $\mathcal{A}$ represents this collection) which gives a set S a "differentiable structure", and is usually denoted $(S, \mathcal{A})$. The details of what the atlas $\mathcal{A}$ contains is left for a course in differential geometry or diffeology.

6. *(Affine Scheme structure) This will be a very important structure on commutative ring that will dominate the readers who will study algebraic geometry. Any commutative ring will induce a natural topological space. A topological space represents some idea of locality on a set; a local piece is called an *open set.* A *sheaf* represents having data on each open set in a topological space in such a way that we can use local data to glue together global data (think of solving a problem by breaking it down into smaller problems and knowing you can glue it back together).

7. *(Model) In section 0.1.1, we have shown how to make logical statements given sets. The full study how sets (usually augmented with other structures) may have statements which are true or provable is the domain of *model theory.* Usually, a set M along with some collection of functions and relations is augmented in such a way to be able to study the properties of logical statements. This would be covered in a class on first order logic or model theory more generally.

[a] In particular, the Lebesgue integral

This textbook is dedicated to the study of algebraic structures. It will sometimes be the case that we encounter ordered structures [30], and only near the end of the book do we encounter topological structures (p. 1008) where it is assumed the reader has taken a course covering point-set topology[31]. The reason for the eventual mixing of structure is because different domains of mathematics start motivating each other in further exploration. For example, many geometers and algebraists found parallels between their fields, and hence algebraic geometry was born. Similarly, it was found that algebra can be used to solve topological problems, and hence algebraic topology was founded.

0.1.7 *Cardinality

(Here in there in he book, cardinality will come up. For ex, in proof that an injective map from $X \to X$ where X is finite is surjective, or when we figure out the algebraic numbers are countable, and other results later in the book)

(classical diagonalization proof?)

(as exercise, do $\{0,1\}^{\mathbb{N}}$ is bijective to $[0,1]$ by thinking of binary representation?)

[30] see section 2.4

[31] Differentiable structure's will be mentioned to motivate the concept of schemes which are based off of manifolds

0.2 Number Theory

Numbers are at the heart of mathematics and algebra. In this section, the basic properties of numbers will be explored. Some history of these concepts will be given as well to give a sense of why we call these special sets numbers.

Numbers started with our intuitive ability to understand discrete quantities (ex. 5 rocks, 2 people). In the past, it was contentious whether 0 was a number (the idea being that numbers are supposed to represent a real quantity), however it is now accepted that 0 represents the notion of "nothingness", "emptyness", or "inaction". Ratios of numbers (e.x 3/4) became important when it was important to be precise on what part of a whole object we want (ex. half the money). Negative numbers became important when considering other peoples debts (you owe 2 sheep, you "have" -2 sheep). In the times of at least the Greeks, it was discovered that there are certain numbers that are irrational (ex. $\sqrt{2}$ cannot be written as $\frac{a}{b}$, where a, b are natural numbers); the square root of any prime number shown to be irrational. The square-root of negative numbers remained undefined for awhile until NAME discovered that they are useful in finding roots of polynomials of degree 3, and it wans't till Euler that square-roots of negative numbers got the symbol $i := \sqrt{-1}$. It wasn't till later that it was understood there were many more irrational numbers than rational numbers; the collection of all rational and irrational numbers form the real numbers. The collection of all numbers $a + bi$, $a, b \in \mathbb{R}$ is known as the *complex numbers*. For reasons that shall be explored later in the book[32], any further addition to the number system beyond the complex numbers start to loose properties of numbers (ex. $ab \neq ba$ or $(ab)c \neq a(bc)$), and so we usually consider the following sets as "numbers" :

$$\mathbb{N} \subseteq \mathbb{Z} \subseteq \mathbb{Q} \subseteq \mathbb{R} \subseteq \mathbb{C}$$

We shall heavily explore the properties of all of these number systems, with the notable exception of $\mathbb{R}$. This is because the real numbers are in many ways the beginning of the exploration of "taming infinities" and start moving beyond algebra. In fact, we shall even show in section E.1 that it is impossible to construct $\mathbb{R}$ in an algebraic way.

0.2.1 GCD and Euclidean Algoritm

Definition 0.2.1: Divisibility

Let a, b be numbers. Then we say that a *divides* b, or b is *divisible by* a if there exists a number k such that

$$ak = b$$

Note that divisibility depends on the set of numbers you are working, In $\mathbb{N}$, 3 is not divisible by 5, however in $\mathbb{Q}$, 3 is divisible by 5, namely let $k = 3/5$.

Example 0.2.2: Divisibility Practice

1. Show that if $x \big| n_1$ and $x \big| n_2$, then $x \big| an_1 + bn_2$ for any $a, b \in \mathbb{Z}$.
2. Show that if $a \big| b$ and $b \big| c$ then $a \big| c$

[32] ref:HERE for details

From now on in this section, we shall limit ourselves to divisibility in $\mathbb{Z}$[33].

One special result about integers is that every number has a prime factorizaiton:

Proposition 0.2.3: Prime Factorization Of Natural Numbers

Let $n \in \mathbb{N}$. Then there exists distinct prime numbers $p_1, p_2, ..., p_k$ and natural numbers $n_1, n_2, ..., n_k$ such that n can be uniquely broken down into:

$$n = p_1^{n_1} p_2^{n_2} \cdots p_k^{n_k}$$

***Proof*:**
exercise (hint: use induction)

This definition can be extended to $\mathbb{Z}$, except we loose uniqueness, the decomposition is unique up to a multiple of -1. This isn't too much of a concern, since -1 has the special property of having a multiplicative inverse, that is we may multiply -1 by a number (in this case, itself) to get to 1:

$$-1 \cdot -1 = 1$$

no other integer (besides trivially 1) can be multiplied by another integer to get to 1. Since all numbers can be broken don into multiples of primes, we can define a ay to compare ho similar to numbers are:

Definition 0.2.4: Greatest Common Divisor

Let $a, b \in \mathbb{Z}$ be to non-zero integers. Then the *greatest common divisor* between these to numbers is the greatest number n such that n divides a and n divides b.
The common notation for this fact is that:

$$n \big| a \qquad n \big| b$$

Similarly, we have the dual concept:

Definition 0.2.5: Least Common Multiple

Let $a, b \in \mathbb{Z}$ be non-zero integers. Then the *least common multiple* is the smallest integer m such that

$$a \big| m \qquad b \big| m$$

Example 0.2.6: Greatest Common Divisor

1. $\gcd(10, 5) = 5$
2. $\gcd(100, 90) = 10$
3. $\gcd(45, 49) = 1$
4. If p, q are distinct prime numbers, then $\gcd(p, q) = 0$

[33] In chapter 10, we shall re-define divisibility in a more general context

It is natural that two prime numbers have 1 as their greatest common divisor for they have no factor in common. However, it is not necessary for numbers to be prime for their gcd to be 1. We give a name to numbers with no common factors:

Definition 0.2.7: Coprime Numbers

Let $a, b \in \mathbb{N}_{>0}$. Then if $\gcd(a, b) = 1$, a and b are said to be *coprime*.

We may find the greatest common multiple by finding the prime factorization, however this is in general very inefficient on large numbers. These values can be found in an algorithmic manner relying on the fact that numbers have a notion of size[34]

Theorem 0.2.8: Euclidean Algorithm

Given two natural numbers a, b we can find their greatest common divisor by the following technic:

1. Let $a > b$. Put them into the equation $a = bk_1 + r_1$, where k is the amount of times b fits into a, and r_1 is the remainder.
2. Once you find these values, all but k shift over to the left, and you repeat the process to find r_2:

$$\begin{aligned} b &= r_1 k_2 + r_2 \\ r_1 &= r_2 k_3 + r_3 \\ r_2 &= r_3 k_4 + r_4 \end{aligned}$$

3. The algorithm continues until $r_n = 0$ (which it will reach since a, b where finite). The final non-zero remainder r_{n-1} is the greatest common divisor.

***Proof*:**
exercise

As an example:

Example 0.2.9: Euclidean Algorithm
Let $a = 20$, $b = 13$. Then:

$$\begin{aligned} 20 &= 1 \cdot 13 + 7 \\ 13 &= 1 \cdot 7 + 6 \\ 7 &= 1 \cdot 6 + 1 \\ 6 &= 6 \cdot 6 + 0 \end{aligned}$$

so $\gcd(20, 13) = 1$, (the number above 0 in the last step). To find the $ax + by = 1$ formula, we

[34]We shall see in chapter 10 that not all systems in which we can define a GCD will allow for a euclidean algorithm

reverse the process by converting the equation above in the following form:

$$7 - 6 = 1$$

so, put the remainder on the right hand side. Then, isolate the next equation in in the appropriate ways to substitute into this one: $7 = 13 - 6$, $6 = 13 - 7$. Since $6 = 13 - 7$, $7 = 13 - (13 - 7) = 7$, so

$$7 - (13 - 7) = 1$$

then, we repeat the same process with the step above: $7 = 20 - 13$, so:

$$20 - 13 - (13 - (20 - 13)) = 1$$

implying, we get

$$2 \cdot 20 - 3 \cdot 13 = 1 \qquad (40 - 39 = 1)$$

giving us our desired equation.

Using the euclidean algorithms, we may always find the solution to the following interesting linear equation:

Theorem 0.2.10: Extended Euclidean Algorithm (Bézout's Identity)

Let $a, b \in \mathbb{Z}$. Then there exists integers $x, y \in \mathbb{Z}$ such that

$$ax + by = \gcd(a, b)$$

***Proof*:**
exercise (reverse the process)

Example 0.2.11: Bézout's Identity
Recall that $\gcd(20, 13) = 1$. To find the $ax + by = 1$ formula, we reverse the process by converting the equation above in the following form:

$$7 - 6 = 1$$

so, put the remainder on the right hand side. Then, isolate the next equation in in the appropriate ways to substitute into this one: $7 = 13 - 6$, $6 = 13 - 7$. Since $6 = 13 - 7$, $7 = 13 - (13 - 7) = 7$, so

$$7 - (13 - 7) = 1$$

then, we repeat the same process with the step above: $7 = 20 - 13$, so:

$$20 - 13 - (13 - (20 - 13)) = 1$$

implying, we get

$$2 \cdot 20 - 3 \cdot 13 = 1 \qquad (40 - 39 = 1)$$

giving us our desired equation.

Try doing the same with 220 and 38.

Bézout's Identity has the following neat interpretation: primes are fundamentally a multiplicative decomposition of numbers: any n is equal to $p_1^{n_1} \cdots p_k^{n_k}$ for appropriate primes p_i and powers n_i. However, they do have an additive relation between them, the Bézout identity. If we have two primes, we can (up to appropriate constants) add them to get to the multiplicative identity 1. In this way, we have a property of primes given via addition.

(should I add other number theory identities? Are there any other useful ones for this book? Perhaps just as exercises? They might be useful way later in algebraic number theory)

Exercise 0.2.1

1. Let $p, q, r \in \mathbb{N}$ be prime numbers greater than 3. Show that 3 divides $p^2 + q^2 + r^2$

0.2.2 Modular Arithmetic's

When counting the hours on a clock-face, we start with 12 o-clock, then add an hour to make it 1 o-clock, then another hour to make it 2 o-clock, until we circle back to 12 o-clock again, at which point we start counting again to 1 o-clock. For the rest of this section, we will say that 12 o-clock is 0 o-clock for reasons that would be explained soon. Notice how the arithmetic's of the clock different from the usual number system where for each $x \in \mathbb{N}$, there exists an n such that $n > x$. Hours on a clock could add to be zero, for example:

$$7 + 5 = 0$$

In this section, we rigorously define this notion

Definition 0.2.12: Modular Arithmetic's

Let $n \in \mathbb{N}_{>0}$. Define the equivalent relation on $\mathbb{Z}$:

$$a \sim_n b \quad \Leftrightarrow \quad a - b = kn \text{ for some } k \in \mathbb{Z}$$

The collection of these equivalences classes is denoted $\mathbb{Z}/n\mathbb{Z}$[a] and is called the *integers modulo* n. The number n is called the *modulus* and if $a \sim_n b$, we say that *a is congruent to b* modulo n. If $a \sim_n b$, then it is denoted

$$a \equiv b \bmod n \qquad \text{or} \qquad a \equiv_n b$$

The equivalence classes may be denoted as:

$$[a],\ [a]_n,\ \bar{a},\ \bar{a}_n$$

where the n is dropped if the context permits it.

[a]This notation shall be elucidated on in chapter 3

Intuitively, this equivalence relation says that the difference of two numbers is a multiple of n. When $n = 12$, then this is the usual clock arithmetic's. When n is any other number, the reader may imagine a clock-face with n digits.

Naturally, it has to be proven that $\sim_n$ is indeed an equivalence relation:

Proposition 0.2.13: Modular Arithmetic's Well-defined

Let $\sim_n$ be defined as in definition 0.2.12. Then $\sim_n$ is indeed an equivalence relation

***Proof*:**
Let $x, y, z \in \mathbb{Z}$ and let $n \in \mathbb{N}_{>0}$ be any positive natural number.

1. $x \equiv x \mod n$ since $x - x = 0 = 0 \cdot n$
2. If $x \equiv y \mod n$ then there exists a k st $x - y = kn$. Then $y - x = -kn = (-k)n$. Then by definition $y \equiv x \mod n$
3. If $x \equiv y \mod n$ and $y \equiv z \mod n$, then $x - y = kn$ and $y - z = \ell n$ for some $k, \ell \in \mathbb{Z}$. Adding these two equations, we get:

$$x - z = x - y + y - z = (k + \ell)n$$

and so $x \equiv z \mod n$

Since all the conditions of an equivalence relation is satisfied, $\sim_n$ is indeed an equivalence relation, as we sought to show.

Example 0.2.14: Modular Arithmetic's
Verify the following:

1. $35 \equiv 11 \mod 12$
2. $2 \equiv -3 \mod 5$
3. If $a \equiv b \mod n$, then $a + t \equiv b + t \mod n$ for any $t \in \mathbb{Z}$
4. If $a \equiv b \mod n$, then $a^t \equiv b^t \mod n$ for any $t \in \mathbb{N}$

Proposition 0.2.15: Properties Of Modular Arithmetic's

Let $n \in \mathbb{N}_{>0}$ and $a_1 \equiv a_2 \mod n$, $b_1 \equiv b_2 \mod n$. Then:

1. $a_1 + b_1 \equiv a_2 + b_2 \mod n$
2. $a_1 - b_1 \equiv a_2 - b_2 \mod n$
3. $a_1 b_1 \equiv a_2 b_2 \mod n$

Furthermore:

1. $a_1 + b_1 \mod n = a_1 \mod n + b_1 \mod n$ (in different notation, $\overline{a_1 + b_1} = \overline{a_1} + \overline{b_1}$)
2. $a_1 b_1 \mod n = (a_1 \mod n)(b_1 \mod n)$ (in different notation, $\overline{a_1 b_1} = \overline{a_1}\overline{b_2}$)

***Proof*:**
This is a good exercise

Proposition 0.2.16: Cancellation Properties Of Modular Arithmetic's

Let $n \in \mathbb{N}_{>0}$. Then:

1. if $a + k \equiv b + k \mod n$ for any integer $k \in \mathbb{Z}$, then $a \equiv b \mod n$
2. if $ka \equiv kb \mod n$ for k coprime with n ($\gcd(k, n) = 1$) then $a \equiv b \mod n$
3. if $ka \equiv kb \mod kn$ for some $\mathbb{N}_{>0}$, then $a \equiv b \mod n$

***Proof*:**
good exercises

Important Results

The following are important properties of the integers modulo n:

Theorem 0.2.17: Fermat's Little Theorem

Let p be prime and $a \in \mathbb{N}_{>0}$ such that $p \nmid a$. Then

$$a^p \equiv a \mod p$$

***Proof*:**
First, we can assume that $0 \leq a \leq p - 1$, for we may always reduce a due to the properties of modular arithmetic's (hint: use the multiplicative properties of modular arithmetic's). Next, we may reduce this assertion to

$$a^{p-1} \equiv 1 \mod p$$

For, if the above hold, then multiplying both sides by a gives the desired result.

Start by list out the first $p - 1$ positive multiples of a:

$$a, 2a, 3a, ..., (p - 1)a \tag{3}$$

I first claim that no two of these are equal $\mod p$. For the sake of contradiction, let's say two were so that $ra \equiv sa \mod p$ for some $1 \leq r, s \leq p - 1$ where $r \neq s$. Then since p is prime and $a > 0$, $\gcd(a, p) = 1$ an hence by proposition 0.2.15:

$$r \equiv s \mod p$$

Since $0 \leq r, s \leq p - 1$, this implies $r = s$; contradiction our earlier claim. Hence, each term in list (3) is congruent (modulo p) to a unique number $1, ..., p - 1$.

Now, multiply the terms in list (3). Then by proposition 0.2.15:

$$(1a)(2a) \cdots ((p - 1)a) \equiv (1)(2) \cdots (p - 1) \mod p$$

which, when simmplifying by combining terms, is:

$$a^{p-1}(p-1)! \equiv (p-1)! \bmod p$$

Since p is prime, each number $1 \leq k \leq p-1$ is coprime to p, and hence by repeated application of proposition 0.2.16:

$$a^{p-1} \equiv 1 \bmod p$$

but now, we may simply multiply both sides by a to get $a^p \equiv a \bmod p$, as we sought to show.

The reader may ask if this can be generalized in some some way to modulo n. The key would be to generlize the step in which each number $1 \leq k \leq p-1$ is coprime with n, and indeed if $\varphi(n)$ represents the number of numbers that are coprime with n, then the identity can be generalized to:

$$a^{\varphi(n)} \equiv 1 \bmod n$$

This proof is best in the context of group theory, and so shall be left alone till then, however the persistent reader has all they need to attempt to directly prove this result.

(There are more cool proofs, but many rely on group theory or field theory. I'd like to incldue them in certain sections of the book: here)

Exercise 0.2.2

1. Let $a \equiv b \bmod n$. If $k \big| a, b$, is it the case that $(a/k) \equiv (b/k) \bmod n$?

0.2.3 Counting Problems

(Also introduce the binomial theorem?) (somehow introduce the binomial theorem. Something else too?)

0.3 Graphs

mention how this is section is not necessary to read for groups, but it will come back up when working with particular perspective on groups. It can be revisited once you make it there (perhaps make the graph intuition of groups a series of exercises?)

(See this link: (commented for compiling purposes))

We first introduce a more concrete definition of a graph in order to be able to work with it unambiguously

Definition 0.3.1: Graph

Let $V = \{v_1, v_2, ..., v_n, ...\}$ be a set of elements representing the *vertices* (also called *nodes* or *points*). Let

$$E = \{\{v_i, v_j\} \mid v_i, v_j \in V\}$$

be the set representing the *edges* of the graph (where there need not be a set $\{v_i, v_j\}$ for every pair of vertices in V). Then $\mathfrak{G} = (V, E)$ is called a *graph*.
If the edges are *pairs* instead of sets, then we call $\mathfrak{G}$ a *direct graph*

Example 0.3.2: Graphs

put different types of graphs as example? infinite, connected, cyclic, tree, completed

1. here + visual
2. here + visual
3. here + visual
4. here + visual

(define endpoints of a vertex, adjacent vertices, and if $e = \{u, v\}$ e is also called the incident of u and v)

(define simple graph: E has no repeated elements. If E can have repeated elements, it's a multigraph)

Definition 0.3.3: Graph Isomorphism

Let $\mathfrak{G} = (V_{\mathfrak{G}}, E_{\mathfrak{G}})$ and $\mathfrak{H} = (V_{\mathfrak{H}}, E_{\mathfrak{H}})$ be two graphs. A function $f : V_{\mathfrak{G}} \to V_{\mathfrak{H}}$ is called a *graph isomorphism* if for every vertex adjacent vertices $v_i, v_j \in V_{\mathfrak{G}}$, we have

$$(v_i, v_j) \in E_{\mathfrak{G}} \;\; \Leftrightarrow \;\; (f(v_i), f(v_j)) \in E_{\mathfrak{H}}$$

that is, f preserves adjacencies between vertices.
If $\mathfrak{G}$ and $\mathfrak{H}$ are isomorphic, we write $\mathfrak{G} \cong \mathfrak{H}$, and we say that $\mathfrak{G}$ is isomorphic to $\mathfrak{H}$

Example 0.3.4: examples of Graph Isomorphisms

here

These will come back in a exercise ref:HERE

0.3.1 Lattices

(These I think are important because they will be periodically mentioned. From the beginning with the partial ordering of subsets via inclusion, to using Zorn's lemma, to group/ring/module lattices, to the correspondence between the latice of divisibility in UFDs and multi-sets of irreducible elements, to the many galois correspondences we shall see, to Quivers)

Grillet has a good section on Lattices, so does Lang.

Exercise 0.3.1

1. tree is a-cyclic
2. probably counting vertices and edges problems?
3. other simple and fun graph theory problems?

0.4 Categories

In the last couple of section, we have been focusing on understanding mathematical objects, in particular, we tried constructing different types of objects (like sets and functions, numbers, graphs, and sets with binary relations) and tried showing some properties of these objects. In this section, we will focus less on the mathematical object, and more on the *relation* between mathematical objects that preserve some of the same "structure". The nouns "structure" and "relation" will be given more precise meaning through the use of *categories*.

Categories have a reputation to be difficult to understand – a reputation I (the author of this book) feel comes from an overly abstract introduction at too early a stage in the pedagogical journey. The hardest part about categories is a shift in perspective in how to think about mathematical objects, and having the sample space to intuit why this shift brings great value. Thus, before getting into the theory of category theory, let's first give an analogy to get a more intuitive understanding in what we are trying to achieve:

> Let's take any Language (ex. English). We can study any given word to get some properties of it (for example, its grammatical properties, syntactic correctness, its phonetic properties in a dialect, and so forth). These give information about words, however what we usually care about is not the properties of a word, but how words *relate* to other words. The meaning behind a sentence doesn't come just from the individual words, but from how the words "come together": This is called semantics [35]. As an example of semantics, we can even see that the meaning of words is defined in relation to other words in a sentence; for example the word *cold* usually means the absence of heat, but in the sentence "those people are giving him a cold stare", I am not referencing to the temperature of their gaze, but some lack of friendliness in their feelings towards him.
>
> If we look at language through the lens of the relation between words, we can get some insights on how language is used. An excellent example of this is translation. Let's say we're trying to translate from English to Sanskrit (or really any two languages). Both English and Sanskrit are languages, and so they have words (i.e. the objects) and rules (i.e. the relation between objects). Don't worry for a moment about exceptions to rule in particular languages (If it helps the reader, pretend they don't exist for the sake of the analogy), the main idea is that English and Sanskrit (and all languages) have words and rules that govern their use. When we compare English to Sanskrit, what we will do is see how the relations of the words in English (ex. conjugations, articles, etc.) link to those in Sanskrit (ex. declension, compounds, etc.). In a sense, we don't care about what the particular words of the respective language are. What we care is to find the word form one language that will create the same relations in the other language.
>
> For example, If I say "dog" in English, then the sentences "The dog barked", "I took the dog for a walk", or "Many dogs like to chase cats" all make sense, while "I dog the took for a walk" is not a sentence that makes sense, since the position of the word "dog" and "take" are wrong given the grammatical rules in the English languages. In Sanskrit, the grammatical rules are different: If we directly translate the sentence "I dog the took for a walk" to Sanskrit, the sentence will actually make sense! That is because the rules of Sanskrit are different than English. In fact, it is a little more subtle than this: There is no "a" (like "a walk") or "the" in Sanskrit – if you're translating from English to Sanskrit,

[35] Semantics is the study of the meaning that sentences (or any discourse) carries

we can't replace "a" with a Sanskrit word (since non exist). Instead, we will make sure that the words we choose in Sanskrit when translating the English sentences will have the appropriate *relations* in order to capture the meaning

The entire point of this analogy is to show that it is useful to understand objects not how they "intrinsically" are, but how they "in relation" to other objects [36]. The analogy showed that an advantage of this approach comes from a better understanding of how to translate sentences. Similarly, thinking about mathematical objects by how they relate to one another will let us be able to compare object by seeing if one mathematical object has the same type of relations with other objects.

In fact, we have already done this before: when we worked with ordered structures (ex. $(\mathbb{N}, \leq)$), we studied objects by how they related to other objects. We can even make definitions based on how objects relate. For example, in $(\mathbb{N}, \leq)$, define the *smallest* to be the object x such that for every other element $y \in \mathbb{N}$, $x \leq y$. Then 0 is the smallest object. Hover, in $(\mathbb{Z}, \leq)$ no object is the smallest object, because for any $x \in \mathbb{Z}$, there exists $x - 1 \in \mathbb{Z}$ such that $x - 1 \leq x$.

In the following section, we will try to more rigorously define a way to talk about the relation between objects

0.4.1 Important Concepts

This section is very much a cursory glance of the elementary concepts of category theory for the sake of exposure rather than for internalizing the definitions and working through problems. Let's start by giving a high-level idea of that a category is. This definition should hold both information about objects and the relation between objects:

Definition 0.4.1: Pre-Definition Of Category

A Category $\mathbf{C}$ is a collection[a] of objects and relations between objects:

1. We denote the collection of objects of $\mathbf{C}$ by $\text{Obj}(b)$.
2. For any to objects $A, B \in \text{Obj}(\mathbf{C})$, $\text{Hom}_{\mathbf{C}}(A, B)$ denotes the collection of relation between objects, which we call *morphisms*. In particular, if $f \in \text{Hom}_{\mathbf{C}}(A, B)$, then f is called a morphism. Morphism are *directional*, meaning that if $f \in \text{Hom}_{\mathbf{C}}(A, B)$, it is *not* the case that $f \in \text{Hom}_{\mathbf{C}}(B, A)$ (unless $A = B$)

   If $f \in \text{Hom}_{\mathbf{C}}(A, B)$, we shall write $f : A \to B$ to emphasize the directionality.
3. If $f \in \text{Hom}_{\mathbf{C}}(A, B)$, and $g \in \text{Hom}_{\mathbf{C}}(B, C)$, then there must exist a morphism $h \in \text{Hom}_{\mathbf{C}}(A, C)$ that we associate g and f; we'll write $h = gf$, $h = g \cdot f$, or $h = g \circ f$, and call h the *composite morphism* of f and g.

[a]the word *collection* is used instead of *set* for a technical reason. More on that after this definition

The fact that we said *collection* instead of *set* is because of a quirk of logic. In particular, once we properly define what a category is, we will show that sets and functions between sets will form a category. Then $\text{Obj}(\mathbf{Set})$ will have to be *a set of all sets*, but it can be proven that no such set exists! This is called *Russel's Paradox*, and is addressed in chapter **??**. The morphisms in $\text{Hom}_{\mathbf{C}}(A, B)$ cannot be chosen arbitrary, they shall have to satisfy the following properties:

[36]The philosophy behind this idea is called *structuralism*

1. (identity morphism) For each $A \in \mathrm{Obj}(\mathbf{C})$, there must at least exists morph $1_A \in \mathrm{Hom}_{\mathbf{C}}(A, A)$ such that for any $f \in \mathrm{Hom}_{\mathbf{C}}(A, X)$ and $g \in \mathrm{Hom}_{\mathbf{C}}(X, A)$ (for any $X \in \mathrm{Obj}(\mathbf{C})$, we have:
$$f\mathrm{id}_X = f \qquad \text{and} \qquad \mathrm{id}_X g = g$$
2. (associativity) For $f \in \mathrm{Hom}_{\mathbf{C}}(A, B)$, $g \in \mathrm{Hom}_{\mathbf{C}}(B, C)$ and $h \in \mathrm{Hom}_{\mathbf{C}}(C, D)$,then if $\alpha = gf$ and $\beta = hg$, then:
$$h\alpha = \beta f \qquad \text{i.e.} \qquad h(gf) = (hg)f$$

The motivation behind these rules will come in chapter 8.

Example 0.4.2: Category of Set

The prototypical examples of a category is **Set** with object as sets and morphisms as functions, namely for any two sets $X, Y \in \mathrm{Obj}(\mathbf{Set})$, $\mathrm{Hom}_{\mathbf{Set}}(X, Y)$ is the collection of all functions from X to Y. For any $X \in \mathrm{Obj}(\mathbf{Set})$, the identity function is the identity morphism, and section 0.1.5 showed functions are associative.

We shall cover more examples when there is more material to cover in chapter 8 and more advanced examples in part XII.

Let us now go over some important concepts that shall come up continuously throughout the entire textbook.

Definition 0.4.3: Isomorphism

Let $\mathbf{C}$ be a category (the reader can keep in mind **Set**). Then we shall say that $X, Y \in \mathrm{Obj}(\mathbf{C})$ are *isomorphic* if there exists morphisms $f \in \mathrm{Hom}_{\mathbf{C}}(X, Y)$ and $g \in \mathrm{Hom}_{\mathbf{C}}(Y, X)$ such that $fg = \mathrm{id}_X$ and $gf = \mathrm{id}_Y$. We shall sometimes abuse notation and simply write $fg = gf = \mathrm{id}$. If X, Y are isomorphic, we shall write:
$$X \cong_{\mathbf{C}} Y$$

Example 0.4.4: Isomorphisms in Set

Let $\mathbf{C} = \mathbf{Set}$. Show that X, Y are isomorphic if and only if $|X| = |Y|$. This shows that the relations in the relations in the category **Set** “capture” or “remember” cardinality information of sets. If we had different morphisms, we can have different types of information.

Recall that what we care about in category theory are the relations between objects and not necessarily the objects themselves. As a consequence, results proven in **Set** shall be true for all sets of the same cardinality. More generally, we shall see that results proven in category for an object X shall be true *up to isomorphism*, that is if they are true for X and $X \cong_{\mathbf{C}} Y$, then they are true for Y.

The reader may think to themselves that cardinality information is not much to preserve, for example we may have two sets of the same cardinality that satisfy different distinct predicates. This is because **Set** is a very simple category and each morphism holds little information[a]. We shall often extend the objects and morphism of **Set** to hold more information.

[a]The knowledgeable category theorist may complain that by Yoneda's lemma, any locally small category can embed into **Set** and so the category is far from being simple. The aforementioned simplicity is from the basic description and isomorphisms classes rather than the complicated subcategories we may produce

The next important idea is that of *commutative diagram*. These can be thought of as defining sets of rules:

Definition 0.4.5: Commutative Diagram

Let $\mathbf{C}$ be a category. Then a *commutative diagram* is a graph where each node is an object $X \in \mathrm{Obj}(\mathbf{C})$ and each edge is a morphism (with appropriate domain and codomain) where for each closed loop in the graph (disregarding orientation), the composite morphisms must agree.

To elucidate the last condition, consider:

$$\begin{tikzcd} & Y \arrow[dr, "g"] & \\ X \arrow[ur, "f"] \arrow[rr, "h"'] & & Z \end{tikzcd}$$

This forms a closed path, and hence composition must agree, and so $gf = h$. Another common example is:

$$\begin{tikzcd} X \arrow[r, "f"] \arrow[d, "h"] & Y \arrow[d, "g"] \\ Z \arrow[r, "i"] & W \end{tikzcd}$$

Then:

$$gf = ih$$

As we care about the relations between objects, commutative diagrams will be the building for most categorical theorems. A lot of results shall come down to saying "the following diagram commutes". Let us apply this to the first concept of category theory: *universal properties*. A universal property shall often be a property that captures the *essence* of a predicate in such a way that we can ask whether an object in different categories have or don't have the property. This is part of where the word *universal* comes from; these shall represent properties that can be described *universally between categories*. The full generality of universal properties is something we shall delay for chapter 8. For now, let us give an important example of the *universal property of quotients*. We shall show what they exist **Set**, and in each part of the book we shall see how the majority of categories we work with shall have objects satisfying this universal property.

The name suggests some notion of "dividing". This is certainly strongly related to dividing in the traditional numerical sense, however the connection to dividing would be too large of a build up for the expository nature of this section, and so it will have to be relegated to chapter 10. Instead, the reader should think of *equivalence relations* of a way of dividing up a set. The universal property of quotients in **Set** shall show that there is an object that represents the equivalence relation:

Theorem 0.4.6: Universal Property Of Quotient Sets

Let A be a set and $\sim$ be an equivalence relation on A. Let $f : A \to Z$ be any set function such that $a \sim b \Rightarrow f(a) = f(b)$. Then there exists a unique map φ from $A/\sim$ to Z such that the following diagram commutes

$$\begin{tikzcd} A/\sim \arrow[r, "F"] & Z \\ A \arrow[u, "\pi"] \arrow[ur, "f"] & \end{tikzcd}$$

***Proof*:**
Let's say $f : A \to Z$ respected where $a \sim b \Rightarrow f(a) = f(b)$. we want to ask if there exists a unique map such that

$$f = F \circ \pi$$

that is

$$f(x) = F(\pi(x))$$

Consider the map $F : (A/\sim) \to Z$, $F([x]) = f(x)$. This map is well-defined since if $[x] = [y]$, then $x \sim y$ so $f(x) = f(y)$.

Furthermore, this map is unique. If $F, F' : (A/\sim) \twoheadrightarrow Z$ were two well-defined maps, then it must be that $F(\pi(x)) = f(x) = F'(\pi(x))$ making $F = F'$, showing uniqueness, completing the proof.

Essentially, given an equivalence relation $\sim$ on a set A it can be "represented" by the object $A/\sim \in \text{Obj}(\textbf{Set})$ which captures the right amount of information such that any other object that tries to captures it is always linked to $A/\sim$. In other words, $A/\sim$ is in a sense always in an object that tries to represent the equivalence classes of A.

This will conclude our quick exposition of category theory. In chapter 8, we shall properly introduce the theory of category theory at the "elementary level". In part XII, we shall properly cover category theory in its full generality and introduce the important major results that students ought to carry throughout their mathematical career.

0.5 Matrices

This section can be comfortably skipped until chapter 13.2. It will also be referenced in section 1.2.5⋆ for examples of groups, but it is skippable without any loss (since it's just one out of over a dozen example of groups that will be presented).

In this book, we will work with many structures. Some of them will be very different from one another, while others share striking similarities. Many of these more similar mathematical structures will be able to be "represented" by matrices. We will define more rigorously for something to be "representable" by matrices later – for now we want to gain some basic appreciation and understanding of how to use matrices.

(History going back to China, this is the early motivator for matrices compared to the "new" one of being a useful classification theorem. Define a matrix. Show it's used to solve systems of linear equations. Explain Gaussian elimination)

(Later, Gauss developed a way to multiply matrices. The intuition shall come in chapter 13)

Part I

Groups

The concept of a group is the culmination of multiple branches of mathematics converging to a single unifying object that works in each domain. Arguably the first branch of mathematics where the notion of a group was used first in the modern sense was in *algebraic equations*, that is the study of polynomials. A young mathematician by the name of Évariste Galois found the answer around 1832[37] to an over 2000 year old problem: does there exist a "quadratic equation" for all polynomials, what mathematicians would call the *solvability of degree n polynomials* problem. The degree 3 (cubic) polynomials and degree 4 (quartic) polynomials case has been solved in the affirmative[38] around 300 years before Galois, but the degree 5 (quintic) polynomial remained a mystery. Galois proved that there exists quintic polynomials with *no* formula, in particular, there exist quintic polynomial $ax^5 + bx^4 + cx^3 + dx^2 + ex + f$ where a root x_0 cannot be written as a sum, difference, product, quotient, or root $(+, -, \times, \text{div}, \sqrt[n]{-})$ of the coefficients! We shall prove this in chapter 20, section 20.2.4.

Around the same time, mathematicians started to discover that there are "more geometries" than just the usual flat geometry of euclidean space (now known as euclidean geometry). It was shown that one of the foundational axiom of geometry ever since the time of the ancient Greeks (the 5th postulate, or parallel postulate) is in fact an assumption that can be removed without causing a contradiction! Near the beginning of these developments, a mathematician by the name of Felix Klein saw that the notion of a group seems to be intimately connected with understanding these new geometries, and a Norwegian mathematician named Sophus Lie (pronounced "Lee") came up with *Lie Groups* which solidified the importance of groups within modern geometry[39].

While all of this was happening, a notion very similar to groups was already being used in *number theory* with mathematicians such as Gauss and Dirichlet noticing inherent symmetries in primes that showed up while solving Diophantine equations. We shall cover these results (in modern algebraic language) in chapter 50.

In these early days, these three disciplines were defining groups in slightly different ways. It was the effort of mathematicians such as Camille Jordan and Ernst Kummer to coalesce these concepts, resulting in the modern definition of a group as we know it today. The formalization of the definition of a group became a huge stepping stone for an enormous amount of disciplines to notice its practicality within their domain. Today, groups have become a major, if not central, object in almost every branch of mathematics, but also have been extensively used in physics, chemistry, cryptography, computer science, and many more fields. The reason groups are a fundamental object in each field is that it can interpreted as representing the *symmetry* of some system.

A modern mathematicians would now consider groups to be a rather universal and "natural" object. However, groups came late in the development of mathematics, partly due to the mathematical build-up that needed to happen and partly due to their abstract nature that does not make them seem natural to a first-time viewer. A consequence of this is that it makes teaching about groups introduce a new level of abstraction that many students have not experienced. The difficulties are further compounded by the fact that visualizations of many groups are scarce or impractical[40]. This is however not a bad thing: a whole new way of learning and "experiencing" mathematical objects is the reward for internalizing groups and all the further objects that got based off of them.

In this Part, we shall cover in detail all the fundamental concepts of groups. Most of these concepts

[37] Évariste Galois accepted a challenge to a dual. The night before the duel he wrote down what is now considered *Galois Theory* before passing away in the duel the next day

[38] see this link for the cubic formula and this link for the quartic

[39] This material belongs in a class in geometry, but we shall cover some important algebraic aspects of it in chapter 54

[40] We shall see that there are natural ways of visualizing groups, but it sometimes worse to think about the visual picture, for example some groups can only be visualized by the symmetries of an over 100'000 dimensional shape

shall be instrumental in the rest of the textbook, and the reader shall find that this part of the book is the nexus for most algebraic tools that we shall cover in the rest of the book.

0.5.1 For Graduate Student The interpretation of a group representing symmetry is in fact very natural: all groups can be represented as a category with one objects whose morphisms are isomorphisms. From this perspective, it is not that it is *an* interpretation, but *the* interpretation.

There is another way that I (currently) personally see groups. They feel like naturally "2nd order" objects. If we are trying to study some object (numbers, shapes, graphs, etc), we will want to understand some property of the object. This property can be local (like all loops based at a certain point, or the functions that exist on a certain subset), or it can be global (ex. all loops that go everywhere up to homotopy, symmetries of the object, etc.). This can usually be identified by defining a certain action on the object of study. Groups arise naturally when considering actions. Thus, groups are a way of capturing a type of algebraic property of an object. There can be other types of properties of an object (ex. curvature, twisting, etc.) which are not represented algebraically, and hence groups are just a part of the puzzle of looking for information about an object.

1

Introduction to Groups

In this chapter, you will cover the basic properties of groups. In the process, we will prove statements that you perhaps thought were axiomatic. For example, if $ab = ac$, does that mean $b = c$? We will classify two very important sub-categories: groups with the axiom of *commutativity* ($ab = ba$), and those without it. These two categories of groups will set the narrative for how we study groups, meaning this one axiom plays an incredibly central role in the study of groups! We will cover many examples of groups to get a taste on what groups are, and define the groups which will be referenced indefinitely throughout the rest of the book. We will then cover how to compare two groups, and define the notion of equality between two groups. We'll then see how we can use groups to "represent" a system of objects. For example, we'll use a group to represent all possible ways of rotating an object. In this case, the system will consist of all possible rotations of the object.

1.1 Basic Properties and Definitions

As mentioned in the preface, algebraic structures are sets along with a function (or functions) which must follow certain rules. The most common type of function used for an algebraic structure is called a *binary operation*:

Definition 1.1.1: Binary Operation

1. A *binary operation* $*$ on a set G is a function $*: G \times G \to G$. For any $a, b \in G$, we shall write $a * b$ or $*(a, b)$. Often the notation is abbreviated to ab.

2. A binary operation $*$ on a set G is *associative* if for every $a, b, c \in G$ we have $a * (b * c) = (a * b) * c$

3. If $*$ is a binary operation on a set G we say elements a and b *commute* if $a * b = b * a$. We say $*$ (or G) is *commutative* if for every $a, b \in G$, $a * b = b * a$

The star of the show in group theory is the binary operation. This function is the key mathematical structure added to a set which we study when talking about groups. When talking about different group structures, we're often talking about different binary operations.

Example 1.1.2: Binary Operations

1. Arithmetic operations are a good source of binary operations. As we've discussed in the preliminary chapter, $+$ and $\times$ are functions that take in two numbers as input, and output one number. If you want to practice proving something is a function, you can try showing that $+ : \mathbb{Z} \times \mathbb{Z} \to \mathbb{Z}$ is a function (and since the domain is the cartesian product of two copies of the codomain, it's a binary operation). Similarly, you can do the same with $- : \mathbb{Z} \times \mathbb{Z} \to \mathbb{Z}$.

2. Is $-$ a binary operation with $\mathbb{N}$ (i.e. is $- : \mathbb{N} \times \mathbb{N} \to \mathbb{N}$ a binary operation)? What about $\times$ and div? Which of these binary operations are associative? Are they commutative? What if you switch $\mathbb{N}$ with $\mathbb{Z}$, $\mathbb{Q}$, or $\mathbb{R}$?

3. When working with sets, it is common to take the union or intersection of sets. These are binary operations. If A is a set and $P(A)$ is its powerset, then $\cup : P(A) \times P(A) \to P(A)$ is a binary operation. Similarly for $\cap$

4. Take $\mathbb{Z}$, and recall the *gcd* and *lcm* from the preliminary chapter (ref:HERE). Then $gcd : \mathbb{Z} \times \mathbb{Z} \to \mathbb{Z}$ is a binary operation. Similarly for *lcm*. Can you replace $\mathbb{Z}$ with $\mathbb{N}$?

5. If you have taken some multi-variable calculus, then the cross product $\times : \mathbb{R}^3 \times \mathbb{R}^3 \to \mathbb{R}^3$ is an example of a binary operation.

6. Take $E : \mathbb{N} \times \mathbb{N} \to \mathbb{N}$ to be $E(x, y) = x^y$. Check that $E(x, E(y, z))$ is not necessarily equal to $E(E(x, y), z)$ (that is $x^{(y^z)}$ is not necessarily equal to $(x^y)^z$). Hence, exponentiation is a non-assotiative operation.

7. Given two functions f and g over the same domain and codomain ($f : X \to X$ and $g : X \to X$), their composition. Thus composition of functions with the same domain and codomain forms a binary operation.

With these intuitions, we can move on to defining a group:

Definition 1.1.3: Group

A *group* is an ordered pair $(G, *)$, where G is a set and $*$ is a binary operation on G satisfying the following axioms:

1. associativity: $(a * b) * c = a * (b * c)$
2. there exists an element e in G, called the *identity* of G, such that for every $a \in G$, we have $e * a = a$
3. For each $a \in G$ there is an element $b \in G$ called an *inverse* of a, such that $ab = ba = e$. We denote this element b by a^{-1}, so that we can equivalently write $a^{-1} * a = e$

The group $(G, *)$ is called *abelian* (or commutative) if $a * b = b * a$ for every $a, b \in G$.

If $(G, *)$ is a group, we say that G is a group under $*$.

1.1.4 Note Some authors like to make it explicit that it's closed under the operation, meaning if $a, b \in G$, then $a * b \in G$. However, that is implied in the definition of binary operation (i.e. a function), since the codomain of $*$ is G.

Example 1.1.5: First Encounter With Groups

In the following examples, when we're working with numbers, think of them only as a set, and define the operations $+$, $\times$ or others in the way they were intuitively defined when learning arithmetic's.

1. $(\mathbb{Z}, +)$, $(\mathbb{Q}, +)$, $(\mathbb{R}, +)$ are groups. We would say, for example, that $\mathbb{Z}$ is a group under $+$. For each of these examples, we would have the identity to be $e = 0$, and for each a, $a^{-1} = -a$. Note that technically, the $+$ in each group is not the same $+$, because the domains and co-domains are different. Note that $\mathbb{N}$ is *not* a group under $+$, since not every element has an inverse: There does not exist an $a \in \mathbb{N}$ such that $2 + a = 0$ (i.e. $-2 \notin \mathbb{N}$). [a]
2. $\mathbb{Z}, \mathbb{Q}, \mathbb{R}$ and $\mathbb{C}$ under $\times$ are *not* groups. $\mathbb{Q} - \{0\}$, $\mathbb{R} - \{0\}$ $\mathbb{C} - \{0\}$, $\mathbb{Q}_{>0}$, $\mathbb{R}_{>0}$ are groups under $\times$, with $e = 1$, and $a^{-1} = \frac{1}{a}$. Why is that? Take a moment to think about why removing the zero will make those sets groups. Is $\mathbb{Z} - \{0\}$ a group under $\times$? Is $(\mathbb{Z} - \{0\})$ a group under $+$?
3. For any $n \in \mathbb{N}$, $\mathbb{Z}/n\mathbb{Z}$ is a group under modulo $+$ $(\mathbb{Z}/n\mathbb{Z}, +)$. Can you show that 0 is the identity? Can you show that every number has an inverse? Note that $((\mathbb{Z}/n\mathbb{Z}), \times)$ under $\times$ *not* a group in general! It *is* a group when p is prime. You can actually pick some non-prime n and try it (ex. $n = 4$).The general proof that $(\mathbb{Z}/n\mathbb{Z}, \times)$ is a group when n is prime will come later (after we introduce theorem 2.3.11)
4. If $(A, \bullet_a)$ and $(B, \bullet_b)$ are groups, we can form a group on their cartesian product $A \times B$ called the *direct product group*, whose elements are those in the Cartesian product

$$A \times B = \{(a, b) | a \in A, b \in B\}$$

and the binary operations is defined point wise:

$$(a, b) * (c, d) = (a \bullet_a c, b \bullet_b d)$$

We can denote this group and its binary operation $(A \times B, (\bullet_a, \bullet_b))$. A common example of a direct product of group is $(\mathbb{R} \times \mathbb{R}, (+, +))$, with the usual notation of $\mathbb{R}^2$. We will dedicate a lot of energy understanding the details of the Cartesian product of groups in chapter 5.

5. Define $+_2 : \mathbb{Z} \times \mathbb{Z} \to \mathbb{Z}$ $x +_2 y = 2 + x + y$. The operation $+_2$ is associative, since

   $$(x +_2 y) +_2 z = (2 + x + y) +_2 z = (2 + x + y) + 2 + z = 2 + x + (2 + y + z) = x +_2 (y +_2 z)$$

   Furthermore, $e = -2$, since
   $$x +_2 -2 = x + 2 - 2 = x$$
   and $x^{-1} = -x - 4$ since
   $$x^{-1} +_2 x = -x - 4 + 2 + x = -2$$
   showing that $(\mathbb{Z}, +_2)$ is indeed a group. Notice that $(\mathbb{Z}, +_2)$ and $(\mathbb{Z}, +)$ are different (for example, their identity is different)[b]. When introducing multiplication to integers, we shall want $+$ and $\cdot$ to behave well together; the usual $+(x, y) = x + y$ will insure this[c].

6. $G = (\mathrm{Mat}_n(\mathbb{R}), \cdot)$ is almost forms a group. A binary operation exists, identity exists, and associativity holds. However, there are non-invertible matrices, meaning that $\forall M \in G$, it's not necessarily the case that $\exists N \in G$ such that $NM = \mathrm{I}$. We will explore this in section 1.2.5⋆

7. Take $\{e\}$ to be the set with a single element, an a binary operation $\cdot$ such that $e \cdot e = e$ (that is, the binary operation that treats e like an identity). Then $(\{e\}, \cdot)$ is a group! This group is called the *trivial group*.

8. Take a set A and its powerset $P(A)$. Is $(P(A), \cup)$ a group? What about $(P(A), \cap)$? answer: [d]

9. Take $(\mathbb{Z}, -)$, where $-$ is the operation of subtraction. Then $(\mathbb{Z}, -)$ is not a group! Can you see which axiom fails?

10. Here is a more complex example which fails for the same reason as the previous one. Take $(\mathbb{R}^3, \times)$ where $\times$ represents the cross product:

    $$\begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix} \times \begin{pmatrix} b_1 \\ b_2 \\ b_3 \end{pmatrix} = \begin{pmatrix} a_2b_3 - a_3b_2 \\ a_1b_3 - a_3b_1 \\ a_1b_2 - a_2b_1 \end{pmatrix}$$

    or, if you've taken a physics or linear algebra class, you might have seen it defined as

    $$a \times b = \det \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix}$$

    where we represent the three components of the vector as $\mathbf{i}$, $\mathbf{j}$, $\mathbf{k}$, and we use the determinant to shorten notation. Then $(\mathbb{R}^3, \times)$ is *not* a group! The cross product is not associative! Take

    $$\begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}, \begin{pmatrix} 2 \\ 3 \\ 4 \end{pmatrix}, \begin{pmatrix} 3 \\ 4 \\ 5 \end{pmatrix}$$

Then

$$\begin{pmatrix}1\\2\\3\end{pmatrix} \times \begin{pmatrix}2\\3\\4\end{pmatrix} = \begin{pmatrix}-1\\2\\-1\end{pmatrix} \qquad \begin{pmatrix}-1\\2\\-1\end{pmatrix} \times \begin{pmatrix}3\\4\\5\end{pmatrix} = \begin{pmatrix}14\\2\\-10\end{pmatrix}$$

and

$$\begin{pmatrix}2\\3\\4\end{pmatrix} \times \begin{pmatrix}3\\4\\5\end{pmatrix} = \begin{pmatrix}-1\\2\\-1\end{pmatrix} \qquad \begin{pmatrix}1\\2\\3\end{pmatrix} \times \begin{pmatrix}-1\\2\\-1\end{pmatrix} = \begin{pmatrix}-8\\-2\\4\end{pmatrix}$$

meaning $(\mathbb{R}^3, \times)$ is *not* associative, meaning it's not a group. The cross product is a different binary operation called the *lie bracket*[e].

11. If you know some complex analysis, then $G = \{z \in \mathbb{C} | z^n = 1 \text{ for some } n \in \mathbb{Z}^+\}$ is a group! Visually, this group looks like points on the unit circle in the complex plane. The main idea here is since $z^n = 1$, then elements in G will be of the form $e^{\frac{2\pi i}{n} \cdot k}$ for $1 \leq k \leq n$. If you're comfortable with multiplying complex numbers and trust me that all the elements of G are of this form, then you can check that this is a group.

    More generally, $G = \{z \in \mathbb{C} \mid \|z\| = 1\}$ is a group with multiplication (i.e. the unit circle in the complex plane). You can verify that $1 \in G$ and for any $z, w \in G$, $\|zw\| = \|z\| \|w\| = 1 \cdot 1 = 1$. Using these two fact, showing there exists an inverse and that G is associative follows from properties of $\mathbb{C}$ (exercise ref:HERE)

12. If you know some complex arithmetic's, show that $G = \{z \in \mathbb{C} \mid 0.1 < |z| < 10\}$ with the usual multiplication structure of $\mathbb{C}$ satisfies associativity, having an identity, and having an inverse, but is *not* a group.

13. Let A be any set. Consider the collection of every bijective function from A to itself:

    $$G = \{f : A \to A \mid f \text{ is bijective}\}$$

    Then with the binary operation of composition, $(G, \circ)$ is a group. What is the identity, and why do inverses always exist? The collection of bijective functions is a very common group that shows up in many different ways in different areas of mathematics and so has lots of notations:

    $$\text{Bij}(A) \qquad \text{Sym}(A) \qquad \text{Aut}_{\mathbf{Set}}(A)$$

Part of the motivation behind learning group theory is in the abundance of areas groups show up in mathematics. The more domains of mathematics you learn, the more types of groups you will encounter. For example, field theory (which we explore in part IV) have *Galois groups* at its heart, Differential Geometry have *lie groups* that allow for a large classification of and construction of "manifolds" (a generalization of the notion of Euclidean space), Algebraic Geometry (which we explore in part V) use *algebraic groups* to represent more complicated objects in more familiar terms, Analysis uses *topological groups* (for example, Amenable and Locally Compact groups) in the study of Fourier Analysis and Functional Analysis, Algebraic topology has the *fundamental group* and *homology group* at its heart, and combinatorics uses *geometric group theory* (for example, the *Kleinian group* or *hyperbolic groups* shows up frequently).

Group can also be thought of as one of the "basic" structures that many other structures will build on or act on. Of the algebraic structures that will be explored in this book, rings, fields, vector-spaces, modules, and (lie and associative) algebras are all also groups.

Overall, it is worth remembering that groups are a very common mathematical object that show up a little bit everywhere.

[a]$\mathbb{N}$ is a weaker structure called a monoid which doesn't require inverses. More on them in section 7.1

[b]I'm careful in saying they are different instead of saying they have *different group structures*. There is a subtly here on how we define two group structures being the same or different. This is addressed in section 1.4

[c]we shall desire distributivity, or for the expert that $\mathbb{Z}$ is representable in its endomorphism ring

[d]the answer is no! Like $\mathbb{N}$, these are example of monoids

[e]Explored in section **??**

In general, when we write a group, the group operation will be clear from context, so instead of writing $(G, *)$, we will write G. We will take on this convention form here on out unless it's ambiguous. We will also assume the standard binary operation of addition for

$$\mathbb{Z}, \mathbb{Q}, \mathbb{R}, \mathbb{C}, \mathbb{Z}/n\mathbb{Z}$$

unless stated otherwise (ex, if you see $\mathbb{Z}$, assume it is $(\mathbb{Z}, +)$). If we want the multiplicative structure of these numbers, we'll add a $\times$ in the superscript and implicitly omit the zero:

$$\mathbb{Z}^\times, \mathbb{Q}^\times, \mathbb{R}^\times, \mathbb{C}^\times (\mathbb{Z}/n\mathbb{Z})^\times$$

These sets are $(\mathbb{Z} - \{0\}, \times)$ [1], $(\mathbb{Q} - \{0\}, \times)$, $(\mathbb{R} - \{0\}, \times)$, and so forth. The sets $\mathbb{R}^+$ and $\mathbb{Q}^+$ shall represent the collection of positive real numbers and positive rational number with the usual multiplication. Furthermore, since we are always working with binary operations, the word "binary" is often omitted for convenience.

1.1.6 Note In many of the previous examples, there is some more structure usually associated with those symbols. For example, in $\mathbb{N}$, $\mathbb{Z}$, $\mathbb{Q}$ and $\mathbb{R}$, there is a notion of some numbers being bigger than others, that is, $1 < 2$, or $n < n+1$. These are *not* part of the definition of a group, and so cannot be used when proving general statements about groups! It will actually be very common that you work with groups with *more* structure than their group structure, so be quite careful that the examples you hold when doing exercises *only* represent the structure you want to work with at that moment

You might have noticed that some of the underlying sets of the above groups are subsets of each other: $\mathbb{Z} \subseteq \mathbb{Q} \subseteq \mathbb{R} \subseteq \mathbb{C}$. Does this imply that there is a relation between their group structure? For example does the usual addition in $\mathbb{Z}$ works as in $\mathbb{Q}$ with usual addition? If you think about this particular example, you'll see that the answer is yes. But being a subset doesn't always means you'll have the same (if any) group structure: $\mathbb{N} \subseteq \mathbb{Z}$, but $\mathbb{N}$ is not even a group! On the other hand, if we take $2\mathbb{Z} \subseteq \mathbb{Z}$, and give $2\mathbb{Z}$ the same operation as $\mathbb{Z}$ (that is $(2\mathbb{Z}, +)$ is our group), then $(2\mathbb{Z}, +)$ acts exactly the same way as $\mathbb{Z}$, with just fewer elements. So, we'll give a name to subsets which inherit group properties of their parent group:

Definition 1.1.7: Subgroup

Let G be a group and $H \subseteq G$. Then H is a subgroup of G if it is still a group under the same operation as G. We denote the subgroup by $H \leqslant G$ [a]

[a]In some textbooks, the symbol for subgroup is $H \preceq G$

[1]Note that $\mathbb{Z}^\times$ is not a group under $\times$ as we've pointed out earlier. However, the binary operations $\times$ is still defined on this set

You might have flinched to it saying "the same operation", since it was explicitly mentioned in the definition that for a group G, the operation is $G \times G \to G$, so H cannot have the same operation since the domain and co-domain are different. This is a common convention in algebra for structures which are similar: if we restrict the operation to only the elements of H, then $(H, *|_H)$ should be a group. In particular, if $* : G \times G \to G$ is the binary operation on G, then we say we *restrict* the binary operation to H if we can define $*|_H : H \times H \to H$, where for any two elements $h_1, h_2 \in H$ $*|_H (h_1, h_2) = *(h_1, h_2)$. If $(H, *|_H)$ is a group, we say it's a subgroup of $(G, *)$. In this way, we say that H has "the same" binary operation as G.

Example 1.1.8: Subgroups

1. Take $2\mathbb{Z} \subseteq \mathbb{Z}$ – the set of all even integers. The identity is there: $0 \in 2\mathbb{Z}$. If you add two even integers, you get an even integer $(2x + 2y = 2(x + y))$ [a]. Furthermore, if $2x \in 2\mathbb{Z}$, then there exists $-2x \in 2\mathbb{Z}$ such that $2x - 2x = 0$, meaning every element has an inverse. Finally, associativity is implied by associativity in $\mathbb{Z}$: for any $2x, 2y, 2z \in 2\mathbb{Z}$, we have that $2x, 2y, 2z \in \mathbb{Z}$, so $2x + (2y + 2z) = (2x + 2y) + 2z$. Thus $2\mathbb{Z} \leqslant \mathbb{Z}$

2. Right before the previous definition, it was mentioned that $\mathbb{Z} \subseteq \mathbb{Q}$ is an example of a subgroup (i.e. $\mathbb{Z} \leqslant \mathbb{Q}$). This is an exercise worth doing to get comfortable with the concept. Mix an match with other types of numbers to get more practice.

3. Let $\mathbb{Z}^\times \subseteq \mathbb{Q}^\times$ (i.e. $Z - \{0\} \subseteq \mathbb{Q} - \{0\}$). Is $\mathbb{Z}^\times$ a subgroup?

4. Let $H \leqslant G$ and $K \leqslant H$. Is $K \leqslant G$?

5. Here is an exercise in understanding the nuance of binary operations: Is $\mathbb{Z}/5\mathbb{Z} \leqslant \mathbb{Z}$? answer: [b]

[a] Note here how I *did* use extra properties that are not necessarily part of a group (I factored the two out). This is fine, since we're working with a *particular* example, this would not be a proof we can do on general groups

[b] It is no, since the binary operation does not "act" the same way $(3 + 3 = 1)$ in $\mathbb{Z}/5\mathbb{Z}$ but $3 + 3 = 6$ in $\mathbb{Z}$

Groups are missing one key property that numbers usually have: commutativity. Though we already talked about them in the definition of a group, I will add it here to emphasize their importance:

Definition 1.1.9: Abelian Group

Given a group G, if $ab = ba$ for all $a, b \in G$, we call G a *commutative* or *abelian* group

Example 1.1.10: Abelian Groups

$\mathbb{Z}, \mathbb{Q}, \mathbb{R}, \mathbb{C}, \mathbb{Z}/n\mathbb{Z}$ are all Abelian Groups [a]. So are $\mathbb{Q}^\times, \mathbb{R}^\times, \mathbb{C}^\times$. In fact, all examples in the example 1.1.5 were abelian groups. We haven't yet introduced groups which are not abelian. The first such group will be the Dihedral group (definition 1.2.25)

[a] This is an example where the binary operation is implied. The binary operations shown in example 1.1.5 are the standard binary operations for these sets

As mentioned earlier, being abelian/commutative is insanely powerful. Of all the properties the reader will cover in basic group theory, a lot of them will be properties a group being "almost" or "kind of" abelian. In this text, we will come across many definitions and sections that are centered around how close they are to being abelian [2]

The reason we relate a lot of conditions to commutativity is because Abelian Groups behave in much more predictable ways as compared to general groups, as shall be explored. The power of commutativity actually permeates a lot of algebraic structures. Because of this, we often split up domains in mathematics in "commutative" and "not-commutative". As an example of this split, some mathematicians like to reference the set of Abelian groups as **Ab** and the set of all groups as **Group** to show how different a general group versus a general abelian group is [3]. There are entire areas of mathematics where the problem has been solved in the "abelian case", while the "nonabelian case" is an active area of research[4].

For the visual reader, the following is a helpful tool to understand the group operation:

Definition 1.1.11: Cayley Table (Multiplication Table)

Let G be a finite group (that is, a group on a finite set). Label the elements of G as $\{g_1, g_2, ..., g_n\}$, where g_1 represents the identity.
Then *Cayley table* or *multiplication table* of G is the $n \times n$ matrix whose i, j entry is the group element $g_i g_j$.

I personally prefer the term Cayley table since not all binary operations are called "multiplication"; it can be confusing to do a multiplication table for $(G, +)$

Example 1.1.12: Cayley Table

Take $G = \mathbb{Z}/4\mathbb{Z}$. Then the Cayley tables for addition and multiplication are:

$+$	0	1	2	3
0	0	1	2	3
1	1	2	3	4
2	2	3	4	0
3	3	4	0	1

$\times$	0	1	2	3
0	0	0	0	0
1	0	1	2	3
2	0	2	0	2
3	0	3	2	1

This can be done for any finite group, and sometimes gives valuable insights instantly. For example, notice that in the Cayley table for multiplication, for row 0, 1, and 3, they contain every element of $\mathbb{Z}/4\mathbb{Z}$, yet the row for 2 doesn't. Notice further that both tables are symmetrical, that is, position ij is equal to position ji.

In general, it can be quite cumbersome to write out an entire Cayley table, especially when the order the group will be higher than 4, if not infinite. So throughout Part I, we develop algebraic tools to deduce information about groups.

Exercise 1.1.1

[2]for reference, see definition 3.1.9, definition 2.2.4, and section 5.3, 6.1, and 6.1

[3]There are some technicalities here in saying it's the *set* of all groups, due to Russel's paradox. We should instead say the *class* of all sets. This is addressed in Chapter **??**

[4]There are a huge number of such examples, but to point out one relevant to algebra class field theory completely classifies abelian extension of fields, while the nonabelian counterpart is a very deeply researched problem and is called *Langland's Program.* Another example that creates a split between the commutative and non-commutative case is within the study of algebraic geometry where the structures are built out of commutative rings, while the non commutative case would require more machinery from analysis such as Fourier Transforms

1. Let $G = \mathbb{Q}$, and the binary operation be average : $\mathbb{Q} \times \mathbb{Q} \to \mathbb{Q}$, where average$(x, y) = \frac{x+y}{2}$. Is $(G, \text{average})$ a group? What if $G = \mathbb{Z}/4\mathbb{Z}$?
2. Can a group structure be put on $\emptyset$ (i.e., can there exist a function $* : \emptyset \times \emptyset \to \emptyset$ such that $(\emptyset, *)$ is a group?) (In prelim chapter, make an exercise saying that $f : \emptyset \to \emptyset$ is a function, while $f : A \to \emptyset$, $|A| > 0$ is *not*, and also that $\emptyset \times \emptyset$ is vacuously well-defined)
3. (If you know some complex analysis) In example 1.1.5[10], it was stated that
$$G = \{z \in \mathbb{C} \mid |z| = 1\}$$
is a group. Prove that G is indeed a group.
4. Let $\text{Bij}(X)$ be the set of all bijections from X to X. Prove that $\text{Bij}(X)$ is a group. In section 1.2.2, we will relabel this as $\text{Sym}(X)$.
5. look at this link to get some ideas
6. (Perhaps this a superficial question, but the identity and inverse condition's can be replaced by functions, and so G would be a set with 1 binary operation, 1 "1-ary" operation, and a 0-ary operation. Introduce this to make groupoids and universal algebras more intuitive way later?
7. perhaps a question on how the identity can be replaced by a monary operation? (i.e. a perspective that aims to be a prelude to Universal Algebras?)

1.1.1 Consequences of Group Axioms

With the definition of groups down, we can deduce a lot of direct consequences form the axioms. One thing common with mathematical constructs is an abundance of propositions. These are vital to be able to prove by oneself, and once comfortable should be treated as knowledge as important as the definition of a group – I would even go so far as to say that not knowing the properties of groups means you don't really know what a group is. For example, do you really know what a frying pan is/can do if you haven't cooked with it before? Do you know the in's an out of what utensils can you use to not damage it, or what type of food will stick to it or not? You will only know once you've practiced using it.

I would strongly suggest that the reader attempts proving *all* of these propositions before looking at the proof, especially if this is the first time the reader has seen algebraic structures! It is crucial moving forward that these properties are second nature, since everything we build form here on out will constantly be relying on these properties.

The first thing to point out is that if we have $a * e = a$, we can deduce $e * a = e$.

Proposition 1.1.13: one-sided identity implies two-sided identity

Let G be a group and e an identity. If $a * e = a$ then $e * a = a$ (or vice-versa)

***Proof*:**
Let's say $e * a = a$. We want to show that $a * e = a$ (1). If we multiply equation (1) on the right

by a^{-1}, we get

$$\begin{aligned} &= (a * e) * a^{-1} & \\ &= a * (e * a^{-1}) & \text{associativity} \\ &= a * a^{-1} & \text{Identity definition} \\ &= e & \text{inverse definition} \end{aligned}$$

Thus, $(a * e) * a^{-1} = e$. Multiplying both sides by a on the right, we get

$$\begin{aligned} ((a * e) * a^{-1}) * a &= e * a \\ \Leftrightarrow \ (a * e) * (a^{-1} * a) &= e * a \\ \Leftrightarrow \ (a * e) * e &= e * a \\ \Leftrightarrow \ a * e &= e * a \end{aligned}$$

as we sought to show.

Some author's put $a * e = e * a = a$ as a definition. This is a matter of preference: since it's a direct consequence of the axioms, it can be put as one of the fundamental rules a group has to follow without worry of constraining the definition. Some authors disagree with this, and would rather keep the definition of group in fuller generality. This preference for generality is a balancing act: the more general you go, the more "complicated" the concept gets. However, if done right, you in general gain a more "nuanced" perspective with generality. In this book, the more general idea was taken, since some early proofs fail for more subtle reasons. For example, $(\mathbb{Z}, -)$ fails to be a group because of associativity, however if we put $a * e = e * a = a$ in the definition, we can also say it fails because $-a = 0 - a \neq a - 0 = a$ (for $a \neq 0$). However, as we've seen in the above proof, this fact *relies on associativity*, and therefore is hiding the fact that $(\mathbb{Z}, -)$ is still failing to be a group because of associativity.

Corollary 1.1.14: one-sided Inverse Implies two-sided Inverse

Let a, a^{-1} be elements of the group G and let e be the identity of G. Then $a * a^{-1} = e$ if and only if $a^{-1} * a = e$

***Proof*:**
Let's say $b = a^{-1}$ so that $a * b = e$. Then $b = b * e = b * a * b$. Multiplying both sies by b^{-1}, we get

$$\begin{aligned} e &= b * b^{-1} & \\ &= (b * a * b) * b^{-1} & \\ &= b * a * (b * b^{-1}) & \\ &= b * a & \\ &= a^{-1} * a & b = a^{-1} \end{aligned}$$

as we sought to show.

Proposition 1.1.15: Basic properties of Group elements

1. The identity of G is unique. Furthermore, if for some $x \in G$, $f * x = x$, then is equal to the identity
2. For each $a, \in G$, a^{-1} is uniquely determined
3. $(a^{-1})^{-1} = a$, for all $a \in G$
4. $(a * b)^{-1} = (b^{-1}) * (a^{-1})$.
5. generalized associative law (induction on associativity).

***Proof*:**
Again, I highly encourage you to attempt all of these before looking at these answers!!

1. Let $e \in G$ be the identity, that is, for all $x \in g$ $e * x = x$. Let's say there is another element $f \in G$ that is also the identity. Then

$$\begin{aligned} e &= e * f && \text{definition of identity} \\ &= f && \text{definition of identity} \end{aligned}$$

therefore, $e = f$, showing that e is unique.

Furthermore, if there was an f such that for only *some* $x \in G$, $f * x = x$, then

$$\begin{aligned} (f * x) * x^{-1} &= (x * x^{-1}) = e \\ \Leftrightarrow f * (x * x^{-1}) &= e \\ \Leftrightarrow f &= e \end{aligned}$$

thus, there cannot be a "partial" identity.

2. exercise

With these in mind, here's some standard notation to simplify writing: One usually write the operation $*$ as $\cdot$ to represent an abstract binary operation, unless otherwise specified; we write $a \cdot b$ as ab. We usually also denote the identity e as 1. We'll also implement the following short hand: [5]

$$xxxxxxx \cdots x = x^n, \qquad x^{-1}x^{-1} \cdots x^{-1} = x^{-n}, \qquad x^0 = 1$$

Don't forget that last one is the identity of G. If we're using a group where specific notation is already defined (like with the $+$ operation), we can use it's specific convention. So $a + a + \cdots + a = na$, $-a + -a + \cdots + -a = -na$, and $0a = 0$ [6]. Furthermore, $a + (-a)$ is sometimes shorthanded to $a - a$.

The next result we'll cover let's us generalize our intuition of pre-abstract algebra. When learning

[5] $x^0 = 1$ since $x^0 = x^{1-1} = x^1x^{-1} = xx^{-1} = 1$

[6] If you have covered rings, this might look familiar to the first property in proposition 9.1.9. It is quite similar in that this shorthand secretly is a ring-action on the group (but not a $\mathbb{Z}$ module since G is not necessarily abelian). This means that the intuition behind the notation is similar to the proof in 9.1.9: $0a = (1-1)a = a - a = 0$

algebra for the first time, you've probably encountered equations of the form:

$$3x = 5$$

and where asked to solve for x. The way you would do this is by dividing both sides by 3 to get the equation:

$$x = \frac{5}{3}$$

This might seem like the obvious thing to do in any situation. However, you can't always cancel out a number on one side and get a unique result! Take a moment to review your proof of uniqueness of identity and uniqueness of the inverse, and consider this thought experiment:

1.1.16 Thought Experiment Let's take a set for which every element does *not* have an inverse. Recall (or prove) that $\mathbb{Z}/6\mathbb{Z}$ is not a group under $\cdot$. In particular, [2] has no multiplicative inverse. Since $[2][2] = [0]$, then:

$$[4] = [2][2] = [2][5] = [10] = [4]$$

However, $[2] \neq [5]$. In other words, I couldn't "cancel" out the [2] on the left side, i.e. for the equation $[2]x = [4]$ we get $x = [2]$ or $x = [5]$.

This cancelling out of a number on the left [or right] side is something you probably took for granted, but it's actually the consequence of the fact that every number in a group must have a unique inverse! This fact is proven in the next proposition:

Proposition 1.1.17: Cancellation Property: Unique solution to monomials

Let G be a group and let $a, b \in G$. The equation $ax = b$ and $ya = b$ have unique solutions for $x, y \in G$. In particular, the left and right cancellation properties hold in G:

1. If $au = av$ then $u = v$
2. if $ub = vb$ and $u = v$

Sometimes, this proposition is called the *Cancellation Law*

***Proof*:**
Simply use uniqueness of inverse. The point of this theorem is that if $ab = e$, for any element $a \in G$ and some element $b \in G$, then $b = a^{-1}$.

This proposition is very powerful, it means we can bring in our elementary algebra knowledge. In this sense, groups are the structure on which we generalize the idea of doing elementary algebra on. They are limited in that they only allow one operation: you probably have worked with equation with both addition and multiplication (ex. $3x^2 - 5 = 0$). We will work with such equations with 2 binary operations once we study *rings* (see part II)

With these properties, I'll take a moment to explain the power of associativity and commutativity. In all type of numbers systems we saw $(\mathbb{N}, \mathbb{Z}, \mathbb{Q}, \mathbb{R})$, for any two numbers a and b:

$$aaabbbbaaabaaaabbbbbaa$$

is unambiguous. and so, we can write

$$a^3b^3a^3ba^4b^5a^2$$

Which is really powerful. If you didn't have associativity, then $aaabbbbaaabaaaabbbbbaa$ would be ambiguous, since it's possible that

$$(aaabbbbaaabaaaabbbbba)a \neq a(aabbbbaaabaaaabbbbbaa)$$

So every pair of elements would have to be bracketed since

$$abc$$

would be ambiguous [7] since it's possible that

$$(ab)c \neq a(bc)$$

and so you would be forced to have an expression like

$$((((aa)b)((bb)(ba)))(((((aa)(ba))a))((a(a(bb)))((b(bb))a))))a$$

with no simpler way of representing the expression – quite a monstrosity. There is more than $2^{22} = 4'194'304$ ways of bracketing this!! The idea is that associativity means that given a sequence of operations that must be performed in a certain order:

$$a_1 * a_2 * a_3 * \cdots * a_n$$

We are allowed to "break up" the problem into smaller problems that can be grouped and operated and then recombined in the right order. For example if we must perform 4 actions in order:

$$a_1 * a_2 * a_3 * a_4$$

Then associativity means that we could have considered $a_3 * a_4 = b_2$ as a new combined actions and $a_1 * a_2 = b_1$ also as a combined action:

$$b_1 * b_2$$

Or we could have taken $a_1 * a_2 = b_1$ and then combine it with the action of a_3, $b_1 * a_3 = b_3$ and then with a_4 and get the same result:

$$a_1 * a_2 * a_3 * a_4 = b_1 * b_2 = b_3 * a_4$$

This can also be thought of as "pre-combining" the steps before performing the result in the same order, which if the operation is associative would give you the same result.

Example 1.1.18: Associativity and The Lack of It

1. An example of real world phenomena where this is not applicable rock-paper-scissors. Let's says the (commutative) binary operation takes in two choices and gives the winner (and if the two input are the same return the input). Then:

   $$(\text{rock} * \text{paper}) * \text{scissors} = \text{paper} * \text{rock} = \text{paper} \neq \text{rock} * \text{scissors} = \text{rock} * (\text{paper} * \text{scissors})$$

2. *Divide and Conquer (Parallel Computing)*: Associativity is key in parallel computing. If a

[7] like $\mathbb{R}^3$ with the cross product

sequence of operations is associative, we don't have to wait for a single processor to calculate from left to right:

$$((((a_1 * a_2) * a_3) * a_4) \dots)$$

Instead, we can dispatch the problem locally to different processors:

(a) Processor 1 solves $a_1 * a_2$ locally.

(b) Processor 2 solves $a_3 * a_4$ locally.

They then combine their local results[a]. Such "reduction" operations must be associative precisely so they can be solved locally on separate servers and combined later without changing the final outcome.

3. Exponentiation is *not* associative: $(a^b)^c$ is not in general equal to $(a)^{b^c}$.

4. The averaging operation is not associative

5. Convince your self that cooking instruction are usually not associative.

[a] Frameworks like Google's MapReduce rely entirely on this

Since functions are associative, most structures in mathematics will be associative, however there are entire domain of study which concentrates on a non-associative structure [8] or drop the condition of associativity[9]. We will briefly see an alternative to associativity for an algebraic structure we will cover in part X, chapter 54, but structures without associativity are outside the scope of this book[10]. As a consequence of associativity, we can unambiguously write:

$$x^{a+b} = x^{b+a}$$

since the order of multiplication doesn't effect the final result. We can also visualize associativity through the use of directed graphs. If we take abc, then we can visualize which tuple of elements we choose to multiply first:

$$\begin{tikzcd}
& ((ab)c) = (a(bc)) & \\
(ab)c \arrow[ur] & & a(bc) \arrow[ul] \\
& abc \arrow[ul] \arrow[ur] &
\end{tikzcd}$$

this is more clear if we have 4 elements, any way we choose to bracket our elements leads to the same

[8] the most famous of is called Lie Algebra, with something called the Jacobi Identity replacing associativity. Another is the logical implications: $(a \to b) \to c$ does not always give the same truth value as $a \to (b \to c)$

[9] Three such examples are *loops*, *Quasigroups*, and *magmas*

[10] See ref:HERE for books covering such topics

result:

$$\begin{tikzcd}
 & & & (abcd) & & & \\
((ab)c)d \arrow[urrr] & & (a(bc))d \arrow[ur] & (ab)(cd) \arrow[u] & a((bc)d) \arrow[ul] & & a(b(cd)) \arrow[ulll] \\
 & (ab)cd \arrow[ul] \arrow[urr] & & a(bc)d \arrow[ul] \arrow[ur] & & ab(cd) \arrow[ull] \arrow[ur] & \\
 & & & abcd \arrow[ull] \arrow[u] \arrow[urr] & & &
\end{tikzcd}$$

Note that adding commutativity makes this even more powerful, since

$$aba = aab = a^2b$$

and so we can group all like terms together, and write

$$aaabbbbaaabaaaabbbbbaa \qquad \text{as} \qquad aaaaaaaaaaaabbbbbbbbbb$$

and collapse it to:

$$a^{12}b^{10}$$

That is, the only information we need is the number of times we multiply by a and b, with no regard to the order we multiply them! This added axiom is so powerful that it will turn out that it will constrain the number of possible (finite and finitely generated) groups, which are relatively few[11] (see section 5.1.1)

or:exercises

(Assuming some basic analysis knowledge) Because of general associativity we have that for any finite number of terms, we can bracket them however we want and will get the same result. Some curious among you might wonder if we can expand this to be able to take an infinite number of terms and have a similar result. However, this immediately becomes tricky. For example, if we take $\mathbb{Z}$, then what is $1+1+1+1+\cdots$? It should be verified that no number in $\mathbb{Z}$ will satisfy this conditions (you can always pick a larger number). Even worse, what if we take $1+1-1+1-1+\cdots$? Due to the infinite nature of this series, it does not have a resolution. We can try to limit ourselves to all infinite series that *converge*. However, there is even a limitation in this case. (move the terms around using commutativity of a non-absolutely convergent sequence). Because of this, the study of figuring out how to "add" infinitely many elements is relegated to another field of mathematics called analysis.

As an exercise to show the power of commutativity, show that if G is abelian, then :$(ab)^{-1} = a^{-1}b^{-1}$. (Once homomorphisms are introduced, show $x \mapsto x^{-1}$ and $x \to x^a$ are also homomorphisms

1.2 Examples of Groups

Now that we covered the basics, it is important to build up a family of common groups to get familiar with the how to recognize groups, and to feel very comfortable manipulating groups. We already

[11]The notion of finitely generated will be covered in section 2.3. This result shall be proved in theorem 5.1.8. As of writing this book, the project of classifying all abelian groups is still a very unsolved problem

showed how many previous mathematical structures were already groups (the canonical examples being $\mathbb{Z}, \mathbb{Q}, \mathbb{R}, \mathbb{C}$). We will now define new groups to show further what is possible with the definition of the group, and to build-up a repertoire of more elaborate examples.

1.2.1 Integers Modulo n and Cyclic Groups

Recall in section 0.2.2 that $\mathbb{Z}/n\mathbb{Z}$ is a collection of equivalences classes satisfying $a \sim_n b$ if and only if there exists a $k \in \mathbb{Z}$ st $a - b = kn$. Given that $\mathbb{Z}$ is a group and the results proposition 0.2.15, it is reasonable to think that it may form a group:

Proposition 1.2.1: Modular Arithmetic's Groups

Let $\mathbb{Z}/n\mathbb{Z}$ be the integesr modulo n. Then $\mathbb{Z}/n\mathbb{Z}$ forms a group under modular addition:

$$(a \bmod n) +_n (b \bmod n) = (a + b) \bmod n$$

with the binary operation denoted as $+_n$, or if the context permits simply as $+$

In almost all cases, the notion $+$ shall be used, for it will be clear from context that the elements will be part of a certain set of modular integers.

***Proof*:**

(Well-defined) By proposition 0.2.15(1), the operation is well-defined

(Idenitty) For any n, there is always $\overline{0}_n \in \mathbb{Z}/n\mathbb{Z}$ for $n - n = 0$. Then for any $\overline{a}_n \in \mathbb{Z}/n\mathbb{Z}$:

$$\overline{0}_n + \overline{a})n = \overline{0 + a}_n$$

Then notice that

$$a - (0 + a) = 0 = kn \qquad (\text{with } k = 0)$$

Thus, $\overline{0 + a}_n = \overline{a}_n$, and so $\overline{0}_n$ is the identity for any $n \in \mathbb{N}_{>0}$

(Inverse) For any $\overline{a}_n \in \mathbb{Z}/n\mathbb{Z}$, pick $\overline{-a}_n$ (i.e., the equivalence class that can be represented by $-a$). Then:

$$(a + -a) - 0 = 0$$

implying that $\overline{a}_n = \overline{-a}_n = \overline{a + (-a)}_n = \overline{0}_n$, showing that $\overline{a}_n$ is indeed the inverse. Usually, a representative between $0 \leq t \leq n - 1$ so that $\overline{-a}_n = \overline{t}_n$.

(Associaivity) let $\overline{a}_n, \overline{b}_n, \overline{b}_n \in \mathbb{Z}/n\mathbb{Z}$. We want to show that

$$(\overline{a}_n + \overline{b}_n) + \overline{c}_n = \overline{a}_n + (\overline{b}_n + \overline{c}_n)$$

expanding the definition of the binary operation, we get:

$$\overline{(a + b) + c}_n = \overline{a + (b + c)}_n$$

In particular, we want to verify that:

$$[(a + b) + c] - [a + (b + c)] = kn$$

which by the associativity of the integers, we get that their difference is 0, which implies both sides are equal, completing the proof.

Definition 1.2.2: Integers Modulo n

The set $\mathbb{Z}/n\mathbb{Z}$ forms a group under modulo addition called *integers modulo n*.

Hence, for each $n \in \mathbb{N}_{>0}$, there is a group with with n elements. It has yet to be explored whether the modulo multiplication is a viable binary operation, namely if:

$$(a \bmod n) \cdot (b \bmod n) = (ab) \bmod n \tag{1.1}$$

can form a binary operation. Some more care must be put in:

Example 1.2.3: Not Always Group

Take $n = 4$. The reader should verify that $\overline{1}_4$ is the identity given the binary operation defined in equation (1.1), and by proposition 1.1.15(1) it is unique. However, notice that $\overline{2}_4$ has *no* multiplicative inverse:

$$\overline{0 \cdot 2}_4 = \overline{0}_4 \qquad \overline{1 \cdot 2}_4 = \overline{2}_4 \qquad \overline{2 \cdot 2}_4 = \overline{0}_4 \qquad \overline{3 \cdot 2}_4 = \overline{2}_4$$

Hence, $(\mathbb{Z}/n\mathbb{Z}, \cdot_4)$ is *not* a group. However, if we pick $n = 3$, notice that this forms a group!

The reader may have observed that 3 is a prime and may wonder if is the key difference. This is in fact close: the important part is to choose all elements that are *coprime* with n:

Definition 1.2.4: Euler Totient

Define $\varphi : \mathbb{N} \to \mathbb{N}$ to be

$$\varphi(n) := |\{k \in \mathbb{N} \mid \gcd(n, k) = 1\}| = \#\{k \in \mathbb{N} \mid \gcd(n, k) = 1\}$$

that is, $\varphi(n)$ counts the number of coprime

The notation φ is often re-used and so shouldn't be considered a "universal" symbol for the Euler Totient.

Definition 1.2.5: Integer Modulo n With Multiplication

Let $\mathbb{Z}/n\mathbb{Z}$ be the integers modulo n and choose representatives for each element so that the n equivalence classes may be represented with $0, ..., n-1$. Take $(\mathbb{Z}/n\mathbb{Z})^\times \subseteq \mathbb{Z}/n\mathbb{Z}$ defined to be

$$(\mathbb{Z}/n\mathbb{Z})^\times = \left\{\overline{t}_n \in \mathbb{Z}/n\mathbb{Z} \;\middle|\; \varphi(t) = 1\right\}$$

Define the binary operation to be:

$$(a \bmod n) \cdot (b \bmod n) = (ab) \bmod n \tag{1.2}$$

Then $(\mathbb{Z}/n\mathbb{Z})^\times$ is a group modulo n with multiplication.

The reader should verify as an exercise that this is indeed a group.

Cyclic Subgroups

Each group G will have something similar to the integers modulo n, which is worth exploring:

Proposition 1.2.6: Creating a Cyclic Subgroup

Let G be a group and take any $x \in G$. Let $\langle x \rangle = \left\{ x^a \middle| a \in \mathbb{Z} \right\}$. Then $\langle x \rangle$ is a subgroup of G

***Proof*:**
I would recommend you do this as an exercise!

Let $\langle x \rangle = \{x^a | a \in \mathbb{Z}\}$. We'll run through the group-axioms:

1. (associativity) Associativity come's from associativity in G. Let $r, s, t \in \langle x \rangle$, and consider

$$(r + s) + t$$

then since $r, s, t \in G$, and G is associative, we get that

$$(r + s) + t = r + (s + t)$$

and since $r, s, t \in \langle x \rangle$ were arbitrary 3 elements, this holds for any elements of $\langle x \rangle$, completing the proof

2. (identity) We need to show that in $\langle x \rangle$, there exists an element e such that $ex^a = x^a$. Since $0 \in \mathbb{Z}$, $x^0 = e \in \langle x \rangle$. And since for any element $y \in x$, $ey = y$, and every x^a is an element of G, $ex^a = x^a e = x^a$

3. (inverse). Let $y \in \langle x \rangle$ be an element of $\langle x \rangle$. Then by definition, for some $a \in \mathbb{Z}$, $y = x^a$. Then since $-a \in \mathbb{Z}$, $x^{-a} \in \langle x \rangle$, and as we've seen earlier:

$$x^a x^{-a} = x^{a-a} = x^0 = e$$

showing that x^a has an inverse for every a.

Thus, $\langle x \rangle$ is a group. Notice that the binary operation is identical to that of G restricted to $\langle x \rangle$, making $\langle x \rangle$ a subgroup of G, completing the proof.

Given any element $x \in G$, we can form a subgroup $\langle x \rangle \leqslant G$. The reason for it being called a cyclic subgroup comes when we consider a finite group G [12]

Example 1.2.7: Why is $\langle x \rangle$ called Cyclic?

Let G be a finite group (i.e. $|G| = n$ for some $n \in \mathbb{N}$), take $x \in G$ and consider $\langle x \rangle \leqslant G$.

1. Show that $\langle x \rangle$ is finite, that is, there is some $k \in \mathbb{N}$ such that $|\langle x \rangle| = k$

2. Show that for any x^{-a}, where $a \in \mathbb{N}$, there exists some number $b \in \mathbb{N}$ such that

$$x^{-a} = x^b$$

[12]More generally, subgroup $\langle x \rangle \leqslant G$ where $|\langle x \rangle|$ is finite

Conclude that $\langle x \rangle$ can be generated by raising number *only* to the power of positive number, i.e.

$$\{x^a \mid a \in \mathbb{Z}\} = \{x^a \mid a \in \mathbb{N}\}$$

3. show that if $|\langle x \rangle| = k$, then $x^k = e$. Use this fact to show that for any $x^a \in \langle x \rangle$, $a \leq k$.

With these facts, we can visualize a finite cyclic subgroup $\langle x \rangle$ using a graph. Let's say 1 is the identity. If we multiply 1 by x $(1 \cdot x)$, we get x. We can represent this visually with

$$1 \xrightarrow{\cdot x} x$$

if we multiply x by x, we get

$$1 \xrightarrow{\cdot x} x \xrightarrow{\cdot x} x^2$$

continuing in this fashion, we will make long "chain":

$$1 \xrightarrow{\cdot x} x \xrightarrow{\cdot x} x^2 \xrightarrow{\cdot x} x^3 \xrightarrow{\cdot x} \cdots$$

until we reach 1, at which point we can circle back to it in the graph:

$$\begin{tikzcd}
1 \arrow[r, "\cdot x"] & x \arrow[r, "\cdot x"] & x^2 \arrow[r, "\cdot x"] & x^3 \arrow[r, "\cdot x"] & \cdots \arrow[r] & x^{k-1} \arrow[lllll, bend right, "\cdot x"']
\end{tikzcd}$$

which more conveniently can be written as

$$\begin{tikzcd}
 & \dots \arrow[dl, "\cdot x"'] & x^3 \arrow[l, "\cdot x"] & \\
x^{k-1} \arrow[dr, "\cdot x"'] & & & x^2 \arrow[ul, "\cdot x"'] \\
 & 1 \arrow[r, "\cdot x"] & x \arrow[ur, "\cdot x"'] &
\end{tikzcd}$$

producing us a cycle!

Since every element can form a cyclic group, and their structure is characterized by the cardinality of $\langle x \rangle$, we will give a special name to this cardinality:

Definition 1.2.8: Order of an element

For a group G and $x \in G$, define the *order of x* to be the smallest positive integer n such that $x^n = 1$, and denote this integer by $|x|$. In this use, x is said to be of order n. If no positive power of x is the identity, the order of x is defined to be infinity and x is said to be of infinite order

Equivalently, we can also say $|x| = |\langle x \rangle|$, that is, the order of x is the order of the group generated by x

Another word for element with finite order is *torsion elements*. Elements that are not torsion elements (i.e. with infinite order) are called *torsion-free elements*.

Example 1.2.9: Order of Elements

1. Every element of $\mathbb{Z}, \mathbb{Q}, \mathbb{R}, \mathbb{C}$ under $+$ has infinite except 0. Is the same true for $\mathbb{R}^\times$ (i.e. $\mathbb{R} - \{0\}$ (answer: [a]) Note the identity always has order 1.
2. Can you find the order of the elements in $\mathbb{Z}/p\mathbb{Z}$ for prime p? What if p is not prime? What are the order of the elements of $\mathbb{Z}/60\mathbb{Z}$?

[a] no, take $-1 \in \mathbb{R}^\times$

1.2.10 Note The order of the element depends on *both* the set and the binary operator. See exercise ref:HERE
Also, the word *order* is sometimes used to mean the cardinality of the group.

I highly suggest you now do a couple of exercises, since these groups will be at the foundation of finding all possible abelian group [13]

Exercise 1.2.1

1. for $n > 1$, $\mathbb{Z}/n\mathbb{Z}$ is not a group under multiplication (hint: think of identity)
2. prove that every element in a finite group has finite order
3. if $x \in G$, G a group, and $|x| = n$, then $x^{-1} = x^{n-1}$.
4. $x \in G$, G a group. Then $|x| = |x^{-1}|$. (hint: $x \cdot x \cdots x = 1$, so keep multiplying by x^{-1} till you get $1 = x^{-1} \cdot x^{-1} \cdots x^{-1}$ If x has infinite order). Though in certain infinite order groups, like $\mathbb{R} - \{0\}$ under multiplication, -1 has order 2, but 1 has order 1. Is this only true when the operation is $+$?
5. 21. is there a catch? why should n be odd?
6. $x, g \in G$, G a group. Then $|x| = |g^{-1}xg|$. Deduce that $|ab| = |ba|$ for all $a, b \in G$. (hint: simply assign an order to x, and expand $g^{-1}xg$. things will cancel out nicely. For the second part, start with ab, and multiply like so $babb^{-1}$. By what you've shown, this has the same order).
7. $x \in G$, $|x| = n < \infty$. If $n = st$ for some positive integers s and t, prove that $|x^s| = t$. (hint: how many times do you need to multiply x^s to get to 1)
8. if $x^2 = 1$ for all $x \in G$ then G is abelian.
9. if x is an element of the group G, then $H = \{x^n | n \in \mathbb{Z}\}$ is a subgroup (cf. show for all $h, k \in H$, $hk \in H$, $h^{-1} \in H$) of G (called the *cyclic subgroup* of G generated by x)
10. $A \times B$ is abelian if and only if A and B is abelian
11. Let $A \times B$ be the *direct product* group formed by the cartesian product of the group A, B as shown in example 1.1.5. Show that $(a, 1)$ and $(1, b)$ commute. Deduce that the order of (a, b)

[13] In particular, finitely generated abelian group, see section ref:HERE

is the least common multiple of $|a|$ and $|b|$.

12. any finite group G of even order contains an element of order 2. (hint: let $t(G)$ be the set $\{g \in G | g \neq g^{-1}\}$. Show that $t(G)$ has even number of elements and every non-identity element of $G - t(G)$ has order 2).

13. Other exercises you can do I'd recommend: 32, do 33, 34, 35, 36

14. Challenging exercise. Let A be a set containing numbers. Define the following

$$\mathcal{A} = \bigoplus_{i=1}^{\infty} A$$

to be the set of infinitely long tuples were *all but finitely many* elements are non zero. For example, if $A = \{1, 2, 3\}$, then

$$(0, 0, 1, 0, 2, 0, 0, 0, 0, 1, 3, 3, 2, 0, ...)$$

where the ... means there are zeros going on forever. Then this tuples is in $\mathcal{A}$. On the other hand

$$(1, 1, 1, 1, 1, ...)$$

where the ... means the ones are going on forever is not in $\mathcal{A}$.

If G is a group, let

$$\mathcal{G} = \bigoplus_{i=1}^{\infty} G$$

be the set of tuples where all but finitely many elements are non-zero, and define the operation on $\mathcal{G}$ to be pointwise:

$$(a_1, a_2, a_3, ...) \cdot (b_1, b_2, b_3, ...) = (a_1b_1, a_2b_2, a_3b_3, ...)$$

Show that $\mathcal{G}$ along with this operation is a group.

15. Let G be a finite group. Show that G can be generated by elements to the power of $\mathbb{N}$ (move into the generators and relations section maybe?)

16. In some textbooks, groups are introduced more through the use of *graphs*. We will present these ideas here. (Google Frucht's Theorem)

17. Let p be a prime number. Show that $(\mathbb{Z}/p\mathbb{Z})$ is a cyclic group.

18. Let $n = 8$. What is $(\mathbb{Z}/8\mathbb{Z})^{\times}$? What if $n = 9$. Conclude that in general, $(\mathbb{Z}/n\mathbb{Z})^{\times}$ is not necessarily cyclic, but it can be.

19. (perhaps a question on the structure of $(\mathbb{Z}/n\mathbb{Z})^{\times}$ in general?)

1.2.2 Symmetric Groups

Let X be any set. As covered in the preliminary chapter, a bijective function $f : X \to X$ always has a (two-sided) inverse function[14], say f^{-1}. Furthermore, function composition is associative, and since

[14] This in fact characterizes bijective functions

the domain and codomain match, the composition of two bijective functions $f, g : X \to X$ give's another bijective function $g \circ f : X \to X$ (as the reader ought to check if they're not convinced). The final property to check for the collection of bijective functions on a set X to form a group is whether there exists a bijective function $\mathrm{id}_X : X \to X$ such that for all bijective functions $f : X \to X$, $\mathrm{id}_X \circ f = f \circ \mathrm{id}_X = f$. The reader should take a moment to think what this function

1.2.11 Moment Taking a moment

The reader may have noticed that this function arises naturally by taking $f \circ f^{-1}$, namely it's the function defined to be $s \mapsto s$. Thus, given any set X, there is a natural group structure on the collection of bijections on X. This set is given a name:

Definition 1.2.12: Symmetric Group

Let X be any set. Then then:

$$S_X := \{f : X \to X \mid f \text{ is a bijection}\}$$

is called the *symmetric group* or *premutation group* of X [with respect to composition $\circ$]. If the cardinality of X is finite, say $|X| = n$, then we denote S_X by S_n.

Other common notations for the symmetric group are:

$$\mathrm{Bij}(A) \qquad \mathrm{Sym}(A) \qquad \mathrm{Aut}_{\mathbf{Set}}(A)$$

If X is finite, then the bijections S_n are determined only by the place to which elements map to and not by any property of the elements, making the bijections only dependent on the number of elements and hence the S_n notation being unambiguous. For this reason, it is often assumed that the set over which the bijections are being taken is given by[15]:

$$\{1, 2, ..., n\}$$

A mention on notation is in order: if we take $f, g \in S_X$, then the current short-hand notation for groups would suggest that the product of f and g ought to be fg and represents the short-hand $f \circ g$. However, the prevailing notation for compositions reads the functions *right-to-left*:

$$f * g = g \circ f \qquad \Rightarrow \qquad (g \circ f)(x) = g(f(x))$$

This notational quirk comes from the fact that the conventions for function composition comes much earlier than the convention for group operations. Mathematically, this shall not effect the underlying theory[16].

Symmetric groups are rather large; the number of bijections between finite sets grows:

$$|S_n| = n!$$

And hence even for $|X| = 100$, $|S_{100}| \gg 10^{100}$. For $n \in \{1, 2\}$, the bijections are rather simple, namely:

$$f_1 : \{1\} \to \{1\},\ f_1(1) = 1 \qquad g_1 : \{1,2\} \to \{1,2\},\ g_1 = \mathrm{id}_{\{1,2\}} \quad g_2(1) = 2\ g_2(2) = 1$$

[15] This set is also sometimes called the *indexing set* and represents labeling each element of some set of size n

[16] It shall complicate notation a little bit when defining left-group actions, which shall be addressed in due time

For larger sets, it may get difficult to track the information a bijection holds. For example, if σ is a bijection on a set with 9 elements:

$$\{1, 2, 3, 4, 5, 6, 7, 8, 9\}$$

then we after the σ permutes, we might get

$$(\sigma(1), \sigma(2), \sigma(3)\sigma(4), \sigma(5), \sigma(6), \sigma(7), \sigma(8), \sigma(9)) = (2, 4, 1, 3, 7, 9, 8, 5, 6)$$

It is cumbersome to write-out the permutation by saying what each value of $\sigma \in S_n$ maps to. Instead, there is the following *matrix notation* as a shorthand:

$$\begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 & 9 \\ 2 & 4 & 1 & 3 & 7 & 9 & 8 & 5 & 6 \end{pmatrix} \tag{1.3}$$

Representing f_1, g_1, g_2 in matrix notation, we would get:

$$\begin{pmatrix} 1 \\ 1 \end{pmatrix} \qquad \begin{pmatrix} 1 & 2 \\ 1 & 2 \end{pmatrix} \qquad \begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix}$$

This is better notation, however we can do even better taking advantage of the fact that a bijection is one-to-one and onto. This implies that every element is only associated with one element and each element must be represented. This allows for the following "cycle" notation. Take equation 1.3, and notice that 1 maps to 2, which maps to 4, which maps to 3, which maps to 1:

$$1 \mapsto 2 \mapsto 4 \mapsto 3 \mapsto 1$$

Furthermore, 5 maps to 7, which maps to 8, which maps to 5:

$$5 \mapsto 8 \mapsto 7 \mapsto 5$$

Finally, 6 maps to 9, and 9 maps to 6:

$$6 \mapsto 9 \mapsto 6$$

This behavior can be represented via the following *cycle notation*:

$$(1\ 2\ 4\ 3)(5\ 8\ 7)(6\ 9)$$

Each "chain" in the above is called a *cycle*, and the combination of each is called the *cycle representaiton* If a cycle contains n elements, it is called an n-cycle. In the above example, there were not any elements that map to itself. If the element 10 is added to the set and we extend σ to $\overline{\sigma}$so that $\overline{\sigma}(10) = 10$, then we would write $(1\ 2\ 4\ 3)(5\ 8\ 7)(6\ 9)(10)$. By convention, any element that maps to itself is dropped in cycle notation, and hence $\overline{\sigma}$ would in cycle notation also be represented as:

$$(1\ 2\ 4\ 3)(5\ 8\ 7)(6\ 9)$$

Hence, there is an inherent ambiguity in cycle notation on the size of the domain/codomain, since any elements that map to themselves is not taken into accost, however it is usually the element that map to a non-trivial element that are of interest and hence this ambiguity does not effect the theory. Note that:

$$(1\,2\,3) = (3\,1\,2) = (2\,3\,1)$$

Since they all actually do the same permutation, you're just starting at different positions when converting to the cycle notation.

Example 1.2.13: Cycle Representation

1. Let τ be a bijective function on a set $\{1, 2, 3, 4, 5\}$ such that:

$$(\sigma(1), \sigma(2), \sigma(3), \sigma(4), \sigma(5)) = (5, 4, 3, 2, 1)$$

   What is the cycle notation for τ

2. Cycles can be used to define permutations. If $X = \{1, 2, 3, 4\}$. Find the bijection represented by the cycle representation:

$$(1\ 2\ 3)(2\ 3\ 4)$$

   Notice that the two cycles overlap. To find τ, first find where 1 maps to in the right-cycle, and the where the resulting element maps to the left cycle; in function notation if f is the left cycle and g is the right cycle then the reader is looking for $f(g(1))$. Proceed with this for all 4 element. Remember that if an element is not in a cycle, then it is considered to map to itself.

3. Is it true that the functions represented by $(1\,2)(3\,4)$ and $(3\,4)(1\,2)$ are equal, that is:

$$(1\ 2)(3\ 4) = (3\ 4)(1\ 2)$$

   More generally, do disjoint cycles commute?

To insure the reader has the right calculation, here is an example of the product of two cycles as reference:

$$(1\ 4\ 5\ 2)(4\ 1\ 3) = (1\ 3\ 5\ 2)$$

With those extra properties of cycles, we will name one more concept for ease of labeling cycles:

Definition 1.2.14: Cycle Type

Given a cycle-representation of a permutation, you can make the cycles disjoint. Then, order them from smallest to largest. Then the cycle representation is the "length" of each cycle in that order.

Example 1.2.15: Cycle-Type

$(1\,2\,4\,3)(5\,7\,8)(6\,9)$ is equivalent to $(6\,9)(5\,7\,8)(1\,2\,4\,3)$ which we can label as a $(2, 3, 4)$-cycle type. If you have a $(1\,2)(3\,4)$ premutation, then it's a $(2, 2)$-cycle type.

A cycle of particular importance are 2-cycles.

Definition 1.2.16: Transposition

2-cycles are usually referred to as ***transpositions***

Using transpositions, any n-cycle can be decomposed into the product of transpositions. For example, take:

$$(1\,2\,4\,3)$$

Then the reader should verify that:

$$(1\,3)(1\,4)(1\,2) = (1\,2\,4\,3)$$

So, the cycle broke up into 3 transpositions. In fact, any n-cycles will break up into $n-1$ transpositions. It will break up to exactly (at most and at least) $n-1$ transpositions.

Theorem 1.2.17: n-Cycle Breakdown To Exactly $n-1$ Transpositions

Let $\sigma \in S_n$ be an n-cycle. Then σ can be broken down to the composition of exactly $n-1$ transpositions

***Proof*:**
exercise

Because of this, given any σ, which can then be written as disjoint cycles, and then into transpositions will always have a fixed number of transpositions in the product, in other words this is an invariant of a cycle σ, which deserves then a name:

Definition 1.2.18: Parity Of Cycle

Given $\sigma \in S_n$, then the parity of the cycle is the parity of the number of transpositions so if we have an n-cycle, where n is even, then it's parity is odd, since it will break down to $n-1$ transpositions. A cycle with even parity is called an *even permutation*, similarly called *odd-permutation* when the parity is odd.

Parity might seem like a weird thing to care about, but it will in fact come in very useful. So useful in fact, that there's a special name for groups based around parity:

Definition 1.2.19: Alternating Group

Let S_n be the permutation group permuting n Elements. Then $A_n \subseteq S_n$ is all cycles with *even* permutations. Note that the compositions of two cycles from A_n is even, and that the inverse of an even permutation is even, thus

$$A_n \leqslant S_n$$

Why this group is interesting requires more algebraic tools, so we'll leave at defining it for now and get back to it to show it's use.

Before moving on to properties of symmetric groups, some reader may find it interesting that at the origins of group theory, groups were defined to be subgroups of symmetric groups! Later, ________and ________have abstracted these properties and formed the modern axiomatic representation. Later in this book, it shall be shown that all groups are in fact subgroups of the appropriate symmetric group (see theorem 4.5.3).

Example 1.2.20: Interesting Exercises on Commutativity

We earlier mentioned how being commutative is a nice property, and that groups which are commutative are quite nice. It was also mentioned that S_n is in a sense the most general group that possible. So, in a sense, if we can establish properties of S_n for particular orders, that can be quite insightful.

Here's one such property worth proving. If we take S_n, $n \geq 3$, then S_n is *not* commutative To prove this, try composing a n-cycles and a 2-cycle which are not disjoint form the left and right and see what you get

Properties of Symmetric Groups

Proposition 1.2.21: Order of element is p iff composition of disjoint p-cycles

Let p be a prime. Show that an element of S_n has order p if and only if its cycle decomposition is a product of (commuting) p-cycles.

***Proof*:**
exercise

Note that if p is not prime, the "if and only if" is not necessarily true.

Example 1.2.22: Counter-Examples

1. Take
$$s = (1\,2)(3\,4\,5\,6) = s_1 s_2$$
Then $(s_1 s_2)^4 = 1$, but $|s_1| = 2$.
2. Take $(1\,2)(3\,4\,5) \in S_5$. Then
$$\begin{aligned}
(1\,2)(3\,4\,5)^2 &= (3\,5\,4)\\
(1\,2)(3\,4\,5)^3 &= (1\,2)\\
(1\,2)(3\,4\,5)^4 &= (3\,4\,5)\\
(1\,2)(3\,4\,5)^5 &= (1\,2)(3\,5\,4)\\
(1\,2)(3\,4\,5)^6 &= id
\end{aligned}$$
but $6 > 5$. Thus, we found a cycle of order 6 in S_5. The trick was to find two disjoint cycles such that the $\mathrm{lcm}(\sigma, \tau) > n$.

Proposition 1.2.23: σ is p-cycle for prime p then σ^k is p-cycle

If $\sigma \in S_n$ has prime order, then σ^k for every $1 \leq k \leq p$ is a p-cycle

***Proof*:**
This is a simple application of proposition 1.2.21, and since we're taking $\langle\sigma\rangle$, we'll use theorem 2.3.11 (the theorem on cyclic groups). Since $|\langle\sigma\rangle| = p$, then $\langle\sigma^k\rangle = p$ for every $k < p$ ($\sigma^p = e$). This means that σ^k has prime order. Thus, by proposition 1.2.21, it is composed of p-cycles. Since $\langle\sigma\rangle = p$, and σ is a bijection (meaning we never go to another element while composing σ), then every σ^k is a p-cycle

Note that if σ is not a p-cycle, this is not necessarily true. For example:

$$(1\,2\,3\,4)\,,(1\,2\,3\,4)^2 = (1\,3)(2\,4)\,,(1\,2\,3\,4) = (1\,4\,3\,2)$$

Notice how σ^2 broke down to two 2-cycles. The way I remember this to recall that if σ is a 4-cycle, then $|\sigma^2| = 2$, so $\sigma^2\sigma^2 = id$, which by what we've covered in the previous proposition (1.2.21), σ must be disjoint 2-cycles!! This holds true for any n-cycle, giving you an easy way of thinking about σ^k for any n-cycle with $1 \leq k \leq n$.

Proposition 1.2.24: Conjugation preserves permutation-type

If $\sigma, \tau \in S_n$, then

$$\sigma\tau\sigma^{-1} = (\sigma(1)\,\sigma(2)\,\cdots\sigma(t_k))$$

i.e., it's like applying the function σ to every element of τ

***Proof*:**
This proof relies on this key observation:

$$(a\,b)(a\,c)(a\,b)^{-1} = (a\,b)(a\,c)(a\,b) = (b\,c)$$

that is, we shifted a to b (so in a sense if $\sigma = (a\,b)$ and $\tau = (a\,c)$, $\sigma\tau\sigma^{-1} = (\sigma(a), \sigma(b))$).

The rest of this proof is now a matter of keeping track of notation [a]

[a] This proof is easier prove once your introduced to group action and conjugacy classes (see definition 4.6.1)

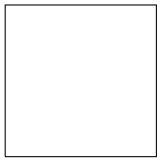
1.2.3 Dihedral Groups

In classical geometry, an n-gon is a 2D shape with all sides of the same length. For example, a triangle is a 3-gon, a square is a 4-gon, a pentagon is a 5-gon, and so forth, while a rectangle and a circle are not n-gons. Each of these shapes can be rotated or reflected in such a way that we get back the same shape: these are called the *rigid-motion symmetries* or just *symmetries* of the n-gon. The term *rigid motion* comes from analysis/linear-algebra where a function $f : X \to X$ is called a *rigid motion* has the property that the distance between each point remains the same after the application of the function. Notationally, this would be represented as:

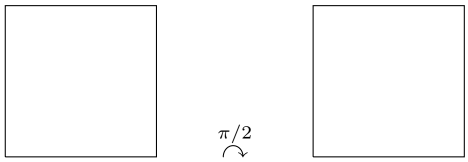
$$\|f(x) - f(y)\| = \|x - y\|$$

where the $\| - \| : X \to \mathbb{R}_{>0}$ is a special kind of function called the *norm* which defines size and distance information on a space[17]. We shall not use rigid motions explicitly, however we shall insure that when transforming n-gons, their they shall never stretch (ex. a square shall not become a rectangle). We shall further place the restriction that the rigid motions of the n-gons have to map to themselves, this is another way of saying that it is a *symmetry*. For example, if we choose a square (a 4-gon):

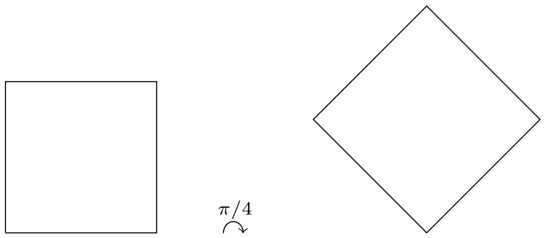
[17] Norms are introduced formally in *Linear Algebra* [17].



and rotated it a quarter turn to the right (90 degrees to the right or $\pi/2$ radians to the right), then we'd get back a square



if we were to rotate it an eighth of a turn, we would not get the exact shape back:



Let's label quarter turns to the right r (for **r**otate). Compounding rotations, the resulting square get's rotated by π, $3\pi/2$, and 2π radians; each of these can be labeled r^2, r^3, and r^4 respectively. Another symmetry of the square would be reflections, namely there is a vertical, horizontal, and two diagonal reflections: $\sigma_h, \sigma_v, \sigma_{d1}, \sigma_{d_2}$. These are not all the symmetries of a square (for example, we transpose the left-edge with the right edge), however it is the only symmetries that are also *rigid motions* (the reader who is already comfortable with norms form linear algebra is free to prove this fact).

More formally, we may represent any n-gon with n-points, with the restrictions that any symmetry (i.e. bijection) f applied to these n-points must further follow the restrictions that the n-gon these points represent must transform as a rigid-motion (all points must remain equi-distant). This special subset of S_n with this added restriciton is called the *dihedral group*:

Definition 1.2.25: Dihedral Groups

Let S_n be the symmetric group on n elements and take the subset $D_n \subseteq S_n$ for which the premutations further follow the restriction of rigid-motions (so that these points represent an n-gon).Then D_n is called the *dihedral group*.

In order for the set D_n to be a group, one ought to prove this is indeed the csae

***Proof*:**

1. Identity comes from the identity function $\mathrm{id}_n \in D_n$ since it is certainly a rigid motion.
2. The inverse come from the fact that if σ is a rigid motion, it's inverse is as well

3. The set is associative since its a collection of functions
4. The composition of two rigid motions is a rigid motion, and hence the binary operation is well-defined

Proposition 1.2.26: Properties of D_n

Let $G = D_n$. Let f represent rotation by by $2\pi/n$, and s represent a reflection about some fixed axis. Then:

1. $1, r, \ldots, r^{n-1}$ are all distinct and $r^n = 1$, so $|r| = n$
2. $|s| = 2$
3. $s \neq r^i$ for any i
4. $sr^i \neq sr^j$ for all $0 \leq i, j \leq n-1$ with $i \neq j$, so
$$D_n = \left\{1, r, r^2, \cdots, r^{n-1}, , sr, sr^2, \cdots, sr^{n-1}\right\}$$
i.e. each element can be written uniquely in the form $s^k r^i$ for some $s = 0$ or 1, and $r = 0, \cdots, n-1$
5. $rs = sr^{-1}$. In particular this shows D_n is not abelian!
6. $r^i s = sr^{-i}$ (use induction)

***Proof*:**
exercise

In particular, we know now any element in D_n can be written as $s^k r^i$, with $k = 0 \vee 1$ and $0 \leq i \leq n-1$. For example, if $n = 12$:

$$(sr^9)(sr^6) = sr^9 sr^6 = sr^9 r^{-6} s = sr^3 s = s^2 r^{-3} = r^{-3} = r^9$$

This group will come up very often since it's a particularly simple.

Proposition 1.2.27: Order of Dihedral Groupq

Let D_n be a dihedral group. Then:
$$|D_n| = 2n$$

***Proof*:**
exercise.

Some authors like to put the order of the Dihedral group in the subscript, that is they write D_{2n}. This convention is rather confusing given the original intuition behind Dihedral groups, and so the D_n notation is used instead for the clarity it brings.

1.2.4 Quaternion Group

The Quaternion (or Hamiltonian) group is a group with 2 fundamental interpretation:

1. The Quaternions are a generalization of the complex numbers! We can think of the fact that $i^2 = -1$ to be the key property which distinguishes the complex numbers from other number systems. If we take $C = \{-1, 1, i, -i\} \subseteq \mathbb{C}$, then
$$\langle C | i^2 = -1 \rangle$$
will form the group which represents the rotation of $\pi/2$ in the cartesian plane. William Hamilton around the 1840s started a project to try an make 3D numbers $(1, i, j)$, the same way $\mathbb{C}$ can be thought of as 2D $(1, i)$. He couldn't manage to make them 3D, but realized the key generalization is making them "4D" $(1, i, j, k)$. He had this flash of inspiration near a bridge in 1843 and wrote down the idea under that very bridge. Today, there is a plaque commemorating the moment of inspiration.

2. It got discovered later that Quaternions efficiently capture the idea of rotating a 3D object in 3D space. At the beginning of vector calculus, some mathematician considered using quaternion instead of vectors as the pedagogical tool used to introduce the concept of 3D rotation, but history went with vector notation. With leaps in quantum mechanics in the late 20th century, quaternions were also used to represent *spin*, a fundamental property of sub-atomic particles.

Let:
$$Q = \{1, -1, i, -i, j, -j, k, -k\}$$
Such that the following relations hold:
$$i^2 = j^2 = k^2 = -1 \tag{1.4}$$
$$1 \cdot a = 1 \cdot a \qquad \forall a \in Q_8 \tag{1.5}$$
$$ij = k \qquad ji = -k \tag{1.6}$$
$$ik = -j \qquad ki = j \tag{1.7}$$
$$jk = i \qquad kj = -i \tag{1.8}$$

Or, in terms of group presentation:
$$Q_8 = \langle Q | i^2 = j^2 = k^2 = ijk = -1 \rangle$$

This group is interesting, for it's the 2nd non-abelian group you've seen with order 8, D_8 being the other group. Furthermore, both of them share a similar intuitive background, being the based on rotation. However, Q_8 rotates 3D objects, while D_8 rotates 2d objects, which actually produce different group structures, a these Cayley tables show:

Example 1.2.28: Comparing D_8 and Q_8

Element	1	-1	i	$-i$	j	$-j$	k	$-k$
1	1	-1	i	$-i$	j	$-j$	k	$-k$
-1	-1	1	$-i$	i	$-j$	j	$-k$	k
i	i	$-i$	-1	1	k	$-k$	$-j$	j
$-i$	$-i$	i	1	-1	$-k$	k	j	$-j$
j	j	$-j$	$-k$	k	-1	1	i	$-i$
$-j$	$-j$	j	k	$-k$	1	-1	$-i$	i
k	k	$-k$	j	$-j$	$-i$	i	-1	1
$-k$	$-k$	k	$-j$	j	i	$-i$	1	-1

Figure 1.1: Q_8 Multiplication Table

While D_8 has

Element	e	a	a^2	a^3	x	ax	a^2x	a^3x
e	e	a	a^2	a^3	x	ax	a^2x	a^3x
a	a	a^2	a^3	e	ax	a^2x	a^3x	x
a^2	a^2	a^3	e	a	a^2x	a^3x	x	ax
a^3	a^3	e	a	a^2	a^3x	x	ax	a^2x
x	x	a^3x	a^2x	ax	e	a^3	a^2	a
ax	ax	x	a^3x	a^2x	a	e	a^3	a^2
a^2x	a^2x	ax	x	a^3x	a^2	a	e	a^3
a^3x	a^3x	a^2x	ax	x	a^3	a^2	a	e

Figure 1.2: Q_8 Multiplication Table

which the reader will notice is different

We will actually be able to show that two groups are not the same once we introduce the concept of isomorphisms (section 1.4).

Exercise 1.2.2

1. Show that there exists 4 homomorphisms $\varphi : Q_8 \to \mathbb{R}^\times$

1.2.5⋆ Matrix Groups

⋆ *Optional. This subsection is an aside; nothing later in the book depends on it.*

This section assumes either some linear algebra knowledge or that you have internalized some of section 0.5 (however, some of these collection of objects goes beyond that section). All prerequisite knowledge assumed in this section shall be covered in great detail in chapter 13 and section 13.2; for the inner-product background behind the orthogonal and unitary groups, see *Linear Algebra* [17]. Let's first recall the notation we use to describe the collection of matrices:

Definition 1.2.29: Matrix Group Under Addition

Let G be an abelian group. Then The set $M_{n\times m}(\mathbb{F})$ forms a group under element-wise operation of G, namely:

$$A * B \equiv (a_{ij} * b_{ij})_{ij}$$

If the reader is familiar with linear algebra, they may notice this group is not actually that interesting: it is equivalent to the group G^{nm}[18]. Sticking to $G = k$ is a field (notice that all fields are Abelian groups under addition), the more interesting binary operation that can be performed on matrices is matrix multiplication, since it combines elements in much more interesting ways. As you might recall, matrix multiplication requires dimensions to line up properly, so for associative to hold, we'll have the dimensions be the same. Furthermore, the identity must be I, the matrix with 1's on the diagonal, and zero's everywhere else. If there are matrices that multiply and *don't* get to I, then they cannot be in the group. Hence, the subset of invertible matrices from $M_n(k)$ must be selected in order to make a group, which the reader may recall from linear algebra implies we are taking the set for which all matrices have a non-zero determinant

Definition 1.2.30: General Linear Group

$$GL_n(F) = (\{A | A \text{ is an } n \times n \text{ matrix with entries from } F \text{ and } \det(A) \neq 0)\}$$

We will also name 3 special subgroups of GL_n since it comes up often enough. Check that these are indeed groups:

Definition 1.2.31: Special Linear Group

$$SL_n(k) = (\{A | A \text{ is an } n \times n \text{ matrix with entries from } F \text{ and } \det(A) = 1)\}$$

If $k \in \{\mathbb{R}, \mathbb{C}\}$, then geometrically these can be thought of as the volume and orientation preserving functions.

Definition 1.2.32: Orthogonal Group

$$O_n(\mathbb{F}) = \big(\{A | A^t A = AA^t = I\big)\}$$

In this group, the transformations are *distance-preserving*, which as an exercise the reader can check that for $O_2(\mathbb{R})$, all transformations are rotations and reflections across lines passing through the

[18]We can actually show this once we get to isomorphisms (definition 1.4.12)

origin[19]. Limiting to orthogonal matrices with determinant 1 gives us the set of all rotations:

Definition 1.2.33: Special Orthogonal Group

$$SO_n(\mathbb{F}) = \{A | A^t A = AA^t = I \,, \det(A) = 1\}$$

If the underlying field is clear, we sometimes write $SO(n)$.

Exercise 1.2.3

1. 6, 7. 11 If you feel like getting comfortable with Heisenberg groups
2. (If you know the dot product) Recall that if v, w are vectors then the dot product can be represented as $v \cdot w = v^T w$ where v^T is the transpose. Show that matrices in $O(\mathbb{R})$ preserve the dot product (namely $Av \cdot Aw = v \cdot w$).

Proposition 1.2.34: Order of $GL_n(\mathbb{F}_p)$

Let $G = GL_n(\mathbb{F}_p)$ where $\mathbb{F}_p$ is a finite field (and thus, has prime order). Then

$$|GL_n(\mathbb{F}_p)| = \prod_{i=0}^{n-1} (p^n - p^i) = (p^n - 1)(p^n - p)(p^n - p^2) \cdots (p^n - p^{n-1})$$

***Proof*:**
The proof for this comes down to remembering that $GL_n(\mathbb{F}_p)$ has linearly independent columns since for any $A \in GL_n(\mathbb{F}_p)$ $\det(A) \neq 0$. We will first show it for $n = 2$. We can separate it in cases where matrices are definitely invertible, and matrices which might not might be invertible. Matrices which only have 1 entry are always non-invertible. These type of matrices are always invertible (since their determinant is always non-zero):

$$\begin{pmatrix} a & 0 \\ 0 & b \end{pmatrix} \quad \begin{pmatrix} 0 & a \\ b & 0 \end{pmatrix} \quad \begin{pmatrix} a & b \\ c & 0 \end{pmatrix} \quad \begin{pmatrix} a & b \\ 0 & c \end{pmatrix} \quad \begin{pmatrix} 0 & a \\ b & c \end{pmatrix} \quad \begin{pmatrix} a & 0 \\ b & c \end{pmatrix}$$

for $a, b, c > 0$ However

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix}$$

Is not invertible if the first column is a scalar multiple of the second.

There are a total of $(p-1)^2$ types of unique columns (corresponding to the $p-1$ possibilities per element in the first column). Since the scalars are in $\mathbb{Z}/p\mathbb{Z}$, then there is $p-1$ unique scalar multiples of any row. Since there are $(p-1)^4$ total non-zero matrices, then we have

$$(p-1)^4 - (p-1)^3$$

invertible matrices with all non-zero entries.

[19] Higher dimensional analogues of "rotation" and "reflection" come from taking this interpretation in lower dimensions and claiming it generalizes to higher dimensions

What's left is to calculate the other types of matrices which are

$$2(p-1)^2$$
$$4(p-1)^3$$

So we have a total of

$$(p-1)^4 - (p-1)^3 + 2(p-1)^2 + 4(p-1)^3$$

possible invertible matrices.

TBD

Exercise 1.2.4

1. Show that there is a bijective correspondence between the platonic solids and the finite subgroups of $SO(3)$ that are not cyclic or dihedral, hence the platonic solids is a special case of the classification "non-trivial" finite subgroups of $SO(3)$, where non-trivial here is a bit more than just the trivial subgroup.

1.3 Generators and Relations

(When dealing with numbers, they can represent anything, for example 5 apples, 5 cars, 5 textbooks, and so forth. However, when we defined the dihedral group. it was specifically linked to the rotation and reflections of n-gons. In this section, we will show a way to abstract away this dependency on a concrete object to define a group).

Notice that D_n can be written with r and s (reflections and rotations). Similarly, $\langle x \rangle \leqslant G$ is an element from which we can find every other element of $\langle x \rangle$. In a sense, they *generated* the group. This notion will be treated rigorously with free groups later (section 2.3.3), but intuitively this idea makes sense. For now, here how we'll define them

Definition 1.3.1: Generators

A subset S of elements of a group G with the property that every element of G can be written as a (finite) product of elements of S and their inverses is called the set of *generators* of G. If $x \subseteq G$ generates x, then we write $\langle x \rangle$. Similarly, if $x, y \in G$ are needed to generate G, we write $\langle x, y \rangle$. More generally, if $A \subseteq G$ generates G we write $\langle A \rangle$.

Example 1.3.2: Generators

1. An easy example would be 1 for $\mathbb{Z}$, since every $x \in \mathbb{Z}$ could be written as adding 1 or it's inverse. So $\mathbb{Z} = \langle 1 \rangle$.

2. By property (4) for D_n, $S = \{r, s\}$ is a set of generators of D_n, so $D_n = \langle r, s \rangle$.

Later, we'll see for finite groups that we don't need the inverse operation to generate finite groups.

For any set of generators, we have a set of relationships that they satisfy[20]. For D_n, they are

$$r^n = 1, \qquad s^2 = 1, \qquad rs = sr^{-1}$$

Why are these the relations? Because any additional relation between elements of groups may be derived from these three (p.40). Note that there can be more than one set of relations for a group (we'll see later that D_n has another set of relations that can be used).

Definition 1.3.3: Presentation of a group

if some group G is generated by a subset S and there is some collection of relations, say $R_1, \cdots, R_m$, which are all equations with elements from $S \cup \{1\}$, such that any relation among the elements of S an be deduced from these, we shall call these generators and relations a *presentation* of G, and write

$$G = \langle S | R_1, \cdots, R_m \rangle$$

Example 1.3.4: Presentation

1. Show that the presentation:

   $$\langle r, s | r^n = 1, s^2 = 1, rs = sr^{-1} \rangle$$

   represents the Dihedral group D_n

2. Let A be any set. Show that

   $$\langle A | ab = 1, \forall a, b \in A \rangle$$

   is group. In particular, show that from any set, we can define the trivial group. In section 1.2.2 and 2.3.3, we'll show two other (those times non-trivial) examples of creating groups from arbitrary sets.

1.3.5 Prelude to Word Problem the right-hand side is, in the general case, a collection of arbitrary equations. It may be hard to determine when two elements in the group are equal! it may also be hard to determine the order of the group. For example

$$\langle x, y | x^2 = y^2 = (xy)^2 = 1 \rangle$$

is a presentation of a group of order 4, whereas

$$\langle x, y | x^3 = y^3 = (xy)^3 = 1 \rangle$$

is a presentation of an infinite group (exercise)

It might be tempting to only work with presentations, or to try and come up with every possible group working with generators. A considerable amount of time has been poured into this problem, and absolutely incredibly, it turns out that you can come up with presentations which are *insoluble*:

[20]With only one exception. When a set of generators has no relationships, we call the group generated by that set a *free group*

Example 1.3.6: Insoluble Presentation

The group representation by this monstrous presentation:

$$\begin{array}{llll}
\langle & a, b, c, d, e, p, q, r, t, k & | & \\
& p^{10}a = ap, & pacqr = rpcaq, & ra = ar, \\
& p^{10}b = bp, & p^2adq^2r = rp^2daq^2, & rb = br, \\
& p^{10}c = cp, & p^3bcq^3r = rp^3cbq^3, & rc = cr, \\
& p^{10}d = dp, & p^4bdq^4r = rp^4dbq^4, & rd = dr, \\
& p^{10}e = ep, & p^5ceq^5r = rp^5ecaq^5, & re = er, \\
& aq^{10} = qa, & p^6deq^6r = rp^6edbq^6, & pt = tp, \\
& bq^{10} = qb, & p^7cdcq^7r = rp^7cdceq^7, & qt = tq, \\
& cq^{10} = qc, & p^8ca^3q^8r = rp^8a^3q^8, & \\
& dq^{10} = qd, & p^9da^3q^9r = rp^9a^3q^9, & \\
& eq^{10} = qe, & a^{-3}ta^3k = ka^{-3}ta^3 & \rangle
\end{array}$$

Figure 1.3: Group for which we cannot write out what the elements are

has elements which we *cannot* write out!

The details of this is are well beyond this course, but it's existence is still interesting to know about

https://en.wikipedia.org/wiki/Word_problem_for_groups

You can get familiar with presentations by doing exercises in the book (p.41,42).

Proposition 1.3.7: Generators For Symmetric Group

$$S_n = \langle (i, i+1) \mid 1 \le i \le n \rangle$$

***Proof*:**

This proof relies on this key observation:

$$(a\, b)(a\, c)(a\, b)^{-1} = (a\, b)(a\, c)(a\, b) = (b\, c)$$

that is, we shifted a to b (so in a sense if $\sigma = (a\, b)$ and $\tau = (a\, c)$, $\sigma\tau\sigma^{-1} = (\sigma(a), \sigma(b))$).

With that in mind, we'll proceed by *induction*. If $n = 2$, then S_2 is generated by this set, so assume it generate S_{n-1} and consider S_n. Take some cycle with n in it:

$$(a_1\, a_2 \ldots a_k\, n)$$

which we've shown earlier can be decomposed as

$$(a_1\, a_2)(a_1\, a_3) \cdots (a_1\, a_k)$$

All the $(a_1\, a_j)$ are in G by the induction hypothesis (they are not in because of the generating set because we don't know if $(a_1\, a_j) = (i\, i+1)$). What's left to show is whether $(a_1\, n)$ is in G. To do this, we use the conjugation fact we brought up earlier:

$$(a_1\, n) = (a_1\, n-1)(n-1\, n)(a_1\, n-1)$$

which since $(n-1\,n) \in S$, and the other two terms are in G by induction hypothesis, it means that $(a_1\,n) \in G$. And Since any cycle can be broken down into the aforementioned form (where if we have a cycle, we can move k to the back, since its the same cycle that represents it), any cycle can be broken down in this fashion, and thus, we completed the proof

As a bonus, if you want to represent such a group:

$$\langle \sigma_i | \sigma_i^2 = 1\,, \sigma_i \sigma j = \sigma_j \sigma i \text{ for } |i-j| > 1\,, (\sigma_i, \sigma_{i+1})^3 = 1 \rangle$$

Though this is not necessary to know, but it can be good to know if you want to show something to be isomorphic to the symmetric group (since if something is isomorphic, it has the same relations).

https://en.wikipedia.org/wiki/Symmetric_group#Generators_and_relations.

There is a way to prove these once we introduce either quotient groups or conjugacy classes (in the group-action section). Thus, we won't prove these here for the nicest way to prove them is once we have those extra tools.

Exercise 1.3.1

1. exercise 7, 16, simply to know how to solve a concrete problem.
2. exercise ,17,18.
3. Do 10 to see that you can find all symmetries
4. Take $(1, i)$ for $2 \leq i \leq n$, that is, the set where you fix one element, then the 2nd position is all other elements.
5. This one is probably the most useful theoretically. You can actually generate a symmetric group given only 2 cycles!! Take any n-cycles from S_n, and a 2-cycle of adjacent elements in the n-cycle!

1.4 Homomorphisms and Isomorphisms

When comparing D_8 and Q_8 in example 1.2.28, we resolved at looking at the Cayley tables to distinguish the two. However, this becomes very tedious to do for larger groups, and downright impossible for infinite groups. In this section, we'll make precise the notion of comparing groups.

What does it mean to compare groups? If we were comparing tuples, $(a, b) \neq (b, a)$, since the order matters. On the other hand, if we're comparing sets, then $\{a, b\} = \{b, a\}$ since the order doesn't matter, as long as the elements are the same. For groups, the star of the show is the binary operation. If two groups G and H shared *exactly* the same binary operation, they'd be identical in terms of the binary operation. We often encounter groups with different elements, but the binary operation acts identically in both groups.

Example 1.4.1: "Identical" groups
Let $G = \{0, 1, 2\}$ and $H = \{a, b, c\}$, and define $(G, +_G)$ and $(H, +_H)$ with binary operation's which have Cayley tables:

$+_G$	0	1	2
0	0	1	2
1	1	2	0
2	2	0	1

and

$+_H$	a	b	c
a	a	b	c
b	b	c	a
c	c	a	b

Then notice we can replace 0 with a, 1 with b, and 2 with c, and the two tables would be identical. In other words, the operation $+_H$ acts exactly the same as $+_G$.

It is sometimes restrictive that the two binary operations act *exactly* the same. sometimes that are similar, but the two groups share a different structure. The take-away is that when working with homomorphisms and isomorphissm, what we care about is the binary operation between two groups behave similarly.

1.4.1 Homomorphisms

Definition 1.4.2: Homomorphism

Let $(G, \cdot)$, and $(H, \bullet)$ be groups. A map $\psi : G \to H$ such that for all $x, y \in G$,

$$\psi(x \cdot y) = \psi(x) \bullet \psi(y)$$

is called a *homomorphism*

Remember that the left side is the domains operation, and the right side is the co-domains operation. A good way of remembering the rule of a homomorphism is that the resulting image $\varphi(G) = \{\varphi(g) \mid g \in G\}$ is also a group (this is proved in proposition 1.4.4). Another way remembering the rule of a homomorphism is that both if there is a homomorphism, then in some way the binary operation must be *preserved*. It is not necessarily perfectly preserved – it's possible that only some of it is preserved. The following examples of homomorphisms may help in gaining an intuition

Example 1.4.3: Homomorphisms

1. Take the integers with the usual addition. Let e represent even integers; o represent odd integers. Then we know that adding even with even is even, even with odd is odd, odd with even is odd, and odd with odd is even:

   $$e + e = e, \ o + o = e, \ o + e = e + o = o$$

   On the other hand, if we take 1 and -1 and find all possible outcomes of multiplication, we get

   $$1 \cdot 1 = 1, \quad (-1) \cdot (-1) = 1, \quad 1 \cdot (-1) = (-1) \cdot 1 = -1$$

   Notice how 1 can represent even integers, and -1 represent odd. In this way, multiplication on -1 and 1 "captures" the idea of adding even and odd numbers.

Let's formalize this idea by making an explicit homomorphism. First, $\{-1, 1\}$ is a group under usual multiplication $\cdot$. The element 1 is the identity since $1 \cdot 1 = 1$, $-1 \cdot 1 = -1$. Furthermore -1 it's own inverse ($-1 \cdot -1 = 1$). For associativity, most cases are essentially immediate to check, with the following case being the non-trivial case:

$$(-1 \cdot -1) \cdot -1 = 1 \cdot -1 = -1 \cdot 1 = -1 \cdot (-1 \cdot -1)$$

Any case with 1 in the question be more easily checked since $a \cdot 1 = 1 \cdot a = 1$, making the cases of $1 \cdot -1 \cdot -1$, $-1 \cdot 1 \cdot -1$, $1 \cdot 1 \cdot -1$, and so forth easy to check. Thus, $\{-1, 1\}$ is a group.

Next, let $\psi : \mathbb{Z} \to \{-1, 1\}$ be a function such that

$$\psi(x) = \begin{cases} 1 & x \text{ is even} \\ -1 & x \text{ is odd} \end{cases}$$

Then if e represents an arbitrary even number and o an arbitrary odd number:

$$\begin{gathered} \psi(e+e) = \psi(e) = 1 = 1 \cdot 1 = \psi(e) \cdot \psi(e) \\ \psi(o+o) = \psi(o) = 1 = (-1) \cdot (-1) = \psi(o) \cdot \psi(o) \\ \psi(o+e) = \psi(e+o) = \psi(o) = -1 = -1 \cdot 1 = 1 \cdot -1 = \psi(e) \cdot \psi(o) = \psi(o) \cdot \psi(e) \end{gathered}$$

Showing that ψ is indeed a homomorphism.

Generalizing the previous result, notice that our group $\{-1, 1\}$ is homomorphic to $\mathbb{Z}/2\mathbb{Z}$ (remember that by default, if we say $\mathbb{Z}/2\mathbb{Z}$ is a group, we mean $(\mathbb{Z}/2\mathbb{Z}, +)$ with modulo 2 addition). In particular

$$f : \{-1, 1\} \to \mathbb{Z}/2\mathbb{Z} \qquad f(1) = 0,\ f(-1) = 1$$

Then one should verify that that f is indeed a homomorphism. It is further a bijection, meaning that every element of the group $\mathbb{Z}/2\mathbb{Z}$ represents exactly one element from $\{-1, 1\}$, showing we can inter-change the two.

Now, take $\varphi : \mathbb{Z} \to \mathbb{Z}/n\mathbb{Z}$, $\varphi(x) = [x]$. Then φ is a homomorphism. Indeed:

$$\begin{aligned} \varphi(x+y) &= [x+y] \\ &= [x] + [y] && \text{property of } \mathbb{Z}/n\mathbb{Z} \\ &= \varphi(x) + \varphi(y) \end{aligned}$$

If $n = 2$, then this will be exactly the homomorphism we defined earlier.

2. If you have done any calculus or analysis, you might have seen before that:

$$\frac{d}{dx}(f+g) = \frac{d}{dx}(f) + \frac{d}{dx}(g) \qquad \int f + g\, dx = \int f, dx + \int g\, dx$$

so $\frac{d}{dx}$ and $\int$ both satisfy this condition! Are they homomorphisms? In fact, they are, with the group being all differentiable functions for the 1st one, and all integrable functions for the second one. These groups require more build-up within Calculus or Analysis, and so we will not cover these operation in more detail. (More words here! Feels incomplete)[a].

3. For those with knowledge of the exponential function e^z (which we'll write as exp), then there is a natural way of relating the addition and multiplication operation of $\mathbb{C}$ and $\mathbb{C}^\times$ (as well as $\mathbb{R}$ and $\mathbb{R}^\times$). The map $\exp : \mathbb{C} \to \mathbb{C}^\times$ is surjective group homomorphism. Restricting the domain and co-domain, we also get an injective (but not surjective) map $\exp : \mathbb{R} \to \mathbb{R}^\times$.

4. Verify that there is no bijective homomorphism from D_8 to Q_8. At the moment, this may be a tedious task, it will become easier to show later with some more properties of bijective homomorphisms.

5. Let $G_1 \times G_2$ be the direct product of two groups G_1 and G_2 as defined in example 1.1.5(4). The map $\pi_1 : G_1 \times G_2 \to G_1$ and $\pi_2 : G_1 \times G_2 \to G_2$ defined by

$$\pi_1((g_1, g_2)) = g_1 \qquad \pi_2((g_1, g_2)) = g_2$$

   are homomorphisms.

6. Let G, H be any group. Then $\varphi : G \to H$ defined by $\varphi(g) = e_H$ where e_H is the identity of H is a homomorphism. This is called the *trivial homomorphism*.

7. (If you know about $\mathbb{C}$) Show that the subset $\{1, i, -1, -i\} \subseteq \mathbb{C}$ is a subgroup of $\mathbb{C}$ and that there is a bijective group homomorphism $\varphi : \mathbb{Z}/4\mathbb{Z} \to \{1, i, -1, -i\}$.

[a]However, $\frac{d}{dx}(fg)$ is not necessarily equal to $\left(\frac{d}{dx}(f)\right)\left(\frac{d}{dx}(g)\right)$ and $\int fg\,dx$ is not necessarily equal to $\left(\int f\,dx\right)\left(\int g\,dx\right)$. The first is studied in a course on Lie Groups and Lie Algebras, while the other is studied in a course in Measure Theory, in particular L^p spaces and Hölder's inequality

We will often simplify notation since lots of information is usually deduced from context. We will drop the binary operation and simply write $\varphi(ab) = \varphi(a)\varphi(b)$, given that it's clear that when we write ab the binary operation happens in the domain, and that when we werite $\varphi(a)\varphi(b)$ the binary operation happens in the codomain.

One more interpretation of a homomorphism that is a little more abstract but is enlightening in it's own way is that a homomorphism f is a function that "commutes" with the binary operation: If $*$ represent any binary operation (of any group, not just one group), then f is a homomorphism if and only if

$$f \circ * = * \circ f$$

Concretely, if we let $*$ for a group G be represented as $*_G$ and $*$ for a group H be represented as $*_H$, then

$$f \circ *_G = *_H \circ f$$

or, if written in terms of elements and with the standard notation for binary operation:

$$f(x *_G y) = f(x) *_H f(y)$$

This again show's how we can think of commutativity as a powerful property. This idea can be captured in a diagram like so:

$$\begin{tikzcd}
G \times G \arrow[r, "\varphi \times \varphi"] \arrow[d, "*_G(a,b)"'] & H \times H \arrow[d, "*_H(\varphi(a),\varphi(b))"] \\
G \arrow[r, "\varphi(ab)"'] & H
\end{tikzcd}$$

which can be read as "start at $G \times G$, then either go down then right, or go right then down, and both paths should give the same result", i.e. $\varphi(ab) = \varphi(a)\varphi(b)$.

Now that homomorphisms are defined, you might ask yourself "given two groups G, and H, how do I know there is a homomorphism between them?" There is always a trivial answer to this, the homomorphism sending everything to the identity always exist [21]. This is called the *trivial homomorphism* (you should check that it's indeed a homomorphism):

$$\varphi : G \to \{e\} \qquad \varphi(g) = e$$

So the better question would be "How can I find the non-trivial homeomorphisms between these two groups?" This in some cases can be a very difficult question. If we just look at them as set-functions from G to H, then if there are n objects in one category and m in the other, there are minimum n^m possible functions! That's a ton to check. We can deduce properties of homomorphisms which will let us diminish that number:

Proposition 1.4.4: Properties of Homomorphisms

Let G, H be groups and φ be a homomorphism. Then:

1. $\varphi(1_G) = 1_H$
2. $\varphi(g^{-1}) = \varphi(g)^{-1}$
3. $\varphi(g^n) = \varphi(g)^n$
4. $\ker \varphi$ is a subgroup of G
5. $\mathrm{Im}(\varphi)$, the image of G under φ is a subgroup of H.
6. if $\psi : H \to K$ is also a homomorphism, then $\psi \circ \varphi : G \to K$ is a homomorphism, i.e., the composition of homomorphisms is a homomorphism.

***Proof*:**

1. $\varphi(1_G) = \varphi(1_G 1_G) = \varphi(1_G)\varphi(1_G)$, so $\varphi(1_G) = \varphi(1_G)\varphi(1_G)$. Then by proposition 1.1.17
$$\varphi(1_G)\varphi(1_G)^{-1} = \varphi(1_G)\varphi(1_G)\varphi(1_G)^{-1}$$
the left hand side becomes 1_H, and the right hand side becomes $\varphi(1_G)1_H = \varphi(1_G)$, giving us
$$\varphi(1_G) = 1_H$$
2. Hint: take $\varphi(1_G) = \varphi(gg^{-1})$, and apply linearity and proposition 1.1.17
3. Hint: $\varphi(g^n) = \varphi(g^{n-1}g)$ and use the fact that φ is a homomorphism
4. Hint: Take the set $\ker(\varphi) := \{g \in G \mid \varphi(g) = 1_H\}$, and check the 3 axioms of a group
5. Hint: Take $\varphi(G) := \{h \in H \mid \exists g \in G$ such that $\varphi(g) = h\}$ and check the 3 axioms of a group
6. Hint: Apply the homomorphism condition twice.

[21] Elaborated in proposition 1.4.5

These propositions can even help us deduce whether [non-trivial] homomorphisms exist. For example, is there a non-trivial homomorphism from $\mathbb{Z}/2\mathbb{Z}$ to $\mathbb{Z}$? Let's say $\varphi : \mathbb{Z}/2\mathbb{Z} \to \mathbb{Z}$ were such a map. By the above proposition we have $\varphi(0) = 0$. Let's say $\varphi(1) = n$. Then

$$0 = \varphi(0) = \varphi(1 + 1) = \varphi(1) + \varphi(1) = n + n$$

giving us $0 = n + n$, which re-arranging is $-n = n$. The only integer equal to its negative is 0, hence $n = 0$. This shows that there is no non-trivial map from $\mathbb{Z}/2\mathbb{Z}$ to $\mathbb{Z}$.

The particular structure of the group that is being preserved depends on the operation and the cardinality of the domain and co-domain. In general, to get an idea of what part of the group is being preserved, you'll have to check by doing a couple of example.

Exercise 1.4.1

1. Find all non-trivial homomorphisms between D_8 and Q_8
2. (If you know some complex analysis) Let S^1 be the circle in the complex plane. show that $f : \mathbb{R} \to S^1$, $f(x) = e^{2\pi ix}$ is a homomorphism.
3. Perhaps another exercise like the one above? Perhaps a $G \to G$ example? I think getting some ideas between which groups are homomorphic to each other to be useful for long-term mathematical career.
4. is there a non-trivial homomorphism between $\mathbb{Z}/n\mathbb{Z}$ and $\mathbb{Z}$? answer:[22]
5. Let $\varphi : G \to H$ be a homomorphism. Show that $\varphi(a + b + c) = \varphi(a) + \varphi(b) + \varphi(c)$. More generally, if $g_1, g_2, ..., g_n \in G$, then $\varphi(g_1 + g_2 + \cdots + g_n) = \varphi(g_1) + \varphi(g_1) + \cdots + \varphi(g_1)$. [hint: recall assocaitivity and induction]
6. Does there exist a surjective homomorphism $\varphi : G \to H$ where G is abelian and H is not?
7. Let S where $|S| = 3$ be a set and let $\text{Sym}(S)$ all bijections on S (i.e. the symmetric group). Show that there exists a homomorphism $\sigma : D_6 \to \text{Sym}(S)$. (it is actually an isomorphism, move this exercise?) (This is a particular example of a *group action*, which we will explore in detail in chapter 4)
8. Define the map $\varphi_g : G \to G$, $\varphi(x) = gx$. Is this map in general a homomorphism? If not in general, in a particular case?
9. Define the map $\psi_g : G \to G$, $\psi(x) = gxg^{-1}$. Is this map in general a homomorphism? If not in general, in a particular case?
10. Let $\varphi : G \to H$ be a homomorphism and $\ker(\varphi) = \{e_G\}$. Prove that φ is injective.
11. Show that there exist an injective homomorphism $\varphi_n : D_n \hookrightarrow O_2(\mathbb{R})$.

Important examples

I'll go over some homomorphisms to build a deeper understanding of the concept

[22] There does not exist a non-trivial homomorphism $\varphi : \mathbb{Z}/n\mathbb{Z} \to \mathbb{Z}$, since every non-identity element of $\mathbb{Z}/n\mathbb{Z}$ ha finite order, but all elements of $\mathbb{Z}$ has infinite order.

The first one is deceptively boring: take $\varphi : G \to \{e\}$ such that $\varphi(g) = e$. The function φ is a homomorphism, since for any $g, h \in G$

$$\varphi(g \cdot h) = e = e \cdot e = \varphi(g) \cdot \varphi(h)$$

This seemingly boring homomorphism has actually a very important property: for *any* group G, there *always* exist a *unique* isomorphism from G to $\{e\}$. This is also true for homomorphisms with $\{e\}$ as the domain! This fact is important enough to make a proposition:

Proposition 1.4.5: Homomorphism To/From Trivial Group

Let G be any group and I represent the trivial group. Then there always exists a unique homomorphism

$$\varphi : G \to I \qquad \psi : I \to G$$

to and from I

***Proof*:**
Let e_G represent the identity of G and e_I represent the element of I. We've already shown φ is a homomorphism, so let's now show uniqueness. Let's say φ' is another homomorphim with the same domain/codomain. Note that

$$\varphi'(e_G) = \varphi'(e_G \cdot e_G) = \varphi'(e_G)\varphi'(e_G) \;\Rightarrow\; \varphi'(e_G) = \varphi'(e_G)\varphi'(e_G)$$

and by proposition 1.1.17, we can multiply each side by$(\varphi')^{-1}$ and get

$$e_I = \varphi'(e_G)$$

meaning the identity *must* map to the identity. But that would then make φ' identical to φ, thus

$$\varphi = \varphi'$$

showing that φ must be unique.

A similar proof works for ψ.

So why is this property important? What can we do with it? In this case with groups, it is not used often [23]. what's more important is the concept that there *always* exists a homomorphism from any group, and that it is a *unique* homomorphism. These properties for the right homomorphisms will come back big-time when we talk about *quotient groups* (chapter 3): For *every* homomorphism there *always* exists a *unique* group (called the quotient group) which will "represent" the homomorphism.

Thus, we'll give a name to groups with this property:

[23] It is used in the context of Category theory, where $\{e\}$ is both the initial and final group in the category of **Group**, making it an intuitive first step towards introducing universal properties. More on this in section 8

Definition 1.4.6: Initial And Final Groups

Let G be a group. If for any other group H, then

1. if there exists a unique group $\varphi : G \to H$, then we say that G is an *initial group* (or *initial object*)
2. if there exists a unique homomorphism $\varphi : H \to G$, then we say that G is an *final group* (or *final object*)
3. if G is either initial or final, we say that G is a *terminal group* (or *terminal object*)

The reason for this vocabulary is clearer is you think of a graph where the edges are homomorphisms and the nodes are groups:

$$\begin{tikzcd}
 & U \arrow[d, shift left] \arrow[d] \arrow[d, shift right] \arrow[dr] \arrow[dr, shift left] & \\
H & L \arrow[l] \arrow[r] \arrow[d] & M \\
G \arrow[u] \arrow[uu, bend left, start anchor=north, end anchor=west] \arrow[ur] \arrow[urr] \arrow[uur] \arrow[r] & K \arrow[ul] \arrow[ur] &
\end{tikzcd} \tag{1.9}$$

You can see how there is only one homomorphism from G to any other group, and M has only one homomorphism from every other group. Thus, G is initial, H is final, and both G and H are terminal.

1.4.7 Note An object *can* be both initial and final. The trivial group is one such example

For now, we do not have the tools to properly analyze this result; we will gain those in chapter 2, and revisit this idea in chapter 3.

Redefining Subgroup

Moving back towards things we can prove, I'll present an important way of using homomorphisms to *define* subgroups, using the fact that subgroups as a similar structure to the original groups, and homomorphisms preserve some structure of a group. It might already be somewhat intuitive that if $H \leqslant G$, then H is pretty similar to G, especially that it must have the same group operation. With homomorphisms, we can formalize this idea:

Proposition 1.4.8: Defining Sub Group With Homomorphism

Let $(G, \cdot)$ and $(H, \bullet)$ be two groups, and $H \subseteq G$. Then H is a subgroup if and only if there exists a inclusion map $i : H \to G$ which is also a homomorphism

***Proof*:**
We'll show that this definition is equivalent to definition 1.1.7, that is, that $(H, \cdot)$ is a group. Let $i : H \to G$ be an homomorphism which is also an inclusion map. All of these will come from proposition 1.4.4.

1. ($e_G = e_H$) We need to how that H has the same identity. This comes for free with proposition 1.4.4

2. (inverse) We need to show that for any $a \in (H, \cdot)$, then there is a $b \in (H, \cdot)$ such that $ab = ba = e_G$. Since $i(a^{-1}) = a^{-1}$, then a^{-1} is a candidate. Indeed, since

$$i(e_H) = i(aa^{-1}) = i(a)i(a^{-1}) = i(a)i(a)^{-1} = aa^{-1} = e_G$$

then a^{-1} will indeed be our desired inverse

3. (associativity) Similarly to how we've down the other two, but the proof is a little more tedious to carry out. One can do this to try proving associativity at least once in their career.

This is probably one of my favourite proposition using homomorphisms: really showing how homomorphism captures the idea that an algebraic structure is preserved by defining subgroups with an inclusion map that is also a homomorphism.

Collection of Homomorphisms

Next, we will address a little more complicated object. We might ask ourselves in what other way can we use homomorphisms to study groups? If a homomorphism preserves group structure, what possible homomorphisms can we have between a group and itself? What would that give us? Let's first label the set of all such homomorphisms:

Definition 1.4.9: Set of Endomorphisms

1. Given a group G, an endomorphism is a homomorphism from an object to itself: $\varphi : G \to G$.

2. Let $\mathrm{End}(G) := \{\varphi : G \to G \mid \forall a, b \in G,\ \varphi(ab) = \varphi(a)\varphi(b)\}$ represent the set of all endomorphisms on G

What $\mathrm{End}(G)$ represent is the set of all possible group structure's that can be derived form G. In the example with $\mathbb{Z}$ being mapped to $\{-1, 1\}$ (example 1.4.3), notice that we can make the codomain all of $\mathbb{Z}$, and the function remains identical. In this way, we would make that function an element of $\mathrm{End}(\mathbb{Z})$, and so this shows that the function in this example captures one of the possible sub structures of $\mathbb{Z}$ (in particular, a special kind of subgroup, we ill explore this in more detail in chapter ref:HERE)

If we can figure out some structure on $\mathrm{End}(G)$, we can start getting a good idea on what are all sub-structures in G. More excitingly, if $\mathrm{End}(G)$ can be proven to be a group (with composition, $\circ$, as the operation), then we can use all the technics of group theory to study it. However, this turns out not to be the case. The problem comes when trying to prove the existence of an inverse, which doesn't necessarily exist (try proving it's a group!).

So you might ask if there is still a way to make $\mathrm{End}(G)$ into a group; perhaps with a nice operation, we can define a [non-trivial] group structure? The one that is usually first attempted is defining addition *pointwise*, that is

$$f, g \in \mathrm{End}(G) \qquad f + g \equiv (f + g)(x) = f(x) + g(x)$$

Where $\equiv$ means we are showing how the ne function "$f+g$" is defined by how it processes every element. With the operation defined, to check that $f+g$ is in $\text{End}(G)$, we need to check if it's a homomorphism, that is

$$(f+g)(x+y) = (f+g)(x) + (f+g)(y)$$

and going through the computation:

$$\begin{aligned}(f+g)(x+y) &= f(x+y) + g(x+y)\\ &= f(x) + f(y) + g(x) + g(y)\end{aligned}$$

and here is were we get stuck, since G is not necessarily commutative.

However, what if G was commutative? Well, the result will actually follow much more nicely!

$$\begin{aligned}&= f(x) + f(y) + g(x) + g(y)\\ &= f(x) + g(x) + f(y) + g(y)\\ &= (f+g)(x) + (f+g)(y)\end{aligned}$$

showing that $(f+g)$ is indeed a homomorphism!! Thus, $f+g$ is a well-defined operation. From here, you can check the axioms of a group to check that $\text{End}(G)$ is indeed a group. It's even an abelian group! This is an example of how being commutativity changes how a group effects.

1.4.10 Remark In fact, more is true. If the domain of the functions are arbitrary sets, then $\text{Hom}(A, G)$ is *still* an abliean group! This fact will be very useful to us much further don the line, in part II, III, and X.

Generalization of 2D and 3D rotations; intuition on 4D

Point Groups? (`https://en.wikipedia.org/wiki/Point_groups_in_two_dimensions`) (and the 3D one too?)

Finally, we will take a moment to detour into trying to use groups to build an intuition for 4+ dimensional objects! As we've seen, D_n represents rotations of 2D polygons, and Q_8 represents rotations of 3D objects. What about rotations and symmetries of higher dimensional objects? To find such a group, we will show that there is a homomorphism $\varphi_D : D_n \Rightarrow SO_3(\mathbb{R})$ and $\varphi_Q : Q_8 \to SO_3(\mathbb{R})$

First, we'll define φ_D:

$$r \to \begin{pmatrix} 0 & 0 & 0\\ 0 & 0 & 0\\ 0 & 0 & 0\end{pmatrix} \qquad d \to \begin{pmatrix} 0 & 0 & 0\\ 0 & 0 & 0\\ 0 & 0 & 0\end{pmatrix}$$

TBD (do same with quaternion. Perhaps take a look at representation theory to figure out Q_8, Generalize to 4D with $SO(4)$)

Example 1.4.11: exercicss
gaining intuition on matrix multiplication

1. matrix multiplication could've been defined component-wise, so why wasn't it? Build up to (using the appropriate vocabulary, not the following one) that given $\dim M = n$, $\dim N = n$, $\text{Hom}_{k-\mathbf{Vect}}(M, N) \cong_{k-\mathbf{Vect}} M_n(k)$, and we'll get $\Psi(f \circ g) = A_f \cdot A_g$ (i.e. composition

becomes matrix multiplication)

2. elaborate if dimensions change?

Exercise 1.4.2

1. Let G be an arbitrary group.
    1. Prove that there always exists a homomorphism from $\mathbb{Z}$ to G
    2. Why does this not contradict the fact that I is the only initial group?

1.4.2 Isomorphisms

While homomorphisms preserve some algebraic structure, isomorphisms preserve the *entire* algebraic structure. An isomorphism can be thought of as the generalization of the concept of equality: If two groups are isomorphic, they are identical (as groups). They will only differ in the symbols which are used in representing them. The idea of an isomorphism extends far beyond group theory. Here's a fun linguistic example: if I say

one, two, three, four, five

or

un, deux, trois, quatre, cinq

Then both are capturing the idea of $1, 2, 3, 4, 5$, except in one I'm writing it in English, and in the other in French. Those two sentences ostensibly capture the same [numerical] ideas, but with different symbols

When trying to capture this idea in groups, we have to consider what it means for two group to be identical. First, the two groups would need to be of the same size, if not we cannot say they are the same. Therefore, the homomorphism must also be bijective. Next, all the relation between the two groups need to be the same. In example ref:HERE, we only preserved part of the relations.

Definition 1.4.12: Isomorphism

The map $\psi : G \to H$ is called an *isomorphism* and G and H are said to be *isomorphic* or of the same *isomorphism* type, written $G \cong H$ if

1. ψ is a homomorphism
2. ψ is a bijection

1.4.13 Note Dummit showed this as the definition of isomorphism. But in a more categorical framework, this is not in general how you would introduce this. You would instead say that there exists an inverse function: $g : H \to G$ which is also homomorphic. It can be shown that this (for groups) implies a bijection. [a]

[a] If you've already studied topology, there's an example where a homomorphic bijection is not an isomorphism

When two groups are isomorphic, it means they only differ by the symbols we choose to use, they are in practice identical. The trivial isomorphism is the identity map $\text{id} : G \to G$. We can also form an equivalence class of isomorphic groups: take $\mathcal{G}$ to the set of groups and let the equivalence relation be isomorphisms.

Example 1.4.14: Isomorphism Examples

Let V_4 (*Vierergruppe*, meaning 4-group, sometimes called Klein 4-group), is the group that represents the symmetries of a [non-square] rectangle. You can define it as

$$V_4 = \langle a, b | a^2 = b^2 = (ab)^2 = e \rangle$$

Show that:

$$V_4 \cong \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$$

Where $\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$ has the same operation defined in the cartesian product example in ref:HERE.

A less obvious example of a group isomorphism is $\psi : \mathbb{R} \to \mathbb{R}^+$, $\exp(x) = e^x$. e^x is invertible, and thus bijective, and $e^{x+y} = e^x e^y$, preserving group structure.

Proposition 1.4.15: Properties Of Isomorphisms

(Motivated by properties of diffeomorphism in John Lee. In particular, I want to say that if $g \circ f$ and f is an isomorphism, [do we need to force g to be either injective or surjective?]so must g be)

***Proof*:**

1. The composition of isomorphism is an isomorphism
2. If $g \circ f$ and g are isomorphisms, then since g is an isomorphism, so if $g^{-1} \circ g \circ f = f$, and similarly if f is an isomorphism: $g \circ f \circ f^{-1} = g$

An interesting exercise to show is if A and B are sets of the same cardinality, and S_A and S_B are two symmetric groups, then $S_A \cong S_B$. The book explains it on p.37, but personally I feel it's more intuitive if you think of it as relabeling since A and B are arbitrary sets.

Going back for a moment to hen e tried to define a group $(\text{End}(G), \circ)$ the fatal flaw as that not all homomorphisms have inverses. However, e have seen that all isomorphisms can be reversed, that is, i $\varphi : G \to H$ is an isomorphism, then there exists a ψ, usually denoted φ^{-1} such that $\varphi^{-1} : H \to G$ is an isomorphism and $\varphi^{-1} \circ \varphi = \varphi \circ \varphi^{-1} = \text{id}$ (here id is either id_G or id_H depending on whether φ is on the right or left in the composition respectively). If e take $\varphi : G \to G$ to e an isomorphism, then φ^{-1} ill *exactly* have an inverse. Excitingly, this in fact define as group!

Definition 1.4.16: Automorphism

1. if $\varphi : G \to G$ is an isomorphism from a group to itself, then φ is called an ***automorphism***
2. Let $\text{Aut}(G) := \{\varphi : G \to G \mid \varphi \text{ is an automorphism}\}$ be the collection of all automorphisms of G

Proposition 1.4.17: $\mathrm{Aut}(G)$ is a Group

Let G be a group and $\mathrm{Aut}(G)$ be the collection of automorphisms. Then $(\mathrm{Aut}(G), \circ)$ forms a group

***Proof*:**
Proving it's a group should be doable at this point

Example 1.4.18: Automorphisms
Simplest example are identity maps. Other examples come form Linear Algebra with bijective linear functions (since $\varphi(a)\varphi(b) = \varphi(ab)$ for a linear function, it *is* an example)

1.4.19 Foreshadowing The concept of automorphisms will not be limited to groups. Many algebraic and non-algebraic structures will have the concept of automorphism defined. Most excitingly, the set of automorphisms for *any* of these objects *always* forms a group! Because of this, we will take a hole chapter to study (TBD chap 4? better transition?)

:exercises

(make sub questions) example 1.1.5[5], we showed that $(\mathbb{Z}, +)$ and $(\mathbb{Z}, +_2)$ had a "different structure" by showing the identity and inverses are different. Show that they are in fact isomorphic. This means that they are not a good example of two different groups on the same set since algebraically, they are identical.

Show that $D_8 \ncong \mathbb{Z}/8\mathbb{Z}$. With this knowledge, conclude that we can define two different groups on the set of 8 elements, giving a proper example of two different groups structure on the same underlying set.

(Challenging) Try to come up with two different structure on the underlying set of $\mathbb{Z}$ (Hint: recall the structure of $\bigoplus_i^\infty(\mathbb{Z}/2\mathbb{Z})$ and show there is a bijection between $\mathbb{Z}$ and this set, so that you can define a group structure on $\mathbb{Z}$ relative to this bijection. Is this group isomorphic to $\mathbb{Z}$?)

In the following section, we will explore classifying all possible groups on n elements where n has small order (larger orders of n will come when we have more tools)

First Use of Isomorphisms

In mathematics, there's a whole set of theorems that we use to show that classes of mathematical objects, like groups, are isomorphic to each other or not. Such theorems are called *classification theorems*. As a simple example, we can classify all non-abelian groups of order 6, in particular:

1.4.20

$$\text{any non-abelian group of order 6 is isomorphic to } S_3$$

Holding the theorem to be true for now, we can conclude that $D_6 \cong S_3$, and $GL_2(\mathbb{F}_2) \cong S_3$ without needing to find an explicit map. Not all groups of order 6 are isomorphic to S_3 because they're not all abelian. However, all abelian groups of order 6 are isomorphic to $\mathbb{Z}/6\mathbb{Z}$! Thus, any group of order 6 is isomorphic to either S_3 or $\mathbb{Z}/6\mathbb{Z}$. This theorem will be proved in exercise ref:HERE.

In general, it is difficult, sometime unfeasible, to find an explicit isomorphism mapping. Here are some necessary, but not sufficient, conditions for isomorphisms:

Proposition 1.4.21: Isomorphism Conditions

Let $\psi : G \to H$ be an isomorphism. Then:

1. $|G| = |H|$
2. G is abelian if and only if H is abelian
3. for all $x \in G$, $|x| = |\psi(x)|$

***Proof*:**
Fairly simple - left as an exercise.

Example 1.4.22
This proposition shows that S_3 and $\mathbb{Z}/6\mathbb{Z}$ are not isomorphic, because one is abelian and the other is not. Additionally, $(\mathbb{R} - \{0\}, \times)$ and $(\mathbb{R}, +)$ are not isomorphic, because $-1 \in \mathbb{R}^\times$ has order 2, and no element in $(\mathbb{R}, +)$ has order 2

There's a cool introduction to how two groups could be homomorphic given the presentation of a group, which will require free groups to prove formally. I'll omit this for now (p.52, bottom).

Example 1.4.23

1. Take D_n and it's presentation $\langle r, s | r^n = s^2 = 1, sr = r^{-1}s \rangle$. Now take $n = mk$, $k \geq 3$ and consider d_{2k}. Then
$$\psi : d_n \to d_{2k} \qquad \text{by} \qquad \psi(r) = r_1, \text{ and } \psi(s) = s_1$$
is a homomorphism. In other words, the operation between Dihedral groups which are "multiples" are the same (think of a triangle and hexagon. two hexagon rotations is 2 triangle rotation, angle wise).
2. an example showing D_6 and S_3 are isomorphic. p.53

Proposition 1.4.24: Isomorphism properties

Let $\psi : G \to H$ be an isomorphism, and $\Psi : G \to H$ an isomorphism (if the context permits it)

1. Let $\psi : G \to H$ is a homomorphism. Prove that the identity maps to the identity
2. $\psi(x^n) = \psi(x)^n$, for all $n \in \mathbb{Z}$.
3. $|\Psi(x)| = |x|$ for all $x \in G$. Deduce that any two isomorphic groups have the same number of elements of order n for each $n \in \mathbb{Z}^+$. Is the result true for ψ? (hint: think of the "trivial homomorphism" which sends every element from G to $\{1\}$.)
4. If Ψ exists, then G is isomorphic if and only if H is isomorphic (easy)
5. If you know where the generators map, you know where all the entries map

***Proof*:**
I'll do one of them:

1. $$1_H\psi(x) = \psi(x) = \psi(1_G \cdot x) = \psi(1_G) \cdot \psi(x) \Rightarrow 1_H = \psi(1_G)$$

The rest is good to do yourself to really understand the proposition.

The last point in proposition 1.4.24 is one to keep in mind. I found it particularly useful when trying to construct isomorphisms between groups and S_n (which you will extensively once you cover group-actions).

Example 1.4.25

Prove that $SL_2(\mathbb{F}_3)$ generated by $\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$ and $\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$ is isomorphic to the quaternion group of order 8

Proof. Label the first matrix A and the second matrix B. To start, bring up the presentation of Q_8 to get an idea of it's structure. See if you can bring it down to two generators. What are the properties of these generators (i.e. representation). Does A and B match it? If so, what's then the isomorphism. □

Prove that a homomorphism on S_n means it will preserve the parity of the cycle

Proof. Check cases □

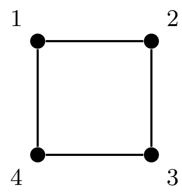
1.5 Symmetry and Transformations: A Prelude to Group Actions

We have so far studied groups as abstract algebraic objects, defined by axioms and explored through subgroups, homomorphisms . But groups did not arise historically from axioms—they arose from *symmetry*. Évariste Galois studied permutations of roots of polynomials. Arthur Cayley studied transformations of space. In every early appearance of what we now call a "group," the central idea was the same: a collection of *reversible transformations* of some object, where transformations can be composed and undone.

In this section, we pause the axiomatic development to revisit this original intuition. The goal is simple: to see that groups *act on things*, and that this perspective is the natural way to think about them. We will make all of this precise in Chapter 4; for now, we build intuition.

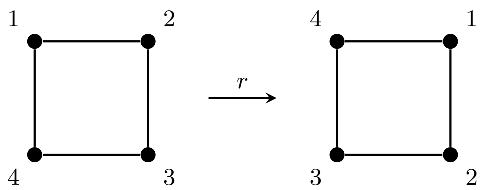
The Symmetries of a Square

Recall from Section 1.2.3 that the dihedral group D_8 consists of the 8 symmetries of a square: four rotations and four reflections. Let us label the corners $1, 2, 3, 4$:



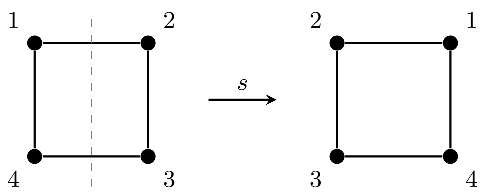
Each symmetry of the square *moves the corners around.* A 90° clockwise rotation r sends each corner to the next one clockwise:

$$1 \longmapsto 2, \quad 2 \longmapsto 3, \quad 3 \longmapsto 4, \quad 4 \longmapsto 1.$$



Similarly, the reflection s across the vertical axis swaps the left and right columns:

$$1 \longmapsto 2, \quad 2 \longmapsto 1, \quad 3 \longmapsto 4, \quad 4 \longmapsto 3.$$

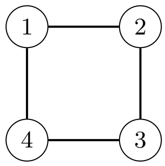


Each element of D_8 determines a specific way of rearranging the four corners—that is, a *bijection* from $\{1, 2, 3, 4\}$ to itself. We can compose these rearrangements: rotating by r twice gives the same result as rotating by r^2. In other words, the group D_8 is *acting* on the set $\{1, 2, 3, 4\}$.

Two observations are worth highlighting. First, every symmetry is *reversible*: if r rotates the square clockwise, then r^{-1} rotates it back. This is why we need a *group* (with inverses) rather than just a monoid. Second, the identity element $e \in D_8$ does nothing to the corners—it is the "trivial symmetry" that leaves everything in place.

Beyond the Square: Symmetry is Everywhere

The group D_8 does not only describe the symmetries of a geometric square. Consider the *cycle graph* C_4 on four vertices:



A *graph automorphism* is a bijection on the vertex set that preserves adjacency. One can verify that the automorphism group $\mathrm{Aut}(C_4)$ is precisely D_8: the same rotations and reflections that preserve the square also preserve which vertices are joined by an edge. The group D_8 captures an abstract *pattern of symmetry* that manifests in both geometry (the square) and combinatorics (the cycle graph). This is the key insight: a single group can describe the symmetries of *many different objects*.

Here are several more instances of this phenomenon, drawn from across mathematics. In each case, a group "acts" on a set by rearranging its elements in a structured, reversible way.

1. **Linear transformations.** The general linear group $GL_n(\mathbb{R})$ acts on $\mathbb{R}^n$: each invertible matrix A transforms a vector $\vec{x}$ to $A\vec{x}$. Composing two transformations corresponds to multiplying the matrices, and the identity matrix leaves every vector fixed. Rotations, reflections, scalings, and shears of n-dimensional space are *all* captured by this single action.

2. **Permutations.** The symmetric group S_n acts on $\{1, 2, \ldots, n\}$ by evaluation: each permutation σ sends i to $\sigma(i)$. This is, in a sense, the "most general" action—every rearrangement of n objects corresponds to some element of S_n. As we will see in Chapter 4, *every* group action can be viewed through the lens of permutations.

3. **The Rubik's Cube.** A standard 3×3 Rubik's Cube has six faces, each consisting of nine colored squares. The center square of each face is fixed (it defines the face's color), leaving $6 \times 8 = 48$ movable *facelets*. Each legal move—a quarter-turn or half-turn of a single face—permutes these 48 facelets, and the set of all finite sequences of legal moves forms a group $\mathcal{R} \leq S_{48}$. This group is enormous:

$$|\mathcal{R}| \;=\; 43{,}252{,}003{,}274{,}489{,}856{,}000 \;\approx\; 4.3 \times 10^{19},$$

   yet it is a *tiny* fraction of $|S_{48}| = 48! \approx 1.2 \times 10^{61}$. Not every conceivable rearrangement of the facelets can be reached by legal moves. The structure of the group $\mathcal{R}$—its subgroups, generators, and quotients—is precisely what determines which configurations are reachable from a solved cube and which are not.

4. **Translations of polynomials.** Consider the set of all real polynomials. The additive group $(\mathbb{R}, +)$ acts on this set by *translation*: given $t \in \mathbb{R}$ and a polynomial f, define

$$(t \cdot f)(x) \;=\; f(x+t).$$

Translating by t and then by s is the same as translating by $t+s$, and translating by 0 changes nothing. This simple action connects algebra to analysis: Taylor's theorem, for instance, describes exactly how a polynomial changes under translation.

What All These Examples Share

In every example above, three things are true:

1. Each element of the group "does something" to every element of the set—it rearranges, transforms, or moves things around.
2. The **identity element does nothing**: it leaves every element of the set exactly where it was.
3. The group operation is **compatible with composition**: performing one element's transformation followed by another's is the same as performing the transformation of their product.

These three observations are the soul of what we will call a *group action*. In Chapter 4, we will distill them into a precise definition—a single homomorphism from the group into a symmetric group—and develop a rich theory around it. But the underlying philosophy is already clear:

A group is not just an abstract set with a binary operation.
It is a collection of symmetries, waiting to act.

Groups Acting on Themselves: A First Glimpse

The examples above involve a group acting on an "external" object: a square, a vector space, a set of facelets. But some of the best applications of group actions come from letting a group act *on its own elements*. There are two natural ways to do this. Given a group G and elements $g, a \in G$:

- Left multiplication: define $g \cdot a = ga$. Each element g "shuffles" the elements of G by multiplying on the left. This will lead us to *Cayley's Theorem*, which asserts that every group is isomorphic to a group of permutations.
- Conjugation: define $g \cdot a = gag^{-1}$. This action detects the internal structure of G: which elements "behave similarly" under the group operation, and how the center $Z(G)$ relates to the rest of the group. It will lead us to the *Class Equation* and, ultimately, to the celebrated *Sylow Theorems*, which describe how a finite group decomposes relative to its prime factors.

Both of these self-actions will be central to Chapter 4. The remarkable fact is that by studying how G transforms *itself*, we can deduce deep structural information about G that is invisible from the axioms alone.

From Actions to Cayley Graphs

We have seen that each element g of a group can be thought of as a transformation: it "moves" every element a to ga. If we focus on a set of generators $\{s_1, s_2, \ldots, s_k\}$, then each generator defines a specific *step* that can be taken from any group element. Starting from the identity e and repeatedly taking these steps, we trace out the entire group.

This is precisely the idea behind a *Cayley graph* (Definition 1.6.1): we place a vertex at each element of G and draw a directed edge from a to $s_i a$ for each generator s_i. The resulting picture is a vivid record of the group's structure, and it is nothing other than a *visual record of the left-multiplication action* restricted to the generators. As we explore Cayley graphs in the next section, keep this action-based perspective in mind—it is the thread that connects the algebra of groups to the geometry of their visual representations, and it is a thread we will pick up again, with full force, in Chapter 4.

1.6 Cayley Graphs

Another way we try to find group-structure is by creating a graph using the generators of a group. This graph gives a visual on how the group elements interact with each other:

Definition 1.6.1: Cayley Graph

Let G be a group and let S be a generating set of G. The *Cayley Graph* $\Gamma = \Gamma(G, S)$ is a colored directed graph constructed as follows:

1. Each element $g \in G$ is assigned a *vertex*: the vertex set $V(\Gamma)$ of Γ bijectivity corresponds with G
2. Each generator $s \in S$ is assigned a colour c_s
3. For any $g \in G$ and $s \in S$, the vertices corresponding to the elements g and gs are joined by a direct edges of colour c_s. Thus, the edge set $E(\Gamma)$ consists of pairs of the form (g, gs), with $s \in S$ providing the colour

some popular, but not necessary, conventions are: [24]

1. The set S is finite
2. $S = S^{-1} = \left\{s^{-1} \middle| s \in S\right\}$
3. $e \notin S$ (i.e., the identity is not in S)

Example 1.6.2: Cayley Graphs

1. Let $G = \mathbb{Z} = \langle 1 \rangle$. Then we can say that $S \cup S = \{1, -1\}$. This gives us the graph:

$$\cdots \qquad -3 \longleftarrow -2 \longleftarrow -1 \longleftarrow 0 \longrightarrow 1 \longrightarrow 2 \longrightarrow 3 \qquad \cdots$$

Take a moment to try and see what the Cayley graph of $\mathbb{Z} \times \mathbb{Z}$ would be (the answer is given

[24] These conventions are usually followed in *geometric group theory*

in a later example)

2. Let $G = S_3 = \langle a, b | a^2 = b^3 = 1, ab = b^2 a \rangle$. Then if b is represented by red edges and a is represented by green vertices, we get:

$$\begin{tikzcd}
1 \arrow[r] \arrow[d, bend right] & b \arrow[r] \arrow[d, bend right] & b^2 \arrow[ll, bend right] \arrow[d, bend right] \\
a \arrow[r] \arrow[u, bend right] & ab \arrow[r] \arrow[u, bend right] & ab^2 \arrow[ll, bend left] \arrow[u, bend right]
\end{tikzcd}$$

3. Let $G = \mathbb{Z}/4\mathbb{Z}$. Then the Cayley graph is:

$$\begin{tikzcd}
0 \arrow[r] & 1 \arrow[r] & 2 \arrow[r] & 3 \arrow[lll, bend right]
\end{tikzcd}$$

4. Let $G = \mathbb{Z} \times \mathbb{Z}$. Then the Cayley graph is:

$$\begin{tikzcd}
 & & & \vdots & & & \\
 & (-2,2) & (-1,2) \arrow[l] & (0,2) \arrow[l] \arrow[r] & (1,2) \arrow[r] & (2,2) & \\
 & (-2,1) \arrow[u] & (-1,1) \arrow[l] \arrow[u] & (0,1) \arrow[l] \arrow[r] \arrow[u] & (1,1) \arrow[r] \arrow[u] & (2,1) \arrow[u] & \\
\cdots & (-2,0) \arrow[u] \arrow[d] & (-1,0) \arrow[l] \arrow[u] \arrow[d] & (0,0) \arrow[l] \arrow[r] \arrow[u] \arrow[d] & (1,0) \arrow[r] \arrow[u] \arrow[d] & (2,0) \arrow[u] \arrow[d] & \cdots \\
 & (-2,-1) \arrow[d] & (-1,-1) \arrow[l] \arrow[d] & (0,-1) \arrow[l] \arrow[r] \arrow[d] & (1,-1) \arrow[r] \arrow[d] & (2,-1) \arrow[d] & \\
 & (-2,-2) & (-1,-2) \arrow[l] & (0,-2) \arrow[l] \arrow[r] & (1,-2) \arrow[r] & (2,-2) & \\
 & & & \vdots & & &
\end{tikzcd}$$

if we let $S = \{\text{here}\}$, then the Cayley graph is

show that there is no twist

5. Let $G = D_\infty = \langle r, s \mid rs = sr^{-1} \rangle$ (i.e this is D_n but with no $r^n = 1$ relation) and let $S = \{r, s, r^{-1}, s^{-1}\}$. Then the Cayley graph is:

show the twist

6. $GL_2(\mathbb{Z}_2)$

Just by looking at the graphs, you might be able to see some interesting patterns. For example, by taking the direct product, we found it produces parallel graphs. We see in S_3 and $\mathbb{Z}/n\mathbb{Z}$ that arrow point *to* the identity represent the *order* of the element. Different Cayley graphs can also give different insights: notice that the Cayley graph for $\mathbb{Z} \times \mathbb{Z}$ with $S = \{\text{here}\}$ was simply two parallel lines with every node being connected with the one above it. On the other hand, the Caley graph for D_∞ looks like the one for $\mathbb{Z} \times \mathbb{Z}$, except it has a “twist” introduced to it[25].

In most of this textbook, Cayley graphs will not play a prominent role. They come back here and there to give a visual way of looking at new concepts introduced[26]. However, they are the foundation for an area of group theory called *geometric group theory.* For the study of geometric group theory, some knowledge of analysis is needed (ex. metrics). For those interested in such explorations, cite:HERE

[25] In chapter 5, we will learn about a way to construct D_∞ from $\mathbb{Z}$ and $\mathbb{Z}/2\mathbb{Z}$, and will denote it as $\mathbb{Z} \rtimes \mathbb{Z}/2\mathbb{Z}$. Some mathematicians will call “$\rtimes$” the *crossed product* because it introduced a “cross” or “twist” in the Cayley graph

[26] For example, they will come back when working with free groups and semi-direct products

2

Sub Groups and Generating Groups

Now that we've studied some fundamental aspects of group theory and important examples, let's dive deeper in the mechanics used in analyzing and finding groups. The first way we'll be studying group is by trying to take smaller groups from the larger group and seeing if we can find information through the pieces of the group.

We will start by trying to find subgroups that exist for *any* group G. These subgroups are sometimes called *natural subgroups*, since they always exist for a group, as though the axioms of the group "naturally" let these subgroups exist. We will then move on to constructing groups given some subset of a group, and show how the basic operations of $\cup$ and $\cap$ interact with groups. Finishing off our exploration of generating groups, we will define the concept of "most general group"; it will turn out that the free groups can be thought of as the groups that "generate" all other possible groups. We will end this chapter by exploring a new way of visualizing the subgroups of finite groups through the use of a graph (in particular a lattice) that will become useful when trying to find how to construct larger groups.

2.1 Basic Definition and Properties

First, a reminder of what it means to take a smaller group from the same group:

Definition 2.1.1: Subgroup

Let G be a group with operation $*$. The subset H of G is a *subgroup* of G if H is a nonempty, closed under the restricted operation $*|_H$, that is, it is closed under products and inverse. We shall write $H \leqslant G$.

Recall that the binary operation $*|_H$ on H is a restriction of the binary operation $*$ on G. In particular, if $* : G \times G \to G$ is the binary operation on G, then we say we *restrict* the binary operation to H if we can define $*|_H : H \times H \to H$, where for any two elements $h_1, h_2 \in H$ $*|_H (h_1, h_2) = *(h_1, h_2)$. If $(H, *|_H)$ is a group, we say it's a subgroup of $(G, *)$. In this way, we say that H has "the same" binary operation as H. We will abuse notation and denote the binary operation of the subgroup $H \leqslant G$ as the same operation as the group G, that is, we will write $(G, \cdot)$ and $(H, \cdot)$ instead of $(G, \cdot)$ and $(H, \cdot|_H)$. If we have $H \leqslant G$ and $H \subsetneq G$, then we will denote this as $H < G$

Example 2.1.2

1. If $H = \{1\}$, then for any G, $H \leqslant G$. H is called the trivial subgroup. Similarly, by definition $G \leqslant G$ always.
2. $\mathbb{Z} \leqslant \mathbb{Q} \leqslant \mathbb{R}$ with the operation of addition.
3. $\mathbb{Z}^+ \nleqslant \mathbb{Z}$ under addition. Can you prove it?
4. Take the set of all symmetries ($\{1, \sigma\} \subseteq D_8$) and all rotations ($\{1, r, r^2, r^3\} \subseteq D_8$). These are both subgroups.

The subgroup H inherits a lot of properties we have proven in section 1.1.1. For example, for any equation of elements in H, it is also an equation of elements in G, and so will have the same solution(s) or lack thereof. Furthermore, the identity will remain the same, since every group must have a unique identity. Similarly for inverses.

For computational purposes, here is a useful proposition that warps up the conditions of a subgroup to be able to more quickly identify whether a subset is or is not a subgroup:

Proposition 2.1.3: The Subgroup Criterion

A subset H of G is a subgroup if and only if

1. $H \neq \emptyset$
2. for all $x, y \in H$, $xy^{-1} \in H$

Furthermore, if H is finite, then it suffices to check that H is non-empty and closed under its operation.

***Proof*:**
Let $H \subseteq G$

($\Rightarrow$) Let $H \leqslant G$. Then $1 \in H$, so $H \neq \emptyset$. Take $x, y \in H$. If $y \in H$, then $y^{-1} \in H$. Since H is closed under its operation, then $xy^{-1} \in H$

($\Leftarrow$) Let $H \subseteq G$ and consider $(H, \cdot|_H)$.

(a) **(associativity)** Associativity comes for free since the binary operation on H is the same, so if $x, y, z \in H$, then $x(yz) = (xy)z$

(b) **(identity)** Since $H \neq \emptyset$, there must exist some $x \in H$. Then by the 2nd condition, $xx^{-1} = 1 \in H$, so $1 \in H$.

(c) **(inverse)** Let $x \in H$. Since $1 \in H$ then $1x^{-1} = x^{-1} \in H$.

(d) **(well-defined)** Let's make sure that if $x, y \in H$ then $xy \in H$ (making sure that the binary operation is still a function). By what we've just shown, $y^{-1} \in H$. Thus $x(y^{-1})^{-1} = xy \in H$ (where $(x^{-1})^{-1} = x$ was proved in proposition 1.1.15). Thus, $\cdot|_H$ is indeed a function

Thus, $H \leqslant G$

Now, let H be a finite group. Then for any $x \in H$, there can only be finitely many distinct elements $x_1, x_2, ..., x_n, ...$, meaning for some $a, b \in \mathbb{N}$, $x^a = x^b$ with $a < b$. This means that $x^{b-a} = 1$, and so for any $x \in H$, x has finite order. Thus, we get that $x^{-1} = x^{n-1}$, where $x^n = 1$. Thus, we automatically get H is a subgroup if H is closed under its binary operation

There are a lot of good exercises if you don't feel comfortable showing a particular subset is a subgroup or not on p.48

Exercise 2.1.1

1. Prove or disprove that the following are groups: (make sub question thing later): $\mathbb{Q}^\times \leqslant \mathbb{R}$, $D_3 \leqslant D_4$
2. Let $H, K \leqslant G$. Is $H \cup K$ always a group? Prove or give a counter-example. If not, then identify what condition(s) is needed for $H \cup K$ to be a group.
3. Prove that G cannot have a subgroup H with $|H| = n - 1$, where $n = |G| > 2$.
4. Let $K \leqslant H$ and $H \leqslant G$. Prove that $K \leqslant G$

2.2 Examples of Subgroups

In this section, we will explore "natural" subgroups as well as subgroups that arise given any homomorphism

2.2.1 Kernel and Image

Given any homomorphism $\varphi : G \to H$, are there subsets that can be created using φ which will be subgroups of either G or H? Naturally, the image of a homomorphism is a subgroup of H ($\varphi(G) \subseteq H$), essentially by definition. We might ask ourselves if there exists some subgroup of G that we can construct using φ?

commented old text here:

For the first subgroup, consider the following set

$$\ker(\varphi) \subseteq G \qquad \ker(\varphi) = \{g \in G \mid \varphi(g) = 1\}$$

then $\ker(\varphi) \leqslant G$

Proof. We will use the subgroup criterion. Take $a, b \in \ker(\varphi)$. Then using properties of homomorphisms:

$$1 = \varphi(ab^{-1}) = \varphi(a)\varphi(b^{-1}) = \varphi(a)\varphi(b)^{-1}$$

and since $\varphi(a) = \varphi(b) = 1$, we get

$$\varphi(a)\varphi(b)^{-1} = 11^{-1} = 11 = 1$$

so $ab^{-1} \in \ker\varphi$

□

the set $\ker(\varphi)$ will thus be given a name:

Definition 2.2.1: Kernel

Let $\varphi : G \to H$ be a homomorphism. Then the set

$$\ker(\varphi) = \{g \in G \mid \varphi(g) = 1\}$$

is the *kernel* of φ

2.2.2 Question to Ponder Let's say $\varphi : G \to H$ and $\psi : G \to H$ are two homomorphisms with the same domain and codomain. If $\ker(\varphi) = \ker(\psi)$, does that imply $\varphi = \psi$?

As we have discussed earlier, $\varphi(G)$ is a group:

$$\varphi(G) \subseteq H, \quad \varphi(G) = \{h \in H \mid \varphi(g) = h \text{ for some } g \in G\}$$

For completeness of this section, we put the proof here:

Definition 2.2.3: Image

Let $\varphi : G \to H$ be a homomorphism. Then the set

$$\varphi(G) = \{h \in H \mid \varphi(g) = h \text{ for some } g \in G\}$$

or

$$\operatorname{im}(\varphi) = \{h \in H \mid \varphi(g) = h \text{ for some } g \in G\}$$

is the *image* of G with φ.

Proof. We will use the subgroup criterion as well as proposition 1.4.4. Since $\varphi(e_G) = e_H$, we have that $\varphi(G)$ is non-empty.

Next, take $a, b \in \varphi(G)$, meaning there exists an $x, y \in G$ such that $\varphi(x) = a$ and $\varphi(y) = b$. Since $\varphi(y^{-1}) = \varphi(y)^{-1}$, we have

$$ab^{-1} = \varphi(x)\varphi(y)^{-1} = \varphi(xy^{-1}) \in \varphi(G)$$

thus, $ab^{-1} \in \varphi(G)$, meaning $\varphi(G)$ passes the subgroup criterion, as we sought to show □

Given that $\varphi(G) \subseteq H$, $\varphi(G)$ has structure of H. However, since $\varphi(G)$ also has some of it's structural information based on G, it would not be surprising if we can somehow characterize $\varphi(G)$ through G. In fact, this will the first link done in the (probably now illusive) chapter 3.

Exercise 2.2.1

1. Do exercises with group actions showing that the kernel of the conjugation action is the centralizer

2.2.2 Center, Centralizers, and Normalizers

Given any group G, does there exist [non-trivial][1] subsets $H \subseteq G$ which will *always* be subgroups (i.e. $H \leqslant G$), regardless of the operation? Not all subsets will form subgroups. For example, take the set

$$T(G) = \{x \in G \mid x^n = 1, \text{ for some } n \in \mathbb{N}_{>0}\}$$

this set is called the *torsion subset* (i.e. the set of all element of finite order). If $T(G) \leqslant G$, then it must pass the subgroup criterion for all possible groups G. However, take

$$G = \langle x, y \mid x^2 = y^2 = 1\rangle$$

both $x, y \in T(G)$. However, $xy^{-1} = xy \notin T(G)$ (note that $y^2 = 1 \Rightarrow y = y^{-1}$) – the element $(xy)^n = xyxyxyxy\cdots$ will never equal to 1, there are no relation that will permit the appropriate cancellation to occur. In particular, note that there is no relation for G which makes $xy = yx$ which would allow the values to commute and cancel. This is rather difficult to prove at this stage in the book; we will come back to this in section 2.3.3. Overall, $T(G) \not\leqslant G$ for all groups G. Notice that if G was abelian then $T(G)$ *will* always be a subgroup (see exercise ref:HERE).

The question arises then if there are properties for which a subset will *always* be a subgroup for any group G regardless of the operation on G. These types of natural subgroups are in general quite hard to find, however they do exist – the rest of this section will be exploring examples of such subsets.

The first one we will explore is a great example of the power of commutativity: Let

$$Z(G) \subseteq G, \; Z(G) = \{g \in G \mid gx = xg \text{ for all } x \in G\}$$

then $Z(G) \leqslant G$

Proof. We will use the subgroup criterion. For this whole proof, let x be an arbitrary element of G. First note that $1x = x1 = x$, and so $1 \in Z(G)$. Next, take any $a, b \in Z(G)$. Since $bx = xb$, multiplying on the left and right by b^{-1} on both sides of the equation:

$$b^{-1}bxb^{-1} = b^{-1}xbb^{-1} \quad \Rightarrow \quad xb^{-1} = b^{-1}x$$

[1]the trivial subset being just the identity $\{e_G\}$, which is always the simplest subgroup which has essentially no "interesting" structure, and hence trivial

and since equality is symmetric, $b^{-1}x = xb^{-1}$, meaning $b^{-1} \in Z(G)$. Finally, we have

$$ab^{-1}x = axb^{-1} = xab^{-1}$$

meaning $ab^{-1} \in Z(G)$, proving it's a subgroup by the subgroup criterion □

This subgroup is special enough to be given a name:

Definition 2.2.4: Center

Let G be an arbitrary group. Define

$$Z(G) := \{g \in G | gx = xg \text{ for all } x \in G\}$$

to be the set of elements commuting with all the elements of G. This subset is called the *center* of G.

2.2.5 trivia The letter Z is because the German word for center is *Zentrum*.

Example 2.2.6: Center of Groups

The provided examples should be verified by the reader as practice

1. If G is abelian, then $Z(G) = G$. Thus $Z(\mathbb{Z}) = \mathbb{Z}$, $Z(\mathbb{R}) = \mathbb{R}$, and so forth
2. $Z(D_4) = \{1, r^2\}$
3. $Z(Q_8) = \{1, -1\}$

The Center of other subgroups we've encountered in section 1.2 will be left as exercises.

This should start giving you an idea of the power of commutativity. In fact, it is even more powerful than we need it to be. The notion of the center can be weakened a bit and still remain a group. Take $A \subseteq G$ (notice that A is just a subset). Then define

$$C_G(A) \subseteq G, \; C_G(A) = \left\{g \in G \mid gag^{-1} = a \text{ for all } a \in A\right\}$$

Note that $gag^{-1} = a$ is equivalent to $ga = ag$. The change in equation is due to a further generalization we will shortly make. Before we get to that, we must first show that $C_G(A)$ is indeed a group

Proof. We will use the subgroup criterion. First, since for any $a \in A$, $1a = a1 = a$, $1 \in C_G(A)$
Next, take $x, y \in C_G(A)$. Since $y \in C_G(A)$, $yay^{-1} = a$ for all $a \in A$. Multiplying on the left by y^{-1} and right by y, we get $a = y^{-1}ay = (y^{-1})a(y^{-1})^{-1}$, showing that $y^{-1} \in C_G(A)$. Finally, to show $xy^{-1}a(xy^{-1})^{-1} = xy^{-1}ayx^{-1} = a$, we use associativity and the fact that $x, y, y^{-1} \in C_G(A)$:

$$\begin{aligned} xy^{-1}axy^{-1} &= xay^{-1}yyx^{-1} \\ &= xax^{-1} \\ &= a \end{aligned}$$

and so $xy^{-1} \in C_G(A)$, showing that $C_G(A) \leqslant G$ by the subgroup criterion □

Since it's a subgroup that will always exist, it is special enough to be given a name:

Definition 2.2.7: Centralizer

Let G be an arbitrary group, and $A \subseteq G$[a]. Define

$$C_G(A) := \{g \in G | gag^{-1} = a \text{ for all } a \in A\}$$

This subset of G is called the *centralizer* of A in G. Since $gag^{-1} = a$ if and only if $ga = ag$, $C_G(A)$ is the set of elements of G which commute with every element of A.
If A is a subgroup of G and $C_G(A) = G$, then A is called a *central subgroup*

[a]Notice A is a *subset*, not necessarily a subgroup

If we take, $A = G$ then $C_G(G) = Z(G)$. Equivalently, notice that:

$$\bigcap_{g \in G} C_G(g) = Z(G)$$

that is, it is the elements which are commutative with *all* elements (see exercise ref:HERE)

If $A \leqslant G$ such that $C_G(A) = G$, that means that every element of G commutes with A. It is left as an exercise (ref:HERE) to show that all central subgroups are contained in the center (hence the name central)

Example 2.2.8: Centralizer of Groups

Let $A = \{1, r, r^2, r^3\} \subseteq D_8$ and $B = \{1, i\} \subseteq Q_8$

1. $C_A(D_8) = A$. Notice how this set is larger than $Z(D_8)$, since $r^3 \in C_A(D_8)$ but $r^3 \notin Z(D_8)$
2. $C_B(Q_8) = \{1, -1, i, -i\}$. Similar to the previous example, notice that $C_B(Q_8)$ is larger than $Z(Q_8)$.

We can generalize this result even further: so far, we have written $ga = ag$ for all $g \in G$. we can re-write this as $g\{a\} = \{a\}g$, where we define $g\{a\} = \{gx \mid x \in \{a\}\}$ and $\{a\}g = \{xg \mid x \in \{a\}\}$ This might seem to be making the situation overly-complex, however the great advantage of this approach is we can now general to taking a general set A

$$gA = Ag \qquad gA = \{ga \mid a \in A\} \;\; Ag = \{ag \mid a \in G\}$$

What this difference does is that it weakens the condition for an element to "directly" commute ($ga = ag$) to "indirectly" commuting, or more precisely commuting with respect to the set A: for every $a \in A$, there exists an $a' \in A$ such that $ga = a'g$. It will not always be the case that if $A \subseteq G$, then $gA = Ag$

Example 2.2.9: Counter-Example and Example

Let $A = \{r, \sigma\} \subseteq D_4$. Take $g = r$. Then:

$$rA = \{r^2, r\sigma\} \qquad Ar = \{r^2, \sigma r\} = \{r^2, r^3\sigma\}$$

notice that $rA \neq Ar$

However, if $A = \{1, r, r^2, r^3\}$ then for any $g \in D_4$, $gA = Ag$. As an example, let $g = \sigma$. Then

$$\sigma A = \{\sigma, \sigma r, \sigma r^2, \sigma r^3\} \qquad A\sigma = \{\sigma, r\sigma, r^2\sigma, r^3\sigma\} = \{\sigma, \sigma r, \sigma r^2, \sigma r^3\}$$

verifying other values of D_4 by hand shows that $gA = Ag$ for all $g \in D_4$

Therefore, we can define the following set of elements that satisfy $gA = Ag$:

$$N_G(A) \subseteq G, \; N_G(A) = \{g \in G \mid gAg^{-1} = A \text{ for all } g \in G\}$$

Again, $N_G(A)$ is a subgroup of G. The proof is almost identical to the proof for $C_G(A)$ (change a for A in the proof). We therefore give a name to this special subgroup:

Definition 2.2.10: Normalizer

Take $gAg^{-1} = \{gag^{-1} | a \in A\}$. Define the *normalizer* of A in G to be the set

$$N_G(A) := \{g \in G | gAg^{-1} = A \text{ for all } g \in G\}$$

If A is a group and $N_G(A) = G$ (that is, $gAg^{-1} = A$ for all $g \in G$) we call A a *normal subgroup* of G

Notice that if we have $g \in C_G(A)$, then $gag^{-1} = a$ with $a \in A$, which means that $gAg^{-1} = A$, and so $g \in N_G(A)$. It follows quickly that $C_G(A) \leqslant N_G(A)$ (see exercise ref:HERE)

Example 2.2.11: Normalizer of Groups

Let $A = \{1, r, r^2, r^3\} \subseteq D_8$ and $B = \{1, i\} \subseteq Q_8$

1. $N_A(D_8) = D_8$. Notice how $Z(G) \subseteq C_A(D_8) \subseteq N_A(D_8)$, that is

$$\{1, r^2\} \subseteq \{1, r, r^2, r^3\} \subseteq D_8$$

Since A is also a subgroup of G, A is also an example of a normal subgroup of D_8

2. $N_B(Q_8) = \{1, -1, i, -i\}$ First, $1 \in N_B(Q_8)$. To show that $i \in N_B(Q_8)$, notice that

$$iB = \{i, -1\} = Bi$$

so $i \in N_B(Q_8)$. Similarly, for -1 and $-i$. For $j, k \in Q_8$, we know that $ij = k$, $ji = -k$ and $ik = -j$, $ki = j$ (see section 1.2.4).

$$jB = \{j, -k\} \neq \{j, k\} = Bj \qquad kB = \{k, j\} \neq \{k, -j\} = Bk$$

meaning $j, k \notin N_B(Q_8)$. Similarly for $-j, -k$. Thus, we are left with $N_B(Q_8) = \{1, -1, i, -i\}$. Notice that $C_B(Q_8) = N_B(Q_8)$, that is, the normalizer doesn't necessarily have to be bigger than the centralizer

3. If $D = \{1, (1\ 2)\} \subseteq S_3$, what is $N_D(S_3)$? What about $C_D(S_3)$?

The notion of the normalizer is not simply generalization for the sake of generalization, it in fact comes up as a very important property: the kernel of a homomorphism has is always normal!

Proposition 2.2.12: Kernels Are Normal Subgroups

Let $\varphi : G \to H$ be a homomorphism, and take $\ker(\varphi) \leqslant G$. Then $\ker(\varphi)$ is a normal subgroup, that is

$$N_G(\ker(\varphi)) = G$$

In other words, for any $g \in G$, $g(\ker(\varphi)) = (\ker(\varphi))g$

***Proof*:**

Let $\varphi : G \to H$, and for notational convenience, let $K = \ker(\varphi)$.

Take any $g \in G$, and $k \in K$, that is, $k \in G$ such that $\varphi(k) = e_H$. We want to show that $gK = Kg$, or equivalently $gKg^{-1} = K$, for all $g \in G$. Take $gkg^{-1} \in gKg^{-1}$. Then

$$\varphi(gkg^{-1}) = \varphi(g)\varphi(k)\varphi(g^{-1}) = \varphi(g)1\varphi(g^{-1}) = \varphi(g)\varphi(g^{-1}) = \varphi(gg^{-1}) = e_H$$

thus, $gkg^{-1} \in K$, so for some k', $k' = gkg^{-1}$. Since this is true for arbitrary $k \in K$, we get $gKg^{-1} = K$, as we sought to show

The fact that kernels are normal subgroups will become very important in chapter 3 This brings us to the end of our generalizations. This section will be closed with a quick remark about normal subgroups. We did not talk about them much here. However, they are at the heart of chapter 3. For now, you can keep them in the back of your mind.

Exercise 2.2.2

1. $C_G(Z(G)) = G$ and deduce that $N_G(Z(G)) = G$
2. If $A, B \subseteq G$, and $A \subseteq B$, then $C_G(B) \leqslant C_G(A)$
3. For each S_3, D_8, Q_8, compute the centralizers of each element and find the center of each group. Does Lagrange's Theorem (ex 19 section 1.7) simplify the work?

   These next 3 subgroups from the next 3 questions will come back when we talk about group actions (and the conjugacy action, section 4.6)
4. Find the centre of $GL_3(\mathbb{C})$
5. Find the centralizer of $\begin{pmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{pmatrix}$ in $GL(3, \mathbb{C})$
6. Find the centralizer of $\begin{pmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 & -1 \end{pmatrix}$ in $SL(5, \mathbb{C})$, the 5×5 complex matrices having determinant equal to 1
7. there are a lot of good exercises in the book
8. . (re-work exercise since $T(G)$ given as an example of why $T(G)$ not considered *natural* torsion

group) Let G be an abelian group. Take $T(G) = \{x \in G \mid \exists n \in \mathbb{N} \text{ such that } x^n = 1\}$. This group is called the *torsion subgroup* of G. Show that $T(G)$ is indeed a group. is $T(G)$ a group is G is not abelian?

9. prove that $\bigcap_{g \in G} C_G(g) = Z(G)$
10. Let $A \leqslant G$ such that $C_G(A) = G$. Prove that $A \leqslant Z(G)$
11. (advanced, maybe move to another section?) Show that $C_G(S) \leqslant N_G(S)$ by showing $C_G(S)$ is the kernel of the homomorphsims $N_G(S) \to \text{Sym}(S)$ by $g \mapsto (x \mapsto gxg^{-1})$ (i.e. $T(g)(x) = T_g(x) = gxg^{-1}$)
12. Maybe do a Galois correspondence question? this wiki-entry made me think of it: here
13. For any group G, and any subset $A \subseteq G$, is it true that $Z(G) \leqslant C_G(A)$?

2.3 Subgroups Generated by Subsets

All groups have an underlying set. We can thus ask ourselves many questions about groups related to it's underlying set. For example, If we have two subgroups $H, K \leqslant G$, is $H \cap K$ or $H \cup K$ also a subgroup? What if we extend to finite or infinite intersection or union? On another hand, we can ask ourselves if there exists subsets $A \subseteq G$ that can in some way generate G using the binary operation. If generating all of G is too strong of a condition, can A always generate a subgroup of G? These are the types of questions that will be asked in this section.

Note that the technics in this section are very common throughout algebra (and many other areas of mathematics). Asking whether a subset of the underlying of an algebraic object "generates" the entire object (or part of the object) will consume our time in many of the following chapters of this book (especially chapter 13).

2.3.1 Cyclic Groups and Cyclic subgroups

We will start our exploration subsets $A \subseteq G$ which are singletons: $|A| = 1$. The first question we will ask ourselves is if there is a subgroup $H \leqslant G$ which contains A. The answer to this is always yes: $G \leqslant G$ is a subgroup and by definition $A \subseteq G$. Though this is a valid answer, this is a very boring answer (in mathematical lingo, we'd say it is the trivial case). Thus, we look for a more interesting question: what is the *smallest* subgroup of G that contains A[2]. If $H \leqslant G$ is the smallest subgroup containing A, H is called a *cyclic subgroup*. If the entire group G is the smallest subgroup containing A, then G is called a *cyclic group*. To find the cyclic subgroup or groups given a subset $\{a\} = A$, a similar strategy will be used from when the order of an element (definition 0.1.14) was discussed. Cyclic groups they pop-up quite often due to their simplicity.

[2]the notion of "smallest" is well-defined if $|G|$ is finite, but get's a bit tricker if $|G|$ is infinite (ex. $2\mathbb{Z} \trianglelefteq \mathbb{Z}$ but $|2\mathbb{Z}| = |\mathbb{Z}|$. We will show in the section 2.3.2 how to be more precise with the notion of smallest

Definition 2.3.1: Cyclic groups

Let G be a group. The group G is *cyclic* if G can be generated by a single element, i.e., there is some element $x \in G$ such that $G = \{x^n | n \in \mathbb{Z}\}$ (where the operation is usually denoted as multiplication)

Sometimes, the notation C_n used to represent arbitrary cyclic groups of order n (with multiplication notations)

We say G is generated by x (or x generated G), and write $G = \langle x \rangle$. It was proven in proposition 1.2.6 that $\langle x \rangle$ is indeed a group. The existence of a C_n for every n is because there exists the cyclic group $\mathbb{Z}/n\mathbb{Z}$ for every n (or $\langle r_n \rangle$ for $r \in D_n$). Later in this section, we will show that $\mathbb{Z}/n\mathbb{Z}$ is the *only* cyclic group of order n up to isomorphism, meaning we will be able to conclude that $C_n \cong \mathbb{Z}/n\mathbb{Z} \cong \langle r_n \rangle$. Since C_n usually has multiplication notation, for this section we will use 1 to represent the identity.

2.3.2 Note The fact that $\langle x \rangle$ is the smallest subgroup containing x forced by construction: if there was a subgroup $H \leqslant G$ such that $H \subsetneq \langle x \rangle$, then there would be number n such that $x^n \in \langle x \rangle$ but $x^n \notin H$. Do you see how this contradicts that H is a subgroup?

Example 2.3.3

1. As mentioned earlier, the usual candidate for this example if $\mathbb{Z}/n\mathbb{Z}$, that is, addition modular n, which can be generated by $\overline{1}$. So $\langle \overline{1} \rangle = \mathbb{Z}/n\mathbb{Z}$. Notice there is some ambiguity here: How can you know if $\overline{1}$ represents the of order $5, 10, 324$? In general, you can't. It is up using the context to determine the order. If the context requires clarity, a subscript will be added: $\langle \overline{1}_n \rangle = \mathbb{Z}/n\mathbb{Z}$ or $\langle [1]_n \rangle = \mathbb{Z}/n\mathbb{Z}$
2. D_n is *not* cyclic. Try finding a single element in D_n which generates all of D_n
3. Is Q_8 cyclic? What about $\mathbb{Z}$? What about $\mathbb{Z} \times \mathbb{Z}$?

Proposition 2.3.4: $|C_n| = |x|$

If $C_n = \langle x \rangle$, then $|C_n| = |x|$ (where if one side of this equality is infinite so is the other).

***Proof*:**
The idea is to show that that if $|x| < \infty$, then if two elements differ in exponent and are smaller than the group, $a \neq b < n$, then $x^a \neq x^b$, which can be shown by contradiction. For the case where the exponent is greater than or equal to n, $t \geq n$, x^t, such an elements can always equal to an element of order less than n. In particular, if $t \geq n$, then $x^t = x^{t-kn}$ for appropriate k for which $0 \leq t - kn < n$. Thus, in the $t \geq n$ case, the proof comes down to the first case that was proved.

For the $|x| = \infty$ case, assume $x^a = x^b$, $a \neq b$, and proceed by contradiction (i.e. $x^{b-a} = 1$).

2.3.5 Note This does not imply that for *any* $x \in C_n$, $|x| = |C_n|$. For example, $\mathbb{Z}/6\mathbb{Z}$ is cyclic since $\mathbb{Z}/6\mathbb{Z} = \langle \overline{1} \rangle$. However, $|\overline{2}| = 3$, meaning $\overline{2}$ cannot generate $\mathbb{Z}/6\mathbb{Z}$. On the other hand, $\overline{1}$ is not the only element of order 6 in $\mathbb{Z}/6\mathbb{Z}$, can you think of a second? Is there more than two?

Corollary 2.3.6: Cyclic $\Rightarrow$ Abelian

If G is cyclic, then G is abelian

***Proof*:**
Let G be a cyclic group. Take some $g \in G$ that generates it. Take some $a, b \in G$ so $g^n = a$ and $g^m = b$ then

$$ab = g^n g^m = g^{n+m} = g^{m+n} = g^m g^n = ba$$

thus, G is also abelian.

You might not be satisfied with this proof: why in the exponent can we commute n and m? The answer comes the associativity of groups. Since

$$a(aa) = (aa)a$$

it makes no difference to write

$$a^{1+2} \qquad a^{2+1}$$

or more generally $a^{n+m} = a^{m+n}$ due to proposition 1.1.15[5].

Since every element in a cyclic is determined by the exponent, having some information about what power of the generator leads to the identity can give us some information on smaller powers of the generator that also lead to the identity.

Proposition 2.3.7: Powers of an Element

Let G be a group, $x \in G$ and let $m, n \in \mathbb{Z}$. If $x^n = 1$ and $x^m = 1$, then $x^d = 1$, where $\gcd(m, n) = d$. In particular, if $x^m = 1$ for some $m \in \mathbb{Z}$, then $|x|$ divides m, $|x|\big|m$

***Proof*:**
This proof relies on Euclid Algorithm (see section 0.2. Let $d = \gcd(a, b)$, so that $m = dm'$ and $n = dn'$. Since $x^m = x^n = 1$, then we have $x^{dm'} = x^{dn'} = 1$. Thus, we have

$$1 = x^{dm'} = \underbrace{x^d x^d \cdots x^d}_{m' \text{ times}}$$

$x^d = 1$ or $x^d = k$ $(k \neq 1)$ so $k^{m'} = 1$. If $x^d = 1$, then we're done, so let's say $x^d = k$. Then, similarly for $n = dn'$:

$$1 = x^{dn'} = \underbrace{x^d x^d \cdots x^d}_{n' \text{ times}}$$

where $x^d = 1$ or $x^d = k$ (same k) so $k^{n'} = 1$. Thus, we get $k^{n'} = k^{m'} = 1$. However, here lies the contradiction. Without loss of generality, let's say $n' < m'$ (if $n' > m'$, simply switch the order of the elements in the proceeding proof) so that $m' = n' + \ell$. Then since

$\gcd(n', m') = 1$, $k^{m'} = k^{n'+\ell} = k^{n'}k^{\ell} = k^{\ell} \neq 1$. If it did, then m' would be a multiple of n', but then $\gcd(m', n') \neq 1$

Thus, $x^{\gcd(n,m)} = 1$, as we sought to show

Corollary 2.3.8

Let G be a group (not necessarily cyclic), let $x \in G$, and let $a \in \mathbb{Z} - \{0\}$.

1. if $|x| = \infty$, then $|x^a| = \infty$
2. if $|x| = n < \infty$, then $|x^a| = \frac{n}{\gcd(n,a)}$. If a divides n, then $|x^a| = \frac{n}{a}$.

***Proof*:**

1. If $|x^a| < \infty$, then there would exist an n such that $(x^a)^n = x^{an} = 1$. (if an is negative, take $-an$). But then x would have finite order, a contradiction
2. Since $x^n = 1$, $1 = 1^a = (x^n)^a = x^{na}$. By proposition 2.3.7, $x^{gcd(a,an)} = 1$

With this idea in mind, let's do what we did before and try to see if we can classify cyclic groups.

For cyclic groups, there's in fact an incredibly simple isomorphism which will be introduced showing that if two cyclic groups have the cardinality (same order), they are in fact isomorphic, and thus can be treated as interchangeable! This means that when dealing with future theorems about cyclic groups, once we found it applies to a cyclic group of a certain order, we know it applies to any cyclic group of the same order! This also means there are only countably many finite cyclic groups, each with a predictable structure.

Theorem 2.3.9: Isomorphism between same-order Cyclic Groups

Any two cyclic groups of the same order are isomorphic. More specifically,

1. if $n \in \mathbb{Z}^+$ and $\langle x \rangle$ and $\langle y \rangle$ are both cyclic groups of order n, then the map
$$\psi : \langle x \rangle \to \langle y \rangle \qquad x^k \mapsto y^k$$
is well defined and is an isomorphism
2. if $\langle x \rangle$ is an infinite cyclic group, the map
$$\psi : \mathbb{Z} \to \langle x \rangle \qquad k \mapsto x^k$$
is well defined and is an isomorphism

This proof is not too difficult and so a guide has being laid out for the reader to deduce the proof

***Proof*:**
First, show the function is well defined, that is, if $x^a = x^b$ (for possibly different a and b), then $y^a = y^b$. This uses proposition 2.3.7. Then show the map is a homomorphism. To prove it's an

isomorphism, you can either

1. Show there exists a homomorphism $\varphi : \langle y \rangle \to \langle x \rangle$ such that $\varphi \circ \psi = \psi \circ \varphi = \text{id}$ (technically, $\text{id}_{\langle x \rangle}$ or $\text{id}_{\langle y \rangle}$ respectively)
2. Show ψ is injective. At which point subjectivity comes from the fact that both groups have the same cardinality and are finite.
3. Show ψ is surjective. At which point injectivity comes from the fact that both groups have the same cardinality and are finite.

For the infinite case, the map is "clearly" well-defined: if $a = b$, then $x^a = x^b [= x^a]$. By the exponent laws, this map is homomorphic. Then to prove this map is an isomorphism, you can either:

1. Show there exists a homomorphism $\varphi : \langle x \rangle \to \mathbb{Z}$ such that $\varphi \circ \psi = \psi \circ \varphi = \text{id}$ (technically, $\text{id}_{\mathbb{Z}}$ or $\text{id}_{\langle x \rangle}$ respectively)
2. Show the map is injective and surjective (not that injectivity (resp. surjectivity) no longer implies surjectivity (resp injectivity) in the infinite case, even if the cardinalities are the same. Can you think of a surjective but not injective map from $\mathbb{Z}$ to $\mathbb{Z}$? An injective but not surjective map from $\mathbb{Z}$ to $\mathbb{Z}$? see exercise ref:HERE (in Cardinality section))

This let's us properly conclude what was stated at the beginning of this section: if G is s cyclic group of order n (ex. $\mathbb{Z}/n\mathbb{Z}$ or $\langle r \rangle$, $r \in D_n$) then

$$\mathbb{Z}/n\mathbb{Z} \cong \langle r \rangle$$

Hence, instead of referencing a specific cyclic group, it is sometimes useful to just write C_n (some authors like Z_n) to represent some arbitrary cyclic group of order n, and use $\mathbb{Z}$ for an infinite cyclic group (instead of C_∞ or Z_∞)

Now that we classified all cyclic groups, we can ask ourselves the question of which elements generate one of these groups. this next proposition leads us to the answer

Now that we know how many cyclic sub-groups there are, and how these groups can be generated, we can explore what kind of sub-groups we get from cyclic groups. Quite naturally, every subgroup of a cyclic sub group is cyclic. It will turn out that if you have a cyclic group of infinite order, it has an infinite number of cyclic subgroups, and if its of finite order, lets say n, then every sub groups is of an order that divides n. That is , if $H \leqslant G$, G is cyclic, $|H| = a$, then $a|n$.

Proposition 2.3.10: x relation to cyclic group H

Let $H = \langle x \rangle$.

1. Assume $|x| = \infty$. Then $H = \langle x^a \rangle$ if and only if $a = \pm 1$.
2. Assume $|x| = n < \infty$. Then $H = \langle x^a \rangle$ if and only if $(a, n) = 1$. In particular, the number of generators of H is $\psi(n)$, where ψ is Euler's totient function. (which gives the number of relatively prime numbers to it).

Proof. Dummit p.57-58 □

Finally, we'll put together all our knowledge into one theorem for easy reference:

Theorem 2.3.11: elements that generate cyclic subgroup

Let $H = \langle x \rangle$ be a cyclic group

1. Every subgroup of H is cyclic. More precisely, if $K \leqslant H$, then either $K = \{1\}$ or $K = \langle x^d \rangle$, where d is the smallest positive integer such that $x^d \in K$
2. If $|H| = \infty$, then for any distinct non negative integers a, b, $\langle x^a \rangle \neq \langle x^b \rangle$. Furthermore, for every integer m, $\langle x^m \rangle = \langle x^{|m|} \rangle$, where $|m|$ denotes the absolute value of m, so that the nontrivial subgroups of h correspond bijectivity with the integers $1, 2, 3, ...$
3. If $|H| = n < \infty$, then for each positive integer a dividing n, there is a unique subgroup of H of order a. This subgroup is the cyclic group $\langle x^d \rangle$, where $d = \frac{n}{a}$. Furthermore, for every integer m, $\langle x^m \rangle = \langle x^{gcd(n,m)} \rangle$, so that the subgroups of H correspond bijectivity with the positive divisors of n.

intuition

***Proof*:**
This proof involves a lot of counting.

1. exercise
2. exercise

Also, once you have Lagrange's Theorem (3.3.1), this proof will become much easier.

Note that the finite group examples of theorem 2.3.11 will eventually easily generalize to perhaps the most use theorem of the course: Lagrange Theorem (theorem 3.3.1), which sates that the order of any subgroup divides the order of a group ($H \leqslant G \Rightarrow |H| \big| |G|$).

The next theorem is in a sense a generalization of part (3) of 2.3.11, since it uses this fact. Once I proof Lagrange, I'll reference this theorem to make sure this makes sense. I do want to introduce this fact here because it feels quite appropriate as a generalization of the previous theorem:

Corollary 2.3.12: if G has order p, then it is cyclic

If G is a group of prime order p, then G is cyclic, hence $G \cong Z_p$.

***Proof*:**
First show that G is cyclic. Let $x \in G$. since $|G| = p$, then $|G| > 1$ so let $x \neq e$. We can always take $\langle x \rangle \subseteq G$. Then by Lagrange theorem (generalization of part 3 of theorem 2.3.11), we know that $\langle x \rangle \big| |G|$, However, $|G|$ is prime, so only 1 and G divide it. Since $x \neq e$, then $|\langle x \rangle| \neq 1$, so $|\langle x \rangle| = p$, meaning that G was generated by a single element, and so is cyclic. By corollary 2.3.6, it is abelian.

To show its isomorphic to $\mathbb{Z}_p$ should now be easy.

Note that this doesn't mean that if $H \leqslant G$ where $|H| = p$ for some prime that $H \in Z(G)$!

Example 2.3.13: : $H \notin Z(G)$
Take D_8. Note that $\langle\sigma\rangle \leqslant D_8$ and $|\langle\sigma\rangle| = 2$, and so it's cyclic and abelian. However. $r\sigma = \sigma r^3$ meaning that $\sigma \notin Z(D_8)$. However, it is in the centralizer $C_{D_8}(\sigma) = \{g \in G | g\sigma = \sigma g\}$ for $\sigma\sigma = e = \sigma\sigma$, which can also be "obvious" since $\langle\sigma\rangle$ is abelian, and so it would be in the centralizer.

The intuition behind this is that for something to be in the center, it must commute with *all* of G. It is possible to take a subgroup that only commutes (as we've seen with the idea of the centralizer). This is why the intersection under definition 2.2.4 is true: it must commute with everything.

2.3.14 Note You cannot take $\langle x\rangle \leqslant G$ and do $G - \langle x\rangle$. G is not necessarily a group. Take D_8, and take $K = \langle\sigma\rangle$. Then consider $H = D_8 - K$. Then $r^3 \cdot r\sigma = \sigma$, showing that H is not closed under multiplication. Thus, subtraction of groups is not well-defined. However, it will turn out that under the right circumstances, *dividing* and *multiplying* groups will be well-defined! This will be the subject of the next chapter (3)

interesting exercise

1. number 9
2. number 13
3. number 16
4. Knowing Cyclic groups, we can look at another great example of a homomorphism: $\psi : \mathbb{Z} \to Z_n$, where $Z_n = \langle x\rangle$,
$$\psi(a) = x^a$$
Prove that it's a homomorphism, then figure out what elements of $\mathbb{Z}$ map to an element of Z_n.
5. the product of two cyclic subgroups is not necessarily cyclic!

2.3.2 Generating Subgroups

We now turn our attention to generating groups or subgroups using more than 1 element. When studying cyclic groups, we asked whether given an element $x \in G$, does there exists a smallest group $H \leqslant G$ such that $x \in H$. It was shown that taking the closure of x by taking all powers of x resulted in such a group, namely $H = \langle x\rangle$. Phrased differently, the subgroup $\langle x\rangle$ is the smallest subgroup of G that contains the set $\{x\}$. What it means to be the *smallest* is if $L \leqslant G$ is another subgroup where and $x \in L$, then $\langle x\rangle \subseteq L$. We now replace the singleton $\{x\}$ with some arbitrary subset $A \subseteq G$.

As mentioned in the beginning of this chapter, generating subgroups from a subset of a group is not unique to Group Theory. In the following chapters studying other algebraic structures (ex. Ideals, rings, modules, vector-spaces, algebras, fields, etc.), the theme of trying to find minimal sub-object $S \leqslant X$ (subgroup, subspace, subfield, subring, etc.) that contain a subset $A \subseteq X$ from the original object (so that $A \subseteq S \leqslant X$) will repeat itself, and so will the general method for finding the minimal subobject. To find a minimal subobject there are two possible general approach:

1. (Generation Method) If $A \subseteq X$ is a subset of the object, take all possible subobjects containing A and intersect them. In the case of groups, if $A \subseteq G$, then take all possible subgroup $H_A \leqslant G$ containing A (so $A \subseteq H_A \leqslant G$ and intersect them all.

2. (Closure Method) The above approach can be thought as the "global approach". Another way of finding the minimal subobject is instead by the "element approach" (or "local approach"): If A is a subset of the object, then let $\overline{A}$ be the set of all possible combinations of elements of A using the binary operation of G. In the case of groups, of $A \subseteq G$, then $\overline{A}$ will be all the possible combination of elements of A with the binary operation of G.

The main result of this section is that these two methods will coincide and form the same subgroup. In proceeding chapters covering subobjects (subrings, ideals, subspaces, etc.), some details will be omitted due to them being covered in this section, so it is advised that this section be well understood.

We start with the first method. We need show that the intersection of subgroups still forms a group:

Proposition 2.3.15: intersection of Groups

Let G be a group and let $\mathcal{H}$ be any collection of subgroups of G. Then

$$\bigcap_{H \in \mathcal{H}} H$$

Is a subgroup of G.

***Proof*:**
Let $K = \bigcap_{H \in \mathcal{H}} H$. First, $e \in H$ for all $H \in \mathcal{H}$, so $e \in K$. Next, take $a, b \in K$. Since $b \in K$, b is in every subgroup H in $\mathcal{H}$. Thus, b^{-1} is in every subgroup H in $\mathcal{H}$, and so $b^{-1} \in K$. Similarly, ab^{-1} would be in every subgroup H in $\mathcal{H}$ (since each subgroup passes the subgroup test), so $ab^{-1} \in K$. Thus, by the subgroup test, K is a subgroup of G

Therefore, the following is well-defined:

Definition 2.3.16: Minimal Subgroup via Generation

let G be a group and $A \subseteq G$. Then

$$\langle A \rangle = \bigcap_{\substack{H \leqslant G \\ A \subseteq H}} H$$

is called the *closure of* A, or the subgroup *generated by* A. If A is a finite set $\{a_1, a_2, ..., a_n\}$, we can write $\langle a_1, a_2, ..., a_n \rangle$. Furthermore, $\langle A, B \rangle := \langle A \cup B \rangle$ for the union of two sets.

We have already been using this notation when $A = \{x\}$ is a singleton, in particular cyclic subgroups are the closure of singletons. Note that $\langle A \rangle$ always exists, since at minimum $A \subseteq G$ (and naturally $G \leqslant G$), so G is part of the sets in be intersection. This is important to point out as there exists sets where $\langle A \rangle = G$:

Definition 2.3.17: Generating Set and Generators

Let $A \subseteq G$. Then if $\langle A \rangle = G$, then A is called a *generating set* for G, and the elements of A are called *generators*.

If $A = \emptyset$, then A is containing in every group, including the trivial group $\{e_G\}$. As the intersection $\{e_G\}$ with any group is $\{e_G\}$, we see that $\langle \emptyset \rangle = \{e_G\}$.

Proposition 2.3.18: Minimal Subgroup using Generation

Let $A \subseteq G$. Then $\langle A \rangle \leqslant G$ is the unique smallest subgroup containing A.

***Proof*:**
If $H' \leqslant G$ and $A \subseteq H'$, then by definition H' is one of the subgroup that were intersected in the aforementioned definition, since $\langle A \rangle$ is equal to this intersection, $\langle A \rangle \subseteq H'$, so $\langle A \rangle$ is minimal.

Next, $\langle A \rangle$ is the unique smallest subgroup containing A (similarly to how $\langle x \rangle$ is unique in the sense that all cyclic subgroups of the same order are isomorphic to it). If there exists another group K' that was also the smallest subgroup containing A, then K' would be in the intersection above, so $\langle A \rangle \subseteq K'$. But since K' is supposed to be the smallest subgroup containing A, it is also the case that $K' \subseteq \langle A \rangle$. Therefore $\langle A \rangle = K'$, showing uniqueness as well, completing the proof.

The limit with this approach is that all the possible subgroups containing A must be found, and that is not always feasible. There isn't a general method to find elements of $\langle A \rangle$ (except, naturally, that $e_G \in A$). The 2nd approach will give a better insight into this problem by changing perspective away from the high-level and starting with the elements of A and trying to "generate" a subgroup containing A. To that end, define the set

$$\overline{A} := \{a_1^{\sigma_1} a_2^{\sigma_2} \cdots a_n^{\sigma_n} \mid n \in \mathbb{Z} \text{ and } a_i \in A,\ \sigma_i = \pm 1, 1 \leq i \leq n\}$$

where if $A = \emptyset$ then $\overline{A} = \{e_G\}$. The set $\overline{A}$ contains all possible finite products (using the binary operation of G) of the elements from A. Note that the a_i's need not be distinct, for example a^2 for $a \in A$ would be written $a^1 a^1$. It is also easy to verify that this set is indeed a subgroup of G using the subgroup criterion (consider that n can vary in any possible length, see exercise 2.3.1(2.3.1 (1))). The interesting question is whether this set is also the smallest minimal subgroup containing A, and indeed it is:

Proposition 2.3.19

Let G be a group and $A \subseteq G$ be a subset of G. Then

$$\langle A \rangle = \overline{A}$$

***Proof*:**
exercise; here is an outline:

1. Let $x \in \overline{A}$ so that they both can be written as:

$$x = a_1^{\sigma_1} \cdots a_n^{\sigma_n}$$

By using closure properties, show that $x \in \langle A \rangle$.

2. Conversely, assume $x \in \langle A \rangle$ so that $x \in H_A$ for each H containing A. For the sake of contradiction, assume that x is not of the above form. Show then that $\overline{A} \subsetneq \langle A \rangle$ contradicting minimality of $\langle A \rangle$.

From now on, the notation $\langle A \rangle$ will be used to represent the subgroup generated by A.

We can ask some basic question about $\langle A \rangle$ given A and G. If G is abelian, then the order the elements are written in doesn't matter, and so if $A = \{a_1, ..., a_k\}$, every element can be written as:

$$a_1^{n_1} a_2^{n_2} \cdots a_k^{n_k}$$

This is not the case if G is not abelian. Consider for example:

$$G = \langle a, b \mid a^2 = b^2 = 1 \rangle \tag{2.1}$$

Notably, $ab \neq ba$. Next, if each $a_i \in A$ has finite order, then if G is abelian we have that G is finite. This is because:

$$(ab)^n = a^n = b^n$$

See exercise 2.3.1(2.3.1 (2)). This is not necessarily the case if G is non-abelian. Consider again the group in equation (2.1) where $|a| = 2$ and $|b| = 2$. Then we see that:

$$abababa....$$

If G is not abelian, then as mentioned in remark 1.3.5 and example 1.3.6, it is exceedingly difficult to determine the structure of G.

We may also ask if there is any "optimal" set of generators for a group, that is a "best" subset $A \subseteq G$ st $\langle A \rangle = G$. Generally, there can be many sets that generate a group: for example $\langle r, \sigma \rangle = \langle r\sigma, \sigma \rangle = D_n$, or $\langle 2, 3 \rangle = \langle 7, 5 \rangle = \mathbb{Z}/10\mathbb{Z}$. Notice that no proper subset of $\{2, 3\}$ generates $\mathbb{Z}/10\mathbb{Z}$, however there do exist smaller subsets of $\mathbb{Z}/10\mathbb{Z}$ that generate $\mathbb{Z}/10\mathbb{Z}$ (namely, $\{1\}$). Thus, the cardinality of the minimal generating subset cannot be determined by simply finding a minimal generating subset: it would have to be $\min \{|A| \mid A \subseteq G,\ \langle A \rangle = G\}$[3].

Notice also that in the case of D_4 with generators $a = \sigma$ and $b = r\sigma$ have order 2, while D_4 has order 8. Using this show how it is not always the case that all elements of D_4 will be written of the form $a^\alpha b^\beta$ $(\alpha, \beta \in \mathbb{Z})$ like in the abelian case, in particular show that aba cannot be written as $a^\alpha b^\beta$ for any $\alpha, \beta \in \mathbb{Z}$.

[3] If G is infinite, it might be that we require the stronger notion of an Infimum: $\inf \{|A| \mid A \subseteq G,\ \langle A \rangle = G\}$

2.3.20 Food for Thought If G is finitely generated, is every subgroup of G finitely generated? In other words, if

$$G = \langle S|R\rangle$$

where S is a set of elements and R is the relation on those elements, then can you find a subgroup $H \leqslant G$ such that you need an infinite number of elements to generate H? There is an answer to this question which will be given as an exercise 2.3.2(2.3.2 (1)), but it can be useful to contemplate this question now as an exercise in the types of questions a mathematicians might ask themselves. You can further ask if $\varphi : G \to H$ is a homomorphsims, and G is finitely generated, is $\varphi(G)$ finitely generated?

Exercise 2.3.1

1. Show that $\overline{A}$ is indeed a subgroup
2. Let G be abelian and $A \subseteq G$ a finite set containing only elements of finite order. Show that $\langle A\rangle$ is finite.
3. Show that every finitely generated subgroup of $\mathbb{Q}$ is cyclic
4. Let (Presentation for V_4. This group is called the Klein 4-group, or *Viergruppe*. Prove that $V_4 \cong \mathbb{Z}_2 \times \mathbb{Z}_2$, showing that a group has a non-unique representation.
5. If $H, K \leqslant G$, is $H \cup K$ a group? If $H \cup K$ is a subgroup in some case, what relation would there be between H and K (does on include the other?)
6. Let $G = k[x, 1/x]$ be the group of polynomials under multiplication.

2.3.3 Free Groups

(somewhere define "formal combination". Will need it for defining polynomial rings and tensors, and other free objects more generally)

So far, when we were generating groups from sets, we either took a subset of G (and so took the relation between elements of G), or we defined the set given a presentation. For example, We can make a group from $\{a, b\}$ by defining [4]

$$\langle a, b|a^n = 1, b^2 = 1, ab = b^{-1}a\rangle$$

what happens if we get rid of one or two of these relations? What about if we get rid of all three – we make a group "free" of relations besides necessary the ones needed to form a group? We explore this question in this section.

Since this group has no restrictions, it can be thought of as the "most general" group on that number of elements. In fact, something more subtle is going on here: since $\{a, b\}$ (or any set A to that matter) is essentially an arbitrary set, we're asking if from *any* arbitrary set, is there a way of defining a group using as little "extra group structure" as possible, or put another way, as free from relations as possible. If we can define such a group given $\{a, b\}$, is there a way of "defining" another group generated by two elements using this more "free group" by adding relations to the "free group"?

[4] This is D_n

The answers to this question lies in exploring how to make a *function* (or to be pedantic a *set-function*) from any set A to a group G

$$f : A \to G$$

be "lifted" to a *group-homomorphism*

$$\varphi : F(A) \to G$$

where $F(A)$ will be our "free group" on A (the notation $F(A)$ is suggestive that this group will indeed be called a free group). The word "lifted" here means that if we restrict $F(A)$ to A, then $\varphi|_A = f$. This can be visualized with the following diagram (called a commutative diagram):

$$\begin{tikzcd} F(A) \arrow[r] & G \\ A \arrow[u] \arrow[ur] & \end{tikzcd}$$

Thus, we turn a set function into a group-homomorphism while still keeping the structure of the set-function. More generally, if we have $f : A \to G_2$, then if there exists a function $f_2 : A \to G_1$ and a homomorphism $\varphi : G_1 \to G_2$ such that $f_2 = \varphi \circ f_1$, then φ is called a lift of f_1. Equivalently, φ is a lift of f_1 is the following diagram commutes:

$$\begin{tikzcd} G_1 \arrow[r, "\varphi"] & G_2 \\ A \arrow[u, "f_1"] \arrow[ur, "f_2"'] & \end{tikzcd}$$

Why are lifts the key concept? It is because not all set-function can be lifted to group homomorphism:

Example 2.3.21: Lifts and non-Lifts

1. Let $A = \{a\}$, $G_1 = \mathbb{Z}/5\mathbb{Z}$, and $G_2 = \mathbb{R}$. Define
   (a) $f_1 : A \to G_1,\ f_1(a) = 0$
   (b) $f_2 : A \to G_2,\ f_2(a) = 4$

   Does A lift to G_1, that is:

$$\begin{tikzcd} \mathbb{Z}/5\mathbb{Z} \arrow[r] & \mathbb{R} \\ \{a\} \arrow[u, "f_1"] \arrow[ur, "f_2"'] & \end{tikzcd}$$

   that is, does there exist a homorphism $\varphi : \mathbb{Z}/5\mathbb{Z} \to \mathbb{R}$ such that $f_2 = \varphi \circ f_1$ If we can find a homomorphism φ such that

$$\begin{tikzcd} \mathbb{Z}/5\mathbb{Z} \arrow[r, "\varphi"] & \mathbb{R} \\ \{a\} \arrow[u, "f_1"] \arrow[ur, "f_2"'] & \end{tikzcd}$$

   then a lift exists. Let's say there was such a φ. Then we have that

$$4 = f_2(a) = (\varphi \circ f_1)(a) = \varphi(0) = 0$$

that is $4 = 0$ in $\mathbb{R}$. Note that since φ is a homomorphism, we are forced to have $\varphi(0) = 0$ as we showed in proposition 1.4.4. However, we already know that $4 \neq 0$. Thus, no such homomorphism can exist. Therefore, these is lift from $\{a\} \xrightarrow{f_1} \mathbb{R}$ to $\mathbb{Z}/5\mathbb{Z} \to \mathbb{R}$

2. What if we swap 0 and 4? Then:

$$0 = f_2(a) = (\varphi \circ f_1(a)) = \varphi(4)$$

Before, we were forced to say $\varphi(0) = 0$. However, this time we are only restricted to saying $\varphi(4) = 0$. There does indeed exist a homomorphism from $\mathbb{Z}/5\mathbb{Z}$ to $\mathbb{R}$ which satisfies this property: $\varphi(x) = 0$ (i.e., the trivial homomorphism). No other homomorphism can exist, since if $\varphi(4) \neq 0$, then $0 = \varphi(0) = \varphi(4 + 4 + 4 + 4 + 4) = 5 \cdot \varphi(4) \neq 0$ – a contradiction.

3. Let's take the same example as before, but let $f_1'(a) = 1$. Like last time, let's assume a homomorphism φ existed such that $f_2 = \varphi \circ f_1'$. Thus

$$1 = f_2(a) = (\varphi \circ f_1')(a) = \varphi(1)$$

Thus, $\varphi(1) = 1$. As a consequence

$$\varphi(1 + 1) = \varphi(1) + \varphi(1) = 1 + 1 = 2 \cdots$$

meaning $\varphi(n) = n$. However, this implies that

$$0 = \varphi([0]) = \varphi([5]) = 5$$

but $0 \neq 5$. Thus, no such homomorphism exists, and so there is no morphism from (f_1', G_1) to (f_2, G_2).

4. Let $G_1 = G_2 = \mathbb{Z}/5\mathbb{Z}$ and let $A = \{a\}$. Let

   (a) $f_1 : \mathbb{Z}/5\mathbb{Z} \to \mathbb{Z}/5\mathbb{Z}$, $f_1(a) = 1$

   (b) $f_2 : \mathbb{Z}/5\mathbb{Z} \to \mathbb{Z}/5\mathbb{Z}$, $f_2(a) = 2$

   (TBD: want to show how there can be more than 1 (non-trivial) morphism between these two objects of $\mathcal{F}^A$)

So what group can take the place of $F(A)$ (if any) that will always satisfy a lift? Perhaps it is a group we have already encountered. We have shown in example 1.3.4 and section 1.2.2 how to define the trivial and symmetric group from arbitrary sets. Let's label the trivial group on A as $T(A)$, and the symmetric group on A as S_A. Though this shows that there is more than one way of defining a group operation on arbitrary sets, this does not help us in finding the most "general" group that can be defined on a set. Let's take the earlier presentation of D_n. The trivial group has no information about this group. The group S_2 has 2 element: the trivial element and (1 2). However, $|D_n| = 2n$, meaning it has many more elements than S_2. Thus, such a group cannot be used as the "most general group on the set $\{a, b\}$"

In terms of intuition in relation to lifts, we see that:

$$A \to G \implies T(A) \to G$$

will not in general be a lift (choose some set $|A| > 1$ and a nontrivial group G). Similarly,

$$A \to G \implies S_A \to G$$

will not be a lift in general.

The problem with both these examples is that they in fact define a lot of structure on their underlying set. Instead, we will try to make a structure which puts the minimal amount of relations between objects to make it a group so that hopefully $F(A) \to G$ can always be a lift! Before I show the answer, I encourage you to stop reading for a moment try to think of another way you can define a group on some arbitrary set A, and perhaps think about how this group can be thought of as the "most general".

The High-level Idea

The first time you'll see this construction, you might think we added a lot of fluff in the way of the answer. However, I assure the reader that the longer build-up will have larger rewards! The construction we're doing here will repeat itself multiple times in the book in subtly different ways. This longer construction allows us to link all these constructions as examples of *free objects* [5], and more generally *universal properties* [6] later down the line. This will be the first example in the book of a universal property, it's the *universal property of free groups.*

Let's say A is an arbitrary set. Our goal is to construct a group, which we will label $F(A)$, from which every other group which contains A can be derived. First, we will be "relating" A to any group G. Let's take all set-functions:

$$f : A \to G$$

Where A is a set that maps into the group G [7]. Let's label this function along with its co-domain G as (f, G). Choosing all possible f and G will make all possible functions from A to every possible group, essentially telling us all ways A relates to any group G. The fact that there is no restriction on the functions from A to G is a way of saying there is *no* group structure on A which limits the types of maps we can form. Remember that when we lift the map $f : A \to G$ to $\varphi : F(A) \to G$, we need that $\varphi|_A = f$, so the fact that the map can be arbitrary means that $F(A)$ must accommodate for any function from A to G. As we have shown in example 2.3.21, not all groups can accommodate this. Let's label the set of all possible (f, G) as $\mathcal{F}^A$:

$$\mathcal{F}^A = \{(f, G) | f : A \to G\}$$

Next, we want to be able to relate the groups in the co-domain. We want to define $F(A)$ to be able to lift from a set-function to a group-homomorphsims. What we will do is define the following relation between any two set's (f_1, G_1), (f_2, G_2): we will say there is a *morphism* between (f_1, G_1), (f_2, G_2)

$$(f_1, G_1) \xrightarrow{\varphi} (f_2, G_2)$$

if the following diagram commutes:

[5] If you're curious to know where else we will encounter free objects, we will see *free monoids* in section 7.1 (which will turn out to be isomorphic to the set of all polynomials!), and *free-modules* in section 12.3.2

[6] You'll see these everywhere, to name a few: theorem 3.1.14, theorem ref:HERE (product), theorem ref:HERE (coproduct), proposition 9.2.4, theorem 9.5.4, theorem 15.3.4, theorem 15.1.12, and more!

[7] It might be more intuition for every f to be *injective*, since this way we'd get that every G contains the set A. We can indeed do this, and we will actually get the same result. For reasons that will be elaborated later in this section, the more general approach is desirable to be able to repeat this trick later in the textbook

$$\begin{tikzcd} G_1 \arrow[r, "\varphi"] & G_2 \\ A \arrow[u, "f_1"] \arrow[ur, "f_2"'] & \end{tikzcd}$$

that is, where $f_2 = f_1 \circ \varphi$. You can think of the morphism φ as a homomorphism that is restricted by f_1 and f_2. In fact, every example in examples 2.3.21 can be re-written to be morphisms from $\mathcal{F}^A$. Take a moment to go back and translate the previous notation to the new one, and think about what $\mathcal{F}^A$ with its objects and morphisms represent. We've essentially taken any group homomorphism between any two groups $G_1 \xrightarrow{\varphi} G_2$ and also made a relation between A and these groups. In fact, as reader's who have read section 0.4 and 8 might have guessed, we have defined a category

Proposition 2.3.22: $\mathcal{F}^A$ is a category

The set $\mathcal{F}^A$ along with the morphisms form a category (we can write $\mathfrak{F}^A S$ as a shorthand to represent $\mathcal{F}^A$ and it's morphisms). In particular:

1. For every $(f, G) \in \mathfrak{F}^A$, there exists an identity morphism $1_{(f,G)} : (f, G) \to (f, G)$
2. If $\varphi, \psi \in \mathfrak{F}^A$ are morphisms

$$\varphi : (f_1, G_1) \to (f_2, G_2) \qquad \psi : (f_2, G_2) \to (f_3, G_3)$$

   then $\psi \circ \varphi$ is a morphism.

***Proof*:**
First, composition of morphisms will be a morphism in $\mathcal{F}$. If $\varphi, \psi \in \mathcal{F}$ are composable morphisms (i.e. the codomain of φ matches the domain of ψ) then

$$\begin{tikzcd} G \arrow[r, "\varphi"] \arrow[rr, bend left, "\psi\circ\varphi"] & H \arrow[r, "\psi"] & K \\ & A \arrow[ul, "f"] \arrow[u, "f'"] \arrow[ur, "f''"'] & \end{tikzcd} \quad \Rightarrow \quad \begin{tikzcd} G \arrow[rr, "\psi\circ\varphi"] & & K \\ & A \arrow[ul, "f"] \arrow[ur, "f''"'] & \end{tikzcd}$$

that is, the morphism $\psi \circ \varphi \in \mathcal{F}^A$ with domain (f, G) and codomain (f'', K) exists

Next, we will check each object $(f, G) \in \mathcal{F}^A$ has an identity morphism. Take an arbitrary $(f, G) \in \mathcal{F}^A$, so that $A \xrightarrow{f} G$. Then the homomorphism $\mathrm{id}_G : G \to G$, $\mathrm{id}_G(x) = x$ satisfies

$$\begin{tikzcd} G \arrow[r, "\mathrm{id}_G"] & G \\ A \arrow[u, "f"] \arrow[ur, "f"'] & \end{tikzcd}$$

If we take any $\varphi \in \mathcal{F}^A$ where $\varphi : (f, G) \to (f', H)$, then $\mathrm{id}_H \varphi = \varphi \mathrm{id}_G = \varphi$, showing that identities exist.

The fact that any compossible triple is associative comes from function composition.

With the category $\mathfrak{F}^A$ finally defined, we can ask the following question

Does $\mathfrak{F}^A$ have an initial object[8]?

That is, does there exists (f^*, G^*) such that for any object $(f, G) \in \mathfrak{F}^A$, there exists a unique morphism φ^* such that

$$\begin{tikzcd} G^* \arrow[r, "\varphi^*"] & G \\ A \arrow[u, "f^*"] \arrow[ur, "f"'] & \end{tikzcd}$$

Notice that this is the same lifting question we have asked about a theoretical "free group" $F(A)$ near the beginning of this section, but re-phrased! You might ask if such a group even exists? If such a group does exist, then have found a group which let's us relate it to any other group. We will call such a group (if it exists) a *free group*. Essentially, $F(A)$ must be able to relate to any other group, making it the "most general" group. But what is $F(A)$? In the following section, we will construct such a group that will indeed be the initial object of $\mathfrak{F}^A$.

Constructions of $F(A)$

Let's first give the answer to this question when $A = \{a\}$. Then the group $F(\{a\})$ is $\mathbb{Z}$. and set map $j : \{a\} \to \mathbb{Z}$ is $j(a) = 1$. Let (f, G) be any element of $\mathcal{F}^A$. Let's say $f(a) = g$. Then $g = f(a) = (\varphi \circ j)(a) = \varphi(j(a)) = \varphi(1)$. That is, the generator of $\mathbb{Z}$ maps to g. By proposition ref:HERE,

$$\varphi(n) = g^n$$

and since this was forced by proposition ref:HERE, φ is unique.

In the case of $\emptyset$, it would be $\{0\}$, which checks out since we saw that $\{0\}$ is the initial objects of groups (proposition 1.4.5). In the case were $A = \{a\}$ (that is, an arbitrary singleton). Then HERE ($\mathbb{Z}$).

For the more general case, we will define a new group.

Let A be an arbitrary set. For every element $a \in A$, let it correspond to some element $b \in B$. For the sake of what's to come, we will instead of label these element a^{-1}, and the set they belong to A'. Then, define a *word* to be any ordered tuple

$$(a_1, a_2, ..., a_n) \qquad a_i \in A \cup A'$$

We usually write a word as the literal putting together of letters side by side:

$$a_1 a_2 \cdots a_n$$

Definition 2.3.23: Word

Let A be some set. Then let $W(A)$ denote all possible combinations of *words*. TBD

[8] see definition 1.4.6

Example 2.3.24: Example

1. here
2. here

Define reduction, define reduced word,

Lemma 2.3.25: reduced word lemma

reduced word lemma

***Proof*:**
here

Proposition 2.3.26: Word Group

$(W(A), R)$ is a group

***Proof*:**

1. (associativity)
2. ($\exists$ identity)
3. ($\exists$ inverse)

Theorem 2.3.27: Universal Property Of Free Groups

The following are each the same statement, just different ways of stating it. Let $f : A \to G$ be a function from a set to a group, $i : A \to F(A)$ to be $i(a) = a$ (i.e. $i : A \hookrightarrow F(A)$), and $F(A) = (W(A), R)$.

1. $(i, F(A))$ is the initial object of $\mathfrak{F}$.
2. The set $(i, F(A))$ satisfies the universal property for free groups on A.
3. Then there exist a unique homomorphism $\varphi : F(A) \to G$ such that f splits with i, that is, $f = i \circ \varphi$, i.e., the following diagram commutes

$$\begin{tikzcd} F(A) \arrow[r, "\varphi"] & G \\ A \arrow[u, "i"] \arrow[ur, "f"'] & \end{tikzcd}$$

we call the group component of $(i, F(A))$ the *free group* on A.

***Proof*:**
We're essentially showing that given map $f : A \to G$, we can extend it to a map $\varphi : F(A) \to G$. So, let f be any map from A to some group G, and i be defined as in the theorem. TO define φ, we can use the information provided by f essentially completely. For any a and $a^{-1} \in A \cup A'$,

$$\varphi(a) = f(a) \qquad \varphi(a^{-1}) = f(a)^{-1}$$

notice that since A doesn't contain A, so the function f doesn't map a^{-1} anywhere. However, by construction, $F(A)$ does have a^{-1}, and so we define a place for it to map. To define φ on words with multiple letters by concatenating the result of the image, that is

$$\varphi(a_1 a_2 \cdots a_n) := \varphi(a_1)\varphi(a_2)\cdots\varphi(a_n)$$

This function is well-defined, since if $w = w'$, then

$$\varphi(w) = \varphi(a_1)\varphi(a_2)\cdots\varphi(a_n) = \varphi(w')$$

furthermore, this functions splits f since for any $a \in A$

$$f(a) = \varphi(i(a)) = \varphi(a)$$

which is true by definition

Now, notice that for any $w \in W(A)$, $\varphi(w) = \varphi(R(w))$. For example, if $w = abb^{-1}a^{-1}a$, then

$$\begin{aligned}
\varphi(w) &= \varphi(abb^{-1}a^{-1}a) \\
&= \varphi(a)\varphi(b)\varphi(b^{-1})\varphi(a^{-1})\varphi(a) \\
&= f(a)f(b)f(b)^{-1}f(a)^{-1}f(a) \\
&= f(a) \\
&= \varphi(a) \\
&= \varphi(R(w))
\end{aligned}$$

With this fact, we have that φ will be a homomorphism If $w, w' \in W(A)$, then

$$\varphi(w \cdot w') = \varphi(R(ww')) = \varphi(ww') = \varphi(w)\varphi(w')$$

establishing that $\varphi : F(A) \to G$ is a homomorphism!

Finally, we need to show that it's unique. This comes down to seeing that by the properties of homomorphisms, the way we've defined is the only way we can define a homomorphism from $F(A)$ to G. Let's say φ' was another homomorphsims from $F(A)$ to G. Then by the commutative diagram, it must be that

$$f(a) = \varphi'(i(a)) = \varphi'(a)$$

but that is exactly the definition of φ. The fact that φ' must split with f forces $\varphi = \varphi'$.

Since f an G was arbitrary, we showed that there always exists a unique homomorphism φ from $F(A)$ to G such that $f = \varphi \circ f$ – as we sought to show

Corollary 2.3.28: Free Group Is Unique

let A and B be sets such that $|A| = |B|$. Then

$$F(A) \cong F(B)$$

***Proof*:**
This comes down to properties of terminal objects in categories (initial objects are *a fortiori* terminal objects).

Since free groups on sets of the same cardinality are isomorphic, we usually write F_n to denote the free group with n generators.

Definition 2.3.29: Freely Generated

Let S be a set. Then we say that S *freely generates* a group G if $G = F(S)$.

Example 2.3.30: Free Groups

1. Earlier we said that $F(A) \cong \mathbb{Z}$. If we take $f : \{a\} \to \mathbb{Z}$, $f(a) = 1$, then by the universal property of free groups, there exists a group-homomorphism: $\varphi : F(A) \to \mathbb{Z}$, where $\varphi(1) = 1$. Then it's true that φ defines an isomorphism between $F(A)$ and $\mathbb{Z}$ (which you should check if it's unclear), and so
$$F(A) \cong \mathbb{Z}$$

2. The free set on 2 elements, say $F(\{x, y\})$ is a group we have encountered. The easiest way to show this group is by showing part of its Cayley graph:

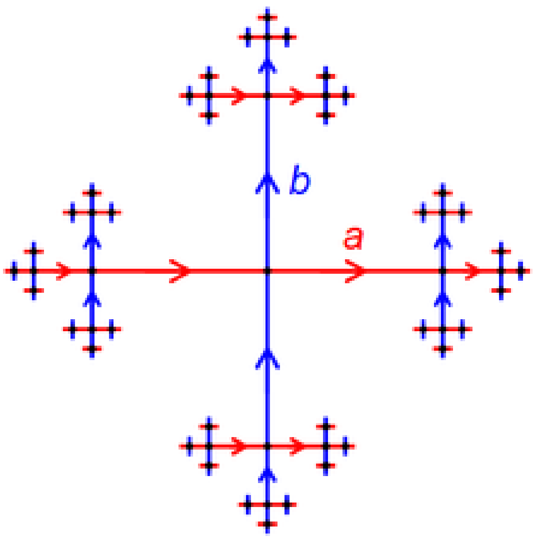


Figure 2.1: Cayley Graph of F_2

this graph in fact goes on forever. The length of the edges where halved every time to make the diagram nicer. [a]

[a]If you have done any algebraic topology, this is the fundamental group of the figure 8!

Some other properties from Clara Löh's book? Geometric group theory p. 17

2.3.4 Free Abelian Group

We can repeat this same construction, except limit our attention to abelian groups. In particular, given a set A an any map $f : A \to G$ to an abelian group, we want to lift it to an abelian group $F^{ab}(A)$ in the domain of the homomorphism $\varphi : F^{ab}(A) \to G$.

The construction is actually basically identical if we impose a commutativity condition on $F(A)$, that is, for any $a, b \in F^{ab}(A)$, $ab = ba$. If you were uncomfortable with the proof of the universal property of free group, it is worth going over it again, then go over it a 2nd time but with only abliean groups and $F^{ab}(A)$ in mind.

What we will be focusing on this section is finding a nicer group that is isomorphic to $F^{ab}(A)$.

As a closing remark on the construction of free and free abelian groups, (How it is plausible that $F(A)$ is empty or a singleton or something smaller than A so this idea of "lift" is not true. However, our construction at the end will show that it is indeed a lift. In many similar constructions in the book (ref:HERE), this construction will still be lifts, however we will encounter some (the first one in chapter 15 in which there is an initial object which is not a lift)

2.3.5 *Free-Forgetful Adjoint

Is this the right place to ask this question? It's asking whether we have really defined the "right" idea of general group. The way we make this idea rigorous is by asking if we have a functor $F : \mathbf{Group} \to \mathbf{Set}$, that forgets the structure, is there a functor $G : \mathbf{Set} \to \mathbf{Group}$ that "regains" some structure? It cannot regain all of it because sets don't have enough information for that, but is there a way in which we can make sure that our free construction is the "mirror-image" of forgetting the structure?

Build up to this: If $L(\_) : \mathbf{Group} \to \mathbf{Set}$ takes a group G and returns the underlying set: $\mathcal{L}((G, \cdot)) = G$, and $\mathcal{F}(\_) : \mathbf{Set} \to \mathbf{Group}$ takes a set and returns a group: $\mathcal{F}(S) = ((F(S), \cdot_{F(S)}),$ then

$$\mathrm{Hom}_{\mathbf{Set}}(S, \mathcal{L}(G)) \cong_{\mathbf{Set}} \mathrm{Hom}_{\mathbf{Group}}(\mathcal{F}(S), G)$$

where we there is $\cong_{\mathbf{Set}}$ in the middle because Hom_{Set} doesn't have a group structure. If it did, we would ask it to preserve that too.

Exercise 2.3.2

1. Let $G = \langle a, b \rangle$ be the free (non-abelian) group generated on two elements. Show that $H = \langle a^n b a^{-n} \mid n \in \mathbb{Z} \rangle$ is an infinitely generated subgroup
2. (harder) Let $G + \mathrm{SL}_2(\mathbb{Z})$. Show it contains a subgroup isomorphic to the free subgroup with two elements.

2.4 Subgroup Lattices

In this section, we will show how to use graphs in order to visually see the subgroup structure of a group.

Before introducing a rigorous definition, let's first do an example: Let's do the subgroup lattice of Q_8. We must first know all the subgroups of Q_8. With some computation, the subgroups that are found are

$$1,\ \langle -1\rangle,\ \langle i\rangle,\ \langle j\rangle,\ \langle k\rangle,\ Q_8$$

where $1 = \{e\}$. By the definition of Q_8, any other set of generators will fall into these groups. For example, notice that $\langle i, j\rangle = \langle k\rangle$ (similarly for i, k and j, k), and $\langle i, j, k\rangle = Q_8$. To construct the lattice, start by imagining Q_8 on the "top" and the identity group on the "bottom"'

$$\begin{array}{c} Q_8 \\ \\ \\ \\ 1 \end{array}$$

then, apply the first rule of doing the lattice:

1. draw a line if one group is a subgroup of another group.

Since $1 \leqslant Q_8$, then

$$\begin{tikzcd} Q_8 \arrow[d, dash] \\ 1 \end{tikzcd}$$

Next, pick one of the groups, let's say it's $\langle -1\rangle$. Now apply the second rule of drawing a lattice:

2. if $H \leqslant K \leqslant G$, then remove the line from H to G (if one is already drawn) and add a line from H to K, and K to G. In other words, only draw a line between subgroups A and B if and only if there is no group H such that $A \leqslant H \leqslant B$.

Since $1 \leqslant \langle -1\rangle \leqslant Q_8$, our diagram will now look like this:

$$\begin{tikzcd} Q_8 \arrow[d, dash] \\ \langle -1\rangle \arrow[d, dash] \\ 1 \end{tikzcd}$$

similarly, we can add $\langle i \rangle$ like so:

$$\begin{tikzcd}
Q_8 \arrow[d, dash] \\
\langle i \rangle \arrow[d, dash] \\
\langle -1 \rangle \arrow[d, dash] \\
1
\end{tikzcd}$$

Next, notice that $\langle i \rangle \not\leqslant \langle j \rangle$, and $\langle j \rangle \not\leqslant \langle i \rangle$, however both $\langle i \rangle$ and $\langle j \rangle$ have $\langle -1 \rangle$ as a subgroup and are both subgroups of Q_8. The same situation occurs for $\langle k \rangle$. Thus, the updated lattice will be:

$$\begin{tikzcd}
 & Q_8 \arrow[dl, dash] \arrow[d, dash] \arrow[dr, dash] & \\
\langle i \rangle \arrow[dr, dash] & \langle j \rangle \arrow[d, dash] & \langle k \rangle \arrow[dl, dash] \\
 & \langle -1 \rangle \arrow[d, dash] & \\
 & 1 &
\end{tikzcd}$$

since we have exhausted our list of possible subgroups, we have finished the subgroup lattice of Q_8.

More generally, the orientation doesn't matter (we could've had 1 on top and Q_8 on bottom, or going from left to right). What matter's is we know how to translate a line into an appropriate subgroup (i.e. If 1 is on top and Q_8 is on bottom, then a line now becomes a supergroup inclusion $H \geqslant G$). Note also that the particular shape the lattice took doesn't really matter, a pretty diamond-like shape was chosen to make the lattice more readable.

If G is an infinite group, then a lattice doesn't always exist. For example, if $G = \mathbb{Q}$, then for any group $1 \leqslant H \leqslant G$, there exists a group K such that $1 \leqslant K \leqslant H$, meaning we cannot follows condition (2). Thus, we usually work with finite groups. Certain infinite groups permit subgroup lattices (see exercise ref:HERE)

With this build-up, we can formally define a subgroup lattice:

Definition 2.4.1: Group Lattice

Let G be a finite group and $\mathcal{G}$ be the collection of subgroups of G. Then $\Gamma = (\mathcal{G}, E_{\leqslant})$ is the graph where there is an edge between a group $A, B \in \mathcal{G}$ if and only if there does not exist a group K such that $A \leqslant K \leqslant B$. When Γ is visualized, we put G on top and the identity group 1 on bottom.

Example 2.4.2: Group Lattices

Many examples, and the problem's that might arise with infinite lattices, and how some infinite groups permit lattices

1. Let $G = \mathbb{Z}/2\mathbb{Z} = C_n$. Then by theorem ref:HERE, the subgroup lattice is easily determined by the divisors of n. Thus ,we get:

$$\begin{tikzcd}
 & & \mathbb{Z}/8\mathbb{Z} \arrow[d, dash] \\
 & \mathbb{Z}/4\mathbb{Z} \arrow[d, dash] & \langle 2 \rangle \arrow[d, dash] \\
\mathbb{Z}/2\mathbb{Z} \arrow[d] & \langle 2 \rangle \arrow[d, dash] & \langle 4 \rangle \arrow[d, dash] \\
0 & 0 & 0
\end{tikzcd}$$

Note that $\mathbb{Z}/2\mathbb{Z} = \langle 1 \rangle$, or $\mathbb{Z}/4\mathbb{Z} = \langle 1 \rangle = \langle 3 \rangle$, or that $\mathbb{Z}/8\mathbb{Z} = \langle 1 \rangle = \langle 3 \rangle = \langle 5 \rangle = \langle 7 \rangle$

2. More generally, if $G = \mathbb{Z}/p^n\mathbb{Z}$, then

$$\begin{tikzcd}
\mathbb{Z}/p^n\mathbb{Z} = \langle 1 \rangle \arrow[d, dash] \\
\langle p \rangle \arrow[d, dash] \\
\langle p^2 \rangle \arrow[d, dash] \\
\vdots \arrow[d, dash] \\
\langle p^n \rangle = 0
\end{tikzcd}$$

3. If $G = \mathbb{Z}/n\mathbb{Z}$ where n is a non-prime power, we can get some interesting diagram. For example:

$$\begin{tikzcd}
 & \mathbb{Z}/20\mathbb{Z} \arrow[dl, dash] \arrow[d, dash] \\
2 \arrow[d, dash] \arrow[dr, dash] & 5 \arrow[d, dash] \\
4 \arrow[dr, dash] & 10 \arrow[d, dash] \\
 & 0
\end{tikzcd}$$

4. Let $G = V_4$ (the Klein 4 group from exercise ref:HERE). Then

$$\begin{tikzcd}
& V_4 \arrow[dl, dash] \arrow[d, dash] \arrow[dr, dash] & \\
\langle a \rangle \arrow[dr, dash] & \langle b \rangle \arrow[d] & \langle c \rangle \arrow[dl, dash] \\
& 0 &
\end{tikzcd}$$

Notice that this is the same lattice as the Quaternions (i.e. Q_8) that we have shown at the beginning. However, $V_4 \ncong Q_8$ (worth doing as an exercise). This shows that having the same group lattice is not a sufficient condition for two groups to be isomorphic (though it is a necessary condition)

Theorem 2.4.3: Subgroup And Sublattices

Let $H \leqslant G$. Then the lattice of H is the sublattice of G of groups contained in H

***Proof*:**
This fact comes straight from the definition of a lattice. Since $H \leqslant G$, then if the subgroup lattice of H where to differ from the subgroup lattice of G whose subgroups are contained in H, then the lattice of G would be wrong – a contradiction.

Example 2.4.4: Example
Give D_4, Q_8, $\mathbb{Z}/n\mathbb{Z}$.

(TBD on phrasing) We can also ask ourselves if homomorphisms preserve some group lattice. In particular, if $\varphi : G \to H$ is a homomorphism, and $\varphi(G)$ is the image of φ, can we say something about the group lattice of $\varphi(G)$ given the group lattice of G? We know that $\varphi(G)$ will be sublattice of H by theorem 2.4.3. However, finding the lattice of $\varphi(G)$ is a little more tricky. We will recieve our answer next chapter – the name of the theorem giving the answer will be the 4th isomorphism theorem.

2.4.5 Foreshadowing $\varphi(G)$ will be have "dual" to the subgroup lattice condition. That is, while subgroup's will have a sublattice of groups contained in H, $\varphi(G)$ will have the subgroup lattice of groups *contained* within a special group (which we will discover in chapter 3

Exercise 2.4.1

1. Let $G = \mathbb{Q}$. If $1 \leqslant H \leqslant G$, show that there always exists a group H' such that $1 \leqslant 1 \leqslant H' \leqslant H$.

2.5 *Universal Property of kernel and coKernels

This section can be comfortably skipped, as its contents will not come back until chapter ref:HERE. This section covers how to think about the kernel as a universal object. As you can probably guess by the fact that there is a universal property, the concept of the kernel of a homomorphism is not limited to groups. They will appear often in other sections. Most importantly, in chapter ref:HERE, (why do we need kernels when working with derived categories)

In the following seciton, $0 : G \to H$ will represent the homomorphism $\varphi(G) = e_H[= 0]$. This is indeed a homomorphism (as we've seen in proposition 1.4.5).

Theorem 2.5.1: Universal Property Of The Kernel

Let $\varphi : G \to H$ be a homomorphism. Then the inclusion map $i : \ker(\varphi) \hookrightarrow H$ is the final object in the category of group-homomorphsism $\alpha : G \to H$ such that $\varphi \circ \alpha$ is the trivial map. In other words, every group homomorphism $\alpha : K \to G$ such that $\varphi \circ \alpha$ is trivial factor through

$$\begin{tikzcd} K \arrow[r, "\alpha"'] \arrow[rr, bend left, "0"] \arrow[dr, "\exists! \alpha'"'] & G \arrow[r, "\varphi"'] & H \\ & \ker(\varphi) \arrow[u, hook, "i"'] & \end{tikzcd}$$

(put this somewhere as a reference to double-check if they got the category right?) Let $C_{\varphi,K}$ be the category who's objects be homomorphsism $\alpha : K \to G$ such that

$$\begin{tikzcd} K \arrow[r, "\alpha"'] \arrow[rr, bend left, "0"] & G \arrow[r, "\varphi"'] & H \end{tikzcd}$$

If (K, α) and (J, β) are two objects in this category, define the morphism $\gamma : (K, \alpha) \to (J, \beta)$ to be a homomorphism from K to J such that

$$\begin{tikzcd} K \arrow[r, "\alpha"'] \arrow[rr, bend left, "0"] \arrow[dr, "\gamma"'] & G \arrow[r, "\varphi"'] & H \\ & J \arrow[u, "\beta"'] & \end{tikzcd}$$

commutes.

***Proof*:**
Let $\alpha : K \to G$ such that $\varphi \circ \alpha = 0$, that is

$$\varphi(\alpha(K)) = 0$$

then $\alpha(K) \subseteq \ker(\varphi)$. If there was a homomorphism $\overline{\alpha} K \to \ker(\varphi)$, it would have to factor through:

$$\alpha = i \circ \overline{\alpha}$$

but this forces $\overline{\alpha}$ to be α restricted to K, forcing uniqueness and existnece.

(Also explain the idea of the dual question, or that it exists), namely

Theorem 2.5.2: Universal Property Of The coKernel

Let $\varphi : G \to H$ be a homomorphism. Then the inclusion map $i : \ker(\varphi) \hookrightarrow H$ is the final object in the category of group-homomorphsism $\alpha : G \to H$ such that $\varphi \circ \alpha$ is the trivial map. In other words, every group homomorphism $\alpha : K \to G$ such that $\varphi \circ \alpha$ is trivial factor through

$$\begin{tikzcd}
G \arrow[r, "\varphi"'] \arrow[rr, bend left, "0"] & H \arrow[r, "\beta"'] \arrow[d, "\pi"] & Q \\
& G/H \arrow[ur, "\exists!\beta'"'] &
\end{tikzcd}$$

which we will anwser in chapter 3?

Also, for cokernel, this useful exact sequence might be helpful:

$$0 \to \ker(f) \to V \xrightarrow{f} W \to \operatorname{coker}(f) \to 0$$

3

Homomorphisms and Quotient Groups

In this chapter, we take a closer look at homomorphisms and how they relate different groups. We start our exploration by getting a better understanding of the image of a homomorphsim. For example, if $\varphi : G \to H$ is a homomorphism, then we saw that $\varphi(G)$ is a group (re-prove this if it is not clear). Since φ is a function, then as we saw in φ making $(G/\sim_\varphi)$ bijective to $\varphi(G)$ (in particular, $x \sim_\varphi y \iff \varphi(x) = \varphi(y)$), which we can see by the φ making $(G/\sim_\varphi)$ bijective to H (in particular, $x \sim_\varphi y \iff \varphi(x) = \varphi(y)$), which we can see by the function:

$$[g] \mapsto \varphi(g)$$

The collection $(G/\sim_\varphi)$ is called the *fibers* of φ[1]. Since there is a bijection between $\varphi(G)$ and $(G/\sim_\varphi)$, and $\varphi(G)$ is a group, can we define a group structure on $(G/\sim_\varphi)$? Furthermore, is there some "minimal" and "unique" group structure that we can put on $(G/\sim_\varphi)$ so that there is a cannonical identification between $\varphi(G)$ and $(G/\sim_\varphi)$; those comfortable with the category theory presented so far will recognize this as asking for the universal properties of quotient objects in the category **Grp**. Due to this universal property, the name of the group structure on $(G/\sim_\varphi)$ satisfying the universal property will be the *quotient group*. The exploration of this question is at the center of the first section of this chapter. Due to the axioms of groups, it will turn out that there is a lot of "regularity" in the collection of fibers which allow for an easy representation of $(G/\sim_\varphi)$.

After finishing the first part, we will have developed the appropriate tools to get back to what we started in section 1.3 and define these notions more formally, i.e., how do we impose relations on groups. about algebraic structures more generally. about algebraic structures mor egenerally.

For the rest of the chapter, we will use our new theory to give us some very powerful results. We will see that when working with finite groups, the order of the group will give us a lot of information

[1]Given $\varphi : G \to H$ and an element $h \in H$, the fiber of h is all elements that map to h by φ

about it's subgroup structure, we will use our understanding of quotient groups to learn how to build some new types of groups, and we will learn 4 isomorphism theorems which give us lots of information on group structure.

We will end this chapter by introducing one of the most fundamental classification theorem in finite group theory: *Jordan Hölder's Program.* In the last few section, we will show how there are groups that can be considered the "building blocks" of any other finite groups, and show how we can put these groups together to form other groups.

3.1 Quotient Groups

Let $\varphi : G \to H$ be a homomorphism. We start by exploring the structure of $\varphi(G)$. Since $\varphi(G)$ is a set, we know by proposition 0.1.28 in the preliminary chapter that there is a bijection between $G/\sim_\varphi$ and $\varphi(G)$ such that for any function $\varphi : G \to H$ where $a \sim b \Leftrightarrow \varphi(a) = \varphi(b)$, we have

$$\begin{tikzcd} G \arrow[r, "\varphi"] \arrow[d, "\pi"'] & H \\ G/\sim \arrow[ur, "\bar{\varphi}"'] & \end{tikzcd}$$

We now try to see if there is an equivalent universal property for quotients in **Grp**. To that end, we would then want $G \xrightarrow{\pi} G/\sim$ to be a homomorphism, that is

$$\pi(a \cdot b) = \pi(a) \bullet \pi(b)$$

or, written in terms of equivalence relations,

$$[a \cdot b] = [a] \bullet [b]$$

We would need to have that $\bullet$ is independent of representation. That is, if we have $x, y \in [a]$, then we want

$$x \sim y \Leftrightarrow xb \sim xb \tag{3.1}$$

since we would also want that

$$[b \cdot a] = [b] \bullet [a]$$

we must have that

$$x \sim y \Leftrightarrow bx \sim by \tag{3.2}$$

These are necessary condition in order for $(G/\sim, \bullet)$ to be a group, since if once of these condition's fail, then we won't have a group. For examle, if $x \sim y \not\Rightarrow xb \sim yb$, then $[x][b] \neq [xb]$.

These are necessary conditions, since if $G/\sim$ was a group, then these must hold. We can ask if they are sufficient conditions, that is, condition's such that if they are followed, then $(G/\sim, \bullet)$ must be a group. These two condition's are also sufficient, meaning defining a group structure on $G/\sim$ is equivalent to the equivalence relation respecting these two properties:

Theorem 3.1.1: $(G/\sim, \bullet)$ is the quotient object of Grp

Let G be a group and $G/\sim$ be the set of equivalence relations given $\sim$. Then $G/\sim$ is a group under $\bullet$ if and only if

$$\forall g \in G,\ (x \sim y \Leftrightarrow gx \sim gy \text{ and } xg \sim yg)$$

Furthermore, If $\varphi : G \to H$ is another homomorphism such that $g \sim g' \Rightarrow \varphi(g) = \varphi(g')$, then there exists a homomorphism φ such that the following diagram commutes:

$$\begin{tikzcd}
G/\sim \arrow[r, "\varphi"] & H \\
G \arrow[u, "\pi"] \arrow[ur, "\varphi"] &
\end{tikzcd}$$

We say that $\sim$ is *compatible* with the group-structure on G

***Proof*:**
The proof of the $\Rightarrow$ direction we have just showed above the theorem, so we'll show that these two equivalence relations are sufficient conditions for to define $(G/\sim, \bullet)$.

First, note that the operation $\bullet : G/\sim \times G/\sim \to G/\sim$ $[a] \bullet [b] = [ab]$ is well-defined, since if we take any other two representative $a' \in [a]$ and $b' \in [b]$, then since

$$b \sim b' \Leftrightarrow ab \sim ab' \text{ and } a'b \sim a'b'$$

and

$$a \sim a' \Leftrightarrow ab \sim a'b \text{ and } ab' \sim a'b'$$

Thus, putting pieces together using transitivity, we get

$$ab \sim ab' \sim a'b' \text{ or } ab \sim a'b \sim a'b'$$

both showing that $ab \sim a'b'$, meaning $\bullet$ is well-defined. Next, we need to show that $(G/\sim, \bullet)$ is indeed a group

1. Let $[a], [b], [c] \in G/\sim$. Then

$$\begin{aligned}
([a][b])[c] &= ([ab])[c] \\
&= [(ab)][c] \\
&= [(ab)c] \\
&= [a(bc)] \\
&= \vdots \\
&= [a]([b][c])
\end{aligned}$$

2. I claim that $[e_G] \in G/\sim$ is the identity. For any $[g] \in G/\sim$,

$$[g][e_G] = [ge_G] = [g] = [e_G g] = [e_G][g]$$

   as we sought to show

3. Finally, for any $[g] \in G/\sim$, I claim that $[g]^{-1} = [g^{-1}]$. Indeed:

$$[g][g^{-1}] = [gg^{-1}] = [e_G]$$

showing that $[g^{-1}]$ is indeed the inverse of G

Thus, $(G/\sim, \bullet)$ is indeed a group.

Finally, we must show that G is universal with respect to homomorphisms $\varphi : G \to G'$ such that $g \sim g' \Rightarrow \varphi(g) = \varphi(g')$. Note that we already have that such a function exists, since φ is also a set function, so we already have that there exists a unique set-funciton $\varphi : G/\sim \to G'$ such that $\varphi([g]) = \varphi(g)$. Thus, we only need to check that φ is a homomorphism. We get this pretty much be definition:

$$\varphi([a][b]) = \varphi([ab]) = \varphi(ab) = \varphi(a)\varphi(b) = \varphi([a])\varphi([b])$$

showing that φ is indeed a homomorphism, and must be unique since there is only one possible set function (and hence homomorphism) that satisfies this property (there are no more functions to check).

Since $\bullet$ is the unique structure on $G/\sim$ that respects the universal property, we will drop it without worry Since $\bullet$ is the unique structure on $G/\sim$ that respects the universal propety, we will drop it without worry that there be any confusion, that is well write

$$[a][b] = [ab]$$

.

Definition 3.1.2: Quotient Group Via Congruence Relation

Let G be a group and $\varphi : G \to H$ a group homomorphism giving the induced equivalence relation $\sim_\varphi$. Then the group $G/\sim_\varphi$ from theorem 3.1.1 is called a *quotient group* and the equivalence relation $\sim_\varphi$ is called a *congrunce relation.*

The fact that equivalence relations that respect 3.1 and 3.2 are the only ones that are The fact that equivalence relations that respect 3.2 and 3.2 are the only ones that are congruence relations is very powerful, since this greatly limits what type of equivalence relations we have. For the 3.1, 3.2, or both, that is we shall find all possible congruence relations, as well as all possible one-sided congruence relations. We start by characterizing the equivalences classes of $G/\sim$ satisfying 3.1. We start by characterizing the equivalences classes of $G/\sim$ satisfying 3.1. there is any information we can figure out about the equivalence classes:

Proposition 3.1.3: Equivalence Classes Using property 3.1

Let $G/\sim$ be a quotient set given $\sim$ with proprety 3.1. Then

1. The equivalence class $K \in G/\sim$ where $e_G \in K$ is a subgroup of G ($K \leqslant G$)
2. Given any other equivalence calss $[g] \in G/\sim$, for any $a, b \in [g]$

$$a \sim b \;\Leftrightarrow\; a^{-1}b \in K \;\Leftrightarrow\; aK = bK$$

where the $aK = \{ak \mid k \in K\}$ and $bK = \{bk \mid k \in K\}$

Proof:

1. Since $K \subseteq G$, we need to show that for any $a, b \in K$, $ab^{-1} \in K$. Since $e_G \in K$, then

$$e_G \sim a$$

since $\sim$ is a congruence relation:

$$a^{-1} \sim e_G$$

using property 3.1. Using property 3.1 again, we get

$$a^{-1}b \sim b \sim e_G$$

where the last part comes from the transitivity of equivalence relations, and so

$$a^{-1}b \sim e_G \Rightarrow a^{-1}b \in K$$

meaning $K \leqslant G$. Note that if we worked with property 3.2, we would instead be showing that $ab^{-1} \in K$.

2. Let $[g]$ be an equivalence class and $a, b \in [g]$ so that $a \sim b$.

Since $\sim$ respects 3.1 $a^{-1}b \sim e_G$, implying $a^{-1}b \in K$. Next, take any element $x \in a^{-1}bK$. Then $x = a^{-1}bk$. Next, since $a^{-1}b \in K$ and $k \in K$, $x \in K$. Thus, $a^{-1}bK \subseteq K$. This implies that $bK \subseteq aK$, since if $x \in bK$, then $x = bk$, and so $x = a(a^{-1}bk)$, and since $a^{-1}b \in K$ and $k \in k$, we get $x = ak'$ for some $k' = a^{-1}bk \in K$, an so $x \in aK$. Using symmetry of $\sim$, we'll get that $aK \subseteq bK$, implying $aK = bK$. Thus, we showed that

$$a \sim b \Rightarrow a^{-1}b \in K \Rightarrow aK = bK$$

Note that if we were using property 3.2, then we'd get $Ka = Kb$ since we'd be multiplying on the right.

To make this chain of implication go full circle, assume $aK = bK$. By setting $k = e_G$, we get $ae_G = a \in bK$, and so $ab^{-1} \in K$. Since K is the equivalence class containing e_G, we have

$$ab^{-1} \sim e_G \Leftrightarrow a \sim b$$

by 3.1, thus showing that $aK = bK \Rightarrow a \sim b$, completing the loop of implication, thus showing that

$$a \sim b \Leftrightarrow a^{-1}b \in K \Leftrightarrow aK = bK$$

as we sought to show

This shows that for any equivalence relation satisfying 3.1 (and similarly 3.2), it's equivalence classes will be of the form aK resp. Ka aK Ka where a is the representative of an equivalence class and K is the equivalence class containing e_G. Thus, an equivalence class will determine a subgroup K which then determines a representation for the equivalence classes of $G/\sim$. equivalence class will determine a subgroup K which then determines the equivalence classes of $G/\sim$.

Corollary 3.1.4: Quotient Set Size

let G be a a finite group and $G/\sim$ be the quotient sets with respect to an equivalence relation $\sim$ satisfying equation 3.1 or 3.2 (or both). Then every quotient set is of the same size, that is

$$|aK| = |bK| \qquad \forall a, b \in G$$

***Proof*:**
exercise (see ref:HERE)

Since these cosets represent our equivalence relations, we will give a special name to these quotient sets:

Definition 3.1.5: Cosets

Let $H \leqslant G$. Then for every $g \in G$, we can define

$$gH = \{gh | h \in H\}$$

we call this a [left]-***coset***. We can also define the ***right***-coset:

$$Hg = \{hg | h \in H\}$$

If additive notation is used for G ($(G, +)$), it is common to denote the coset as

$$g + H \qquad H + g$$

The set of left-cosets is represented as G/H, an the set of right-cosets is represented as $G \setminus H$.

What equivalence relations that satisfies poposition 3.1.3? In other words, we're asking for a converse to proposition 3.1.3[1], which stated that if $\sim$ respected property 3.1 (or 3.2), then the equivalence class containing e_G is a subgroup. The answer is yes:

Proposition 3.1.6: Subgroups And Equivalence Classes

Let $H \leqslant G$ be a subgroup of G. Define the equivalence relation

$$a \sim_L b \Leftrightarrow a^{-1}b \in H$$

then $\sim_L$ respects property 3.1

***Proof*:**
Let's first show that $\sim_L$ is indeed an equivalence relation

1. If $a \sim a$, then $a^{-1}a = e \in H$, which is the case since H is a group
2. Let $a \sim b$. Then $a^{-1}b \in H$, implying $(a^{-1}b)^{-1} = b^{-1}a \in H$, implying $b \sim a$
3. Let $a \sim b$ and $b \sim c$. Then $a^{-1}b, b^{-1}c \in H$. Then $a^{-1}bb^{-1}c = a^{-1}c \in H$, implying $a \sim c$.

Thus, $\sim_L$ is indeed an equivalence relation.

To show it satisfies property 3.1, notice that for any $g \in G$

$$a \sim b \to a^{-1}b \in Ha^{-1}(g^{-1}g)b \in H \Rightarrow (ga)^{-1}gb \in H \Rightarrow ga \sim_L gb$$

showing that $\sim_L$ satisfies property 3.1

This proposition shows that any subgroup of H will define an equivalence relation $\sim_L$ respecting property 3.1, and thus by proposition 3.1.3[2] partition G (even better, equally partition G due to corollary 3.1.4). Similarly, we can show that the relatoin $a \sim_R b \Leftrightarrow ab^{-1} \in H$ gives an equivalence relation which satisfies property 3.2. Note that since H partitioned G, only $eH = H$ is a subgroup of G since only it contains the identity.

Combining proposition 3.1.3 and 3.1.6 we get the following one to one correspondence:

$$\{H | H \leqslant G\} \leftrightarrow \left\{ \begin{array}{c} \text{equivalence relations } \sim_L \\ \text{on } G \text{ satisfying} \\ \text{property 3.1} \\ \\ \\ \end{array} \right\}$$

Similarly:

$$\{H | H \leqslant G\} \leftrightarrow \left\{ \begin{array}{c} \text{equivalence relations } \sim_R \\ \text{on } G \text{ satisfying} \\ \text{property 3.2} \\ \\ \\ \end{array} \right\}$$

3.2, showing that the subgroups of G give us all possible such equivalence relations, that is all one-sided congruence relations. Because of this, we can without ambiguity write G/H to represent $G/\sim$ with an equivalence relation $\sim$ satisfying property 3.1, and similarly for $G \setminus H$. 3.2, showing that the subgroups of G give us all possible such equivalence relations. Because of this, we can without ambiguity write G/H to represent $G/\sim$ with an equivalence relation $\sim$ satisfying property 3.1, and similarly for $G \setminus H$.

$G \setminus H$ satisfying $\sim_R$? The answer is actually no: G/H satisfying $\sim_R$? The answer is actually no:

3.1.7 counter-example Take $G = D_6$. Then $H = \langle \sigma \rangle$ is a subgroup, and has left cosets $eH(= H), rH, r^2H$. giving us

$$\{e, \sigma\}, \{r, r\sigma\}, \{r^2, r^2\sigma\}$$

and the right-cosets are

$$\{e, \sigma\}, \{r, \sigma r\}, \{r^2, \sigma r^2\}$$

using the fact that $r\sigma = \sigma r^2$ and $r^2\sigma = \sigma r^2$, we can re-write the right cosets as

$$\{e, \sigma\}, \{r, r^2\sigma\}, \{r^2, r\sigma\}$$

showing that they partition G differently than the left cosets!
This means that $\sim_L$ defined by H does not respect property 3.2. If it did, then

$$r \sim_L r\sigma \Rightarrow r(r\sigma)^{-1} = r\sigma r^2 = rr\sigma \in H$$

however, $r^2\sigma \notin H$, showing that it does not satify the propety

Remember that we showed in proposition 3.1.3 that *both* the properties need to be satisfied in order for G/H to be a group. Since every $\sim$ with these properties corresponds to a subgroup H, it will come down to a special subgroup H which has both properties

(build-up normal subgroups somehow!) We have in fact already encountered the property which will always make H induce both $\sim_L$ and $\sim_R$

(I don't like this "we've already encountered". Would be better to build up the same way we did before, with the "neccessary and sufficient" narrative from before)

Proposition 3.1.8: When H Induces Both $\sim_L$ And $\sim_R$

The relatoins $\sim_L$ and $\sim_R$ induce the same cosets corresponding to H if and only if H is normal

***Proof*:**
If H is normal, then $gH = Hg$, meaning the left and right cosets are the same and so H induces both $\sim_L$ and $\sim_R$. Conversely, if $\sim_L$ and $\sim_R$ are both respected, and take $x \in aH$, so $xa^{-1} \in H$, meaning $xa^{-1} = h$. Manipulating this, we get that $ax^{-1} = h^{-1}$ so $x^{-1} = a^{-1}h^{-1}$, and so $x^{-1} \in a^{-1}H$. Then since $\sim_R$ is respected, $x \sim a$. Since $\sim_L$ is repsected then $x^{-1}a \in H$, meaining $x^{-1}a = h$ or $x^{-1} \in Ha^{-1}$. Then by similar reasoning we get $x \in Ha$, showing that $aH \subseteq Ha$. doing the same argument backwards we get that $Ha \subseteq aH$ giving us $aH = Ha$, which is exactly the condition of normality

(TBD, the converse of my proof is very messy)

Definition 3.1.9: Normal Subgroup

Let $H \leqslant G$ such that $aH = Ha$. Then H is a normal subgroup, and we denoted it $H \trianglelefteq G$

We've already encountered normal subgroups before

Example 3.1.10: Normal Subgroups

1. for any group G, $Z(G) \leqslant G$. Take any $y \in G$ and consider $yx \in yZ(G)$. Then by definition of $Z(G)$, $yx = xy$, and so $[xy] = yx \in Z(G)y$, showing that $Z(G)$ is normal
2. (another such example?)

(after seeing how important normal subgroups are, here are a couple of properties about them:)

Proposition 3.1.11: properties of normal sub-groups

Let N be a normal sub-group of a group G. Then the following are equivalent

1. $N \trianglelefteq G$
2. $N_G(N) = G$
3. $gN = Ng$
4. the operation on left cosets of N in G as described in the previous proposition turns the set of cosets into a group.
5. $gNg^{-1} \subseteq N$ for all $g \in G$
6. if G is abelian and $H \leqslant G$ then $H \trianglelefteq G$

Proposition 3.1.8 is very important. The equivalence relatoin $\sim$ corresponding to H has both property 3.1 and 3.2 by proposition 3.1.6 and so by proposition 3.1.3 G/H is a group! Since normal subgroups are so important, we'll use the notation $\trianglelefteq$ instead of $\leqslant$ to say that H is a normal subgroup of G ($H \trianglelefteq G$) Given our new found knowledge, we will give a new definition of quotient group

Definition 3.1.12: Quotient Group

Let $H \trianglelefteq G$ be a normal subgroup of G. The *quotient group* G/H, sometimes written G mod H (and read G modulo H or G mod H for short) is the group $(G/\sim, \bullet)$ where $\sim$ is defined in proposition 3.1.6, and $\bullet$ is defined by

$$aHbH := (ab)H$$

An easy way to remember that this operation is defined when is H is normal is by remembering $a(Hb)H = abHH = abH$. For notation, we can write $aH = \bar{a} = [a]$ for some representative u and $G/H = \bar{G}$. Thus, going all the way back to $\pi : G \to G/\sim$, and use our knowledge to re-write this as Thus, going all the way back to $\pi : G \to G/\sim$, and use our knowledge to re-write this as

$$\pi : G \to H \qquad \pi(g) = gH$$

where π is a homomorphism if $H \trianglelefteq G$. If $H \ntrianglelefteq G$, then π is *not* a homomorphism.

Since π is a homomorphism, it's kernel is a subgroup:

$$\ker(\pi) = \{g \in G \mid \pi(g) = eH\}$$

and since $eH = H$, this means that $\ker(\pi) = H$. Since H can be any normal subgroup, this means that H is always the kernel of some map

$$\text{Normal} \Rightarrow \text{Kernel}$$

Is the converse true, that is, the kernel is always normal? If so, that would be really powerful, since the kernel would then always be a normal subgroup of G, and so any homomorphism $\varphi : G \to H$ would induce a quotient group G/K, where $K = \ker(\varphi)$. Kernels are always normal subgroups!

Proposition 3.1.13: Kernel Then Normal

Let $\varphi : G \to H$ be a homomorphism and let $K = \ker(\varphi)$. Then $K \trianglelefteq G$

***Proof*:**
Let K be the kernel of $\varphi : G \to H$, and take any $g \in g$. then

$$\varphi(gkg^{-1}) = \varphi(g)\varphi(k)\varphi(g^{-1}) = \varphi(g)\varphi(k)\varphi(g)^{-1} = \varphi(g)\varphi(g)^{-1} = e_H$$

and so $gkg^{-1} \in K$, showing that $gK = Kg$, meaning $K \trianglelefteq G$

This means that given any homomorphism $\varphi : G \to H$, we can *always* define an equivalence relation on G given the kernel of φ since $\ker(\varphi)$ is normal!

Since any normal subgroup will be the kernel of some homomorphism, and the kernel of any homomorphism will be normal, we get that the two are essentially synonymous in group theory:

$$\text{Normal} \Leftrightarrow \text{Kernel}$$

(A word how with this fact, we can find the structure of G/H by relating it to the image of $\varphi(G)$?)

With our new knowledge on quotient groups, we re-visit the universal property of quotients (theorem 3.1.1) with the intend on being able

(words here on bringing back intuition from uni. prop. of set, but now in Grp it's more prevalent, since we're linking group structures. Notice that since $H \subseteq \ker(\varphi)$, that means H will make a "finer" as compared to $\ker(\varphi)$, and since both will respect the corresponding $\sim_H$ induced by H, we know by theorem 3.1.1 that it must factor through G/H. THis result is important enough to be given a name:)

Theorem 3.1.14: The First Isomorphism Theorem

Let $\varphi : G \to H$ be a homomorphism and let $K = \ker(\varphi)$. If $N \trianglelefteq G$ where $N \subseteq \ker(\varphi)$, then there exists a unique *homomorphism* $\Phi : G/N \to H$ such that the following diagram commute:

$$\begin{tikzcd} G/N \arrow[r, "\Phi"] & H \\ G \arrow[u, "\pi"] \arrow[ur, "\varphi"'] & \end{tikzcd}$$

where $\varphi = \Phi \circ \pi$, and π is the natural projection

***Proof*:**
We already know this universal property in terms of congruence relations. We simply must translate the information of this theorem to the one required for theorem 3.1.1

So far, the question is saying that if

$$N \subseteq \ker(\varphi) \Leftrightarrow (\forall n \in N,\ \varphi(n) = e_H) \Rightarrow \varphi(ab^{-1}) = e_H$$

for $a, b \in N$ moving things around we get

$$\varphi(ab^{-1}) = \varphi(a)\varphi(b)^{-1} = e_G \ \Leftrightarrow\ \varphi(a) = \varphi(b)$$

thus giving us

$$(\forall a, b \in N),\ ab^{-1} \in N \Rightarrow \varphi(a) = \varphi(b)$$

and, since $ab^{-1} \in N$ is the condition defining the induced equivalence relation by N (proposition 3.1.6), we get

$$a \sim b \Rightarrow \varphi(a) = \varphi(b)$$

which put's us exactly at the inition condition for theorem 3.1.1, which gives us that the diagram commutes, as we sought to show.

What theorem 3.1.14 (the universal property of quotient groups) shows is that all the information a homomorphism φ gives us can be captured via the quotient group G/K (TBD?)

Corollary 3.1.15: Canonical Decomposition For Groups

Let $\varphi : G \to H$ be any group homomorphism. Then it may be decomposed into

$$\begin{tikzcd} G \arrow[r, two heads, "\pi"'] \arrow[rrr, bend left, "\varphi"] & G/\ker(\varphi) \arrow[r, "\cong", "\Phi"'] & \operatorname{im}(G) \arrow[r, hook, "\iota"'] & H \end{tikzcd}$$

where Φ is an isomorphism

***Proof*:**
This proof is simply piecing together all the results with figured out so far. Since $\operatorname{im}(G) \leqslant H$, then there exists an inclusion map ι which is a homomorphism. We've established that $\ker(\varphi)$ is normal, and that any normal subgroup will induce a quotient group $G/\ker(\varphi)$, meaning π (the natural projection) is a well-defined homomorphism.

To show $\Phi : G/\ker(\varphi) \to \operatorname{im}(G)$, is an isomorphism, note that we already have that Φ is given by the canonical decomposition of quotient sets. By the universal property of quotient groups, we have that it's a homomorphism. Since bijective homomorphisms are isomorphisms, we have that it's an isomorphism, completing the proof.

Corollary 3.1.16: Surjective Homomorphisms

Let $\varphi : G \to H$ be a surjective homomorphism, and let $K = \ker(\varphi)$. Then

$$G/K \cong H$$

***Proof*:**
For $G/K \cong H$, notice that $H = \text{im}(G)$ in corollary 3.1.15.

This last corollary is very useful and one you will use constantly in both theoretical and practical context. It is saying that if we want to know the structure of G/K, if we can find a surjective homomorphism from G to H with kernel K, then $G/K \cong H$, i.e., H will represent the structure!

(Integarte this somehow?? Maybe later)

$$1 \xrightarrow{\iota_1} K \xhookrightarrow{\iota_K} G \xrightarrow{\varphi} H \xrightarrow{\pi} 1$$

where 1 is the trivial group, ι_1 and ι_K are the inclusion homomorphisms, and π is the projection of H onto the trivial group.

The 2nd form presented in corollary 3.1.16 lets us nicely split up the pieces of any given homomorphism. This representation is called a *short exact sequence*. We will return to short exact sequences soon to flesh out many intersting results when precieving which groups can be put in the 3 non-trivial positions, that is

$$1 \xrightarrow{\iota_1} A \hookrightarrow B \twoheadrightarrow C \xrightarrow{\pi} 1$$

(bunch of examples: perhaps a reminder that if $H \leqslant G$ the $i : H \to G$ shows that G contains H??)

Exercise 3.1.1

Throughout these exercises, let $\varphi : G \to H$ be a homomorphism between the two groups G and H, and $K = \ker(\varphi)$

1. Show that $\varphi^{-1}(Q)$ (the pre-image of Q) is a group. If Q was a normal subgroup, $Q \trianglelefteq H$, show that $\varphi^{-1}(Q)$ is also normal. Is it true that if $H \trianglelefteq G$ then $\varphi(H) \trianglelefteq \varphi(G)$?
2. Let $h \in H$. We call $\varphi^{-1}(h)$ a *fiber* of φ over h. Let $F_h = \varphi^{-1}(H)$. Show that
    1. For any $u \in F_h$, $F_h = uK$
    2. For any $u \in F_h$, $F_h = Ku$
3. Let $aKbK = \{akbk' \mid a, b \in G, k, k' \in K\}$ be a set, and $K \trianglelefteq G$. Show that
$$aKbK = abK$$
where $abK = \{abk \mid a, b \in G, k \in K\}$ [hint: for $aKbK \subseteq abK$, recall that $aK = Ka$ since K is normal. For $aKbK \supseteq abK$, note that $1 \in K$ so $1 \cdot a \in Ka$. Take advantage of normality to complete the proof]
4. Let $R \leqslant G$. Show that the cosets of G/R are all of equal size.
5. Show that if $aB = 1B$, then $a \in B$

6. Let $H, K \trianglelefteq G$. Show that $H \cap K \trianglelefteq G$
7. Generalizing the previous result, If $\{H_i\}_{i\in I}$ is some collection of normal subroups of G show that $\cap_{i\in I} H_i \trianglelefteq G$
8. Let G be finitely generated and let $N \trianglelefteq G$ be finitely generated. Prove that G/N is finitely generated.
9. For any group G, show that $Z(G) \trianglelefteq G$
10. Show that $n\mathbb{Q} = \mathbb{Q}$, where we're abusin' notation in writing $n\mathbb{Q}$ to mean $\{nq \mid q \in \mathbb{Q}\}$ even though multiplication is not defined on $\mathbb{Q}$ (Groups that have this property are called *divisible* or *injective* groups. More on them in section 17.3) [maybe any divisible group has a natural 2nd group structure? cause in a sense, every element can be "divided", and so it respects the property of inverse, which is also unique by the fact that the element is a unique group element, and the other axioms follow naturally]
11. Let $G \cong H$, $N \trianglelefteq G$ and $K \trianglelefteq H$. If $N \cong K$ is it true that $G/N \cong H/K$? Prove or show a counter example. hint: [2]
12. (Useful in Lie Groups) Let $GL_n(\mathbb{R})$ be the set of invertible $n \times n$ matrices over $\mathbb{R}$ (more generally, if you know linear algebra, over a field $\mathbb{F}$ with characteristic $\neq 2$ or 3 if $n = 2$). What are the normal subgroups of $GL_n(\mathbb{R})$?
13. We've shown that if we have a map $f : G \to H$ and $N \trianglelefteq H$, we can define a map $\overline{f} : G \to H/N$. Is the converse true? That is, if we have a map $f : G \to H/N$, do we have a map $F : G \to H$ such that $f = \pi \circ F$? [Hint: Consider maps with domains and codomains of $\mathbb{Z}$ and $\mathbb{Z}/n\mathbb{Z}$, $n > 1$]
14. Show that every subgroup of Q_8 is normal. Hence, every subgroup being normal does not imply the group is abelian[3]

3.2 Presentations

We have seen in section 2.3.3 that if we have any set map from a set S to a group G, $f : S \to G$, then it can be extended it to a homomorphism from the group $F(S)$ to G: $\varphi : F(S) \to G$, $\varphi|_S = f$. We've also seen that if we have a surjective homomorphism $G \to H$, then $G/N \cong H$ (where N is the kenrel of the homomorphism). We can combine these results: Converting to $(G, *)$ notation for groups for a moment, given an identity map $\mathrm{id} : G \to (G, *)$, which is a prefectly valid surjective set function from the set G to the underlying $\varphi : (F(G), *_F) \to (G, *)$. By the first isomorphism theorem, if N is the kernel, then $(F(G)/N, (*_F)_\varphi) \cong (G, *)$. Thus, $\varphi : F(G) \to (G, *)$. By the first isomorphism theorem, if N is the kernel, then $F(G)/N \cong (G, *)$. Thus, every group is the quotient of a free group! We now go back to our usual convention of groups being written as G where we drop the binary operation, where we will consider it clear from context when a map is a set map and when it is a homomorphism.

[2]reverse text? This might be giving the answer? Consider $G = H = N = \mathbb{Z},\ K = 2\mathbb{Z}$

[3]On the other hand, Q_8 is special: Every non-abelian group for which all subgroups are normal is of the form $G \cong Q_8 \times B \times D$ where B is something called an elementary abelian 2-group and D is called a torsion abelian group whose elements are of odd order. A group where all subgroups are normal is called a Dedekind group, and a non-abelian Dedekind group is called a Hamiltonian group.

Let's generalize this result a bit: Let's say $S \subseteq G$ such that $\langle S \rangle = G$, and consider $f : S \to G$. Then by the universal property of free groups, we have that $\varphi : F(S) \to G$ is a group homomorphism. Since $\langle S \rangle = G$, and all possible words are in $F(S)$, then by construction of $F(S)$ and φ, the map φ is a surjective group homomorphism. Thus, for the appropriate kernel N, $F(S)/N \cong G$.

Since $F(S)$ is free from relations, but G has relations, φ somehow imposed relations onto $F(S)$. This means that N captured the idea of the relations imposed on $F(S)$ to make $F(S)/N \cong G$. As observed in section 2.3, any relations of a group can be written as

$$r_1 = r_2 = \cdots = r_k = 1$$

If we let $r_i = \prod_{i \in I_{r_i}} s_i$ to be some arbitrary finite[4] product of elements of S and it's inverses. Then if If we let $r_i = \prod_{i \in I_{r_i}} s_i$ to be some arbitrary product of elements of S and it's inverses. Then if we let $R = \{r_1, r_2, ..., r_k\}$, we can write $G = \langle S | r_1 = r_2 = \cdots = r_k = 1 \rangle$ as

$$G = \langle S | R \rangle$$

for short hand. Since $F(S)/N \cong G$, it must be that $R \subseteq N$. We might be tempted to say $\langle R \rangle \subseteq N$, however the closure *need not be normal*, while N must be a normal subset. Instead, we'll define the noraml closure:

Definition 3.2.1: Normal Closure

Let $S \subseteq G$ be a subset of a group G. Then define the intersection of all normal subgroups of G containing S

$$\langle S \rangle^{\trianglelefteq} = \bigcap_{\substack{N_i \trianglelefteq G \\ S \subseteq N_i}} N_i$$

to be *normal closure of* S.

$\langle S \rangle^{\trianglelefteq}$ is indeed a normal subgroup by proposition ref:HER. Since $r_1 = r_1 = \cdots = r_k = 1$ in G, then $[r_1] = [r_2] = \cdots = [r_k] = [1] \in F(S)/N$, and so $r_1, r_2 ... r_k \in N$. We thus have that $\langle R \rangle^{\trianglelefteq} \subseteq N$.

Next, let's say we have some $x \in N$. Then that means then $[x] = [1]$, so for some $n' \in N$, $xn' = 1$. But then, xn' must be some combinations of elements $r_1, r_2, \cdots, r_k$, since by the presentation of G, those are the elements that can equal to 1. Thus $N \subseteq \langle R \rangle^{\trianglelefteq}$. Therefore $N = \langle R \rangle^{\trianglelefteq}$

From this, we can formally define a presentation

Definition 3.2.2: Presentation

Let G be any group. Then a *presentation* of G is a pair of sets (S, R) where $\langle R \rangle^{\trianglelefteq}$ is the kernel of the homomorphism $F(S) \to G$. The elements of S are called the *generators* of G, and the elements of R are called the *relations* of G:

$$\langle S | R \rangle := F(S)/\langle R \rangle^{\trianglelefteq}$$

We say G is finitely generated if given a presentation (S, R), $|S|$ is finite. We say G is *finitely presented* if both $|S|$ and $|R|$ are finite.

[4] Infinite products would require more care to be well-defined, see EYNTKA real analysis I

Notice how this statement is like the dual equivalent of Cayley's theorem (4.5.3): In Cayley's theorem, every group is the subgroup of a symmetric group. Here, every group is the quotient of a free group.

Example 3.2.3

1. $\langle A|\emptyset\rangle$ is the presentation of the free group $F(A)$
2. dihedral group

3.2.4 Note Let $N \trianglelefteq G$. Clearly, $\langle N\rangle^{\trianglelefteq} = N$. However, it is rather difficult to determine whether $S \subseteq N$ is a smallest set that satisfies $\langle S\rangle^{\trianglelefteq} = N$. (More words about the word problem?)

Corollary 3.2.5: Finitely Generated Groups

Let G be a group. Then G is finitely generated if and only if G is isomorphic to the quotient group of a finitely generated free group.

***Proof*:**
Let $F(S)/N \cong G$. If $F(S)$ is finitely generated, then $F(S)/N$ is finitely generated (see exercise ref:HERE) (Though in that exercise, $N \trianglelefteq G$ is finitely generated given the regular closure, not the normal closure... must think about that)

On the other hand, let G is finitely generated, say $S \subseteq G$ such that $\langle S\rangle = G$ and $|S| = n$. Then there exists an inclusion map $S \hookrightarrow G$. By the universal property of free groups, this extends to a map $\varphi : F(S) \twoheadrightarrow G$. This map is surjective since S is a generating set for G. Then by the 1st isomorphism theorem, $F(S)/\ker(\varphi) \cong G$, fulfilling our requirement.

(ref:HERE if you do geometric group theory or if this is referenced in other places)

In this book, this will be the end our theoretical journey at looking at groups as $F(S)/N \cong G$. A whole field of study has emerged because of this universal property called *geometric group theory*, which we do some exposition for in the appendix (see ref:HERE). For those interested in geometric group theory, I recommend (ref:HERE, perhaps Clara Löh's book and that othe guy whose book is harder).

3.3 Application to Finite Groups

As mentioned earlier, thinking about cosets in themselves is also very useful. In fact, a theorem which you will you to death in this course relies on cosets. With the fact that cosets partition your group, you can deduce this strikingly important result:

Theorem 3.3.1: Lagrange Theorem

If G is a finite group and H is a subgroup of G, then the order of H divides the order of G (i.e., $|H|\,\big|\,|G|$), and the number of left cosets of H in G equals $\frac{|G|}{|H|}$

***Proof*:**
Let $H \leqslant G$. Then you know that H partitions G into equal-sized cosets by proposition ref:HERE (it is an exercise), and so if $|G| = n$, and $|H| = k$, then you know $k|n$, implying $|H|\big|G$. Furthermore, since if $H \trianglelefteq G$, then

$$|G/H| = |G|/|H|$$

Example 3.3.2

1. As a quick reminder, we used this theorem in (*corollary* 2.3.12)to get a useful result a bit early. You can now go back and perhaps check your intuition to make sure it makes sense.
2. Try going back to (*theorem* 2.3.11) and proving it using Lagrange's Theorem. You will find it much easier, almost trivial even.

As the examples show, Lagrange is incredibly powerful as it let's us relate the order of a group and it's subgroup, which is a fact used extensively.

Definition 3.3.3: Index

If G is a group (possibly infinite) and $H \leq G$, the number of left cosets of H in G is called the index of H in G and is denoted by $|G : H|$.

With this notation, we can re-write Lagrange's Theorem as

$$|G| = |H||G : H|$$

Corollary 3.3.4: x to the order of G

If G is a finite group and $x \in G$, then the order of x divides the order of G. In particular $x^{|G|} = 1$ for all $x \in G$.

Corollary 3.3.5: index of H is 2 $\Rightarrow$ H is normal

If The $H \leqslant G$ and the index of H is 2, then H is normal

***Proof*:**
Since $[G : H] = 2$, we know that there are only 2 cosets: eH and xH for the correct $x \in G$. We want to show that H is normal, that is, for every element $g \in G$, $gH = Hg$ (which in our case there is only e and x which form distinct cosets) . Since $eH = He$, by commutativity of identity element from last homework, it sufices to show that

$$xH = Hx$$

for this, we'll take advantage of the index of H. Since the index of H is 2, we know that

$$H \cup xH = G \qquad \text{and} \qquad H \cup Hx = G$$

since the cosets of H partition G, as we showed in part 1 of this homework. Since $xH \subset G$, we know that

$$xH \subset H \cup Hx$$

therefore

$$\forall h \in H, \text{ either } xh = h'x \qquad \text{or} \qquad xh = h' \qquad \text{for some appropriate } h' \in H$$

let's say that for the sake of contradiction $xH \neq Hx$, so we can rule out the first option for a moment, leaving us with $xh = h'$. However, that would imply that $xh \in H$, which is means xh is in the first coset. However, we've shown in the first question of this homework that cosets are distinct, making this a contradiction. Therefore, we're forced to say $xh = h'x$ for appropriate h', however that implies that $xh \in Hx$. Since this holds for any $h \in H$, and we can repeat the same process for Hx, we have that

$$xH = Hx$$

meaning that all cosets "commute", and so G is normal, as we sought to show.

3.3.6 Why this doesn't necessarily work when index greater than 2

Counter-example: Take $G = D_3$, i.e. the triangle group, and $H = e, \sigma$. Then

$$\{e, \sigma\} \cup \{r, r\sigma\} \cup \left\{r^2, r^2\sigma\right\} = G$$

and

$$\{e, \sigma\} \cup \{r, \sigma r\} \cup \left\{r^2, \sigma r^2\right\} = G$$

i.e. the union of the left and right cosets both must equal G. This idea is very important as it forces a relation between left and right cosets. now consider rH. Since the unions of both sets must match, we know that

$$rH \subseteq G \Rightarrow rH \subseteq H \cup Hr \cup Hr^2$$

we knwo that $rH \not\subseteq H$ since that would make the two cosets *not* distinct, and so we have

$$rH \subseteq Hr \cup Hr^2$$

which is true! test it out (while recalling that $r\sigma = \sigma r^2$). In the case we since the index being 2 situation, that would make $xH \subseteq Hx$, and vice versa for Hx too, and so it forces normality.

Because of the power of Lagrange, one might ask themselves if the converse is true: if a number divides the order of a group $n \big| |G|$, does that mean there's a subgroup $H \leqslant G$ such that $|H| = n$? That turns out to be not true:

Example 3.3.7: : Counter-Example

We'll show that $6 \big| |A_4| \not\Rightarrow H \leqslant A_4, |H| = 6$. First, let's figure out the elements of A_4:

1. the identity
2. all even permutation with max order of 4: $(\cdot\cdot)(\cdot\cdot)$
3. Either the 2 transpositions are disjoint, or they overlap with 1 element. They cannot overlap

with 2 elements because then $(\cdot\,\cdot)(\cdot\,\cdot) = (\cdot\,\cdot)$, making it an odd-permutation.

4. If they overlap with 1 element, then $(\cdot\,\cdot)(\cdot\,\cdot) = (\cdot\ \cdot\ \cdot)$, thus we have all 3-cycles, and all permutation represented by 2 disjoint 2-cycles

5. By counting, we can find 8 3-cycles, and 3 $(2,2)$-cycle types, for a total of 12 cycles (when including the identity), meaning $6\big||A_4|$

Let's now suppose there exists a subgroup $H \leqslant A_4$ of order 6. To analyze it we'll take advantage of the fact that the intersection of two subgroups is a subgroup (2.3.15). If you take all $(2,2)$ cycle type elements, plus the identity, you will get a group $(K \leqslant A_4)$ [a]. Take

$$H \cap K = H' \qquad \Rightarrow \qquad H' \leqslant H\,, H' \leqslant K$$

By Lagranges Theorem, $|H'|\big||H|$ and $|H'|\big||K|$ which implies the order is either 1 or 2. If $|H'| = 1$, then it only has the identity. That would mean that H and K are practically disjiont, meaning that if we take 2 non-identity elements $h \in H, k \in K$, then $h \cdot k = g$ we cannot produce an element $g \in H$ or $g \in K$ since then by closure, we'd get $k \in H$ and $h \in K$. Doing this for the case of H:

$$\begin{aligned} hk &\in H \\ \Rightarrow h^{-1}hk &\in H \\ \Rightarrow k &\in H \\ \Rightarrow H \cap K &\supseteq \{e\} \end{aligned}$$

Thus, we'd get $|H| \cdot |K|$ total elements. But $|H| \cdot |K| = 6 \cdot 4 = 24$, which is greater than the number of elements we have: $|A_4| = 12$. Thus, $|H'| \neq 1$, so we must have that $|H'| = 2$. This must also imply that H has an element $h \in H$ that is both a product of two disjoint cycles, while the other 4 elements are 3-cycles.

If $|H'| = 2$, then we'll show that it will also lead to a contradiction. Note that since $|H| = 6$, then $|A_4 : H| = 2$, so by corollary (3.3.5), $H \trianglelefteq A_4$, meaning $\forall \sigma \in A_4$, $\sigma H = H\sigma$, or equivalently $\sigma H \sigma^{-1} = H$

Now, take take the cycle of type $(2,2)$ in H from earlier: $h = (i\,j)(k\,l)$ with $\{i,j,k,l\} = \{1,2,3,4\}$ and the 3-cycle $\sigma = (i\,j\,k) \in A_4$. Then we have that

$$\sigma h \sigma^{-1} = (j\,k)(i\,l) \neq (i\,j)(k\,l)$$

And since H is normal, $\sigma k \sigma^{-1} \in H$. However, that would imply that there is a *second* cycle of type $(2,2)$, meaning that $|H \cap V| \geq 3$, again a contradiction.

Therefore, there cannot be a subgroup of order 6 in A_4, as we sought to show

[a] Trivia: This group is isomorphic to the Klein 4-group V_4

We showed how the converse is not true, but are there partial converses that are true? There are actually two famous theorems which talk about partial converses: *Cauchy's Theorem* and *Sylow's Theorem*. Both these theorems rely on prime-number properties to give us a partial converse. Sylow's theorem goes a step further in telling us how many groups of a particular order we can find (so more than there exists one). However, by the way we'll present Sylow's theorem, it will require group-actions, the topic of the next section. However, Cauchy's theorem technically doesn't. There is an exercises in the book for doing this theorem, so I'll link it here for now for when I have time to do it:

Theorem 3.3.8: Cauchy's Theorem

If G is a finite group and p is prime diving $|G|$, then G has an element of order p

***Proof*:**
proof outlined in exercise 9.

There is a simpler proof which is related to another theorem called Sylow's Theorem (4.8.5) in a similar way that the Inverse Function Theorem and Implicit function Theorem are related: proving one of them is relatively hard, but then proving the other from the first is a super easy consequence. Since Sylow's theorem *is* proved and covered extensively, I proved this theorem using Sylow's theorem once it is introduced (see 4.8.6)

Cauchy's theorem is super important. It pop's a lot when trying to find groups of smaller order:

Example 3.3.9
If G is a finite abelian group with no normal subroups (besides 0 and G)[a], then $G \cong Z_p$.

Proof. Since G is abelian, if $H \leqslant G$, then $H \trianglelefteq G$ (since $aH = Ha$ by use of commutative property). However, G is simple, so either $H = \{e\}$ or $H = G$. By Cauchy's theorem, we also know that if $p \big| |G|$ (which once must as $|G|$ is composed of prime-factors), then there exists an element of order p (meaning a subgroup of order p). However, that's a contradiction as we've just shown. Therefore, either $G = \{e\}$ or $G = \langle x \rangle$ where $|G| = p$. Either way, G, is cyclic and abelian, which by 2.3.9 means that

$$G \cong Z_p$$

□

[a]maybe introduce simple groups earlier?

Note that G was a general finite group. If we have that G is abelian, then the converse is actually *completely* true! This turns out to simply be by the fact that if $\langle x \rangle = p$ and $\langle y \rangle = q$ for prime p, q then $\langle xy \rangle = pq$

Corollary 3.3.10: Cauchy's Theorem For Abelian Groups

If G is a finite abelian group, and n divides G, then there is a group of order n

First, we prove it for if $n = p$ is prime

***Proof*:**
p.102 in book

Now let's prove it in general

***Proof*:**
We'll for the first time proceed by induction! If $|G| = 1$, we're done, so let's say $|G| = n$, and if there is a an element $m < n$ such that $m \big| |G|$ then there exists an $H \leqslant G$ such that $|H| = m$. The edge cases are the same as Cauchy's theorem, so they'll be omitted.

By Cauchy's theorem, we know that if $p \big| |G|$ then $P \leqslant G$ such that $|P| = p$. Since $|G| > 1$, we can assume such a p exists. Then since G is abelian, $P \trianglelefteq G$, so G/P is defined. In particular

$$|G/P| = n/p$$

Which means we found our smaller group G/P. Now for any $m \big| |G/P|$, there exists a $\overline{H}$ such that $\overline{H} = m$. By the 4th Isomorphism theorem, we know there exists a $H \leqslant G$ such that

$$H = H/N$$

meaning that $|H| = \frac{n}{p} \cdot p = n$, showing that there is group of order H, as we sought to show.

Here's also another proof I found

Let G be a finite abelian group. By Cauchy's theorem, it has a subgroup of order p for any prime divisor of $|G|$. We use mathematical induction on the factors of $|G| = n$. *Suppose* that there was a subgroup of order k, for some factor k of n.

We will show that if $kp \mid n$, then there is a subgroup of order kp, for a prime number p. By the induction hypothesis, there is a subgroup H of order k. Now, since $|G/H| = |G|/k$ is still divisible by p, G must have another element of order p not contained in H. Otherwise the quotient

$$G/H$$

would have order divisible by p and yet no element of order p! (Notice that the order of an element $\overline{g} \in G/H$ must divide the order $g \in G$ so no other element can give us an element of order p.)
This shows that there is an element x of order p not in H. But then

$$\langle x \rangle H$$

has order $|\langle x \rangle H| = |x||H| = kp$ (since the group is abelian) and so there is a subgroup of order kp. This completes the inductive step.

Figure 3.1: Alternative proof

This theorem should show you the power of commutativity! If a group is commutative, you have an an incredibly simple group structure! If you know the order of the group, you know the entire substrucutres of the group, giving you a lot of information! This very fact will be key when categorizing every abelian group in chapter 5! This means that a good chunk of the groups studied in group theory are non-commutative groups, since the structure of commutative groups is pretty simple and well-understood.

Be mindful that Lagrange's theorem gives you information about *finite* groups. In general, arithmetics on the infinite won't be defined in a meaningful way for us. In fact, we can get quite some weird results, like subgroups being isomorphic to their parent group!

Example 3.3.11: Subgroups Of $\mathbb{Z}$
Take the group $\mathbb{Z}$, and any subgroup $n\mathbb{Z} \leqslant \mathbb{Z}$ for $n \in \mathbb{N}_+$. Show that

$$n\mathbb{Z} \cong \mathbb{Z}$$

that is, every (non-trivial) subgroup of $\mathbb{Z}$ is isomorphic to $\mathbb{Z}$!

3.4 Internal Direct Product

Though it might seem inconspicuous at first, the introduction of a normal subgroup will let us be able to define the multiplication of subgroups that from subgroups in themselves!! The idea behind this idea is to decompose a group into smaller subgroups, giving an insight on its structure. In chapter 5, we will explore in greater detail the results of this decomposition. For now, we will explore a couple of properties of internal direct product.

We'll first define some notation:

Definition 3.4.1: HK (Internal Direct Product)

Let H, K be subgroups of a group and define

$$HK = \{hk|h \in H, k \in K\}$$

3.4.2 Note This is *not* necessarily a group! Take $\langle(123)\rangle = H \leqslant S_4$ and $\langle(3\,4)\rangle = K \leqslant S_4$. Then

$$HK = \{\text{id}, (123), (132), (123)(34), (132)(34), (34)\}$$

and $(132)(34) = (3421)$, but $(3421)^2 \notin HK$, meaning HK is not closed!

Proposition 3.4.3: cardinality of HK

Let H, K be finite subgroups of a group. Then

$$|HK| = \frac{|H||K|}{|H \cap K|}$$

***Proof*:**
Notice that HK is the union of left-cosets of K:

$$HK = \bigcup_{h \in H} hK$$

Thus, it suffices to find the number of distinct left-cosets hK.

If $h_1K = h_2K$ then $h_2^{-1}h_1K = K \Rightarrow h_2^{-1} \in K$. Since $h_2^{-1}h_1 \in H$, this means that

$$h_2^{-1}h_1 \in H \cap K \Rightarrow h_2^{-1}h_1H \cap K = H \cap K \Rightarrow h_1H \cap K = h_2H \cap K$$

And so we reduced the problem to counting the number of distinct $H \cap K$ cosets. This is useful, since $|H \cap K| \big| H$, and so by Lagrange's theorem

$$|H : H \cap K| = \frac{|H|}{|H \cap K|}$$

Thus, there are that many left cosets of K. Since K partitioned HK, it means that every element of the cosets are unique, so to find $|HK|$ we now get

$$|HK| = \frac{|H|}{|H \cap K|}|K| = \frac{|H||K|}{|H \cap K|}$$

as we sought to show

Proposition 3.4.4: When is HK a group

If H and K are subgroups of a group, HK is a subgroup if and only if $HK = KH$

***Proof*:**
Let $H, K \leqslant G$.

($\Rightarrow$) Let's say $HK \leqslant G$. We want to show that $HK = KH$, that is, for any $hk \in HK$, there exists an $h' \in H$, $k' \in K$ such that $hk = k'h'$. Since HK is a group, it is closed under its operation. Thus, for any $h_1k_1, h_2k_2 \in HK$, $h_1k_1h_2k_2 \in HK$. Since $HK = \langle hk | h \in H, k \in K \rangle$, there is a $h_3 \in H$ and $k_3 \in K$ such that

$$h_1k_1h_2k_2 = h_3k_3$$

re-arranging, we get that

$$h_3h_1k_1 = k_3k_2^{-1}h_2$$

By letting $h = h_3h_1$, $k = k_1$, $h' = h_2$, and $k' = k_3k_2^{-1}$, we get that

$$hk = k'h'$$

as we sought to show (more words on how we can choose these values and still let h, h', k, k' be "arbitrary")

($\Leftarrow$) Let's say $HK = KH$. We want to conclude that HK is a subgroup of G. We'll show that HK is closed under its binary operation – the other axioms follows straightforwardly.

Corollary 3.4.5: Condition on K when Internal Direct Product is Present

If H and K are subgroups of G and $H \leq N_G(K)$, then HK is a subgroup of G. In particular, if $K \leqslant G$ then $HK \leqslant G$ for any $H \trianglelefteq G$. Furthermore, if H and K are normal subgroups, then $HK \trianglelefteq G$.

***Proof*:**
The first part of the corrollary was proven in the last proposition. What is left to prove if both H and K are normal subgroups, then $HK \trianglelefteq G$. (TBD – is this true?)

Defining when HK is a group is intersting, but it would be even more useful if we can figure out the structure of HK. For example, Can we say that $HK \cong H \times K$? The answer is not always! In fact, we will find a way of exactly figuring out the structure of HK in chapter 5 once we have learned the appropriate tools in chapter 4

3.4.6 Foreshadowing Thinking about HK like in corollary 3.4.5 will become the defining feature when we discuss *semi-direct* products in chapter 5.

Since we have now defined how we can take the "product" of two groups, we can ask the following intersting question: If $\mathcal{N}$ is some subset of normal subgroup G that is "closed" under multiplication, does it form a group? The reader should fiddle around with this before the answer is given. The same question will be asked in the next part of the book where we will have the equivalent of normal subgroup for rings (an algebraic structure with 2 binary operations). It with the "normal" objects of rings. The answer for groups shall be presented in chapter 7. with the "normal" objects of rings. In the case for groups, such a collection will not form a group.

Thus, we will leave internal direct product alone, with only an exception made for the next section, until we have accumulated more tools to discuss them again.

3.5 Isomorphism Theorems

We know turn back to how quotient groups are inseparably related to homomorphisms. Mathematicians who were studying homomorphisms wanted to understand the relation between groups given certain properties, for then we can gain new perspective into problems being solved. For example, if we have a group $HK \leqslant G$, and $K \trianglelefteq G$, can we always determine what HK/K is? Is there *always* an answer to this question, no matter what group HK we pick? Another question is what if we "recusively" take quotients: If $H \trianglelefteq K \trianglelefteq G$, what is $(G/H)/(K/H)$? Is there a simplification that will make this more understandable? Yet another such question is how much can we deduce of the subgroups of G/N (i.e. the lattice of G/N) given what we know about G?

In this section, we'll show these results:

1. We'll sum up the results of of theorem 3.1.14, but give it the name that is more commonly used
2. We will look at equivalent ways of looking at internal direct products, since they pop up a lot (especially chapter 5)

3. An easier way of thinking of quotient groups whose quotient is a quotient group.

4. the subgroup lattice of G and the subgroup lattice of G/N

We first recall theorem 3.1.14, but given it its classical name:

Theorem 3.5.1: First Isomorphism Theorem

If $\Phi : G \to H$ is a homomorphism of groups, then $\ker(\Phi) \trianglelefteq G$ and $G/\ker\Phi \cong \Phi(G)$. Similarly, if $N \trianglelefteq G$ and $N \subseteq \ker(\Phi)$, there exists a unique *homomorphism* $\varphi : G/N \to H$ such that the following diagram commutes:

$$\begin{tikzcd} G \arrow[r, "\pi"] \arrow[dr, "\Phi"'] & G/N \arrow[d, "\varphi"] \\ & H \end{tikzcd}$$

where $\Phi = \varphi \circ \pi$, and π is the natural projection

Recall that the powerful part of this theorem is that it is the universal property of quotient groups: for *any* homomorphism $\varphi : G \to H$, there exists a unique group G/N (which represents the equivalence class, and thus the homomorphism, within its group structure) along with the unique homomoprhism π [5]which let's us link G to G/N such that $\Phi = \pi \circ \varphi$, meaning we can study the results of the homomorphism Φ by looking at group information of G/N.

Also, remember that if we take $\Phi(G) \leqslant H$, and $N = \ker(\Phi)$ so that

$$\begin{tikzcd} G \arrow[r, "\pi"] \arrow[dr, "\Phi"'] & G/\ker(\Phi) \arrow[d, "\varphi"] \\ & \Phi(G) \end{tikzcd}$$

then

$$G/\ker(\Phi) \cong \Phi(G)$$

Example 3.5.2

1. we can get some information about S_n. We can define the function $\epsilon : S_n \to \{-1, 1\}$ whether if σ is even then $\epsilon(\sigma) = 1$, and if σ is odd then $\epsilon(\sigma) = -1$. Note that we can treat $\{-1, 1\}$ as a group if $-1 \cdot -1 = 1, 1 \cdot -1 = -1, -1 \cdot 1 = 1, 1 \cdot 1 = 1$. Then ϵ would actually be a homomorphism. Then $\ker(\epsilon) = A_n$, that is, the cycles of even permutation, also known as the alternating groups. By the first isomorphism theorem, we then know that

$$S_n/A_n \cong \epsilon(S_n) = \{\pm 1\}$$

[5] In particular, it's a *natural homomorphism*. The word natural here has a technical meaning which we explore in section **??**. We will explore the concept once we explore the non-natural choice for basis for vector-spaces, which will let us contrast what it means to have a "natural" versus a "non-natural" function

and so, using Lagrange, we get that

$$2 = |S_n/A_n| = \frac{|S_n|}{|A_n|} \qquad \Rightarrow \qquad |A_n| = \frac{1}{2}|S_n|$$

2. Let's say we have two groups G, H with $\varphi : G \to H$ being a surjective homomorphism. Then we know that

$$G/\ker(\varphi) = H$$

in other words, if we have a surjective homomorphism, we can always view the structure of H through the perspective of the quotient group $G/\ker(\varphi)$!

Corollary 3.5.3: properties as consequence

Let $\varphi : G \to H$ be a group homomorphism

1. φ is injective if and only if $\ker \varphi = 1$
2. $|G : \ker \varphi| = |\varphi(G)|$ [a]

[a] If you have taken linear algebra, this is the Rank-Nullity theorem equivalent for Group Theory

The second Isomorphism theorem let's us view the quotients of internal direct products (definition 3.4.1) form a different angle.

Theorem 3.5.4: The Second or Diamond Isomorphism Theorem

Let G be a group, let A and B be subgroups of G and assume $A \leq N_G(B)$. Then AB is a subgroup of G, $B \trianglelefteq AB$, $A \cap B \trianglelefteq A$, and $AB/B \cong A/A \cap B$.

The name diamond comes form this diagram where the arrow represent subgroups, and a red arrow represents norma subgroup:

$$\begin{tikzcd}
 & G \arrow[d] & & & & G \arrow[d] & \\
 & AB \arrow[dl] \arrow[dr, "\trianglelefteq"] & & & & AB \arrow[dl] \arrow[dr] & \\
A \arrow[dr] & & B \arrow[dl] & \Rightarrow & A \arrow[dr, "\trianglelefteq"] & & B \arrow[dl] \\
 & A \cap B \arrow[d] & & & & A \cap B \arrow[d] & \\
 & \{e\} & & & & \{e\} &
\end{tikzcd}$$

and

$$\frac{AB}{B} \cong \frac{A}{A \cap B}$$

The reason this theorem is intersting is because we often enough want to take the internal direct prodcut of groups, and deduce some structural information form it with quotients, and this theorems gives us a a tool do do so (see exercise ref:HERE)

***Proof*:**
Since $B \trianglelefteq G$, by proposition 3.4.4, $AB \leqslant G$. Another way of seeing this is that AB is the collection of cosets Ab. Then if $\pi : G \to G/A$ is the cannonical projection, $AB = \pi^{-1}(\pi(B))$. The image of a subgroup is a group (ref:HERE) and the pre-iamge of a subgroup is a group (see exercise ref:HERE). Thus AB is a subgroup.

To show $B \trianglelefteq AB$, we need to show that for any $ab \in AB$, $abB(ab)^{-1} = abBb^{-1}a^{-1} = B$. To do so, recall that B is normal, so that $kBk^{-1} = B$, and so

$$abBb^{-1}a^{-1} \stackrel{!}{=} aBa^{-1} = B$$

where $\stackrel{!}{=}$ comes from the fact that B is closed. Since ab was a general element of AB, it follows that $B \trianglelefteq AB$.

Finally, to show that $\frac{B}{A\cap B} \cong \frac{AB}{B}$, define the map $\varphi : A \to AB/B$, where $\varphi(a) = aB$. We'll show this is a surjective homomorphism. This function is indeed a homomorphism, since

$$\varphi(ab) = abB \stackrel{!}{=} aBbB = \varphi(a)\varphi(b)$$

where $\stackrel{!}{=}$ comes from exercise ref:HERE. Alternativly, Since $B \trianglelefteq AB$, $\pi : AB \to AB/B$ is a well-defined group-homomorphism. Then one can show that restricting to $\pi|_A\, A \to AB/B$ is also a group-homomorphism.

To show that φ is surjective, notice that every element of AB/B can be written as abB. Since $b \in B$, then

$$abB = aB$$

it follows that an arbitrary coset abB has a map to it (i.e. $\varphi(a) = aB = abB$), showing that φ is surjective. Thus, by corollary 3.1.16, $A/\ker(\varphi) \cong AB/B$. To find $\ker(\varphi)$, we simply put together the information given to us:

$$\ker(\varphi) = \{a \in A \mid \varphi(a) = 1B = B\} = \{a \in A \mid aB = B\} \stackrel{!}{=} \{a \in A \mid a \in B\} = A \cap B$$

where $\stackrel{!}{=}$ comes from exercise ref:HERE. Thus, substituing $\ker(\varphi)$ by $A \cap B$, we get

$$\frac{A}{A\cap B} \cong \frac{AB}{B}$$

as we sought to show

One way to remember this theorme is through the following mnemonic:

If we have $\frac{AB}{B}$, then add a $A \cap \_$ to the top and bottom, giving us

$$\frac{A\cap AB}{A\cap B} = \frac{A}{A\cap B}$$

Another way to think about the 2nd isomorphism theorem is to consider the case where $A \cap B = \{e\}$.

Then this theorem shows that

$$\frac{AB}{B} \cong A \qquad \frac{AB}{A} \cong B$$

Heuristically, we can think in this situation that we are "cancelling out" the B's. However, be mindful to take this as a heuristic – this is *not* what is actually happening!

The third Isomorphism theorem lets quickly simplify quotient groups quotiented *by* a quotient group:

Theorem 3.5.5: The Third Isomorphism Theorem

Let G be a group and let H and K be normal subgroups of G with $H \leq K$. Then K/H, G/H and

$$(G/H)/(K/H) \cong G/K$$

If we denote the quotient by H with a bar, this can be written

$$\bar{G}/\bar{K} \cong G/K$$

***Proof*:**
Since $H \leqslant K$, then $K/H \trianglelefteq G/H$. To show this, first note that K/H is indeed a subgroup of G/H. For any $[x],[y] \in K/H$, $[x][y]^{-1} = [xy^{-1}] \in K/H$ since $xy^{-1} \in K$. Furthermore, for any $[g] \in G/H$, and any coset $[k] \in K/H$, $[g][k] = [k'][g]$, $k' \in K$ (i.e. $[g](K/H) = (K/H)[g]$), since both K and H are normal. Thus $K/H \trianglelefteq G/H$.

Now, define the map

$$\varphi : G/H \to G/K \qquad gH \mapsto gK$$

This map is well-defined: If $g_1H = g_2H$, then $g_1h_1 = g_2h_2$. Since $H \leqslant K$, then $h_1, h_2 \in K$, so $g_1h_1 = g_2h_2$ still holds in G/K for any $h_i \in H$. Thus $\varphi(g_1H) = \varphi(g_2H)$, showing the function is well-defined.

Now, notice this map is a homomorphism and surjective. It is a homomorphism since

$$\varphi(g_1Hg_2H) \overset{!}{=} \varphi(g_1g_2H) = g_1g_2K \overset{!}{=} g_1Kg_2K = \varphi(g_1H)\varphi(g_2H)$$

where the $\overset{!}{=}$ equalities come from exercise ref:HERE. For surjectivity, since $H \leqslant K$, then for any gK, we know there is at least the coset gH that maps to it (there might be more, since $H \leqslant K$). Hence, by corrollary 3.1.16

$$(G/H)/\ker(\varphi) \cong G/K$$

to find $\ker(\varphi)$, we put together the information we have so far:

$$\begin{aligned}\ker(\varphi) &= \{gH \in G/H \mid \varphi(gH) = K\}\\ &= \{gH \in G/H \mid gK = K\}\\ &= \{gH \in G/H \mid g \in K\}\\ &= K/H\end{aligned}$$

Thus, replacing $\ker(\varphi)$ with K/H we get

$$(G/H)/(K/H) \cong G/K$$

as we sought to show

An easy heuristic to remember $(G/H)/(K/H) \cong G/K$ is to think about how to simply fraction's of fractions, that is, the "flip and cancel" trick

$$\frac{\frac{1}{2}}{\frac{4}{2}} = \frac{1}{2}\frac{2}{4} = \frac{1}{4}$$

Note however that this is *not* waht's happening here – it is just a useful heuristic to remember the 3rd Isomorphism Theorem. Note too that for the simplification to occur, you need that H is in the denominator of both quotients, you cannot have a different value.

Example 3.5.6: Example
Using the 3rd isomorphism theorem?

Finally, we will relate the subgroup structure (the lattice) of G/N with G. It will actually be precisely a sublattice of G that contains N! This means that going to G/N will preserve subgroup relations:

Theorem 3.5.7: The Fourth or Lattice Isomorphism Theorem (Correspondence Theorem)

Let G be a group and let N be a normal subgroup of G. Then there is a bijection from the set of subgroups H of G which contain N onto the set of subgroups $\bar{H} = H/N$ of G/N.

$$\{H | H \leqslant G,\ H \text{ contains } N\} \leftrightarrow \left\{\overline{H} \leqslant \overline{G}\right\}$$

In particular, every subgroup of $\bar{G}$ is of the form H/N for some subgroup H of G containing N (namely, its pre-image in G under the natural projection homomorphism from G to G/N). This bijection has the following properties between the two lattices

***Proof*:**
Let $\overline{G} = G/N$. Let $\overline{H} \leqslant \overline{G}$. Then since the pre-image of a subgroup is a group, take $H = \pi^{-1}(\overline{H})$. Then by definition, $\pi(H) = \overline{H}$, meaning for every subgroup of $\overline{G}$, there is a subgroup of G that maps to it.

Furthermore, $N \leqslant H$. Since $\pi : H \to H/N$, $\pi^{-1}(1N) \subseteq H$ (note that it's a subset and not element since it's possible that the pre-image contains multiple elements). By definition

$$\pi^{-1}(1N) = \{h \in H \mid \pi(h) = N\} = \{h \in H \mid hN = N\} = \{h \in H \mid h \in N\}$$

and since $\pi^{-1}(1N) \subseteq H$, it must be that $N \subseteq H$, as we sought to show

This fact has many very nice consequences:

Corollary 3.5.8: Equivalent way of looking at 4th Isomorphism Theorem

1. $A \leqslant B$ if and only if $\overline{A} \leqslant \overline{B}$
2. if $A \leqslant B$ then $|B : A| = |\overline{B} : \overline{A}|$
3. $\overline{\langle A, B\rangle} = \langle \overline{A}, \overline{B}\rangle$
4. $\overline{A \cap B} = \overline{A} \cap \overline{B}$
5. $A \trianglelefteq G$ if and only if $\overline{A} \trianglelefteq \overline{G}$

***Proof*:**
Each of these can be proved using a simialr technic as the proof for the 4th Isomorphism Theorem

Another way of looking at the first part of the Correspondence Theorem is in the following slight generalization: Let G be a group and N a normal subgroup so that G/N is well-defined. If $\mathcal{G}$ contains all subgroups of G, for any $H \in \mathcal{G}$, the map $\pi : H \to G/N$, $\pi(h) = hN$ is well-defined. So, H maps to a subgroup of G/N. If $\mathcal{N}$ represents all subgroups of G/N, then there is a surjective map

$$\mathcal{G} \twoheadrightarrow \mathcal{N}$$

and if we take $\mathcal{G}|_N$ to be all subgroups that contain N, then there is a bijection between $\mathcal{G}|_N$ and $\mathcal{N}$:

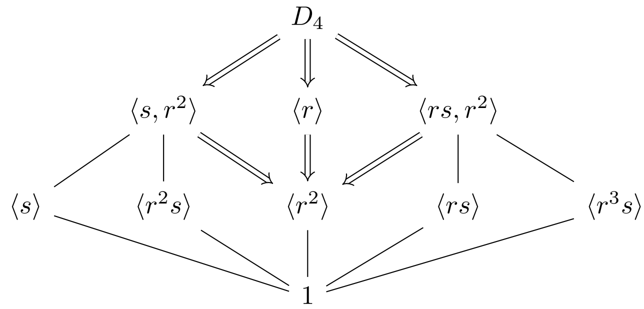
$$\mathcal{G}|_N \leftrightarrow \mathcal{N}$$

Example 3.5.9: Using Correspondence Theorem

1. Let $G = Q_8$, and take $\langle -1\rangle \trianglelefteq Q_8$. Then in the following lattice, the double lines represent the sublattice of Q_8 which is an isomorphic copy of the lattice of $Q_8/\langle -1\rangle$

$$\begin{tikzcd}
& Q_8 \arrow[dl, Rightarrow] \arrow[d, Rightarrow] \arrow[dr, Rightarrow] & \\
\langle i \rangle \arrow[dr, Rightarrow] & \langle i \rangle \arrow[d, Rightarrow] & \langle i \rangle \arrow[dl, Rightarrow] \\
& \langle i \rangle \arrow[d, dash] & \\
& \{1\} &
\end{tikzcd}$$

2. Let $G = D_4$ and let $\langle r^2 \rangle \trianglelefteq D_4$. Then with the same definitions as the last example, we get:



Notice that several subgroup of G (for example $\langle s \rangle$) become elements in a larger subgroup (in this case $\langle s, r^2 \rangle$). Thus, every subgroup of $\overline{G}$ might have multiple subgroups of G that map to it, but a unique subgroup which contains N.

A good example how the map of subgroup of G to G/N is surjective is by the following example: Notice that $Q_8/\langle -1 \rangle \cong D_8/\langle r^2 \rangle \cong \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$, and $\langle -1 \rangle \cong \langle r^2 \rangle$, but $Q_8 \not\cong D_8$. We will explore this idea in more depth in the next section.

Exercise 3.5.1

1. In the following questions, we're going to prove what's sometimes called the 5th Isomorphism Theorem (also known as the *Butterfly Lemma*), which for our purposes will offer an alternative proof to the main result of section 3.7. HERE

3.6 Extensions

Recall (or revisit) that if $H \leqslant G$ (and G is finite), then the subgroup lattice of H is the sublattice of G of all groups *contained* in H (theorem 2.4.3), so *a fortiori* if $K \trianglelefteq G$ is a normal subgroup then the subgroup lattice of K is the sublattice of G of all subgroups contained in K. Even hen G is infinite, the set of subgroup of K is the set o subgroup of G which are contained in K. The Correspondence Theorem has shown us that the "dual" is true for quotient groups (except it orks for *all* groups, not just finite): If $K \trianglelefteq G$, then the subgroup lattice of G/K is the sublattice of G of all groups *containing* K. In this way, K and G/K together capture the subgroup structure of a group! In this section, we will explore how we can find a group structure given *only* the information from K and G/K.

More succinctely, Given $K \trianglelefteq G$, the group K and G/K contain the subgroup structure of G. However, K and G/K don't capture the full structure of a group: in the process of quotienting, G/K losses some information:

Example 3.6.1: Subgroup And Quotient Group Limits

Take $G = D_8$ and $K = \{1, r, r^2, r^3\} \cong C_4$. Then $D_8/K \cong C_2$ – let's label this K'. Then given just K and K', e can't guarantee that $K' = D_8/K$, other groups can take the place of D_8. For example:

$$H = C_4 \times C_2 \qquad \varphi : C_4 \times C_2 \to C_2,\ \varphi((a,b)) = ((0,b))$$

Here, H is another group here the kernel is C_4 and the quotient is C_2.

It must be noted though that there is still information within cosets: for any $x \in G$, then $[x] = xK \in G/K$, so for any $y \in xK$, $y = xk$ for appropriate $k \in K$ (proposition 3.1.3). This means for every coset, there is some information about how the elements relate. An easy example of this can be seen if $K = Z(G)$.

Example 3.6.2: Relation Within Cosets

Take $G/Z(G)$ (as an asside, try proving that $Z(G) \trianglelefteq G$ alway, exercise ref:HERE). Take any $[x] \in G/Z(G)$, and any $a, b \in [x]$. Then $a = xz_a$ and $b = xz_b$, so $ab = xz_axz_b = xz_bxz_a = ba$ (since $z_a, z_b \in Z(G)$). Thus, within any coset of $G/Z(G)$, the elements commute! This is not nesesarily the case if the elements are *not* within the same coset (ex. $G = D_7$).

However, as we saw in example 3.6.1, there can be more than one group (say G') we can construct from K and K' (where $G'/K = K'$). We will call groups which can be built in this fashion *extension*:

Definition 3.6.3: Extension

Let A and B be groups. If $G/A \cong B$, then we say *G is an extension of A by B.*

The idea behind this name is that we want to be able to restrict G to A (i.e. A must be able to map injectively and homomorphically into G; $A \hookrightarrow G$), similarly to how we can restrict functions (the dual of restricting functions is also called *extending* functions, where the extended function must be exactly the same as the orginal funciton when restricting).

As we saw at the beginning of this sectino, quotient groups are the prefect candidate for extension of groups since every group can be decomposed into a [normal] subgroup and quotient group (TBD – perhaps put this earlier)

Since all normal subgroups are also kernels, we can re-write what it means to be an extension in terms of homomorphisms. In the following definition, let I represent the trivial group.

Definition 3.6.4: Short Exact Sequence

Let A, B, C be group. If there exists an injective homomorphism f and a surjective homomorphism g such that

$$I \xrightarrow{\iota} A \xrightarrow{f} B \xrightarrow{g} C \xrightarrow{\pi} I$$

where $\mathrm{im}(f) = \ker(g)$, ι is the trivial injective $(0 \mapsto 0)$ map and π is the total projective map $(c \mapsto 0)$, then we say that *B is an extension of A by C.*

Notice that $\mathrm{im}(\iota) = \ker(f)$ since f is injective, $\mathrm{im}(g) = \ker(\pi)$ since g is surjective, and $\mathrm{im}(f) = \ker(g)$ by definition. The fact that the image of the previous function is the kernel of the next function is no coincidence – this property is called *exactness* [6]. If we make a longer chain of homomorphisms with this property (ex. $X_1 \xrightarrow{f_1} X_2 \xrightarrow{f_2} X_3 \xrightarrow{f_3} X_4 \xrightarrow{f_4} \cdots$), then such a chain is called an *exact sequence* (we will return to these in chapter 17$\star$). The exact sequence in definition 3.6.4 is the

[6] You might have seen the term *exact vector-field*. This concept is in fact the same, however it would take us a long time before we can link the two (see ref:HERE)

shortest non-trivial sequence with the trivial group at the beginning and end, and so it is called a short exact sequence.

For our purposes, the fact that $\mathrm{im}(f) = \ker(g)$ means that $B = C/A$, meaning that B is an extension of A by C. (exercise: does $G \to H$ imply a short exact sequence? If no, can it)

Example 3.6.5: Short Exact Sequences

In the following examples, let 1 represent the trivial group

1. If $\varphi : G \to H$ is a homomorphism and $K = \ker(\varphi)$, we always have
$$1 \to K \hookrightarrow G \xrightarrow{\varphi} \varphi(G) \to 1$$
is a short exact sequence. If φ is surjective, then
$$1 \to K \hookrightarrow G \xrightarrow{\varphi} H \to 1$$
In this way, every surjective homomorphism represents an extension G of K by H.

2. Let A and B be two arbitrary groups. Then
$$1 \to A \hookrightarrow A \times B \xrightarrow{(a,b)\mapsto(0,b)} B \to 1$$
is a short exact sequence. In this way, any group can be extended by any other group to the extensions $A \times B$. This is sometimes called the *trivial extension* [a]

3. Take $A = B = 2$. Then
$$1 \to C_2 \xrightarrow{f} C_4 \xrightarrow{g} C_2 \to 1$$
where $f(x) = 2x$ and $g(0)$ and $g(2)$ both map to 0, while $g(1)$ and $g(3)$ map to 1. It is easy to verify that f and g are indeed homomorphism and that the above sequence is exact.

4. Similarly to the last example, Let $A = C_2$ and $B = C_8$ and $C = C_4$. Can you find the appropriate maps such that
$$1 \to C_2 \to C_8 \to C_4 \to 1$$
is a short exact sequence? Is the following
$$1 \to C_2 \to D_4 \to C_4 \to 1$$
also a short exact sequence with the appropriate maps?

[a] In fact, once we cover homological algebra in chapter 17⋆, the idea of this being the trivial extension will become rigorous

An extension of an X group by an X group is X. When is this statement true when X is:

1. cyclic
2. If M and A are abelian, G need not be abelian
3. finite
4. finitely generated

5. Noetherian

It is in general difficult to tell when when given a short exact sequence if the sequence splits. In section 17.2, we will classify which *abelian groups* in a short exact sequence of abelian groups split[7] (abelian groups which always split in a short exact sequence of abelian groups be called *projective groups*). For general groups, The best tool we have is trying to identify when C acts on B, (mabye there are some identification of when this is possible??)

(maybe make HNN extensions an exericise? Or more it to an Algebraic topology section if one ever exists?)

(

You may also think of the short exact sequence:

$$1 \to H \to G \to G/H \to 1$$

Then since H is the kernel, we can break down G into zH for some representative $z \in G/H$. It is natural to ask if we can think of as G as "$(G/H) \cdot H$". However, this is not so simple. It sometimes works, it is called a split extension, and in general you require the fundamental theorme of extension of groups (which shows that G is a subgroup of the wearth product of H and G/H. We alos use homology to see how much G "fails" to be a direct product of H and G/H)

)

Exercise 3.6.1

1. Let $0 \to \mathbb{Z} \to B \to A \to 0$ be an extension, where B is an abelian group and A is free. Show that B splits. [Interestingly, The converse of this statement (known as Whitehead's Problem) is undecidable in ZFC, that is, it is a statement that could neither be true nor false! This idea would be explored with Gōdel's Incompleteness Theorems]

2. Is it save to say that if $0 \to A \to G \to B \to 0$ where $|B|$ is finite, then G is A-by-finite (and so virtually A)? That would be another cool way of thinking about the B term in a short exact sequence!

3.7 Classifying all Finite Groups: Hölder Program

With the isomorphism theorems under our belt, we are ready to tackle one of the central questions posed by group-theorists: How can we classify all groups? This sections start by building up the mental models used in order to comprehend the way classification efforts.

We can approach the problem of classification in many different ways. We briefly mentioned the idea trying to construct arbitrary presentation's of groups to try and create groups, but we've mentioned that this can create "insoluble presentation", meaning presentaion in which we cannot represent every element. In general, it is very hard to find groups in that manner. Perhaps instead we can start with the groups we knwo and see if there is a way to "decompose" them. The first thing you might try is decomposing the group into cyclic sub-groups. This is certainly possible (take an element,

[7] It will turn out that all abelian groups are $\mathbb{Z}$-modules, and so this will be a consequence of classifying when modules split with the space case of when the module is over $\mathbb{Z}$

generate a group from it, remove those elements from your origianl group, continue doing this till you've exhauted the group [8]). But how would you then put them back together? For example, we know that $\langle r\rangle, \langle s\rangle \leqslant D_4$. With what we've learnt, is there a way to represent D_4 using these? In fact, we have just seen that we can but them back together using short exact sequences! The group D_4 is an extension of C_2 by C_4.

However, we must be more careful using extensions when we are trying to classify groups; are cyclic subgroups really the way to go? In fact, this is not the case: there exists cyclic groups which are an extension of smaller groups. We would want to be able to have some fixed subgroups we can use as our "building puzzles" which are not extensions of any other groups. In this section we will define these "smalles puzzle pieces" or "atoms of groups". Try taking a moment to see if you can come up with a more suitable definition of "smallest puzzle piece" for groups and have an idea of an example of a group following your definition.

We will end this section by being able to declare a theorem that took group theorists the better part of the 20th century to prove – the classification of all finite groups! As a quick note before proceeding, this classification effort is in fact much simpler for *abelian groups*. In chapter 6.1, section ref:HERE, we will continue this exploration in the finite abelian case.

Defining our Terms

First, we will have to define what would be our "puzzle piece". Since we are using extensions, we know there is a non-trivial homomorphism from a group G to another group H if G has a non-trivial normal subgroup. If G has *only* trivial subgroups, the structure of G is not "composed" of other possible structure but is as "simple" as possible. This motivates the folloing definition.

Definition 3.7.1: Simple Groups

A (finite or infinite) group G is called *simple* if $|G| > 1$ and the only normal subgroups of G are 1 and G.

These groups are called "simple" because there are no more group-structures within them [9]. To undersatnd what "no more group-structures within them" means, recall the 4th isomorphism theorem. If N is a noraml subgroup, then ther is a direct correspondence between the subgroups of G/N and the subgroups of G containing N. If there is no $N \trianglelefteq G$ besides $\{1\}$ and G, then the lattice of G can't be split into G/N and G. So in a sense, that lattice is as small as it gets. (comment on infinite case?)

Example 3.7.2: Simple Groups

We know that by Lagrange's theorem that if G is prime, then it's only subgroups must be 1 and G, and so groups of order p are examples of simple groups. If G is abelian, then it can in fact be shown that every simple subgroup is isomorphic to $\mathbb{Z}/p\mathbb{Z}$ for appropriate prime p, since those are the only subgroups which no non-trivial subgroups (and so no normal subgroups)

There are also examples of non-abelian simple Groups, both infinite and finite (the smallest finite

[8] This naturally works well in groups of *countable* size, one has to be more careful for groups of *uncountable* size – the axiom of choice is involved (see chapter A for details on the axiom of choice)

[9] The term simple is one that comes up often enough in matheamtics. It has a sligthly more nuanced meaning than just the simplest object (ex. the trivial group is not a simple group, it's "too simple to be simple". See this nlab article on simple objects

one (of order 60) will be explored in detail in chapter 4, section ref:HERE).

Looking at simple groups a little more abstractly for a moment, they play an analogous role to primes in arithmetic's of $\mathbb{N}$ as in they cannot be "factored" into smaller [meaningful] pieces[10] (i.e. N and G/N). There is even a "unique factorization theorem" based on this idea. First, we'll define what is meant by "factorize" a group

(link to extensions)

Definition 3.7.3: Subnormal Series

Let G be a group. Then

$$G = G_1 \geqslant G_2 \geqslant G_3 \geqslant \cdots \geqslant G_n = \{1\}$$

is called a *series* (or *tower*) for the group G. If every $G_i \trianglelefteq G$, then G the series is called a *normal series*. If $G_i \trianglelefteq G_{i-1}$ for every $1 \leq i \leq n$, then we call the series a *subnormal series*:

$$G = G_1 \trianglerighteq G_2 \trianglerighteq \cdots \trianglerighteq G_n = \{1\}$$

Sometimes, [normal/subnomral] series are written in the ascending, rather than descending order:

$$\{1\} = G_0 \leqslant G_1 \leqslant \cdots \leqslant G_n = G$$

I chose to write them in descending order since down the line we will be comparing some types of series to other types, and if everything is written in descending order, then we'd have a scendario were we start writing $G_{n-i} \trianglelefteq B_i$ ineated of $G_i \trianglelefteq B_i$, which I feel will make the simple relations a little more notationaly cumbersome than it needs to be.

You might ask whether this a subnormal series would imply a noraml series, that is, that $N_i \trianglelefteq G$? However, that doesn't need to be the case:

Example 3.7.4: Normality *not* transitive

Here's an example of how normality is *not* transitive. Let $G = D_8$. Let $H \trianglelefteq G$ such that $H = \langle r\sigma, \sigma r\rangle$. Next, take $K \trianglelefteq H$ such that $K = \langle r\sigma \rangle$. Note that

$$K \trianglelefteq H \trianglelefteq G$$

but $K \ntrianglelefteq G$

thus a subnormal series need not be a noraml series. However, a normal series is a subnormal series: if $N_{i+1} \trianglelefteq G$, then $gN_{i+1}g^{-1} = N_{i+1}$ for every $g \in G$. If we restrict to every $n \in N_i$, then since $N_i \subseteq G$, it is already the cse that $nN_{i+1}n^{-1} = N_{i+1}$. Thus $N_{i+1} \trianglelefteq N_i$. Thus:

$$\text{normal series} \Rightarrow \text{subnormal series} \qquad \text{subnormal series} \not\Rightarrow \text{normal series}$$

Thus, subnormal series can be thought of as the decomposition of a group.

[10] meaningful here meaning there is a notion of combining groups using short exact sequence, unlike general groups can't be

Example 3.7.5: Example

here. In particualr, give a two subnromal series for a group, one more refined than the other

In (reference previous example), the subnormal serise cannot be refined further, since every N_i/N_{i+1} is simple. We give a name to such a serise:

Definition 3.7.6: Composition Series

In a group G, a sequence of groups

$$1 = N_0 \leqslant N_1 \leqslant N_2 \leqslant \cdots \leqslant N_{k-1} \leqslant N_k = G$$

is called a *composition series* if $N_i \trianglelefteq N_{i+1}$ and N_{i+1}/N_i is a simple group, for all $0 \leq i \leq k-1$. If the above sequence is a composition series, the quotient group N_{i+1}/N_i are called the *composition factors* of G.

Every finite group has a composition series. Since the group is finite, there exists a maximal subgroup [11], so that $N \trianglelefteq G$ so that G/N is simple. If $N = \{1\}$, then G is already simple, since the only proper normal subgroup of G is $\{1\}$, meaning the only other normal subgroup is G. If the group G/N was not simple, then there would exist a group $N'/N \trianglelefteq G/N$. Then by the 4th isomorphism theorem, $N' \trianglelefteq G$, where N' is a group containing N. However, N is a maximal subgroup, so it must be that $N = N'$, so we'd get $N'/N = N/N \cong \{1\} \trianglelefteq G/N$, and so G/N is simple. We can then continue the same process with N, taking $N_1 \trianglelefteq N$ to be the maximal normal subgroup. Since G was finite this process will be end.

Example 3.7.7: Some Composition Series

Take $\mathbb{Z}_p \times \mathbb{Z}_q$. Then

$$\{e\} \trianglelefteq \mathbb{Z}_p \times \{e\} \trianglelefteq \mathbb{Z}_p \times \mathbb{Z}_q$$
$$\{e\} \trianglelefteq \{e\} \times \mathbb{Z}_q \trianglelefteq \mathbb{Z}_p \times \mathbb{Z}_q$$

The quotient of any two will produce simple groups

Take Q_8 Then

$$\{e\} \trianglelefteq \langle -1 \rangle \trianglelefteq \langle i \rangle \trianglelefteq Q_8$$
$$\{e\} \trianglelefteq \langle -1 \rangle \trianglelefteq \langle j \rangle \trianglelefteq Q_8$$
$$\{e\} \trianglelefteq \langle -1 \rangle \trianglelefteq \langle k \rangle \trianglelefteq Q_8$$

Where I figured out the composition series with the help of the lattice of subgroups to trace possible paths and figuring out whether they're normal or not. The quotient so any of these factors will be isomorphic to $\mathbb{Z}_2$, which is simple, and thus there are composition series.

another example is:

$$1 \trianglelefteq \langle s \rangle \trianglelefteq \langle s, r^2 \rangle \trianglelefteq D_8 \qquad \text{and} \qquad 1 \trianglelefteq \langle r^2 \rangle \trianglelefteq \langle r \rangle \trianglelefteq G$$

[11] note that maximal subgroups by definition are proper subgroups

Note that not all infinite groups admit a compositions series

Example 3.7.8: Example

I think $M_2(\mathbb{Q})$ or $GL_2(\mathbb{Q})$ is an example.

However, some groups do have compositions series. Take the group

$$D_\infty = \langle x, y | y^2 = 1,\ xy = y^{-1}x \rangle$$

this is called the *infinite dihedral group*. Notice that $\langle x \rangle \trianglelefteq D_8$: if we take some $g \in \langle x \rangle$, then since $\langle x \rangle$ is cyclic, then $g = x^a$ for some a. Conjgating x^a, we get $yx^ay^{-1} = yyx^a = x^a$, meaning $yxy^{-1} \in \langle x \rangle$. Using this, we can show that $\langle x \rangle \trianglelefteq D_\infty$. Thus, $D_1 = D_\infty / \langle x \rangle$ is well-defined. We need to show that D_1 is Let's take some $g \notin \langle x \rangle$ so that $[g] \neq [e]$. (buidlling up to show that)

$$C_2 \cong D_\infty / \langle x \rangle$$

Which is simple. Further more, C_2 is itself a simple abelian group. Thus, our composition seris is

$$D_\infty \trianglerighteq \langle x \rangle \trianglerighteq \{1\}$$

Some of infinite groups are close to being composition series, in the sense that $G = \cup_{i=1}^{\infty} G_i$ where G_i / G_{i+1}. Such groups will be addressed further [book ref:HERE]

(A word on how composition series are just simple extensions:

We saw that a finite group G always has a composition series, and hence a subnormal series:

$$G = G_0 \trianglerighteq G_1 \trianglerighteq G_2 \trianglerighteq \cdots \trianglerighteq G_n = \{1\}$$

Then notice that every subnormal series can be broken down into short exact sequences:

$$1 \longrightarrow G_{n-1} \longrightarrow G_{n-2} \longrightarrow G_{n-2}/G_{n-1} \longrightarrow 1$$

$$1 \longrightarrow G_i \longrightarrow G_{i-1} \longrightarrow G_{i-1}/G_i \longrightarrow 1 \quad 1 \longrightarrow G_1 \longrightarrow G_0 \longrightarrow G_0/G_1 \longrightarrow 1$$

thus, every subnormal series can be thought of as a series of series of extensions G_0/G_1, G_1/G_2, ... G_{n-1}.)

One might ask about if a composition series is supposed to be unique factorization, what about there being 3 composition series for Q_8, and what about the re-arranging we've done of $\mathbb{Z}_p \times Z_q$? The *quotients* of each N_{i+1}/N_i will be unique *up to rearrangement*! That is the content of the next theorem:

Theorem 3.7.9: Jordan-Hölder's Theorem

Let G be a finite group with $G \neq 1$. Then

1. G has a composition series
2. The composition factors in a composition series are unique, namely, if $1 = N_0 \leqslant N_1 \leqslant \cdots \leqslant N_r = G$, and $1 = M_0 \leqslant M_1 \leqslant \cdots \leqslant M_s = G$, are two composition series of G, then $r = s$ and there is some permutation π of $\{1, 2, ..., r\}$ such that

$$M_{\pi(i)}/M_{\pi(i)-1} \cong N_{\pi(i)}/N_{\pi(i)-1} \qquad 1 \leq i \leq r$$

***Proof*:**
apparently doable. This material wasn't covered in class and there's a test coming up so I'll get back to it another time

With that definition of factorization, we can now ask one of the biggest questions in group theory:

Theorem 3.7.10: The Hölder Program

1. Classify *all* finite simple groups
2. Find all ways of "putting simple groups together" to from other groups

***Proof*:**
The proof for the 1st point took around 100 years of work, with the effort of around 100 mathematicians, writing between 300-500 papers, totalling 5000-10000 pages. It is left as some light reading for the reader ☺

The second point is yet to be complete.

The result of all that work means the following single incompassing theorem can confidentily be write down:

Theorem 3.7.11: Types of Finite Groups

There is a list consisting of 18 (infinite) families of simple groups and 26 (or 27 [a]) simple groups not belonging to these families (the *sporadic* simple groups) such that every finite simple group is isomorphic to one of the groups in this list

[a] some include Tits group

Example 3.7.12
Here are a couple such family of groups:

1. $\{Z_p \big| p \text{ is prime }\}$
2. $\{SL_n(\mathbb{F})/Z(SL_n(\mathbb{F})) \big| n \in \mathbb{Z}^+, n \geq 2, \text{ and } \mathbb{F} \text{ is a finite field}\}$
3. A_n, $n \geq 5$.

> Examples after this start to get more complicated. I found this wikiepdia article outlining more of the history of this process.
>
> here

(somehow mention "generalization of abelian group section", where I can expan on how nilpotent and solvable groups give us some nice properties to make some groups easier to reprsent)

:exercise Let $H \trianglelefteq G$ such that there does not exist a $K \trianglerighteq H$ such that $H \trianglelefteq K \trianglelefteq G$. In other words if $K \trianglerighteq H$, then either $K = H$ or $K = G$. Such a normal subgroup is called the maximal normal subgroup. Show that H is a maximal normal subgroup if and only if G/H is a simple group

3.8 Quotient Objects

Link to Epimorphisms, link the idea of the "dual" is sub-object. Some intuition then perhaps on how the sub and quotient object together capture the idea of the entire group G. Leads to the idea of extensions.

In the previous chapter, we introduced the equivalent definition of subgroup in proposition 1.4.8. In it, we had an injective map $i : H \to G$. We will now ask what mathematicians call the natural *dual* question: What happens if we instead consider a surjective function $j : G \to H$ which is also a homomorphism?

(Link to Ker Object?)

4

Group Actions

group action is functor $* \to X$

In Section 1.5, we saw that groups arise naturally as *symmetries*: the dihedral group captures the symmetries of a polygon, the symmetric group captures every possible rearrangement of a finite set, and the general linear group captures every invertible linear transformation of a vector space. In each case, the group *acted* on something.

Historically, groups *were* symmetry groups before they were axiomatized: Galois studied permutations of roots. Cayley studied transformations of coordinates. Klein, in his 1872 *Erlangen Programme*, went so far as to propose that *geometry itself* should be defined by specifying a group of transformations and studying the properties they leave invariant (see the aside below). The abstract axioms we introduced in Section 1.1 are a *distillation* of what all these concrete symmetry groups have in common.

In this chapter, we reverse the process. Having built the abstract theory we now reconnect it to its roots by developing the theory of *group actions*: the study of how a group can act on a set (or on itself). We will show:

1. The Orbit-Stabilizer Theorem (§4.3). If a group G acts on a set X, then X decomposes into orbits, and the size of each orbit is controlled by a subgroup of G (the stabilizer). This is, in spirit, a generalization of Lagrange's Theorem (theorem 3.3.1). We shall see that if the action is *transitive*, then X can be represented as cosets of G, leading the way to motivate why it is in fact natural (perhaps better to say more universal) to study actions of a group on itself

2. Cayley's Theorem (§4.5). Every group is isomorphic to a subgroup of a symmetric group. In other words, every abstract group *is* a group of permutations—the symmetry perspective is not just a heuristic, it is a theorem.

3. The Class Equation (§4.6). By letting a group act on itself by conjugation, we obtain an

arithmetic identity that splits $|G|$ into contributions from the center $Z(G)$ (definition 2.2.4) and the non-central conjugacy classes. This will immediately imply that every p-group has a nontrivial center.

4. The Sylow Theorems (§4.8). For any prime p dividing $|G|$, Sylow's three theorems guarantee the *existence* of subgroups of maximal p-power order, prove they are all *conjugate* (hence isomorphic), and tightly constrain their *number*. These theorems are among the most powerful tools in finite group theory.

Along the way, we will see several important *types* of actions—faithful, transitive, free, and regular—and see how each captures a different facet of the relationship between a group and the object it acts on.

A recurring theme will be that the most important applications of group actions come not from letting a group act on some external set, but from letting a group act *on itself* or on its own substructure (elements, subsets, cosets). When a group acts on itself by left multiplication, we get Cayley's Theorem. When it acts on its cosets, we get structural information about normal subgroups. When it acts on itself by conjugation, we get the Class Equation. When it acts on the set of its own Sylow subgroups, we get the Sylow counting theorem. Each of these is an instance of the same machine; what changes is the *choice of action*, and each choice reveals different information.

If there is one philosophical takeaway from this chapter, it is this:

To understand a group, watch how it acts.

4.0.1 Applied Aside: Klein's Erlangen Programme In 1872, Felix Klein presented a new vision of geometry at the University of Erlangen. Rather than defining geometries axiomatically (as Euclid had done), Klein proposed that a *geometry* is a set X together with a group G acting on it, and that the "geometric properties" of X are precisely those invariant under the action of G.

Under this lens, different choices of group yield different geometries:

Group G	**Set** X	**Geometry**
Euclidean isometries	$\mathbb{R}^2$	Euclidean geometry
Affine transformations	$\mathbb{R}^2$	Affine geometry
Projective transformations	$\mathbb{RP}^2$	Projective geometry

Euclidean geometry studies properties like distance and angle (invariant under rigid motions); affine geometry studies parallelism (invariant under the larger group of affine maps); projective geometry studies incidence (invariant under the still larger projective group). More symmetry means fewer invariants, hence a "coarser" geometry.

Klein's Programme was one of the first explicit formulations of the idea that *group actions define structure*. It remains a guiding principle: in differential geometry, in topology, and throughout modern mathematics, the question "what is the symmetry group, and how does it act?" is often the first one asked.

4.1 Definitions and First Examples

In Section 1.5, we observed informally that a group can "act" on a set by rearranging its elements. Our goal now is to make this precise. We will give two equivalent formulations of the definition—one emphasizing the homomorphism, the other emphasizing axioms on the action map—and prove that they are interchangeable.

The Homomorphism Definition

Let us return to the motivating example from the prelude. The dihedral group D_{2n} (definition 1.2.25) acts on the set of vertices $\{1, 2, \dots, n\}$ of a regular n-gon. Each element $g \in D_{2n}$ determines a bijection of the vertex set: a rotation cyclically permutes the labels, a reflection swaps them. Different elements yield different (or sometimes the same) bijections, and—crucially—performing one symmetry followed by another corresponds to composing the associated bijections.

In other words, there is a map ρ that assigns to each group element $g \in D_{2n}$ a permutation $\rho(g) \in \mathrm{Sym}(\{1, \dots, n\})$, and this map *respects the group structure*: $\rho(gh) = \rho(g) \circ \rho(h)$. That is, ρ is a homomorphism (definition 1.4.2). This observation is the definition.

Definition 4.1.1: Group Action

Let G be a group and let X be a set. A *group action* of G on X is a group homomorphism

$$\rho\colon G \to \mathrm{Sym}(X),$$

where $\mathrm{Sym}(X)$ is the symmetric group on X (definition 1.2.12), i.e. the group of all bijections $X \to X$ under composition.
We write $G \curvearrowright X$ (read "*G acts on X*") to indicate that such a homomorphism has been specified. The set X, together with the action ρ, is called a ***G-set***.

Let us unpack what this definition gives us concretely.

- For each $g \in G$, the image $\rho(g)$ is an element of $\mathrm{Sym}(X)$—that is, a *bijection* from X to X. We can *evaluate* this bijection at any point $x \in X$ to obtain an element $\rho(g)(x) \in X$. One should read "$\rho(g)(x)$" as: *"the result of applying the permutation $\rho(g)$ to the element x."*
- Since ρ is a homomorphism, it respects composition: $\rho(gh) = \rho(g) \circ \rho(h)$ for all $g, h \in G$.
- Since ρ is a homomorphism, it sends the identity to the identity (proposition 1.4.4): $\rho(e) = \mathrm{id}_X$, the identity permutation.

It is convenient to introduce a shorthand that suppresses the homomorphism ρ:

$$g \cdot x \;:=\; \rho(g)(x) \qquad \text{for all } g \in G,\ x \in X.$$

In this notation, the two properties above translate into:

1. Compatibility: $(gh) \cdot x \;=\; g \cdot (h \cdot x) \qquad$ for all $g, h \in G$ and $x \in X$.
2. Identity: $e \cdot x \;=\; x \qquad$ for all $x \in X$.

These are not new axioms—they are automatic consequences of ρ being a homomorphism. The natural question is: are they *sufficient*? That is, if we have a rule "$g \cdot x$" satisfying just these two properties, does it necessarily come from a homomorphism to $\mathrm{Sym}(X)$? The answer is yes.

The Axiomatic Characterization

Many textbooks take the following as the *definition* of a group action, and then derive the homomorphism as a consequence. Since we have gone the other direction, we state it as a theorem establishing the equivalence.

Theorem 4.1.2: Equivalence of Definitions

Let G be a group and X a nonempty set. There is a bijective correspondence between:

(a) group homomorphisms $\rho\colon G \to \mathrm{Sym}(X)$, and

(b) functions $\cdot\colon G \times X \to X$, written $(g, x) \mapsto g \cdot x$, satisfying

(A1) $(gh) \cdot x = g \cdot (h \cdot x)$ for all $g, h \in G$ and $x \in X$, and

(A2) $e \cdot x = x$ for all $x \in X$.

The correspondence is given by $g \cdot x = \rho(g)(x)$.

***Proof*:**
(a)⇒(b). Given a homomorphism $\rho\colon G \to \mathrm{Sym}(X)$, define $g \cdot x := \rho(g)(x)$. We check the two axioms.

For (A1): since ρ is a homomorphism, $\rho(gh) = \rho(g) \circ \rho(h)$, so

$$(gh) \cdot x \;=\; \rho(gh)(x) \;=\; \big(\rho(g) \circ \rho(h)\big)(x) \;=\; \rho(g)\big(\rho(h)(x)\big) \;=\; g \cdot (h \cdot x).$$

For (A2): since ρ sends identities to identities (proposition 1.4.4), $\rho(e) = \mathrm{id}_X$, so

$$e \cdot x \;=\; \rho(e)(x) \;=\; \mathrm{id}_X(x) \;=\; x.$$

(b)⇒(a). Given a function $\cdot\colon G \times X \to X$ satisfying (A1) and (A2), we must construct a homomorphism $\rho\colon G \to \mathrm{Sym}(X)$. For each $g \in G$, define the function

$$\sigma_g\colon X \to X, \qquad \sigma_g(x) = g \cdot x.$$

We must show two things: that each σ_g is a bijection (hence an element of $\mathrm{Sym}(X)$), and that the map $g \mapsto \sigma_g$ is a group homomorphism.

Step 1: Each σ_g is a bijection. We claim that $\sigma_{g^{-1}}$ is a two-sided inverse for σ_g. For any $x \in X$:

$$(\sigma_g \circ \sigma_{g^{-1}})(x) = \sigma_g(g^{-1} \cdot x) = g \cdot (g^{-1} \cdot x) \overset{\text{(A1)}}{=} (gg^{-1}) \cdot x = e \cdot x \overset{\text{(A2)}}{=} x,$$

so $\sigma_g \circ \sigma_{g^{-1}} = \mathrm{id}_X$. An identical computation gives $\sigma_{g^{-1}} \circ \sigma_g = \mathrm{id}_X$. Thus σ_g is a bijection, and $\sigma_g \in \mathrm{Sym}(X)$.

Step 2: $g \mapsto \sigma_g$ is a homomorphism. Define $\rho\colon G \to \mathrm{Sym}(X)$ by $\rho(g) = \sigma_g$. For any $g, h \in G$ and $x \in X$:

$$\rho(gh)(x) = \sigma_{gh}(x) = (gh) \cdot x \overset{(A1)}{=} g \cdot (h \cdot x) = \sigma_g(\sigma_h(x)) = (\sigma_g \circ \sigma_h)(x) = \big(\rho(g) \circ \rho(h)\big)(x).$$

Since this holds for every $x \in X$, we have $\rho(gh) = \rho(g) \circ \rho(h)$, so ρ is a homomorphism.

Finally, these two constructions are inverse to each other: starting from ρ, forming $g \cdot x = \rho(g)(x)$, and then recovering the homomorphism $g \mapsto \sigma_g$ gives back $\sigma_g = \rho(g)$; conversely, starting from $\cdot$, forming $\rho(g) = \sigma_g$, and then recovering the action gives $\rho(g)(x) = \sigma_g(x) = g \cdot x$. Thus the correspondence is bijective.

4.1.3 Remark: Currying The equivalence above is an instance of a general phenomenon in mathematics (and computer science) called *currying*. A function of two variables $\cdot\colon G \times X \to X$ can be equivalently viewed as a function of one variable $\rho\colon G \to X^X$ that returns a *function* $X \to X$. The group action axioms are exactly the conditions ensuring that ρ lands in $\mathrm{Sym}(X) \subseteq X^X$ and is a homomorphism. This point of view—recasting a "two-variable" structure as a "one-variable" homomorphism—will reappear when we study modules and representations.

Thanks to theorem 4.1.2, we may define group actions using whichever formulation is most convenient. In practice:

- To define an action on a concrete set, it is usually easiest to write down a rule $g \cdot x =$ (some formula) and verify axioms (A1) and (A2).
- To reason abstractly about actions (e.g. to apply the isomorphism theorems), it is usually more powerful to work with the homomorphism $\rho\colon G \to \mathrm{Sym}(X)$.

We give a name to this homomorphism when it arises from an action:

Definition 4.1.4: Permutation Representation

If $G \curvearrowright X$ is a group action, the associated homomorphism $\rho\colon G \to \mathrm{Sym}(X)$ is called the *permutation representation* of the action. We say the action *affords* or *induces* the permutation representation ρ.

Since ρ is a homomorphism, all the tools we developed for homomorphisms apply: its kernel $\ker(\rho)$ is a normal subgroup of G (proposition 2.2.12), its image $\rho(G)$ is a subgroup of $\mathrm{Sym}(X)$ (proposition 1.4.8), and the First Isomorphism Theorem (theorem 3.1.14) gives $G/\ker(\rho) \cong \rho(G) \leq \mathrm{Sym}(X)$. We will exploit these facts heavily starting in Section 4.2.

We close this subsection by recording why we chose the homomorphism formulation as our leading definition. This is a matter of taste, but the reasoning is forward-looking:

- A group action is a homomorphism $G \to \mathrm{Sym}(X)$.
- A linear representation (Chapter X, if applicable) will be a homomorphism $G \to GL(V)$.
- A module (Part III) will be defined via a ring homomorphism $R \to \mathrm{End}(M)$.

In each case, the pattern is the same: the algebraic object "acts" by mapping homomorphically into the automorphisms of the thing it acts on. Remembering this single pattern—*action = homomorphism into an automorphism group*—will serve you well across all of algebra.

A Menagerie of Examples

We now build fluency with the definition by working through a collection of examples. In each case, we specify the group G, the set X, and the action rule $g \cdot x$, and we either verify the axioms or indicate that the verification is a straightforward exercise.

Example 4.1.5: Fundamental Group Actions

1. *The tautological action of S_n.* Let $G = S_n$ and $X = \{1, 2, \ldots, n\}$. Define

$$\sigma \cdot i \;=\; \sigma(i).$$

The permutation representation is the identity map $\rho = \mathrm{id}\colon S_n \to \mathrm{Sym}(\{1, \ldots, n\}) = S_n$, which is trivially a homomorphism. This is the most elementary group action: each permutation simply *does what it says*.

2. *Dihedral symmetries of a regular n-gon.* Let $G = D_{2n}$ (definition 1.2.25) and let $X = \{1, 2, \ldots, n\}$ represent the vertices of a regular n-gon, labeled clockwise. The rotation r acts by $r \cdot i = i + 1 \pmod{n}$, and the reflection s acts by $s \cdot i = n + 1 - i \pmod{n}$ (with an appropriate labeling convention). The permutation representation $\rho\colon D_{2n} \to S_n$ is an injective homomorphism, embedding D_{2n} as a subgroup of S_n.

3. *Matrix groups acting on vectors.* Let $G = GL_n(\mathbb{R})$ (definition 1.2.30) and $X = \mathbb{R}^n$. Define

$$A \cdot \vec{x} \;=\; A\vec{x} \qquad \text{(matrix–vector multiplication).}$$

Axiom (A1) is the associativity of matrix multiplication: $(AB)\vec{x} = A(B\vec{x})$. Axiom (A2) is the identity matrix property: $I_n\vec{x} = \vec{x}$. Thus this defines a group action.

This action is the foundation of linear algebra: every invertible linear transformation $\mathbb{R}^n \to \mathbb{R}^n$ is encoded by an element of $GL_n(\mathbb{R})$, and composition of transformations corresponds to matrix multiplication.

4. *Two actions of $\mathbb{R}^\times$ on $\mathbb{R}^2$.* Let $G = (\mathbb{R}^\times, \cdot)$, the multiplicative group of nonzero reals, and $X = \mathbb{R}^2$. Here are two *different* actions of the same group on the same set:

$$\textbf{Scaling:}\quad r \cdot \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} rx \\ ry \end{pmatrix}, \qquad\qquad \textbf{Hyperbolic:}\quad r \cdot \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} rx \\ r^{-1}y \end{pmatrix}.$$

Both satisfy (A1) and (A2) (exercise—note that commutativity of $\mathbb{R}^\times$ is used in the hyperbolic case). The scaling action dilates every vector uniformly, while the hyperbolic action stretches the x-axis and compresses the y-axis (or vice versa), preserving the hyperbolas $\{xy = c\}$.

This example makes an important point: *a group action is not determined by the group and the set alone.* One must specify the rule.

5. *Left multiplication.* Let G be any group, and let $X = G$ (the underlying set of the group itself). Define
$$g \cdot a \;=\; ga \qquad \text{(group multiplication)}.$$
Axiom (A1) is exactly the associativity of the group operation: $(gh) \cdot a = (gh)a = g(ha) = g \cdot (h \cdot a)$. Axiom (A2) is the identity property: $e \cdot a = ea = a$.

   Note that for a given g, the map $a \mapsto ga$ is *not* a group homomorphism (unless $g = e$)—it is only a bijection $G \to G$. This action is our first example of a group acting on *itself*, and it will lead to Cayley's Theorem in Section 4.5.

6. *Conjugation.* Let G be any group, and again let $X = G$. Define
$$g \cdot a \;=\; gag^{-1}.$$
Let us verify the axioms. For (A1):
$$(gh) \cdot a = (gh)\,a\,(gh)^{-1} = g\,h\,a\,h^{-1}g^{-1} = g \cdot (hah^{-1}) = g \cdot (h \cdot a).$$
For (A2): $e \cdot a = eae^{-1} = a$.

   Unlike left multiplication, conjugation "preserves" much of the group's internal structure: it fixes the identity, sends subgroups to (isomorphic) subgroups, and preserves the order of elements. This action will lead to the Class Equation in Section 4.6.

7. *Rotations of a tetrahedron.* Let $G = A_4$ (definition 1.2.19), the alternating group on four symbols, and let $X = \{1, 2, 3, 4\}$ represent the vertices of a regular tetrahedron. The 12 elements of A_4 correspond to the 12 rotational symmetries of the tetrahedron:
   - e: the identity (no rotation).
   - 8 rotations by 120° and 240° about axes through a vertex and the center of the opposite face, e.g. $(1\,2\,3)$ and $(1\,3\,2)$ fix vertex 4.
   - 3 rotations by 180° about axes connecting the midpoints of opposite edges, e.g. $(1\,2)(3\,4)$ swaps the two pairs.

   The action is $\sigma \cdot i = \sigma(i)$, i.e. the restriction of the tautological action of S_4 to A_4.

8. *Translation of polynomials.* Let $G = (\mathbb{R}, +)$ and let $X = \mathbb{R}[x]$, the set of all polynomials with real coefficients. Define
$$t \cdot f \;=\; f_t, \qquad \text{where } f_t(x) = f(x + t).$$
For (A1): $\big((s+t) \cdot f\big)(x) = f(x + s + t)$ and $\big(s \cdot (t \cdot f)\big)(x) = (t \cdot f)(x + s) = f(x + s + t)$. For (A2): $(0 \cdot f)(x) = f(x + 0) = f(x)$.

   This action connects algebra to analysis: Taylor's theorem describes how the coefficients of f_t depend on the coefficients of f and the shift t.

4.1.6 Applied Aside: Symmetry in Chemistry and Crystallography The symmetry group of a molecule—the collection of all rotations and reflections that map the molecule to itself—is called its *point group*. These point groups are finite subgroups of the orthogonal group $O(3)$ (definition 1.2.32). Remarkably, there are only 32 distinct crystallographic point groups compatible with the periodic structure of a crystal lattice.

Group actions enter directly: a point group acts on the set of atoms in a molecule, and the *orbits* of this action (a concept we formalize in Section 4.2) determine which atoms are "chemically equivalent." In spectroscopy, *selection rules*—which govern which transitions between energy levels are allowed—are derived from the representation theory of these point groups. In crystallography, the 32 point groups combine with translational symmetries to yield the *230 space groups*, which classify all possible crystal structures in three dimensions.

A Word on Left and Right Actions

What we have defined above is, more precisely, a *left* group action: the group element appears on the *left* in the expression "$g \cdot x$." There is an equally valid notion of a right action.

Definition 4.1.7: Right Group Action

A *right action* of a group G on a set X is a function $X \times G \to X$, written $(x, g) \mapsto x \cdot g$, satisfying:

(R1) $(x \cdot g) \cdot h = x \cdot (gh)$ for all $g, h \in G$ and $x \in X$, and

(R2) $x \cdot e = x$ for all $x \in X$.

Equivalently, a right action is a group homomorphism $\rho\colon G^{\mathrm{op}} \to \mathrm{Sym}(X)$, where G^{op} is the *opposite group*[a] of G.

[a] The opposite group G^{op} has the same underlying set as G but with the multiplication reversed: $a \cdot_{\mathrm{op}} b = ba$.

Observe the key difference in (R1): the parentheses associate to the *left* ($(x \cdot g) \cdot h$), and the group product is gh in its original order. Compare with the left-action axiom (A1), where the parentheses associate to the *right* ($g \cdot (h \cdot x)$).

Left and right actions are completely interchangeable in the following sense:

Proposition 4.1.8: Converting Between Left and Right Actions

If $\cdot_L\colon G \times X \to X$ is a left action, then

$$x \cdot_R g \;:=\; g^{-1} \cdot_L x$$

defines a right action. Conversely, if $\cdot_R$ is a right action, then $g \cdot_L x := x \cdot_R g^{-1}$ defines a left action.

***Proof*:**
We verify (R1). For any $g, h \in G$ and $x \in X$:

$$(x \cdot_R g) \cdot_R h = h^{-1} \cdot_L (x \cdot_R g) = h^{-1} \cdot_L (g^{-1} \cdot_L x)$$
$$= (h^{-1}g^{-1}) \cdot_L x = (gh)^{-1} \cdot_L x = x \cdot_R (gh).$$

For (R2): $x \cdot_R e = e^{-1} \cdot_L x = e \cdot_L x = x$. The converse is analogous.

Because of this correspondence, there is no essential loss of generality in working with one type exclusively.

4.1.9 Convention For the remainder of this book, "group action" means *left* group action unless explicitly stated otherwise.

4.2 Orbits, Stabilizers, and the Kernel

Given a group action $G \curvearrowright X$, two fundamental questions arise:

1. From the perspective of X: Given a point $x \in X$, where can x be "moved to" by the elements of G?
2. From the perspective of G: Given a point $x \in X$, which elements of G "leave x in place"?

The answers to these questions—*orbits* and *stabilizers*—are two of the most important concepts in the theory of group actions. As we will see, they are intimately related, and much of the power of group actions comes from playing them off against each other.

Orbits

Consider the special orthogonal group $SO(3)$ (definition 1.2.33) acting on the unit sphere $S^2 = \{(x, y, z) \in \mathbb{R}^3 : x^2 + y^2 + z^2 = 1\}$ by matrix multiplication. If we start at the north pole $N = (0, 0, 1)$ and apply every rotation in $SO(3)$, we reach *every* point on the sphere: $SO(3) \cdot N = S^2$. Now restrict to the subgroup $SO(2) \leq SO(3)$ of rotations about the z-axis. From the north pole, $SO(2)$ goes nowhere: $SO(2) \cdot N = \{N\}$. But from a point P at latitude θ, $SO(2)$ traces out the entire latitude circle through P. The latitude circles—together with the two poles—form a family of non-overlapping subsets that cover the sphere.

This is the geometric prototype of a general phenomenon: under any group action, the set X decomposes into non-overlapping "layers" of points that can be reached from one another.

Definition 4.2.1: Orbit

Let $G \curvearrowright X$. For any $x \in X$, the ***orbit*** of x under the action of G is the set

$$G \cdot x \;=\; \{g \cdot x \big| g \in G\}.$$

In words, $G \cdot x$ is the set of all points in X that x can be "sent to" by some element of G.

The notation $\mathrm{Orb}_G(x)$ or $\mathcal{O}_x$ is also common in the literature; we will use $G \cdot x$ throughout.

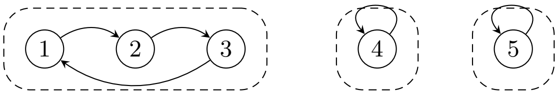
Example 4.2.2: Orbits Under a Cyclic Group

Let $G = \langle(1\ 2\ 3)\rangle = \{e, (1\ 2\ 3), (1\ 3\ 2)\} \cong \mathbb{Z}/3\mathbb{Z}$, viewed as a subgroup of S_5, and let $X = \{1, 2, 3, 4, 5\}$ with the tautological action $\sigma \cdot i = \sigma(i)$. Then:

$$\begin{aligned} G \cdot 1 &= \{e(1), (1\,2\,3)(1), (1\,3\,2)(1)\} = \{1, 2, 3\}, \\ G \cdot 4 &= \{e(4), (1\,2\,3)(4), (1\,3\,2)(4)\} = \{4\}, \\ G \cdot 5 &= \{5\}. \end{aligned}$$

Note that $G \cdot 1 = G \cdot 2 = G \cdot 3$ (the three points can all reach each other), and the set X decomposes as the disjoint union

$$X \;=\; \{1, 2, 3\} \;\sqcup\; \{4\} \;\sqcup\; \{5\}.$$



$G \cdot 1$ $\qquad$ $G \cdot 4$ $\qquad$ $G \cdot 5$

The decomposition into orbits holds for any group action.

Proposition 4.2.3: Orbits Partition the Set

Let $G \curvearrowright X$. Define a relation $\sim$ on X by

$$x \sim y \quad \iff \quad y = g \cdot x \text{ for some } g \in G.$$

Then $\sim$ is an equivalence relation on X, and the equivalence class of x is precisely the orbit $G \cdot x$. Consequently, the orbits of the action partition X:

$$X \;=\; \bigsqcup_{i \in I} G \cdot x_i,$$

where $\{x_i\}_{i \in I}$ is any set of orbit representatives (one element chosen from each orbit).

***Proof*:**

We verify the three properties of an equivalence relation.

- *Reflexive:* $x = e \cdot x$, so $x \sim x$. (Uses axiom (A2).)
- *Symmetric:* If $x \sim y$, then $y = g \cdot x$ for some $g \in G$. Applying g^{-1}:

  $$g^{-1} \cdot y = g^{-1} \cdot (g \cdot x) \overset{\text{(A1)}}{=} (g^{-1}g) \cdot x \overset{\text{(A2)}}{=} e \cdot x = x,$$

  so $x = g^{-1} \cdot y$, which gives $y \sim x$. (Uses the existence of inverses in G.)
- *Transitive:* If $x \sim y$ and $y \sim z$, then $y = g \cdot x$ and $z = h \cdot y$ for some $g, h \in G$. Then

  $$z = h \cdot y = h \cdot (g \cdot x) \overset{\text{(A1)}}{=} (hg) \cdot x,$$

so $x \sim z$. (Uses closure of the group operation.)

The equivalence class of x is $[x] = \{y \in X : y \sim x\} = \{g \cdot x : g \in G\} = G \cdot x$. Since equivalence classes partition a set, the orbits partition X.

The margin annotations in the proof are worth reflecting on: *reflexivity* uses the identity axiom, *symmetry* uses the existence of inverses, and *transitivity* uses closure. In other words, the group axioms are precisely what is needed to make the orbit relation an equivalence relation. A monoid action (where inverses may not exist) would give reflexivity and transitivity but *not* symmetry, and the orbits would generally fail to partition the set.

We introduce notation for the *set of orbits*:

Definition 4.2.4: Set of Orbits

If $G \curvearrowright X$, the set of all orbits is denoted

$$X/G \;=\; \{G \cdot x : x \in X\}.$$

This is called the ***orbit space*** (or *set of orbits*) of the action.

Counting orbits will become a central theme once we prove Burnside's Lemma (§4.3).

Stabilizers and Fixed Points

We now turn to the second question: which group elements "do nothing" to a given point?

Definition 4.2.5: Stabilizer

Let $G \curvearrowright X$ and let $x \in X$. The ***stabilizer*** (or *isotropy subgroup*) of x in G is

$$G_x \;=\; \{g \in G \big| g \cdot x = x\}.$$

The notation $\mathrm{Stab}_G(x)$ is also standard; we use the subscript notation G_x for brevity.

Proposition 4.2.6: The Stabilizer is a Subgroup

For any $x \in X$, the stabilizer G_x is a subgroup of G.

***Proof*:**
We apply the subgroup criterion (proposition 2.1.3).

- *Non-empty:* $e \cdot x = x$ by axiom (A2), so $e \in G_x$.
- *Closed under products and inverses:* Let $g, h \in G_x$. Then

$$(gh^{-1}) \cdot x \overset{\text{(A1)}}{=} g \cdot (h^{-1} \cdot x).$$

Since $h \in G_x$, we have $h \cdot x = x$, and so $h^{-1} \cdot x = h^{-1} \cdot (h \cdot x) = (h^{-1}h) \cdot x = e \cdot x = x$. Therefore $(gh^{-1}) \cdot x = g \cdot x = x$, giving $gh^{-1} \in G_x$.

The fact that G_x is a *subgroup* (not just a subset) is crucial: it means we can bring the full weight of our subgroup theory to bear. In particular, G_x has an index in G given by Lagrange's Theorem (theorem 3.3.1), and—as we will see in the next section—this index is precisely the size of the orbit $G \cdot x$.

Example 4.2.7: Computing Stabilizers

Consider $G = D_{2\cdot 4} = D_8$ acting on the vertices $X = \{1, 2, 3, 4\}$ of a square, with the labeling from Section 1.5. We compute the stabilizer of vertex 1. An element $g \in D_8$ stabilizes 1 if and only if g maps vertex 1 to itself. Among the 8 elements of D_8:

- The identity e fixes 1.
- The rotations r, r^2, r^3 all move vertex 1.
- Among the four reflections, exactly one fixes vertex 1: the reflection sr^3 across the diagonal through vertex 1 (depending on the labeling convention).

Thus $G_1 = \{e, sr^3\} \cong \mathbb{Z}/2\mathbb{Z}$, a subgroup of order 2. Note that $|G_1| = 2$ and $|G\cdot 1| = |\{1, 2, 3, 4\}| = 4$, so $|G_1| \cdot |G \cdot 1| = 2 \cdot 4 = 8 = |D_8|$. This is the orbit-stabilizer theorem, proved below.

There is a dual notion to the stabilizer that will be useful later:

Definition 4.2.8: Fixed-Point Sets

Let $G \curvearrowright X$.

1. For a single element $g \in G$, the *fixed-point set* of g is
$$X^g \;=\; \{x \in X \,|\, g \cdot x = x\}.$$
2. The *fixed-point set* of the entire group G is
$$X^G \;=\; \{x \in X \,|\, g \cdot x = x \text{ for all } g \in G\} \;=\; \bigcap_{g\in G} X^g.$$

Observe the duality: the stabilizer G_x is a subset of G (the group elements that fix a *given point*), while the fixed-point set X^g is a subset of X (the points fixed by a *given group element*). Both notions have the same underlying condition, "$g \cdot x = x$," but viewed from opposite sides:

	Lives in...	**Defined by fixing...**
Stabilizer G_x	G	the point $x \in X$
Fixed-point set X^g	X	the element $g \in G$

4.2.9 A Mnemonic for Orbits and Stabilizers Here is a useful way to keep the two concepts straight. Both G_x (stabilizer) and $G \cdot x$ (orbit) have an element of X in their notation:

- If G_x is *large* (many elements of G fix x), then $G \cdot x$ is *small* (the point x is not moved very far).
- If G_x is *small* (few elements of G fix x), then $G \cdot x$ is *large* (the point x is moved to many different places).

In the extreme: if $G_x = G$ (everything fixes x), then $G \cdot x = \{x\}$ (a singleton orbit). Conversely, if $G_x = \{e\}$ (only the identity fixes x), then the orbit is as large as possible. This inverse relationship will be made precise by the Orbit-Stabilizer Theorem in Section 4.3.

We close this subsection with two remarks connecting stabilizers to concepts from Chapter 2 that will be developed fully later in this chapter.

1. Left multiplication (item 5). When G acts on itself by $g \cdot a = ga$, the stabilizer of any $a \in G$ is
$$G_a = \{g \in G : ga = a\} = \{e\}.$$
Every stabilizer is trivial: no element other than the identity can "fix" a point under left multiplication.
2. Conjugation (item 6). When G acts on itself by $g \cdot a = gag^{-1}$, the stabilizer of a is
$$G_a = \{g \in G : gag^{-1} = a\} = \{g \in G : ga = ag\} = C_G(a),$$
the centralizer of a (definition 2.2.7). Thus the orbit-stabilizer framework gives us a direct link between orbits (conjugacy classes) and centralizers. We will exploit this connection in Section 4.6.

The Kernel of an Action

The stabilizer G_x captures which group elements fix a *single* point. We now ask: which group elements fix *every* point?

Definition 4.2.10: Kernel of a Group Action

Let $G \curvearrowright X$ with permutation representation $\rho\colon G \to \mathrm{Sym}(X)$. The ***kernel*** of the action is
$$\ker(G \curvearrowright X) \;=\; \{g \in G \,|\, g \cdot x = x \text{ for all } x \in X\} \;=\; \bigcap_{x \in X} G_x.$$

An element in the kernel "acts trivially"—it is a group element that induces the identity permutation on X. This means $\rho(g) = \mathrm{id}_X$, and so the kernel of the *action* is exactly the kernel of the *homomorphism* ρ:
$$\ker(G \curvearrowright X) \;=\; \ker(\rho).$$

Proposition 4.2.11: The Kernel is a Normal Subgroup

$$\ker(G \curvearrowright X) \trianglelefteq G$$

***Proof*:**
Since $\rho\colon G \to \mathrm{Sym}(X)$ is a group homomorphism, its kernel $\ker(\rho)$ is a normal subgroup of G by proposition 2.2.12.

The kernel measures how much "information about G" is lost by the action. If the kernel is large, many group elements are "wasted"—they all induce the same (trivial) permutation. If the kernel is trivial, then distinct group elements always induce distinct permutations. This motivates:

Definition 4.2.12: Faithful Action

A group action $G \curvearrowright X$ is called *faithful* (or *effective*) if its kernel is trivial:

$$\ker(G \curvearrowright X) = \{e\}.$$

Equivalently, the action is faithful if and only if the permutation representation $\rho\colon G \to \mathrm{Sym}(X)$ is *injective*.

In a faithful action, distinct group elements always produce distinct permutations of X: if $g \neq h$, then there exists some $x \in X$ with $g \cdot x \neq h \cdot x$. The group G is thus "faithfully represented" as a group of permutations of X.

Example 4.2.13: A Non-Faithful Action

Let $G = \mathbb{Z}/4\mathbb{Z} = \{\bar{0}, \bar{1}, \bar{2}, \bar{3}\}$ and $X = \{1, 2\}$. Define an action by $\bar{n} \cdot i = \tau^n(i)$, where $\tau = (1\ 2) \in S_2$. Then:

$$\bar{0} \cdot i = i, \quad \bar{1} \cdot i = \tau(i), \quad \bar{2} \cdot i = \tau^2(i) = i, \quad \bar{3} \cdot i = \tau^3(i) = \tau(i).$$

The elements $\bar{0}$ and $\bar{2}$ both act as the identity on X, so

$$\ker(\rho) \ = \ \{\bar{0}, \bar{2}\} \ \cong \ \mathbb{Z}/2\mathbb{Z}.$$

The action is *not* faithful. By the First Isomorphism Theorem (theorem 3.1.14),

$$G/\ker(\rho) \ \cong \ \rho(G) \ \leq \ \mathrm{Sym}(X) \ = \ S_2,$$

so $(\mathbb{Z}/4\mathbb{Z})/(\mathbb{Z}/2\mathbb{Z}) \cong \mathbb{Z}/2\mathbb{Z}$ embeds into S_2—consistent with the fact that the *effective* part of the action is just "swap or don't swap."

The example above illustrates a general principle: even if an action is not faithful, we can always *make* it faithful by passing to the quotient.

Proposition 4.2.14: Quotient by the Kernel Acts Faithfully

Let $G \curvearrowright X$ with kernel $K = \ker(\rho)$. Then the quotient group G/K acts faithfully on X via

$$(gK) \cdot x \;=\; g \cdot x.$$

***Proof*:**
We first check well-definedness: if $gK = hK$, then $h^{-1}g \in K$, so $(h^{-1}g) \cdot x = x$ for all x, whence $g \cdot x = h \cdot x$ for all x. Thus the action of gK does not depend on the choice of representative.

The axioms (A1) and (A2) are inherited directly from the action of G. Finally, if gK is in the kernel of this new action, then $g \cdot x = x$ for all $x \in X$, so $g \in K$, meaning $gK = eK$. The kernel of the G/K-action is trivial, so the action is faithful.

In the language of the permutation representation: ρ factors through the quotient as

$$G \twoheadrightarrow G/K \xhookrightarrow{\bar{\rho}} \mathrm{Sym}(X),$$

where $\bar{\rho}$ is injective. This is simply the First Isomorphism Theorem (theorem 3.1.14) applied to ρ.

We summarize the key objects introduced in this section for quick reference.

Object	**Notation**	**Lives in**	**Definition**
Orbit of x	$G \cdot x$	subsets of X	$\{g \cdot x : g \in G\}$
Stabilizer of x	G_x	subgroups of G	$\{g \in G : g \cdot x = x\}$
Fixed-point set of g	X^g	subsets of X	$\{x \in X : g \cdot x = x\}$
Kernel of the action	$\ker(\rho)$	normal subgroups of G	$\bigcap_{x \in X} G_x$

4.3 The Orbit-Stabilizer Theorem

In the previous section, we observed an inverse relationship between orbits and stabilizers: the larger the stabilizer, the smaller the orbit, and vice versa. We also noticed, in the D_8 example (example 4.2.7), that $|G_1| \cdot |G \cdot 1| = |D_8|$. The Orbit-Stabilizer Theorem makes this relationship precise, and reveals it to be a direct consequence of Lagrange's Theorem.

Statement and Proof

Theorem 4.3.1: The Orbit-Stabilizer Theorem

Let G be a group acting on a set X, and let $x \in X$. There is a bijection

$$G/G_x \longleftrightarrow G \cdot x, \qquad gG_x \longmapsto g \cdot x,$$

between the set of left cosets of the stabilizer G_x in G and the orbit of x. In particular,

$$|G \cdot x| \ = \ [G : G_x].$$

If G is finite, this gives $|G \cdot x| = |G| \,/\, |G_x|$, or equivalently,

$$|G| \ = \ |G \cdot x| \cdot |G_x|.$$

***Proof*:**
Define the map

$$\varphi\colon G/G_x \to G \cdot x, \qquad \varphi(gG_x) = g \cdot x.$$

We verify that φ is a well-defined bijection.

Well-defined. Suppose $gG_x = hG_x$. Then $h^{-1}g \in G_x$, so $(h^{-1}g) \cdot x = x$. By axiom (A1):

$$g \cdot x = h \cdot \big((h^{-1}g) \cdot x\big) = h \cdot x.$$

Thus $\varphi(gG_x) = \varphi(hG_x)$, and the map does not depend on the choice of coset representative.

Injective. Suppose $\varphi(gG_x) = \varphi(hG_x)$, i.e. $g \cdot x = h \cdot x$. Then

$$(h^{-1}g) \cdot x = h^{-1} \cdot (g \cdot x) = h^{-1} \cdot (h \cdot x) = (h^{-1}h) \cdot x = e \cdot x = x,$$

so $h^{-1}g \in G_x$, hence $gG_x = hG_x$.

Surjective. Every element of $G \cdot x$ has the form $g \cdot x$ for some $g \in G$, and $g \cdot x = \varphi(gG_x)$.

Since φ is a bijection, $|G \cdot x| = |G/G_x| = [G : G_x]$. When G is finite, Lagrange's Theorem (theorem 3.3.1) gives $[G : G_x] = |G|/|G_x|$.

4.3.2 Remark The formula $|G| = |G \cdot x| \cdot |G_x|$ has exactly the same shape as Lagrange's Theorem: $|G| = [G : H] \cdot |H|$. It *is* Lagrange's Theorem, applied to the subgroup $H = G_x$. The content of the Orbit-Stabilizer Theorem is the bijection φ that identifies the abstract index $[G : G_x]$ with the concrete, countable object $G \cdot x$.

When X is finite, we can combine the Orbit-Stabilizer Theorem with the fact that orbits partition X (proposition 4.2.3) to obtain a useful counting formula.

Corollary 4.3.3: The Orbit-Counting Formula

Let G be a finite group acting on a finite set X, and let $x_1, \ldots, x_k$ be representatives from the distinct orbits. Then

$$|X| \;=\; \sum_{i=1}^{k} |G \cdot x_i| \;=\; \sum_{i=1}^{k} [G : G_{x_i}].$$

***Proof*:**

Since the orbits partition X (proposition 4.2.3), we have $|X| = \sum_{i=1}^{k} |G \cdot x_i|$. Replacing each $|G \cdot x_i|$ by $[G : G_{x_i}]$ via theorem 4.3.1 yields the result.

This formula will be the key ingredient in the Class Equation (Section 4.6), where we apply it to the conjugation action and split the sum according to whether orbit representatives lie in the center.

Examples

Example 4.3.4: Orbits of Polynomials Under S_4

Let $R = \mathbb{Z}[x_1, x_2, x_3, x_4]$, the ring of polynomials in four variables with integer coefficients, and let S_4 act on R by permuting the indices:

$$\sigma \cdot p(x_1, x_2, x_3, x_4) \;=\; p(x_{\sigma(1)}, x_{\sigma(2)}, x_{\sigma(3)}, x_{\sigma(4)}).$$

We compute the orbit of $f = x_1 + x_2$ under this action.

Rather than applying all 24 elements of S_4 to f, we first identify the stabilizer. A permutation σ fixes $x_1 + x_2$ if and only if the set $\{\sigma(1), \sigma(2)\} = \{1, 2\}$ (since addition is commutative, the order within the pair does not matter). This means σ must permute $\{1, 2\}$ among themselves and $\{3, 4\}$ among themselves. The stabilizer is therefore

$$(S_4)_f \;=\; \langle (1\ 2),\, (3\ 4) \rangle \;\cong\; \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z},$$

a copy of the Klein four-group, with $|(S_4)_f| = 4$. By the Orbit-Stabilizer Theorem,

$$|S_4 \cdot f| = \frac{|S_4|}{|(S_4)_f|} = \frac{24}{4} = 6.$$

We can now enumerate the orbit with confidence that we are looking for exactly 6 polynomials. Choosing one representative from each coset of $(S_4)_f$ yields:

$$S_4 \cdot (x_1 + x_2) = \{x_1 + x_2,\ x_1 + x_3,\ x_1 + x_4,\ x_2 + x_3,\ x_2 + x_4,\ x_3 + x_4\}.$$

The Orbit-Stabilizer Theorem saved us from checking all 24 permutations: knowing the stabilizer immediately told us the orbit size, and we only needed to find 6 distinct images.

Example 4.3.5: S_n Acting on $\{1, \ldots, n\}$

Consider S_{10} acting on $\{1, \ldots, 10\}$ by $\sigma \cdot i = \sigma(i)$. For any $i, j \in \{1, \ldots, 10\}$, there exists a permutation sending i to j (for instance, the transposition $(i\ j)$), so the action has a single orbit: $S_{10} \cdot i = \{1, \ldots, 10\}$ for every i.

The stabilizer of, say, 1 consists of all permutations that fix 1:

$$(S_{10})_1 = \{\sigma \in S_{10} : \sigma(1) = 1\} \cong S_9.$$

The Orbit-Stabilizer Theorem confirms: $|S_{10} \cdot 1| = |S_{10}|/|(S_{10})_1| = 10!/9! = 10$.

Example 4.3.6: D_8 Acting on Vertices, Revisited

The group D_8 acts on $X = \{1, 2, 3, 4\}$ (the vertices of a square). Since any vertex can be sent to any other by a suitable rotation, the action has a single orbit of size 4. From example 4.2.7, the stabilizer of vertex 1 has order 2. The Orbit-Stabilizer Theorem gives $|G \cdot 1| = 8/2 = 4$, as expected.

Now consider D_8 acting on the set of two diagonals, $\mathcal{D} = \{d_1, d_2\}$, where $d_1 = \{1, 3\}$ and $d_2 = \{2, 4\}$. A rotation by 90° swaps the two diagonals, while a rotation by 180° preserves both. Here $|G \cdot d_1| = 2$ and $|G_{d_1}| = 8/2 = 4$. The stabilizer G_{d_1} consists of the symmetries that preserve the diagonal d_1 as a set: $G_{d_1} = \{e,\ r^2,\ sr,\ sr^3\}$, a copy of the Klein four-group.

Burnside's Lemma

The Orbit-Stabilizer Theorem gives the size of each individual orbit. But in many applications—particularly in combinatorics—what we really want to know is the *number* of orbits. The following classical result answers this question.

Theorem 4.3.7: Burnside's Lemma

Let G be a finite group acting on a finite set X. Then the number of orbits is

$$|X/G| \;=\; \frac{1}{|G|} \sum_{g \in G} |X^g|,$$

where $X^g = \{x \in X : g \cdot x = x\}$ is the fixed-point set of g (definition 4.2.8).

In words: the number of distinct orbits equals the average number of fixed points, averaged over all group elements.

4.3.8 Historical Note Despite its name, this result was known to Cauchy (1845) and proved in full generality by Frobenius (1887). Burnside included it in his influential 1897 textbook, citing Frobenius, and subsequent authors attributed it to Burnside. For this reason, some authors call it the *Cauchy–Frobenius Lemma*. We follow the (ahistorical but widespread) convention of calling it Burnside's Lemma.

***Proof*:**
The proof is a double-counting argument. Consider the set of pairs

$$\mathcal{S} \;=\; \{(g,x) \in G \times X \,\big|\, g \cdot x = x\}.$$

We count $|\mathcal{S}|$ in two ways.

Counting by group elements (summing over $g \in G$). For each g, the number of $x \in X$ with $g \cdot x = x$ is $|X^g|$. Therefore

$$|\mathcal{S}| = \sum_{g \in G} |X^g|. \tag{4.1}$$

Counting by set elements (summing over $x \in X$). For each x, the number of $g \in G$ with $g \cdot x = x$ is $|G_x|$. Therefore

$$|\mathcal{S}| = \sum_{x \in X} |G_x|. \tag{4.2}$$

We now simplify (4.2) by grouping the sum according to orbits. Let $\mathcal{O}_1, \ldots, \mathcal{O}_k$ be the distinct orbits. For any x in an orbit $\mathcal{O}_i$, the Orbit-Stabilizer Theorem (theorem 4.3.1) gives $|G_x| = |G|/|\mathcal{O}_i|$. Summing over the $|\mathcal{O}_i|$ elements of the orbit:

$$\sum_{x \in \mathcal{O}_i} |G_x| \;=\; |\mathcal{O}_i| \cdot \frac{|G|}{|\mathcal{O}_i|} \;=\; |G|.$$

Each orbit contributes exactly $|G|$ to the sum, and there are $k = |X/G|$ orbits, so

$$\sum_{x \in X} |G_x| \;=\; |X/G| \cdot |G|.$$

Equating (4.1) and (4.2):

$$\sum_{g \in G} |X^g| \;=\; |X/G| \cdot |G|,$$

and dividing by $|G|$ yields the result.

The double-counting technique at the heart of this proof is worth internalizing: whenever a condition involves both a group element and a set element (here, "g fixes x"), counting the pairs from both sides often yields a clean identity.

Let us see Burnside's Lemma in action.

Example 4.3.9: Counting Colorings of a Square

How many distinct ways can we color the vertices of a square using 2 colors (say, black and white), if two colorings are considered "the same" when one can be obtained from the other by a symmetry of the square?

Let X be the set of all 2-colorings of 4 labeled vertices, so $|X| = 2^4 = 16$. The symmetry group $G = D_8$ acts on X by permuting the vertices (and hence the colors assigned to them). The distinct colorings "up to symmetry" are precisely the orbits, so we seek $|X/G|$.

We compute $|X^g|$ for each $g \in D_8$. Using the vertex labeling 1 (top-left), 2 (top-right), 3 (bottom-right), 4 (bottom-left), the eight elements act as the following permutations:

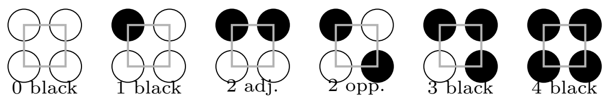
g	Permutation	Cycle structure	$\|X^g\|$
e	$()$	four fixed points	$2^4 = 16$
r	$(1\ 2\ 3\ 4)$	one 4-cycle	$2^1 = 2$
r^2	$(1\ 3)(2\ 4)$	two 2-cycles	$2^2 = 4$
r^3	$(1\ 4\ 3\ 2)$	one 4-cycle	$2^1 = 2$
s	$(1\ 2)(3\ 4)$	two 2-cycles	$2^2 = 4$
sr	$(2\ 4)$	one 2-cycle, two fixed	$2^3 = 8$
sr^2	$(1\ 4)(2\ 3)$	two 2-cycles	$2^2 = 4$
sr^3	$(1\ 3)$	one 2-cycle, two fixed	$2^3 = 8$

The rule for each row: a coloring is fixed by g if and only if all vertices within each cycle of g receive the same color. So $|X^g| = 2^{(\text{number of cycles of } g)}$, counting fixed points as 1-cycles.

Burnside's Lemma gives:

$$|X/G| \;=\; \frac{1}{8}(16 + 2 + 4 + 2 + 4 + 8 + 4 + 8) \;=\; \frac{48}{8} \;=\; 6.$$

The six distinct colorings are:



4.3.10 Applied Aside: Counting Molecular Structures The same technique applies in chemistry. Consider a benzene ring C_6H_6, where each hydrogen atom can be replaced by a substituent (say, chlorine). How many distinct chlorinated benzene derivatives exist? The symmetry group of the hexagonal ring is D_{12}, acting on the six positions. Burnside's Lemma (or its refinement, Pólya's Enumeration Theorem) answers this question systematically. For instance, with two possible substituents (H or Cl) at each of 6 positions, there are

$$\frac{1}{12}(2^6 + \cdots) = 13$$

distinct molecules, a count that matches experimental chemistry. This interplay between symmetry and enumeration is foundational in chemical graph theory.

4.4 Types of Actions

Not all group actions are created equal. Some actions allow different group elements to "look the same" on X; others ensure that every group element acts distinctly. Some actions can move any point to any other; others split X into multiple unreachable regions. In this section, we introduce a taxonomy of four properties—faithful, transitive, free, and regular—that classify the most important qualitative features of an action.

Faithful Actions

We have already defined this notion in definition 4.2.12: an action $G \curvearrowright X$ is *faithful* if $\ker(\rho) = \{e\}$, i.e. if the only element that fixes every point of X is the identity. Equivalently, the permutation representation $\rho\colon G \to \mathrm{Sym}(X)$ is injective, so G embeds into $\mathrm{Sym}(X)$.

Faithfulness is a *global* condition: it says nothing about any individual stabilizer G_x, only about their intersection $\bigcap_{x \in X} G_x$. Individual stabilizers may well be nontrivial.

Example 4.4.1: Faithful Actions

1. The tautological action of S_n on $\{1, \dots, n\}$ is faithful: any nontrivial permutation must move at least one element.
2. The action of D_{2n} on the vertices of a regular n-gon is faithful for $n \geq 3$: distinct symmetries of the polygon induce distinct permutations of the vertices. Note, however, that stabilizers are nontrivial—the stabilizer of a vertex is the order-2 subgroup generated by the reflection fixing that vertex.
3. The conjugation action of G on itself (item 6) has kernel
$$\ker(\rho) = \{g \in G : gag^{-1} = a \text{ for all } a \in G\} = Z(G),$$
the center of G (definition 2.2.4). This action is faithful if and only if $Z(G) = \{e\}$, i.e. if and only if G has trivial center.

As we saw in proposition 4.2.14, any action can be made faithful by passing to the quotient $G/\ker(\rho)$. In this sense, the "faithful part" of an action is always available; the kernel measures the redundancy.

Transitive Actions

Definition 4.4.2: Transitive Action

A group action $G \curvearrowright X$ is called *transitive* if it has exactly one orbit, i.e. $G \cdot x = X$ for every $x \in X$. Equivalently, for any $x, y \in X$, there exists some $g \in G$ such that $g \cdot x = y$.

In a transitive action, every point can be reached from every other point. No point is "isolated" from the rest.

Example 4.4.3: Transitive and Non-Transitive Actions

1. The action of S_n on $\{1, \ldots, n\}$ is transitive: for any i, j, the transposition $(i\ j)$ sends i to j.
2. The action of D_8 on the four vertices of a square is transitive: r cyclically permutes all four vertices.
3. The action of $\langle(1\ 2\ 3)\rangle$ on $\{1, 2, 3, 4, 5\}$ from example 4.2.2 is *not* transitive: it has three orbits.
4. The conjugation action of G on itself is transitive if and only if every pair of elements is conjugate. For finite groups, this can only happen when $|G| = 1$ (since the identity is conjugate only to itself, so if $|G| > 1$, the singleton $\{e\}$ is always an orbit separate from any non-identity element).

Transitive actions enjoy a useful structural property: the stabilizers of any two points are conjugate subgroups.

Proposition 4.4.4: Stabilizers of Points in the Same Orbit

Let $G \curvearrowright X$. If $y = g \cdot x$ for some $g \in G$, then

$$G_y = g\, G_x\, g^{-1}.$$

In particular, if the action is transitive, then the stabilizers of any two points in X are conjugate subgroups of G.

***Proof*:**
Let $h \in G_y$, so that $h \cdot y = y$. Since $y = g \cdot x$:

$$g \cdot x = y = h \cdot y = h \cdot (g \cdot x) = (hg) \cdot x,$$

so $(g^{-1}hg) \cdot x = g^{-1} \cdot (g \cdot x) = x$, giving $g^{-1}hg \in G_x$, i.e. $h \in gG_xg^{-1}$. This shows $G_y \subseteq gG_xg^{-1}$. Conversely, if $h \in gG_xg^{-1}$, write $h = gkg^{-1}$ for some $k \in G_x$. Then

$$h \cdot y = (gkg^{-1}) \cdot (g \cdot x) = g \cdot (k \cdot x) = g \cdot x = y,$$

so $h \in G_y$. Thus $gG_xg^{-1} \subseteq G_y$.

This proposition has an important consequence: in a transitive action, knowing the stabilizer of *any single point* determines (up to conjugacy) the stabilizer of every point.

Even more is true. A transitive action is completely determined, up to a natural notion of equivalence, by the stabilizer of any chosen point. To make this precise, we need a brief definition.

Definition 4.4.5: Isomorphism of G-sets

Let $G \curvearrowright X$ and $G \curvearrowright Y$ be two actions of the same group. A bijection $\varphi\colon X \to Y$ is an *isomorphism of G-sets* (or a *G-equivariant* bijection) if

$$\varphi(g \cdot x) = g \cdot \varphi(x) \qquad \text{for all } g \in G,\ x \in X.$$

If such a bijection exists, we write $X \cong_G Y$ and say the two G-sets are *isomorphic*.

In other words, φ is a relabeling of X as Y that is compatible with the group action: it does not matter whether we act first and then relabel, or relabel first and then act.

Theorem 4.4.6: Structure Theorem for Transitive Actions

Let $G \curvearrowright X$ be a transitive action, and fix any point $x_0 \in X$. Then the bijection

$$\varphi\colon G/G_{x_0} \to X, \qquad gG_{x_0} \mapsto g \cdot x_0,$$

from the Orbit-Stabilizer Theorem (theorem 4.3.1) is an isomorphism of G-sets, where G acts on G/G_{x_0} by left multiplication: $h \cdot (gG_{x_0}) = (hg)G_{x_0}$.
In particular, every transitive G-set is isomorphic to G/H for some subgroup $H \leq G$.

***Proof*:**
We already know from theorem 4.3.1 that φ is a well-defined bijection (using transitivity so that $G \cdot x_0 = X$). It remains to verify equivariance. For any $h \in G$ and any coset gG_{x_0}:

$$\varphi(h \cdot gG_{x_0}) = \varphi((hg)G_{x_0}) = (hg) \cdot x_0 = h \cdot (g \cdot x_0) = h \cdot \varphi(gG_{x_0}).$$

Thus φ respects the action, and the two G-sets are isomorphic.

This theorem is one of the most clarifying results in the theory: it says that, up to relabeling, *every* transitive action is a coset action. We will study coset actions in detail in Section 4.5.

Free Actions

Faithful actions require the *intersection* of all stabilizers to be trivial. Free actions require something much stronger: every *individual* stabilizer must be trivial.

Definition 4.4.7: Free Action

A group action $G \curvearrowright X$ is called *free* if

$$G_x = \{e\} \qquad \text{for every } x \in X.$$

Equivalently, if $g \cdot x = x$ for some $x \in X$, then $g = e$.

There is a useful reformulation: an action is free if and only if distinct group elements never produce the same result at any point.

Proposition 4.4.8: Characterization of Free Actions

An action $G \curvearrowright X$ is free if and only if for all $g, h \in G$ and all $x \in X$,

$$g \cdot x = h \cdot x \implies g = h.$$

***Proof*:**
If $g \cdot x = h \cdot x$, then $(h^{-1}g) \cdot x = x$, so $h^{-1}g \in G_x$. If the action is free, $G_x = \{e\}$, so $h^{-1}g = e$, giving $g = h$. Conversely, if the implication holds for all g, h, x, then taking $h = e$: $g \cdot x = x$ implies $g = e$, so every stabilizer is trivial.

Example 4.4.9: Free Actions

1. The left-multiplication action $G \curvearrowright G$ (item 5) is free: if $ga = a$, then $g = e$ by cancellation (proposition 1.1.17). This holds for any group G.
2. The action of $(\mathbb{Z}, +)$ on $\mathbb{R}$ by $n \cdot x = x + n$ is free: if $x + n = x$, then $n = 0$.
3. The action of D_8 on the vertices of a square is *not* free: the stabilizer of each vertex has order 2.
4. The action of $\mathbb{Z}/2\mathbb{Z} = \{e, \sigma\}$ on $\{1, 2, 3, 4\}$ via $\sigma = (1\ 2)(3\ 4)$ is free: σ has no fixed points. It is not transitive, however (the orbits are $\{1, 2\}$ and $\{3, 4\}$).

Free actions interact cleanly with the Orbit-Stabilizer Theorem.

Corollary 4.4.10: Orbit Sizes in Free Actions

If G is a finite group acting freely on X, then every orbit has size $|G|$. In particular, $|G|$ divides $|X|$.

***Proof*:**
For every $x \in X$, $|G_x| = 1$, so $|G \cdot x| = |G|/|G_x| = |G|$ by theorem 4.3.1. Since the orbits partition X (proposition 4.2.3), $|X|$ is a sum of copies of $|G|$.

Note that every free action is faithful: if $G_x = \{e\}$ for all x, then $\ker(\rho) = \bigcap_x G_x = \{e\}$. The converse is false, as the example of D_8 on the vertices of a square shows (faithful, but not free).

Regular Actions

The strongest condition arises when we ask for an action that is simultaneously free and transitive.

Definition 4.4.11: Regular Action

A group action $G \curvearrowright X$ is called *regular* if it is both free and transitive.

Regular actions have a particularly elegant characterization.

Proposition 4.4.12: Characterization of Regular Actions

An action $G \curvearrowright X$ is regular if and only if for every pair $x, y \in X$, there exists a *unique* $g \in G$ such that $g \cdot x = y$.

***Proof*:**
Transitivity says "at least one such g exists." Freeness says "at most one such g exists" (by proposition 4.4.8: if $g \cdot x = y = h \cdot x$, then $g = h$). Together, they give exactly one.

Conversely, if for every x, y there exists a unique g with $g \cdot x = y$, then existence gives transitivity, and uniqueness gives freeness.

Example 4.4.13: Regular Actions

1. The left-multiplication action $G \curvearrowright G$ is regular: it is free (example 4.4.9) and transitive (given $a, b \in G$, the element $g = ba^{-1}$ satisfies $g \cdot a = ba^{-1}a = b$). This is the *canonical* regular action.

2. The action of $(\mathbb{Z}, +)$ on $\mathbb{R}$ by translation $(n \cdot x = x + n)$ is free but not transitive ($\frac{1}{2}$ is not in the orbit of 0), hence not regular.

3. The action of D_8 on the four vertices of a square is transitive but not free, hence not regular.

By the Orbit-Stabilizer Theorem, a regular action of a finite group G on a set X satisfies $|X| = |G \cdot x| = |G|$ for every x: the set and the group have the same cardinality. In fact, the Structure Theorem for Transitive Actions (theorem 4.4.6) shows precisely what every regular action looks like:

Corollary 4.4.14: Regular Actions are Left Multiplication

If G acts regularly on X, then $X \cong_G G$ as G-sets. That is, up to isomorphism, the left-multiplication action of G on itself is the *only* regular G-action.

***Proof*:**
By theorem 4.4.6, any transitive G-set is isomorphic to G/G_{x_0} for some $x_0 \in X$. Since the action is also free, $G_{x_0} = \{e\}$, so $X \cong_G G/\{e\} = G$.

Summary: A Taxonomy of Group Actions

We collect the four types and their interrelationships in a single reference table.

Type	*Condition*	*Via Orbit-Stabilizer*	*Canonical example*
Faithful	$\bigcap_{x \in X} G_x = \{e\}$	$\rho\colon G \hookrightarrow \mathrm{Sym}(X)$	$D_{2n} \curvearrowright$ vertices
Transitive	$G \cdot x = X$ for all x	$\lvert X\rvert = [G : G_x]$	$G \curvearrowright G/H$
Free	$G_x = \{e\}$ for all x	$\lvert G \cdot x\rvert = \lvert G\rvert$ for all x	$G \curvearrowright G$ by left mult.
Regular	Free + Transitive	$\lvert X\rvert = \lvert G\rvert$ and $G_x = \{e\}$	$G \curvearrowright G$ by left mult.

The logical implications among the four types are:

$$\text{Regular} \implies \text{Free} \implies \text{Faithful}, \qquad \text{Regular} \implies \text{Transitive}.$$

None of the converse implications hold in general, as the following examples illustrate:

- Faithful $\not\Rightarrow$ Free: $D_8 \curvearrowright \{1, 2, 3, 4\}$ (faithful, but $|G_1| = 2$).
- Free $\not\Rightarrow$ Transitive: $\mathbb{Z}/2\mathbb{Z} \curvearrowright \{1, 2, 3, 4\}$ via $(1\ 2)(3\ 4)$ (free, but two orbits).
- Transitive $\not\Rightarrow$ Faithful: The conjugation action of S_3 on $\{1, 2, 3\}$ by relabeling is transitive, but if we instead consider $\mathbb{Z}/6\mathbb{Z}$ acting on $\{1, 2, 3\}$ via $\bar{1} \mapsto (1\ 2\ 3)$, the action is transitive with kernel $\{\bar{0}, \bar{3}\} \cong \mathbb{Z}/2\mathbb{Z}$.
- Transitive $\not\Rightarrow$ Free: $D_8 \curvearrowright \{1, 2, 3, 4\}$ (transitive, but not free).

4.5 A Group Acting on Itself

So far, our group actions have involved a group acting on an external set: vertices of a polygon, vectors in $\mathbb{R}^n$, polynomial rings. But as we saw, this often leads to G-isomorphisms between the G-set X and some quotient of G. Thus, we turn our attention to a group acting on *itself* (or on structures derived from itself, such as cosets). These self-actions will produce some of the key results in the chapter.

Action by Left Multiplication

The simplest self-action is left multiplication: each group element shuffles the entire group by multiplying on the left.

Let G be a group and let $X = G$ (the underlying set). The action is

$$g \cdot a = ga \qquad \text{for all } g, a \in G.$$

We verified in item 5 that this is a valid group action. We can now classify it using the taxonomy of Section 4.4:

Proposition 4.5.1: Left Multiplication is a Regular Action

The left-multiplication action $G \curvearrowright G$ is regular (free and transitive).

***Proof*:**
The action is free: if $g \cdot a = a$, then $ga = a$, so $g = e$ by cancellation (proposition 1.1.17). The action is transitive: given any $a, b \in G$, taking $g = ba^{-1}$ gives $g \cdot a = ba^{-1}a = b$.

Since the action is faithful (every free action is), the permutation representation $\rho\colon G \to \mathrm{Sym}(G)$ is injective. Let us see what ρ looks like concretely.

Example 4.5.2: Permutation Representation of the Klein Four-Group
Let $G = \{e, a, b, c\}$ be the Klein four-group, with multiplication defined by $a^2 = b^2 = c^2 = e$ and $ab = c$, $ac = b$, $bc = a$. Label $e \leftrightarrow 1$, $a \leftrightarrow 2$, $b \leftrightarrow 3$, $c \leftrightarrow 4$. For each $g \in G$, we compute the permutation $\sigma_g(i) = j$ where $g \cdot g_i = g_j$:

g	$g \cdot e$	$g \cdot a$	$g \cdot b$	$g \cdot c$	σ_g
e	e	a	b	c	$()$
a	a	e	c	b	$(1\ 2)(3\ 4)$
b	b	c	e	a	$(1\ 3)(2\ 4)$
c	c	b	a	e	$(1\ 4)(2\ 3)$

The image $\rho(G) = \{(),\ (1\ 2)(3\ 4),\ (1\ 3)(2\ 4),\ (1\ 4)(2\ 3)\}$ is a subgroup of S_4 isomorphic to the Klein four-group. One can verify directly that $g \mapsto \sigma_g$ is a homomorphism, and injectivity is visible from the table: distinct rows give distinct permutations.

Cayley's Theorem

The example above embeds a group of order 4 into S_4. The same construction works for any group.

Theorem 4.5.3: Cayley's Theorem

Every group is isomorphic to a subgroup of a symmetric group. More precisely, if G is a group, then the permutation representation $\rho\colon G \to \mathrm{Sym}(G)$ of the left-multiplication action is an injective homomorphism. If $|G| = n$, then G is isomorphic to a subgroup of S_n.

***Proof*:**
The left-multiplication action $G \curvearrowright G$ is free (proposition 4.5.1), hence faithful (definition 4.2.12). Thus the permutation representation $\rho\colon G \to \mathrm{Sym}(G)$ has trivial kernel, so ρ is injective. By the First Isomorphism Theorem (theorem 3.1.14), $G \cong \rho(G) \leq \mathrm{Sym}(G)$.

If $|G| = n$, we may label the elements of G as $\{g_1, \ldots, g_n\}$ and identify $\mathrm{Sym}(G) \cong S_n$, so G embeds into S_n.

Cayley's Theorem is one of the most philosophically significant results in group theory. It shows that the abstract axioms (closure, associativity, identity, inverses) do not create anything genuinely "new" beyond permutation groups. Every group, no matter how abstractly defined, *is* a group of symmetries—a subgroup of a symmetric group. The Erlangen Programme's vision (see the aside in box 4.0.1) is vindicated: groups and symmetries are one and the same.

4.5.4 Historical Note When group theory began in the early 19th century, a "group" simply *meant* a group of permutations. Cayley was among the first to propose the abstract axiomatic definition. His 1854 theorem then showed that the abstraction lost nothing: the original permutation-based intuition remains fully applicable to every abstract group.

That said, the embedding $G \hookrightarrow S_n$ is rarely practical. The symmetric group S_n has order $n!$, which is vastly larger than n for even modest values: if $|G| = 100$, then $|S_{100}| = 100! \approx 9.3 \times 10^{157}$. Working inside S_n is typically no easier than working directly with G. The theorem's value is *conceptual*: it guarantees that any general statement proved about subgroups of symmetric groups automatically applies to all groups.

The following example illustrates a subtler limitation: Cayley's bound $n = |G|$ is sometimes optimal—you genuinely cannot embed the group into any smaller symmetric group.

Example 4.5.5: The Minimum Faithful Degree of Q_8

The quaternion group $Q_8 = \{\pm 1, \pm i, \pm j, \pm k\}$ has order 8, so Cayley's Theorem gives $Q_8 \hookrightarrow S_8$. We claim that Q_8 cannot act faithfully on any set X with $|X| \leq 7$.

Suppose $Q_8 \curvearrowright X$ with $|X| \leq 7$. For any $x \in X$, the Orbit-Stabilizer Theorem (theorem 4.3.1) gives

$$|G_x| = \frac{|Q_8|}{|Q_8 \cdot x|} = \frac{8}{|Q_8 \cdot x|} \geq \frac{8}{7} > 1,$$

so $G_x \neq \{e\}$ for every x. Now, a particular property of Q_8 is that every nontrivial subgroup contains -1 (one can verify this by inspecting the subgroup lattice of Q_8, cf. example 1.2.28). Since G_x is a nontrivial subgroup for every x, we have $-1 \in G_x$ for every x. Therefore $-1 \in \bigcap_{x \in X} G_x = \ker(\rho)$, and the action is not faithful.

Thus, the Cayley embedding into S_8 is the smallest symmetric group that can faithfully contain Q_8.

Action on the Coset Space

Left multiplication acts on the full group G. We now generalize by letting G act on the set of left cosets of a subgroup.

Let $H \leq G$, and let $G/H = \{gH : g \in G\}$ denote the set of left cosets (this is a set, not necessarily a group, since H need not be normal). Define

$$g \cdot (aH) = (ga)H \qquad \text{for all } g \in G,\ aH \in G/H.$$

This is well-defined: if $aH = bH$, then $b^{-1}a \in H$, so $(gb)^{-1}(ga) = b^{-1}a \in H$, giving $(ga)H = (gb)H$. The two axioms are immediate: $(gh) \cdot aH = (gh)aH = g \cdot (haH)$ and $e \cdot aH = aH$.

Note that when $H = \{e\}$, the cosets of H are simply the elements of G, and this action reduces to left multiplication. In this sense, the coset action *generalizes* the construction of the previous subsection.

The following theorem extracts the structural information that the coset action provides.

Theorem 4.5.6: Properties of the Coset Action

Let $H \leq G$, and let G act on G/H by left multiplication. Let $\pi_H \colon G \to \mathrm{Sym}(G/H)$ be the associated permutation representation. Then:

1. The action is transitive.
2. The stabilizer of the coset eH is H:
$$G_{eH} = H.$$
3. The kernel of the action is the largest normal subgroup of G contained in H:
$$\ker(\pi_H) = \bigcap_{g \in G} gHg^{-1}.$$

***Proof*:**

1. Given any two cosets $aH, bH \in G/H$, take $g = ba^{-1}$. Then $g \cdot aH = (ba^{-1})aH = bH$. Since a and b were arbitrary, the action is transitive.
2. The stabilizer of eH is $\{g \in G : g \cdot eH = eH\} = \{g \in G : gH = H\}$. Now $gH = H$ if and only if $g \in H$ (recall that $gH = H \Leftrightarrow g \in H$). Thus $G_{eH} = H$.
3. The kernel is $\ker(\pi_H) = \bigcap_{aH \in G/H} G_{aH}$. The stabilizer of an arbitrary coset aH is:
$$\begin{aligned} G_{aH} &= \{g \in G : g \cdot aH = aH\} = \{g \in G : gaH = aH\} \\ &= \{g \in G : a^{-1}ga \in H\} = \{g \in G : g \in aHa^{-1}\} = aHa^{-1}. \end{aligned}$$
Therefore
$$\ker(\pi_H) = \bigcap_{a \in G} aHa^{-1}.$$
It remains to show this is the *largest* normal subgroup of G contained in H. Since $\ker(\pi_H) \leq G_{eH} = H$ (part 2), the kernel is contained in H. It is normal in G since it is the kernel of a homomorphism (proposition 2.2.12).

   Conversely, let $N \trianglelefteq G$ with $N \leq H$. For any $a \in G$: $N = aNa^{-1} \leq aHa^{-1}$ (since $N \leq H$ implies $aNa^{-1} \leq aHa^{-1}$). Thus $N \leq \bigcap_{a \in G} aHa^{-1} = \ker(\pi_H)$, confirming maximality.

Example 4.5.7: D_8 Acting on Cosets of $\langle s \rangle$

Let $G = D_8$ and $H = \langle s \rangle = \{e, s\}$. The index $[D_8 : H] = 8/2 = 4$, so there are four left cosets:

$$C_1 = \{e,\, s\}, \quad C_2 = \{r,\, sr^3\}, \quad C_3 = \{r^2,\, sr^2\}, \quad C_4 = \{r^3,\, sr\}.$$

The coset action gives a homomorphism $\pi_H \colon D_8 \to S_4$. To find the image, we compute where r and s send each coset:

$$\begin{aligned} r\colon\ & C_1 \mapsto C_2,\ C_2 \mapsto C_3,\ C_3 \mapsto C_4,\ C_4 \mapsto C_1 && \Longrightarrow \quad \sigma_r = (1\,2\,3\,4), \\ s\colon\ & C_1 \mapsto C_1,\ C_2 \mapsto C_4,\ C_3 \mapsto C_3,\ C_4 \mapsto C_2 && \Longrightarrow \quad \sigma_s = (2\,4). \end{aligned}$$

The image $\pi_H(D_8) = \langle(1\ 2\ 3\ 4),\ (2\ 4)\rangle$ is a subgroup of S_4 isomorphic to D_8. In particular, $\ker(\pi_H) = \{e\}$, so the action is faithful: D_8 embeds into S_4. This is smaller than the Cayley embedding into S_8, and it recovers the natural representation of D_8 as the symmetry group of a square.

theorem 4.5.6 predicted this: the kernel is $\bigcap_{g\in G} g\langle s\rangle g^{-1}$. Since the only conjugates of $\langle s\rangle$ are $\{e, s\}$ and $\{e, sr^2\}$, their intersection is $\{e\}$.

Example 4.5.8: Recovering the Index-2 Theorem
If $[G : H] = 2$, the coset action gives a homomorphism $\pi_H : G \to S_2 \cong \mathbb{Z}/2\mathbb{Z}$. Since $|S_2| = 2$, the kernel has index at most 2 in G. But $\ker(\pi_H) \le H$ and $[G : H] = 2$, so $\ker(\pi_H) = H$. Since kernels are normal, $H \trianglelefteq G$.

We have thus re-derived corollary 3.3.5 as a special case of the coset action.

Subgroups of Smallest Prime Index

The observation in the previous example generalizes far beyond index 2. The key insight is that if $[G : H] = p$ for a small prime p, then the coset action maps G into S_p, and S_p is "small enough" to force the kernel to be large.

Theorem 4.5.9: Subgroups of Smallest Prime Index are Normal

Let G be a finite group and let p be the smallest prime dividing $|G|$. If $H \le G$ with $[G : H] = p$, then $H \trianglelefteq G$.

***Proof*:**
The coset action gives a homomorphism $\pi_H : G \to \mathrm{Sym}(G/H) \cong S_p$. Let $K = \ker(\pi_H)$. By theorem 4.5.6, $K \trianglelefteq G$ and $K \le H$. We will show $K = H$.

By the First Isomorphism Theorem (theorem 3.1.14), $G/K \cong \pi_H(G) \le S_p$, so $|G/K|$ divides $|S_p| = p!$. Since $K \le H$, the index decomposes as

$$[G : K] = [G : H] \cdot [H : K] = p \cdot [H : K].$$

Therefore $p \cdot [H : K]$ divides $p!$, which gives $[H : K]$ divides $(p - 1)!$.

On the other hand, $[H : K]$ divides $|H|$ by Lagrange's Theorem (theorem 3.3.1), and $|H|$ divides $|G|$, so every prime factor of $[H : K]$ is also a prime factor of $|G|$. Since p is the *smallest* prime dividing $|G|$, every prime factor of $[H : K]$ is at least p.

But every prime factor of $(p - 1)!$ is strictly less than p. The only positive integer all of whose prime factors are simultaneously $\ge p$ and $< p$ is 1. Therefore $[H : K] = 1$, so $K = H$, and $H = \ker(\pi_H) \trianglelefteq G$.

Example 4.5.10: Applications of the Index-p Theorem

1. If $|G| = 2m$ with m odd, the smallest prime dividing $|G|$ is 2. Any subgroup of index 2 is normal. This recovers corollary 3.3.5.

2. Let $|G| = 2 \cdot 3 \cdot 5 = 30$. The smallest prime dividing $|G|$ is 2. If G has a subgroup of index 2 (i.e. order 15), it is automatically normal. Any subgroup of index 3 need not be normal, since 3 is not the smallest prime dividing 30.

3. Let $|G| = 3 \cdot 7 = 21$. The smallest prime is 3. If G has a subgroup of index 3 (i.e. order 7), it is normal.

4. Let $|G| = 5 \cdot 11 \cdot 13 = 715$. The smallest prime is 5. A subgroup of index 5 (order 143), should one exist, is normal.

4.5.11 Remark A group of order n need not have a subgroup of every possible index. For instance, A_4 has order 12 and the smallest prime dividing 12 is 2, but A_4 has no subgroup of order 6 (cf. the counterexample to the converse of Lagrange's Theorem, example 3.3.7). The theorem says that *if* a subgroup of index p exists, it must be normal; it does not guarantee existence. The question of existence will be addressed by the Sylow Theorems in Section 4.8.

We pause to take stock. Starting from the simple idea of a group multiplying its own elements, we have obtained:

- Cayley's Theorem: every group embeds into a symmetric group.
- The coset action: a versatile tool that converts subgroup information ($H \leq G$) into homomorphism information ($G \to S_p$).
- The index-p theorem: a powerful normality criterion that requires nothing more than knowing $[G : H]$ and the prime factorization of $|G|$.

In the next section, we will see that a different self-action—conjugation—yields an entirely different class of structural results.

4.6 The Conjugation Action and the Class Equation

In Section 4.5, a group acted on itself by left multiplication, and the payoff was Cayley's Theorem: every group is a group of permutations. We now let a group act on itself by *conjugation*, and the payoff will be of a completely different character: arithmetic information about the internal structure of the group, encoded in an identity called the *Class Equation*.

Conjugation on Elements

Let G be a group and let $X = G$. The *conjugation action* is defined by

$$g \cdot a \;=\; gag^{-1} \qquad \text{for all } g, a \in G.$$

We verified in item 6 that this is a valid group action. Let us now identify its orbits and stabilizers.

Definition 4.6.1: Conjugacy Class

Two elements $a, b \in G$ are *conjugate* in G if $b = gag^{-1}$ for some $g \in G$, i.e. if they lie in the same orbit of the conjugation action. The orbit

$$\mathrm{cl}(a) \;=\; G \cdot a \;=\; \{gag^{-1} : g \in G\}$$

is called the *conjugacy class* of a.

Since orbits partition the set (proposition 4.2.3), conjugacy classes partition G: every element belongs to exactly one conjugacy class.

The stabilizer of the conjugation action has a familiar name:

$$G_a = \{g \in G : gag^{-1} = a\} = \{g \in G : ga = ag\} = C_G(a),$$

the centralizer of a in G (definition 2.2.7). The Orbit-Stabilizer Theorem (theorem 4.3.1) therefore gives:

$$|\,\mathrm{cl}(a)| \;=\; [G : C_G(a)]. \tag{4.3}$$

An element a has a singleton conjugacy class, $\mathrm{cl}(a) = \{a\}$, if and only if $gag^{-1} = a$ for all $g \in G$, i.e. if and only if a commutes with every element of G. This is precisely the condition $a \in Z(G)$ (definition 2.2.4). Thus:

The singleton conjugacy classes are exactly the elements of the center.

Example 4.6.2: Conjugacy Classes of Small Groups

1. If G is abelian, then $gag^{-1} = a$ for all g, a, so every conjugacy class is a singleton: $\mathrm{cl}(a) = \{a\}$ for all a. The conjugation action is trivial.

2. Let $G = Q_8 = \{\pm 1, \pm i, \pm j, \pm k\}$. The center is $Z(Q_8) = \{1, -1\}$ (example 1.2.28). The non-central elements form three conjugacy classes:

$$\mathrm{cl}(i) = \{i, -i\}, \qquad \mathrm{cl}(j) = \{j, -j\}, \qquad \mathrm{cl}(k) = \{k, -k\}.$$

   To verify: $jij^{-1} = ji(-j) = ji \cdot j^{-1}$. Using $ji = -k$ and $(-k)(-j) = kj = -i$, so $jij^{-1} = -i$. Each conjugacy class has size $[Q_8 : C_{Q_8}(i)] = 8/4 = 2$, since $C_{Q_8}(i) = \langle i \rangle$ has order 4.

3. Let $G = D_8$. The center is $Z(D_8) = \{e, r^2\}$. The non-central conjugacy classes are:

$$\mathrm{cl}(r) = \{r, r^3\}, \qquad \mathrm{cl}(s) = \{s, sr^2\}, \qquad \mathrm{cl}(sr) = \{sr, sr^3\}.$$

   Each has size 2, consistent with $[D_8 : C_{D_8}(r)] = 8/4 = 2$, etc.

The Class Equation

We now translate the orbit decomposition of the conjugation action into an arithmetic identity.

Theorem 4.6.3: The Class Equation

Let G be a finite group, and let $g_1, \ldots, g_r$ be representatives of the distinct conjugacy classes of G that are *not* contained in $Z(G)$. Then

$$|G| \;=\; |Z(G)| \;+\; \sum_{i=1}^{r} [G : C_G(g_i)].$$

Each term $[G : C_G(g_i)]$ in the sum is strictly greater than 1 and divides $|G|$.

***Proof*:**
The conjugacy classes partition G, so $|G| = \sum |\operatorname{cl}(a)|$ as a ranges over a set of orbit representatives. We split this sum into two parts:

- Elements $a \in Z(G)$ have $|\operatorname{cl}(a)| = 1$. There are $|Z(G)|$ such elements.
- Elements $a \notin Z(G)$ have $|\operatorname{cl}(a)| = [G : C_G(a)] > 1$ (since $C_G(a) \neq G$ when $a \notin Z(G)$). Let $g_1, \ldots, g_r$ be representatives of these non-central classes.

Combining:

$$|G| = \sum_{a \in Z(G)} 1 \;+\; \sum_{i=1}^{r} |\operatorname{cl}(g_i)| = |Z(G)| + \sum_{i=1}^{r} [G : C_G(g_i)].$$

Each $[G : C_G(g_i)]$ divides $|G|$ by Lagrange's Theorem (theorem 3.3.1), and is > 1 since $g_i \notin Z(G)$.

Example 4.6.4: The Class Equation in Practice

1. For $G = Q_8$: $|Z(Q_8)| = 2$ and the three non-central classes each have size 2, giving $8 = 2 + 2 + 2 + 2$.
2. For $G = D_8$: $|Z(D_8)| = 2$ and the three non-central classes each have size 2, giving $8 = 2 + 2 + 2 + 2$.
3. For $G = S_3$: $Z(S_3) = \{e\}$, and the conjugacy classes are $\{e\}$, $\{(1\ 2), (1\ 3), (2\ 3)\}$, $\{(1\ 2\ 3), (1\ 3\ 2)\}$, giving $6 = 1 + 3 + 2$.
4. If G is abelian, the Class Equation reads $|G| = |G|$, which is vacuously true but uninformative. The Class Equation is a tool for *non-abelian* groups.

p-Groups Have Nontrivial Center

The Class Equation becomes especially powerful when $|G|$ is a prime power. The following theorem is the first result that students typically cannot prove *without* group actions.

Theorem 4.6.5: p-Groups Have Nontrivial Center

Let p be a prime and let G be a group of order p^α for some $\alpha \geq 1$. Then $Z(G) \neq \{e\}$; more precisely, p divides $|Z(G)|$.

***Proof*:**
The Class Equation gives

$$p^\alpha \;=\; |Z(G)| + \sum_{i=1}^{r} [G : C_G(g_i)].$$

For each i, the centralizer $C_G(g_i)$ is a proper subgroup of G (since $g_i \notin Z(G)$), so $[G : C_G(g_i)] > 1$. Since $C_G(g_i) \leq G$ and $|G| = p^\alpha$, Lagrange's Theorem gives $|C_G(g_i)| = p^{\beta_i}$ for some $\beta_i < \alpha$. Therefore $[G : C_G(g_i)] = p^{\alpha-\beta_i}$, which is divisible by p.

The left-hand side p^α is divisible by p, and every term in the sum is divisible by p. Therefore $|Z(G)|$ is divisible by p, so $|Z(G)| \geq p > 1$.

The argument is clean and short, but notice how much machinery is involved beneath the surface: the conjugation action, the Orbit-Stabilizer Theorem (to get $|\operatorname{cl}(g_i)| = [G : C_G(g_i)]$), Lagrange's Theorem (to deduce $p \big| [G : C_G(g_i)]$), and the orbit partition (to write the Class Equation). This is a vivid demonstration of the power of group actions as a *proof technique*.

Corollary 4.6.6: Groups of Order p^2 Are Abelian

If $|G| = p^2$ for some prime p, then G is abelian. More precisely, $G \cong \mathbb{Z}/p^2\mathbb{Z}$ or $G \cong \mathbb{Z}/p\mathbb{Z} \times \mathbb{Z}/p\mathbb{Z}$.

***Proof*:**
By theorem 4.6.5, $p \big| |Z(G)|$. Since $Z(G) \leq G$ and $|G| = p^2$, Lagrange gives $|Z(G)| \in \{p, p^2\}$.

If $|Z(G)| = p^2$, then $Z(G) = G$ and G is abelian.

If $|Z(G)| = p$, then $|G/Z(G)| = p^2/p = p$, so $G/Z(G)$ has prime order and is therefore cyclic (corollary 2.3.12). We claim this forces G to be abelian, contradicting $Z(G) \neq G$.

Indeed, let $G/Z(G) = \langle gZ(G) \rangle$. Then every element of G can be written as $g^k z$ for some $k \in \mathbb{Z}$ and $z \in Z(G)$. For any two elements $g^{k_1} z_1$ and $g^{k_2} z_2$:

$$(g^{k_1} z_1)(g^{k_2} z_2) = g^{k_1} g^{k_2} z_1 z_2 = g^{k_2} g^{k_1} z_2 z_1 = (g^{k_2} z_2)(g^{k_1} z_1),$$

where we used that powers of g commute with each other, that $z_1, z_2 \in Z(G)$ commute with everything, and that $Z(G)$ is abelian. Thus G is abelian, a contradiction.

Therefore $|Z(G)| = p^2$, and G is abelian of order p^2. By the classification of finite abelian groups (or by direct argument using corollary 2.3.12), the only abelian groups of order p^2 are $\mathbb{Z}/p^2\mathbb{Z}$ and $\mathbb{Z}/p\mathbb{Z} \times \mathbb{Z}/p\mathbb{Z}$.

4.6.7 Remark The key step—"if $G/Z(G)$ is cyclic, then G is abelian"—is worth remembering in its own right. More generally, if $G/Z(G)$ is cyclic, the same argument shows every pair of elements commutes, so G is abelian and $Z(G) = G$. This means $G/Z(G)$ is trivial, a stronger conclusion than cyclic. In other words: *$G/Z(G)$ can never be a nontrivial cyclic group.*

Conjugation on Subsets and Subgroups

So far, conjugation has acted on *elements* of G. There is a natural extension to *subsets*: let $\mathcal{P}(G)$ denote the power set of G, and define

$$g \cdot S \;=\; gSg^{-1} \;=\; \{gsg^{-1} : s \in S\} \qquad \text{for } g \in G,\ S \subseteq G.$$

This defines a group action $G \curvearrowright \mathcal{P}(G)$. (Verification of the axioms is identical to the element case.)

Definition 4.6.8: Conjugate Subsets

Two subsets $S, T \subseteq G$ are *conjugate* in G if $T = gSg^{-1}$ for some $g \in G$.

When we restrict attention to the set of all *subgroups* of G, conjugation sends subgroups to (isomorphic) subgroups, so this restriction is also a valid action.

The stabilizer and orbit of a subset S under this action connect to familiar concepts:

- The *stabilizer* of S is
$$G_S = \{g \in G : gSg^{-1} = S\} = N_G(S),$$
the normalizer of S in G (definition 2.2.10).
- The *orbit* $G \cdot S = \{gSg^{-1} : g \in G\}$ is the set of all conjugates of S.

The Orbit-Stabilizer Theorem immediately yields:

Corollary 4.6.9: Number of Conjugates

The number of distinct conjugates of a subset $S \subseteq G$ is $[G : N_G(S)]$. In particular, S is the only member of its conjugacy class if and only if $N_G(S) = G$, i.e. if and only if $S \trianglelefteq G$ (when S is a subgroup) or S is a union of conjugacy classes (when S is an arbitrary subset).

This corollary will be indispensable in the proof of Sylow's Theorems (Section 4.8), where we count the number of conjugates of a Sylow p-subgroup.

Example 4.6.10: Conjugates of a Subgroup

Let $G = S_4$ and $H = \langle (1\ 2) \rangle = \{e, (1\ 2)\}$. The normalizer $N_{S_4}(H)$ consists of all $\sigma \in S_4$ with $\sigma H \sigma^{-1} = H$, i.e. $\sigma(1\ 2)\sigma^{-1} \in \{e, (1\ 2)\}$. Since conjugation preserves cycle type (proposition 1.2.24), $\sigma(1\ 2)\sigma^{-1}$ is always a transposition. It equals $(1\ 2)$ precisely when σ fixes the set $\{1, 2\}$, so

$$N_{S_4}(H) = \{\sigma \in S_4 : \{\sigma(1), \sigma(2)\} = \{1, 2\}\} = \{e,\ (1\ 2),\ (3\ 4),\ (1\ 2)(3\ 4)\}.$$

Therefore H has $[S_4 : N_{S_4}(H)] = 24/4 = 6$ conjugates. These are the six subgroups $\langle (i\ j) \rangle$ for $1 \le i < j \le 4$, one for each transposition.

Conjugacy in S_n and Cycle Types

Since S_n plays a distinguished role throughout this chapter, we take a moment to characterize its conjugacy classes completely. The answer is as clean as one could hope: conjugacy in S_n is determined entirely by cycle type.

Theorem 4.6.11: Conjugacy Classes in S_n

Two permutations in S_n are conjugate if and only if they have the same cycle type (definition 1.2.14).

***Proof*:**
One direction was established in proposition 1.2.24: if $\tau = \sigma\pi\sigma^{-1}$, then τ has the same cycle type as π, because conjugation by σ sends the cycle $(a_1\ a_2\ \cdots\ a_k)$ to $(\sigma(a_1)\ \sigma(a_2)\ \cdots\ \sigma(a_k))$, preserving the cycle length.

For the converse, suppose π and τ have the same cycle type. Write them in disjoint cycle notation (including 1-cycles for fixed points):

$$\pi = (a_1\ \cdots\ a_{k_1})(b_1\ \cdots\ b_{k_2})\cdots(z_1\ \cdots\ z_{k_m}),$$
$$\tau = (c_1\ \cdots\ c_{k_1})(d_1\ \cdots\ d_{k_2})\cdots(w_1\ \cdots\ w_{k_m}),$$

where the cycle lengths $k_1, k_2, \ldots, k_m$ are the same in both (this is what "same cycle type" means). Define $\sigma \in S_n$ by

$$\sigma(a_i) = c_i, \quad \sigma(b_j) = d_j, \quad \ldots, \quad \sigma(z_l) = w_l$$

for all indices. Since both decompositions account for every element of $\{1, \ldots, n\}$ exactly once, σ is a well-defined bijection. Then

$$\sigma\pi\sigma^{-1} = (\sigma(a_1)\ \cdots\ \sigma(a_{k_1}))(\sigma(b_1)\ \cdots\ \sigma(b_{k_2}))\cdots = (c_1\ \cdots\ c_{k_1})(d_1\ \cdots\ d_{k_2})\cdots = \tau.$$

So π and τ are conjugate.

Example 4.6.12: Conjugacy Classes of S_4
The partitions of 4 are $4 = 4 = 3+1 = 2+2 = 2+1+1 = 1+1+1+1$, giving five cycle types and hence five conjugacy classes:

Cycle type	*Example*	*Description*	*Class size*	$\lvert C_{S_4}(\cdot)\rvert$
(1^4)	e	identity	1	24
$(2, 1^2)$	$(1\ 2)$	transpositions	6	4
(2^2)	$(1\ 2)(3\ 4)$	double transpositions	3	8
$(3, 1)$	$(1\ 2\ 3)$	3-cycles	8	3
(4)	$(1\ 2\ 3\ 4)$	4-cycles	6	4

The class sizes sum to $1+6+3+8+6 = 24 = |S_4|$. The centralizer sizes in the last column satisfy

$|\operatorname{cl}(\pi)| \cdot |C_{S_4}(\pi)| = 24$ by (4.3).

Since $Z(S_4) = \{e\}$ (for $n \geq 3$, the center of S_n is trivial), the Class Equation reads:

$$24 = 1 + 6 + 3 + 8 + 6.$$

More generally, there is a formula for the size of each conjugacy class in S_n. If a permutation has cycle type $(1^{a_1}\, 2^{a_2}\, \cdots\, n^{a_n})$ (meaning a_k cycles of length k, with $\sum k\, a_k = n$), then the size of its centralizer is

$$|C_{S_n}(\pi)| \;=\; \prod_{k=1}^{n} k^{a_k} \cdot a_k! \tag{4.4}$$

and the conjugacy class has size $n!/|C_{S_n}(\pi)|$. (The factor k^{a_k} accounts for rotations within each k-cycle, and $a_k!$ accounts for permuting cycles of the same length.) We leave the verification as an exercise.

4.7 Automorphisms and Inner Automorphisms

In Section 4.6, the conjugation action $g \cdot a = gag^{-1}$ gave us the Class Equation by viewing each g as a *permutation* of the underlying set of G. But conjugation does more than permute elements: it preserves the group operation. That is, the map $a \mapsto gag^{-1}$ is not just a bijection $G \to G$ but an *automorphism*. Upgrading the permutation representation from $\operatorname{Sym}(G)$ to $\operatorname{Aut}(G)$ will yield one of the most important structural results in the chapter.

The Conjugation Homomorphism

Fix an element $g \in G$ and consider the map

$$\varphi_g \colon G \to G, \qquad \varphi_g(x) = gxg^{-1}.$$

Proposition 4.7.1: Conjugation by g is an Automorphism

For each $g \in G$, the map φ_g is an automorphism of G.

***Proof*:**
We verify that φ_g is a bijective homomorphism.

Homomorphism. For any $x, y \in G$:

$$\varphi_g(xy) = g(xy)g^{-1} = (gxg^{-1})(gyg^{-1}) = \varphi_g(x)\,\varphi_g(y).$$

Bijection. The map $\varphi_{g^{-1}}$ is a two-sided inverse:

$$(\varphi_g \circ \varphi_{g^{-1}})(x) = g(g^{-1}xg)g^{-1} = x = g^{-1}(gxg^{-1})g = (\varphi_{g^{-1}} \circ \varphi_g)(x).$$

Thus $\varphi_g \in \operatorname{Aut}(G)$ (definition 1.4.16).

Since each φ_g is an automorphism, we can ask whether the assignment $g \mapsto \varphi_g$ is compatible with the group structure.

Proposition 4.7.2: The Conjugation Map is a Homomorphism

The map
$$\Phi\colon G \to \mathrm{Aut}(G), \qquad \Phi(g) = \varphi_g,$$
is a group homomorphism, with $\ker(\Phi) = Z(G)$.

***Proof*:**
For any $g, h \in G$ and $x \in G$:
$$\Phi(gh)(x) = \varphi_{gh}(x) = (gh)x(gh)^{-1} = g(hxh^{-1})g^{-1} = \varphi_g(\varphi_h(x)) = (\varphi_g \circ \varphi_h)(x).$$
Since this holds for all x, we have $\Phi(gh) = \Phi(g) \circ \Phi(h)$, so Φ is a homomorphism.

The kernel consists of those g for which φ_g is the identity automorphism:
$$\ker(\Phi) = \{g \in G : gxg^{-1} = x \text{ for all } x \in G\} = \{g \in G : gx = xg \text{ for all } x \in G\} = Z(G).$$

Let us pause to compare this with the conjugation action from Section 4.6. There, we had a homomorphism $\rho\colon G \to \mathrm{Sym}(G)$ (the permutation representation of the conjugation action). Now we have a homomorphism $\Phi\colon G \to \mathrm{Aut}(G)$. Since $\mathrm{Aut}(G) \leq \mathrm{Sym}(G)$ (proposition 1.4.17), these are compatible: Φ is simply ρ with its codomain refined from $\mathrm{Sym}(G)$ to $\mathrm{Aut}(G)$. The kernel is the same in both cases—$Z(G)$—but the refined codomain gives us access to the richer structure of $\mathrm{Aut}(G)$.

Inner Automorphisms and the Isomorphism $G/Z(G) \cong \mathrm{Inn}(G)$

Definition 4.7.3: Inner Automorphism

The image of the conjugation homomorphism $\Phi\colon G \to \mathrm{Aut}(G)$ is called the group of *inner automorphisms* of G:
$$\mathrm{Inn}(G) \;=\; \Phi(G) \;=\; \{\varphi_g : g \in G\} \;\leq\; \mathrm{Aut}(G).$$
An automorphism that is *not* inner is called an *outer automorphism.*

Since Φ is a homomorphism with kernel $Z(G)$ and image $\mathrm{Inn}(G)$, the First Isomorphism Theorem (theorem 3.1.14) immediately yields:

Theorem 4.7.4: $G/Z(G) \cong \mathrm{Inn}(G)$

For any group G,
$$G/Z(G) \;\cong\; \mathrm{Inn}(G).$$

***Proof*:**
Apply the First Isomorphism Theorem to $\Phi\colon G \to \mathrm{Aut}(G)$:

$$G/\ker(\Phi) \;\cong\; \Phi(G), \qquad \text{i.e.,} \qquad G/Z(G) \;\cong\; \mathrm{Inn}(G).$$

Short as it is, this theorem encodes a great deal of structural information. It says that the "non-abelian part" of G—measured by the quotient $G/Z(G)$—is faithfully captured by the inner automorphisms. The larger the center, the fewer inner automorphisms there are, and vice versa.

Example 4.7.5: Inner Automorphisms of Familiar Groups

1. If G is abelian, then $Z(G) = G$, so $\mathrm{Inn}(G) \cong G/G = \{e\}$. Every automorphism of G is outer.
2. For $G = S_3$: $Z(S_3) = \{e\}$, so $\mathrm{Inn}(S_3) \cong S_3/\{e\} \cong S_3$. Since $|\mathrm{Aut}(S_3)| = 6 = |S_3|$ (one can verify that every automorphism of S_3 is inner), we have $\mathrm{Inn}(S_3) = \mathrm{Aut}(S_3)$.
3. For $G = D_8$: $Z(D_8) = \{e, r^2\}$, so $\mathrm{Inn}(D_8) \cong D_8/\{e, r^2\}$, a group of order 4. Since $D_8/\{e, r^2\}$ is non-cyclic (it is generated by two elements of order 2), $\mathrm{Inn}(D_8)$ is isomorphic to the Klein four-group. In fact, $|\mathrm{Aut}(D_8)| = 8$, so $[\mathrm{Aut}(D_8) : \mathrm{Inn}(D_8)] = 2$: there are automorphisms of D_8 that cannot be realized by conjugation.

The construction above has a natural generalization. Instead of conjugating G by itself, we can conjugate any subgroup H by elements that normalize it.

Theorem 4.7.6: The N/C Theorem

Let $H \le G$. The map

$$\Psi\colon N_G(H) \to \mathrm{Aut}(H), \qquad \Psi(g)(h) = ghg^{-1},$$

is a well-defined homomorphism with $\ker(\Psi) = C_G(H)$. Consequently,

$$N_G(H)\,/\,C_G(H) \;\hookrightarrow\; \mathrm{Aut}(H).$$

In particular, $N_G(H)/C_G(H)$ is isomorphic to a subgroup of $\mathrm{Aut}(H)$.

***Proof*:**
The map $\Psi(g) = \varphi_g|_H$ (conjugation by g, restricted to H) is well-defined because $g \in N_G(H)$ means $gHg^{-1} = H$ (definition 2.2.10), so φ_g sends H to H. That $\varphi_g|_H$ is an automorphism of H follows from proposition 4.7.1 by restriction.

The verification that Ψ is a homomorphism is identical to the proof of proposition 4.7.2. The kernel is

$$\ker(\Psi) = \{g \in N_G(H) : ghg^{-1} = h \text{ for all } h \in H\} = C_G(H) \cap N_G(H) = C_G(H),$$

where the last equality uses $C_G(H) \le N_G(H)$ (if g commutes with every element of H, then certainly $gHg^{-1} = H$). The embedding follows from the First Isomorphism Theorem (theorem 3.1.14).

When $H \trianglelefteq G$, we have $N_G(H) = G$, so the theorem gives $G/C_G(H) \hookrightarrow \mathrm{Aut}(H)$. When $H = G$ (which is always normal in itself), we recover $G/Z(G) \cong \mathrm{Inn}(G)$ as a special case.

4.7.7 Foreshadowing: Semi-Direct Products The N/C Theorem has an important special case. Suppose $H \trianglelefteq G$ and $K \leq G$. Then $K \leq N_G(H) = G$, so restricting Ψ to K gives a homomorphism

$$\psi\colon K \to \mathrm{Aut}(H), \qquad k \mapsto (h \mapsto khk^{-1}).$$

This homomorphism ψ encodes how K "twists" H by conjugation, and it will be the key ingredient in the definition of the *semi-direct product* (chapter 5). Knowing H, K, and ψ is enough to reconstruct the group G (under suitable conditions).

4.7.1 Outer Automorphisms

To define the quotient $\mathrm{Aut}(G)/\mathrm{Inn}(G)$, we first verify that $\mathrm{Inn}(G)$ is normal in $\mathrm{Aut}(G)$.

Proposition 4.7.8: $\mathrm{Inn}(G) \trianglelefteq \mathrm{Aut}(G)$

For any group G, $\mathrm{Inn}(G)$ is a normal subgroup of $\mathrm{Aut}(G)$.

***Proof*:**
Let $\sigma \in \mathrm{Aut}(G)$ and $\varphi_g \in \mathrm{Inn}(G)$. For any $x \in G$:

$$(\sigma \circ \varphi_g \circ \sigma^{-1})(x) = \sigma\big(g \cdot \sigma^{-1}(x) \cdot g^{-1}\big) = \sigma(g) \cdot x \cdot \sigma(g)^{-1} = \varphi_{\sigma(g)}(x).$$

Thus $\sigma \circ \varphi_g \circ \sigma^{-1} = \varphi_{\sigma(g)} \in \mathrm{Inn}(G)$, and $\mathrm{Inn}(G) \trianglelefteq \mathrm{Aut}(G)$.

Definition 4.7.9: Outer Automorphism Group

The *outer automorphism group* of G is the quotient

$$\mathrm{Out}(G) \;=\; \mathrm{Aut}(G)\,/\,\mathrm{Inn}(G).$$

The outer automorphism group measures how much of $\mathrm{Aut}(G)$ is *not* explained by conjugation. A group with $\mathrm{Out}(G) = \{e\}$ is called *complete* (when it also has trivial center); in such a group, every automorphism is inner.

Example 4.7.10: Outer Automorphisms

1. If G is abelian, then $\mathrm{Inn}(G) = \{e\}$, so $\mathrm{Out}(G) \cong \mathrm{Aut}(G)$. All automorphisms are outer.
2. $\mathrm{Out}(S_n) = \{e\}$ for $n \neq 6$: every automorphism of S_n is inner. The case $n = 6$ is exceptional: $|\mathrm{Out}(S_6)| = 2$, giving S_6 a unique outer automorphism (up to composition with inner ones).
3. $\mathrm{Out}(D_8) \cong \mathbb{Z}/2\mathbb{Z}$: there is one nontrivial coset of $\mathrm{Inn}(D_8)$ in $\mathrm{Aut}(D_8)$.

We record for later use the short exact sequence that encodes the relationship between the center, the group, and its inner automorphisms:

$$1 \longrightarrow Z(G) \hookrightarrow G \xrightarrow{\Phi} \mathrm{Inn}(G) \longrightarrow 1 \tag{4.5}$$

This sequence says: $Z(G)$ measures how far G is from being faithfully represented by its inner automorphisms. $\mathrm{Inn}(G)$ captures the "non-central" part of G; the center is precisely the part that acts trivially by conjugation.

4.7.2 Characteristic Subgroups

Normal subgroups are defined by invariance under *inner* automorphisms: H is normal in G if $\varphi_g(H) = H$ for every $\varphi_g \in \mathrm{Inn}(G)$. A natural strengthening is to require invariance under *all* automorphisms.

Definition 4.7.11: Characteristic Subgroup

A subgroup $H \le G$ is *characteristic* in G, written $H \operatorname{char} G$, if

$$\sigma(H) = H \qquad \text{for every } \sigma \in \mathrm{Aut}(G).$$

Proposition 4.7.12: Properties of Characteristic Subgroups

1. If $H \operatorname{char} G$, then $H \trianglelefteq G$.
2. If H is the unique subgroup of G of a given order, then $H \operatorname{char} G$.
3. If $H \operatorname{char} K$ and $K \trianglelefteq G$, then $H \trianglelefteq G$.

***Proof*:**

1. Since $\mathrm{Inn}(G) \le \mathrm{Aut}(G)$, invariance under all automorphisms implies invariance under inner automorphisms: $gHg^{-1} = \varphi_g(H) = H$ for all $g \in G$.
2. Let $\sigma \in \mathrm{Aut}(G)$. Since σ is an isomorphism, $\sigma(H)$ is a subgroup of G with $|\sigma(H)| = |H|$. By uniqueness, $\sigma(H) = H$.
3. Let $g \in G$. Since $K \trianglelefteq G$, the inner automorphism φ_g restricts to an automorphism $\varphi_g|_K \in \mathrm{Aut}(K)$ (because $gKg^{-1} = K$). Since $H \operatorname{char} K$, this automorphism preserves H:

   $$gHg^{-1} = \varphi_g(H) = (\varphi_g|_K)(H) = H.$$

   Since this holds for every $g \in G$, $H \trianglelefteq G$.

Item 3 is the principal reason characteristic subgroups are useful. Recall that normality is *not* transitive: $H \trianglelefteq K$ and $K \trianglelefteq G$ does *not* imply $H \trianglelefteq G$ in general (example 3.7.4). Characteristic subgroups provide a partial remedy: if the inner link is characteristic (rather than only normal), transitivity is restored.

Example 4.7.13: Characteristic Subgroups

1. The center $Z(G)$ is characteristic in G: if $\sigma \in \mathrm{Aut}(G)$ and $z \in Z(G)$, then for any $x \in G$,

$$\sigma(z)\,x = \sigma(z)\,\sigma(\sigma^{-1}(x)) = \sigma(z \cdot \sigma^{-1}(x)) = \sigma(\sigma^{-1}(x) \cdot z) = x\,\sigma(z),$$

so $\sigma(z) \in Z(G)$. Thus $\sigma(Z(G)) \subseteq Z(G)$, and applying the same argument to σ^{-1} gives equality.

2. The commutator subgroup $G' = \langle xyx^{-1}y^{-1} : x, y \in G\rangle$ is characteristic in G: any automorphism σ sends commutators to commutators ($\sigma(xyx^{-1}y^{-1}) = \sigma(x)\sigma(y)\sigma(x)^{-1}\sigma(y)^{-1}$), so $\sigma(G') = G'$.

3. Every Sylow p-subgroup of a group G is normal if and only if it is the *unique* Sylow p-subgroup. By part 2 of proposition 4.7.12, such a subgroup is automatically characteristic. We will use this in Section 4.8.

4.7.14 Remark: Normality vs. Characteristic The logical chain is:

$$\text{characteristic} \implies \text{normal} \implies \text{subgroup},$$

and neither implication reverses. A normal subgroup that is not characteristic: consider $G = \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$ and $H = \{(0,0),(1,0)\}$. Then $H \trianglelefteq G$ (since G is abelian), but the automorphism swapping the two coordinates sends H to $\{(0,0),(0,1)\} \neq H$. So H is normal but not characteristic.

We summarize the automorphism-related groups and their relationships in a single diagram:

$$\begin{tikzcd}
1 \arrow[r] & Z(G) \arrow[r, hook] & G \arrow[r, two heads, "\Phi"] & \mathrm{Inn}(G) \arrow[r, hook] \arrow[d, phantom, sloped, "\trianglelefteq"] & \mathrm{Aut}(G) \arrow[r, two heads] & \mathrm{Out}(G) \arrow[r] & 1 \\
 & & & \mathrm{Aut}(G) & & &
\end{tikzcd}$$

The top row is exact: $Z(G)$ is the kernel of Φ, $\mathrm{Inn}(G)$ is its image (hence the kernel of the projection to $\mathrm{Out}(G)$), and $\mathrm{Out}(G)$ is the cokernel. This picture captures the hierarchy at a glance: $Z(G)$ measures commutativity, $\mathrm{Inn}(G)$ measures conjugation, and $\mathrm{Out}(G)$ measures the "exotic" automorphisms that conjugation cannot produce.

4.8 Sylow's Theorems

Lagrange's Theorem (theorem 3.3.1) shows that if $H \leq G$, then $|H|$ divides $|G|$. The converse is false in general: A_4 has order 12 but no subgroup of order 6 (example 3.3.7). Sylow's Theorems provide a remarkable *partial converse*: for every prime power p^α dividing $|G|$, a subgroup of order p^α not only exists but is essentially unique up to conjugation, and the number of such subgroups is tightly constrained. These are among the most powerful and frequently used results in finite group theory, and their proofs are an applications of everything we have built in this chapter: the Orbit-Stabilizer Theorem, the Class Equation, and the coset action.

p-Groups and Sylow Subgroups

Definition 4.8.1: p-Group

Let p be a prime. A group P is called a *p-group* if $|P| = p^k$ for some $k \geq 0$. A subgroup $H \leq G$ that is a p-group is called a *p-subgroup* of G.

Definition 4.8.2: Sylow p-Subgroup

Let G be a finite group and write $|G| = p^\alpha m$ where $\alpha \geq 0$ and $p \nmid m$ (so p^α is the largest power of p dividing $|G|$). A subgroup $P \leq G$ of order p^α is called a *Sylow p-subgroup* of G. We write $\mathrm{Syl}_p(G)$ for the set of all Sylow p-subgroups of G, and $n_p = n_p(G) = |\mathrm{Syl}_p(G)|$ for their number.

Example 4.8.3: Sylow Subgroups of Small Groups

1. $|S_3| = 6 = 2 \cdot 3$. Sylow 2-subgroups have order 2: these are $\langle(1\ 2)\rangle$, $\langle(1\ 3)\rangle$, $\langle(2\ 3)\rangle$. Sylow 3-subgroups have order 3: only $\langle(1\ 2\ 3)\rangle$. So $n_2 = 3$ and $n_3 = 1$.
2. $|A_4| = 12 = 2^2 \cdot 3$. Sylow 2-subgroups have order 4; Sylow 3-subgroups have order 3.
3. If $|G| = p^\alpha$ (a p-group), then G itself is its only Sylow p-subgroup, so $n_p = 1$.
4. $|\mathbb{Z}/n\mathbb{Z}| = n$. For each prime p dividing n, the unique subgroup of order p^α (where $p^\alpha \| n$) is the unique Sylow p-subgroup. (Here uniqueness follows from the fact that $\mathbb{Z}/n\mathbb{Z}$ has exactly one subgroup of each order dividing n.)

Sylow's Theorems

Before proving Sylow's Theorems, we isolate a lemma that will be used repeatedly. It says that when a p-group acts on a finite set, the number of fixed points is congruent to the total size of the set modulo p.

Lemma 4.8.4: Fixed-Point Congruence for p-Group Actions

Let P be a finite p-group acting on a finite set X. Then

$$|X| \equiv |X^P| \pmod{p},$$

where $X^P = \{x \in X : g \cdot x = x \text{ for all } g \in P\}$ is the set of fixed points (definition 4.2.8).

***Proof*:**
The orbits of P on X partition X. By the Orbit-Stabilizer Theorem (theorem 4.3.1), each orbit has size $[P : P_x] = |P|/|P_x|$, which divides $|P| = p^k$, so every orbit size is a power of p. An orbit has size 1 (i.e. p^0) if and only if it is a fixed point. Every other orbit has size p^j with $j \geq 1$,

hence divisible by p. Therefore

$$|X| = |X^P| + \sum_{\substack{\text{non-fixed}\\ \text{orbits}}} p^{j_i} \equiv |X^P| \pmod{p}.$$

This lemma is the mechanism behind many counting arguments in p-group theory. For instance, theorem 4.6.5 (p-groups have nontrivial center) is a special case: applying the lemma to the conjugation action $G \curvearrowright G$ with $P = G$ gives $|G| \equiv |Z(G)| \pmod{p}$, so $p \big| |Z(G)|$.

Theorem 4.8.5: Sylow's Theorems

Let G be a finite group with $|G| = p^\alpha m$, where p is prime and $p \nmid m$.

1. *(Existence.)* G has a Sylow p-subgroup, i.e. a subgroup of order p^α.
2. *(Conjugacy.)* Any two Sylow p-subgroups of G are conjugate. More generally, if P is a Sylow p-subgroup and Q is any p-subgroup of G, then $Q \le gPg^{-1}$ for some $g \in G$.
3. *(Counting.)* The number n_p of Sylow p-subgroups satisfies:
   (a) $n_p = [G : N_G(P)]$ for any $P \in \mathrm{Syl}_p(G)$,
   (b) n_p divides m,
   (c) $n_p \equiv 1 \pmod{p}$.

We prove each part in turn.

***Proof of Sylow I*:**
We proceed by strong induction on $|G|$. If $|G| = 1$ (so $\alpha = 0$), the trivial subgroup has order $p^0 = 1$ and there is nothing to prove. Suppose $|G| > 1$ and that the result holds for all groups of order less than $|G|$. Write $|G| = p^\alpha m$ with $\alpha \ge 1$ and $p \nmid m$.

Consider the Class Equation (theorem 4.6.3):

$$|G| = |Z(G)| + \sum_{i=1}^{r} [G : C_G(g_i)].$$

Case 1: $p \big| |Z(G)|$. Since $Z(G)$ is abelian, Cauchy's Theorem for abelian groups (corollary 3.3.10) gives an element $z \in Z(G)$ of order p. Let $N = \langle z \rangle$. Since $N \le Z(G)$, we have $N \trianglelefteq G$, and $|G/N| = p^{\alpha-1}m < |G|$.

By the induction hypothesis, G/N has a subgroup $\overline{P}$ of order $p^{\alpha-1}$. By the Correspondence Theorem (theorem 3.5.7), there exists $P \le G$ with $N \le P$ and $P/N = \overline{P}$. Then

$$|P| = |P/N| \cdot |N| = p^{\alpha-1} \cdot p = p^\alpha.$$

Case 2: $p \nmid |Z(G)|$. Since $p \big| |G|$ and the Class Equation gives $|Z(G)| = |G| - \sum [G : C_G(g_i)]$, if p divided every term $[G : C_G(g_i)]$, then p would divide $|Z(G)|$, a contradiction. So there exists some g_j with

$$p \nmid [G : C_G(g_j)].$$

Since $|G| = |C_G(g_j)| \cdot [G : C_G(g_j)]$ and $p^\alpha \big| |G|$ while $p \nmid [G : C_G(g_j)]$, we conclude $p^\alpha \big| |C_G(g_j)|$. But $g_j \notin Z(G)$, so $C_G(g_j) \neq G$ and $|C_G(g_j)| < |G|$.

By the induction hypothesis, $C_G(g_j)$ has a subgroup P of order p^α. Since $P \leq C_G(g_j) \leq G$ and $|P| = p^\alpha$, P is a Sylow p-subgroup of G. $\square$

***Proof of Sylow II*:**
Let $P \in \mathrm{Syl}_p(G)$ and let Q be any p-subgroup of G. Consider the set of left cosets G/P, and let Q act on G/P by left multiplication: $q \cdot gP = (qg)P$.

Since $|G/P| = [G : P] = m$ and $p \nmid m$, the Fixed-Point Congruence (lemma 4.8.4) gives

$$m = |G/P| \equiv |(G/P)^Q| \pmod{p}.$$

Since $p \nmid m$, we have $(G/P)^Q \neq \varnothing$: there is at least one fixed point.

A coset gP is fixed by all $q \in Q$ if and only if $qgP = gP$ for all $q \in Q$, i.e. $g^{-1}qg \in P$ for all $q \in Q$, i.e. $g^{-1}Qg \leq P$, i.e. $Q \leq gPg^{-1}$.

So there exists $g \in G$ with $Q \leq gPg^{-1}$. If Q is itself a Sylow p-subgroup, then $|Q| = p^\alpha = |gPg^{-1}|$, and the inclusion forces $Q = gPg^{-1}$: all Sylow p-subgroups are conjugate. $\square$

***Proof of Sylow III*:**
Let $\mathcal{S} = \mathrm{Syl}_p(G)$ and $n_p = |\mathcal{S}|$.

Part (a): $n_p = [G : N_G(P)]$. The group G acts on $\mathcal{S}$ by conjugation: $g \cdot P_i = gP_ig^{-1}$. By Sylow II, this action is transitive (any two Sylow p-subgroups lie in the same orbit). The stabilizer of P under this action is

$$G_P = \{g \in G : gPg^{-1} = P\} = N_G(P).$$

By the Orbit-Stabilizer Theorem (theorem 4.3.1), $n_p = |\mathcal{S}| = [G : N_G(P)]$.

Part (b): $n_p \big| m$. Since $P \leq N_G(P) \leq G$, the index decomposes as

$$[G : P] = [G : N_G(P)] \cdot [N_G(P) : P].$$

Thus $n_p = [G : N_G(P)]$ divides $[G : P] = m$.

Part (c): $n_p \equiv 1 \pmod{p}$. Fix $P_1 \in \mathcal{S}$ and let P_1 act on $\mathcal{S}$ by conjugation. By the Fixed-Point Congruence (lemma 4.8.4),

$$n_p \equiv |\mathcal{S}^{P_1}| \pmod{p},$$

where $\mathcal{S}^{P_1}$ is the set of Sylow p-subgroups fixed by every element of P_1.

We claim $\mathcal{S}^{P_1} = \{P_1\}$. Suppose $P_j \in \mathcal{S}^{P_1}$, so $P_1 \leq N_G(P_j)$. Since $P_j \trianglelefteq N_G(P_j)$ and $P_1 \leq N_G(P_j)$, the product P_1P_j is a subgroup of $N_G(P_j)$ (proposition 3.4.4). Its order is

$$|P_1P_j| = \frac{|P_1| \cdot |P_j|}{|P_1 \cap P_j|} = \frac{p^\alpha \cdot p^\alpha}{|P_1 \cap P_j|} = \frac{p^{2\alpha}}{|P_1 \cap P_j|},$$

which is a power of p (since $P_1 \cap P_j$ is a p-group). So P_1P_j is a p-subgroup of G containing P_1. Since P_1 is a *maximal* p-subgroup (it has order p^α), we must have $P_1P_j = P_1$, which gives $P_j \leq P_1$. Since $|P_j| = |P_1| = p^\alpha$, it follows that $P_j = P_1$.

Thus $|\mathcal{S}^{P_1}| = 1$, and $n_p \equiv 1 \pmod{p}$. $\square$

Cauchy's Theorem

With Sylow's Theorems in hand, we can immediately prove the full version of Cauchy's Theorem for arbitrary finite groups, completing the proof that was deferred in theorem 3.3.8.

Theorem 4.8.6: Cauchy's Theorem

If G is a finite group and p is a prime dividing $|G|$, then G has an element of order p.

***Proof*:**
Write $|G| = p^\alpha m$ with $\alpha \geq 1$ and $p \nmid m$. By Sylow I, G has a Sylow p-subgroup P of order p^α. Let $x \in P$ with $x \neq e$. Then $|\langle x \rangle|$ divides $|P| = p^\alpha$ (theorem 3.3.1), so $|x| = p^\beta$ for some $1 \leq \beta \leq \alpha$. The element $y = x^{p^{\beta-1}}$ satisfies

$$|y| = \frac{|x|}{(|x|,\, p^{\beta-1})} = \frac{p^\beta}{p^{\beta-1}} = p,$$

where we used theorem 2.3.11. Since $y \in P \leq G$, G has an element of order p.

Note the logical structure: the abelian case of Cauchy's Theorem (corollary 3.3.10) was used *in the proof of Sylow I*, and the full Cauchy's Theorem is now derived *from* Sylow I. There is no circular reasoning.

Consequences

The following corollary collects the most frequently used structural consequences.

Corollary 4.8.7: Characterizations of a Unique Sylow Subgroup

Let $P \in \mathrm{Syl}_p(G)$. The following are equivalent:

1. $n_p = 1$ (i.e. P is the unique Sylow p-subgroup of G).
2. $P \trianglelefteq G$.
3. $P \operatorname{char} G$.

***Proof*:**
1 $\Rightarrow$ 3: If P is the unique subgroup of G of order p^α, then P is characteristic by proposition 4.7.122.

3 $\Rightarrow$ 2: Characteristic implies normal (proposition 4.7.121).

2 $\Rightarrow$ 1: If $P \trianglelefteq G$ and Q is any other Sylow p-subgroup, then by Sylow II, $Q = gPg^{-1}$ for some $g \in G$. Since P is normal, $gPg^{-1} = P$, so $Q = P$. Thus $n_p = 1$.

In particular, to show that a Sylow subgroup is normal, it suffices to show $n_p = 1$. The congruence and divisibility conditions from Sylow III often force this conclusion.

Example 4.8.8: Using Sylow's Theorems

1. Let $|G| = 15 = 3 \cdot 5$. Then $n_3 \mid 5$ and $n_3 \equiv 1 \pmod 3$, giving $n_3 \in \{1\}$. Similarly, $n_5 \mid 3$ and $n_5 \equiv 1 \pmod 5$, giving $n_5 \in \{1\}$. Both Sylow subgroups are normal and unique, each cyclic of prime order.

2. Let $|G| = 12 = 2^2 \cdot 3$. Then $n_3 \mid 4$ and $n_3 \equiv 1 \pmod 3$, giving $n_3 \in \{1, 4\}$. Also, $n_2 \mid 3$ and $n_2 \equiv 1 \pmod 2$, giving $n_2 \in \{1, 3\}$. Both possibilities genuinely occur: for A_4, $n_3 = 4$ and $n_2 = 1$; for $\mathbb{Z}/12\mathbb{Z}$, $n_3 = 1$ and $n_2 = 1$.

3. Let $|G| = 99 = 9 \cdot 11$. Then $n_{11} \mid 9$ and $n_{11} \equiv 1 \pmod{11}$. The divisors of 9 are $1, 3, 9$, and only $1 \equiv 1 \pmod{11}$. So $n_{11} = 1$ and the Sylow 11-subgroup is normal.

4.8.9 A Strategy for Sylow Problems Many problems in finite group theory reduce to the following systematic procedure:

1. Write $|G| = p^\alpha m$ with $p \nmid m$ for each prime p dividing $|G|$.

2. For each p, list the candidates for n_p: they must simultaneously divide m and be congruent to $1 \pmod p$.

3. Use the candidates to deduce structural information: if $n_p = 1$, the Sylow p-subgroup is normal; if $n_p > 1$, count the elements of order p^k across all Sylow subgroups and check for contradictions.

We will apply this strategy systematically in Section 4.9.

4.8.10 Applied Aside: Noether's Theorem in Physics While Sylow's Theorems concern finite groups, the underlying philosophy—that symmetries constrain structure—extends far beyond algebra. In 1918, Emmy Noether proved that every continuous symmetry of a physical system corresponds to a conserved quantity:

Symmetry (group action)	*Conservation law*
Time translation	Energy
Spatial translation	Momentum
Rotational symmetry	Angular momentum

Noether's Theorem uses *Lie groups* (continuous groups acting smoothly on manifolds), but the conceptual root is identical to everything in this chapter: a group acts on a space, and the structure of the action—its orbits, its stabilizers, its kernel—reveals deep invariants. The conservation laws of physics are, in a precise sense, consequences of group actions.

4.9 Applications of Sylow's Theorems

We now put the Sylow machinery to work. The goal is to demonstrate, through a sequence of increasingly sophisticated examples, how the existence, conjugacy, and counting theorems combine to constrain—and sometimes completely determine—the structure of a finite group from its order alone.

Groups of Order pq

Let $p < q$ be distinct primes and let $|G| = pq$. This is the simplest case beyond prime and prime-squared orders, and Sylow's Theorems give a nearly complete classification.

Theorem 4.9.1: Classification of Groups of Order pq

Let $p < q$ be primes and $|G| = pq$.

1. G has a unique (hence normal) Sylow q-subgroup.
2. If $p \nmid (q-1)$, then G is cyclic: $G \cong \mathbb{Z}/pq\mathbb{Z}$.
3. If $p \big| (q-1)$, then there are exactly two isomorphism classes of groups of order pq: the cyclic group $\mathbb{Z}/pq\mathbb{Z}$, and a non-abelian group (which can be constructed as a semi-direct product, see Section 5).

***Proof*:**
Part I. By Sylow III, $n_q \big| p$ and $n_q \equiv 1 \pmod q$. The divisors of p are 1 and p. Since $p < q$, we have $p \not\equiv 1 \pmod q$ (as $1 < p < q$). Therefore $n_q = 1$, and the unique Sylow q-subgroup Q is normal in G (corollary 4.8.7).

Part II. By Sylow III, $n_p \big| q$ and $n_p \equiv 1 \pmod p$. Since q is prime, $n_p \in \{1, q\}$. If $n_p = q$, then $q \equiv 1 \pmod p$, i.e. $p \big| (q-1)$. Contrapositively, if $p \nmid (q-1)$, then $n_p = 1$, and the Sylow p-subgroup P is also normal.

Now $P \cong \mathbb{Z}/p\mathbb{Z}$ and $Q \cong \mathbb{Z}/q\mathbb{Z}$ (both have prime order, hence are cyclic by corollary 2.3.12). Since $P \cap Q$ is a subgroup of both P and Q, its order divides $\gcd(p, q) = 1$, so $P \cap Q = \{e\}$. By proposition 3.4.3, $|PQ| = |P||Q|/|P \cap Q| = pq = |G|$, so $PQ = G$.

We claim G is abelian. For any $x \in P$ and $y \in Q$, the commutator $xyx^{-1}y^{-1}$ lies in Q (since $Q \trianglelefteq G$ gives $xyx^{-1} \in Q$, so $xyx^{-1}y^{-1} \in Q$) and also in P (since $P \trianglelefteq G$ gives $yx^{-1}y^{-1} \in P$, so $xyx^{-1}y^{-1} \in P$). Thus $xyx^{-1}y^{-1} \in P \cap Q = \{e\}$, giving $xy = yx$.

Since every element of $G = PQ$ is a product of an element of P and an element of Q, and these all commute, G is abelian. Let x generate P and y generate Q. Then $|xy| = \operatorname{lcm}(p, q) = pq$ (since $\gcd(p, q) = 1$), so $G = \langle xy \rangle \cong \mathbb{Z}/pq\mathbb{Z}$.

Part III. If $p \big| (q-1)$, then $n_p = q$ is possible, and a non-abelian group of order pq exists. Its construction requires the semi-direct product, which we defer to Section 5. One can show (using the semi-direct product theory) that the non-abelian group is unique up to isomorphism.

Example 4.9.2: Concrete Cases

1. $|G| = 15 = 3 \cdot 5$. Here $p = 3, q = 5$, and $3 \nmid 4$. So $G \cong \mathbb{Z}/15\mathbb{Z}$: there is only one group of order 15.
2. $|G| = 35 = 5 \cdot 7$. Here $p = 5, q = 7$, and $5 \nmid 6$. So $G \cong \mathbb{Z}/35\mathbb{Z}$.
3. $|G| = 6 = 2 \cdot 3$. Here $p = 2, q = 3$, and $2 \big| 2$. There are two groups: $\mathbb{Z}/6\mathbb{Z}$ and the non-abelian group S_3.
4. $|G| = 21 = 3 \cdot 7$. Here $p = 3, q = 7$, and $3 \big| 6$. There are two groups: $\mathbb{Z}/21\mathbb{Z}$ and a non-abelian group of order 21.

Groups of Order p^2q: A Worked Example

We illustrate the general strategy for $|G| = p^2q$ ($p \neq q$ primes) through the concrete case $|G| = 12 = 2^2 \cdot 3$.

Example 4.9.3: Sylow Analysis of Groups of Order 12
Let $|G| = 12$. The Sylow constraints give:

$$\begin{aligned} n_3 \big| 4 \text{ and } n_3 \equiv 1 \pmod 3 &\implies n_3 \in \{1, 4\}, \\ n_2 \big| 3 \text{ and } n_2 \equiv 1 \pmod 2 &\implies n_2 \in \{1, 3\}. \end{aligned}$$

Case $n_3 = 4$. There are 4 Sylow 3-subgroups, each cyclic of order 3 with 2 non-identity elements. Since distinct Sylow 3-subgroups intersect trivially (their intersection has order dividing 3 and is a proper subgroup of each, so it is $\{e\}$), there are $4 \times 2 = 8$ elements of order 3. The remaining $12 - 8 = 4$ elements (including e) must form the unique Sylow 2-subgroup. Thus $n_2 = 1$ and the Sylow 2-subgroup P is normal.

Since P has order 4, $P \cong \mathbb{Z}/4\mathbb{Z}$ or $P \cong V_4 \cong \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$ (corollary 4.6.6). The group A_4 is an example of this case, with $P \cong V_4$ and $n_3 = 4$.

Case $n_3 = 1$. The unique Sylow 3-subgroup $Q \cong \mathbb{Z}/3\mathbb{Z}$ is normal. The Sylow 2-subgroup P (order 4) acts on Q by conjugation, giving a homomorphism $P \to \mathrm{Aut}(Q) \cong \mathbb{Z}/2\mathbb{Z}$ (by theorem 4.7.6). The structure of G depends on this homomorphism: different choices yield the cyclic group $\mathbb{Z}/12\mathbb{Z}$, the dihedral group D_{12}, and the dicyclic group Dic_3 (among others, depending on whether $P \cong V_4$ or $P \cong \mathbb{Z}/4\mathbb{Z}$).

The key technique on display is *element counting*: when n_p is large, the Sylow p-subgroups consume many elements, leaving fewer elements for Sylow subgroups at other primes. This often forces $n_q = 1$ at a different prime, yielding a normal subgroup.

The same strategy applies to any order p^2q. In general:

1. Compute the Sylow constraints for n_p and n_q.
2. If both $n_p = 1$ and $n_q = 1$ are forced, both Sylow subgroups are normal, and G is a semi-direct product (or direct product) of a p-group and a q-group.

3. If $n_q > 1$ is possible, use element counting to check whether $n_p = 1$ is forced. If so, the normal Sylow p-subgroup provides structural leverage.
4. The full classification typically requires the semi-direct product machinery of Section 5.

4.9.1 Simplicity of A_5

We now prove one of the most celebrated results in group theory: the alternating group A_5 is simple. This fact is significant far beyond group theory—it is the algebraic reason that the general quintic equation cannot be solved by radicals, as Galois discovered.

We begin by determining the conjugacy classes of A_5.

Lemma 4.9.4: Conjugacy Classes of A_5

The group A_5 has five conjugacy classes, with sizes 1, 12, 12, 15, 20.

***Proof*:**
The even permutations in S_5 have four possible cycle types:

Cycle type	*Example*	*Count in A_5*
(1^5)	e	1
$(3, 1^2)$	$(1\ 2\ 3)$	$\binom{5}{3} \cdot 2 = 20$
$(2^2, 1)$	$(1\ 2)(3\ 4)$	$\frac{1}{2}\binom{5}{2}\binom{3}{2} = 15$
(5)	$(1\ 2\ 3\ 4\ 5)$	$5!/5 = 24$

Total: $1 + 20 + 15 + 24 = 60 = |A_5|$.

In S_5, permutations of the same cycle type form a single conjugacy class (theorem 4.6.11). In A_5, an S_5-conjugacy class of even permutations either remains a single A_5-class or splits into two classes of equal size. The criterion is: a class *splits* in A_5 if and only if the centralizer in S_5 consists entirely of even permutations.

- 3-*cycles:* $C_{S_5}((1\ 2\ 3))$ contains the odd permutation $(4\ 5)$. The class does not split: one A_5-class of size 20.
- $(2^2, 1)$-*type:* $C_{S_5}((1\ 2)(3\ 4))$ contains the odd permutation $(1\ 2)$. The class does not split: one A_5-class of size 15.
- 5-*cycles:* $C_{S_5}((1\ 2\ 3\ 4\ 5)) = \langle(1\ 2\ 3\ 4\ 5)\rangle \cong \mathbb{Z}/5\mathbb{Z}$, all even. The class *splits* into two A_5-classes, each of size $24/2 = 12$.

The representatives of the two 5-cycle classes are $(1\ 2\ 3\ 4\ 5)$ and $(1\ 3\ 5\ 2\ 4)$ (or equivalently $(1\ 2\ 3\ 4\ 5)$ and $(1\ 2\ 3\ 5\ 4)$—any 5-cycle not conjugate to the first within A_5). The five class sizes are therefore $1, 12, 12, 15, 20$.

Theorem 4.9.5: A_5 is Simple

The alternating group A_5 has no nontrivial proper normal subgroups.

***Proof*:**
Suppose $N \trianglelefteq A_5$ with $N \neq \{e\}$. Since N is normal, it is a union of conjugacy classes and must include $\{e\}$. The class sizes are $1, 12, 12, 15, 20$. Since $|N|$ divides $|A_5| = 60$ (theorem 3.3.1), we check all possible unions:

Sum including 1	*Divides* 60*?*
$1 + 12 = 13$	No
$1 + 15 = 16$	No
$1 + 20 = 21$	No
$1 + 12 + 12 = 25$	No
$1 + 12 + 15 = 28$	No
$1 + 12 + 20 = 33$	No
$1 + 15 + 20 = 36$	No
$1 + 12 + 12 + 15 = 40$	No
$1 + 12 + 12 + 20 = 45$	No
$1 + 12 + 15 + 20 = 48$	No
$1 + 12 + 12 + 15 + 20 = 60$	Yes (all of A_5)

No proper nontrivial union of conjugacy classes (including $\{e\}$) has order dividing 60. Therefore $N = \{e\}$ or $N = A_5$, and A_5 is simple.

4.9.6 Remark: The Quintic and Simplicity The simplicity of A_5 is the key obstruction in Galois's proof that there is no formula (using radicals) for the roots of a general polynomial of degree ≥ 5. The rough idea: the "solvability" of a polynomial equation by radicals corresponds to a property of its Galois group called *solvability* (which we will define in Section 6.1.1), and A_5 is the smallest non-solvable group. The chain of normal subgroups required by solvability cannot exist in a simple group with more than one element. Since A_5 arises as the Galois group of certain quintic polynomials, those polynomials are not solvable by radicals.

Exercise 4.9.1

1. Let $|G| = pqr$ where $p < q < r$ are distinct primes. Show that G has a normal Sylow subgroup for at least one of the primes p, q, r.

   Hint: Apply Sylow III to all three primes. If $n_r > 1$ and $n_q > 1$, count the elements of orders r and q (they contribute $(n_r)(r - 1) + (n_q)(q - 1)$ non-identity elements from distinct cyclic subgroups). Show that if all three $n_p, n_q, n_r > 1$, the total exceeds pqr.

2. Let $|G| = 30 = 2 \cdot 3 \cdot 5$.

1. Show that $n_5 \in \{1, 6\}$ and $n_3 \in \{1, 10\}$.
2. Using 4.9.1 (1), conclude that G has a normal Sylow 5-subgroup or a normal Sylow 3-subgroup (or both).
3. Deduce that G has a normal subgroup of order 15.

   Hint: If $Q \trianglelefteq G$ is a Sylow 5-subgroup, consider G/Q and use the result on groups of order 6. If the Sylow 3-subgroup R is also normal, argue using $|QR| = 15$.

3. Let $|G| = 36 = 2^2 \cdot 3^2$.

   1. Show that $n_3 \in \{1, 4\}$.
   2. If $n_3 = 4$, show that the action of G on $\mathrm{Syl}_3(G)$ by conjugation gives a homomorphism $\pi \colon G \to S_4$. Conclude that $\ker(\pi)$ is a nontrivial normal subgroup of G.

      Hint: $|G| = 36 > 24 = |S_4|$, so π cannot be injective.

   Deduce that every group of order 36 has a nontrivial proper normal subgroup (and hence is not simple).

4. Show that the only simple group of order 60 is A_5 (up to isomorphism).

   Hint: Let G be simple of order 60. Show $n_5 = 6$, giving $[G : N_G(P)] = 6$ for $P \in \mathrm{Syl}_5(G)$. The action of G on the 6 cosets of $N_G(P)$ gives an injective homomorphism $G \hookrightarrow S_6$. Argue that the image lies in A_6, and identify it with A_5.

5. Let p be an odd prime. Show that every Sylow p-subgroup of D_{2n} is cyclic and normal.

   Hint: The cyclic subgroup $\langle r \rangle \trianglelefteq D_{2n}$ has index 2.

4.10 Summary

This chapter developed the theory of group actions from first principles and applied it to obtain structural results about finite groups. We collect the main threads here for reference.

The Core Framework

A *group action* of G on a set X is a homomorphism $\rho \colon G \to \mathrm{Sym}(X)$ (definition 4.1.1), equivalently a map $G \times X \to X$ satisfying the compatibility and identity axioms (theorem 4.1.2). Every action decomposes X into *orbits* (definition 4.2.1), and the size of each orbit is governed by the *stabilizer* via the Orbit-Stabilizer Theorem (theorem 4.3.1):

$$|G \cdot x| = [G : G_x].$$

The number of orbits can be computed by Burnside's Lemma (theorem 4.3.7): $|X/G| = \frac{1}{|G|} \sum_{g \in G} |X^g|$.

Types of Actions

Type	*Defining condition*	*Consequence*
Faithful	$\ker(\rho) = \{e\}$	$G \hookrightarrow \mathrm{Sym}(X)$
Transitive	One orbit: $G \cdot x = X$	$X \cong_G G/G_{x_0}$ for any x_0
Free	All stabilizers trivial: $G_x = \{e\}$	Every orbit has size $\lvert G \rvert$
Regular	Free + Transitive	$X \cong_G G$ (left multiplication)

The Key Self-Actions

The most powerful applications came from letting G act on itself or on structures derived from itself. The following table summarizes the three principal self-actions and their consequences.

Action	*Set X*	*Orbit of x*	*Stabilizer G_x*	*Kernel*	*Key result*
Left mult. $g \cdot a = ga$	G	G (one orbit)	$\{e\}$	$\{e\}$	Cayley's Thm
Conjugation $g \cdot a = gag^{-1}$	G	$\mathrm{cl}(a)$	$C_G(a)$	$Z(G)$	Class Equation
Coset action $g \cdot aH = gaH$	G/H	G/H (one orbit)	H (at eH)	$\bigcap_g gHg^{-1}$	Index-p Thm

Two further actions played essential roles:

- *Conjugation on subgroups* ($g \cdot S = gSg^{-1}$): the stabilizer is the normalizer $N_G(S)$, and the number of conjugates of S is $[G : N_G(S)]$ (corollary 4.6.9).
- *Conjugation on Sylow subgroups*: transitivity (Sylow II) gives $n_p = [G : N_G(P)]$, and the Fixed-Point Congruence (lemma 4.8.4) gives $n_p \equiv 1 \pmod p$.

The Major Theorems

1. *Orbit-Stabilizer Theorem* (theorem 4.3.1): $\lvert G \cdot x \rvert = [G : G_x]$.
2. *Burnside's Lemma* (theorem 4.3.7): $\lvert X/G \rvert = \frac{1}{\lvert G \rvert} \sum_g \lvert X^g \rvert$.
3. *Cayley's Theorem* (theorem 4.5.3): Every group embeds into a symmetric group.
4. *The Class Equation* (theorem 4.6.3): $\lvert G \rvert = \lvert Z(G) \rvert + \sum [G : C_G(g_i)]$.
5. *p-Groups have nontrivial center* (theorem 4.6.5).
6. $G/Z(G) \cong \mathrm{Inn}(G)$ (theorem 4.7.4).
7. *The N/C Theorem* (theorem 4.7.6): $N_G(H)/C_G(H) \hookrightarrow \mathrm{Aut}(H)$.

8. *Sylow's Theorems* (theorem 4.8.5): Existence, conjugacy, and counting of Sylow p-subgroups.

9. *Cauchy's Theorem* (theorem 4.8.6): If $p \big| |G|$, then G has an element of order p.

Looking Ahead

The idea that a group *acts* on something—and that the structure of the action reveals the structure of the group—is not a technique that ends with this chapter. In the chapters ahead, group actions will reappear in new forms:

- In the study of *semi-direct products* (Section 5), the N/C Theorem and the conjugation action will be the key ingredients for constructing new groups from old ones.
- In *Galois theory*, a Galois group acts on the roots of a polynomial and on intermediate field extensions; the Fundamental Theorem of Galois Theory is, at its core, a correspondence between subgroups (stabilizers) and orbits (field extensions).
- In *module theory* (Part III), the group action framework generalizes to rings acting on abelian groups. The single condition "ρ is a homomorphism" will become the defining property of a module.

The unifying philosophy of this chapter bears repeating one last time:

> *Groups are not static objects. They are agents of symmetry, and the way to understand them is to watch how they act—on sets, on themselves, and on each other.*

4.11⋆ Group Actions in Advanced Mathematics

⋆ *Optional. This section is an aside; nothing later in the book depends on it.*

This section is addressed to the mathematically curious reader who wishes to see where the ideas of this chapter lead. No proofs are given, and nothing here is prerequisite for the rest of the book.

The theory of group actions developed in this chapter—homomorphisms to symmetric groups, orbits, stabilizers, counting—is the finite, discrete case of a theme that pervades nearly every branch of modern mathematics. In each new setting, the pattern is the same: a group acts on a structured object, and the interplay between the action and the structure produces deep theorems. We sketch three such settings.

4.11.1 Representation Theory

In this chapter, a group action was a homomorphism $\rho\colon G \to \mathrm{Sym}(X)$, where X was only a *set*. What happens if we upgrade X to a *vector space*?

A *(linear) representation* of a group G is a homomorphism

$$\rho\colon G \to GL(V),$$

where V is a vector space over a field $\mathbb{F}$ and $GL(V)$ is the group of invertible linear maps $V \to V$. Each group element acts on V not as a bijection alone, but as an *invertible linear transformation*: it respects addition and scalar multiplication.

Every group action $G \curvearrowright X$ on a finite set gives rise to a representation in a canonical way. Let $V = \mathbb{F}^X$ be the vector space with basis $\{e_x : x \in X\}$, one basis vector per element of X. Then G acts on V by permuting the basis: $g \cdot e_x = e_{g\cdot x}$. This is the *permutation representation* in the linear sense, and it converts our set-theoretic theory into linear algebra.

The power of representation theory lies in the tools of linear algebra that become available: eigenvalues, traces, direct-sum decompositions. The *character* of a representation is the function $\chi\colon G \to \mathbb{F}$ defined by $\chi(g) = \mathrm{tr}(\rho(g))$. Characters satisfy remarkable orthogonality relations, and for finite groups over $\mathbb{C}$, the irreducible characters form a basis for the space of class functions (functions constant on conjugacy classes). The number of irreducible representations equals the number of conjugacy classes—a striking connection to the Class Equation of Section 4.6.

Two landmark results illustrate the depth of the theory:

- *Maschke's Theorem (theorem 53.1.9):* If $\mathrm{char}(\mathbb{F}) \nmid |G|$, then every representation decomposes as a direct sum of irreducible representations. This is the analogue of diagonalizability for group actions.
- *Burnside's p^aq^b Theorem (theorem 53.2.24):* Every group of order p^aq^b (two prime factors) is solvable. Remarkably, the only known proof for over a century used *representation theory*, not pure group theory—a vivid demonstration that "linearizing" group actions can unlock results inaccessible by other means. (A purely group-theoretic proof was found by Thompson and others only much later.)

4.11.2 Algebraic Topology

Group actions enter topology through the theory of *covering spaces*.

Let X be a topological space and $\widetilde{X} \to X$ a covering map. The *deck transformations* (or covering transformations) are the homeomorphisms $\varphi\colon \widetilde{X} \to \widetilde{X}$ that commute with the projection to X. These form a group $\mathrm{Deck}(\widetilde{X}/X)$ that acts on $\widetilde{X}$ by homeomorphisms. When the covering is *normal* (or *regular*—note the terminology!), the deck group acts *freely and transitively* on each fiber. This is precisely a regular action in the sense of definition 4.4.11, and the base space is recovered as the orbit space:

$$X \;\cong\; \widetilde{X} \,/\, \mathrm{Deck}(\widetilde{X}/X).$$

The fundamental group $\pi_1(X, x_0)$ acts on the fiber $p^{-1}(x_0)$ by *monodromy*: lifting a loop in X to a path in $\widetilde{X}$ and tracking where the endpoint lands. This action encodes the topology of the covering:

- The *orbits* of the monodromy action correspond to the connected components of $\widetilde{X}$ (if the covering is connected, there is one orbit—the action is transitive).
- The *stabilizer* of a point in the fiber corresponds to the image of $\pi_1(\widetilde{X})$ in $\pi_1(X)$—the “surviving” loops.
- The *orbit-stabilizer correspondence* becomes the classical bijection between connected coverings of X (up to isomorphism) and conjugacy classes of subgroups of $\pi_1(X)$—a topological analogue of the Structure Theorem for Transitive Actions (theorem 4.4.6).

More broadly, whenever a group G acts on a topological space X, one can study *equivariant topology*: homology and cohomology theories that respect the action. The *equivariant cohomology* $H^*_G(X)$ combines the topology of X with the algebra of G, and its computation often reduces to understanding the fixed points and stabilizers of the action—exactly the tools developed in this chapter.

4.11.3 Category Theory

Category theory provides the most general and conceptually clean framework for understanding group actions.

A group G can be viewed as a category $\mathbf{B}G$ with a single object $*$ and one morphism $* \to *$ for each element of G, with composition given by the group operation. (The invertibility of morphisms makes $\mathbf{B}G$ a *groupoid*—a category in which every morphism is an isomorphism.)

From this viewpoint:

Algebraic concept	*Categorical translation*	*Target category*
Group action on a set	Functor $\mathbf{B}G \to \mathbf{Set}$	Sets and functions
Linear representation	Functor $\mathbf{B}G \to \mathbf{Vect}_{\mathbb{F}}$	Vector spaces
Module over a group ring	Functor $\mathbf{B}G \to \mathbf{Ab}$	Abelian groups
Topological G-space	Functor $\mathbf{B}G \to \mathbf{Top}$	Topological spaces

In each row, the pattern from definition 4.1.1 is visible: a group action is a homomorphism into the automorphism group of some object. The categorical perspective replaces "automorphism group" with "endomorphism monoid" and "homomorphism" with "functor," but the essential idea is identical.

The collection of all G-sets itself forms a category, denoted G-**Set**, whose morphisms are the G-equivariant maps (definition 4.4.5). The Structure Theorem for Transitive Actions (theorem 4.4.6) becomes the statement that every transitive G-set is isomorphic to a coset space G/H in this category. Burnside's Lemma can be reinterpreted as a statement about the *colimit* of the functor $\mathbf{B}G \to \mathbf{Set}$.

Perhaps most strikingly, the passage from group actions to module theory—which will occupy much of Part III—is simply the passage from functors into **Set** to functors into **Ab**. The group ring $\mathbb{Z}[G]$ arises as the *free abelian group* on the morphisms of $\mathbf{B}G$, and a $\mathbb{Z}[G]$-module is exactly a functor $\mathbf{B}G \to \mathbf{Ab}$. In this way, the entire theory of modules over group rings—and by extension, much of representation theory—is a natural continuation of the ideas introduced in this chapter.

A group action is a functor from a one-object groupoid.
Change the target category, and the whole of algebra unfolds.

5

Composing Groups: Direct and Semi-Direct Products

(Current intuition: We've learned that any finite group is an extension of simple groups. How now do we concretely know the structure of G given H and $G/H = K$? This can in general be a very hard question, however there are cases in which this question is much more managable: the *direct produc* and *semi-direct product*. As it turns out, if G is an extention of H by K, then the relations of G can purely be described by H and K by something called the *wreath product* which a sort of action on H given by K including the semi-direct product; this is the *Krasner–Kaloujnine universal embedding theorem*. This theorem is more difficult and of little use to the general project of introducing algebra in this book, but the spirit of undersatnding the structure of G given H and K is going to be a theme that shall crop up a lot, and deeply explored in part XI)

I got a couple of intuition on direct products

1. Adding group structure to products. There is a notion of products with sets: X, Y can give $X \times Y$. It is very easy to show that if X and Y have group structures $((X, *_X), (Y, *_Y))$, then the product of these form a very natural group. This product also doesn't break under infinitely many products too [1]

2. Sort of the dual to the previous intuition: instead of building up groups, we're going to *decompsoe* groups into direct product of groups. We've learnt how to make subgroups out of groups by looking for elements with special properties (ex. commutativity) or generating subgroups with a subset of elements. We've found smaller groups by learning about how to properly define the idea of quotienting two groups with the help of homomorphisms. Now, we'll sort of combine the two. We are going to try and decompoes groups (ex. G) to into smaller

[1] If you've seen topology, you know that this is not always true in every Category. In the **Top** Category, you will/have seen that the product topology on infinitely many spaces need some special conditions

groups whose direct product would be G (ex. $G = H \times K$), or when we *almost* have a direct product ($G = H \rtimes K$)

3. This intuition overlaps with the last point. If we have $H, K \trianglelefteq G$, so that we don't have to think about $H \cap K$, then we can form $HK = G$ (or if $H \cap K = \{e\}$, $H \times K \cong G$). We can generalize this, and have $H \trianglelefteq G$ and $K \leqslant G$, an we will from here define a new product $H \rtimes K = G$ to be able to construct some differnet groups. Remebmer how I said a lot of stuff in math defines how you're *almost* abelian? Here is another example of the idea of being commutative as being the point from which we are trying to generalize. This one will heavily rely on intuition presented about the automorphisms! TBD

5.1 Classical Definitions And Properties

You might recall example 1.1.5[4], we showed we can create a group-structure given two groups. We revisit this definition here in generality:

Definition 5.1.1: Direct Product

The ***direct product*** of $G_1 \times G_2 \times \cdots$ of groups $G_1, ..., G_n, ...$ with the operations $*_1, ..., *_n, ...$ respectively is the set of sequences $(g_1, ..., g_n, ...)$ (or n-tuples if the number of groups is finite) where $g_9 \in G_i$ with the operation defined componentwise:

$$(g_1, ..., g_n, ...) * (h_1, ..., h_n, ...) = (g_1 *_1 h_1, ..., g_n *_n h_n, ...)$$

Sometimes, the direct product is called the *external direct product* to differentiate it from the *internal direct product* (see definition 3.4.1)

Example 5.1.2

1. Think of $\mathbb{R}^2 = \mathbb{R} \times \mathbb{R}$ with component-wise addition: $(\mathbb{R} \times \mathbb{R}, (+, +))$ or $(\mathbb{R}^2, (+, +))$.
2. The groups can also be arbitrary. Let $G_1 = \mathbb{Z}$, $G_2 = S_3$ and $G_3 = GL_2(\mathbb{R})$. Then $G_1 \times G_2 \times G_3$ with the operation defined component wise is indeed a group

For the direct product, there are a few general properties you can prove as an exercise:

Proposition 5.1.3: properties of Direct Products

Let every G_i be a group.

1. $|\prod_i G_i| = \prod_i |G_i|$
2. For each fixed i, the set of elements of G which have the identity of G_j in the j^{th} position for all $j \neq i$ and arbitrary elements of G_i in position i is a subgrou pof G isomorphic to G_i:
$$G_i \cong (1_{g_1}, ..., g_i, 1_{g_{i+1}}, ...)$$
3. For each fixed i, define $\pi_i : G \to G_i$ by
$$\pi_i((g_1, ..., g_n)) = g_i$$
then π_i is a surjective homomorphsim with
$$\begin{aligned}\ker(\pi_i) &= \{(g_1, ..., g_{i-1}, 1, g_{i+1}, ..., g_n) | g_j \in G_j \text{ for all } j \neq i\} \\ &\cong G_1 \times \cdots \times G_{i-1} \times G_{i+1} \times G_n\end{aligned}$$
4. Under the identificatiosn in part (1), if $x \in G_i$ and $y \in G_j$ for some $i \neq j$, then $xy = yx$

***Proof*:**
exercise

(Comment on that last part of the proposition with $xy = yx$. Since $(X, 0)$ and $(0, Y)$ are "independent" from each other (i.e., they both kind of stay in their own lanes, when multiplying we just do component wise), it comes at the "cost", or a necessary consequence of it, is that we must treat those element having an *abelian* relation. This is another example of how something being abelian makes thing very powerful – if we want those two to be "independent", we impose an abeilan condition, we impose an abeilan condition)

Exercise 5.1.1

1. Does there exist a group such that $G \times G \cong G$? hint: [2]

   Let G be a finitely generated abelian group. Show that every subgroup is also a finitely generated abelian group.

In definition 5.1.1, the components of the direct product need not be the same. If they are the same, we can introduce the following notation:

$$G^4 = G \times G \times G \times G$$

which repesents the direct product of the group G 4 times.

We can equivalently write it this way

is equivalent to $\big\{f : \{0, 1, 2, 3\} \to G \big| \ f \text{ is a function}\big\}$ where f is any function! (Poof of this here).

[2]Think about $G = \prod_{n=1}^{\infty} \mathbb{Z}$

Example 5.1.4: Example
here

Show how this is useful in generalizing to (for example):

$$(\mathbb{Z}/2\mathbb{Z})^{\mathbb{R}} = \{f : \mathbb{R} \to \mathbb{Z}/2\mathbb{Z}\}$$

more generally, we can write A^B!

Maybe as a bonus (so stared section) show the link between A^B and the universal property of the product.

Example 5.1.5: Exercies

1. If $H \leqslant G_1 \times G_2$, is it true that there always exists subgroups $H_1 \leqslant G_1$ and $H_2 \leqslant G_2$ such that
$$H \cong H_1 \times H_2$$
if yes prove it, if not give a counter-example.

5.1.1 Fundamental theorem of Finitely Generated Abelian Groups

This theorem is stated, but not proved till till much later (chapter 14, theorem 14.1.9). The main idea that is trying to be presented here is the power of of being abelian: If we have an abelian, it is *such* a nice group that we can classify them all with isormophism to $\mathbb{Z}$ and $\mathbb{Z}_{p^\alpha}$ for appropriate power-prime. It will also help in our indeavour to classify groups. With this representation, it becomes much much easier. Here are two terms we introduce now to present the theorem:

Definition 5.1.6: Finitely Generated Group

A group is finitely generated if $\exists A \subseteq G$ such that $\langle A \rangle = G$.

Definition 5.1.7: Free Abelian Group

$\mathbb{Z}^r = \mathbb{Z} \times \cdots \times \mathbb{Z}$. $\mathbb{Z}^r$ is called the *free abelian group of rank r*.

Theorem 5.1.8: Fundamental Theorem of Finitely Generated Abelean Groups

Let G be a finitely generated group G. Then:

$$G \cong \mathbb{Z}^r \times \mathbb{Z}_{n_1} \times \cdots \mathbb{Z}_{n_s}$$

for some integers $r, n_1, ..., n_s$ with the following conditions:

1. $r \geq 0$ and $n_j \geq 2$ for all j, and
2. $n_{i+1} | n_i$ for $1 \leq i \leq s-1$

Furthermore, the expression is unique (p.173 for clarification).

For those interested in a proof of this theorem for finite group see section ref:HERE [3]

***Proof*:**
We prove this in theorem 14.1.9. We will give a purely group-theoretical proof of this fact in ref:HERE.

What we'll do for the rest of this section is give the vocabulary to talk about abelian groups.

Notice also that it's for *finitely generated abelian groups*. What about infinitely generated abelian groups? We can simplify for a moment and ask about countably generated abelian groups. We have solved this for p-groups, but we have yet to solve it in the general case:

https://math.stackexchange.com/questions/3927834/classification-of-countably-infinite-abelian-groups

Definition 5.1.9: free rank and invariant factors

r is called the *free rank* or *betti number*, while $n_1, ..., n_s$ are called the *invariant factors*. The whole expression is called the *invariant factor decomposition of G*.

will finish later

Exercise 5.1.2

1. Let $A \subseteq B$ two bwo finitely generated free abelian groups each of rank n. Show that B/A has rank 0.
2. Let $G = (\mathbb{Z}/n\mathbb{Z})^\times$. Decompose G into a product of cyclic groups with additive notation.

5.2 Recognizing Direct Products

Let's say we have a (not necessarily abelian) group G. Can G be decomposed into the direct product of two smaller groups? This question is what we'll be exploring in this section.

The 4th identity of proposition 5.1.3 gives us a very valuable insight on how to think of direct products. If we have two groups G_1 and G_2, we showed that creating a set of 2-tulpes with component-wise operation creates a new group. With the fact that components commute ($xy = yx$, if $x \in G_1$, $y \in G_2$), we can also define the direct product in terms of the following:

$$G_1 \times G_2 = \langle x, y | xy = yx, x \in G_1, y \in G_2 \rangle$$

where the operation is done when everything is "moved to the appropriate place"

Example 5.2.1: presentation example
Let $G_1 = S_3$ and $G_2 = \mathbb{Z}_5$. Take $x = (1\ 2)4$ and $y = 4(2\ 3)2$. Then

$$xy = (1\ 2)44(2\ 3)2 = (1\ 2)(2\ 3)442 = (1\ 2\ 3)2$$

[3]For those who want to see how the difficulties of generalizing this theorem to infinitely (particularly countably) generated abelian groups and want to see a particular generalization, google Prüfer Theorems

You might ask yourself what's the point of this 2nd perspective? The idea is that we can now ask ourselves when can a group be decomposed into smaller groups. In other words, if we had a group G, can we split it into two parts G_1, G_2 such that $G \cong G_1 \times G_2$?

Thus, we will explore exactly how far a two subgroups (or more generally subsets) of a group are from being commutative, and then use that information to find out when we can decompose groups.

We start with defining a way to capture *how commutative* a groups/subgroups/elements are:

Definition 5.2.2: Commutator and Commutator Group

Let G be a group, let $x, y \in G$, and let A, B be nonemtpy subset of G

1. Define $[x, y] = x^{-1}y^{-1}xy$. $[x, y]$ is called the *commutator* of x and y
2. Define $[A, B] = \langle [a, b] \big| a \in A,\ b \in B \rangle$, the group geenrated by the commutators of elements from A and B
3. Define $G' = \langle [x, y] | x, y \in G \rangle$ to be the subgroup of G generated by commutators of all element sfrom G. G' is called the *commutator subgroup* of G

Notice that if x and y commute, then $[x, y] = 1$. If all the element of G commute, then $G' = \{1\}$. Thus, the commutator captures an idea of how much the elements of G commute

Proposition 5.2.3: Commutator Subgroup Properties

Let G be a group, $x, y \in G$, and $H \leqslant G$. Then

1. $xy = yx[x, y]$
2. $H \trianglelefteq G$ if and only if $[H, G] \leqslant H$
3. for any automorphism σ of G, $\sigma([x, y]) = [\sigma(x), \sigma(y)]$, that is, $G' \mathrm{char} G$. As a consequence, G/G' is abelian
4. G/G' is the largest abelian quotient of G. That is, if $H \trianglelefteq G$ and G/H is abelian, then $G' \leqslant H$. Conversely, if $G' \leqslant H$, then $H \trianglelefteq G$ and G/H is abelian
5. (Universal Abelian Quotient) if $\varphi : G \to A$ is any homomorphism of G into an abelian group A, then φ factors through G', i.e., $G' \leqslant \ker(\varphi)$ and the following diagram commutes:

$$\begin{tikzcd} G \arrow[r] \arrow[dr, "\varphi"'] & G/G' \arrow[d] \\ & A \end{tikzcd}$$

(A word on how:) These proofs are a good review of many previous concepts covered in this textbook, and so I highly recommend you try to do all of these yourself before doing the proof – you'll gain a much stronger intuition behind commutator subgroups.

***Proof*:**

1. $xy = yxx^{-1}y^{-1}xy = xy$
2. Let $H \trianglelefteq G$. Then $gH = Hg$, meaning every element of H, $gh = h'g$ for some $h, h' \in H$. Thus
$$[H, G] = \langle h^{-1}g^{-1}hg | h \in H, g \in G\rangle = \langle hh' | h, h' \in H\rangle$$
thus, $[H, G]$ is generated by elements of H, meaning $[H, G] \leqslant H$. Conversely, if $[H, G] \leqslant G$, then for every $h^{-1}g^{-1}hg \in [H, G]$, there is an h' such that
$$h^{-1}g^{-1}hg = h' \Rightarrow g^{-1}hg = hh'$$
and since this is true for any $h \in H$, $g^{-1}Hg = H$, or if we replace g with g^{-1}, we get the more conventional $gHg^{-1} = H$, meaning $H \trianglelefteq G$.
3. Let $[x, y] \in G'$. If $\sigma \in \mathrm{Aut}(G)$, then
$$\sigma([x, y]) = \sigma(x^{-1}y^{-1}xy) = \sigma(x)^{-1}\sigma(y)^{-1}\sigma(x)\sigma(y) = [\sigma(x), \sigma(y)]$$
and since $\sigma(x), \sigma(y) \in G$, that means' σ maps commutators to commutators. Since σ is bijective (i.e., has a two-sided inverse):
$$\sigma(G') = G$$
for any σ, and so $G'\mathrm{char}G$, and so $G' \trianglelefteq G$ (proposition 4.7.12).
Since G/G' is well-defined, notice that for $[x], [y] \in G/G'$,
$$[xy] = [xyy^{-1}x^{-1}yx] = [yx]$$
and so, G/G' is abelian.
4. Let $H \trianglelefteq G$ such that G/H is abelian. Then for any $[x], [y] \in G/H$, $[xy] = [yx]$, implying
$$[xyx^{-1}y^{-1}] = [1] \Rightarrow xyx^{-1}y^{-1} \in H$$
showing that $G' \leqslant H$. Conversely, if $G' \leqslant H$, then for any $g \in G$ and any $h \in H$
$$ghg^{-1}h^{-1} \in H \Rightarrow ghg^{-1}h^{-1} = h'$$
for some $h' \in H$, but then $ghg^{-1} = h'h$. Since h, h', g were all arbitrary, that means that $gHg^{-1} = H$ for any g, and so H is normal, and so G/H is well-defined. Furthermore, for any $[x], [y] \in G/H$,
$$[xy] = [xyy^{-1}x^{-1}yx] = [yx]$$
since $y^{-1}x^{-1}yx \in H$. Thus, G/H is also abelian.
Another way of proving this is by taking advantage of isomorphism theorems (take a moment to think about it if you didn't see it this way as good practice!).
If G/G' is abelian, then every subgroup is normal, and since $G' \leqslant H$, $H/G' \trianglelefteq G/G'$ is well-defined. By the 4th isomorphism theorem, $H \trianglelefteq G$, and so G/H is well-defined. Then, by the 3rd isomorphism theorem:
$$G/H \cong (G/G')(H/G')$$
and since G/G' is abelian, so is it's quotient, and so G/H is abelian.

5. This is another way of writing (4), but in terms of homomorphisms.

   Let $\varphi : G \to A$ be a homomorphism from G to any abelian group G, and let $\ker(\varphi)$ be the kernel. We need to find a map from G to G/G' and a map G/G' to A such that the diagram commutes, that is, if π is the map from G ot G/G' and ψ is the map from G/G' to A, then $\varphi = \psi \circ \pi$.

   Since G is abelian, $G/\ker(\varphi)$ is abelian. Thus, by part (4), $G' \leqslant \ker(\varphi)$. Since G' is also a normal subgroup of G, G/G' is well-defined, the natrual projection map exists and is well-defined:

   $$\pi : G \to G/G' \qquad \pi(g) = gG'$$

   Now, for the map from G/G' to A, we need to make sure that it splits with φ, meaning it is dependent on where φ sends it's elements, and that it is independent of the representatives of G/G', that is, if

   $$[g] = [gx^{-1}y^{-1}xy] \Rightarrow \psi(g) = \psi(gx^{-1}y^{-1}xy)$$

   In fact, φ with the co-domain of G/G' will do just that. Let $\psi = \overline{\varphi}$ represent the function:

   $$\overline{\varphi} : G/G' \to A \qquad \overline{\varphi}([g]) = \varphi(g)$$

   this function is well-defined, since if $[g] = [gx^{-1}y^{-1}xy]$ for any $x, y \in G$

   $$\begin{aligned}
   \overline{\varphi}(gx^{-1}y^{-1}xy) &= \varphi(gx^{-1}y^{-1}xy) && \text{by definition}\\
   &= \varphi(g)\varphi(x^{-1})\varphi(y^{-1})\varphi(x)\varphi(y) && \varphi \text{ is a homomorphism}\\
   &= \varphi(g)\varphi(x)\varphi(x^{-1})\varphi(y)\varphi(y^{-1}) && A \text{ is abelian}\\
   &= \varphi(g)\varphi(xx^{-1})\varphi(yy^{-1}) && \varphi \text{ is a homomorphism}\\
   &= \varphi(g)\varphi(1)\varphi(1) &&\\
   &= \varphi(g) && \varphi(1) = 1\\
   &= \overline{\varphi}([g]) && \text{by definition}
   \end{aligned}$$

   Thus, this function is indeed well-defined! It came down to φ already being defined for us, and A being abelian. One should also check that $\overline{\varphi}$ is indeed a homomorphism (exercise).

   Finally, we need to show that $\varphi = \overline{\varphi} \circ \pi$. If $g \in g$. Then

   $$(\overline{\varphi} \circ \pi)(g) = \overline{\varphi}(\pi(g)) = \overline{\varphi}([g]) = \varphi(g)$$

   and since this is true for all $g \in G$, then $\varphi = \overline{\varphi} \circ \pi$, meaning φ does indeed split! Thus, the diagram does indeed commute:

   $$\begin{tikzcd} G \arrow[r] \arrow[dr, "\varphi"] & G/G' \arrow[d] \\ & A \end{tikzcd}$$

(develop universal property and commutative groups)

Example 5.2.4

1. If G is abelian, then for any $x, y \in G$, $[x, y] = xyx^{-1}y^{-1} = xx^{-1}yy^{-1} = 1$, and so $G' = 1$, and $G/G' \cong G$.
2. Example with D_8
3. Example with D_n
4. Example with S_5
5. Example with F_2

With this, we can answer now give criterion to recognize direct products. First, a lemma:

Lemma 5.2.5: Representation's of HK

Let H and K be subgrousp of a group G. The number of distinct ways of writing each element of the set HK in the form hk, for some $h \in H$ and $k \in K$ is $|H \cap K|$. In particualr, if $H \cap K = 1$, then each element of HK can be written uniquely as a product of hk, for some $h \in H$ and $k \in K$.

***Proof*:**
see proposition 3.4.3

Theorem 5.2.6: Recognition Theorem For Direct Products

Let G. If H, K are subgroups such that

1. H and K are normal in G, and
2. $H \cap K = \{1\}$

Then

$$HK \cong H \times K$$

Conersly, if $HK \cong H \times K$, then $H \cap K = \{1\}$ and both H and K are normal subgroups of G.

If $HK = G$, then $G \cong H \times K$. Similarly, If G is isomorphic to the diret product of M and N (where M and N need not be subgroups of G) if $H \cong M$, $K \cong N$, and $G = HK$.

The propety that $H \cap K = 1$ has a name:

Definition 5.2.7: Complement

Let H be a subgroup of G. A subgroup K of G is called a *complement* for H in G if $G = hK$ and $H \cap K = 1$.

Thus, we can rephrase the previous theroem as a group is a derict product if there exists a normal complement for some proper normal subgroup of G.

***Proof*:**
We'll first prove a lemma that we'll use for both direction: If HK is a group and $|H \cap K| = 1$, then $hk = kh$.

Proof. Since $H \trianglelefteq G$, $khk^{-1} \in H$, and so $(khk^{-1})h^{-1} \in H$. Similarly, $k(hk^{-1}h^{-1}) \in H$. Since $H \cap K = 1$, we have

$$khk^{-1}h^{-1} = 1 \Rightarrow kh = hk$$

showing that every element of H commutes with every element of K. Note that does does not show that HK is abelian. For example, we did not show that for any $k, k' \in HK$, $kk' = k'k$. $\square$

If $HK \cong H \times K$, we know that $|HK| = |H \times K| = |H||K|$, implying that $|H \cap K| = 1$, fulfilling the first requirement (ONLY IN THE FINITE CASE!!). Next, since HK is indeed a group (being isomorphis to $H \times K$), then by corolalry 3.4.5, at least one of H or K are normal. We need to show that both H and K are normal. We know by proposition 5.1.3 that $(H, 1)$ and $(1, K)$ are normal. We can't just push H or K through to $(H, 1)$ or $(1, K)$ since we don't have control over what isomorphism we have, and we can't force H to map to $(H, 1)$ [a]. Since $HK \cong H \times K$ their pre-image are also normal groups. Let's say φ is the isomorphism. Then since both of these preimage are isomorphic to H and K repsectively, then we can create an isomorphism $\psi : HK \to HK$ where $\psi(hk) = \varphi^{-1}(h, k)$. Since φ^{-1} is an isomorphism, this map is well-defined and bijective. THis map is also a homomorphism since:

$$\begin{aligned}
\psi(h_1k_1h_2k_2) &= \psi(h_1h_2k_1k_2) && \text{lemma at the beginning} \\
&= \varphi^{-1}((h_1h_2, k_1k_2)) \\
&= \varphi^{-1}((h_1, k_1) \cdot (h_2, k_2)) \\
&= \varphi^{-1}(h_1, k_1)\varphi^{-1}(h_2, k_2) \\
&= \psi(h_1k_1)\psi(h_2'k_2)
\end{aligned}$$

Thus, H and K can be asociated with $(H, 1)$ and $(1, K)$, and must also be normal – completing this direciton.

For the other way, we need to find an isomorphism from HK to $H \times K$. We'll first establish some properties of HK to show such a function exists.

Since H and K are normal, then *a fortiori*, we met the condition for HK to be a subgroup by proposition 3.4.4, and so by the lemma done earlier, every element of H commutes with elements of K.

Now, by lemma 5.2.5, the elements of HK can be uniqueley written as hk, and so we can define the function

$$\varphi : HK \to H \times K \qquad hk \mapsto (h, k)$$

This function is a homomorphism, since $h_1k_1h_2k_2 = h_1h_2k_1k_2$. Since the right hand side is of the form $h'k'$ for some $h' \in H$, $k' \in K$, this is the unique way of representing it, and so:

$$\begin{aligned}
\varphi(h_1k_1h_2k_2) &= \varphi(h_1h_2k_1k_2) \\
&= (h_1h_2, k_1k_2) \\
&= (h_1, k_1)(h_2k_2) \\
&= \varphi(h_1k_1)\varphi(h_1k_2)
\end{aligned}$$

From here, φ is a bijection, since every element of $H \times K$ must uniquely be mapped to by an element hk in HK, and so φ is an isomrophism.

[a]Ex. Take $(\mathbb{Z}/6\mathbb{Z})(\mathbb{Z}/6\mathbb{Z}) \to \mathbb{Z}/6\mathbb{Z} \times \mathbb{Z}/6\mathbb{Z}$ and map the element order 2 and 3 into different places in the image

Another way of thinking about this theorems is by saying that the *interal direct product* (definition 3.4.1) and *external direct product* (definition 5.1.1) of two subgroups H and K of G are equivalent if the two conditions in theorem 5.2.6 match. If $H \cap K \neq \{1\}$ or one of them is not normal, then $HK \ncong H \times K$. In the next section, we'll explore a generalization of the direct product, the *semi-direct product*, and show that HK can *always* be written as $HK \cong H \rtimes K$, where $\rtimes$ represents the semi-direct product

Note that if we have an abelian group G, finitely generated or not, (theorem 5.1.8 can't be used), and we have $H, K \leqslant G$, (really $\trianglelefteq G$ since G is abelian), then it's always the case that $HK \cong H \times K$. This is another example of the niceties of being commutative.

5.2.1 A word on Perfect Groups

We saw that if a group G is abelian, then $G' = \{1\}$. What if we had the complete opposite: if $G = G'$? Such a group has a name:

Definition 5.2.8: Perfect Group

Let G be a group and G' be the commutator subgroup. If $G = G'$, then G is called a *prefect group*

Example 5.2.9: Perfect Groups

1. A_5 is an example of a perfect group.

I'm not sure yet how yet they are important. They seemed to be linked to solvable groups.

5.3 Semi-Direct Products

As we mentioned earlier, not all groups can be broken down into [non-trivial] direct products:

Example 5.3.1: $G \ncong H \times K$

Let $G = D_6 = \langle \sigma, r | \sigma^2 = r^3 = 1,\ \sigma r = r^{-1}\sigma \rangle$. By Lagrange, we know non-trivial proper subgroups of D_6 would have order 2 or 3. We can computaitonly determine that $\langle \sigma \rangle$ and $\langle r \rangle$ are the only subgroups of order 2 and 3 respectively. Let $\langle \sigma \rangle$ represent the subgroup of order 2, and $\langle r \rangle$ represent the subgroup of order 3. Since $|D_6| = 6$, then $\langle r \rangle \trianglelefteq D_6$ (corollary 3.3.5). Since one of the subgroups are normal, by proposition 3.4.4 $\langle r \rangle \langle \sigma \rangle$ is a group, and by proposition 3.4.3, the cardinality of the group is 6. Since $|D_6| = 6$, it must be that $\langle r \rangle \langle \sigma \rangle \cong D_6$. However, we saw under the definition of 3.1.12 that $\langle 2 \rangle \ntrianglelefteq D_6$, and so by the recognition theorem for direct products:

$$D_6 \ncong \langle \sigma \rangle \times \langle r \rangle$$

If there were an isomorphism between these two, then no matter what isomorphism we chose, we will encounter a problem from the fact that $rs = sr^{-1}$. let's say φ was some isomorphism from $\langle s\rangle\langle r\rangle$ to D_6. Then $\varphi(ab) = (s, r)$ for appropriate a, b where $a \in \{e, \sigma\}$ and $b = \{e, r, r^2\}$ Then:

$$(e, r^2) = (s, r) \cdot (s, r) = \varphi(abab) = \varphi(aab^{-1}b) = \varphi(e) = (e, e)$$

but $(e, r^2) \neq (e, e)$, and therefore this isomrophism *must* fail, no matter how it's defined.

Since, $\langle\sigma\rangle \cong \mathbb{Z}/2\mathbb{Z}$ and $\langle r\rangle \cong \mathbb{Z}/3\mathbb{Z}$, we can swtich over to that notation. Then we can say that

$$D_6 \ncong \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/3\mathbb{Z}$$

The question might arise on whether there is a suitable generalization of the direct product to classify the above group. More generally, is there a way to capture the structure of HK, where H and K are *arbitrary* groups (i.e. not subgroups of a larger group G). That question is explored in this section. Note that if G was abelian then automatically, $K \trianglelefteq G$, since every subgroup of an abelian group is normal. Thus, this distinction becomes interesting only when G is non-abelian.

The way we'll do this is by trying to fix the fact that elements of the group H and K will no longer always commute with each other. Note that we always have that $HK = KH$ (proposition 3.4.4), but that didn't imply $hk = kh$. In the dihedral example, we had $rs = sr^{-1}$. If we were to try to write this as a product, we'd get

$$(r, 1)(1, s) = (1, s)(r^{-1}, 1)$$

Another example would be S_3. The group S_3 is another group of order 6 with subgroups of order 2 and 3, for example: $(1\ 2)$ and $(1\ 2\ 3)$). Since $|S_3| = 6$, we again have $|(1\ 2\ 3)| \trianglelefteq S_3$ and $\langle(1\ 2\ 3)\rangle\langle(1\ 2)\rangle \cong S_3$. However, one can easily check that $(1\ 2) \ntrianglelefteq S_3$ (try taking $g = (1\ 2\ 3)$ and see if $g\langle(1\ 2)\rangle g^{-1} = \langle(1\ 2)\rangle$). We again get that $ab \neq ba$, and instead, they commute in a different manner. let $a = (1\ 2)$ and $b = (1\ 2\ 3)$. Then you can verify that:

$$ab = b^2 a$$

meaning when a and b comute, b becomes b^2. Thus, we can write

$$S_3 = \langle a, b | a^2 = b^3 = 1,\ ab = b^2 a\rangle$$

If were to try to write this as a product, we'd get that

$$(a, 1)(1, b) = (1, b^2)(a, 1)$$

So the question is "Is there a more general way to capture how elements in a group commute"?

As a way to motivate the coming definition, Let's start by having a larger group G, and recall that to define HK as a group, we need $H \trianglelefteq G$ and $K \leqslant G$.

Since HK is a group, we already have that $(h_1k_1)(h_2k_2) = h_3k_3$ for some $h_3k_3 \in HK$ (similarly to how in a direct product, $(a_1, b_1)(a_2b_2) = (a_1a_2, b_1b_2) = (a_3, b_3)$). More precicely, we take $h_1k_1, h_2k_2 \in HK$, we can re-write $h_1k_1h_2k_2$ as $h_3k_3 \in HK$. We showed this in proposition 3.4.4. Doing it again here, notice exactly how we get to write $h_1k_1h_2k_2$ as h_3k_3:

$$\begin{aligned}(h_1k_1)(h_2k_2) &= h_1k_1h_2(k_1^{-1}k_1)k_2 \\ &= h_1(k_1h_2k_1^{-1})k_1k_2 \\ &= (h_1(k_1h_2k_1^{-1}))k_1k_2 \\ &= h_3k_3\end{aligned}$$

where $h_3 = h_1(k_1 h_2 k_1^{-1})$ (since $H \trianglelefteq G$, $gh_1g^{-1} \in H$) and $k_3 = k_1k_3$.

To get to this construction, we heavily relied on *starting* with the group G and all the properties of the subgroups H, and K (ex. $H \cap K = \{e\}$, $H \trianglelefteq G$). Relying so heavily on a bigger group G does not help in our generalization of the direct product: when defining an arbitrary direct product $G_1 \times G_2$, we did not rely on some bigger group G from which G_1 and G_2 were normal subgroups. Thus, we have to figure out a way to do this construction by only relying on information on H and K.

To do this, we must be able to describe the operation between $(h_1k_1) \cdot (h_2k_2)$ without using the normality of $H \trianglelefteq G$. It is easy for the k elements since $k_3 = k_1k_2$, so the question about about defining $k_1h_2k_1^{-1}$.

We have in fact already encountered a solution to this problem. Since H is normal in G, the group K *acts* on H by conjugation:

$$k \cdot h = khk^{-1} \qquad \text{for } h \in H, k \in K$$

Since H is normal, $khk^{-1} = h' \in H$, meaning if we know how every k acts on every h, we can drop the necessity for a larger group G. In fact, all this information *can* be captured with a group-action:

$$\varphi : K \to \text{Aut}(H)$$

meaning we now only rely on the product of K, H, and the homomorphism φ which captures how H and K interact. Importantly for defining a group with this, $\text{Aut}(H) = \text{Aut}_{\mathbf{Grp}}(H)$, not $\text{Aut}_{\mathbf{Set}}(H)$, and so

$$\varphi(x)(ab) = \varphi(x)(a)\varphi(x)(b)$$

Thus, we've eliminated the need for a larger group G!

With this idea, we can define a our new product:

Definition 5.3.2: Semi-Direct Product

Let H and K be groups and let φ be a homomorphism form K into $\text{Aut}(H)$. Let $\cdot$ denote the (left) action of K on H determined by φ. Let G be the set of ordered pairs (h, k) with $h \in H$ and $k \in K$ and define the following multiplication on G:

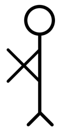
$$(h_1, k_1)(h_2, k_2) = (h_1k_1 \cdot h_2, k_1k_2)$$

we usually denote $G = H \rtimes_\varphi K$, or $H \rtimes K$ if the context permits it

Notice how we are not using a bigger group, but a lot of intuition can be derived from thinking about H and K as subgroups of a larger group, showing how subgroups still play an important role in the analysis and conception of semi-direct products, but it is certainly not necessary as we will point out in the examples.

The $\rtimes$ notation is to remember that H is the normal subgruop of $H \rtimes K$ (as we'll prove soon). It might be evident that in most cases $H \rtimes K \ncong K \rtimes H$, since we'd be defining different actions. More on this when we show some examples.

Another way of remembering which group is acting on which is with a pop-culture reference (curtosy of one of my professors): if you are aware of what a "Kamehameha" is from dragon ball, then the symbol "$\rtimes$" can be thought of as Goku's hand preforming the Kamehameha.



If you're unaware of the cultural reference, here's a youtube video I found (or you can google it):

https://www.youtube.com/watch?v=_ct3ccrU4AI

As silly as it may be, this turned out to be the most efficient way for me to remember which groups acts on which.

Before we get too hasty, we should prove that this is indeed a group and that all the properties we were talking about earlier hold:

Proposition 5.3.3: Properties Of Semi-Direct Product

1. This multiplication makes G into a group of order $|G| = |H||K|$
2. The sets $\{(h,1)|h \in H\}$ and $\{(1,k)|k \in K\}$ are subgroups of G and the maps $h \mapsto (h,1)$ and $k \mapsto (1,k)$ for $k \in K$ are isomorphisms of these subgroups with the groups H and K respectively:
$$H \cong \{(h,1)|h \in H\} \qquad K \cong \{(1,k)|k \in K\}$$
3. $H \trianglelefteq G$
4. $H \cap K = \{e\}$
5. for all $h \in H$ and $k \in K$, $khk^{-1} = k \cdot h = \varphi(k)(h)$

***Proof*:**

1. To show $H \rtimes K$ is a group, we need to show it obeys the axioms of a group
   (a) **associativity**. Let $(a,x),(b,y),(c,z) \in H \rtimes Z$. Then
$$\begin{aligned}((a,x)(b,y))(c,z) &= (a\ x \cdot b, xy)(c,z)\\ &= (a\ x \cdot b\ (xy) \cdot c, xyz)\\ &= (a\ x \cdot b\ x \cdot (y \cdot c), xyz)\\ &= (a\ \varphi(x)(b)\ \varphi(x)(y \cdot c), xyz)\\ &= (a\ \varphi(x)(b\ y \cdot c), xyz)\\ &= (a\ x \cdot (b\ y \cdot c), xyz)\\ &= (a,x)(by \cdot c, yz)\\ &= (a,x)((b,y),(c,z))\end{aligned}$$
   where I switched over to homomorphism notation to show that since $\varphi(x)$ is a homomorphism (note that $\varphi(x) : K \to K$, not $\varphi : K \to K$), then $\varphi(x)(st) = \varphi(x)(s)\varphi(x)(t)$.

(b) **Identity**: I claim that $(1,1)$ is an identity. In particular, for any $(x,y) \in H \rtimes K$

$$(1,1)(x,y) = (1\ 1 \cdot x, 1y) = (x,y) = (x,y)(1,1)$$

(c) **Inverse**: I claim that for any $(h,k) \in H \rtimes K$, $(h,k)^{-1} = (k^{-1} \cdot h^{-1}, k^{-1})$ is an inverse. Indeed:

$$\begin{aligned}(h,k)(k^{-1} \cdot h^{-1}, k^{-1}) &= (hk \cdot (k^{-1} \cdot h^{-1}), kk^{-1}) \\ &= (h(kk^{-1}) \cdot h^{-1}, 1) \\ &= (h\ 1 \cdot h^{-1}, 1) = (hh^{-1}, 1) \\ &= (1,1)\end{aligned}$$

and similarly for $(h,k)^{-1}(h,k)$.

Thus, $H \rtimes K$ is a group.

2. Let $\hat{H} = \{(h,1) | h \in H\}$ and $\hat{K} = \{(1,k) | k \in K\}$. Notice that for any $a, b \in H$,

$$(a,1)(b,1) = (ab \cdot 1, 1) = (ab, 1)$$

and for any $a, b \in K$

$$(1,a)(1,b) = (1a \cdot 1, ab) = (1, ab)$$

where $a \cdot 1 = 1$ was shown in ref:HERE (in the Aut**Grp**(G) section should be said we can now do)

$$\varphi(x)(a+b) = \varphi(x)(a) + \varphi(x)(b)$$

leading us to this fact.

Thus, we can see that the inculsion map is a homomorphism, and that the map from H to $\hat{H}$ and K to $\hat{K}$ is isomorophic. By this identification,

3. Since $K \cong \hat{K}$ and $H \cong \hat{H}$, it's clear then that $H \cap K = \{1\}$.

4. If we identify elements k with $(1,k)$ and h with $(h,1)$ Then notice that

$$\begin{aligned}khk^{-1} &= (1,k)(h,1)(1,k)^{-1} \\ &= ((1,k)(h,1))(1,k^{-1}) \\ &= (k \cdot h, k)(1, k^{-1}) \\ &= (k \cdot h\ k \cdot 1, kk^{-1}) \\ &= (k \cdot h, 1) \\ &= k \cdot h\end{aligned}$$

i.e., we have that $khk^{-1} = k \cdot h$. In other words $kh = \varphi(k)(h)k$, so commuting kh made k act on h.

5. Apparnelty, $\hat{K} \leqslant N_G(\hat{H})$. Since $G = HK$ (I guess from earliers intuition?) then "certainly" $H \leqslant N_G(H)$, so we have $N_G(H) = G<$ i.e, $H \trianglelefteq G$, completing (3)

Point (4) in particular is worth pointing out since this is were we generalized our intuition. In the coming example, play close attention to exactly how the action commute elements.

Remember that the semi-direct products are a generalization of direct products. Exactly how is what this next theorem identifies:

Proposition 5.3.4: When The Semi-Direct Product is a Direct Product

Let H and K be groups and let $\varphi : K - \text{Aut}(H)$ be a homomorphism. Then the following are equivalent:

1. The identity (set) map between $H \rtimes K$ and $H \times K$ is a group homomorphism hence an isomrophism
2. φ is the trivial homomorphism form K into $\text{Aut}(H)$
3. $K \trianglelefteq H \rtimes K$

***Proof*:**
p.178

Example 5.3.5: Semi-Direct Products

1. Let H be any abelian group (even of infinite order) and let $K = \langle x \rangle \cong Z_2$. be the group of order 2. Define $\varphi : K \to \text{Aut}(H)$ by mapping x to the automorphism of inversion on H, so that the associated action is $x \cdot h = h^{-1}$, or, written in homomorphism form:

$$\varphi(x)(h) = h^{-1} \qquad \varphi(e)(h) = h$$

Note that this is an automorphism, for

$$f_x : H \to H, \qquad f_x(h) = h^{-1} \qquad f_x(xy) = (xy)^{-1} = x^{-1}y^{-1} = f_x(x)f_x(y)$$

since H is abelian, the map is it's own inverse (making it bijective).

Thus, we can define $H \rtimes_\varphi K$. To distinguish between the group-action and the group operation, I'll ues the $+$ symbol to represent the operation for both groups. To get comfortable with this, let's do a computation:

$$(a_1, x)(a_2, x) = (a_1 + x \cdot a_2, x + x) = (a_1 a_2^{-1}, 0)$$

$$\begin{aligned} xhx^{-1} &= (0, x)(h, 0)(0, x^{-1}) \\ &= (0 + x \cdot h, 0 + x)(0, x^{-1}) = (-h, x)(0, x) \\ &= (-h + x \cdot 0, x + x) \\ &= (-h - h, 0) \\ &= (-h, 0) \\ &= h^{-1} \end{aligned}$$

Thus, we have:

$$xhx^{-1} = h^{-1} \Rightarrow xh = h^{-1}x$$

in the special case where $H = Z_n$, then $\mathbb{Z}/n\mathbb{Z} \rtimes \mathbb{Z}/2\mathbb{Z} \cong D_n$!!! if $H = \mathbb{Z}$, then we'd say $G = D_\infty$.

2. Let $H = \mathbb{R}^2$ and $K = SO(2)$.

3. Since simple groups have no normal subgroups, simple groups cannot be decomposed into semi-direct products, showing further how they are like "prime numbers" for groups.

More examples in the book p.178

With the idea of the semi-direct product in mind and a couple of examples, we end this section by given sufficient and necessary conditions to recognize semi-direct products:

Theorem 5.3.6: Recognition Theorem For Semi-Direct Products

Suppose G is a group with subgrosup H and K such that

1. $H \trianglelefteq G$
2. $H \cap K = 1$

Let $\varphi : K \to \mathrm{Aut}(H)$ be the homomorphism define dby mapping $k \in K$ to the automrophis mof left conjugation by k on H. Then

$$HK \cong H \rtimes K$$

In particualr, if $G = HK$, with H and K satisfying the aforementioend conditions, then

$$G \cong H \rtimes K$$

Put another way, a group is a semi-derict product if there exists a complement for some proper normal subgroup of G.

***Proof*:**
p. 180

(A word on how not all groups are semi-direct products. Show example. So Semi-direct product is actually a special kind of extension (will show in exercises). This type of extensions (called a split extension) will come back again later with modules (part III, chapter 17⋆).

There is a product that does represent any extension by the two groups on the left and right (sort of); it is called teh wreath product, and it comes along with the universal embedding theorem.

https://en.wikipedia.org/wiki/Wreath_product

Definition 5.3.7: Split Sequecne

Given a short exact sequence

$$1 \longrightarrow A \longrightarrow B \longrightarrow C \longrightarrow 1$$

we say that it *splits* if there eexists a homomorphism $\psi : H \to G$ such that $\psi \circ f = \mathrm{id}_G$ (so in a sense, also embedd C in B). We can write this as the commutative diagram:

$$\begin{tikzcd}
1 \arrow[r] & A \arrow[r] & B \arrow[r, "f"] & C \arrow[r] & 1 \\
& & & C \arrow[u, "\mathrm{id}"'] \arrow[ul, "\psi"] &
\end{tikzcd}$$

Point out how with this we can make $B \cong C \rtimes_\alpha \ker(\varphi) = C \rtimes A$, where the action $\alpha : C \to \mathrm{Aut}(A)$, $\varphi(c)(a) = c \cdot a = \psi(c)^{-1} a \psi(c) [\in A]$.

If the action is trivial, then the extension can be represented as a direct product

$$1 \longrightarrow A \longrightarrow A \times C \longrightarrow C \longrightarrow 1$$

So is every short exact sequence split? The answer is unfortunately no

Example 5.3.8: Counter-Example

Take

$$1 \longrightarrow C_2 \longrightarrow C_4 \longrightarrow C_2 \longrightarrow 1$$

With the first component mapping $0 \mapsto 0$ and $1 \mapsto 2$. while in the 2nd map, $0, 2$ map to 0 and $1, 3$ map to 1.

5.4 A Categorical approach of the Product

The idea of being able to take the "product" of two objects is not unique to sets and groups. As we continue to explore more and more algebraic structures, we will keep asking ourselves a similar question: Does algebraic structure X have the flexibility/restriction to define products? [4] What we'll be doing then is finding a property which we can use to define what a product is in different algebraic settings.

Let's pick two arbitrary groups A and B. We defined product in the classical definition as the new group $(A \times B, (\cdot_A, \cdot_B))$. Let's denote this new group C. The key idea is that C is a group that contains groups A and B, and you can "recover" A and B from C. The way we can recover these two groups from C is simply by projcting to them:

$$\pi_A : C \to A,\ (a, b) \mapsto a \qquad \pi_B : C \to B,\ (a, b) \mapsto b$$

[4]As usual with category theory, the question of whether the product exists goes beyond algebra: Topological spaces, uniform spaces, graphs, ordered sets, and other domains of mathematics will also have products!

It should be checked that π_A and π_B are both surjective homomorphisms. Notice that $\pi_A(C) \cong A$ and $\pi_A(C) \cong B$. In this way, we have "recovered" A and B from C. Furthermore, C contains no more information than information about A and B. So essentially, if there exists a group C that most efficienly stores A and B, in particular if there are two functions (in our case π_A, π_B) which let's us recover A and B, then we can say C is the product of A and B. let's write this group and these two functions (C, π_A, π_B)

But is C the most efficient group for storing information about the group A and B? We can answ3re this by relating C to any other group, as is common in category theory. Let's say we have a group $G \in \mathrm{Obj}(Grp)$ were we have a homomorphism $G \xrightarrow{f_A} A$ and $G \xrightarrow{f_B} B$ (so some structe of G is in A and B). Then does there exist a homomorphism $f : G \to C$ which can compose with π_A and π_B and produce f_1 and f_2? In other words: (Rushed)

$$\begin{tikzcd}
& G \arrow[d, "f"] \arrow[ddl, bend right, "f_A"'] \arrow[ddr, bend left, "f_B"] & \\
& C \arrow[dl, "\pi_A"] \arrow[dr, "\pi_B"'] & \\
A & & B
\end{tikzcd}$$

In a sense, it's saying that whatever structure G has of A and B, C must have it too, since we can pass through C wihtout loosing that structure.

We can now foramlize this idea with categories: Let $C_{A,B}$ be a category. The objects are

$$\begin{tikzcd}
& X \arrow[dl, "f_A"'] \arrow[dr, "f_B"] & \\
A & & B
\end{tikzcd}$$

and it can be represented as (X, f_A, f_B). If (X, f_A, f_B) and (Y, f_A, f_B) are objects, then a morphism $f : (X, f_A, f_B) \to (Y, f_A, f_B)$ exists if the following diagram commutes:

$$\begin{tikzcd}
& X \arrow[d, "f"] \arrow[ddl, bend right, "f_A"'] \arrow[ddr, bend left, "f_B"] & \\
& Y \arrow[dl, "\pi_A"] \arrow[dr, "\pi_B"'] & \\
A & & B
\end{tikzcd}$$

> **Example 5.4.1: Morphissm In $C_{A,B}$**
> here. Examples and counter-examples

Proposition 5.4.2: $C_{A,B}$ Froms A Category

Let $C_{A,B}$ be defined as above. Then $C_{A,B}$ forms a category

***Proof*:**
Left as an exercise that everything works out.

We can ask whether $C_{A,B}$ has *final* objects, That is, an object $(G, f_A, f_B) \in C_{A,B}$ where for any $(H, g_A, g_B) \in C_{A,b}$, $\text{Hom}((H, g_A, g_B), (G, f_A, F_B))$ is a singleton. In fact, (C, π_A, π_B) is such an object. In other words, we'll show that (C, π_A, π_B) must be the only group with this property since the axioms of groups and homomorphisms forces it to be.

Theorem 5.4.3: Universal Property Of The Product For Groups

Let A and B be groups in G. Then $A \times B$ is a group, $\pi_A : A \times B \to A$ and $\pi_B : A \times B \to B$ are homomorphisms. For any group H, if $f_A : H \to A$ and $f_B : H \to B$, then there exists a unique f such that $f_A = \pi_A \circ f$ and $f_B = \pi_B \circ f$. The morphism f is denoted $f_A \times f_B$.

***Proof*:**
We showed that $(A \times B, \cdot_A, \cdot_B)$ in proposition ref:HERE. Next, let $(A \times B, \pi_A, \pi_B)$ be as above and let $(H, f_A, f_B) \in C_{A,B}$. Define

$$f : H \to A \times B \qquad f(h) = (f_A(h), f_B(h))$$

Then we can see that $f_A(h) = (\pi_A \circ f)(h) = \pi_A(f_A(h), f_B(h)) = f_A(h)$, and similalry for f_B, showing that the diagram indeed comutes, and so f is indeed a morphism in $C_{A,B}$.

To show that f is unique, let's assume there is another morphism f'. Then

$$f_A(h) = (\pi_A \circ f')(h) = \pi_A(f'(h)) = \pi_A(f'(h)_A, f'(h)_B) = f'(h)_A$$

where $f'(h)_A$ and $f'(h)_B$ represent the components of $f'(h)$. Similarly, $f'(h)_B = f_B(h)$. Therefore, $f'(h) = (f_A(h), f_B(h))$. However, that is exactly what f does:

$$f(h) = (f_A(h), f_B(h)) = f'(h)$$

thus $f = f'$, showing uniqueness. Thus (C, π_A, π_B) is the final object of $C_{A,B}$. In other words, $A \times B$ respects the universal property of the product in Group.

We can further generalize this for an arbitrary direct product, that is if $(G_i)_{i \in I}$ is a family of groups, then their direct product $G = \prod_{i \in I} G_i$ along with $\pi_i : G \to G_i$ for every $i \in I$ will be a final object in $C_{(G_i)_{i \in I}}$.

Theorem 5.4.4: Universal Property For Arbitrary Direct Product Of Groups

For every group H, and every I-indexed family of morphisms $f_i : H \to G_i$, there exists a unique morphism $f : Y \to X$ such that the following diagrams commute for every $i \in I$:

$$\begin{tikzcd} H \arrow[d, "f"] \arrow[dr, "f_i"] & \\ G \arrow[r, "\pi_i"] & G_i \end{tikzcd}$$

***Proof*:**
Similar to 5.4.4. All the steps generalize without a problem.

Exercise 5.4.1

1. Given that you've seen the universal property of products for groups, what do you think it would be for sets? What would be the generalization for an aribtary category? (For a hint, see the preliminary chapter or section 5.4)
2. let $1 \to A \to B \to C \to 1$ be an exact sequence. Show there exists an action $\rho : C \to \mathrm{Out}(A)$

5.4.1 The dual of the Product

What happens if you flip the arrows? Instead of looking at $C \xrightarrow{\pi_A} A$ and $C \xrightarrow{\pi_B} B$, we look at $A \xrightarrow{i'_A} C$ and $B \xrightarrow{i'_A} C$, and for all groups G, if we have $A \to G$ and $B \to G$, then there exists a homomorphism $f : C \to G$ such that the following diagram commutes

$$\begin{tikzcd}
A \arrow[dr, "i'_A"] \arrow[ddr, bend right, "f'_A"'] & & B \arrow[dl, "i'_B"'] \arrow[ddl, bend left, "f'_B"] \\
& C \arrow[d, "F"] & \\
& G &
\end{tikzcd}$$

Does $(A \times B, i_A, i_B)$ where $i_A(a) = (a, 1)$ and $i_B(b) = (1, b)$ work? Let's break down the restrictions of this commutative diagram to check. By the above commutative diagram, we need that $f'_A(a) = F(i_A(a)) = F((a, 1))$ and that $f'_B(b) = F(i_B(b)) = F((1, b))$. So we have

$$F((a, 1))F((1, b)) = f'_A(a)f'_B(b)$$

Since F must be a homomorphism, then:

$$F((a, 1))F((1, b)) = F((a, 1)(1, b)) = F((a, b)) = F((1, b)(a, 1)) = F((1, b))F((a, 1))$$

that is $f'_A(a)f'_B(b) = f'_B(b)f'_A(a)$, that is, the image of A and B must always commute. However, we can see that there will exist a group G and homomorphisms f'_A f'_B where $f'_A(a)f'_B(b) \neq f'_B(b)f'_A(a)$:

Example 5.4.5: Counter-example
Let $A = \mathbb{Z}_2$ and $B = D_6$ so that we get

$$\begin{tikzcd}
\mathbb{Z}_2 \arrow[dr, "i_A"] & & D_6 \arrow[dl, "i_B"] \\
& \mathbb{Z}_2 \times D_6 &
\end{tikzcd}$$

Let $G = D_6$, $f'_A : \mathbb{Z}_2 \to D_6$, $f'_A(x) = \sigma^x$, $f'_B : D_6 \to D_6$, $f'_B(x) = x$ (i.e. f'_B is the identity map).

Then is it true that there exists a homomorphism F such that:

$$\begin{tikzcd}
\mathbb{Z}_2 \arrow[dr, "i_A"] \arrow[ddr, bend right, "x\mapsto\sigma^x"'] & & D_6 \arrow[dl, "i_B"'] \arrow[ddl, bend left, "\mathrm{id}"] \\
& \mathbb{Z}_2 \times D_6 \arrow[d, "F"] & \\
& D_6 &
\end{tikzcd}$$

By definition $F((a,1)) = F(i_A(a)) = f'_A(a) = \sigma^a$ and $F((1,b)) = F(i_B(b)) = f'_B(b) = b$, so

$$F((a,b)) = \sigma^a b$$

and by what we just saw, we must also have that $F((a,b)) = b\sigma^a$ for every $a \in A$ and $b \in B$. However

$$F((1,2)) = \sigma r^2 \neq r^2\sigma$$

showing that F is *not* a well-defined homomorphism!

It should be noted that if we *only* stuck to abelian group, then this wouldn't be a problem, since we always have that $ab = ba$, so $F((a,b))$ is well-defined. In fact, it is worth checking that if A and B are abelian groups, $((A \times B), i_A, i_B)$ *does* respect the commutative diagram, and is the initial object in the category who's objects are triples (G, f_A, f_B) (where G *has* to be abelian) and morphsism are the aformentioned commutaive diagram.

The problem here came that $A \times B$ actually added structure on it. As we saw in proposition 5.1.3[5], for any $x, y \in A \times B$, $xy = yx$. (Talk about how then the product is a way to store information by somehow adding conditions to let the storage happen??)

So does there exist a group and maps will be the initial object in this category? The answer to this question is actually yes! It is a group we have not yet encountered. The main idea here is that this group is supposted to be the most generic way of having A and B (so as little relations btween A and B) so that the homomorphism from this group to any G does not preserve these properties. We have encountered a similar situation like this before when defining the *free group* (see section 2.3.3). Something analogous is defined called the *free product*.

Definition 5.4.6: Free Product

Let H and K be groups, S_H be a generating set for H and S_K be a generating set for K, R_H and R_K be the relations given the generating sets respectively (at this point, I already showed presentations, so I should be able to introduce R_H and R_K rigorously). Then

$$H * K = \langle S_H \cup S_k | R_H \cup R_K \rangle$$

Example 5.4.7: Free Products

Take $\mathbb{Z}/2\mathbb{Z}$ and $\mathbb{Z}/2\mathbb{Z}$. Mention that this is the group mentioned in section re:HERE (chap 2 near beginning) that as a group for which the torsion subset as not a group

Exercise 5.4.2

1. Show that $(H * K, i_H, i_K)$ respects the universal property of the co-product in **Grp**.

Infinite abelian case

It should be mentioend again that if we restrict to **Ab**, then the universal property of the coproduct is fulfilled by $H \times K$ (notice that $H * K = H \times K$). Is this also true in the infinite case: given some groups $G_1, G_2, \cdots$, does $G_1, G_2 \times ...$ respect the universal property of the coproduct? The answer is no!

Example 5.4.8: Example

Let's say $G = \prod_{i=1}^{\infty} \mathbb{Z}$. Let's say that every component group is represented by G_i. For every G_i, define the inclusion map $G_i \xrightarrow{\iota_i} G$. Let's say $G_i \xrightarrow{f_i} \mathbb{Z}$, $f_i(x) = x$ Then if G respects the universal property of the coproduct, then there exists a function F such that

$$f_i(g_i) = F(0, 0, ..., g_i, ..., 0, ...)$$

This is true for every component of F. Now here is the question: for some elements $1 \in G_1$, $1 \in G_2 ... 1 \in G_i, ...$, what is:

$$F(1, 1, ..., 1, ...)$$

By linearity, we can see that

$$F(1, 1, ..., 1, ...) = 1 + F(0, 1, 1, ..., 1, ...)$$

and we can keep going this way. However, the problem is that we will continue indefenetily. More precicely, there is no number $y \in \mathbb{Z}$ such that $y = F(1, 1, 1, ..., 1, ...)$. Thus, F is *not* a well-defined function!

The problem came from the fact that with the flexibility of having an infinite sum, we run into the possibility of never terminating our sum, that is, we can create a scenario where we have series which do not converge (TBD?).

A group still exists which can fix our scenario and respct the universal property for coproducts in the infinite case:

Definition 5.4.9: Direct Sum Of Groups

Let $\{A_i\}_{i \in I}$ be a family of groups. Then the *direct product*

$$A = \bigoplus_{i \in I} A_i$$

is a subset of $\prod_{i \in I} A_i$ consisting of all elements $(x_i)_{i \in I}$ with $x_i \in A_i$ such that $x_i = 0$ for all but finitely many indices i (cofinitely many indices).

The proof that this is a group was an exercise ref:HERE (in section 1.1). Note that in the finite case, $\bigoplus_{i=1}^{n} G_i = \prod_{i=1}^{n} G_i$. It only differs once we reach the infinite case

Theorem 5.4.10: Universal Property Of Coproducts In Ab

Let I be an index for the family of groups $G_{i\in I}$. Then $G = \bigoplus_{i\in I} G_i$ respects the universal property of the corpoduct in **Ab**.

***Proof*:**
This proof is very similar to the infinte case one and the one for the usual universal property of the direct product.

5.4.2 Universal Property of Semi-Direct Products?

I want to bring in the intuition of a product that when elements commute it introduces a twist. Show this through Cayley graphs

here

Then generalize the how the universal property of products with coslice categories to fibered categories.

However, it also seems like there isn't a universal property for semi-direct products, since it's not unique (saw that wiki entry where many semi-direct products shared the same structure, so I'm not sure where this yet goes in the "global landscape of mathematics"

https://en.wikipedia.org/wiki/Semidirect_product#Generalizations

Also, to show lack of uniqueness:

$$(C_3 \rtimes D_8) \cong (Q_{12} \rtimes C_2) \cong (D_{12} \rtimes C_2) \cong (V \rtimes D_6)$$

There is also no necessary and sufficient conditions that are known for semi-direct products, although they can be characterzied by being a right-split exact sequence.

Exercise 5.4.3

1. Show that $\mathrm{GL}(F) \cong \mathrm{SL}(F) \rtimes F^{\times}$.

6

Further Topic in Group Theory

Miscelaneous things I don't know how to classify yet

More on J.H. program? More on solvable groups? introduce metabelian groups? nilpotent goups? (Have to mention Krasner–Kaloujnine universal embedding theorem and make exercises defining the wreath product)

6.1 Generalization of Abelian Groups

So far we saw that

$$\text{cyclic} \Rightarrow \text{Abelian}$$

We found a way to find how close agroup is to being abelian using commutator groups. If $G' = \{1\}$, then G is abelian. Otherwise, G is not abelian. We'll develop further generalization's from this concept.

Definition 6.1.1: Metabelian Group

Let G be a group. If $G^{(1)} \neq \{1\}$, but $G^{(2)} = \{1\}$, we sy that G is a *metablean* group

Proposition 6.1.2: Properties Of Metabelian Groups

Let G is metabelian, then

1. any subgroup is metabelian
2. If $\varphi : G \to H$ is a homomorphism, then $\varphi(G)$ is metabelian

***Proof*:**

1. Let $H \leqslant G$ be a subgroup of G, and consider H''. If, $H'' \subseteq G'' = \{1\}$, and so H is metabelian as well. This fact can easily be shown direclty. Notice that $H' \subseteq G'$, since if $x = aba^{-1}b^{-1} \in G$, then this is also a commutator for the group G, and so $x \in G'$. By the same logic, the group H'' will be a subset of G'', meaning H'' is trivial as well.

2. Let $\varphi : G \to H$ be a homomorphism and G a metabelian group, and consider $\varphi(G) \leqslant H$. We'll show that homomorphisms preserve commutator subgroups to show that $\varphi(G)$ is metabelian. In paticualr, if $N \subseteq G$, then $\varphi(N') = \varphi(N)'$.

   (a) ($\subseteq$) Let $x \in \varphi(N')$. Then $x = \varphi(xyx^{-1}y^{-1}) = \varphi(x)\varphi(y)\varphi(x)^{-1}\varphi(y)^{-1}$. And so by definition, $x \in \varphi(N)'$.

   (b) ($\supseteq$) Let $x \in \varphi(N)'$. Then $x = \varphi(x)\varphi(y)\varphi(x)^{-1}\varphi(y)^{-1} = \varphi(xyx^{-1}y^{-1})$. And so by definition $x \in \varphi(N')$.

   Thus
   $$\varphi(G)'' = \varphi(G')' = \varphi(G'') = \varphi(\{1\}) = \{1\}$$
   and so $\varphi(G)$ is also metabelian.

Example 6.1.3: Metabelian Groups

1. Any abelian group is always metabelian
2. if A and B are abelian groups, and $A \rtimes B$ is well-defined, then $A \rtimes B$ is metabelian.

Proposition 6.1.4: Extension of Abelian Groups

Let G be a group. If $H \trianglelefteq G$ is abelian and G/H is abelian, then G is metabelian. More generally, if G extends an abelian group N by an abelian group M, then G is metabliean, i.e, the following is a short exact sequence

$$1 \longrightarrow A \longrightarrow G \longrightarrow M \longrightarrow 1$$

Furthermore, the converse is true too, namely if $H \trianglelefteq G$ and G/H are abelian, then $G'' = \{1\}$, showing this is an equivalent definition

***Proof*:**
Let $A \trianglelefteq G$ and G/A be abelian. Then by proposition ref:HERE, $G' \subseteq A$, meaning $G' = \{1\}$. Evidentally then, $G'' = \{1\}$. Conversly, if $G'' = \{1\}$, then G' the elements of G' commute. Note that this does not imply $G' \trianglelefteq G$ immediately, since we did not esatblish those elements commute with every element of G, and so we must show that $G' \trianglelefteq G$. Let $g \in G$ and $h \in G'$. By definition, $ghg^{-1}h^{-1} \in G'$. Since $h \in G'$, $ghg^{-1}h^{-1}h = ghg^{-1} \in G'$, and thus $G' \trianglelefteq G$. Thus, by proposition ref:HERE, G/G' is also abelian completing the other direction.

Thus this is an equivalent definition of metabelian.

6.1.1 Solvable Groups

Metabelian groups are a simple example of a group called a *solvable* group. Instead of looking at the 2nd commutator subgroup being trivial, we can ask if the nth commutator subgroup will be trivial:

Definition 6.1.5: nth Commutator Group

Let G be a group and $G' = G_{(1)}$.

$$G_{(n)} = [G_{(n-1)}, G_{(n-1)}] = \left\{aba^{-1}b^{-1} \middle| a, b \in G_{(n-1)}\right\} = (G_{(n-1)})'$$

represents the nth commutator group.

Definition 6.1.6: Solvable Groups

A group G is *solvable* if there is a subnormal series with commutator subgroup:

$$G = G_0 \trianglerighteq G_{(1)} \trianglerighteq G_{(2)} \trianglerighteq \cdots \trianglerighteq G_{(n)} = \{1\}$$

such a subnormal series is called the *derived series*. I $G_{(n)} = 1$, we say G has *solvable length n*.

Example 6.1.7: Solvable Groups

1. Every abelian group is solvable, since $1 \trianglelefteq G$ is a normal series.
2. Any metabelian group is solvable, with the series is called the *derived series of G*.

$$1 = G'' \trianglelefteq G' \trianglelefteq G$$

Since G/G' is abelian, and $G'/G'' \cong G'$ which is abelian.

There are equivalent definitions of solvability that are worth keeping in mind to have a variety of intuitions.

Lemma 6.1.8: characteristic of nth commutator

Let G be a group. Then $G_{(i+1)}\text{char}G_{(i)}$ for any $i \in \mathbb{N}$. Consequently, $G_{(i+1)} \trianglelefteq G_{(i)}$.

***Proof*:**
We have already proved this in proposition 5.2.3. Then normality comes from proposition 4.7.12.

We can also prove normality directly. We want to show that for any $g \in G_{(i)}$, $gG_{(i+1)}g^{-1} = G_{(i+1)}$. Take $h \in G_{(i+1)}$. Then by definition of $G_{(i+1)}$, $ghg^{-1}h^{-1} \in G_{(i+1)}$. Since $h \in G_{(i+1)}$,

$$ghg^{-1}h^{-1}h = ghg^{-1} \in G_{(i+1)}$$

implying $gG_{(i+1)}g^{-1} = G_{(i+1)}$, and so $G_{(i+1)} \trianglelefteq G_i$. since i was arbitrary, we see that this holds true for any nth commutator subgroup, as we sought to show.

Theorem 6.1.9: Equivalent Definitions Of Solvable group

The following are equivlent

1. G is solvable
2. There exists a subnormal series

$$G = A_0 \trianglerighteq A_1 \trianglerighteq A_2 \trianglerighteq \cdots A_n = \{1\}$$

such that A_i/A_{i+1}. Such a subnormal series is called a *subnormal series with abelian factors.*

***Proof*:**
$(1 \Rightarrow 2)$ We want to construct a derived series. However, we already have one: since $G_{(i+1)} \trianglelefteq G_{(i)}$, let $A_i = G_i$.

$(2 \Rightarrow 1)$ Since $G/A_{(1)}$ is abelian, then by proposition 5.2.3 $G_{(1)} \leqslant A_1$. Then inductively, let $G_{(i)} \leqslant A_i$. then since A_i/A_{i+1} is abelian, there exists a group $G^* \leqslant A_{i+1}$ such that A_i/G^* is abelian. Then, take $f : G_{(i)} \to A_i/G^*$, $x \mapsto [x]$. Consider $[x], [y] \in f(G_{(i)})$. Then since $f(G_{(i)})/G^*$ is a subgroup of A_i/A_{i+1}, it is also abelian, so

$$[xy] = [yx] \Rightarrow xyg = yx \qquad g \in G*$$

moving variables around we get

$$xyx^{-1}y^{-1} = [x, y] = g$$

since x^{-1}, y^{-1} was arbitrary, we see that $G_{(i+1)} \leqslant G* \leqslant A_{i+1}$, that is

$$G_{(i+1)} \trianglelefteq A_i$$

Thus, once we reach A_n, we see that

$$G_{(n)} \trianglelefteq A_n = \{1\}$$

thus $G_{(n)} = 1$, as we sought to show.

The use of the 2nd definition is to show that finding any solvable series for a group is enough to show a group is solvable. The advantage of the 1st definition is that it is the most refined solvable series of a group, since the commutator subgroup is smallest subgroup which makes the quotients A_i/A_{i+1} abelian.

Proposition 6.1.10: Properties Of Solvable Groups

Let G be solvable. Then

1. subgroups of solvable groups are solvable
2. if $\varphi : G \to H$ is a homomorphism, then $\varphi(G)$ is solvable
3. if M is an extension of a solvable group by a solvable group, then M is solvable.

***Proof*:**
For (1) and (2), the commutator subgroup definition makes these proofs much easier. To get used to both definitions, we'll prove the statement using both:

1. Let $G_{(n)} = \{1\}$, and consider $H \leqslant G$. Then as we've seen, $H_{(n)} \leqslant G_{(n)}$ meaning H is solvable.

   Using the other definition, we'll construct a solvable series for H. Let $G \trianglerighteq G_1 \trianglerighteq G_2 \trianglerighteq \cdots \trianglerighteq G_n$ be a solvalbe series for G. Take the homomorphism

   $$\varphi_i : H \cap G_i \to G_i/G_{i+1} \qquad \varphi(x) = xG_{i+1}$$

   which has kernel $(H \cap G_i) \cap G_{i+1} = H \cap G_{i+1}$. Since it's a kernel, it is a normal subgruop of $H \cap G_i$, and the quotient $H \cap G_i/H \cap G_{i+1}$ maps injectivly into G_i/G_{i+1}. Since it is commutative, then so is $H \cap G_{i+1}$. Thus

   $$H \trianglerighteq H \cap G_1 \trianglerighteq H \cap G_2 \trianglerighteq \cdots \trianglerighteq H \cap G_n$$

   is a solvable series for H.

2. We've shown that $\varphi(G'') = \varphi(G')' = \varphi(G)''$ (proposition 6.1.2). Continuing inductively, we'd get $\varphi(G_{(n)}) = \varphi(G)_{(n)}$. We can see that the derived series will be preserved.

   Using the other definition, let $\overline{G}$ be the quotient gorup of G, and let $\overline{G}_i$ be the image of G_i into $\overline{G}$. Then

   $$\overline{G} \trianglerighteq \overline{G}_1 \trianglerighteq \cdots \trianglerighteq \overline{G}_n = \{1\}$$

   is a solvable series for $\overline{G}$

3. Let $N \trianglelefteq G$ and G/N be solvable. We'll show then that G has a series. Since N and $\overline{G}$ is solvable, then they both have a solvable series:

   $$\overline{G}\overline{G}_1 \trianglerighteq \overline{G}_2 \trianglerighteq \cdots \trianglerighteq \overline{G}_n = \{1\}$$

   $$N \trianglerighteq N_1 \trianglerighteq N_2 \trianglerighteq \cdots \trianglerighteq N_m = \{1\}$$

   Thus, by the 3rd isomrophism theorem $\overline{G}_i$ in G we get $G_i/G_{i+1} \cong \overline{G}_i/\overline{G}_{i+1}$. Combining these series, we get

   $$G \trianglerighteq G_1 \trianglerighteq \cdots \trianglerighteq G_n = N \trianglerighteq N_1 \trianglerighteq \cdots \trianglerighteq N_m = \{1\}$$

   which is a solvable series for G, completing the proof.

Corollary 6.1.11: Finite P-Groups

Let G be a finite p-group. Then G is solvable.

***Proof*:**
Let's use induction on the order of G. The base case is trivial, so let's consider an order $|G| > 1$. Then by hw exexerice (or corollary 4.16 in Milne Group Theory book) $Z(G)$ has non-trivial center. Since $Z(G)$ is normal in G, $G/Z(G)$ is well-defined. Since $Z(G)$ is non-trivial $G/Z(G)$ has smaller order. Thus, by the induction hypothesis, $G/Z(G)$ is solvable. Since $Z(G)$ as a group is commutative, by proposition 6.1.10(2), G is solvable.

There is actually one more incredibly powerful theorem for which groups are solvable:

Theorem 6.1.12: Feit-Thompson's Theorem

Let G be a finite group of odd order. Then G is solvable.

***Proof*:**
Huge proof requiring 255 pages.

From this statement, we can state that either a finite group is solvable, or it has an element of order 2 (see exercice 1.2.1)!

Theorem 6.1.13: Solvability Condition

The finite group is solvable if and only if for every divisor n of $|G|$ such that $\left(n, \frac{|G|}{n}\right) = 1$, G has a subgroup of order n .

***Proof*:**
very difficult proof not in textbook.

6.1.2 Nilpotent Groups

(put somehere ho to classify finite abelian groups) We've defined solvable groups using nth commutator groups to give a notion of generalization from being commutative. What if we used the center instead? SDince $Z(G) \trianglelefteq G$, $G/Z(G)$ is well-defined. To generalize, we can define:

$$Z^2(G) = Z(G/Z(G))$$

that is, the 2nd center is the center of the quotient $G/Z(G)$. If we can continue this chain with $Z^n(G)$, and eventually get a commutative group $z^n(G) = G$, we call such a group a *nilpotent group*:

Definition 6.1.14: Nilpotent Groups

Let G be a group. If $Z^m(G) = G$. for some $m \in \mathbb{N}$, then G is said to be nilpotent. The smallest such m is called the *(nilpotency) class of G*.

Example 6.1.15: Nilpotent Groups

1. If G is abelian, then $Z(G) = G$, meaning all abelian groups are nilpotent
2. If G is meta-abelian, Then $G/Z(G)$ is abelian (since $G' \subseteq Z(G)$), meaning $Z(G/Z(G)) = G/Z(G)$, and so is nilpotent with nilpotency class 2.
3. Every p-groups are an example of nilpotent groups. Let G be a finite p-group, so that $|G| = p^a$ for some $a \in \mathbb{N}$. If $Z(G) = G$, we're done. If not, then since $Z(G)$ is non-trivial (proposition ref:HERE), $|Z(G)|||G|$, so $|Z(G)| = p^{a_2}$ for some $a_2 < a$. But then $Z(G)$ is again a p-group, so we can continue this process. Thus, we get an ever shrinking sequence of nubmers $a > a_2 > a_3 \cdots a_n$, till eventually it must be that either $a_n = 1$ or $a_n = 0$. If $a_n = p$, then the associated group is cyclic, implying it's abelian, and we're done. If $a_n = 0$, then we have the trival subgroup, which is trivially abelian.
4. This is not so much an example as future motivation: Nilpotent groups make an appearance in the classification of Lie Groups.

Notice that for any $[g] \in Z^2(G)$, $[g, x] \in Z(G)$ for any $x \in G$. To see this, pick $[x] \in G/Z(G)$ (i.e., the coset which can have x as it's reprsentative). Then since $[g] \in Z^2(G)$, $[gx] = [xg]$. Then $gxz = xgz'$ for some $z, z' \in Z(G)$. Then$g^{-1}x^{-1}gx = z'z^{-1} \in Z(G)$. Thus, $Z^2(G) \subseteq Z(G)$. This holds true for any $Z^n(G)$ (defined inductively). Thus, can also get a subnormal series, called the *upper central series*:

$$\{1\} \subseteq Z(G) \trianglelefteq Z^2(G) \cdots Z^n(G) \cdots$$

If this series terminates with $Z^n(G) = G$, then we say that G is a *nilpotent group*.

Proposition 6.1.16: Equivalent Definition

The following are equivalent:

1. G is nilpotent (i.e. there exists an *upper central series*)
2. There exists a *lower central series*
$$G = G_0 \trianglerighteq G_1 \trianglerighteq G_2 \trianglerighteq \cdots \trianglerighteq G_n = \{1\}$$
where $G_i = [G, G_{i-1}] = \left\{ghg^{-1}h^{-1} \middle| g \in G, h \in G_{i-1}\right\}$.
3. There exists a *central series*
$$G = G_0 \trianglerighteq G_1 \trianglerighteq \cdots \trianglerighteq G_n = \{1\}$$
where $G_i/G_{i+1} \leqslant Z(G/G_i)$, or equivalently, $[G, G_{i+1}] \leqslant G_i$

***Proof*:**
(1) $\Rightarrow$ (2) Let G be nilpotent. Then we saw that if $g \in Z^{i+1}(G)$, then $[g,x] \in Z^i(G)$, meaning $[G, Z^{i+1}(G)] \leqslant Z^i(G)$. Then decomposing recusively, we see that $G_{i+1} = [G, G_i] \trianglelefteq Z^i(G)$. Showing that every G_i exists, and will have to terminate at G_n. Furthermore, by the same justification as $G_{(i)}$, $G_{i+1} \trianglelefteq G_i$. Thus, they form a subnormal series

(1) $\Rightarrow$ (3) easy to do

(2) $\Rightarrow$ (1)

With the lower central series, we can gain an intuition on why the group is called nilpotent. Let G be any group and $g \in g$ be any element. Define the *adjoint action*

$$\mathrm{ad}_g : G \to G \qquad \mathrm{ad}_g(x) = [g, x]$$

Then if G has nilpotency class n, then $(\mathrm{ad}_g)^n(x) = e$, for all $x \in G$. In this sense, the adjoint action is nilpotent. Note that if this is true for each element, this does not imply that G is nilpotent

Example 6.1.17: Example
here

Such groups are called *Engel groups*, and come up in Lie Algebra.

Proposition 6.1.18: Nilpotent Then Solvable

Let G be a nilpotent group. Then G is solvable

***Proof*:**
Notice that $G_{(n)} \trianglelefteq G_n$, meaning the there must be a solvable length of G, since for some n, $G_{(n)} \subseteq G_n = \{1\}$, implying $G_{(n)} = 1$, showing that G is solvable.

Note that the converse is not true

Example 6.1.19: Solvable, but not Nilpotent
let

$$G = \left\{ \begin{pmatrix} a & b \\ 0 & c \end{pmatrix} \middle| a, b, c \in \mathbb{R}, ab \neq 0 \right\} \leqslant GL_2(\mathbb{R})$$

Then I encourage you to check that $Z(G) = \{aI | a \neq 0\}$ where I is the identity matrix. Furthermore $Z^2(G) = Z(G/Z(G)) \cong \{1\}$, i.e., we don't resolve to the an abelian group but a trivial group, and so G is not nilpotent.

However, G is solvable. We can form a solvable series using $H = \left\{ \begin{pmatrix} 1 & a \\ 0 & 1 \end{pmatrix} | a \in \mathbb{R} \right\}$. It should be verified that $H \trianglelefteq G$, and that $G/H \cong \mathbb{R}^\times \times \mathbb{R}^\times$ (meaning it's abelian), and that $H \cong (\mathbb{R}, +)$, meaning H is abelian, so $G'' = H' = \{1\}$, showing it is indeed solvable.

Another example would be S_3. Since $S_3 \cong \mathbb{Z}_3 \rtimes \mathbb{Z}_2$, and both $\mathbb{Z}_3$ and $\mathbb{Z}_2$ are nilpotent (since their abelian), then the semi-direct product of nilpotent groups is not necessarily nilpotent. More generally, the extension of a nilpotent group by a nilpotent group is not necessarily nilpotent.

Thus, we have the folowing chain of implications cyclic $\Rightarrow$ abelian $\Rightarrow$ nilpotent $\Rightarrow$ solvable $\Rightarrow$ finitely generated (show A_5 not solvable)

So nilpotent groups don't interact well with extensions. However, the fac that their slightly stronger than extensions, but more general then abelian groups, brings us some useful properties:

Proposition 6.1.20: Properties Of Nilpotent Groups

Let G be a nipotent group. Then

1. If H is a proper subgruop of G, then H is a proper normal subgruop of $N_G(H)$. We call this property the *normalizer property*, and we sometimes say that the "normalizer grows"
2. Every Sylow subgroup of G is normal
3. G is the direct product of its Sylow subgroups
4. if d divides the oder of G, then G has a noraml subgroup of order d

In fact, (4) implies that G is nilpotent, giving us another definition of nilpotency.

***Proof*:**

1. Let G be a nilpotent group and $H \leqslant G$. We want to show that $H \trianglelefteq N_G(H)$. We'll do this by induction on the order of G. This is clearly true when $|G| = 1$, so let $|G| = n$. IF G is abelian, then for any $H \leqslant$, $N_G(H) = G$, so we get our result trivially. So let's say G was not abelian.

   if $Z(G) \not\leqslant H$, then
   $$here$$
   If $Z(G) \leqslant H$, then $H/Z(G)$ is contained in $G/Z(G)$. By proposition ref:HERE, $G/Z(G)$ is nilpoent. Thus, there es a subgroup of $G/Z(G)$ which normalizer $H/Z(G)$ and $H/Z(G)$ is a proper subgroup. Thus, by the correspondence theorem, $H \trianglelefteq N_G(H)$. TBD

2. Let P be a Sylow p-subgroup, and consider $N = N_G(P)$. Since $P \trianglelefteq N$, by proposition 4.8.7, $P\text{char}N$. Since
   $$P\text{char}N \trianglelefteq N_G(N)$$
   Then by transitivity of characteristic subgruops, we have that $P \trianglelefteq N_G(N)$. Then by Sylow's Theorem, $N_G(N) \trianglelefteq N$. By part (1), we get that $N = G$, showing that $P \trianglelefteq G$, that is, Sylow p-subgroups are normal subgroups of G

3. Let $p_1, p_2, ..., p_n$ represent distinct pimres dividing the order of G

https://en.wikipedia.org/wiki/Nilpotent_group#Properties

Exercise 6.1.1

1. Prove that any nilpotent group of class 3 or lower is metabelian
2. Show that if $\text{Aut}(G)$ is abelian, then G is nilpotent of class 2

6.1.3 Polycyclic Groups

We've extended a abelian group using *central extensions* and *abelian extensions* (TBD). In this section, we'll once again be extending an abelian group, but this time we'll ask what happens if we extend a cyclic group in such a way that G_i/G_{i+1} is cyclic?

Definition 6.1.21: Polycyclic Group

Let G be a group. Then if

$$G = G_0 \trianglerighteq G_1 \trianglerighteq \cdots \trianglerighteq G_n = \{1\}$$

st G_i/G_{i+1} is cyclic, Then G is said to be polycyclic. The series G admits is called a *subnormal series with cyclic factors*. If $G_n = \{1\}$, then we say that G is polycyclic with length n.
If $n = 2$, then we say that G is a *metacyclic group*.

Example 6.1.22: polycyclic groups
By the fundamental theroem of finitely generated group, you can prove that all finitely generated abelian group are in fact polycyclic (exercice).

word on saying *the* subnormal series with cyclic factors? (up to isomorphism?)

Proposition 6.1.23: Polycyclic Then Finitely Generated

Let G be a polycyclic group. Then G is finitely generated.

***Proof*:**
Let G be polycyclic, so that

$$G = G_0 \trianglerighteq G_1 \trianglerighteq \cdots \trianglerighteq G_n = \{1\}$$

is the subnormal series with cyclic factors. Since G_0/G_1 is cyclic, let' say $[g_0] \in G_0/G_1$ generates G_0/G_1. Similarly, let's say $[g_i] \in G_i/G_{i+1}$ such that $\langle g_i \rangle = G_i/G_{i+1}$. Eventually we'll get that $\langle g_{n-1} \rangle = G_{n-1}/G_n \cong G_{n-1}$, meaning G_{n-1} is cyclic. Then we get that

$$1 \to G_{n-1} \to G_{n-2} \to G_{n-2}/G_{n-1} \to 1$$

i.e., we extend a cyclic group by a cyclic group. Notice if we take any $x \in G_{n-2}$ then we get $[x] \in G_{n-2}/G_{n-1}$, so $[x] = [g_{n-2}^{\alpha}] = [g_{n-2}]^{\alpha}$ so

$$x = (g_{n-2}g_{n-1})^{\alpha}$$

and since x was arbitrary, this mean that $\langle g_{n-2}, g_{n-1} \rangle = G_{n-2}$. continuing inductively, we'll get that

$$\langle g_i | 1 \leq i \leq n \rangle = G$$

showing that G is finitely generated.

Note that the converse is false:

Example 6.1.24: Finitely Generated, But Not Polycyclic
Let $G = A_5$, which is a simple non-cyclic group. Since it simple, it's subnormal series is only

$$A_5 \trianglerighteq \{1\}$$

and so, it cannot be polycyclic.

Then it can be checked that if we restrict to finitely generated groups, we get:

cyclic $\Rightarrow$ abelian $\Rightarrow$ nilpotent $\Rightarrow$ Polycyclic $\Rightarrow$ solvable $\Rightarrow$ finitely generated (show A_5 not solvable)

6.2 Non-finitely generated abelian groups

(Prüfer's Theroems!)

6.3 Linear Groups

Perhaps add a section here on linear groups and some more complicated properties because of how important they are to Lie Algebra and Invariant Theory?

7

Generalization of Groups

In this chapter, I will cover common algebraic objects which are generalizations of groups.

7.1 Monoid

Having a unique inverse is one of the more powerful axioms you can add to an algebraic structure. Recall how the uniqueness of a linear solution is bassed off of the fact that the invere exists and is unique (proposition 1.1.17). This property is also called the *cancellation property.* There are many mathematical structures which do not have this ($\mathbb{N}$ would be a group if inverses existed). For this reason, the study of this structure is also of interest to many

Definition 7.1.1: Monoid

A *Monoid* is an ordered pair $(M, *)$ where M is a set and $*$ is a binary operation on M satisfying the following axioms

1. associativity: $(a * b) * c = a * (b * c)$
2. there exists an element e in G, called the *identity* of G such that for every $a \in G$, we have $a * e = e * a = a$ [a]

The Monoid $(M, *)$ is called *abelian* (or commutative) if $a * b = b * a$ for every $a, b \in M$

[a]it can in fact be proven that if $a * e = a$ then $e * a = a$ (or vice-versa); an indispensable exercise worth doing

Example 7.1.2
Naturally, all groups are monoids. I'll give non-group example of monoids:

1. The natural numbers $\mathbb{N}$
2. Given a set A and it's powerset $P(A)$ (set of all substs), then $(P(A, \cup))$ is a monoid with $\emptyset$ as the identity.
3. Given a set A and it's powerset $P(A)$, then $(P(A), \cap)$ is a monoid with A as the identity.
4. The set $\{0, 1\}$ with any boolean operation ($\Rightarrow$, NOR, AND, etc.) is a monoid. More commonly, the set is relabled as $\{\text{False}, \text{True}\}$
5. The set of strings, given an alphabet is a free monoid.
6. Given a set A, the set of all functions $f : A \to A$ is a monoid! This is incredibly powerful, since we can now put an algebraic structure on sets themselves!! Any set has a default algebraic structure! Further wil be true soon: When we study rings, rings will be an abelian group, along with a second operation which will be a monoid!!

   There are far reaching consequences of this when relating algebra to other fields. If you can put a abliean group structure on something, then taking the set of all functions will now produce you a ring! In algebraic topology, this is exact trick is used to be able to compare rings with topologies, which all comes down to
7. Given the collection of all normal subgroups of a group, their multiplications form a monoid structure! functions having a natural monoid structure!

Recall that $\text{Aut}(G)$ is a group, but $\text{End}(G)$ was not since not every element has an inverse (section 1.4)? This then makes $\text{End}(G)$ a monoid

Proposition 7.1.3: G Abelian Implies End(G) Monoid

here

Proof. Let G be an abliean group and consider $\text{End}(G)$.

Associativity holds by

The identity naturally exists, since $1 : G \to G$, $1(g) = g$ is definable and $\forall r \in \text{End}(G)$, $r1 = 1r = r$.

□

Example 7.1.4
exercice: Show that if $G = D_6$, then $\text{End}(G)$ is not a group.

There is a lot of intereting theory for monoids which I will simply sum up for now to get to the main reason I wanted to bring them up

1. Recall that I said that groups have the cancellation property because of the unique inverse of every element. It is possible that there are monoids with this property that are not groups. Such monoids are called monoids *with the cancellation property.* The natural numbers is an

example of such a monoid. Monoids with the cancellation property can be extended to form groups (called the Grothendieck construction). If the monoid is finite, then it implies that it's a group

2. Recall that the set of all homomorphisms between a group and itself forms a group. With a monoid, we can weaken the condition of the functions being homomorphisms. Given a set S and every set-function $f : S \to S$ with the operation defined point-wise ($(f+g)(x) = f(x) + g(x)$) is also a monoid.

3. The same way we defined group-actions with groups, we can define monoid-actions with the same axioms. These pop-up in computer-science in the study of automata.

4. In functional programming, monads is a central tool in programming. Monads are monoids! The objects on which they are monoids is a more complicated subject, and will be left for the appendix on category theory [1]

5. free-monoids play a pivitol role in modeling the behavior of machines (ex. Turing machines). (DEVELOP ON THIS)

Though all these reaon are good reasons to look into monoids, I bring them up for their use in Category Theory. More particularly, we can define a monoid to be a category with a single element!

The Grothendieck construction named in (1) above is worth carrying out in full, because it is the mechanism by which a number of later invariants are built. Whenever a class of objects carries a notion of direct sum, its isomorphism classes form an abelian monoid, and the invariant one actually wants is the group this construction produces: the Grothendieck group of vector bundles, of finitely generated projective modules, and of a C^*-algebra are each defined this way. The construction is stated for an arbitrary abelian monoid, since cancellation is exactly what it does *not* need.

Definition 7.1.5: Grothendieck Completion

Let $(M, +)$ be an abelian monoid. Define a relation on $M \times M$ by

$$(m,n) \sim (m',n') \quad \text{if and only if} \quad m + n' + p = m' + n + p \ \text{ for some } p \in M$$

The *Grothendieck completion* of M is the quotient $M^{-1}M = (M \times M)/\sim$ with addition induced coordinatewise, together with the monoid homomorphism

$$\iota \colon M \to M^{-1}M, \qquad m \mapsto [(m,0)]$$

The class of (m,n) is written $[m] - [n]$, which the group structure below justifies.

The auxiliary p in the relation is what makes $\sim$ transitive without assuming cancellation, and it is exactly the obstruction measured in part (3) below.

[1] For those who are impatient or know some Category theory, a monoid on a category of functors which are derived given by a particular adjoint

Proposition 7.1.6: Properties of the Grothendieck Completion

Let M be an abelian monoid. Then:

1. $M^{-1}M$ is an abelian group, and every element of it has the form $[m]-[n]$ for some $m, n \in M$.
2. $\iota(m) = \iota(n)$ if and only if $m + p = n + p$ for some $p \in M$.
3. ι is injective if and only if M has the cancellation property.
4. For every abelian group A, composition with ι is a bijection
$$\mathrm{Hom}_{\mathbf{Grp}}(M^{-1}M, A) \xrightarrow{\sim} \mathrm{Hom}_{\mathbf{Mon}}(M, A)$$
natural in M and A; that is, $M \mapsto M^{-1}M$ is left adjoint to the forgetful functor from abelian groups to abelian monoids.

***Proof*:**
Step 1: $\sim$ is an equivalence relation. Reflexivity and symmetry are immediate. For transitivity, suppose $m + n' + p = m' + n + p$ and $m' + n'' + q = m'' + n' + q$. Adding the two equations gives
$$m + n'' + (n' + m' + p + q) = m'' + n + (n' + m' + p + q)$$
so $(m, n) \sim (m'', n'')$ with the witness $r = n' + m' + p + q$. Note that no cancellation was used, which is the point of carrying p at all.

Step 2: the group structure. Coordinatewise addition respects $\sim$, since adding a fixed pair to both sides of the defining equation preserves it. Associativity and commutativity are inherited from M, the class $[(0,0)]$ is an identity, and $[(n,m)]$ is inverse to $[(m,n)]$ because $(m+n,\ n+m) \sim (0,0)$ with $p = 0$. So $M^{-1}M$ is an abelian group, and $[(m,n)] = \iota(m) - \iota(n)$, which is (1).

Step 3: parts (2) and (3). By definition $\iota(m) = \iota(n)$ means $(m, 0) \sim (n, 0)$, that is $m + p = n + p$ for some p, which is (2). Then ι is injective exactly when $m + p = n + p$ forces $m = n$, which is the cancellation property, giving (3).

Step 4: the universal property. Let $f\colon M \to A$ be a monoid homomorphism into an abelian group. If $(m, n) \sim (m', n')$, say $m + n' + p = m' + n + p$, then applying f and cancelling in A, which is legitimate because A is a group, gives $f(m) - f(n) = f(m') - f(n')$. So
$$\tilde{f}([m] - [n]) := f(m) - f(n)$$
is well defined, and it is a homomorphism because addition in $M^{-1}M$ is coordinatewise. It satisfies $\tilde{f} \circ \iota = f$. For uniqueness, any homomorphism g with $g \circ \iota = f$ is determined on the elements $\iota(m)$, and these generate $M^{-1}M$ by (1), so $g = \tilde{f}$. Naturality in M and A is immediate from the formula, completing the proof.

Example 7.1.7: Grothendieck Completions

1. $\mathbb{N}$ under addition is cancellative, so ι is injective, and $\mathbb{N}^{-1}\mathbb{N} = \mathbb{Z}$. This is the motivating case named at the start of the section.

2. Let $M = \{0,1\}$ with $1+1=1$. Cancellation fails, since $1+1=0+1$ while $1 \neq 0$, and indeed $M^{-1}M = 0$: taking $p=1$ shows $(1,0) \sim (0,0)$, so $\iota(1) = 0$.

3. More generally, if M has an absorbing element ω, meaning $\omega + m = \omega$ for every m, then $M^{-1}M = 0$: for any m, taking $p = \omega$ gives $m + \omega = 0 + \omega$. This is why the construction is applied to *finitely generated* objects rather than arbitrary ones, an infinite direct sum being absorbing.

Exercise 7.1.1

1. standard monoid questions
2. In the following exercices, we'll test your understanding of the idea of defining quotient objects by building up to defining a quotient monoid
   a) Recall the difference between an equivalence relation and a congruence relation. What condition do we need in order that $[x][y] = [xy]$ in a monoid? (I saw that the regular one for groups fails, and it works if there's an inverse – investigate this further for it seems to be an interesting question to do!!)

7.2 Groupoid

Recall how at the very begining under definition 1.1.3, I remarked how some authors prefer to say groups are not closed under multiplication? Some authors introduce this convention to be able to later drop this axiom in order to define the more general notion of *groupoid*. A groupoid is a group which is not closed under the binary operation! [2]

Definition 7.2.1: Groupoid

A groupoid is a set G with a unary operation $^{-1} : G \to G$ and a patial function $* : G \times G \rightharpoonup G$ where the following axioms hold:

1. Where $a * b$ and $b * c$ are defined, $(a*b)*c = a*(b*c)$. Conversaly, if $(a*b)*c$ and $a*(b*c)$ is defined, then so are both $a*b$ and $b*c$ and $(a*b)*c = a*(b*c)$
2. $a^{-1} * a$ and $a * a^{-1}$ are always well-defined
3. if $a*b$ is defined, then $a*b*b^{-1} = a$ and $a^{-1}*a*b - b$

Lot's of cool stuff I'll fill in later

Semi-groups? If you introduce wreath product, can be intersting to talk about in conjuction with automata?

[2]Technically, this means it's not a binary operation, because binary operations are functions, which need to be closed, but the spirit of the sentence is correct

8

Category of Group

Throughout the book, we'll be introducing more and more algebraic structures of varying levels of similarity and differences[1]. It would be very convenient to be able to have a way to compare these algebraic structures. In this section, we elaborate more on the concept of *Categories* which were introduced in section 0.4 . When we introduce new algebraic structures, we will show how the algebraic structure also forms a *category*, and by doing so we will open up the ability to compare and contrast all sorts of properties. For example, we saw in example 1.1.5 that if G and H are groups we can define $G \times H$ to be a group under point-wise operations. Will the same type of construction be possible for algebraic structures in coming chapters? For rings? Group-actions? Fields? Vector-space? Will all of these objects have a notion similar to subgroup? Will they all have a notion of homomorphism/isomorphism, or a similar notion to group action? The answer will be yes to some and no to others! In order for these structure to have or not have a property, say for example the ability to do a "direct product", there must be a way of saying "the direct product of groups follows the same principle as the direct product of rings/group-actions/fields/vector spaces/etc." This will be accomplished with the language of category theory.

As mentioned in section 0.4, the core concept of categories is that we want to be able to have a structure that holds *objects* (ex. groups), and a particular way to *relate* these objects (ex. homomorphisms) – see the analogy in the aforementioned section for an analogous point of reference. Different categories can have the same objects but different relations, or the same relations but different objects, or different objects and relations. Commonly, a relation between two objects will be a way to change from one object to another while preserving some of the original structure of the object. These types of relations are called *structure preserving* relations. As an example of relations between objects that preserve some structure, take homomorphism, which relates two groups if there is some common group-structure. The fact that homomorphisms preserve (some or all of) the structure can also be

[1]In this book, there will be 15 algebraic objects that can be considered "key objects" to algebra, while we also work with 6 others non-algebraic objects throughout the book (ex. sets, graphs, ordered sets, etc.)

found in the etymology of the word: the word *homomorphism* come from *homo-* meaning "same" [2] and *morphism* meaning shape. Since homomorphsims preserve (some or all) group structure, they in a sense change a shape from one group to another while keeping some of the "sameness" between the two. Many different mathematical objects have a relation between them that preserve some of it's structure (topologies with topology-preserving functions, graphs and graph-preserving functions, or ordered sets with order-preserving functions to name few), which is why when categories were first introduced, the word *morphism* came to be the name used to describe relations between objects (i.e. if A and B are related in a category, there is a morphism from A to B). The word morphism is a generalization of the word homomorphism to capture the idea of relation's between objects.

I was purposefully being vague in the start by saying that a category has a "particular way to relate objects" instead of just jumping to saying that morphisms preserve some of the structure of the objects. At the beginning of category theory in the 1940s, the types of relations morphisms represented were relations that preserved (some or all of) the structure of the objects. As time went on, more mathematical objects were shown to form categories using morphisms that didn't preserve structure in a way we saw with homomorphisms, and the idea that some of the structure is preserved in some context can be a little of a stretch (see exercise ref:HERE for an example (order on sets)). This is good to know early on to not get the wrong idea of categories, however in the undergraduate level, 99% of the time morphism in categories will be about preserving some/all the structure of a particular object.

With this very high-level idea, let's give some concrete examples before introducing the definition of a category. If we take all the groups and all the possible homomorphism between groups; that will in fact from the category of groups, sometimes denoted **Grp** or **Group**. Another example would be the collection of all sets and all function between sets; that will form the category of sets, sometimes denoted **Set**. We'll soon bring in the definition of a category to prove that it will follow the axioms of a category, for now we will build-up some terminology in order to talk about these collections.

Definition 8.0.1: Category Of Grp and Set

1. Let **Grp** represent the collection of all groups and homomorphism between Groups and **Set** the collection of all sets and functions between sets.

2. Let Obj(**Grp**) represent all the objects in the category of **Grp**. Similarly for Obj(**Set**).

3. For any $G, H \in \text{Obj}(\mathbf{Grp})$, Let $\text{Hom}_{\mathbf{Grp}}(G, H)$ represent all homomorphisms from G to H. Similarly, if $A, B \in \text{Obj}(\mathbf{Set})$, let $\text{Hom}_{\mathbf{Set}}(A, B)$ represent all function form A to B.

In many ways, properties of these collections was the starting point for the first category theorists definition of a category, so it is worth taking a closer look at the properties. To not clutter the analysis, we will stick with **Grp** to build-up to the definition of a category.

When we relate two group using homomorphsims, we always have a domain and a co-domain:

$$\varphi : G \to H$$

This means that relations are *directional*: a homomorphism from G to H is different from a homomorphism from H to G. In some structure, direction of the relation doesn't matter (for example, in equivalence relations, $x \sim y$ if and only if $y \sim x$). In categories, we pay careful attention at the

[2] Other words with homo- would be homeostasis (same state), homochromatic (same colour), or homogeneous (same kind/nature, or uniform structure). It's antonym would be hetero- (heterostatis, heterochromatic, heterogenous)

direction of the arrow, and so when generalizing, a morphism from an object A to B is not the same from a morphism from an object B to A.

When we work with homomorphism, we showed in proposition 1.4.4 that the composition of two homomorphisms is a homomorphism. Particularly, if have a homomorphism from G to H, and a homomorphism from H to K, then there will always exist a homomorphism from G to K:

$$\begin{tikzcd} G \arrow[r, "f"] \arrow[rr, bend right, "g\circ f"'] & H \arrow[r, "g"] & K \end{tikzcd}$$

So, if there is a way to get from G to K through homomorphism, there must be a way to "get there directly" too. Morphisms will have the same property.

Finally, notice that for every group there always exists a "trivial" homomorphism:

$$1_G : G \to G \qquad 1_G(g) = g$$

that is, the homomorphism takes G to itself. In a sense, what is being said here is that every object trivially relates to itself. Another way of thinking that 1_G is trivial is that when composed with other homomorphisms (with appropriate domain or co-domain), it does not change them. If:

$$\alpha : G \to H, \ \beta : F \to G$$

were homomorphsims, then

$$\alpha \circ 1_G = \alpha \qquad 1_G \circ \beta = \beta$$

We want this same property to exist for categories: That every object in a category has a *trivial morphism* which can compose with morphisms (with appropriate domain or co-domain) and not affect it

You might have noticed that all of these properties are about the morphisms of a category. This is a common philosophy in category theory: What's important isn't the objects, but the how an objects relates to other object! It is arguable that the more mathematics you will encounter, the more it will be clear that it is the way objects can relate to object objects that i the defining feature of many objects. In fact, many algebraic objects that we will see will be defined by how they relate to other objects.

With this, we will finally define a category:

Definition 8.0.2: Category

A ***category*** C consists of[a]

1. A collection of *objects*, denoted $\text{Obj}(C)$
2. A collection of *morphisms*, denoted $\text{Mor}C$

where

- Each morphism has a specified ***domain*** and ***codomain*** object. Sometimes, this is written as $f : X \to Y$, where f is the morphism with domain X and codomain Y [b]. For any $A, B \in \text{Obj}(C)$, $\text{Hom}_C(A, B)$ denotes all morphisms form A to B.
- Each object has an ***identity morphism*** $1_X : X \to X$
- For each pair of morphisms f, g with the codomain of f equal to the domain of g (ex. $f \in \text{Hom}C(A, B)$, $g \in \text{Hom}_C(B, C)$), there exists a specified ***composite morphism*** gf (or $g \cdot f$)[c] whose domain is equal to the domain of f and whose codomain is equal to the codomain of g, i.e.

$$f : X \to Y \qquad g : Y \to Z \qquad \Rightarrow \qquad gf : X \to Z$$

such that

1. For any $f : X \to Y$, the composites $1_Y f$ and $f 1_X$ are both equal to f (i.e. unital with the identity homomorphism)
2. For any composable triple of morphisms f, g, h, the composites $h(gf)$ and $(hg)f$ are equal (i.e. associative), and so we can write hgf unambiguously.

[a]The word *collection* being used instead of *set* was purposeful; we will get to this
[b]Note that f **need not be a function**! Namely, the morphism need not have a unique codomain
[c]the notation is purposefully not $\circ$ so as to not draw too much of an analogy of gf with the process of composing functions to demonstrate that in some cases gf is *not* function composition

Example 8.0.3: Examples of Categories

1. It was mentioned that groups along with homomorphisms from a category. Indeed, by proposition 1.4.4 the composition of two homomorphsims is a homomorphism, so $gf = g \circ f$. Homomorphsims are associative since function composition is associative (proposition 0.1.22), and for any $G \in \text{Obj}(Grp)$, the map $\text{id} :\to G$, $\text{id}(g) = g$ is the identity morphisms since for any $f : F \to G$ and $h : G \to H$, $f \circ \text{id} = f$ and $\text{id} \circ h = h$.
2. If we take all groups, and restrict to taking only the isomorphisms, we will again have a category; check that all the axioms are still satisfied. Such a category is called a *groupoid* (the reason for this name will be made evident in exercise ref:HERE (show with 1 element is group, then define more general structure groupoid with no identity, which we expand more upon in section 7.2))
3. If we take all abelian groups and use homomorphsims, this will again be a category. This category is significant enough to deserve it's own notation: this category will denoted **Ab**.

4. If we take all sets and all functions between sets, we will get the category of **Set**.

   Notice that every morphism in **Grp** can be thought of as a morphism in **Set** if we "forget" the group structure of the domain and co-domain, and the homomorphism condition of the morphism of **Grp**. In This way, the category of **Grp** and the category of **Set** are related if we "forget" some of the additional structure of **Grp**. Is it possible to have the reverse? That is, starting with **Set**, is it possible to somehow "upgrade" it to **Grp**? Is there more than one way of doing this? Is there a "natural" way of doing it, that is, is there a way of performing this upgrade in such a way that it follows from the properties of groups and sets without needing to add any more conditions? These questions will be further explored in section 2.3.3. generally.

5. If we take the collection partially ordered sets, that the collection of $(S, \leq)$ where $S \in \text{Obj}(Set)$ and $\leq$ is a possible partial order on S, then we can define a notion similar to homomorphisms that preserve (some or all of) the structure of the order, i.e. we can define a $f : (S, \leq_S) \to (R, \leq_R)$ where f is the appropriate morphism. Can you think of what this morphism should be?

6. In section 0.3, we introduced graphs and graph isomorphisms. These also form a category! There is also a notion of "graph homomorphisms" mapping vertices to adjacent vertices and a generalization of the notion of *graph colouring*. For those interested in this topic, see ref:HERE.

Before continuing with the language of category, we will add one more example of a category which is a little more abstract than the previous categories but one we will constantly use. Namely, we can take categories themselves to be objects and define a notion of morphism between categories! This category will be denoted **Cat**. Since morphisms relate objects, the morphisms in the category of **Cat** allows us to find commonalities between different categories! We have in fact already described a category morphism in example 8.0.3[4]: If we take **Grp**, we can forget the group structure of every group and the homomorphism property of every homomorphsims and we will end up with **Set**. In effect, we did $(G, *) \to G$ and mapped the homomorphism f where $f : (G, *_G) \to (H, *_H)$ to the function f $f : G \to H$ where the function f is identical to the homomorphsims f except it no longer requires the homomorphism property (in fact, it is impossible for it to have the homomorphism property since sets don't have a binary operation). We will call category morphisms *Functors* (as a generalization of the word function)

Definition 8.0.4: [Covariant] Functor

Let C, D be two categories. Then a *Functor* F will consist of:

1. a function between $\text{Obj}(C)$ and $\text{Obj}(D)$. If $a \in C$ is an object, then $F(a) \in D$ denotes the object in D.
2. a function between $\text{Mor}C$ and $\text{Mor}D$. If $f \in C$ is a morphism in C, then $F(f)$ denotes the morphism in D.

such that

1. F must preserves the domain and codomain of each morphism: $f : A \to B$ must map to $F(f) : F(A) \to F(B)$
2. F must map identities to identities: $F(\text{id}_f) = \text{id}_{F(f)}$
3. F must preserve composition: $F(g \cdot_C f) = F(g) \cdot_D F(f)$.

Example 8.0.5: Functor

1. For any category C, there is always a trivial functor sending objects to themselves and functions to themselves
2. The relation above between **Group** and **Set** is indeed a functor: $F : \textbf{Group} \to \textbf{Set}$ Such a functor is called a *forgetful functor* since it forgets some of the structure of the category. In the coming chapters, many of the objects are augmentation of sets. All of these categories will have a forgetful functor $F : \mathbf{C} \to \textbf{Set}$.

Some authors prefer to be more specific about the second condition and define it instead as the following: for any two object $a, b \in \text{Obj}(C)$ there exists a function $\text{Hom}_C(a, b) \to \text{Hom}_D F(a), F(b)$. We will see many more examples of functors throughout this book. For now, it is left as an optional exercise to prove that categories and functors together form a category[3].

So far, all the categories we have dealt were all in some fundamental sense based on sets: each object was either a set with an augmentation (ex. $(G, *)$ or $(S, \leq)$) or a collection of sets following certain restrictions (ex. graphs). Because of this, every object in every category so far has the notion of having "elements". However, there is nothing in the definition of an object in a category that restricts objects to needing elements:

Example 8.0.6: Non-Set Category

Let $\mathbb{N}$ be the set of all natural numbers and $\leq$ be the usual order. Then we define a category C whose objects are the numbers of $\mathbb{N}$ and whose morphism is $\leq$, that is for every $x \in \mathbb{N}$, $\leq$ is the morphism that relates x to numbers greater than or equal to number y, which we will label $x \leq y$. Notice that $\leq$ is not a function, since x is not "mapped" to a single element but to every element greater than it. Furthermore, $\leq$ is the only morphism in in this category. Composition will return $\leq$ (since it's the only morphism) and we see that if $x \leq y$ and $y \leq z$ then their composition results

[3]There is a technicality about the fact we are taking "the category of all categories" in the sense that this is self-referential creating a contradiction usually called Russel's Paradox. There is a definition of *small category* which circumnavigates this problem which was not introduced so as to no overload new terminology

in $x \leq z$.

This is indeed a category: $\leq$ has a domain and co-domain (if you like you can picture $\leq : x \to y$ as long as it is kept in mind that $\leq$ is not a function). For any $x \in \mathbb{N}$, x has an identity morphism, namely the only morphism $\leq$, (you can picture $\leq \cdot \leq = \leq$).

More generally, any set S with some partial order forms a category. For example, if X is a set, then $(P(X), \subseteq)$ is a category whose objects are all subsets of X and the morphism is the partial order induced by $\subseteq$ (i.e. $U \subseteq V$ then $U \to V$).

(Another would be to simply draw diagrams!!)

The set $\mathbb{N}$ was an arbitrary choice in the previous construction: we could have chosen $\mathbb{R}$ or $\{0\}$ or an arbitrary set with an arbitrary ordering. We can visualize the category in the example by imagining a directed graph where every edge represents the morphism between the domain and codomain:

$$\begin{tikzcd}
0 \arrow[r] \arrow[rr, bend right] \arrow[rrr, bend right] & 1 \arrow[r] \arrow[rr, bend left] & 2 \arrow[r] & 3 & \cdots
\end{tikzcd}$$

Such a directed graph is called a *commutative diagram*. The full reason behind the name is a little abstract without the proper build-up[4], however we can still understand the etymology given the following illustrative example: imagine we start at the number 1, and every horizontal arrow represents multiplying on the right by x, and every vertical arrow represents multiplying on the right by y, and we form a square like so:

$$\begin{tikzcd}
1 \arrow[r, "\cdot x"] \arrow[d, "\cdot y"] & x \arrow[d, "\cdot y"] \\
y \arrow[r, "\cdot x"] & yx = xy
\end{tikzcd}$$

Similarly, we could have started with $G \times G$ in the top left, let the right arrow be the swap operation $(x, y) \mapsto (y, x)$, the bottom arrow being the identity $(x, y) \mapsto (x, y)$, and then both arrow to the bottom right being the group multiplication arrows.

This square shape is so common that the any directed graph represented morphisms came to be called a commutative diagram. In section 8.2, we shall further explore the use of commutative diagram to understand properties of categories.

8.1 Basic properties of Morphisms

I clearly don't have enough knowledge to write this yet:

https://en.wikipedia.org/wiki/Monomorphism.

Read this and come back.

When we studied function, we started by classifying injective and surjective functions due to their simplicity and universality. It would be nice if there are some nice equivalent, or at least similar, notions of injectivity and subjectivity for morphisms in general categories. Since objects need not

[4] We will often care about morphisms between categories commuting in some "natural" way. This will lead to the study of natural transformation from which the name commutative diagram will takes it's name

have elements, the definition has to take this generality into account. Fortunately, there is already a property of injective and surjective functions that was characterized in terms of functions in proposition 0.1.26 and 0.1.28: it was shown that all injective functions have a left inverse and all surjective functions have a right-inverse. These notions can be generalized in the following way:

Definition 8.1.1: Monomorphism

Let $f : A \to B$ be a morphism. Then if for any morphism $\alpha : Z \to A$, $\beta : Z \to A$, if

$$f \circ \alpha = f \circ \beta \Rightarrow \alpha = \beta$$

then we call f a monomorphism

Intuitively, monomorphisms should take you to a "unique place" similarly to how injective functions have a single element to represent them.

Proposition 8.1.2: In Set, monomorphism if and only if Injective

Let $f : A \to B$ be a set function. Then f is injective if and only if it is a monomorphism

***Proof*:**

($\Rightarrow$) Let's say f is injective, and that $f \circ \alpha = f \circ \beta$. We want to show that $\alpha = \beta$. By proposition 0.1.26, there exists a left-inverse f^{-1}. Now, notice that

$$\begin{aligned}
(f^{-1} \circ f) \circ \alpha &= f^{-1} \circ (f \circ \alpha) && \text{associativity of composition} \\
&= f^{-1} \circ (f \circ \beta) && \text{by assumption} \\
&= (f^{-1} \circ f) \circ \beta && \text{associativity of composition}
\end{aligned}$$

meaning we get $\mathrm{id}_A \circ \alpha = \mathrm{id}_A \circ \beta$, and so

$$\alpha = \beta$$

as we sought to show

($\Leftarrow$) Let's say f is a monomorphism so that $f \circ \alpha = f \circ \beta$ implies $\alpha = \beta$. Let's say f wasn't injective, that is, for some $a, b \in A$, $f(a) = f(b)$ but $a \neq b$. Choose $B' = f(B)$ to be the domain of α and β where α and β both agree where to send elements with the exception of α sending $f(a)$ to a and β sending $f(a)$ to b. Then

$$f \circ \alpha = f \circ \beta$$

but $\alpha \neq \beta$ a contradiction. Since those elements were arbitrary, it must be the case that,

$$f(a) = f(b) \Rightarrow a = b$$

showing injectivity.

One can also prove this directly by letting $Z = \{p\}$ (i.e., a singleton) and using the monomorphism property to conclude that

$$f(a) = f(b) \Rightarrow a = b$$

Similarly, we can define an epimorphism:

Definition 8.1.3: Epimorphism

Let $f : A \to B$ be a morphism.

If for any function $\alpha : Z \to B$, $\beta : Z \to B$, if

$$\alpha \circ f = \beta \circ f \Rightarrow \alpha = \beta$$

then we call f a epimorphism

Proposition 8.1.4: In Set, epimorphism if and only if Surjective

Let $f : A \to B$ be a set function. Then f is surjective if and only if it is a epimorphism

***Proof*:**
Exercise

We have mentioned earlier that in category theory, we define objects based on how they relate to other objects. We have in fact done this before: When we defined a subgroup based on a group. We mentioned how this more complicated definition generalizes we

Definition 8.1.5: Subobject

Let C be a category and $X \in C$ be an object in this category. We say $Y \in C$ is a *suboject* of X is there exists a monomorphism

$$Y \to X$$

(maybe details with the "up to isomorphism" thing)

To demonstrate what I mean, Let's take an arbitrary set X, and an arbitrary set Y. When can I say that Y is a subset of X, or vice versa? You probably guessed it: if one can be part of included in another, one is a subset of the other. Written in terms of functions: if there exists an inclusion map $i : Y \to X$, then Y is a subset of X ($Y \subseteq X$). We then can define a set being a subset as follows:

If there exists an inclusion map $i : X \to Y$, then X is a subset of Y ($Y \subseteq X$).

If both inclusion maps exist, then there is two injections, and so by Schröder–Bernstein theorem (see appendix A), there exists a bijection between X and Y ($X = Y$). In this way, we can actually *define* a set to be a *subset* of another set if there exists an inclusion map from Y *to* the bigger set. If we weaken our need for i to be an inclusion map, to i being an injective map, we can start comparing sets with different elements [5]

We can apply this now to groups! First, recall (or revisit) what we did in proposition 1.4.8. Since we have proven the proposition is equivalent to the definition of subgroup, we will bring it here, but label it as a second definition of subgroup:

[5]More philosophically, it doesn't actually matter what are the elements of a set; what matters is that there *are* arbitrary distinct elements in a set. So being an inclusion or injective is actually "equivalent" in a broader sense. How so? This question will be answered in appendix **??** with more build-up to the greater philosophy of mathematics

Since the same technic is used in defining both concepts, we can say each are similar concepts or types of "objects". Thus, we can say what we have here is applied the more general idea of *sub-object* to both situations.

Notice how for sets, if there are is an injection going both way, we say that the two sets are bijective and $X = Y$? This is actually *exactly* the same with groups!

Example 8.1.6: sub-objects

1. **subobjects in Grp**: Recall in proposition 1.4.8 where we showed that if $(H, \bullet)$ is a group and $H \subseteq G$, then H is a subgroup if and only if the inclusion map $i : H \to G$ is a homomorphism. This proposition was the proof that **Grp** had subojects!
2. **subobjects in Set**:
3. sub-order?

(maybe mention the question of switching the monomorphism for an epimorphism and how that let's you define something we'll call *quotient objects*, which we'll get too soon?)

(build-up to the idea of isomorphism in category theory)

Definition 8.1.7: Isomorphism In C

Let C be a category, and f be a morphism. Then f is said to be an *isomorphism* if HERE.

Example 8.1.8: isomorphism in different Categories

1. In **Grp**, HERE
2. In **Set**, HERE
3. In the category of ordered sets, HERE

Proposition 8.1.9: Isomorphism Generalized

Let G, H be groups. Then if there exists an injective homomorphism $G \to H$ *and* and injective homomorphism $H \to G$, then we say that G and H are *isomorphic* and we write $G \cong H$.

Notice how I didn't say that $G = H$. That is because I said we have an *injective* homomorphism. I did not say that it's necessarily an *inclusion* map. If it were, we can justify saying $G = H$ completely. However, group theoretically, there is no difference between G and H, making them "equivalent". In this way $G \sim H$ in terms of equality $=$. So, combining the symbols $\sim$ and $=$ gets us $G \cong H$.

The great advantage of defining subgroup/subset in this way is that we can now define what it means to be a sub-object in many other areas of mathematics! We can even ask whether it is *possible* to define the notion of sub-object; there are some areas in mathematics where the concept of an injective map cannot be defined!![6]

[6] The category of posets $(P, \leq)$ is an example. See appendix **??**

Also, perhaps define Hom, Aut, and End here? Perhaps Also give the category of **Set**? To be able to give an idea of what a category is and to bet able to define $\text{Aut}_{\textbf{Set}}(A)$ for group actions. Introduce the notation

$$\text{Sym}(X) := \text{Aut}_{\textbf{Set}}(X)$$

to use it later?

Show how for any category C, and any object $X \in C$, then $\text{Aut}_C(X)$ forms a group! In this way, groups appear everywhere. The fact that $\text{Aut}_C(G)$ is a group will come back big-time when studying *group actions*.

Next talk about initial and final objects. This leads to universal properties. Bring up how $\{e\}$ was an initial and final object in **Group**. Bring up how initial and final objects are unique, so once we proved on exists, we proved it's the only one up to isomorphism.

8.2 Commutative diagrams

As mentioned countless times, categories try to capture the idea of relations between objects. Very often, this is done by forcing some relation between morphisms. We've already seen an example of this in the category of **Set** and **Grp** by saying the composition of two functions (or homomorphisms) is again a function (or homomorphism). We represented this by the diagram:

$$\begin{tikzcd} X \arrow[r, "f"] \arrow[rr, bend right, "h"'] & Y \arrow[r, "g"] & Z \end{tikzcd}$$

where we usually denote h as $g \circ f$. We can form all sorts of relations between objects and enforce the fact that compositions of functions must lead to other arrow (TBD). We call this a commutative diagram

(category can be projected down to a graph. I you only project (PROPER PROPERTY HERE), then you get a special graph called a *quiver*, which try to be a generalization of algebraic structures (TBD).)

Definition 8.2.1: Commutative Diagram

Give a nice definition here

Example 8.2.2: Example

Must re-phrase

1. Recall that a group homomorphsims is a function $\varphi : G \to H$ which preserves the group action. If we take G, and H represent the underlying set, and $(G, \cdot_G)$, $(H, \cdot_H)$ represent the set with the underlying operation, then we can what's the most *natural* requirement for the operation to be preserved? Note that if we drop the operation preserving part of φ, we can think of the inputs of the binary operation as [a]

$$(\varphi \times \varphi) : G \times G \to H \times H \qquad (\varphi \times \varphi)(a, b) = (\varphi(a), \varphi(b))$$

Then, we can combine this with the operations on the group to get the following diagram:

$$\begin{tikzcd} G \times G \arrow[r, "\varphi\times\varphi"] \arrow[d, "\cdot_G"] & H \times H \arrow[d, "\cdot_H"] \\ G \arrow[r, "\varphi"] & H \end{tikzcd}$$

Thus, from a categorical perspective, the best way to define an operation to be preserving is if this diagram *commutes*, meaning:

$$\begin{tikzcd} (a,b) \arrow[d, mapsto] \arrow[r, dotted, dash] & {} \arrow[d, dotted, dash] \\ a \cdot b \arrow[r, mapsto] & \varphi(a \cdot b) \end{tikzcd} \qquad \begin{tikzcd} (a,b) \arrow[r, mapsto] \arrow[d, dotted, dash] & (\varphi(a), \varphi(b)) \arrow[d, mapsto] \\ {} \arrow[r, dotted, dash] & \varphi(a) \cdot \varphi(b) \end{tikzcd}$$

which in it's familiar form would be $\varphi(a \cdot b) = \varphi(a)\varphi(b)$

2. maybe bring up how 1.9 is not necessarily a commutative diagram since the arrow's aren't necessarily restricted through composition.

[a]The reason for this construction, as we shall soon see, relies on the *universal property of products*. For now, since it's something all readers would know at this point, we leave it as intuitive

8.3 More abstract examples

(Give an example of a slice category or comma category, and a fibered category. Explain how the point of these categories and many like them is to capture certain relations between objects.

8.4 Summary

A category (definition 8.0.2) supplies objects, morphisms between them, and a composition law; a functor (definition 8.0.4) is the corresponding map between categories; and a commutative diagram (definition 8.2.1) is the notation in which an equality between composites is asserted. Against that backdrop, each notion below replaces a statement about *elements* by a statement about *morphisms*. What licenses the replacement in each case is that the two agree in **Set**, where the element-level notion is the familiar one.

*Notion in **Set***	*Categorical form*	*Agreement in **Set***
Injective function	Monomorphism (definition 8.1.1)	proposition 8.1.2
Surjective function	Epimorphism (definition 8.1.3)	proposition 8.1.4
Subset $Y \subseteq X$	Subobject (definition 8.1.5)	inherited, the definition being via a monomorphism Y -

The pattern is the one every later categorical definition in this book follows: a property phrased by quantifying over elements is re-phrased by quantifying over morphisms into or out of the object, and the re-phrasing is checked against **Set** before being adopted elsewhere.

Part II

Rings

Now that we've the essentials covered groups, we can move on to structures which have 2 binary operations on the set. These structures are called *rings* [7] As with commutative groups, commutative rings will be very different from rings, so much so that one can think of ring theory as really being 2 seperate domains put under one label: commutative rings and non-commutative rings. However, similarly to how all groups share some common structure (a notion of homomorphism, isomorphism theorem, direct product, free group, group-action, etc.), so do rings, and so we will be building on the foundation of ring theory.

We may wonder how to interpret adding a second binary operation to an abelian group? Does this restrict our structure? Does this limit what we can prove? Does it give more tools to work with when proving? The way I like to perceive adding this second operation is that there are more properties that we have at our disposal to work with. The "type" of new property we add also determines what properties we work with (for example, as is briefly addressed in appendix C, section C.4, we can add a topological structure to groups which gives us different tools we can use to study groups). In the case of adding a new binary operations, we gain more "algebraic" tools to work with. For example, we have an understanding of how the prime numbers of $\mathbb{N}$ interact when we add them[8] or when we multiply them, but we have little knowledge of the interaction of prime numbers between these two binary operations. Furthermore, for any abelian group, we can always add at least one new binary operation that satisfies the axioms of a ring[9] (which we will present near the end of section 9.1.1). In that way, all groups can be made into rings, and we can study them as rings.

For the 1st operation, we will always use additive notation. For example, if we're taking the internal direct product of two groups in this notation, we will write $H + K$, since we will reserve the multiplicative notation for the 2nd binary operation.

[7] The term *ring* was probably a contraction of David Hilbert's term *Zahlring*, literally translated as *number ring*. The word ring was probably coined because of the cyclic nature of the numbers Hilbert was working with. More details on this discussion can be found here: https://math.stackexchange.com/questions/61497/why-are-rings-called-rings

[8] All be it to a very limited degree

[9] Similarly to how all abelian groups are topological groups with the discrete topology

9

Introduction To Rings

With Groups extensively covered, what's "basic" has now changed. We need not so extensively elaborate intuitions for homomorphisms, quotients, products, or actions. Instead, we'll show how these concept grow when considering *rings*. Fundamentally, a ring is a set with two operators (hence, is a triple $(R, +, \cdot)$ where $+$ and $\cdot$ satisfy certain axioms[1]). We will show some really interesting examples which we gain when we introduce a second operator, and take advantage of some of the categorical language we have built-up to shed some more light on the nature of rings (which will be noticably different than that of groups).

As with groups homomorphisms, "ring homomorphisms" will play a central role in the study of rings. There will be an equivalent to "normal groups" which we will extensively study: the equivalent for rings is called *ideals*. Manipulating and combining ideals is at the core of ring theory. After exploring ideals, we will take a moment to look at a way to embedd any ring R into a larger ring that permits the notion of "division" (as we'll see, unlike groups, not all elements of a ring will have an inverse with the second operation). Through this construction, we will show how to take $\mathbb{Z}$ and create $\mathbb{Q}$.

[1] These axioms can be said to be "natural" in some way. These exploration's will be done later in the chapter, see ref:HERE

9.1 Basic Properties and Definitions

Definition 9.1.1: Ring

A *ring* R is a set together with two binary operations $+$ and $\cdot$ (called addition and multiplication), $(R, +, \cdot)$ satisfying the following axioms:

1. $(R, +)$ is an *abelian* group, where the identity is denoted by 0
2. **(Mult. Associativity)** $(a \cdot b) \cdot c = a \cdot (b \cdot c)$ for all $a, b, c \in R$
3. **(Mult. Identity)** There exists an $e. \in R$ such that for every $r \in R$, $e. \cdot r = r \cdot e. = r$. The element $e.$ is usually labeled 1.
   - Sometimes, we use *rng* to denote a ring without multiplicative identity.
4. **(Distributivity)** For all $a, b, c \in R$

$$a \cdot (b + c) = (a \cdot b) + (a \cdot c) \qquad \text{and} \qquad (a + b) \cdot c = (a \cdot c) + (b \cdot c)$$

   The two equations separately are called left and right distributively respectively.

If $a \cdot b = b \cdot a$ for all $a, b \in R$, we call the ring a *commutative ring*.

We'll usually write $a \cdot b$ as ab for convenience. The convention that the multiplicative identity is 1 and additive identity is 0 is essentially universal, and so it is safe enough to have it as part of the definition of a ring.

9.1.2 Ring always has identity? Some author's do not include 1 in the definition of a ring (we called rings without a (multiplicative) identity *rng*); in this case, a ring with a multiplicative identity is usually called a unital ring. Any *rng* R can be extended to a ring by taking $\mathbb{Z} \oplus R$ with identity $(1, 0)$ (What is the multiplication here that makes $\mathbb{Z} \oplus \mathbb{R}$ into a ring?)[a]. The most common place in mathematics where mathematicians usually don't want rings to have identities is in analysis; for those inclined towards examples see [42]. In Algebra, it is now pretty rare to find a textbook that define a ring without a multiplicative identity, however some classical textbooks have done so (for example [28] and [33])

[a] More generally, taking the direct sum with any [unital] ring will do, however different choices might lead to better result in different fields, for those interested see exercise ref:HERE

Just like associativity was a powerful axiom, so is distributivity for linking the $+$ operation and the $\times$ operation. If you have any arbitrary expression:

$$(a + b)(c + d + e(f + g(i + j)kl(m + n)op))q$$

It can be simplified to the sum of monomials:

$$\begin{gathered} ac + bc + ad + bd + aefq + befq + aegiklmopq + begiklmopq + aegjklmopq \\ +begjklmopq + aegiklnopq + begiklnopq + aegjklnopq + begjklnopqq \end{gathered}$$

There are non-distributive algebraic structures. However, the study of non-distributive structures seems to be much scarser than that of non-associative structures or structures without inverse. [2]. Distribuitivity seems to be how most structure link the two operations; at the minimum it usually would be distributive on only one side [3].

9.1.3 Note R being an abelian group is unavoidable given the distributive property

$$a+b+a+b=(1+1)(a+b)=a+a+b+b \Rightarrow a+b=b+a$$

Another way to think about G being abelian is that we want to concentrate on the *ring* operation now, so if we can make the group-operation very-well behaved (as we've shown commutativity does in chapter 5), it will not obtrude our study of the ring operation. This is a common technic.

Example 9.1.4: Ring Examples

1. Taking any abelian group G, and define $a \cdot b = 0$. This is the *trivial ring.* If If $G = \{0\}$, then we call this the *zero ring.* In fact, all $ab = 0$ for all $a, b \in R$ if and only if R is the zero ring:

   Proof. If R is the zero ring, then we already have that $0 \cdot 0 = 0$. Thus, we must prove the converse

   Let's say $\forall a, b \in R$, $ab = 0$. Then $1 \cdot 1 = 0$, meaning $1 = 0$

   Then, notice that

   $$\begin{aligned} &\Leftrightarrow\ 0\cdot a = (0+0)\cdot a = 0\cdot a + 0\cdot a \\ &\Leftrightarrow\ 0\cdot a = 0\cdot a + 0\cdot a \\ &\Leftrightarrow\ 0\cdot a - 0\cdot a = 0\cdot a + 0\cdot a - 0\cdot a && \text{additive inverse} \\ &\Leftrightarrow 0 = 0\cdot a \end{aligned}$$

   Thus, we have $0 = 0 \cdot a = 1 \cdot a = a$ for all a, and so all element equal 0, showing it's a zero ring. □

   In fact, this proof has a very powerful corollary: if $1 = 0$, then we have the zero ring! This property usually breaks many parts of many proofs, so we usually say that $1 \neq 0$ in most theorems and propositions. In some textbooks, this is even part of the axioms of a ring

2. The integers $\mathbb{Z}$ is the canonical example of a ring; when ring theory started, a lot of study was with trying to characterize properties of integers. Note that $ab = ba$, making $\mathbb{Z}$ a *commutative ring.*

3. $\mathbb{Z}/n\mathbb{Z}$ is a [commutative] ring with identity

4. $\mathbb{R}, \mathbb{Q}, \mathbb{C}$ are all [commutative] rings with identity. Naturally, $\mathbb{N}$ is not [a]

5. If R and S are rings, then $R \times S$ is also a ring with operation component wise. [b]. What is the identity of $R \times S$?

[2] I found this paper: generalized De Morgan algebra
[3] Something called a *Quasifields* is an example

6. If you have taken Linear Algebra, then all fields are an example of [commutative] rings

7. Recall the group Q_8 (see 1.2.4). Using it, we will define the set $\mathbb{H} = \{a + bi + cj + dk \mid a, b, c, d \in \mathbb{R}\}$ where the elements bi, cj, an dk are distinct elements of $\mathbb{H}$, and are not another element in $\mathbb{R}$ or Q_8 (think of a, b, c, d as coefficients). Then we can make $\mathbb{H}$ into a ring by defining addition to be ponitwise:

$$(a + bi + cj + dk) + (a' + b'i + c'j + d'k) = (a + a') + (b + b')i + (c + c')j + (d + d')k$$

and multiplication through repeated use of the distributive property, which distributes to:

$$\begin{aligned}(a + bi + cj + dk)(a' + b'i + c'j + d'k) = &(aa' - bb' - cc' - dd') \\ &+ (ab' + ba' + cd' - cd')i \\ &+ (ac' - bd' + ca' + db')j \\ &+ (ad' + bc' - cb' + da')k\end{aligned}$$

Then $\mathbb{H}$ is called the *(real) Hamiltonians* (real since we could have defined them over $\mathbb{C}$). They form a non-commutative ring, and can be considered an extension of the complex numbers $\mathbb{C}$ (there are in fact two copies of $\mathbb{C}$ inside $\mathbb{H}$, more in that in exercise ref:HERE)

8. If X is a set and A is a ring, then the collection of all function $f : X \to A$ are a ring with function addition and multiplication being done point-wise: $(f + g)(x) = f(x) + g(x)$, $(fg)(x) = f(x)g(x)$. All ring structure on this function space comes from A. A concrete example of this is if $X = [0, 1]$, $A = \mathbb{R}$, and f is continuous $f : [0, 1] \to \mathbb{R}$. These types of rings show up a lot in analysis with the collection of function from a space X (a topological space, or measurable space, a manifold, etc.) to $\mathbb{R}$ or $\mathbb{C}$ giving many properties of the domain of the functions.

9. The set $2\mathbb{Z}$ is not a ring since it does not have an identity. It is thus a *rng*.

10. From section 1.2.5⋆, we saw that $(M_n(G), \cdot)$ is not necessarily a group since inverses were not not always defined. However, this is not a requirements for ring multiplication! It is very tedious and a bit hard at this stage to check that associativity and distributivity hold (there will be a simple proof of this in corrollary 13.2.7), but they are verifiable and show that $(M_n(R), +, \cdot)$ is a ring. Note that $M_n(R)$ is not commutative. For example

$$\begin{pmatrix} 1 & 0 \\ 1 & 0 \end{pmatrix} \cdot \begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix} \neq \begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 1 & 0 \end{pmatrix}$$

Matrix rings will be explored in more detail soon, and will have a whole chapter (in particular chapter 53) dedicated to the study of when matrix rings "act" on finite groups (in a similar way groups acted on sets in group-actions).

11. Recall when I asked if $P(X)$ is a group under $\cup$? Here is another similar question with rings. Take $P(X)$ again, and define addition and multiplication as

$$A + B = (A - B) \cup (B - A) \qquad A \cdot B = A \cap B$$

prove that it's a [commutative ring]. Another thing you can show is that for any $Y \in P(X)$, $Y^2 = Y$. If a ring has this property, it is said to be a *boolean ring.* [c]

12. Let G be any abelian group. Then $(\text{End}_{\mathbf{Grp}}(G), +, \circ)$ is a ring. In this way, any ring is formed from an abelian group. [d]

[a]$\mathbb{N}$ is a semi-ring with $+$ and $\cdot$

[b]Though this might feel "trivially true", the fact that taking the direct product of two rings will produce the same algebraic structure, in this case a ring, is not always true! See 9.1.27 for a counter-example

[c]This ring comes up in *measure theory*, where the set of measurable sets can be made into a boolean ring

[d]The general term for taking [abelian] objects (ex. abelian groups) from an category C and working with $\text{End}(X)$ as the new object and morphisms that also preserve $\circ$ is **Ab**-enriched category given the abelian category C

When working with rings, we will assume the usual ring structure for $\mathbb{Z}, \mathbb{Q}, \mathbb{R}$ and $\mathbb{Z}/n\mathbb{Z}$, and use the same notation if it's clear in context. The notation $\mathbb{Z}^{\times}, \mathbb{Q}^{\times}, ...$ will soon be repurposed for another use later in this section.

Note that every ring is naturally an abelian group as well as a monoid (see section 7.1). When working with homomorphisms for rings, both structure need to be preserved in some way for the image of a funciton to be a ring. There is in fact one more sneaky criterion that needs to be added in order for a the image of a function be be a ring, and hence a *ring homomorphism*:

Definition 9.1.5: Ring Homomorphism

Let R and S be rings. Then:

1. *ring homomorphism* is a map $\psi : R \to S$ satisfying
   (a) $\varphi(a + b) = \varphi(a) + \varphi(b)$
   (b) $\varphi(a \cdot b) = \varphi(a) \cdot \varphi(b)$
   (c) $\varphi(1_R) = 1_S$
2. If $\varphi : R \to S$ is a ring homomorphism, and there exists a ring homomorphism $\psi : S \to R$, then φ is a ring isomorphism

Note that just like for group homomorphisms, it can be proven that definition 9.1.5[2] is equivalent to a ring homomorphism being bijective.

Note that the reason we add the condition that $\varphi(1_R) = \varphi(1_S)$ is so that the image $\varphi(R)$ is also a ring. If a ring is not defined to have a multiplicative identity (as is done for *rngs*), then the condition of $\varphi(1_R) = \varphi(1_S)$, is not necessary. Furthermore, where $\psi(e_G) = \psi(e_H)$ is a provable property given that ψ is a group-homomorphism and e_G, e_H are the respective identities of the domain and codomain, the same is not true for rings, that is, the first 2 conditions of a ring homomorphism can be satisfied, however the resulting image will not be a ring (take $f : \mathbb{Z} \to \mathbb{Z}$, $x \mapsto 2x$, see exercise ref:HERE).

Many times, mathematical objects will have multiples different structures defined on them, but we sometimes want to only work with some of them[4]. Thus, a group homomorphism between rings will only preseve group structure, a while ring homomorphism preserves both. Ring homomorphisms will be further explored in section 9.3, where we will show how to relate the structure of $\varphi(R) \subseteq S$ to R similarly to how it was done with quotient groups.

From definition 9.1.5, we can define a sub-ring:

[4]An example with full of different structures is $\mathbb{R}$ which can come with a group, ring, and order and structure, but also structures that we haven't covered, some of which are non-algebraic (ex. completeness (see uniformly continuous functions and uniform spaces), or topological)

Definition 9.1.6: Sub-Ring

Let R be a ring and $S \subseteq R$ a subset. Then S is a *subring* of R if it is still a ring under the same operations as R. In particular, a subring of a ring R is a subset $S \subseteq R$ with a ring operation such that the inclusion map $i : S \to R$ is a ring homomorphism [a].

[a] See proposition 1.4.8

Corollary 9.1.7: sub-ring condition

a *subring* of the ring R is a subgroup of R that is closed under multiplication

***Proof*:**
see exercise ref:HERE

It is simpler to prove a subset is a subring as compared to subgroups: you need to show it' an abelian sub-group that is still closed under multiplication. Notice that there is no special symbol for a subring like there as for subgroup. This will be a recurring pattern as we continue to explore algebra: there is no special symbol for "subobjects". We will always say "Let S be a subring of R" or "Let $S \subseteq R$ be a subring of R". Another consideration to keep in mind is that the identity of a subring need not be the same (since an inclusion homomorphism $S \hookrightarrow R$ need not map the identity to itslef): If $R \times S$ is a ring, $R \times \{0\}$ is a subring. What is it's [multiplicative] identity?

One subring is available in every ring and measures how far the multiplication is from being commutative. It is the ring-theoretic analogue of definition 2.2.4, and unlike the group case it is a subring rather than merely a subgroup.

Definition 9.1.8: [Ring] Center

Let R be a ring. The *center* of R is

$$Z(R) := \{c \in R \mid cx = xc \text{ for all } x \in R\}$$

the set of elements commuting with every element of R. A ring is commutative exactly when $Z(R) = R$.

That $Z(R)$ is a subring is immediate from corollary 9.1.7: it contains 0 and 1, it is closed under subtraction since $(c - c')x = cx - c'x = xc - xc' = x(c - c')$, and it is closed under multiplication since $(cc')x = c(c'x) = c(xc') = (cx)c' = x(cc')$. It is commutative, being a set of pairwise commuting elements, so the center of any ring is a commutative ring.

Proposition 9.1.9: Properties of Rings

Let R be a ring. Then:

1. $0a = a0 = 0$ for all $a \in R$
2. $(-a)b = a(-b) = -(ab)$ for all $a, b \in R$ (recall $-a$ is the additive inverse)
3. $(-a)(-b) = ab$ for all $a, b \in R$
4. The multiplicative identity is unique and $-a = (-1)a$.
5. Generalized Distribuitivity: Distributivity can be extended inductively.

***Proof*:**
All of these proofs take advantage of the abelian structure of R and distributivity. For example:

1. Using the fact that 0 is an additive identity and distribuitivity, we get:
$$0 \cdot a = (0 + 0)a = 0 \cdot a + 0 \cdot a$$
Since $0 \cdot a \in R$, there exists an additive inverse, thus
$$0 \cdot a - 0 \cdot a = 0 \cdot a + 0 \cdot a - 0 \cdot a$$
implying $0 = 0 \cdot a$. A similar argument can be made for $0 = a \cdot 0$

the other 3 proofs continue in a similar fashion.

:exercise Show that $(-1)^2 = 1$.

Consequences of no Multiplicative Inverse

Now that we know what is needed to be a ring, let's put names to common rings with properties added on. For each of these properties, I would recommend to go back and see which rings we covered have these properties. If you want to verify your intuition, you can check glossary (see part XIV).

The first thing is that the multipication operation is missing one of the axioms of a group: elements need not have a multiplicative inverse. If every element has a multiplicative inverse, we will give a name to such a ring:

Definition 9.1.10: Division Ring

A ring R with identity 1, where $1 \neq 0$ is called a *division ring* (or a *skew field*) if every non-zero element $a \in R$ has a multiplicative inverse, $b \in R$ such that $ab = 1$. It can be proven this implies that $ba = 1$

If furthermore, multipication is commutative, the resulting ring is a commutative division ring, whcih is called *field*:

Definition 9.1.11: Field

Let F be a divison ring. If F is also *commutative*, then F is called a *field*.

Sometimes, the symbol k (from German Körperor[5]) or $\mathbb{F}$ will be used for a field The field is the "smallest" mathematical structure in which we can perform all the standard arithmetic operations $+, -, \times, \div$ in the usual way. If R is a field, then it is as though there is two compatible abelian group structures (both are associatve, have an identity, have an inverese, both commute), the two operations being linked through distributivity. In particular, it's like $(R, +)$ and $(R - \{0\}, \cdot)$ are combined through distributivity. . If R is a field, we can call the sub-rings of R a sub-field. For example, $\mathbb{Q}$ is a subfield of $\mathbb{R}$.

(TBD revise this in light of the following fact: If F and K are field, $(F, +) \cong_{\mathbf{Grp}} (K, +)$ and $F^\times \cong_{\mathbf{Grp}} K^*$, then is it true that $F \cong_{\text{Ring}} K$? I saw that no, and in principle there can be uncountably many combinations of possible field structures)

Example 9.1.12: Exercise

Prove that any subfield of $\mathbb{R}$ must contain $\mathbb{Q}$

Proof. Let $K \subseteq \mathbb{F}$ be a subifeld. First show the identity of $\mathbb{R}$ is the identity of K. Then $\mathbb{Z}$ is there (why?). Then $\mathbb{Q}$ is there (why?) $\square$

define a quandle (an exampleis matrices over non-commutative rings and $(a^b)^c = (a^c)^{(b^c)}$. Show that conjugation of a group defines a quandle. This is more useful in set theory

Zero divisors and Units

If a ring is not a division ring, then the cancellation property does not necessarily hold (see proposition 1.1.17). In groups, we constantly take advantage of the fact that $ab = ac \Rightarrow b = c$. This proof relied on the [unique] inverse of elements. However, there are rings in which this does not happen

Example 9.1.13

Let $R = \mathbb{Z}/6\mathbb{Z}$ with the usual addition and multiplication. First, show that it's a ring.

Then, notice that $[2]$ times any number in $\mathbb{Z}/6\mathbb{Z}$ is not $[1]$ (i.e. there does not exist $[a]$ such that $[2][a] = [1]$):

$$[2][1] = [2]\,, [2][2] = [4]\,, [2][3] = [0]\,, [2][4] = [2]\,, [2][5] = [4]$$

Furthermore, there exists an element ($[3]$) such that $[2]$ times it gives the additive identity: $[2][3] = [0]$

So it breaks down! You might wonder if for any $a \in R$ if there exists an element b such that $ab = 0$ or $ab = 1$, but that need not be the case:

[5]some words about whatever German Mathematician pioneered fields that lead to this notation

Example 9.1.14: No Multiplicative Inverses

In $\mathbb{Z}$ with the usual addition and multiplication, take any $a \in \mathbb{Z}$, $a \notin \{0, 1\}$. Then there does not exist an element b such that $ab = 1$, which is shown by $\mathbb{Z}$ not having multiplicative inverse, but also $ab = 0$ cannot happen in $\mathbb{Z}$ for any $b \in \mathbb{Z}$

Thus, one cannot imply the other in general

You might further wonder if for any $a \in R$ that only $ab = 0$ happens or only $av = 1$ happens, that is, if there exists an element $b \in R$ such that $ab = 0$, then there does *not* exist an element $v \in R$ such that $av = 1$ (or vice versa). However, this again is not the case! What if $ab = 0$, or $ab = 1$, does that means $ba = 0$ or $ba = 1$? Also no (in the non-commutative case)!

Example 9.1.15

1. Take S to be the set of all infinite sequences. This set, under component-wise addition, is a group. Then take $\mathrm{End}(S)$, making function composition multiplication. Then this is a ring with identity.
2. Prove that

$$\begin{aligned} P(a_1, a_2, ..., a_n, ...) &= (a_1, 0, ..., 0, ...) && (9.1)\\ L(a_1, a_2, ..., a_n, ...) &= (a_2, a_3, ..., a_{n-1}, ...) && (9.2)\\ R(a_1, a_2, ..., a_n, ...) &= (0, a_1, ..., a_{n+1}, ...) && (9.3)\\ 0(a_1, a_2, ..., a_n, ...) &= (0, 0, ..., 0, ...) && (9.4)\\ 1(a_1, a_2, ..., a_n, ...) &= (a_1, a_2, ..., a_n, ...) && (9.5)\\ & && (9.6) \end{aligned}$$

are all elements in $\mathrm{End}(S)$ ($L, R, P, 0, 1 \in \mathrm{End}(S)$)
3. Show that $LP = 0$, but $LR = 1$. Then, show that $PL \neq 0$ (so $LP = 0$, but $PL \neq 0$)
4. You just showed that $LR = 1$. Show that $RL \neq 1$.

Since it's not commutative, it's not imediately clear that since we found an example where $ab = 0$ but $ac = 1$, then we can find the same but multiplying from the left side. However, we can construct exactly such an example from the previous example:

1. Define $\mathrm{End}(S)^{op}$ to be the object where function composition is reversed:

$$f \cdot g = g \circ f$$

show that $\mathrm{End}(S)^{op}$ is still a ring. Convince yourself that all the previous elements $(1) - (5)$ are in $\mathrm{End}(S)^{op}$.
2. Show that $PL = 0$, but $RL = 1$. Then, show that $PL \neq 0$ (so $LP = 0$, but $PL \neq 0$)

So, we cannot carry over our intuition of the cancellation property into general rings! However, thanks to associatvity, there is *one* relation we can prove: if $ab = 0$, then there does not exist a c such that $ca = 1$. That is, if there exists an element b such that $ab = 0$, then there does not exist a c

such that $ca = 1$:

Proposition 9.1.16: Partial Result for ab and ca

Let $a \in R$ such that there exists an element $b \in R$ such that $ab = 0$ (or $ba = 0$). Then there does not exist an element c such that $ca = 1$ (or $ac = 1$), that is:

$$\begin{array}{c} ab = 0 \\ ca \neq 1 \end{array} \quad \text{and} \quad \begin{array}{c} ab = 1 \\ ca \neq 0 \end{array}$$

***Proof*:**
Take $a \in R$ as in the proposition. Let's say there exists a $b \in R$, $b \neq 0$, such that $ab = 0$. Let's say there is a c such that $ca = 1$, $c \neq 0$. Then:

$$b = 1b = (ca)b = c(ab) = c(0) = 0$$

a contradiction since $b \neq 0$

Let's say there exists a $b \in R$, $b \neq 0$, such that $ba = 1$. Let's say there is a c such that $ac = 0$, $c \neq 0$.

$$c = 1c = (ba)c = b(ac) = b \cdot 0 = 0$$

a contradiction since $c \neq 0$

Note that as a consequence of this theorem, if an element a has a left-sided inverse l and ar right-sided inverse r, then $l = r$, since:

$$l = l1 = l(ar) = (la)r = 1r = r$$

Since multiplicative inverses can either multiply to 1, to 0 or to neither, we will take a moment to name elements that can multiply into 1, 0, or neither:

Definition 9.1.17: Zero Divisors and Units

Let R be a ring.

1. Let $a \in R$ be a nonzero element. Then a is called a *left* (resp. *right*) *zero divisor* if there is a nonzero element $b \in R$ such that ab (resp. ba) equals to 0. If $ab = ba = 0$, then a is called a *two-sided zero divisor*, or simply a *zero-divisor*

2. Assume $1 \neq 0$. An element u of R is called a left (resp. right) *unit* in R if there is some v in R such that vu (resp. uv) equals to 1. If $uv = vu = 1$. , then u is called a *two-sided unit*, or simply *unit* for short

If R is commutative, then there is no distinction between left and right zero-divisors/units.

The term zero-divisors comes from the following observation: if $ab = n$ in a commutative ring for simplicity, then we may say that $a|n$ and $b|n$ (i.e. a, b divide n). Then if $ab = 0$, then "$a\big|0$" and "$b\big|0$", so a, b divide 0 (they are *zero-divisors*). This notion of divisibility shall further be explored in chapter 10. Note that if u is a unit of R so that there is some v such that $uv = vu = 1$, then v is unique (think of the same proof for groups). The same is not the case for zero-divisors (try playing around in $\mathbb{Z}/10\mathbb{Z}$). Since any unit must be a two-sided inverse, we can define the following:

Proposition 9.1.18: Group of Units

Let R be a commutative ring, and let $R^\times$ denote the set of all units of R. Then $R^\times$ is a group under $\cdot$, and is called the *group of units of R*

***Proof*:**
Associativity comes from associativity of multiplication, multiplicative identity comes from the fact that 1 is a unit, and inverses come from the definition of units.

It only remains to show that $\cdot$ is indeed a well-defined operation; does it map units of R to units of R. Take $a, b \in R^\times$. Since both are units, then there exists a u and v st $au = ua = bv = vb = 1$. Then I claim that vu is the element such that $(vu)(ab) = 1$. Indeed:

$$(vu)(ab) = v(ua)b = v \cdot 1 \cdot b = v \cdot b = 1$$

Thus $ab \in R^\times$, showing it is indeed well-defined as we sought to show

Sometimes, the group of units of R is denoted $U(R)$, R^*, or $E(R)$ (from German *Einheit*, meaning "oneness") Notice how this let's us characterize division rings and fields in a new way: if every non-zero element of a ring (resp. commutative ring) is a unit, the ring is then a division ring (resp. field). Note too that it is not always obvious when a ring can be characterized as a field in this way:

Example 9.1.19: Every Element is a Unit
Let D be a not-a-perfect square rational number [a] Then take $\mathbb{Q}(\sqrt{(D)}) \subseteq \mathbb{C}$,

$$\mathbb{Q}(\sqrt{(D)}) := \left\{ a + b\sqrt{(D)} \;\middle|\; a, b \in \mathbb{Q} \right\}$$

Note that if $D > 0$, then $\mathbb{Q}(\sqrt{D}) \subseteq \mathbb{R}$. This set is indeed a ring since

$$\begin{aligned} &\forall x, y \in \mathbb{Q}(\sqrt{5}) \qquad a + b\sqrt{D} - c - d\sqrt{D} = (a - c) + (b - d)\sqrt{D} \\ &\forall x, y \in \mathbb{Q}(\sqrt{5}) \qquad (a + b\sqrt{D})(c + d\sqrt{D}) = (ac + bdD) + (ad + bc)\sqrt{D} \end{aligned}$$

showing it's indeed a [commutative] ring.Note that D being not-a-square implies every element can be written uniquely. If it were a square, say 9, then

$$2\sqrt{9} = 6$$

meaning two elements were written the same way! Because of this, if a, b are non-zero, then $a^2 - b^2 D$ is non-zero. For example, if $D = 4$, then

$$4^2 - 2^2 4 = 16 - 16 = 0$$

but Since D is not-a-square,

$$a^2 - b^2 D = 0 \Rightarrow a^2 = b^2 D \Rightarrow \frac{a^2}{b^2} = D \Rightarrow \frac{a}{b} = \sqrt{D}$$

which since D is not-a-square, it's square-root is not a rational number [b], hence a contradiction.

Thus, $a^2 - b^2 D$ is non-zero if a, b are non-zero. The point of this is to notice that

$$(a + b\sqrt{D})(a - b\sqrt{D}) = a^2 - b^2 D$$

then $a + b\sqrt{D}$, $a - b\sqrt{D}$ must both be non-zero! Thus, we can define

$$\frac{a - b\sqrt{D}}{a^2 - b^2 D}$$

to be the inverse of $a + b\sqrt{D}$! This shows that every non-zero element of $\mathbb{Q}(\sqrt{D})$ is a unit, and hence it's a field!

[a] perfect square number: $x \in X$ such that $\sqrt{x} \in X$
[b] A similar proof to that of $\sqrt{2}$ is not rational!

This also means that since every element in a field has a multiplicative inverse, a field has *no* zero-divisors.

As we saw, if x is *not* a zero divisor, this does not imply that x *is* a unit, and vice-versa, as we have seen with the integers. However, in the finite case, this *does* hold

Proposition 9.1.20: In finite ring, not zero divisor ⇒ unit

If R is a finite ring, and $x \in R$ is *not* a zero-divisor, then x is a unit

***Proof*:**
The key idea of finiteness is that we can take advantage of forcing injective maps being surjective. Withought loss of generality, when there are zero divisors, we take the left zero divisors. The proof for right zero-divisors works identically by switching the order of the elements.

Let's say $x \in R$ is not a left zero divisor so that for no element $y \in R$ there is $xy = 0$. If $x = 1$, we're done, so let's say $x \neq 1$. Define the map $y \mapsto xy$. [a]. Since for no y there is $xy = 0$, then this map is injective. Since R is finite and the map is from R to R, then this map is also surjective. Thus, there exists a y' such that $xy' = 1$. It remains to show that $y'x = 1$.

The element x can either be a right zero-divisor or it can not be. If it is not a right-zero divisor, then the map the map $y \mapsto yx$ would be injective, and hence surjective, and we can find an element y'' such that $y''x = 1$. So, let's say x was a right zero-divisor. If it was a right zero-divisor, then there would exist a $z \in R$ such that $zx = 0$, and x would be a right, but not a left zero-divisor. However, there exists a y' such that $xy' = 1$, which is a contradiction by proposition 9.1.16. Therefore, in the finite case, if x is a right zero-divisor, it would also have to be a left zero-divisor

Therefore, x is not a zero-divisor, so we can again define the map $y \mapsto yx$ which is also injective, and so there exists a y'' such that $y''x = 1$. Finally, notice that:

$$y' = 1 \cdot y' = (y''x) \cdot y' = y'' \cdot 1 = y''$$

showing that $y' = y''$ and so $xy' = y'x = 1$, fulfilling the condition for x being a unit, as we sought to show.

[a] Tangentially, note that this function is not in general a ring homomorphism – can you see why?

No Zero-Divisors: Integral Domains

Numbers as they are generally known do not have zero divisors (ex. $\mathbb{N}, \mathbb{Z}, \mathbb{Q}, \mathbb{R}, \mathbb{C}$). A large part of the study of rings is the study of number theory, and so defining rings which do not have zero divisors is important for ease of communication:

Definition 9.1.21: Integral Domain

A commutative ring with identity $1 \neq 0$ is called *integral domain* if it has a no zero divisors

Example 9.1.22: Integral Domains

1. $\mathbb{Z}$ is the canonical example, even the name-sake
2. $\mathbb{Q}$ and $\mathbb{R}$ are integral domains. Note that every (nonzero) elements are also units
3. $(\mathrm{GL}_n(\mathbb{F}), +, \cdot))$ is an integral domain
4. $M_n(F), +, \cdot)$ is not an integral domain, since

$$\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \cdot \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$$

5. Let $F = \mathrm{Hom}_{\mathbf{Set}}(X, R)$ be a collection of functions where $|X| > 2$ $R \neq \{0\}$. As noted in example 9.1.4 this is a ring. This ring is not an integral domain. Let $a, b \in X$, $a \neq b$, and $x \in R$, $x \neq 0$. Then define a function where $f(a) = 0, f(b) = x$, and the rest of the elements of f map to zero, and define $g(a) = x, g(b) = 0$, and the rest of the elements of g map to zero. Note that f and g are not the zero function since they are not identically zero, but $fg = 0$. Importantly

$$fg(a) = f(a)g(a) = 0x = 0 \qquad fg(b) = f(b)g(b) = x0 = 0$$

The set of functions from a set X to a ring R is rarely an integral domain when $|X| > 0$ and $R \neq \{0\}$, since it would have to be the case that $fg = 0$ implies that either f is zero everywhere or g is zero everywhere. However, they do exist (an important example in complex analysis is the ring of power-series, or more generally the ring of analytic functions).

Sometimes the term *domain* used for the same type of ring without the commutative property:

Definition 9.1.23: Domain

A ring in which there are no zero-divisors is called a domain

The nicest property of integral domains is that the cancellation law holds!

Proposition 9.1.24: Cancellation Law for Integral Domains

Assume a, b, c are elements of any ring with a not a zero divisor. If $ab = ac$, then either $a = 0$ or $b = c$.

***Proof*:**

$$\begin{aligned} ab &= ac \\ \Leftrightarrow a(b-c) &= 0 \\ \Leftrightarrow a = 0 \text{ or } b-c &= 0 \\ \Leftrightarrow a = 0 \text{ or } b &= c \end{aligned}$$

With this property of integral domains, proposition 9.1.20 can be expanded upon:

Corollary 9.1.25: Finite Integral Domain is a Field

If R is a finite integral domain, then R is a field

Just like in proposition 9.1.20, the trick in this proof is a neat use of function to capture the idea of how an element acts on every other element in an object, not unlike how we can think when working with group actions:

***Proof*:**
Let R be a finite integral domain. and let a be a nonzero element of R. By the cancellation law the map $x \mapsto ax$ is an injective function. To see this, let's say it was not injective so $ax = ay$ but $x \neq y$. Then

$$ax - ay = 0 \quad \Rightarrow \quad a(x-y) = 0$$

Since a is non-zero and there are no zero divisors, it must be that $x - y = 0$, so $x = y$.

Now, since the map is injctive, then $|\varphi(R)| = |R|$, meaning it must be surjcetive [a]. In particular, there is some $b \in R$ such that $ab = 1$. A similar procedure can be done to show that there exists a c such that $ca = 1$. Since units are unique (see exercise ref:HERE) $c = b$, so a is a unit in R. Since a was arbitrary nonzero element, R is a field.

[a]Note that for finite sets, it's impsosible to have a set with "less" elements, but be smaller cardinality, while it is possible for infinite sets, like $2\mathbb{Z} \subseteq \mathbb{Z}$, but $|2\mathbb{Z}| = |\mathbb{Z}$

9.1.26 Note If we drop commutativity and add that every element is a unit (getting a divsion ring), then this theorem still holds (partly by proving that every finite division ring is necessarily commutative)! The proof of this is relegated to Rerpresentation theory in part X since it requires a lot more build-up, but it's interesting to note.

Though integral domains are nice, they loose a every important property: If R is an integral domain, then $R \times R$ is *not* an ingeral domain

Example 9.1.27: product of integrals domains is not an integral domain
Let R be an arbitrary integral domain. Then take $R \times R$, with addition and multiplication done point-wise. Then

$$(1,0) \cdot (0,1) = (0,0)$$

meaning there are zero-divisors, and so $R \times R$ is not a integral domain A particular consequence of this is that the direct product of fields is not a field (since fields are integral domains) More generally, integral domains do not satisfy the universal property of products.

A summary of all these properties is in section 9.1.2.

Exercise 9.1.1

1. An element $e \in R$ is called *idempotent* if $e^2 = e$. Does there exist a ring R that has exactly 3 idempotent elements? Find one or show it's not possible.

2. We'll prove that every finite integral domain is a ring using the pigeonhole principle

   a) Show... click here for this proof

3. We will define the complex numbers. Take the real numbers $\mathbb{R}$, and let i be some variable. Define
$$\mathbb{C} := \langle\{\mathbb{R}, i\} \mid i^2 = -1\rangle$$
That is, we will take all possible combinations of the elements from $\mathbb{R}$ and the element i, and impose the relation $i^2 = -1$ on i. To get an idea of what's in $\mathbb{C}$:
$$i \in \mathbb{C},\ ci \in \mathbb{C},\ a + bi \in \mathbb{C} \cdots$$
Prove that $\mathbb{C}$ is a ring. We will later give a slightly more nuanced definition of of $\mathbb{C}$, see ref:HERE.

4. (why not extend to 3 binary operations) (this exercise shows that if you want even more Yoneda compatibilities then you crash out and get trivial structures)

5. Let' say $s \neq 1$ and $s^2 = s$. This propety is called idempotence. Show that the only idempotent element in an integral domain is 0 and 1

6. Let R be a ring with $1 \neq 0$. Is $\varphi : R \to R$, $\varphi(r) = 0$ a ring-homomorphism? If so, prove it, and if not, would it be a rng-homomorphism (i.e. would the image be a rng)?

7. Show that the only idempotent elemnt in a field' is 0 or 1 (there exists a proof using properties of a field but not properties of integral domain)

8. A field is said to be characteristic n if there exists a number n st $\underbrace{1 + 1 + 1 \cdots + 1}_{n \text{ times}} = 0$. If no such number exists, we say that the field has characteristic 0. Give an

9. Let $u \in R$ be a unit and $R \to S$ a ring homomorphism. Then $\varphi(u)$ is a unit. Is the same true for zero-divisors? What about non-zero-divisors? example (or examples) of fields with characteristic 0, 2, and 5.

10. Show that if $\mathbb{F}$ is a field of characteristic 0, then $\mathbb{Q} \subseteq \mathbb{F}$.

11. (For those who know some Analysis and want examples of *rngs*) In the process of defining the Fourier series, it is necessary to define the convolution product between two continuous functions (on compact support, or more generally L^2)

12. (ibid last comment) Let X be a locally compact space (i.e. for every $x \in X$, there exists a compact set $K \subseteq X$ such that $x \in K$, that is every point has a compact neighbourhood). Let $C_0(X)$ be the set of continuous complex functions $f : X \to \mathbb{C}$ that vanish at infinity (for every $\epsilon > 0$, there exists a K such that $|f(x)| < \epsilon$ for all $x \in X - K$). Then $(C_0(X), +, \cdot)$ with function addition and multiplicatoin is a rng, but not a ring (namely, the function $1(x)$ does not vanish at infinity). This rng is used quite often in analysis: it comes up in Fourier

analysis, in defining Radon Measures, and in the study of C^*-algebra to name a few examples (see Folland or ref:HERE).

13. (do the example in Commutative algebra 2 (Rings, ideals, modules) at timestamp 13:33)

    (somehow make this a comment??) Loosely speaking, rngs correspond to locally compact spaces, while rings correspond to compact spaces. For analysists, working over compact spaces is much to restrictive ($\mathbb{R}^n$ is not compact), while locally compact spaces are much more natural to work with ($\mathbb{R}^n$ is locally compact). Furthermore, recall we have mentioned we can turn a rng into a ring by doing $\mathbb{Z} \oplus R$ to some rng R. If we were interested in keeping nice analytical properties, we would instead take $\mathbb{R} \oplus R$. (ex. $\mathbb{R} \oplus C_0(X)$). This would correspond to the 1-point compatification of X: $C_0(X \cup \{\infty\}) \equiv \mathbb{R} \oplus C_0(X)$!

14. Show that $(F, +, \cdot)$ be a ring. Show that $(F, +, \cdot)$ is a field if and only if $(F, +)$ is an abelian group and the set $F^\times$ is equal to $F - \{0\}$.

15. Let R be a ring and R^{op} denote the ring where the underlying set and addition operation is the same, and the multiplication is defined as $x * y = yx$ where yx happens in R. Show that $\text{id} : R \to R^{\text{op}}$ is not necessarily a ring isomorphism (it is sometimes called an *anti-isomorphism*). When is it a ring isomorphism? Show that if R is a division ring, then the map $x \mapsto x^{-1}$ *is* a ring isomorphism.

16. show that if $x^n = 0$, then $(x - 1)$ is an invertible element. An element where $x^n = 0$ is called a *nilpotent element*.

17. recall that $(x + y)^n = \sum_{k=1}^n \binom{n}{k} x^n y^{n-k}$. Show that the binomial formula holds if and only if $xy = yx$.

18. I the set of all left (resp. right) invertible elements a group? How that $Z(R)$ is a subring of R.

19. let G be a cyclic group of order n. For each $k \in \mathbb{Z}$, define $f : G \to G$, $f_k(g) = kg$. The n the map $k \mapsto f_k$ defines a ring isomorphism $\mathbb{Z}/n\mathbb{Z} \cong_{\mathbf{Ring}} \text{End}_{\mathbf{Ring}}(G)$ and a group isomorphism $(\mathbb{Z}/n\mathbb{Z})^* \cong_{\mathbf{Grp}} \text{Aut}_{\mathbf{Grp}}(G)$

9.1.1 Category of Ring

We now approach rings a 2nd time, but this time, we will take a more birds-eye view of the category or rings. We will see if this category has some properties different than the category of groups, and whether there is a general way of extending groups to form rings.

Proposition 9.1.28: Category of Ring

The collection of Rings along with ring homomorphisms form a category. This category will be denoted **Ring**.

***Proof*:**
See exercise ref:HERE

Since **Ring** is a category, it will be verified what are the initial/final objects, and whether there are product, sub-, and quotient objects, and whether quotient objects can be classified through kernels. We can further ask about the relation between **Group** and **Ring**, and ask whether there is an analogous concept of ring-actions. In this section we will drop the subscript and write $\text{Hom}(R, S)$ or $\text{End}(R)$ instead of $\text{Hom}_{\mathbf{Ring}}(R, S)$ or $\text{End}_{\mathbf{Ring}}(R)$

The fact that sub-objects and product objects exist in rings (that is, subrings and product rings) has already been explored in definition 9.1.6 and example 9.1.4[5], though not in categorical rigour (which the reader can practice in exercise ref:HERE if they are so inclined). We will mention some more about product of rings and the consequences of integral domains not having products as an interesting example of category with no products. Quotient objects will be revisited in more detail in section 9.3.

Rings and $\mathbb{Z}$

The first thing to check is whether **Ring** has initial and final objects. In definition 1.4.6, we introduced the concept of initial and final group. We found there to not really be an intersting initial and final group: only the identity group. The same is true for the final object of rings. However, the zero ring cannot be the initial: if R is the zero ring, and we have a homomorphism $R \to S$, then S must be the zero ring (see exercise ref:HERE). There is still an initial object in the category or rings: we have our first example of a non-trivial initial object!

Proposition 9.1.29: $\mathbb{Z}$ is an Initial Ring

The ring $\mathbb{Z}$ is an intial ring

***Proof*:**
Let R be any ring (with identity) and let $\varphi : \mathbb{Z} \to R$ such that

$$\varphi(z) = 1 + 1 + 1 \text{ (z times) } = z \cdot 1$$

this is a ring homomorphism. For

$$\varphi(a + b) = (a + b) \cdot 1 = a + b = \varphi(a)\varphi(b) \qquad \varphi(ab) = (ab) \cdot 1 = a(b \cdot 1) = (a \cdot 1)(b \cdot 1) = \varphi(a)\varphi(b)$$

Furthermore, if $\varphi' : \mathbb{Z} \to R$ was another ring homomorphism, then by proposition 9.1.9

$$\varphi'(1) = 1\,, \varphi'(2) = 1 + 1\,, \cdots$$

that is, by the fact that the identity *must* be distributed in this fashion, we are forced that

$$\varphi = \varphi'$$

making it unique

Note that if rings are not defined to have a multiplicative identity, then there is no unique ring wihtout a multiplicative identity, see exercise ref:HERE (exercise 2.16 in Aluffi part III)

So how can we use this result? The way I think about it is something "integer-like" in every ring [with identity]. It might not be commutative, it might not be injective (the codomain might even be finite), or it might be uncountably big (like $\mathbb{R}$ or $M_n(\mathbb{C})$) but there is still something integer like.

For this reason, asking questions like "what properties of the integers do rings have" is actually a very sensible question! In fact, that will be essentially what we do in chapter 10!

For example, if we take $x \in \mathbb{Z}$, then $x = p_1 p_2 \cdots p_n$, that is x has a *unique prime factorization*. In the next chapter, we will study rings which have this similar property (they will be called *unique factorization domains*). Similarly, to find the $gcd(a, b)$, we can use the euclidean algorithm which algorithmically shows how much the set of divisors of the two numbers intersect. This is very handy since we can in very few steps find the common divisors of two numbers of extremely large numbers. Rings in which the notion of the euclidean algorithm is well-dfined will be called *euclidean domains*. Finally, a little more difficult but important concept, any number can be decomposed into *unique* prime numbers in $\mathbb{Z}$. We will find the equivalent of "numbers" in rings to be "ideals" (the original name was "ideal numbers" of rings). If every ideal can be generated by a single element $x \in R$, the ideal is called a *principal ideal*. Thus, a ring which is has "numbers" (ideals) which are generated by one element (just like a number just needs itself to describe, well, itself) will be called a *principal ideal domain*.

Thus, in the folloing chapters, some basic theory will extensively be used. Take a moment to review section 0.2 for a refresher on some basic tools of number theory.

Rings and $\mathrm{End}(G)$

You might ask how should we choose which of the axioms are "nice" axioms to add to our structure. One way of doing so is by thinking about the multiplication action being permutations of the elements, not unlike ho the group operation can be thought of as the permutation of elements (i.e. any group $(G, +)$ is a subgroup of $\mathrm{Sym}(G)$) [7]

We saw in definition 1.4.9 that if G is an abelian group that $\mathrm{End}(G)$ is an agelian group under pointwise addition. Notice that there is actually already a 2nd operation, since function can compose:

$$(\mathrm{End}(G), +, \circ)$$

Then, it can be checked that composition here will make this into a ring (as shown in example 9.1.4)

You might then ask converse: given ring $(R, +, \cdot)$, can we find a ring isomorphism to $\mathrm{End}((R, +))$? The answer is yes! This will actually turn out to be the Cayley's theorem for rings (theorem 4.5.3)! We will not cover the ring version of Cayley's theorem, for a more general theorem called Yonneda's Lemma cover's the idea of using objects to "represent" other objects (ref:HERE)

If you want to check, the natural ring embedding from $(R, +, \cdot)$ to $\mathrm{End}(R, +)$ is

$$r \mapsto (x \mapsto xr) \qquad \text{or} \qquad r \mapsto (x \mapsto rx)$$

depending on the convention on composition.

:exercse

Let $\mathrm{Aut}_{\mathbf{Ab}}(R)$ be the set of group-automorphisms on R. Show that $U(R) \cong_{\mathbf{Ab}} \mathrm{Aut}_{\mathbf{Ab}}(R)$

let R be a commutative ring. Show that $\lambda : R \to \mathrm{End}_{\mathbf{Ab}}(R)$ by $r \mapsto (y \mapsto ry)$ is indeed a ring homomorphism. If R is not commutative, show that it's almost abelian.

[7] Research **Ab**-enriched categories for information about this process

Products of Rings

Rings have products: Given any to rings A and B, there exists a ring $A \times B$ along ring homomorphisms $\pi_A : A \times B \to A$ and $\pi_B : A \times B \to B$ such that for any C here $f_A : C \to A$ and $f_B : C \to B$, there exists a unique $f! : C \to A \times B$ such that the folloing diagram commutes:

$$
\begin{tikzcd}
 & C \arrow[d, "f!"] \arrow[ddl, bend right, "f_A"'] \arrow[ddr, bend left, "f_B"] & \\
 & A \times B \arrow[dl, "\pi_A"] \arrow[dr, "\pi_B"'] & \\
A & & B
\end{tikzcd}
$$

In this way, products in **Ring** are like products in **Grp**. A similar situation occurs for coproducts.

However, as mentioned earlier, if we retirct to *integral domains* and take the category whose objects are integral domains (this is indeed still a category as can be verified), they are no longer products, as demonstrated in exmaple 9.1.27!

Since since all fields are integral domains, this means that products are not defined for fields! As a consequence of this, we cannot repeat the same trick as in chapter 5 in creating larger groups using products. Thus, extensions of fields (similar to how they were defined for groups in section 3.6) will have to be used to study the creation of larger fields. To study field extensions, we must first study ring homomorphisms more extensively; the particular way in which ring homomorphissm will be used in constructing larger fields requires more build-up, and so the study of fields is relegated to part IV.

9.1.2 Summary

Keeping track of these properties and their relation to everything can be a bit to take in. It might help to have a visual too:

$$
\begin{tikzcd}
 & \text{field} & \\
\text{integral domain} \arrow[ur, "\forall a \exists b \ st \ ab{=}1 \ (+1)"] & & \text{division ring} \arrow[ul, "ab{=}ba"'] \\
\text{commutative ring} \arrow[u, "\forall a \nexists b \ st \ ab{=}0"] & & \text{domain} \arrow[ull, "ab{=}ba"] \arrow[u, "\forall a \exists b \ st \ ab{=}1 \ (+1)"'] \\
 & \text{ring} \arrow[ul, "ab{=}ba"] \arrow[ur, "\forall a \nexists b \ st \ ab{=}0"'] & \\
 & \text{rng} \arrow[u, dashed, "\text{add identity } 1"] &
\end{tikzcd}
$$

The *rng* can actually have more arrow's extending from it to commutative ring, domain, and integral domain, but the diagram would be cumbersome, so I opted to make it a dashed line from *rng* to *ring*, since, it is not needed to "upgrade" those rings until we get to division ring and fields. I added a (+1) where we really need the multiplicative identity to define the next concept.

Notice that if a commutative ring only has units, it is then a commutative division ring, and hence a field. The line as not drawn so as to not clutter the diagram.

look at comment:

The following categorical results are also noted:

1. $\mathbb{Z}$ is the initial object of rings
2. All rings are subrings of $\operatorname{End}((R, +))$
3. Products exist for rings, but not for integral domains (and so not for fields).

Exercise 9.1.2

1. Since $\mathbb{Z}$ is initial, for any ring R and natural homomorphism $\mathbb{Z} \to \mathbb{R}$, $\operatorname{im}(\mathbb{Z})$ exists is a subring. Show that $\operatorname{im}(\mathbb{Z})$ is the *smallest* subring of any ring. Show further that $\operatorname{im}(\mathbb{Z})$ is either $\mathbb{Z}_n$ for any $n > 0$ or $\mathbb{Z}$. If $|\operatorname{im}(\mathbb{Z})| < \infty$, then we say the ring R has *characteristic n*, and if $\operatorname{im}(\mathbb{Z}) \cong \mathbb{Z}$ then R has *characteristic 0*.
2. Let R be a ring and define R^{op} to be the ring with the same underlying set and addition operation, and multiplication defined by $x * y = yx$, where yx happens in R. Show that R^{op} is a ring, and that $(\_)^{\mathrm{op}} : \mathbf{Ring} \to \mathbf{Ring}$ is a functor (where do ring homomorphisms map to?). Is it covariant or contravariant?
3. Let $a, b \in R$ such that $ab = ba$. Show that ab is invertible if and only if a and b are invertible. Generlize your result to show that if $ab \neq ba$, then if ab invertible, then a is left-invertible, and b is right-invertible.
4. Show that $\operatorname{End}_{\mathbf{Ab}}(\mathbb{Z}) \cong_{\mathbf{Ring}} \mathbb{Z}$.

9.2 Examples of Rings

The following are rings that will show up time and time again, and being aquainted with them is very useful. Many of these rings will be the starting point to a new field of study in mathematics. For example, Polynomial rings are pivotal in Commutative Algebra (see part V), and when the polynomial ring takes on coefficients from a field (i.e. a polynomial ring over a field), then they form the foundation for field theory (see part IV). Matrix rings are a common the "representation objects", the same way $\operatorname{Sym}(X)$ was an object that can be used to "represent" groups (see part X chapter 51). Group rings can be thought of as the dual of the group of unity of rings, and also appear frequently in representation theory (see part X chapter 53) and group cohomology (see part XI chapter 57).

There is also an optional section on formal power series covering the elementary operations that can be performed on formal power series. These results are important for a class in complex analysis, however the formal power series ring will come up as examples for certain concepts introduced later in the book as well. These examples of formal power series will not be critical in understanding any material, and so they can be comfortably skipped.

9.2.1 Polynomial Rings

Polynomial Rings are one of the first examples we'll give of a [commutative] extension of a ring. Let R be a commutative ring [8]. What we'll be doing is extending R by an arbitrary unkown element x. In a sense, what we're doing is saying x is now an element of R, an then we figure out how to define this new structure (which we label $R[x]$) so that it's also a ring. Since $x \in R[x]$, then $x \cdot x \in R[x]$. Since we know nothing of x, we can only say that $x \cdot x = x^2$ is another element. We can continue this process inductively getting us every x^k for all $k \in \mathbb{N}$. For any other element r: $r \cdot x = rx$ where rx is a new element of $R[x]$. A similar reasoning to creating new elements applies to adding: $r + x$ is a new element. However, it need not be the case that an element has an inverse, so we do not define an x^{-1} such that $xx^{-1} = 1$[9]. All of these properties together give us the following definition:

Definition 9.2.1: Polynomial Ring

Let R be a ring and x be an indeterminante. Then the sum

$$a_n x^n + a_{n-1}x^{n-1} + \cdots + a_1 x + a_0$$

with $n \geq 0$ and each $a_i \in R$ is called a *polynomial* in x with coefficients a_i in R. The set of all such polynomials is called the ring of *polynomial in the variable x with coefficients in R* (or *polynomial ring* if context permits) and will be denoted $R[x]$. Addition is defined as

$$(a_n x^n + \cdots + a_1 x + a_0) + (b_n x^n + \cdots + b_1 x + b_0) = (a_n + b_n)x^n + \cdots (a_1 + b_1)x + (a_0 + b_0)$$

where a_n or b_n may be zero in order for addition of polynomials of different degrees to be defined. Multiplication is perfomed by defining $(a_i x^i)(b_j x^j) = a_i b_i x^{i+j}$ and then extending to all polynomials by the distributive law; this is called the *convolution product*
The following are pertinent definitions with regards to polynomials.

1. If $a_n \neq 0$, then the polynomial is said to be of *degree n*, and $a_n x^n$ is called the *leading coefficient* (where the leading coefficeint of the zero polynomial is taken to be 0).
2. The polynomial is *monic* if $a_n = 1$.

For reference, here is the general formula for polynomial multiplication (i.e. the convolution product)

$$\left(\sum_{i=0}^{\deg(p)} p_i x^i\right) \cdot \left(\sum_{i=0}^{\deg(q)} q_j x^j\right) = \sum_{k=0}^{\deg(p)+\deg(q)} \sum_{l=0}^{k} p_l q_{k-l} x^k$$

9.2.2 Remark If you have done Fourier Analysis (or intend to in the future), then the convolution of two continuous (more generally measurable) functions follow's the exact same properties as polynomial multiplication. In fact, polynomial multiplication would be a discrete convolution.

[8] In the exercises, there are questions concerning the study of non-commutative R forming a polynomial ring and why it is often preferred for R to be commutative

[9] polynomial rings where the indeterminate x has an inverse are called the *ring of fractions of R[x]*. see section 9.5

Note that R is always a subring of $R[x]$, which we can see through the natural inclusion map [10] $R \hookrightarrow R[x]$ of $r \mapsto r$.

Usually, when we manipulate polynomials we treat them as functions. It must be stressed that we are *not* treating the elements of $R[x]$ as functions. The study of the elements of $R[x]$ as functions will be done in part V, which will establish the largest connection between algebra and geometry in the 20th century.

If $R = F$ is a field, then every coefficient is a unit, meaning we can divide each coefficient a polynomial to make it *monic*, that is,

$$a_n x^n + a_{n-1}x^{n-1} + \cdots + a_1 x + a_0 \overset{1/a_n}{\mapsto} x^n + a'_{n-1}x^{n-1} + \cdots + a'_1 x + a'_0$$

where $a'_i = \frac{a_i}{a_n}$. We shall see that often-times many results *up to multiplication by unit* (meaning if the result holds on p, it will hold on cp, where c is a unit). For this reason, it is worth keeping the term *monic* in mind.

We can extend the definition of a polynomial ring to a polynomial ring over several variables by simply making $R[x]$ a polynomial ring over y, that is

$$R[x, y] = (R[x])[y] = R([y])[x] = R[y, x]$$

Continuing inductively, we can define a ring over a finite number of variable this way:

$$R[x_1, x_2, ..., x_{n+1}] = (R[x_1, x_2, ..., x_n])[x_{n+1}]$$

This process can be extended "indefinitely"[11] to define $R[x_1, x_2, ..., x_n, ...]$, in which every polynomial (i.e. elements of $R[x_1, x_2, ...,])$ has finitely many terms, and each term can have multiples of any of variables of $R[x_1, x_2, ..., x_n, ...]$, for example:

$$x_{30}^2 x_7^{23523} x_2 - r x_{100} x^{39} \in R[x_1, x_2, ...]$$

Multiplication in $R[x_1, x_2, ...]$ is the same as the multiplication in polynomial rings over finitely many variables, where we know the value for any term be expanding out the multiplication far enough to group like-terms. If $x_1^{d_1} x_2^{d_2} \cdots x_k^{d_k}$ is a term, then the *total degree* of the term is

$$d = d_1 + d_2 + \cdots + d_k$$

and the tuple $(d_1, d_2, ..., d_k)$ is the multi-degree of the term.

Properties of Polynomial Rings

To motivate the next theorem, let's say we have two polynomials of degree 1, $ax + b, cx + d \in R[x]$. We can ask what is the degree of $(ax + b)(cx + d) = acx^2 = (ad + bc)x + bd$. We might expect the degree to be 2. However, it is possible that a and c are zero-divisors: take $4x + 1, 4x + 1 \in (\mathbb{Z}/8\mathbb{Z})[x]$. In fact, in the example just stated, $4x + 1$ is in fact a unit in $(\mathbb{Z}/8\mathbb{Z})[x]$! If R is an integral domain, then some of its structure will be simpler:

[10] the map is natural since there is no other information needed to define this map besides R and $R[x]$. For a more rigorous definition of natural, appendix **??** needs to be read to reach section **??**

[11] For those uncomfortable with how the fact that the above process of defining a polynomial ring will always only produce polynomial rings with finitely many variables, the construction of $R[x_1, x_2, ...]$ is done using a *limit*, see ref:HERE

Proposition 9.2.3: Degree Of Polynomials

Let R be an integral domain and let $p(x), q(x)$ be nonzero elements of $R[x]$. Then

1. degree $p(x)q(x)$ = degree $p(x)$ + degree $q(x)$
2. the units of $R[x]$ are the units of R
3. $R[x]$ is an integral domain

***Proof*:**

1. This comes from the definition of multiplication of polynomials and that R is an integral domain, as the highests order term will be $x^{\deg(p(x)+\deg(q(x)))}$ and there is no zero-divisors to cancelt out the term.
2. Let $u \in R[x]$ be a unit so that there exists a $v \in R[x]$ such that $uv = 1$. Then by the degree formula, $\deg(uv) = \deg(u) + \deg(v)$ [a]. Since $\deg(uv) = 0$, then it must be that $\deg(u) = \deg(v) = 0$, showing that $v \in R$
3. Let $p(x), q(x) \in R[x]$ such that $p(x)q(x) = 0$. Then $0 = \deg(0) = \deg(p(x)q(x)) = \deg(p(x) + \deg(q(x))$, and so both $p(x), q(x)$ must be of degree zero and so in R. Since R is an integral domain, then it must be that $p(x) = q(x) = 0$, showing that $R[x]$ is an integral domain.

[a] notice again that R must be an integral domain for this to work

The defining property of commutative rings is that they are the free objects of commutative rings.

Theorem 9.2.4: Universal Property of Polynomial Rings

Let S be a commutative ring and let

$$\varphi : R \to S$$

be any ring homomorphism and let $s \in S$ be any element of S. Then there is a unique ring homomorphism

$$\psi : R[x] \to S$$

such that $\varphi(x) = s$ and which makes the following diagram commute

$$\begin{tikzcd} R \arrow[r, "\varphi"] \arrow[d, "f"] & S \\ R[x] \arrow[ur, "\psi"'] & \end{tikzcd}$$

A nuance that must be mentioned is that polynomial rings may be over different rings. We thus may thus ask how a ring S is generated with respect to a ring R. As demonstrated earlier, $\mathbb{Z}$ is the initial object of **Ring**, and so in the arbitrary case $f : X \to S$ for some set X, we shall have a polynomial ring on the elements X over $\mathbb{Z}$. For those who are categorically inclined, try figuring out what is the category involved behind the scenes (this is similar to the construction for free groups in section 2.3.3)

***Proof*:**
The proof becomes easy to see once you realize a polynomial:

$$a_n x^n + a_{n-1}x^{n-1} + \cdots a_1 x + a_0$$

becomes

$$\varphi(a_n)s^n + \varphi(a_{n-1})s^{n-1} + \cdots \varphi(a_1)s + \varphi(a_0)$$

A particular case of this is when $\varphi : R \to R$ is an identity map. In this case, we may link the notion of a [formal] polynomial to a polynomial function:

Definition 9.2.5: Evalutation Map

Let R be a ring and let $\alpha \in R$. The natural ring homomorphism

$$\varphi : R[x] \to R$$

which acts as the identity on R and which sends x to α is called the *evaluation at* α and is often denoted ev_α.
We say that α is a *zero* (or root) of $f(x)$ if $f(x)$ is in the kernel of ev_α.

The evaluation map can come in quite handy in many proofs that will be seen later.

Example 9.2.6
Move this to somewhere else

1. Take $\mathbb{Q}[x,y]$ $(= (\mathbb{Q}[x])[y]$, i.e. a polynomial ring over $\mathbb{Q}[x]$ in y). Is (x) a prime ideal? Take $\mathbb{Q}[x,y] \to \mathbb{Q}[x]$ by $p(x,y) \mapsto p(x,0)$. This is the evaluation map at 0.

   This maps is clearly surjective, since for every $p(x)$, $p(x,0)$ will map to it. This map is also clearly linear since

   $$\varphi(p(x,y) + q(x,y)) = \varphi\left(\sum_{i=1}^{n}(a_i)x^i + \text{y terms}\right) = \sum_{a_i}^{n} a_i = \varphi(p(x,y)) + \varphi(q(x,y))$$

   and since it's linear on every term, to check the multiplicative operation, it suffices to check on monomials, which is easy:

   $$\varphi(ax^n bx^m) = \varphi(abx^{n+m}) = abx^{n+m} = ax^n bx^m = \varphi(ax^n)\varphi(bx^m)$$

   and if y is aterm, y geos to zero so mulitplicativity for those terms is automatic. Therefore, φ is a ring homomorphism. It's kernel is clealry (y), meaning

   $$\mathbb{Q}[x,y]/(y) \cong \mathbb{Q}[x]$$

   and since $\mathbb{Q}[x]$ is an integral domain (y) is a prime ideal.

When working with functions, they are determined by the value at each point of evaluation. It must be stressed that in general, a polynomial is *not* determined by it's evaluation

> **Example 9.2.7: Polynomial Not Determined By Evaluation**
> [Polynomial Not Determined By EvaluationPolynomial Not Determined By Evaluation] Take $x^2 - x \in (\mathbb{Z}/2\mathbb{Z})[x]$. Does the evaluation of this polynomial uniquely determine $x^2 - x$?

9.2.2⋆ Formal Power Series

⋆ *Optional. This subsection is an aside; nothing later in the book depends on it.*

(A realisation i had that maybe should be integrated?: I just realised. So we know from complex that [complex] polynomial behaves like waves (since they are harmonic). We know that multiplication of polynomials introduces the operator of convolution. We know Fourier somehow gives you a notion finding the fourier-coefficients, i.e. finding the waves. Well, if F is the fourier transform, the fact that $F(f * g) = F(f) \cdot F(g)$ feels like it makes more intuitive sense!)

Any polynomial is a finite linear combination of the terms x^n. If we allow for infinitely many terms, we call such an object a *formal power series*, and write it like so:

$$f(x) = \sum_{i=1}^{\infty} a_i x^i \qquad a_i \in R$$

The reader may ask what does it mean to take an infinite linear combination, as we have defined the + notation to only take in two inputs, and if we recurse over the inputs we may have a finite linear combination. These are valid concerns which will be addressed in *Commutative Algebra* [10] (completion). For now, we may consider $f(x)$ in the above equation as "actually" being an infinite tuple[12]

$$(a_0, a_1, ..., a_n) \in \prod_{i=0}^{\infty} R$$

however we should take this identification as simply insuring that $f(x)$ is well-defined; we shall not write $f(x)$ in the notation of tuples and from now on shall assume we can take infinite sums. The collection of all formal power series are certainly closed under termwise addition. We must make sure multiplication is still well-defined. For multipilcation to be well-defined, each coefficient must be well-defined. Fortunately, this is the case. If $p(x)$ and $q(x)$ where

$$\begin{aligned}
p(x)q(x) &= \left(\sum_{i=0}^{\infty} a_i x^i\right)\left(\sum_{j=0}^{\infty} b_j x^j\right) \\
&= a_0b_0 + (a_0b_1 + a_1b_0)z + (a_0b_2 + 2a_1b_1 + a_2b_0)z^2 + \cdots \\
&= \sum_{n=0}^{\infty}\left(\sum_{i=0}^{n} a_i b_{n-i}\right) x^n \\
&= \sum_{n=0}^{\infty} (A_n * B_n)\, x^n
\end{aligned}$$

[12] this may push the concern in asking how $\prod_{i=0}^{\infty}$ is defined. For those with this concern, they may skip to section 25 or they may take for granted that we may always make a countable number of choices and remember the order

where $A_n = \sum_{i=0}^{n} a_i$ and $B_n = \sum_{i=0}^{n} b_i$ and $A_n * B_n$ is equal to the the summand above and is called the *convolution* of the two finite sequences. Since addition and multiplication is well-defined, we have a ring:

Definition 9.2.8: Formal Power Series

The collection of all infinite R-linear combinations of $\{1, x, x^2, \ldots\}$ forms a ring known as the ring of *formal power series* over R, and is denoted $R[[x]]$

The term "formal" comes from how we often associate to an object $f \in R[[x]]$ the evaluation function. If we had a function with a domain and codomain for which $r \mapsto \sum_{i=0}^{\infty} a_i r^i$, we would call this function a *power series*. However, for such a function to be defined, we require the ability to unambiguously define summing infinitely many elements of a ring. Such considerations require the introduction of a topology on our ring, something we shall not cover. We shall remark that it is very commmon in compex analysis to work with power series and choose $R \in \{\mathbb{R}, \mathbb{C}\}$ with the usual euclidean topology, leading to many very important results about complex-differentiable functions. There is one value we may unambiguously define without the need for any convergence considerations:

$$f(0) := a_0$$

Proposition 9.2.9: Formal Power Series Integral Domain

Let $R[[x]]$ be the formal power series over R. Then $R[[x]]$ is an integral domain

***Proof*:**
proof here

Given that $R[[x]]$ is the extension of $R[x]$, it may see intriguing to think that these two rings share very similar properties. However, the "infinitude" of $R[[x]]$ will make the ring act rather differently. The following may be the most striking change when extending to infinite polynomials:

Lemma 9.2.10: Existence of More Units

Let $R[[x]]$ be the formal power series over a ring R. Then $(1 - x)$ has an inverse in $R[[x]]$

***Proof*:**
Notice that

$$(1 - x)(1 + x + x^2 + \cdots + x^n + \cdots) = 1$$

and hence by definition that means that $1 - x$ has an inverse, namely $\sum_{n=0}^{\infty} x^n$.

By convention, we denote inverses using quotient notation, and hence we would write:

$$\frac{1}{1-x} := \sum_{n=0}^{\infty} x^n$$

For those who have taken calculus, they may be familiar with an "analysis" version of this proof relying on absolute convergence. The above shows that this result is purely algebraic. In fact, we can fully classify all units in a formal power series

Proposition 9.2.11: Units In Formal Power Series

Let $f \in k[[x]]$. Then f is a unit (i.e. f has an inverse) if and only if $a_0 \neq 0$

***Proof*:**
The condition is certainly necessary since if

$$g(x) = \sum_n b_n x^n \qquad f(x)g(x) = 1$$

then we must have $a_0 b_0 = 1$, so $a_0 \neq 0$. If conversely we have $a_0 \neq 0$, we shall show that the formal power series $(a_0)^{-1} f(x) = f_1(x)$ has an inverse $g_1(x)$, and hence $(a_0)^{-1} g_1(x)$ is the inverse of $f(x)$. Decompose $f_1(x)$ into:

$$f_1(x) = 1 - u(x)$$

then substitute $u(x)$ for y in the proof of lemma 9.2.10 to get that $1 - u(x)$ has an inverse. But then we may conclude that $f(x)$ has an inverse, thus completing the proof.

Another question we may ask is the following: Given two polynomials $f(x), g(x)$, we may take $f(g(x))$, (replace x with $g(x)$). Can the same be done for power series? The following shows when this is possible:

Definition 9.2.12: Formal Derivative

Let $f \in k[[x]]$ be a formal power series. Then if $f(x) = \sum_{k=0}^{\infty} a_k x^k$, the formal derivative of f is the value

$$\frac{d}{dx}(f) = \sum_{k=0}^{\infty} k a_k x^{k-1}$$

we usually denote the derivative of $f(x)$ as $f'(x)$, and for higher derivatives we write $f^{(n)}(x)$ to represent taking recursively the derivative of f n times.

We may use the formal derivative recursively to find the coefficients of f. Namely, $f(0) = a_0$, $f'(0) = a_1$, ..., $f^{(n)}(0) = a_n$, and so forth.

Let f, g be power series. We may wonder when $g(f(z))$ is well-defined:

Proposition 9.2.13: Composing Power Series

Let f, g be formal power series. Then $g(f(z))$ is well-defined if $b_0 = 0$.

***Proof*:**
When composing power series, we get:

$$a_0 + a_1(b_1 z + b_2 z^2 + \cdots) + a_2(b_1 z = b_2 z^2 + \cdots)^2 + \cdots$$

where each coefficient is an infinite sum, and so we must ask for their convergence conditions. One thing we can do to guarantee this is by setting $b_0 = 0$, which will guarantee that all the coefficients are finite sums and hence well-defined.

We often denote the composition as $g \circ f$.

Proposition 9.2.14: Associativity Of Composition

Let $f, g, h \in k[[x]]$ be power series. Then

$$(h \circ g) \circ f = h \circ (g \circ f)$$

***Proof*:**
Henri Cartan Elementary Theory of Analytic functions p. 13

Define $I(x) = x$. Notice that for any $f \in k[[x]]$, $I \circ f = f \circ I = f$.

Proposition 9.2.15: Composition Has Inverse

Let $f \in k[[x]]$ and $I(x) = x$. Then a right inverse exists, $f \circ g$ if and only if $f(0) = 0$ and $f'(0) \neq 0$ (i.e. $a_1 \neq 0$). For g to also be a left-inverse, we require $g(0) = 0$, in which case

$$f \circ g = g \circ f = I$$

9.2.16 remark (If you know some calculus) Notice how the condition $f'(0) \neq 0$ bears resemblence to the inverse function theorem

***Proof*:**
We want to a g such that $f(g(z)) = z$. We want:

$$a_0 + a_1(b_1 z + b_2 z^2 + \cdots) + a_2(b_1 z = b_2 z^2 + \cdots)^2 = z$$

expanding and using the method of undetermined coefficients expanding the first two terms we get:

$$a_0 = 0 \qquad a_1 b_1 = 1$$

This shows us that $a_0 = 0$ and $a_1 = f'(0) \neq 0$ are necessary conditions. To show they are sufficient conditions, we can see that we may deduce all other coefficeints given these conditions.For example, we may deduce the coefficient of z^n as $a_0 + a_1 g(z) + \cdots + a_n g(z)^n$, so

$$a_1 b_n + P_n(a_2, ..., a_n, b_1, ..., b_{n-1}) = 0$$

Thus, we would start by doing $b_1 = 1/a_1$, and $b_2, b_3, ...$ are defined recursively (this resulting polynomials is called *Bells polynomial*). With our construction, we have that g satisfies $b_0 = 0$ and $b_1 \neq 0$. Thus, we can do the same thing to g, that is $g(f_1(w)) = w$. These inverses are equal:

$$f_1 = \text{id} \circ f_1 = (f \circ g) \circ f_1 = f \circ (g \circ f_1) = f \circ \text{id} = f$$

where we assumed the associativity of composition, something that can be checked. Finally, for the radius of convergence, we shall delay such considerations for now, noting that when we show that holomorphic functions are analytic we can use the inverse function theorem to estimate the values.

Example 9.2.17: Formal Power Series

Some classical examples of power series include:

1. $\exp(x) = e^x = \sum_{k=0}^{\infty} \frac{x^k}{k!}$
2. $\cos(x) =$
3. $\sin(x) =$
4. $\log(x) =$

9.2.3 Matrix Rings

As was briefly mentioned in the preliminary chapter (section 0.5), many structures can be "represented as matrices", that is, instead of studying all these structure as separate algebraic objects, they all follow the same rules as matrix rings.

Definition 9.2.18: Matrix Ring

Let R be a ring. Then $M_n(R)$ is the collection of all $n \times n$ matrices with coefficients in R. Addition is defined term-wise and multiplication is defined via matrix-multiplication as given in section 0.5. A matrix may be represented by:

$$(a_{ij})_{ij} \in M_n(R)$$

where a_{ij} represents the value at position ij of the matrix. The identity of $M_n(R)$ is denoted I_n.

In the above notation, matrix multiplication AB of two matrices can be represented as

$$(c)_{ij} = \left(\sum_{k=1}^{n} a_{ik} b_{kj} \right)_{ij}$$

Note that $M_1(R) \cong R$, and $M_2(R)$ is *not* commutative, and always has zero divisors, for:

$$\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \cdot \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \neq \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \cdot \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$$

R can be embedded into R by taking by taking mapping it to the diagonal:

$$a \mapsto (a)_{ii}$$

where the matrix $(a)_{ii}$ has zero in all other entries. It should be check that the subring of diagonal entries with all the same value is a subring, sometimes denoted $\iota(R) \subseteq M_n(R)$ to emphasize that $\iota(R)$ is an embedding, or RI_n. It should be check that $I_n = \iota(1)$. If R is commutative, then:

$$Z(M_n(R)) = RI_n$$

that is, the center of the matrix ring is the diagonals with the same coefficients. Slightly more generally, the subset of diagonal matrices also form a subring., espec Diagonal matrices are rather common, and so they are often denoted

$$\mathrm{diag}(a_1, a_2, ..., a_n) = (a_i)_{ii}$$

Using the diagonal embedding, we can define a functor $M_n(-) : \mathbf{Ring} \to \mathbf{Ring}$ mapping R to the diagonal and φ to $M_n(\varphi)$ that maps $(a_{ij})_{ij} \mapsto (\varphi(a_{ij}))_{ij}$ (the reader should feel free to check this is indeed a functor). Thus, there is the following diagram:

$$\begin{tikzcd} R \arrow[r, "\varphi"] \arrow[d, hook] & S \arrow[d, hook] \\ M_n(R) \arrow[r, "M_n(\varphi)"'] & M_n(S) \end{tikzcd}$$

Thus, if $S \subseteq R$, then $M_n(S) \subseteq M_n(R)$.

Matrix rings shall appear in many context, notably in the case when $R = k$ is a field in the theory of vector-spaces and when R is a division ring in representation theory.

9.2.4 Group Rings

We have seen how to take a set and turn it into a group in many ways (ex. trivial group $T(X)$, permutation group S_X, free group $F(X)$). Of these methods the free-group construction gave us a way to lift a set map to a group map. If you have read section 2.3.5, **Set** and **Grp** form a free-forgetful adjoint, which tells us that than the "lifting" of set functions to group homomorphisms happened in a "natural way". We can repeat this process with groups and rings, forming a "free ring" with respect to groups. This "free ring" will not be free with respect to *sets*, and so will differ from free groups[13]. However, after going through free rings with respect to groups, the reader might start seeing why both constructions are called *free*.

Definition 9.2.19: Group Ring

Let R be a commutative ring and let G be any group, where $g_0 = e$ is the identity, with the operation written multiplicity. Then define the *group ring* $R[G]$ (or RG) of G with respect to R to be the set of all formal sums:

$$\sum_{g_i \in G} a_i g_i$$

where only finitely many a_i are non-zero where addition is done point wise and the product is defined per monomial as $(ag_i)(bg_j) = (ab)g_k$ where ab is performed in R and $g_i g_j = g_k$, and then extended this is extended to multiplication of the formal sums using convolution.

Since $g_0 = e$, we often write $a_0 g_0 = a_0$. In this way, we can identify $R \subseteq R[G]$, and R in as a subring of $R[G]$ has it's original ring operations. Similarly, taking $a_{ij} = \delta_{ij}$[14] where

$$\delta_{ij} = \begin{cases} 0 & i \neq j \\ 1 & i = j \end{cases}$$

[13] We will cover free rings later
[14] this is called the Kronecker delta

we can identify $G \subseteq R[G]$, and G as a subgroup of $R[G]$ has it's original group operation. Even better, $G \leqslant R[G]^{\times}$. It is an easy exercise to check that $R[G]$ is indeed a ring, and that $R[G]$ is commutative if and only if G is commutative. Another simple exercise to check is that R (identified as a subring of $R[G]$) is in the center of $R[G]$ ($R \subseteq Z(G)$), and that $1 \in R$ (identified as a subring of $R[G]$) is the identity of $R[G]$

The reader may see that group rings are indeed the free objects with respect to the forgetful functor **Ring** $\to$ **Grp** by repeating the construction given for free groups: Take the collection of set maps $f : G \to R$ where only finitely many $f(g)$ are nonzero (such maps are said to have *finite support*[15]). Then two elements are added pointwise, $h \mapsto f(h) + g(h)$, and multiplication has the following clever interpretation:

$$h \mapsto \sum_{ab=h} f(a)g(b) = \sum_{a \in G} f(a)g(a^{-1}h)$$

Notice this sum is indeed finite since f, g have finite support.

Example 9.2.20: Group Rings

1. Let $R = \mathbb{Z}$ and $G = D_8$. Then, for example $3r^2 - s, rs + 9 \in \mathbb{Z}[D_8]$.
2. If G is any finite subgroup, and $S \subseteq R$ is a subring, then $S[G] \subseteq R[G]$ is a subring.
3. (If you know about the Hamiltonian's $\mathbb{H}$), then $\mathbb{R}[Q_8] \neq \mathbb{H}$, even though they both conains Q_8. Can you see why?
4. Let $R = \mathbb{Q}$ and $G = S_3$. Let $a = 3(2\ 3) - 4(1\ 2\ 3)$ and $b = 1/2(1\ 2) + 5(3\ 2\ 1)$. Compute:

$$a + b \qquad ab$$

Note that if G has an element of finite order, then the group ring always has zero divisors: let $g \in G$ such that $m = |g| > 1$. Then

$$(1 - g)(1 + ... + g^{m-1}) = 1 - g^m = 1 - 1 = 0$$

Exercise 9.2.1

1. Is $R[x]$ isomorphic to $\oplus_{i=1}^{\infty} R$? Is $R[[x]]$ isomorphic to $\prod_{i=1}^{\infty} R$?)
2. Let $f \big| g$. Does this imply $\deg(f) \big| \deg(g)$?
3. Let R be a **non**-commmutative ring. Define the appropriate ring so that for any set function $f : A \to S$, there is a unique map $\iota : A \to R$ to the ring $F : R \to S$ such that the diagram commutes. That is, define the free object in the general case so that it is left-adjoint to the forgetful functor **Ring** $\to$ **Set**.
4. Take the full sub-category **Poly-Ring** $\subseteq$ **Ring**. What is the inital object of **Poly-Ring**?
5. Let $p(x) \in R[x]$ be a polynomial such that for all $r \in R$,

$$p(r) = a_n r^n + \cdots + a_1 r + a_0 \neq 0$$

[15]For those who know some calculus/analysis, this is similar to how many smooth functions are defined on compact support

Show that there is an extension $S \supseteq R$ such that there exists $s \in S$ where $p(s) = 0$. In particular, there exists an S such that there exists a natural inclusion $\iota : R \hookrightarrow S$, $p(x)$ naturally includes in S and some element $s \in S$ is a root to $p(x)$:

$$p(s) = a_n s^n + \cdots + a_1 s + a_0 = 0$$

6. Show that $R[\mathbb{N}] \cong R[x]$. Show that $R[\mathbb{Z}] \cong R[x, x^{-1}]$
7. Is $\mathbb{Z}[x, x^{-1}] \cong \mathbb{Z}[[x]]$

9.3 Ring Homomorphisms and Quotient Rings

In this section, we explore the structure of the quotient objects of rings, that is, if $\varphi : R \to S$ is a ring-homomorphism, we aim to find a way to talk about the structure of $\varphi(R) \subseteq S$ given structure for R. We first recall the definition of ring homomorphisms from the first section:

Definition 9.3.1: Ring Homomorphism Remainder

Let R and S be rings. Then:

1. *ring homomorphism* is a map $\psi : R \to S$ satisfying
   (a) $\varphi(a + b) = \varphi(a) + \varphi(b)$
   (b) $\varphi(a \cdot b) = \varphi(a) \cdot \varphi(b)$
   (c) $\varphi(1_R) = 1_S$
2. The *kernel* of the ring homomorphism φ, denoted $\ker(\varphi)$ is the set of elements of R that map to 0 in S
3. A ring homomorphism with an inverse that is a ring homomorphism is a *ring isomorphism*.

If the context is clear, "homomorphism" will be used instead of "ring homomorphism". Also, if A, B are rings, then $A \cong B$ will always be interpreted as an isomorphism between the two rings.

Example 9.3.2

1. $\pi : Z \to \mathbb{Z}/2\mathbb{Z}$, here $\pi(x) = [x]$ is the natural projection map, is a ring homomorphism
2. $\varphi_n : \mathbb{Z} \to \mathbb{Z}$, $\varphi_n(x) = nx$ is *not* a ring-homomorphism.
3. $\varphi : \mathbb{Q}[x] \to \mathbb{Q}$, $\varphi(p(x)) = p(0)$ is a ring homomorphism, here $\ker \varphi = \{a_n x^n + \cdots + a_1 x\}$. Notice that every elements in the kernel has an x. We can also think of this as $p(x) \cdot x$ for any $p(x) \in \mathbb{Q}[x]$. This way of looking at it will become the main way we look at the kernel of a ring homomorphism shortly.
4. As shown earlier, if $S \subseteq R$ has a ring-structure, then if the inclusion map $i : S \to \mathbb{R}$ is a [ring] homomoprhism, then S is a subring.

5. The inclusion map $\mathbb{Z} \hookrightarrow \mathbb{Q}$ is a ring homomorphism. Is there a non-trivial homomorphim from $\mathbb{Q}$ to $\mathbb{Z}$?

6. Embedd $k[x] \hookrightarrow k[[x]]$ by mapping $k[x]$ to be all formal power series with only finitely many nonzero elements. Then by proposition 9.2.11, any polynomial $p(x) \in k[x]$ such that $p(0) \neq 0$ under the evaluation map is invertible in $k[[x]]$. Hence, there is a power series that can be identified by $1/p(x)$. Thus, we may define polynomial quotients in $k[[x]]$

$$\frac{p(x)}{q(x)} \qquad q(0) \neq 0$$

7. Is the derivative map $\frac{d}{dx} : k[[x]] \to k[[x]]$ defined by

$$\frac{d}{dx}\left(\sum_{k=0}^{\infty} a_k x^k\right) = \sum_{k=0}^{\infty} k a_k x^{k-1}$$

   a ring homomorphism? what is $\frac{d}{dx}(fg)$?

8. Let $k[[x]]$ and $k[[y]]$ be two formal power series over a field k. Let $T \in k[[y]]$ such that $b_0 = 0$. If we define $f \circ g$ to be (HERE). Then $f \mapsto f \circ T$ is a ring homomorphism from $k[[x]]$ to $k[[y]]$. What happens if $b_0 \neq 0$?

Just as the study of group-homomorphisms is equivalent to finding a normal subgroup H and studying G/H, so too will there be a link between ring-homomorphisms and a special subset I of R for which R/I represents the homomorphism. Just like for group-homomorphisms, the set I is based on the kernel of the ring-homomorphism, and R/I will be isomorphic to the image of φ:

Proposition 9.3.3: Image and Kernel of Ring Homomorphism

Let R, S be rings and $\psi : R \to S$ be a homomorphism.

1. The image of ψ is a sub-ring of S (as should be clear as it is one of the "properties" that are sought after for defining a homomorphism)

2. the kernel of ψ is, in general, *not* a subring of R. Rather, the kernel of ψ retains the abelian structure of R (i.e. is closed under adition an $a - b \in I$), and is closder under multiplication (left/right multiplication)[a] from elements of R (for all $r \in R$ and $i \in I$, $ri \in \ker(\psi)$ and $ir \in \ker(\psi)$).

[a]note that it was specified that it was both left and right multiplication because the multiplication is not necessarily commutative, while addition is

As an example of a kernel, take $\varphi : \mathbb{Z}[x] \to \mathbb{Z}[x]$, $p(x) \mapsto p(0)$. Then $\ker(\varphi) = x\mathbb{Z}[x]$, which is not a ring as $1 \notin x\mathbb{Z}[x]$. Kernels are the more general object *rng*. Note that if we define a ring to not have an identity, then the kernel would indeed be a sub-ring of R. Some authors define rings to not have an identity for this very reason. However, as a consequence, almost every theorem will have to be prefaced with "let R be a ring with identity", meaning the study of ring theory will ultimately be the study of a special case of rings with identity, which begs the question whether broadening the definition to not have an identity is worth the trade-off. However one sees it, it is simply a mathematical reality that kernels are different objects than rings (as the proposition suggests), and this author belives it is better to treat them as such[16]. Kernels will soon be defined as *ideals* (definition 9.3.6) since they have different properties from rings.

***Proof*:**

1. This should be an easy exercise to make sure homomorphisms can be comfortably worked with

2. Let $I = \ker\varphi$ and take $a, b \in I$. Then

$$\varphi(a-b) = \varphi(a) - \varphi(b) = 0 \Rightarrow a - b \in I$$

showing that I is closed addition, and in particular is an abelian group.

Now take $r \in R$. We want to show that for any $i \in I$, $ri, ir \in I$. The key to this proof is the 1st property of property 9.1.9. Let $I = \ker\varphi$. Then

$$\varphi(ri) = \varphi(r)\varphi(i) = \varphi(r) \cdot 0 = 0 \Rightarrow ri \in I$$

and similarly

$$\varphi(ir) = \varphi(i)\varphi(r) = 0 \cdot \varphi(r) = 0 \Rightarrow ir \in I$$

Thus, I is closed under left-right multiplication of R.

9.3.4 Remark As a consequence of kernels not being subrings, the concept of extensions as discussed in section 3.6 don't apply so simply to rings! A group extension relied on the fact that both N and G/N were groups. Since ideals are not rings, the same concept does not apply for rings! There is an analogous concept of a "ring extension"[a], which generalize the notion of an extension by just requiring a ring extension S of R to be that there exists an injective ring homomorphism $\varphi : R \to S$ (similarly to how if $N \trianglelefteq G$ then we always have that $N \hookrightarrow G$). This notion of ring extension will be central to us in Part V [b], but the "block-building" story that happened for groups does not play out in the same way way for rings. However, in part III, we will see that any ring R is a natural R-modules, and modules will have extensions defined on them [c]. Thus, we may study the extensions of rings in a group-theoretic way in the context of module theory.

[a]not to be confused with ring extensions that extend a ring by an abelian group as described in Nagata Masayoshi's book *Local Rings* (see Wikipedia article)

[b]It will be the fundamental way we look at varieties and number fields

[c]The category of R-module is closed under kernels

This does not mean that we cannot define $R/\ker(\varphi)$. In fact, $R/\ker(\varphi)$ will be a well-defined ring, as we explore now. Note that since R is an abelian group, and φ is also always a group-homomorphism, $\ker\varphi$ is always be a normal group. If we label $I = \ker(\varphi)$ then R/I already has a group structure, that is:

$$(r + I) + (s + I) = (r + s) + I$$

is well defined. The question becomes if we can also add a multiplicative structure to the quotient

[16]The fact that the kernel is not a ring means that the category of **Ring** does not have kernel objects

group, that is:

$$(r+I)\cdot(s+I) = (rs)+I \qquad \text{(in equivalence class notation: } [x\cdot y] = [x]\cdot[y])$$

With I being the kernel of a ring homomorphism, this result is immediate (i.e. it is well-defined). For $i,k \in I$ and $r,s \in R$:

$$\varphi((r+i)(s+k)) = \varphi(r+i)\varphi(s+k) = (\varphi(r)+\varphi(i))(\varphi(s)+\varphi(k)) = \varphi(r)\varphi(s)+0+0+0 = \varphi(r)\varphi(s)$$

where we used both the additive and multiplicative linearity condition. Labelling $[x] = x+I$, we call $(R/I, +_\sim, \cdot_\sim)$ with $+_\sim$ and $\cdot_\sim$ being defined through φ the *quotient ring* [17]. Thus, I being the kernel of a ring is a sufficient condition to make R/I into a ring. Is it a necessary condition? That is, can there be weaker structures that still preserves the multiplicative structure as shown above? For example, can I simply be a normal subgroup and always preserve the multiplicative structure? The answer is no:

Example 9.3.5: : Quotient group without preserving multiplicative property

Take $\mathbb{Q}/\mathbb{Z}$ as a quotient group. Since $\mathbb{Q}$ is abelian, $\mathbb{Q}/\mathbb{Z}$ is a well-defined quotient group. Let's say it was a well-defined quotient ring. Consider

$$\left(\frac{a}{n}+\mathbb{Z}\right) = (a+\mathbb{Z})\left(\frac{1}{n}=\mathbb{Z}\right) = (0+\mathbb{Z})\left(\frac{1}{n}+\mathbb{Z}\right) = [0]$$

so, every element greater than 1 is zero. Through this, we can make the rest of the elements of $\mathbb{Q}/\mathbb{Z}$ zero. Take $\frac{1}{n}$. Then

$$[0] = \left[0\cdot\frac{1}{n}\right][0]\left[\frac{1}{n}\right] = [1]\left[\frac{1}{n}\right] = \left[1\cdot\frac{1}{n}\right] = \left[\frac{1}{n}\right]$$

Which means that multiplication forced $(\frac{a}{b}+\mathbb{Z}) = (0+\mathbb{Z})$, meaning that

$$(r+\mathbb{Z})(s+\mathbb{Z}) \neq (rs+\mathbb{Z})$$

unless $(r+I) = [0]$ or $(s+I) = 0$, meaning there is no salvageable sub-ring we can which would make this work.

So we must figure out what properties I requires for closure of multiplication. Let's say that I is a normal subgroup, and let's work with the cosets R/I. Let's say that multiplication of cosets is well defined. That would imply that

$$[r+i][s+j] = [rs] \qquad i,j\in I$$

that is, multiplication is independent of representation. If $r=s=0$, then that means that

$$(i+I)(j+I)\in I \Rightarrow ij \in I$$

meaning that I is closed under multiplication. Next, what if $s=0$ and r is arbitrary. Then that

[17] as before we'll delay defining the quotient object till we find the sufficient conditions on I independent of a ring homomorphism

would lead to $[r][0] = [0]$ by proposition 9.1.9, which, written in terms of cosets, will be:

$$\begin{aligned} &((r+i)+I)\cdot(j+I) = I \\ \Rightarrow &(rj+ij)+I = I \\ \Rightarrow &(ri)+I = I \\ \Rightarrow &rb \in I \end{aligned}$$

and since r is any arbitrary element $r \in R$, then I must be closed under left multiplication of any element of R. Similarly if $r = 0$, meaning $is \in I$.

Conversely, let's say I is closed under multiplication from the left and right-side: $\forall i \in I$, $ri \in I$, $ir \in I$ $\forall r \in R$. Then

$$\begin{aligned} (r+i)+I\cdot(s+j)+I &= (r+i)(s+j)+I \\ &= (rs+rj+is+ij)+I \\ &= (rs+I)+(rj+I)+(is+I)+(ij+I) \end{aligned}$$

and since I is closed under multiplication, $ij \in I$, and since I is closed under left and right multiplication by any element of R, $rj, is \in I$, meaning we are left with:

$$= (rs)+I$$

meaning we are once again representative independent. Thus, we showed that left and right R-closure of I is a necessary *and* sufficient condition for the cosets defined by I to have a 2nd operation.

As with cosets, the equivalence classes induced by ring homomorphisms are given a special name:

Definition 9.3.6: Ideals

Let R be a ring, and let I be a subset of R, and let $r \in R$. Then:

1. $rI = \{ra | a \in I\}$, and $Ir = \{ar | a \in I\}$
2. a subset I of R is a *left ideal* if I is closed under *left* multiplication by elements from R ($rI \subseteq I$ for all $r \in R$)
3. Similarly, I is a *right* ideal if I is closed under *right* multiplication by elements from R ($Ir \subseteq I$ for all $r \in R$)
4. A subset I that is *both* a left and right ideal is called an *ideal* (or, for added emphasis, a *two-sided* ideal) of R

Example 9.3.7

1. Take $m \in \mathbb{Z}$. Then $m\mathbb{Z} \subseteq \mathbb{Z}$ is an ideal. Similarly for $\mathbb{Z}m$.
2. Take $m\mathbb{Z} \subseteq \mathbb{R}$. Then $m\mathbb{Z}$ is not not an ideal (since $\frac{1}{2m}mk \notin m\mathbb{Z}$).
3. In $\mathbb{Z}_4 \times \mathbb{Z}_4$, $\langle(1,1)\rangle$ is a sub-ring, but not an ideal
4. Take any product of two rings $R \times S$. Then $R \times 0 \subseteq \times S$ is an ideal, since for any $(a, 0) \in R \times 0$, $(R \times S)\cdot(a, 0) \subseteq R \times 0$. Notice that $R \times 0$ is also a subring of $R \times S$ meaning it contains an

identity. This is the *only* exception in which a subring an ideal can contain an identity.

(A word somewhere on how ideals define the relatinships in rings, similarly to how normal groups did for rings. We do not use the same angle-braket notation $\langle\ \rangle$ as we've done for groups since this notation will be used for a different purpose (note to self: in constructing free-algebras. This genelizes better the angle-bracket notation since all ring's are subsumed as an algebra by being $\mathbb{Z}$-algebras))

9.3.8 Trivia Why is it called an ideal? In number theory, people tried finding more information about polynomials like

$$x^2 - d = 0 \Rightarrow x = \sqrt{d}$$

And we would construct the field $\mathbb{Q}(\sqrt{d}) = \left\{a + b\sqrt{d} | a, b \in \mathbb{Q}\right\}$ which was called $\mathbb{Q}$ with $\sqrt{d}$ adjoint. This ring (in fact field) has a sub-ring $\mathbb{Z}[\sqrt{d}]$ and Q, making the diagram:

$$\begin{tikzcd}
\mathbb{Q}[\sqrt{d}] \arrow[d] \arrow[dr] & \\
\mathbb{Z}[\sqrt{d}] \arrow[dr] & \mathbb{Q} \arrow[d] \\
& \mathbb{Z}
\end{tikzcd}$$

So, with this construction, we have the question of developping the concept of prime number in $\mathbb{Z}$ to it in $\mathbb{Z}[\sqrt{d}]$. If you try to take the same idea (a element that has no divisors in the ring except the element and ± 1). But it turns that if you try to define an element of that, elements don't necessarily have a unique factorization!
Example
This is so fundamental that this was dropped as a good definition as prime element in $\mathbb{Z}[\sqrt{d}]$. Insead, the proper generalization of this concept will turn out to be ideals in $\mathbb{Z}$ which will be called *prime ideals*, and then look at their natural extension into $\mathbb{Z}[\sqrt{d}]$. The word "ideal" here comes from how these elements would be the *ideal numbers*. The word number will eventaully get dropped, leaving us with ideal.

9.3.9 Computational Shortcut To prove that I is an ideal, we must show that I is non-empty, that it's closed under subtraction ($\forall r \in R$), closed under multiplication ($\forall r \in R$), and that it's a normal subgroup. If R has a 1, then you can replace "closed under subtraction" to "closed under multiplication" since $(-1)a = -a$ will cover the negative case

Another reason to treat ideals differently than rings is the following:

> Another way of how ideals are different form rings is that being a ideal is "ring-dependent". For example $(x) = \mathbb{R}[x]x$ is an ideal of $\mathbb{R}[x]$, but not an ideal of $\mathbb{C}[x]$, since $ix + 1$ notin $(x) = \mathbb{R}[x]x$. This points again to how ideals are different.

This discussion is summarized in the following proposition:

Definition 9.3.10: Quotient Ring

When I is an ideal of R. The ring R/I with the natural multiplicative operation is called the *quotient ring* of R by I.

Proposition 9.3.11: Quotient groups and Ring Homomorphism

Let R be a ring and let I be an ideal of I. Then the (additive) quotient group R/I is a ring under the binary operation:

$$(r+I)+(s+I)=(r+s)+I \qquad \text{and} \qquad (r+I)\cdot(s+I)=(r\cdot s)+I$$

for all $r, s \in \mathbb{R}$. Conversely, if I is any subgroup such that the above operations are well defined, then I is an ideal of R.

Just like for groups, we can ask if every ideal is the kernel of a ring homomorphism, essentially asking for the 1st isomorphism theorem for rings (i.e. the universal property of quotient rings). This turns out to be the case:

Theorem 9.3.12: The first isomorphism theorem for Rings

1. If $\varphi : R \to S$ is a homomorphism of rings, then the kernel of φ is an ideal of R, the image of φ is a subring of S and $R/\ker(phi)$ is isomorphic as a ring of $\varphi(R)$.
2. If I is any ideal of R, then the map

$$R \to R/I \qquad \text{defined by} \qquad r \mapsto r+I$$

is a surjective ring homomorphism with kernel I (this homomorphism is called the *natural projection* of R onto R/I). Thus every ideal is the kernel of a ring homomorphism and vice versa.

***Proof*:**
This proof is similar to the fisrt isomorphism theorem (theorem 3.1.14), and we already did a good chunk of it in the preceeding paragarphs. The proof of the natural projection follows very similar lines to theorem 3.1.14.

As with groups, the bar notation $\overline{r}$ or $r \bmod I$ notation is used to represent the equivalence class $[r]$.

Example 9.3.13

1. R and $\{0\}$ are ideals. An ideal is proper if $I \neq R$. The set with only zero is called the *trivial* ideal.
2. $n\mathbb{Z}$ is an ideal of $\mathbb{Z}$.
3. Take $R = \mathbb{Z}[x]$, and let I be the subset whose polynomials are of degree at least 2. Note that I is closed under addition and multiplication, and thus an ideal. The cosets consist of

the elements with the same 0th and 1st term, so

$$\overline{1+2x+x^2} = \overline{1+2x-5x^3} = \overline{1+2x+x^6+87x^{1231}}$$

Note that $\overline{x}\overline{x} = \overline{x^2} = \overline{0}$, thus R/I has a zero divisor, even though $\mathbb{Z}[x]$ does not .

4. We saw $\mathbb{Q}[x]/(x)$. Let's try another polynomial: $\mathbb{Q}[x]/(x^2-2)$. Since we can divide polynomials, we can divide any polynomial by x^2-2. We can in fact keep dividing it till we get a polynomial of lesser degree. We will actually then get that

$$\mathbb{Q}[x]/(x^2-2) = \{ax+b | a,b \in \mathbb{Q}\}$$

Note that x^2-2, so $x^2-2=0 \Rightarrow x = \pm\sqrt{2}$. So, since $[x^2-2]=[0]$, this will make

$$\mathbb{Q}[x]/(x^2-2) \cong \mathbb{Q}(\sqrt{2}) = \left\{x+y\sqrt{2} | x,y \in \mathbb{Q}\right\}$$

Or, more specifically, if we take the ring-homomorphism $\varphi : \mathbb{Q}[x] \to \mathbb{Q}(\sqrt{2})$, $p(x) \mapsto p(\sqrt{2})$, then This is a isomorphism.

5. If we take $k[x,y]$, then $k[x,y]/(x) \cong k[y]$

6. Let $G = \langle g \rangle$ be a cyclic group and k be any field. Define:

$$\varphi : k[G] \to k[x] \qquad \varphi(ag^k) = ax^k \qquad \forall a \in k$$

Show that:

$$k[G] \cong \frac{k[x]}{x^n-1}$$

Example 9.3.14

Let's say you want to figure out all integer solutions to

$$x^2+y^2=3z^2$$

Suppose such integers exist. Observe first that we may assume x, y and z have no factors in common, since otherwise we could divide through this equation by the square of this common factor and obtain another set of integer solutions smaller than the initial ones. This equation simply states a relation between these elements in the ring $\mathbb{Z}$. As such, the same relation must also hold in any quotient ring as well. In particular, this relation must hold in $\mathbb{Z}/n\mathbb{Z}$ for any integer n. The choice $n=4$ is particularly efficacious, for the following reason: the squares mod 4 are just 0^2, 1^2, 2^2, 3^2, i.e., 0, 1 (mod 4). Reading the above equation mod 4 (that is, considering this equation in the quotient ring $\mathbb{Z}/4\mathbb{Z}$), we must have

$$\begin{Bmatrix}0\\1\end{Bmatrix} + \begin{Bmatrix}0\\1\end{Bmatrix} = + \begin{Bmatrix}0\\3\end{Bmatrix} \qquad (\text{mod} 4)$$

where the $\begin{Bmatrix}0\\1\end{Bmatrix}$ indicate that either a 0 or a 1 may be taken. The only possibility is if they're all even, but we assumed they weren't, thus there is no solution.

Note that this technique has limitation. The following equation has solution integers modulo any n, but no integer solutions (which is hard to prove).

$$3x^3 + 4x^3 + 5z^3 = 0$$

Eventually, we'll be using a similar technique to factor polynomial.

There is also the 2nd, 3rd, and 4th isomorphisms theorems. Each of these are quite easy to show: We already know they're true because R is an *additive group* (or *correspondence of groups*, in the case of the Fourth Isomorphism theorem). It only needs to be show that the group isomorphism is also a multiplicative map, and thus a ring-isomorphism.

Beforehand, we had $H, K \leqslant G$ and defined HK. Since we are now writing the group operation additively, we will convert our notation to saying $H + K$. If I, J are subrings of R, then since they're both normal (even abelian) subgroups, then $I + J$ is well-defined.

Theorem 9.3.15: 2nd, 3rd, 4th Isomorphism Theorem for Rings

(2) Let A be a subring and B be an ideal of R. Then $A + B$ is a subring of R, $A \cap B$ is an idea of A, and $\frac{A+B}{B} \cong \frac{A}{A\cap B}$

(3) Let I, J be ideals of R iwth $I \subseteq J$. Then J/I is an ideal of R/I and $(R/I)/(J/I) = R/I$.

(4) Let I be an ideal of F. The correspondence $A \leftrightarrow A/I$ is an inclusion preserving bijetion between the set of subrings A of R that contian I and the set of subrings of R/I. Furthermore, A (a subring containin I) is an ideal of R if and only if A/I is an ideal of R/I

***Proof*:**
The proof of each of these is an upgrade from the proof for the group version. Proving the group-isomorphism is immediate, then proving multiplicative linearity, it's an immediate consequence of the definition of the quotient ring.

(A word on simple ring, i.e. rings with no *two-sided* right-sided ideals? A word on $M_n(k)$ is simple, since no two-sided ideals (since two ided of ideals of $M_n(k)$ are of the form $M_n(I)$), but there exist left-sided ideals (ex. set of matrices with some fixed zero column) and right-sided ideals (set of with fixed zero row))

9.3.16 Remark simpleCommRingIffFieldRmk
We can define a *simple ring* to be a ring whose only two-sided ideals are R and itself[a], which mirrors the definition of simple groups. Then fields *are* all simple commutative rings[b]. For the non-commutative case, it is harder to find simple rings. (see exercise ref:HERE for an example, or proposition 51.2.5).

[a] the reason for sticking to two-sided ideals is because we can quotient R by two-sided ideals to get a quotient ring, but we cannot quotient by by 1-sided ideals and get a quotient ring

[b] The story is a bit different for *rngs*; see comment in document

Exercise 9.3.1

1. Let R, S be rings and consider the product ring $R \times S$. What are the ideals of $R \times S$?
2. Let I be an ideal of the ring R and let $(I) = I[x]$ denote the ideal of $R[x]$ generated by I (the set of polynomials with coefficient I). Then

$$R[x]/(I) \cong (R/I)[x]$$

   In particular, if I is a prime ideal of R then (I) is a prime ideal of $R[x]$. Conclude that $\mathbb{Z}[x]/(n\mathbb{Z}[x]) \cong (\mathbb{Z}/n\mathbb{Z})[x]$
3. Show that the set of upper-triangular matrices is a subring.

9.4 Properties and Further Structures of Ideals

Since the kernel of ring homomorphisms are ideals, they form the core of much of ring theory. Recall how when working with groups, they can be thought of as $F(A)/N$ for an appropriate normal set N which represents the *relations* on the generators A. Ideals will play a role of similar importance: Ideals will represent the relations. There will be an analogous notion of free ring, however the study of generation of free rings shall be delayed to part III since the two constructions are identical.

This is not to say that thinking of ideals as adding relations is un-useful! For example, since $\mathbb{Z}$ is initial in rings, it is of great interest to see how what relations ideals of $\mathbb{Z}$ can give us, what properties do they have, and how they translate with ring homomorphisms. (We will focus a lot on this in number theory).

Furthermore, closure is not well-defined on subrings, that is the "closure" of a set $A \subseteq R$, that is the smallest largest set that makes $+$ and $\cdot$ well-defined, need not be a ring. Take for example $\{2\} \subseteq \mathbb{Z}$, then if $\overline{\{2\}}$ is the "ring closure" of $\{2\}$, then $\overline{\{2\}} = (2)$, which is not a ring. Another example is $\{1 + x\} \subseteq \mathbb{Z}[x]$. Then $\overline{\{1 + x\}}$ is not a ring, since $1 \notin \overline{\{1 + x\}}$[18], and it is not an ideal since $(x + 2)(x + 1) \notin \overline{\{x + 1\}}$[19]. It is a rng, and so it makes senes to define finitely generated rngs, however due to the reason stated in remark ref:HERE, we continue to focus on rings.

9.4.1 Operations and Generation on Ideals

The first thing we might think of is way of extending how to manipulates ideals through binary operations (in essense, we find different ways of combining what relations we impose on a ring). We give a list of common operations of ideals

[18]This is actually a bit difficult to prove, that no polynomial $p(y)$ exists such that $p(1+x) = 0$, and so no polynomial $q = p + 1$for which $q(x + 1) = 1$. The full answer will be given in part IV when discussing transcendental extensions.

[19]To see this, we require the fact that every polynomials factors uniquely into irreducibles, and so it's impossible that $p(x + 1) = (x + 2)(x + 1)$ for any $p(y)$. We will show this in chapter 11

Definition 9.4.1: Operations on Ideals

Let R be a ring and $I, J \subseteq R$ be (left, right, or two-sided) ideals. Then:

1. $I \cap J$ is the intersection of two ideals. More generally,
$$\bigcap_{i \in \mathcal{I}} I_i$$
is the intersection of ideals I_i indexed by $\mathcal{I}$.

2. Define the *sum* of I and J by $I + J = \{a + b | a \in I, b \in J\}$. More generally,
$$\sum_{i \in \mathcal{I}} I_i = \left\{ \sum_{i \in N} a_i \;\middle|\; N \subseteq \mathcal{I},\ |N| \in \mathbb{N},\ a_i \in I_i \right\}$$
is the sums of ideals I_i indexed by $\mathcal{I}$.

3. Define the *product* of I and J, denoted IJ, to be the set of all finite sums of elements of he form ab wit $a \in I$ and $b \in J$.
$$\left\{ \sum_{i=1}^{n} a_i b_i \middle| a_i \in I,\ b_i \in J \right\}$$
Note that we must take finite sums since the ideal must be closed under addition. More generally:
$$\prod_{i \in \mathcal{I}} I_i = \left\{ \sum_{i=1}^{m} a_{i_1} \cdots a_{i_n} \;\middle|\; a_{i_1}, ..., a_{i_n} \in \mathcal{I}, n, m \in \mathbb{N} \right\}$$

4. For any $n \geq 1$, define the n^{th} power of I, denoted I^n to be the set consisting of all finite sums of elements of the form $a_1 a_2 \cdots a_n$ with $a_i \in I$ for all $i>$ Equivalently, I^n is defined inductively by defining $I^1 = I$, and $I^n = II^{n-1}$ for $n = 2, 3, ...$ By convention $I^0 = R$.

5. Let R be commutative. Define the *ideal quotient* $(I : J)$ to be:
$$(I : J) := \{x \in R \mid xI \subseteq J\}$$

6. Let R be commutative. Define the *radical* of I to be
$$\sqrt{I} := \{r \in R \mid r^n \in I \text{ for some } n \geq 1\}$$
An ideal satisfying $I = \sqrt{I}$ is called a *radical ideal.*

Note that if I, J are *both* ideals, then $I + J$ is an ideal (this set is closed under sums and $r(a + b) = ra + rb \in I + J$). It should be emphasized that the set IJ is the *finite sum* of elements of the form ab. This condition is required so that IJ is closed under addition, which is necessary for IJ to be an ideal. Just like groups and rings, the union of two ideals is not an ideal (for example, $2\mathbb{Z} \cup 3\mathbb{Z}$ does not contain 5, however $2 \in 2\mathbb{Z}$, $3 \in 3\mathbb{Z}$). Can you show that the quotient ideal $(I : J)$ is indeed an ideal? In some very lose way, if $a \in (x)$, then that means that x is a "factor" of a in the sense that $a = rx$ for some $r \in R$)

Example 9.4.2

1. Let $I = 6\mathbb{Z}$ and $J = 10\mathbb{Z}$. Then $I + J$ is the ideal of all integers of the form $6x + 10y$ for $x, y \in \mathbb{Z}$. Since 6 an 10 are divisible by 2, meaning that $6x + 10y = 2(3x + 5y)$, so that $2\mathbb{Z} \supseteq 6\mathbb{Z} + 10\mathbb{Z}$. Conversely, we see that $2 = 6(2) + 10(-1)$, so $2x = 6(2x) + 10(-x)$, meaning that every element of $2\mathbb{Z}$ can be written of the form $6\mathbb{Z} + 10\mathbb{Z}$, so $2\mathbb{Z} \subseteq 6\mathbb{Z} + 10\mathbb{Z}$. Thus $2\mathbb{Z} = 6\mathbb{Z} = 10\mathbb{Z}$. We can actually generalize this to $n\mathbb{Z} + m\mathbb{Z} = d\mathbb{Z}$ where $d = (m, n)$, in the next section. Similarly, it is worth checking that $6\mathbb{Z}10\mathbb{Z} = 60\mathbb{Z}$.

2. Take $I \subseteq \mathbb{Z}[x]$ to be the ideal whose constant term is even. Then (2) and (x) are in subsets of I. Thus, $4 = 2 \cdot 2$ and $x^2 = x \cdot x$ are in I^2, as is their sum $x^2 + 4$. However, $x^2 + 4$ cannot be written as a product $p(x)q(x)$ of two elements of I.

3. the ideal $((0) : I)$ known as

4. Let $(2), (6) \subseteq \mathbb{Z}$. Then $((2) : (6)) = (3)$, showing how the quotient ideal generlizes some notion of division. *annihilator of* I and consists of all elements of R such that $rx = 0$ for all $x \in I$. Usually, for any ideal $I = (r)$ generated by a single element, we write $(r : I)$ instead of $((r) : I)$. Using this, can you deduce what $(0 : R)$ is equal to?

The radical needs a word of its own, since unlike the other five operations it is not built from sums and products of elements already in the ideals, and that it is an ideal at all is a small argument rather than an observation.

Proposition 9.4.3: The Radical Is an Ideal

Let R be a commutative ring and $I \subseteq R$ an ideal. Then $\sqrt{I}$ is an ideal of R containing I, and $\sqrt{\sqrt{I}} = \sqrt{I}$.

***Proof*:**

Every $r \in I$ satisfies $r^1 \in I$, so $I \subseteq \sqrt{I}$; in particular $0 \in \sqrt{I}$.

For closure under the ring action, let $r \in \sqrt{I}$ with $r^n \in I$ and let $a \in R$. Since R is commutative, $(ar)^n = a^n r^n$, which lies in I because I absorbs multiplication, so $ar \in \sqrt{I}$.

For closure under addition, let $r^n \in I$ and $s^m \in I$. By the binomial theorem, valid in any commutative ring,

$$(r+s)^{n+m-1} = \sum_{i=0}^{n+m-1} \binom{n+m-1}{i} r^i s^{n+m-1-i}$$

Consider a single term. If $i \geq n$ then $r^i = r^n r^{i-n} \in I$; and if $i \leq n-1$ then $n+m-1-i \geq m$, so $s^{n+m-1-i} \in I$ by the same argument. Either way the term lies in I, and I is closed under addition, so $(r+s)^{n+m-1} \in I$ and $r + s \in \sqrt{I}$. Together with the previous paragraph this makes $\sqrt{I}$ an ideal.

Finally, if $r \in \sqrt{\sqrt{I}}$ then $r^n \in \sqrt{I}$ for some n, so $(r^n)^m = r^{nm} \in I$ for some m, giving $r \in \sqrt{I}$; the reverse inclusion is the first paragraph applied to $\sqrt{I}$, completing the proof.

We can also ask oursevles about the generation of ideals:

Definition 9.4.4: Generating Ideals

1. (A) is the smallest ideal containing A. Then

$$(A) = \cap_{A \subseteq I} I$$

   where I is an ideal containing A.

2. RA is he set of all finite sums of elements of the form ra between R and A.

$$\{r_1 a_1 + \cdots + r_k a_k | r_i \in R \, , a_i \in A\}$$

   Similarly for AR and RAR.

3. An ideal generated by a single element is a *principal ideal.*

4. an ideal generated by a finite set is called a *finitely generated ideal.*

If R is commutative, then $RA = AR = RAR = (A)$.

9.4.5 Moment To Ponder You might ask yourself, with addition and multiplication defined on the ideals, does the set of ideals form a ring? Try it out! Is this a way to study ideals? More on studying the ideal structure of a ring when studying Modules in part III (actually, wouldn't referencing Picard's group Pic(R) be a better reference here)

In the following chapter's, ideals generate by a singleton (a) will be of keen interest. As a prelude, notice that if $b \in (a)$, then $b = ra$ for some $r \in R$. Phrased differently, a divides b. Also, $b \in (a)$ if and only if $(b) \subseteq (a)$. In other words, being an element inside the ideal gives some idea of "divisibility" or "factorization".

Example 9.4.6

1. (0) and R are both principal ideals, since (0) is generated by 0, and $R = (1)$ (since we assumed R has an identity).

2. Every ideal of $\mathbb{Z}$ is a principal ideal

   Proof. We'll show that $(m) + (n) = gcd(m, n)$

   ($\Rightarrow$) Let $x \in (m) + (n)$. Let's show that $x \in (m, n)$. Since $x \in (m) + (n)$, then:

$$x = ma + nb \qquad a, b \in \mathbb{Z}$$

   letting $p = (m, n)$, we see that

$$ma + nb = (m'a + n'b)p = cp \in (m, n)$$

   where m', n' are the remainders, and $c = m'a + n'b$, showing that $cp \in (m, n)$

($\Leftarrow$) Let $p = gcd(m, n)$. Without loss of generality, let's say $m > n$. We will take advantage of the euclidean algorithm. First, start with:

$$m = a_1 n - r_1$$

where r_1 is the remaidner by dividng m by n and a is the number of times n can be divided into m. This is the extended euclidean algorithm covered in MAT240. We can work our way down till we get

$$r_k = a_{k+1} p - 0$$

where $p = gcd(m, n)$. Then, we can use Bézouts identity from the results to get that:

$$p = k_1 m - k_2 n$$

with appropriate k_1, k_2. This then implies that

$$cp = ck_1 m - ck_2 n$$

for any $c \in \mathbb{Z}$, and so cp can be written of the form $am + bn$, meaning $cp \in (m) + (n)$, as we sought to show.

We can actually use this to show that *any* (A) is a principle ideal in $\mathbb{Z}$. Since $\mathbb{Z}$ is commutative, we can think of $(A) = RA$. Then we can do the same greatest common denominator trick to get down to our desied number such that $d\mathbb{Z}$ is equal to $\mathbb{Z}A$. We can use this to show that every ideal of $\mathbb{Z}$ are principal ideals. Since $\mathbb{Z}$ is also an integral domain, meaning it's also a domain, we will end up calling $\mathbb{Z}$ a *principal ideal domain*.

However, it's not always true that if (a, b) can be reduced to (c) that a ring is a principal ideal domain. There is a sub-class of rings called *Bézout Domains* for which every pair of number can be reduced to a principal ideal, but the ring itself is *not* a principal ideal domain. More on that in chapter 10. $\square$

3. Let $R = \mathbb{Z}[x]$. Then $(2, x)$ is *not* a principal ideal. Let

$$(2, x) = \{2p(x) + xq(x) | p(x), q(x) \in \mathbb{Z}[x]\}$$

that is, all polynomials with *even* constants (and hence, a proper ideals, since it doesn't containt 1). Let's say for some $a(x) \in \mathbb{Z}[x].(2, x) = (a(x))$. Since $2 \in (a(x))$, there must be some $p(x)$ such that $2 = p(x)a(x)$. By proposition 9.2.3, we know that degree $p(x)a(x) =$ degree $p(x)+$ degree$a(x)$, and since degree(2) is 0, then $a(x)$ and $p(x)$ must *both* be constant polynomials. Since 2 is a prime number, the only possibilities are $p(x), a(x) \in \{\pm 1, \pm 2\}$. If $a(x) = \pm 1$, then $1 \in (a(x))$, and so by proposition 9.4.10, $(a(x)) = \mathbb{Z}[x]$. But that's a contradiction to $(2, x)$ being proper. Thus, it must be that $a(x) = \pm 2$. But then, $x \in (\pm 2)$, and so $x = p(x)(\pm 2)$. But this is impossible; we'd need $\frac{1}{2}x$ to be $p(x)$, but $\frac{1}{2} \notin \mathbb{Z}$. Thus, $(2, x)$ is *not* a principal ideal. Thus $\mathbb{Z}[x]$ is *not* a principal ideal domain.

I like to thinke of elements of (p, q) for some $R[x]$ as elements in which every term has p or q as a factor. This is because when taking $\alpha p + \beta q$ for $\alpha, \beta \in R[x]$, you can have α or β be 0, meaning you just need elements which have p or q as a factor.

4. If we take $\mathbb{Q}[x]$ instead of $\mathbb{Z}[x]$, then this would be solved since $(2) = \mathbb{Q}[x]$ since $\frac{1}{2}2 = 1$, maning (2) has a unit, meaning $(2, x)$ has a unit. In fact, $\mathbb{Q}[x]$ will become a principal ideal domain because of this! Try playing around with the following ideals to see they are indeed principal

- $(2, x)$
- (x, x^2+1)
- $(p(x), q(x))$ for $p, q \in \mathbb{Q}[x]$.

5. Take $R = k[x]$ and take $I = \{p(x) \in k[x] | p(q) = 0\}$ for some arbitrarry $q \in \mathbb{F}$. One can check that it is an ideal. More interestingly though, we know from high-school math that if $p(x) = 0$, then we can factor out $p(x) = (x-q)r(x)$ for any polynomial $r(x) \in \mathbb{F}[x]$. This will thus make $I = ((x-q))$, making these ideals principal ideals! However, if $I = \{p(x) \in \mathbb{F}[x] | p(a) = p(b) = 0, a \neq b\}$ for two arbitrary $a, b \in F$, $a \neq b$, then I is *not* a principal ideal.

As the intuition for ideals suggest, they are a sort of generalization of properties of $\mathbb{Z}$. For example, if $a \in R$ and (a) is an ideal, then $R/(a)$ is well-defined. Now, take $\bar{b} \in R/(a)$ and generate $(\bar{b})$. Then it should be verified that

$$(R/(a)/(\bar{b}) = R/(a, b)$$

Furthermore, notice that in $\mathbb{Z}$, the ideal (a, b) is equal to the ideal $(\gcd(a, b))$. For this reason, ideals capture some more general sense of divisibility for rings! Exploring the notion of divisibility further will be the goal of chapter 10.

Since ideals are so important, it is often the case that studying rings with simpler ideal structure are of much interest. Two particular cases are especially important due to the abundance of finiteness:

Definition 9.4.7: Noetherian rings

Let R be a commutative ring. Then if every ideal[a] $I \subseteq R$ is finitely generated, then R is called a *Noetherian ring*.

[a]Be sure not to mistake ideal for ring, for finitely generated rings have not been defined

Noetherian rings are particularly nice in that they are finite locally and globally. If we just require that the ring is finitely generated, it may be that a subring is non-finitely generated. Conversely. We shall have lots of experience with Noetherian rings later, for now it is useful to know these rings since since being Noetherian is stable under taking the quotient, and is also stable under the "converse" of taking the quotient (if you've covered section 3.6, the Noetherian condition is stable under extensions). For the next proposition, note that a finitely generated ideal is defined identically to a finitely generated ring (that it is generated by finitely many elements):

Proposition 9.4.8

let R be a Noetherian ring and I be any ideal of R. Then R/I is finitely generated. Conversely, if I and R/I is finitely generated, then R is finitely generated.

***Proof*:**
Make it a guided exercise, since it is easy to follow

One of the simplest Noetherian rings are those for which every ideal is not only finitely generated, but also principal. If the ideal is furthermore an integral domain (to avoid pathologies from zero-divisors) then it is a principal ideal domain:

Definition 9.4.9: Principal Ideal Domain

Let R be an integral domain. Then if every ideal $I \subseteq R$ is a principal ideal, then R is called a *Principal Ideal Domain.*

As a simple example, $\mathbb{Z}$ is a PID (can you see why?). It should be noted that *every* ideal is principal, even those that may seem infinitely-generated. In fact, there are domain in which only finitely-generated ideals are principle (such rings are called *Bézout* rings).

Another way of looking at principal ideal domains is that they are "locally free". Here, locality is taken to mean that any proper submodule is free. Since R is an integral domain, it is torsion free, and since R is principle, if $I \subseteq R$ is a submodule (i.e. ideal), then

$$I = (a) \qquad a \in R$$

This fact shall make PID's very powerful, and in fact the idea that PID's have "local freeness" shall be further explore in chapter $17\star$[20].

(TBD – somehow incorporate this?) *Principal Ideal Domains* (PIDs) will have the property of having "prime elements" well-defined and lend itself useful in many classification theorems (for example, study of Jordan Canonical Forms in chapter 14 will heavily rely on this type of ring). (The point is to introduce them later in the chapter on factorization)

Exercise 9.4.1

1. Michael Atiyah has some excellent elementary exercises!! Some of his propositions are also great: the sequence of propositions on radical ideals can totally be exercises here! p. 6-9.
2. $(I \cap J)(I + J) = IJ$

9.4.2 Ideals and Units

As briefly mentioned, some ideals seem to be related to a notion of divisibility. This intuition more clear with the following theorem:

Proposition 9.4.10: units in Ideals and ideals in fields

1. $I = R$ if and only if I contains a unit
2. R is a division ring if and only if it's only left- and right-sided ideals are 0 and itself. If R is commutative, then if the only ideals of R are 0 and itself, then R is a field.

***Proof*:**

1. one inclusion is easy. The key part to the other inclusion is that I must be closed under multiplication from all elements of R. If $I = R$ then I contains the unit 1. If I contains a unit u, then given the inverse $v \in I$ and any element $r \in R$
$$r = r \cdot 1 = r(uv) = (rv)u \in I$$

[20] A projective module over a PID is in fact a free module

Since I is an ideal.

2. If R is a division ring, then every element is a unit, and thus for any set A $RA, AR \subseteq R$ must contain units, and so $I = R$. Conversly, if the only left and right-sided ideals are R and 0, then for any $u \in R$, $Ru = uR = R$ (since it's the only ideal). Then $1 \in (u)$ meaning $1 = uv = vu$ for some $v \in R$. Since this holds for every element, then R is also a division ring. If R is also commutative, it is a commutative division ring, and thus a field.

9.4.11 Note As a consequence, if $1 \in I$, then $I = R$, since 1 is a unit!! This means that in a proper ideal, $1 \notin I$, and so, they cannot be sub-rings with identity $1 \neq 0$.

It should be pointed out it was necessary to have the condition that both let and right-sided ieals are trivial, instead of just picking one or saying that only two-sided ideals are trivial.

Example 9.4.12: Two-Sided Ideals Trivial, but non-trivial left/right-sided ideals

Let k be a field and $M_2(k)$ the matrix ring over k. Then it should be checked that $M_2(k)$ has non-trivial left and right-sided ideals (think of columns and rows of $M_2(k)$). However, $M_2(k)$ has *no* two-sided non-trivial ideals.

Now, $M_2(k)$ is not a division ring: the element

$$\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$$

does not have an inverse (as can be shown using some lienar algebra).

(make an exercise somewhere finding the ideals of $M_n(R)$, namely $M_n(I)$)

Since any two-sided ideal is a left and right ideal, then division ring's naturally have no non-trivial two-sided ideals.

Corollary 9.4.13: Division Ring and Ring-Homomorphism

If R is a division ring, then any non-zero ring-homomorphism $\varphi : R \to S$ to any other ring must be injective.

***Proof*:**
Let φ be a ring-homomrophism. The kernel of φ is a two-sided ideal of R, but R is a divison ring, meaning the only ideals of R are 0 and R. Since φ is non-trivial, $\ker \varphi \neq R$, and so $\ker \varphi = 0$, making φ injective.

This might make one think that studying division rings, or more particularly fields, in terms of homomorphisms is less useful. However, we will *still* use ring-homomorphisms on fields, since mapping a field injectively into another field will turn out to be quite useful: we can study how fields *embedd* into one another (this will be called a *field extensions*, which will be defined in part IV, and an *integral extension* for more general rings like division rings, which will be defined in part V).

If a ring does only have 0 and R as two-sided ideals, we have the analogous concept to simple groups

Definition 9.4.14: Simple Rings

Let R be a ring. Then R is said to be simple of it's only [left/right/two-sided] ideals are the 0 ideal and itself

One might think that from here, there will be a similar story as in group theory: A lot of time and effort spent on classifying all rings (and finite rings) given this idea of a simple ring. It turns out though that the classification of finite rings are in fact much much easier: Every finite simple ring is isomorphic to the ring $M_n(\mathbb{F}_q)$ over a finite field of order q (i.e., intimately related to matrix rings)! In fact, the study of simple rings is the setting of representation theory! We'll cover simple ring in part X

9.4.3 Maximal Ideals

Many times, we might want if an ideals is *as large as possible* in a ring. That is, if I is an ideal, then there does not exist another J such that $I \subseteq J$ unless $J = R$

Definition 9.4.15: Maximal ideal

An ideal M in an arbitrary ring S is called a *maximal ideal* if $M \neq S$ and the only ideals containing M are M and S.

Example 9.4.16: Ring with maximal ideal

Take $\mathbb{Z}$ and $6\mathbb{Z}$. Since $6 = 2(x)$, $6\mathbb{Z} \subseteq 2\mathbb{Z}$. However, since 2 is a prime, we cannot repeat this same trick twice. Thus, $2\mathbb{Z}$ is a maximal ideal. In general, $p\mathbb{Z}$ for prime p is a maximal ideal

All rings have a maximal ring with the exception of a ring with the trivial ring structure:

Example 9.4.17: Ring without maximal ideal

Take any group with a non-maximal group (ex $\mathbb{Q}$) and take the trivial ring structure ($ab = 0$).

The only restriction on a ring that requires it to have a maximal ideal is that $1 \neq 0$. The way we will prove this will require a form of "super-induction". The use of this proof technique requires the axiom of choice. For those who want to have an idea of Zorn's lemma, see section A.3, and for those who want to see how it links to the axiom of choice, see appendix A

Proposition 9.4.18: Ring with $1 \neq 0$ have Maximal ideals

Let R be a non-trivial ring and $I \subsetneq R$ be a proper ideal. Then I is contained in a maximal ideal

***Proof*:**
Since $I \subsetneq R$, $1 \neq 0$), and $1 \notin I$ (or else $I = R$), a maximal ideal can exist. To show that it is the case, we will us Zorn's Lemma (axiom A.2.3). Let $\mathcal{I}$ be the collection of all proper ideals of R containing I, and let it be ordered by inclusion. Then $(\mathcal{I}, \subseteq)$ is a partial order. If e take any

chain $\mathcal{C} \subseteq \mathcal{S}$, then

$$J = \bigcup_{A \in \mathcal{C}} A$$

is the ideal which is the upper-bound for $\mathcal{C}$. For this to be the case, J must be an ideal.

1. J is non-empty, since any ideal must contain 0, thus $0 \in J$
2. Take $a, b \in J$ and consider $a - b$. Then for some $A, B \in \mathcal{C}$, $a \in A$ and $b \in B$. Since $\mathcal{C}$ is a chain, $A \subseteq B$ or $B \subseteq A$. Witout loss of generality, let's assume $A \subseteq B$ and say $a, b \in B$ (if e pick $B \subseteq A$ do $a, b \in A$). Then since B is an ideal, $a - b \in B$, so $a - b \in J$, showing that J is closed under subtraction
3. Take $a \in J$ and consider $r \cdot a$ and $a \cdot r$. Then since $a \in J$, a is in some ideal, let's say $a \in A$. Then $r \cdot a \in A$ and $a \cdot r \in A$, so $r \cdot a, a \cdot r \in J$, showing it's closed uner left and right multiplication

Thus, J is an ideal. It is further a proper ideal. If it ere not, then $1 \in J$, meaning for some ideal A, $1 \in A$. But then $A = R$, showing it is *not* a proper ideal, a contradiction to the definition of J. Thus, J must also be a proper ideal.

Therefore, by Zorn's Lemma, $\mathcal{I}$ must have a maximal element M, that is, if there is an element M' such that $M \subseteq M'$, then it must be that $M' \subseteq M$, so $M = M'$, which fulfills the definition of a maximal ideal, as e sought to show.

Note that we took the union of ideals within the proof. In general, if $I, J \subseteq R$, then $I \cup J$ is not an ideal (consider $2\mathbb{Z} \cup 3\mathbb{Z}$). The reason it worked for this proof is because the ideals are in a chain:

$$I_1 \subseteq I_2 \subseteq \cdots \subseteq I_n \subseteq \cdots$$

It should be pointed out that it is possible to have an infinitely long chain of the form $(x_1) \subsetneq (x_1, x_2) \subsetneq \cdots$ of which the union is not a maximal ideal, hence being maximal is a different condition that such a chain terminating in finite itterations, or even countably-infinite many iterations. This is part of the importance of using Zorn's lemma in the proof.

Since Zorn's Lemma was used, there is no way to computationally figure out what the maximal ideal is by just using this proposition. If a ring is also commutative, we can more easily characterize it with the structure of their quotient ring

Proposition 9.4.19: Maximal Ideals in Commutative Rings

Let R be a commutative ring with $1 \neq 0$. The ideal M is a maximal ideal if and only if the quotient R/M is a field

***Proof*:**
This follow's from the Correspondence Theorem and proposition 9.4.10. Let M be a maximal ideal so that there is no proper I such that $M \subsetneq I \subsetneq R$. Then by the Correspondence Theorem, the ideals I containing M correspond bijectively to the ideals of R/M, and since no proper ideal contains M, the only ideals of R/M are (0) and R/M, thus M is maximal if and only if the only ideals of R/M are (0) and R/M. By proposition 9.4.10, a ring is a field if and only if it's only

ideals or (0) and itself. Thus, M is an ideal if and only if R/M is a field, as we sought to sho

The use of the Correspondence theorem in proposition 9.4.19 can be visualized with such a diagram:

$$\begin{tikzcd} R \arrow[d, dash] & R/M \arrow[d, dash] \\ M & 0 \end{tikzcd}$$

Notice that if R was not commutative, then left ideals are not necessarily right ideals, and so the theorem does not apply: if we pick any one-sided ideal (say a left-ideal), and perform proposition 9.4.18 to show the existence of a maximal left-ideal (and hence, we have a non-empty collection of maximal left-ideals), it is not guarnateed that the set of maximal left-ideals will contain any proper right-ideals[21]. A different approach must be taken to study R/I when R is not commutative and I is a maximal left/right ideal, which will lead to many fruitful results in part X. This case will require some more module theory, and so will be left alone for now.

In the commutative case, this let's us study properties of R by studying the field R/M for all maximal ideal M. If R has a single maximal ideal (say $\mathfrak{m}$) then $R/\mathfrak{m}$ is called the *residue field*. An example of a local ring would be $\mathbb{Z}/4\mathbb{Z}$, with single maximal ideal $\{\overline{0}, \overline{2}\}$. Such are called local rings for a reason; they show up intensively in commutative algebra, and will be central in chapter 35 in part V.

This last characterization of maximal ideals through quotienting for commutative rings will come to be very useful in the construction of some fields. For example this will be used to construct all finite fields by taking quotient of the ring $\mathbb{Z}[x]$ by maximal ideals. In a more specific case, we will be construction the "extension" of a field k by first defining a polynomial ring $k[x]$, and quotienting it by a maximal ideal $M \subseteq k[x]$. This will be the foundation of *field Theory*, and will be covered in part IV

Example 9.4.20

1. We've hand-waved a bit before how $p\mathbb{Z}$ is a maximal field. Since $\mathbb{Z}$ is commutative, we can now rigorously state this by noticing that $\mathbb{Z}/p\mathbb{Z}$ is a field
2. The ideal $(2, x)$ is a *maximal* in $\mathbb{Z}[x]$, since $\mathbb{Z}[x]/(2, x) \cong \mathbb{Z}/2\mathbb{Z}$, and $\mathbb{Z}/2\mathbb{Z}$ is a field
3. Let $R = \mathcal{F}([0, 1], \mathbb{R})$ be the ring of all functions from $[0, 1]$ to $\mathbb{R}$, and let M_a be the kernel of evaluation at a ($M_a = \{x \in [0, 1] | f(x) = a\}$). Show that M_a is maximal.

There are two more cool examples I'll get to when I see their importance

Note that a ring can have multiple maximal ideals. For example, $\mathbb{Z} \times \mathbb{Z}$ is a ring, and $I = ((1, 0), (0, 2))$ is an ideal generated by $(1, 0)$ and $(0, 2)$. Then $(\mathbb{Z} \times \mathbb{Z})/I \cong \mathbb{Z}/2\mathbb{Z}$, meaning I is a maximal ideal. Similarly, $J = ((2, 0), (0, 1))$ is an ideal and $(\mathbb{Z} \times \mathbb{Z})/J \cong \mathbb{Z}/2\mathbb{Z}$. Thus, both I and J are maximal. Note too that the two quotients need not be isomorphic (see exercise ref:HERE, thinking of $\mathbb{Z}$, with the fact that $\mathbb{Z}/3\mathbb{Z}$ and $\mathbb{Z}/5\mathbb{Z}$ are not isomorphic).

[21] Indeed, we can modify example 9.4.12 to see that $M_2(k)/(0)$ is not a field, and hence a counter-example

9.4.4 Prime Ideals

The first definition given of prime numbers is that a number is prime if and only if it's only divisors are 1 and itself, that is the only numbers such that $a|p$ are 1 and p. This definition works well for integers, but does it generalize well when we try to talk about divisibility over more general rings. This is due to a sneaky "global" assumption when working with primes, namely that an element is prime if *any* factorization that may include the prime element *has* to include the element as a factor. The following example should illustrate how this isn't necessarily the case:

Example 9.4.21: The Need For Globalness

[The Need For Globalness] Take $R = \mathbb{Z}[2i]$. Then $4 = (-2i)(-2i)$, and $-2i$ can no longer be factor[a]. However, we also have $4 = 2 \cdot 2$, and 2 can no longer factor (which in this case can be verified by taking a few cases). Thus, even though 2 was a factor of 4, we may split 4 differently, showing that there are "local" factorizations.

[a] up to multiplying by a unit, but factorizaiton is only defined up to unt, think $2 \cdot 3 = (-2) \cdot (-3)$

This subtly becomes very important, and shall be explored in depth in chapter 10.

To capture this idea, consider a prime number p. Then p is a prime number and if p divides ab, $a, b \in \mathbb{Z}$, then p divides a or p divides b. For example, if $3|15$, then either $3|3$ or $3|15$. If p was not a prime number, this is not the case: $8|16[= 4 \cdot 4]$, but $8 \nmid 4$ and $8 \nmid 4$. The prime ideals for the integers are the sets that contain all the multiples of a given prime number

Definition 9.4.22: Prime Ideal

Assume R is commutative. An ideal P is called a ***prime ideal*** if $P \neq R$ and whenver the product ab of two element $a, b \in R$ is an element of P, then at least one of the a an b is an element of P.

Since R is commutative, another way of saying the condition is that if A, B where ideals and $AB \subseteq P$, then $A \subseteq P$ or $B \subseteq P$. The non-commutative case does not generalize well, however we discuss a remedy for this case in chapter 50[22].

Example 9.4.23: Prime Ideals

The origins of the idea come from generalizing the notion of primes in $\mathbb{Z}$, so we'll do the same. The prime numbers p generate the prime-ideals (p).

Let's say $ab \in (p)$, meaning $ab = rp$ for some $r \in \mathbb{Z}$ (note that $\mathbb{R}$ is commutative so $(p) = Rp$). Since $ab = rp$ that impleis that $p|ab$ (p divides ab): If you think about what it means for a number x to divide y ($x|y$), then that means there exists another number a such that $ax = y$. For example $3|6$, so $2\dot{3} = 6$.

Now, since $p|ab$, then $p|a$ or $p|b$. If $p \nmid a$ and $p \nmid b$, then ab wouldn't have it as a prime number under their factorization, but then $p \nmid ab$ – a contradiction. So, $p|a$ or $p|b$. But that would mean $r_a p = a$ or $r_b p = b$ for appropriate r_a or r_b. But then, $a \in (p)$ or $b \in (p)$.

Here's a concrete exmpale. Take $3\mathbb{Z}$, an ideal of $\mathbb{Z}$. (can think of $\varphi : \mathbb{Z} \to \mathbb{Z}/3\mathbb{Z}$, $\varphi(x) = [x]$ to convince yourself it's an ideal). Now let's say $ab \in 3\mathbb{Z}$. That means that $3|ab$, meaning either $3|a$ or $3|b$. In either case, that would imply $a \in 3\mathbb{Z}$ or $b \in 3\mathbb{Z}$, showing that $3\mathbb{Z}$ is a prime ideal. On

[22] namely, the ideal group is abelian for certain non-abelian

the other hand, if we have $6\mathbb{Z}$, Then $2 \cdot 3 \in 6\mathbb{Z}$, but $2 \notin 6\mathbb{Z}$ and $3 \notin 6\mathbb{Z}$, meaning $6\mathbb{Z}$ is *not* a prime ideal.

Note that this is an *inclusvie* or, not exlusive. If $p = 2$, and $a = b = 2$, then $2|4$, and both $2|2$ and $2|2$.

Another way I like to think about is is that we already know that if $a \in I$ and $b \in I$, then $ab \in I$. However, it's not clear the converse is true $ab \in I$, $a \in I$ and $b \in I$. This condition is a little too strong: Since $1 \cdot x \in I$ then $1 \in I$, meaning $I = R$. So what if we loosen the condition a bit , and say $ab \in I$ then $a \in I$ *or* $b \in I$. In other words, we can to a limited degree work backwards from *any* combination of elements $ab \in I$ to find one of them keeps needing to be in I. For example, if $abc \in I$ (say we braketed $(ab)c$) then either $ab \in I$ or $c \in I$. Since $ab \in I$, then either $a \in I$ or $b \in I$. This idea will come back in section 10.1, when we talk about factorizing elements in an integral domain, and will be the key idea to a couple of proofs[23].

(Is all I said earlier a fancy way to say that prime ideals are multiplicatively closed?)

This interpretation has some validity if we consdider the following scenario: Take a ring R, and a prime element $p \in R$. Then let's say there was an element $r \in R$ such that $r|p$. This would imply that $p = rr'$ for some $r' \in R$. Since the two values are the same $(rr') = (p)$, implying $rr' \in (p)$. Here, we get to use the property of p being prime: either $r' \in (p)$ or $r \in (p)$, meaning $p|r$ or $p|r'$. If $p|r$, then we have $r|p$ and $p|r$, meaning $p = r$. Be careful here, this is not always true. If r, r' where unit, then we always have $r|r'$ and $r'|r$, but not awlays $r = r'$. For example, in $\mathbb{R}$, $1|2$, $2|1$ but $1 \neq 2$. However, it works here because HERE. So, we are forced to say $r' = u$, where u is a unit. For the othe way around, HERE.

Take $\mathbb{Z}$ again and let's say I was an ideal with this condition. then for some $x \in I$, $px \in I$ for every prime p. By our condition, $p \in I$. This will hold true for every prime-number, meaning I now has all but the identity. I'm not sure if this will always happen, but it's worth pondering.

Another thought: Take (x) and consider $ab \in (x)$. Then $ab = cx$ for appropriate $c \in R$. Then we no get the question $cx \in (x)$ too. This though is "obvious" since $x \in (x)$. But ab is not obvious. So the "representation" of the value that both ab and cx represents is important: it's a question about hether the constituent components of a represetnation is in there (a or b or c or x)

Here are some more example of prime ideals:

Example 9.4.24: Further Examples of Prime Ideals

1. Take $C([0,1])$, that is, all continuous function to/from $[0,1]$. Let $I \subseteq C([0,1])$ be all polynomials with roots at $\frac{1}{2}$ and $\frac{1}{3}$ ($f(\frac{1}{2}) = f(\frac{1}{3}) = 0$). Is I an ideal? Is it a prime ideal? Hint [a].

2. As an exercise to show that you can no longer "go backards", prove that if R is a commutative ring, and P is a prime ideal with no zero-divisors, then R is an integral domain. (Hint [b]).

[a] Think about two polynomials with single roots such that when multiplied, you get an element of I

[b] Assume there is a zero-divisor

This last example is important enough to be referenced many times, and so we make it a proposition:

[23] for example, proposition 10.1.32 and ref:HERE

Proposition 9.4.25: Prime Ideals And Integral Domains

Assume R is commmutative. Then the ideal P is a prime ideal in R if and only if the quotient ring R/P is an integral domain

Corollary 9.4.26: Maximal Ideal Then Prime Ideal

Assume R is commutative. Then every maximal ideal of R is a prime ideal

***Proof*:**
If M is a maximal idea, then by proposition 9.4.19 R/M is a field. A field is an integral domain, so the corollary folllows from proposition 9.4.25

Example 9.4.27

1. The principal ideals generated by (p) where p is prime are both prime and maximal. Not all prime ideals are maximal: (0) is a prime ideal, but not maximal.
2. The ideal (x) is a prime ideal in $\mathbb{Z}[x]$ since $\mathbb{Z}[x]/(x) \cong \mathbb{Z}$, but not maximal since $\mathbb{Z}$ is not a field. Or, you can diretly see that $(x) \subseteq (2, x)$, meaning that (x) is not maximal.

Exercise 9.4.2

1. Let R, S be rings and $R \times S$ be the product ring. What are the prime ideals of $R \times S$(answer: $R \times P$ and $Q \times S$ for primes P and Q. In general, only one component can be the ideal)
2. Let $a, b \subseteq R$ be ideals. Show that $(a + b)(a \cap b) \subseteq ab$ and $ab \subseteq a \cap b$. Show that if $(a, b) = (1)$, then $a \cap b = ab$.
3. exercise on quotient ideals $(a : b)$
4. Let $f : R \to S$ be a ring homomorphism and $p \subseteq S$ a prime ideal. Show that $f^{-1}(p)$ is a prime ideal. If $m \subseteq S$ is maximal, is $f^{-1}(m)$ maximal? (Consider $R = \mathbb{Z}$ and $S = \mathbb{Q}$)
5. Perhaps, if possible at this stage how if if $a_1, a_2, ..., a_n$ are ideals and $\cap_i a_i$ is contained in the prime ideal p, then p contains one of the a_i, and if $p = \cap_i a_i$, then $p = a_i$ for on of them (shows how prime genelize numbers, so to speak).
6. Notice that we have defined what I^k is. Let's say we started with an ideal J; is there a rasonable way of defining $\sqrt{J}$? Define $\sqrt{J} = \{a \in R \mid \exists n \in \mathbb{N}_{>0},\ a^n \in J\}$. Show that $\sqrt{J}$ is an ideal.

9.5 Ring of Fractions: Localization

In this section, we'll take a moment to discuss how we can take any ring with no zero divisors, and extend the ring to one in which every element has a multiplicative inverse! More generally, given any non zero-divisor element $d \in R$ (where R will have to be commutative), we can construct a ring

R_d which will where $d^{-1} \in R_d$. This concept was applied in the field of *algebraic geometry* for the purposes of looking at a ring "locally". What this means and how to think about this consruction "geometrically" will be discussed much later in part V in section 23.4. In this section, the full generality of this construction is not particularly useful. What's more interesting at the moment is that this gives us a way to add units to our ring!

We'll start with a practical example by using $\mathbb{Z}$ to construct $\mathbb{Q}$! Taking $\mathbb{Z}$, we can make an equivalence class

$$(a, b) \sim (c, d) \Leftrightarrow ad = bc$$

where $b, c \neq 0$. Then $[(a, b)]$ represents a fraction $\frac{a}{b}$, with the point of the equivalence relation is to make all fractions with similar divisors be the same:

$$\frac{1}{2} = \frac{2}{4} = \frac{3}{6} = \cdots$$

I'll stick with fraction notation for these equivalence relationships. We can see that we can define

$$\frac{a}{b} + \frac{c}{d} = \frac{ad + bc}{bd} \qquad \text{and} \qquad \frac{a}{b} \cdot \frac{c}{d} = \frac{ac}{bd}$$

It should be checked that these operations are independent of representative. With this construction, we can identity $\mathbb{Z}$ with $\left\{\frac{a}{1} | a \in \mathbb{Z}\right\}$ [24], showing how $\mathbb{Z}$ is a sub-ring of $\mathbb{Q}$.

With that intuition, we can ask about general commutative rings R. The problem comes when we allow $d = 0$ or be a zero divisor. Say $d \neq 0$ $bd = 0$. Then:

$$d = \frac{d}{1} = \frac{bd}{b} = \frac{0}{b} = 0$$

We'll avoid this by taking a set $R^\times$ which represents all the units of R

Theorem 9.5.1: Ring of Fractions

Let R be a commutative ring R (not necessarily with identity). Let D be any nonempty subset of R that does not contain 0, does not contain any zero-divisors and is closed under multiplication (i.e. $ab \in D$, for every $a, b \in D$)[a]. Then there is a commutative ring Q with 1 such that Q contains R as as a subring and every element of D is a unit in Q. The ring Q has the following additional properties

1. Every element of Q is of the form rd^{-1} for some $r \in R$ nd $d \in D$. In particualr if $D = R - \{0\}$ then Q is a field
2. (uniqueness of Q) The ring Q is the "smallest" ring containing R in which al elements of D become units. That is, if S is any commutative ring with identity and let $\varphi : R \to S$ be any injective ring homomorphism such that $\varphi(d)$ is a unit in S for every $d \in D$. Then there is an injective homomorphism $\Phi : Q \to S$ such that $\Phi|_R = \varphi$. In other words, any ring containing an isomorphic copy of R in which which all the elements of D become units must also contain an isomrophic copy of Q.

[a] such a set is called a *semi-group*

[24] meaning construct a isomorphism

***Proof*:**
We'll start like we started with $\mathbb{Z}$. Let $F = \{(r,d)|r \in R, d \in D\}$ (d in the 2nd component because we don't want zero-divisors in the bottom, and we need the denominator to be closed to define multiplication). Then, we define the equivalence relation

$$(r,d) \sim (s,e) \text{ if and only if } re = sd$$

It is easy to show that $\sim$ is an equivalence relation [a]
We commonly denote (r,d) by $\frac{r}{d}$. Now, let

$$Q = \left\{[(a,b)]\middle| a \in R, d \in D\right\}$$

be the set of equivalence relations. Note that $\frac{r}{d} = \frac{re}{de}$ for any $e \in D$, as we would expect.
We can define addition and multiplication as

$$\begin{aligned} [(a,b)] + [(c,d)] &= [(ad+bc, bd)] &\Rightarrow&\quad \frac{a}{b} + \frac{c}{d} = \frac{ad+bc}{bd} \\ [(a,b)] \cdot [(c,d)] &= [(ac, bd)] &\Rightarrow&\quad \frac{a}{b} \cdot \frac{c}{d} = \frac{ac}{bd} \end{aligned}$$

Then the following properties need to be checked. I'll show that addition is well-defined, and the rest is not hard to show that it follows:

1. The operations are well-defined (not dependent on representatives)

   Proof. We'll show addition is well-defined. Let

   $$\frac{a}{b} = \frac{a'}{b'}\ (ab' = a'd) \qquad \frac{c}{d} = \frac{c'}{d'}\ (c'd = cd')$$

   then

   $$\frac{ad+bc}{bd} = \frac{a'd'+b'c'}{b'd'}$$

   which equivalently means:

   $$\begin{aligned} (ad+bc)(b'd') &= adb'd' + bcb'd' && \text{distributivity in } R \\ &= ab'dd' + cd'bb' && \text{commutativity in } R \\ &= ab'dd' + cd'bb' && ab' = a'b,\ cd' = c'd \\ &= (a'd' + b'c')(bd) \end{aligned}$$

   thus, the two are equivalent. A similar procedure works for multiplication ☐

2. Q is an abliean group under addition, where the aditive identity is $\frac{0}{d}$ for any $d \in D$ and the additive inverse of $\frac{a}{d}$ is $\frac{-a}{d}$
3. Multiplication is associative, distributive, commutative,
4. Q has an identity $\frac{d}{d}$ for any $d \in D$

Next, we show that R can indeed be embedded into Q. Let

$$i : R \to Q \qquad r \mapsto \frac{rd}{d}$$

Since $\frac{rd}{d} = \frac{re}{e}$, this function is independent of choice of d. Since d is closed under multiplication:

$$i(rs) = \frac{rsd}{d} = \frac{rs}{1} = \frac{r}{1}\frac{s}{1} = \frac{rd}{d}\frac{sd}{d} = i(r)i(s)$$

finally, for injectivity of the inclusion, notice that

$$i(r) = 0 \Leftrightarrow \frac{rd}{d} = \frac{0}{d} \Leftrightarrow rd^2 = 0 \Leftrightarrow r = 0$$

meaning the kernel is trivial, and i is injective.

Next, we show that every element $q \in Q$ can be written as rd^{-1}. Since $d = \frac{d}{1}$, then $d^{-1} = \frac{1}{d}$ since $\frac{d}{1}\frac{1}{d} = \frac{1}{1}$. Then by definition of an element in Q, we define an elemnent as $\frac{r}{d} \in Q$, and so we have that $q = rd^{-1}$.

Note that if $D = R - \{0\}$, then every element has a multiplicative inverse, and so that makes Q into a field.

Finally, we must show this extension is unique. To do this, let's say $\varphi : R \to S$ is an injective ring-homomorphism such that $\varphi(d)$ is a unit in S. In this way, S acts as "another" extension of R (since we shows that R can be imbedded into Q). Since $q = rd^{-1}$ for every $q \in Q$, extend φ to Φ by definining

$$\Phi(rd^{-1}) = \varphi(s)\varphi(d)^{-1}$$

for all $r, d \in R$. Note that $\varphi(d)^{-1}$ is defined since $\varphi(d)$ is a unit in S. This map is well-defined. If we have $rd^{-1} = re^{-1}$, then multiplying around we get $re = rs$, so by the property of ring homomrphisms $\varphi(r)\varphi(e) = \varphi(s)\varphi(d)$ and so

$$\Phi(rd^{-1}) = \varphi(r)\varphi(d)^{-1} = \varphi(s)\varphi(e)^{-1} = \Phi(se^{-1})$$

showing it is indeed well-defined. Checking Φ is a ring homomorphism is strait forward using commutativity of Q.

Last step is to show Φ is injective. There is a sneaky trick you can do to pull this off. Let's say $rd^{-1} \in \ker(\Phi)$. Then that means that $r \in \ker(\varphi)$ since we'd need $\Phi(rd^{-1}) = \varphi(r)\varphi(d)^{-1} = 0$ and $\varphi(d)^{-1}$ cannot be zero since d is a unit. However, $\ker(\varphi)$ is trivial, so $r = 0$, but then $rd^{-1} = 0d^{-1} = 0$, meaning Φ is also injective! Thus, we forced the S to have Q embedded inside of it, meaning *any* extension of R to S would inevitably have Q. This makes Q the smallest extension of R and the unique extension of R.

[a] Reflexivity and symmetry come immediately. For transitivity, combining the two equations appropriately and realise you can factor by $d \in D$, which is not zero or a zero-divisor, which will force the equation you want to be zero, forcing the equivalence for transitivity

We'll give Q a name:

Definition 9.5.2: Ring Of Fraction

Let R be a commutative ring, D be the appropriate subset and Q be the extension as in the above theorem. Then

1. The ring Q is called the *ring of fractions* of D with respect to R and is dneoted $D^{-1}R$
2. If R is an integral domain an $D = R - \{0\}$, Q is called the *field of fractions* or *quotient field* of R.

we often want all units of a ring R. We denote all units as $\mathbb{R}^{\times}$.

Here are some examples of ring of fractions

Example 9.5.3

1. the trivial example would a a field $\mathbb{F}$, which would be it's own field of fractions.
2. the construction we've done at the beginning is a special case of our construction. So, if we take $R = \mathbb{Z}$, and take $D = \mathbb{Z} - \{0\}$, then $D^{-1}\mathbb{Z} = \mathbb{Q}$
3. Let $R = \mathbb{Z}$ and take $D = \{1\}$. Then the quotient ring would be $\mathbb{Z}$ again!
4. $R = \mathbb{Z}$, and take $D = \left\{19, 19^2, ...\right\}$. This is a valid D, since it doesn't contain 0, doesn't contain zero divisors, and is closed under multiplicationso what is $D^{-1}\mathbb{Z}$?
5. The ring $2\mathbb{Z}$ is alsmot an integral domain; it doens't have an identity. However, $(R^{\times})^{-1}(2\mathbb{Z}) = \mathbb{Q}$ as well! If you think about it, being able to divide will give us the identity, as well as $\frac{1}{2}$, which will produce the rest
6. $Z[p]$ should be elaborated on
7. $\mathbb{F}[x]$ becomes $\mathbb{F}(x)$, that is, elements of the form $\frac{p(x)}{q(x)}$ where $q(x)$ is not the zero-polynomial (Since $\mathbb{F}[x]$ is an integral domain, it's the only value we would need to worry about). Notice the square-brakets became parenthesis. In this way, we might see the motivation to write $\mathbb{Q}(\sqrt{D})$, since we are allowed to divide those values as well! The difference here being that $x, x^2, x^3, ..., x^n, ...$ are all distinct elements, non of them being an element of their original field. However, this same chain doens't appear for $Q(\sqrt{D})$: $\sqrt{D}, \sqrt{D}^2 = D$. This is because $\sqrt{D}$ is not an arbitrary indeterminate value, giving us this relation, similar to what we saw with group-rings.
8. Take $\mathbb{Z}[i]$ and $D = R^{\times} = R - \{0\}$. Then the ring of fractions is $\mathbb{Q}[i]$! Notice it's not something like $\mathbb{Z}(i)$. This comes down to how multiplication is defined in this ring, which is much more akin to multiplying complex numbers.
9. what is the field of fractions of $\mathbb{Q}[\sqrt{2}]$?
10. What if we have $R = \mathbb{Z}/4\mathbb{Z}$ and $D = \{1, 3\}$?

Theorem 9.5.4: Universal Property Of Field Of Fractions

If $h : R \to F$ is an injective homomorphism from R into a field F, then there exists a unique ring homomorphism $g : Q \to F$ which extends h

***Proof*:**
Use the uniqueness of the extension to show that $\Phi(Q)$ will be in any sub-field.

The universal property on $D^{-1}R$ will in fact still work if we generalize our construction to letting D have zero-divisors. The motivation for this generalization is something that (ref:HERE TBD). This topic will be exlpored in part V, section 23.4.

(Another perspective would be that we are giving a solution to all equations of the form $bx - a = 0$, for $a, b \in R$. Thus, for $R = \mathbb{Z}$, we are doing:

$$\frac{\mathbb{Z}[x_{1/2}, x_{1/3}, ..., x_{1/p}, ...]}{(2x_{1/2} - 1, 3x_{1/3} - 1, ...,)} =: \mathbb{Q}$$

Though this way of looking at it seems more complicated, it fits better with the idea that we are adding relations to $\mathbb{Z}$. In particular, it shows that we are adding all *linear* relations (i.e. solutions) to $\mathbb{Z}$. We will in later section's show how we add non-linear solution.

Note that we construct $\mathbb{Z}$ from $\mathbb{N}$ by adding all monomial-linear polynomials. We then construct $\mathbb{Q}$ by taking all linear equations. We shall further explore in *Commutative Algebra* [10] how to think of non-monic polynomials with non-unit leading coefficients.)

9.5.5 Foreshadowing By taking the "infinite" version of polynomials, we managed to make all the non-zero-divisors become units. In this section, another way of producing using by formally inverting them. These two processes shall seem very different, however they are related, and the connection will be made in section 25.

9.5.1 Constructing $\mathbb{R}$?

In ref:HERE, we mentioned how we can think of $\mathbb{N}$ being a free-monoid on 1 element (say x), so in a sense, it was constructed from 1 element and the rules of a monoid. We then showed in ref:HERE how we can construct $\mathbb{Z}$ from $\mathbb{N}$ by adding the condition that $x^2 = x$ (i.e. $1 \cdot 1 = 1$) so that x is the multiplicative identity. We just showed how we can construct $\mathbb{Q}$ from $\mathbb{Z}$. We have also give an exercise that constructs $\mathbb{C}$ from $\mathbb{R}$. The reader might feel like the natural next step up would be to construct $\mathbb{R}$ from $\mathbb{Q}$. There is in fact a few constructions of the real numbers (two popular ones are the *Dedekind cut's construction* and the *Cauchy-limit construction*). However, it turns out that there is no "algebraic" construction of the real numbers! That is, we can't start off with some set (say $\mathbb{Q}$) and extended using binary (more generally n-ary) operations with certain rules on creating new elements from old ones (like we did for $\mathbb{N}$, $\mathbb{Z}$, and $\mathbb{Q}$)! The closest we can get is the *real closure* of $\mathbb{Q}$, denoted $\overline{\mathbb{Q}}^{\mathbb{R}}$ that is covered in section 21.1.

The fact that there isn't an algebraic construction of the reals is a fascinating result, and it will have to wait till we cover ________to fully appreciate. (ref:HERE).

Found this really interesting link!

https://math.stackexchange.com/questions/3761716/
is-there-an-algebraic-way-to-construct-the-reals

It seems to go down fundamentally that an algebraic construction tries to use the elements from current object to construct the bigger object *by means of finite-tuples* (because of the way we think of functions!!), so if we were to try to do $\mathbb{Q} \rightsquigarrow \mathbb{R}$, then it would be representing elements from $\mathbb{R}$ with finite tuples from $\mathbb{Q}$. But $\mathbb{Q}$ is countable, and the number of finite tuples from $\mathbb{Q}$ is countable, but $\mathbb{R}$ has uncountably many elements. Not sure about the subtlies yet, Löwenheim-Skolem's Theorem is beyond my comprehension right now, but I think this idea is good enough for now to appreciate!

Exercise 9.5.1

1. Let F be a field of characteristic 0. Show that there exists a ring R such that $\text{Frac}(R) = F$. (If F is not characteristic 0, then there are a couple of complications. The equivalent exercise is in ref:HERE (under valuations in comm alg, since it requires valuations))

9.6 Chinese Remainder Theorem

Let's say we have a ring R and a collection of ideals $I_1, I_2, ..., I_k$. Then given

$$a_1 \bmod I_1, \cdots, a_k \bmod I_k$$

does there exist a $a \in R$ such that

$$a \equiv a_1 \bmod I_1, \cdots, a \equiv a_k \bmod I_k$$

Think about it for a moment before looking at the answer

9.6.1 Taking a moment aking a moment to think about the answer.

The answer depends on the choice of I_i. The reader should have tried this with $R = \mathbb{Z}$ and noticed it is not always possible, for example if $R = \mathbb{Z}$, $a \equiv 1 \bmod 2$, $a \equiv 2 \bmod 4$ does not have a solution.

The Chinese Remainder Theorem (CRT) is a way of finding an to this question. It's name sake most likely comes from a book *Sun Tzu Suan Ching*. It is said that it was originally developed either for accounting purposes or for dividing troops.

Before we get into the actual theorem, we'll first explore a direct application of Chinese Remainder Theorem:

Example 9.6.2: Concrete Use
simple form

$$\text{if } gcd(m,n) = 1 \text{ and } 0 \le a < m\,, 0 \le b < n$$

then

$$\exists!\ k,\ 0 \le k < mn \text{ such that } k \equiv a \bmod m\,, k \equiv b \bmod n$$

Let's say: $m = 6$, $n = 5$, $a = 4$, $b = 2$. Then $k = 22$.

Proof. We have
$$k \equiv 4 \bmod 6,\ k \equiv 2 \bmod 5$$
so using the classical method, we first find
$$6 \cdot 5 = 30$$
from this, we find $30/6 = 5$ and $30/5 = 6$ to be the values such that and kno that we get the solution by solving the equivalent problem of:
$$k \equiv 5 \cdot 1 \bmod 30,\ k \equiv 6 \cdot 1 \bmod 30$$
meaning the solution is $(4 \cdot 5 \cdot 5) + (2 \cdot 6 \cdot 6) =$ □

When generalizing this result, we can no longer ask for numbers to be co-prime. The equivalent concept in rings are co-maximal ideals [25]

Definition 9.6.3: Comaximal Ideals

The ideals A and B of the ring R are said to be *comaximal* if $A + B = (1)$

The intuition on when to spot something is co-maximal is analogous to two numbers being prime. If $a, b \in \mathbb{Z}$ are prime, then by Bézout's identity, there exists numbers x, y such that $xa + yb = 1$, meaning $1 \in (a) + (b)$, meaning $(a) + (b) = \mathbb{Z}$ by proposition 9.4.10. In the next chapter, we'll generalize the concept of gcd for more general rings, meaning where the gcd can be defined, this theorem has a useful application [26]

Lemma 9.6.4: Characterizing Products of Coprime Ideals

Let $I_1, I_2, ..., I_n$ be coprime ideals. Then:
$$\bigcap_{i=1}^{n} I_i = \prod_{i=1}^{n} I_i$$

***Proof*:**
We shall do induction so consdier first when $n = 2$ so that $I, J \subseteq R$ such that $I + J = (1)$. It is always the case that $IJ \subseteq I \cap J$ so we have to show that $I \cap J \subseteq IJ$ when $I + J = (1)$. Given $I + J = (1)$, there exists $a \in I$, $b \in J$, such that $a + b = 1$. Now, if $r \in I \cap J$, then
$$r = r \cdot 1 = r(a + b) = ra + ra \in IJ$$
Since $r \in J$ and $a \in I$, $ra \in IJ$, an similalry $rb \in IJ$ since $r \in I$ and $b \in J$.

Then the inductive step follows immediately.

[25] why not co-prime ideals? We'll show how prime and maximal ideals will match in special rings in chapter 50, section 50.2, here we would usually apply such a result. However, we stay general since we can prove it in more general contexts

[26] For example, we'll use this theorem when looking at polynomial rings over fields

Lemma 9.6.5: Recursive Comaximal Of Products

Let $I_1, I_2, ..., I_n \subseteq R$ be ideals such that $I_i + I_n = (1)$ for all $i \in \{1, ..., n-1\}$. Then $(I \cdots I_{n-1}) + I_n = (1)$

***Proof*:**
By assumption, there exists $a_i \in I_k$ and $1 - a_k \in I_i$ such that

$$(1 - a_1) \cdots (1 - a_{n-1}) \in I_1 \cdots I_{n-1}$$

And

$$1 - (1 - a_1) \cdots (1 - a_{n-1}) \in I_n$$

since it is a comination of the elements $a_1, a_2, ..., a_{n-1} \in I_n$

Theorem 9.6.6: Chinese Remainder Theorem

Let $A_1, A_2, ..., A_k$ be ideals in R. The map

$$R \to R/A_1 \times \cdots \times R/A_k \qquad \text{defined by} \qquad r \mapsto (r + A_1, \cdots, r + A_k)$$

If for each $i, j \in \{1, 2, ..., k\}$ with $i \neq j$ the ideals A_i and A_j are comaximal, then this map is surjective and so

$$R/(A_1 A_2 \cdots A_k) = R(A_1 \cap A_2 \cap \cdots \cap A_k) \cong R/A_1 \times R/A_2 \times \cdots \times R/A_k$$

***Proof*:**
Take

$$R \to \bigoplus_{i=1}^{n} R/A_i \qquad a \mapsto (a \bmod A_1, ..., a \bmod A_n)$$

Then this is easily checked to be a ring homomorphism with kernel $A = \bigcap_i A_i$. Assuming now the ideals are comaximal, we shall show surjectivity. We shall do a proof by induction. If $n = 1$ where done, so let for let's say

$$R/(A_1 \cdots A_{n-1}) \cong R/A_1 \times \cdots \times R/A_{n-1}$$

Then by lemma 9.6.5, $(A_1 \cdots A_{n-1}) + A_n = (1)$, and hence this is reduced to two ideals. By comaximality there exists $a_1 \in A_1$, $a_2 \in A_2 \cdots A_{n-1}$ such that $1 = a_1 + a_2$. Then writting

$$x = x_2 a_1 + x_1 a_2$$

we get that $x \equiv x_1 \bmod A_1$ and $x \equiv x_2 \bmod A_2 \cdots A_n$. But then $x \mapsto (x_1 \bmod A_1, x_2 \bmod A_2 A_3 \cdots A_{n-1})$, completing the proof.

A useful consequence to remember is that if there are elements $r_i \in R$ where $r_i \bmod A_i$, then there exists an element a

$$a \equiv r_i \bmod A_i$$

The ability to solve such a system mod A_i is sometimes called the *Weak Approximation Theorem*.

Corollary 9.6.7: Chinese Remainder Theorem For Integers

Let $R = \mathbb{Z}$ $a_1, a_2, ..., a_n \in \mathbb{Z}$ be elements such that $\gcd(a_i, a_j) = 1$ for all $i \neq j$. Then if $a = a_1 \cdots a_n$, The function

$$\varphi : \mathbb{Z}/a\mathbb{Z} \to \mathbb{Z}/a_1\mathbb{Z} \times \cdots \times \mathbb{Z}/a_k\mathbb{Z}$$

given in the natural way is an isomorphism

***Proof*:**
This is an immediate consequene of the chinese remainder theorem since $\gcd(a, b) = 1$ if and only if $(a, b) = (a) + (b) = 1$

Exercise 9.6.1

1. Show that if $P + Q = R$, then for any $n, m > 0$, $P^n + Q^m = R$. Conclude the CRT can be generalized to ideals of the form $A_1^{n_1}, ..., A_k^{n_k}$.

9.7⋆ Monoidal Objects

⋆ *Optional. This section is an aside; nothing later in the book depends on it.*

Maybe have this here to be able to introduce R-algebras as the monoidal objects in the R-**Mod** Category?

10

Exploring Integral Domains: Factorization and Valuation

This chapter will assume basic number theory knowledge from the preliminary chapter (section 0.2)

Integral domains are a huge overarching category of rings, being in a meaningful way the generalization of integers. As they do not having zero-divisors makes it possible to talk about partially inverting elements (by proposition 9.1.24). There are many different consequences of this result especially in the right specializations, for example divisibility, size, and even the euclidean algorithm shall all be definable on the appropriate integral domains[1]. These topics all interact greatly with each other and there is no linear way of addressing these ideas. However, books are inherently written with a total ordering and we must learn content in some order, and hence this chapter has been written with the idea of exploration the notion of divisibility (which the reader can think of the best that can be done to check invertibility if an element is not a unit). As this exploration naturally progresses, all other pertinent notions about integral domains shall become natural questions and will be answered.

The following diagram sums up the results of this chapter:

[1]There is also a more geometric interpretation, which is that integral domains will correspond to "irreducible" geometric objects (think of how $R \times S$ is no longer integral. This idea is explored in section 32.3

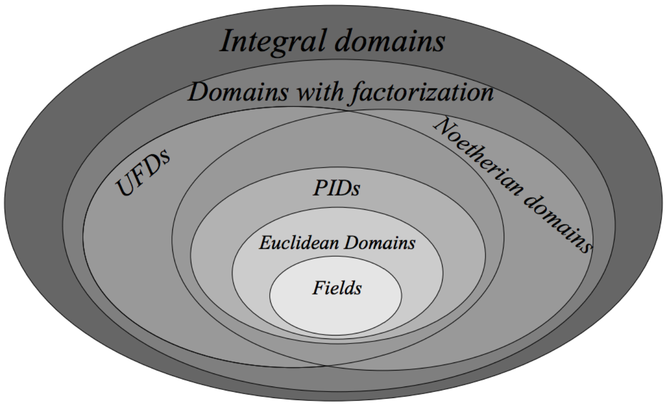


Figure 10.1: Image form Alufi's textbook

At the end of this chapter, we shall explore how these abstract notions were in fact developed to solve number-theoretic problems. We shall see our first example of finding the solution to a Diophantine equation, namely

$$x^2 + y^2 = p$$

In other words, which primes can be written as sums of two squares. As equations are the fundamental way in which we determine the properties of objects, this will be the first glimpse into the properties of prime numbers. This exploration is greatly generalized in chatper 50, section 50.4.6.

10.0.1 Note Some results in this chapter apply beyond integral domains. It will be made extra-clear when that is.

Goal

In fields, every element has an inverse. This is not so for general rings, as we have seen when studying units and zero-divisors. If not every element is a unit, there is a new idea that is introduced when learning some basic number theory: *factorization*. In this way, we can "decompose" an element into some basic building blocks. This idea of factorizing is one the reader is already familiar with: any number $x \in \mathbb{Z}$ can be factored, even uniquely, into prime numbers:

$$x = u p_1^{k_1} p_2^{k_2} \cdots p_n^{k_n}$$

where $u \in \{1, -1\}$ and each p_i is a prime. Is there a generelizable notion of being factorizable in more general rings? Is this idea related to prime ideals? Some types of rings do admit an idea of factorization, or even some stronger conditions (like the euclidean algorithm being well-defined, or the factorization being unique), while other rings have a weaker condition (like the factorization not being unique or even existing). The first major focus of this chapter are these ideas.

The overarching idea in factorization is centered around the idea of divisibility like we have seen in $\mathbb{Z}$. For example:

$$2|8 \qquad 32|17728 \qquad \text{but} \qquad 3 \not| 8 \qquad 5 \not| 7$$

When an element is divisible, let's say $2\big|8$, then there exists some $x \in \mathbb{Z}$ such that $2x = 8$, that is, 8 got subdivided into other elements; in this way, factorization can be thought of as a "partial division" or "partial inverse" condition. This idea will be at the heart of our following exploration.

The result will give 3 new classes of rings and develop on 2 previously mentioned classes or rings which are very often studied.

1. *Atomic Domains* (or Domains with Factorization) are will be rings in which element can be factored into elements that can no longer be factored (they will be called irreducible elements), though not necessarily uniquely
2. *Noetherian domains* are similar to Unique Factorization domain as in every element can be factorzied, but it need not be unique. The reader may recall these as rings in which every ideal may be finitely generated.
3. *Unique Factorization Domains* (UFDs) will allow for every element to have a unique factorization, and let's us define the notion of there being a greatest common divisor between elements.
4. *Principal Ideal Domains* (PIDs) will be stronger than PIDs in that they allow for unique factorization and will have a more rigid structure. The reader may recall these as rings in which every ideal may be generated by a single element.
5. *Euclidean Domains* will have the property of the euclidean division algorithm applying to them to quickly find the greatest common divisor. These will allow for algorithmic decomposition of elements into prime elements.

Since factorization is heavily inspired by the properties of $\mathbb{Z}$, all rings in this chapter will be *integral domains* unless specified otherwise.

10.1 Factorization

We will first define more rigorously the notion of elements being factorizable

Definition 10.1.1: Divisibility

Let R be a commutative ring and $a, b \in R$ be elements of R. Then we say that a *divides* b (or b *is divisible by* a.) if there exists an element c such that

$$ca = b$$

Equivalently, $ca = b$ if and only if $b \in (a)$. Notationly, we write $a\big|b$ to mean "a divides b".

If R was not commutative, we can define left and right divisibility, though we do not cover the non-commutative case (see exercise ref:HERE for some exploration of the concept). The opposite of a being divisible by b is b being a *multiple* of a – this terminology is used much less often.

The equivalent condition with ideals are very insightful way of understanding divisibility. Here are the 4 equivlent condition's of divisibility for ease of reference:

1. $b|a$ if and only if

2. $a = bb'$ if and only if
3. $a \in (b)$ if and only if
4. $(a) \subseteq (b)$

It is unfortunate that the notation $b|a$ stuck, since you will have to swap at some point the order of the a and the b given that it's read "b divides a" and not "a is divided by b". It's a matter of getting used to.

The notation for divisibility using ideals is particularly insightful here; notice that if $u \in R$ is a unit, then for any element $r \in R$ $(r) = (ur)$. Both inclusion are easy and should be proven before continuing to read. Because of this property, it is immediate that $a\big|b$ if and only if $a\big|ub$, that is, if an element is divided (or not divided) by another, then multiplying it by a unit let's it remain divisible (or not divisible) by the other element. This is in fact something the reader is already is familiar with; notice that $6 = -1 \cdot -6$. Then $2\big|6$, and $2\big|-6$, but also $5 \nmid 6$ and $5 \nmid -6$. Thus, when we think of divisibility, it is important to think of a number being divisible "up to a unit":

Definition 10.1.2: Associative

Let R be a commutative ring and let $a, b \in R$ be two elements. Then we say that a and b are *associates* if the ideals generated by a and b are equal, that is:

$$(a) = (b)$$

It is important to check that this is equivalent to saying that a and b are associates if $a\big|b$ and $b\big|a$, showing how this links to divisibility. Notice that in the previous definition, R was not an integral domain, but just a commutative ring, which allows zero-divisors. This in fact breaks the intuition of associativity we just showed:

Example 10.1.3: associativity in non-integral domains

Let $R = \mathbb{Z}/6\mathbb{Z}$. Then $(\overline{2}) = (\overline{4})$, since

$$(\overline{2}) = \{\overline{2}, \overline{4}, \overline{0}\} = \{\overline{4}, \overline{2}, \overline{0}\} = (4)$$

Furthermore, $\overline{4} = \overline{2} \cdot \overline{2}$ but $\overline{2}$ is *not* a unit.

For this reason, restricting to integral domains let's us keep this "up to a unit" intuition for for divisibility using ideals:

Proposition 10.1.4: Associativity in Integral Domains

Let R be an integral domain. Then a and b are associates if and only if $a = ub$ for all units $u \in R$

***Proof*:**

Let $(a) = (b)$, so that for some $r, s \in R$, $ra = b$ and $a = sb$. Then

$$sra = sb = a$$

and since the cancellation property holds for integral domains, we have $sr = 1$, showing that r

and s are units.

Conversely, if $a = ub$ for all units $u \in R$, then it's clear that $(a) \subseteq (b)$ since $u \cdot b = a$ and $(a) \subseteq (b)$ since $u^{-1} \cdot a = b$.

Incidentally, notice that in example 10.1.3, the units of $\mathbb{Z}/6\mathbb{Z}$ is only $\overline{5}$ and that $\overline{2} \cdot \overline{5} = \overline{4}$, meaning the previous proposition actually holds for $\mathbb{Z}/6\mathbb{Z}$. This is true for a special subset of commutative rings usually called *rings with harmless zero-divisors* (harmless since the intuition of divisibility is preserved). These are not covered in this book, but can be explored in (reference something: here). Since the intuition for associativity holds true for all integral domains, and most work being done which includes divisibility in rings focuses on integral domains (due to the cancellation property), we shall do the same restriction here and work over integral domains rather than generalize the following proofs into much more (arguably needlessly) complex arguments to account for the case of rings with harmless zero-divisors.

With a notion of divisibility up to unit established, we can start thinking of what are the building blocks, the "fundamental elements", into which we will factorize an element:

Definition 10.1.5: Irreducible and Prime Elements

Let R be an intergral domain

1. Let $r \in R$ be a nonzero element that is not a unit. Then r is called *irreducible* in R if whenever $r = ab$, at least one of a or b must be a unit in R. Otherwise, r is said to be *reducible*.

2. The nonzero element $p \in R$ is called *prime* in R if the ideal (p) generated by p is a prime ideal. In other words, a nonzero element p is a prime if it is not a unit and whenver $p|ab$ for any $a, b \in R$ then either $p|a$ or $p|b$.

Example 10.1.6: Irreducible and Prime elements

Let $R = \mathbb{Z}$, and take $6 \in \mathbb{Z}$. Then we can see that $2 \cdot 3 = 6$. Since 2 and 3 are both not units, that means that 6 is a reducible element in $\mathbb{Z}$. On the otherhand, 2 and 3 are not reducible elements. If there existed a a, b such that $2 = ab$, then by our knowledge that 2 is a prime number (or by some elementary number theory argument), the only number that divide it are 1 and 2, so it must be that either

$$1 \cdot 2 = 2 \quad \text{or} \quad -1 \cdot -2 = 2$$

In both cases, a is a unit, and so 2 is irreducible (similarly for 3). Furthermore, 2 and -2 are associates, since $2 = -1 \cdot -2$, i.e., the differ by a unit. in $\mathbb{Z}$, every n are associates to it's negative $-n$.

As pointed out in example 9.4.23, 2 is also a prime element since (2) is a prime ideal. In fact, in the previous exercise, we used the fact that 2 is a prime integer (which corresponds to the prime ideals of $\mathbb{Z}$) in order to show it's irreducible. Though 2 is both an irreducible and a prime element in $\mathbb{Z}$, it need not be the case in general integral domains that an element is both!

As an example, take $\mathbb{F}[x]$ to be the polynomials over a field, and take $F_{0x} \subseteq \mathbb{F}[x]$ to be all the polynomials such that the coefficient of x is zero. As an example of elements in F_{0x}:

$$ax^3 + b \in F_{0x} \qquad cx^3 + dx \notin F_{0x} \qquad \forall a, b, c, d \in \mathbb{F}$$

It is easy to check that F_{0x} is an integral domain (as you should check), and so we can try to find irreducible and prime elements. Take x^6. It has two factorizations: $x^6 = x^2x^2x^2$ and $x^6 = x^3x^3$. The elements x^2 and x^3 are irreducible, but not prime.

To show x^2 and x^3 are irreducible, let's say

$$x^2 = p(x)q(x)$$

then since we're working with polynomials over an integral domain, by proposition 9.2.3:

$$\deg(x^2) = \deg(p(x)) + \deg(q(x))$$

meaning $2 = \deg(p(x)) + \deg(q(x))$. Since both $p(x)$ and $q(x)$ cannot be one (since $ax \notin F_{0x}$, for all $a \in \mathbb{F}$) then one of them has to be of degree zero, but since x^2 is monic this implies it must be 1, meaning it's a unit, meaning x^2 is irreducible. Similarly for x^3. Notice that this means that x^6 had two decompositions into irreducible factors, meaning the decomposition was not unique.

Now, take (x^2), i.e. the ideal generated by x^2. Notice that $(x^3)(x^3) \in (x^2)$ because $x^4 \cdot x^2 \in (x^2)$. However, $x^3 \notin (x^2)$. If it were, then there would exist a polynomal $p(x)$ such that $p(x)x^2 = x^3$. By the degree formula, $p(x)$ would have to be of degree 1. However, no degree 1 polynomial exists in F_{0x}. Therefore, (x^2) is not prime! This harkens back to when we talked about prime-ideals, where being prime in some way means we can work backwards. This "working backwards" let's us guarantee we decompose our number into unique elements, as we'll soon see.

Though not all irreducible elements are prime, the converse is true, that is, being "globally divisible"' is a strong enough condition that it guarantee's irreducibility:

Proposition 10.1.7: Prime Then Irreducible In Integral Domain

Let R be an integral domain. Then prime elements in R are always irreducible

***Proof*:**
Supose $(p) \subseteq R$ is a nonzero prime ideal, and that $p = ab$. Since (p) is a prime ideal, then either $a \in (p)$ or $b \in (p)$. If $b \in (p)$. Then $b = rp$ for some $r \in R$, and if $a \in (p)$ then $a = pr$ for some $r \in R$. Without loss of generality, let's say $a \in (p)$. Then

$$p = ab = prb$$

and since R is an integral domain, the cancellation property holds (proposition 9.1.24), and we get

$$1 = rb$$

implying r, b are units, and so b is a unit. The same argument applies if we chose b. Thus, p is irreducible.

Thus, being prime is stronger than being an irreducible. The natural question to ask now is whether all elements in an integral domain can be decomposed into irreducible or prime factors (if it is factored into prime factors, then automatically it is factored into irreducible factors). It might already be clear that some Integral domains we have explored have this property (ex. $\mathbb{Z}$, and if you're familiar with some manipulation of polynomials, then $\mathbb{F}[x]$). This is however not the case for all integral domains:

Example 10.1.8: Domain with no Factorization

Let $R = \overline{\mathbb{Z}} \subseteq \mathbb{C}$, where $\overline{\mathbb{Z}}$ consists of all elements in $r \in \mathbb{C}$ such there exists a monic polynomial with integer coefficients

$$a_0 + a_1x + a_2x^2 + \cdots + x^n \qquad a_i \in \mathbb{Z}$$

such that

$$a_0 + a_1r + a_2r^2 + \cdots + r^n = 0$$

In other words, $\overline{\mathbb{Z}}$ is the set of elements which are the root of a monic polynomial with integer coefficients (i.e. $\mathbb{Z}[x]$). The ring $\overline{\mathbb{Z}}$ is called the *integral closure* of $\mathbb{Z}$ over $\mathbb{C}$ and will be extensively studied in chapter 50 (algebraic number theory). For now, we just need to show that it is indeed an integral domain, and there exists elements that do not factor.

Since $\overline{\mathbb{Z}}$ is a subring of $\mathbb{C}$, we can use the subring test, that is, see if $\overline{\mathbb{Z}}$ is closed under multiplication and passes the subgroup test. Is passes the subgroup test, since for any $a, b \in \overline{\mathbb{Z}}$, there exists polynomials

$$a_0 + a_1x + \cdots + x^n \qquad b_0 + b_1x + \cdots + x^n$$

(where some coefficients might be zero to match the length) such that

$$a_0 + a_1 \cdot a + \cdots + a^n = 0 \qquad b_0 + b_1 \cdot b + \cdots + b^n = 0$$

then

$$(a_0 - b_0) + (a_1 - b_1)x + \cdots + x^n$$

is a polynomial that has root $a - b$. By a similar argument, $\overline{\mathbb{Z}}$ using the binomial theorem is closed under multiplication.

Note that $\overline{\mathbb{Z}}$ is an integral domain since $\mathbb{C}$ has no zero divisors (hence $\overline{\mathbb{Z}}$ can't have any either), however it is not a field. Notice that $2 \in \overline{\mathbb{Z}}$ (2 is the root of $x - 2$). However, $1/2 \notin \overline{\mathbb{Z}}$. If it were, then there would be a monic polynomial with root $1/2$. However, any monic polynomial would be of the form

$$x^n + a_{n-1}x^{n-1} + \cdots + a_1x + a_0$$

if we plug in $1/2$, we get

$$\left(\frac{1}{2}\right)^n + a_{n-1}\left(\frac{1}{2}\right)^{n-1} + \cdots + a_1\left(\frac{1}{2}\right) + a_0$$

with a little more work (which I suggest the reader try out), this can be shown to never has a root (start with $x + b$ to see why it doesn't have a solution, and try to generalize from there).

However, $\overline{\mathbb{Z}}$ has elements (in fact all non unit elements) that do not factor! Take $r \in \overline{\mathbb{Z}}$. Then there exists integers $a_0, a_1, ..., a_{n-1} \in \mathbb{Z}$ such that

$$a_0 + a_1r + a_2r^2 + \cdots + a_nr^n = 0$$

Since $\overline{\mathbb{Z}} \subseteq \mathbb{C}$, the square root of any number is well-defined (see exercise ref:HERE (in Introduction to Rings)). Thus $\sqrt{r}$ is well-defined. Further, $\sqrt{r} \in \overline{\mathbb{Z}}$ since

$$a_0 + a_1\sqrt{r}^2 + a_2\sqrt{r}^4 + \cdots + a_n\sqrt{r}^{2n} = 0$$

Since $1/\sqrt{r} \notin \overline{\mathbb{Z}}$ (as similar argument to why $1/2$ is not in $\overline{\mathbb{Z}}$) it is not a unit, and hence $r = \sqrt{r}\sqrt{r}$ factors into two non-units. Now here is the key: we can repeat this process for $\sqrt{r}$, showing that

$\sqrt{r} = \sqrt{\sqrt{r}} \cdot \sqrt{\sqrt{r}}$, and so on and so forth. This process never terminates, showing that $\overline{\mathbb{Z}}$ has elements that do not factor!

For more examples, see:

https://math.stackexchange.com/questions/96790/integral-domain-that-is-not-a-factorization-domain

For this reason, we must define a special subclass of integral domains which do admit a factorization into irreducibles for every element:

Definition 10.1.9: Atomic Domains

Let R be an integral domain. Then R is an *atomic domain* (or *domain with factorization*) if for every non-unit nonzero element $r \in R$, there exists irreducible elements $q_1, q_2, ..., q_n$ such that

$$r = q_1 q_2 \cdots q_n$$

that is, every element has a *factorization* (or *decomposition*) into irreducible

The name atomic is due to every element being decomposable into element that can no longer be subdivided, similar to atoms, where the word atom comes from the Greek *atomos* meaning "not cuttable". Note that the q_i's need not be distinct (for example, we will show that $\mathbb{Z}$ is an atomic domain, and that $4 = 2 \cdot 2$). Usually, we are more interested in domains where the factorization into irreducible elements is unique:

Definition 10.1.10: Unique Factorization Domain (UFD)

Let R be an integral domain. Then R is a *Unique Factorization Domain* (commonly abbreviated to UFD) if for every non-unit nonzero element $r \in R$, there exists irreducible elements $q_1, q_2, ..., q_n$ and unit u_r such that

$$r = u_q q_1 q_2 \cdots q_n$$

and if there is another set of irreducible element $q_1', q_2', ..., q_m'$ such that $r = u_{q'} q_1' q_2' \cdots q_m'$, then $n = m$, and the two products differ by a unit after some reshuffling (i.e. they are associates), that is, for appropriate $u \in R$

$$q_1 q_2 \cdots q_n = u q_{\sigma(1)}' q_{\sigma(2)}' \cdots q_{\sigma(n)}'$$

where $\sigma(i)$ is the appropriate permutation

The following corollary is stated for the sake of reference:

Corollary 10.1.11: Unique Factorization Domain ⇒ Atomic Domain

Let R be a Unique Factorization Domain. Then R is an Atomic domain

***Proof*:**
By definition, every element in a Unique Factorization Domain has a factorization into irreducibles, and so is an Atomic Domain

Note that "unique" means unique up to unit, as we pointed out earlier. The factorization in both definitions could have had the unit dropped and just have been written $r = q_1, q_2, ..., q_n$ since it is a the result is an associate to $u^{-1}r$, and so it is harmless to drop the unit. However, I chose to keep the unit in the definition since it may lead to certain unfortunate mistakes when factoring concrete elements.

Example 10.1.12: Unique Factorization Domains

1. Every field is trivially a Unique Factorization Domain, since every nonzero element is a unit, so there are no elements for which the two properties must be verified. In some textbooks, fields are purposefully omitted as Unique Factorization Domains to omit mentioning them as exceptions in proofs. They are still considered Unique Factorization Domains in this book since they fit well in the diagram presented at the beginning of this chapter. However, in a sense, this means that in a field there is no "unique factorization", or at least it's vacuously true that all elements can be uniquely factorized since no nonzero elements are non-units.

2. As we saw in example 11.2.5, $F_{0x} \subsetneq \mathbb{F}[x]$ which has x coefficient of 0 is not a Unique Factorization Domain. We saw that x^2 and x^3 are irreducible. If we take x^6, then has two factorizations: $x^6 = x^2x^2x^2$ and $x^6 = x^3x^3$, and hence the factorization is not unique.

3. The ring $\mathbb{Z}[2i]$ is *not* a Unique Factorization Domain. $4 = 2 \cdot 2 = ((-2i) \cdot 2i)$. The elements 2 and $2i$ are irreducible, but not associates, since $2 = 2 \cdot i$, but $i \notin \mathbb{Z}[2i]$. They are furthermore not prime, since $\mathbb{Z}[2i]/(2i) \cong \mathbb{Z}/4\mathbb{Z}$ which is not an integral domain (see proposition 9.4.25)

4. In the ring $\mathbb{Z}[i]$, 2 and $2i$ *are* associates. We have $i \in \mathbb{Z}[i]$, and $i \cdot i^3 = 1$ showing that i is a unit, and hence 2 and $2i$ are indeed associated. We'll soon show that $\mathbb{Z}[i]$ is also a Unique Factorization Domain (see section 10.3)

5. We shall prove in chapter 11 that if R is a UFD, then $R[x]$ is a UFD. Note how this contrasts with PID's ($\mathbb{Z}$ is a PID but $\mathbb{Z}[x]$ is not, see example 9.4.6[3]).

All of these are examples of Unique Factorization Domains. Atomic Domains which are not Unique Factorization Domains are rather difficult to find. Fortunately, there is an easy criterion that allows us to find (a subset of) common atomic domains

10.1.1 Example of Atomic Domains: Noetherian Domains

The goal of this section is to show that Noetherian domains are atomic domains, and that there exists noetherian domains which are not Unique Factorization Domain's. Let's first remember the definition of a Noetherian Domain, or more generally a Noetherian ring:

Definition 10.1.13: Noetherian Rings

Let R be ring. Then if every ideal $I \subseteq R$ is finitely generated, then R is called a *Noetherian Ring*. If R is an integral domain, we call R a *Noetherian Domain*

Most results that will be presented apply to Noetherian rings in general, and so we will work with Noetherian rings. The first thing to show is how to relate Noetherian rings to the idea of factorization. Emmy Noether discoverd in HERE a condition which let's us characterize this property[2]:

Proposition 10.1.14: Ascending Chain Condition

The following are equivalent:

1. R is a Noetherian ring
2. for every chain of ascending left-ideals of R
$$I_1 \subseteq I_2 \subseteq \cdots \subseteq I_n \subseteq \cdots$$
there exists an number m such that for all $n > m$, $I_n = I_{n+1} = \cdots$
3. Every nonempty family of left-ideals of R partially ordered by inclusion has a maximal left-ideal

Note that replacing left by right or two-sided ideals will not break the theorem. If R is an integral domain, then there is no distinction between these concepts

***Proof*:**

(1) $\Rightarrow$ (2) Let R be a Noetherian ring. If $I_1 = R$ we're done, so assume I_1 is a proper ideal of R, and consider any ascending chain of left ideals

$$I_1 \subseteq I_2 \subseteq I_3 \subseteq \cdots \subseteq I_n \subseteq \cdots$$

Take
$$I = \bigcup_{i=1}^{\infty} I_i$$
which is an ideal for a similar reasoning as presented in the proof that every proper ideal is contained in a maximal ideal (see proposition 9.4.19). Since R is a Noetherian domain and $I \subseteq R$, then I is finitely generated. Let's say $I = (a_1, a_2, ..., a_n)$. Then these generators must each be in some ideal I_m, meaning that for all $n > m$, $I_{n+1} = I_{n+2} = \cdots$, showing that R respects the ascending chain condition

(2) $\Rightarrow$ (3) Let R respect the ascending chain condition on it's left ideals, and let $\mathcal{F}$ be a family of left-ideals of R. Pick some random $I_1 \in \mathcal{F}$. If I_1 is maximal in $\mathcal{F}$, then we'd be done, so let's say I_1 is not maximal in $\mathcal{F}$. Then there exists an I_2 such that $I_1 \subset I_2$. If I_2 is maximal in $\mathcal{F}$, then we're done. If not, then we can

[2] more on the generalization of it??

continue this process. If every element we pick is not maximal, we can create an infinitely long chain[a]

$$I_1 \subsetneq I_2 \subsetneq I_3 \cdots \subsetneq I_n \subsetneq \cdots$$

However, that contradicts the ascending chain condition, meaning for some n, the chain must stabilize. Thus, $\mathcal{F}$ must have a maximal element

(3) $\Rightarrow$ (1) Let $I \subseteq R$ be a left-ideal. Take $\mathcal{I}$ to be the set of all finitely-generated left-ideals contained in I. The set $\mathcal{I}$ is nonempty since $\{0\} \in \mathcal{I}$. It is thus a family of left-ideals of R and so by the 3rd condition, $\mathcal{I}$ admits a maximal,/Then if say I' [b]. If $I \neq I'$, then there exists an element $x \in I$ such that $x \notin I'$. But then $(I' \cup x)$ is a finitely generated left-ideal larger than I', contradicting the maximality of I'. Hence, $I = I'$, showing that I is finitely generated.

[a] technically, to create this infinitely long chain, we used the axiom of choice

[b] Note that Zorn's lemma can't be used here since it is not known whether every chain will have an upper bound in this set, their union potentially no longer being finitely generated

10.1.15 Remark This proof will become a special case of the equivalent proof for modules in section 12.3.4.

For our purposes, this let's us think of Noetherian domains in terms of the ascending chain condition. You probably already see how this might start related to Factorization domains: If every element in the ring is finitely generated, it must be decomposable into a finite number of factors. This is essentially true:

Proposition 10.1.16: A.C.C. and Factorization

Let R be an integral domain, and let r be a nonzero nonunit element of R. Then if every ascending chain of *principal ideals*

$$(r) \subseteq (r_1) \subseteq (r_2) \subseteq \cdots \subseteq (r_n) \subseteq$$

stabilizes (there exists an m such that for all $l > m$, $(r_l) = (r_{l+1}) = \cdots$), then r can be factored into irreducible elements. In particular, if R respects the A.C.C. on principal ideals, then R is an atomic domain.

Re-writing these ideal using the divisibility notation, it may be more evident: if $r_1 \big| r$, $r_2 \big| r_1$, and so on eventually stabalizes, then r can be factored into irreducible elements.

***Proof*:**
Let's prove this using the contrapositive ($\neg B \Rightarrow \neg A$), so that we'll show if r does not factor into irreducible then there exists an ascending chain of principal ideals that does not stabalize.

Since r is not irreducible, then there exists two nonunit non zero element r_1, s_2 suc that $r = r_1 s_2$. Since r does not have a factorization, either r_1 or s_1 are reducible. Without loss of generality, let's say r_1 is reducible. Then $(r) \subseteq (r_1)$ since $s_2 \cdot r_1 = r$, and $r_1 = r_2 s_2$. The same being applied to r_1, we get $(r) \subseteq (r_1) \subseteq (r_2)$. Continuing on indefinitely, we get

$$(r) \subseteq (r_1) \subseteq (r_2) \subseteq \cdots \subseteq (r_n) \subseteq \cdots$$

which never stabilizes since r does not factor, as we sought to show.

Notice how similar this it to Noetherian rings: In a Noetherian ring, the ascending chain condition works for *all* left/right/two-sided ideals. Therefore, it certainly works when restricted to only it's *principal ideals*. We therefore get this immediate corollary

Corollary 10.1.17: Noetherian Domain $\Rightarrow$ Atomic Domain

Let R be a Noetherian Domain. Then it is an Atomic Domain

***Proof*:**
By proposition 10.1.14, R respects the ascending chain condition on any left ideals. Since any element $r \in R$ forms a principal ideal, any ascending chain of principal ideals must stabalize. Thus, by proposition 10.1.16 r must have a factorization. Since r was arbitrary, every element has a factorization, and so R is an Atomic Domain as we sought to show

It might seem like the converse is clearly true: If R is an atomic domain, then R respects the ascending chain condition on principal ideals. P.M.Cohn made this same assertion in 1968 during his research on rings[3], however it was ultimately proven to not be the case: there are element for which there are infinitely many factorizations. Though each factorization is finite, there are factorizations for every natural number (see exercise ref:HERE, maybe?). This theorem is still useful to our endeavour, since it let's us link a finite generation property (A.C.C. forces all principal ideals to be contained in at most a finitely generated ideal[4]) to a factorization condition (a condition for how to decompose elements into simpler elements).

Unfortunately, there isn't a simple counter-example I know of until we prove the famous Hilbert's Basis Theorem in section 23.3. For the reader who wants to know the counter-example, they can either skip to Hilbert's basis theorem in *Commutative Algebra* [10] (you have everything you need to know to understand it now), or take on faith that "if R is a Noetherian Domain then $R[x_1, x_2, ..., x_n]$ is a Noetherian Domain". The following is a counter-example:

Example 10.1.18: Atomic Domain $\not\Rightarrow$ Noetherian Domain

Let $R = \mathbb{Z}[x_1, x_2, ..., x_n, ...]$ be a polynomial ring over an infinite number of variables (see section 9.2). Notice that $\mathbb{Z}$ is a Noetherian Domain: As we have shown, $\mathbb{Z}$ a Principal Ideal Domain (example 9.4.6, and so any ideal of $\mathbb{Z}$ is generated by one element, and so it is Noetherian. By Hilbert's Basis Theorem, $\mathbb{Z}[x_1, x_2, ..., x_n]$ is a Noetherian Domain as well.

Now, any polynomial $f \in R$ has finitely many terms by definition, and so is also an element of the subring $\mathbb{Z}[x_{\sigma(1)}, x_{\sigma(2)}, ..., x_{\sigma(n)}]$ where $\sigma(i)$ is some appropriate permutation function so that the terms of f are in the aforementioned subring. But then it is an element of a Noetherian Domain, which means it could be factored by proposition 10.1.16. Since $\mathbb{Z}[x_{\sigma(1)}, x_{\sigma(2)}, ..., x_{\sigma(n)}] \hookrightarrow R$, the same factorization happens in R, and since f was arbitrary, R is an Atomic Domain

However, R is not a Noetherian Domain. The ideal $(x_1, x_2, ..., x_n, ...)$ of all of the terms of R is not finitely generated (it does not respect the ascending chain condition).

This means that in the image presented in the beginning of the chapter, the strict inclusion of Noetherian Domains in Atomic Domains is justified. Similarly, Being an Atomic Domain does not

[3]P.M. Cohn, Bézout rings and their subring; Proc. Camb. Phil.Soc. 64 (1968) 251–264
[4]assuming the Axiom of Choice

mean the factorization is unique

Example 10.1.19: Atomic Domain $\not\Rightarrow$ Unique Factorization Domain
Take the ring F_{0x} from example 10.1.6. Take any element $f \in F_{0x}$, and form a ascending chain of principal ideals. For the sake of contradiction, let's say the chain never stabilizes: $(f) \subsetneq (f_1) \subseteq (f_2) \subsetneq \cdots \subseteq (f_n) \subsetneq \cdots$. Then since $(f_i) \subsetneq (f_j)$, and every degree zero element is a unit (recall that $F_{0x} \subseteq \mathbb{F}[x]$ for a field $\mathbb{F}$) then the degree of f_j must be *less* than that of f_i. This means we have a decreasing chain:

$$\deg(f) > \deg(f_1) > \deg(f_2) > \cdots > \deg(f_n) > \cdots$$

Notice importantly that this chain *must* terminate since there are no polynomials of degree 0. Since f was arbitrary, every element has a factorization and so it is an Atomic Domain.

However, as we have shown in example 10.1.6, $x^6 = x^2x^2x^2 = x^3x^3$ are two factorizations of x^6 where x^2 and x^3 are not associates, showing it is not a Unique Factorization Domain.

Notice the ring F_{0x} is not a Noetherian Domain in the previous example. Using Hilbert's Basis Theorem, we can also find a Noetherian Domain which is not a UFD:

Example 10.1.20: Noetherian Domain $\not\Rightarrow$ Unique Factorization Domain
Let

$$R = \frac{\mathbb{C}[x, y, z, w]}{(xw - yz)}$$

be the polynomial ring over 4 variables quotiented by the ideal $(xw - yz)$ [a]. By Hilbert's Basis Theorem, $\mathbb{C}[x, y, z, w]$ is a Noetherian ring. By proposition 9.4.8, $\frac{\mathbb{C}[x,y,z,w]}{(xw-yz)} = R$ is a noetherian ring. It is also an integral domain, since $(xw - yz)$ is a prime ideal (see exercise ref:HERE. It is therefore a Noetherian Domain (and hence an Atomic Domain).

However, it is not a Unique Factorization Domain. The element x, y, z, w are all irreducible, and not associates. Since $xw - yz = 0$, we have $r = xw = yz$ has two distinct factorization, showing that R is not a Unique Factorization Domain.

[a] In chapter 31, we will see this ring be called the *quadratic cone in* $\mathbb{A}^4$

The converse is not true either: being a Unique Factorization Domain does not imply it is a Noetherian Domain. In fact, the ring $\mathbb{Z}[x_1, x_2, ..., x_n, ...]$ from example 10.1.18 is also a Unique Factorization Domain that is not a Noetherian Domain. This proof relies a theorem similar to Hilbert's Basis Theorem (in particular a generalization of theorem 11.2.3 on page 363). However, this theorem requires much longer build-up, and so for now the reader can take it on faith that such a ring is a counter-example (see exerise ref:HERE for a way to prove this using gcd domains).

Thus, we have shown (or claimed) that

$$\begin{array}{rcl@{\qquad}rcl}
\text{Noetherian Domain} & \Rightarrow & \text{Atomic Domain} & \text{Atomic Domain} & \not\Rightarrow & \text{Noetherian Domain} \\
\text{UFD} & \Rightarrow & \text{Atomic Domain} & \text{Atomic Domain} & \not\Rightarrow & \text{UFD} \\
\text{UFD} & \not\Rightarrow & \text{Noetherian Domain} & \text{Noetherian Domain} & \not\Rightarrow & \text{UFD}
\end{array}$$

This completes some of the layers in the diagram presented at the beginning of this chapter.

:exercise

Exercise 10.1.1

1. Let R be a finitely generated ring. Does that imply R is Noetheria
2. Show that $\mathbb{C}[x, y, z, w]/(xw - yz)$ is an integral domain.
3. Show that if $M \subseteq R$ is a maximal ideal then M is Noetherian. In fact, if the prime ideals are finitely generated, then R is Noetherian.

10.1.2 Unique Factorization Domains

We know continue exploring Atomic Domains in which the factorization of each element is unique (up to unit). As a reminder, here is the definition of a UFD:

Definition 10.1.21: Unique Factorization Domain (UFD)

Let R be an integral domain. Then R is a *Unique Factorization Domain* (commonly abbreviated to UFD) if for every non-unit nonzero element $r \in R$, there exists irreducible elements $q_1, q_2, ..., q_n$ such that

$$r = q_1 q_2 \cdots q_n$$

and if there is another set of irreducible element $q'_1, q'_2, ..., q'_m$ such that $r = q'_1 q'_2 \cdots q'_m$, then $n = m$, and the two products differ by a unit after some reshuffling (i.e. they are associates), that is, for appropriate $u \in R$

$$q_1 q_2 \cdots q_n = u q'_{\sigma(1)} q'_{\sigma(2)} \cdots q'_{\sigma(n)}$$

where $\sigma(i)$ is the appropriate permutation

Perhaps the most intuitive way of thinking about elements in Unique Factorization Domains is by linking to every element of the UFD a multiset (a set that allow's multiplicity) of irreducible factors. Units and 0 are both associated to empty multisets. The multiplication structure can also be thought of as forming a free abelian monoid structure with all prime elements as generators.

Lemma 10.1.22: Irreducible and Multisets

Let R be a Unique Factorization Domain and let $a, b, c \in R$ be nonzero elements of R. Then

1. $(a) \subseteq (b)$ if and only if the multiset of irreducible elements of a is contained in the multiset of irreducible elements of b
2. a and b are associates if and only if their corresponding multisets are equal
3. if $a = bc$, then the corresponding multiset of a is the union of the multisets of b and c

This lemma essentially shows that working with elements of a Unique Factorization Domain comes down to working with their corresponding multiset, making the analysis of their structure more often than not being more set theoretical or number theoretical. As an example of this, Unique Factorization Domains make it very easy to define the concept of *greatest common divisor*.

Greatest Common Divisor

The set of sets of multi-sets thus has a natural lattice paralleled on rings given by the divisibility. In fact, meet and joins are well-defined[5]:

Definition 10.1.23: Greatest Common Divisors In Commutative Rings

Let R be a commutative ring and let $a, b \in R$ with $b \neq 0$. Then the *greatest common divisor* of a and b is a nonzero element d such that

1. $d|a$ and $d|b$,
2. if $d'|a$ and $d'|b$ then $d'|d$

A greatest common divisor of a and b will be denoted $gcd(a, b)$ or (a, b)

Notice that in the definition, R was a commutative ring, since divisibility was defined for commutative rings in general. It should be noted that the gcd of two elements is unique *up to associativity*. it is helpful to put the definition of gCD in ideal-interpretation of divisibility:

Definition 10.1.24: GCD Using Principal Ideals

Let R be a commutative ring and $(a), (b) \subseteq R$. Then $\gcd(a, b)$ is an ideal $I \subseteq R$ such that $(a, b) \subseteq I$ and I is the smallest ieal that contains (a, b). In other words

$$\gcd(a, b) = (a) + (b)$$

As an exercise, see that this does indeed match the previous definition of gcd, and test it in $\mathbb{Z}$. There do exist rings for which the gcd between two elements do not exist (in example 10.1.6, take the ring R_{0x} from the second part of the example. Prove that the $\gcd(x^5, x^6)$ does not exist, see exercise ref:HERE). For Unique Factorization Domains, every element has a gcd:

Proposition 10.1.25: GCD in Unique Factorization Domain

let R be a Unique Factorization Domain. Then for any $a, b \in R$, $\gcd(a, b)$ exists. Furthermore, given

$$a = up_1^{n_1} p_2^{n_2} \cdots p_k^{n_k} \qquad b = vp_1^{m_1} p_2^{m_2} \cdots p_k^{m_k}$$

is the factorization of a and b, that is, every p_i is irreducible, u and v are associates, every p_i is not associates to p_j, and some n_i's or m_i's can be zero. Then the gcd (up to a unit) is:

$$\gcd(a, b) = p_1^{\min(n_1, m_1)} p_2^{\min(n_2, m_2)} \cdots p_k^{\min(n_k, m_k)}$$

***Proof*:**
This is an almost immediate consequence of using the factorization of a and b. Let d equal the above equation. Then clearly $d|a$ an $d|b$. If there is some c that also divides a and b ($c|a$ and $c|b$), then $a = ca'$ and $b = cb'$ for appropriate a' and b'. By lemma 10.1.22, c must be contained

[5] see section 0.3.1

in both the multisets of a and b, and therefore

$$c = wp_1^{\ell_1} p_2^{\ell_2} \cdots p_k^{\ell_k}$$

where w is a unit and $\ell_i \leq n_i$ and $\ell_i \leq m_i$. Then $\ell_i \leq \min(n_i, m_i)$ for all i, and so $(c) \subseteq (d)$, and by lemma 10.1.22 $c \big| d$, as we sought to show

10.1.26 Relation Between Lattice's and UFDs It should be noted that there are integral domains which are not Unique Factorization Domain's but do have the concept of gcd defined for every element. The class of all integral domains which have the concept of gcd defined for every element is broader than Unique Factorization Domain's and are not always Noetherian, and so such domains need their own name and are called *gcd domains*. In a sense, gcd domains are a generalization of Unique Factorization Domains: A gcd domain is Unique Factorization Domain if and only if the gcd domain satisfies the ascending chain condition on principal ideals (similarly to how Atomic domains do). However, a gcd domain isn't always an Atomic domain, and hence fits a bit awkwardly in our current hierarchy. In essence, this is because gcd domains hold information on information between elements, while atomic domains hold element-wise information.
For future purposes, gcd domains do fit into an important hierarchy of domains: *normal domains* shall be introduced in *Commutative Algebra* [10], definition 26.1.3, and all gcd are normal domains:

$$\text{normal domains} \supseteq \text{gcd domains} \supseteq \text{UFD's}$$

The definition of GCD's using ideals suggests that the GCD, when it exists, ought to be unique in general, and indeed it is:

Proposition 10.1.27: Uniqueness Of Greatest Common Divisor (In Integral Domain)

Let R be an integral domain. If (d) and (d') are both greatest common divisors of then $(d) = (d')$.

***Proof*:**
If d and d' are greatest common divisors, then they must each divide each other (condition 2 of definition 10.1.23), and so we'll get $(d') \subseteq (d)$ and $(d') \supseteq (d)$, meaning $(d') = (d)$,

Just like for the integers, we may also define the dual of the GCD, the least common multiple. In the following, the LCM will be defined in terms of principle ideals, the similar definition should be translated into elements:

Definition 10.1.28: Least Common Multiple (lCM)

Let R be a commutative ring and $(a), (b) \subseteq R$. Then the *least common muliple* is the ideal I such that $I \subseteq (a)$ and $I \subseteq (b)$ and is the largest such ideal. Namely

$$\operatorname{lcm}((a), (b)) = (a) \cap (b)$$

Properties of UFD's

As we have shown in example 10.1.6 and 10.1.20 being an irreducible element in an integral domain does not imply that the element is a prime element. This was because prime elements are "stronger" than irreducible in that they not only have to be able to be divisible, but they have to be "globally divisible" for any factorization. However, since factorization is unique in a Unique Factorization Domain, it should come as no surpise that there is no distinction between prime elements and irreducible elements:

Proposition 10.1.29: Prime If And Only If Irreducible In UFD

Let R be a Unique Factorization Domain. Then a nonzero element is a prime if and only if it is irreducible

***Proof*:**
The fact that prime elements are irreducible was proven in proposition 10.1.7, so we'll show the converse. Let a be an irreducible element. Then a is not a unit, and if $a = bc$ then either b or c is a unit. Now, assume $bc \in (a)$ so that $d \cdot a = bc$. Then $(bc) \subseteq (a)$, and by lemma 10.1.22, the set of the irreducible factors of a (i.e. a itself) is contained in the set of irreducible factors of b and/or c. Therefore, it must be that a is a factor of b, or c, or both. If it's factors are within the factors of b, then $b \in (a)$, and if it's factors are within the factors of c, then $c \in (a)$. If in both, then $b, c \in (a)$. In any case, at least one of b or c are in (a), showing that (a) is prime.

Thus, if we see every irreducible element being prime as representing "global" irreducibility (or divisibility), then if we further assume the A.C.C. on principle ideals, we see that this ought to fully characterize UFDs. The following theorem demonstrate exactly this:

Theorem 10.1.30: Unique Factorization Domain Equivalence

Let R be an integral domain. Then R is a Unique Factorization Domain if and only if

1. The ascending chain condition for principal ideal holds in R
2. Every irreducible element is a prime element

***Proof*:**

($\Rightarrow$) Let R be a Unique Factorization Domain. Then by proposition 10.1.29 every irreducible element is a prime element, and by corrollary 10.1.11 R is an atomic domain, and so by proposition 10.1.16 R respects the ascending chain condition (this can also be proven directly using the corresponding multisets to elements)

($\Leftarrow$) Let R satisfy the A.C.C. for principal ideals and let every irreducible be prime. By proposition 10.1.16, R is an Atomic domain. It remains to show that R factors uniquely. To show this, let n be a nonunit and

$$n = p_1 p_2 \cdots p_n = q_1 q_2 \cdots q_m$$

be two factorizations of n into irreducibles. Then since $q_2 q_3 \cdots q_n \cdot q_1 = p_1 p_2 \cdots p_n$, we have $p_1 p_2 \cdots p_n \in (q_1)$. Since q_1 is irreducible, by the 2nd condition it is a prime, and so for some i, $p_i \in (q_1)$. Without loss of generality, we can say $p_1 \in (q_1)$ by just rearranging and relabelling the factors. Since $p_1 \in (q_1)$, then $u_1 \cdot q_1 = p_1$. Since p_1 is irreducible and q_1 is not a unit, necessarily u_1 is a unit. Then, we can re-write the equation as

$$u_1 q_1 p_2 \cdots p_n = q_1 q_2 \cdots q_m$$

Cancelling out q_1 from each side and replacing p_2 by $u_1 p_2$ produces us

$$p_2 p_3 \cdots p_n = q_2 q_3 \cdots q_m$$

from here, we can continue this process for the rest of the p_i's, where we will end up with

$$u_1 u_2 \cdots u_n q_1 q_2 \cdots q_m = q_1 q_2 \cdots q_m$$

leting $u = u_1 u_2 \cdots u_n$, we get $uq_1 q_2 \cdot q_n = q_1 q_2 \cdots q_n$. Note that $n = m$ since if not then at some point the remaininng part of the left hand side will be equal to 1, a contradiction to n not being a unit. Thus, the two factorization are equal up to associates, as we sought to show.

Notice how the uniqueness of the factorization allowed us to make the A.C.C. for principal ideals a defining feature of Unique Factorization Domains, in contrast to ho the A.C.C. for principal ideals was insufficient to define Atomic Domains.

This theorem is also very useful in easily remembering how Unique Factorization Domain's and Noetherian Domains are different: If we start with a domain with the A.C.C. for principal ideals (i.e. the types of Atomic domains we ill work with) strengthen the A.C.C. condition to work on all ideals, we get a Noetherian domain. Conversely, if we are working with a Noetherian domain and we loosen the A.C.C. condition to principal ideals but add that all irreducible are primes, then we get a Unique Factorization Domain.

10.1.3 Important UFD: Principle Ideal Domains

Principal Ideal Domains are stronger than Noetherian Domains in the sense that every ideal is generated by a single element (rather than finitely many element for Noetherian Domains). They are therefore immediately Atomic Domains by proposition 10.1.16. By theorem 10.1.30, if a Principal Ideal Domain also has that every irreducible element is prime, then it is automatically a Unique Factorization Domain. Are irreducible elements prime in a Principal Ideal Domain? The answer is yes; hence PIDs form a subclass of UFDs. Take a moment to think about why irreducible elements are prime in PIDs before proceeding to the next proposition (hint: can you prove prime ideals *must* be maximal ideals in a PID?). From this proof, there will now be a new set of examples for Unique Factorization Domains similarly to how all Noetherian domains are an easy set of examples for Atomic domains

10.1.31 Taking a moment

As the hint suggests, we first prove the following important intermediary result:

Proposition 10.1.32: In PID, Prime Ideal iff Maximal Ideal

Let R be a Principal Ideal Domain and take a nonzero ideal $(r) \subseteq R$. Then (r) is a prime ideal if and only if it is a maximal ideal.

Intuitively, since

***Proof*:**
Let R be a Principal Ideal Domain. If an ideal of R is maximal, by corollary 9.4.26 that ideal is prime. Thus we'll prove the converse.

Let $(p) \subseteq R$ be a prime ideal, and assume that $I \supseteq (p)$ is an ideal that contains (p). We must show that $I = (p)$ or $I = R$. Since R is a Principal Ideal Domain, there exists an $m \in R$ such that $(m) = I$. Since $(m) \supseteq (p)$, $p \in (m)$, and so $p = rm$. Since p is a prime element, then either $r \in (p)$ or $m \in (p)$. If $m \in (p)$, then $(m) = (p)$, fulfilling the $I = (p)$ case. If $r \in (p)$ then $r = sp = ps$ and so

$$p = rm = psm$$

And by the cancellation property of integral domains, $1 = sm$, and so $1 \in (m)$ implying that $(m) = R$, fulfilling the $I = R$ condition. That means that (p) fulfills the conditions of being a maximal ideal, as we sought to show.

The main idea is that any ideal requries only 1 element to repersent it, which as it will turn out will be key to why elements will factor uniquely.

Proposition 10.1.33: Prime If And Only If Irreducible In PID

Let R be a Principal Ideal Domain. Then a nonzero element is a prime if and only if it is irreducible. Along with the fact that Principal Ideal Domain are Noetherian Domains (and hence respecting the A.C.C. on principal ideals) we have

$$\text{PID} \Rightarrow \text{UFD}$$

***Proof*:**
Since a Principal Ideal Domain is an integral domain, we've just shown in proposition 10.1.7 that prime elements are irreducible, so we'll show the converse.

Let's say p is an irreducible element of R. To show it's prime, we must how (p) is a prime ideal. Since we're working in a Principal Ideal Domain, by 10.1.32, it's equivalent to show that (p) is a maximal ideal. So, for the sake of contradiction, let's assume (p) is not maximal, and so there exists a strictly larger ideal that is the maximal ideal containing (p) $(p) \subsetneq (m) \neq R$ for appropriate $m \in R$. Then

$$p = am$$

for some $a \in R$. Since p is irreducible, then either a or m is a unit in R. If M is a unit, then $m^{-1}m = 1 \in (m)$ implying $(m) = (1) = R$, but then it's not a maximal ideal, since maximal ideals are proper ideals. Conversely, if a is a unit, then by proposition p and m are associates, meaning $(p) = (m)$. But since (m) was maximal, that makes (p) maximal. Thus, (p), by proposition 10.1.32, (p) is a prime ideal.

Therefore, since all Principal Ideal Domain are Noetherian Domains and Noetherian respect the A.C.C. on principal ideals, by thereom 10.1.30, all Principal Ideal Domain are Unique Factorization Domains, as we sought to show.

There is also a very simple proof I like I found in Eisenbud: (Eisenbud) We know that if R is a Unique Factorization Domain, then it respects the A.C.C. on principal ideals. Use that to find a maximal ideal, which must be generated by one element since it's a Principal Ideal Domain, but then it must be that for one of the elements in the chain, it plateau's, and so it fulfills the ascending chain condition, and so it is a Unique Factorization Domain.

Corollary 10.1.34: $\mathbb{Z}$ is a UFD (Fundamental Theorem of Arithmetics)

The integers ($\mathbb{Z}$) as a ring are a Unique Factorization Domain.

***Proof*:**
As shown in example ref:HERE (under defn of pid), $\mathbb{Z}$ is a Principal Ideal Domain, and so by proposition 10.1.32 is a Unique Factorization Domain.

The converse of this theorem is not true, that is if an integral domain is a Unique Factorization Domain, this does not imply that it is a PID, justifying the strict inclusion of the image at the very beginning. Recall from example 9.4.6 that the ideal $(2, x)$ from the ring $\mathbb{Z}[x]$ is not a principal ideal. However, the ring $\mathbb{Z}[x]$ is a Unique Factorization Domain. This is rather difficult to show directly: if we have any any polynomial $p(x) \in \mathbb{Z}[x]$, how can we guarantee it factors and that it's unique up to units, that is that $\mathbb{Z}[x]$ respects the ascending chian conditions on principal ideals and that all irreducibles are primes? On page 363, we shall show this result.

(Also something about begging the question now on finding qualities that allow for PIDs? For example, in $\mathbb{Z}$, we figured out it's a pid because for every number, we can make it a linear combination of other numbers. Is that so for every pid?)

([perhaps make this an exercise?] As a final comparison between UFDs and PIDs, the Chinese Remainder Theorem (CRT). Recall that in a pid, $\gcd(a, b) = 1$ if and only if $(a, b) = (1)$. This is *not* the case for UFD's, take for example the natural map

$$\mathbb{Z}[x] \to \mathbb{Z}[x]/(2) \times \mathbb{Z}[x](x)$$

Then this map is *not* surjective, though $\gcd(2, x) = 1$, even when the kernel, $(2x)$, is still the product of the two ideals.)

Proposition 10.1.35: Sufficient Condition for Greatest Common Divisor

If a and b are nonzero elements in the commutative ring R such that the ideal gnerated by a and b is a principal ideal (d), then d is a greatest common divisor of a and b

***Proof*:**
p.274

This proposition explains the (a, b) notation for the gcd of a and b.

Note that an integral domain in which every ieal generated by 2 elements (a, b) is a principal ideal is called a *Bézout Domain* (more generally, if every *finitely generated* ideal is principal, then we have a

Bézout Domain. For Principal Ideal Domains, every ideal is principal). Note that it's not a *necessary* condition.

For future reference, we also prove the following[6]:

Proposition 10.1.36: UFD And Maximal, Then PID

Let R be a Unique Factorization Domain where every prime ideal is maximal. Then R is a Principal Ideal Domain

***Proof*:**
Let R be a Unique Factorization Domain where every prime ideal is maximal. First, note that if $I \subseteq R$ is prime, then I is principle. Indeed, for some $r \in I$, then (up to unit) we have $r = p_1, p_2, ..., p_n$. By primality, we have that $p_i \in I$. Since p_i is prime, by assumption it is maximal, and furthermore $(p_i) \subseteq I$. But then $(p_i) = I$.

Now, let's N be the set of all non-principal ideals. For the sake of contradiction let's say it's non-empty. Then N is partially ordered by inclusion and every chain has an upper bound: if $J = \bigcup_i J_i$, then if $J = (x)$, then $x \in J_i$ for some i and $(x) = J$; contradicting non-principality.

Hence, by Zorn's lemma there exists a maximal element $\mathfrak{n} \in N$. Then by what we pointed out earlier $\mathfrak{n}$ cannot be prime, or else it would be principal. Therefore there exists $a, b \in R$, $a, b \notin \mathfrak{n}$ but $ab \in \mathfrak{n}$. Consdier $\mathfrak{n} + (a)$ and $\mathfrak{n} + (b)$. By maximality of $\mathfrak{n}$ hese must be principal: $\mathfrak{n} + (a) = (u)$ and $\mathfrak{n} + (b) = (v)$. But then:

$$\mathfrak{n} = \mathfrak{n} + (ab) = (\mathfrak{n} + (a))(\mathfrak{n} + (b)) = (u)(v) = (uv)$$

but that's a contradiction to $\mathfrak{n} \in N$, showing that it must have been that $N = \emptyset$, completing the proof.

(also thought this was cool here)

Exercise 10.1.2

1. Let $R[x]$ be a Principal Ideal Domain. Then R is a UFD. Conclude that if R is not a field, then $R[x]$ is not a Principal Ideal Domain!

   Since $\mathbb{Z}$ is not a field, then $\mathbb{Z}[x]$ is not a Principal Ideal Domain! In example 9.4.6[3] it was manually shown that $(2, x)$ is not a principal ideal in $\mathbb{Z}[x]$; this theorem makes it easier to prove.

   The converse is true to: If F is a field, $F[x]$ is a Principal Ideal Domain. In fact, it's even a Euclidean Domain. We'll cover this result when talking about polynomials over fields in section 11.1.

2. (Challenging) Let R be an integral domain. Then if every prime ideal is principle, R is a PID (In this way, UFDs are almost PIDs, however not all ideals are principle)

3. (If you know Complex Analysis) Show that the ring of entire functions is a gcd domain. Show that the ring of real analytic functions is a Bézout domain.

[6] This will be useful in number theory

10.2 Euclidean Domains

So far, we have been exploring divisibility in terms of being able to directly factor elements. However, it is in general difficult to establish when an element is divisible by another (try it out yourself with relatively large elements of some of the more complicated UFDs that were presented). If an element is not divisible by another element is there some way to measure to what "degree" an element is not divisible by another? Can we quantify the lack of divisibility given by the gcd.

In the integers, there is the *euclidean algorithm* that gives the answer. The existence of an algorithm to find the GCD cannot be under-estimated: If given two integers with over a thousand digits, it will be impossible to find the greatest common divisor in a reasonable amount of time, that is, before the heat death of the universe (until a faster way of factorizing is found). The problem is about our initial choice for divisors. If I say R is a Unique Factorization Domain and $x \in R$, then how would you start finding it's prime factorization? If you have some a and b such that $ab = x$, How do you start decomposing them? Even for the integers, it is in general very difficult to find prime factors of an element x[8].

For the integers, Euclid in approximately the year 300BC discovered that there is a simple algorithm for finding the GCD that quantifies exactly how divisible two numbers are. The key observation is that if $a, b \in \mathbb{Z}$, where $a < b$ (notice the user of the total order, $<$, of $\mathbb{Z}$) then we may write

$$b = ka + r$$

where $r < a$. If $a|b$, then $r = 0$. If $a \nmid b$, then r is the "torsion" or "remainder" that is left-over. In terms of the GCD, if $a|b$, then $\gcd(a, b) = a$, and if $a \nmid b$, then $\gcd(a, b) < a$ and in the "worst case scenario" $\gcd(a, b) = 1$. The problem when generalizing this to general rings is that we require a notion of "size", namely it was crucial to use the fact that $a < b$ and it is imperative that $r < a$ to have a reasonable notion of such a factorization. In this section, we shall endavour to find a function, say $v : R \to \mathbb{N}_{\geq 0}$ that will allow us to assign a natural number to each ring element so that we may recapture this notion from the integers.

Definition 10.2.1: [Ring] Valuation

Any function $v : R \to \mathbb{Z}$ with $v(0) = 0$ is called a *valuation* on the ring R. If $v(a) > 0$ for $a \neq 0$, then v is said to be a *positive valuation*.

10.2.2 Note The notion of valuation shall be defined later in these notes on fields as *real [Archimedean] valuation* or *absolute value* (definition 21.2.1) and generalized to *G-valuation* (definition 27.1.1)[a]. The notion of a G-valuation is a generalization of the absolute value, however it is often presented with the axiom that $v(0) = \infty$. Valuations can just as well be defined with $v(0) = 0$, and hence the use of the term "valuation" here is justified.

[a] More specifically non-Archimedean valuations shall be generalized to G-valuations

[8] RSA encryption is based off of the fact that prime factorization is a every difficult task if you don't know a priori what primes you are working with

10.2.3 Note There are at least two common uses of the word "valuation" in contemporary abstract algebra: an *absolute-valuation* or more commonly an *absolute value* and a *G-valuation* (definition 27.1.1). These two concepts are non-compatible[a], notably G-valuations have axiom dictating that $v(0) = \infty$ while absolute values has an axiom dictating that $v(0) = 0$. The above use of the term valuation is closer to an *absolute value*.

[a]Though an absolute value may induce a G-valuation, this is shown when the concepts are introduced

If you know some analysis, you may wonder if the valuation is related to the one introduced on $\mathbb{R}$ or $\mathbb{C}$ (or normed vector spaces more generally). These concepts are indeed related, and we shall even see more explicit connection in chapter 21⋆.

Returning to the issue at hand, we shall require a valuation N to be defined on an integral domain R that satisfies the following property:

Definition 10.2.4: Euclidean Valuation and Euclidean Domain

Let R be an integral domain and $N : R \to \mathbb{N}_{\geq 0}$ a valuation. Then v is a *euclidean valuation* on R if for any two elements a, b of R with $b \neq 0$ there exists elements q, r in R such that

$$a = qb + r \qquad \text{with } r = 0 \text{ or } N(r) < N(b)$$

The element q is called the *quotient* and the element r the *remainder* of the division if a euclidean valuation can be defined on a ring, then the ring is said to be a *Euclidean Domain* (or posses a *division algorithm*)

The main thing we can do in euclidean domains is run the division algorithm to keep "dividing" our numbers

$$\begin{aligned} a &= q_0 b + r_0 \\ b &= q_1 r_0 + r_1 \\ r_0 &= 2_0 r_1 + r_2 \\ &\vdots \\ r_{n-1} &= q_{n+1} r_n \end{aligned}$$

Such a sequence will terminate since $N(b) > N(r_0) > \cdots > N(r_n)$, and it will eventually hit 0.

Example 10.2.5: Euclidean Domains

1. Every field with the euclidean valuation is a euclidean domain! In fact, any valuation will work!

   Proof. Let $x \in \mathbb{F}$ be an element of a field. Then since every element in a field has a unit, then

   $$x = xb + 0$$

   where $b = x^{-1}$.

   □

Thus, we see that if a division rings was also commutative, then it'd clearly be a Euclidean domain.

2. In general, every unit will trivially satisfy this condition, since if $u \in R$ is a unit, then

$$u = uu^{-1} + 0$$

and so terminates instantly. So in a sense, Euclidean domains say that if we have elements that are *not* units, how many properties of divisible things can we put on them?

Note importantly that euclidean domains must be domains (meaning no zero-divisors)! So oddities with multiplying not increasing the "size" of elements will not happen.

3. Ring of fractions in general are not euclidean domains, since they might have zero-divisors. So even though rings of fractions can be thought of as a way of enlarging a ring to make it's elements more divisible, it cannot do so if it has zero-divisors.

4. The integers $\mathbb{Z}$ are a euclidean domain with $N(a) = |a|$. The division algorithm is your usual euclidean algorithm. If you take $a, b \in \mathbb{Z}$, and $a < b$ then $b = aq + r$. We see that $N(r) = |r| < |b| = N(b)$. This will always be the case with $b \neq 0$! Thus, it is a Euclidean Domain.

5. If $\mathbb{F}$ is a field, then $\mathbb{F}[x]$ is also a Euclidean domain with $N(p(x)) = deg(p(x))$. The division algorithm is your usual polynomial division. The reason is simple for valuation: polynomial division *must* produce a remainder r with degree *less* than b, by definition! This makes $\mathbb{F}[x]$ into a Euclidean Domain. We'll prove this more rigorously in section 11.1.

6. Let

$$\mathbb{Z}[i] = \{a + bi | a, b \in \mathbb{Z}\}$$

And let

$$N(a + bi) = (a + bi)(a - bi) = a^2 + b^2$$

which is the usual valuation on $\mathbb{C}$. Intuitively, this is because we have subrings $\mathbb{Z}[i] \subseteq \mathbb{C}$. These are called the *Gaussian Integers*. We'll show that Gaussian integers ($\mathbb{Z}[i]$) is a Euclidean Domain.

Consider $a, b \in \mathbb{Z}[i]$. We need to show that

$$a = kb + r \qquad N(r) < N(b)$$

for appropriate k, r. Since $\mathbb{Z}[i] \subseteq \mathbb{Q}(i)$, take

$$\frac{a}{b} = \frac{ac + bd}{c^2 + d^2} + \left(\frac{bc - ad}{c^2 + d^2}\right) = r + si$$

this gives us an idea of how close a is to b by having by looking at their fraction. Let p be the closest (or one of the 2 closest) integers to r and q be the closest (or one of the 2 closest) integers to s so that $|r - p|, |s - q| \leq \frac{1}{2}$. Then we'll show that

$$a = (p + qi)b + r$$

Whatis r? Notice that $t = (r-p) + (s-q)i$, so that $r = tb$, giving us $r = a - (p+qi)b$, showing that $r \in \mathbb{Z}[i]$. With this, we can calcualte the valuations quite easily! Notice that $N(t) = (r-p)^2 + (s-q)^2$ would at most

$$N(t) \leq \frac{1}{4} + \frac{1}{4} = \frac{1}{2}$$

and so, by the multiplicity of this valuation (TBD move that part earlier so that I can use it)

$$N(tb) = N(t)N(b) \leq \frac{1}{2}N(b) \Rightarrow N(t) < N(b)$$

showing that $\mathbb{Z}[i]$ is indeed a Euclidean Domain.

7. Another class of Euclidean domains come from polynomial rings over fields. The proof is a little more elaborate and so we shall leave it till page 358.

Notice how central defining a valuation is to showing a ring is a euclidean domain. This algorithm *almost* produces a divisibility condition, making Euclidean domains "close" to fields. However, they are not fields, since $\mathbb{Z}$ is a euclidean domain. So instead of every element having an inverse, for every pair of elements (a, b), we can find a number(s) c such that we have a notion of $c|a$ and $c|b$. If every pair of numbers has this condition, we have a Euclidean domain!

(Add here a proof of how the gcd works for Euclidean Domains)

We have not seen many examples of Euclidean domains. The following is a class of rings that shall provide a testing ground for finding more Euclidean domains and will come up a lot in chapter 50:

Example 10.2.6: Rings of Integers of Quadratic Extenions

Let D be a squarefree integer (it's prime-factorization has exactly one factor for each prime that appears in it, so $10 = 2 \cdot 5$ is square free, but $18 = 2^2 \cdot 3$ is not). Then we can define $\mathbb{Z}[\sqrt{D}] = \left\{a + b\sqrt{D} | a, b \in \mathbb{Z}\right\}$, which is a ring, even a subring, of the quadratic field $\mathbb{Q}(\sqrt{D})$ (that we defined in the first examples). If we take $D \equiv 1 \bmod 4$, then consider the slighty larger subset

$$\mathbb{Z}\left[\frac{1+\sqrt{D}}{2}\right] = \left\{a + b\frac{1+\sqrt{D}}{2} | a, b \in \mathbb{Z}\right\}$$

is also a subring. With this, define

$$\mathcal{O} = \mathcal{O}_{\mathbb{Q}(\sqrt{(D)})} = \mathbb{Z}[\omega] = \{a + b\omega | a, b \in \mathbb{Z}\}$$

where

$$\omega = \begin{cases} \sqrt{D} & \text{If } D \equiv 2, 3 \bmod 4 \\ \frac{1+\sqrt{D}}{2} & \text{If } D \equiv 1 \bmod 4 \end{cases}$$

called the *ring of integers* in the quadratic field $\mathbb{Q}(\sqrt{D})$. The idea being that the elements of $\mathcal{O}$ of the field $\mathbb{Q}(\sqrt{D})$ have many properties analogous to those of the subring of integers $\mathbb{Z}$ in the field of rational numbers. Or, if we have a "field extension", which in a diagram we can think of as:

$$\begin{tikzcd} \mathbb{F} \arrow[r, phantom, "\supseteq"] & O_{\mathbb{F}} \\ \mathbb{Q} \arrow[u] \arrow[r, phantom, "\supseteq"] & \mathbb{Z} \arrow[u] \end{tikzcd}$$

where the arrows represent inclusions, then we want $\mathbb{F}$ to share properties that $\mathbb{Z}$ has Given an extension field $\mathbb{F}$ for $\mathbb{Q}$, we identity rings of integers $O_{\mathbb{F}}$ a subrging of F analogous to $\mathbb{Z} \subset \mathbb{Q}$

Note also that $\mathbb{Z}[\omega] = \mathcal{O}_{\sqrt{(D)}}$ is an integral domain, and that it's field of fraction is $\mathbb{Q}(\sqrt{D})$, giving us another way of thinking of ho $\mathbb{Z}[\omega]$ is sort of the "natural" integral domain to choose.

In the special case where $= -1$, we get $\mathbb{Z}[i]$, called the *Gausian integers*, used to study powers of numbers in the complex plane.

The reason for defining this is because we can define the concept of a *field norm*. Let $N : \mathbb{Q}(\sqrt{D}) \to Q$ by

$$N(a + b\sqrt{D}) = (a + b\sqrt{D})(a - b\sqrt{D}) = a^2 - Db^2 \in \mathbb{Q}$$

which is non-zero if $a + b\sqrt{D} \neq 0$. This gives an idea of "size" in the field $\mathbb{Q}(\sqrt{D})$. For instance, when $D = -1$, then the valuation of $a + bi$ is $a^2 + b^2$ (as we know it from Complex Analysis!), which is the square of the length of this complex number considered as a vector in the compelx plane.

We can see that N is multiplicative, that is, $N(\alpha\beta) = N(\alpha)N(\beta)$ for all $\alpha, \beta \in \mathbb{Q}(\sqrt{D})$ given by

$$N(a + b\omega) = (a + b\omega)(a - b\overline{\omega}) = \begin{cases} a^2 - b^2 D & \text{if } D \cong 2, 3 \bmod 4 \\ a^2 + ab + \frac{1-\sqrt{D}}{2} & \text{if } D \cong 1 \bmod 4 \end{cases}$$

where

$$\overline{\omega} = \begin{cases} -\sqrt{D} & \text{if } D \equiv 2, 3 \bmod 4 \\ \frac{1-\sqrt{D}}{2} & \text{if } D \equiv 1 \bmod 4 \end{cases}$$

It follows that $N(\alpha) \in \mathbb{Z}$, meaning we can relate properties of $\mathbb{Z}$ to the field norm!

For example, we can use this norm to characetrize the units of $\mathbb{Z}[\omega]$. If $N(\alpha) = \pm 1$, then $(a + b\omega)^{-1} = (a + b\overline{\omega})$, which is an element of $\mathbb{Z}[\omega]$, and so α is a unit. Suppose, conversely, that α is a unit in $\mathbb{Z}[\omega]$, so $\exists \beta$ such that $\alpha\beta = 1$. Then since N is multiplicative,

$$N(\alpha)N(\beta)N(\alpha\beta) = N(1) = \pm 1$$

and since $N(\alpha)$, $N(\beta)$ must be integers, they must be ± 1, and so

$$\text{the element } \alpha \text{ is a unit in } \mathcal{O} \text{ if and only if } N(\alpha) = \pm 1$$

Thus, the field norm also gives us an easy way to find the units of a given ring of integers.

We'll be using the field norm to how for certain values of D, $\mathbb{Z}[\sqrt{D}]$ will be a Euclidean Domain, and not for others (ex. we've already shown that $\mathbb{Z}[\sqrt{-1}] = \mathbb{Z}[i]$ is a Euclidean Domain).

Euclidean domain ⇒ PID

As mentioned earlier, Euclidean Domains will be useful to factorizing elements in UFDs in which they are defined. Not all UFDs may have a valuation satisfying the euclidean algorithm, namely since being able to define such a va implies a good deal of structure on the ring. One important consequence of the existence of such a norm is that every Eucildean domain is a PID

Proposition 10.2.7: Euclidean Domain $\Rightarrow$ Pid

Given a Euclidean domain R, then every ideal of R is a principal ideal, meaning R is also a PID. In particular, if I is an ideal, then $I = (d)$, where d is any nonzero element of I of minimum norm

***Proof*:**
If I is the zero ideal, then we're done, so let $I \neq (0)$. This means that we can choose an element $d \in I$ such that $N(d)$ is a minimum element. This exists since $\{N(a) | a \in I\}$ has a minimum element by the well-ordering principle of $\mathbb{Z}$. We know show that $(d) = I$

($\subseteq$) it's clear that $(d) \subseteq I$ by the definition of ideals

($\supset$) Let $a \in I$ be any element of our ideal. Then since R is a euclidean domain, we can write a as :

$$a = qd + r$$

where $r = 0$ or $N(r) < N(d)$. Since $r = a - qd$, and $a, qd \in I$, then $r \in I$. But then, it's impossible that $N(r) < N(d)$ since $N(d)$ is already the minimal norm!! Thus $r = 0$, implying $a = qd$. But then $a = qd \in (d)$, as we sought to show

As usual, the contra-positive should be kept in mind: R is *not* an Euclidean Domain if it is *not* a PID.

Example 10.2.8: Implications

1. Since $\mathbb{Z}$ is a Euclidean domain, it is A Principal Ideal Domain. Note that we didn't show that PID $\Rightarrow$ Euclidean domain, we'll present a counter-example soon, so this was only a hint in the right direction
2. $\mathbb{Z}[x]$ is *not* a PID since $(2, x)$ is a non-principal ideal, and so not a euclidean domain.
3. $\mathbb{Q}[x]$ is a PID, and can be shown to be the euclidean domain with $N(p(x)) = \deg(p(x))$
4. $\mathbb{Z}[\sqrt{5}]$ is not a PID, and so not a Euclidean Domain

You might wonder if *all* PID's are euclidean domains. That need not be the case. To prove this, we'll first prove a another way of finding Euclidean domains:

Definition 10.2.9: Side Divisor

Let $\overline{R} = R^{\times} \cup \{0\}$ be the collection of units of R together with 0. An element $u \in R - \overline{R}$ is called a *universal side divisor* if for evry $x \in R$ there is some $z \in \overline{R}$ such that u divides $x - z$ in R, i.e., there is a type of "division algorithm" for u: every x may be written as $x = qu + z$ where z is either zero or a unit.

The existence of universal side divisors is weakening of the Euclidean condition, since we're saying that for any element $x \in R$, we can "divide" it by u and have a remainder of a unit z. These conditions are not as general as the conditions for the euclidean algorihtm (where z doesn't need to be constrained to be $N(z) = 1$). With this, we prove the following equivalent condition:

Proposition 10.2.10: Euclidean Domain Implies Universal Side Divisors

Let R be an integral domain that is not a field. If R is a euclidean domain, then there are universal side divisors in R

***Proof*:**
Suppose R is Euclidean with respect to some norm N and let u be an element $R - \overline{R}$ (which is non-empty since F is not a field), of minimal norm. For any $x \in R$ write $x = qu + r$ where r is etiher 0 or $N(r) < N(u)$. In either case the minimality of u implies $r \in \overline{R}$. Hence u is a univerasl side divisor in R

With this, we can find

Example 10.2.11: : PID but not euclidean domain
We'll take the following ring of integers.

$$\mathcal{O}_{Q(\sqrt{-19}} = \mathbb{Z}\left[\frac{1+\sqrt{-19}}{2}\right]$$

with the field norm:

$$N(a + b\sqrt{-19}) = (a + b\frac{1+\sqrt{-19}}{2})(a - b\frac{1+\sqrt{-19}}{2}) = a^2 + ab + 5b^2$$

This ring is a PID; a fact we'll show in the next chapter. However, it is not a Euclidean domain, for there does not exist a universal side divisor.

We can find that $\overline{R} = \{0, \pm 1\}$ are the unit of R union 0. Now suppose $u \in R$ is a universal side divisor. Note that if $b \neq 0$, then

$$a^2 + ab + 5b^2 = (a + \frac{b}{2})^2 + \frac{19}{4b^2} \geq 5$$

and so the smallest values of N on R are 1 (for units ± 1) and 4 (for $a = \pm 2$). Let's say $x = 2$. Then, we have

$$2 = qu + z \Rightarrow 2 = 2u + 0 \qquad \text{or} \qquad 2 = u \pm 1$$

or, written in division form, we see that $u|2 - 0$ or $u|2 \pm 1$.

Let's say $z = 0$, so $2 = qu$. Then $N(2) = 4 = N(q)N(u)$. TBD (it's a tedious example)

It is tempting to think that PIDs are close to being Euclidean domains as they share a notion of gcd and lcm, unique factorization. The following measure the degree of seperation of a Principal Ideal Domain being a Euclidean Domain:

Definition 10.2.12: Dedikend-Hasse Valuation

Define N to be the *Dedekind-Hasse valuation* if N is a positive norm and for every nonzero $a, b \in R$ either a is an element of the ideal (b) or there is a nonzero elemtn of the ideal (a, b) of norm strictly smaller than the norm of b (i.e., either b divides a in R or their exists $s, t \in R$ such that $0 < N(sa - tb) < N(b)$

Note that R is Euclidean with respect to a positive norm N if it is always possible to satisfy the Dedekind-Hasse norm with $s = 1$, so this is indeed a weakening of our condition for Euclidean Domains.

Proposition 10.2.13: Principal Ideal Domain If And Only If Dedekind-Hasse Norm

The integral domain R is a PId if and only if R has a Dedekind-Hasse nrom

***Proof*:**
p.281

Corollary 10.2.14: Defning Dedekind-Hasse Norm

let R be a Principal Ideal Domain. Then there exists a multiplicative Dedekin-Hasse norm.

***Proof*:**
p.289

Example 10.2.15
Showing that $\mathbb{Z}[\sqrt{-19}]$ is a Principal Ideal Domain but not a Euclidean Domain.

10.3 Application: Sums of Primes and Gaussian Integers

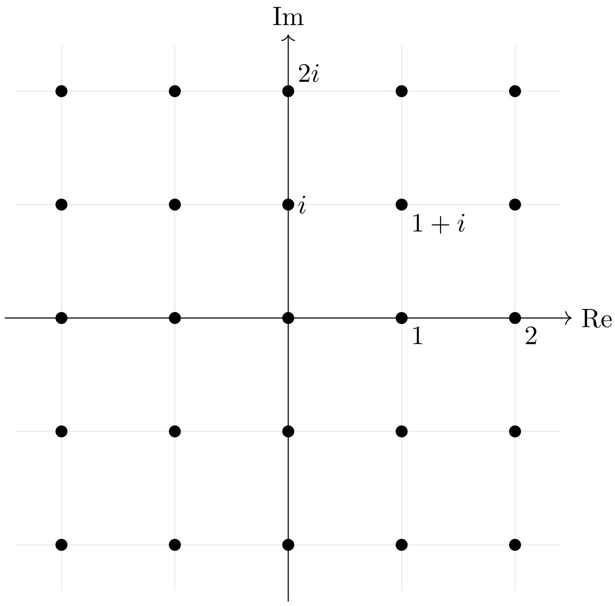
Let us use the theory we've developed to get our first result about prime numbers, namely let us classify all prime numbers that can be written as the sum of two squares $p = x^2 + y^2$. [9]

Recall that the Gaussian integers are of the form:

$$\mathbb{Z}[i] \subseteq \mathbb{C}$$

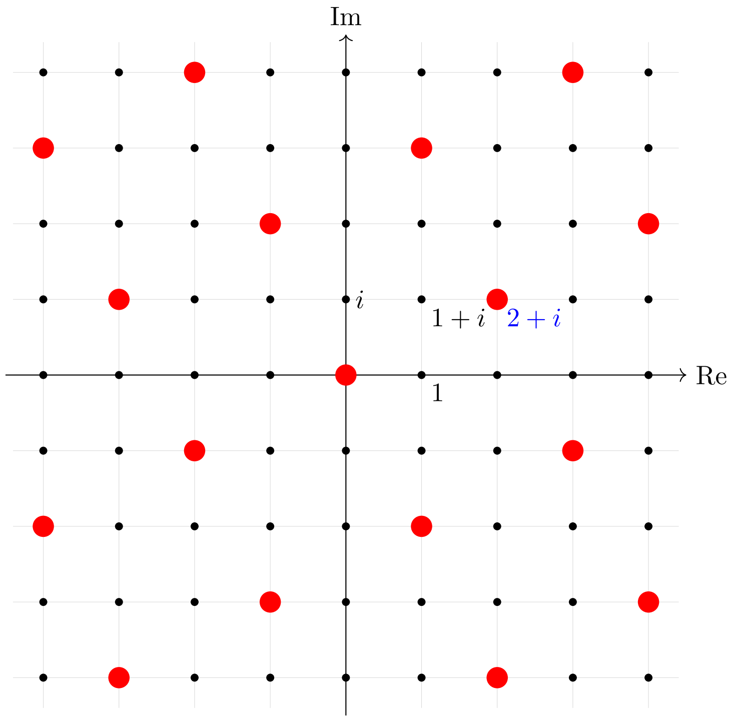
which can be thought of as the smallest ring containing $\mathbb{Z}$ and i. The Gaussian integers have a pretty geometric picture as a lattice:

[9] Naturally, a prime by definition cannot be the square of a natural number, however it can be $\mod q$ under the right condition; this is called *quadratic reciprocity* and is covered in section 50.4.6



The Gaussian Integers $\mathbb{Z}[i]$

This gives us a geometric picture of this ring! For example, we may visualize principal ideals in this grid, for example the ideal $(2 + i)$ is the point:



The Gaussian Integers $\mathbb{Z}[i]$ with $(2+i)$ highlighted

Recall how we introduced the field-norm on on quadratic integers, where $\mathbb{Z}[i] = \mathbb{Z}[\sqrt{-1}]$ is an example of quadratic integers. With the above visuals, it is more clear to see that the field norm has similar geometric information to the norm presented in real or complex analysis, namely:

$$N(a+bi) = (a+bi)(a-bi) = a^2 + b^2 = \|a+bi\|^2$$

Using the fact that the codomain of the field norm is $\mathbb{N}_{\geq 0}$, we can see that the norm is in fact a euclidean valuation

Lemma 10.3.1: Gaussian Integers Are Euclidean Domain

Let $R = \mathbb{Z}[i]$ and N be the field norm. Then N is a Euclidean valuation. Furthermore, N is multiplicative so that for every $z, w \in \mathbb{Z}[i]$,

$$N(zw) = N(z)N(w)$$

***Proof*:**
Multiplicity is immediate as:

$$N(zw) = (zw)(\overline{zw}) = (z\overline{z})(w\overline{w}) = N(z)N(w)$$

The proof that it is a euclidean valuation is straight-forward if not a little verbose, and so is left as an exercise.

What could be more insightful is to understand using the lattice picture the significant of the field norm on $\mathbb{Z}[i]$ (this was demonstrated in Aluffi's textbook see [2]). Take $z, w \in \mathbb{Z}[i]$ with $w \neq 0$ nd consider the lattice given by (w). Then either z lies on a point in this lattice, in which case $z = qw$ for appropriate q, or z is inside a "box" of the lattice. Pick some points qw that is a vertex of the box containing z, and set $r = z - qw$ so that $z = qw + r$. By visual inspection, we see that the length of r is ells than that of r, namely:

$$N(r) < N(w)$$

This trick shall not work for all quadratic fields, as we shall demonstrates after finishing elaborating on Gaussian integers.

It is a good exercise to show that if u is a unit then $N(u) = \pm 1$, from which we can conclude that the only possible units are $1, -1, i, -i$. We can use the norm to also detect irreducible elements. To see this, let' say $p \in \mathbb{Z}$ is a prime. Then since $\mathbb{Z}$ is an integral domain, p is irreducible, and so $p = ab$ if and only if a or b is a unit. Let's now take $\pi \in \mathcal{O}$ such that $p = N(\pi)$. If $\pi = \alpha \cdot \beta$, then:

$$p = N(\pi) = N(\alpha)N(\beta)$$

meaning either $N(\alpha)$ or $N(\beta)$ is a unit, meaning it's value is ± 1. Thus, we have that

If $N(\alpha)$ is $\pm p$ for a prime $p \in \mathbb{Z}$, then α is irreducible (even prime) in $\mathbb{Z}[i]$

The converse is not necessarily true: we can have a prime in $\mathbb{Z}[i]$ but it's norm is not p. Fortunately, this is a feature of primes in $\mathbb{Z}[i]$ and demonstrates a deeper insight on the structure of primes in $\mathbb{Z}[i]$:

Lemma 10.3.2: Gaussian Integers Norm of Primes

Let $\pi \subseteq \mathbb{Z}[i]$ be prime where. Then

$$N(\pi) \in \{p, p^2\}$$

for appropriate prime $p \in \mathbb{Z}$.

Fermat's theorem of sum of square (theorem 10.3.3) shall show that both p and p^2 show up as norms of primes.

***Proof*:**
Let $\pi \in \mathbb{Z}[i]$ be a prime element, that is, (π) is a prime ideal in $\mathbb{Z}[i]$. As π is not a unit, $N(\pi) \neq 1$. Then $(\pi) \cap \mathbb{Z}$ is also a prime ideal: if for $a, b \in \mathbb{Z}$, we have $ab \in (\pi)$ and $ab \in \mathbb{Z}$[a] then either a or b is an element of (π), so a or b is in $(\pi) \cap \mathbb{Z}$. Thus, there exists a prime p such that $(p) = (\pi) \cap \mathbb{Z}$.

Now, take $(p) \subseteq \mathbb{Z}[i]$. Then $(p) \subseteq (\pi)$ so that $xp = \pi$ for some $x \in \mathbb{Z}[i]$ (which may be a unit, as p may be an associate of π). Then $N(p) = p^2$, and since the norm is multiplicative

$$N(\pi) \big| N(p) = p\overline{p} = pp = p^2$$

from which it can be immediately deduced that $N(\pi) \in \{p, p^2\}$, completing the proof.

[a] if $a, b \in \mathbb{Z}$, automatically $ab \in \mathbb{Z}$

Note that irreducible does not imply prime, as it was shown with $\mathbb{Z}[\sqrt{-5}]$, however if the same trick is done as in the above proof on quadratic rings more generally it suggests that irreducible elements are associated to prime elements in $\mathbb{Z}$ via the norm map (see exercise ref:HERE). Under the right framework, this connection can be made more precise, which will be part [o]f the aim of chapter 50.

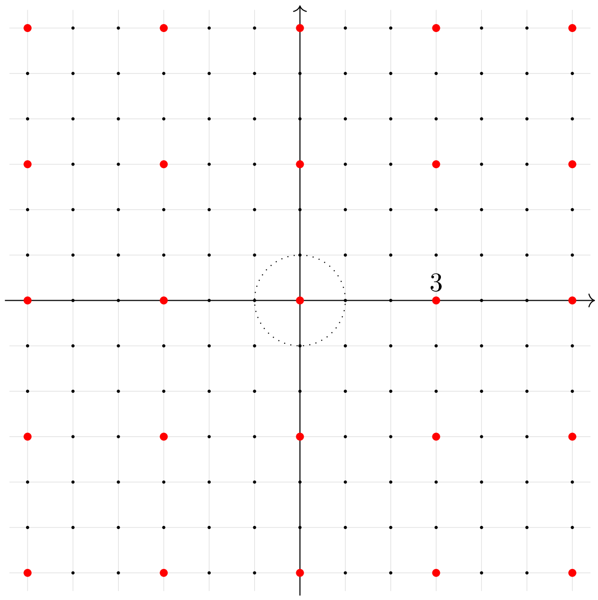
Let us find some examples of primes with norms p and p^2. Let's take $\alpha = a + bi \in \mathbb{Z}[i]$ to be prime. Then $N(\alpha) = \alpha\overline{\alpha} = a^2 + b^2 \in \{p, p^2\}$. If p is reducible in $\mathbb{Z}[i]$,then

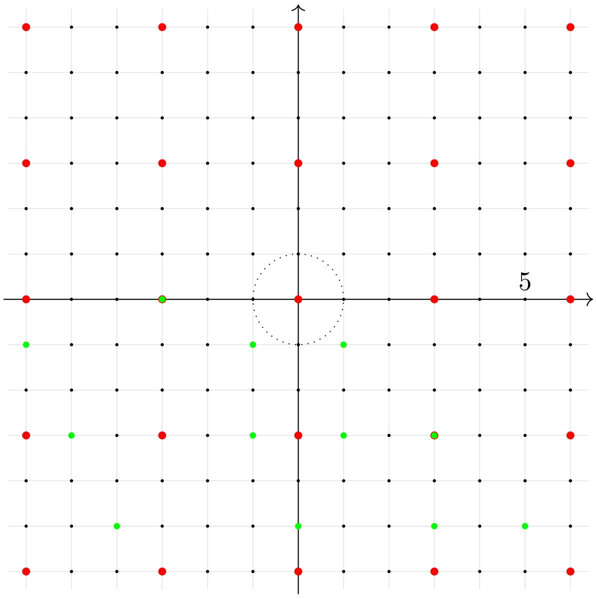
$$p = \pi\pi' \qquad \Rightarrow \qquad N(p) = N(\pi)N(\pi') = p^2$$

Thus, p will factor in $\mathbb{Z}[i]$ into two irreducibles if and only if $p = a^2 + b^2$, that is, p is the sum of two integers squares. If p is not the sum of two integer squares, it will remain irreducible. Take or example $3 \in \mathbb{Z}[i]$. Then $N(3) = 9$, and if 3 factored we would need an element $a + bi$ such that $N(a+bi) = a^2+b^2 = 3$ for natural numbers $a, b \in \mathbb{N}$. However, it can be quickly verified that no such numbers exist, and so 3 is prime in $\mathbb{Z}[i]$. For another example, take 2. Then as $2 = 1^2 + 1^2$, 2 factors in $\mathbb{Z}[i]$! In particular, $2 = (1 + i)(1 - i) = -i(1 + i)^2$ (note that $N(-i) = 1$ and so these two are associates). This will be the only case were conjugate irreducibles $a + bi$ and $a - bi$ can be associate. Let us next consider 5, so we need to see if there is an element $a + bi$ such that $N(a + bi) = 5$, and indeed:

$$N(2 + 1 \cdot i) = 2^2 + (1)^2 = 5$$

from which we find $5 = (2 + i)(2 - i)$. This factorization picture can be seen more clearly when viewing this elements via their ideals in the lattice. In the case of 3, there is no possible further refinement that can be done to the lattice, while for 5 we may further refine the lattice (without it being the entire lattice):





Let us now find a condition that shall let us immediately tell if a prime can splits in $\mathbb{Z}[i]$. First, any number in $\mathbb{Z}$ is congruent to $0, 1, 2, 3 \bmod (4)$. Therefore, the square of any number is congruent to either 0 or 1. The possible sums of square are thus $0, 1, 2 \bmod 4$. If the sum is $0, 2 \bmod 4$, the reader should check that if $p = a^2 + b^2$ then p would not be prime. Thus an odd prime (so $p > 2$) in $\mathbb{Z}$ that is the sum of two squares must be congruent to 1 mod 4 (ex. 5). Therefore, if $p \equiv 3 \bmod 4$ in $\mathbb{Z}$, then p is *not* the sum of squares, and remains irreducible (and thus prime) in $\mathbb{Z}[i]$.

Theorem 10.3.3: Fermat's Theorem on Sums Of Squares

The prime p is the sum of two integer squares, $p = a^2 + b^2$, $a, b \in \mathbb{Z}$ if and only if $p = 2$ or $p \equiv 1 \bmod 4$. Except for interchanging a and b or changing the sign of a and b, the representation of p as a sum of two squares is unique.

Furthermore, the prime elements in the Gaussian integers $\mathbb{Z}[i]$ are as follows:

1. $1 + i$ (which has norm 2)
2. the primes $p \in \mathbb{Z}$ with $p \equiv 3 \bmod 4$ (which have norm p^2)
3. $(a + bi)$, $(a - bi)$, the distinct irreducible factors of $p = a^2 + b^2 = (a + bi)(a - bi)$ for the primes $p \in \mathbb{Z}$ with $p \equiv 1 \bmod 4$ (both of which have norm p).

In other words, each prime $p \equiv 1 \bmod 4$ can be written as the sum of two squares:

$$p = a^2 + b^2$$

***Proof*:**
The case for $1+i$ has been done, so consider an odd prime $p \in \mathbb{Z}$. Then verify the following chain of implications:

$$\begin{aligned} p \text{ splits in } \mathbb{Z}[i] &\Leftrightarrow \frac{\mathbb{Z}[i]}{p\mathbb{Z}[i]} \text{ is not an integral domain} \\ &\Leftrightarrow \frac{\mathbb{Z}/p\mathbb{Z}[x]}{(x^2+1)} \text{ is not an integral domain} \\ &\Leftrightarrow x^2+1 \text{ is not irreducible in } \mathbb{Z}/p\mathbb{Z}[x] \\ &\Leftrightarrow x^2+1 \text{ has a root in } \mathbb{Z}/p\mathbb{Z}[x] \\ &\Leftrightarrow \exists\, n \in \mathbb{Z} \text{ such that } n^2 \equiv -1 \bmod p \end{aligned}$$

Thus, the splitting of p has been reduced to solving the congruence equation $n^2 \equiv -1 \bmod p$. To verify this condition, recall that $(\mathbb{Z}/p\mathbb{Z})^*$ is as p is prime, and it has order $p-1$. As p is odd, $p-1$ is even, so there is a generator $g \in (\mathbb{Z}/p\mathbb{Z})^*$ such that $|g| = 2\ell$ for some appropriate natural number ℓ. Then write $[n] = g^m$ for appropriate m. Now, certainly $[-1] \in (\mathbb{Z}/p\mathbb{Z})^*$ creates an (unique) order 2 subgroup, and g^ℓ also creates an order 2 subgroup, and so:

$$g^\ell = [-1]$$

Then $n^2 \equiv -1 \bmod p$ if and only if $g^{2m} = g^\ell$, that is if and only if

$$2m \equiv \ell \bmod 2\ell$$

Now this linear congruence relation is readily solvable: it is true if and only if ℓ is even, and so when $(p-1) = 2\ell$ is a multiple of 4, or shuffled around:

$$p \equiv 1 \bmod 4$$

giving us an exact condition for when p can be factored, equivalently when p can be written as the sum of two squares, completing the proof.

Lemma 10.3.4: Writing p as n^2+1

The prime number $p \in \mathbb{Z}$ divides an integer of the form n^2+1 if and only if p is either 2 or is an odd prime congruent to 1 mod 4.

***Proof*:**
By Fermat sum of square's theorem, we know that there exists an n such that $n^2+1 \equiv 0 \bmod p$, which is equivalent to saying p divides n^2+1.

Though this condition is equivalent to Fermat' sum of Square's Theorem, it is not as informative as $5|3^1+1=10$, however $3+i$ and $3-1i$ are *not* prime (for example, $3+i = (1+i)(2-i)$). There is however something very interesting that was demonstrated in this entire section: we have answered a purely number theoretic question about which primes can be expressible as sums of squares (a fact that should at first not be so obvious) by taking advantage of a lot of simple theoretical properties built-up about factorization and of cyclic groups!

10.3.5 Foreshadowing Fermat's Theorem of sums of two squares will be generalized to the factorization of any polynomial over any finite extension of a ring (theorem 50.4.7), though the strategy of writing a prime p as some polynomial expression is in general more difficult.

Exercise 10.3.1

1. I think I can do an exercise finding solutions to $y^2 = y^3 - 1$ eh? I must've made a typo here, I'll revisit

11

Exploring Polynomial Rings

When first studying algebra, we often looked to find solutions to linear and quadratic equations:

$$ax + b = 0 \qquad ax^2 + bx + c = 0$$

This naturally generalizes to polynomial of higher degree. In pre-abstract algebra, solutions to such equations where usually limited to $\mathbb{R}$, probably $\mathbb{C}$. We have now developed the abstraction to consider the value (or root) of such a polynomial for an arbitrary ring R, that is for an arbitrary polynomial in $R[x]$, does there exist an element $r \in R$ such that

$$p(r) = 0$$

As is the usual strategy in algebra, we generalize a problem into an algebraic object, in this case the polynomial ring $R[x]$, and look for interesting properties of the objects and tools we can use to manipulate the object. Such polynomials rings are paramount to study. To give an idea of their importance:

1. they arbitrarily approximate continuous functions on compact neighborhoods
2. all complex differentiable function can be locally be represented as infinite polynomials (i.e. they are analytic)
3. All relations in rings are represented by polynomial (similarly to how all relations in a group is represented by a monomial)

The last point shows that polynomial rings are used to "locally" represent rings, namely through the universal property of [commutative] universal rings (see theorem 9.2.4). It may thus seem natural to consider studying polynomials rings themselves to see what properties we may deduce from them.

One necessary limitation of the exploration of polynomial rings in chapter is that we shall not look at finding solutions to polynomials, we shall rather look at the structure of polynomial rings. The reason for this is that it is in fact a much more difficult task to find solutions requiring more build-up that starts venturing into the domain of algebraic geometry. The beginnings of this effort shall be presented in chapter 31.

It must be noted that we shall further develop on poomial rings in *Commutative Algebra* [10]. This exploration is left to that book since it is not necessary to cover some results until much later.

Note that this chapter shall use some exercises from exercise 9.3.1, most importantly:

Proposition 11.0.1: Ideals of Polynomial Ring wrt. R

Let $I \subseteq R$ be an ideal. Then:

$$(R[x])/(IR[x]) \cong (R/I)[x]$$

***Proof*:**
exercise 9.3.1[3]

11.1 Polynomial Rings over Fields

We'll now explore polynomial rings with coefficients from a field $\mathbb{F}$. The first question we may ask if what type of ring is $\mathbb{F}[x]$? It's not a field, since we already saw that x is not invertible (meaning it's also not a division ring). The next strongest ring we know is Euclidean domains. If $\mathbb{F}[x]$ is a euclidean domain, there must be a norm we define, which we'll use $N(p(x)) = \deg(p(x))$ (the usual polynomial norm). Then $\mathbb{F}[x]$ is indeed a Euclidean Domain:

Theorem 11.1.1: Field Then $\mathbb{F}[x]$ Is Euclidean Domain

Let $\mathbb{F}$ be a field. The $\mathbb{F}[x]$ is a Euclidean Domain. In particular, if $a(x)$ and $b(x)$ are two polynomials in $\mathbb{F}[x]$ with $b(x)$ nonzero, then there are *unique* $q(x)$ and $r(x)$ such that

$$a(x) = q(x)b(x) + r(x_0) \qquad \text{with} \qquad r(x) = 0 \text{ or } \deg(r(x)) < \deg(b(x))$$

***Proof*:**
First, the edge cases:

If $a(x)$ is the zero polynomial, then let $q(x) = r(x) = 0$. So let $a(x)$ be a nonzero polynomial. Let $b(x)$ be an arbitrary other polynomial. If $\deg(a(x)) < \deg(b(x))$, then let $q(x) = 0$ and $r(x) = a(x)$.

Now, take the nonzero polynomial $a(x)$ such that $\deg(a(x)) > \deg(b(x))$. We'll do a proof by induction on the degree of $a(x)$ ($n = \deg(a(x))$).

For visualization, we can denote $a(x)$ and $b(x)$ as:

$$a(x) = a_n x^n + a_{n-1}x^{n-1} + \cdots + a_0 \qquad b(x) = b_m x^m + b_{m-1}x^{m-1} + \cdots + b_0$$

We want to apply the induction hypothesis. Let $a'(x) = a(x) - \frac{a_n}{b_m}x^{n-m}b(x)$, which purposely defined to eliminate the monomial of degree n from $a(x)$. Then by the induciton hypothesis:

$$a'(x) = q'(x)b(x) + r(x) \qquad r(x) = 0 \text{ or } \deg(r(x)) < \deg(p(x))$$

Then, we "reverse" the step of taking $a'(x)$ by doing $q(x) = q'(x) + \frac{a_n}{b_n}x^{n-m}$, which will give us

$$a(x) = q(x)b(x) + r(x) \qquad r(x) = 0 \text{ or } \deg(r(x)) < \deg(p(x))$$

showing we can factor two polynomials.

It remains to show that this factorization is unique. Suppose $q(x), r(x)$, and $q_0(x)$, $r_0(x)$ are both polynomials that satisfy the euclidean condition. Then

$$a(x) - q(x)b(x) \qquad a(x) - q_0(x)b(x)$$

are both of degree less than m. The difference between these two polynomials then:

$$b(x)(q(x) - q_0(x))$$

is also less than m. But the degree of product of polynomials over an integral domain is the sum of their degree $(m > \deg(b(x)(q(x)-q_0(x))) = \deg(b(x)+\deg(q(x)-q_0(x))$. So, $q(x)-q_0(x) = 0$, implying $q(x) = q_0(x)$. This then implies $r(x) = r_0(x)$, completing the proof of uniqueness

Corollary 11.1.2: Field Then $\mathbb{F}[x]$ PID And UFD

Let $\mathbb{F}$ be a field. The $\mathbb{F}[x]$ is a Principal Ideal domain, and a Euclidean Domain

***Proof*:**
By the previous theorem (theorem 11.1.1), we know $\mathbb{F}[x]$ is a euclidean domain. By 10.2.7, it is a Principal Ideal Domain, and since PID's are UFD's, it is also a Unique Factorization Domain

Example 11.1.3

1. $\mathbb{Q}[x]$ is a Principal Ideal Domain since $\mathbb{Q}$ is a field.
2. $\mathbb{Q}[x, y]$ is not a Principal Ideal Domain, since $\mathbb{Q}[x]$ is not a field (using the contra-positive).

One final fact we'll introduce as a result of what we talked about:

Corollary 11.1.4: Kernel Of Evaluation Map

Let $\mathbb{F}$ be a field and let α be an element of $\mathbb{F}$. Then the kernel of ev_α is the ideal $(x - \alpha)$

***Proof*:**
Let $I = \ker(\text{ev}_\alpha)$. By definition of ev_α, $x - \alpha \in I$. Now, since $\mathbb{F}$ is a field, $\mathbb{F}[x]$ is a Euclidean Domain, and thus a Principal Ideal Domain, so I is a principal ideal. Let's say $I = (p)$ for some $p \in \mathbb{F}[x]$. Then by definition, $x - \alpha = up$, meaning $p | x - \alpha$. Note that p cannot have degree

higher than 2, since then $x - \alpha \notin I$. So, if p has degree one, then the two elements are associates, meaning $I = (p) = (up) = (x - \alpha)$. If p has degree 0, then it must be a constant. However, p has a root at α, so if p were a constant, it would have to be 0. But then $x - \alpha \notin (0)$ – a contradiction. Therefore $(x - \alpha) = \ker(\mathrm{ev}_\alpha)$.

Here is also a fun result I want to put down:

Example 11.1.5: generalization of Famous infinite primes proof
Let F be a finite field. Then there are infinitely many prime elements in $F[x]$

Proof. Let's assume there is a finite amount. Then $p_1, ..., p_n$ is our list of irreducible polynomials. Note that $\deg(p_i) \geq 1$ for each polynomial. Then take $P = p_1 p_1 \cdots p_n + 1$. Then $\deg(P) \geq 1$, and so P is not a unit or 0. Therefore, P must be divisible by some p_i, meaning $p_i | P$, which implies $qp_i = P$, or equivalently $P \in (p_i)$. Next, by closure of ideals, $-p_1 p_2 \cdots p_n \in (p_i)$. Then, by closure under addition:

$$P - p_1 p_2 \cdots p_n = p_1 p_2 \cdots p_n + 1 - p_1 p_2 \cdots p_n = 1 \in (p_i)$$

meaning $up_i = 1$ for appropriate $u \in F[x]$, implying p_i is a unit! However, prime elements are *not* units. Since p_i was arbitrary, *no* p_i can divide P. But then, P will have non of the factor's in it's factorization, a contradiction to $\deg > 0$ and the p_i's being the only irreducibles! Therefore, P must be a new prime number!

Therefore, there is an infinite number of prime numbers. □

We finish off with an application of the Chinese remainder theorem with the new fact that $k[x]$ is a Euclidean domain, and hence a UFD:

Proposition 11.1.6: Chinese Remainder On Polynomials

Let $g(x)$ be a nonconstant monic element of $k[x]$ and let

$$g(x) = f_1(x)^{n_1} f_2(x)^{n_2} \cdots f_k(x)^{n_k}$$

be its factorization into irreducibles, where the $f_i(x)$ are distinct. Then we have the following isomorphism of rings:

$$k[x]/(g(x)) \cong k[x]/(f_1(x)^{n_1}) \times k[x]/(f_2(x)^{n_2}) \times \cdots \times k[x]/(f_k(x)^{n_k})$$

***Proof*:**
This follows from the Chinese Remainder Theorem (theorem 9.6.6), since $(f_i(x)^{n_i})$ and $(f_i(x)^{n_i})$ are comaximal if $f_i(x)$ and $f_j(x)$ are distinct (they are relatively prime in the Euclidean Domain $k[x]$, hence $gcd(f_i^{n_i}, f_j^{n_j}) = 1$, and so $p(x)f_i(x)^{n_i} + q(x)f_j(x)^{n_j} = 1$, so $1 \in (f_i^{n_i}) + (f_j^{n_j})$ implying $(f_i^{n_i}) + (f_j^{n_j}) = k[x]$ by propositin 9.4.10).

Exercise 11.1.1

1. Let $R[x]$ be a Principal Ideal Domain. Show that $R[x]$ is a Euclidean Domain.

11.2 Polynomial Rings over UFD's

We've seen R is an integral domain then $R[x]$ is an integral domain. We've also seen that if R is an integral domain, than we can embed R into a field of fraction F (see section 9.5). This means that $R[x] \subseteq F[x]$. Furthermore, since F is a field, then $F[x]$ is a euclidean domain. With this, we can do lots of our computation in $F[x]$ instead of $R[x]$, at the expense of allowing fractional coefficients. The question then arises if we can use information from $F[x]$ to gain information about $R[x]$!

For example, take a polynomial $p(x) \in R[x]$. Since $p(x) \in k[x]$, and $k[x]$ is a Unique Factorization Domain, then $p(x)$ factors in $k[x]$ into irreducibles. We can then ask if $p(x)$ will factor in $R[x]$. The answer is no in general, since $R[x]$ is not always a Unique Factorization Domain, since that would imply R is a Unique Factorization Domain since the coefficients of $R[x]$ must also factor uniquely. However, if R were a unique factorization domain, then this will actually imply that $R[x]$ is Unique Factorization Domain! Hence, factoring in $k[x]$ gives information on how we factor in $R[x]$. The way we'll prove this is first by uniquely factoring in $k[x]$, then "clear denominators" to obtain unique factorization in $R[x]$!

The first thing to do is make precise the notion of comparing factorization of a polynomial in $k[x]$ and a factorization in $R[x]$

Proposition 11.2.1: Gauss's Lemma

Let R be a Unique Factorization Domain with field of fraction K and let $p(x) \in R[x]$. If $p(x)$ is reducible in $K[x]$ then $p(x)$ is reducible in $R[x]$. More precisely, if $p(x) = A(x)B(x)$ for some non constant polynomials $A(x), B(x) \in K[x]$, then there are nonzero elements $r, s \in F$ such that $rA(x) = a(x)$ and $sB(x) = b(x)$, such that $a(x), b(x) \in R[x]$ and $p(x) = a(x)b(x)$ is a factorization in $R[x]$

A concrete case that can be kept in mind is with the polynomials $p(x), q(x) \in \mathbb{Q}[x]$, then $rp(x), sq(x) \in \mathbb{Z}[x]$ for appropriates $r, s \in \mathbb{Q}$. Note that it is possible that $q(x) \in \mathbb{Q}$, i.e. $q(x)$ is a constant polynomial[1].

***Proof*:**
Let's say $p(x) \in k[x]$ and is reducible $p(x) = A(x)B(x)$. Then the coefficients of $p(x)$ are quotients from R. Since R is a Unique Factorization Domain, let's take d a number common betewen all denominators for all the coefficients of $A(x)B(x)$, so that we can obtain

$$dp(x) = a'(x)b'(x)$$

where $a'(x), b'(x)$ are now in $R[x]$ since we multiplied away the denominators with d! If d is a unit, then

$$a(x) = d^{-1}a'(x) \qquad b(x) = b'(x)$$

which completes the proof, so let's assume d is not a unit. Since $d \in R$, and R is a Unique Factorization Domain, then there is a unique factorization into irreducibles:

$$d = p_1 \cdots p_n$$

[1] In textbooks where the contra-positive of this statement is used, it is often specified that $p(x)$ is *primitive*, meaning for all principal prime $\mathfrak{p} \subseteq R$, $p(x) \notin \mathfrak{p}R[x]$; in the UFD case, this amounts to the gcd of the coefficients being 1

Since p_1 is an irreducible element, and R is a Unique Factorization Domain, then $(p_1) = p_1R[x]$ (since $R[x]$ is commutative $(a) = aR$) is prime (exercise), and so $R[x]/p_1R[x]$ is an integral domain (proposition 9.4.25) and $R[x]/p_1R[x] \cong (R/p_1R)[x]$ (exercise).

Now, reducing $dp(x) = a'(x)b'(x)$ modulo p_1, we get

$$\overline{0} = \overline{a'(x)} \cdot \overline{b'(x)}$$

Since $(R/p_1R)[x]$ is an integral domain, then either $\overline{a'(x)}$ or $\overline{b'(x)}$ is zero, say $\overline{a'(x)} = 0$. This would imply $p_1|a'(x)$, and so every coefficient of $a'(x)$ is divisible by p_1, meaning $\frac{1}{p_1}a'(x)$ also has coefficients from R! So, we can divide both sides by p_1 (that is d by p_1 and either $a'(x)$ or $b'(x)$ by p_1). But then, the factor d on the left hand side has one fewer irreducible element! We can continue this procedure till we cancel out all of the factors, leaving us with $p(x) = a(x)b(x)$, with $a(x), b(x) \in R[x]$, $a(x)$ and $b(x)$ being F-mutiples of $A(x)$ and $B(x)$!

Note that $A(x)$ and $B(x)$ aren't necessarily R-multiples, nor are they necessarily unique. For example, $x^2 \in Q[x]$ is reducible, and so reducible in $\mathbb{Z}[x]$. One factorization would be $A(x) = 2x$ and $B(x) = \frac{1}{2}x$. However, $\frac{1}{2} \notin \mathbb{Z}$: no integer multiple give's us a factorization of $A(x)$ and $B(x)$. What the theroem will tell us is that $\frac{1}{2}A(x) \in \mathbb{Z}[x]$, and $2B(x) \in \mathbb{Z}[x]$.

Since k is a field, the elements of R becomes units in $k[x]$ (the units in $k[x]$ being the non-zero constants). For example, $7x$ factors in $\mathbb{Z}[x]$ and produces two irreducible 7 and x, so $7x$ is *not* irreducible in $\mathbb{Z}[x]$. However, in $\mathbb{Q}[x]$, $7x$ is a unit 7 times an irreducible, so $7x$ *is* irreducible in $\mathbb{Q}[x]$. However, this is the *only* difference between the irreducible elements in $R[x]$ and $k[x]$

Corollary 11.2.2: Irreducibility In $R[x]$ And $k[x]$

Let R be a UFD, let $k = K(R)$ be its field of fractions, and let $p(x) \in R[x]$. Suppose the greatest common divisor of the coefficients of $p(x)$ is 1. Then $p(x)$ is irreducible in $R[x]$ if and only if it is irreducible in $K[x]$. In particular, if $p(x)$ is a monic polynomial that is irreducible in $R[x]$, then $p(x)$ is irreducible in $K[x]$.

***Proof*:**
We'll proceed by the contrapositive: $p(x)$ is reducible in $R[x]$ if and only if $p(x)$ is reducible $K[x]$. The ($\Leftarrow$) direction is Gauss's Lemma, so let's go ($\Rightarrow$). Let's say $p(x)$ is reducible in $R[x]$ and the greatest comon divisor of the coefficients is 1. That means that when we reduce $p(x) = a(x)b(x)$, neither $a(x)$ or $b(x)$ can be constants! Note that polynomials which aren't constants in $K[x]$ are not units, meaning $p(x) = a(x)b(x)$ is also a factorization in $K[x]$, completing the proof!

This theorem is already incredibly powerful, it means that the task of finding the irreducible elements of $R[x]$ reduces to finding the irreducible elements of $k[x]$ (which we know is a euclidean domain) up to a multiple of R[2]. We shall later find more techniques for finding solutions to polynomials over fields, which by our above results immediately give us a way to translate these results for integral domains whose field of fraction is the given polynomial ring over a field.

Yet another result is the following which shows that extending a ring to form a polynomial ring preserves UFD!

[2]For those keen on doing algebraic geometry, this process of reducing the problem of factoring in $R[x]$ to factoring in $k[x]$ can be thought of as a form of using the stalk at 0 to find information about the elements of $R[x]$, that is it is taking advantage of local information for global results TBD

Theorem 11.2.3: R UFD If And Only If $R[x]$ UFD

R is a Unique Factorization Domain if and only if $R[x]$ is a Unique Factorization Domain

***Proof*:**
If $R[x]$ is a Unique Factorization Domain, then the constants of $R[x]$ must uniquely factor, meaning R is a Unique Factorization Domain.

So let's suppose R is a Unique Factorization Domain, and take $p(x) \in R[x]$. If $p(x)$ is a constant, then it's uniquely factorizable, so let's say $\deg(p(x)) > 0$. Let d be the g.c.d. of the coefficients of $p(x)$, so that $p(x) = dp'(x)$, meaning the g.c.d. of the coefficients of $p'(x)$ is 1. Note that this factorization is unique up to a unit multiple of d (i.e. up to a unit in R by proposition 9.2.3), and since d can be factored uniquely in R (and these are also irreducibles in $R[x]$ too), it suffices now to prove that $p'(x)$ can be uniquely factored into irreducibles in $R[x]$.

Let k be the field of fractions of R. Then since $k[x]$ is a Unique Factorization Domain (proposition 11.1.1), $p(x)$ can be factored uniquely into irreducibles in $k[x]$. By Gauss's Lemma, such a factorization implies a factorization in $R[x]$ whose factors are F-multiples of the factors in $k[x]$. Since the g.c.d. of $p'(x)$ is 1, the g.c.d. of the coefficients of each of these factors in $R[x]$ must be 1. By corollary 11.2.2, each of these factors are irreducible in $R[x]$! Hence $p(x)$ can be written as the product of irreducibles in $R[x]$.

The uniqueness of the factorization of $p(x)$ follows from the uniqueness in $k[x]$. Suppose

$$p'(x) = q_1(x)q_2(x)\cdots q_n(x) = q_1'(x)q_2'(x)\cdots q_m'(x)$$

Since the g.c.d. of the coefficients of $p'(x)$ is 1, so to it is for both factorizations. By corollary 11.2.2, each $q_i(x)$ and $q_j'(x)$ is an irreducible in $k[x]$. By uniqueness of factorization on $k[x]$, $n = m$, and the up to rearrangement, the two factorizations are the same up to associates, that is $q_i(x) = uq_i'(x)$ for a unit $u \in k[x]$. It remains to show they are associates in $R[x]$. By proposition 9.2.3, the untis of $k[x]$ are the units of F, so

$$q_i(x) = \frac{a}{b}q_i'(x)$$

for $a, b \in R$. Thus, $bq_i(x) = aq_i'(x)$. Thinking about it another way, the g.c.d. of the coefficients of the polynomial on the left hand side is b, and a on the right hand side respectively. Since in a Unique Factorization Domain, the coefficients of a nonzero polynomial is unique up to units ($a = ub$ for a unit in R), then $q_i(x) = uq_i'(x)$, and so $q_i(x)$ and $q_i'(x)$ are associates in R as well, completing the proof!

Corollary 11.2.4: Polynomial Rings in Several Variables over a UFD are UFDs

If R is a Unique Factorization Domain, then a polynomial ring in an arbitrary number of variables with coefficients in R is also a UFD

***Proof*:**
For the finitely many case, apply induction on theorem 11.2.3, since $R[x, y] = R[x][y]$, and $R[x]$ is a Unique Factorization Domain. The general case follwos fromthe definition of a polynomial ring in an arbitrar ynumber of variables as the union of polynomial rings in finitely many variables.

Example 11.2.5

1. $\mathbb{Z}[x], \mathbb{Z}[x, y]$, and so on are Unique Factorization Domains. The ring $\mathbb{Z}[x]$ now gives a good example of a Unique Factorization Domain that is *not* a Principal Ideal Domain!
2. Similarly, $\mathbb{Q}[x]$, $\mathbb{Q}[x, y]$, and so forth are Unique Factorization Domains.

Finally, we'll ask if we can weaken the condition's and still get the same result. What if R was just an integral domain? Can we follow a similar construction? Generalizing in a different direction, if R is a Unique Factorization Domain, does that imply that $R[[x]]$ is a Unique Factorization Domain? The answer to both is in general no

Example 11.2.6: Counter-Example

1. Let $R = \mathbb{Z}[2i]$, and let $p(x) = x^2 + 1$. Then the field of fractions of R is $F = \mathbb{Q}[2i]$. The polynomial $p(x)$ factors uniquely into a product of two linear facrors in $k[x] : x^2 = (x-i)(x+1)$. so in particular $p(x)$ *is* reducible in $\mathbb{Q}[2i]$. However, Neither of these factors lies in $R[x]$ (because $i \notin R$) so $p(x)$ is *irreducible* in $R[x]$. In particualr, by corollary 11.2.2, $\mathbb{Z}[2i]$ is *not* a Unique Factorization Domain! If it were, then it should've been reducible!
2. It is a rather technical project to show that if R is a UFD, then $R[[x]]$ is a UFD. The somewhat canonical counter-example is

$$R = \frac{k[x, y, z]}{(x^2 + y^3 + z^7)}$$

See (citation commented) for an in-depth explanation[a].

[a] this paper does require some algebraic knowledge introduced in part V

Exercise 11.2.1

1. Let $x^5 - x + 1 \in \mathbb{Q}[x]$. Is this polynomial irreducible? (Hint: use Gauss's Lemma)
2. Recall that if R Is a UFD, it is not necessarily the case that $R[[x]$. Show that $\mathbb{R}[[x]]$ is a UFD

11.3 Irreducibility of Polynomials

Now that we've shown R, $R[x]$, $R[x, y]$ and so forth are Unique Factorization Domains if and only if R is a Unique Factorization Domain and that irreducibility over $R[x]$ is symmetrically reflected to irreducibility over $k[x]$, we now raise the question about finding irreducible elements in such a ring (i.e. the prime elements – the "building blocks"). A polynomial $p(x)$ is reducible if we can find two polynomials of smaller degree such that $p(x) = a(x)b(x)$. A polynomial is irreducible if no such polynomials exist. In general this is difficult, which is why we look for nicer irreducibility criteria.

Throughout these proofs, we shall be taking advantage of the fact that the field of fractions is a UFD, hence the polynomial ring over the field is a UFD. We shall not even require the ring that we are using to construct the field of fraction to be a UFD, simply an integral domain. To demonstrate, consider the following key fact:

Proposition 11.3.1: Finding Linear Factors

Let R be an integral domain and let $p(x) \in R[x]$. Then $p(x)$ has a factor of degree one if and only if $p(x)$ has a root in F, i.e., there is an $r \in R$ such that $p(\alpha) = 0$

***Proof*:**
Let $p(x) = (x - r)q(x)$ for some $r \in R$, $q(x) \in R[x]$. Then certainly

$$p(r) = (r - r)q(r) = 0q(r) = 0$$

Hence, r is a root. Conversely, if $p(r) = 0$ for some $r \in R$, take the field of fractions and consider $k[x]$. Then by the division algorithm in $k[x]$ we may write

$$p(x) = q(x)(x - a) + r$$

Since $p(r) = 0$, then r must be 0. Finally, multiply $q(x)$ by an appropriate α to clear the denominators so that $\alpha q(x) = Q(x) \in R[x]$, and hence we get

$$p(x) = Q(x)(x - r)$$

as we sought to show.

Corollary 11.3.2: Maximum Number Of Roots

[Maximum Number Of Roots] Let R be an integral domain and $p(x) \in R[x]$ of degree n. Then $p(x)$ has *at most* n roots.

***Proof*:**
Let $k = K(R)$ be the field of fraction of R. Then the number of roots of $p(x) \in [R]x$ is certainly less than or equal to the number of roots in $p(x) \in k[x]$. Now, *a fortiori* k is a UFD, hence by theorem 11.2.3 $k[x]$ is a UFD, and so each root corresponds to an (irreducible) factor of degree 1. But then, f can have at most n factors of degree 1 by unique factorization, completing the proof.

Note that this statement can fail if R is not an integral domain!

Example 11.3.3: Polynomial With More Roots And Its Degree
[Polynomial With More Roots And Its Degree] Let $x^2 + x \in (\mathbb{Z}/6\mathbb{Z})[x]$. Show that this polynomial has 6 roots!

However, using proposition 11.3.1, we may show that polynomials over infinite integral domains are determined by their roots!

Corollary 11.3.4: Polynomials Detremiend By Evaluation

[Polynomials Detremiend By Evaluation] Let R be an infinite integral domain and $f, g \in R[x]$. Then $f = g$ if and only if

$$r \mapsto f(r) \qquad \text{and} \qquad r \mapsto g(r)$$

agree for all $r \in R$

***Proof*:**
Two functions agree if every $r \in R$ is a root of $f - g<$ but a nonzero polynomial over R cannot have infinitely many roots by corollary 11.3.2, hence they must be equal as we sought to show.

Continuing from this, we may ask about the irreducibility of degree 2 and 3 polynomials, which is an immediate consequence of the above results:

Proposition 11.3.5: Degree 2 Or 3 Polynomials

A polynomial of degree two or three over a field F is reducible if and only if it has a root in F.

***Proof*:**
This follow's immediately from the previous proposition, see exercise ref:HERE

The reason for this is that if we have a degree higher than 4, then we can reduce to two irreducible polynomials of degree 2, like $(x^2 + 1)(x^2 + 1)$ in $\mathbb{Q}[x]$. For higher degree polynomials, we may have more solutions depending on what type of field we are working over. For example, we shall introduce in chapter 19, section 19.5.1, the existence of field for which only linear polynomials are irreducible ($\mathbb{C}$ being a canonical example of such a field). Yet another example is that any polynomial of degree ≥ 3 over $\mathbb{R}$ is reducible[3]. For more general rings, it is in general much more difficult to determine if a polynomial is reducible. There are however some tricks that help in arbitrary situations, especially if R is a UFD (for then we may use the gcd)

This next proposition let's us limit the number of "rational" solutions of a polynomial over a UFD.

Proposition 11.3.6: Rational Roots Test

Let R be a UFD and $k = K(R)$. Let $p(x) = a_n x^n + a_{n-1}x^{n-1} + \cdot + a_1 x + a_0$ be a polynomial of degree n with coefficients in the UFD R. If $r/s \in k$ with $gcd(r, s) = 1$, and r/s is a root of $p(x)$ ($p(r/s) = 0$), then r divides the constant term and s divides the leading coefficient of $p(x)$: $r \big| a_0$ and $s \big| a_n$. In particular, if $p(x)$ is a monic polynomial with integer coefficients and $p(d) \neq 0$ for all integers d dividing the constant term of $p(x)$, then $p(x)$ has no roots in $k[x]$.

[3]This is exercise ref:HeRE (closure of fields, 3rd exercise). An astute student would be able to work out this exercise

***Proof*:**
Let's say $p(r/s) = 0$. Then

$$p(r/s) = 0 = a_n(r/s)^n + a_{n-1}(r/s)^{n-1} + \cdots + a_1(r/s) + a_0$$

multiplying all by s^n, we get:

$$0 = a_n r^n + a_{n-1}r^{n-1}s + \cdots + a_1 r s^{n-1} + a_0 s^n$$

so

$$a_n r^n = s(-a_{n-1}r^{n-1} - \cdots - a_1 r s^{n-1} - a_0 s^{n-1})$$

thus, $s \big| a_n r^n$. $gcd(r, s) = 1$, then it must be that $s|a_n$. Doing the same trick for a_0:

$$a_0 s^n = r(-a_n r^{n-1} - a_{n-1}r^{n-2} - \cdots - a_1 s^{n-1})$$

then $r|a_0$.

If $a_n = 1$, then we'd have $s|1$, so $r/s = r/1 = r$. This means we only need to check the values r which divide a_0. If every value that dividse a_0 is not a root of $p(x)$, then $p(x)$ has no roots.

This gives us a really easy way of testing irreducibility of low-degree polynomials:

Example 11.3.7: Irreducibility Of Low-Degree Polynomials

1. The polynomial $p(x) = x^3 - 3x - 1$ is irreducible in $\mathbb{Z}[x]$. Since the polynomial is monic, by corollary 11.2.2, it must be reducible in $\mathbb{Q}[x]$. By the rational number test, we are given the candidates ± 1. Plugging both of them in, we get no roots of $\mathbb{Q}[x]$, and so by 11.3.5, this polynomial is irreducible, and so is irreducible in $\mathbb{Z}[x]$.

2. For any prime p, $x^2 - p$ and $x^3 - p$ are irreducible in $\mathbb{Q}[x]$. We have ± 1 and $\pm p$ as candidates. Both of them are not roots, and therefore, these polynomials are not reducible!

3. The polynomial $x^2 + 1$ is *reducible* in $(\mathbb{Z}/2\mathbb{Z})[x]$, namely $(1)^2 + 1 = 1 + 1 = 0$. It's factors are $(x + 1)(x + 1) = x^2 + 1$. The rational roots test applies since TBD.

4. The polynomial $x^2 + x + 1$ is *irreducible* in $(\mathbb{Z}/2\mathbb{Z})[x]$, since there are no roots.

5. Similarly for $x^3 + x + 1$ in $(\mathbb{Z}/2\mathbb{Z})[x]$

All the polynomials in the aforementioned examples are all of degree less than or equal to 3. This is because when factoring polynomials of degree greater than or equal to 4, it's possible that we get 2nd degree polynomials as irreducible factors (like $(x^2 + 1)(x^2 + 1)$ in $\mathbb{Q}[x]$, which has no factors).

We use another trick for these type of polynomials: We can use 11.0.1 ($R[x]/I[x] \cong (R/I)[x]$) to generalize the rational root test:

Proposition 11.3.8: Reduction Test

Let I be a proper ideal in the integral domain R and let $p(x)$ be a non-constant *monic* polynomial in $R[x]$. If the image of $p(x)$ in $(R/I)[x]$ cannot be factored in $(R/I)[x]$ into two polynomials of smaller degree, then $p(x)$ is irreducible in $R[x]$.

***Proof*:**
For the sake of contradiction, let's supose $p(x)$ cannot be factored in $(R/I)[x]$, but $p(x)$ is reducible in $R[x]$. Then there must be two polynomials such that $p(x) = a(x)b(x)$. Note that both $a(x)$ and $b(x)$ must be monic.

Reducing the coefficients modulo I gives a factorization in $R[x]/I[x]$, but by 11.0.1, this is equivalent to $(R/I)[x]$ – contradiction our assumpiton.

The power here comes from the fact that R/I might have much simpler structure for factoring (ex. $\mathbb{Z}$ vs $\mathbb{Z}/n\mathbb{Z}$!) For example, in exercises 11.2.1, we may prove that $x^5 - x + 1$ using this trick rather quickly! So, If we reduce a polynomial into $(R/I)[x]$, and show it cannot be factored, then it is irreducible in $R[x]$ as well! However, the converse (if $p(x)$ is irreducible in $R[x]$, then $p(x)$ is irreducible in every $(R/I)[x]$) is not true.

Example 11.3.9: Limitation of reduction method

Take $x^4 + 1 \in \mathbb{Z}[x]$. This polynomial is irreducible in $\mathbb{Z}[x]$, (hint: recall that $\sqrt{2}$ is irrational), but is reducible for every prime. Verify this by considering cases of

$$p \equiv 1, 3, 5, 7 \mod 8$$

since the case of $p = 2$ can quickly be manually verified.

So, if you get that a polynomial is reducible in $(R/I)[x]$, that unfortunately does not tell you it's reducible in $R[x]$.

A special case of proposition 11.3.8 that is quite useful is called Eisenstein's Criterion (or Eisenstein-Schönenmann Criterion, since Schönenmann discovered it first)

Proposition 11.3.10: Eisenstein's Criterion

Let P be a prime ideal of the commutative ring R and let $f(x) = x^n + a_{n-1}x^{n-1} + \cdots + a_1 x + a_0$ be a polynomial in $R[x]$ (where $n \geq 1$). Suppose $a_{n-1}, ..., a_0 \in P$, but $a_0 \notin P^2$. Then $p(x)$ is irreducible.

***Proof*:**
Let's take $p(x)$ like in the proposition. For the sake of contradiction, let's say $p(x)$ is reducible: $p(x) = a(x)b(x)$, where $a(x)$ and $b(x)$ are non-constant.Then by proposition 11.3.8 (the contrapositive of it), it is reducible in $(R/P)[x]$. Since every non-leading coefficient is in P:

$$\overline{p(x)} = \overline{x^n} = \overline{a(x)} \cdot \overline{b(x)}$$

Since P is prime, R/I is an integral domain, and so the constant terms of $a(x)$ and $b(x)$ must be in P, or else we'll get $\overline{x} \cdot \overline{y} = \overline{0}$. But then, multiplying those terms means $a_0 \in P^2$, a contradiction!

For the purposes of this book, the following is the most immediate use of Eisenstein's Criterion:

Corollary 11.3.11: Eisenstein's Criterion For $\mathbb{Z}[x]$

Let p be a prime in $\mathbb{Z}$ and let $f(x) = x^n + a_{n-1}x^{n-1} + \cdots a_1 x + a_0$. If every a_i is divisible by p, but a_0 is not divisible by p^2, then $f(x)$ is irreducible in both $\mathbb{Z}[x]$ and $\mathbb{Q}[x]$.

***Proof*:**
This is proposition 11.3.10 and using corollary 11.2.2.

Here are some examples using Eisenstein's criterion, and some more tricks when finding irreducible polynomials:

Example 11.3.12: Irreducibility Tricks

1. When solving polynomials of degree 3 or less, it'd suggest to always start with checking roots: it's the simpliest criterion.
2. The polymoial $x^4 + 10x + 5$ in $\mathbb{Z}[x]$ is irreducible, if we take $p = 5$. Since $5|10$, $5|5$, but $5^2 \nmid 5$, then Eisntein's criterion holds.
3. For any integer a such that $p\big|a$ but $p^2 \nmid a$, then $x^n - a$ is irreducible in $\mathbb{Z}[x]$. If $a = p$ a prime, then $x^n - p$ is irreducible for all positive integers n, so for $n \geq 2$, the n^{th} root of p are not rational numbers (i.e. has no roots in $\mathbb{Q}[x]$)
4. We cannot apply the same tricks we did before to the polynomial $x^4 + 1$ in $\mathbb{Z}[x]$, since nothing divides prime, and any proper ideal will not contain 1. Instead, we'll use a trick called *substitution.*

   Consider the polynomial $f(x+1) = (x+1)^4 + 1 = x^4 + 4x^3 + 6x^2 + 4x + 2$. On this polynimal, Eisntein's criterion works, since all non-leading coefficients are divisible by 2, and $2^2 \nmid a_0$. From this, it follows that $f(x)$ is irreducible, since any factorization of $f(x)$ will be a factorization for $f(x+1)$ (replace x by $x+1$ in the factorization!) Remember this trick for polynomials which seem at first glance to not get an application of Eisntein's criterion.
5. (cyclotomic polynomials) Perhaps the most famous application of Eisenstein's criterion is on polynomials of the following form:

   $$1 + x + x^2 + \cdots = x^{p-1} \in \mathbb{Z}[x]$$

   Eisenstein's criterion cannot be directly applied, however as before we can do the $x \mapsto x+1$ trick to get

   $$p(x+1) = 1 + (x+1) + \cdots + (x+1)^{p_1} = x^{p-1}\binom{p}{p-1}x^{p-1} + \cdots \binom{p}{2}x + \binom{p}{1}$$

   where $\binom{p}{1} = p$.It suffices to show that $p\Big|\binom{p}{k}$ for $1 \leq k \leq p-1$ to conclude the result.
6. On $\mathbb{Q}[x][X]$?

7. Notice that in the proof, we reduced to $(R/P)[x]$. If $R = \mathbb{F}$ was a field, then the only ideals are 0 and $\mathbb{F}$, meaning $(\mathbb{F}/P)[x] \cong 0[x] \cong 0$. Therefore, Einsteins criterion doesn't work will with such polynomials.

   For example, is $x^4 + 1$ reducible in $\mathbb{F}_5[x]$ ($\mathbb{F}_5$ is the field with 5 elements, and is isomorphic ot $\mathbb{Z}/p\mathbb{Z}$)? If if were over $\mathbb{Z}[x]$, we already mentioned how it is irreducible (using a tirck we which will be explained in chapter ref:HERE). It also has no roots, as can be verified by plugging in $0, ..., 4$.

   We can't use einstein's criteron, but since the degree is 4, we know if a factorization did exist, they would have to be of degree 2, that is

   $$x^4 + 1 = (x^2 + ax + b)(x^2 + cx + d) = x^4 + (a + c)x^4 + (b + ac + d)x^2 + (ab + bc)x + bd$$

   note that the coefficients of x^2 are one since the coeficient of x^4 is one. We can then deduce that
   $$a + c = 0, b + ac + d = 0, ad + bc = 0, bd = 1$$

   Manipulating a bit, we can get the following information:

   $$\begin{aligned} c &= -a \\ b + a(-a) + d &= 0 \\ ad + b(-a) &= 0 \\ a^2 &= b + d \\ ad &= ab \end{aligned}$$

   Let's take cases: let's say $a \neq 0$. Then $b = d$, $a^2 = 2b$, $b^2 = 1$. (TBD). So, we get a contradiction.

   Let's check $a = 0$. Then $062 = 0 = b + d$, implying $b = -d$, and $b^2 = -bd = -1 = 4$, so $b = 2$. implying $d = 2$ and $a = 0$, $c = 0$. No contradiction was found! Thus, we can plug in, and factor to get

   $$p(x) = x^4 + 1 = (x^2 - 3)(x^2 - 2) = (x^2 + 2)(x^2 + 3)$$

   Thus, $p(x)$ is reducible in $\mathbb{F}_5[x]$.

There are further irreducibiltiy algorithms that were programmed. One algorithm is Berlekamp Algorithm. there are others TBD.

Finally, we give a few important results that shall be very handy tools. The first tells us that factorization of a polynomial is invariant to field extensions, namely if $F \hookrightarrow E$ is an injective map, then the factor of $p(x)$ in E are the same factors (if not further reduced) in $E(x)$:

Proposition 11.3.13: Invariant Under Field Extensions

If you extend from a field F to E, then by the uniqueness of the euclidean algorithm, everything in $E[x]$ will still factor the same way as before (in $k[x]$): it cannot inroduce new solutions.

***Proof*:**
exercise

This next one is of particular interest for the structure of the multiplicative group of a field!

Theorem 11.3.14: Subgroup of Multiplicative Group is Cyclic

A finite subgroup of the multiplicative group of a field is cyclic. In particular, if F is a finite field, then the multiplicative group $F^\times$ of nonzero elements of F is a cyclic group.

Another way of saying this is that $F^\times$ is locally cyclic (being "locally P" means that every finite subgroup has the property P)

***Proof*:**
We will use the fundamental theorem for finitely generated abliean groups!

Let $A \subseteq F^\times$ be a finitely generated abelian group. Then:

$$A \cong \mathbb{Z}/n_1\mathbb{Z} \times \mathbb{Z}/n_2\mathbb{Z} \times \cdots \times \mathbb{Z}/n_k\mathbb{Z}$$

where $n_k \big| n_{k-1}, n_{k-1} \big| n_{k-2}, ..., n_2 \big| n_1$. Since n_k divides the order of each of the cyclic groups in the direct product, it follows that each direct factor contains n_k elements of order dividing n_k (by prop ref:HERE). If k were greater than 1, there would therefore be a total of more than n_k such elements. But then there would be more than n_k roots of the polynomial $x^{n_k} - 1$ in the field F, contradicting proposition 11.3.1!! Hence, $k = 1$, and the group is cyclic!

Corollary 11.3.15: $(\mathbb{Z}/p\mathbb{Z})^\times$ is cyclic

Let p be a prime. The multiplicative group $(\mathbb{Z}/p\mathbb{Z})^\times$ of nonzero residue classes mod p is cyclic

***Proof*:**
Since $\mathbb{Z}/p\mathbb{Z}$ is a field, apply proposition 11.3.14

These corollaries will be useful when proving Kronnecker-Weber's Theorem (theorem 50.5.4):

Corollary 11.3.16: Isomorphisms Of Multiplicative Groups

Let $n \geq 2$ be an integer with factorization $n = p_1^{\alpha_1} p_2^{\alpha_2} \cdots p_n^{\alpha_n}$ in $\mathbb{Z}$, where $p_1, ..., p_n$ are distinct primes. Then we have the following isomorphsism of (multiplicative) groups:

1. $(\mathbb{Z}/n\mathbb{Z})^\times = (\mathbb{Z}/p_1^{\alpha_1})^\times \times (\mathbb{Z}/p_2^{\alpha_2})^\times \times \cdots \times (\mathbb{Z}/p_n^{\alpha_n})^\times$
2. $(\mathbb{Z}/2^\alpha\mathbb{Z})^\times$ is the direct product of a cyclic group of order 2 and a cyclic group of order $2^{\alpha-2}$, for all $\alpha \geq 2$
3. $(\mathbb{Z}/p^\alpha\mathbb{Z})^\times$ is a cyclic group of order $p^{\alpha-1}(p-1)$ for all odd primes p

***Proof*:**
exercise (comment)

Exercise 11.3.1

1. Show that there are irreducible polynomial in $\mathbb{Z}[x]$ and $\mathbb{Q}[x]$ for every degree.
2. Let $f(x)$ be an irreducible monic polynomial of $\mathbb{Z}[x]$. Show that $f(x) \bmod p$ has a root for infinitely many primes. The opposite is true too, that there are infinitely many primes for which $f(x) \bmod p$ does *not* have a root, but as far as I am aware that requirse some analytic number theory[4]
3. Show that $t^2 + t + 1 \in \mathbb{F}_2[t]$ is irreducible. Use this to show that $\mathbb{F}_2[t]/(t^2 + t + 1)$ is a field with 4 elements.

[4] see [39]

Part III

Modules

When we explored groups, we saw that the kernel of groups were groups (in particular a special subset of a group called the normal subgroup), and if we have an abelian group, every abelian group is the kernel of some group.

The story for rings is more complicated. As we mentioned at the beginning of the part on rings, some authors include an identity in the definition, others do not. If we include an identity, then the kernel of ring homomorphisms are not rings. Even if we take exclude identities as part of the definition, notice that practically non of the ideals we talked about were previously mentioned rings, essentially because almost all rings we talk about have an identity, but if an ideal has an identity, it is the entire ring. The cokernels is also not as well-behaved (recall that $i : \mathbb{Z} \hookrightarrow \mathbb{Q}$ is an epimorphism). In this way, the kernel of rings and rings are very different objects, and so a lot goes into studying ideals as though they are a seperate object in ring theory. In particular, we do not focus a lot of attention of constructing rings based on other rings (similar to how we constructed polyabelian or poly-cyclic groups). Overall, the category of **Ring** and **Rng** are not the best behaved categories.

The concept of Modules can be thought of as a fix to this. Modules are more similar to abliean groups than they are to rings, for example if you have any sub-module, then the quotient will always be well-defined. Furthermore, all Rings can be thought as a module, making the study of module subsumes the study of rings. As we shall see,

We shall see that an R-module M will be an abelian group M with an ring action by R. The reason M is an abelian group and not just a set like in group actions is because it is a generalization of vector-spaces, not an addition to group actions. What I mean by this is that you have to think of the reasoning from the point of view that vector-spaces came first, and then there were many instances where the field acting on the group was too restrictive, which lead to module.

Over the 19th and 20th century, many mathematical objects were studied that all turn out to be modules. To those who want the punchline, I'll point out a few right now:

1. All abelian groups are $\mathbb{Z}$-modules
2. All linear transformations are $F[x]$-modules
3. All rings are modules (If R is a ring, then it is an R-module over itself)
4. All ideals are modules (if $I \subseteq R$ is an ideal, it is an R-module)
5. All vector-spaces over k are k-modules
6. All vector-bundles are projective modules

12

Introduction to Modules

A lot of the algebraic structures we've covered so far share a similar story: we ask how maps between them can be studied (leading us to explore sub-objects and quotient-objects), and we ask how to generate them. Modules will have the same story at the beginning. [Question: Why didn't group actions have the same story? no sub/quotient actions]. When we'll be building up modules, they'll more closely resemble groups and rings. In groups, simple groups and free groups played a bigger role than they did in ring theory. Same in Module theory (TBD).

12.1 Basic Definitions and Examples

Definition 12.1.1: [left]-Module

Let R be a ring (more generally a rng). A *left R-module* or a *left module over R* is a set M together with

1. a binary operation $+$ on M under which M is an **abelian group**, and
2. an action of R on M (that is, a map $R \times M \to M$) denoted by rm, for all $r \in R$ and for all $m \in M$ which satisfies
   (a) $(r+s)m = rm + sm$, for all $r, s \in R$, $m \in M$
   (b) $(rs)m = r(sm)$, for all $r, s \in R$, $m \in M$
   (c) $r(m+n) = rm + rn$ for all $r \in R$, $m, n \in M$

   If the rng R is a ring (i.e. has a 1), we impose the additional axiom

   (d) $1m = m$, for all $m \in M$

   And we sometimes call such a module a *unital* R-module.

Right modules are defined similarly. If the R ring is commutative, and M is an left $R-module$, we can turn it into a right $R - module$ by defining $rm = mr$. To distinguish when different rings may be acting on an abelian group M, the symbol $\curvearrowright$ (read *acts on*) may be used to show that a ring R is acting on M: $R \curvearrowright M$

Unless stated otherwise, the term "module" will reference "left-module" Note that for a module M, we usually denote what ring it is over (or what ring acts upon it) by saying it is an *[ring symbol here]-module* (ex. R-module, $F[x]$-module, $[M_{m\times n}(\mathbb{F})]$-module). We might also say R acts on M where R is the ring and M is the abelian group. We will always be working *rings* actions, however as noted in the definition, we can also work with *rng* actions. Note too that we will often say that "R acts on the module M" to mean that we define a ring-action of R on the abelian group M.

When working over rings, the distributivity condition forces M to be abelian: let $r = s = 1$:

$$(1+1)\cdot(m+n) = (1+1)\cdot m + (1+1)\cdot n = 1\cdot m + 1\cdot m + 1\cdot n + 1\cdot n = m+m+n+n$$

$$(1+1)\cdot(m+n) = 1\cdot(m+n) + 1\cdot(m+n) = 1\cdot m + 1\cdot n + 1\cdot m + 1\cdot n = m+n+m+n$$

which implies $m+n = n+m$ for all $m, n \in M$. Notice how this when working with group actions, the group acting on the set did not effect any strucutre of set, namely since the set has no additional structure. Be mindeful that when talking about a module, we will usually call M the module. For example we might say "Let $\mathbb{Z}$ act on $\mathbb{Q}$ by $r \cdot a = ra$; then $\mathbb{Q}$ has the following properties". When talking about $\mathbb{Q}$ here, we're talking about $\mathbb{Q}$ as a module, namely a $\mathbb{Z}$-module, not just as abelian group or a ring.

If R is commutative then we defined a two-sided action $r \cdot m = m \cdot r$. If R is not commutative, then defining this property will have problems with $2(b)$: Let R be non-commutative, let $r \cdot m = m \cdot r$, and take $r, s \in R$ such that $rs \neq sr$ and $rs \cdot m \neq sr \cdot m$ (an example of such a module will be presented in the examples 12.1.7). Then:

$$(rs)m = m(rs) = (mr)s = (rm)s = sr(m) = (sr)m$$

however, $sr \cdot m \neq rs \cdot m$ so this statement is not necessarily always true. Note how it may be possible that even though $rs \neq sr$ that $rs \cdot m = sr \cdot m$, for example, the trivial module structure $r \cdot m = m$ for any $r \in R$ will produce this result, however as we'll soon see in example 12.1.7, every module will always induce a *faithful* module (the word faithful being defined equivalently to the group-action case[1]), and so making a distinction between left and right modules is very important over non-commutative rings.

We now have many operators to keep track of: the group operation of M, the two rings operations of R, and the ring action of R on M. When working with modules, we often focus on the ring-action, and use group and rings structures that were previously covered.

One might ask themselves if modules and group-rings might be similar concept, boths somehow being a ring "acting" on a group. If this suspicions crossed your mind, it is good idea to check how these two structure's differ (for example, if $|G| > 1$, recall a group-ring always has zero-divisors. Find an example in the list of examples bellow that does not have this property).

Since a module is an action on an abelian group, it would make sense if there is an equivalent to a permutation representation for module's as there is for group-actions. Recall that any group action can be seen as a group-homomorphism:

$$\rho : G \to \mathrm{Sym}(S)$$

The exact same idea works for defining an R-module as an action. In order to do so, we need an a notion of an R-module homomorphism that preserves some of the structure. Before looking at the definition bellow, see if you can come up with the definition of an R-module homomorphism.

Definition 12.1.2: R-module homomorphism

Let R be a ring and let M and N be R-module A map $\varphi : M \to N$ is an *R-module homomorphism* if it respects the R-module structure of M and N, i.e.

1. $\varphi(x + y) = \varphi(x) + \varphi(y)$ for all $x, y \in M$
2. $\varphi(r \cdot x) = r\varphi(x)$, for all $r \in R$, $m \in M$

Since M and N are both abelian groups, R-module homomorphisms are sometimes called *R-linear group homomorphisms*, where the map is R-linear since the ring actions passes through the homomorphism. Notice that both M and N both have the same ring acing on them: it is imperative that an R-module homomorphism be betewen R-modules.

Returning to the discussion on treating R-modules as an action, it is a good exercise to check ones understanding of the axioms of modules that the "axiomatic representation" of an R-module is equivalent to defining an R-module as a ring-homomorphism

$$\rho : R \to \mathrm{End}_{\mathbf{Ab}}(M)$$

where $\mathrm{End}_{\mathrm{Ab}}(M)$ is the collection of all group-homomorphisms from M to itself. Notice that $\mathrm{End}(M)$ is an [abelian] group because M is an abelian group, while the ring operation of composition, $\circ$,

[1] That is, there must exist an $m \in M$ st $r \cdot m = s \cdot m$ implies $r = s$

automatically satisfies the condition to make $\mathrm{End}_{\mathbf{Ab}}(M)$ a ring, which should be checked (for example, $\mathrm{id} : M \to M$ is an element of $\mathrm{End}_{\mathbf{Ab}}(M)$, hence it has a multiplicative identity). It will sometimes useful to define an R-module as a ring homomorphsim $R \to \mathrm{End}_{\mathbf{Ab}}(M)$, if we want to manipulate with the ring action on it (see exercise ref:HERE (how $R \times S$ can act on an R-module, easy to see with this definition))

As with rings and and with groups, we can define a category: given a ring R, the collection of R-modules and the collection of R-module homomorphisms forms a category:

Definition 12.1.3: R-Mod

The collection of all R-modules and R-module homomorphisms forms a category, which we'll denote R-**Mod** or $\mathbf{Mod}_R$.

This category has some of the nice properties that the other categories that were extensively worked in so far have:

Definition 12.1.4: Classical definition of Submodule

Let R be a ring and let M be an R-module. An R-submodule of M is a subgroup N of M which is closed under the action of ring elements, i.e., $rn \in N$, for $r \in R$, $n \in N$

Definition 12.1.5: Categorical Definition of Submodule

Let $N \subseteq M$. Then if there exists a R-module inclusion map $\varphi : N \to M$ that is also an R-module homomorphism, then N is a submodule of M

Proposition 12.1.6: The sub-module Criterion

Let R be a ring and let M be a R-module. A subset of N of M is a submodule of M if and only if

1. $N \neq \emptyset$, and
2. $x + ry \in N$ for all $r \in R$, and for all $x, y \in N$

***Proof*:**
Easy to prove both ways (recall that M is an abelian group, and for the other way, simply set the values of r and x to some appropriate values)

Example 12.1.7: Modules

1. Let R be any ring. Then $M = R$ is a left R-module with the ring multiplication as the ring action: $r \cdot s = rs$. Take a moment to think about what are the submodules of R? The answer will be revieled in the next sentence. the sub-modules of M are the left-ideals of R! This is because ideals of R are closed under multiplication of R, and the axioms of modules require closure under "scalar multiplication" by the ring elements. Is R/I for any (two-sided) ideal of R also an R-module? Take a moment to think about it. The answer is that R/I

is indeed an R-module! We will soon show that the kernel of module-homomorphisms will be submodules, showing that the axioms of a module "fix" the category of **Ring** by being closed under kernels!

From now on, if we say that R is a [left/right] R-module over itself, it will be taken that the R-action on R is [left/right] multiplication by R, with a bias towards *left* R-modules if R is non-commutative and which side of the action was not specified. We will also sometimes say "let R have the natural R-module structure" to mean the one induced by the ring operation of R.

2. Let R be a ring with 1 and let $n \in \mathbb{Z}^+$. Then since R is a group, we can define the direct product:
$$R^n = \{(a_1, ..., a_n) | a_i \in R, \text{ for all } i\}$$
We can turn R^n to an R-module by defining addition in R^n component wise, and the action being componentwise:
$$r(a_1, ..., a_n) = (ra_1, ..., ra_n)$$
Such a module R^n is called a *free module* [a]. More on these in section 12.3.2. If $R = \mathbb{R}$, then we have the usual algebraic structure of $\mathbb{R}^n$ common in analysis and linear algebra.

3. If M is an R-module, and S is a sub-ring of R, then M is automatically an S-module (i.e. the set M is also a module under the ring action with S). Since if the module axioms hold for R, i.e. the ring being closed under the ring action and the rules holding, then it will still hold for a sub-ring. In this way $\mathbb{R}$ is an $\mathbb{R}$-module, a $\mathbb{Q}$-module, and a $\mathbb{Z}$-module with the ring action being defined by regular multiplication. Notice though that the $\mathbb{R}$-submodules, $\mathbb{Q}$-submodules, and $\mathbb{Z}$-submodules differ, showing that having a different ring act on $\mathbb{R}$ greatly changes its module structure.

4. Let M be an R-module. If $I \subseteq R$ is a two-sided ideal, is it always true that M is an R/I-module? Under what condition's is it an R/I-module? The following is an exploration of when this is the case: Let I be a 2-sided ideal of R such that $am = 0$, for all $a, \in I, m \in M$. It should be checked that I is indeed an ideal (it is usually denoted $\text{Ann}_R(M)$, and called the *annihilator* of M [over R]). Then we say that M is *annihilated* by I. In this situation, we can make M into an (R/I)-module by defining an action of the quotient ring R/I on M as follows: for each $m \in M$, and $(r + I) \in R/I$, let
$$(r + I)m = rm$$
This is well defined since for any $a \in I$, $(r + a)m = rm + am = rm$. Check that the axioms of a module still hold. Furthermore, The R/I action is a *faithful* action, since if $rx = sx$, then $r = s$. To see this, consider any $x \in M$. Then since M and R are groups, define the group-homomorphism:
$$\varphi : R/I \to M \qquad \varphi(\overline{r}) = \overline{r} \cdot x$$
The function φ is indeed a group-homomorphism, since $\varphi(\overline{r} + \overline{s}) = (\overline{r} + \overline{s}) \cdot x = \overline{r} \cdot x + \overline{s} \cdot x = \varphi(\overline{r}) + \varphi(\overline{s})$. The kernel of this action is all elements such that $\varphi(\overline{r}) = \overline{r} \cdot x = (r + I) \cdot x = 0$. By definition of I,
$$I \cdot x = \{i \cdot x \mid i \in I\} = \{0\}$$
so it follows that if $\overline{r} \in \ker(\varphi)$, $\overline{r} = \overline{0}$, showing that $\ker(\varphi) = \overline{0}$ (Notice the parallell with the Orbit-Stabalizer, theorem 4.3.1)

If, conversely, if $I = \mathrm{Ann}_R(M)$, and we have defined an R/I-action on a module M, can we define an R-action on M? More generally, given two rings R and S acting on M, when can we say that the module structure they induce on M is "comparable"' in some way? One way two types of modules can be equivalent is if they are something called "Morita equivalence", which is settled in chapter 18$\star$[b])

5. What is and isn't a well defined R-module isn't always intuitive. Take for example the group ring $R = \mathbb{Z}[G]$ where $G = \{1, \sigma\}$ is a group of order 2. For the abelian group, take the module $M = \mathbb{Z}e_1 + \mathbb{Z}e_2$, where e_1, e_2 are generators for $\mathbb{R}^2$ (for example, $(1,0)$ and $(0,1)$ generate $\mathbb{R}^2$ as an abelian group), and the product $\mathbb{Z}e_i = \{ze_i | z \in \mathbb{Z}\}$.

   Define the action as:
   $$\sigma(e_1) = e_1 + 2e_2 \qquad\qquad \sigma(e_2) = -e_2$$
   Then M is a $\mathbb{Z}[G]$-module! The big one to check is $r_1(r_2 m) = (r_1 r_2)m$ – the other conditions follow more easily. This is an example of how the limit on what can be defined as a module isn't easy, since an example like this one exists, checking when a module is defined can be quite useful.

6. Let R be any ring and let $F(A, R)$ be the set of all [set]-functions from A to R:
   $$f \in F(A, R) \Rightarrow f : A \to R$$
   Then define the action by R to be
   $$r \cdot f = rf \qquad r \cdot f(a) := rf(a) \qquad \forall a \in A$$
   since $f(a) \in R$ Band $r \in R$. Define the group operation to be the new function $f + g$ where
   $$(f + g)(a) := f(a) + g(a) \qquad \forall a \in A$$
   Note that these functions are *not* R-module homomorphisms, since the set A has no structure! This R-module is is often dentoed as R^A (to symbolise all functions from A to R)[c].

   Very often in Analysis and Algebraic Geometry, the set A will be all continuous functions on a topological space X, denoted $C(X)$, and the ring R would be the real or complex numbers $\mathbb{R}$ or $\mathbb{C}$. The study of these spaces is done commonly in functional analysis and classical algebraic geometry, usually requiring some analysis knowledge. In this textbook, these types of R-module will come back when discussing free-modules in theorem 12.3.11.

7. Let X be any set and $P(X)$ the powerset.

8. (An example with a ring acting on a module in two different ways, non trivially)

[a]not dissimilar to free-groups we studied in 2.3.3

[b]For the categorically inclined, R-mod and S-mod are Morita equivalent if they are equivalent as categories. Are R-mod and R/I-mod equivalent?

[c]The reason for this notation in fact very interesting and is covered in ref:HERE (in the category theory chapter, categorification of natural numbers, see Emily Reil Category Theory in Context)

When we worked with group-actions, we can take any group G and make it act on any set S. This is not the case with modules (i.e. given some fixed ring R and abelian group M, there does not always exist a ring-homomorphism $\varphi : R \to \mathrm{End}_{\mathbf{Ab}}(M)$):

Example 12.1.8: Impossible Module

Consider $M = \mathbb{Z}$ and $R = \mathbb{Q}$. Is there always a $\mathbb{Q}$ action over $\mathbb{Z}$? If it was, then:

$$\frac{1}{2} \cdot 1 = z$$

must be an element of $\mathbb{Z}$ ($z \in \mathbb{Z}$). Then:

$$z + z = \frac{1}{2} \cdot 1 + \frac{1}{2} \cdot 1 = (\frac{1}{2} + \frac{1}{2}) \cdot 1 = 1 \cdot 1 = 1$$

as a direct consequence of the module axioms. However, this implies

$$z + z = 1$$

which is not possible for any $z \in \mathbb{Z}$ – and thus a contradiction! This is also an example where you can have a $\mathbb{Z} \subseteq \mathbb{Q}$ and a defined $\mathbb{Z}$-module, but there is no extension of M to a $\mathbb{Q}$-module for which the previous module structure is embedded within it (opposite to how if we had a $\mathbb{Q}$-module, it would automatically be a $\mathbb{Z}$-module)[a]

[a]There is in fact still a notion of extending a module along a ring map, however the original module need not embed in the result; see section 15.3, and corollary 15.3.8 for exactly when it does

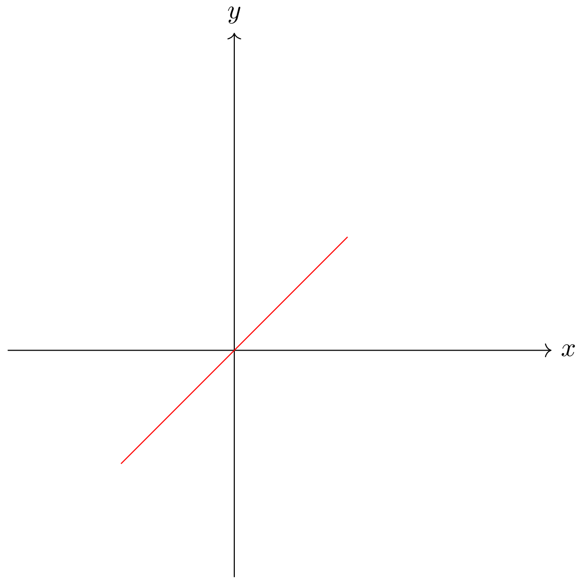
12.1.1 Vector-spaces

If the ring is a field, we give a special name to such a module:

Definition 12.1.9: Vector Space

Let M be an R-module. If $R = k$ is a field, then we call M a *vector-space* over k. An arbitrary field is often denoted by k or $\mathbb{F}$.

Due to historical reasons $\mathbb{F}$-modules homomorphisms are called *linear transformations*. The motivation for this will be more clearly seen in chapter 13, but the gist is that when $\mathbb{F} = \mathbb{R}$, the geometric intuition of the images and kernels of linear transformations will all be hyper-planes in the domain and codomain. A simple example would be $f : \mathbb{R} \to \mathbb{R} \times \mathbb{R}$ with $f(x) = (x, x)$. The resulting hyper-plane (simply called a line since it's 1 dimensional), looks like so:



If you are comfortable with linear algebra, then you are probably familiar of how simple the structure of vector-spaces are (if not, you may skip this remark: all that will be discussed here is covered in chapter 13). Essentially, a vector space V over k can be completely determined by it's dimension. For modules over a general ring R, this is sometimes far from the case: sometimes R will not even embed itself in V! However, there are ways in which we can systematically loosen the conditions on the ring k acting on V and get similar, but weaker, results. Thus, if it is intuitive to think of modules as generlizations of vector-spaces, which historically is how many mathematicians thought of modules, then these chapters on module theory can be thought of as what happens if the action on a vector-space no longer forces to give a basis.

Example 12.1.10: Vector Spaces

1. The most common example is when $R = \mathbb{R}$ and $M = R^n = \mathbb{R}^n$. This vector-space is commonly reffered to as *euclidean space* due to the classical geometrical properties that can be defined on it(ex. a notion of distance, length, angle, translation and rotation)

2. If I is a maximal ideal in a commutative ring, and $IM = 0$, then M is a vector space over R/I (since R/I is a field)

3. Take $\mathbb{F}[x]$ to be a polynomial ring over a field $\mathbb{F}$, an let $f(x)$ be any polynomial of degree $n \geq 1$. Then $V = \mathbb{F}[x]/(f(x))$ is another polynomial ring. Let $\mathbb{F}$ act on V by $a \cdot \overline{f} = \overline{af}$. This will be a vector-space. If $\mathbb{F}$ is a finite field of order p, then we can show that V has p^n.

4. (If you know some calculus). Consider $\mathbb{R}[x]$. Then it is a vector-space over $\mathbb{R}$ ($\mathbb{R}[x]$ is an abelian group, and the action is simply the multiplication in $\mathbb{R}[x]$, since $\mathbb{R} \subseteq \mathbb{R}[x]$). Recall that the derivative has the following two properties:

$$\frac{d}{dx}(p(x) + q(x)) = \frac{d}{dx}p(x) + \frac{d}{dx}q(x)$$
$$\frac{d}{dx}(cf(x)) = c\left(\frac{d}{dx}f(x)\right)$$

This shows that the derivative is in fact a linear transformation (i.e. $\mathbb{R}$-module homomorphism) from $\mathbb{R}[x]$ to itself![a]

5. If R is a ring where the only two-sided ieals are (0) and R, then R is called a *simple ring* (similarly to how simple groups are groups whose only normal subgroup is $\{0\}$ and the entire group). Then $Z(R)$ (the center of the ring) *has* to be a field (can you show this?). If $k = Z(R)$, then any simple ring is a vector-space over its center.

[a]For those who know some Linear algebra and Fourier Analysis, the "diagonalization" of the derivative is the Fourier transform of a function! For details, see the Gelfand Transformation of unbounded operators

If $R = \mathbb{F}$ is a field, we give a special name to sub-modules:

Definition 12.1.11: Subspace

If $W \subseteq V$ is a submodule of a vector space, then we call it a *subspace*

12.1.12 Remark (If you know linear algebra) When determining where a linear transformation maps, it is sufficient to know where the basis maps to. General R-modules need not have a basis well-defined on them. However, *cyclic* modules will always have a basis, since $M = R \cdot x$ for some $x \in M$. Thus, R is always a cyclic R-module over itself since $R \cdot 1 = R$. General R-modules need not have a basis, and so R as an R-module a particularly nice module.
As a consequence of this, for any R-module homomorphism: $\varphi : R \to N$, it suffices to know where $\varphi(1)$ maps to. In general, it is much harder to determine a map $\varphi : M \to N$ given where only certain elements are mapped to.

Exercise 12.1.1

1. Let $\langle m \rangle = M$ be an R-module generated by a single element (i.e. if $x \in M$, $x = a_i m$ for some $a_i \in R$)[2]. Show that $\mathrm{Ann}(M) := \{m \in M \mid \exists r \in R \text{ such that } rm = 0\}$ forms an ideal of R.
2. Let M be as in the previous question. Show that $M/\mathrm{Ann}(M) \cong HERE$ (I.e. trying to show how we can study ideals in the context of modules, and how ideals are studied as R/I R-modules
3. (practice with categories). Let A be a set and $F(A)$ be the group freely generated by A (definition 2.3.29).
    1. Prove that the collection of all vector-spaces along with linear transformations form a category. Label it k-**Vect**.
    2. Show that $F(A)$ can have a vector-space structure over k put on it such that $F : \mathbf{Set} \to k\text{-}\mathbf{Vect}$ is a functor. (Hint: See the functor defined in section 2.3.5)
4. I feel like I should add this somehow link to Ab-enriched categories
5. What is the initial object of Mod_R (if any)?

[2]We'll explore module generation in section 12.3

12.1.2 Algebras

Recall that if S is a ring, then it is always an abelian group, and there is always a unique homomorphism $\mathbb{Z} \to S$. Using this homomorphism, we can define a $\mathbb{Z}$-action given by $z \cdot s = \varphi(z)s$, and so every ring is automatically a $\mathbb{Z}$-module. More generally, any ring map $R \to S$ where R is commutative produces an R-module structure on the ring S that is compatible with the multiplication of S. To encapsulate this extra structure, we shall define a new term known as an *R-algebra* that represents a ring that also has a ring action on it.

The definition is simple if constrained to commutative rings, but there are many important cases where we must work with a non-commutative codomain (particularly in Part X), and so the above intuition must be made more precise. Given any ring homomorphism $\varphi : R \to S$, we can always define an R-module structure on S by defining

$$r \cdot s = \varphi(r)s$$

that is, the action is multiplication in S. The axioms of an R-module is immediate given the axioms of a ring and that φ is a homomorphism. Since we can multiply, we run into situations like the following: If $s_1 = r_1 \cdot x$ and $s_2 = r_2 \cdot y$, then:

$$s_1 s_2 = (r_1 \cdot x)(r_2 \cdot y)$$

Such an element is not necessarily equal to $(r_1 r_2) \cdot xy$ (similarly to how with rings, if the underlying set is not abelian, distributivity would not hold). This would be desirable, as in we would want $\varphi : R \to S$ to be a ring homomorphism that was also compatible with the R-module structure in the following way:

$$(r_1 \cdot x)(r_2 \cdot y) = (r_1 r_2) \cdot xy$$

As a commutative diagram:

$$\begin{tikzcd} (R \times S) \times (R \times S) \arrow[r, "a \times a"] \arrow[d, "\ell"'] & S \times S \arrow[d, "m"] \\ R \times S \arrow[r, "a"'] & S \end{tikzcd}$$

where the left map is

$$\ell\big((r_1, x), (r_2, y)\big) = (r_1 r_2,\ xy), \qquad \ell = (\mu_R \times m) \circ \sigma,$$

with σ the shuffle $\big((r_1, x), (r_2, y)\big) \mapsto \big((r_1, r_2), (x, y)\big)$.

Getting back to the homomorphism $\varphi : R \to S$ and the R-module structure on S, if we further require that R be *commutative* and that $\varphi(R) \subseteq Z(S)$ (the multiplicative center), then the left and right R-module structure will coincide (i.e. left and right multiplication produce the same module structure), and the ring operation of S will be compatible with that of R, that is

$$(r_1 r_2) \cdot (s_1 s_2) = \varphi(r_1 r_2)(s_1 s_2) = \varphi(r_1)\varphi(r_2)(s_1 s_2) = \varphi(r_1) s_1 \varphi(r_2) s_2 = (r_1 \cdot s_1)(r_2 \cdot s_2)$$

Boxing these properties we define an algebra:

Definition 12.1.13: R-algebra

Let S be a ring, R a commutative ring. Then S is called an (associative) R-algebra if S is also an R-module satisfying:

$$r \cdot (xy) = (r \cdot x)y = x(r \cdot y)$$

Equivalently, S is an R-algebra if there exists ring homomorphism $\varphi : R \to S$ where $\varphi(R) \subseteq Z(S)$. Furthermore:

1. If S is a division ring, then the R-algebra S is called a *division algebra*
2. If S is commutative, then the R-algebras S is called a *commutative algebra*

Note that S has has a natural R-module structure by forgetting the multiplication so that the action

$$r \cdot a[= a \cdot r] = \varphi(r)a$$

is the R-module structure. Unless otherwise stated, this will be the assumed R-module structure of an R-algebra S.

Algebra homomorphisms are defined similarly to how they've been defined before (can you figure it out without looking at the following definition?)

Definition 12.1.14: R-Algebra Homomorphism and Isomorphism

If A and B are R-algebra, then $\varphi : R \to S$ is an R-algebra homomorphism if φ is a ring-homomorphism and

$$r \cdot \varphi(x) = \varphi(r \cdot x)$$

If there exists an inverse function φ that is also an R-algebra homomorphism, then A and B are *algebra isomorphic*, or just *isomorphic* to each other, and we write $A \cong B$ if the context permits.

With this definition, the collection of R-algebra's along with R-algebra homomorphism forms the category R-**Alg**.

12.1.15 For the Categorically Inclined: Monoidal Object in R-module
Another way of saying that an R-algebra adds the notion of multiplication to an R-module in a "consistent" way is by noticing that R-algebra are the monoidal objects in the category of R-mod. Since all rings are $\mathbb{Z}$-modules, the discussion in section 9.7$\star$ shows that rings are the monoidal objects in $\mathbb{Z}$-Mod (Hence, the category $\mathbb{Z}$-Alg is the category **Ring**).

$$\begin{tikzcd}
(R \otimes S) \otimes (R \otimes S) \arrow[r, "a \otimes a"] \arrow[d, "\cong"', "1_R \otimes \tau_{S,R} \otimes 1_S"] & S \otimes S \arrow[dd, "m"] \\
(R \otimes R) \otimes (S \otimes S) \arrow[d, "\mu_R \otimes m"'] & \\
R \otimes S \arrow[r, "a"'] & S
\end{tikzcd}$$

The word "associative" was put in parenthesis in the definition of an algebra since there is a notion of an algebra that is not associative: some author's say the object in definition 12.1.13 is an *associative algebra*, and define algebras more generally as follows: Take an R-module M, and equip it with a binary operation such that the binary operation is left and right distributive (for all $a, b, c \in M$, $a \cdot (b + c) = a \cdot b + a \cdot c$, an $(b + c) \cdot a = b \cdot a + c \cdot a$), and that it is compatible with scalars (for all $a, b \in R$ and $x, y \in M$, $(ab) \cdot (xy) = (a \cdot x)(b \cdot y)$), then M is called an R-algebra. If the binary operation is associative and an identity exists, then it would be a ring operation it will be called an associative algebra.

There is a good reason adding the prefix "associative" in the greater context of mathematics: a large domain of study in mathematics is the study of *lie groups* and *lie algebras*, which focus on understanding the structure of smooth manifolds and representations of algebraic groups. Notably, Lie algebras are *not associative*, satisfying instead a relation called the *Jacobi identity*. However, since they are non-associative, they are not *rings*. They are thus quite different objects from R-algebras as we've defined them, and so don't encompass our current discussion. Their study take up many books (references to some are HERE) which the reader with a good background in linear algebra has the algebraic background to understand[3]. For the curious, we will briefly discuss (representations of) Lie Algebra in chapter 54.

From now on, when we say algebra or R-algebra, we will mean an associative algebra as defined in definition 12.1.13.

If $R = k$ is a field, then using the terminology of vector-spaces, a k-algebra would be defining a way of multiplying vectors. It is non-trivial to always define a multiplication structure on an R-module (I suggest you try it on $\mathbb{R}^n$!). Even harder is to ask whether there is a "natural" way of creating a multiplication structure, or more broadly to make an R-module M big enough so that we can define an R-algebra in a "natural" way[4]. We will in fact define such a notion in section 16.3.

[3]Though a good basis in differential geometry would go a long way for intuition
[4]natural meaning it satisfies the appropriate universal property

Example 12.1.16: R-Algebras

1. If R has positive characteristic p ($\#(\mathrm{im}\mathbb{Z}) = p < \infty$), then it is more natural to define a $\mathbb{Z}/p\mathbb{Z}$-algebra, namely since as a $\mathbb{Z}$-algebra the kernel of the map $\mathbb{Z} \to R$ would be $p\mathbb{Z}$, and hence by the 1st isomorphism theorem it homomorphism commutes with $\mathbb{Z}/p\mathbb{Z} \to R$.
2. If S is commutative, then any subring of $R \subseteq S$ will define R-algebra on S (take $\iota : R \hookrightarrow S$). If S is not commutative, any subring of the center will define an R-algebra.
3. Let $R[x_1, x_2, ..., x_n]$ be the polynomial ring over several variables. Then it is an R-algebra through $r \cdot p(x) = rp(x)$.
4. Let $k[G]$ be a group ring over the field k. Then $k[G]$ is naturally a k-algebra.
5. If D is a divisor ring, then $Z(D)$ is always a *field*. If $k = Z(D)$, then any division ring D has a k-algebra structure on it, which also implies it is a vector-space over k.
6. Let $R = k^n$ for some field k. Then R is a k-algebra, with multiplication done point-wise.
7. Let $R = M_n(k)$ over a field k. Then R is a k-algebra. More generally, If A is a k-algebra, then $M_n(A)$ is a k-algebra.

   Compare this example to the previous one, which we'll label $S = k^{n^2}$. As vector-spaces over k, $R \cong_k S$. However, as k-algebras, $R \ncong_{k\text{-}\mathbf{Alg}} S$, since the multiplication structure is different between the two (one has nilpotent elements, the other does not).
8. If A, B are k-algebra, so is $A \times B$. By the previous part, that implies
   $$M_n(A) \times M_n(B)$$
   is also a k-algebra.
9. At the heart of real analysis is the $\mathbb{R}$-algebra $\mathbb{R}$, and at the heart of complex analysis is the $\mathbb{C}$-algebra $\mathbb{C}$. Note that $\mathbb{C}$ is also a $\mathbb{R}$-algebra.
10. Let $R = \mathbb{R}^2$. Then we know that $\mathbb{R}^2$ has point-wise multiplication, but we can also define the following:

    define complex number multiplication

    then this also forms an $\mathbb{R}$-algebra. In fact, this is $\mathbb{R}$-algebra isomorphic to $\mathbb{C}$ (over $\mathbb{R}$)!
11. (If you know linear algebra) (example with defining multiplication in vector-spaces, Also thought the "Structure Coefficients" section here was really cool! `https://en.wikipedia.org/wiki/Algebra_over_a_field`)
12. One might ask themselves how many $\mathbb{R}$-algebra's exist (i.e. classify all $\mathbb{R}$-algebras). A little more generally, one might ask how many finite-dimensional associatvive divison algebras there are, that is, algebra on which you can has division and asociativity, and the notion of finite-dimension is well-defined. Only $\mathbb{R}$, $\mathbb{C}$, and $\mathbb{H}$ exist, with dimension 1, 2, and 4! This is called *Frobenius' Theorem*. In this way, the real, compelx, and quaternions are kind of special. [a]

There are a large number of areas that algebras appear in, to name a notable few:

1. (is the cross product an R-algebra) Let define a $\mathbb{R}^3$-algebra with $(a,b,c)\cdot(x,y,z) =$
2. commutative algebras in algebraic geometry
3. Banach-Algebra's in real analysis
4. C^*-algebra or $*$-algebras more generally in functional analysis
5. Though Lie algebra's are not [associative] algebras, there are some interesting intersections between the two TBD

[a] https://math.stackexchange.com/questions/368456/frobenius-from-hurwitzs-theorem

Theorem 12.1.17: Universal Property of Free R-algebras

$R[A]$ is a free commutative R-algebra over a [finite set] A

Notice how this is similar to the universal property of commutative polynomial rings. In fact, that is a special case of this theorem by taking $R = \mathbb{Z}$

***Proof*:**
Let S be any commutative R-algebra and $\alpha : A \to S$ be any set-function. Since S is an commutative R-algebra, it is accompanied by a function $f : R \to S$ defining its multiplication.

Since $A = \{a_1, a_2, ..., a_n\}$ is finite, $R[A] = R[x_1, x_2, ..., x_n]$. Similarly to how the universal property of polynomial rings was proven (theorem 9.2.4), we will extend $f : R \to S$ to $f_1 : R[x_1] \to S$ by mapping $f_1(x_1) = \alpha(a_1)$ and extending linearly. f_1 is still an R-algebra homomorphism since $f_1(n_0 + n_1x + n_2x^2 + \cdots + n_kx^k) = f(n_0) + f(n_1)\alpha(a_1) + f(n_2)\alpha(a_1)^2 + \cdots + f(n_k)\alpha(a_1)^k$, and Similarly for multiplying two polynomials (using the binomial theorem and both the additive and multiplicative linearity of f_1). We can again extend f_1 to f_2 by defining $f_2(a_2) = \alpha(a_2)$ so that $f_2 : (R[x_1])[x_2] \to S$ (usually written as $f_2 : R[x_1, x_2] \to S$) is an R-algebra homomorphism. Continuing inductively will get us an R-algebra homomorphism $F : R[x_1, x_2, ..., x_n] \to S$. Thus, picking $\iota : A \to R[x_1, x_2, ..., x_n]$ will show that $\alpha : A \to S$ clearly lifts uniquely to $F : R[A] \to S$, as we sought to show.

As a consequence of this, all R-algebras are quotient of polynomial rings!! That is, if S is a finitely genererarted S algebra

$$R[x_1, x_2, ..., x_n] \twoheadrightarrow S$$

12.1.3 Abelian Groups as Modules

Some entire domains of study are done on particular modules; we have actually already studied a particular collection modules extensively! The most immediate example is that all rings are R-modules over themselves with the action defined by:

$$r \cdot x = rx$$

Then submodules are precisely the ideals of R, and so the study of all R-modules includes the study of the ring R.

In this section, we show another important classes of R-modules: let $R = \mathbb{Z}$, and let $M = A$ be *any* abelian group (finite or infinite), and write the operation of A as $+$. Make A into a $\mathbb{Z}$-module by defining the ring action as :

$$na = \begin{cases} a + a + \cdots + a \ (n \text{ times}) & \text{if } n > 0 \\ 0 & \text{if } n = 0 \\ -a - a - \cdots - a \ (n \text{ times}) & \text{if } n < 0 \end{cases}$$

Then M is a $\mathbb{Z}$-module. In fact, this ring action is the only possible ring action on M (i.e. abelian groups):

Proposition 12.1.18: Abelian Group iff $\mathbb{Z}$-module

Every abelian group is a $\mathbb{Z}$-module, and every $\mathbb{Z}$-module is an abelian group

***Proof*:**
for $\Rightarrow$, we can always define the previous ring-action and so it's always a $\mathbb{Z}$-module

For $\Leftarrow$, it is simply a consequence of the existence of 1 and the distributivity property, i.e., the ring action $\mathbb{Z}$ performs on abelian groups is "stuck" since we must play out the consequence of the axioms.

By definition

$$n \cdot 1 = n$$

and therefore

$$n \cdot 2 = n \cdot (1 + 1) = n \cdot 1 + n \cdot 1 = n + n$$

and continuity inductively

$$n \cdot k = n \cdot (1 + \cdots + 1) = n + \cdots + n$$

k times. This result is exactly the ring action we defined earlier. And so, forced by the axioms of groups and modules, it must be that every $\mathbb{Z}$ action on an abelian foms this particular $\mathbb{Z}$-module.

This proposition has a very intersting consequence: for every R-module, $R \neq \mathbb{Z}$, then it always has at least one other module structure, namely the natural $\mathbb{Z}$-module structure! An intuitive way to remember this fact is that numbers as we have learnt them earlier in our life usually have a $\mathbb{Z}$-module structure ($\mathbb{Z}$, $\mathbb{Q}$, $\mathbb{R}$, $\mathbb{C}$). The natrual numbers form an abelian monoid, so this intuition doesn't directly apply to $\mathbb{N}$, but they indirectly apply since the same proof works for ring-actions on monoids (TBD?).

This example can also make us think of the class of all abelian groups as a subclass of modules. Abelian groups are also a subclass of groups. However, not every group is a module (and vice versa). The following diagram I'm about to draw shouldn't be taken as a commutative diagram, but as an "analogy":

$$\begin{tikzcd} \mathbf{Grp} & & R\text{-}\mathbf{Mod} \\ & \mathbf{Ab} \arrow[ul] \arrow[ur] & \end{tikzcd}$$

So in a sense, we are "switching gears" when studying modules. We are taking a different approach to the study of algebraic structures as compared to studying groups.

Exercise 12.1.2

1. Recall that for any $\alpha \in R$ for a ring R, $f_\alpha : R[x] \to R$ mapping $p(x) \mapsto p(a)$ is a ring-homomorphism. Is it an R-module homomorphism?

12.2 Homomorphism and Quotients of Modules

In this section we explore R-module homomorphisms and how the image of R-module homomorphisms can be represented by a particular quotient of the domain (as we've seen with groups and rings). As R-module homomorphsism are explored, they should be constrasted to rings in that they will have many nicer properties. They are in fact comparable to abelian groups; every subgroup of an abelian group is normal, and so too every submodule of a R-module will be "normal".

R-Module Homomorphisms

Definition 12.2.1: R-module homomorphism Terminology

Let R be a ring, M and N be R-module, and $\varphi : M \to N$ be a function. Then:

1. φ is called an *R-module homomorphism* if:
   (a) $\varphi(x + y) = \varphi(x) + \varphi(y)$ for all $x, y \in M$
   (b) $\varphi(r \cdot x) = r\varphi(x)$, for all $r \in R$, $m \in M$

   If furthermore there exists a φ^{-1} such that $\varphi \circ \varphi^{-1} = \varphi^{-1} \circ \varphi = \text{id}$ and φ^{-1} is an R-module homomorphism, then φ is called an *R-module isomorphism* and we write $M \cong_{\mathbf{R\text{-}Mod}} N$, or $M \cong N$ if it is unambiguous

   If $R = k$ is a field, then k-module homomorphisms are called *linear maps*, *linear functions*, or *linear transfomration*, and k-module isomorphisms are called *linear isomorphisms* (due to historical precedence)

2. Denote
   (a) $\ker(\varphi) = \{m \in M \mid \varphi(m) = 0\}$
   (b) $\text{im}(\varphi) = \varphi(M) = \{n \in N | \varphi(m) = n\}$

3. Define $\text{Hom}_R(M, N)$ be the set of all R-module homomorphisms from M into N, $\text{End}_R(M)$ the set of all R-module homomorphism from M to itself, and $\text{Aut}_R(M)$ the set of all all R-module isomorphisms from M to itself. A function $\varphi \in \text{End}_R(M)$ is called an *R-module enormophism*, and a function $\varphi \in \text{Aut}_R(M)$ is called an *R-module automorphism*.

Notice how it's key that the two structures be R-modules; If M is an R-module and N is an S-module, then $\varphi : M \to N$ is not an R-module or S-module homomrophsim unless either R has a compatible S-module structure of N has a compatible R-module structure that is preserved under φ (this is similar to how G-equivariant functions are betwen two G-sets)

Many of the properties demonstrated of group and ring homomorphisms apply for R-modules

Lemma 12.2.2: R-Module Homomorphism Equivalence

Let $\varphi : M \to N$ be a function. Then φ is an R-module homomorphism if and only if for every $r \in R$ and $x, y \in M$,

$$\varphi(x + ry) = \varphi(x) + r\varphi(y)$$

***Proof*:**
exercise

Lemma 12.2.3: R-Module Isomorphism Equivalence

Let $\varphi : M \to N$ be an R-module homomorphism. Then φ is an R-module isomorphism if and only if φ is an R-module homomorphism and bijective

***Proof*:**
Imitate the proof in proposition ref:HERE

Lemma 12.2.4: Composition Of R-Module Homomorphism/Isomorphism

Let M, N, L be R-modules. If $\varphi \in \mathrm{Hom}_R(L, M)$ and $\psi \in \mathrm{Hom}_R(M, N)$ then $\psi \circ \varphi \in \mathrm{Hom}_R(L, N)$

***Proof*:**
exercise

Lemma 12.2.5: R-module Automorphism Group

Let $\mathrm{Aut}_R(M)$ be the such set of R-module automorphism on M. Then $(\mathrm{Aut}_R(M), \circ)$ forms a group under composition.

***Proof*:**
exercise

Example 12.2.6: Module Homomorphisms

1. if a ring is an R-module over itself with the natural ring action, then R-module homomorphisms need not be ring homomorphisms (even from R to itself), and ring homomorphisms need not be R-module homomorphisms. For example, $\mathbb{Z}$ act on itself in the natural way,

and let $\varphi(x) = 2x$. φ is not a ring homomorphism [a], but φ is an R-module homomorphism. For the example going the other way, $F[x]$ act on itself in the natural way, and let $\varphi : f(x) \mapsto f(x^2)$. Then φ is not an $F[x]$-module homomorphism [b], but it is a ring homomorphism.

2. Let R be a ring and $M = R^n$ be an R-module with the natural ring action. Then the projective map is naturally an R-module homomorphism.

3. If we take $\mathbb{R}$ to be an $\mathbb{R}$ module over $\mathbb{R}$, then $\mathbb{R}$ has no non-trivial submodules (since $\mathbb{R}$ is a field, it's only ideals are 1 and itself). We call such a module a simple or an irreducible module. Thus, the quotient structure of $\mathbb{R}$ as an $\mathbb{R}$-module is trivial. However, if we consider $\mathbb{R}$ as a $\mathbb{Q}$-module (which we can). Then $\mathbb{Q}$ is a submodule of $\mathbb{R}$ (since $\mathbb{Q}$ is clearly closed under the ring action of $\mathbb{Q}$)! $\mathbb{R}$ as a $\mathbb{Q}$-module will be an infinite-dimensional vector-space [c]. Therefore, it becomes meaningful in this situation to look at $\mathbb{R}/\mathbb{Q}$ under the $\mathbb{Q}$ action!!

4. It should be done as an exercise that if $\varphi : M \to N$ is an R-module homomorphism, then the image and kernel are both R-modules.

5. (p.346)

[a] $\varphi(xy) = 2xy \neq 2x2y = \varphi(x)\varphi(y)$
[b] $x^2 = \varphi(x) = \varphi(x \cdot 1) = x\varphi(1) = x$, a contradiction
[c] In fact, it will even be *uncountably* infinite-dimensional

Recall in remark HomGrpIfAbRmk that $(\mathrm{Hom}_{\mathbf{Set}}(A, G), +)$ (where $+$ is defined pointwise: $f + g(x) = f(x) + g(x)$) can be given a group structure if G is abelian, and is not a group if G was not abelian (i.e. it not important what the domain was for the structure of the hom-set). Since every R-module is always over an abelian group, $\mathrm{Hom}_R(M, N)$ is *always* an abelian group. In the following proposition we further the structure of $\mathrm{Hom}_R(M, N)$

Proposition 12.2.7: Properties of $\mathrm{Hom}_R(M, N)$

Let M, N and L be R-module. Then

1. $(\mathrm{Hom}_R(M, N), +)$ is an abelian group with $+$ being defined pointwise:

$$(\varphi + \psi)(m) = \varphi(m) + \psi(m) \qquad \forall \varphi, \psi \in \mathrm{Hom}_R(M, N),\ \forall m \in M$$

   In particular $\varphi + \psi \in \mathrm{Hom}_R(M, N)$

2. If R is a commutative ring, then for $r \in R$ define $r\varphi$ by

$$(r\varphi)(m) = r(\varphi(m)) \qquad \text{for all } m \in M$$

   Then $r\varphi \in \mathrm{Hom}_R(M, N)$, and with this action $\mathrm{Hom}_R(M, N)$ is an R-module.

3. With addition defined as above, $(\mathrm{End}_R(M), +, \circ)$ is a ring. When R is commutative, $\mathrm{End}_R(M)$ is an R-algebra.

***Proof*:**
The first statement was proven above remark HomGrpIfAbRmk.

For an R-action to be defined on $\mathrm{Hom}_R(M, N)$ we would need the following to be equal:

$$(rs\varphi)(m) = rs \cdot \varphi(m)$$
$$(r(s\varphi))(m) = r \cdot (s\varphi)(m) = sr \cdot \varphi(m)$$

showing R must be commutative. The other axioms follow quickly

Finally, $(\mathrm{End}_R(M), +, \circ)$ follows the ring axioms. If R is commutative then for ring homomorphism $f : R \to \mathrm{End}_R(M)$, since f is commutative, $f(R)$ is always abelian, and hence $f(R) \subseteq Z(\mathrm{End}_R(M))$, and so $\mathrm{End}_R(M)$ is always an R-algebra, completing the proof.

Note that $(\mathrm{Aut}_R(M), +)$ is *not* a group structure, since the identity *does not exist* (the zero homomorphism is not an automorphism)![5]

Like with groups and rings, we want to know what conditions are needed on a sub-module so that:

$$(x + N) + (y + N) = (x + y) + N \qquad r(x + N) = (rx) + N$$

are both well defined. Since Modules are already abelian groups, than M/N is already well-defined as an abelian group! What's left is to see what condition's are needed so that a submodule N can be used to define an R-module structure on the abelian group M/N. Like abelian groups, every sub-module is "normal"!

Proposition 12.2.8: sub-modules are "normal" sub-modules

Let R be a ring and let M be an R-module and let N be a submodule of M. The (additive, abeliean) quotient group M/N can be made into an R-module by definin an action of elements of R by

$$r(x + N) = (rx) + N, \qquad \text{for all } r \in R,\ x + N \in M/N$$

The natural projection $\pi : M \to M/N$ defined by $\pi(x) = x+N$ is an R-module homomorphism with kernel N.

***Proof*:**
First, M/N is already an abelian quotient group. We want the ring action to be well defined, that is

$$x + N = y + N$$

which equivalently means that $x - y \in N$. Since N is a submodule, $r(x - y) \in N$, thus

$$rx + N = ry + N$$

therefore, the R-action on M naturally induces an action on M/N. The rest of the module axioms can now e easily verified.

Next, we must show the implication goes the other way, that is, if the elements of the quotient groups are formed by the cosets $x + N$, then N is the kernel of some R-module homomorphis (i.e. the natural projection). The natural projection map π described above is already the natural

[5] semi-groups are groups without an identity. Thus, $(\mathrm{Aut}_R(M)+)$ is a natural example of a semi-group(!), with a compatible group structure with $\circ$

projection of the abelian group M into the abelian group M/N. It only remains to show that it is a R-module homomorphism:

$$\begin{aligned}\pi(rm) &= rm + N \\ &= r(m+N) && \text{(by definition of the action of } R \text{ on } M/N) \\ &= r\pi(m)\end{aligned}$$

Thus, the natural projection an R-module homomorphism, completing the proof.

As with groups and rings, R-modules also have the isomorphism theorems. Since R-modules are already abelian groups, half the proof is already done. What is left to check would be if the action still preserved under the isomorphisms. To define the 2nd isomorphism theorem, we define what it means to take sum of two modules:

Definition 12.2.9: Sum Of Modules

Let A, B be submodules of an R-module M. Then the *sum* of A and B is the set

$$A + B = \{a + b | a \in A, b \in B\}$$

As before, $A + B$ is the smallest module containing A and B (as you sould check)

Theorem 12.2.10: Isomorphism Theorems For Modules

1. Let M, N be R-modules and let $\varphi : M \to N$ be an R-module homomorphism. Then $\ker(\varphi)$ is a submodule of M and

$$M/\ker(\varphi) \cong \varphi(M)$$

2. Let A, B be submodules of the R-module M. Then

$$\frac{A+B}{B} \cong \frac{A}{A \cap B}$$

3. Let M be an R-module, and let A and B be submodules of M, with $A \subseteq B$. Then

$$(M/A)/(B/A) \cong M/B$$

4. Let N be a submodule of the R-module M. There is a bijection beetween the submodules of M which contain N and the submodules of M/N. The correspondence is give nby $A \leftrightarrow A/n$, an for all $A \supseteq N$. The correspondence commutes with the processes of taking sums and intersection (i.e., is a lattice isomorphism between the lattice of submodules of M/N an dthe lattice of submodules of M which contain N

***Proof*:**
Very similar to previous isomorphism proofs.

12.2.1 Extensions and Exactness

We conclude this section with a quick word on extensions and where they will show up. Similarly to groups (and rings), we may wonder under what conditions is a module M an extension of two modules A and B, that is $M \cong B/A$, or in terms of a short exact sequence:

$$0 \to A \to M \to B \to 0$$

Section 3.7 showed that mathematicians managed to find the building blocks of finite group extensions, known as simple groups, and classify all finite simple groups, and section 5.4 showed that semi-direct products can be represented by split exact sequences. The classification of simple modules depends on the ring that's acting, R, and is much simpler since the underlying group is abelian. Similar to groups, there exist classification project for R-modules for some particular rings R, at least four of which will be covered in this book[6].

We also endeavour to study R-modules in which the extension property works nicely. This is further explored in Part X, chapter 51. Since all modules are abelian groups, the semi-direct product is simply the direct product. However, split exact sequences are still an important concept since it gives us a way of recognizing direct products using homomorphisms. A particularly interesting collection of modules are those for which every short exact sequence is a split exact sequence, or phrased differently that every short exact sequence splits. This endeavour is taken up in chapter 17⋆, and explored in full length in Part XI.

The vocabulary for a sequence of R-module homomorphisms that represent extensions is quite important, and so we define the term here for future use:

Definition 12.2.11: Exact Chain Complex

Let $M_0, M_1, M_2, ..., M_n, ...$ is a collection of R-modules $d_i : M_i \to M_{i-1}$ be a collection of R-module homomorphisms where

$$\cdots \xleftarrow{d_0} M_0 \xleftarrow{d_1} M_1 \xleftarrow{d_2} \cdots \xleftarrow{d_n} M_n \xleftarrow{d_{n+1}} \cdots$$

and $\ker(d_i) = \operatorname{im}(d_{i-1})$. Then the tuple $(M_i, d_i)_{i=-\infty}^{\infty}$ is called an *exact chain complex*, *exact chain*, or *exact sequence*. Often, we abbreviate and write $(M_\bullet, d_\bullet)$. If the arrows are pointing the opposite direction and $\ker(d_i) = \operatorname{im}(d_{i+1})$ then the resulting tuple $(M_i, d_i)_{i=-\infty}^{\infty}$ is called an *exact cochain*. If the chain has only finite length, (i.e. only finitely many of the modules are non-zero), then the resulting exact (co)chain is called a *finite exact (co)chain*.

A single homomorphism already sits inside such a chain, once one more module is named.

Definition 12.2.12: Cokernel

Let $\varphi \colon M \to N$ be a homomorphism of R-modules. The *cokernel* of φ is the quotient module

$$\operatorname{coker}(\varphi) := N/\operatorname{im}(\varphi)$$

The kernel measures how far φ is from being injective and the cokernel how far it is from being surjective: φ is injective exactly when $\ker(\varphi) = 0$, and surjective exactly when $\operatorname{coker}(\varphi) = 0$. Both

[6] When R is a Field, a Principal Ideal Domain, a Dedekind Domain, and Semi-Simple

statements are packaged in the one exact sequence

$$0 \to \ker(\varphi) \xhookrightarrow{\iota} M \xrightarrow{\varphi} N \xrightarrow{\pi} \operatorname{coker}(\varphi) \to 0$$

which is exact at $\ker(\varphi)$ because ι is injective, at M because $\ker(\varphi)$ is by definition $\operatorname{im}(\iota)$ and is what φ kills, at N because $\ker(\pi) = \operatorname{im}(\varphi)$ is exactly what the quotient divides out, and at $\operatorname{coker}(\varphi)$ because π is surjective.

(examples and some explanation here, important because it comes back in at least the next two chaptesr)

Exercise 12.2.1

1. Let M be an R-module. Prove that $\operatorname{Hom}_R(R, M) \cong M$
2. Let R^{op} be the opposite ring of R (see exercise ref:HERE), and let $\operatorname{End}_R(R)$ be all R-module homomorphisms from R to itself. Considering $(\operatorname{End}_R(R), +, \circ)$ as a ring show that $R^{\text{op}} \cong_{\mathbf{Ring}} \operatorname{End}_R(R)$ [Hint ; think of Yoneda's lemma]

12.3 Generation of Modules

We now extend the notion of generating algebraic structures to subsets of a module. We will start by generalizing the results of section 2.3.2: we take a subset of an R-module, and permit all possible combinations and R-actions of those elements to form an R-module, or equivaelntly we take the intersection of all R-submodules containing the elements of the set.

Then, as with groups and rings, the easiest way of creating a larger module from smaller ones is to take the direct sum or direct product. Due to their simplicity, it is worth taking the time to find out under what condition's can a module be decomposed into direct products or sums (just like for groups in section 5.2). We then take the approaching of trying to define the "most free" module we can put on an arbitrary set, similar to what we've done in section 2.3.4, that is we are looking to find the universal property of free-modules. As compared to groups, we will spend more time looking at minimal generating sets. Recall that for groups, it is possible that minimal generating sets can differ in cardinatliy. With modules, we will ask if there are modules where there exists a subset $A \subseteq M$ such that A generates M and if any other subset B generated A, then $|A| \leq |B|$. It will turn out that some modules have this property, some will be close, and other will not have this property at all.

After having tackled the question of generating modules "freely" and how to recognize free modules, we will then turn our attention to finitely generated modules, and modules where all submodules are finitely generated. Similalrly to how we can classify all finitely generated abelian groups, finitely generated and Noetherian modules have more "regularly", or equivalently have less "pathological" properties.

A new approach to studying structure's of objects we will take is considering the collection of homomorphisms on modules. As we saw in proposition 12.2.7, $\operatorname{Hom}_R(M, N)$ is always an abelian group, and if R is commutative it is always R-module. Thus, the "symmetries" of an R-module are themselves and R-module, meaning we would like to explore what properties M or N can have given the properties of $\operatorname{Hom}_R(M, N)$ have as an R-module, or even better that $\operatorname{End}_R(M)$ has as an R-algebra. This exploration is very fruitful, and will be of key importance in the theory of representation (of finite groups, lie algebras, etc.). In fact, if this intrigues the reader, after reading chapter 13 and chapter 14, the reader can skip to part X and read chapter 51 (only requiring knowledge from section 19.5 for the last 2 propositions of that chapter). A particular example of the above is when $N = R$ so that we consider $\mathrm{Hom}_R(M, R)$. This module is called the *dual module* of M, and will be very important in the study of M, even outside the field of algebra[7].

How does the generation of modules compare to the generation of groups and rings? With general groups, we saw that it was very difficult to show what the elements in an arbitrarily generated group look like (it is considered one of the hardest problems in mathematics, known as the *word problem for groups*). However, if G is abelian and finitely generated, then every element can be written in a very simple form due to the fundamental theorem of finitely generated abelian groups (theorem 5.1.8) With rings, we had that the objects that we quotient by (ideals) were no longer rings, and so took the time to study generation of ideals, but we have not studied any structure theorems on rings.

With modules, the story of the exploration of module structures is much closer to that of abelian groups, namely because every module *is* an abelian group. One major advantage of studying module is that many of the previous structure's we've studied can be represented as modules: since all abelian groups are $\mathbb{Z}$-modules, the study of the structure of abelian group is subsumed in the study of the structure of modules (we prove the Fundamental Theorem of Finitely Generated Abelian Groups in greater generality in chapter 14), and since all rings are module over themselves, the study of the structure of modules is also the study of the structure of rings! Due to the fact that all module are abelian groups, we can always collect "like terms":

$$r_1a_1 + r_2a_2 + s_1a_1 = (r_1 + s_1)a_1 + a_2$$

due to this, the order of the generating elements does not affect their sum! It becomes meaningful to ask if the ring acting on a module somehow determines some (or even all!) of the structure of the module. This question is a very meaningful question, and the answer will vary depending on what type of ring is acting on the module: chapters 13, 14, and 53 will each focus modules over particular types of rings and show how the answer to this question changes depending on what type of ring is acting on the module.

12.3.1 Direct Sum of Modules

We first update the terminology from the generation of groups and rings for the generation of modules:

[7]dual modules are central to Functional analysis and Harmonic Analysis

Definition 12.3.1: Generating Modules

Let M be an left R-module and let $(N_i)_{i\in I}$ be the submodules of M where I is an indexing set.

1. The *sum* of $(N_i)_{i\in I}$ is the smallest submodule of M containing each N_i, and is the set of all finite sums of elements from N_i:

$$\sum_i N_i := \{a_1 + \cdots + a_k | a_i \in N_i \text{ for all } i\}$$

If I is finite with $|I| = k$, then it is usually written as

$$N_1 + \cdots + N_k$$

2. For any subset $A \subseteq M$, then RA is the set:

$$RA = \left\{r_1 a_1 + \cdots + r_m a_m | r_1, ..., r_m \in R, a_1, ..., a_m \in A, m \in \mathbb{Z}^+\right\}$$

(where by convention $RA = \{0\}$ if $A = \emptyset$) is the smallest [left] submodule of M generated by the elements of A, an is called the *[left] submodule generated by* A. If A is finite the set $\{a_1, a_2, ..., a_n\}$ we shall write

$$Ra_1 + \cdots Ra_n$$

If N is a submodule of M (possibly $N = M$) and $N = RA$, for some subset, then A is called the *set of generators* or *generating set for* N, and we say N is generated by A and is sometimes written as $\langle A\rangle = N$.

3. A submodule N of M (possibly $N = M$) is *finitely generated* if there is some finite subset A of M such that $N = RA$.

4. A submodule N of M (possibly $N = M$) is *cyclic* if there exists an element $a \in M$ such that $N = Ra$, that is, if N is generated by one element

$$N = Ra = \{ra|\ r \in R\}$$

and is sometimes written as $\langle a\rangle = N$.

It is an easy starter exercise to show that the structures in definition 12.3.1 are indeed modules. If R is a *rng*, then it is not necessarily the case that $A \subseteq RA$.

Example 12.3.2: Example

1. Let $R = \mathbb{Z}$ and let M be any $\mathbb{Z}$-module. Take any $a \in M$. Then $\mathbb{Z}a$ is a cyclic sub-module. It is in fact a cyclic subgroup! More generally, A generates M as a $\mathbb{Z}$-module if and only if A generates M as an [abelian] group. That is, generating M through A as a group (using $+$ and $-$ in groups) will produce as many elements as generating M through A as a module (which also allows $\mathbb{Z}$-actions). Thus, the definition of a finitely generated $\mathbb{Z}$-module is identical to the definition of a finitely generated group (definition 5.1.6).

2. Let R be a ring and let $M = R$ be the [left] R-module of R acting on itself. Then R is finitely generated as a module, even if R is infinitely generated as a ring! Even better: it is cyclic, since $R \cdot 1 = R$. Thus, in terms of generation, R is the "simplest" R-module.

3. If R is an R-module over itslef, then $I \subseteq R$ is a finitely generated R-module if and only if it is a finitely generated *left-ideal.* If R was a right-module, then I would be a finitely generated *right-ideal*, and if R was commutative, I would be a finitely generated *two-sided ideal.*

4. Let M be a finitely generated R-module. Then it is not the case that every submodule of M is finitely generated! Take $P[x_1, x_2, ...] = P[X]$ (i.e. the polynomial ring over countably many coefficients) which is a $P[X]$-module over itself. Then $P[X]$ is cyclic as a $P[X]$-module, and hence finitely generaed, however $(x_2, x_4, x_6, ...) \subseteq P[X]$ is a $P[X]$-submodule of $P[X]$ (being a left-ideal), however, it is an *infinitely generated* left-ideal. Finitely generated modules in which all submodules are finitely generated will be called *Noetherian.*

5. Let R^k be an R-module with the action being pointwise multiplication. Then R^k is finitely generaed by
$$e_i = (\underbrace{0, ..., 0}_{i-1 \text{ times}}, 1, \underbrace{0, ..., 0}_{k-i \text{ times}})$$
hence, for any $(r_1, r_2, ..., r_k) \in R^k$,
$$(r_1, ..., r_k) = \sum_i r_i e_i$$

6. Here is a neat way of writing down M if $A = M/B$. Then $M \cong A' + B$, since any element in M is some element of A (namely A' is a set of coset representative when taking $b = 0$), and an element of B:
$$\frac{M}{B} = \frac{A' + B}{B} \cong A$$
Note that is in general smmimply a *subset* of M, we can define the operation on A' by taking:
$$a_1 + a_2 = a_3 \qquad a_3 \text{ is the appropriate representative from } [a_1 + a_2] \text{ that belongs in } A'$$
In this way, $A' \not\subseteq M$ for as an R-module since the binary operation is different. Note too that $M = A' + B$ does not imply that $M \cong A \oplus B$. For example, consider $A = \mathbb{Z}/6\mathbb{Z}$ with $M = \mathbb{Z}$, and $B = 6\mathbb{Z}$. Letting $A' = \{0, 1, 2, 3, 4, 5\}$, we get that as sets:
$$M = A' + B = \{0, 1, 2, 3, 4, 5\} + 6\mathbb{Z} = \mathbb{Z}$$
for example, $10 = 4 + 6$, and $101 = 5 + 96$. However, giving A' the group structure of $\mathbb{Z}/6\mathbb{Z}$:
$$\mathbb{Z} \ncong_R (\mathbb{Z}/6\mathbb{Z}) + 6\mathbb{Z}$$
For the right hand side has torsion elements, while the left does not. For the $+$ to be upgraded to a $\oplus$, the operation of the quotient must in some natural way be compatible with the operation in M, that is A must in not only be the quotient M/B, but must also be considered as $A \subseteq M$. This is explore in chapter 17⋆.

Naturally, we would want to find when direct sums of modules behave like direct products, or more specifically external direct sums

Definition 12.3.3: Direct Product and Sum of Modules

Let $(M_i)_{i\in I}$ be a collection of R-modules where I is an indexing set. Then the collection of tuples $(m_i)_{i\in I}$ where $m_i \in M_i$ with addition and action of R defined componentwise is called the *direct product* of $(M_i)_{i\in I}$, denoted

$$\prod_{i\in I} M_i$$

If I is finite where $|I| = n$, then it is usually denoted:

$$M_1 \times \cdots \times M_k$$

If we take the subset of $\prod_{i\in I} M_i$ where only finitely many of the terms are nonzero, then this this set is called the *external direct sum*, and is written as:

$$\bigoplus_{i\in I} M_i$$

If I is finite where $|I| = n$, then it is usually denoted:

$$M_1 \oplus M_2 \oplus \cdots \oplus M_n$$

If I is infinite and $R \neq 0$, then $\bigoplus_{i\in I} M_i \subsetneq \prod_{i\in I} M_i$.

12.3.4 Remark It might be tempting to say that:

$$M \cong N_1 \oplus N_2 \quad \Leftrightarrow \quad M = N_1 + N_2$$

however that is not the case: Take $\mathbb{Z}$ as a $\mathbb{Z}$-module over itself, $N_1 = 2\mathbb{Z}$ an $N_2 = (0)$. Then $2\mathbb{Z} + 0 = 2\mathbb{Z}$, and:

$$\mathbb{Z} \cong 2\mathbb{Z} \qquad \text{but} \qquad \mathbb{Z} \neq 2\mathbb{Z}$$

which we can see since, $1 \notin 2\mathbb{Z}$

If $M \cong A \times B$ for some modules M, A, and B, then for any $m \in M$, there exist unique pair of elements $a \in A, b \in B$ such that $m \mapsto (a, 0) + (0, b)$. Thus, any generating set (or more specifically minimal generating set) of M breaks down to a question of a generating set (or minimal generating gset) of A and B. The key here is that any element $m \in M$ can be uniquely represented by elements a and b (namely since these two modules are isomorphic).

Let's say conversely that we want to find submodules $A, B \subseteq M$ for which $M = A + B$. If a module M is finitely generated, by Zorn's lemma there must exist a minimal nonnegative integer d such that M is generated by d elements (and no less)[8]. If M was not finitely generated, this need not be the case. If M can be generated by d elements, it need not be the case that there is a unique set of generating elements, nor the case that the R-linear combination of these elements be unique:

[8]Note that minimal doesn't mean smallest! There can be multiple different $d \in \mathbb{N}$ that satisfy this property, see the discussion in section 2.3

Example 12.3.5: Non-Unique Sets And R-Linear Combinations

here

It will turn out that we almost never get a unique minimal generating set [a]. Take for example R over itself. Let $u \in R$ be a unit of R. Since u is a unit, let $v \in R$ such that $uv = 1$. Then $R \cdot 1 = R \cdot (vu) = (Rv)u = R \cdot u$ (since $u \in R$, $uv = 1 \in Rv$, and from that we can show that $(Rv)u = Ru$).

[a] If you have already taken a course in linear algebra, this is equivalent to saying that there is no unique basis

Though unique minimal generated sets are rare to come accorss, uniqe R-linear combinations are much more common. For example, taking R over itself so that $\{1\}$ generated R as an R-module. Then every R-linear combination is unique, that is

$$r \cdot 1 = s \cdot 1 \Leftrightarrow r = s$$

We can see this by taking the homomorphism $\varphi : R \to M$, $r \mapsto r \cdot 1$ and look at the kernel. This property is very desirable, since it will allow us to show that M can split into a direct sum of submodules. If the R-linear combinations are unique, then we need only concern ourselves with any minimal generating set when considering the structure of such a module!

For the rest of the chapter, we identify equivalent properties to saying R-linear combinations are unique, and identify modules which have this property.

Proposition 12.3.6: Equivalent Conditions for Linear Combinations

Let $N_1, ..., N_k$ be (not necessarily distinct) submodules of R-module M. Then the following are equivelent

1. the map $\pi : \bigoplus_{i \in I} N_i \to \sum_i N_i$ defined by

$$(n_i)_{i \in I} \mapsto \sum_i n_i$$

is an R-module isomorphim. If $|I| < k$ is finite, we write:

$$N_1 + N_2 + \cdots + N_k \cong N_1 \times N_2 \times \cdots \times N_k$$

2. $N_j \cap (N_1 + N_2 + \cdots + N_{j-1} + N_{j+1} + \cdots + N_k) = 0$ for all $j \in \{1, 2, ..., k\}$
3. Every $x \in N_1 + N_2 + \cdots + N_k$ can be written *uniquely* in the form $a_1 + a_2 + \cdots + a_k$ with $a_i \in N_i$

***Proof*:**

For (1) implies (2). let's assume that the intersection is not empty. Then there exists some a_j such that

$$a_j = a_1 + a_2 + \cdots + a_{j-1} + a_{j+1} + \cdots + a_k$$

However, this implies that

$$\pi(0, ..., a_j, ..., 0) = \pi(a_1, a_2, ..., a_{j-1}, a_{j+1}, ..., a_k)$$

but
$$(0, ..., a_j, ..., 0) \neq (a_1, a_2, ..., a_{j-1}, a_{j+1}, ..., a_k)$$
contradiction injectivy, since π is an isomorphism

For (2) implies (3). Let's say there are two linear combinations
$$a_1 + a_2 + \cdots + a_k = b_1 + b_2 + \cdots + b_k$$
then we can isolate any jth components:
$$b_j - a_j = (b_1 - a_1) + \cdots + (b_{j-1} - a_{j-1}) + (b_{j+1} - a_{j+1}) + \cdots + (b_k - a_k)$$
which implies by our previous proposition
$$b_j - a_j \in N_j \cap (N_1 + N_2 + \cdots + N_{j-1} + N_{j+1} + \cdots + N_k) = 0$$
thus, $b_j - a_j = 0$ and so $b_j = a_j$ for every j, showing that the linear combination is unique!

For (3) implies (1), Notice that the function is surjective construction: for any element $a_1 + a_2 + \cdots + a_k$, the element $(a_1, a_2, ..., a_k)$ would map to it. What (2) does is show injectivity, since if two sums are the same, then the direct product of elements must also be the same!. This map is also easily checked to be an R-module homomorphism. For example
$$\begin{aligned} \pi(r \cdot (a_1, a_2, ..., a_k) &= \pi(ra_1, ra_2, ..., ra_k) && (12.1)\\ &= ra_1 + ra_2 + \cdots + ra_k && (12.2)\\ &= r \cdot (a_1 + a_2 + \cdots + a_k) && (12.3)\\ &= r\pi(a_1, a_2, ..., a_k) && (12.4)\end{aligned}$$
where line 3 comes from induction on $r \cdot (m + n) = r \cdot m + r \cdot n$, and linearity comes easily too.

This show's us how we may split up a module into "direct products" which satisfy the universal property of direct products. We now introduce a notion that is slightly stronger than the external direct product and of sums of modules:

Definition 12.3.7: Internal Direct Sum

Let $(N_i)_{i \in I}$ be a collection of submodules of M modules respecting one of the 3 condition's in proposition 12.3.6. If:
$$\sum_i N_i = M$$
then we will replace the summation symbol $\sum_i$ with $\bigoplus_i$. If the indexing set is finite, then:
$$N_1 + \cdots + N_k = N_1 \oplus \cdots \oplus N_k$$
Thus, we are *overloading* the notation of $\oplus$ to represent the fact that the sum $\sum_i N_i$ is isomorphic to $\oplus_i N_i$ *and* that $\sum_i N_i = M$

We have already shown in remark where M is an R-module that is isomorphic to a direct sum of submodules $(N_i)_{i \in I}$ but $\sum_i N_i \neq M$, namely by taking $\mathbb{Z}$ as a $\mathbb{Z}$-module and $N_1 = 2\mathbb{Z}$, $N_2 = (0)$.

Then:

$$\mathbb{Z} \cong 2\mathbb{Z} \oplus (0) \qquad \mathbb{Z} \neq 2\mathbb{Z} \oplus (0)$$

In chapter 17⋆ and 51, we will further explore how to find when a module M can be decomposed into direct products.

One case is settled immediately, and is used often enough to record. When the *ring* is a product, every module over it splits, and the splitting is forced by the two idempotents $(1,0)$ and $(0,1)$. The upshot is that module theory over $R \times S$ is nothing more than module theory over R and over S carried out side by side.

Proposition 12.3.8: Modules over a Product Ring

Let R and S be rings and put $e = (1,0)$ and $f = (0,1)$ in $R \times S$, so that e and f are central idempotents with $e + f = 1$ and $ef = 0$. Then every $(R \times S)$-module M decomposes as an internal direct sum

$$M = eM \oplus fM$$

where eM is an R-module and fM is an S-module. The assignments $M \mapsto (eM, fM)$ and $(M_1, M_2) \mapsto M_1 \oplus M_2$ are mutually inverse, and give an equivalence of categories

$$(R \times S)\textbf{-Mod} \;\simeq\; R\textbf{-Mod} \times S\textbf{-Mod}$$

Under it, a module is finitely generated, or projective, exactly when both of its components are. Freeness is *not* componentwise: $M_1 \oplus M_2$ is free over $R \times S$ if and only if M_1 and M_2 are free of the *same* rank, since a free $(R \times S)$-module is $(R \times S)^n \cong R^n \oplus S^n$.

***Proof*:**
Step 1: the decomposition. For $m \in M$ we have $m = 1 \cdot m = (e + f)m = em + fm$, so $M = eM + fM$. If $x \in eM \cap fM$, write $x = em' = fm''$; then $x = ex = e(fm'') = (ef)m'' = 0$, using that e is idempotent and $ef = 0$. So the sum is direct.

Step 2: the components are modules over the factors. The subgroup eM is stable under $R \times S$, and f acts on it as $fe = 0$, so the action factors through $(R \times S)/(0 \times S) \cong R$, making eM an R-module. Symmetrically fM is an S-module.

Step 3: the two constructions are inverse. Given R-module M_1 and S-module M_2, let $R \times S$ act on $M_1 \oplus M_2$ componentwise. Then $e(M_1 \oplus M_2) = M_1$ and $f(M_1 \oplus M_2) = M_2$, recovering the pair. Conversely Step 1 rebuilds M from (eM, fM). Both assignments send a homomorphism to its restrictions, and a map commutes with the action of e and f exactly when it preserves the two summands, so the correspondence is functorial in both directions and is an equivalence.

Step 4: the listed properties. Finite generation and projectivity are checked componentwise because the equivalence carries $R \times S$ to the pair (R, S) and direct sums to direct sums. A generating set for M projects to generating sets of eM and fM and conversely, giving finite generation; and M is a summand of a free $(R \times S)$-module exactly when each component is a summand of a free module over its factor, giving projectivity.

Freeness is the exception, and the equivalence is what shows why. A free $(R \times S)$-module of rank n corresponds to the pair (R^n, S^n), so M is free exactly when its two components are free *of a common rank.* For $R = S = k$ a field, $M = k \oplus k^2$ has both components free, of ranks 1 and 2, and is projective; but it is not free, since that would force $1 = 2$. This completes the proof.

To conclude, we re-phrase the above results in the language of universal properties. The properties of $M \oplus N$ can be described in terms of maps $\varphi : L \to M \oplus N$ where this map is uniquely determined by the maps $\varphi : L \to M$ and $\varphi : L \to N$, that is:

Proposition 12.3.9: Universal Property of Products, Modules

Let M, N be two R-modules. Then $M \oplus N$ as constructed above satisfies the universal property products: for any R-module L, if $\varphi :_M L \to M$ and $\varphi :_N L \to N$ are two R-module homomorphisms, then there exists a unique R-module homomorphism $\varphi : L \to M \oplus N$ such that the following diagram commutes:

$$\begin{tikzcd} & L \arrow[d, dashed, "\varphi"] \arrow[ddl, bend right, "\varphi_M"'] \arrow[ddr, bend left, "\varphi_N"] & \\ & M \oplus N \arrow[dl, "\pi_M"'] \arrow[dr, "\pi_N"] & \\ M & & N \end{tikzcd}$$

More generally, if M_i is a collection of R-modules for an indexing set I, for any R-module L and collection of R-module homomorphisms $\varphi_i : L \to M_i$, there exists a unique R-module homomorphism $\varphi : L \to \oplus_{i \in I} M_i$ such that all of the following diagrams commutes:

$$\begin{tikzcd} L \arrow[d, "\varphi"'] \arrow[dr, "\varphi_i"] & \\ \oplus_{i \in I} M_i \arrow[r, "\pi_{M_i}"] & M_i \end{tikzcd}$$

***Proof*:**
The reader should do this exercise as a good work-through of how to show a universal property, and to be comfortable with commutative diagrams for proofs in later chapters.

12.3.2 Free Modules

Given a ring R and a an arbitrary abelian group A, we can always define the trivial module via the action $r \cdot a = 0$: let's label this module $T(A)$ and call it the trivial R-module. Just like trivial groups, trivial modules offer little in structure[9]. It would be desirable if we can find a better struture on A, in particular if A represents the generating set of some R-module M, then we would like to define a "free R-module" $F(A)$ (or more particularly $F^R(A)$ to remember the ring R) which represents an R-module with no relationship, and then find a R-submodule $N \subseteq F(A)$ such that $F(A)/N \cong M$ (i.e. the submodule N woule represent adding relationships to $F(A)$). This mirrors the development of the free group from section 2.3.3 and 3.2.

Let's say we were to define a R-module $F^R(A)$ on the generating set A of M with only the bare minimum relations to make it into an R-module ($1 \cdot m = m$, $(rs) \cdot m = r \cdot (s \cdot m)$, and so forth).

[9]When dealing with groups, we also defined $\mathrm{Sym}(A)$. As a thought exercise, can you see why we cannot parallel the same idea with module? Hint: Think of $\mathbb{Z} \hookrightarrow \mathbb{Q}$

Then similarly to free groups, we would want that for any element of $x \in F^R(A)$, which is some linear combination:

$$x = \sum_{a \in B \subseteq A} r_a \cdot a \quad \text{for some subset } B \subseteq A$$

to be equal to 0 if and only if $r_a = 0$ for all a. This is similar to free groups, where any relation between generating elements can be move to the left hand side and the right hand side equals to the identity element. By proposition 12.3.6, this means that $F^R(A) = R^{|A|}$ is our desired free R-module. To see this more clearly, let's say that some $r_a \neq 0$. Then that means we have added some relation into $F^R(A)$, that is the function $\varphi : R^{|A|} \to F^R(A)$ would have a non-trivial kernel, going against the idea that $F^R(A)$ is a free-module.

If $F^R(A)$ is a free module on A, just like for free groups, any set-function from A to any R-module N, $f : A \to N$, should lift to an R-module homomorphism $F : F^R(A) \to N$, i.e. there is a map $\iota : A \hookrightarrow F^R(A)$ that injects A into $F^R(A)$ and $f = F \circ \iota$. In particular, given any arbitrary map $f : A \to N$, no matter where f maps A, whether f is injective or not, we can find a map F such that $F|_A = f$, so F must map any element of the set A exactly as how f would map it, hence F is an extension of f that has the restriction of being an R-module homomorphism. Furthermore, N is any R-module, meaning any $f : A \to N$ can be lifted to $F : F^R(A) \to N$ (see how this is not the case for an arbitrary domain M, or check out section 2.3.3 for the parallel problem in group theory). A consequence of this result is that we can determine how $F^R(A)$ maps to an R-module N simply by determining where A maps to, that is we have the universal property of free modules.

12.3.10 On Recovering A We would have to already know of the set A in order to do the above strategy. The notation $F^R(A)$ makes it seem obvious that we know about the set A, namely since we will soon construct $F^R(A)$ with respect to A, but it is not so obvious that if we are given a module M that we say is a free R-module, can we find a subset A such that $F^R(A) \cong_R M$. In fact, depending on the ring R acting on M, it will sometimes be possible to "recover" A, while other times it will be impossible[a]. We will explore this in chapter 13

[a]It will be possible to say there exists such a set using the axiom of choice, but it will be impossible to give a description of the elements

We now prove that $F^R(A) = R^{|A|}$ does indeed respect the universal property of free R-modules.

Theorem 12.3.11: Universal Property of Free Modules

For any set A, there exists a free R-module $F^R(A)$ on the set A where $F^R(A)$ satisfies the following *universal property*: If M is any R-module and $\varphi : A \to M$ is any map of sets, then there is a unique R-module homomorphism $\psi : F(A) \to M$ such that $\psi|_A = \varphi$, that is, the following diagram commutes

$$\begin{tikzcd} S \arrow[r, hook, "\text{incl.}"] \arrow[dr, "\varphi"'] & F(S) \arrow[d, "\psi"] \\ & M \end{tikzcd}$$

When A is a finite set $\{a_1, a_2, ..., a_n\}$, $F^R(A) = Ra_1 \oplus \cdots + \oplus RA_n \cong R^n$, and if A has any cardinality, then:

$$F^R(A) = \bigoplus_{a \in A} R_a$$

where A is treated as an indexing set.

***Proof*:**
if $A = \emptyset$, then let $F(A) = \{0\}$. Then this trivially satisfies:

$$\begin{tikzcd} \emptyset \arrow[r, hook, "\text{incl.}"] \arrow[dr, "\varphi"'] & \{0\} \arrow[d, "\Phi"] \\ & M \end{tikzcd}$$

since $\emptyset$ is included in any set, φ is the vacuous map, and $\Phi(0) = 0$ makes Φ into an R-module homomorphism which restircted to $\emptyset$ is itself again.

So let $A \neq \emptyset$. What we'll do is construct funtions with the domain as a set and the co-domain as the ring R. These functions will be the mechanism by which set A is given an R-module structure! Let $F^R(A)$ be the collection of all set functions $f : A \to R$ such that $f(a) = 0$ for all but finitely many $a \in A$:

$$F^R(A) = \{f \in \mathrm{Hom}_{\mathbf{Set}}(A, R) \mid f(a) \neq 0 \text{ for only finitely many } a \in A\}$$

Make $F^R(A)$ into an R-module by defining pointwise addition and pointwise multiplication for the action:

$$(f + a)(a) = f(a) + g(a)$$
$$(rf)(a) = r(f(a))$$

for all $a \in A$, $r \in R$, $f, g \in F^R(A)$. Notice how $F^R(A)$ is isomorphic to $R^|A|$, meaning this construction matches definition 12.3.12.

Let $\iota : A \to F^R(A)$ vis $a \mapsto f_a$, where f_a is the function which is 1 at a and zero everywhere else. We can thus identify a set $A \leftrightarrow A' \subseteq F^R(A)$ as the subset containing all of f_a. Notice that since any f has only finitely many non-zero elements, we can write any $f \in F^R(A)$ as

$$f = r_1 f_{a_1} + r_2 f_{a_2} + \cdots + r_n f_{a_n}$$

which by our identification of A' with A, represents:

$$r_1a_1 + r_2a_2 + \cdots + r_na_n$$

where f takes on the value of r_i at a_i and is zero at all other elements of A. Even better, by construction, every element of $F^R(A)$ has a *unique* expression as such a formal sum, since by construction, non of the a_i have any relation to each other, there can't be a way of cancelling them out!

To establish the universal property, of $F^R(A)$, Let's say we have a function $\varphi : A \to M$, where M is an R-module. We need to show this function raises to an R-module homomorphism $\Phi : F^R(A) \to M$. Let Φ be defined by

$$\sum_{i=1}^{n} r_1a_i \mapsto \sum_{i=1}^{n} r_i\varphi(a)$$

by the uniqueness fo the expression of the elements of $F^R(A)$ as linear combinations of a_i, we see that Φ is indeed a well dfiend R-module homomrophsim!!

check here

By definition, the restriction of $\Phi|_A$ is φ. For uniqueness of $F^R(A)$, since $F^R(A)$ is generated by A, once we know the values of an R-module homomorphism on A, tis value on every element of $F^R(A)$ are uniquely determined (by force of construction), so Φ is the unique extension of Φ to all of $F^R(A)$.

If A is finite ($\{a_1, a_2..., a_n\}$), proposition 12.3.6

$$F^R(A) = Ra_1 \oplus Ra_2 \oplus \cdots \oplus Ra_n$$

Since $R \cong Ra_i$ (under $r \mapsto ra_i$) by proposition 12.3.6 again, the direct sum is isomorphic to R^n.

Definition 12.3.12: Free Module

An R-module F is said to be *free* on the subset A of F if for every nonzero element x of F, there exists a unique nonzero elements $r_1, ..., r_n$ of R and unique $a_1, ..., a_n$ in A such that $x = r_1a_1 + \cdots r_na_n$, for some $n \in \mathbb{Z}^+$, which by proposition 12.3.6 implies that:

$$F = \bigoplus_{i \in I} R$$

If R is a commutative ring, the cardinality of A is called the *free rank* of F, or just *rank* if unambiguous.

Note that A need not be finite, but the linear combinations of elements of A need to be finite (notice that we say $a_1, ..., a_n \in A$, and not $a_1, ..., a_n, ... \in A$). From now on, we will write R^n to mean the free R-module $R \oplus R \oplus \cdots \oplus R$. So $(\mathbb{Z}/2\mathbb{Z})^3$ is a free $(\mathbb{Z}/2\mathbb{Z})$-module. R^1 is a free R-module of rank 1 (to be distinguished from R which we will treat as a ring). If the superscript is infinite, say $N = |\mathbb{N}|$ or $N = |\mathbb{R}|$, then R^N is considered an infinite direct sum, not a direct product. In fact, non-trivial infinite direct product of modules (modules of the form R^N for an N of infinite cardinality) are far from being free (see exercise ref:HERE[10]). Furthermore, the notion of rank is not necessarily well-defined if R is not commutative. We shall explore more of this in chapter 13.

Example 12.3.13: Free Modules

1. Let R be any ring with identity. Then R over itself is a free-module! The element 1 will always generated all of R since 1 is a unit (and so $ab \neq 0$ for any $b \neq 0$) and $r \cdot 1 = r$, and so it's a linearly independent generating set!

2. The "freeness" is dependent on the ring you choose, as well as the module (i.e., the abelian group). So $\mathbb{Z}/n\mathbb{Z}$ over itself is free (by the logic provided above), but not over $\mathbb{Z}$ (which we already know is a well-defined module), since

$$n \cdot 3 = 0$$

   In general, one criterion for freeness is that $r \cdot a \neq 0$ for any r. This criterion is called being *torsion-free*, and if $r \cdot a = 0$, then r is said to be a *torsion element* of the R-module M. However, a module being torsion free does not imply the module is free.

3. Since $\mathbb{Z}/n\mathbb{Z}$ overt $\mathbb{Z}$ is not a free-module, it does not have a basis. If we try to choose $A \subseteq \mathbb{Z}/n\mathbb{Z}$ to be a basis, it will always have torsion elements. $\mathbb{Z}/n\mathbb{Z}$ over $\mathbb{Z}$ is also an example of a module *without* any linearly independent sets!

4. Let $\mathbb{C}[x, y]$ act on itself. Then $\mathbb{C}[x, y]$ is a free module. As stated in example 12.1.7, ideals of modules over themselves are sub-modules. Take $(x, y) \subseteq \mathbb{C}[x, y]$. Then

$$y \cdot x - x \cdot y = 0$$

   showing that (x, y) is *not* free! However, (x, y) is certainly torsion-free, since $\mathbb{C}[x, y]$ is an integral domain.

   Thus, freeness is not preserved by taking sub-modules

5. Let $\mathbb{Z}$ act on itself. Then $\mathbb{Z}/n\mathbb{Z}$ is a quotient $\mathbb{Z}$-module as we've seen in section 12.2. However, as we've seen in the 2nd example, $\mathbb{Z}/n\mathbb{Z}$ over $\mathbb{Z}$ is not free.

   Thus, freeness is not preserved when taking quotient modules.

6. Let's say $S = \{1, 2, 3\}$, and $M = \mathbb{R}^3$ (over $\mathbb{R}$). Take any $f : S \to M$. For our concrete example, let's say:

$$f(1) = (\varphi, \sqrt{2}, 4) \qquad f(2) = (0, 0, 0) \qquad f(3) = (0, 5753, 9)$$

   I chose particularly random values to show that f is indeed any set function map. Now if you consider $F^R(S)$ (over $\mathbb{R}$), we'd have some uncountable set of elements, which most importantly are all distinct! It's that last part that let's us say that

$$\psi : F^R(S) \to \mathbb{R}^3$$

   will be the unique homomorphism where $\psi|_S = f$. Consider

[10] Inspired by the link here

$$\psi(x) = \psi(\sum_{i=1}^{3} r_i s_i) = \sum_{i=1}^{3} r_i \varphi(s_i)$$

This function is indeed well defined since every x is *unique* in it's represetnation from S, since if it wasn't, and we consider

$$x = y(\sum_{i=1}^{3} r_i s_i = \sum_{i=1}^{3} r_i' s_i)$$

then it is not true in general that

$$\psi(x) = \psi(y)$$

what let's use insure that this will always be true is precicely the linear independence, or free property, of the module!

Finally, this is indeed a [unique] homomorphism. it is easy to check that

$$\psi(x + ry) = \psi(x) + r\psi(y)$$

and the uniqueness will come from free modules elements having no relation to each other (if they did, you might be able to have a second homomrophism).

7. You can have a module with linearly independent sets, but that always fail to generate the module, meaning it is not a free module. For example, $\mathbb{Z}^{\mathbb{N}}$ has a great multitude of linearly independent sets (even infinite ones), however it is not a free-module. This is a rather complicated proof, and is relegated to exercise ref:HERE in chapter ref:HERE.

8. Take $\mathbb{C} \times \mathbb{C}$ over itself. This is free module (its basis is $(1, 1)$). Then $\mathbb{C} \times \{0\}$ is a submodule, but not a free module: Any basis of $\mathbb{C} \times \{0\}$ would consist of elements of the form $(0, a)$. If A is the colleciton of all elements in the 2nd component of a basis, then:

$$\sum_{a \in A} (a, 0) \cdot (0, a) = \sum_{a \in A} (0, 0) = (0, 0)$$

meaning a non-trivial linear combination summed to zero.

9. Let R be a Principal Ideal Domain acting over itself. Then any submodule N of R is an ideal, and since it's principal $N = (a)$ for some $a \in R$. Since $(a) = Ra$, it follows that N is torsion-free (see ref:HERE). Thus, any submodule of R is torsion-free. (TBD, not sure, trying to get to if R is not principal, then (x, y) is never torsion free. Maybe move this to Mod over PID sec? exercice 6 in section 12.1 Dummit)

10. In chapter 13, we'll show that $\mathbb{R}$ is a $\mathbb{Q}$-vector space with uncountable basis! Note that even though the basis is uncountable, any element $x \in \mathbb{R}$ will still be a $\mathbb{Q}$-linear combination of *finitely many* basis elements.

11. On the other hand $\mathbb{Q}$ over $\mathbb{Z}$ is *not* a free module (i.e. $\mathbb{Q}$ is not a free abelian group). To see this, notice that for any $\frac{a}{b}, \frac{c}{d} \in \mathbb{Q}$, there exists $m, n \in \mathbb{Z}$ (namely $m = bc,\ n = -da$) such that $(bc)\frac{a}{b} + (-da)\frac{c}{d} = 0$, and hence any two elements are linearly independent, meaning if $\mathbb{Q}$ was a free $\mathbb{Z}$-module, it would have to be cyclic. As we've seen in section 2.3.4, the only free cyclic $\mathbb{Z}$-module is $\mathbb{Z}$. Can you see why $\mathbb{Z} \ncong_{\mathbb{Z}} \mathbb{Q}$?

Take note of the fact that whether a module is free or not is highly dependent on what ring is acting on it. In $\mathbb{Z}$-**Mod**, $\mathbb{Z}/n\mathbb{Z}$ is certainly not a free (it is not a free group). However, we've just seen that $\mathbb{Z}/n\mathbb{Z}$ is a free-module over itself! The intuition here behind being "free" comes down to the ring-action being "free".

Corollary 12.3.14: Consequences Of Universal Property

1. If F_1 and F_2 are free modules on the same set A, there is a unique isomorphism between F_1 and F_2 which is the identity map on A
2. if F is any free R-module a free-generating set A, then $F \cong F^R(A)$. In particular, F enjoys the same universal property with respect to A as $F^R(A)$ does in theorem 12.3.11.

***Proof*:**
exercise

12.3.15 Foreshadowing Though a free R-module has no relations, it is not the case that any submodule has no relation, that is the submodule of a free module *need not be free*! In other words, though it may at first seem counter-intuitive, the process of taking a submodule can add relations (see exercsie ref:HERE)! This contrasts to how the subgroup of a free group is always free (see appendix B). There are certain rings for which the R-submodule are always free, while there are other rings where submodules are far from being free (ex. some submodules will be torsion modules, which will be far away from being free). We will explore these concepts further in chapter 13 and 17$\star$.

Measuring degree of Freeness

As we've seen, not all modules are free modules. However, we may still like to find collection of elements that are linearly independent (i.e. that are "free" of relations). We explore these notions here.

Definition 12.3.16: [Free] Rank

For any ring R, the *rank* of an R-module M is the maximum number of R-linearly independent elements of M, that is if M has rank n, there exists elements $y_1, y_2, ..., y_n$ such that

$$r_1 y_1 + r_1 y_2 + \cdots + r_n y_n = 0 \quad \Leftrightarrow \quad r_1 = r_2 = \cdots = r_n = 0$$

If for all $n \in \mathbb{N}$, there exists a R-linearly independent set, then M is said to be of infinite rank.

Due to the importance of the notion of rank for modules in geometry, the rank is also sometimes called the *Betti Number*[11]. The rank of a module is similar to the size of the generating set of a module, and it will in fact match exactly the idea of rank for modules over fields, but this is not necessarily the case for modules over weaker rings:

[11]The kth Betti number loosely refers to the number of k-dimensional holes ona a topological surface

Example 12.3.17: Different Rank and Size of Generators

Let $R = \mathbb{Z}[x]$ and let $M = (2, x)$ be an R-module. If we take $2 \in M$, then since R is an R an integral domain and M is an ideal of R, there does not exist a r such that

$$ra = 0$$

thus, $\{2\}$ is a linearly independent set from M, meaning it's at least rank 1. What if we two elements? Let's take $a, b \in M$. Since $a, b \in R$, in particular $-a \in R$

$$ba - ab = 0$$

meaning 2 elements will never be linearly independent. This will naturally generalize to n elements, so M is of rank 1.

However, M is generated by 2 elements, showing that the rank and the number of generators does not match.

Sometimes, authors define the minimal set of generators to be the rank of the matrix, and rank as was defined above would be called the *free rank*. Since the size of the generating sets rarely comes up outside the context of free generating sets, this book decided to stick with "rank" to mean the maximal number of linearly independent elements.

Not that it is yet to be established if there is a unique maximal rank (just like in a tree, there may be many maximal heights). For modules over integral domains, the maximal height will alwaays be unique, and under appropriate finiteness conditions it shall also be unique, however there are counter-examples over more general rings; this shall be further explore in chapter 13.

Using the notion of rank, we can see how a module might fail to be free. In the "worst case scenario", we can single out elements fail to be linearly independent:

Definition 12.3.18: Torsion Element

Let M be an R-module and $m \in M$. Then m is called a *torsion element* if m is linearly *de*pendent, in particular there exists an $r \in R$ such that

$$r \cdot m = 0$$

The collection of all torsion elements of M is denoted $\mathrm{Tor}_R(M)$ or $\mathrm{Tor}(M)$ is the context is clear. An element that is not torsion is called *torsion-free*.

Elements that are torsion-free can be thought of as being "locally" free. Being is a property of modules, while being torsion-free is a property of elements. If every element has this property, then we give a name to such a module:

Definition 12.3.19: Torsion-Free Modules

Let M be an R-module. Then if $\mathrm{Tor}_R(M) = \{0\}$, then M is said to be *torsion-free* or to be a *torsion-free module*.

Being torsion free is an interesting example of the fact that it is not always enough that every element have a property for the global module to have the property. In this case, even if every element itself is free, this does not imply the module itself is free

Example 12.3.20: Torsion-free, But Not Free Module

Take $\mathbb{C}[x, y]$ over itself. Then since it's an integral domain, it is torsion free and each submodule is torsion free. However, (x, y) is not free.

Looking instead at the ring action instead of the module element, we can also find all elements of the ring that fail to be linearly dependent with *all* module elements:

Definition 12.3.21: Annihilator

Let M be an R-module. Then the set

$$\mathrm{Ann}_R(M) = \{r \in R \mid r \cdot m = 0 \ \forall m \in M\}$$

is called the *annihilator* of M.

Recall that we have talked about this set in example 12.1.7(4) when discussing modules where the annihilator is trivial; such a module is given a name:

Definition 12.3.22: Faithful Module

Let M be an R-module. Then if $\mathrm{Ann}_R(M) = \{0\}$, M is said to be a *faithful R-module*, or that the ring R is acting faithfully on M

If R does not act faithfuully on M, when we may always induce a faithful action along the lines of example 12.1.7(4). The same cannot be said about non-torsion modules, instead we may ask if a module M can be split into a torsion-free part and a torsion part. Being trosion free is much more flexible than being free:

Proposition 12.3.23: Extensions And Submodules Of Torsion-Free Modules

Let M be an R-module. Then:

1. If M is torison-free, then any submodule is torsion free
2. the direct sum of torsion free modules is torsion-tree
3. if M is an extension of torsion-free modules, that is $0 \to A \to M \to B \to 0$ is a short exact sequence with A and B being torsion-free, then M is torsion-free.
4. If R is an integral domain M is free, then M is torsion-free (i.e. free modules over integral domains are torsion-free)

***Proof*:**

1. This is immediate after realising $\mathrm{Tor}_R(N) \subseteq \mathrm{Tor}_R(M)$
2. this is also almost immediate givn a similar type of reasoning as in part (1)
3. If $m \in M$, where $r \cdot m = 0$, then $r \cdot \overline{m} = \overline{0}$, but since $r \in\in \mathrm{Tor}_R(B) = \{0\}$ $r = 0$, and so $\mathrm{Tor}_R(M) = 0$.

4. Since the direct of free modules is free, and the integral domain R as a module over itself is torsion free, M is torsion-free.

In chapter 14, we shall further explore the decomposition of the torsion and torsion-free part of a module over nicer enough rings so that the decomposition is possible.

If N is not a torsion module, then $\text{Ann}(N) = 0$. It is possible that N is a torsion module, but $\text{Ann}(N) = 0$:

Example 12.3.24: Torsion Module With No Annihilators

Take

$$Z = \bigoplus_k \mathbb{Z}/2^k\mathbb{Z}$$

then Z is a $\mathbb{Z}$-torsion module, since for every $z \in \mathbb{Z}$, we can take 2^k, where we pick k relative to the largest nonzero component. However,

$$\text{Ann}(Z) = 0$$

Since if $r \in \mathbb{Z}$ was an annihilator, then $r \cdot z = 0$ for every z, but z can take any value of $\mathbb{N}$, leading us to a contradiction

If the module is finitely generated the annihilator will always be non-empty, since if we annihilate the generators, we annihilate the entire module.

Exercise 12.3.1

1. Let M be an R-module, and define $\text{Ann}_R(M) = \{r \in R \mid r \cdot m = 0, \ \forall m \in M\}$.
    1. Show that $\text{Ann}_R(M)$ is a two-sided ideal
    2. If $\text{Ann}_R(M) = 0$, then M is said to be a faithful module. Show that all free modules are faithful, but there are faithful non-free modules
2. Let R be a commutative ring and an R-module over itself. Then show that an ideal $I \subseteq R$ is a free R-module if and only if it is principal and generated by a nonzero divisor.
3. Show that $\mathbb{Z}^{\mathbb{N}}$ (a direct product of countably many $\mathbb{Z}$ is *not* a free abelian group

12.3.3 Presentations and Resolutions

We claimed that all R-modules are a quotient of a free R-module, just how all groups are quotient of free groups. We will demonstrate this here, and then explore a bit furter how we can think of relations and presentations of modules:

Proposition 12.3.25: Constructing Modules using Free Modules

Any R-module M is the quotient of a free module $F^R(A)$ for appropriate set A:

$$F^R(A)/K \cong_R M$$

***Proof*:**
Let M be an R-module and $\mathcal{F} = F^R(M)$. Define the map:

$$\varphi : \mathcal{F} \to M \qquad (r_i m_i)_{i\in M} \mapsto \sum_{i\in M} r_i m_i$$

where any element on the left hand side is a tulpe indexed by M where all but finitely many of the terms are nonzero, and hence the right hand side is well-defined. This map is evidentely surjective (since the tulpe of m in the first element maps to m), and so if K is the kernel of this map, by the first isomrophism theorem:

$$F^R(M)/K \cong_R M$$

as we sought to show.

Hence, the kernel of a free module plays the role of codifying all the relations in a module. When working with groups, we encoded these relations writting a group as either, for example:

$$\langle x, y \ : \ xy - yx = 0 \rangle \qquad \text{or} \qquad \frac{F(x,y)}{xy - yx}$$

The former notaiton being a convienient notation that can be easier to read before the introduciton of quotienting. In the next definition, we will define the presentaiton of a module. Going off of the fact that the kernel is the central object defining the relation, we incorporate this idea into the definition:

Definition 12.3.26: Presentation Of Module

Let M be an R-module. Then a *presentation* of M is a an exact sequence:

$$R^n \to R^m \to M \to 0$$

If n is finite (i.e. if $n \in \mathbb{N}$), then M is said to be *finitely presented*, or a *finitely presented module*.

Note that m and n need not be natural numbers, and that the above is *not* equivalent to

$$0 \to R^n \to R^m \to M \to 0 \tag{12.5}$$

i.e. R^n need not map injectively into R^m. The easiest way to show this is if we classify all modules for which equation (12.5) holds, which turn out to be all modules over PID's, and hence modules that are *not* over PID's (for example, $\mathbb{Z}[x]$ over itself) cannot be written down in the form presented in equation (12.5). In chapter 14, we shall show that a module over a PID shall imply the existence of a chain as given in equation (eq:modOverPidAsPresentaiton), that is it is a sufficeint condition that R is a PID. It is easier to show that it is a necessary condition; we shall show this at the end of this seciton, for now we will assume the complete result to point out some interesting observations.

It is informative to the reader to see how presentations of modules compares to presentations of groups. Note that $\mathbb{Z}$ is a Principal Ideal Domain, and so for any [finitely generated?] abelian group (i.e. any $\mathbb{Z}$-module), we have the exact sequence:

$$0 \to \mathbb{Z}^n \to \mathbb{Z}^m \to G \to 0$$

Furthermore, any subgroup of a free group is free (see appendix B), and so more generally for any group G and appropriate sets A and B:

$$0 \to F(A) \to F(B) \xrightarrow{\varphi} G \to 0$$

where $F(A) = \ker(\varphi)$. Hence, the fact that there are modules that do not admit a chain like in equation (12.5) is a property that distinguishes groups from modules. In fact, there are even modules for which there is no finite length chain of free modules. If we interpret equation (12.5) as saying that the relations on a module is free, then if we extend the length by 1:

$$0 \to R^n \to R^m \to R^p \to M \to 0$$

can interpret the extra homomorphism as the relations of the relations, where the "relations of the relations" are free. Some modules do not have a finite length chain of this sort!! We give a name to such a chain

Definition 12.3.27: Free Resolution

Let R be a ring and M an R-module. A *free resolution* of M is an *exact* chain complex (definition 12.2.11)

$$\cdots \to F_2 \to F_1 \to F_0 \to M \to 0$$

in which every F_i is a free R-module. The resolution has *finite length* k if $F_i = 0$ for every $i > k$, so that it reads

$$0 \to F_k \to F_{k-1} \to \cdots \to F_1 \to F_0 \to M \to 0$$

and M is then said to have a *finite length free resolution* of length k.

Exactness is what makes the resolution say something about M: at F_0 it says that the relations among the chosen generators are exactly the image of F_1, at F_1 that the relations among *those* relations are the image of F_2, and so on.

By definition, all free modules have a finite length free resolution of length zero since

$$0 \to R^{n_0} \to M \to 0$$

i.e. $R^{n_0} \cong M$. By the universal property of free modules, any R-module M has a free resolution (can you see why). The resolution is called a *free* resolution since we are using free-modules. If we use a different type of module, we will get a different type of resolution[12].

Example 12.3.28: Free Resolution

Let G be a finitely generated abelian group. Then by the fundamental theorem for such groups (theorem 5.1.8),

$$G \cong \mathbb{Z}^n \oplus \mathbb{Z}/n_1Z \oplus \mathbb{Z}/n_2\mathbb{Z} \oplus \cdots \oplus \mathbb{Z}/n_k\mathbb{Z}$$

As pointed out, all free-modules have a resolution of length 1, namely

$$0 \to \mathbb{Z}^n \to \mathbb{Z}^n \to 0$$

[12] In chapter 17⋆, we will introduce projective, injective, and flat modules, and then in part XI, we will introduce projective, injective, and flat resolutions

On the other hand, any cyclic torsion group has a resolution of length 2:

$$0 \to \mathbb{Z} \xrightarrow{\cdot n} \mathbb{Z} \xrightarrow{\pi} \mathbb{Z}/n\mathbb{Z} \to 0$$

These results can be combined to show that G always has a resolution of length 2. Free resolution that must be at least of length 3 will be given in part XI.

Finally, as promised earlier, we shall show that it is a necesary conditions that R-modules admitting resolutions of length two must be over PID's:

Proposition 12.3.29: Free Resolution Of Length 2, Then Pid

Let R be an integral domain such that every R-module M admits a resolution of length 2. Then R is a PID. In particular, for any resolution:

$$R^n \to M$$

There exists an R^m such that

$$0 \to R^m \to R^n \to M$$

***Proof*:**
Choose any $I \subseteq R$. Then R/I is an R-module and

$$R \to R/I$$

is an R-module homomorphism. By the property of R, the above homomorphism can be extended to a free-resolution of length 2. Since the kernel of the above homomorphism is I, we get:

$$0 \to I \to R \to R/I$$

where we must assume I is a free R-module. Since R is of rank 1, I cannot be greater than rank 1 (see exercise ref:HERE (above direct sums and homomorphisms)). But then, since I is free, I must be generated by this single element, making it free. Since this is true for an arbitrary ideal $I \subseteq R$, this implies that every ideal of R is principal making R a PID, as we sought to show.

12.3.4 Finitely Generated Modules and Noetherian Modules

As with many structures in mathematics, the "infinite" case (in our case infinitely generated objects) have different behavior than the finite case and require some different tools to analyze (possibly some analysis to use the notion of limits). We thus study the case where modules are finitely generated or all submodules are finitely generated.

Let R be a ring and let M be a left R-module. We first define what types of modules are *finitely generated*, and some consequences of being finitely generated,

Definition 12.3.30: Finitely Generated R-Module

Let M be an R-module. Then M is said to be finitely generated if there exists an $n \in \mathbb{N}_{>0}$ such that $M \cong_R R^n/K$ for some R-module K, that is there exists a map φ where

$$R^n \xrightarrow{\varphi} M \to 0$$

is exact so that $\ker(\varphi) = K$ and $R^n/K \cong_R M$.

Though we have defined finitely generated modules, we usually require the stronger condition that all submodules of a finitely generated R-module to be finitely generated (which is not true for every finitely generated modules)

Example 12.3.31: Finitely Generated But A Non-finitely Generated Submodule

Take $R = k[x_1, x_2, ..., x_n, ...]$ be the ring with infinitely many indeterminates over a field. Then R is cyclic over itself, generated by 1, but the R-submodule, i.e. ideal, $(x_1, x_2, ..., x_n, ...) \subseteq R$ is not finitely generated as an R-module.

The main inhibitor to this is We will phrase this condition a little differently and show it's equivalent to the previous statement:

Definition 12.3.32: Noetherian R-Module and Ring

Let M be a left R-module Then:

1. M is said to be *Noetherian R-module* or to satisfy the *ascending chain condition on submodules* (or A.C.C. On submodules), that is there are no infinite increasing chains of submodules, i.e., whenever
$$M_1 \subseteq M_2 \subseteq M_3 \subseteq \cdots$$
is an increasing chain of submodules of M, then there is a positive integer m such that for all $k \geq m$, $M_k = M_m$ (so the chain becomes stationary at stage m: $M_m = M_{m+1} = M_{m+2} = \cdots$)
2. The ring R is said to be Neotherian if it is Neotherian as a left-module over itself, i.e., if there are no infinite increasing chains of left-ideals in R.

One can have an analogous definition of A.C.C. for right and two-sided ideals in (possibly noncommutative) ring R. For noncommutative rings, left and right A.C.C need not relate (you may be left A.C.C but not right, and vice-versa, see exercise ref:HERE)

Theorem 12.3.33: Equivalent Conditions To Noetherian

Let R be a ring and let M be a left R-module. Then the following are equivalent

1. M is a Noetherian R-module
2. Every nonempty set of submodules of M contains a maximal element under inclusion
3. Every submodule of M (including M) is finitely generated

***Proof*:**
(1) ⇒ (2). Let S be any collection of submodules. Let's pick some sub-module $M_1 \in S$. If S is maximal, then we're done. If not, then M_1 is not maximal, meaning there exists an M_2 such that $M_1 \subseteq M_2$. If M_2 is maximal, we're done. If not, pick M_3 in a similar fashion. Keep picking sub-modules M_i. If none of them are maximal, then by the axiom of choice, we can pick an infinitely long chain of submodules

$$M_1 \subseteq M_2 \subseteq \cdots \subseteq M_k \subseteq \cdots$$

which is a contradiction to the statement that M is Noetherian.

For (2) ⇒ (3), Let M be a module. Pick $x \in M$. If $\langle x_1 \rangle = M$, then M is finitely generated. If $\langle x_1 \rangle = N_1 \subsetneq M$, then pick another element $x_2 \in M - N_1$. If $\langle x_1, x_2 \rangle = M$, we're done. If $\langle x_1, x_2 \rangle = N_2 \subsetneq M$, then pick another element $x_3 \in M - N_2$. Either this stabalizes, or we can craete an infinite chain:

$$N_1 \subseteq N_2 \subseteq \cdots \subseteq N_k \cdots$$

This is a collection of submomdules of M, and so by the 2nd condition there must exist a maximal module, say N. Since every module in this chain is finitely generated, N mut be finitely generated, and since it's maximal adding one more element to N, $N + x$ and taking it's closure, $\overline{N + x} = N'$ must make the new larger module N' equal to M. Hence, M is finitely genearted.

For (3) ⇒ (1), Take any ascending chain of modules

$$N_1 \subsetneq N_2 \subsetneq \cdots \subsetneq N_k \subsetneq \cdots$$

then $N = \bigcup_{i=1}^{\infty} N_i$ is a module, and by assumption it is finitely generated, say by $\{x_1, x_2, ..., x_n\}$. By definition of N, for each x_i, there exists a j such that $x_i \in N_j$. Since N is the union of the modules N_j, it must be that at some finite distance into the chain, a module contains all the generators. But that means the chain stabalizes.

Note that if M is a Noetherian R-module, then R need not be Noetherian. If R is noetherian, then M is a Noetherian R-module? (Use the fact that the direct sum and quotient of a noetherian ring is noetherian, an the uni prop of free modules).[13]. In proposition 9.4.18, it was shown that rings where $1 \neq 0$ have maximal ideals. Since all rings are modules over themsleves, it might seem that all rings are Noetherian modules due to the second conditions. However, this is certainly not the case, consider $R[x_1, x_2, ..., x_n, ...]$ over itself and the chain $(x_1) \subsetneq (x_1, x_2,) \subsetneq \cdots$. The subtly lies in the fact that the maximal element must lie in the collection of submodules, while the maximal element in the previous example is the union of all the submodules.

Many textbooks introduce Noetherian modules as modules where each submodule is finitely generated. The reason the A.C.C. was chosen for this textbook was for it's greater applicability in many other fields. Notions of being "Noetherian" are pervasive in mathematics, but not all structure are "Noetherian" by having every substructure be finitely generated (sometimes, the notion of substructure is not even well-defined)[14]. The A.C.C. allows us to still consider those structures.

Since Principal Ideal Domain are *a fortiori* Noetherian rings, finitely generated modules over PID's are are also Noetherian, and all the results we discussed earlier apply to such modules (including PID's over themsleves).

[13]This mirrors a more general theme in which a property of the underlying ring "becomes" the property of all modules over that ring. Can you see how we can (in a sense) call a field a "free ring"?

[14]See Noetherian topologies for an interesting example

Finite Generation vs. Finite Type

Being finitely generated as an R-module and finitely generated as an R-algebra are very differnet:

Example 12.3.34: Finite Generation: Module vs. Algebra
Take $R = k[x]$ as an k-algebra and consider the natural k-module structure. Then it is an *infinitely generated* k-module generated (for example by $\{1, x, x^2, x^3, ...\}$ however it is a *finitely generated* k-algebra generated by $\{1, x\}$.

Mo generally, contrast an finitely generated R-module M to a finitely generated R-algebra M:

$$R^{\oplus A} \twoheadrightarrow M \quad R[A] \twoheadrightarrow M$$

Definition 12.3.35: Finite Type

Let A be an R-algebra. Then A is of *finite type* (over R) if A is finitely generated as an R-algebra.

Often, instead of finite type the terms "finitely generated as an R-algebra" or even "finitely generated R'algebra" are often used interchangeably.

Exercise 12.3.2

1. Is a subring of a Noetherian ring Noetherian?
2. (If you know some complex analysis) In this exercise, we try to show how the Noetherian property is not stable by taking subrings and show demonstrate a correspondence between being Noetherian and how the zero of functions are in some way "well-behaved". Consider the following chain of subrings:

$$\begin{gathered} \mathbb{R}[x] \\ \subseteq \\ \text{Analytic functions on } \mathbb{R} \\ \subseteq \\ \text{Analytic functions on } [0,1] \\ \subseteq \\ \text{Analytic functions on } (0,1) \\ \subseteq \\ \text{Analytic functions at } 0 \\ \subseteq \\ \text{Smooth functions near } 0 \\ \subseteq \\ \text{Former Power series} \end{gathered}$$

   which of these rings are Noetherian, and which are not?

3. Show that a finitely generated R-module over a Noetherian ring R is itself Noetherian
4. Let M be an R-module over an integral domain R. Then if M is of rank n, any submodule $N \subseteq M$ can be of rank at most $n \leq m$ (hint: take the ring of fractions).

12.3.5 Direct Sums and Homomorphisms

For the first time in this book, we will pay closer attention to hom-sets. The structure of hom-sets of modules will allo us to determine many important properties of modules.

Proposition 12.3.36: Direct Sums And Homomorphisms

Let $(M_i)_{i\in I}$ and $(N_j)_{j\in J}$ be a collection of modules with I and J indexing sets. Then:

$$\mathrm{Hom}_R\left(\bigoplus_{i\in I} M_i, N\right) = \prod_{i\in I} \mathrm{Hom}_R(M_i, N)$$

and

$$\mathrm{Hom}_R\left(M, \prod_{j\in J} N_j\right) = \prod_{j\in J} \mathrm{Hom}_R(M, N_j)$$

***Proof*:**
let $\varphi : \oplus_{i\in I} M_i \to N$. Then for any $(\alpha_i)_{i\in I} \in \oplus_{i\in I} M_i$, there are only finitely many non-zero entries. For ease of visualization, lkets represent the element $(\alpha_i)_{i\in I}$ as

$$(\alpha_1, \alpha_2, ..., \alpha_n, 0, ...)$$

This represetnation can be confusing on two terms: we have implicitly chosen an "order" of the elements and re-labelled the indices of the finite eleents accordingly, and it makes the set I look like it is *countable*, which is not necessarily the case. These two remarks should be kept in mind while working with this represetnation of the element.

Since there are n elements, write each of them as:

$$\alpha_i = \underbrace{0 + \cdots + 0}_{i-1 \text{ times}} + \alpha_i + \underbrace{0 + \cdots + 0}_{n-i \text{ times}}$$

Now, the finiteness condition was key here so that we may use linearity inductively, so that for any $\varphi \in \mathrm{Hom}_R\left(\bigoplus_{i\in I} M_i, N\right)$

$$\varphi((\alpha_i)_{i\in I}) = \varphi(\alpha_1) + \cdots + \varphi(\alpha_n)$$

Now, it is immediate that each function on the right hand side is naturally isomorphic to a function $\varphi_i \mathrm{Hom}_R(M_i, N)$. Labelling eahc of thes functions a φ_i, we get:

$$\varphi \to (\varphi_i)_{i\in I}$$

which is clearly an isomorphism.

The direct product case is done almost identically, except we don't need to worry about finite additivity, since for any

$$\varphi(\alpha) = (\beta_i)_{i \in I}$$

we can define:

$$\varphi_i(\alpha) = \beta_i$$

which will give us the desired isomorphism, completing the proof.

As a consequence of this, we can simplify:

$$\mathrm{Hom}_R\left(\bigoplus_i M_i, \bigoplus_j N_j\right) \cong \prod_{i,j} \mathrm{Hom}_R(M_i, N_j)$$

If you are comfortable with linear algebra, then in the case where both direct sums have only finitely many components (also known as *summands*), then if φ_{ji} represent's the component function taking in the ith component in the domain and is in the jth component of the domain, then: the there is an isomrophism mapping $\varphi \mapsto (\varphi_{ij})$ where the maps is defined as:

$$\varphi\begin{pmatrix} s_1 \\ s_2 \\ \vdots \\ s_n \end{pmatrix} = \begin{pmatrix} \varphi_{11} & \varphi_{12} & \cdots & \varphi_{1n} \\ \varphi_{21} & \varphi_{22} & \cdots & \varphi_{2n} \\ \vdots & \vdots & \ddots & \cdots \\ \varphi_{n1} & \varphi_{n2} & \cdots & \varphi_{nn} \end{pmatrix} \begin{pmatrix} s_1 \\ s_2 \\ \vdots \\ s_n \end{pmatrix}$$

Showing that there is a natural relation between $\mathrm{Hom}_R(\oplus_i^m M_i, \oplus_j^n N_j)$ and matrix rings, especially if $M_i = N_j$. More on that in chapter 53, section 51.

Notice how important linearity is to split up the homomorphsim in direct products in the *domain*. To make sure that no misconceptions are being formed, here is a counter-example when a function is not linear:

Example 12.3.37: Hom's And Products Not Commuting in Domain

Let $f(x, y) = xy$. Then $f(x, 0) = 0$ and $f(0, y) = 0$, showing this breakdown does not happen. This was a particualr example, not a proof of the non-existence of the isomorphism. However, we see with f that the output of f must always depend on both x and y, so any two functions f_1 and f_2 solely dependning on the x or y variable cannot be used to reconstruct f using a direct product TBD.

On the other hand, if $f : M \to A \oplus B$ was some function, then by the universal property of the product we have that $f_a : M \to A$ and $f_b : M \to B$ exists. (Think more on why this exists but the other doesn't so nicely.. think about what it means for the uni prop of the co-product and how it relates to hom-sets.. TBD)

Exercise 12.3.3

1. Show that if M is an R-module, then $\mathrm{Hom}_R(R, M) \cong M$
2. Show that $\mathrm{Hom}_R(M \oplus M, R) \cong$(want to show the case where direct product becomes matrices! i.e. bringing out the direct product makes a $M_2(R)$ matrix, whatever should be there instead of R)

3. Let $\varphi : A \to B$ be an R-module isomorphism. Show that $\Phi : \mathrm{End}_R(A) \to \mathrm{End}_R(B)$ where $\Phi(f) = \varphi \circ f \circ \varphi^{-1}$ is a ring isomorphism, and if R is commutative an R-algebra isomrophism.

4. show that if $I \subsetneq R$ is a proper ideal, $I \cong S$ as an R-module isomorphism $\varphi(e) = 1$ a,d $e^2 = e$, then φ is in fact a ring-isomrophism, so I has the ring structure of S (though it cannot have the ring structure of R, since it cannot contain a unit).

13

Modules over Fields – Vector Spaces and Linear Algebra I

This chapter focuses on the study of modules where the ring are fields (i.e. [unital] commutative division ring). We'll use the letter k (from German Körperor) or the symbol $\mathbb{F}$ to represent an arbitrary field. As we've mentioned in definition 12.1.9, a module over a field is called a vector-space due to the historical precedence of the terminology which is now widely accepted in and beyond mathematics (for example, the term vector-space is used in physics, biology, and chemistry). This precedence of modules over fields comes from the fact that historically, fields were studied before general rings. Many problems in physics in the 19th century turned out to requires vector-spaces (ex. solutions to ODEs formed vector spaces, many important spaces of functions formed vector-spaces, the Cartesian plane formed a vector space), leading to precedence in nomenclature for modules over fields. Mathematically, giving another name to modules over fields is not unreasonable: as we'll see, having a field act on a module is so restrictive (or powerful) that we will only need to keep track of two pieces of information to fully understand the structure and the classification of vector-spaces: the underlying field k, and a new number called the *dimension* of the vector-space[1]. No algebraic structure we've encountered requires so little data to be able to classify them (for example, the classification of finite groups required tens of thousands of paper's over almost a century, while the classification of vector-space is regularly taught in a 1st or 2nd year undergraduate class).

Due to the simplicity of vector-space structures, vector space homomorphisms (k-module homomorphisms) will be completely classified. (A further word on this before talking a bit about dual theory and geometry)

As a consequence of the historical precedence of vector-spaces, a separate vernacular was developed

[1]This is particularly true for finite dimensional vector-spaces. In the infinite dimensional case, there is usually more information present than in the finite dimensional case, and so we require more information to sufficiently classify infinite vector-spaces, see ref:HERE

Terminology over any Ring	**Terminology over Fields**
M is an R-module	M is a vector space over R
m is an element of M	m is a vector in M
N is a submodule of M	N is a subspace of M
M/N is a quotient module	M/N is a quotient space
M is a free module of rank n	M is a vector space of dimension n
M is a nonzero cyclic module	M is a 1-dimensional vector space
$\varphi : M \to N$ is an R-module homomorphism	$\varphi : M \to N$ is a (R) linear transformation
M and N are isomorphic as R-modules	M and N are isomorphic vector spaces
the subset A of M generates M	The subset A of M spans M
$M = RA$	each element of M is a linear combination of elements of A, i.e., $M = \text{Span}(A)$

for modules over fields that is now considered standard:

13.1 Vector Spaces and their Properties

Vector spaces are famous for having a very simple structure: Given a vectorspace V over k, then V is isomorphic to k^N for appropriate N. In a linear algebra class, we say that every vector space has a basis. This N will be called the *dimension* of the vector-space. In many linear algebra classes, the attention is limited to $N \in \mathbb{N}$. However, this section will allow N to be of any cardinality (ex. $|\mathbb{N}|$ or $|\mathbb{R}|$). This is to show some of the power of vector-spaces: the (algebraic) structure doesn't get more complicated when dealing with infinite dimensions[2]

Before these results are shown, for those who gain an intuition via examples, here are a couple more example's of vector-spaces

Example 13.1.1: Vector Spaces
We already gave 4 examples in example 12.1.10. We give a few more here

1. We can generlize k^n as follows: notice that we can define k^n as

$$k^n = \{f \mid f : \{1, ..., n\} \to \mathbb{R}\}$$

   where addition is defined pointwise, and since k is commutative we can define a natural k-action on k^n (can you see it?). Any function $f \in k^n$ will tell us what value we should have in the component i. For example, if $f(1) = \pi$, $f(2) = 3$, an $f(3) = \sqrt{10}$, then this function is equivalent to:

$$(\pi, 3, \sqrt{10})$$

   Thus, We can consider:

$$k^{\mathbb{N}} = \{f \mid f : \mathbb{N} \to \mathbb{R}\}$$

   To be the $\mathbb{N}$-dimensional vector-space, or $k^{\mathbb{R}} = \{\mathbb{R} \mid f : \mathbb{R} \to \mathbb{R}\}$ to be the $\mathbb{R}$-dimensional vector space

2. Generalizing the above example, let k be a field, X a set, and $V = \text{Hom}_{\textbf{Set}}(X, k)$ (another common notation is k^X). Then we can make V into a vector-space via point-wise operation in the codomain: for any $f, g \in V$, $c \in k$, $f + g$ is defined to be:

$$(f + g)(x) = f(x) + g(x) \qquad \forall x \in X$$

   while fg and cf is defined as:

$$(fg)(x) = f(x)g(x) \qquad (cf)(x) = cf(x) \qquad \forall x \in X$$

   These definitions are possible since the image of any function in V lands in a field, where all the above operations are well-defined. If we take the collection of lienar transformation from V to k (where V is a vector space), then we often denote that as V^*. Note that since k is also a ring, we can also define $(fg)(x) = f(x)g(x)$, making $\text{Hom}_{\textbf{Set}}(X, k)$ into a k-algebra.

   Often, the set X is given its own properties (a popular example are X is a locally compact Hausdorff space, or X is a smooth manifold). Then we often try to deduce more information about X given the structure of hom-set $\text{Hom}_{\textbf{Set}}(X, k)$.

[2]On the other hand, the set-theoretical and analytic structure (if there is a topology present) gets more complicated, and arguably non-trivial, in infinite dimensions. For example, in infinite dimensions, surjective does not imply injective (similar to the proof for infinite sets), and there will exist linear transformations that are not continuous (for the appropriate topology).

3. Let $R = k[x]$ for some field x. Then $k[x]$ is a k vector-space with action $c \cdot p(x) = cp(x)$. More generally, if $R = k[x_1, x_2, ..., x_n, ...]$ is some polynomial ring over several variables, then R is a k vector-space with $c \cdot p(x_1, x_2, ..., x_n, ...) = cp(x_1, x_2, ..., x_n, ...)$.
4. Let $k = \mathbb{C}$ an $V = \mathbb{C}^n$. Then $\mathbb{C}^n$ is a complex vector space with the action:
$$z \cdot (z_1, ..., z_n) = (zz_1, ..., zz_n)$$
There is a second natural action, known as the *conjugate action*:
$$z \cdot (z_1, ..., z_n) = (\overline{z}z_1, ..., \overline{z}z_n)$$
More generally, if V is a vector-space over $\mathbb{C}$ (also known as a complex vector space), then $\overline{V}$ denote the same space with the element being conjugated before the action is applied.
5. (If you know some ODE's) Given a linear ordinary differential equation, the set of all solutions form's a vector-space. Take for example $\varphi : C^\infty(\mathbb{R}) \to C^\infty(\mathbb{R})$ where:
$$f \mapsto f'' + f$$
Consider the subspace V where
$$V := \{f \in C^\infty(\mathbb{R}) \mid f'' + f \equiv 0\}$$
Then using elementary ODE theory, we find that:
$$V = \text{span}_{\mathbb{R}}(\{e^t, e^{-t}\})$$
The fact that all solutions to the ODE $f'' + f = 0$ is a linear combination of these two solutions is a remarkable property of *linear* ODE's. This simple property of linearity can be taken very far: in the world of PDE's (Parital Differential Equations), if you have something called a *linear PDE*, which means that if you have two solutions, then add them and/or scale the terms, the result is another solution. The power we get from this is that we can then see if this is generalizable to taking *limits* of some very simple solutions to the PDE. As a common example, let's say the solution is of the form $\cos(\alpha x)$ or $\sin(\beta x)$ for any α, β. Then under the right conditions[a], the collection of these (after being closed under addition) is *dense* in $L^2(S^1)$, meaning we can use limits to construct *any* $L^2(S^1)$ function and find another solution to our PDE. A common place this particular example is used is *Fourier Analysis*[b]
6. (If you know Differential Geometry) Let M be a smooth n-manifold and consider the collection of all $\varphi : C^\infty(M) \to \mathbb{R}$. Then the collection of all *derivations at p*[c] for some point $p \in M$, denoted $\text{der}_p(M) \subseteq \text{Hom}_{\textbf{Smooth}}(C^\infty(M), \mathbb{R})$ forms a vector-space of dimension n, and is a common definition of the *tangent space* of the manifold M at the point p[d]

[a] See a textbook covering Fourier Analysis, ex. Pugh or Folland
[b] See EYNTKA Analysis, [45]
[c] linear function where $\varphi(fg) = \varphi(f)g + f\varphi(g)$
[d] See EYNTKA Differential Geometry, [41]

Now, we shall develop a more sophisitaced notion of generating set:

Definition 13.1.2: Linear Independence And Basis

Let M be an R-module.

1. A subset $S \subseteq M$ is said to be *linearly indepenent*, or a set of *linearly independent elements*, if for any equation

$$a_1 m_1 + a_2 m_2 + ... + a_n m_n = 0$$

with $a_1, a_2, ..., a_n \in R$ and $m_1, m_2, ..., m_n \in S$, then $a_1 = a_2 = \cdots = a_n = 0$. If this is not the case, the elements are called *linearly dependent*. In other words, if $I = \{m_1, m_2, ..., m_n\} \subseteq M$, so that $\iota : I \to M$ is injective, then:

$$\begin{tikzcd} F^R(I) \arrow[r, "\varphi"] & M \\ I \arrow[u, "j"] \arrow[ur, "\iota"'] & \end{tikzcd} \tag{13.1}$$

commutes[a] and φ is injective. Note that I can be an arbitrary indexing set, that is for every $i \in I$, $m_i \in M$ corresponds to it, and then we would have an injection $\iota : I \to M$ which lifts to $\varphi : F^R(I) \to M$. If $R = k$, then the elements are called *linearly independent vectors*.

2. A *basis* for M is an *set* $\{a_1, a_2, ..., a_n\}$ of elements that is linearly independent and which generate sM (i.e. span M).

3. An *ordered basis* for M is an *ordered set* (a tuple) $(a_a, a_2, ..., a_n)$ of linearly independent elements which generates M (i.e. span M).

[a]The diagram commuting is the universal property of free modules

The point of an *ordered basis* is so that $A \subseteq M$ gives a *unique* isomorphism:

$$\varphi : F^R(A) \cong M$$

i.e. The φ in equation (13.1) is both injective and surjective. In particular, two ordered bases will be considered different even if one is a rearrangement of the other (hence why it's ordered).

13.1.3 Convention In this section, all bases shall implicitly be ordered basis. The convention shall be dropped when working with tensor products of vector spaces.

Note that the φ from the commutative diagram in equation (13.1) has as domain all *finite* combinations of elements of $\{m_i\}_{i \in I}$, so that when writing:

$$\sum_{i \in I} r_i m_i$$

only finitely many r_i are nonzero. If φ is injective, then $\sum_{i \in I} r_i m_i = 0$ if and only if $r_i = 0$ for all $i \in I$. Sometimes to emphasize the fact that it's ordered, a basis is called an *ordered basis*. In terms of indexing sets, the ordered elements are generalized to elements having different indexes. Notice how being linearly independent is different from proposition 12.3.6(3), where if $x \in \oplus_{i \in I} N_i$ implies there exists unique elements $v_i \in N_i$ such that $x = \sum_i v_i$. We are adding another requirement on $\bigoplus_i v_i$, namely that the function φ in the above definition is injective.

A module M having a basis is in fact a very restricting condition:

Lemma 13.1.4: Basis iff Free

Let M be an R-module. Then M is free if and only if it admits a basis, in paritcular $B \subseteq M$ is a basis if and only if the natural homomorphism $R^{\oplus B} \to M$ is an isomorphism

***Proof*:**
This is immedaite from the definition: if $B \subseteq M$ is linearly independent and generates M, then $\varphi : R^{\oplus B} \to M$ is injective and surjective, and if $\varphi : R^{\oplus B} \to M$ is an isomorphism, then $\varphi(B)$ is a subset of M which generates it and is linearly independent.

Example 13.1.5: Linearly Dependent and Independent Sets

1. Let $M = \mathbb{Z}/n\mathbb{Z}$ be a $\mathbb{Z}$-module. Then:

$$n \cdot [1]_n = 0$$

   but naturally $n \neq 0$. More generally, if R is a ring with a zero divisor, and $(z) \subseteq R$ is the sub-module generated by the zero-divor (z), then if $zz' = 0$, $z' \cdot z = 0$,showing that the singleton z is linearly dependent.

2. Let $R = \mathbb{C}[x, y]$ be a module over itself. This ring has no zero divisors, hence the above does not apply. However, that there are still linearly dependent elements, for example:

$$y \cdot x + (-x) \cdot y = xy - xy = 0$$

   showing that x, y are *not* linearly independnet over $\mathbb{C}[x, y]$. However, if we consider $\mathbb{C}[x, y]$ over $\mathbb{C}$, then $\{x, y\}$ *are* linearly independent, since $\mathbb{C}[x, y]$ is the free $\mathbb{C}$-algebra. Hence:

$$F^{\mathbb{C}}(\{x, y\}) = \mathbb{C}^2 \to \mathbb{C}[x, y]$$

   is an injective function. However, $\{x, y\}$ is not a basis since the above homomorphsim is not surjective, since nothing maps to xy. Is there any finite that will create a surjective map?

3. If R is any ring (which includes rings with torsion elements), then $(1) = R$, and $r \cdot 1 = r1$ if and only if $r = 0$. Hence,

$$R \mapsto R$$

   is an injective homomorphism (even surjective, making it a basis). This shows that the choice of subset of the module effects whether the set if linearly independent or a basis.

   Though R has a basis when over itself, show that $\mathbb{Z}/2\mathbb{Z} \oplus \mathbb{Z}/4\mathbb{Z}$ over $\mathbb{Z}$ does *not* have a basis, or even a linearly independent set.

As we saw in the above example, some modules have linearly independent sets, some don't, and others have a basis. By a simple applications of Zorn's lemma and some manipulation of definition, we can show that every module has a maximally linearly independent set:

Lemma 13.1.6: Maximally Linearly Independent Set

Let M be an R-module and let $S \subseteq M$ be a lienarly independnet set. Then there exsits a maximal linearly independent subset of M containing S.

***Proof*:**
This is an application of Zorn's lemma: Let $\mathcal{L}$ be the collection of linearly independent subset of M containing S ordered by inclusion. Since $S \in \mathcal{L}$, $\mathcal{L} \neq \emptyset$. If we take any chain $\mathcal{C} \subseteq \mathcal{L}$, and take $C = \cup\mathcal{C}$, then C is linearly independent, since it it were linearly dependent, then only finitely many linear element would be used in the equation, but then they all belong to some $C_i \in \mathcal{C}$, which by assumption is linearly independnet. Hence, by Zorn's lemma $\mathcal{L}$ contains a maximally linearly independent set containing S.

13.1.7 Trivia The Axiom of Choice and the above lemma are in fact equivalent, hence it is impossible to prove the above statement without using Zorn's lemma.

If S is a maximally linearly independent set containing $\emptyset$, we shall simply say that S is maximally linearly independent. Note that being maximally linearly independent does not guarantee a unique size or that it generates the entire module, for exmaple $\{2\} \subseteq \mathbb{Z}$ is maximally linearly independent but does not generate $\mathbb{Z}$. In the special case where $R = k$ is a field so that we have a vector-space V, then a maximally linearly independent set is indeed a basis!

Theorem 13.1.8: Vector Space has Basis

Let V be a vector space over k and let B be a maixmally lienarly independnet set. Then B is a basis of V, hence V is a free k-module. Furthermore, if S is a liearly independent set, then S is contained in a basis B.

***Proof*:**
We need to show that B generates V. Let $v \in V$, $v \notin B$. Then by maximality of B, $B \cup \{v\}$ is linearly dependent, hence there exists $v_1, v_2, ..., v_n$ and $c_0, c_1, ..., c_n$ such that

$$c_0 v + c_1 v_1 + \cdots + c_n v_n = 0$$

with not all $c_i = 0$. Note that $c_0 \neq 0$ or else we would get $c_1 v_1 + \cdots + c_n v_n = 0$ with not all $c_i \neq 0$, contradicting linearly independence of elements in B. Since k is a field, c_0 is a unit, hence:

$$v = (-c_0^{-1} c_1) v_1 + \cdots + (-c_0^{-1} c_n) v_n$$

But then, $v \in \text{span}(B)$, showing that B generated V. By lemma 13.1.4, V is a free k-module.

Furthermore, if S is lienarly independent, then S is contained in a maximally linearly independent set, say B. But by our above reasoning B is a basis, hence S is contained in a basis, as we sought to show.

Hence, every vector-space is free. Before continuing our exploration of bases, a quick comment must be made: we can define the dual notion of *minimally spanning set* and have proved the same statment in the reverse direction namely:

Proposition 13.1.9: Vector space, Minimally Spaning then Basis

Let V be a k-vector space and let B be a minimally spanning set. Then B is a basis for V, making V a free k-module. Furthermore, if S is any spanning set,then S containins a minimal spanning set B.

***Proof*:**
exercise

Corollary 13.1.10: Subspaces Of Vector-Spaces Are Free

Let V be a vector-space and $W \subseteq V$ be a subspace. Then W is free

***Proof*:**
Let $W \subseteq V$ be a subspace. Since W is a k-module, by theorem 13.1.8, it is free.

Note that this is not true for arbitrary free-modules: there exists submodule of free modules which are *not* free[3]. In fact, since any ring R is a module over itself, then it is easy to find counter-examples; any polynomial rings over several variables will have non-free submodules, see exercise ref:HERE. Another simpler example is as follows

Example 13.1.11: Non Free Submodule
Take $\mathbb{Z}_6$ over itself, and consider the ideal (2). Show that it is not free[a].

[a] Hint: it's torsion

In later chapters, we shall see that submodules of free modules over slightly more general rings, namely PIDs, will still be free[4], however this generlization is limited in scope and does not further generlize over more general rings, as we shall discuss later.

The key in theorem 13.1.8 is that every element in a field is a unit; commutativity was not important. Thus, this same proof can work for division rings more generally[5]. Though this set is the maximal linearly independent set, since the partial order on $P(A)$ forms a lattice, there may be many different cardinalities of sets which form maximal generating sets (see exercise ref:HERE for an example where this happens for some rings and groups). We thus ask the question of when do we get bases with different cardinalites, or if there are conditions for which different basis always have the same cardinality. For vector-spaces, every basis *always* has the same cardinality! The following theorem is the key result for the proof:

[3] Though it is true for free groups, see appendix B
[4] see corollary 14.1.7
[5] And will be used in chapter 53, section 51

Theorem 13.1.12: Replacement Theorem

Assume $A = \{a_1, a_2, \cdots, a_n\}$ is a basis for V containing n elements and $B = \{b_1, b_2, ..., b_n\}$ is a set of linearly independent vectors in V. Then there is an order $a_1, a_2, ..., a_n$ st for each $k \in \{1, 2, ..., m\}$, the set $\{b_1, b_2, ..., b_k, a_{k+1}, a_{k+2}, ..., a_n\}$ is a basis of V. In other words, the elements $b_1, b_2, ..., b_m$ can be used to successively replace the elements of the basis A, and A will still be a basis.

***Proof*:**
Let the set A and B be as inthe statement of the theorem. We proceed with induction on k. If $k = 0$, we're done, so let's say $\mathcal{B}$ is a basis for V and

$$(b_1, b_2..., b_k, a_{k+1}, ..., a_n)$$

is a basis for V. Since it's a basis, it's a spanning set, and so for b_{k+1}:

$$b_{k+1} = \beta_1 b_1 + \cdots + \beta_k b_k + \alpha_{k+1} a_{k+1} + \cdots + \alpha_n a_n \tag{13.2}$$

Notice not all the α_i can be zero, since then b_{k+1} is a linear combination of elements in B, but B is linearly independent. Without loss of generailty, we can re-arrange the above equation so that $\alpha_{k+1} \neq 0$. Then re-arranging, we get:

$$a_{k+1} = \frac{1}{\alpha_{k+1}}(\beta_1 b_1 + \cdots + b_{k+1} + \alpha_{k+2} a_{k+2} + \cdots + \alpha_n a_n)$$

Thus, we get that:

$$\text{span}(\{b_1, ..., b_k, a_{k+1}, ..., a_n\}) = \text{span}(\{b_1, ..., b_{k+1}, a_{k+2}, ..., a_n)$$

it not suffice to show that that $\{b_1, ..., b_k, a_{k+1}, ..., a_n\}$ is linearly independent. Let's say:

$$\beta_1 b_1 + \cdots + \beta_{k+1} b_{k+1} + \alpha_{k+2} a_{k+2} + \cdots + \alpha_n a_n = 0 \tag{13.3}$$

By equation (13.2), we can make equation (13.3) a linear combination of elements of $(b_1, b_2..., b_k, a_{k+1}, ..., a_n)$:

$$(\beta_1 + \beta_{k+1}\beta_1')b_1 + \cdots + (\beta_k + \beta_{k+1}\beta_k')b_k + (\beta_{k+1}\alpha_{k+1}')a_{k+1} + \cdots + (\alpha_n + \beta_{k+1}\alpha_n')a_n = 0$$

In particular, we can multily this entire equation by $\beta_{k+1}(\alpha_{k+1}')^{-1}\beta_{k+1}^{-1}$ [a] so that the coefficient of α_{k+1} is β_{k+1}. Since this is a basis by the induction hypothesis, all the coefficients must be zero, including β_{k+1}. Thus, we get

$$\begin{aligned} 0 &= \beta_1 b_1 + \cdots + 0 b_{k+1} + \alpha_{k+2} a_{k+2} + \cdots + \alpha_n a_n \\ &= \beta_1 b_1 + \cdots + \beta_k b_k + \alpha_{k+2} a_{k+2} + \cdots + \alpha_n a_n \\ &= \beta_1 b_1 + \cdots \beta_k b_k + 0 a_{k+1} + \alpha_{k+2} a_{k+2} + \cdots + \alpha_n a_n \end{aligned}$$

where the last line is a basis by the induction hypothesis, and so all other coefficients must be zero, and so the set must be linearly independent, as we sought to show.

[a] The reason for writing the scalar like this is so show that the proof works even if we limit ourselves to division rings

Since the cardinality of the basis is independent of the choice of elements that make the basis, we can define it as an invariant on vector-spaces:

Definition 13.1.13: Dimension

Let V be a vector space over F. Then the cardinality of the basis is called the *dimension* of V, and is denoted $\dim_F(V)$ or $\dim(V)$ if context permits. If V is not finitely generated, then we say that V is infinite dimensional, and we write $\dim V = \infty$.

The key take-away of the Replacement Theorem is that there is a direct relation in cardinality between sets that are linearly independent and sets that are spanning:

Corollary 13.1.14: Steinitz exchange lemma

1. Suppose V has a finite basis with n elements. Any set of linearly independent vector has less than or equal to n elements. Any spanning set has greater than or equal to n elements

2. If V has some finite basis then any two bases of V have the same cardinality

***Proof*:**

1. A be the basis. If a set is linearly independent, then we can use the replacement theorem to systematically replace the basis with the elements of the linearly indepennet set. Thus, it cannot have more than n elements, since a basis has maximum n elements that are linearly independent. If a set is a spanning set, then by corollary 13.1.14, the set contains a basis, say B. Thus, $|B| \leq |A|$.

   Since B is a linearly independent, we can use the replacement theorem to replace the elements of A with B. If $|B| < |A|$, that would contradict the minimality of the set A since A is a basis, and hence a minimal spanning set. Thus, it must be that $|A| \leq |B|$.

2. This is an immediate consequence of the first part, since both both conditions have to match.

As we saw in example 13.1.5, a singleton set with a single element which has torsion is always linearly dependent. In an integral domain, we have some more power:

Corollary 13.1.15: Module Linearity

Let R be an integral domain and let M be a free R-module of rank $n < \infty$. Then any $n+1$ elements of M are R-linearly dependent, i.e., for any $y_1, y_2, ..., y_{n+1} \in M$, there are elements $r_1, r_2 ..., r_{n+1} \in R$ not all nonzero such that

$$r_1 y_1 + r_2 y_2 + \cdots + r_{n+1} y_{n+1} = 0$$

***Proof*:**
There is a trick similar to Gauss's lemma. We can embed R in its quotient field (since R is an integral domain, this is indeed an embedding), say k is its quotient field. Since M is a free module,

$$M \cong \underbrace{R \oplus R \oplus \cdots R \oplus R}_{n \text{ times}}$$

and so

$$M \subseteq \underbrace{k \oplus k \oplus \cdots \oplus k}_{n \text{ times}}$$

Since the latter is an n dimensional vector space over F, so any $n+1$ dimensional elements of M are F-linearly dependent. By clearing the denominators of the scalars (ex. multiplying through by the product of all the denominaotrs), we get an R-linear dependence realtion among $n+1$ elements of M.

An important consequence of the existence of a basis is that linear transformation $f : V \to W$ can be defined by first determining where a basis of V maps to, and then extending linearly (that is, if $v = \sum_{\beta_i \in \mathcal{B}} b_i \beta_\in W$, then $f(v) = f\left(\sum_{\beta_i \in \mathcal{B}} b_i \beta_i\right) = \sum_{\beta_i \in \mathcal{B}} b_i f(\beta_i)$). Note however that the choice basis is important, we shall get back to this in section 13.2, and that vector spaces may have many different bases (of the same cardinality by the replacement theorem):

Example 13.1.16: Different Basis
As we've defined in example 12.3.2the set

$$e_i = (\underbrace{0, ..., 0}_{i-1 \text{ times}}, 1, \underbrace{0, ..., 0}_{k-i \text{ times}}) \qquad 1 \le i \le n$$

is a basis for $\mathbb{Q}^n$. There are countably many different basis for $\mathbb{Q}^n$; can you think of a few? If you know about inner product spaces, can you think of an orthonormal [hamel][a] basis?

[a]The term orthonormal basis is usually reserved for *countable* linear combination for *Hilbert space*. Since the linear combination is not finite, it requires some analysis to define; this would be covered in a text on *Functional Analysis*. If you know about Hilbert spaces, then taking $\mathbb{C}[x]$ as the closure of $\mathbb{C}[x]$, is there an orthonormal basis for it?

13.1.1 Quotient Spaces and Subspaces

Now that we've established the struture of a vector-space, we can some questions about subspaces the structure of quotient spaces, with particular interest in finite vector-spaces to be able to take advantage of counting arguments. We already saw that subspace of vector spaces are free k-module.

Proposition 13.1.17: Dimension Of Quotient Space

Let V be a vector space over k and let W be a vector space. Then V/W is a vector space with $\dim V = \dim W + \dim V/W$. If one term is finite, all terms are finite

***Proof*:**
Certainly V/W is a vector space being a quotient k-module. How would you prove $\dim V = \dim W + \dim V/W$? Why should this be intuitive?

Naturally, since we are working with quotients, we have an associated linear transformation The following is the "homomorphism-equivalent" of the previous proposition:

Corollary 13.1.18: Rank-Nullity Theorem

Let $\varphi : V \to W$ be a linear transfomration of vector spacees over k and $\dim(V) < \infty$. Then $\ker \varphi$ is a subspace of V, $\varphi(V)$ is a subpace of W, and

$$\dim_k(V) = \dim_k(\ker(\varphi)) + \dim_k(\varphi(V))$$

***Proof*:**
follows from previous theorem

Since the dimensions are of the subspaces are invariant to basis, we can give a name to the dimension of the image and null space of a linear transformation

Definition 13.1.19: Rank And Nullity

If $\varphi : V \to W$ is a linear transformation of vector spaces over k, $\ker(\varphi)$ is called the *null space* of φ ,and the dimension of $\ker(\varphi)$ is called the *nullity* of φ. The dimension of $\varphi(V)$ is called the *rank* of φ. If $\ker(\varphi) = 0$, the transformatoin is said to be *nonsingular*.

The reasoning behind the name *nonsingular* is due to the fact that very often in (differentil) geometry, we associated to shapes a vector-space at each point (such an object is called a *vector bundle*) and if the curve "drops rank" this would correspond to a particular function having $\dim_k(\ker(\varphi)) > 0$[6]. For example, the curve $x^3 = y^2$ "drops rank" at $(0, 0)$.

Corollary 13.1.20: Properties of [finite dim.] Linear Transformations

Let $\varphi : V \to W$ be linear transfomration of vector spaces of the same finite dimenison. Then the following are equivalent

1. φ is an isomorphism
2. φ is injective, i.e., $\ker(\varphi) = 0$
3. φ is surjective, i.e. $\varphi(V) = W$
4. φ sends a basis of V to a basis in W

Notice that this proof is saying that if $f : V \to W$ is a k-homomorphism where $\dim_k(V) = \dim_k(W) = n$, then given f is injective, it is surejctive. This fails to be the case even for maps between free moduels over PIDs, for exmaple $f : \mathbb{Z} \to \mathbb{Z}$ mapping $z \mapsto 2z$ is an injective $\mathbb{Z}$-module homomorphism between free $\mathbb{Z}$-modules, but is certainly not surjective! This also fails for infinite dimensional vectors

[6] In particular, if $T_p f$ is the tangent map at p, then $T_p f$ would drop rank

spaces as will be demonsarted by a counter-example after the proof. Hence, the fact that it's sufficient to check injectivity shows how linear maps between finite vector-spaces are akin to maps between finite sets[7].

***Proof*:**
Counting dimension arguments

Note that finiteness of dimension is important in the previous corrollaries: these need not hold in infinite dimension

Example 13.1.21: Infinite Dimensinoal Linear Transformations
Take $V = \ell^2$, the set of all infinite sequences of real numbers satisfying the property:

$$\left(\sum_{i\in\mathbb{N}}^{\infty} |x_i|^2\right)^{1/2} < \infty$$

Then define $f : V \to V$ via:

$$(x_1, x_2, ..., x_n, ...) \mapsto (x_1, \frac{x_2}{2}, ..., \frac{x_n}{n})$$

Then this function is injective, but not surjective. Even more is true; V and f belong to some of the nicest infinite dimensional spaces and functions (V is a Hilbert space and f is a bounded self-adjoint linear operator), and hence this property is not easily rectified.

We now take a look at subspaces. We have already established that all subspaces of a vector-spaces are free. Another importnant property is that every subspace always has a *compliment*, that is if $W \subseteq V$, then there exists a $U \subseteq V$ such that $U \cap W = \{0\}$ and $W + U = V$. This is certainly *not* true for general R-modules, simply let $M = \mathbb{Z}/4\mathbb{Z}$ be a $\mathbb{Z}$-module and take $\{0, 2\} \subseteq \mathbb{Z}/4\mathbb{Z}$. Then we would require $U = \{0, 1, 3, 4\}$, which is not a subgroup. In fact, *no* finite abelian subgroup has this property[8]. However, we may always find such a compliment for subspaces of vector-spaces, given the space is finite dimensional:

Proposition 13.1.22: Existence Of Compliment

Let V be a finite dimensional vector space and $W \subseteq V$ be a subspace. Then there exists a subspace $U \subseteq V$ such that

1. $W \cap U = \{0\}$
2. $W + U = V$

If these two conditions are satisfied, we will write $W \oplus V$ and call it the direct sums (just like in definition 12.3.3).

[7] In fact, this parallels is runs more deeply than it may at first seem, see a text on K-theory for some interesting elaboration on this notion

[8] More particularly, no torsion $\mathbb{Z}$-module has this property. In the Category of **Grp**, there are finite groups that "spilt". See section 17.2 for a generalized notion of splitting and section 5.4.2 for identifying groups that split

***Proof*:**
Let $U \subseteq V$ be a subspace. Choose basis $\{e_1, ..., e_k\}$. Then this basis can be extended to a basis $\{e_1, ..., e_k, ..., e_n\}$ for V. Define the map $\pi_U : V \to V$ given by:

$$\sum_i^n a_i e_i \mapsto \sum_i^k a_i e_i$$

This map is certainly a linear transformation, and $\mathrm{im}(\pi) = U$. The kernel of this map is:

$$\ker(\pi_U) = \langle \sum_{i=k+1}^n a_i e_i | a_i \in k \rangle$$

Notice that $v \in \ker(\pi_U)$ if and only if $v \in \mathrm{im}(I - \pi_U)$, namely:

$$v \in \ker(\pi_U) \Leftrightarrow \pi_U(v) = 0 \Leftrightarrow v - \pi_U(v) = v \Leftrightarrow v \in \mathrm{im}(I - \pi_U)$$

I claim that $v = \pi_U(v) + (I - \pi_U)(v)$. Indeed if $v = \sum_i a_i e_i$:

$$\sum_i^n a_i e_i = \sum_i^k a_i e_i + \sum_{i=k+1}^n a_i e_i = \pi_U(v) + (I - \pi_U)(v)$$

and furthermore $\ker(\pi_U) \cap \mathrm{im}(\pi_U) = 0$, hence setting $W = \ker(\pi_U)$ and $V = \mathrm{im}(\pi_U)$ we get:

$$V = \ker(\pi_U) \oplus \mathrm{im}(\pi_U) = U \oplus W$$

as we sought to show.

The vector space being finite dimensional is key to the proof, as this does *not* generalize to general infinite dimensional vector-spaces (it can be thought of as characterizing *Hilbert spaces*).

Corollary 13.1.23: Finite Generation of Subspace

Let $W \subseteq V$ be a subspace of a finite dimensional vector space. Then W is finitely generated

***Proof*:**
use proposition 13.1.22.

The map used in the above proof is important enough to be given a name:

Definition 13.1.24: Projection Map

Let $\pi : V \to V$ be a linear transformation $U \subseteq V$. Given $V = U \oplus W$, then the *projection map* (with respect to U) is the map $\pi_U : V \to V$ such that $\pi(v) = \pi(u + w) = u$.

Notice $\pi \circ \pi = \pi$, that is, applying the map twice gives back the same result: $\pi(\pi(v)) = \pi(v)$. This in fact characterizes such a map:

Proposition 13.1.25: Characterizing Projection Maps

Let π be linear operator over a [not necessarily finite dimensional] vector space such that $\pi \circ \pi = \pi$. Then π is a projection map.

***Proof*:**
Let $U = \operatorname{im}(\pi)$ and $W = \operatorname{im}(I - \pi)$. Then for any $v \in V$:

$$\pi(v) + v - \pi(v) = v$$

hence $U + W = V$. If $x \in U \cap W$. Since $x \in \operatorname{im}(\pi)$, $x = \pi(y)$ for some $y \in V$, and so:

$$\pi(x) = \pi(\pi(y)) = \pi(y) = x$$

hence, $\pi(x) = x$. Then:

$$\begin{aligned}
x &= \pi(x) && x \in U \\
&= (I - \pi)(x) && x \in W \\
&= \pi((I - \pi)(x)) && (I - \pi)(x) = x \in U \\
&= \pi(x - \pi(x)) \\
&= \pi(x - x) && \pi(x) = x \\
&= \pi(0) \\
&= 0
\end{aligned}$$

Hence $x = 0$, and so $U \cap W = \{0\}$. With the above, we get that

$$V = U \oplus W = \ker(\pi) \oplus \operatorname{im}(\pi)$$

as we sought to show.

Notice how the above proof did not rely on any structure of V (notably, it wasn't important that V had a basis). Thus, if $\varphi : M \to M$ is any R-module homomorphism such that $\varphi \circ \varphi = \varphi$, then

$$M = \ker(\varphi) \oplus \operatorname{im}(\varphi)$$

For this reason, it is much more common to define projection maps as maps that satisfy the property of $\varphi^2 = \varphi$ (i.e., they are *idempotent*).

Exercise 13.1.1

1. Consider $\mathbb{R}[x]$ as an $\mathbb{R}$ module. Take $\mathbb{R}_n[x]$ to be all polynomial where $\deg(p(x)) \leq n$. Show that (evalutaion polynomials has $\left\{(x-a), (x-a)^2, ..., (x-a)^n\right\}$ as a basis). (This might at seem counter-intuitive at fisrt after doing some rings, but it's really cool!)
2. Let R be a commutative ring that is *not* a field. Show that not all R-modules are free, i.e. find a counter-example [Hint: Look back at the examples given in section 12.1]
3. (More challenging) Let R be any ring (not necessarily commutative). Show that if any left (or right) R-module is free, then R is a divison ring.

4. Show that $(x, y) \subseteq \mathbb{C}[x, y]$ is not a free $\mathbb{C}[x, y]$-submodule.
5. Show that any uncountable set of $\mathbb{C}[x]$ is linearly dependent.
6. (Right shift operator)
7. Let $R = k[x_1, ..., x_n]$ and $I = (x_1, ..., x_n)$. Show that I^m is a k-vector space with basis $\left\{x_{i_1}^{r_1} x_{i_2}^{r_2} \cdots x_{i_s}^{r_s} \mid \sum r_s = m, 1 \leq i_j \leq n\right\}$ (that is, all forms of degree m).

A lot of exercices in linear algebra are about determining if a set is linearly independent, spanning, (more?). These exercices are to practice these skills:

13.2 Linear Transformations and Matrix Representation

As mentioned at the beginning of the chapter, k-module homomorphisms are called linear transformation in the theory of vector-spaces. In this section, we will show how every linear transformation between finite dimensional vector-spaces can be represented as a matrix. Since all vector-spaces are free with an invariant basis number, when defining a linear transformation we only need to be concerned with where the basis get's mapped to. If $\varphi : V \to W$, where V and W are finite dimensional vector-spaces over k, and $\varphi \in \operatorname{Hom}_k(V, W)$. Let $\beta_V = \{v_1, v_2, ..., v_n\}$ and $\beta_W = \{w_1, w_2, ..., w_m\}$ be (ordered) bases for V and W respectively. Then we are only concerned where the v_i's map to, that is, what $\varphi(v_j) = w_j$ is. Since they map into the vector-space W, $w_j = \sum_{i=1}^m b_{ij} w_i$, $b_{ij} \in k$,

$$\varphi(v_j) = \sum_{i=1}^{m} b_{ij} w_i$$

Let $M_{\beta_V}^{\beta_W}(\varphi)$ be the $m \times n$ matrix whose ij entry is b_{ij} (i.e. the coefficients of w_i given v_j). This matrix is given a name:

Definition 13.2.1: Matrix Representation

Let $\varphi : V \to W$ be a linear transformation between two finite dimensional vector spaces with dimension n and M and basis β_V and β_W respecitvely. Then the *matrix representation* $M_{\beta_V}^{\beta_W}$ of φ (also written as $M_{\beta_V}^{\beta_W}(\varphi)$) is

$$\begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{pmatrix}$$

where each entry ij is the coefficient of the ith basis element of β_W in the linear combination of where the element v_j maps.
If a matrix A is a matrix representation of a linear transformation φ, we say that A *represents* φ *with respect to the basis* β_V, β_W. Similarly, φ would be the linear transformation represented by A with respect to the bases β_V, β_W.

With the matrix representation of φ, we can re-write $\varphi(v) = w$ as

$$\begin{pmatrix} c_{11} & c_{12} & \cdots & c_{1n} \\ c_{21} & c_{22} & \cdots & c_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ c_{m1} & c_{m2} & \cdots & c_{mn} \end{pmatrix} \begin{pmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{pmatrix} = \begin{pmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{pmatrix} \tag{13.4}$$

where $v = \sum_{i=1}^n a_i v_i$ and $\varphi(v) = \sum_{i=1}^m b_i w_i$. Sometimes, the following notation is used as a shorthand for (13.4): $[\varphi(v)]_{\beta_W} = M_{\beta_V}^{\beta_W}[v]_{\beta_V}$.

Example 13.2.2: Matrix Representation

1. Let $V = \mathbb{R}^3$, $W = \mathbb{R}^2$, and take $\beta_V = \{(1,0,0),(0,1,0),(0,0,1)\}$ and $\beta_W = \{(1,0),(0,1)\}$ (i.e., the standard basis for $\mathbb{R}^3$ and $\mathbb{R}^2$). Let

$$\varphi(x,y,z) = (4x + 2z, x - y + 5z)$$

Then $\varphi(1,0,0) = (4,1)$, $\varphi(0,1,0) = (0,-1)$, and $\varphi(0,0,1) = (2,5)$. Thus, the matrix representation of φ with respect to β_V and β_W is

$$\begin{pmatrix} 4 & 0 & 2 \\ 1 & -1 & 5 \end{pmatrix}$$

Theorem 13.2.3: Matrix Representation Correspondence

Let V and W be a finite dimensional vector space over F with dimension n and m, as well as basis β_V and β_W respectively. Then the map $\Psi : \mathrm{Hom}_F(V,W) \to M_{mn}(F)$, where $\varphi \mapsto M_{\beta_V}^{\beta_W}(\varphi)$ is a [vector-space] isomorphism.

Note that the operation on $\mathrm{Hom}_F(V,W)$ if point-wise function addition and $M_{mn}(F)$ is matrix addition.

***Proof*:**
To show Ψ is F-linear, it comes down to the fact that every $\varphi \in \mathrm{Hom}_F(V,W)$ is F-linear. Since every column of $M_{\beta_V}^{\beta_W}$ represents $\varphi(v_i)$, and $(\varphi + \psi)(v) = \varphi(v) + \psi(v)$, then we get that the map Ψ is also F-linear.

The map Ψ is surjective, since given a matrix $A \in M_{mn}(F)$, the map φ that maps to it can be defined by reversing the process that was done in equation (13.4). This map is injective since if $A = B$, then $A = M_{\beta_V}^{\beta_W}(\varphi) = M_{\beta_V}^{\beta_W}(\psi) = B$. Since a basis uniquely defines a linear transformation and both representations are with respect to the same bases, it must be that $\varphi = \psi$, proving injectivity.

Thus, Ψ is an isomorphism, as we sought to show.

13.2.4 Note Different choices of bases produces different isomorphisms. There is no "natural choice" in the same way that there is no natural choice for a basis for a vector-space

Corollary 13.2.5: Dimension Of $\mathrm{Hom}_F(V, W)$

The dimension of $\mathrm{Hom}_F(V, W)$ is $\dim(V) \cdot \dim(W)$

***Proof*:**
Since $\mathrm{Hom}_F(V, W)$ is isomorphic to $M_{mn}(F)$, and $\dim(M_{mn}(F)) = mn$, then $\dim(\mathrm{Hom}_F(V, W)) = mn = \dim(V) \cdot \dim(W)$

We've shown that function addition is compatible with matrix addition. When it comes to function composition, the story becomes a little trickier, since we need the co-domain of one function to line-up with the domain of the next. Thus, to that end, let U, V, W be finite dimensional vector-spaces with dimension k, n, m and bases $\beta_U = \{u_1, u_2, ..., u_k\}$, $\beta_V = \{v_1, v_2, ..., v_n\}$, and $\beta_W = \{w_1, w_2, ..., w_m\}$ respectively. Let $\varphi : U \to V$ and $\psi : V \to W$ be two linear transformations. Their composition $\psi \circ \varphi$ is a linear fransformation from U to W, and so there exists a representation $M_{\beta_U}^{\beta_W}(\psi \circ \varphi)$ given by

$$(\psi \circ \varphi)(u_j) = \sum_{i=1}^{m} c_{ij} w_i \qquad c_{ij} \in F$$

Since φ and ψ are linear transfomation, they too have a matrix representation:

$$\varphi(u_j) = \sum_{p=1}^{n} a_{pj} v_p \qquad \psi(v_p) = \sum_{i=1}^{m} b_{ip} w_i$$

from this, we can find c_{ij} in terms of the coefficients a_{pj} and b_{ip}:

$$\begin{aligned}
(\psi \circ \varphi)(u_j) &= \psi\left(\sum_{p=1}^{n} a_{pj} v_p\right) \\
&= \sum_{p=1}^{n} a_{pj} \psi(v_p) \\
&= \sum_{p=1}^{n} a_{pj} \sum_{i=1}^{m} b_{ip} w_i \\
&= \sum_{p=1}^{n} \sum_{i=1}^{m} a_{pj} b_{ip} w_i \\
&= \sum_{i=1}^{m} \sum_{p=1}^{n} a_{pj} b_{ip} w_i
\end{aligned}$$

Thus, if we let $c_{ij} = \sum_{p=1}^{n} a_{pj} b_{ip}$, we get

$$= \sum_{i=1}^{m} c_{ij} w_i$$

Take a moment to look over this and think about how this would be represented as multiplying the matrix $M_{\beta_U}^{\beta} V(\varphi)$ and $M_{\beta_V}^{\beta_W}(\psi)$. In fact, the coefficents of $M_{\beta_U}^{\beta_W}(\psi \circ \varphi)$ are exactly what you get by multiplying the other two matrices! This is summed up in the following theorem:

Theorem 13.2.6: Matrix Representation And Composition

Let U, V, W be finite dimensional vector-spaces with bases β_U, β_V, β_W, of size k, m, n respectively. Let $\varphi : U \to V$ and $\psi : V \to W$ be two linear transformations. Then

$$M_{\beta_U}^{\beta_W}(\psi \circ \varphi) = M_{\beta_V}^{\beta_W}(\psi) M_{\beta_U}^{\beta_V}(\varphi)$$

Another way of seeing this theorem is by saying the following diagram commutes:

$$\begin{tikzcd}
M_{mp}(F) \times M_{mp}(F) \arrow[r] \arrow[d] & M_{mn}(F) \arrow[d] \\
\mathrm{Hom}_F(F^p, F^m) \times \mathrm{Hom}_F(R^p, R^m) \arrow[r] & \mathrm{Hom}_F(F^n, F^m)
\end{tikzcd}$$

Corollary 13.2.7: Matrix Multiplication Properties

Let $M_{mn}(F)$ be the set of $m \times n$ matrices over F. Then given appropriate matrices, matrix multiplication is associative and distributive

Note how this is different from the associativity and disctributivity of $GL_n(F)$. Namely, all the matrices of $GL_n(F)$ are square matrices. We have not yet proven that non-square matrices are associative and distributive, particularly because they do not always line up properly, and their result can pass from one set of matrices to another (ex. $A \in M_{ab}(F)$, $B \in M_{bc}(F)$, $AB = M_{ac}(F)$)

***Proof*:**
We'll show the proof for associativity – the proof for distribuity uses the same technic. Let A, B, C be matrices such that $(AB)C$ and $A(BC)$ are defined. By theorem 13.2.6, (AB) represents $S \circ T$ for appropriate linear transformation S, T. Multiplying on the right side by C, $(AB)C$, the resulting matrix represents $(S \circ T) \circ R$ for appropriate R. Similalry, $A(BC)$ is represented by $S \circ (T \circ R)$. Since the composition of functions is assosiative, Then

$$\Psi((AB)C) = (S \circ T) \circ R = S \circ (T \circ R) = \Psi(A(BC))$$

and since Ψ is injective, $(AB)C = A(BC)$. A similar proof follows for distributivity.

Corollary 13.2.8: Matrix Representation Isomorphisms

1. If β is a basis of the n-dimensional vector space V, the map $\varphi \mapsto M_\beta^\beta(\varphi)$ is a ring isomorphism and a vector-space isomorphism of $\mathrm{Hom}_F(V, V)$ (i.e. $\mathrm{End}(V)$) onto $M_n(F)$:

$$\mathrm{End}(V) \cong_{\mathbf{Ring}} M_n(F) \qquad \mathrm{End}(V) \cong_{F\text{-}\mathbf{Vect}} M_n(F)$$

2. If we restrict to automorphisms, then $\mathrm{Aut}(V) \cong GL_n(F)$ as vector-spaces[a]. In particular

$$\mathrm{Aut}_{\mathbf{Vect}}(V) \cong_{F\text{-}\mathbf{Vect}} GL_n(F)$$

[a] $\mathrm{Aut}(V)$ is not a ring

***Proof*:**
In both cases, we've already shown that it is isomorphic as F-vector-spaces, so we'll show they're isomorphic as rings

(1) Using corollary 13.2.7, we see that $M_n(F)$ is a ring (since multiplying two $n \times n$ matrices get's us an $n \times n$ matrix). We've shown this map is bijective in theorem 13.2.3. theorem 13.2.6 shows that this map also preserves the ring structure of $\text{End}(V)$, showing that the map is a ring-isomorphism.

(2) This is the same proof as (1) (Dummit addded: since it sends units to units)

Finally, we convert the powerful result of corollary 13.1.20 into for matrices, in particular we showed that if f is injective between two finite dimensional vector space of the same dimension, then it is surjective. This result is important enough that the matrix representation of such an f should be given a name

Definition 13.2.9: Nonsingular Matrix

An $m \times n$ matrix A is called *nonsingular* if $Ax = 0$ with $x \in F^n$ implies $x = 0$

For some intuition band the name, see remark ref:HERE (text under definition 13.1.19)

Proposition 13.2.10: Nonsingular Square Matrix

Let A be an $n \times n$ matrix. Then A is invertible if and only if A is nonsingular

***Proof*:**
If $\dim_k(V) = \dim_k(W)$, then $V \cong_k W$, so without loss of generailty we may say $f : V \to V$ is a linear map from V to itself. Let's say A is invertible and $Ax = 0$. then

$$0 = Ax = A^{-1}Ax = x \quad \Rightarrow \quad x = 0$$

giving us that A is nonsingular.

Conversely, let A be nonsingular. Then A represents some linear transformation $\varphi : V \to V$ for some finite dimensional vector-space V with respect to basis β and γ for the domain and codomain. Since $\varphi(x) = 0$ if and only if $x = 0$, then φ is injective, and so by corollary 13.1.20, φ is an isomorphism. Thus, there exists a linear transfomration φ^{-1} such that $\varphi^{-1} \circ \varphi = \text{id}$. By using theorem 13.2.3, let B be the representation of φ^{-1} with respect to γ and β. Using theorem 13.2.6 we have

$$AB = M_\beta^\gamma(\varphi)M_\gamma^\beta(\varphi^{-1}) = M_\gamma^\gamma(\varphi \circ \varphi^{-1}) = M_\gamma^\gamma(I) = I$$

thus $AB = I$. similarly $BA = I$. Thus, $B = A^{-1}$, as we sought to show.

(dummit here put's a throw-away definition and comment about the row rank and column rank, and that the rank of φ is the row rank of it's representation, and that the row and column rank match. Once I have a more solid grasp on whats going on here, I'll come back to it; for now my intuition relates this to kernel an co-kernel, or more generally injectivity/surjectivity test)

Exercise 13.2.1

1. We know that in general, if R is a ring and $x \in R$ is not a unit, that doesn't imply that x is a zero-divisor (ex. take the integers). Show that for the matrix ring $M_n(k)$, if $A \in M_n(k)$ is not a unit, then A is a zero-divisor.

2. Show that any polynomial (over a subfield of $\mathbb{C}$) of degree n is uniquely determined by $n+1$ points (You may freely use the fact that a polynomial of degree n can be broken down into a product of n linear terms).

3. (If you know some complex arithmetics) Let $f : \mathbb{C} \to \mathbb{C}$ be a $\mathbb{C}$-linear function. Recall that $\mathbb{C}$ is a 2 dimensional $\mathbb{R}$-vector space, and so f may be represented as a real matrix. Show that if $f(z) = \alpha z = (a+bi)z$, then the matrix representation of f in the standard basis will be of the form:
$$[f] = \begin{pmatrix} a & b \\ -b & a \end{pmatrix}$$

13.2.1 Change of Basis

While representing vectors in a vector space in terms of coordinates simplifies their representation, and linear transformations and their compositions can be seen as matrices and matrix multiplication, we are *always* dependent on a choice of basis (i.e. a choice of isomorphism $\varphi : k^n \to V$). As pointed out earlier, there is no "canonical" choice for a basis – there are many choices of bases which will produce isomorphic vector spaces[9]. Thus, in this section we will explore how to change from one basis to another to be able to compare representations. We will be working with vector-spaces, but we will only be using properties of free-modules meaning the theory is true for free-modules in general.

Let V be an F-vector space of dimension n. Let β_A and β_B be two basis for V, meaning we have the two isomorphissm
$$F^{\oplus \beta_A} \xrightarrow{\varphi} V \qquad F^{\oplus \beta_B} \xrightarrow{\psi} V$$

Then we will get the following isomorphism:
$$F^{\oplus \beta_A} \xrightarrow{\psi^{-1} \circ \varphi} F^{\oplus \beta_B}$$

Then by theorem 13.2.3, there is a matrix $N_{\beta_A}^{\beta_B}$ that corresponds to $(\psi^{-1} \circ \varphi)$. In particular, the jth column of $N_{\beta_A}^{\beta_B}$ is the image of the jth element of A viewed as a column vector in $f^{\oplus \beta_B}$.

If $\beta_A = (a_1, a_2, ..., a_n)$ and $\beta_B = (b_1, b_2, ..., b_n)$ then
$$(\psi^{-1} \circ \varphi)(a_j) = \sum_{i=1}^{n} n_{ij} b_i$$

We give this matrix a name:

[9]There are ways to make it "canonical" by augmenting the vector-space to have further structure which may naturally induce a basis (for example, an inner product). For such examples, see *Linear Algebra* [17].

Definition 13.2.11: Change Of Basis

Let V be a F-vector space of dimension n, and let β_A and β_B be two bases corresponding to the isomorphisms φ and ψ. Then the *change of basis matrix* is the matrix representation of $(\psi^{-1} \circ \varphi)$.
We can represent this matrix as $N_{\beta_A}^{\beta_B}$ or $N_{\beta_A}^{\beta_B}(V)$

Corollary 13.2.12: Properties Of Change Of Basis

Let $N_{\beta_A}^{\beta_B}$ and $N_{\beta_B}^{\beta_C}$ be two change of basis for a vector-space V. Then

1. $(N_{\beta_A}^{\beta_B})^{-1} = N_{\beta_B}^{\beta_A}$
2. $N_{\beta_A}^{\beta_B} N_{\beta_B}^{\beta_C} = N_{\beta_A}^{\beta_C}$

***Proof*:**
Writing out the isomorphism explicitely will immediately give the result, and so it is left as an exercise

With the change of basis for a vector space taken care of, we know check how to perform a change of basis for a matrix representation of a linear transformation. The idea behind this comes down to the following visual:

$$\begin{tikzcd}
F^{\oplus \beta_A} \arrow[dr, "\varphi_1"] \arrow[dd, "\nu_{\beta_A}^{\beta_B} = \varphi_2^{-1} \circ \varphi_1"'] & & & F^{\oplus \beta_C} \arrow[dl, "\varphi_2"'] \arrow[dd, "\mu_{\beta_C}^{\beta_D} = \psi_2^{-1} \circ \psi_1"] \\
& V \arrow[r, "\alpha"] & W & \\
F^{\oplus \beta_B} \arrow[ur, "\psi_1"'] & & & F^{\oplus \beta_D} \arrow[ul, "\psi_2"]
\end{tikzcd}$$

Let N_A^B be the matrix representation of $\varphi_2^{-1} \circ \varphi_1$ and M_C^D be the matrix representation of $\psi_2^{-1} \circ \psi_1$. If we choose the basis β_A for V and β_C for W, then we have:

$$\varphi_2^{-1} \circ \alpha \circ \varphi_1 : F^{\oplus \beta_A} \to F^{\oplus \beta_C} \tag{13.5}$$

Let P be the matrix representation of $\varphi_2^{-1} \circ \alpha \circ \varphi_1$. If we choose the basis β_B for V and β_D for W, then we have:

$$\psi_2^{-1} \circ \alpha \circ \psi_1 : F^{\oplus \beta_B} \to F^{\oplus \beta_D} \tag{13.6}$$

now with some manipulation, we can transform equation 13.6 into equation 13.5 like so

$$\psi_2^{-1} \circ \alpha \circ \psi_1 = (\mu_{\beta_C}^{\beta_D} \circ \varphi_2^{-1}) \circ \alpha \circ (\varphi_1 \circ (\nu_{\beta_A}^{\beta_B})^{-1}) = \mu_{\beta_C}^{\beta_D} \circ (\varphi_2^{-1} \circ \alpha \circ \varphi_1) \circ (\nu_{\beta_A}^{\beta_B})^{-1}$$

And using the matrix correspondence, this is equivalent to

$$M_{\beta_C}^{\beta_D} P (N_{\beta_A}^{\beta_B})^{-1} = M_{\beta_C}^{\beta_D} P N_{\beta_B}^{\beta_A} = Q$$

This should be able to be read quite intuitively: $N_{\beta_B}^{\beta_A}$ converts our vector represented by a linear combination of vectors in β_B into linear combination of vectors in β_A, then P transforms the result and outputs it as a linear combination of vectors in β_C, and finally $M^{\beta_D}_{\beta_C}$ converst the result into a linear representation of vectors in β_D. W

We sum up these results into a proposition:

Proposition 13.2.13: Change Of Basis For Linear Transformation

Let $\varphi : V \to W$ be a linear transformation from two finite dimensional vector spaces, and let P be the matrix representation of φ with respect to bassi β_A and β_C for V and W respectively. Then the matrix representing linear transformation φ with respect to basis β_B and β_D is

$$M^{\beta_D}_{\beta_C} P N^{\beta_A}_{\beta_B}$$

with the notation given above.

More generally, if M and N are any invertible matrix, they represent a change of basis, so MPN would be a change of basis (given the appropriate dimensions).

Definition 13.2.14: Equivalence

Let $P, Q \in M_{mn}(F)$. Then we say P and Q are *equivalent* if they represent the same linear transformation $\varphi : V \to W$ up to a chocie of basis. If that's the case, there exists matrices M and N such that

$$Q = MPN$$

It should be shown that equivalence is an equivalence class, and hence there is an entire equivalence class of matrices that can represent a linear transformation, where any representative of the equivalence class equates to a choice of basis for the domain and codomain. This brings up an interesting question: is there a particular representative (i.e. choices of bases for P, Q) that will make it "easier" to see what the linear transformation is doing? This is a very interesting question, and in the next section shall be explored to great success, in particular, all linear transformation can be represented in "reduced form" and which can be algorithmically found.

13.2.2 Similarity

Before proceeding, there is one more important concept that must be covered, namely the special case where $V = W$ is the same vector-space:

$$\mathrm{Hom}_R(V, V) = \mathrm{End}_R(V)$$

that is, we shall look at linear transformations from V to itself. Such endomorphisms are in fact extremely common; they can be thought of as representing the "change in state" of a system. As the reader may surmise, this means they have applications all other the physical science, however they shall amply show up in pure mathematics. In fact, chapter 14 is essentially the theoretical build-up for studying such functions.

For this discussion, we shall tacitly assume $R = k$ is a field, however for most of the discussion having R simply be an integral domain shall work. The reader should keep in mind though that many results work in broader generality want to make this more the responsibility of the author rather than the reader

We shall simply start by asserting that the vector-space V in $\mathrm{Hom}_R(V,V)$ shall be considered to be the exact same space, not two isomorphic copies of these spaces as generally is considered. In this situation, an element $\alpha \in \mathrm{Hom}_R(V,V) = \mathrm{End}_R(V)$ is usually called a *linear operator* to emphasize the fact that the domain and codomain are "identical". With this in mind, there are new questions that could be asked, such as if there exists $v \in V$ such that $\alpha(v) = \lambda v$ for some $\lambda \in R$, or if a subspace $W \subseteq V$ maps into itself.

Another important consequence of identifying the domain and codomain is that when choosing a basis, the *same* basis must be chosen for both of them (as α is considered to be mapping V to itself). This means that there is a more specific notion of equivalence as only one matrix is chosen:

Definition 13.2.15: Similarity

Let $A, B \in M_n(R)$. Then these two matrices are *similar* if they represent the same linear operator $\alpha : V \to V$ up to choice of basis for V. Equivalently, two endomorphism α, β are similar if there exists an automorphism φ such that $\beta = \varphi \circ \alpha \circ \varphi^{-1}$
Similarity is an equivalence relation[a], and so we say $A \sim B$ if and only if A and B are similar.

[a]check if you are not convinced

An immediate consequence of staring at this diagram:

$$\begin{tikzcd}
R^n \arrow[dd, "\nu^{\beta_B}_{\beta_A}=\varphi^{-1}\circ\varphi"'] \arrow[dr, "\varphi"] & & & R^n \arrow[dl, "\varphi"'] \arrow[dd, "\mu^{\beta_D}_{\beta_C}=\psi^{-1}\circ\psi"] \\
& V \arrow[r, "\alpha"] & V & \\
R^n \arrow[ur, "\psi"'] & & & R^n \arrow[ul, "\psi"]
\end{tikzcd}$$

we see that $A \sim B$ if and only if there exists an invertible $P \in \mathrm{GL}_n(R)$ such that

$$B = PAP^{-1}$$

which explains the definition for similarity of endomorphisms, though the definition for endomorphisms generalizes to non-finite cases. When working over a finite-dimensional vector spaces, then each endomorphisms has a matrix representation.

The reader may have noticed that $A \sim B$ if B is the conjugate of A via an invertible matrix. In the language of group theory, this shows that $\mathrm{GL}(V)$ naturally acts on $\mathrm{End}_R(V)$ via conjugation[10], hence $A \sim B$ iff A, B are in the same orbit in this action.

Just like for equivalence, we may ask for when two matrices are similar, or if there is a nice representation in the orbit. This question is a bit trickier because the domain and codomain have are the same, and so there is less flexibility in choosing representatives. In chapter 14, we shall show how to find a nice basis to find a nice representation of α.

13.2.3 Gaussian Elimination: Finding a Nice Basis

Let $\varphi : V \to W$ be a linear transformation, and P be some matrix representation given by choosing a basis for V and W. Then P belongs to some equivalence class representing all possible representatives

[10]Recall that conjugation is *the* natural group-action

of φ. If we multiply P on the left or right by an invertible matrix, then by by proposition 13.2.13, P remains in the same equivalence class. It would thus be nice if given P, we may simplify P as much as possible to get an "nice" looking represetnative for φ. To do this, we will find nice sets of invertible matrices that do "elementary" operations to a marix which will allow the matrix P to be put in as simple a form as possible. In particular, we will find invertible matrice that will allow us to perfom the following "elementary" operations:

1. switching two rows (resp. columns) or P. For example, let's say we want to switch the 1st and 2nd row of a matrix. This can be done by multiplying on the right (resp. left) by:

$$\begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

and the identity on the left (resp. right). In particular, we swapped the rows of the above matrix.

2. multiply all elements in a row (resp. column) of P by a unit of R by multiplying on the left (resp. right) by a matrix of the form:

$$\begin{pmatrix} u & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

This particular matrix will scale the first row (resp. column) by a factor of u when multiplying on the left (resp. right).

3. add a multiple of one row (resp. column) to another row (resp. column) of P. For example, to add a multiple of the first row to the second,we would multiply on the right (resp. left) by:

$$\begin{pmatrix} 1 & 3 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

This particular matrix will add 3 copies of the 2nd row (resp. column) to the 1st row (resp. column) when multiplying on the left (resp. right).

It should be checked that these matrices are indeed invertible matrices (can you find their inverse?), and that the side on which the matrix is multiplie by effects either columns or rows. The following proposition summarizes the above discussion, generalizing it to general commutative rings R:

Proposition 13.2.16: Equivalence An Elementary Matrices

Let $P, Q \in M_{m,n}(R)$ where R is a commutative ring. Then if Q may be obtained from a sequence of elementary operations, then P an Q are equivalent.

***Proof*:**
Verify this proposition.

Due to these matrices being the transition matrices between two equivalent matrices, we give matrices that perfom elementary operations a name:

Definition 13.2.17: Elementary Matrices

Elementary matrices are matrices such that when mutliplying a matrix from the left or right side,the reuslting matrix has had an elementary operation performed on it.

Notice that proposition 13.2.16 is not an "if and only if", simple an "if". This is because it is possible that P and Q are equivalent, but the invertible matrices that "connects" the two are not the product of elementary matrices. If R has certain extra properties, then it will be the case that all invertible matrices can be written as a product of elementary matrices, as we'll explore now. To make this easier to exlpore, let's give a name to all invertible linear matrices

Definition 13.2.18: General Linear Matrices

Let R be a ring and $M_n(R)$ the set of all matrices of of dimension n over R. Then:

$$\mathrm{GL}_n(R) := (M_n(R))^\times$$

that is, $\mathrm{GL}_n(R)$ is the set of all units of $M_n(R)$, i.e. the set of all invertible matrices of dimension n over R.

The word "dimension" is used to indicate the number of rows and columns (which are both equal in this case, and hence why a single number is named). The term "general" is used since there other set of invertible matrices commonly studied, usually given a different prefix ("special", "orthogonal", "orthonormal", "unitary", and so forth). This set of invertible matrices is called "general" since it ill encompass *all* of these other sets of invertible matrices. For who may have come across these terms before, notice that this set was already mentioned in definition 1.2.30. One thing that as difficult to prove directly with matrices but easy to prove now is that $(\mathrm{GL}_n(R), \cdot)$ is associative: every invertible matrix represents a function, and function composition is associative, and hence $\cdot$ is an associative operation.

As mentioned, there can be matrices in $\mathrm{GL}_n(R)$ which are not a product of elementary matrices, that is, $\mathrm{GL}_n(R)$ need not be generated by elementary matrices. However, if $R = k$ is a field, then this is in fact the caes, and the "if" in proposition 13.2.16 becomes and "iff"

Proposition 13.2.19: General Linear Group Of Field

Let $R = k$ and $\mathrm{GL}_n(k)$ be the general linear group over a field. Then $\mathrm{GL}_n(k)$ is generated by elementary matrices.

***Proof*:**

The idea is as follows: take advantage that all entries are units. Make a_{11} 1 and then eliminate all a_{i1} and a_{1j} (i.e. make them zero). Then you have made a $(n-1) \times (n-1)$ matrix ith a 1 in the top-left. Repeat this process until you get the identity:

$$I_n = M \cdot P \cdot N$$

Since M and N are invertible, we wget:

$$(NM)^{-1} = M^{-1}I_nN^{-1} = P$$

registering all the operations will produce you product of elementary matrices, an (NM) is still the prouct of invertible matrices, and so $(NM)^{-1} = P$ is the product of invertible matrices, completing the proof.

The above method can be used more generally to find the inverse of a matrix, and we only need to multiply *on the left hand side* or the *right hand side*. The method is common enough to be given a name: *Gaussian Elimination.* [11]. Since Gaussian elimination usually reffers to operations on the left, we may think of this as a form of "row"-equivalence. However, as we saw we may as easily multiply the elementary matrices only on the right and get "column"-equivalence. This shows us that the two forms of equivalence are the same. Some authors define the notion of *row rank* and *column rank* for matrices which correspond to the dimension of a vector-space spanned by the rows and column respectfully, and proceed to show that the row and column rank are the same number. For reference, we put the followign definition in a box:

Definition 13.2.20: [matrix] Rank

Let A be a matrix. Then the *row rank* is the dimension spanned by its rows, the *column rank* is the dimension spanned by its colun. By the above remark, the row-rank and the column-rank are equal, and so the single numer is called the *rank* of the matrix. If the rank is maximal, the matrix is said to be of *full rank*.

If a matrix is full rank, then the linear map it represents is surjective.

Corollary 13.2.21: Equivalence Of Matrices

Let $R = K$ be a field and $P \in M_{m,n}(R)$ be any m trix. Then P is equivalent to:

$$\left(\begin{array}{c|c} I_k & 0 \\ \hline 0 & 0 \end{array}\right)$$

for appropriate k, $1 \leq k \leq \min(m, n)$, here 0 is the represents that all of the entries are 0 (for the appropriate size).

***Proof*:**
This is just combining the previous results.

Thus, two matrices that have a different value of r in the above matrix will *not* be equivalent. This also shows that there are only *finitely many* equivalence classes matrix representations of linear transformation's between two finite dimensional vector-spaces, in particular there is one for each dimension$1 \leq k \leq n$. This also show's that if we look at a linear transformation in the right way (i.e. pick the right basis), a linear transformation *always* looks like it some projection eliminating certain

[11]Though Gauss has most likely invented the above method when working with matrices, it has been known under slightly different symbols and theory at least as far back as Han China

dimensions while embedding the rest of the space into the codomain! See proposition 13.1.22 to see how this intuition and the existence of compliments of subspaces work together.

To finish off this section, we can ask if the same logic can be done for the notion of similarity. Unfortunately, similarity is more complicated: since only one matrix is being picked (i.e. only one basis is being picked) the rows and column operations must be done "simultaneously". Not all hope is lost: using properties of PID's will show us how to find a good matrix representation for similar matrices in the next chapter.

Gaussian Elimination over Euclidean Domain

From the rings we explored earlier, the euclidean domain is a good start, and indeed with some cleverness a matrix over a euclidean domain can be reduced, though not intirely to an identity matrix in a sub-matrix of the original.

To demonstrate the idea, let R be a euclidean domain with valuation ν, and take any 2×2 matrix:

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix}$$

then use elementary matrices to put the element with the lowest valuation at position $(1, 1)$. Then use the euclidean algorithm to find:

$$b = aq + r$$

where either $r = 0$ or $\nu(r) < \nu(a)$. Then do the appropriate elementary operation to get:

$$\begin{pmatrix} a & r \\ c & d - aq \end{pmatrix}$$

if $r = 0$, then we have eliminated one of the components. If $r \neq 0$, then repeat the same process, moving the term with the lowest valuation to the $(1, 1)$ position using elementary matrices. Since each term has a finite valuation, it must be that eventually $r = 0$. With minor modifications, the same thing can be done in the $(2, 1)$ position, getting us:

$$\begin{pmatrix} a' & 0 \\ 0 & d' \end{pmatrix}$$

Next, notice that we may find an a', d' such that $a' \big| d'$. For if not, we may consider:

$$\begin{pmatrix} a' & d' \\ 0 & d' \end{pmatrix}$$

and repeat the process until we get to the same state, and keep repeating until either $d' = 0$ or $a' \big| d'$. Thus, we produce a 2×2 matrix where the diagonal divides each other.

From this, it is not too hard to see that this is easily generelizable to an $n \times n$ matrix:

Theorem 13.2.22: Smith Normal Form

Let R be a euclidean domain and $P \in M_{m,n}(R)$. Then P is equivalent to a matrix of the form

$$\begin{pmatrix} [ccc|c]d_1 & \cdots & 0 & 0 \\ \vdots & \ddots & 0 & 0 \\ 0 & \cdots & d_k & 0 \\ \hline 0 & \cdots & 0 & 0 \end{pmatrix}$$

where

$$d_1 \big| \cdots \big| d_k$$

The resulting matrix is called the *smith normal form*.

***Proof*:**
exercise

Corollary 13.2.23: Finding Inverse Of Matrix

Let $A \in \mathrm{GL}_n(k)$. Then A^{-1} can be found by performing Gaussian Elimination.

***Proof*:**
This is taking the strategy done in theorem 13.2.22 and keeping track of the elementary matrices used to produce the inverse. Multiplying all of them gives the desired result.

Using smith normal forms, we may prove that $\mathrm{GL}_n(R)$ is generated by elementary matrices. Unfortunately, there does exist a counter if R is a PID; there is exists a 2×2 counter-example over $\mathbb{Q}(\sqrt{19})$ (see cite:HERE, under books/math/papers/ the GL2 paper). On the other hand, we can put any matrix in $\mathrm{GL}_n(R)$ into smith normal form using the gcd and some non-elementary matrix operations. This shall be further explored in chapter 14.

13.2.4 Geometric Result: Systems of Linear Equations

Vector spaces and linear maps have part of their origins in geometry, where the term *linear* originates. The rigorous interpretation of a concept being "geometrical" is something that would be explored in a class on [differential] geometry, and so for the uses of this book geometrical shall mean "locally or globally looks like the solutions to equations":

$$\begin{aligned} f_1(\vec{x}) &= a_1 \\ f_2(\vec{x}) &= a_2 \\ &\vdots \\ f_n(\vec{x}) &= a_n \end{aligned}$$

Often, the constant is moved to the left modifying the equations to be:

$$
\begin{aligned}
f_1(\vec{x}) - a_1 &= 0 \\
f_2(\vec{x}) - a_2 &= 0 \\
&\vdots \\
f_n(\vec{x}) - a_n &= 0
\end{aligned}
$$

This understanding of geometrical is in fact very potent, for example the reader may have already encountered many geometrical objects that are defined in this way, for example:

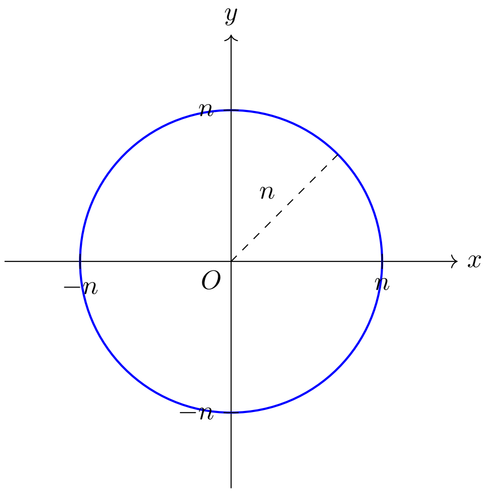


Figure 13.1: solution to $x^2 + y^2 - n^2 = 0$

For the particular explorations of this section, we shall show that vector spaces can be interpreted as the of solutions to *linear equations*:

Definition 13.2.24: Linear Equation and System of Linear Equations

Let V be a k-vector space of dimension n and $k[x_1, x_2, ..., x_n]$ be a polynomial ring over the field k with n variables. Then a *linear equation* is a polynomial, possibly over several variables, where the degree of each variable is at most 1:

$$a_1x_1 + a_2x_2 + \cdots + a_nx_n$$

A collection of linear equations is called a *system of linear equations.*

Given a (system of) linear equation(s), we can define *linear functions*: if $\sum_i a_ix_i$ is a linear equation then we can define:

$$L_1 : k^n \to k \qquad L(x_1, x_2, ..., x_n) = a_1x_1 + a_2x_2 + \cdots a_nx_n$$

and similarly a system of linear functions is a collection of linear functions $L_1, ..., L_m$. Since these functions are linear, they can be represented by a matrix, namely given the standard basis:

$$A = [L_1] = \begin{pmatrix} a_1 & a_2 & \cdots & a_n \end{pmatrix}$$

so that if $\vec{x}$ is an $n \times 1$ matrix with with x_i coefficients, then plugging in a value from k^n is represented as matrix multiplication by

$$A\vec{x} = \begin{pmatrix} a_1 & a_2 & \cdots & a_n \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix} = a_1x_1 + a_2x_2 + \cdots + a_nx_n$$

Equivalently, any system of linear equation can be combined to form a linear equation:

$$L : k^n \to k^m \qquad L = \begin{bmatrix} L_1 \\ L_2 \\ \vdots \\ L_m \end{bmatrix} \tag{13.7}$$

Let's now say we want to find the solution to the system:

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 + \cdots a_{1n}x_n &= b_1 \\ a_{21}x_1 + a_{22}x_2 + \cdots a_{2n}x_n &= b_n \\ &\vdots \\ a_{m1}x_1 + a_{m2}x_2 + \cdots a_{mn}x_n &= b_m \end{aligned}$$

Then if $\vec{b} = (b_{k1})$, then what we are solving for is:

$$A\vec{x} = \vec{b}$$

If A is invertible and $n = m$, then the answer is simple:

$$\vec{x} = A^{-1}\vec{b}$$

where A^{-1} is found using corollary 13.2.23 (how does this relate properties of linear maps that have been explored earlier?). If A is not invertible or $n \neq m$, the next best thing that can be done is find its smith normal form giving us matrices M, N such that

$$M \cdot A \cdot N = \left(\begin{array}{cccc} [ccc|c]d_1 & \cdots & 0 & 0 \\ \vdots & \ddots & 0 & 0 \\ 0 & \cdots & d_k & 0 \\ \hline 0 & \cdots & 0 & 0 \end{array}\right)$$

then if $\vec{c} = M\vec{b}$ and $y = (y_k)$ we get:

$$(M \cdot A \cdot N) \cdot \vec{y} = \vec{c}$$

and producing the system:

$$d_1 y_1 = c_1 \tag{13.8}$$
$$d_2 y_2 = c_2 \tag{13.9}$$
$$\vdots \tag{13.10}$$
$$d_r y_r = c_r \tag{13.11}$$
$$0 = c_{r+1} \tag{13.12}$$
$$\ddots \tag{13.13}$$
$$= c_m \tag{13.14}$$

where $d_i \big| c_i$. Conversely, if there is a set of linear equation of the form given in equation 13.8, then it has a solution of $d_i \big| c_i$.

Where this links to geometry is when considering the space of solution to systems of linear equations. First, notice that without loss of generality the linear equations may equal to 0, for this would simply translate the resulting set of solutions in the vector-space, hence it suffices to consider:

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 + \cdots a_{1n}x_n &= 0 \\ a_{21}x_1 + a_{22}x_2 + \cdots a_{2n}x_n &= 0 \\ &\vdots \\ a_{m1}x_1 + a_{m2}x_2 + \cdots a_{mn}x_n &= 0 \end{aligned}$$

That is, it suffices to consider the kernel of a system of linear equations, equivalently the kernel of L as defined in equation (13.7).

Now, to endeavor exploring the shape of the kernel of linear functions, let's start with $k = \mathbb{R}$, $L : \mathbb{R}^3 \to \mathbb{R}$, and always use the standard basis. It is easy to visualize $\mathbb{R}, \mathbb{R}^2, \mathbb{R}^3$, and higher dimensional analogues can be fathomed. Naturally, if $L(x) = 0$, then $\ker = \mathbb{R}^3$. It is more interesting if L is non-trivial, which for linear functions implies L is surjective. Then by the Rank-Nullity Theorem $\dim \ker = 3 - 1 = 2$, i.e. it is a plane. A plane that is one-dimension less than the ambient space (in this case $\mathbb{R}^3$) is called a *hyper-plane*. Since it is a subspace, it must contain 0 and hence passes through the origin. Since the plane is two dimension, it is spanned by two vectors. Using Gaussian elimination, we can deduce it is spanned by two vectors:

$$\left\{ \begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix}, \begin{pmatrix} b_1 \\ b_2 \\ b_3 \end{pmatrix} \right\}$$

Example 13.2.25: Finding A Hyperplane

Take $L(x, y, z) = 3x + 2y - 3z$. Find a basis for the kernel. The kernel when embedded in $\mathbb{R}^3$ looks like so:

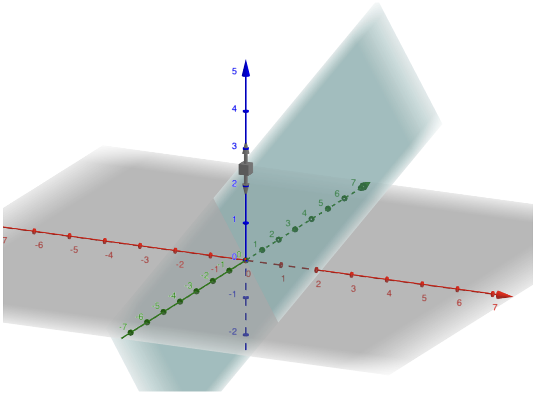


Figure 13.2: $3x + 2y - 3z = 0$

Notice that since the image is one dimensional, the hyper-plane is defined by only one equation, however it is a two dimensional object. More generally, for a surjective linear function $L : \mathbb{R}^n \to \mathbb{R}$, the hyperplane shall be $(n-1)$ dimensional and be defined by a single equation.

Next, let's consider the case where $L : \mathbb{R}^3 \to \mathbb{R}^2$, that is there are two linear equations $L_1 : \mathbb{R}^3 \to \mathbb{R}$ and $L_2 : \mathbb{R}^3 \to \mathbb{R}$. Let's again assume that L is surjective, or full-rank, for if L was one dimension less it would be equivalent to the case of $L : \mathbb{R}^3 \to \mathbb{R}$. Then by the Rank-Nullity theorem we have that $\ker L = 3 - 2 = 1$. Intuitively, this is because there are more restrictions for $L(x) = 0$, namely both L_1 and L_2 must equal 0. Geometrically, this can be thought of as the *intersection* of the kernel of L_1 and the kernel of L_2. Since L is full rank, the intersection is not parallel (the geometric word for this is that the intersection is *transversal*). More generally, $L : \mathbb{R}^n \to \mathbb{R}^2$ would have $\dim \ker L = n - 2$, and would be defined by two linear equations.

Extended the pattern further, given a full-rank linear function $L : \mathbb{R}^n \to \mathbb{R}^m$, the resulting plane given by the kernel would be $n - m$ dimensional and would represent the intersection of m hyper-plane of dimension $n - 1$, each new hyper-plane restricting the dimension of the kernel. The orientation of the kernel is completely dependent on the linear map in question. Nothing in these results required to work specifically with $k = \mathbb{R}$, it was simply convenient for visual purposes. These results can be summarized into the following:

Theorem 13.2.26: Classification Of Planes

Let V be a k-vector-space of dimension n. Then the sub-spaces of V correspond to the kernel of full-rank linear maps $L : V \to k^m$, where $1 \leq m \leq n$.

***Proof*:**
Fill in the blanks from the above discussion

Given that L is surjective, notice that as m increases in size, the dimension of the kernel decreases in size; in a sense it is the "co-dimension" of the domain. This observation is not lost on mathematicians, and the notion of codimension is quite important in geometry: the dimension of an object represents the degree of freedom an object may have within a space (ex. a hyper-plane is $n - 2$ dimensional) while the codimension represents the degree of restriction an object may have within a space (ex. a hyper-plane is 1 codimensional).

(Also, a geometric interpretation of first is iso theorem: when solving system of Lin. Eq, look for kernel. But notice the solution to $Ax+c=b$ is just an affine translation of the kernel! All elementary row operations fix the kernel and permute the cosets, when the elementary operations generate $\mathrm{GL}_n(R)$.)

Example 13.2.27: Exercices
Something with transpose. Perhaps even with conjugate? Get started with $\langle A, A*\rangle$ stuff?

Exercise 13.2.2

1. (from dummit and foote) (note that chagne of absis for a lienar transformation from V ot itself is the same as conjugation by some element of the group $GL(V)$ of nonsingualr linear transformation of V to V. In particular , the relation "similarity" is an equivalence relation whose equivalence classes are the orbits of $GL(V)$ acting by conjugation on $\mathrm{Hom}_F(V,V)$.)

13.2.5 Use Case: Reading Presentations

Recall from section 12.3.3 that a presentation of a [finitely generated] R-module can be represented as an exact sequence:

$$R^n \to R^M \to M \to 0$$

Using the above results, we can more easily read the presentation. Let $\varphi : R^n \to R^m$. Then by exactness, we have that $M \cong \mathrm{coker}(\varphi)$. Hence, if we find the image of φ, we found the relations of M, and and if R is a field of euclidean domain, this is task is simplified using Gaussian elimination:

Example 13.2.28: Finding A Presentation
Let $R=\mathbb{Z}$ and let φ be represented by the matrix:

$$\begin{pmatrix} 1 & 3 \\ 2 & 3 \\ 5 & 9 \end{pmatrix}$$

which means that $\varphi : \mathbb{Z}^3 \to \mathbb{Z}^2$ and this matrix determines an abelian group G (since abelian groups have free resolution of length at most 2 as mentioned in section 12.3.3). To find G, let us first simplify the above matrix. The reader should check that the matrix reduces to:

$$\begin{pmatrix} 1 & 0 \\ 0 & 3 \\ 0 & 0 \end{pmatrix}$$

Hence, G is isomorphic to the cokenrel of $\varphi : \mathbb{Z}^2 \to \mathbb{Z}^3$ that maps $(1,0)$ to $(1,0,0)$ and $(0,1)$ to $(0,3,0)$. In particular, since the image is $\mathbb{Z} \oplus 3\mathbb{Z} \oplus 0$, we get

$$G \cong \mathrm{coker}\varphi \cong \frac{\mathbb{Z} \oplus \mathbb{Z} \oplus \mathbb{Z}}{\mathbb{Z} \oplus 3\mathbb{Z} \oplus 0} \cong (\mathbb{Z}/3\mathbb{Z}) \oplus \mathbb{Z}$$

Showing how a matrix can uniquely represent an abelian group! In the next chapter, we shall show how a matrix can more generally represent a finitely generated PID.

13.2.29 Remark Notice that we may eliminate first row and column if the leading term is 1 (or more generally a unit) since in the quotient the summand shall be completely quotiented.

13.3 Dual Spaces and Forms

Let's say you are studying some mathematical object, say the set of linear transformations. We may want to find numerical properties. For example of linear transformations, we can quantify the size of a matrix, or give some numerical value to similarity, or the dimension of the image. We can define a function from $L(V)$ to a field, say $\mathbb{R}$, that shall give us this quantification. The collection of all these functions are called the *functionals* on $L(V)$. Depending on what type of property we want want the invariances that are being sought after, we will have different types of functionals. For example, when working in analysis, we shall often want our functional to be continuous or perhaps bounded in some way [12]. In Algebra, we shall want our functionals to be homomorphisms, and depending on what other property we want we shall add more properties to our functionals (ex. it shall be important in section 31.1 that the functional preserves solutions to a special set of polynomial equations).

The set of functionals on a vector space V can be thought of as containing all possible quantifiable "information" about a space (with the limits of the information provided by the properties of the functionals). Most of the time, the space of functionals is larger than the original space[13]. In some special cases, the spaces of all functionals is isomorphic to the space over which we are taking the functionals (as we shall see, all linear functionals on a finite dimensional vector space V is isomorphic to V). In this special case, we can think of our space being characterized by the property the functionals represent! This realization was the origin of *dual theory*, which we shall start exploring in this section. Dual theory is normally explored in earnest in Real and Functional analysis, however it shall show up in some important ways in section 32.1, and certain key functionals shall give a lot of information on linear transformations, namely the trace and the determinant.

13.3.1 Dual Spaces

In mathematics, there is frequently two "opposing" or "dual" ways of describing objects: Let's say X is our object of study

1. internally: Using properties internal to your object to describe that object. Often, maps *to* the object will give us internal information ($f : A \to X$ for some appropriate object A). This can may also be thought of as local information
2. externally: Using some exterior property (ex. be it by putting it an ambient space, or a sapce in which the object X "resonates" or "interacts" with[14]). Ofen, maps *from* the object will give us external information ($f : X \to A$ for osme appropriate object A). This can also often be thought as global information

As an example of this, when we were working with group theory, we found two way of looking at all possible groups:

[12] See Lebesgue Spaces in [42] for more details

[13] This is a guiding principle behind many generalizations in Mathematics, Distribution Theory being the prototypical example

[14] If you know Fourier Analysis, think of phase space

1. Using quotients of free groups, we managed to make normal subgroups of free groups represent relations between elements. In the process of doing this, we defined a surjective homomorphism:

$$\varphi : F(A) \to G$$

2. Using Cayley's theorem, we manage to embed any group into a subgroup of S_A. In the process of doing this, we defined an invective map

$$\psi : G \to S_A$$

Notice how the first one defined us what's happening between elements in a sort of "local" way, while in the 2nd example we defined it by using properties of the larger group S_A and how elements of G embed into S_A. Another example of this principle can be seen with the universal property of products and coproducts of groups. In a sense, the universal property of products only requires the product group to have groups A and B "locally" in the most efficient way, leading to the direct product of A and B, while the corpoduct group require it to have A and B "globally" in the most efficient way, leading to the free product of A and B.

Often when studying an object X, we will study if from these two perspectives: from maps going *into* X and maps going *out of* X. One remarkable phenomena that occurs often in mathematics is that there is often a [contravariant] *dual object* in which the maps *into* and *out of* X are naturally associated to maps *out of* and *into* a dual object X^*. Note the reversal in the directions of the maps! It is often very fruitful to look at objects from the perspective of both X and X^* to gain more information.

To demonstrate a simple but powerful example of "dualness" in mathematics (or the pervasiveness of dualness), we require knowledge of fields outisde of algebra. Usually, I (the author of this book) would decide to leave out any such example to insure no prerequisites are required to learn the material. However, I have decided to break this rule here for those who know a bit of topology since it is an example so powerful that it changed my view of duality, and requires elementary topology knowledge. The following example is *not* necessary for understanding any material of this book and can be considered motivation for dual spaces and dual theory in general for those who know some topology (if you need a refresher on topology, see appendix C). In this example, we will take a moment and venture outisde of algebra and into the realm of analysis and and geometry to give an interesting example.We will give a common phenomena in which "algebraic" object have "geometric" objects as their dual. In particular, unlike the duality of a group in its represetnation in terms of the quotient of a free group or an embedding into S_G, both of which are algebraic ways of looking at groups, we will show that dual objects are sometimes found in other fields of mathematics, allowing us to connect these two fields of mathematics proof-strategies and techniques!

First, define a *Boolean Algebra* to be a $\mathbb{F}_2$-algebra (where $\mathbb{F}_2 = \mathbb{Z}/2\mathbb{Z}$). Since $\mathbb{F}_2$ is a field, a Boolean Algebra B is module-isomorphic to $\mathbb{F}_2^n$ for some cardinality n. For $0, 1$ in a Boolean algebra B:

$$0+0=0 \quad 0+1=1+0=0 \quad 1+1=0 \qquad 0\cdot 0=0\cdot 1=1\cdot 0=0 \quad 1\cdot 1=0$$

We usually also define the additional operation of $\neg$:

$$\neg(x) = 1 - x$$

If you have done some computer science or logic, then you might notice that 1 and 0 can represent the values of *true* and *false* respectively, while $+$ and $\cdot$ represent XOR and AND respectively (hence the name Boolean to this algebra). This algebra underlines most of the logic behind computer science, making us want to find properties of boolean algebras. One way we may study these properties is by looking for maps going *into* a boolean algebra B. Dually, by a result known as *Stone Duality*, we can instead study all the maps *out of* a totally disconnected[15] Hausdorff spaces B^*! Thus, the "dual" of Boolean Algebra is all continuous maps from B^* to $\mathbb{R}$! More generally, locally compact[16] topological spaces (which we can consider as a geometric object) are the dual of a type of algebra known as a C^*-algebras[17] (which we consider as algebraic objects): this is known as *Gelfand Duality*[18]. The study of these objects that are related to each other by "duality" is called *Dual Theory*.

Coming back to algebra and the study of vector-spaces, we will see the very beginnings of Dual Theory by studying the simplest case of dual objects, *dual vector-spaces*. The way we will define the dual of a vector-space will be through "external" means. When we defined vector spaces using a basis, that is a "local" property. In a sense, we showed that the map $f : F(A) \to V$ can define a vector space where A is a basis. What we'll now do is the *dual* equivalent: we'll define a vector space by mapping V into a different space.

The question arrises as to what space might we might we map to? When working with groups, it was not immediately obvious that S_A would be the right group to map. Fortunately for us, there is an "obvious" space to map to: for any vector-sapce over k, we are always given two algebraic objects, namely the abelian group V and the field k. Since k is always a vector-sapce over itself,we can consider linear maps $f : V \to k$! The collection of all such maps is called the *dual space* of V.

Definition 13.3.1: Dual Space

Let V be a vector space over k. Then $V^* = \mathrm{Hom}_k(V, k)$ is the space of all linear transformations from V to f, and is called the *dual space* of V.

By proposition 12.2.7, $\mathrm{Hom}_k(V, k)$ is a vector-space over k (if you're not convinced, it is recommended to reprove this before continuing). The notation $V^\vee$ is also often used to define the dual space and will be used interchangibly. Elements inside the dual space V^* are sometimes called *covectors*, paralleling to how elements inside a vector space V is called *vectors*. In physics and analysis, there is seomtimes a different convention for how to write the evalutation of if $f \in V^*$ at $x \in V$: instead of writing $f(x)$, to mean "evaluate f at the point x", it is sometimes written $\langle f, x\rangle$.

Elements of V^* are given a name:

Definition 13.3.2: Linear Functionals

Let V^* be the dual space of V. Then $f \in V^*$ is called a *linear functional* or *linear form*.

[15]A topological space in which the only connected sets are singletons
[16]a topological space in which for any point x, there exists a compact neighbourhood
[17]A Banach Algebra over $\mathbb{C}$ with a map known as an involution satisfying the properties of an adjoins
[18]This duality requires a good deal more topology and analysis than is part of the scope of this book. However, in section 31.3, we will see the Zariski Duality which studies a geometric-algebraic duality without the need for much analysis or topology

Example 13.3.3: Linear Functional and Dual Spaces

1. (If you know some calculus) Perhaps the most famous linear functional is $\int_a^b \_\_\_ dx : L^1(\mathbb{R}) \to \mathbb{R}$, where $L^1(\mathbb{R})$ will denote the set of all (either Riemann or Lebesgue)[a] integrable functions $f : \mathbb{R} \to \mathbb{R}$ where $\int_{\mathbb{R}} |f| dx < \infty$ (so that the integral is well-defined). Note that $L^1(\mathbb{R})$ is a vector-space with addition and scalar multiplication defined pointwise since the codomain is $\mathbb{R}$:

$$f + g \Leftrightarrow (f+g)(a) = f(a) + (g)$$
$$c \cdot f \Leftrightarrow (c \cdot f)(a) = cf$$

   Now, $\int_a^b \_\_\_ dx$ takes in a function $f : \mathbb{R} \to \mathbb{R}$, an gives th number $\int_a^b f dx \in \mathbb{R}$. By basic properties of integarls, we know that:

$$\int_a^b (f+g) dx = \int_a^b f dx + \int_a^b g dx$$
$$\int_a^b cf dx = c \int_a^b f dx \qquad \forall a \in \mathbb{R}$$

   and hence $\int_a^b \_\_\_ dx$ is indeed a linear functional! Hence $\int_a^b \_\_\_ dx \in L^1(\mathbb{R})^*$, for any $a, b \in \mathbb{R}$[b]

   The study of Analysis focuses a lot on the study of the properties of these functionals, and would be covered in many Analysis courses. For textbooks, see ref:HERE.

2. Let $C([a,b])$ be the collection of all continuous maps from $[a,b]$ to $\mathbb{R}$, $f : [a,b] \to \mathbb{R}$, with addition and scalar multiplication defined pointwise. Then define: $\text{ev}_x : C([a,b]) \to \mathbb{R}$ to be

$$f \mapsto f(x)$$

   Then it can be shown that ev_x is a linear transformation, called the *evaluation map*, and is thus a linear functional. More generally, as long as the codomain is a field $\mathbb{R}$, then the evaluation map is a linear functional.

3. A more advanced case of the above example occurs if we consider the set $C^1(\mathbb{R})$ of all differentiable functions[c] forms a vector-space with point-wise addition and scalar multiplicaiton. The operator $D_a : C^1(\mathbb{R}) \to \mathbb{R}$ takes in a differentiable function f and gives back $Df(a)$ (which in first year calculus is more commonly denoted $f'(a)$).

4. (If you know geometry or analysis) The following examples are not examples of dual spaces as we've defined them, but the generlization of them. In this generlization, the dual space we've defined would be called a *linear* dual space, or the dual space of in the category of $\mathbf{Vect}_k$. More generally, if X is an object in a category, then given a *dualizing* object D (often it is $\mathbb{R}$, $\mathbb{C}$, $\mathbb{C}^\times$, $\mathbb{Z}/n\mathbb{Z}$, or $\mathbb{R}/\mathbb{Z}$ depending on the context), then $\text{Hom}_C(X, D)$ is the dual space. In the following example, let D be either $\mathbb{R}$ or $\mathbb{C}$. Let's say X is either a vector-space over $\mathbb{R}$ or $\mathbb{C}$ (you should have function spaces in mind) or a manifold over $\mathbb{R}$ or $\mathbb{C}$. Then $C_0(X)$, $C^k(X)$, $L^p(X)$, an so forth are all subspace of the dual space X^* in the appropriate category.

Another place in which [ring] funcitonals shall become of paramount importance is in section 31.1 and section 32.5, where a duality is set up between algebraic and geometric objects.

[a]For the sake of this example, it doesn't matter, though L^1 usually denote *Lebesuge* measurable functions where $\int |f| < \infty$

[b](if you know some analysis) one classical result known as Riesz Representation Theorem (for measures) is that all functionals $f \in L^1(\mathbb{R})^*$ will be of the form $f \mapsto \int fg$ where $g \in L^\infty(\mathbb{R})$ and the essential support of g is bounded.

[c]in particular, continuously differentiable functions. This distinction is of no importance for this example

In the above examples, all the vector-spaces are infinite dimensional. The vector-space being infinite dimensional will in fact greatly complicate the structure of dual spaces[19]. In general, more structure is added to a vector-space (ex. a norm) to have better "control" of infinite dimensional vector-spaces. Such discussion is left to a course on functional analysis[20]. We will mainly focus on the theory of finite dimensional vector spaces, since the results of this theory is in it of itself quite fruitful and provides a good basis for its study of dual spaces more generally.

Given a basis for V, it straight-forward to find a basis for $V^\vee$:

Definition 13.3.4: Dual Basis

Let $\beta = \{v_1, v_2, ..., v_n\}$ be a basis for the finite dimensional vector-space V. Then define $v_i^* \in V^*$ to be

$$v_i^*(v_j) = \begin{cases} 1 & \text{if } i = j \\ 0 & \text{if } i \neq j \end{cases}$$

for $1 \leq i, j \leq n$. Let β^* be the set of dual-basis elements

Since β^* has the word basis in it's name, you might guess that it is ineed a basis for the dual space:

Proposition 13.3.5: Dual Basis

Let $\beta*$ be defined as in definition 13.3.4. Then β^* is a basis for V^*

***Proof*:**
exercise

As a consequence, V^* has the same dimension as V

Proposition 13.3.6: Reflectivity Of Finite Dimensional Vector Spaces

Let V be a finite dimensional vector space. Then $V \cong V^*$.

***Proof*:**
By proposition 13.3.5, since V and V^* are of the same dimension, $V \cong V^*$.

An intuitive way of viewing the reflectively product to reference the intuition presented in the introduction of this section mentioning how functional represent *information*. Each linear function

[19]This is one of the analytic properties that was discussed in the footnote in page 427

[20]Give references to books HERE, ex. Folland

provides some information about a vector-space (prototypical examples would be the projection functions $\pi_{\beta,i} : V \to k$ picking a component of a vector given a choice of basis β). The type of information provided is restricted by the properties of the functions on V, in this case the co-domain is always k, and the functional is always linear. In finite dimension, these functions can be completely characterized by elements in V by the existence of the above isomorphism. The choice of the isomorphism is still an important factor to consider, since there is not natural basis or extra information that allows for favoring one isomorphism over another.

Since the structure of V isomorphic to V^*, it would be interesting if each map $f : V \to W$ can be associated to an appropriate "dual" function f^* where composition is respected, hopefully giving the existence of the functor $(-)^*$ from Vect_k to itself. To start, there is a notion of a dual function:

Definition 13.3.7: Dual map (Transpose)

Let $f : V \to W$ be a linear transformation. Then the *transpose* of f, denoted f^*, is defined as:

$$f^* : W^* \to V^* \qquad f^*(\varphi) = \varphi \circ f$$

Unfortunately, $(-)^*$ can't so easily be defined as each choice of isomorphism $V \cong V^*$ for each vector-space $V \in \mathbf{Vect}_k$ may not be compatible, see exercise ref:HERE. Note that the transpose of a function insures that maps that used to "out of" the space V will now "go into" the space V^* (and similarly for maps that go into V). Thus, this definition matches with our goal for V^* to be the dual of V, and furthermore is independent of any choice of basis. The name transpose comes from the matrix representation of f^*

Proposition 13.3.8: Matrix Representation Of Dual Map

Let $f : V \to W$ be a linear transformation and $A = (a_{ij})$ be the matrix representation of f given some basis β and γ for V and W respectively. Then if $f^* : W^* \to V^*$ is the dual map of f, where W^* and V^* have as basis β^* and γ^*, then the change of basis matrix A^* is:

$$A^* = [f]_\beta^\gamma = [f^*]_{\gamma^*}^{\beta^*} = (a_{ji}) = A^t$$

Intuitively, in the matrix representation, each entry a_{ij} transposed with entry a_{ji}. For this reason, the map f^* is sometimes called the *transpose* of f.

***Proof*:**
exercise

As a corollary, if A is a nonsingular map $f : V \to W$, then the matrix representation for the map $(f^{-1})^* : V^* \to W^*$ is $(A^{-1})^t$.

Similarly to how the kernel of a linear transformation f provides the needed information to deduce the structure of the image of f, we get the dual equivalent for f^*, given an appropriate "dual" for the kernel:

Definition 13.3.9: Annihilator Of Dual Map

$U \subseteq V$ be a subspace of V. Then the *annihilator* of U is:

$$U^o := \{f \in V^* \mid f(U) = 0\}$$

Note that $V^0 = \{0\}$ and $\{0\}^0 = V$. Another common notation is $U^\perp$, which is pronounced "U perp", which shows how U^0 is related to the orthogonal complement of U. That relation needs an inner product, and is carried out in *Linear Algebra* [17].

Lemma 13.3.10: Properties Of Annihilator

Let $U \subseteq V$ be a subspace. Then:

1. $U^0 \subseteq V^*$ is a subspace
2. if $\dim(V) < \infty$, then $\dim(V) = \dim(U) + \dim(U^0)$

***Proof*:**
exercise

Proposition 13.3.11: Properties Of Dual Maps

Let V and W be (not necessarily finite dimensional) vector space over k, $f \in \mathrm{Hom}_k(V, W)$ so that $f^* \in \mathrm{Hom}_k(W^*, V^*)$ is its dual map. Then:

1. $\ker(f^*) = (\mathrm{im}(f))^0$ (implying f^* injective if and only if f surjective)
2. $\mathrm{im}(f^*) \subseteq (\ker(f))^0$ is a subspace
3. If $\dim(V), \dim(W) < \infty$, then $\mathrm{im}(f^*) = (\ker(f))^0$ (implying f^* surjective if and only if f injective)

***Proof*:**
(prove these instead of exercise? The adjoint generalization is in *Linear Algebra* [17].)

The first point in proposition 13.3.11 is particularly useful in that in infinite dimensions, f being injective *does not* imply f is surjective. In the right situation, checking the injectivity of a linear transformation is *much* easier than checking surjectivity, and so checking the injectivity of f^* is often advantageous.

The next topic we will cover might at first seem abstract, however it is an important demonstration of the concept of *naturality*. Since we have taken $\mathrm{Hom}_k(V, k)$ and have shown it is a vector-space, we can take the dual of it again:

Definition 13.3.12: Double Dual

Let V be a vector-space and V^* it's dual. Then the *double-dual* of V is the dual of V^*, denoted V^{**}

The double dual has the property of being *naturally isomorphic* to V. To explain the idea of a natural isomorphism, we will show how V is *not* naturally isomorphic to V^*:

Example 13.3.13: Double Dual Motivation

Let V be a finite-dimensional vector space over k and $V^* = \text{Hom}(V, \mathbb{F})$. If $\beta = \{e_1, ..., e_n\}$ is a basis for V, and $\beta^* = \{e_1^*, ..., e_n^*\}$ is the dual basis for V^* where:

$$e_j^*(e_i) = \begin{cases} 1 & i = j \\ 0 & i \neq j \end{cases}$$

Then the maps $e_i \mapsto e_i^*$ are bijective and linear, meaning V and V^* are isomorphic ($V \cong V^*$).

We can further consider $V^{**} = \text{Hom}(\text{Hom}(V, \mathbb{F}), \mathbb{F})$ being the set of all linear maps mapping all elements form $\text{Hom}(V, \mathbb{F})$ to $\mathbb{F}$. We can in similar fashion make an isomorphism $V^* \cong V^{**}$ letting us conclude that $V \cong V^{**}$.

This whole construction started with a choice of basis form V, which need not be unique, and so it's not a "property" of V. However, we can still define an isomorphism between V and V^{**} *without* any reference to a basis. For any $v \in V$, we can define a map

$$(f \mapsto f(v)) : V^* \to \mathbb{F} \qquad (f \mapsto f(v))(g) = g(v)$$

that is, every functional $g \in V^*$ evaluates the value and gets us $g(v)$. This map is sometimes called the evaluation function. Let's switch up the notation for simplicity:

$$ev_v : V^* \to \mathbb{F} \qquad ev_v(g) = g(v)$$

where ev_v stands for evaluate at v. Now, defining map $V \to \text{Hom}(V^*, \mathbb{F})$, $v \mapsto ev_v$ is an *isomorphism*. Bijectivity is left to check, for linearity, let φ be the function in question, so that for $k \in \mathbb{F}$ and $v, w \in V$:

$$ev(k \cdot v) = ev(kv) = ev_{kv} = k \cdot ev_v = kev(v)$$

$$ev(v + w) = ev_{v+w} = ev_v + ev_w = ev(v) + ev(w)$$

where kv represents the vector we get when applying the ring action on v. Note that we essentially used the lineary of the object we are mapping to in order to prove linearity.

Thus, we defined this isomorphism with *no* reference to a basis on V. We can't do the same trick for $V \to V^*$, since if we try to do $v \mapsto f(v)$, it is ambiguous which f we are talking about.

The intuition behind the double-dual having a natural isomorphism is that there is more information to work with; the reader should ponder on what extra information has been provided.

13.3.2 Prelude to Adjoints of Vector Space

must also foreshadow Yoneda's lemma here and also foreshadow, if possible (potentially in a titledbox) Z-valued Points.

Now that we have the basic structure of (finite dimensional) dual spaces, we may return to the discussion of defining V internally or externally. Let's say that you have been given $\text{Hom}_k(V, k)$, but not the structure of V. Is the information in $\text{Hom}_k(V, k)$ sufficient to recover the structure of V?

The answer is almost: the following proposition shows how much information we can recover:

Proposition 13.3.14: Recovering V From it's Dual

Let V be some vector-space and V^* it's dual. Then if $f(v) = f(w)$ for all $f \in V^*$, then

$$v = w$$

***Proof*:**
Exercise. This can be proven with and without the use of a basis for V.

13.3.3 Functor of Points for Vector Spaces

Another way to bring in the intuition from the introduction is to take the dual of a functional $V \to k$, namely to think of elements of V as functions $k \to V$. The reader should check that all linear transformations $k \to V$ are of the form:

$$t \to tx$$

for some $x \in V$. Represent these transformations as $g_x : k \to V$. Taking the dual of g_x we get:

$$\begin{aligned} g_x^*(\varphi) &= \varphi \circ g_x \\ &\equiv \varphi(g_x(t)) && \forall t \in k \\ &= \varphi(tx) && \forall t \in k \\ &= t\varphi(x) && \forall t \in k \\ &\equiv g_{\varphi(x)} \end{aligned}$$

Hence g_x^* maps φ to $g_{\varphi(x)} : k \to k$ where $t \mapsto t\varphi(x)$. With the identification between elements of V and the linear transformations $k \to V$, we may interpret this as a functional $V^* \to k$ where:

$$\varphi \mapsto \varphi(x)$$

Given $V \cong V^*$, then if β is the basis defining the isomorphism, each $\varphi \in V^*$ is associated to a vector $v \in V$, namely the vector for which:

$$\begin{pmatrix} \beta_1 \\ \beta_2 \\ \vdots \end{pmatrix} \mapsto \sum_i v_i \beta_i$$

where $\sum_i$ is a finite sum. Then the linear functional $V \to k$ is defined by:

$$\begin{pmatrix} v_1 \\ v_2 \\ \vdots \end{pmatrix} \mapsto \sum_i v_i x_i$$

where again $\sum_i$ is a finite sum since only finitely many v_i (and x_i) are nonzero Let's label this functional f_x. The element $x \in V$ may be interpreted, functionally, as $t \to tx$ or $v \to \sum_i v_i x_i$. Geometrically, the first function can loosely be thought of as defining a magnitude and direction. This is *not* a correct interpretation in infinite dimensions in general, since a magnitude requires a norm and a bare vector space carries none, however it is valid in finite dimension[21]. The second function can is the "dual" notion, where instead of focusing on x, the focus is on how x manipulates every other element. The reader may want to ponder how this relates to the "internal" and "external" way of describing object mentioned at the beginning of section 13.3. This also shows that V always *injects* into V^*.

As mentioned, it is not always the case that $V \cong V^*$. This immediately falls apart if V is infinite dimensional. To demonstrate this, let us consider the duality presented above where each $x \in V$ is associated with the functional f_x. The key is that given any basis β for V, $[x]_\beta$ has only finitely many components. Hence, the following valid function $F \in V^*$ is not associated to any f_x:

$$[v]_\beta \to \sum_i v_i$$

where the sum is finite since v when represented as an infinite tuple must only have finitely many nonzero components. The functional F is associated to the sequence $(1, 1, ..., 1, ...)$, namely the sequence where 1 is in each position. No f_x can be associated to F since by construction f_x, when represented as an infinite tuple, must have all but finitely position zero. This immediately shows that V^* is much larger, and hence contains much more information than the elements of V can represent. In fact, V^* contains even more interesting functionals. Consider:

$$(v_1, v_2, ..., v_n, ...) \mapsto (0, v_1, ..., v_{n-1}, ...)$$

Notationally, this may be represented as $f(\beta_i) = \beta_{i+1}$ given a basis β for V. This is called he *right shift operator*. This linear transformation can only be defined since there are infinite dimensions, hence each component may shift over by 1.

There are in fact so many linear functionals on infinite dimensional vector spaces that the study of such spaces is almost always accompanied with additional structure to bring some more regularity to the elements. The core problem stems from the extra flexibility working with infinitely many dimensions entails. Hence, some form of "finiteness" is usually imposed via the introduction of a *norm*, or a particular powerful norm called an *inner product*. In the case where there is an inner product on a vector space V, the isomorphism $V \cong V^*$ is recovered where the isomorphism is an inner-product space isomorphism. The inner product gives even more information. As shown in exercise ref:HERE, $(-)^*$ is not a functor since there is no choice of basis that shall naturally allow for composition of linear transformations. In the case of there being an inner product, this issue will be resolved, namely for each matrix representation $[L^*] = [L]^t$, there shall exist a linear transformation L^t for which $L^t : W \to V$ is a linear transformation that shall make $(-)^*$ (resp. $(-)^t$) a functor! This new linear transformation L^t shall be called the *adjoint* of L. The exploration of these notions is very fruitful even in finite dimensions, and it is carried out in *Linear Algebra* [17].

Exercise 13.3.1

1. Some things with infinite dimensions
2. Let $U \subseteq V$ be a subspace. Show that $U^{00} \cong U$. If $\widehat{(\_\_\_)}$ is the natural embedding of V into V^{**}, then $\hat{U} = U^{00}$
3. Let U_1, U_2, V be finitely dimensional vector spaces. Show that:
    1. $(U_1 + U_2)^0 = U_1^0 \cap U_2^0$

[21]This interpretation holds true in Hilbert Spaces

2. $(U_1 \cap U_2)^0 = U_1^0 + U_2^0$
3. $U^{00} = U$
4. If $W \subseteq V$, then $(V/W)^* = W^*$

4. Consider the category whose only object is a vector-space V and morphisms are all linear transformations from V to itself. Choosing a basis for V, show that $(-)^*$ is a contravariant-functor.

13.3.4 Multilinear Forms

Studying linear functionals allow us to give some scalar value to vectors. However, this is sometimes not enough: it is very common in mathematics that one of the two scenarios occur (we will see examples of these in this and coming section[22])

1. We want to study many functional on spaces simultaneously (ex. both V and V^*)
2. We require many copies of the same space to have enough "data" to define the desired functional

In this section, we elaborate on the theory allowing us to do just that, give some examples elaborating the above intuitions, and in the coming section we will see how these are of great use in defining some invariants for linear transformations and bringing in geometry into vector-spaces:

Definition 13.3.15: Bilinear and Multilinear Maps

Let V and W be vector spaces over k. Then a map $\varphi : V \times W \to k$ is called *bilinear* if

1. for any $v \in V$, the map $\varphi(v, \cdot) : W \to k$ mapping $w \mapsto \varphi(v, w)$ is linear
2. for any $w \in W$, the map $\varphi(\cdot, w) : V \to k$ mapping $v \mapsto \varphi(v, w)$ is linear.

More generally, if $V_1, V_2, ..., V_n$ are vector-space sover k, then $\varphi : V_1 \times V_2 \times \cdots \times V_n \to k$ is *multilinear* (or n-linear if the number of variables is known) if for all i, $1 \leq i \leq n$, given fixed $v_1, v_2, ..., v_{i-1}, v_{i+1}, ..., v_n$, the map

$$\varphi(v_1, v_2, ..., v_{i-1}, \cdot, v_{i+1} ..., v_n)$$

is linear. Two important types of multilinear maps are:

- (Symmetric) for any $v_i, v_j \in V$:

$$\varphi(v_1, ..., v_i, ..., v_j, ..., v_n) = \varphi(v_1, ..., v_j, ..., v_i, ..., v_n)$$

- (Alternating) for any $v_i, v_j \in V$:

$$\varphi(v_1, ..., v_i, ..., v_j, ..., v_n) = -\varphi(v_1, ..., v_j, ..., v_i, ..., v_n)$$

[22]There are also many examples in differential geometry, such as covariant/contravariant/mixed tensor fields on a manifold

13.3.16 WARNING Since we are now dealing with multilinear maps, these are *no longer morphisms in the category of* Vect_k! Hence, multilinear maps are a different object of study. In chapter 15, we will show how we can study multilinear maps via linear algebra.

Example 13.3.17: Bilinear and Multilinear Maps

1. Let $\varphi : V \times V^* \to k$ be defined by $(v, f) \mapsto f(v)$. Then φ is bilinear
2. Let $V = \mathbb{R}^n$ be a vector-space over $\mathbb{R}$ and take $\varphi : \mathbb{R}^n \times \mathbb{R}^n \to \mathbb{R}$ to be defined as:
$$\varphi\left(\begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix}, \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{pmatrix}\right) = \sum_{i=1}^{n} x_i y_i$$
Then φ is bilinear, and is called the *dot product* or *scalar product* of two vectors $x, y \in \mathbb{R}^n$
3. Let $\mathbb{R}$ be a vector-space over itself, and take $V : \mathbb{R} \times \mathbb{R} \times \cdots \times \mathbb{R} \to \mathbb{R}$. Then the map that multiplies all the elements:
$$V(x_1, x_2, ..., x_n) = x_1 x_2 \cdots x_n$$
is a n-linear. As a paticualr example, consider $\varphi : \mathbb{R} \times \mathbb{R} \times \mathbb{R} \to \mathbb{R}$, and imagine any 3-tulpe (x_1, x_2, x_3) represents the length, width, and height of a shape. Then $V(x_1, x_2, x_3) = x_1 x_2 x_3$ will give the volume of a rectangular prism (or a cube if $x_1 = x_2 = x_3$). The fact that multiplication produces a natural multi-linear map means that multilienar maps show up a lot in geometry. it is for this reason that multilinear maps are sometimes called *volume forms*. We will see more of these in section 16.3.
4. (If you know some calculus) Let $V = C([0, 1], \mathbb{R})$ be the set of all continuous functions from $[0, 1]$ to $\mathbb{R}$. Then define $\varphi : V \times V \to \mathbb{R}$ to be:
$$\varphi(f, g) = \int_0^1 f(t)g(t)\, dt$$
Then φ is bilinear. This is the infinite dimensional equivalent of the dot product.

In the rest of this section, we will focus on Bilinear forms: the results can easily be generlized using induction to Mutilinear forms. The first thing worth trying to find is wether there is still a reasonable notion of matrix representaiton of bilinear forms, since they are no longer linear maps. The following definition and lemma give us jut that:

Definition 13.3.18: Bilinear Form Matrix Representation

Let V and W be finite dimensional vector spaces over k of dimension n and m and with basis β, γ respectively. Let $\varphi : V \times W \to k$ a bilinear form. Then define the *matrix representation* of φ to be:

$$[\varphi]_{\beta,\gamma} := (\varphi(\beta_i, \gamma_j))_{\substack{1\leq i\leq n\\ 1\leq j\leq m}}$$

Lemma 13.3.19: Bilinear Form Matrix Representation

Let V and W be finite dimensional vector spaces over k of dimension n and m and with basis β, γ respectively. Let $\varphi : V \times W \to k$ a bilinear form. Then

$$\varphi(v, w) = [v]_\beta^t [\varphi]_{\beta,\gamma}[w]_\gamma$$

***Proof*:**
Write $v = \sum_{i=1}^n a_i\beta_i$ and $w = \sum_{j=1}^m b_j\gamma_j$, so that $[v]_\beta$ has entries a_i and $[w]_\gamma$ has entries b_j. Bilinearity of φ expands each argument separately:

$$\varphi(v, w) = \varphi\Big(\sum_i a_i\beta_i,\ \sum_j b_j\gamma_j\Big) = \sum_{i=1}^n\sum_{j=1}^m a_ib_j\,\varphi(\beta_i, \gamma_j)$$

On the other side, the (i, j) entry of $[\varphi]_{\beta,\gamma}$ is $\varphi(\beta_i, \gamma_j)$, so the matrix product is $[v]_\beta^t[\varphi]_{\beta,\gamma}[w]_\gamma = \sum_i\sum_j a_i\,\varphi(\beta_i, \gamma_j)\,b_j$. The two sums agree term by term.

Definition 13.3.20: Orthogonal

Let $\varphi : V \times V \to k$ be a bilinear form and $S \subseteq V$ a *subset*. Then the set:

$$S^\perp := \{y \in V \mid \varphi(x, y) = 0 \text{ for every } x \in S\}$$

is called the *orthogonal complement* of S. If $\varphi(x, y) = 0$, then x is *perpendicular* to y, written $x \perp y$. If $x \perp y$ for all $y \in S$, then $x \perp S$.

The orthogonal complement is always a subspace, whatever the subset S was.

Lemma 13.3.21: The Orthogonal Complement Is a Subspace

Let $\varphi\colon V \times V \to k$ be a bilinear form and $S \subseteq V$ any subset. Then $S^\perp$ is a subspace of V, and $S^\perp = (\operatorname{span} S)^\perp$.

***Proof*:**
Linearity of φ in its second argument gives $\varphi(x, 0) = 0$ for every x, so $0 \in S^\perp$. If $y, y' \in S^\perp$ and $a \in k$, then for every $x \in S$

$$\varphi(x, y + ay') = \varphi(x, y) + a\,\varphi(x, y') = 0 + a \cdot 0 = 0$$

so $y + ay' \in S^\perp$, and $S^\perp$ is a subspace.

For the second claim, $S \subseteq \operatorname{span} S$ forces $(\operatorname{span} S)^\perp \subseteq S^\perp$. Conversely let $y \in S^\perp$ and let $x = \sum_{i=1}^m a_i x_i$ with $x_i \in S$; linearity in the first argument gives $\varphi(x,y) = \sum_i a_i \varphi(x_i, y) = 0$, so $y \in (\operatorname{span} S)^\perp$, as we sought to show.

Taking S to be all of V produces the subspace on which the form carries no information at all, and the form deserves a name according to whether that subspace is trivial.

Definition 13.3.22: The Radical and Nondegeneracy

Let $\varphi\colon V \times V \to k$ be a bilinear form. The *radical* of φ is the subspace

$$\operatorname{rad}(\varphi) := V^\perp = \{y \in V \mid \varphi(x,y) = 0 \text{ for every } x \in V\}$$

The form φ is *nondegenerate* if $\operatorname{rad}(\varphi) = 0$, and *degenerate* otherwise.

For a symmetric form the radical is the same computed on either side, so no ambiguity arises from having used the second argument. The radical is detected by the matrix of the form in any single basis, which is the form in which nondegeneracy is checked in practice.

Lemma 13.3.23: Nondegeneracy and the Matrix of the Form

Let V be finite dimensional with basis β, let $\varphi\colon V \times V \to k$ be a bilinear form, and let $A = [\varphi]_{\beta,\beta}$ be its matrix (definition 13.3.18). Then

$$\operatorname{rad}(\varphi) = \left\{ y \in V \;\middle|\; A^t [y]_\beta = 0 \right\}$$

and consequently φ is nondegenerate if and only if A is invertible. The criterion does not depend on which basis β was chosen, since the left-hand side does not.

***Proof*:**
By lemma 13.3.19, $\varphi(x,y) = [x]_\beta^t A [y]_\beta$ for all $x, y \in V$. Fix y. Then $y \in \operatorname{rad}(\varphi)$ means $[x]_\beta^t A [y]_\beta = 0$ for every $x \in V$, and as x ranges over V the coordinate vector $[x]_\beta$ ranges over all of k^n. A vector $u \in k^n$ satisfies $w^t u = 0$ for every $w \in k^n$ if and only if $u = 0$, so the condition is $A[y]_\beta = 0$; transposing the defining relation gives the displayed form. Hence $\operatorname{rad}(\varphi) = 0$ if and only if A has trivial kernel, which for a square matrix is invertibility.

The final sentence needs no comparison of bases: $\operatorname{rad}(\varphi)$ was defined without reference to any basis, so if A is invertible for one choice of β the radical is zero, and then the same equivalence run backwards makes the matrix invertible for every other choice, completing the proof.

Exercise 13.3.2

1. Let $f\colon V \times W \to k$ be a bilinear form on a pair of spaces, and write $W^\perp := \{v \in V \mid f(v,w) = 0 \text{ for all } w \in W\}$ and $V^\perp := \{w \in W \mid f(v,w) = 0 \text{ for all } v \in V\}$ for its two kernels. Show that f descends to a well-defined bilinear form $\bar{f}\colon V/W^\perp \times W/V^\perp \to k$, and that $\bar{f}$ is non-degenerate in each variable.

13.4 Linear Operators Invariants: Trace and Determinant

In this section, we will go over to common functionals on $M_n(k)$ (equivalently, $\text{End}_k(V)$ for an n-dimensional vector-space) that give more insight on linear transformations. A good chunk of the following theory works well over more general rings, and so we will sometimes work over a general commutative ring R.

The first functional on $M_n(k)$ we will define is called the *trace* and will allow us to detect matrices are similar to each other (i.e. the two are related by a change of basis):

Definition 13.4.1: Trace

Let $M_n(R)$ be the collection of all $n \times n$ matrices over k. Then $\text{tr} : M_n(R) \to R$ is the map:

$$\text{tr}(A) = \sum_{i=1}^{n} a_{ii}$$

The trace is studied for the following very useful properties:

Proposition 13.4.2: Properties of Trace

Let $A, B \in M_n(R)$. Then:

1. $\text{tr}(A) = \text{tr}(A^t)$
2. $\text{tr}(AB) = \text{tr}(BA)$

***Proof*:**
computational exercise

As a consequence, if A and B are equivalent so that there exists a P such that $A = PBP^{-1}$, then:

$$\text{tr}(A) = \text{tr}(PBP^{-1}) = \text{tr}(P(BP^{-1})) = \text{tr}((BP^{-1})P) = \text{tr}(BPP^{-1}) = \text{tr}(B)$$

Hence the trace is invariant under change of basis, making it a good tool to study linear maps without needing to pick a basis

Definition 13.4.3: Trace Of Linear Map

Let $f : V \to W$ be a linear map and let β, γ be basis for V and W respectively. Then define $\text{tr}(f)$ to be

$$\text{tr}(f) := \text{tr}([f]_\beta^\gamma)$$

Corollary 13.4.4: Trace Of Dual

Let $f : V \to W$ and $f^* : W^* \to V^*$ be the dual map of f. Then:

$$\text{tr}(f) = \text{tr}(f^*)$$

***Proof*:**
exercise

(check this out to: geometric interpretation of trace)

The next map is a very common multilinear form that will be used extensively: this form will also be independnet of basis, but with the extra information giving a very clear geometric picture of how a linear transformation effects a 1-cube in $\mathbb{R}^n$:

Definition 13.4.5: Determinant

Let R be a commutative ring and $A = (a_{ij}) \in M_n(R)$ a square matrix. then the *determinant* of A is a map $\det : M_n(R) \to R$:

$$\det(A) := \sum_{\sigma \in S_n} (-1)^\sigma \prod_{i=1}^{n} a_{i\sigma(i)} \in R$$

Proposition 13.4.6: Properties Of The Determinant

Let R be a commutative ring and $A \in M_n(R)$. Then:

1. $\det(I_n) = 1$.
2. $\det(A) = \det(A^t)$, here $(a_{ij})^t = (a_{ji})$.
3. (multilinearity) det is R-linear in each column separately: if the j-th column of A is $cu + c'u'$ for $c, c' \in R$ and column vectors u, u', and if A_u, $A_{u'}$ denote the matrices agreeing with A outside column j and having u, respectively u', in column j, then
$$\det(A) = c\det(A_u) + c'\det(A_{u'})$$
By (2) the same holds for rows.
4. (alternating) If two columns of A are equal then $\det(A) = 0$; the same holds for rows. Combined with (3), if one column is a scalar multiple of another then $\det(A) = 0$.
5. (permuting rows) If A_τ denotes the matrix whose i-th row is the $\tau(i)$-th row of A, for $\tau \in S_n$, then $\det(A_\tau) = (-1)^\tau \det(A)$.

***Proof*:**
All five are read off the defining sum.

1. For $A = I_n$ the product $\prod_i \delta_{i\sigma(i)}$ vanishes unless $\sigma(i) = i$ for every i, so only $\sigma = \mathrm{id}$ contributes, and it contributes 1.
2. By definition $\det(A^t) = \sum_\sigma (-1)^\sigma \prod_i a_{\sigma(i)i}$. Re-indexing the product by $j = \sigma(i)$ gives $\prod_i a_{\sigma(i)i} = \prod_j a_{j\sigma^{-1}(j)}$, and $(-1)^\sigma = (-1)^{\sigma^{-1}}$ since a permutation and its inverse have the same parity. As σ runs over S_n so does σ^{-1}, so the two sums agree term for term.
3. In the term indexed by σ, exactly one factor is drawn from column j, namely $a_{\sigma^{-1}(j)\,j}$.

Each term is therefore R-linear in column j with the other columns held fixed, and a sum of such terms is too.

4. Suppose columns $j \neq k$ of A are equal, and let $\tau = (j\ k)$. Since $a_{ij} = a_{ik}$ for all i, one has $a_{i\,\tau(m)} = a_{im}$ for every i and m, and hence

$$\prod_i a_{i\,\tau\sigma(i)} = \prod_i a_{i\sigma(i)}$$

while $(-1)^{\tau\sigma} = -(-1)^{\sigma}$. The assignment $\sigma \mapsto \tau\sigma$ is an involution of S_n without fixed points, so it partitions S_n into pairs whose two terms are negatives of one another. The sum is therefore 0. Note that this argument cancels terms exactly and so does not require 2 to be invertible in R. The statement for rows follows by (2), and the statement for proportional columns by pulling the scalar out with (3).

5. By definition $\det(A_\tau) = \sum_\sigma (-1)^\sigma \prod_i a_{\tau(i)\sigma(i)}$. Substituting $j = \tau(i)$ turns the product into $\prod_j a_{j\,\sigma\tau^{-1}(j)}$. Writing $\rho = \sigma\tau^{-1}$, so that $(-1)^\sigma = (-1)^\rho(-1)^\tau$, and letting ρ run over S_n as σ does,

$$\det(A_\tau) = (-1)^\tau \sum_{\rho\in S_n} (-1)^\rho \prod_j a_{j\rho(j)} = (-1)^\tau \det(A)$$

completing the proof.

The determinant is multiplicative, and this holds for all square matrices over a commutative ring, not only invertible ones. The proof is a direct expansion of the defining sum, with proposition 13.4.6(4) killing every term that is not indexed by a permutation.

Proposition 13.4.7: Multiplicativity of the Determinant

Let R be a commutative ring and let $A, B \in M_n(R)$. Then

$$\det(AB) = \det(A)\det(B)$$

***Proof*:**
Write $A = (a_{ij})$ and $B = (b_{ij})$, so that $(AB)_{ij} = \sum_k a_{ik}b_{kj}$. Expanding the defining sum,

$$\det(AB) = \sum_{\sigma\in S_n} (-1)^\sigma \prod_{i=1}^{n} \left(\sum_{k=1}^{n} a_{ik}b_{k\sigma(i)}\right)$$

Expanding the product over i amounts to choosing an index k for each i, that is, a function $f\colon \{1,\dots,n\} \to \{1,\dots,n\}$, so

$$\det(AB) = \sum_f \left(\prod_{i=1}^{n} a_{if(i)}\right)\left(\sum_{\sigma\in S_n} (-1)^\sigma \prod_{i=1}^{n} b_{f(i)\sigma(i)}\right)$$

the sum being over all n^n such functions. For fixed f, let B_f be the matrix whose i-th row is the $f(i)$-th row of B, so that $(B_f)_{ij} = b_{f(i)j}$. The inner sum is exactly $\det(B_f)$.

If f is not injective, say $f(i) = f(i')$ with $i \neq i'$, then B_f has two equal rows and $\det(B_f) = 0$ by proposition 13.4.6(4). Only the injective f survive, and an injective self-map of a finite set is a permutation, so the sum runs over $f = \tau \in S_n$. For such an f, B_τ is B with its rows permuted by τ, so $\det(B_\tau) = (-1)^\tau \det(B)$ by proposition 13.4.6(5). Therefore

$$\det(AB) = \sum_{\tau \in S_n} \left(\prod_{i=1}^{n} a_{i\tau(i)} \right) (-1)^\tau \det(B) = \left(\sum_{\tau \in S_n} (-1)^\tau \prod_{i=1}^{n} a_{i\tau(i)} \right) \det(B) = \det(A)\det(B)$$

as we sought to show.

A block diagonal matrix has the determinant one would hope for. This is the computation that makes determinants of matrices assembled out of smaller ones tractable.

Proposition 13.4.8: Determinant of a Block Diagonal Matrix

Let R be a commutative ring and let $M \in M_n(R)$ be block diagonal with square diagonal blocks $B_1, \ldots, B_m$, so that $M_{uv} = 0$ whenever u and v lie in different blocks. Then

$$\det(M) = \prod_{t=1}^{m} \det(B_t)$$

***Proof*:**
Index the rows and columns of M by pairs (t, u), where t names the block and u the position inside it, so that $M_{(t,u),(t',u')}$ is $(B_t)_{uu'}$ when $t = t'$ and 0 otherwise. A permutation σ of the index set contributes a nonzero term to the defining sum only if σ maps each block into itself, since otherwise some factor $M_{(t,u),\sigma(t,u)}$ has $t' \neq t$ and vanishes. Such a σ is precisely a tuple $(\sigma_1, \ldots, \sigma_m)$ of permutations of the individual blocks, acting as $\sigma(t, u) = (t, \sigma_t(u))$.

These block permutations move disjoint sets of indices, so σ is the product of the commuting permutations they determine and $(-1)^\sigma = \prod_t (-1)^{\sigma_t}$, the sign being a homomorphism. Hence

$$\det(M) = \sum_{(\sigma_1,\ldots,\sigma_m)} \prod_{t=1}^{m} (-1)^{\sigma_t} \prod_{u} (B_t)_{u\sigma_t(u)} = \prod_{t=1}^{m} \left(\sum_{\sigma_t} (-1)^{\sigma_t} \prod_{u} (B_t)_{u\sigma_t(u)} \right) = \prod_{t=1}^{m} \det(B_t)$$

the middle step being the distributive law, since the tuples range independently over the factors, completing the proof.

Determinents will have the incredible property of being able to detect nonsingualr matrice. To see this, let's take a moment study the determinent of serval maps of the form $f : \mathbb{R}^2 \to \mathbb{R}^2$ (HERE, show the geometric intuition)

With this geometric intuition, we will now proceed with how to capture it algebraically:

Proposition 13.4.9: Elementary Matrices And Determinant

Let A be a matrix then

1. (sitching rows) $\det(A') = -\det(A)$
2. (adding copies of one ro to another) $\det(A') = \det(A)$
3. (multiplying one ro by a unit c) $\det(A') = c\det(A)$

***Proof*:**
here

Proposition 13.4.10: Invertability And Determinants

Let R be a commutative ring. Then:

1. $A \in M_n(R)$ is invertible if and only if $\det(A)$ is a unit in R
2. the determinante is a homomorphism $\det : \mathrm{GL}_n(R) \to R^\times$, that is, for any $A, B \in \mathrm{GL}_n(R)$:
$$\det(A \cdot B) = \det(A) \cdot \det(B)$$

***Proof*:**
(if $R = k$ is a field)

(words here)

***Proof*:**
(if R is a general commutative ring)

We will finish off this section an application of determinents to solve systems of linear equations. This will become particularly useful for a few proofs in the book[23].

Proposition 13.4.11: Cramer's Rule

Let R be a commutative ring, $A \in M_n(R)$, and $\det(R)$ be a unit of R. Let $A^{(j)}$ be the matrix obtined by replacing the jth column of A by the column vector b. Then:

$$x_j = \det(A)^{-1}\det(A^{(j)})$$

***Proof*:**

(Also really want to put down the "matrix invertible if determant maps to units" result)

The final topic covered in this section will be that of defining the *volume*:

[23] In particular, proposition 26.1.4

Definition 13.4.12: Volume

Let V be an inner product space and $e = (e_1, e_2, ..., e_n)$ an orthonormal basis for V. Then if $v = (v_1, v_2, ..., v_n)$ is a set of vectors given by a change of basis matrix $v = Ae$, *volume* of v is

$$\mathrm{vol}(v) = |\det(A)|$$

The absolute value around $\det(A)$ is to eliminate orientation information. Geometrically, this can be seen as the volume of a parallellapided when $v_1, v_2, ..., v_n$ are linearly independent, namely if

$$P = \left\{ \sum a_i v_1 \;\middle|\; 0 \leq x_i < 1 \right\}$$

Then the intuitive notion of volume of P agrees with our definition. The case where $v = (e_1, e_2, ..., e_n)$ shows that the volume of a vector of unit lengths is always 1.

Exercise 13.4.1

1. Let $A \in M_n(k)$, $k \subseteq \mathbb{C}$. Show that $\det(\overline{A}) = \overline{\det(A)}$ where we define $\overline{A}$ component-wise to be $(\overline{A})_{ij} = \overline{a_{ij}}$ with $\overline{a_{ij}}$ being the complex conjugate $(\overline{x + iy} = x - iy)$
2. Let $A \in M_n(k)$. Define the *adjucate* of A to be $\mathrm{Ajd}(A) = C^T$ where C is the cofactor expansion matrix of A. show that if $\det(A) \neq 0$, then $A^{-1} = \det(A)^{-1}\,\mathrm{Adj}(A)$.
3. (If you konw multi-variable calculus) Treating $M_n(\mathbb{R})$ as $\mathbb{R}^{n^2}$, show that the dervative of the determinant map $\det : M_n(\mathbb{R}) \to \mathbb{R}$ at the identity I is the trace tr: $(\det)'(I) = \mathrm{tr}$. This gives a surprisingly interesting link between the two concepts, and is important in the beginning of the study of *Lie Algebras* of many matrix groups such as $\mathrm{GL}_n(k)$, $\mathrm{SL}_n(k)$, $O_n(k)$, and so forth, where $k \in \{\mathbb{R}, \mathbb{C}\}$.
4. More generally, show that $\det : M_n(\mathbb{R}) \to \mathbb{R}$ has a derivative at any point A, namely: $(D_A \det)(X) = \mathrm{tr}(\mathrm{Adj}(A^t)X)$.
5. Let R be an integral domain and $f : M \to N$ a R-module homomorphism between free R-modules of rank n, m respectively (note M, N satisfy the IBN property). Choosing a basis for M, N and a matrix A representing f, show that if $\det(A) = 0$ Then A is not invertible. Show furthermore A is not linearly dependent.
6. Take the same setup as in question (13.4.1 (5)). Show that the converse is not true: $\det(A) \neq 0$ does *not* imply A is invertible (hint: consider $R = \mathbb{Z}$).

13.5 Eigenvectors and Characteristic Polynomial

After having found the trace and determinant, the reader may wonder if there are more invariant functionals on $\mathrm{End}_k(V)$ for an n-dimensional vector space. As it turns out, these are both special cases of a collection of n invariant given by the coefficients of the following polynomial:

Definition 13.5.1: Characteristic Polynomial

Let M be a free R-module and let $\alpha \in \mathrm{End}_R(M)$. Let $I \in \mathrm{End}_R(M)$ denote the identity operator. Then the characteristic polynomial is defined to be:

$$P_\alpha(t) := \det(tI - \alpha) \in \mathrm{End}_R(M)$$

which when α has a basis representation A, then $P_\alpha(t)$ is an element of $R[t]$, R-polynomials with indeterminate t.

The origins of this seemingly complicated looking polynomial comes from its link to eigenvectors and eigenvalues (namely scalars and vector of a linear operator that satisfy $\alpha(v) = \lambda v$ for appropriate $\lambda \in R$ and $v \neq 0$) which shall be explored soon. Some important properties are established first:

Proposition 13.5.2: Properties Of The Characteristic Polynomial

Let $P_\alpha(t)$ be the characteristic polynomial of the linear operator $\alpha : M \to M$ for a free R-module of rank n. Then:

1. $P_\alpha(t)$ is monic of degree n
2. The coefficient of t^{n-1} term is $-\mathrm{tr}(\alpha)$
3. The constant term is $(-1)^n \det(\alpha)$
4. If α and β are similar, then $P_\alpha = P_\beta$

***Proof*:**

1. Expanding the definition immediately reveals the result. Let $A = (a_{ij})$ be a representative of α given some choice of basis. Then:

$$P_\alpha(t) = \det \begin{pmatrix} t - a_{11} & -a_{12} & \cdots & -a_{1n} \\ t - a_{21} & t - a_{22} & \cdots & -a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ -a_{n1} & t - a_{n2} & \cdots & t - a_{2n} \end{pmatrix}$$

2. Choose A as above. Then notice that when expanding $P_\alpha(t)$, the values of the t^{n-1} term only come from terms on the diagonal, namely we get:

$$\sum_i^n t \cdot t \cdot \cdots \cdot t \cdot a_{ii} \cdot t \cdot \cdots \cdot t \cdot t$$

giving us the desired result

3. Setting $t = 0$ and using multi-linearity gives the desired result

4. Assuming $\alpha \sim \beta$, there exists a ρ such that

$$\alpha = \rho \circ \beta \circ \rho^{-1}$$

Then by using properties of functions, we see that:

$$tI - \alpha = \rho \circ (tI - \beta) \circ \rho^{-1}$$

But then, the determinant of both sides are the same. Choosing a matrix representation P for ρ and B for $tI - \beta$ we see that by proposition 13.4.10

$$\det(PBP^{-1}) = \det(P)\det(B)\det(P)^{-1} = \det(B)\det(P)\det(P)^{-1} = \det(B)$$

hence

$$P_\alpha(t) = \det(tI - \alpha) = \det(\rho \circ (tI - \beta) \circ \rho^{-1}) = \det(tI - \beta) = P_\beta(t)$$

completing the proof.

Hence, the characteristic polynomial is of the form:

$$t^n - (\mathrm{tr}\alpha)t^{n-1} + \cdots + (-1)^n(\det \alpha)$$

As far as I am aware, the other coefficients of this polynomial don't have a special name like the trace and determinant, however on the other hand these coefficients are special since they always appear in any dimension (they are equal when $n = 1$ for trivial reasons, the determinant is zero if α is not surjective, and the trace is zero if α is a commutator, see exercise ref:HERE not done yet see this). Furthermore, note that all coefficients are invariant under similarity, meaning that if the trace or determinant between two linear operators differ, they are *not* similar! Unfortunately, the converse does not hold:

Example 13.5.3: Similar Matrices

Let:

$$A = \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix} \qquad B = \begin{pmatrix} 2 & 1 \\ 0 & 2 \end{pmatrix}$$

These each have a characteristic polynomial of $t^2 - 4$. However, they are not similar: Notice that A is the identity, and conjugating A by any invertible matrix P would give the identity, and hence never B.

As mentioned earlier, the characteristic polynomial naturally arises when studying eigenvector and values. Let us start by formally defining these terms:

Definition 13.5.4: Eigenvalue and Spectrum

Let M be a free R-module over a commutative ring R and let $\alpha \in \mathrm{End}_R(M)$ be a linear operator on M. Then a scalar $\lambda \in R$ is called an *eigenvalue* of α if there exists a nonzero $v \in M$ such that

$$\alpha(v) = \lambda v$$

The collection of all eigenvalues is called the *spectrum* of α.

The existence of eigenvalues of an operator α means that in some subspace $N \subseteq M$ the transformation $\alpha|_N$ is simply scaling. Over $\mathbb{C}$[24], we shall see that under a suitable choice of basis that scaling represents one of the two possible behaviors of a linear operator, in particular a linear operator can

[24] so that there is no "algebraic obstruction", an intuition that shall be covered in great detail in chapter 19

only either scale or *shear*. In example 13.5.3, matrix B scaled the first basis vector and sheared the second basis vector. Scaling is the easier of the two behaviours as the algebraic invariant for detecting scaling is the characteristic polynomial, while detecting and characterizing shearing requires more complicated invariants: shearing shall be explored in chapter 14, section 14.3.

Eigenvalues are extremely important, having a wide range of applications in quantum mechanics (the term *spectrum of an atom* comes from finding the spectrum of an endomorphism) functional analysis (K-theory has its origin in the spectrum of infinite linear operators), differential equations (in finding a basis to the solutions of an ODE and in defining the Fourier Transform), to number theory and Field theory (as we shall see later in the book), and so much more! Due to the prominence of eigenvalues, some may claim that the study of endomorphisms of vector-spaces is one of the fundamental concepts of mathematics or the sciences. In *Algebraic Geometry* [5] it is shown that any commutative ring has a spectrum associated to (though by the definition of a spectrum of a ring, the connection shall not be obvious[25]).

The eigenvalue is invariant under similarity, which the reader should attempt to prove before looking at the following: Letting α, β be similar using ρ ($\alpha = \rho \circ \beta \circ \rho^{-1}$) and
$beta(v) = \lambda v$, then $\alpha \circ \rho = \rho \circ \beta$ and

$$\alpha(\rho(v)) = (\rho \circ \beta \circ \rho^{-1})(\rho(v)) = \rho \circ \beta(v) = \rho(\lambda v) = \lambda \rho(v)$$

since $\rho(v) \neq 0$ if $\lambda \neq 0$ (namely, it is invertible), we see that λ is an eigenvalue of α if and only if λ is an eigenvalue of β.

Now, as promised, the following shows why the characteristic polynomial is a natural concept based on its invariance in similarity:

Proposition 13.5.5: Eigenvalues And Characteristic Polynomial

Let $P_\alpha(t)$ be the characteristic polynomial of the linear operator α. Then λ is an eigenvalue if and only if λ is a root of $P_\alpha(t)$. In other words, the set of roots of $P_\alpha(t)$ is exactly the spectrum of α.

***Proof*:**
Using the properties of determinants:

$$\begin{aligned}\lambda \text{ is an eigenvalue} &\Leftrightarrow \exists v \neq 0 \text{ s.t. } \alpha(v) = \lambda v \\ &\Leftrightarrow \exists v \neq 0 \text{ s.t. } (\lambda I - \alpha)(v) = 0 \\ &\Leftrightarrow (\lambda I - \alpha) \text{ is not injective} \\ &\Leftrightarrow \det(\lambda I - \alpha) = 0 \\ &\Leftrightarrow P_\alpha(\lambda) = 0\end{aligned}$$

as we sought to show.

[25] The curious reader may checkout EYNKTA functional analysis where this connection is fully fleshed out, as the connection is fundamentally a functional analysis result and would be too far out of the scope of an algebra textbook to be covered

Example 13.5.6: Eigenvalues And Characteristic Polynomials

1. If α is not injective, then 0 is an eigenvalue, otherwise 0 is not an eigenvalue.
2. Take
$$\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$$
This matrix has a characteristic polynomial of $t^2 + 1$. If $R = \mathbb{R}$, then this characteristic polynomial has *no* roots. Similarly, the linear operator this matrix represents has no eigenvalues. Geometrically, this matrix can be seen as rotating $\mathbb{R}^2$ by $\pi/2$ degrees, which is why it has no eigenvalues over $\mathbb{R}$. If $R = \mathbb{C}$, then the polynomial has two roots, namely $i, -i$. The reader should verify that these are eigenvalues of a linear operator represented by this matrix. Intuitively, multiplying by i can be thought of as representing rotation by $\pi/2$ degree's which is why it is an eigenvalue.
3. Take
$$\begin{pmatrix} 0 & 2 \\ 1 & 0 \end{pmatrix}$$
This matrix has a characteristic polynomial of $t^2 - 2$. If $R = \mathbb{R}$, then this polynomial has roots $\pm\sqrt{2}$, while if $R = \mathbb{Q}$, this polynomial has no roots, which equivalently implies that there is no eigenvalues when $R = \mathbb{Q}$. Geometrically, this transformation can be seen as reflecting the plane across a line passing through the origin with an irrational slope. Since the slope is irrational, there is no nonzero irrational scalar to a vector, and hence cannot be a rational eigenvalue.

Geometrically, eigenvectors correspond to the linear operator scaling in the appropriate direction. In the simplest case, an n-dimensional linear operator α can be represented in a basis for which $[\alpha]$ just scales appropriately in each direction. We take a moment to explore these type of matrices.

Since roots have the notion of multiplicity (the number of roots sharing the same value), it is natural to contemplate the multiplicity of eigenvalues and see what information this gives us:

Definition 13.5.7: Algebraic Multiplicity

Let λ be an eigenvalue of a linear operator α over a finitely generated free module. Then the *algebraic multiplicity* is its multiplicity as a root of $P_\alpha(t)$.

Example 13.5.8: Algebraic Multiplicity

Let M be a finitely generated free module over an integral domain R.

1. the identity operator I on M has a single eigenvalue, namely 1, which has algebraic multiplicity equal to the rank of M.
2. (give a non-trivial example)

Notice that the sum of the algebraic multiplicities is bounded by the rank of the free module M (resp. the dimension of the vector-space V). If R is *algebraically closed* (meaning every polynomial of degree n over R has exactly n roots[26]), then the sum of the algebraic multiplicities of the eigenvalues always equals the rank of M. The prototypical algebraically closed field is $\mathbb{C}$, the complex numbers.

The definition has been careful to call the multiplicity "algebraic". There is another notion of multiplicity which corresponds to the geometric behavior of multiplicity, namely there is a special subspace that is linked to each eigenvalue:

Definition 13.5.9: Eigenspace And Eigenvector

Let λ be an eigenvalue associated to the linear operator α over a finitely generated free R-module M. Then the sub-module

$$E_\lambda = \ker(\lambda I - \alpha)$$

is called the *eigenspace* associated to λ, and the elements of E_λ are called *eigenvectors* and represent the elements of M such that

$$\alpha(v) = \lambda v$$

Technically, it is more correct to call these spaces "eigen-modules" rather than eigenspace, however the study of eigen-modules over vector-spaces greatly precedes the generalization to eigen-modules over R-modules, giving us our current vocabulary.

Definition 13.5.10: Geometric Multiplicity

Let E_λ be an eigenspace corresponding to an eigenvalue λ. Then the rank of E_λ is the *geometric multiplicity* of λ.

It should be shown that the eigenspace is invariant under similarity, meaning it a special subspace that is invariant under α (we shall later call spaces that are invariant under an operator α-stable, where α is the operator in question).

Example 13.5.11: Geometric And Algebraic Multiplicity

Take

$$\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$$

Then the characteristic polynomial is $(t-1)^2$, and hence 1 is an eigenvalue with algebraic multiplicity 2. To find it's geometric multiplicity, by considering

$$\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}\begin{pmatrix} v_1 \\ v_2 \end{pmatrix} = \begin{pmatrix} v_1 + v_2 \\ v_2 \end{pmatrix}$$

we see that the right hand side is equal to $1v$ if and only if $v_2 = 0$, hence the eigenspace is

$$\operatorname{span}\begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

which is 1 dimensional, hence the geometric multiplicity is 1.

[26] See definition 19.5.1 and the lemma under it for more details

A result that should stand out to the reader given the above example is the following

Proposition 13.5.12: Algebraic And Geometric Multiplicity

Let λ be an eigenvalue associated to the linear operator α over a finitely generated free R-module M. Then the geometric multiplicity is always less than or equal to the algebraic multiplicity

***Proof*:**
exercise

One of the main results comes in the algebraic multiplicities add up to the rank of the free module and when the algebraic multiplicity matches the geometric multiplicity. Then there exists a basis P making α a bunch of scalars, namely:

Theorem 13.5.13: Diagonalizable

Let α be a linear operator over a finitely generated free R-module M. Then if the algebraic multiplicities of the eigenvalues sum to n and the algebraic multiplicities of each eigenvalue equal to their respective geometric multiplicities, there exists an invertible matrix $P \in \mathrm{GL}_n(R)$ such that for any representation A of α:

$$A = P \begin{pmatrix} \lambda_1 & 0 & \cdots & 0 \\ 0 & \lambda_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \lambda_n \end{pmatrix} P^{-1}$$

where $\lambda_1, \lambda_2, ..., \lambda_n$ are the eigenvalues of α, and α (resp. A) is said to be *diagonalizable*. If $m = V$ is a vector-space, α is diagonalizable if and only if there exists a basis for V consisting of eigenvectors.

***Proof*:**
exercise.

Definition 13.5.14: Diagonalizable

Let α be a linear operator over a finitely generated free R-module M. Then if α admits a representation with only nonzero diagonal entries, α is said to be *diagonalizable*. Similarly, a matrix that is similar to a diagonal matrix is said to be *diagonalizable*.

If α is diagonalizable, then M can be expressed in the following useful decomposition:

Theorem 13.5.15: Spectral Decomposition

Let $\alpha : M \to M$ be diagonalizable. Then M admits the following decomposition into direct sums

$$M \cong \bigoplus_{\text{eigenvalue } \lambda} E_\lambda$$

***Proof*:**
exercise

For future reference, let's give a name to the above decomposition:

Definition 13.5.16: Spectral Decomposition

Let $\alpha : M \to M$ be diagonalizable. Then the decomposition in the above theorem is called the *spectral decomposition* of M.

Given the spectral decomposition of M, α can simply be thought of as scaling each eigenspace E_λ by the eigenvalue, making the "action" of α on the space M completely transparent!

If the algebraic multiplicity does not sum to the rank of M or the geometric multiplicity does not match the algebraic multiplicity, then the matrix is *not* diagonalizable. If the algebraic multiplicity does not sum to the rank of M, then if we are working over field[27] we can take something called the *algebraic closure* of the field, which as the name implies means it is *algebraically closed*. To rigorously go over the algebraic closure of a field is too much work at this stage, and so will just be taken for granted when it is brought up, the curious reader may checkout part IV chapter 19, section 19.5.1. With the reader's current knowledge, most of the section can be parsed.

If the geometric multiplicity does not match the algebraic multiplicity, then it is a harder to find a form. However, in the next chapter, we shall see that there exists a representation that is *almost* diagonalizable: it shall be diagonalizable with possibly a 1 in the super-diagonal (the position right above the diagonal). In fact, *all* square matrices will be able to have this almost diagonal form! This shall be the aforementioned *shearing* behaviour that is the only other behaviour besides scaling that a matrix shall represent when choosing an appropriate basis.

13.6 Geometry and Inner Product

A vector space over an arbitrary field carries no notion of length and no notion of angle, and none can be manufactured from the field axioms alone. Producing one takes two things a general field does not have: an ordering, so that a quantity can be called nonnegative, and a conjugation, so that $\langle v, v \rangle$ can be forced to be real. The real and complex numbers supply both, and over them the whole geometric apparatus follows, namely inner products and the Cauchy-Schwarz inequality, orthonormal bases and the Gram-Schmidt process, orthogonal complements and projections, the Riesz representation of a linear functional, adjoints, and the spectral theorem for normal and self-adjoint operators.

That development belongs to a book about $\mathbb{R}$ and $\mathbb{C}$ rather than to one about modules over a general ring, and it is carried out in *Linear Algebra* [17]. Nothing later in this book depends on it.

[27] more generally, over integral domains, since the ring of fractions of an integral domains is a field

14

Modules Over PIDs: Linear Algebra II

Modules over fields have some of the simplest possible structures. Because they are always free, they admit a trivial short exact sequence:

$$0 \to R^n \to M \to 0$$

In the language of free resolutions (definition 12.3.27), every module over a field admits a free resolution of length 0 (recall indexing starts at 0). It is natural to want to study modules that admit a free resolution of length 1. Specifically, if R is a ring and M is a finitely generated module with a surjection

$$R^n \to M \to 0$$

we ask when the kernel is also free, meaning there exists an R^m such that

$$0 \to R^m \to R^n \to M \to 0$$

is exact. As was foreshadowed in section 12.3.3, all finitely generated modules over PIDs admit free resolutions of length 1, and hence they shall be the focus of this chapter. Having a free resolution of length 0 (i.e., being free) implied the existence of a basis, and even over PIDs, matrix representations of linear transformations admitted Smith Normal Forms. A free resolution of length 1 offers slightly less rigid structural results; however, it remains incredibly powerful, introducing us to *Jordan Canonical Forms* and *Rational Canonical Forms* and expands the number of modules for which we have a structure theorem. These forms will provide a crucial new way of representing linear transformations, giving us deep invariants and information (namely ref:HERE).

The reader may naturally ask if it is interesting to further push the length of the free resolution and explore modules that admit a free resolution of length 2, 3, or n. While modules over polynomial rings always admit a finite length resolution (Hilbert's Syzygy Theorem; theorem 57.2.5), most interesting examples of such modules become a good deal more complex and require a build-up in algebraic geometry and homological algebra. For the interested reader, these lengths are intimately linked to the *Krull dimension* of a commutative ring[1]. The essential concepts for understanding this link will be covered later in this book (specifically, chapter 34 and chapter 57 section 57.2), and further reading can be done in [CITATION].

Goal

The goal of this section is to show finitely generated modules over PID's have the following structure:

$$M \cong R^k \oplus R/(a_1) \oplus \cdots \oplus R/(a_m)$$

We'll define the terms a_m and how to find k soon. Essentially, M shall be split into a part which is "as linearly dependent as possible", and the rest of M will be the elements which cannot be linearly independent (over R). These "rest of elements" of M can be identified with the simplest quotient rings of R, in particular a cyclic module.

Since PID's are UFD's, we will be able to find a 2nd representation given appropriate prime elements:

$$M \cong R^k \oplus R/(p_1) \oplus \cdots \oplus R/(p_n)$$

where p_i will turn out to represent prime elements of R (and the k will be the same).

A consequence of this would be the proof of the Fundamental Theorem of finitely Generated abelian groups (theorem 5.1.8). Since $\mathbb{Z}$ is a finitely generated Principal Ideal Domain, if M is a $\mathbb{Z}$-module (and so an abelian group), it will be a consequence of the following theoretical build-up that

$$M \cong \mathbb{Z}^k \oplus \mathbb{Z}/n_1\mathbb{Z} \oplus \cdots \oplus \mathbb{Z}/n_k\mathbb{Z}$$

Another consequence will be the ability to classify all linear maps from a vector-space to itself (up to similarity). This means that the structure of $M_n(k)$ and $\mathrm{GL}_n(k)$ will be eminently understandable, having huge consequence in representation theory (part X) and many areas beyond algebra (Differential Geometry, K-theory, Optimization, ODEs, and so forth). In particular, if V is a vector space over $\mathbb{F}$ and T is a linear transformation, then like in example ref:HERE, we can form an $F[x]$-module over V where representing T with the action:

$$p(x) \cdot v = (a_n x^n + a_{n-1}x^{n-1} + \cdots a_1 x + a_0) \cdot v = a_n T(v)^n + \cdots + a_1 T(v) = a_0$$

then we can use the fact that $F[x]$ is a Principal Ideal Domain to decompose T into a nicer representation. Since every linear transformation can be represented as a matrix, then we can also decompose matrices in this fashion (which will be called Rational Canonical form and Jordan Canonical Form).

Note that finiteness is an important condition: we shall go over an example where if you have an infinitely generated Principal Ideal Domain over a module, then the decomposition is not unique, among other issues [2]

[1]The link is not perfect; often some additional conditions must be placed on the ring, such as R needing to be a regular local ring or Cohen–Macaulay

[2]wikipedia: Structure theorem for finitely generated modules over a principal ideal domain, Non-finitely generated modules

14.1 Over PID, Submodule of Free Module is Free

Theorem 13.1.12 used the fact that a field k was composed of only units (and 0) to show dimension of a vector-space was well-defined. Not all free R-modules have this property. Fortunately, this will not be an issue for PID's as the following much more general result shows:

Lemma 14.1.1: Invariant Basis Well-defined Over Commutative Ring

Let R be a nonzero commutative ring and let $n, m \in \mathbb{N}$. If $R^n \cong R^m$ as R-modules, then $n = m$

This result can be proven for arbitrary cardinalities, which is left as an exercise. In particular a vector space over a field has a well-defined dimension, finite or not.

***Proof*:**
As R is commutative, there exists a maximal ideal, say $\mathfrak{m}$. Then it is easy to verify that if $R^n \cong R^m$ then:

$$(R/\mathfrak{m})^n \cong (R/\mathfrak{m})^m$$

this is then a vector-space which by theorem 13.1.12 implies $n = m$, completing the proof.

Definition 14.1.2: Invariant Basis Number

Let R be a ring. Then R has the *invariant basis number property* (IBN property) if $R^n \cong R^m$ if and only if $n = m$.

A free R-module where R has the IBN property has the notion of *rank* well-defined.

If R is not commutative, this need not be the case:

Example 14.1.3: Ring Without IBN Property
Let $V = k^{\oplus \mathbb{N}}$ and $R = \mathrm{End}_k(V)$. Prove that:

$$R \cong R^4$$

Thus, there is no ambiguity for the notion of rank when working with free-modules over commutative rings.

Recall that a sub-module of a free module need not be free. For example, if we take $\mathbb{C}[x, y]$ over itself, then it is free with rank 1, but the sub-module (ideal) (x, y) is not free. If it were of rank 1[3], then (x, y) would be generated by a single element, but it has been shown earlier this is not possible. However, if R is a PID, then just like for modules over fields, submodules of free modules over a PID are free. For free modules, this was because every module over a field is free, however there do exist modules over PID's that are not free (consider any torsion abelian group). To prove this result, we shall show that if F is a free module over a PID and $M \subseteq F$, we can iteratively split off a cyclic direct summand $\langle y \rangle$ from $M \subseteq F$, which shall result in separating the free part from the torsion part. Furthermore, y can be chosen in such a way that it is a scalar multiple of a basis element of F: $y = ax$ for appropriate $a \in R$; think of submodules of $\mathbb{Z}^n$ as a canonical example of this behavior:

[3] Why can't the rank be ≥ 1? hint: use field of fractions

Lemma 14.1.4: Splitting Modules Over PID

Let R be a PID, F a finitely generated free module over R, and $M \subseteq F$ be a nonzero submodule. Then there exists elements $a \in R$, $x \in F$, $y \in M$, and submodules $F' \subseteq F$ and $M' \subseteq M$ such that

1. $y = ax$
2. $M' = F' \cap M$
3. $F = \langle x \rangle \oplus F'$, $M = \langle y \rangle \oplus M'$

***Proof*:**
Since $F \cong R^n$, it is perhaps tempting to think that we can our factor may be one of the components of R^n, however this implicitly chooses a particular representation (i.e. a basis for the isomorphism $\cong$). Since we don't know that M is free yet, we cannot know how to pick a basis representation of F that matches well with a basis representation of M, we cannot pick a basis β for F in such a way that the free submodule M is a projection. In that case, the splitting in the lemma would be obtained by choosing a component, or in terms of functions using the projection map:

$$\pi(r_1, r_2, ..., r_n) = r_1$$

Instead, we shall employ the following trick that has been used in theorem ref:HERE to make it choice-independent: we shall take the see of all possible homomorphism $\varphi : F \to R$ and use Zorn's lemma to choose a maximal element.

Thus, define the following collection of ideals

$$\mathscr{F} = \{\varphi(M) \mid \varphi \in \mathrm{Hom}_R(F, R)\} = \{\varphi(M) \mid \varphi \in F^*\}$$

This set contain non-trivial functions since the projections $\pi_k \in \mathscr{F}$ and M, F are nonzero. Thus since R is a PID, namely it is Noetherian, by theorem 12.3.33 there exists a $\alpha : F \to R$ such that $\alpha(M)$ is maximal. Since $\mathscr{F}$ is nontrivial, it must be that $\alpha(M) \neq 0$. Since R is a PID, there exists an $a \in R$ such that $\alpha(M) = a$. Thus, there exists a $y \in M$ such that $\alpha(y) = a$.

We shall see that this y and a are the desired values. To find the desired x, we shall show that a divides $\varphi(y)$ for all $\varphi \in F^*$. To see this, let b_φ be the element such that

$$(b_\varphi) = (a, \varphi(y))$$

Then there exists $r, s \in R$ such that

$$b = ra + s\varphi(y)$$

Using this, define the homomorphism $\psi := r\alpha + s\varphi$. Since $a \in (b)$, $\alpha(M) \subseteq (b)$, and by maximality $\psi(M) \subseteq \alpha(M)$, hence $\alpha(M) = \psi(M)$. But then

$$(a) = (b) = (a, \varphi(y))$$

namely, $\varphi(y) \in (a)$, hence a divides $\varphi(y)$. Now, since F is a finitely generated free module, $F \cong R^n$. Given this representation of F, we may write:

$$y = (s_1, s_2, ..., s_n)$$

and define the projection $\pi_i : F \to R$, $\pi_i(y) = s_i$. Since $\pi_i \in \mathscr{F}$, a divides s_i for each s_i, namely there exists $r_1, ..., r_n$ such that $s_i = ar_i$. Letting x be the element associated to the element

$$(r_1, ..., r_n)$$

in the isomorphism, we get that

$$y = ax$$

Given us our desired x.

Let's now show that x, y have the desired properties. First, notice that

$$a = \alpha(y) = \alpha(ax) = a\alpha(x)$$

Since R is an integral domain, $\alpha(x) = 1$. Next, let $F' = \kappa(\alpha)$ and $M' = F' \cap M$. Notice that for every $z \in f$,

$$z = \alpha(z)x + (z - \alpha(z)x)$$

which by linearity:

$$\alpha(z - \alpha(z)x) = \alpha(z) - \alpha(z)\alpha(x) = \alpha(z) - \alpha(z) = 0$$

Thus,

$$z - \alpha(z)x \in \ker \alpha$$

implying that $F = \langle x \rangle + F'$. Next, take $rx \in \langle x \rangle \cap F'$. Then $\alpha(rx) = 0$, and by linearity $r\alpha(x) = 0$, but $\alpha(x) = 1$ so $r = 0$, and so $\langle x \rangle \cap F' = 0$, which by definition implies:

$$F = \langle x \rangle \oplus F'$$

Giving part of the 3rd claim. Finally, if $z \in M$, then $a \big| \alpha(z)$, and indeed

$$\alpha(z) \in \alpha(M) = (a)$$

Writing $\alpha(z) = ca$, we get $\alpha(z)x = cax = cy$. Splitting z as we've done before, notice that:

$$z - \alpha(z)x = z - cy \in M \cap F' = M'$$

and so, repeating the same logic as before, we get:

$$M = \langle y \rangle \oplus M'$$

completing the last part of the proof.

With this, we may prove one of the main results:

Theorem 14.1.5: Submodule Of Free Module Over PID

Let R be a PID, let F be a free R-module of finite rank n, and let $M \subseteq F$. Then M is free, of rank at most n

***Proof*:**
If $M = 0$, we are done, so let $M \neq 0$. by lemma 14.1.4 there exists a $y_1 \in M$ and a $M^1 \subseteq M$ such that
$$M = \langle y_1 \rangle \oplus M^1$$
If $M^1 = 0$, we are done. Otherwise, since $M^1 \subseteq F$, we may keep applying lemma 14.1.4 producing a y_2 and M^2. Since the set $\{y_1, y_2, ..., y_m\}$ is linearly independent and F has finite rank, this process must terminate, leaving us with a $M^m = 0$ for some $m \leq n$, at which point:
$$M = \langle y_1 \rangle \oplus \langle y_2 \rangle \oplus \cdots \oplus \langle y_m \rangle$$
as we sought to show.

The finite rank hypothesis is not needed for the conclusion, only for this proof: what makes the argument stop is that a linearly independent subset of F cannot have more than n elements. Removing the hypothesis costs exactly one thing, namely a well-ordering in place of the counter m, and the following proof is the same argument with that substitution made. It is worth reading the finite case first, because the well-ordering conceals how simple the mechanism is: peel off one basis vector, split what the submodule contributes in that coordinate, and recurse.

Theorem 14.1.6: Submodule Of Free Module Over PID, General Case

Let R be a PID, let F be a free R-module of any rank, and let $M \subseteq F$ be a submodule. Then M is free

***Proof*:**
Let $(e_\lambda)_{\lambda \in \Lambda}$ be a basis of F and well-order Λ, which is possible by theorem A.2.4. For each λ write
$$F_{<\lambda} = \bigoplus_{\mu < \lambda} Re_\mu, \qquad F_{\leq\lambda} = \bigoplus_{\mu \leq \lambda} Re_\mu,$$
and set $M_{<\lambda} = M \cap F_{<\lambda}$ and $M_{\leq\lambda} = M \cap F_{\leq\lambda}$.

Step 1: one coordinate at a time. Every $x \in F_{\leq\lambda}$ has a unique expression $x = y + re_\lambda$ with $y \in F_{<\lambda}$ and $r \in R$, so
$$\pi_\lambda \colon F_{\leq\lambda} \to R, \qquad \pi_\lambda(y + re_\lambda) = r$$
is a well defined R-module homomorphism with $\ker(\pi_\lambda) = F_{<\lambda}$. Its image on $M_{\leq\lambda}$ is a submodule of R, that is an ideal, and R is a principal ideal domain, so
$$\pi_\lambda(M_{\leq\lambda}) = (a_\lambda) \qquad \text{for some } a_\lambda \in R$$
Let $\Lambda' = \{\lambda \in \Lambda \mid a_\lambda \neq 0\}$, and for $\lambda \in \Lambda'$ choose $x_\lambda \in M_{\leq\lambda}$ with $\pi_\lambda(x_\lambda) = a_\lambda$.

Step 2: the splitting at λ. If $a_\lambda = 0$ then $M_{\leq\lambda} \subseteq \ker(\pi_\lambda)$, so $M_{\leq\lambda} = M_{<\lambda}$. If $a_\lambda \neq 0$ then
$$M_{\leq\lambda} = M_{<\lambda} \oplus Rx_\lambda$$
For the sum: given $m \in M_{\leq\lambda}$, write $\pi_\lambda(m) = ra_\lambda$; then $\pi_\lambda(m - rx_\lambda) = 0$, so $m - rx_\lambda \in M \cap F_{<\lambda} = M_{<\lambda}$. For directness: if $rx_\lambda \in M_{<\lambda}$ then $0 = \pi_\lambda(rx_\lambda) = ra_\lambda$, and R is a domain with $a_\lambda \neq 0$, so $r = 0$. The same computation shows $Rx_\lambda \cong R$ is free of rank one.

Step 3: the x_λ span M. Put $N = \sum_{\lambda \in \Lambda'} R x_\lambda \subseteq M$ and suppose $N \neq M$. Every element of F has finite support, so every $m \in M$ lies in $M_{\leq \lambda}$ for λ the largest index in its support; hence the set

$$S = \{\lambda \in \Lambda \mid M_{\leq \lambda} \not\subseteq N\}$$

is nonempty, and being a subset of a well-ordered set it has a least element λ_0. Choose $m \in M_{\leq \lambda_0}$ with $m \notin N$. By Step 2, $m = m' + r x_{\lambda_0}$ with $m' \in M_{< \lambda_0}$, where $r = 0$ in the case $a_{\lambda_0} = 0$. Now m' has finite support contained in $\{\mu \mid \mu < \lambda_0\}$, so $m' \in M_{\leq \mu}$ for some $\mu < \lambda_0$, and $M_{\leq \mu} \subseteq N$ by minimality of λ_0. Then $m = m' + r x_{\lambda_0} \in N$, a contradiction. So $N = M$.

Step 4: the x_λ are independent. Suppose $r_1 x_{\lambda_1} + \cdots + r_n x_{\lambda_n} = 0$ with $\lambda_1 < \cdots < \lambda_n$ in Λ'. Each x_{λ_i} with $i < n$ lies in $F_{\leq \lambda_i} \subseteq F_{< \lambda_n} = \ker(\pi_{\lambda_n})$, so applying π_{λ_n} leaves $0 = \pi_{\lambda_n}(r_n x_{\lambda_n}) = r_n a_{\lambda_n}$, whence $r_n = 0$. Repeating with λ_{n-1}, and so on down, gives $r_i = 0$ for every i.

So $(x_\lambda)_{\lambda \in \Lambda'}$ is a basis of M, and M is free.

The finite case does not require the divisibility result proved above. Using it, we can get a more precise result giving us something similar to a smith normal form for PIDs:

Corollary 14.1.7: Basis For Submodules Of Free Modules

Let R be a PID, F a finitely-generated free R-module, and $M \subseteq F$. Then there exists a basis $(x_1, x_2, ..., x_n)$ for F and nonzero elements $a_1, a_2, ..., a_m \in R$ with $m \leq n$ such that

$$(a_1 x_1, ..., a_m x_m)$$

is a basis for M. Furthermore, $a_1 | a_2 | \cdots | a_m$

***Proof*:**
By theorem 14.1.5 M is free and by inductively applying lemma 14.1.4 the basis for M can be chosen so that it is of the form in the lemma.

To show the coefficients divide each other, the end of the above proof shows that $a_1 \big| a_2$ since then the argument can be repeated inductively.

Corollary 14.1.8: If PID, Then Free Resolution Of Length 2

Let R be a PID. Then every finitely generated R-module M and every surjective homomorphism:

$$R^{m_0} \xrightarrow{\pi_0} M \to 0$$

there exists a free R-module R^{m_1} and a homomorphism $\pi_1 : R^{m_1} \to R^{m_0}$ such that the sequence

$$0 \to R^{m_1} \xrightarrow{\pi_1} R^{m_0} \xrightarrow{\pi_0} M \to 0$$

is exact. Along with proposition 12.3.29, we get that R is a PID if and only if every R-module admits a free resolution of length 2.

***Proof*:**
Since R^{m_0} is an R-module over a PID, $\ker(\pi_0)$ is is free by theorem 14.1.5, and since R^{m_0} is finitely generated, $\ker(\pi_0) \cong R^{m_1}$ for some $m_1 \in \mathbb{N}$. Hence choose the natural embedding $\pi_1 : \ker(\pi_0) \hookrightarrow R^{m_0}$ giving the desired exact sequence, completing the proof.

Since all modules over PID's admit a free resolution of length 2, they are characterized by a matrix (up to equivalence), namely if

$$0 \to R^m \xrightarrow{\varphi} R^n \xrightarrow{\pi_0} M \to 0$$

Then $M = \text{coker}(\varphi)$, and by exactness we know that φ injects into R^n (or, if the reader rather not use exactness, φ injects since $R^m = \ker(\pi_0)$). Then by corollary 14.1.7, given a basis $(x_1, x_2, ..., x_n)$ of R^n, there exists nonzero elements $a_1, ..., a_m \in R$ where $a_1 \big| a_2 \big| \cdots \big| a_m$ and $(a_1x_1, a_2x_2, ..., a_mx_m)$ forms a basis for $R^m \cong \ker(\pi_0)$. Hence, M is presented by the matrix:

$$\begin{pmatrix} a_1 & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & a_m \\ 0 & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & 0 \end{pmatrix}$$

By remark (readingPresentationsAndUnits),any row/column which contains an a_i that is a unit may be removed. Without loss of generality, let's say that $a_1, a_2, ..., a_m$ are non-units. Then:

$$M \cong (R/a_1) \oplus \cdots (R/a_m) \oplus R^{(n-m)}$$

Notice what just happened: a finitely generated module over M is isomorphic to a rather simple structure, that is such modules lend themselves to a classification theorem! The following theorem summarizes this result and proves a couple more important facts:

Theorem 14.1.9: Invariant Factor Form of Modules over PID's

Let R be a Principal Ideal Domain and let M be a finitely generated R-module. Then:

1. M is isomorphic to a direct sum of finitely many cyclic modules. More precisely,

$$M \cong_R R^r \oplus R/(a_1) \oplus R/(a_2) \oplus \cdots \oplus R/(a_m)$$

for some integer $r \geq 0$ and nonzero element $a_1, a_2, ..., a_m$ of R which are not units in R and which satisfy the divisibility relations

$$a_1 \mid a_2 \mid \cdots \mid a_m$$

2. M is torsion-free if and only if M is free

3. In the decomposition of (1)

$$\text{Tor}(M) \cong_R R/(a_1) \oplus R/(a_2) \oplus \cdots \oplus R/(a_m)$$

in particular M is a torsion module if and only if $r = 0$ and in this case the annihilator of M is the ideal (a_m).

In ref:HERE, we shall prove the uniqueness of this decomposition up to isomorphism, allowing us to say it is *the* direct sum of finitely many cyclic modules.

***Proof*:**
Let's first state more succinctly the above analysis to prove the existence of part (1). Let M be a finitely generated module over a PID R, say $x_1, x_2, ..., x_n$ is a generating set of minimal cardinality. Then given R^n with basis $b_1, b_2, ..., b_n$:

$$\pi : R^n \to M \qquad \pi(b_i) = x_i$$

is a surjective R-module homomorphism with kernel $\ker(\pi)$ which is free by theorem 14.1.5. Then by corollary 14.1.7 choose a basis $y_1, y_2, ..., y_n$ for R^n such that $a_1y_1, a_2y_2, ..., a_my_m$ is a basis for $\ker(\pi)$ with $a_1 \big| a_2 \big| \cdots \big| a_m$. Then:

$$M \cong R^n / \ker(\pi) = (y_1R \oplus y_2R \oplus \cdots \oplus y_nR)/(a_1y_1R \oplus a_2y_R2 \oplus \cdots \oplus a_my_mR))$$

which the reader may verify (exercise ref:HERE) that is isomorphic to:

$$R/(a_1) \oplus R/(a_2) \oplus \cdots \oplus R/(a_m) \oplus R^{n-m}$$

This immediately gives us the other two conclusions:

(2) Since $R/(a_i)$ is torsion and the direct product of torsion modules is torsion, then M is torsion free if and only if $M \cong R^n$, which implies it's free (hence being "element-wise" free implies being "globally" free)

(3) This follows from the fact that the direct product of torsion modules is torsion, hence

$$\text{Tor}_R(M) \cong_R R/(a_1) \oplus \cdots \oplus R/(a_m)$$

In order to prove the uniqueness of this decomposition, we shall show how we can systematically "embed" parts of M into a vector-space and take advantage of the invariant basis number (IBN) of finitely dimensional isomorphic vector spaces, that is if two vector-spaces are isomorphic they are the same dimension.

First, since R is a PID the annihilator of M is a principal ideal. We start by finding the annihilator of M when it is a torsion module:

Lemma 14.1.10: Annihilator of Module Over PID

Let M be a finitely generated torsion module over a PID R with decomposition:

$$R/(a_1) \oplus \cdots R/(a_m)$$

with $a_1 \Big| \cdots \Big| a_m$. Then $\mathrm{Ann}_R(M) = (a_m)$.

***Proof*:**
$0 \neq r \in \mathrm{Ann}_R(M)$ so that

$$0 = r \cdot (1, ..., 1) = (r, ..., r)$$

which by the given decomposition implies:

$$r \equiv 0 \bmod (a_m) \qquad \Leftrightarrow \qquad r \in (a_m)$$

Hence, $\mathrm{Ann}_R(M) \subseteq (a_m)$. Conversely, if $r \in (a_m)$ and $y \in M$, decomposing y so that

$$y = (y_1, ..., y_m) \qquad y_i \in R/(a_i)$$

since $r \in (a_m) \subseteq (a_i)$, we see that for each a_i:

$$ry_i = 0$$

hence $ry = 0$, and so $r \in \mathrm{Ann}_R(M)$, as we sought to show.

Next, since R is a UFD, we may find a decomposition, we have $a_m = up_1^{k_1} p_2^{k_2} \cdots p_r^{k_r}$. Since $\gcd(p_i^{k_i}, p_j^{k_j}) = 1$, by exercise ref:HERE (rings, under CRT)

$$(p_i^{k_i}) + (p_j^{k_j}) = (1)$$

Hence, by the Chinese Remainder Theorem: $R/(a_m) \cong R/(p_1^{k_1}) \oplus \cdots \oplus R/(p_r^{k_r})$ Doing this to every a_i gives the following decomposition:

Definition 14.1.11: Elementary Divisor Form Of Modules Over PID's

Let M be a finitely generated torsion module over a PID R. Then given the above decomposition, the isomorphism:

$$M \cong R^r \oplus R/(p_1^{k_1}) \oplus \cdots \oplus R/(P_s^{k_s})$$

is called a[a] elementary divisor form of M with each (not necessarily unique) p_i being primes are R and are called the *elementary divisors.*

[a]we can't yet say "the"

Given M is torsion, since M can be decomposed into modules of the form $R/(p_k^{k_i})$, we may group together these modules to form a decomposition:

$$M \cong N_1 \oplus \cdots N_s$$

where the elements of N_i are annihilated by a power of a distinct prime p_i.

Theorem 14.1.12: Primary Decomposition Theorem

Let R be a PID and let M be a nonzero torsion R-module (not necessarily finitely generated) with nonzero annihilator a. Suppose the factorization of a into distinct primer powers in R is

$$a = up_1^{\alpha_1} p_2^{\alpha_2} \cdots p_n^{\alpha_n}$$

and let $N_i = \{x \in M | p_i^{\alpha_i} x = 0\}$, $1 \leq i \leq n$. Then N_i is a submodule of M with annihilator $p_i^{\alpha_i}$ and is the submodule of M of all elements annihilated by some power of p_i. Thus, we get

$$M = N_1 \oplus N_2 \oplus \cdots \oplus N_n$$

If M is finitely generated, then each N_i is the direct sum of finitely many cyclic modules whose annihilators are divisors of $p_i^{\alpha_i}$

***Proof*:**
The finite case has already been proved. For the general case, modify the CRT to modules and take advantage of the fact that ideals of the form $(p_i^{k_i})$ are coprime (since R is a PID) to get the desired result

We give a name to the N_i modules

Definition 14.1.13: p_i-Primary Components

The submodules N_i in the previous theorem are called *p_i-primary components* of M.

With this, we come to the key result:

Lemma 14.1.14: Embedding Module over PID into Vector Space

Let R be a PID and let p be a prime in R. Let F denote the field $R/(p)$. Then:

1. if $M = R^t$, then $M/pM \cong F^t$
2. if $M = R/(a)$ where a is a nonzero element of R, then

$$M \cong \begin{cases} F & \text{if } p \text{ divides } a \text{ in } R \\ 0 & \text{if } p \text{ does not divides } a \text{ in } R \end{cases}$$

3. if $M = R/(a_1) \oplus R/(a_2) \oplus \cdots \oplus R/(a_k)$ where each a_i is divisible by p, then $M/pM \cong F^k$

***Proof*:**

1. Let $M = R^t$. Define the map $\varphi : R^t \to (R/(p))^t$ by:

$$(r_1, ..., r_t) \mapsto (r_1 \bmod (p), ..., r_t \bmod (p))$$

Then certainly $\ker(\varphi) = pR^t = pM$, φ is surjective, and thus $M/pM \cong F^t$

2. Let $M = R/(a)$. Let $\pi : R \to R/(a)$ be the quotient map. Then

$$\pi((p)) = pM = p(R/(a)) = ((p) + (a))/(a)$$

Since R is a PID,

$$(p) + (a) = \gcd(a, p)$$

If p divides a, then $\gcd(a, p) = p$, and is 1 otherwise. Applying the appropriate isomorphism theorems gives the desired result.

3. Let $M = R/(a_1) \oplus \cdots \oplus R/(a_k)$. This follows the same procedure as theorem 14.1.9 with the modification that each a_i is divisible by p, giving us $M/pM \cong F^k$

Theorem 14.1.15: Fundamental Theorem of Uniqueness of Factors

Let R be a Principal Ideal Domain. Then two finitely generated R-modules M_1 and M_2 are isomorphic if and only if they have the same rank and the same list of invariant factors (resp. list of elementary divisors).

***Proof*:**
If M_1 and M_2 have the same rank and list of invariant factors (resp. Elementary divisors), then they are clearly isomorphic, hence consider the converse: assume $M_1 \cong_R M_2$. Then any isomorphism must map torsion to torsion, and hence $\mathrm{Tor}_R(M_1) \cong_R \mathrm{Tor}_R(M_2)$. Then:

$$R^{t_1} \cong_R M_1/\mathrm{Tor}_R(M_1) \cong_R M_2/\mathrm{Tor}_R(M_2) \cong_R R^{t_2}$$

that is $R^{t_1} \cong_R R^{t_2}$. Choosing any prime $p \in R$, we may apply lemma 14.1.14 $F^{t_1} \cong_F F^{t_2}$ (can you see why this is an F-module isomorphism where $F = R/(p)$?). Then as we saw before,

$t_1 = t_2$, giving the invariant rank.

Hence, we may quotient by the free-part of each module and show that $\mathrm{Tor}_R(M_1) \cong_R \mathrm{Tor}_R(M_2)$ implies the invariant factors (resp. elementary divisors) are the same. Starting by showing the elementary divisors are the same, it suffices to show that for any fixed prime p, the elementary divisors which are a power of p are the same for M_1 and M_2. Hence, we may reduce to the case where $M_1 \cong_R M_2$ have annihilators which are a power of p.

We do this by induction on the power of p for the annihilator of $M - 1$. If the power is 0, then $M_1 = M_2 = 0$ and hence they are clearly isomorphic. Thus ,assume M_1 (thus M_2) have nontrivial elementary divisors. Say that the elementary divisors of M_1 and M_2 are:

$$M_1 : \underbrace{p, p, ..., p}_{a \text{ times}}, p^{k_1}, p^{k_2}, ..., p^{k_s}$$
$$M_2 : \underbrace{p, p, ..., p}_{b \text{ times}}, p^{r_1}, p^{r_2}, ..., p^{r_t}$$

Then the submodules pM_1, pM_2 would eliminate the p elementary divisors and reduce each k_i, r_i by 1 (i.e. they would have power $k_i - 1$ and $r_i - 1$). Since $M_1 \cong_R M_2$, $pM_1 \cong_R pM_2$. By the induction hypothesis, the elementary divisors of pM_1 are the same as those of pM_2, that is $s = t$ and $k_i - 1 = r_i - 1$, and hence $k_i = r_i$. What's left is to show that $a = b$. By what we've just shown, we have the following isomorphism

$$M_1/pM_1 \cong_R M_2/pM_2$$

which without the fact that the elementary divisors of pM_1 and pM_2 are the same would not necessarily be true (recall that $\mathbb{Z} \cong_{\mathbb{Z}} 2\mathbb{Z}$ but $\mathbb{Z}/2\mathbb{Z} \ncong_{\mathbb{Z}} 2\mathbb{Z}/2\mathbb{Z}$). Then by lemma 14.1.14 we have $F^{a+s} \cong_F F^{b+t}$ implying $a + s = b + t$ which since $s = t$ we get $a = b$, giving us that the elementary divisors match!

Finally, for the invariant factors, invariant factors $a_1 \big| \cdots \big| a_m$ and $b_1 \big| \cdots \big| b_n$ for M_1 and M_2 respectively, take the set of elementary factors for M_1 and notice that a_m is the product of the largest of the prime powers among these elementary divisors, a_{m_1} is the product of the largest prime powers once the factors of a_m have been removed, and so forth. Doing the same procedure to the set of b's, we see that these must produce the same elements, and hence the invariant factors must match, completing the proof.

Finally, we can now talk about *the* invariant factor form and *the* elementary divisor form of a finitely generated R-module M for R a PID.

Corollary 14.1.16: Computation Shortcut For Invariant Factors

Let R be a Principal Ideal Domain and let M be a finitely generated R-module. Then:

1. The elementary divisors of M are the prime power factors of the invariant factors of M.
2. The largest invariant factor of M is the product of the largest of the distinct prime powers among the elementary divisors of M, the next largest invariant factor is the product of the largest of the distinct prime powers among the remaining elementary divisors of M, and so on

***Proof*:**
Fallows by the uniqueness given from 14.1.15

Corollary 14.1.17: The Fundamental Theorem of f.g. Abelian Groups

Let G be a finitely generated group G. Then:

$$G \cong \mathbb{Z}^r \times \mathbb{Z}_{n_1} \times \cdots \mathbb{Z}_{n_s}$$

for some integers $r, n_1, ..., n_s$ with the following conditions:

1. $r \geq 0$ and $n_j \geq 2$ for all j, and
2. $n_{i+1} | n_i$ for $1 \leq i \leq s - 1$

***Proof*:**
Take $R = \mathbb{Z}$ in theorem 14.1.9. Note that the invariant factors are listed in reverse order in Chapter 5 for ease of computation.

Summary

Any finitely generated module over a PID is fully characterised by a number $n \in \mathbb{N}$ and a choice of non-units ideals $(a_1), ..., (a_n)$ such that

$$(a_1) \supseteq (a_2) \supseteq \cdots \supseteq (a_n)$$

If M is torsion, (a_n) is the annihilator of M. The elements $a_1, a_2, ..., a_n$ may be combined to a single ideal that holds all the information about he module, the ideal has a name:

Definition 14.1.18: Characteristic Ideal

Let M be a finitely generated module over a PID. Then the collection of invariant factors forms an ideal:

$$(a_1, ..., a_m)$$

known as the *characteristic ideal* of M.

The name "characteristic ideal" is given due to its link to the characteristic polynomial explored in section 13.5.

14.2 Application: Linear Algebra revisited

With the classification established, it can be put to good use in better understanding linear operators between vector spaces. Recall in section 13.2.2 that it was mentioned that chapter 14 is essentially the theoretical build-up for studying linear operators. We now make good on this promise. In this section, we define a $k[x]$-module structure on vector-spaces given by a linear operator α, and we shall define a new invariant similar to the characteristic polynomial known as the *minimal polynomial*.

To define this new module, we'll have to juggle lot of components. Let k be a field, let x be an indeterminate and let R be the polynomial ring $k[x]$. Let V be an k-module (i.e. a vector space over k). Notice that we're using the same field for V and for $k[x]$. Finally, let α be a linear operator on V, that is a linear operator from V to V where we consider the domain and codomain to be the same object. We shall extend the k-module V (i.e. a vector-space) into an $k[x]$-module using α. To accomplish this, we'll take advantage of nice properties of linear operators. First, define

$$\alpha^0 = I \qquad \alpha^n = \alpha \circ \alpha \circ \cdots \circ \alpha \qquad (n \text{ times})$$
$$a_n\alpha^n + a_{n-1}\alpha^{n-1} + \cdots + a_1\alpha + a_0 \in \mathrm{End}_k(V)$$

where I is the identity map from V to V and $\circ$ is function composition (which is well-defined since α maps to V). We'll now define the ring action of $k[x]$; let $p(x) \in k[x]$ be a polynomial. Then define an action of the ring element $p(x)$ on the module element b by:

$$\begin{aligned} p(x) \cdot v &= (a_n\alpha^n + \cdots + a_1\alpha + a_0)(v) \\ &= a_n\alpha^n(v) + \cdots + a_1\alpha(v) + a_0 v \end{aligned}$$

That is, $p(x)$ acts by substituting the linear operator α for x in $p(x)$ and applying the resulting linear operator to v. One should check that these rules satisfy the module axioms that make V into an $k[x]$-module. Note that k is a subring of $k[x]$, and this new module with constant polynomials acting on it is simply the structure of the vector space. Thus the definition of $k[x]$-module *extends* the action of k to an action of the larger ring $k[x]$.

The key is that $k[x]$ is a PID when k is a field (recall theorem 11.1.1 and its corollaries) and so we have:

$$V \cong_{k[x]} (k[x])^n \oplus (k[x]/(f_1)) \oplus \cdots \oplus (k[x]/(f_m))$$

Since V is finite k-dimensional, the action $k[x] \curvearrowright V$ must have rank 0, since if it had rank $n \geq 1$, then the invariant factor $(k[x])^n$ would be nonzero, and it is k-isomorphic to:

$$k[x] \cong k \oplus k \oplus \cdots \oplus k \oplus \cdots$$

which would make V infinite k-dimensional. Thus,

$$V \cong_{k[x]} (k[x]/(f_1)) \oplus \cdots \oplus (k[x]/(f_m))$$

Since the action of $k[x]$ is completely dependent on α, the above proposition may be interpreted as there being a special basis for V that will make the action of α on V rather transparent!

Example 14.2.1: Actions of this Module

1. If $\alpha = 0$, that is, the function maps every element to the identity in V (recall that V is an [abelian] group). Then

$$p(x)v = a_n \cdot 0 + \cdots + a_1 \cdot 0 + a_0 v = a_0 v$$

Thus, with such a linear operator, the $k[x]$-module structure is just like the k-module structure.

2. Another simple example is $\alpha = I$, that is, the identity. Thus
$$p(x)v = a_n v + \cdots + a_0 v = (a_n + \cdots + a_0)v$$
Or simply the sum of coefficient of $p(x)$ act on v

3. Here's a more specific example. Take V to be the affine n-space k^n and let α be the "shift operator"
$$\alpha(x_1, ..., x_n) = (x_2, x_3, ..., x_n, 0)$$
Since V is a vector-space, let e_i be a choice of basis vectors. Then applying this linear operator to only the basis, we get:
$$\alpha^k(e_i) = \begin{cases} e_{i-k} & \text{if } i > k \\ 0 & \text{if } i \leq k \end{cases}$$
So that if $m < n$
$$(a_m x^m + a_{m-1}x^{m-1} + \cdots + a_0)e_n = (0, 0, ..., 0, a_m, a_{m-1}, ..., a_0)$$

4. under what condition is $W \subseteq V$ a $k[x]$-submodule of V with the $k[x]$-action? Since $k[x]$ extends the actions of k, it is necessary that W be a subspace. By the subspace criterion we have that it is necessary that $\alpha(W) \subseteq W$. Not all subspaces satisfy this conditions, if they do they are called α-stable subspaces.

Now, notice that we may take the annihilator of the $k[x]$-module V, that is all elements $p(x) \in k[x]$ such that $p(\alpha) = 0$ for all $\alpha \in V$. Since $k[x]$ is a PID, the annihilator is a principal ideal. This ideal shall be important in contexts where R need not be a field, and hence we define it more generally:

Definition 14.2.2: Annihilator Ideal and Minimal Polynomial

Let M be a finitely generated free R-module where R is an integral domain, and $\alpha \in \text{End}_R(M)$. Then the ideal:
$$\mathfrak{M}_\alpha = \text{Ann}_R(M)$$
is called the *annihilator ideal* of α. If K is the field of fraction of R, then α can be viewed as an element of $\text{End}_K(K^n)$ and $\mathfrak{M}_\alpha^{(K)} \subseteq k[x]$ is the annihilator of α over K, where since $k[x]$ is a PID we get:
$$\mathfrak{M}_\alpha^{(K)} = (m_\alpha) \qquad m_\alpha \in k[x]$$
the resulting polynomial is called the *minimal polynomial* of α

Recall that $k[x]$ is a PID if and only if k is a field, hence it is natural to work in the setting of $R = k$ is a field to use the above module structure and the classification theorem. The term "minimal polynomial" is because it is the smallest polynomial (with respect to divisibility) that solves the equation $f(\alpha) = 0$. The reader may ask if the characteristic polynomial shares the same property, that is if
$$P_\alpha(\alpha) = 0$$
This shall indeed be the case and shall be called the *Cayley-Hamilton Theorem* and will be proven

in due time[4] (the reader may try to use definition 14.1.18 to find a connection between these two polynomials and hence a proof of the above equation). The annihilator ideal shares the same nice invariance properties as the characteristic polynomial:

Lemma 14.2.3: Annihilator Ideal And Similarity

Let α be similar to β. Then $\mathfrak{M}_\alpha = \mathfrak{M}_\beta$

***Proof*:**
Let $\rho \in \mathrm{GL}_R(M)$ such that $\beta = \rho \circ \alpha \circ \rho^{-1}$. Then since when composing conjugates:

$$\beta^k = \rho \circ \alpha^k \circ \rho^{-1}$$

we get that for any $f(x) \in R[x]$,

$$f(\beta) = \rho \circ f(\alpha) \circ \rho^{-1}$$

Hence, $f(\alpha) = 0$ if and only if $f(\beta) = 0$, as we sought to show.

The converse is not necessarily, the case: if $\mathfrak{M}_\alpha = \mathfrak{M}_\beta$, it does not imply that $\alpha \sim \beta$:

Example 14.2.4: Same Annihilators, Not Similar
(put a matrix in JcF that makes this evident. Also ask if then we can condlue that the char poly and min poly is enough to deduce similarity for dimension $n \leq 3$)

On the other hand, going back to example 13.5.3, we see that $x - 1$ is the annihilator of the identity matrix, while it is not the annihilator of:

$$A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$$

Coming back to the $k[x]$-module on V, we may use it to finally have a converse to establishing two linear operators are similar:

Lemma 14.2.5: Similarity And New Module Structure

Let α, β be linear opereators on a free R-module M. Then the corresponding $R[x]$-module structures on M are isomorphic if and only if $\alpha \sim \beta$

***Proof*:**
Let M_α, M_β be the corresponding $R[x]$-modules given by α, β respectively. Assume $\alpha \sim \beta$ so that there exists a $\rho \in \mathrm{GL}(M)$ such that

$$\beta = \rho \circ \alpha \circ \rho^{-1} \Rightarrow \beta \circ \rho = \rho \circ \alpha$$

I claim that $\rho : M_\alpha \to M_\beta$ is an $R[x]$-linear map. It is certainly a R-linear map, hence to see if

[4] the curious reader may see exercise ref:HERE to prove this result with judicious application of Cramer's rule

it $R[x]$ linear consider the action of x in M_α and M_β. Then:

$$\rho(x \cdot v) = \rho \circ \alpha(v) = \beta \circ \rho(v) = x \cdot \rho(v)$$

which then may be generalized to show that it is indeed $R[x]$-linear. Furthermore, ρ is certanly an invertible $R[x]$-linear map (being bijective and an $R[x]$-module homomorphism) and hence $M_\alpha \cong_{R[x]} M_\beta$, as we sought to show.

Corollary 14.2.6: Similarity And Isomorphism Classes

Let α, β be two linear operators on a free R-module M. Then there is a one-to-one correspondence between the similarity classes of R-linear operators of a M and the isomorphism classes of of $R[x]$-module structures on M.

***Proof*:**
an immediate consequence of the above. If M has finite rank, then it is a bijection.

Exercise 14.2.1

1. Show that the $k[x]$-module on V can be constructed using the universal property of polynomials

14.3 Rational and Jordan Canonical Form

From now on, $R = F$ is a field and V is an n dimensional vector-space over F. We shall use the above results to find anice representation of a lienar operator α, namely we shall find a matrix representation for the action of x $R[x] \curvearrowright V$.

Take the following decomposition of V:

$$V \cong k[x]/(a_1(x)) \oplus k[x]/(a_2(x)) \oplus \cdots \oplus k[x]/(a_m(x))$$

where $a_1(x), a_2(x), ..., a_m(x)$ are polynomial in $F[x]$ of degree at least 1 with the divisibility condition

$$a_1(x)|a_2(x)| \cdots |a_m(x)$$

These invariant factors are determined up to units, which in $F[x]$ is F, meaning we can safely say they are all monic. Since the annihilator of V is the ideal $(a_m(x))$ (lemma 14.1.10), we get:

$$a_1(x)|a_2(x)| \cdots |m_A(x)$$

we will make a proposition of this for easy reference:

Proposition 14.3.1: Finding Minimal Polynomial

The minimal polynomial of $m_T(x)$ is the largest invariant factor of V. All the invariant vactors of V divide $m_T(x)$

***Proof*:**
direct application of (theroem 14.1.9)

Be mindful that since everything divides $m_A(t)$, $a_i(x)|m_A(x)$, then $(m_A(x)) \subseteq (a_i(x))$, so it's the *smallest* ideal of all the invariant factors. This should make sense, as since everything divides it, all of the other ideals should be bigger. Be careful then when thinking of the word "minimal", for it's actually the *biggest* polynomial relative to all the invariant factors. It is smallest when it comes to being the *smallest* polynomial which can annihilate V over $F[x]$ (i.e $m_T(T) \equiv 0$) All the other invariant factors will not manage to do this.

To calculate V with respect to $F[x]/(a_i)$ we will do the following:

For every $F[x]/(a_i)$, we can treat it as a vector-space over $F[x]$, and get a basis

$$1, \overline{x}, \overline{x}^2, ..., \overline{x}^{k-1}$$

where $a(x) = x^k + b_{k-1}x^{k-1} + \cdots + b_1 x + b_0$ is any monic polynomial in $F[x]$ and $\overline{x} = x \bmod (a(x))$.

Next recall that:

$$V \cong F[x]/(a_1(x)) \oplus F[x]/(a_2(x)) \oplus \cdots \oplus F[x]/(a_m(x))$$

Applying T on the left-side get's transalte to multiplying by x on the right-side.

$$T(V) \cong x \cdot F[x]/(a_1(x)) \oplus x \cdot F[x]/(a_2(x)) \oplus \cdots \oplus x \cdot F[x]/(a_m(x))$$

we we concentrate on on $F[x]/(a_i(x))$ and it's basis we get

$$1 \mapsto \overline{x},\ \overline{x} \mapsto \overline{x}^2,\ \overline{x}^2 \mapsto \overline{x}^2,\ \cdots \overline{x}^{k-1} \mapsto \overline{x}^k = -b_0 - b_1\overline{x} - \cdots - b_{k-1}\overline{x}^{k-1}$$

where we got the last value since

$$a(\overline{x}) = 0 \Rightarrow \overline{x}^k + b_{k-1}\overline{x}^{k-1} + \cdots + b_1\overline{x} + b_0 = 0$$

With respect to the aformentioned basis, the linear transformation T (i.e. the matrix multiplication by x) is therefore

$$\begin{pmatrix} 0 & 0 & \cdots & \cdots & \cdots & -b_0 \\ 1 & 0 & \cdots & \cdots & \cdots & -b_1 \\ 0 & 1 & \cdots & \cdots & \cdots & -b_2 \\ 0 & 0 & \ddots & & & \vdots \\ \vdots & \vdots & & \ddots & & \vdots \\ 0 & 0 & \cdots & \cdots & 1 & -b_{k-1} \end{pmatrix}$$

If we we do this for $F[x]/(m_A(x))$, then we can already see that if we do $m_A(V)$, we will annihilate V

Example 14.3.2: Sanity Check

Let's say $m_T(x) = x^5 + x^4 + 5x^3 - 2x^2 + 3$. Then, we we just consider $F[x]/(m_T(x))$, and think of $T(V)$ as multiplying the basis of $F[x](m_T(x))$ by x, we get the matrix:

$$\begin{pmatrix} 0 & 0 & 0 & -3 \\ 1 & 0 & 0 & 2 \\ 0 & 1 & 0 & -5 \\ 0 & 0 & 1 & -1 \end{pmatrix}$$

If we do $m_T(V)$ we get (test)

$$\begin{pmatrix}0&0&0&-3\\1&0&0&2\\0&1&0&-5\\0&0&1&-1\end{pmatrix}^5+\begin{pmatrix}0&0&0&-3\\1&0&0&2\\0&1&0&-5\\0&0&1&-1\end{pmatrix}^4+5\begin{pmatrix}0&0&0&-3\\1&0&0&2\\0&1&0&-5\\0&0&1&-1\end{pmatrix}^3-2\begin{pmatrix}0&0&0&-3\\1&0&0&2\\0&1&0&-5\\0&0&1&-1\end{pmatrix}^2$$
$$+3\begin{pmatrix}0&0&0&-3\\1&0&0&2\\0&1&0&-5\\0&0&1&-1\end{pmatrix}$$
$$=\begin{pmatrix}0&0&0&0\\0&0&0&0\\0&0&0&0\\0&0&0&0\end{pmatrix}$$

that is, m_T indeed annihilated V, as we sought to show. This doesn't tell us yet what the other invariant factors should be, or even how to find the minimal polynomial, but it does confirm that matrices of this type do what we would hope they do.

we give this matrix a name:

Definition 14.3.3: Companion Matrix

Let $a(x) = x^k + b_{k-1}x^{k-1} + \cdots + b_1 + b_0$ be any monic polynomial in $F[x]$. The *companion matrix* of $a(x)$ is the $k \times k$ matrix whith 1's down the first subdiagonal, $-b_0, -b_1, .., -b_{k-1}$ down the alst column and zeros elsewhere. The companion matrix of $a(x)$ will be denoted $C_{a(x)}$

We can do this process to each cyclic module $F[x]/(a_i(x)$. Let β_i be the elements of V that map to the basis of $F[x]/(a_i(x))$. Then by definitin, the linear transformation T acts on β_i by the companion matrix for $a_i(x)$, as we have seen by how the right hand side is multiplies by x. If we take the union of every β_i,$\beta = \cup_i \beta_i$, then this forms a basis for V (with over $F[x]$) since the sum on the RHs is a direct sum, and with respect to this basis the linear transformation T has as matrix the *direct sum* of the companion matrices, which by ref:HERE [5] we saw can be represented by:

$$\begin{pmatrix}C_{a_1(x)} & & & \\ & C_{a_2(x)} & & \\ & & \ddots & \\ & & & C_{a_m(x)}\end{pmatrix}$$

As we can see, this matrix is determined by the invariant factors of $F[x]$-module V, in fact is uniquely determined by theorem 14.1.15. Thus, we can give this matrix a name:

[5] showing this eventually using the universal property?

Definition 14.3.4: Rational Cannonical Form

1. A matrix is said to be in *rational canonical form* if it is the direct sum of companion matrices for monic polynomials $a_1(x), a_2(x)..., a_m(x)$ of degree at least one with $a_1(x)|a_2(x)|\cdots|a_m(x)$. The polynomial $a_i(x)$ are called the *invariatn factors* of the martrix. Such a matrix is also said to be a *block diagonal* matrix with blocks the companion matrice for $a_i(x)$

2. A *rational canonical form* for a linear transformation T is a matrix represetning T which is in rational canonical form

Note that we proved that the RCF is unique, meaning if

1. we are given monic polynomial $a_1(x), a_2(x), ..., a_m(x)$ of degree at least 1 such that $a_i(x)|a_{i+1}(x)$ for all i

2. We have a linear transformation $T : V \to V$, with a matrix representation A

3. we have a basis β of V such that $[A]_\beta$ is the direct sum of companion matrices

4. Then V is the direct sum of T-stable subspaces D_i, one for each $b_i(x)$, such that the amtrix of T on each D_i is the compantion matrix of $b_i(x)$. Some more words

which prove the following result:

Theorem 14.3.5: Rational Canonical Form

Let V be a finite dimensional vector sapce over the field F and let T be a linear transformation of V

1. There is a basis for V with respect to which the matrix T is in rational canonical form, i.e., is a block diagonal matrix whose diagonal blocks are the companion matrices for monic polynomials $_1(x), a_2(x), ..., a_m(x)$ of degree at least one with $a_1(x)|a_2(x)|\cdots|a_m(x)$

2. The factors $a_1, ..., a_m$ are called the *invariant factors* of α.

Corollary 14.3.6: RCF And Similarity

Let $\alpha, \beta : V \to V$ be two linear on a finitely dimensional vector space V. Then $\alpha \sim \beta$ if they have the same RCF.

***Proof*:**
combine everything we did earlier.

why it's called rational: if we go in a bigger field, the values don't change. In this way it's rational Here is a theorem linking to that

Corollary 14.3.7: RCF Over Extended Fields

Let A and B be two $n \times n$ matrices over a field F and suppose F is a subfield of the field K ($F \subseteq K$, or sometimes written K/F) Then

1. The RCF of A is the same whether it is computed over K or over F. The minimal and characteristic polynomials and the invariatn factors of A are the same whether A is considered as a matrix over F or as a matrix over K.

2. The matrices A and B are similar over K if and only if they are similar over F, i.e., there exists an invertible $n \times n$ matrix p with entries from K such that $B = P^{-1}AP$ if and only if there exists an (in general different) invertible $n \times n$ matrix Q with entries from F such that $B = Q^{-1}AQ$

***Proof*:**
Let M be the rational canonical form of a matrix A over the field F ($\subseteq K$). Since M is in rational canonical form, by 14.3.5, it is unique. Thus, if we were to extend the field to K, then we will get the same matrix. Thus, the invariant factors are the same as well. In particular, since the minimal polynomial is the largest invariant factor of A, it also does not depend on the field over which A is viewed.

Lemma 14.3.8: Characteristic Polynomial For Block Matrix

Let $a(x) \in F[x]$ be any monic polynomial

1. The characteristic polynomial is the product of the invariant factors:

$$P_\alpha(x) = a_1(x)a_2(x)\cdots a_m(x)$$

2. If M is the block diagonal matrix

$$M = \begin{pmatrix} A_1 & 0 & \cdots & 0 \\ 0 & A_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & A_k \end{pmatrix}$$

given by the direct sum of matrices $A_1, A_2, ..., A_k$, then the characteristic polynomial of M is the product of the characteristic polynomial of $A_1, A_2, ..., A_k$

***Proof*:**
A direct application of the determinant formula.

This gives the promised Cayley-Hamilton Theorem:

Proposition 14.3.9: Cayley-Hamilton Theorem

Let A be an $n \times n$ matrices over the field F. Then the minimal polynomial of A divides the characteristic polynomial of A.

Proof:
This is immediate by the above lemma.

This will come back in Field theory where finding a polynomial which is used to represent an extension will exactly be the polynomial found using Cayley-Hamilton (build up, and eventually ref:HERE into field theory)

14.3.1 Converting a square matrix to Rational Canonical Form

Let A be an $n \times n$ matix with entries from a field F

1. First, diagonalize the matrix $xI - A$ over $F[x]$ (same process as described when finding Eigenspaces in ref:HERE), while keeping track of the process so that you can apply them to step (2) (you can do them on the same time). The matrix should reach the Smith Normal from form theorem ref:HERE

   Let $d_1, d_2, ..., d_m$ be the degrees ofthe monic nonconstant polynomials $a_1(x), ..., a_m(x)$ that are in the respective diagonals

2. Let P' be an $n \times n$ identity matrix. apply the operations from (1) to P'.

3. Once you have diagonalize $xI - A$ to the Smith Normal Form, the first $n - m$ columns of the matrix P' should be 0, and the remaining m columns of P' are nonzero. For each $i = 1, 2, ...m$, multiply the i^{th} nonzero column of P' successively by $A^0 = I, A^1, A^2, ..., A^{d_i - 1}$, where d_i is the integer in (1) aboce and use the resulting column vectors (in this order) as the next d_i column of an $n \times n$ matrix $P>$ Then $P^{-1}AP$ is in rational canonical form (whose diagonal blocks are the companion matrices for the polynomials $a_1(x), ..., a_m(x)$ in (1))

Example 14.3.10: Finding Rational Canonical Form

1. Let the following 3×3 matrices be over $\mathbb{Q}$:

$$A = \begin{pmatrix} 2 & -2 & 14 \\ 0 & 3 & -7 \\ 0 & 0 & 2 \end{pmatrix} \quad B = \begin{pmatrix} 0 & -4 & 85 \\ 1 & 4 & -30 \\ 0 & 0 & 3 \end{pmatrix} \quad C = \begin{pmatrix} 2 & 2 & 1 \\ 0 & 2 & -1 \\ 0 & 0 & 3 \end{pmatrix}$$

   We can see that each matrix has the same characteirstic polynomial

$$c_A(x) = c_B(x) = c_C(x) = (x-2)^2(x-3)$$

   Since the minimal and characteristic polynomial have the same roots, the only possibilities for the minimal polynomials are $(x-2)(x-3)$ or $(x-2)^2(x-3)$. Since there are only two options, we it will either be one minimal polynomial or the other. Thus, with some computation, we see that $(A - 2I)(A - 3I) = 0$, $(B - 2I)(B - 3I) \neq 0$ (the $1,1$-entry is nonzero) and $(C-2I)(C-3I) \neq 0$ (the 1, 2-entry is nonzero). Thus, we get

$$m_A(x) = (x-2)(x-3), \qquad m_B(x) = m_C(x) = (x-2)^2(x-3)$$

since other invariant factors would need to divide $m_B(c)/m_C(x)$, there must be no other invariatn facors for B and C. Since the invariant factors for A divide the minimal polynomial and have as a product the characteristic polynomial, we see that A has for invariant factors the polynomials, $x-2, (x-2)(x-3) = x^2-5x+6$. (For 2×2 and 3×3 matrices the determination of the characteristic and minimal polynomials determine all the invariant factors, due to ________).

Thus, we can conclude that B and C are similar, and both are not simialr to A. The rational canonical forms are(note $(x-2)^2(x-3) = x^3-7x^2+16x-12$)

$$\begin{pmatrix} 2 & 0 & 0 \\ 0 & 0 & -6 \\ 0 & 1 & 5 \end{pmatrix} \quad \begin{pmatrix} 0 & 0 & 12 \\ 10 & 0 & -16 \\ 0 & 1 & 7 \end{pmatrix} \quad \begin{pmatrix} 0 & 0 & 12 \\ 1 & 0 & -16 \\ 0 & 1 & 7 \end{pmatrix}$$

2. In example (1), we managed to conclude the rational canonical form from the minimal polynomials, which we can always do for 3×3 matries and lower. However, that is not the case for matries of higher degree. Before doing such an example, let's practice by going the long way on matrix A from example (1).

   (a) We first start by reducint $xI - A$ in $\mathbb{Q}[x]$

$$xI - A = \begin{pmatrix} x-2 & 2 & -14 \\ 0 & x-3 & 7 \\ 0 & 0 & x-2 \end{pmatrix}$$

   which we can do by

   here

   so, our final result is

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & x-2 & 0 \\ 0 & 0 & x^2-5x+6 \end{pmatrix}$$

   This gives us the invariant factors of $x-2$ and $x^2-5x+6 = (x-2)(x-3)$ for this matrix. Now, let V be a 3 dimensional vector space over $\mathbb{Q}$ with standard basis e_1, e_2, e_3 and let T be the corresponding linear operator (which defines the action of x on V):

$$\begin{aligned} xe_1 &= T(e_1) = 2e_1 \\ xe_2 &= T(e_2) = -2e_1 + 3e_2 \\ xe_3 &= T(e_3) = 14e_1 - 7e_2 + 2e_3 \end{aligned}$$

   taking the row operations from earlier

   here

   and applying them to this basis we'll get

   here

with a final result of

$$[-e_1 - (x-3)(e_2 e_1), e_3 + 7(e_2 - e_1), e_2 - e_1]$$

and pplying the result of T from eraliaer, we'll get

$$[0, -7e_1 + 7e_2 + e_3, -e_1 + e_2]$$

which correspond to the elements 1, $x-2$, and $x^2 - 5x + 6$ we found earlier. Since the first invariant factor has degree 0, we can ignore it. So, we have $f_1 = -7e_1 + 7e_2 + e_3$, and $f_2 = -e_1 + e_2$. These are the $\mathbb{Q}[x]$-module generators for the two cyclic factors of V in its invariant factor decomposition as $\mathbb{Q}[x]$-modules. For the corresponding $\mathbb{Q}$-vector space bases for theset two factros, since $x-2$ has degree 1, and $x^2 - 5x + 6$ has degree 2, we get f_1, f_2, xf_2 as a basis, where $xf_2 = T(f_2) = -4e_1 + 3e_2$, giving us

$$P = \begin{pmatrix} -7 & -1 & -4 \\ 7 & 1 & 3 \\ 1 & 0 & 0 \end{pmatrix}$$

which we can use to find P^{-1} and get

$$P^{-1}AP = \begin{pmatrix} 2 & 0 & 0 \\ 0 & 0 & -6 \\ 0 & 1 & 5 \end{pmatrix}$$

which is the same result we had in (1).

(b) We can also convert strait into Rational cannonical form. Given the operation for the diagonalization of A, we use them on the identity matrix:

$$here$$

Since we have $d_1 = 1$ and $d_2 = 2$, we can take the 2nd row the following for P:

$$\begin{pmatrix} -7 \\ 7 \\ 1 \end{pmatrix} \qquad \text{and} \qquad \begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix} A \begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix} = \begin{pmatrix} -4 \\ 3 \\ 0 \end{pmatrix}$$

which gives the same P as before.

Exercise 14.3.1

1. Let$A \in M_n(k)$ be a square matrix. Then A is similar to its trnaspose (hint: is transpose similarity invariant?)

14.4 Jordan Canonical Form

The RCF was found by taking using the invariant factors. Let us now find a form based on the elementary divisors when the characteristic polynomial factors completely. If the characteristic polynomial does not factor completely, then the field can be enlarged so that every polynomial factors completely (as mentioned in section 13.5 the avid reader may checkout section 19.5.1 for more details on algebraic closure). As we've outlined earlier, the similarity of two linear transformations is independent of the base field, and hence this pursuit does lead to a useful matrix form for deducing similarity.

Given a $\alpha \in \text{End}_k(V)<$ we may obtain the elementary divisor decomposition of the correspondign $k[x]$-module to be:

$$V = \bigoplus_{i,j} \frac{k[x]}{p_i(x)^{r_{ij}}}$$

Then it is an immediate consequence that the characteristic polynomial $P_\alpha(x)$ is equal to the product:

$$\prod_{i,j} p_i(x)^{r_{ij}}$$

This gives us the following results:

Lemma 14.4.1: Elementary Divisors And Characteristic Polynomial

Let α be a linear operators on a finitely dimensional vector space V, and assume its characteristic polynomial $P_\alpha(x)$ factor completely so aht

$$P_\alpha(x) = \prod_{i}^{s} (x - \lambda_i)^{m_i}$$

where λ_i are the distinct eigenvalues and m_i are teh algebraic multiplicities. Then:

$$p_i(x) = (t - \lambda_i) \qquad m_i = \sum_j r_{ij}$$

And the minimal polynomial is:

$$m_\alpha(x) = \prod_{i=1}^{s} (x - \lambda_i)^{\max_j \{r_{ij}\}}$$

***Proof*:**
This is an exercise in putting together what was covered so far.

With this, let's start finding the matrix representation of α. Just like before, let's tart with the cyclic factor, $F[x]/((x-\lambda)^k)$. We'll purposefully be choosing our basis so that our representation is particularly simple:

$$(\overline{x} - \lambda)^{k-1}, (\overline{x} - \lambda)^{k-2}, ..., (\overline{x} - \lambda), (\overline{x} - \lambda^)0 = 1$$

this is indeed a basis, since expanding out, simplifying, we get these elements to be a linear combination of $\overline{x}^{k-1}, \overline{x}^{k-2}, ..., \overline{x}, 1$, meaning they do form a basis. Then, we can associate the linear operator T to

x, and get

$$\begin{cases} (x-\lambda)^{r-1} \mapsto x(x-\lambda)^{r-1} = \lambda(x-\lambda)^{r-1} + (x-\lambda)^r = \lambda(x-\lambda)^{r-1} \\ (x-\lambda)^{r-2} \mapsto x(x-\lambda)^{r-2} = (x-\lambda)^{r-1} + \lambda(x-\lambda)^{r-1} \\ \vdots \\ 1 \mapsto x = (x-\lambda) + \lambda \end{cases}$$

which in matrix form is be represented as:

$$\begin{pmatrix} \lambda & 1 & & & \\ & \lambda & 1 & & \\ & & \ddots & \ddots & \\ & & & \lambda & 1 \\ & & & & \lambda \end{pmatrix}$$

where the blank spots are all zero. We give a name to this matrix

Definition 14.4.2: Jordan Block

The $k \times k$ matrix with λ along the main diagonal and 1 along the first superdiagonal depicted avoce is called the *$k \times k$ elementary Jordan matrix with eigenvalue* λ, or the *Jordan block of size k with eigenvalue* λ

As before, we may combine all of these to get a full representation of α:

Definition 14.4.3: Jordan canonical Form (JCF)

Let α be a linear operator on a finitely dimensional vector space V. Then it's *Jordan Canonical Form* (JCF) is the combination of all the Jordan blocks:

$$\begin{pmatrix} [c|c|c] J_{\lambda_1,(r_{1,j})} & & \\ \hline & \ddots & \\ \hline & & J_{\lambda_s,(r_{s,j})} \end{pmatrix}$$

where $(x-\lambda_i)^{r_{ij}}$ are the elementary divisors of α.

14.4.4 An ambiguity in Uniqueness While the RCF is is completely unique up to isomorphism, since there is no unique way of arranging the eigenvalues, there is no unique way of arranging the Jordan blocks (unlike the invariant blocks which are arranged by the divisibility of the invariant factors)

So far the Jordan canonical form has only been described. What has not yet been said is when an operator *has* one: unlike the rational canonical form, which exists over any field, the Jordan form requires the relevant polynomials to factor into linear pieces. The elementary divisor decomposition of chapter 14 settles the question, and the proof also explains where the superdiagonal 1s come from.

Theorem 14.4.5: Existence of the Jordan Canonical Form

Let k be a field, let V be a finite dimensional k-vector space and let α be a linear operator on V whose minimal polynomial factors into linear factors over k (which holds for every α when k is algebraically closed). Then there is a basis of V in which the matrix of α is in Jordan canonical form, and the multiset of Jordan blocks appearing is uniquely determined by α. Equivalently, a matrix $A \in M_n(k)$ whose minimal polynomial splits over k is similar to a matrix in Jordan canonical form, unique up to the order of its blocks.

***Proof*:**
Regard V as a $k[x]$-module with x acting as α. It is finitely generated, being finite dimensional over k, and it is torsion, since the minimal polynomial $m(x)$ annihilates it. As $k[x]$ is a principal ideal domain, definition 14.1.11 gives a decomposition

$$V \cong \bigoplus_{i=1}^{s} k[x]/\big(p_i(x)^{r_i}\big)$$

with each p_i irreducible, and the multiset of the $p_i^{r_i}$ is uniquely determined by V; there is no free part because V is torsion.

Step 1: the p_i are linear. The polynomial $m(x)$ annihilates V, hence annihilates each summand $k[x]/(p_i^{r_i})$, and therefore $p_i^{r_i}$ divides $m(x)$. By hypothesis $m(x)$ is a product of linear factors, and $k[x]$ is a unique factorization domain, so every irreducible factor of m is linear. Hence each p_i is linear, say $p_i = x - \lambda_i$.

Step 2: each summand is a Jordan block. Fix a summand $W = k[x]/\big((x-\lambda)^r\big)$ and, for $1 \le j \le r$, let $e_j \in W$ be the class of $(x-\lambda)^{r-j}$. These r elements form a k-basis of W, since $1, (x-\lambda), \dots, (x-\lambda)^{r-1}$ is a k-basis of $k[x]/\big((x-\lambda)^r\big)$: the polynomials $(x-\lambda)^d$ for $0 \le d < r$ have distinct degrees $0, 1, \dots, r-1$, so they are a basis of the space of polynomials of degree less than r, which maps isomorphically onto the quotient. Writing $x = \lambda + (x-\lambda)$ and setting $e_0 := 0$, which is consistent because $(x-\lambda)^r = 0$ in W,

$$\alpha(e_j) = x \cdot (x-\lambda)^{r-j} = \lambda(x-\lambda)^{r-j} + (x-\lambda)^{r-j+1} = \lambda e_j + e_{j-1}$$

So the matrix of $\alpha|_W$ in the ordered basis $(e_1, \dots, e_r)$ has λ in every diagonal entry and 1 in every entry of the first superdiagonal, which is the Jordan block $J_{\lambda,r}$ of definition 14.4.2.

Step 3: assembly. Concatenating the bases of the summands gives a basis of V in which the matrix of α is block diagonal with the blocks of Step 2, which is the Jordan canonical form of definition 14.4.3. Uniqueness of the multiset of blocks is the uniqueness of the elementary divisors, since the block $J_{\lambda,r}$ corresponds to the elementary divisor $(x-\lambda)^r$, as we sought to show.

Only the order of the blocks is undetermined, which is the ambiguity recorded in box 14.4.4.

An interesting proeprty the JCF gives us is that it distinguishes algebraic and geometric multiplicity of eigenvalues:

Proposition 14.4.6: Geometric An Algebraic Multiplicity Of Eigenvalues Using JCF

Let α be a linear operator on a finitely dimensional vector space V. Then the geometric multiplicity of an eigenvalue λ of α equals the number of Jordan blocks corresponding to λ in the Jordan canonical form of α, and the algebraic multiplicity equals the number of times the eigenvalue shows up in the Jordan canonical form.

***Proof*:**
The fact that the algebraic multiplicity of the eigenvalue corresponds to the number or times it shows up in the JCF is immediate, hence let's show the number of Jordan blocks equasl the geometric multiplicity. Let:

$$J =: \begin{pmatrix} \lambda & 1 & \cdots & 0 & 0 \\ 0 & \lambda & \cdots & 0 & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \cdots & \lambda & 1 \\ 0 & 0 & \cdots & 0 & \lambda \end{pmatrix}$$

be a Jordan block and $v = \begin{pmatrix} v_1 \\ \vdots \\ v_r \end{pmatrix}$ be an eigenvector corresponding to λ. Then:

$$\lambda \begin{pmatrix} v_1 \\ v_2 \\ \vdots \\ v_r \end{pmatrix} = J \begin{pmatrix} v_1 \\ v_2 \\ \vdots \\ v_r \end{pmatrix} = \begin{pmatrix} \lambda v_1 + v_2 \\ \lambda v_2 + v_3 \\ \vdots \\ \lambda v_r \end{pmatrix}$$

which solving gives us:

$$v_2 = \cdots = v_r = 0$$

Hence, the eigenspace of λ is generated by:

$$v = \begin{pmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix}$$

completing the proof.

Corollary 14.4.7: Diagonalizability And Minimal Polynomial

Assume the characteristic polynomial of $\alpha \in \mathrm{End}_k(V)$ factors completely over k. Then α is diagonalizable iff the minimal polynomial of α has no multiple roots

***Proof*:**
exercise

Naturally, the diagonalizability of a matrix also depends on the base field, for example the rotation by $\pi/2$ degrees for $\mathbb{R}^2$ is not diagonalizable, while it is diagonalizable over $\mathbb{C}$

14.4.1 Changing From one Canonical Form to Another

see D& F

14.4.2 Computation: Convert to JCF

HERE

15

Tensor Products Of Modules: Linear Algebra III

The results of chapter 13 and chapter 14 describe modules and the linear maps between them, and several of the objects those chapters produced are not linear maps. The determinant of section 13.4 takes n column vectors and is linear in each one separately, not in all of them together. An inner product (section 13.6) takes two vectors and is linear in each. The consequence is that such maps are not morphisms in Vect_k, so no theorem about morphisms applies to them. Multiplication in a ring is the same phenomenon at its most basic. The map $R \times R \to R$ sending (a, b) to ab is additive in each variable when the other is held fixed, and it is not additive in the pair: the product $(a + a')(b + b')$ is $ab + ab' + a'b + a'b'$, not $ab + a'b'$.

This chapter removes the obstruction by changing the domain rather than the map in a natural way. From two modules M and N over a ring R it constructs a single module $M \otimes_R N$, together with a bilinear map $M \times N \to M \otimes_R N$ through which every bilinear map out of $M \times N$ factors, uniquely, as a homomorphism out of $M \otimes_R N$. A question about bilinear maps on $M \times N$ then becomes a question about homomorphisms on $M \otimes_R N$, where the machinery of the previous two chapters applies. Iterating the construction on a single module produces the tensor algebra $\mathcal{T}(M)$ and its two quotients, the symmetric algebra $\mathcal{S}(M)$ and the exterior algebra $\bigwedge(M)$. The determinant reappears in section 16.4 as the map an endomorphism induces on the top exterior power.

15.1 The Tensor Product of Modules

Suppose for a moment that the module $M \otimes_R N$ has been built, along with a map $\iota\colon M \times N \to M \otimes_R N$ whose value at (m, n) is written $m \otimes n$. Asking that ι be additive in each variable forces

$$(m_1 + m_2) \otimes n = m_1 \otimes n + m_2 \otimes n, \qquad m \otimes (n_1 + n_2) = m \otimes n_1 + m \otimes n_2$$

giving relations any construction must impose. The condition governing scalars needs more care. Take R commutative first, with M, N and the target L all R-modules. A map $\varphi\colon M \times N \to L$ that is R-linear in each variable satisfies

$$\varphi(rm, n) = r\varphi(m, n) = \varphi(m, rn)$$

so a scalar may be moved from one argument to the other. Over a non-commutative ring that same conditin is nearly vacuous. Suppose M, N and L are left R-modules and φ is R-linear in each variable. Evaluating $\varphi(rm, sn)$ two ways, once by extracting s from the second argument and once by extracting r from the first, gives

$$(sr)\varphi(m, n) = s\varphi(rm, n) = \varphi(rm, sn) = r\varphi(m, sn) = (rs)\varphi(m, n)$$

so that $(rs - sr)\varphi(m, n) = 0$ for every $m \in M$ and $n \in N$. Any such φ is annihilated by every commutator in R, and over a ring with few central elements this leaves almost nothing.

The repair is to weaken the condition in two places. The target drops from an R-module to an abelian group, so that "pulling out a scalar" is no longer available and cannot be contradicted. And the two modules are taken on opposite sides, M a right R-module and N a left one, so that the surviving requirement $\varphi(mr, n) = \varphi(m, rn)$ moves a scalar across the comma without ever moving it outside. This is the condition the construction can enforce.

Definition 15.1.1: R-Balanced Map

Let R be a ring, let M be a right R-module, let N be a left R-module, and let L be an abelian group written additively. A map $\varphi\colon M \times N \to L$ is *R-balanced*, or *middle linear over R*, if for all $m, m_1, m_2 \in M$, all $n, n_1, n_2 \in N$ and all $r \in R$,

$$\begin{aligned}
\varphi(m_1 + m_2, n) &= \varphi(m_1, n) + \varphi(m_2, n)\\
\varphi(m, n_1 + n_2) &= \varphi(m, n_1) + \varphi(m, n_2)\\
\varphi(mr, n) &= \varphi(m, rn)
\end{aligned}$$

The third axiom is the only one that mentions R at all, and it is weaker than linearity in either variable, since it never asserts what happens to a scalar that is pulled out of the map, namely as L has no module structure.

Example 15.1.2: R-Balanced Maps

1. Let R be commutative and let M, N, L be R-modules, with $mr := rm$ making M a right R-module. Every map that is R-linear in each variable is R-balanced: $\varphi(mr, n) = \varphi(rm, n) = r\varphi(m, n) = \varphi(m, rn)$. So over a commutative ring the balanced condition is implied by, and strictly weaker than, bilinearity.

2. For any ring R and any right R-module M, the action map $M \times R \to M$ sending (m, r) to mr is R-balanced, where R is viewed as a left module over itself. Additivity in each variable is a module axiom, and the third axiom is associativity of the action: $(ms)r = m(sr)$.

3. Take $R = \mathbb{Z}$. Every $\mathbb{Z}$-module is an abelian group and every abelian group is a $\mathbb{Z}$-module (section 12.1), so a $\mathbb{Z}$-balanced map is exactly a map additive in each variable: the third axiom, $\varphi(mk, n) = \varphi(m, kn)$ for $k \in \mathbb{Z}$, already follows from the first two. Balanced maps over $\mathbb{Z}$ therefore impose no condition beyond biadditivity.

4. The determinant of a 2×2 matrix, read as a map $F^2 \times F^2 \to F$ sending the pair of columns $\big((a,b),(c,d)\big)$ to $ad - bc$, is F-bilinear (section 13.4) and hence F-balanced. It is one of the maps this chapter exists to bring inside linear algebra.

The construction now writes down the freest object carrying such a map. Start with a group in which the pairs (m,n) satisfy nothing whatsoever, then impose the three axioms of definition 15.1.1 by hand. Write $F^{\mathbb{Z}}(A)$ for the free $\mathbb{Z}$-module on a set A (theorem 12.3.11), that is, the free abelian group with basis A.

Applied to the underlying *set* of $M \times N$, this group is enormous. If $R = \mathbb{R}$, $M = \mathbb{R}^2$ and $N = \mathbb{R}^3$, the basis of $F^{\mathbb{Z}}(M \times N)$ is the whole of $\mathbb{R}^2 \times \mathbb{R}^3$, so distinct pairs (m,n) and (m',n') are independent generators even when m and m' differ by a scalar. Nothing that holds in M or in N survives into $F^{\mathbb{Z}}(M \times N)$; every relation the tensor product satisfies has to be put in by quotienting.

Definition 15.1.3: Tensor Product

Let R be a ring, let M be a right R-module and let N be a left R-module. Let $K \subseteq F^{\mathbb{Z}}(M \times N)$ be the subgroup generated by all elements of the three forms

$$\begin{aligned}
&(m_1 + m_2,\, n) - (m_1, n) - (m_2, n)\\
&(m,\, n_1 + n_2) - (m, n_1) - (m, n_2)\\
&(mr,\, n) - (m, rn)
\end{aligned}$$

taken over all $m, m_1, m_2 \in M$, all $n, n_1, n_2 \in N$ and all $r \in R$. The *tensor product of M and N over R* is the quotient group

$$M \otimes_R N := F^{\mathbb{Z}}(M \times N)/K$$

The coset of the basis element (m,n) is written $m \otimes n$ and is called a *simple tensor*; the subscript R is dropped when the ring is clear from context.

Killing K is exactly what makes the three defining identities hold in the quotient, so that for all $m, m_i \in M$, $n, n_i \in N$ and $r \in R$,

$$\begin{gathered}
(m_1 + m_2) \otimes n = m_1 \otimes n + m_2 \otimes n \qquad m \otimes (n_1 + n_2) = m \otimes n_1 + m \otimes n_2\\
mr \otimes n = m \otimes rn
\end{gathered} \tag{15.1}$$

These three are the only relations available. Anything else true in $M \otimes_R N$ has to be derived from them, and the next lemma is the derivation used most often.

Lemma 15.1.4: Simple Tensors Generate

Let M be a right R-module and N a left R-module. Then

1. $0 \otimes n = 0$ and $m \otimes 0 = 0$ for all $m \in M$, $n \in N$;
2. every element of $M \otimes_R N$ is a finite sum $\sum_{i=1}^{k} m_i \otimes n_i$ of simple tensors;
3. two group homomorphisms $M \otimes_R N \to L$ that agree on every simple tensor are equal.

***Proof*:**

1. By the first identity in (15.1) applied to $m_1 = m_2 = 0$, one has $0 \otimes n = (0+0) \otimes n = 0 \otimes n + 0 \otimes n$, and subtracting $0 \otimes n$ from both sides in the group $M \otimes_R N$ gives $0 \otimes n = 0$. The argument for $m \otimes 0$ is the same using the second identity.

2. The group $F^{\mathbb{Z}}(M \times N)$ is generated by its basis, so every element of it is a finite sum $\sum_i k_i(m_i, n_i)$ with $k_i \in \mathbb{Z}$, and the quotient map is surjective, so every element of $M \otimes_R N$ is a finite sum $\sum_i k_i(m_i \otimes n_i)$. It therefore suffices to see that an integer multiple of a simple tensor is again a simple tensor. For $k \geq 0$ induction on k using the first identity in (15.1) gives $k(m \otimes n) = (km) \otimes n$, the base case $k = 0$ being part (1). For $k < 0$ it suffices to treat $k = -1$, and

$$(-m) \otimes n + m \otimes n = (-m + m) \otimes n = 0 \otimes n = 0$$

gives $(-m) \otimes n = -(m \otimes n)$, so $k(m \otimes n) = (km) \otimes n$ for every $k \in \mathbb{Z}$.

3. Let ψ, ψ' be two such homomorphisms. The set on which two group homomorphisms agree is a subgroup of the domain, and by (2) the simple tensors generate $M \otimes_R N$, so that subgroup is everything, completing the proof.

Part (2) says that every element of the tensor product is a finite sum of simple tensors; it does not say that every element *is* a simple tensor, and that distinction is the most common source of error in computations with tensor products; box 15.1.6 exhibits an element that is not simple.

The definition of $M \otimes_R N$ was engineered to make ι balanced and to impose nothing further. The following theorem is the statement that it succeeded, and it is the only property of the tensor product used from here on: every later result in this chapter is proved from it rather than from the quotient construction.

Theorem 15.1.5: Universal Property of the Tensor Product

Let R be a ring, let M be a right R-module and let N be a left R-module. The map

$$\iota \colon M \times N \to M \otimes_R N, \qquad \iota(m, n) = m \otimes n$$

is R-balanced, and it is universal among such maps: for every abelian group L and every R-balanced map $\varphi \colon M \times N \to L$ there is a unique group homomorphism $\psi \colon M \otimes_R N \to L$ with $\varphi = \psi \circ \iota$, that is, making

$$\begin{tikzcd} M \otimes_R N \arrow[r, "\psi"] & L \\ M \times N \arrow[u, "\iota"] \arrow[ur, "\varphi"'] & \end{tikzcd}$$

commute.

***Proof*:**
That ι is R-balanced is precisely the three identities (15.1), which hold because the corresponding elements of $F^{\mathbb{Z}}(M \times N)$ were placed in K.

Existence. Regard $M \times N$ as a set and φ as a map of sets into the abelian group L, which is a $\mathbb{Z}$-module. By the universal property of free modules (theorem 12.3.11, applied with $R = \mathbb{Z}$) there is a group homomorphism $\tilde{\varphi}\colon F^{\mathbb{Z}}(M \times N) \to L$ with $\tilde{\varphi}\big((m,n)\big) = \varphi(m,n)$ on each basis element. Evaluating $\tilde{\varphi}$ on the three families of generators of K gives

$$\begin{aligned}
\tilde{\varphi}\big((m_1+m_2,n)-(m_1,n)-(m_2,n)\big) &= \varphi(m_1+m_2,n)-\varphi(m_1,n)-\varphi(m_2,n) = 0\\
\tilde{\varphi}\big((m,n_1+n_2)-(m,n_1)-(m,n_2)\big) &= \varphi(m,n_1+n_2)-\varphi(m,n_1)-\varphi(m,n_2) = 0\\
\tilde{\varphi}\big((mr,n)-(m,rn)\big) &= \varphi(mr,n)-\varphi(m,rn) = 0
\end{aligned}$$

each vanishing by the corresponding axiom of definition 15.1.1. A subgroup is generated by any set whose elements it contains together with their sums and inverses, and $\ker\tilde{\varphi}$ is a subgroup containing all three families, so $K \subseteq \ker\tilde{\varphi}$. Since $F^{\mathbb{Z}}(M \times N)$ is abelian, K is normal in it, so theorem 3.1.14 applies and yields a unique map $\psi\colon F^{\mathbb{Z}}(M\times N)/K \to L$ with $\tilde{\varphi} = \psi \circ \pi$, where π is the quotient map. It is a group homomorphism, since for $x, y \in F^{\mathbb{Z}}(M \times N)$,

$$\psi\big(\pi(x)+\pi(y)\big) = \psi\big(\pi(x+y)\big) = \tilde{\varphi}(x+y) = \tilde{\varphi}(x)+\tilde{\varphi}(y) = \psi(\pi(x))+\psi(\pi(y))$$

and π is surjective. Reading the identity $\tilde{\varphi} = \psi\circ\pi$ on the basis element (m,n) gives $\psi(m\otimes n) = \varphi(m,n)$, which is $\varphi = \psi \circ \iota$.

Uniqueness. A homomorphism ψ' with $\varphi = \psi' \circ \iota$ satisfies $\psi'(m\otimes n) = \varphi(m,n) = \psi(m\otimes n)$ for every simple tensor, so $\psi' = \psi$ by lemma 15.1.4, completing the proof.

The theorem is what converts the tensor product from a construction into a tool. To produce a homomorphism out of $M \otimes_R N$ one never returns to cosets of a free group: one writes down a balanced map on $M \times N$, checks three identities, and reads off the homomorphism. To prove two homomorphisms out of $M \otimes_R N$ equal, one checks them on simple tensors. Both moves are used repeatedly below, starting immediately.

15.1.6 Warning: Not Every Tensor Is Simple Let F be a field, let $V = F^2$ with standard basis e_1, e_2, and consider

$$t = e_1 \otimes e_1 + e_2 \otimes e_2 \in V \otimes_F V$$

Then t is not a simple tensor. Suppose $t = v \otimes w$. For linear functionals $f, g \in V^*$ (section 13.3) the map $V \times V \to F$ sending (x,y) to $f(x)g(y)$ is F-bilinear, hence F-balanced, so by theorem 15.1.5 below it induces a homomorphism $\theta_{f,g}\colon V \otimes_F V \to F$ with $\theta_{f,g}(x\otimes y) = f(x)g(y)$. Evaluating on both expressions for t,

$$f(v)g(w) = \theta_{f,g}(t) = f(e_1)g(e_1) + f(e_2)g(e_2)$$

Taking (f,g) to be (e_1^*, e_1^*), then (e_2^*, e_2^*), then (e_1^*, e_2^*) gives in turn $v_1w_1 = 1$, $v_2w_2 = 1$ and $v_1w_2 = 0$, where $v = v_1e_1 + v_2e_2$ and $w = w_1e_1 + w_2e_2$. The first two force v_1 and w_2 to be nonzero, contradicting the third.

A consequence worth recording is that the pair $(M \otimes_R N, \iota)$ is pinned down by the theorem, so the arbitrary choices in definition 15.1.3 leave no trace.

Corollary 15.1.7: Uniqueness of the Tensor Product

Let T be an abelian group and $j\colon M \times N \to T$ an R-balanced map with the property that for every abelian group L and every R-balanced $\varphi\colon M \times N \to L$ there is a unique homomorphism $\psi\colon T \to L$ with $\varphi = \psi \circ j$. Then there is a unique isomorphism $\theta\colon M \otimes_R N \to T$ with $\theta \circ \iota = j$.

***Proof*:**
Applying theorem 15.1.5 to the balanced map j gives a homomorphism $\theta\colon M \otimes_R N \to T$ with $\theta \circ \iota = j$, and applying the hypothesis on T to the balanced map ι gives $\eta\colon T \to M \otimes_R N$ with $\eta \circ j = \iota$. Then $\eta \circ \theta$ is an endomorphism of $M \otimes_R N$ satisfying $(\eta \circ \theta) \circ \iota = \iota$, and so is the identity; by the uniqueness clause of theorem 15.1.5 applied to $\varphi = \iota$, the two agree. Symmetrically $\theta \circ \eta = \mathrm{id}_T$, using the uniqueness clause assumed for T. Hence θ is an isomorphism, and it is the unique one satisfying $\theta \circ \iota = j$ by that same uniqueness clause, as we sought to show.

So far $M \otimes_R N$ is only an abelian group, and definition 15.1.1 was designed to make that unavoidable: nothing in the data of a right module M and a left module N produces an R-action on the tensor product. A module structure has to come from somewhere else, and the smallest extra hypothesis that supplies one is a second, compatible action on one of the factors.

Definition 15.1.8: (S, R)-Bimodule

Let R and S be rings. An abelian group M is an (S, R)-*bimodule* if it is a left S-module, a right R-module, and the two actions commute:

$$s(mr) = (sm)r \qquad \text{for all } s \in S,\ r \in R,\ m \in M$$

Example 15.1.9: Bimodules

1. Any ring R is an (R, R)-bimodule over itself, with both actions given by ring multiplication; the compatibility axiom is associativity, $s(mr) = (sm)r$.

2. Let $f\colon R \to S$ be a ring homomorphism. Then S is an (S, R)-bimodule: the left S-action is multiplication in S, and the right R-action is $s \cdot r := sf(r)$. Compatibility reads $s'(sf(r)) = (s's)f(r)$, which is associativity in S. This is the structure that makes extension of scalars along f possible, and it is taken up in section 15.3.

3. As a special case of (2), let $I \subseteq R$ be a two-sided ideal and $f\colon R \to R/I$ the quotient map. Then R/I is an $(R/I, R)$-bimodule.

4. Let R be commutative and M an R-module. Setting $mr := rm$ makes M an (R, R)-bimodule, since the compatibility axiom $s(mr) = (sm)r$ reads $(sr)m = (rs)m$. This is called the *standard* bimodule structure on M, and it is the structure meant whenever a module over a commutative ring is treated as a bimodule without further comment.

Proposition 15.1.10: Module Structure on the Tensor Product

Let M be an (S, R)-bimodule and let N be a left R-module. Then $M \otimes_R N$ is a left S-module under the action determined by

$$s \cdot (m \otimes n) = (sm) \otimes n$$

on simple tensors.

***Proof*:**
Fix $s \in S$ and consider $\lambda_s \colon M \times N \to M \otimes_R N$ defined by $\lambda_s(m, n) = (sm) \otimes n$. It is R-balanced. Additivity in the first variable holds because $s(m_1 + m_2) = sm_1 + sm_2$ and $\otimes$ is additive in its first argument; additivity in the second is immediate. For the third axiom, the bimodule condition gives $s(mr) = (sm)r$, so

$$\lambda_s(mr, n) = \big(s(mr)\big) \otimes n = \big((sm)r\big) \otimes n = (sm) \otimes rn = \lambda_s(m, rn)$$

the middle equality being the third relation of (15.1). By theorem 15.1.5 there is therefore a unique group endomorphism Λ_s of $M \otimes_R N$ with $\Lambda_s(m \otimes n) = (sm) \otimes n$. Define $s \cdot x := \Lambda_s(x)$ for $x \in M \otimes_R N$; this is additive in x because Λ_s is a homomorphism.

The remaining module axioms are equalities between group homomorphisms $M \otimes_R N \to M \otimes_R N$, so by lemma 15.1.4 each may be checked on simple tensors alone. On $m \otimes n$,

$$\begin{aligned}
\Lambda_{s+s'}(m \otimes n) &= \big((s + s')m\big) \otimes n = (sm + s'm) \otimes n \\
&= (sm) \otimes n + (s'm) \otimes n = (\Lambda_s + \Lambda_{s'})(m \otimes n) \\
\Lambda_{ss'}(m \otimes n) &= \big((ss')m\big) \otimes n = \big(s(s'm)\big) \otimes n \\
&= \Lambda_s\big((s'm) \otimes n\big) = (\Lambda_s \circ \Lambda_{s'})(m \otimes n) \\
\Lambda_1(m \otimes n) &= (1m) \otimes n = m \otimes n
\end{aligned}$$

using in each case only a left S-module axiom for M. Hence $\Lambda_{s+s'} = \Lambda_s + \Lambda_{s'}$, $\Lambda_{ss'} = \Lambda_s \circ \Lambda_{s'}$ and $\Lambda_1 = \mathrm{id}$, which are exactly the axioms making $M \otimes_R N$ a left S-module, completing the proof.

The proof is a template for everything that follows: a map is defined on simple tensors, shown balanced, promoted to a homomorphism by theorem 15.1.5, and its properties are then checked on simple tensors by lemma 15.1.4. No later argument in this chapter returns to the free group.

Specializing to a commutative ring recovers the situation the chapter opened with, in which both factors and the target are modules over the same ring and the universal property is stated for bilinear maps. That case needs the module-theoretic form of bilinearity, which definition 13.3.15 gave only for vector spaces with target the base field.

Definition 15.1.11: R-Bilinear Map

Let R be a commutative ring and let M, N and L be R-modules. A map $\varphi \colon M \times N \to L$ is *R-bilinear* if for each fixed $n \in N$ the map $\varphi(-, n) \colon M \to L$ is an R-module homomorphism, and for each fixed $m \in M$ the map $\varphi(m, -) \colon N \to L$ is an R-module homomorphism.

Theorem 15.1.12: Universal Property Of Tensor Product (Over Commutative Rings)

Let R be a commutative ring and let M, N be R-modules, each carrying its standard bimodule structure. Then $M \otimes_R N$ is an R-module, the map $\iota\colon M \times N \to M \otimes_R N$ is R-bilinear, and for every R-module L and every R-bilinear map $\varphi\colon M \times N \to L$ there is a unique R-module homomorphism $\psi\colon M \otimes_R N \to L$ with $\varphi = \psi \circ \iota$.

***Proof*:**
The R-module structure is proposition 15.1.10 applied with $S = R$, using the standard bimodule structure on M from example 15.1.9(4); explicitly, $r(m \otimes n) = (rm) \otimes n$. That ι is R-bilinear now follows from (15.1) together with this action: it is additive in each variable, $r(m \otimes n) = (rm) \otimes n$ by definition of the action, and $r(m \otimes n) = (mr) \otimes n = m \otimes rn$ by the third relation.

Let φ be R-bilinear. By example 15.1.2(1) it is R-balanced, so theorem 15.1.5 supplies a unique group homomorphism ψ with $\varphi = \psi \circ \iota$. It remains to see that ψ is R-linear. Fix $r \in R$. The two maps $x \mapsto \psi(rx)$ and $x \mapsto r\psi(x)$ are group homomorphisms $M \otimes_R N \to L$, and on a simple tensor

$$\psi\big(r(m \otimes n)\big) = \psi\big((rm) \otimes n\big) = \varphi(rm, n) = r\varphi(m, n) = r\psi(m \otimes n)$$

so they agree on all simple tensors and hence are equal by lemma 15.1.4. Thus ψ is R-linear. Uniqueness is inherited: an R-module homomorphism satisfying $\varphi = \psi \circ \iota$ is in particular a group homomorphism satisfying it, and there is only one of those, as we sought to show.

Two computations show what the construction does to familiar modules, and they point in opposite directions. The first says that tensoring against the ring itself changes nothing; the second says that tensoring two nonzero modules can annihilate both.

Example 15.1.13: Tensoring Against the Ring

Let R be a ring and N a left R-module. Then $R \otimes_R N \cong N$ as left R-modules, where R carries its (R, R)-bimodule structure from example 15.1.9(1).

The action map $\mu\colon R \times N \to N$ with $\mu(r, n) = rn$ is R-balanced: additivity in each variable is a module axiom, and $\mu(rs, n) = (rs)n = r(sn) = \mu(r, sn)$. By theorem 15.1.5 it induces a group homomorphism $\bar{\mu}\colon R \otimes_R N \to N$ with $\bar{\mu}(r \otimes n) = rn$. In the other direction let $\nu\colon N \to R \otimes_R N$ be $\nu(n) = 1 \otimes n$, which is additive. Then $\bar{\mu}(\nu(n)) = \bar{\mu}(1 \otimes n) = n$, so $\bar{\mu} \circ \nu = \mathrm{id}_N$. For the other composite, on a simple tensor

$$\nu\big(\bar{\mu}(r \otimes n)\big) = \nu(rn) = 1 \otimes rn = 1 \cdot r \otimes n = r \otimes n$$

the third equality being the third relation of (15.1) read backwards. So $\nu \circ \bar{\mu}$ agrees with the identity on simple tensors and is the identity by lemma 15.1.4. Finally $\bar{\mu}$ is R-linear, since $\bar{\mu}\big(s(r \otimes n)\big) = \bar{\mu}\big((sr) \otimes n\big) = (sr)n = s(rn) = s\,\bar{\mu}(r \otimes n)$ on simple tensors.

Example 15.1.14: Tensor Products of Cyclic Groups

1. $\mathbb{Z}/2\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Z}/3\mathbb{Z} = 0$. Since $3 \equiv 1 \pmod 2$, one has $\bar{a} = 3\bar{a}$ in $\mathbb{Z}/2\mathbb{Z}$ for every a, and therefore

$$\bar{a} \otimes \bar{b} = (3\bar{a}) \otimes \bar{b} = \bar{a} \otimes (3\bar{b}) = 0$$

using $3\bar{b} = \bar{0}$ in $\mathbb{Z}/3\mathbb{Z}$. So every simple tensor vanishes, and by lemma 15.1.4(2) the whole group is zero.

2. $\mathbb{Z}/2\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Z}/2\mathbb{Z} \cong \mathbb{Z}/2\mathbb{Z}$. The same manipulation is unavailable here, and in fact $\bar{1} \otimes \bar{1} \neq 0$. To see it, the multiplication map $\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z} \to \mathbb{Z}/2\mathbb{Z}$ sending $(\bar{a}, \bar{b})$ to $\overline{ab}$ is $\mathbb{Z}$-balanced, so it induces Φ with $\Phi(\bar{1} \otimes \bar{1}) = \bar{1} \neq 0$; a homomorphism sends 0 to 0, so $\bar{1} \otimes \bar{1} \neq 0$. Conversely every simple tensor is $\bar{a} \otimes \bar{b}$ with $a, b \in \{0, 1\}$, hence is 0 or $\bar{1} \otimes \bar{1}$, so by lemma 15.1.4(2) the group $\mathbb{Z}/2\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Z}/2\mathbb{Z}$ is cyclic, generated by $t = \bar{1} \otimes \bar{1}$. That generator satisfies $2t = (2\bar{1}) \otimes \bar{1} = \bar{0} \otimes \bar{1} = 0$, so it has order dividing 2, and it is nonzero, so its order is exactly 2. Hence $\mathbb{Z}/2\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Z}/2\mathbb{Z} \cong \mathbb{Z}/2\mathbb{Z}$, with Φ the isomorphism.

The contrast between the two computations is the first real content of the construction. Both $\mathbb{Z}/2\mathbb{Z}$ and $\mathbb{Z}/3\mathbb{Z}$ are nonzero, and their tensor product is not; so $\otimes_R$ is not a construction that remembers its inputs. The mechanism is visible in the computation: the relation $mr \otimes n = m \otimes rn$ lets a scalar be moved from a factor where it acts invertibly to one where it acts as zero. section 15.3 turns this from an obstruction into a tool, and the systematic account of which sequences of modules survive tensoring is the subject of section 17.5.

Example 15.1.15: Tensors as Multilinear Maps

Outside algebra, and particularly in physics and in analysis, the word "tensor" usually names a multilinear map rather than an element of a tensor product. In that convention a k-tensor on a vector space V over F *is* a multilinear map $T\colon V^k \to F$, and the tensor product of a k-tensor T and an ℓ-tensor S is defined directly by the formula

$$(T \otimes S)(x_1, \ldots, x_k, y_1, \ldots, y_\ell) = T(x_1, \ldots, x_k) \cdot S(y_1, \ldots, y_\ell)$$

which is again multilinear, now in $k + \ell$ arguments.

The two usages agree when V is finite dimensional. A multilinear map $V^k \to F$ is the same thing as a linear functional on $V^{\otimes k}$, which is the universal property of theorem 15.2.11 below, and for finite-dimensional V the space of such functionals is identified with $(V^*)^{\otimes k}$ by the basis computation of corollary 15.2.6; under that identification the displayed formula is the image of $T \otimes S$ in the sense of definition 15.1.3. The reason for the analyst's convention is that a multilinear map has a domain and a target, so conditions like continuity, differentiability, or being C^r can be imposed on it; an element of an abstract quotient group cannot carry such conditions until it is realized as a function. This is the form in which tensors appear in *Calculus on Manifolds* [8].

Exercise 15.1.1

1. Let R be a ring and M a right R-module, with R carrying its (R, R)-bimodule structure. Show that $M \otimes_R R \cong M$, and identify the isomorphism on simple tensors.

2. Let $m, n \geq 1$ and let $d = \gcd(m, n)$. Show that

$$\mathbb{Z}/m\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Z}/n\mathbb{Z} \cong \mathbb{Z}/d\mathbb{Z}$$

(Suggestion: since $\mathbb{Z}$ is a principal ideal domain, $(d) = (m) + (n)$ by definition 10.1.24, so $d = um + vn$ for some $u, v \in \mathbb{Z}$; use this to show first that d annihilates the left-hand side.)

3. Let R be a ring, M a right R-module and N a left R-module, and suppose there is an $r \in R$

with $Mr = 0$ and $rN = N$. Show that $M \otimes_R N = 0$. Deduce that $\mathbb{Q} \otimes_{\mathbb{Z}} A = 0$ for every finite abelian group A, and that $\mathbb{Q}/\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Q}/\mathbb{Z} = 0$.

4. Let R be commutative, let M be generated as an R-module by a set X and let N be generated by a set Y. Show that $M \otimes_R N$ is generated as an R-module by the simple tensors $x \otimes y$ with $x \in X$ and $y \in Y$. Give an example showing that these need not be independent.
5. Let M be a right R-module, N a left R-module and L an abelian group. Show that $\varphi \mapsto \psi$ is a bijection from the set of R-balanced maps $M \times N \to L$ onto $\mathrm{Hom}_{\mathbb{Z}}(M \otimes_R N, L)$, and that it commutes with composition by a homomorphism $L \to L'$.
6. Let R be commutative, let A, B, M be R-modules and let $f\colon A \to M$ and $g\colon B \to M$ be R-linear. Show that $(a, b) \mapsto f(a) + g(b)$ is an R-module homomorphism $A \oplus B \to M$, and exhibit f and g for which it is not R-bilinear. Conclude that $A \oplus B$ and $A \otimes_R B$ solve different universal problems.

15.2 Properties of the Tensor Product

With one exception, named where it occurs, no proof below returns to the free abelian group of section 15.1. Everything is derived from theorem 15.1.5 and lemma 15.1.4, by the two moves those results supply: to produce a map out of a tensor product, write down a balanced map and invoke the universal property; to prove two maps out of a tensor product equal, check them on simple tensors. What follows records how $\otimes_R$ interacts with each construction the previous chapters supplied, in the order in which the proofs depend on one another: homomorphisms, direct sums, free modules, and then the two structural isomorphisms, commutativity and associativity, that make an unbracketed $M_1 \otimes \cdots \otimes M_n$ meaningful. The section closes with the ring structure carried by a tensor product of algebras.

proposition 15.1.10 produced a left module structure from a second action on the left-hand factor. A second action on the right-hand factor does the mirror-image job, and the two are independent, so both can be imposed at once. The right action is what theorem 15.2.8 needs, since there the middle factor of a triple product is acted on from both sides.

Proposition 15.2.1: Two-Sided Module Structure on a Tensor Product

Let R, S, T be rings, let M be an (S, R)-bimodule and let N be an (R, T)-bimodule. Then $M \otimes_R N$ is an (S, T)-bimodule, with actions determined on simple tensors by

$$s \cdot (m \otimes n) = (sm) \otimes n \qquad \text{and} \qquad (m \otimes n) \cdot t = m \otimes (nt)$$

***Proof*:**
The left S-module structure is proposition 15.1.10, which uses only that M is an (S, R)-bimodule and N a left R-module.

For the right T-action, fix $t \in T$ and consider $\rho_t\colon M \times N \to M \otimes_R N$ given by $\rho_t(m, n) = m \otimes (nt)$. It is R-balanced: additivity in m is additivity of $\otimes$ in its first argument; additivity in n follows because $n \mapsto nt$ is additive and $\otimes$ is additive in its second argument; and for the third axiom the (R,T)-bimodule condition on N gives $r(nt) = (rn)t$, so

$$\rho_t(mr, n) = (mr) \otimes (nt) = m \otimes r(nt) = m \otimes (rn)t = \rho_t(m, rn)$$

the middle equality being the third relation of (15.1). By theorem 15.1.5 there is a unique group endomorphism P_t of $M \otimes_R N$ with $P_t(m \otimes n) = m \otimes nt$; set $x \cdot t := P_t(x)$, which is additive in x because P_t is a homomorphism.

The remaining right-module axioms are equalities of group homomorphisms, so by lemma 15.1.4(3) each may be checked on simple tensors:

$$\begin{aligned} P_{t+t'}(m \otimes n) &= m \otimes n(t+t') = m \otimes (nt + nt') = m \otimes nt + m \otimes nt' = (P_t + P_{t'})(m \otimes n) \\ P_{tt'}(m \otimes n) &= m \otimes n(tt') = m \otimes (nt)t' = P_{t'}(m \otimes nt) = (P_{t'} \circ P_t)(m \otimes n) \\ P_1(m \otimes n) &= m \otimes n1 = m \otimes n \end{aligned}$$

The second line gives $x(tt') = (xt)t'$, which is the associativity axiom for a *right* action, the order of composition being reversed exactly as it should be. Finally the two actions commute: $x \mapsto s(xt)$ and $x \mapsto (sx)t$ are group homomorphisms, and on a simple tensor both give $(sm) \otimes (nt)$, so they agree by lemma 15.1.4(3), completing the proof.

Homomorphisms of the two factors induce a homomorphism of the tensor product, and the assignment respects composition. This is what makes $\otimes_R$ usable inside diagrams rather than only on individual modules.

Theorem 15.2.2: The Tensor Product of Two Homomorphisms

Let $\varphi\colon M \to M'$ be a homomorphism of right R-modules and let $\psi\colon N \to N'$ be a homomorphism of left R-modules.

1. There is a unique group homomorphism $\varphi \otimes \psi\colon M \otimes_R N \to M' \otimes_R N'$ with

$$(\varphi \otimes \psi)(m \otimes n) = \varphi(m) \otimes \psi(n)$$

for all $m \in M$ and $n \in N$.

2. If M and M' are (S,R)-bimodules and φ is also a homomorphism of left S-modules, then $\varphi \otimes \psi$ is a homomorphism of left S-modules. In particular, if R is commutative and all four modules carry their standard structures, then $\varphi \otimes \psi$ is R-linear.

3. $\mathrm{id}_M \otimes \mathrm{id}_N = \mathrm{id}_{M \otimes_R N}$, and if $\lambda\colon M' \to M''$ and $\mu\colon N' \to N''$ are further homomorphisms of right and left R-modules respectively, then

$$(\lambda \otimes \mu) \circ (\varphi \otimes \psi) = (\lambda \circ \varphi) \otimes (\mu \circ \psi)$$

***Proof*:**

1. The map $M \times N \to M' \otimes_R N'$ sending (m,n) to $\varphi(m) \otimes \psi(n)$ is R-balanced. Additivity in each variable follows from additivity of φ and ψ together with additivity of $\otimes$ in each argument. For the third axiom, φ is a homomorphism of right R-modules and ψ of left

R-modules, so

$$\varphi(mr) \otimes \psi(n) = \big(\varphi(m)r\big) \otimes \psi(n) = \varphi(m) \otimes r\psi(n) = \varphi(m) \otimes \psi(rn)$$

the middle equality being the third relation of (15.1) in $M' \otimes_R N'$. theorem 15.1.5 now gives the homomorphism, and its uniqueness.

2. Both $x \mapsto (\varphi \otimes \psi)(sx)$ and $x \mapsto s(\varphi \otimes \psi)(x)$ are group homomorphisms $M \otimes_R N \to M' \otimes_R N'$, so by lemma 15.1.4(3) it suffices to compare them on a simple tensor, where

$$(\varphi \otimes \psi)\big(s(m \otimes n)\big) = (\varphi \otimes \psi)\big((sm) \otimes n\big) = \varphi(sm) \otimes \psi(n) = \big(s\varphi(m)\big) \otimes \psi(n) = s\big(\varphi(m) \otimes \psi(n)\big)$$

using S-linearity of φ in the third step and proposition 15.1.10 in the fourth. When R is commutative and the structures are standard, an R-module homomorphism is S-linear for $S = R$, so the statement applies with $S = R$.

3. All three claims are equalities of group homomorphisms out of $M \otimes_R N$, so lemma 15.1.4(3) reduces each to a check on simple tensors. There $(\mathrm{id} \otimes \mathrm{id})(m \otimes n) = m \otimes n$, and

$$(\lambda \otimes \mu)\big((\varphi \otimes \psi)(m \otimes n)\big) = (\lambda \otimes \mu)\big(\varphi(m) \otimes \psi(n)\big) = \lambda(\varphi(m)) \otimes \mu(\psi(n))$$

which is the value of $(\lambda \circ \varphi) \otimes (\mu \circ \psi)$ on $m \otimes n$, completing the proof.

Parts (1) and (3) say precisely that fixing a left R-module N makes $- \otimes_R N$ a functor from right R-modules to abelian groups, and symmetrically for $M \otimes_R -$. The question of how this functor treats exact sequences is left open here and settled in section 17.5; the computation $\mathbb{Z}/2\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Z}/3\mathbb{Z} = 0$ of example 15.1.14 is already a warning that it does not treat them well.

The next two results compute a tensor product one summand at a time.

Theorem 15.2.3: Tensor Products of Direct Sums

Let M, M' be right R-modules and let N, N' be left R-modules. There are unique group isomorphisms

$$(M \oplus M') \otimes_R N \cong (M \otimes_R N) \oplus (M' \otimes_R N)$$
$$M \otimes_R (N \oplus N') \cong (M \otimes_R N) \oplus (M \otimes_R N')$$

determined on simple tensors by $(m, m') \otimes n \mapsto (m \otimes n, m' \otimes n)$ and $m \otimes (n, n') \mapsto (m \otimes n, m \otimes n')$ respectively. If M and M' are (S, R)-bimodules, the first is an isomorphism of left S-modules; if R is commutative and all modules carry their standard structures, both are isomorphisms of R-modules.

***Proof*:**
Consider $\alpha\colon (M \oplus M') \times N \to (M \otimes_R N) \oplus (M' \otimes_R N)$ given by $\alpha\big((m, m'), n\big) = (m \otimes n, m' \otimes n)$. It is R-balanced, each axiom holding in both coordinates at once; for the third,

$$\alpha\big((m, m')r, n\big) = \big(mr \otimes n,\ m'r \otimes n\big) = \big(m \otimes rn,\ m' \otimes rn\big) = \alpha\big((m, m'), rn\big)$$

since the action on $M \oplus M'$ is componentwise. By theorem 15.1.5 there is a group homomorphism $\bar{\alpha}$ on $(M \oplus M') \otimes_R N$ with the stated values.

For the inverse, the maps $\beta_1\colon M \times N \to (M \oplus M') \otimes_R N$ with $\beta_1(m,n) = (m,0) \otimes n$ and $\beta_2\colon M' \times N \to (M \oplus M') \otimes_R N$ with $\beta_2(m',n) = (0,m') \otimes n$ are R-balanced, because $m \mapsto (m,0)$ and $m' \mapsto (0,m')$ are homomorphisms of right R-modules; so they induce group homomorphisms $\bar\beta_1$ and $\bar\beta_2$. Define

$$\beta\colon (M \otimes_R N) \oplus (M' \otimes_R N) \to (M \oplus M') \otimes_R N, \qquad \beta(x,y) = \bar\beta_1(x) + \bar\beta_2(y)$$

which is a group homomorphism.

On a simple tensor,

$$\beta\big(\bar\alpha((m,m') \otimes n)\big) = (m,0) \otimes n + (0,m') \otimes n = \big((m,0) + (0,m')\big) \otimes n = (m,m') \otimes n$$

so $\beta \circ \bar\alpha = \mathrm{id}$ by lemma 15.1.4(3). In the other direction, lemma 15.1.4(2) applied in each summand shows that $(M \otimes_R N) \oplus (M' \otimes_R N)$ is generated as a group by the elements $(x,0)$ and $(0,y)$ with x and y simple tensors, so it is enough to evaluate $\bar\alpha \circ \beta$ on those. On $(m \otimes n, 0)$,

$$\bar\alpha\big(\bar\beta_1(m \otimes n)\big) = \bar\alpha\big((m,0) \otimes n\big) = (m \otimes n,\, 0 \otimes n) = (m \otimes n,\, 0)$$

by lemma 15.1.4(1), and the computation on $(0, m' \otimes n)$ is the same. Hence $\bar\alpha \circ \beta = \mathrm{id}$ and $\bar\alpha$ is an isomorphism.

When M, M' are (S,R)-bimodules, so is $M \oplus M'$, and S-linearity of $\bar\alpha$ is again an equality of group homomorphisms checked on simple tensors, where both sides give $\big((sm) \otimes n, (sm') \otimes n\big)$. Uniqueness in all cases is lemma 15.1.4(3).

The second isomorphism is the mirror image of the first: interchange the two factors throughout, replacing each appeal to additivity in the first argument of $\otimes$ by additivity in the second, and proposition 15.1.10 by proposition 15.2.1. No new step enters, completing the proof.

Theorem 15.2.4: Arbitrary Direct Sums Commute with Tensor Products

Let $\{M_i\}_{i \in I}$ be a family of right R-modules and let N be a left R-module. Then there is a unique group isomorphism

$$\left(\bigoplus_{i \in I} M_i\right) \otimes_R N \cong \bigoplus_{i \in I} (M_i \otimes_R N)$$

sending $(m_i)_i \otimes n$ to $(m_i \otimes n)_i$. If each M_i is an (S,R)-bimodule it is an isomorphism of left S-modules, and if R is commutative it is an isomorphism of R-modules.

***Proof*:**
Write $M = \bigoplus_{i \in I} M_i$, whose elements are the families $(m_i)_i$ with $m_i = 0$ for all but finitely many i. The assignment $\alpha\big((m_i)_i, n\big) = (m_i \otimes n)_i$ lands in the direct sum rather than the product, since $m_i = 0$ forces $m_i \otimes n = 0$ by lemma 15.1.4(1), so all but finitely many entries vanish. It is R-balanced for the same componentwise reason as in theorem 15.2.3, and so induces $\bar\alpha$ on $M \otimes_R N$.

Let $\kappa_j\colon M_j \to M$ be the inclusion of the j-th summand, a homomorphism of right R-modules. Each $\beta_j\colon M_j \times N \to M \otimes_R N$ given by $\beta_j(m,n) = \kappa_j(m) \otimes n$ is therefore R-balanced, inducing

$\bar{\beta}_j$. Define

$$\beta\colon \bigoplus_{i\in I}(M_i \otimes_R N) \to M \otimes_R N, \qquad \beta\big((x_i)_i\big) = \sum_{i\in I} \bar{\beta}_i(x_i)$$

a finite sum because $(x_i)_i$ has finite support, and a group homomorphism because each $\bar{\beta}_i$ is. On a simple tensor, writing $(m_i)_i = \sum_i \kappa_i(m_i)$ as the finite sum it is,

$$\beta\big(\bar{\alpha}((m_i)_i \otimes n)\big) = \sum_i \kappa_i(m_i) \otimes n = \left(\sum_i \kappa_i(m_i)\right) \otimes n = (m_i)_i \otimes n$$

using additivity of $\otimes$ in its first argument, so $\beta \circ \bar{\alpha} = \mathrm{id}$. Conversely $\bigoplus_i (M_i \otimes_R N)$ is generated as a group by the elements $\kappa'_j(x)$ with x a simple tensor of $M_j \otimes_R N$, where κ'_j is the inclusion of the j-th summand, and

$$\bar{\alpha}\big(\beta(\kappa'_j(m \otimes n))\big) = \bar{\alpha}\big(\kappa_j(m) \otimes n\big) = \kappa'_j(m \otimes n)$$

since every entry of $\kappa_j(m)$ other than the j-th is zero and therefore contributes $0 \otimes n = 0$. Hence $\bar{\alpha} \circ \beta = \mathrm{id}$. The bimodule claims and uniqueness are as in theorem 15.2.3, completing the proof.

Theorem 15.2.4 decomposes a direct sum in the *left* argument, and theorem 15.2.3 handles two summands on either side. The remaining case, an arbitrary direct sum in the right argument, is what a free presentation of a module needs, so it is recorded separately.

Corollary 15.2.5: Arbitrary Direct Sums in the Right Argument

Let M be a right R-module and $\{N_i\}_{i\in I}$ a family of left R-modules. Then there is a unique group isomorphism

$$M \otimes_R \Big(\bigoplus_{i\in I} N_i\Big) \cong \bigoplus_{i\in I}(M \otimes_R N_i)$$

sending $m \otimes (n_i)_i$ to $(m \otimes n_i)_i$. If M is an (S,R)-bimodule it is an isomorphism of left S-modules.

***Proof*:**
Write $N = \bigoplus_{i\in I} N_i$. The assignment $\alpha\big(m, (n_i)_i\big) = (m \otimes n_i)_i$ lands in the direct sum rather than the product, since $n_i = 0$ for all but finitely many i and $m \otimes 0 = 0$ by lemma 15.1.4(1). It is additive in each variable componentwise, and R-balanced because

$$\alpha\big(mr, (n_i)_i\big) = (mr \otimes n_i)_i = (m \otimes rn_i)_i = \alpha\big(m, r \cdot (n_i)_i\big)$$

so it induces $\bar{\alpha}\colon M \otimes_R N \to \bigoplus_i (M \otimes_R N_i)$.

For the inverse, let $\kappa_j\colon N_j \to N$ be the inclusion of the j-th summand, a map of left R-modules. Each $\beta_j\colon M \times N_j \to M \otimes_R N$ given by $\beta_j(m,n) = m \otimes \kappa_j(n)$ is R-balanced, hence induces $\bar{\beta}_j$ on $M \otimes_R N_j$, and these assemble into a map β on the direct sum.

The composites are the identity on simple tensors, which suffices by lemma 15.1.4. Writing $(n_i)_i = \sum_{j\in F} \kappa_j(n_j)$ over the finite set F of nonzero entries,

$$\beta\big(\bar{\alpha}(m \otimes (n_i)_i)\big) = \sum_{j\in F} m \otimes \kappa_j(n_j) = m \otimes \sum_{j\in F} \kappa_j(n_j) = m \otimes (n_i)_i$$

and conversely $\bar{\alpha}\big(\beta(\kappa'_j(m \otimes n))\big) = \bar{\alpha}\big(m \otimes \kappa_j(n)\big) = \kappa'_j(m \otimes n)$, every entry of $\kappa_j(n)$ other than the j-th being zero and so contributing $m \otimes 0 = 0$. The S-linearity claim holds because the left S-action on both sides is $s \cdot (m \otimes n) = (sm) \otimes n$ by proposition 15.2.1, which $\bar{\alpha}$ respects, completing the proof.

Applying theorem 15.2.4 to a free module, where every summand is a copy of the ring, computes the tensor product of two free modules completely. This is the result that makes tensor products calculable in linear algebra: it says that a basis of M and a basis of N determine a basis of $M \otimes_R N$, indexed by pairs.

Corollary 15.2.6: Tensor Products of Free Modules

Let R be a commutative ring, let M be a free R-module with basis $\{m_i\}_{i \in I}$ and let N be a free R-module with basis $\{n_j\}_{j \in J}$. Then $M \otimes_R N$ is a free R-module with basis

$$\{m_i \otimes n_j\}_{(i,j) \in I \times J}$$

In particular $R^s \otimes_R R^t \cong R^{st}$, and if V, W are finite-dimensional vector spaces over a field F then

$$\dim_F(V \otimes_F W) = \dim_F(V) \cdot \dim_F(W)$$

***Proof*:**
Since $\{m_i\}$ is a basis, $M = \bigoplus_{i \in I} Rm_i$ with each $Rm_i \cong R$ as an R-module, the isomorphism carrying m_i to 1 (theorem 12.3.11). Applying theorem 15.2.4, and then example 15.1.13 to each summand after transporting it along $Rm_i \cong R$ by theorem 15.2.2, gives R-module isomorphisms

$$M \otimes_R N \cong \bigoplus_{i \in I} \big(Rm_i \otimes_R N\big) \cong \bigoplus_{i \in I} N$$

under which $m_i \otimes n$ corresponds to the family whose i-th entry is n and whose other entries are 0. Writing $N = \bigoplus_{j \in J} Rn_j \cong \bigoplus_{j \in J} R$ in the same way and substituting,

$$M \otimes_R N \cong \bigoplus_{i \in I} \bigoplus_{j \in J} R \cong \bigoplus_{(i,j) \in I \times J} R$$

and following $m_i \otimes n_j$ through the three isomorphisms shows that it corresponds to the element of $\bigoplus_{I \times J} R$ with 1 in position (i, j) and 0 elsewhere. Those elements form the standard basis of $\bigoplus_{I \times J} R$, and an isomorphism of R-modules carries a basis to a basis, so $\{m_i \otimes n_j\}$ is a basis of $M \otimes_R N$.

The two special cases follow by counting: $I \times J$ has st elements when $|I| = s$ and $|J| = t$, and the dimension of a vector space is the cardinality of any basis, as we sought to show.

This closes the identification promised in example 15.1.15: for finite-dimensional V, a basis of V^* and one of W^* produce a basis of $V^* \otimes_F W^*$, matching the basis of the space of bilinear forms on $V \times W$ under the correspondence of theorem 15.1.12.

Two structural isomorphisms remain. Both are needed before an expression such as $M_1 \otimes M_2 \otimes M_3$ can be written without brackets.

Proposition 15.2.7: Commutativity of the Tensor Product

Let R be a commutative ring and let M, N be R-modules with their standard structures. Then there is a unique R-module isomorphism

$$\tau\colon M \otimes_R N \to N \otimes_R M, \qquad \tau(m \otimes n) = n \otimes m$$

***Proof*:**
The map $M \times N \to N \otimes_R M$ sending (m, n) to $n \otimes m$ is R-bilinear: it is additive in each variable, and for $r \in R$,

$$n \otimes (rm) = (nr) \otimes m = r(n \otimes m)$$

using the third relation of (15.1) and the description of the action in theorem 15.1.12, with the same computation in the other variable. So theorem 15.1.12 supplies a unique R-module homomorphism τ with $\tau(m \otimes n) = n \otimes m$. The same construction with the roles of M and N exchanged gives $\tau'\colon N \otimes_R M \to M \otimes_R N$ with $\tau'(n \otimes m) = m \otimes n$, and both composites fix every simple tensor, so both are the identity by lemma 15.1.4(3), as we sought to show.

Theorem 15.2.8: Associativity of the Tensor Product

Let M be a right R-module, let N be an (R, T)-bimodule and let L be a left T-module. Then there is a unique isomorphism of abelian groups

$$(M \otimes_R N) \otimes_T L \cong M \otimes_R (N \otimes_T L)$$

carrying $(m \otimes n) \otimes l$ to $m \otimes (n \otimes l)$. If M is an (S, R)-bimodule it is an isomorphism of left S-modules; if $R = T$ is commutative and all three modules carry their standard structures, it is an isomorphism of R-modules.

***Proof*:**
By proposition 15.2.1, $M \otimes_R N$ is a right T-module with $(m \otimes n)t = m \otimes (nt)$, and $N \otimes_T L$ is a left R-module with $r(n \otimes l) = (rn) \otimes l$. Both tensor products in the statement are therefore defined.

Step 1: a homomorphism for each l. Fix $l \in L$. The map $M \times N \to M \otimes_R (N \otimes_T L)$ sending (m, n) to $m \otimes (n \otimes l)$ is R-balanced. It is additive in m; it is additive in n because $n \mapsto n \otimes l$ is additive and $\otimes$ is additive in its second argument; and

$$(mr) \otimes (n \otimes l) = m \otimes r(n \otimes l) = m \otimes \big((rn) \otimes l\big)$$

which is the third axiom. By theorem 15.1.5 there is a unique group homomorphism $\theta_l\colon M \otimes_R N \to M \otimes_R (N \otimes_T L)$ with $\theta_l(m \otimes n) = m \otimes (n \otimes l)$.

Step 2: assembling over l. The map $(M \otimes_R N) \times L \to M \otimes_R (N \otimes_T L)$ sending (x, l) to $\theta_l(x)$ is T-balanced. It is additive in x because θ_l is a homomorphism. For additivity in l, both $\theta_{l+l'}$ and $\theta_l + \theta_{l'}$ are group homomorphisms out of $M \otimes_R N$, and on a simple tensor both give $m \otimes (n \otimes l) + m \otimes (n \otimes l')$, so they are equal by lemma 15.1.4(3). For the third axiom, $x \mapsto \theta_l(xt)$ and $x \mapsto \theta_{tl}(x)$ are group homomorphisms, and on a simple tensor

$$\theta_l\big((m \otimes n)t\big) = \theta_l(m \otimes nt) = m \otimes \big((nt) \otimes l\big) = m \otimes \big(n \otimes (tl)\big) = \theta_{tl}(m \otimes n)$$

the third equality being the third relation of (15.1) inside $N \otimes_T L$. So theorem 15.1.5 supplies a group homomorphism Θ on $(M \otimes_R N) \otimes_T L$ with $\Theta\big((m \otimes n) \otimes l\big) = m \otimes (n \otimes l)$.

Step 3: the inverse. The same two steps run with the outer factors exchanged. Fixing $m \in M$, the map $N \times L \to (M \otimes_R N) \otimes_T L$ sending (n, l) to $(m \otimes n) \otimes l$ is T-balanced, giving $\eta_m : N \otimes_T L \to (M \otimes_R N) \otimes_T L$ with $\eta_m(n \otimes l) = (m \otimes n) \otimes l$; and $(m, y) \mapsto \eta_m(y)$ is then R-balanced by the same three checks. This yields a group homomorphism Ξ on $M \otimes_R (N \otimes_T L)$ with $\Xi\big(m \otimes (n \otimes l)\big) = (m \otimes n) \otimes l$.

Step 4: the composites. By lemma 15.1.4(2) every element of $(M \otimes_R N) \otimes_T L$ is a finite sum of terms $x \otimes l$ with $x \in M \otimes_R N$, and each such x is itself a finite sum of simple tensors, so by additivity of $\otimes$ in its first argument every element is a finite sum of terms $(m \otimes n) \otimes l$. Since $\Xi \circ \Theta$ is a group homomorphism fixing each such term, it is the identity; the same argument on the other side gives $\Theta \circ \Xi = \mathrm{id}$.

The S-linearity of Θ, when M is an (S, R)-bimodule, is once more an equality of group homomorphisms verified on the generating terms $(m \otimes n) \otimes l$, where both sides give $(sm) \otimes (n \otimes l)$; and uniqueness holds because a homomorphism is determined by its values on those terms, completing the proof.

15.2.9 Foreshadowing: Monoidal Structure on R-Mod For a commutative ring R, the three isomorphisms

$$(M \otimes_R N) \otimes_R L \cong M \otimes_R (N \otimes_R L), \qquad R \otimes_R M \cong M, \qquad M \otimes_R N \cong N \otimes_R M$$

of theorem 15.2.8, example 15.1.13 and proposition 15.2.7 are not three unrelated facts. They are the associator, the unit isomorphism and the braiding of a structure called a *symmetric monoidal category*, carried by R-Mod with $\otimes_R$ as its product and R as its unit. In that language the R-algebras of definition 12.1.13 are exactly the monoid objects of R-Mod, and the compatibility conditions the three isomorphisms satisfy with one another are the subject of a coherence theorem. The structure is developed in *Category Theory* [9]; nothing in this chapter depends on it.

Associativity licenses the notation $M_1 \otimes_R \cdots \otimes_R M_n$: define it to mean the left-bracketed product $(\cdots(M_1 \otimes_R M_2) \otimes_R \cdots) \otimes_R M_n$, and write $m_1 \otimes \cdots \otimes m_n$ for the corresponding element. Repeated application of theorem 15.2.8 produces an isomorphism from any other bracketing onto this one matching the corresponding simple tensors, which is what makes the bracket-free notation harmless. With that in place the universal property extends from two factors to n.

Definition 15.2.10: Multilinear Maps

Let R be a commutative ring and let $M_1, \ldots, M_n$ and L be R-modules with their standard structures. A map $\varphi : M_1 \times \cdots \times M_n \to L$ is *R-multilinear*, or *n-linear*, if for each index i and each choice of fixed elements $m_k \in M_k$ for $k \neq i$, the map

$$M_i \to L, \qquad m \mapsto \varphi(m_1, \ldots, m_{i-1}, m, m_{i+1}, \ldots, m_n)$$

is an R-module homomorphism. For $n = 2$ and $n = 3$ one says *bilinear* and *trilinear* rather than 2-linear and 3-linear.

This is definition 13.3.15 with the target allowed to be any R-module rather than the base field, and with any number of arguments; for $n = 2$ it is definition 15.1.11.

Theorem 15.2.11: Universal Property for Multilinear Maps

Let R be a commutative ring and let $M_1, \dots, M_n$ and L be R-modules. For every R-multilinear map $\varphi\colon M_1 \times \cdots \times M_n \to L$ there is a unique R-module homomorphism

$$\psi\colon M_1 \otimes_R \cdots \otimes_R M_n \to L, \qquad \psi(m_1 \otimes \cdots \otimes m_n) = \varphi(m_1, \dots, m_n)$$

***Proof*:**
Induction on n. The case $n = 1$ is the identity and $n = 2$ is theorem 15.1.12, so suppose $n \geq 3$ and the statement holds for $n - 1$. Write $P = M_1 \otimes_R \cdots \otimes_R M_{n-1}$, so that by definition of the left-bracketed product the n-fold tensor product is $P \otimes_R M_n$.

First note that P is generated as an R-module by the simple tensors $m_1 \otimes \cdots \otimes m_{n-1}$: this holds for $n - 1 = 1$ trivially, and the inductive step is lemma 15.1.4(2) applied to $P' \otimes_R M_{n-1}$ together with additivity of $\otimes$ in its first argument.

Fix $m_n \in M_n$. The map $(m_1, \dots, m_{n-1}) \mapsto \varphi(m_1, \dots, m_n)$ is $(n-1)$-linear, since fixing all but one of its arguments fixes all but one argument of φ. By the inductive hypothesis there is a unique R-module homomorphism $\psi_{m_n}\colon P \to L$ with $\psi_{m_n}(m_1 \otimes \cdots \otimes m_{n-1}) = \varphi(m_1, \dots, m_n)$.

The map $P \times M_n \to L$ sending (x, m_n) to $\psi_{m_n}(x)$ is R-bilinear. Linearity in x holds because ψ_{m_n} is an R-module homomorphism. For linearity in m_n, the maps $\psi_{m_n + m_n'}$ and $\psi_{m_n} + \psi_{m_n'}$ are R-module homomorphisms out of P agreeing on every $m_1 \otimes \cdots \otimes m_{n-1}$, by linearity of φ in its last argument, and those generate P; so the two are equal. The same argument gives $\psi_{rm_n} = r\psi_{m_n}$.

theorem 15.1.12 therefore supplies a unique R-module homomorphism ψ on $P \otimes_R M_n$ with $\psi(x \otimes m_n) = \psi_{m_n}(x)$, and evaluating at $x = m_1 \otimes \cdots \otimes m_{n-1}$ gives the stated formula. Uniqueness follows because the simple tensors $m_1 \otimes \cdots \otimes m_n$ generate $M_1 \otimes_R \cdots \otimes_R M_n$, by the generation statement above and lemma 15.1.4(2), completing the proof.

One fact about $\otimes_R$ cannot be obtained from the universal property, because it is a statement about how an equation in $M \otimes_R N$ arises rather than about maps out of it. Lemma 15.1.4(2) says every element is a finite sum of simple tensors; the lemma below says that when such a sum vanishes, the vanishing is already visible inside a finitely generated piece of M. Its proof is the one place in this section that returns to the construction in definition 15.1.3.

Lemma 15.2.12: Vanishing Is Witnessed Finitely

Let R be a ring, M a right R-module and N a left R-module, and suppose

$$\sum_{i=1}^{k} m_i \otimes n_i = 0 \qquad \text{in } M \otimes_R N$$

Then there is a *finitely generated* submodule $M_0 \subseteq M$ containing $m_1, \ldots, m_k$ such that already

$$\sum_{i=1}^{k} m_i \otimes n_i = 0 \qquad \text{in } M_0 \otimes_R N$$

The same holds with the roles of M and N exchanged.

***Proof*:**
By definition 15.1.3 the group $M \otimes_R N$ is $F^{\mathbb{Z}}(M \times N)/K$, so the hypothesis says that the element $\sum_i (m_i, n_i)$ of $F^{\mathbb{Z}}(M \times N)$ lies in K. Since K is generated by the three displayed forms, there are finitely many generators $g_1, \ldots, g_t$ of K and integers $c_1, \ldots, c_t$ with

$$\sum_{i=1}^{k} (m_i, n_i) = \sum_{j=1}^{t} c_j\, g_j$$

Each g_j mentions only finitely many elements of M: a generator of the first form mentions m, m' and $m + m'$, one of the second mentions a single m, and one of the third mentions mr and m. Let $S \subseteq M$ be the finite set consisting of $m_1, \ldots, m_k$ together with every element of M mentioned by some g_j, and let M_0 be the submodule of M generated by S. It is finitely generated and contains each m_i.

Now form $F^{\mathbb{Z}}(M_0 \times N)$ and the subgroup K_0 generated by the three forms of definition 15.1.3 taken over M_0, N and R, so that $M_0 \otimes_R N = F^{\mathbb{Z}}(M_0 \times N)/K_0$. Every g_j lies in $F^{\mathbb{Z}}(M_0 \times N)$, since all of its M-coordinates belong to $S \subseteq M_0$; and g_j is a generator of K_0, because M_0 is a submodule and so is closed under the addition and the R-action that the three forms use. Hence the displayed identity holds in $F^{\mathbb{Z}}(M_0 \times N)$ and exhibits $\sum_i (m_i, n_i)$ as an element of K_0, which is to say $\sum_i m_i \otimes n_i = 0$ in $M_0 \otimes_R N$.

The statement with M and N exchanged is the same argument applied to the second coordinate.

The lemma does not say that $M_0 \otimes_R N \to M \otimes_R N$ is injective, and in general it is not. What it says is that a *relation* that holds downstairs already holds upstairs, which is what makes it possible to reduce a question about an arbitrary module to the finitely generated case. Theorem 17.5.7 is the use it was recorded for.

A tensor product of quotients of a commutative ring is again a quotient, and the ideal one quotients by is the sum. This is the computation behind most small examples.

Proposition 15.2.13: Tensor Product of Quotient Rings

Let R be a commutative ring and let I, J be ideals of R. Then there is an R-module isomorphism
$$R/I \otimes_R R/J \cong R/(I+J), \qquad \overline{a} \otimes \overline{b} \mapsto \overline{ab}$$

***Proof*:**
The map $R/I \times R/J \to R/(I+J)$ sending $(\overline{a}, \overline{b})$ to $\overline{ab}$ is well defined: if $a - a' \in I$ and $b - b' \in J$ then
$$ab - a'b' = (a-a')b + a'(b-b') \in I + J$$
It is R-bilinear, so theorem 15.1.12 induces an R-module homomorphism Φ with $\Phi(\overline{a} \otimes \overline{b}) = \overline{ab}$.

In the other direction set $\Psi(\overline{r}) = \overline{r} \otimes \overline{1}$. Throughout, a bar in the left-hand factor denotes a class in R/I and a bar in the right-hand factor a class in R/J. To see that Ψ is well defined, let $r - r' \in I + J$ and write $r - r' = a + b$ with $a \in I$ and $b \in J$, so that
$$\overline{r - r'} \otimes \overline{1} = \overline{a} \otimes \overline{1} + \overline{b} \otimes \overline{1}$$
The first term vanishes because $\overline{a} = \overline{0}$ already in R/I. For the second, moving the scalar b across by the third relation of (15.1),
$$\overline{b} \otimes \overline{1} = (b \cdot \overline{1}) \otimes \overline{1} = \overline{1} \otimes (b \cdot \overline{1}) = \overline{1} \otimes \overline{b} = 0$$
since $\overline{b} = \overline{0}$ in R/J. So Ψ is well defined, and it is R-linear.

The composite $\Phi \circ \Psi$ sends $\overline{r}$ to $\overline{r}$. For the other composite,
$$\Psi\big(\Phi(\overline{a} \otimes \overline{b})\big) = \overline{ab} \otimes \overline{1} = (\overline{a} \cdot b) \otimes \overline{1} = \overline{a} \otimes b\overline{1} = \overline{a} \otimes \overline{b}$$
so $\Psi \circ \Phi$ fixes every simple tensor and is the identity by lemma 15.1.4(3), as we sought to show.

Taking $R = \mathbb{Z}$, $I = (m)$ and $J = (n)$ recovers $\mathbb{Z}/m\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Z}/n\mathbb{Z} \cong \mathbb{Z}/\gcd(m,n)\mathbb{Z}$, since $(m) + (n) = (\gcd(m,n))$ by definition 10.1.24; that is the second exercise of section 15.1, and both computations of example 15.1.14 are the cases $(m,n) = (2,3)$ and $(m,n) = (2,2)$.

When the two factors are R-algebras rather than bare modules, their multiplications combine into one on the tensor product. This is the construction that will produce the tensor algebra in section 16.3, and it is the source of the coproduct description that closes this section.

Theorem 15.2.14: Tensor Product of R-Algebras

Let R be a commutative ring and let A and B be R-algebras (definition 12.1.13). Then there is a unique R-bilinear multiplication on $A \otimes_R B$ satisfying
$$(a \otimes b)(a' \otimes b') = (aa') \otimes (bb')$$
and it makes $A \otimes_R B$ into an R-algebra with identity $1_A \otimes 1_B$. If A and B are both commutative, so is $A \otimes_R B$.

***Proof*:**
Step 1: the multiplication exists. The map $A \times B \times A \times B \to A \otimes_R B$ sending (a, b, a', b') to $(aa') \otimes (bb')$ is R-multilinear: in each of its four arguments the other three are fixed, and the resulting map is a composite of multiplication in A or in B, which is R-linear in each variable by definition 12.1.13, with $\otimes$, which is R-linear in each argument by theorem 15.1.12. By theorem 15.2.11 it induces an R-module homomorphism

$$\mu' \colon A \otimes_R B \otimes_R A \otimes_R B \to A \otimes_R B, \qquad \mu'(a \otimes b \otimes a' \otimes b') = (aa') \otimes (bb')$$

By theorem 15.2.8 there is an isomorphism $(A \otimes_R B) \otimes_R (A \otimes_R B) \cong A \otimes_R B \otimes_R A \otimes_R B$ matching $(a \otimes b) \otimes (a' \otimes b')$ with $a \otimes b \otimes a' \otimes b'$; composing it with μ' gives an R-module homomorphism μ on $(A \otimes_R B) \otimes_R (A \otimes_R B)$ with $\mu\big((a \otimes b) \otimes (a' \otimes b')\big) = (aa') \otimes (bb')$. Setting $xy := \mu(x \otimes y)$ defines a map $(A \otimes_R B) \times (A \otimes_R B) \to A \otimes_R B$ which is R-bilinear, being the composite of the R-bilinear ι with the R-linear μ, and which satisfies the displayed formula on simple tensors. It is the unique such: two R-bilinear maps agreeing on pairs of simple tensors induce, by theorem 15.1.12, two homomorphisms on $(A \otimes_R B) \otimes_R (A \otimes_R B)$ agreeing on the simple tensors $(a \otimes b) \otimes (a' \otimes b')$, which generate it, so they coincide.

Step 2: the ring axioms. Distributivity is additivity of the multiplication in each variable, which is part of bilinearity. For associativity, both $(x, y, z) \mapsto (xy)z$ and $(x, y, z) \mapsto x(yz)$ are R-trilinear, being built from bilinear maps, so by theorem 15.2.11 each corresponds to a homomorphism on the triple tensor product, and the two agree if and only if they agree on triples of simple tensors. There

$$\big((a{\otimes}b)(a'{\otimes}b')\big)(a''{\otimes}b'') = \big((aa')a''\big){\otimes}\big((bb')b''\big) = \big(a(a'a'')\big){\otimes}\big(b(b'b'')\big) = (a{\otimes}b)\big((a'{\otimes}b')(a''{\otimes}b'')\big)$$

by associativity in A and in B. For the identity, $x \mapsto (1_A \otimes 1_B)x$ and the identity map are both additive and agree on simple tensors, so they are equal by lemma 15.1.4(3), and likewise on the other side.

Step 3: the R-algebra structure. The R-module structure is the one $A \otimes_R B$ already carries, and the compatibility $r(xy) = (rx)y = x(ry)$ required by definition 12.1.13 is R-bilinearity of the multiplication. Finally, if A and B are commutative then $(x, y) \mapsto xy$ and $(x, y) \mapsto yx$ are R-bilinear and agree on pairs of simple tensors, since $(aa') \otimes (bb') = (a'a) \otimes (b'b)$, so they are equal by the uniqueness argument of Step 1, completing the proof.

Example 15.2.15: Tensoring Two Copies of $\mathbb{C}$ over $\mathbb{R}$
The base ring matters, and comparing $\mathbb{C} \otimes_{\mathbb{C}} \mathbb{C}$ with $\mathbb{C} \otimes_{\mathbb{R}} \mathbb{C}$ shows how much. The first is $\mathbb{C}$ itself by example 15.1.13. The second has $\mathbb{R}$-dimension 4 by corollary 15.2.6, since $\mathbb{C}$ is free of rank 2 over $\mathbb{R}$, and it is not even a domain:

$$\mathbb{C} \otimes_{\mathbb{R}} \mathbb{C} \cong \mathbb{C} \times \mathbb{C}$$

as $\mathbb{R}$-algebras. To see this, define $\Phi \colon \mathbb{C} \otimes_{\mathbb{R}} \mathbb{C} \to \mathbb{C} \times \mathbb{C}$ on simple tensors by

$$\Phi(z \otimes w) = (zw,\ z\overline{w})$$

This is well defined because $(z, w) \mapsto (zw, z\overline{w})$ is $\mathbb{R}$-bilinear, complex conjugation being $\mathbb{R}$-linear, so theorem 15.1.12 applies. It is a ring homomorphism: on simple tensors

$$\Phi\big((z{\otimes}w)(z'{\otimes}w')\big) = \Phi(zz'{\otimes}ww') = \big(zz'ww',\ zz'\overline{ww'}\big) = \big(zw{\cdot}z'w',\ z\overline{w}{\cdot}z'\overline{w'}\big) = \Phi(z{\otimes}w)\Phi(z'{\otimes}w')$$

and both $(x, y) \mapsto \Phi(xy)$ and $(x, y) \mapsto \Phi(x)\Phi(y)$ are $\mathbb{R}$-bilinear, so agreeing on simple tensors makes them equal. Also $\Phi(1 \otimes 1) = (1, 1)$.

For surjectivity, compute Φ on three elements:

$$\Phi(1 \otimes 1) = (1, 1), \qquad \Phi(i \otimes 1) = (i, i), \qquad \Phi(i \otimes i) = (i \cdot i,\ i\bar{i}) = (-1, 1)$$

The image of Φ is both an $\mathbb{R}$-subspace and a subring. It contains $(1, 1) - (-1, 1) = (2, 0)$ and $(1, 1) + (-1, 1) = (0, 2)$, hence $(1, 0)$ and $(0, 1)$; and being closed under multiplication it then contains $(1, 0)(i, i) = (i, 0)$ and $(0, 1)(i, i) = (0, i)$. Those four elements form an $\mathbb{R}$-basis of $\mathbb{C} \times \mathbb{C}$, so Φ is surjective. Both sides have $\mathbb{R}$-dimension 4, and a surjective linear map between vector spaces of equal finite dimension is an isomorphism.

So a tensor product of two fields over a common subfield need not be a field, and need not even be an integral domain: $\mathbb{C} \times \mathbb{C}$ has the zero divisors $(1, 0)$ and $(0, 1)$.

The last result of this section reads the tensor product of algebras backwards, as the answer to a universal problem rather than as a construction. Recall that in R-Mod the direct sum $M \oplus N$ is simultaneously the product and the coproduct of M and N; this is a feature of *finite* families only, since for an infinite index set the product $\prod_i M_i$ and the coproduct $\bigoplus_i M_i$ differ, the latter consisting of the finitely supported families. In commutative R-algebras the two come apart even for two factors. The product is the direct product $A \times B$ with componentwise operations, and the coproduct is the tensor product.

Lemma 15.2.16: Generation of $A \otimes_R B$ as an Algebra

Let R be a commutative ring and A, B be R-algebras. Then $A \otimes_R B$ is generated as an R-algebra by the set $\{a \otimes 1 | a \in A\} \cup \{1 \otimes b | b \in B\}$.

***Proof*:**
By theorem 15.2.14, $(a \otimes 1)(1 \otimes b) = (a \cdot 1) \otimes (1 \cdot b) = a \otimes b$, so every simple tensor is a product of two elements of the stated set. By lemma 15.1.4(2) every element of $A \otimes_R B$ is a finite sum of simple tensors, completing the proof.

Theorem 15.2.17: Universal Property of the Coproduct in R-Alg

Let R be a commutative ring and let A and B be commutative R-algebras, with structure maps $i_A \colon R \to A$ and $i_B \colon R \to B$. Then the maps

$$\iota_A \colon A \to A \otimes_R B, \quad a \mapsto a \otimes 1 \qquad \text{and} \qquad \iota_B \colon B \to A \otimes_R B, \quad b \mapsto 1 \otimes b$$

are R-algebra homomorphisms, and $(A \otimes_R B, \iota_A, \iota_B)$ is the coproduct of A and B in the category of commutative R-algebras: for every commutative R-algebra Z and every pair of R-algebra homomorphisms $f \colon A \to Z$ and $g \colon B \to Z$ there is a unique R-algebra homomorphism $h \colon A \otimes_R B \to Z$ with $h \circ \iota_A = f$ and $h \circ \iota_B = g$:

$$\begin{tikzcd}
R \arrow[r, "i_B"] \arrow[d, "i_A"'] & B \arrow[d, "\iota_B"] \arrow[ddr, bend left, "g"] & \\
A \arrow[r, "\iota_A"] \arrow[drr, bend right, "f"'] & A \otimes_R B \arrow[dr, dashed, "h"] & \\
 & & Z
\end{tikzcd}$$

***Proof*:**
That ι_A is an R-algebra homomorphism is immediate from theorem 15.2.14: it is additive and R-linear because $\otimes$ is R-linear in its first argument, multiplicative because $(a \otimes 1)(a' \otimes 1) = (aa') \otimes 1$, and unital because $\iota_A(1_A) = 1_A \otimes 1_B$. The same holds for ι_B.

Existence. Define $\Phi \colon A \times B \to Z$ by $\Phi(a, b) = f(a)g(b)$. It is R-bilinear: it is additive in each variable because f and g are additive and multiplication in Z distributes, and for $r \in R$,

$$\Phi(ra, b) = f(ra)g(b) = \big(rf(a)\big)g(b) = r\big(f(a)g(b)\big) \qquad \Phi(a, rb) = f(a)\big(rg(b)\big) = r\big(f(a)g(b)\big)$$

using R-linearity of f and of g and the compatibility $r(xy) = (rx)y = x(ry)$ of definition 12.1.13 in Z. By theorem 15.1.12 there is a unique R-module homomorphism $h \colon A \otimes_R B \to Z$ with $h(a \otimes b) = f(a)g(b)$.

The map h is multiplicative. Both $(x, y) \mapsto h(xy)$ and $(x, y) \mapsto h(x)h(y)$ are R-bilinear, and on a pair of simple tensors

$$\begin{aligned}
h\big((a \otimes b)(a' \otimes b')\big) &= h(aa' \otimes bb') = f(aa')g(bb') \\
&= f(a)f(a')g(b)g(b') = f(a)g(b)f(a')g(b') \\
&= h(a \otimes b)\, h(a' \otimes b')
\end{aligned}$$

the fourth equality using that Z is commutative. Two R-bilinear maps agreeing on pairs of simple tensors are equal, by the argument in Step 1 of theorem 15.2.14, so h is multiplicative. It is unital, since $h(1_A \otimes 1_B) = f(1)g(1) = 1$, and therefore an R-algebra homomorphism. Finally $h(\iota_A(a)) = h(a \otimes 1) = f(a)g(1) = f(a)$ and likewise $h \circ \iota_B = g$, so the diagram commutes.

Uniqueness. Suppose h' is an R-algebra homomorphism with $h' \circ \iota_A = f$ and $h' \circ \iota_B = g$. Then h and h' agree on every $a \otimes 1$ and every $1 \otimes b$. By lemma 15.2.16 those elements generate $A \otimes_R B$ as an R-algebra, and two R-algebra homomorphisms agreeing on a generating set are equal, since the set on which they agree is a subalgebra. Hence $h' = h$, completing the proof.

Every commutative ring is an algebra over $\mathbb{Z}$ in exactly one way, which turns the theorem into a statement about rings with no base ring chosen in advance.

Corollary 15.2.18: Universal Property of the Coproduct in CRing

Let A and B be commutative rings. Then $A \otimes_{\mathbb{Z}} B$, with $\iota_A(a) = a \otimes 1$ and $\iota_B(b) = 1 \otimes b$, is the coproduct of A and B in the category **CRing** of commutative rings.

***Proof*:**
By proposition 9.1.29 there is exactly one ring homomorphism $\mathbb{Z} \to A$ for every ring A, so every commutative ring carries exactly one $\mathbb{Z}$-algebra structure, and its image lies in the centre because A is commutative. Moreover every ring homomorphism $\varphi\colon A \to B$ of commutative rings is automatically a $\mathbb{Z}$-algebra homomorphism: the two ring homomorphisms $\varphi \circ i_A$ and i_B from $\mathbb{Z}$ to B coincide by that same uniqueness. Hence the category of commutative $\mathbb{Z}$-algebras and **CRing** have the same objects and the same morphisms, and theorem 15.2.17 applied with $R = \mathbb{Z}$ gives the claim, completing the proof.

The asymmetry between the two constructions is worth recording. The product $A \times B$ is formed with no reference to a base ring, while the coproduct is formed over one, and in **CRing** the base is forced to be the initial object $\mathbb{Z}$. Over a general base R, the object $A \otimes_R B$ solves the coproduct problem for R-algebras, which is the same as a pushout of $A \leftarrow R \to B$ in commutative rings. Reversing every arrow turns a pushout of rings into a pullback of spaces, and this is the computation that makes $\otimes_R$ the algebraic form of the fibre product of affine schemes in *EYNTKA Algebraic Geometry* [5].

Exercise 15.2.1

1. Show that $\varphi \otimes \mathrm{id}$ need not be injective when φ is. (Take $\varphi\colon \mathbb{Z} \to \mathbb{Z}$ to be multiplication by 2 and tensor with $\mathbb{Z}/2\mathbb{Z}$ over $\mathbb{Z}$.) Conclude that the functor of theorem 15.2.2 does not preserve injections.

2. Let R be commutative, let M be an R-module and let I be an ideal of R. Show that $M \otimes_R R/I \cong M/IM$, and recover proposition 15.2.13 as the case $M = R/J$.

3. Show that every element of $\mathbb{Q} \otimes_{\mathbb{Z}} \mathbb{Q}$ is a simple tensor, and deduce that $\mathbb{Q} \otimes_{\mathbb{Z}} \mathbb{Q} \cong \mathbb{Q}$. (Note that example 15.1.13 does not apply: the left factor is $\mathbb{Q}$, not the base ring $\mathbb{Z}$.)

4. Let R be commutative and A an R-algebra. Show that the isomorphism $A \otimes_R R \cong A$ of example 15.1.13 is an isomorphism of R-algebras.

5. Using the isomorphism of example 15.2.15, find the two nontrivial idempotents of $\mathbb{C} \otimes_{\mathbb{R}} \mathbb{C}$ explicitly as combinations of simple tensors, and verify directly that each squares to itself.

6. Let R be commutative and let M, N be finitely generated R-modules, generated by s and t elements respectively. Show that $M \otimes_R N$ is generated by st elements. Show also that the bound is attained but not always sharp, by computing a case where $M \otimes_R N$ needs fewer.

7. Let F be a field and let V, W be finite-dimensional F-vector spaces. Show that

$$V^* \otimes_F W \to \mathrm{Hom}_F(V, W), \qquad f \otimes w \mapsto \big(v \mapsto f(v)w\big)$$

is an isomorphism of F-vector spaces, and identify where finite-dimensionality is used.

8. Combine theorem 15.2.8 and proposition 15.2.7 into an isomorphism $(A \otimes_R B) \otimes_R (A \otimes_R B) \cong (A \otimes_R A) \otimes_R (B \otimes_R B)$ carrying $(a \otimes b) \otimes (a' \otimes b')$ to $(a \otimes a') \otimes (b \otimes b')$. Use it, together with the maps $A \otimes_R A \to A$ and $B \otimes_R B \to B$ induced by multiplication, to give a second construction of the multiplication of theorem 15.2.14.

15.3 Extension of Scalars

The previous two sections held the ring fixed and varied the modules. This one does the opposite. Given a ring homomorphism $f: R \to S$, there are two ways to move a module between the two rings, and they are not symmetric: passing from S to R costs nothing, while passing from R to S requires a construction. That construction is the tensor product, applied to the bimodule of example 15.1.9(2), and everything in this section is a consequence of theorem 15.1.5 rather than a new construction.

The cheap direction first.

Definition 15.3.1: Restriction of Scalars

Let $f: R \to S$ be a ring homomorphism and let L be a left S-module. The *restriction of scalars* of L along f is the left R-module L_R with the same underlying abelian group and action

$$r \cdot l := f(r)\, l$$

An S-module homomorphism is in particular an R-module homomorphism $L_R \to L'_R$, so restriction of scalars sends S-modules to R-modules and S-linear maps to R-linear maps.

The module axioms hold because f is a ring homomorphism, exactly as in the construction of an R-algebra from a ring map (section 12.1.2); nothing is created and nothing is destroyed, only forgotten. The other direction cannot work this way. An R-module N carries no S-action to forget, and there is in general no way to define one on the same underlying set: if $R = \mathbb{Z}$, $S = \mathbb{Q}$ and $N = \mathbb{Z}/2\mathbb{Z}$, then any $\mathbb{Q}$-action on N would make multiplication by 2 invertible on a group killed by 2. A new module has to be built, and the tensor product builds it.

Definition 15.3.2: Extension of Scalars

Let $f: R \to S$ be a ring homomorphism and let N be a left R-module. Regard S as an (S, R)-bimodule as in example 15.1.9(2), with left action given by multiplication in S and right action $s \cdot r := sf(r)$. The *extension of scalars* of N along f is the left S-module

$$S \otimes_R N$$

whose S-action, supplied by proposition 15.1.10, is determined on simple tensors by $s' \cdot (s \otimes n) = (s's) \otimes n$.

No verification is owed here: the module structure was proved once in proposition 15.1.10 and the only input is the bimodule structure on S, which is example 15.1.9(2). The relation that does the work is the third one in (15.1), which now reads

$$sf(r) \otimes n = s \otimes rn$$

so a scalar of R may be moved out of the module and into S, where it becomes invertible whenever $f(r)$ is.

Example 15.3.3: Extension of Scalars of a Free Module

Let $f\colon R \to S$ be a ring homomorphism. Then

$$S \otimes_R R^n \cong S^n$$

as left S-modules, and more generally $S \otimes_R R^{(I)} \cong S^{(I)}$ for a free module on any index set I. Indeed $R^{(I)} = \bigoplus_{i\in I} R$, so theorem 15.2.4 applied on the right-hand factor gives

$$S \otimes_R \bigoplus_{i\in I} R \cong \bigoplus_{i\in I} \big(S \otimes_R R\big) \cong \bigoplus_{i\in I} S$$

the second isomorphism being example 15.1.13 in each summand. Following a basis element e_i through, $1 \otimes e_i$ corresponds to the i-th standard generator of $S^{(I)}$. So extension of scalars carries a free R-module of rank n to a free S-module of the same rank: a basis survives, and only the ring it is a basis over changes.

Free modules therefore lose nothing. In general something is lost, and the map that measures the loss is the following.

The assignment $\iota\colon N \to S \otimes_R N$ with $\iota(n) = 1 \otimes n$ is additive, and it is R-linear when $S \otimes_R N$ is regarded as an R-module by restriction:

$$\iota(rn) = 1 \otimes rn = 1 \cdot f(r) \otimes n = f(r)\big(1 \otimes n\big) = r \cdot \iota(n)$$

using the third relation and then the definition of the S-action. It need not be injective. For $R = \mathbb{Z}$, $S = \mathbb{Q}$ and N any finite abelian group, $\mathbb{Q} \otimes_{\mathbb{Z}} N = 0$ by the third exercise of section 15.1, so ι is the zero map on a nonzero module.

The next theorem says that although N may not sit inside $S \otimes_R N$, every R-linear map from N into an S-module factors through it, uniquely and S-linearly. It is the defining property of extension of scalars and the reason the construction is the right one.

Theorem 15.3.4: Universal Property of Extension of Scalars

Let $f\colon R \to S$ be a ring homomorphism, let N be a left R-module and let $\iota\colon N \to S \otimes_R N$ be the R-module homomorphism $\iota(n) = 1 \otimes n$. Then for every left S-module L and every R-module homomorphism $\varphi\colon N \to L_R$ there is a unique S-module homomorphism $\psi\colon S \otimes_R N \to L$ with $\varphi = \psi \circ \iota$, that is, making

$$\begin{tikzcd}
S \otimes_R N \arrow[r, "\psi"] & L \\
N \arrow[u, "\iota"] \arrow[ur, "\varphi"'] &
\end{tikzcd}$$

commute.

***Proof*:**
Existence. The map $S \times N \to L$ sending (s, n) to $s\varphi(n)$ is R-balanced. Additivity in s holds because the S-action on L distributes, and additivity in n because φ is additive. For the third axiom, the right R-action on S is $s \cdot r = sf(r)$ and the R-action on L_R is $r \cdot l = f(r)l$, so

$$(s \cdot r)\varphi(n) = \big(sf(r)\big)\varphi(n) = s\big(f(r)\varphi(n)\big) = s\big(r \cdot \varphi(n)\big) = s\,\varphi(rn)$$

the last equality because φ is R-linear. By theorem 15.1.5 there is a unique group homomorphism $\psi\colon S \otimes_R N \to L$ with $\psi(s \otimes n) = s\varphi(n)$.

It is S-linear. Fix $s' \in S$; the maps $x \mapsto \psi(s'x)$ and $x \mapsto s'\psi(x)$ are group homomorphisms, and on a simple tensor

$$\psi\big(s'(s \otimes n)\big) = \psi\big((s's) \otimes n\big) = (s's)\varphi(n) = s'\big(s\varphi(n)\big) = s'\psi(s \otimes n)$$

so they agree by lemma 15.1.4(3). Finally $\psi(\iota(n)) = \psi(1 \otimes n) = 1 \cdot \varphi(n) = \varphi(n)$, so $\varphi = \psi \circ \iota$.

Uniqueness. As an S-module, $S \otimes_R N$ is generated by the image of ι, since $s \otimes n = s(1 \otimes n) = s\,\iota(n)$ and every element is a finite sum of simple tensors by lemma 15.1.4(2). An S-module homomorphism is determined by its values on a generating set, and any ψ' with $\varphi = \psi' \circ \iota$ agrees with ψ on $\iota(N)$, so $\psi' = \psi$, completing the proof.

Restated without diagrams, the theorem says that R-linear maps out of N into an S-module are the same thing as S-linear maps out of $S \otimes_R N$. That is an adjunction, and it is worth naming as one, because it is the form in which the construction is used downstream.

Corollary 15.3.5: Extension of Scalars Is Left Adjoint to Restriction

Let $f\colon R \to S$ be a ring homomorphism. For every left R-module N and every left S-module L the map

$$\mathrm{Hom}_S\big(S \otimes_R N,\, L\big) \to \mathrm{Hom}_R\big(N,\, L_R\big), \qquad \psi \mapsto \psi \circ \iota$$

is an isomorphism of abelian groups. A pair of functors related this way is called an *adjunction*, and this one is expressed by saying that $S \otimes_R -$, from left R-modules to left S-modules, is *left adjoint* to restriction of scalars. The general theory is section 62.1; only the displayed bijection is used below.

***Proof*:**
The map is additive, since $(\psi + \psi') \circ \iota = \psi \circ \iota + \psi' \circ \iota$, and it lands in $\mathrm{Hom}_R(N, L_R)$ because ι is R-linear and ψ is S-linear, hence R-linear after restriction. It is surjective by the existence half of theorem 15.3.4 and injective by the uniqueness half: if $\psi \circ \iota = \psi' \circ \iota$ then both are maps induced by the same φ, so they coincide, as we sought to show.

Two computations show what the construction does. The first is the standard way a real vector space is turned into a complex one.

Example 15.3.6: Complexification
Take $R = \mathbb{R}$, $S = \mathbb{C}$ and f the inclusion. For a real vector space V the *complexification* is the complex vector space

$$V_{\mathbb{C}} := \mathbb{C} \otimes_{\mathbb{R}} V$$

Every element of $V_{\mathbb{C}}$ has the form $1 \otimes v + i \otimes w$ for unique $v, w \in V$. A simple tensor already does, since for $z = x+iy$ with $x, y \in \mathbb{R}$ the third relation of (15.1) gives $z \otimes v = x \otimes v + iy \otimes v = 1 \otimes xv + i \otimes yv$; and a sum of two such collapses to one by additivity of $\otimes$ in its second argument. If V has $\mathbb{R}$-basis $e_1, \ldots, e_n$ then $V_{\mathbb{C}}$ has $\mathbb{C}$-basis $1 \otimes e_1, \ldots, 1 \otimes e_n$ by example 15.3.3, so

$$\dim_{\mathbb{C}}(V_{\mathbb{C}}) = \dim_{\mathbb{R}}(V)$$

A linear map $\varphi \colon V \to W$ extends to $\mathrm{id}_{\mathbb{C}} \otimes \varphi \colon V_{\mathbb{C}} \to W_{\mathbb{C}}$ by theorem 15.2.2, and on the bases above it has the same matrix as φ; the entries are simply now read as complex numbers. This is what licenses the standard move of computing eigenvalues of a real matrix over $\mathbb{C}$.

One warning. Complexifying a space that is already complex does not return it. Reading $\mathbb{C}$ itself as a real vector space and complexifying gives $\mathbb{C} \otimes_{\mathbb{R}} \mathbb{C} \cong \mathbb{C} \times \mathbb{C}$ by example 15.2.15, which has $\mathbb{C}$-dimension 2, not 1. The construction depends on the ring the tensor product is taken over, not only on the two spaces.

Example 15.3.7: Induced Representations

Let K be a field, let G be a finite group and let $H \leq G$ be a subgroup. The group rings $K[H]$ and $K[G]$ of definition 9.2.19 are related by the inclusion $K[H] \hookrightarrow K[G]$, so extension of scalars along it carries $K[H]$-modules to $K[G]$-modules. Since a $K[G]$-module is the same data as a representation of G over K, this manufactures a representation of G from one of H.

The dimension is forced. Choosing coset representatives $g_1, \ldots, g_k$ for G/H, where $k = [G : H]$, gives $K[G] = \bigoplus_{i=1}^{k} g_i K[H]$ as a right $K[H]$-module, so $K[G]$ is free of rank k over $K[H]$. If V is a representation of H of dimension d, then theorem 15.2.4 gives

$$K[G] \otimes_{K[H]} V \cong \bigoplus_{i=1}^{k} \big(g_i K[H] \otimes_{K[H]} V\big) \cong V^{\oplus k}$$

as K-vector spaces, so the induced representation has dimension kd. Concretely, $\mathbb{Z}/3\mathbb{Z}$ sits inside S_3 with index 2, so a 10-dimensional representation of $\mathbb{Z}/3\mathbb{Z}$ induces a 20-dimensional representation of S_3.

The construction is the *induced representation*, and its theory, including Frobenius reciprocity, which is corollary 15.3.5 read in this setting, is developed in *Core Representation Theory* [11].

Returning to the map ι, the universal property settles exactly when it is injective, and the answer identifies what part of N can survive into any S-module at all.

Corollary 15.3.8: When the Unit Is Injective

Let $f \colon R \to S$ be a ring homomorphism, let N be a left R-module and let $\iota \colon N \to S \otimes_R N$ be as above. Then every R-module homomorphism from N into (the restriction of) a left S-module vanishes on $\ker \iota$. Consequently N embeds as an R-submodule of some left S-module if and only if ι is injective, in which case $N \cong \iota(N) \subseteq S \otimes_R N$.

***Proof*:**
Let L be a left S-module and $\theta\colon N \to L_R$ an R-module homomorphism. By theorem 15.3.4 there is an S-module homomorphism ψ with $\theta = \psi \circ \iota$, so θ kills $\ker \iota$.

If ι is injective it is itself such an embedding, with image $\iota(N)$. If conversely some $\theta\colon N \to L_R$ is injective, then $\ker \iota \subseteq \ker \theta = 0$ by the previous paragraph, so ι is injective, completing the proof.

For a finite abelian group N and $\mathbb{Z} \hookrightarrow \mathbb{Q}$ this recovers a fact already visible: ι is the zero map, so no nonzero element of N survives into any $\mathbb{Q}$-vector space, which is another way of saying that a torsion element cannot live in a space where every nonzero integer acts invertibly.

The adjunction of corollary 15.3.5 is one instance of a general relationship between $\otimes$ and Hom. Stating it here rather than alongside its applications keeps every property of the tensor product in one place; it is used in section 17.5 to compare tensoring with the Hom functors, and in *Homological Algebra* [15] to produce derived functors.

Theorem 15.3.9: Hom-Tensor Adjunction

Let R and S be rings, let A be a right R-module, let B be an (R,S)-bimodule and let C be a right S-module. Give $\operatorname{Hom}_S(B,C)$ the right R-action $(g\cdot r)(b) := g(rb)$. Then there is an isomorphism of abelian groups

$$\operatorname{Hom}_S\big(A \otimes_R B,\, C\big) \cong \operatorname{Hom}_R\big(A,\, \operatorname{Hom}_S(B,C)\big)$$

sending φ to the map $a \mapsto \big(b \mapsto \varphi(a\otimes b)\big)$.

***Proof*:**
First, the objects are what the statement claims. Every right R-module is a $(\mathbb{Z},R)$-bimodule, so proposition 15.2.1, applied with $\mathbb{Z}$ in the role of the left-hand ring and S in the role of the right-hand one, makes $A\otimes_R B$ a right S-module with $(a\otimes b)s = a\otimes(bs)$. The stated action on $\operatorname{Hom}_S(B,C)$ is a right action, since $\big((g\cdot r)\cdot r'\big)(b) = (g\cdot r)(r'b) = g(rr'b) = \big(g\cdot(rr')\big)(b)$.

The forward map. Given $\varphi \in \operatorname{Hom}_S(A\otimes_R B, C)$, define $\Lambda(\varphi)(a)(b) = \varphi(a\otimes b)$. For fixed a the map $\Lambda(\varphi)(a)$ is S-linear: it is additive because $\otimes$ is additive in its second argument, and

$$\Lambda(\varphi)(a)(bs) = \varphi\big(a\otimes bs\big) = \varphi\big((a\otimes b)s\big) = \varphi(a\otimes b)\, s = \Lambda(\varphi)(a)(b)\, s$$

using S-linearity of φ. The map $\Lambda(\varphi)$ is R-linear: it is additive because $\otimes$ is additive in its first argument, and

$$\Lambda(\varphi)(ar)(b) = \varphi(ar\otimes b) = \varphi(a\otimes rb) = \Lambda(\varphi)(a)(rb) = \big(\Lambda(\varphi)(a)\cdot r\big)(b)$$

the second equality being the third relation of (15.1). Finally Λ is additive in φ, directly from the definition.

The backward map. Given $\Psi \in \operatorname{Hom}_R(A, \operatorname{Hom}_S(B,C))$, the map $A\times B \to C$ sending (a,b) to $\Psi(a)(b)$ is R-balanced: it is additive in a because Ψ is additive, additive in b because each $\Psi(a)$ is, and

$$\Psi(ar)(b) = \big(\Psi(a)\cdot r\big)(b) = \Psi(a)(rb)$$

which is the third axiom. By theorem 15.1.5 there is a unique group homomorphism $\Gamma(\Psi)\colon A\otimes_R B \to C$ with $\Gamma(\Psi)(a\otimes b) = \Psi(a)(b)$, and it is S-linear because $x\mapsto \Gamma(\Psi)(xs)$ and $x \mapsto \Gamma(\Psi)(x)s$

are group homomorphisms agreeing on simple tensors, where both give $\Psi(a)(bs)$. Again Γ is additive.

The two are mutually inverse. For $\Psi \in \mathrm{Hom}_R(A, \mathrm{Hom}_S(B, C))$ and all $a \in A$, $b \in B$,

$$\Lambda\big(\Gamma(\Psi)\big)(a)(b) = \Gamma(\Psi)(a \otimes b) = \Psi(a)(b)$$

so $\Lambda(\Gamma(\Psi))(a) = \Psi(a)$ for every a, giving $\Lambda \circ \Gamma = \mathrm{id}$. In the other direction, $\Gamma(\Lambda(\varphi))$ and φ are group homomorphisms out of $A \otimes_R B$ agreeing on every simple tensor, since $\Gamma(\Lambda(\varphi))(a \otimes b) = \Lambda(\varphi)(a)(b) = \varphi(a \otimes b)$, so they are equal by lemma 15.1.4(3). Hence Λ is an isomorphism of abelian groups, completing the proof.

Taking $B = S$, viewed as an (R, S)-bimodule along $f\colon R \to S$, and using $\mathrm{Hom}_S(S, C) \cong C_R$ recovers corollary 15.3.5 in its right-handed form: extension of scalars is the case of the Hom-tensor adjunction in which the bimodule is the ring itself.

Exercise 15.3.1

1. Let $f\colon R \to S$ be a ring homomorphism and let $I \subseteq R$ be an ideal. Show that $S \otimes_R R/I \cong S/f(I)S$ as S-modules, where $f(I)S$ is the ideal of S generated by $f(I)$.

2. Let A be a finitely generated abelian group, so that $A \cong \mathbb{Z}^r \oplus T$ with T finite by chapter 14. Show that $\mathbb{Q} \otimes_{\mathbb{Z}} A \cong \mathbb{Q}^r$ as $\mathbb{Q}$-vector spaces, so that extension of scalars along $\mathbb{Z} \hookrightarrow \mathbb{Q}$ destroys exactly the torsion and $r = \dim_{\mathbb{Q}}(\mathbb{Q} \otimes_{\mathbb{Z}} A)$.

3. Let $R \subseteq S$ be an inclusion of rings and suppose S is free as a right R-module with a basis containing 1. Show that $\iota\colon N \to S \otimes_R N$ is injective for every left R-module N. Where does the argument fail for $\mathbb{Z} \subseteq \mathbb{Q}$?

4. Viewing S as an (R, S)-bimodule along $f\colon R \to S$, show that $\mathrm{Hom}_R(S, L)$ is a left S-module for every left R-module L, and that
$$\mathrm{Hom}_R\big(M_R,\, L\big) \cong \mathrm{Hom}_S\big(M,\, \mathrm{Hom}_R(S, L)\big)$$
for every left S-module M. (So restriction of scalars has a right adjoint as well as a left one; this is *coextension of scalars*.)

5. Let V be a real vector space. Show that $V_{\mathbb{C}} \cong V \oplus V$ as real vector spaces, and describe the complex structure that this decomposition carries in terms of the two summands.

6. Let $f\colon R \to S$ and $g\colon S \to T$ be ring homomorphisms. Show that extension of scalars along $g \circ f$ agrees with extension along f followed by extension along g, that is $T \otimes_S (S \otimes_R N) \cong T \otimes_R N$ naturally in N.

7. Show that theorem 15.3.9 is natural in C: for an S-module homomorphism $h\colon C \to C'$, the two ways of passing from $\mathrm{Hom}_S(A \otimes_R B, C)$ to $\mathrm{Hom}_R(A, \mathrm{Hom}_S(B, C'))$ agree.

15.4 Application: The Kronecker Product

chapter 13 and chapter 14 computed with matrices: a linear map became an array once bases were chosen, and questions about the map became questions about the array. Nothing in this chapter has yet been written down that way. The apparatus for doing so is already in place: corollary 15.2.6 says that bases of V and W determine a basis of $V \otimes_R W$, and theorem 15.2.2 says that maps of the factors determine a map of the tensor product. Putting the two together produces a matrix, and it has a name.

Definition 15.4.1: Kronecker Product

Let R be a commutative ring, let $A = (a_{ij}) \in M_{m\times n}(R)$ and let $B \in M_{p\times q}(R)$. The *Kronecker product* $A \otimes B \in M_{mp\times nq}(R)$ is the block matrix

$$A \otimes B := \begin{pmatrix} a_{11}B & a_{12}B & \cdots & a_{1n}B \\ a_{21}B & a_{22}B & \cdots & a_{2n}B \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1}B & a_{m2}B & \cdots & a_{mn}B \end{pmatrix}$$

Equivalently, indexing rows by pairs (i, k) with $1 \le i \le m$, $1 \le k \le p$ and columns by pairs (j, l) with $1 \le j \le n$, $1 \le l \le q$, both ordered lexicographically,

$$(A \otimes B)_{(i,k),(j,l)} = a_{ij} b_{kl}$$

The lexicographic indexing is part of the definition rather than a convenience: a different ordering of the pairs produces a matrix conjugate to this one by a permutation, which is the content of proposition 15.4.3(4) below. Every statement in this section is made with respect to that ordering.

Proposition 15.4.2: The Kronecker Product Represents the Tensor Product of Maps

Let R be a commutative ring and let V, V', W, W' be free R-modules with ordered bases $(e_1, \dots, e_n)$, $(e'_1, \dots, e'_m)$, $(f_1, \dots, f_q)$ and $(f'_1, \dots, f'_p)$ respectively. Let $\varphi\colon V \to V'$ be represented by A and $\psi\colon W \to W'$ by B. Give $V \otimes_R W$ the basis $(e_j \otimes f_l)$ and $V' \otimes_R W'$ the basis $(e'_i \otimes f'_k)$ of corollary 15.2.6, each ordered lexicographically. Then $\varphi \otimes \psi$ is represented by $A \otimes B$.

***Proof*:**
By theorem 15.2.2 and the bilinearity of $\otimes$,

$$(\varphi \otimes \psi)(e_j \otimes f_l) = \varphi(e_j) \otimes \psi(f_l) = \left(\sum_{i=1}^{m} a_{ij} e'_i\right) \otimes \left(\sum_{k=1}^{p} b_{kl} f'_k\right) = \sum_{i=1}^{m} \sum_{k=1}^{p} a_{ij} b_{kl} \, (e'_i \otimes f'_k)$$

By definition 13.2.1 the entry of the representing matrix in row (i, k) and column (j, l) is the coefficient of $e'_i \otimes f'_k$ in the image of $e_j \otimes f_l$, which the display gives as $a_{ij} b_{kl}$. That is the entry prescribed by definition 15.4.1, completing the proof.

Every property of the Kronecker product below is therefore a property of $\otimes$ read in coordinates, and the two viewpoints are used interchangeably from here on.

Proposition 15.4.3: Arithmetic of the Kronecker Product

Let R be a commutative ring.

1. (mixed product rule) $(A\otimes B)(C\otimes D) = (AC)\otimes(BD)$ whenever AC and BD are defined.
2. $I_m \otimes I_p = I_{mp}$.
3. If $A \in M_m(R)$ and $B \in M_p(R)$ are invertible then so is $A \otimes B$, with $(A \otimes B)^{-1} = A^{-1} \otimes B^{-1}$.
4. There is a permutation matrix $P \in M_{mp}(R)$, depending only on m and p, with
$$P(A \otimes B)P^{-1} = B \otimes A$$
for all $A \in M_m(R)$ and $B \in M_p(R)$.

***Proof*:**

1. Both sides are indexed by pairs, and summing over the intermediate index pair (r, s),
$$\begin{aligned}\big((A \otimes B)(C \otimes D)\big)_{(i,k),(j,l)} &= \sum_r \sum_s a_{ir} b_{ks}\, c_{rj} d_{sl} \\ &= \left(\sum_r a_{ir} c_{rj}\right)\left(\sum_s b_{ks} d_{sl}\right) = (AC)_{ij}(BD)_{kl}\end{aligned}$$
the middle step using commutativity of R to separate the two sums. The right-hand side is the $(i, k), (j, l)$ entry of $(AC) \otimes (BD)$.
2. $(I_m \otimes I_p)_{(i,k),(j,l)} = \delta_{ij}\delta_{kl}$, which is 1 if $(i, k) = (j, l)$ and 0 otherwise.
3. By (1) and (2), $(A \otimes B)(A^{-1} \otimes B^{-1}) = (AA^{-1}) \otimes (BB^{-1}) = I_m \otimes I_p = I_{mp}$, and the same on the other side.
4. Both $\{(i, k)\}$ and $\{(k, i)\}$ index sets of size mp, so the slot-exchanging bijection $\pi(i, k) = (k, i)$ between them has a permutation matrix P, determined by $Pe_{(i,k)} = e_{(k,i)}$ on standard basis vectors. For any M this gives $\big(PMP^{-1}\big)e_{\pi(y)} = PMe_y = \sum_x M_{xy}e_{\pi(x)}$, that is $\big(PMP^{-1}\big)_{\pi(x),\pi(y)} = M_{x,y}$. Applying this with $M = A \otimes B$,
$$\big(P(A \otimes B)P^{-1}\big)_{(k,i),(l,j)} = (A \otimes B)_{(i,k),(j,l)} = a_{ij}b_{kl} = b_{kl}a_{ij} = (B \otimes A)_{(k,i),(l,j)}$$
again using commutativity, completing the proof.

Part (4) is proposition 15.2.7 in coordinates: the isomorphism $V \otimes W \cong W \otimes V$ becomes, once bases are fixed on both sides, conjugation by the permutation that exchanges the two index slots. The matrix P is the reason the ordering convention in definition 15.4.1 costs nothing.

The trace and the determinant of a Kronecker product are determined by those of the factors. Both formulas hold over an arbitrary commutative ring.

Theorem 15.4.4: Trace and Determinant of a Kronecker Product

Let R be a commutative ring, $A \in M_m(R)$ and $B \in M_p(R)$. Then

$$\operatorname{tr}(A \otimes B) = \operatorname{tr}(A)\operatorname{tr}(B) \qquad \text{and} \qquad \det(A \otimes B) = \big(\det A\big)^p \big(\det B\big)^m$$

***Proof*:**
Trace. The diagonal entries of $A \otimes B$ are those indexed by $(i,k) = (j,l)$, that is by $i = j$ and $k = l$, and there the entry is $a_{ii}b_{kk}$. Hence

$$\operatorname{tr}(A \otimes B) = \sum_{i=1}^{m}\sum_{k=1}^{p} a_{ii}b_{kk} = \left(\sum_{i=1}^{m} a_{ii}\right)\left(\sum_{k=1}^{p} b_{kk}\right) = \operatorname{tr}(A)\operatorname{tr}(B)$$

Determinant. By the mixed product rule and proposition 15.4.3(2),

$$A \otimes B = (AI_m) \otimes (I_pB) = (A \otimes I_p)(I_m \otimes B)$$

so $\det(A \otimes B) = \det(A \otimes I_p)\det(I_m \otimes B)$ by proposition 13.4.7. The second factor is immediate: $I_m \otimes B$ is block diagonal with m diagonal blocks, each equal to B, so $\det(I_m \otimes B) = (\det B)^m$ by proposition 13.4.8.

For the first factor, proposition 15.4.3(4) gives a permutation matrix P with $P(A \otimes I_p)P^{-1} = I_p \otimes A$. Determinants are unchanged by conjugation, since proposition 13.4.7 gives $\det(P)\det(P^{-1}) = \det(PP^{-1}) = \det(I) = 1$ and hence $\det(PMP^{-1}) = \det(M)$. Now $I_p \otimes A$ is block diagonal with p diagonal blocks each equal to A, so

$$\det(A \otimes I_p) = \det(I_p \otimes A) = (\det A)^p$$

by proposition 13.4.8 again. Multiplying the two factors gives the claim, as we sought to show.

Note the asymmetry of the exponents: each determinant is raised to the size of the *other* factor. A quick check is $A = B = I$, where both sides are 1, and a sharper one is $m = p = 2$ with $\det A = \det B = 2$, giving $\det(A \otimes B) = 16$ rather than 4.

Rank behaves as simply, though here a field is needed: the argument passes through a complement, and over a general commutative ring a submodule need not be a direct summand.

Proposition 15.4.5: Rank of a Kronecker Product

Let F be a field, $A \in M_{m\times n}(F)$ and $B \in M_{p\times q}(F)$. Then

$$\operatorname{rank}(A \otimes B) = \operatorname{rank}(A) \cdot \operatorname{rank}(B)$$

***Proof*:**
Let $\varphi\colon V \to V'$ and $\psi\colon W \to W'$ be the linear maps represented by A and B, so that $A \otimes B$ represents $\varphi \otimes \psi$ by proposition 15.4.2, and rank is the dimension of the image (definition 13.1.19).

Choose complements $V' = \operatorname{im}\varphi \oplus U$ and $W' = \operatorname{im}\psi \oplus U'$, which exist because F is a field.

Applying theorem 15.2.3 in each variable,

$$V' \otimes_F W' \cong \big(\mathrm{im}\varphi \otimes \mathrm{im}\psi\big) \oplus \big(\mathrm{im}\varphi \otimes U'\big) \oplus \big(U \otimes \mathrm{im}\psi\big) \oplus \big(U \otimes U'\big)$$

and under this decomposition the natural map $\mathrm{im}\varphi \otimes_F \mathrm{im}\psi \to V' \otimes_F W'$ is the inclusion of the first summand, hence injective.

The image of $\varphi \otimes \psi$ is exactly that summand. It is contained in it: every element of $V \otimes_F W$ is a finite sum of simple tensors by lemma 15.1.4(2), and $(\varphi \otimes \psi)(v \otimes w) = \varphi(v) \otimes \psi(w)$ lies in $\mathrm{im}\varphi \otimes \mathrm{im}\psi$. It contains it, because those same elements $\varphi(v) \otimes \psi(w)$ span $\mathrm{im}\varphi \otimes \mathrm{im}\psi$, again by lemma 15.1.4(2) applied there. Therefore

$$\mathrm{rank}(\varphi \otimes \psi) = \dim_F \big(\mathrm{im}\varphi \otimes_F \mathrm{im}\psi\big) = \dim_F(\mathrm{im}\varphi) \cdot \dim_F(\mathrm{im}\psi) = \mathrm{rank}(\varphi)\mathrm{rank}(\psi)$$

the middle equality being corollary 15.2.6, as we sought to show.

Eigenvalues multiply as well. One half of the statement needs nothing at all.

Proposition 15.4.6: Eigenvectors of a Kronecker Product

Let F be a field, $A \in M_m(F)$, $B \in M_p(F)$, and suppose $Av = \lambda v$ and $Bw = \mu w$ with $v \neq 0$ and $w \neq 0$. Then $v \otimes w \neq 0$ and

$$(A \otimes B)(v \otimes w) = \lambda\mu\,(v \otimes w)$$

so $\lambda\mu$ is an eigenvalue of $A \otimes B$.

***Proof*:**
Reading $A \otimes B$ as $\varphi \otimes \psi$ by proposition 15.4.2 and applying theorem 15.2.2,

$$(\varphi \otimes \psi)(v \otimes w) = \varphi(v) \otimes \psi(w) = (\lambda v) \otimes (\mu w) = \lambda\mu\,(v \otimes w)$$

using bilinearity to extract both scalars. For $v \otimes w \neq 0$, extend v to a basis of F^m and w to a basis of F^p; then $v \otimes w$ is one of the basis elements of $F^m \otimes_F F^p$ furnished by corollary 15.2.6, and a basis element is nonzero, completing the proof.

That accounts for the products of eigenvalues that come from actual eigenvectors. The full spectrum, counted with multiplicity, needs a triangular form, and theorem 14.4.5 supplies one exactly when the minimal polynomials split.

Theorem 15.4.7: Spectrum of a Kronecker Product

Let F be a field and let $A \in M_m(F)$ and $B \in M_p(F)$ have minimal polynomials that split into linear factors over F, which holds for all A and B when F is algebraically closed. List the eigenvalues of A as $\lambda_1, \ldots, \lambda_m$ and those of B as $\mu_1, \ldots, \mu_p$, each repeated according to its algebraic multiplicity. Then the characteristic polynomial of $A \otimes B$ is

$$\prod_{i=1}^{m} \prod_{k=1}^{p} (x - \lambda_i \mu_k)$$

so the eigenvalues of $A \otimes B$, with algebraic multiplicity, are the mp products $\lambda_i \mu_k$.

***Proof*:**
Two general facts are used repeatedly, so both are recorded first.

The determinant of an upper triangular matrix is the product of its diagonal entries. In the defining sum a term $\prod_i t_{i\sigma(i)}$ is nonzero only if $i \leq \sigma(i)$ for every i, and since $\sum_i \big(\sigma(i) - i\big) = 0$ this forces $\sigma(i) = i$ for all i; only the identity term survives. Consequently an upper triangular T has characteristic polynomial $\prod_i (x - t_{ii})$, since $xI - T$ is again upper triangular.

Similar matrices have the same characteristic polynomial, because $\det\big(xI - PMP^{-1}\big) = \det\big(P(xI - M)P^{-1}\big) = \det(xI - M)$ by proposition 13.4.7, applied over the commutative ring $F[x]$.

Now by theorem 14.4.5 there are invertible P and Q with $A = PJ_AP^{-1}$ and $B = QJ_BQ^{-1}$, where J_A and J_B are in Jordan canonical form and hence upper triangular. Combining the two facts, the characteristic polynomial of A is $\prod_i (x - (J_A)_{ii})$, so the diagonal entries of J_A are exactly the eigenvalues $\lambda_1, \ldots, \lambda_m$ of A listed with algebraic multiplicity, and likewise for J_B and the μ_k.

By the mixed product rule and proposition 15.4.3(3),

$$A \otimes B = \big(PJ_AP^{-1}\big) \otimes \big(QJ_BQ^{-1}\big) = (P \otimes Q)\,(J_A \otimes J_B)\,(P \otimes Q)^{-1}$$

so $A \otimes B$ is similar to $J_A \otimes J_B$ and the two have the same characteristic polynomial.

The matrix $J_A \otimes J_B$ is upper triangular for the lexicographic ordering. Its $(i,k),(j,l)$ entry is $(J_A)_{ij}(J_B)_{kl}$, which vanishes unless $i \leq j$ and $k \leq l$; and if (i,k) comes strictly after (j,l) lexicographically then either $i > j$, or $i = j$ and $k > l$, and in both cases one of the two factors is 0. Its diagonal entries are $(J_A)_{ii}(J_B)_{kk} = \lambda_i \mu_k$.

So $J_A \otimes J_B$ is upper triangular with diagonal entries $\lambda_i \mu_k$, and by the first recorded fact its characteristic polynomial is $\prod_{i,k}(x - \lambda_i \mu_k)$, as we sought to show.

Example 15.4.8: Computing with the Kronecker Product
Take $R = \mathbb{Q}$ and

$$A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}, \qquad B = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$$

so that $\det A = -2$, $\mathrm{tr} A = 5$, $\det B = 1$ and $\mathrm{tr} B = 0$. Assembling the blocks $a_{ij}B$,

$$A \otimes B = \begin{pmatrix} 0 & 1 & 0 & 2 \\ -1 & 0 & -2 & 0 \\ 0 & 3 & 0 & 4 \\ -3 & 0 & -4 & 0 \end{pmatrix}$$

Its diagonal entries are all 0, so its trace is $0 = 5 \cdot 0 = \mathrm{tr}(A)\mathrm{tr}(B)$, as theorem 15.4.4 requires. That theorem also predicts

$$\det(A \otimes B) = (\det A)^2 (\det B)^2 = (-2)^2 \cdot 1^2 = 4$$

Both A and B are invertible, so both have rank 2, and proposition 15.4.5 predicts $\mathrm{rank}(A \otimes B) = 4$, consistent with the nonzero determinant.

Replacing A by the rank one matrix

$$A' = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$$

gives $\mathrm{rank}(A' \otimes B) = 1 \cdot 2 = 2$, so $A' \otimes B$ is a 4×4 matrix of rank 2; correspondingly $\det(A' \otimes B) = (\det A')^2 (\det B)^2 = 0$. Note that the rank drops to 2 rather than to 3: a Kronecker product cannot have rank 3 at all, since by proposition 15.4.5 its rank is a product of two integers each at most 2.

One identification remains, and it explains a phenomenon recorded without explanation in section 15.1. For finite-dimensional spaces the tensor product of a dual space with another space is the space of linear maps between them.

Theorem 15.4.9: $V^* \otimes W$ and Linear Maps

Let F be a field and let V, W be finite-dimensional F-vector spaces. Then

$$\Theta \colon V^* \otimes_F W \to \mathrm{Hom}_F(V, W), \qquad \Theta(f \otimes w)(v) = f(v)\, w$$

is an isomorphism of F-vector spaces. Under Θ:

1. the simple tensors correspond exactly to the homomorphisms of rank at most one;
2. the least number of simple tensors whose sum is a given $t \in V^* \otimes_F W$ equals $\mathrm{rank}\Theta(t)$.

***Proof*:**

The map $(f, w) \mapsto \big(v \mapsto f(v)w\big)$ is F-bilinear, so theorem 15.1.12 supplies the F-linear map Θ.

Let $(e_1, \dots, e_n)$ be a basis of V with dual basis $(e_1^*, \dots, e_n^*)$ (section 13.3) and $(w_1, \dots, w_p)$ a basis of W. By corollary 15.2.6 the np elements $e_i^* \otimes w_k$ form a basis of $V^* \otimes_F W$. Their images are the maps $e_j \mapsto \delta_{ij} w_k$, whose matrices with respect to the two bases are the np elementary matrices having a single 1 in position (k, i); those form a basis of $\mathrm{Hom}_F(V, W)$. An F-linear map carrying a basis onto a basis is an isomorphism, so Θ is one.

(1). The image of $\Theta(f \otimes w)$ is contained in Fw, so its rank is at most one. Conversely, let θ have rank at most one. If $\theta = 0$ it is $\Theta(0 \otimes 0)$. Otherwise $\mathrm{im}\theta = Fw$ for some $w \neq 0$, and writing $\theta(v) = f(v)w$ defines a map $f \colon V \to F$ which is linear because θ is and w is a basis of its image;

then $\theta = \Theta(f \otimes w)$.

(2). If $t = \sum_{j=1}^{r} f_j \otimes w_j$ then $\Theta(t) = \sum_j \Theta(f_j \otimes w_j)$ is a sum of r maps of rank at most one. The image of a sum is contained in the sum of the images, and $\dim(X+Y) \le \dim X + \dim Y$, so $\operatorname{rank}\Theta(t) \le r$. Conversely let $\operatorname{rank}\Theta(t) = r$ and choose a basis $u_1, \dots, u_r$ of $\operatorname{im}\Theta(t)$. Every v has $\Theta(t)(v) = \sum_{j=1}^{r} g_j(v) u_j$ for unique scalars $g_j(v)$, and each $g_j \colon V \to F$ so defined is linear because $\Theta(t)$ is and the u_j are independent. Then $\Theta\big(\sum_j g_j \otimes u_j\big) = \Theta(t)$, and Θ is injective, so $t = \sum_{j=1}^{r} g_j \otimes u_j$ is a sum of r simple tensors, completing the proof.

Part (2) settles the question left open in box 15.1.6. Taking $W = V$ of dimension n, the element $\sum_{i=1}^{n} e_i^* \otimes e_i$ corresponds under Θ to the identity map, which has rank n; so it needs exactly n simple tensors and is not itself simple once $n \ge 2$. The witness exhibited there, $e_1 \otimes e_1 + e_2 \otimes e_2$, is this element for $n = 2$ under the identification of V with V^* by the dual basis, and the ad hoc argument given there is replaced by a count of rank.

Exercise 15.4.1

1. Show that $A \otimes B$ and $B \otimes A$ are always similar, and give explicit 2×2 matrices A and B for which they are not equal.
2. Show that $(A \otimes B)^k = A^k \otimes B^k$ for every $k \ge 1$, and deduce that if $A^r = 0$ then $(A \otimes B)^r = 0$ for every B. Conclude that a Kronecker product with a nilpotent factor is nilpotent, with index at most that of the factor.
3. Deduce from the mixed product rule that $A \mapsto A \otimes I_p$ is an injective ring homomorphism $M_m(R) \to M_{mp}(R)$, and that its image commutes elementwise with the image of $B \mapsto I_m \otimes B$.
4. Identify the step in the proof of proposition 15.4.5 that requires a field. Then show it genuinely fails over $\mathbb{Z}$: with $\varphi \colon \mathbb{Z} \to \mathbb{Z}$ multiplication by 2 and ψ the identity on $\mathbb{Z}/2\mathbb{Z}$, show that the natural map $\operatorname{im}\varphi \otimes_{\mathbb{Z}} \mathbb{Z}/2\mathbb{Z} \to \mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Z}/2\mathbb{Z}$ is not injective.
5. Let V and W be arbitrary vector spaces over F, not assumed finite dimensional. Show that Θ of theorem 15.4.9 is still injective and that its image is exactly the set of homomorphisms of finite rank. Deduce that Θ is an isomorphism if and only if V or W is finite dimensional.
6. Show that under the isomorphism $V^* \otimes_F V \cong \operatorname{End}_F(V)$ of theorem 15.4.9, the trace of definition 13.4.3 corresponds to the evaluation map determined by $f \otimes v \mapsto f(v)$. (This is the basis-free description of the trace.)

15.5 Summary

Everything in this chapter rests on one universal property: bilinear maps out of $M \times N$ are homomorphisms out of a single module $M \otimes_R N$. The tables below state what follows from it, each entry pointing at the result that establishes it.

Representing objects and adjunctions

Each class of map has a representing object, and in every case it is a tensor product. The representing object is determined up to unique isomorphism by its property (corollary 15.1.7), so the particular construction used in definition 15.1.3 is not part of the data.

Maps	On	Represented by	Reference
R-balanced, into an abelian group	$M \times N$	$M \otimes_R N$	theorem 15.1.5
R-bilinear, into an R-module	$M \times N$	$M \otimes_R N$	theorem 15.1.12
R-multilinear	$M_1 \times \cdots \times M_n$	$M_1 \otimes_R \cdots \otimes_R M_n$	theorem 15.2.11
R-linear, into an S-module	N	$S \otimes_R N$	theorem 15.3.4
Pairs of R-algebra maps	A, B	$A \otimes_R B$	theorem 15.2.17

The second and third rows require R commutative and the standard structures of example 15.1.9(4); the fifth requires A and B commutative. Two facts are used constantly with these: simple tensors generate, so a homomorphism out of a tensor product is determined by its values on them (lemma 15.1.4), and not every element is itself simple (box 15.1.6).

Two of these are adjunctions, which is the form in which downstream books invoke them.

Left adjoint	Right adjoint	Reference
$S \otimes_R -$	restriction of scalars along $R \to S$	corollary 15.3.5
$- \otimes_R B$	$\mathrm{Hom}_S(B, -)$	theorem 15.3.9

The first is the case $B = S$ of the second.

Module structure

The tensor product of a right module and a left module is only an abelian group. Any further structure comes from a second action on one of the factors.

Hypothesis	Structure on $M \otimes_R N$	Reference
M right R-, N left R-module	abelian group only	definition 15.1.3
M an (S, R)-bimodule	left S-module	proposition 15.1.10
M an (S, R)-, N an (R, T)-bimodule	(S, T)-bimodule	proposition 15.2.1
R commutative, standard structures	R-module	theorem 15.1.12
A, B R-algebras	R-algebra	theorem 15.2.14

What is preserved, and what is not

Statement	Reference
$R \otimes_R N \cong N$	example 15.1.13
$(\bigoplus_i M_i) \otimes_R N \cong \bigoplus_i (M_i \otimes_R N)$	theorem 15.2.4
Free $\otimes$ free is free, on the basis $\{m_i \otimes n_j\}$	corollary 15.2.6
$M \otimes_R N \cong N \otimes_R M$	proposition 15.2.7
$(M \otimes_R N) \otimes_T L \cong M \otimes_R (N \otimes_T L)$	theorem 15.2.8
$(\lambda \otimes \mu) \circ (\varphi \otimes \psi) = (\lambda \circ \varphi) \otimes (\mu \circ \psi)$	theorem 15.2.2
$\mathrm{rank}(A \otimes B) = \mathrm{rank}(A)\mathrm{rank}(B)$ over a field	proposition 15.4.5

The first, fourth and fifth rows are the unit, commutativity and associativity constraints of a symmetric monoidal structure on R-Mod (box 15.2.9).

Fails	**Witness**	**Reference**
Being nonzero	$\mathbb{Z}/2\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Z}/3\mathbb{Z} = 0$	example 15.1.14
Injectivity of a map	multiplication by 2 against $\mathbb{Z}/2\mathbb{Z}$	section 17.5
Containing the original module	ι has a kernel	corollary 15.3.8
Being an integral domain	$\mathbb{C} \otimes_{\mathbb{R}} \mathbb{C} \cong \mathbb{C} \times \mathbb{C}$	example 15.2.15

The second row is the one with consequences beyond this chapter: which modules D make $D \otimes_R -$ preserve injections is the definition of flatness, taken up in section 17.5.

In coordinates

Once bases are fixed, corollary 15.2.6 turns each of the constructions above into a matrix identity.

Object	**Coordinate form**	**Reference**
$\varphi \otimes \psi$	the Kronecker product $(a_{ij}B)$	proposition 15.4.2
$\mathrm{tr}(A \otimes B)$	$\mathrm{tr}(A)\mathrm{tr}(B)$	theorem 15.4.4
$\det(A \otimes B)$	$(\det A)^p(\det B)^m$	theorem 15.4.4
Eigenvalues of $A \otimes B$	the products $\lambda_i \mu_k$	theorem 15.4.7
$\mathrm{Hom}_F(V, W)$	$V^* \otimes_F W$	theorem 15.4.9

Computations worth reusing

Identity	**Reference**
$\mathbb{Z}/m\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Z}/n\mathbb{Z} \cong \mathbb{Z}/\gcd(m,n)\mathbb{Z}$	proposition 15.2.13
$R/I \otimes_R R/J \cong R/(I+J)$	proposition 15.2.13
$S \otimes_R R^n \cong S^n$	example 15.3.3
$V_{\mathbb{C}} = \mathbb{C} \otimes_{\mathbb{R}} V$, with $\dim_{\mathbb{C}} V_{\mathbb{C}} = \dim_{\mathbb{R}} V$	example 15.3.6
$A \otimes_{\mathbb{Z}} B$ is the coproduct in **CRing**	corollary 15.2.18

16

Multilinear Algebra

chapter 15 represented the bilinear maps on a product of two modules. Fixing a single module M and allowing more arguments produces further classes of map on M^k: the k-multilinear maps, those left unchanged by permuting the arguments, and those that vanish whenever two arguments coincide. Each class has its own representing object, and the three objects are quotients of one another.

Summing over k turns each family of representing objects into a ring. The tensor algebra $\mathcal{T}(M)$, the symmetric algebra $\mathcal{S}(M)$ and the exterior algebra $\bigwedge(M)$ are graded by the number of arguments, and each carries a universal property of its own, now for homomorphisms into an algebra rather than into a module. The determinant returns at the end as the map an endomorphism induces on the top exterior power, recovering section 13.4 from a universal property rather than from the defining sum.

16.1 Multilinear, Symmetric and Alternating Maps

The k-multilinear maps on M already have a representing object. definition 15.2.10 defined the class and theorem 15.2.11 produced the module through which every such map factors: an R-linear map out of the k-fold tensor product of M with itself is the same thing as a k-multilinear map on M^k.

Two subclasses of these maps occur constantly, and both are cut out by requiring the map to be insensitive to something. A symmetric map does not see the order of its arguments; an alternating map does not survive a repetition among them. Each subclass is again represented by a module, and in each case that module is a quotient of the k-fold tensor product by exactly the elements the subclass is required to kill. That is the pattern of this section, applied three times.

Definition 16.1.1: Tensor Powers

Let R be a commutative ring and M an R-module. For $k \geq 1$ the *k-th tensor power* of M is

$$\mathcal{T}^k(M) := \underbrace{M \otimes_R M \otimes_R \cdots \otimes_R M}_{k \text{ factors}}$$

bracketed to the left as in section 15.2, with $\mathcal{T}^0(M) := R$ and $\mathcal{T}^1(M) = M$. Its elements are called *k-tensors*.

In this notation theorem 15.2.11 reads: the map $M^k \to \mathcal{T}^k(M)$ sending $(m_1, \ldots, m_k)$ to $m_1 \otimes \cdots \otimes m_k$ is k-multilinear, and every k-multilinear map on M^k factors through it uniquely by an R-module homomorphism. Nothing further is needed about $\mathcal{T}^k(M)$ here; the two quotients are what this section constructs.

Definition 16.1.2: Symmetric Multilinear Maps

Let R be a commutative ring and let M, N be R-modules. A k-multilinear map $\varphi \colon M^k \to N$ is *symmetric* if

$$\varphi\big(m_{\sigma(1)}, \ldots, m_{\sigma(k)}\big) = \varphi(m_1, \ldots, m_k)$$

for every $\sigma \in S_k$ and all $m_1, \ldots, m_k \in M$.

Definition 16.1.3: Alternating Multilinear Maps

With R, M, N as above, a k-multilinear map $\varphi \colon M^k \to N$ is *alternating* if

$$\varphi(m_1, \ldots, m_k) = 0 \qquad \text{whenever } m_i = m_j \text{ for some } i \neq j$$

The determinant of definition 13.4.5, read as a function of the n columns of a matrix, is the standard example of an alternating n-multilinear map $(R^n)^n \to R$: it is multilinear in the columns and vanishes when two of them agree, which is proposition 13.4.6(3) and (4). A symmetric bilinear form, such as an inner product (section 13.6), is the standard example of a symmetric map.

Alternating maps are often introduced by a sign rule rather than by the vanishing condition. The two agree except in characteristic 2, and the direction that always holds is worth isolating, since it is what makes the exterior power computable.

Lemma 16.1.4: Equivalence Of Alternating Map

Let R be a commutative ring and let $\varphi \colon M^k \to N$ be k-multilinear.

1. If φ is alternating then for every $\sigma \in S_k$,

$$\varphi\big(m_{\sigma(1)}, \ldots, m_{\sigma(k)}\big) = (-1)^{\sigma}\, \varphi(m_1, \ldots, m_k)$$

2. If 2 is a unit in R, the converse holds: a k-multilinear map satisfying the displayed sign rule is alternating.

***Proof*:**
(1). For $\sigma \in S_k$ write $\varphi^\sigma(m_1, \dots, m_k) := \varphi(m_{\sigma(1)}, \dots, m_{\sigma(k)})$. Substituting shows $(\varphi^\sigma)^\tau = \varphi^{\tau\sigma}$, so if the identity $\varphi^\sigma = (-1)^\sigma \varphi$ holds for two permutations it holds for their product:

$$\varphi^{\tau\sigma} = (\varphi^\sigma)^\tau = \big((-1)^\sigma \varphi\big)^\tau = (-1)^\sigma \varphi^\tau = (-1)^\sigma (-1)^\tau \varphi = (-1)^{\tau\sigma} \varphi$$

Since the transpositions generate S_k, it therefore suffices to treat a transposition $\tau = (a\ b)$ with $a < b$.

Fix all arguments except those in positions a and b, and let $f(x, y)$ denote the value of φ with x in position a and y in position b. Then f is bilinear, and $f(x, x) = 0$ for every x because φ is alternating. Expanding,

$$0 = f(x + y,\, x + y) = f(x, x) + f(x, y) + f(y, x) + f(y, y) = f(x, y) + f(y, x)$$

so $f(y, x) = -f(x, y)$, which is exactly the claim for τ.

(2). Suppose the sign rule holds and $m_i = m_j$ with $i \neq j$. Applying the rule to the transposition $\tau = (i\ j)$, the left-hand side is φ evaluated at the same arguments, since interchanging two equal entries changes nothing, while the right-hand side is $-\varphi(m_1, \dots, m_k)$. Hence $2\,\varphi(m_1, \dots, m_k) = 0$, and multiplying by the inverse of 2 gives $\varphi(m_1, \dots, m_k) = 0$, completing the proof.

The hypothesis in (2) cannot be dropped. Over $R = \mathbb{Z}/2\mathbb{Z}$ every sign $(-1)^\sigma$ equals 1, so the sign rule says only that the map is symmetric; and the multiplication map $\varphi\colon \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z} \to \mathbb{Z}/2\mathbb{Z}$ with $\varphi(x, y) = xy$ is symmetric and bilinear but has $\varphi(\overline{1}, \overline{1}) = \overline{1} \neq \overline{0}$, so it is not alternating. In characteristic 2 the vanishing condition of definition 16.1.3 is strictly stronger, and it is the one the exterior power is built from.

Each subclass is now represented, by quotienting $\mathcal{T}^k(M)$ by the elements a map in the subclass is obliged to kill.

Definition 16.1.5: Symmetric Powers

Let $C^k(M) \subseteq \mathcal{T}^k(M)$ be the submodule generated by all elements

$$m_1 \otimes \cdots \otimes m_k - m_{\sigma(1)} \otimes \cdots \otimes m_{\sigma(k)} \qquad \sigma \in S_k,\ m_i \in M$$

The *k-th symmetric power* of M is $\mathcal{S}^k(M) := \mathcal{T}^k(M)/C^k(M)$, and the image of $m_1 \otimes \cdots \otimes m_k$ is written $m_1 m_2 \cdots m_k$. Since $C^0(M) = C^1(M) = 0$, one has $\mathcal{S}^0(M) = R$ and $\mathcal{S}^1(M) = M$.

Theorem 16.1.6: Universal Property of the Symmetric Power

Let R be a commutative ring and M an R-module. The map

$$\varsigma\colon M^k \to \mathcal{S}^k(M), \qquad \varsigma(m_1, \dots, m_k) = m_1 m_2 \cdots m_k$$

is symmetric and k-multilinear, and it is universal among such maps: for every R-module N and every symmetric k-multilinear $\varphi\colon M^k \to N$ there is a unique R-module homomorphism $\overline{\varphi}\colon \mathcal{S}^k(M) \to N$ with $\varphi = \overline{\varphi} \circ \varsigma$.

***Proof*:**
The map ς is the composite of the universal k-multilinear map of theorem 15.2.11 with the quotient map $\pi\colon \mathcal{T}^k(M) \to \mathcal{S}^k(M)$, hence k-multilinear. It is symmetric because the difference $m_1 \otimes \cdots \otimes m_k - m_{\sigma(1)} \otimes \cdots \otimes m_{\sigma(k)}$ lies in $C^k(M)$ by construction and so maps to 0.

Let φ be symmetric and k-multilinear. By theorem 15.2.11 there is a unique R-module homomorphism $\psi\colon \mathcal{T}^k(M) \to N$ with $\psi(m_1 \otimes \cdots \otimes m_k) = \varphi(m_1, \ldots, m_k)$. On a generator of $C^k(M)$,

$$\psi\big(m_1 \otimes \cdots \otimes m_k - m_{\sigma(1)} \otimes \cdots \otimes m_{\sigma(k)}\big) = \varphi(m_1, \ldots, m_k) - \varphi(m_{\sigma(1)}, \ldots, m_{\sigma(k)}) = 0$$

precisely because φ is symmetric. A homomorphism vanishing on a generating set vanishes on the submodule it generates, so $C^k(M) \subseteq \ker\psi$ and ψ factors through the quotient, giving $\overline{\varphi}$ with $\overline{\varphi} \circ \varsigma = \varphi$.

For uniqueness, the products $m_1 \cdots m_k$ generate $\mathcal{S}^k(M)$, being the images of the simple tensors, which generate $\mathcal{T}^k(M)$; two homomorphisms agreeing on a generating set are equal, as we sought to show.

Definition 16.1.7: Exterior Powers

Let $A^k(M) \subseteq \mathcal{T}^k(M)$ be the submodule generated by all simple tensors

$$m_1 \otimes \cdots \otimes m_k \qquad \text{with } m_i = m_j \text{ for some } i \neq j$$

The *k-th exterior power* of M is $\bigwedge^k(M) := \mathcal{T}^k(M)/A^k(M)$, and the image of $m_1 \otimes \cdots \otimes m_k$ is written $m_1 \wedge \cdots \wedge m_k$, the operation being called the *wedge* or *exterior product*. Since $A^0(M) = A^1(M) = 0$, one has $\bigwedge^0(M) = R$ and $\bigwedge^1(M) = M$.

Theorem 16.1.8: Universal Property of the Exterior Power

Let R be a commutative ring and M an R-module. The map

$$\alpha\colon M^k \to \bigwedge^k(M), \qquad \alpha(m_1, \ldots, m_k) = m_1 \wedge \cdots \wedge m_k$$

is alternating and k-multilinear, and it is universal among such maps: for every R-module N and every alternating k-multilinear $\varphi\colon M^k \to N$ there is a unique R-module homomorphism $\overline{\varphi}\colon \bigwedge^k(M) \to N$ with $\varphi = \overline{\varphi} \circ \alpha$.

***Proof*:**
As before α is k-multilinear, being the composite of the universal k-multilinear map with the quotient map, and it is alternating because a simple tensor with a repeated entry lies in $A^k(M)$ by construction.

Given an alternating k-multilinear φ, let $\psi\colon \mathcal{T}^k(M) \to N$ be the homomorphism supplied by theorem 15.2.11. Each generator of $A^k(M)$ is a simple tensor $m_1 \otimes \cdots \otimes m_k$ with $m_i = m_j$ for some $i \neq j$, and ψ sends it to $\varphi(m_1, \ldots, m_k)$, which is 0 because φ is alternating. So $A^k(M) \subseteq \ker\psi$ and ψ factors through $\bigwedge^k(M)$. Uniqueness holds because the elements $m_1 \wedge \cdots \wedge m_k$ generate

$\bigwedge^k(M)$, completing the proof.

The sign rule now transfers from maps to elements.

Proposition 16.1.9: Anticommutativity of the Wedge Product

Let M be an R-module. In $\bigwedge^k(M)$,

$$m_1 \wedge \cdots \wedge m_k = 0 \quad \text{whenever } m_i = m_j \text{ for some } i \neq j$$
$$m_{\sigma(1)} \wedge \cdots \wedge m_{\sigma(k)} = (-1)^\sigma \, m_1 \wedge \cdots \wedge m_k$$

for every $\sigma \in S_k$. In particular $m \wedge m' = -\, m' \wedge m$ for all $m, m' \in M$.

***Proof*:**
The first identity is the statement that the generators of $A^k(M)$ die in the quotient. The second is lemma 16.1.4(1) applied to the alternating map α of theorem 16.1.8, whose values are precisely the elements $m_1 \wedge \cdots \wedge m_k$. Taking $k = 2$ and σ the transposition gives the last claim, completing the proof.

Both $\mathcal{S}^k(M)$ and $\bigwedge^k(M)$ have been built as quotients of $\mathcal{T}^k(M)$. They can also be found inside it, as submodules, provided $k!$ is invertible. The comparison rests on an action of the symmetric group.

Proposition 16.1.10: The Symmetric Group Acts on Tensor Powers

Let M be an R-module and $k \geq 1$. For each $\sigma \in S_k$ there is a unique R-module homomorphism $\sigma \cdot (-)$ of $\mathcal{T}^k(M)$ with

$$\sigma \cdot (m_1 \otimes \cdots \otimes m_k) = m_{\sigma^{-1}(1)} \otimes \cdots \otimes m_{\sigma^{-1}(k)}$$

and these assemble into a left action of S_k on $\mathcal{T}^k(M)$ by R-module automorphisms.

***Proof*:**
For fixed σ the assignment $(m_1, \ldots, m_k) \mapsto m_{\sigma^{-1}(1)} \otimes \cdots \otimes m_{\sigma^{-1}(k)}$ is k-multilinear, since each argument appears in exactly one slot of the output, so theorem 15.2.11 supplies the homomorphism and its uniqueness.

For the action law, evaluate on a simple tensor. The i-th slot of $\tau \cdot (m_1 \otimes \cdots \otimes m_k)$ holds $m_{\tau^{-1}(i)}$, so the i-th slot of $\sigma \cdot \big(\tau \cdot (m_1 \otimes \cdots \otimes m_k)\big)$ holds $m_{\tau^{-1}(\sigma^{-1}(i))} = m_{(\sigma\tau)^{-1}(i)}$, which is the i-th slot of $(\sigma\tau) \cdot (m_1 \otimes \cdots \otimes m_k)$. Two homomorphisms agreeing on the simple tensors are equal, so $\sigma \cdot (\tau \cdot z) = (\sigma\tau) \cdot z$ for all z. The identity permutation acts as the identity, so each σ acts invertibly, completing the proof.

The inverse in the formula is what makes the assignment a left action rather than a right one; without it the composition law would come out reversed.

Definition 16.1.11: Symmetric and Alternating Tensors

Let M be an R-module and $k \geq 1$. A tensor $z \in \mathcal{T}^k(M)$ is

1. a *symmetric k-tensor* if $\sigma \cdot z = z$ for every $\sigma \in S_k$;
2. an *alternating k-tensor* if $\sigma \cdot z = (-1)^\sigma z$ for every $\sigma \in S_k$.

These form submodules of $\mathcal{T}^k(M)$, written $\mathcal{T}^k_{\mathrm{sym}}(M)$ and $\mathcal{T}^k_{\mathrm{alt}}(M)$, each being an intersection of kernels of R-module homomorphisms.
The *symmetrization* and *antisymmetrization* operators on $\mathcal{T}^k(M)$ are

$$\mathrm{Sym}(z) := \sum_{\sigma \in S_k} \sigma \cdot z \qquad \text{and} \qquad \mathrm{Alt}(z) := \sum_{\sigma \in S_k} (-1)^\sigma \, \sigma \cdot z$$

Proposition 16.1.12: Symmetric and Alternating Tensors When $k!$ Is Invertible

Let M be an R-module, $k \geq 1$, and suppose $k!$ is a unit in R. Then $\frac{1}{k!}\mathrm{Sym}$ and $\frac{1}{k!}\mathrm{Alt}$ are idempotent endomorphisms of $\mathcal{T}^k(M)$ with images $\mathcal{T}^k_{\mathrm{sym}}(M)$ and $\mathcal{T}^k_{\mathrm{alt}}(M)$, and they induce isomorphisms of R-modules

$$\mathcal{S}^k(M) \xrightarrow{\sim} \mathcal{T}^k_{\mathrm{sym}}(M) \qquad \text{and} \qquad \bigwedge^k(M) \xrightarrow{\sim} \mathcal{T}^k_{\mathrm{alt}}(M)$$

In particular $\mathcal{T}^k(M) = \mathcal{T}^k_{\mathrm{sym}}(M) \oplus \ker \mathrm{Sym}$, and likewise for the alternating part.

***Proof*:**
The symmetric case. Reindexing the sum shows $\tau \cdot \mathrm{Sym}(z) = \mathrm{Sym}(z)$ for every τ, since $\sigma \mapsto \tau\sigma$ permutes S_k; so Sym lands in $\mathcal{T}^k_{\mathrm{sym}}(M)$. If z is already symmetric then every term of the sum equals z, so $\mathrm{Sym}(z) = k!\, z$ and $\frac{1}{k!}\mathrm{Sym}$ restricts to the identity there. Hence $\frac{1}{k!}\mathrm{Sym}$ is idempotent with image exactly $\mathcal{T}^k_{\mathrm{sym}}(M)$, and an idempotent endomorphism splits its domain as image plus kernel.

Writing $\rho = \sigma^{-1}$, the generator $m_1 \otimes \cdots \otimes m_k - m_{\sigma(1)} \otimes \cdots \otimes m_{\sigma(k)}$ of $C^k(M)$ is $z - \rho \cdot z$ for the simple tensor z, and $\mathrm{Sym}(z - \rho \cdot z) = \mathrm{Sym}(z) - \mathrm{Sym}(z) = 0$ by the reindexing above. So $\frac{1}{k!}\mathrm{Sym}$ kills $C^k(M)$ and induces $\Theta\colon \mathcal{S}^k(M) \to \mathcal{T}^k_{\mathrm{sym}}(M)$.

Its inverse is the composite $\mathcal{T}^k_{\mathrm{sym}}(M) \hookrightarrow \mathcal{T}^k(M) \xrightarrow{\pi} \mathcal{S}^k(M)$. Indeed, for a simple tensor z one has $\pi(\sigma \cdot z) = \pi(z)$, since $z - \sigma \cdot z \in C^k(M)$, so

$$\pi\left(\tfrac{1}{k!}\mathrm{Sym}(z)\right) = \tfrac{1}{k!} \sum_{\sigma \in S_k} \pi(\sigma \cdot z) = \pi(z)$$

and both sides are homomorphisms, so this holds for all z; that is one composite. In the other order, $\frac{1}{k!}\mathrm{Sym}$ is the identity on $\mathcal{T}^k_{\mathrm{sym}}(M)$, which is the second. So Θ is an isomorphism.

The alternating case. The same reindexing gives $\tau \cdot \mathrm{Alt}(z) = (-1)^\tau \mathrm{Alt}(z)$, so Alt lands in $\mathcal{T}^k_{\mathrm{alt}}(M)$; and if z is alternating then $(-1)^\sigma \sigma \cdot z = (-1)^\sigma(-1)^\sigma z = z$ for every σ, so $\mathrm{Alt}(z) = k!\, z$. Idempotence and the splitting follow as before.

That Alt kills $A^k(M)$ needs the pairing argument. Let $z = m_1 \otimes \cdots \otimes m_k$ have $m_a = m_b$ with

$a \neq b$ and set $\tau = (a\ b)$. Then $\tau \cdot z = z$, because the two interchanged entries are equal. Pairing each σ with $\sigma\tau$ gives $(\sigma\tau) \cdot z = \sigma \cdot (\tau \cdot z) = \sigma \cdot z$ while $(-1)^{\sigma\tau} = -(-1)^{\sigma}$, so the two terms cancel and $\mathrm{Alt}(z) = 0$. Hence $\frac{1}{k!}\,\mathrm{Alt}$ induces a map on $\bigwedge^k(M)$.

For the inverse, let $\pi_A : \mathcal{T}^k(M) \to \bigwedge^k(M)$ be the quotient map. By proposition 16.1.9, $\pi_A(\sigma \cdot z) = (-1)^{\sigma}\pi_A(z)$ on simple tensors, so

$$\pi_A\left(\tfrac{1}{k!}\,\mathrm{Alt}(z)\right) = \tfrac{1}{k!} \sum_{\sigma \in S_k} (-1)^{\sigma}(-1)^{\sigma}\pi_A(z) = \pi_A(z)$$

using $(-1)^{\sigma}(-1)^{\sigma} = 1$; and $\frac{1}{k!}\,\mathrm{Alt}$ is the identity on $\mathcal{T}^k_{\mathrm{alt}}(M)$. So the two maps are mutually inverse, as we sought to show.

Defining an invariant element by averaging over a finite group, which is what $\frac{1}{k!}\,\mathrm{Sym}$ does, is a technique that recurs whenever the order of the group is invertible in the ring; it is the mechanism behind Maschke's theorem (theorem 53.1.9), where averaging over a finite group produces an invariant complement to a subrepresentation. When $k!$ is not invertible the comparison genuinely fails, and the exercises give a witness.

Exercise 16.1.1

1. Show that $C^k(M)$ is already generated by the elements $z - \tau \cdot z$ with z a simple tensor and τ a transposition, so that symmetry of a multilinear map need only be checked on transpositions.
2. Let R have characteristic 2. Show that a k-multilinear map satisfies the sign rule of lemma 16.1.4 if and only if it is symmetric, and exhibit a symmetric trilinear map over $\mathbb{Z}/2\mathbb{Z}$ that is not alternating.
3. Show that if M is generated by fewer than k elements then $\bigwedge^k(M) = 0$.
4. Let $\varphi \colon M \to N$ be a surjection of R-modules. Show that the induced maps $\mathcal{S}^k(M) \to \mathcal{S}^k(N)$ and $\bigwedge^k(M) \to \bigwedge^k(N)$ are surjective. (Use theorem 15.2.2 on $\mathcal{T}^k$ first, then check that generators map to generators.)
5. Take $R = \mathbb{Z}$ and $M = \mathbb{Z}/2\mathbb{Z}$. Compute $\mathcal{T}^2(M)$, $\mathcal{S}^2(M)$ and $\bigwedge^2(M)$.
6. Show that $\mathrm{Alt} \circ \mathrm{Sym} = 0$ on $\mathcal{T}^k(M)$ for every $k \geq 2$, over any commutative ring.
7. Continuing the previous exercise with $R = \mathbb{Z}$ and $M = \mathbb{Z}/2\mathbb{Z}$, where $2!$ is not a unit, compute $\mathcal{T}^2_{\mathrm{alt}}(M)$ and compare it with $\bigwedge^2(M)$. Conclude that the invertibility hypothesis in proposition 16.1.12 cannot be dropped: the two modules are not merely joined by a different isomorphism, they are not isomorphic at all.

16.2 Graded Rings and Graded Modules

The tensor powers of definition 16.1.1 come with a bookkeeping device that has so far gone unused. Associativity (theorem 15.2.8) identifies $\mathcal{T}^k(M) \otimes_R \mathcal{T}^l(M)$ with $\mathcal{T}^{k+l}(M)$, so a k-tensor times an l-tensor is a $(k+l)$-tensor: the index adds. A ring carrying an index of this kind is called graded, and the three algebras of section 16.3 are all built by taking a graded ring apart and putting it back together modulo relations that respect the index. Grading is therefore developed first, in its own right.

Definition 16.2.1: Graded Ring

A ring S is a *graded ring* if it comes with a decomposition

$$S = \bigoplus_{k \geq 0} S_k$$

of its underlying additive group into subgroups such that

$$S_k S_l \subseteq S_{k+l} \qquad \text{for all } k, l \geq 0$$

An element of S_k is *homogeneous of degree* k, and S_k is the *homogeneous component of* S *of degree* k. Every $s \in S$ has a unique expression $s = \sum_k s_k$ with $s_k \in S_k$ and all but finitely many s_k zero; the s_k are the *homogeneous components* of s.

The base ring plays no role in the definition, and the letter S is used for the graded ring throughout to keep it distinct from a ring R that may be acting. Note also that S_k is only required to be an additive subgroup: the multiplicative structure is carried entirely by the containments $S_k S_l \subseteq S_{k+l}$. What those containments force is the following.

Proposition 16.2.2: The Degree-Zero Part

Let $S = \bigoplus_{k \geq 0} S_k$ be a graded ring. Then

1. $1 \in S_0$, and S_0 is a subring of S;
2. each S_k is an S_0-module, and S is an S_0-algebra when S_0 lies in the centre of S;
3. $S_+ := \bigoplus_{k > 0} S_k$ is a two-sided ideal of S, and $S/S_+ \cong S_0$ as rings.

***Proof*:**

1. Write $1 = \sum_k e_k$ with $e_k \in S_k$. For $x \in S_l$ the product $x \cdot 1 = \sum_k x e_k$ has $x e_k \in S_{l+k}$, so comparing homogeneous components of degree l on both sides of $x = x \cdot 1$ gives $x = x e_0$. Additivity extends this from homogeneous x to all of S, and the same argument on the other side gives $e_0 x = x$. So e_0 is a two-sided identity, and an identity is unique, so $1 = e_0 \in S_0$. Now $S_0 S_0 \subseteq S_0$ makes S_0 closed under multiplication, it is an additive subgroup by construction, and it contains 1; so it is a subring.
2. $S_0 S_k \subseteq S_k$ and $S_k S_0 \subseteq S_k$, so each S_k is a module over the subring S_0 on either side, and S is their direct sum. The algebra statement is definition 12.1.13, whose hypothesis is exactly that the image of S_0 is central.
3. S_+ is an additive subgroup, and for $x \in S_l$ and $y \in S_k$ with $k > 0$ one has $xy \in S_{k+l}$ with $k + l > 0$; additivity extends this to all of S, on both sides, so S_+ is a two-sided ideal. The projection $S \to S_0$ sending s to its degree-zero component is additive, sends 1 to 1 by (1), and is multiplicative because the degree-zero component of a product st is $s_0 t_0$, every

other contribution $s_k t_l$ with $k + l = 0$ and $k, l \geq 0$ being absent. Its kernel is exactly S_+, completing the proof.

Note that $S_i \subseteq S_j$ is not to be expected for $i < j$: the components are summands, not a filtration. In $R[x]$ graded by degree, $S_1 = Rx$ and $S_2 = Rx^2$ are disjoint apart from 0.

Definition 16.2.3: Graded Modules, Ideals and Homomorphisms

Let $S = \bigoplus_{k \geq 0} S_k$ and $T = \bigoplus_{k \geq 0} T_k$ be graded rings.

1. A left S-module M is *graded* if it comes with a decomposition $M = \bigoplus_{k \geq 0} M_k$ into additive subgroups with $S_k M_l \subseteq M_{k+l}$ for all k, l. Elements of M_k are *homogeneous of degree* k.
2. An ideal $I \subseteq S$ is a *graded ideal*, or a *homogeneous ideal*, if
$$I = \bigoplus_{k \geq 0} (I \cap S_k)$$
Writing $I_k := I \cap S_k$, a graded ideal is in particular a graded S-module.
3. A ring homomorphism $\varphi \colon S \to T$ is *graded* if $\varphi(S_k) \subseteq T_k$ for every k.

Example 16.2.4: Graded Rings

1. Any ring S is graded with $S_0 = S$ and $S_k = 0$ for $k > 0$. This is the *trivial* grading, and it shows the definition imposes nothing by itself.
2. For any ring R, the polynomial ring $R[x]$ is graded by degree:
$$R[x] = \bigoplus_{k \geq 0} Rx^k$$
the degree-k component being the free R-module on the single monomial x^k, and $Rx^k \cdot Rx^l \subseteq Rx^{k+l}$. It is worth being precise about what the components are, since the powers $(x)^k$ of the ideal (x) are *not* a grading: they are nested, $(x) \supseteq (x)^2 \supseteq \cdots$, so their sum is not direct.
3. More generally $R[x_1, \ldots, x_n]$ is graded with k-th component the free R-module on the monomials $x_1^{a_1} \cdots x_n^{a_n}$ with $a_1 + \cdots + a_n = k$; these are the *homogeneous polynomials* of degree k. Here $S_0 = R$, and S_+ is the ideal $(x_1, \ldots, x_n)$ of polynomials with zero constant term.
4. Let R be commutative and $I \subseteq R$ an ideal. The *Rees algebra* of R with respect to I is
$$\mathrm{Rees}_R(I) := \bigoplus_{k \geq 0} I^k = R \oplus I \oplus I^2 \oplus \cdots$$
with multiplication induced by that of R. It is graded because $I^k I^l = I^{k+l}$, so in particular $I^k I^l \subseteq I^{k+l}$. The summands are indexed copies of the ideals I^k, kept formally separate; the sum inside R would not be direct, for the same reason as in (2).[a]

[a] Rees algebras are the algebraic form of a blow-up in algebraic geometry, and are taken up alongside the Artin-Rees lemma in *Commutative Algebra* [10].

Quotienting by a graded ideal is exactly the operation that keeps the grading.

Proposition 16.2.5: Quotient of a Graded Ring by a Graded Ideal

Let $S = \bigoplus_k S_k$ be a graded ring and $I \subseteq S$ a graded ideal, and write $I_k = I \cap S_k$. Then S/I is a graded ring with

$$(S/I)_k \cong S_k/I_k$$

and the quotient map $S \to S/I$ is a graded ring homomorphism.

***Proof*:**
Let $\pi\colon S \to S/I$ be the quotient map and let $(S/I)_k := \pi(S_k)$, an additive subgroup. These generate S/I additively, since π is surjective and the S_k generate S. Their sum is direct: if $\sum_k \pi(s_k) = 0$ with $s_k \in S_k$ and almost all zero, then $\sum_k s_k \in I$, so every $s_k \in I$ by definition 16.2.3(2), whence $\pi(s_k) = 0$ for each k. The containment $(S/I)_k(S/I)_l \subseteq (S/I)_{k+l}$ follows by applying π to $S_kS_l \subseteq S_{k+l}$. So $S/I = \bigoplus_k (S/I)_k$ is a graded ring, and $\pi(S_k) = (S/I)_k$ says exactly that π is graded.

For the displayed isomorphism, the restriction $\pi|_{S_k}\colon S_k \to (S/I)_k$ is a surjective homomorphism of additive groups with kernel $S_k \cap I = I_k$, so $S_k/I_k \cong (S/I)_k$ by theorem 3.1.14, completing the proof.

The same bookkeeping applies to a graded homomorphism, whose kernel and image decompose degree by degree.

Proposition 16.2.6: Kernel and Image of a Graded Homomorphism

Let $\varphi\colon S \to T$ be a graded ring homomorphism and write $\varphi_k := \varphi|_{S_k}\colon S_k \to T_k$. Then

$$\ker\varphi = \bigoplus_{k\geq 0} \ker\varphi_k \qquad \text{and} \qquad \varphi(S) = \bigoplus_{k\geq 0} \varphi(S_k)$$

so $\ker\varphi$ is a graded ideal of S and $\varphi(S)$ is a graded subring of T.

***Proof*:**
If $s \in \ker\varphi$ has homogeneous components s_k, then $\sum_k \varphi(s_k) = 0$ with $\varphi(s_k) \in T_k$, so each $\varphi(s_k) = 0$ by uniqueness of the decomposition in T; hence $s \in \bigoplus_k \ker\varphi_k$. The reverse inclusion is clear, and the displayed equality shows $\ker\varphi$ is graded.

For the image, $\varphi(S) = \sum_k \varphi(S_k)$ because φ is additive and the S_k span S, and the sum is direct because $\varphi(S_k) \subseteq T_k$ and the T_k are independent. Closure under multiplication is $\varphi(S_k)\varphi(S_l) \subseteq \varphi(S_{k+l})$, which holds because φ is multiplicative, completing the proof.

Those two results give the last of several equivalent ways to recognize a graded ideal.

Proposition 16.2.7: Characterizing Graded Ideals

Let $S = \bigoplus_k S_k$ be a graded ring and $I \subseteq S$ an ideal. The following are equivalent.

1. I is a graded ideal.
2. For every $s \in I$, each homogeneous component of s lies in I.
3. I is generated by homogeneous elements.
4. I is the kernel of a graded ring homomorphism out of S.

***Proof*:**
(1) ⇒ (2). If $I = \bigoplus_k (I \cap S_k)$ then every $s \in I$ is a finite sum $\sum_k t_k$ with $t_k \in I \cap S_k$. That expression is also the decomposition of s into homogeneous components, which is unique, so each component $s_k = t_k$ lies in I.

(2) ⇒ (3). By (2) the set H of homogeneous elements of I is contained in I, and every $s \in I$ is the sum of its homogeneous components, all of which lie in H. So H generates I.

(3) ⇒ (1). Let I be generated by homogeneous elements h_α, say $h_\alpha \in S_{d_\alpha}$. An element $s \in I$ is a finite sum $s = \sum_\alpha a_\alpha h_\alpha$ with $a_\alpha \in S$. Decomposing each $a_\alpha = \sum_j a_{\alpha,j}$ into homogeneous components,

$$s = \sum_\alpha \sum_j a_{\alpha,j} h_\alpha$$

and each $a_{\alpha,j} h_\alpha$ is homogeneous of degree $j + d_\alpha$ and lies in I. Collecting the terms of each fixed degree therefore exhibits the homogeneous components of s as elements of I, so $s \in \bigoplus_k (I \cap S_k)$. The reverse inclusion is automatic, giving (1).

(1) ⇒ (4). By proposition 16.2.5 below, S/I is a graded ring and the quotient map is a graded homomorphism; its kernel is I.

(4) ⇒ (2). Let $\varphi\colon S \to T$ be graded with kernel I, and let $s \in I$ have homogeneous components s_k. Then $0 = \varphi(s) = \sum_k \varphi(s_k)$ with $\varphi(s_k) \in T_k$, and the decomposition of 0 into homogeneous components of T is unique, so every $\varphi(s_k) = 0$ and every $s_k \in I$, completing the proof.

For prime ideals the grading cuts the work down: only homogeneous elements need to be tested.

Corollary 16.2.8: Prime Graded Ideals

Let S be a graded commutative ring and $I \subseteq S$ a graded ideal with $I \neq S$. Then I is prime if and only if for all *homogeneous* $f, g \in S$ with $fg \in I$, either $f \in I$ or $g \in I$.

***Proof*:**
If I is prime the condition holds, since it is a special case of the definition.

Conversely, suppose the condition holds and let $f, g \in S$ with $fg \in I$ but $f \notin I$ and $g \notin I$. By proposition 16.2.7(2) some homogeneous component of f lies outside I, and only finitely many components are nonzero, so there is a largest index a with $f_a \notin I$; choose b likewise for g.

Consider the homogeneous component of fg in degree $a + b$, namely $\sum_{i+j=a+b} f_i g_j$. In any term with $i > a$ one has $f_i \in I$ by maximality of a, so $f_i g_j \in I$; and in any term with $i < a$ one has $j = a + b - i > b$, so $g_j \in I$ and again $f_i g_j \in I$. Every term except $f_a g_b$ therefore lies in I. Since $fg \in I$ and I is graded, the whole degree-$(a+b)$ component lies in I by proposition 16.2.7(2), and subtracting the other terms gives $f_a g_b \in I$. As f_a and g_b are homogeneous, the hypothesis forces $f_a \in I$ or $g_b \in I$, contradicting the choice of a and b. So I is prime, completing the proof.

Finally, the operations on ideals of definition 9.4.1 preserve gradedness.

Proposition 16.2.9: Arithmetic of Homogeneous Ideals

Let S be a graded commutative ring and let $I, J \subseteq S$ be graded ideals. Then each of

$$I + J, \qquad IJ, \qquad I \cap J, \qquad \sqrt{I}$$

is a graded ideal.

***Proof*:**
Throughout, proposition 16.2.7 is used to switch between the definition and the two convenient criteria.

$I + J$ and IJ. By criterion (3) each of I and J has a generating set of homogeneous elements, say H_I and H_J. Then $H_I \cup H_J$ generates $I + J$ and consists of homogeneous elements, and the products $\left\{hh' \middle| h \in H_I,\ h' \in H_J\right\}$ generate IJ and are homogeneous, since $S_k S_l \subseteq S_{k+l}$. Criterion (3) applies to both.

$I \cap J$. Let $s \in I \cap J$. Each homogeneous component of s lies in I by criterion (2) applied to I, and in J by the same criterion applied to J, hence in $I \cap J$. Criterion (2) gives the claim.

$\sqrt{I}$. That $\sqrt{I}$ is an ideal is proposition 9.4.3. For criterion (2), let $s \in \sqrt{I}$ and argue by induction on the number of nonzero homogeneous components of s. If that number is 0 there is nothing to prove. Otherwise let d be the largest degree with $s_d \neq 0$, and choose n with $s^n \in I$. The homogeneous component of s^n in the top degree nd is s_d^n, because every other product of n components has total degree strictly less than nd. Since $s^n \in I$ and I is graded, criterion (2) gives $s_d^n \in I$, so $s_d \in \sqrt{I}$. Then $s - s_d \in \sqrt{I}$, as $\sqrt{I}$ is an ideal, and it has one fewer nonzero homogeneous component, so the inductive hypothesis puts all of its components in $\sqrt{I}$. Those together with s_d are the components of s, completing the proof.

Exercise 16.2.1

1. Show that a graded ring S has $S_0 \neq 0$ unless $S = 0$, and give an example of a graded ring in which $S_1 = 0$ but $S_2 \neq 0$.

2. Show that an ideal of $R[x_1, \ldots, x_n]$ is graded if and only if it is generated by homogeneous polynomials, and use this to show that $(1 + x) \subseteq \mathbb{Z}[x]$ is not graded.

3. Show that $(y - x^2) \subseteq k[x, y]$ is not a graded ideal for the standard grading, while $(y,\, y - x^2)$ is. (For the second, identify the ideal.)

4. Let S be a graded ring. Show that S_+ is the unique largest graded ideal I with $I \cap S_0 = 0$, and that S is generated as an S_0-algebra by any set of homogeneous elements whose images generate S_+/S_+^2 as an S_0-module.

5. Show that a graded ideal I is radical if and only if $s_d^n \in I$ implies $s_d \in I$ for every homogeneous s_d and every $n \geq 1$.

6. Show that $I = (x^2 - y^2) \subseteq k[x, y]$ is a graded ideal that is not prime, and exhibit homogeneous $f, g \notin I$ with $fg \in I$. (By corollary 16.2.8 such a homogeneous witness must exist whenever a graded ideal fails to be prime.)

16.3 The Tensor, Symmetric and Exterior Algebras

section 16.1 produced one module for each degree k. Adding them up produces a ring. The multiplication is concatenation of tensors, which raises the degree by exactly the amount it should, so the result is a graded ring in the sense of definition 16.2.1; and the sum acquires a universal property that the individual degrees do not have, one about homomorphisms into an algebra rather than into a module. The symmetric and exterior algebras are then quotients of it, and the last thing to check is that quotienting the whole and quotienting degree by degree agree.

Definition 16.3.1: Tensor Algebra

Let R be a commutative ring and M an R-module. The *tensor algebra* of M is the R-module

$$\mathcal{T}(M) := \bigoplus_{k \geq 0} \mathcal{T}^k(M) = R \oplus M \oplus \mathcal{T}^2(M) \oplus \cdots$$

equipped with the multiplication determined on homogeneous elements by the composite

$$\mathcal{T}^k(M) \times \mathcal{T}^l(M) \xrightarrow{\iota} \mathcal{T}^k(M) \otimes_R \mathcal{T}^l(M) \xrightarrow{\sim} \mathcal{T}^{k+l}(M)$$

where the second map is the associativity isomorphism of theorem 15.2.8 when $k, l \geq 1$, and is example 15.1.13 when $k = 0$ or $l = 0$, and extended to all of $\mathcal{T}(M)$ by additivity in each variable.

On simple tensors the multiplication is concatenation,

$$(m_1 \otimes \cdots \otimes m_i)(m_1' \otimes \cdots \otimes m_j') = m_1 \otimes \cdots \otimes m_i \otimes m_1' \otimes \cdots \otimes m_j'$$

which is what makes it computable. Note that this is not the multiplication of theorem 15.2.14, which was $(a \otimes b)(a' \otimes b') = aa' \otimes bb'$ and required both factors to be algebras already; here M is only a module, and the product of two elements of M lands not in M but one degree higher.

Theorem 16.3.2: $\mathcal{T}(M)$ is an R-algebra

Let R be a commutative ring and M an R-module.

1. $\mathcal{T}(M)$ is an associative R-algebra with identity $1 \in R = \mathcal{T}^0(M)$, and it is graded, with $\mathcal{T}^k(M)\mathcal{T}^l(M) \subseteq \mathcal{T}^{k+l}(M)$. It contains M as the homogeneous component of degree 1.

2. (universal property) For every R-algebra A and every R-module homomorphism $\varphi\colon M \to A$ there is a unique R-algebra homomorphism $\Phi\colon \mathcal{T}(M) \to A$ with $\Phi|_M = \varphi$:

$$\begin{tikzcd} \mathcal{T}(M) \arrow[r, "\Phi"] & A \\ M \arrow[u, "\iota"] \arrow[ur, "\varphi"'] & \end{tikzcd}$$

***Proof*:**

1. The multiplication is R-bilinear on each pair of homogeneous components, being a composite of the R-bilinear ι with an R-module isomorphism, and it is extended biadditively, so it distributes over addition and commutes with the R-action; that is the compatibility definition 12.1.13 asks for. The containment $\mathcal{T}^k(M)\mathcal{T}^l(M) \subseteq \mathcal{T}^{k+l}(M)$ holds by construction, which is the grading.

   For associativity, both $(x, y, z) \mapsto (xy)z$ and $(x, y, z) \mapsto x(yz)$ are R-trilinear, so by theorem 15.2.11 each is determined by its values on triples of simple tensors, where both are the concatenation

   $$m_1 \otimes \cdots \otimes m_i \otimes m'_1 \otimes \cdots \otimes m'_j \otimes m''_1 \otimes \cdots \otimes m''_h$$

   since concatenation of finite sequences is associative. For the identity, $1 \in R = \mathcal{T}^0(M)$ acts on $\mathcal{T}^k(M)$ through the isomorphism $R \otimes_R \mathcal{T}^k(M) \cong \mathcal{T}^k(M)$ of example 15.1.13, which is the identity map. Finally $\mathcal{T}^1(M) = M$ by definition 16.1.1.

2. Fix A and φ. For each $k \geq 1$ the map

   $$M^k \to A, \qquad (m_1, \ldots, m_k) \mapsto \varphi(m_1)\varphi(m_2)\cdots\varphi(m_k)$$

   is k-multilinear: φ is R-linear and multiplication in A is R-linear in each variable by definition 12.1.13. So theorem 15.2.11 supplies a unique R-module homomorphism $\Phi_k\colon \mathcal{T}^k(M) \to A$ with $\Phi_k(m_1 \otimes \cdots \otimes m_k) = \varphi(m_1)\cdots\varphi(m_k)$. Let $\Phi_0\colon R \to A$ be the structure map of the R-algebra A, and let $\Phi := \sum_k \Phi_k$ on the direct sum, which is a well-defined R-module homomorphism because every element has only finitely many nonzero components.

   Φ is multiplicative: both $(x, y) \mapsto \Phi(xy)$ and $(x, y) \mapsto \Phi(x)\Phi(y)$ are R-bilinear, and on a pair of simple tensors both give $\varphi(m_1)\cdots\varphi(m_i)\varphi(m'_1)\cdots\varphi(m'_j)$, so they agree by theorem 15.1.12. It sends 1 to 1 because Φ_0 does. Restricted to $\mathcal{T}^1(M) = M$ it is φ.

   For uniqueness, $\mathcal{T}(M)$ is generated as an R-algebra by M: the simple tensors $m_1 \otimes \cdots \otimes m_k$ are products of elements of M, and they generate $\mathcal{T}^k(M)$ as an R-module by lemma 15.1.4. An R-algebra homomorphism is determined by its values on an algebra generating set, so any Φ' agreeing with φ on M equals Φ, completing the proof.

Part (2) says that $\mathcal{T}(-)$ converts modules into algebras in the freest possible way: prescribing an algebra homomorphism out of $\mathcal{T}(M)$ costs no more than prescribing a module homomorphism out of M. In categorical language $\mathcal{T}(-)$ is left adjoint to the functor forgetting the multiplication of an R-algebra, which is the same shape of statement as corollary 15.3.5; the general theory is section 62.1.

The two quotients of interest are cut out by degree-two relations. Before taking them, one compatibility has to be settled, since section 16.1 defined $\mathcal{S}^k(M)$ and $\bigwedge^k(M)$ by quotienting $\mathcal{T}^k(M)$ by a *submodule*, while the algebras are obtained by quotienting $\mathcal{T}(M)$ by an *ideal*. The two operations agree, but that is a statement to prove rather than to observe.

Lemma 16.3.3: The Defining Ideals Are Graded, Degree by Degree

Let R be a commutative ring and M an R-module. Let $\mathcal{C}(M) \subseteq \mathcal{T}(M)$ be the two-sided ideal generated by all $m \otimes m' - m' \otimes m$ with $m, m' \in M$, and let $\mathcal{A}(M) \subseteq \mathcal{T}(M)$ be the two-sided ideal generated by all $m \otimes m$ with $m \in M$. Then both are graded ideals, and for every $k \geq 0$

$$\mathcal{C}(M) \cap \mathcal{T}^k(M) = C^k(M), \qquad \mathcal{A}(M) \cap \mathcal{T}^k(M) = A^k(M)$$

where $C^k(M)$ and $A^k(M)$ are the submodules of definition 16.1.5 and definition 16.1.7.

***Proof*:**
Both ideals are generated by homogeneous elements of degree 2, so both are graded by proposition 16.2.7(3), and it remains to identify the degree-k parts. Write $\mathcal{C}(M)_k$ for $\mathcal{C}(M) \cap \mathcal{T}^k(M)$.

The symmetric case, $C^k(M) \subseteq \mathcal{C}(M)_k$. By definition $C^k(M)$ is generated by the differences $z - \sigma \cdot z$ for simple tensors z and $\sigma \in S_k$. Since adjacent transpositions generate S_k, and

$$z - (\tau_1 \cdots \tau_r) \cdot z = \sum_{j=1}^{r} \Big((\tau_{j+1} \cdots \tau_r) \cdot z - (\tau_j \tau_{j+1} \cdots \tau_r) \cdot z \Big)$$

telescopes with each summand of the form $w - \tau \cdot w$ for a simple tensor w and an adjacent transposition τ, it suffices to treat $\tau = (i\ i+1)$. There

$$z - \tau \cdot z = (m_1 \otimes \cdots \otimes m_{i-1}) \big(m_i \otimes m_{i+1} - m_{i+1} \otimes m_i\big) (m_{i+2} \otimes \cdots \otimes m_k)$$

a product in $\mathcal{T}(M)$ of an element of $\mathcal{T}^{i-1}(M)$, a generator of $\mathcal{C}(M)$, and an element of $\mathcal{T}^{k-i-1}(M)$, hence an element of $\mathcal{C}(M)$; and it is homogeneous of degree k.

The symmetric case, $\mathcal{C}(M)_k \subseteq C^k(M)$. An element of $\mathcal{C}(M)$ is a finite sum of products $x\,(m \otimes m' - m' \otimes m)\,y$ with $x, y \in \mathcal{T}(M)$. Expanding x and y into homogeneous components and then into simple tensors, and keeping only the terms of total degree k, every such term is of the form $w - \tau \cdot w$ for a simple tensor $w \in \mathcal{T}^k(M)$ and an adjacent transposition τ, by the display above read backwards. Each lies in $C^k(M)$, and $C^k(M)$ is a submodule, so the sum does too.

The alternating case. The inclusion $\mathcal{A}(M)_k \supseteq A^k(M)$ needs slightly more, since a generator of $A^k(M)$ is a simple tensor whose repeated entries need not be adjacent. Note first that for $u, v \in M$,

$$u \otimes v + v \otimes u = (u+v) \otimes (u+v) - u \otimes u - v \otimes v \in \mathcal{A}(M)$$

so interchanging two adjacent factors of a simple tensor changes it by an element of $\mathcal{A}(M)$, up to sign. Given $z = m_1 \otimes \cdots \otimes m_k$ with $m_a = m_b$ and $a < b$, repeatedly interchanging adjacent factors moves m_b into position $a+1$, so that $z \equiv \pm z'$ modulo $\mathcal{A}(M)$, where z' is a simple tensor with two equal adjacent entries. Then z' factors as a product of an element of $\mathcal{T}^{a-1}(M)$, the generator $m_a \otimes m_a$, and an element of $\mathcal{T}^{k-a-1}(M)$, so $z' \in \mathcal{A}(M)$ and hence $z \in \mathcal{A}(M)$. Being homogeneous of degree k, it lies in $\mathcal{A}(M)_k$.

The reverse inclusion $\mathcal{A}(M)_k \subseteq A^k(M)$ follows as in the symmetric case: the degree-k part of a product $x\,(m \otimes m)\,y$ expands into simple tensors each having two equal adjacent entries, and those are generators of $A^k(M)$, completing the proof.

Corollary 16.3.4: The Quotients Are Graded

With the notation above, $\mathcal{T}(M)/\mathcal{C}(M)$ and $\mathcal{T}(M)/\mathcal{A}(M)$ are graded rings, and their homogeneous components of degree k are $\mathcal{S}^k(M)$ and $\bigwedge^k(M)$ respectively.

***Proof*:**
By lemma 16.3.3 both ideals are graded, so proposition 16.2.5 applies and gives graded quotients with degree-k component $\mathcal{T}^k(M)/(\mathcal{C}(M) \cap \mathcal{T}^k(M))$, which is $\mathcal{T}^k(M)/C^k(M) = \mathcal{S}^k(M)$ by the same lemma and definition 16.1.5; the alternating case is identical, completing the proof.

Definition 16.3.5: Symmetric Algebra

The *symmetric algebra* of an R-module M is the quotient R-algebra

$$\mathcal{S}(M) := \mathcal{T}(M)/\mathcal{C}(M)$$

where $\mathcal{C}(M)$ is the two-sided ideal generated by the elements $m \otimes m' - m' \otimes m$ for $m, m' \in M$. By corollary 16.3.4 it is graded with $\mathcal{S}(M)_k = \mathcal{S}^k(M)$, and the product of the images of $m_1, \dots, m_k$ is written $m_1 m_2 \cdots m_k$.

Theorem 16.3.6: The Symmetric Algebra Is the Free Commutative Algebra

Let R be a commutative ring and M an R-module. Then $\mathcal{S}(M)$ is a commutative R-algebra, and for every *commutative* R-algebra A and every R-module homomorphism $\varphi\colon M \to A$ there is a unique R-algebra homomorphism $\Phi\colon \mathcal{S}(M) \to A$ restricting to φ on $M = \mathcal{S}^1(M)$.

***Proof*:**
The algebra $\mathcal{S}(M)$ is generated by the image of M, since $\mathcal{T}(M)$ is generated by M and the quotient map is surjective. The images of any two elements of M commute, because $m \otimes m' - m' \otimes m \in \mathcal{C}(M)$. An associative algebra generated by a set of pairwise commuting elements is commutative: the set of elements commuting with everything in the generating set is a subalgebra, so it is the whole algebra, and then the same argument applied a second time gives commutativity outright. So $\mathcal{S}(M)$ is commutative.

Let A be commutative and $\varphi\colon M \to A$ an R-module homomorphism. By theorem 16.3.2 there is a unique R-algebra homomorphism $\Psi\colon \mathcal{T}(M) \to A$ with $\Psi|_M = \varphi$. On a generator of $\mathcal{C}(M)$,

$$\Psi(m \otimes m' - m' \otimes m) = \varphi(m)\varphi(m') - \varphi(m')\varphi(m) = 0$$

because A is commutative. An algebra homomorphism vanishing on a set of generators of a two-sided ideal vanishes on the ideal, so $\mathcal{C}(M) \subseteq \ker \Psi$ and Ψ factors through $\mathcal{S}(M)$, giving Φ. Uniqueness holds because $\mathcal{S}(M)$ is generated as an R-algebra by the image of M, as we sought to show.

Definition 16.3.7: Exterior Algebra

The *exterior algebra* of an R-module M is the quotient R-algebra

$$\bigwedge(M) := \mathcal{T}(M)/\mathcal{A}(M)$$

where $\mathcal{A}(M)$ is the two-sided ideal generated by the elements $m \otimes m$ for $m \in M$. By corollary 16.3.4 it is graded with $\bigwedge(M)_k = \bigwedge^k(M)$, and the image of $m_1 \otimes \cdots \otimes m_k$ is written $m_1 \wedge \cdots \wedge m_k$.

Proposition 16.3.8: The Exterior Algebra Is the Free Algebra on Squares Zero

Let R be a commutative ring and M an R-module. For every R-algebra A and every R-module homomorphism $\varphi\colon M \to A$ satisfying $\varphi(m)^2 = 0$ for all $m \in M$, there is a unique R-algebra homomorphism $\Phi\colon \bigwedge(M) \to A$ restricting to φ on $M = \bigwedge^1(M)$.

***Proof*:**
As in theorem 16.3.6, theorem 16.3.2 supplies $\Psi\colon \mathcal{T}(M) \to A$ with $\Psi|_M = \varphi$, and $\Psi(m \otimes m) = \varphi(m)^2 = 0$ kills every generator of $\mathcal{A}(M)$, hence the ideal. The induced Φ is unique because $\bigwedge(M)$ is generated as an R-algebra by the image of M, completing the proof.

The wedge product is not commutative, and it is not anticommutative either; the sign depends on the degrees. proposition 16.1.9 gave the rule for elements of degree 1, and the general rule follows from it.

Proposition 16.3.9: Graded Commutativity of the Exterior Algebra

Let $a \in \bigwedge^p(M)$ and $b \in \bigwedge^q(M)$. Then

$$a \wedge b = (-1)^{pq}\, b \wedge a$$

In particular elements of odd degree anticommute and elements of even degree are central among homogeneous elements; the identity $ab = -ba$ fails as soon as one of the two degrees is even.

***Proof*:**
Both $(a, b) \mapsto a \wedge b$ and $(a, b) \mapsto (-1)^{pq} b \wedge a$ are R-bilinear on $\bigwedge^p(M) \times \bigwedge^q(M)$, so by theorem 15.1.12 it suffices to compare them on pairs of products of degree-one elements, which generate each factor. Take $a = m_1 \wedge \cdots \wedge m_p$ and $b = m'_1 \wedge \cdots \wedge m'_q$. Moving m'_1 leftwards past each of $m_1, \ldots, m_p$ in turn costs a factor of -1 each time by proposition 16.1.9, so p sign changes; doing this for each of the q elements of b in turn gives $(-1)^{pq}$ in total and rearranges $a \wedge b$ into $b \wedge a$. Hence $a \wedge b = (-1)^{pq} b \wedge a$.

For the last sentence, take $p = q = 1$ to get $m \wedge m' = -m' \wedge m$, and $p = 2$, $q = 1$ to get $a \wedge m = m \wedge a$, which is not $-m \wedge a$ unless $2a \wedge m = 0$, completing the proof.

Example 16.3.10: Tensor Algebras

1. Let $R = \mathbb{Z}$ and $M = \mathbb{Q}/\mathbb{Z}$. Then $(\mathbb{Q}/\mathbb{Z}) \otimes_{\mathbb{Z}} (\mathbb{Q}/\mathbb{Z}) = 0$ by the third exercise of section 15.1, so $\mathcal{T}^k(M) = 0$ for every $k \geq 2$ and
$$\mathcal{T}(\mathbb{Q}/\mathbb{Z}) = \mathbb{Z} \oplus \mathbb{Q}/\mathbb{Z}$$
Addition is componentwise, and the multiplication is
$$(r, \overline{p})(s, \overline{q}) = (rs,\ r\overline{q} + s\overline{p})$$
since the product of two degree-one elements lands in $\mathcal{T}^2 = 0$. This is the *square-zero extension* of $\mathbb{Z}$ by $\mathbb{Q}/\mathbb{Z}$: the degree-one part is an ideal whose square vanishes.

2. Let $R = \mathbb{Z}$ and $M = \mathbb{Z}/n\mathbb{Z}$. Then $(\mathbb{Z}/n\mathbb{Z}) \otimes_{\mathbb{Z}} (\mathbb{Z}/n\mathbb{Z}) \cong \mathbb{Z}/n\mathbb{Z}$ by proposition 15.2.13, and inductively $\mathcal{T}^k(M) \cong \mathbb{Z}/n\mathbb{Z}$ for every $k \geq 1$, so
$$\mathcal{T}(\mathbb{Z}/n\mathbb{Z}) = \mathbb{Z} \oplus (\mathbb{Z}/n\mathbb{Z}) \oplus (\mathbb{Z}/n\mathbb{Z}) \oplus \cdots \cong \mathbb{Z}[x]/(nx)$$
the isomorphism sending the degree-k generator to x^k; the relation $nx = 0$ records that the degree-one part is killed by n, and every higher degree inherits it.

3. Take $M = \mathbb{Q}$ as a module over itself. Then $\mathcal{T}^k(\mathbb{Q}) \cong \mathbb{Q}$ for every k by example 15.1.13, so $\mathcal{T}(\mathbb{Q}) \cong \mathbb{Q}[x]$ as graded $\mathbb{Q}$-algebras. Here the tensor algebra happens to be commutative, because M has rank one; for a module of rank at least two it never is, as section 16.4 will show.

Exercise 16.3.1

1. Show that $\mathcal{T}(M)$ is commutative if and only if $\mathcal{S}(M) = \mathcal{T}(M)$, and that this happens if and only if $m \otimes m' = m' \otimes m$ for all $m, m' \in M$.

2. Show that $\mathcal{S}(M)$ satisfies the universal property of theorem 16.3.6 with respect to R-algebra homomorphisms into non-commutative algebras as well, provided the image of M is required to consist of pairwise commuting elements.

3. Let $I \subseteq \mathcal{T}(M)$ be the two-sided ideal generated by $\mathcal{T}^{\geq d}(M) := \bigoplus_{k \geq d} \mathcal{T}^k(M)$ for some $d \geq 1$. Show that $I = \mathcal{T}^{\geq d}(M)$, that it is graded, and identify the quotient $\mathcal{T}(M)/I$ as an R-module.

4. Show that $\bigwedge(M)$ is commutative when $2 = 0$ in R, and give an example over $\mathbb{Z}$ where it is not. (For the example, use the determinant to produce an alternating bilinear map on $\mathbb{Z}^2$ and deduce that $e_1 \wedge e_2$ has infinite order.)

5. Let $\varphi \colon M \to N$ be an R-module homomorphism. Show that it induces graded R-algebra homomorphisms $\mathcal{T}(\varphi)$, $\mathcal{S}(\varphi)$ and $\bigwedge(\varphi)$ between the corresponding algebras, and that each is surjective when φ is.

6. Show that $\mathcal{S}(M \oplus N) \cong \mathcal{S}(M) \otimes_R \mathcal{S}(N)$ as R-algebras, using theorem 16.3.6 together with

theorem 15.2.17, which identifies the right-hand side as the coproduct of commutative R-algebras.

16.4 Bases, Ranks and the Determinant

Nothing so far has required M to be free. When it is, all three graded algebras become computable: a basis of M determines a basis of each homogeneous component, and the ranks are the three counts one would guess from thinking of the components as words, unordered words, and sets. The exterior case then delivers the determinant, which reappears not as the defining sum of definition 13.4.5 but as the action of an endomorphism on a rank-one module.

Theorem 16.4.1: Bases of the Tensor Powers

Let R be a commutative ring and let M be a free R-module with basis $\{e_i\}_{i\in I}$. Then $\mathcal{T}^k(M)$ is free with basis

$$\left\{\, e_{i_1} \otimes e_{i_2} \otimes \cdots \otimes e_{i_k} \;\middle|\; (i_1, \ldots, i_k) \in I^k \,\right\}$$

indexed by all k-tuples. If I is finite of size n, then $\mathcal{T}^k(M)$ is free of rank n^k.

***Proof*:**
Induction on k. For $k = 0$ and $k = 1$ the claim is $\mathcal{T}^0(M) = R$ and $\mathcal{T}^1(M) = M$. For the inductive step, $\mathcal{T}^k(M) = \mathcal{T}^{k-1}(M) \otimes_R M$ by definition 16.1.1, and $\mathcal{T}^{k-1}(M)$ is free on the $(k-1)$-tuples by hypothesis, so corollary 15.2.6 makes $\mathcal{T}^k(M)$ free on the products of a basis element of each factor, which are exactly the k-tuples. The rank count is $|I^k| = n^k$, completing the proof.

In particular $\mathcal{T}(M)$ is the free associative R-algebra on the set $\{e_i\}$: its elements are the R-linear combinations of words in the e_i, multiplied by concatenation. For $|I| \geq 2$ it is not commutative, since $e_1 \otimes e_2$ and $e_2 \otimes e_1$ are distinct basis elements, which settles the question raised in example 16.3.10(3).

The symmetric algebra is the polynomial ring, and it is quickest to identify the whole algebra at once and read off the components afterwards.

Theorem 16.4.2: The Symmetric Algebra of a Free Module

Let R be a commutative ring and M free with basis $e_1, \ldots, e_n$. Then there is an isomorphism of graded R-algebras

$$\mathcal{S}(M) \;\cong\; R[x_1, \ldots, x_n]$$

carrying e_i to x_i. Consequently $\mathcal{S}^k(M)$ is free with basis the monomials

$$e_{i_1} e_{i_2} \cdots e_{i_k}, \qquad 1 \leq i_1 \leq i_2 \leq \cdots \leq i_k \leq n$$

and has rank $\binom{n+k-1}{k}$.

***Proof*:**
The polynomial ring is a commutative R-algebra, so the R-module homomorphism $M \to R[x_1, \dots, x_n]$ sending e_i to x_i induces, by theorem 16.3.6, an R-algebra homomorphism

$$\Phi\colon \mathcal{S}(M) \longrightarrow R[x_1, \dots, x_n], \qquad \Phi(e_i) = x_i$$

Conversely the universal property of polynomial rings (theorem 9.2.4) gives an R-algebra homomorphism $\Psi\colon R[x_1, \dots, x_n] \to \mathcal{S}(M)$ with $\Psi(x_i) = e_i$, since $\mathcal{S}(M)$ is commutative by theorem 16.3.6.

Each composite is an R-algebra endomorphism fixing a set of algebra generators, hence the identity: $\mathcal{S}(M)$ is generated by the image of M, and $R[x_1, \dots, x_n]$ by the x_i. So Φ is an isomorphism. It is graded because it sends the degree-one generators e_i to the degree-one generators x_i and both algebras are generated in degree one, so it carries a product of k generators to a product of k generators.

Restricting to degree k, $\mathcal{S}^k(M) = \mathcal{S}(M)_k$ by corollary 16.3.4, and the degree-k component of $R[x_1, \dots, x_n]$ is free on the monomials of total degree k (example 16.2.4(3)). Their preimages are the products $e_{i_1} \cdots e_{i_k}$ with indices weakly increasing, one for each monomial. Counting monomials of degree k in n variables is the standard problem of distributing k identical items into n labelled boxes, whose answer is $\binom{n+k-1}{k}$, completing the proof.

The exterior powers need a separate argument, because there is no ambient ring to compare against. Spanning is immediate; independence is produced by exhibiting, for each proposed basis element, an alternating map that detects it. Those maps are minors, and the fact that a determinant is alternating is proposition 13.4.6. There is no circularity: det is the sum of definition 13.4.5, defined without any reference to exterior algebra, and it is only in theorem 16.4.7 below that the two are identified.

Theorem 16.4.3: Bases of the Exterior Powers

Let R be a commutative ring and M free with basis $e_1, \dots, e_n$. Then $\bigwedge^k(M)$ is free with basis

$$\{\, e_{i_1} \wedge e_{i_2} \wedge \cdots \wedge e_{i_k} \mid 1 \le i_1 < i_2 < \cdots < i_k \le n \,\}$$

so it has rank $\binom{n}{k}$; in particular $\bigwedge^k(M) = 0$ for $k > n$, and $\bigwedge^n(M)$ is free of rank one on $e_1 \wedge \cdots \wedge e_n$. The whole exterior algebra $\bigwedge(M)$ is then free of rank 2^n.

***Proof*:**
Spanning. By theorem 16.4.1 the tensors $e_{i_1} \otimes \cdots \otimes e_{i_k}$ span $\mathcal{T}^k(M)$, so their images span $\bigwedge^k(M)$. If two indices coincide the image is 0 by proposition 16.1.9; otherwise the same proposition rearranges the factors into strictly increasing order at the cost of a sign. So the displayed set spans.

Independence. Fix a strictly increasing $I = (i_1 < \cdots < i_k)$ and define

$$\varphi_I\colon M^k \to R, \qquad \varphi_I(v_1, \dots, v_k) = \det\Big((v_j)_{i_a} \Big)_{1 \le a, j \le k}$$

the determinant of the $k \times k$ matrix whose (a, j) entry is the i_a-th coordinate of v_j in the basis $e_1, \dots, e_n$. Each coordinate is R-linear in v_j, and the determinant is R-linear in each column by proposition 13.4.6(3), so φ_I is k-multilinear; and if $v_j = v_{j'}$ for $j \neq j'$ then the matrix has two equal columns, so φ_I vanishes by proposition 13.4.6(4). Thus φ_I is alternating, and theorem 16.1.8 induces $\overline{\varphi_I}\colon \bigwedge^k(M) \to R$.

Evaluate $\overline{\varphi_I}$ on a proposed basis element $e_J = e_{j_1} \wedge \cdots \wedge e_{j_k}$ with J strictly increasing. The matrix in question has (a,b) entry δ_{i_a,j_b}. If $I = J$ it is the identity matrix and the determinant is 1 by proposition 13.4.6(1). If $I \neq J$ then, both being strictly increasing k-tuples, some i_a lies outside J, and then row a is zero, so the determinant is 0. Hence $\overline{\varphi_I}(e_J) = \delta_{IJ}$.

Now suppose $\sum_J c_J e_J = 0$ is a relation among the proposed basis elements, with $c_J \in R$. Applying $\overline{\varphi_I}$ gives $c_I = 0$ for every I. So the set is independent, and with spanning it is a basis.

Counts. The strictly increasing k-tuples from $\{1, \ldots, n\}$ are the k-element subsets, of which there are $\binom{n}{k}$; for $k > n$ there are none, so $\bigwedge^k(M) = 0$, and for $k = n$ there is exactly one. Summing over k gives $\sum_{k=0}^{n} \binom{n}{k} = 2^n$, completing the proof.

Example 16.4.4: Exterior Algebras of Small Vector Spaces

Let F be a field and V an F-vector space.

1. If $\dim V = 1$ with basis v, then theorem 16.4.3 gives $\bigwedge^0(V) = F$, $\bigwedge^1(V) = V$, and $\bigwedge^i(V) = 0$ for every $i \geq 2$, since there are no strictly increasing i-tuples from a one-element set. So
$$\bigwedge(V) = F \oplus V$$
with every product of two elements of V equal to zero.

2. Now let $\dim V = 2$ with basis v, v'. The only strictly increasing pairs give $\bigwedge^2(V) = F\,(v \wedge v')$, and $\bigwedge^i(V) = 0$ for $i \geq 3$. Computing a general product of two vectors,
$$\begin{aligned}(av + bv') \wedge (cv + dv') &= ac\,(v \wedge v) + ad\,(v \wedge v') + bc\,(v' \wedge v) + bd\,(v' \wedge v') \\ &= (ad - bc)\, v \wedge v'\end{aligned}$$
using $v \wedge v = v' \wedge v' = 0$ and $v' \wedge v = -v \wedge v'$. The coefficient is the determinant of the matrix with those columns, which is theorem 16.4.7 in the case $n = 2$.

3. If $\dim V = 3$ with basis $\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3$, then $\bigwedge^2(V)$ has rank $\binom{3}{2} = 3$, the same as V itself, and
$$v \wedge w = (v_1 w_2 - v_2 w_1)(\mathbf{e}_1 \wedge \mathbf{e}_2) + (v_2 w_3 - v_3 w_2)(\mathbf{e}_2 \wedge \mathbf{e}_3) + (v_1 w_3 - v_3 w_1)(\mathbf{e}_1 \wedge \mathbf{e}_3)$$
Identifying the basis of $\bigwedge^2(V)$ with that of V turns $\wedge$ into the cross product of vector calculus. The coincidence is exactly $\binom{3}{2} = 3$ and happens in no other dimension, which is why the cross product is a three-dimensional phenomenon.

4. Let $R = \mathbb{Z}[x,y]$ and $I = (x,y)$, both regarded as R-modules. The inclusion $\varphi\colon I \hookrightarrow R$ induces $\bigwedge^2(\varphi)\colon \bigwedge^2(I) \to \bigwedge^2(R)$. Now R is free of rank one, so $\bigwedge^2(R) = 0$ by theorem 16.4.3, while $x \wedge y \neq 0$ in $\bigwedge^2(I)$. For the latter, every $f \in I$ has zero constant term, so its coefficients f_x and f_y of x and y are well defined; the map $(f,g) \mapsto f_x g_y - f_y g_x$ into $R/I = \mathbb{Z}$ is R-bilinear, because multiplying f by $h \in R$ scales both coefficients by $h(0,0)$, and it is alternating. So theorem 16.1.8 induces $\bigwedge^2(I) \to \mathbb{Z}$ sending $x \wedge y$ to 1. Hence $\bigwedge^2(\varphi)$ is not injective even though φ is: the exterior power of an injection need not be an injection, which is the same failure of exactness recorded for $\otimes$ in section 15.5.

Exterior powers turn direct sums into a graded tensor product, in the same way symmetric powers turn them into an ordinary one. The formula below is what makes the exterior algebra computable on a module built from simpler pieces, and it is the tool behind the fact that a stably free module of rank one is free.

Theorem 16.4.5: Exterior Powers of a Direct Sum

Let R be a commutative ring and M, N two R-modules. Then for every $k \geq 0$ there is an isomorphism of R-modules

$$\bigwedge^k(M \oplus N) \;\cong\; \bigoplus_{i+j=k} \bigwedge^i(M) \otimes_R \bigwedge^j(N)$$

natural in M and N, carrying $\alpha \otimes \beta$ to $\alpha \wedge \beta$ when $\bigwedge M$ and $\bigwedge N$ are read inside $\bigwedge(M \oplus N)$.

***Proof*:**
Step 1: a twisted product on the tensor product. Put $T = \bigwedge(M) \otimes_R \bigwedge(N)$, graded by total degree, so that T_k is the right-hand side. Define a multiplication on homogeneous elements by

$$(\alpha \otimes \beta) \cdot (\alpha' \otimes \beta') = (-1)^{\deg\beta\,\deg\alpha'}\,(\alpha \wedge \alpha') \otimes (\beta \wedge \beta')$$

and extend bilinearly. This is well defined, both wedge products being bilinear, and it is associative: comparing the two bracketings of a triple product, each produces $(\alpha \wedge \alpha' \wedge \alpha'') \otimes (\beta \wedge \beta' \wedge \beta'')$ with sign exponent $\deg\beta(\deg\alpha' + \deg\alpha'') + \deg\beta' \deg\alpha''$. The element $1 \otimes 1$ is a unit, so T is an R-algebra.

Step 2: a map out of $\bigwedge(M \oplus N)$. Consider the R-module map $\varphi\colon M \oplus N \to T$ sending (m, n) to $m \otimes 1 + 1 \otimes n$, landing in degree one. Then

$$\varphi(m,n)^2 = (m \wedge m) \otimes 1 + m \otimes n + (-1)^{1\cdot 1} m \otimes n + 1 \otimes (n \wedge n) = 0$$

the two middle terms coming from $(m \otimes 1)(1 \otimes n)$ and $(1 \otimes n)(m \otimes 1)$ with signs $(-1)^0$ and $(-1)^1$, and the outer terms vanishing since $v \wedge v = 0$. So proposition 16.3.8 supplies a unique R-algebra map $\Phi\colon \bigwedge(M \oplus N) \to T$ restricting to φ.

Step 3: a map back. The inclusions $M \hookrightarrow M \oplus N$ and $N \hookrightarrow M \oplus N$ induce algebra maps $\iota_M\colon \bigwedge M \to \bigwedge(M \oplus N)$ and ι_N, again by proposition 16.3.8, since the composites $M \to \bigwedge(M \oplus N)$ and $N \to \bigwedge(M \oplus N)$ square to zero. The assignment $(\alpha, \beta) \mapsto \iota_M(\alpha) \wedge \iota_N(\beta)$ is R-bilinear, hence induces $\Psi\colon T \to \bigwedge(M \oplus N)$ on the tensor product, and Ψ respects the twisted multiplication: moving $\iota_N(\beta)$ past $\iota_M(\alpha')$ costs exactly $(-1)^{\deg\beta\,\deg\alpha'}$ by proposition 16.3.9.

Step 4: the two composites are the identity. Both $\Psi \circ \Phi$ and $\Phi \circ \Psi$ are algebra maps, and on degree-one elements they are the identity: $\Psi(\Phi(m,n)) = \Psi(m \otimes 1) + \Psi(1 \otimes n) = m + n = (m, n)$, and similarly in the other direction. Since $\bigwedge(M \oplus N)$ is generated as an algebra by its degree-one part, the uniqueness clause of proposition 16.3.8 makes $\Psi \circ \Phi$ the identity; and T is generated by the elements $m \otimes 1$ and $1 \otimes n$, which gives the other. Both maps preserve total degree, so restricting to degree k gives the displayed isomorphism, as we sought to show.

Corollary 16.4.6: A Stably Free Module of Rank One Is Free

Let R be a commutative ring and P a finitely generated projective R-module with $P_{\mathfrak{p}} \cong R_{\mathfrak{p}}$ for every prime $\mathfrak{p}$. If $P \oplus R^n \cong R^{n+1}$ for some $n \geq 0$, then $P \cong R$.

***Proof*:**
First, $\bigwedge^i(P) = 0$ for $i \geq 2$. Exterior powers are quotients of tensor powers, so they commute with localization; hence $\left(\bigwedge^i P\right)_{\mathfrak{p}} \cong \bigwedge^i(P_{\mathfrak{p}}) \cong \bigwedge^i(R_{\mathfrak{p}})$, which vanishes for $i \geq 2$ by theorem 16.4.3. A module vanishing at every prime is zero.

Now apply theorem 16.4.5 with $M = P$ and $N = R^n$ in degree $n+1$. The summand indexed by (i, j) vanishes unless $i \leq 1$ and $j \leq n$, the latter by theorem 16.4.3; since $i + j = n + 1$, the only survivor is $(i, j) = (1, n)$. Hence

$$R \cong \bigwedge^{n+1}(R^{n+1}) \cong \bigwedge^{n+1}(P \oplus R^n) \cong \bigwedge^1(P) \otimes_R \bigwedge^n(R^n) \cong P$$

using theorem 16.4.3 for the two free top powers, as we sought to show.

The top exterior power of a free module of rank n is free of rank one, so an endomorphism of M acts on it by multiplication by a scalar. That scalar is the determinant, and this is the basis-free description of it.

Theorem 16.4.7: The Determinant as the Top Exterior Power

Let R be a commutative ring, let M be free of rank n, and let $\varphi \colon M \to M$ be an R-module homomorphism with matrix A with respect to some basis. Then

$$\bigwedge^n(\varphi) \colon \bigwedge^n(M) \to \bigwedge^n(M)$$

is multiplication by $\det(A)$. Consequently $\det(A)$ does not depend on the chosen basis, and for endomorphisms φ, ψ of M,

$$\det(\varphi \circ \psi) = \det(\varphi)\det(\psi)$$

***Proof*:**
Let $e_1, \ldots, e_n$ be the basis and $A = (a_{ij})$, so $\varphi(e_j) = \sum_i a_{ij} e_i$. By theorem 16.4.3 the module $\bigwedge^n(M)$ is free of rank one on $\omega := e_1 \wedge \cdots \wedge e_n$, so it suffices to compute $\bigwedge^n(\varphi)(\omega) = \varphi(e_1) \wedge \cdots \wedge \varphi(e_n)$.

Expanding each factor and using multilinearity of the wedge,

$$\varphi(e_1) \wedge \cdots \wedge \varphi(e_n) = \sum_f \left(\prod_{j=1}^n a_{f(j)\,j} \right) e_{f(1)} \wedge \cdots \wedge e_{f(n)}$$

the sum running over all functions $f \colon \{1, \ldots, n\} \to \{1, \ldots, n\}$. If f is not injective then two factors coincide and the wedge vanishes by proposition 16.1.9; an injective self-map of a finite set is a permutation, so only $f = \sigma \in S_n$ contributes, and there $e_{\sigma(1)} \wedge \cdots \wedge e_{\sigma(n)} = (-1)^\sigma \omega$ by the same proposition. Therefore

$$\bigwedge^n(\varphi)(\omega) = \left(\sum_{\sigma\in S_n}(-1)^\sigma\prod_{j=1}^n a_{\sigma(j)\,j}\right)\omega = \det(A^t)\,\omega = \det(A)\,\omega$$

the last step by proposition 13.4.6(2). Since $\bigwedge^n(\varphi)$ is defined from φ alone, with no basis involved, the scalar it multiplies by is an invariant of φ; so any two bases give matrices with the same determinant.

Finally $\bigwedge^n$ is functorial by theorem 15.2.2, so $\bigwedge^n(\varphi\circ\psi) = \bigwedge^n(\varphi)\circ\bigwedge^n(\psi)$, and composing two multiplications by scalars multiplies the scalars, completing the proof.

The multiplicativity just obtained is proposition 13.4.7 again, now with a reason attached rather than a computation: determinants multiply because $\bigwedge^n$ is a functor into rank-one modules, and endomorphisms of a rank-one free module compose by multiplying their scalars. The Leibniz sum of definition 13.4.5 is what the scalar looks like once a basis is fixed.

Corollary 16.4.8: Invertibility and the Top Exterior Power

Let M be free of rank n over a commutative ring R and φ an endomorphism of M. Then φ is an isomorphism if and only if $\det(\varphi)$ is a unit of R.

***Proof*:**
If φ is an isomorphism with inverse ψ, then $\det(\varphi)\det(\psi) = \det(\mathrm{id}_M) = 1$ by theorem 16.4.7 and proposition 13.4.6(1), so $\det(\varphi)$ is a unit. The converse is proposition 13.4.10(1) applied to the matrix of φ, completing the proof.

Exercise 16.4.1

1. Let M be free of rank n and let $v_1,\dots,v_k\in M$ be linearly dependent. Show that $v_1\wedge\cdots\wedge v_k = 0$ in $\bigwedge^k(M)$.
2. Let F be a field with $-1\neq 1$ and V a finite-dimensional F-vector space. Show that $V\otimes_F V\cong \mathcal{S}^2(V)\oplus\bigwedge^2(V)$, and check the dimensions against theorem 16.4.2 and theorem 16.4.3.
3. Verify that $\mathcal{T}^k$, $\mathcal{S}^k$ and $\bigwedge^k$ of a free module of rank n have ranks n^k, $\binom{n+k-1}{k}$ and $\binom{n}{k}$, and show that for $k=2$ the last two add up to the first. Then show the analogous identity fails for $k=3$ by computing all three ranks at $n=2$.
4. Let M be free of rank n and φ an endomorphism. Show that $\bigwedge^k(\varphi)$ acts on the basis of theorem 16.4.3 by the $k\times k$ minors of the matrix of φ, and deduce that φ has rank less than k if and only if $\bigwedge^k(\varphi)=0$ when R is a field.
5. Let M be free of finite rank n. Show that $\bigwedge(M)$ is finitely generated as an R-module, of rank 2^n, while $\mathcal{S}(M)$ is not finitely generated as an R-module unless $M=0$.
6. Show that theorem 16.4.3 needs freeness. Take $R=\mathbb{Z}$ and $M=\mathbb{Z}/2\mathbb{Z}\oplus\mathbb{Z}/2\mathbb{Z}$, compute $\mathcal{T}^2(M)$ using theorem 15.2.3, identify the submodule $A^2(M)$, and conclude that $\bigwedge^2(M)\cong\mathbb{Z}/2\mathbb{Z}$. Compare with what $\binom{2}{2}=1$ would predict for a free module of rank 2.

16.5⋆ The Koszul Complex

⋆ *Optional. This section is an aside; nothing later in the book depends on it.*

The exterior powers of a free module assemble into a complex, and when the ring elements defining it are well enough behaved that complex is exact. The result is the standard supply of free resolutions in commutative algebra, and it is the first place where the exterior algebra does work that has nothing to do with volume or orientation.

Nothing later in this book uses it. It is here because the books that do use it are downstream: *Commutative Algebra* [10] and *Homological Algebra* [15] both build on the complex constructed below, and this is the last point in the prerequisite chain at which the exterior powers are available and the statement can be made.

Two definitions are needed, and neither has appeared so far.

Definition 16.5.1: Regular Sequence

Let R be a commutative ring. A sequence $a_1, \dots, a_n \in R$ is a *regular sequence* if

1. a_1 is not a zero divisor in R, and for each $i \geq 2$ the image of a_i is not a zero divisor in $R/(a_1, \dots, a_{i-1})$;
2. $(a_1, \dots, a_n) \neq R$.

An ideal generated by a regular sequence is called a *regular ideal.*

Condition (1) is the one that does the work, and it says the a_i impose genuinely new conditions one after another: each is a non-zero-divisor on what survives the previous ones. In $R = k[x_1, \dots, x_n]$ the variables themselves form a regular sequence, since $k[x_1, \dots, x_n]/(x_1, \dots, x_{i-1}) \cong k[x_i, \dots, x_n]$ is a domain in which $x_i \neq 0$. Note that an initial segment $a_1, \dots, a_m$ of a regular sequence is again one, directly from the definition; this is what makes induction on n available below.

The object the complex will produce is a free resolution in the sense of definition 12.3.27: an exact chain complex $\cdots \to F_1 \to F_0 \to M \to 0$ with every F_i free. Every module admits one, since a module is a quotient of a free module and the kernel is again a module; the content of a particular resolution is that it is short, explicit, or both. Over a principal ideal domain chapter 14 produced very short ones, because a submodule of a free module over a PID is free, so the process stops after one step. Over a general ring the process need not stop, and the Koszul complex is an explicit finite resolution available whenever the defining elements form a regular sequence.

The differential is contraction against a linear functional, and it exists for the same reason everything in this chapter exists: it is an alternating map, so theorem 16.1.8 builds it.

Proposition 16.5.2: Contraction Against a Functional

Let R be a commutative ring, F an R-module and $\lambda \in \operatorname{Hom}_R(F, R)$. For each $r \geq 1$ there is a unique R-module homomorphism

$$d_r\colon \bigwedge^r(F) \longrightarrow \bigwedge^{r-1}(F)$$

with

$$d_r(v_1 \wedge \cdots \wedge v_r) = \sum_{j=1}^{r} (-1)^{j-1} \lambda(v_j)\, v_1 \wedge \cdots \wedge \widehat{v_j} \wedge \cdots \wedge v_r$$

where $\widehat{v_j}$ means that the factor is omitted. Setting $d_0 = 0$, these satisfy the graded Leibniz rule

$$d(u \wedge w) = d(u) \wedge w + (-1)^p\, u \wedge d(w) \qquad \text{for } u \in \bigwedge^p(F)$$

***Proof*:**
Existence. The right-hand side, read as a function of $(v_1, \ldots, v_r)$, is r-multilinear: each v_j appears once in each term, linearly, since λ is linear and $\wedge$ is multilinear.

It is alternating. Suppose $v_a = v_b =: v$ with $a < b$. Any term with $j \notin \{a, b\}$ still contains both v_a and v_b among its factors, so it vanishes by proposition 16.1.9. The two remaining terms are

$$(-1)^{a-1}\lambda(v)\, W_a \qquad \text{and} \qquad (-1)^{b-1}\lambda(v)\, W_b$$

where W_a omits the factor in position a and W_b the one in position b. In W_a the surviving copy of v sits in position $b-1$, and in W_b it sits in position a; moving it from one to the other is $b-1-a$ interchanges of adjacent factors, so $W_b = (-1)^{b-1-a} W_a$ by proposition 16.1.9. The sum of the two terms is therefore

$$\big((-1)^{a-1} + (-1)^{b-1}(-1)^{b-1-a}\big)\lambda(v)W_a = \big((-1)^{a-1} + (-1)^{a}\big)\lambda(v)W_a = 0$$

So the map is alternating, and theorem 16.1.8 supplies d_r and its uniqueness.

Leibniz rule. Both sides are R-bilinear in (u, w), so it suffices to check them on products of degree-one elements, which generate each factor. For $u = v_1 \wedge \cdots \wedge v_p$ and $w = v_{p+1} \wedge \cdots \wedge v_{p+q}$, the defining formula for $d(u \wedge w)$ has $p + q$ terms. The first p of them are exactly the terms of $d(u) \wedge w$; the last q are the terms of $u \wedge d(w)$, except that in $d(u \wedge w)$ the term omitting v_{p+j} carries the sign $(-1)^{p+j-1}$ while in $u \wedge d(w)$ it carries $(-1)^{j-1}$. The two differ by $(-1)^p$, which is the stated factor, completing the proof.

Definition 16.5.3: Koszul Complex

Let R be a commutative ring and $a_1, \ldots, a_n \in R$. Let $F = R^n$ with basis $e_1, \ldots, e_n$ and let $\lambda\colon F \to R$ be the R-module homomorphism with $\lambda(e_i) = a_i$. The *Koszul complex* $K_\bullet(a_1, \ldots, a_n)$ is

$$0 \longrightarrow \bigwedge^n(F) \xrightarrow{d_n} \bigwedge^{n-1}(F) \xrightarrow{d_{n-1}} \cdots \xrightarrow{d_2} \bigwedge^1(F) \xrightarrow{d_1} \bigwedge^0(F) = R$$

with d_r the contraction of proposition 16.5.2. On basis elements,

$$d_r\big(e_{i_1} \wedge \cdots \wedge e_{i_r}\big) = \sum_{j=1}^{r} (-1)^{j-1} a_{i_j}\, e_{i_1} \wedge \cdots \wedge \widehat{e_{i_j}} \wedge \cdots \wedge e_{i_r}$$

Proposition 16.5.4: The Koszul Complex Is a Complex

With the notation above, $d_{r-1} \circ d_r = 0$ for every $r \geq 1$, so $\operatorname{im}(d_r) \subseteq \ker(d_{r-1})$.

***Proof*:**
Both composites are R-linear, so it suffices to evaluate on a basis element $e_I = e_{i_1} \wedge \cdots \wedge e_{i_r}$ with $i_1 < \cdots < i_r$. Applying d_r and then d_{r-1} produces a sum of terms indexed by ordered pairs of distinct positions (j, k), the term being $\pm\, a_{i_j} a_{i_k}$ times the wedge with both factors omitted. Fix $j < k$ and collect the two terms that omit those two positions.

Removing position j first contributes the sign $(-1)^{j-1}$; in the shortened wedge the factor originally at position k now sits at position $k-1$, so removing it contributes $(-1)^{k-2}$, for a total of $(-1)^{j+k-3} = (-1)^{j+k-1}$. Removing position k first contributes $(-1)^{k-1}$, and the factor at position j is unmoved, so removing it contributes $(-1)^{j-1}$, for a total of $(-1)^{j+k-2} = (-1)^{j+k}$. The two signs are opposite and the coefficients $a_{i_j} a_{i_k}$ agree because R is commutative, so the two terms cancel. Every term cancels in such a pair, completing the proof.

Exactness is where the hypothesis on the a_i enters, and it is the reason regular sequences were defined.

Theorem 16.5.5: Exactness of the Koszul Complex

Let R be a commutative ring and let $a_1, \ldots, a_n$ be a regular sequence, with $I = (a_1, \ldots, a_n)$. Then $\ker(d_r) = \operatorname{im}(d_{r+1})$ for every $r \geq 1$, and $R/\operatorname{im}(d_1) = R/I$. Consequently

$$0 \longrightarrow \bigwedge^n(F) \longrightarrow \cdots \longrightarrow \bigwedge^1(F) \longrightarrow R \longrightarrow R/I \longrightarrow 0$$

is a finite free resolution of R/I, of length n.

***Proof*:**
The statement about $R/\mathrm{im}(d_1)$ holds for any a_i: $d_1 = \lambda$ has image $\lambda(F) = (a_1, \ldots, a_n) = I$. Freeness of each term is theorem 16.4.3. So only exactness at $r \geq 1$ is at issue, and it is proved by induction on n.

Base case $n = 1$. The complex is $0 \to R \xrightarrow{a_1} R$, and exactness at degree 1 says that multiplication by a_1 is injective, which is the statement that a_1 is not a zero divisor.

Inductive step. Let $n \geq 2$, let $F' \subseteq F$ be free on $e_1, \ldots, e_{n-1}$ and let $K'_\bullet$ be the Koszul complex of $a_1, \ldots, a_{n-1}$ on F', with differentials d'. An initial segment of a regular sequence is regular, so the inductive hypothesis applies to K'.

By theorem 16.4.3 every basis element of $\bigwedge^r(F)$ either involves e_n or does not, which gives

$$\bigwedge^r(F) = \bigwedge^r(F') \;\oplus\; \Big(\bigwedge^{r-1}(F') \wedge e_n\Big)$$

so every element is uniquely $x + y \wedge e_n$ with $x \in \bigwedge^r(F')$ and $y \in \bigwedge^{r-1}(F')$. The Leibniz rule of proposition 16.5.2, together with $d(e_n) = a_n$, computes the differential in these coordinates:

$$d_r\big(x + y \wedge e_n\big) = \Big(d'_r(x) + (-1)^{r-1} a_n\, y\Big) + \Big(d'_{r-1}(y)\Big) \wedge e_n \tag{16.1}$$

Suppose now $r \geq 1$ and $d_r(x + y \wedge e_n) = 0$. Comparing the two summands,

$$d'_{r-1}(y) = 0 \qquad \text{and} \qquad d'_r(x) + (-1)^{r-1} a_n y = 0$$

Producing v with $d'_r(v) = y$. If $r \geq 2$ then $r - 1 \geq 1$, so the inductive hypothesis gives exactness of K' at degree $r - 1$ and $y = d'_r(v)$ for some $v \in \bigwedge^r(F')$. If $r = 1$ then $y \in \bigwedge^0(F') = R$ and the first equation is vacuous, so a different argument is needed. There the second equation reads $\lambda'(x) + a_n y = 0$ with $\lambda' = d'_1$, so $a_n y \in (a_1, \ldots, a_{n-1})$. Because $a_1, \ldots, a_n$ is a regular sequence, a_n is not a zero divisor on $R/(a_1, \ldots, a_{n-1})$, so $y \in (a_1, \ldots, a_{n-1})$; writing $y = \sum_{i<n} c_i a_i$ and setting $v = \sum_{i<n} c_i e_i \in \bigwedge^1(F')$ gives $d'_1(v) = y$.

Producing u. From $d'_r(x) = -(-1)^{r-1} a_n y = (-1)^r a_n\, d'_r(v)$ one gets

$$d'_r\big(x - (-1)^r a_n v\big) = 0$$

and $r \geq 1$, so the inductive hypothesis gives $x - (-1)^r a_n v = d'_{r+1}(u)$ for some $u \in \bigwedge^{r+1}(F')$.

Conclusion. Apply (16.1) in degree $r + 1$ to $u + v \wedge e_n$:

$$d_{r+1}\big(u + v \wedge e_n\big) = \Big(d'_{r+1}(u) + (-1)^r a_n v\Big) + d'_r(v) \wedge e_n = x + y \wedge e_n$$

So every cycle in degree r is a boundary, which is exactness, completing the proof.

Example 16.5.6: The Koszul Complex of the Variables
Take $R = k[x_1, \ldots, x_n]$ over a field k and the regular sequence $x_1, \ldots, x_n$, so that $I = (x_1, \ldots, x_n)$ and $R/I \cong k$. The theorem produces a free resolution of k by free R-modules of ranks $\binom{n}{k}$, of total length n:

$$0 \to R \to R^n \to R^{\binom{n}{2}} \to \cdots \to R^n \to R \to k \to 0$$

For $n = 1$ this is $0 \to k[x] \xrightarrow{x} k[x] \to k \to 0$, which is the observation that x is not a zero divisor. For $n = 2$ it is

$$0 \to R \xrightarrow{\binom{-x_2}{x_1}} R^2 \xrightarrow{(x_1 \quad x_2)} R \to k \to 0$$

and exactness in the middle says precisely that a pair (f, g) with $fx_1 + gx_2 = 0$ is a multiple of $(-x_2, x_1)$: the only relations among x_1 and x_2 are the obvious ones. Regularity is exactly what makes that true, and it fails for a non-regular sequence, as the exercises show.

The construction is the algebraic origin of several later ones. In *Homological Algebra* [15] the Koszul complex computes the derived functors that measure how far a sequence is from being regular, and the same complex reappears in intersection theory, where the exactness above expresses that n hypersurfaces meeting in the expected dimension impose independent conditions.

Exercise 16.5.1

1. Show that x, y is a regular sequence in $k[x, y]$, and that x, x is not. Write out the Koszul complex for x, x and exhibit an element of $\ker(d_1)$ that is not in $\operatorname{im}(d_2)$, so that theorem 16.5.5 genuinely needs its hypothesis.
2. Show that a permutation of a regular sequence need not be regular, using $a_1 = x$, $a_2 = y(1-x)$, $a_3 = z(1-x)$ in $k[x, y, z]$. (Check that a_1, a_2, a_3 is regular but a_2, a_3, a_1 is not.)
3. Show that $d_n \colon \bigwedge^n(F) \to \bigwedge^{n-1}(F)$ is injective for a regular sequence, and identify $\bigwedge^n(F)$ and the image of d_n explicitly when $n = 2$.
4. Show that the Koszul complex of $a_1, \ldots, a_n$ is, up to isomorphism, the tensor product over R of the n two-term complexes $0 \to R \xrightarrow{a_i} R \to 0$, in the sense that its degree-r term is the direct sum of the products of r of the left-hand copies with $n-r$ of the right-hand ones. (Only the module structure is asked for, not the differential.)
5. Write out the Koszul complex of x, y, z in $k[x, y, z]$ explicitly, giving the matrices of d_3, d_2 and d_1 with respect to the bases of theorem 16.4.3, and verify $d_2 d_3 = 0$ and $d_1 d_2 = 0$ by direct computation.

16.6 Summary

One module M, four classes of map on its powers, and four objects representing them. The tables below state what was established, each entry pointing at the result that establishes it.

The representing objects

Each class of multilinear map on M^k is represented by a module, and each representing object is a quotient of the tensor power by exactly what the class is required to kill.

Maps on M^k	Represented by	Universal property	Rank
k-multilinear	$\mathcal{T}^k(M)$	theorem 15.2.11	n^k
symmetric k-multilinear	$\mathcal{S}^k(M)$	theorem 16.1.6	$\binom{n+k-1}{k}$
alternating k-multilinear	$\bigwedge^k(M)$	theorem 16.1.8	$\binom{n}{k}$

The ranks are for M free of rank n, and are theorem 16.4.1, theorem 16.4.2 and theorem 16.4.3 respectively. In particular $\bigwedge^k(M) = 0$ once $k > n$, and $\bigwedge^n(M)$ is free of rank one, which is what makes the determinant a scalar.

The graded algebras

Summing the powers over k turns each family into a ring, graded in the sense of definition 16.2.1, and each acquires a universal property about homomorphisms into an algebra.

Algebra	Construction	Universal for maps into	Reference
$\mathcal{T}(M)$	$\bigoplus_k \mathcal{T}^k(M)$	any R-algebra	theorem 16.3.2
$\mathcal{S}(M)$	$\mathcal{T}(M)/\mathcal{C}(M)$	commutative R-algebras	theorem 16.3.6
$\bigwedge(M)$	$\mathcal{T}(M)/\mathcal{A}(M)$	algebras where $\varphi(m)^2 = 0$	proposition 16.3.8

That the two routes to $\mathcal{S}^k(M)$ and $\bigwedge^k(M)$ agree, one quotienting $\mathcal{T}^k(M)$ by a submodule and the other quotienting $\mathcal{T}(M)$ by an ideal, is lemma 16.3.3 and corollary 16.3.4. For M free of rank n the three algebras are the free associative algebra on n generators, the polynomial ring $R[x_1, \ldots, x_n]$ (theorem 16.4.2), and a free module of rank 2^n (theorem 16.4.3).

Graded rings

Statement	Reference
$1 \in S_0$; S_0 is a subring; S_+ is an ideal with $S/S_+ \cong S_0$	proposition 16.2.2
Graded $\Leftrightarrow$ components in I $\Leftrightarrow$ homogeneous generators $\Leftrightarrow$ a graded kernel	proposition 16.2.7
S/I is graded, with $(S/I)_k \cong S_k/I_k$	proposition 16.2.5
$\ker\varphi$ and $\mathrm{im}\varphi$ decompose degree by degree	proposition 16.2.6
Primality is testable on homogeneous elements	corollary 16.2.8
$I + J$, IJ, $I \cap J$ and $\sqrt{I}$ are graded	proposition 16.2.9

Signs

The exterior algebra is governed by signs, and three separate statements are involved, which are easy to conflate.

Statement	Reference
Alternating $\Rightarrow$ the sign rule; the converse needs 2 invertible	lemma 16.1.4
$m_{\sigma(1)} \wedge \cdots \wedge m_{\sigma(k)} = (-1)^\sigma m_1 \wedge \cdots \wedge m_k$	proposition 16.1.9
$a \wedge b = (-1)^{pq} b \wedge a$ for $a \in \bigwedge^p$, $b \in \bigwedge^q$	proposition 16.3.9
$\mathcal{S}^k(M)$ and $\bigwedge^k(M)$ sit inside $\mathcal{T}^k(M)$ when $k!$ is invertible	proposition 16.1.12

In characteristic 2 the first row is strict: a symmetric bilinear form satisfies the sign rule vacuously without being alternating, so the vanishing condition of definition 16.1.3 is the one the exterior power is built from.

The determinant, and what fails

Statement	**Reference**
$\bigwedge^n(\varphi)$ is multiplication by $\det(\varphi)$	theorem 16.4.7
det is independent of the basis, and multiplicative	same
φ is an isomorphism $\Leftrightarrow$ $\det(\varphi)$ is a unit	corollary 16.4.8
$\bigwedge^k$ does not preserve injections	example 16.4.4(4)
$\bigwedge^k(M)$ need not be free when M is not	theorem 16.4.3

The second row is the conceptual reading of multiplicativity: determinants multiply because $\bigwedge^n$ is a functor landing in rank-one modules, where composition is multiplication of scalars. The elementary proof from the defining sum is proposition 13.4.7.

The exterior powers of a free module also assemble into a complex, which resolves $R/(a_1, \ldots, a_n)$ whenever the a_i are regular. That construction is section 16.5$\star$, marked as skippable for a reader of this book, and is the form in which *Commutative Algebra* [10] and *Homological Algebra* [15] take up the exterior algebra.

17⋆

Projective, Injective and Flat Modules

⋆ *Optional. This chapter is an aside; nothing later in the book depends on it.*

Given two R-modules A and C, which modules M fit into a short exact sequence

$$0 \to A \to M \to C \to 0\,?$$

This is the *extension problem.* Section 3.6 asked it for groups and left it open, and theorem 3.7.9 is exactly the reason it matters: the Jordan-Hölder theorem pins down the composition factors of a finite group without saying how they are assembled. For modules there is always at least one answer, $M = A \oplus C$, and for $A = \mathbb{Z}$ and $C = \mathbb{Z}/n\mathbb{Z}$ there are exactly two.

Exactness itself is the tool. A short exact sequence records how M is assembled from A and C, and the three constructions already available ($\mathrm{Hom}_R(D, -)$, $\mathrm{Hom}_R(-, D)$ and $- \otimes_R D$) each turn such a sequence into one that is exact everywhere except at a single spot. Whether the failure at that spot vanishes depends on the module D, and the modules for which it does are the projective, injective and flat modules of the title. What measures the failure when it does not vanish is a module $\mathrm{Ext}^1_R(C, A)$, whose elements turn out to be in bijection with the answers to the question above.

17.1 Exact Sequences, Splitting, and the Extension Problem

Definition 12.2.11 already said what it means for a chain of R-module homomorphisms to be exact: the image of each map is the kernel of the next. Two degenerate cases fix how short an exact sequence can be before it stops carrying information.

Proposition 17.1.1: Injective and Surjective Maps in Exact Sequences

Let A, B and C be R-modules.

1. The sequence $0 \to A \xrightarrow{\psi} B$ is exact at A if and only if ψ is injective.
2. The sequence $B \xrightarrow{\varphi} C \to 0$ is exact at C if and only if φ is surjective.

***Proof*:**

1. The only homomorphism $\alpha\colon 0 \to A$ sends $0 \mapsto 0$, so $\operatorname{im}(\alpha) = 0$. Exactness at A says $\operatorname{im}(\alpha) = \ker(\psi)$, which is therefore the assertion $\ker(\psi) = 0$, and a homomorphism is injective exactly when its kernel is trivial.
2. The only homomorphism $\beta\colon C \to 0$ sends every c to 0, so $\ker(\beta) = C$. Exactness at C says $\operatorname{im}(\varphi) = \ker(\beta) = C$, which is the assertion that φ is surjective.

Consequently $0 \to A \xrightarrow{f} B \to 0$ is exact precisely when f is both injective and surjective, so A and B are isomorphic and nothing has been learned. The first length that carries information is one longer.

Definition 17.1.2: Short Exact Sequence

A *short exact sequence* of R-modules is an exact sequence

$$0 \to A \xrightarrow{\psi} B \xrightarrow{\varphi} C \to 0$$

In this situation B is called an *extension of C by A*.

Unwinding proposition 17.1.1, the displayed sequence is short exact if and only if ψ is injective, φ is surjective and $\operatorname{im}(\psi) = \ker(\varphi)$. The first isomorphism theorem (theorem 12.2.10) then gives $C \cong B/\psi(A)$, while ψ identifies A with the submodule $\psi(A)$. An extension of C by A is therefore a module B containing a copy of A whose quotient by that copy is C, packaged in a form that names the two maps along with the three modules.

17.1.3 The Two Extension Conventions Definition 3.6.3 called G an extension of A by B when $G/A \cong B$, which is the order used in classical group theory and is the reverse of the order fixed above. The convention attached to short exact sequences is the one used for the rest of this chapter: the module being extended is the quotient C, and the module extending it is the submodule A. Both orders are current in the literature, and the safe reading is always the sequence itself.

Example 17.1.4: Short Exact Sequences

1. For any A and C the sequence
$$0 \to A \xrightarrow{\iota} A \oplus C \xrightarrow{\pi} C \to 0, \qquad \iota(a) = (a, 0), \quad \pi(a, c) = c$$
is short exact, since ι is injective, π is surjective, and $\operatorname{im}(\iota) = A \oplus 0 = \ker(\pi)$. Every pair of

modules therefore admits at least one extension.

2. Any homomorphism $\varphi\colon B \to C$ yields

$$0 \to \ker(\varphi) \hookrightarrow B \xrightarrow{\varphi} \operatorname{im}(\varphi) \to 0$$

and when φ is surjective the right-hand term is C itself.

3. Let M be an R-module and S a generating set for M. The homomorphism $\varphi\colon F(S) \to M$ extending the inclusion of S (theorem 12.3.11) is surjective, so with $K = \ker(\varphi)$,

$$0 \to K \hookrightarrow F(S) \xrightarrow{\varphi} M \to 0$$

is short exact. Every module is thus a quotient of a free module, and $F(S)$ is an extension of M by the module K of relations holding among the chosen generators.

Exactness is not confined to modules, nor to length three.

Example 17.1.5: The Centre and the Outer Automorphisms
Section 4.7.2 described the centre $Z(G)$ of a group and its outer automorphism group $\operatorname{Out}(G)$ as opposite ends of one construction, and exactness states this precisely. Let $\sigma\colon G \to \operatorname{Aut}(G)$ send g to conjugation by g, so that $\operatorname{im}(\sigma) = \operatorname{Inn}(G)$ (definition 4.7.3) and $\ker(\sigma) = Z(G)$, and let ρ be the quotient map by $\operatorname{Inn}(G)$, which is normal in $\operatorname{Aut}(G)$ by proposition 4.7.8. Then

$$1 \longrightarrow Z(G) \hookrightarrow G \xrightarrow{\sigma} \operatorname{Aut}(G) \xrightarrow{\rho} \operatorname{Out}(G) \longrightarrow 1$$

is exact: at $Z(G)$ because the inclusion is injective, at G because $\ker(\sigma) = Z(G)$, at $\operatorname{Aut}(G)$ because $\ker(\rho) = \operatorname{Inn}(G) = \operatorname{im}(\sigma)$, and at $\operatorname{Out}(G)$ because ρ is surjective. Cutting the sequence in the middle recovers theorem 4.7.4, the isomorphism $G/Z(G) \cong \operatorname{Inn}(G)$.

Example 17.1.4(1) makes $A \oplus C$ available for every pair. It is not always the only answer.

Example 17.1.6: Two Extensions of $\mathbb{Z}/n\mathbb{Z}$ by $\mathbb{Z}$
Fix $n \geq 2$. Two short exact sequences of $\mathbb{Z}$-modules with the same outer terms are

$$0 \to \mathbb{Z} \xrightarrow{\iota} \mathbb{Z} \oplus \mathbb{Z}/n\mathbb{Z} \xrightarrow{\pi} \mathbb{Z}/n\mathbb{Z} \to 0 \qquad \text{and} \qquad 0 \to \mathbb{Z} \xrightarrow{\cdot n} \mathbb{Z} \xrightarrow{p} \mathbb{Z}/n\mathbb{Z} \to 0$$

where $\iota(z) = (z, \overline{0})$ and $\pi(z, \overline{a}) = \overline{a}$, while $\cdot n$ is multiplication by n and p is reduction modulo n. The first is short exact by example 17.1.4(1). The second is short exact because multiplication by n is injective on $\mathbb{Z}$, reduction is surjective, and $\operatorname{im}(\cdot n) = n\mathbb{Z} = \ker(p)$.

The two middle terms are not isomorphic, since $\mathbb{Z} \oplus \mathbb{Z}/n\mathbb{Z}$ has an element of order n and $\mathbb{Z}$ has no element of finite order other than 0. So $\mathbb{Z}/n\mathbb{Z}$ has at least two genuinely different extensions by $\mathbb{Z}$.

The same phenomenon occurs among finite modules. Both

$$0 \to \mathbb{Z}/2\mathbb{Z} \xrightarrow{\overline{a} \mapsto \overline{2a}} \mathbb{Z}/4\mathbb{Z} \xrightarrow{\text{mod } 2} \mathbb{Z}/2\mathbb{Z} \to 0$$

$$0 \to \mathbb{Z}/2\mathbb{Z} \xrightarrow{\iota} \mathbb{Z}/2\mathbb{Z} \oplus \mathbb{Z}/2\mathbb{Z} \xrightarrow{\pi} \mathbb{Z}/2\mathbb{Z} \to 0$$

are short exact (for the first, the injection has image $\{\overline{0}, \overline{2}\}$, which is exactly the kernel of reduction modulo 2), and $\mathbb{Z}/4\mathbb{Z} \ncong \mathbb{Z}/2\mathbb{Z} \oplus \mathbb{Z}/2\mathbb{Z}$.

17.1.1 Maps Between Short Exact Sequences

Two extensions of the same pair can differ, and there is more than one reasonable sense in which two of them are the same.

Definition 17.1.7: Homomorphism and Equivalence of Short Exact Sequences

Let
$$0 \to A \xrightarrow{\psi} B \xrightarrow{\varphi} C \to 0 \qquad \text{and} \qquad 0 \to A' \xrightarrow{\psi'} B' \xrightarrow{\varphi'} C' \to 0$$
be short exact sequences of R-modules.

1. A *homomorphism of short exact sequences* from the first to the second is a triple (α, β, γ) of R-module homomorphisms making
$$\begin{tikzcd} 0 \arrow[r] & A \arrow[r, "\psi"] \arrow[d, "\alpha"'] & B \arrow[r, "\varphi"] \arrow[d, "\beta"'] & C \arrow[r] \arrow[d, "\gamma"'] & 0 \\ 0 \arrow[r] & A' \arrow[r, "\psi'"'] & B' \arrow[r, "\varphi'"'] & C' \arrow[r] & 0 \end{tikzcd}$$
commute, that is $\beta \circ \psi = \psi' \circ \alpha$ and $\gamma \circ \varphi = \varphi' \circ \beta$.

2. The homomorphism is an *isomorphism of short exact sequences* if α, β and γ are all isomorphisms, and the extensions B and B' are then called *isomorphic*.

3. The two sequences are *equivalent* if $A = A'$ and $C = C'$ and there is an isomorphism as in (2) with $\alpha = \mathrm{id}_A$ and $\gamma = \mathrm{id}_C$. The extensions B and B' are then called *equivalent* extensions.

Equivalence is strictly stronger than isomorphism, and the gap between them is what the classification in section 17.6 counts. An isomorphism is free to rebuild the outer terms by any automorphisms it likes; an equivalence holds the outer terms fixed and asks only that the middles be matched compatibly. Two extensions of one fixed pair can accordingly be isomorphic as diagrams while remaining inequivalent as extensions.

Example 17.1.8: Homomorphisms, Isomorphisms and Equivalences

1. Let $n|m$ with $n, m > 1$ and put $k = m/n$. Take the second sequence of example 17.1.6 on top and, beneath it, the sequence built from $\ell(\overline{a}) = \overline{na}$ and reduction modulo n:
$$\begin{tikzcd} 0 \arrow[r] & \mathbb{Z} \arrow[r, "\cdot n"] \arrow[d, "\alpha"'] & \mathbb{Z} \arrow[r, "p"] \arrow[d, "\beta"'] & \mathbb{Z}/n\mathbb{Z} \arrow[r] \arrow[d, "\mathrm{id}"'] & 0 \\ 0 \arrow[r] & \mathbb{Z}/k\mathbb{Z} \arrow[r, "\ell"'] & \mathbb{Z}/m\mathbb{Z} \arrow[r, "p'"'] & \mathbb{Z}/n\mathbb{Z} \arrow[r] & 0 \end{tikzcd}$$
with α and β the reductions modulo k and modulo m. The bottom row is short exact: ℓ is well defined because $a \equiv a' \pmod{k}$ gives $na \equiv na' \pmod{m}$, injective because $na \equiv 0$

(mod $m = nk$) gives $a \equiv 0 \pmod{k}$, and $\operatorname{im}(\ell) = n\mathbb{Z}/m\mathbb{Z} = \ker(p')$. Both squares commute, since $\ell(\alpha(a)) = \overline{na} = \beta(na)$ and $p'(\beta(z)) = \overline{z} = \operatorname{id}(p(z))$.

2. Let p be a prime and, for each $c \in \{1, \ldots, p-1\}$, let

$$\mathcal{E}_c\colon \qquad 0 \to \mathbb{Z}/p\mathbb{Z} \xrightarrow{\psi} \mathbb{Z}/p^2\mathbb{Z} \xrightarrow{\varphi_c} \mathbb{Z}/p\mathbb{Z} \to 0, \qquad \psi(\overline{x}) = \overline{px}, \quad \varphi_c(\overline{y}) = \overline{cy}$$

Each is short exact: ψ is well defined and injective because $px \equiv 0 \pmod{p^2}$ forces $x \equiv 0 \pmod{p}$; φ_c is surjective because c is invertible modulo p; and $\operatorname{im}(\psi) = p\mathbb{Z}/p^2\mathbb{Z} = \ker(\varphi_c)$.

Any two of them are isomorphic. Taking $\alpha = \operatorname{id}$, $\beta = \operatorname{id}$ and γ multiplication by $c'c^{-1}$ makes both squares of definition 17.1.7 commute, and γ is an automorphism of $\mathbb{Z}/p\mathbb{Z}$ because $c'c^{-1}$ is a unit modulo p.

No two of them are equivalent. Every endomorphism β of $\mathbb{Z}/p^2\mathbb{Z}$ is multiplication by some t, so $\beta \circ \psi = \psi$ forces $tp \equiv p \pmod{p^2}$, that is $t \equiv 1 \pmod{p}$. Then $\varphi_{c'} \circ \beta$ is multiplication by $c't \equiv c' \pmod{p}$, and requiring it to equal φ_c forces $c' \equiv c \pmod{p}$.

So $\mathcal{E}_1, \ldots, \mathcal{E}_{p-1}$ are $p-1$ pairwise inequivalent extensions of $\mathbb{Z}/p\mathbb{Z}$ by $\mathbb{Z}/p\mathbb{Z}$, all isomorphic to one another, and none of them is $\mathbb{Z}/p\mathbb{Z} \oplus \mathbb{Z}/p\mathbb{Z}$.

3. The two extensions of $\mathbb{Z}/n\mathbb{Z}$ by $\mathbb{Z}$ in example 17.1.6 are not even isomorphic, since an isomorphism of short exact sequences carries B to B' isomorphically and the two middle terms are not isomorphic.

A homomorphism of short exact sequences constrains its own three components: the two outer maps already determine a great deal about the middle one.

Proposition 17.1.9: Short Five Lemma

Let (α, β, γ) be a homomorphism of short exact sequences, in the notation of definition 17.1.7. Then

1. if α and γ are injective, so is β;
2. if α and γ are surjective, so is β;
3. if α and γ are isomorphisms, so is β, and the two sequences are isomorphic.

***Proof*:**
This is the first diagram chase in the book and is written out element by element. A chase is an algorithm with three moves: push an element along an available arrow, use commutativity of a square to identify the two routes around it, and use exactness of a row to move *backwards* along an arrow once the element is known to lie in that arrow's image. The bookkeeping is done by redrawing the relevant square with elements in place of modules, and $\mapsto$ in place of $\to$.

Step 1: β is injective. Let $b \in B$ with $\beta(b) = 0$; the goal is $b = 0$. Chase b around the right-hand square:

$$\begin{tikzcd} b \arrow[r, mapsto, "\varphi"] \arrow[d, mapsto, "\beta"'] & \varphi(b) \arrow[d, mapsto, "\gamma"] \\ 0 \arrow[r, mapsto, "\varphi'"'] & 0 \end{tikzcd}$$

Down then right gives $\varphi'(\beta(b)) = \varphi'(0) = 0$, and right then down gives $\gamma(\varphi(b))$. The square commutes, so $\gamma(\varphi(b)) = 0$; and γ is injective, so $\varphi(b) = 0$, that is $b \in \ker(\varphi)$.

Exactness is what makes the next move available. The top row is exact at B, so $\ker(\varphi) = \operatorname{im}(\psi)$, and there is an $a \in A$ with $\psi(a) = b$. Knowing only that φ kills b, the chase has produced a preimage in A, which is where the hypothesis on α can be used. Now chase a around the left-hand square:

$$\begin{tikzcd} a \arrow[r, mapsto, "\psi"] \arrow[d, mapsto, "\alpha"'] & b \arrow[d, mapsto, "\beta"] \\ \alpha(a) \arrow[r, mapsto, "\psi'"'] & 0 \end{tikzcd}$$

Commutativity gives $\psi'(\alpha(a)) = \beta(\psi(a)) = \beta(b) = 0$. The map ψ' is injective by proposition 17.1.1 applied to the bottom row, so $\alpha(a) = 0$, and α is injective by hypothesis, so $a = 0$. Therefore $b = \psi(a) = \psi(0) = 0$.

Step 2: β is surjective. Let $b' \in B'$; the goal is to produce a preimage under β. Both surjectivity hypotheses supply a pullback. Since γ is surjective there is a $c \in C$ with $\gamma(c) = \varphi'(b')$, and since φ is surjective by proposition 17.1.1 there is a $b \in B$ with $\varphi(b) = c$. Chasing b gives

$$\begin{tikzcd} b \arrow[r, mapsto, "\varphi"] \arrow[d, mapsto, "\beta"'] & c \arrow[d, mapsto, "\gamma"] \\ \beta(b) \arrow[r, mapsto, "\varphi'"'] & \varphi'(b') \end{tikzcd}$$

where the bottom right entry is computed by commutativity:

$$\varphi'(\beta(b)) = \gamma(\varphi(b)) = \gamma(c) = \varphi'(b')$$

Hence $\varphi'(b' - \beta(b)) = 0$, so $b' - \beta(b) \in \ker(\varphi') = \operatorname{im}(\psi')$ by exactness of the bottom row, and $b' - \beta(b) = \psi'(a')$ for some $a' \in A'$. Since α is surjective, $a' = \alpha(a)$ for some $a \in A$, and the left square gives

$$b' - \beta(b) = \psi'(\alpha(a)) = \beta(\psi(a))$$

Therefore $b' = \beta(b) + \beta(\psi(a)) = \beta\big(b + \psi(a)\big)$, so b' lies in the image of β.

Step 3. If α and γ are isomorphisms then Steps 1 and 2 both apply, so β is injective and surjective, hence an isomorphism, and (α, β, γ) is an isomorphism of short exact sequences by definition 17.1.7(2).

Neither chase used the R-action, only the underlying additive groups, and the single subtraction in Step 2 can be written multiplicatively as $b'\beta(b)^{-1}$. The proposition and its proof therefore hold verbatim for short exact sequences of groups, abelian or not.

The next result relates two short exact sequences by splicing their kernels and cokernels into one long exact sequence. Its proof runs the same three moves, and from here on a chase states which move it is making rather than tracking every element.

Proposition 17.1.10: Mini Snake Lemma

Let (α, β, γ) be a homomorphism of short exact sequences, in the notation of definition 17.1.7. Then there is an exact sequence of R-modules

$$0 \to \ker(\alpha) \xrightarrow{\overline{\psi}} \ker(\beta) \xrightarrow{\overline{\varphi}} \ker(\gamma) \xrightarrow{d} \operatorname{coker}(\alpha) \xrightarrow{\overline{\psi}'} \operatorname{coker}(\beta) \xrightarrow{\overline{\varphi}'} \operatorname{coker}(\gamma) \to 0$$

in which $\overline{\psi}$ and $\overline{\varphi}$ are the restrictions of ψ and φ, the maps $\overline{\psi}'$ and $\overline{\varphi}'$ are induced by ψ' and φ' on cokernels, and d is the *connecting homomorphism* constructed in the proof.

***Proof*:**
The whole statement is read off the following diagram, in which the two middle rows are the given ones and the outer rows are the sequence being constructed. The connecting homomorphism d is the one arrow not drawn: it runs from the top right corner $\ker(\gamma)$ to the bottom left corner $\operatorname{coker}(\alpha)$, and the route taken to build it (down the right column, left along the bottom row, then up the left column) is the snake the lemma is named for.

$$\begin{tikzcd}
 & \ker(\alpha) \arrow[r, "\overline{\psi}"] \arrow[d, hook] & \ker(\beta) \arrow[r, "\overline{\varphi}"] \arrow[d, hook] & \ker(\gamma) \arrow[d, hook] & \\
0 \arrow[r] & A \arrow[r, "\psi"] \arrow[d, "\alpha"'] & B \arrow[r, "\varphi"] \arrow[d, "\beta"'] & C \arrow[r] \arrow[d, "\gamma"'] & 0 \\
0 \arrow[r] & A' \arrow[r, "\psi'"] \arrow[d, two heads] & B' \arrow[r, "\varphi'"] \arrow[d, two heads] & C' \arrow[r] \arrow[d, two heads] & 0 \\
 & \operatorname{coker}(\alpha) \arrow[r, "\overline{\psi}'"'] & \operatorname{coker}(\beta) \arrow[r, "\overline{\varphi}'"'] & \operatorname{coker}(\gamma) &
\end{tikzcd}$$

Step 1: the four induced maps exist. If $a \in \ker(\alpha)$ then $\beta(\psi(a)) = \psi'(\alpha(a)) = 0$, so ψ carries $\ker(\alpha)$ into $\ker(\beta)$; the same computation with the right square carries $\ker(\beta)$ into $\ker(\gamma)$. Dually $\psi'(\operatorname{im}(\alpha)) \subseteq \operatorname{im}(\beta)$, because $\psi'(\alpha(a)) = \beta(\psi(a))$, so ψ' descends to $\overline{\psi}' : \operatorname{coker}(\alpha) \to \operatorname{coker}(\beta)$; the same computation gives $\overline{\varphi}'$.

Step 2: construction of d. Let $c \in \ker(\gamma)$. Pull back along φ, which is surjective, to get $b \in B$ with $\varphi(b) = c$. Commutativity of the right square gives $\varphi'(\beta(b)) = \gamma(\varphi(b)) = \gamma(c) = 0$, so $\beta(b) \in \ker(\varphi') = \operatorname{im}(\psi')$ by exactness of the bottom row. Since ψ' is injective there is a *unique* $a' \in A'$ with $\psi'(a') = \beta(b)$. Define

$$d(c) := a' + \operatorname{im}(\alpha) \in \operatorname{coker}(\alpha)$$

The three steps just taken, namely pulling back along φ, crossing by β and pulling back along ψ', are the three segments of the route described above.

Step 3: d is a well defined homomorphism. The only choice made was the lift b. If $\varphi(b_1) = c$ as well then $b_1 - b \in \ker(\varphi) = \mathrm{im}(\psi)$, say $b_1 - b = \psi(a)$, so

$$\psi'(a_1') - \psi'(a') = \beta(b_1) - \beta(b) = \beta(\psi(a)) = \psi'(\alpha(a))$$

and injectivity of ψ' gives $a_1' - a' = \alpha(a) \in \mathrm{im}(\alpha)$. The two answers agree in $\mathrm{coker}(\alpha)$. Additivity and R-linearity follow because a lift of $c_1 + rc_2$ may be taken to be $b_1 + rb_2$, and ψ' and β are homomorphisms.

Step 4: exactness, at each of the six interior spots.

At $\ker(\alpha)$. The map $\overline{\psi}$ is a restriction of the injective ψ, so it is injective.

At $\ker(\beta)$. Since $\varphi \circ \psi = 0$ the composite $\overline{\varphi} \circ \overline{\psi}$ is zero, giving $\mathrm{im}(\overline{\psi}) \subseteq \ker(\overline{\varphi})$. Conversely let $b \in \ker(\beta)$ with $\varphi(b) = 0$. Exactness of the top row gives $b = \psi(a)$, and $\psi'(\alpha(a)) = \beta(\psi(a)) = \beta(b) = 0$, so $\alpha(a) = 0$ by injectivity of ψ'. Thus $a \in \ker(\alpha)$ and $b = \overline{\psi}(a)$.

At $\ker(\gamma)$. If $c = \overline{\varphi}(b)$ with $b \in \ker(\beta)$, then b is an admissible lift of c in Step 2 and $\beta(b) = 0 = \psi'(0)$, so $d(c) = 0 + \mathrm{im}(\alpha) = 0$. Conversely suppose $d(c) = 0$, so the element a' of Step 2 is $\alpha(a)$ for some $a \in A$. Then $\beta(b) = \psi'(\alpha(a)) = \beta(\psi(a))$, so $b - \psi(a) \in \ker(\beta)$, while $\varphi(b - \psi(a)) = \varphi(b) = c$. Hence $c = \overline{\varphi}(b - \psi(a))$.

At $\mathrm{coker}(\alpha)$. With c, b and a' as in Step 2,

$$\overline{\psi}'(d(c)) = \psi'(a') + \mathrm{im}(\beta) = \beta(b) + \mathrm{im}(\beta) = 0$$

so $\mathrm{im}(d) \subseteq \ker(\overline{\psi}')$. Conversely let $a' + \mathrm{im}(\alpha)$ lie in $\ker(\overline{\psi}')$, meaning $\psi'(a') = \beta(b)$ for some $b \in B$. Then $\gamma(\varphi(b)) = \varphi'(\beta(b)) = \varphi'(\psi'(a')) = 0$, so $c := \varphi(b)$ lies in $\ker(\gamma)$, and b is a lift of c exhibiting $d(c) = a' + \mathrm{im}(\alpha)$.

At $\mathrm{coker}(\beta)$. Since $\varphi' \circ \psi' = 0$ the composite $\overline{\varphi}' \circ \overline{\psi}'$ is zero. Conversely let $b' + \mathrm{im}(\beta)$ lie in $\ker(\overline{\varphi}')$, so $\varphi'(b') = \gamma(c)$ for some $c \in C$, and write $c = \varphi(b)$ using surjectivity of φ. Then $\varphi'(b' - \beta(b)) = \gamma(c) - \gamma(\varphi(b)) = 0$, so $b' - \beta(b) = \psi'(a')$ for some a', and $b' + \mathrm{im}(\beta) = \overline{\psi}'(a' + \mathrm{im}(\alpha))$.

At $\mathrm{coker}(\gamma)$. The map $\overline{\varphi}'$ is induced by the surjective φ', so it is surjective.

The name records that this is the three-module case. The general snake lemma replaces each of the three modules in a row by a whole chain complex and each kernel and cokernel by homology, and the six-term sequence above becomes a long exact sequence continuing in both directions; that version is [15, the snake lemma]. The connecting homomorphism is what turns a short exact sequence into a longer one, and section 17.6 uses it to build the long exact sequences of Ext.

17.1.2 Split Sequences

Among all extensions of C by A, the direct sum $A \oplus C$ carries no information beyond its two ends. Example 17.1.6 showed that other extensions exist, so the ones that reduce to the direct sum deserve a name.

Definition 17.1.11: Split Short Exact Sequence

A short exact sequence $0 \to A \xrightarrow{\psi} B \xrightarrow{\varphi} C \to 0$ of R-modules *splits* if $\psi(A)$ has a complement in B, that is if there is a submodule $C' \subseteq B$ with

$$B = \psi(A) \oplus C'$$

The extension B is then called a *split extension* of C by A.

The complement is automatically a copy of C, so the definition need not demand it. If $B = \psi(A) \oplus C'$, then φ restricted to C' is injective because $C' \cap \ker(\varphi) = C' \cap \psi(A) = 0$, and surjective because $\varphi(B) = \varphi(\psi(A)) + \varphi(C') = \varphi(C')$. Hence $B \cong \psi(A) \oplus C \cong A \oplus C$.

Definition 17.1.12: Section and Retraction

Let $0 \to A \xrightarrow{\psi} B \xrightarrow{\varphi} C \to 0$ be a short exact sequence of R-modules.

1. A *section* of φ is an R-module homomorphism $\mu\colon C \to B$ with $\varphi \circ \mu = \mathrm{id}_C$. It is also called a *splitting homomorphism* for the sequence.
2. A *retraction* of ψ is an R-module homomorphism $\lambda\colon B \to A$ with $\lambda \circ \psi = \mathrm{id}_A$.

Proposition 17.1.13: Splitting Criteria

For a short exact sequence $0 \to A \xrightarrow{\psi} B \xrightarrow{\varphi} C \to 0$ of R-modules the following are equivalent.

1. The sequence splits.
2. φ has a section $\mu\colon C \to B$.
3. ψ has a retraction $\lambda\colon B \to A$.

***Proof*:**
(1) $\Rightarrow$ (2). Let $B = \psi(A) \oplus C'$. As noted above φ restricts to an isomorphism $\varphi|_{C'} : C' \to C$, so its inverse followed by the inclusion $C' \subseteq B$ is a homomorphism $\mu\colon C \to B$, and $\varphi(\mu(c)) = c$ for every $c \in C$.

(2) $\Rightarrow$ (1). Let μ be a section and put $C' := \mu(C)$, a submodule of B. For any $b \in B$,

$$b = \big(b - \mu(\varphi(b))\big) + \mu(\varphi(b))$$

and $\varphi(b - \mu(\varphi(b))) = \varphi(b) - \varphi(b) = 0$, so the first summand lies in $\ker(\varphi) = \psi(A)$ while the second lies in C'. The sum is direct: if $\mu(c) \in \psi(A) = \ker(\varphi)$ then $0 = \varphi(\mu(c)) = c$, so $\mu(c) = 0$. Hence $B = \psi(A) \oplus C'$.

(1) $\Rightarrow$ (3). Let $B = \psi(A) \oplus C'$ and let $\pi_{\psi(A)}$ be the projection onto the first summand. The map ψ is an isomorphism onto $\psi(A)$ by proposition 17.1.1, so $\lambda := \psi^{-1} \circ \pi_{\psi(A)}$ is defined, and $\lambda(\psi(a)) = \psi^{-1}(\psi(a)) = a$.

(3) ⇒ (1). Let λ be a retraction and put $C' := \ker(\lambda)$. For any $b \in B$,

$$b = \psi(\lambda(b)) + \big(b - \psi(\lambda(b))\big)$$

where the first summand lies in $\psi(A)$ and the second lies in C', since $\lambda(b-\psi(\lambda(b))) = \lambda(b)-\lambda(b) = 0$. The sum is direct: if $b = \psi(a)$ and $\lambda(b) = 0$ then $a = \lambda(\psi(a)) = \lambda(b) = 0$, so $b = 0$. Hence $B = \psi(A) \oplus C'$.

Example 17.1.14: Split and Non-split Sequences

1. The sequence $0 \to A \xrightarrow{\iota} A \oplus C \xrightarrow{\pi} C \to 0$ splits, with section $\mu(c) = (0, c)$.
2. Of the two sequences in example 17.1.6, the first splits and the second does not. A section of the second would be a nonzero homomorphism $\mathbb{Z}/n\mathbb{Z} \to \mathbb{Z}$, and the image z of $\overline{1}$ would satisfy $nz = 0$, forcing $z = 0$ and hence the whole map to be zero.
3. The sequence $0 \to \mathbb{Z}/2\mathbb{Z} \to \mathbb{Z}/4\mathbb{Z} \to \mathbb{Z}/2\mathbb{Z} \to 0$ does not split, since a splitting would give $\mathbb{Z}/4\mathbb{Z} \cong \mathbb{Z}/2\mathbb{Z} \oplus \mathbb{Z}/2\mathbb{Z}$, and only the left-hand side has an element of order 4.
4. Over $R = k[x]$ the sequence

$$0 \to (x) \hookrightarrow k[x] \to k[x]/(x) \to 0$$

does not split. A section would embed $k[x]/(x)$ as a nonzero submodule of $k[x]$ annihilated by x, and $k[x]$ is an integral domain, so no nonzero element is annihilated by x.

Splitting is a property of the sequence alone, and it is carried along by any isomorphism of sequences; equivalence is not needed.

Proposition 17.1.15: Splitting Is an Isomorphism Invariant

Let (α, β, γ) be an isomorphism of short exact sequences, in the notation of definition 17.1.7. Then the top sequence splits if and only if the bottom one does.

***Proof*:**
Suppose the top sequence splits and let $\mu\colon C \to B$ be a section of φ, which exists by proposition 17.1.13. Set

$$\mu' := \beta \circ \mu \circ \gamma^{-1} \colon C' \to B'$$

a homomorphism because γ^{-1} is one. Commutativity of the right square reads $\varphi' \circ \beta = \gamma \circ \varphi$, so

$$\varphi' \circ \mu' = \varphi' \circ \beta \circ \mu \circ \gamma^{-1} = \gamma \circ \varphi \circ \mu \circ \gamma^{-1} = \gamma \circ \gamma^{-1} = \mathrm{id}_{C'}$$

Hence μ' is a section of φ' and the bottom sequence splits, again by proposition 17.1.13.

For the converse, note that $(\alpha^{-1}, \beta^{-1}, \gamma^{-1})$ is an isomorphism of short exact sequences from the bottom one to the top one: composing $\beta \circ \psi = \psi' \circ \alpha$ with β^{-1} on the left and α^{-1} on the right gives $\psi \circ \alpha^{-1} = \beta^{-1} \circ \psi'$, and the right square is handled the same way. Applying the first paragraph to it gives the converse.

The vocabulary is now sufficient to state the question this chapter opened with. Given R-modules A and C, write $\mathcal{E}(C, A)$ for the set of equivalence classes of extensions of C by A, in the sense of definition 17.1.7(3). This set is never empty, and all split extensions lie in a single class: if μ is a section of φ, then $B = \psi(A) \oplus \mu(C)$ by the proof of proposition 17.1.13, so $\beta(a, c) := \psi(a) + \mu(c)$ is an isomorphism $A \oplus C \to B$, both ψ and μ being injective. It satisfies $\beta(\iota(a)) = \psi(a)$ and $\varphi(\beta(a, c)) = c = \pi(a, c)$, so $(\mathrm{id}_A, \beta, \mathrm{id}_C)$ is an equivalence with the direct-sum sequence. By example 17.1.8(2) the set can have more than one element.

The *extension problem* for modules asks for $\mathcal{E}(C, A)$. Section 17.6 answers it by constructing a module $\mathrm{Ext}^1_R(C, A)$ together with a bijection $\mathrm{Ext}^1_R(C, A) \leftrightarrow \mathcal{E}(C, A)$ carrying 0 to the class of the split extension.

Exercise 17.1.1

1. For each of the following short exact sequences of $\mathbb{Z}$-modules, decide whether it splits. Exhibit a section when one exists, and prove that none exists otherwise.

   1. $0 \to \mathbb{Z} \xrightarrow{\cdot 3} \mathbb{Z} \to \mathbb{Z}/3\mathbb{Z} \to 0$
   2. $0 \to \mathbb{Z}/2\mathbb{Z} \to \mathbb{Z}/2\mathbb{Z} \oplus \mathbb{Z}/3\mathbb{Z} \to \mathbb{Z}/3\mathbb{Z} \to 0$, with the evident injection and projection
   3. $0 \to \mathbb{Z}/3\mathbb{Z} \to \mathbb{Z}/9\mathbb{Z} \to \mathbb{Z}/3\mathbb{Z} \to 0$, with $\overline{a} \mapsto \overline{3a}$ and reduction modulo 3
   4. $0 \to \mathbb{Z}/2\mathbb{Z} \to \mathbb{Z}/6\mathbb{Z} \to \mathbb{Z}/3\mathbb{Z} \to 0$, with $\overline{a} \mapsto \overline{3a}$ and reduction modulo 3

2. Let p be a prime. Show that every extension of $\mathbb{Z}/p\mathbb{Z}$ by $\mathbb{Z}/p\mathbb{Z}$ has a middle term of order p^2, hence isomorphic to $\mathbb{Z}/p^2\mathbb{Z}$ or to $\mathbb{Z}/p\mathbb{Z} \oplus \mathbb{Z}/p\mathbb{Z}$ by theorem 5.1.8, and that in the second case the sequence splits. In the first case, show the sequence is equivalent to one of the $\mathcal{E}_c$ of example 17.1.8(2); you may use that $\mathbb{Z}/p^2\mathbb{Z}$ has a unique subgroup of order p. Conclude that $\mathcal{E}(\mathbb{Z}/p\mathbb{Z}, \mathbb{Z}/p\mathbb{Z})$ has exactly p elements.

3. Let $m, n \geq 1$. Take both rows of definition 17.1.7 to be $0 \to \mathbb{Z} \xrightarrow{\cdot m} \mathbb{Z} \to \mathbb{Z}/m\mathbb{Z} \to 0$ and take α, β and γ all to be multiplication by n. Verify that this is a homomorphism of short exact sequences, identify all six terms of the sequence of proposition 17.1.10 in terms of $g = \gcd(m, n)$, and compute the connecting homomorphism d explicitly. Then specialise to $m = 4$ and $n = 6$.

4. Let (α, β, γ) be a homomorphism of short exact sequences.

   1. Show that if β is injective then so is α.
   2. Show that if β is surjective then so is γ.
   3. Give an example in which β is an isomorphism while α is not surjective and γ is not injective. *Hint*: one of the two rows may end in the zero module.

5. Show that a short exact sequence $0 \to A \to B \xrightarrow{\varphi} C \to 0$ of R-modules splits whenever C is a free module. *Hint*: choose a basis of C, lift each basis element along φ, and use theorem 12.3.11.

6. Show that if $0 \to A \xrightarrow{\psi} B \xrightarrow{\varphi} C \to 0$ splits then for every R-module D there are isomorphisms of abelian groups
$$\mathrm{Hom}_R(D, B) \cong \mathrm{Hom}_R(D, A) \oplus \mathrm{Hom}_R(D, C)$$
$$\mathrm{Hom}_R(B, D) \cong \mathrm{Hom}_R(A, D) \oplus \mathrm{Hom}_R(C, D)$$

Hint: use the section and the retraction supplied by proposition 17.1.13.

7. *(The five lemma.)* Consider a commutative diagram of R-modules

$$\begin{tikzcd} A_1 \arrow[r] \arrow[d, "f_1"'] & A_2 \arrow[r] \arrow[d, "f_2"'] & A_3 \arrow[r] \arrow[d, "f_3"'] & A_4 \arrow[r] \arrow[d, "f_4"'] & A_5 \arrow[d, "f_5"'] \\ B_1 \arrow[r] & B_2 \arrow[r] & B_3 \arrow[r] & B_4 \arrow[r] & B_5 \end{tikzcd}$$

with exact rows. Prove that if f_2 and f_4 are isomorphisms, f_1 is surjective and f_5 is injective, then f_3 is an isomorphism. *Hint*: run the two chases of proposition 17.1.9 one column further in each direction.

17.2 Projective Modules

Example 17.1.14 produced short exact sequences that do not split, so the extension problem has content. The instrument for measuring the failure is already available: fix an R-module D, apply $\mathrm{Hom}_R(D, -)$ to a short exact sequence, and see how much of the sequence survives.

Throughout, ψ_* denotes the map induced on $\mathrm{Hom}_R(D, -)$ by a homomorphism ψ, reserving the primes of definition 17.1.7 for the second row of a diagram.

Proposition 17.2.1: Left Exactness of $\mathrm{Hom}_R(D, -)$

Let D be an R-module.

1. For every R-module homomorphism $\psi\colon L \to M$, the assignment

$$\psi_*\colon \mathrm{Hom}_R(D, L) \to \mathrm{Hom}_R(D, M), \qquad f \mapsto \psi \circ f$$

is a homomorphism of abelian groups.

2. If $0 \to L \xrightarrow{\psi} M \xrightarrow{\varphi} N \to 0$ is a short exact sequence of R-modules, then

$$0 \to \mathrm{Hom}_R(D, L) \xrightarrow{\psi_*} \mathrm{Hom}_R(D, M) \xrightarrow{\varphi_*} \mathrm{Hom}_R(D, N)$$

is exact, for every D. In this situation one says that $\mathrm{Hom}_R(D, -)$ is *left exact*.

Note what part (2) does *not* claim: that φ_* is surjective, so that the sequence may be closed off with a $\to 0$ on the right. That claim is false in general, and the whole section grows out of the failure.

***Proof*:**
(1) For $f, g \in \mathrm{Hom}_R(D, L)$ and $d \in D$,

$$\big(\psi \circ (f + g)\big)(d) = \psi\big(f(d) + g(d)\big) = \psi(f(d)) + \psi(g(d)) = \big(\psi \circ f + \psi \circ g\big)(d)$$

using that ψ is a homomorphism, so $\psi_*(f + g) = \psi_*(f) + \psi_*(g)$.

(2) *Exactness at* $\mathrm{Hom}_R(D, L)$, which by proposition 17.1.1 is the injectivity of ψ_*. Suppose $\psi \circ f = 0$. Then $\psi(f(d)) = 0$ for every $d \in D$, so $f(d) \in \ker(\psi) = 0$ because ψ is injective, and $f = 0$.

Exactness at $\mathrm{Hom}_R(D, M)$, that is $\mathrm{im}(\psi_*) = \ker(\varphi_*)$.

($\subseteq$) For $f \in \mathrm{Hom}_R(D, L)$ we have $\varphi_*(\psi_*(f)) = \varphi \circ \psi \circ f$, and $\varphi \circ \psi = 0$ because $\mathrm{im}(\psi) = \ker(\varphi)$. So $\varphi_*(\psi_*(f)) = 0$.

($\supseteq$) Let $F \in \ker(\varphi_*)$, so $\varphi \circ F = 0$. For each $d \in D$ we have $\varphi(F(d)) = 0$, hence $F(d) \in \ker(\varphi) = \mathrm{im}(\psi)$, so there is an $l \in L$ with $\psi(l) = F(d)$; and l is unique because ψ is injective. Define $F' \colon D \to L$ by letting $F'(d)$ be that unique l, so that $\psi \circ F' = F$ by construction.

It remains to check that F' is a homomorphism, which is the only step here with content. For $d, e \in D$ and $r \in R$,

$$\psi\big(F'(d) + rF'(e)\big) = \psi(F'(d)) + r\psi(F'(e)) = F(d) + rF(e) = F(d + re) = \psi\big(F'(d + re)\big)$$

and ψ is injective, so $F'(d + re) = F'(d) + rF'(e)$. Hence $F' \in \mathrm{Hom}_R(D, L)$ and $F = \psi_*(F') \in \mathrm{im}(\psi_*)$.

Now the failure, computed before anything is named after it. The sequence at hand is the non-split extension of example 17.1.6 with $n = 2$.

Example 17.2.2: $\mathrm{Hom}_R(D, -)$ Need Not Be Right Exact
Take $R = \mathbb{Z}$, the short exact sequence

$$0 \to \mathbb{Z} \xrightarrow{\cdot 2} \mathbb{Z} \xrightarrow{\pi} \mathbb{Z}/2\mathbb{Z} \to 0$$

and $D = \mathbb{Z}/2\mathbb{Z}$. Then $\mathrm{Hom}_{\mathbb{Z}}(\mathbb{Z}/2\mathbb{Z}, \mathbb{Z}) = 0$: a homomorphism h sends $\overline{1}$ to some $z \in \mathbb{Z}$ with $2z = h(\overline{2}) = h(\overline{0}) = 0$, so $z = 0$. On the other hand $\mathrm{Hom}_{\mathbb{Z}}(\mathbb{Z}/2\mathbb{Z}, \mathbb{Z}/2\mathbb{Z}) \cong \mathbb{Z}/2\mathbb{Z}$, since such a homomorphism is determined by the image of $\overline{1}$ and both choices are legitimate. So applying $\mathrm{Hom}_{\mathbb{Z}}(D, -)$ gives

$$0 \to 0 \xrightarrow{(\cdot 2)_*} 0 \xrightarrow{\pi_*} \mathbb{Z}/2\mathbb{Z}$$

which is exact, as proposition 17.2.1 promised, and which cannot be extended by $\to 0$ on the right, since π_* is the zero map into a nonzero group.

The obstruction has a name in the language of diagrams. The element of $\mathrm{Hom}_{\mathbb{Z}}(\mathbb{Z}/2\mathbb{Z}, \mathbb{Z}/2\mathbb{Z})$ that is not hit is $\mathrm{id}_{\mathbb{Z}/2\mathbb{Z}}$, and saying it is not hit is saying that the dashed arrow below does not exist:

$$\begin{tikzcd}
& \mathbb{Z}/2\mathbb{Z} \arrow[dl, dashed, "g\,?"'] \arrow[d, "\mathrm{id}"] & \\
\mathbb{Z} \arrow[r, "\pi"'] & \mathbb{Z}/2\mathbb{Z} \arrow[r] & 0
\end{tikzcd}$$

Indeed the only homomorphism $g \colon \mathbb{Z}/2\mathbb{Z} \to \mathbb{Z}$ is the zero map, and $\pi \circ 0 = 0 \neq \mathrm{id}_{\mathbb{Z}/2\mathbb{Z}}$.

A homomorphism out of D that fails to lift is exactly what stops $\mathrm{Hom}_R(D, -)$ from being exact. The modules for which this never happens are the subject of this section.

Definition 17.2.3: Projective Module

An R-module P is *projective* if the functor $\mathrm{Hom}_R(P, -)$ is exact, that is if for every short exact sequence $0 \to L \to M \to N \to 0$ of R-modules the sequence

$$0 \to \mathrm{Hom}_R(P, L) \to \mathrm{Hom}_R(P, M) \to \mathrm{Hom}_R(P, N) \to 0$$

is again short exact.

By proposition 17.2.1 everything in that sequence is automatic except surjectivity on the right, so the definition asks for one thing only: that every homomorphism out of P lift along every surjection. The theorem below makes that precise and adds two further descriptions, one about splitting and one about free modules.

Theorem 17.2.4: Equivalencies for Projective Modules

Let P be an R-module. The following are equivalent.

1. P is projective.
2. *(Lifting.)* For all R-modules M and N, if $\varphi \colon M \to N$ is surjective, then every $f \in \mathrm{Hom}_R(P, N)$ has a lift $g \in \mathrm{Hom}_R(P, M)$ with $\varphi \circ g = f$:

$$\begin{tikzcd} & P \arrow[dl, dashed, "g"'] \arrow[d, "f"] & \\ M \arrow[r, "\varphi"'] & N \arrow[r] & 0 \end{tikzcd}$$

3. *(Splitting.)* Every short exact sequence $0 \to L \to M \to P \to 0$ of R-modules splits.
4. *(Summand.)* P is isomorphic to a direct summand of a free R-module.

***Proof*:**

(1) $\Rightarrow$ (2). Let $\varphi \colon M \to N$ be surjective and put $L = \ker(\varphi)$, so that $0 \to L \to M \xrightarrow{\varphi} N \to 0$ is short exact by example 17.1.4(2). Condition (1) makes $\varphi_* \colon \mathrm{Hom}_R(P, M) \to \mathrm{Hom}_R(P, N)$ surjective, so a given f equals $\varphi_*(g) = \varphi \circ g$ for some g.

(2) $\Rightarrow$ (1). Proposition 17.2.1 gives exactness at the first two spots of the sequence in definition 17.2.3, and (2) applied to the surjection φ says precisely that φ_* is surjective.

(2) $\Rightarrow$ (3). Let $0 \to L \xrightarrow{\psi} M \xrightarrow{\varphi} P \to 0$ be short exact. Then φ is surjective, so applying (2) to $f = \mathrm{id}_P$ produces $\mu \in \mathrm{Hom}_R(P, M)$ with $\varphi \circ \mu = \mathrm{id}_P$. That is a section of φ, so the sequence splits by proposition 17.1.13.

(3) $\Rightarrow$ (4). By example 17.1.4(3), applied with $S = P$ as a generating set, there is a short exact sequence

$$0 \to K \hookrightarrow F \xrightarrow{\pi} P \to 0$$

with F free. By (3) it splits, so $F = K \oplus C'$ for some submodule C', and $\pi|_{C'} : C' \to P$ is an isomorphism by the remark following definition 17.1.11. Hence P is isomorphic to the direct summand C' of F.

(4) $\Rightarrow$ (2). Suppose $F = P' \oplus K$ with F free and $P' \cong P$; replacing P by P', we may assume P itself is a summand, and write $\iota\colon P \to F$ for the inclusion and $\rho\colon F \to P$ for the projection, so $\rho \circ \iota = \mathrm{id}_P$.

Let $\varphi\colon M \to N$ be surjective and $f \in \mathrm{Hom}_R(P, N)$. Let $(e_s)_{s \in S}$ be a basis of F. For each s the element $f(\rho(e_s))$ lies in N, so by surjectivity of φ we may choose $m_s \in M$ with $\varphi(m_s) = f(\rho(e_s))$. By theorem 12.3.11 there is a unique homomorphism $G\colon F \to M$ with $G(e_s) = m_s$ for every s, and then $\varphi \circ G$ and $f \circ \rho$ are homomorphisms $F \to N$ agreeing on a basis, hence equal.

Now set $g := G \circ \iota\colon P \to M$. It is a homomorphism, and

$$\varphi \circ g = \varphi \circ G \circ \iota = f \circ \rho \circ \iota = f \circ \mathrm{id}_P = f$$

which is the required lift.

17.2.5 Summand, Not Sum Condition (4) says direct **summand** of a free module, which is weaker than being a free module. A direct summand of M is any submodule admitting a complement, so if $M = A \oplus B \oplus C$ then

$$A,\ B,\ C,\ A \oplus B,\ A \oplus C,\ B \oplus C,\ A \oplus B \oplus C$$

are all direct summands of M, and only the last is M itself. Reading (4) as "direct sum" would collapse the notion of projective onto the notion of free, and the two are genuinely different, as the two examples below show.

Corollary 17.2.6: Free Modules Are Projective

Every free R-module is projective.

***Proof*:**
A free module F is a direct summand of itself, with complement 0, so theorem 17.2.4(4) applies.

The converse fails, and it fails for a reason worth seeing twice. Both witnesses come from splitting a ring into a direct sum of ideals; the summands are then projective by construction and too small to be free.

Example 17.2.7: Projective But Not Free

1. Let $R = \mathbb{Z}/6\mathbb{Z}$. The Chinese remainder theorem gives a ring isomorphism $R \cong \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/3\mathbb{Z}$, which is in particular an isomorphism of R-modules, so the ideal $P = \overline{3}R = \{\overline{0}, \overline{3}\}$ is a direct summand of R, with complement $\overline{4}R = \{\overline{0}, \overline{4}, \overline{2}\}$. Since R is free of rank one over itself, P is projective by theorem 17.2.4(4).

   It is not free. A free R-module of finite rank k has 6^k elements and P has 2, while a free module of infinite rank is infinite; and $2 \neq 6^k$ for every $k \geq 0$.

2. Let F be a field, $n \geq 2$, and $R = M_n(F)$ the ring of $n \times n$ matrices. Let $C = F^n$ be the module of column vectors, a left R-module by matrix multiplication. Writing E_{jj} for the

matrix with a 1 in position (j, j) and zeros elsewhere,

$$R = \bigoplus_{j=1}^{n} RE_{jj}$$

since RE_{jj} consists of the matrices supported in column j and every matrix is the sum of its columns. Each RE_{jj} is isomorphic to C as a left R-module, by reading off column j. So C is a direct summand of the free module R and is projective.

It is not free. Counting dimensions over F, a free R-module of finite rank k has F-dimension kn^2, and one of infinite rank is infinite dimensional; meanwhile $\dim_F C = n$, which is finite, and $n = kn^2$ has no solution $k \in \mathbb{Z}_{\geq 0}$ once $n \geq 2$.

Condition (4) is structural but awkward to verify, since it requires producing a free module and a complement. The following reformulation is the working criterion, because it asks only for elements of A and homomorphisms out of A, both of which can be exhibited.

Proposition 17.2.8: Dual Basis Lemma

An R-module A is projective if and only if there exist a family $(a_i)_{i\in I}$ of elements of A and a family $(f_i)_{i\in I}$ of homomorphisms $f_i \in \mathrm{Hom}_R(A, R)$ such that, for every $a \in A$,

1. $f_i(a) = 0$ for all but finitely many $i \in I$, and
2. $a = \sum_{i\in I} f_i(a)\, a_i$.

Such a pair of families is called a *dual basis* for A. It may be taken finite exactly when A is finitely generated.

***Proof*:**
($\Rightarrow$) Let A be projective. By example 17.1.4(3) there is a free module F with basis $(e_i)_{i\in I}$ and a surjection $\pi\colon F \to A$, and by theorem 17.2.4(3) the sequence $0 \to \ker(\pi) \to F \xrightarrow{\pi} A \to 0$ splits, so proposition 17.1.13 supplies $\sigma\colon A \to F$ with $\pi \circ \sigma = \mathrm{id}_A$.

Put $a_i := \pi(e_i)$, and let $f_i(a)$ be the i-th coordinate of $\sigma(a)$ in the basis (e_i), so that $\sigma(a) = \sum_i f_i(a)e_i$. Each f_i is a homomorphism $A \to R$, being σ followed by a coordinate function, and (1) holds because an element of F has only finitely many nonzero coordinates. Applying π,

$$a = \pi(\sigma(a)) = \pi\Big(\sum_i f_i(a)e_i\Big) = \sum_i f_i(a)\pi(e_i) = \sum_i f_i(a)a_i$$

which is (2).

($\Leftarrow$) Given the two families, let F be free on a basis $(e_i)_{i\in I}$ and define $\pi\colon F \to A$ by $\pi(e_i) = a_i$, using theorem 12.3.11. Define $\sigma\colon A \to F$ by $\sigma(a) = \sum_i f_i(a)e_i$, which is a well defined element of F by (1) and is a homomorphism because each f_i is. Then (2) says exactly that $\pi(\sigma(a)) = a$, so $\pi \circ \sigma = \mathrm{id}_A$. In particular π is surjective, and σ is a section of π, so $0 \to \ker(\pi) \to F \xrightarrow{\pi} A \to 0$ splits by proposition 17.1.13 and A is a direct summand of F by definition 17.1.11. Theorem 17.2.4(4) makes A projective.

For the last sentence: if the families are finite then (2) exhibits $a_1, \ldots, a_n$ as generators of A. Conversely if A is generated by n elements, take $F = R^n$ in the forward direction, so that I is finite.

Chapter 15 produced an isomorphism $V^* \otimes_F W \cong \mathrm{Hom}_F(V, W)$ for finite dimensional vector spaces (theorem 15.4.9). Over a general commutative ring the same map exists, and the dual basis says exactly when it is an isomorphism.

Corollary 17.2.9: Evaluation and Finitely Generated Projectives

Let R be a commutative ring, A an R-module, and $A^* := \mathrm{Hom}_R(A, R)$. Let

$$\theta\colon A^* \otimes_R A \to \mathrm{End}_R(A), \qquad \theta(f \otimes b)(a) = f(a)b$$

Then A is finitely generated and projective if and only if θ is an isomorphism, and in fact if and only if θ is surjective.

***Proof*:**
The map θ exists because $(f, b) \mapsto \big(a \mapsto f(a)b\big)$ is R-bilinear, so theorem 15.1.12 produces it.

Suppose first that θ is surjective. Then $\mathrm{id}_A = \theta(t)$ for some $t \in A^* \otimes_R A$, and every element of a tensor product is a finite sum of simple tensors, say $t = \sum_{i=1}^n f_i \otimes a_i$. Unwinding, $a = \mathrm{id}_A(a) = \sum_{i=1}^n f_i(a)a_i$ for every $a \in A$. That is a finite dual basis, so A is finitely generated and projective by proposition 17.2.8.

Conversely let A be finitely generated and projective, with a finite dual basis $(a_i, f_i)_{i=1}^n$ from proposition 17.2.8. Two observations do the work. The first is that

$$g = \sum_{i=1}^n g(a_i)\, f_i \qquad \text{for every } g \in A^*$$

since evaluating the right-hand side at a gives $\sum_i g(a_i)f_i(a) = g\big(\sum_i f_i(a)a_i\big) = g(a)$, using R-linearity of g.

Surjectivity. Given $\alpha \in \mathrm{End}_R(A)$, put $t = \sum_i f_i \otimes \alpha(a_i)$. Then for every a,

$$\theta(t)(a) = \sum_i f_i(a)\, \alpha(a_i) = \alpha\Big(\sum_i f_i(a)a_i\Big) = \alpha(a)$$

so $\theta(t) = \alpha$.

Injectivity. Let $t = \sum_{j=1}^m g_j \otimes b_j$ satisfy $\theta(t) = 0$. Expanding each g_j by the displayed identity and collecting,

$$t = \sum_j \Big(\sum_i g_j(a_i)f_i\Big) \otimes b_j = \sum_i f_i \otimes \Big(\sum_j g_j(a_i)b_j\Big) = \sum_i f_i \otimes \theta(t)(a_i) = 0$$

where the middle step moves the scalars $g_j(a_i) \in R$ across the tensor sign, which is legitimate because the tensor product is over R.

Example 17.2.10: Projective Modules

Throughout, proposition 12.3.36 is what lets Hom_R be computed one summand at a time.

1. If F is a field then every F-module is free (theorem 13.1.8), so every vector space is projective.
2. Any ring R is free of rank one over itself and hence projective. Directly: $\mathrm{Hom}_R(R, M) \cong M$ by $h \mapsto h(1)$, so applying $\mathrm{Hom}_R(R, -)$ to a short exact sequence returns that sequence, which is exact.
3. No nonzero finite abelian group is a projective $\mathbb{Z}$-module. A projective is a direct summand of a free module by theorem 17.2.4(4), hence a submodule of one; a free $\mathbb{Z}$-module has no nonzero element of finite order; and every nonzero element of a nonzero finite group has finite order.

   For $\mathbb{Z}/n\mathbb{Z}$ with $n \geq 2$ the same conclusion is visible in the functor. Applying $\mathrm{Hom}_{\mathbb{Z}}(\mathbb{Z}/n\mathbb{Z}, -)$ to $0 \to \mathbb{Z} \xrightarrow{\cdot n} \mathbb{Z} \xrightarrow{\pi} \mathbb{Z}/n\mathbb{Z} \to 0$ gives

   $$0 \to 0 \to 0 \to \mathbb{Z}/n\mathbb{Z}$$

   since $\mathrm{Hom}_{\mathbb{Z}}(\mathbb{Z}/n\mathbb{Z}, \mathbb{Z}) = 0$ by the argument of example 17.2.2 and $\mathrm{Hom}_{\mathbb{Z}}(\mathbb{Z}/n\mathbb{Z}, \mathbb{Z}/n\mathbb{Z}) \cong \mathbb{Z}/n\mathbb{Z}$. The last map is not surjective.
4. $\mathbb{Q}/\mathbb{Z}$ is not a projective $\mathbb{Z}$-module, being a nonzero torsion module and so not a submodule of a free one. Correspondingly

   $$0 \to \mathbb{Z} \hookrightarrow \mathbb{Q} \to \mathbb{Q}/\mathbb{Z} \to 0$$

   does not split: a section would exhibit $\mathbb{Q}/\mathbb{Z}$ as a submodule of $\mathbb{Q}$, and $\mathbb{Q}$ has no nonzero element of finite order while $\mathbb{Q}/\mathbb{Z}$ is nonzero and torsion.
5. $\mathbb{Q}$ is not a projective $\mathbb{Z}$-module either, and the reason is different: it is torsion-free, so the argument above does not apply. Instead, *every* homomorphism $h\colon \mathbb{Q} \to \mathbb{Z}$ is zero. For if $h(x) \neq 0$ for some x, then for every $n \geq 1$

   $$h(x) = h\Big(n \cdot \frac{x}{n}\Big) = n\, h\Big(\frac{x}{n}\Big)$$

   so n divides $h(x)$ for every $n \geq 1$, forcing $h(x) = 0$.

   Consequently every homomorphism from $\mathbb{Q}$ into a free $\mathbb{Z}$-module $F = \bigoplus_{i \in I} \mathbb{Z}$ is zero, since composing with each coordinate projection gives a homomorphism $\mathbb{Q} \to \mathbb{Z}$. So $\mathbb{Q}$ is not isomorphic to a submodule of a free $\mathbb{Z}$-module, in particular not to a direct summand of one, and theorem 17.2.4(4) fails.
6. Let $0 \to L \xrightarrow{\psi} M \xrightarrow{\varphi} N \to 0$ be short exact with N projective, and write $X^* = \mathrm{Hom}_R(X, R)$. Then

   $$0 \to N^* \xrightarrow{\varphi^*} M^* \xrightarrow{\psi^*} L^* \to 0$$

   is short exact, where $\varphi^*(h) = h \circ \varphi$ and $\psi^*(g) = g \circ \psi$.

   What makes this work is *not* that the modules are projective as arguments of $\mathrm{Hom}_R(-, R)$; projectivity of L, M and N says nothing directly about a contravariant Hom. It is that N projective forces the sequence to split (theorem 17.2.4(3)), and a split sequence is preserved by any additive functor. Explicitly, let μ be a section of φ and λ a retraction of ψ (proposition 17.1.13). Then λ^* is a section of ψ^*, since $\psi^*(\lambda^*(g)) = g \circ \lambda \circ \psi = g$, so ψ^* is surjective; and μ^* is a retraction of φ^*, since $\mu^*(\varphi^*(h)) = h \circ \varphi \circ \mu = h$, so φ^* is injective. Exactness in the middle holds because $\psi^*\varphi^*(h) = h \circ \varphi \circ \psi = 0$, and conversely a $g \in M^*$ with $g \circ \psi = 0$ kills $\psi(L) = \ker(\varphi)$ and so factors through $M/\ker(\varphi) \cong N$ by theorem 12.2.10.

7. If P_1 and P_2 are projective then so is $P_1 \oplus P_2$: writing $F_1 = P_1 \oplus K_1$ and $F_2 = P_2 \oplus K_2$ with F_i free, the module $F_1 \oplus F_2$ is free and contains $P_1 \oplus P_2$ as a direct summand with complement $K_1 \oplus K_2$.

Over one class of rings the gap between projective and free closes entirely.

Proposition 17.2.11: Projective Modules Over a PID

Let R be a principal ideal domain. Then every projective R-module is free. Together with corollary 17.2.6, projective and free coincide over a PID.

***Proof*:**
Let P be projective. By theorem 17.2.4(4) there is a free R-module F and a submodule $P' \subseteq F$ with $P \cong P'$, namely a direct summand. By theorem 14.1.6 the submodule P' is free, and hence so is P.

No finiteness hypothesis is needed, which is exactly what theorem 14.1.6 was proved for; the finite rank case (theorem 14.1.5) would give the statement only for finitely generated P. Example 17.2.7 shows that some hypothesis on R is unavoidable, since $\mathbb{Z}/6\mathbb{Z}$ and $M_n(F)$ are not domains. A second class where the two notions agree is the local rings, where every finitely generated projective module is free; that result belongs with the theory of local rings and is proved in *Commutative Algebra* [10, projective modules over a local ring].

Two different presentations of the same module by projectives differ by no more than a free summand. This is the result that makes an invariant computed from a projective resolution independent of the resolution chosen, and section 17.6 depends on it.

Proposition 17.2.12: Schanuel's Lemma

Let M be an R-module and let

$$0 \to K_1 \hookrightarrow P_1 \xrightarrow{\pi_1} M \to 0 \qquad \text{and} \qquad 0 \to K_2 \hookrightarrow P_2 \xrightarrow{\pi_2} M \to 0$$

be short exact sequences with P_1 and P_2 projective. Then

$$K_1 \oplus P_2 \cong K_2 \oplus P_1$$

***Proof*:**
Let

$$X = \{(p_1, p_2) \in P_1 \oplus P_2 \mid \pi_1(p_1) = \pi_2(p_2)\}$$

which is a submodule of $P_1 \oplus P_2$, being the kernel of the homomorphism $(p_1, p_2) \mapsto \pi_1(p_1) - \pi_2(p_2)$. Let q_1 and q_2 be the two coordinate projections restricted to X.

q_2 is surjective with kernel K_1. Given $p_2 \in P_2$, surjectivity of π_1 produces $p_1 \in P_1$ with $\pi_1(p_1) = \pi_2(p_2)$, and then $(p_1, p_2) \in X$ maps to p_2. Its kernel consists of the pairs of the form $(p_1, 0)$ that lie in X, and $(p_1, 0) \in X$ says $\pi_1(p_1) = \pi_2(0) = 0$, so the kernel is $K_1 \oplus 0$, which is isomorphic to K_1.

So $0 \to K_1 \to X \xrightarrow{q_2} P_2 \to 0$ is short exact, and P_2 is projective, so it splits by theorem 17.2.4(3), giving $X \cong K_1 \oplus P_2$ by the remark following definition 17.1.11. Exchanging the roles of the two sequences, the same argument applied to q_1 gives $X \cong K_2 \oplus P_1$.

Exercise 17.2.1

1. For each pair, decide whether the module is projective over the indicated ring, and justify the answer.
    1. $\mathbb{Z}/3\mathbb{Z}$ over $\mathbb{Z}/6\mathbb{Z}$
    2. $\mathbb{Z}/2\mathbb{Z}$ over $\mathbb{Z}/4\mathbb{Z}$
    3. $2\mathbb{Z}$ over $\mathbb{Z}$
    4. $\mathbb{Z}[x]/(x)$ over $\mathbb{Z}[x]$
    5. $\mathbb{Z}/2\mathbb{Z} \oplus \mathbb{Z}/3\mathbb{Z}$ over $\mathbb{Z}/6\mathbb{Z}$
2. Let $R = \mathbb{Z}/6\mathbb{Z}$ and $P = \overline{3}R$, the projective module of example 17.2.7(1). Exhibit a dual basis for P consisting of a single pair (a_1, f_1), and verify both conditions of proposition 17.2.8 directly. Then do the same for the complement $\overline{4}R$.
3. Show that the evaluation map θ of corollary 17.2.9 fails to be surjective in each of the following cases, and identify which hypothesis fails.
    1. $R = \mathbb{Z}$ and $A = \mathbb{Q}$
    2. $R = \mathbb{Z}$ and $A = \bigoplus_{k=1}^{\infty} \mathbb{Z}$
4. Take $R = \mathbb{Z}$, $P = \mathbb{Z}$, $M = \mathbb{Z}$, $N = \mathbb{Z}/6\mathbb{Z}$ with φ the reduction map, and $f\colon \mathbb{Z} \to \mathbb{Z}/6\mathbb{Z}$ the homomorphism with $f(1) = \overline{5}$. Determine every lift g of f along φ, and show that the set of lifts is a coset of $\mathrm{Hom}_{\mathbb{Z}}(\mathbb{Z}, 6\mathbb{Z})$ inside $\mathrm{Hom}_{\mathbb{Z}}(\mathbb{Z}, \mathbb{Z})$.
5. Let $(P_i)_{i \in I}$ be a family of R-modules. Show that $\bigoplus_{i \in I} P_i$ is projective if and only if every P_i is projective. *Hint*: for one direction, a direct summand of a direct summand is a direct summand.
6. Let $0 \to L \to M \to N \to 0$ be a short exact sequence of R-modules.
    1. Show that if L and N are projective then so is M.
    2. Give an example with M projective and L not projective. *Hint*: a ring of the form $\mathbb{Z}/p^2\mathbb{Z}$ has a non-projective ideal.
7. Call an R-module M *finitely presented* if there is an exact sequence $R^s \xrightarrow{\alpha} R^t \to M \to 0$ for some $s, t \in \mathbb{N}$. Use proposition 17.2.12 to show that if M is finitely presented then *every* surjection $\varphi\colon R^r \to M$ from a finitely generated free module has finitely generated kernel. Deduce that for a finitely generated M the converse holds too, so that finite presentation may be tested on any one such surjection. *Hint*: the kernel of $R^t \to M$ is $\mathrm{im}(\alpha)$, and a direct summand of a finitely generated module is finitely generated.

17.3 Injective Modules

Section 17.2 fixed a module D and asked what $\operatorname{Hom}_R(D,-)$ does to a short exact sequence. The other slot of Hom_R asks the same question with the arrows reversed: fix D and apply $\operatorname{Hom}_R(-,D)$, which turns a homomorphism $M \to N$ into a homomorphism $\operatorname{Hom}_R(N,D) \to \operatorname{Hom}_R(M,D)$ by composition. A short exact sequence $0 \to L \xrightarrow{\psi} M \xrightarrow{\varphi} N \to 0$ therefore produces

$$0 \longrightarrow \operatorname{Hom}_R(N,D) \xrightarrow{\varphi^*} \operatorname{Hom}_R(M,D) \xrightarrow{\psi^*} \operatorname{Hom}_R(L,D) \longrightarrow 0$$

with the outer terms of the original sequence exchanged. The superscript stars record that the assignment reverses arrows, in contrast to the lower stars of section 17.2.

Proposition 17.3.1: Left Exactness of $\operatorname{Hom}_R(-,D)$

Let D be an R-module.

1. For every R-module homomorphism $\varphi\colon M \to N$, the assignment

$$\varphi^*\colon \operatorname{Hom}_R(N,D) \to \operatorname{Hom}_R(M,D), \qquad f \mapsto f \circ \varphi$$

is a homomorphism of abelian groups.

2. If $0 \to L \xrightarrow{\psi} M \xrightarrow{\varphi} N \to 0$ is a short exact sequence of R-modules, then

$$0 \to \operatorname{Hom}_R(N,D) \xrightarrow{\varphi^*} \operatorname{Hom}_R(M,D) \xrightarrow{\psi^*} \operatorname{Hom}_R(L,D)$$

is exact, for every D. In this situation one says that $\operatorname{Hom}_R(-,D)$ is *left exact*, the word being applied to the sequence as displayed, after the reversal.

As with proposition 17.2.1, part (2) does not claim that the last map is surjective. The two propositions fail at opposite ends of the original sequence: $\operatorname{Hom}_R(D,-)$ can miss homomorphisms into the *right* hand term N, and $\operatorname{Hom}_R(-,D)$ can miss homomorphisms out of the *left* hand term L.

***Proof*:**
(1) For $f, g \in \operatorname{Hom}_R(N,D)$ and $m \in M$,

$$\big((f+g)\circ\varphi\big)(m) = f(\varphi(m)) + g(\varphi(m)) = \big(f\circ\varphi + g\circ\varphi\big)(m)$$

so $\varphi^*(f+g) = \varphi^*(f) + \varphi^*(g)$.

(2) *Exactness at* $\operatorname{Hom}_R(N,D)$, which is the injectivity of φ^*. Suppose $f \circ \varphi = 0$. Every $n \in N$ is $\varphi(m)$ for some m, since φ is surjective, so $f(n) = f(\varphi(m)) = 0$ and $f = 0$.

Exactness at $\operatorname{Hom}_R(M,D)$, that is $\operatorname{im}(\varphi^*) = \ker(\psi^*)$.

($\subseteq$) For $f \in \operatorname{Hom}_R(N,D)$, $\psi^*(\varphi^*(f)) = f \circ \varphi \circ \psi = 0$, since $\varphi \circ \psi = 0$ by $\operatorname{im}(\psi) = \ker(\varphi)$.

($\supseteq$) Let $g \in \operatorname{Hom}_R(M,D)$ with $g \circ \psi = 0$, so g vanishes on $\operatorname{im}(\psi) = \ker(\varphi)$. Let $\pi\colon M \to M/\ker(\varphi)$ be the quotient map. By theorem 12.2.10 the map g factors as $g = \overline{g} \circ \pi$ for a unique $\overline{g}\colon M/\ker(\varphi) \to D$, and φ induces an isomorphism $\overline{\varphi}\colon M/\ker(\varphi) \to N$ with $\overline{\varphi} \circ \pi = \varphi$. Put $f := \overline{g} \circ \overline{\varphi}^{-1} \in \operatorname{Hom}_R(N,D)$. Then

$$\varphi^*(f) = f \circ \varphi = \overline{g} \circ \overline{\varphi}^{-1} \circ \overline{\varphi} \circ \pi = \overline{g} \circ \pi = g$$

so $g \in \operatorname{im}(\varphi^*)$.

The failure at the other end is visible in the same sequence that example 17.2.2 used, with the roles of the two slots exchanged.

Example 17.3.2: $\operatorname{Hom}_R(-, D)$ Need Not Be Right Exact

Take $R = \mathbb{Z}$, the short exact sequence

$$0 \to \mathbb{Z} \xrightarrow{\cdot 2} \mathbb{Z} \xrightarrow{\pi} \mathbb{Z}/2\mathbb{Z} \to 0$$

and $D = \mathbb{Z}/2\mathbb{Z}$. A homomorphism out of $\mathbb{Z}$ is determined by the image of 1, so $\operatorname{Hom}_{\mathbb{Z}}(\mathbb{Z}, \mathbb{Z}/2\mathbb{Z}) \cong \mathbb{Z}/2\mathbb{Z}$, and likewise $\operatorname{Hom}_{\mathbb{Z}}(\mathbb{Z}/2\mathbb{Z}, \mathbb{Z}/2\mathbb{Z}) \cong \mathbb{Z}/2\mathbb{Z}$. Applying $\operatorname{Hom}_{\mathbb{Z}}(-, D)$ gives

$$0 \to \mathbb{Z}/2\mathbb{Z} \xrightarrow{\pi^*} \mathbb{Z}/2\mathbb{Z} \xrightarrow{(\cdot 2)^*} \mathbb{Z}/2\mathbb{Z}$$

Here $\pi^*(\mathrm{id}_{\mathbb{Z}/2\mathbb{Z}}) = \pi \neq 0$, so π^* is injective and therefore, both groups having two elements, bijective. And $(\cdot 2)^*$ is the zero map: for $g \in \operatorname{Hom}_{\mathbb{Z}}(\mathbb{Z}, \mathbb{Z}/2\mathbb{Z})$ and $x \in \mathbb{Z}$,

$$\big(g \circ (\cdot 2)\big)(x) = g(2x) = 2\,g(x) = 0$$

because every element of $\mathbb{Z}/2\mathbb{Z}$ is killed by 2. The displayed sequence is thus exact, as proposition 17.3.1 promised, and it cannot be closed off with a $\to 0$ on the right, since $(\cdot 2)^*$ has image 0 inside a group of order two.

In the language of diagrams, the homomorphism not in the image of $(\cdot 2)^*$ is π itself, and saying it is not in the image says that π does not *extend* along the injection $\cdot 2$:

$$\begin{tikzcd}
0 \arrow[r] & \mathbb{Z} \arrow[r, hook, "\cdot 2"] \arrow[d, "\pi"'] & \mathbb{Z} \arrow[dl, dashed, "F\,?"] \\
& \mathbb{Z}/2\mathbb{Z} &
\end{tikzcd}$$

Any such F would satisfy $F(2x) = 2F(x) = 0$ while $\pi(x) \neq 0$ for odd x.

The contrast with section 17.2 is worth naming precisely, because the two words are easy to swap. A projective module is one out of which every homomorphism *lifts* along a surjection. An injective module is one into which every homomorphism *extends* along an injection.

Definition 17.3.3: Injective Module

An R-module Q is *injective* if the functor $\operatorname{Hom}_R(-, Q)$ is exact, that is if for every short exact sequence $0 \to L \to M \to N \to 0$ of R-modules the sequence

$$0 \to \operatorname{Hom}_R(N, Q) \to \operatorname{Hom}_R(M, Q) \to \operatorname{Hom}_R(L, Q) \to 0$$

is again short exact.

Theorem 17.3.4: Injective Modules Equivalences

Let Q be an R-module. The following are equivalent.

1. Q is injective.
2. *(Extension.)* For all R-modules L and M, if $\psi\colon L \to M$ is injective, then every $f \in \operatorname{Hom}_R(L, Q)$ has an extension $F \in \operatorname{Hom}_R(M, Q)$ with $F \circ \psi = f$:

$$\begin{tikzcd} 0 \arrow[r] & L \arrow[r, hook, "\psi"] \arrow[d, "f"'] & M \arrow[dl, dashed, "F"] \\ & Q & \end{tikzcd}$$

3. *(Splitting.)* Every short exact sequence $0 \to Q \to M \to N \to 0$ of R-modules splits. Equivalently, whenever Q is a submodule of an R-module M, it is a direct summand of M.

***Proof*:**

$(1) \Rightarrow (2)$. Let $\psi\colon L \to M$ be injective and put $N = M/\psi(L)$, so that $0 \to L \xrightarrow{\psi} M \to N \to 0$ is short exact by proposition 17.1.1 and theorem 12.2.10. Condition (1) makes $\psi^*\colon \operatorname{Hom}_R(M, Q) \to \operatorname{Hom}_R(L, Q)$ surjective, so a given f is $\psi^*(F) = F \circ \psi$ for some F.

$(2) \Rightarrow (1)$. Proposition 17.3.1 gives exactness at the first two spots of the sequence in definition 17.3.3, and (2) applied to the injection ψ says precisely that ψ^* is surjective.

$(2) \Rightarrow (3)$. Let $0 \to Q \xrightarrow{\psi} M \xrightarrow{\varphi} N \to 0$ be short exact, so ψ is injective. Applying (2) to $f = \mathrm{id}_Q$ gives $\lambda \in \operatorname{Hom}_R(M, Q)$ with $\lambda \circ \psi = \mathrm{id}_Q$, a retraction of ψ, so the sequence splits by proposition 17.1.13.

$(3) \Rightarrow (2)$. Let $\psi\colon L \to M$ be injective and $f \in \operatorname{Hom}_R(L, Q)$. The construction that follows glues Q and M along L. Let

$$W = \big\{\big(f(l), -\psi(l)\big) \bigm| l \in L\big\} \subseteq Q \oplus M$$

which is a submodule, being the image of the homomorphism $l \mapsto (f(l), -\psi(l))$, and set $X = (Q \oplus M)/W$. Write $\iota\colon Q \to X$ for $q \mapsto \overline{(q, 0)}$ and $j\colon M \to X$ for $m \mapsto \overline{(0, m)}$, both homomorphisms.

The map ι is injective: if $(q, 0) \in W$ then $q = f(l)$ and $\psi(l) = 0$ for some l, and ψ is injective, so $l = 0$ and $q = 0$. Thus $\iota(Q)$ is a submodule of X isomorphic to Q, and by (3) it is a direct summand, say $X = \iota(Q) \oplus C$. Let $\rho\colon X \to Q$ be the projection onto the first summand followed by ι^{-1}, so that $\rho \circ \iota = \mathrm{id}_Q$.

Put $F := \rho \circ j\colon M \to Q$. For $l \in L$ the element $(f(l), -\psi(l))$ lies in W, so $\overline{(0, \psi(l))} = \overline{(f(l), 0)}$ in X, that is $j(\psi(l)) = \iota(f(l))$. Hence

$$F(\psi(l)) = \rho\big(j(\psi(l))\big) = \rho\big(\iota(f(l))\big) = f(l)$$

so $F \circ \psi = f$.

The name records condition (2): the property concerns extending along injections, exactly as *projective* records lifting along surjections. Condition (3) is the concrete reading: an injective module cannot be enlarged without splitting off, so it is as far from the inside of a bigger module as a submodule can be.

For projective modules theorem 17.2.4 offered a fourth description, as a direct summand of a free module, which reduces every question to free modules. Nothing that simple is available here, and the classification of injective modules over a general ring is genuinely harder. What replaces it is a test that only has to be run on the ideals of R.

Theorem 17.3.5: Baer's Criterion

Let R be a ring and Q an R-module.

1. Q is injective if and only if, for every left ideal I of R, every R-module homomorphism $g\colon I \to Q$ extends to an R-module homomorphism $G\colon R \to Q$.
2. Suppose R is a principal ideal domain. Then Q is injective if and only if $rQ = Q$ for every *nonzero* $r \in R$.

The hypothesis $r \neq 0$ in (2) is not cosmetic: $0 \cdot Q = 0$, so without it the condition would fail for every nonzero module.

***Proof*:**
(1) ($\Rightarrow$) The inclusion $I \hookrightarrow R$ is injective, so theorem 17.3.4(2) extends any $g\colon I \to Q$ to $G\colon R \to Q$.

($\Leftarrow$) Assume the extension property for ideals. It suffices to verify theorem 17.3.4(2), so let $\psi\colon L \to M$ be injective and $f \in \mathrm{Hom}_R(L, Q)$. Replacing L by $\psi(L)$ and f by $f \circ (\psi|^{\psi(L)})^{-1}$, we may assume L is a submodule of M and ψ the inclusion.

Step 1: a poset of partial extensions. Let

$$\mathcal{L} = \{(L', f') \mid L \subseteq L' \subseteq M,\ f' \in \mathrm{Hom}_R(L', Q),\ f'|_L = f\}$$

ordered by $(L', f') \leq (L'', f'')$ when $L' \subseteq L''$ and $f''|_{L'} = f'$. This is a partial order, and $\mathcal{L}$ is nonempty because $(L, f) \in \mathcal{L}$.

Step 2: chains have upper bounds. Let $\{(L_i, f_i)\}_{i \in \mathcal{I}}$ be a chain. The union $\mathfrak{L} = \bigcup_i L_i$ is a submodule of M, since any two of its elements lie in a common L_i by totality of the chain. Define $\mathfrak{f}\colon \mathfrak{L} \to Q$ by $\mathfrak{f}(x) = f_i(x)$ for any i with $x \in L_i$; this is unambiguous because the chain is compatible, and it is a homomorphism because any two elements lie in a common L_i. Then $(\mathfrak{L}, \mathfrak{f}) \in \mathcal{L}$ bounds the chain.

Step 3: a maximal element. By Zorn's lemma (theorem A.2.3) there is a maximal $(M', F) \in \mathcal{L}$.

Step 4: $M' = M$. Suppose not, and choose $m \in M$ with $m \notin M'$. Set

$$I = \{r \in R \mid rm \in M'\}$$

which is a left ideal of R: it contains 0; it is closed under addition because M' is; and if $r \in I$ and $s \in R$ then $(sr)m = s(rm) \in M'$. Define $g\colon I \to Q$ by $g(r) = F(rm)$, which is a homomorphism since F is and since $r \mapsto rm$ is. By hypothesis g extends to $G\colon R \to Q$.

Now let $M'' = M' + Rm$, a submodule of M strictly containing M', and define

$$F' \colon M'' \to Q, \qquad F'(m' + rm) = F(m') + G(r)$$

This is well defined. If $m_1' + r_1 m = m_2' + r_2 m$ then $(r_1 - r_2)m = m_2' - m_1' \in M'$, so $r_1 - r_2 \in I$ and therefore

$$G(r_1) - G(r_2) = G(r_1 - r_2) = g(r_1 - r_2) = F\big((r_1 - r_2)m\big) = F(m_2') - F(m_1')$$

which rearranges to $F(m_1') + G(r_1) = F(m_2') + G(r_2)$. That F' is a homomorphism is immediate from the formula, and taking $r = 0$ gives $F'|_{M'} = F$, so in particular $F'|_L = f$ and $(M'', F') \in \mathcal{L}$.

Since $m \in M''$ and $m \notin M'$, the pair (M'', F') is strictly larger than (M', F), contradicting maximality. Hence $M' = M$ and F is the required extension.

(2) ($\Rightarrow$) Let $x \in Q$ and $r \in R$ with $r \neq 0$. Define $g \colon (r) \to Q$ by $g(sr) = sx$. This is well defined because R is an integral domain: if $sr = s'r$ with $r \neq 0$ then $s = s'$. It is a homomorphism, and by part (1) it extends to $G \colon R \to Q$. Then

$$x = g(r) = G(r) = r\,G(1)$$

so $x \in rQ$. As x was arbitrary, $rQ = Q$. Only the domain hypothesis was used here, not that ideals are principal, so this direction holds over any integral domain.

($\Leftarrow$) Assume $rQ = Q$ for every nonzero r, and verify the criterion of part (1). Let I be an ideal of R and $g \colon I \to Q$ a homomorphism. Since R is a principal ideal domain, $I = (r)$ for some r. If $r = 0$ then $I = 0$ and $G = 0$ extends g. If $r \neq 0$, then $g(r) \in Q = rQ$, so there is a $y \in Q$ with $ry = g(r)$. Define $G \colon R \to Q$ by $G(s) = sy$, a homomorphism. For any element sr of I,

$$G(sr) = sry = s\,g(r) = g(sr)$$

so $G|_I = g$. By part (1), Q is injective.

Corollary 17.3.6: Quotients of Injectives Over a PID

Let R be a principal ideal domain, Q an injective R-module and $N \subseteq Q$ a submodule. Then Q/N is injective.

***Proof*:**
For $r \neq 0$ we have $rQ = Q$ by theorem 17.3.5(2), hence

$$r(Q/N) = (rQ + N)/N = (Q + N)/N = Q/N$$

and theorem 17.3.5(2) applies again in the other direction.

The condition appearing in theorem 17.3.5(2) is worth a name of its own, both because it is the only practical test for injectivity over $\mathbb{Z}$ and because it is the source of every concrete injective module in this chapter.

Definition 17.3.7: Divisible Group

Let R be an integral domain. An R-module M is *divisible* if $rM = M$ for every nonzero $r \in R$; equivalently, if for every $x \in M$ and every nonzero $r \in R$ there is a $y \in M$ with $ry = x$. For $R = \mathbb{Z}$ this says that an abelian group G is divisible when every $x \in G$ is n times some element of G, for every $n \geq 1$. Such a G is called a *divisible group.*

In this language, theorem 17.3.5(2) reads: over a principal ideal domain, injective and divisible are the same condition. Over an integral domain that is not principal only half survives, namely that injective modules are divisible; the reverse implication used principality when it wrote $I = (r)$, and it genuinely fails in general.

Example 17.3.8: Divisible Groups

1. $\mathbb{Q}$ is divisible: given $\frac{a}{b} \in \mathbb{Q}$ and $n \geq 1$, the rational $\frac{a}{nb}$ satisfies $n \cdot \frac{a}{nb} = \frac{a}{b}$. So $\mathbb{Q}$ is an injective $\mathbb{Z}$-module. More generally the field of fractions of any integral domain R is a divisible R-module.

2. Any quotient of a divisible module is divisible, since $r(M/N) = (rM + N)/N = M/N$. In particular $\mathbb{Q}/\mathbb{Z}$ is divisible, hence an injective $\mathbb{Z}$-module. Note that $\mathbb{Q}/\mathbb{Z}$ is a torsion group, so divisibility does not force torsion-freeness; and in a torsion group the element y with $ry = x$ is not unique.

3. Fix a prime p and let
$$\mathbb{Z}_{p^\infty} = \{[x] \in \mathbb{Q}/\mathbb{Z} \mid p^n[x] = 0 \text{ for some } n \geq 0\} = \left\{ \left[\tfrac{a}{p^n}\right] \in \mathbb{Q}/\mathbb{Z} \;\middle|\; a \in \mathbb{Z},\ n \geq 0 \right\}$$
the p-primary part of $\mathbb{Q}/\mathbb{Z}$, called the *Prüfer p-group*[a]. The two descriptions agree: $p^n \left[\frac{a}{p^n}\right] = [a] = 0$ gives one inclusion, and if $p^n[x] = 0$ with $[x] = \left[\frac{c}{d}\right]$ in lowest terms then $d \big| p^n$, so $[x]$ has the displayed form. It is a subgroup, being the set of elements killed by some power of p.
It is divisible. Let $[x] = \left[\frac{a}{p^n}\right]$ and let $m \geq 1$, and write $m = p^k u$ with u coprime to p. Since u is invertible modulo p^{n+k}, choose v with $uv \equiv 1 \pmod{p^{n+k}}$, say $uv = 1 + p^{n+k}w$, and set $y = \left[\frac{av}{p^{n+k}}\right] \in \mathbb{Z}_{p^\infty}$. Then
$$my = \left[\frac{p^k uav}{p^{n+k}}\right] = \left[\frac{a(1+p^{n+k}w)}{p^n}\right] = \left[\frac{a}{p^n} + ap^k w\right] = [x]$$
since $ap^k w \in \mathbb{Z}$. So $\mathbb{Z}_{p^\infty}$ is an injective $\mathbb{Z}$-module.

4. A direct product $\prod_i M_i$ of divisible modules is divisible, since the required y may be chosen coordinatewise, and so is a direct sum. Conversely if $M \oplus N$ is divisible then so are M and N: given $x \in M$ and $r \neq 0$, some (y, z) has $r(y, z) = (x, 0)$, and then $ry = x$.

5. No nonzero finite abelian group is divisible: if $|G| = m$ then $mg = 0$ for every g, so $mG = 0 \neq G$. Combined with theorem 5.1.8, no nonzero finitely generated abelian group is divisible, since a free summand $\mathbb{Z}$ is not divisible either.

[a] It is also $\varinjlim \mathbb{Z}/p^k\mathbb{Z}$ along the inclusions.

Before collecting examples of injective modules, one closure property is needed, and it is the one that fails to be symmetric with the projective case.

Proposition 17.3.9: Summands and Products of Injectives

Let R be a ring.

1. A direct summand of an injective R-module is injective.
2. A direct product $\prod_{i\in I} Q_i$ of R-modules is injective if and only if every Q_i is injective. In particular a finite direct sum of injective modules is injective.

***Proof*:**
(1) Let $Q = E \oplus C$ with Q injective, and let $\iota\colon E \to Q$ and $\rho\colon Q \to E$ be the inclusion and projection, so $\rho \circ \iota = \mathrm{id}_E$. Given an injection $\psi\colon L \to M$ and $f \in \mathrm{Hom}_R(L, E)$, the map $\iota \circ f$ extends to some $H \in \mathrm{Hom}_R(M, Q)$ by theorem 17.3.4(2). Then $F := \rho \circ H$ satisfies $F \circ \psi = \rho \circ \iota \circ f = f$, so theorem 17.3.4(2) holds for E.

(2) ($\Leftarrow$) Let every Q_i be injective, let $\psi\colon L \to M$ be injective and let $f \in \mathrm{Hom}_R(L, \prod_i Q_i)$. Each component $\pi_i \circ f$ extends to $F_i \in \mathrm{Hom}_R(M, Q_i)$, and $F := (F_i)_i$ satisfies $\pi_i \circ F \circ \psi = \pi_i \circ f$ for every i, hence $F \circ \psi = f$.

($\Rightarrow$) Each Q_j is a direct summand of $\prod_i Q_i$, with complement $\prod_{i\neq j} Q_i$, so part (1) applies.

Example 17.3.10: Injective Modules

1. $\mathbb{Z}$ is not an injective $\mathbb{Z}$-module: it is not divisible, since $2y = 1$ has no solution in $\mathbb{Z}$. Equivalently, the sequence $0 \to \mathbb{Z} \xrightarrow{\cdot 2} \mathbb{Z} \to \mathbb{Z}/2\mathbb{Z} \to 0$ does not split (example 17.1.14(2)), so theorem 17.3.4(3) fails. Since $\mathbb{Z}$ is free, free does not imply injective, in contrast with corollary 17.2.6.
2. $\mathbb{Q}$, $\mathbb{Q}/\mathbb{Z}$ and $\mathbb{Z}_{p^\infty}$ are injective $\mathbb{Z}$-modules, by example 17.3.8 and theorem 17.3.5(2). So is $\mathbb{Q} \oplus \mathbb{Q}/\mathbb{Z}$, by proposition 17.3.9(2).
3. No nonzero finitely generated $\mathbb{Z}$-module is injective, by example 17.3.8(5).
4. If $R = F$ is a field then every F-module is injective. Indeed every F-module is free (theorem 13.1.8), and any short exact sequence of F-modules $0 \to Q \to M \to N \to 0$ has N free, hence projective, hence the sequence splits by theorem 17.2.4(3); theorem 17.3.4(3) follows. Over a field, projective, free and injective all coincide.
5. $\mathbb{Z}/2\mathbb{Z}$ is not an injective $\mathbb{Z}/4\mathbb{Z}$-module. It occurs as the ideal $(\overline{2}) \subseteq \mathbb{Z}/4\mathbb{Z}$, and $\mathbb{Z}/4\mathbb{Z}$ is indecomposable as a module over itself, since its only proper nonzero submodule is $(\overline{2})$ and a complement would have to be a second one. So $(\overline{2})$ is not a direct summand and theorem 17.3.4(3) fails. Compare exercise set 17.2.1, question 1: the same module is not projective either.

17.3.1 Embedding a Module in an Injective

Example 17.1.4(3) says every module is a quotient of a free, hence of a projective, module. The dual statement is that every module is a submodule of an injective one. It is true, but it is not formal: there is no dual of a free module to hand, and the proof goes through $\mathbb{Z}$ and back.

Lemma 17.3.11: $\mathbb{Z}$-module Submodules of Injective Modules

Every $\mathbb{Z}$-module is a submodule of an injective $\mathbb{Z}$-module.

***Proof*:**
Let M be a $\mathbb{Z}$-module and A a set of generators, so that theorem 12.3.11 gives a surjection $\pi\colon \bigoplus_{a\in A}\mathbb{Z} \to M$ from the free $\mathbb{Z}$-module on A. Writing $K = \ker(\pi)$, theorem 12.2.10 gives $M \cong \bigoplus_{a\in A}\mathbb{Z}/K$.

Now $\bigoplus_{a\in A}\mathbb{Z}$ sits inside $V := \bigoplus_{a\in A}\mathbb{Q}$, the direct sum of copies of $\mathbb{Q}$ indexed by A, which is divisible by example 17.3.8(1) and (4), hence injective. Since $K \subseteq \bigoplus_{a\in A}\mathbb{Z} \subseteq V$, the quotient V/K is defined, and it is divisible by example 17.3.8(2), hence injective. Finally

$$M \cong \bigoplus_{a\in A}\mathbb{Z}/K \subseteq V/K$$

exhibits M as a submodule of an injective $\mathbb{Z}$-module.

Passing from $\mathbb{Z}$ to an arbitrary ring is done by a change of rings, and the mechanism is the adjunction of theorem 15.3.9. The proof below writes the correspondence out explicitly rather than invoking naturality.

Lemma 17.3.12: Inducing Injective R-modules

Let $f\colon S \to R$ be a homomorphism of commutative rings and let Q be an injective S-module. Regard R as an S-module through f, and give

$$\mathrm{Hom}_S(R, Q)$$

the R-module structure $(r\cdot h)(x) := h(xr)$. Then $\mathrm{Hom}_S(R, Q)$ is an injective R-module.

***Proof*:**
Write $f_*(N)$ for an R-module N regarded as an S-module through f, so that $s \cdot n = f(s)n$.

Step 1: the correspondence. For an R-module N define

$$\alpha \mapsto \widetilde{\alpha}, \quad \widetilde{\alpha}(n) = \alpha(n)(1) \qquad \text{and} \qquad \beta \mapsto \widehat{\beta}, \quad \widehat{\beta}(n)(r) = \beta(rn)$$

between $\mathrm{Hom}_R\big(N, \mathrm{Hom}_S(R, Q)\big)$ and $\mathrm{Hom}_S\big(f_*(N), Q\big)$. These are the two directions of theorem 15.3.9 applied with N in the role of A, the ring R in the role of the bimodule B and Q in the role of C, using $N \otimes_R R \cong N$; the verifications are short enough to give directly.

That $\widetilde{\alpha}$ is S-linear: $\alpha(n) \in \mathrm{Hom}_S(R, Q)$ is S-linear on R, whose S-action is $s \cdot x = f(s)x$, so

$\alpha(n)(f(s)) = s\,\alpha(n)(1)$; combining with R-linearity of α,

$$\widetilde{\alpha}(s \cdot n) = \alpha(f(s)n)(1) = \big(f(s) \cdot \alpha(n)\big)(1) = \alpha(n)(f(s)) = s\,\widetilde{\alpha}(n)$$

That $\widehat{\beta}(n)$ is S-linear: $\widehat{\beta}(n)(s \cdot r) = \beta(f(s)rn) = \beta(s \cdot (rn)) = s\,\widehat{\beta}(n)(r)$. That $\widehat{\beta}$ is R-linear: $\widehat{\beta}(r'n)(r) = \beta(rr'n) = \widehat{\beta}(n)(rr') = \big(r' \cdot \widehat{\beta}(n)\big)(r)$.

The two assignments are mutually inverse:

$$\widehat{\widetilde{\alpha}}(n)(r) = \widetilde{\alpha}(rn) = \alpha(rn)(1) = \big(r \cdot \alpha(n)\big)(1) = \alpha(n)(r), \qquad \widetilde{\widehat{\beta}}(n) = \widehat{\beta}(n)(1) = \beta(n)$$

and both are visibly compatible with restriction along a submodule inclusion $L \subseteq N$, in the sense that $\widetilde{\alpha|_L} = \widetilde{\alpha}|_L$.

Step 2: the extension property. Verify theorem 17.3.4(2) for $\mathrm{Hom}_S(R, Q)$. Let $L \subseteq M$ be R-modules and $\alpha \in \mathrm{Hom}_R(L, \mathrm{Hom}_S(R, Q))$. Then $\widetilde{\alpha} \in \mathrm{Hom}_S(f_*(L), Q)$, and $f_*(L) \subseteq f_*(M)$ is still an inclusion of S-modules, since restriction of scalars changes only the action. As Q is an injective S-module, theorem 17.3.4(2) gives $\beta \in \mathrm{Hom}_S(f_*(M), Q)$ with $\beta|_L = \widetilde{\alpha}$.

Put $A := \widehat{\beta} \in \mathrm{Hom}_R(M, \mathrm{Hom}_S(R, Q))$. Then

$$A|_L = \widehat{\beta|_L} = \widehat{\widetilde{\alpha}} = \alpha$$

so A extends α, and $\mathrm{Hom}_S(R, Q)$ is injective.

Corollary 17.3.13: Inducing Injectives Using Divisible Groups

Let R be a commutative ring and D a divisible abelian group. Then $\mathrm{Hom}_{\mathbb{Z}}(R, D)$ is an injective R-module.

***Proof*:**
The group D is an injective $\mathbb{Z}$-module by theorem 17.3.5(2), and there is a unique ring homomorphism $\mathbb{Z} \to R$, so lemma 17.3.12 applies with $S = \mathbb{Z}$.

Lemma 17.3.14: R-module Submodules of Injective Modules

Let R be a commutative ring and M an R-module. Then M is a submodule of an injective R-module.

***Proof*:**
By lemma 17.3.11 there is an injective $\mathbb{Z}$-module D with $M \subseteq D$ as $\mathbb{Z}$-modules; by theorem 17.3.5(2) the group D is divisible. Now compose three injections of R-modules:

$$M \xrightarrow{\sim} \mathrm{Hom}_R(R, M) \hookrightarrow \mathrm{Hom}_{\mathbb{Z}}(R, M) \hookrightarrow \mathrm{Hom}_{\mathbb{Z}}(R, D)$$

The first sends m to the map $x \mapsto xm$, and is an isomorphism of R-modules for the action $(r \cdot h)(x) = h(xr)$, its inverse being $h \mapsto h(1)$. The second is the inclusion of the R-linear maps among the additive ones, which respects that action. The third is composition with $M \subseteq D$, injective because a map into M is zero exactly when it is zero into D. By corollary 17.3.13 the target $\mathrm{Hom}_{\mathbb{Z}}(R, D)$ is an injective R-module.

Definition 17.3.15: Injective Resolution

Let R be a ring and M an R-module. An *injective resolution* of M is an exact chain complex (definition 12.2.11)

$$0 \longrightarrow M \longrightarrow Q^0 \longrightarrow Q^1 \longrightarrow Q^2 \longrightarrow \cdots$$

in which every Q^i is injective. It has *finite length* k if $Q^i = 0$ for every $i > k$.

Corollary 17.3.16: Existence of Injective Resolutions

Every module over a commutative ring admits an injective resolution.

***Proof*:**
Embed M in an injective Q^0 by lemma 17.3.14, embed the quotient Q^0/M in an injective Q^1, embed $Q^1/(Q^0/M)$ in an injective Q^2, and continue. The composites $Q^i \to Q^i/(\text{previous image}) \hookrightarrow Q^{i+1}$ have image the kernel of the next map by construction, which is exactness.

17.3.2 Injective Hulls

Every module embeds in an injective one, but lemma 17.3.14 produces a wasteful embedding: nothing stops Q from being far larger than M. Asking for the smallest such Q turns out to have a clean answer, and the condition that pins it down is that M should meet every nonzero piece of Q.

Definition 17.3.17: Essential Extension

Let $M \subseteq E$ be R-modules. Then M is *essential* in E, and E is an *essential extension* of M, if

$$N \cap M \neq 0 \qquad \text{for every nonzero submodule } N \subseteq E$$

Equivalently, for every $x \in E$ with $x \neq 0$ there is an $r \in R$ with $rx \in M$ and $rx \neq 0$.

The two formulations agree because Rx is a nonzero submodule whenever $x \neq 0$, and any nonzero N contains such an Rx.

Lemma 17.3.18: Essential Extensions Compose

Let $M \subseteq E \subseteq F$ be R-modules with M essential in E and E essential in F. Then M is essential in F.

***Proof*:**
Let $N \subseteq F$ be a nonzero submodule. Then $N \cap E \neq 0$ because E is essential in F, and $N \cap E$ is a nonzero submodule of E, so $(N \cap E) \cap M \neq 0$ because M is essential in E. That intersection is contained in $N \cap M$.

Theorem 17.3.19: Universal Property of Injective Hulls

Let R be a commutative ring and M an R-module.

1. There is an injective R-module E containing M as an essential submodule. Such an E is called an *injective hull*, or *injective envelope*, of M, and is written $E(M)$.
2. If E and E' are both injective hulls of M, there is an isomorphism $E \to E'$ restricting to the identity on M.
3. If $M \subseteq Q$ with Q injective, there is an injective homomorphism $E(M) \to Q$ restricting to the identity on M, and its image is a direct summand of Q. So $E(M)$ is the smallest injective module containing M.

***Proof*:**
(1) By lemma 17.3.14 choose an injective Q with $M \subseteq Q$.

Step 1: a maximal essential extension inside Q. Let

$$\Sigma = \{E \mid M \subseteq E \subseteq Q \text{ and } M \text{ is essential in } E\}$$

partially ordered by inclusion. It contains M, so it is nonempty. A chain $\{E_i\}$ in Σ has upper bound $\bigcup_i E_i$, which is a submodule of Q containing M, and M is essential in it: a nonzero x in the union lies in some E_i, and $Rx \cap M \neq 0$ there. By Zorn's lemma (theorem A.2.3) there is a maximal $E \in \Sigma$.

Step 2: a maximal complement. Let $\mathcal{C}$ be the set of submodules $C \subseteq Q$ with $C \cap E = 0$, ordered by inclusion. It contains 0, and a chain's union again meets E trivially, so Zorn's lemma gives a maximal $C \in \mathcal{C}$.

Step 3: E is essential in Q/C. First, $E \cap C = 0$, so E maps injectively into Q/C. Let X/C be a nonzero submodule of Q/C, so $C \subsetneq X \subseteq Q$. By maximality of C we have $X \cap E \neq 0$; choose $0 \neq e \in X \cap E$. Then $e \notin C$, since $E \cap C = 0$, so the class $e + C$ is a nonzero element of X/C lying in the image of E.

Step 4: E is a direct summand of Q. The sum $E + C$ is direct, so there is a projection $p\colon E \oplus C \to E \subseteq Q$. Since $E \oplus C$ is a submodule of Q and Q is injective, theorem 17.3.4(2) extends p to a homomorphism $g\colon Q \to Q$ with $g|_E = \mathrm{id}_E$ and $g(C) = 0$.

Then $\ker(g) \cap E = 0$, because g is the identity on E, and $C \subseteq \ker(g)$; so maximality of C forces $C = \ker(g)$. Hence g induces an isomorphism $Q/C \to \mathrm{im}(g)$, carrying the image of E onto E itself. By Step 3, E is essential in $\mathrm{im}(g)$, and M is essential in E, so M is essential in $\mathrm{im}(g)$ by lemma 17.3.18. Thus $\mathrm{im}(g) \in \Sigma$ and $E \subseteq \mathrm{im}(g)$, and maximality of E gives $E = \mathrm{im}(g)$.

Now $g\colon Q \to Q$ has image E and restricts to the identity on E, so $g \circ g = g$. An idempotent endomorphism splits its domain: every q is $g(q) + (q - g(q))$ with the first term in $\mathrm{im}(g)$ and the second in $\ker(g)$, and an element of $\mathrm{im}(g) \cap \ker(g)$ is $g(y)$ with $0 = g(g(y)) = g(y)$. Therefore $Q = \mathrm{im}(g) \oplus \ker(g) = E \oplus C$.

Step 5. By proposition 17.3.9(1) the summand E of the injective module Q is injective, and M is essential in E by construction.

(2) Let E and E' be injective hulls of M. Since E' is injective and $M \subseteq E$, theorem 17.3.4(2) extends the inclusion $M \hookrightarrow E'$ to a homomorphism $h\colon E \to E'$. Then $\ker(h) \cap M = 0$, because

h is injective on M, and M is essential in E, so $\ker(h) = 0$ and h is injective.

The image $h(E) \cong E$ is injective, so $h(E)$ is a direct summand of E' by theorem 17.3.4(3), say $E' = h(E) \oplus X$. Since $M = h(M) \subseteq h(E)$, the submodule X meets M trivially, and M is essential in E', so $X = 0$. Hence h is an isomorphism, and it is the identity on M by construction.

(3) Let $M \subseteq Q$ with Q injective. Extending the inclusion $M \hookrightarrow Q$ along $M \subseteq E(M)$ gives $h\colon E(M) \to Q$ with $h|_M$ the inclusion. As in (2), $\ker(h) \cap M = 0$ and M essential in $E(M)$ force h to be injective, and $h(E(M)) \cong E(M)$ is injective, hence a direct summand of Q by theorem 17.3.4(3).

Combining (1) and (3): every injective module containing M contains a copy of $E(M)$ over M, and $E(M)$ is injective, so $E(M)$ is a smallest one. Exercise set 17.3.1 below computes $E(\mathbb{Z}) = \mathbb{Q}$ and $E(\mathbb{Z}/p\mathbb{Z}) = \mathbb{Z}_{p^\infty}$.

One consequence of injectivity is worth recording before leaving the section, because it is what makes duality work for abelian groups at all. Section 13.3 defined the dual of a vector space as $V^* = \operatorname{Hom}_k(V, k)$ and found $V \cong V^{**}$ naturally in finite dimensions. Over $\mathbb{Z}$ the same recipe collapses: $\operatorname{Hom}_{\mathbb{Z}}(\mathbb{Z}/n\mathbb{Z}, \mathbb{Z}) = 0$ for every $n \geq 2$, by the argument of example 17.2.2, so dualizing against $\mathbb{Z}$ destroys every torsion group. Replacing $\mathbb{Z}$ by the injective module $\mathbb{Q}/\mathbb{Z}$ repairs it, since a homomorphism $\mathbb{Z}/n\mathbb{Z} \to \mathbb{Q}/\mathbb{Z}$ sends $\overline{1}$ to an element killed by n, and those form the cyclic group $\frac{1}{n}\mathbb{Z}/\mathbb{Z}$, so

$$\operatorname{Hom}_{\mathbb{Z}}(\mathbb{Z}/n\mathbb{Z},\, \mathbb{Q}/\mathbb{Z}) \cong \mathbb{Z}/n\mathbb{Z}$$

The module $\mathbb{Q}/\mathbb{Z}$ is the *dualizing module* for abelian groups. Injectivity is exactly what makes this work: it is what keeps $\operatorname{Hom}_{\mathbb{Z}}(-, \mathbb{Q}/\mathbb{Z})$ exact, so that the dual of a submodule is a quotient of the dual. The isomorphism displayed above is not natural, there being no canonical generator of a cyclic group, but the double dual is; the systematic development, in which these homomorphisms become characters, is section 53.2.

Exercise 17.3.1

1. For each pair, decide whether the module is injective over the indicated ring, and justify the answer.
   1. $\mathbb{Q}$ over $\mathbb{Z}$
   2. $\mathbb{Z}/6\mathbb{Z}$ over $\mathbb{Z}/6\mathbb{Z}$
   3. $\mathbb{Z}/2\mathbb{Z}$ over $\mathbb{Z}/4\mathbb{Z}$
   4. $\mathbb{Q}/\mathbb{Z}$ over $\mathbb{Z}$
   5. $F[x]$ over $F[x]$, for a field F
2. Let $g\colon 6\mathbb{Z} \to \mathbb{Q}/\mathbb{Z}$ be the homomorphism of $\mathbb{Z}$-modules determined by $g(6) = \left[\frac{1}{4}\right]$. Find every extension $G\colon \mathbb{Z} \to \mathbb{Q}/\mathbb{Z}$ of g, and say how many there are. Then explain why theorem 17.3.5 guarantees at least one without any computation.
3. Essential extensions.
   1. Show that $\mathbb{Z}$ is essential in $\mathbb{Q}$ as a $\mathbb{Z}$-module.
   2. Show that $\mathbb{Z}/p\mathbb{Z}$ is essential in $\mathbb{Z}/p^2\mathbb{Z}$.
   3. Show that $\mathbb{Z} \oplus 0$ is *not* essential in $\mathbb{Z} \oplus \mathbb{Z}/2\mathbb{Z}$.

4. Deduce from (a) that $E(\mathbb{Z}) \cong \mathbb{Q}$.

4. Show that every nonzero subgroup of $\mathbb{Z}_{p^\infty}$ contains an element of order p, and deduce that $\mathbb{Z}/p\mathbb{Z}$ is essential in $\mathbb{Z}_{p^\infty}$. Conclude that $E(\mathbb{Z}/p\mathbb{Z}) \cong \mathbb{Z}_{p^\infty}$ as $\mathbb{Z}$-modules. *Hint*: the subgroups of $\mathbb{Z}_{p^\infty}$ are totally ordered by inclusion.

5. Let $0 \to L \to M \to N \to 0$ be a short exact sequence of R-modules.

   1. Show that if L is injective the sequence splits, so $M \cong L \oplus N$.
   2. Show that if L and N are injective then so is M.
   3. Give an example with M injective and N not injective. *Hint*: take $R = \mathbb{Z}/4\mathbb{Z}$, and use example 17.3.10(5). Why does no such example exist over $\mathbb{Z}$?

6. *(The dual of Schanuel's lemma.)* Let

$$0 \to M \to Q_1 \to C_1 \to 0 \qquad \text{and} \qquad 0 \to M \to Q_2 \to C_2 \to 0$$

   be short exact sequences of R-modules with Q_1 and Q_2 injective. Prove that $C_1 \oplus Q_2 \cong C_2 \oplus Q_1$. *Hint*: run the proof of proposition 17.2.12 backwards, replacing the submodule of $P_1 \oplus P_2$ by the quotient of $Q_1 \oplus Q_2$ by the copy of M embedded as $m \mapsto (\iota_1(m), -\iota_2(m))$.

7. Compute injective resolutions.

   1. Show that $0 \to \mathbb{Z} \to \mathbb{Q} \to \mathbb{Q}/\mathbb{Z} \to 0$ is an injective resolution of $\mathbb{Z}$ as a $\mathbb{Z}$-module, of length 1.
   2. Show that multiplication by p on $\mathbb{Z}_{p^\infty}$ is surjective with kernel the subgroup of order p, and deduce that $0 \to \mathbb{Z}/p\mathbb{Z} \to \mathbb{Z}_{p^\infty} \xrightarrow{\cdot p} \mathbb{Z}_{p^\infty} \to 0$ is an injective resolution of $\mathbb{Z}/p\mathbb{Z}$, again of length 1.
   3. Conclude that every $\mathbb{Z}$-module has an injective resolution of length at most 1. *Hint*: corollary 17.3.6.

17.4⋆ Divisible Groups

⋆ *Optional. This section is an aside; nothing later in the book depends on it.*

Example 17.3.8 exhibited $\mathbb{Q}$, $\mathbb{Q}/\mathbb{Z}$ and the Prüfer groups $\mathbb{Z}_{p^\infty}$ as divisible, and theorem 17.3.5(2) made each of them an injective $\mathbb{Z}$-module. This section proves that the list is complete: every divisible group is a direct sum of copies of those, in essentially one way. Since divisible and injective coincide over $\mathbb{Z}$, that is a full classification of the injective $\mathbb{Z}$-modules.

Everything here is about abelian groups, so *group* means *abelian group* throughout, written additively, and $\mathrm{Tor}(G)$ denotes the torsion subgroup of definition 12.3.18.

A subgroup of a divisible group need not be divisible: $\mathbb{Z} \subseteq \mathbb{Q}$ is the standard example. It is therefore not obvious that a group has a largest divisible subgroup, and the first thing to establish is that it does.

Definition 17.4.1: Divisible Subgroup

Let G be an abelian group. Write dG for the subgroup of G generated by all the divisible subgroups of G, that is, the set of finite sums $x_1 + \cdots + x_k$ with each x_i lying in some divisible subgroup. If $dG = 0$, the group G is called *reduced*.

Lemma 17.4.2: dG Is the Largest Divisible Subgroup

Let G be an abelian group. Then dG is divisible, and every divisible subgroup of G is contained in dG.

***Proof*:**
Let $x \in dG$ and $n \geq 1$. Write $x = x_1 + \cdots + x_k$ with $x_i \in D_i$ for divisible subgroups $D_i \subseteq G$. Each D_i is divisible, so there is a $y_i \in D_i$ with $ny_i = x_i$. Then $y := y_1 + \cdots + y_k$ lies in dG and

$$ny = ny_1 + \cdots + ny_k = x_1 + \cdots + x_k = x$$

so $n\,dG = dG$. The second claim holds by construction, since every divisible subgroup is one of the D_i available in the generating set.

Lemma 17.4.3: Splitting Off the Divisible Part

Let G be an abelian group. Then there is a subgroup $H \subseteq G$ with

$$G = dG \oplus H$$

and every such H is reduced, and isomorphic to G/dG.

***Proof*:**
By lemma 17.4.2 the subgroup dG is divisible, hence an injective $\mathbb{Z}$-module by theorem 17.3.5(2), so the short exact sequence

$$0 \to dG \hookrightarrow G \to G/dG \to 0$$

splits by theorem 17.3.4(3). This gives $G = dG \oplus H$ for some subgroup H, with $H \cong G/dG$ by the remark following definition 17.1.11.

Such an H is reduced. The subgroup dH is a divisible subgroup of H, hence a divisible subgroup of G, so $dH \subseteq dG$ by lemma 17.4.2. Also $dH \subseteq H$. Therefore $dH \subseteq dG \cap H = 0$.

So every abelian group is a divisible group plus a reduced one, and the two halves do not interact. Classifying the divisible groups is thus a genuine piece of the classification of abelian groups, and it is the piece that has a clean answer. The route splits G once more, into its torsion part and the rest.

Lemma 17.4.4: The Torsion Subgroup of a Divisible Group

Let G be a divisible abelian group. Then $\mathrm{Tor}(G)$ is divisible, and

$$G \cong \mathrm{Tor}(G) \oplus G/\mathrm{Tor}(G)$$

***Proof*:**
Let $x \in \mathrm{Tor}(G)$ and $n \geq 1$. Since G is divisible there is a $y \in G$ with $ny = x$. Since x is a torsion element there is an $m \geq 1$ with $mx = 0$, and then

$$(mn)y = m(ny) = mx = 0$$

so y is a torsion element and $y \in \mathrm{Tor}(G)$. Hence $n\mathrm{Tor}(G) = \mathrm{Tor}(G)$ for every $n \geq 1$, and $\mathrm{Tor}(G)$ is divisible.

Being divisible, $\mathrm{Tor}(G)$ is an injective $\mathbb{Z}$-module by theorem 17.3.5(2), so the short exact sequence $0 \to \mathrm{Tor}(G) \to G \to G/\mathrm{Tor}(G) \to 0$ splits by theorem 17.3.4(3), and the complement is isomorphic to $G/\mathrm{Tor}(G)$.

The two factors are now treated separately. The quotient turns out to carry a structure it had no right to expect.

Lemma 17.4.5: Structure of $G/\mathrm{Tor}(G)$

Let G be a divisible abelian group. Then $G/\mathrm{Tor}(G)$ carries a unique structure of $\mathbb{Q}$-vector space compatible with its addition, and consequently

$$G/\mathrm{Tor}(G) \cong \bigoplus_{i \in I} \mathbb{Q}$$

for some index set I.

***Proof*:**
Write $A = G/\mathrm{Tor}(G)$.

Step 1: A is torsion-free. This holds for every abelian group G. If $n\overline{x} = \overline{0}$ with $n \geq 1$, then $nx \in \mathrm{Tor}(G)$, so $m(nx) = 0$ for some $m \geq 1$; then $(mn)x = 0$ with $mn \geq 1$, so $x \in \mathrm{Tor}(G)$ and $\overline{x} = \overline{0}$.

Step 2: A is divisible, being a quotient of the divisible group G (example 17.3.8(2)).

Step 3: division is unambiguous. If $ny = ny'$ with $n \geq 1$ then $n(y - y') = 0$, and A is torsion-free by Step 1, so $y = y'$. Combined with Step 2: for every $x \in A$ and $n \geq 1$ there is exactly one $y \in A$ with $ny = x$.

Step 4: the action. For $\frac{a}{b} \in \mathbb{Q}$ written with $b \geq 1$, and $x \in A$, define $\frac{a}{b} \cdot x$ to be the unique $y \in A$ with $by = ax$, which exists and is unique by Step 3.

This does not depend on the representative. If $\frac{a}{b} = \frac{a'}{b'}$ with $b, b' \geq 1$, so $ab' = a'b$, let y and y' be the elements produced by the two fractions. Then

$$bb'y = b'(by) = b'ax = a'bx = b(b'y') = bb'y'$$

and $bb' \geq 1$, so $y = y'$ by Step 3.

Step 5: the axioms. Write $y = \frac{a}{b}x$, $u = \frac{c}{d}x$ and $y_i = \frac{a}{b}x_i$, so $by = ax$, $du = cx$ and $by_i = ax_i$.

- Additivity in x: $b(y_1 + y_2) = ax_1 + ax_2 = a(x_1 + x_2)$, so $\frac{a}{b}(x_1 + x_2) = y_1 + y_2$.
- Additivity in the scalar: $bd(y + u) = d(by) + b(du) = dax + bcx = (ad + bc)x$, so $\frac{a}{b}x + \frac{c}{d}x = \frac{ad+bc}{bd}x$.
- Associativity: let $v = \frac{a}{b}u$, so $bv = au$. Then $bd\,v = d(bv) = dau = a(du) = acx$, so $\frac{a}{b}\left(\frac{c}{d}x\right) = \frac{ac}{bd}x$.
- Unit: $1 \cdot x$ is the unique y with $y = x$.

So A is a $\mathbb{Q}$-vector space, and the structure is unique because Step 3 leaves no choice about what $\frac{a}{b} \cdot x$ can be.

Step 6. By theorem 13.1.8 the vector space A has a basis, indexed by some set I, so $A \cong \bigoplus_{i \in I} \mathbb{Q}$.

The torsion factor is where the Prüfer groups appear, and the argument has three stages: break the torsion into primary pieces, check each piece is still divisible, and identify a divisible p-group.

Lemma 17.4.6: Structure of $\mathrm{Tor}(G)$

Let G be a divisible abelian group. Then

$$\mathrm{Tor}(G) \cong \bigoplus_{p \text{ prime}} \bigoplus_{i \in I_p} \mathbb{Z}_{p^\infty}$$

for index sets I_p.

***Proof*:**

Write $T = \mathrm{Tor}(G)$, which is divisible by lemma 17.4.4. For a prime p let

$$T_p = \left\{ x \in T \;\middle|\; p^k x = 0 \text{ for some } k \geq 0 \right\}$$

the *p-primary component*, a subgroup of T since $p^k x = p^l y = 0$ gives $p^{k+l}(x - y) = 0$.

Step 1: $T = \bigoplus_p T_p$. This holds for any torsion group. For the sum, let $x \in T$ have order $n = p_1^{n_1} \cdots p_k^{n_k}$ with the p_i distinct primes and $n_i \geq 1$, and set $q_i = n/p_i^{n_i}$. No prime divides every q_i, since p_j does not divide q_j, so $\gcd(q_1, \ldots, q_k) = 1$ and there are integers h_i with

$$h_1 q_1 + h_2 q_2 + \cdots + h_k q_k = 1 \tag{17.1}$$

Put $x_i = h_i q_i x$. Then $p_i^{n_i} x_i = h_i n x = 0$, so $x_i \in T_{p_i}$, and multiplying (17.1) by x gives $x = x_1 + \cdots + x_k$.

For directness, suppose $x \in T_p \cap \sum_{q \neq p} T_q$. Then $p^k x = 0$ for some k, so the order of x is a power of p. On the other hand $x = y_1 + \cdots + y_r$ with $y_j \in T_{q_j}$ for primes $q_j \neq p$, and each y_j has order a power of q_j, so the order of x divides a product of powers of primes different from p. A number that is both a power of p and coprime to p is 1, so $x = 0$.

Step 2: each T_p is divisible. Let $x \in T_p$ and $n \geq 1$. Since T is divisible there is a $y \in T$ with $ny = x$. Decompose $y = \sum_q y_q$ by Step 1. Then $x = ny = \sum_q n y_q$ with $n y_q \in T_q$, and x lies in

T_p, so uniqueness of the decomposition forces $ny_q = 0$ for $q \neq p$ and $ny_p = x$. As $y_p \in T_p$, this shows $nT_p = T_p$.

Step 3: a nonzero divisible p-group contains a copy of $\mathbb{Z}_{p^\infty}$. Let D be a nonzero divisible group in which every element has order a power of p, and pick $g \neq 0$, of order p^k with $k \geq 1$. Then $x_1 := p^{k-1}g$ has order exactly p. Using divisibility choose x_2 with $px_2 = x_1$, then x_3 with $px_3 = x_2$, and so on.

Each x_r has order exactly p^r: repeatedly applying $px_{i+1} = x_i$ gives $p^{r-1}x_r = x_1 \neq 0$ and $p^r x_r = px_1 = 0$. Also $x_r = px_{r+1} \in \langle x_{r+1}\rangle$, so the cyclic groups $\langle x_1\rangle \subseteq \langle x_2\rangle \subseteq \cdots$ form a chain, and $Z := \bigcup_r \langle x_r\rangle$ is a subgroup of D.

Define $\theta\colon Z \to \mathbb{Z}_{p^\infty}$ by $\theta(ax_r) = \left[\frac{a}{p^r}\right]$. It is well defined: if $ax_r = bx_s$ with $r \leq s$, then $x_r = p^{s-r}x_s$, so $(ap^{s-r} - b)x_s = 0$, so p^s divides $ap^{s-r} - b$, and therefore $\left[\frac{a}{p^r}\right] = \left[\frac{ap^{s-r}}{p^s}\right] = \left[\frac{b}{p^s}\right]$. It is a homomorphism by the same computation, surjective because every element of $\mathbb{Z}_{p^\infty}$ has the form $\left[\frac{a}{p^r}\right]$ (example 17.3.8(3)), and injective because $\theta(ax_r) = 0$ says $p^r \big| a$, which says $ax_r = 0$. So $Z \cong \mathbb{Z}_{p^\infty}$.

Step 4: a divisible p-group is a direct sum of copies of $\mathbb{Z}_{p^\infty}$. Let D be such a group and let $\mathcal{T}$ be the set of families of subgroups of D, each member of each family isomorphic to $\mathbb{Z}_{p^\infty}$, whose sum inside D is direct. Order $\mathcal{T}$ by inclusion of families. It contains the empty family. A chain has the union of its families as an upper bound, which is again in $\mathcal{T}$ because directness is a condition on finitely many members at a time, and any finitely many members of the union already lie in one family of the chain. By Zorn's lemma (theorem A.2.3) there is a maximal family $\{Z_j\}_{j\in J}$; put $H = \bigoplus_{j\in J} Z_j \subseteq D$.

The group H is divisible, being a direct sum of divisible groups (example 17.3.8(4)), hence injective, so $D = H \oplus K$ for some K by theorem 17.3.4(3). The complement K is divisible: given $x \in K$ and $n \geq 1$, divisibility of D gives $d = h + k$ with $nd = x$, and comparing components in $H \oplus K$ gives $nh = 0$ and $nk = x$. Every element of K has order a power of p, so if $K \neq 0$ then Step 3 supplies a subgroup $Z \subseteq K$ isomorphic to $\mathbb{Z}_{p^\infty}$; since $H \cap K = 0$, the family $\{Z_j\}_{j\in J} \cup \{Z\}$ is again in $\mathcal{T}$ and strictly larger, contradicting maximality. So $K = 0$ and $D = H$.

Applying Steps 2 and 4 to each T_p and assembling with Step 1 gives the statement.

Theorem 17.4.7: Classification of Divisible Groups

Let G be a divisible abelian group. Then there are index sets I and I_p, one for each prime p, with

$$G \cong \left(\bigoplus_{p \text{ prime}} \bigoplus_{i\in I_p} \mathbb{Z}_{p^\infty}\right) \oplus \bigoplus_{i\in I} \mathbb{Q}$$

The cardinalities of these index sets are determined by G, namely

$$|I| = \dim_{\mathbb{Q}}\big(G/\mathrm{Tor}(G)\big) \qquad \text{and} \qquad |I_p| = \dim_{\mathbb{F}_p} G[p], \quad \text{where } G[p] = \{g \in G \mid pg = 0\}$$

Since divisible and injective agree over $\mathbb{Z}$ (theorem 17.3.5(2)), this classifies the injective $\mathbb{Z}$-modules.

***Proof*:**
Existence. By lemma 17.4.4, $G \cong \mathrm{Tor}(G) \oplus G/\mathrm{Tor}(G)$. Lemma 17.4.6 identifies the first summand and lemma 17.4.5 the second.

The invariants. Suppose G is written as displayed. Each $\mathbb{Z}_{p^\infty}$ is a torsion group and each copy of $\mathbb{Q}$ is torsion-free, so the torsion subgroup of the right-hand side is exactly the first bracket, and

$$G/\mathrm{Tor}(G) \cong \bigoplus_{i \in I} \mathbb{Q}$$

which is a $\mathbb{Q}$-vector space with a basis indexed by I. Since G determines $\mathrm{Tor}(G)$ and hence $G/\mathrm{Tor}(G)$ up to isomorphism, and the cardinality of a basis is an invariant of a vector space (lemma 14.1.1 and the remark following it), $|I|$ is determined by G.

For the primary invariants, note that $G[p]$ is killed by p and so is a vector space over $\mathbb{F}_p$, and that forming $G[p]$ commutes with direct sums, an element of a direct sum being killed by p exactly when each of its components is. Now

$$\mathbb{Q}[p] = 0, \qquad \mathbb{Z}_{q^\infty}[p] = 0 \text{ for } q \neq p, \qquad \mathbb{Z}_{p^\infty}[p] = \left\{ \left[\tfrac{a}{p}\right] \;\middle|\; a \in \mathbb{Z} \right\} \cong \mathbb{Z}/p\mathbb{Z}$$

The first holds because $\mathbb{Q}$ is torsion-free; the second because every element of $\mathbb{Z}_{q^\infty}$ has order a power of q, and no nonzero such order is divisible by p; the third because $p\left[\frac{a}{p^n}\right] = 0$ forces $p^{n-1} \big| a$, so the element is $\left[\frac{b}{p}\right]$ for some b. Therefore

$$G[p] \cong \bigoplus_{i \in I_p} \mathbb{Z}/p\mathbb{Z}$$

an $\mathbb{F}_p$-vector space with a basis indexed by I_p. As G determines $G[p]$, the cardinality $|I_p|$ is determined by G.

Exercise 17.4.1

1. Compute dG, the largest divisible subgroup, for each of the following.
    1. $G = \mathbb{Z}$
    2. $G = \mathbb{Q}$
    3. $G = \mathbb{Q} \oplus \mathbb{Z}$
    4. $G = \mathbb{Z}/6\mathbb{Z}$
    5. $G = \mathbb{Q}/\mathbb{Z}$
2. Decide which of the following are divisible, and justify each answer.
    1. $\mathbb{R}$ under addition
    2. $\mathbb{R}/\mathbb{Z}$ under addition
    3. $\mathbb{C}^\times$ under multiplication
    4. $\mathbb{Z}[1/p]$ under addition, for a prime p

5. $\mathbb{Q}^\times$ under multiplication

3. For each of the following divisible groups, compute the invariants $|I|$ and $|I_p|$ of theorem 17.4.7.
   1. $\mathbb{Q}/\mathbb{Z}$
   2. $\mathbb{Q} \oplus \mathbb{Q}/\mathbb{Z}$
   3. $\mathbb{Z}_{p^\infty} \oplus \mathbb{Z}_{p^\infty}$
   4. $\mathbb{R}$ under addition. *Hint*: compare cardinalities.

4. Show that $\mathbb{Z}_{p^\infty} \cong \mathbb{Z}[1/p]/\mathbb{Z}$, and that the proper subgroups of $\mathbb{Z}_{p^\infty}$ are exactly the finite cyclic groups $\langle \left[\frac{1}{p^n}\right] \rangle \cong \mathbb{Z}/p^n\mathbb{Z}$ for $n \geq 0$, totally ordered by inclusion. Deduce that $\mathbb{Z}_{p^\infty}$ has no maximal subgroup.

5. Show that every homomorphism $f: G \to H$ of abelian groups satisfies $f(dG) \subseteq dH$, and that $G \mapsto dG$ is therefore a functor $\mathbf{Ab} \to \mathbf{Ab}$. *Hint*: the image of a divisible group is divisible.

6. Show that a nonzero divisible group has no maximal subgroup. *Hint*: if M were maximal, G/M would be a simple abelian group, hence $\mathbb{Z}/p\mathbb{Z}$ for some prime p; but G/M is divisible. Deduce that no nonzero divisible group is finitely generated, recovering example 17.3.8(5).

17.5 Flat Modules

The third functor available on a short exact sequence is the tensor product. Fix a right R-module D and apply $D \otimes_R -$, which theorem 15.2.2 turns into a covariant functor from left R-modules to abelian groups, landing in left S-modules when D is an (S, R)-bimodule. Everything below is stated for $D \otimes_R -$; the functor $- \otimes_R D$ on right modules behaves identically, with left and right exchanged throughout.

This functor fails at the opposite end from the two in section 17.2 and section 17.3. Both Hom functors are left exact and can lose surjectivity; tensoring is *right* exact and can lose injectivity. The reason is the Hom-tensor adjunction of theorem 15.3.9: $D \otimes_R -$ is a left adjoint, and left adjoints preserve the constructions that sit at the right-hand end of a sequence.

Theorem 17.5.1: $D \otimes_R -$ Is Right Exact

Let D be a right R-module and let

$$0 \to L \xrightarrow{\psi} M \xrightarrow{\varphi} N \to 0$$

be a short exact sequence of left R-modules. Then

$$D \otimes_R L \xrightarrow{1 \otimes \psi} D \otimes_R M \xrightarrow{1 \otimes \varphi} D \otimes_R N \to 0$$

is an exact sequence of abelian groups, and of left S-modules when D is an (S, R)-bimodule. The map $1 \otimes \psi$ need not be injective.

***Proof*:**
Surjectivity of $1 \otimes \varphi$. By lemma 15.1.4(2) every element of $D \otimes_R N$ is a finite sum $\sum_i d_i \otimes n_i$. Each n_i is $\varphi(m_i)$ for some m_i, since φ is surjective, so the element is the image of $\sum_i d_i \otimes m_i$.

$\mathrm{im}(1 \otimes \psi) \subseteq \ker(1 \otimes \varphi)$. By theorem 15.2.2 the composite $(1 \otimes \varphi) \circ (1 \otimes \psi)$ is $1 \otimes (\varphi \circ \psi)$, and $\varphi \circ \psi = 0$ because $\mathrm{im}(\psi) = \ker(\varphi)$.

$\ker(1 \otimes \varphi) \subseteq \mathrm{im}(1 \otimes \psi)$. Write $I = \mathrm{im}(1 \otimes \psi)$. By the previous paragraph $1 \otimes \varphi$ kills I, so it factors as

$$\overline{\varphi}\colon (D \otimes_R M)/I \to D \otimes_R N, \qquad \overline{\varphi}\big(\overline{d \otimes m}\big) = d \otimes \varphi(m)$$

by theorem 12.2.10. It suffices to produce an inverse, for then $\overline{\varphi}$ is injective, which says exactly that $\ker(1 \otimes \varphi) = I$.

Define $\beta\colon D \times N \to (D \otimes_R M)/I$ by $\beta(d, n) = \overline{d \otimes m}$ for any m with $\varphi(m) = n$. This does not depend on the choice: if $\varphi(m) = \varphi(m')$ then $m - m' \in \ker(\varphi) = \mathrm{im}(\psi)$, say $m - m' = \psi(l)$, and then

$$d \otimes m - d \otimes m' = d \otimes \psi(l) = (1 \otimes \psi)(d \otimes l) \in I$$

The map β is R-balanced in the sense of definition 15.1.1: it is additive in d because $\otimes$ is; it is additive in n because a lift of $n + n'$ may be taken to be $m + m'$; and $\beta(dr, n) = \overline{dr \otimes m} = \overline{d \otimes rm} = \beta(d, rn)$, since rm is a lift of rn and by the third relation of (15.1).

By theorem 15.1.5 there is a homomorphism $\overline{\beta}\colon D \otimes_R N \to (D \otimes_R M)/I$ with $\overline{\beta}(d \otimes n) = \overline{d \otimes m}$. Both composites fix every simple tensor, hence are the identity by lemma 15.1.4(3), so $\overline{\varphi}$ is an isomorphism.

Now the failure at the left, computed before anything is named after it, on the same sequence that example 17.2.2 and example 17.3.2 used.

Example 17.5.2: $D \otimes_R -$ Need Not Be Left Exact

1. Take $R = \mathbb{Z}$, the short exact sequence $0 \to \mathbb{Z} \xrightarrow{\cdot 2} \mathbb{Z} \xrightarrow{\pi} \mathbb{Z}/2\mathbb{Z} \to 0$, and $D = \mathbb{Z}/2\mathbb{Z}$. Since $A \otimes_{\mathbb{Z}} \mathbb{Z} \cong A$ for every A, and $\mathbb{Z}/2\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Z}/2\mathbb{Z} \cong \mathbb{Z}/2\mathbb{Z}$ by proposition 15.2.13, applying $D \otimes_{\mathbb{Z}} -$ gives

$$\mathbb{Z}/2\mathbb{Z} \xrightarrow{1 \otimes (\cdot 2)} \mathbb{Z}/2\mathbb{Z} \xrightarrow{1 \otimes \pi} \mathbb{Z}/2\mathbb{Z} \to 0$$

The first map is zero: for $\overline{a} \in \mathbb{Z}/2\mathbb{Z}$ and $x \in \mathbb{Z}$,

$$\overline{a} \otimes 2x = \overline{a} \cdot 2 \otimes x = \overline{0} \otimes x = 0$$

So the sequence is exact where theorem 17.5.1 promises and fails to be exact at the left, the first map being zero on a group of order two.

2. The sharpest version replaces the middle term. The inclusion $\mathbb{Z} \hookrightarrow \mathbb{Q}$ is injective, but $\mathbb{Z}/2\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Q} = 0$, since

$$\overline{a} \otimes q = \overline{a} \otimes 2 \cdot \tfrac{q}{2} = \overline{a} \cdot 2 \otimes \tfrac{q}{2} = 0$$

for every q. So $1 \otimes \iota$ maps $\mathbb{Z}/2\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Z} \cong \mathbb{Z}/2\mathbb{Z}$ to the zero group, and injectivity fails as badly as it can.

Proposition 17.5.3: Conditions for $A \otimes_R -$ to Be Exact

Let A be a right R-module. The following are equivalent.

1. For every short exact sequence $0 \to L \xrightarrow{\psi} M \xrightarrow{\varphi} N \to 0$ of left R-modules, the sequence
$$0 \to A \otimes_R L \xrightarrow{1 \otimes \psi} A \otimes_R M \xrightarrow{1 \otimes \varphi} A \otimes_R N \to 0$$
is short exact.
2. For every injective homomorphism $\psi \colon L \to M$ of left R-modules, the map $1 \otimes \psi \colon A \otimes_R L \to A \otimes_R M$ is injective.

***Proof*:**

(1) $\Rightarrow$ (2). Given ψ injective, put $N = M/\psi(L)$, so that $0 \to L \xrightarrow{\psi} M \to N \to 0$ is short exact by proposition 17.1.1 and theorem 12.2.10. Condition (1) makes $1 \otimes \psi$ injective.

(2) $\Rightarrow$ (1). Theorem 17.5.1 supplies exactness at $A \otimes_R M$ and at $A \otimes_R N$, and (2) supplies injectivity of $1 \otimes \psi$, which is exactness at $A \otimes_R L$ by proposition 17.1.1.

Definition 17.5.4: Flat Module

A right R-module A is *flat* if it satisfies the equivalent conditions of proposition 17.5.3, that is if the functor $A \otimes_R -$ is exact. A left R-module is *flat* when the corresponding statement holds for $- \otimes_R A$.

The side matters only because $\otimes_R$ does. Over a commutative ring the two notions coincide, by proposition 15.2.7, and the qualification can be dropped.

Corollary 17.5.5: Projective Modules Are Flat

Every free R-module is flat, and more generally every projective R-module is flat.

***Proof*:**

First, R itself is flat: the map $R \otimes_R L \to L$ sending $r \otimes l$ to rl is an isomorphism with inverse $l \mapsto 1 \otimes l$, and under it $1 \otimes \psi$ becomes ψ, so $1 \otimes \psi$ is injective whenever ψ is.

Next let $F = \bigoplus_{i \in I} R$ be free. By theorem 15.2.4 there are isomorphisms $F \otimes_R L \cong \bigoplus_{i \in I}(R \otimes_R L) \cong \bigoplus_{i \in I} L$, and similarly for M, carrying $1 \otimes \psi$ to the map $\bigoplus_i \psi$ acting coordinatewise. A direct sum of injective maps is injective, so F is flat.

Finally let P be projective. By theorem 17.2.4(4) there is a free module F and a module K with $F \cong P \oplus K$. By theorem 15.2.3,
$$F \otimes_R L \cong (P \otimes_R L) \oplus (K \otimes_R L)$$
compatibly with M in place of L, and under these identifications $1_F \otimes \psi$ is $(1_P \otimes \psi) \oplus (1_K \otimes \psi)$. A direct sum of maps is injective only if each summand is, so injectivity of $1_F \otimes \psi$ forces injectivity of $1_P \otimes \psi$.

So free implies projective implies flat, and the second implication is strict in a way the first is not obviously: example 17.5.8 exhibits a flat $\mathbb{Z}$-module that is not projective. Over a principal ideal domain there is a criterion for flatness as concrete as Baer's is for injectivity.

Proposition 17.5.6: Flat Implies Torsion-Free

Let R be an integral domain and A a flat R-module. Then A is torsion-free.

***Proof*:**
Let $r \in R$ be nonzero. Multiplication by r is an injective R-module homomorphism $R \to R$, since R is a domain, so flatness makes $1 \otimes (\cdot r) \colon A \otimes_R R \to A \otimes_R R$ injective. The map $A \otimes_R R \to A$ sending $a \otimes r$ to ar is an isomorphism, with inverse $a \mapsto a \otimes 1$, and it carries $1 \otimes (\cdot r)$ to multiplication by r on A. Hence $ra = 0$ forces $a = 0$, and A has no nonzero torsion element.

Theorem 17.5.7: Flat Equals Torsion-Free Over a PID

Let R be a principal ideal domain and A an R-module. Then A is flat if and only if A is torsion-free.

***Proof*:**
One direction is proposition 17.5.6, which needs only that R is a domain.

For the converse let A be torsion-free, let $\psi \colon L \to M$ be an injective homomorphism, and let $\xi \in A \otimes_R L$ satisfy $(1 \otimes \psi)(\xi) = 0$. Write $\xi = \sum_{i=1}^{k} a_i \otimes l_i$ using lemma 15.1.4(2), so that

$$\sum_{i=1}^{k} a_i \otimes \psi(l_i) = 0 \qquad \text{in } A \otimes_R M$$

Step 1: pass to a finitely generated piece. By lemma 15.2.12 there is a finitely generated submodule $A_0 \subseteq A$ containing $a_1, \ldots, a_k$ with

$$\sum_{i=1}^{k} a_i \otimes \psi(l_i) = 0 \qquad \text{in } A_0 \otimes_R M$$

Step 2: A_0 is flat. A submodule of a torsion-free module is torsion-free, so A_0 is a finitely generated torsion-free module over a principal ideal domain, hence free by theorem 14.1.9(2), hence flat by corollary 17.5.5.

Step 3: conclude. The map $1_{A_0} \otimes \psi \colon A_0 \otimes_R L \to A_0 \otimes_R M$ is therefore injective, and Step 1 says it kills the element $\sum_i a_i \otimes l_i$ of $A_0 \otimes_R L$. So that element is zero in $A_0 \otimes_R L$, and its image under the map $A_0 \otimes_R L \to A \otimes_R L$ induced by the inclusion is ξ. Hence $\xi = 0$, and $1 \otimes \psi$ is injective.

Example 17.5.8: Flat Modules

Throughout, $R = \mathbb{Z}$ unless stated otherwise, so theorem 17.5.7 applies and flat means torsion-free.

1. $\mathbb{Z}$ is flat, being free. For $n \geq 2$ the module $\mathbb{Z}/n\mathbb{Z}$ is not flat, having $\overline{1}$ as a torsion element; example 17.5.2(1) is that failure for $n = 2$, seen directly.

2. $\mathbb{Q}$ is a flat $\mathbb{Z}$-module, being torsion-free. It is *not* projective, by example 17.2.10(5). So the implication projective $\Rightarrow$ flat is strict, and flatness is genuinely weaker than the other two conditions of this chapter.

3. $\mathbb{Q}/\mathbb{Z}$ is an injective $\mathbb{Z}$-module (example 17.3.10(2)) but is not flat, being a nonzero torsion group. So injective does not imply flat.

4. $\mathbb{Q} \oplus \mathbb{Z}$ is torsion-free, hence flat. It is not projective: a direct summand of a projective module is projective, being a summand of a summand of a free module, and $\mathbb{Q}$ is not projective. It is not injective either, since a direct summand of an injective module is injective by proposition 17.3.9(1) and $\mathbb{Z}$ is not injective (example 17.3.10(1)).

5. A direct sum $\bigoplus_i A_i$ is flat if and only if every A_i is, by the computation in the proof of corollary 17.5.5 applied to theorem 15.2.4.

6. Over a field every module is free (theorem 13.1.8), hence flat. Over a field, free, projective, injective and flat all coincide.

7. Let R be a principal ideal domain and $D \subseteq R \setminus \{0\}$ a multiplicatively closed set containing 1. The ring of fractions $D^{-1}R$ of section 9.5 sits inside the field of fractions of R, which is torsion-free as an R-module, so $D^{-1}R$ is a torsion-free and therefore flat R-module. That localization is flat holds over any commutative ring, and is taken up in *Commutative Algebra* [10].

Corollary 17.5.9: Tensor Products of Projective Modules

Let R be a commutative ring and let P and Q be projective R-modules. Then $P \otimes_R Q$ is projective.

***Proof*:**

Choose P' and Q' with $P \oplus P' = F$ and $Q \oplus Q' = G$ free, using theorem 17.2.4(4). Then $F \otimes_R G$ is free by corollary 15.2.6, and theorem 15.2.3 expands it as

$$F \otimes_R G \cong (P \otimes_R Q) \oplus (P \otimes_R Q') \oplus (P' \otimes_R Q) \oplus (P' \otimes_R Q')$$

So $P \otimes_R Q$ is a direct summand of a free module and theorem 17.2.4(4) applies.

A ring homomorphism $R \to S$ making S flat as an R-module is called a *flat extension*, and the condition is what allows a construction performed over R to be transported to S without losing exactness. Example 17.5.8(7) is the first example. Flat extensions, the local criterion that detects flatness one prime at a time, and the geometric reading of flatness as a family of fibres varying continuously all belong with commutative algebra and are developed in *Commutative Algebra* [10].

Exercise 17.5.1

1. Decide whether each module is flat over the indicated ring, and justify.
    1. $\mathbb{Z}/6\mathbb{Z}$ over $\mathbb{Z}$
    2. $\mathbb{Q}$ over $\mathbb{Z}$
    3. $\mathbb{Z}[1/2]$ over $\mathbb{Z}$
    4. $\mathbb{Z}/2\mathbb{Z}$ over $\mathbb{Z}/6\mathbb{Z}$
    5. $k[x]/(x)$ over $k[x]$, for a field k
2. Let $m, n \geq 1$ and $d = \gcd(m, n)$. Using $\mathbb{Z}/m\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Z}/n\mathbb{Z} \cong \mathbb{Z}/d\mathbb{Z}$ (proposition 15.2.13), show directly that $\mathbb{Z}/2\mathbb{Z}$ is not a flat $\mathbb{Z}$-module, by tensoring the injection $\mathbb{Z}/2\mathbb{Z} \to \mathbb{Z}/4\mathbb{Z}$, $\overline{a} \mapsto \overline{2a}$, with $\mathbb{Z}/2\mathbb{Z}$ and computing both sides.
3. Show that a direct sum $\bigoplus_{i \in I} A_i$ of right R-modules is flat if and only if every A_i is flat.
4. Let R be a principal ideal domain.
    1. Show that every submodule of a flat R-module is flat.
    2. Show that a quotient of a flat R-module need not be flat, by exhibiting a surjection from a flat $\mathbb{Z}$-module onto a non-flat one.
5. Let R be a commutative ring and let A and B be flat R-modules. Show that $A \otimes_R B$ is flat. *Hint*: theorem 15.2.8 identifies $(A \otimes_R B) \otimes_R L$ with $A \otimes_R (B \otimes_R L)$.
6. Show that every element of $M \otimes_R N$ lies in the image of $M_0 \otimes_R N_0 \to M \otimes_R N$ for some finitely generated submodules $M_0 \subseteq M$ and $N_0 \subseteq N$. *Hint*: lemma 15.1.4(2). Explain why this is weaker than lemma 15.2.12, and why the weaker statement would not suffice in the proof of theorem 17.5.7.
7. Assemble the three strictness witnesses of this chapter into one picture. Exhibit $\mathbb{Z}$-modules, or modules over a ring of your choice, showing that each of the implications
$$\text{free} \Rightarrow \text{projective} \Rightarrow \text{flat}$$
is strict, and that neither implication holds between injective and flat in either direction.

17.6 Ext and Tor

Section 17.1 asked for $\mathcal{E}(C, A)$, the set of equivalence classes of extensions of C by A, and everything since has been about the three functors that a short exact sequence can be pushed through. Each is exact at one end and can fail at the other. Writing $0 \to A \to B \to C \to 0$ for a short exact sequence and D for a fixed module, proposition 17.2.1, proposition 17.3.1 and theorem 17.5.1 say

$$0 \to \mathrm{Hom}_R(D, A) \to \mathrm{Hom}_R(D, B) \to \mathrm{Hom}_R(D, C)$$
$$0 \to \mathrm{Hom}_R(C, D) \to \mathrm{Hom}_R(B, D) \to \mathrm{Hom}_R(A, D)$$
$$A \otimes_R D \to B \otimes_R D \to C \otimes_R D \to 0$$

and in each case the missing end is missing for a reason: something is there, and it is not zero. This section names it. The construction is given in full and the computations below are unconditional, but three of the theorems are stated without proof, because their proofs need machinery that this book does not build. Where that happens it is said explicitly, and the reason is given.

17.6.1 Complexes and Their Homology

Definition 12.2.11 defined an *exact* chain complex. What is needed here is the same object without the exactness, together with a measure of how far from exact it is.

Definition 17.6.1: Chain Complex and Homology

A *chain complex* of R-modules is a family $(M_i)_{i\in\mathbb{Z}}$ together with homomorphisms $d_i\colon M_i \to M_{i-1}$ satisfying $d_i \circ d_{i+1} = 0$ for every i, equivalently $\operatorname{im}(d_{i+1}) \subseteq \ker(d_i)$. Its *$i$-th homology* is the quotient module

$$H_i(M_\bullet) := \ker(d_i)/\operatorname{im}(d_{i+1})$$

A *cochain complex* is the same data with the arrows reversed, $d^i\colon M^i \to M^{i+1}$ and $d^{i+1}\circ d^i = 0$, and its *i-th cohomology* is $H^i(M^\bullet) := \ker(d^i)/\operatorname{im}(d^{i-1})$.

So a complex is exact at M_i exactly when $H_i(M_\bullet) = 0$, and the homology records the failure one degree at a time. This is the whole of the apparatus needed below; the theory of complexes proper is *Homological Algebra* [15].

17.6.2 The Construction

Fix an R-module M and choose a projective resolution of it, which exists by example 17.1.4(3) and corollary 17.2.6: an exact complex

$$\cdots \to P_2 \xrightarrow{d_2} P_1 \xrightarrow{d_1} P_0 \xrightarrow{\varepsilon} M \to 0$$

with every P_i projective. Deleting M leaves the complex $P_\bullet$, which is no longer exact at P_0; its homology there is M itself. Now apply one of the three functors to $P_\bullet$ and take homology.

Definition 17.6.2: Tor

Let R be a ring, M a right R-module with a chosen projective resolution $P_\bullet$, and N a left R-module. The complex $P_\bullet \otimes_R N$ has differentials $d_i \otimes 1$ and satisfies $(d_i \otimes 1)(d_{i+1} \otimes 1) = (d_i d_{i+1}) \otimes 1 = 0$ by theorem 15.2.2, so it is a chain complex. Define

$$\operatorname{Tor}_i^R(M,N) := H_i\big(P_\bullet \otimes_R N\big) = \ker(d_i \otimes 1)/\operatorname{im}(d_{i+1} \otimes 1)$$

Definition 17.6.3: Ext

With M, $P_\bullet$ as above and N any R-module, the family $\operatorname{Hom}_R(P_i, N)$ with the maps d_{i+1}^* is a cochain complex, since $d_{i+1}^* d_i^* = (d_i d_{i+1})^* = 0$. Define

$$\operatorname{Ext}^i_R(M,N) := H^i\big(\operatorname{Hom}_R(P_\bullet, N)\big) = \ker(d_{i+1}^*)/\operatorname{im}(d_i^*)$$

17.6.4 Two Meanings of Tor The symbol Tor now carries two jobs. Definition 12.3.18 wrote $\mathrm{Tor}_R(M)$, or $\mathrm{Tor}(M)$, for the torsion submodule of M, and definition 17.6.2 writes $\mathrm{Tor}_i^R(M, N)$ for the module just constructed. The subscript decides: an *integer* index means the functor of this section, a *ring* means the torsion submodule, and the functor always takes two arguments. The collision is standard in the literature and is not an accident of notation: proposition 17.6.7 shows that $\mathrm{Tor}_1^{\mathbb{Z}}(\mathbb{Z}/n\mathbb{Z}, A)$ is the subgroup of A killed by n, so the derived functor really is measuring torsion.

The definitions depend on a choice, and nothing so far says the answer does not. It does not, and that is the first of the three deferred theorems.

Theorem 17.6.5: The Derived Functors Are Well Defined

Let R be a ring and M, N be R-modules.

1. $\mathrm{Tor}_i^R(M, N)$ and $\mathrm{Ext}_R^i(M, N)$ do not depend on the choice of projective resolution of M, up to isomorphism.
2. For Tor, a resolution by flat modules (definition 17.5.4) may be used in place of a projective one, with the same result.
3. $\mathrm{Tor}_i^R(M, N)$ may equally be computed from a projective resolution of the second argument N, and $\mathrm{Ext}_R^i(M, N)$ from an *injective* resolution (definition 17.3.15) of N, in both cases giving the same module.

The proofs are omitted here, and the reason is a genuine gap rather than an economy: all three rest on the comparison theorem, which lifts a map of modules to a map of resolutions uniquely up to chain homotopy, and on the fact that chain homotopic maps induce the same map on homology. Neither statement is available in this book, both being about complexes rather than modules, and they are proved in *Homological Algebra* [15]. Proposition 17.2.12 is the visible shadow of the comparison theorem in degree one: two projective presentations of the same module differ by a free summand, which is why the degree-one answer cannot depend on the presentation.

Every computation below fixes one explicit resolution, so each is a complete calculation on its own terms; theorem 17.6.5 is what upgrades it from a statement about that resolution to a statement about M.

Proposition 17.6.6: Degree Zero

Let M and N be R-modules. Then

$$\mathrm{Tor}_0^R(M, N) \cong M \otimes_R N \qquad \text{and} \qquad \mathrm{Ext}_R^0(M, N) \cong \mathrm{Hom}_R(M, N)$$

***Proof*:**

Let $P_\bullet \to M$ be a projective resolution. Exactness at P_0 says $P_1 \xrightarrow{d_1} P_0 \xrightarrow{\varepsilon} M \to 0$ is exact.

For Tor, apply $- \otimes_R N$. By theorem 17.5.1 the sequence $P_1 \otimes N \to P_0 \otimes N \to M \otimes N \to 0$ is exact, so $\varepsilon \otimes 1$ identifies $(P_0 \otimes N)/\mathrm{im}(d_1 \otimes 1)$ with $M \otimes_R N$. Since $d_0 = 0$ in the deleted complex, $\ker(d_0 \otimes 1) = P_0 \otimes N$ and the left-hand side is exactly $\mathrm{Tor}_0^R(M, N)$.

For Ext, apply $\mathrm{Hom}_R(-, N)$. By proposition 17.3.1 the sequence $0 \to \mathrm{Hom}_R(M, N) \xrightarrow{\varepsilon^*} \mathrm{Hom}_R(P_0, N) \xrightarrow{d_1^*} \mathrm{Hom}_R(P_1, N)$ is exact, so ε^* is an isomorphism onto $\ker(d_1^*)$. Since $d_0^* = 0$, that kernel is exactly $\mathrm{Ext}^0_R(M, N)$.

Proposition 17.6.7: Ext and Tor of a Cyclic Group

Let $n \geq 1$ and let A be an abelian group. Write $A[n] = \{a \in A \mid na = 0\}$. Then, as abelian groups,

$$\mathrm{Ext}^i_{\mathbb{Z}}(\mathbb{Z}/n\mathbb{Z}, A) \cong \begin{cases} A[n] & i = 0 \\ A/nA & i = 1 \\ 0 & i \geq 2 \end{cases} \qquad \mathrm{Tor}_i^{\mathbb{Z}}(\mathbb{Z}/n\mathbb{Z}, A) \cong \begin{cases} A/nA & i = 0 \\ A[n] & i = 1 \\ 0 & i \geq 2 \end{cases}$$

***Proof*:**

The sequence $0 \to \mathbb{Z} \xrightarrow{\cdot n} \mathbb{Z} \xrightarrow{\pi} \mathbb{Z}/n\mathbb{Z} \to 0$ of example 17.1.6 is a projective resolution of $\mathbb{Z}/n\mathbb{Z}$, since $\mathbb{Z}$ is free: take $P_0 = P_1 = \mathbb{Z}$, $d_1 = \cdot n$, and $P_i = 0$ for $i \geq 2$. The deleted complex is

$$0 \to \mathbb{Z} \xrightarrow{\cdot n} \mathbb{Z} \to 0$$

concentrated in degrees 1 and 0.

Every homomorphism $\mathbb{Z} \to A$ is determined by the image of 1, so $\mathrm{Hom}_{\mathbb{Z}}(\mathbb{Z}, A) \cong A$ by $h \mapsto h(1)$, and under this identification $d_1^* = (\cdot n)^*$ is multiplication by n on A, since $(h \circ (\cdot n))(1) = h(n) = n\, h(1)$. Likewise $\mathbb{Z} \otimes_{\mathbb{Z}} A \cong A$ by $x \otimes a \mapsto xa$, and $d_1 \otimes 1$ becomes multiplication by n. Both functors therefore turn the deleted complex into

$$0 \to A \xrightarrow{\cdot n} A \to 0$$

with A in degrees 1 and 0 for Tor, and in degrees 0 and 1 for Ext, the arrows of a cochain complex running the other way.

Reading off homology: the kernel of multiplication by n is $A[n]$ and its cokernel is A/nA, and every other module in the complex is zero. For Tor this gives $\mathrm{Tor}_1 = \ker(\cdot n) = A[n]$ and $\mathrm{Tor}_0 = A/nA$; for Ext it gives $\mathrm{Ext}^0 = \ker(\cdot n) = A[n]$ and $\mathrm{Ext}^1 = A/nA$. All higher terms vanish because $P_i = 0$ for $i \geq 2$.

The two degree-zero answers agree with proposition 17.6.6: $\mathrm{Hom}_{\mathbb{Z}}(\mathbb{Z}/n\mathbb{Z}, A)$ is the set of elements of A killed by n, and $\mathbb{Z}/n\mathbb{Z} \otimes_{\mathbb{Z}} A \cong A/nA$ by proposition 15.2.13. Specialising the degree-one answers,

$$\mathrm{Ext}^1_{\mathbb{Z}}(\mathbb{Z}/m\mathbb{Z},\, \mathbb{Z}/n\mathbb{Z}) \cong \mathbb{Z}/\gcd(m,n)\mathbb{Z}, \qquad \mathrm{Ext}^1_{\mathbb{Z}}(\mathbb{Z}/n\mathbb{Z},\, \mathbb{Z}) \cong \mathbb{Z}/n\mathbb{Z}, \qquad \mathrm{Ext}^1_{\mathbb{Z}}(\mathbb{Z}/n\mathbb{Z},\, \mathbb{Q}) = 0$$

the last because $\mathbb{Q} = n\mathbb{Q}$.

Corollary 17.6.8: Vanishing

Let $i \geq 1$.

1. If P is projective then $\mathrm{Ext}^i_R(P, N) = 0$ and $\mathrm{Tor}^R_i(P, N) = 0$ for every N.
2. If F is flat then $\mathrm{Tor}^R_i(M, F) = 0$ for every M.

***Proof*:**
(1) A projective module is its own projective resolution: take $P_0 = P$, $\varepsilon = \mathrm{id}_P$ and $P_i = 0$ for $i \geq 1$, which is exact. The deleted complex has a single nonzero term in degree 0, so every functor applied to it has vanishing homology in degrees $i \geq 1$.

(2) Let $P_\bullet \to M$ be a projective resolution, so the complex $\cdots \to P_1 \to P_0 \to M \to 0$ is exact. Since F is flat, applying $- \otimes_R F$ preserves that exactness, by proposition 17.5.3 applied at each spot. So $P_\bullet \otimes_R F$ is exact except in degree 0, which is to say $\mathrm{Tor}^R_i(M, F) = 0$ for $i \geq 1$.

Part (2) is the sense in which flatness is the condition that makes Tor disappear, just as projectivity kills Ext in the first argument. The dual statement, that $\mathrm{Ext}^i_R(M, Q) = 0$ for Q injective, needs theorem 17.6.5(3): compute Ext from an injective resolution of Q, which may be taken to be Q itself.

17.6.3 What They Repair

Theorem 17.6.9: Long Exact Sequences

Let $0 \to A \to B \to C \to 0$ be a short exact sequence of R-modules and let D be an R-module. Then there are homomorphisms δ_i, δ'_i and ∂_i, called *connecting homomorphisms*, making the following sequences exact.

$$0 \to \mathrm{Hom}_R(D, A) \to \mathrm{Hom}_R(D, B) \to \mathrm{Hom}_R(D, C) \xrightarrow{\delta_0} \mathrm{Ext}^1_R(D, A) \to \cdots$$
$$\cdots \to \mathrm{Ext}^i_R(D, A) \to \mathrm{Ext}^i_R(D, B) \to \mathrm{Ext}^i_R(D, C) \xrightarrow{\delta_i} \mathrm{Ext}^{i+1}_R(D, A) \to \cdots$$

$$0 \to \mathrm{Hom}_R(C, D) \to \mathrm{Hom}_R(B, D) \to \mathrm{Hom}_R(A, D) \xrightarrow{\delta'_0} \mathrm{Ext}^1_R(C, D) \to \cdots$$
$$\cdots \to \mathrm{Ext}^i_R(C, D) \to \mathrm{Ext}^i_R(B, D) \to \mathrm{Ext}^i_R(A, D) \xrightarrow{\delta'_i} \mathrm{Ext}^{i+1}_R(C, D) \to \cdots$$

$$\cdots \to \mathrm{Tor}^R_i(C, D) \xrightarrow{\partial_i} \mathrm{Tor}^R_{i-1}(A, D) \to \mathrm{Tor}^R_{i-1}(B, D) \to \mathrm{Tor}^R_{i-1}(C, D) \to \cdots$$
$$\cdots \to A \otimes_R D \to B \otimes_R D \to C \otimes_R D \to 0$$

These are the sequences the section exists to produce: each short exact sequence, which the three functors truncate, is restored to an exact sequence by the derived functors, running off to infinity in the direction the functor was failing. The proofs are again omitted, and again for a reason that can be named: the connecting homomorphisms are built by the snake lemma applied to a short exact sequence of *complexes* rather than of modules, which is the general form of proposition 17.1.10, and that is developed in *Homological Algebra* [15].

Notice how the first sequence recovers a fact already proved. If D is projective then $\mathrm{Ext}^1_R(D, A) = 0$ by corollary 17.6.8, so the first sequence reads $0 \to \operatorname{Hom}_R(D, A) \to \operatorname{Hom}_R(D, B) \to \operatorname{Hom}_R(D, C) \to 0$, which is definition 17.2.3. In the same way $\operatorname{Tor}_1^R(C, D) = 0$ for D flat turns the third into the short exact sequence of proposition 17.5.3(1). The derived functors do not replace the three module classes; they say what happens when a module is not in them.

17.6.4 The Extension Problem, Answered

Section 17.1 defined $\mathcal{E}(C, A)$ and showed it has at least one element, the class of the split extension, and sometimes more.

Theorem 17.6.10: Ext^1 Classifies Extensions

Let R be a ring and let A and C be R-modules. Then there is a bijection

$$\operatorname{Ext}_R^1(C, A) \longleftrightarrow \mathcal{E}(C, A)$$

between the underlying set of $\operatorname{Ext}_R^1(C, A)$ and the set of equivalence classes of extensions of C by A, under which 0 corresponds to the class of the split extension $A \oplus C$. The addition of $\operatorname{Ext}_R^1(C, A)$ transports to an operation on $\mathcal{E}(C, A)$, the *Baer sum*, making it an abelian group.

This is the theorem the chapter opened with, and it is the last of the three whose proof is omitted. Half of the construction is short, and it uses the following description of Ext^1, which also isolates why the theorem is plausible.

Proposition 17.6.11: Ext^1 as a Cokernel

Let $0 \to K \hookrightarrow P_0 \xrightarrow{\varepsilon} M \to 0$ be exact with P_0 projective, and let N be any R-module. Then

$$\operatorname{Ext}_R^1(M, N) \cong \operatorname{coker}\Big(\operatorname{Hom}_R(P_0, N) \xrightarrow{\text{restrict}} \operatorname{Hom}_R(K, N)\Big)$$

***Proof*:**
Extend the presentation to a projective resolution by choosing a projective P_1 with a surjection $d\colon P_1 \twoheadrightarrow K$ and setting $d_1 = \iota \circ d$, where $\iota\colon K \to P_0$ is the inclusion, and continuing with a resolution of $\ker(d)$.

Since d is surjective, $d^*\colon \operatorname{Hom}_R(K, N) \to \operatorname{Hom}_R(P_1, N)$ is injective by proposition 17.3.1, and its image is $\ker(d_2^*)$: the sequence $P_2 \to P_1 \xrightarrow{d} K \to 0$ is exact, so applying the same proposition identifies $\operatorname{Hom}_R(K, N)$ with the kernel of $\operatorname{Hom}_R(P_1, N) \to \operatorname{Hom}_R(P_2, N)$. Under this identification $\operatorname{im}(d_1^*)$ becomes the image of the restriction map $\operatorname{Hom}_R(P_0, N) \to \operatorname{Hom}_R(K, N)$, because $d_1 = \iota \circ d$ gives $d_1^* = d^* \circ \iota^*$. Hence

$$\operatorname{Ext}_R^1(M, N) = \ker(d_2^*)/\operatorname{im}(d_1^*) \cong \operatorname{Hom}_R(K, N)/\operatorname{im}(\iota^*)$$

which is the stated cokernel.

Now the construction. An extension $0 \to A \to B \xrightarrow{\beta} C \to 0$ together with a projective presentation $0 \to K \to P \xrightarrow{\varepsilon} C \to 0$ produce, by projectivity of P applied to the surjection β, a map $P \to B$ over

C; it carries K into A, and the class of the resulting $K \to A$ in the cokernel of proposition 17.6.11 is the invariant attached to the extension. What is genuinely work is that the assignment is well defined, bijective and additive, and that is carried out in *Homological Algebra* [15].

The theorem closes the loop with example 17.1.8(2), where the extensions of $\mathbb{Z}/p\mathbb{Z}$ by $\mathbb{Z}/p\mathbb{Z}$ were counted by hand. Proposition 17.6.7 gives

$$\operatorname{Ext}^1_{\mathbb{Z}}(\mathbb{Z}/p\mathbb{Z},\, \mathbb{Z}/p\mathbb{Z}) \cong (\mathbb{Z}/p\mathbb{Z})/p(\mathbb{Z}/p\mathbb{Z}) \cong \mathbb{Z}/p\mathbb{Z}$$

which has p elements, matching the $p-1$ pairwise inequivalent nonsplit extensions found there together with the split one. Likewise $\operatorname{Ext}^1_{\mathbb{Z}}(\mathbb{Z}/n\mathbb{Z}, \mathbb{Z}) \cong \mathbb{Z}/n\mathbb{Z}$ says there are exactly n equivalence classes of extensions of $\mathbb{Z}/n\mathbb{Z}$ by $\mathbb{Z}$, while example 17.1.6 exhibited only two non-isomorphic middle terms. There is no conflict: equivalence is finer than isomorphism, exactly as definition 17.1.7 warned, and Ext^1 counts the finer relation.

Finally, $\operatorname{Ext}^1_{\mathbb{Z}}(\mathbb{Z}/n\mathbb{Z}, \mathbb{Q}) = 0$ says every extension of $\mathbb{Z}/n\mathbb{Z}$ by $\mathbb{Q}$ splits, which is theorem 17.3.4(3) for the injective module $\mathbb{Q}$; and $\operatorname{Tor}^{\mathbb{Z}}_1(\mathbb{Z}/n\mathbb{Z}, A) = A[n]$ says the failure of $- \otimes \mathbb{Z}/n\mathbb{Z}$ to preserve an injection is measured by the n-torsion, which is theorem 17.5.7 read one module at a time.

Exercise 17.6.1

1. Using proposition 17.6.7, compute each of the following.
    1. $\operatorname{Ext}^1_{\mathbb{Z}}(\mathbb{Z}/4\mathbb{Z}, \mathbb{Z}/6\mathbb{Z})$ and $\operatorname{Tor}^{\mathbb{Z}}_1(\mathbb{Z}/4\mathbb{Z}, \mathbb{Z}/6\mathbb{Z})$
    2. $\operatorname{Ext}^1_{\mathbb{Z}}(\mathbb{Z}/n\mathbb{Z}, \mathbb{Q}/\mathbb{Z})$ and $\operatorname{Tor}^{\mathbb{Z}}_1(\mathbb{Z}/n\mathbb{Z}, \mathbb{Q}/\mathbb{Z})$
    3. $\operatorname{Ext}^1_{\mathbb{Z}}(\mathbb{Z}/n\mathbb{Z}, \mathbb{Z}_{p^\infty})$, for a prime p

2. Use theorem 17.6.10 and proposition 17.6.7 to say how many equivalence classes of extensions of $\mathbb{Z}/4\mathbb{Z}$ by $\mathbb{Z}$ there are. Then show that the possible middle terms are $\mathbb{Z} \oplus \mathbb{Z}/4\mathbb{Z}$, $\mathbb{Z}$ and $\mathbb{Z} \oplus \mathbb{Z}/2\mathbb{Z}$, so that strictly fewer isomorphism types occur than there are classes. *Hint*: the extension attached to k has middle term $(\mathbb{Z} \oplus \mathbb{Z})/\langle (4, -k) \rangle$.

3. Show directly from definition 17.6.3 that $\operatorname{Ext}^i_R(M, N) = 0$ for all $i \geq 1$ whenever M admits a projective resolution of length 0, and that $\operatorname{Ext}^i_R(M, N) = 0$ for all $i \geq 2$ whenever M admits one of length 1. Deduce that $\operatorname{Ext}^i_{\mathbb{Z}}(M, N) = 0$ for every $i \geq 2$ and every pair of abelian groups M, N. *Hint*: theorem 14.1.6.

4. Let R be a principal ideal domain. Show that $\operatorname{Tor}^R_1(M, N) = 0$ whenever either argument is torsion-free, and give an example over $R = \mathbb{Z}$ where $\operatorname{Tor}^{\mathbb{Z}}_1(M, N) \neq 0$.

5. Verify the first long exact sequence of theorem 17.6.9 by hand in one case: take $0 \to \mathbb{Z} \xrightarrow{\cdot 2} \mathbb{Z} \to \mathbb{Z}/2\mathbb{Z} \to 0$ and $D = \mathbb{Z}/2\mathbb{Z}$, compute every term using proposition 17.6.7, and check that the sequence you obtain is exact. Identify the connecting map δ_0.

6. Show that $\operatorname{Ext}^1_R(M, N) = 0$ for every N if and only if M is projective. *Hint*: one direction is corollary 17.6.8; for the other, take a projective presentation $0 \to K \to P \to M \to 0$ and use the description of Ext^1 as $\operatorname{coker}\big(\operatorname{Hom}_R(P, N) \to \operatorname{Hom}_R(K, N)\big)$ with $N = K$.

7. Explain why $\operatorname{Tor}_R(M)$ and $\operatorname{Tor}^R_1(M, N)$ are not the same kind of object, and compute both for $M = \mathbb{Z}/6\mathbb{Z}$ over $R = \mathbb{Z}$, taking $N = \mathbb{Z}/4\mathbb{Z}$ in the second.

17.7 Summary

One short exact sequence, three functors that truncate it, three classes of module that stop the truncation, and two families of modules that measure it when it happens. The tables below state what was established, each entry pointing at the result that establishes it.

Short exact sequences

Statement	**Reference**
$0 \to A \to B$ exact $\Leftrightarrow$ injective; $B \to C \to 0$ exact $\Leftrightarrow$ surjective	proposition 17.1.1
Short five lemma: α, γ injective (surjective) $\Rightarrow \beta$ is	proposition 17.1.9
Mini snake lemma: kernels and cokernels splice into one exact sequence	proposition 17.1.10
Splits $\Leftrightarrow$ a section exists $\Leftrightarrow$ a retraction exists	proposition 17.1.13
Splitting is carried along by any isomorphism of sequences	proposition 17.1.15
Equivalence of extensions is strictly finer than isomorphism	example 17.1.8(2)

The last row is what makes the extension problem a counting problem rather than a classification of middle terms, and it is what theorem 17.6.10 counts.

The three functors against exactness

Each functor below is exact at one end of a short exact sequence and can fail at the other. Naming the modules for which it does not fail is what the chapter is for. The first and third are covariant and the second contravariant.

Functor	**Always**	**Exact when**	**Reference**
$\mathrm{Hom}_R(D, -)$	left exact	D projective	proposition 17.2.1, definition 17.2.3
$\mathrm{Hom}_R(-, D)$	left exact	D injective	proposition 17.3.1, definition 17.3.3
$D \otimes_R -$	right exact	D flat	theorem 17.5.1, definition 17.5.4

For the contravariant row, *left exact* refers to the sequence after the outer terms have been exchanged, which is how proposition 17.3.1 states it. The two Hom functors can lose surjectivity; tensoring can lose injectivity. The failures are at opposite ends because $D \otimes_R -$ is a left adjoint and $\mathrm{Hom}_R(D, -)$ its right adjoint (theorem 15.3.9).

The three classes

Class	Equivalent descriptions	Reference
Projective P	maps out of P lift along surjections; every $0 \to L \to M \to P \to 0$ splits	theorem 17.2.4
	P is a direct summand of a free module; P has a dual basis	proposition 17.2.8
Injective Q	maps into Q extend along injections; every $0 \to Q \to M \to N \to 0$ splits	theorem 17.3.4
	every $g\colon I \to Q$ on a left ideal extends to R	theorem 17.3.5
Flat A	$1 \otimes \psi$ is injective for every injective ψ	proposition 17.5.3

Projective modules have a structural description and injective modules do not, which is why Baer's criterion exists: it replaces a quantifier over all modules by one over the ideals of R.

How the classes relate, and where the implications are strict

Statement	Witness	Reference
free $\Rightarrow$ projective $\Rightarrow$ flat	(always)	corollary 17.2.6, corollary 17.5.5
projective $\not\Rightarrow$ free	$\mathbb{Z}/2\mathbb{Z}$ over $\mathbb{Z}/6\mathbb{Z}$; F^n over $M_n(F)$	example 17.2.7
flat $\not\Rightarrow$ projective	$\mathbb{Q}$ over $\mathbb{Z}$	example 17.5.8(2)
injective $\not\Rightarrow$ flat	$\mathbb{Q}/\mathbb{Z}$ over $\mathbb{Z}$	example 17.5.8(3)
flat $\not\Rightarrow$ injective	$\mathbb{Z}$ over $\mathbb{Z}$	example 17.3.10(1)
free $\not\Rightarrow$ injective	$\mathbb{Z}$ over $\mathbb{Z}$	example 17.3.10(1)
Projective, injective and flat pass to direct summands	(always)	proposition 17.3.9

Over particular rings the classes collapse.

Ring	Collapse	Reference
A field	free = projective = injective = flat	example 17.3.10(4)
A PID	projective = free	proposition 17.2.11
A PID	injective = divisible	theorem 17.3.5(2)
A PID	flat = torsion-free	theorem 17.5.7
An integral domain	injective $\Rightarrow$ divisible, and not conversely	theorem 17.3.5(2)

Resolutions and hulls

Every module is a quotient of a projective one and a submodule of an injective one, so both kinds of resolution exist. The injective side needed real work, having no dual of a free module to hand.

Statement	**Reference**
Every module is a quotient of a free module	example 17.1.4(3)
Every module is a submodule of an injective module	lemma 17.3.14
Injective resolutions exist	corollary 17.3.16
Every module has an injective hull, unique over the module	theorem 17.3.19
Two projective presentations differ by a free summand	proposition 17.2.12
Divisible groups: $\bigoplus_p \bigoplus_{I_p} \mathbb{Z}_{p^\infty} \oplus \bigoplus_I \mathbb{Q}$, invariants determined	theorem 17.4.7

The last row is the classification of injective $\mathbb{Z}$-modules, since divisible and injective agree over $\mathbb{Z}$. It is the one starred division of the chapter, and nothing outside it depends on it.

Ext, Tor, and the answer

Statement	**Reference**
$\mathrm{Tor}_i^R(M,N) = H_i(P_\bullet \otimes_R N)$ and	definition 17.6.2,
$\mathrm{Ext}^i_R(M,N) = H^i(\mathrm{Hom}_R(P_\bullet, N))$	definition 17.6.3
$\mathrm{Tor}_0 = M \otimes_R N$ and $\mathrm{Ext}^0 = \mathrm{Hom}_R(M,N)$	proposition 17.6.6
$\mathrm{Ext}^i_{\mathbb{Z}}(\mathbb{Z}/n\mathbb{Z}, A) = A[n],\ A/nA,\ 0$ and	proposition 17.6.7
$\mathrm{Tor}_i^{\mathbb{Z}}(\mathbb{Z}/n\mathbb{Z}, A) = A/nA,\ A[n],\ 0$	
$\mathrm{Ext}^i(P,N) = 0$ for P projective; $\mathrm{Tor}_i(M,F) = 0$ for F flat	corollary 17.6.8
Independent of the resolution; flat resolutions suffice for Tor	theorem 17.6.5
Long exact sequences in each variable	theorem 17.6.9
$\mathrm{Ext}^1_R(C,A) \leftrightarrow \mathcal{E}(C,A)$, with 0 the split class	theorem 17.6.10

The last row answers the question the chapter opened with, and the row above it is what turns the answer into a computation: $\mathrm{Ext}^1_{\mathbb{Z}}(\mathbb{Z}/p\mathbb{Z}, \mathbb{Z}/p\mathbb{Z}) \cong \mathbb{Z}/p\mathbb{Z}$ has p elements, which is the count example 17.1.8(2) arrived at by hand.

Three of those statements are proved elsewhere. Theorem 17.6.5 and theorem 17.6.9 need the comparison theorem and chain homotopies, and theorem 17.6.10 needs the full construction of the correspondence; all three are in *Homological Algebra* [15], which takes up the three module classes of this chapter in the setting of an abelian category and takes the derived functors as its subject.

$18\star$

Morita Theory

$\star$ *Optional. This chapter is an aside; nothing later in the book depends on it.*

The rings R and $M_n(R)$ agree in almost nothing. One may be commutative; the other is not, once $n \geq 2$. Their unit groups are unrelated, and so are their underlying additive groups. Yet every statement about left R-modules translates into a statement about left $M_n(R)$-modules and back again, because the two module categories are equivalent. Chapter 12 met this phenomenon, named it *Morita equivalence*, and deferred it. This chapter settles it.

The result is Morita's theorem: every equivalence R-**Mod** $\to$ S-**Mod** is given by tensoring with a bimodule, and the bimodules that occur are exactly those that are finitely generated, projective, and large enough to reach every module. Most of what the proof needs is already available. Chapter 15 constructed the tensor product of bimodules and proved the Hom-tensor adjunction (theorem 15.3.9); chapter $17\star$ supplied projective modules and the dual basis lemma (proposition 17.2.8). One notion is new, and section 18.1 isolates it before section 18.2 proves the theorem and section 18.3 works out what an equivalence preserves and what it destroys.

Nothing in Part IV uses this chapter, which is why it is marked optional. Its consumers are further out in the series: Morita equivalence is the reduction that makes the representation theory of finite-dimensional algebras tractable, in *Core Representation Theory* [11], and the invariance that makes K_0 computable, in *Algebraic K-Theory* [6].

18.1 Generators and Progenerators

An equivalence of categories carries an object to one with the same categorical description, so the first question is which description singles out R among left R-modules. Three properties do, and each is phrased purely in terms of maps: R is projective (definition 17.2.3), it is finitely generated, and every left R-module is a quotient of a direct sum of copies of R. The first two have names already.

The third does not.

Definition 18.1.1: Generator

Let R be a ring. A left R-module P is a *generator* of R-**Mod** if for every left R-module M there is an index set I and a surjection

$$P^{(I)} := \bigoplus_{i \in I} P \twoheadrightarrow M$$

The module R is a generator: any M is the image of the map $R^{(M)} \to M$ sending the basis element indexed by m to m. So definition 18.1.1 asks of P exactly what makes that argument work for R.

Checking the definition directly means quantifying over every module and every index set, which is unworkable. The next definition gathers the images of all maps $P \to R$ into a single subset of R, and theorem 18.1.4 shows that this one subset already decides the question.

Definition 18.1.2: Trace Ideal

Let P be a left R-module. The *trace ideal* of P is

$$\mathrm{Tr}(P) := \sum_{f \in \mathrm{Hom}_R(P,R)} f(P) \subseteq R$$

the additive subgroup of R generated by the elements $f(p)$ for $f \in \mathrm{Hom}_R(P,R)$ and $p \in P$.

Lemma 18.1.3: The Trace Ideal Is Two-Sided

For every left R-module P, the set $\mathrm{Tr}(P)$ is a two-sided ideal of R.

***Proof*:**
It is an additive subgroup by construction, so only the two multiplications need checking, and since $\mathrm{Tr}(P)$ is generated additively by the elements $f(p)$ it is enough to check them on such an element.

For left multiplication, let $r \in R$. Then $r\,f(p) = f(rp)$ because f is R-linear, and $rp \in P$, so $r\,f(p) \in f(P) \subseteq \mathrm{Tr}(P)$.

For right multiplication, fix $a \in R$ and define $g\colon P \to R$ by $g(x) = f(x)a$. This g is additive, and for $r \in R$ and $x \in P$,

$$g(rx) = f(rx)a = \big(r f(x)\big)a = r\big(f(x)a\big) = r\,g(x)$$

so $g \in \mathrm{Hom}_R(P,R)$. Hence $f(p)a = g(p) \in \mathrm{Tr}(P)$, completing the proof.

Theorem 18.1.4: Characterizations of a Generator

Let P be a left R-module. The following are equivalent.

1. P is a generator of R-**Mod**.
2. $\mathrm{Tr}(P) = R$.
3. R is isomorphic to a direct summand of P^n for some finite $n \geq 1$.
4. The functor $\mathrm{Hom}_R(P, -)\colon R\text{-}\mathbf{Mod} \to \mathbf{Ab}$ is faithful.

***Proof*:**
(1) $\Rightarrow$ (2). Apply definition 18.1.1 to $M = R$, giving a surjection $\pi\colon P^{(I)} \to R$. Write $\iota_i\colon P \to P^{(I)}$ for the inclusion of the i-th summand and set $f_i := \pi \circ \iota_i \in \mathrm{Hom}_R(P, R)$. Since π is surjective there is $x \in P^{(I)}$ with $\pi(x) = 1$, and an element of a direct sum has only finitely many nonzero coordinates, so $x = \sum_{i\in F} \iota_i(p_i)$ for a finite $F \subseteq I$ and elements $p_i \in P$. Then

$$1 = \pi(x) = \sum_{i\in F} f_i(p_i) \in \mathrm{Tr}(P)$$

By lemma 18.1.3 the set $\mathrm{Tr}(P)$ is an ideal, and an ideal containing 1 is the whole ring, so $\mathrm{Tr}(P) = R$.

(2) $\Rightarrow$ (3). By (2) there are $f_1, \dots, f_n \in \mathrm{Hom}_R(P, R)$ and $p_1, \dots, p_n \in P$ with $\sum_{i=1}^n f_i(p_i) = 1$. Define

$$\varphi\colon P^n \to R, \quad \varphi(x_1, \dots, x_n) = \sum_{i=1}^n f_i(x_i), \qquad \sigma\colon R \to P^n, \quad \sigma(r) = (rp_1, \dots, rp_n)$$

Both are R-linear: φ because each f_i is, and σ because the action on P^n is coordinatewise. For every $r \in R$,

$$\varphi(\sigma(r)) = \sum_{i=1}^n f_i(rp_i) = r \sum_{i=1}^n f_i(p_i) = r$$

so σ is a section of φ. In particular φ is surjective, and proposition 17.1.13 applied to $0 \to \ker(\varphi) \to P^n \xrightarrow{\varphi} R \to 0$ shows that this sequence splits, so $P^n \cong \sigma(R) \oplus \ker(\varphi)$ with $\sigma(R) \cong R$.

(3) $\Rightarrow$ (4). Let $f\colon M \to N$ be a nonzero map of left R-modules; the claim is that $f_* = \mathrm{Hom}_R(P, f)$ is nonzero. Choose $m \in M$ with $f(m) \neq 0$ and let $g\colon R \to M$ be the R-linear map $g(r) = rm$, so $f(g(1)) = f(m) \neq 0$ and hence $f \circ g \neq 0$. By (3) there are R-linear maps $\iota\colon R \to P^n$ and $\rho\colon P^n \to R$ with $\rho \circ \iota = \mathrm{id}_R$. Then

$$(f \circ g \circ \rho) \circ \iota = f \circ g \circ (\rho \circ \iota) = f \circ g \neq 0$$

so $f \circ g \circ \rho\colon P^n \to N$ is nonzero. A map out of a finite direct sum vanishes if and only if its restriction to every summand vanishes, so there is a coordinate inclusion $\kappa_j\colon P \to P^n$ with $f \circ (g \circ \rho \circ \kappa_j) \neq 0$. Setting $h := g \circ \rho \circ \kappa_j \in \mathrm{Hom}_R(P, M)$ gives $f_*(h) \neq 0$.

(4) $\Rightarrow$ (1). Let M be a left R-module and put

$$T := \sum_{h\in \mathrm{Hom}_R(P,M)} h(P) \subseteq M$$

a submodule of M, being a sum of the submodules $h(P)$. Let $q\colon M \to M/T$ be the quotient map. Every $h \in \mathrm{Hom}_R(P, M)$ has $h(P) \subseteq T$, so $q \circ h = 0$; that is, q_* is the zero map on $\mathrm{Hom}_R(P, M)$. Since $\mathrm{Hom}_R(P, -)$ is faithful and q_* agrees with the map induced by the zero homomorphism, it follows that $q = 0$, hence $M/T = 0$ and $T = M$. The map

$$P^{(\mathrm{Hom}_R(P,M))} \to M$$

whose restriction to the summand indexed by h is h therefore has image $T = M$, so it is surjective. As M was arbitrary, P is a generator, completing the proof.

Condition (2) is the one to check in practice, since it trades a quantifier over all modules for a computation inside R. Condition (3) is the form section 18.2 uses: it presents R as a retract of a finite power of P, which is what allows an argument about R to be transported to P. Condition (4) is the categorical reading, that P is a generator exactly when no nonzero map between modules is invisible to P.

Definition 18.1.5: Progenerator

A left R-module P is a *progenerator* if it is finitely generated, projective, and a generator of R-**Mod**.

These are the three properties of R listed at the start of the section, so R is a progenerator. Theorem 18.2.8 will show that the progenerators are precisely the modules that can take its place.

Example 18.1.6: Progenerators and Two Near Misses

1. For any ring R and any $n \geq 1$, the module R^n is a progenerator. It is free, hence projective by corollary 17.2.6, and finitely generated; the coordinate projection $R^n \to R$ is surjective, so $\mathrm{Tr}(R^n) = R$.

2. *Finitely generated and projective, but not a generator.* Let $R = \mathbb{Z}/6\mathbb{Z}$ and $e = \overline{3}$, which satisfies $e^2 = \overline{9} = \overline{3} = e$. Then $R = Re \oplus R(1-e)$, so $P := Re = \{\overline{0}, \overline{3}\}$ is a direct summand of R, hence projective by theorem 17.2.4(4), and it is generated by one element. Every $f \in \mathrm{Hom}_R(P, R)$ satisfies
$$f(e) = f(e \cdot e) = e\, f(e) \in eR$$
and R is commutative here, so $f(P) = f(Re) = Rf(e) \subseteq Re$, giving $\mathrm{Tr}(P) = Re \neq R$. By theorem 18.1.4 the module P is not a generator. Concretely, every element of a direct sum of copies of P is annihilated by $\overline{2}$, while $\overline{1} \in R$ is not, so no such sum surjects onto R.

3. *A generator that is not projective.* Let $R = \mathbb{Z}$ and $P = \mathbb{Z} \oplus \mathbb{Z}/2\mathbb{Z}$. The projection onto the first coordinate lies in $\mathrm{Hom}_{\mathbb{Z}}(P, \mathbb{Z})$ and has image $\mathbb{Z}$, so $\mathrm{Tr}(P) = \mathbb{Z}$ and P is a generator by theorem 18.1.4. It is not projective: over the principal ideal domain $\mathbb{Z}$ projective and free coincide (proposition 17.2.11), and P has the nonzero torsion element $(0, \overline{1})$, so it is not free.

The two near misses show that neither condition in definition 18.1.5 implies the other, so both are genuine requirements on the bimodule that Morita's theorem produces.

Exercise 18.1.1

1. Let P be a left R-module and $n \geq 1$. Show that $\mathrm{Tr}(P^n) = \mathrm{Tr}(P)$, and deduce that P is a generator if and only if P^n is.
2. Show that every nonzero free left R-module is a generator of R-**Mod**, and that any module with a nonzero free direct summand is a generator.
3. Let $R = \mathbb{Z}/6\mathbb{Z}$. Decide, for each of the R-modules $\mathbb{Z}/2\mathbb{Z}$, $\mathbb{Z}/3\mathbb{Z}$, $\mathbb{Z}/6\mathbb{Z}$ and $\mathbb{Z}/2\mathbb{Z} \oplus \mathbb{Z}/3\mathbb{Z}$, whether it is a progenerator, justifying each answer using theorem 18.1.4.
4. Let P be a projective left R-module. Using the dual basis lemma (proposition 17.2.8), show that $\mathrm{Tr}(P)$ is idempotent, that is $\mathrm{Tr}(P)\,\mathrm{Tr}(P) = \mathrm{Tr}(P)$.
5. Let P be a generator of R-**Mod**. Show that $\mathrm{Hom}_R(P, M) = 0$ implies $M = 0$. Then show that this consequence does not characterize generators, by exhibiting a ring R and a non-generator P for which it still holds.

18.2 Morita's Theorem

The statement to be proved compares two module categories, so it needs the vocabulary for comparing categories. Functors are available (definition 8.0.4); the maps between functors are not, and the notion of two categories being "the same" rests on them.

Definition 18.2.1: Natural Transformation and Natural Isomorphism

Let $F, G\colon \mathcal{C} \to \mathcal{D}$ be functors. A *natural transformation* $\eta\colon F \Rightarrow G$ assigns to each object C of $\mathcal{C}$ a morphism $\eta_C\colon F(C) \to G(C)$ of $\mathcal{D}$ such that for every morphism $f\colon C \to C'$ of $\mathcal{C}$ the square

$$\begin{tikzcd} F(C) \arrow[r, "\eta_C"] \arrow[d, "F(f)"'] & G(C) \arrow[d, "G(f)"] \\ F(C') \arrow[r, "\eta_{C'}"'] & G(C') \end{tikzcd}$$

commutes. It is a *natural isomorphism*, written $F \cong G$, if every η_C is an isomorphism.

Definition 18.2.2: Equivalence of Categories

An *equivalence* between categories $\mathcal{C}$ and $\mathcal{D}$ is a pair of functors $F\colon \mathcal{C} \to \mathcal{D}$ and $G\colon \mathcal{D} \to \mathcal{C}$ together with natural isomorphisms $GF \cong \mathrm{id}_{\mathcal{C}}$ and $FG \cong \mathrm{id}_{\mathcal{D}}$. The functor G is a *quasi-inverse* of F, and $\mathcal{C}$ and $\mathcal{D}$ are *equivalent*.

An equivalence is bijective on morphism sets: F induces a bijection $\mathrm{Hom}_{\mathcal{C}}(C, C') \to \mathrm{Hom}_{\mathcal{D}}(FC, FC')$, with inverse obtained by applying G and conjugating by the natural isomorphism $GF \cong \mathrm{id}$. Every property of an object stated purely in terms of morphisms therefore transfers across an equivalence, which is what makes theorem 18.1.4(4) the useful characterization of a generator.

Definition 18.2.3: Morita Equivalence

Rings R and S are *Morita equivalent* if R-**Mod** and S-**Mod** are equivalent categories.

Now fix a left R-module P. Two rings act on it: R on the left, and its own endomorphism ring. Writing endomorphisms on the right makes the second action a right action, and that is the convention the whole section uses.

Proposition 18.2.4: The Bimodules Attached to a Module

Let P be a left R-module and set $S := \mathrm{End}_R(P)^{\mathrm{op}}$ and $Q := \mathrm{Hom}_R(P, R)$.

1. P is an (R, S)-bimodule under $p \cdot s := s(p)$.
2. Q is an (S, R)-bimodule under $(s \cdot q)(p) := q(s(p))$ and $(q \cdot r)(p) := q(p)r$.

***Proof*:**

1. Multiplication in S is $s_1 s_2 = s_2 \circ s_1$, so $p \cdot (s_1 s_2) = (s_2 \circ s_1)(p) = s_2(s_1(p)) = (p \cdot s_1) \cdot s_2$, and $p \cdot \mathrm{id}_P = p$; the action is additive in each variable because endomorphisms are additive. Compatibility with the left action is R-linearity of s:
$$(rp) \cdot s = s(rp) = r\, s(p) = r(p \cdot s)$$
2. First, $s \cdot q$ and $q \cdot r$ lie in Q: the map $p \mapsto q(s(p))$ is a composite of R-linear maps, and $(q \cdot r)(ap) = q(ap)r = a\, q(p)\, r = a\, (q \cdot r)(p)$. The left action is an action because
$$\big((s_1 s_2) \cdot q\big)(p) = q\big((s_1 s_2)(p)\big) = q\big(s_2(s_1(p))\big) = (s_2 \cdot q)(s_1(p)) = \big(s_1 \cdot (s_2 \cdot q)\big)(p)$$
using $(s_1 s_2)(p) = s_2(s_1(p))$ from the opposite multiplication. The right action is one because $(q \cdot (r_1 r_2))(p) = q(p) r_1 r_2 = ((q \cdot r_1) \cdot r_2)(p)$. They commute:
$$\big((s \cdot q) \cdot r\big)(p) = (s \cdot q)(p)\, r = q(s(p))\, r = (q \cdot r)(s(p)) = \big(s \cdot (q \cdot r)\big)(p)$$
completing the proof.

With P an (R, S)-bimodule and Q an (S, R)-bimodule, the two tensor products $P \otimes_S Q$ and $Q \otimes_R P$ are an (R, R)-bimodule and an (S, S)-bimodule respectively (proposition 15.2.1). Each carries an evaluation map to the ring on the corresponding side, and the entire theorem is the assertion that both are isomorphisms.

Proposition 18.2.5: The Morita Maps

With P, S and Q as in proposition 18.2.4, there are well-defined maps

$$\mu\colon P\otimes_S Q\to R,\quad \mu(p\otimes q)=q(p),\qquad \tau\colon Q\otimes_R P\to S,\quad \tau(q\otimes p)=\big[x\mapsto q(x)p\big]$$

the first a homomorphism of (R,R)-bimodules and the second of (S,S)-bimodules, and they satisfy the *associativity identities*

$$\mu(p\otimes q)\,p'=p\cdot\tau(q\otimes p'),\qquad q\cdot\mu(p\otimes q')=\tau(q\otimes p)\cdot q'$$

for all $p,p'\in P$ and $q,q'\in Q$.

***Proof*:**
Well-definedness. The assignment $(p,q)\mapsto q(p)$ is additive in each variable and satisfies $\mu(p{\cdot}s,q)=q(s(p))=(s\cdot q)(p)=\mu(p,s\cdot q)$, so it is S-balanced and induces μ on $P\otimes_S Q$. For τ, the map $x\mapsto q(x)p$ is R-linear since $q(ax)p=a\,q(x)p$, so it is an element of S; the assignment is additive in each variable and

$$\tau(q\cdot r,p)(x)=(q\cdot r)(x)p=q(x)rp=\tau(q,rp)(x)$$

so it is R-balanced and induces τ.

Bimodule linearity. For μ: $\mu(rp\otimes q)=q(rp)=r\,q(p)$ and $\mu(p\otimes q\cdot r)=(q\cdot r)(p)=q(p)r$. For τ, recall $s_1s_2=s_2\circ s_1$ in S. On the left,

$$\tau(s\cdot q\otimes p)(x)=(s\cdot q)(x)\,p=q(s(x))\,p=\big(\tau(q\otimes p)\circ s\big)(x)=\big(s\,\tau(q\otimes p)\big)(x)$$

and on the right, using R-linearity of s,

$$\tau(q\otimes p\cdot s)(x)=q(x)\,s(p)=s\big(q(x)p\big)=\big(s\circ\tau(q\otimes p)\big)(x)=\big(\tau(q\otimes p)\,s\big)(x)$$

The identities. For the first, $\mu(p\otimes q)p'=q(p)p'$, while $\tau(q\otimes p')$ is the endomorphism $x\mapsto q(x)p'$, so $p\cdot\tau(q\otimes p')=\tau(q\otimes p')(p)=q(p)p'$. For the second, evaluate both sides at $x\in P$:

$$\big(q\cdot\mu(p\otimes q')\big)(x)=q(x)\,q'(p),\qquad \big(\tau(q\otimes p)\cdot q'\big)(x)=q'\big(\tau(q\otimes p)(x)\big)=q'\big(q(x)p\big)=q(x)\,q'(p)$$

as we sought to show.

The two maps measure exactly the two conditions of definition 18.1.5, one each.

Lemma 18.2.6: What the Morita Maps Detect

With notation as above:

1. $\operatorname{im}\mu=\operatorname{Tr}(P)$, so μ is surjective if and only if P is a generator.
2. τ is surjective if and only if P is finitely generated and projective.

***Proof*:**

1. The image of μ is the additive subgroup generated by the elements $q(p)$ for $q \in \mathrm{Hom}_R(P, R)$ and $p \in P$, which is definition 18.1.2 verbatim. The equivalence with being a generator is theorem 18.1.4, (1) $\Leftrightarrow$ (2), since a subgroup of R equals R exactly when the map onto it is surjective.

2. ($\Rightarrow$) If τ is surjective, then $\mathrm{id}_P = \tau(t)$ for some $t \in Q \otimes_R P$, and every element of a tensor product is a finite sum of simple tensors, say $t = \sum_{i=1}^{n} q_i \otimes p_i$. Evaluating at $x \in P$,

$$x = \mathrm{id}_P(x) = \sum_{i=1}^{n} q_i(x)\, p_i$$

which is a finite dual basis, so P is finitely generated and projective by proposition 17.2.8.

($\Leftarrow$) If P is finitely generated and projective, proposition 17.2.8 supplies a finite dual basis $(p_i, q_i)_{i=1}^{n}$, and the displayed identity says $\tau\big(\sum_i q_i \otimes p_i\big) = \mathrm{id}_P$. The image of τ is a two-sided ideal of S, being the image of an (S, S)-bimodule map, and an ideal containing the identity is the whole ring, completing the proof.

Surjectivity of the two maps is therefore exactly the progenerator hypothesis. What is not yet clear is injectivity, and the next lemma is the reason the theorem needs no separate argument for it: each map is forced to be injective by its own surjectivity, through the associativity identities that tie it to the other.

Lemma 18.2.7: Surjective Forces Bijective

If μ is surjective then μ is injective. If τ is surjective then τ is injective.

***Proof*:**
Suppose μ is surjective and let $t = \sum_{i=1}^{m} p_i \otimes q_i$ lie in $\ker \mu$, so that $\sum_i q_i(p_i) = 0$. Surjectivity gives elements $a_j \in P$ and $b_j \in Q$ with $1 = \sum_{j=1}^{k} \mu(a_j \otimes b_j)$. Then

$$t = \sum_i \Big(\sum_j \mu(a_j \otimes b_j)\Big) p_i \otimes q_i = \sum_{i,j} \big(\mu(a_j \otimes b_j)\, p_i\big) \otimes q_i = \sum_{i,j} \big(a_j \cdot \tau(b_j \otimes p_i)\big) \otimes q_i$$

using the first associativity identity of proposition 18.2.5. Moving the element of S across the tensor and applying the second identity,

$$t = \sum_{i,j} a_j \otimes \big(\tau(b_j \otimes p_i) \cdot q_i\big) = \sum_{i,j} a_j \otimes \big(b_j \cdot \mu(p_i \otimes q_i)\big) = \sum_j a_j \otimes \Big(b_j \cdot \sum_i \mu(p_i \otimes q_i)\Big)$$

and the inner sum is $\mu(t) = 0$, so $t = 0$. The argument for τ is the same with the roles of the two identities and of (P, R) and (Q, S) interchanged, completing the proof.

Theorem 18.2.8: Morita's Theorem

Let P be a progenerator in R-**Mod**, and set $S = \mathrm{End}_R(P)^{\mathrm{op}}$ and $Q = \mathrm{Hom}_R(P, R)$. Then

$$\mu\colon P \otimes_S Q \xrightarrow{\sim} R, \qquad \tau\colon Q \otimes_R P \xrightarrow{\sim} S$$

are isomorphisms of bimodules, and the functors

$$Q \otimes_R -\colon R\text{-}\mathbf{Mod} \to S\text{-}\mathbf{Mod}, \qquad P \otimes_S -\colon S\text{-}\mathbf{Mod} \to R\text{-}\mathbf{Mod}$$

are mutually quasi-inverse equivalences. In particular R and S are Morita equivalent.

***Proof*:**
Since P is a generator, μ is surjective by lemma 18.2.6(1); since P is finitely generated and projective, τ is surjective by lemma 18.2.6(2). Both are then injective by lemma 18.2.7, hence bijective, and they are bimodule maps by proposition 18.2.5.

For the functors, let N be a left S-module. Associativity of the tensor product (theorem 15.2.8) and the isomorphism just established give

$$Q \otimes_R \big(P \otimes_S N\big) \cong \big(Q \otimes_R P\big) \otimes_S N \cong S \otimes_S N \cong N$$

and symmetrically, for a left R-module M,

$$P \otimes_S \big(Q \otimes_R M\big) \cong \big(P \otimes_S Q\big) \otimes_R M \cong R \otimes_R M \cong M$$

Each isomorphism in these chains is natural in the module argument, since theorem 15.2.8 is natural and the remaining two steps are the unit isomorphisms of the tensor product. The two chains are therefore natural isomorphisms $(Q\otimes_R -)\circ(P\otimes_S -) \cong \mathrm{id}$ and $(P\otimes_S -)\circ(Q\otimes_R -) \cong \mathrm{id}$, which is definition 59.5.5, completing the proof.

The theorem manufactures an equivalence from every progenerator. The converse says that this is the only source: an equivalence cannot arise any other way, and the progenerator producing it is recovered as the image of the ring.

Theorem 18.2.9: Every Equivalence Is Tensoring with a Progenerator

Let $F\colon R\text{-}\mathbf{Mod} \to S\text{-}\mathbf{Mod}$ be an equivalence with quasi-inverse G. Put $Q := F(R)$ and $P := G(S)$. Then Q carries a right R-action making it an (S, R)-bimodule, P carries a right S-action making it an (R, S)-bimodule,

$$P \otimes_S Q \cong R \ \text{ as } (R, R)\text{-bimodules}, \qquad Q \otimes_R P \cong S \ \text{ as } (S, S)\text{-bimodules}$$

and $F \cong Q \otimes_R -$. Moreover P is a progenerator in R-**Mod**.

***Proof*:**
Step 1: the right action on Q. For $r \in R$ let $\rho_r\colon R \to R$ be $x \mapsto xr$, which is left R-linear. Since $\rho_{r_1 r_2} = \rho_{r_2} \circ \rho_{r_1}$ and F is a functor, $F(\rho_{r_1 r_2}) = F(\rho_{r_2}) \circ F(\rho_{r_1})$. Setting $q \cdot r := F(\rho_r)(q)$ therefore defines a right R-action on Q, and it commutes with the left S-action because each

$F(\rho_r)$ is a map of S-modules. The same construction gives the right S-action on P.

Step 2: F preserves direct sums and cokernels. An equivalence is bijective on morphism sets, as recorded after definition 59.5.5, and it is essentially surjective, every object of S-**Mod** being isomorphic to $F(G(N))$. A cocone is a direct sum, or a cokernel, exactly when a certain map of morphism sets is bijective for every test object; bijectivity on morphism sets transports the condition forward and essential surjectivity supplies every test object on the far side. So F carries direct sums to direct sums and cokernels to cokernels.

Step 3: $F \cong Q \otimes_R -$. For a left R-module M and $m \in M$ let $\lambda_m \colon R \to M$ be $r \mapsto rm$. The assignment $(q, m) \mapsto F(\lambda_m)(q)$ is additive in each variable and R-balanced, since $\lambda_{rm} = \lambda_m \circ \rho_r$ gives $F(\lambda_{rm})(q) = F(\lambda_m)(q \cdot r)$. It therefore induces $\theta_M \colon Q \otimes_R M \to F(M)$, and θ is natural in M because $f \circ \lambda_m = \lambda_{f(m)}$ for any $f \colon M \to M'$.

At $M = R$ the map θ_R is the isomorphism $Q \otimes_R R \cong Q = F(R)$, since $\lambda_r = \rho_r$. Both $Q \otimes_R -$ and F preserve direct sums, the first by corollary 15.2.5 and the second by Step 2, so $\theta_{R^{(I)}}$ is an isomorphism for every index set I. Now take any M and a free presentation $R^{(J)} \to R^{(I)} \to M \to 0$, which exists because every module is a quotient of a free one and the kernel of that quotient is again such a quotient. Both functors are right exact, again by theorem 17.5.1 and Step 2, so the rows of

$$\begin{tikzcd}
Q \otimes_R R^{(J)} \arrow[r] \arrow[d, "\theta_{R^{(J)}}"'] & Q \otimes_R R^{(I)} \arrow[r] \arrow[d, "\theta_{R^{(I)}}"'] & Q \otimes_R M \arrow[r] \arrow[d, "\theta_M"] & 0 \\
F(R^{(J)}) \arrow[r] & F(R^{(I)}) \arrow[r] & F(M) \arrow[r] & 0
\end{tikzcd}$$

are exact and the square commutes by naturality. The two left vertical maps are isomorphisms, and a cokernel is determined up to unique isomorphism by the map it is taken of, so θ_M is an isomorphism.

Step 4: the bimodule isomorphisms. By Step 3 applied to F and to G, the composite $GF \cong P \otimes_S (Q \otimes_R -) \cong (P \otimes_S Q) \otimes_R -$ using theorem 15.2.8. Since $GF \cong \mathrm{id}$, evaluating at R gives $(P \otimes_S Q) \otimes_R R \cong R$, that is $P \otimes_S Q \cong R$; the isomorphism is one of (R, R)-bimodules because the natural isomorphism $GF \cong \mathrm{id}$ commutes with the maps ρ_r, whose images generate the right action. Symmetrically $Q \otimes_R P \cong S$.

Step 5: P is a progenerator. From $P \otimes_S Q \cong R$: the tensor product is generated as an abelian group by simple tensors, so the map $\bigoplus_{q \in Q} P \to P \otimes_S Q$ sending the summand indexed by q to $p \mapsto p \otimes q$ is surjective, exhibiting R as a quotient of a direct sum of copies of P. Hence P is a generator. From $Q \otimes_R P \cong S$: the element of $Q \otimes_R P$ corresponding to 1_S is a *finite* sum $\sum_{i=1}^n q_i \otimes p_i$, and unwinding the isomorphism as in lemma 18.2.6(2) yields $x = \sum_i q_i(x) p_i$ for all $x \in P$, a finite dual basis. So P is finitely generated and projective by proposition 17.2.8, completing the proof.

Taken together the two theorems say that Morita equivalence is not an exotic relation to be checked category by category: R and S are Morita equivalent precisely when $S \cong \mathrm{End}_R(P)^{\mathrm{op}}$ for some progenerator P over R. The next example is the case that motivated the whole discussion.

Example 18.2.10: A Ring and Its Matrix Rings

Let R be any ring and $n \geq 1$, and take $P = R^n$, the module of columns. By example 18.1.6(1) it is a progenerator. Its endomorphism ring is computed one summand at a time: an R-linear map $R \to R$ is right multiplication by its value at 1, so $\mathrm{End}_R(R) \cong R^{\mathrm{op}}$, and therefore

$$\mathrm{End}_R(R^n) \cong M_n\big(\mathrm{End}_R(R)\big) \cong M_n(R^{\mathrm{op}}) \cong M_n(R)^{\mathrm{op}}$$

the last isomorphism being transposition, which reverses the order of multiplication. Hence

$$S = \mathrm{End}_R(R^n)^{\mathrm{op}} \cong M_n(R)$$

and theorem 18.2.8 gives an equivalence R-**Mod** $\simeq M_n(R)$-**Mod**. Unwinding it, $Q = \mathrm{Hom}_R(R^n, R) \cong R^n$ as rows, the equivalence R-**Mod** $\to M_n(R)$-**Mod** sends M to the column space M^n with $M_n(R)$ acting by matrix multiplication, and its quasi-inverse sends N to eN where $e = E_{11}$ is the idempotent matrix with a single 1 in the top-left entry.

Nothing about R is recovered by $M_n(R)$ beyond its module category: for $n \geq 2$ the ring $M_n(R)$ is never commutative, so a commutative R is Morita equivalent to noncommutative rings. Section 18.3 makes precise which properties survive.

Exercise 18.2.1

1. Let P be a progenerator over R and $S = \mathrm{End}_R(P)^{\mathrm{op}}$. Show that $\mathrm{Hom}_R(P, -)$ and $Q \otimes_R -$ are naturally isomorphic as functors R-**Mod** $\to S$-**Mod**, where $Q = \mathrm{Hom}_R(P, R)$.
2. Show that Morita equivalence is an equivalence relation on rings.
3. Show that if R and S are Morita equivalent, then so are $M_n(R)$ and $M_n(S)$ for every $n \geq 1$.
4. Show that a commutative ring R and the matrix ring $M_n(R)$ have isomorphic centres for every $n \geq 1$, and explain why example 18.2.10 does not contradict this.
5. Let D be a division ring. Show that every progenerator over D is isomorphic to D^n for some $n \geq 1$, and deduce that the rings Morita equivalent to D are exactly the matrix rings $M_n(D)$.
6. Give an example of two rings that are Morita equivalent but not isomorphic, and an example of two rings that are not Morita equivalent even though both are commutative and have the same cardinality.
7. Let P be a progenerator over R. Show that P is also a progenerator when regarded as a right S-module, where $S = \mathrm{End}_R(P)^{\mathrm{op}}$.

18.3 Morita Invariants and the Basic Algebra

Example 18.2.10 showed that R and $M_n(R)$ have the same module category while differing in nearly every element-level respect. A property of a ring that transfers across every Morita equivalence is called a *Morita invariant*, and the question this section answers is which properties those are. The test is uniform: a property is a Morita invariant when it can be stated in terms of the module category alone, and it fails to be one as soon as a single equivalence moves it.

Commutativity is the basic failure. For $n \geq 2$ the ring $M_n(R)$ is noncommutative whatever R is, so no property implying commutativity survives. What does survive is the part of the multiplication that is visible categorically, and theorem 18.3.2 identifies it exactly: the center.

Proposition 18.3.1: The Center as Endomorphisms of the Identity

Let R be a ring. Sending $c \in Z(R)$ to the family of maps $\eta^c_M \colon M \to M$, $\eta^c_M(m) = cm$, is a ring isomorphism from $Z(R)$ (definition 9.1.8) to the ring of natural transformations $\mathrm{id}_{R\text{-}\mathbf{Mod}} \Rightarrow \mathrm{id}_{R\text{-}\mathbf{Mod}}$, under pointwise addition and composition.

***Proof*:**
The map is well defined. For $c \in Z(R)$ each η^c_M is R-linear, since $\eta^c_M(rm) = c(rm) = (cr)m = (rc)m = r(cm) = r\,\eta^c_M(m)$, which is exactly where centrality is used. It is natural: for $f \colon M \to N$ and $m \in M$,

$$f\big(\eta^c_M(m)\big) = f(cm) = c\,f(m) = \eta^c_N\big(f(m)\big)$$

Injectivity. If $\eta^c = \eta^{c'}$ then in particular $c = \eta^c_R(1) = \eta^{c'}_R(1) = c'$.

Surjectivity. Let η be a natural transformation of the identity and put $c := \eta_R(1)$. For $r \in R$, R-linearity of η_R gives $\eta_R(r) = \eta_R(r \cdot 1) = r\,\eta_R(1) = rc$. Naturality applied to the right multiplication $\rho_r \colon R \to R$, $x \mapsto xr$, which is a map of left R-modules, gives $\eta_R(xr) = \eta_R(x)r$; at $x = 1$ this reads $\eta_R(r) = cr$. Comparing the two expressions, $rc = cr$ for every $r \in R$, so $c \in Z(R)$.

It remains to see that $\eta = \eta^c$ on every module, not only on R. For $m \in M$ let $\lambda_m \colon R \to M$ be $r \mapsto rm$. Naturality gives

$$\eta_M(m) = \eta_M\big(\lambda_m(1)\big) = \lambda_m\big(\eta_R(1)\big) = \lambda_m(c) = cm = \eta^c_M(m)$$

Ring structure. Addition is pointwise on both sides. For multiplication, $\eta^c_M\big(\eta^{c'}_M(m)\big) = c(c'm) = (cc')m = \eta^{cc'}_M(m)$, so composition corresponds to multiplication in $Z(R)$, and $\eta^1 = \mathrm{id}$, completing the proof.

The right-hand side of proposition 18.3.1 mentions only the category R-**Mod**, never the ring R. That is the whole content of the next theorem.

Theorem 18.3.2: The Center Is a Morita Invariant

If R and S are Morita equivalent, then $Z(R) \cong Z(S)$ as rings.

***Proof*:**
Let $F \colon R\text{-}\mathbf{Mod} \to S\text{-}\mathbf{Mod}$ be an equivalence with quasi-inverse G, and let $\alpha \colon GF \cong \mathrm{id}$ and $\beta \colon FG \cong \mathrm{id}$ be the natural isomorphisms of definition 59.5.5. Given a natural transformation η of $\mathrm{id}_{R\text{-}\mathbf{Mod}}$, define

$$\Phi(\eta)_N := \beta_N \circ F\big(\eta_{G(N)}\big) \circ \beta_N^{-1} \colon N \to N \qquad (N \in S\text{-}\mathbf{Mod})$$

Each component is a composite of S-linear maps. It is natural in N: for $g \colon N \to N'$, naturality of β gives $g \circ \beta_N = \beta_{N'} \circ FG(g)$, and naturality of η at $G(g)$ gives $\eta_{G(N')} \circ G(g) = G(g) \circ \eta_{G(N)}$, so applying F and conjugating,

$$g \circ \Phi(\eta)_N = \beta_{N'} \circ F\big(G(g) \circ \eta_{G(N)}\big) \circ \beta_N^{-1} = \beta_{N'} \circ F\big(\eta_{G(N')} \circ G(g)\big) \circ \beta_N^{-1} = \Phi(\eta)_{N'} \circ g$$

So Φ carries natural transformations of $\mathrm{id}_{R\text{-}\mathbf{Mod}}$ to natural transformations of $\mathrm{id}_{S\text{-}\mathbf{Mod}}$. It preserves addition because F is additive on morphism sets and conjugation is additive, and it preserves composition because the conjugating isomorphisms cancel in the middle:

$$\Phi(\eta)_N \circ \Phi(\eta')_N = \beta_N \circ F\big(\eta_{G(N)}\big) \circ F\big(\eta'_{G(N)}\big) \circ \beta_N^{-1} = \beta_N \circ F\big(\eta_{G(N)} \circ \eta'_{G(N)}\big) \circ \beta_N^{-1}$$

The same construction applied to G gives Ψ in the opposite direction, and $\Psi\Phi = \mathrm{id}$ follows from α being a natural isomorphism by the same cancellation. Hence Φ is a ring isomorphism, and proposition 18.3.1 on each side identifies its source and target with $Z(R)$ and $Z(S)$, completing the proof.

Two consequences follow at once, and together they say that Morita equivalence is a coarser relation than isomorphism on noncommutative rings while collapsing to isomorphism on commutative ones.

Corollary 18.3.3: Commutative Rings Are Morita Rigid

Let R and S be commutative rings. Then R and S are Morita equivalent if and only if $R \cong S$.

***Proof*:**
If $R \cong S$ then the isomorphism induces an equivalence of module categories by restriction of scalars along it. Conversely, if R and S are Morita equivalent then $Z(R) \cong Z(S)$ by theorem 18.3.2, and commutativity gives $Z(R) = R$ and $Z(S) = S$ by definition 9.1.8, so $R \cong S$, as we sought to show.

Example 18.3.4: Morita Equivalence Is Coarser than Isomorphism
Take R any ring and $n \geq 2$. Then R and $M_n(R)$ are Morita equivalent by example 18.2.10 but not isomorphic whenever R is commutative, since $M_n(R)$ is not commutative: the matrix units E_{11} and E_{12} satisfy $E_{11}E_{12} = E_{12}$ while $E_{12}E_{11} = 0$. Consistently with theorem 18.3.2, the center is preserved: $Z(M_n(R))$ consists of the scalar matrices cI with $c \in Z(R)$, so $Z(M_n(R)) \cong Z(R)$.

By corollary 18.3.3 no two non-isomorphic commutative rings are Morita equivalent, so every instance of the phenomenon involves a noncommutative ring.

The remaining business of the section is the construction that produces those equivalences in practice. Example 18.1.6(2) showed that Re is projective and finitely generated for an idempotent e, and computed its trace ideal in a special case; the general computation is what decides when Re is a progenerator.

Definition 18.3.5: Full Idempotent

An idempotent $e \in R$, so $e^2 = e$, is *full* if $ReR = R$, that is if the two-sided ideal generated by e is the whole ring.

Theorem 18.3.6: The Corner Ring of a Full Idempotent

Let $e \in R$ be an idempotent. Then

1. Re is a finitely generated projective left R-module with $\mathrm{Tr}(Re) = ReR$, so Re is a progenerator if and only if e is full;
2. $\mathrm{End}_R(Re)^{\mathrm{op}} \cong eRe$, the *corner ring* of e.

Consequently, if e is full then R and eRe are Morita equivalent.

***Proof*:**

1. Since $e^2 = e$, the element $1 - e$ is also idempotent and $R = Re \oplus R(1 - e)$: every r decomposes as $r = re + r(1-e)$, and if x lies in the intersection, written both as $x = re$ and as $x = r'(1 - e)$, then multiplying on the right by e gives $x = re \cdot e = xe = r'(1 - e)e = 0$, the last step because $(1 - e)e = e - e^2 = 0$. So Re is a direct summand of R, hence projective by theorem 17.2.4(4), and it is generated by the single element e.

   For the trace ideal, take $f \in \mathrm{Hom}_R(Re, R)$. Then $f(e) = f(e \cdot e) = e\, f(e)$, so $f(e) \in eR$, and since every element of Re has the form re,

   $$f(Re) = f(Re \cdot e) = R\, f(e) \subseteq ReR$$

   Conversely, for each $a \in R$ the map $f_a \colon Re \to R$ given by $f_a(x) = xa$ is left R-linear with image Rea. Summing over $a \in R$ gives $\mathrm{Tr}(Re) \supseteq \sum_a Rea = ReR$. Hence $\mathrm{Tr}(Re) = ReR$, and theorem 18.1.4 makes Re a generator exactly when $ReR = R$. Combined with the previous paragraph, Re is a progenerator exactly when e is full.

2. A map $\varphi \in \mathrm{End}_R(Re)$ is determined by $\varphi(e)$, since $\varphi(re) = r\varphi(e)$, and $\varphi(e) = \varphi(e \cdot e) = e\,\varphi(e)$ shows $\varphi(e) \in eRe$ (it already lies in Re). Conversely each $x \in eRe$ gives $\varphi_x(re) = rex = rx$, which is left R-linear and lands in Re because $x = ex'e \in Re$. The two assignments are mutually inverse and additive.

   For the multiplication, let $\varphi(e) = ebe$ and $\psi(e) = eae$. Then

   $$(\varphi \circ \psi)(e) = \varphi(eae) = \varphi\big((ea) \cdot e\big) = (ea)\,\varphi(e) = ea\,ebe = (eae)(ebe) = \psi(e)\,\varphi(e)$$

   using $e^2 = e$ in the last-but-one step. So $\varphi \mapsto \varphi(e)$ reverses multiplication, giving $\mathrm{End}_R(Re) \cong (eRe)^{\mathrm{op}}$ and therefore $\mathrm{End}_R(Re)^{\mathrm{op}} \cong eRe$.

For the final claim, if e is full then Re is a progenerator by (1), so theorem 18.2.8 applied to $P = Re$ gives an equivalence between R-**Mod** and $\mathrm{End}_R(Re)^{\mathrm{op}}$-**Mod** $= eRe$-**Mod** by (2), completing the proof.

Theorem 18.3.6 is the form in which Morita theory is used downstream. Passing from R to eRe for a full idempotent shrinks the ring while keeping the module category, and choosing e well makes the smaller ring markedly more tractable. The matrix case is the choice $e = E_{11}$ in $R = M_n(A)$, where $eRe \cong A$ recovers example 18.2.10. In the representation theory of finite-dimensional algebras the same construction selects one idempotent from each isomorphism class of indecomposable projectives and produces the *basic algebra*, over which every algebra's representation theory can be read; that development, which needs the semisimple structure theory this book does not carry, is in *Core Representation Theory* [11].

Example 18.3.7: The Corner Ring of a Non-Full Idempotent

Fullness is not automatic. Let $R = \mathbb{Z}/6\mathbb{Z}$ and $e = \overline{3}$, as in example 18.1.6(2). Then $ReR = Re = \{\overline{0}, \overline{3}\}$, which is not R, so e is not full. The corner ring is $eRe = \{\overline{0}, \overline{3}\} \cong \mathbb{Z}/2\mathbb{Z}$, and $\mathbb{Z}/6\mathbb{Z}$ is not Morita equivalent to $\mathbb{Z}/2\mathbb{Z}$: both are commutative and not isomorphic, so corollary 18.3.3 rules it out. The construction still produces a ring, but without fullness it produces the wrong one.

One invariant is worth recording separately, because it is the reason *Algebraic K-Theory* [6] cares about this section at all.

Corollary 18.3.8: Finitely Generated Projectives Correspond

Let $F\colon R\textbf{-Mod} \to S\textbf{-Mod}$ be an equivalence. Then F restricts to a bijection between the isomorphism classes of finitely generated projective R-modules and those of finitely generated projective S-modules, and this bijection preserves direct sums.

***Proof*:**

Projectivity is preserved: M is projective exactly when $\mathrm{Hom}_R(M, -)$ carries surjections to surjections (definition 17.2.3), F is bijective on morphism sets, and F carries surjections to surjections because a map is surjective exactly when its cokernel vanishes and F preserves cokernels by theorem 18.2.9, Step 2.

Being finitely generated is preserved because $F \cong Q \otimes_R -$ with $Q \otimes_R R \cong Q$ (theorem 18.2.9), so F carries R^n to Q^n and a summand of R^n to a summand of Q^n. That Q^n is itself finitely generated over S is theorem 18.2.9 applied with the roles of F and G exchanged, which makes $Q = F(R)$ a progenerator in $S\textbf{-Mod}$; the same argument then runs backwards through G. Direct sums are preserved by Step 2 of the same proof, and the quasi-inverse G supplies the inverse bijection on isomorphism classes, completing the proof.

Since K_0 of a ring is built from the isomorphism classes of finitely generated projectives under direct sum, corollary 18.3.8 says that K_0 cannot distinguish Morita equivalent rings. That statement, and the group it is about, belong to *Algebraic K-Theory* [6].

Exercise 18.3.1

1. Show that the center of $M_n(R)$ consists of the scalar matrices rI with $r \in Z(R)$, and deduce $Z(M_n(R)) \cong Z(R)$ directly, without using theorem 18.3.2.
2. Decide which of the following are Morita invariants, with proof or counterexample: being a division ring; being commutative; being an integral domain (definition 9.1.21); having exactly two two-sided ideals.
3. Show that if R is commutative then an idempotent $e \in R$ is full if and only if $e = 1$, and give a noncommutative ring with a full idempotent $e \neq 1$.
4. Let $R = M_2(\mathbb{Z})$ and $e = E_{11}$. Verify directly that e is full, identify Re and eRe explicitly, and describe the equivalence $R\textbf{-Mod} \simeq \mathbb{Z}\textbf{-Mod}$ it produces.

5. Show that if e and f are idempotents of R with $Re \cong Rf$ as R-modules, then $eRe \cong fRf$ as rings.

6. Let R be a ring in which $1 = e_1 + \cdots + e_n$ for orthogonal idempotents e_i, meaning $e_i e_j = 0$ for $i \neq j$. Show that $R \cong \bigoplus_i Re_i$ as left R-modules, and that e_1 is full whenever $Re_1 \cong Re_i$ for every i.

Part IV

Fields

When studying classical algebra, a lot of time and energy is dedicated to solving equations. Famously, for any equation of the form $ax^2 + bx + c = 0$ has solution

$$x = \frac{-b - \sqrt{b^2 - 4ac}}{2a}$$

which was known since at least the time of ancient Egypt (though not in the current algebraic form[1])After a long gap, Niccolò Tartaglia discovers the general equation for the cubic equation in terms by using an "imaginary" number i as a trick to solve the problem of $\sqrt{-n}$ coming up (it will take some more time before i is considered a number and not simply a tool to solve equations). Using this equation, a solution for quartic equation with respect to x was also found. The mission was on in finding a solution to the quintic equation. However, over 200 years, no such equation was found. Abel and Galois have in fact proved that only using $+, -, \times, \div$ and $\sqrt{}$, it is *impossible* to isolate x!

This is the origin of the study of field theory and Galois theory, and will be the driving motivation in this part of the book, however the study of fields have in modern times moves past this and are now used as a primary tool in the study of algebraic number theory. In chapter 50, we will use the language of field theory as the language of algebraic number theory.

[1]It was a geometric proof involving fitting squares of different sizes together

19

Field Theory

A reminder that a field F is a unital commutative division ring (definition 9.1.11). Another way of thinking of a field is that it's a set with two binary operations both acting like abelian groups. Yet another way of seeing them is that $F^\times = F - \{0\}$ is an abelian group. In this chapter, we'll be mostly thinking about fields as unital commutative divsion rings, in particular, we're going to develop how the trivial ideal structure of fields (proposition 9.4.10) lets us uniquely define some extensions of a field. We will denote $\mathbb{F}_p := \mathbb{Z}/p\mathbb{Z}$ and usually use the letters k and K to refer to fields.

Furthermore, the complex numbers $\mathbb{C}$ will be referenced for it's nice algebraic properties (ex. $i^2 = 1$ or $z = a + bi$ for any $z \in \mathbb{C}$). If the reader is unfamiliar with these notion, there are many resources online explaining these concepts[1].

19.1 Properties of Fields

First, all fields are rings, and so we can form the category **Fld** with fields as objects and ring homomorphisms as morphisms. We will use the term field homomorphsisms sometimes since the objects of study are fields with the knowledge that they are just ring homomorphisms. In general, the term ring homomorphism will be reserved for when rings will be brought into the discussion as will sometimes happen. Since the objects are fields, it is important to remember that all homomorphisms must be *injective* (see corollary 9.4.13)

[1]A famous book explaining these notions is *Althor's Complex Analysis*, in which the first chapter explores complex numbers

Proposition 19.1.1: Field Homomorphisms

Let $\varphi : k \to K$ be a ring homomorphism of fields. Then φ is either identically 0 or is injective, so that the image of φ is either 0 or isomorphic to k. Since the set 0 is not a field (by definition, $0 \neq 1$), φ in **Fld** can only be injective.

***Proof*:**
This is proposition 9.4.10 (2) updated so that we only care about fields.

Thus, If a homomorphism between fields $\varphi : k \to K$ exists, $\varphi(k) \cong k$. In this way, K can be thought of as extending k:

Definition 19.1.2: Field Extension

Let k and K be fields. If there exists a homomorphism $\varphi : k \to K$ (which as we've shown must be injective), then we say that K *extends* k, or K is a *field extension* of k. We will denote this by $k \subseteq K$, K/k, or

$$\begin{array}{c} F \\ | \\ K \end{array}$$

the field from which we are extending is sometimes called the *base field* or *ground field.*

Note that since fielsd are commutative, field extension $\varphi : k \to F$ makes F a k-algebra.

Example 19.1.3: Field Extensions

1. For any field F, F/F is trivially a field extension, since the identity map is a field homomorphism.

2. Perhaps the most famous field extension is that of $\mathbb{C}/\mathbb{R}$, since $\mathbb{R}$ is directly embedded into $\mathbb{C}$. A perhaps easy mistake to make when first learning about field extensions is that $\mathbb{R}^2/\mathbb{R}$ is not a field extension since $\mathbb{R}^2$ is simply not a field. On the other hand, $\mathbb{C}/\mathbb{Q}$ is also a field extension, since the natural inclusion map $\iota : \mathbb{Q} \to \mathbb{C}$ is a field homomorphism. Similarly, $\mathbb{R}/\mathbb{Q}$ is a field extension.

3. In example 9.1.19, we saw that $\mathbb{Q}(\sqrt{D})$ is a field where D is not a perfect square (i.e. there does not an element $d \in \mathbb{Q}$ s.t. $d^2 \in \mathbb{Q}$). For a concrete examples, let's say $D = 2$. Then $\mathbb{Q}(\sqrt{2})/\mathbb{Q}$ is a field extension. If $D = -1$, then we usually write $\sqrt{-1}$ as i, and we get that $\mathbb{Q}(i)/\mathbb{Q}$ is a field extension. We also have that $\mathbb{C}/\mathbb{Q}(i)$ is a field extension, with the map $\iota : \mathbb{Q}(i) \to \mathbb{C}$ given by mapping $a + bi \mapsto a + bi$. Notice that the map $a + bi \mapsto a - bi$ is also a valid field homomorphism, showing that there need not be a unique map.

   We will soon find a way to define $\mathbb{Q}(\sqrt{D})$ without needing to reference set-builder notation.

4. Let k be a field and $k(x) = \mathrm{Frac}(k[x])$ be the field of fractions of $k[x]$. Then $k(x)/k$ is a field extension.

The notation K/k may be read as "K over k". It does *not* represent the quotient K by k. This can be consider an overload of notation, but since the kernel of a ring homomorphism between fields must *always* be trivial, it is safe to see that K/k would not represent a quotient of the field K by k.

Another thing we can observe is that every field homomorphism $\varphi : F \to K$ can be extended into a ring homomorphism $f^{\varphi} : F[x] \to K[x]$ through

$$\sum_i a_i x^i \mapsto \sum_i \varphi(a_i) x^i$$

the fact that this is a ring homomorphisms comes straight from the properties of φ. Thus, we can consider this process to be a functor $f^{-} : \mathbf{Fld} \to \mathbf{Ring}$.

If F/k is any field extension, then we can naturally make F into a *vector-space over k* by making k act on F by multiplication (since $k \subseteq F$, this action is well-defined). Since every field has a prime subfield, every field is at least a vector-space over it's prime-subfield. The dimension of the vector-space is given a name:

Definition 19.1.4: Degree

The *degree* (or *relative degree* or *index*) of a field of extension F/k, denoted $[F : k]$, is the dimension of F as a vector-space over k (i.e. $[F : k] = \dim_k(F)$). The extension is said to be *finite* if $[F : k]$ is finite and is said to be *infinite* otherwise.

Definition 19.1.5: Finite Extension

Let F/k be a field extension. Then F/k is called a *finite extension* if $[F : k]$ is finite, and an *infinite extension* otherwise.

In example 19.1.3, $\mathbb{C}/\mathbb{R}$ is an extension of degree 2, the basis of $\mathbb{C}$ over $\mathbb{R}$ consiss of $\{1, i\}$, and hence a finite extension. The extension $k(x)/k$ is an extension if infinite degree: If it were finite, then any rational function can be written as a sum of a finite number of polynomials up to a constant $c \in k$, but then this collection must have a highest degree term, say n. Then any polynomial of degree $n + 1$ cannot be written as a linear combination of the basis – a conradiction. Hence, it is an infinite extension. Note too that as was demonstrated in theorem 13.1.8, every vector-space has a basis, and so any element $x \in F$ can be represented as a finite linear combination:

$$a_1\beta_1 + a_2\beta_2 + \cdots + a_n\beta_n \qquad a_i \in k, \ \beta_i \in \mathcal{B}$$

We will soon show how to find such a basis.

Since F/k can also be represented as a vector-space over k, we might ask if any field homomorphism/isomorphism can is also a linear transformation. The answer is yes, if we are a bit careful. Let's say that F/k and L/k are two fields extending k, and $\varphi : F \to L$ is a field homomorphism/isomorphism. We will look for a condition under which φ is also a k-vector space homomorphism. Linearly comes for free. For homogeneity of the scalar, we see that for any $r \in k$ and $m \in F$:

$$\varphi(rm) = \varphi(r)\varphi(m) \overset{!}{=} r\varphi(m)$$

where $\overset{!}{=}$ is the condition we need so that φ is a k-vector space homomorphism. In particular, this conditions say's that for any element $r \in k$, $\varphi(r) = r$. This is stronger than saying φ

maps k to itself (i.e. $\varphi(k) = k$), rather it is saying that φ is the *identity* on k. Since such field homomorphisms/isomorphsisms are also vector-space homomorphisms, they are given a name:

Definition 19.1.6: Fixed Homomorphisms

Let F/k and L/k be two field extensions of k and $\varphi : F \to L$ be a field homomorphism (resp. isomorphism). Then if φ is the identity on k, φ is said to be a *k-homomorphsm* (resp. *k-isomorphism*). If there exists a k-isomorphisms $A \to B$, then we'll say that A is k-isomorphic to B and write $A \cong_k B$.

Note that a fixed homomorhism is in fact a k-algebra homomorphism: if L/k, then there is a natural k-module structure on L given by multiplication.

As was shown in section 13.2, a linear transformation is determined by where it maps it's basis to. Hence, if we have a k-homomorphism, then can deduce where the element map (or resp. construct a k-homomrophism) by finding where the basis maps to (resp. by only choosing where the basis maps to). Even better, this linear transformation respects the multiplicative structure of F (that is $\varphi(ab) = \varphi(a)\varphi(b)$), that if a basis β can be generated multiplicatively and additively from a set S, $\langle S \rangle = \beta$, then a k-homomorphism is only determined by where it sends S. Even better, this makes any k-homomorphism a k-algebra homomrophism, and a field F that extends k (F/k) is a k-algebra.

One more word of warning is in order: Let's say we find two field extensions A/k, B/k such that $A \cong_k B$. Then this implies that $\dim_k A = \dim_k B$, however it does not imply $A = B$. This might seem obvious, however it is an mistake that is easy to make after working with field extensions for awhile (this nuance will be particularly important in theorem 19.5.13).

In most of this and the next chapter, we will focus on finite extensions. Some results for extensions of infinite degree will arise as a natural generalization of the results of finite extensions, however finiteness is used in some key proofs to construct bijections (most importantly, in proposition 20.1.12, TBD). We will touch on infinite extensions in TBD (don't yet know when they will become central, i.e. when transcendental degrees will come in play).

Characteristic

The following is is perhaps the simplest invariant of a field k (that is, if $\varphi : k \to K$ is a homomorphism, the property we're about to discuss is preserved). Every field has a multiplicative identity $1 \in k$. Since k is a field, it is also a ring, and hence there exists a unique ring homomorphism from $\mathbb{Z}$ to k mapping 1 to 1 (see proposition 9.1.29):

$$\mathbb{Z} \to k,\ n \mapsto \begin{cases} \underbrace{1 + 1 \cdots + 1}_{n \text{ times}} & n > 0 \\ -(\underbrace{1 + 1 \cdots + 1}_{n \text{ times}}) & n < 0 \\ 0 & n = 0 \end{cases}$$

We will use multiplication notation $n \cdot 1$ as a shorthand [2]. Recall $\operatorname{im}(\mathbb{Z})$ of the natural homomorphism is always the smallest subring of k (exercise ref:HERE (ex:ringCatExercise)). Recall further from exercise ref:HERE (ibid) the following definition:

[2]Note that we can also have defined it as a $\mathbb{Z}$-module on k

Definition 19.1.7: Characteristic

The *characteristic* of a field F, dentoed $ch(F)$, is the order of $\mathrm{im}(\mathbb{Z})$. If $|\mathrm{im}(\mathbb{Z})| < \infty$, we say that F has a *finite characteristic*, and characteristic 0 otherwise. In other words, it is defined as the smallest positive integer n such that $n \cdot 1 = 0$ if such a n exists, and is defined to be 0 otherwise

If n exists, then n needs to be prime. If it weren't, then

$$n = n \cdot 1 = (a \cdot 1)(b \cdot 1) = 0$$

meaning either $a \cdot 1$ or $b \cdot 1$ must be zero-divisors – a contradiction since a field has no zero-divisors! Notice that we only need to have an integral domain for the previous proposition, and that the characteristic of the integral on an integral domain will be the same as it's field of fractions. Since $\mathbb{Z}/p\mathbb{Z}$ is a field (since $\mathbb{Z}/p\mathbb{Z}^\times = \mathbb{Z}/p\mathbb{Z} - \{0\}$, see exercise ref:HERE (ex:introToRings)). However, $\mathbb{Z}$ is not a field. Using theorem 9.5.1, we know the smallest field containing $\mathbb{Z}$ would be $\mathbb{Q}$! In both cases, the field this map injects to must be the *smallest* field that contains 1 (which can also be thought of as the field *generated* by 1 in k)

Definition 19.1.8: Prime Subfield

The *prime subfield* of a field k is the subfield k generated by the multiplicative identity 1_F of k. It is (isomorphic to) either $\mathbb{Q}$ (if $\mathrm{ch}(k) = 0$) or k_p (if $\mathrm{ch}(k) = p$).

19.1.9 Note A shorthand of writing the characteristic of a field is the following: if $p \cdot 1 = 0$, then it is sometimes written $p = 0$ to mean a field with characteristic p.

Notice that for a field k of characteristic n ($n \geq 0$), $k/\mathbb{F}_p$ (resp. $k/\mathbb{Q}$) is always a field extension, since we can always map $\mathbb{F}_p$ or $\mathbb{Q}$ to k through the identity map.

Note that for fields of characteristic 0, field homomorphism are automatically a vector-space homomorphism over $\mathbb{Q}$: linearity holds automatically, and for any scalar $c \in \mathbb{Q}$:

$$\varphi(cx) = \varphi(c)\varphi(x) \stackrel{!}{=} c\varphi(x)$$

where $\stackrel{!}{=}$ comes from how a field homomorphism must map $\mathbb{Q}$ to itself.

The claim at the beginning of this section should now be checked, namely that if $\mathrm{ch}(k) = n$, $n \geq 0$, and $\varphi : k \to K$ is a homomorphism, then $\mathrm{ch}(K) = n$ (see exercise ref:HERE). Since charactersitic is preserved, we can take the subcategory of $\mathbf{Fld}_0$ or $\mathbf{Fld}_p$ where we restric to fields of characteristic p or 0. We won't really do this, however it is inisghtful that doing so will now make these categories have initial objects, namely the respective prime subfield (can you prove this? Can you see why $\mathbf{Fld}$ has no initial objects?). More generally, since homomorphisms can only be injective, we can take $\mathbf{Fld}_k$ to be the category ring homomorphisms as morphisms and all fields that extend k as objects! In most of the following studies, we will fix a field k and study extensions from it, and so field theory can be thought of as studying particular $\mathbf{Fld}_k$ categories

Torsion in Fields

As we have shown in theorem 11.3.14, $F^\times$ is locally cyclic. We recall the theorem here for convinience:

Theorem 19.1.10: $F^\times$ is Locally Cyclic

A finite subgroup of the multiplicative group of a field is cyclic. In particular, if F is a finite field, then the multiplicative group $F^\times$ of nonzero elements of F is a cyclic group.

Note that this does not mean that for every element $a \in F$, there exists a power n such that $a^n = 1$. For exmaple, in $\mathbb{R}$, no power of 2 will result in $2^n = 1$. If this is the case for an element, we will give it a name:

Definition 19.1.11: Root of Unity

Let F be a field and $r \in F$. If there exists an $n \in \mathbb{N}$ such that $r^n = 1$, then we call r an *nth root of unity*.

Example 19.1.12

In $\mathbb{C}$, all elements of the form $e^{\frac{2\pi ik}{n}}$ for $0 \leq k \leq n-1$ are nth roots of unity. In fact, they are all the roots of unity of $\mathbb{C}$. The real numbers $\mathbb{R}$ have only 2 roots of unity of degree 1 and 2 respectively: ± 1.

Notice that not all $e^{\frac{2\pi ik}{n}}$ are generators for the cyclic subgroup they belong to. The generators of these cyclic subgroups are given a name:

Definition 19.1.13: Primitive Root Of Unity

Let F be a field. Then a *primitive nth root of unity* is a generator of the cyclic multiplicative subgroup of all nth roots of unity.

We will return to the study of the roots of unity later; they will be the "building blocks" for [finitely generated] abelian groups in field theory (and Representation theory)[3]

Exercise 19.1.1

1. (Do some exercises showing relations beween roots of unity and other identites? Ex. I think $\zeta_8 + \zeta_8^7 = \sqrt{2}$. Important later. Also elaborate on $\mathbb{C}$ for all future purposes, since it comes up a lot)
2. (Analysis Problem) Show that the roots of unity are dense in $S^1 = \{z \in \mathbb{C} \mid |z| = 1\}$.

Generating Fields

Let's say $k \subseteq F$ is a subfield of F, and take some subset $A \subseteq F$. We can construct a field using k and A in the same way as we have done all the way back in section 2.3.2: Take all subfield of F containing k and A, and intersect them. Similarly to section 2.3.2, this is equivalent to "closing" the field k with A by taking all possible combinations. There is always at least exists the field K with

[3]Remember by the fundamental theory of finitely generated abelian groups that any finitely generated abelian group is the direct product of a free part $\mathbb{Z}^r$ and a torsion part? Notice how the roots of unity can represent any cyclic subgroup?

this which contains this property. Since the intersection of fields is a field (this should be done as an exercise), then we can intersect all fields with this property to get the smallest such subfield.

Definition 19.1.14: Field Generated By Elements Over F

Let F/k be a field extension and S be a set of elements of F. Then the smallest subfield of F containing both k and the elements of S is denoted $k(S)$ and is called *the field generated by S over k*. If S is finite so that $S = \{\alpha_1, \alpha_2, ..., \alpha_n\}$, we usually denote $k(S)$ by $k(\alpha_1, \alpha_2, ..., \alpha_n)$ and call it the field *generated by $\alpha, \alpha_2, ..., \alpha_n$ over k.*

If we have a field extension F/k, and F can be generated by a single element, then we call that a simple extension:

Definition 19.1.15: Simple Extension

Let F/k be a field extension. Then if there exists an element $\alpha \in F$ such that $F = k(\alpha)$, we say that F is a *simple extension*, and call α the *primitive element* of the extension. The field $k(\alpha)$ is sometimes called "k adjoin α".

Be mindful to not confuse the terms "finite generation over k", "finite type k-algebra" [4], or "finite degree over k" (for example, $k(t)/k$ is a simple extension, but certainly not finite as was discussed under the definition of degree).

A similar definition works if we broaden to rings: instead of taking the intersection of all fields containing α and F, we take all rings containing α and F. It is easy to generlize the proof that the intersection of fields is a field to rings. We will usually denote this smallest ring $F[S]$, and if $S = \{\alpha_1, \alpha_2, ..., \alpha_n\}$ is finite, then $F[S] = F[\alpha_1, \alpha_2, ..., \alpha_n]$.

Proposition 19.1.16: Characterizing $F(S)$ and $F[S]$

Let K/F be a field extensions, $S \subseteq K$, and consider both $F(S)$ and $F[S]$. Then:

1. $F[S]$ can be characterized as the set of all finite linear combinations with coefficients in K of finite products of powers of elements of S, that is, polynomials with variables taken from S
2. $F(S)$ can be characterized as the set of all $ab^{-1} \in K$ with $a, b \in F[S]$ with $b \neq 0$. Furthermore, $F(S) \cong \text{Frac}(F[S])$

Note that in the "polynomials" mentioned above, the varaibles S are *not* indeterminates, since they might have relations in K (for example, it might be that for some $s \in S$, $s^2 + 1 = 0$)

***Proof*:**
These proofs are similar to those of section 2.3.2 with appropriate modifications for the added operation of multiplication and inversing.

[4] It is important to point out that there is actually a deep connection between those two concepts which we'll explore in part V. As foreshadowing, try seeing if you can think of a finitely generated field extension F/k that is not a finite-type k-algebra (or vice-versa). The answer to this question is given by Zariski's Lemma

To close off this section, we introduce a generalization to notion of the smallest field K/k generated by some elements to the smallest field generated with all the elements of another field:

Definition 19.1.17: Composite Field

Let K_1 and K_2 be two subfield of K. Then the *composite field* of K_1 and K_2, denoted K_1K_2, is the smallest subfield of K containing both K_1 and K_2. Similarly, the composite of any collection of subfields of K is the smallest subfield containign all the subfields.

Note that like in section 2.3.2, we can describe K_1K_2 as the intersection of all subfields contianing K_1 and K_2 (or any number of subfields).

Example 19.1.18: Composite Fields

The composite field $\mathbb{Q}(\sqrt{2})\mathbb{Q}(\sqrt[3]{2})$ is $\mathbb{Q}(\sqrt[6]{2})$, since thie field contains both these fields, and conversely any field containing both $\sqrt{2}$ and $\sqrt[3]{2}$ contains their quotient, namely $\sqrt[6]{2}$

Exercise 19.1.2

1. Sometimes, we want to talk about field extensions without an underlying field. Let E, F be two fields of the same characteristic, say p. Show that $E \otimes_{\mathbb{F}_p} F$ (resp. $E \otimes_{\mathbb{Q}} F$) as a $\mathbb{F}_p$ (resp. $\mathbb{Q}$)-algebra is a field if and only if $[EF : \mathbb{F}_p] = [E : \mathbb{F}_p][F : \mathbb{F}_p]$ (resp. $[EF : \mathbb{Q}] = [E : \mathbb{Q}][F : \mathbb{Q}]$)

19.2 Field Extensions

We now continue our exploration of field extensions. As we saw in proposition 19.1.16, any field K/F such that $K = F(S)$ is the field of fractions of all "polynomials" with variables S and coefficients F. In particular, these polynomials have "solutions" or "roots" characterizing their relations. Similarly to how generators in a group where characterized by a monomial, (for example, D_4 is characterized with the generators a, b satisfying the monomails $ab^{-1}ab = 1$ and $a^2 = b^4 = 1$), generators in fields are characterized by polynomial relationships (for example, $\mathbb{C}/\mathbb{R}$ is characterized by being generated by $\mathbb{R}$ and i where $i^2 = -1$). Just like how any relation of elements within a group can be shifted so all the elements are on the left hand side (ex. writing $a^{-1}bab = 1$ instead of $ab = b^{-1}a$), we can do the same with polynomial relationships (ex. $i^2 + 1 = 0$). In proposition 19.2.24, we will show why it only matters to characterize polynomials, and not rational polynomials (i.e. elements of $p(S)/q(S)$)

As we've seen in chapter 11, every polynomial $p(x) \in F[x]$ can be broken down into irreducible polynomials. We will start finding an easy way of extending any field k which will give us a large class of field and field extensions to work with using irreducible polynomials (recall that x^2+1 has no solutions in $\mathbb{R}$, but does in $\mathbb{C}$, so is there an extension from $\mathbb{R}$ to $\mathbb{C}$?). We will find that the method used to construct fields will almost create "universal fields" (the same way free groups are sort of the universal groups since any group is the quotient of a free group). This endevaour to create a universal property will be very close in obtaining a universal property for fields, however it will fail in one important way. Due to that failure, it will be important carry around some more information to take into account the lake of universality. In fact, the lack of universal property may be seen less as a failure and more as a peek into the further structure that underly field extensions. Figuring all of this out is the goal of this section

Field Extensions using Polynomial

If we take our field to be $\mathbb{R}$, then $x^2 + 1 = 0$ does not have a solution in $\mathbb{R}$. The question then is whether there is a larger field K for which $x^2 + 1 = 0$ *has* a solution. More generally, given any irreducible polynomial $p(x) \in k[x]$, does there exists a field K such that K contains a root of $p(x)$[5]?. The answer is yes! We actually have all the tools already to answer this question!! I suggest to take a moment and try to find the larger field K which satisfies this!!

19.2.1 Taking a moment

Theorem 19.2.2: Kronecker's Theorem

Let k be a field and let $p(x) \in k[x]$ be an irreducible polynomial. Then there exists a field K containing an isomorphic copy of k in which $p(x)$ has a root. Identifying k with this isomorphic copy through i shows that there exists an extension of k in which $p(x)$ has a root. Furthermore, if L/k is a field extension in which $p(x)$ has a root, then there exists a k-homomorphism $j : K \to L$ such that

$$\begin{tikzcd} K \arrow[r, hook, "j"] & L \\ k \arrow[u, hook, "\iota"] \arrow[ur, hook, "i"'] & \end{tikzcd}$$

commutes. Due to this, the field K produced in the following construction is sometimes called the *Kronecker field for* $p(x)$ *over* k.

If in the Kronecker construction we use a *reducible* polynomials, then the resulting object is no longer a field (it contains zero-divisors), but remains to be a k-algebra.

***Proof*:**
Consider

$$K = k[x]/(p(x))$$

Since $k[x]$ is a Principal Ideal Domain (corrollary 11.1.2) and since $p(x)$ is irreducible (so prime by proposition 10.1.33), then $p(x)$ is maximal (proposition 10.1.32). Then $k[x]/(p(x))$ is a field by proposition 9.4.19. If we take the cannonical projection $\pi : k[x] \to K(= k[x]/(p(x)))$, then we can restrict it to F ($\pi|_F : F \to K$) to get a homomorphism from F to K. Since this cannot be identically 0 or else K would be the zero field, so by proposition 19.1.1, F maps injectively into K, so $\varphi(F)(\cong F)$ is an isomorphic copy of F contained in K. We identify F with its isomorphic image in K and view K as a *subfield* of K. If $\overline{x} = \pi(x)$ denotes the image of x in the quotient K, then

$$\begin{aligned} p(\overline{x}) &= \overline{p(x)} && \text{since } \pi \text{ is a homomorphism} \\ &= p(x) \bmod (p(x)) && \text{in } k[x]/(p(x)) \\ &= 0 \end{aligned}$$

so that K does indeed contain a root of the polynomial $p(x)$! Thus, K is an extension of F in

[5]We can assume that $p(x)$ is irreducible in $F[x]$, since if any factors of $p(x)$ has a root, then so does $p(x)$

which the polynomial $p(x)$ has a root

For the second assertion, let's say L/k has a root to $p(x)$, say $p(\alpha) = 0$. Then the evalutation map $\text{ev}_\alpha : k[x] \to L$ (which we have proved is a ring-homomorphism in theorem 9.2.4), $g(x) \mapsto g(\alpha)$ must map $p(x)$ to 0. Hence, $p(x) \in \ker(\text{ev}_\alpha)$ meaning $(p(x)) \subseteq \ker(\text{ev}_\alpha)$, and by maximality of $(p(x))$ $(p(x)) = \ker(\text{ev}_\alpha)$. Since ideals correspond one-to-one with homomorphisms, we have a homomorphism:

$$j : \frac{k[x]}{(p(x))} \to L$$

and since $\frac{k[x]}{(p(x)}\cong K$, we have:

$$j : K \to L$$

Furthermore, for any $r \in k$, $j(r) = r$. Thus j clearly commutes in the aformentioned diagram, completing the proof.

From now on, when we talk about irreducible polynomial, we will specifically mean *non-linear irreducible polynomials*, since linear irreducible polynomials produce trivial field extensions (i.e. k being extended to sovle for $x - \alpha \in k[x]$ is just k) unless stated otherwise.

The field extension to the Kronecker field K/k is *almost* universal with respect to finding roots to the polynomial $p(x)$. If it were universal, then if there was another field extension F/k such that F has a root of the irreducible polynomial $p(x) \in k[x]$, there would exist a *unique* homomorphism $j : F \to K$ such that the diagarm commutes. However, the homomorphism j need not be unique.

Example 19.2.3: Lack Of Uniqueness

Take $K = \mathbb{Q}[x]/(x^2 - 2)$ and $\mathbb{R}$. Then the polynomial $x^2 - 2$ has a root in $\mathbb{R}$, hence we know there exists a homomorphism:

$$K = \mathbb{Q}[x]/(x^2 - 2) \hookrightarrow \mathbb{R}$$

However, there exists two homomorphisms mapping K to $\mathbb{R}$. Since K extends $\mathbb{Q}$, any homomorphism can be represented by where the roots map: we can map $1 \mapsto 1$ and $\overline{x} \mapsto \sqrt{2}$, or map $1 \mapsto 1$ and $\overline{x} \mapsto -\sqrt{2}$. Both of these are valid homomorphisms, and so adjoints are not universal with respect to roots.

The universality can be recovered if we keep track of which root we want to map to (ex. $\sqrt{2}$), however we would have to make sure that there is only 1 homomorphism given some root (how do we know in more general cases that there can't be 2 homomorphisms mapping $\overline{x}$ to the same root?). This concern is solved soon in theorem 19.2.27.

Since the aforementioned diagram commutes, notice that for any $r \in k$:

$$r \overset{!}{=} i(r) = j(\iota(r)) = j(r)$$

where ι is the natural inclusion into K, and the image of i is identified with the domain, justifying the $\overset{!}{=}$. Thus, j must *always* fix the base field, i.e. j is a k-homomorphism/isomorphism.

You might be concerned about *which* root of an irreducible polynomial does $\overline{x}$ represent in the previous theorem. For example, $x^2 - 2$ is irreducible over $\mathbb{Q}$, with $\pm\sqrt{2}$. So which root should we pick to represent $\overline{x}$? In fact, we can pick $\overline{x}$ to be represented *any* root of this polynomial: the field extending to contain $\sqrt{2}$ or $-\sqrt{2}$ will be isomorphic. More generally, if $p(x)$ is any irreducible polynomial, then $k[x]/(p(x))$ can be representing *any* choice of the possible roots of $p(x)$[6] and extending k to contain that root. To make the choice concrete, we can choose a particular root of $p(x)$, say α (which we define by saying α satisfies some polynomial equation), and isomorphically identify $K \xrightarrow{\sim} k(\alpha)$ (which we will show in more detail in theroem 19.2.8).

Note that to find the root of a polynomial, we required that polynomial to be irreducible. This technique will not obtain us a root of $(x^2-2)(x^3-2) = x^6 - 5x^2 + 6$, namely becaue $(x^6 - 5x^2 + 6)$ is not even a prime ideal (the resulting quotient contains zero-divisors). In section 19.5.2, we will explore how to find extensions for which *any* polynomial has all of its roots, essentially by finding the roots of every irreducible polynomial and progressively extending our field by adjoning roots.

Note that we haven't shown a way to "find" the root. For example, if we were looking at irreducible polynomial over $\mathbb{R}$ and we pick a root $\alpha \in \mathbb{C}$ that satisfies the polynomial equation, we don't have a computational way of finding it:

Example 19.2.4: Existence, but not construction

Let's say say F is a field extension $F/\mathbb{Q}$. Consider the irreducible polynomial $x^3 - 3x - 1$. over $\mathbb{Q}$ (it has no rationals roots by prosposition 11.3.5). Thus, $F = \mathbb{Q}[x]/(x^3 - 3x - 1)$ is a field which has a root! The fact that all the roots of $x^3 - 3x - 1$ are real (and hence being able to identify F with a subfield of $\mathbb{R}$) can be considerd as taken for granted for now; we will have the tools to show that by the end of section 20.1.1. Visually, if we plot $x^3 - 3x - 1$, notice that all the roots are real, and hence $\mathbb{Q}$ can only by roots of $x^3 - 3x - 1$ which are real.

Let's say α is the root, so $\alpha \in \mathbb{R}$. We might hope that we can find which α in $\mathbb{R}$ satisfies the polynomial given that we know α is a root, however there is no *algebraic* way of finding this root! Using Newton-Raphsons approximation method, we can find that it is a real number $x \approx 1.97938...$. Finding these roots is a sub-branch of numerical analysis known as *root-finding algorithms*

For most common roots, we have already developped a symbol to represent them, namely $\sqrt[n]{m}$ is the root of $x^n - m$. More generally, if we extend $\mathbb{Q}[x]/(x^n - m)$ to solve another polynomial of a similar form, say $x^k - l$:

$$\frac{(\mathbb{Q}[x]/(x^n - m))[y]}{(y^k - l)}$$

we can "nest" as many square roots as we want. Since adding and multiplying is closed in a field, any extension of such polynomials produce elements of the form:

$$\sqrt[3]{\frac{3 + \sqrt[9]{40}}{\sqrt[8]{\sqrt{2}\sqrt{10}}}}$$

However, most more complicated polynomials are not given a special symbol, a notable exception being the golden ratio φ which is the root of $x^2 - x - 1$[7]. In section 20.2.4, we will further explore extensions which only add radical elements.

With the extension of a field through an irreducible polynomial constructed, it would be advantageous to be able to easily work with it's elements. Since K is a subfield of k, we can describe K as a k-vector-space!

[6]What it means to "choose" a root here is a bit ambiguous, since we have yet to say there exists a field that makes $p(x)$ factor into linear terms. Such a field is constructed in section 19.5.1

[7]Though, this is not much of an exception since $\varphi = \frac{1+\sqrt{5}}{2}$

Theorem 19.2.5: Basis For Kronecker Field

Let $p(x) \in F[x]$ be an irreducible polynomial of degree n over the field F and let K be the field $F[x]/(p(x))$. Let $\theta = x \bmod (p(x)) \in K$. Then the elements

$$1, \theta, \theta^2, ..., \theta^{n-1}$$

are a basis for K as a vector space over F, so the degree of the extension is n, i.e., $[K : F] = n$. Thus

$$K = \left\{a_0 + a_1\theta + a_2\theta^2 + \cdots + a_{n-1}\theta^{n-1} | a_0, a_1, ..., a_{n-1} \in F\right\}$$

consists of all polynomials of degree $< n$ in θ. In particular, all field extensions of this form have finite degree.

***Proof*:**
Take $K = F[x]/(p(x))$, and consider $a(x) \in F[x]$ to be any polynomial. Then, by the Euclidean algorithm, we get that

$$a(x) = q(x)p(x) + r(x) \qquad q(x), r(x) \in F[x] \text{ with } \deg(r(x)) < n$$

then in K, we get that $[a(x)] = [r(x) + q(x)p(x)] = [r(x)]$, so any polynomial can be represented by it's residue polynomial, which will always be of degree $< n$. Thus, $1, \theta, \theta^2, ..., \theta^{n-1}$ (the image of $1, x, x^2, ..., x^{n-1}$) span the quotient. It remains to show they are linearly independent to conclude they are a basis.

If they weren't, then for some $b_i \in F$ (not all 0)

$$b_0 + b_1\theta + \cdots + b_{n-1}\theta^{n-1} = 0$$

that is

$$b_0 + b_1x + \cdots + b_{n-1}x^{n-1} \equiv 0 \bmod (p(x))$$

meaning $p(x) \big| b_0 + b_1x + \cdots + b_{n-1}x^{n-1}$. However, that's a contradiction, since the degree of $p(x)$ is higher than the polynomial on the right hand side (which is at most $n-1$)! Thus, these must be linearly independent. Since they also span, they fulfill the condition of a basis.

Since a basis is generated as a k-algebra by θ, any k-homomorphism $\varphi : k(\theta) \to F$ is determined by where it maps θ, making the map particularly easy to determine if we determine where θ maps to (the criterion for which we do in corollary 19.2.29).

As a reminder of arithmetics in $F[x]/(p(x))$:

Corollary 19.2.6: Shortcut On Arithmetics In K

Let K be as in theorem 19.2.2, and let $a(\theta), b(\theta) \in K$ be two polynomials of degree $< N$ in θ. Then addition in K is defined simply bu usual polynomial addition and multiplicatin in K is defined by

$$a(\theta)b(\theta) = r(\theta)$$

where $f(x)$ is the remainder (of degree $< n$) obtaiend after dividing the polynomial $a(x)b(x)$ by $p(x)$ in $F[x]$

***Proof*:**
An exercice in quotient ring of polynomials. Addition multiplication is done as in any quotient ring:

1. addition is usual addition of polynomials, since it doesn't change the degree
2. For multiplication, one can multiply the terms, then find a representative in the coset with degree $< n$

Note that what is described in corrollary 19.2.6 works in any quotient field, including over polynomials which are *not* irreducible. However, the result is not a field – not even an integral domain [8].

Example 19.2.7: Field Extension Through Irreducible Polynomials

1. Let $F = \mathbb{R}$ and $p(x) = x^2 + 1$. Then we know that $x^2 + 1$ does not have a root in $\mathbb{R}$. Thus, consider:
$$K = \mathbb{R}[x]/(x^2 + 1)$$
is an extension field (of degree 2) that contains a root to this polynomial. Let $\theta = \overline{x}$. By theorem 19.2.5, $K/\mathbb{R}$ has degree 2 and any element $t \in K$ can be written as:
$$t = a + b\theta \qquad a, b \in \mathbb{R}$$
If you are comfortable with the arithmetics of $\mathbb{C}$, show that:
$$\mathbb{R}[x]/(x^2 + 1) \cong \mathbb{C}$$
2. Let $F = \mathbb{Q}$ and $p(x) = x^2 + 1$. Then by substitution with Eisntein's Criteron, $x^2 + 1$ is irreducible, hence $K = \mathbb{Q}[x]/(x^2 + 1)$ is a field extension of $\mathbb{Q}$. Since the base field is now $\mathbb{Q}$, any element $t \in K$ is of the form:
$$t = a + b\theta \qquad a, b \in \mathbb{Q}$$
3. Let $F = \mathbb{Q}$ and $p(x) = x^2 - 2$. Then $x^2 - 2$ is irreducible by Eisenstein's, so $F = \mathbb{Q}[x]/(x^2 - 2)$ is a field extension of $\mathbb{Q}$. Letting $K = \mathbb{Q}[x](x^2 + 1)$, show that:
$$F \ncong K$$
that is, extending by different roots of different irreducible polynomial produces distinct fields.

[8] Recall that if $p(x) = a(x)b(x)$, $a(x)$, $b(x)$ not being units, then in $F[x]/(p(x))$, $\overline{a(x)}, \overline{b(x)}$ are both nonzero, but $\overline{a(x)} \cdot \overline{b(x)} = \overline{0}$, meaning $F[x]/(p(x))$ has zero-divisors

4. Let $F = \mathbb{F}_2$ (the finite field with 2 elements) and $p(x) = x^2 + x + 1$. Then $p(x)$ is irreducible since $0^2 + 0 + 1 = 1^1 + 1 + 1 = 1$. Hence, $K = \mathbb{F}_2[x]/(x^2 + x + 1)$ is irreducible of degree 2. Can you make the multiplication table for this field?

Classifying Simple and Transcendental Extensions

Let's now turn our attention to classifying field extensions, given $k \subseteq K$ where we consider k the base field. If $A \subseteq K$, then we already have a notion of field extensions: $k \subseteq k(A)$. We will focus on the special cae where $A = \{\alpha\}$, that is, simple extension with $\alpha \in K$. As an example of this, imagine taking $\mathbb{R} \subseteq \mathbb{C}$, $\alpha = i$, and considering $\mathbb{R}(i)$. Then it is a known fact that $i^2 + 1 = 0$. We might ask if in general, given some $\alpha \in K$, can we find a polynomial that sovles it? In this way, we can characterize the field $k(\alpha)$ without needing to resort to taking the intersection of all subfields of K containing F and α. The answer is in fact, half the time. All simple extensions can be characterized by either being isomorphic to the rational functions over the base field or in terms of a quotient of a polynomial ring, giving us an easy classification of simple extensions:

Theorem 19.2.8: Classifying Simple Extensions

Let K/k be a field, $\alpha \in K$, and $k(\alpha)$. Consider $\mathrm{ev}_\alpha : k[x] \to k(\alpha)$ mapping $f(x) \mapsto f(\alpha)$. Then if ev_α is injective, $k(\alpha)$ is an infinite extension and $k(\alpha) \cong k(x)$, and if ev_α is not injective, then there exists a polynomial $p(x)$ such that $k[x]/(p(x)) \cong k(\alpha)$. Furthermore, $p(x)$ is the polynomial of *minimal degree* that has α as a root.

Note that in this theorem, we're saying that *any* field over K extending k in which $p(x)$ contains a root in K and is irreducible in k contains a subfield isomorphic to the extension of k as we constructed in theorem 19.2.2 and that this field is (up to isomorphism) the smallest extension of k containing such a root.

***Proof*:**
Take the natural homomorphism

$$\mathrm{ev}_\alpha : F[x] \to F(\alpha)(\subseteq K) \qquad a(x) \mapsto a(\alpha)$$

which maps the polynomial $a(x)$ to the polynomial evaluated at α. Note that $\mathrm{ev}_\alpha(F[x])$ is an integral domain, since $F(\alpha)$ is integral domain.

If ev_α is injective, then $F[x]$ can be identified with an subring of $F(\alpha)$ that is an integral domain. By the universal property of field of fraction (theorem 9.5.4), ev_α uniquely extends to a homomorphism

$$\varphi : F(x) \to F(\alpha)$$

Since $\varphi(F(x))$ is a field containing F and α, $\varphi(F(x)) \supseteq F(\alpha)$, and since φ is a function $\varphi(F(x)) \subseteq F(\alpha)$, hence $\varphi(F(x)) = F(\alpha)$. To find the degree $[F(\alpha) : F]$, notice that $1[= \alpha^0], \alpha, \alpha^2, ...$ are all linearly independent over F in $F(\alpha)$ (since $1[x^0], x, x^2, ...$ are all linearly independent all linearly indepndent over F in $F(x)$), hence any basis must at least contain countably many linearly independent elements, and hence the degree must be infinite.

On the otherhand, if ev_α is not injective, then $\ker(\mathrm{ev}_\alpha) \neq 0$ and ev_α is clearly surjective. Since ev_α is surjective and $F(\alpha)$ is a field, $\ker(\mathrm{ev}_\alpha)$ must be a maximal ideal, and since $F[x]$ is a euclidean domain (and hence a PID), any polynomial of minimal degree in $\ker(\mathrm{ev}_\alpha)$ can represent the ideal, that is $(m_{\alpha,k}(x)) = \ker(\text{ev}_\alpha)$ for some minimal degree polynomial $m_{\alpha,k}(x) \in \ker(\text{ev}_\alpha)$. In fact, this minimal polynomial must be unique up to unit: if there where two minimal polynomials where $(m_{\alpha,k}) = (m'_{\alpha,k})$, then they must be equal up to unit, i.e. up to a multiple of k. Hence, we have that

$$\frac{k[x]}{(m_{\alpha,k}(x))} \cong F(\alpha)$$

By theorem 19.2.5, $[F(\alpha) : F] = \deg m_{\alpha,k}(x)$, and so the extension is finite, completing the proof.

This theorem motivates the following vocabulary:

Definition 19.2.9: Algebraic and Transcendental Elements

Let F be a field and let K be an extension of F. Then:

1. The element $\alpha \in K$ is said to be *algebraic* over F if α is a root of some nonzero polynomial $f(x) \in F[x]$.
2. If α is not algebraic over F (i.e., is not the root of any nonzero polynomial with coefficeints in F) then α is said to be *transcendental over F*.

Dealing with more general transcendental extensions is a straight-forward generlization of the previous thoerem; it naturally generlizes for $k(\alpha_1, ..., \alpha_n) \subseteq K$ where each α_i is transcendental. We first define the following term for the corollary:

(are all monomorphisms injective? I replaced the word monomorphism with injective in my notes for definition 19.2.10)

Definition 19.2.10: Algebraically Independent

Let R be a k-algebra. Then $\alpha_1, ..., \alpha_n \in R$ are *algebraically independent* over k if the k-algebra homomorphism ev_k (i.e., the evaluation map) which fixes k:

$$\text{ev}_k : k[x_1, ..., x_n] \to R \qquad \text{ev}_k(x_i) = \alpha_i$$

is injective. The set is called *algebraically dependent* otherwise.

The definition is stated for a finite list, and that is the only case a polynomial relation can see: a relation involves finitely many indeterminates however large the set. So the notion extends to arbitrary subsets without any new content.

Definition 19.2.11: Algebraically Independent Set

Let K/k be a field extension. A subset $S \subseteq K$ is *algebraically independent* over k if every finite list of distinct elements of S is algebraically independent over k in the sense of definition 19.2.10. The empty set is algebraically independent.

The analogy with linear algebra now runs further than it first appears. An algebraically independent set behaves like a linearly independent one, a set over which K is algebraic behaves like a spanning one, and the two notions meet in an object that deserves the name basis.

Definition 19.2.12: Transcendence Basis

Let K/k be a field extension. A subset $B \subseteq K$ is a *transcendence basis* of K over k if B is algebraically independent over k and K is an algebraic extension of $k(B)$.

Existence follows the same route as for vector-space bases: take a maximal independent set. Maximality is exactly what forces the algebraicity.

Theorem 19.2.13: Existence of a Transcendence Basis

Let K/k be a field extension and let $S \subseteq K$ be a subset with K algebraic over $k(S)$. Let $A \subseteq S$ be algebraically independent over k. Then there is a transcendence basis B of K/k with $A \subseteq B \subseteq S$. In particular, taking $S = K$ and $A = \emptyset$, every field extension has a transcendence basis.

***Proof*:**
Order by inclusion the collection of algebraically independent subsets of S containing A; it is nonempty, containing A. A chain in this collection has its union as an upper bound, and that union is algebraically independent because a dependence relation involves finitely many elements and would therefore already lie in some member of the chain. By Zorn's lemma there is a maximal such subset B.

It remains to see that K is algebraic over $k(B)$. Let $s \in S$. If $s \in B$ this is clear, and otherwise $B \cup \{s\}$ is a subset of S containing A that strictly contains B, so by maximality it is algebraically dependent: some finite list of distinct elements of $B \cup \{s\}$ satisfies a nontrivial polynomial relation over k. That list must involve s, since B is independent, so collecting the relation as a polynomial in s with coefficients in $k[B]$ gives a nonzero polynomial over $k(B)$ killing s; hence s is algebraic over $k(B)$.

So every element of S is algebraic over $k(B)$, whence $k(S)$ is algebraic over $k(B)$, and K is algebraic over $k(S)$ by hypothesis. Algebraicity is transitive in towers, so K is algebraic over $k(B)$ and B is a transcendence basis, as we sought to show.

The substance is that the size of a transcendence basis does not depend on the basis chosen. As in linear algebra, this rests on an exchange argument comparing an independent set against a set over which everything is algebraic.

Lemma 19.2.14: Independent Sets Are No Larger Than Algebraic Generating Sets

Let K/k be a field extension, let $S = \{s_1, \ldots, s_n\} \subseteq K$ be finite with K algebraic over $k(S)$, and let $A \subseteq K$ be algebraically independent over k. Then A is finite with $|A| \leq n$.

***Proof*:**
It suffices to show that no $n+1$ distinct elements of A can be algebraically independent. Suppose $a_1, \ldots, a_{n+1} \in A$ are distinct. The argument replaces the s_j by the a_i one at a time, and the

claim to be maintained after r steps is that, after renumbering the s_j, the field K is algebraic over $k(a_1, \ldots, a_r, s_{r+1}, \ldots, s_n)$.

Step 1: the base case and the replacement. For $r = 0$ the claim is the hypothesis. Suppose it holds for some $r < n + 1$, so K is algebraic over $F_r = k(a_1, \ldots, a_r, s_{r+1}, \ldots, s_n)$. Then a_{r+1} is algebraic over F_r, so there is a nonzero polynomial relation

$$f\big(a_{r+1};\, a_1, \ldots, a_r,\, s_{r+1}, \ldots, s_n\big) = 0$$

with f a nonzero polynomial over k in which the first slot occurs. The relation must involve at least one s_j with $j > r$: otherwise it exhibits a nontrivial algebraic relation among $a_1, \ldots, a_{r+1}$ over k, contradicting the independence of A. Renumber so that s_{r+1} occurs. Reading the relation as a nonzero polynomial in s_{r+1} over $k(a_1, \ldots, a_{r+1}, s_{r+2}, \ldots, s_n)$ shows s_{r+1} is algebraic over that field, so F_r is algebraic over $F_{r+1} = k(a_1, \ldots, a_{r+1}, s_{r+2}, \ldots, s_n)$, and by transitivity K is too. This advances the claim to $r + 1$.

Step 2: the contradiction. Running the replacement n times gives that K is algebraic over $k(a_1, \ldots, a_n)$. Then a_{n+1} is algebraic over $k(a_1, \ldots, a_n)$, which is a nontrivial algebraic relation among distinct elements of A, contradicting independence. Hence A has at most n elements, completing the proof.

Theorem 19.2.15: Transcendence Degree Is Well-Defined

Let K/k be a field extension admitting a finite transcendence basis. Then every transcendence basis of K/k is finite, and any two have the same number of elements.

***Proof*:**
Let B be a finite transcendence basis, of size n, and let B' be any transcendence basis. Since K is algebraic over $k(B)$ and B' is algebraically independent, lemma 19.2.14 gives that B' is finite with $|B'| \leq n$. Exchanging the roles of B and B', which is legitimate because B' is now known to be finite and K is algebraic over $k(B')$, gives $n \leq |B'|$. Hence $|B'| = n$, as we sought to show.

Definition 19.2.16: Transcendence Degree

Let K/k be a field extension admitting a finite transcendence basis. The *transcendence degree* $\operatorname{trdeg}_k K$ is the common number of elements of its transcendence bases. An extension is algebraic exactly when $\operatorname{trdeg}_k K = 0$, and $\operatorname{trdeg}_k k(x_1, \ldots, x_n) = n$ for indeterminates x_i.

Transcendence degree adds along towers, which is what makes it usable as a notion of dimension. The proof is the statement that a transcendence basis of the top over the middle, together with one of the middle over the bottom, is a transcendence basis of the top over the bottom.

Proposition 19.2.17: Transcendence Degree Is Additive in Towers

Let $k \subseteq K \subseteq L$ be fields with L/k admitting a finite transcendence basis. Then

$$\operatorname{trdeg}_k L = \operatorname{trdeg}_k K + \operatorname{trdeg}_K L$$

***Proof*:**
Let B be a transcendence basis of K/k and C a transcendence basis of L/K; both are finite by lemma 19.2.14, since L is algebraic over $k(B \cup C)$ once the two are combined. The sets B and C are disjoint, an element of C being transcendental over $K \supseteq k(B)$.

Algebraicity. L is algebraic over $K(C)$, and K is algebraic over $k(B)$, so $K(C)$ is algebraic over $k(B)(C) = k(B \cup C)$; by transitivity L is algebraic over $k(B \cup C)$.

Independence. Suppose a nontrivial polynomial relation over k holds among distinct elements of $B \cup C$. Collect it as a polynomial in the elements of C with coefficients in $k[B] \subseteq K$. Those coefficients are not all zero, since the relation is nontrivial and B is algebraically independent over k; so the relation is a nontrivial algebraic relation among elements of C over K, contradicting the independence of C over K.

So $B \cup C$ is a transcendence basis of L/k, and counting gives $\operatorname{trdeg}_k L = |B| + |C| = \operatorname{trdeg}_k K + \operatorname{trdeg}_K L$, completing the proof.

Corollary 19.2.18: Classifying Finitely Generated Transcendental Extensions

Let K/k be a field extensions such that $K = k(\alpha_1, ..., \alpha_n)$ where each element is trancsendental. Then if $k(x_1, ..., x_n)$ is the field of rational functions over the indeterminates x_i[a], then, the set $\alpha_1, ..., \alpha_n$ is algebraically independent, and:

$$k(x_1, ..., x_n) \cong_k k(\alpha_1, ..., \alpha_n)$$

[a]sometimes called *formal variables*

The case for arbitrarily many transcendental elements is the same proof with more book-keeping on notation.

***Proof*:**
This is an immediate generalization of theorem 19.2.8

This shows that in terms of fields, algebraically independent sets don't offer much to our analysis of classification; they act more like a basis with essentially no relation, a resemblance made precise in definition 19.2.12 below. This is not to say that they don't offer any use in Galois theory, where we will correspond group for transcendental extensions, but in terms of field theory their structure is very simple. Thus, we will take more time stuyding algebraically dependent sets, especially since we can use theorem 19.2.2 to study study irreducible polynomials that aquire a root.

Going back to the minimal polynomial that we found in the proof,

Definition 19.2.19: Minimal Polynomial

Let $k(\alpha)$ be a finite simple extension as in theorem 19.2.8, and $m_{\alpha,k}(x)$ be the polynomial given in the proof. Then $m_{\alpha,k}(x)$ is called the *minimal polynomial* of α over k .

it can be shown that $m(x)$ is unique *up to base field* by taking advantage of the euclidean algorithm for $F[x]$ (see exercise ref:HERE), and we have an easy way of linking minimal polynomial between fields:

Proposition 19.2.20

If L/F is an extension of fields and α is algebraic over both F and L, then $m_{\alpha,L}(x)$ divides $m_{\alpha,F}(x)$ in $L[x]$.

***Proof*:**
Since both $m_{\alpha,L}(x)$ and $m_{\alpha,F}(x)$ are polynomials in $L[x]$, we can divide $m_{\alpha,F}(x)$ by $m_{\alpha,L}(x)$ by each other. If we consider:

$$m_{\alpha,L}(x) = m_{\alpha,F}(x)g(x) + r(x)$$

then since α is a root:

$$0 = m_{\alpha,L}(\alpha) = m_{\alpha,F}(\alpha)g(\alpha) + r(\alpha) = 0g(x) + r(\alpha)$$

which implies $r(\alpha) = 0$, and has degree smaller than $m_{\alpha,F}(x)$ – a contradiction to the minimality of the degree of $m_{\alpha,F}(x)$ unless $r(x) = 0$. Thus:

$$m_{\alpha,L}(x) = m_{\alpha,F}(x)g(x) \qquad g(x) \in L[x]$$

However, since $m_{\alpha,L}(x)$ is the minimal polynomial over $L[x]$ that has a root α, it must be that $g(x) = l \in L$. However, if we proceed with the same proof dividing $m_{\alpha,F}(x)$ by $m_{\alpha,L}(x)$:

$$m_{\alpha,F}(x) = m_{\alpha,L}(x)g(x) \qquad g(x) \in L[x]$$

then this does not contradict minimality, showing that $m_{\alpha,L}$ divides $m_{\alpha,F}$ in $L[x]$.

to obtain (using the euclidean algorithm):

$$m_{\alpha,F}(x) = m_{\alpha,L}(x)g(x) + r(x)$$

where $\deg r(x) < \deg m_{\alpha,F}(x)$. Then:

$$0 = m_{\alpha,L}(\alpha) = m_{\alpha,F}(\alpha)g(\alpha) + r(\alpha) = 0g(x) + r(\alpha)$$

which implies $r(\alpha) = 0$, and has degree smalelr than $m_{\alpha,F}(x)$ – a contradiction to the minimality of the degree of $m_{\alpha,F}(x)$ unless $r(x) = 0$. Thus:

$$m_{\alpha,L}(x) = m_{\alpha,F}(x)g(x) \qquad g(x) \in L[x]$$

completing the proof.

The miniaml polynomial will always be irreducible: If it was reducible, then $m_{\alpha,k}(x) = g(x)h(x)$ would produce a polynomial where either $g(\alpha) = 0$ or $h(\alpha) = 0$, contradicting the minimality of $m_{\alpha,k}(x)$. To find the minimal polynomial we can do (ex. 20 in the book), it is surprisngly linked to finding a particular characteritic polynomial!!

Proposition 19.2.21: Finding Minimal Polynomial

Let F/k be a finite field extension. Then there exists a minimal polynomial $m_{\alpha,k}$ such that $m_{\alpha,k}(\alpha) = 0$.

***Proof*:**
Let F/k be a finite field extension.

1. For any $\alpha \in F$, prove that α acting by left multiplication on F is an k-linear transfomration of F, label it L_α.
2. Show α is a root of the characteristic polynomial for L_α. This gives an effective procedure for determining an equation of degree n satisfied by an element α in the extension of k of degree n.

Alternitavely, we can use what find a Gröbner bases(see section 23.3)

Example 19.2.22: Minimal Polynomials

1. The minimal polynomial for $\sqrt{2}$ over $\mathbb{Q}$ is $x^2 - 2$ and $\sqrt{2}$ is of degree 2 over $\mathbb{Q}$: $[\mathbb{Q}(\sqrt{2}), \mathbb{Q}] = 2$
2. The minimal polynomial for $\sqrt[3]{2}$ over $\mathbb{Q}$ is $x^3 - 2$ and $\sqrt[3]{2}$ is of degree 3 over $\mathbb{Q}$: $[\mathbb{Q}(\sqrt[3]{2}), \mathbb{Q}] = 3$
3. similarly, for any $n > 1$, the polynomial $x^n - 2$ is irreducible (by Eisenstein's Criterion).
4. The minimal polynomial and the degree of an element α are dependent on the base field! For example, if $F = \mathbb{R}$, then the element $\sqrt[n]{2}$ has degree 1, with minimal polynomial
$$m_{\sqrt[n]{2},\mathbb{R}}(x) = x - \sqrt[n]{2}$$
5. the minimal polynomial for the golden ratio φ is $x^2 - x - 1$.
6. In general, most algebraic elements do not have a special symbol representing them, and so we would have to say that some $\alpha \in K$ has minimal polynomial (say) $x^5 - x - 1$

As a result of theorem 19.2.8, it also makes sense to ask the context of the existence of an irreducible polynomial what is the degree of an α over F: given $p(x)$ is the polynomial for which α is a root, then it's the degree of $\dim_F F[x]/(p(x)) = \dim_F F(\alpha)$. We may write $\deg \alpha$ (or $\deg_F \alpha$) for the degree of α (over F). If $\deg \alpha = 1$, then $\alpha \in F$.

Combining the classification of finite simple extensions with corrollary 19.2.6, we get to easily represent $F(\alpha)$

Corollary 19.2.23: Representing $F(\alpha)$

Suppose in theorem 19.2.8 that $p(x)$ is of degree n. Then

$$F(\alpha) = \left\{a_0 + a_1\alpha + a_2\alpha^2 + \cdots + a_{n-1}\alpha^{n-1} | a_0, a_1, ..., a_{n-1} \in F\right\} \subseteq K$$

Proposition 19.2.24: Finite Simple Extensions Elements

Let $k(\alpha)/k$ be a finite simple extension. Then:

$$k[\alpha] = k(\alpha)$$

***Proof*:**
exercise

This justifies why we only need to care about polynomials, and not rational polynomials.

Example 19.2.25: Simple Extension

1. In $\mathbb{Q}(\sqrt{2})$, $\dim_{\mathbb{Q}} \mathbb{Q}(\sqrt{2}) = 2$, and we can write $\mathbb{Q}(\sqrt{2}) = \left\{a + b\sqrt{2} \mid a, b \in \mathbb{Q}\right\}$. We could also have chosen the other root of $x^2 - 2$, namely $-\sqrt{2}$. However, these two fields are isomrophic: $\mathbb{Q}(\sqrt{2}) \cong \mathbb{Q}(-\sqrt{2})$, showing they are algebraically indistinguishable.
2. In $\mathbb{Q}(\sqrt[3]{2})$, $\dim_{\mathbb{Q}} \mathbb{Q}(\sqrt[3]{2}) = 3$, and we can write $\mathbb{Q}(\sqrt[3]{2}) = \left\{a + b\sqrt[3]{2} + c(\sqrt[3]{2})^2 \mid a, b, c \in \mathbb{Q}\right\}$.
3. Over $\mathbb{R}$, $x^3 - 2$ has no more solutions than the last example. However, over $\mathbb{C}$, there are two more solutions:
$$\zeta_2 = \sqrt[3]{2}\left(\frac{-1 + i\sqrt{3}}{2}\right) \qquad \zeta_3 = \sqrt[3]{2}\left(\frac{-1 - i\sqrt{3}}{2}\right)$$
Thus, $\mathbb{Q}(\zeta_2)$ and $\mathbb{Q}(\zeta_3)$ are field extensions.

Thus, we have fully classified all simple extensions of a field and dealt with transcendental extensions. In following sections, we will show how to classify fields generated by finitely many elements (ex. $k(\alpha, \beta)$) and infinitely many elements. Before continuing into that endeavour, isomorphisms between fields should be given some more attention to be more cear on how to classify fields.

The fields in example 19.2.25 need not be isomorphic (as fields) if the roots solve different polynomials, meaning different irreducible polynomials will produce different extensions:

Example 19.2.26: $\mathbb{Q}(\sqrt{2}) \ncong \mathbb{Q}(\sqrt{5})$

Take $\mathbb{Q}(\sqrt{2})$ and $\mathbb{Q}(\sqrt{5})$. Let's say φ is an isomorphism. Then we know that

$$\varphi(1) = 1$$

since 1 is the only element of order 1 in both fields. Then:

$$\varphi(\sqrt{2}) = a + b\sqrt{5}$$

for some non-zero value in the codomain. Then

$$\varphi(2) = a^2 + 2ab\sqrt{5} + 5b^2$$

but we also know that

$$\varphi(1) = 1 \qquad \varphi(2) = \varphi(1+1) = \varphi(1) + \varphi(1) = 1 + 1 = 2$$

so

$$2 = a^2 + 2ab\sqrt{5} + 5b^2 \qquad a, b \in \mathbb{Q}$$

If $a = b = 0$, $2 = 0$, a contradiction. If $a, b \neq 0$, then:

$$\frac{2 - a^2 - 5b^2}{2ab} = \sqrt{5}$$

implying that $\sqrt{5}$ is a rational number, a contradiction. If $a = 0, b \neq 0$, then:

$$\frac{\sqrt{2}}{\sqrt{5}} = b$$

but b is a rational number, a contraidction. If $a \neq 0, b = 0$, then:

$$\sqrt{2} = a$$

but a is a rational number, a contradiction. Thus, these two fields cannot be isomorphic.

However, notice that the extension of to $\mathbb{Q}(\sqrt{2})$ and $\mathbb{Q}(-\sqrt{2})$ were isomorphic, and so was all the extensions of $\mathbb{Q}$ by roots of $x^3 - 2$:

$$\mathbb{Q}(\sqrt[3]{2}) \cong \mathbb{Q}\left(\sqrt[3]{2}\left(\frac{-1 + i\sqrt{3}}{2}\right)\right) \cong \mathbb{Q}\left(\sqrt[3]{2}\left(\frac{-1 - i\sqrt{3}}{2}\right)\right)$$

even though those 3 fields are not equal. In particular, that means that non of these fields contain *all* of the roots of $x^3 - 2$. One of the large motifs in the following study of extensions is to find extensions in which for any irreducible (or more generally any polynomial) will contain *all* of the roots.

19.2.1 Isomorphism of Fields Extensions

As we have seen in theorem 19.2.2, If we know that an irreducible polynomial $p(x) \in k[x]$ has a root in a field F. Then there exists a [not necessarily unique] isomorphism φ such that the following diagram commutes:

$$\begin{tikzcd}
k[x]/(p(x)) \arrow[r, "\varphi"] & F \\
k \arrow[u, hook, "\iota"] \arrow[ur, hook, "\iota"'] &
\end{tikzcd}$$

The fact that φ is not unique gives raise to several interesting results. As we have shown in example 19.2.3, $\mathbb{Q}(\sqrt{2})$ has two homomorphisms into $\mathbb{R}$. In fact, since $-\sqrt{2} \in \mathbb{Q}(\sqrt{2})$, we have that:

$$\begin{tikzcd}
\mathbb{Q}(\sqrt{2}) \arrow[r, "\varphi"] & \mathbb{Q}(-\sqrt{2}) \\
\mathbb{Q} \arrow[u, hook, "\iota"] \arrow[ur, hook, "\iota"']
\end{tikzcd}$$

If we extend the bottom to be the identity map form k to k, then we get a square:

$$\begin{tikzcd}
\mathbb{Q}(\sqrt{2}) \arrow[r, "\sim"] & \mathbb{Q}(-\sqrt{2}) \\
\mathbb{Q} \arrow[u] \arrow[r, "\mathrm{id}"] & \mathbb{Q} \arrow[u]
\end{tikzcd}$$

Showing the fields with the two roots of $x^2 - 2$ are isomorphic. In fact,$-\sqrt{2} \in \mathbb{Q}(\sqrt{2})$, and so this is in fact a field automorphism (though in general it doesn't have to be, like $\mathbb{Q}(\sqrt[3]{2})$ and $\mathbb{Q}(\sqrt[3]{2}\zeta_3)$). We have claimed earlier than two fields that have two different roots of the same irreducible polynomial adjoined to them are algebraically indistinguishable. This shows it in the case for $x^2 - 2$ over $\mathbb{Q}$, and the following theorem shows the generlizatoin of this result. Note how by keeping track of the root we have in fact recovered a sense of uniqueness:

Theorem 19.2.27: Extending Isomorphisms to Adjoint Fields

Let $k_1 \subseteq F_1 = k_1(\alpha_1)$, $k_2 \subseteq F_2 = k_2(\alpha_2)$ be simple extensions and let $\varphi : k_1 \to k_2$ be an isomorphism of fields. Let $p(x) \in k_1[x]$ be an irreducible polynomial and let $p'(x) \in k_2[x]$ be the irredcibble polynomial obtained by applying the map φ to the coefficients of $p(x)$. Let α_1 be a root of $p(x)$ (in some extension of k_1) and let α_2 be a root of $p'(x)$ (in some extension k_2). Then there is an isomorphism:

$$\sigma : k_1(\alpha_1) \to k_2(\alpha_2) \qquad \alpha_1 \mapsto \alpha_2$$

mapping α_1 to α_2 and extending φ, .i.e., such that σ restricted to F is the isomorphism φ.

In other words, if we have

$$\begin{tikzcd}
k_1(\alpha) & k_2(\beta) \\
k_1 \arrow[u, hook] \arrow[r, "\sim"] & k_2 \arrow[u, hook]
\end{tikzcd}$$

and $\iota(m_{\alpha,k_1})(x)) = m_{\beta,k_2}(x)$ (i.e. it maps the coefficients of the first minimal polynomial to the second one), then we can *uniquely* extend ι to an isomorphism j between the two adjoint fields:

$$\begin{tikzcd}
k_1(\alpha) \arrow[r, dashed, "\sim"] & k_2(\beta) \\
k_1 \arrow[u, hook] \arrow[r, "\sim"] & k_2 \arrow[u, hook]
\end{tikzcd}$$

such that $j(\alpha) = \beta$. Note that j being an extension of ι means the elements of k_1 are fixed in j.

***Proof*:**
Let $\varphi : k_1 \to k_2$ be an isomorphism of fields. The map φ induces a ring isomorphism:

$$f^\varphi : k_1[x] \to k_2[x]$$

defined by applying φ to the coefficeints of a polynomial in $k_1[x]$. Let $p(x) \in k_1[x]$ be an irreducible polynomial, and let $p'(x) \in k_2[x]$ be the polynomial obtained by applying the map φ to the coefficients of $p(x)$, i.e., the image of $p(x)$ under f^φ. The isomorphism f^φ maps the maximal ideal $(p(x))$ to the ideal $(p'(x))$, so the ideal is also maximal, which shows that $p'(x)$ is also irreducible in $k_2[x]$.

Taking the quotinet by these ideals of the map, we get the following isomorphism of fields

$$k_1[x]/(p(x)) \to k_2[x]/(p'(x))$$

By theorem 19.2.8, the left side is isomorphic to $k_1(\alpha_1)$, and the right side to $k_2(\alpha_2)$, so composing these maps will produce a map for σ! It should also be clear now that the restriction of this isomorphism to k_1 is φ.

This isomorphism will come back big time when we address galois theory. It is sometimes represented in as in the following diagram:

$$\begin{tikzcd} \sigma : & k_1(\alpha_1) \arrow[r, "\cong"] \arrow[d, dash] & k_2(\alpha_2) \arrow[d, dash] \\ \varphi : & k_1 \arrow[r, "\cong"] & k_2 \end{tikzcd}$$

If we let $\mathrm{id} : k \to k$ be our k-isomorphism, and $m_{\alpha,k}(x)$ be our minimal polynomial with roots α, β, then $\varphi : k(\alpha) \xrightarrow{\sim} k(\beta)$ is also an isomorphism, showing that the roots of a minimal polynomials are algebraically indistinguishable. Furthermore, we can upgrade Kronecker's Theorem so that it gives us a universaly property: instead of L being a field that has a root of $p(x)$, upgrade that to $\alpha \in L$ being a root of $p(x)$. Then there exists a *unique* homomorphism mapping $\overline{x}$ to α (with uniqueness established by the above theorem).

Through this theorem, we get another important insight. If we have a non-trivial finite field extension $k(\alpha)/k$ which has roots of a minimal polynomial $m_{k,\alpha}(x)$, then if any other roots of $m_{k,\alpha}(x)$ are in $k(\alpha)$, there is a k-automorphisms of $k(\alpha)$ (can you see why a k-automorphism is determined by where it sends the roots?). This means we can find elements of the group $\mathrm{Aut}_k(k(\alpha))$ (where the subscript means we fix k), and hence the structure of $k(\alpha)$ [9]. More conretely, $k(\alpha)/k$ represents adding a new element α to the field k which satisfies an algebraic relation $m_{\alpha,k}(\alpha)$ (a polynomial). Finding information about the new field $k(\alpha)$ by looking at the k-automorphisms is essentially studying the properties of the polynomial $m_{\alpha,k}(x)$.

Even better, we will show that this characterizes *all* k-automorphisms, giving us a complete description of these symmetries. These symmetries are actually at the heart of studying fields in galois theory. When Galois originally formulated his theory, he looked different ways to permute the roots and noticed that there are often limitation to the permutations of the roots since the roots may be related

[9] Recall that we already used the automorphism group to deduce information about groups, for example $\mathrm{Out}(G) \cong_{\mathbf{Grp}} \mathrm{Aut}(G)$ if and only if G is abelian

to each other[10]. For example if α is a root of $x^3 - x - 1$, then the other two roots are:

$$\frac{1}{1-\alpha} \qquad 1 + \frac{1}{\alpha}$$

Similarly, if ζ is a root of $x^n - 1$, then the other roots of $x^n - 1$ are $\zeta^2, \zeta^3, ..., \zeta^{n-1}, 1$. Thus, we take some time to define our terms more precisely, and elaborate further what the automorphisms look like:

Definition 19.2.28: Automorphism and fixed Element

Let K be a field and $\mathrm{Aut}(K)$ be the collection of automorphisms (isomorphisms from k to itself). Then:

1. An auotmorphism $\sigma \in \mathrm{Aut}(K)$ is said to *fix* an element $\alpha \in K$ if $\sigma\alpha = \alpha$. If F is a subset of K, then an automorphism σ is said to *fix* F if it fixes all the elements of F

$$\sigma a = a \qquad \forall a \in F$$

2. If F/k be an extension of fields, then $\mathrm{Aut}(F/k)$ or $\mathrm{Aut}_k(F)$ is the collection of all automorphsism of F which fix k.

the previous theorem gives us the following link between $\mathrm{Aut}_k(F)$ and F given that F is finite and simple:

Corollary 19.2.29: Order of Fixed Automorphism Group

Let F/k be a simple finite extension (so $F = k(\alpha)$ for some $\alpha \in F$), and let $m_{k,\alpha}(x) \in k[x]$ be the minimal polynomial of α. Then $|\mathrm{Aut}_k(F)|$ is equal to the number of *distinct* roots of $m_{k,\alpha}(x)$ in F. In particular,

$$|\mathrm{Aut}_k(F)| \leq [k(\alpha) : k] \qquad |\mathrm{Aut}_k(F)| = [k(\alpha) : k]_d$$

where $[k(\alpha) : k]_d$ represents the number of *distinct* roots $k(\alpha)$ contains of the minimal polynomial $m_{k,\alpha})(x)$. Equality holds if and only if $m_{k,\alpha}(x)$ factors over F as a product of *distinct linear* polynomials

This result will be called the *fixed root criterion* when we cover Galois theory.

***Proof*:**
Let $\sigma \in \mathrm{Aut}_k(F)$ be an automorphism fixing k (so for any $t \in k$, $\sigma(t) = t$). Then since every element $t \in F = k(\alpha)$ can be written as a polynomial $t = f(\alpha) \in k[\alpha]$ (recall that $k(\alpha)$ is finite, and so has a basis of the form $\alpha, \alpha^2, ..., \alpha^n$ for some appropriat n), and σ fixes k, all the elements are determined by where α maps to, that is $\sigma(\alpha)$. Then since σ is an automorphism (hence ring homomorphism), given

$$x^n + a_{n-1}x^{n-1} + \cdots + a_1 x + a_0$$

[10]Later we shall see that the roots are almost never related to each other, that is most Galois groups are S_n, but it is a good insight to notice that the roots of small polynomials are often related

is the minimal polynomial with coefficients from k so that

$$\alpha^n + a_{n-1}\alpha^{n-1} + \cdots + a_1\alpha + a_0 = 0$$

then, applying any automorphism σ to this polynomial will get us

$$\begin{aligned} \sigma(\alpha^n + a_{n-1}\alpha^{n-1} + \cdots + a_1\alpha + a_0) &= \sigma(0) \\ \sigma(\alpha)^n + a_{n-1}\sigma(\alpha)^{n-1} + \cdots + a_1\sigma(\alpha) + \sigma(a_0) &= 0 \qquad \sigma \text{ fixes } a_i \text{ for each } i \end{aligned}$$

since $a_i \in k$ for each i, showing that $\sigma(\alpha)$ is also a root of the minimal polynomial, i.e. σ must map α to some other root of $m_{k,\alpha}$.

As α is the generator, all roots are determined by where ' maps to, hence there can be at most $\deg(m_{k,\alpha}(x))$ automorphisms.

Conversely, for any root β of $p(x)$ in $k(\alpha)$, $\beta \in k(\alpha)$, we can take the isomorphism $\mathrm{id} : k \to k$, which by theorem 19.2.27, there exists a unique lift to a root $k(\alpha) \to k(\beta)$ that fixes k. Since $k(\alpha) \cong k(\beta)$, then we can use this isomorphism to define an automorphism from $k(\alpha)$ to $k(\alpha)$, showing us that $\mathrm{Aut}_k(F)$ must have at least as many elements as there are roots of the minimal polynomial in F. If F contains all the roots of $p(x)$ and each root of $p(x)$ is distinct, then $|\mathrm{Aut}_k(F)| = [F : k]$, completing the proof.

Interpreting the above example as a group actions on the set of roots R_F of $m_{\alpha,k}(x)$ in F, $\mathrm{Aut}_k(F) \curvearrowright R_F$, we have that $\mathrm{Aut}_k(F)$ acts *faithfully and transitively* on the set of roots of $p(x)$ in F (the group $\mathrm{Aut}_k(F)$ can be identified with some transitive subgroup of the symmetric group acting on the roots of $p(x)$ in F). This correlation hints at the connection beween fields and groups at the heart of the galois theory, since any collection of automorphisms will always form a group[11]. The main impediment to directly translating this result of more general [finite] field extensions[12] is that the minimal polynomial might not factor into *linear* and *distinct* terms over the simple extension $k(\alpha)/k$[13]. For example, in $\mathbb{Q}(\sqrt[3]{2})$, the polynomial $x^3 - 1$ doesn't fully factor. The next two sections will encompass finding field extensions in which these conditions are met. In particular, we shall take some time to find fields in which:

1. All polynomials factor (splitting fields)
2. If $f(x) \in k[x]$ is irreducible, then in F/k, either $f(x)$ factors completely or $f(x)$ remains irreducible (normal extensions)
3. Given a polynomial $f(x)$, F/k factors completely (Splitting field)
4. If $f(x)$ is irreducible in k, then in F/k, $f(x)$ factors into linear terms of degree 1 (Separable extensions)

As a final word on relating isomorphisms and extensions, note that if $\dim_k A = \dim_k B$, this does not imply $[B : A] = 1$ or that $A \cong_k B$:

[11] If you are not convinced, now is good time to check this
[12] infinite extension are more finicky to deal with, and are relegated to chapter ref:HERE
[13] It is possible that a irreducible polynomial factors linearly, but the factors are not distinct, see section 19.6

Example 19.2.30: Nuance On Dimension

Let $A = \mathbb{Q}(\sqrt{3})$ and $B = \mathbb{Q}(\zeta_3\sqrt{3})$. Then the two are of the same dimension, both extend $\mathbb{Q}$ and $A \cong_k B$, but $[B : A]$ is not even well defined. To visualize:

$$\begin{tikzcd}
\mathbb{Q}(\sqrt{3}) & \cong_k & \mathbb{Q}(\zeta_3\sqrt{3}) \\
& \mathbb{Q} \arrow[ul] \arrow[ur] &
\end{tikzcd}$$

Exercise 19.2.1

1. Let $\text{ch}(k) = n$, $n \geq 0$, and $\varphi : k \to K$ be a homomorphism. Then $\text{ch}(K) = n$.
2. Show that the intersection of any collection of subfields of a field is a field. Is the same true for the union? What if the fields where given a partial order through inclusion?
3. (personal idea, not sure if valid yet) Want to give another intuition using the fact that $R[x]$ is a free monoid, so that in a sense, we can get a "presenation" when we do $R[x]/p(x)$. If $p(x)$ is also irreducible then we get a field as a result.
4. Let $N \subseteq M_{2\times 2}(\mathbb{R})$ be a subset where
$$N = \left\{ \begin{pmatrix} a & b \\ -b & a \end{pmatrix} \;\middle|\; a, b \in \mathbb{R} \right\}$$
Then $N \cong \mathbb{C}$.
5. Show that If Let E/k and F/k be field extensions. Show that if E/k is finite, then so is EF/F. Show that if E/k and F/k are finite, so is EF/F.
6. Suppose that $f(x) \in k[x]$ where $f(x)$ factors into $f(x) = p_1(x)p_2(x)\cdots p_n(x)$ where $p_i(x) \neq p_j(x)$. Show that $k[x]/(f(x))$ is a product of fields (Hint: Chinese Remainder Theroem). If one of the terms had multiplicity greater than 1, would the result still be a product of fields?
7. (If you know Calculus) The $\mathbb{R}$-algebra $\mathbb{R}[x]/(x^2)$ is sometimes the *dual numbers*. Let any element of this degree 2 extension of $\mathbb{R}$ be written in the form $a + b\epsilon$. If $p'(x)$ is the derivtive of $p(x)$, show that $p(a + b\epsilon) = P(a) + bP'(a)\epsilon$, that is, the derivitve "automatically" shows up.
8. Let m, n be square free and $n \neq m$. Show that $\mathbb{Q}(\sqrt{n}) \not\cong_{\mathbb{Q}} (\sqrt{m})$.
9. Let ζ_n be a root of unity where n is odd. Show that $\mathbb{Q}(\zeta_n) \cong_{\mathbb{Q}} \mathbb{Q}(\zeta_{2n})$

19.3 Algebraic Extensions

We'll now focus some attention on the algebraic elements of an extension. We first give name to elements for which all elements are algebraic or non-algebraic:

Definition 19.3.1: Algebraic and Transcendental Extensions

Let F be a field and let K be an extension of F. Then:

1. The element $\alpha \in K$ is said to be *algebraic* over F if α is a root of some nonzero polynomial $f(x) \in F[x]$.
2. If α is not algebraic over F (i.e., is not the root of any nonzero polynomial with coefficeints in F) then α is said to be *transcendental over* F.
3. The extension K/F is said to be *algebraic* or an *algebraic extension* if every element of K is algebraic over F.
4. The extension K/F is said to be *transcendental* or a *transcendental extension* if every element of K not in F is transcendental over F.

Note that if an element α is algebraic over a field F, then it is algebraic over *any* extension field L of F (if $f(x)$ having α as a root has coefficients in F, then it also has coefficients in L). Note too that the polynomial for which α must be a root need not be irreducible, however it is an immediate consequence that if α is the root of a polynomial $p(x) = (a_1(x))^{n_1} \cdots (a_k(x))^{n_k}$, it is a root for some irreducible polynomial $a_i(x)$.

Example 19.3.2: Algebraic extensions

1. Take $\mathbb{Q}(\sqrt{2})/\mathbb{Q}$. Then every element in $\mathbb{Q}(\sqrt{2})$ is a solution to a polynomial over $\mathbb{Q}$, thus $\mathbb{Q}(\sqrt{2})/\mathbb{Q}$ is algebraic: any element $a + b\sqrt{2}$ is the root of $\left(\frac{x-a}{b}\right)^2 - 2$. If you take $\mathbb{Q}(\sqrt{2})$ as a vector-space over $\mathbb{Q}$, then we see that 1 and $\sqrt{2}$ generate $\mathbb{Q}(\sqrt{2})$. Note that $\sqrt{2}$ is not in $\mathbb{Q}$, but but $\sqrt{2}$ is algebraic over $\mathbb{Q}$).

   More generally, is it true that any finite simple extension $F(\alpha)$ of F is an algebraic extension? Is every element of $F(\alpha)$ the root of a polynomial with coefficients in F?
2. $\mathbb{C}$ as an $\mathbb{R}$-vector space is generated by 1 and i. Both of these are root to polynomials over $\mathbb{R}$, in particular $x - 1$ and $x^2 + 1$. It is a famous result that any $c \in \mathbb{C}$ is a solution of a polynomial over $\mathbb{R}$ (i.e., with coefficients from $\mathbb{R}$). Thus, $\mathbb{C}/\mathbb{R}$ is algebraic.

   The degree of $[\mathbb{C} : \mathbb{R}] = 2$
3. Consider $\mathbb{C}$ again. For any $\alpha \in \mathbb{C}$, is there a polynomial $p(x) \in \mathbb{C}[x]$ for which $p(\alpha) = 0$? In a sense, we're asking if $\mathbb{C}$ is algebraic over itself. [a]
4. For a moment, take for granted that there is no polynomial in $\mathbb{Q}$ such that π is a root (TBD? appendix?). Then π is *transcendental* over $\mathbb{Q}$, and so $\mathbb{R}/\mathbb{Q}$ is *not* an algebraic extension.

   Here is an idea to ponder: What is $[\mathbb{R} : \mathbb{Q}]$? The answer will be given in example 19.3.15

[a] The answer is yes! We call a field such that every element of it has a root over itself *algebraically closed*

In these examples, the algebraic extensions all had a finite degree. Is it possible to have an algebraic extension with infinite degree? If K is algebaric over F, then every element of K is a roof a polynomial over F? What about the converse: If K/F is finite, is K algebraic over F? (take a moment to ponder here)

Proposition 19.3.3: degree of Algebraic Extension

The element α is algebraic over F if and only if the simple extension $F(\alpha)/F$ is finite. More precisely, if α is an element of an extension of degree n over F, then α satisfies a polynomial of degree at most n over F and if α satisfies a polynomial of degree n over F, then the degree of $F(\alpha)$ over F is at most n.

***Proof*:**
If α is algebraic over F (so for some $p(x) \in F[x]$, $p(\alpha) = 0$), then the degree of the extension $F(\alpha)/F$ is the degree of the minimal polynomial, making it an extension of finite degree at most n.

Conversly, let α be an element of degee n over F (ex. $[F(\alpha) : F] = n$). Then by 19.2.5, we know that $n + 1$ elements will be linearly dependent:

$$c_0 + c_1\alpha + c_2\alpha^2 + \cdots + c_n\alpha^n = 0$$

with $c_0, c_2, ..., c_n \in F$ which are not all 0. Thus, α must be the root of some non-zero polynomial with coefficients from F (of degree $\leq n$), showing α is algebraic over F.

Corollary 19.3.4: Finite Extension Then Algebraic Extension

If the extension K/F is finite, then it is algebaric

***Proof*:**
let K/F be finite and take any $\alpha \in K$. Consider the simple extension $F(\alpha)$. Consider K as an F-vector space, then since the extension is finite, $\dim_F K = n$. Now, $F(\alpha) \subseteq K$ would be a subspace of K ($F(\alpha)$ also being an F-vector sapce), so

$$\dim_F F(\alpha) \leq \dim_F K \qquad \text{or} \qquad [F(\alpha) : F] \leq [K : F]$$

showing the extension is finite, and so by proposition 19.3.3, α is algebraic over F. since α was an artibrary element of K, this means that K is algebaric over F, as we sought to show.

Though not important to us in this chapter, we also have the following characterisation of an algegraic extension more useful in algebraic number theory:

Corollary 19.3.5: Algebraic Extension and Algebras

K/F is algebraic if and only if every sub F-algebra is a field

***Proof*:**
exercise

Note that the converse of corrollary 19.3.4 is not true: If K is algebraic over F, this does not mean that K/F has finite degree

Example 19.3.6: Algebraic and infinite degree

We require corollary 19.3.14 which is introduced in a couple of pages, but we can still state the case with the added assumption here to get the idea:

Take $\mathbb{C}/\mathbb{Q}$, and take the set $\overline{\mathbb{Q}}$ to be the set of all algebraic elements over $\mathbb{Q}$. Then by the aforementioned corollary, this is a subfield of $\mathbb{C}$. Since $\sqrt[n]{2}$ is algebraic over $\mathbb{Q}$ (root of the polynomial $x^n - 2$), we have

$$[\mathbb{Q}(\sqrt[n]{2}) : \mathbb{Q}] = n$$

for every n. Since $\mathbb{Q}(\sqrt[n]{2}) \subseteq \overline{\mathbb{Q}}$, it follows that $\overline{\mathbb{Q}}$ is an algebraic extension of infinite degree.

However, a weaker version of this direction does have a converse, and we will build up to it now. Let's first show that the finite extension of a finite extension remains a finite extension (if you like, the composition of finite extensions is a finite extension)

Theorem 19.3.7: Multiplicity of Finite Extensions (Tower Law)

Let $k \subseteq E \subseteq F$ be fields. Then

$$[F : k] = [F : E][E : k]$$

that is, extension degree multiply each other, where if one side of the equation is infinite, so is the other.

***Proof*:**

The infinite case is clear, so we will deal with the finite case. First, if F is finite dimensional over k, then the subspace E is also finite dimensional. Furthermore, any linear combination of elements of F over the field k naturally is a linear combination over the larger field E. Thus, if $[F : k]$ is finite, so is $[F : E]$ and $[E : k]$.

Conversely, suppose $[F : E]$ and $[E : k]$ are both finite. Let $(e_1, e_2, ..., e_m)$ be a basis for E over k and $(f_1, f_2, ..., f_n)$ be a basis for F over E. I claim that

$$(e_1 f_1, e_1 f_2, ..., e_1 f_n, e_2 f_1, ..., e_m f_n)$$

is a basis for F over k.

Let $g \in F$. Then for appropriate $d_i \in E$,

$$g = \sum_j d_j f_j$$

Next, for each d_j, for appropriate $c_{ij} \in k$, we have

$$d_j = \sum_i c_{ij} e_i$$

Thus, we have:

$$g = \sum_i \sum_j c_{ij} e_i f_j$$

Therefore, the products $e_i f_j$ span F, showing that F must at most be finite dimensional.

To show they form a basis, assume

$$\sum_{i,j} \lambda_{ij} e_i f_j = 0$$

Then:

$$\sum_j \left(\sum_i \lambda_{ij} e_i \right) f_j = 0$$

which, since the the elemnts f_j are linearly independent over E, we have that $\sum_i \lambda_{ij} e_i = 0$. But then, the elements e_i are linearly indepenent over e_i,and so each $\lambda_{ij} = 0$, showing that the original linear combination is also linearly independent, as we sought to show.

Notice how this corollary mirrors Lagranges theorem with respect to the order of groups. This parrallel is actually quite a deep one which will capture our attention in the next chapter. The following corollary deepends this connection:

Corollary 19.3.8: Degree Extension Divisibility

Suppose L/F is a finite extension and let K be any subfield of L contianing F, $F \subseteq K \subseteq L$. Then $[K : F]$ divides $[L : F]$

***Proof*:**
Since

$$[L : F] = [L : K][K : F]$$

this result holds.

This result gives us a similar power that Lagrange's theorem gives us

Example 19.3.9: Use Of Tower Law

1. We can quickly rule out that $\sqrt{2} \notin \mathbb{Q}(\sqrt[n]{2})$ if n is odd. To see this, if $\sqrt{2} \in \mathbb{Q}(\sqrt[n]{2})$, then $\mathbb{Q}(\sqrt{2}) \subseteq \mathbb{Q}(\sqrt[n]{2})$. Then by what we've just shown:

$$[\mathbb{Q}(\sqrt[n]{2}) : \mathbb{Q}] = [\mathbb{Q}(\sqrt[n]{2}) : \mathbb{Q}(\sqrt{2})][\mathbb{Q}(\sqrt{2}) : \mathbb{Q}]$$

   However, that would imply that an even number divides an odd numbers – a contradiction.

2. More generally, we can conclude the following: If K/k is an extension of odd degree and $\alpha \in K$, then α can be written as a polynomial in α^2, that is $\alpha = p(\alpha^2) = \sum_i k_i(\alpha^2)^i$ for $k_i \in k$.

   To see this, we first notice that through simple set theory we must have

$$k \subseteq k(\alpha^2) \subseteq k(\alpha)$$

   and that $k(\alpha)/k(\alpha^2)$. To find the degree d of this extension, notice that over $k(\alpha^2)$, we get that $x^2 - \alpha^2 \in k(\alpha^2)[x]$, so $d \leq 2$. On the other hand, d must divide $[k(\alpha) : k]$ which is odd, forcing us to have $d = 1$. Thus $k(\alpha^2) = k(\alpha)$, and so $\alpha \in k(\alpha^2)$.

3. Let's say K/k and $[K : k]$ is prime. Then K contains *no* fields that contain k. This is similar to how all finite groups of prime order are simple.

4. Take $\mathbb{Q} \subseteq \mathbb{R}$ and consider $\mathbb{Q}/(x^3-3x-1) \cong \mathbb{Q}(\alpha)$ as in example 19.2.4. Since $[\mathbb{Q}(\sqrt{2}) : \mathbb{Q}] = 2$, and $[\mathbb{Q}(\alpha) : \mathbb{Q}] = 3$, and $2 \not| 3$, then $\mathbb{Q}(\sqrt{2}) \not\subseteq \mathbb{Q}(\alpha)$, so $\sqrt{2} \notin \mathbb{Q}(\alpha)$! It is much more difficult to directly show $\sqrt{2}$ cannot be written as a $\mathbb{Q}$-linear combination of $1, \alpha, \alpha^2$.

5. More in Dummit and Foote?

Continuing to strive towards a partial converse, we define the following generlization of a simple extension. The key feature about finitely generated fields is that they can be decomposed into simple extensions:

Lemma 19.3.10: Recursive Definition Of Extension Field

$F(\alpha, \beta) = (F(\alpha))(\beta)$, i.e., the field generated over F by α and β is the field generated by β over the field $F(\alpha)$ generated by α.

***Proof*:**
This proof takes complete advantage that by definition, field extensions are the *smallest* field containing F and α (or α and β)

The field $F(\alpha, \beta)$ is the smallest field to contain F, α, β, so $F(\alpha, \beta) \subseteq (F(\alpha))(\beta)$. Conversely, $F((\alpha))(\beta)$ is the smallest field to contain $F(\alpha)$ and β. Since $F(\alpha) \subseteq F(\alpha, \beta)$ and $\beta \in F(\alpha, \beta)$, $(F(\alpha))(\beta) \subseteq F(\alpha, \beta)$, implying

$$F(\alpha, \beta) = (F(\alpha))(\beta)$$

With this, we can break down any finitely generated:

$$K = F(\alpha_1, \alpha_2, ..., \alpha_k) = (F(\alpha_1, \alpha_2, ..., \alpha_{k-1}))(\alpha_k) = ((\cdots F(\alpha_1))(\alpha_2)) \cdots (\alpha_n)$$

Thus, we get the following chain of extensions:

$$F = F_0 \subseteq F_1 \subseteq F_2 \subseteq \cdots \subseteq F_k = K$$

where

$$F_{i+1} = F_i(\alpha_{i+1}) \qquad i = 0, 1, ..., k-1$$

Importantly, this reduces everything down to lots of simple extensions! Now, suppose that the elements $\alpha_1, \alpha_2, ..., \alpha_k$ are algebraic over F of degrees $n_1, n_2, ..., n_k$

$$[F(\alpha_1) : F] = n_1,\ [F(\alpha_2) : F] = n_2 \cdots [F(\alpha_k) : F] = n_k$$

By proposition 19.2.20, we can replace F with F_i. Then these can all be characterized in terms of simple extensions, meaning we can use proposition 19.3.3. The relative extension degree $[F_{i+1} : F_i]$ is equal to the degree of the minimal polynomoial of α_{i+1} over F_i, which is at most n_{i+1} (and is equal to n_{i+1} if and only if the minimal polynomial of α_{i+1} over F remains irreducible over F_i). So, by the Tower Law:

$$[K : F] = [F_k : F_{k-1}][F_{k-1} : F_{k-2}] \cdots [F_1 : F_0] \leq [F(\alpha_1) : F][F(\alpha_2) : F] \cdots [F(\alpha_k) : F] \qquad (19.1)$$

And so $[K : F]$ must be a *finite* extension of degree less than or equal to $n_1 n_2 \cdots n_k$! We sum up this entire discussion of finite degrees in the following theorem:

Theorem 19.3.11: Degree of Algebraic Extension

The extension K/F is finite if and only if K is generaed by a finite number of algebraic elements over F. More precisely, a field generated over F by a finite number of algebraic elements of degrees $n_1, n_2, ..., n_k$ is algegraic of degree $\leq n_1, n_2, ..., n_k$

Before, we hade

$$\alpha \text{ algebraic over } F \Leftrightarrow F(\alpha)/F \text{ finite}$$

and when we try to generlize, we loose one side of the implication

$$K \text{ algebraic over } F \Leftarrow K/F \text{ finite}$$

with the counter example 19.3.6 for $\Rightarrow$.

We now restricted the hypothesis to give us the if and only if again:

$$K \text{ generated by finite number of algebraic elements over } F \Leftrightarrow K/F \text{ finite}$$

***Proof*:**
Let the extension K/F be finite, so $a_1, a_2, ..., a_n$ forms a basis for K over F. Then since $F(a_i) \subseteq K$ as a F-vector-space, $[F(a_i) : F] \leq [K : F]$, meaning it has finite degree, and so by 19.3.3, a_i is algebraic. Since a_i was arbitrary, this means that every a_i is algebraic. Thus K is generated by a finite number of algebraic elements over F.

For the converse, let's say K is generated by a finite number of algebraic elements over F. Let

$$\alpha_1, \alpha_2, ..., \alpha_n$$

all be algebraic over F and generate K. Since they generated K, $K = F(\alpha_1, \alpha_2, ..., \alpha_n)$. Then by what we've seen in equation (19.1), we have that K/F is finite.

Using what we've found, it is easier to compute the degree of non-simple extensions:

Example 19.3.12: Computing Degree of non-simple Extension

1. The extension $\mathbb{Q}(\sqrt[6]{2}, \sqrt{2})$ is $\mathbb{Q}(\sqrt[6]{2})$ since $\sqrt{2} \in \mathbb{Q}(\sqrt[6]{2})$. Another way of showing this is that the degree d of $\sqrt{2}$ over $\mathbb{Q}(\sqrt[6]{2})$ is 1. Thus, even though the degree of $\sqrt{2}$ is 2 and the degree of $\sqrt[6]{2}$ is 6, the degre of $\mathbb{Q}(\sqrt[6]{2}, \sqrt{2})$ is 6, not 12.

2. Consider $\mathbb{Q}(\sqrt{2}, \sqrt{3})$. We'll proceed to find the degree of this extensions over $\mathbb{Q}$. Since $\sqrt{3}$ is of degree 2 over $\mathbb{Q}$, then the degree of the extension $\mathbb{Q}(\sqrt{2}, \sqrt{3})/\mathbb{Q}(\sqrt{2})$ is at most 2 ($\sqrt{3}$ can at most add another dimension), and *is* 2 if and only if $x^2 - 3$ is irreducible over $\mathbb{Q}(\sqrt{2})$. To see if it is, let's suppose it is reducible. Since $x^2 - 3$ is of degree 2, by proposition 11.3.5, it is irreducible if and only if it has a root, that is, if $\sqrt{3} \in \mathbb{Q}(\sqrt{2})$.

   If $\sqrt{3} \in \mathbb{Q}(\sqrt{2})$, $\sqrt{3} = a + b\sqrt{2}$. From here, it's a lot of manipulation (squaring both sides and checking cases). It will turn out that every possibility wil make either $\sqrt{2}$, $\sqrt{3}$ or $\sqrt{6}$ rational – a contradiction.

Thus, $\deg_{\mathbb{Q}(\sqrt{2})} \mathbb{Q}(\sqrt{2}, \sqrt{3}) = 2$, so

$$\deg_{\mathbb{Q}} \mathbb{Q}(\sqrt{2}, \sqrt{3}) = 4$$

and we can find the basis for it by "closing" the set $1, \sqrt{2}, \sqrt{3}$ to produce $1, \sqrt{2}, \sqrt{3}, \sqrt{6}$, giving us

$$\mathbb{Q}(\sqrt{2}, \sqrt{3}) = \left\{ a + b\sqrt{2} + c\sqrt{3} + d\sqrt{6} | a, b, c, d \in \mathbb{Q} \right\}$$

Corollary 19.3.13: Algebraic Elements are Closed

Suppose α and β are algebraic over F. Then $\alpha \pm \beta$, $\alpha\beta$, α/β (for $\beta \neq 0$, are all algebraic (since α/β is algebraic $1/\alpha$ is also algebraic)

***Proof*:**
All these elements are part of $F(\alpha, \beta)$, which is finite over F by theorem 19.3.11, and so are algebraic by corollary 19.3.4 (in particular, $F(\alpha + \beta)$ is a simple extension, and since $\alpha + \beta \in F(\alpha, \beta)$(and $F \subseteq F(\alpha, \beta)$), we have that $F(\alpha + \beta) \subseteq F(\alpha, \beta)$ by definition, so it is a finite extension, so by corollary 19.3.4 every element will be a root of a polynomial over F).

Corollary 19.3.14: Algebraic Elements Form a Field

Let L/k be an arbitrary extension. Then the collection of elements of k that are algebraic over F form a subfield K of F.

***Proof*:**
By the previous corollary (19.3.13), we get that the collection is closed under $+$ and $\times$. From here, it is easy to verify any extra axiom we want.

Example 19.3.15: Algebraic Closure

1. Consider the extension $\mathbb{C}/\mathbb{Q}$, and let $\overline{\mathbb{Q}}$ represent the subfield of $\mathbb{C}$ containing all elements that are algebraic over $\mathbb{Q}$. Then $\sqrt[n]{2} \in \overline{\mathbb{Q}}$ for all $n \in \mathbb{N}$ (taking the positive roots), so $[\overline{\mathbb{Q}} : \mathbb{Q}] \geq n$ for every $n > 1$. Thus, $\overline{\mathbb{Q}}$ is an *infinite* algebraic extension of $\mathbb{Q}$, called the field of *algebraic numbers*. We have now proved what we have assumed in example 19.3.6.

2. Take $\mathbb{R}$ and the subfield $\overline{\mathbb{Q}}$ consisting of all algebraic elements over $\mathbb{Q}$, and consider $\mathbb{R} \cap \overline{\mathbb{Q}}$. This set is countable: $\mathbb{Q}$ is countable, and $\mathbb{Q}[x]$ consists of the countable union of polynomials of degree n ($S_0 \cup S_1 \cup \cdots$ where S_n contains all polynomials of degree n), so $\mathbb{Q}[x]$ is countable. $\overline{\mathbb{Q}}$ must have a cardinality at least less than $\mathbb{Q}[x]$, and so is also countable.

   Since $\mathbb{R}$ is uncountable, we see that there must be elements in $\mathbb{R}$ which are *not* algebraic (i.e. transcendental) over $\mathbb{Q}$. In fact, the vast majority of elements are transcendental. The subfield $\overline{\mathbb{Q}} \cap \mathbb{R}$ is a *proper* subfield of $\mathbb{R}$.

   Though the majority of elements[a] are transcendental, it is extremely difficult to show that an element *is* transcendental. For example, it was proved that $\pi = 3.1415926...$ was transcendental in 1880, and $e = 2.171828...$ is transcendental in 1870. Proving these are transcendental is very involved. Even combining these elements is tricky to prove: e^π was proven to be transcendental in 1934, however as of writing this book, it is unknown whether π^e is transcendental. In fact, it is also not known whether $\pi + e$ or πe are transcendental.

We actually have a small list of numbers which we *know* to be transcendental, even though the vast majority of them are transcendental [b]. As of writting this, there are approximately 24 classes of known transcendental numbers, including numbers like e^a for an algebraic number a, a^b where a is albebraic $\neq 0, 1$ and b is irrational (ex. $2^{\sqrt{2}}$), $\sin(a)$ for an algebraic number a, and so on (commented here link to wikipedia, since it was not liking some characters in it)

[a] In terms of probability, 100% of the elements

[b] If you've taken some statistics/measure theory, then if $\mathbb{A}$ are the algebraic numbers over $\mathbb{Q}$, $\int \chi_{\mathbb{A}} = 0$, i.e., the chances of picking an algebraic number from the set of real number is 0!

We can extend the results of theorem 19.3.11 a bit further to algebraic extensions (finite or infinite)

Theorem 19.3.16: Composing Algebraic extensions

Let $k \subseteq E \subseteq F$ be field extensions. Then F/k is algebraic if and only if F/E and E/k are algebraic.

***Proof*:**
If F/k is algebraic, then *a fortiori* F/E is algebraic (If $\alpha \in F$ are all roots of some polynomial $k[x]$ and $k \subseteq E$, then they are certainly roots in $E[x]$). Since $E \subseteq F$, *a priori* E/k.

For the other way, assume E/k and F/E are algebraic, and consider $\alpha \in F$. Since F/E is algebraic, there exists a $p(x) \in E[x]$ such that

$$p(\alpha) = \alpha^n + e_{n-1}\alpha^{n-1} + \cdots + e_1\alpha + e_0 = 0$$

We thus have that α is algebraic over the subfield $k(e_0, ..., e_{n-1}) \subseteq E$. Hence, by theorem 19.3.11

$$k(e_1, ..., e_{n-1}) \subseteq k(e_1, ..., e_{n-1}, \alpha)$$

is a finite extension. Furthermore, by theorem 19.3.11 again, we have that

$$k \subseteq k(e_1, ..., e_{n-1})$$

is also a finite extension, since each e_i is in E, so is algebraic over k by assumption that E/k is algebaric. Hence, by the Tower Law,

$$k \subseteq k(e_1, ..., e_n, \alpha)$$

is a finite extension as well, and hence α is algebraic over k. Since α was arbitrary, F is algebraic over k, as we sought to show.

To close off this section, we show how our previous discussion applies to composite fields:

Proposition 19.3.17: Degree Of Composite Field

Let K_1 and K_2 be two finite extensions of a field F contained in K. Then

$$[K_1K_2 : F] \leq [K_1 : F][K_2 : F]$$

with equality if and only if an F-basis for one of the fields remains linearly independend over the over field. If $\alpha_1, \alpha_2, ..., \alpha_n$ and $\beta_1, \beta_2, ..., \beta_n$ are bases for K_1 and K_2 over F respectively, then the elements $\alpha_i\beta_j$ for $i = 1, 2, ..., n$ and $j = 1, 2, ..., m$ span K_1K_2 over F

***Proof*:**
exercise

summing up the result in a diagram:

$$\begin{tikzcd}
 & K_1K_2 \arrow[dl, dash, "\leq n"'] \arrow[dr, dash, "\leq m"] & \\
K_1 \arrow[dr, dash, "n"'] & & K_2 \arrow[dl, dash, "m"] \\
 & F &
\end{tikzcd}$$

Corollary 19.3.18: Degree of Composite Field and Equality

Suppose that $[K_1 : F] = n$, $[K_2 : F] = m$ in proposition 19.3.17, where n and m are relatively prime $(n, m) = 1$. Then $[K_1K_2 : F] = [K_1 : F][K_2 : F] = nm$

***Proof*:**
In general, the extension is divisible by n and m since K_1 and K_2 are subfields of K_1K_2 (so divisible by its least common multiple). Since $(m, n) = 1$, then $[K_1K_2 : F]$ is divisible by nm. Since by proposition 19.3.17, $[K_1K_2 : F] \leq nm$, we get $[K_1K_2 : F] = nm$

Exercise 19.3.1

1. (practice quadratic extensions here, see Dummit and Foote example)
2. Show that If Let E/k and F/k be field extensions. Show that if E/k is algebraic, then so is EF/F. Show that if E/k and F/k are algebraic, so is EF/F.

19.4 Historical Detour: Straightedge and Compass Construction

For millennia in mathematical history, mathematicians as far back as ancient greeks tried to construct shapes or compute numbers through geometric means by using a straight edge and compass. For example, using a straight edge and compass, there is a way to compute sums, products, quotient (which would have been known as ratios to the ancient greeks), and square-roots. Geometry, mathematicians have figured out how to double the area of the square, take exactly half the angle, and navigate using the stars [14]. These all conformed with the rules of Euclid's geometry, and so straight-edge and compass constructions seemed like a powerful algorithm for geoemtry and arithmetics. However, a few problems were open since at least since the times of the ancient greeks around 400 BC, namely:

1. Given a square, using a straight-edge and compass, can we find a circle with the same area
2. Given a cube, using a straight-edge and compass, can we find another cube that's double it's volume
3. Given an angle, using a straight-edge and compass, can we triscect the angle (i.e. split the angle into 3 equa angles)

These problems don't have much use practically. However the simpliciy in their statement, the ubiquity of using rulers and compasses in many fields of engeneering, navigation, and so on, as well as the belief that the foundation of the world's geometry is inherently euclidean (in particular, that geometric reality followed euclidean rules and were provable using euclid's simple axioms), made these problems the central attention of a huge plethora of mathematicians for over 2 millennia.

With the new tools of field theory, we can characterize the types of numbers a straight-edge and compass can give us:

1. using a straight-dge construction, the 4 basic operations $(+, -, \times, \text{div})$ can be derivd. Thus, it will form a field.
2. the compass constructions allows the performance of square-roots. Translating this into the cartesian plane, it will be the intersction of a straight-line and a circle, which will sometimes gives polynomials which will have solutions *not* inside the base field, and so we would use an algebraic extension. Such polynomials will always be of degree 2. Hence, if we keep trying to extend these fields, we will always get multiples of 2 as the degree of the extension

From this comes the theorem that feels so mind-blowing that I can state after literal millenia of people trying to solve this problem:

Theorem 19.4.1: Straightedge And Compass Imposibilities

The 3 classical greek straight-edge and compass problems:

1. **Doubling the square** Using only straight edge and compasss, construct a cube with precisely twice hte volume of a given cube
2. **Triscecting an Angle** Given a straightedge and compass, triseced any given angle θ
3. **Squaring the circle** Given a straightedge and compass, construct a square whose area is precisely the area of a given circle.

are all answered in the negative

[14] and an additional instrument to measure the angle between the person and the horizon, and the north star

***Proof*:**
Comes down to the aforementioned intuition and a bit more: Dummit and Foote p. 553

19.5 Closures of Fields

Through theorem 19.2.2, we saw that if we have a irreducible polynomial $f(x) \in k[x]$, we can find a larget field which gives $f(x)$ a root: $f(x) = f'(x)(x-\alpha)$. However, we have only established result for 1 polynomial, and for only 1 root. It would be nice if the set of all roots of every polynomials formed a field, and even better if that field existed for any field k and was unique (up to isomorphism). In such a field, every polynomial would *fully factor* or *split* into linear terms. Through corollary 19.3.14, we found that given K/k the set of algebraic elements $\alpha \in K$ over k forms a field, namely the set of elements from the larger field K which are algebraic over k. In example (19.3.15) (1), we where implicitely using the fact that every polynomial over $\mathbb{C}$ reduces to linear terms, allowing us to say that $\overline{\mathbb{Q}}$ to be algebraically closed. It would be much better if the definition would be independent of any larger field, that is, find a field $\overline{k}$ such that the only irreducible polynomials in $\overline{k}[x]$ are of degree 1. Furthermore, if $\overline{k}$ existed for every field and was unique, we can call it *the* algebraic closure of k. This is the first thing proven in this section. Then, we take a closer look at fields for which only some subset $S \subseteq k[x]$ have polynomials that split. These fields will be at the heart of Galois theory, for as we saw in corollary 19.2.29, the group of k-automorphism $\text{Aut}_k(F)$ depends on the number of roots of a polynomial in F. Thus, we take some time constructing such a field.

19.5.1 Algebraic Closure

We start off this section by reminding the reader what an algebraically closed field is:

Definition 19.5.1: Algebraically Closed

Let K be a field. Then K is called algebraically closed if every polynomial in $K[x]$ factors into a product of linear terms.

There are many equivalent ways of defining algebraic closure:

Lemma 19.5.2: Algebraic Closure Equivalences

Let K be a field. then the following are equivalent:

1. K ia algebraically closed
2. every maximal ideal in $K[x]$ is of the form $(x-c)$ for all $c \in K$
3. There does not exist nontrivial algebraic extensions of K
4. If L/K is some field extension, and $\alpha \in L$ is algebraic over K, then $\alpha \in K$

***Proof*:**
exercise

These show that being an algebraically closed field K means that there are no more algebraic extensions L/K, besides trivial ones. Given some field k, there can be many field extensions that are algebraically closed. For example, $\overline{\mathbb{Q}}/\mathbb{Q}$, $\mathbb{C}/\mathbb{Q}$, and $\mathbb{C}(t)/\mathbb{Q}$ are all field extensions of $\mathbb{Q}$ to algebraically closed fields. However, the first extensions seems in some way more simple or "minimal" than the other in that it is defined through only algebraic elements (recall form example (19.3.15)(1) that all the elements of $\overline{\mathbb{Q}}$ are algebraic over $\mathbb{Q}$), while $\mathbb{C}$ contains transcendental elements (we've mentioend in example (19.3.15)(2) how π and e are not algebraic).

Definition 19.5.3: Algebraic Closure

Let k be any field. Then an *algebraic closure* of k is a field $\overline{k}$ where $\overline{k}/k$ is an *algebaric extension* and $\overline{k}$ is algebraically closed.

Soon, the statement can be re-written with "the algebraic closure" and "the field" instead of "an algebraic closure" and "a field". The field $\overline{k}$ will be the smallest field in which every nonconstant polynomial $p(x) \in k[x]$ will split into linear factors, and since $\overline{k}$ is algebraically closed, the subfield of $\overline{k}$ that contains all the roots of all non-constant polynomials in $k[x]$ is $\overline{k}$ itself. To show the existence of such a field, we first prove that there exists a field extension where every polynomial in $k[x]$ has *at least* one root, and then recursively apply the theorem to get a long chain of field extensions, each containing at least one more root to every polynomial.

Lemma 19.5.4: Adding One Root

Let k be a field. Then there exists a field K such that every non-constant polynomial of $k[x]$ admits at least 1 root.

The following proof is credited to Emil Artin

***Proof*:**
Let $\mathcal{F}$ be a set that is in bijection with all non-constant monic polynomials $F \subseteq k[x]$: $\mathcal{F} = \{t_f\}_{f\in F}$, and let $k[\mathcal{F}]$ be a polynomial ring with indeterminates $t_f \in \mathcal{F}$. Let $I \subseteq k[\mathcal{F}]$ be the set of all polynomials generated by all $f(t_f)$ (that is, the polynomial f that takes in the indeterminate t_f that it is in bijective correspondence).

I claim that I is a proper ideal of $k[\mathcal{F}]$. Indeed, if it weren't, then we would have:

$$1 = \sum_i^n a_i f(t_{f_i})$$

with $a_i \in k[\mathcal{F}]$. The identity cannot be possible, for that would lead to imply $1 = 0$. Let's take $k \subseteq F$ where each polynomial $f_1(x), ..., f_n(x)$ has a root $\alpha_1, ..., \alpha_n$ (apply theorem 19.2.2 n times). Then viewing the previous identity to be in $F[\mathcal{F}]$ we get:

$$1 = \sum_i^n a_i f(t_{f_i}) = \sum_i^n a_i f(\alpha_i) = \sum_i^n a_i \cdot 0 = 0$$

a contradiction.

Now, since I is proper, it is contained in a maximal ideal m (proposition 9.4.18). Thus, we have the field extension:

$$k \subseteq K := \frac{k[\mathcal{F}]}{m}$$

By construction every non-constant monic (and so every nonconstant) polynomial $f(x) \in k[x]$ must have a root in K, namely $\overline{t_f}$

Note that we had to use Zorn's lemma to find the maximal ideal m. Very commonly, a statement using Zorn's lemma is equivalent to it (ex. instead of using Zorn's lemma to prove the existence of a maximal ideal, we can use the existence of a maximal ideal to prove Zorn's lemma). We might ask if the existence if the previous result is also equivalent to Zorn's lemma, however it turns out that this is not the case (it is apparently due to the *compactness theorem of first order logic*, which is known to be weaker than the axiom of choice)

We can now find a field K_1 in which every polynomial in $k[x]$ has at least 1 root. We can continue this process by applying the theroem on K_1, and defining the field K_2 for which every polynomial of K_1 has at least 1 root in $K_1[x]$ (and every polynomial of degree 2 in $k[x]$ must not split into linear factors). Continuing this process, we get a chain:

$$k \subseteq K_1 \subseteq K_2 \subseteq \cdots$$

Now, take $L = \bigcup_i K_i$[15]. L can easily be turned into a field through the operation of $a + b$ and $a \cdot b$ where we choose K_i in which $a, b \in K_i$.

Lemma 19.5.5

Let L be defined as above. Then L is algebraically closed

***Proof*:**
Let $f(x) \in L[x]$ be a nonconstant polynomial. Then since $f(x)$ has finitely many coefficients, there exists some field K_i for some i which has all the coefficients of $f(x)$. Thus, $f(x)$ has a root in $K_{i+1} \subseteq L$: $f(x) = f'(x)(x - \alpha)$. Since $f'(x) \in L[x]$ (the coefficients of $f'(x)$ must be in some K_j, $j \geq i$), this process repeats until f is completely split into linear terms. Since this is true for arbitrary $f(x) \in L[x]$, L is algebraically closed, completing the proof.

Thus, for any field k, there exists an algebreaically closed field L such that L/k. Note that L need not be algebraic closure, since L/k need not be an algebraic extension. However, with an algebraically closed field, we can very simply construct an algebraic closure of k:

Theorem 19.5.6: Existence Of Algebraic Closure

Let k be a field and L/k be a field extension where L is algebraically closed. Then

$$\overline{k} := \{\alpha \in L \mid \alpha \text{ is algebraic over } k\ \}$$

is an algebraic closure of k

Note how this mimicked $\overline{\mathbb{Q}}$ in example (19.3.15)(2) assuming $\mathbb{C}$ is algebraically closed.

[15]This is an example of a *direct limit*

***Proof*:**
First, by corrollary 19.3.14, $\overline{k}$ is a field, and *a prior* $\overline{k}$ is an algebraic extension of k. To show that k is algeraically closed, let α be algebraic over $\overline{k}$. Then

$$k \subseteq \overline{k} \subseteq \overline{k}(a)$$

is a composition of albegraic extensions, hence $\overline{k}(a)$ is algebraic (theorem 19.3.16), thus α is algebraic over k. But then by definition of $\overline{k}$, $\alpha \in \overline{k}$. Thus, by lemma 19.5.2$\overline{k}$ is algebraically closed, completing the proof

Next, we strive for uniqueness. As we establisehd in theorem 19.2.2, being a field extension is not a universal property, and so the best we an strive for is the existence of a homomorphism, though not necesarrily a unique one. We first establish a lemma:

Lemma 19.5.7: Embedding Into Algebraic Extension

Let L/k be any field extension where L is algebraically closed over k. Let F/k be an algebraic extensions. Then there exists a [not necessarily unique] homomorphism $i : F \to L$ fixing k.

Note that *a-priori*, L need not have any relation to F (ex. we don't know if F is a subset of L, they might only intersect on k).

***Proof*:**
This is a Zorn's lemma argument similar to Baer's Criterion (theorem 17.3.5). Let S be the set of homomorphisms of the form

$$i_K : K \to L$$

The set S is nonempty since $i_k : k \to L \in S$. Define a patial orer on S $i_K \preceq i_{K'}$ if $K \subseteq K' \subseteq L$ and $i_{K'}|_K = i_K$. To apply Zorn's lemma, we need to show that every chain has an upper bound. Let $C \subseteq S$ be a chain, and let K_C be the union of all the domains of elements $i_K \in C$. Then K_C is clearly a field (lemma ref:HERE), and for every $\alpha \in K_C$ define $i_{K_C}(\alpha)$ to be $i_K(\alpha)$ for appropriate K where $\alpha \in K$. Then i_{K_C} is clearly well-defined, not being dependent on the choice of K, and so we have a homomorphism $i_{K_C} : K_C \to L$ where $i_K \preceq i_{K_C}$ for all $i_K \in C$, showing that i_{K_C} is indeed an upper bound for C. Thus, by Zorn's lemma, Z has an upper bound $i_G : G \to L$ where $k \subseteq G \subseteq L$.

I claim that $G = F$, which will prove the statement. For the sake of contradiction, let's assume this is not the case, so that there exists an element $\alpha \in F \setminus G$. Consider $G(\alpha)/G$. Since $\alpha \in F$ is algebraic over k, α is algebraic over G, and since $\alpha \notin G$, it is the root of an irreducible polynomial $g(x) \in G[x]$. Let $H = i_G(G)$. Then as discussed in the first pages of section 19.2 i_G induces a ring homomorphism

$$f^{i_G} : G[x] \to H[x]$$

Let $h(x) = f^{i_G}(g(x))$. Then $h(x)$ is irreducible over H, and since L is algebraically closed, it must have a root $\beta \in L$.

We are now setup in a situation to apply theorem 19.2.27: we have an isomorphism $i_G : G \to H$ and two simples extensions $G(\alpha), H(\beta)$ with α and β being roots of $g(x)$ and $h(x)$ respectively (with $h(x) = i_G(g(x))$). Thus, i_G lifts to an isomorphism:

$$i_{G(\alpha)} G(\alpha) \to H(\beta) \subseteq L$$

which sends α to β. Then $i_G \preceq i_{G(\alpha)}$, contradicting the maximality of i_G.

Thus, it must be that $G = F$, and so $i_F : F \to L$ is a field homomorphism, as we sought to show.

The fact that the isomorphism is not unique will become important in representing algebraic closures. We first finish off by showing that the algebraic closure is indeed unique up to isomorphism (given a fixed field k)

Theorem 19.5.8: Uniqueness Of Algebraic Closure

Let $k \subseteq \overline{k_1}$ and $k \subseteq \overline{k_2}$ be two algebraic closures of k. Then $\overline{k_1} \cong \overline{k_2}$ where the isomorphism extends the identity on k, or equivalently k_1 is isomorphic to k_2 as extensions. In terms of a commutative diagram, in the following

$$\begin{tikzcd} k_1 \arrow[rr, "\sim"] & & k_2 \\ & k \arrow[ul, hook] \arrow[ur, hook] & \end{tikzcd}$$

there always exists an isomorphism from k_1 to k_2

the second condition for the isomorphism is so that the two fields represent the algebraic closure over the common field k. Sometimes, such a isomorphism (resp. homomorphism) is called a k-isomorphism (resp. k-homomorphism)

***Proof*:**

Since $\overline{k_1}/k$ is algebraic and $\overline{k_2}$ is algebraically closed over k, by lemma 19.5.7, there exists a homomorphism $i : \overline{k_1} \to \overline{k_2}$. This homomorphism is injective, being a field homomorphism, and surjective since otherwise, $\overline{k_1} \subseteq \overline{k_2}$ would be a non-trivial algebraic extension, contradicting lemma 19.5.2 since $\overline{k_1}$ is algebraically closed. Thus, i must be an isomorphism as we sought to show.

The fact that the isomorphism is not necessarily unique has some important consequences: There is no "cannonical" way of representing the algebraic closure. At first, it might seem harmless that the isomorphism is not unique, as long as the two field are isomorphic. However, this means there is no real cannonical "choice" when it comes to representing algebraic closures.

Take for example $\overline{\mathbb{R}}$, which we know to be $\mathbb{C}$. We can write $\overline{\mathbb{R}}$ as $\mathbb{R}[i]$, which for many mathematicians is "the" algebraic closure of $\mathbb{R}$. However, if we go over to an electrical engineer, we will see that they have also constructed an algebraic closure for $\mathbb{R}$ to represent electrical charge and circuits where they use the letter j (since i is already used for something else), and write it as $\mathbb{R}[j]$. We might think it to be harmless to make the isomorphism $1 \mapsto 1$ and $i \mapsto j$, however the electrical engineers have decided that electrical charge should be represented by a -1. Does this means we should instead have

$$1 \mapsto 1 \qquad i \mapsto -j?$$

Mathematically, there is no good reason to pick one representation over another. In other words, if one person constructs a splitting field L for k and another person constructs a algebraic closure L'

for k, then we know that we get:

$$\begin{tikzcd} k_1 \arrow[rr, "\sim"] & & k_2 \\ & k \arrow[ul, hook] \arrow[ur, hook] & \end{tikzcd}$$

however, there is no "natural" way to associate an element of L to an element of L'. This often means that we should keep track of the construction of the splitting field we have done, since if we don't we might not be able to communicate to other mathematicians who have done a different construction (how can you compare your work if you can't correspond the results you have come up with with the other mathematicians).

The idea there there is no natural choice of isomorphism can be made more rigorous if we ask it in the language of category theory. If we map every field to it's algebraic closure $k \to \overline{k}$, does this define a functor? In order for it to define a functor, we would need that every map $k \to l$ of field defines a map of algebraic closures

$$\begin{tikzcd} k \arrow[r] \arrow[d] & \overline{k} \arrow[d, dashed] \\ l \arrow[r] & \overline{l} \end{tikzcd}$$

However, it is not clear what map should be chosoen here[16]. There is in fact a way to reduce this ambiguity, which I don't fully undertand yet intuity, but it was mentioned in Richard Borcherds video: here.

19.5.2 Splitting Field and Normal Extensions

Having established that every field has an algebraic closure, meaning we can unambigurously talk about roots of polynomials, we now return to analyzing algebraic extensions. We start off by zoom in on fields that only split some subset of polynomials $S \subseteq k[x]$. In the following sections, we will be working over the case where S is finite [17]. Since S is finite, say $S = \{f_1(x), ..., f_n(x)\}$, then having each $f_i(x)$ split is equivalent to having $f_1(x)f_2(x)\cdots f_n(x)$, thus, we usually have $S = \{f(x)\}$ be one polynomial

Definition 19.5.9: Splitting Field

Let k be a field and $f(x) \in k[x]$ be a polynomial where $\deg(f(x)) = d$. Then a *splitting field* for $f(x)$ over k is a minimal extension F/k such that

$$f(x) = c\prod_{i}^{d}(x - \alpha_i)$$

i.e. $f(x)$ splits completely in $F[x]$, and $f(x)$ does not factor in any proper subfield of F.

Furthermore, by minimality, $F = k(\alpha_1, ..., \alpha_n)$ (i.e. F is generated by the roots of $f(x)$ in F)

[16] perhaps you can force through a construction using the axiom of choice, but it is certainly not a natural choice
[17] The infinite case of this I'm not yet sure is not studied TBD

The fact that a splitting field exists is immediate from the fact that $\overline{k}$ exists, namely if $f(x)$ is a polynomial, then $F \subseteq \overline{k}$ generated by the roots of $f(x)$ in $\overline{k}$. Furthermore, the splitting field is in fact unique (up to isomorphism), justifying replacing any "a splitting field" in the definition to "the splitting field"

Theorem 19.5.10: Uniqueness Of Splitting Field

Let k be a field and let $f(x) \in k[x]$. Then the splitting field F for $f(x)$ over k is unique up to isomorphism, and $[F : k] \leq (\deg(f))!$ (i.e. the factorial of the degree of f). Furthermore, if $i : k \to k'$ is an isomorphism, $f(x) \in k[x]$ and $g(x) = i(f(x))$, then i extends to an isomorphism from the splitting field of $g(x)$ over k' to the splitting field of $f(x)$ over k:

$$\begin{tikzcd} F \arrow[r, dashed, "\sim"] & F' \\ k \arrow[u, hook] \arrow[r, "\sim"] & k' \arrow[u, hook] \end{tikzcd}$$

***Proof*:**
We will first construct a splitting field explicitely with $[F : k] \leq \deg(f)$, and then use that field to show uniqueness. For existence, this is an application of induction on the degree of the polynomial. Degree 1 is trivial, covering the base-case, so let's say the splitting field was constructed for all fields and all polynomials of degree $\deg(f) - 1$. Let $f(x)$ be some polynomial that is not yet fully factored into linear terms; let $q(x)$ be some irreducible non-linear factor over k. Then

$$k \subseteq F' := \frac{k[x]}{(q(x))}$$

is an extension of degree $\deg q \leq \deg f$ in which $q(x)$ has a root (and so $f(x)$ has a root), say α, and so $q(x) = q'(x)(x - \alpha)$. Then we can define the polynomial $h(x) = f(x)/(x - \alpha) \in F'[x]$ of degree $\deg(f) - 1$. Thus, by the induction hypothesis, there exists a splitting field F for $h(x)$ over F' and

$$[F : F'] \leq (\deg(f) - 1)!$$

Since the factors of $f(x)$ over F are $(x - \alpha)$ and the factors of $h(x)$, it follows that F is a splitting field for $f(x)$, and

$$[F : k] = [F : F'][F' : k] \leq (\deg(f) - 1)!(\deg(f)) = (\deg(f))!$$

To prove that an isomorphism $i : k \to k'$ extends to isomorphisms of splitting fields as stated, let F be the splitting field for $f(x)$, and consider $k \to k' \subseteq \overline{k'}$. Since the extension $k \subseteq F$ is algebraic, by lemma 19.5.7 there exists a k-homomorphism $i : F \to \overline{k'}$. Since F is generated over k by the roots of $f(x)$ in F, we have

$$i(F) = k'(\alpha_1, ..., \alpha_d) \subseteq \overline{k'}$$

where the α_i's are the roots of $g(x) = i(f(x))$ in $\overline{k'}$. Therefore, $L := i(F)$ is independent of the given isomorphism i and the splitting field F. Applying this to i and to the identity $k' \to k'$ completes the proof (since both F and G are isomorphic to L)

Example 19.5.11: Splitting Fields

1. The splitting field of x^2-2 over $\mathbb{Q}$ is $\mathbb{Q}(\sqrt{2})$, since the two roots are $\pm\sqrt{2}$ and $-\sqrt{2} \in \mathbb{Q}(\sqrt{2})$. The splitting field of $(x-2)^n$ is also $\mathbb{Q}(\sqrt{2})$ for all $n \in \mathbb{N}_{>0}$, since we only need to add the root $\sqrt{2}$ (resp. $-\sqrt{2}$) for this polynomial to split.

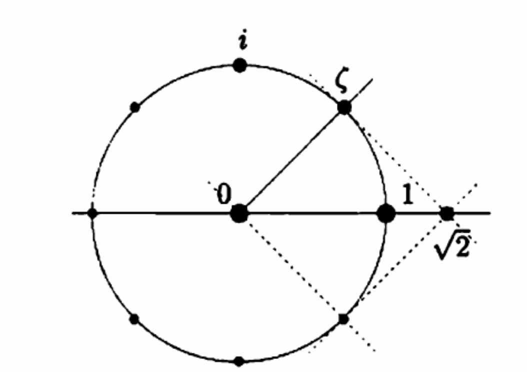
2. Splitting field for $(x^2-2)(x^2-3)$ over $\mathbb{Q}$ is $\mathbb{Q}(\sqrt{2}, \sqrt{3})$. Note that $\mathbb{Q}(\sqrt{2}, \sqrt{3})$ is equal to the composite field $\mathbb{Q}(\sqrt{2})\mathbb{Q}(\sqrt{3})$, i.e. $\mathbb{Q}(\sqrt{2})\mathbb{Q}(\sqrt{3}) = \mathbb{Q}(\sqrt{2}, \sqrt{3})$.

3. More generally, If $f(x)$ is a polynomial over k such that
$$f(x) = \prod g_i(x)$$
where each $g_i(x)$ is irreduible over k, then the splitting field of $f(x)$ is the composite field of the splitting fields for each $g_i(x)$ since each splitting field is the smallest field containing all the roots of g_i, and the composite field is the smallest field containing all the splitting fields.

4. splitting field of x^3-2 over $\mathbb{Q}$ is not $\mathbb{Q}(\sqrt[3]{2})$, since we are missing two roots, namely:
$$\sqrt[3]{2}\left(\frac{-1+i\sqrt{3}}{2}\right) \qquad \sqrt[3]{2}\left(\frac{-1-i\sqrt{3}}{2}\right)$$
Thus, the splitting field is $\mathbb{Q}(\sqrt[3]{2}, i\sqrt{3}) = \mathbb{Q}(\sqrt[3]{2}, \sqrt{-3})$.

5. The splitting field of x^2+1 over $\mathbb{Q}$ is $\mathbb{Q}(i)$, and the splitting field of the same polynomial x^2+1 over $\mathbb{R}$ is $\mathbb{C}$, showing that the splitting field is dependent on the base field.

6. Let F be the splitting field of x^8-1 over $\mathbb{Q}$. With some knowledge of complex analysis, we see that all the roots are 8th root of unit, that is,
$$\zeta = e^{\frac{2\pi i}{8}}$$
is a generator for the roots (i.e. the primite elements), and so $F = \mathbb{Q}(\zeta)$. Visually, these roots look like so:



In fact, ζ is also the root of x^4+1 (since $\zeta^4 = -1$), and so is the root of the smaller irreducible polynomial x^4+1 over $\mathbb{Q}$. There is no smaller minimal polynomial for which ζ is a root (since smaller powers of ζ is not in $\mathbb{R}$) and so:
$$[\mathbb{Q}(\zeta) : \mathbb{Q}] = 4$$

showing that the degree of the splitting field is much less than $8! = 40'320$. To better undersatnd this field, notice that $\zeta^2 = i$, so $\sqrt{2} = \zeta + \zeta^7$. Thus:

$$\mathbb{Q}(i, \sqrt{2}) \subseteq \mathbb{Q}(\zeta)$$

Conversely, $\zeta = \frac{\sqrt{2}}{2}(1+i) \in \mathbb{Q}(i, \sqrt{2})$, showing that:

$$\mathbb{Q}(i, \sqrt{2}) \supseteq \mathbb{Q}(\zeta)$$

showing that they are in fact equal.

As an exercise, show that $\mathrm{Aut}_{\mathbb{Q}}(\mathbb{Q}(i, \sqrt{2})) = \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$

7. Let $p(x) = x^4 - 1$. this polynomial in fact factors over $\mathbb{Q}$:

$$x^4 - 1 = (x-1)(x+1)(x^2+1)$$

and so the splitting field of $x^4 - 1$ is the splitting field of $x^2 + 1$, which is $\mathbb{Q}(i)$

8. Let $p(x) = x^4 + 2$. This polynomial is irreducible by Eisenstein. Given $\zeta = e^{\frac{2\pi i}{8}}$, then the roots of $p(x)$ are:

$$\sqrt[4]{2}\zeta,\ \sqrt[4]{2}\zeta^3,\ \sqrt[4]{2}\zeta^5,\ \sqrt[4]{2}\zeta^7$$

and so the splitting field is $K = \mathbb{Q}(\sqrt[4]{2}\zeta,\ \sqrt[4]{2}\zeta^3,\ \sqrt[4]{2}\zeta^5,\ \sqrt[4]{2}\zeta^7)$. We can simply the adjoint elements with the following observations:

$$K \subseteq \mathbb{Q}(\zeta, \sqrt[4]{2}) \subseteq \mathbb{Q}(i, \sqrt{2}, \sqrt[4]{2}) = \mathbb{Q}(i, \sqrt[4]{2})$$

so $K \subseteq \mathbb{Q}(i, \sqrt[4]{2})$. For the $\supseteq$ direaction, notice that:

$$\sqrt{2} = \frac{(\sqrt[4]{2}\zeta)^3}{\sqrt[4]{2}\zeta^3} \in K \qquad i = \frac{(\sqrt[4]{2})^2}{\sqrt{2}} \in K$$

So $\zeta = \frac{\sqrt{2}}{2}(1+i) \in K$ and $\sqrt[4]{2} = \frac{\sqrt[4]{2}\zeta}{\zeta} \in K$, showing that $K \supseteq \mathbb{Q}(i, \sqrt[4]{2})$. With a little bit of computation, we see that:

$$[\mathbb{Q}(i, \sqrt[4]{2})] = 8$$

From Dummite and Foote. There are in fact a ton of good example that I can put here, check them out! Overall, small variation in polynomials can create quite large differences in splitting fields (as we've seen with x^4+2, x^4+1, x^4-1). This shows how precise the tools of field theory are in the study of polynomials.

Given a polynomial $f(x)$ over k, we can find a field F in which $f(x)$ factor completely over F. In fact, we have also gained a slightly stronger result for free: if $g(x)$ is irreducible in k has a root in F, then it in fact *also* splits. The idea is that the automorphism group acts transitively on all the roots of an irreducible polynomials (recall the discussion under corollary 19.2.29). If all the roots are inside the field, then the automorphism group will shuffle around all the roots within the field. Hence, if we are given that a single root is inside a field, and that the image of each automorphism in the automorphism group is inside the field, we would have that all the roots are within the field. As the action of the automorphism group $\mathrm{Aut}_k(F) \curvearrowright R_F$ is a evaluation action $\sigma \cdot \alpha = \sigma(\alpha)$ (often called the *galois conjugate* or just *conjugate)*, another way of saying it is that the field is closed under conjugation of the automorphism group. To understand this phenomenon better, we give a name to fields with this property:

Definition 19.5.12: Normal Extension

let F/k be a field extension. Then F/k is called a *normal extension* if for every polynomial $f(x) \in k[x]$, $f(x)$ has a root if and only if $f(x)$ splits as a product of linear factors over F

Some authors like to say that F/k needs to also be *algebraic*. This so that we need not worry about any transcendental elements in the following proofs. It is no harm on making F/k possibly not algebraic, since we work with finite extensions in the majority of cases

A normal extension is *not necessarily* the algebraic closure since not necessarily all irreducible polynomials have a root, however the irreducible polynomials over the base field become the polynomials that split completely in the normal extension, constant polynomials, and irreducible polynomials that have no roots in the normal extension (or, they can be thought of as polynomials that have yet to have a root added to the extension).

The name normal is given to this type of extension is due to the fact that most of the nice properties of automorphism groups we have discussed in corollary 19.2.29 apply to normal extensions[18]

Theorem 19.5.13: Finite Normal Extension Equivalences

Let F/k be a field extension. Then the following are equivalent:

1. F is finite and normal
2. F is the splitting field of some polynomial $f(x) \in k[x]$
3. For all k-homomorphisms, $F \xrightarrow{\varphi} \overline{F}$ where $\varphi(k) = k$, $\varphi(F) = F$

***Proof*:**
We'll first show $2 \Leftrightarrow 3$ since it's simple. For $2 \Rightarrow 3$, let F be the splitting fifeld for some polynomial $f(x) \in k[x]$, then if $F \xrightarrow{\varphi} \overline{F}$ is a k-homomorpism, $F \xrightarrow{\varphi} \varphi(F)$ is a k-isomorphism. Hence, φ must map roots of $f(x)$ to roots of $f(x)$: If α is a root of $f(x)$, then we know that $m_{\alpha,k}(x)\big|f(x)$, and by corollary 19.2.29, $\varphi(\alpha)$ is another root of $m_{\alpha,k}$, and hence $f(x)$. Since F is a splitting field, it contains all the roots of $f(x)$, and so $\varphi(\alpha) \in F$. Since the k-homomorphism is determined by where it maps the roots of the generators of F (namely the roots of $f(x)$), $\varphi(F) \subseteq F$. Since φ must be injective, $F \subseteq \varphi(F)$, so $F = \varphi(F)$.

For $3 \Rightarrow 2$, since all k-homomorphisms $F \xrightarrow{\varphi} \overline{F}$ map $\varphi(F) = F$. If F/k is purely transcendental (i.e., no elements of $\alpha \in F - k$ is the the root of a polynomial), Then then clearly $\varphi(F) = F$ or else φ would not be injective. Thus, if F/k is not purely transcendental, choose an algebraic element $\alpha \in F$ over k, so $m_{\alpha,k}(x)$ is it's minimal polynomial. Since φ is defined by where it maps roots of minimal polynomials over $k[x]$, $\varphi(\alpha)$ is another root of $m_{\alpha,k}(x)$. Since $\varphi(F) = F$, $\varphi(\alpha) \in F$. Using corollary 19.2.29, we can see that there are as many automorphisms as distinct roots of $m_{\alpha,k}(x)$, and so every root of $m_{\alpha,k}(x)$ is in F. Since this is true for every algebraic element, we see that $\varphi(F) = F$.

[18] In fact, normal subgroups got their name from normal extensions! The connection will be made in the chapter 20 when discussing Galois Theory in theorem 20.1.23

Next, we show $1 \Rightarrow 2$. Assume F is finite and normal. Since it's finite, $F = k(\alpha_1, \alpha_2, ..., \alpha_n)$, where each α_i is algebraic over k. Next, let $m_i(x)$ be the minimal polynomial of α_i. Since F is normal, and each m_i has at least one root (namely α_i), $m_i(\alpha)$ splits. Then $f(x) = m_1(x) \cdots m_n(x)$ is a polynomial that splits over F. But then, F is the smallest field in which $f(x)$ splits (since $F = k(\alpha_1, ..., \alpha_n)$), and so F is the splitting field for $f(x)$.

Finally, we show $2 \Rightarrow 1$: assume F is the splitting field for some polynomial $f(x) \in k[x]$, and assume that $p(x) \in k[x]$ is an irreducible polynomial over k such that F contains a root α for $p(x)$. If $p(x)$ has degree less than or equal to 2, we're done, so let's say $\deg p(x) \geq 3$. Since $F \subseteq \overline{k}$, let $\beta \in \overline{k}$ be a different root of $p(x)$. We want to show that $\beta \in F$. This will show that F is normal, and by theorem 19.5.10, F must be finite.

Since $\mathrm{id} : k \to k$ is a field homomorphism, by theorem 19.2.27, there is an isomorphism $i : k(\alpha) \to k(\beta)$ sending $\alpha \mapsto \beta$. Note that $F(\beta) \subseteq \overline{k}$. Visually, we can imagine all of this in the commutative diagram:

$$\begin{tikzcd}
 & k(\alpha) \arrow[r, hook] \arrow[dd, "\iota"] & F \arrow[r, hook] & \overline{k} \\
k \arrow[ur, hook] \arrow[dr, hook] & & & \\
 & k(\beta) \arrow[r, hook] & F(\beta) \arrow[r, hook] & \overline{k}
\end{tikzcd}$$

Note that we cannot draw the identity map from $\overline{k}$ to $\overline{k}$, since ι maps $\alpha \mapsto \beta$. Now, notice that F is the *splitting field* $f(x)$ over $k(\alpha)$. Indeed, F contains all the roots of $p(x)$ and is generated over k (and hence $k(\alpha)$) by the roots of $p(x)$. Similarly, $F(\beta)$ is the splitting field for $f(x)$ over $k(\beta)$. Then by the uniqueness of the splitting field (theorem 19.5.10), ι extends to an isomorphism $\iota' : F \to F(\beta)$. Now, since ι restricts to the identity on k, ι' is an isomorphism of k-vector spaces, in particular $\dim_k F = \dim_k F(\beta)$ (since splitting fields are always finite extensions). In particular, we get to take advantage of facts about liner algebra, in particular that if the dimensions of two vector-spaces are the same, then any injective linear transformation is in fact an isomrophism.

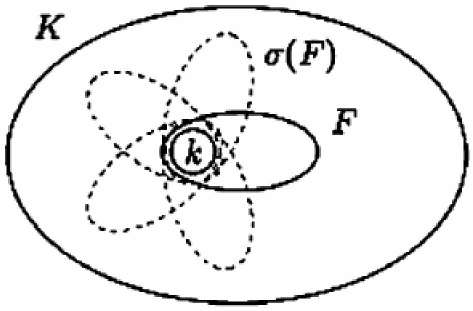
Now, we also have the natural inclusion map $\iota : F \to F(\beta)$ which is both a k-homomorphism and a k-vector spaces. Since F and $F(\beta)$ are k-vector spaces of the same [finite!] dimension, i must *also* be an isomorphism, i.e.

$$[F(\beta) : F] = 1$$

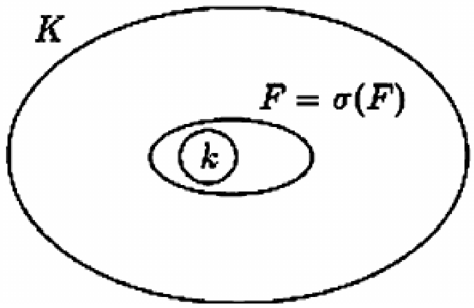
Thus, $\beta \in F$, completing the proof.

Note that it can be proven that F is normal (not necessarily finite) if and only if for any k-homomorphism $\varphi : F \to \overline{F}$, $\varphi(F) = F$. The proof naturally generlizes from the proof of $2 \Leftrightarrow 3$ since we can apply the results to each irreducible polynomial and their roots.

19.5.14 Remark Notice how condition (3) of theorem 19.5.13 allows us to talk about "the" embedding of a Normal extension F/k into $\overline{k}$, since $\varphi(F) = F$ for any F. Recall that $\mathbb{Q}(\sqrt[3]{2})/\mathbb{Q}$ is not normal, and that we have shown that $\mathbb{Q}(\sqrt[3]{2})$ is isomorphic to 3 subfields of $\overline{\mathbb{Q}}$ which do not all overlap:



However, if the extensoin is normal, then there is a notion of "the" embedding since they all overlap:



In this way, having a [algebraic][a] normal extension F/k can be thought of as being more "precise" than a non-normal extension, since any other isomorphic copy of F is just an automorphism of F.

[a]This is to ensure that $F \subseteq \overline{k}$. The algebraic elements of F is also what we care about when we look at k-homomorphisms $F \to \overline{k}$

(there is also a cool proof the professor gave in class, the Jan 20th lecture)

Example 19.5.15: Consequences

Let's say $p(x)$ is an irreducible polynomial over $\mathbb{Q}$ with a complex root that may be expressed as a polynomial in i and $\sqrt[4]{2}$ over $\mathbb{Q}$. Then *all* roots of $p(x)$ may be expressed as a polynomial in i and $\sqrt[4]{2}$. As we have shown in example 19.5.11, $\mathbb{Q}(\sqrt[4]{2}, i)$ is a splitting field over $\mathbb{Q}$, and hence a normal extension.

Unlike algebraic and finite extensions, normal extensions are more finicky when it comes to composing normal extensions:

Proposition 19.5.16: Tower Law On Normal Extensions

If $L/K/k$ and L/k is normal, then L/K is normal

***Proof*:**
Let $f(x) \in K[x]$ be irreducible with a root $\alpha \in L$. Then $f(x) = m_{\alpha,K}(x)$ is the minimal polynomial over K. Note that

$$m_{\alpha,K}(x) | m_{\alpha,k}(x)$$

And since $m_{\alpha,k}(x)$ splits over L (since L/k is normal), $m_{\alpha,K}$ splits over L, as we sought to show.

Note that this result has little wiggle room:

Example 19.5.17: Low Tolerance for Normal Extension
Consider $L/K/k$ where:

$$\begin{array}{c} \mathbb{Q}(\sqrt[3]{2}, \zeta_3) \\ | \\ \mathbb{Q}(\sqrt[3]{2}) \\ | \\ \mathbb{Q} \end{array}$$

where σ_3 is one of the two other roots of $x^3 - 2$. Then while L/k is normal, K/k is *not* normal here (we have yet to add enough roots). Furthermore, if L/K and K/k is normal, this does not imply that L/k is normal. For example:

$$\begin{array}{c} \mathbb{Q}(\sqrt[4]{2}) \\ | \\ \mathbb{Q}(\sqrt{2}) \\ | \\ \mathbb{Q} \end{array}$$

To see this, notice that each extension is of degree 2, and hence is normal (exercise). However, $\mathbb{Q}(\sqrt[4]{2})/\mathbb{Q}$ is not normal, since 2 of the roots of $x^4 - 2$ are complex, while $\mathbb{Q}(\sqrt[4]{2})$ is contained in the reals, and hence cannot have all the roots.

Notice how this behavior is mimicked by normal subgroups.

Definition 19.5.18: Normal Closure

Let K/k be an algebraic extenion. Then the *normal closure* of K given k is an algebraic extension N/K such that

1. N/k is normal
2. If $N \supseteq N' \supseteq k$, and N'/k is normal, then $N = N'$

Proposition 19.5.19: Existence Of Normal Closure

Let K/k be a finite extension. Then there exists a unique normal closure N up to isomorphism fixing k.

Note the finiteness condition given. It is true for algebraic ones too, but we did not yet prove that

***Proof*:**
Since K is finite, let $K = k(\alpha_1, \alpha_2, ..., \alpha_n)$ where each $\alpha_i \in K$ is algebraic over k. Let

$$f(x) = \prod m_{\alpha_i,k}(x) \in k[x]$$

and let N be the splitting field of $f(x)$ over K. Then $[N : K] < \infty$. But then, N is the splitting field of $f(x)$ over k as $K = k(\alpha_1, \alpha_2, ..., \alpha_n)$. But this implies N/k is normal.

Finally, let's say $k \subseteq N' \subseteq N$ and N'/k is normal. Then by normality, N' has to split each $m_{\alpha_i,k}(x)$, and so splits $f(x)$, and so N/k *is* a normal closure.

Exercise 19.5.1

1. (Splitting Field proof using induction)
2. Show that If Let E/k and F/k be field extensions. Show that if E/k is normal, then so is EF/F. Show that if E/k and F/k are normal, so is EF/F.
3. Show that every polynomial of $\mathbb{R}[x]$ of degree ≥ 3 is reducible. Show that every irreducible polynomial is linear or is of degree two with negative discriminant, that is if $ax^2 + bx + c = 0$, then $b^2 - 4ac = 0$.
4. Let L/K be an algebraic field extension, and $\varphi_K : K \to \overline{F}$ be a homomorphism to an algebraically closed field. Then φ_K extends to a homomorphism $\varphi_L : L \to \overline{F}$ (use Zorn's lemma and a proof similar to lemma 19.5.7 with Zorn being applied to (E, φ_E) where E/K is a subextension of L/K and $\varphi_E : E \to \overline{F}$ extends φ_K)

19.6 Separable Extensions

We've now shown that there exists a field K over F such that a polynomial $f(x)$ splits (or a field $\overline{F}$ such that all polynomials in $F[x]$ split). A pattern has been occuring which you might have not noticed: every time we split an irreducible polynomial completely into linear factors, each factor was *unique*, that is, we had that the multiplicity of every linear term was 1. Does that need to be the case for any polynomial over any field? The answer is no:

Example 19.6.1: Higher Multiplicity

Let p be a prime number, and take the field $\mathbb{F}_p(t)$. Consider the polynomial:

$$p(x) = x^p - t \in \mathbb{F}_p(t)[x]$$

Then $p(x)$ is irreducible: by Eisenstein's criterion, it is irreducible in $\mathbb{F}_p[t][x]$ (since (t) is prime in $\mathbb{F}_p[t]$), and hence it is prime in $\mathbb{F}_p(t)[x]$ by corollary 11.2.2. Let u be a root of $p(x)$ in the extension $L/\mathbb{F}_p(t)$ (L can be the algebraic closure, or the splitting field). Then notice that:

$$x^p - t = (x-u)^p$$

Simply expand the right hand side of the equation. Thus, u has multiplicity p, i.e., the minimal polynomial over $\mathbb{F}_p(t)$ for $u \in L$ vanishes p times at u[a].

[a] $p(x)$ must be the irreducible polynomial, since if the degree was smaller then $p(x)$ would be reducible

Thus, in general, a polynomial $f(x) \in k[x]$ can factorize over a splitting field as:

$$f(x) = (x-\alpha_1)^{n_1}(x-\alpha_2)^{n_2}\cdots(x-\alpha_k)^{n_k}$$

where $\alpha_1, \alpha_2, ..., \alpha_k$ are distinct elements in the splitting fields, and $n_1, n_2, ..., n_k$ is the multiplicity of every root. However, this seems to be a rare occurence given the fields we have worked with. In fact, most field we will work with do not have this property. It will also become important in Galois theory that a polynomial would have to to linear polynomials with multiplicity 1[19]; for those who are curious, see what hapens to the automorphism group $\mathrm{Aut}_k(F)$ of the splitting field F if $(x-\alpha)^n$ is the linear factorization of $f(x)$ over k. Hence, we give a name to polynomials that factor with multiplicity 1:

Definition 19.6.2: Separable Polynomial

A polynomial over k is called *separable* if it factors into linear terms with multiplicity 1 in it's splitting field. Equivalently, it has no multiple roots (i.e., all its' roots are distinct). A polynomial which is not separable is called *inseparable*.

My intuition behind the name is that the roots cannot be completely "separated" from each other if the polynomial is *inseparable*. In Algebraic Geometry, polynomials that have roots with multiplicity higher than 1 (say multiplicity m) shall correspond to a collection of points that shall all be arbitrarily close to each other while not being equal, ostensibly being "inseparable"[20]. Naturally, the choice of splitting field does not affect whether a polynomial is separable since all splitting fields for a polynomial are isomorphic, and isomorphisms preserves roots (theorem 19.2.27). Even better, we can use *any* field in which the chosen polynomial $f(x)$ splits into linear factors (since there will be a homomorphism from the splitting field F to the field K in which $f(x)$ splits). For our purposes, that means that it is just as good to work over the algebraic closure $\overline{k}$.

[19] (the correspondence Galois group is trivial otherwise)
[20] See the plane with two origins for an example of such a geometric object

Definition 19.6.3: Separable Extension

Let F/k be a field extension and $\alpha \in F$. Then:

1. The element α is called *seprable* if α is algebraic and the minimal polynomial corresponding to α is seperable. Otherwise, the element α is called *inseparable*

2. F/K is called a *separable extension* if every element $\alpha \in F$ is separable, and an *inseparable extension* otherwise.

19.6.4 Remark Note that it is not immediately obvious that if α is separable over k that $k(\alpha)/k$ is a separable extension (what if $1 + \alpha$ is inseparable?). We will answer this question after we develop more tools to work with separable elements.

Using the fact that separability is independent of the field in which it splits, we can use a clever trick in finding whether a polynomial is separable using a tool usually defined in calculus:

Definition 19.6.5: Derivative

Given a polynomial $f(x) \in k[x]$ where

$$f(x) = a_n x^n + a_{n-1}x^{n-1} + \cdots + a_1 x + a_0$$

then its *derivative* or *formal derivative* is the image of the function $D_x : k[x] \to k[x]$ where:

$$D_x f(x) = n a_n x^{n-1} + (n-1)a_{n-1}x^{n-2} + \cdots + 2a_2 x + a_1$$

or more succinctly

$$D_x\left(\sum_i^n a_i x^i\right) = \sum_i^n i a_i x^{i-1} \tag{19.2}$$

Notice that this derivative acts exactly like the derivative seen in a regular calculus course [21]. In a calculus course, we derive this formula using a limit construction. However, if we just think about the derivative (restricted to polynomials) as a function, then we get a purely algebraic definition of the derivative. In fact, the derivative (and even its generalization with he external operator on differential forms) can be studied purely algebraically; see [5] a deeper exploration of these ideas[22].

Note that the i in equation (19.2) is well-defined since we always have the unique homomorphism $\iota : \mathbb{Z} \to k[x]$ since $\mathbb{Z}$ is initial in **Ring**, and so i is the element associated with $\iota(i)$. The *linearity* and *Leibniz property* of the derivative can both be proven with just this definition:

$$D_x(f(x) + g(x)) = D_x f(x) = D_x g(x)$$
$$D_x(f(x)g(x)) = (D_x f(x))g(x) + f(x)(D_x g(x))$$

We will often abbreviate notation and write $f'(x) := D_x f(x)$ when it is not ambiguous.

[21]A good resources to learn this would be Spivak's Analysis

[22]For example, algebraic differential forms shall be useful to determine the separability of algebraic maps, which is the generalization of the idea that an extension is separable if each element has a nonzero derivative

Proposition 19.6.6: Separable Polynomial Test

A polynomial $f(x)$ has multiple roots α if and only if α is also a root of $D_x f(x)$, i.e., $f(x)$ and $D_x f(x)$ are both divisible by the minimal polynomial for α. In particular, $f(x)$ is separable if and only if it is relatively prime to its derivative

$$(f(x), f'(x)) = 1$$

Noet how $f(x)$ is *not necessarily irreducible*. This means that the derivative test is incredibly powerful, telling us that if $(f(x), f'(x)) = 1$, then any irreducible factor of $f(x)$ but all be distinct and have distinct roots.

***Proof*:**
First, let's say $f(x)$ is inseparable so that it has multiple roots in a splitting field F:

$$f(x) = (x-\alpha)^n g(x)$$

where $n \geq 2$ and $g(x) \in F[x]$. Then:

$$f'(x) = n(x-\alpha)^{n-1}g(x) + (x-\alpha)^n g'(x)$$

showing that $\gcd(f(x), f'(x)) \neq 1$ in $F[x]$. By corrollary 11.2.2 to Gauss's Lemma, $\gcd(f(x), f'(x)) \neq 1$ in $k[x]$
Conversely, Assume that $f(x)$ and $f'(x)$ share a root (say $\alpha \in \overline{k}$), so that $\gcd(f(x), f'(x)) \neq 1$ in $k[x]$. Since $f(x)$ has a root, we can write it as:

$$f(x) = (x-\alpha)h(x)$$

for $h(x) \in k[x]$ (or $\overline{k}[x]$?). Then the derivative is:

$$f'(x) = h(x) + (x-\alpha)h'(x)$$

Since $f(x)$ and $f'(x)$ share a root, it must be that $(x-\alpha)\big|h(x)$. But then:

$$(x-\alpha)^2\big|f(x)$$

showing that $f(x)$ has multiple roots, hence inseparable, as we sought to show.

Using this theorem, we can see that $p(x) = x^p - t$ is inseparable since $p'(x) = px^{p-1} = 0$ (since the characteristic of $\mathbb{F}_p(t)$ is p). Thus, $\gcd(x^p - t, 0) = x^p - t \neq 1$. In fact, this captures a bigger result:

Proposition 19.6.7: Separability and irreducible polynomial

Let k be a field and $f(x) \in k[x]$ be an *inseparable and irreducible* polynomial. Then

$$f'(x) = 0$$

***Proof*:**
Since $f(x)$ is inseparable, by proposition 19.6.6, $f(x)$ and $f'(x)$ share a common irreducible factor $q(x)$, that is $f(x) = q(x)g(x)$ for non-constant $g(x) \in k[x]$. However, f is irreducible, and so $q(x)$ must be an associate of $f(x)$ (and hence be of equal degree). However, that implies that it has degree higher than $f'(x)$ if $f'(x) \neq 0$. Since $q(x)\big|f'(x)$, the only possibility left is that $f'(x) = 0$, as we sought to show.

Corollary 19.6.8: Separability Over Characteristic 0

let k be a field of characteristic 0 and $f(x) \in k[x]$ an irreducible polynomial. Then $f(x)$ is seperable.

***Proof*:**
Since f is irreducible, it must have degree at least 2, and so $f'(x) \neq 0$, showing it is separable by proposition 19.6.7

Thus, every algebraic extension of a characteristic 0 field is a separable extension, and fields in which an irreducible polynomial is inseperable must have characteristic p. Even better, given an inseperable polynomial:

$$f(x) = \sum_i^n a_i x^i$$

since $f'(x) = 0$:

$$f'(x) = \sum_i^n i a_i x^{i-1} = 0$$

thus, $ia_i = 0$ for all i, so ia_i is a multiple of p. If i itself is a multiple of p, then certainly $ia_i = 0$, and so it must be that $a_i = 0$ for all indicies that are *not* multiples of p. Therefore, we have reduced inseparable polynomials to polynomials of the form:

$$f(x) = a_0 + a_p x^p + a_{2p} + x^{2p} + \cdots + a_{np} x^{np} = \sum_{k=1}^n a_k (x^p)^k$$

Thus, an inseparable polynomial $f(x)$ can be thought of as a polynomial in x^p, $f(x) = g(x^p)$. Using this, we can define the following homomorphism which will be useful to characterize fields which have inseparable polynomials:

Definition 19.6.9: Frobenius Homomorphism

Let k be a field such that $p = \mathrm{ch}(k) > 0$. Then the *Frobenius homomorphism* is a map $k \to k$ defined by

$$x \mapsto x^p$$

It should be verified that this map is indeed a field homomorphism. the Frobenius must be injective, being a homomorphism between fields, however it need not be surjective. Using this homomorphism, we will find all fields in which being separable is identical to being algebraic:

Definition 19.6.10: Perfect Field

Let k be a field. Then k is called a *perfect field* if all irreducible polynomials in $k[x]$ are seperable, that is, every algebraic extension is a separable extension.

Proposition 19.6.11: Frobenius Homomorphism And Separability

Let k be a field. Then k is perfect if and only if either:

1. if $\mathrm{ch}(k) = 0$
2. if $\mathrm{ch}(k) > 0$ and the Frobenius homomorphism is surjective.

***Proof*:**
If $\mathrm{ch}(k) = 0$, then by corrollary 19.6.8, any irreducible polynomial is already seperable, so let's say $\mathrm{ch}(k) = p$ for some prime p.

($\Leftarrow$) By our discussion from earlier, we know an inseparable irreducible polynomial must be of the form:

$$f(x) = \sum_{i=1}^{n} a_i (x^p)^i$$

Now, since the Frobenius homomorphism is surjective, for everry a_i, there exists a b_i such that $b_i \mapsto a_i$, that is $(b_i)^p = a_i$. Thus, using the fact that this map is indeed a ring homomorphism, we get:

$$f(x) = \sum_{i=1}^{n} b_i^p (x^p)^i = \sum_{i=1}^{n} (b_i x^i)^p = \left(\sum_{i=1}^{n} b_i x^i \right)^p = g(x)^p$$

where $g(x) = \sum_{i=1}^{n} b_i x^i$. But then $f(x)$ has become reducible! This contradicts our assumption of the irreducibility of $f(x)$, showing us that no such polynomial can exist.

($\Rightarrow$) Take the contra-positive, that $\mathrm{ch}(k) \neq 0$ (say $\mathrm{ch}(k) = p$) and that the Frobenius homomorphism is not surjective. Let's say that φ is the Frobenius homomorphism and that $t \in k \setminus \varphi(k)$. I claim that $x^p - t$ does not split into linear factors. Indeed, emulating what we have done in example 19.6.1, we get $x^p - t$ is irreducible, and that $(x - u)^p = x^p - t$

Here is another one: Suppose that T has characteristic $p > 0$, and that $x \mapsto x^p$ is not an automorphism. Since a ring homomorphism from a field into itself is injective or zero, then $x \mapsto x^p$ is not surjective.

Let $a \in T$ be such that $a \neq b^p$ for every $b \in T$. Let $f(x) = x^p - a$. We prove that f is irreducible and inseparable. If α is a root of $x^p - a$, then $x^p - a = (x - a)^p$, so α is a multiple root of f (with multiplicity p), hence f is inseparable. Now, let $g(x)$ be an irreducible factor of $f(x)$. Note that $\alpha \notin T$, otherwise $a = \alpha^p$, contrary to the assumption. Then $g(x) = (x - \alpha)^k$ for some $1 < k \leq p$. Using the binomial theorem, we have

$$g(x) = (x - \alpha)^k = x^k - k\alpha x^{-1} + \cdots + (-\alpha)^k$$

Therefore, $k\alpha \in T$. Since $\alpha \notin T$, then $k = p$, so $g = f$. Hence, f is irreducible.

Corollary 19.6.12: Finite Fields Are Perfect

let k be a finite field. Then k is perfect

***Proof*:**
Since a finite field must have a $\mathrm{ch}(k) = p > 0$, the Frobenius homomorphism is well-defined. Furthermore, Since k is a field, the homomorphism is injective. But an injective map between two finite sets is surjective, hence the Frobenius homomorphism is surjective, and so by proposition 19.6.11 k is perfect.

Thus, we've shown that over most fields we have worked with so far, they are separable. Inseparable extensions do not behave so well (as we'll see, their galois group is [almost] trivial, and hence we get little information about them through galois theory), and so more focus is given in understanding the more common subclass of seperable extensions of a field.

19.6.1 Separability and Embeddings

It should be noted that there *are* separable extensions even if a field is not perfect. For example $x^6 - tx^3 + t \in \mathbb{F}_3(t)[x]$ is irreducible over $\mathbb{F}_3(t)$ and factors into unique linear terms in the splitting field, however the field is not perfect since $x^6 - x^3 + 1$ is irreducible and inseparable. Therefore, we need a better way of finding separable polynomial without relying on nice properties of the ground field. This will be part of the goal of this section.

To characterize separable extensions without relying on k being perfect, we will take advantage that a k-homomorphism is determined by where it maps the roots of minimal polynomials of elements of F. In this way, we get an idea of how many distinct roots a minmial polynomial contains. Recall that we have been able to characterize normal fields through k-homomorphisms: if $\sigma : F \to \overline{k}$ is a k-homomorphism and F/k is normal, then $\sigma(F) = F$, which as we talked about in remark 19.5.14 shows that normal extensions shows that we can characterize all k-homomorphisms as in fact being k-automorphisms (in the sense that all isomorphic copies of F are just symmetries of F). In general, separable extensions need not be normal, and hence they might have multiple embeddings into $\overline{k}$. However, understanding all possible k-homomorphisms still gives us a lot of information about our field as was just pointed out. Studying these embeddings will give us some more information about separable extensions, and will allow us to answer the question of whether an extension $k(\alpha)/k$ is separable if α is separable over k, and to see that any algebraic extension can further be broken down into a separable extension followed by a "purely inseparable" extension: F_{sep}/k and F/F_{sep}.

Definition 19.6.13: Separable Degree

Let F/k be a field extension. Then the number of k-homomorphism $F \to \overline{k}$ called the *separable degree* and is denoted

$$[F : k]_s = \mathrm{Hom}_k(F, \overline{k}) = \mathrm{Emb}_k(F)$$

where $\mathrm{Emb}_k(F)$ is shortand for all possible k-embeddings of F into the algebraic closure of F [a]

[a] Note that for any two algebraic closures $\overline{k}_1, \overline{k}_2$, $\mathrm{Hom}(F, \overline{k}_1) \cong \mathrm{Hom}(F, \overline{k}_2)$, making the definition independent of particular algebraic closure, and hence the notation $\mathrm{Emb}_k(F)$ unambiguous

For the curious reader who would want to see the connection between separability and the separable degree and wants a hint before looking at the solution, they may want to think about what happens to the root, since the following theorem is a generalization of theorem 19.2.27.

19.6.14 moment to ponder

Proposition 19.6.15: Separable Degree of Simple Extensions

Let $k \subseteq k(\alpha)$ be a simple algebraic extension. Then $[k(\alpha) : k]_s$ is equal to the number of distincts roots in $\overline{k}$ of the minimal polynomial of α. In particular,

$$[k(\alpha) : k]_s \leq [k(\alpha) : k]$$

with equality if and only if α is separable over k

Notice how this proposition gives us a way to check the separability of the element α without any appeal to k (i.e., without knowing whether k is perfect or not). In fact more is true: $[k(\alpha) : k]_s \mid [k(\alpha) : k]$. The proof of this fact is a bit more laborious than just showing the inequality since it requires the notion of *separable closure*, which is simply the intersection of all fields that are separable, and so we will leave it as an exercise (I think I'll actually incorporate this material).

***Proof*:**
This proof will be very similar to that of corrollary 19.2.29. Consider the collection of k-homomorpihism $\{\iota_r : k(\alpha) \to \overline{k}\}$. By corrollary 19.2.29, we can associate each of these homomorphisms with where a root r of the minimal polynomial $m_{\alpha,k}(x)$ maps to, i.e. $\iota_r(\alpha) = r$. We will start by showing that this correspondence is bijective:

$$\{\iota_r : k(\alpha) \to \overline{k}\} \leftrightarrow \{r\}$$

Injectivity comes easily since ι_r fixes k, and so it is completely determined by where α maps to, which necessarily is a root of the minimal polynomial for α. To show surjectivity, consider $\beta \in \overline{k}$ which is some root of the minimal polynomial $m_{\alpha,k}(x)$. Then by lemma 19.5.7, $k(\beta) \subseteq \overline{k}$ exists, and by theorem 19.2.27, there exists a unique isomorphism $k(\alpha) \xrightarrow{\sim} k(\beta)$. Hence, combining with $k(\beta) \subseteq \overline{k}$, we get an extension $k(\alpha) \subseteq \overline{k}$ that maps α to the root β, showing surjectivity.

Thus, we have establisehd a bijection between the roots of the minimal polynomial and the number of embeddings of a simple algebraic extension. Since the degree of α is the degree of the minimal polynomial, which is less than or equal to the number of roots the minimal polynomial contains, we have that:

$$[k(\alpha) : k]_s \leq [k(\alpha) : k]$$

If furthermore $k(\alpha)$ is separable, then the number of roots is the degree of the polynomial, and hence:

$$[k(\alpha) : k]_s = [k(\alpha) : k]$$

completing the proof.

Corollary 19.6.16: Tower Law For Finite Separable Extensions

Furthermore, if $F/E/k$ is a sequence of algebraic extensions, $[F:k]_s$ is finite if and only if $[F:E]_s$ and $[E:k]_s$ is finite, and furthermore:

$$[F:k]_s = [F:E]_s[E:k]_s$$

***Proof*:**
We'll show that if $[E:k]_s$ or $[F:E]_s$ are infinite, so must $[F:k]_s$, and then show that if they are both finite, so must $[F:k]_s$ be. Let $\varphi: E \to \overline{k}$ be a k-homomorphism. Then since F/E exists and is algebraic over E (and hence k), by lemma 19.5.7, there exists an exension of φ to $\Phi: F \to \overline{k}$ fixing E (i.e. $\Phi|_E = \varphi$). Furthermore, an extension F into $\overline{E} = \overline{k}$ fixing E extends *a fortiori* the identity on k. Thus, if either $[E:k]_s$ or $[F:E]_s$ is infinite, so must $[F:k]_s$ be.

Conversely if $[F:E]_s$ and $[E:k]_s$ is finite, we'll show $[F:k]_s$ is finite and the multiplicity formula holds. This is almost immediate: for any embedding $F \to \overline{k}$ that extends k, we can can first find all extensions from k to E fixing k, of which there are $[E:k]_s$ possible ways of doing so, and then and then extend any chosen embedding from F to $\overline{E} = \overline{k}$ $(F \to \overline{E})$ fixing E in $[F:E]_s$ different ways, giving us a total of $[F:E]_s[E:k]_s$ total possible extensions, completing the proof.

Proposition 19.6.17: Finite Separable Extension Equivalence

Let F/k be a finite extension. Then $[F:k]_s \leq [F:k]$, and the following are equivalent:

1. $F = k(\alpha_1, ..., \alpha_n)$ where each α_i, is separable over k
2. F/k is a separable extension
3. $[F:k]_s = [F:k]$

***Proof*:**
We'll first generlize proposition 19.6.15 for finite extensions to show that $[F:k]_s \leq [F:k]$. Since F/k is finite, F is finitely generated over k, so let $F = k(\alpha_1, ..., \alpha_n)$. Using proposition 19.6.15, corollary 19.6.16, and theorem 19.3.7:

$$\begin{aligned}
[F:k]_s &= [k(\alpha_1, ..., \alpha_{n-1})(\alpha_n) : k(\alpha_1, ..., \alpha_{n-1})]_s \cdots [k(\alpha_1):k]_s \\
&\leq [k(\alpha_1, ..., \alpha_{n-1})(\alpha_n) : k(\alpha_1, ..., \alpha_{n-1})] \cdots [k(\alpha_1):k] \\
&= [F:k]
\end{aligned}$$

showing the inequality

$(1 \Rightarrow 3)$ Since each α_i is separable over k, it is separable over $k(\alpha_1, ..., \alpha_{i-1})$ (the minimal polynomial of α_i over the intermediate field divides the minimal polynomial over the ground field), and so the inequalites in the previous argument are in fact all equalities by corollary 19.6.16

(3 ⇒ 2) Let $[F:k]_s = [F:k]$, take $\alpha \in F$, and consider $k \subseteq k(\alpha) \subseteq F$. Since F is a finite extension, by corollary 19.6.16,

$$[F:k(\alpha)]_s[k(\alpha):k]_s = [F:k]_s = [F:k] = [F:k(\alpha)][k(\alpha):k]$$

Thus, we have two numbers, say a, b such that $a \leq A$, $b \leq B$ and $ab = AB$. This forces $a = A$ and $b = B$, and so we can cancel $[F:k(\alpha)]_s$ and $[F:k(\alpha)]$ on both sides to get:

$$[k(\alpha):k]_s = [k(\alpha):k]$$

which, by proposition 19.6.15 shows that α is separable. Since $\alpha \in F$ was arbitrary, this shows that all elements of F are separable, and hence F/k is a separable extension.

(2 ⇒ 1) Since F/k is separable and F is finite, $F = k(\alpha_1, ..., \alpha_n)$, and each α_i must be separable by definition.

We can finally answer the question posed in remark ref:RemarkSeparableEx: if α is separable over k, then so is $1+\alpha$, α^2, and any other combination of α, since $k(\alpha)/k$ is separable.

Proposition 19.6.18: Tower Law For Separable Extensions

Let $F/E/k$ be a sequence of algebraic extensions. Then F/k is separable if and only if F/E and E/k is separable.

***Proof*:**
Suppose first that F/k is separable. Then E/k is separable, since every element of E lies in F. And for $\alpha \in F$ the minimal polynomial $m_{\alpha,E}$ divides $m_{\alpha,k}$ in $E[x]$, because $m_{\alpha,k}$ lies in $E[x]$ and has α as a root; a divisor of a polynomial with no repeated roots has none either, so $m_{\alpha,E}$ is separable and F/E is a separable extension.

Conversely, let F/E and E/k both be separable and take $\alpha \in F$. The point is to replace E, which may be enormous, by the finite piece of it that the minimal polynomial of α sees. Write

$$m_{\alpha,E}(x) = x^n + c_{n-1}x^{n-1} + \cdots + c_0 \qquad c_j \in E$$

and put $K = k(c_0, ..., c_{n-1})$. Each c_j lies in E and so is separable over k, whence K/k is a finite separable extension by proposition 19.6.17.

Now $m_{\alpha,E}$ has all its coefficients in K, so α is algebraic over K and $m_{\alpha,K}$ divides $m_{\alpha,E}$ in $K[x]$. The latter is separable because F/E is, so $m_{\alpha,K}$ is separable and $K(\alpha)/K$ is a finite separable extension. Multiplying separable degrees along $K(\alpha)/K/k$ with corollary 19.6.16, and ordinary degrees with theorem 19.3.7,

$$[K(\alpha):k]_s = [K(\alpha):K]_s\,[K:k]_s = [K(\alpha):K]\,[K:k] = [K(\alpha):k]$$

using proposition 19.6.17 on each of the two factors. That same proposition, read backwards, makes $K(\alpha)/k$ separable, so α is separable over k. As $\alpha \in F$ was arbitrary, F/k is separable, as we sought to show.

Note that along with the fact that an extension is normal (implying the minimal polynomial splits), we get a bijection between the roots of the minimal polynomial and the automorphisms fixing the base field. For this reason, this will be one of the definitions (out of many equivalent definitions) of a galois extension.

Exercise 19.6.1

1. Let $K/\mathbb{Q}$ be a finite extension. Show that there exists an injective ring homomorphism $K \hookrightarrow \mathbb{R}^{r_1} \oplus \mathbb{C}^{r_2}$ for appropriate r_1, r_2. This is called the *Archimedean embedding* and the codomain is called the *Minkowski space*; both are taken up in [7, the Archimedean embedding].

19.6.2 Separability and Simple Extensions

Finally, we show that all finite-degree separable fields (of infinite cardinality) are in fact simple. This means that when we were working with the finite extension F/k, we can always find an α where $F = k(\alpha)$. We will show the case for fields of finite cardinality in the next section.

Proposition 19.6.19: Algebraic And Simple Extensions

Let F/k be an algebraic extension. Then F is simple if and only if the number of distinct intermediate fields $k \subseteq E \subseteq F$ is finite

***Proof*:**

Let F/k be simple, so that $F = k(\alpha)$. Since F is algebraic, let $m_{\alpha,k}(x)$ be the minimal polynomial over k. Embed $k(\alpha)$ into an algebraic closure $\overline{k}$. If E is an intermediary field $k \subseteq E \subseteq F$, then $k(\alpha) = F = E(\alpha)$ is also a simple algebraic extension. Denote $m_{E,\alpha}(x)$ to be the minimal polynomial for $E(\alpha)$ over E. Since $m_{k,\alpha}(x) \in E[x]$ for all E and $m_{k,\alpha}(\alpha) = 0$, we know that each $m_{E,\alpha}(x)$ is a factor of $m_{k,\alpha}(x)$.

I claim that E is determined by $m_{E,\alpha}(x)$. Since $m_{k,\alpha}(x)$ has finitely many factors in $\overline{k}$, this will prove there are only finitely many intermediate fields, completing one direction of the proof.

Proceeding with proving the claim, I will show that E is generated (over k) by the coefficients of $m_{E,\alpha}(x)$. To see this, let E' be the set generated by k and the coefficients of $m_{E,\alpha}(x)$. Then $m_{E,\alpha}(x) \in E'[x]$, and since $m_{E,\alpha}(x)$ is irreducible in E, it is irreducible in E', and in fact minimal. Note too that $E'(\alpha) = k(\alpha) = E(\alpha)$, so $[k(\alpha) : E'] = \deg(m_{E,\alpha}(x)) = [k(\alpha) : E]$

Now, we have $E' \subseteq E \subseteq k(\alpha)$, therefore by the tower law:

$$\deg(m_{E,\alpha}(x)) = [k(\alpha) : E'] = [k(\alpha) : E][E : E'] = \deg(m_{E,\alpha}(x))[E : E']$$

Thus, it must be that $[E : E'] = 1$, hence $E = E'$, showing that E is indeed completely determined by $m_{E,\alpha}(x)$, hence there can only be finiely many intermediate fields.

Conversely, let's assume there are finitely many intermediate field $k \subseteq E \subseteq F$. Then it must be that the extension F/k is finite, or else we could construct an infinite sequene of sub-extensions $k \subseteq k(\alpha_1) \subseteq k(\alpha_1, \alpha_2) \subseteq \cdots F$. Thus, F/k is algebraic. We will deal with the case of F being a finite field in corrollary 19.7.15. Thus, for now, assume k is a field of infinite cardinality. We have to show that every $F = k(\alpha_1, ..., \alpha_n)$ is in fact simple.

Arguing induictively, it suffices to show that $k(\alpha, \beta)$ is simple. For every $c \in k$, we have:

$$k \subseteq k(c\alpha + \beta) \subseteq k(\alpha, \beta)$$

Since there are only finitely many intermediate fields, it must be that for some two $c, c' \in k$, $c \neq c$:

$$k(c\alpha + \beta) = k(c'\alpha + \beta)$$

But then:

$$\alpha = \frac{(c'\alpha + \beta) - (c\alpha + \beta)}{c' - c} \in k(c\alpha + \beta)$$
$$\beta = (c\alpha + \beta) - c\alpha \in k(c\alpha + \beta)$$

Thus, $k(\alpha, \beta) \subseteq k(c\alpha + \beta)$. Since $k(c\alpha + \beta) \subseteq k(\alpha, \beta)$, we have an equality, showing that it is indeed simple, completing the proof.

Theorem 19.6.20: Primitive Element Theorem

Let F/k be a finite separable extension. Then F is also a *simple* extension.

***Proof*:**
Arguing inductively, we may take $F = k(\alpha, \beta)$ with α and β separable over k (and hence algebraic). As before, the finite case will be dealt with in corrollary 19.7.15.

Let I be the number of k-homomorphisms $\iota : F \to \overline{k}$ (so $|I| = [F : k]_s$). Choose two $\iota, \iota' \in I$ where $\iota \neq \iota$. If x is an indeterminate, then the polynomials

$$\iota(\alpha)x + \iota(\beta) \quad \iota'(\alpha)x + \iota'(\beta)$$

are different (or else $\iota'(\alpha) = \iota(\alpha)$ and $\iota'(\beta) = \iota(\beta)$ so ι, ι' would act in the same way on the whole of $k(\alpha, \beta)$ forcing $\iota = \iota'$)

Therefore, define:

$$f(x) = \prod_{\iota \neq \iota'} ((\iota(\alpha)x + \iota(\beta)) - (\iota'(\alpha)x + \iota'(\beta))) \in \overline{k}[x]$$

The polynomial $f(x)$ cannot be identically 0 since k is infinite. Thus, take $c \in k$ such that $f(c) \neq 0$. So, a distinct map $\iota \in I$ will map $\gamma = c\alpha + \beta$ to distinct elements $\iota(\alpha)c + \iota(\beta)$ (recall $\iota(c) = c$). Since $|I| = [F : k]_s$, and each $\iota(\gamma)$ is a root of the minimal polynomial of γ over k, we have:

$$[F : k]_s \leq [k(\gamma) : k] \leq [F : k]$$

By assumption, the extension is separable, so we have equalities above:

$$[F : k]_s = [k(\gamma) : k] = [F : k]$$

Thus:

$$F = k(\gamma)$$

as we sought to show.

Using this, we can extend corollary 19.2.29 from simple to finite extensions:

Corollary 19.6.21: Order of $\mathrm{Aut}_k(F)$ Of Finite Extension

Let F/k be a finite seprable extension. Then:

$$|\mathrm{Aut}_k(F)| \leq [F : k]$$

with equlaity if and only if F/k is a finite separable normal extension.

***Proof*:**
By theorem 19.6.20, if K is finite and separable, it is simple, and so the inequality immediately holds by corollary 19.2.29. Equality holds if the minimal polynomial splits, which it does over it's splitting field, which is a normal extension by theorem 19.5.13. Conversely, if K/F is normal, then $f(x)$ splits completly in F and has distinct roots since α is separable over k, establishing equalitly once again, completing the proof.

Note that the separability condition is necessary:

Example 19.6.22: Nonseparable Nonsimple Extension
Let $F = \mathbb{F}_p(u, v)$ be the field with two indeterminate variables over $\mathbb{F}_p$, and subfield $k = \mathbb{F}_p(s, t)$ such that $s = u^p$ and $t = v^p$. Then F/k is an algebraic extension; the minimal polynomial of u over k is $x^p - s$, and the minimal polynomial of v over $k(u)$ is $y^p - t$. Thus:

$$[F : k] = p^2$$

Arguing like before, As c ranges in k, we get the intermediate fields:

$$k \subseteq k(cu + v) \subseteq F$$

If any two chocies c, c' get give the same intermediate field, then we can deduce $k(u, v) = k(cu + v)$ as we've argued earlier. This would then give us:

$$[k(cu + v) : k] = p^2$$

However, the minimal polynomial of $cu + v$ has degree p:

$$(cu + v)^p = c^p + st \in k(s, t) = k$$

Thus, since $k = \mathbb{F}_p(s, t)$ is infinite, there are infinitely many intermediate fields, and it follows that F is *not* a simple extension of k by proposition 19.6.19

As a final result, we state the following that shall be handy for chapter 33:

Corollary 19.6.23: Lüroh's Theorem

Let $k(t)/k$ be a simple extension and $E \subseteq k(t)$ an intermediary field containing k. Then there exists a polynomial g such that
$$E = k(g(t))$$

***Proof*:**
Exercise. Sketch of proof: use the primitive element theorem, use Gauss's lemma, and find the minimal polynomial of the primitive element.

Exercise 19.6.2

1. Show that If E/k and F/k be field extensions, E/k is separable, then so is EF/F. Show that if E/k and F/k are separable, so is EF/F.
2. (exercise on automatic differentiation? $k[x,\epsilon]/(\epsilon^2)$)
3. Let F/k is simple so that $F = k(\alpha)$. then if L is the splitting field for a [not necessarily separable] minimal polynomial $m_{\alpha,k}(x)$, $L = \prod_{\sigma \in \mathrm{Emb}_k(F)} \sigma(F)$. If F is normal, then $L = F$, and if F separable, we have an idea of the number of σ exists, namely the degree of $k(\alpha)$ over k)

19.6.3 Important Functional: Resultant and Discriminant

One very useful functional on $R[x]$ that allows us to detect when a polynomial has a root even R need not contain the roots is the following:

Theorem 19.6.24: Resultant

Let R be an integral domain and $f(x), g(x) \in R[x]$ be two polynomials:

$$f(x) = a_n x^n + a_{n-1}x^{n-1} + \cdots + a_1 x + a_0$$
$$g(x) = b_m x^m + b_{m-1}x^{m-1} + \cdots + b_1 x + b_0$$

where $a_n, b_m \neq 0$ so that their degree is the highest order term. Then f and g have a common root if and only if:

$$\operatorname{Res}(f,g) := \det \begin{pmatrix} a_n & 0 & \cdots & 0 & b_m & 0 & \cdots & 0 \\ a_{n-1} & a_n & \cdots & 0 & b_{m-1} & b_m & \cdots & 0 \\ a_{n-2} & a_{n-1} & \cdots & 0 & b_{m-2} & b_{m-1} & \ddots & 0 \\ \vdots & \vdots & \ddots & a_n & \vdots & \vdots & \ddots & b_m \\ a_0 & a_1 & \cdots & \vdots & b_0 & b_1 & \cdots & \vdots \\ 0 & a_0 & \ddots & \vdots & 0 & b_0 & \ddots & \vdots \\ \vdots & \vdots & \ddots & a_{n-1} & \vdots & \vdots & \ddots & b_1 \\ 0 & 0 & \cdots & a_0 & 0 & 0 & \cdots & b_0 \end{pmatrix} = 0$$

is zero.

***Proof*:**
Let $f, g \in R[x]$ be (not necessarily monic) polynomials with degree n and m respectively. If either $f = 0$ or $g = 0$, there is nothing to prove, so assume both are nonzero. Let $\mathcal{P}_n$, $\mathcal{P}_m$ be the free R-modules of polynomial up to degree strictly less than n and strictly less than m respectively; then $\mathcal{P}_n, \mathcal{P}_m$ have dimension (rank)[a] n, m, so $\mathcal{P}_n \times \mathcal{P}_m$ has dimension $n + m$. Define:

$$\varphi : \mathcal{P}_m \times \mathcal{P}_n \to \mathcal{P}_{n+m} \qquad \varphi(p,q) = fp + gq$$

As R is an integral domain, $\deg(fp + gq) = n + m$, otherwise it may be of a smaller degree and we may get unwanted zero-polynomials. The function φ is linear as:

$$\begin{aligned} \varphi((p_1, q_1) + (p_2, q_2)) &= \varphi(p_1 + q_2, q_1 + q_2) \\ &= f(p_1 + p_2) + g(q_1 + q_2) \\ &= fp_1 + fp_2 + gq_1 + gq_2 \\ &= (fp_1 + gq_1) + (fp_2 + gq_2) \\ &= \varphi(p_1, q_1) + \varphi(p_2, q_2) \end{aligned}$$

Hence, this transformation has a matrix representation. Choosing the (ordered) basis $(x^{n-1}, x^{n-2}, ..., x, 1)$ for $\mathcal{P}_n$ and the (ordered) basis $(y^{m-1}, y^{m-2}, ..., y, 1)$ for $\mathcal{P}_m$, we get the matrix representation:

$$\begin{pmatrix} a_n & 0 & \cdots & 0 & b_m & 0 & \cdots & 0 \\ a_{n-1} & a_n & \cdots & 0 & b_{m-1} & b_m & \cdots & 0 \\ a_{n-2} & a_{n-1} & \cdots & 0 & b_{m-2} & b_{m-1} & \ddots & 0 \\ \vdots & \vdots & \ddots & a_n & \vdots & \vdots & \ddots & b_m \\ a_0 & a_1 & \cdots & \vdots & b_0 & b_1 & \cdots & \vdots \\ 0 & a_0 & \ddots & \vdots & 0 & b_0 & \ddots & \vdots \\ \vdots & \vdots & \ddots & a_{n-1} & \vdots & \vdots & \ddots & b_1 \\ 0 & 0 & \cdots & a_0 & 0 & 0 & \cdots & b_0 \end{pmatrix}$$

Note that $f \notin \mathcal{P}_n$ and $g \notin \mathcal{P}_m$ or else there would always exist $\varphi(-g, f) = -fg + fg = 0$, which would ruin the following argument using the determinant. Let M represent this matrix.

If f, g share a common $\alpha \in \overline{K}$ where $K = \mathrm{Frac}(R)$, then there exists an $h \in \overline{K}[x]$ such that $h(x)f_0(x) = f(x)$ and $h(x)g_0(x) = g(x)$ for appropriate factors $f_0, g_0 \in \overline{K}[x]$. Then simply take $\varphi(-g_0, f_0) = 0$ so $\det(M) = 0$ (see exercise 13.4.1(13.4.1 (5)))

Conversely, suppose $\det(M) = 0$. Then there exists p_0, q_0 such that

$$fp_0 + gq_0 = 0 \qquad \Rightarrow \qquad fp_0 = -gq_0 \tag{19.3}$$

We'll use this to show that f, g are not relatively prime. Suppose $(f, g) = 1$ so that there exists s, t such that $ft + sg = 1$. Then $ftp_0 + sgp_0 = p_0$. Substituting $fp_0 = -gq_0$ from equation (19.3) we get:

$$gq_0p_0 + sgp_0 = g(q_0p_0 + sp_0) = p_0$$

As R is an integral domain, $g \big| p_0$ so $p_0 = gh$. Similarly, $q_0 = fr$.

Both of these cause a contradiction. For example, considering $p_0 = gh$, then:

$$m > \deg(p_0) = \deg(gh) = \deg(g) + \deg(h) = m + \epsilon$$

where $\epsilon \in \mathbb{N}$. Thus (f, g) is *not* relatively prime. Then if $K = \mathrm{Frac}(R)$ we have that $\overline{K}[x]$ is a PID and it must be that there exists a unit $h \in \overline{K}[x]$

$$(h) = (f, g)$$

showing f, g have a common root.

[a]Note that as this is over finite dimensions and over commutative rings, the rank is invariant and hence there is no issue when we use the specific rank later in the proof

Definition 19.6.25: Resultant

The determinant in theorem 19.6.24 $\mathrm{Res}(f, g)$ is called the *resultant*. If the base-ring is not clear, we often denote it as $\mathrm{Res}_R(f, g)$ for the desired ring R, and if the indeterminate element is not clear we may denote it as $\mathrm{Res}_x(f, g)$.

The form above is useful as it can be defined over an integral domain R and is better suited for computational purposes as no field extensions are required. The following is often useful when strictly working over algebraically closed fields:

Proposition 19.6.26: Characterizing Resultant using Algebraically Closed Fields

Let $f, g \in R[x]$ be (not necessarily monic) polynomial over an integral domain R, and let $\overline{K}$ be the algebraic closure of $K = \text{Frac}(R)$. Then if $\alpha_1, ..., \alpha_n$ are the roots of f and $\beta_1, ..., \beta_m$ the roots of g respectively:

$$\text{Res}(f, g) = a_n^m b_n^m \prod_{i.j} (\alpha_i - \beta_j)$$

***Proof*:**
exercise.

One of the more important application is to find whether a polynomial has a multiple root:

Definition 19.6.27: Discriminant

Let $f \in R[x]$ be a polynomial over an integral domain. Then:

$$\Delta(f) = \text{disc}(f) = \frac{(-1)^{n(n-2)/2}}{a_n} \text{Res}(f, f')$$

If f is monic,

$$\Delta(f) = \text{disc}(f) = (-1)^{n(n-2)/2} \text{Res}(f, f')$$

Proposition 19.6.28: Discriminant As A Product

Let $f \in R[x]$ be polynomial over an integral domain, $K = \text{Frac}(R)$, and let L/K be the splitting field of f. Then if $\alpha_1, ..., \alpha_n$ are the roots of f,

$$\Delta(f) = \prod_{1 \leq i \leq j \leq n} (\alpha_i - \alpha_j)^2$$

***Proof*:**
exercise

As is immediate after proposition 19.6.6:

Proposition 19.6.29: Environment Title

Let $f \in \mathbb{R}[x]$ be a polynomial. Then if $\Delta(f) = 0$, f has a multiple root in the splitting field.

***Proof*:**
immediate.

here)

Separability has a numerical measure that is often more useful than the definition. A finite extension F/k makes F a finite-dimensional k-vector space, so every element of F acts on it by multiplication and has a trace in the sense of definition 13.4.3. That assignment is a k-linear form on F, and the symmetric bilinear form it induces detects separability exactly.

Definition 19.6.30: Trace of a Finite Field Extension

Let F/k be a finite field extension and let $x \in F$. Multiplication by x is a k-linear endomorphism $m_x : F \to F$, and the *trace* of x is

$$\mathrm{tr}_{F/k}(x) \;:=\; \mathrm{tr}(m_x) \in k$$

the trace of that endomorphism in the sense of definition 13.4.3. The *trace form* of F/k is the symmetric k-bilinear form

$$\langle x, y\rangle \;:=\; \mathrm{tr}_{F/k}(xy)$$

Proposition 19.6.31: The Trace Form of a Separable Extension Is Nondegenerate

Let F/k be a finite separable field extension. Then the trace form of F/k is nondegenerate: if $x \in F$ satisfies $\mathrm{tr}_{F/k}(xy) = 0$ for all $y \in F$, then $x = 0$. Equivalently, for any k-basis $e_1, \dots, e_n$ of F there is a dual basis $e_1^*, \dots, e_n^*$ with $\mathrm{tr}_{F/k}(e_i e_j^*) = \delta_{ij}$.

***Proof*:**
Step 1: reduce to a computation in one element. By theorem 19.6.20 the extension is simple, say $F = k(\alpha)$ with α of degree n over k and minimal polynomial m. Then $1, \alpha, \dots, \alpha^{n-1}$ is a k-basis of F, and it suffices to show that the Gram matrix $M = \big(\mathrm{tr}_{F/k}(\alpha^{i+j})\big)_{0\le i,j\le n-1}$ of the trace form in that basis is invertible: by lemma 13.3.23 a bilinear form on a finite-dimensional space is nondegenerate (definition 13.3.22) exactly when its matrix in a basis is invertible, and the criterion is independent of which basis is used.

Step 2: traces are power sums of the roots. In the basis $1, \alpha, \dots, \alpha^{n-1}$ the matrix of m_α is the companion matrix of m, so the characteristic polynomial of m_α is m itself. Let $\alpha_1, \dots, \alpha_n$ be the roots of m in a splitting field; they are distinct because F/k is separable, so m is a separable polynomial (definition 19.6.3, proposition 19.6.7). The eigenvalues of the companion matrix are exactly the α_r, so those of m_α^j are the α_r^j, and therefore

$$\mathrm{tr}_{F/k}(\alpha^j) \;=\; \mathrm{tr}\big(m_\alpha^j\big) \;=\; \sum_{r=1}^{n} \alpha_r^j$$

for every $j \ge 0$.

Step 3: a Vandermonde determinant. Let $V = (\alpha_r^i)_{0\le i\le n-1,\, 1\le r\le n}$ be the Vandermonde matrix of the roots. By Step 2,

$$M_{ij} \;=\; \sum_{r=1}^{n} \alpha_r^i \alpha_r^j \;=\; \big(VV^\mathsf{T}\big)_{ij}$$

so $M = VV^\mathsf{T}$ and $\det M = (\det V)^2 = \prod_{r<s}(\alpha_r - \alpha_s)^2$. The roots are distinct, so this is nonzero

and M is invertible. The dual basis is then obtained by inverting M, completing the proof.

The dual basis is what makes the trace form a finiteness tool: an element whose products with a fixed basis all have trace in a subring is pinned inside the free module spanned by the dual basis. That is the mechanism behind the finiteness of integral closure in *Commutative Algebra* [10], and behind the finiteness of the ring of integers of a number field in *Algebraic Number Theory* [7].

Both of those applications need one more thing before the trace form can be brought to bear: the extension in question has to be separable over the subfield chosen as a base. A transcendence basis makes K algebraic over $k(B)$, but says nothing about separability, and in characteristic p the two can come apart. Over a perfect field they can always be reconciled.

Definition 19.6.32: Separating Transcendence Basis

Let K/k be a field extension. A transcendence basis B of K/k is *separating* if K is a separable algebraic extension of $k(B)$. The extension K/k is *separably generated* if it admits one.

Theorem 19.6.33: Perfect Base Fields Give Separating Transcendence Bases

Let k be a perfect field and let $K = k(a_1, \dots, a_n)$ be a finitely generated field extension. Then some subset of $\{a_1, \dots, a_n\}$ is a separating transcendence basis of K over k. In particular every finitely generated extension of a perfect field is separably generated.

***Proof*:**
If $\operatorname{char} k = 0$ every algebraic extension is separable (corollary 19.6.8), so any transcendence basis extracted from the generators by theorem 19.2.13 is separating. Assume $\operatorname{char} k = p > 0$ and induct on n, the case $n = 0$ being trivial.

Step 1: dispose of the independent case. If $a_1, \dots, a_n$ are algebraically independent over k they form a transcendence basis and $K = k(a_1, \dots, a_n)$ is separable over itself, so the basis is separating. Otherwise there is a nonzero $f \in k[X_1, \dots, X_n]$ with $f(a_1, \dots, a_n) = 0$, and we may take f irreducible, replacing it by an irreducible factor that still vanishes.

Step 2: perfectness forces a nonzero partial derivative. Suppose every X_i occurred in f only through p-th powers, so that $f = \sum_\alpha c_\alpha X^{p\alpha}$. Since k is perfect, every c_α is a p-th power, say $c_\alpha = d_\alpha^p$, and then

$$f = \sum_\alpha d_\alpha^p X^{p\alpha} = \Big(\sum_\alpha d_\alpha X^\alpha\Big)^p$$

using that raising to the p-th power is a ring homomorphism in characteristic p (proposition 19.6.11). That contradicts irreducibility of f. So some variable occurs in f outside a p-th power, which is to say $\partial f/\partial X_i \neq 0$ for some i; fix such an i.

Step 3: the i-th generator is separable over the others. Put $F = k(a_1, \dots, \widehat{a_i}, \dots, a_n)$, the subfield generated by the remaining generators. Viewing f as a polynomial in X_i with coefficients in $k[X_1, \dots, \widehat{X_i}, \dots, X_n]$, it is nonconstant in X_i (otherwise $\partial f/\partial X_i = 0$) and irreducible, so up to a unit it is the minimal polynomial of a_i over F. Its derivative in X_i is nonzero and of smaller degree, hence not divisible by the minimal polynomial, so the minimal polynomial has no repeated root by proposition 19.6.6. Thus a_i is separable algebraic over F, and $K = F(a_i)$ is a finite separable extension of F.

Step 4: induct. The field F is generated over k by $n-1$ elements, so by induction some subset B of $\{a_1, \ldots, \widehat{a_i}, \ldots, a_n\}$ is a separating transcendence basis of F/k. Then K is algebraic over F, which is algebraic over $k(B)$, so K is algebraic over $k(B)$ and B is a transcendence basis of K/k. Separability passes up the tower $k(B) \subseteq F \subseteq K$ by proposition 19.6.18, both steps being separable, so $K/k(B)$ is separable and B is separating, completing the proof.

The hypothesis is not decorative. Over an imperfect field the theorem fails: if $t \in k$ is not a p-th power, the extension $k(u)$ with $u^p = t$ adjoined to a rational function field produces a finitely generated extension with no separating transcendence basis, and the trace form degenerates accordingly.

Exercise 19.6.3

1. Show that if f, g are polynomials of degree n, m respectively then
$$\begin{aligned}(x^{m-1}f(x)) \wedge (x^{m-2}f(x)) \wedge \cdots \wedge f(x) \wedge (x^{n-1}g(x) \wedge \cdots \wedge g(x)\\ = \operatorname{Res}(f,g)x^{n+m-1} \wedge x^{n+m-2} \wedge \cdots \wedge 1\end{aligned}$$
2. Let $f, g \in R[x]$. Show that $\operatorname{Res}(f,g) = 0$ iff f, g have a common divisor in $\operatorname{Frac}(R)$.
3. Show that $\operatorname{Res}(f(x+a), g(x+b)) = \operatorname{Res}(f(x), g(x))$ for $a, b \in R$.
4. Show that $\Delta(fg) = \Delta(f)\Delta(g)\operatorname{Res}(f,g)^2$.
5. Let $f(x) = ax^2 + bx + c$. Show that $\Delta(f) = b^2 - 4ac$.
6. (a bunch about the resultant under ring homomorphism!)
7. (There are a ton of cool exercises that can be put here! Importantly, how the resultant changes when the polynomials change)
8. (on homogeneous polynomials; and homogeneity of Res in general)
9. (A note on how it does *not* generalize nicely to multiple polynomials.

19.7 Application: Cyclotomic Extensions and Finite Fields

In this section, we use the tools given to us to gain some important examples of field extensions and analyse the corresponding $\operatorname{Aut}_k(F)$, and generalize theorem 19.6.20 to show finite separable extensions of finite fields are in fact simple.

We start by using the tools that were developed to study the important class of polynomials of the form $x^n - 1$; such extensions will be called *Cyclotomic extensions*. These polynomials represent the "torsion units" and shall show up everywhere when considering solutions to equations as they often add the necessary "symmetry" to find all the solutions to an algebraic equation. They are also important for the continuation of Field Theory to Kummer Theory, which is the study of the case where $\mathrm{Aut}_k(F)$ is an abelian group, and they form an important first step towards Fermat's Last theorem through their application to solutions of Elliptic curves[23].

After having classified Cyclotomic extensions, we shall finish the section with the study of extensions of finite fields. As it turns out, all finite extensions will be splitting fields of polynomials of the form $x(x^{p^d}-1)$, meaning they will have a very simple structure and automorphism group.

19.7.1 Cyclotomic Polynomials and Fields

We start off with some reminders:

Definition 19.7.1: Roots Of Unity Reminder

We will denote $\zeta_n = e^{\frac{2\pi i}{n}}$, and the cyclic subgroup they generated to be μ_n. Note that:

$$\zeta_n^n = e^{\frac{2\pi i k}{k}} = e^0 = 1$$

If $\langle \zeta_n^k \rangle = \mu_n$, then this element is called the *primitve root of unity.*

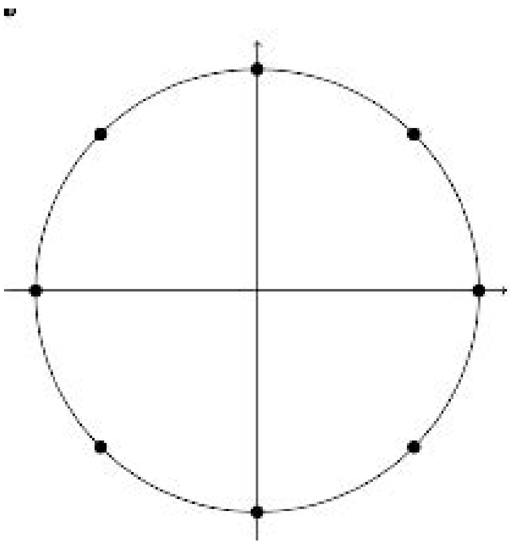
The element ζ_n is always a root of $x^n - 1$, and each ζ_n^k, $1 \leq k \leq n$ where $\gcd(n,k) = 1$ consist of the rest of the roots of $x^n - 1$, for

$$(\zeta_n^k)^n = 1$$

and if $\gcd(n,k) = \ell$, then:

$$(\zeta_n^k)^\ell = 1$$

It is an easy mistake to think that $\zeta_2 = i$, however $\zeta_2 = -1$ and $\zeta_4 = i$. These elements always form a group under multiplication, which we labeled μ_n. This group always lies on the unit circle, for example:



[23]if $\mathrm{ch}(F) \notin 2,3$, then an elliptic curve is a polynomial of the form $y^2 = x^3 + ax + b$, $a, b \in F$. A more technical definition is given when $\mathrm{ch}(F) \in \{2,3\}$ that would be presented in a course on elliptic curves

If you know about the euler representation of complex numbers:

$$re^{i\theta} = r(\cos(\theta) + i\sin(\theta))$$

Then $e^{\frac{2\pi i}{n}} = \zeta_n = \cos(2\pi/n) + i\sin(2\pi/n)$. Other identities we get are:

1. $\zeta_3 = \frac{-1+i\sqrt{3}}{2}$
2. $\zeta_5 = \frac{\sqrt{5}-1}{4} + \frac{\sqrt{10+2\sqrt{5}}}{4}i$
3. $\zeta_6 = \frac{1+i\sqrt{3}}{2}$ (to not be confused with ζ_3)
4. $\zeta_8 = \frac{\sqrt{2}}{2} + i\frac{\sqrt{2}}{2}$

These can be deduced using the results from section 19.4. In paticular Gauss proved that primitive nth roots of unity ζ_n can be constructed using the square-roots, additions, subtraction, multiplication, and division of rational numbers, which we saw implies that n is a power of two, or in our case we would require that n is the product of two discint *fermat primes*.

Every root of unity of a root of $x^n - 1$ for some n, however this polynomial is not irreducible, namely $x^n - 1 = (x-1)p(x)$. The polynomial $p(x)$ will turn out to be irreducible, and hence is given a name:

Definition 19.7.2: Cyclotomic Polynomials

Let:

$$\Phi_n(x) = \prod_{\zeta \text{ primitive } n\text{th root of } 1} (x - \zeta) = \prod_{\substack{1\le m\le n \\ \gcd(m,n)=1}} (x - \zeta_n^m)$$

Then the resulting polynomial is called the *nth cyclotomic polynomial*

As we have seen in the section on number theory (section 0.2), there are $\varphi(n)$ primitive roots of 1, where φ is Euler's totient function. The term cyclotomic comes from the greek *cyclo* (circle) and *tomos* (divide), owing to how the roots of unity divide the circle into regular sections. Though not obvious, this polynomial has rational coefficients and is irreducible over $\mathbb{Q}$ (which we'll show).

Example 19.7.3: Cyclotomic Polynomials

In the case of $n = p$ is a prime, it is simple to see it's irreducible. Every non-identity element of μ_p is a generator, i.e. every pth root of 1 is primitive except 1. In fact, 1 is a root of $x^p - 1$ ($1^p - 1 = 0$), and so we get:

$$\Phi_p(x) = \frac{x^p - 1}{x - 1} = x^{p-1} + x^{p-2} + \cdots + x + 1$$

which is irreducible over $\mathbb{Q}$ using Eisenstein's trick along with shifting (noting that p divides $\binom{n}{k}$). Furthermore, $\mu_p = C_p = \mathbb{Z}/p\mathbb{Z}$.

Aside: for those who know Taylor series and the exponential function, notice how close $\Phi_p(x)$ resembles the pariatal sum for the taylor expansion of e^x:

$$1 + x + \frac{x^2}{2} + \frac{x^3}{3!} + \cdots + \frac{x^p}{p!}$$

the main difference being the coefficient of the partial taylor expansion of e^x being $1/n!$. This reflects on the "rotation" nature of e^x, as well as it's link to roots of unity. However, note that e^x has no roots (even when expended to $\mathbb{C}$, since $e^z \cdot e^{-z} = 1$). Hence, though the partial expansion may have root, even roots that resemble the roots of unity, taylor expansion, the "infinite polynomial representation" need not have roots!

For ease of reference, we record what happens when n is not prime:

Lemma 19.7.4: Cyclotomic Polynomial divisor Representation

For all $n > 0$,

$$x^n - 1 = \prod_{1 \leq d \big| n} \Phi_d(x)$$

***Proof*:**
For all ζ_n^k, $0 \leq k \leq n-1$, $(\zeta_n^k) - 1 = 0$. Hence, $x^n - 1$ splits into products

$$x^n - 1 = \prod_k^n (x - \zeta_k)$$

If $n = dk$ then $1 = \zeta_n^n = (\zeta_n^d)^k$, hence every dth root of unity is an nth root of unity. In particular, every primitive dth root of unity is an nth root of unity. Conversely, if $\zeta_n^d \in \mu_n$, then ζ_n^d forms cyclic subgroup, say μ_d where d is a divisor of n. But then ζ_n^d is a primitive dth root of unity for some divisor d. This establishes:

$$x^n - 1 = \prod_{1 \leq d \big| n} \left(\prod_{\substack{\zeta \text{ primitive} \\ d\text{th root of } 1}} (x - \zeta) \right) = \prod_{1 \leq d \big| n} \Phi_d(x)$$

as we sought to show.

Corollary 19.7.5: Integer Coefficients

The cyclotomic polynomials $\Phi_n(x)$ have integer coefficients

***Proof*:**
This is a proof by [strong] induction. The case of $\Phi_1(x) = x - 1$ is immediate, so assume that for $m < n$, Φ_m is a monic polynomial with integer coefficients. Then:

$$f(x) = \prod_{\substack{1 \leq d | n \\ d < n}} \Phi_d(x)$$

is a monic polynomial with degree $< n$. Since $f(x), x^n - 1 \in \mathbb{Z}[x]$ are two monic polynomials with integer ceofficients, we may use the euclidean algorithm to find $q(x), r(x) \in \mathbb{Z}[x]$, $\deg(r(x)) < \deg(f(x))$ or $r(x) = 0$ where

$$x^n - 1 = f(x)q(x) + r(x)$$

Then by lemma 19.7.4 we have
$$x^n - 1 = f(x)\Phi_n(x)$$
Hence, we get
$$f(x)(q(x) - \Phi_n(x)) = r(x)$$
But that implies $\deg(r(x)) > \deg(f(x))$, a contradiction. Thus $r(x) = 0$, giving us $f(x)q(x) = f(x)\Phi_n(x)$, which implies $\Phi_n(x) = q(x) \in \mathbb{Z}[x]$, as we sought to show.

Example 19.7.6: Cyclotomic Polynomial

1. Let $p(x) = x^4 - 1 = \Phi_1(x)\Phi_2(x)\Phi_4(x)$. Then:
$$\Phi_4(x) = \frac{x^4 - 1}{x^2 - 1} = x^2 + 1$$
2. Let $p(x) = x^6 - 1 = \Phi_1(x)\Phi_2(x)\Phi_3(x)\Phi_6(x)$. Then:
$$\Phi_6(x) = \frac{x^6 - 1}{(x^2 - 1)(x^2 + x + 1)} = x^2 - x + 1$$
3. Let $p(x) = x^{12} - 1 = \Phi_1(x)\Phi_2(x)\Phi_3(x)\Phi_4(x)\Phi_6(x)\Phi_{12}(x)$. Then:
$$\Phi_{12}(x) = \frac{x^{12} - 1}{(x^6 - 1)(x^2 + 1)} = x^4 - x^2 + 1$$

It might seem that all the coefficients are 0 or ± 1. However, the first counter-example to this problem is when $n = 105$, and in general the coefficients can be as large as we might want

Proposition 19.7.7: Irreducibility Of Cyclotomic Polynomials

For all $n \in \mathbb{N}_{>0}$, $\Phi_n(x) \in \mathbb{Z}[x]$ is irreducible over $\mathbb{Q}$

Note that since $\Phi_n(x)$ is monic, by Gauss's lemma, irreducibility in $\mathbb{Q}[x]$ is equivalen to ireducibility in $\mathbb{Z}[x]$.

***Proof*:**
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Definition 19.7.8: Cyclotomic Field

The splitting field $\mathbb{Q}(\zeta_n)$ for the polynomial $x^n - 1$ over $\mathbb{Q}$ is the nth *cyclotomic field*

Proposition 19.7.9: Cyclotomic Field Automorphism

$$\mathrm{Aut}_{\mathbb{Q}}(\mathbb{Q}(\zeta_n)) \cong (\mathbb{Z}/n\mathbb{Z})^\times$$

***Proof*:**
First, the cardinality of both are the same since cyclotomic fields are a splitting field with degree $\varphi(n)$ (Euler's Totient) while $(\mathbb{Z}/n\mathbb{Z})^\times$ has order $\varphi(n)$ as well. Define

$$j : (\mathbb{Z}/n\mathbb{Z})^\times \to \mathrm{Aut}_{\mathbb{Q}}(\mathbb{Q}(\zeta_n)) \qquad [m]_n \mapsto (\zeta_n \mapsto \zeta_n^m)$$

This map is certainly independent of representative and sends roots of polynomials of $m_{\zeta_n,\mathbb{Q}}(x)$ to themselves making it an automorphism.

19.7.2 Finite Fields

Let F be a finite field of characteristic p. Then as we remarked under definition 19.1.8,

$$F/\mathbb{F}_p$$

where $\mathbb{F}_p = \mathbb{Z}/p\mathbb{Z}$. Since F is finite, let $d = [F : \mathbb{F}_p]$. Since F is a d-dimensional vector-space over $\mathbb{F}_p$, $F \cong_{\mathbf{Vect}} \mathbb{F}_p^d$, hence the order of $|F|$ is a power of p: $|F| = p^d$.

With everything we've learned, we can now classify all finite fields. Interestingly, we get essentially the best possible result we might ask for when classifying objects: for every order p^d, we get a unique field up to isomorphism:

Theorem 19.7.10: Classifying Finite Fields

Let $q = p^d$ be a power of the prime number p. Then the polynomial $x^q - x$ is separable over $\mathbb{F}_p$, and the splitting field of the polynomial $x^q - x$ over $\mathbb{F}_p$ is a field with exactly q elements. Conversely, let F be a field with exactly q elements. Then F is a splitting field for $x^q - x$ over $\mathbb{F}_p$

Hence, all finite extension of finite fields is abelian, and even better cyclic.

***Proof*:**
Let F be the splitting field of $x^q - x$ over $\mathbb{F}_p$. Then necessarily F is finite, and hence it's perfect by corollary 19.6.12, and hence it's a separable extension, meaning F contains q distinct roots of $x^q - x$. We need to show that every element of F is a root (i.e. $|F| = q$). Let R be the set of roots of $x^q - x$ from F. Then I claim that R is a field. Since F is the smallest field containing all the roots, it follows that $R = F$.

We simply need to show it's a subring, and since all nonzero elements are units, it is a field. For any $a, b \in R$, since $a^q - a = 0$, $b^q - b = 0$, then $a^q = a$ and $b^q = b$. Then using the binomial theorem and the characteristic of F, we get:

$$(a - b)^q = a^q + (-1)^q b^q = a - b$$

where $(-1)^q = -1$ if p is odd, and $(-1)^q = 1 = -1$ if $p = 2$. Furthermore, if $b \neq 0$, then:

$$(ab^{-1})^q = a^q(b^q)^{-1} = ab^{-1}$$

showing that R is closed under both operations, hence it is a subfield, hence $R = F$

On the other hand, let F be a field with q elements. Then $|F^\times| = q - 1$, and so by Lagrange

Theorem, for any $0 \neq a \in F^{\times}$, $|a| \big| q-1$ Thus:

$$a \neq 0 \Rightarrow a^{q-1} = 1 \Rightarrow a^q - a = 0$$

showing that a *must* be a root of $x^q - x$. Furthermore, $0^q - 0 = 0$, showing that $x^q - x$ has q roots in F (i.e. all elements of F). But then, F is the splitting field of $x^q - x$ by the previous logic, as we sought to show.

Corollary 19.7.11: Unique Finite Field For Every Order

For every $q = p^d$, p a prime and $d \in \mathbb{N}_{>0}$, there exists a unique field of order q

***Proof*:**
this follows from theorem 19.7.10 and the uniqueness of the splitting field (theorem 19.5.10)

Definition 19.7.12: Galois Field

Let F be a field of order $q = p^d$. Then F is called the *Galois Field* of order q, and we usually denote it as $\mathbb{F}_q$ or $\mathbb{F}_{p^d}$.

Example 19.7.13: Galois Fields
let us show hat x^4+1 is reducible over *any* field $\mathbb{F}_p$, and hence over any finite field $F/\mathbb{F}_p$. First, when $p = 2$, $x^4+1 = (x+1)^4$, hence this is certainly the case when $p = 2$. If p is odd, I claim $x^4+1 \big| x^{p^2} - x$. Indeed, $n^2 \equiv 1 \bmod 8$ for any odd number, thus $8 \big| (p^2-1)$, hence $x^8 - 1 \big| x^{p^2-1} - x$ (exercise). Thus:

$$(x^4+1) \big| (x^8-1) \big| \big| (x^{p^2-1} - x) \big| (x^{p^2} - x)$$

Thus, x^4+1 factors completely in the splitting field of $x^{p^2} - x$, that is in $\mathbb{F}_{p^2}$. Thus, if α is a root of x^4+1, :

$$\mathbb{F}_p \subseteq \mathbb{F}_p(\alpha) \subseteq \mathbb{F}_{p^2}$$

Thus $[\mathbb{F}_p(\alpha) : \mathbb{F}_p] \big| [\mathbb{F}_{p^2} : \mathbb{F}_p] = 2$, thus α has either degree 1 or 2 over $\mathbb{F}_p$. But then, the minimal polynomial must be a factor of x^4+1 of degree 1 or 2, thus x^4+1 must be reducible.

Corollary 19.7.14: Finite Field Extensions

Let p be a prime and let $d \leq e$ be positive integers. Then there is an extension $\mathbb{F}_{p^d} \subseteq \mathbb{F}_{p^e}$ if and only if $d \big| e$. If $d \big| e$, then there is exactly one such extension, in the sense that $\mathbb{F}_{p^e}$ conains a unique copy of $\mathbb{F}_{p^d}$.

Furthermore, all extensions $\mathbb{F}_{p^d} \subseteq \mathbb{F}_{p^e}$ are simple

***Proof*:**
First, let's show that extensions are only dependent on the divisibility of the order. Let's say there is an extension $\mathbb{F}_P \subseteq \mathbb{F}_{p^d} \subseteq \mathbb{F}_{p^e}$. Then $[F_{p^d} : \mathbb{F}]$ divides $[\mathbb{F}_{p^e} : \mathbb{F}]$, meaning $d \big| e$.

Conversely, let's say $d \leq e$ such that $d \big| e$. Since:

$$p^e - 1 = (p^d - 1)((p^d)^{\frac{e}{d}-1} + (p^d)^{\frac{e}{d}-\frac{e}{d}})$$

we see that $p^d - 1$ divides $p^e - 1$, and so $x^{p^d-1} - 1$ divides $x^{p^e-1} - 1$ (Aluffi gave this as exercsie V.2.13). Therefore:

$$(x^{p^d} - x)\big|(x^{p^e} - x)$$

Since $\mathbb{F}_{p^e}$ is a splitting field for the second polynomial by theorem 19.7.10, it must contain a unique copy of the splitting field for the first plynomial (exercsie 4.5 in Aluffi, p. 442).

Finally, to show the extension is simple, recall that any finite field is necessarily cyclic, as we recalled in theorem 19.1.10 ($F^\times$ is locally cyclic, and since F is finite, $F^\times$ is cyclic). Let's say $\alpha \in F_{p^e}$ generated the multiplicative group of $\mathbb{F}_{p^e}$. Then clearly, α will generated $\mathbb{F}_{p^e}$ over any subfield, so if $d|e$, $\mathbb{F}_{p^e} = \mathbb{F}_{p^d}(\alpha)$, showing that $\mathbb{F}_{p^e}/\mathbb{F}_{p^e}$ is simple, completing the proof.

Corollary 19.7.15: Separable Finite Extensions Of Finite Fields

Let k be a finite field and F/k a finite separable extension. Then F is a simple extension.

***Proof*:**
Most of the proof of this was already done by proposition 19.6.19. If k is a finite field, then so is F, and the extensions are automatically simple by corollary 19.7.14.

Using these results, we can be more gain a lot of information about polynomials over finite fields:

Corollary 19.7.16: Finite Fields And Irreducible Polynomials

Let F be a finite field. Then for all integers $n \geq 1$, there exists irreducible polynomials of degree n in $F[x]$

***Proof*:**
Let $n \geq 1$ and $F = \mathbb{F}_{p^d}$. Then

$$\mathbb{F}_{p^d} \subseteq \mathbb{F}_{p^{dn}}$$

is a finite simple extension, say $\mathbb{F}_{p^{dn}} = \mathbb{F}_{p^d}(\alpha)$. Then sine

$$[\mathbb{F}_{p^{dn}} : \mathbb{F}_{p^d}] = n$$

and α has a minimal polynomial over $\mathbb{F}_{p^d} = F$, we see that we can find an irreducible polynomial of degree n in $F[x]$.

Note how this proof doesn't necessarily generalize for infinite fields, since we don't have so much control over the degree of the extension (worth pondering over).

Corollary 19.7.17: Factorizing $x^{q^n} - x$

here p. 443 aluffi

Proof:

Example 19.7.18: Factorizing Polynomials over Finite Fields
p. 443 Aluffi

Another result that finiteness brings is that the structure of $\text{Aut}_{\mathbb{F}_p}(\mathbb{F}_{p^d})$ is very simple. We already know that any finite field is perfect, hence all algebraic extensions are separable, and so

$$|\text{Aut}_{\mathbb{F}_p}(\mathbb{F}_{p^d})| = d$$

However, we can say even more about the structure of the group

Proposition 19.7.19: Structure of Aut Of Finite Fields

Let $\mathbb{F}_{p^d}/\mathbb{F}_p$ be a field extension of degree d. Then $\text{Aut}_{\mathbb{F}_p}(\mathbb{F}_{p^d})$ is cyclic, and generated by the Frobenius isomorphism.

Proof:
Let $\varphi : \mathbb{F}_{p^d} \to \mathbb{F}_{p^d}$ be the map such that $x \mapsto x^p$. Since finite fields are perfect, φ is surjective, and hence an isomorphism. For any $x \in \mathbb{F}_p$, by Fermat's theorem (theorem 10.3.3)

$$x^p \mod p$$

and so φ is in fact a $\mathbb{F}_p$-isomorphism, meaning $\varphi \in \text{Aut}_{\mathbb{F}_p}(\mathbb{F}_{p^d})$. Since $|\text{Aut}_{\mathbb{F}_p}(\mathbb{F}_{p^d})| = d$, we need to show that $|\varphi| = d$.

Since φ is in a finite group, there exists an e such that $\varphi^e = \text{id}$, so $x^{p^e} = x$ for all $x \in \mathbb{F}_{p^d}$. Then $x^{p^e} - x$ has p^d roots in $\mathbb{F}_{p^d}$ (since φ makes all elements of $\mathbb{F}_{p^d}$ satisfy this relation). Then since the number of roots of a polynomial cannot exeed the highes degree, we have that $p^d \leq p^e$, which implies $d \leq e$. Thus, in terms of order of groups, we get that:

$$|\text{Aut}_{\mathbb{F}_p}(\mathbb{F}_{p^d})| \leq |\varphi|$$

but by Lagrange's theorem, $|\varphi| \big| |\text{Aut}_{\mathbb{F}_p}(\mathbb{F}_{p^d})|$,

This result is rather interesting: it means that given $(\mathbb{Z}/m\mathbb{Z})^*$ we have a natural relation $p \mod m$ associated to the Frobenius morphism $x \mapsto x^p$. This relation to prime being modded and the Frobenius homomorphism shall go far in algebraic number theory (chapter 50) and will fully be realized in class field theory (see [39]).

(A word on a way to construct an algebraic closure of finite fields, and on how the category of finite fields seems to be close to another known category)

To close off this section, we saw that all finite extensions of finite fields is a cyclic extension. This has not been shown for finite extension for $\mathbb{Q}$. However, it turns out that this is *almost* the case, namely all finite extensions of $\mathbb{Q}$ embed into a cyclic extensions. This requires a lot more build-up, and is easier to prove in a course on Class Field Theory[24]. These notes shall indeed prove this fact not using class field theory as the "reward" for the material covered in chapter 50 (see theorem 50.5.4).

[24]See CITE:HERE for my book on it

Exercise 19.7.1

1. Let $f(x)$ be an irreducible polynomial of degree n over $\mathbb{F}_p$. Show that any field extension of $f(x)$ splits and is isomorphic to $x^{p^n} - x$
2. Let $k = \mathbb{F}_p$. Show that $f(x^p) = f(x)^p$.
3. Let L be a finite division algebra (or "noncommutative field") with center $\mathbb{F}_p$. Show that L is a field (hint: use class equation, theorem 4.6.3)

20

Galois Theory

For eons, mathematicians looked for relations between coefficients and roots of a polynomial. The answer is "trivial" for linear polynomials, given:

$$ax + b = 0 \qquad \text{with solution } \alpha \text{, then} \quad \alpha = -ba^{-1}$$

For a quadratic $x^2 + bx + c$ with roots α and β, then $\alpha + \beta = -b$ and $\alpha\beta = c$[1], and each individual root is given by the quadratic formula:

$$r = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

More generally, a lot of focus was put into finding the roots of polynomials given their coefficients[2], eventually finding the cubic and quartic equation relating the coefficients to roots of cubic and quartic polynomials respectively. However, the effort to generalize to higher degree polynomials brought limited results, in particular even for the next degree (the quintic) it was unclear how its roots can written in terms of it's coefficients and the arithmetic operations of $+, -, \times, \div$ and $\sqrt{-}$. It wasn't until the turn of the 18th century when serious progress has been made on this front. The key insight was to treat the coefficients of a polynomials as determining a dynamical system (that is, the dynamic of how the roots of a polynomial change as we move around the coefficients). In chapter 19, the right context to consider is the algebraic closure of a field; for pedagogical purposes we stick to $\overline{\mathbb{Q}}$ or $\mathbb{C}$ namely since this is the theories starting point and generalizations to $\mathbb{F}_{q^n}$ and other fields/rings is the natural progression.

To demonstrate these dynamics, consider $x^2 + x - 1 = 0$. This equation (by using the quadratic

[1] See section 20.2.3 for much more detail about these relations

[2] This is a little anachronistic, for when the quadratic formula was first discovered in Babylonia and more succinctly by Brahmagupta, the "modern" notion of polynomial and variables where not used, but rather it was shown by relating areas of circles and rectangles

equations) has solutions:

$$x_1 = \frac{-1+\sqrt{5}}{2} x_2 = \frac{-1-\sqrt{5}}{2}$$

If the b and c coefficients are swapped to get $x^2 - x + 1 = 0$, then the solutions are:

$$x_1 = \frac{1+i\sqrt{3}}{2} x_2 = \frac{1-i\sqrt{3}}{2}$$

In other words if (a, b, c) represents the coefficients, then the permutation action $(2\ 3) \in S_3$ does not leave the roots invariant, that is this is *not* invariant to coefficient permutation. Note these dynamics are "discrete", however explain the connection of the action here! Does it natural correspond to a sort of continuous action, up to some homotopy thing?.

Now, given a general $ax^2 + bx + c = 0$, factoring as $(x-\alpha)(x-\beta)$, then by commutativity:

$$(x-\alpha)(x-\beta) = (x-\beta)(x-\alpha)$$

In other words, applying an S_2 action on the roots leaves the coefficients invariant[3 4].

The first immediate introduction of dynamics into polynomials is by applying group-actions to the roots. The realization that links roots to coefficients is that, given we can completely split a polynomial:

$$x^n + \cdots + \alpha_1 x + \alpha_0 = (x-a_1)(x-a_2)\cdots(x-a_n)$$

each coefficient of the original polynomial can be represented as a *symmetric function* (see section 20.2.3).

$$\begin{aligned}
\alpha_{n-1} &= a_1 + a_2 + \cdots + a_n \\
\alpha_{n-2} &= a_1a_2 + a_1a_3 + \cdots + a_1a_n + a_2a_3 + \cdots a_{n-1}a_n \\
&\cdots \\
\alpha_0 &= a_1a_2\cdots a_n
\end{aligned}$$

These form a natural relationship between the roots and the coefficients. It was noticed that these equation are invariant under permutation action of S_n; if $A = \{A_0, ..., A_{n-1}\}$ represent the above n equations and the action is given by $(r\ s) \cdot a_r = a_s$, then $\sigma \cdot A_i = A_i$ for each A_i and σ, hence the action $S_n \curvearrowright A$ is the trivial action. Thus, there is always a natural group-action.

As any roots of polynomial induce such equations and induce this trivial action, these don't give enough information about polynomials. It shall be shown that roots of a generic polynomial satisfy no further equation, and in these cases there is nothing more that can be done to understand a polynomial via a group-action on its roots. However, there are polynomials whose roots shall have *more* relations the roots satisfy, limiting what group can act on them (i.e. a subgroup of S_n will act trivially, but S_n will not). This chapter shall make rigorous this realization and translate it into the modern language of field theory.

This realization almost managed to show the general quintic is unsolvable:

> "The quintic was almost proven to have no general solutions by radicals by Paolo Ruffini in 1799, whose key insight was to use permutation groups, not just a single permutation. His

[3]extending the action to be continuous, this just makes a coefficient move around in a small circle make this precise, this is jut from the animation form the video

[4]As a consequence, there is no continuous function that can take in coefficients and return roots

> solution contained a gap, which Cauchy considered minor, though this was not patched until the work of the Norwegian mathematician Niels Henrik Abel, who published a proof in 1824, thus establishing the Abel–Ruffini theorem."[5]

This theory was established with the aim to show there is an unsolvable quintic, but not how to find whether a given quintic is solvable. This is where the work of Évariste Galois comes in. He noticed that the key insight was to work with all possible relationships of the roots of the polynomial, assuming all the roots exist.

Modern uses of Galois Theory

Galois groups are central in both algebraic geometry and number theory, where they serve different purposes in each domain. The galois group of a polynomial:

1. establishes the algebraic relations between the roots and creates a "covering space" that has some similar properties to the covering spaces seen in algebraic topology (see remark 20.0.1),
2. establishes the invariance necessary to find a solution over a lower field, in particular if L/K is a field extension and a root of an $L[x]$ polynomial has a root in K, then the root is invariant under the $\mathrm{Gal}_K(L)$-action.

From action on roots to field theory

Take the polynomial:

$$x^4 - 2 = 0$$

which is irreducible over $\mathbb{Q}$ by Eisenstein's criterion (proposition 11.3.10). Adjoining $\sqrt[4]{2}$, $\mathbb{Q}(\sqrt[4]{2})$, the polynomial splits to:

$$(x - \sqrt[4]{2})(x + \sqrt[4]{2})(x^2 + \sqrt{2})$$

Adjoining $\sqrt[4]{2}i$, the polynomial splits completely:

$$(x - \sqrt[4]{2})(x + \sqrt[4]{2})(x - \sqrt[4]{2}i)(x + \sqrt[4]{2}i) \tag{20.1}$$

Already, there is some interesting phenomena at play, for example, another way of getting this splitting is by adjoining $\sqrt[4]{2}$ and then adjoining i (or adjoining i then $\sqrt[4]{2}$). In fact, by just adjoining $\sqrt[4]{2} + i$, we immediately get this splitting. Indeed, certainly $\mathbb{Q}(\sqrt[4]{2} + i) \subseteq \mathbb{Q}(\sqrt[4]{2}, i)$. Conversely, to show $i \in \mathbb{Q}\sqrt[4]{2} + i$, letting $a = \sqrt[4]{2} + i$, then:

$$\begin{aligned} a - i &= \sqrt[4]{2} \\ \Leftrightarrow\ (a-i)^4 &= 2 \\ \Leftrightarrow\ (a-i)^4 &= 2 \\ \Leftrightarrow\ a^4 - 4a^3 i - 6a^2 + 4ai + 1 &= 2 \\ \Leftrightarrow\ i(4a - 4a^3 i) &= 1 - a^4 + 6a^2 \\ \Leftrightarrow\ i &= \frac{1 - a^4 + 6a^2}{4a - 4a^3 i} \end{aligned}$$

[5]taken from Wikipedia for Galois Theory

showing that $i \in \mathbb{Q}(\sqrt[4]{2}+i)$, and so $a - i = \sqrt[4]{2} \in \mathbb{Q}(\sqrt[4]{2}+i)$. More generally for any finite separable extension $K/\mathbb{Q}$, such an element always exists and is constructible by the primitive element theorem (theorem 19.6.20).

Let us understand the equations the roots form given by symmetric functions. As $x^4 - 2 = x^4 + 0x^3 + 0x^2 + 0x - 2$ The only meaningful relation obtained by multiplying out the roots in equation (20.1) is:

$$(\sqrt[4]{2})(-\sqrt[4]{2})(\sqrt[4]{2}i)(-\sqrt[4]{2}i) = 2 \qquad \text{equiv.} \qquad (\sqrt[4]{2})(-\sqrt[4]{2})(\sqrt[4]{2}i)(-\sqrt[4]{2}i) - 2 = 0$$

This equation is invariant of the action of S_4, namely if

$$(a_1, a_2, a_3, a_4) = (\sqrt[4]{2}, \sqrt[4]{2}i, -\sqrt[4]{2}, -\sqrt[4]{2}i)$$

represent the 4 roots, and $A_0 = (\sqrt[4]{2})(-\sqrt[4]{2})(\sqrt[4]{2}i)(-\sqrt[4]{2}i) - 2 = 0$, then:

$$\sigma \cdot A_0 = A_0 \qquad \forall \sigma \in S_4$$

However, there are more equations that can be formed with these roots (using the operations of a field):

$$\begin{aligned} A_1 &: \ \sqrt[4]{2} - \sqrt[4]{2} = 0 \\ A_2 &: \ \sqrt[4]{2}i - \sqrt[4]{2}i = 0 \end{aligned}$$

These equations are *not* invariant under the permutation action of S_4:

$$(1\ 2) \cdot A_1 \neq A_1 \qquad (\text{concretely: } (1\ 2) \cdot (\sqrt[4]{2} - \sqrt[4]{2}) = (\sqrt[4]{2}i - \sqrt[4]{2}) \neq A_1)$$

A subgroup of S_4 must be chosen, namely notice that $D_4 \cong \langle (1\ 2\ 3\ 4), (1\ 3) \rangle \subseteq S_4$[6] keeps these equations invariant:

$$D_4 \cdot A_0 = A_0 \qquad D_4 \cdot A_1 = A_1 \qquad D_4 \cdot A_2 = A_2$$

Are there any other equations that we can be considered which would further restrict the choice of invariant permutation action? To answer this, the above is re-framed in the language of field extensions.

Let us say f is a polynomial over $\mathbb{Q}$ and that α is a root of f. Then the algebraic extension $\mathbb{Q}(\alpha)/\mathbb{Q}$ contains *all* possible polynomial equations that can be generated with elements from $\mathbb{Q}$ and α (recall $\mathbb{Q}(\alpha) = \mathbb{Q}[\alpha]$, see proposition 19.2.24). If any of these polynomials happen to be another root of f, that is $p(\alpha) = \beta$ where $f(\beta) = 0$ and $p(x) \in \mathbb{Q}[x]$, then $\mathbb{Q}(\alpha)/\mathbb{Q}$ contains this relationship between the roots. Let's say one of these relationships is represented as :

$$P(x) = p(x) - \beta \text{ so that } P(\alpha) = p(\alpha) - \beta = 0$$

Then a $\mathbb{Q}$-automorphism (in this case nothing more than a field automorphism) would have to preserve this relationship (it must map roots to roots). Hence, the set of $\mathbb{Q}$-automorphism captures the relationships between α and all possible roots that are $\mathbb{Q}$-polynomials of α. If $K/\mathbb{Q}$ is the splitting field of K, then the $\mathbb{Q}$-automorphisms of K would be restricted to automorphism that must satisfy all possible relationships between roots. Given $K/\mathbb{Q}$ is a splitting field for f, as any $\mathbb{Q}$-automorphisms σ

[6]The reader may recall exercise set 1.3.1 asks to show S_n is generated by a two cycle and an n-cycle; it was important the 2-cycle was an *adjacent* cycle

is determined by where the roots map to, it is associated to a permutation in S_n. If there were no relationships (you need to add n roots to split f), then the automorphism group would be isomorphic to S_n. If there is any other type of group, that means there are some relationships between the coefficients (just like the example of $\mathbb{Q}(\sqrt[4]{2}+i)/\mathbb{Q}$).

The automorphism group of the splitting field above is called the *galois group* and it is the aim of this chapter define the galois group for any field extension L/K, to be able to find some galois groups, to find how frequent certain galois groups are, to show the correspondence between field extension and galois groups, and answer some long-pending mathematical questions such as the solvability of [general] quintic using the newly developed theory.

Another way of thinking about galois group is they are a rough classification tool for field extensions: if two field extensions have different Galois groups they cannot be the same extensions. It is similar in roughness to the fundamental group $\pi_1(X)$ as is often encountered in algebraic topology. To see this roughness in classification, as the splitting field $\mathbb{Q}(\sqrt[4]{2}, i)$ of $x^4 - 2$ is equal to $\mathbb{Q}(\sqrt[4]{2}+i)$, and the minimal polynomial of $\sqrt[4]{2}+i$ is:

$$x^8 + 4x^6 + 2x^4 + 28x^2 + 1$$

then they share the same restrictions on their roots as they have the same galois group, but they certainly also have many differences (for example, these two polynomials have different discriminants, see definition 19.6.27).

20.0.1 Galois Group as Fundamental Group

(Assuming strong familiarity of algebraic geometry and algebraic topology) Under the right framework in algebraic geometry, galois groups can be interpreted as *étale fundamental group* for Spec K,

$$\pi_1^{\text{ét}}(\operatorname{Spec}(K)) \cong \operatorname{Gal}_F(\overline{F})$$

The name *fundamental group* is well-deserved for if X is a connected $\mathbb{C}$-scheme of finite type (i.e. "essentially" a connected variety) and $X(\mathbb{C})$ is it's analytification, then:

$$\pi_1^{\text{ét}}(X) \cong \widehat{\pi_1^{\text{top}}(X(\mathbb{C}))}$$

where $\pi_1^{\text{top}}(-)$ is the usual fundamental group and $\widehat{(-)}$ is the profinite completion with respect to its normal subgroups. See [5][add the locator] for the details.

20.1 Galois Extension and Galois Group

First, let's the recall the definition of an automorphism and define a new term:

Definition 20.1.1: Automorphism and fixed Element

1. An isomorphism σ of F with itself is called an *automorphism* of F. The collection of automorphisms of F is denoted $\text{Aut}(F)$. If $\alpha \in F$, we write $\sigma\alpha$ for $\sigma(\alpha)$
2. An auotmorphism $\sigma \in \text{Aut}(F)$ is said to *fix* an element $\alpha \in F$ if $\sigma\alpha = \alpha$. If k is a subset of F, then an automorphism σ is said to *fix* F if it fixes all the elements of F

$$\sigma a = a \qquad \forall a \in k$$

3. Let F/k be an extension of fields. Then $\text{Aut}(F)$ is the collection of automorphissm of F, and $\text{Aut}_k(F)$ is collection of automorphsism of F which fix F.

Example 20.1.2: Automorphism and Fixed F

The identity map *id* is always an automorphism.

Furthermore, the prime field of any field K is always fixed. If P is the prime-field of F, then for any $\sigma \in \text{Aut}(K)$, $\sigma(1) = 1$, since $\sigma(0) = 0$. Thus, $\sigma(a) = a$ for every $a \in P$. This means that $\mathbb{Q}$ and $\mathbb{F}_p$ have only trivial automorphisms

$$\text{Aut}(\mathbb{Q}) = \{\text{id}\}\,, \text{Aut}(\mathbb{F}_p) = \{\text{id}\}$$

and if P is a prime field of F, then

$$\text{Aut}_P(F) = \text{Aut}(F)$$

Next, we recall a very useful criterion for finding automorphisms of algebraic extensions K/F

Proposition 20.1.3: Fixed Roots Criterion

Let F/k be a field extension and let $\alpha \in F$ be algebraic over k. Then for any $\sigma \in \text{Aut}(F/k)$, $\sigma\alpha$ is a root of the minimal polynomial for α over k, i.e., $\text{Aut}(F/k)$ permutes the roots of irreducible polynomials. Equivalently, any polynomial whith coefficeints in k having α as a root also has $\sigma\alpha$ as a root.
Furthermore, $\text{Aut}_k(k(\alpha))$ acts *transitively* on the set of roots of $m_{k,\alpha}(x)$ (i.e. if β_1, β_2 are roots of $m_{k,\alpha}(x)$, then there exists a $\sigma \in \text{Aut}_k(k(\alpha))$ such that $\sigma\beta_1 = \beta_2$)

***Proof*:**
Let

$$x^n + a_{n-1}x^{n-1} + \cdots + a_1 x + a_0$$

be the minimal polynomial with coefficients from F so that

$$\alpha^n + a_{n-1}\alpha^{n-1} + \cdots + a_1\alpha + a_0 = 0$$

then, applying any automorphism σ to this polynomial will get us

$$\begin{aligned} \sigma(\alpha^n + a_{n-1}\alpha^{n-1} + \cdots + a_1\alpha + a_0) &= \sigma(0) \\ \sigma(\alpha)^n + a_{n-1}\sigma(\alpha)^{n-1} + \cdots + a_1\sigma(\alpha) + \sigma(a_0) &= 0 \qquad \sigma \text{ fixes } a_i \text{ for each } i \end{aligned}$$

since $a_i \in F$ for each i, showing that $\sigma(\alpha)$ is also a root of the minimal polynomial

Transitivty is left as an exercise (recall the appropriate theorem)

Example 20.1.4: Automorphism Groups

1. Take $K = \mathbb{Q}(\sqrt{2})$. Then $\text{Aut}(\mathbb{Q}(\sqrt{2})) = \text{Aut}(\mathbb{Q}(\sqrt{2})/\mathbb{Q})$. If $\sigma \in \text{Aut}(\mathbb{Q}(\sqrt{2}))$, then by the fixed root criterion, it must be that

$$\sigma(\sqrt{2}) = \pm\sqrt{2}$$

and since σ must fix everything else in place, this means that we get

$$\sigma(a + b\sqrt{2}) = a \pm b\sqrt{2}$$

in other words, we've determined were every other element must map to! Thus, we have the maps

$$\sqrt{2} \mapsto \sqrt{2} \qquad \sqrt{2} \mapsto \sqrt{-2}$$

If we let the first map be represetned by 1 and the second by σ, then we get

$$\text{Aut}(\mathbb{Q}(\sqrt{2})) = \{1, \sigma\} = C_2$$

that is, $\text{Aut}(\mathbb{Q}(\sqrt{2}))$ is a cyclic group of order 2.

2. Let $K = \mathbb{Q}(\sqrt[3]{2})$. By theorem 19.2.5, every element of $\mathbb{Q}$ can be written as

$$a + b\sqrt[3]{2} + c(\sqrt[3]{2})^2$$

If we consider any $\sigma \in \text{Aut}(\mathbb{Q}(\sqrt[3]{2}))$, then we get

$$a + b\sigma(\sqrt[3]{2}) + c\sigma(\sqrt[3]{2}) \qquad a, b, c \in \mathbb{Q}$$

and since $\mathbb{Q}(\sqrt[3]{2})$ has non of it's other roots (recall example 19.3.12). Then that means that the only automorphism is the identity, meaning

$$\text{Aut}(\mathbb{Q}(\sqrt[3]{2})) = \{1\}$$

3. Now let $K = \mathbb{Q}(\sqrt[3]{2}, \sqrt{-3})$, be the splitting field of $x^3 - 2$ in example (19.5.11), which unlike the previous example will insure we have interesting automorphisms since K contains all the roots of $x^3 - 2$. Let's label these roots $\sqrt[3]{2}, \omega, \omega^2$. Then:

$$[\mathbb{Q}(\sqrt[3]{2}, \sqrt{-3}) : \mathbb{Q}] = [\mathbb{Q}(\sqrt[3]{2}, \sqrt{-3}) : \mathbb{Q}(\sqrt[3]{2})] : [\mathbb{Q}(\sqrt[3]{2}) : \mathbb{Q}] = 2 \cdot 3 = 6$$

We know that $\mathbb{Q}(\sqrt[3]{2})$ contains one root of $x^3 - 2$ over $\mathbb{Q}$ ($\sqrt[3]{2} \in \mathbb{Q}(\sqrt[3]{2})$), and that $\mathbb{Q}(\sqrt[3]{2}, \omega)$ also contains the other two roots over $\mathbb{Q}$, and in particular over $\mathbb{Q}(\sqrt[3]{2})$ ($\omega, \omega^2 \in \mathbb{Q}(\sqrt[3]{2}, \omega)$.

Next, Since the degree of K is 6, we know it has a basis consisting of 6 elements, namely:

$$1,\ \sqrt[3]{2},\ \sqrt[3]{4},\ \omega,\ \omega\sqrt[3]{2},\ \omega\sqrt[3]{4}$$

Since any $\mathbb{Q}$-automorphism of K is a $\mathbb{Q}$-linear automorphism on K, it suffices to characterise these automorphisms by where by map these basis elements. By the fixed roots criterion, a $\mathbb{Q}$-automorphism σ on K can map $\sqrt[3]{2}$ to:

$$\sqrt[3]{2},\ \omega\sqrt[3]{2},\ \omega^2\sqrt[3]{2}$$

and map ω to:

$$\omega,\ \omega^2$$

Using this information, we can carry through computation's to see that $\mathrm{Aut}_{\mathbb{Q}}(K)$ is S_3. (I think we will encounter theorems that will make this easier later)

4. More on computing automorphisms groups here: here

Since any field homomorphism must be injective, $\mathrm{Aut}_k(-)$ can in fact respect composition, making it a [contravariant] functor[7]

Lemma 20.1.5: Field Automorphism Is Functorial

let $\mathbf{Field}_k$ be all fields such that any $F \in \mathbf{Field}_k$ has k as a subfield: $k \subseteq F$. Then $\mathrm{Aut}_k(-) : \mathbf{Field}_k \to \mathbf{Grp}$ sending F to $\mathrm{Aut}_k(F)$ and $F \to L$ to $\mathrm{Aut}_k(L) \to \mathrm{Aut}_k(F)$ via $\varphi \mapsto \varphi|_F$ is a contraviant functor

***Proof*:**
simple definition pushing.

Though $\mathrm{Aut}_k(-)$ being functorial is quite powerful and a fact that shall be often used, it actually perserves even more structure, namely the latice of groups and interemetiate fields is [contravariantly] preserved. The following propositions builds up to this result. Since $\mathrm{Aut}_k(F)$ has a relatively easy to find structure, any connections between the structure of $\mathrm{Aut}_k(F)$ (which is a finite group if $[F : k] < \infty$) and F/k would allow us to use the full power of group theory (and better still finite group theory) to analise field extensions. We endevaour to deepen this connection in this chapter. One of the most important connection is that intermediate fields $k \subseteq E \subseteq F$ correspond to subgroups of $\mathrm{Aut}(F/k)$! This is called the *Galois correspondence*, and is what we will be building up for the rest of this section. We start by showing how subgroups of $\mathrm{Aut}_k(F)$ correspond to an intermediate field $k \subseteq E \subseteq F$, and conversely intermediate extension $k \subseteq E \subseteq F$ correspond a subgroup fo $\mathrm{Aut}_k(F)$

Proposition 20.1.6: Fixing F with automorphisms

Let $H \leqslant \mathrm{Aut}_k(F)$ be a subgroup of automorphisms of F (or $H \subseteq \mathrm{Aut}_k(F)$ more generally). Then the collection F^H of elements of F fixed by all the elements of H is a subfield of F extending k, $k \subseteq F^H \subseteq F$. Conversely, Let E be an intermediate field $k \subseteq E \subseteq F$. Then $\mathrm{Aut}_E(F) \leqslant \mathrm{Aut}_k(F)$.

[7] Note that $\mathrm{Aut}_{\mathbf{Grp}}(-)$ is *not* a functor; there are counter-examples, see exercise ref:HERE

***Proof*:**
Let F^H be the collection of fixed elements by element $h \in H$. Then by definition, every element of k is fixed by H, so F^H is nonempty and $k \subseteq F^H$. Next, for any $a, b \in F^H$

$$h(a \pm b) = h(a) \pm h(b) = a \pm b \qquad h(ab) = h(a)h(b) = ab \qquad h(a^{-1}) = h(a)^{-1} = a^{-1}$$

that is, F^H is closed under the operations of F, and so $F^H \subseteq F$, showing that $k \subseteq F^H \subseteq F$.

For the other direction, let $\mathrm{Aut}_E(F)$ be the set of automorphisms that fix E. This subset is nonempty since $\mathrm{id}_F \in \mathrm{Aut}_E(F)$. If $\sigma \in \mathrm{Aut}_E(F)$, clearly $\sigma^{-1} \in \mathrm{Aut}_E(F)$. Finally, for any $\sigma, \tau \in \mathrm{Aut}_E(F)$, given $e \in E$,

$$\sigma \circ \tau^{-1}(e_1) = \sigma(\tau^{-1}(e)) = \sigma(e) = e$$

showing that $\sigma \circ \tau^{-1} \in \mathrm{Aut}_E(F)$, an so $\mathrm{Aut}_E(F) \leqslant \mathrm{Aut}_k(F)$, completing the proof.

Definition 20.1.7: Fixed Field

Let $H \subseteq \mathrm{Aut}_k(F)$. Then the subfield of F fixed by all the elements of H is called the *fixed field* of H, and is denoted F^H.

Example 20.1.8: Fixed Field

1. The fixed field of $\mathrm{Aut}_{\mathbb{Q}}(\mathbb{Q}(\sqrt{2}))$ is $\mathbb{Q}$. As we have seen in example 20.1.4.
2. The fixed field of $\mathrm{Aut}_{\mathbb{Q}}(\mathbb{Q}(\sqrt[3]{2}))$ is $\mathbb{Q}(\sqrt[3]{2})$, since $\mathrm{Aut}_{\mathbb{Q}}(\sqrt[3]{2}) = \{\mathrm{id}\}$. If we add the other roots of $x^3 - 2$, then the fixed field of $\mathrm{Aut}_{\mathbb{Q}}(\mathbb{Q}(\sqrt[3]{2}, \omega))$ is $\mathbb{Q}$.

Using this vocabulary, we can make the following correspondence betweeen F/k and its intermediate fields with $\mathrm{Aut}_k(F)$ and it's subgroups:

Proposition 20.1.9: Galois Correspondence

Let $k \subseteq F$ be a fields extension. Then there exists a correspondence:

$$\left\{\begin{matrix}\text{Intermediate fields } E \\ k \subseteq E \subseteq F\end{matrix}\right\} \underset{F^H}{\overset{\mathrm{Aut}_E(F)}{\rightleftarrows}} \left\{\begin{matrix}\text{subgroups } H \\ H \subseteq \mathrm{Aut}_k(F)\end{matrix}\right\}$$

sending an intermediate field E to the subgroup $\mathrm{Aut}_E(F)$, and a subgroup H of $\mathrm{Aut}_k(F)$ to the fixed field F^H. Furthermore, we have the following properies:

1. The correspondence is inclusion reversing, that is:
   - if $k \subseteq E_1 \subseteq E_2 \subseteq F$ are two subfield of F extending k , then $\mathrm{Aut}_{E_2}(F) \leqslant \mathrm{Aut}_{E_1}(F)$
   - if $H_1 \leqslant H_2 \leqslant \mathrm{Aut}_k(F)$ are two subgroups of automorphsism with associated fixed fields F^{H_1} and F^{H_2} respectively, then $F^{H_2} \subseteq F^{H_1}$.
2. For all subgroups $H \leqslant \mathrm{Aut}_k(F)$ and intermediate fields $k \subseteq E \subseteq F$
   - $E \subseteq F^{\mathrm{Aut}_E(F)}$
   - $H \subseteq \mathrm{Aut}_{F^H}(F)$
3. If E_1E_2 is the smallest subfield of F containing the two intermediate the E_1 and E_2, and $\langle G_1, G_2\rangle$ denotes the smallest subgroup of $\mathrm{Aut}_k(F)$ containing G_1 and G_2, then:

$$\mathrm{Aut}_{E_1E_2}(F) = \mathrm{Aut}_{E_1}(F) \cap \mathrm{Aut}_{E_2}(F)$$
$$F^{\langle G_1, G_2\rangle} = F^{G_1} \cap F^{G_2}$$

***Proof*:**
The Galois correspondence is an immediate consequence of proposition 20.1.6.

For the inclusion-reversal, consider the extension, $k \subseteq E_1 \subseteq E_2 \subseteq F$. Then $\mathrm{Aut}_{E_2}(F)$ *a fortiori* fixes all the elements of E_1, since $E_1 \subseteq E_2$, and so can only be *smaller* as the number of elements in Aut_{E_2} depend on the number of roots that can be premuted (as shown in the root permutation Criterion). A similar argument can be made for $H_1 \leqslant H_2 \leqslant \mathrm{Aut}_k(F)$.

For the subsets, Take the intermediate field E, $k \subseteq E \subseteq F$. Then $F^{\mathrm{Aut}_E(F)}$ is the set of elements That are fixed by the automorphisms that fix E. Then it is immediate that $E \subseteq F^{\mathrm{Aut}_E(F)}$. Similarly for $H \subseteq \mathrm{Aut}_{F^H}(F)$.

For the minimality statements, If E_1E_2 is the smallest of F containing both E_1 and E_2, then the $\subseteq$ inclusion is immediate (since $\mathrm{Aut}_{E_1}(F) \cap \mathrm{Aut}_{E_2}(F)$ contains only elements that fix E_1 and E_2), while the $\supseteq$ comes from the fact that $\mathrm{Aut}_{E_1}(F)$ and $\mathrm{Aut}_{E_2}(F)$ are in themselves the minimal subgroups of $\mathrm{Aut}_k(F)$ that fix all elements of E_1 and E_2. A similar argument can be made for the second assertion.

It is natural to ask whether the galois correspondence is bijective. Before showing when it is, we will show that it is possible that it is not:

Example 20.1.10: non-bijective Galois Correspondence

Consider $Q(\sqrt[3]{2})/\mathbb{Q}$. Then we know that $[\mathbb{Q}(\sqrt{3})/\mathbb{Q}] = 3$, which is prime, and so the only intermediate fields are $\mathbb{Q}$ and $\mathbb{Q}(\sqrt[3]{2})$. For $\mathrm{Aut}_{\mathbb{Q}}(\mathbb{Q}(\sqrt[3]{2}))$, recall that $\mathbb{Q}(\sqrt[3]{2}) \subseteq \mathbb{R}$, and that $\sqrt[3]{2}$ is the only root of the minimal polynomial within $\mathbb{R}$. Thus, since automorphisms of fields permute roots, $\mathrm{Aut}_{\mathbb{Q}}(\mathbb{Q}(\sqrt[3]{2}))$ can only have one element:

$$\mathrm{Aut}_{\mathbb{Q}}(\mathbb{Q}(\sqrt[3]{2})) = \{e\}$$

Hence, we have:

$$\{\mathbb{Q}, \mathbb{Q}(\sqrt[3]{2})\} \rightleftarrows \{\,\{e\}\,\} = \{\mathrm{Aut}_{\mathbb{Q}}(\mathbb{Q}(\sqrt[3]{2}))\}$$

showing that the function from intermediate fields to subgroups of the automorphism group is not injective with respect to intermediate fields in general[a] (for every intermediate field, we may map it to the same subgroup of $\mathrm{Aut}_k(F)$). In this case, $\mathbb{Q} \subseteq F^{\mathrm{Aut}_{\mathbb{Q}}(\mathbb{Q}(\sqrt[3]{2}))}$ and $\mathbb{Q}(\sqrt[3]{2}) \subseteq F^{\mathrm{Aut}_{\mathbb{Q}}(\mathbb{Q}(\sqrt[3]{2}))}$

[a] Or similarly, the correspondence is not surjective with respect to subgroups of $\mathrm{Aut}_k(F)$

This examples shows that it possible that $E \subsetneq F^{\mathrm{Aut}_E(F)}$. Fortunately, in the finite case, we always have that for any $H \leqslant \mathrm{Aut}_k(F)$, there exists an intermediate field (namely F^H) such that $H = \mathrm{Aut}_{F^H}(F)$ (and hence, the correspondence is at least always surjective with respect to intermediate fields). To prove this result we need the following lemma which shows that the field F^H is essentially as nice as a field as we want. The high-level idea is that automorphisms in $H = \mathrm{Aut}_E(F)$ fix E, and F^H is *defined* to be the field that is fixed by H:

Lemma 20.1.11: Properties Of Fixed Field

Let F/k be a finite extension, and let $H \leqslant \mathrm{Aut}_k(F)$. Then F/F^H is normal and separable. It is furthermore finite and simple.

Note that we are not given that F^H or k is a perfect field, and so separability is a powerful result. Similarly, we do not know beforehand that F is a normal extension of k, yet F is a normal extension of F/F^H. Note too that we can take $H = \{e\}$ so that $F/F^{\{e\}}$ has all the aforementioned properies. As we've seen in the previous example, it is not necessarily the case that $F^{\{e\}} = k$. (in the previous example, we had $F^{\{e\}} = \mathbb{Q}(\sqrt[3]{2})$)

***Proof*:**

First, F/F^H is finite since F/k is finite, so by the tower law:

$$[F:k] = [F:F^H][F^H:k]$$

Next, to show F/F^H is a separable extension, we'll show any $\alpha \in F$ is separable over F^H. To do so, we use the following clever trick: Let $\alpha \in F$ be any element of F (possibly an element of F^H). We'll construct the minimal polynomial of α, and show that it is separable. Take $\sigma \in H$. Since F is finite over k, α is the root of some minimal polynomial, $m_{\alpha,k}(x)$. By the fixed roots criterion, $h\alpha$ must be a root of the minimal polynomial $m_{\alpha,k}(x)$ of α over k (which might also imply $h\alpha = \alpha$ if no other root are in F). Since $m_{\alpha,k}(x)$ has only finitely many roots, the H-orbit of α (i.e. all possibles distinct images of α under all possible $\sigma \in H$) consists of finitely many elements $\alpha_1, ..., \alpha_n$. Now, the group H acts on the H-orbit by premuting its elements. Therefore, every element of H keeps the polynomial:

$$q_\alpha(x) = (x-\alpha_1)(x-\alpha_2)\cdots(x-\alpha_n)$$

fixed (where $\alpha_1 = \alpha$). In other words, the coefficients of $q_\alpha(x)$ *must* be in the fixed field F^H: $q_\alpha(x) \in F^H[x]$! Next, we show that $q_\alpha(x) = m_{\alpha,F^H}(x)$, which will show that the minimal polynomial corresponding to α is separable, showing α is separable. To see this, let $g(x)$ be any other polynomial having α as a root. Then:

$$g(\alpha) = b_m\alpha^m + \cdots + b_1\alpha + b_0 = 0$$

and by definition, any $\sigma \in H$ will map α to another root α_i, so:

$$g(\alpha_i) = b_m\alpha_i^m + \cdots + b_1\alpha_i + b_0 = 0$$

meaning $g(x)$ is divisible by $x - \alpha_i$ for every i, and hence $q_\alpha(x)\big|g(x)$. Since the minimal polynomial of α over F^H also has α as a root, we have that $q_\alpha(x)\big|m_{\alpha,f^H}(x)$, and as we've shown before $m_{\alpha,F^H}(x)\big|q_\alpha(x)$. Thus, $q_\alpha(x) = m_{\alpha,F^H}(x)$ up to a constant. Thus, α is separable and since α was arbitrary, F/F^H is a separable extension (Note that it's possible that $\alpha \in F^H$, in which case $q_\alpha(x) = x - \alpha$, which is certainly the minimal polynomial of α over F^H and is separable).

Next, since F/F^H is finite and separable, then by theorem 19.6.20, it simple. Let's say $F = F^H(\beta)$, so $F^H(\beta)/F^H$ is simple. Let $q_\beta(x)$ be define as we've done above (note that $q_\beta(x) = m_{\beta,F^H}(x)$ and that it is not possible that $q_\beta(x) = x - \beta$ unless $F = F^H$)

Finally, since $q_\beta(x) \in F^H[x]$ splits in F, and F is generated over F^H (namely, F is generated by F and β), F is the splitting field of $q_\alpha(x)$ over F^H, and so $F^H \subseteq F$ is a normal extension by theorem 19.5.13, completing the proof (notice again how finiteness was important for this part of the proof)

(checkout other proofs of this that do not require using the fact it's a simple extension, since the primitive roots theorem is sometimes proven using Galois Theory)

This proposition provides us with a couple more results: The polynomial $q_\alpha(x)$ (say of degree n) can be expanded to:

$$q_\alpha(x) = x^n - s_1x^{n-1} + s_2x^{n-1} - \cdots + (-1)^{n-1}s_{n-1}x + (-1)^n s_n$$

where each s_i are functions $s_i(\alpha_1, ..., \alpha_n)$ and represent:

$$s_1(\alpha_1, ..., \alpha_n) = \alpha_1 + \alpha_2 + \cdots + \alpha_n = \sum_i \alpha_i$$

$$s_2(\alpha_1, ..., \alpha_n) = \sum_{i<j} \alpha_i\alpha_j$$

$$s_3(\alpha_1, ..., \alpha_n) = \sum_{i<j<k} \alpha_i\alpha_j\alpha_k$$

$$\vdots$$

$$s_n(\alpha_1, ..., \alpha_n) = \alpha_1\alpha_2\cdots\alpha_n$$

Since $q_\alpha(x) \in F^H$, then each of these coefficients are in F^H. Furthermore, $\deg(q_\alpha(x)) \leq |H|$

Proposition 20.1.12: $H = \mathrm{Aut}_{F^H}(F)$

Let F/k be a *finite* extension, and let H be a subgroup of $\mathrm{Aut}_k(F)$. Then $|H| = [F : F^H]$ and

$$H = \mathrm{Aut}_{F^H}(F)$$

In particular, the Galois correspondence is surejctive for a finite field (for any subgroup $H \leqslant \mathrm{Aut}_k(F)$, there exists an intermediate field (namely F^H) such that $H = \mathrm{Aut}_{F^H}(F)$)

Intuitively, we want to show that $\mathrm{Aut}_{F^H}(F)$ *only* fixes F^H, and so $H = \mathrm{Aut}_{F^H}(F)$.

***Proof*:**
Let $H \leqslant \mathrm{Aut}_k(F)$. By proposition 20.1.9, we have that $H \leqslant \mathrm{Aut}_{F^H}(F)$, so $|H| \leq |\mathrm{Aut}_{F^H}(F)|$. Since $\mathrm{Aut}_{F^H}(F)$ is finite, verifying that $\geq$ holds will complete the proof.

To see this, recall that we've shown in lemma 20.1.11 that $F = F^H(\alpha)$ for some $\alpha \in F$. Thus, by corollary 19.2.29, $|\mathrm{Aut}_{F^H}(F)|$ equals the number of distinct roots of $m_{\alpha,F^H}(\alpha)$. Though we do not know the minimal polynomial $m_{\alpha,F^H}(x)$, we do know that α is a root of the polynomial $q_\alpha(x)$ defined in lemma 20.1.11. By construction, the number of roots of $q_\alpha(x)$ must be less than or equal $|H|$, giving us:

$$|\mathrm{Aut}_{F^H}(F)| \leq |H|$$

And since $|\mathrm{Aut}_{F^H}(F)| \geq |H|$, we get $|\mathrm{Aut}_{F^H}(F)| = |H|$ and since $H \subseteq \mathrm{Aut}_{F^H}(F)$, then $H = \mathrm{Aut}_{F^H}(F)$, giving the desired equality.

Finally, to show that $[F : F^H] = H$, since F/F^H is finite, separable, and normal, by corollary 19.6.21, $[F : F^H] = |\mathrm{Aut}_{F^H}(F)|$, completing the proof.

A particular consequence of this is that $\mathrm{Aut}_{F^{\mathrm{Aut}_E(F)}}(F) = \mathrm{Aut}_E(F)$, since the intermediate field that maps to $\mathrm{Aut}_{F^{\mathrm{Aut}_E(F)}}(F)$ is $F^{\mathrm{Aut}_E(F)}$, and so the group that it maps to is $\mathrm{Aut}_E(F)$.

Note that F/k being a *finite* extension was necesary for the proof. The Galois correspondence is in fact not necessarily surjective in the infinite case. This correspondence can be recovered given a suitable topology known as the *Krull topology* on $\mathrm{Aut}_k(F)$, which is used to limit the galois correspondence to *closed* subgroups of the automorphism group, which will regain surjectivity. More on this in appendix HERE (or another chapter?)

20.1.1 Galois Correspondence

We finally have reached a point where we can define a Galois extension. the following theorem is sometime also called the *Galois Extension Equivalences*, but it will be called by it's more famous name:

Theorem 20.1.13: Fundamental Theorem of Galois Theory I

Let F/k be a finite field extension. Then the following are equivalent:

1. F is the splitting field of a separable polynomial $f(x) \in k[x]$ over k
2. F/k is separable and normal
3. $|\mathrm{Aut}_k(F)| = [F : k]$
4. $k = F^{\mathrm{Aut}_k(F)}$ is the fixed field of $\mathrm{Aut}_k(F)$
5. F/k is separable, and if K/F is an algebraic extension and $\sigma \in \mathrm{Aut}_k(K)$, then $\sigma(F) = F$
6. The Galois correspondence for F/k is a bijection

***Proof*:**
Many of these have already been shown to be equivalent. We have shown:

- $1 \Leftrightarrow 2$ in theorem 19.5.13
- $2 \Leftrightarrow 3$ in corrollary 19.6.21
- $3 \Leftrightarrow 4$ follows from proposition 20.1.12, since $[F : F^{\mathrm{Aut}_k(F)}] = |\mathrm{Aut}_k(F)|$, and since $k \subseteq F^{\mathrm{Aut}_k(F)}$, we have that

$$k = F^{\mathrm{Aut}_k(F)} \text{ if and only if } |\mathrm{Aut}_k(F)| = [F : k]$$

- $2 \Leftrightarrow 5$ follows from the 3rd equivalence of theorem 19.5.13 along with the fact that finite separable extensions are simple (theorem 19.6.20)

Thus, we'll show $6 \Rightarrow 4$ and $1 \Rightarrow 6$, which will complete the chain of implications.

Starting with $6 \Rightarrow 4$, assume the galois correspondence is bijective, and consider $E = F^{\mathrm{Aut}_k(F)}$. By proposition 20.1.12, $\mathrm{Aut}_{F^{\mathrm{Aut}_k(F)}}(F) = \mathrm{Aut}_k(F)$, and so E and $F^{\mathrm{Aut}_k(F)}$ map to the same subgroup: $\mathrm{Aut}_k(F) = \mathrm{Aut}_E(F)$. Since the correlation is bijective, it must be that $k = E$, implying 4.

Next, we'll show $1 \Rightarrow 6$. Since F/k is a finite extension, by proposition 20.1.12, we already know the Galois correspondence is surjective (i.e., has a right-inverse), so it suffice to show it's injective, or equivalently that it has a left-inverse. To that end, we need to show that every intermediate field E equals the fixed field of the corresponding subgroup $\mathrm{Aut}_E(F)$, that is $E = F^{\mathrm{Aut}_E(F)}$. So, pick some E such that $k \subseteq E \subseteq F$. Since F is the splitting field for some separable polynomial $f(x) \in k[x] \subseteq E[x]$, we see that F is the splitting field of the separable polynomial $f(x)$ over E (i.e., condition (1) holds for F/E too). Since $1 \Leftrightarrow 4$, then $E = F^{\mathrm{Aut}_E}(F)$, thus $1 \Rightarrow 6$ completing the proof.

It can be helpful to have the following visual to keep track of where intermediate fields are with respect to their galois group:

$$\begin{aligned}\mathrm{Aut}_E(F) &= G \\ E &= F^G\end{aligned}$$

Definition 20.1.14: Galois Extension

Let F/k be a finite extension. Then F is called *Galois* or F/k is called a *Galois extension* if it satisfies any of the conditions in theorem 20.1.13.

Note that we have limited the definition to *finite galois extensions*, due to the requirement to bring in more tools to study the infinite equivalence. Nothing is lost in doing so, since infinite galois extensions will naturally generalize and improve upon the results we'll present. From now on, we will take galois extension to mean finite galois extension.

Corollary 20.1.15: Galois Extension Of Char 0

Let F/k be an extension of characteristic 0. Then F is galois over k if and only if F is finite and normal over k

Note that the finite condition was only added since we are focusing on finite Galois extensions.

***Proof*:**
If F is galois then it is *a priori* normal. Conversely, if F is finite and normal, then by corollary 19.6.8, any algebraic extension of F is separable, and hence by theorem 20.1.13, it is Galois.

Definition 20.1.16: Galois Group

Let F/k be a Galois extension. Then the corresponding automrophsim group $\mathrm{Aut}_k(F)$ is called the *galois group* of the extension. We will denote it by $\mathrm{Gal}_k(F)$ or $\mathrm{Gal}(F/k)$

The symbol $\mathrm{Gal}_k(F)$ is simply to emphasize we are working over a Galois extension, and doesn't reflect any additional structrue or data given to $\mathrm{Aut}_k(F)$. Furthermore, (word on $\mathrm{Gal}_k(F)$ acting on $k[x]$ or $F[x]$?)

Definition 20.1.17: Galois Group Of A Polynomial

Let $f(x)$ be a separable polynomial over $k[x]$. Then the *galois group* $\mathrm{Gal}_k(f(x))$ is the galois group of the splitting field of $f(x)$).

Definition 20.1.18: Abelian and Cyclic Extensions

Let F/k be a galois extension. Then F is called a *cyclic* (resp. *abelian*) extension if it's corresponding Galois group is cyclic (resp. abelian)
More generally, if the Galois group G has some property (ex. nilpotent, solvable, free), then the respetive extension is said to have that same property (ex. nilpotent extension, solvable extension, free extension)

Note that any vocabulary already used (like simple extension) will be prioritized over the group theory vocabulary (a simple extension is not related to simple group)

The equivalent conditions for Galois extensions gives us many details on results we have encountered earlier:

20.1.19 Remark

1. For any $H \leqslant \mathrm{Aut}_k(F)$, F/F^H is a Galois extension, and every intermediate field $k \subseteq E \subseteq F$ appears as a fixed field for a unique $H \leqslant \mathrm{Aut}_k(F)$ (If F/k is not Galois, this is not the case, like for $\mathbb{Q}(\sqrt[3]{2})/\mathbb{Q}$). This also means that by the 4th condition of Galois extension equivlences:
$$F^H = F^{\mathrm{Aut}_{F^H}(F)}$$
and since every intermediate field appears as a fixed field, we get that:
$$E = F^{\mathrm{Aut}_E(F)}$$

2. Let F/k be any finite extension. Then $|\mathrm{Aut}_k(F)|$ divides $[F:k]$. Taking any $G = \mathrm{Aut}_k(F)$, then
$$[F:k] = [F:F^G][F^G:k] \stackrel{!}{=} |\mathrm{Aut}_{F^{\mathrm{Aut}_k(F)}}(F)|[F^G:k] = |\mathrm{Aut}_k(F)|[F^G:k]$$
where $\stackrel{!}{=}$ comes from theorem 20.1.13(3). If F/k is Galois, then $F^G = k$, meaninig $|\mathrm{Aut}_k(F)|[F^G:k] = |\mathrm{Aut}_k(F)|[k:k] = |\mathrm{Aut}_k(F)|$.

3. Let F/k be a Galois extension. Then $|\mathrm{Aut}_k(F)|$ divides $|\mathrm{Aut}_{F^H}(F)|$. Taking any $H \leqslant \mathrm{Aut}_k(F)$ and $G = \mathrm{Aut}_k(F)$, then
$$|\mathrm{Aut}_k(F)| = [F:k] = [F:F^H][F^H:k] = |\mathrm{Aut}_{F^H}(F)|[F^H:k]$$
Switching up the notation, this means that $|G| = |H|[F^H:k]$, meaning $[F^H:k]$ is the index of H in G: $[F^H:k] = [G:H]$. The fact that it equals to the index means that it corresponds bijectively with the cosets $g\mathrm{Aut}_k(F^H)$ where $g \in \mathrm{Aut}_k(F)$, however it is yet to be shown how these two sets can correspond, only that they are the same size. In theroem 20.1.23, we'll see when this set of cosets forms a group.

4. It may be really difficult to figure out if we have found all intermediate fields of an infinite in general, namely since in general, there are infinitely many possible choices. However, it is relatively easy to find the subgroups of a finite group! Hence, using finite group theory allows us to bring in the powerful tools having that finiteness brings (including Sylow's theorems).

5. Note how the 5th condition is almost the condition for an extension being normal, where just also assumed that the extension is separable. Note that the 5th condition still works if the extension is not separable. This can motivate the idea that we can somhow make separability a superfluous condition given some appropriate generalization, and indeed we can. If we replace Galois Group with an appropriate *algebraic group*, then we can recover the bijective correspondence[a]

6. Given the base field is perfect, for any extension F/k, we can always extend it to its normal closure $N/F/k$, which is normal over k, and so is Galois. Hence, many field theory problems over $\mathbb{Q}$ can be translated "wlog" to Galois theory

[a]algebraic groups are schemes (with with a group structure), for which one of the motivations for their invention was to distinguish roots higher multiplicities at a point

If $E = F^H$, the last observation can be seen visually as:

$$\begin{array}{ccc} F & & \\ | & \} & {\scriptstyle |H|=[F:E]} \\ E & & \\ | & \} & {\scriptstyle |G:H|=[E:k]} \\ k & & \end{array}$$

Example 20.1.20: Galois Groups

1. Let $K = \mathbb{Q}(\sqrt{2})/\mathbb{Q}$. Then since $[K : \mathbb{Q}] = 2.$ and $|\mathrm{Aut}(K/\mathbb{Q})| = 2$ (as we've seen in example 20.1.4), then K is galois with galois group

$$\mathrm{Gal}(\mathbb{Q}(\sqrt{2})/\mathbb{Q}) = \{1, \sigma\} \cong \mathbb{Z}/2\mathbb{Z}$$

With the identity automorphism and

$$a + b\sqrt{2} \mapsto a - b\sqrt{2}$$

2. More generally, let K be any quadratic extension of F ($K = F(\sqrt{D})$). If $\mathrm{ch}(K) \neq 2$, then K is Galois.

3. The extension $\mathbb{Q}(\sqrt[3]{2})/\mathbb{Q}$ is *not* Galois, since $|\mathrm{Aut}(\sqrt[3]{2})| = 1 < 3 = [\mathbb{Q}(\sqrt[3]{2}) : \mathbb{Q}]$. Notice that $\mathbb{Q} \subseteq \mathbb{Q}(\sqrt[3]{2})$ is not a fixed field, since the only fixed field is $\mathbb{Q}(\sqrt[3]{2})$.

4. Let $K = \mathbb{Q}(\sqrt{2}, \sqrt{3})$ be an extension of $\mathbb{Q}$. Then since K is the splitting field of $(x^2-2)(x^2-3)$, it Galois. To determine its Galois group, we know from example (19.3.12) that $[K : \mathbb{Q}] = 4$, giving us the amount of automorphisms to look for. Since $\sqrt{2} \notin \mathbb{Q}(\sqrt{3})$, by the fixed roots criterion, we see that any automorphism $\sigma \in \mathrm{Aut}_{\mathbb{Q}}(K)$ must map $\sqrt{2}$ to $\pm\sqrt{2}$, and since $\sqrt{3} \notin \mathbb{Q}(\sqrt{2})$, we see that any automorphism $\tau \in \mathrm{Aut}_{\mathbb{Q}}(K)$ must map $\sqrt{3}$ map $\pm\sqrt{3}$.

   Going through defining a σ and a τ by permuting $\sqrt{2}$ and $\sqrt{3}$ we see that:

$$\mathrm{Gal}(K/\mathbb{Q}) = \{1, \sigma, \tau, \sigma\tau\} \cong V_4 \cong \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$$

Next, we can use proposition 20.1.9 to find the associated fixed field given the subgroups of $\mathrm{Gal}(K/\mathbb{Q})$. Using some arithmetic, we see that:

$$\begin{array}{cc} \{1\} & \mathbb{Q}(\sqrt{2},\sqrt{3}) \\ \{1,\sigma\} & \mathbb{Q}(\sqrt{3}) \\ \{1,\sigma\tau\} & \mathbb{Q}(\sqrt{6}) \\ \{1,\tau\} & \mathbb{Q}(\sqrt{2}) \\ \{1,\sigma,\tau,\sigma\tau\} & \mathbb{Q} \end{array}$$

which exhausts all possible intermediate fields of $\mathbb{Q}(\sqrt{2},\sqrt{3})/\mathbb{Q}$

5. Take the splitting field of x^4-2 (say we choose adjoints $\mathbb{Q}(i,\sqrt[4]{2})$). Since $\mathrm{ch}(\mathbb{Q})=0$, the polynomial must be separable, and hence $\mathbb{Q}(i,\sqrt[4]{2})$ must be Galois. Then we know that:

$$[\mathbb{Q}(i,\sqrt[4]{2}):\mathbb{Q}]=[\mathbb{Q}(i,\sqrt[4]{2}):\mathbb{Q}(\sqrt[4]{2})][\mathbb{Q}(\sqrt[4]{2}):\mathbb{Q}]=4\cdot 2=8$$

Telling us the Galois group has 8 elements. Since $i^n\sqrt[4]{2}$ are the 4 roots of x^4-2, we see that

$$\mathrm{Gal}_{\mathbb{Q}}(\mathbb{Q}(i,\sqrt[4]{2}))\leqslant S_4$$

Since $|\mathrm{Gal}_{\mathbb{Q}}(\mathbb{Q}(i,\sqrt[4]{2}))|=2\cdot 4$, $\mathrm{Gal}_{\mathbb{Q}}(\mathbb{Q}(i,\sqrt[4]{2}))$ has a Sylow 2-subgroup of S_4. By Sylow's 2nd theorem (theorem 4.8.5) this group then *must* be D_8.

This makes finding the automorphisms quite easily. Notice that the automorphism σ:

$$\begin{cases} \sqrt[4]{2}\longrightarrow i\sqrt[4]{2} \\ i\longrightarrow i \end{cases}$$

is cyclic of order 4, and the automorphism τ:

$$\begin{cases} \sqrt[4]{2}\longrightarrow \sqrt[4]{2} \\ i\longrightarrow -i \end{cases}$$

is cyclic of order 2, and that:

$$\sigma\tau=\tau^{-1}\sigma$$

Which gives us that $\mathrm{Gal}_{\mathbb{Q}}(\mathbb{Q}(i,\sqrt[4]{2}))\cong D_8$! We correspond the subgroups with the intermediate fields in example 20.1.22.

6. As we've seen with cyclotomic fields, by proposition 19.7.9,
$$\mathrm{Aut}_{\mathbb{Q}}(\mathbb{Q}(\zeta_n)) \cong (\mathbb{Z}/n\mathbb{Z})^\times$$
giving us a Galois group for every $n \in \mathbb{N}_{>0}$.

7. By theorem 19.7.10, all finite field are Galois, since they are the splitting field over the separable polynomial $x^{p^d} - x$. As we've seen in proposition 19.7.19, the galois group is cyclic of order d:
$$\mathrm{Aut}_{\mathbb{F}_p}(\mathbb{F}_{p^d}) \cong \mathbb{Z}/d\mathbb{Z}$$
and is generated by the frobenius k-automorphism $x \mapsto x^p$

8. (An example of a normal, but not separable extension, is $F_2(x^{1/2})/F_2(x)$ (since $x^2 - t$ is inseparable over $\mathbb{F}_2(x)$), and so it's not Galois)

The correspondence by theorem 20.1.13 is in fact even stronger: we can correspond the intermediate lattice to the subgroup lattice, and we can (Something about how the III gal fund thm feels like it's defininga group-action). The first extension of the theorem is to give the correspondence between the lattices of subgroups of the galois group and of the field extensions. To show two lattices corresond, we need to show that if A_1 and A_2 are two objects, then A_1A_2 (i.e. the smallest object containing both A_1 and A_2) corresponding to B_1B_2, or if the relation is contravariant, A_1A_2 corresponds to $B_1 \cap B_1$. This is what we show with subgroups and intermediate fields:

Theorem 20.1.21: Fundamental Theorem Of Galois Theory II

Let F/k be a Galois extension. Then the Galois correspondence is an inclusion-reversing isomorphism between the lattice of intermediate subfield of F/k with the lattice of subgroups of $\mathrm{Aut}_k(F)$.

In particular, If E_1, E_2 are intermediate fields and G_1, G_2 are the correspondong subgroups of $\mathrm{Aut}_k(F)$, then $E_1 \cap E_2$ coresponds to $\langle G_1, G_2 \rangle$ and E_1E_2 corresponds to $G_1 \cap G_2$

***Proof*:**
From theorem 20.1.13, we get the bijection between the intermediate fields and the subgroups of $\mathrm{Gal}_k(F)$, and from the 3rd property of the Galois Correspondence (proposition 20.1.9), we get that:
$$\mathrm{Aut}_{E_1E_2}(F) = G_1 \cap G_2 \qquad F^{\langle G_1, G_2 \rangle} = E_1 \cap E_2$$
completing the proof.

We usually write the lattice of fields with k on the bottom and F on top, while the lattice of groups usualy has $\{e\}$ on top and $\mathrm{Aut}_k(F)$ on bottom to more easily associate the two lattices:

$$\begin{tikzcd}
F \arrow[d, dash] & \{e\} \arrow[d, dash] \\
E \arrow[d, dash] & \mathrm{Aut}_E(F) \arrow[d, dash] \\
k & \mathrm{Aut}_k(F)
\end{tikzcd}$$

Some authors mark the Galois group though their own edges:

$$\begin{tikzcd}
F \arrow[d, dash, "\mathrm{Aut}_E(F)"'] \arrow[dd, dash, bend left, "\mathrm{Aut}_k(F)"] \\
E \arrow[d, dash] \\
k
\end{tikzcd}$$

Example 20.1.22: Lattice Correspondence

1. As we've shown in example (20.1.20)(3), where we corresponded the subgroups of $\mathrm{Gal}_{\mathbb{Q}}(\mathbb{Q}(\sqrt{2}, \sqrt{3}))$ with the intermediate fields $\mathbb{Q} \subseteq E \subseteq \mathbb{Q}(\sqrt{2}, \sqrt{3})$, we found that $\mathrm{Gal}_{\mathbb{Q}}(\mathbb{Q}(\sqrt{2}, \sqrt{3})) \cong \mathbb{Z}_2 \times \mathbb{Z}_2$ and intermediate fields for each subgroup. We can make this result more precise. If the subgroup lattice is:

$$\begin{tikzcd}
& \langle (0,0) \rangle \arrow[dl, dash] \arrow[d, dash] \arrow[dr, dash] & \\
\langle (1,0) \rangle \arrow[dr, dash] & \langle (1,1) \rangle \arrow[d, dash] & \langle (0,1) \rangle \arrow[dl, dash] \\
& \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z} &
\end{tikzcd}$$

then the corresponding intermediate field lattice is:

$$\begin{tikzcd}
& \mathbb{Q}(\sqrt{2}, \sqrt{3}) \arrow[dl, dash] \arrow[d, dash] \arrow[dr, dash] & \\
\mathbb{Q}(\sqrt{2}) \arrow[dr, dash] & \mathbb{Q}(\sqrt{6}) \arrow[d, dash] & \mathbb{Q}(\sqrt{3}) \arrow[dl, dash] \\
& \mathbb{Q} &
\end{tikzcd}$$

We can futhermore deduce that there are *no* further intermediate extensions $k \subseteq E \subseteq F$.

2. As we've shown in example (20.1.20)(4), $x^4 - 2$ has splitting field $\mathbb{Q}(i, \sqrt[4]{2})$ with Galois group

$\mathrm{Gal}_{\mathbb{Q}}(\mathbb{Q}(i, \sqrt[4]{2})) = D_8$. Then:

$$
\begin{tikzcd}
 & & D_8 = \langle a, b\rangle \arrow[dl, dash] \arrow[d, dash] \arrow[dr, dash] & & \\
 & \langle a^2, b\rangle \arrow[dl, dash] \arrow[d] \arrow[dr, dash] & \langle a\rangle \arrow[d, dash] & \langle a^2, ab\rangle \arrow[dl, dash] \arrow[d, dash] \arrow[dr, dash] & \\
\langle b\rangle \arrow[drr, dash] & \langle a^2 b\rangle \arrow[dr, dash] & \langle a^2\rangle \arrow[d, dash] & \langle ab\rangle \arrow[dl, dash] & \langle a^3 b\rangle \arrow[dll, dash] \\
 & & \{e\} & &
\end{tikzcd}
$$

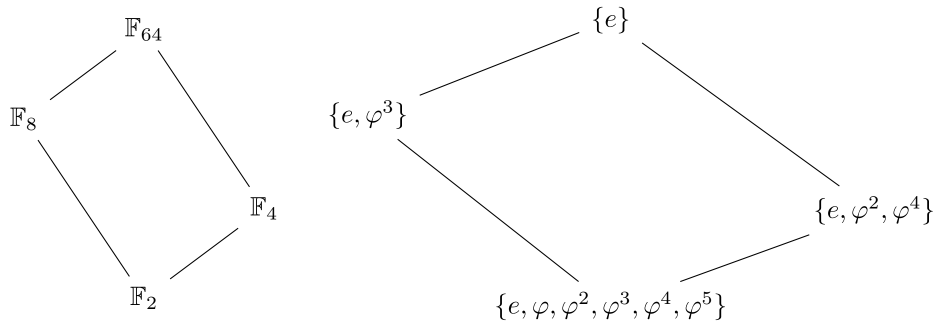
with corresponding intermediate field lattice:

$$
\begin{tikzcd}
 & & \mathbb{Q} \arrow[dl, dash] \arrow[d, dash] \arrow[dr, dash] & & \\
 & \mathbb{Q}(\sqrt[4]{4}) \arrow[dl, dash] \arrow[d] \arrow[dr, dash] & \mathbb{Q}(i) \arrow[d, dash] & \mathbb{Q}(\sqrt[4]{4}i) \arrow[dl, dash] \arrow[d, dash] \arrow[dr, dash] & \\
\mathbb{Q}(\sqrt[4]{2}) \arrow[drr, dash] & \mathbb{Q}(\sqrt[4]{2}i) \arrow[dr, dash] & \mathbb{Q}(i, \sqrt[4]{4}) \arrow[d, dash] & \mathbb{Q}(\sqrt[4]{2}(1+i)) \arrow[dl, dash] & \mathbb{Q}(\sqrt[4]{2}(1-i)) \arrow[dll, dash] \\
 & & \mathbb{Q}(i, \sqrt[4]{2}) & &
\end{tikzcd}
$$

3. Take $\mathbb{F}_{64}/\mathbb{F}_2$, which is a degree 6 extension with Galois group $\mathrm{Aut}_{\mathbb{F}_2}(\mathbb{F}_{64}) \cong \mathbb{Z}/6\mathbb{Z}$. the subgroup lattice is:

$$
\begin{tikzcd}
 & \{0\} \arrow[dl, dash] \arrow[ddr, dash] & \\
\{0, 3\} \arrow[ddr, dash] & & \\
 & & \{0, 2, 4\} \arrow[dl, dash] \\
 & \mathbb{Z}/6\mathbb{Z} &
\end{tikzcd}
$$

which corresponds to:



The next result comes from recalling that if $k \subseteq E \subseteq F$ is a series of field extensions where F/k is normal, this does not imply E/k is normal (as we have discussed in proposition 19.5.16 and example 19.5.17). Intuitively, this can be thought of as E having some but not all the required roots so that F is normal (ex. $\mathbb{Q} \subseteq \mathbb{Q}(\sqrt[3]{2}) \subseteq \mathbb{Q}(\sqrt[3]{2}, \varphi)$). However, it is sometimes the case that the intermediate fields are normal (ex. $Q \subseteq \mathbb{Q}(\sqrt{2}) \subseteq \mathbb{Q}(\sqrt{2}, \sqrt{3})$). We may ask when more generally is an extension E/k also normal, or more generally Galois? The following theorem answers that question:

Theorem 20.1.23: Fundamental Theorem Of Galois Theory III

Let F/k be a galois extension, and let E be an intermediate field. Then E/k is Galois if and only if $\text{Aut}_E(F)$ is *normal* in $\text{Aut}_k(F)$. If this is the case, then

$$\text{Aut}_k(E) = \text{Aut}_k(F)/\text{Aut}_E(F)$$

Hence, *normal* subgroups arise from *normal* extensions. In fact, the term "normal" originate in field theory, and the term normal subgroup was coined after it's field theory counterpart.

The proof essentially comes down to a couple of key observations about the cosets of $\text{Aut}_k(F)/\text{Aut}_E(F)$ and how we can surmise information about them through group actions! We have already observed in remark 20.1.19 that E/k has the same cardinality as $\{g\text{Aut}_E(F) \mid g \in \text{Aut}_k(F)\}$. We will make this correspondence more precise.

(We have seen in theorem 19.5.13 that E is normal if and only if $\iota(E) = E$ for all ι. We are building up the consequences of this I think)

Let F/K be a Galois extension. Identify $F \subseteq \overline{k}$ within some algebraic closure of $\overline{k}$. Let $k \subseteq E \subseteq F$ be some intermediate field, and consider:

$$I = \left\{ \iota : E \to \overline{k} \;\middle|\; \iota \text{ is a } k\text{-homomorphism } \right\} = \text{Hom}_k(E, \overline{k})$$

We can equivalently see E/k as an arbitrary extension of k where for some $\iota \in I$, $\iota(E) \subseteq F$.

Claim 1: for all $\iota \in I$, $\iota(E) \subseteq F$

Proof. Let ι be a $\iota \in I$ such that $\iota(E) \subseteq F$. Since E is separable (since $\iota(E) \subseteq F$ and F is separable) and finite, is is simple by theorem 19.6.20, Thus $E = k(\alpha)$ for some $\alpha \in \overline{k}$ and α is the root some minimal polynomial over k, $m_{\alpha,k}(x)$. Since $\iota(\alpha)$ determines ι, looking at where it maps is all that matters. Since $\iota(\alpha)$ is *some* root of the minimum polynomial $m_{\alpha,k}(x)$, and F is normal, F contains *all* roots of the minimal polynomial. But then, for any ι, $\iota(E) \subseteq F$, as we sought to show. □

Notice that

Claim 2: $\mathrm{Aut}_k(F)$ acts on I: for $g \in \mathrm{Aut}_k(F)$ and $\iota \in I$, $g \circ \iota \in I$

Proof. Since $\iota(E) \subseteq F$, and $g : F \to F$, we can intermret ι as $\iota : E \to F$ and compose the functions □

claim 3: $\mathrm{Aut}_k(F)$ is transitive on I (i.e. on $\mathrm{Hom}_k(E, \overline{k})$)

Proof. We need to show that for any $\iota_1, \iota_2 \in I$, $g \circ \iota_1 = \iota_2$. This follows from uniqueness of the splitting field. Since $\iota_1(E)$ and $\iota_2(E)$ each contain k and are each contained in F, and F is a splitting field over k for some polynomial, say $f(x)$ (since F is Galois, it follows from theorem 20.1.13). Hence, *a fortiori*, F must be the splitting field of $f(x)$ over $\iota_1(E)$ and over $\iota_2(E)$. Note that:

$$\iota_2 \circ \iota_1^{-1} : \iota_1(E) \to \iota_2(E)$$

is an k-isomorphism, and hence by the uniqueness of the splitting field, this isomorphism is extended to a k-isomorphism:

$$g : F \to F$$

By construction, $g|_{\iota_1(E)} = \iota_2 \circ \iota_1^{-1}$, or equivalently:

$$g \circ \iota_1 = \iota_2$$

as we sought to show. □

Thus, the galois group $\mathrm{Gal}_k(F)$ always acts transitively on $\mathrm{hom}_k(E, \overline{k})$ (do we know how it linkes corresponding embeddings more precisely? Is that the III thm?)

Now, choose an $\iota \in I$ and identify E with the intermediate field $k \subseteq \iota(E) \subseteq F$. Then $\mathrm{Aut}_E(F)[= \mathrm{Aut}_{\iota(E)}(F)]$ is a subgroup of $\mathrm{Aut}_k(F)$ of elements that restrict to the identity on $E[= \iota(E)]$. More particularly, $\mathrm{Aut}_E(F)$ consists of the *stabalizers of* I (i.e. for all $g \in \mathrm{Aut}_E(F)$, $g \circ \iota = \iota$). Thus, by the Orbit-Stabalizer Thoerem, we can deduce the following result:

Let $G = \mathrm{Aut}_k(F)$. Then consdering I as a G-set, I is isomorphic to the set of left-cosets of $\mathrm{Aut}_E(F)$ in $\mathrm{Aut}_k(F)$

20.1.24 key Isomorphism

$$\mathrm{hom}_k(E, \overline{k}) \quad \cong_G \quad \{g\mathrm{Aut}_E(F) \mid g \in \mathrm{Aut}_k(F)\}$$

(Aluffi says he views this statement as a key statement in Galois Theory, so it's worth pondering over it more. I believe it's becasue the bijection is not just a bijection, but also a G-isomorphism, and we see that through this result.)

***Proof*:**
of theorem 20.1.23

Let $G = \mathrm{Aut}_k(F)$ and $I = \{\iota : E \to \overline{k} \mid \iota \text{ is a } k\text{-homomorphism}\}$. Then as we've just seen, I is a G-set, and it's stabalizer is $\mathrm{Aut}_E(F)$. By proposition ref:HERE (I don't think I've proven this, it's proposition II.9.12 in Aluffi), if ι_1, ι_2 are both in the stabalizer, they are conjugates: there exists a $g \in \mathrm{Aut}_k(F)$ such that $g \circ \iota_1 \circ g^{-1} = \iota_2$, in particular, ι and id are conjugates for any $\iota \in I$. Thus, if $\mathrm{Aut}_E(F)$ is normal:

$$\mathrm{Aut}_{\iota(E)}(F) = \mathrm{Aut}_E(F)$$

which, by the bijectivity of the Galois correspondence for F/k, we have that $\iota(E) = E$. But then E/k satisfies the 5th equivalence of Galois Extensions in theorem 20.1.13, meaning E/k is Galois

Conversely, if E/k is galois, then $\iota(E) = E$ for all ι (again by the 5th equivalence of Galois extensions in theorem 20.1.13). Define the homomorphism ρ which takes $g \in \mathrm{Aut}_k(F)$ and restricts it to $E[= \iota(E)]$:

$$\rho : \mathrm{Aut}_k(F) \to \mathrm{Aut}_k(E)$$

This homomorphism is surjective (since the G-action acts transitively on I), and $\ker(\rho)$ is all automorphism that restrict to the identity on E, that is, $\mathrm{Aut}_E(F)$. But then, we know that the kernel of a group homomorphism is normal, and by the 1st isomorphism theorem for groups:

$$\mathrm{Aut}_k(F)/\mathrm{Aut}_E(F) \cong \mathrm{Aut}_k(E)$$

completing the proof.

("It sends subgroup to subgroups by conjugation", i.e. if E is associatd to $H = \mathrm{Aut}_E(F)$, then for any $\sigma \in \mathrm{Aut}_k(F)$, $\sigma(E)$ is associated to $\mathrm{Aut}_{\sigma(E)}(F) = \sigma H \sigma^{-1}$)

(There is another really powerful result. Let's say $F/E/k$ is a sequence of extensions, F/k and E/k both Galois (and hence F/E is Galois. Then the map

$$\mathrm{Gal}_E(F) \times \mathrm{Gal}_k(E) \to \mathrm{Gal}_k(F) \qquad (\sigma, \tau) \mapsto \sigma \circ \tau$$

is is well defined and bijective (namely, since the identity must map to the identity, and since E/k is normal $\mathrm{Gal}_k(F)$ is determined by what happens to E, and then fixing E, what happens to F. Thus, constructing Galois groups is much easier in this case!!! Note, however, that this is a homomorphism if and only if $E \cap F = k$. Once I read through Dummit's and foote on Composite fields, this will be incorporated))

Corollary 20.1.25: Isomorphism Of Categories

Let $\mathbf{Fld}_k^E$ be the collection of fields F such that $k \subseteq F \subseteq E$ whose morphisms are k-homomorphisms and $\mathcal{O}_G$ where $G = \mathrm{Aut}_k(E)$ be the category whose objects are left G-sets G/H for every $H \subseteq G$ and the morphisms are G-equivariant maps $G/H \to G/K$. Then as categories, $\mathcal{O}_G^{\mathrm{op}} \cong \mathbf{Flb}_k^E$

***Proof*:**
see p.21 of Category theroy in Context if you need some help

(more in dummit and foote on Galois Composite fields! Really cool!)

Example 20.1.26: Application To Lattices
(Draw two latties, and show the normal extensions)

20.1.27 Foreshadowing Can give the connection between Galois group/extensions to Algebraic geometry with regular covers and deck transformations

Exercise 20.1.1

1. Let $L = \mathbb{Q}(\zeta_p)$ where p is an odd prime. Show there exists a subfield $K \subseteq L$ such that $K = \mathbb{Q}(\sqrt{p'})$ where
$$p' = \begin{cases} p & p \equiv 1 \bmod 4 \\ -p & p \equiv 2, 3 \bmod 4 \end{cases}$$
2. Let $K/\mathbb{Q}$ be an odd Galois Extension, Show that L must be totally real.

20.1.2 Composite Fields

We now take what we proved earlier and do two things with it: we extend the results for composite fields and we classify all cyclic Galois Extensions (given an appropriate restriction on the base field) In the previous chapters, there has consistenly been a question working with composite fields (exercise ref:HERE), in particular, it was asked to be shown that:

> If E/k and F/k are field extensions, and E/k is (finite/algebraic/normal/separable resp.), then EF/F is (finite/algebraic/normal/separable resp). If E/k and F/k are (finite/algebraic/normal/separable resp.) then EF/k is (finite/algebraic/normal/separable resp.)

Since [finite] Galois extensions are finite, normal, and separable, we see immediately that if E/k is Galois, then EF/F is Galois, and if E/k and F/k are Galois, so is EF/k. The natural question to ask is what's the galois group of a composite field and of EF/F:

Proposition 20.1.28: Galois Composite Fields I

Suppose E/k is Galois and F/k is any field extension. Then EF/F is Galois, and $\text{Aut}_F(EF) \cong \text{Aut}_{E\cap F}(E)$. Visually, we see that:

$$\begin{tikzcd}
& KF \arrow[dl, dash] \arrow[dr, dash, "G"] & \\
F \arrow[dr, dash, "G"'] & & K \arrow[dl, dash] \\
& F \cap K \arrow[d] & \\
& k &
\end{tikzcd}$$

The reader should ponder why does F/k need to be finite for this theorem?

***Proof*:**
Using the Galois extension equivalences, we can quickly prove that EF/k is Galois, namely since E/k is Galois, it is the splitting field of a separable polynomial $p(x) \in k[x] \in F[x]$. The roots of $p(x)$ generate E over k ,and so they generate EF over F, i.e., EF is the splitting field of $p(x)$ over F, thus EF/F is a Galois Extension

To show that $\text{Aut}_F(EF) \cong \text{Aut}_{E\cap F}(E)$, we'll first show we can embedd $\text{Aut}_F(EF)$ into $\text{Aut}_k(E)$, and then find that the subgroup of $\text{Aut}_k(E)$ associated to this embedding is $\text{Aut}_{E\cap F}(E)$. For the embedding, consider first any F-automorphism $\sigma : EF \to EF$ (and hence k-automorphism). Then we can restrict σ to E, giving us $\sigma : E \to \sigma(E)$. Since E is Galois, $\sigma(E) = E$. Thus, any element of $\text{Aut}_F(EF)$ get's mapped to an element of $\text{Aut}_k(E)$, that is, we have a natural group homomorphism:

$$\rho : \text{Aut}_F(EF) \to \text{Aut}_k(E)$$

This homomorphism is injective: If the only element that maps to the identity is the identity, then we're done. Indeed, if $\sigma \in \text{Aut}_k(E)$ is the identity on E, since any element $\text{Aut}_F(EF)$ is the identity on F, the coresponding σ in $\text{Aut}_F(EF)$ must be the identity on EF, showing injectivity.

Thus, $\text{Aut}_F(EF)$ embeds into $\text{Aut}_k(E)$, and so it is associated with some $H \leqslant \text{Aut}_k(E)$. By proposition 20.1.12, $H = \text{Aut}_{E^H}(E)$. It suffices to show that $E^H = E \cap F$. For the $\supseteq$ direction, since every $\sigma \in \text{Aut}_F(EF)$ fixes F, the restriction $\rho(\sigma)$ fixes $E \cap F$, hence $E^H \supseteq E \cap F$. For the $\subseteq$ direction, take $\alpha \in E^H$. Then $\alpha \in E$. By definition, $F(\alpha)$ is in the fixed field of $\text{Aut}_F(EF)$, which is F sinc F is Galois over EF/F (1st Galois Theorem (20.1.13)(4)). Thus, $\alpha \in F$, and so $E^H \subseteq E \cap F$. Therefore, $H = E \cap F$, and:

$$\text{Aut}_F(EF) \cong \text{Aut}_{E\cap F}(E)$$

as we sought to show

Example 20.1.29: Composite Fields Determining Abelian Extensions

1. As we have seen in section 19.7.1, if ζ_n is the nth root of unity, $\mathbb{Q}(\zeta_n)$ is the splitting field of $x^n - 1$ over $\mathbb{Q}$, and hence is Galois since $\mathrm{ch}(\mathbb{Q}) = 0$. We have shown in proposition ref:HERE that $\mathrm{Aut}_{\mathbb{Q}}(\mathbb{Q}(\zeta_n)) \cong (\mathbb{Z}/n\mathbb{Z})^{\times}$.

   Now, let k be any field of characteristic 0. Then the splitting field K/k for $x^n - 1$ is the composite of k and $\mathbb{Q}(\zeta)$, namely $K = k(\zeta)$. Since $\mathbb{Q}/(\zeta)/\mathbb{Q}$ is Galois and $k/\mathbb{Q}$ is a field extension, by proposition 20.1.28, we get that:

   $$\mathrm{Aut}_k(k(\zeta)) \cong \mathrm{Aut}_{\mathbb{Q}(\zeta)\cap k}(\mathbb{Q}(\zeta)) \leqslant \mathrm{Aut}_{\mathbb{Q}}(\mathbb{Q}(\zeta)) \cong (\mathbb{Z}/n\mathbb{Z})^{\times}$$

   showing that $\mathrm{Aut}_k(k(\zeta))$ is some subgroup of $(\mathbb{Z}/n\mathbb{Z})^{\times}$. in particular, *any* such extension *must* be abelian.

   The fact that every extension of the form $k(\zeta)$ is abelian is quite powerful. In fact, they are relatively easy to classify. The study of Galois extensions that are abelian(which also must contain n distinct roots of unity for some $n \in \mathbb{N}$) is called *Kummer Theory*. We touch a bit of Kummer theory in the following, since we need it for showing no quintic is unsolvable, and will later show that all finite abelian group show up as a field extension of $\mathbb{Q}$ in section 20.4.

2. Generalizing the results from the exercises, we know that if E/k and F/k are Galois, so is EF/k. Can you find a way to link $\mathrm{Gal}_k(EF)$ with $\mathrm{Gal}_k(E)$ and $\mathrm{Gal}_k(F)$? (Hint: Consider $\mathrm{Gal}_k(EF) \to \mathrm{Gal}_k(E) \times \mathrm{Gal}_k(F)$, $\sigma \mapsto (\sigma|_E, \sigma|_F))$

Corollary 20.1.30: Order of Galois Composite

Let E/k be a galois extension and F/k be any finite extension. Then:

$$[EF : k] = \frac{[E : k][F : k]}{[E \cap F : k]}$$

***Proof*:**
exercise

Note that this formula does not hold in general if niether of the extensions is Galois.

Example 20.1.31: Failure of Formula with no Galois Extension

Let $k = \mathbb{Q}$, $E = \mathbb{Q}(\sqrt[3]{2})$, and $F = \mathbb{Q}(\zeta_3\sqrt[3]{2}))$, since $EF = \mathbb{Q}(\zeta_3\sqrt[3]{2}), \sqrt{3}$ (with minimal polynomial $x^3 - 2$, and hence degree 6 extension) and:

$$6 = [\mathbb{Q}(\zeta_3\sqrt[3]{2})] \neq \frac{[\mathbb{Q}(\sqrt[3]{2}) : \mathbb{Q}][\mathbb{Q}(\zeta_3\sqrt[3]{2}) : \mathbb{Q}]}{[\mathbb{Q} : \mathbb{Q}]} = 9$$

Using this, we may answer the arguably more intesesting questions of the composite and intersection of fields being galois over the same base field:

Proposition 20.1.32: Galois Composite Fields II

Let $E/k, F/k$ be galois extensions of F. Then:

1. $E \cap F$ is galois over k
2. EF is galois over k. It's galois group is isomorphic to:
$$G = \{(\sigma, \tau) \mid \sigma|_{E\cap F} = \tau|_{E\cap F}\} \leqslant \mathrm{Gal}_k(E) \times \mathrm{Gal}_k(F)$$
3. If $E \cap F = k$, then:
$$\mathrm{Gal}_k(EF) \cong \mathrm{Gal}_k(E) \times \mathrm{Gal}_k(F)$$

***Proof*:**
D&F p.593

Corollary 20.1.33: Galois Closure

Let F/k be a finite separable extension. Then there exists a minimal finite galois extension E/k such that $E/F/k$.

***Proof*:**
Simply take the composite of the splitting fields of the minimal polynomials for the basis of F over k (which is separable since F/k is separable), nad intersect all the galois extensions of k containing F.

Definition 20.1.34: Galois Closure

Let F/K be a finite separable extension. Then the smallest extension $E/F/k$ such that E/k is galois is called the *galois closure* of F over k.

Exercise 20.1.2

1. Show there exists fields E, F such that $[EF : F] \neq [F : E \cap F]$

20.1.3 A bit of Character Theory and Kummer Theory

Let F/K be a galois extension. Then each $\sigma \in \mathrm{Gal}_K(F)$ is also a K-linear isomorphism $F \to F$ of n-dimensional K-vector spaces. Thus, does σ have eigenvalues? Can it be diagonalized? How does each $\sigma \in \mathrm{Gal}_K(F)$ relate to each other? To start:

Lemma 20.1.35: Eigenvalues Are Roots of Unity

Let F/K be a galois extension and $\sigma \in \mathrm{Gal}_K(F)$. Then if σ has an eigenvalue λ, then λ is a root of unity

***Proof*:**
Let σ be as in the statement and assume $\sigma(\alpha) = \lambda\alpha$ where $\lambda \in K$. Then as $\text{Gal}_K(F)$ is a finite group, the exists an n such that $\sigma^n = \text{id}$. Thus:

$$\alpha = \text{id}(\alpha) = \sigma^n(\alpha) = \lambda^n \alpha$$

Then $\lambda^n = 1$, hence λ is a root of unity.

Thus, σ is diagonalizable if K contains enough roots of unity; such an extension is called a *Kummer extension* (see definition 20.4.2, the definition is delayed due to the need to consider the behaviour over different characteristic which is covered in section 20.4).

Next, we would like to know how each automorphism relates to each other. We shall use a concept from Character theory to frame the discussion:

Definition 20.1.36: Group Character

Let G be a group and L a field. Then a *[linear] character* (denoted χ) is a homomorphism from G to the multiplicative group of L:

$$\chi : G \to L^\times$$

Example 20.1.37: Group Character

For our purposes, any field homomorphism $\sigma : F \to K$ induces a character $\chi_\sigma : F^\times \to K^\times$ (since σ maps the multiplicative group of F into the multiplicative group of K). Note that χ_σ captures essentially all the mapping information of σ, since χ only omits mapping 0 from F to 0 from K, but we can simply deduce that. Thus, if we are working with χ_σ, we are showing that we do not want to remember that the additive structure is preserved.

Definition 20.1.38: Character Linearly Independent

The characters $\chi_1, ..., \chi_n$ of G are *linearly independent* over L if they are linearly independent as functions on G, i.e. for any $a_i \in L$

$$a_1\chi_1 + a_2\chi_2 + \cdots + a_n\chi_n = 0 \quad \text{if and only if} \quad a_i = 0, 1 \leq i \leq n$$

Note that the expression on the left hand side of the above equality is a large function, and hence the it means that characters are linearly independent if for all $g \in G$:

$$a_1\chi_1(g) + a_2\chi_2(g) + \cdots + a_n\chi_n(g) = 0 \quad \text{if and only if} \quad a_i = 0, 1 \leq i \leq n$$

Theorem 20.1.39: Artin's Lemma

Let $\chi_1, ..., \chi_n$ be distinct characters of G with values in L. Then they are linearly independent over L

***Proof*:**
Suppose that the characters are linearly dependent, that is, there exists (possibly many) sets of coefficients coefficients $a_1, ..., a_n \in L$ where not all are zero and

$$a_1\chi_1 + a_2\chi_2 + \cdots + a_n\chi_n = 0$$

To construct a minimality argument, choose a set of not-all-zero coefficients of minimal cardinally that satisfies the above equation (such coefficients exists since these character are linearly dependent). Thus, for any $g \in G$:

$$a_1\chi_1(g) + a_2\chi_2(g) + \cdots + a_n\chi(g) = 0 \tag{20.2}$$

Let g_0 be an element of G such that $\chi_1(g_0) \neq \chi_n(g_0)$ (which exists since $\chi_1 \neq \chi_n$). Since equation (20.2) holds for every element of G, we have that:

$$a_1\chi_1(g_0g) + a_2\chi_2(g_0g) + \cdots + a_n\chi_n(g_0g) = 0 \tag{20.3}$$

i.e.

$$a_1\chi_1(g_0)\chi_1(g) + a_2\chi_2(g_0)\chi_2(g) + \cdots + a_n\chi_n(g_0)\chi_n(g) = 0 \tag{20.4}$$

Now, multiplying equation (20.2) by $\chi_n(g_0)$ and subtracting by equation (20.4), we get:

$$[\chi_n(g_0) - \chi_1(g_0)]a_1\chi_1(g) + \cdots + [\chi_n(g_0) - \chi_{n-1}(g_0)]a_{n-1}\chi_{n-1}(g) = 0$$

which holds for all $g \in G$. However, the first coefficient is nonzero, and this is a relation with fewer nonzero coefficients than the minimal choice we have made earlier – a contradiction.

Corollary 20.1.40: linear independence of Automorphisms (Dedekind's Lemma)

Let $\sigma_1, \sigma_2, ..., \sigma_n$ be distinct embeddings of K into L. Then they are linearly independent as functions on K. In particular, distinct automorphism of a field K are linearly independent as functions on K.

***Proof*:**
This is an immediate consequence of theorem 20.1.39

Proposition 20.1.41: Galois And Cyclic Extensions

Let F/k be an extension of degree m. Assume that k contains a primitive mth root of 1 and $\mathrm{char}(k)$ does not divide m. Then F/k is Galois and cyclic if and only if $F = k(\delta)$ with $\delta^m \in k$.

***Proof*:**
Let $\zeta \in k$ be a primitive m-th root of 1.

($\Leftarrow$) Let's say $F = k(\delta)$ with $\delta^m = c \in k$. Then all m roots of the polynomial $x^m - c$ are in F, in particular,

$$c, \zeta\delta, \zeta^2\delta, ..., \zeta^{m-1}\delta \in F$$

Furthermore, F must be generated by them. Hence F is the splitting field of the separable polynomial $x^m - c$ in k (separable since $\mathrm{ch}(k)$ does not divide m). Thus, F is Galois

Next, to see that $\mathrm{Aut}_k(F)$ is cyclic, note that for any $\varphi \in \mathrm{Aut}_k(F)$, it is determined by $\varphi(\delta)$, which his necessarily a root of $x^m - c$, and so $\varphi(\delta) = \zeta^i \delta$ for some i determined up to a multiple of m. Then this can be used to define an isomorphism of $\mathrm{Aut}_k(F)$ with $\mathbb{Z}/m\mathbb{Z}$.

($\Rightarrow$) Conversely, assume that F/k is Galois and $\mathrm{Aut}_k(F)$ is cyclic (say, generated by φ). Since the automorphisms φ^i $(1 \leq i \leq m-1)$ are pairwise distinct, by corollary 20.1.40, they are linearly independent. Therefore, there exists an $\alpha \in F$ such that

$$\delta := \alpha + \zeta^{-1}\varphi(\alpha) + \cdots + \zeta^{-(m-2)}\varphi^{m-2}(\alpha) + \zeta^{-(m-1)}\varphi^{m-1}(\alpha) \neq 0$$

Then note that :

$$\varphi(\delta) := \varphi(\alpha) + \zeta^{-1}\varphi^2(\alpha) + \cdots + \zeta^{-(m-2)}\varphi^{m-1}(\alpha) + \zeta^{-(m-1)}\alpha = \zeta\delta$$

since $\zeta \in k$ so $\varphi(\zeta) = \zeta$. Then δ is *not* fixed by any nonidentity element of $\mathrm{Aut}_k(F)$, that is $\mathrm{Aut}_{k(\delta)}(F) = \{e\}$, so $k(\delta) = F$. Furthermore:

$$\varphi(\delta^m) = \varphi(\delta)^m = \zeta^m\delta^m = \delta^m$$

so, δ^m is fixed by φ, and hence by every element of $\mathrm{Aut}_k(F)$. Since F/k is Galois, it must be that $\delta^m \in k$, completing the proof.

20.1.42 Kummer Theory This proposition shall have an alternative proof (theorem 20.4.6) in section 20.4.

Proposition 20.1.43: Kummer Extension and Eigenvalue

Let $F = K(\sqrt[n]{a})$ and let K have enough roots of unity and $\mathrm{char}(K) \nmid n$. Let σ be a generator for $\mathrm{Gal}_K(F)$. Then $\sqrt[n]{a}$ is the eigenvector of σ.

***Proof*:**
exercise.

Exercise 20.1.3

1. Let $k(t)/k$ be an infinite extension. Determine $\mathrm{Aut}_k(k(t))$
2. Show that for all n, there is an extension F of $\mathbb{Q}$ such that $F/\mathbb{Q}$ is not Galois. Similarly, show that for all n, there exists an extension L such that $\mathrm{Gal}_{\mathbb{Q}}(L) \cong \mathbb{Z}/n\mathbb{Z}$ (Hint: show that for all n, there exists a p such that $p \equiv 1 \bmod n$)

20.2 Application of Galois Theory

20.2.1 Fundamental Theorem of Algebra

(Some build-up with properties of $\mathbb{C}$ that are given a exercsie 5.23)

Theorem 20.2.1: Fundamental Theorem of Algebra

$\mathbb{C}$ is Algebraically closed

***Proof*:**
Take $f(x) \in \mathbb{C}[x]$ to be a nonconstant polynomial. If $f(x)$ is irreducible, then since conjugation is a field-homomorphism, $f(x) \to \overline{f(x)}$, then $f(x)\overline{f(x)}$ is also irreducible. Since $f(x)\overline{f(x)} \in \mathbb{R}[x]$, we can assume without loss of generality that all coefficents are real. Let F be the splitting field of $f(x)$ over $\mathbb{R}$, and embed $F \subseteq \overline{\mathbb{R}}$ (we want to show that $\overline{\mathbb{R}} = \mathbb{C}$).

Now, consider the extension:

$$\mathbb{R} \subseteq F(i)$$

this extension is Galois: it is the splitting field of the square-free part of $f(x)(x^2+1)$ (any irreducible polynomial over $\mathbb{R}$ of degree ≥ 2 will have such a square free part, as should be verified TBD). Let $G = \mathrm{Aut}_{\mathbb{R}}(F(i))$ By Sylow's Theorem, G has a 2-sylow subgroup H (even if $2 \nmid |G|$, since then it would be $\{e\}$). Then the index $[G : H]$ is oddand so by the Galois correspondence, it corresponds to a finite separable extension $\mathbb{R} \subseteq E$ with $[E : \mathbb{R}]$ odd. Since every polynomial of odd degree has a root by elementary calculus [a], it must be that $\mathbb{R} \subseteq E$ is *trivial* (since TBD, was exercsie VII.5.23 in Aluffi which stated that if $k[x]$ had no irreducible polynomials of degree $n > 0$, then it has no separable extensions). Thus, $[G : H] = 1$, proving that G is a 2-group. Since $\mathbb{C} = \mathbb{R}(i) \subseteq F(i)$, the Galois group ofthe Galois extension $|C \subseteq F(i)$ is a subgroup of G, and therefore it is a 2-group: $|\mathrm{Aut}_{\mathbb{C}}(F(i))| = 2^n$ for some $n \geq 0$.

To finish the proof, we must show that $n = 0$. By proposition ref:HERE, for any $1 \leq m \leq n$, a group of order p^n contains a subgroup of order p^m. For this case, we have that $\mathrm{Aut}_{\mathbb{C}}(F(i))$ contains a subgroup of order 2^{n-1}. However, by exercise 5.23 again, $\mathbb{C}$ contains *no* quadratic exensions of $\mathbb{C}$, since there are no irreducible polynomials in $\mathbb{C}[x]$.

Thus, $n = 0$, so $\mathrm{Aut}_{\mathbb{C}}(F(i))$ is trivial, and so $F(i) = \mathbb{C}$, in particular $\mathbb{C}$ contains the roots of $f(x)$.

[a] Namely, the intermediate value theorem

This proof can also be accomplished without Sylow's theorem's by HERE. However, since the tools were developped, it is good practice to use them.

Corollary 20.2.2: No Field Structure On $\mathbb{R}^n$, $n \geq 3$

There does not exist a field structure on $\mathbb{R}^n$ ($n \geq 3$) such that $\mathbb{R}$ includes into $\mathbb{R}^n$ (i.e. $\mathbb{R}^n$ is a field extension of $\mathbb{R}$)

***Proof*:**
For the sake of contradiction, assume $\mathbb{R} \subseteq \mathbb{R}^n$ is a field extension for $n \geq 3$. Since $\mathbb{R}^n$ is an n-dimensional vector-space, it is a degree n algebraic extension. Hence, by lemma 19.5.7, there is an (injective) homomorphism $\mathbb{R}^n \to \mathbb{C}$ fixing $\mathbb{R}$. However, the dimension of the right hand side is greater than 2, and hence no such homomorphism exists, a contradiction.

Note that we had to be careful in how we state the above corollary. It is not true that there is no field structure on $\mathbb{R}^n$ ($n \geq 3$). Since $\mathbb{R}^n$ and $\mathbb{R}$ have the same cardinality, we may (by the axiom of choice) find a bijection from $\mathbb{R}^n$ to $\mathbb{R}$. Then we may put a field structure on $\mathbb{R}$ through the bijection. However, the field structure on $\mathbb{R}^n$ would certainly not have $\mathbb{R}$ as a subfield, and it impossible to determine the product of any two elements[8].

Theorem 20.2.3: Frobenius Theorem

There are only 3 finite-dimensional associative division algebras, namely

1. $\mathbb{R}$, the real numbers
2. $\mathbb{C}$, the complex numbers
3. $\mathbb{H}$, the Quaternions

If we drop associativity, there is furthermore:

4. $\mathbb{O}$, the Octonions

***Proof*:**
This is beyond this book; for a proof see for example [43, Bott Periodicity]. A high level summary is the existence of a division algebras on $\mathbb{R}^n$ is tied to whether or not the tangent bundle S^{n-1} is parallelizable (which as it turns out only TS^0, TS^1, TS^3, TS^7 are parallelizable). This result is dependent on the classical result in algebraic topology known as Bott Periodicity.

As an exercise, show that $\mathbb{H}$ is not a $\mathbb{C}$-algebra. The quaternions are also the Clifford algebra of the negative definite form $\langle -1, -1 \rangle$ on $\mathbb{R}^2$, which is example 22.7.6 in section 22.7.

20.2.2 Construction of Regular n-gon

20.2.3 Symmetric Functions

If $t_1, ..., t_n$ are indeterminates, then the polynomial:

$$P_n(x) := (x - t_1)(x - t_2) \cdots (x - t_n) \in \mathbb{Z}[t_1, ..., t_n][x]$$

is universal in the sense thati every polynomial of degree n with coefficeints in (say) an intergral domain may be obtained from $P_n(x)$ given a suitable choice for $t_1, ..., t_n$, choices form a possibly larger ring. Expanding this polynomial, we get that:

$$P_n(x) = x^n - s_1(t_1, ..., t_n)x^{n-1} + \cdots + (-1)^n s_n(t_1, ..., t_n)$$

[8]loosely speaking, if we would be able to explicitly find a bijection from $\mathbb{R}^n$ to $\mathbb{R}$, but that would be a "countable process", but $\mathbb{R}^n$ and $\mathbb{R}$ is uncountable

where each s_i are the *elementary functions.* If $n = 3$, then:

$$\begin{aligned} s_1(t_1, t_2, t_3) &= t_1 + t_2 + t_3 \\ s_2(t_1, t_2, t_2) &= t_1t_2 + t_1t_3 + t_2t_3 \\ s_3(t_1, t_2, t_3) &= t_1t_2t_3 \end{aligned}$$

The function s_n are called symmetric since they are invariant under permutation (essentially, since $P_n(x)$ is invariant under permutation). The name "elementary" comes because of the following theorem

Theorem 20.2.4: Fundamental Theorem On Symmetric Functions

Let K be a field and let $\varphi \in K(t_1, ..., t_n)$. Then φ is invariant under permutation (i.e. symmetric) if and only if it is a rational function (with coefficient in K) of the elementary symmetric functions $s_1, ..., s_n$.

The functions s_i are viewed as elements of $K[t_1, ..., t_n]$ which we can do unambiguously since $\mathbb{Z}$ is initial in **Ring**.

***Proof*:**
Let $F = K(t_1, ..., t_n)$ and $k = K(s_1, ..., s_n)$ be the subfield generated by the elementary symmetric functions over K. Then F is a splitting field of the separable polynomial $P_n(x)$ over k, i.e. F is Galois and:

$$|\mathrm{Aut}_k(F)| = [F : k] \leq n!$$

Notice that the symmetric group S_n acts (faithfully) on F by permuting $t_1, ..., t_n$ and the action is the identity on $k = K(s_1, ..., s_n)$ since each s_i is symmetric. Thus, S_n can be identified with some subgroup of $\mathrm{Aut}_k(F)$. However,

$$n! = |S_n| \leq |\mathrm{Aut}_k(F)| \leq n!$$

showing that $\mathrm{Aut}_k(F) \cong S_n$

Now, since F/k is Galois, we get $k = F^{S_n}$. This in fact completes the rpoof: if $\varphi \in K(t_1, .., t_n)$, then φ is invariant under the S_n action on $t_1, ..., t_n$ if and only if $\varphi \in k \in K(s_1, ..., s_n)$, as we sought to show.

The theorem above is about rational functions of finitely many variables. Downstream work with formal characteristic-class calculi needs the same statement for formal power series over $\mathbb{Q}$, which does not follow from it, so it is proved here alongside.

The elementary symmetric functions are one basis for the symmetric polynomials, and for counting problems they are the wrong one: the classes that arise geometrically are indexed by partitions, and expressing them in the s_i hides that indexing behind a determinant. The Schur polynomials are the basis adapted to it, and they are recorded here because [16] builds its degeneracy-locus formulas out of them.

Definition 20.2.5: Schur Polynomial

A *partition* $\lambda = (\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_n \geq 0)$ of length at most n is a weakly decreasing tuple of non-negative integers; write $|\lambda| = \sum_i \lambda_i$. Put $\delta = (n-1, n-2, \dots, 1, 0)$. The *Schur polynomial* attached to λ in the variables $t_1, \dots, t_n$ is the bialternant

$$s_\lambda(t_1, \dots, t_n) = \frac{\det\left(t_i^{\lambda_j + n - j}\right)_{1 \leq i,j \leq n}}{\det\left(t_i^{n-j}\right)_{1 \leq i,j \leq n}}$$

the denominator being the Vandermonde determinant $\prod_{i<j}(t_i - t_j)$.

Proposition 20.2.6: Schur Polynomials Are a Basis

Each s_λ is a symmetric polynomial, homogeneous of degree $|\lambda|$, and the s_λ for λ of length at most n form a $\mathbb{Z}$-basis of the ring of symmetric polynomials in $t_1, \dots, t_n$. For the single-row partition $\lambda = (m)$ one has $s_{(m)} = h_m$, the complete homogeneous symmetric polynomial, and for the single-column partition $\lambda = (1^m)$ one has $s_{(1^m)} = s_m$, the m-th elementary symmetric polynomial.

***Proof*:**
The numerator of definition 20.2.5 is alternating: transposing two variables transposes two rows of the matrix and changes the sign of the determinant. The Vandermonde is alternating for the same reason, so the quotient is symmetric. It is a polynomial because an alternating polynomial vanishes on every hyperplane $t_i = t_j$ and is therefore divisible by each $t_i - t_j$, hence by their product, these being pairwise coprime. Degrees subtract to $|\lambda| + \binom{n}{2} - \binom{n}{2} = |\lambda|$.

For the basis claim, order partitions lexicographically. The monomial t^λ occurs in s_λ with coefficient 1 and every other monomial of s_λ is lexicographically smaller, so the transition matrix from $\{s_\lambda\}$ to the monomial symmetric functions $\{m_\lambda\}$ is unitriangular over $\mathbb{Z}$, hence invertible; and the m_λ are a $\mathbb{Z}$-basis by inspection. The two special cases are the standard evaluations of the bialternant, the second being the observation that $\det(t_i^{\lambda_j+n-j})$ for $\lambda = (1^m)$ factors as the Vandermonde times s_m.

Theorem 20.2.7: Jacobi–Trudi Formulas

For a partition λ with conjugate partition λ',

$$s_\lambda = \det\left(h_{\lambda_i - i + j}\right)_{1 \leq i,j \leq \ell}, \qquad s_\lambda = \det\left(s_{\lambda'_i - i + j}\right)_{1 \leq i,j \leq \ell'}$$

where h_m is the complete homogeneous and s_m the elementary symmetric polynomial, with the conventions $h_0 = s_0 = 1$ and $h_m = s_m = 0$ for $m < 0$, and ℓ, ℓ' are the lengths of λ and λ'.

***Proof Cited*:**
Not reproduced here (QC #3(a)). The clean proof is the Lindström–Gessel–Viennot lattice-path argument: both determinants count families of non-intersecting paths, and the sign-reversing involution on intersecting families cancels every off-diagonal term. That is combinatorics rather than algebra, and belongs with the lattice-path methods of the combinatorics volume; the identity is recorded here because this is where the symmetric functions live.

What the identity is for: it converts a partition-indexed basis element into a determinant in classes one can compute with. In [16] the h_m and s_m become Chern classes of a bundle, and the determinant becomes the Giambelli and Thom–Porteous formulas.

Lemma 20.2.8: Symmetric Power Series in the Chern Roots

Let $\Phi \in \mathbb{Q}[[a_1, \ldots, a_r]]$ be invariant under every permutation of $a_1, \ldots, a_r$. Then there is a unique $\widetilde{\Phi} \in \mathbb{Q}[[y_1, \ldots, y_r]]$ with

$$\Phi(a_1, \ldots, a_r) = \widetilde{\Phi}\big(e_1(a), \ldots, e_r(a)\big)$$

If Φ is homogeneous of degree d, then $\widetilde{\Phi}$ is a polynomial, isobaric of weight d when y_i is assigned weight i.

***Proof*:**
Existence. Each homogeneous component of Φ is symmetric, and an isobaric polynomial of weight d in the y_i contributes only in degree d after substituting $y_i = e_i$; so it suffices to treat a symmetric homogeneous polynomial P of degree d and assemble the results over d. Order the monomials $a^\nu = a_1^{\nu_1} \cdots a_r^{\nu_r}$ lexicographically, and let a^ν be the largest monomial occurring in a nonzero P, with coefficient λ. Then $\nu_1 \geq \nu_2 \geq \cdots \geq \nu_r$: if $\nu_j < \nu_{j+1}$ for some j, the transposed monomial also occurs in P, by symmetry, and is lexicographically larger. The product

$$e_1^{\nu_1-\nu_2} e_2^{\nu_2-\nu_3} \cdots e_{r-1}^{\nu_{r-1}-\nu_r} e_r^{\nu_r}$$

is symmetric and homogeneous of degree $\sum_j \nu_j = d$, and its lexicographic leading monomial is a^ν with coefficient 1, since the leading monomial of e_k is $a_1 a_2 \cdots a_k$ and the exponent of a_j in the displayed product is $\sum_{k \geq j}(\nu_k - \nu_{k+1}) = \nu_j$. Subtracting λ times it from P leaves a symmetric homogeneous polynomial of degree d whose leading monomial is strictly smaller. There are finitely many monomials of degree d, so the process terminates and expresses P as an isobaric polynomial of weight d in the e_i.

Uniqueness. It suffices to show that $e_1, \ldots, e_r$ are algebraically independent, since a difference of two representations of Φ is a power series R with $R(e_1, \ldots, e_r) = 0$, whose isobaric components of distinct weights land in distinct degrees in the a_j and so vanish separately. Let $Q = \sum_m \kappa_m y^m$ be a nonzero polynomial. The leading monomial of $e^m = e_1^{m_1} \cdots e_r^{m_r}$ is $a_1^{m_1+\cdots+m_r} a_2^{m_2+\cdots+m_r} \cdots a_r^{m_r}$, and the assignment $m \mapsto (m_1 + \cdots + m_r,\, m_2 + \cdots + m_r,\, \ldots,\, m_r)$ is injective, so distinct exponent vectors m produce distinct leading monomials. Among the finitely many m with $\kappa_m \neq 0$, take the one whose leading monomial is lexicographically largest. Every monomial of every other term is strictly smaller than it, so it survives in $Q(e_1, \ldots, e_r)$ with coefficient $\kappa_m \neq 0$, and $Q(e_1, \ldots, e_r) \neq 0$, as we sought to show.

One important result that was proven in the previous theorem is noted for future reference:

Lemma 20.2.9: S_n Extension

Let K be a field, $t_1, ..., t_n$ indeterminates, and let $s_1, ..., s_n$ be the elementary symmetric functions on $t_1, ..., t_n$. Then the extension:

$$K(s_1, ..., s_n) \subseteq K(t_1, ..., t_n)$$

is Galois, with Galois group S_n

From this, we can prove that every finite group is represented as a galois group:

Theorem 20.2.10: Finite Group Represented by Galois Group

Let G be a finite group. Then there exists a Galois extension F/k such that $\text{Aut}_k(F) \cong G$

***Proof*:**
exercsie

Note that it is not certain if every finite group G appears as an extension of F over $\mathbb{Q}$. This is called the *inverse Galois Problem.* more here! Can also checkout Aluffi p.473

Corollary 20.2.11

Let D be the discriminant of the field extension. Then:

$$K(s_1, ..., s_n)(\sqrt{D}) \subseteq K(t_1, ..., t_n)$$

is Galois, with Galois group A_n.

20.2.4 Solvability By Radicals

We are finally in a position to prove that an arbitrary degree 5 polynomial cannot be written in terms of it's coefficients using the operations $+, -, \times, \div, \sqrt{}$. Is is famously known, any quadratic polynomial over k ($\text{char}(k) \neq 2$) can be written in terms of:

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

and similar equations exists for degree 3 and 4 polynomials, though they get horrendously messy.

20.2.12 Goal There exists irreducible polynomials of degree $n \geq 5$ that are unsolvable

To avoid complications, we will assume in this section that $\text{char}(k) = 0$.

Definition 20.2.13: Radical Extension

Let F/k be a field extension. Then if $F = k(\alpha_1, \alpha_2, ..., \alpha_n)$ where

$$k \subseteq k(\alpha_1) \subseteq k(\alpha_1, \alpha_2), \subseteq \cdots \subseteq k(\alpha_1, ..., \alpha_n)$$

where for each $i \in \{1, ..., n\}$, there exists an n_i such that $\alpha^{n_i} \in k(\alpha_1, ..., \alpha_{i-1})$, then F/k is called a *radical extension*

Elements of radical extensions are easy to picture: if $k = \mathbb{Q}$, then simply nest as many nth-roots as you please, for example:

$$\sqrt[7]{3 + \sqrt[2]{19}^5 \sqrt[11]{3+2}}$$

it is also clear by definition that a radical extension is the composition of cyclic exensions provided that the appropriate roots of 1 are in k. The next result insures us that any [separable] radical extension is contained in a radical extensions that is *Galois*:

Lemma 20.2.14: Separable And Galois Radical Extensions

Let F/k be a separable radical extension. Then F is conatiend in a Galois radical extension

***Proof*:**
Let F/K be a separable radical extension. Then F is a separable finite extension of K, and so by the primitive root theorem (theorem 19.6.20), there exists an $\alpha \in F$ such that $F = k(\alpha)$. Let L be the splitting field of $k(\alpha)$. Then L is equal to the composite of all the field $\sigma(F)$ for every σ that embeds F into $\overline{k}$:

$$L = \sigma_1(F) \cdots \sigma_n(F)$$

Then the composition of radical extensions is radical, showing that L is radical. Thus, L is the splitting field of a separable polynomial, and so it is Galois, completing the proof.

Lemma 20.2.15: Solvability In Char 0

Let k be a field such that $\text{char}(k) = 0$, and let $f(x) \in k[x]$ be an irreducible polynomial. Then the roots of $f(x)$ can be written in terms of field operations and radicals if and only if the splitting field fo $f(x)$ is contined in a Galois Radical extension

***Proof*:**

($\Leftarrow$) If the splitting field $f(x)$ is contained in a radical extension (not necessarily Galois), then *a fortiori* the roots can be written using field operations and radicals.

($\Rightarrow$) Conversely, assume the roots of the irreducible polynomial $f(x)$ can be written in terms of field operations and radicals . Then any formula for a root t_1 of $f(x)$ can be turned into a formula for t_i by applying an element σ_i of $\text{Aut}_k(\overline{k})$ sending t_1 to t_i (this automorphism exists since $f(x)$ is irreducible). Thus, if the formula exists for one root, a formula exists for all roots. Thus, $k(t_1), ..., k(t_n)$ are each radical extensions and

$$L = k(t_1)k(t_2) \cdots k(t_n)$$

is a radical extension that contains all the roots of $f(x)$. Thus, by minimality of the splitting field, it is contained in a radical extension of k. By lemma 20.2.14, it must also be contained in a galois radical extension, as we sought to show.

Thus, we can characterize when a polynomial is solvable by radicals by when it is contained in a Galois radical extension. Since the extension is Galois, we have a corresponding Galois group to analyze. We will soon see that the appropriate type of group that is needed for $f(x)$ to be solvable is one we have encountered a long time ago:

Definition 20.2.16: Solvable Extension

Let F/K be a Galois extension. Then F/K is a *solvable extension* if $\text{Aut}_K(F)$ is a solvable group

Recall that if G is a solvable group and finite, then G can be decomposed into a subnormal series with cyclic factors:

$$G = G_0 \trianglerighteq G_1 \trianglerighteq G_2 \trianglerighteq \cdots G_n = \{1\}$$

notice how this parrallels the fact that any radical extension is a sequence of cyclic extensions.

Lemma 20.2.17: Relating Radical And Solvable Extensions

Let F/K be a galois extension with $\text{char}(k) = 0$. If F/K has enough roots of 1, then F/K is radical if and only if it is solvable

What is meant by "enough roots of 1" is that K contains primitive mth roots of 1 for a common multiple m of the order of the corresponding cyclic factors.

***Proof*:**
This is a generalization of proposition 20.1.41.

Let F/K be a radical extension, so that:

$$K \subseteq k(\delta_1) \subseteq \cdots \subseteq K(\delta_1, ..., \delta_r) = F$$

where $\delta_i^{m_i} \in K(\delta_1, ..., \delta_{i-1})$. By assumption, K contains primitive m_ith roots of 1 for each i. Next, consider the corresponding subgroups of $G = \text{Aut}_k(F)$:

$$G = G_0 \supseteq G_1 \supseteq \cdots \supseteq G_r = \{e\}$$

where $G_i = \text{Aut}_{K(\delta_1,...,\delta_i)}(F)$. Next, note That F is Galois over each intermediate field (by proposition 19.5.16), and each extension:

$$K(\delta_1, ..., \delta_{i-1}) \subseteq K(\delta_1, ..., \delta_i)$$

satisfies proposition 20.1.41, and hence each G_i is cyclic. By theorem 20.1.23, each G_i is normal (since the automorphism group is abelian) in the previous G_{i-1} with a cyclic quotient, and hence G is solvable.

Conversely, the same argument works in reverse using the converse implication in proposition 20.1.41 to show that an extension with enough roots of 1 is radical.

We were dependent on each extension containing enough roots of unity. In the following, we show that we can drop them:

Proposition 20.2.18: Solvable In Char 0

Let K be a field where $\text{char}(K) = 0$, and let F/K be a Galois extension. Then F/K is solvable if and only if it is containd in a Galois Radical extension

Technically, we only need that $\text{char}(K)$ not divide the relevant degrees of the extensions, however for our purposes it is safe to assume $\text{char}(K) = 0$.

***Proof*:**

($\Rightarrow$) Assume $\text{Aut}_K(F)$ is solvable. Let M be the common multiple of the cyclic quotients and let ζ be a mth root of 1 in an algebraic closure $\overline{K}$ of K. The splitting field $K(\zeta)$ of $x^M - 1$ is Galois over K since $\text{char}(K) = 0$. By proposition 20.1.28, the extension $K(\zeta) \subseteq F(\zeta)$ is also Galois, with Galois group isomorphic to $\text{Aut}_{K(\zeta)\cap F}(F)$. Since $K(\zeta) \cap F$ is Galois over K (Aluffi says exercise 6.13), this subgroup is normal, and it follows (corollary IV.3.13) that $\text{Aut}_{K(\zeta)}(F(\zeta))$ is solvable.

Since $K(\zeta) \subseteq F(\zeta)$ has enough roots of 1, by lemma 20.2.17, it is radical. Thus, so is $K \subseteq F(\zeta)$ since $K \subseteq K(\zeta)$ is by trivially radical. Thus, F/K is contained in a radical extension.

($\Leftarrow$) Conversely, assume that F/K is Galois and contained in a radical Galois extension L/K. Let M be a common multiple of the order of the cyclic factors in this extensions,and let ζ be a primitive mth root of 1. The composite $L(\zeta)$ is Galois and radical over $K(\zeta)$ (show that if E/K is radical and T/K is any extension, then ET/K is a radical extension), and it has enough roots of 1.

By lemma 20.2.17, $L(\zeta)/K(\zeta)$ is solvable. Then we actually have that $L(\zeta)/K$ is solvable. Indeed, this extension is Galois since it's a composition of galois (exercise 6.13). Applying the fundamental theorem to:

$$K \subseteq K(\zeta) \subseteq L(\zeta)$$

to established that $\text{Aut}_K(K(\zeta))$ is isomorphic to the quotient of $\text{Aut}_K(L(\zeta))$ by its normal subgroup $\text{Aut}_{K(\zeta)}(L(\zeta))$. Both $\text{Aut}_{K\zeta)}(L(\zeta))$ and $\text{Aut}_K(K(\zeta))$ are solvable (being abelian), and it follows that $\text{Aut}_K(L(\zeta))$ is solvable.

Since F is an intermediate field and Galois over T, by the fundamental theorem, $\text{Aut}_K(F)$ is a homomorphic image of $\text{Aut}_K(L(\zeta))$. This shows that $\text{Aut}_K(F)$ is a homomorphic image of a solvable group, hence it is itself solvable, and so we're done.

Corollary 20.2.19: Galois Criterion

Let K be a field such that $\text{char}(K) = 0$ and let $f(x) \in K[x]$ be an irreducible polynomial. Then f(x) is solvable by radicals if and only if its galois group is solvable

***Proof*:**
This is an immediate consequence of lemma 20.2.15 and proposition 20.2.18.

Theorem 20.2.20: Abel's Theorem

Let $f(x)$ be a generic polynomial of degree 5 or higher. Then it admits no solutions by radicals

***Proof*:**
By lemma 20.2.9, the Galois group of the generic polynomial of degree n is S_n. The group S_n is not solvable for $n \geq 5$ (see ref:HERE), and hence the statement follows from the Galois Criterion.

Furthermore, we know that S_3 and S_4 *are* solvable, with subnormal series:

$$\{e\} \trianglelefteq \mathbb{Z}/2\mathbb{Z} \trianglelefteq D_6 = S_3$$

$$\{e\} \trianglelefteq V_4 \trianglelefteq A_4 \trianglelefteq S_4$$

hence all degree less than or equal to 4 polynomial have a root using radicals.

Exercise 20.2.1

1. Let $\alpha \in F$ for an algebraic extension $F/\mathbb{Q}$. In particular, we may think of F as being embedded in $\mathbb{C}$ so that $\alpha = a + bi$. Does there always exist an N such that $\alpha^N \in \mathbb{R}$?

20.3 Galois Groups of Polynomials

In this section, we shall focus on techniques to compute the galois group of polynomial. So far, much of the focus was on showing what properties can be deduced given properties of the galois group, there have not been many technics to find galois groups (for example, can you come up with a non-solvable degree 5 polynomial?). Hence, this section is a compilation of interesting facts and techniques useful for working with galois groups of poylnomials.

Lemma 20.3.1: Transitivity Of Galois Group

Let $f(x) \in k[x]$ be a separable irreucible polynomial of degree n. Then $\mathrm{Gal}_k(f(x))$ acts transitively on the set of roots of $f(x)$ in $\overline{k}$. In particular, $\mathrm{Gal}_k(f(x))$ may be identified with a *transitive* subgroup of the symmetric group S_n.

***Proof*:**
exercise

By corrollary 19.2.29, given α a root of $m_{\alpha,k}(x)$, then the elements of $\mathrm{Gal}_k(m_{\alpha,k}(x))$ are determined by where this root maps. This motivates the following definition:

(In general)

$$\mathrm{Gal}_k(fh) \leqslant \mathrm{Gal}_k(f) \times \mathrm{Gal}_k(h) \leqslant S_{n_1} \times S_{n_2} \leqslant S_n$$

(but more relations can be added: here)

This gives us the limits of the possible subgroups of galois groups:

1. The only possible group for degree 2 irreducible polynomials is: $\mathbb{Z}/2\mathbb{Z}$
2. The only possible groups for degree 3 irreducible polynomials is:
$$\mathbb{Z}/3\mathbb{Z} \cong A_3, S_3$$
3. The only possible groups for degree 4 irreducible polynomials is:
$$S_4, A_4, D_8, \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}, \mathbb{Z}/4\mathbb{Z}$$

Definition 20.3.2: Galois Conjugate

Let $\alpha \in L$ be an algebraic element of L over k. Then the set of roots of the minimal polynomial $m_{\alpha,k}(x)$ is called the *galois conjugates*, *algebraic conjugates*, or *conjugate elements* of α over k.

The motivation comes from the case of $\alpha \in \mathbb{C}$ which are algebraic over $\mathbb{R}$. In this case, the minimal polynomial is either $x - \alpha$ (in which case it has one root), or it is of degree 2 with galois conjugate $\overline{\alpha}$. We thus see that the galois conjugate is the generalization of the complex conjugate.

For the next result, we recall the following result of the discriminant;

Definition 20.3.3: Discriminant Reminder

Let $f(x) \in k[x]$ be a separable polynomial with roots $\alpha_1, ..., \alpha_n$ in its splitting field. Then the *discriminant* of $f(x)$ is the element $D = \Delta^2$ where
$$\Delta = \prod_{1 \leq i < j \leq n} (\alpha_i - \alpha_j)$$

Every permutation of the roots fixes D, so D is fixed by the whole Galois group, therefore $D \in k$. Odd permutations move Δ (if $\mathrm{ch}(k) \neq 2$) and even permutations fix it. Tus, Δ is fixed by the Galois group G (i.e. $\Delta \in k$) if and only if $G \subseteq A_n$

Lemma 20.3.4: Discriminant And Galois Group

Let k be a field where $\mathrm{ch}(k) \neq 2$ and let $f(x) \in k[x]$ be a separable polynomial with discriminant D. Then the Galois group o f $f(x)$ is contaiend in the alternating group A_n if and only if D is a quare in k

Example 20.3.5: Discriminant

1. Let $f(x) = x^3 + ax^2 + bx + c$ be a general monic cubic. By shifting by $a/3$, we get
$$f(x - a/3) = x^3 + px + q$$

for appropriate p, q. Then it can be computed that:

$$D = -4p^3 - 27q^2$$

Then by our above lemma, if D is a square then the galois group is S_3, and A_3 otherwise. For example, the galois group of $x^3 - 2$ is S_3 since $D = -108$ is not a square in $\mathbb{Q}$, while $x^3 - 3x + 1$ has galois group $A_3 \cong \mathbb{Z}/3\mathbb{Z}$ since $D = 81 = 9^2$.

2. Let $f(x) \in \mathbb{Q}[x]$ be of degree p for prime p with $p-2$ real roots and 2 non-real complex roots. Then $\mathrm{Gal}_{\mathbb{Q}}(f(x)) = S_p$. Prove this, then show that the galois group of $x^5 - 5x - 1$ over $\mathbb{Q}$ is S_5.

Other useful discriminants are:

$$\begin{aligned} \mathrm{D}(x^3 + ax + b) &= -4a^3 - 27b^2 \\ \mathrm{D}(x^4 + ax + b) &= -27a^4 + 256b^3 \\ \mathrm{D}(x^4 + ax^2 + b) &= 16b(a^2 - 4b)^2 \end{aligned}$$

(a word on the inverse galois problem? Sovled for every finite abelian group?)

Exercise 20.3.1

1. Let $K/\mathbb{Q}$ be Galois. Show that if $\mathrm{disc}(K)$ is square-free, then K has no intermediate fields

20.4 Kummer Theory and Artin-Schreier Extensions

As we saw an section 20.2.4, if a field extension F/K is a radical extension, then the Galois group is solvable. The converse is not necessarily true:

Example 20.4.1: Solvable, But Not Radical Extension

Consider $K = \mathbb{Q}(\sqrt{5})$ and $M = K(\zeta_3)$; recall the minimal polynomial of ζ_3 is $x^2 + x + 1$. Then $[M : K] = 2$, and it is easy to see then that if M/K were a radical extension then $M = K[\sqrt{\alpha}]$ for some $\alpha \in K$ but no such element can exist.

By doing multiple examples like these, the reader would notice that this seems to be more of a problem of there being a lack of roots of unity in K to insure there are enough solutions to $x^n - a$. In fact, there is no additional issues that arise when the characteristic of K is 0, and the issues in finite characteristic only happen for certain bad primes. We thus explore this converse: that under these mild extra conditions, if the galois group is solvable then the extension is a radical extension. As solvable groups are towers of cyclic groups, it shall suffice to show that if the galois groups in the tower of fields is cyclic, each corresponding field extension is given by a radical $\sqrt[n]{\alpha}$. An interesting consequence of this is that the galois groups of Kummer extensions shall be characterized by certain units of the base field. This shall in fact be true for all abelian extensions (not just Kummer extensions), but the proof of this result is a lot more complicated and was one of the crowning achievements 20th century number theory proved in 1930; the result is called the *Artin Reciprocity* and its proof can be found in [39].

Definition 20.4.2: Kummer Extensions and G-Extensions

Let L/K be a field extension. Then L/K is called a *Kummer Extension* if:

1. for some n, K contains μ_n (the roots of $x^n - 1$)
2. L/K is an Abelian extension with exponent n[a]

If in particular we want extensions with Galois groups G, we shall call such extensions G-extensions.

[a]The least common multiple of the orders of the elements

Example 20.4.3: Kummer Extensions

1. Any quadratic extension of $\mathbb{Q}$ is a Kummer extension of degree 2, since $\mathbb{Q}$ have $1, -1$ and has abelian extension. More generally, any multi-quadratic extension is a Kummer extension of degree 2
2. There is no Kummer extensions of degree 3 over $\mathbb{Q}$ since $\mathbb{Q}$ does not have a 3rd root of unity, since it's complex.
3. More generally if K contains μ_n (meaning the characteristic of K does not divide n), then adjoining any nth root of $a \in K$, $L = K(\sqrt[n]{a})$ is a Kummer extension (of degree m dividing n, since the minimal polynomial divides $x^n - a$)

So let's now say that L/K is a Kummer $\mathbb{Z}/n\mathbb{Z}$-extension where the characteristic of K does not divide n. In order to show $L = K(\sqrt[n]{a})$, we must find an appropriate $a \in K^*$. Let us find some necessary conditions. As $\text{Gal}_K(L) = \mathbb{Z}/n\mathbb{Z}$, there is a generator σ such that $\sigma^n = 1$. Then given $x^n = a$, we have the roots:

$$\sqrt[n]{a},\ \ \sqrt[n]{a}\zeta_n, ..., \ \sqrt[n]{a}\zeta_n^{n-1}$$

note all these roots of unity are distinct and in K by the characteristic condition. Thus, by field theory we know that σ must map $\sqrt[n]{a}$ to another one of these roots, and that this conditions characterizes σ; without loss of generality we can say $\sigma(\sqrt[n]{a}) = \zeta\sqrt[n]{a}$. Importantly, interpreting L as a K-vector space, $\sqrt[n]{a}$ becomes an *eigenvector* of σ. Note that as $\sigma^n = 1$, the possible eigenvalues must be $1, \zeta_n, ..., \zeta_n^{n-1}$. Thus if σ is diagonalizable, we can find such an a such that $a^n = \alpha \in K$. Exercise 20.4.1(20.4.1 (1)) gives a very intuitive computational approach to finding this. We present the following more abstract way of finding it that works more generally so that we can use it later in number theory and cohomology theory in this book:

Theorem 20.4.4: Hilbert's 90th Theorem

Let K be a field of characteristic 0 and L/K be an abelian extension such that

$$\mathrm{Gal}_K(L) \cong \mathbb{Z}/n\mathbb{Z}$$

If σ generate $\mathrm{Gal}_K(L)$, then:

1. For $\beta \in L^\times$ such that $\mathrm{Nm}_{L/K}(\beta) = 1$, there exists an $\alpha \in L^\times$ such that

$$\beta = \frac{\sigma(\alpha)}{\alpha}$$

2. Let $\beta \in L$ such that $\mathrm{tr}(\beta) = 0$. Then there exists an $\alpha \in L$ such that

$$\beta = \alpha - \sigma(\alpha)$$

***Proof*:**
Since L/K is a finite separable extension, pick t so that $L = K(t)$. Let

$$S = \beta + \beta\sigma(\beta) + \beta\sigma(\beta)\sigma^2(\beta) + \cdots = \sum_{i=1}^{n-1} \prod_{j}^{i} \sigma^j(\beta)$$

If $S \neq 0$, then:

$$\frac{\sigma(S)}{S} = \beta^{-1} \implies \frac{\sigma(S^{-1})}{S^{-1}} = \beta$$

Thus, we would need to guarantee such a nonzero S exists.

To see this, notice that solving $\beta = \frac{\alpha}{\sigma(\alpha)}$ is equivalent to $\beta\sigma(\alpha) = \alpha$. Notice that

$$\beta\sigma(-) : L \to L$$

mapping $\alpha \mapsto \beta\sigma(\alpha)$ is a K-linear map. If we show that it has an eigenvector with eigenvalue 1, then we solved our problem, for if α is such an eigenvector then (if we represent our function as f)

$$\alpha = f(\alpha) = \beta\sigma(\alpha)$$

We shall verify this by extending scalars from K to L. Recall by corollary 14.3.7 that extending scalars does not change the eigenvectors. We may thus extend our map to:

$$\begin{aligned} \mathrm{id} \otimes_K \beta\sigma(-) : L \otimes_K L &\to L \otimes_K L \\ \ell \otimes_K \ell' &\mapsto \ell \otimes_K \beta\sigma(\ell') \end{aligned}$$

Since L/K is finite and separable, there exists a γ such that $L = K(\gamma)$. Then γ has minimal polynomial $m_{\gamma,K}(x) \in K[x]$, and we have the isomorphisms:

$$(L \otimes_K L) \cong \left(L \otimes_K \frac{K[x]}{m_{\gamma,K}(x)}\right) \cong \frac{L[x]}{m_{\gamma,K}(x)} \cong L^n$$

Since any $\ell' \in L$ can be written as $\ell' = p(\gamma)$ for some polynomial $p(x) \in K[x]$, we have:

$$(\ell \otimes_K p(\gamma)) \mapsto \ell(p(\gamma), p(\sigma(\gamma)), ..., p(\sigma^{n-1}(\gamma)))$$

Thus, the map $\beta\sigma(-)$ becomes interpreted as $L^n \to L^n$ as:

$$\ell(p(\gamma), ..., p(\sigma^{n-1}(\gamma))) \mapsto \ell\left(\beta p(\sigma^{n-1}(\gamma)), \sigma(\beta p(\gamma)), ..., \sigma^{n-1}(\beta p(\sigma^{n-2}(\gamma)))\right)$$

or perhaps a bit more clearly:

$$(\ell_1, \ell_2, ..., \ell_n) \mapsto (\beta\ell_n, \sigma(\beta\ell_1), ..., \sigma^{n-1}(\beta\ell_{n-1}))$$

Since $\mathrm{Nm}_{L/k}(\beta) = 1$,

$$\left(1, \sigma(\beta), \sigma(\beta)\sigma^2(\beta), ..., \sigma(\beta)\sigma^2(\beta)\cdots\sigma^{n-1}(\beta)\right)$$

is an eigenvector with eigenvalue 1, hence our original transformation had an eigenvector of eigenvalue 1, as we sought to show.

For the trace argument, say $\beta \in L$ such that $\mathrm{tr}(\beta) = 0$. Take

$$S = \beta 2\sigma(\beta) + 3\sigma^2(\beta) + \cdots n\sigma^{n+1}(\beta)$$

Then:

$$\sigma(S) = \sigma(\beta) + 2\sigma^2(\beta) + \cdots + n\sigma^n(\beta)$$

since $\sigma^n = \mathrm{id}$, $\sigma^n(\beta) = \beta$, so we get:

$$S - \sigma(S) = -n\beta$$

Thus, set $\alpha = -\beta/n$, completing the proof.

20.4.5 Foreshadowing In part XI, we shall reformulate this theorem in purely Homological terms, namely it says that $H^1(G, L^*) = 0$

We now look at some properties of Kummer extensions. First, by some Galois theory, we see that if $\mathbb{Q}(\zeta_n) \subseteq K$, $L = K(\alpha)$ where $\alpha^n = \beta \in K$, then L/K is an abelian Galois extension, for:

$$\varphi : \mathrm{Gal}_K(L) \xrightarrow{\sim} \mu_n \qquad \tau \mapsto \frac{\tau(\alpha)}{\alpha}$$

which is well-defined, and any nth root of β gives the same isomorphism. Note how important $\mu_n \subseteq K$ is for this isomorphism to work! This is certainly not true in general (is $\mathbb{Q}(\sqrt[n]{2})/\mathbb{Q}$ galois for $n > 2$?). Using Hilbert's 90th problem, the converse of this statement is also true:

Theorem 20.4.6: Primitive Element For Cyclic Kummer Extension

Let $\mathrm{char}(K) = 0$, L/K be a Galois extension and $\mathrm{Gal}_K(L) \cong \mathbb{Z}/n\mathbb{Z}$. Assume K contains μ_n, the group of roots of unity generated by ζ_n. Then there exists an $\alpha \in L^\times$ such that

$$L = K(\alpha) \qquad \alpha^n \in K^\times$$

***Proof*:**
Let ζ_n be a primitive nth root of unity. Recall that $\mathrm{Nm}_{L/K}(\zeta_n) = \zeta_n^n = 1$. Thus, by Hilbert's 90th, there exists $\alpha \in L$ such that $\sigma(\alpha) = \zeta_n \alpha$. Since σ is a generator, we see that α is not fixed by any σ, hence:

$$L = K(\alpha)$$

Furthermore,

$$\sigma(\alpha^n) = \sigma(\alpha)^n = \zeta_n^n \alpha^n = \alpha^n$$

Hence, each σ fixes α^n, meaning $\alpha^n = \beta \in K$, hence

$$L = k(\sqrt[n]{\beta})$$

as we sought to show.

With the existence established, let us now established to what extent it is unique. Say $K(\sqrt[n]{a}) \cong K(\sqrt[n]{b})$. Then we know $\sqrt[n]{a}$ and $\sqrt[n]{b}$ are both eigenvectors, but they may have different eigenvalues. This is not an issue as we may raise one of them to the appropriate power, say $\sqrt[n]{b}$ so that the eigenvalue is the same root of unity as $\sqrt[n]{a}$ and it is still of order n (which we can do as $\gcd(n, n-1) = 1$). We may thus simply assume they have the same eigenvalue. But then:

$$\sigma\left(\frac{\sqrt[n]{a}}{\sqrt[n]{b}}\right) = \frac{\sqrt[n]{a}}{\sqrt[n]{b}}$$

and hence it is fixed for all σ^i, and hence is in K. We thus get the following correspondence:

Corollary 20.4.7: Cyclic Kummer Extension Correspondence

Let $\mu_n \subseteq K$. Then we have the following bijective correspondence

$$\left\{\begin{matrix} L/K \\ \mathrm{Gal}_K(L) \cong \mathbb{Z}/n\mathbb{Z}\end{matrix}\right\} \underset{\psi}{\overset{\varphi}{\rightleftarrows}} \left\{\begin{matrix} G \subseteq K^\times/(K^\times)^n \\ G \cong \mathbb{Z}/n\mathbb{Z}\end{matrix}\right\}$$

***Proof*:**
First finding ψ, pick $\overline{\beta} \in K^\times/(K^\times)^n$. Pick $\beta \in K^\times$ which represents $\overline{\beta}$. Then define

$$\psi(\langle \overline{\beta} \rangle) = K(\sqrt[n]{\beta}) = L$$

this is well-defined for if we have any other coset representative, $\beta' = \beta\gamma^n$, $K(\sqrt[n]{\beta}) = K(\sqrt[n]{\beta'})$. Certainly $\mathrm{Gal}_K(L) \cong \mathbb{Z}/m\mathbb{Z}$ and $m \big| n$. We need to show that $n = m$. For the sake of contradiction, suppose not. Then if σ generates $\mathrm{Gal}_K(L)$, then if we let $\alpha = \sqrt[n]{\beta}$, we have

$$\sigma(\alpha) = \zeta_m \alpha \implies \sigma(\alpha^m) = \zeta_m^m \alpha^m = \alpha^m$$

hence $\alpha^m \in K^\times$. Then $\beta = \alpha^n = (\alpha^m)^{\frac{n}{m}} \in (K^\times)^{n/m}$, a contradiction.

Conversely, to define φ, suppose we have an L where $\langle \sigma \rangle = \mathrm{Gal}_K(L)$. By theorem 20.4.6, pick $\alpha \in L^\times$ such that $\sigma(\alpha)/\alpha = \zeta_n$. Then α is well-defined up to being an element of $K^\times$. In particular, $[\alpha^n] \in K^\times/(K^\times)^n$ is well-defined. Thus, take the map:

$$\varphi(L) = \langle [\alpha^n] \rangle \subseteq K^\times/(K^\times)^n$$

It remains to show these maps are inverses of each other (important exercise!)

Corollary 20.4.8: Pontryagin Dual For Kummer Extensions

Let $\mathbb{Q}(\zeta_n) \subseteq K$, and $K_n \subseteq \overline{K}$ the maximal finite Kummer extension, that is the maximal finite abelian extension with $\text{ord}(\text{Gal}_K(K_n))\big|n$ (every element of the group has order that divides n). Then there exists a natural isomorphism:

$$\text{Gal}_K(K_n) \cong \text{Hom}(K^\times/(K^\times)^n, \mu_n)$$

$$\sigma \mapsto \left(\overline{\beta} \mapsto \frac{\sigma(\sqrt[n]{\beta})}{\sqrt[n]{\beta}}\right)$$

***Proof*:**
exercise

We can do all cyclic extensions at once by taking advantage of the infinite galois theory developed earlier

Theorem 20.4.9: Kummer Pairing

Let K contain all roots of unity, let M/K be the extension of K by all possible roots, and let μ_n be all nth roots of unity. Then there exists a map:

$$\varphi : \text{Gal}_K(M) \times (K^*/(K^*)^n) \to \mu_n \qquad \varphi(\sigma, a) = \frac{\sigma(a)}{a}$$

***Proof*:**
exercise.

Definition 20.4.10: Kummer Pairing

Let K contain all roots of unity, let M/K be the extension of K by all possible roots, and let μ_n be all nth roots of unity. Then the map

$$\varphi : \text{Gal}_K(M) \times (K^*/(K^*)^n) \to \mu_n \qquad \varphi(\sigma, a) = \frac{\sigma(a)}{a}$$

is called the *Kummer Pairing*.

This shows that the galois group and the units are almost dual to each other. A point was made to put μ_n instead of $\mathbb{Z}/n\mathbb{Z}$; these two groups are group isomorphic, however there as a different group-action on μ_n, namely the galois group acts on it non-trivially. The fact there is a non-trivial action is often called the *Tate Twist*.

Artin-Schreier Extensions

If $\text{char}(K)\big|n$, then $x^n - a$ has multiple roots and hence the extension is not Galois, and furthermore if L/k is cyclic, all eigenvalues of σ are 1, loosing the method to find the element α. Note that there do exist such galois extensions, for example $\mathbb{F}_{p^p}/\mathbb{F}_p$, and hence this is a valid problem.

The key is to use generalized eigenvalues.

I'll definitely type all of this up, I'll leave it blank for now but it is pretty neat to finish off the correspondence between solvable groups and field extensions

Definition 20.4.11: Artin-Schreier Extensions

Let L/K be a cyclic extension of order n where $\text{char}(K)\big|n$. Then L is an *Artin-Schreier* extension $L = K(\alpha)$ where $\sigma(\alpha) = \alpha + 1$ and $\alpha^p - \alpha \in K$ and α is the solution to $x^p - x - a$ (as compared to $x^n - a$ for Kummer extensions).

There is a similar Artin-Schreier pairing as well.

Exercise 20.4.1

1. Show that we can find α such that $L = K(\alpha)$ where L is a Kummer $\mathbb{Z}/n\mathbb{Z}$-extension showing all elements can be written as the sum of eigenvalues. (hint: `https://www.youtube.com/watch?v=UaeJNQ5x17g&list=PL8yHsr3EFj53Zxu3iRGMYL_89GDMvdkgt&index=15`)

21⋆

More Structures on Fields

⋆ *Optional. This chapter is an aside; nothing later in the book depends on it.*

The most common fields the reader will encounter are $\mathbb{Q}, \mathbb{R}$, and $\mathbb{C}$. Each of these fields are natural in some way; $\mathbb{Q}$ is initial in $\mathbf{Fld}_0$, $\mathbb{R}$ is the smallest complete ordered field[1], and $\mathbb{C}$ is the smallest complete and algebraically closed field. A key property all three of these fields share is that there is a notion of *size*. For $\mathbb{Q}$ and $\mathbb{R}$, there is even the stronger notion of *ordering* for all the elements. For $\mathbb{R}$ and $\mathbb{C}$, there is a notion of *differentiability*. In this chapter, we explore these properties

Recall that an order can be defined on a set (definition 0.1.14). In this section, we shall explore fields with orders, or fields for which these is a function that maps to an ordered field.

This chapter shall also incorporate some topology from chapter C as it is beneficial for a deeper understanding of the subjects presented in this chapter.

21.1 Ordered and Real Fields

Definition 21.1.1: Ordered Field

Let k be a field. Then k is a *totally ordered field* (or *ordered field* for short) if there exists a (total) order $\leq$ on k such that for all $x, y, z \in k$:

1. $x < y$, then $x + z < y + z$
2. If $z > 0$, then if $x < y$, $zx < zy$

If a field k admits a total ordering, then k is said to be a *formally real field*.

[1] More technically, it is the unique Dedekind-complete ordered field up to isomorphism

Famously, $\mathbb{Q}$ and $\mathbb{R}$ are ordered fields. The reader should prove that $\mathbb{C}$ is *not* an ordered field.

Proposition 21.1.2: Properties Of Ordered Fields

Let k be an ordered field and $x \in k$. Then

1. $x > 0$ if and only if $-x < 0$ (similarly, $x < 0$ if and only if $-x > 0$)
2. $x > y$ if and only if $x - y > 0$
3. $x < y$ if and only if $-x > -y$
4. $x > y > 0$, then $y^{-1} > x^{-1} > 0$
5. If $x < y$ and $z < 0$, then $xz = (-x)(-z) > (-y)(-z) = yz$
6. $x^2 > 0$ (hence $1 > 0$ always since $1 = 1^2 > 0$)

Furthermore, ordered fields have characteristic zero

***Proof*:**
All of these should be done as an exercise. For the last one, note that

$$0 < 1 < 1 + 1 < 1 + 1 + 1 < \cdots$$

Note how this proposition immediately shows that $\mathbb{C}$ is not ordered since $-1 = i^2 \not> 0$, and that it can be shown that all finite fields are not ordered. In fact, this is a characterizing property of non-ordered fields

Proposition 21.1.3: Artin-Schreier Conditions

Let k be a field. Then k can be an ordered field. Then the following are equivalent:

1. k is an ordered field
2. -1 is *not* the sum of squares
3. 0 is *not* a (non-zero) sum of square

***Proof*:**

($1 \Rightarrow 2$) see previous proposition.

($2 \Leftrightarrow 3$) If $-1 = \sum_i^n x_i^2$, then $0 = 1 + \sum_i^n x_i^2 = 1^2 + \sum_i^n x_i^2$. Conversely, if $0 = \sum_i^n x_i^2$ for nonzero x_i, then

$$-x_n^2 = \sum_i^{n-1} x_i^2 \quad \Leftrightarrow \quad -1 = \sum_i^{n-1} (\frac{x_i}{x_n})^2$$

Then by proposition 21.1.2, squares of elements cannot be negative, however $-1 < 0$.

(2 $\Leftrightarrow$ 1) Assume -1 is not a sum of squares. Let S be the set of all elements that are (non-zero) sums of squares. Notably, $0, -1 \notin S$, S is closed under addition, and since k is commutative S is also closed under multiplication. Furthermore, it is a multiplicative subgroup of $k \setminus \{0\})$ for if $s \in S$ then:

$$\frac{1}{s} = \frac{s}{s^2} = \frac{\sum_i x_i^2}{s^2} = \sum_i \frac{x_i^2}{s^2} = \sum_i \left(\frac{x_i}{s}\right)^2$$

Now take the collection of sets that include S and that are closed under addition, multiplication, and don't contain -1 and 0. These sets are closed under inclusion and are certainly closed under chains, hence by Zorn's lemma there exists a maximal set $M \subseteq k$ that is closed under addition is a multiplicative subgroup of $k \setminus \{0\}$, which also cannot contain $-1, 0 \in M$. Then $M, \{0\}$, and $-M$ must be disjoint. I claim that

$$k = -M \cup \{0\} \cup M$$

Given this is proven, k then must be ordered, where $x < y$ iff $y - x \in M$. Suppose $0 \neq a \in k$ such that $-a \notin M$. Define:

$$M' = \{x + ay \mid x, y \in M \cup \{0\}, (x \neq 0) \vee (x \neq y)\}$$

Certainly $M \subseteq M'$ and M' is closed under addition, and is closed under multiplication for if $x, y, z, t \in M \cup \{0\}$, Then:

$$(x + ay)(z + at) = (xz + a^2yt) + a(yz + xt)$$

where $(xz + a^2yt), yz + xt \in M \cup \{0\}$ (note that $a^2 \in S \subseteq M'$). Furthermore, it should be clear that $1 \in M'$ $1 \in S \subseteq M \subseteq M'$) and $0 \notin M'$ (or else $a \in -M$). Finally, if $t \in M'$, then:

$$t^{-1} = \frac{t}{t^2} = \frac{x}{t^2} + a\frac{y}{t^2} \in M'$$

since $t^2 \in M$. Thus, M' is a multiplicative subgroup of $k \setminus \{0\}$, and hence by maximality $M = M'$, giving $a \in M$.

Since any ordered field is characteristic zero, it must contain a subfield

$$Q = \left\{ \frac{m1}{n1} \;\middle|\; m, n \in \mathbb{Z} \right\}$$

field-isomorphic to $\mathbb{Q}$. In fact, Q is order-field isomorphic, for $n1 > 0$ in Q when $n > 0$ in $\mathbb{Z}$, and $m1/n1 > 0$ in Q when $m/n > 0$ in $\mathbb{Q}$, and so $a1/b1 > c1 > d$ in Q iff $a/b > c/d$ in $\mathbb{Q}$. Thus, we usually identify Q with $\mathbb{Q}$ as an ordered subfield. A property that we may take for granted that an ordered field may have is the following:

Definition 21.1.4: Archimedean Field

Let k be an ordered field. Then k is an *Archimedean field* if every positive element of k is less than a positive integer

Though it may seem like all fields have this property, there do exists field without it, most notably the *hyperreals numbers* which can be thought of as the "order completion" of the reals (think of adding an element ϵ st $0 < \epsilon < x$ for all $x > 0$ and a ω st $0 < x < \omega$ for all $x > 0$). Fortunately, all the ordered fields we shall work with will be Archimedean due to the following classification result:

Theorem 21.1.5: Classification Of Archimedean Fields

Let k be an Archimedean field. Then k is isomorphic to a field:

$$\mathbb{Q} \subseteq k' \subseteq \mathbb{R}$$

***Proof*:**
Grillet p.233

Let us focus on formally real fields. If k is formally real, then so is $k(x)$. For an algebraic extension

Proposition 21.1.6: Algebraic Extension Of Formally Real Field

Let k be a formally real field. Then if $\alpha^2 \in k$ such that $\alpha^2 > 0$, $k(\alpha)$ is a formally real field

***Proof*:**
exercise

If an algebraic extension of a formally real field is formally real, then it is called a *real extension*.

Proposition 21.1.7: Odd Extension Of Formally Real Field

Let k be a formally real field. Then every finite extension of odd degree is a formally real field.

***Proof*:**
Grillet p.234

Using this, we can find the real closure of a formally real field:

Definition 21.1.8: Closed Real Field

Let k be a formally real field. Then k is said to be *closed* if there is no real algebraic extension L/k.

Theorem 21.1.9: Detecting Real Closure

Let k be a formally real field. Then k is real closed if and only if:

1. every positive element of k is a square in k
2. every polynomial of odd degree has a root in k

From these condition, we get that $\overline{k} = k(i)$ where $i^2 = -1$.

***Proof*:**
Grillet p.235

Due to this, a real closed field can be made into an ordered field in a unique way, namely $x \leq y$ if and only if $y - x = a^2$ for some $a \in k$. It should also be clear that the field of all algebraic real numbers is certainly a real closed field.

Corollary 21.1.10: Irreducible Elements In Real Closed Field

Let k be a real closed field. Then $f \in k[x]$ is irreducible if and only if f has degree 1 or f has degree 2 with no roots

***Proof*:**
exercise

Finally, we have:

Theorem 21.1.11: Real Closure

Every formally real field has a real closure

***Proof*:**
Grillet p.236

The reader should verify that any two real closures of k are order-field isomorphic.

Finally, we state without proof the following characterization of real-closed fields that show a naturalness to the choice of $\mathbb{R}$

Theorem 21.1.12: Artin-Schreier Theorem

Let $k \neq \overline{k}$ be a field. Then the following are equivalent:

1. k is real closed
2. $[\overline{k} : k]$ is finite
3. there is an upper bound for the degrees of irreducible polynomials in $k[x]$

> ***Proof*:**
> Grillet p.236-7

A variant of the theorem is worth recording, also without proof. Let G_K denote the absolute Galois group of a field K, and let $\sigma \in G_K$ be an involution, so that $\sigma \neq \mathrm{id}$ and $\sigma \circ \sigma = \mathrm{id}$. Then the fixed field $\overline{K}^{\sigma}$ is real closed. The two statements carry the same content, since the fixed field of an involution sits at degree 2 below $\overline{K}$ and theorem 21.1.12 then applies to it.

21.1.1 The Fields That Carry a Geometry

A vector space over an arbitrary field has no length and no angle, and section 13.6 recorded what producing them takes: an ordering, so that $\langle v, v\rangle$ can be called nonnegative, and a conjugation, so that $\langle v, v\rangle$ can be forced into the ordered field underneath. Both were described there as structure that $\mathbb{R}$ and $\mathbb{C}$ happen to carry. The results of this section say where that structure comes from, and how little freedom there was in it.

Whether a field can support a notion of positivity at all is decided by the field itself. By proposition 21.1.3, k admits a total order exactly when -1 is not a sum of squares, a condition stated in the ring operations alone with no order mentioned.

One step further up, the order and the conjugation both stop being choices. Let k be real closed. Every positive element of k is a square by theorem 21.1.9, and every nonzero square is positive by proposition 21.1.2(6), so the positive elements are exactly the nonzero squares. Hence $x \leq y$ holds if and only if $y - x$ is a square, and the order is recovered from the field operations rather than imposed on them. The same theorem gives $\overline{k} = k(i)$. Since -1 is not a square in a formally real field this extension has degree 2; it is the splitting field of $x^2 + 1$ over k, hence normal, and separable because an ordered field has characteristic zero (proposition 21.1.2). So $k(i)/k$ is Galois, $\mathrm{Gal}(\overline{k}/k)$ has order 2, and its one nontrivial element is $a + bi \mapsto a - bi$. A real closed field therefore comes with a unique order and a unique conjugation, both determined by k.

Corollary 21.1.10 supplies the third property. Over a real closed k every irreducible polynomial has degree 1 or 2, so a characteristic polynomial factors into linear and quadratic pieces over k itself and not only over $\overline{k}$.

Theorem 21.1.12 says that no other configuration of this shape exists. If $k \neq \overline{k}$ and $[\overline{k} : k]$ is finite, then k is real closed and the degree is 2. In particular no field has an algebraic closure of degree 3: a field sits at finite distance below its algebraic closure in exactly one way, and that way is the real closed one.

What none of this pins down is $\mathbb{R}$. The field of real algebraic numbers is real closed as well and satisfies every condition above, so the separation between the two is order-completeness rather than an algebraic property. Theorem 21.1.5 gives the algebraic half of that separation by placing every Archimedean field between $\mathbb{Q}$ and $\mathbb{R}$. The geometry built on the order and the conjugation (inner products, orthogonality, adjoints, the spectral theorem) is carried out over $\mathbb{R}$ and $\mathbb{C}$ in *Linear Algebra* [17].

21.2 Absolute Values on General Fields

The reader [ought to have] showed that $\mathbb{C}$ is not an ordered field. The next best thing is some notion of "size". check this! Intuitively, size is a totally ordered concept, and hence the absolute value shall be a function that will capture the intuition of size with codomain $\mathbb{R}$:

Definition 21.2.1: Absolute Value

Let k be a field. Then an *absolute value real valuation* on k is a function $v : k \to \mathbb{R}$ whose image is denoted $|x|_v$ such that

1. (positivity) $|x|_v \geq 0$ for all $x \in k$ and $|x|_v = 0$ if and only if $x = 0$
2. (multiplicative) $|xy|_v = |x|_v|y|_v$
3. (triangle inequality) $|x + y|_v \leq |x|_v + |y|_v$

If the absolute value is unambiguous, $|x|_v$ will be denoted $|x|$. If the image of v is a finitely generated abelian group, then $| - |_v$ is called a *discrete absolute value*

21.2.2 Note The reader should be careful with the definition *real valuation* as there is a different concept called *valuation* that is not a generalization o specialization of the above (see definition 27.1.1).

If the reader is familiar with norms, then this can be thought of as the field-equivalent as norms are defined on vector-spaces. In fact, taking a field to be a vector-space over itself, the existence of a absolute value induces a norm. If the reader is familiar with some Lebesgue space theory, they may recall Hölder's inequality stating that:

$$\|fg\|_1 \leq \|f\|_p\|g\|_q \qquad \frac{1}{p} + \frac{1}{q} = 1$$

which contrasts with the strict equality result given above. Recall that equality happens if and only if $\alpha|f|^p = \beta|g|^q$ for $\alpha, \beta \neq 0$, that is they are "power multiples" of each other. This is certainly the case for fields. Furthermore, there is only one absolute value instead of the 3 norms present in Hölder's inequality. A norm that satisfies the 2nd condition (on an algebra!) is called a *multiplicative norm*, and implies certain geometric invariances in the space.

Example 21.2.3: Absolute Values

1. The usual absolute values on $\mathbb{Q}, \mathbb{R}, \mathbb{C}$ all satisfy the above definition
2. For any field k, there always exists the trivial valuation

$$|x|_t = 1 \qquad x \neq 0$$

   This real valuation shall be an example of a *nonarchimedean* real valuation.

3. Let k be any field and take $k(x)$, the simple transcendental extension of k. Then there are

two interesting valuations that may be defined. The first is called the infinity valuation:

$$|f/g|_\infty = \begin{cases} 2^{\deg f - \deg g} & f \neq 0 \\ 0 & f = 0 \end{cases}$$

and the 0 valuation:

$$|f/g|_0 = \begin{cases} 2^{\operatorname{ord} f - \operatorname{ord} g} & f \neq 0 \\ 0 & f = 0 \end{cases}$$

where $\operatorname{ord}(f)$ for a polynomial $f(x) = a_0 + a_1x_1 + \cdots + a_nx^n$ to be the smallest $k \geq 0$ such that $a_k \neq 0$. The choice of 2 was arbitrary in both and can be replaced with any positive constant.

From the 0-valuation, we may define:

$$\left|\frac{f(x)}{g(x)}\right|_a = \left|\frac{f(x-a)}{g(x-a)}\right|_0$$

These valuations when restricted to k are all trivial.

With positivity and the triangle inequality, the absolute value can define a metric (see chapter C) and hence a metric topology on a field:

$$d(x, y) = |x - y|_v$$

For the p-adic case, the idea behind this metric is that $|x - y|$ is small is $x \equiv y \mod p^n$ for a large n, that is a large power of p divides the difference.

Proposition 21.2.4: Equivalent Topoloies

Let v, w be two absolutes values on a field k. Then the following are equivalent:

($\Rightarrow$) v, w induce the same topology on k

($\Rightarrow$) Given $x \in k$, $|x|_v < 1$ if and only if $|x|_w < 1$

($\Rightarrow$) There exists a $n \in k$ such that $|x|_w = |x|_v^n$ for all $x \in k$

***Proof*:**
exercise

The following can be htougt of as the CRT for real valutions

Proposition 21.2.5: Approximation Theorem

Let $|-|_1, ..., |-|_n$ be pairwise inequivalent valuations for a field K, and let $a_1, ..., a_n \in K$ be some choice elements. Then for every $\epsilon > 0$,there exists a $x \in K$ such that

$$|x - a_i|_i < \epsilon \qquad \forall 1 \leq i \leq n$$

***Proof*:**
exercise, use the negation of the above proposition (or Neukrich p.118)

As mentioned earlier, the trivial valuation is non-Archimedean. The following show that non-Archimedean valuations can be thought of as absolute values induced by *ultra-metrics*

Definition 21.2.6: Archimedean Valuation

Let v be a real valuation on a field k. Then if for all $n \in \mathbb{N}$, there exists a $x \in k$ such that

$$|n|_v \leq |x|_v$$

the valuation is called an *Archimedean valuation.* On the other hand, if there exists an x such that for all $n \in \mathbb{N}$:

$$|n|_v \leq |x|_v \qquad \forall n \in \mathbb{N}$$

Then v is called a *non-Archimedean valuation.*

Proposition 21.2.7: Characterizing Non-Archimedean Valuation

Let v be a real valuation on a field k. Then the following are equivalent:

($\Rightarrow$) v is non-Archimedean

($\Rightarrow$) $|n|_v \leq 1$ for all $n \in \mathbb{N}$

($\Rightarrow$) $|n|_v \leq 1$ for all $n \in \mathbb{Z}$

($\Rightarrow$) for all $x, y \in k$,
$$|x+y|_v \leq \max(|x|_v, |y|_v)$$

The final inequality upgrading the triangle inequality is called the *strong triangle inequality* or *ultra-metric inequality.*

Note that if $|x| \neq |y|$ in an ultra-metric , then $|x+y| = \max(|x|, |y|)$!

***Proof*:**

(1) $\Rightarrow$ (2) Let v be a non-Archimedean valuation. For the sake of contradiction, suppose that for some $n \in \mathbb{N}$, $|n|_v > 1$. Then by multiplicity we may have an $|n|_v$ with an arbitrarily large valuation:

$$|n^k| = |n|_v^k \xrightarrow{k \to \infty} \infty$$

Which would imply there is no $x \in k$ such that $|n|_v \leq |x|_v$ for all $n \in \mathbb{N}$.

(2) $\Rightarrow$ (3) show that $|-1| = |1|$ and use this to conclude.

(3) $\Rightarrow$ (4) Take:

$$\begin{aligned} |x+y|^n &= \left|\sum_k^n \binom{n}{k} x^k y^{n-k}\right| \\ &\leq \sum_k^n \left|\binom{n}{k}\right| |x|^k |y|^{n-k} \\ &\leq \sum_k^n |x|^k |y|^{n-k} \\ &\leq (n+1)\max(|x|^n, |y|^n) \end{aligned}$$

Hence

$$|x+y| \leq (n+1)^{1/n} \max(|x|, |y|)$$

As this is true for all n, and $(n+1)^{1/n} \xrightarrow{n\to\infty} 1$, we get the desired result

(4) $\Rightarrow$ (1) by induction we get:

$$|n| = |1+1+\cdots+1| \leq |1| = 1$$

giving the desired result.

Corollary 21.2.8: Values On different characteristic

Let k be a finite with characteristic $p \neq 0$. Then there does not exist an Archimedean valuation on k.

If k is a finite field, the only valuation is the trivial valuation.

***Proof*:**
exercise.

In the case of characteristic 0, given $k = \mathbb{Q}$, all the absolute values may be classified. Define p-valuation to be:

$$v_p(m/n) = \begin{cases} p^{-k} & \frac{m}{n} = p^k \frac{m'}{n'} \neq 0,\ p \nmid m', n' \\ 0 & \frac{m}{n} = 0 \end{cases}$$

Then

Theorem 21.2.9: Classification Of Absolute Values On Rationals

Let $k = \mathbb{Q}$. Then every non-trivial valuation on k is either a p-valuation or the usual absolute value.

This result is due to Ostrowski.

***Proof*:**
Let v be a nonarchemidian valuation. Then $|n|_v \leq 1$ for all $n \in \mathbb{Z}$. If $|n|_v = 1$ for all $0 \neq n \in \mathbb{Z}$ then

$$|m/n|_v |n|_v = |m|_v = 1$$

which can be sued to show that $|x|_v = 1$ for all $x \in \mathbb{Q}$. Thus, the following set is non-empty:

$$P = \{x \in \mathbb{Z} \mid |n|_v < 1\}$$

This can be shown to be an prime ideal of $\mathbb{Z}$ (exercise), and since $\mathbb{Z}$ is a PID we have that $P = (p)$ for appropriate prime $p \in \mathbb{Z}$. Let

$$c = |p|_v < 1$$

Now, if

$$\frac{m}{n} = p^k \frac{m'}{n'} \neq 0$$

where $p \nmid m', n'$, then by definition of the prime ideal:

$$|m'|_v = |n'|_v = 1 \qquad \Rightarrow \qquad n \left| \frac{m}{n} \right| = c^k$$

This valuation is not equivalent to v_p. Note that v_p and v_q are not equivalent as that would imply $|p|_v = 1$ (exercise).

The Archimedean is left as a real-analysis exercise, or see Grillet p.242,

In *Commutative Algebra* [10] (valuation rings and DVRs), many of these concepts shall be generalized to more general valuations, namely we have been working with *real* valuations, however there are more generally G-valuations for totally ordered groups G.

In an analysis class, the absolute value is used to define the completion of a ring. If k is a field with an absolute value, then the completion $\widehat{k}$ of k is the collection of Cauchy-sequences under the appropriate quotient where k is dense in $\widehat{k}$, and the absolute value n extends naturally to $\widehat{v}$ in $\widehat{k}$.

21.2.1 Extending Valuation

Eventually here

21.2.2 Ostrowski Theorem

Show that if $(K, |-|)$ is a field with an Archimedean absolute value, then

$$\widehat{K} \underset{\sim}{\in} \{\mathbb{R}, \mathbb{C}\}$$

Neukirch p.124

21.3 Derivation on Field

Derivations are the algebraic form of differentiation, and the module of relative differential forms packages every derivation of a ring into a single universal one. The construction is pure commutative algebra and belongs here rather than in any geometry book: it is consumed by [27] for the canonical class of a curve, by [5] for the sheaf of differentials, and by [4] for invariant differentials on a group variety.

Definition 21.3.1: A-derivation

Let A be a commutative ring, B an A-algebra, and M a B-module. Then an A-derivation of B into M is a map $d : B \to M$ such that

1. $d(b+b') = d(b) + d(b')$
2. $d(bb') = d(b)b' + bd(b')$
3. $d(a) = 0$ for all $a \in A$

Definition 21.3.2: Module of Relative Differential Forms

Let A be a commutative ring, B an A-algebra. Then a module of relative differential form is a B-module $\Omega_{B/A}$ with an A-derivation $d : B \to \Omega_{B/A}$ which satisfies the universal property of relative differential forms: if M is another B-module and $d' : B \to M$ an A-derivation, then there is a unique B-module homomorphism $f : \Omega_{B/A} \to M$ such that the following diagram commutes:

$$\begin{tikzcd} B \arrow[r, "d"] \arrow[dr, "d'"'] & \Omega_{B/A} \arrow[d, "f"] \\ & M \end{tikzcd}$$

Certainly $\Omega_{B/A}$ exists via the usual construction: take the free B-module F generated by the primitives $d(b)$ for $b \in B$ and quotient by the submodule generated by the relationships:

$$d(b+b') - d(b) - d(b') \quad \text{and} \quad d(bb') - d(b)b' - bd(b') \quad \text{and} \quad d(a)$$

Then the map $d : B \to \Omega_{B/A}$ is given by $b \mapsto d(b)$. Then it is an exercise in algebra to see that this satisfies the universal property and $\left(\Omega_{B/A}, d\right)$ is unique up to unique isomorphism. The following gives an alternative insightful construction:

Proposition 21.3.3: Natural Differential Structure

Let B be an A-algebra, $\delta : B \otimes_A B \to B$ the diagonal map given by $f(b \otimes b') = bb'$, and let $I = \ker \delta$. Then I/I^2 has the structure of a left B-module, and we can define a map

$$d : B \to I/I^2 \qquad d(b) = 1 \otimes b - b \otimes 1 \bmod I^2$$

In which case $(I/I^2, d)$ is a module of relative differential for B/A.

***Proof*:**
Write $\delta \colon B \otimes_A B \to B$ for the multiplication map and $I = \ker \delta$. Give I/I^2 the B-module structure coming from the left factor, $b \cdot (x \otimes y) = (bx) \otimes y$; the two possible structures agree modulo I^2, since $b \otimes 1 - 1 \otimes b \in I$.

d is an A-derivation. Additivity is clear. For the Leibniz rule, compute in $B \otimes_A B$:

$$d(bb') - b\,d(b') - b'\,d(b) = \big(1 \otimes bb' - bb' \otimes 1\big) - b\big(1 \otimes b' - b' \otimes 1\big) - b'\big(1 \otimes b - b \otimes 1\big)$$

which expands to $-(1 \otimes b - b \otimes 1)(1 \otimes b' - b' \otimes 1)$, a product of two elements of I and so zero in I/I^2. And $d(a) = 1 \otimes a - a \otimes 1 = 0$ for $a \in A$, the tensor being over A.

Universality. Let $d' \colon B \to M$ be an A-derivation. Define $B \otimes_A B \to M$ by $x \otimes y \mapsto x\,d'(y)$, which is A-bilinear and hence well defined. It kills I^2: for $u = \sum x_i \otimes y_i$ and $v = \sum x'_j \otimes y'_j$ in I, meaning $\sum x_i y_i = 0$ and $\sum x'_j y'_j = 0$, expanding uv and applying the Leibniz rule for d' leaves $\big(\sum x_i y_i\big)\big(\sum x'_j d'(y'_j)\big) = 0$. So it descends to $f \colon I/I^2 \to M$, and $f(d(b)) = f(1 \otimes b - b \otimes 1) = d'(b) - b\,d'(1) = d'(b)$ since $d'(1) = 0$. Uniqueness holds because the $d(b)$ generate I/I^2: any $\sum x_i \otimes y_i \in I$ equals $\sum x_i(1 \otimes y_i - y_i \otimes 1)$ modulo nothing, using $\sum x_i y_i = 0$.

So $(I/I^2, d)$ has the universal property of definition 21.3.2, completing the proof.

Example 21.3.4: Relative Differential

1. Take $A = \mathbb{R}$ and $B = \mathbb{R}[x]$. Then $I = \ker \delta$. Then as $\mathbb{R}[x]$ is an integral domain, it can be shown that any element is a linear combination of elements of the form:

$$f(x) \otimes g(x) - f(x)g(x) \otimes 1 \in I$$

Then it should be verified any element in I can be written as:

$$f(x)(1 \otimes g(x) - g(x) \otimes 1)$$

The reader should then verify that I/I^2 is generated by $f'(x)(x \otimes 1 - 1 \otimes x)$[a]. Let dx represent this element. Notice that this is the image of $d(x)$. Now, let's consider:

$$d(f(x) = d\left(\sum_i a_i x^i\right) = \sum_i a_i d(x^i)$$

Hence, let's compute $d(x^i)$. For $i = 2$, we get:

$$d(x^2) = d(x \cdot x) = xd(x) - xd(x) = 2xdx$$

Repeating this pattern, we can see that:

$$d(x^n) = nx^{n-1}dx$$

2. For example, taking $B = \mathbb{R}[x_1, ..., x_n]$, Then $\Omega_{B/A}$ is a free B-module generated by $dx_1, ..., dx_n$.

3. In proposition 21.3.10, we shall see that if E/F is a finitely generated separable extension (of characteristic 0), then

$$\Omega_{B/A} = \bigoplus_{b \in T} B db$$

where T is a transcendental A-basis for B.

4. What if we take $B = A[\epsilon]/(\epsilon^2)$?

5. Let $A = \mathbb{R}$ and $B = C^\infty(M)$ for some smooth n-manifold M. Then $C^\infty(M)$. Then $\Omega_{C^\infty(M)/\mathbb{R}}$ is a locally free $C^\infty(M)$-module (based off of the patches of M), consisting of the space of sections of the cotangent bundle T^*M.

[a]Hint: what is $(1 \otimes x^2 - x^2 \otimes 1)$. Repeat for higher powers

Proposition 21.3.5: Base Change For Differentials

Let A', B be A-algebras and $B' = B \otimes_A A'$. Then:

$$\Omega_{B'/A'} \cong \Omega_{B/A} \otimes_B B'$$

Furthermore, if S is a multiplicative subset of of S, then:

$$\Omega_{S^{-1}B/A} \cong S^{-1}\Omega_{B/A}$$

***Proof*:**
Both statements are checked against the universal property of definition 21.3.2.

For the base change, a B'-module M receives an A'-derivation from B' exactly when it receives an A-derivation from B that is A'-linear on the second factor, since $B' = B \otimes_A A'$ is generated over A' by the image of B. So $\mathrm{Der}_{A'}(B', M) \cong \mathrm{Der}_A(B, M)$ naturally, and representing both sides gives $\Omega_{B'/A'} \cong \Omega_{B/A} \otimes_B B'$.

For the localization, an A-derivation $d\colon B \to M$ into an $S^{-1}B$-module extends uniquely to $S^{-1}B$ by the quotient rule $d(b/s) = \big(s\,d(b) - b\,d(s)\big)/s^2$, which is forced by the Leibniz rule applied to $b = (b/s) \cdot s$. So $\mathrm{Der}_A(S^{-1}B, M) \cong \mathrm{Der}_A(B, M)$ for such M, and representing gives $\Omega_{S^{-1}B/A} \cong S^{-1}\Omega_{B/A}$, completing the proof.

Proposition 21.3.6: Differentials and Exact Sequence I

Let $A \to B \to C$ be a (not necessarily exact nor a complex) sequence of rings. Then there is an exact sequence of C-modules:

$$\Omega_{B/A} \otimes_B C \to \Omega_{C/A} \to \Omega_{C/B} \to 0$$

***Proof*:**
The maps are the natural ones: $db \otimes c \mapsto c\,db$ and $dc \mapsto dc$, the latter making sense because an A-derivation of C is in particular one killing the image of A.

Exactness is checked by applying $\mathrm{Hom}_C(-, M)$ for an arbitrary C-module M and using definition 21.3.2, which turns the sequence into

$$0 \to \mathrm{Der}_B(C, M) \to \mathrm{Der}_A(C, M) \to \mathrm{Der}_A(B, M)$$

This is exact: a derivation of C over A kills the image of B exactly when it is a derivation over B, which is exactness at the middle term and injectivity on the left. Since $\mathrm{Hom}_C(-, M)$ is left exact and the resulting sequence is exact for every M, the original sequence is exact at $\Omega_{C/A}$ and at $\Omega_{C/B}$, completing the proof.

Proposition 21.3.7: Differentials and Exact Sequence II

Let B be an A-algebra, I be an ideal of B, and $C = B/I$. Then there is a natural exact sequence of C-modules

$$I/I^2 \xrightarrow{\delta} \Omega_{B/A} \otimes_B C \to \Omega_{C/A} \to 0,$$

where for any $b \in I$, if $\bar{b}$ is its image in I/I^2, then $\delta\bar{b} = db \otimes 1$. In particular, I/I^2 has a natural structure of a C-module, and δ is a C-linear map, even though it is defined via the derivation d.

***Proof*:**
First, δ is well defined and C-linear. For $b, b' \in I$ the Leibniz rule gives $d(bb') = b\,db' + b'\,db$, which lies in $I\,\Omega_{B/A}$ and so vanishes in $\Omega_{B/A} \otimes_B C$; hence δ kills I^2. C-linearity follows because multiplying b by an element of I again lands in I^2.

For exactness, apply $\mathrm{Hom}_C(-, M)$ as in proposition 21.3.6. The sequence becomes

$$0 \to \mathrm{Der}_A(C, M) \to \mathrm{Der}_A(B, M) \to \mathrm{Hom}_C(I/I^2, M)$$

An A-derivation of B into M, where M is a C-module and so killed by I, descends to $C = B/I$ exactly when it vanishes on I; and its restriction to I is precisely the composite with δ. That is exactness at the middle and injectivity on the left, and as before it holds for every M, giving the stated sequence, completing the proof.

Corollary 21.3.8: Finite Generation of Differential Forms

If B is a finitely generated A-module or a localization of a finitely generated A-module, then $\Omega_{B/A}$ is a finitely generated B-module

***Proof*:**
As B is a finitely generated A-module, it is the quotient of polynomial rings. Hence, it follows from our direct computations from example 21.3.4 and, proposition 21.3.7, and the localization result comes from proposition 21.3.5.

For finite extensions the module of differentials sees exactly one thing, namely separability, and it sees it sharply: a nonzero derivation exists precisely when some element of the extension fails to be separable. The mechanism is already visible on a simple extension $k(\alpha)$, where a k-derivation is pinned down by its value on α and that value is constrained only through the single scalar $f'(\alpha)$.

Lemma 21.3.9: Derivations Detect Separability

Let F/k be a finite field extension. Then the following are equivalent:

1. F/k is a separable extension
2. $\Omega_{F/k} = 0$
3. the only k-derivation of F into F is the zero map

***Proof*:**

($1 \Rightarrow 2$) Let M be any F-module and $D : F \to M$ a k-derivation. Since F/k is finite and separable, theorem 19.6.20 gives $F = k(\beta)$ for some $\beta \in F$, and β is separable over k by definition 19.6.3. Write $g(x) = m_{\beta,k}(x) = \sum_j c_j x^j$. Additivity and the Leibniz rule give $D(\beta^j) = j\beta^{j-1}D(\beta)$, and D kills each $c_j \in k$, so

$$0 = D\big(g(\beta)\big) = \sum_j c_j D(\beta^j) = g'(\beta)D(\beta)$$

Now β is a root of the separable polynomial g, hence not a multiple root, so $g'(\beta) \neq 0$ by proposition 19.6.6. As $g'(\beta)$ is a unit in the field F, this forces $D(\beta) = 0$. Every element of F is a polynomial in β with coefficients in k by proposition 19.2.24, so additivity and the Leibniz rule propagate $D(\beta) = 0$ to $D = 0$. Thus $\mathrm{Der}_k(F, M) = 0$ for every F-module M.
Apply this to $M = \Omega_{F/k}$: the universal derivation d of definition 21.3.2 is itself zero. Both the identity and the zero map $\Omega_{F/k} \to \Omega_{F/k}$ are F-module homomorphisms f with $f \circ d = d$, so the uniqueness clause of definition 21.3.2 forces them to agree, giving $\Omega_{F/k} = 0$.

($2 \Rightarrow 3$) By definition 21.3.2 the k-derivations of F into F are in bijection with the F-module homomorphisms $\Omega_{F/k} \to F$, and the zero module admits only the zero homomorphism.

(3 ⇒ 1) Take the contrapositive: assume F/k is inseparable and produce a nonzero k-derivation of F into F. Since F/k is finite, write $F = k(\alpha_1, ..., \alpha_n)$ and set $F_0 = k$ and $F_i = F_{i-1}(\alpha_i)$, so that $F_n = F$.

First, some α_i is inseparable over F_{i-1}. If instead every α_i were separable over F_{i-1}, then $[F_i : F_{i-1}]_s = [F_i : F_{i-1}]$ for each i by proposition 19.6.15, and multiplying these together using corollary 19.6.16 on the left and theorem 19.3.7 on the right would give $[F : k]_s = [F : k]$, whence F/k is separable by proposition 19.6.17, contrary to assumption. So let i be the *largest* index for which α_i is inseparable over F_{i-1}, and abbreviate

$$E = F_{i-1} \qquad \alpha = \alpha_i \qquad L = F_i = E(\alpha)$$

By maximality of i, each α_j with $j > i$ is separable over F_{j-1}; running the same product over the tower $L = F_i \subseteq \cdots \subseteq F_n = F$ gives $[F : L]_s = [F : L]$, so F/L is a finite separable extension, again by proposition 19.6.17.

A nonzero E-derivation of L. Let $f(x) = m_{\alpha,E}(x)$. It is irreducible, and it is inseparable because α is inseparable over E (definition 19.6.3), so proposition 19.6.7 gives $f'(x) = 0$. Reading off the leading term of f', this already forces $\mathrm{ch}(k) = p > 0$ and $f \in E[x^p]$, exactly as in the discussion preceding definition 19.6.9; in characteristic 0 there is nothing to prove, since corollary 19.6.8 rules out inseparable irreducibles outright. Regard L as an $E[x]$-module through the evaluation map $\mathrm{ev}_\alpha : E[x] \to L$ of theorem 19.2.8, whose kernel is (f), and consider

$$\partial : E[x] \to L \qquad \partial(h) = h'(\alpha)$$

with h' the formal derivative of definition 19.6.5. It is additive, it kills E, and the Leibniz rule for the formal derivative gives

$$\begin{aligned}\partial(h_1h_2) = (h_1h_2)'(\alpha) &= h_1'(\alpha)h_2(\alpha) + h_1(\alpha)h_2'(\alpha)\\ &= \partial(h_1)h_2(\alpha) + h_1(\alpha)\partial(h_2)\end{aligned}$$

so ∂ is an E-derivation in the sense of definition 21.3.1. It kills the ideal (f): for any $h \in E[x]$,

$$\partial(fh) = (fh)'(\alpha) = f'(\alpha)h(\alpha) + f(\alpha)h'(\alpha) = 0$$

because $f' = 0$ and $f(\alpha) = 0$. Hence ∂ factors through ev_α as an additive map $D : L \to L$, and the displayed Leibniz identity descends, since both sides depend only on the values $h_1(\alpha), h_2(\alpha)$. So D is an E-derivation of L into L with

$$D(\alpha) = \partial(x) = 1$$

and in particular $D \neq 0$.

Extending D across F/L. Since F/L is finite and separable, theorem 19.6.20 gives $F = L(\beta)$ with β separable over L. Put $g(x) = m_{\beta,L}(x)$; as in (1 ⇒ 2), $g'(\beta) \neq 0$ by proposition 19.6.6. For $h(x) = \sum_j c_j x^j \in L[x]$ write $h^D(x) = \sum_j D(c_j)x^j$; expanding a product coefficientwise and applying the Leibniz rule for D to each term gives

$$(h_1h_2)^D = h_1^D h_2 + h_1 h_2^D$$

Set
$$t = -\frac{g^D(\beta)}{g'(\beta)} \in F$$
and regard F as an $L[x]$-module through evaluation at β. Define
$$\Delta : L[x] \to F \qquad \Delta(h) = h^D(\beta) + h'(\beta)\, t$$
Then Δ is additive, and combining the two product rules just recorded,
$$\begin{aligned}\Delta(h_1 h_2) &= h_1^D(\beta)h_2(\beta) + h_1(\beta)h_2^D(\beta) + \big(h_1'(\beta)h_2(\beta) + h_1(\beta)h_2'(\beta)\big)t \\ &= \Delta(h_1)h_2(\beta) + h_1(\beta)\Delta(h_2)\end{aligned}$$
On a constant $c \in L$ we get $\Delta(c) = D(c)$, so Δ vanishes on E and a fortiori on k. By the choice of t,
$$\Delta(g) = g^D(\beta) + g'(\beta)t = 0$$
and therefore $\Delta(gh) = \Delta(g)h(\beta) + g(\beta)\Delta(h) = 0$ for every $h \in L[x]$, since $g(\beta) = 0$. So Δ kills the kernel (g) of evaluation at β and descends, by theorem 19.2.8, to an additive map $\widetilde{D} : F \to F$; as before the Leibniz identity descends with it. Thus $\widetilde{D}$ is a k-derivation of F into F restricting to D on L, and
$$\widetilde{D}(\alpha) = D(\alpha) = 1 \neq 0$$
So F carries a nonzero k-derivation into itself, which is the contrapositive of $(3 \Rightarrow 1)$, completing the proof.

Note that the primitive element theorem is used only where it is legitimate, on the separable extension F/L; it says nothing about F/k itself, which by example 19.6.22 need not be simple once separability fails. The whole content of $(3 \Rightarrow 1)$ is that an inseparable step in a tower is a step at which a derivation may be defined freely, and the remaining steps are separable and so carry the derivation along uniquely.

Proposition 21.3.10: Differentials and Field Extensions

Let K/k be a finitely generated field extension. Then:
$$\dim_K \Omega_{K/k} \geq \operatorname{trdeg} K/k$$
where equality holds iff K is separably generated over k.

Proof Key ideas:
This is the separability criterion for field extensions in differential form, and both implications reduce to the exact sequences above: for a separably generated extension a transcendence basis gives a basis of Ω, and conversely a nonzero differential of a purely inseparable generator is killed by the p-th-power relation. A full account is in [31, chapter II.8]; the statement is recorded here because [27] and [5] both use it to detect smoothness.

The finite algebraic case is the one this book uses most often, and it is not left to a citation:

lemma 21.3.9 proves that a finite extension K/k has $\Omega_{K/k} = 0$ exactly when it is separable. Note that the transcendence-degree statement above genuinely needs the finitely generated hypothesis, whereas the finite case does not.

Proposition 21.3.11: Local Ring and Relative Differential

Let B be a local ring containing a field k isomorphic to $B/\mathfrak{m}$. Then given the map δ from proposition 21.3.7:

$$\delta : \mathfrak{m}/\mathfrak{m}^2 \xrightarrow{\cong} \Omega_{B/k} \otimes_B k$$

***Proof*:**
By proposition 21.3.7 $\operatorname{coker}(\delta) = 0$, so δ is surjective. To show that δ is injective, we shall show the following dual vector space map is surjective:

$$\delta' : \operatorname{Hom}_k(\Omega_{B/k} \otimes k, k) \to \operatorname{Hom}_k(\mathfrak{m}/\mathfrak{m}^2, k)$$

The term on the left is isomorphic to $\operatorname{Hom}_B(\Omega_{B/k}, k)$, which by definition of the differentials, can be identified with the set $\operatorname{Der}_k(B, k)$ of k-derivations of B to k. If $d : B \to k$ is a derivation, then $\delta'(d)$ is obtained by restricting to $\mathfrak{m}$, and noting that $d(\mathfrak{m}^2) = 0$. Now, to show that δ' is surjective, let $h \in \operatorname{Hom}(\mathfrak{m}/\mathfrak{m}^2, k)$. For any $b \in B$, we can write $b = \lambda + c$, $\lambda \in k$, $c \in \mathfrak{m}$, in a unique way. Define $db = h(\bar{c})$, where $\bar{c} \in \mathfrak{m}/\mathfrak{m}^2$ is the image of c. Then one verifies immediately that d is a k-derivation of B to k, and that $\delta'(d) = h$. Thus δ' is surjective, as required.

We finish off this brief summary of the essential results about relative differentials by showing the relation between regular local rings and relative differentials. We require the following lemma:

Lemma 21.3.12: Characterizing Free Modules using Residue and Quotient Field

Let A be a Noetherian local integral domain, with residue field k and quotient field K. Then if M is a finitely generated A-module and $\dim_k M \otimes_A k = \dim_k M \otimes_A K = r$, then M is free of rank r.

***Proof*:**
Since $\dim_k M \otimes k = r$, by Nakayama's lemma M can be generated by r elements. Hence, there is a surjective map $\varphi : A^r \to M$. Let R be its kernel. Then we obtain an exact sequence

$$0 \to R \otimes K \to K^r \to M \otimes K \to 0,$$

As $\dim_K M \otimes K = r$, we have $R \otimes K = 0$. But R is torsion-free, so $R = 0$, and M is isomorphic to A^r, completing the proof.

22⋆

Bilinear, Quadratic and Alternating Forms over a Field

⋆ *Optional. This chapter is an aside; nothing later in the book depends on it.*

A bilinear form is a piece of data attached to a vector space, and section 13.3 said what that data is: a matrix, once a basis is fixed. It did not say when two such data are the same. Two forms should count as the same when some change of basis carries one to the other, and the matrix statement of that relation is $A \mapsto P^t AP$ rather than the $A \mapsto P^{-1}AP$ that governs linear operators. The two relations are genuinely different, and the invariants that survive the first are not the eigenvalues.

This chapter classifies bilinear forms over a field up to that relation, in the two cases that occur: the symmetric forms, where the answer is subtle and field-dependent, and the alternating ones, where it is not. The answer is controlled by a single group built from the field, the square classes $k^\times/(k^\times)^2$, and this is why the subject sits after Galois theory: for a field of characteristic other than 2 that group is exactly what Kummer theory at $n = 2$ (section 20.4) classifies. Over $\mathbb{C}$ the group is trivial and the classification collapses to the rank; over $\mathbb{R}$ it has two elements and one further invariant appears; over a general field it can be arbitrarily complicated, and the bookkeeping of what survives is the Witt group.

22.1 Quadratic Forms and Symmetric Bilinear Forms

The forms of interest are the symmetric ones, and over a field where 2 is invertible a symmetric bilinear form carries exactly the same information as the single-variable function $x \mapsto \varphi(x, x)$. That function is easier to write down and is what one meets in practice, so the correspondence between the two is recorded first. It fails in characteristic 2, and the failure is not a technicality to be waved past: it is why the whole theory below carries the hypothesis $\operatorname{char} k \neq 2$.

Definition 22.1.1: Quadratic Form over a Field

Let k be a field and V a finite-dimensional k-vector space. A *quadratic form* on V is a function $q\colon V \to k$ such that

1. $q(\lambda v) = \lambda^2 q(v)$ for all $\lambda \in k$ and $v \in V$;
2. the function $b_q\colon V \times V \to k$ defined by
$$b_q(x, y) := q(x + y) - q(x) - q(y)$$
is bilinear.

The form b_q is the *polar form* of q. It is symmetric, since the defining expression is unchanged by exchanging x and y.

Two consequences of the definition are used constantly and are worth extracting at once. Setting $y = x$ in the polar form and applying (1) gives
$$b_q(x, x) = q(2x) - 2q(x) = 4q(x) - 2q(x) = 2q(x)$$
so the polar form recovers q after division by 2, and only then. And if φ is any symmetric bilinear form, the function $q_\varphi(x) := \varphi(x, x)$ satisfies both conditions above, with $b_{q_\varphi} = 2\varphi$ by the same computation. The two constructions are therefore mutually inverse exactly when 2 is invertible.

Proposition 22.1.2: Quadratic Forms and Symmetric Bilinear Forms Correspond

Let k be a field with $\operatorname{char} k \neq 2$ and let V be a finite-dimensional k-vector space. The assignments
$$\varphi \longmapsto q_\varphi, \qquad q \longmapsto \tfrac{1}{2} b_q$$
are mutually inverse bijections between the set of symmetric bilinear forms on V and the set of quadratic forms on V.

***Proof*:**
Step 1: the assignments are well defined. Let φ be symmetric bilinear and put $q_\varphi(x) = \varphi(x, x)$. Then $q_\varphi(\lambda v) = \varphi(\lambda v, \lambda v) = \lambda^2 \varphi(v, v) = \lambda^2 q_\varphi(v)$, which is condition (1). For condition (2), expanding by bilinearity and using symmetry,
$$b_{q_\varphi}(x, y) = \varphi(x + y, x + y) - \varphi(x, x) - \varphi(y, y) = 2\varphi(x, y)$$
which is bilinear. So q_φ is a quadratic form and $b_{q_\varphi} = 2\varphi$. Conversely, if q is a quadratic form then b_q is symmetric bilinear by definition 22.1.1, hence so is $\frac{1}{2} b_q$, using $\operatorname{char} k \neq 2$ to divide.

Step 2: the composites are the identity. Starting from a symmetric φ, Step 1 gives $\frac{1}{2} b_{q_\varphi} = \frac{1}{2} \cdot 2\varphi = \varphi$. Starting from a quadratic form q and writing $\varphi = \frac{1}{2} b_q$, the computation recorded before the proposition gives $b_q(x, x) = 2q(x)$, so
$$q_\varphi(x) = \varphi(x, x) = \tfrac{1}{2} b_q(x, x) = \tfrac{1}{2} \cdot 2q(x) = q(x)$$
Both composites are the identity, so the assignments are mutually inverse bijections, as we sought to show.

Because of the proposition, the words *quadratic form* and *symmetric bilinear form* are used interchangeably below whenever $\operatorname{char} k \neq 2$, and disc, orthogonality and the matrix of a form are attached to either at will. The next example shows what is lost when the hypothesis is dropped, and it is the reason the hypothesis is carried explicitly rather than assumed silently.

Example 22.1.3: Polarization Fails in Characteristic Two

Let $k = \mathbb{F}_2$ and $V = k^2$ with coordinates (x, y).

The map $q \mapsto b_q$ is not injective. Take $q(x, y) = x^2$. Then $q(\lambda v) = \lambda^2 q(v)$ holds, and

$$b_q\big((x, y), (x', y')\big) = (x + x')^2 - x^2 - x'^2 = 2xx' = 0$$

since $2 = 0$ in k. So q and the zero form have the same polar form, yet $q(1, 0) = 1 \neq 0$. More generally $b_q(v, v) = 2q(v) = 0$ for every quadratic form in characteristic 2, so every polar form is alternating and the diagonal information in q is invisible to b_q. Throughout this example *alternating* means the vanishing condition of definition 16.1.3, that $\varphi(v, v) = 0$ for every v. In characteristic 2 that is strictly stronger than the sign rule appearing in definition 13.3.15, since there $-1 = 1$ makes the sign rule say nothing beyond symmetry.

The map $q \mapsto b_q$ is not surjective either. The form $\varphi\big((x, y), (x', y')\big) = xx'$ is symmetric bilinear, but $\varphi\big((1, 0), (1, 0)\big) = 1 \neq 0$, so φ does not vanish on the diagonal and is therefore not the polar form of anything.

Orthogonal bases can fail to exist. Let $\psi\big((x, y), (x', y')\big) = xy' + x'y$, with matrix $\left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$ in the standard basis. It is symmetric, nondegenerate, and nonzero, since $\psi(e_1, e_2) = 1$. But $\psi(v, v) = 2xy = 0$ for every v. Were w_1, w_2 an orthogonal basis, the matrix of ψ in it would be $\operatorname{diag}\big(\psi(w_1, w_1), \psi(w_2, w_2)\big) = \operatorname{diag}(0, 0)$, forcing $\psi = 0$ and contradicting $\psi(e_1, e_2) = 1$. So ψ admits no orthogonal basis at all. Contrast $\varphi(u, v) = xx'$ above, which is already diagonal in the standard basis: failing to be a polar form and failing to be diagonalizable are different failures, and ψ exhibits the second.

All three failures trace to the division by 2 in proposition 22.1.2. In characteristic 2 the quadratic and the bilinear theories separate, and the correct objects are quadratic forms together with their alternating polar forms, and that theory is not developed here.

The classification question can now be posed precisely. Fixing a basis turns a form into a matrix (definition 13.3.18), and changing the basis changes the matrix in a way that must be pinned down before any invariant can be extracted.

Theorem 22.1.4: Change of Basis for the Matrix of a Form

Let V be a finite-dimensional k-vector space with bases $\beta = (\beta_1, \ldots, \beta_n)$ and $\beta' = (\beta'_1, \ldots, \beta'_n)$, and let $\varphi\colon V \times V \to k$ be a bilinear form. Let P be the $n \times n$ matrix whose j-th column is $[\beta'_j]_\beta$, the coordinate vector of β'_j in the basis β (this is the change of basis matrix of definition 13.2.11). Then P is invertible, $[v]_\beta = P[v]_{\beta'}$ for every $v \in V$, and

$$[\varphi]_{\beta'} = P^t \, [\varphi]_\beta \, P$$

***Proof*:**
The columns of P are the coordinate vectors of a basis, hence linearly independent, so P is invertible. Writing $v = \sum_j c_j \beta'_j$, that is $[v]_{\beta'} = (c_j)_j$, linearity of the coordinate map gives

$$[v]_\beta = \Big[\sum_j c_j \beta'_j\Big]_\beta = \sum_j c_j [\beta'_j]_\beta = P[v]_{\beta'}$$

which is the second claim. For the third, write $A = [\varphi]_\beta$. By lemma 13.3.19, for all $v, w \in V$

$$\varphi(v,w) = [v]_\beta^t A [w]_\beta = \big(P[v]_{\beta'}\big)^t A \big(P[w]_{\beta'}\big) = [v]_{\beta'}^t \big(P^t A P\big)[w]_{\beta'}$$

Taking $v = \beta'_i$ and $w = \beta'_j$ makes the coordinate vectors the standard basis vectors e_i and e_j, so the displayed expression is the (i,j) entry of P^tAP. On the other hand that value is $\varphi(\beta'_i, \beta'_j)$, which is the (i,j) entry of $[\varphi]_{\beta'}$ by definition 13.3.18. The two matrices agree entrywise, completing the proof.

Definition 22.1.5: Congruent Matrices

Two matrices $A, B \in M_n(k)$ are *congruent* if $B = P^tAP$ for some invertible $P \in M_n(k)$. Congruence is an equivalence relation: it is reflexive by $P = I$, symmetric because $A = (P^{-1})^t B P^{-1}$, and transitive because $(PQ)^t A (PQ) = Q^t(P^tAP)Q$.

Theorem 22.1.4 says that the matrices of one form in the various bases of V form exactly one congruence class, so a property of matrices is a property of the form precisely when it is congruence-invariant. Similarity, the relation $A \mapsto P^{-1}AP$ appropriate to linear operators, is a different relation, and the invariants of a form are not those of an operator: the determinant, for instance, is not congruence-invariant, but the next definition extracts from it something that is.

Definition 22.1.6: The Square Classes of a Field

Let k be a field. The squares $(k^\times)^2 = \{c^2 \mid c \in k^\times\}$ form a subgroup of the abelian group $k^\times$, and the quotient

$$k^\times/(k^\times)^2$$

is the group of *square classes* of k. Every element has order dividing 2, so it is a vector space over $\mathbb{F}_2$. The class of $a \in k^\times$ is written $a\,(k^\times)^2$, and two elements lie in the same class exactly when their ratio is a square.

The square classes are where the classification lives, and their size measures how far the field is from having all square roots. For $k = \mathbb{C}$, or any algebraically closed field, every element is a square and the group is trivial. For $k = \mathbb{R}$ it has order 2, the classes being the positive and the negative reals. For $k = \mathbb{Q}$ it is infinite. The relevance to forms is immediate from theorem 22.1.4.

Definition 22.1.7: Discriminant of a Nondegenerate Form

Let φ be a nondegenerate symmetric bilinear form (definition 13.3.22) on a finite-dimensional k-vector space V, and let β be any basis. The *discriminant* of φ is the square class

$$\operatorname{disc}(\varphi) := \det\big([\varphi]_\beta\big)\,(k^\times)^2 \in k^\times/(k^\times)^2$$

Proposition 22.1.8: The Discriminant Is Well Defined and Is an Invariant

The class $\operatorname{disc}(\varphi)$ of definition 22.1.7 does not depend on the basis β, and congruent forms have equal discriminants.

***Proof*:**
Nondegeneracy makes $\det([\varphi]_\beta) \neq 0$ by lemma 13.3.23, so the class is taken in $k^\times$ and the definition makes sense. If β' is a second basis then $[\varphi]_{\beta'} = P^t[\varphi]_\beta P$ by theorem 22.1.4, and the multiplicativity of the determinant together with $\det(P^t) = \det(P)$ gives

$$\det\big([\varphi]_{\beta'}\big) = (\det P)^2 \det\big([\varphi]_\beta\big)$$

The two determinants differ by the square $(\det P)^2$ of the nonzero scalar $\det P$, hence lie in the same square class. The same computation, read as a statement about congruent matrices rather than about two bases, gives the second claim, as we sought to show.

22.1.9 Convention: the discriminant is unsigned, and the Witt group will twist it
Definition 22.1.7 takes the determinant with no sign attached. That is the right convention here and the wrong one for the Witt group, and the two sections genuinely differ, so the choice is recorded rather than left implicit.
At a fixed rank n the factor $(-1)^{n(n-1)/2}$ is a constant and carries no information, while multiplicativity over orthogonal sums, $\operatorname{disc}(\varphi \perp \psi) = \operatorname{disc}(\varphi)\operatorname{disc}(\psi)$, holds exactly as stated and is what the classification over a finite field uses.
The Witt group of section 22.6 identifies φ with $\varphi \perp \langle 1, -1\rangle$, and the unsigned discriminant does not survive that identification: adding $\langle 1, -1\rangle$ multiplies the determinant by -1. The *signed* discriminant $(-1)^{n(n-1)/2}\det$ does survive, since the rank rises by two and the sign changes to compensate; the price is that it is multiplicative only up to $(-1)^{mn}$. Each section therefore takes the convention its own problem needs, and section 22.6 states the twist when it introduces it.

The discriminant is the first invariant of the subject that is invisible over $\mathbb{R}$ and $\mathbb{C}$: there the square class group has at most two elements, so the discriminant carries at most one bit, whereas over a general field it is as large as $k^\times/(k^\times)^2$. The notion of sameness it is invariant under deserves its own name.

Definition 22.1.10: Isometry and the Orthogonal Group

Let (V, φ) and (V', φ') be finite-dimensional k-vector spaces equipped with symmetric bilinear forms. An *isometry* $f\colon (V,\varphi) \to (V',\varphi')$ is a linear isomorphism $f\colon V \to V'$ with

$$\varphi'\big(f(x), f(y)\big) = \varphi(x,y) \qquad \text{for all } x, y \in V$$

The two forms are *isometric*, written $\varphi \cong \varphi'$, if such an f exists. The isometries of (V, φ) to itself form a group under composition, the *orthogonal group* $O(\varphi)$ of the form.

Choosing bases turns an isometry into a congruence: if f is an isometry and β is a basis of V, then the matrix of φ' in the basis $f(\beta)$ is $[\varphi]_\beta$, so two forms on spaces of the same dimension are isometric exactly when their matrices are congruent. The classification of forms up to isometry and the classification of symmetric matrices up to congruence are therefore one problem in two languages. The chapter's opening example is a form the reader has already met.

Example 22.1.11: The Trace Form as a Quadratic Form

Let F/k be a finite separable extension with $\operatorname{char} k \neq 2$. The trace form $\langle x, y\rangle = \operatorname{tr}_{F/k}(xy)$ of definition 19.6.30 is a symmetric k-bilinear form on F, viewed as a k-vector space, and it is nondegenerate by proposition 19.6.31. Its associated quadratic form under proposition 22.1.2 is

$$q(x) = \operatorname{tr}_{F/k}(x^2)$$

so F carries a canonical quadratic form built from nothing but its field structure, and definition 22.1.7 attaches to it a square class in $k^\times/(k^\times)^2$.

Take $k = \mathbb{Q}$ and $F = \mathbb{Q}(\sqrt{d})$ with d a squarefree integer, $d \neq 1$. In the basis $\beta = (1, \sqrt{d})$ one computes $\operatorname{tr}(1) = 2$, since multiplication by 1 is the identity of a two-dimensional space. Next $\operatorname{tr}(\sqrt{d}) = 0$, since multiplication by $\sqrt{d}$ sends $1 \mapsto \sqrt{d}$ and $\sqrt{d} \mapsto d$, so its matrix in that basis has both diagonal entries zero. Finally $\operatorname{tr}(d) = 2d$. Hence

$$[\langle\,,\rangle]_\beta = \begin{pmatrix} 2 & 0 \\ 0 & 2d \end{pmatrix}, \qquad \det = 4d$$

and since 4 is a square,

$$\operatorname{disc} = 4d\,(\mathbb{Q}^\times)^2 = d\,(\mathbb{Q}^\times)^2$$

The discriminant of the trace form recovers the square class of d, which is precisely the datum distinguishing the fields $\mathbb{Q}(\sqrt{d})$ from one another. The invariant is doing real work already: over $\mathbb{Q}$ the square class group is infinite, so this single class separates infinitely many non-isometric forms, where over $\mathbb{R}$ or $\mathbb{C}$ it could not have separated more than two.

Exercise 22.1.1

1. Let q be a quadratic form with polar form b_q. Prove the parallelogram identity $q(x+y)+q(x-y) = 2q(x) + 2q(y)$ directly from definition 22.1.1, using no hypothesis on the characteristic. Then explain why the identity carries no information when $\operatorname{char} k = 2$, and say what that has to do with example 22.1.3.

2. Exhibit two matrices in $M_2(\mathbb{Q})$ that are congruent but not similar. Then show that when the

change of basis matrix P of theorem 22.1.4 satisfies $P^t = P^{-1}$ the two relations produce the same matrix, so congruence and similarity can differ only through changes of basis that are not of this kind.

3. Let φ and ψ be nondegenerate symmetric bilinear forms on V and W, and let $\varphi \perp \psi$ denote the form on $V \oplus W$ given by $(\varphi \perp \psi)\big((v,w),(v',w')\big) = \varphi(v,v') + \psi(w,w')$. Show that $\varphi \perp \psi$ is nondegenerate and that $\operatorname{disc}(\varphi \perp \psi) = \operatorname{disc}(\varphi)\operatorname{disc}(\psi)$.
4. Let $F = \mathbb{F}_{q^2}$ and $k = \mathbb{F}_q$ with q odd. Compute the trace form of F/k in a basis of your choosing and determine its discriminant as an element of $k^\times/(k^\times)^2$. Is it the trivial class?
5. Let φ be a symmetric bilinear form on V with radical $R = \operatorname{rad}(\varphi)$ (definition 13.3.22). Show that φ descends to a well-defined form $\bar{\varphi}$ on V/R, that $\bar{\varphi}$ is nondegenerate, and that $\operatorname{disc}(\bar{\varphi})$ is therefore defined even though $\operatorname{disc}(\varphi)$ is not.

22.2 Diagonalization and Orthogonal Bases

A symmetric matrix is congruent to a diagonal one, and the diagonal entries are visible invariants only up to squares. Both halves of that sentence are proved here, and together they reduce the classification problem to a question about the square class group of the field.

Theorem 22.2.1: Every Symmetric Form Has an Orthogonal Basis

Let k be a field with $\operatorname{char} k \neq 2$, let V be a k-vector space of finite dimension n, and let φ be a symmetric bilinear form on V. Then V has a basis $w_1, \ldots, w_n$ with $\varphi(w_i, w_j) = 0$ whenever $i \neq j$. Equivalently, every symmetric matrix over k is congruent to a diagonal matrix.

***Proof*:**
Induct on n. For $n = 0$ there is nothing to prove, and for $n = 1$ any basis is orthogonal because the condition $i \neq j$ is vacuous.

Step 1: either φ vanishes identically or some vector is non-isotropic. Suppose $\varphi(v,v) = 0$ for every $v \in V$. Then for all $x, y \in V$,

$$0 = \varphi(x+y, x+y) = \varphi(x,x) + 2\varphi(x,y) + \varphi(y,y) = 2\varphi(x,y)$$

and since $\operatorname{char} k \neq 2$ this forces $\varphi(x,y) = 0$, so φ is the zero form and every basis is orthogonal. This is the only place the characteristic hypothesis is used, and it cannot be dropped: the form ψ of example 22.1.3 is a nonzero symmetric form over $\mathbb{F}_2$ with $\psi(v,v) = 0$ for every v, so it falsifies the implication just proved, and that example shows it admits no orthogonal basis at all.

Otherwise choose $w_1 \in V$ with $a_1 := \varphi(w_1, w_1) \neq 0$.

Step 2: split off w_1. Let $W = \{w_1\}^\perp$, a subspace by lemma 13.3.21. Every $x \in V$ decomposes as

$$x = \frac{\varphi(w_1, x)}{a_1} w_1 + \left(x - \frac{\varphi(w_1, x)}{a_1} w_1 \right)$$

and the bracketed vector lies in W, since applying $\varphi(w_1, -)$ to it gives $\varphi(w_1, x) - \frac{\varphi(w_1,x)}{a_1} a_1 = 0$. So $V = kw_1 + W$. The sum is direct: if $\lambda w_1 \in W$ then $0 = \varphi(w_1, \lambda w_1) = \lambda a_1$, and $a_1 \neq 0$ forces $\lambda = 0$. Hence $V = kw_1 \oplus W$ and $\dim W = n - 1$.

Step 3: induct. The restriction $\varphi|_{W\times W}$ is a symmetric bilinear form on W, so by the inductive hypothesis W has an orthogonal basis $w_2, \ldots, w_n$. Then $w_1, w_2, \ldots, w_n$ is a basis of V by Step 2, and it is orthogonal: $\varphi(w_i, w_j) = 0$ for $2 \leq i \neq j \leq n$ by the choice of the w_i, and $\varphi(w_1, w_j) = 0$ for $j \geq 2$ because $w_j \in W = \{w_1\}^{\perp}$.

For the matrix formulation, a symmetric $A \in M_n(k)$ is the matrix of a symmetric form on k^n in the standard basis, and an orthogonal basis for that form has diagonal matrix, and by theorem 22.1.4 the two matrices are congruent, completing the proof.

In an orthogonal basis the form is determined by the n scalars $a_i = \varphi(w_i, w_i)$, and the standard notation records exactly that: write

$$\varphi \cong \langle a_1, a_2, \ldots, a_n \rangle$$

for a form admitting an orthogonal basis with those diagonal entries, so that the associated quadratic form is $q(x) = a_1 x_1^2 + \cdots + a_n x_n^2$ in the corresponding coordinates. The notation is an assertion about an isometry class, not about a chosen basis, and the next proposition says how far the entries themselves are from being invariants.

Proposition 22.2.2: Rescaling the Entries of a Diagonalization

Let $\operatorname{char} k \neq 2$. For any $c_1, \ldots, c_n \in k^{\times}$,

$$\langle a_1, \ldots, a_n \rangle \cong \langle a_1 c_1^2, \ldots, a_n c_n^2 \rangle$$

so each diagonal entry is well defined only up to multiplication by a nonzero square. A diagonal entry that is zero stays zero, and the number of zero entries is the dimension of the radical, hence is an invariant.

***Proof*:**
Let $w_1, \ldots, w_n$ be an orthogonal basis realizing $\langle a_1, \ldots, a_n \rangle$ and put $w_i' = c_i w_i$. Each c_i is nonzero, so $w_1', \ldots, w_n'$ is again a basis, and it is orthogonal because $\varphi(w_i', w_j') = c_i c_j \varphi(w_i, w_j) = 0$ for $i \neq j$. Its diagonal entries are $\varphi(w_i', w_i') = c_i^2 a_i$, which gives the displayed isometry. Since $c_i^2 \neq 0$, the entry $c_i^2 a_i$ vanishes exactly when a_i does.

For the last claim, in an orthogonal basis a vector $x = \sum_i x_i w_i$ lies in $\operatorname{rad}(\varphi)$ if and only if $\varphi(w_j, x) = a_j x_j = 0$ for every j, that is if and only if $x_j = 0$ for every j with $a_j \neq 0$. So the radical is spanned by those w_j with $a_j = 0$, and its dimension is the number of zero entries, which is independent of the diagonalization because the radical is, completing the proof.

Corollary 22.2.3: The Discriminant from a Diagonalization

Let $\varphi \cong \langle a_1, \ldots, a_n \rangle$ be nondegenerate, so every $a_i \neq 0$. Then

$$\operatorname{disc}(\varphi) = a_1 a_2 \cdots a_n \, (k^{\times})^2$$

***Proof*:**
In an orthogonal basis the matrix of φ is $\operatorname{diag}(a_1, \ldots, a_n)$, whose determinant is $a_1 \cdots a_n$. Apply definition 22.1.7, whose independence of the basis is proposition 22.1.8. Consistently with proposition 22.2.2, replacing each a_i by $a_i c_i^2$ multiplies the product by $(c_1 \cdots c_n)^2$ and so does not change the class, as we sought to show.

Two fields where the square class group is completely known illustrate how much of the classification the discriminant sees. The first is the degenerate case, in the sense that the group is trivial and nothing beyond the rank survives.

Example 22.2.4: Forms over an Algebraically Closed Field
Let k be algebraically closed with $\operatorname{char} k \neq 2$. Every $a \in k^\times$ has a square root, so $(k^\times)^2 = k^\times$ and the square class group of definition 22.1.6 is trivial. Given any symmetric form, diagonalize it by theorem 22.2.1 as $\langle a_1, \ldots, a_r, 0, \ldots, 0\rangle$ with the $a_i \neq 0$, and apply proposition 22.2.2 with $c_i = a_i^{-1/2}$ to obtain

$$\varphi \cong \langle \underbrace{1, \ldots, 1}_{r}, \underbrace{0, \ldots, 0}_{n-r} \rangle$$

So a form over an algebraically closed field is determined up to isometry by n and r, and r is the rank of any matrix of φ. The discriminant of a nondegenerate such form is the class of 1 and carries no information, as it must, the group having one element.

Example 22.2.5: Forms over a Finite Field
Let $k = \mathbb{F}_q$ with q odd. The group $\mathbb{F}_q^\times$ is cyclic by theorem 11.3.14, of even order $q - 1$, so the squaring map has kernel $\{\pm 1\}$ of order 2 and image of index 2. Hence

$$\mathbb{F}_q^\times / (\mathbb{F}_q^\times)^2 \cong \mathbb{Z}/2\mathbb{Z}$$

with the two classes the nonzero squares and the non-squares. Unlike the algebraically closed case, the discriminant now carries a genuine bit, and it obstructs isometries that the rank alone would permit.

Take $q = 5$, where the squares in $\mathbb{F}_5^\times = \{1, 2, 3, 4\}$ are $1 = 1^2$ and $4 = 2^2$, so the non-squares are 2 and 3. The forms $\langle 1, 1\rangle$ and $\langle 1, 2\rangle$ on $\mathbb{F}_5^2$ are both nondegenerate of rank 2, but by corollary 22.2.3 their discriminants are the classes of 1 and of 2 respectively, which are distinct in $\mathbb{F}_5^\times/(\mathbb{F}_5^\times)^2$. By proposition 22.1.8 an isometry would force those classes to agree, so

$$\langle 1, 1\rangle \ncong \langle 1, 2\rangle$$

Rank does not classify forms over $\mathbb{F}_5$. Whether rank together with the discriminant does classify them is a genuine question, and it is answered affirmatively in section 22.6 once enough is known about which values a form represents.

Exercise 22.2.1

1. Exhibit two different orthogonal bases for the same form on $\mathbb{Q}^2$ whose diagonal entries are not related by the rescaling of proposition 22.2.2 entry by entry, and reconcile this with that proposition.

2. Let $k = \mathbb{R}$. Determine the square class group, and show that every nondegenerate form is isometric to $\langle 1, \ldots, 1, -1, \ldots, -1\rangle$. Explain why the discriminant alone does not classify these, and identify the invariant that does. (This book does not prove that the counts are well defined; that is Sylvester's law of inertia, and it needs an order on the scalars.)

3. Over $k = \mathbb{Q}$, show that $\langle 1, 1\rangle$ and $\langle -1, -1\rangle$ have the same rank and the same discriminant and are nevertheless not isometric, by asking which nonzero rationals each represents. Conclude that rank and discriminant together do not classify forms over $\mathbb{Q}$, in contrast with example 22.2.4. Show also that $\langle 1, 1\rangle \cong \langle 2, 2\rangle$ over $\mathbb{Q}$, so a difference of diagonal entries is not by itself evidence of non-isometry.

4. Let $\varphi \cong \langle a_1, \ldots, a_n\rangle$ be nondegenerate over k and let $c \in k^\times$ be a value represented by φ, meaning $c = q(v)$ for some $v \in V$. Show that $\varphi \cong \langle c, b_2, \ldots, b_n\rangle$ for some b_i, and deduce that the set of represented values is a union of square classes.

22.3 Alternating Forms, the Linear Darboux Theorem, and Lagrangians

Section 22.1 and section 22.2 treated the symmetric forms, where the classification runs through the square classes of k and is not settled by the rank alone. The alternating forms are the second class of bilinear forms for which a classification is available, and there the answer does not depend on the field: a single integer, the codimension of the radical, determines the form up to isometry, and neither the statement nor its proof carries a hypothesis on the characteristic.

The classification is proved in the shape of a basis in which the matrix of the form is one fixed block matrix. That basis then makes visible the subspaces on which an alternating form vanishes identically. When the form is nondegenerate such a subspace has dimension at most half that of the ambient space, and the ones attaining the bound are the Lagrangian subspaces.

Definition 22.3.1: Alternating Bilinear Form

Let k be a field, with no hypothesis on its characteristic, and let V be a finite-dimensional k-vector space. A bilinear form $\varphi\colon V \times V \to k$ (definition 13.3.15) is *alternating* if

$$\varphi(v, v) = 0 \qquad \text{for every } v \in V$$

This is the vanishing condition of definition 16.1.3 read for a 2-multilinear map, and not the sign rule listed in definition 13.3.15. The two conditions differ in characteristic 2, and throughout this section *alternating* means the displayed vanishing condition.

Two alternating forms (V, φ) and (V', φ') are *isometric* if there is a linear isomorphism $f\colon V \to V'$ with $\varphi'\big(f(x), f(y)\big) = \varphi(x, y)$ for all $x, y \in V$. This is the condition of definition 22.1.10 with the hypothesis that the two forms be symmetric dropped, and it is used in that generality below.

One expansion carries the whole section. For $x, y \in V$ the vanishing condition applied to $x + y$ gives

$$0 = \varphi(x+y, x+y) = \varphi(x,x) + \varphi(x,y) + \varphi(y,x) + \varphi(y,y) = \varphi(x,y) + \varphi(y,x)$$

so every alternating form satisfies

$$\varphi(y,x) = -\varphi(x,y) \qquad \text{for all } x,y \in V \tag{22.1}$$

which is lemma 16.1.4(1) for the transposition in S_2, its hypotheses holding here because a field is a commutative ring and both V and k are k-modules. The converse needs 2 to be invertible, and that is lemma 16.1.4(2): over a field with $\operatorname{char} k \neq 2$ a bilinear form satisfying (22.1) is alternating, since setting $y = x$ there gives $2\varphi(x,x) = 0$. Over such a field, therefore, *alternating* and *skew-symmetric* name the same forms, and a form that is both symmetric and alternating is zero: symmetry and (22.1) together give $\varphi(x,y) = -\varphi(x,y)$, hence $2\varphi(x,y) = 0$, hence $\varphi(x,y) = 0$. The classification below and the one in section 22.2 therefore overlap only in the zero form. In characteristic 2 neither statement survives, and $-1 = 1$ makes (22.1) say only that φ is symmetric.

The condition is visible in the matrix of the form. Fix a basis $\beta = (\beta_1, \ldots, \beta_n)$ of V and let $A = (a_{ij}) = [\varphi]_\beta$ be the matrix of definition 13.3.18. If φ is alternating then $a_{ii} = \varphi(\beta_i, \beta_i) = 0$ and, by (22.1), $a_{ji} = -a_{ij}$. Conversely, suppose A has zero diagonal and satisfies $A^t = -A$. For $v = \sum_i x_i \beta_i$, lemma 13.3.19 gives

$$\varphi(v,v) = \sum_{i,j} x_i x_j a_{ij} = \sum_i x_i^2 a_{ii} + \sum_{i<j} x_i x_j \big(a_{ij} + a_{ji}\big) = 0$$

so φ is alternating. Both conditions on A are needed: in characteristic 2 the matrix $\left(\begin{smallmatrix} 1 & 0 \\ 0 & 0 \end{smallmatrix}\right)$ satisfies $A^t = -A$ and has a nonzero diagonal entry.

Example 22.3.2: The Determinant Form, and the Overlap in Characteristic Two

1. Let $V = k^2$ and
$$\varphi(x,y) = x_1 y_2 - x_2 y_1$$
which is the determinant of the matrix with columns x and y. It is bilinear in each column and $\varphi(x,x) = x_1x_2 - x_2x_1 = 0$, so it is alternating. Its matrix in the standard basis is $\left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right)$, of determinant 1, so φ is nondegenerate by lemma 13.3.23.

2. Over $k = \mathbb{F}_2$ the form in (1) reads $\varphi(x,y) = x_1y_2 + x_2y_1$, which is exactly the form ψ of example 22.1.3. That form is symmetric, nonzero and alternating at once, which is possible only because $\operatorname{char} k = 2$; the example also shows it admits no orthogonal basis, so the conclusion of theorem 22.2.1 genuinely fails for it.

3. Still over $\mathbb{F}_2$, the form $\varphi(x,y) = x_1y_1$ is symmetric, hence satisfies (22.1) because $-1 = 1$, but $\varphi(e_1, e_1) = 1 \neq 0$. So the sign rule does not imply the vanishing condition when 2 is not invertible, and the hypothesis in lemma 16.1.4(2) cannot be dropped.

The proof of theorem 22.2.1 split off one dimension at a time, and the entry it left behind was pinned down only up to a square (proposition 22.2.2). For an alternating form there is no such entry to leave behind, since $\varphi(v,v) = 0$ for every v, and the smallest piece that carries any information is two-dimensional. On such a piece the one scalar available can be scaled to 1 outright, which is why no square classes enter the answer. The resulting normal form deserves a name before it is proved to exist.

Definition 22.3.3: Symplectic Basis and Symplectic Space

Let k be a field, with no hypothesis on its characteristic, let V be a finite-dimensional k-vector space, and let φ be an alternating bilinear form on V (definition 22.3.1). A *symplectic basis* for φ is a basis

$$e_1, f_1, e_2, f_2, \dots, e_m, f_m$$

of V, necessarily of even cardinality $2m$, such that $\varphi(e_i, f_i) = 1$ for every i, while

$$\varphi(e_i, f_j) = 0 \quad (i \neq j), \qquad \varphi(e_i, e_j) = 0, \qquad \varphi(f_i, f_j) = 0$$

for all i and j. A pair (V, φ) with φ alternating and nondegenerate (definition 13.3.22) is called a *symplectic space*.

Example 22.3.4: The Standard Symplectic Space k^{2m}

Let k be any field and let $e_1, f_1, \dots, e_m, f_m$ be the standard basis of k^{2m}, listed in that order. Define

$$\varphi\Big(\sum_i (x_i e_i + y_i f_i),\ \sum_i (x_i' e_i + y_i' f_i)\Big) = \sum_{i=1}^m (x_i y_i' - y_i x_i')$$

Each summand is the form of example 22.3.2(1) in the coordinate pair (x_i, y_i), so φ is bilinear and vanishes when its two arguments agree, hence is alternating. Evaluating on basis vectors gives $\varphi(e_i, f_i) = 1$, while $\varphi(e_i, f_j) = 0$ for $i \neq j$ and $\varphi(e_i, e_j) = \varphi(f_i, f_j) = 0$ for all i and j, because the i-th summand involves only the i-th coordinate pair. So the standard basis is symplectic. The matrix of φ in this basis is block diagonal with m blocks $\left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right)$, so its determinant is 1 and φ is nondegenerate by lemma 13.3.23.

Listing the same basis as $e_1, \dots, e_m, f_1, \dots, f_m$ changes the matrix to $\left(\begin{smallmatrix} 0 & I_m \\ -I_m & 0 \end{smallmatrix}\right)$. The two orderings differ by a permutation matrix P, so by theorem 22.1.4 the matrices differ by $A \mapsto P^t A P$ and the determinant is unchanged, being multiplied by $(\det P)^2 = 1$. This second ordering identifies k^{2m} with $k^m \oplus k^m$ and writes the form as

$$\varphi\big((x, y), (x', y')\big) = x^t y' - y^t x' \tag{22.2}$$

for column vectors $x, y, x', y' \in k^m$, which is the shape used for computations below.

A degenerate example is obtained by padding. On k^{2m+r} with basis $e_1, f_1, \dots, e_m, f_m, z_1, \dots, z_r$, extend φ by declaring $\varphi(z_j, -) = \varphi(-, z_j) = 0$ for every j. The result is alternating, and its radical is $\operatorname{span}(z_1, \dots, z_r)$. To see this, write $v = \sum_i (x_i e_i + y_i f_i) + \sum_j c_j z_j$; then $\varphi(e_i, v) = y_i$ and $\varphi(f_i, v) = -x_i$ and $\varphi(z_j, v) = 0$, and since φ is linear in its first argument, v lies in the radical (definition 13.3.22) exactly when every x_i and every y_i vanishes.

Every alternating form is one of the forms just written down. This is the classification, and it holds over every field.

Theorem 22.3.5: Linear Darboux Theorem

Let k be a field, with no hypothesis on its characteristic, let V be a k-vector space of finite dimension n, and let φ be an alternating bilinear form on V (definition 22.3.1).

1. There are integers $m, r \geq 0$ with $n = 2m + r$ and a basis

$$e_1, f_1, \ldots, e_m, f_m, z_1, \ldots, z_r$$

of V such that $\varphi(e_i, f_i) = 1$ for every i, while every other value of φ on an ordered pair of basis vectors is 0 except $\varphi(f_i, e_i) = -1$.

2. For any such basis, $\mathrm{rad}(\varphi) = \mathrm{span}(z_1, \ldots, z_r)$. Hence $r = \dim \mathrm{rad}(\varphi)$ and $2m = n - \dim \mathrm{rad}(\varphi)$, so both integers are determined by φ and do not depend on the basis chosen in (1).

3. φ is nondegenerate if and only if $r = 0$, and in that case $n = 2m$ is even and the basis of (1) is a symplectic basis (definition 22.3.3). In particular a symplectic space has even dimension.

4. Let φ' be an alternating form on a k-vector space V' with $\dim V' = \dim V$. Then φ and φ' are isometric if and only if $\dim \mathrm{rad}(\varphi) = \dim \mathrm{rad}(\varphi')$. In particular, for each even n there is exactly one nondegenerate alternating form on an n-dimensional space up to isometry, and none at all when n is odd.

***Proof*:**

1. *Step 1: a pair e, f with $\varphi(e, f) = 1$.* Assume $\varphi \neq 0$ and choose $u, w \in V$ with $c := \varphi(u, w) \neq 0$. Put $e := u$ and $f := c^{-1}w$, so that $\varphi(e, f) = c^{-1}\varphi(u, w) = 1$, and $\varphi(f, e) = -1$ by (22.1). The two are linearly independent: if $\alpha e + \beta f = 0$ then applying $\varphi(e, -)$ gives $\alpha\varphi(e, e) + \beta\varphi(e, f) = \beta = 0$, and applying $\varphi(f, -)$ gives $\alpha\varphi(f, e) = -\alpha = 0$. Write $P = \mathrm{span}(e, f)$, a two-dimensional subspace.

   Step 2: $V = P \oplus P^\perp$. The set $P^\perp$ is a subspace and equals $\{e, f\}^\perp$, both by lemma 13.3.21 applied to the subset $\{e, f\}$, whose span is P. Given $v \in V$, set

$$v' := v - \varphi(e, v)\, f + \varphi(f, v)\, e$$

   Then, using $\varphi(e, e) = \varphi(f, f) = 0$, $\varphi(e, f) = 1$ and $\varphi(f, e) = -1$,

$$\varphi(e, v') = \varphi(e, v) - \varphi(e, v)\cdot 1 + \varphi(f, v)\cdot 0 = 0 \qquad \varphi(f, v') = \varphi(f, v) - \varphi(e, v)\cdot 0 + \varphi(f, v)\cdot(-1) = 0$$

   so $v' \in P^\perp$, and $v = \big(\varphi(e, v)f - \varphi(f, v)e\big) + v'$ exhibits v in $P + P^\perp$. The sum is direct: if $\alpha e + \beta f \in P^\perp$ then $\varphi(e, \alpha e + \beta f) = \beta = 0$ and $\varphi(f, \alpha e + \beta f) = -\alpha = 0$. Hence $V = P \oplus P^\perp$ and $\dim P^\perp = n - 2$.

   Step 3: induction on n. For $n = 0$ take the empty basis and $m = r = 0$. Let $n \geq 1$ and assume the statement for all spaces of smaller dimension. If $\varphi = 0$, any basis of V serves with $m = 0$ and $r = n$. Otherwise Steps 1 and 2 produce $V = P \oplus P^\perp$ with $\dim P^\perp = n - 2$.

The restriction of φ to $P^\perp \times P^\perp$ is bilinear and satisfies the vanishing condition, so it is alternating, and the inductive hypothesis gives a basis $e_2, f_2, \dots, e_m, f_m, z_1, \dots, z_r$ of $P^\perp$ with the stated table of values. Set $e_1 := e$ and $f_1 := f$. The concatenated list is a basis of V by Step 2. Its values are as claimed: within $P^\perp$ by the inductive hypothesis, on the pair (e_1, f_1) by Step 1, and for a pair with one entry in $\{e_1, f_1\}$ and the other in $P^\perp$ because $\varphi(e_1, v) = \varphi(f_1, v) = 0$ for $v \in P^\perp$ by the definition of $P^\perp$, whence also $\varphi(v, e_1) = \varphi(v, f_1) = 0$ by (22.1). Counting the list gives $n = 2m + r$.

2. Let $v = \sum_i a_i e_i + \sum_i b_i f_i + \sum_j c_j z_j$. Since φ is linear in its first argument, $v \in \operatorname{rad}(\varphi)$ holds if and only if $\varphi(x, v) = 0$ for x running over the basis. The table of values in (1) gives

$$\varphi(e_i, v) = b_i, \qquad \varphi(f_i, v) = -a_i, \qquad \varphi(z_j, v) = 0$$

for all i and j. So $v \in \operatorname{rad}(\varphi)$ if and only if every a_i and every b_i vanishes, that is if and only if $v \in \operatorname{span}(z_1, \dots, z_r)$. Those z_j are part of a basis, hence independent, so $\dim \operatorname{rad}(\varphi) = r$, and $2m = n - r$ follows from $n = 2m + r$. The radical was defined in definition 13.3.22 without reference to a basis, so r and m are the same for every basis as in (1).

3. By definition 13.3.22, φ is nondegenerate exactly when $\operatorname{rad}(\varphi) = 0$, which by (2) happens exactly when $r = 0$. Then $n = 2m$, and the list of (1) is a basis satisfying the conditions of definition 22.3.3.

4. Suppose first that $\dim \operatorname{rad}(\varphi) = \dim \operatorname{rad}(\varphi')$. By (2) the two forms have bases as in (1) with the same m and the same r, since $\dim V = \dim V'$. Let $f\colon V \to V'$ be the linear map carrying each basis vector of V to the correspondingly named basis vector of V'. It is an isomorphism, being a bijection on bases, and the two bilinear maps $(x, y) \mapsto \varphi'(f(x), f(y))$ and $(x, y) \mapsto \varphi(x, y)$ agree on every ordered pair of basis vectors by construction, hence agree everywhere by bilinearity. So f is an isometry in the sense of definition 22.3.1.

   Conversely let f be an isometry. Its inverse is one as well: putting $x = f^{-1}(x')$ and $y = f^{-1}(y')$ in the defining identity gives $\varphi\big(f^{-1}(x'), f^{-1}(y')\big) = \varphi'(x', y')$. Now let $v \in \operatorname{rad}(\varphi)$ and let $x' \in V'$. Since f is onto, $x' = f(x)$ for some x, and then $\varphi'(x', f(v)) = \varphi(x, v) = 0$. Hence $f\big(\operatorname{rad}(\varphi)\big) \subseteq \operatorname{rad}(\varphi')$, and the same argument for f^{-1} gives $f^{-1}\big(\operatorname{rad}(\varphi')\big) \subseteq \operatorname{rad}(\varphi)$, that is $\operatorname{rad}(\varphi') \subseteq f\big(\operatorname{rad}(\varphi)\big)$. So f restricts to an isomorphism between the two radicals, and their dimensions agree.

   For the last sentence, take φ and φ' nondegenerate of the same dimension n: both radicals are zero, so they are isometric, and the standard form of example 22.3.4 supplies one such φ when n is even. When n is odd, (3) forbids a nondegenerate alternating form outright, completing the proof.

The symmetric case behaves differently over the very same field. Example 22.2.5 exhibits two nondegenerate symmetric forms of rank 2 over $\mathbb{F}_5$ that are not isometric, the discriminant separating them, while part (4) makes any two nondegenerate alternating forms on a space of the same dimension isometric, over $\mathbb{F}_5$ or over any other field. No invariant of the discriminant's kind can exist for alternating forms. Indeed, if φ is nondegenerate alternating with symplectic basis β_0, then $[\varphi]_{\beta_0}$ is block diagonal with m blocks $\left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right)$ by definition 22.3.3, so $\det[\varphi]_{\beta_0} = 1$. Any other basis β gives

$[\varphi]_\beta = P^t[\varphi]_{\beta_0}P$ by theorem 22.1.4, whence

$$\det[\varphi]_\beta = (\det P)^2$$

So the determinant of every matrix of φ is a nonzero square, and the square class that definition 22.1.7 extracts in the symmetric case is trivial here for every form and separates nothing.[1]

In odd dimension the theorem is a nonexistence statement: there is no nondegenerate alternating form at all, and every alternating form on a space of odd dimension n has a radical of odd dimension $n - 2m$. Symmetric forms are unconstrained in this respect, since $\langle 1, 1, 1\rangle$ has the identity as its matrix and is nondegenerate in dimension 3 by lemma 13.3.23.

The basis of theorem 22.3.5 exhibits two subspaces of dimension m on which φ vanishes identically, namely $\mathrm{span}(e_1, \ldots, e_m)$ and $\mathrm{span}(f_1, \ldots, f_m)$. Subspaces with that property have a name, and it is given for an arbitrary bilinear form since the symmetric case needs it too.

Definition 22.3.6: Totally Isotropic Subspace

Let k be a field, with no hypothesis on its characteristic, let V be a finite-dimensional k-vector space, and let φ be a bilinear form on V. A subspace $W \subseteq V$ is *totally isotropic* for φ if

$$\varphi(x, y) = 0 \qquad \text{for all } x, y \in W$$

equivalently if $W \subseteq W^\perp$ in the notation of definition 13.3.20. The shorter term *isotropic subspace* is used below for the same condition.

Example 22.3.7: Totally Isotropic Subspaces in Two Standard Forms

1. Let φ be alternating on V over any field. Every line $W = kv$ is totally isotropic, since $\varphi(\lambda v, \mu v) = \lambda\mu\, \varphi(v, v) = 0$. So an alternating form has isotropic subspaces in abundance, whatever the field.

2. In the standard symplectic space k^{2m} of example 22.3.4, the subspaces $E = \mathrm{span}(e_1, \ldots, e_m)$ and $F = \mathrm{span}(f_1, \ldots, f_m)$ are totally isotropic, since $\varphi(e_i, e_j) = 0$ and $\varphi(f_i, f_j) = 0$ for all i, j and the condition is bilinear in the two arguments. Each has dimension m, half of $\dim k^{2m}$.

3. Let $k = \mathbb{R}$ and let $\varphi = \langle 1, 1\rangle$ on $\mathbb{R}^2$, that is $\varphi(x, y) = x_1y_1 + x_2y_2$, with associated quadratic form $q(x) = x_1^2 + x_2^2$ under proposition 22.1.2. If W is totally isotropic and $v \in W$, then $q(v) = \varphi(v, v) = 0$, and a sum of two real squares vanishes only when both do, so $v = 0$. The only totally isotropic subspace is 0. So the abundance in (1) is a feature of the alternating case and not of bilinear forms in general.

The subspaces in item (2) have exactly half the dimension of the ambient space, and item (1) produces nothing larger there. Half is in fact the maximum whenever the form is nondegenerate, for symmetric and alternating forms alike, and the bound follows from a dimension count for orthogonal

[1] The name of the theorem is by analogy with Darboux's theorem in symplectic geometry, which states that a closed nondegenerate differential 2-form on a smooth manifold admits, around each point, coordinates in which it is the constant form displayed in example 22.3.4. That theorem is about manifolds and is not proved in this book; the statement here is its pointwise linear-algebra counterpart.

complements. Degeneracy destroys the bound at once: the zero form on any V is alternating, and V itself is then totally isotropic.

Proposition 22.3.8: The Dimension Bound for Isotropic Subspaces

Let k be a field, with no hypothesis on its characteristic, let V be a k-vector space of finite dimension n, and let φ a nondegenerate bilinear form on V (definition 13.3.22).

1. For every subspace $W \subseteq V$,

$$\dim W + \dim W^{\perp} = n$$

2. If W is totally isotropic (definition 22.3.6) then $\dim W \leq \frac{1}{2}n$, with equality if and only if $W = W^{\perp}$.
3. Assume in addition that φ is alternating, so that $n = 2m$ is even by theorem 22.3.5(3). Then every totally isotropic subspace of V is contained in a totally isotropic subspace of dimension m. Such subspaces exist, and the maximal totally isotropic subspaces of V are exactly those of dimension m.

***Proof*:**

1. Let $d = \dim W$ and define $\Psi\colon V \to W^*$ by $\Psi(y)(x) = \varphi(x, y)$ for $x \in W$. Each $\Psi(y)$ is linear on W and Ψ is linear in y, both by bilinearity of φ. By definition 13.3.20, $\Psi(y) = 0$ says exactly that $y \in W^{\perp}$, so $\ker \Psi = W^{\perp}$.

   Ψ is surjective. First, the map $\Theta\colon V \to V^*$ with $\Theta(y) = \varphi(-, y)$ has kernel $\mathrm{rad}(\varphi)$, which is 0 by definition 13.3.22 and the nondegeneracy hypothesis. The dual basis of proposition 13.3.5 is a basis of V^*, so $\dim V^* = \dim V$, while corollary 13.1.18 gives $\dim V = \dim \ker \Theta + \dim \Theta(V) = \dim \Theta(V)$. Hence the image $\Theta(V)$ is a subspace of V^* of the full dimension, so $\Theta(V) = V^*$. Second, every $g \in W^*$ extends to V: choose a basis $w_1, \ldots, w_d$ of W, extend it to a basis $w_1, \ldots, w_n$ of V, and let $\tilde{g} \in V^*$ be the linear map with $\tilde{g}(w_i) = g(w_i)$ for $i \leq d$ and $\tilde{g}(w_i) = 0$ for $i > d$; then $\tilde{g}$ and g are linear maps on W agreeing on a basis of W, so $\tilde{g}|_W = g$. Combining, given $g \in W^*$ pick y with $\Theta(y) = \tilde{g}$; then $\Psi(y) = \tilde{g}|_W = g$.

   Now corollary 13.1.18 applied to Ψ gives $n = \dim \ker \Psi + \dim \Psi(V) = \dim W^{\perp} + \dim W^*$, and $\dim W^* = \dim W$ again by proposition 13.3.5, this time for the space W.
2. If W is totally isotropic then $W \subseteq W^{\perp}$ by definition 22.3.6, so $\dim W \leq \dim W^{\perp} = n - \dim W$ by (1), which rearranges to $2 \dim W \leq n$. Equality holds exactly when $\dim W = \dim W^{\perp}$, and a subspace contained in another of the same finite dimension equals it, so equality is equivalent to $W = W^{\perp}$.
3. Let W be totally isotropic with $d = \dim W < m$. By (1), $\dim W^{\perp} = 2m - d > d = \dim W$, and $W \subseteq W^{\perp}$, so there is a vector $v \in W^{\perp}$ with $v \notin W$. Put $W' = W + kv$, of dimension $d + 1$. It is totally isotropic: for $w, w' \in W$ and $\lambda, \mu \in k$,

$$\varphi(w + \lambda v,\ w' + \mu v) = \varphi(w, w') + \mu\,\varphi(w, v) + \lambda\,\varphi(v, w') + \lambda\mu\,\varphi(v, v)$$

and the four terms vanish in turn because W is totally isotropic, because $v \in W^\perp$, because $\varphi(v, w') = -\varphi(w', v) = 0$ by (22.1) and $v \in W^\perp$, and because φ is alternating. Repeating the step $m - d$ times produces a totally isotropic subspace of dimension m containing W.

Existence follows by applying this to $W = 0$, or directly from theorem 22.3.5(3): the span of $e_1, \ldots, e_m$ in a symplectic basis is totally isotropic of dimension m. Finally, a totally isotropic subspace of dimension m is maximal among totally isotropic subspaces, since any subspace properly containing it has dimension exceeding the bound in (2); and one of dimension $d < m$ is not maximal, since the construction above properly enlarges it. So maximality is equivalent to having dimension m, as we sought to show.

Part (3) fails for symmetric forms: example 22.3.7(3) gives a nondegenerate symmetric form of rank 2 over $\mathbb{R}$ whose only totally isotropic subspace is 0, so there the maximal one has dimension 0 rather than 1. The alternating case is uniform because $\varphi(v, v) = 0$ holds for every v by definition: enlarging an isotropic W needs a vector of $W^\perp$ outside W on which the form vanishes, and the vanishing is automatic, so the dimension count in (1) supplies the vector by itself. The subspaces attaining the bound carry a name.

Definition 22.3.9: Lagrangian Subspace

Let k be a field, with no hypothesis on its characteristic, and let (V, φ) be a symplectic space of dimension $2m$, so φ is alternating and nondegenerate (definition 22.3.3). A *Lagrangian subspace* of (V, φ) is a totally isotropic subspace $L \subseteq V$ with $\dim L = m$.

By proposition 22.3.8(2) this is the same as requiring $L = L^\perp$, and by proposition 22.3.8(3) it is the same as requiring L to be maximal among the totally isotropic subspaces of V. That same part guarantees Lagrangian subspaces exist in every symplectic space.

Example 22.3.10: Lagrangian Subspaces of the Standard Symplectic Space

Work in $k^{2m} = k^m \oplus k^m$ with the form (22.2), $\varphi\big((x, y), (x', y')\big) = x^t y' - y^t x'$.

1. For $S \in M_m(k)$ let $L_S = \{(x, Sx) \mid x \in k^m\}$, the graph of S. It is a subspace of dimension m, the map $x \mapsto (x, Sx)$ being linear and injective. For $x, x' \in k^m$,
$$\varphi\big((x, Sx), (x', Sx')\big) = x^t Sx' - (Sx)^t x' = x^t Sx' - x^t S^t x' = x^t \big(S - S^t\big) x'$$
Taking x and x' to be the i-th and j-th standard basis vectors of k^m turns the right-hand side into the (i, j) entry of $S - S^t$, so the expression vanishes for all x, x' if and only if $S = S^t$. Hence L_S is Lagrangian exactly when S is symmetric. Taking $S = 0$ recovers $E = \operatorname{span}(e_1, \ldots, e_m)$.

2. Conversely, every Lagrangian L with $L \cap F = 0$, where $F = \{(0, y) \mid y \in k^m\}$, is of this form. The projection $\pi(x, y) = x$ restricts to a linear map $L \to k^m$ with kernel $L \cap F = 0$, hence injective, and $\dim L = m$ forces it to be bijective by corollary 13.1.18. Let σ be its inverse and write $\sigma(x) = (x, Sx)$, where S is linear because σ is. Then $L = L_S$, and S is symmetric by (1).

3. For $m = 1$ every line in k^2 is Lagrangian: a line is totally isotropic by example 22.3.7(1) and has dimension $1 = m$. Over $k = \mathbb{F}_q$ the lines of k^2 number $(q^2 - 1)/(q - 1) = q + 1$, so

there are $q+1$ Lagrangians. For $m \geq 2$ having dimension m is not sufficient: inside k^4 the subspace $\operatorname{span}(e_1, f_1)$ has dimension $2 = m$ and is not totally isotropic, since $\varphi(e_1, f_1) = 1$.

Items (1) and (2) together put the Lagrangians meeting F trivially in bijection with the symmetric $m \times m$ matrices, the assignment $S \mapsto L_S$ being injective because L_S determines Sx as the second component of its element with first component x. The symmetric matrices form a k-vector space of dimension $\binom{m+1}{2}$. The two counts agree where they overlap: for $m = 1$ every 1×1 matrix is symmetric, so over $\mathbb{F}_q$ that bijection accounts for q Lagrangians, and the one remaining line, F itself, brings the total to the $q+1$ of item (3).

Exercise 22.3.1

1. Over $k = \mathbb{F}_2$, show that every symmetric bilinear form on k^2 satisfies the sign rule $\varphi(x, y) = -\varphi(y, x)$. Then determine which of the eight symmetric bilinear forms on $\mathbb{F}_2^2$ are alternating in the sense of definition 22.3.1, and say how the count is consistent with lemma 16.1.4(2).

2. Let $\operatorname{char} k \neq 2$ and let φ be a nondegenerate alternating form on a k-vector space of dimension n, with matrix A in some basis. Using $\det(A^t) = \det(A)$ and $A^t = -A$, show directly that n is even. Then explain which step of your argument fails over $\mathbb{F}_2$, and why the conclusion nevertheless still holds there.

3. Let k be any field. Show that a bilinear form φ on k^n is alternating if and only if there are scalars a_{ij} for $1 \leq i < j \leq n$ with
$$\varphi(x, y) = \sum_{i<j} a_{ij}\big(x_i y_j - x_j y_i\big)$$
and that the a_{ij} are then determined by φ. Deduce that over $k = \mathbb{F}_q$ there are exactly $q^{n(n-1)/2}$ alternating forms on k^n.

4. Let $\operatorname{char} k \neq 2$. Show that every bilinear form φ on V is uniquely $\varphi = s + a$ with s symmetric and a alternating, and identify s and a in terms of φ and $(x, y) \mapsto \varphi(y, x)$. Then show the statement fails over $\mathbb{F}_2$ by exhibiting a bilinear form on $\mathbb{F}_2^2$ that is not such a sum.

5. Let (V, φ) be a symplectic space of dimension $2m$ and let L be a Lagrangian subspace with basis $e_1, \ldots, e_m$. Show that there exist $f_1, \ldots, f_m \in V$ making $e_1, f_1, \ldots, e_m, f_m$ a symplectic basis of (V, φ). (Suggestion: for each i apply proposition 22.3.8(1) to the span of the e_j with $j \neq i$ to produce u_i with $\varphi(e_j, u_i) = 1$ when $i = j$ and 0 otherwise, then replace u_i by $u_i + \sum_j c_{ij} e_j$ for a suitable choice of scalars, noting that this leaves the previous values unchanged.) Deduce that for any two Lagrangian subspaces L and L' of (V, φ) there is an isometry of (V, φ) carrying L onto L'.

6. Let φ be an alternating form on V with radical $R = \operatorname{rad}(\varphi)$. Show that $\bar{\varphi}(x+R, y+R) := \varphi(x, y)$ is a well-defined alternating form on V/R, that it is nondegenerate, and that $\dim V/R$ is even. Compare the resulting statement with theorem 22.3.5(2).

7. Let (V, φ) be a symplectic space of dimension $2m$. Show that there is a chain of subspaces $0 = W_0 \subset W_1 \subset \cdots \subset W_m$ with $\dim W_i = i$ and each W_i totally isotropic, and that W_m is then a Lagrangian subspace. Show also that no such chain can be continued to a totally isotropic W_{m+1}.

22.4 Isotropy, the Hyperbolic Plane, and Witt Decomposition

A diagonalization records the values a form takes on n chosen vectors and says nothing about the vectors the form kills. This section organizes a symmetric form around those vectors instead, the v with $\varphi(v, v) = 0$. Two cases are extreme. A form may have no such vector other than 0; or it may be spanned by two of them paired against each other, and then it is a single two-dimensional form which is the same over every field. The theorem of this section splits an arbitrary symmetric form into pieces of exactly those two kinds together with its radical, and so isolates the one summand in which the arithmetic of k is still visible.

The number of two-dimensional pieces and the isometry class of the remaining piece are determined by the form, but the argument for that is Witt cancellation and is deferred to section 22.5. What is available here without it is a statement about the subspaces on which the form vanishes identically: the largest dimension such a subspace can have is an invariant by construction, and it counts the two-dimensional pieces in a decomposition built from a subspace attaining it.

Every statement below breaks V into mutually orthogonal subspaces, and the notation for that is fixed first. It has already been used in box 22.1.9.

Definition 22.4.1: Orthogonal Sum of Forms

Let k be a field.

1. Let φ and ψ be bilinear forms on finite-dimensional k-vector spaces V and W. Their *orthogonal sum* is the bilinear form $\varphi \perp \psi$ on $V \oplus W$ given by

$$(\varphi \perp \psi)\big((v, w), (v', w')\big) := \varphi(v, v') + \psi(w, w')$$

   It is symmetric when φ and ψ are.

2. Let φ be a bilinear form on V and let $U_1, U_2 \subseteq V$ be subspaces with $V = U_1 \oplus U_2$ and $\varphi(u_1, u_2) = 0$ for all $u_1 \in U_1$ and $u_2 \in U_2$. Then V is the *internal orthogonal sum* of U_1 and U_2, written $V = U_1 \perp U_2$.

Both constructions iterate to finitely many summands.

The two halves of the definition describe one situation from outside and from inside. If $V = U_1 \perp U_2$ internally, then $(u_1, u_2) \mapsto u_1 + u_2$ is an isometry from $\varphi|_{U_1} \perp \varphi|_{U_2}$ to φ in the sense of definition 22.1.10, because expanding $\varphi(u_1 + u_2,\, u_1' + u_2')$ by bilinearity leaves only $\varphi(u_1, u_1')$ and $\varphi(u_2, u_2')$, the two cross terms being zero by hypothesis. Concatenating a basis β of U_1 with a basis γ of U_2 gives a basis of V in which every entry $\varphi(\beta_i, \gamma_j)$ vanishes, so by definition 13.3.18 the matrix of φ is block diagonal with blocks $[\varphi|_{U_1}]_\beta$ and $[\varphi|_{U_2}]_\gamma$. The determinant of an orthogonal sum is therefore the product of the determinants of its summands, and by lemma 13.3.23 the sum is nondegenerate exactly when both summands are. The diagonal notation of section 22.2 is an orthogonal sum of lines,

$$\langle a_1, \ldots, a_n\rangle = \langle a_1\rangle \perp \cdots \perp \langle a_n\rangle$$

so theorem 22.2.1 says exactly that every symmetric form is an orthogonal sum of one-dimensional forms.

Definition 22.4.2: Isotropic and Anisotropic Vectors

Let k be a field, let V be a finite-dimensional k-vector space, let φ be a symmetric bilinear form on V, and write $q(v) := \varphi(v, v)$ for the associated quadratic form. A nonzero vector $v \in V$ is *isotropic* if $q(v) = 0$, and *anisotropic* if $q(v) \neq 0$. The form φ is *isotropic* if V contains an isotropic vector, and *anisotropic* otherwise, that is if

$$q(v) = 0 \implies v = 0$$

The zero vector is excluded deliberately: $q(0) = 0$ for every form, so admitting it would make every form isotropic. An anisotropic form is automatically nondegenerate, since a nonzero $v \in \operatorname{rad}(\varphi)$ (definition 13.3.22) satisfies $\varphi(v, x) = 0$ for every x and in particular $q(v) = 0$, which is an isotropic vector. Restricting an anisotropic form to a subspace leaves it anisotropic, since the defining implication is inherited by subsets, and hence leaves it nondegenerate.

A subspace on which φ vanishes identically, $\varphi(w, w') = 0$ for all $w, w' \in W$, is *totally isotropic* (definition 22.3.6). When $\operatorname{char} k \neq 2$ it suffices to check the quadratic form on W: if $q(w) = 0$ for every $w \in W$, then for $w, w' \in W$,

$$0 = q(w + w') = q(w) + 2\varphi(w, w') + q(w') = 2\varphi(w, w')$$

and dividing by 2 gives $\varphi(w, w') = 0$. A totally isotropic subspace is thus one all of whose nonzero vectors are isotropic.

Example 22.4.3: Isotropy of Diagonal Forms

Whether a form is isotropic depends on the field and not only on the diagonal entries.

1. $\langle 1, 1\rangle$ over $\mathbb{R}$ is anisotropic: $x^2 + y^2 = 0$ forces $x^2 = -y^2$, and a sum of two nonnegative reals vanishes only when both do.

2. $\langle 1, 1\rangle$ over $\mathbb{F}_5$ is isotropic. The vector $(2, 1)$ is nonzero and $2^2 + 1^2 = 5 = 0$ in $\mathbb{F}_5$. Over $\mathbb{F}_3$ the squares are 0 and 1, so $x^2 + y^2$ takes only the values $0, 1, 2$ and vanishes only at $x = y = 0$: the same diagonal entries give an anisotropic form.

3. $\langle 1, -1\rangle$ is isotropic over every field, since $q(1, 1) = 1 - 1 = 0$ and $(1, 1) \neq 0$.

4. Let d be a squarefree integer with $d \neq 1$ and let $F = \mathbb{Q}(\sqrt{d})$. By example 22.1.11 the trace form of $F/\mathbb{Q}$ is $\langle 2, 2d\rangle$ in the basis $(1, \sqrt{d})$, with quadratic form $q(x, y) = 2x^2 + 2dy^2$. A zero with $y = 0$ forces $x = 0$, so an isotropic vector has $y \neq 0$, and dividing $x^2 = -dy^2$ by y^2 gives $-d = (x/y)^2$. The trace form is therefore isotropic exactly when $-d$ is a square in $\mathbb{Q}$. A positive squarefree integer is a rational square only when it equals 1, so this happens for $d = -1$ alone: of all the quadratic fields, only $\mathbb{Q}(i)$ has isotropic trace form, and $q(1, 1) = 2 - 2 = 0$ exhibits the vector.

Item (3) is the smallest isotropic form, and it is worth presenting in coordinates that make its two isotropic directions visible at once. In place of one distinguished isotropic vector it carries two, paired against each other, and that pairing is what keeps the form nondegenerate.

Definition 22.4.4: The Hyperbolic Plane

Let k be a field and let φ be a symmetric bilinear form on a k-vector space U. A *hyperbolic pair* in U is a pair of vectors $e, f \in U$ with

$$\varphi(e,e) = \varphi(f,f) = 0, \qquad \varphi(e,f) = 1$$

The form φ is a *hyperbolic plane* if $\dim U = 2$ and U has a basis which is a hyperbolic pair, so that the matrix of φ in that basis is $\left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$. A hyperbolic plane is written H. A subspace of (V, φ) is *a hyperbolic plane in* V when the restriction of φ to it is one, and the orthogonal sum of m copies,

$$mH := \underbrace{H \perp \cdots \perp H}_{m}$$

is the *hyperbolic space* of rank $2m$, with $0H := 0$.

Writing H for *the* hyperbolic plane is legitimate because any two are isometric: if (e, f) and (e', f') are hyperbolic pairs spanning U and U', then the linear isomorphism carrying $e \mapsto e'$ and $f \mapsto f'$ gives the two forms the same matrix in the corresponding bases, and by lemma 13.3.19 a form is recovered from its matrix in a basis, so the isomorphism is an isometry. That matrix has determinant $-1 \neq 0$, so a hyperbolic plane is nondegenerate by lemma 13.3.23, and its discriminant (definition 22.1.7) is the square class of -1.

Example 22.4.5: The Hyperbolic Plane as a Diagonal Form

Let $\operatorname{char} k \neq 2$ and let H have hyperbolic pair (e, f). Put

$$x := e + \tfrac{1}{2} f, \qquad y := e - \tfrac{1}{2} f$$

Then $x + y = 2e$ and $x - y = f$, and since $2 \neq 0$ these recover e and f, so (x, y) is again a basis of H. Expanding by bilinearity and using $\varphi(e,e) = \varphi(f,f) = 0$ together with $\varphi(e,f) = 1$,

$$q(x) = 2 \cdot \tfrac{1}{2} \varphi(e,f) = 1, \qquad q(y) = -2 \cdot \tfrac{1}{2} \varphi(e,f) = -1$$

$$\varphi(x,y) = -\tfrac{1}{2}\varphi(e,f) + \tfrac{1}{2}\varphi(f,e) = 0$$

so (x, y) is an orthogonal basis and

$$H \cong \langle 1, -1 \rangle$$

The product of the diagonal entries is -1, which agrees through corollary 22.2.3 with the determinant computed above. Over $\mathbb{R}$ this is the indefinite plane $x^2 - y^2$. Over a field containing an i with $i^2 = -1$, proposition 22.2.2 applied with $c_1 = 1$ and $c_2 = i$ gives $\langle 1, -1 \rangle \cong \langle 1, 1 \rangle$; combined with example 22.2.4, every nondegenerate binary form over an algebraically closed field of characteristic other than 2 is a hyperbolic plane.

The hyperbolic plane was assembled from a pair of isotropic vectors. The next result runs the construction in reverse: a single isotropic vector of a nondegenerate form always has a partner, and the plane the two span splits off as an orthogonal summand. The splitting is what makes the construction iterable, and iterating it is the proof of the decomposition theorem.

Proposition 22.4.6: Every Isotropic Vector Lies in a Hyperbolic Plane

Let k be a field with $\operatorname{char} k \neq 2$, let V be a finite-dimensional k-vector space, and let φ be a nondegenerate symmetric bilinear form on V. Let $v \in V$ be isotropic. Then there is an $f \in V$ such that (v, f) is a hyperbolic pair, and, writing $H_v := \operatorname{span}(v, f)$,

$$V = H_v \perp H_v^{\perp}$$

where H_v is a hyperbolic plane in V and $\varphi|_{H_v^{\perp}}$ is nondegenerate. In particular a nondegenerate symmetric form is anisotropic if and only if V contains no hyperbolic plane.

***Proof*:**
Step 1: producing the partner. An isotropic vector is nonzero, so $v \notin \operatorname{rad}(\varphi) = 0$ and there is a $w_0 \in V$ with $\varphi(v, w_0) \neq 0$. Replacing w_0 by $w := \varphi(v, w_0)^{-1} w_0$ gives $\varphi(v, w) = 1$. Set

$$f := w - \frac{q(w)}{2}\, v$$

which uses $\operatorname{char} k \neq 2$ to divide. Using $q(v) = \varphi(v, v) = 0$,

$$\varphi(v, f) = \varphi(v, w) - \frac{q(w)}{2}\varphi(v, v) = 1$$

and, expanding by bilinearity and symmetry,

$$\varphi(f, f) = \varphi(w, w) - 2 \cdot \frac{q(w)}{2}\varphi(w, v) + \frac{q(w)^2}{4}\varphi(v, v) = q(w) - q(w) = 0$$

so (v, f) is a hyperbolic pair. The two vectors are linearly independent: a relation $f = \lambda v$ would give $1 = \varphi(v, f) = \lambda\varphi(v, v) = 0$, and $v \neq 0$. Hence $\dim H_v = 2$, the matrix of $\varphi|_{H_v}$ in the basis (v, f) is $\left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$, and H_v is a hyperbolic plane in V, nondegenerate by lemma 13.3.23.

Step 2: the splitting. Let $x \in V$ and put

$$x' := x - \varphi(x, f)\, v - \varphi(x, v)\, f$$

Using the values of φ on the pair (v, f),

$$\varphi(x', v) = \varphi(x, v) - \varphi(x, f)\varphi(v, v) - \varphi(x, v)\varphi(f, v) = \varphi(x, v) - \varphi(x, v) = 0$$

$$\varphi(x', f) = \varphi(x, f) - \varphi(x, f)\varphi(v, f) - \varphi(x, v)\varphi(f, f) = \varphi(x, f) - \varphi(x, f) = 0$$

so x' is orthogonal to v and to f, hence to their span by lemma 13.3.21, that is $x' \in H_v^{\perp}$. Since $x - x' \in H_v$, this gives $V = H_v + H_v^{\perp}$. The sum is direct: an element $u \in H_v \cap H_v^{\perp}$ lies in H_v and is orthogonal to every element of H_v, hence lies in $\operatorname{rad}(\varphi|_{H_v})$, which is zero by Step 1. The two summands are orthogonal by the definition of $H_v^{\perp}$ (definition 13.3.20), so $V = H_v \perp H_v^{\perp}$ in the sense of definition 22.4.1.

Step 3: the complement is nondegenerate. Let $x \in \operatorname{rad}\big(\varphi|_{H_v^{\perp}}\big)$, so $\varphi(x, y) = 0$ for every $y \in H_v^{\perp}$. Since $x \in H_v^{\perp}$ and φ is symmetric, $\varphi(x, y) = 0$ for every $y \in H_v$ as well. By Step 2 each element of V is a sum of one vector of each kind, so $\varphi(x, -)$ vanishes on V and $x \in \operatorname{rad}(\varphi) = 0$.

Step 4: the equivalence. If φ is isotropic, Steps 1 to 3 produce a hyperbolic plane in V. Conversely, a hyperbolic plane in V contains a vector e with $\varphi(e,e) = 0$ and $e \neq 0$, the latter because $\varphi(e,f) = 1$ forbids $e = 0$, so φ is isotropic. Hence V contains a hyperbolic plane precisely when φ is isotropic, completing the proof.

The proposition converts one-dimensional information, the existence of a single vector with $q(v) = 0$, into a two-dimensional summand of known isometry type. What is left of φ is the restriction $\varphi|_{H_v^\perp}$, again nondegenerate and of dimension two smaller, so the exchange can be repeated until no isotropic vector remains. In dimension two there is nothing left to repeat, and the statement becomes a criterion.

Example 22.4.7: Binary Forms and Hyperbolic Planes

Let $\operatorname{char} k \neq 2$ and let $\varphi \cong \langle a, b\rangle$ be a nondegenerate form of rank 2, so that $a, b \in k^\times$ by proposition 22.2.2. If φ is isotropic then proposition 22.4.6 puts a hyperbolic plane H_v inside V, and $\dim H_v = 2 = \dim V$ forces $H_v = V$, so $\varphi \cong H$. A nondegenerate binary form is therefore either anisotropic or a hyperbolic plane.

The entries determine which of the two occurs. An isotropic vector for $q(x,y) = ax^2 + by^2$ has both coordinates nonzero, since $y = 0$ forces $ax^2 = 0$ and hence $x = 0$, and symmetrically $x = 0$ forces $y = 0$. Multiplying $ax^2 = -by^2$ by b and dividing by x^2 gives

$$-ab = \left(\frac{by}{x}\right)^2$$

Conversely, if $-ab = c^2$ then $c \neq 0$ because $ab \neq 0$, and the nonzero vector (b, c) satisfies $ab^2 + bc^2 = ab^2 - ab \cdot b = 0$. So $\langle a, b\rangle$ is a hyperbolic plane exactly when $-ab$ is a square, which by corollary 22.2.3 says exactly that $\operatorname{disc}(\varphi)$ is the class of -1. For instance $\langle 1, 1\rangle$ over $\mathbb{F}_5$ has $-ab = -1 = 4 = 2^2$, so it is the hyperbolic plane, whereas over $\mathbb{F}_3$ the element $-1 = 2$ is not a square and the form is anisotropic, matching example 22.4.3.

Iterating proposition 22.4.6 splits a hyperbolic plane off a nondegenerate form as long as one isotropic vector can be found, and the process stops only when none can. A degenerate form has to be handled first: the nonzero vectors of its radical are isotropic and orthogonal to all of V, so no partner exists for them and proposition 22.4.6 does not apply. Splitting the radical off and then iterating gives the decomposition.

Theorem 22.4.8: Witt Decomposition

Let k be a field with $\operatorname{char} k \neq 2$, let V be a finite-dimensional k-vector space and let φ be a symmetric bilinear form on V. Then there are an integer $m \geq 0$, subspaces $H_1, \dots, H_m \subseteq V$ each of which is a hyperbolic plane in V, and a subspace $V_{\mathrm{an}} \subseteq V$ on which φ is anisotropic, such that

$$V = \operatorname{rad}(\varphi) \perp H_1 \perp \cdots \perp H_m \perp V_{\mathrm{an}}$$

Writing $r = \dim \operatorname{rad}(\varphi)$, this reads

$$\varphi \cong \underbrace{\langle 0, \dots, 0\rangle}_{r} \perp mH \perp \varphi|_{V_{\mathrm{an}}}$$

***Proof*:**
Step 1: splitting off the radical. Write $R := \mathrm{rad}(\varphi)$ and choose a basis of R. By theorem 13.1.8 it extends to a basis of V; let V' be the span of the added vectors, so that $V = R \oplus V'$. Every element of R is orthogonal to every element of V, in particular to every element of V', so $V = R \perp V'$ in the sense of definition 22.4.1, and $\varphi|_R = 0$.

The restricted form $\varphi|_{V'}$ is nondegenerate. Let $x \in V'$ satisfy $\varphi(x, y) = 0$ for all $y \in V'$. Since also $\varphi(x, y) = 0$ for all $y \in R$, and $V = R+V'$, the map $\varphi(x, -)$ vanishes on V, so $x \in R \cap V' = 0$.

Step 2: splitting hyperbolic planes off a nondegenerate form. It suffices to prove that a nondegenerate symmetric form ψ on a space U decomposes as $U = H_1 \perp \cdots \perp H_m \perp U_{\mathrm{an}}$ with each H_i a hyperbolic plane in U and $\psi|_{U_{\mathrm{an}}}$ anisotropic; applying this to $\varphi|_{V'}$ and appending Step 1 gives the theorem, because every summand of V' is orthogonal to R.

Induct on $\dim U$. If ψ is anisotropic, take $m = 0$ and $U_{\mathrm{an}} = U$. This covers $\dim U = 0$, where there is nothing to decompose, and $\dim U = 1$: an isotropic vector v would span U, and then $\psi(v, \lambda v) = \lambda\psi(v, v) = 0$ for every λ would put v in $\mathrm{rad}(\psi)$, contradicting nondegeneracy. Otherwise ψ has an isotropic vector, and proposition 22.4.6 supplies a hyperbolic plane $H_1 \subseteq U$ with

$$U = H_1 \perp H_1^{\perp}$$

and $\psi|_{H_1^{\perp}}$ nondegenerate. Since $\dim H_1^{\perp} = \dim U - 2$, the inductive hypothesis applies to $H_1^{\perp}$ and gives $H_1^{\perp} = H_2 \perp \cdots \perp H_m \perp U_{\mathrm{an}}$ with each H_i a hyperbolic plane and $\psi|_{U_{\mathrm{an}}}$ anisotropic. Every one of these subspaces lies in $H_1^{\perp}$ and is therefore orthogonal to H_1, so the displayed decomposition of U refines to $U = H_1 \perp H_2 \perp \cdots \perp H_m \perp U_{\mathrm{an}}$.

Step 3: the diagonal reading. Take the chosen basis of R, a hyperbolic pair in each H_i, and an orthogonal basis of V_{an}, which exists by theorem 22.2.1. Their concatenation is a basis of V, and by the block-diagonal description following definition 22.4.1 the matrix of φ in it is the block sum of the zero matrix of size r, m copies of $\left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$, and a diagonal matrix. That is the asserted isometry, completing the proof.

The three summands answer three different questions about φ. The radical is where the form carries no information at all, and its dimension is already known to be an invariant by proposition 22.2.2. The hyperbolic part carries no information beyond its rank, since any two hyperbolic planes over k are isometric, whatever k is. Everything that distinguishes φ from other forms of the same rank is therefore concentrated in $\varphi|_{V_{\mathrm{an}}}$, and how large that summand can be is a property of the field. Over an algebraically closed field it has dimension at most 1: a two-dimensional subspace of an anisotropic space is nondegenerate, hence a hyperbolic plane by example 22.4.5, hence isotropic, which contradicts anisotropy. This recovers example 22.2.4, where the classification collapsed to the rank. Over a finite field the bound is one step weaker, and a counting argument gives it.

Example 22.4.9: Isotropy over a Finite Field
Let $k = \mathbb{F}_q$ with q odd.

Every equation $ax^2+by^2 = c$ with $a, b \in k^{\times}$ and $c \in k$ has a solution. As recorded in example 22.2.5, the squaring map on $\mathbb{F}_q^{\times}$ has kernel $\{\pm 1\}$, which has two elements because q is odd, so there are $(q-1)/2$ nonzero squares and the set $\left\{x^2 \;\middle|\; x \in k\right\}$ has $(q+1)/2$ elements. Multiplication by a is a bijection of k, so

$$S := \{ax^2 \mid x \in k\}, \qquad T := \{c - by^2 \mid y \in k\}$$

each have $(q+1)/2$ elements. Their sizes add to $q + 1 > q = |k|$, so S and T are not disjoint, and a common element gives x, y with $ax^2 = c - by^2$.

Every nondegenerate form of rank at least 3 over $\mathbb{F}_q$ is isotropic. Diagonalize by theorem 22.2.1 as $\varphi \cong \langle a_1, \ldots, a_n \rangle$ in an orthogonal basis $w_1, \ldots, w_n$; nondegeneracy makes every a_i nonzero by the last claim of proposition 22.2.2. With $n \geq 3$, solve $a_1 x^2 + a_2 y^2 = -a_3$ by the previous paragraph and set $v := xw_1 + yw_2 + w_3$, which is nonzero because its third coordinate is 1. Then $q(v) = a_1 x^2 + a_2 y^2 + a_3 = 0$.

Consequently, in the Witt decomposition of a form over $\mathbb{F}_q$ the anisotropic summand has dimension at most 2. Both values occur: $\langle 1 \rangle$ is anisotropic of rank 1, and if $\epsilon \in \mathbb{F}_q^\times$ is a non-square then $\langle 1, -\epsilon \rangle$ has $-ab = \epsilon$ a non-square, so it is anisotropic of rank 2 by example 22.4.7.

Theorem 22.4.8 produces some number m of hyperbolic planes, and nothing in its proof says that a different sequence of choices produces the same number. A quantity that is visibly independent of all choices is available instead, and the rest of the section relates the two.

Definition 22.4.10: The Witt Index

Let k be a field, let V be a finite-dimensional k-vector space and let φ be a symmetric bilinear form on V. The *Witt index* of φ is

$$\nu(\varphi) := \max \{\dim W \mid W \subseteq V \text{ a totally isotropic subspace}\}$$

The maximum exists: the zero subspace is totally isotropic, so the set of dimensions is nonempty, and it is bounded above by $\dim V$.

The Witt index is an invariant of φ by construction, since it is defined without reference to a basis or a decomposition, and isometries carry totally isotropic subspaces to totally isotropic subspaces of the same dimension. It is zero exactly when φ is anisotropic, since a one-dimensional totally isotropic subspace is the span of an isotropic vector. The next result computes it against theorem 22.4.8.

Proposition 22.4.11: The Witt Index Counts Hyperbolic Planes

Let k be a field with $\operatorname{char} k \neq 2$, let V be a finite-dimensional k-vector space of dimension n and let φ be a nondegenerate symmetric bilinear form on V.

1. Let $W \subseteq V$ be totally isotropic with $\dim W = d$. Then there are hyperbolic planes $H_1, \dots, H_d$ in V, with hyperbolic pairs (e_i, f_i) such that $e_1, \dots, e_d$ is a basis of W, and a subspace U with $\varphi|_U$ nondegenerate, such that
$$V = H_1 \perp \cdots \perp H_d \perp U$$
In particular $2d \leq n$, which is also proposition 22.3.8(2) applied to the totally isotropic W.
2. In the situation of (1), the form $\varphi|_U$ is anisotropic if and only if W is maximal among the totally isotropic subspaces of V.
3. $\nu(\varphi) \leq n/2$; some decomposition as in theorem 22.4.8 has exactly $\nu(\varphi)$ hyperbolic planes; and every such decomposition has at most $\nu(\varphi)$ of them.

***Proof*:**

1. Induct on d. For $d = 0$ take $U = V$, and there is nothing to prove. Let $d \geq 1$ and choose a basis $w_1, \dots, w_d$ of W. Since W is totally isotropic and $w_1 \neq 0$, the vector w_1 is isotropic, so proposition 22.4.6 gives an f_1 making (w_1, f_1) a hyperbolic pair, with $H_1 = \operatorname{span}(w_1, f_1)$ a hyperbolic plane, $V = H_1 \perp H_1^\perp$ and $\varphi|_{H_1^\perp}$ nondegenerate.

   Consider the linear map $\lambda \colon W \to k$ defined by $\lambda(x) = \varphi(x, f_1)$. It is nonzero, since $\lambda(w_1) = \varphi(w_1, f_1) = 1$, so $\ker \lambda$ has dimension $d-1$. Moreover $\ker \lambda = W \cap H_1^\perp$: every $x \in W$ satisfies $\varphi(x, w_1) = 0$ because W is totally isotropic and $w_1 \in W$, so x lies in $H_1^\perp$ exactly when in addition $\varphi(x, f_1) = 0$, using lemma 13.3.21 to reduce orthogonality to H_1 to orthogonality to the two spanning vectors.

   Set $W_1 := W \cap H_1^\perp$, a totally isotropic subspace of $H_1^\perp$ of dimension $d-1$. The inductive hypothesis, applied to the nondegenerate form $\varphi|_{H_1^\perp}$, gives hyperbolic planes $H_2, \dots, H_d$ in $H_1^\perp$ with hyperbolic pairs (e_i, f_i) whose vectors $e_2, \dots, e_d$ form a basis of W_1, and a subspace U with $\varphi|_U$ nondegenerate and $H_1^\perp = H_2 \perp \cdots \perp H_d \perp U$. All of these lie in $H_1^\perp$ and are therefore orthogonal to H_1, so $V = H_1 \perp H_2 \perp \cdots \perp H_d \perp U$. Finally $w_1 \notin W_1$, because $\lambda(w_1) = 1$ while λ vanishes on W_1, so $w_1, e_2, \dots, e_d$ are d linearly independent vectors of W and hence a basis of it; setting $e_1 := w_1$ gives the statement. The dimension count is $n = 2d + \dim U \geq 2d$.

2. Write $\mathcal{H} := H_1 \perp \cdots \perp H_d$, so $V = \mathcal{H} \oplus U$ and $W = \operatorname{span}(e_1, \dots, e_d)$ by (1).

   Suppose first that $\varphi|_U$ is anisotropic, and let $W' \supseteq W$ be totally isotropic. Take $x \in W'$ and write $x = \sum_i (a_i e_i + b_i f_i) + u$ with $u \in U$. For each j the vector e_j lies in $W \subseteq W'$, so $\varphi(x, e_j) = 0$; on the other hand the only summand of x not orthogonal to e_j is $b_j f_j$, whence $\varphi(x, e_j) = b_j$. So every b_j vanishes and $x = \sum_i a_i e_i + u$. Expanding $q(x)$ by bilinearity, the terms $\varphi(e_i, e_j)$ vanish for all i, j and the terms $\varphi(e_i, u)$ vanish because $e_i \in \mathcal{H}$ and $u \in U$, leaving $q(x) = q(u)$. As $x \in W'$ is isotropic or zero, $q(u) = 0$, so $u = 0$ by anisotropy and $x \in W$. Hence $W' = W$ and W is maximal.

Suppose instead that $\varphi|_U$ is isotropic, say $u \in U$ is nonzero with $q(u) = 0$. Then $W + ku$ is totally isotropic: expanding $\varphi(w + au, w' + a'u)$ leaves $\varphi(w, w') = 0$, the cross terms $\varphi(w, u) = 0$ and $\varphi(u, w') = 0$ since $W \subseteq \mathcal{H}$ and $u \in U$, and $aa'q(u) = 0$. It strictly contains W, because $u \notin \mathcal{H} \supseteq W$, so W is not maximal. The two implications together give the stated equivalence.

3. Applying (1) to a totally isotropic subspace of dimension $\nu(\varphi)$ gives $2\nu(\varphi) \leq n$. That subspace is maximal, being of largest dimension, so (2) makes the corresponding U anisotropic and
$$V = H_1 \perp \cdots \perp H_{\nu(\varphi)} \perp U$$
is a decomposition as in theorem 22.4.8, with radical zero and exactly $\nu(\varphi)$ hyperbolic planes. Conversely, let $V = H_1' \perp \cdots \perp H_m' \perp V_{\text{an}}$ be any such decomposition, with hyperbolic pair (e_i', f_i') in H_i'. The vectors $e_1', \ldots, e_m'$ lie in distinct summands of a direct sum and are nonzero, hence are linearly independent, and their span is totally isotropic: $\varphi(e_i', e_j') = 0$ for $i \neq j$ by orthogonality of the summands and $\varphi(e_i', e_i') = 0$ by the definition of a hyperbolic pair. So $m \leq \nu(\varphi)$, completing the proof.

Part (3) does not say that every decomposition has $\nu(\varphi)$ hyperbolic planes. It produces one that does and bounds all the others by that number, and no argument given above closes the gap: comparing two decompositions of the same space requires an isometry between them, and producing one is what Witt cancellation supplies. Section 22.5 proves it, and with it every decomposition of a nondegenerate φ in theorem 22.4.8 has $m = \nu(\varphi)$ and an anisotropic summand determined up to isometry. Nondegeneracy is needed there: the Witt index of a degenerate form counts its radical as well, and running the same statement on a complement of the radical gives $m = \nu(\varphi) - \dim \operatorname{rad}(\varphi)$ instead, by exercise 22.4.1 (4). What is established here is that $\nu(\varphi)$ bounds the number of hyperbolic planes in any decomposition, and that the bound is attained when φ is nondegenerate. Over $\mathbb{R}$ that number can be computed directly from a diagonalization.

Example 22.4.12: The Witt Index over the Reals

Let φ be nondegenerate on a real vector space V of dimension n. By theorem 22.2.1 and proposition 22.2.2 applied with $c_i = |a_i|^{-1/2}$, there is a basis $u_1, \ldots, u_p, w_1, \ldots, w_r$ of V, orthogonal for φ, with $q(u_i) = 1$ and $q(w_j) = -1$ and $p + r = n$.

Pairing the vectors two at a time, $\operatorname{span}(u_i, w_i)$ carries the form $\langle 1, -1 \rangle$, which is a hyperbolic plane by example 22.4.5. Doing this for $i \leq \min(p, r)$ and collecting the remaining $|p - r|$ basis vectors, which all have the same value of q, gives
$$\varphi \cong \min(p, r)\, H \perp \langle \pm 1, \ldots, \pm 1 \rangle$$
with $|p - r|$ entries in the last summand, all equal to $+1$ if $p > r$ and all to -1 if $r > p$. That summand is anisotropic, a sum of $|p - r|$ squares being zero only when each is.

The Witt index is $\min(p, r)$. It is at least that, by the span of the $u_i + w_i$ for $i \leq \min(p, r)$, which are linearly independent and satisfy $q(u_i + w_i) = 1 - 1 = 0$ and $\varphi(u_i + w_i, u_j + w_j) = 0$ for $i \neq j$. For the reverse bound let W be totally isotropic and let $P := \operatorname{span}(u_1, \ldots, u_p)$. A nonzero $x \in P$ has $q(x) = \sum_i x_i^2 > 0$, so x is not isotropic and $W \cap P = 0$, forcing $\dim W \leq n - p = r$. The same argument with $N := \operatorname{span}(w_1, \ldots, w_r)$, where $q(y) = -\sum_j y_j^2 < 0$ for $y \neq 0$, gives $\dim W \leq n - r = p$. Hence $\nu(\varphi) = \min(p, r)$, in agreement with proposition 22.4.11(3).

Since $\nu(\varphi)$ and n depend only on φ, the unordered pair $\{p, r\}$ does too. Which of the two is p is not settled by this argument, and that remaining bit is Sylvester's law of inertia, which needs the order on $\mathbb{R}$ in a way the present chapter does not use.

Exercise 22.4.1

1. Let $\operatorname{char} k \neq 2$ and let $a \in k^\times$. Show that $\langle 1, -a\rangle$ is isotropic if and only if a is a square in k, and that in that case $\langle 1, -a\rangle \cong H$. Using that $\mathbb{Q}^\times/(\mathbb{Q}^\times)^2$ is infinite, deduce that $\mathbb{Q}$ carries infinitely many pairwise non-isometric anisotropic binary forms.
2. Let $\operatorname{char} k \neq 2$. Show that the hyperbolic plane represents every scalar: for each $c \in k$ there is $x \in H$ with $q(x) = c$. Deduce that a nondegenerate isotropic form on any V represents every element of k.
3. Let $\operatorname{char} k \neq 2$ and let $\varphi \cong \langle a, b\rangle$ be nondegenerate of rank 2. Using example 22.4.7, decide for which pairs (a, b) over $k = \mathbb{Q}$ the form is a hyperbolic plane, and exhibit one anisotropic and one hyperbolic example.
4. Let φ be a symmetric bilinear form on V with radical $R = \operatorname{rad}(\varphi)$, and let $\bar\varphi$ be the induced nondegenerate form on V/R from exercise 22.1.1 (5). Show that $\nu(\varphi) = \dim R + \nu(\bar\varphi)$.
5. Let φ be nondegenerate on V and let $W \subseteq V$ be totally isotropic. Show that W is maximal among totally isotropic subspaces if and only if every isotropic vector of $W^\perp$ already lies in W.
6. Show that $\nu(\varphi \perp \psi) \geq \nu(\varphi) + \nu(\psi)$ for symmetric bilinear forms φ and ψ, and give an example over $\mathbb{R}$ where the inequality is strict.
7. Let q be odd and let φ be a nondegenerate symmetric bilinear form of odd rank n over $\mathbb{F}_q$. Show that $\nu(\varphi) = (n-1)/2$, so the Witt index over a finite field is forced by the rank alone when the rank is odd.
8. Let $V = \mathbb{F}_7^3$ with φ the standard dot product, so $q(x, y, z) = x^2+y^2+z^2$. Verify that $v = (3, 2, 1)$ is isotropic, run the construction in the proof of proposition 22.4.6 with $w = (0, 0, 1)$ to produce a partner f, compute a spanning vector of $H_v^\perp$, and write out the resulting decomposition $V = H_v \perp \langle a\rangle$. Is a a square in $\mathbb{F}_7$?

22.5 Witt Cancellation and Witt's Extension Theorem

Theorem 22.4.8 splits a form into an anisotropic part, a sum of hyperbolic planes and the radical, and each step of that construction rests on a choice: which isotropic vector to complete to a hyperbolic plane, and which complement to continue in. Nothing proved so far forces two runs of the construction to return the same number of hyperbolic planes, or isometric anisotropic parts. This section proves that they do. The result that delivers it is a cancellation law for orthogonal sums: a nondegenerate summand common to both sides of an isometry may be deleted. The section then proves Witt's extension theorem, which says that an isometry between two subspaces of a nondegenerate space is always the restriction of an isometry of the whole space. Both rest on one construction, the reflection in an anisotropic vector, which this book has not yet defined.

Throughout, $q(x) = \varphi(x, x)$ is the quadratic form attached to a symmetric bilinear form φ by proposition 22.1.2, and $\varphi \perp \psi$ is the orthogonal sum of definition 22.4.1. One further fact about that operation is used repeatedly below: the bijections $\big(v, (w, x)\big) \mapsto \big((v, w), x\big)$ and $(v, w) \mapsto (w, v)$ leave the defining expression unchanged, so they are isometries, and an orthogonal sum of several forms may be regrouped and reordered freely without changing its isometry class.

Every theorem below produces an isometry of a space with itself, so those isometries need a name.

Definition 22.5.1: The Orthogonal Group of a Form

Let k be a field, let V be a finite-dimensional k-vector space and let φ be a symmetric bilinear form on V. The *orthogonal group* of φ is

$$O(\varphi) = \{\sigma \colon V \to V \mid \sigma \text{ is an isometry of } (V, \varphi) \text{ with itself}\}$$

where isometry is meant in the sense of definition 22.1.10, and the operation is composition of maps.

That this is a group takes nothing beyond the definition: id_V is an isometry; a composite of linear bijections preserving φ is again a linear bijection preserving φ; and if σ is an isometry then so is σ^{-1}, since $\varphi(\sigma^{-1}x, \sigma^{-1}y) = \varphi(\sigma\sigma^{-1}x, \sigma\sigma^{-1}y) = \varphi(x, y)$ for all $x, y \in V$.

The construction used throughout the section takes an anisotropic vector (definition 22.4.2) and produces the isometry that negates it and fixes everything orthogonal to it. Over $\mathbb{R}$ with the dot product this is the familiar mirror reflection in the hyperplane $\{v\}^\perp$, and the formula below is the one from that setting, written so that it makes sense over any field in which $q(v)$ is invertible.

Definition 22.5.2: Reflection in an Anisotropic Vector

Let k be a field with $\operatorname{char} k \neq 2$, let V be a finite-dimensional k-vector space with a symmetric bilinear form φ, and let $v \in V$ be anisotropic, that is $q(v) = \varphi(v, v) \neq 0$. The *reflection in* v is the map $\tau_v \colon V \to V$ given by

$$\tau_v(x) = x - \frac{2\varphi(v, x)}{\varphi(v, v)}\, v$$

The denominator is nonzero precisely because v is anisotropic, which is why the construction is available for anisotropic vectors and for no others.

The properties that make τ_v useful are all direct computations with the defining formula.

Lemma 22.5.3: Reflections Are Involutive Isometries

Let k be a field with $\operatorname{char} k \neq 2$, let V be a finite-dimensional k-vector space with a symmetric bilinear form φ, and let $v \in V$ be anisotropic. Then:

1. $V = kv \oplus \{v\}^\perp$;
2. τ_v is linear, $\tau_v(v) = -v$, and $\tau_v(x) = x$ for every $x \in \{v\}^\perp$;
3. τ_v preserves φ and satisfies $\tau_v \circ \tau_v = \mathrm{id}_V$; in particular $\tau_v \in O(\varphi)$ and $\tau_v^{-1} = \tau_v$;
4. $\tau_{\lambda v} = \tau_v$ for every $\lambda \in k^\times$;
5. $\det \tau_v = -1$. In particular $\tau_v \neq \mathrm{id}_V$, so τ_v has order exactly 2 in $O(\varphi)$.

***Proof*:**
Write $a = q(v) = \varphi(v, v) \neq 0$ throughout. The set $\{v\}^\perp$ is a subspace by lemma 13.3.21.

1. Every $x \in V$ decomposes as
$$x = \frac{\varphi(v,x)}{a} v + \left(x - \frac{\varphi(v,x)}{a} v\right)$$
and applying $\varphi(v, -)$ to the bracketed vector gives $\varphi(v,x) - \frac{\varphi(v,x)}{a} a = 0$, so that vector lies in $\{v\}^\perp$ and $V = kv + \{v\}^\perp$. The sum is direct: if $\lambda v \in \{v\}^\perp$ then $0 = \varphi(v, \lambda v) = \lambda a$, and $a \neq 0$ forces $\lambda = 0$.

2. Linearity holds because $x \mapsto \varphi(v,x)$ is linear and τ_v subtracts a scalar multiple of the fixed vector v. Substituting $x = v$ gives $\tau_v(v) = v - \frac{2a}{a} v = v - 2v = -v$, and if $\varphi(v,x) = 0$ then the subtracted term vanishes, so $\tau_v(x) = x$.

3. For $x, y \in V$, expanding by bilinearity and symmetry,
$$\varphi\big(\tau_v(x), \tau_v(y)\big) = \varphi(x,y) - \frac{2\varphi(v,y)}{a}\varphi(x,v) - \frac{2\varphi(v,x)}{a}\varphi(v,y) + \frac{4\varphi(v,x)\varphi(v,y)}{a^2}\, a$$
The last three terms are $-2c$, $-2c$ and $+4c$ with $c = \varphi(v,x)\varphi(v,y)/a$, so they cancel and $\varphi(\tau_v(x), \tau_v(y)) = \varphi(x,y)$. For the involution, first
$$\varphi\big(v, \tau_v(x)\big) = \varphi(v,x) - \frac{2\varphi(v,x)}{a}\, a = -\varphi(v,x)$$
so
$$\tau_v\big(\tau_v(x)\big) = \tau_v(x) - \frac{2\varphi(v, \tau_v(x))}{a}\, v = x - \frac{2\varphi(v,x)}{a}\, v + \frac{2\varphi(v,x)}{a}\, v = x$$
Hence τ_v is a linear bijection preserving φ, which is an isometry of (V, φ) with itself in the sense of definition 22.1.10, so $\tau_v \in O(\varphi)$ by definition 22.5.1 and $\tau_v^{-1} = \tau_v$.

4. For $\lambda \neq 0$ the vector λv is anisotropic, since $q(\lambda v) = \lambda^2 a \neq 0$, and
$$\tau_{\lambda v}(x) = x - \frac{2\varphi(\lambda v, x)}{\lambda^2 a}\, \lambda v = x - \frac{2\lambda^2\varphi(v,x)}{\lambda^2 a}\, v = \tau_v(x)$$

5. By (1) choose a basis $v, z_2, \ldots, z_n$ of V with $z_2, \ldots, z_n$ a basis of $\{v\}^\perp$. By (2) the matrix of τ_v in this basis is $\operatorname{diag}(-1, 1, \ldots, 1)$, whose determinant is -1. Since $\operatorname{char} k \neq 2$ this is not 1, so $\tau_v \neq \mathrm{id}_V$, and $\tau_v^2 = \mathrm{id}_V$ from (3) then makes its order exactly 2, completing the proof.

Example 22.5.4: Reflections of $\langle 1, 1\rangle$ over $\mathbb{Q}$

Let $k = \mathbb{Q}$ and let φ be the form on $V = \mathbb{Q}^2$ with matrix the identity in the standard basis e_1, e_2, so $q(x_1, x_2) = x_1^2 + x_2^2$ and $\varphi \cong \langle 1, 1\rangle$.

Take $v = e_1$, with $q(e_1) = 1$. Then $\tau_{e_1}(x) = x - 2\varphi(e_1, x)e_1 = x - 2x_1 e_1$, so $\tau_{e_1}(e_1) = -e_1$ and $\tau_{e_1}(e_2) = e_2$, and the matrix of τ_{e_1} is $\operatorname{diag}(-1, 1)$.

Take instead $v = (1,1)$, with $q(v) = 2 \neq 0$. Here $\tau_v(x) = x - \frac{2\varphi(v,x)}{2}v = x - \varphi(v,x)v$, so

$$\tau_v(e_1) = (1,0) - (1,1) = (0,-1), \qquad \tau_v(e_2) = (0,1) - (1,1) = (-1,0)$$

and the matrix of τ_v is $\left(\begin{smallmatrix} 0 & -1 \\ -1 & 0 \end{smallmatrix}\right)$. Both matrices have determinant -1 and square to the identity, as lemma 22.5.3 requires.

Their product is

$$\begin{pmatrix} 0 & -1 \\ -1 & 0 \end{pmatrix}\begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$$

an isometry of determinant 1 whose square is $-\mathrm{id}$ and whose fourth power is id. So $O(\langle 1,1\rangle)$ over $\mathbb{Q}$ contains an element of order 4, which is not a reflection: reflections have order 2 and determinant -1. Products of reflections are therefore strictly more general than reflections, and corollary 22.5.10 will say how much more general.

Reflections are used here for one property: they move a vector to any other vector of the same nonzero value under q. That is how every theorem below produces the isometry it needs, and it takes at most two reflections.

Lemma 22.5.5: Two Reflections Move a Vector to Any Other of the Same Value

Let k be a field with $\operatorname{char} k \neq 2$, let V be a finite-dimensional k-vector space with a symmetric bilinear form φ, and let $u, w \in V$ satisfy $q(u) = q(w) \neq 0$. Then there is a $\sigma \in O(\varphi)$ with $\sigma(u) = w$, and σ can be taken to be a product of at most two reflections.

Proof:
Expanding both terms by bilinearity and using $q(w) = q(u)$,

$$q(u-w) + q(u+w) = \big(q(u) - 2\varphi(u,w) + q(w)\big) + \big(q(u) + 2\varphi(u,w) + q(w)\big) = 4q(u)$$

Since $\operatorname{char} k \neq 2$, the scalar $4 = 2^2$ is nonzero, so $4q(u) \neq 0$ and $q(u-w)$ and $q(u+w)$ do not both vanish.

Case 1: $q(u-w) \neq 0$. Put $v = u - w$, an anisotropic vector. Then

$$q(v) = q(u) - 2\varphi(u,w) + q(w) = 2\big(q(u) - \varphi(u,w)\big) = 2\varphi(v,u)$$

the last equality because $\varphi(v,u) = \varphi(u,u) - \varphi(w,u) = q(u) - \varphi(u,w)$. Hence $2\varphi(v,u)/q(v) = 1$, and

$$\tau_v(u) = u - \frac{2\varphi(v,u)}{q(v)}v = u - v = w$$

so $\sigma = \tau_v$ works, and it lies in $O(\varphi)$ by lemma 22.5.3.

Case 2: $q(u-w) = 0$. Then $q(u+w) = 4q(u) \neq 0$ by the displayed identity. The vectors u and $-w$ satisfy $q(-w) = q(w) = q(u) \neq 0$ and $q\big(u - (-w)\big) = q(u+w) \neq 0$, so Case 1 applied to the pair $u, -w$ gives $\tau_{u+w}(u) = -w$. Also $q(w) \neq 0$, so τ_w is defined, and $\tau_w(-w) = -\tau_w(w) = w$ by lemma 22.5.3(2). Therefore

$$(\tau_w \circ \tau_{u+w})(u) = \tau_w(-w) = w$$

and $\sigma = \tau_w \circ \tau_{u+w}$ is a product of two reflections, lying in $O(\varphi)$ because $O(\varphi)$ is closed under composition (definition 22.5.1), as we sought to show.

Both cases occur. In example 22.5.4 the vectors $u = e_1$ and $w = e_2$ have $q(u) = q(w) = 1$ and $q(u-w) = q(1,-1) = 2 \neq 0$, so Case 1 applies and the single reflection $\tau_{(1,-1)}$ carries e_1 to e_2; indeed $\tau_{(1,-1)}(e_1) = e_1 - \frac{2 \cdot 1}{2}(1,-1) = (0,1)$. Case 2 is not an artifact of the proof: over $\mathbb{Q}$ with $\varphi \cong \langle 1, 1, -1 \rangle$ the vectors $u = e_1$ and $w = (1,1,1)$ both have $q = 1$ while $q(u-w) = q(0,-1,-1) = 0$, and there no single reflection suffices: an equality $\tau_v(u) = w$ reads $u - w = \frac{2\varphi(v,u)}{q(v)} v$, and $w \neq u$ makes the coefficient nonzero, so $q(u-w)$ would be a nonzero square multiple of $q(v) \neq 0$ and in particular nonzero, contradicting $q(u-w) = 0$.

Lemma 22.5.5 carries no nondegeneracy hypothesis on φ, which matters immediately: the cancellation theorem is proved for arbitrary φ_1 and φ_2, whose radicals are not controlled, and only the summand being cancelled is required to be nondegenerate. The proof reduces that summand to one dimension, where the lemma does the work.

Theorem 22.5.6: Witt Cancellation

Let k be a field with $\operatorname{char} k \neq 2$. Let ψ be a *nondegenerate* symmetric bilinear form on a finite-dimensional k-vector space, and let φ_1, φ_2 be symmetric bilinear forms on finite-dimensional k-vector spaces. If

$$\psi \perp \varphi_1 \cong \psi \perp \varphi_2$$

then $\varphi_1 \cong \varphi_2$.

***Proof*:**
Step 1: reduction to a one-dimensional ψ. Let ψ live on U with $\dim U = m$. By theorem 22.2.1 the space U has an orthogonal basis $u_1, \ldots, u_m$, so $\psi \cong \langle c_1, \ldots, c_m \rangle$ with $c_i = \psi(u_i, u_i)$. By the last claim of proposition 22.2.2 the number of indices with $c_i = 0$ is $\dim \operatorname{rad}(\psi)$, which is 0 because ψ is nondegenerate; hence every $c_i \neq 0$. Regrouping the orthogonal sum,

$$\psi \perp \varphi_j \cong \langle c_1 \rangle \perp \big(\langle c_2, \ldots, c_m \rangle \perp \varphi_j \big) \qquad (j = 1, 2)$$

So if the theorem holds whenever ψ is one-dimensional and nondegenerate, then cancelling $\langle c_1 \rangle$ from the hypothesis gives $\langle c_2, \ldots, c_m \rangle \perp \varphi_1 \cong \langle c_2, \ldots, c_m \rangle \perp \varphi_2$, and $m-1$ further cancellations, formally an induction on m with the case $m = 0$ reading $\varphi_1 \cong \varphi_2$, give the theorem. It therefore suffices to treat $\psi = \langle c \rangle$ with $c \in k^\times$.

Step 2: the one-dimensional case. Let φ_j live on W_j and realize $\langle c \rangle \perp \varphi_j$ on $V_j = k e_j \oplus W_j$, where $q(e_j) = c$, $\varphi(e_j, W_j) = 0$, and the form restricted to W_j is φ_j. Write $\varphi^{(j)}$ for the form on V_j. First,

$$W_j = \{e_j\}^\perp \text{ inside } V_j$$

since $W_j \subseteq \{e_j\}^\perp$ by construction, and conversely if $x = \lambda e_j + y$ with $y \in W_j$ satisfies $\varphi^{(j)}(e_j, x) = 0$, then that value is λc, and $c \neq 0$ forces $\lambda = 0$.

Let $f \colon V_1 \to V_2$ be an isometry. Put $u = f(e_1) \in V_2$. Then $q^{(2)}(u) = q^{(1)}(e_1) = c \neq 0$ and $q^{(2)}(e_2) = c$, so lemma 22.5.5 applied inside $(V_2, \varphi^{(2)})$ supplies $\sigma \in O(\varphi^{(2)})$ with $\sigma(u) = e_2$. The composite $g = \sigma \circ f \colon V_1 \to V_2$ is again an isometry, and $g(e_1) = e_2$.

Step 3: g carries W_1 onto W_2. For $x \in V_1$,

$$\varphi^{(2)}\big(e_2, g(x)\big) = \varphi^{(2)}\big(g(e_1), g(x)\big) = \varphi^{(1)}(e_1, x)$$

so $x \in \{e_1\}^{\perp}$ if and only if $g(x) \in \{e_2\}^{\perp}$. Since g is bijective, g maps $\{e_1\}^{\perp}$ onto $\{e_2\}^{\perp}$, which by Step 2 says $g(W_1) = W_2$. The restriction $g|_{W_1} : W_1 \to W_2$ is a linear isomorphism preserving the forms, that is an isometry $\varphi_1 \cong \varphi_2$, completing the proof.

The theorem is what makes the decomposition of theorem 22.4.8 independent of the choices made in producing it.

Corollary 22.5.7: The Anisotropic Part and the Witt Index Are Invariants

Let k be a field with $\operatorname{char} k \neq 2$ and let φ be a nondegenerate symmetric bilinear form on a finite-dimensional k-vector space. Suppose

$$\varphi \cong \varphi_1 \perp m_1 H \qquad \text{and} \qquad \varphi \cong \varphi_2 \perp m_2 H$$

with φ_1 and φ_2 anisotropic and $m_1, m_2 \geq 0$. Then $m_1 = m_2$ and $\varphi_1 \cong \varphi_2$.

***Proof*:**
By definition 22.4.4 a hyperbolic plane H has a basis e, f with $q(e) = q(f) = 0$ and $\varphi(e, f) = 1$, so its matrix in that basis is $\left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$, of determinant $-1 \neq 0$; by lemma 13.3.23 the form H is nondegenerate. The matrix of an orthogonal sum in the union of two bases is block diagonal with the two matrices as blocks, so its determinant is the product of theirs, and mH is nondegenerate for every $m \geq 0$ by the same lemma. Also e is a nonzero isotropic vector of H.

Assume without loss of generality that $m_1 \leq m_2$. Regrouping the orthogonal sum,

$$\varphi_1 \perp m_1 H \cong \varphi \cong \varphi_2 \perp m_2 H \cong \big(\varphi_2 \perp (m_2 - m_1)H\big) \perp m_1 H$$

Since $m_1 H$ is nondegenerate, theorem 22.5.6 cancels it from the two ends and gives

$$\varphi_1 \cong \varphi_2 \perp (m_2 - m_1)H$$

If $m_2 > m_1$, the right-hand form has the nonzero isotropic vector e of any one of its hyperbolic summands, and the preimage of e under the isometry is nonzero and isotropic for φ_1, since an isometry is injective and preserves q; that contradicts the hypothesis that φ_1 is anisotropic. Hence $m_1 = m_2$, the term $(m_2 - m_1)H$ is the form on the zero space, and $\varphi_1 \cong \varphi_2$, as we sought to show.

So the Witt index of definition 22.4.10 deserves its name. Proposition 22.4.11(3) computes $\nu(\varphi)$ as the number of hyperbolic planes in the decomposition produced by the proof of theorem 22.4.8, and corollary 22.5.7 says that number is the same in every decomposition of that shape; so $m = \nu(\varphi)$ in *any* of them, and the anisotropic part is determined up to isometry by φ alone. Two nondegenerate forms $\varphi \cong \varphi_1 \perp mH$ and $\varphi' \cong \varphi'_1 \perp m'H$, with φ_1 and φ'_1 anisotropic, are therefore isometric if and only if $m = m'$ and $\varphi_1 \cong \varphi'_1$: if $\varphi \cong \varphi'$ then φ has the two decompositions $\varphi \cong \varphi_1 \perp mH$ and $\varphi \cong \varphi'_1 \perp m'H$, so corollary 22.5.7 gives $m = m'$ and $\varphi_1 \cong \varphi'_1$, while conversely $m = m'$ and $\varphi_1 \cong \varphi'_1$ give $\varphi \cong \varphi_1 \perp mH \cong \varphi'_1 \perp m'H \cong \varphi'$. That biconditional is the reduction the classification of forms over a given field uses.

Cancellation deletes a summand from both sides of an isometry that is already given. The extension theorem is the statement that isometries are plentiful: any isometry defined on a subspace already extends to the whole space, with no compatibility condition beyond preserving the form. The subspace is allowed to be degenerate, and that is the substance of the theorem; for a subspace containing an anisotropic vector, the reduction to a smaller space is the argument already used twice above, whereas a totally isotropic subspace gives that argument nothing to work with. The following lemma is what repairs that case: it manufactures, for an isotropic vector inside a totally isotropic subspace, a partner making it half of a hyperbolic plane, chosen orthogonal to the rest of the subspace.

Lemma 22.5.8: Hyperbolic Partners for Isotropic Vectors

Let k be a field with $\operatorname{char} k \neq 2$, let φ be a nondegenerate symmetric bilinear form on a finite-dimensional k-vector space V, let $U \subseteq V$ be a subspace with $\varphi(x,y) = 0$ for all $x, y \in U$, let $u \in U$ be nonzero, and let $U_1 \subseteq U$ be a subspace with $U = ku \oplus U_1$. Then there exists $u^* \in V$ with

$$q(u^*) = 0, \qquad \varphi(u, u^*) = 1, \qquad \varphi(u^*, x) = 0 \text{ for all } x \in U_1$$

***Proof*:**
Let $u_1, \dots, u_m$ be a basis of U_1, so that $u, u_1, \dots, u_m$ is a basis of U and in particular a linearly independent family in V. Consider the linear map

$$\Phi \colon V \to k^{m+1}, \qquad \Phi(y) = \big(\varphi(u,y), \varphi(u_1,y), \dots, \varphi(u_m,y)\big)$$

Suppose Φ were not surjective. Its image is then a proper subspace of k^{m+1}, hence contained in the kernel of some nonzero linear functional (extend a basis of the image to a basis of k^{m+1} and take the member of the dual basis of proposition 13.3.5 attached to a vector outside the image): there are $\lambda_0, \dots, \lambda_m \in k$, not all zero, with $\lambda_0\varphi(u,y) + \sum_{i=1}^m \lambda_i \varphi(u_i, y) = 0$ for every $y \in V$. Bilinearity rewrites this as $\varphi(z,y) = 0$ for every $y \in V$, where $z = \lambda_0 u + \sum_i \lambda_i u_i$. As φ is symmetric, this says $z \in \operatorname{rad}(\varphi)$ (definition 13.3.22), which is zero by nondegeneracy; and $u, u_1, \dots, u_m$ being linearly independent then forces every $\lambda_j = 0$, contradicting the choice of the λ_j. So Φ is surjective.

Choose $y \in V$ with $\Phi(y) = (1, 0, \dots, 0)$, so $\varphi(u,y) = 1$ and $\varphi(u_i, y) = 0$ for each i; by bilinearity the latter gives $\varphi(x,y) = 0$ for every $x \in U_1$. Since $\operatorname{char} k \neq 2$ the scalar $q(y)/2$ is defined, and we may set

$$u^* := y - \frac{q(y)}{2}\, u$$

Because $u \in U$ and φ vanishes identically on U, we have $q(u) = 0$ and $\varphi(u,x) = 0$ for $x \in U_1$. Hence

$$\varphi(u,u^*) = \varphi(u,y) - \frac{q(y)}{2} q(u) = 1 - 0 = 1$$

$$q(u^*) = q(y) - 2 \cdot \frac{q(y)}{2}\varphi(y,u) + \frac{q(y)^2}{4} q(u) = q(y) - q(y) + 0 = 0$$

using $\varphi(y,u) = \varphi(u,y) = 1$ by symmetry. Finally, for $x \in U_1$ both $\varphi(y,x) = 0$ and $\varphi(u,x) = 0$, so $\varphi(u^*, x) = 0$, completing the proof.

Theorem 22.5.9: Witt's Extension Theorem

Let k be a field with $\operatorname{char} k \neq 2$, let φ be a nondegenerate symmetric bilinear form on a finite-dimensional k-vector space V, let $U \subseteq V$ be a subspace, and let $f\colon U \to V$ be an injective linear map with

$$\varphi\big(f(x), f(y)\big) = \varphi(x, y) \qquad \text{for all } x, y \in U$$

Then there is an $F \in O(\varphi)$ with $F|_U = f$.

***Proof*:**
Induct on $\dim V$. If $\dim V = 0$ then $U = 0$ and $F = \mathrm{id}_V$ serves. So let $\dim V = n \geq 1$ and assume the theorem for every nondegenerate symmetric bilinear form on a space of dimension smaller than n. If $U = 0$, take $F = \mathrm{id}_V$; assume $U \neq 0$.

Step 1: reduction to the case where φ does not vanish identically on U. Suppose $\varphi(x, y) = 0$ for all $x, y \in U$. Choose $u \in U$ nonzero and a subspace U_1 with $U = ku \oplus U_1$, and let $u_1, \dots, u_m$ be a basis of U_1. Lemma 22.5.8 supplies $u^* \in V$ with $q(u^*) = 0$, $\varphi(u, u^*) = 1$ and $\varphi(u^*, U_1) = 0$.

Put $u' = f(u)$ and $U_1' = f(U_1)$. Since f is injective and preserves φ, the subspace $f(U)$ satisfies $\varphi(x, y) = 0$ for all $x, y \in f(U)$, the vector u' is nonzero, and $f(U) = ku' \oplus U_1'$. A second application of lemma 22.5.8, to the subspace $f(U)$ with the vector u' and the complement U_1', supplies $u'^* \in V$ with $q(u'^*) = 0$, $\varphi(u', u'^*) = 1$ and $\varphi(u'^*, U_1') = 0$.

Because $\varphi(u, u^*) = 1$ while $\varphi(u, x) = 0$ for every $x \in U$, the vector u^* is not in U, so $\tilde{U} := U \oplus ku^*$ has $u, u^*, u_1, \dots, u_m$ as a basis. Define a linear map $\tilde{f}\colon \tilde{U} \to V$ by $\tilde{f}|_U = f$ and $\tilde{f}(u^*) = u'^*$.

The map $\tilde{f}$ preserves φ. By bilinearity it is enough to compare φ on pairs of basis vectors with φ on the pairs of their images, and the six comparisons are:

$$q(u) = 0 = q(u'), \qquad q(u^*) = 0 = q(u'^*), \qquad \varphi(u, u^*) = 1 = \varphi(u', u'^*)$$

$$\varphi(u, u_i) = 0 = \varphi\big(f(u), f(u_i)\big), \qquad \varphi(u^*, u_i) = 0 = \varphi(u'^*, f(u_i)), \qquad \varphi(u_i, u_j) = \varphi\big(f(u_i), f(u_j)\big)$$

the fourth and sixth because f preserves φ, the fifth by the two orthogonality conditions on u^* and u'^*.

The map $\tilde{f}$ is injective. Let $x = \lambda u + \mu u^* + w$ with $w \in U_1$ and $\tilde{f}(x) = 0$, that is $\lambda u' + \mu u'^* + f(w) = 0$. Applying $\varphi(u', -)$ and using $\varphi(u', u') = 0$, $\varphi(u', u'^*) = 1$ and $\varphi(u', f(w)) = \varphi(f(u), f(w)) = \varphi(u, w) = 0$ gives $\mu = 0$. Then $f(\lambda u + w) = 0$, so $\lambda u + w = 0$ by injectivity of f, so $\lambda = 0$ and $w = 0$ because $U = ku \oplus U_1$.

Finally $q(u + u^*) = q(u) + 2\varphi(u, u^*) + q(u^*) = 2 \neq 0$, so φ does not vanish identically on $\tilde{U}$. Any $F \in O(\varphi)$ with $F|_{\tilde{U}} = \tilde{f}$ satisfies $F|_U = f$, so it suffices to prove the theorem for the pair $(\tilde{U}, \tilde{f})$, which is of the kind treated next.

Step 2: the case where φ does not vanish identically on U. Some $u \in U$ has $q(u) \neq 0$: otherwise q vanishes on U and, for $x, y \in U$, $0 = q(x + y) = q(x) + 2\varphi(x, y) + q(y) = 2\varphi(x, y)$, which with $\operatorname{char} k \neq 2$ makes φ vanish identically on U.

Then $q(f(u)) = q(u) \neq 0$, so lemma 22.5.5 supplies $\sigma \in O(\varphi)$ with $\sigma(f(u)) = u$. The map $f' := \sigma \circ f$ is again an injective linear map $U \to V$ preserving φ, and $f'(u) = u$.

Set $V_1 = \{u\}^\perp$ and $U' = U \cap V_1$. Four facts about these:

1. $V = ku \oplus V_1$ with $\varphi(u, V_1) = 0$, by lemma 22.5.3(1) applied to (V, φ) and the anisotropic vector u; in particular $\dim V_1 = n - 1$.
2. $U = ku \oplus U'$, by the same part of the same lemma applied to $(U, \varphi|_U)$, whose orthogonal complement of u inside U is $U \cap \{u\}^\perp = U'$.
3. $\varphi|_{V_1}$ is nondegenerate: if $y \in V_1$ satisfies $\varphi(y, V_1) = 0$, then also $\varphi(y, u) = 0$, so $\varphi(y, V) = 0$ by (1) and bilinearity, whence $y \in \operatorname{rad}(\varphi) = 0$.
4. $f'(U') \subseteq V_1$: for $x \in U'$, $\varphi(u, f'(x)) = \varphi(f'(u), f'(x)) = \varphi(u, x) = 0$.

So $f'|_{U'} : U' \to V_1$ is an injective linear map from a subspace of V_1 into V_1 preserving $\varphi|_{V_1}$, and $\dim V_1 = n - 1$. The inductive hypothesis gives $G \in O(\varphi|_{V_1})$ with $G|_{U'} = f'|_{U'}$.

Step 3: assembling F. Define $F' : V \to V$ by $F'(\lambda u + y) = \lambda u + G(y)$ for $\lambda \in k$ and $y \in V_1$; this is well defined and bijective by (1) and the bijectivity of G. It preserves φ: for $\lambda, \mu \in k$ and $y, z \in V_1$, using $G(y), G(z) \in V_1$ and $\varphi(u, V_1) = 0$ twice,

$$\varphi\big(\lambda u + G(y), \mu u + G(z)\big) = \lambda\mu\, q(u) + \varphi\big(G(y), G(z)\big) = \lambda\mu\, q(u) + \varphi(y, z) = \varphi(\lambda u + y, \mu u + z)$$

So $F' \in O(\varphi)$. It restricts to f' on U: writing $x \in U$ as $x = \lambda u + x'$ with $x' \in U'$ by (2),

$$F'(x) = \lambda u + G(x') = \lambda u + f'(x') = \lambda f'(u) + f'(x') = f'(x)$$

Set $F = \sigma^{-1} \circ F'$, an element of $O(\varphi)$. Then $F|_U = \sigma^{-1} \circ f' = \sigma^{-1} \circ \sigma \circ f = f$, completing the proof.

The hypothesis that φ be nondegenerate on the ambient space is not removable, and it enters the proof twice: in Step 1 through lemma 22.5.8, and in Step 2 as fact (3), where $\operatorname{rad}(\varphi) = 0$ is what makes $\varphi|_{V_1}$ nondegenerate and so licenses the inductive hypothesis. What goes wrong without it is direct: an isometry of V carries $\operatorname{rad}(\varphi)$ onto itself, so a form-preserving injection sending a vector outside the radical into the radical extends to no isometry, and exercise 22.5.1 (6) exhibits one. Nondegeneracy of the *subspace* U, on the other hand, is nowhere assumed, and Step 1 is the work of doing without it.

Taking U to be a line already extends lemma 22.5.5 past its restriction to anisotropic vectors. Let φ be nondegenerate, let $c \in k$, and let v, v' be nonzero vectors with $q(v) = q(v') = c$. The map $f : kv \to V$ with $f(\lambda v) = \lambda v'$ is linear and injective, and $\varphi(f(\lambda v), f(\mu v)) = \lambda\mu\, q(v') = \lambda\mu\, q(v) = \varphi(\lambda v, \mu v)$, so theorem 22.5.9 extends it to an isometry of V carrying v to v'. So $O(\varphi)$ acts transitively on the nonzero vectors of each fixed value of q, including the value 0, where no reflection was available. The last consequence counts how many reflections it takes to build an arbitrary isometry.

Corollary 22.5.10: Reflections Generate the Orthogonal Group

Let k be a field with $\operatorname{char} k \neq 2$ and let φ be a nondegenerate symmetric bilinear form on a k-vector space V of finite dimension n. Then every $\sigma \in O(\varphi)$ is a product of at most $2n$ reflections τ_v in anisotropic vectors $v \in V$, where the empty product is read as id_V. In particular $O(\varphi)$ is generated by its reflections. The bound $2n$ is not the sharp one; see the discussion after the proof.

***Proof*:**
Induct on n. For $n = 0$ the group $O(\varphi)$ is trivial and id_V is the empty product. Let $n \geq 1$ and assume the statement for nondegenerate forms on spaces of dimension $n-1$.

First, V contains an anisotropic vector: if $q(x) = 0$ for every $x \in V$, then for all $x, y \in V$

$$0 = q(x+y) = q(x) + 2\varphi(x,y) + q(y) = 2\varphi(x,y)$$

and $\operatorname{char} k \neq 2$ makes φ the zero form, so $\operatorname{rad}(\varphi) = V \neq 0$, contradicting nondegeneracy. Fix such a u, with $q(u) \neq 0$.

Let $\sigma \in O(\varphi)$. Then $q(\sigma(u)) = q(u) \neq 0$, so lemma 22.5.5, applied to the pair $\sigma(u)$ and u, supplies $\pi \in O(\varphi)$ with $\pi(\sigma(u)) = u$ and π a product of at most two reflections. Put $\rho = \pi \circ \sigma$, so $\rho(u) = u$.

Set $V_1 = \{u\}^\perp$. As in the proof of theorem 22.5.9, $V = ku \oplus V_1$ with $\varphi(u, V_1) = 0$ by lemma 22.5.3(1), $\dim V_1 = n-1$, and $\varphi|_{V_1}$ is nondegenerate because a vector of V_1 orthogonal to V_1 is also orthogonal to u, hence to V, hence zero. Moreover $\rho(V_1) = V_1$: for $y \in V_1$, $\varphi(u, \rho(y)) = \varphi(\rho(u), \rho(y)) = \varphi(u, y) = 0$, so $\rho(V_1) \subseteq V_1$, and ρ is injective, so the inclusion is an equality by dimension count.

By the inductive hypothesis $\rho|_{V_1}$ is a product $\tau_{v_1}^{V_1} \cdots \tau_{v_r}^{V_1}$ of $r \leq 2(n-1)$ reflections of $(V_1, \varphi|_{V_1})$, with each $v_i \in V_1$ anisotropic. Each v_i is anisotropic in V as well, and the reflection τ_{v_i} of V satisfies $\tau_{v_i}(u) = u$, since $\varphi(v_i, u) = 0$, and restricts on V_1 to $\tau_{v_i}^{V_1}$, the defining formula being the same. Hence $\tau_{v_1} \cdots \tau_{v_r}$ and ρ agree on u and on V_1, so they agree on $V = ku \oplus V_1$, and ρ is a product of at most $2(n-1)$ reflections of V.

Finally $\sigma = \pi^{-1} \circ \rho$, and π^{-1} is a product of at most two reflections, since each reflection is its own inverse by lemma 22.5.3(3) and the inverse of $\tau_v \tau_w$ is $\tau_w \tau_v$. So σ is a product of at most $2 + 2(n-1) = 2n$ reflections, as we sought to show.

The sharp form of this count is the Cartan-Dieudonné theorem, proved in Artin's *Geometric Algebra*: over a field of characteristic other than 2, every isometry of an n-dimensional nondegenerate space is a product of at most n reflections. The bound proved above loses a factor of two at Case 2 of lemma 22.5.5, where two reflections are used in place of one. Recovering the sharp bound requires a separate analysis of the isometries σ for which $q(x - \sigma(x)) = 0$ for every $x \in V$, so that no single reflection returns $\sigma(x)$ to x. Such isometries exist: on $H \perp H$, with the two hyperbolic bases e_1, f_1 and e_2, f_2, the map fixing e_1 and e_2 and sending $f_1 \mapsto f_1 + e_2$ and $f_2 \mapsto f_2 - e_1$ is one. For them the reflection count is governed by $\dim \operatorname{im}(\sigma - \mathrm{id}_V)$ rather than by n. That analysis is not carried out in this book.

Exercise 22.5.1

1. Let φ be a nondegenerate symmetric bilinear form on V with $\dim V = n \geq 1$ and $\operatorname{char} k \neq 2$, and let $\sigma \in O(\varphi)$. Show that if A is the matrix of φ and S the matrix of σ in one common basis, then $S^t AS = A$ (use lemma 13.3.19), and deduce $\det \sigma = \pm 1$ with the help of lemma 13.3.23. Using $\det \tau_v = -1$ from lemma 22.5.3, deduce that $SO(\varphi) := \{\sigma \in O(\varphi) \mid \det \sigma = 1\}$ is a subgroup of index exactly 2, and that the parity of the number of reflections in a factorization of σ given by corollary 22.5.10 does not depend on the factorization.

2. Let $\operatorname{char} k \neq 2$, let φ be a symmetric bilinear form on V, and let $v, w \in V$ be anisotropic. Show that $\tau_v = \tau_w$ if and only if $kv = kw$, so that $v \mapsto \tau_v$ is a bijection from the set of anisotropic lines of V onto a set of involutions in $O(\varphi)$. Then count the reflections in $O(\langle 1, 1\rangle)$ over $\mathbb{F}_5$, by listing the six lines of $\mathbb{F}_5^2$ and deciding which are anisotropic.

3. Show that the hypothesis that ψ be nondegenerate can be removed from theorem 22.5.6: for arbitrary symmetric bilinear forms over a field with $\operatorname{char} k \neq 2$, $\psi \perp \varphi_1 \cong \psi \perp \varphi_2$ implies $\varphi_1 \cong \varphi_2$. Reduce, using theorem 22.2.1, to cancelling $\langle 0\rangle$, and for that case use that an isometry carries the radical of one form onto the radical of the other, together with the description of the radical in an orthogonal basis given by proposition 22.2.2.

4. Over $k = \mathbb{Q}$, show that $\langle 1, 1\rangle$ and $\langle -1, -1\rangle$ are not isometric by asking which nonzero rationals each represents. Deduce from theorem 22.5.6 that $\langle 1, 1, 5\rangle \not\cong \langle -1, -1, 5\rangle$, and check that these two forms have the same rank and the same discriminant, so that neither invariant of section 22.1 could have separated them.

5. Let φ be nondegenerate on V with $\operatorname{char} k \neq 2$, and recall the totally isotropic subspaces of definition 22.3.6. Call one *maximal* if no strictly larger subspace of V is totally isotropic. Using theorem 22.5.9, show that any two maximal totally isotropic subspaces of V have the same dimension. Then explain why lemma 22.5.5 gives nothing about isotropic vectors, by exhibiting two nonzero isotropic vectors u, w of $H \perp H$ with $q(u - w) = q(u + w) = 0$.

6. Let $\operatorname{char} k \neq 2$ and let $V = k^3$ carry the form whose matrix in the standard basis is $\operatorname{diag}(1, -1, 0)$. Show that $f\colon k(e_1 + e_2) \to V$ with $f(e_1 + e_2) = e_3$ is an injective linear map preserving the form, and that no isometry of V restricts to it. Conclude that the nondegeneracy hypothesis in theorem 22.5.9 cannot be dropped.

7. Let $\operatorname{char} k \neq 2$ and let $a \in k^\times$. Show that $\langle a, -a\rangle \cong H$, by exhibiting an isotropic vector and applying proposition 22.4.6. Then show that the isometry class of $\langle a, -a\rangle$ does not depend on a, even though the square class of a may vary.

8. Let $\operatorname{char} k \neq 2$ and let V be four-dimensional with basis e_1, f_1, e_2, f_2 and the symmetric bilinear form determined by $\varphi(e_1, f_1) = \varphi(e_2, f_2) = 1$, all other products of two basis vectors being zero, so that $V \cong H \perp H$. Define σ by $\sigma(e_i) = e_i$, $\sigma(f_1) = f_1 + e_2$ and $\sigma(f_2) = f_2 - e_1$. Show that $\sigma \in O(\varphi)$, that $x - \sigma(x)$ lies in the totally isotropic subspace spanned by e_1 and e_2 for every $x \in V$, and hence that $q(x - \sigma(x)) = 0$ for every x. Deduce that no reflection τ_v satisfies $\tau_v(\sigma(x)) = x$ for any x with $\sigma(x) \neq x$, so that Case 2 of lemma 22.5.5 is unavoidable for this σ.

22.6 The Witt Group

Theorem 22.4.8 writes a nondegenerate form as an anisotropic form orthogonal to a number of hyperbolic planes, and corollary 22.5.7 says that the anisotropic part is determined up to isometry. Two forms with isometric anisotropic parts therefore differ only in how many hyperbolic planes are attached to them, and this section discards that number. What remains is an abelian group under $\perp$, and a commutative ring once the tensor product of forms is available.

The construction needs no group completion. The inverse of a class is again the class of a form, because $\varphi \perp (-\varphi)$ is a sum of hyperbolic planes, and that computation is the first item below. The group is then computed for the two families of fields whose square classes are known: the algebraically closed fields, where it has two elements, and the finite fields, where it has four. The finite case answers the question left open at the end of example 22.2.5.

Throughout, k is a field with $\operatorname{char} k \neq 2$ and every form is a symmetric bilinear form on a finite-dimensional k-vector space, identified with its quadratic form through proposition 22.1.2 whenever that is convenient. The orthogonal sum $\varphi \perp \psi$ is definition 22.4.1, and everything used below about it is established there: the block-diagonal Gram matrix, the multiplicativity $\det(\varphi \perp \psi) = \det(\varphi)\det(\psi)$, that the sum is nondegenerate exactly when both summands are, and that $\perp$ descends to isometry classes and is commutative and associative up to isometry. Write $-\varphi$ for the form $(x, y) \mapsto -\varphi(x, y)$ on V. In the diagonal notation of section 22.2,

$$\langle a_1, \dots, a_m\rangle \perp \langle b_1, \dots, b_n\rangle = \langle a_1, \dots, a_m, b_1, \dots, b_n\rangle$$

and $-\langle a_1, \dots, a_n\rangle = \langle -a_1, \dots, -a_n\rangle$. Finally mH denotes the orthogonal sum of m copies of the hyperbolic plane H (definition 22.4.4), with $0H$ the zero form on the zero space.

The identification that drives everything below is that a form and its negative add up to nothing but hyperbolic planes.

Lemma 22.6.1: A Form and Its Negative Sum to Hyperbolic Planes

Let k be a field with $\operatorname{char} k \neq 2$.

1. For every $a \in k^\times$ one has $\langle a, -a\rangle \cong H$. In particular $H \cong \langle 1, -1\rangle$, so $\det H = -1$.
2. If φ is a nondegenerate form of rank n on V, then $\varphi \perp (-\varphi) \cong nH$.

***Proof*:**

1. By example 22.4.7 a nondegenerate binary form $\langle a, b\rangle$ is a hyperbolic plane exactly when $-ab$ is a square in $k^\times$. Here $b = -a$, so $-ab = a^2$ is a square and $\langle a, -a\rangle \cong H$. Taking $a = 1$ gives $H \cong \langle 1, -1\rangle$, whose matrix in an orthogonal basis is $\operatorname{diag}(1, -1)$, of determinant -1.
2. Diagonalize $\varphi \cong \langle a_1, \dots, a_n\rangle$ by theorem 22.2.1. Nondegeneracy forces every $a_i \neq 0$: by the last claim of proposition 22.2.2 the number of zero entries is the dimension of the radical, which is 0 here by definition 13.3.22. Then $-\varphi \cong \langle -a_1, \dots, -a_n\rangle$, so

$$\varphi \perp (-\varphi) \cong \langle a_1, \dots, a_n, -a_1, \dots, -a_n\rangle \cong \langle a_1, -a_1\rangle \perp \cdots \perp \langle a_n, -a_n\rangle$$

the second isometry being the permutation of an orthogonal basis that places each a_i next to its negative. Applying part (1) to each summand gives $\varphi \perp (-\varphi) \cong nH$, as we sought to show.

Every form of rank n is isometric to a form on k^n, obtained by transporting it along a choice of basis, so the isometry classes of nondegenerate forms over k constitute a set rather than a proper class. That set carries $\perp$ as a commutative associative operation with the zero form as a neutral element, and the lemma says that after hyperbolic planes are discarded every element acquires an inverse.

Definition 22.6.2: Witt Equivalence and the Witt Group

Let k be a field with $\operatorname{char} k \neq 2$. Two nondegenerate symmetric bilinear forms φ and ψ over k, possibly on spaces of different dimensions, are *Witt equivalent*, written $\varphi \sim \psi$, if there are integers $m, n \geq 0$ with

$$\varphi \perp mH \cong \psi \perp nH$$

The *Witt group* $W(k)$ is the set of equivalence classes, the class of φ being written $[\varphi]$, equipped with the operation

$$[\varphi] + [\psi] := [\varphi \perp \psi]$$

That $\sim$ is an equivalence relation and that the operation is well defined are proved in theorem 22.6.3.

Two small computations show what the relation does. Over any k one has $\langle 1 \rangle \perp \langle -1 \rangle \cong H$ by lemma 22.6.1, so $\langle 1, -1 \rangle \sim 0$: the hyperbolic plane is Witt trivial, which is the whole content of the definition. Consequently $\langle 1, 1, -1 \rangle \sim \langle 1 \rangle$, taking $m = 0$ and $n = 1$ against $\psi = \langle 1 \rangle$, since $\langle 1 \rangle \perp H \cong \langle 1, 1, -1 \rangle$. Forms of different ranks can therefore be Witt equivalent, and rank is not an invariant of a Witt class.

Theorem 22.6.3: The Witt Classes Form an Abelian Group

Let k be a field with $\operatorname{char} k \neq 2$. Witt equivalence is an equivalence relation on nondegenerate forms over k, the operation of definition 22.6.2 is well defined on classes, and $W(k)$ is an abelian group whose neutral element is the class $[0] = [H]$ of the zero form and in which $-[\varphi] = [-\varphi]$.

***Proof*:**
Step 1: $\sim$ is an equivalence relation. Reflexivity is the case $m = n = 0$, and symmetry is immediate from the symmetry of $\cong$. For transitivity, suppose $\varphi \perp mH \cong \psi \perp nH$ and $\psi \perp pH \cong \chi \perp qH$. Adding pH to the first isometry and nH to the second, which is legitimate because orthogonal sums of isometric forms are isometric, and rearranging summands by commutativity and associativity of $\perp$, gives

$$\varphi \perp (m+p)H \cong \psi \perp (n+p)H \cong \chi \perp (q+n)H$$

so $\varphi \sim \chi$.

Step 2: the operation is well defined. Let $\varphi \sim \varphi'$ and $\psi \sim \psi'$, say $\varphi \perp mH \cong \varphi' \perp m'H$ and $\psi \perp nH \cong \psi' \perp n'H$. Taking the orthogonal sum of the two isometries and rearranging summands,

$$(\varphi \perp \psi) \perp (m+n)H \cong (\varphi' \perp \psi') \perp (m'+n')H$$

so $\varphi \perp \psi \sim \varphi' \perp \psi'$. Both $\varphi \perp \psi$ and $\varphi' \perp \psi'$ are nondegenerate, since orthogonal sums of nondegenerate forms are, so the sum of two classes is again a class.

Step 3: the group axioms. Commutativity and associativity of $+$ are those of $\perp$, transported to classes by Step 2. The zero form 0 on the zero space is nondegenerate, its radical being the zero space, and $\varphi \perp 0 = \varphi$, so $[0]$ is neutral. Taking $m = 0$ and $n = 1$ in definition 22.6.2 gives $H \perp 0H \cong 0 \perp 1H$, that is $H \sim 0$, so $[H] = [0]$. Finally, if φ has rank n then $\varphi \perp (-\varphi) \cong nH$ by lemma 22.6.1, and $nH \sim 0$ directly from definition 22.6.2, taking the two integers there to be 0 and n, so

$$[\varphi] + [-\varphi] = [\varphi \perp (-\varphi)] = [nH] = [0]$$

which exhibits $[-\varphi]$ as the inverse of $[\varphi]$, completing the proof.

The group is built from forms alone: no formal differences are introduced, and no group completion of the monoid of isometry classes is taken, because the negative of a form is a form and lemma 22.6.1 makes the two cancel. What the construction has cost is the ability to see the rank, since φ and $\varphi \perp H$ have become equal. The next result says that nothing else was lost, and it is where Witt cancellation enters: cancellation plays no part in the group axioms above, and its role is to identify the classes with objects one can list.

Proposition 22.6.4: Each Witt Class Contains Exactly One Anisotropic Form

Let k be a field with $\operatorname{char} k \neq 2$ and let φ, ψ be nondegenerate forms over k, with Witt decompositions $\varphi \cong \varphi_{\mathrm{an}} \perp m_\varphi H$ and $\psi \cong \psi_{\mathrm{an}} \perp m_\psi H$ as in theorem 22.4.8. Then

$$\varphi \sim \psi \quad \text{if and only if} \quad \varphi_{\mathrm{an}} \cong \psi_{\mathrm{an}}$$

Consequently $[\varphi] \mapsto \varphi_{\mathrm{an}}$ is a bijection from $W(k)$ to the set of isometry classes of anisotropic forms over k, and $[\varphi] = 0$ if and only if $\varphi \cong m_\varphi H$.

***Proof*:**
First note that an anisotropic form is nondegenerate: a nonzero vector v of the radical satisfies $\varphi(v, v) = 0$, hence is an isotropic vector (definition 22.4.2), so the radical of an anisotropic form is zero and definition 13.3.22 applies. Both φ_{an} and ψ_{an} are therefore nondegenerate, as is H, which is what the cancellation theorem requires.

Step 1: isometric anisotropic parts give Witt equivalence. If $\varphi_{\mathrm{an}} \cong \psi_{\mathrm{an}}$ then

$$\varphi \perp m_\psi H \cong \varphi_{\mathrm{an}} \perp (m_\varphi + m_\psi)H \cong \psi_{\mathrm{an}} \perp (m_\varphi + m_\psi)H \cong \psi \perp m_\varphi H$$

so $\varphi \sim \psi$ by definition 22.6.2.

Step 2: Witt equivalence gives isometric anisotropic parts. Suppose $\varphi \perp mH \cong \psi \perp nH$. Substituting the two Witt decompositions and writing $a = m_\varphi + m$ and $b = m_\psi + n$,

$$\varphi_{\mathrm{an}} \perp aH \cong \psi_{\mathrm{an}} \perp bH$$

This is exactly the situation corollary 22.5.7 settles: two Witt decompositions of isometric forms have isometric anisotropic parts and equal hyperbolic counts. Applying it gives $a = b$ and $\varphi_{\mathrm{an}} \cong \psi_{\mathrm{an}}$.

Step 3: the two consequences. The two steps say exactly that $[\varphi] \mapsto \varphi_{\mathrm{an}}$ is well defined and injective on $W(k)$; it is surjective because an anisotropic form χ is nondegenerate with $\chi_{\mathrm{an}} = \chi$ and $m_\chi = 0$. For the last claim, $[\varphi] = [0]$ holds if and only if φ_{an} is isometric to the zero form on the zero space, that is if and only if $\varphi \cong m_\varphi H$, completing the proof.

So $W(k)$ is the set of anisotropic forms over k, given an addition that is not orthogonal sum: to add two anisotropic forms, take their orthogonal sum and discard the hyperbolic planes that appear in it. The integer m_φ discarded in the process is the Witt index of definition 22.4.10, the two agreeing by proposition 22.4.11(3), and the pair (anisotropic part, Witt index) is a complete isometry invariant of a nondegenerate form, while the class in $W(k)$ retains only the first half of that pair. The first field to compute is one over which every anisotropic form has rank at most one.

Example 22.6.5: The Witt Group of an Algebraically Closed Field
Let k be algebraically closed with $\operatorname{char} k \neq 2$; the case $k = \mathbb{C}$ is the one to keep in mind. Fix $c \in k$ with $c^2 = -1$, which exists because $x^2 + 1$ has a root.

No anisotropic form has rank 2 or more. Indeed, by example 22.2.4 a nondegenerate form of rank n is isometric to $\langle 1, \dots, 1\rangle$, and if $n \geq 2$ the vector $v = (c, 1, 0, \dots, 0)$ is nonzero and satisfies

$$q(v) = c^2 + 1 = -1 + 1 = 0$$

so the form is isotropic. The anisotropic forms are therefore the zero form and the rank one form $\langle 1\rangle$, the latter being anisotropic because $x^2 = 0$ forces $x = 0$. By proposition 22.6.4,

$$W(k) = \{\, [0],\ [\langle 1\rangle] \,\}$$

has two elements. Its group law is forced: $2[\langle 1\rangle] = [\langle 1, 1\rangle]$, and $\langle 1, 1\rangle \cong \langle 1, -1\rangle$ by proposition 22.2.2, since $-1 \cdot c^2 = 1$ rescales the second entry by the square c^2, so $\langle 1, 1\rangle \cong H$ by lemma 22.6.1 and $2[\langle 1\rangle] = 0$. Hence $W(k) \cong \mathbb{Z}/2\mathbb{Z}$, the isomorphism sending $[\varphi]$ to the rank of φ modulo 2.

Addition is not the only operation available on forms. The tensor product of the underlying spaces carries a form built from the two given ones, and it is compatible with $\perp$ and with Witt equivalence, which upgrades $W(k)$ from a group to a ring. In diagonal coordinates the operation is multiplication of the diagonal entries in all pairs, which is what makes it computable.

Proposition 22.6.6: The Tensor Product of Forms and the Witt Ring

Let k be a field with $\operatorname{char} k \neq 2$, let φ be a symmetric bilinear form on V and ψ one on W.

1. There is exactly one bilinear form $\varphi \otimes \psi$ on $V \otimes_k W$ satisfying

$$(\varphi \otimes \psi)(v \otimes w,\ v' \otimes w') = \varphi(v, v')\,\psi(w, w')$$

   for all $v, v' \in V$ and $w, w' \in W$, and it is symmetric.

2. If $\varphi \cong \langle a_1, \dots, a_m\rangle$ and $\psi \cong \langle b_1, \dots, b_n\rangle$, then $\varphi \otimes \psi$ is isometric to the diagonal form whose mn entries are the products $a_i b_j$. In particular $\varphi \otimes \psi$ is nondegenerate when φ and ψ are, and isometric inputs produce isometric outputs.

3. The rule $[\varphi] \cdot [\psi] := [\varphi \otimes \psi]$ is well defined and makes $W(k)$ a commutative ring with identity $[\langle 1\rangle]$.

***Proof*:**

1. Fix bases $v_1, \dots, v_m$ of V and $w_1, \dots, w_n$ of W. By corollary 15.2.6 the mn vectors $v_i \otimes w_j$ form a basis of $V \otimes_k W$. Let M be the $mn \times mn$ matrix indexed by pairs, with entry $\varphi(v_i, v_{i'})\psi(w_j, w_{j'})$ in position $\big((i, j), (i', j')\big)$, and define $B(x, y) := [x]^t M[y]$ using coordinates in that basis. Taking coordinates is linear and matrix multiplication is bilinear in the two coordinate vectors, so B is a bilinear form.

   To see that B has the stated values, write $v = \sum_i x_i v_i$, $v' = \sum_{i'} x'_{i'} v_{i'}$, $w = \sum_j y_j w_j$ and $w' = \sum_{j'} y'_{j'} w_{j'}$. The map $(v, w) \mapsto v \otimes w$ is bilinear by theorem 15.1.12, so $v \otimes w = \sum_{i,j} x_i y_j\,(v_i \otimes w_j)$, which is the coordinate vector of $v \otimes w$ in the chosen basis. Hence

$$\begin{aligned} B(v \otimes w, v' \otimes w') &= \sum_{i,j,i',j'} x_i y_j x'_{i'} y'_{j'}\, \varphi(v_i, v_{i'})\psi(w_j, w_{j'}) \\ &= \Big(\sum_{i,i'} x_i x'_{i'} \varphi(v_i, v_{i'})\Big)\Big(\sum_{j,j'} y_j y'_{j'} \psi(w_j, w_{j'})\Big) \\ &= \varphi(v, v')\,\psi(w, w') \end{aligned}$$

   the last line by bilinearity of φ and of ψ. For uniqueness, the simple tensors span $V \otimes_k W$ by lemma 15.1.4, and two bilinear forms agreeing on all pairs of vectors from a spanning set agree everywhere, by expanding both arguments. Symmetry follows the same way: the bilinear forms $(x, y) \mapsto B(x, y)$ and $(x, y) \mapsto B(y, x)$ agree on pairs of simple tensors because φ and ψ are symmetric, hence agree.

2. Choose the bases in (1) to be orthogonal bases realizing the two diagonalizations, which exist by theorem 22.2.1, so that $\varphi(v_i, v_{i'}) = 0$ for $i \neq i'$ with $\varphi(v_i, v_i) = a_i$, and likewise for ψ. Then for $(i, j) \neq (i', j')$,

$$(\varphi \otimes \psi)(v_i \otimes w_j,\ v_{i'} \otimes w_{j'}) = \varphi(v_i, v_{i'})\psi(w_j, w_{j'}) = 0$$

   since at least one of the two factors has distinct indices, while the value at $(i, j) = (i', j')$ is $a_i b_j$. So the basis $\{v_i \otimes w_j\}$ is orthogonal for $\varphi \otimes \psi$ with the mn diagonal entries $a_i b_j$, which is the claim. If all a_i and all b_j are nonzero then so are all a_ib_j, so the matrix of $\varphi\otimes\psi$ in this basis is diagonal with nonzero entries, hence invertible, and lemma 13.3.23 gives nondegeneracy. Finally, if $\varphi' \cong \varphi$ and $\psi' \cong \psi$, then $\varphi' \cong \langle a_1, \ldots, a_m\rangle$ and $\psi' \cong \langle b_1, \ldots, b_n\rangle$ for the very same entries, since the diagonal notation asserts an isometry class and not a chosen basis. Running the argument just given for those diagonalizations exhibits $\varphi' \otimes \psi'$ and $\varphi \otimes \psi$ as isometric to one and the same diagonal form, so isometric inputs produce isometric outputs. The multiset $\{a_ib_j\}$ is not itself an invariant of the pair of isometry classes, being pinned down only up to multiplying its entries by squares, but the isometry class it names is.

3. Write $\varphi \cong \langle a_1, \ldots, a_m\rangle$, $\psi \cong \langle b_1, \ldots, b_n\rangle$ and $\chi \cong \langle c_1, \ldots, c_p\rangle$, all nondegenerate. By (2) the products a_ib_j and b_ja_i agree, giving $\varphi\otimes\psi \cong \psi\otimes\varphi$; the products $(a_ib_j)c_l$ and $a_i(b_jc_l)$ agree, giving associativity; the entries of $\varphi\otimes(\psi\perp\chi)$ are the a_ib_j together with the a_ic_l, which are the entries of $(\varphi\otimes\psi)\perp(\varphi\otimes\chi)$, giving distributivity over $\perp$; and $\langle 1\rangle\otimes\varphi \cong \varphi$ because the products $1\cdot a_i$ are the a_i.

   It remains to check that the rule respects Witt equivalence. First, $\langle a\rangle\otimes H \cong \langle a\rangle\otimes\langle 1,-1\rangle \cong \langle a,-a\rangle \cong H$, using lemma 22.6.1 at both ends and (2) in the middle, so by distributivity

$$\psi\otimes H \cong (\langle b_1\rangle\otimes H)\perp\cdots\perp(\langle b_n\rangle\otimes H) \cong nH$$

   Now suppose $\varphi \sim \varphi'$, say $\varphi\perp pH \cong \varphi'\perp p'H$. Tensoring with ψ and using that isometric inputs give isometric outputs, then expanding by distributivity and the previous display,

$$(\varphi\otimes\psi)\perp pnH \cong (\varphi'\otimes\psi)\perp p'nH$$

   so $\varphi\otimes\psi \sim \varphi'\otimes\psi$. Commutativity gives the same conclusion in the second variable, so the product of two classes does not depend on the representatives. The ring axioms are the isometries established in the previous paragraph, read on classes, completing the proof.

$W(k)$ is called the *Witt ring* when the multiplication is in play, and the group and the ring have the same underlying set. The multiplication restricts usefully to rank one forms: $\langle a\rangle\otimes\langle b\rangle \cong \langle ab\rangle$ by part (2), so the square classes of k act on $W(k)$ through $[\langle a\rangle]$, and each $[\langle a\rangle]$ is a unit since $\langle a\rangle\otimes\langle a\rangle \cong \langle a^2\rangle \cong \langle 1\rangle$ by proposition 22.2.2.

Two invariants from section 22.1 can now be tested against Witt equivalence. Rank fails, as the computation after definition 22.6.2 shows, but rank modulo 2 survives because a hyperbolic plane has rank 2. The discriminant also fails, and for a reason worth isolating: $\det(\varphi\perp H) = -\det(\varphi)$ by lemma 22.6.1(1), so $\operatorname{disc}(\varphi\perp H)$ and $\operatorname{disc}(\varphi)$ differ by the class of -1, which is nontrivial in most fields. A sign attached to the determinant repairs this, and box 22.1.9 announced it.

Definition 22.6.7: The Signed Discriminant

Let φ be a nondegenerate symmetric bilinear form of rank n over a field k with $\operatorname{char} k \neq 2$. The *signed discriminant* of φ is the square class

$$d_{\pm}(\varphi) := (-1)^{n(n-1)/2}\operatorname{disc}(\varphi) \in k^\times/(k^\times)^2$$

with disc as in definition 22.1.7. For $n = 0$ the determinant of the empty matrix is 1, so $d_{\pm}$ of the zero form is the trivial class.

The definition depends on nothing but the isometry class of φ, since the rank does not and disc does not by proposition 22.1.8. For a diagonal form the sign multiplies the product of the entries: $d_\pm\langle a_1, \dots, a_n\rangle = (-1)^{n(n-1)/2} a_1 \cdots a_n$ modulo squares, by corollary 22.2.3. Thus $d_\pm\langle a\rangle = a$, $d_\pm\langle a, b\rangle = -ab$ and $d_\pm\langle a, b, c\rangle = -abc$.

Proposition 22.6.8: Rank Modulo Two and the Signed Discriminant Are Witt Invariants

Let k be a field with $\operatorname{char} k \neq 2$.

1. The assignment $e\big([\varphi]\big) := \dim\varphi + 2\mathbb{Z}$ is a well defined surjective ring homomorphism $e\colon W(k) \to \mathbb{Z}/2\mathbb{Z}$.
2. For nondegenerate φ of rank m and ψ of rank n,
$$d_\pm(\varphi \perp \psi) = (-1)^{mn}\, d_\pm(\varphi)\, d_\pm(\psi)$$
3. Witt equivalent forms have equal signed discriminants, so $d_\pm$ is a well defined map $W(k) \to k^\times/(k^\times)^2$.
4. Restricted to the classes of even rank, $d_\pm$ is a group homomorphism from $\big(\ker e, +\big)$ to $k^\times/(k^\times)^2$.

***Proof*:**

1. If $\varphi \perp mH \cong \psi \perp nH$ then comparing dimensions gives $\dim\varphi + 2m = \dim\psi + 2n$, so $\dim\varphi \equiv \dim\psi \pmod 2$ and e is well defined. It is additive because $\dim(\varphi \perp \psi) = \dim\varphi + \dim\psi$, and multiplicative because $\dim(V \otimes_k W) = \dim V \cdot \dim W$ by corollary 15.2.6. It sends the identity $[\langle 1\rangle]$ to $1 + 2\mathbb{Z}$, which also gives surjectivity.
2. Both sides are computed from determinants and ranks. The rank of $\varphi \perp \psi$ is $m + n$ and its determinant is $\det(\varphi)\det(\psi)$, by the block form recorded at the start of this section. So the two sides differ only in the power of -1, and
$$\frac{(m+n)(m+n-1)}{2} - \frac{m(m-1)}{2} - \frac{n(n-1)}{2} = \frac{2mn}{2} = mn$$
after expanding the numerator as $m^2 + 2mn + n^2 - m - n - m^2 + m - n^2 + n$. Raising -1 to that difference gives the stated factor.
3. By lemma 22.6.1(1) the hyperbolic plane has rank 2 and determinant -1, so $d_\pm(H) = (-1)^{2(2-1)/2} \cdot (-1) = (-1)(-1) = 1$ in $k^\times/(k^\times)^2$. Applying (2) with $\psi = H$, where $n = 2$ makes $(-1)^{mn} = 1$,
$$d_\pm(\varphi \perp H) = d_\pm(\varphi)\, d_\pm(H) = d_\pm(\varphi)$$
Iterating, $d_\pm(\varphi \perp mH) = d_\pm(\varphi)$ for every $m \geq 0$. Now if $\varphi \perp mH \cong \psi \perp nH$, then the two sides have equal rank and equal discriminant, the latter by proposition 22.1.8, hence equal signed discriminant; combining the three equalities gives $d_\pm(\varphi) = d_\pm(\psi)$.
4. If m and n are both even then $(-1)^{mn} = 1$, so (2) reads $d_\pm(\varphi \perp \psi) = d_\pm(\varphi) d_\pm(\psi)$. Since addition in $W(k)$ is induced by $\perp$ and the target group is written multiplicatively, this is the homomorphism property on $\ker e$, as we sought to show.

The restriction in (4) is not an artifact of the proof. Over $\mathbb{Q}$ one has $d_\pm\langle 1\rangle = 1$ while $d_\pm\big(\langle 1\rangle \perp \langle 1\rangle\big) = d_\pm\langle 1, 1\rangle = -1$, and -1 is not a square in $\mathbb{Q}$, so $d_\pm$ is not additive on all of $W(\mathbb{Q})$. The subgroup where it is additive has a name.

Definition 22.6.9: The Fundamental Ideal

Let k be a field with $\operatorname{char} k \neq 2$. The *fundamental ideal* of $W(k)$ is

$$I(k) := \ker\big(e\colon W(k) \to \mathbb{Z}/2\mathbb{Z}\big)$$

the set of Witt classes of even rank. It is an ideal of the Witt ring, being the kernel of the ring homomorphism of proposition 22.6.8(1), and $W(k)/I(k) \cong \mathbb{Z}/2\mathbb{Z}$.

The classes $[\langle 1, -a\rangle]$ for $a \in k^\times$ lie in $I(k)$ and have signed discriminant $-1 \cdot (1 \cdot (-a)) = a$, so $d_\pm$ maps $I(k)$ onto the whole square class group. Rank modulo 2 and the signed discriminant are thus two homomorphisms out of $W(k)$ and out of $I(k)$ respectively, and over a finite field they turn out to separate every class from every other.

Proposition 22.6.10: Rank and Discriminant Classify Forms over a Finite Field

Let q be an odd prime power and fix a non-square $\epsilon \in \mathbb{F}_q^\times$.

1. For all $a, b \in \mathbb{F}_q^\times$ and all $c \in \mathbb{F}_q$ there exist $x, y \in \mathbb{F}_q$ with $ax^2 + by^2 = c$.
2. Every nondegenerate form over $\mathbb{F}_q$ of rank at least 3 is isotropic, and every nondegenerate form of rank at least 2 represents every element of $\mathbb{F}_q^\times$.
3. Two nondegenerate forms over $\mathbb{F}_q$ are isometric if and only if they have the same rank and the same discriminant.
4. Up to isometry the anisotropic forms over $\mathbb{F}_q$ are exactly the zero form, $\langle 1\rangle$, $\langle \epsilon\rangle$ and $\langle 1, -\epsilon\rangle$.

***Proof*:**

1. This is proved in example 22.4.9, by the counting argument on $S = \{ax^2 \mid x \in \mathbb{F}_q\}$ and $T = \{c - by^2 \mid y \in \mathbb{F}_q\}$, each of size $(q+1)/2$ in a set of size q.
2. The first claim is example 22.4.9, which deduces it from (1). For the second, let φ be nondegenerate of rank $n \geq 2$ and diagonalize it by theorem 22.2.1 as $\varphi \cong \langle a_1, \ldots, a_n\rangle$ in an orthogonal basis $w_1, \ldots, w_n$, every a_i being nonzero by the last claim of proposition 22.2.2. Given $c \in \mathbb{F}_q^\times$, part (1) supplies $x, y \in \mathbb{F}_q$ with $a_1 x^2 + a_2 y^2 = c$. Put $v := xw_1 + yw_2$; then $q(v) = a_1 x^2 + a_2 y^2 = c$, which is nonzero, so $v \neq 0$ and φ represents c.
3. Isometric forms have the same rank and, by proposition 22.1.8, the same discriminant. For the converse, induct on the common rank n. For $n = 0$ both forms are the zero form on the zero space. For $n = 1$, write $\varphi \cong \langle a\rangle$ and $\psi \cong \langle b\rangle$; equality of the discriminants means $b = ac^2$ for some $c \in \mathbb{F}_q^\times$ by corollary 22.2.3 and definition 22.1.6, and then $\langle a\rangle \cong \langle b\rangle$ by proposition 22.2.2.

   Let $n \geq 2$. Diagonalizing φ produces a vector v with $c := \varphi(v, v) \neq 0$, and by (2) the form

ψ also represents c, say $\psi(v', v') = c$. A vector of nonzero square splits off: for x in the space of φ,

$$x = \frac{\varphi(v,x)}{c} v + \left(x - \frac{\varphi(v,x)}{c} v \right)$$

where applying $\varphi(v, -)$ to the bracket gives $\varphi(v,x) - \frac{\varphi(v,x)}{c} c = 0$, so the bracket lies in $\{v\}^\perp$ (definition 13.3.20), a subspace by lemma 13.3.21; and if $\lambda v \in \{v\}^\perp$ then $\lambda c = 0$, so $\lambda = 0$. Hence the space is the direct sum of kv and $\{v\}^\perp$, these are orthogonal, and $\varphi \cong \langle c \rangle \perp \varphi'$ with φ' the restriction to $\{v\}^\perp$. The same argument gives $\psi \cong \langle c \rangle \perp \psi'$. Both φ' and ψ' have rank $n-1$ and are nondegenerate, since $c \det(\varphi') = \det(\varphi) \neq 0$ and likewise for ψ. Their discriminants agree: in the square class group, $\operatorname{disc}(\varphi') = c^{-1} \operatorname{disc}(\varphi) = c^{-1} \operatorname{disc}(\psi) = \operatorname{disc}(\psi')$. By the inductive hypothesis $\varphi' \cong \psi'$, and taking the orthogonal sum with the identity of $\langle c \rangle$ gives $\varphi \cong \psi$.

4. An anisotropic form has rank at most 2 by (2). The zero form is anisotropic vacuously. In rank 1, every $\langle a \rangle$ is anisotropic, and by (3) the isometry class is determined by the discriminant, which is the class of a; the two square classes of $\mathbb{F}_q$ are those of 1 and ϵ by example 22.2.5. In rank 2, the form $\langle a, b \rangle$ is isotropic exactly when $-ab$ is a square: given $ax^2 + by^2 = 0$ with $(x, y) \neq (0, 0)$, the case $y = 0$ would force $ax^2 = 0$ and hence $x = 0$, so $y \neq 0$ and $-ab = a^2(x/y)^2$ is a square; conversely if $-ab = t^2$ then $(x, y) = (t, a)$ is nonzero and $at^2 + ba^2 = a(t^2 + ab) = 0$. So $\langle a, b \rangle$ is anisotropic exactly when $-ab$ is a non-square, that is when $\operatorname{disc}\langle a, b \rangle = ab$ is the class of $-\epsilon$. By (3) that single condition on the discriminant pins down one isometry class in rank 2, and $\langle 1, -\epsilon \rangle$ is in it, completing the proof.

Part (3) is the affirmative answer promised at the end of example 22.2.5: over $\mathbb{F}_q$ the rank and the discriminant are a complete system of invariants, so the two forms $\langle 1, 1 \rangle$ and $\langle 1, 2 \rangle$ exhibited there over $\mathbb{F}_5$ fail to be isometric for the only reason available. Part (4) turns proposition 22.6.4 into a count.

Example 22.6.11: The Witt Group of a Finite Field

Let q be odd and $\epsilon \in \mathbb{F}_q^\times$ a non-square. By proposition 22.6.4 the elements of $W(\mathbb{F}_q)$ correspond to isometry classes of anisotropic forms, and proposition 22.6.10(4) lists four of them, so

$$W(\mathbb{F}_q) = \{ [0],\ [\langle 1 \rangle],\ [\langle \epsilon \rangle],\ [\langle 1, -\epsilon \rangle] \}$$

has exactly four elements. Which group of order four it is depends on q modulo 4, through whether -1 is a square.

The case $q \equiv 1 \pmod 4$. Then -1 is a square, say $-1 = c^2$: the group $\mathbb{F}_q^\times$ is cyclic by theorem 11.3.14, of order $q - 1$, which is divisible by 4 in this case, so it has an element of order 4, whose square has order 2 and hence equals -1, the unique element of order 2. For any $a \in \mathbb{F}_q^\times$, proposition 22.2.2 gives $\langle a \rangle \cong \langle ac^2 \rangle = \langle -a \rangle$, so every form satisfies $\varphi \cong -\varphi$ after diagonalizing, and

$$2[\varphi] = [\varphi] + [-\varphi] = 0$$

by theorem 22.6.3. A group of order 4 in which every element has order dividing 2 is $\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$.

The case $q \equiv 3 \pmod 4$. Then -1 is a non-square, so $\epsilon = -1$ is an admissible choice, and $\langle 1, 1 \rangle$ is anisotropic by the criterion in the proof of proposition 22.6.10(4), since $-1 \cdot 1 = -1$ is a non-square. Hence $2[\langle 1 \rangle] = [\langle 1, 1 \rangle] \neq 0$ by proposition 22.6.4. On the other hand $\langle 1, 1, 1, 1 \rangle$ and

$2H \cong \langle 1,-1,1,-1\rangle$ both have rank 4 and discriminant the class of 1, so they are isometric by proposition 22.6.10(3), giving $4[\langle 1\rangle] = [2H] = 0$. So $[\langle 1\rangle]$ has order 4 in a group of order 4, and $W(\mathbb{F}_q) \cong \mathbb{Z}/4\mathbb{Z}$.

The invariants separate the four classes. Writing each class through its anisotropic representative and applying proposition 22.6.8, the rank modulo 2 and the signed discriminant take the values

	$[0]$	$[\langle 1\rangle]$	$[\langle \epsilon\rangle]$	$[\langle 1,-\epsilon\rangle]$
e	0	1	1	0
$d_\pm$	1	1	ϵ	ϵ

the last entry because $d_\pm\langle 1,-\epsilon\rangle = -(1\cdot(-\epsilon)) = \epsilon$. The four pairs are distinct, so a Witt class over a finite field is determined by its rank modulo 2 together with its signed discriminant.

A worked instance. Take $q = 5$. The squares in $\mathbb{F}_5^\times$ are 1 and 4, so $\epsilon = 2$ is a non-square and $-1 = 4 = 2^2$ is a square, putting this in the first case: $W(\mathbb{F}_5) \cong \mathbb{Z}/2\mathbb{Z}\times\mathbb{Z}/2\mathbb{Z}$, with anisotropic representatives 0, $\langle 1\rangle$, $\langle 2\rangle$ and $\langle 1,-2\rangle = \langle 1,3\rangle$. The two forms of example 22.2.5 land in different classes: $\langle 1,1\rangle \cong \langle 1,-1\rangle \cong H$ is Witt trivial, while $[\langle 1,2\rangle] = [\langle 1\rangle] + [\langle 2\rangle]$ has signed discriminant $-2 = 3$, a non-square, so it is not the class of 0.

Exercise 22.6.1

1. Take $k = \mathbb{F}_7$. List the square classes, determine the four anisotropic forms over $\mathbb{F}_7$ up to isometry, and identify $W(\mathbb{F}_7)$ as an abstract group. Which of the four classes is $3[\langle 1\rangle]$?
2. Let $\operatorname{char} k \neq 2$ and let φ be nondegenerate over k. Show that $[\varphi] = 0$ in $W(k)$ if and only if $\varphi \cong mH$ for some $m \geq 0$, and deduce that a nondegenerate form of odd rank is never Witt trivial. Conclude that $W(k)$ has at least two elements for every such k.
3. Let φ and ψ be nondegenerate forms over k with $[\varphi] = [\psi]$ and $\dim\varphi = \dim\psi$. Show that $\varphi \cong \psi$. Give an example over some field showing that the hypothesis on the dimensions cannot be dropped.
4. Suppose -1 is a square in k, where $\operatorname{char} k \neq 2$. Show that $\varphi \cong -\varphi$ for every form over k, hence that $2[\varphi] = 0$ for every $[\varphi] \in W(k)$, so that $W(k)$ is a vector space over $\mathbb{F}_2$. Check that example 22.6.5 is an instance.
5. Show that $a \mapsto [\langle a\rangle]$ induces a well defined group homomorphism from $k^\times/(k^\times)^2$ to the group of units of the Witt ring, and that it is injective for every field k with $\operatorname{char} k \neq 2$. (For injectivity, identify the anisotropic part of $\langle a\rangle$.)
6. Show that $d_\pm\colon I(k) \to k^\times/(k^\times)^2$ is surjective for every k with $\operatorname{char} k \neq 2$. Then compute $I(\mathbb{F}_q)$ for q odd and show that $d_\pm$ is an isomorphism of groups $I(\mathbb{F}_q) \xrightarrow{\sim} \mathbb{F}_q^\times/(\mathbb{F}_q^\times)^2$.
7. Show that $I(k)$ is generated as an abelian group by the classes $[\langle 1,-a\rangle]$ with a ranging over $k^\times$. (Start by expressing $[\langle a\rangle] - [\langle 1\rangle]$ as one of these generators up to sign.)
8. Show that the form $\langle 1,1,\ldots,1\rangle$ of rank n is anisotropic over $\mathbb{Q}$ for every n, and deduce that the classes $n[\langle 1\rangle]$ for $n \geq 0$ are pairwise distinct, so that $W(\mathbb{Q})$ is infinite and $[\langle 1\rangle]$ has infinite order. Contrast this with example 22.6.11.

22.7 The Clifford Algebra

Every construction so far has compared one form with another. This section attaches to a single form an associative algebra, built so that the form becomes the operation of squaring: inside it the equation $v^2 = q(v)$ holds for every $v \in V$. That equation is meaningless in V, where there is no product, so the object carrying it must be an algebra containing V, and section 16.3 is where algebras containing a given vector space are constructed.

The construction below is a quotient of $\mathcal{T}(V)$ by one family of relations, and for $q = 0$ it is the exterior algebra. What is proved here is that the quotient has dimension 2^n rather than collapsing, that maps out of it are prescribed by their restriction to V, and that although the $\mathbb{Z}$-grading of $\mathcal{T}(V)$ does not survive, its reduction modulo 2 does.

Definition 16.3.7 formed $\bigwedge(V)$ by killing the elements $v \otimes v$ of $\mathcal{T}(V)$, which forces $v^2 = 0$ in the quotient. Killing $v \otimes v - q(v)$ instead forces $v^2 = q(v)$. Assume $q \neq 0$ and choose v with $q(v) \neq 0$. The element $v \otimes v - q(v)$ is then a sum of a nonzero term of degree 2 and a nonzero term of degree 0, so it is not homogeneous, and the ideal generated by these elements is not a graded ideal in the sense of definition 16.2.3: were it graded, the two homogeneous components of $v \otimes v - q(v)$ would lie in it separately, so the invertible scalar $q(v)$, and with it 1, would lie in it and the quotient would be 0, contradicting theorem 22.7.5. For $q \neq 0$ the images of the $\mathcal{T}^d(V)$ therefore do not make the quotient a graded ring (definition 16.2.1); only in the excluded case $q = 0$, where the ideal is the one of definition 16.3.7 and is graded, does the $\mathbb{Z}$-grading survive (example 22.7.2). Proposition 22.7.7 says exactly how much of the grading is left in general.

Definition 22.7.1: Clifford Algebra

Let k be a field with $\operatorname{char} k \neq 2$, let V be a finite-dimensional k-vector space, and let q be a quadratic form on V (definition 22.1.1). Let $I(q) \subseteq \mathcal{T}(V)$ be the two-sided ideal of the tensor algebra (definition 16.3.1) generated by the elements

$$v \otimes v - q(v) \cdot 1 \qquad (v \in V)$$

where $q(v) \cdot 1$ lies in $\mathcal{T}^0(V) = k$. The *Clifford algebra* of q is the quotient k-algebra

$$C(q) := \mathcal{T}(V)/I(q)$$

and $\iota \colon V \to C(q)$ denotes the composite of the inclusion $V = \mathcal{T}^1(V) \subseteq \mathcal{T}(V)$ with the projection $\pi \colon \mathcal{T}(V) \to C(q)$.

Write φ for the symmetric bilinear form attached to q by proposition 22.1.2, so that $\varphi(v, v) = q(v)$. Two identities in $C(q)$ are used in everything below. The first is the defining relation

$$\iota(v)^2 = q(v)$$

which holds because π is an algebra homomorphism, so that $\iota(v)^2 = \pi(v \otimes v) = \pi\big(q(v) \cdot 1\big) = q(v)$, the middle equality holding because the two arguments differ by a generator of $I(q)$. The second identity is obtained by applying the first to $v + w$. Bilinearity of φ gives $q(v+w) = \varphi(v+w, v+w) = q(v) + 2\varphi(v, w) + q(w)$, while expanding the square in $C(q)$ gives

$$\iota(v+w)^2 = \big(\iota(v) + \iota(w)\big)^2 = \iota(v)^2 + \iota(v)\iota(w) + \iota(w)\iota(v) + \iota(w)^2$$

using that ι is linear. Equating the two and cancelling $\iota(v)^2 = q(v)$ and $\iota(w)^2 = q(w)$ leaves

$$\iota(v)\iota(w) + \iota(w)\iota(v) = 2\varphi(v, w) \tag{22.3}$$

Since $\operatorname{char} k \neq 2$, the right-hand side vanishes exactly when v and w are orthogonal (definition 13.3.20), so orthogonality of vectors becomes anticommutation of their images. This is the mechanism that turns a choice of orthogonal basis, which theorem 22.2.1 always supplies, into a presentation of $C(q)$ by generators and relations.

Example 22.7.2: The Clifford Algebra of the Zero Form

Let k be a field with $\operatorname{char} k \neq 2$ and take $q = 0$ on a k-vector space V of dimension n. The generators of $I(0)$ are the elements $v \otimes v - 0 = v \otimes v$ for $v \in V$, which are precisely the generators of the ideal $\mathcal{A}(V)$ of definition 16.3.7. The two ideals therefore coincide, and

$$C(0) = \mathcal{T}(V)/\mathcal{A}(V) = \bigwedge(V)$$

not merely up to isomorphism but as the same quotient. Its dimension is

$$\dim_k \bigwedge(V) = \sum_{r=0}^{n} \binom{n}{r} = 2^n$$

since $\bigwedge^r(V)$ is free of rank $\binom{n}{r}$ by theorem 16.4.3 and vanishes for $r > n$. Note that nothing in definition 22.7.1 asked q to be nondegenerate, and for $V \neq 0$ the zero form has radical all of V.

The definition determines the algebra but leaves two questions open: how large it is, and how to produce a homomorphism out of it. The second is answered first, and the answer then supplies the first, because the way to bound the dimension of $C(q)$ from below is to map it onto something of known dimension.

Proposition 22.7.3: The Universal Property of the Clifford Algebra

Let k be a field with $\operatorname{char} k \neq 2$, let V be a finite-dimensional k-vector space, and let q be a quadratic form on V.

1. For every k-algebra A and every k-linear map $f\colon V \to A$ satisfying $f(v)^2 = q(v) \cdot 1_A$ for all $v \in V$, there is a unique k-algebra homomorphism $F\colon C(q) \to A$ with $F \circ \iota = f$.
2. Let q' be a quadratic form on V', with associated symmetric bilinear forms φ and φ', and let $g\colon (V, \varphi) \to (V', \varphi')$ be an isometry (definition 22.1.10). Then there is a unique k-algebra isomorphism $C(g)\colon C(q) \to C(q')$ with $C(g)\big(\iota(v)\big) = \iota'\big(g(v)\big)$ for all $v \in V$. In particular isometric forms have isomorphic Clifford algebras.

Proof:

1. *Existence.* Take $R = k$ and $M = V$ in theorem 16.3.2(2): its hypotheses ask for a k-algebra and a k-module homomorphism from V into it, which are A and f. So it supplies a unique k-algebra homomorphism $\Psi\colon \mathcal{T}(V) \to A$ with $\Psi|_V = f$. On a generator of $I(q)$,

$$\Psi\big(v \otimes v - q(v) \cdot 1\big) = \Psi(v)\Psi(v) - q(v)\Psi(1) = f(v)^2 - q(v) \cdot 1_A = 0$$

by the hypothesis on f. Every element of $I(q)$ is a finite sum of products $x\gamma y$ with $x, y \in \mathcal{T}(V)$ and γ one of those generators, and $\Psi(x\gamma y) = \Psi(x)\Psi(\gamma)\Psi(y) = 0$, so $I(q) \subseteq \ker\Psi$. Hence

$$F\big(x + I(q)\big) := \Psi(x)$$

is well defined: two representatives of the same class differ by an element of $I(q)$, on which Ψ vanishes. It is k-linear, multiplicative and unital because Ψ is and because $\pi\colon \mathcal{T}(V) \to C(q)$ is a surjective algebra homomorphism with $F \circ \pi = \Psi$. Finally $F \circ \iota = \Psi|_V = f$.

Uniqueness. The algebra $C(q)$ is generated as a k-algebra by $\iota(V)$: the simple tensors $v_1 \otimes \cdots \otimes v_m$ generate $\mathcal{T}^m(V)$ as a k-module by lemma 15.1.4, hence together with 1 they span $\mathcal{T}(V)$, and π is surjective and sends $v_1 \otimes \cdots \otimes v_m$ to $\iota(v_1)\cdots\iota(v_m)$. Now if F_1, F_2 are algebra homomorphisms out of $C(q)$ agreeing on $\iota(V)$, the set on which they agree is closed under sums, products and scalars and contains 1, so it is a subalgebra containing a generating set, hence all of $C(q)$. So F is unique.

2. An isometry satisfies $\varphi'(g(v), g(v)) = \varphi(v, v)$, that is $q'(g(v)) = q(v)$. So the k-linear map $f := \iota' \circ g\colon V \to C(q')$ satisfies

$$f(v)^2 = \iota'\big(g(v)\big)^2 = q'\big(g(v)\big) = q(v)$$

and part (1) gives a unique algebra homomorphism $C(g)$ with $C(g) \circ \iota = \iota' \circ g$. The inverse map g^{-1} is again an isometry, since applying $\varphi'(g(-), g(-)) = \varphi(-, -)$ to the vectors $g^{-1}(x), g^{-1}(y)$ gives $\varphi'(x, y) = \varphi(g^{-1}x, g^{-1}y)$, so the same construction produces $C(g^{-1})\colon C(q') \to C(q)$. The composite $C(g^{-1}) \circ C(g)$ is an algebra homomorphism $C(q) \to C(q)$ carrying $\iota(v)$ to $\iota(g^{-1}g(v)) = \iota(v)$, and the identity of $C(q)$ carries $\iota(v)$ to $\iota(v)$ as well. Both therefore solve the problem of part (1) for $A = C(q)$ and $f = \iota$, whose hypothesis $\iota(v)^2 = q(v)$ holds, so uniqueness forces $C(g^{-1}) \circ C(g) = \mathrm{id}_{C(q)}$. Exchanging the roles of g and g^{-1} gives $C(g) \circ C(g^{-1}) = \mathrm{id}_{C(q')}$, so $C(g)$ is an isomorphism, completing the proof.

Part (1) is the statement that prescribing an algebra homomorphism out of $C(q)$ costs exactly one k-linear map $V \to A$ whose values square to the scalars $q(v)$; the relations impose nothing further. Part (2) is what makes $C(q)$ an invariant of the isometry class rather than of a chosen presentation, so it may be computed from any convenient diagonalization $\langle a_1, \ldots, a_n\rangle$ produced by theorem 22.2.1. The smallest case already shows that the answer depends on the field.

Example 22.7.4: Clifford Algebras of a Line

Let k be a field with $\operatorname{char} k \neq 2$, let $\dim_k V = 1$, say $V = kv$, and let $a := q(v)$, so that $q \cong \langle a\rangle$. Then

$$C(q) \cong k[x]/(x^2 - a)$$

To see this, write $A := k[x]/(x^2 - a)$ and let $\bar{x}$ be the class of x. The map $f\colon V \to A$ sending $\lambda v \mapsto \lambda\bar{x}$ is k-linear, and $f(\lambda v)^2 = \lambda^2\bar{x}^2 = \lambda^2 a = q(\lambda v)$ by definition 22.1.1(1), so proposition 22.7.3(1) gives an algebra homomorphism $F\colon C(q) \to A$ with $F(\iota(v)) = \bar{x}$. In the other direction, evaluation at $\iota(v)$ is an algebra homomorphism $k[x] \to C(q)$, since the powers of $\iota(v)$ commute with one another and k is central in $C(q)$; it kills $x^2 - a$ because $\iota(v)^2 = q(v) = a$, so it factors through a homomorphism $G\colon A \to C(q)$ with $G(\bar{x}) = \iota(v)$. Now $F \circ G$ is an algebra endomorphism of A fixing $\bar{x}$, and the elements it fixes form a subalgebra, which therefore contains the algebra generated by $\bar{x}$, namely all of A; so $F \circ G = \mathrm{id}_A$. Likewise $G \circ F$ is an algebra endomorphism of $C(q)$ fixing $\iota(V)$ pointwise, hence equals $\mathrm{id}_{C(q)}$ by the uniqueness clause of proposition 22.7.3(1). So F is an isomorphism.

Three cases occur, according to the square class of a (definition 22.1.6).

1. $a = 0$. Then $C(q) \cong k[x]/(x^2)$, the algebra of dual numbers, in agreement with example 22.7.2, which identifies it with $\bigwedge(V)$.
2. $a = c^2$ with $c \in k^\times$. Then $x^2 - a = (x-c)(x+c)$ is a product of two distinct monic irreducibles, distinct because $c \neq -c$ when $\operatorname{char} k \neq 2$, so proposition 11.1.6 gives

$$C(q) \cong k[x]/(x-c) \times k[x]/(x+c) \cong k \times k$$

   an algebra with zero divisors: $(\bar{x}-c)(\bar{x}+c) = 0$.
3. $a \in k^\times$ is not a square. Then $C(q)$ is a field, the quadratic extension of k obtained by adjoining a square root of a. Indeed $(u + w\bar{x})(u - w\bar{x}) = u^2 - w^2 a$ for $u, w \in k$, and this is nonzero unless $u = w = 0$: if $w \neq 0$ then $u^2 = w^2 a$ would make $a = (u/w)^2$ a square, and if $w = 0$ then $u \neq 0$ gives $u^2 \neq 0$. So every nonzero element of $C(q)$ has the inverse $(u - w\bar{x})/(u^2 - w^2 a)$.

Over $k = \mathbb{R}$ the three cases are $\mathbb{R}[x]/(x^2)$, $\mathbb{R} \times \mathbb{R}$ and $\mathbb{C}$, the last being $C(\langle -1 \rangle)$.

All three algebras above have dimension $2 = 2^1$, and the same count holds in every dimension. Proving it is the one place where the construction has to be shown not to collapse: nothing said so far rules out $C(q) = 0$, since a quotient by a large ideal can kill everything. The proof below produces an explicit $C(q)$-module of dimension 2^n, and the independence of the monomials of $C(q)$ follows from the independence of that module's basis.

Theorem 22.7.5: The Monomial Basis of $C(q)$

Let k be a field with $\operatorname{char} k \neq 2$, let V be a k-vector space of dimension n, let q be a quadratic form on V, and let $e_1, \ldots, e_n$ be an orthogonal basis of V for the associated symmetric bilinear form φ, with $a_i := q(e_i)$. For a subset $S = \{i_1 < i_2 < \cdots < i_r\}$ of $\{1, \ldots, n\}$ write

$$e_S := \iota(e_{i_1})\iota(e_{i_2}) \cdots \iota(e_{i_r}) \in C(q), \qquad e_\emptyset := 1$$

Then the 2^n elements e_S form a k-basis of $C(q)$. In particular $\dim_k C(q) = 2^n$ and $\iota\colon V \to C(q)$ is injective.

***Proof*:**
An orthogonal basis exists by theorem 22.2.1, whose hypotheses are the standing ones here. Throughout, ι is suppressed from the notation for products, so e_i denotes $\iota(e_i) \in C(q)$.

Step 1: the relations. By (22.3) applied to $v = w = e_i$ and to $v = e_i$, $w = e_j$ with $i \neq j$,

$$e_i^2 = q(e_i) = a_i, \qquad e_i e_j = -e_j e_i \quad (i \neq j)$$

the second because $\varphi(e_i, e_j) = 0$ for $i \neq j$ by the choice of basis. Also, for $v = \sum_i c_i e_i$, expanding

φ by bilinearity and using orthogonality gives

$$q(v) = \varphi(v, v) = \sum_i c_i^2 a_i \tag{22.4}$$

Step 2: the e_S span $C(q)$. As established in the uniqueness half of the proof of proposition 22.7.3, $C(q)$ is spanned as a k-vector space by 1 together with the products $\iota(v_1)\cdots\iota(v_m)$ with $v_t \in V$. Expanding each v_t in the basis $e_1, \ldots, e_n$ and multiplying out, which is legitimate because multiplication in $C(q)$ is k-bilinear, shows that the products $e_{j_1}e_{j_2}\cdots e_{j_m}$ of basis vectors already span.

It remains to reduce such a product to a scalar multiple of some e_S, which is done by induction on m. For $m = 0$ the product is $1 = e_\emptyset$. If the indices $j_1, \ldots, j_m$ are pairwise distinct, then finitely many interchanges of adjacent factors sort them into increasing order, each interchange multiplying the product by -1 by Step 1, so the product equals $\pm e_S$ with $S = \{j_1, \ldots, j_m\}$. If two indices coincide, choose a pair $j_s = j_t = i$ with $s < t$ and $t - s$ minimal; then $j_u \neq i$ for $s < u < t$, so moving the factor e_{j_t} leftwards past those $t - s - 1$ factors changes the product only by a sign and makes the two occurrences of e_i adjacent. Replacing e_ie_i by the scalar a_i produces a scalar multiple of a product of length $m - 2$, which the inductive hypothesis handles. Hence the e_S span, and $\dim_k C(q) \leq 2^n$.

Step 3: a module of dimension 2^n. Let M be the k-vector space with basis $\{u_S \mid S \subseteq \{1, \ldots, n\}\}$, so $\dim_k M = 2^n$. For an index i and a subset S write $S\triangle i$ for the symmetric difference, that is $S \cup \{i\}$ if $i \notin S$ and $S \setminus \{i\}$ if $i \in S$, and set

$$d(i, S) := \#\{l \in S \mid l < i\}, \qquad \sigma(i, S) := (-1)^{d(i,S)}, \qquad c_i(S) := \begin{cases} a_i & i \in S \\ 1 & i \notin S \end{cases}$$

Define $E_i \in \operatorname{End}_k(M)$ on the basis by

$$E_i(u_S) := c_i(S)\,\sigma(i, S)\,u_{S\triangle i}$$

Three facts about these coefficients are what the computation uses. First, $\sigma(i, S\triangle j) = \sigma(i, S)$ whenever $j \geq i$, since $d(i, -)$ counts only elements smaller than i and is therefore unchanged by inserting or deleting j. Second, $\sigma(j, S\triangle i) = -\sigma(j, S)$ whenever $i < j$, since $d(j, -)$ then changes by exactly one. Third, $c_i(S\triangle j) = c_i(S)$ for $j \neq i$, since membership of i is unchanged, while $c_i(S)c_i(S\triangle i) = a_i$, one of the two factors being a_i and the other 1.

Applying E_i twice and using the first and third facts,

$$E_i^2(u_S) = c_i(S)\sigma(i, S)\,c_i(S\triangle i)\sigma(i, S\triangle i)\,u_S = a_i\,\sigma(i, S)^2\,u_S = a_i\,u_S$$

so $E_i^2 = a_i\,\mathrm{id}_M$. For $i < j$, the same facts give

$$E_iE_j(u_S) = c_j(S)\sigma(j, S)\,c_i(S\triangle j)\sigma(i, S\triangle j)\,u_{S\triangle\{i,j\}} = c_i(S)c_j(S)\sigma(i, S)\sigma(j, S)\,u_{S\triangle\{i,j\}}$$

whereas, the second fact now contributing a sign,

$$E_jE_i(u_S) = c_i(S)\sigma(i, S)\,c_j(S\triangle i)\sigma(j, S\triangle i)\,u_{S\triangle\{i,j\}} = -\,c_i(S)c_j(S)\sigma(i, S)\sigma(j, S)\,u_{S\triangle\{i,j\}}$$

so $E_iE_j = -E_jE_i$ for $i \neq j$.

Step 4: the e_S are independent. Let $\rho\colon V \to \mathrm{End}_k(M)$ be the k-linear map with $\rho(e_i) = E_i$. For $v = \sum_i c_i e_i$, expanding the square and grouping the cross terms in pairs,

$$\rho(v)^2 = \sum_i c_i^2 E_i^2 + \sum_{i<j} c_i c_j \big(E_i E_j + E_j E_i\big) = \Big(\sum_i c_i^2 a_i\Big)\mathrm{id}_M = q(v)\,\mathrm{id}_M$$

by Step 3 and (22.4). So proposition 22.7.3(1), applied to the k-algebra $A = \mathrm{End}_k(M)$, yields an algebra homomorphism $\Phi\colon C(q) \to \mathrm{End}_k(M)$ with $\Phi(e_i) = E_i$.

Evaluate $\Phi(e_S)$ at $u_\emptyset$ for $S = \{i_1 < \cdots < i_r\}$, by descending induction on t on the claim that $E_{i_{t+1}} \cdots E_{i_r}(u_\emptyset) = u_T$ with $T = \{i_{t+1}, \ldots, i_r\}$. For $t = r$ the product is empty and the claim reads $u_\emptyset = u_\emptyset$. Given it for t, every element of T exceeds i_t, so $d(i_t, T) = 0$ and $\sigma(i_t, T) = 1$, while $i_t \notin T$ gives $c_{i_t}(T) = 1$; hence $E_{i_t}(u_T) = u_{T\cup\{i_t\}}$, which is the claim for $t-1$. At $t = 0$ it reads $\Phi(e_S)(u_\emptyset) = u_S$. Now if $\sum_S \lambda_S e_S = 0$ in $C(q)$, applying Φ and evaluating at $u_\emptyset$ gives $\sum_S \lambda_S u_S = 0$ in M, and the u_S are a basis, so every $\lambda_S = 0$.

The e_S therefore span and are independent, so they are a basis and $\dim_k C(q) = 2^n$. The images $\iota(e_1), \ldots, \iota(e_n)$ are the basis elements e_S with $\#S = 1$, hence linearly independent, so the linear map ι carries a basis of V to an independent set and is injective, completing the proof.

Two consequences are recorded before any further computation. First, ι being injective, V is identified with its image from here on and the symbol ι is dropped: a vector $v \in V$ is treated as an element of $C(q)$ with $v^2 = q(v)$. Second, $C(q)$ and $\bigwedge(V)$ have the same dimension for every q, by theorem 22.7.5 and example 22.7.2, and the same monomials index a basis of each; the multiplications differ, since squares of vectors are 0 in one and $q(v)$ in the other. For $n \geq 2$ neither algebra is commutative: $e_1e_2 = -e_2e_1$ in both, and $e_1e_2 \neq 0$ because it is a basis element (theorem 22.7.5 and theorem 16.4.3), so $e_1e_2 = e_2e_1$ would give $2e_1e_2 = 0$ against $\operatorname{char} k \neq 2$.

Example 22.7.6: The Quaternions as a Clifford Algebra

Let $V = \mathbb{R}^2$ and $q = \langle -1, -1\rangle$, with orthogonal basis e_1, e_2 and $q(e_1) = q(e_2) = -1$. By theorem 22.7.5 the algebra $C(q)$ has basis $1, e_1, e_2, e_1e_2$. Writing $\omega := e_1e_2$ and using $e_2e_1 = -e_1e_2$ to move factors past one another,

$$e_1^2 = e_2^2 = -1, \qquad \omega^2 = e_1e_2e_1e_2 = -e_1e_1e_2e_2 = -(-1)(-1) = -1$$

$$e_1e_2 = \omega, \qquad e_2\omega = e_2e_1e_2 = -e_1e_2e_2 = e_1, \qquad \omega e_1 = e_1e_2e_1 = -e_1e_1e_2 = e_2$$

These are the relations $i^2 = j^2 = k^2 = -1$, $ij = k$, $jk = i$, $ki = j$ satisfied by the quaternion units, so $C(q)$ and the quaternions $\mathbb{H}$ are the same algebra; $\mathbb{H}$ appears in theorem 20.2.3 as one of the three associative real division algebras, though that theorem supplies no basis or multiplication rule, so both are recalled here. To see it as an isomorphism rather than a coincidence of tables, recall that $\mathbb{H}$ is the 4-dimensional $\mathbb{R}$-algebra with basis $1, i, j, k$ and $ij = k = -ji$. Those two relations already force the rest of the table: putting $k = ij$ and using associativity,

$$jk = j(ij) = (ji)j = -ij^2 = i, \qquad ki = (ij)i = i(ji) = -i^2 j = j$$

and in the same way $ik = i(ij) = i^2 j = -j$ and $kj = (ij)j = ij^2 = -i$, so the products computed in $C(q)$ above are exactly the quaternion products, and they agree with the relations displayed for Q_8 in section 1.2.4. Now let $f\colon V \to \mathbb{H}$ be the $\mathbb{R}$-linear map with $f(e_1) = i$ and $f(e_2) = j$. For $v = c_1e_1 + c_2e_2$,

$$f(v)^2 = c_1^2 i^2 + c_1c_2(ij + ji) + c_2^2 j^2 = -(c_1^2 + c_2^2) = q(v)$$

so proposition 22.7.3(1) gives an algebra homomorphism $F\colon C(q) \to \mathbb{H}$ with $F(e_1) = i$ and $F(e_2) = j$. Its image is a subalgebra containing i, j and hence $ij = k$, so it contains the spanning set $1, i, j, k$ and F is surjective; both algebras have dimension 4 over $\mathbb{R}$, so

$$C(\langle -1, -1\rangle) \cong \mathbb{H}$$

Compare example 22.7.4, where $C(\langle -1\rangle) \cong \mathbb{C}$: the first two entries of a negative definite real form produce the complex numbers and then the quaternions.

The $\mathbb{Z}$-grading of $\mathcal{T}(V)$ was lost when the inhomogeneous relations were imposed, but the parity of a monomial survived the reduction in Step 2 of theorem 22.7.5: cancelling a repeated index removes two factors at a time. The following proposition makes that observation into a decomposition of the algebra, obtained not from the monomials but from the universal property, which is what makes it independent of the chosen orthogonal basis.

Proposition 22.7.7: The Parity Grading and the Even Subalgebra

Let k be a field with $\operatorname{char} k \neq 2$, let V be a k-vector space of dimension n, and let q be a quadratic form on V.

1. There is a unique k-algebra automorphism $\alpha\colon C(q) \to C(q)$ with $\alpha(v) = -v$ for all $v \in V$, and $\alpha^2 = \mathrm{id}$.
2. Setting $C^0(q) := \{x \in C(q) \mid \alpha(x) = x\}$ and $C^1(q) := \{x \in C(q) \mid \alpha(x) = -x\}$,
$$C(q) = C^0(q) \oplus C^1(q), \qquad C^\epsilon(q)\, C^\delta(q) \subseteq C^{\epsilon+\delta}(q)$$
the superscripts being read modulo 2. In particular $C^0(q)$ is a subalgebra, the *even part* of $C(q)$, and $V \subseteq C^1(q)$.
3. With the notation of theorem 22.7.5, $C^0(q)$ has basis the e_S with $\#S$ even and $C^1(q)$ has basis the e_S with $\#S$ odd. For $n \geq 1$ both have dimension 2^{n-1}.

Proof:

1. The map $f\colon V \to C(q)$ with $f(v) = -v$ is k-linear and satisfies $f(v)^2 = (-v)^2 = v^2 = q(v)$, so proposition 22.7.3(1) with $A = C(q)$ gives a unique algebra homomorphism α with $\alpha(v) = -v$ on V. Then $\alpha \circ \alpha$ is an algebra homomorphism $C(q) \to C(q)$ restricting on V to $v \mapsto -(-v) = v$, and $\mathrm{id}_{C(q)}$ restricts to that same map; the uniqueness clause of the same proposition, applied to $f = \iota$, forces $\alpha^2 = \mathrm{id}$. Hence α is bijective, so it is an automorphism.
2. For $x \in C(q)$ write
$$x = \tfrac{1}{2}\big(x + \alpha(x)\big) + \tfrac{1}{2}\big(x - \alpha(x)\big)$$
which is legitimate since $\operatorname{char} k \neq 2$. Applying α and using $\alpha^2 = \mathrm{id}$ shows the first summand is fixed by α and the second is negated, so $C(q) = C^0(q) + C^1(q)$. If x lies in both then $x = -x$, so $2x = 0$ and $x = 0$; the sum is direct. For the product rule, let $x \in C^\epsilon(q)$ and $y \in C^\delta(q)$. Then $\alpha(xy) = \alpha(x)\alpha(y) = (-1)^\epsilon(-1)^\delta xy = (-1)^{\epsilon+\delta}xy$, which is the stated containment. Taking $\epsilon = \delta = 0$ shows $C^0(q)$ is closed under multiplication, and $\alpha(1) = 1$ puts 1 in it, so it is a subalgebra. Finally $\alpha(v) = -v$ for $v \in V$ is the statement $V \subseteq C^1(q)$.

3. Since α is an algebra homomorphism, $\alpha(e_S) = \alpha(e_{i_1}) \cdots \alpha(e_{i_r}) = (-1)^r e_S$ for S of size r. Writing an arbitrary $x = \sum_S \lambda_S e_S$ in the basis of theorem 22.7.5,

$$\alpha(x) = \sum_S (-1)^{\#S} \lambda_S e_S$$

and comparing coefficients, $\alpha(x) = x$ holds if and only if $\lambda_S = 0$ for every S of odd size, while $\alpha(x) = -x$ holds if and only if $\lambda_S = 0$ for every S of even size. So the even monomials are a basis of $C^0(q)$ and the odd ones a basis of $C^1(q)$. Their numbers are equal for $n \geq 1$, since

$$\sum_{r \text{ even}} \binom{n}{r} - \sum_{r \text{ odd}} \binom{n}{r} = (1-1)^n = 0$$

by the binomial theorem, and they sum to 2^n, so each is 2^{n-1}, as we sought to show.

The decomposition is a $\mathbb{Z}/2\mathbb{Z}$-grading, and $C^0(q)$ is a subalgebra of half the dimension of $C(q)$. In example 22.7.6 it has basis $1, e_1e_2$ with $(e_1e_2)^2 = -1$, so evaluation at e_1e_2 is a surjective algebra homomorphism $\mathbb{R}[x] \to C^0$ killing $x^2 + 1$, and comparing dimensions gives

$$C^0(\langle -1, -1 \rangle) \cong \mathbb{R}[x]/(x^2+1) \cong \mathbb{C}$$

sitting inside $\mathbb{H}$ as the subalgebra spanned by 1 and one imaginary unit. The general rank-two computation, which recovers $\operatorname{disc}(q)$ of definition 22.1.7 from the even part, is the third exercise below.

Exercise 22.7.1

1. Let $\operatorname{char} k \neq 2$, let q be a quadratic form on V and regard $V \subseteq C(q)$ as in theorem 22.7.5. Show that a vector $v \in V$ is invertible in $C(q)$ if and only if $q(v) \neq 0$, and that its inverse is then $q(v)^{-1}v$. Deduce that if $q(v) = 0$ for some nonzero $v \in V$, then $C(q)$ is not a division algebra.

2. Let $\operatorname{char} k \neq 2$ and let $q = \langle 1, -1 \rangle$ on k^2. Show that $C(q) \cong M_2(k)$. (Send e_1 and e_2 to two explicit matrices, check the hypothesis of proposition 22.7.3(1) for every vector rather than only for e_1 and e_2, and finish with theorem 22.7.5.)

3. Let $\operatorname{char} k \neq 2$ and let $q \cong \langle a, b \rangle$ be nondegenerate of rank 2. Show that $C^0(q) \cong k[x]/(x^2+ab)$, and that this algebra is a field exactly when $-ab$ is not a square in k. Using corollary 22.2.3, express the condition in terms of $\operatorname{disc}(q)$, and check the answer against example 22.7.6.

4. Let $\operatorname{char} k \neq 2$ and let $q \cong \langle a_1, \ldots, a_n \rangle$ be nondegenerate, so every $a_i \neq 0$, and set $\omega := e_1e_2 \cdots e_n$ for an orthogonal basis as in theorem 22.7.5. Show that $\omega e_i = (-1)^{n-1} e_i \omega$ for every i, and deduce that ω lies in the centre of $C(q)$ if and only if n is odd.

5. Let $\operatorname{char} k \neq 2$ and let $C(g)$ be as in proposition 22.7.3(2). Show that $C(\mathrm{id}_V) = \mathrm{id}_{C(q)}$ and that $C(h \circ g) = C(h) \circ C(g)$ for isometries g and h. Deduce that $g \mapsto C(g)$ is a group homomorphism from the orthogonal group $O(\varphi)$ of definition 22.1.10 to the group of k-algebra automorphisms of $C(q)$, and that every automorphism in its image commutes with the α of proposition 22.7.7 and therefore preserves $C^0(q)$ and $C^1(q)$.

6. Let $\operatorname{char} k \neq 2$ and let $C(q)^{\mathrm{op}}$ denote the k-algebra with the same underlying vector space as $C(q)$ and multiplication $x * y := yx$. Show that there is a unique k-algebra homomorphism

$t\colon C(q) \to C(q)^{\mathrm{op}}$ with $t(v) = v$ for all $v \in V$, that the resulting map on $C(q)$ satisfies $t(xy) = t(y)t(x)$ and $t^2 = \mathrm{id}$, and that $t(e_{i_1}e_{i_2}\cdots e_{i_r}) = (-1)^{r(r-1)/2}e_{i_1}e_{i_2}\cdots e_{i_r}$ for $i_1 < \cdots < i_r$.

7. Let $\operatorname{char} k \neq 2$ and let q be a quadratic form on V. Show that $C^0(q)$ is spanned as a k-vector space by the products $v_1v_2\cdots v_{2m}$ of an even number of vectors of V, and hence that it is generated as a k-algebra by the products vw with $v, w \in V$.

Part V

Commutative Algebra

In this part of the book, we are no longer studying a new object as the central focus. Rather, it is the basic language that deals with *algebraic number theory* and *algebraic geometry*; this is the language of commutative algebra. Commutative algebra is an extension of ring theory where it is always assumed that the ring R is commutative. As shall be shown soon, this theory will prove to be very useful in developing two major studies of mathematics:

1. *Algebraic Geometry*: At the heart of Algebraic Geometry is the realization that all commutative rings have a corresponding *geometric object* (known as an *affine scheme*). We have already encountered the simplest of schemes in section 13.2.4 with the classification of hyper-planes of a finitely dimensional vector-space. Recall that each hyper-plane is associated the solution (or kernel) of linear equations. Schemes generalize this notion a great deal by working with polynomials over general commutative rings instead of restriting to linear equations (which are polynomials of degree 1), and by allowing the resulting geometric object to be "glued together" from kernels of polynomials instead of being the solution to polynomials. In *Algebraic Geometry* [5] the reader can see how to associate a scheme to every commutative ring, and the theory behind that correspondence.

2. *Algebraic Number Theory*: Studying fields allowed us to study the solutions to polynomials over fields. However, fields are very restrictive (dually very versatile) algebraic structures that allowed for some very powerful results. When trying to replicate the same strategies for solving polynomial equations over more general rings, many of these strategies fall apart. To properly generlize this, we shall need to invent a new theory, namely *algebraic number theory* is exactly this theory. In chapter 50, we shall explore some results such as quadratic reciprocity and classifying all Abelian extensions (i.e. Kronnecker-Weber's Theorem). Most of the time, we focus on solutions to equation over $\mathbb{Z}$, as $\mathbb{Z}$ is the initial object of ring, or more concretely $\mathbb{Z}$ is an object that appears very naturally in the world.

In this chapter, some basic topological notions will be used. The geometric intuition presented in this chapter will be lost on those who have not taken or learned some basic topology (up to the separation axioms, in particular T_1 and T_2, and compactness/connectedness). Appendix C serves as a review of what topological concepts will be used in this chapter.

chapter 31 and chapter 50 can be thought of as covering the more "classical" versions of heir respective field. For example, modern algebraic geometry shall start with schemes, while algebraic number theory is developed over local fields and uses techniques from complex analysis. Most of the motivation for these developments come form more concrete questions that naturally arose when studying polynomials and their solutions. The reader shall certainly gain a lot by understanding the core ideas behind these two chapters before tackling their more abstract modernization's. The reader wishing to follow up these chapters with a more in-depth covering can checkout [5] for the continuation of this book towards Algebraic Geometry and [39] for a continuation of this book towards Class Field Theory.

Note that algebraic geometry is *not* algebraic topology. In algebraic topology, we usually start with a topological space (or manifold) and associate to it an algebraic object to find properties of the topological/geometrical object. In Algebraic Geometry, we start with a commutative ring (or a collection of commutative rings), and create a geometric object that can be used to study the ring.

23

Noetherian Rings, Hilbert's Basis Theorem, and Localization

As with groups an modules, all R-algebras are quotient of a polynomial ring over many variables. If the polynomial ring is over finitely many variables. The fact that all algebras are quotient of polynomial rings will let the study of geometry and algebra.

Next, we recall some important definition and results that shall be central to the following section:

23.1 Noetherian Rings and Modules

Definition 23.1.1: Noetherian Ring

Let R be a commutative ring. Then if R satisfies the ascending chain condition on Ideals, then R is Noetherian. In particular, if $\mathcal{I}$ is a collection of ideals ordered by inclusion $(I_1 \subseteq I_2 \subseteq \cdots I_k \cdots)$, then there exists an m such that for all $n \geq m$, $I_m = I_{m+1} = \cdots$.

Proposition 23.1.2: Extension of Noetherian Rings

Let R be a Noetherian ring. Then R/I is Noetherian for any ideal I. Conversely, if I and R/I is Noetherian, then R is Noetherian.

***Proof*:**
Ideals of R/I are exactly the ideals of R containing I, so an ascending chain in R/I pulls back to an ascending chain in R, which stabilizes; thus R/I is Noetherian. Conversely, suppose I and R/I are Noetherian and let $J_1 \subseteq J_2 \subseteq \cdots$ be ideals of R. The chains $J_n \cap I$ (in I) and $(J_n + I)/I$ (in R/I) both stabilize, say for $n \geq m$. For such n, if $x \in J_{n+1}$ write $x = j + a$ with $j \in J_n$, $a \in I$ (possible since $J_n + I = J_{n+1} + I$); then $a = x - j \in J_{n+1} \cap I = J_n \cap I \subseteq J_n$, so $x \in J_n$. Hence $J_n = J_{n+1}$ and R is Noetherian.

As a consequence the image of any Noetherian ring is Noetherian.

Proposition 23.1.3: Equivalence of Noetherian Ring

Let R be a commutative ring. Then the following are equivalent

1. R is Noetherian
2. every nonempty collection of ideals of R has a maximal element under inclusion
3. every ideal of R is finitely generated

***Proof*:**
(1) $\Rightarrow$ (2): a nonempty collection of ideals with no maximal element would yield a strictly ascending chain, contradicting the ascending chain condition. (2) $\Rightarrow$ (3): given an ideal J, the collection of finitely generated subideals of J has a maximal element J_0; if $J_0 \neq J$, then for $x \in J \setminus J_0$ the ideal $J_0 + (x)$ is finitely generated and strictly larger, a contradiction, so $J = J_0$ is finitely generated. (3) $\Rightarrow$ (1): for an ascending chain $J_1 \subseteq J_2 \subseteq \cdots$, the union $J = \bigcup_n J_n$ is an ideal, hence finitely generated; its finitely many generators lie in some J_m, so $J = J_m$ and the chain stabilizes.

Example 23.1.4: Exercise
Show that if R is a Noetherian ring and M is a finitely generated R-module, then M is Noetherian (Hint: remember the universal property of free-modules)

The maximal-element form of the condition, part (2) above, is the one that does the work in practice. Here is a first instance, and the finiteness statement the factorization theory of Dedekind domains rests on: an ideal of a Noetherian ring need not factor into primes, but it always *contains* a product of them.

Proposition 23.1.5: Every Proper Ideal Contains a Product of Primes

Let R be a Noetherian ring and let $I \subsetneq R$ be a proper ideal. Then there are finitely many prime ideals $\mathfrak{p}_1, \ldots, \mathfrak{p}_r$, each containing I, with

$$\mathfrak{p}_1\mathfrak{p}_2 \cdots \mathfrak{p}_r \subseteq I$$

***Proof*:**
Let Σ be the collection of proper ideals of R admitting no such family of primes, and suppose $\Sigma \neq \emptyset$. Since R is Noetherian, Σ has a maximal element I by proposition 23.1.3(2).

Step 1: the maximal counterexample is not prime. Were I prime, $r = 1$ and $\mathfrak{p}_1 = I$ would satisfy the conclusion. So there are $a, b \in R \setminus I$ with $ab \in I$. Put $J = I + (a)$ and $J' = I + (b)$; both strictly contain I. Since $ab \in I$ and I is an ideal,

$$JJ' = (I + (a))(I + (b)) \subseteq I \cdot I + I(b) + (a)I + (ab) \subseteq I$$

Step 2: both are proper. If $J = R$ then $J' = JJ' \subseteq I$, contradicting $J' \supsetneq I$; symmetrically $J' \neq R$. So J and J' are proper ideals strictly containing I.

Step 3: descend. By maximality of I in Σ, neither J nor J' lies in Σ, so there are primes $\mathfrak{p}_1, \dots, \mathfrak{p}_s \supseteq J$ with $\mathfrak{p}_1 \cdots \mathfrak{p}_s \subseteq J$ and primes $\mathfrak{q}_1, \dots, \mathfrak{q}_t \supseteq J'$ with $\mathfrak{q}_1 \cdots \mathfrak{q}_t \subseteq J'$. Then

$$\mathfrak{p}_1 \cdots \mathfrak{p}_s \mathfrak{q}_1 \cdots \mathfrak{q}_t \subseteq JJ' \subseteq I$$

and each of these primes contains J or J', hence contains I. So $I \notin \Sigma$, a contradiction, and $\Sigma = \emptyset$ as we sought to show.

Over a domain the primes so produced are automatically nonzero, since they contain I. That is the form the theory of Dedekind domains actually consumes:

Corollary 23.1.6: Nonzero Ideals Contain Products of Nonzero Primes

Let R be a Noetherian domain and let $I \subsetneq R$ be a nonzero proper ideal. Then I contains a product of finitely many *nonzero* prime ideals, each containing I.

***Proof*:**
Take $\mathfrak{p}_1, \dots, \mathfrak{p}_r$ as in proposition 23.1.5. Each $\mathfrak{p}_i$ contains I, and $I \neq 0$, so each $\mathfrak{p}_i$ is nonzero.

We will soon encounter rings that satisfy the descending chain condition (commonly abbreviated to D.C.C). Such rings are called *Artinian rings*. For the following result, let us upgrade definition 3.7.6 for modules:

Definition 23.1.7: Chain of Modules and Composition Series

Let M be a module. Then a *chain* of submodule of M is a sequence (M_i) of submodules st

$$M = M_0 \supsetneq M_1 \supsetneq \cdots \supsetneq M_n = 0$$

where the length of the chain is the number of $\supsetneq$. A maximal chain is called a *composition series* of M (equivalently, each M_{i-1}/M_i is *simple*. IF the chain is of infinite length, we shall say that M does not have a composition series.

Proposition 23.1.8: Finite Length, then Length is Invariant

Let M have a composition series of length n. Then every composition series of M has length n, and every chain in M can be extended to a composition series.

***Proof*:**
Let $\ell(M)$ denote the least length of composition series of a module M To get the desired result, let us prove a few results:

1. $N \subseteq M \Rightarrow \ell(N) \leq \ell(M)$, with equality iff $N = M$. Let (M_i) be a composition series of M of minimum length, and take $N_i = N \cap M_i$. Then $N_{i-1}/N_i \subseteq M_{i-1}/M_i$, which is simple, so each N_{i-1}/N_i is 0 or all of M_{i-1}/M_i; dropping repetitions gives a composition series of N of length $\leq \ell(M)$, and $\ell(N) = \ell(M)$ forces $N_i = M_i$ for all i, i.e. $N = M$

2. each $M_i \subsetneq M$ have length $\leq \ell(M)$. To see this, we may make a new chain with M_0 as the starting point:
$$M = M_0 \supsetneq M_1 \supsetneq \cdots \supsetneq M_k = 0$$
which by part 1 give us:
$$\ell(M) = \ell(M_0) \geq \ell(M_1) \geq \cdots \geq \ell(M_k) = 0$$

3. Take any composition series of M. Then if it has length k, it must be that $k \leq \ell(M)$ by part (2). As $\ell(M)$ is the least length of a composition series, also $k \geq \ell(M)$, so $k = \ell(M)$. Hence, all composition series have the same length.

Finally, consider any chain for M. If its length is $\ell(M)$, by (2) it must be a composition series, if its length is less than $\ell(M)<$ it cannot be a composition series, and hence cannot be maximal. There does exists a lengthening by a new term, which as the number of possible terms that can be added is of finite length we may continue until we have a series of length $\ell(M)$, completing the proof.

23.2 Hilbert Basis Theorem

Recall from theorem 11.1.1 that if F is a field, then $F[x]$ is a Euclidean domain. By corollary 11.2.4, $F[x_1, x_2, \ldots]$ is a Unique Factorization Domain. However, of all the condition's on rings we have covered they are not necessarily stronger than Unique Factorization Domains (with the counter-example being example 11.2.5 where $\mathbb{Z}[x, y]$ is a UFD but not a PID). If we loosen the condition from the ideals be generated by a single element to being generated by *finitely many* elements (i.e. a Noetherian ring), then we in fact get that extending by an indeterminate x does not affect the finiteness condition on ideals.

Theorem 23.2.1: Hilbert's Basis Theorem (HBT)

Let R be a Noetherian ring. Then $R[x]$ is a Noetherian ring

If you know some linear algebra, the term "basis" is not like the term basis in linear algerba. It is because of an archaic use of the word basis when Hilbert formulated the theorem near the end of the 19th century, where it had a closer to "generation" than basis.

***Proof*:**

Recall that if $p(x) = a_nx^n + a_{n-1}x^{n-1} + \cdots + a_1x + a_0$, then a_n is the leading coefficient of $p(x)$. This terminology will often be used in this proof.

Let $I \subseteq R[x]$ be an ideal. The trick is to somehow make I linked to an ideal of R (which is finitely generated since R is Noetherian). Let L be the set of all leading coefficients of the ideal I. Then L is an ideal. First, $0 \in I$, so $0 \in L$. Next, take $f, g \in I$ be two polynomials in I so that $f = ax^n + a_{n-1}x^{n-1} + \cdots + a_1x + a_0$ and $g = bx^m + b_{n-1}x^{m-1} + \cdots + b_1x + b_0$ with leading coefficients a and b and degrees n and m respectively, with $n \geq m$, so that $a, b \in L$. Multiplying the lower-degree polynomial by a power of x to equalize degrees, $rx^{n-m}g \in I$ has degree n and leading coefficient rb, so $f - rx^{n-m}g \in I$ has degree-n coefficient $a - rb$; hence $a - rb \in L$ whenever it is nonzero. Thus L is closed under subtraction and absorbs multiplication by R, so L is an ideal. Now, since R is a Noetherian ring and L is an ideal of R, L is finitely generated, say $L = \langle a_1, a_2, ..., a_n \rangle$. Then for each a_i, there is a polynomial f_i that has a_i as it's coefficient. Let's say e_i is the degree of f_i, so that the set $e_1, e_2, ..., e_n$ represents the degree's of the polynomials representing the generators of L. Let $N = \max\{e_1, e_2, ..., e_n\}$.

Next, we will construct a couple of smaller ideals. Let $d \in \{0, 1, 2, ..., N-1\}$ (i.e. d is strictly smaller than the largest degree N) and define $L_d \subseteq L$ to be the set of all leading coefficients of polynomials of I of degree d, together with 0. Then L_d is also an ideal of R as can be checked by a similar method as last proof. Since R is Noetherian, there exists a set of generators for L_d, say $L_d = \{b_{d,1}, b_{d,2}, ..., b_{d,k_d}\}$. For each generator of L_d, there is a polynomial $f_{d,i} \in I$ of degree d that has the generator $b_{d,i}$ as its leading coefficient.

Now, we will show that

$$I = \langle f_1, f_2, ..., f_n, f_{1,1}, f_{1,2}, ...f_{1,k_1}, f_{2,1}, ..., f_{N-1,k_{N-1}} \rangle$$

that is, I is generated by the polynomial f_i and the polynomials $f_{d,i}$. Since there are finitely many of them, I would be finitely generating completing the proof. To show this, let I' represent the right hand side. Then $I' \subseteq I$, since every generator of I' is in I, and I' is the smallest set that contains those generators. So, let's prove $I \subseteq I'$, and take a $f \in I$.

If $\deg(f) \geq N$, then since $f \in I$, it's leading coefficient (say a) is in L: $a \in L$. Then since L is finitely generated, it is an R-linear combination of the generators of L: $r_1a_1 + r_2a_2 + \cdots + r_na_n$. From this, we can define a new polynomial $g_1 = r_1x^{d-e_1}f_1 + r_2x^{d-e_2}f_2 + \cdots + r_nx^{d-e_n}f_n$ with the same leading coefficient a of f that is in I'. Then

$$f - g_1$$

is a polynomial in I of degree less than $\deg(f)$. If $\deg(f - g_1)$ is still greater than or equal to N, then repeat the same process, making the polynomial $(f - g_1) - g_2$. Continue on this way till we get a polynomial of degree less than N:

$$f - g_1 - g_2 - \cdots - g_\ell = f - (g_1 + g_2 + \cdots + g_\ell)$$

Let's label this above polynomial f'. Note that the sum of the g_i's can be re-written as an R-linear combination of elements of I' since they all belong to I':

$$f - (\alpha_1 f_1 + \alpha_2 f_2 + \cdots + \alpha_n f_n)$$

Now, $\deg(f') < N$. Since $f' \in I'$, it's leading coefficient (say a) is in L_d: $a \in L_d$. Then since L_d is finitely generated, it is an R-linear combination of the generators of L_d: $a = r_1 b_{d,1} + r_2 b_{d,2} + \cdots + r_{k_d} b_{d,k_d}$. From this, we can define a new polynomial $h_1 = r_1 f_{d,1} + r_2 f_{d,2} + \cdots + r_{k_d} f_{d,k_d}$ with the same leading coefficient as f' that is in I'. Then

$$f' - h_1$$

Is of degree less than f'. Just as before, we can continue this procedure, which will terminate once the degree reaches 0:

$$f' - h_1 - h_2 - \cdots - h_p = f' - (h_1 + h_2 + \cdots + h_p)$$

Note that $\sum_i h_i$ can be decomposed into a sum of polynomials with leading coefficients from L_d:

$$f' - (\beta_{1,1} f_{1,1} + \beta_{1,2} f_{1,2} + \cdots + \beta k, k_d f_{k,k_d})$$

Since our algorithm terminated after we have subtacted the constant (that is, the leading term of L_0), we are left with:

$$f - (g_1 + g_2 + \cdots + g_\ell + h_1 + h_2 + \cdots + h_p) = 0$$

which implies $f = g_1 + g_2 + \cdots + g_\ell + h_1 + h_2 + \cdots + h_p$. Re-writing slightly, we get that

$$f = \alpha_1 f_1 + \alpha_2 f_2 + \cdots + \alpha_n f_n + \beta_{1,1} f_{1,1} + \beta_{1,2} f_{1,2} + \cdots + \beta k, k_d f_{k,k_d}$$

which shows that f is a combination of elements of I' and so $f \in I'$. Since f was arbitrary, we cna conclude that $I \subseteq I'$.

Along with the quick assertion earlier that $I' \subseteq I$, that shows us that $I = I'$, showing that I was finitely generated. Since I was an arbitrary ideal, we are left with saying that every ideal of $R[x]$ is finitely generated, showing that $R[x]$ is also Noetherian, as we sought to show.

The result is not necessarily true if R is not Noetherian

In the exercises there will be the outlining of an alternative proof which will give a different insight in the exercises.

Corollary 23.2.2: HBT on multi-variable Polynomials

Let R be a Noetherian ring. Then $R[x_1, x_2, ..., x_n]$ is a Noetherian ring

***Proof*:**
Since R is Noetherian, by the Hilbert Basis Theorem, $R[x_1]$ is Noetherian. Since $R[x_1]$ is Noetherian, $(R[x_1])[x_2] = R[x_1, x_2]$ is Noetherian. Continuing inductively, we see that $R[x_1, x_2, ..., x_n]$ is Noetherian, as we sought to show

Corollary 23.2.3: f.g. K-algebra then Finitely Presented

Let R be a finitely generated k-algebra. Then R is *finitely presented*, meaning

$$R \cong \frac{k[x_1, ..., x_n]}{(f_1, ..., f_m)} = \frac{k[x_1, ..., x_n]}{I}$$

where I is finitely generated.

***Proof*:**
As R is finitely generated as a k algebra:

$$R \cong \frac{k[x_1, ..., x_n]}{I}$$

for some finitely many variable $x_1, ..., x_n$ and an ideal $I \subseteq R$. By corollary 23.2.2 R is Noetherian, hence I is *finitely generated*: $I = (f_1, ..., f_m)$. But then R is finitely presented, completing the proof.

You can now return to example 10.1.18 to fully appreciate how there exist Atomic Domains which are not Noetherian Domains.

This theorem is very powerful. Since $R[x]$ is an ideal of itself, it is also finitely generated and the polynomials which generate it finitely can be characterized by their leading coefficients. In section 23.3, we will expand this theory further find a computational way of getting these polynomials.

Since a field is Noetherian (only ideal is (1) and (0), both clearly finite), Hilbert's Basis Theorem and induction immediately give:

Corollary 23.2.4: HBT on Polynomial Rings of Several Variables

Every ideal in the polynomial ring $F[x_1, x_2, ..., x_n]$ with coefficients from a field F is finitely generated

***Proof*:**
A field is Noetherian, so by corollary 23.2.2 the ring $F[x_1, \dots, x_n]$ is Noetherian; every ideal is therefore finitely generated by proposition 23.1.3.

Recall that finitely generated R-algebras are quotient of polynomial rings R. By HBT, we have the following

Corollary 23.2.5: Finitely Generated Algebras Are Noetherian

Let R be Noetherian and let A be a finitely generated R-algebra. Then A is Noetherian.

***Proof*:**
Write $A \cong R[x_1, \dots, x_n]/I$ as a quotient of a polynomial ring. By corollary 23.2.2 the polynomial ring is Noetherian, and by proposition 23.1.2 so is its quotient A.

Polynomial rings being over a Noetherian ring being Noetherian is central to to the study of algebras thanks to the universal property of Algebras

(A word on how Artinian rings are the opposite with D.C.C. We won't study them till chapter 28, since they sort of seem like a "oh wow, Noetherian rings can do so much, what about the opposite?" and then we realise that Artinian ring implies Noetherian ring, and it has so much structure that we can easily classify them all with no further fought, so we'll just leave that for later. All in all, the Noetherian condition turns out to be "more interesting")

23.2.6 Without Assuming Noetherian Let $R = k[x_1, ..., x_n, ...]$ and $A = R/(x_1, ..., x_n, ...)$. Then A is finitely generated, but the ideal is not finitely generated. In algebraic geometry, there is often the need to work with rings that look like quotients of polynomial rings (ex. polynomial rings over local rings). Thus, it common to work with *finitely presented* rings.

Exercise 23.2.1

1. Alternative HBT: (the one uses Diksons lemma)
2. Consider the chain $k[[x]]] \subseteq k[[x^{1/2}]] \subseteq k[[x^{1/6}]] \subseteq \cdots \subseteq k[[x^{1/n!}]]$. Taking the union of all of these forms an object called the *Puisseaux series*. Is the Puisseaux seris a ring, and if so is the Puisseaux series Noetherian?
3. Let S be a finitely generated R-algebra, where R is Noetherian. Prove that S is Noetherian.
4. We know that ideals of $k[x]$ are generated by 1 element if k is a field. By Hilbert's basis theorem, we know that every ideal of $k[x, y]$ is finitely generated. Is it true that ideals of $k[x, y]$ is generated by at most 2 elements if k is a field?
5. Which of the following are Noetherian Rings:
    1. $\mathbb{Z}[x, 1/x]$ as a subring of $\mathbb{Q}$
    2. $R := \mathbb{R} + x\mathbb{C}[x]$
    3. $R := k + xL[x]$ where $[L : k] < \infty$ is a finite field extension
    4. $R := k + xk[x, y]$ as a subring of $k[x, y]$, k a field.
6. Let M be an R-module over a Noetherian ring R. Show that there exists a $x \in M$ such that $\mathrm{Ann}_R(x)$ is a prime ideal (large hint: consider the set $\mathcal{S} = \{\mathrm{Ann}_A(x) \mid x \in M\}$ of ideals, and use the fact that R is Noetherian to conclude $\mathcal{S}$ has a maximal element. Is this element prime?)

23.3 Gröbner Bases

Hilbert's basis theorem guarantees that every ideal of $k[x_1, \ldots, x_n]$ is finitely generated, but its proof is not constructive: it exhibits no canonical generating set and no way to decide whether a given polynomial lies in an ideal. *Gröbner bases* repair both defects. They provide, for each ideal and each choice of monomial order, a distinguished finite generating set against which a multivariate division algorithm gives unique remainders, turning ideal membership into a finite computation. This is the computational basis of commutative algebra and algebraic geometry; the standard reference is Cox, Little, and O'Shea, *Ideals, Varieties, and Algorithms.*

Fix the polynomial ring $S = k[x_1, \dots, x_n]$ over a field k, and write a monomial as $x^\alpha = x_1^{\alpha_1} \cdots x_n^{\alpha_n}$ for $\alpha \in \mathbb{N}^n$.

Definition 23.3.1: Monomial Order

A *monomial order* on S is a total order $\preceq$ on the monomials of S (equivalently on $\mathbb{N}^n$) such that

1. $\preceq$ is a well-order: every nonempty set of monomials has a least element; and
2. it is compatible with multiplication: if $x^\alpha \preceq x^\beta$ then $x^\gamma x^\alpha \preceq x^\gamma x^\beta$ for all $\gamma \in \mathbb{N}^n$.

The prototype is the *lexicographic order*: $x^\alpha \succ x^\beta$ when the leftmost nonzero entry of $\alpha - \beta$ is positive. The *graded* and *graded reverse* lexicographic orders first compare total degree $|\alpha| = \sum_i \alpha_i$ and break ties lexicographically; the latter is usually fastest in practice. Fixing a monomial order, every nonzero $f \in S$ has a largest monomial.

Definition 23.3.2: Leading Term

Let $\preceq$ be a monomial order and $0 \neq f \in S$. The *multidegree* $\operatorname{multideg}(f)$ is the $\preceq$-largest α with x^α occurring in f; the corresponding term is the *leading term* $\operatorname{LT}(f)$ and its monomial the *leading monomial* $\operatorname{LM}(f)$. For an ideal I, its *leading term ideal* is $\operatorname{LT}(I) = (\operatorname{LT}(f) : 0 \neq f \in I)$, a monomial ideal.

Compatibility makes leading terms multiplicative, $\operatorname{LT}(fg) = \operatorname{LT}(f)\operatorname{LT}(g)$, and the well-order axiom is exactly what makes the following algorithm terminate.

Theorem 23.3.3: Multivariate Division Algorithm

Fix a monomial order and an ordered tuple $(g_1, \dots, g_s)$ of nonzero polynomials. Then every $f \in S$ can be written

$$f = q_1 g_1 + \cdots + q_s g_s + r$$

where no monomial of the remainder r is divisible by any $\operatorname{LM}(g_i)$. It is produced by repeatedly subtracting the appropriate multiple of some g_i whenever $\operatorname{LM}(g_i)$ divides the current leading monomial, and otherwise moving the leading term into r.

***Proof*:**
Each step strictly lowers the $\preceq$-largest monomial not yet moved into r, so by the well-order property the process terminates; the stated condition on r holds by construction.

The remainder generally depends on the *order* of the g_i, and $r = 0$ is sufficient but not necessary for $f \in (g_1, \dots, g_s)$. Both defects vanish for a special generating set.

Definition 23.3.4: Gröbner Basis

Let $I \subseteq S$ be a nonzero ideal and $\preceq$ a monomial order. A finite set $G = \{g_1, \ldots, g_s\} \subseteq I$ is a *Gröbner basis* of I if the leading terms of G generate the leading term ideal:

$$\mathrm{LT}(I) = (\mathrm{LT}(g_1), \ldots, \mathrm{LT}(g_s))$$

Theorem 23.3.5: Existence and Membership

Every nonzero ideal $I \subseteq S$ has a Gröbner basis for any monomial order, and any Gröbner basis of I generates I. Dividing $f \in S$ by a Gröbner basis of I gives a remainder that is independent of the order of the divisors and is zero if and only if $f \in I$.

***Proof*:**
The monomial ideal $\mathrm{LT}(I)$ is finitely generated by finitely many $\mathrm{LT}(f_i)$ with $f_i \in I$ (*Dickson's lemma*, equivalently theorem 23.2.1); these f_i are a Gröbner basis. If G is a Gröbner basis and $f \in I$, division gives $f = \sum q_i g_i + r$ with $r \in I$ and no monomial of r divisible by any $\mathrm{LM}(g_i)$; were $r \neq 0$, then $\mathrm{LT}(r) \in \mathrm{LT}(I) = (\mathrm{LM}(g_i))$ would be divisible by some $\mathrm{LM}(g_i)$, a contradiction. Hence $r = 0$ and $f \in (G)$, so G generates I and membership is detected by the remainder. Two remainders of the same f differ by an element of I with no monomial divisible by a leading term, hence by 0; the remainder is unique.

Thus a Gröbner basis solves the *ideal membership problem*: $f \in I$ iff its Gröbner-basis remainder is 0. To compute one, we measure the failure of a generating set to be Gröbner through cancellations of leading terms.

Definition 23.3.6: S-polynomial

For nonzero $f, g \in S$ with $L = \mathrm{lcm}(\mathrm{LM}(f), \mathrm{LM}(g))$, the *S-polynomial* is

$$S(f,g) = \frac{L}{\mathrm{LT}(f)} f - \frac{L}{\mathrm{LT}(g)} g,$$

built so the leading terms of the two products cancel.

Theorem 23.3.7: Buchberger's Criterion

A finite generating set $G = \{g_1, \ldots, g_s\}$ of an ideal I is a Gröbner basis if and only if, for every pair $i \neq j$, the remainder of $S(g_i, g_j)$ on division by G is zero.

***Proof*:**
Necessity is immediate, since $S(g_i, g_j) \in I$ has zero remainder by theorem 23.3.5. For sufficiency, the content is that every cancellation of leading terms among the g_i is an S-polynomial syzygy; when all S-polynomial remainders vanish this forces $\mathrm{LT}(I) = (\mathrm{LM}(g_i))$. The details are in Cox, Little, and O'Shea.

The criterion is a decision procedure, and completing a set to satisfy it is an algorithm.

Theorem 23.3.8: Buchberger's Algorithm

Given generators $\{f_1, \dots, f_m\}$ of I, start with $G = \{f_1, \dots, f_m\}$; repeatedly compute the remainder r of each $S(g_i, g_j)$ on division by G, adjoining r to G whenever $r \neq 0$; stop when all S-polynomial remainders are zero. This terminates and returns a Gröbner basis of I.

***Proof*:**
Each adjoined remainder r has leading monomial outside the current $(\mathrm{LM}(g))$, so the leading term ideals strictly increase; by the ascending chain condition in the Noetherian ring S (theorem 23.2.1) this stabilizes, so finitely many elements are added and the loop halts. On termination all S-polynomial remainders vanish, so G is a Gröbner basis by theorem 23.3.7.

Discarding redundant generators and scaling leading coefficients to 1 yields the unique *reduced* Gröbner basis of I for a given order, a genuine canonical form: two ideals coincide iff their reduced Gröbner bases (for the same order) are equal.

Gröbner bases make ideal membership *decidable*, but not cheap. In the worst case the degrees occurring in a reduced Gröbner basis, and hence the running time of Buchberger's algorithm, are doubly exponential in the number of variables, and ideal membership for $k[x_1, \dots, x_n]$ is EXPSPACE-complete (Mayr and Meyer). The abstract finiteness handed to us by Hilbert's basis theorem thus conceals real computational hardness: knowing that an ideal is finitely generated is a long way from being able to compute with it.

23.4 Localization

Localization shall be the generalization of section 9.5 to commutative rings and modules in general (including ones with zero-divisors). With the introduction of the tensor product, localization shall be shown to have the important property of producing *flat module*, which shall have important geometric consequences in algebraic geometry. From this, we shall define the notion of local properties and the famous Nakayama lemma which shall allow many proofs to be done "locally". Concluding this section, we shall wrap up the correspondence between primes of a ring R and primes in its integral extension.

The term "localization" comes from algebraic geometry, where this process of localizing mirrors the process of "localizing" for manifolds. If you have done some differential geometry, recall that the tangent space at a point p has *cotangent* space $\mathfrak{m}_p/\mathfrak{m}_p^2$, where $\mathfrak{m}_p$ is the maximal ideal of germs vanishing at p (the tangent space is its dual). Localizing shall correspond to this process, where "inverting" elements around a point is the same as a function on a manifold being invertible on a small enough local neighborhood where $f(p) \neq 0$. The reader may question this correspondence, for in differential geometry we are working with smooth functions over a [smooth] manifold, while in the algebraic case we simply have a [commutative] ring. However, the correspondence is more equivalent than it may at first seem. It shall be particularly clear by the end of section 32.5, where it is shown that there is a way of thinking of all commutative rings as a set of "functions" on a special geometric space (the geometric space is called an [affine] *scheme*), hence there is a direct parallel between smooth functions on the geometric object of a manifold and commutative rings where each element is interpreted as a function on a scheme.

23.4.1 Localization of Rings

This is a generalization of the constructing of ring of fraction from theorem 9.5.1 by allowing the set that's closed under multiplication D to have zero divisors. This will imply that R need not be embedded into $D^{-1}R$, but the universal property is still satisfied. Furthermore, we will have to solve the problem of elements that "should" be zero being equal to nonzero elements. If $a \in R$ and $d \in D$ where $as = 0$, then:

$$a = \frac{a}{1} = \frac{as}{s} = \frac{0}{s} = 0$$

this issue will be addressed in the proof:

Theorem 23.4.1: Universal Property of Localization

Let R be a commutative ring and D a multiplicatively closed subset containing 1. Then there is a ring $D^{-1}R$ and a homomorphism $\iota\colon R \to D^{-1}R$ with $\iota(D) \subseteq (D^{-1}R)^{\times}$ such that for every ring S and homomorphism $\psi\colon R \to S$ with $\psi(D) \subseteq S^{\times}$, there is a *unique* homomorphism $\Psi\colon D^{-1}R \to S$ making the following diagram commute:

$$\begin{tikzcd} D^{-1}R \arrow[dr, "\Psi"] & \\ R \arrow[u, "\iota"] \arrow[r, "\psi"'] & S \end{tikzcd}$$

The pair $(D^{-1}R, \iota)$ is universal among all such (S, ψ).

***Proof*:**
A lot of this proof is similar to theorem 9.5.1 (Ring of Fractions) and so it will be quickly presented pointing out where the generalization of D having zero divisors makes a difference.

Let $R \times D$ have the equivalence relation

$$(a, b) \sim (c, d) \;\Leftrightarrow\; x(ad - bc) = 0 \text{ for some } x \in D$$

Which is essentially the same as in theorem 9.5.1. The difference between the proof in the case where D does not have zero divisors' is with the fact that the terms are scaled by some $x \in D$. This is to make sure elements that are zero-divisors will belong to the [0] (as we'll show in the proceeding corollary). Usually, the equivalence class $[(r, d)]$ is denoted $\frac{r}{s}$ or r/s so that

$$\frac{a}{b} = \frac{c}{d} \text{ if and only if } x(ad - bc) = 0 \qquad \text{for some } x \in D$$

Define

$$\frac{r}{d} + \frac{s}{e} := \frac{re + sd}{de} \qquad \frac{r}{d}\frac{s}{e} := \frac{rs}{de}$$

Then these two operations form a well-defined commutative ring $(R \times D)/\sim$; this ring is denoted $D^{-1}R$. Note that $\frac{d}{1}$ is a unit in $D^{-1}R$ for every $d \in D$, with inverse $\frac{1}{d}$.

Define $\iota : R \to D^{-1}R$ by $\iota(r) = \frac{r}{1}$. This map defines a natural ring homomorphism. Using this map, if the map $\varphi : R \to S$ maps D to units, we can define the map $\Psi : D^{-1}R \to S$ by defining:

$$\Psi(\frac{r}{d}) = \psi(r)\psi(d)^{-1}$$

This map is well-defined: Notice if $\frac{r}{d} = \frac{s}{e}$ then by definition of the equivalence relation, for some $x \in D$, $x(re - sd) = 0$ Thus $\varphi(x(re - sd)) = \varphi(x)(\varphi(re) - \varphi(sd)) = 0$. Since x is a unit in S and S is a commutative ring, s is not a zero-divisor, and so $\varphi(re) - \varphi(sd) = 0$ Moving values around, we get

$$\varphi(r)\varphi(d)^{-1} = \varphi(s)\varphi(e)^{-1}$$

Thus, Ψ is a well-defined ring homomorphism. Furthermore, by construction, $\Psi \circ \iota = \varphi$

Finally, Ψ is unique. If there was another Ψ' such that $\Psi' \circ \iota = \varphi$, then:

$$\varphi(r) = \Psi'(\iota(r)) = \Psi'\left(\frac{r}{1}\right)$$

Thus Ψ' maps $\frac{r}{1}$ in the same way as Ψ. For the $\frac{1}{d}$ terms, notice that

$$\frac{1}{d} = \left(\frac{d}{1}\right)^{-1}$$

and so $\varphi(d) = \Psi'(\frac{d}{1})$, and so, by properties of ring homomorphisms

$$\Psi'\left(\frac{d}{1}\right)^{-1} = \Psi'\left(\frac{1}{d}\right)$$

showing that $\Psi' = \Psi$, fulfilling the last condition needed to prove for the universal property

As before, $D^{-1}R$ is still called the ring of fractions. However, there is a more common name with a geometric intuition:

Definition 23.4.2: Localization of R at D

Let R be a commutative ring and D be a multiplicatively closed subset of R. Then the ring $D^{-1}R$ produced in the previous theorem is the *ring of fractions of R with respect to D*, or the *localization of R at D*. This ring is also sometimes denoted $R[D^{-1}]$.
If $D = \{f^n \mid n \in \mathbb{Z}_{\geq 0}\}$ for some $f \in R$, we usually denote $D^{-1}R = R_f$.

If D is not a multiplicatively closed set, we define $R[D^{-1}] := R[\overline{D}^{-1}]$ where $\overline{D^{-1}}$ is the multiplicative closure of D.

The fact that D can contain zero divisors means that ι can now have non-trivial kernel:

Corollary 23.4.3: Kernel of ι

Let ι be the natural inclusion of R into $D^{-1}R$ for some appropriate D. Then

1. $\ker(\iota) = \{r \in R \mid xr = 0 \text{ for some } x \in D\}$. In particular, ι is an injection if and only if D doesn't contain 0 or zero-divisors
2. $D^{-1}R = 0$ if and only if D contains 0 or a nilpotent element (since D is multiplicatively closed)

***Proof*:**

1. If $\iota(r) = \frac{r}{1} = 0$, then we have that $(r, 1) \sim (0, 1)$ and so $x(1r - 0) = xr = 0$ for some $x \in D$. Thus, ι is an injection if and only if D does not contain 0 or zero-divisors
2. $D^{-1}R = 0$ if and only if $r/1 = 0/1$ for all r i.e. $(r, 1) \sim (0, 1)$ if and only if $(1, 1) \sim (0, 1)$ if and only if $x \cdot 1 = 0$. If $x = 0$, we're done, and if not then as D is multiplicatively closed then $x^n \in D$ for each $n \in \mathbb{N}$ and so if $x^n = 0$ then x is nilpotent and $x^n \cdot 1 = 0$, completing the proof.

This let's us get a sense of how adding zero-divisors to D changes the resulting localization of R at D.

Example 23.4.4: Localization
All examples in section 9.5 are valid examples, including the trivial one of any field $\mathbb{F}$, $\mathbb{Z}$ to $\mathbb{Q}$ with $D = \mathbb{Z} - \{0\}$, and $\mathbb{F}[x]$ to $\mathbb{F}(x)$ with $D = F[x] - \{0\}$. We shall give some more interesting examples here (all rings are commutative):

1. If R is an integral domain, then we can always take $D = R - \{0\}$. The ring $D^{-1}R$ is then always a field, and is called the *field of fractions of* R, and is sometimes denoted Q. Then for any smaller multiplicatively closed subset E, $E^{-1}R$ is a subring of Q with elements $r/e \in Q$ with $r \in R$ and $e \in E$
2. Let R be a ring and $f \in R$. Define $D = \{f^n \mid n \in \mathbb{Z}_{\geq 0}\}$. Then we'll denote $D^{-1}R = R_f$. Then R_f is a ring in which the element f has become a unit. In a sense, it is now possible to solve the question
$$fx - 1 = 0$$
In fact, this intuition can be made more rigorous showing that
$$R_f \cong \frac{R[x]}{fx - 1}$$
This is similar to how we worked with field extensions, and in fact is an important link to connect the intuition of localization to a geometric intuition. Keep this example in the back of your mind for now.
3. Consider $R = k[x, y]/(xy)$, take $\overline{x} \in R$, and consider $R_{\overline{x}}$. Then the the map $\iota : R \to R_{\overline{x}}$ is no longer injective as:
$$\overline{y} = \frac{\overline{xy}}{\overline{x}} = \frac{\overline{0}}{\overline{x}} = \overline{0}$$
4. This example is called *localizing at a prime*. Let R be a ring and P a prime ideal of R. Then the set $D = R - P$ is multiplicatively closed. To see that, let $x, y \in D$. If it were the case that $xy \in P$, then either $x \in P$ or $y \in P$. But then one of them wouldn't be in D, a contradiction. Thus, D is indeed multiplicatively closed. The localization $D^{-1}R$ is called the *localization of* R *at* P and is commonly denoted as R_P. In this ring, all elements not in P become units. As a simple example, take $R = \mathbb{Z}$ and $P = (p)$ for some prime p. Then by the nice divisibility properties of integers, the set $\mathbb{Z}_{(p)}$ can with a little work be shown to be:
$$\mathbb{Z}_{(p)} = \left\{ \frac{a}{b} \in \mathbb{Q} \;\middle|\; p \nmid b \right\}$$

that is, it is all elements of $\mathbb{Q}$ for which the denominator is not divisible by p. Can you prove that $\mathbb{Z}_{(p)}$ has only one maximal ideal (and hence, is a local ring)? Is this true of all localization at primes?

It is tempting to say that if $R_{(p)}$ is the localization of a ring R at the prime p and $a/b \in R_{(p)}$ if $b \notin (p)$, however this is not necessarily the case as we are dealing with an equivalence class! The reader may then ask if we can take a better representative (like the representative a/b in $\mathbb{Z}$ such that $\gcd(a, b) = 1$), however R is not necessarily a UFD and so this may not be well-defined.

5. The following examples is a way to "localise" functions. Let k be a field and V be any set. Define R to be functions from V to k containing the constant functions ($F \subseteq \mathrm{Hom}^{+\text{ constants}}_{\mathbf{Set}}(V, k)$) with $+$ and $\cdot$ defined by component-wise addition and multiplication on the image: $R = (F, +, \cdot)$. For example, take all continuous functions from $[0, 1]$ to $\mathbb{R}$. For any $a \in V$, let $M_a \subseteq R$ be the set of all functions that vanish at a: $f(a) = 0$. Then M_a is the kernel of the evaluation map from R to k:
$$\mathrm{ev}_a : R \to k \qquad f \mapsto f(a) \qquad \ker(\mathrm{ev}_a) = M_a$$
Since R contains constant functions, ev_a is surjective and so $k \cong R/M_a$. By proposition 9.4.19, M_a is a maximal idea, and hence a prime ideal. Thus, we can define the localization of R at M_a to be:
$$R_{M_a} = \left\{ \frac{f}{g} \;\middle|\; f, g \in R,\ g(a) \neq 0 \right\}$$
where ev_a is extended as:
$$(f/g)(a) = f(a)/g(a)$$
This then makes R_{M_a} a way of defining rational functions.

6. Let $R = \mathbb{C}[x]$, and take $p = (x - a)$. Since $\mathbb{C}[x]$ is a Principal Ideal Domain and $(x - a)$ is irreducible, $(x - a)$ is prime, and so we can define
$$\mathbb{C}[x]_{(x-a)} = \left\{ p(x)/q(x) \;\middle|\; x - a \nmid q(x) \right\}$$
which can be thought of all rational functions which do not vanish at a.

Localization has many nice geometric interpretations. As we will see in section 32.4, all algebraic varieties (i.e. irreducible algebraic sets) are related to the prime ideals. Hence, understanding how ideals of R correspond to ideals of $D^{-1}R$ will bring us to understanding a geometric connection.

For an ideal $I \subseteq R$, its *extension* ${}^{\iota}I$ is the ideal of $D^{-1}R$ generated by $\iota(I)$; since every such element can be written as a/d with $a \in I$ and $d \in D$, the notation $D^{-1}I$ is also used. For an ideal $J \subseteq D^{-1}R$, its *contraction* ${}^{\pi}J = \iota^{-1}(J)$ is an ideal of R, being the preimage of an ideal under the ring homomorphism ι.

Proposition 23.4.5: Ideals of Localization

Let R be a ring and $D^{-1}R$ be the localization of R at D for appropriate D. Then:

1. For every ideal $J \subseteq D^{-1}R$,
$$J = J^{ce}$$
As a consequence, every ideal of $D^{-1}R$ is an extension of an ideal in R, and every ideal of D^{-1} contracts to a distinct ideal in R

2. If $I \subseteq R$ is an ideal of R, then:
$$^{\pi}(^{\iota}I) = \{r \in R \mid dr \in I \text{ for some } d \in D\}$$
As a consequence, $I^e = D^{-1}R$ if and only if $I \cap D \neq \emptyset$

3. There is a bijection between the set of prime ideals with $P \cap D = \emptyset$ and the set of prime ideals of $D^{-1}R$:
$$\left\{\begin{matrix}\text{prime ideals of } R \text{ with} \\ P \cap D = \emptyset\end{matrix}\right\} \underset{(-)^c}{\overset{(-)^e}{\rightleftarrows}} \left\{\text{prime ideals of } D^{-1}R\right\}$$

4. If R is Noetherian, then $D^{-1}R$ is Noetherian. Similarly if R is Artinian

***Proof*:**

1. Notice that by construction, $^{\iota}(^{\pi}J) \subseteq J$. For the other direction, notice that for any $a/d \in J$ it is always the case that $a/1 = d(a/d) \in J$, that is there exists a a/d such that $d(a/d) = a/1$. Since $\iota(\pi(a/d)) = a/1$. and so by definition, of $^{\iota}(^{\pi}J)$, $(1/d)(a/1) = a/d \in {}^{\iota}(^{\pi}J)$ (since it closed by multiplication of elements from $D^{-1}R$.

2. Let $I' = \{r \in R \mid dr \in I \text{ for some } d \in D\}$ be the set in the question. We'll show $I' = {}^{\pi}(\lim I)$. For the $\subseteq$ direction, take $r \in I'$. Then by definition there exists a $d \in D$ such that $dr = a \in I$. Then $r/1 = a/d \in \lim I$, so contracting we get $r \in {}^{\pi}(\lim I)$. For the $\supseteq$ direction, take $r \in {}^{\pi}(\lim I)$. Then $r/1 = a/d \in \lim I$ for some $a \in I$ and $d \in D$. Then by the definition of the equivalence class, we have $x(dr - a) = 0$ for some $x \in D$, and hence $xdr = xa$ where $xa \in I$. Since $xd \in D$, $r \in I'$. The rest of the assertions of (2) follow quickly.

3. To map prime ideals from $D^{-1}R$ to R, recall that preimages of prime ideals are prime. Thus, by construction, the contraction must map prime ideals of $D^{-1}R$ to prime ideals that are disjoint from D.

   For the converse, let P be a prime ideal of R that is disjoint from D, and consider $Q = {}^{\iota}P$. We'll show Q is prime. Suppose $(a/d_1)(b/d_2) = (ab/d_1d_2) \in Q$. Then $ab/d_1d_2 = c/d$ for some $c \in P$ and $d \in D$. Then by definition of the equivalent relation, $x(dab - d_1d_2c) = 0$ for some $x \in D$. Since $xdab - xd_1d_2c = 0 \in P$ and P is closed under subtraction and left-multiplication of elements of R, it must be that $xdab = (xd)(ab) \in P$. Since P is disjoint from D, it must be that $ab \in P$. Since P is prime, either a or b is an element of P. Thus, either a/d_1 or b/d_2 is an element of Q, showing that Q is a prime ideal. Thus, extending prime ideals that are disjoint from D map to prime ideals in $D^{-1}R$.

Finally, by (2), since P is disjoint from D we have $P = {}^{\pi}({}^{\iota}P)$, showing that extending and contracting are inverses of each other, finishing the proof for (3).

4. Given some ascending chain of ideals in $D^{-1}R$, they contract to ascending chain of ideals in R. Since R is Noetherian, they stabilize. Thus, the ideals that get contracted must also stabilize. Similarly for Artinian Rings.

The ideal in part (2) of proposition 23.4.5 is given a name:

Definition 23.4.6: Saturated Ideal

Let R be a commutative ring, $I \subseteq R$ is an ideal and $D^{-1}R$ is the localization of R at D for appropriate D. Then *saturation* of the ideal I with respect to D is ${}^{\pi}(\lim I)$. If $I = {}^{\pi}(\lim I)$, then I is said to be *saturated* or *D-saturated* with respect to D

Geometrically, the prime-ideal bijection of proposition 23.4.5(3) identifies $\operatorname{Spec}(D^{-1}R)$ with the subspace of $\operatorname{Spec} R$ consisting of the primes disjoint from D. For $D = \{f^n \mid n \in \mathbb{Z}_{\geq 0}\}$ this is the distinguished open set $D(f) = \{\mathfrak{p} \in \operatorname{Spec} R \mid f \notin \mathfrak{p}\}$, so $\operatorname{Spec} R_f \cong D(f)$; dually $\operatorname{Spec} R/(f) \cong V(f)$, the closed set of primes containing f. Thus $\operatorname{Spec} R = D(f) \sqcup V(f)$ as sets, the two pieces being homeomorphic to $\operatorname{Spec} R_f$ and $\operatorname{Spec} R/(f)$ as subspaces (though $\operatorname{Spec} R$ is their topological coproduct only when $D(f)$ is also closed).

Proposition 23.4.7: Localization Property Preservation

Let R be a ring, $D \subseteq R$ a multiplicative set with $0 \notin D$, and $D^{-1}R$ the localization. Then the following properties of R are preserved in $D^{-1}R$:

1. no zero divisors
2. Noetherian
3. Integrally closed

For (3), R is understood to be an integrally closed domain, so that (1) makes $D^{-1}R$ a domain with the same fraction field.

***Proof*:**

1. Suppose $(a/s)(b/t) = 0$ in $D^{-1}R$. Then $uab = 0$ for some $u \in D$ by corollary 23.4.3, and $u \neq 0$ because $0 \notin D$, so $ab = 0$ in the domain R and hence $a = 0$ or $b = 0$. Thus $a/s = 0$ or $b/t = 0$.

2. Let $J \subseteq D^{-1}R$ be an ideal and put $I = \iota^{-1}(J)$, where $\iota\colon R \to D^{-1}R$. By proposition 23.4.5 every ideal of the localization is the extension of its contraction, so $J = D^{-1}I$. Since R is Noetherian, $I = (a_1, \ldots, a_n)$ for finitely many a_i, and then J is generated by $a_1/1, \ldots, a_n/1$: an element of J is a/s with $a \in I$, and writing $a = \sum r_i a_i$ gives $a/s = \sum (r_i/s)(a_i/1)$. Every ideal of $D^{-1}R$ is therefore finitely generated.

3. Write $K = \operatorname{Frac}(R)$, which by (1) is also the fraction field of $D^{-1}R$. Let $z \in K$ be integral

over $D^{-1}R$, say
$$z^n + \frac{a_{n-1}}{s_{n-1}} z^{n-1} + \cdots + \frac{a_1}{s_1} z + \frac{a_0}{s_0} = 0$$
with $a_i \in R$ and $s_i \in D$. Put $s = s_0 s_1 \cdots s_{n-1} \in D$ and multiply the relation by s^n, collecting powers of sz:
$$(sz)^n + \sum_{i=1}^{n} \frac{a_{n-i}\, s^i}{s_{n-i}} (sz)^{n-i} = 0$$
Each coefficient lies in R, since $s^i/s_{n-i} = s^{i-1} \prod_{j \neq n-i} s_j$ and $i \geq 1$. So sz is integral over R, hence $sz \in R$ because R is integrally closed, and therefore $z = (sz)/s \in D^{-1}R$, as we sought to show.

Exercise 23.4.1

1. Let R be a ring, f be an element of R ,and R_f be the localization of R at $\{f^n \mid n \in \mathbb{Z}_{\geq 0}\}$. Then
   1. $$R_f \cong \frac{R[x]}{fx-1}$$
   2. $R_f \cong R_{f^n}$ for $n \geq 1$
   3. If f is nilpotent, then $R_f = 0$
2. Let R be a normal domain and S a multiplicatively closed set. Then $S^{-1}R$ is a normal domain

23.4.2 Localization of Modules

Since every ring is a module over itself, we can see how localization works for this special case of modules. We extend the concept of localization now to more general modules. The construction is essentially the same thing: if M is an R-module, and $D \subseteq R$ is multiplicatively closed, define the same equivalence class on $M \times D$ and the operation's on $(M \times D)/\sim$ as
$$\frac{m}{d} + \frac{m'}{d'} = \frac{d'm + dm'}{dd'} \quad \text{and} \quad r \cdot \frac{m}{d} = \frac{r \cdot m}{d}$$
Then this is indeed a module with the proof proceeeding essentially the same as theorem 23.4.1, and so is left as a sanity check. Just like $R[D^{-1}]$ is the localization of R at D, we will denote $M[D^{-1}]$ to be the localization of M at D. Thus, an R-module M is extended to an $D^{-1}R$-module $M[D^{-1}]$.

Proposition 23.4.8: Zeros of $M[D^{-1}]$

Let R be a ring, $D \subseteq R$ be a multiplicatively closed subset, and M be a module over R. Then m goes to 0 in $M[U^{-1}]$ if and only if m is annihilated by an element $d \in D$. If M is finitely generated, then $M[U^{-1}] = 0$ if and only if M is annihilated by an element in D.

If R lies in a field K and M is an R-module that is contained in a k-vector space, then $\iota : M \to D^{-1}M$ is injective iff $M \xrightarrow{\cdot s} M$ is injective for all $s \in D$.

***Proof*:**
Exercise

Localization modules works well with homomorphisms: let $\varphi : M \to N$ be an R-module homomorphism. Then by the universal property, we see that this is extended to an $D^{-1}R$-module homomorphism $D^{-1}\varphi : M[D^{-1}] \to N[D^{-1}]$ by mapping:

$$D^{-1}\varphi\left(\frac{m}{d}\right) = \frac{\varphi(m)}{d}$$

Furthermore, $D^{-1}(\varphi \circ \psi) = (D^{-1}\varphi) \circ (D^{-1}\psi)$, i.e. $D^{-1}(-)$ is a functor from Mod_R to $\text{Mod}_{D^{-1}R}$. We may ask for certain properties of this functor, most basic perhaps would be if $D^{-1}(-)$ is an exact functor, that it preserves kernels and cokernels. This is the case:

Proposition 23.4.9: Localization Is Exact

Let M_1, M_2, M_3 be R-modules such that $0 \to M_1 \xrightarrow{f} M_2 \xrightarrow{g} M_3 \to 0$ is a short exact sequence. Then

$$0 \to M_1[D^{-1}] \xrightarrow{D^{-1}f} M_2[D^{-1}] \xrightarrow{D^{-1}g} M_3[D^{-1}] \to 0$$

is a short exact sequence, that is $D^{-1}(\underline{\hspace{2em}})$ is an exact functor.

***Proof*:**
Since $g \circ f = 0$, $D^{-1}g \circ D^{-1}f = D^{-1}(0) = 0$, and so $\text{im}(D^{-1}f) \subseteq \ker(D^{-1}g)$. For the reverse inclusion, let $m/d \in \ker(D^{-1}g)$ so that $g\left(\frac{m}{d}\right) = \frac{g(m)}{d} = 0$ in $M_3[D^{-1}]$. Thus, there exists an s such that $sg(m) = 0$ in M_3. Since g is an R-module homomorphism, $sg(m) = g(sm)$, and so $sm \in \ker(g) = \text{im}(f)$, implying for some $m' \in M_1$, $f(m') = sm$. Putting it all together, we get that in $M_2[D^{-1}]$:

$$\frac{m}{d} = \frac{f(m')}{ds} = (D^{-1}f)\left(\frac{m'}{ds}\right) \in \text{im}(D^{-1}f)$$

showing that $\ker(D^{-1}g) \subseteq \text{im}(D^{-1}f)$, which along with the fact that $\text{im}(D^{-1}f) \subseteq \ker(D^{-1}g)$ show's that

$$\ker(D^{-1}g) = \text{im}(D^{-1}f)$$

For exactness at the ends: if $(D^{-1}f)(m/d) = f(m)/d = 0$ then $sf(m) = f(sm) = 0$ for some $s \in D$, and injectivity of f gives $sm = 0$, so $m/d = 0$; thus $D^{-1}f$ is injective. And any m_3/d equals $(D^{-1}g)(m_2/d)$ where $g(m_2) = m_3$ (surjectivity of g), so $D^{-1}g$ is surjective. Hence the localized sequence is short exact, completing the proof.

Notice that as we are defining a $D^{-1}R$-module, we are extending scalars. If we use section 17.5, then we can use the following observation to arrive at this result more quickly:

Proposition 23.4.10: Characterization of Localization

Let M be an R-module and $M[D^{-1}]$ a localiation. Then:

$$M[D^{-1}] \cong_R D^{-1}R \otimes_R M$$

***Proof*:**
Define the map:

$$(r/d) \times m \mapsto \frac{r \cdot m}{d}$$

then it is easy to that this map is R-bilinear, and hence by the universal property of the tensor-product, there exists an R-module homomorphism:

$$f : D^{-1}R \otimes M \to M[D^{-1}]$$

It is clear that f is surjective and is uniquely defined with respect to the universal property. It remains to check that f is injective. First, show that for any element in $D^{-1}R \otimes M$, it can be written as:

$$\frac{1}{d} \otimes m$$

for some $d \in D$ and $m \in M$. Let's say $f(1/d \otimes m) = m/d = 0$. Then $sm = 0$ for some $s \in D$. Then:

$$\frac{1}{d} \otimes m = \frac{s}{ds} \otimes m = \frac{1}{ds} \otimes sm = \frac{1}{ds} \otimes 0 = 0$$

showing that f is injective, and hence an R-module isomorphism, completing the proof.

As a direct conssequence, $D^{-1}R$ is a *flat R-module.*

Corollary 23.4.11: Localization Commuting With Tensor

Let M and N be R-modules and $D \subseteq R$ multiplicatively closed. Then there is a unique isomorphism:

$$\varphi : M[D^{-1}] \otimes_{D^{-1}R} N[D^{-1}] \to (M \otimes_R N)[D^{-1}] \qquad (m/d) \otimes (n/s) \mapsto (m \otimes n)/ds$$

In particular, if $\mathfrak{p}$ is any prime ideal:

$$M_{\mathfrak{p}} \otimes_{R_{\mathfrak{p}}} N_{\mathfrak{p}} \cong (M \otimes_R N)_{\mathfrak{p}}$$

***Proof*:**
By proposition 23.4.10, localizing is base change along $R \to D^{-1}R$, so $M[D^{-1}] = D^{-1}R \otimes_R M$. Associativity of the tensor product then gives

$$M[D^{-1}] \otimes_{D^{-1}R} N[D^{-1}] \cong D^{-1}R \otimes_R (M \otimes_R N) = (M \otimes_R N)[D^{-1}],$$

and the prime-ideal case is the special case $D = R \setminus \mathfrak{p}$.

(Some more cool results from Micheal's p. 43)

23.4.3 Local Properties

Localization allows us to diminish our problem to a ring much closer to a field, in particular if we want to study a family of prime ideals contained in some prime ideal p. In fact, there are many results about rings that can be deduced by studying rings *locally*, namely by studying rings at each prime ideal[1]. This motivates the following concept:

Definition 23.4.12: Local Properties

Let P be a property of a ring (ex. commutative, free, etc.). Then a property P of a ring R (resp. of an R-module M) is called a *local property* if it respects the following:

$$R \text{ (resp. } M\text{) has property } P \Leftrightarrow R_p \text{ (resp. } M_p\text{) has property } P \text{ for each prime ideal } p \subseteq R$$

In otherwords, a property is local if we can move back and forth between studying any ring (resp. module) localized at a prime. Here are some examples of local properties:

Proposition 23.4.13: Zero Module Is A Local Property

Let M be an R-module. Then the following are equivalent:

1. $M = 0$
2. $M_p = 0$ for all prime ideals $p \subseteq R$
3. $M_m = 0$ for all maximal ideal $m \subseteq R$

***Proof*:**
Certainy (1) $\Rightarrow$ (2) $\Rightarrow$ (3). For (3) $\Rightarrow$ (1), let $0 \neq x \in M$ be a nonzero element of an R-module M with $0 \neq 1$. Let $I = \text{Ann}(x)$. Then I is contained in some maximal ideal m: $I \subseteq m$. Consider M_m. Since $M_m = 0$, $x/1 = 0$, implying $rx = 0$ for some $r \in R - m$. Howver $\text{Ann}(x) \subseteq m$, meaning no such element can exist, a contradiction, showing that $M = 0$ as we sought to show.

Another property that only needs to be verified locally is the injectivity/surjectivity of a function:

Proposition 23.4.14: Injectivity/Surjectivity/Isomorphism Is A Local Property

Let $\varphi : M \to N$ be an R-module homomorphism. Then the following are equivalent:

1. φ is an injection/surjection/isomorphism
2. $\varphi_p : M_p \to N_p$ is injection/surjection/isomorphism for all prime ideals $p \subseteq R$
3. $\varphi_m : M_m \to N_m$ is injection/surjection/isomorphism for all maximal ideals $m \subseteq R$

***Proof*:**
The proof shall be doen for injectivity, surjectivitiy follows easly and isomorphism is combining the two.

(1) $\Rightarrow$ (2) Since $0 \to M \xrightarrow{\varphi} N$ is exact, by proposition 23.4.9, $0 \to M_p \xrightarrow{\varphi_p} N_p$ is exact, hence φ_p is injective for any prime ideal $p \subseteq R$.

(2) $\Rightarrow$ (3) Every maximal idael is prime

[1] This observation is at the basis of interpreting rings as geometric objects

(3) $\Rightarrow$ (1) Let $K = \ker(\varphi)$. Then the sequence $0 \to K \to M \to N$ is exact, and hence proposition 23.4.9 $0 \to K_m \to M_m \to N_m$ is exact, therefore $K'_m \cong \ker(\varphi_m) = 0$ since φ_m is injective. Thus, by proposition 23.4.13 φ is injective.

to show surjectivity, simply reverse the arrows.

Corollary 23.4.15: Exactness Is A Local Property

Let $A \xrightarrow{f} B \xrightarrow{g} C$ be a sequence of R-module homomorphisms. Then it is exact if and only if $A_{\mathfrak{m}} \xrightarrow{f_{\mathfrak{m}}} B_{\mathfrak{m}} \xrightarrow{g_{\mathfrak{m}}} C_{\mathfrak{m}}$ is exact for every maximal ideal $\mathfrak{m} \subseteq R$.

***Proof*:**
Exactness at B says the homology $H = \ker g/\mathrm{im} f$ vanishes. Localization is exact (proposition 23.4.9), so it commutes with kernels, images, and quotients, giving $H_{\mathfrak{m}} = \ker g_{\mathfrak{m}}/\mathrm{im} f_{\mathfrak{m}}$. By proposition 23.4.13, $H = 0$ iff $H_{\mathfrak{m}} = 0$ for every maximal $\mathfrak{m}$, which is exactness of the localized sequences.

A final example that will be useful is that faltness is a local property:

Proposition 23.4.16: Flatness Is A Local Property

Let M be an R-module. Then the following are equivalent:

1. M is a flat R-module
2. M_p is a flat R_p-module for each prime ideal $p \subseteq R$
3. M_m is a flat R_m-module for each maximal ideal $m \subseteq R$

***Proof*:**
(1) $\Rightarrow$ (2) comes from pervious lemmas and (2) $\Rightarrow$ (3) is immediate. For (3) $\Rightarrow$ (1), let $N \to P$ be any R-module homomorphism and $m \subseteq R$ any maximal idea. Then:

$N \to P$ is injective	$\Rightarrow N_m \to P_m$ is injective	proposition 23.4.14
	$\Rightarrow N_m \otimes_{R_m} M_m \to P_m \otimes_{R_m} M_m$ is injective	flatness property
	$\Rightarrow (N \otimes_R M)_m \to (P \otimes_R M)_m$ is injective	corollary 23.4.11
	$\Rightarrow N \otimes_R M \to P \otimes_R M$ is injective	proposition 23.4.14

Hence, M is flat.

Proposition 23.4.17: Integral Closure Is A Local Property

Let R be an integral domain. Then the following are equivalent:

1. R is integrally closed
2. R_p is integrally closed for each prime ideal $p \subseteq R$
3. R_m is integrally closed for each maximal ideal $m \subseteq R$

***Proof*:**
exercise.

The following is a way of proving results locally:

Proposition 23.4.18: Local Properties On Modules

Let R be a subring of a field k, and let M be an R-module contained in a k-vector space V (equiv. the map $M \to M \otimes_R k$ is injective). Then:

$$M = \bigcap_{\mathfrak{m}} M_{\mathfrak{m}} = \bigcap_{\mathfrak{p}} M_{\mathfrak{p}}$$

***Proof*:**
Certainly $M \subseteq \cap_{\mathfrak{m}} M_{\mathfrak{m}}$. For the other direciton, take $x \in \cap_{\mathfrak{m}} M - \mathfrak{m}$. For each maximal ideal $\mathfrak{m}$, $x = m/s$ for some $m \in M$ and $s \in R - \mathfrak{m}$. Hence $sx \in M$. Define the ideal

$$I_a = \{a \in R \mid ax \in M\}$$

Then $as \in I_a$. But by construction $s \notin \mathfrak{m}$,hence $I_a \not\subseteq \mathfrak{m}$. But then it must be that $I_a = R$, hence $x = 1 \cdot x \in M$, the giving the result

Extending to prime ideals is left as an exercise.

Corollary 23.4.19: Local Properties Of Ideals

Let R be an integral domain. Then every ideal $I \subseteq R$ is equal to the intersection of its localizations at the maximal ideals of R, and also to the intersections of its localization at the prime ideals of R

Hence, to show a property of an ideal I in an integral domain R, it suffices to show that the properly holds for all its localization $I_{\mathfrak{p}}$ and it is preserved under intersections.

Exercise 23.4.2

1. Show that M is projective if and only if each M_p is free.

23.4.4 Local Rings

Recall that the localization of $\mathbb{Z}$ at any prime ideal (p), $\mathbb{Z}_{(p)}$, has a unique maximal ideal. It was left as an exercise to see whether any localization at a prime, R_p left R with only a single maximal ideal. This is indeed the case, meaning there is a unique field associated to R_p, letting us study R in terms of the collection of field induced by R_p. Thus, for any integral domain R, we can always produce a collection of rings that only have a single maximal ideal $\mathfrak{m}_\mathfrak{p}$. Once we interpret a ring as a geometric object $\operatorname{Spec}(R)$, we shall see that the rings $R_\mathfrak{p}$ representing the local properties of $\operatorname{Spec}(R)$, not unlike how we have stalks at a point in Differential Geometry. Due to this, it is worth-while studying rings with a unique maximal ideal:

Definition 23.4.20: Local Ring

Let R be a commutative ring. Then R is a *local ring* if it has a unique maximal ideal.

Every field is a local ring, its unique maximal ideal being (0). The definition is enlightening: if R is local with maximal ideal $\mathfrak{m}$ and $x \notin \mathfrak{m}$, then x must be a unit, since otherwise (x) is a proper ideal and so lies in a maximal ideal, forcing $x \in \mathfrak{m}$. Conversely, if the non-units of R form an ideal then that ideal is the unique maximal ideal, so R is local.

Example 23.4.21: Local Rings

1. Every field is trivially a local ring.
2. $\mathbb{Z}/p^n\mathbb{Z}$ is a local ring with $n > 1$ with maximal ideal $\overline{p}$
3. if $f \in k[x]$ is irreducible then $k[x]/(f(x)^n)$ is a local ring with maximal ideal $\overline{(f(x))}$
4. (geometric example) Let M be a smooth manifold (real or complex). Then define $\mathcal{O}_{M,x}$ to be the germ of smooth functions at x, that is, two smooth functions $f, g \in C^\infty(M)$ are in the same equivalence relation $f \sim g$ (i.e. $[f] = [g]$ are the same germ) if there exists some open set $U \subseteq M$ containing x such that $f|_U = g|_U$. Clearly, $\mathcal{O}_{M,x}$ is a ring (using the ring operation of the codomain $\mathbb{R}$ or $\mathbb{C}$). The maximal ideal of $\mathcal{O}_{M,x}$ are those functions that are 0 at x

In fact, this last example will have a direct parallel in scheme theory, see [5].

Proposition 23.4.22: Local Ring Equivalences

Let R be a commutative ring with $1 \neq 0$. Then the following are equivalent:

1. R is a local ring.
2. the sum of any two non-units is a non-unit
3. for every $x \in R$, either x is a unit or $1 - x$ is a unit
4. if a finite sum is a unit, then one of its terms is a unit

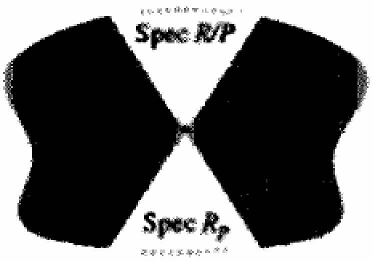
***Proof*:**
(1) $\Rightarrow$ (2): if R is local with maximal ideal $\mathfrak{m}$, an element is a non-unit iff it generates a proper ideal iff it lies in $\mathfrak{m}$, so the non-units are exactly $\mathfrak{m}$, which is closed under addition.

(2) $\Rightarrow$ (4): if a finite sum is a unit but no term is, then every term is a non-unit, so by (2) the sum is a non-unit, a contradiction.

(4) $\Rightarrow$ (3): apply (4) to $x + (1 - x) = 1$.

(3) $\Rightarrow$ (1): let $\mathfrak{m}$ be the set of non-units; it absorbs multiplication by R. If $x, y \in \mathfrak{m}$ had a unit sum $u = x + y$, then $1 = u^{-1}x + u^{-1}y$, so by (3) one of $u^{-1}x, u^{-1}y$ is a unit, making x or y a unit, a contradiction. Hence $\mathfrak{m}$ is closed under addition and is an ideal. Every proper ideal consists of non-units, hence lies in $\mathfrak{m}$, so $\mathfrak{m}$ is the unique maximal ideal and R is local.

Using localization and quotienting, we have a large amount of control over the prime ideals we want to work with: given a (commutative) ring R, if we take R_p, we eliminates all ideals except those contained in p, while R/p eliminates except those contained not contained in p (i.e. containing p). This can be visualized like so:



Hence, R_p/q^e (resp. $(R/p)_{\overline{q}}$) will only keep all the ideals between p and q, and if $p = q$, then the result is a field, known as the *residue field*:

Definition 23.4.23: Residue Field

Let R be a commutative local ring with maximal ideal m. Then R/m is called the *residue field*.

The quotient $R/\mathfrak{p}$ and the ring $R_\mathfrak{p}$ are canonically related: the ring $R_\mathfrak{p}$ can be thought of as the localization of $R/\mathfrak{p}$:

Proposition 23.4.24: Localization and Quotient Commute

Let $\mathfrak{p} \subseteq R$ be prime ideal of a ring R, and let $\mathfrak{m}_\mathfrak{p}$ be the unique maximal ideal of $R_\mathfrak{p}$. Then localizing at $\mathfrak{p}$ and then taking the field of fractions is isomorphic to taking quotient at $\mathfrak{p}$ and then localizing (at $\overline{fp} = \overline{0}$) Then there is a canonical embedding:

$$R/\mathfrak{p} \hookrightarrow R_\mathfrak{p}/\mathfrak{m}_\mathfrak{p}$$

that maps $R_\mathfrak{p}/\mathfrak{m}_\mathfrak{p}$ into the field of fractions of $R/\mathfrak{p}$

This may be visualized as the following diagram:

$$\begin{tikzcd}
 & R/\mathfrak{p} \arrow[dr, "\text{localize}"] & \\
R \arrow[ur, "\text{quotient}"] \arrow[dr, "\text{localize}"'] & & K(R/\mathfrak{p}) \cong R_\mathfrak{p}/\mathfrak{p}R_\mathfrak{p} \\
 & R_\mathfrak{p} \arrow[ur, "\text{quotient}"'] &
\end{tikzcd}$$

***Proof*:**
exercise

23.4.5 Nakayama's Lemma

(Atiyah Michael in Comm alg has some interesting results that feel are worth putting in here: p.21 on fin. gen modules)

As mentioned at the beginning, localization let's us get geometric information on a local level. Nakayama's lemma will be one way we translate that information to a global level. In this section, we will restrict to finitely generated modules (the infinite case is treated in the exercises).

Let's say M is a finitely generated module over R. If we were to localize M at a prime, we would get a local ring $R[D^{-1}]$. For simplicity, let's denote this ring G. As we've shown, G has only one maximal ideal m. Modding G by m, we get the field G/m. M over G/m is a vector-space, and so it has a basis. The heart of Nakayama's lemma is that this can be reversed: Using the basis of M over G/m, we can find a basis for M over G!

We will guild this up in three steps, which are collectively called Nakayama's Lemma:

Definition 23.4.25: Jacobson Radical

Let R be a ring. Then the *jacobson radical* of R, denoted $\text{Jac}(R)$ is the intersection fo all maximal ideals of R:

$$\text{Jac}(R) = \bigcap_{\substack{M \subseteq R \\ \text{maximal}}} M$$

(analogous to Fratini subgroup)

Proposition 23.4.26: Properties Of Jacobson Radical

Let $J \subseteq R$ be the Jacobson radical of a commutative ring R. Then:

1. If $I \subsetneq R$ is a proper ideal, then the ideal generated by I and J is also proper: $(I, J) \subsetneq R$
2. If R is Artinian, then the Jacobson radical is nilpotent: $J^n = 0$ for some n
3. $x \in J$ if and only if $1 - rx$ is a unit for every $r \in R$

***Proof*:**

1. If I is maximal, then $J \subseteq I$ and we're done, and if not then $I \subseteq M$ for some maximal ideal so $(I, J) \subseteq M \subsetneq R$

2. Let J be the jacobson radical. By the D.C.C. the following chain stabalizes
$$J \supseteq J^2 \supseteq \cdots \supseteq J^n = J^{n+1} = \cdots$$
Consider the set of all annihilators of J^n:
$$I = \{x \in R \mid xJ^n = 0\}$$
I claim that $I = (1)$, which will show that $1J^n = J^n = 0$ (i.e. J is nilpotent). For the sake of contradiction, let's say it is not I. Then pick $I' \supsetneq I$ that is minimal (i.e. there is no ideal between I' and I). Such an ideal exists by the D.C.C. condition. Then by minimality, it must be that I' is generated by adding just one more element, that is
$$I' = I + Rx \qquad x \in R,\ x \notin I$$
Thus, I'/I is a simple R-module by minimality (i.e. has no other ideals in it). However, it is easy to see that any simple R-module is of the form R/m for some maximal ideal m. Thus,
$$I'/I \cong R/m$$
multiplying both sidses by m, we "kill" the right hand side and get
$$mI' \subseteq I$$
Thus $JI' \subseteq I.$, so $xJ \in I$, and so by definition of I,
$$xJ^{n+1} = 0$$
but by assumption $xJ^{n+1} = J^n$, but that implies $x \in I$. However, $x \notin I$, a contradiction.

3. Doing the contra-positive, let's say $x \notin \text{Jac}(R)$. Then $x \notin M'$ for some maximal ideal M'. By maximality, $M + (x) = (1)$, so $y + rx = 1$, implying $1 - rx = y \in M'$, showing that $1 - rx$ cannot be a unit (this does not show "for all $r \in R$"). Conversely, if $1 - rx$ is not a unit, $1 - rx \in M$ for some M. However, $1, rx \notin M$, since if they were then $1 \in M$. Thus, $rx \notin \text{Jac}(R)$ for all r, but then $x \notin \text{Jac}(R)$.

Lemma 23.4.27: Nakayama's Lemma I

Let M be a finitely generated R-module, and suppose $\alpha \subseteq R$ is an ideal that is contained in all maximal ideals of R (i.e. $\alpha \subseteq \text{Jac}(R)$). Then:
$$\text{if } \alpha M = M \text{ then } M = \{0\}$$

***Proof*:**
We will proceed by induction on the number of generators of M. If $n = 1$, then $\langle x_1 \rangle = M$. If $\alpha M = M$, we have that:

$$x_1 = a_1 x_1 \text{ for some } a_1 \in \alpha$$

we get that $(1 - a_1)x_1 = 0$, meaning $1 - a_1$ annihilates x_1.

However, $(1 - a_1)$ is in fact a unit (as will be shown in a moment). Since it's a unit, there exists a b such that

$$0 = b \cdot 0 = b(1 - a_1)x_1 = x_1$$

Thus it must be that $x_1 = 0$, showing that $M = \{0\}$. So it's left to show that $1 - a_1$ is a unit. Let's say $(1 - a_1)$ was a proper ideal. Then it must lie in some maximal ideal. Since $a_1 \in \alpha$ and α is a subset of all maximal ideals, it must be that $1 - \alpha_1 + \alpha_1 = 1$ is in a maximal ideal, however that would make the ideal a non-proper ideal, a contradiction.

For the case of $n = k$, let $x_1, x_2, ..., x_k$ generate M, and suppose $\alpha M = M$. Then

$$x_k = a_1 x_1 + a_2 x_2 + \cdots + a_k x_k$$

so $(1 - a_k)x_k = a_1 x_1 + \cdots + a_{k-1} x_{k-1} \in \langle x_1, \ldots, x_{k-1} \rangle$. As in the base case $1 - a_k$ is a unit, hence $x_k \in \langle x_1, \ldots, x_{k-1} \rangle$, so $M = \langle x_1, \ldots, x_{k-1} \rangle$ is generated by $k-1$ elements and still satisfies $\alpha M = M$. By the induction hypothesis $M = \{0\}$, as we sought to show

If we restrict to $R = A$ being a local ring and $\alpha = m$ the maximal ideal of A, we will have the first step in constructing a basis for M over A:

Lemma 23.4.28: Nakayama's Lemma II

Let M be a finitely generated A-module over a local ring A with maximal ideal m, and let F be a submodule of M. Then:

$$\text{if } M = F + mM \text{ then } M = F$$

***Proof*:**
Take $M/F = (F + mM)/F = mM/F$ so that $M/F = mM/F$. Then by lemma 23.4.27, $M/F = \{0\}$, so that means that M moded by F makes everything 0, and so

$$M = F$$

Finally, we put these two together to create create a basis for a module over a local ring using a basis for a smaller module!

Lemma 23.4.29: Nakayama's Lemma III

Let M be a A-module over a local ring A and m the maximal ideal of A. If $x_1, x_2, ..., x_n$ generated M/mM (M mod mM), then they also generators for M

***Proof*:**
Let $F = \langle x_1, x_2, ..., x_n \rangle \subseteq M$ be the submodule generated by the x_i. Then

$$M = F + mM$$

Then by lemma 23.4.28, $M = F$, and so M has $\langle x_1, x_2, ..., x_n \rangle$ as generators, as we sought to show

Nakayama's lemma let's us reverse some constructions. As we saw in chapter 14, a [finitely generated?] projective module over a PID is free. Using Nakayama's lemma, we can show that a finitely generated module over a local ring is also free!

Theorem 23.4.30: Projective Module over Local Ring

Let M be a finitely generated projective module over a local ring A. Then M is free. In particular, If if $\{x_1, ..., x_n\}$ are elements of M who's residue class $\{\overline{x_1}, ..., \overline{x_1}\}$ form a basis for M/mM over A/m, then $x_1, ..., x_n$ is a *basis* for M over A.

Furthermore, if $\overline{x_1}, ..., \overline{x_n}$ are linearly independent over A/m, then this collection can be completed to a basis for M over A

***Proof*:**
First, Note that $M/\mathfrak{m}M$ over $A/\mathfrak{m}$ is indeed a vector space with action $(a + \mathfrak{m})(m + \mathfrak{m}M) = (am + \mathfrak{m}M)$ being well-defined.

Since $M/\mathfrak{m}M$ is a finite-dimensional $A/\mathfrak{m}$-vector space, choose $x_1, \dots, x_n \in M$ lifting a basis of it (so by Nakayama's lemma III they generate M). Let F be a free module with basis $e_1, e_2, ..., e_n$, and consider the homomorphism $f : F \to M$, $f(e_i) = x_i$. We'll show that f is in fact an isomorphism. Since $x_1, ..., x_n$ generate M mod $\mathfrak{m}M$, by Nakayama's lemma III, they generate M, and so f is surjective. Next, since M is projective, f splits, i.e., $F = P_0 \oplus P_1$ where $P_0 = \ker(f)$ and P_1 and $P_1 \cong M$ through f. We'll show $P_0 = 0$.

Take some $\sum a_i x_i \in \ker(f)$ so that $\sum a_i x_i = 0$. Then reducing to the mod, we have $\sum a_i x_i + \mathfrak{m}M = 0 + \mathfrak{m}M = \sum (a_i \mathfrak{m}) \cdot (x_i + \mathfrak{m}M) = 0 + \mathfrak{m}M$. Since the $\overline{x}_i$ form a basis for $M/\mathfrak{m}M$ over $A/\mathfrak{m}A$, it must be that $a_i \in \mathfrak{m}$. Thus, we have $\sum a_i e_i \in \mathfrak{m}F$. Therefore:

$$\ker(f) \subseteq \mathfrak{m}F = \mathfrak{m}\ker(f) \oplus \mathfrak{m}M$$

Thus, $\mathfrak{m}\ker(f) = \ker(f)$. Since $\ker(f)$ is the direct summand of finitely generated modules, it is itself finitely generated. Thus, by Nakayama's lemma, $\ker(f) = 0$, and so f is an isomorphism! Therefore, we have $F \cong M$, and so M is free!

Notice that this immediately implies that the $x_1, ..., x_n$ are linealy independent since they form a basis for M over A (since f is an isomorphism and $f(e_i) = x_i$). Finally, if we extend $\overline{x}_1, ..\overline{x}_r$ to $\overline{x}_1, ..., \overline{x}_r, ...\overline{x}_n$ to be a basis for $M/\mathfrak{m}M$, then we just showed they will indeed form a basis for M.

A cool result of this theorem is a way to show some properties f based on f on the respective local modules. Let M be module over a local ring A. Then there is an associated module $M(\mathfrak{m}) = M/\mathfrak{m}M$

Then $\_\_\_(\mathfrak{m})$ induces a functor mapping M to $M(\mathfrak{m})$ and

$$f : M \to N \quad \mapsto \quad f_{(\mathfrak{m})} : M(\mathfrak{m}) \to N(\mathfrak{m})$$

Notice that if f is surjective, then $f_{(\mathfrak{m})}$ is surjective. The magic of Nakayama's lemma is that the converse is true too!

Proposition 23.4.31: Local Homomorphisms

Let M, N be finitely generated modules over a local ring A and $f : M \to N$ be an A-module homomorphism. Then:

1. f is surjective if and only if $f_{(\mathfrak{m})}$ is surjective
2. Assume f is injective. Then if $f_{(\mathfrak{m})}$ is surjective, f is an isomorphism
3. If M and N are free and $f_{(\mathfrak{m})}$ is injective, then f is injective; and f is an isomorphism if and only if $f_{(\mathfrak{m})}$ is an isomorphism

***Proof*:**

1. Let $f : M \to N$ be an R-module homomorphism and let the inducd $f_{(m)}$ be surjective so that $f(M \bmod \mathfrak{m}M) = N \bmod \mathfrak{m}N$. Since N is finitely generated, so is any quotient of it, and so $N \bmod \mathfrak{m}N$ is finitely generated by $\{\overline{x}_1, \overline{x}_2, ..., \overline{x}_n\}$ By Nakayama's lemma, $\{x_1, x_2, ..., x_n\}$ generated N. Since $x_1, ..., x_n \in f(M)$, f is also surjective

   More consisely, since $f_{(\mathfrak{m})}$ is surjective, $N = \text{im}(f) + \mathfrak{m}N$. Thus, by Nakayama's lemma, $N = \text{im}(f)$.

2. By part (1), surjectivity of $f_{(\mathfrak{m})}$ makes f surjective; together with the assumed injectivity of f, this makes f an isomorphism.

3. Assume M and N are free and $f_{(\mathfrak{m})}M/\mathfrak{m}M \to N/\mathfrak{m}N$ is injective. Since M is free, let $\{x_1, ..., x_n\} \subseteq M$ so that $\{\overline{x}_1, ..., \overline{x}_n\}$ is a basis for $M/\mathfrak{m}M$. Then by theorem 23.4.30, $x_1, ..., x_n$ forms a basis for M. Suppose now that $f(a_1x_1 + \cdots + a_nx_n) = 0$. By definition of $f_{(\mathfrak{m})}$, $f_{(\mathfrak{m})}(\overline{a}_1\overline{x}_1 + \cdots + \overline{a}_n\overline{x}_n) = \overline{0}$. Since $f_{(\mathfrak{m})}$ is injective, we have $\overline{a}_1\overline{x}_1 + \cdots + \overline{a}_n\overline{x}_n = 0$, and since $\overline{x}_1, ..., \overline{x}_n$ is a basis, we have that the coefficients are all $\overline{0}$, that is $a_i \in \mathfrak{m}$.

   To show that $\ker(f) = 0$, we'll use Nakayama's lemma, so it suffices to show that $\ker(f) \subseteq \mathfrak{m}\ker(f)$. By what we've just shown, we have $\ker(f) \subseteq \mathfrak{m}M$. If $f_{(\mathfrak{m})}$ were surjective, then this proof is rather quick. By part (1), we have that f is surjective. Thus the following short exact sequence splits:

   $$0 \to \ker(f) \to M \to N \to 0$$

   since N is free, So $M \cong \ker(f) \oplus N$, and $\ker(f) \subseteq \mathfrak{m}M \cong \ker(f) \oplus N$, and so $\ker(f) \subseteq \mathfrak{m}\ker(f)$, and so by Nakayama's lemma $\ker(f) = 0$.

   If $f_{(\mathfrak{m})}$ is not surjective, then we need to do a little more work. We'll take advantage of the basis lifting property we proved in the previous theorem. Let $\{\overline{x}_1, ..., \overline{x}_n\}$ be a basis for $M/\mathfrak{m}M$ over $A/\mathfrak{m}A$ which induces a basis $\{x_1, .., x_n\}$ for M over A.

$\langle f_{(\mathfrak{m})}(\overline{x}_1), ..., f_{(\mathfrak{m})}(\overline{x}_n)\rangle = \text{im}(f_{(\mathfrak{m})})$ is a submodule of $N/\mathfrak{m}N$ which have that generating set as a basis (since $f_{(\mathfrak{m})}$ is injective). Thus, if:

$$\overline{a}_1 f_{(\mathfrak{m})}(\overline{x}_1) + \cdots + \overline{a}_n f_{(\mathfrak{m})}(\overline{x}_n) = 0$$

then since f is a homomorphism, we have

$$f_{(\mathfrak{m})}(\overline{a}_1\overline{x}_1 + \cdots + \overline{a}_1\overline{x}_1) = 0$$

and since these elements form a basis and f is injective, $\overline{a}_i = 0$ for all i. Now, extend this to a basis for $N/\mathfrak{m}N$ over $A/\mathfrak{m}A$:

$$\{f_{(\mathfrak{m})}(\overline{x}_1), ..., f_{(\mathfrak{m})}(\overline{x}_n), \overline{y}_1, ..., \overline{y}_m\}$$

Since N is free, this basis lifts to a basis for N over A, in particular:

$$\{f(x_1), ..., f(x_n), y_1, ..., y_m\}$$

since $f_{(\mathfrak{m})}(\overline{x}_i) = f_{(\mathfrak{m})}(x_i + \mathfrak{m}M) = f(x_i) = \mathfrak{m}N$. With this, we can now get our conclusion: since the $f(x_i)$ are linearly independent, we have that:

$$a_1 f(x_1) + \cdots + a_n f(x_n) = f(a_1x_1 + \cdots + a_nx_n) = 0$$

then it must be that the $a_i = 0$ for all the i, but then this is the condition for a function to be injective, and so f is injective, as we sought to show

Proposition 23.4.32: Projective Modules Determined by Their Reduction

Let P and P' be finitely generated projective modules over a local ring A. Then $P \cong P'$ if and only if $P/\mathfrak{m}P \cong P'/\mathfrak{m}P'$.

***Proof*:**
By theorem 23.4.30, P and P' are free, with ranks $\dim_{A/\mathfrak{m}}(P/\mathfrak{m}P)$ and $\dim_{A/\mathfrak{m}}(P'/\mathfrak{m}P')$; these ranks agree iff $P/\mathfrak{m}P \cong P'/\mathfrak{m}P'$, in which case $P \cong P'$. The converse is immediate.

Recall that if R is a finite integral domain, then it is necessarily a field (corollary 9.1.25). Naturally, this is not necessarily the case for infinite rings ($\mathbb{Z}$ being the prototypical example). We may thus take $Q = \text{Frac}(R)$, and consdier the R-module Q

Corollary 23.4.33: Infinite Generation Of Ring Of Fractions

Let R be an integral domain that is not a field, and $Q = \text{Frac}(R)$ its field of fractions. Then Q, considered as an R-module, is not finitely generated.

***Proof*:**
Suppose Q were finitely generated as an R-module. Since R is not a field, choose a nonzero maximal ideal $\mathfrak{m}$. As every nonzero element of R is already a unit in Q, we have $Q_{\mathfrak{m}} = Q$, still finitely generated over the local ring $R_{\mathfrak{m}}$. For any $0 \neq a \in \mathfrak{m}$ and $q \in Q$ we have $q = a(q/a)$ with

$a \in \mathfrak{m}R_{\mathfrak{m}}$, so $\mathfrak{m}R_{\mathfrak{m}} \cdot Q = Q$. By lemma 23.4.27, $Q = 0$, contradicting $R \subseteq Q$.

Recall that when dealing with infinite functions, we may have some strange results, like a surjection $f : M \to M$ that is not an injection. Nakayama's lemma insures we get some sanity here:

Corollary 23.4.34: Surjective Implies Injective

Let M be a finitely generated R-module. Then if $f : M \to M$ is a surjective R-module homomorphism, it is injecive.

Note that no Noetherian assumption on R or M is required.

***Proof*:**
Make M a module over $R[x]$ by letting x act as f, i.e. $x \cdot m = f(m)$. Surjectivity of f gives $(x)M = xM = M$. Since M is a finitely generated $R[x]$-module with $(x)M = M$, the determinant trick (the Cayley-Hamilton form of Nakayama's lemma) yields $q(x) \in (x)$ with $(1 - q(x))M = 0$. Write $q(x) = x\,h(x)$. For $m \in \ker f$ we have $xm = f(m) = 0$, hence $m = q(x)m = h(x)(xm) = 0$. Thus $\ker f = 0$ and f is injective.

24

Primary Decomposition and Associated Primes

The integer n factors uniquely into prime powers, and the ideal $(n) \subseteq \mathbb{Z}$ inherits the factorization: writing $n = p_1^{a_1} \cdots p_k^{a_k}$ gives $(n) = (p_1^{a_1}) \cap \cdots \cap (p_k^{a_k})$, a finite intersection of ideals each built from a single prime. Over a general ring the unique factorization of elements fails, and the factorization of ideals into primes fails with it. What survives, in every Noetherian ring, is the weaker statement that an ideal is a finite intersection of *primary* ideals, the ideal-theoretic analogue of a prime power. This is the content of the Lasker-Noether theorem, and it is the decomposition facet of the finiteness thread: the same Noetherian hypothesis that produced Hilbert's basis theorem produces here the existence and finiteness that make the theory run.

The primes attached to such a decomposition are not artifacts of the factorization but invariants of the ideal, its *associated primes*. They split into two kinds. The minimal ones record the irreducible components and are uniquely determined; the *embedded* ones sit inside those components and can shift from one decomposition to another. Read through the correspondence between primes and points, the minimal primes will become the irreducible components of a variety and the embedded primes its embedded components, a reading developed downstream in [27]. Here the account is purely ring-theoretic, and its natural home is a module: the associated primes $\mathrm{Ass}(M)$ and the support $\mathrm{Supp}(M)$ are the invariants the later chapters, and the geometry and homological volumes, reach back for.

24.1 Primary Ideals

A prime ideal $\mathfrak{p}$ is one for which $xy \in \mathfrak{p}$ forces $x \in \mathfrak{p}$ or $y \in \mathfrak{p}$. The primary ideals relax the second alternative to a power.

Definition 24.1.1: Primary Ideal

A proper ideal $\mathfrak{q} \subsetneq R$ is *primary* if, whenever $xy \in \mathfrak{q}$ with $x \notin \mathfrak{q}$, some power y^n lies in $\mathfrak{q}$.

Passing to the quotient makes the condition transparent: $\mathfrak{q}$ is primary exactly when $R/\mathfrak{q} \neq 0$ and every zero-divisor of $R/\mathfrak{q}$ is nilpotent. Indeed $xy \in \mathfrak{q}$ with $x \notin \mathfrak{q}$ says that $\bar{y}$ kills the nonzero element $\bar{x}$ of $R/\mathfrak{q}$, and $y^n \in \mathfrak{q}$ says $\bar{y}$ is nilpotent.

Proposition 24.1.2: The Radical of a Primary Ideal

If $\mathfrak{q}$ is primary then $\mathfrak{p} = \sqrt{\mathfrak{q}}$ is prime, and it is the smallest prime containing $\mathfrak{q}$. One says $\mathfrak{q}$ is $\mathfrak{p}$-*primary*.

***Proof*:**
The radical of any ideal is the intersection of the primes containing it, so $\sqrt{\mathfrak{q}}$ is contained in every prime over $\mathfrak{q}$; it suffices to show it is prime. Suppose $xy \in \sqrt{\mathfrak{q}}$, say $(xy)^m \in \mathfrak{q}$, and $x \notin \sqrt{\mathfrak{q}}$. Then no power of x lies in $\mathfrak{q}$, so from $x^m y^m \in \mathfrak{q}$ and primarity applied to $x^m \cdot y^m$ we get $(y^m)^k \in \mathfrak{q}$ for some k, whence $y \in \sqrt{\mathfrak{q}}$.

The radical being prime does *not*, by itself, make an ideal primary; the converse needs the radical to be maximal.

Lemma 24.1.3: Ideals with Maximal Radical are Primary

If $\sqrt{\mathfrak{q}} = \mathfrak{m}$ is maximal, then $\mathfrak{q}$ is primary. In particular every power $\mathfrak{m}^n$ of a maximal ideal is $\mathfrak{m}$-primary.

***Proof*:**
In $R/\mathfrak{q}$ the image of $\mathfrak{m}$ is the nilradical, since $\mathfrak{m} = \sqrt{\mathfrak{q}}$, and it is also maximal. Thus $R/\mathfrak{q}$ has a single prime ideal, so every element is either a unit or nilpotent, and every zero-divisor is nilpotent. For the last claim, $\sqrt{\mathfrak{m}^n} = \mathfrak{m}$.

Example 24.1.4: Primary and Non-Primary Ideals

1. In $\mathbb{Z}$ the primary ideals are (0) and the prime powers (p^n); this is the factorization $(n) = \bigcap (p_i^{a_i})$ read one prime at a time.
2. In $R = k[x, y]$ the ideal $\mathfrak{q} = (x, y^2)$ is primary: $R/\mathfrak{q} \cong k[y]/(y^2)$, whose only zero-divisors are the multiples of $\bar{y}$, all nilpotent. Its radical is (x, y), and $(x, y)^2 \subsetneq \mathfrak{q} \subsetneq (x, y)$, so a primary ideal need not be a power of its radical prime.
3. A power of a prime need not be primary. In $R = k[x, y, z]/(xy - z^2)$ let $\mathfrak{p} = (\bar{x}, \bar{z})$, which is prime because $R/\mathfrak{p} \cong k[y]$ is a domain. Then $\bar{x}\bar{y} = \bar{z}^2 \in \mathfrak{p}^2$, yet $\bar{x} \notin \mathfrak{p}^2$ and $\bar{y} \notin \sqrt{\mathfrak{p}^2} = \mathfrak{p}$, so $\mathfrak{p}^2$ is not primary although its radical $\mathfrak{p}$ is prime.

24.2 Primary Decomposition

Definition 24.2.1: Primary Decomposition

A *primary decomposition* of an ideal I is an expression

$$I = \mathfrak{q}_1 \cap \mathfrak{q}_2 \cap \cdots \cap \mathfrak{q}_n$$

with each $\mathfrak{q}_i$ primary. It is *minimal* if the radicals $\mathfrak{p}_i = \sqrt{\mathfrak{q}_i}$ are pairwise distinct and no $\mathfrak{q}_i$ contains the intersection of the others. An ideal admitting a primary decomposition is called *decomposable.*

Any primary decomposition can be trimmed to a minimal one, once we know that primary ideals with a common radical combine.

Lemma 24.2.2: Intersections of $\mathfrak{p}$-Primary Ideals

A finite intersection $\mathfrak{q} = \mathfrak{q}_1 \cap \cdots \cap \mathfrak{q}_m$ of $\mathfrak{p}$-primary ideals (same radical $\mathfrak{p}$) is again $\mathfrak{p}$-primary.

***Proof*:**
Radicals commute with finite intersections, so $\sqrt{\mathfrak{q}} = \bigcap \sqrt{\mathfrak{q}_i} = \mathfrak{p}$. Suppose $xy \in \mathfrak{q}$ with $y \notin \sqrt{\mathfrak{q}} = \mathfrak{p}$. For each i, $xy \in \mathfrak{q}_i$ and $y \notin \sqrt{\mathfrak{q}_i}$, so no power of y lies in $\mathfrak{q}_i$; primarity of $\mathfrak{q}_i$ then gives $x \in \mathfrak{q}_i$. Hence $x \in \mathfrak{q}$, and $\mathfrak{q}$ is primary.

Given any primary decomposition, group the $\mathfrak{q}_i$ by their radical and replace each group by its intersection, which lemma 24.2.2 keeps primary; then discard any term containing the intersection of the rest. The result is minimal, so *every decomposable ideal has a minimal primary decomposition.* The point of the next two sections is that the data surviving this trimming is intrinsic to I.

24.3 Associated Primes

The invariants are cleanest for modules, where they will also serve the later theory of depth and support.

Definition 24.3.1: Associated Prime

Let M be an R-module. A prime $\mathfrak{p}$ is an *associated prime* of M if $\mathfrak{p} = \operatorname{Ann}(x)$ for some $x \in M$; equivalently, if $R/\mathfrak{p}$ embeds in M. The set of associated primes is written $\operatorname{Ass}(M)$. The associated primes of an ideal I mean those of the module R/I.

The two forms match because a copy of $R/\mathfrak{p}$ inside M is generated by an element whose annihilator is $\mathfrak{p}$, and conversely $\operatorname{Ann}(x) = \mathfrak{p}$ makes $r \mapsto rx$ an embedding $R/\mathfrak{p} \hookrightarrow M$.

Lemma 24.3.2: Zero-Divisors and Annihilators

Let R be Noetherian and $M \neq 0$ a finitely generated module. A maximal element of the family $\{\mathrm{Ann}(x) : 0 \neq x \in M\}$ is prime, so it lies in $\mathrm{Ass}(M)$; in particular $\mathrm{Ass}(M) \neq \emptyset$.

***Proof*:**
The family is nonempty and, R being Noetherian, has a maximal member $\mathfrak{p} = \mathrm{Ann}(x)$. It is proper since $x \neq 0$. If $ab \in \mathfrak{p}$ with $b \notin \mathfrak{p}$, then $bx \neq 0$ and $\mathrm{Ann}(bx) \supseteq \mathfrak{p} + (a) \supseteq \mathfrak{p}$; maximality forces $\mathrm{Ann}(bx) = \mathfrak{p}$, so $a \in \mathrm{Ann}(bx) = \mathfrak{p}$. Thus $\mathfrak{p}$ is prime.

The associated primes are exactly the radicals appearing in a minimal decomposition.

Theorem 24.3.3: First Uniqueness Theorem

Let R be Noetherian and let I have a minimal primary decomposition $I = \mathfrak{q}_1 \cap \cdots \cap \mathfrak{q}_n$ with radicals $\mathfrak{p}_i = \sqrt{\mathfrak{q}_i}$. Then

$$\{\mathfrak{p}_1, \ldots, \mathfrak{p}_n\} = \mathrm{Ass}(R/I),$$

and in particular the set of radicals does not depend on the chosen minimal decomposition.

***Proof*:**
Work in R/I; annihilators there are the ideals $(I : x)/I$ for $x \in R$, and $\sqrt{(I : x)}$ measures which components miss x. For $x \in R$ one computes $(I : x) = \bigcap_i (\mathfrak{q}_i : x)$, and $(\mathfrak{q}_i : x) = R$ when $x \in \mathfrak{q}_i$ while $\sqrt{(\mathfrak{q}_i : x)} = \mathfrak{p}_i$ when $x \notin \mathfrak{q}_i$ (primarity: if $y(x) \in \mathfrak{q}_i$ with $x \notin \mathfrak{q}_i$ then $y \in \mathfrak{p}_i$). Hence

$$\sqrt{(I : x)} = \bigcap_{x \notin \mathfrak{q}_i} \mathfrak{p}_i$$

If this radical is itself prime it must equal some $\mathfrak{p}_i$, since a prime containing a finite intersection contains one of the factors and each $\mathfrak{p}_i$ here contains the intersection. Thus every prime of the form $\sqrt{(I : x)}$ is one of the $\mathfrak{p}_i$. Conversely, minimality provides for each i an element x_i lying in every $\mathfrak{q}_j$ with $j \neq i$ but not in $\mathfrak{q}_i$, and then $\sqrt{(I : x_i)} = \mathfrak{p}_i$. The primes among $\sqrt{(I : x)}$ are therefore exactly $\mathfrak{p}_1, \ldots, \mathfrak{p}_n$. For $\mathrm{Ass}(R/I) \subseteq \{\mathfrak{p}_i\}$: if $\mathfrak{p} \in \mathrm{Ass}(R/I)$ then $\mathfrak{p} = (I : x)/I$ is an annihilator, and being prime it equals its radical $\sqrt{(I : x)}$, hence some $\mathfrak{p}_i$. For the reverse inclusion, fix i and use minimality to pick $x_i \in \bigcap_{j \neq i} \mathfrak{q}_j$ with $x_i \notin \mathfrak{q}_i$; then $(I : x_i) = (\mathfrak{q}_i : x_i)$, which is $\mathfrak{p}_i$-primary (for $x \notin \mathfrak{q}$ with $\mathfrak{q}$ being $\mathfrak{p}$-primary, $(\mathfrak{q} : x)$ lies between $\mathfrak{q}$ and $\mathfrak{p}$, and $ab \in (\mathfrak{q} : x)$ with $a \notin (\mathfrak{q} : x)$ gives $(ax)b \in \mathfrak{q}$, $ax \notin \mathfrak{q}$, so $b \in \mathfrak{p}$). Thus the cyclic submodule $R\bar{x}_i \cong R/(\mathfrak{q}_i : x_i)$ of R/I has all its zero-divisors in $\mathfrak{p}_i$; a maximal annihilator in it is prime (lemma 24.3.2), lies in $\mathfrak{p}_i$, and, containing $(\mathfrak{q}_i : x_i)$, contains $\sqrt{(\mathfrak{q}_i : x_i)} = \mathfrak{p}_i$, so it equals $\mathfrak{p}_i$. Since annihilators in a submodule are annihilators in R/I, this gives $\mathfrak{p}_i \in \mathrm{Ass}(R/I)$, and hence equality.

24.4 The Lasker-Noether Theorem

Existence rests on two Noetherian facts: that ideals are finite intersections of irreducible ones, and that over a Noetherian ring irreducible means primary. Call a proper ideal *irreducible* if it is not the intersection of two strictly larger ideals.

Lemma 24.4.1: Irreducible Implies Primary

In a Noetherian ring every irreducible ideal is primary.

***Proof*:**
Passing to R/I, it suffices to show that the zero ideal, if irreducible, is primary; that is, that a zero-divisor of a ring in which (0) is irreducible is nilpotent. Let $xy = 0$ with $y \neq 0$. The annihilators $\mathrm{Ann}(x) \subseteq \mathrm{Ann}(x^2) \subseteq \cdots$ stabilize, say $\mathrm{Ann}(x^n) = \mathrm{Ann}(x^{n+1})$. Then $(x^n) \cap (y) = 0$: if $ax^n = by$ then $ax^{n+1} = bxy = 0$, so $a \in \mathrm{Ann}(x^{n+1}) = \mathrm{Ann}(x^n)$ and $ax^n = 0$. Since (0) is irreducible and $(y) \neq 0$, we must have $(x^n) = 0$, so x is nilpotent.

Theorem 24.4.2: Lasker-Noether

In a Noetherian ring every proper ideal is a finite intersection of irreducible ideals, and hence is decomposable. Consequently every proper ideal has a minimal primary decomposition.

***Proof*:**
Let Σ be the set of ideals that are *not* finite intersections of irreducible ideals. If $\Sigma \neq \emptyset$ it has a maximal member I by the ascending chain condition. Now I is not irreducible (an irreducible ideal is such an intersection, of length one), so $I = J \cap K$ with $J, K \supsetneq I$. By maximality J and K are each finite intersections of irreducibles, hence so is I, contradicting $I \in \Sigma$. Thus $\Sigma = \emptyset$. Each irreducible ideal is primary by lemma 24.4.1, giving a primary decomposition, which is trimmed to a minimal one by grouping common radicals and discarding redundant terms.

The finiteness of the associated primes is now immediate.

Corollary 24.4.3: Finiteness of Associated Primes

Over a Noetherian ring, $\mathrm{Ass}(R/I)$ is finite for every ideal I; more generally $\mathrm{Ass}(M)$ is finite for every finitely generated module M.

***Proof*:**
The finitely generated case includes every R/I, so we prove it. Two standard facts suffice. First, a short exact sequence $0 \to M' \to M \to M'' \to 0$ gives $\mathrm{Ass}(M) \subseteq \mathrm{Ass}(M') \cup \mathrm{Ass}(M'')$: if $\mathfrak{p} = \mathrm{Ann}(x)$ then $Rx \cong R/\mathfrak{p}$, and either $Rx \cap M' \neq 0$, so a nonzero element of the domain $R/\mathfrak{p}$ lies in M' and (being torsion-free) still has annihilator $\mathfrak{p}$, whence $\mathfrak{p} \in \mathrm{Ass}(M')$; or $Rx \cap M' = 0$, so Rx injects into M'' and $\mathfrak{p} \in \mathrm{Ass}(M'')$. Second, applying lemma 24.3.2 repeatedly to split off a submodule $\cong R/\mathfrak{p}$ and pass to the quotient produces, by the ascending chain condition, a finite filtration $0 = M_0 \subsetneq \cdots \subsetneq M_r = M$ with $M_j/M_{j-1} \cong R/\mathfrak{p}_j$. Subadditivity along it gives $\mathrm{Ass}(M) \subseteq \{\mathfrak{p}_1, \ldots, \mathfrak{p}_r\}$, a finite set; for $M = R/I$ with I proper, theorem 24.3.3 identifies it precisely as the radicals $\{\mathfrak{p}_i\}$.

24.5 Support and the Isolated Components

The associated primes live inside a coarser invariant, the support, whose minimal members are the geometrically visible components.

Two consequences are used constantly and are recorded here, the first because it is invoked by name later in this book and in [15] without ever having been stated.

Proposition 24.5.1: Prime Avoidance

Let I be an ideal of a ring R and $\mathfrak{p}_1, \dots, \mathfrak{p}_n$ prime ideals. If $I \subseteq \mathfrak{p}_1 \cup \cdots \cup \mathfrak{p}_n$ then $I \subseteq \mathfrak{p}_i$ for some i. Equivalently, if I is contained in no $\mathfrak{p}_i$, then I contains an element lying in no $\mathfrak{p}_i$.

***Proof*:**
Induct on n, the case $n = 1$ being trivial. Discarding redundant terms, assume the covering is irredundant, so that for each i there is $a_i \in I$ with $a_i \notin \mathfrak{p}_j$ for $j \neq i$; if some a_i also avoided $\mathfrak{p}_i$ we would be done, so assume $a_i \in \mathfrak{p}_i$ for every i. Consider

$$b = a_1 + a_2 a_3 \cdots a_n$$

It lies in I. If $b \in \mathfrak{p}_1$ then, since $a_1 \in \mathfrak{p}_1$, the product $a_2 \cdots a_n \in \mathfrak{p}_1$, so some $a_j \in \mathfrak{p}_1$ with $j \geq 2$, contradicting the choice of a_j. If $b \in \mathfrak{p}_i$ for some $i \geq 2$ then, since $a_2 \cdots a_n \in \mathfrak{p}_i$, we get $a_1 \in \mathfrak{p}_i$, again a contradiction. So b lies in I and in no $\mathfrak{p}_i$, contradicting the assumed covering; hence the covering was not irredundant and induction finishes the proof.

Corollary 24.5.2: The Zero-Divisors Are the Union of the Associated Primes

Let R be Noetherian and $M \neq 0$ a finitely generated module. Then

$$\{ z \in R : z \text{ is a zero-divisor on } M \} = \bigcup_{\mathfrak{p} \in \operatorname{Ass}(M)} \mathfrak{p}$$

In particular an element lying outside every associated prime of M is a non-zero divisor on M.

***Proof*:**
If $\mathfrak{p} \in \operatorname{Ass}(M)$ then $\mathfrak{p} = \operatorname{Ann}(x)$ for some $x \neq 0$, so every element of $\mathfrak{p}$ kills x and is a zero-divisor. Conversely let $zx = 0$ with $x \neq 0$, so $z \in \operatorname{Ann}(x)$. The family of annihilators $\operatorname{Ann}(y)$ for $0 \neq y \in M$ that contain $\operatorname{Ann}(x)$ is nonempty and, R being Noetherian, has a maximal element $\operatorname{Ann}(y_0)$. It is maximal in the whole family as well: an annihilator strictly containing it also contains $\operatorname{Ann}(x)$, so lies in the subfamily. By lemma 24.3.2 it is therefore prime, so $\operatorname{Ann}(y_0) \in \operatorname{Ass}(M)$, and it contains z.

Definition 24.5.3: Support

The *support* of an R-module M is $\operatorname{Supp}(M) = \{\mathfrak{p} \in \operatorname{Spec} R : M_{\mathfrak{p}} \neq 0\}$.

Proposition 24.5.4: Support of a Finitely Generated Module

If M is finitely generated then $\operatorname{Supp}(M) = V(\operatorname{Ann} M)$. In particular $\operatorname{Supp}(R/I) = V(I)$.

***Proof*:**
For finitely generated $M = (x_1, \ldots, x_m)$, the localization $M_{\mathfrak{p}}$ vanishes iff each generator dies, that is iff $\operatorname{Ann}(x_i) \not\subseteq \mathfrak{p}$ for all i. Since $\mathfrak{p}$ is prime, a containment $\bigcap_i \operatorname{Ann}(x_i) \subseteq \mathfrak{p}$ would force some $\operatorname{Ann}(x_i) \subseteq \mathfrak{p}$ (a prime containing a finite intersection contains a factor); so $M_{\mathfrak{p}} = 0$ iff $\operatorname{Ann}(M) = \bigcap_i \operatorname{Ann}(x_i) \not\subseteq \mathfrak{p}$. Thus $M_{\mathfrak{p}} \neq 0$ iff $\operatorname{Ann}(M) \subseteq \mathfrak{p}$, which is $\mathfrak{p} \in V(\operatorname{Ann} M)$.

Proposition 24.5.5: Associated Primes Sit in the Support

For finitely generated M over a Noetherian ring, $\operatorname{Ass}(M) \subseteq \operatorname{Supp}(M)$, and the two sets have the same minimal elements. For an ideal I these common minimal elements are the minimal primes over I.

***Proof*:**
If $\mathfrak{p} = \operatorname{Ann}(x) \in \operatorname{Ass}(M)$ then $R/\mathfrak{p} \hookrightarrow M$ localizes to $(R/\mathfrak{p})_{\mathfrak{p}} = \kappa(\mathfrak{p}) \neq 0 \hookrightarrow M_{\mathfrak{p}}$, so $\mathfrak{p} \in \operatorname{Supp}(M)$. For the minimal elements: $\operatorname{Supp}(M) = V(\operatorname{Ann} M)$ by proposition 24.5.4, whose minimal elements are the minimal primes over $\operatorname{Ann}(M)$. Each such minimal prime $\mathfrak{p}$ has $M_{\mathfrak{p}} \neq 0$ finitely generated over the Noetherian local ring $R_{\mathfrak{p}}$, so $\operatorname{Ass}(M_{\mathfrak{p}}) \neq \emptyset$ by lemma 24.3.2; but $\mathfrak{p}$ is minimal in $\operatorname{Supp}(M)$, so $\mathfrak{p}R_{\mathfrak{p}}$ is the only prime of $R_{\mathfrak{p}}$ in the support, forcing $\operatorname{Ass}(M_{\mathfrak{p}}) = \{\mathfrak{p}R_{\mathfrak{p}}\}$ and hence $\mathfrak{p} \in \operatorname{Ass}(M)$ (associated primes localize, as in corollary 24.6.2). So minimal elements of $\operatorname{Supp}(M)$ lie in $\operatorname{Ass}(M)$, and being minimal in the larger set they are minimal in $\operatorname{Ass}(M)$; the reverse inclusion of minimal elements is immediate from $\operatorname{Ass}(M) \subseteq \operatorname{Supp}(M)$.

The associated primes thus divide in two: the *isolated* (or minimal) primes, which are the minimal primes over I, and the *embedded* primes, which properly contain another associated prime. The isolated components are intrinsic; the embedded ones are not.

Theorem 24.5.6: Second Uniqueness Theorem

Let $I = \mathfrak{q}_1 \cap \cdots \cap \mathfrak{q}_n$ be a minimal primary decomposition with $\mathfrak{p}_i = \sqrt{\mathfrak{q}_i}$. If $\mathfrak{p}_i$ is isolated (minimal among the $\mathfrak{p}_j$), the component $\mathfrak{q}_i$ is uniquely determined by I: it is the contraction of $IR_{\mathfrak{p}_i}$ to R. Embedded components need not be unique.

***Proof*:**
Localize at an isolated prime $\mathfrak{p} = \mathfrak{p}_i$. For $j \neq i$, $\mathfrak{p}_j \not\subseteq \mathfrak{p}$ by isolation, so $\mathfrak{q}_j$ meets $R \setminus \mathfrak{p}$ and $\mathfrak{q}_j R_{\mathfrak{p}} = R_{\mathfrak{p}}$; for the index i, $\mathfrak{q}_i R_{\mathfrak{p}}$ is $\mathfrak{p}R_{\mathfrak{p}}$-primary with contraction $\mathfrak{q}_i$ (see proposition 24.6.1). Intersecting, $IR_{\mathfrak{p}} = \mathfrak{q}_i R_{\mathfrak{p}}$, so $\mathfrak{q}_i = IR_{\mathfrak{p}} \cap R$ depends only on I and $\mathfrak{p}$. Non-uniqueness of embedded components is exhibited by example 24.5.7.

Example 24.5.7: Non-Uniqueness of Embedded Components
In $R = k[x,y]$ the ideal $I = (x^2, xy)$ has radical (x) and the two minimal primary decompositions

$$I = (x) \cap (x,y)^2 = (x) \cap (x^2, y),$$

both with radicals (x) and (x,y). The isolated (x)-component is (x) in each, as theorem 24.5.6 requires. The embedded (x,y)-component is $(x,y)^2 = (x^2, xy, y^2)$ in the first and (x^2, y) in the second, so it is genuinely not determined by I.

24.6 Localization and Primary Decomposition

The behavior under localization used above is the following, and it is what recovers the isolated components.

Proposition 24.6.1: Primary Ideals under Localization

Let $S \subseteq R$ be multiplicative and $\mathfrak{q}$ a $\mathfrak{p}$-primary ideal.

1. If $S \cap \mathfrak{p} \neq \emptyset$ then $S^{-1}\mathfrak{q} = S^{-1}R$.
2. If $S \cap \mathfrak{p} = \emptyset$ then $S^{-1}\mathfrak{q}$ is $S^{-1}\mathfrak{p}$-primary and its contraction to R is $\mathfrak{q}$.

***Proof*:**
If $s \in S \cap \mathfrak{p}$ then $s^n \in \mathfrak{q}$ for some n, and s^n is a unit in $S^{-1}R$, giving (1). For (2), assume $S \cap \mathfrak{p} = \emptyset$. Since localization is exact, $S^{-1}R/S^{-1}\mathfrak{q} \cong \bar{S}^{-1}(R/\mathfrak{q})$, the localization of $R/\mathfrak{q}$ at the image $\bar{S}$ of S. The zero-divisors of $R/\mathfrak{q}$ are nilpotent, hence lie in $\mathfrak{p}/\mathfrak{q}$; as $S \cap \mathfrak{p} = \emptyset$, the elements of $\bar{S}$ avoid $\mathfrak{p}/\mathfrak{q}$ and are non-zero-divisors of $R/\mathfrak{q}$. A zero-divisor of $\bar{S}^{-1}(R/\mathfrak{q})$ then has a zero-divisor of $R/\mathfrak{q}$ for numerator, hence is nilpotent; so $S^{-1}\mathfrak{q}$ is primary, with radical $S^{-1}\mathfrak{p}$. Its contraction is $\mathfrak{q}$: if $sx \in \mathfrak{q}$ with $s \in S$, then $s \notin \mathfrak{p} = \sqrt{\mathfrak{q}}$ forces $x \in \mathfrak{q}$ by primarity.

Corollary 24.6.2: Associated Primes Localize

For a multiplicative set S and finitely generated M over a Noetherian ring,

$$\operatorname{Ass}(S^{-1}M) = \{\, S^{-1}\mathfrak{p} : \mathfrak{p} \in \operatorname{Ass}(M),\ \mathfrak{p} \cap S = \emptyset \,\}$$

Consequently, if $I = \bigcap \mathfrak{q}_i$ is a minimal decomposition, localizing at an isolated prime $\mathfrak{p}_i$ deletes every component except $\mathfrak{q}_i$, recovering $\mathfrak{q}_i = IR_{\mathfrak{p}_i} \cap R$.

***Proof*:**
For $\supseteq$: an embedding $R/\mathfrak{p} \hookrightarrow M$ with $\mathfrak{p} \cap S = \emptyset$ localizes, by exactness (proposition 23.4.9), to $S^{-1}R/S^{-1}\mathfrak{p} \hookrightarrow S^{-1}M$, and $S^{-1}\mathfrak{p}$ is prime, so $S^{-1}\mathfrak{p} \in \operatorname{Ass}(S^{-1}M)$. For $\subseteq$: let $P = \operatorname{Ann}(x/s) \in \operatorname{Ass}(S^{-1}M)$ and $\mathfrak{p} = P \cap R$, a prime with $\mathfrak{p} \cap S = \emptyset$. As R is Noetherian, $\mathfrak{p} = (a_1, \ldots, a_k)$; each $a_j/1 \in P$ kills x/s, so $t_j a_j x = 0$ in M for some $t_j \in S$, and with $t = t_1 \cdots t_k$ we get $\mathfrak{p} \subseteq \operatorname{Ann}(tx)$. Conversely, if $r \in \operatorname{Ann}(tx)$ then $r \cdot (x/s) = rtx/(ts) = 0$, so $r \in P \cap R = \mathfrak{p}$; hence

$\mathfrak{p} = \mathrm{Ann}(tx) \in \mathrm{Ass}(M)$ and $P = S^{-1}\mathfrak{p}$. The recovery statement is proposition 24.6.1(2) applied to $S = R \setminus \mathfrak{p}_i$, exactly as in the proof of theorem 24.5.6.

Read geometrically, $\mathrm{Supp}(R/I) = V(I)$ decomposes into irreducible pieces indexed by the minimal primes over I, and the isolated primary components cut out those pieces with their natural scheme structure while the embedded primes mark where that structure thickens; this translation is the subject of [27], which cites the present chapter for the algebra.

25

Completion

The next type of construction we'll do is called a *completion*. These will be in the appropriate sense "stronger" then localizations, which will result in them being more restrictive, though also more powerful. A completion of a ring in many ways parallel's the completion of the rationals numbers $\mathbb{Q}$ to the real numbers $\mathbb{R}$. In fact, defining the appropriate metric, this process is a special case of the cauchy-sequence completion of the rational numbers to the reals. Namely, the completion of a ring is the equivalence class of all Cauchy-sequences which converge to the same point. Due to the topological background needed for this perspective, we shall relegate the equivalence of what shall be explicated in this section and Cauchy-sequences to appendix C.5.

There were rings that we have observed earlier that are completions of simpler rings; we will use completions to construct rings like the ring of *formal power series* $R[[x]]$ from $R[x]$, the ring $R[x_1, x_2, ...]$, the ring with *infinitely many monomial* from $R[x_1, ..., x_n]$, and the *p-adic integers* $\mathbb{Z}_p$ from $\mathbb{Z}$..

Completions will be handy for us in many ways. One important use that we will see when looking into algebraic geometry is looking at the "singularities" of schemes (or more particularly algebraic curves). Singularities will be points that don't be had as nicely as most other points, for example in the following image (which is the zero set of the equation $y^2 - x^2\,(x+1) = 0$) does not behave well at $(0, 0)$:

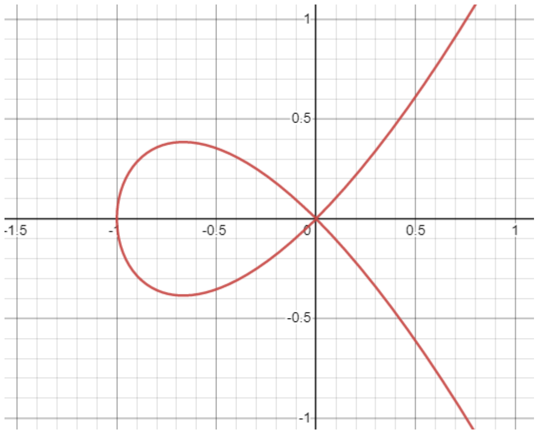


Figure 25.1: zero set of $y^2 - x^2(x+1) = 0$

This may seem similar to the geometric notion of localization, and this is indeed the case. Indeed, in the case of $\mathbb{Z}_{(p)}$ the p-adic integers are the completion of this ring, namely $\mathbb{Z}_p$ is the completion of $\mathbb{Z}_{(p)}$.

Completions shall also play a very important role in algebraic number theory. In this book, a more "classical" approach to algebraic number theory was taken, which shall give the reader great insights on the fundamental of algebraic number theory. However, a large part of *class number theory* shall rely on special completions of rings.

In this section, topological notion's will be introduced, and so an acquaintance of the material in appendix C is recommended. It should be noted that the topologies that will be discussed will be very different from that of Euclidean topologies, a new notion of size that is rather different from Euclidean distance shall be developed. The reader that is familiar with the construction of the real numbers using Cauchy sequences shall find it much easier to follow this section.

25.1 Constructing Completions

We will first define what is a "sequence". Let's say we have a collection of countable rings $R_1, R_2, ..., R_n$ and homomorphisms:

$$\cdots \xrightarrow{f_3} R_3 \xrightarrow{f_2} R_2 \xrightarrow{f_1} R_1$$

Then a sequence, or a *coherent sequence* is an element $(x_1, x_2, ...) \in \prod_{i=1}^{\infty} R_i$ where:

$$f_i(x_{i+1}) = x_i$$

If we take the special case of $I \subseteq R$ being an ideal and

$$\cdots \xrightarrow{f_3} R/(I^3) \xrightarrow{f_2} R/(I^2) \xrightarrow{f_1} R/I$$

then we will define a form of completion from these maps. For ease of notation, the sequence is usually written in the reverse order, as the reader shall understand once examples will be presented

Definition 25.1.1: Completion Of A Ring

Let R be a ring and $I \subseteq R$ be an ideal, and:

$$R/I \xleftarrow{f_1} R/(I^2) \xleftarrow{f_2} R/(I^3) \leftarrow \cdots$$

Then:

$$\hat{R} := \left\{ (a_1, a_2, ...,) \in \prod_{i=1}^{\infty} R_i \;\middle|\; f_i(a_{i+1}) = a_i \right\}$$

which is often denoted as:

$$\varprojlim R/I^n := \hat{R}$$

and is called the *limit* (or in older textbooks the *inverse limit* or *projective limit*). The collection $\{R/I^n\}$ is called the *inverse system*, with the function usually taken as implicit

A couple of remarks are in order:

1. There is a natural topology and a metric in which these elements can be thought of as Cauchy sequences and $\hat{R}$ is indeed the completion of R. If $\{R/\mathfrak{a}^n R\}$ is an inverse system with ideal $\mathfrak{a} \subseteq R$, and any $x \in R$, define a neighborhood basis (or local basis) $\{x + \mathfrak{a}^n R\}_{n=1}^{\infty}$. Then this forms a basis for a topology. This topology is called the $\mathfrak{a}$-*adic topology*, U is open in this topology if and only if for each $x \in U$, there exists a positive integer n such that

$$x + \mathfrak{a}^n R \subseteq U$$

   Then R is Hausdorff if and only if $\cap \mathfrak{a}^n R = 0$. If this is the case, define the metric $d(x, y) = 2^{-n}$ whenever $x \neq y$, where n is the largest integer with $x - y \in \mathfrak{a}^n R$ (equivalently $x - y \in \mathfrak{a}^n R \setminus \mathfrak{a}^{n+1} R$). The constant 2 does not matter, it can be replaced with any $C > 1$ and it will give the same topology.

   This exact same process can be repeated on an arbitrary R-module, where the inverse system would be $\{M/a^n M\}$, and for a given ideal $I \subseteq R$, we would define the I-adic topology on M.

2. This is a special case of a more general construction of a *limit* in the category of **Ring**. Namely, it is the limit of a special form of the diagram:

$$\cdots \to \bullet \to \bullet \to \bullet$$

   where each arrow has the coherence condition stated above.

3. The transition maps $R/\mathfrak{a}^{n+1} \to R/\mathfrak{a}^n$ of the systems built from powers of an ideal are surjective; such inverse systems are called *surjective systems*.

4. If R is a Noetherian integral domain and $I \subsetneq R$ is a proper ideal, then $\bigcap_n I^n = 0$, and hence $R \subseteq \hat{R}$. For those who shall read appendix C.4, this shows that inegral Noetherian domains are Hausdorff in the I-adic topology.

Example 25.1.2: Completion of Rings

1. Let $R = k[x]$ and $I = (x)$. Then:
$$\varprojlim (k[x]/(x)^n) = k[[x]]$$
To see this, consider:
$$\cdots k[x]/(x^3) \to k[x]/(x^2) \to k[x]/(x) = k$$
Notice that in each $k[x]/(x^k)$, can be thought of the collection of all polynomial of degree up to $k-1$. Hence, a coherent sequence $(a_1, a_2, ...)$ can be thought of as an "infinitely long polynomial" and would usually be denoted
$$\sum_k^\infty a_k x^k$$
Similarly, the reader may verify that $\widehat{k[x,y]} = k[[x,y]]$ with $I = (x,y)$
2. Let $R = \mathbb{Z}$ and $I = (10)$. Then $\varprojlim \mathbb{Z}/(10)^n$ is represented by all infinitely long numbers. Just like before, each $\mathbb{Z}/(10)^k$ can be thought of as as holding information up to the $k-1$th position. So if we take:
$$Z/(10) \leftarrow Z/(100) \leftarrow \mathbb{Z}/(1000) \leftarrow$$
and construct a coherent chain, we may chose elements like 5, 25, 425, ... and so on. The reader may see that this would represent an infinitely long number. The resulting ring is often denoted $\mathbb{Z}_{10}$ and is called the 10-adic integers. Any element of $\mathbb{Z}_{10}$ could formally be represented as
$$\sum_k^\infty a_k 10^k$$
3. Let $R_n = k[x_1, ..., x_n]$ with transition maps setting the last variable to 0. The limit $\varprojlim R_n$ strictly contains the polynomial ring $k[x_1, x_2, \dots]$: it also has elements of infinite support, such as the coherent sequence $(x_1,\, x_1 + x_2,\, \dots)$ representing $\sum_i x_i$

Completions are a way to construct rings. It would be nice if exactness interact nicely with it. If $\{A_n\}$, $\{B_n\}$, $\{C_n\}$ are three inverse systems where:

$$\begin{tikzcd}
0 \arrow[r] & A_{n+1} \arrow[r, "\psi"] \arrow[d, "\alpha"] & B_{n+1} \arrow[r, "\varphi"] \arrow[d, "\beta"] & C_{n+1} \arrow[r] \arrow[d, "\gamma"] & 0 \\
0 \arrow[r] & A'_n \arrow[r, "\psi'"] & B'_n \arrow[r, "\varphi'"] & C'_n \arrow[r] & 0
\end{tikzcd}$$

Then we'll denote it as $0 \to \{A_n\} \to \{B_n\} \to \{C_n\} \to 0$. Then we naturally get the induced homomorphisms:
$$0 \to \varprojlim A_n \to \varprojlim B_n \to \varprojlim C_n \to 0$$
however this is not always exact. The following proposition shows us when it is:

Proposition 25.1.3: Limits and Exactness for inverse Systems

Let $0 \to \{A_n\} \to \{B_n\} \to \{C_n\} \to 0$ be an exact sequence of inverse systems. Then:

$$0 \to \varprojlim A_n \to \varprojlim B_n \to \varprojlim C_n$$

is always exact. If $\{A_n\}$ is a surjective system, then:

$$0 \to \varprojlim A_n \to \varprojlim B_n \to \varprojlim C_n \to 0$$

is always exact

Hence, when working with completions induced by surjective systems, the completion functor $\widehat{(-)}$ preserves injective and surjective functions[1].

***Proof*:**
This is an application of Snakes lemma. Let $A = \prod_{n=1}^{\infty} An$, and

$$d^A : A \to A \qquad d^A(a_n) = a_n - f_{n+1}(a_{n+1})$$

Then by construction

$$\ker d^A = \varprojlim A_n$$

We may define B, C via d^B and d^C respectively. Now, as we have an exact sequence of inverse systems, we get the following diagram:

$$\begin{tikzcd}
0 \arrow[r] & A \arrow[r] \arrow[d, "d^A"] & B \arrow[r] \arrow[d, "d^B"] & C \arrow[r] \arrow[d, "d^C"] & 0 \\
0 \arrow[r] & A \arrow[r] & B \arrow[r] & C \arrow[r] & 0
\end{tikzcd}$$

Then by proposition 17.1.10 we get:

$$0 \to \ker d^A \to \ker d^B \to \ker d^C \to \text{coker} d^A \to \text{coker} d^B \to \text{coker} d^C \to 0$$

Finally, if the system is a surjective system, we can solve inuctively the equation:

$$x_n - f_{n+1}(x_{n+1}) = a_n$$

to show that d^A is surjective, completing the proof.

The following result shall be stated as it gives a very fascinating result, however for the full background the reader should read appendix C.5:

[1] In Part XI, this shall imply that the completion functor has trivial cohomology, while general inverse systems shall have non-trivial cohomology

Corollary 25.1.4: Completion of Group Exact Sequences

Let $0 \to G' \to G \xrightarrow{p} G'' \to 0$ be an exact sequence of groups. Let G have a topology given by the subgroups $\{G_n\}$, and induce the topology on G', G'' via the subgroups $\{G' \cap G_n\}$ and $\{p(G_n)\}$. Then:

$$0 \to \hat{G}' \to \hat{G} \to \hat{G}'' \to 0$$

is exact

***Proof*:**
Apply proposition 25.1.3 to

$$0 \to \frac{G'}{G' \cap G_n} \to \frac{G}{G_n} \to \frac{G''}{pG_n} \to 0$$

Corollary 25.1.5: Quotient in Completion

Let $\hat{G}_n \leqslant \hat{G}$. Then:

$$\hat{G}/\hat{G}_n \cong G/G_n$$

***Proof*:**
The quotient G/G_n carries the discrete topology (its induced filtration is eventually 0), so it equals its own completion. Applying the exact completion functor (proposition 25.1.3) to $0 \to G_n \to G \to G/G_n \to 0$ gives $0 \to \hat{G}_n \to \hat{G} \to G/G_n \to 0$, whence $\hat{G}/\hat{G}_n \cong G/G_n$.

As a consequence, in particular interest is the case of $\mathbb{Z}_{10}$ where

$$\mathbb{Z}_{10} \cong \mathbb{Z}_2 \oplus \mathbb{Z}_5$$

Hence $\mathbb{Z}_{10}$ is decomposeable to smaller rings. These rings are also in many ways nicer: notice that as a consequence of the above isomorphism that $\mathbb{Z}_{10}$ has zero divisors (any product of two non-trivial rings has zero-divisors)[2], showing that in general the completion may introduce zero-divisors, while $\mathbb{Z}_2$ and $\mathbb{Z}_5$ do not introduce zero-divisors. Furthermore, $\mathbb{Z}_p$ introduces more units (this was more evident with $k[[x]]$ where the units were explicitly found in proposition 9.2.11). For this reason, we would concentrate on the p-adic integers $\mathbb{Z}_p$. Any element of $\mathbb{Z}_p$ could be represented as

$$\sum_{k}^{\infty} a_k p^k$$

We may generalize the result for finding units in completions:

Lemma 25.1.6: Units In Completion

Let R be a ring and $\hat{R}$ a completion of R via p. Then if $u \in R/p^n$ is a unit, $u \in R/p^{2n}$ is a unit.

[2] Can you find two explicit elements?

***Proof*:**
Let u be a unit so $1 + us \in p^n$. Squaring, we get:

$$1 + u(2s + us^2) \in p^{2n}$$

Hence, it is a unit in R/p^{2n}.

The reader should use this to show that u is a unit in $\hat{R}$. This also generalizes proposition 9.2.11, for this directly shows that any power series with nonzero constant is a unit. In the case where $p = \mathfrak{m}$ is a maximal ideal (as in the case for p-adic integers and $k[x]$), we get the following result:

Lemma 25.1.7: Completion At Maximal Ideals

Let R be a ring and $\mathfrak{m}$ a maximal ideal. Then $\varprojlim R/\mathfrak{m}^n$ is a local ring, that is, it contains a single maximal ideal.

***Proof*:**
Let $z = (z_n) \in \varprojlim R/\mathfrak{m}^n$ have nonzero residue $z_1 \in R/\mathfrak{m}$. Since $\mathfrak{m}/\mathfrak{m}^n$ is the unique maximal ideal of $R/\mathfrak{m}^n$, each z_n has nonzero residue mod $\mathfrak{m}$ and so is a unit there, and the inverses z_n^{-1} are coherent; hence z is a unit. Thus the non-units are exactly $\ker(\varprojlim R/\mathfrak{m}^n \to R/\mathfrak{m})$, an ideal, so $\varprojlim R/\mathfrak{m}^n$ is local.

Thus, there is a natural map:

$$R_{\mathfrak{m}} \to \varprojlim R/\mathfrak{m}^n$$

where $R_{\mathfrak{m}}$ is the localization. Recall that in $R_{\mathfrak{m}}$, the elements are constructed by taking formal inverses, hence completion at a maximal ideal shows a "dual" way of creating units by finding them in some way "at infinity". Note that it is important that $\mathfrak{m}$ be maximal, for this will fail if an ideal is only prime (however, one may localize the ring at the prime ideal to make it maximal and then take the completion).

When working with localizaiton, we saw that a localization of an integral domain is an integral domain. This is no longer true for completion:

Example 25.1.8: Completion Looses Integral Domain
Let

$$R = \frac{k[x, y]}{(y^2 - x^3 - x^2)}$$

with k a field of characteristic 0. Then completing this ring, since completion is exact, we get:

$$\hat{R} = \frac{k[[x, y]]}{(y^2 - x^3 - x^2)} \cong \frac{k[[x, y]]}{(y - x\sqrt{1+x})(y + x\sqrt{1+x})}$$

where $\sqrt{1+x}$ is the short-hand for the power-series associated to $\sqrt{1+x}$. Then it is easy to see that this ring has zero-divisors. This shall eventually be made rather clear when a geometric perspective is taking. The ring R shall in a concrete sense have the curve $y^2 = x^3 + x^2$ associated to it:

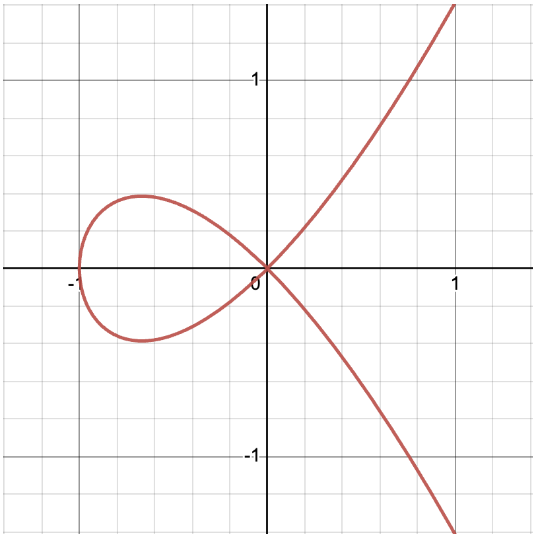


Figure 25.2: $y^2 = x^3 + x^2$

This curve shall be shown to be *irreducible* in a concrete way. The localization shall represent this curve minus some finite amount of points. The completion at (x, y) shall represent "zooming in" at the point (x, y), at which point geometric object shall roughly correspond to:

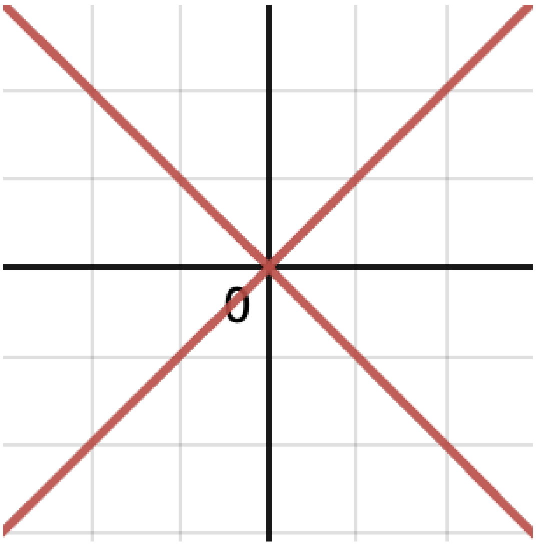


Figure 25.3: Approximate Geometric picture corresponding to $\varprojlim R/(x, y)^n$

Then this shall be shown to be decomposable in a rigorous sense into two components (the two strokes of the "x" in the above picture).

Definition 25.1.9: Separated and Complete

Let M be an R-module with the $\mathfrak{a}$-adic filtration, and let $\iota\colon M \to \hat{M} = \varprojlim M/\mathfrak{a}^n M$ be the natural map to its completion. Then M is

- *separated* if ι is injective, equivalently $\bigcap_n \mathfrak{a}^n M = 0$;
- *complete* if ι is surjective;

and *separated and complete* (usually abbreviated to just *complete*) if ι is an isomorphism, i.e. $M \cong \hat{M}$. A ring R is complete for the $\mathfrak{a}$-adic topology if it is so as a module over itself.

The two conditions genuinely differ: over a non-Noetherian ring a module can be complete without being separated. The completion $\hat{M}$ is always separated and complete, and it is the universal such object.

Proposition 25.1.10: Universal Property of Completion

Let R carry the $\mathfrak{a}$-adic topology with completion $\iota\colon R \to \hat{R}$. Then $\hat{R}$ is separated and complete, and ι is universal among continuous homomorphisms into separated complete rings: for every separated complete ring S and continuous homomorphism $f\colon R \to S$ there is a unique continuous homomorphism $\hat{f}\colon \hat{R} \to S$ with $\hat{f} \circ \iota = f$.

***Proof*:**
Continuity of f gives, after reindexing, $f(\mathfrak{a}^n) \subseteq \mathfrak{b}^n$ for a defining ideal $\mathfrak{b}$ of S, so f induces compatible maps $R/\mathfrak{a}^n \to S/\mathfrak{b}^n$. Passing to limits yields $\hat{f}\colon \hat{R} \to \hat{S} = S$ (using that S is separated and complete), and $\hat{f} \circ \iota = f$ holds on each quotient. Uniqueness follows because $\iota(R)$ is dense in $\hat{R}$ and S is separated.

Taking $\mathfrak{a}^n M$ as the filtration, a homomorphism $f\colon M \to N$ satisfies $f(\mathfrak{a}^n M) \subseteq \mathfrak{a}^n N$, so it induces maps on the quotients and lifts to $\hat{f}\colon \hat{M} \to \hat{N}$; completion is thus a functor.

25.2 Filtrations

The following construction is a more general version of an inverse system that drops many of the algebraic properties and is a more "pure topological" definition. It is being presented here for its use in the study of graded algebras, as well as it's use when defining *spectral sequences* in chapter 58

Definition 25.2.1: Filtration

Let M be an R-module. Then a *filtration* of M is an infinite chain:

$$M = M_0 \supseteq M_1 \supseteq \cdots \supseteq M_n \supseteq \cdots$$

where each M_n is a submodule of M. The chain is usually denoted (M_n). If furthermore $\mathfrak{a} \subseteq R$ is an ideal, then an $\mathfrak{a}$-filtration of M is a filtration st

$$\mathfrak{a}M_n \subseteq M_{n+1} \qquad \forall n \in \mathbb{N}$$

An $\mathfrak{a}$-filtration (M_n) is called *stable* (or $\mathfrak{a}$-*stable*) if there exists an N such that $\mathfrak{a}M_n = M_{n+1}$ for all $n \geq N$.

An $\mathfrak{a}$-stable filtration can be thought of as the generalization of a graded module which is in some way finite. The following lemma will be important to compare two $\mathfrak{a}$-stable filtrations:

Lemma 25.2.2: Comparing Stable $\mathfrak{a}$-Filtartions

Let $(M_n), (M'_n)$ be stable $\mathfrak{a}$-filtrations of M. Then they have bounded difference, that is, there exists an integer n_0 such that $M_{n+n_0} \subseteq M'_n$ and $M'_{n+n_0} \subseteq M_n$ for all $n \geq 0$.

This will imply that all stable $\mathfrak{a}$-filtrations determine the same topology on M (The $\mathfrak{a}$-topology)

***Proof*:**
It suffices to compare each stable $\mathfrak{a}$-filtration with the $\mathfrak{a}$-adic filtration $(\mathfrak{a}^n M)$. For any $\mathfrak{a}$-filtration, $\mathfrak{a}M_n \subseteq M_{n+1}$ gives $\mathfrak{a}^n M \subseteq M_n$ by induction. Stability provides an N with $M_{n+1} = \mathfrak{a}M_n$ for $n \geq N$, so $M_{n+N} = \mathfrak{a}^n M_N \subseteq \mathfrak{a}^n M$. Thus $\mathfrak{a}^n M \subseteq M_n$ and $M_{n+N} \subseteq \mathfrak{a}^n M$; doing the same for (M'_n) with some N' and combining, with $n_0 = \max(N, N')$, gives $M_{n+n_0} \subseteq M'_n$ and $M'_{n+n_0} \subseteq M_n$.

The key use of this lemma comes when defining the α-adic topology given in appendix C.4, namely all stable $\mathfrak{a}$-filtrations produce the same topology.

Exercise 25.2.1

1. Show that $\mathbb{Z}_p \cap \mathbb{Q} \cong \mathbb{Z}_{(p)}$
2. Show that $\hat{\hat{R}} \cong \hat{R}$

25.3 More on Graded Algebras

Graded algebras were initially explored in section 16.3. In this section, we expand further on graded-algebras, explore their relation to the Neotherian property, and look at some important associated graded algebras given an ideal I or module M. This exploration shall bear fruitful results that will be important in the development of *projective algebraic geometry* and *homological algebra*.

First, recall that a ring R is a graded ring if $R = \bigoplus_i R_i$ where $R_i R_j \subseteq R_{i+j}$, which makes R into an R_0-algebra, and each R_i is an R_0-module. A graded R-module is a module $M = \bigoplus_0^\infty M_i$ such that $R_m M_n \subseteq M_{m+n}$. If M is also a graded ring, then M is an R-graded algebra. Elements of M_n are called *homogeneous*, and an ideal $I \subseteq R$ is called homogeneous if $I = \bigoplus_i^\infty (R_i \cap I)$. A ring or R-module homomorphism is called graded if it respects the graded structure ($f(M_n) \subseteq N_n$). Each graded ring has an ideal $R_+ = \bigoplus_{n>0} R_n$. The elements of R_n for each n are called the *homogeneous elements*.

Proposition 25.3.1: Noetherian Graded Rings

Let R be a graded ring. Then the following are equivalent:

1. R is Noetherian
2. R_0 is Noetherian and R is a finitely generated R_0-algebra

***Proof*:**

(1) $\Rightarrow$ (2) As $R_0 \cong R/R_+$, R_0 is Noetherian. As $R_+ \subseteq R$ is an ideal of R, it is finitely generated, say $x_1, ..., x_s$. Then these elements can be used to construct homogeneous s element, let's use the same label and say they are of order $k_1, ..., k_s$ (all necessarily order greater than 0). Let $R' \subseteq R$ be the ring generated as an R_0 algebra over the homogeneous elements $x_1, ..., x_s$. Then I claim that $R = R'$, which will the proof. First, let' show $A_n \subseteq A'$ using induction. It is clear if $n = 0$. For $n > 0$, taking $y \in A_n$, as $y \in A_+$, y is a linear combination of the x_i

$$y = \sum_i^s a_i x_i$$

with $a_i \in A_{n-k_i}$ (and $A_m = 0$ of $m < 0$). Then by the inductive hypothesis, each a_i is a polynomial in the x's with coefficients in A_0. Hence, by the principle of induction, the same holds for y, we have that $A_n \subseteq A'$. As $A' \subseteq A$ by definition, we have $A = A'$, as we sought to show

(2) $\Rightarrow$ (1) Follow's from HBT

Any ring R induces a graded ring by taking $\mathfrak{a} \subseteq R$ and taking $R^* = \bigoplus_n^\infty \mathfrak{a}^n$. Similarly, any R-module M induces a graded R^*-module $M^* = \bigoplus_n M_n$ where M_n is the $\mathfrak{a}$-filtration. The "canonical" grading comes from $S = R[x_1, ..., x_n]$ with $\mathfrak{a} = (x_1, ..., x_n)$ which will produce the homogeneous grading of S.

In the case where R is Noetherian and $\mathfrak{a}$ is finitely generated (say by $x_1, ..., x_s$), then R^* is a quotient of the polynomial ring $R[x_1, ..., x_s]$ (with the x_i in degree 1), hence Noetherian by proposition 25.3.1.

Proposition 25.3.2: Topology of Noetherian Graded Module

Let R be a Noetherian ring, M a finitely generated R-module, (M_n) an $\mathfrak{a}$-filtration of M. Then the following are equivalent:

1. M^* is a finitely generated R^*-module
2. The filtration (M_n) is stable

***Proof*:**
This is about creating the right conditions to see the filter is stable. Take $Q_n = \bigoplus_{r=0}^{n} M_r$. As each M_r is finitely generated, Q_n is finitely generated. Then Q_n forms a subgroup of M^* but not in general a R^*-submodule. We may however use it to generate one:

$$M_n^* = M_0 \oplus \cdots M_n \oplus \mathfrak{a}M_n \oplus \mathfrak{a}^2 M_n \oplus \cdots \oplus \mathfrak{a}^r M_n \oplus \cdots$$

Now, as Q_n is finitely generated as an R-module, M_n^* is finitely generated as an R^*-module, and the M_n^* form an ascending chain whose unions is M^*.

Finally, we now have that as R^* is Noetherian ,M^* is finitely generated as an R^*-module if and only if the chain stops ($M^* = M_{n_0}^*$) if and only if the filtration is stable.

Corollary 25.3.3: Graded Noetherian then Component Noetherian

Let R be graded and Noetherian, and suppose M is a finitely generated graded R-module. Then M_n is a finitely-generated R_0-module for all $n \in \mathbb{N}$

***Proof*:**
Let

$$d_j = \deg m_j$$

Then every element of $m \in M_n$ can be written as:

$$m = \sum_j f_j(x) m_j \qquad f_j(x) \in R$$

where f_j is homogeneous of degree $n - d_j$ (and is zero if $n < d_j$). Hence, M_n is finitely generated as an R_0-module generated by all $g_j(x)m_j$ where $g_j(x)$ is a monomial in x_i of total degree $n - d_j$, as we sought to show.

25.4 Artin-Rees Lemma

The Artin-Rees lemma is a central result that will allow us to compare the induced and $\mathfrak{a}$-adic topology on a submodule, giving us insight on their topologies.

Theorem 25.4.1: Artin-Rees Lemma

Let R be a Noetherian ring, $\mathfrak{a} \subseteq R$ an ideal, M a finitely generated R-module, and (M_n) a stable $\mathfrak{a}$-filtration of M. Then if $M' \subseteq M$ is a submodule of M, $(M' \cap M_n)$ is a stable $\mathfrak{a}$-filtration of M'

***Proof*:**
As $\mathfrak{a}(M' \cap M_n) \subseteq \mathfrak{a}M' \cap \mathfrak{a}M_n \subseteq M' \cap M_{n+1}$, $(M' \cap M_n)$ is an $\mathfrak{a}$-filtration, and so it defines a graded R^*-module which is a submodule of M^*, and so it is finitely generated, letting us apply by proposition 25.3.2 to completing the proof.

Corollary 25.4.2: Graded Sub-module, Computational Shortcut

Let R be a Noetherian ring, $\mathfrak{a} \subseteq R$ an ideal, M a finitely generated R-module, and $M' \subseteq M$ a submodule. Then there exists a k such that

$$(\mathfrak{a}^n M) \cap M' = \mathfrak{a}^{n-k}\left((\mathfrak{a}^k M) \cap M'\right)$$

for all $n \geq k$

***Proof*:**
By the Artin-Rees lemma (theorem 25.4.1), $(\mathfrak{a}^n M \cap M')$ is a stable $\mathfrak{a}$-filtration of M', so there is a k with $\mathfrak{a}(\mathfrak{a}^n M \cap M') = \mathfrak{a}^{n+1} M \cap M'$ for all $n \geq k$; iterating gives $\mathfrak{a}^n M \cap M' = \mathfrak{a}^{n-k}(\mathfrak{a}^k M \cap M')$ for $n \geq k$.

Theorem 25.4.3: Submodule Adic Topology Equivalence

Let R be a Noetherian ring, $\mathfrak{a} \subseteq R$ an ideal, M a finitely generated R-module, $M' \subseteq M$ a submodule. Then the filtrations

$$\mathfrak{a}^n M' \qquad \text{and} \qquad (\mathfrak{a}^n M) \cap M'$$

have bounded difference, hence the $\mathfrak{a}$-topology of M' coincides with the induced topology given by the $\mathfrak{a}$-topology of M

***Proof*:**
Direct result of corollary 25.4.2 and theorem 25.4.1.

Corollary 25.4.4: Module Completion Preserves Exactness

Let

$$0 \to M' \to M \to M'' \to 0$$

be a short exact sequence of finitely generated R-modules over a Noetherian ring R. Then if $\mathfrak{a} \subseteq R$ is an ideal, the $\mathfrak{a}$-adic completion functor preserves exactness:

$$0 \to \hat{M}' \to \hat{M} \to \hat{M}'' \to 0$$

***Proof*:**
Theorem 25.4.3 provides the condition needed to use corollary 25.1.4.

Next, as we have a natural homomorphism $R \to \hat{R}$, we can think of $\hat{R}$ as an R-algebra, and hence we have an extension of scalars of an R-module M given to an $\hat{R}$-module $\hat{R} \otimes_R M$. This gives a natural sequence of maps that we may want to understand:

$$\hat{R} \otimes_R M \to \hat{R} \otimes_R \hat{M} \to \hat{R} \otimes_{\hat{R}} \hat{M} = \hat{M}$$

In general, these maps are neither injective nor surjective. Under nicer conditions, we have a better control:

Proposition 25.4.5: Module Completion as Tensor Product

Let R be any ring. Then if$M$ is finitely generated,

$$\hat{R} \otimes_R M \twoheadrightarrow \hat{M}$$

is surjective. If furthermore R is Neotherian, then:

$$\hat{R} \otimes_R M \cong \hat{M}$$

***Proof*:**
Let $F \cong R^n$ so that $M \cong F/N$ for appropriate N. The conclusion can be drawn from the following commutative diagram:

$$\begin{tikzcd}
 & \hat{R} \otimes_R N \arrow[r] \arrow[d, "\gamma"] & \hat{R} \otimes_R F \arrow[r] \arrow[d, "\beta"] & \hat{R} \otimes_R M \arrow[r] \arrow[d, "\alpha"] & 0 \\
0 \arrow[r] & \hat{N} \arrow[r] & \hat{F} \arrow[r, "\delta"] & \hat{M} \arrow[r] & 0
\end{tikzcd}$$

where the top line is given by the right-exactness of the tensor product, and the bellow line is given by corollary 25.1.4. Note that as completion commutes with finite direct sums (use corollary 25.1.4), we have that $\hat{R} \otimes_R F \cong \hat{F}$, and so α is surjective, giving us the first part. If R is Noetherian, hen N is finitely generated and so γ is surjective, and so we get exactness of the bottom line. Then by chasing the diagram we will get hat α is injective, and hence an isomorphism, completing the proof.

From this, we get that when R is Noetherian, the extension of scalar functor $\widehat{(-)}$ is equivalent to the functor $\hat{R} \otimes_R (-)$ from the category of R-modules to the category of $\hat{R}$-modules is exact. As a consequence:

Corollary 25.4.6: Flatness of Completion

Let R be a Neotherian ring, $\mathfrak{a} \subseteq R$ an ideal, $\hat{R}$ the $\mathfrak{a}$-adic completing of R. Then $\hat{R}$ is a flat R-algebra.

***Proof*:**
direct result of above discussion.

Proposition 25.4.7: Properties of Adic Completion

Let R be Noetherian and $\hat{R}$ its $\mathfrak{a}$-adic completion. Then:

1. $\hat{\mathfrak{a}} = \hat{R}\mathfrak{a} = \hat{R} \otimes_R \mathfrak{a}$
2. $\widehat{\mathfrak{a}^n} = \hat{\mathfrak{a}}^n$
3. $\mathfrak{a}^n/\mathfrak{a}^{n+1} \cong \hat{\mathfrak{a}}^n/\hat{\mathfrak{a}}^{n+1}$
4. $\hat{\mathfrak{a}}$ is contained in the Jacobson radical of $\hat{R}$

***Proof*:**
exercise (or see Atiyah p.109)

We shall show the last one. By (2) and the fact that $\hat{\hat{R}} \cong \hat{R}$, we see that $\hat{R}$ is complete in the $\mathfrak{a}$-adic topology. Hence for any $x \in \hat{\mathfrak{a}}$,

$$(1-x)^{-1} = 1 + x + x^2 + \cdots$$

converges in $\hat{R}$, and so $1-x$ is a unit. Then this implies that $\hat{\mathfrak{a}}$ is contained in the Jacobson radical of $\hat{R}$, completing the proof.

Let us next characterize the kernel of the map $M \to \hat{M}$

Theorem 25.4.8: Krull's Theorem

LEt R be a Noetherian ring, $\mathfrak{a} \subseteq R$ an ideal, M a finitely generated R-module, and $\hat{M}$ the $\mathfrak{a}$-completion of M, Then the kernel

$$E = \bigcap_{n=1}^{\infty} \mathfrak{a}^n M$$

given by the map $M \to \hat{M}$ consist of all $x \in M$ that are annihilated by some element $1 + \mathfrak{a}$.

***Proof*:**
It suffices to show that $(1-\alpha)E = 0$. IT is clear that if $(1-\alpha)x = 0$

$$x = \alpha x = \alpha^2 x = \cdots \in \bigcap_{n=1}^{\infty} \mathfrak{a}^n M = E$$

Hence we shall show the converse. By the Artin-Rees consequence (corollary 25.4.2) applied to $E \subseteq M$, there is a k with $\mathfrak{a}^n M \cap E = \mathfrak{a}^{n-k}(\mathfrak{a}^k M \cap E)$ for $n \geq k$. Since $E \subseteq \mathfrak{a}^n M$ for all n, the left-hand side is E and the right-hand side is $\mathfrak{a}^{n-k}E$, so $E = \mathfrak{a}^{n-k}E \subseteq \mathfrak{a}E$, giving $\mathfrak{a}E = E$. As M is finitely generated over the Noetherian ring R, E is finitely generated, so the determinant form of Nakayama's lemma yields $\alpha \in \mathfrak{a}$ with $(1-\alpha)E = 0$, completing the proof.

Proposition 25.4.9: Localization is A Subring of Completion

Let R be Noetherian and let $S = 1 + \mathfrak{a}$. Then

$$S^{-1}R \hookrightarrow \hat{R}$$

***Proof*:**
The map $R \to \hat{R}$ and $R \to S^{-1}R$ have the same kernel, and for any $\alpha \in \hat{\mathfrak{a}}$ we have

$$(1-\alpha)^{-1} = 1 + \alpha + \alpha^2 + \cdots$$

converges in $\hat{A}$, and hence all S become a unit in $\hat{A}$. By the universal property of $S^{-1}A$, there is a natural homomorphism $S^{-1}A \to \hat{A}$ which is injective by theorem 25.4.8.

The Noetherian condition is necessary, an easy example lies within the realm of analysis:

Example 25.4.10: Localization and Completion Without Noetherian Property
Take R to be the collection of smooth functions on $\mathbb{R}$, $C^\infty(\mathbb{R})$. Let $\mathfrak{a}$ be the ideal of all f which vanish at 0. This ideal is maximal as $R/\mathfrak{a} \cong \mathbb{R}$. Then $\mathfrak{a}$ is generated by x (if you know some multi-variable calculus prove that any f that vanishes at 0 can be written as $f(x) = xg(x)$ for g smooth, it is a variation of Hadamard's lemma), and $E = \bigcap_{n=1}^{\infty} \mathfrak{a}^n$ is the set of all $f \in R$ such that all derivatives vanish at the origin. On the other hand, f is annihilated by some element $1+\alpha$ for $\alpha \in \mathfrak{a}$ iff f vanishes identically in some neighborhood of 0. Now, the function e^{-1/x^2} has vanishing derivative at 0, but is not identically zero near 0. Hence the kernels of the $R \to \hat{R}$ and $\mathbb{R} \to S^{-1}R$ with $S = 1 + \mathfrak{a}$ do not coincide.

This can also be a way of showing that R cannot be Noetherian.

Corollary 25.4.11: Krull Theorem, Noetherian Domain

Let R be a Noetherian domain and $(1) \neq \mathfrak{a} \subseteq R$ an ideal. Then

$$\bigcap \mathfrak{a}^n = 0$$

***Proof*:**
By Krull's theorem (theorem 25.4.8), $\bigcap \mathfrak{a}^n$ consists of the x annihilated by some $1+\alpha$ with $\alpha \in \mathfrak{a}$. Since $\mathfrak{a}$ is proper, $-1 \notin \mathfrak{a}$, so $1+\alpha \neq 0$; as R is a domain, $(1+\alpha)x = 0$ forces $x = 0$, whence $\bigcap \mathfrak{a}^n = 0$.

Corollary 25.4.12: Krull Theorem, Hausdorff Topology

Let R be a Noetherian ring, $\mathfrak{a} \subseteq R$ an ideal contained in the Jacobson radical, and M a finitely generated R-module. Then the $\mathfrak{a}$-topology of M is Hausdroff, i.e.

$$\bigcap \mathfrak{a}^n M = 0$$

***Proof*:**
Since $\mathfrak{a} \subseteq \text{Jac}(R)$, every element of $1+\mathfrak{a}$ is a unit (proposition 23.4.26). By Krull's theorem (theorem 25.4.8), each element of $\bigcap \mathfrak{a}^n M$ is annihilated by such a unit, hence is 0.

Corollary 25.4.13: Krull Theorem, Kernel Into Localization

Let R be a Noetherian ring, $\mathfrak{p} \subseteq R$ a prime ideal. Then the intersection of all $\mathfrak{p}$-primary ideals of A is the kernel of $A \to A_{\mathfrak{p}}$

***Proof*:**
Write $K = \ker(A \to A_{\mathfrak{p}}) = \{a \in A \mid sa = 0 \text{ for some } s \notin \mathfrak{p}\}$. Every $\mathfrak{p}$-primary ideal $\mathfrak{q}$ contains K: if $sa = 0 \in \mathfrak{q}$ with $s \notin \mathfrak{p} = \sqrt{\mathfrak{q}}$, then $a \in \mathfrak{q}$; hence $K \subseteq \bigcap\{\mathfrak{p}\text{-primary}\}$. Conversely the symbolic powers $\mathfrak{p}^{(n)} = (\mathfrak{p}^n A_{\mathfrak{p}}) \cap A$ are $\mathfrak{p}$-primary, and since $\mathfrak{p}A_{\mathfrak{p}}$ lies in the Jacobson radical of $A_{\mathfrak{p}}$, corollary 25.4.12 gives $\bigcap_n \mathfrak{p}^n A_{\mathfrak{p}} = 0$, whence $\bigcap_n \mathfrak{p}^{(n)} = K$. Thus $\bigcap\{\mathfrak{p}\text{-primary}\} \subseteq \bigcap_n \mathfrak{p}^{(n)} = K$, and the two inclusions give equality.

25.5 Associated Graded Rings: Noetherian Closed under Completion

The following will let us show another way of characterizing complete rings, and let us prove that the completion of a Noetherian ring is Noetherian.

Definition 25.5.1: Associated Graded Ring

Let R be a ring and $\mathfrak{a} \subseteq R$ an ideal. Then the *associated graded ring* is

$$G_{\mathfrak{a}}(R) = \bigoplus_{n=0}^{\infty} \mathfrak{a}^n/\mathfrak{a}^{n+1}$$

where $\mathfrak{a}^0 = R$. If unambiguous, $G(R)$ will be written instead of $G_{\mathfrak{a}}(R)$. For an R-module M with an $\mathfrak{a}$-filtration (M_n), the graded $G(R)$-module is given by:

$$G(M) = \bigoplus_{n=0}^{\infty} M_n/M_{n+1} = \bigoplus_{n=0}^{\infty} G_n(M)$$

Multiplication is certainly well-defined: for each $x_n \in \mathfrak{a}^n$, $\overline{x}_n$ is the image of x_n in $\mathfrak{a}^n/\mathfrak{a}^{n+1}$. Then $\overline{x}_m\overline{x}_m$ is $\overline{x_m x_n}$ which is the image of $x_m x_n$ in $\mathfrak{a}^{m+n}/\mathfrak{a}^{m+n+1}$ (if the reader is not convinced, check to see if this is well-defined).

Proposition 25.5.2: Properties of Associated Graded Rings

Let R be a Noetherian ring and $\mathfrak{a} \subseteq R$ an ideal. Then:

1. $G_{\mathfrak{a}}(R)$ is Noetherian
2. $G_{\mathfrak{a}}(R)$ is isomorphic to $G_{\hat{\mathfrak{a}}}(\hat{R})$ as graded rings
3. If M is a finitely generated R-module and (M_n) is a stable $\mathfrak{a}$-filtration of M, then $G(M)$ is a finitely generated graded $G_{\mathfrak{a}}(R)$-module

***Proof*:**

1. $G_{\mathfrak{a}}(R)$ is generated as an algebra over the Noetherian ring $R/\mathfrak{a}$ by the images of generators of $\mathfrak{a}$ in $\mathfrak{a}/\mathfrak{a}^2$, hence is Noetherian by HBT (theorem 23.2.1).
2. By proposition 25.4.7(3), $\mathfrak{a}^n/\mathfrak{a}^{n+1} \cong \hat{\mathfrak{a}}^n/\hat{\mathfrak{a}}^{n+1}$ in each degree, and these degreewise isomorphisms respect multiplication, giving $G_{\mathfrak{a}}(R) \cong G_{\hat{\mathfrak{a}}}(\hat{R})$.
3. Pick N with $\mathfrak{a}M_n = M_{n+1}$ for $n \geq N$ (stability). Then $G(M)$ is generated over $G_{\mathfrak{a}}(R)$ by the finitely many pieces M_n/M_{n+1} with $n \leq N$, each finitely generated over $R/\mathfrak{a}$ (corollary 25.3.3); hence $G(M)$ is finitely generated over $G_{\mathfrak{a}}(R)$.

The most important case of an associated graded ring is the one where the ideal is generated by a regular sequence: there the associated graded ring is as simple as it could be, a polynomial ring, and the geometric consequence is that the normal cone of a regularly embedded subscheme is a vector bundle rather than a general cone.

Theorem 25.5.3: Associated Graded Ring of a Regular Sequence

Let A be a Noetherian ring and let $x_1, \ldots, x_d \in A$ be a regular sequence ([15, Regular Sequence], a co-requisite of this book: x_i is a nonzerodivisor on $A/(x_1, \ldots, x_{i-1})$ for each i, and $A/(x_1, \ldots, x_d) \neq 0$). Let $\mathfrak{a} = (x_1, \ldots, x_d)$. Then the $(A/\mathfrak{a})$-algebra homomorphism

$$(A/\mathfrak{a})[T_1, \ldots, T_d] \longrightarrow G_{\mathfrak{a}}(A), \qquad T_i \longmapsto \overline{x_i} \in \mathfrak{a}/\mathfrak{a}^2$$

is an isomorphism of graded rings. In particular $\mathfrak{a}^n/\mathfrak{a}^{n+1}$ is a free $A/\mathfrak{a}$-module on the monomials of degree n in the $\overline{x_i}$.

***Proof*:**
Surjectivity. The ideal $\mathfrak{a}^n$ is generated by the degree-n monomials in the x_i, so $\mathfrak{a}^n/\mathfrak{a}^{n+1}$ is generated as an $A/\mathfrak{a}$-module by their classes, which are the images of the degree-n monomials in the T_i. Surjectivity in each degree follows.

Injectivity, by induction on d. For $d = 0$ the ideal is zero and both sides are A.

Suppose $d \geq 1$ and the statement holds for shorter regular sequences. Let $F \in (A/\mathfrak{a})[T_1, \ldots, T_d]$ be homogeneous of degree n with $F(\overline{x}) \in \mathfrak{a}^{n+1}$; the claim is that all coefficients of F vanish in $A/\mathfrak{a}$. Lift the coefficients to A and write F for a lift, so $F(x_1, \ldots, x_d) \in \mathfrak{a}^{n+1}$.

Group F by its T_d-degree, $F = \sum_j T_d^j G_j(T_1, \ldots, T_{d-1})$ with G_j homogeneous of degree $n - j$. Reducing modulo x_d kills every term with $j > 0$, so $G_0(\overline{x_1}, \ldots, \overline{x_{d-1}}) \in \overline{\mathfrak{a}}^{n+1}$ in $A/(x_d)$, where $x_1, \ldots, x_{d-1}$ is again a regular sequence and generates $\overline{\mathfrak{a}}$. By the inductive hypothesis applied in $A/(x_d)$, every coefficient of G_0 lies in $\mathfrak{a} + (x_d) = \mathfrak{a}$, which is the required vanishing for the $j = 0$ part.

Hence $F - G_0 = T_d \cdot H$ for a homogeneous H of degree $n-1$, and $x_d \cdot H(x) = F(x) - G_0(x) \in \mathfrak{a}^{n+1}$. Since $G_0(x) \in \mathfrak{a}^{n+1}$ we get $x_d H(x) \in \mathfrak{a}^{n+1}$. Now $\mathfrak{a}^{n+1} = x_d \mathfrak{a}^n + \mathfrak{a}'^{n+1}$ where $\mathfrak{a}'$ is generated by $x_1, \ldots, x_{d-1}$; writing $x_d H(x) = x_d a + b$ with $a \in \mathfrak{a}^n$ and $b \in \mathfrak{a}'^{n+1}$ gives $x_d(H(x) - a) = b$. Because x_d is a nonzerodivisor modulo $\mathfrak{a}'$ and $b \in \mathfrak{a}'^{n+1} \subseteq \mathfrak{a}'$, this forces $H(x) - a \in \mathfrak{a}^n$, hence $H(x) \in \mathfrak{a}^n$. The inductive hypothesis in degree $n-1$, applied to H, then gives that all coefficients of H lie in $\mathfrak{a}$, completing the induction and the proof.

Lemma 25.5.4: Associated Graded Ideal and Completion Map Relation

Let $\varphi : A \to B$ be a homomorphism of filtered groups ($\varphi(A_n) \subseteq B_n$), and let $G(\varphi) : G(A) \to G(B)$, $\hat{\varphi}\hat{A} \to \hat{B}$ be the induced homomorphisms of the associated graded group and complete group respectively. Then:

1. $G(\varphi)$ is injective $\Rightarrow$ $\hat{\varphi}$ is injective
2. $G(\varphi)$ is surjective $\Rightarrow$ $\hat{\varphi}$ is surjective

***Proof*:**
The key is to look at the following commutative diagram of exact sequences:

$$\begin{tikzcd} 0 \arrow[r] & A_n/A_{n+1} \arrow[r] \arrow[d, "G_n(\varphi)"] & A/A_{n+1} \arrow[r] \arrow[d, "\alpha_{n+1}"] & A/A_n \arrow[r] \arrow[d, "\alpha_n"] & 0 \\ 0 \arrow[r] & B_n/B_{n+1} \arrow[r] & B/B_{n+1} \arrow[r] & B/B_n \arrow[r] & 0 \end{tikzcd}$$

which by the snake lemma gives us the usual:

$$0 \to \ker G_n(\varphi) \to \ker \alpha_{n+1} \to \ker \alpha_n \to \text{coker} G_n(\varphi) \to \text{coker}\alpha_{n+1} \to \text{coker}\alpha_n \to 0$$

Now, by induction on n, we see that:

1. $\ker \alpha_n = 0$
2. $\text{coker}\alpha_n = 0$

taking the limit of the homomorphisms α_n and applying proposition 25.1.3 completes the proof.

Proposition 25.5.5: Partial Converse of proposition 25.5.2(3)

Let R be a ring, $\mathfrak{a} \subseteq R$ an ideal, M an R-module, and (M_n) an $\mathfrak{a}$-filtration of M. Suppose R is complete in the $\mathfrak{a}$-topology, M is Hausdorff in its filtration topology ($\bigcap_n M_n = 0$), and $G(M)$ is a finitely generated $G(R)$-module.

Then M is a finitely generated R-module.

***Proof*:**
As $G(M)$ is finitely generated, pick some finite set of generators. Split them up into the homogeneous components, say ζ_i (with $1 \leq i \leq \nu$) where ζ_i has degree $n(i)$, and is the image of some $x_i \in M_{n(i)}$. Let F^i be the module with the stable $\mathfrak{a}$-filtration given by $F^i_K = \mathfrak{a}^{k+n(i)}$ and let $F = \bigoplus_{i=1}^r F^i$. Then the mapping that takes the generator 1 of each F^i to x_i gives a homomorphism of filtered groups

$$\varphi : F \to M$$

Hence it induces a homomorphism of $G(A)$-modules. By construction this map is surjective. Hence by lemma 25.5.4 $\hat{\varphi}$ is surjective. Now consider the diagram:

$$\begin{tikzcd} F \arrow[r, "\varphi"] \arrow[d, "\alpha"'] & M \arrow[d, "\beta"] \\ \hat{F} \arrow[r, "\hat{\varphi}"] & \hat{M} \end{tikzcd}$$

As F is free and $A = \hat{A}$, α is an isomorphism. Next, as M is Hausdorff, β is injective. . As $\hat{\varphi}$ is surjective, φ is surjective, and so $x_1, ..., x_r$ generate M as an A-module completing the proof.

Corollary 25.5.6: Partial Converse in Noetherian Case

Let R be a ring, $\mathfrak{a} \subseteq R$ an ideal, M an R-module, and (M_n) an $\mathfrak{a}$-filtration of M. Suppose R is complete in the $\mathfrak{a}$-topology, M is Hausdorff in its filtration topology ($\bigcap_n M_n = 0$), and $G(M)$ is a finitely generated $G(R)$-module.

Then if $G(M)$ is a Noetherian $G(R)$-module, then M is a Noetherian R-module.

***Proof*:**
If we show that every submodule $M' \subseteq M$ is finitely generated, we're done. Let $M'_n = M' \cap M_n$. Then (M'_n) is an $\mathfrak{a}$-filtration of M', and the embedding $M'_n \to M_N$ induces an injective homomorphism:

$$M'_n/M'_{n+1} \to M_n/M_{n+1}$$

Hence we gt a embedding of $G(M')$ in $G(M)$, $G(M)$ is Noetherian, $G(M')$ is finitely generated, and so M' is Hausdroff. As $\cap M'_n \subseteq \cap M_N = 0$, we get that M' is finitely generated, as we sought to show.

Theorem 25.5.7: Noetherian Closed Under Completion

Let R be a Noetherian ring and $\mathfrak{a} \subseteq R$ an ideal. Then the $\mathfrak{a}$-completion $\hat{R}$ of R is Noetherian

***Proof*:**
As:

$$G_{\mathfrak{a}}(R) = G_{\hat{\mathfrak{a}}}(\hat{R})$$

is Noetherian, apply corollary 25.5.6 to the complete ring $\hat{R}$ and taking $M = \hat{R}$.

Corollary 25.5.8: Power Series Ring is Noetherian

Let R be a Noetherian ring, and let $S = R[[x_1, ..., x_n]]$ be the power series ring in n variables. Then S is Noetherian.

***Proof*:**
The ring $R[x_1, ..., x_n]$ is Noetherian by Hilbert's Basis Theorem, and S is the completion for the $(x_1, ..., x_n)$-adic topology.

Completion also leaves the dimension of a Noetherian local ring unchanged, which is how dimension is often computed by passing to the simpler complete ring.

Corollary 25.5.9: Completion Preserves Dimension

Let $(R, \mathfrak{m})$ be a Noetherian local ring with $\mathfrak{m}$-adic completion $\hat{R}$. Then $\dim \hat{R} = \dim R$.

***Proof*:**
By proposition 25.5.2(2), $G_{\mathfrak{m}}(R) \cong G_{\hat{\mathfrak{m}}}(\hat{R})$ as graded rings, and corollary 25.1.5 gives $R/\mathfrak{m}^n \cong \hat{R}/\hat{\mathfrak{m}}^n$; hence the Hilbert-Samuel functions $n \mapsto \ell(R/\mathfrak{m}^n)$ and $n \mapsto \ell(\hat{R}/\hat{\mathfrak{m}}^n)$ coincide. Since the Krull dimension of a Noetherian local ring is the degree of its Hilbert-Samuel polynomial (chapter 29), $\dim R = \dim \hat{R}$.

25.6 The Cohen Structure Theorem

Completion gives the local structure of a ring coordinates: a complete Noetherian local ring containing its residue field is a quotient of a power series ring, the algebraic form of a formal coordinate chart around a point.

Theorem 25.6.1: Cohen Structure Theorem, Equicharacteristic Case

Let $(R, \mathfrak{m})$ be a complete Noetherian local ring containing a field k with $R/\mathfrak{m} = k$. Then $R \cong k[[x_1, \ldots, x_n]]/I$ for an ideal I, where $n = \dim_k \mathfrak{m}/\mathfrak{m}^2$ is the minimal number of generators of $\mathfrak{m}$.

***Proof*:**
Choose $y_1, \ldots, y_n \in \mathfrak{m}$ whose residues form a k-basis of $\mathfrak{m}/\mathfrak{m}^2$; by Nakayama's lemma (lemma 23.4.27) they generate $\mathfrak{m}$. The composite $k \hookrightarrow R \twoheadrightarrow R/\mathfrak{m} = k$ is the identity, making R a k-algebra, so we may define

$$\varphi \colon k[[x_1, \ldots, x_n]] \to R, \qquad \sum_\alpha c_\alpha x^\alpha \mapsto \sum_\alpha c_\alpha y^\alpha,$$

the sum on the right converging because $y^\alpha \in \mathfrak{m}^{|\alpha|}$ and R is $\mathfrak{m}$-adically complete (definition 25.1.9); φ is a k-algebra homomorphism. For surjectivity, reduce modulo $\mathfrak{m}^m$. Each graded piece $\mathfrak{m}^j/\mathfrak{m}^{j+1}$ is spanned over k by the degree-j monomials in the residues $\bar{y}_i$, since multiplication gives a surjection $\mathrm{Sym}^j(\mathfrak{m}/\mathfrak{m}^2) \twoheadrightarrow \mathfrak{m}^j/\mathfrak{m}^{j+1}$ and $\mathfrak{m}/\mathfrak{m}^2$ is spanned by the $\bar{y}_i$. Hence $R/\mathfrak{m}^m$, whose composition factors are the $\mathfrak{m}^j/\mathfrak{m}^{j+1}$ for $j < m$, is spanned over k by monomials in the y_i of degree $< m$, all in the image of φ; so φ is surjective modulo every $\mathfrak{m}^m$. As $R = \varprojlim R/\mathfrak{m}^m$ and $k[[x_1, \ldots, x_n]]$ is complete, φ is surjective. Thus $R \cong k[[x_1, \ldots, x_n]]/I$ with $I = \ker \varphi$, and $n = \dim_k \mathfrak{m}/\mathfrak{m}^2$ is minimal by the choice of the y_i.

The hypothesis $R/\mathfrak{m} = k$ is the equicharacteristic case with trivial residue extension, the form deformation theory over a field uses. In general Cohen's theorem first produces a *coefficient field* $k \hookrightarrow R$ lifting the residue field (equicharacteristic case) or, in mixed characteristic, a complete discrete valuation coefficient ring; that mixed-characteristic theory, through Witt vectors, is developed in [19].

25.7 Weierstrass Preparation and Division

The Cohen structure theorem realizes a complete local ring as a quotient of a power series ring; the *Weierstrass* theorems describe the internal structure of the power series ring itself, showing that division by a sufficiently generic series behaves like polynomial division. Fix a complete local ring $(R, \mathfrak{m})$ (for instance $R = k[[x_1, \ldots, x_{n-1}]]$) and one further variable x.

Definition 25.7.1: Regular Power Series

A power series $f = \sum_{i \geq 0} a_i x^i \in R[[x]]$ is *regular of order* d in x if $a_0, \ldots, a_{d-1} \in \mathfrak{m}$ and a_d is a unit; equivalently, its image in $(R/\mathfrak{m})[[x]]$ is x^d times a unit.

Theorem 25.7.2: Weierstrass Division Theorem

Let $(R, \mathfrak{m})$ be complete and $f \in R[[x]]$ regular of order d in x. Then every $g \in R[[x]]$ can be written uniquely as

$$g = qf + r, \qquad q \in R[[x]], \quad r \in R[x], \ \deg r < d$$

Theorem 25.7.3: Weierstrass Preparation Theorem

Let $(R, \mathfrak{m})$ be complete and $f \in R[[x]]$ regular of order d in x. Then $f = u\,\omega$ uniquely, where $u \in R[[x]]$ is a unit and

$$\omega = x^d + b_{d-1}x^{d-1} + \cdots + b_0, \qquad b_i \in \mathfrak{m},$$

is a monic *Weierstrass polynomial.*

***Proof*:**
Preparation follows from division: apply theorem 25.7.2 to $g = x^d$, writing $x^d = qf + r$ with $\deg r < d$; reducing modulo $\mathfrak{m}$ shows q is a unit and $\omega := x^d - r$ is a Weierstrass polynomial, so $f = u\omega$ with $u = q^{-1}$. Division itself is proved by successive approximation: writing $f = P + x^d U$ with $P = \sum_{i<d} a_i x^i \in \mathfrak{m}[x]$ and $U = a_d + a_{d+1}x + \cdots$ a unit, the operator returning the low-degree remainder of $g - (\text{quotient part})f$ is a contraction for the $\mathfrak{m}$-adic topology, and completeness of R makes the resulting series for q and r converge; uniqueness holds because a Weierstrass polynomial is a non-zero-divisor. See Zariski-Samuel or Eisenbud, Ch. 7.

Iterating in $x_1, \ldots, x_n$ pins down the local structure of $k[[x_1, \ldots, x_n]]$: it is a regular local ring of dimension n in which, after a generic linear change of coordinates, ideals are generated by Weierstrass polynomials, the algebraic counterpart of resolving a formal germ into a finite branched cover of a coordinate subspace. This underlies formal and analytic local geometry, intersection multiplicities, and Puiseux expansions used downstream.

26

Ring Extension and Integral Extension

In this section, we generalize the notion of extension from fields to commutative rings. Just like how $k \subseteq F$ can be seen as a field extension, if we let k and R be rings instead, we will et the accompying notion of ring extension:

26.1 Ring and Integral Extensions

Definition 26.1.1: Ring Extension

Let S be a commutative ring and $R \subseteq S$ a subring. Then we say that S is a *ring extension* of R. Equivalently, if there exists an injective ring homomorphism $\varphi : R \to S$, then S is a ring extension of R.

Unlike fields, we must specify the homomorphism to be injective, since ring homomorphisms need not be injective. The notation S/R is also used just as in field, though more scarcely. Just like how any field extension F/k makes F into a k-vector-space and we have special k-homomorphisms letting k-commute between the homomorphism, a ring extension S/R have an analogue: in fact, a k-homomorphism is *precisely* a k-algebra homomorphism!

Recall that any *R-algebra* can be defined as a ring homomorphism $\varphi : R \to S$, *provided* that $\varphi(R) \subseteq Z(S)$. Since we are working with commutative rings, this is *always* the case, and so we can always thing of a ring extension S over R is an R-algebra! This is why we focus on *commutative rings*. There is a similar notion of ring extension for non-commutative which is closer to the notion of group extension (i.e. a short exact sequence extending the object), however ring extensions don't break down into nice building blocks since kernels of rings are not rings, which means we would have to extend a ring R by an abelian group (i.e. two-sided ideal) I. This very much changes how we would study rings. However, we can always study the structure of any commutative ring S by how behaves as an R-algebra similarly to how we studied fields, which is what we'll do in these chapters.

26.1.2 Advanced Intuition Integral extensions shall correspond to *proper maps* of morphisms. Non-integral extensions shall never be proper, thus geometrically exhibiting greatly different behaviour. In particular, taking base change if necessary, the image of closed sets (essentially zero-sets of polynomials) need not be closed.

Perhaps the first notion we might want to genelize is the equivalent of an *algebraic element*, since they capture the relations within a ring extension with respect to the base ring. The following is the exactly this generlization:

Definition 26.1.3: Integral Elements

Let $R \subseteq S$.

1. An element $s \in S$ is called an *integral over R* if s is the root of a monic polynomial over $R[x]$:
$$s^n + r_{n-1}s^{n-1} + \cdots + r_1 s + r_0 = 0 \qquad r_i \in R$$
2. S is called *integral over R* or an *integral extension of R* if every element of S is integral over R. The integral closure of an integral domain is also called the *normalization of R*. Sometimes, the integral extension will be written as S/R
3. The set $\overline{R}^S$ containing all the elements of S that are integral over R is called the *integral closure of R*
4. If $R = \overline{R}^S$, then R is said to be *integrally closed over S*.
5. If R is an integral domain that is integrally closed in it's field of fractions F, $\overline{R}^F$, then R is called *normal domain.*

The fact that the polynomial it satisfies has to be monic can be thought of as not wanting to add relations that produce "units" or unlit-like elements (ex. $3x - 1 = 0$ implies there is a new element that behaves like $1/3$). For a more specific motivation, we can think of the following: If $f(x) \in R[x]$ is a polynomial of minimal degree with root r, and $h(x) \in R[x]$ is another monic polynomials with minimal degree with root r, then we would want that $h(x)|f(x)$, just like for minimal polynomials over fields. In the process of doing so, we would require the coefficient of the leading terms to divide. Since they can be arbitrary, we would require the coefficients of these polynomial to be units, or just 1, in order for the division to follow through. It follows that it is a natural condition that all polynomials be monic for an integral extension.

Another motivation comes from Gauss's lemma: over a UFD R with fraction field K, a primitive polynomial of $R[x]$ is irreducible in $R[x]$ if and only if it is irreducible in $K[x]$. Monic polynomials are primitive, so factorization of monic polynomials over R mirrors factorization over K; this parallel between $R[x]$ and $K[x]$ becomes important in [7].

Integral elements share many of the same properties as algebraic elements:

Proposition 26.1.4: Integral Element Equivalences

Let $R \subseteq S$ be a subring of S. Then the following are equivalent:

1. s is integral over R
2. $R[s]$ is a finitely generated R-module
3. The element s is in T for some subring T where $R \subseteq T \subseteq S$, that is a finitely generated R-module

***Proof*:**

$1 \Rightarrow 2$ Recall that $R[s]$ can be expressed as all R-linear combinations of powers of s. Since s is integral over R:

$$s^n + r_n s^{n-1} + \cdots + r_1 s + r_0 = 0$$

in particular:

$$s^n = -(r_n s^{n-1} + \cdots + r_1 s + r_0)$$

It follows that for all $m \geq n$, s^m can be expressed in terms of $1, s, \ldots, s^{n-1}$ (for the same reasons as we've shown for algebraic elements). Hence:

$$R[s] = R + Rs + \cdots + Rs^{n-1}$$

Notice here how if the polynomial was not monic, we might not have been able to isolae s^n, causing complications.

$2 \Rightarrow 3$ Let $T = R[s]$. Then the reuslt immediately follows

$3 \Rightarrow 1$ We take advantage of rational canonical forms and Cramer's rule (remember that if R is an integarl domain, we can take it's field of fractions, and the resulting matri multiplication remains the same). Since T is finitely generated, let $v_1, v_2, ..., v_n$ be a finite generating set. Since $s \in T$ and T is a ring, $sv_i \in T$ for all $1 \leq i \leq n$. Thus:

$$sv_i = \sum_j a_{ij} v_j$$

Giving us the equation n:

$$\sum_j (\delta_{ij} s - a_{ij}) v_j = 0 \qquad 1 \leq i \leq n$$

where δ_{ij} is one when $i = j$ and 0 otherwise (i.e. the Kronecker delta). Letting B be the matrix such that the entries are $B_{ij} = \delta_{ij} s - a_{ij}$ and letting v be the $n \times 1$ matrix whose entries are $v_1, ..., v_n$, we get:

$$Bv = 0$$

Thus, by Cramer's rule:

$$(\det B) v_i = 0 \qquad 1 \leq i \leq n$$

Now, since 1 is an R-linear combination of $v_1, v_2, ..., v_n$,

$$\det B = (\det B)1 = (\det B)\sum_i a_i v_i = \sum a_i((\det B)v_i) = 0$$

i.e. $\det B = 0$. Now, B is by definition $B = sI - A$ (where $A_{ij} = a_{ij}$). Thus, s is the root of a monic polynomial $\det(xI - A) \in R[x]$, namely the characteristic polynomial of A, and so is the root of a monic polynomial with coefficients in R, giving (1)

Corollary 26.1.5: Transitivity of Integrality

Let $R \subseteq S \subseteq T$ be rings. If T is integral over S and S is integral over R, then T is integral over R.

***Proof*:**
Let $t \in T$, a root of a monic $x^n + a_{n-1}x^{n-1} + \cdots + a_0 \in S[x]$. Each a_i is integral over R, so in the tower $R \subseteq R[a_0] \subseteq \cdots \subseteq R[a_0, \ldots, a_{n-1}] =: R'$ every step is module-finite (proposition 26.1.4), hence R' is a finitely generated R-module. As t is integral over R', $R'[t]$ is module-finite over R', hence over R; by proposition 26.1.4 (applied to the subring $R'[t]$), t is integral over R.

Corollary 26.1.6: Integral Elements Form Rings

Let $R \subseteq S$ be a subring of S, and $s, t \in S$. Then:

1. If s and t are integral over R, so is $s \pm t$ and st
2. $\overline{R}^S$ is a subring of S containing R, and is integrally closed[a]

[a] If we were working over rng's, $\overline{R}^S$ need not be integrally closed

***Proof*:**
Let s, t be integral over R, sy that by proposition 26.1.4 $R[s]$ and $R[t]$ are finitely generated R-modules, say:

$$R[s] = Rs_1 + \cdots + Rs_n$$
$$R[t] = Rt_1 + \cdots + Rt_m$$

Then:

$$R[s, t] = Rs_1t_1 + \cdots + Rs_it_j + \cdots + Rs_nt_m$$

is a ring containing $s \pm t$ and st that is also a finitely generated R-module, so $s \pm t$ and st are integral over R. Thus $\overline{R}^S$ is a subring of S containing R; it is integrally closed in S, since any $s \in S$ integral over $\overline{R}^S$ is integral over R by transitivity (corollary 26.1.5), hence lies in $\overline{R}^S$.

In particular, taking the integral closure is idempotent: $\overline{(\overline{R}^S)}^S = \overline{R}^S$, since integrality is transitive (corollary 26.1.5).

Note that integral closure is dependent on the ring over which we are closing; we do not have the equivalent notion of "integral closure of R" without specifying a ring over which it is integrally closed (Why?)

Corollary 26.1.7: Module-Finite versus Finite-Type-and-Integral

Let S be an R-algebra. Then S is a finitely generated R-module if and only if S is of finite type over R and integral over R.

Notice how if S is not integral, then it is possible that S is of finite type over R but not finitely generated over R, the most cannonical example being $R[x]$ over R (since $R[x]$ is finitely generated as an R-algebra by 1 and x, but $1, x, x^2, \dots$ is an infinite set that is a basis for $R[x]$)

***Proof*:**
If S is a finitely generated R-module it is of finite type, and every element is integral (proposition 26.1.4). Conversely, let $S = R[s_1, \dots, s_n]$ be of finite type with each s_i integral over R. In the tower

$$R \subseteq R[s_1] \subseteq R[s_1, s_2] \subseteq \cdots \subseteq R[s_1, \dots, s_n] = S$$

each step $R[s_1, \dots, s_{i-1}][s_i]$ is module-finite, since s_i is integral over the smaller ring (proposition 26.1.4); a module-finite extension of a module-finite extension is module-finite, so S is a finitely generated R-module.

Before giving examples, we'll show that we have been studying special types of integral closures when we worked with field exensions in Part IV:

Proposition 26.1.8: Integral Extension over Fields

Let S/R be an integral extension where S is an integral domain. Then R is a field if and only if S is a field

Note that if S was not an integral domain or commutative, this theorem does not work: Take $M_2(k)$ over any field. Then $M_2(k)$ is a 4-dimensional k-algebar, but certianly not a field (it contains zero-divisors, and is not commutative).

***Proof*:**

($\Rightarrow$) Let R be a field and take $0 \neq s \in S$. Since s is integral over R, we have:

$$s^n + a_{n-1}s^{n-1} + \cdots + a_1 s + a_0 = 0 \qquad a_i \in R$$

Without loss of generality, let $a_0 \neq 0$; otherwise, factor an s, and using the fact that S is an integral domain, we see that either $s = 0$ or $s^{n-1} + \cdots + a_1 = 0$, and since $s \neq 0$, then we simply choose $s^{n-1} + \cdots + a_1 = 0$ to be our equation. Thus, moving a_0 and factoring s we get:

$$s(s^{n-1} + \cdots + a_1) = -a_0$$

Since R is a field, $\frac{-1}{a_0} \in R$, so:

$$s \cdot \big((-1/a_0)(s^{n-1} + \cdots + a_1)\big) = 1$$

showing that s has a multiplictative inverse. Since s was arbitrary, we have that S is a field.

($\Leftarrow$) Let S be a field and take $0 \neq r \in R$ (automatically implying that R is an integral domain). Since $r \in R$, $r \in S$, and so $r^{-1} \in S$. Since S is an integral extension of R:

$$r^{-n} + a_{n-1}r^{-(n-1)} + \cdots + a_1 r^{-1} + a_0 = 0$$

Multiply the entire expression by r^{n-1}, and move all the terms except the r^{-1} terms to the right hand side to get:

$$r^{-1} = -(a_{n-1} + \cdots + a_1 r^{n-2} + a_0 r^{n-1})$$

which is a linear combination of elements in R, showing that $r^{-1} \in R$, and so R is a field.

Thus, any algebraic field extension is an integral extension.

Example 26.1.9: Integral Extensions

1. As we've just shown, all algebraic extensions of fields are integral extensions.

2. Let R be a Unique Factorization Domain and let $F = \text{Frac}(R)$. Then $\overline{R}^F = R$. The $\supseteq$ direction is obvious, so consider the $\subseteq$ direciton: suppose that $a/b \in \overline{R}^F$ with $\gcd(a, b) = 1$. Then by definition
$$\left(\frac{a}{b}\right)^n + r_{n-1}\left(\frac{a}{b}\right)^{n-1} + \cdots + r_1\frac{a}{b} + r_0 = 0$$
Then:
$$a^n = b(-r_{n-1}a^{n-1} - \cdots - r_1 ab^{n-2} - r_0 b^{n-1})$$
so $b\big|a^n$, and hence $b\big|a$. Since $\gcd(a, b) = 1$, it must thus be that b is a unit, and hence we have that $a/b \in R$, showing that $\overline{R}^F = R$. Hence, all UFD's are normal domains. This should make intuitive sense, since any element $a/b \in \text{Frac}(R)$ would satisfy the equation $bx - a = 0$, which is not-monic.

   More generally, any polynomial ring over a UFD R, $R[x_1, ..., x_n]$ is integrally closed over it's field of fractions $R(x_1, ..., x_n)$ ([25]), showing every polynomial ring over an Unique Factorization Domain is a normal domain.

3. Let S be integral over R and $I \subseteq S$ be an ideal. Then S/I is integral over $R/(R \cap I)$. To see this, notice that a monic polynomial over R where $s \in S$ is a root, when reduced modulo I gives a monic polynomial over $R/(R \cap I)$ where $\overline{s} \in S/I$ is a root.

4. Let k be any field and take $k[x, y]$ (which is integrally closd since it's a UFD). Take the prime ideal $(x^2 - y^3) \subseteq k[x, y]$ (see Dummit and Foote, §9.1, Exercise 14). Consider the integral domain
$$R = k[x, y]/(x^2 - y^3) = k[\overline{x}, \overline{y}]$$
Notice that R is *not* normal: the element $\overline{x}/\overline{y}$ is the solution to the equation
$$\left(\frac{\overline{x}}{\overline{y}}\right)^3 - \overline{x} = 0$$

but evidently $\overline{x}/\overline{y} \notin R$. Thus, R cannot be a Unique Factorization Domain.

If you wish to practice, another example is $\mathbb{Z}[\sqrt{-3}]$. It's integral closure over it's field of fraction's is:

$$\mathbb{Z}\left[\frac{1+\sqrt{-3}}{2}\right]$$

Corollary 26.1.10: Normalization Commutes with Localization

Let R be a domain with fraction field K, $D \subseteq R$ a multiplicative set, and $\overline{R}$ the integral closure of R in K. Then, computing integral closures inside K, $\overline{D^{-1}R} = D^{-1}\overline{R}$.

***Proof*:**
If $x \in \overline{R}$ satisfies $x^n + a_{n-1}x^{n-1} + \cdots + a_0 = 0$ with $a_i \in R$, the same relation holds over $D^{-1}R$, so $D^{-1}\overline{R} \subseteq \overline{D^{-1}R}$. Conversely, if $y \in K$ is integral over $D^{-1}R$, then clearing the denominators in its integral relation shows sy is integral over R for some $s \in D$, so $sy \in \overline{R}$ and $y = (sy)/s \in D^{-1}\overline{R}$. In particular R is normal iff $R_{\mathfrak{m}}$ is normal for every maximal ideal $\mathfrak{m}$.

Locality shows where to compute an integral closure but not whether the answer is manageable. The integral closure of a Noetherian domain in its fraction field can fail to be a finite module, and can even fail to be Noetherian, so finiteness is a genuine theorem and not a formality. For the rings that geometry actually produces, finitely generated algebras over a field, it holds; this is what makes the normalization of a variety a variety again rather than an unwieldy limit.

Theorem 26.1.11: Finiteness of Integral Closure

Let k be a field, let A be a domain that is finitely generated as a k-algebra, and let L be a finite field extension of $\mathrm{Frac}(A)$. Then the integral closure of A in L is a finite A-module. In particular it is a Noetherian ring and is finitely generated as a k-algebra, and taking $L = \mathrm{Frac}(A)$, the normalization $\overline{A}$ of A is a finite A-module.

***Proof*:**
The argument is given in full for k perfect, which is the case every consumer in this series needs; the imperfect case is discussed after the proof.

Step 1: reduce to a polynomial ring. By Noether normalization (theorem 29.1.1) there is a polynomial subalgebra $B = k[x_1, \dots, x_d] \subseteq A$ with A finite, hence integral, over B. An element of L integral over A is then integral over B by transitivity of integral dependence, and conversely $B \subseteq A$, so the integral closures of A and of B in L coincide; it suffices to prove the theorem for B. Note that $\mathrm{Frac}(B) \subseteq \mathrm{Frac}(A) \subseteq L$ with both extensions finite, so $L/\mathrm{Frac}(B)$ is finite. The ring B is a unique factorization domain and hence normal (example 26.1.9).

Step 2: arrange separability. Because k is perfect, the finitely generated field extension L/k is separably generated: some subset of any finite generating set is a transcendence basis $x_1, \dots, x_d$ for which L is a finite separable extension of $k(x_1, \dots, x_d)$. This is proved in *Core Algebra* [25]. Choosing the Noether normalization of Step 1 compatibly with such a basis, we may assume

$L/\mathrm{Frac}(B)$ is separable.

Step 3: the trace form pins the integral closure inside a free module. Let C denote the integral closure of B in L and put $F = \mathrm{Frac}(B)$. Choose an F-basis $e_1, \ldots, e_n$ of L consisting of elements integral over B; this is possible because any F-basis can be scaled into C, an element of L being algebraic over F and hence integral after multiplication by a suitable element of B clearing denominators in its minimal polynomial. Since L/F is separable, the trace form is nondegenerate and there is a dual basis $e_1^*, \ldots, e_n^*$ of L over F with $\mathrm{tr}_{L/F}(e_i e_j^*) = \delta_{ij}$, by *Core Algebra* [25].

Let $c \in C$ and write $c = \sum_j \lambda_j e_j^*$ with $\lambda_j \in F$. Then $\lambda_j = \mathrm{tr}_{L/F}(c\, e_j)$. Now $c\, e_j$ is a product of elements integral over B, hence integral over B, and the trace of an element integral over B lies in B: the trace is, up to sign, a coefficient of the characteristic polynomial of multiplication by that element, which is a power of its minimal polynomial, and the minimal polynomial of an integral element has coefficients in B because B is integrally closed in F. So $\lambda_j \in B$ for every j, and

$$C \;\subseteq\; Be_1^* + \cdots + Be_n^*$$

Step 4: conclude. The right-hand side is a free B-module of rank n, and B is Noetherian by Hilbert's basis theorem, so its submodule C is a finite B-module. Since B is a finitely generated k-algebra, C is a finitely generated k-algebra and a Noetherian ring, as we sought to show.

26.1.12 Note: The Imperfect Case Step 2 is the only place perfectness is used, and it cannot be avoided there: over an imperfect field a finitely generated extension need not be separably generated (*Core Algebra* [25] records a counterexample), so the trace form can degenerate and the argument above stops. The theorem is nonetheless true over any field. The standard route replaces k by a finite purely inseparable extension over which the situation becomes separably generated, and then descends; see Matsumura [37], or [51, Tag 0335], where the result appears in the form that a finite type algebra over a field is a Nagata ring, hence Japanese, which is exactly the finiteness asserted here. No book in this series currently consumes the imperfect case.

26.2 Extension and Contraction of Ideals

Probably the biggest difference between fields and rings is that ideals are nontrivial. Thus, it is meaningful to ask what happens to ideals in (general) ring extensions. Recall that if $f : R \to S$ is a ring homomorphism, and $J \subseteq S$ is an ideal, $f^{-1}(J)$ is an ideal. However, if $I \subseteq R$ is an ideal, $f(I)$ need not be an ideal (take $\mathbb{Z} \hookrightarrow \mathbb{Q}$ with any non-trivial ideal of $\mathbb{Z}$). The next best thing would be to define the smallest ideal of S containing $f(I)$:

Definition 26.2.1: Extension and Contraction

Let $\varphi : R \to S$ be a homomorphism between commutative rings and let $I \subseteq R$, $J \subseteq S$.

1. The *extension* of I to S is the ideal $\varphi(I)S$, commonly denoted I^e
2. The *contraction* of J in R is the ideal $\varphi^{-1}(J)$, commonly denoted J^c

In the case where φ is the natural inclusion $\iota : R \to S$, then the extension of I is IS, and the contraction of J is $J \cap R$. If $I \subseteq R$, there are multiple extension of R, say S/R and T/R, then the extension of I is sometiems denoted I_S or I_T to emphasize to what ring I is being extended too S or T respectively.

Proposition 26.2.2: Properties of Extensions and Contractions

1. $I \subseteq IS \cap R$, or more generally that I is contained in the contraction of its extension to S ($I \subseteq I^{ec}$)
2. $(J \cap R)S \subseteq J$, or more genrally that J is contained in the extension of its contraction in R ($J \supseteq J^{ce}$)
3. $I^e = I^{ece}$
4. $J^c = J^{cec}$
5. If J is prime, J^c is prime. If I is prime, I^e need not be prime (consider $\mathbb{Z} \hookrightarrow \mathbb{Q}$). If $\varphi : R \to S$ is surjective and I is a prime ideal containing $\ker\varphi$, then I^e is prime.
6. If J is maximal, J^c need not be maximal, and similarly if I is maximal then I^e need not be maximal.

***Proof*:**
(1) holds since $I \subseteq \varphi^{-1}(\varphi(I)S) = I^{ec}$, and (2) since $I^{ce} = \varphi(\varphi^{-1}(J))S \subseteq J$. Applying (1) to I^e gives $I^e \subseteq I^{ece}$, and applying $(-)^e$ to (2) gives $I^{ece} \subseteq I^e$, so (3) holds; (4) is dual. For (5): $J^c = \varphi^{-1}(J)$ is prime as a preimage of a prime; over $\mathbb{Z} \hookrightarrow \mathbb{Q}$ the nonzero prime (p) has $(p)^e = \mathbb{Q}$, not a proper ideal; and if φ is surjective with prime $I \supseteq \ker\varphi$, then $I^e = \varphi(I)$ and $S/I^e \cong R/I$ is a domain. (6) is $\mathbb{Z} \hookrightarrow \mathbb{Q}$ again, where (0) is maximal in $\mathbb{Q}$ but not in $\mathbb{Z}$.

Lemma 26.2.3: Extension And Contraction With Common Operations

Let $\mathfrak{a}_1, \mathfrak{a}_2 \subseteq A$ be ideals of A and $\mathfrak{b}_1, \mathfrak{b}_2 \subseteq B$ be ideals of B. Then:

$$
\begin{aligned}
(\mathfrak{a}_1+\mathfrak{a}_2)^e &= \mathfrak{a}_1^e+\mathfrak{a}_2^e & (\mathfrak{b}_1+\mathfrak{b}_2)^c &\supseteq \mathfrak{b}_1^c+\mathfrak{b}_2^c \\
(\mathfrak{a}_1\cap\mathfrak{a}_2)^e &\subseteq \mathfrak{a}_1^e\cap\mathfrak{a}_2^e & (\mathfrak{b}_1\cap\mathfrak{b}_2)^c &= \mathfrak{b}_1^c\cap\mathfrak{b}_2^c \\
(\mathfrak{a}_1\mathfrak{a}_2)^e &= \mathfrak{a}_1^e\mathfrak{a}_2^e & (\mathfrak{b}_1\mathfrak{b}_2)^c &\supseteq \mathfrak{b}_1^c\mathfrak{b}_2^c \\
(\mathfrak{a}_1:\mathfrak{a}_2)^e &\subseteq (\mathfrak{a}_1^e:\mathfrak{a}_2^e) & (\mathfrak{b}_1:\mathfrak{b}_2)^c &\subseteq (\mathfrak{b}_1^c:\mathfrak{b}_2^c) \\
\sqrt{\mathfrak{a}}^e &\subseteq \sqrt{\mathfrak{a}^e} & \sqrt{\mathfrak{b}}^c &= \sqrt{\mathfrak{b}^c}
\end{aligned}
$$

***Proof*:**
Each identity is routine from the definitions of extension and contraction (Atiyah-Macdonald, Chapter 1); the inclusions that are not equalities are strict already for $\mathbb{Z} \hookrightarrow \mathbb{Q}$.

If $\varphi : R \to S$ is a ring homomorphism, we may decompose φ into the following:

$$R \xrightarrow{p} \varphi(R) \xhookrightarrow{j} S$$

where p maps R surjectively onto $\varphi(R)$, and j injects into S. Relating the ideals of R to the ideals of $\varphi(R)$ is very easy:

Proposition 26.2.4: Correspondence Between Ideals And Image Of Ideals

Let $\varphi : R \to S$ be a ring homomorphism. Then there exists a bijective correspondence:

$$\{I \subseteq \varphi(R)\} \underset{\varphi(-)}{\overset{(-)^c}{\rightleftarrows}} \{\ker(\varphi) \subseteq J \subseteq R\}$$

Furthermore, prime ideals of $\varphi(R)$ correspond to prime ideals of R containing $\ker(\varphi)$.

***Proof*:**
Since $\varphi(R) \cong R/\ker\varphi$, this is the correspondence theorem for the quotient $R \twoheadrightarrow R/\ker\varphi$: ideals of $R/\ker\varphi$ correspond to ideals of R containing $\ker\varphi$ via image and preimage, and primes correspond to primes.

On the other hand, the correspondence between $\varphi(R)$ and S is much harder, and is covered in section 26.3 when S/R is an integral extension. As shown in proposition 26.2.2, every prime ideal "lies over" another prime ideal. This motivates the following vocuabilary:

Definition 26.2.5: Lying Over

Let $I \subseteq S$ be an ideal and S/R an integral extension. If $J = I \cap R$, I is said to be *lying over* J.

Example 26.2.6: Extending ideals
Take $\varphi : \mathbb{Z} \to \mathbb{Z}[i]$ to be the inclusion. Then a prime ideal $(p) \subseteq \mathbb{Z}$ may or may not remain prime when extended to $\mathbb{Z}[i]$. Using Fermat's theorem of sums of squares (theorem 10.3.3), it can be deduced that:

1. $(2)^e = (1+i)^2$
2. If $p \equiv 1 \bmod 4$, then p^e is the product of two distinct prime ideals (ex. $(5)^e = (2+i)(2-i)$)
3. if $p \equiv 3 \bmod 4$, then p^e is prime in $\mathbb{Z}[i]$

In general, it is rather hard to determine when $\varphi(p)$ is prime when p is prime. If $\varphi : R \to S$ is an integarl extension, then the situation becomes much better, however it will require the notion of *localization* which shall be covered in the next section. One special case is note mentioning:

Proposition 26.2.7: Maximal Ideals Lying Over

Let S/R be an integral extension and $\mathfrak{m} \subseteq S$ a prime ideal. Then $\mathfrak{m} \cap R$ is maximal if and only if $\mathfrak{m}$ is maximal.

***Proof*:**
Consider the integral domain $S/\mathfrak{m}$. Then $S/\mathfrak{m}$ is an integral extension of $R/(\mathfrak{m} \cap R)$. Then by proposition 26.1.8, $S/\mathfrak{m}$ is a field if and only if $R/(\mathfrak{m} \cap R)$ is a field. But then $\mathfrak{m}$ is maximal if and only if $\mathfrak{m} \cap R$ is maximal.

26.3 Integral Extensions and Prime Ideals

We return to the question of the extension of ideals, in particular prime ideals, given S/R. The going-down half rests on the following, where integrality over an *ideal* I means being a root of a monic $x^n + a_{n-1}x^{n-1} + \cdots + a_0$ with all $a_i \in I$.

Lemma 26.3.1: Integral over an Ideal

Let R be a normal domain with fraction field K, let $S \supseteq R$ be a domain, and let $I \subseteq R$ be an ideal. If $x \in S$ is integral over I, then x is algebraic over K and the non-leading coefficients of its minimal polynomial over K lie in $\sqrt{I}$.

***Proof*:**
The integral relation shows x is algebraic over K; let $m(t) = t^r + c_{r-1}t^{r-1} + \cdots + c_0$ be its minimal polynomial over K. In a splitting field the conjugate roots $x = x_1, \ldots, x_r$ of m each satisfy the same relation with coefficients in I, so all x_j are integral over I; hence so are the c_i, being (up to sign) elementary symmetric functions of the x_j. Each $c_i \in K$ is integral over R, so $c_i \in R$ since R is normal; and $c_i \in R$ integral over I satisfies $c_i^m + b_{m-1}c_i^{m-1} + \cdots + b_0 = 0$ with $b_j \in I$, giving $c_i^m \in I$, i.e. $c_i \in \sqrt{I}$.

Theorem 26.3.2: Integral Extension Prime Correspondence (Cohen-Seidenberg Theorem)

Let S be an integral extension of the integral domain R

1. (**Lying-over Property**) Let $\mathfrak{p} \subseteq R$ be a prime ideal. Then there is a prime ideal $P \subseteq S$ such that $\mathfrak{p} = P \cap R$. Furthermore, $\mathfrak{p}$ is maximal if and only if P is maximal.
2. (**Going-up Property**) Let $p_1 \subseteq p_2 \subseteq \cdots \subseteq p_n$ be a chain of prime ideals of R, and suppose there exists a chain $q_1 \subseteq q_2 \subseteq \cdots \subseteq q_m$, $(m < n)$ such that $p_i = q_i \cap R$. Then there exists prime ideals $q_{m+1}, ..., q_n$ that complete the chain: $q_m \subseteq q_{m+1} \subseteq \cdots \subseteq q_n$, $p_i = q_i \cap R$ for all i.
3. (**Going-down Property**) Suppose in addition that S is a domain. If R is a normal domain, $p_1 \supseteq p_2 \supseteq \cdots \supseteq p_n$ be a chain of prime ideals of R, and suppose there exists a chain $q_1 \supseteq q_2 \supseteq \cdots \supseteq q_m$, $(m < n)$ such that $p_i = q_i \cap R$. Then there exists prime ideals $q_{m+1}, ..., q_n$ that complete the chain: $q_m \supseteq q_{m+1} \supseteq \cdots \supseteq q_n$, $p_i = q_i \cap R$ for all i.
4. (**Incomparability property**) Let q_1 and q_2 lie over p. Then either $q_1 = q_2$, or they do not contain each other: $q_1 \not\subseteq q_2$ and $q_2 \not\subseteq q_1$

***Proof*:**

1. We'll take advantage of the corresponence theorem for rings and their localization. Let $D = R - P$. Then as we saw, D is multiplicatively closed, and so $R[D^{-1}]$ is well-defined. Since D is also a multiplicatvely closed set of S, $S[D^{-1}]$ is also well-defined. Furthermore, $S[D^{-1}]/R[D^{-1}]$ is an integral extensions of $R[D^{-1}]$: for any $s \in S[D^{-1}]$, if $s \in S$, clearly S is still integral. If $t = \frac{s}{d}$ for some $s \in S$ and $d \in D$, then letting
$$s^n + a_{n-1}s^{n-1} + \cdots + a_1 s + a_0 = 0$$
This equation is still zero if we multiply by d^{-n}:
$$\frac{s^n}{d^n} + \frac{a_{n-1}s^{n-1}}{d^n} + \cdots + \frac{a_1 s}{d^n} + \frac{a_0}{d^n} = 0$$
and shuffling the coefficients we get:
$$\frac{s^n}{d^n} + \frac{da_{n-1}s^{n-1}}{d^{n-1}} + \cdots + \frac{d^{n-1}a_1 s}{d} + \frac{a_0}{d^n} = 0$$
and replacing each $d^{n-i}a_i = b_i$, and $s/d = t$,
$$t^n + b_{n-1}t^{n-1} + \cdots + b_1 t + b_0 = 0$$
showing that t is integral over $R[D^{-1}]$, and so $S[D^{-1}]$ is an integral extension of $R[D^{-1}]$. Let $m \subseteq S[D^{-1}]$ be a maximal ideal. Then by the corresponence of ideals from a ring and it's localization, we have that $m \cap R_p$ is a maximal ideal in R_p, and $m \cap R_p$ is the extension of P to the local ring R_p, so it contracts to p. It might be easier to see if we

have a visual:

$$\begin{array}{ccc} S & \xrightarrow{\pi} & S[D^{-1}] \\ {\scriptstyle\iota}\uparrow & & {\scriptstyle\iota}\uparrow \\ R & \xrightarrow{\pi} & R[D^{-1}] = R_p \end{array}$$

The maximal ideal m is maximal in $S[D^{-1}]$, it's pre-image is a maximal ideal of $R[D^{-1}]$, and the pre-image of that map is p, showing that $p = m \cap R$, as we sought to show.

2. This can be reducd to showing that if $p_1 \subseteq p_2$ and q_1 lies above p_1 such that $p_1 = q_1 \cap R$, then there exists a q_2 that lies above p_2 ($q_2 = p_2 \cap R$) and $q_1 \subseteq q_2$. First, let $\overline{S} = S/q_1$ and $\overline{R} = R/p_1$. Then $\overline{S}/\overline{R}$ is an integral extension by similar reasoning to what we've shown in part 1. By the ideal-correspondence for the quotient $R \to R/p_1$ (proposition 26.2.4), p_2/p_1 is a prime ideal of $\overline{R}$. Then by part (1), we have that there exists an ideal $q_2/q_1 \subseteq \overline{S}$ that lies abouve p_2/p_1, in particular:
$$q_2/q_1 \cap \overline{R} = p_2/p_1$$
At this point we are essentially done: taking the pre-image of q_2/q_1, we get q_2, and by the correspondence theorem we get
$$q_2 \cap R = p_2$$

3. It suffices to find, given $p_1 \supseteq p_2$ in R and q_1 over p_1, a prime $q_2 \subseteq q_1$ over p_2. Localize S at the multiplicative set $S \setminus q_1$; it is enough that $p_2 S_{q_1} \cap R = p_2$, for then a prime of S_{q_1} over p_2 contracts to the desired $q_2 \subseteq q_1$. The inclusion $\supseteq$ is clear. For $\subseteq$, take $x \in p_2 S_{q_1} \cap R$ and write $x = y/s$ with $y \in p_2 S$, $s \in S \setminus q_1$. Now $y \in p_2 S$ is integral over p_2, so by lemma 26.3.1 its minimal polynomial over $K = \mathrm{Frac}(R)$ has non-leading coefficients in $\sqrt{p_2} = p_2$: say $y^r + u_{r-1} y^{r-1} + \cdots + u_0 = 0$ with $u_i \in p_2$. Then $s = y/x$ has minimal polynomial $s^r + v_{r-1} s^{r-1} + \cdots + v_0$ with $v_i = u_i / x^{r-i} \in R$ (as s is integral over the normal R), and $x^{r-i} v_i = u_i \in p_2$. Were $x \notin p_2$, each $v_i \in p_2$, so $s^r = -(v_{r-1} s^{r-1} + \cdots + v_0) \in p_2 S \subseteq q_1$, forcing $s \in q_1$, a contradiction. Hence $x \in p_2$, giving $p_2 S_{q_1} \cap R = p_2$.

4. For the sake of contradiction, let $q_1 \subseteq q_2$, and let both lie over p: $q_1 \cap R = q_2 \cap R = p$. Consider $\overline{S} = S/q_1$ and $\overline{R} = R/p$. Then as we've shown earlier, $\overline{S}/\overline{R}$ is an integral extension, and $q_2/q_1 \subseteq \overline{S}$ is a prime ideal. Consider:
$$(q_2/q_1) \cap \overline{R} = (q_2 \cap R)/(R \cap q_1) = (R \cap q_1)/(R \cap q_1) = \{\overline{0}\}$$
Now $\overline{S}$ is a domain integral over the domain $\overline{R}$, and the prime q_2/q_1 meets $\overline{R}$ only in $\{0\}$. If $q_2/q_1 \neq 0$, pick $0 \neq \bar{x} \in q_2/q_1$ satisfying an integral relation $\bar{x}^n + \overline{a}_{n-1} \bar{x}^{n-1} + \cdots + \overline{a}_0 = 0$ of least degree; then $\overline{a}_0 = -\bar{x}(\bar{x}^{n-1} + \cdots + \overline{a}_1) \in (q_2/q_1) \cap \overline{R} = \{0\}$, so $\overline{a}_0 = 0$, and cancelling $\bar{x}$ (valid in the domain $\overline{S}$) contradicts minimality. Hence $q_2/q_1 = \{0\}$, i.e. $q_2 = q_1$.

Exercise 26.3.1

1. Let $I \subseteq R$ be an ideal and R a ring. Then if we consider integral extension of *rngs*, show that I^{n+1}/I^n is an integral extension.

2. Let $I \subseteq R$, $f : R \to S$ an integral extension and $I^e \subseteq S$ the extension of S. Then:

$$S/I^e \cong S \otimes_R R/I$$

27

Valuation Ring and Local Rings

Recall from the theory of absolute values ([19]) that an absolute value defines a notion of size on a field. On a finite field the only absolute value is the trivial one ($|x| = 1$ for $x \neq 0$), and by Ostrowski's theorem the only absolute values on $\mathbb{Q}$ up to equivalence are the p-adic $|-|_p$ and the usual $|-|_\infty$. These were defined using a constant c^{-m}; the choice of constant does not affect the topology, so what really matters is the exponent m. This section defines a function capturing that exponent, a useful link between geometric and algebraic intuition on certain fields.

27.1 Valuations

Definition 27.1.1: $\mathbb{R}$-Valuation

Let K be a field. Append ∞ to $\mathbb{R}$ and preserve the total ordering by adding the properties that $g \leq \infty$ for all $g \in \mathbb{R}$ and $g + \infty = \infty + g = \infty + \infty = \infty$. Then a function $\nu : K \to \mathbb{R} \cup \{\infty\}$ is called a $\mathbb{R}$-*valuation* if

1. $\nu(K^\times) \subseteq \mathbb{R}$ and $\nu|_{K^\times} : K^\times \to \mathbb{R}$ is a group-homomorphism, that is $\nu(x \cdot y) = \nu(x) + \nu(y)$.
2. $\nu(x + y) \geq \min\{\nu(x), \nu(y)\}$
3. $\nu(x) = \infty$ if and only if $x = 0$

It may be more clear that $\mathbb{R}$-valuations come as a generalization of non-Archimedean absolute values from the original definition given by Emil Artin (and hence why these are called valuations). To make sense of this definition, we shall introduce the slightly more general notion of a totally ordered abelian group.

Definition 27.1.2: Totally Ordered [Abelian] Group

Let G be an abelian group. Then G is said to be a *totally ordered group*, or simply an *ordered group* if $\leq$ is a total ordering on G that is translation invariant, that is for all $x, y, z \in G$ where $x \leq y$:

$$zx \leq zy$$

An abelian group admits a translation-invariant total order if and only if it is torsion-free. General ordered groups genuinely extend the real-valued case, but for our purposes valuations into $\mathbb{Z}$ or $\mathbb{R}$ suffice; the general notion mainly serves to connect $\mathbb{R}$-valuations to non-Archimedean absolute values ([19]):

Example 27.1.3: Value Groups

The field of *Puiseux series* $\bigcup_{n\geq 1} k((x^{1/n}))$ carries a valuation with value group $\mathbb{Q}$, given by $\nu(x^a) = a$. More generally, Hahn series over a totally ordered group Γ realize Γ as a value group.

Now, in definition 27.1.1, the $(\mathbb{R}, +)$ can be replaced with $(G, *)$. The following shows how this links to absolute values:

Definition 27.1.4: G-Valuation Equivalent Definition

Let K be a field. Append 0 to G and preserve the total ordering by adding the properties that $0 \leq g$ for all $g \in G$ and $0 * g = g * 0 = 0 * 0 = 0$. Then a function $\nu : K \to G \cup \{0\}$ is called a *G-valuation* if

1. $\nu(K^\times) \subseteq G$ and $\nu|_{K^\times} : K^\times \to G$ is a group-homomorphism, that is $\nu(x \cdot y) = \nu(x) * \nu(y)$.
2. $\nu(x + y) \leq \max\{\nu(x), \nu(y)\}$
3. $\nu(x) = 0$ if and only if $x = 0$

The image of $\nu(K^\times)$ is called the *value group*.

If $G = \mathbb{Z}$, or equivalently in the $\mathbb{R}$-valuation definition, when $v(K^*) \subseteq \mathbb{Z}$, a special name is given:

Definition 27.1.5: Discrete Valuation

Let K be a field. Then a function $\nu : K \to \mathbb{Z} \cup \{\infty\}$ is called a *discrete valuation* if

1. $\nu(x \cdot y) = \nu(x) + \nu(y)$
2. $\nu(x + y) \geq \min\{\nu(x), \nu(y)\}$
3. $\nu(x) = \infty$ if and only if $x = 0$

Example 27.1.6: Valuations

1. Given any non-Archimedean absolute value v_p with values in $(\mathbb{R}^+, \cdot)$, we get a $\mathbb{R}^+$ valuation. Hence, each v_p is a valuation on $\mathbb{Q}$ in the equivalent definition, and v_∞, v_0 are valuations on $k(x)$ for any field k.

Given a field k, it may be asked what valuations may be put on it. This is in correspondence with certain sub-rings of k:

Definition 27.1.7: Valuation Ring

Let ν be a valuation on K. Then

$$\mathcal{O}_K = \{x \in K \mid \nu(x) \geq 0\}$$

forms a ring, called the *Valuation Ring.* If ν is a discrete valuation, the ring is called a *discrete valuation ring.*

The reader should verify this is indeed a ring.

Proposition 27.1.8: Valuation Ring Properties

Let k be a field with valuation ν and let $\mathcal{O}_k$ be the valuation ring associated to it. Then:

1. Given $x \in k \setminus \{0\}$, either $x \in \mathcal{O}_k$ or $x^{-1} \in \mathcal{O}_k$ (where both are possible at once). As a consequence, k is a field of fractions of $\mathcal{O}_k$. [a]
2. The group of units is $\mathcal{O}_K^* = \{x \in k \mid \nu(x) = 0\}$ (equiv. $\nu(x) = 1$)
3. $\mathcal{O}_k$ has exactly one maximal ideal $\mathfrak{m}_v = \{x \in k \mid v(x) > 0\} = \mathcal{O}_k \setminus \mathcal{O}_k^*$ (equiv $v(x) < 1$)
4. The ideal of $\mathcal{O}_k$ form a total ordering under $\subseteq$
5. The ring $\mathcal{O}_k$ is integrally closed (i.e. a normal domain)

[a] Note that the either-or implies a general Γ-valuation!

***Proof*:**
(1) For $x \neq 0$, either $\nu(x) \geq 0$ or $\nu(x^{-1}) = -\nu(x) \geq 0$, so $x \in \mathcal{O}_k$ or $x^{-1} \in \mathcal{O}_k$; hence $k = \mathrm{Frac}(\mathcal{O}_k)$. (2) x is a unit iff $x, x^{-1} \in \mathcal{O}_k$ iff $\nu(x) = 0$. (3) By (2) the non-units form $\mathfrak{m}_\nu = \{x \mid \nu(x) > 0\}$, which is an ideal ($\nu(x+y) \geq \min(\nu(x), \nu(y)) > 0$ and $\nu(rx) \geq \nu(x) > 0$); being exactly the non-units, it is the unique maximal ideal.

For 4, first show it on principal ideals, namely if $(a), (b) \subseteq \mathcal{O}_K$, then $(a) \subseteq (b)$ or $(b) \subseteq (a)$. Do this by noticing that either $ab^{-1} \in \mathcal{O}_K$ or $a^{-1}b \in \mathcal{O}_K$. For general ideals, if $\mathfrak{a}$ and $\mathfrak{b}$ are ideals that are not ordered, then there exists $a \in \mathfrak{a} \setminus \mathfrak{b}$ and $b \in \mathfrak{b} \setminus \mathfrak{a}$. Consider $(a), (b)$ which must be ordered namely $(a) \subseteq (b)$ or $(b) \subseteq (a)$. Say $(a) \subseteq (b)$. Then $(a) \cap (b) = (a)$. But then $a \in (b)$ meaning $a = bx$ for some $x \in \mathcal{O}_K$. But as $bx \in \mathfrak{b}$, this would imply $a \notin \mathfrak{a} \setminus \mathfrak{b}$, a contradiction.

We shall show that $\mathcal{O}_K$ is integrally closed. Say that $x \in K$ is integral over $\mathcal{O}_K$, for the sake of contradiction say $x \notin \mathcal{O}_K$. As it is integral:

$$x^n + a_1 x^{n-1} + \cdots + a_n = 0$$

with $a_i \in \mathcal{O}_K$. If $x \notin \mathcal{O}_K$ then $\nu(x) < 0$, so $x^{-1} \in \mathcal{O}_K$; dividing the relation by x^{n-1} gives

$$x = -(a_1 + a_2 x^{-1} + \cdots + a_n x^{-(n-1)}),$$

whose right-hand side lies in $\mathcal{O}_K$ (each $x^{-j} \in \mathcal{O}_K$), forcing $x \in \mathcal{O}_K$, a contradiction, completing the proof.

These properties in fact characterize valuation rings:

Proposition 27.1.9: Characterizing Valuation Ring

Let k be a field, $R \subseteq k$ a sub-ring, and R^* the group of units of R. Then the following are equivalent:

1. R is a valuation ring of k
2. $k = \mathrm{Frac}(R)$ and the ideals of R form a chain under inclusion
3. for every $x \in k \setminus \{0\}$, either $x \in R$ or $x^{-1} \in R$

Then $G = (k \setminus \{0\})/R^*$ is a totally ordered abelian group and

$$\nu_R : x \mapsto xR^*$$

is a valuation on k. The induced valuation ring is R. As a consequence, two valuations on the same field are equivalent if and only if they have the same valuation ring.

***Proof*:**
(1) $\Rightarrow$ (3) is proposition 27.1.8(1), and (1) $\Rightarrow$ (2) its parts (1) and (4). For (3) $\Rightarrow$ (2): (3) gives $k = \mathrm{Frac}(R)$, and any two ideals $\mathfrak{a}, \mathfrak{b}$ are comparable, for if $a \in \mathfrak{a} \setminus \mathfrak{b}$ and $b \in \mathfrak{b}$ then $a/b \notin R$ (else $a \in (b) \subseteq \mathfrak{b}$), so $b/a \in R$ and $b \in (a) \subseteq \mathfrak{a}$, giving $\mathfrak{b} \subseteq \mathfrak{a}$. For (2) $\Rightarrow$ (3): for $x = a/b \in k^\times$ the principal ideals $(a), (b)$ are comparable, so $a/b \in R$ or $b/a \in R$. Given (3), $G = k^\times/R^*$ is totally ordered by $xR^* \leq yR^* \Leftrightarrow y/x \in R$, and $\nu_R(x) = xR^*$ is a valuation whose ring is $\{x \in k \mid \nu_R(x) \geq 0\} = R$; a valuation is determined up to equivalence by its ring.

27.2 Discrete Valuation Ring

Limiting to discrete valuation rings, then there is a smallest positive value s such that:

$$\nu(K^*) = s\mathbb{Z}$$

We can normalize this result and get an equivalent ν where $\nu(K^*) = \mathbb{Z}$ and $\mathcal{O}_K$, $\mathcal{O}_K^*$, and $\mathfrak{p}$ are the same. Then there must be an element such that

$$\nu(\pi) = 1$$

Labelling this element as π, we can use it to show that any $x \in K \setminus \{0\}$ can be written as :

$$x = u\pi^m$$

for a unit $u \in K^\times$ $(\nu(u) = 0)$. To see this, let

$$\nu(x) = m \Rightarrow \nu(x\pi^{-m}) = 0 \Rightarrow u = x\pi^{-m} \in \mathcal{O}_K^*$$

which gives the result, where $m = \nu(x) \in \mathbb{Z}$. This in fact characterizes DVRs:

Proposition 27.2.1: Characterizing DVR

Let R be an integral domain and $k = \mathrm{Frac}(R)$. Then R is the valuation ring of a discrete valuation ν on k if and only if R is a PID with a unique nonzero prime ideal.

***Proof*:**
($\Rightarrow$) Normalize ν so $\nu(k^\times) = \mathbb{Z}$, with uniformizer π ($\nu(\pi) = 1$). Every nonzero $x \in R$ is $u\pi^n$ with u a unit and $n = \nu(x) \geq 0$, so every nonzero ideal equals $(\pi^n) = \mathfrak{m}^n$ for the least n occurring; thus R is a PID with unique nonzero prime $\mathfrak{m} = (\pi)$. ($\Leftarrow$) If R is a PID with unique nonzero prime (π), each nonzero $x \in R$ factors uniquely as $u\pi^n$; setting $\nu(x) = n$ and extending by $\nu(a/b) = \nu(a) - \nu(b)$ gives a discrete valuation on k with valuation ring R.

The DVRs are exactly the "nicest" one-dimensional local rings, characterized among Noetherian local domains of dimension one in several equivalent ways:

Theorem 27.2.2: Characterizations of a DVR

Let $(R, \mathfrak{m})$ be a Noetherian local domain of Krull dimension 1 with fraction field k. The following are equivalent:

1. R is a discrete valuation ring;
2. R is integrally closed (normal);
3. $\mathfrak{m}$ is principal;
4. every nonzero ideal of R is a power of $\mathfrak{m}$;
5. R is a regular local ring, i.e. $\dim_{R/\mathfrak{m}} \mathfrak{m}/\mathfrak{m}^2 = 1$.

***Proof*:**
(1) $\Rightarrow$ (2) is proposition 27.1.8(5), and (3) $\Leftrightarrow$ (5) holds since, by Nakayama (lemma 23.4.29), $\mathfrak{m}$ is principal iff $\dim_{R/\mathfrak{m}} \mathfrak{m}/\mathfrak{m}^2 = 1$. For (2) $\Rightarrow$ (3): pick $0 \neq a \in \mathfrak{m}$; as $\dim R = 1$, $\mathfrak{m}$ is the only prime over (a), so $\mathfrak{m}^n \subseteq (a)$ for some least n. Choose $b \in \mathfrak{m}^{n-1} \setminus (a)$ and put $t = a/b \in k$; then $b\mathfrak{m} \subseteq \mathfrak{m}^n \subseteq (a)$ gives $t^{-1}\mathfrak{m} = (b/a)\mathfrak{m} \subseteq R$. If $t^{-1}\mathfrak{m} \subseteq \mathfrak{m}$, the finitely generated faithful module $\mathfrak{m}$ would be stable under t^{-1}, making t^{-1} integral over R (determinant trick) and hence $t^{-1} \in R$ by normality, contradicting $b \notin (a)$. So $t^{-1}\mathfrak{m} = R$, i.e. $\mathfrak{m} = tR$ is principal. (3) $\Rightarrow$ (4): writing $\mathfrak{m} = (\pi)$, Krull's intersection theorem (corollary 25.4.11) gives $\bigcap_n \mathfrak{m}^n = 0$, so any nonzero ideal is $(\pi^n) = \mathfrak{m}^n$ for the largest n with the ideal $\subseteq \mathfrak{m}^n$. (4) $\Rightarrow$ (1): defining $\nu(x) = n$ where $(x) = \mathfrak{m}^n$ gives a discrete valuation on k with valuation ring R.

Example 27.2.3: Discrete Valuation Ring

1. The ring of p-adic integers $\mathbb{Z}_p$ is a discrete valuation ring.
2. For any prime p, $\mathbb{Z}_{(p)}$ the localization of $\mathbb{Z}$ at the prime (p) is a discrete valuation ring.

3. The ring $k[[x]]$ of formal power series is a DVR with unique non-zero prime ideal (x).
4. (If you know some analysis) The ring of convergent power series over $\mathbb{R}$ or $\mathbb{C}$ (those with a positive radius of convergence) is a DVR.
5. (If you know some complex analysis) Let X be a connected Riemann surface and let $M(X)$ be the field of meromorphic functions $X \to \mathbb{C} \cup \{\infty\}$. For each point $p \in X$, define a discrete valuation on $M(X)$ where
$$\nu_p(f) = i$$
where i is the largest number st
$$\frac{f(z)}{(z-p)^i}$$
can be extended to a holomorphic function (i.e. the order of the root/pole).

It shall be seen in algebraic geometry ([5]) that DVRs appear naturally in the study of smooth projective algebraic curves".

Proposition 27.2.4: Representation Of Elements Of a Complete DVR

Let k be a field with *complete* DVR $\mathcal{O}_k$, uniformizer π, and let $\mathcal{T} \subseteq \mathcal{O}_k$ contain 0 and exactly one element of every coset of $\mathfrak{m}$. Then every element $x \in \mathcal{O}_k$ is a unique convergent series
$$\sum_{i \geq 0} r_i \pi^i$$
with coefficients $r_i \in \mathcal{T}$, and every nonzero $x \in k$ is a unique Laurent series $\sum_{i \geq m} r_i \pi^i$ with $r_i \in \mathcal{T}$ for all $i \geq m$ and $r_m \neq 0$

***Proof*:**
By completeness, build the series by repeatedly subtracting coset representatives: let $r_0 \in \mathcal{T}$ represent x mod $\mathfrak{m}$, so $x - r_0 \in \mathfrak{m} = (\pi)$; write $x - r_0 = \pi x_1$ and repeat with x_1. The partial sums $\sum_{i<N} r_i \pi^i$ are Cauchy and converge to x, and the r_i are forced as coset representatives, giving uniqueness. The Laurent case follows by factoring out π^m with $m = \nu(x)$.

Valuation rings are the maximal objects in the domination order, which yields a clean description of integral closure.

Proposition 27.2.5: Domination and Existence of Valuation Rings

Let k be a field. Every local subring $A \subseteq k$ is dominated by a valuation ring $\mathcal{O}$ of k (that is, $A \subseteq \mathcal{O}$ and $\mathfrak{m}_{\mathcal{O}} \cap A = \mathfrak{m}_A$). Consequently, for a domain A with fraction field k, the integral closure of A in k is the intersection of the valuation rings of k containing A:
$$\overline{A} = \bigcap_{A \subseteq \mathcal{O}} \mathcal{O},$$
so A is integrally closed iff it is an intersection of valuation rings.

***Proof*:**
Order the local subrings of k dominating A by domination; the union of a chain dominates A, so by Zorn there is a maximal one $\mathcal{O}$. It is a valuation ring: suppose $x \in k$ with $x, x^{-1} \notin \mathcal{O}$. Not both of $\mathfrak{m}_{\mathcal{O}}\mathcal{O}[x]$ and $\mathfrak{m}_{\mathcal{O}}\mathcal{O}[x^{-1}]$ are the unit ideal, for otherwise $1 = \sum_{i \leq n} a_i x^i = \sum_{j \leq m} b_j x^{-j}$ with $a_i, b_j \in \mathfrak{m}_{\mathcal{O}}$ and n, m minimal, and eliminating the top power of x between the two relations lowers one of the degrees, a contradiction. Say $\mathfrak{m}_{\mathcal{O}}\mathcal{O}[x] \neq \mathcal{O}[x]$ and let $\mathfrak{n}$ be a maximal ideal of $\mathcal{O}[x]$ containing it; then $\mathcal{O}[x]_{\mathfrak{n}}$ is a local ring strictly dominating $\mathcal{O}$ (it contains $x \notin \mathcal{O}$), contradicting maximality. Every valuation ring is integrally closed (proposition 27.1.8), so $\overline{A} \subseteq \bigcap \mathcal{O}$. Conversely, if $x \in k$ is not integral over A, then x^{-1} is a non-unit of $A[x^{-1}]$, hence lies in a maximal ideal $\mathfrak{n}$; the localization $A[x^{-1}]_{\mathfrak{n}}$ is dominated by a valuation ring $\mathcal{O} \supseteq A$ with $x^{-1} \in \mathfrak{m}_{\mathcal{O}}$, so $\nu(x) < 0$ and $x \notin \mathcal{O}$. Thus $\bigcap \mathcal{O} \subseteq \overline{A}$.

Proposition 27.2.6: Extension of Valuations

Let $K \subseteq L$ be a field extension and ν a valuation on K with valuation ring $\mathcal{O}_\nu$. Then ν extends to a valuation on L.

***Proof*:**
The local ring $\mathcal{O}_\nu \subseteq L$ is dominated by a valuation ring $\mathcal{O}$ of L (proposition 27.2.5). Then $\mathcal{O} \cap K$ is a valuation ring of K dominating $\mathcal{O}_\nu$, hence equals $\mathcal{O}_\nu$, so the valuation of $\mathcal{O}$ restricts to ν on K.

Exercise 27.2.1

1. Show that an abelian group admits a translation-invariant total order if and only if it is torsion-free.
2. Let $\mathcal{O}$ be a DVR with unique prime $\mathfrak{p}$. Show that $\mathfrak{p}^n/\mathfrak{p}^{n+1} \cong \mathcal{O}/\mathfrak{p}$ for $n \geq 0$
3. Let $x \in \mathfrak{m}$ be a non-unit of a complete DVR $\mathcal{O}$. Show that $1 - x$ is a unit, its inverse being the convergent series $\sum_{n \geq 0} x^n$.

28

Artinian Rings

Being Noetherian brings a consistent notion of finiteness to rings, letting us define concepts like dimension. It is natural to ask what the dual *descending chain condition* yields. This chapter studies rings and modules satisfying the D.C.C.; it turns out to be far more restrictive than the A.C.C., forcing an Artinian ring to be zero-dimensional and, in fact, a finite product of local Artinian rings.

28.1 Artinian Rings and Modules

Definition 28.1.1: Artinian module and ring

Let M be a R-module. Then if M satisfies the *descending chain condition* on submodules, it is an *artinian module*. More precisely, for any collection of submodules such that

$$M_1 \supseteq M_2 \supseteq \cdots \supseteq M_k \supseteq \cdots$$

then for some n, we get $M_n = M_{n+1} = \cdots$.
If we let $M = R$ be a module over itself, then we say that R is an *Artinian ring*.

Naturally, an Artinian ring has the D.C.C. condition on its ideals.

Proposition 28.1.2: Artinian Module Equivalent

M is Artinian if and only if every non-empty set of submodules of M contains a minimal element under inclusion

***Proof*:**
If M satisfies the D.C.C. and a nonempty set $\mathcal{S}$ of submodules had no minimal element, choose $M_1 \in \mathcal{S}$ and inductively $M_{i+1} \subsetneq M_i$ in $\mathcal{S}$, an infinite strictly descending chain, a contradiction. Conversely, if every nonempty set has a minimal element, a descending chain $M_1 \supseteq M_2 \supseteq \cdots$ has a minimal M_n, whence $M_n = M_{n+1} = \cdots$.

Just like Noetherian modules, a quotient of an Artinian module is Artinian.

Lemma 28.1.3: Artinian Module Closed Under Quotient

Let M be an Artinian module and $N \subseteq M$. Then M/N is also Artinian

***Proof*:**
A descending chain of submodules of M/N pulls back, via the correspondence for $M \twoheadrightarrow M/N$, to a descending chain of submodules of M containing N, which stabilizes; hence the chain in M/N stabilizes.

Example 28.1.4: Artinian, But Not Noetherian Module
Take $\mathbb{Z}[1/2]/\mathbb{Z}$ as a $\mathbb{Z}$-module. Its submodules form the increasing chain

$$0 \subsetneq \tfrac{1}{2}\mathbb{Z}/\mathbb{Z} \subsetneq \tfrac{1}{4}\mathbb{Z}/\mathbb{Z} \subsetneq \cdots, \qquad \tfrac{1}{2^n}\mathbb{Z}/\mathbb{Z} \cong \mathbb{Z}/2^n\mathbb{Z},$$

and these are all of its submodules. Any descending chain among them is eventually constant, so the module is Artinian, while the increasing chain shows it is not Noetherian. (It is a module, not a ring.)

However, unlike Noetherian rings, Artinian rings need to satisfy stronger properties on their ideals:

Proposition 28.1.5: Artinian, Prime $\Rightarrow$ Maximal

Let R be an Artinian ring and $P \subseteq R$ be a prime ideal. Then P is maximal

Notice how this differs from Noetherian rings: $\mathbb{Z}[x]$ is Noetherian, (x) is prime, but not maximal since $(x) \subsetneq (2, x)$. Thus, being Artinian immediately imposes stronger conditions on its ideals.

***Proof*:**
Consider R/P, which is also artinian by lemma 28.1.3. We'll show $R/P = F$ is a field. COnsider $x \in F \setminus \{0\}$. Then by the descendign chain condition:

$$(x) \supseteq (x^2) \supseteq \cdots$$

must staxalize, that is for some n:

$$(x^n) = (x^{n+1})$$

which implies $x^n = x^{n+1}c$. By the Cancellion law, we get $1 = xc$, showing that x is in fact a unit. Since every element is a unit in F, we have that F is in fact a field, and so P is maximal, as we sought to show.

Theorem 28.1.6: Structure Theorem for Artinian Rings

Let R be an Artinian ring. Then:

1. There are only finitely many maximal ideals
2. The quotient $R/(\text{Jac}(R))$ is the direct product of a finite number of fields, where if there are n maximal ideals, we have a direct product of n fields R/M_i.
3. Every prime ideal of R is maximal. Thus $\text{kdim}(R) = 0$. As a consequence, $\text{Jac}(R) = \text{Nil}(R)$ and is a nilpotent ideal, $(\text{Jac}(R))^n = 0$ for some $n \geq 1$
4. R is isomorphic to a direct product of a finite number of local artinian rings
5. Every Artinian ring is Noetherian

Note that the last conditions is not true for Artinian Modules.

***Proof*:**

1. Consider the chain:
$$m_1 \supseteq m_1 \cap m_2 \supseteq \cdots$$
I first claim that each inclusion of this chain is proper. To see this, recall that if m_i and m_j are non-equal maximal ideal, it must be that $m_i + m_j = 1$. Thus, any collection of maximal ideals has this pair-wise condition. By the Chinese Remainder Theorem:
$$R/(m_1 \cap \cdots \cap m_n) \cong R/m_1 \times R/m_2 \times \cdots \times R/m_n$$
Now simply see that the element $(\overline{0}, ..., \overline{0}, \overline{1})$ is non-zero, and hence not contain in $m_1 \cap \cdots \cap m_n$, showing that each inclusion *must* be proper! Thus, by the D.C.C., the above chain must stabalize, meaning there can only be finitely many maximal ideals.
2. The proof follows immediately from the previous observation of the CRT
3. First, $J := \text{Jac}(R)$ is nilpotent: by the D.C.C. the chain $J \supseteq J^2 \supseteq \cdots$ stabilizes, say $J^k = J^{k+1}$. If $J^k \neq 0$, the nonempty set of ideals I with $IJ^k \neq 0$ has a minimal element I_0; pick $a \in I_0$ with $aJ^k \neq 0$, so $(a) \subseteq I_0$ is in the set and $(a) = I_0$ by minimality. Then $((a)J)J^k = (a)J^{k+1} = (a)J^k \neq 0$ and $(a)J \subseteq (a)$, so $(a)J = (a)$ by minimality, and Nakayama (lemma 23.4.27, as $J \subseteq \text{Jac}(R)$) forces $(a) = 0$, a contradiction. Hence $J^k = 0$, so $J \subseteq \sqrt{(0)} = \text{Nil}(R)$; the reverse inclusion always holds, giving $\text{Jac}(R) = \text{Nil}(R)$. Now let P be prime. As $P \supseteq \text{Nil}(R) = J$, its image in $R/J \cong k_1 \times \cdots \times k_n$ is a prime; since ideals of a product are products of ideals, that image is $k_1 \times \cdots \times 0 \times \cdots \times k_n$, which is maximal, so P is maximal by the correspondence theorem. Thus every prime is maximal and $\text{kdim} R = 0$.
4. Let $m_1, ..., m_n$ be the maximal ideals of R and let k be such that $(\text{Jac}(R))^k = 0$. Then
$$\prod_i^n M_i^k \subseteq \left(\prod_i^n M_i\right)^k \subseteq (\text{Jac}(R))^k = 0$$

Thus, by the CRT
$$R \cong (R/M_1^k) \times \cdots \times (R/M_n^k)$$
Notice that each R/M_i^k is an artinian with unique maximal ideal M_i/M_i^k.

5. By part (4) it suffices to show local Artinian rings are Noetherian. Let $\mathfrak{m}$ be the maximal ideal; then $\mathfrak{m} = \mathrm{Jac}(R)$ and $\mathfrak{m}^k = 0$ for some k (part 3), giving the filtration $R \supseteq \mathfrak{m} \supseteq \mathfrak{m}^2 \supseteq \cdots \supseteq \mathfrak{m}^k = 0$. Each quotient $\mathfrak{m}^i/\mathfrak{m}^{i+1}$ is a vector space over the field $R/\mathfrak{m}$ satisfying the D.C.C., hence finite-dimensional (lemma 28.1.11) and of finite length; concatenating their composition series gives one for R, so R is Noetherian.

Corollary 28.1.7: Uniqueness of the Local Decomposition

The decomposition of an Artinian ring $R \cong \prod_{i=1}^n R_i$ into local Artinian rings is unique up to reordering: the factors are the localizations $R_i \cong R_{\mathfrak{m}_i}$ at the maximal ideals (which here coincide with the completions $\hat{R}_{\mathfrak{m}_i}$), corresponding to the primitive idempotents of R.

***Proof*:**
In a product $\prod R_j$ of local rings the idempotents are the tuples of 0s and 1s, the primitive ones being the standard vectors e_i; hence the factors $R_i = e_i R$ are intrinsic. Localizing at $\mathfrak{m}_i$ kills every e_j with $j \neq i$ (each $1 - e_j \notin \mathfrak{m}_i$ becomes a unit and $e_j(1-e_j) = 0$), leaving R_i, so $R_{\mathfrak{m}_i} \cong R_i$; and R_i, being local Artinian, is complete.

Corollary 28.1.8: Reduced Artinian Rings

An Artinian ring is reduced if and only if it is a finite product of fields.

***Proof*:**
A finite product of fields is reduced. Conversely, if R is reduced Artinian then $\mathrm{Nil}(R) = 0$, so by the structure theorem $R \cong \prod_i R/\mathfrak{m}_i^k$ with $\bigcap_i \mathfrak{m}_i^k = \mathrm{Nil}(R) = 0$; reducedness forces each local factor to have zero nilradical, i.e. $\mathfrak{m}_i^k = 0$ collapses to $\mathfrak{m}_i = 0$ there, so each factor is the field $R/\mathfrak{m}_i$.

Corollary 28.1.9: Artinian Ring Equivalent Condition I

Let R be a ring. Then R is Artinian if and only if R is Noetherian and $\dim(R) = 0$

***Proof*:**
R Artinian implies R Noetherian (theorem 28.1.6) with all primes maximal, i.e. $\dim R = 0$. For the converse, let R be Noetherian with $\dim R = 0$. Its minimal primes $\mathfrak{m}_1, \ldots, \mathfrak{m}_n$ are finite in number (Noetherian) and all maximal ($\dim R = 0$), and $\mathrm{Nil}(R) = \mathfrak{m}_1 \cap \cdots \cap \mathfrak{m}_n$ is nilpotent (finitely generated), so $(\mathfrak{m}_1 \cdots \mathfrak{m}_n)^k \subseteq \mathrm{Nil}(R)^k = 0$ for some k. By the Chinese Remainder Theorem $R \cong \prod_i R/\mathfrak{m}_i^k$, each factor Noetherian local with nilpotent maximal ideal, hence of finite length and Artinian; so R is Artinian.

Corollary 28.1.10: Artinian Ring Equivalent Condition II

Let k be a field and R a finitely generated k-algebra. Then R is artinian if and only if R is finitely generated as a k-module.

***Proof*:**
If R is a finite-dimensional k-vector space, its ideals are k-subspaces, so the D.C.C. holds and R is Artinian. Conversely, if R is Artinian then $\dim R = 0$ (theorem 28.1.6), and a finitely generated k-algebra of dimension 0 is a finite k-module (chapter 29).

For artinian modules over a field $R = k$, there is a clean characterization:

Lemma 28.1.11: Artin+Noetherian Module

Let M be a vector space over k. Then the following are equivalent:

1. M satisfies the A.C.C.
2. M satisfies the D.C.C.
3. M is finite-dimensional

***Proof*:**
Naturally, (3) implies both (1) and (2). Conversely, assume $\dim_k(M) = \infty$, so pick some infinite collection of linearly independent vectors. Then we have an infinite strict chain:

$$\langle e_1 \rangle \subsetneq \langle e_1, e_2 \rangle \cdots$$

and an infinite descending chain:

$$\langle e_1, e_2, \ldots \rangle \supsetneq \langle e_2, e_3, \ldots \rangle \supsetneq \cdots$$

completing the proof

Example 28.1.12: Artinian Rings

1. Let $N > 1$ be an integer and $R = \mathbb{Z}/N\mathbb{Z}$. Since R is finite it is Artinian. If $N = p_1^{\alpha_1} \cdots p_r^{\alpha_r}$ for distinct primes p_i and $\alpha_i \geq 1$, then by the Chinese Remainder Theorem

$$\mathbb{Z}/N\mathbb{Z} \cong (\mathbb{Z}/p_1^{\alpha_1}\mathbb{Z}) \times (\mathbb{Z}/p_2^{\alpha_2}\mathbb{Z}) \times \cdots \times (\mathbb{Z}/p_r^{\alpha_r}\mathbb{Z})$$

   where each $\mathbb{Z}/p_i^{\alpha_i}\mathbb{Z}$ is a local Artinian ring with maximal ideal $(p_i)/(p_i^{\alpha_i})$. The Jacobson radical of R is $(p_1 p_2 \cdots p_r)$.

2. Let R be a finite dimensional k-algebra over some field k (it need not be commutative!). Then R is an Artinian ring, since any descending chain of R is a descending chain of k-subspces of R, which is bounded by $\dim_k(R)$. Note that if R is not commutative, R is not the finite direct product of local artinian rings, but the finite product of indecomposable artinian rings (i.e. artinian rings which cannot be split into direct sum of submodules).

3. Let $k[x]$ be the polynomial ring over a field k and $f \in k[x]$ be a nonzero polynomial. Then if $f(x) = f_1^{n_1}(x) \cdots f_k^{n_k}(x)$, by the CRT we have:

$$k[x]/(f(x)) \cong k[x]/(f_1^{n_1}(x)) \times \cdots \times k[x]/(f_k^{n_k}(x))$$

which is artinian, since $k[x]/(f(x))$ is a finite dimensional k-module (each $k[x]/(f_i^{n_i}(x))$ is a local artinian ring, with maximla ideal $(f_i(x))/(f_i^{n_i}(x))$). If $n_i = 1$ for all i, $k[x]/(f(x))$ is a product of fields, and if furtheremore there was only 1 n_i (i.e. $i \in \{1\}$), then $k[x]/(f(x))$ would be a field extension

We shall give names to modules that satisfy both the Artinian and Noetherian condition:

Definition 28.1.13: Modules of Finite Length

Let M be an R-module that is Noetherian and Artinian. Then M is an R-module of *finite length*.

By proposition 23.1.8 each finite-length module has a well-defined length $\ell(M) \in \mathbb{N}$. In fact the composition factors themselves are determined:

Theorem 28.1.14: Jordan-Hölder

Let M be a module of finite length. Then any two composition series of M have the same length and the same multiset of simple quotients, up to isomorphism and reordering.

***Proof*:**
Length invariance is proposition 23.1.8. Induct on $\ell(M)$ for the factors: given two composition series with maximal submodules M_1, M_1', either $M_1 = M_1'$ (apply the inductive hypothesis to M_1), or $M_1 + M_1' = M$, whence $M/M_1 \cong M_1'/(M_1 \cap M_1')$ and $M/M_1' \cong M_1/(M_1 \cap M_1')$; inserting a composition series of $M_1 \cap M_1'$ into each shows the two factor multisets agree.

Example 28.1.15: Module of Finite Length
The simplest example would be when $R = k$ is a field, which gives vector-spaces. In this case, for any V

$$\ell(V) = \dim_k(V)$$

Proposition 28.1.16: Finite Length Additive Under Extensions

Let

$$0 \to M' \to M \to M'' \to 0$$

Be a short exact sequence of finite length modules. Then:

$$\ell(M) = \ell(M') + \ell(M'')$$

***Proof*:**
Concatenating a composition series $0 = M'_0 \subsetneq \cdots \subsetneq M'_{\ell(M')} = M'$ of M' with the preimage in M of a composition series of $M'' = M/M'$ yields a composition series of M of length $\ell(M') + \ell(M'')$; by invariance of length (proposition 23.1.8) this is $\ell(M)$.

For a finite-length module the length is computed locally, and the module splits as a direct sum over its (finitely many, maximal) associated points.

Proposition 28.1.17: Length, Support, and Local Decomposition

Let M be a finite-length R-module. Then $\mathrm{Supp}(M)$ is a finite set of maximal ideals, the natural map $M \to \bigoplus_{\mathfrak{m} \in \mathrm{Supp}(M)} M_{\mathfrak{m}}$ is an isomorphism, and

$$\ell_R(M) = \sum_{\mathfrak{m} \in \mathrm{Supp}(M)} \ell_{R_{\mathfrak{m}}}(M_{\mathfrak{m}})$$

***Proof*:**
A composition series of M has simple factors $R/\mathfrak{m}_i$ with the $\mathfrak{m}_i$ maximal, so $\mathrm{Supp}(M) = \{\mathfrak{m}_i\}$ is a finite set of maximal ideals and some product $\mathfrak{m}_1^{a_1} \cdots \mathfrak{m}_r^{a_r}$ annihilates M. Thus M is a module over the Artinian ring $R/\operatorname{Ann}(M) \cong \prod_i R/\mathfrak{m}_i^{a_i}$, which gives the direct-sum decomposition $M \cong \bigoplus_i M_{\mathfrak{m}_i}$. Localizing a composition series at $\mathfrak{m}$ deletes the factors $R/\mathfrak{m}'$ with $\mathfrak{m}' \neq \mathfrak{m}$ and preserves those with $\mathfrak{m}' = \mathfrak{m}$, so $\ell_{R_{\mathfrak{m}}}(M_{\mathfrak{m}})$ counts the factors isomorphic to $R/\mathfrak{m}$; summing over $\mathfrak{m}$ gives $\ell_R(M)$.

Proposition 28.1.18: Length Along a Flat Local Homomorphism

Let $\varphi\colon (A, \mathfrak{m}) \to (B, \mathfrak{n})$ be a flat local homomorphism of Noetherian local rings whose closed fibre $B/\mathfrak{m}B$ has finite length over B, and let M be an A-module of finite length. Then $M \otimes_A B$ has finite length over B, and

$$\ell_B(M \otimes_A B) \;=\; \ell_A(M) \cdot \ell_B(B/\mathfrak{m}B)$$

In particular, if A is Artinian then $\ell_B(B) = \ell_A(A) \cdot \ell_B(B/\mathfrak{m}B)$.

***Proof*:**
Induct on $n = \ell_A(M)$. If $n = 0$ then $M = 0$ and both sides vanish. If $n = 1$ then M is simple, so $M \cong A/\mathfrak{m}$ because A is local, and $M \otimes_A B \cong B/\mathfrak{m}B$, whose length is $\ell_B(B/\mathfrak{m}B)$ by hypothesis; this is the claimed product since $\ell_A(M) = 1$.

For $n > 1$, choose a submodule $M' \subseteq M$ with $\ell_A(M') = n - 1$ and $M/M' \cong A/\mathfrak{m}$, which exists by taking a composition series. Tensoring

$$0 \to M' \to M \to A/\mathfrak{m} \to 0$$

with B over A leaves it exact, since B is flat over A. Length is additive on short exact sequences (proposition 28.1.16), so

$$\ell_B(M \otimes_A B) = \ell_B(M' \otimes_A B) + \ell_B(B/\mathfrak{m}B) = (n-1)\ell_B(B/\mathfrak{m}B) + \ell_B(B/\mathfrak{m}B)$$

using the inductive hypothesis, and the right side is $n \cdot \ell_B(B/\mathfrak{m}B)$, as we sought to show.

The multiplicativity is the algebraic content behind the geometric statement that a flat morphism pulls the fundamental cycle of the base back to the fundamental cycle of the total space: the multiplicity with which a component occurs is multiplied by the length of the fibre, uniformly.

Proposition 28.1.19: Finiteness over an Artinian Ring

Let R be an Artinian ring and M an R-module. Then M is finitely generated iff it has finite length iff it is Noetherian iff it is Artinian. In particular R has finite length over itself.

***Proof*:**
Finite length trivially implies the other three. If M is finitely generated it is a quotient of some R^n; since R has finite length over itself (its composition series comes from the filtration by the $\mathfrak{m}_i^j$), so do R^n and its quotient M. A Noetherian module is finitely generated, hence of finite length by the above. An Artinian module M over the Artinian R also has finite length: with $J = \mathrm{Jac}(R)$ and $J^k = 0$, the filtration $M \supseteq JM \supseteq \cdots \supseteq J^k M = 0$ has each piece $J^i M/J^{i+1}M$ a module over the product of fields R/J satisfying the D.C.C., hence a finite direct sum of simple modules.

Exercise 28.1.1

1. Let M be a Noetherian and Artinian A-module, $\varphi : M \to M$ an A-module homomorphism. Show that there exists an $n \geq 0$ such that $M = \ker(\varphi^n) \oplus \mathrm{im}(\varphi^n)$.
2. Let M be an R-module. Then M is of finite length iff it has a composition series.

29

Dimension Theory

How large is a ring? For the coordinate ring of a variety the answer should be the variety's dimension, and the algebra offers three ways to measure it, each capturing a different intuition. One counts how many independent parameters the ring carries: the length of the longest chain of prime ideals, since primes correspond to irreducible closed subsets and a chain of them is a tower of nested subvarieties. A second counts how many equations it takes to cut the ring down to something finite: a zero-dimensional, Artinian quotient. A third reads the dimension off a growth rate, the rate at which $R/\mathfrak{m}^n$ grows with n. The central result of the chapter, the *dimension theorem*, is that these three numbers coincide. It is the payoff of the finiteness and locality threads: the Noetherian hypothesis makes the growth rate polynomial, localization concentrates the question at a single prime, and Noether normalization ties the whole theory back to the transparent case of a polynomial ring.

The geometric cash-out (that the Krull dimension of a coordinate ring is the topological dimension of its variety, that a hypersurface drops dimension by one) is developed in [27], which cites this chapter for the ring theory.

29.1 Noether Normalization

The theory rests on the fact that a finitely generated algebra over a field is only finitely far from a polynomial ring.

Theorem 29.1.1: Noether Normalization

Let k be a field and $A = k[r_1, \ldots, r_n]$ a finitely generated k-algebra. Then there are elements $y_1, \ldots, y_q \in A$, algebraically independent over k, such that A is a finite (module-finite) extension of $k[y_1, \ldots, y_q]$.

***Proof*:**
Induct on n, the number of generators. If $r_1, \ldots, r_n$ are algebraically independent, take $y_i = r_i$ and $A = k[y_1, \ldots, y_n]$. Otherwise a nonzero $f \in k[x_1, \ldots, x_n]$ satisfies $f(r_1, \ldots, r_n) = 0$; let d exceed every exponent appearing in f, and set

$$x_i' = x_i - x_n^{d^i} \quad (1 \le i \le n-1)$$

Rewriting f in the variables $x_1', \ldots, x_{n-1}', x_n$, each monomial $x_1^{a_1} \cdots x_n^{a_n}$ of f contributes a top x_n-power $x_n^{a_1 d + a_2 d^2 + \cdots + a_{n-1} d^{n-1} + a_n}$; base-$d$ uniqueness makes these exponents distinct across monomials, so f becomes, up to a nonzero scalar from k, monic in x_n of some degree N with coefficients in $k[x_1', \ldots, x_{n-1}']$. Putting $s_i = r_i - r_n^{d^i}$, the relation $f(r_1, \ldots, r_n) = 0$ exhibits r_n as a root of a monic polynomial over $B = k[s_1, \ldots, s_{n-1}]$, so r_n is integral over B; each $r_i = s_i + r_n^{d^i}$ is then integral over $B[r_n]$, and by transitivity A is integral, hence (being finitely generated) module-finite, over B. As B is generated by $n-1$ elements, induction gives a polynomial subalgebra $k[y_1, \ldots, y_q] \subseteq B$ over which B, and therefore A, is module-finite.

The number q is an invariant: it equals the transcendence degree of A over k when A is a domain, as theorem 29.6.2 will record once dimension is in hand.

29.2 Krull Dimension

Definition 29.2.1: Krull Dimension and Height

The *height* of a prime $\mathfrak{p}$, written $\operatorname{ht}\mathfrak{p}$, is the supremum of lengths r of chains $\mathfrak{p}_0 \subsetneq \mathfrak{p}_1 \subsetneq \cdots \subsetneq \mathfrak{p}_r = \mathfrak{p}$ of primes descending from $\mathfrak{p}$. The *Krull dimension* $\dim R$ is the supremum of $\operatorname{ht}\mathfrak{p}$ over all primes $\mathfrak{p}$, equivalently the supremum of lengths of chains of primes in R.

A field has dimension 0; a principal ideal domain that is not a field has dimension 1, its chains being $0 \subsetneq (\pi)$ for irreducible π. For a local ring $(R, \mathfrak{m})$ the dimension is $\operatorname{ht}\mathfrak{m}$, since $\mathfrak{m}$ contains every prime. The chain $0 \subsetneq (x_1) \subsetneq (x_1, x_2) \subsetneq \cdots \subsetneq (x_1, \ldots, x_n)$ in $k[x_1, \ldots, x_n]$ shows $\dim k[x_1, \ldots, x_n] \ge n$; equality is theorem 29.6.2.

Dimension is controlled locally: $\dim R = \sup_{\mathfrak{m}} \dim R_{\mathfrak{m}}$ over maximal ideals, because a chain of primes below $\mathfrak{p}$ survives in $R_{\mathfrak{p}}$. The substance of the theory is therefore the local case, to which the next two sections are devoted.

29.3 The Hilbert-Samuel Function

Throughout this section $(R, \mathfrak{m})$ is a Noetherian local ring and $\mathfrak{q}$ an $\mathfrak{m}$-*primary* ideal, so that $\mathfrak{m}^r \subseteq \mathfrak{q} \subseteq \mathfrak{m}$ for some r and $R/\mathfrak{q}$ is Artinian. We measure R through the lengths $\ell(R/\mathfrak{q}^n)$. The tool is the associated graded ring $\operatorname{gr}_{\mathfrak{q}} R = \bigoplus_{n \ge 0} \mathfrak{q}^n/\mathfrak{q}^{n+1}$, a Noetherian graded ring, finitely generated over the Artinian ring $R/\mathfrak{q}$ by the finitely many degree-one elements $\mathfrak{q}/\mathfrak{q}^2$.

Lemma 29.3.1: Hilbert Polynomial of a Graded Module

Let $S = S_0[t_1, \ldots, t_s]$ be a graded ring, finitely generated over an Artinian ring S_0 by degree-one elements t_i, and $N = \bigoplus_n N_n$ a finitely generated graded S-module. Then $\ell(N_n)$ is finite for each n and agrees, for $n \gg 0$, with a polynomial in n of degree $\leq s-1$ (rational coefficients).

***Proof*:**
Each N_n is a finitely generated S_0-module, hence of finite length. Induct on s. If $s = 0$ then $S = S_0$ and N, being finitely generated, is concentrated in finitely many degrees, so $\ell(N_n) = 0$ for $n \gg 0$ (a polynomial of degree ≤ -1). For the step, multiplication by t_s gives an exact sequence of $S_0[t_1, \ldots, t_{s-1}]$-modules in each degree,

$$0 \to K_n \to N_n \xrightarrow{t_s} N_{n+1} \to (N/t_sN)_{n+1} \to 0,$$

where $K = (0 :_N t_s)$ and N/t_sN are finitely generated and killed by t_s, hence graded modules over $S_0[t_1, \ldots, t_{s-1}]$. Additivity of length gives $\ell(N_{n+1}) - \ell(N_n) = \ell\big((N/t_sN)_{n+1}\big) - \ell(K_n)$, which by induction agrees with a polynomial of degree $\leq s-2$ for $n \gg 0$. A function whose first difference is eventually a polynomial of degree $\leq s-2$ is itself eventually a polynomial of degree $\leq s-1$.

Proposition 29.3.2: The Hilbert-Samuel Polynomial

For a finitely generated R-module M, the *characteristic function* $\chi^M_{\mathfrak{q}}(n) = \ell(M/\mathfrak{q}^nM)$ agrees for $n \gg 0$ with a polynomial in n of degree $\leq s$, where s is the least number of generators of $\mathfrak{q}$. Moreover $\deg \chi^M_{\mathfrak{q}} = \deg \chi^M_{\mathfrak{m}}$, so the degree is independent of the $\mathfrak{m}$-primary ideal $\mathfrak{q}$.

***Proof*:**
The associated graded module $\mathrm{gr}(M) = \bigoplus_n \mathfrak{q}^nM/\mathfrak{q}^{n+1}M$ is a finitely generated graded module over $\mathrm{gr}_{\mathfrak{q}} R$, itself generated over the Artinian ring $R/\mathfrak{q}$ by the s degree-one generators of $\mathfrak{q}/\mathfrak{q}^2$. By lemma 29.3.1, $\ell(\mathfrak{q}^nM/\mathfrak{q}^{n+1}M)$ is eventually a polynomial of degree $\leq s-1$. Since $\chi^M_{\mathfrak{q}}(n) = \ell(M/\mathfrak{q}^nM) = \sum_{r=0}^{n-1} \ell(\mathfrak{q}^rM/\mathfrak{q}^{r+1}M)$, summing raises the degree by one: $\chi^M_{\mathfrak{q}}$ is eventually a polynomial of degree $\leq s$. For the last claim, $\mathfrak{m}^r \subseteq \mathfrak{q} \subseteq \mathfrak{m}$ gives $\mathfrak{m}^{rn} \subseteq \mathfrak{q}^n \subseteq \mathfrak{m}^n$, hence $\chi^M_{\mathfrak{m}}(n) \leq \chi^M_{\mathfrak{q}}(n) \leq \chi^M_{\mathfrak{m}}(rn)$ for all n; the outer terms are polynomials of equal degree, so the middle one shares it.

Write $d(M) = \deg \chi^M_{\mathfrak{m}}(n)$ and $d(R) = \deg \chi_{\mathfrak{m}}(n)$ where $\chi_{\mathfrak{m}}(n) = \ell(R/\mathfrak{m}^n)$. The next proposition is the inductive engine.

Proposition 29.3.3: Dropping the Degree by a Non-Zero-Divisor

If $x \in \mathfrak{m}$ is not a zero-divisor on M, then $d(M/xM) \leq d(M) - 1$.

***Proof*:**
Set $M' = M/xM$. Additivity of length in $0 \to (xM + \mathfrak{q}^n M)/\mathfrak{q}^n M \to M/\mathfrak{q}^n M \to M'/\mathfrak{q}^n M' \to 0$ gives $\chi_{\mathfrak{q}}^{M'}(n) = \chi_{\mathfrak{q}}^{M}(n) - g(n)$, where $g(n) = \ell\big(xM/(xM \cap \mathfrak{q}^n M)\big)$. As x is a non-zero-divisor, $M \xrightarrow{x} xM$ is an isomorphism carrying $\mathfrak{q}^n M$ onto $\mathfrak{q}^n(xM)$, so $\ell(xM/\mathfrak{q}^n(xM)) = \chi_{\mathfrak{q}}^{M}(n)$. Now $\mathfrak{q}^n(xM) \subseteq xM \cap \mathfrak{q}^n M$, and by the Artin-Rees lemma (theorem 25.4.1) the filtration $\{xM \cap \mathfrak{q}^n M\}$ is $\mathfrak{q}$-stable, so $xM \cap \mathfrak{q}^n M \subseteq \mathfrak{q}^{n-c}(xM)$ for some c and all $n \geq c$. Taking lengths,

$$\chi_{\mathfrak{q}}^{M}(n-c) \leq g(n) \leq \chi_{\mathfrak{q}}^{M}(n)$$

Both bounds are $\chi_{\mathfrak{q}}^{M}$ evaluated at $n-c$ and n, so g agrees for large n with a polynomial of degree $d(M)$ and the same leading coefficient as $\chi_{\mathfrak{q}}^{M}$. Hence $\chi_{\mathfrak{q}}^{M'} = \chi_{\mathfrak{q}}^{M} - g$ has degree $\leq d(M) - 1$.

29.4 The Dimension Theorem

Theorem 29.4.1: Dimension Theorem

For a Noetherian local ring $(R, \mathfrak{m})$ the following three numbers coincide:

$$d(R) = \delta(R) = \dim R,$$

where $d(R) = \deg \ell(R/\mathfrak{m}^n)$, and $\delta(R)$ is the least number of generators of an $\mathfrak{m}$-primary ideal.

***Proof*:**
We show $d(R) \leq \delta(R) \leq \dim R \leq d(R)$.

$d(R) \leq \delta(R)$. If $\mathfrak{q}$ is $\mathfrak{m}$-primary and generated by $s = \delta(R)$ elements, then $\deg \chi_{\mathfrak{q}} \leq s$ by proposition 29.3.2, and $\deg \chi_{\mathfrak{q}} = \deg \chi_{\mathfrak{m}} = d(R)$; so $d(R) \leq \delta(R)$.

$\dim R \leq d(R)$. Induct on $d(R)$. If $d(R) = 0$ then $\ell(R/\mathfrak{m}^n)$ is eventually constant, so $\mathfrak{m}^n = \mathfrak{m}^{n+1}$ for some n; Nakayama's lemma (lemma 23.4.27) gives $\mathfrak{m}^n = 0$, so R is Artinian and $\dim R = 0$. Suppose $d(R) = d > 0$ and let $\mathfrak{p}_0 \subsetneq \cdots \subsetneq \mathfrak{p}_r$ be a chain of primes; we bound r. We may take $\mathfrak{p}_0$ minimal, and choosing $x \in \mathfrak{p}_1 \setminus \mathfrak{p}_0$, the ring $\bar{R} = R/\mathfrak{p}_0$ is a domain in which x is a nonzero, hence non-zero-divisor, element. By proposition 29.3.3, $d(\bar{R}/x\bar{R}) \leq d(\bar{R}) - 1 \leq d(R) - 1$ (the last as $\bar{R}$ is a quotient of R, whose characteristic polynomial dominates). The chain $\mathfrak{p}_1 \subsetneq \cdots \subsetneq \mathfrak{p}_r$ descends to a chain of length $r-1$ in $\bar{R}/x\bar{R}$, so by induction $r - 1 \leq d(R) - 1$, i.e. $r \leq d(R)$. Hence $\dim R \leq d(R)$.

$\delta(R) \leq \dim R$. Let $d = \dim R$; we build an $\mathfrak{m}$-primary ideal from d elements. Choose $x_1, \ldots, x_d \in \mathfrak{m}$ so that x_i lies in no minimal prime of $(x_1, \ldots, x_{i-1})$; this is possible by prime avoidance, as there are only finitely many such minimal primes (corollary 24.4.3), unless one of them is $\mathfrak{m}$ itself, in which case $(x_1, \ldots, x_{i-1})$ is already $\mathfrak{m}$-primary and fewer elements suffice. An induction then shows every prime $\mathfrak{p} \supseteq (x_1, \ldots, x_i)$ has $\operatorname{ht} \mathfrak{p} \geq i$: such $\mathfrak{p}$ contains a minimal prime $\mathfrak{p}'$ of $(x_1, \ldots, x_{i-1})$, which has $\operatorname{ht} \mathfrak{p}' \geq i - 1$ by induction, and $\mathfrak{p} \neq \mathfrak{p}'$ because $x_i \in \mathfrak{p} \setminus \mathfrak{p}'$, so $\operatorname{ht} \mathfrak{p} \geq \operatorname{ht} \mathfrak{p}' + 1 \geq i$. At $i = d$ every prime over $(x_1, \ldots, x_d)$ has height $\geq d = \operatorname{ht} \mathfrak{m}$; since $\mathfrak{m}$ is the only prime of that height in the local ring R, the ideal $(x_1, \ldots, x_d)$ is $\mathfrak{m}$-primary. Thus $\delta(R) \leq d$.

Krull's height theorem, proved next, is now a corollary of the dimension theorem, with no circularity:

its proof invokes only the equalities just established.

29.5 Systems of Parameters and Krull's Principal Ideal Theorem

Definition 29.5.1: System of Parameters

A *system of parameters* for a Noetherian local ring $(R, \mathfrak{m})$ of dimension d is a set of d elements generating an $\mathfrak{m}$-primary ideal. The dimension theorem guarantees one exists, and d is the least possible number of such generators.

Theorem 29.5.2: Krull's Height Theorem

Let R be Noetherian and $\mathfrak{a} = (x_1, \dots, x_c)$ generated by c elements. Then every prime $\mathfrak{p}$ minimal over $\mathfrak{a}$ has $\operatorname{ht}\mathfrak{p} \le c$. In particular (the *Hauptidealsatz*, $c = 1$) a prime minimal over a single nonunit x has height at most 1.

***Proof*:**
Localizing at $\mathfrak{p}$, we may assume $(R, \mathfrak{p})$ is local and $\mathfrak{p}$ is minimal over $\mathfrak{a}$, so that $\mathfrak{a}$ is $\mathfrak{p}$-primary and $\operatorname{ht}\mathfrak{p} = \dim R$. Then $\mathfrak{a}$ is an $\mathfrak{p}$-primary ideal with c generators, so $\dim R = \delta(R) \le c$ by the inequality $\dim R \le d(R) \le \delta(R)$ established in theorem 29.4.1; here only $\dim R \le d(R)$ and $d(R) \le \delta(R)$ are used, and $\delta(R) \le c$ because $\mathfrak{a}$ itself is $\mathfrak{p}$-primary with c generators.

A converse rounds out the picture: in a Noetherian ring a prime of height c is minimal over some ideal generated by c elements, so height is exactly the least number of generators needed to cut out a prime as a component. The Hauptidealsatz is the algebraic form of the geometric principle that one equation cuts dimension by one: passing to $R/(x)$ for x a non-unit non-zero-divisor lowers dimension by exactly one,

$$\dim R/(x) = \dim R - 1,$$

the inequality $\ge$ from theorem 29.5.2 and $\le$ from proposition 29.3.3.

29.6 Dimension and Transcendence Degree

Noether normalization now identifies the dimension of a geometric ring with a transcendence degree, and pins down the dimension of affine space. The engine is that a proper prime quotient of a finitely generated domain loses transcendence degree.

Lemma 29.6.1: Transcendence Degree Drops on a Prime Quotient

Let A be a finitely generated domain over a field k and $\mathfrak{p} \ne 0$ a prime of A. Then $\operatorname{trdeg}_k \operatorname{Frac}(A/\mathfrak{p}) < \operatorname{trdeg}_k \operatorname{Frac}(A)$.

***Proof*:**
Throughout, transcendence bases and the transcendence degree of a field extension are those of *Core Algebra* [25], where in particular the degree is shown to be well-defined.

Put $d = \operatorname{trdeg}_k A$ and $e = \operatorname{trdeg}_k A/\mathfrak{p} \leq d$, and suppose $e = d$. Choose $x_1, \dots, x_d \in A$ whose images in $A/\mathfrak{p}$ form a transcendence basis; they are algebraically independent in A, hence (as $d = \operatorname{trdeg} A$) a transcendence basis of A as well, so every element of A is algebraic over $B = k[x_1, \dots, x_d]$. Pick $0 \neq a \in \mathfrak{p}$. An algebraic relation for a over $\operatorname{Frac}(B)$, cleared of denominators and divided by the largest power of a that factors out (valid as A is a domain), becomes $c_0 + c_1 a + \cdots + c_m a^m = 0$ with $c_i \in B$ and $c_0 \neq 0$; then $c_0 = -a(c_1 + \cdots + c_m a^{m-1}) \in \mathfrak{p}$. But $\mathfrak{p} \cap B = 0$, since a nonzero element of $\mathfrak{p} \cap B$ would be a nontrivial algebraic relation among the algebraically independent images $\bar{x}_i$ in $A/\mathfrak{p}$. This contradicts $c_0 \neq 0$, so $e < d$.

Theorem 29.6.2: Dimension of Finitely Generated Algebras

Let A be a finitely generated algebra over a field k that is a domain. Then

$$\dim A = \operatorname{trdeg}_k \operatorname{Frac}(A),$$

and in particular $\dim k[x_1, \dots, x_n] = n$. More generally the number q in Noether normalization (theorem 29.1.1) equals $\dim A$.

***Proof*:**
By theorem 29.1.1, A is module-finite, hence integral, over a polynomial subalgebra $k[y_1, \dots, y_q]$ with the y_i algebraically independent, so $q = \operatorname{trdeg}_k \operatorname{Frac}(A)$; integral extensions preserve Krull dimension (theorem 26.3.2 supplies lying-over, going-up, and incomparability), so $\dim A = \dim k[y_1, \dots, y_q]$. We prove the general claim $\dim B \leq \operatorname{trdeg}_k \operatorname{Frac}(B)$ for every finitely generated domain B over k, by induction on transcendence degree. Given a chain $0 = \mathfrak{p}_0 \subsetneq \mathfrak{p}_1 \subsetneq \cdots \subsetneq \mathfrak{p}_r$ in B, the tail descends to a chain of length $r - 1$ in the finitely generated domain $B/\mathfrak{p}_1$, whose transcendence degree is at most $\operatorname{trdeg} B - 1$ by lemma 29.6.1; induction gives $r - 1 \leq \operatorname{trdeg} B - 1$, so $r \leq \operatorname{trdeg} B$. Applied to $B = k[y_1, \dots, y_q]$ this yields $\dim k[y_1, \dots, y_q] \leq q$, while the chain of definition 29.2.1 gives $\geq q$. Hence $\dim A = \dim k[y_1, \dots, y_q] = q = \operatorname{trdeg}_k \operatorname{Frac}(A)$.

The equality $\dim A = \operatorname{trdeg}_k \operatorname{Frac}(A)$ is the algebraic form of the statement that dimension counts independent coordinates; in [27] it becomes the assertion that the dimension of an irreducible affine variety is the transcendence degree of its function field.

29.7 Regular Local Rings

The dimension theorem produced three numbers and showed them equal. A fourth is always available for a local ring and is never smaller: the least number of generators of the maximal ideal, which by Nakayama is the dimension of $\mathfrak{m}/\mathfrak{m}^2$ as a vector space over the residue field. Rings for which the fourth number agrees with the other three are the regular ones, and they are the subject of this section.

The inequality has a geometric reading that fixes the terminology. For a variety X and a point p, the vector space $(\mathfrak{m}_p/\mathfrak{m}_p^2)^\vee$ is the tangent space at p, so the inequality says a variety has at least as many tangent directions at a point as it has dimensions, and regularity is the case where it has exactly as many. That reading, and the identification of regular local rings with the smooth points of a variety, is [27]; what follows is the ring theory it cites.

Corollary 29.7.1: Dimension and Embedding Dimension

Let $(R, \mathfrak{m})$ be a Noetherian local ring with residue field $k = R/\mathfrak{m}$. Then

$$\dim R \leq \dim_k \mathfrak{m}/\mathfrak{m}^2$$

***Proof*:**
Put $s = \dim_k \mathfrak{m}/\mathfrak{m}^2$ and choose $x_1, \ldots, x_s \in \mathfrak{m}$ whose images form a k-basis of $\mathfrak{m}/\mathfrak{m}^2$. By Nakayama's lemma (lemma 23.4.27) they generate $\mathfrak{m}$, which is $\mathfrak{m}$-primary, so $\delta(R) \leq s$. The dimension theorem (theorem 29.4.1) gives $\dim R = \delta(R) \leq s$, as we sought to show.

The same argument identifies $\dim_k \mathfrak{m}/\mathfrak{m}^2$ with the least number of generators of $\mathfrak{m}$: Nakayama makes any lift of a basis a generating set, and no smaller set can generate, since a generating set spans $\mathfrak{m}/\mathfrak{m}^2$. This is the number called the *embedding dimension*, and corollary 29.7.1 says it bounds the dimension from above.

Definition 29.7.2: Regular Local Ring

A Noetherian local ring $(R, \mathfrak{m})$ with residue field k is *regular* if

$$\dim R = \dim_k \mathfrak{m}/\mathfrak{m}^2,$$

equivalently if $\mathfrak{m}$ can be generated by $\dim R$ elements. Such a generating set is a *regular system of parameters*: it is a system of parameters (definition 29.5.1) that generates $\mathfrak{m}$ itself rather than merely an $\mathfrak{m}$-primary ideal.

Regularity is therefore the statement that the maximal ideal is as small as the dimension permits. The next theorem replaces that numerical condition by a structural one, and the structural form is what every later argument uses: the associated graded ring, which for a general local ring records how the powers of $\mathfrak{m}$ shrink, is for a regular local ring a polynomial ring on the regular system of parameters. Following this chapter's convention it is written $\operatorname{gr}_{\mathfrak{m}} R$; it is the ring called $G_{\mathfrak{m}}(R)$ in definition 25.5.1.

Theorem 29.7.3: Characterizations of a Regular Local Ring

Let $(R, \mathfrak{m})$ be a Noetherian local ring of dimension d with residue field k. The following are equivalent.

1. R is regular.
2. $\mathfrak{m}$ is generated by d elements.
3. $\operatorname{gr}_{\mathfrak{m}} R \cong k[t_1, \ldots, t_d]$ as graded k-algebras.

***Proof*:**

(1) $\iff$ (2) The least number of generators of $\mathfrak{m}$ is $\dim_k \mathfrak{m}/\mathfrak{m}^2$, as recorded after corollary 29.7.1, and that number is at least d by the same corollary. So $\mathfrak{m}$ is generated by d elements exactly when $\dim_k \mathfrak{m}/\mathfrak{m}^2 = d$, which is definition 29.7.2.

(2) $\Rightarrow$ (3) Let $x_1, \dots, x_d$ generate $\mathfrak{m}$ and let

$$\varphi\colon k[t_1, \dots, t_d] \to \operatorname{gr}_{\mathfrak{m}} R, \qquad t_i \mapsto x_i + \mathfrak{m}^2 \in \mathfrak{m}/\mathfrak{m}^2$$

be the graded k-algebra map they determine. It is surjective, because $\mathfrak{m}^n$ is generated by the degree-n monomials in the x_i, so $\mathfrak{m}^n/\mathfrak{m}^{n+1}$ is spanned over k by their classes. If $d = 0$ then $\mathfrak{m} = \mathfrak{m}^2$, Nakayama gives $\mathfrak{m} = 0$, and both sides are k; assume $d \geq 1$.

Suppose $\ker\varphi \neq 0$. The kernel of a graded map is graded, so it contains a nonzero homogeneous F of some degree e. For $n \geq e$ multiplication by F is an injection $k[t]_{n-e} \hookrightarrow k[t]_n$, the polynomial ring being a domain, and $(\operatorname{gr}_{\mathfrak{m}} R)_n$ is a quotient of $k[t]_n/F\,k[t]_{n-e}$. Hence

$$\dim_k \mathfrak{m}^n/\mathfrak{m}^{n+1} \leq \binom{n+d-1}{d-1} - \binom{n-e+d-1}{d-1}$$

Both binomials are polynomials in n of degree $d-1$ with the same leading coefficient $1/(d-1)!$, so their difference has degree at most $d-2$.

Since $R/\mathfrak{m} = k$, the length $\ell(\mathfrak{m}^r/\mathfrak{m}^{r+1})$ is its k-dimension, and

$$\ell(R/\mathfrak{m}^n) = \sum_{r=0}^{n-1} \ell(\mathfrak{m}^r/\mathfrak{m}^{r+1})$$

Summing a function that is eventually bounded by a polynomial of degree at most $d-2$ produces one eventually bounded by a polynomial of degree at most $d-1$; the finitely many terms with $r < e$ contribute a constant and do not affect the degree. So $d(R) \leq d-1$. But $d(R) = \dim R = d$ by theorem 29.4.1, a contradiction. Therefore $\ker\varphi = 0$ and φ is an isomorphism.

(3) $\Rightarrow$ (1) Comparing degree-one pieces, $\mathfrak{m}/\mathfrak{m}^2 \cong k^d$, so $\dim_k \mathfrak{m}/\mathfrak{m}^2 = d = \dim R$.

This closes the cycle, completing the proof.

Criterion (3) is the one that travels. Everything below is extracted from it by the same mechanism: a property of the polynomial ring $\operatorname{gr}_{\mathfrak{m}} R$ is transferred back to R along the filtration by powers of $\mathfrak{m}$, and the transfer works because Krull's theorem makes the filtration separated. The first instance is that a regular local ring has no zero-divisors, which is not visible from the numerical definition at all.

Lemma 29.7.4: A Graded Domain Lifts

Let $(R, \mathfrak{m})$ be a local ring with $\bigcap_{n\geq 0} \mathfrak{m}^n = 0$. If $\operatorname{gr}_{\mathfrak{m}} R$ is an integral domain, then so is R.

***Proof*:**
Let $a, b \in R$ be nonzero. Because $\bigcap_n \mathfrak{m}^n = 0$, neither lies in every power of $\mathfrak{m}$, so there are largest integers $r, s \geq 0$ with $a \in \mathfrak{m}^r$ and $b \in \mathfrak{m}^s$ (reading $\mathfrak{m}^0 = R$); thus $a \notin \mathfrak{m}^{r+1}$ and $b \notin \mathfrak{m}^{s+1}$. Their classes $a^* = a + \mathfrak{m}^{r+1}$ and $b^* = b + \mathfrak{m}^{s+1}$ are then nonzero in the graded pieces $(\operatorname{gr}_{\mathfrak{m}} R)_r$ and $(\operatorname{gr}_{\mathfrak{m}} R)_s$.

Since $\operatorname{gr}_{\mathfrak{m}} R$ is a domain, $a^*b^* \neq 0$ in $(\operatorname{gr}_{\mathfrak{m}} R)_{r+s}$, which says exactly that $ab \notin \mathfrak{m}^{r+s+1}$. In particular $ab \neq 0$, completing the proof.

Corollary 29.7.5: A Regular Local Ring Is a Domain

Every regular local ring is an integral domain.

***Proof*:**
Let $(R, \mathfrak{m})$ be regular of dimension d. In a local ring $\mathfrak{m}$ is the Jacobson radical, so corollary 25.4.12 applies and gives $\bigcap_n \mathfrak{m}^n = 0$. By theorem 29.7.3(3) the ring $\operatorname{gr}_{\mathfrak{m}} R$ is a polynomial ring over a field, hence a domain, and lemma 29.7.4 transfers this to R, as we sought to show.

Regularity is also insensitive to completion, for the same reason: passing to $\hat{R}$ changes neither the associated graded ring nor the dimension, and criterion (3) sees nothing else. This is what makes it legitimate to test regularity after completing, where the Cohen structure theorem (theorem 25.6.1) puts formal coordinates on the ring.

Proposition 29.7.6: Regularity and Completion

Let $(R, \mathfrak{m})$ be a Noetherian local ring and $\hat{R}$ its $\mathfrak{m}$-adic completion. Then R is regular if and only if $\hat{R}$ is regular.

***Proof*:**
The ring $\hat{R}$ is Noetherian by theorem 25.5.7, and it is local with maximal ideal $\hat{\mathfrak{m}}$ and the same residue field, since $R/\mathfrak{m}^n \cong \hat{R}/\hat{\mathfrak{m}}^n$ for every n. By proposition 25.5.2(2) the two associated graded rings agree, $\operatorname{gr}_{\mathfrak{m}} R \cong \operatorname{gr}_{\hat{\mathfrak{m}}} \hat{R}$, and by corollary 25.5.9 the two dimensions agree. Criterion (3) of theorem 29.7.3 is a statement about those two data alone, so it holds for R exactly when it holds for $\hat{R}$, completing the proof.

Example 29.7.7: Regular and Singular Local Rings

1. *Formal power series.* Let $R = k[[x_1, \ldots, x_n]]$, the completion of $A = k[x_1, \ldots, x_n]_{(x_1, \ldots, x_n)}$ (example 25.1.2). Then $\mathfrak{m} = (x_1, \ldots, x_n)$ and $\mathfrak{m}/\mathfrak{m}^2$ has the n classes $\bar{x}_i$ as a basis, so the embedding dimension is n. The dimension is also n: $\dim A = n$ because $\dim k[x_1, \ldots, x_n] = n$ by theorem 29.6.2 and localizing at a maximal ideal of height n retains it, and $\dim R = \dim A$ by corollary 25.5.9. So R is regular, with $x_1, \ldots, x_n$ a regular system of parameters, and $\operatorname{gr}_{\mathfrak{m}} R \cong k[t_1, \ldots, t_n]$ recovers the fact that a power series has a well-defined lowest-degree homogeneous part.

2. *Discrete valuation rings.* A Noetherian local domain of dimension 1 is regular exactly when

$\mathfrak{m}$ is principal, which by theorem 27.2.2 is exactly when it is a discrete valuation ring. So in dimension one, regular, normal and discrete-valuation all coincide; this is the case [27] uses to identify the smooth points of a curve.

3. *The cusp.* Let $A = k[x, y]/(y^2 - x^3)$ and $R = A_{\mathfrak{m}}$ with $\mathfrak{m} = (\bar{x}, \bar{y})$. Then $\dim R = 1$, since A is a one-dimensional domain by theorem 29.6.2 ($\mathrm{Frac}(A) = k(t)$ under $\bar{x} \mapsto t^2$, $\bar{y} \mapsto t^3$) and $\mathfrak{m}$ is maximal. But the defining relation $y^2 - x^3$ lies in $(x, y)^2$, so it imposes no linear condition: writing $\mathfrak{n} = (x, y) \subseteq k[x, y]$, one has $\mathfrak{m}/\mathfrak{m}^2 \cong \mathfrak{n}/(\mathfrak{n}^2 + (y^2 - x^3)) = \mathfrak{n}/\mathfrak{n}^2$, which is two-dimensional. So the embedding dimension is $2 > 1 = \dim R$ and R is not regular. This is the algebraic signature of the cusp of the curve $y^2 = x^3$ at the origin.

4. *A regular ring of dimension zero.* If $\dim R = 0$ then regularity forces $\mathfrak{m} = \mathfrak{m}^2$, hence $\mathfrak{m} = 0$ by Nakayama, so the regular local rings of dimension zero are exactly the fields. Contrast the Artinian local ring $k[x]/(x^2)$, which has dimension 0 and embedding dimension 1, so it is not regular; by corollary 29.7.5 it could not be, having the zero-divisor $\bar{x}$.

29.8 Regular Local Rings Are Factorial

Section 29.7 characterized regularity three ways and drew one consequence from the graded criterion, that a regular local ring is a domain. The theorem of this section is deeper and is not suggested by any of the three criteria: a regular local ring has unique factorization. It is due to Auslander and Buchsbaum, and the proof is an induction on dimension that leaves the ring at every step, passing to a localization in which the dimension has dropped.

Two criteria carry the induction, and both are worth having on their own. The first converts factoriality into a statement about height-one primes, which is what makes it accessible to dimension theory at all. The second says factoriality may be checked after inverting a single prime element. Read geometrically the theorem says that on a smooth variety every codimension-one subvariety is cut out near each point by one equation, which is how [27] uses it.

Lemma 29.8.1: Factoriality by Height-One Primes

Let R be a Noetherian domain. Then R is a unique factorization domain if and only if every prime of height one is principal.

***Proof*:**
($\Rightarrow$) Let $\mathfrak{p}$ have height one and pick $a \in \mathfrak{p}$ nonzero; a is a nonunit because $\mathfrak{p}$ is proper. Factor $a = p_1 \cdots p_r$ into irreducibles. As $\mathfrak{p}$ is prime, some $p_i \in \mathfrak{p}$, and in a unique factorization domain an irreducible element is prime (proposition 10.1.29), so (p_i) is a nonzero prime with $0 \subsetneq (p_i) \subseteq \mathfrak{p}$. A strict inclusion $(p_i) \subsetneq \mathfrak{p}$ would exhibit a chain of length two descending from $\mathfrak{p}$, contradicting $\operatorname{ht}\mathfrak{p} = 1$. Hence $\mathfrak{p} = (p_i)$.

($\Leftarrow$) A Noetherian ring satisfies the ascending chain condition on ideals, in particular on principal ideals, so by theorem 10.1.30 it suffices to show every irreducible element is prime. Let a be irreducible. Then (a) is a proper nonzero ideal, so it has a minimal prime $\mathfrak{p}$. Krull's height theorem (theorem 29.5.2) with $c = 1$ gives $\operatorname{ht}\mathfrak{p} \leq 1$, and $\mathfrak{p} \neq 0$ since $0 \neq a \in \mathfrak{p}$, so $\operatorname{ht}\mathfrak{p} = 1$. By hypothesis $\mathfrak{p} = (p)$ for some p, and $a \in (p)$ gives $a = pb$. Here p is a nonunit, so irreducibility of a forces b to be a unit; therefore $(a) = (p) = \mathfrak{p}$ is prime and a is a prime element, completing the proof.

The criterion is what ties factoriality to the rest of this chapter: height is a dimension-theoretic invariant, and Krull's height theorem is exactly what produced the height-one prime in the second half. The next lemma is the descent step. Inverting one prime element destroys only the primes containing it, and those form a single height-one class that can be checked by hand.

Lemma 29.8.2: Nagata's Criterion

Let R be a Noetherian domain and $x \in R$ a prime element. If the localization $R_x = R[1/x]$ is a unique factorization domain, then so is R.

***Proof*:**
Both R and R_x are Noetherian domains, the latter by proposition 23.4.7(1),(2), so lemma 29.8.1 applies to each. Let $\mathfrak{p} \subseteq R$ be a prime of height one; it suffices to show $\mathfrak{p}$ is principal.

Case $x \in \mathfrak{p}$. Then $0 \subsetneq (x) \subseteq \mathfrak{p}$ with (x) prime, and $\operatorname{ht}\mathfrak{p} = 1$ forces $\mathfrak{p} = (x)$.

Case $x \notin \mathfrak{p}$. The primes of R_x are the extensions of the primes of R not meeting $\{x^n \mid n \geq 0\}$, and the correspondence preserves inclusions (proposition 23.4.5), so $\mathfrak{p}R_x$ is a prime of height one. As R_x is a unique factorization domain, lemma 29.8.1 makes $\mathfrak{p}R_x$ principal. Multiplying a generator by a power of the unit x we may take the generator to be some $a \in R$, and then $a \in \mathfrak{p}R_x \cap R = \mathfrak{p}$, the contraction of the extension of $\mathfrak{p}$ being $\mathfrak{p}$ again since $\mathfrak{p}$ misses the powers of x.

We may further assume $x \nmid a$. If $a = xa'$ then $a' \in \mathfrak{p}$, because $a \in \mathfrak{p}$, $x \notin \mathfrak{p}$ and $\mathfrak{p}$ is prime; and a' still generates $\mathfrak{p}R_x$, since x is a unit there. This replacement cannot repeat indefinitely: it would place a in $\bigcap_n (x^n)$, which is 0 by corollary 25.4.11, while $a \neq 0$.

Now $\mathfrak{p} = (a)$. One inclusion is $a \in \mathfrak{p}$. For the other, let $b \in \mathfrak{p}$. Then $b \in \mathfrak{p}R_x = aR_x$, so $x^n b = ac$ for some $c \in R$ and $n \geq 0$. If $n \geq 1$ then x divides ac; as x is prime and $x \nmid a$, it divides c, and cancelling x from both sides (legitimate in a domain) lowers n by one. Iterating gives $b \in (a)$. So every height-one prime of R is principal, and lemma 29.8.1 makes R a unique factorization domain, completing the proof.

What remains is to produce a generator for a height-one prime after inverting x, and this is where the induction's homological input enters. The prime will be visibly principal at every localization, and the problem is to pass from that local information to a single global generator. The next two lemmas do so; the first is elementary, and the second is the one place this section imports a fact it does not prove.

Lemma 29.8.3: A Finite Free Resolution Makes a Projective Module Stably Free

Let S be a ring and P a finitely generated projective S-module admitting a resolution $0 \to F_n \to \cdots \to F_0 \to P \to 0$ by finitely generated free modules. Then P is *stably free*: $P \oplus S^a \cong S^b$ for some $a, b \geq 0$.

***Proof*:**
Induct on n. If $n = 0$ then $P \cong F_0$ is free. For $n \geq 1$ put $K = \ker(F_0 \to P)$. Since P is projective the sequence $0 \to K \to F_0 \to P \to 0$ splits, so $F_0 \cong P \oplus K$; in particular K is a direct summand of a finitely generated free module, hence finitely generated projective, and $0 \to F_n \to \cdots \to F_1 \to K \to 0$ is a free resolution of length $n-1$. By induction $K \oplus S^a \cong S^b$ for some a, b. Then

$$P \oplus S^b \cong P \oplus K \oplus S^a \cong F_0 \oplus S^a$$

which is free, as we sought to show.

Lemma 29.8.4: A Locally Principal Ideal of Finite Global Dimension Is Principal

Let S be a Noetherian domain of finite global dimension and $I \subseteq S$ a nonzero ideal such that $I_\mathfrak{q}$ is a principal ideal of $S_\mathfrak{q}$ for every prime $\mathfrak{q}$ of S. Then I is principal.

***Proof*:**
Step 1: I is invertible, hence projective. Work inside $K = \mathrm{Frac}(S)$ and put $I^{-1} = \{z \in K \mid zI \subseteq S\}$, an S-submodule of K containing S. The product II^{-1} is an ideal of S; it equals S, because for each prime $\mathfrak{q}$ the localization $I_\mathfrak{q}$ is generated by one element $t_\mathfrak{q} \neq 0$, so $t_\mathfrak{q}^{-1} \in (I^{-1})_\mathfrak{q}$ and $(II^{-1})_\mathfrak{q} \ni t_\mathfrak{q} t_\mathfrak{q}^{-1} = 1$, and an ideal that is the unit ideal at every prime is the unit ideal (proposition 23.4.13 applied to S/II^{-1}). Choose $a_1, \ldots, a_r \in I$ and $b_1, \ldots, b_r \in I^{-1}$ with $\sum_i a_i b_i = 1$. Then $x \mapsto (b_1 x, \ldots, b_r x)$ is an S-module map $I \to S^r$, and composing with $(s_1, \ldots, s_r) \mapsto \sum_i s_i a_i$ returns x. So I is a direct summand of S^r, hence finitely generated projective, and $I_\mathfrak{q} \cong S_\mathfrak{q}$ at every prime.

Step 2: I is stably free. Finite global dimension gives I a finite resolution by finitely generated free modules, so lemma 29.8.3 applies: $I \oplus S^a \cong S^b$ for some a, b. Localizing at any prime and comparing ranks gives $1 + a = b$, so the relation reads $I \oplus S^a \cong S^{a+1}$.

Step 3: rank one and stably free give free. This is corollary 16.4.6, applied to I: it is finitely generated projective with $I_\mathfrak{q} \cong S_\mathfrak{q}$ at every prime, and Step 2 supplies the required relation. So $I \cong S$ as an S-module, and the image of 1 under such an isomorphism generates I, completing the proof.

Theorem 29.8.5: Regular Local Rings Are Factorial

Every regular local ring is a unique factorization domain.

***Proof*:**
Let $(R, \mathfrak{m})$ be regular of dimension d; induct on d.

Base cases. If $d = 0$ then R is a field (example 29.7.7), which has no nonzero nonunits and is trivially factorial. If $d = 1$ then R is a Noetherian local domain whose maximal ideal is principal, hence a discrete valuation ring (theorem 27.2.2); its only height-one prime is $\mathfrak{m}$, which is principal, so lemma 29.8.1 applies.

Producing a prime element. Let $d \geq 2$. By corollary 29.7.5 R is a domain. Choose $x \in \mathfrak{m} \setminus \mathfrak{m}^2$. Its class is nonzero in $\mathfrak{m}/\mathfrak{m}^2$, so it extends to a basis, and lifting gives a minimal generating set $x, x_2, \ldots, x_d$ of $\mathfrak{m}$. Then $\mathfrak{m}/(x)$ is generated by the $d-1$ classes of $x_2, \ldots, x_d$, while

$\dim R/(x) = d-1$ because x is a nonzero nonunit in a domain (the consequence of theorem 29.5.2 recorded after it). So $R/(x)$ is regular by theorem 29.7.3(2), hence a domain by corollary 29.7.5, and therefore x is a prime element of R.

Reduction to a localization. By lemma 29.8.2 it is enough that $S = R_x$ is a unique factorization domain, and by lemma 29.8.1 it is enough that every height-one prime of S is principal. Note S is a Noetherian domain by proposition 23.4.7.

Global dimension is finite along the way. The Auslander-Buchsbaum-Serre theorem (theorem 57.2.8) makes $\operatorname{gldim} R$ finite, R being regular. Global dimension does not rise under localization: an S'-module over a localization S' of R is the localization of itself viewed as an R-module, and localizing an R-projective resolution gives an S'-projective resolution of the same length, localization being exact (proposition 23.4.9) and carrying projectives to projectives. Hence $\operatorname{gldim} S \leq \operatorname{gldim} R < \infty$, and likewise for every $R_{\mathfrak{p}}$.

The localizations are smaller regular rings. Let $\mathfrak{q}$ be a prime of S and $\mathfrak{p}$ the corresponding prime of R, so $S_{\mathfrak{q}} \cong R_{\mathfrak{p}}$ and $x \notin \mathfrak{p}$. Then $\mathfrak{p} \subsetneq \mathfrak{m}$, so $\dim R_{\mathfrak{p}} = \operatorname{ht} \mathfrak{p} < d$. Since $\operatorname{gldim} R_{\mathfrak{p}}$ is finite, the Auslander-Buchsbaum-Serre theorem in the other direction makes $R_{\mathfrak{p}}$ regular, and the induction hypothesis makes it a unique factorization domain.

Conclusion. Let $\mathfrak{P}$ be a height-one prime of S. For a prime $\mathfrak{q}$ of S, either $\mathfrak{P} \not\subseteq \mathfrak{q}$, in which case $\mathfrak{P}S_{\mathfrak{q}} = S_{\mathfrak{q}}$ is principal, or $\mathfrak{P} \subseteq \mathfrak{q}$, in which case $\mathfrak{P}S_{\mathfrak{q}}$ is a height-one prime of the unique factorization domain $S_{\mathfrak{q}}$ and is principal by lemma 29.8.1. So $\mathfrak{P}$ is a nonzero locally principal ideal of a Noetherian domain of finite global dimension, and lemma 29.8.4 makes it principal. Therefore S is a unique factorization domain, and lemma 29.8.2 carries this back to R, completing the proof.

Factoriality is the strongest of the standard finiteness conditions a local ring can carry, and the two weaker ones follow from it at once. The first is integral closure.

Corollary 29.8.6: A Unique Factorization Domain Is Integrally Closed

Every unique factorization domain is integrally closed in its fraction field.

***Proof*:**
Let R be a unique factorization domain with fraction field K and let $z \in K$ be integral over R. Write $z = a/b$ with $a, b \in R$, $b \neq 0$, sharing no irreducible factor, which unique factorization permits. From $z^n + c_{n-1}z^{n-1} + \cdots + c_0 = 0$ with $c_i \in R$, multiplying by b^n gives

$$a^n = -b\left(c_{n-1}a^{n-1} + c_{n-2}a^{n-2}b + \cdots + c_0 b^{n-1}\right)$$

So every irreducible factor p of b divides a^n, and p is prime (proposition 10.1.29), so p divides a, contrary to the choice of a and b. Hence b has no irreducible factor at all, so it is a unit and $z = ab^{-1} \in R$, as we sought to show.

Corollary 29.8.7: A Regular Local Ring Is Normal

Every regular local ring is a normal domain, and in particular is regular in codimension one.

***Proof*:**
Combine theorem 29.8.5 with corollary 29.8.6; the ring is a domain by corollary 29.7.5. For the last clause, a localization of a normal domain is normal by proposition 23.4.7(3), completing the proof.

Example 29.8.8: Normal but Not Factorial
Neither implication in

$$\text{regular} \Rightarrow \text{factorial} \Rightarrow \text{normal}$$

reverses. Let k have characteristic $\neq 2$ and let $B = k[u^2, uv, v^2] \subseteq k[u, v]$, the subring of polynomials of even total degree. Sending $x \mapsto u^2$, $y \mapsto v^2$, $z \mapsto uv$ identifies B with $k[x, y, z]/(xy - z^2)$, so B is the coordinate ring of the quadric cone.

B is normal. Let σ be the involution $u \mapsto -u$, $v \mapsto -v$ of $k[u, v]$. Its fixed ring is the even part, which is B, and its fixed field inside $k(u, v)$ is $\mathrm{Frac}(B) = k(u^2, u/v)$. Let $w \in \mathrm{Frac}(B)$ be integral over B. Then w is integral over $k[u, v]$, which is a unique factorization domain (corollary 11.2.4) and hence integrally closed by corollary 29.8.6; so $w \in k[u, v]$. Being in $\mathrm{Frac}(B)$, w is fixed by σ, so $w \in k[u, v]^\sigma = B$.

B is not factorial. In B the identity $u^2 \cdot v^2 = (uv)^2$ gives two factorizations of one element into irreducibles: each of u^2, v^2, uv has degree two, which is the least positive degree occurring in B, so none of them factors further in B, and no two of them are associates ($B^\times = k^\times$).

B is not regular. Localize at $\mathfrak{m} = (u^2, uv, v^2)$. The fraction field $k(u^2, u/v)$ has transcendence degree two over k, so $\dim B_\mathfrak{m} = 2$ by theorem 29.6.2, while $\mathfrak{m}/\mathfrak{m}^2$ has the three classes of u^2, uv, v^2 as a basis, $\mathfrak{m}^2$ being spanned by the monomials of degree four. So the embedding dimension is $3 > 2$.

The three computations are consistent with the displayed implications and show both are strict: $B_\mathfrak{m}$ is normal and not factorial, and by theorem 29.8.5 a non-factorial ring cannot be regular, which the third computation confirms directly.

29.9 Normal Domains in Codimension One

A normal domain is determined by its behaviour in codimension one. That is the content of this section, and it is what makes normality a condition one can check by looking only at the height-one primes: the ring is the intersection of the discrete valuation rings sitting at them. The result is used by the homological characterizations of normality in *Homological Algebra* [15] and by the divisor theory of [27] and [5].

Everything rests on one fact about principal ideals, which is where normality enters.

Lemma 29.9.1: Associated Primes of a Principal Ideal Have Height One

Let R be a Noetherian normal domain and $b \in R$ a nonzero nonunit. Then every associated prime of (b) has height one.

***Proof*:**
Let $\mathfrak{p} \in \operatorname{Ass}(R/(b))$. Associated primes localize (corollary 24.6.2), so $\mathfrak{p}R_\mathfrak{p} \in \operatorname{Ass}\big(R_\mathfrak{p}/bR_\mathfrak{p}\big)$, and $R_\mathfrak{p}$ is again a Noetherian normal domain by proposition 23.4.7. Since $\operatorname{ht}\mathfrak{p} = \operatorname{kdim} R_\mathfrak{p}$, it suffices to treat the case where $(R, \mathfrak{m})$ is local with $\mathfrak{m} \in \operatorname{Ass}(R/(b))$ and to show $\operatorname{kdim} R = 1$.

By definition 24.3.1 there is $a \in R$ with $\mathfrak{m} = \operatorname{Ann}(\overline{a}) = (b : a)$, where $\overline{a}$ is the class of a in $R/(b)$; in particular $a \notin (b)$, since otherwise the annihilator would be all of R. Put $x = a/b \in \operatorname{Frac}(R)$, so $x \notin R$. For every $r \in \mathfrak{m}$ one has $ra \in (b)$, hence $rx = ra/b \in R$: that is, $x\mathfrak{m} \subseteq R$, and $x\mathfrak{m}$ is an R-submodule of R, so an ideal.

Case 1: $x\mathfrak{m} \subseteq \mathfrak{m}$. Then $x^n\mathfrak{m} \subseteq \mathfrak{m}$ for every n, so $R[x]\mathfrak{m} \subseteq \mathfrak{m} \subseteq R$. Choose $0 \neq c \in \mathfrak{m}$, which exists as $b \in \mathfrak{m}$ is nonzero. Then $cR[x] \subseteq R$, so $R[x] \subseteq c^{-1}R$, a cyclic R-module; as R is Noetherian, the submodule $R[x]$ is a finitely generated R-module, and proposition 26.1.4(2) makes x integral over R. Normality gives $x \in R$, contradicting $x \notin R$.

Case 2: $x\mathfrak{m} \not\subseteq \mathfrak{m}$. Then the ideal $x\mathfrak{m}$ is not contained in the unique maximal ideal, so $x\mathfrak{m} = R$. Pick $t \in \mathfrak{m}$ with $xt = 1$. For any $p \in \mathfrak{m}$ we have $xp \in R$ and $p = t\,(xp)$, so $\mathfrak{m} = tR$ is principal, and it is nonzero. A Noetherian local domain whose maximal ideal is principal and nonzero is a discrete valuation ring by theorem 27.2.2, hence of dimension one.

Case 1 is impossible, so Case 2 holds and $\operatorname{kdim} R = 1$, as we sought to show.

Theorem 29.9.2: A Normal Domain Is the Intersection of Its Height-One Localizations

Let R be a Noetherian normal domain with fraction field K. Then $R_\mathfrak{p}$ is a discrete valuation ring for every prime $\mathfrak{p}$ of height one, and

$$R = \bigcap_{\operatorname{ht}\mathfrak{p}=1} R_\mathfrak{p}$$

the intersection taken inside K.

***Proof*:**
For $\operatorname{ht}\mathfrak{p} = 1$ the ring $R_\mathfrak{p}$ is a Noetherian local domain of dimension one, normal by proposition 23.4.7, hence a discrete valuation ring by theorem 27.2.2.

The inclusion $R \subseteq \bigcap R_\mathfrak{p}$ is immediate. For the reverse, let x lie in every $R_\mathfrak{p}$ and write $x = a/b$ with $a, b \in R$ and $b \neq 0$. If b is a unit then $x \in R$, so assume otherwise and suppose $a \notin (b)$, aiming at a contradiction.

The class $\overline{a} \in R/(b)$ is then nonzero, and $I = \operatorname{Ann}(\overline{a}) = (b : a)$ is a proper ideal every element of which kills $\overline{a}$, hence is a zero-divisor on $R/(b)$. By corollary 24.5.2 the zero-divisors on $R/(b)$ are the union of the associated primes of (b), of which there are finitely many (corollary 24.4.3), so prime avoidance (proposition 24.5.1) places $I \subseteq \mathfrak{q}$ for a single $\mathfrak{q} \in \operatorname{Ass}(R/(b))$. By lemma 29.9.1, $\operatorname{ht}\mathfrak{q} = 1$.

Now $x \in R_\mathfrak{q}$ by hypothesis, so $a/b = c/s$ for some $c \in R$ and $s \notin \mathfrak{q}$, whence $sa \in bR$ and $s \in (b : a) = I \subseteq \mathfrak{q}$. That contradicts $s \notin \mathfrak{q}$. Therefore $a \in (b)$ and $x \in R$, completing the proof.

The theorem is the reason divisor theory works on a normal variety: a rational function regular at every codimension-one point is regular, so a function is controlled by its orders of vanishing along the height-one primes alone. That reading, and the class group it supports, is [27].

30

Flatness and Faithfully-Flat Descent

Flatness is the algebraic form of continuous variation. A flat module is one against which tensoring preserves injections, so that submodule relations survive base change; for a ring map $R \to S$ it says the fibers of $\operatorname{Spec} S \to \operatorname{Spec} R$ vary without jumping. When in addition the base change loses no information (when it is *faithfully flat*), one can argue in reverse: a property that holds after tensoring up already held below. This is *descent*, and it is the reason the local models of a space determine the space. The chapter records the ring-theoretic core that the geometry books cash out as faithfully-flat and fppf descent, closing with the local criterion that detects flatness one fiber at a time, the tool deformation theory runs on.

Basic flatness is assumed from [25]; the homological criterion and its local refinement are proved in [15] and cited here.

30.1 Flat Modules

Recall from [25] that an R-module M is *flat* if $- \otimes_R M$ is exact, equivalently if it preserves injections; a ring map $R \to S$ is flat if S is flat as an R-module. Flatness is preserved under base change and composition, and is detected locally: M is flat over R iff $M_{\mathfrak{p}}$ is flat over $R_{\mathfrak{p}}$ for every prime $\mathfrak{p}$ (proposition 23.4.16). Localizations are flat, and for a Noetherian local ring the completion is flat (corollary 25.4.6). The homological measure of the failure of flatness is Tor: M is flat iff $\operatorname{Tor}_1^R(M, N) = 0$ for all N, and it suffices to test $N = R/\mathfrak{a}$ on finitely generated ideals; this is proved in [15] and used freely below.

30.2 Faithfully Flat Modules and Algebras

Definition 30.2.1: Faithfully Flat

An R-module M is *faithfully flat* if it is flat and, for every R-module N, the vanishing $N \otimes_R M = 0$ forces $N = 0$. A ring map $R \to S$ is faithfully flat if S is faithfully flat as an R-module.

Proposition 30.2.2: Characterizations of Faithful Flatness

For a flat R-module M the following are equivalent:

1. M is faithfully flat;
2. a sequence of R-modules is exact iff it stays exact after $- \otimes_R M$;
3. $\mathfrak{m}M \neq M$ for every maximal ideal $\mathfrak{m}$, i.e. $M \otimes_R \kappa(\mathfrak{m}) \neq 0$ for all $\mathfrak{m}$.

For a flat ring map $R \to S$ these are equivalent to $\operatorname{Spec} S \to \operatorname{Spec} R$ being surjective.

***Proof*:**
(1) $\Rightarrow$ (2): flatness already preserves exactness; conversely if $C_\bullet \otimes M$ is exact then $H(C_\bullet) \otimes M = H(C_\bullet \otimes M) = 0$ (a flat functor commutes with homology), so $H(C_\bullet) = 0$ by (1). (2) $\Rightarrow$ (3): the sequence $0 \to R/\mathfrak{m} \to 0$ is not exact, so by (2) it remains non-exact after $- \otimes M$, forcing $M/\mathfrak{m}M = (R/\mathfrak{m}) \otimes M \neq 0$. (3) $\Rightarrow$ (1): let $N \neq 0$ and $0 \neq x \in N$, so $Rx \cong R/\mathfrak{a}$ with $\mathfrak{a} = \operatorname{Ann}(x) \subseteq \mathfrak{m}$ for some maximal $\mathfrak{m}$. Tensoring with the flat M: the inclusion $Rx \hookrightarrow N$ gives an injection $Rx \otimes M \hookrightarrow N \otimes M$, while the surjection $R/\mathfrak{a} \to R/\mathfrak{m}$ gives a surjection $Rx \otimes M = M/\mathfrak{a}M \twoheadrightarrow M/\mathfrak{m}M \neq 0$; hence $Rx \otimes M \neq 0$, so $N \otimes M \neq 0$. For a flat ring map these are further equivalent to surjectivity of $\operatorname{Spec} S \to \operatorname{Spec} R$: (1) gives $S \otimes_R \kappa(\mathfrak{p}) \neq 0$ for every prime $\mathfrak{p}$, so every fibre is nonempty and the map is surjective; conversely a surjective map hits every maximal ideal, which is (3).

Corollary 30.2.3: Faithfully Flat Maps are Injective

A faithfully flat ring map $R \to S$ is injective, and $\mathfrak{a} = \mathfrak{a}S \cap R$ for every ideal $\mathfrak{a}$ of R.

***Proof*:**
Apply faithful flatness to the kernel K of $R \to S$: tensoring $0 \to K \to R \to S$ with the flat S gives $K \otimes_R S = \ker(S \to S \otimes_R S)$, the map being $s \mapsto 1 \otimes s$, which is split by the multiplication $S \otimes_R S \to S$; so $K \otimes S = 0$, whence $K = 0$. For the second claim, the map $R/\mathfrak{a} \to (R/\mathfrak{a}) \otimes_R S = S/\mathfrak{a}S$ has kernel $(\mathfrak{a}S \cap R)/\mathfrak{a}$; since $S/\mathfrak{a}S$ is faithfully flat over $R/\mathfrak{a}$ (base change of a faithfully flat map), it is injective by the first part, so $\mathfrak{a}S \cap R = \mathfrak{a}$.

Localization $R \to R_f$ is flat but rarely faithfully flat (its image in $\operatorname{Spec} R$ omits $V(f)$); by contrast a nonzero free module is faithfully flat, a finite free ring extension is faithfully flat, and for a Noetherian local ring the completion $R \to \hat{R}$ is faithfully flat.

30.3 Faithfully-Flat Descent

Faithful flatness lets a property established upstairs be brought back down.

Proposition 30.3.1: Descent of Module Properties

Let $R \to S$ be faithfully flat and M an R-module. Then M is finitely generated (respectively finitely presented, flat, or finitely generated projective) over R iff $M \otimes_R S$ has the corresponding property over S.

***Proof*:**
Necessity is base change. For sufficiency: if $M \otimes S$ is generated by finitely many elements, clear denominators so they are images of $m_i \otimes 1$; the cokernel C of $R^n \xrightarrow{(m_i)} M$ satisfies $C \otimes S = 0$, so $C = 0$ and M is finitely generated. Given that, finite presentation descends by applying the same to the kernel of $R^n \to M$ (finitely generated as $\ker \otimes S = \ker(S^n \to M \otimes S)$ is). Flatness descends because $\mathrm{Tor}_1^R(M, N) \otimes S = \mathrm{Tor}_1^S(M \otimes S, N \otimes S) = 0$ for all N, whence $\mathrm{Tor}_1^R(M, N) = 0$ by faithful flatness. Finitely generated projective descends since it equals finitely presented plus flat.

The deeper statement recovers a module, not merely a property, from data upstairs. Write $S^{\otimes 2} = S \otimes_R S$ and $S^{\otimes 3} = S \otimes_R S \otimes_R S$, with the two coprojections $p_1, p_2 \colon S \to S^{\otimes 2}$.

Theorem 30.3.2: Faithfully-Flat Descent for Modules

Let $R \to S$ be faithfully flat. Then $M \mapsto M \otimes_R S$ is an equivalence from R-modules to the category of *descent data*: pairs (N, φ) with N an S-module and $\varphi \colon p_1^* N \xrightarrow{\sim} p_2^* N$ an isomorphism over $S^{\otimes 2}$ satisfying the cocycle condition over $S^{\otimes 3}$. The inverse sends (N, φ) to the equalizer of φ-twisted coprojections $N \rightrightarrows N \otimes_S S^{\otimes 2}$.

***Proof*:**
The heart is the exactness, for every R-module M, of the *Amitsur complex*

$$0 \to M \to M \otimes_R S \to M \otimes_R S^{\otimes 2},$$

the second map being $m \otimes s \mapsto m \otimes (p_1 - p_2)(s)$. It suffices to check this after the faithfully flat base change $S \otimes_R -$, where the augmented complex acquires the contracting homotopy $s_0 \otimes s_1 \otimes \cdots \mapsto (s_0 s_1) \otimes s_2 \otimes \cdots$ (multiplication of the first two tensor factors), which shows exactness. Faithful flatness then gives exactness over R. Applying this with the canonical datum shows $M \otimes S$ recovers M as the stated equalizer, so the functor is fully faithful; the cocycle condition is exactly what makes the equalizer of a general (N, φ) an R-module whose base change is (N, φ), giving essential surjectivity.

This is the affine case of the descent that lets sheaves and torsors be glued along faithfully flat (fppf) covers, developed in [23].

30.4 The Local Criterion for Flatness

For finitely generated modules over a Noetherian local ring, flatness collapses to a single Tor, and this is what makes flatness checkable in families.

Theorem 30.4.1: Local Criterion for Flatness

Let $(R, \mathfrak{m})$ be a Noetherian local ring with residue field κ and M a finitely generated R-module. Then

$$M \text{ flat} \iff M \text{ free} \iff \mathrm{Tor}_1^R(\kappa, M) = 0$$

More generally, for a local homomorphism $(R, \mathfrak{m}) \to (S, \mathfrak{n})$ of Noetherian local rings and a finitely generated S-module M, one has M flat over R iff $\mathrm{Tor}_1^R(\kappa, M) = 0$; over an Artinian local base this is the criterion by which flatness of a deformation is tested one infinitesimal step at a time.

The equivalences and the general criterion are proved homologically in [15, Local Criterion for Flatness], which owns this theorem (DECISIONS 2026-08-17); here the point is that they reduce flatness (a condition on all modules) to the vanishing of one Tor against the residue field, and it is in this form that [23] invokes flatness over Artinian bases.

30.5 Fitting Ideals

The Fitting ideals of a finitely presented module refine the notion of rank: they cut out the loci where the number of generators jumps, commute with base change, and are the standard tool behind flattening stratifications.

Definition 30.5.1: Fitting Ideals

Let M be a finitely presented R-module with presentation $R^m \xrightarrow{A} R^n \to M \to 0$, so A is an $n \times m$ matrix. For $i \geq 0$ the *i-th Fitting ideal* $\mathrm{Fitt}_i(M)$ is generated by the $(n-i) \times (n-i)$ minors of A, with the conventions $\mathrm{Fitt}_i(M) = R$ when $n - i \leq 0$ and $\mathrm{Fitt}_i(M) = 0$ when $n - i > m$.

Proposition 30.5.2: Properties of Fitting Ideals

The Fitting ideals are independent of the chosen presentation, and:

1. $\mathrm{Fitt}_0(M) \subseteq \mathrm{Fitt}_1(M) \subseteq \cdots$, with $\mathrm{Fitt}_i(M) = R$ for $i \geq n$;
2. base change: $\mathrm{Fitt}_i(M \otimes_R R') = \mathrm{Fitt}_i(M)R'$ for any $R \to R'$;
3. $\mathrm{Fitt}_0(M) \subseteq \mathrm{Ann}(M)$ and $\mathrm{Ann}(M)^n \subseteq \mathrm{Fitt}_0(M)$;
4. for a prime $\mathfrak{p}$, the minimal number of generators of $M_{\mathfrak{p}}$ is $> d$ iff $\mathrm{Fitt}_d(M) \subseteq \mathfrak{p}$.

***Proof*:**
Any two finite presentations of M are related by adding free summands and invertible row/column operations, under which the ideal of $(n-i)$-minors is unchanged (a standard determinantal computation), giving well-definedness; base change is immediate since the minors of A map to those of $A \otimes R'$. The chain and the values for $i \geq n$ are clear. For (3), Cramer's rule applied to A gives $\mathrm{Fitt}_0(M)\, M = 0$, and expanding determinants shows $\mathrm{Ann}(M)^n \subseteq \mathrm{Fitt}_0(M)$. For (4), by Nakayama the minimal number of generators of $M_{\mathfrak{p}}$ equals $\dim_{\kappa(\mathfrak{p})} M \otimes \kappa(\mathfrak{p}) = n - \mathrm{rank}(A \bmod \mathfrak{p})$, and $\mathrm{Fitt}_d(M) \not\subseteq \mathfrak{p}$ says exactly that some $(n-d)$-minor is a unit mod $\mathfrak{p}$, i.e. $\mathrm{rank}(A \bmod \mathfrak{p}) \geq n - d$.

Thus the rank-jump locus $V(\mathrm{Fitt}_d(M)) = \{\mathfrak{p} \mid M_{\mathfrak{p}} \text{ needs} > d \text{ generators}\}$ is closed and base-change compatible, giving the flattening stratification used to build Quot and Hilbert schemes ([23]); $V(\mathrm{Fitt}_0(M))$ cuts out the support of M (with its Fitting scheme structure; the scheme-theoretic support proper is $V(\mathrm{Ann}\, M)$, with the same underlying closed set). Over a domain, a finitely presented M is locally free of rank r iff $\mathrm{Fitt}_{r-1}(M) = 0$ and $\mathrm{Fitt}_r(M) = R$, i.e. iff $\mathrm{Fitt}_r(M) = R$ and A has constant rank $n - r$.

30.6 Generic Flatness

Away from a proper closed subset of the base, a coherent family becomes flat. This Grothendieck finiteness underlies dimension counts in families and the construction of moduli.

Theorem 30.6.1: Generic Flatness

Let R be a Noetherian integral domain, S a finitely generated R-algebra, and M a finitely generated S-module. Then there is a nonzero $f \in R$ such that M_f is a free R_f-module (in particular flat over R_f).

***Proof*:**
We sketch the standard argument (Eisenbud, *Commutative Algebra*, Theorem 14.4; equivalently the Stacks Project generic-flatness lemma). By dévissage a finite filtration of M with quotients $S/\mathfrak{p}_i$ ($\mathfrak{p}_i \subseteq S$ prime) reduces the claim to $M = S/\mathfrak{p}$, since a module filtered by pieces that are free after localizing is itself free after localizing. If $\mathfrak{q} := \ker(R \to S/\mathfrak{p})$ is nonzero, then $\mathfrak{q}$ annihilates M, so $M_f = 0$ is free for any $0 \neq f \in \mathfrak{q}$. Otherwise $R \hookrightarrow S/\mathfrak{p}$ is a domain of finite type over R; by Noether normalization $(S/\mathfrak{p}) \otimes_R K$ is a *finite* module (not in general free) over a polynomial ring $K[t_1, \ldots, t_d]$, where $K = \mathrm{Frac}(R)$, and spreading out gives $0 \neq f_0$ with S_{f_0} finite over $R_{f_0}[t_1, \ldots, t_d]$. A Noetherian induction on the relative dimension d then trivializes, after inverting finitely many further elements of R, the torsion pieces of this finite module, producing a single f with M_f free over R_f.

Part VI

Affine Varieties

A shape cut out by polynomial equations is determined by the polynomial functions on it. This Part makes that statement into a dictionary: to each algebraic set in affine space is attached a ring, its coordinate ring, and the geometry of the set is recoverable from the algebra of the ring. Points become maximal ideals, irreducible subsets become primes, and the maps between varieties become homomorphisms between the rings. The dictionary is global: every entry in it is an invariant of the whole variety, and Part VII takes up the invariants attached to a single point.

The ground field is algebraically closed throughout, from chapter 32 onward; the Nullstellensatz, which is what makes the dictionary exact rather than approximate, is false without it. The commutative algebra the Part rests on is cited to *Commutative Algebra* [10] rather than rebuilt.

31

Algebraic Sets and the Zariski Topology

The dictionary this book is about begins with a single construction: to a set of polynomials, associate the points at which they all vanish. This chapter sets up that construction and its converse, which sends a set of points to the polynomials vanishing on it, and finds that the two are almost inverse to each other. The gap between "almost" and "exactly" is what chapter 32 closes.

Along the way the vanishing sets turn out to be the closed sets of a topology, the Zariski topology, which is coarse enough to be unlike any topology from analysis and fine enough to carry the whole theory. The chapter ends with the maps that respect it.

31.1 Affine Space and Regular Functions

In this whole chapter, R is a commutative ring and k be a field.

So far when polynomials were encountered, they were treated as a commutative ring with an indeterminate extension (or equivalently, they can be thought of as a ring over a free monoid). There is a natural way of thinking about any such polynomial as a function (which is probably how most first encounter polynomials): if $f \in k[x_1, x_2, ..., x_n]$, then if $(a_1, a_2, ..., a_n) \in k^n$, let $f(a_1, a_2, ..., a_n) \in k$ be the element which you get when replacing each x_i with a_i. Then we can make f into a function through the following map[1,2]:

$$k[x_1, ..., x_n] \to \mathrm{Hom}_{\mathbf{Set}}(k^n, k) \qquad f \mapsto ((a_1, ...a_n) \mapsto f(a_1, ..., a_n)) \tag{31.1}$$

where we shall consider the constants $c \in k[x_1, ..., x_n]$ to map to the functions that shall return the value of $c \in k$:

$$c \mapsto ((a_1, ..., a_n) \mapsto c)$$

Note that the codomain of the function in (31.1) is be the hom-set over the category **Set**, and not any other category we've extensively worked over so far (**Grp**, **Ring**, **Alg**, etc.); this was done since there are many non-linear functions representing non-linear polynomials (a simple example would be $x^2 \in k[x]$). On the other hand, the image of the above function is not just a collection of all set functions as they must all satisfy polynomials relations (can you find a set-function that is not given/representable by a polynomial?[3] The subset given by the function in (31.1) shall be called the set of *regular functions*: $\mathrm{Hom}_{\mathbf{Reg}}(k^n, k)$. Before properly boxing this definition, we shall take some more time pondering what the domain of the hom-set should be, namely we shall want the domain to represent Cartesian space without the extra algebraic structure that is usually implied by the notation k^n. For example, k^n usually denotes the n-dimensional vector-space over k with an origin, and more generally we think of k^n as having addition and scalar multiplication satisfying the axioms of a vector-space. Vector-spaces are "defined" for linear polynomials, with subspaces that are naturally "linear" (recall that subspaces are given by solutions of linear polynomials, see theorem 13.2.26)[4]. We shall want a new type of space for which subspaces are solutions to general polynomial equations.

Definition 31.1.1: Affine Space

Let $(k, +, \cdot)$ be any field. Then the collection of the n-tuples of k^n as a set shall be called *affine space*. Its elements shall be called *points* and the space shall be denoted $\mathbb{A}^n_k$.

Hence, a field can be thought of as $(\mathbb{A}^1_k, +, \cdot)$. The space $\mathbb{A}^n_k$ is called *affine* space rather than just "space" as there shall be an equally important (or arguably more important) space called *projective space* that shall be introduced in chapter 38[5]. The term affine can also be thought of as emphasizing the lack of importance of the origin (recall that a linear transformation that doesn't need to map the origin to itself was called affine).

With affine space established, let us now define the image of the function in (31.1):

---

[1] For those familiar with algebraic geometry, this map can be thought of as selecting the global sections of $\mathbb{A}^n_k$. By the Nullstellensatz, $k^n = \mathrm{Spec}\,(k[x_1, .., x_n])$, and by the representability of affine schemes $\Gamma(\mathrm{Spec}\,(R)) \cong \mathrm{Hom}_{\mathrm{Reg}}(\mathrm{Spec}\,(R), A^1_k)$

[2] advanced comment: if k is not algebraically closed, we are missing points, and hence this technically is *not* a representable functor and is not the sheaf obtains from embedding Var_k into $\mathbf{Sch}_k$

[3] Over finite fields, and only over finite fields, every set function is given by a polynomial: interpolate, using that $a^{|k|} = a$ for every $a \in k$.

[4] In [26], we shall show that the vector-space structure can be recovered using Hopf Algebras, but this adds more structure to the algebraic set

[5] More generally, we shall see that affine space will be the representation space for schemes which generalizes affine space, something which chapter 49 points the reader toward

Definition 31.1.2: Regular Function [on Affine Space]

Let F represent the function in (31.1):

$$F : k[x_1, ..., x_n] \to \mathrm{Hom}_{\mathbf{Set}}(k^n, k) \qquad f \mapsto ((a_1, ...a_n) \mapsto f(a_1, ..., a_n))$$

Then the collection $\mathrm{im}(F)$ is called the *regular functions*, and is usually denoted $\Gamma(\mathbb{A}^n_k)$, $k[\mathbb{A}^n_k]$, or $\mathrm{Hom}_{\mathbf{Reg}}(\mathbb{A}^n_k, \mathbb{A}^1_k)$.

Each of these notations for the collection of regular functions have its use. The $\mathrm{Hom}_{\mathbf{Reg}}(\mathbb{A}^n_k, \mathbb{A}^1_k)$ notion makes it clear what the domain and codomain is, which shall be particularly important once we take different codomain. The $k[\mathbb{A}^n_k]$ notation is particularly useful when we shall take subsets $V \subseteq \mathbb{A}^n_k$ and denote all regular functions on V as $k[V]$; in this case this notation emphasizes that we are working over the particular field k. This shall be useful for there are circumstances where we change which field we are working over[6]. The $\Gamma(\mathbb{A}^n_k)$ notation comes from modern algebraic geometry where gamma is used hint that these are the *global section* over $\mathbb{A}^n_k$ (see [5] for a full explanation). As this chapter is gearing up the reader for modern algebraic geometry, we shall preferentially use the $\Gamma(\mathbb{A}^n_k)$ notation to acclimate the reader. It should be pointed out that another common notation is $\mathcal{O}_{\mathbb{A}^n_k}(\mathbb{A}^n_k)$ which is more useful when it is important to keep track of regular functions on open subsets of $\mathbb{A}^n_k$ (once we define a topology on $\mathbb{A}^n_k$) in which case we shall have the collection of regular function on $U \subseteq \mathbb{A}^n_k$ be written as $\mathcal{O}_{\mathbb{A}^n_k}(U)$. This is not useful to us in this chapter, and so shall not be used.

The set $\Gamma(\mathbb{A}^n_k)$ forms a natural k-algebra structure given by point-wise operations:

$$f + g \equiv f(a) + g(a) \qquad fg \equiv f(a)g(a) \qquad c \cdot f = cf(a) \qquad \forall a \in k \tag{31.2}$$

With this k-algebra structure, it can be shown that $k[x_1, ..., x_n] \cong_k \mathrm{Hom}_{\mathbf{Reg}}(\mathbb{A}^n_k, \mathbb{A}^1_k)$, and so the two sets are often used interchangeably. If the distinction needs to be made clear between the element x_i and the function x_i, we shall write

$$k[X_1, ..., X_n] = \mathrm{Hom}_{\mathbf{Reg}}(\mathbb{A}^n_k, \mathbb{A}^1_k) = k[\mathbb{A}^k_n]$$

where each X_i is a regular function that is the projection onto the ith coordinate, and call an element of $k[X_1, ..., X_n]$ a *polynomial function* or *polynomial map*. This distinction shall become more important once there can be multiple polynomials that will be associated to the same regular function, which shall be the case in section 32.1. When it is clear from context, we may denote a polynomial or a regular function by $f(x)$ and $f(X)$ respectively where x is understood to be the indeterminate variables and X is understood to be the regular functions $X_1, ..., X_n$.

Due to this isomorphism, we can define a function from $k[x_1, ..., x_n]$ to k directly by directly substituting the values for the indeterminates with a point in the affine space. Namely, for each $a \in \mathbb{A}^n_k$, we take:

$$\mathrm{ev}^k_a : k[x_1, ..., x_n] \to k \qquad f(x) \mapsto f(a)$$

where x and a is understood to be n-tuples of indeterminates and values respectively. These are all k-algebra homomorphisms, for

$$\mathrm{ev}^k_a(f(x) + cg(x)) = \mathrm{ev}^k_a(f(x)) + c \cdot \mathrm{ev}^k_a(g(x)) \qquad \mathrm{ev}^k_a(f(x)g(x)) = \mathrm{ev}^k_a(f(x)\mathrm{ev}^k_a(g(x))$$

[6] A good example of the use of group cohomology is measure how much an isomorphism $k[V] \cong k[W]$ is preserved/fails when consider $\overline{k}[V]$ and $\overline{k}[W]$

The subscript of the evaluation function represents the point of evaluation, while the superscript represents the values we choose to plug for every indeterminate, where the values are usually the same as the base field, however they may differ, for example it is common to take $\mathbb{Z}$ or $\mathbb{Q}$[7]. Thus, a regular function can be thought of as the collection of evaluation functions on a polynomial.

31.1.3 Foreshadowing If you know elementary analysis or calculus, then you are probably aware that the types of functions we most commonly use are *continuous*. Continuous functions are a very important tool in analysing the properties of the space we're studying (ex. $\mathbb{R}$, $\mathbb{R}^n$, $L^2(\mathbb{R})$, etc.), continuous functions can be thought of as the natural types of functions that preserve some local properties of the space. In Algebra, we can think of polynomial functions as the natural types of functions. This comparison between continuous functions mapping to $\mathbb{R}$ and polynomial functions mapping to k will in fact be incredibly precise, as we will see in the last section of this chapter.

31.2 Algebraic Sets and Vanishing Ideals

31.2.1 Quick Side-Note A common misconception, born from a strong reliance on graphs in most first-year undergraduate calculus/analysis classes, is to intuitively associate the zero-set of a function with the *graph* of a function.
For example, $f(x) = x$ has a zero-set containing only the point 0, while the function $F(x, y) = x - y$ has a zero-set that forms the graph $\Gamma(f)$. This distinction may seem trivial, but it is surprisingly easy to conflate the two concepts when moving into higher dimensions.

Graphs can be thought of as the collection of the solutions of $f(x_1, \ldots, x_n) = y_0$ as the constant term y_0 varies.

From such polynomials functions, we can create many interesting shapes depending on the zeros', or locus, of polynomial:

[7]These shall be called $\mathbb{Z}$-valued and $\mathbb{Q}$-valued points of the polynomial, or more precisely the variety (as we shall soon see)

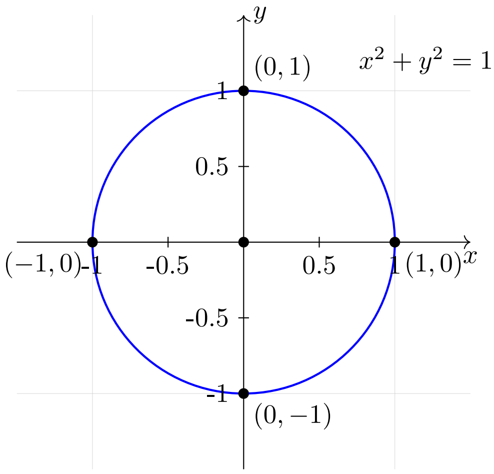
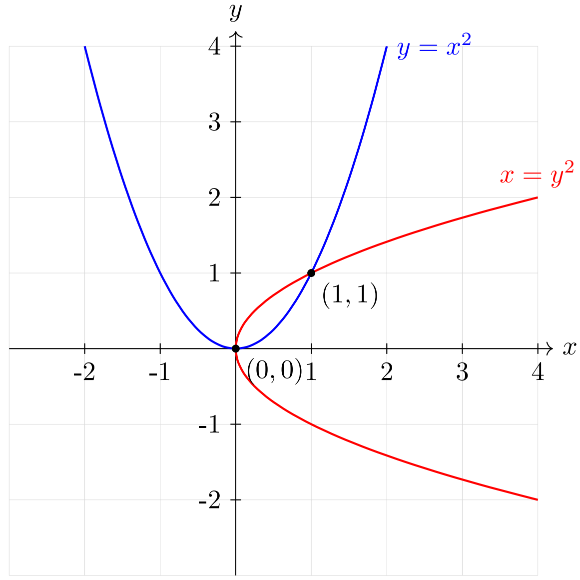


Figure 31.1: Circle represented by the equation $x^2+y^2-1=0$

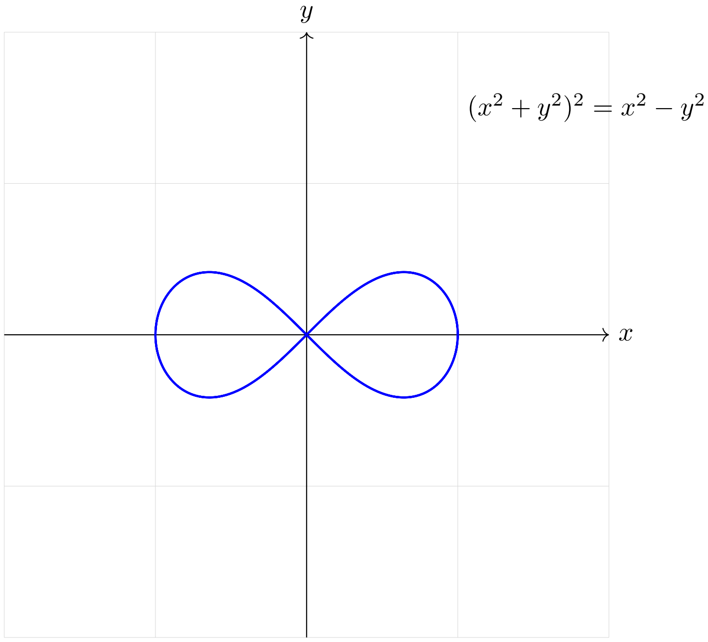
Example 31.2.2: Zeros' of Polynomials

1. The polynomial $x^2+y^2-1 \in \mathbb{R}[x,y]$ gives the circle of fig. 31.1. More precisely, the polynomial shall be associated to an element of $\Gamma(\mathbb{A}_k^n)$, where plugin in any point on from $\mathbb{A}_{\mathbb{R}}^2$ that is on the circle (ex. $(1,0)$ or $(\sqrt{2}/2, \sqrt{2}/2)$ shall give 0.

2. Take $y-x^2 \in \mathbb{R}[x,y]$. Then this shall give the usual parabola $y=x^2$. If we take $x-y^2 \in \mathbb{R}[x,y]$, we shall get a parabola rotated 90 degree's (or $\pi/2$ radians) clockwise. If we take the polynomial $(y-x^2)(x-y^2) \in \mathbb{R}[x,y]$, then the zero set of this new polynomial shall be the zero-set of the two multiplied polynomials:



3. Take $(x^2+y^2)^2-(x^2-y^2) \in \mathbb{R}[x,y]$. This shape is called the *Bernoulli Lemniscate*:

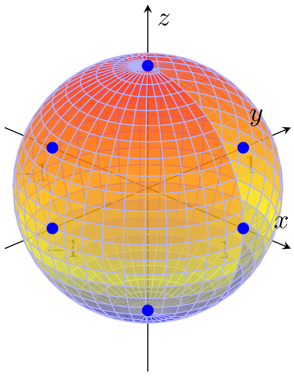


Notice that the shape crosses over itself. This shows that zero-sets of polynomials differ from graphs, and also differ from the usual smooth manifolds from differential geometry.

4. Take $p(x) \in k[x]$ over any field k. Then the shape associated to $p(x)$ are the *roots* of the polynomials. For example, taking $x^2-2 \in \mathbb{Q}(\sqrt{2})[x]$ shall give $\{\sqrt{2}, -\sqrt{2}\} \subseteq \mathbb{A}^1_{\mathbb{Q}(\sqrt{2})}$ as these are the solution to $x^2-2=0$. More generally, the shape given by a polynomial in one variable is always the collection of roots that is present in the underlying field. This means that $x^2-2 \in \mathbb{Q}[x]$ would give the empty set, as there are no rational roots. By the fundamental theorem of algebra, any polynomial $p(x) \in \mathbb{C}[x]$ of degree d shall have d roots, and hence the shape associated to $p(x)$ shall have d points.

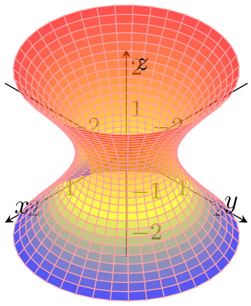
5. Take $x^2+y^2+z^2-1 \in \mathbb{R}[x,y,z]$. Then this is the classical unit sphere centered at the origin:

$$x^2 + y^2 + z^2 = 1$$

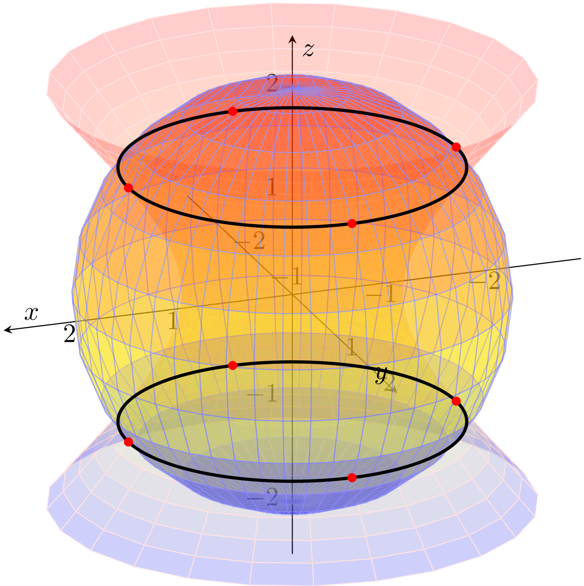


What polynomial would you choose to center the unit sphere at different points of $\mathbb{A}^3_{\mathbb{R}}$? How would you change the radius of the sphere?

6. Let us now consider taking two shapes: the sphere given by $x^2 + y^2 + z^2 - 4$ and the hyperboloid given by $x^2 + y^2 - z^2 - 1$. The hyperboloid looks like so:

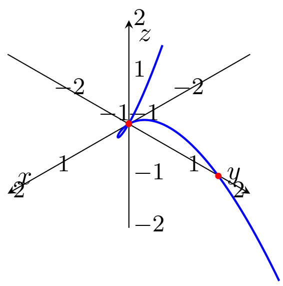


Then the intersection of these two shapes shall produce two circles, visualized like so:



This produce *two disconnected components* (in the usual euclidean topology). We shall show later that no single polynomial can create these two circles, hence we shall often have the case where we get interesting shapes by interesting or taking the union of two zero-sets.

7. The following is an infamous example of a shape that looks like it could be the zero-set of a single polynomial, but can't be: consider the following shape given by $t \mapsto (t, t^2, t^3)$:



It can be proven that this shape is *not* the zero-set of a single polynomial $p(x, y, z) \in k[x, y, z]$. On the other hand, this shape, is the intersection of two zero-sets of the surfaces $y = x^2$ and $z = xy$. This shape is called the *twisted cubic.*

More generally, given any polynomial in $k[x_1, x_2, ..., x_n]$, we can associate with it a geometric shape based on its zero set!

Definition 31.2.3: [Affine] Algebraic Set

Let $\mathbb{A}^n$ be an affine n-space over a field k, and let $S \subseteq \Gamma(\mathbb{A}^n)$. Then:

$$\mathcal{Z}(S) \subseteq \mathbb{A}^n \quad \mathcal{Z}(S) := \{(a_1, a_2, ..., a_n) \mid f(a_1, a_2, ..., a_n) = 0, \text{ for all } f \in S\}$$

If for $V \subseteq \mathbb{A}^n$, there exists an $S \subseteq \Gamma(\mathbb{A}^n)$ such that $V = \mathcal{Z}(S)$, then V is called an *affine algebraic set*, *locus*, or *zero set*. If it is unambiguous, the set may be called an *algebraic set*[a].

[a] In some textbooks, this set is called a *variety*. The term variety shall be reserved for definition 32.3.1

From now on, when taking a subset of $\mathbb{A}^n$, $A \subseteq \mathbb{A}^n$, it will be assumed that A is an algebraic set unless specified otherwise (i.e. it would be stated “for some subset $A \subseteq \mathbb{A}^n$)

Example 31.2.4: Affine Algebraic Sets

All the above are algebraic sets. Let's go over a few more examples:

1. If $n = 1$, then the zeros of a polynomial are always points, that is, the roots of the polynomial. When k is infinite the only polynomial with infinitely many zeros is 0 (more precisely, the regular function associated to 0), which has all of k as its vanishing locus. Infiniteness is the right hypothesis rather than characteristic 0: item (5) below exhibits the failure over $\mathbb{F}_2$, while $\overline{\mathbb{F}_p}$ has characteristic p and the claim still holds there. Furthermore, the polynomial 1 vanishes nowhere, and hence the corresponding algebraic set is $\emptyset$. Therefore, in $\mathbb{A}^1$, the affine sets are

$$\emptyset \qquad \text{finite sets} \qquad k$$

   This also shows that not all sets are algebraic sets (for example, if $k = \mathbb{R}$, then $[0, 1]$ is not an algebraic set in $\mathbb{A}^1_{\mathbb{R}}$)

2. If $n = 2$, Then take $f \in \Gamma(\mathbb{A}^2)$ such that $f(x, y) = x^2 + y^2 - 1$. Then

$$\mathcal{Z}(f) = \left\{(x, y) \in \mathbb{A}^2 \;\middle|\; x^2 + y^2 = 1\right\}$$

   If $k = \mathbb{R}$, then $\mathcal{Z}(f)$ is infinite, so it is not a product of two algebraic subsets of $\mathbb{A}^1$: those are finite or all of $\mathbb{A}^1$, and neither product is a circle.

   Furthermore, this examples shows that there are lot's more zero sets in $\mathbb{A}^2$ than there are in $\mathbb{A}^1 \times \mathbb{A}^1$. This is in stark contrast to how every element in the vector space $\mathbb{R}^2$ can be written as an element in $\mathbb{R}$ and an element in $\mathbb{R}$, i.e. $\mathbb{R}^2 \cong \mathbb{R} \times \mathbb{R}$ as vector spaces.

3. For any $\mathbb{A}^n$, can you always find a set of functions $F \subseteq \Gamma(\mathbb{A}^n)$ such that $|V(F)| = 1$ (i.e. is every singleton an algebraic set for every $\mathbb{A}^n$ over any field?) This shall be answered in section 32.4.

4. What is the vanishing set of linear polynomials?

5. In the first example, we produced the entire space using the zero polynomial. Is it necessary that we use the zero-polynomial to make the zero set the entire space? The answer is no: simply take $k = \mathbb{Z}/2\mathbb{Z}$ and $p(x) = x^2 + x$. However, this is a problem exclusive to finite fields. Can you prove that if k is an infinite field, $p \in k[x_1, x_2, ..., x_n]$ and $p(a_1, a_2, ..., a_n) = 0$ for all $a_i \in k$ then $p \equiv 0$? Freely use that a nonzero one-variable polynomial has finitely many roots, and induct on the number of variables.

6. Let k be a finite field and take any subset $S \subseteq \mathbb{A}_k^n$. Show that S is always an algebraic set. Hence, being algebraic gives no additional information in the finite case.

7. Let $f \in k[x_1, x_2, ..., x_n]$. Then $\mathcal{Z}(f)$ is called a *hyper-surface*. For example $S^1 = \mathcal{Z}(x^2 + y^2 - r^2)$ represents a circle of radius r. Another example would be $\mathcal{Z}(xy - 1)$ which represents the hyperbola $y = 1/x$. More generally, any conic section is a hyper-surface. In higher dimensions, $\mathcal{Z}(x^2 + y^2/4 + z^2/9 - 1)$ represents an ellipsoid in $\mathbb{R}^3$. The twisted cubic is *not* a hyper-surface.

8. For any polynomial function $f : k \to k$, we can define $F : k^2 \to k$ where $F(x, y) = f(x) - y$. Then $\mathcal{Z}(F)$ consists of all points where $f(x) - y = 0$, i.e. $f(x) = y$, which is the pair of points $\{(a, b) \mid f(a) = b\}$, i.e. the graph of f! More generally, if $f : k^n \to k$ is a polynomial, the polynomial $F : k^{2n} \to k$ gives rise to the polynomial $F : (x, y) = f(x) - y$ whose zero set is the graph of f (recall the convention that x, y represent multiple variables when it is clear from context).

Proposition 31.2.5: Properties of Affine Algebraic Sets

Let $S, T \subseteq \Gamma(\mathbb{A}^n)$. Then

1. If $S \subseteq T$ then $\mathcal{Z}(T) \subseteq \mathcal{Z}(S)$ (i.e. $\mathcal{Z}$ is *contravariant* in respect to inclusion)

2. If $(S) = I \subseteq \Gamma(\mathbb{A}^n)$ is the ideal of S, then $\mathcal{Z}(I) = \mathcal{Z}(S)$.

3. For any arbitrary collection of algebraic sets $\{S_i\}_{i\in I}$,
$$\bigcap_{i\in I} \mathcal{Z}(S_i) = \mathcal{Z}\left(\bigcup_{i\in I} S_i\right)$$
that is, the arbitrary intersection of algebraic sets is an algebraic sets, and is equal to the algebraic set of the intersections of the $\{S_i\}_{i\in I}$

4. For any finite collection of algebraic sets $\{S_i\}_{i=1}^n$,
$$\bigcup_{i=1}^{n} \mathcal{Z}(S_i) = \mathcal{Z}\left(\prod_{i=1}^{n} S_i\right)$$
that is, the finite union of algebraic sets is an algebraic set, and is equal to the product of the S_i as ideals (from part 2, we can assume $(S_i) = S_i$ without loss of generality).

5. If $0, 1 \in \Gamma(\mathbb{A}^n)$ denote the zero and constant function, then
$$\mathcal{Z}(0) = \mathbb{A}^n \qquad \mathbb{Z}(1) = \emptyset$$

6. If I is an ideal of $\Gamma(\mathbb{A}^n)$, then
$$\mathcal{Z}(I) = \mathcal{Z}(I^m)\ \forall m \in \mathbb{N}_{>0}$$
that is, $\mathcal{Z}$ is invariant to taking higher powers of the ideals

These are all straight-forwards proofs, and so should all be done before reading these proofs for the best learning experience.

***Proof*:**

1. Let $S \subseteq T$, and consider $x \in \mathcal{Z}(T)$. Then $f(x) = 0$ for every $f \in T$. Since $S \subseteq T$, in particular $g(x) = 0$ for every $g \in S$. Thus $x \in \mathcal{Z}(S)$, establishing $\mathcal{Z}(T) \subseteq \mathcal{Z}(S)$.

2. Let $S \subseteq \Gamma(\mathbb{A}^n)$ be any set and let $I = (S)$. We want to show that $\mathcal{Z}(I) = \mathcal{Z}(S)$. Since $S \subseteq I$, the $\mathcal{Z}(I) \subseteq \mathcal{Z}(S)$ conclusion comes from the first part. For the reverse, consider $x\mathcal{Z}(S)$. Then for all $t \in S$, $t(x) = 0$. If we pick any two $f, g \in S$, then $(f+g)(x) = f(x) + g(x) = 0$. If we pick any $c(x) \in \Gamma(\mathbb{A}^n)$ and any $f \in S$, then $(cf)(x) = c(x)f(x) = c(x)0 = 0$. Thus, if we enlarge S to be closed under addition and multiplication by elements of $k^[\mathbb{A}^n]$, those polynomial will also be zero on x, and so $\mathcal{Z}(S) \subseteq \mathcal{Z}(I)$, establishing $\mathcal{Z}(S) = \mathcal{Z}(I)$

3. We want to show that $\bigcap_{i \in I} \mathcal{Z}(S_i) = \mathcal{Z}\left(\bigcup_{i \in I} S_i\right)$ for an arbitrary indexing set I. This is just some set theory: by definition:

$$x \in \mathcal{Z}(\cup_{i \in I} S_i) = \{y \in \mathbb{A}^n \mid f(y) = 0,\ \forall f \in \cup_{i \in I} S_i\} = \mathcal{Z}\left(\sum_\alpha I_\alpha\right)$$

   which is true if and only if x is the solution for polynomials in every set S_i. The collection of such elements is:

$$\bigcap_{i \in I} \mathcal{Z}(S_i)$$

4. We want to show that $\bigcup_{i=1}^n \mathcal{Z}(S_i) = \mathcal{Z}\left(\prod_{i=1}^n S_i\right)$. By definition:

$$x \in \mathcal{Z}\left(\prod_{i=1}^n S_i\right) = \left\{ y \in \mathbb{A}^n \;\middle|\; f(y) = 0\ \forall f \in \prod_{i=1}^n S_i \right\}$$

   where by definition $f(x) = \sum_{j=1}^m (f_{1_j} f_{2_j} \cdots f_{n_j})(x) = \sum_{j=1}^m f_{1_j}(x) f_{2_j}(x) \cdots f_{n_j}(x) = 0$. Notice that for a particular term in the sum to be zero, we only need on of the f_{i_j} to be zero. Therefore, the sum is zero if and only if one polynomial in every term of the sum is zero. Since this also applies to sums with only one term in it, then if x is the solution to one polynomial in S_i, then it will be part of the zero's of $\prod_{i=1}^n S_i$. But this is just another way of saying:

$$x \in \bigcup_{i=1}^n \mathcal{Z}(S_i)$$

5. Notice that $0(x) = 0$ for all $x \in \mathbb{A}^n$ and $1(x) = 1 \neq 0$ for all $x \in \mathbb{A}^n$, and so

$$\mathcal{Z}(0) = \mathbb{A}^n \qquad \mathcal{Z}(1) = \emptyset$$

6. Since $I^m \subseteq I$, the first part gives $\mathcal{Z}(I) \subseteq \mathcal{Z}(I^m)$. For the reverse inclusion, let $x \in \mathcal{Z}(I^m)$ and let $f \in I$ be arbitrary. Then f^m is a product of m elements of I, so $f^m \in I^m$, and therefore

$$f(x)^m = f^m(x) = 0$$

   As k is a field it has no nonzero nilpotents, so $f(x) = 0$. Since $f \in I$ was arbitrary, $x \in \mathcal{Z}(I)$, giving $\mathcal{Z}(I^m) \subseteq \mathcal{Z}(I)$ and hence $\mathcal{Z}(I) = \mathcal{Z}(I^m)$, as we sought to show.

A couple of remarks are in order. From (2), we can without loss of generality restrict $\mathcal{Z}$ be a function from the ideals of $k[x_1, ..., x_n]$ to the algebraic sets in $\mathbb{A}^n$:

$$\mathcal{Z} : \{\text{ideals in } k[x_1, ..., x_n]\} \to \{\text{affine algebraic sets in } \mathbb{A}^n\}$$

Furthermore, $\mathcal{Z}$ is a surjection from the ideals of $k[x_1, x_2, ..., x_n]$ and the algebraic sets (the ideals I^m for each $m \geq 1$ map to the same algebraic set). Next, the finiteness condition on the product of ideals is a necessary condition for closure of algebraic sets:

Example 31.2.6: Not Closed Under Arbitrary Unions

Take $\mathbb{A}^1_{\mathbb{R}}$ and the sets $V(x-a)$ for $a \in \mathbb{Z}$. Then prove that the union of these sets is *not* an algebraic set (you may freely use the fact that polynomials have at most finite roots, and recall that $\mathbb{R}[x]$ is Noetherian).

Next, we took the product of ideal $\prod_i^n I_i$ instead of the intersection. Since it always the case that $IJ \subseteq I \cap J$ then in the cases where $IJ \subsetneq I \cap J$, $\mathcal{Z}(I \cap J) \subsetneq \mathcal{Z}(I) \cup \mathcal{Z}(J)$[8]. However, since $\mathcal{Z}(I^2) = \mathcal{Z}(I)$, the distinction is invisible to $\mathcal{Z}$, and $\mathcal{Z}(IJ) = \mathcal{Z}(I \cap J)$ always

Perhaps most importantly for this chapter, notice that these properties are the properties of a topology (see appendix C):

Definition 31.2.7: Zariski Topology

Let $\mathbb{A}^n$ be an n-affine space over k. Then the set

$$\mathcal{T}_{\mathcal{Z}} = \{\mathcal{Z}(I) \mid \text{I is an ideal of } \Gamma(\mathbb{A}^n)\}$$

from the set of closed sets in the *Zariski topology* $(\mathbb{A}^n, \mathcal{T}_{\mathcal{Z}})$

We can therefore try and study this topological space! This is essentially the foundation for studying geometrical concepts in algebra, and vice-versa, it is the foundation for studying algebraic concepts in geometry, and so we will spend a lot of time establishing the link between the ideals of $\Gamma(\mathbb{A}^n)$ and the Zariski topology.

Before such topological considerations, we shall first see if we can find some function that is the inverse of $\mathcal{Z}$, that is we start from an algebraic set in $\mathbb{A}^n$ (or more generally any subset of $\mathbb{A}^n$) and associate it to an ideal in $\Gamma(\mathbb{A}^n)$, letting us strive towards some way of corresponding the two in a manner similar to Galois Theory[9]. Towards the goal of finding an inverse for $\mathcal{Z}$, notice that we already have that $\mathcal{Z}$ is a surjection of ideals of $\Gamma(\mathbb{A}^n)$ to the algebraic sets. Thus, it is meaningful to ask about having a function from the set of algebraic sets to the set of ideals of $k[x_1, x_2, ..., x_n]$. To understand this question, we start off with a slightly more general question (just like we started with subsets of $k[x_1, ..., x_n]$ instead of ideals):

[8]The condition $IJ = I \cap J$ is more delicate than it looks: it holds exactly when $\mathrm{Tor}_1^R(R/I, R/J) = 0$, for which see [15].

[9]In fact,the order reversing nature of this correspondence will end up making this correspondence a *Galois correspondence*, which name is inspired by how the Galois correspondence between intermediate fields and subgroups was one of the first well-known order-reversing correspondences

Definition 31.2.8: Vanishing Ideal

Let $V \subseteq \mathbb{A}^n$ be a subset of the affine n-space. Then

$$\mathcal{I}(V) := \{f \in \Gamma(\mathbb{A}^n) \mid f(v) = 0, \ \forall v \in V\}$$

is the *vanishing ideal* of V.

Notice that $\mathcal{I}(V)$ was not defined as an ideal, however it is indeed an ideal; it is important for the sake of intuition that the reader should prove this at least once. Even better, $\mathcal{I}(V)$ is the *largest* ideal of $\Gamma(\mathbb{A}^n)$ that corresponds to the set V. Just like $\mathcal{Z}$, $\mathcal{I}$ can be interpreted as a function

$$\mathcal{I} : \{V \subseteq \mathbb{A}^n_k\} \to \{\text{ideals of } \Gamma(\mathbb{A}^n)\}$$

Example 31.2.9: Vanishing Ideal

1. Let $\mathbb{A}^2$ be an affine space over the field $\mathbb{R}$ and let $X \subseteq \mathbb{A}^2$ be the x-axis. Then $\mathcal{I}(X) = (y) \subseteq \mathbb{R}[x, y]$
2. Let k be infinite. Then $\mathcal{I}(\mathbb{A}^n) = 0$.
3. Find the vanishing ideals if of subsets $\mathbb{A}^n_k$ if k is finite.

Just like $\mathcal{Z}$, $\mathcal{I}$ shares similar properties:

Proposition 31.2.10: Properties of Vanishing Ideals

let $A, B \subseteq \mathbb{A}^n$ be two algebraic sets. Then

1. If $A \subseteq B$, then $\mathcal{I}(B) \subseteq \mathcal{I}(A)$ (i.e. $\mathcal{I}$ is *contravariant* with respect to inclusion)
2. $\mathcal{I}(A \cup B) = \mathcal{I}(A) \cap \mathcal{I}(B)$
3. $\mathcal{I}(\emptyset) = \Gamma(\mathbb{A}^n)$. If k is infinite, $\mathcal{I}(\mathbb{A}^n) = 0$

***Proof*:**

1. Let $A \subseteq B$ and take $f \in \mathcal{I}(B)$, so $f(b) = 0$ for every $b \in B$. Every $a \in A$ lies in B, so $f(a) = 0$, giving $f \in \mathcal{I}(A)$.
2. A polynomial vanishes on $A \cup B$ exactly when it vanishes on A and vanishes on B, which is the assertion $\mathcal{I}(A \cup B) = \mathcal{I}(A) \cap \mathcal{I}(B)$.
3. The condition defining $\mathcal{I}(\emptyset)$ is vacuous, so every polynomial belongs to it and $\mathcal{I}(\emptyset) = \Gamma(\mathbb{A}^n)$. For the second claim, suppose k is infinite and f vanishes at every point of $\mathbb{A}^n$. Induct on n. For $n = 1$ a nonzero polynomial has at most $\deg f$ roots, while k is infinite, so $f = 0$. For $n > 1$ write $f = \sum_i g_i(x_1, \ldots, x_{n-1})x_n^i$. Fixing any $a \in k^{n-1}$ makes $f(a, x_n)$ a one-variable polynomial vanishing on all of k, so each $g_i(a) = 0$ by the base case. As a was arbitrary, each g_i vanishes on $\mathbb{A}^{n-1}$, and by induction $g_i = 0$. Hence $f = 0$, so $\mathcal{I}(\mathbb{A}^n) = 0$, as we sought to show.

The hypothesis that k be infinite in (3) is not removable: over $\mathbb{F}_q$ the polynomial $x^q - x$ vanishes at every point of $\mathbb{A}^1$, so $\mathcal{I}(\mathbb{A}^1_{\mathbb{F}_q}) \neq 0$.

Notice that we have omitted showing what $\mathcal{I}(A \cap B)$ is. One problem is that $\mathcal{I}(A) \cup \mathcal{I}(B)$ is not necessarily an ideal (since the union of two ideals is not necessarily an ideal). It might be tempting to say that since

$$\langle \mathcal{I}(A) \cup \mathcal{I}(B) \rangle = \mathcal{I}(A) + \mathcal{I}(B)$$

we can conclude that:

$$\mathcal{I}(A \cap B) = \mathcal{I}(A) + \mathcal{I}(B)$$

However, this is not generally the case (consider the algebraic sets given by the ideals $(x^2 - y)$ and (x)). Proving the reverse inclusion needs the Nullstellensatz, but it can already be shown from the definitions that $\mathcal{I}(A \cap B) \supseteq \mathcal{I}(A) + \mathcal{I}(B)$, since a function vanishing on A or on B vanishes on $A \cap B$. With the properties of $\mathcal{I}$ established, we can start getting an idea of what ideals of $\Gamma(\mathbb{A}^n)$ are necessary to restrict $\mathcal{Z}$ to make it a bijection:

Proposition 31.2.11: Zero Sets and Coordinate Ideals

Let $\Gamma(\mathbb{A}^n)$ be the set of polynomial functions from $\mathbb{A}^n$ to k. Then

1. If $V \subseteq \mathbb{A}^n$, then $V \subseteq \mathcal{Z}(\mathcal{I}(V))$. If $I \subseteq \Gamma(\mathbb{A}^n)$ is an ideal of $\Gamma(\mathbb{A}^n)$, then $I \subseteq \mathcal{I}(\mathcal{Z}(I))$.
2. If $V = \mathcal{Z}(I)$ is an algebraic set, then $V = \mathcal{Z}(\mathcal{I}(V))$. Similarly, if $I = \mathcal{I}(V)$ is a vanishing ideal, then $I = \mathcal{I}(\mathcal{Z}(I))$. In other words, for every subset $V \subseteq \mathbb{A}^n$ and every ideal $I \subseteq \Gamma(\mathbb{A}^n)$,
$$\mathcal{I}(\mathcal{Z}(\mathcal{I}(V))) = \mathcal{I}(V) \qquad \mathcal{Z}(\mathcal{I}(\mathcal{Z}(I))) = \mathcal{Z}(I)$$

***Proof*:**

1. For $V \subseteq \mathcal{Z}(\mathcal{I}(V))$, since $\mathcal{I}(V)$ is the set of all functions such that $f(V) \equiv 0$ (including at least the zero polynomial), then V is certainly part of its zero set.

   Conversely, for $I \subseteq \mathcal{I}(\mathcal{Z}(I))$, the coordinate ideal of $\mathcal{Z}(I)$ certainly contains I since for any $f \in I$ and $x \in \mathcal{Z}(I)$, $f(x) = 0$. Thus, $I \subseteq \mathcal{I}(\mathcal{Z}(I))$
2. Let $V = \mathcal{Z}(I)$. By part 1 it suffices to prove $V \supseteq \mathcal{Z}(\mathcal{I}(V))$. Take $x \in \mathcal{Z}(\mathcal{I}(V))$, so that $f(x) = 0$ for every $f \in \mathcal{I}(V)$. Every $f \in I$ vanishes on $\mathcal{Z}(I) = V$ and so lies in $\mathcal{I}(V)$, giving $I \subseteq \mathcal{I}(V)$; hence $f(x) = 0$ for every $f \in I$, which says exactly that $x \in \mathcal{Z}(I) = V$. Therefore $V = \mathcal{Z}(\mathcal{I}(V))$.

   The second claim is the same argument with the roles exchanged. If $I = \mathcal{I}(V)$ then part 1 gives $I \subseteq \mathcal{I}(\mathcal{Z}(I))$; conversely if $g \in \mathcal{I}(\mathcal{Z}(I))$ then g vanishes on $\mathcal{Z}(\mathcal{I}(V))$, which we have just shown equals V, so $g \in \mathcal{I}(V) = I$. Applying $\mathcal{I}$ and $\mathcal{Z}$ to these two identities gives the displayed equations, completing the proof.

What the 2nd part of proposition 31.2.11 shows is that if we restrict to just the algebraic sets and vanishing ideals of $\mathbb{A}^n$ and $\Gamma(\mathbb{A}^n)$ respectively, then $\mathcal{Z}$ and $\mathcal{I}$ are indeed inverses of each other. The goal then would be to be able to characterize vanishing ideals, that is ideals that are of the form $I = \mathcal{I}(V)$ for some algebraic set V. It is in fact a rather involved process to find the nature of the ideal $\mathcal{I}(V)$; a lot of section 32.2 will be building up to get to the theorem which will tell us the answer: *Hilbert's Nullstellensatz*. In the process of building up the theory to answer this question, there are a few other loose ends that we will need to tie up. For example, what properties must the Zariski topology have, and what does that imply for the corresponding ring $\Gamma(\mathbb{A}^n)$? Another important point to address is that as $k[x_1, ..., x_n]$ is Noetherian, hence every ideal is finitely generated: $I = (f_1, ..., f_m) = (f_1) \cup (f_2) \cup \cdots \cup (f_m)$. By the properties of zero sets:

$$\mathcal{Z}(I) = \mathcal{Z}(f_1) \cap \mathcal{Z}(f_2) \cap \cdots \cap (f_m)$$

In other words, there seem to be a natural theory of decomposition of algebraic sets. Furthermore, we have been shown theoretically what properties algebraic sets have, but how is one computed in practice? These questions will receive answers in the following section.

Exercise 31.2.1

1. Show that $z \mapsto \overline{z}$ is not a complex polynomial.
2. Show that $V = \left\{(z, w) \in \mathbb{A}^2_{\mathbb{C}} \;\middle|\; |z|^2 + |w|^2 = 1\right\}$ is not an algebraic subset of $\mathbb{A}^2_{\mathbb{C}}$, but that the same set, viewed in $\mathbb{A}^4_{\mathbb{R}}$, is algebraic.
3. Let k be a perfect field with algebraic closure $\overline{k}$ and Galois group $\mathrm{Gal}_k(\overline{k})$, acting on $\mathbb{A}^n_{\overline{k}}$ coordinatewise. Show that
$$\mathbb{A}^n_k \cong \left\{p \in \mathbb{A}^n_{\overline{k}} \;\middle|\; p^\sigma = p \text{ for all } \sigma \in \mathrm{Gal}_k(\overline{k})\right\}$$
and say where perfectness is used.
4. Let $V = \mathcal{Z}(x^2 + y^2) \subseteq \mathbb{A}^2$. Show that V is irreducible over $\mathbb{Q}$ but reducible over $\mathbb{Q}(i)$, so that irreducibility depends on the field of definition. Why does this phenomenon disappear once the field is algebraically closed?
5. Here is a way of recovering $A = k[x_1, \ldots, x_n]$ from $V = k^n$. Write $V^* = \mathrm{Hom}_k(V, k)$ and recall the symmetric algebra (definition 16.3.5):
$$\mathrm{Sym}^\bullet(V^*) = k \oplus V^* \oplus \mathrm{Sym}^2(V^*) \oplus \mathrm{Sym}^3(V^*) \oplus \cdots$$
Show that $\mathrm{Sym}^\bullet(V^*) \cong A$. Then let $\mathfrak{m} \subseteq A$ be the maximal ideal of a point and show $\mathfrak{m}/\mathfrak{m}^2 \cong V^*$, so that the linear structure is recovered at each point. This is the *Zariski tangent space* of chapter 35.

31.3 The Zariski Topology

In this section, we will be assuming basic topological knowledge from the reader (see appendix C)

As the reader (ought to have) showed in in example 31.2.4, if $k = \mathbb{F}_q$ is finite, then every subset of $\mathbb{A}^n_{\mathbb{F}_q}$ is algebraic, hence the Zariski topology is isomorphic to the discrete topology, showing us that there is no "interesting" behaviour in the finite case. Thus, we shall usually focus on the case where k is not-finite. This doesn't mean that finite fields don't play an important role in algebraic geometry, however their importance shall not be visible until the development of schemes[10] or Class

[10] In particular, for any Scheme X, there is always a scheme morphism $X \to \mathrm{Spec}(\mathbb{Z})$ a there is always a ring homomorphism $\mathbb{Z} \to \mathcal{O}_X(X)$; the fibers of this scheme X_p are $\mathbb{F}_p$-schemes. Furthermore, $\mathbb{F}_q$-schemes shall be important in number theory

Field Theory[11] and so they shall have negligible importance this chapter.

As the topology was defined using closed sets (sets for which a collection of polynomials is zero on that set), a moment should be given to give a geometric interpretation to open sets. Open sets are those for which a collection of polynomials are *not all zero* at once on those points. If $S = \{f\} \in \Gamma(\mathbb{A}_k^n)$ is a singleton, then it will be precisely the set of points where the function f does not vanish. This makes all open sets of $\mathbb{A}_k^1$ (where k is not finite) the *cofinite sets* and hence it has the cofinite topology. If $n > 1$, the topology is certainly no longer cofinite. Open sets shall become importance when we look to define *rational polynomials* and *rational functions* for they give us the domain on which a function is non-zero. (see definition 31.3.5).

Let us now look at the separatedness of the Zariski topology. Notice that $\mathcal{T}_{\mathcal{Z}}$ is rarely Hausdorff (also known as T_2): if $\mathbb{A}^1 = k$ is any infinite field, as we've established that the only algebraic sets (and hence closed sets) are the empty set, finite sets, and $\mathbb{A}_k^1$ and so the only non-trivial open sets in $\mathcal{T}_{\mathcal{Z}}$ would be co-finite sets (i.e. $\mathcal{T}_{\mathcal{Z}}$ is the co-finite topology). Then as we can see, if we pick any two points $a, b \in \mathbb{A}^1$, we cannot separate them by two open disjoint sets. However, over any $\mathbb{A}^1 = k$, the singleton's are always algebraic sets (giving a hint to those who were working on example 31.2.4), and so it is always a Fréchet space (or T_1)[12].

This result generalizes for any k^n

Proposition 31.3.1: Zariski Topology Always T_1

Let k be any field and give $\mathbb{A}_k^n$ the Zariski topology. Then:

1. $\mathbb{A}_k^n$ is T_1, that is, every singleton is closed.
2. If k is infinite, $\mathbb{A}_k^n$ is not Hausdorff.

The two halves need different hypotheses, and the second genuinely fails without its one: over a finite field every subset is algebraic (example 31.2.4), so the topology is discrete and therefore Hausdorff.

***Proof*:**

1. For any $a = (a_1, \ldots, a_n) \in k^n$ let $I = (x_1 - a_1, \ldots, x_n - a_n)$. A point b lies in $\mathcal{Z}(I)$ exactly when $b_i - a_i = 0$ for every i, so $\mathcal{Z}(I) = \{a\}$ and singletons are closed. No hypothesis on k is used.
2. Let k be infinite and suppose U, V are nonempty open sets. Write $U = \mathbb{A}^n \setminus \mathcal{Z}(I)$ and $V = \mathbb{A}^n \setminus \mathcal{Z}(J)$ with $\mathcal{Z}(I), \mathcal{Z}(J) \neq \mathbb{A}^n$, so neither I nor J lies in $\mathcal{I}(\mathbb{A}^n) = 0$ (proposition 31.2.10); choose $f \in I$ and $g \in J$ nonzero. Then fg is nonzero, since $\Gamma(\mathbb{A}^n) \cong k[x_1, \ldots, x_n]$ is an integral domain, so again by proposition 31.2.10 the product fg does not vanish identically on $\mathbb{A}^n$. Any point where fg is nonzero has both f and g nonzero, hence lies in $U \cap V$. So any two nonempty open sets meet, and $\mathbb{A}_k^n$ is not Hausdorff, as we sought to show.

Two topological concepts which should be immediately introduced to be explored as the chapter develops are the following:

[11] In particular, the behaviour of unramified field extensions between local fields will be captured via the automorphisms of finite fields

[12] For those familiar with algebraic topology, schemes are almost never T_1, as prime ideals that are not maximal are not closed points

Definition 31.3.2: Zariski Dense And Closure

Let $A \subseteq \mathbb{A}^n$ be any subset of the affine space $\mathbb{A}^n$. Then the *zariski closure* of A is the smallest algebraic set containing A, denoted $\overline{A}$. Geometrically, it is the smallest set $\mathcal{A}$ for which every point in A is the zero set of a collection of polynomials.

If $\overline{A} = \mathbb{A}^n$, then A is said to be *zariski dense*. More generally if $A \subseteq V \subseteq \mathbb{A}^n$ such that $\overline{A} = V$, then A is said to be zariski dense in V.

As an example, if we give $\mathbb{R}$ the Zariski topology, the closure of any finite set is itself, while the closure of any infinite set is all of $\mathbb{R}$. The Zariski closure of a set is in fact quite easy to characterize using $\mathcal{Z}$ and $\mathcal{I}$ and follows quickly using proposition 31.2.11:

Corollary 31.3.3: Zariski Closure

Let $A \subseteq \mathbb{A}^n$. Then:
$$\overline{A} = \mathcal{Z}(\mathcal{I}(A))$$

***Proof*:**
The set $\mathcal{Z}(\mathcal{I}(A))$ is algebraic and contains A by proposition 31.2.11(1), so it is one of the algebraic sets whose intersection defines $\overline{A}$, giving $\overline{A} \subseteq \mathcal{Z}(\mathcal{I}(A))$.

Conversely, let W be any algebraic set with $A \subseteq W$. Applying $\mathcal{I}$ reverses the inclusion, so $\mathcal{I}(W) \subseteq \mathcal{I}(A)$, and applying $\mathcal{Z}$ reverses it again:
$$\mathcal{Z}(\mathcal{I}(A)) \subseteq \mathcal{Z}(\mathcal{I}(W)) = W$$
the last equality because W is algebraic (proposition 31.2.11(2)). So $\mathcal{Z}(\mathcal{I}(A))$ is contained in every algebraic set containing A, hence in $\overline{A}$, as we sought to show.

Sticking to $k = \mathbb{R}$ or $k = \mathbb{C}$ for a moment, we can compare the euclidean topology on $\mathbb{R}^n$ with the Zariski topology on $\mathbb{A}^n_{\mathbb{R}}$. (Coarser, and I think is the corsest topology on which all polynomial functions are continuous)

Another important topological consideration would be to find a topological basis for the Zariski topology. Recall a subset of the topology on a space X is a basis B if any open set of X is the union of a family of sets from B.

Proposition 31.3.4: Basis For Zariski Topology

For $f \in \Gamma(\mathbb{A}^n)$ define
$$D(f) = \mathbb{A}^n_k \setminus \mathcal{Z}(f) = \{a \in \mathbb{A}^n_k \mid f(a) \neq 0\}$$
Then the collection $\{D(f) \mid f \in \Gamma(\mathbb{A}^n)\}$ forms a [topological] basis for the Zariski topology on $\mathbb{A}^n$.

The letter D can be thought of as a mnemonic for *doesn't* vanish.

***Proof*:**
It suffices to show that for any open U and any $a \in U$ there is an f with $a \in D(f) \subseteq U$, since then U is the union of those $D(f)$. As U is open, $U = \mathbb{A}_k^n \setminus \mathcal{Z}(I)$ for some ideal $I \subseteq \Gamma(\mathbb{A}^n)$. Now $a \notin \mathcal{Z}(I)$ says precisely that $f(a) \neq 0$ for *some* $f \in I$; fix such an f, so that $a \in D(f)$. Since $(f) \subseteq I$,

$$\mathcal{Z}(I) \subseteq \mathcal{Z}(f)$$

and taking complements gives:

$$a \in D(f) \subseteq U$$

and since a was arbitrary, we see that U is the union of sets $D(f)$, as we sought to show.

The collection of $D(f)$ for each $f \in k[x_1, ..., x_n]$ forms the coarsest basis:

Definition 31.3.5: Distinguished Open Sets

Let $f \in k[x_1, ..., x_n]$. Then the set

$$D(f) = \mathbb{A}_k^n \setminus \mathcal{Z}(f) = \{a \in \mathbb{A}_k^n \mid f(a) \neq 0\}$$

is called he distinguished open set.

Example 31.3.6: Distinguished Open Sets

1. In $\mathbb{A}_k^1$ take $f = x$. Then $D(x) = \mathbb{A}_k^1 \setminus \{0\}$, the punctured line. It is not closed, but it is itself an algebraic set in disguise: the map

$$D(x) \to \mathcal{Z}(xy - 1) \subseteq \mathbb{A}_k^2 \qquad a \mapsto (a, 1/a)$$

   is a bijection with inverse $(a, b) \mapsto a$, and both are given by polynomials on their domains. So the punctured line is the hyperbola.

2. The same works in general. For $f \in k[x_1, \ldots, x_n]$,

$$D(f) \;\cong\; \mathcal{Z}\big(y\, f(x_1, \ldots, x_n) - 1\big) \;\subseteq\; \mathbb{A}_k^{n+1}$$

   by $a \mapsto (a, 1/f(a))$. Every distinguished open set is therefore an algebraic set in one more variable, which is why they are the right basic open sets: the topology has a basis of sets that are themselves affine.

3. The distinguished opens do form a basis. Any open set is $\mathbb{A}_k^n \setminus \mathcal{Z}(I)$, and

$$\mathbb{A}_k^n \setminus \mathcal{Z}(I) = \mathbb{A}_k^n \setminus \bigcap_{f \in I} \mathcal{Z}(f) = \bigcup_{f \in I} D(f)$$

4. Not every open set is distinguished. In $\mathbb{A}_k^2$ the set $\mathbb{A}_k^2 \setminus \{(0, 0)\}$ is open, but $D(f)$ misses the whole curve $\mathcal{Z}(f)$, which is infinite for nonconstant f, and equals $\mathbb{A}_k^2$ for constant $f \neq 0$. Neither omits exactly one point.

The next property we get about the Zariski topology is that it's always Noetherian, that is, if there is an ascending chain of open sets, it must stabilize:

Proposition 31.3.7: Zariski Topology Is Noetherian

Let $\mathbb{A}_k^n$ have the Zariski topology. Then $\mathbb{A}_k^n$ is Noetherian.

***Proof*:**
Let $U_1 \subsetneq U_2 \subsetneq \cdots$ be an ascending chain of open sets, so there is a corresponding

$$X_1 \supsetneq X_2 \supsetneq \cdots$$

descending chain of closed sets. Then there is a corresponding

$$I(X_1) \subsetneq I(X_2) \subsetneq \cdots$$

ascending chain of ideals. Then we know that $\Gamma(\mathbb{A}^n)$ is *noetherian*, and so this chain must stabilize at some n:

$$I(X_n) = I(X_{n+1}) = \cdots$$

meaning the respective closed chain, and hence open chain, must stabilize, as we sought to show.

Noetherianity already forces compactness, and by the same argument every subspace of $\mathbb{A}_k^n$ is compact too: given an open cover of a subspace, the closed sets complementary to finite subcovers form a descending chain, which must stabilize. Proposition 32.5.1 nonetheless gives a second proof, via the Nullstellensatz, because that argument produces the explicit partition of unity $1 = \sum g_i f_i$ rather than the bare existence of a finite subcover, and that identity is what later arguments use[13].

31.4 Regular Maps and Continuity

The next thing we should consider the continuous functions. As the topology is the weakest in which the zero-sets is closed, it may not surprise the reader that it is the coarsest topology in which polynomials are continuous

Definition 31.4.1: Regular Map on Affine Space

A map $\varphi : \mathbb{A}_k^n \to \mathbb{A}_k^m$ is called a *regular map* if there exists polynomials $\varphi_1, \varphi_2, ..., \varphi_m \subseteq k[x_1, x_2, ..., x_n]$ such that

$$\varphi(a_1, a_2, ..., a_n) = (\varphi_1(a_1, a_2, ..., a_n), \varphi_2(a_1, a_2, ..., a_n)..., \varphi_m(a_1, a_2, ...a_n))$$

If $m = 1$, a regular map is often called a *regular function*.

[13] Compactness is cheap here and correspondingly weak: almost every scheme is compact, so the useful strengthening is *properness*, which chapter 40 develops for projective varieties.

Lemma 31.4.2: Coordinate Ring and Regular Map

The map $\varphi : \mathbb{A}_k^n \to \mathbb{A}_k^m$ is regular if and only if the induced map $\varphi^\# : \Gamma(\mathbb{A}_k^m) \to \Gamma(\mathbb{A}_k^n)$ given by $\varphi^\#(f) = f \circ \varphi$ is a k-algebra homomorphism

***Proof*:**
Let $\varphi : \mathbb{A}_k^n \to \mathbb{A}_k^m$ be a regular map so that each component is given by a polynomial. Then $\varphi^\#$ maps a function $f \in \Gamma(\mathbb{A}_k^m)$ to $f \circ \varphi \in \Gamma(\mathbb{A}_k^n)$, namely $f \circ \varphi$ is a polynomial map. If $f, g \in \Gamma(\mathbb{A}_k^m)$, then:

$$(f+g)\circ\varphi = f\circ\varphi + g\circ\varphi$$
$$(f\cdot g)\circ\varphi = f\circ\varphi\cdot g\circ\varphi$$
$$(k\cdot f)\circ\varphi = k\cdot(f\circ\varphi)$$

thus it is a k-algebra homomorphism.

Conversely, let $\varphi^\# : \Gamma(\mathbb{A}_k^m) \to \Gamma(\mathbb{A}_k^n)$ is a k-algebra homomorphism that maps:

$$f \mapsto \varphi^\#(f) = f\circ\varphi \in \Gamma(\mathbb{A}_k^n)$$

Let $y_1, ..., y_n : \mathbb{A}_k^m \to k$ be the coordinate functions in $\Gamma(\mathbb{A}_k^m)$. If one of the components of φ where not a polynomial, then take $f = x_i$, so that the composition $x_i \circ \varphi = \varphi_i$ is not a polynomial, giving a contradiction. Hence φ must be a regular map, completing the proof.

Lemma 31.4.3: Ring homomorphism Induces Regular map

Let $f : \Gamma(\mathbb{A}_k^m) \to \Gamma(\mathbb{A}_k^n)$ be a k-algebra homomorphism. Then it induces a map $\varphi : \mathbb{A}_k^n \to \mathbb{A}_k^m$ such that $\varphi^\# = f$.

***Proof*:**
Let $k[y_1, ..., y_m] \cong \Gamma(\mathbb{A}_k^m)$ and $k[x_1, ..., x_n] \cong \Gamma(\mathbb{A}_k^n)$. Then as f is a k-algebra homomorphism, it is enough to map each y_i to some polynomial $f(y_i) = p(x_1, ..., x_m)$ and extend to form a map on all of $\Gamma(\mathbb{A}_k^m)$. Let φ_i be the function determine by $f(y_i)$. Then it is a matter of course to check that $\varphi^\# = f$, completing the proof.

Note that the maps must be k-algebra homomorphism, not just ring homomorphisms. For example, if $n = m = 1$, then $f : \mathbb{C}[\mathbb{A}_k^1] \to \mathbb{C}[\mathbb{A}_k^1]$ sending $p(x) \mapsto \overline{p(x)}$ (i.e. the complex conjugate) is not a $\mathbb{C}$-algebra homomorphism, and the reader could check there can't be a corresponding φ such that $\varphi^\# = f$.

Lemma 31.4.4: Regular Map is Continuous

A regular map $\varphi : \mathbb{A}_k^n \to \mathbb{A}_k^m$ is continuous.

***Proof*:**
It suffices to check that the pre-image of every closed set is closed. If $V = \mathcal{Z}(I) \subseteq \mathbb{A}_k^m$ is an algebraic set, we need to show there exists an ideal J such that

$$\mathcal{Z}(J) = \varphi^{-1}(V)$$

To that end:

$$\begin{aligned}
\varphi^{-1}(V) &= \{a \in \mathbb{A}_k^n \mid \varphi(a) \in V\} \\
&= \{a \in \mathbb{A}_k^n \mid f(\varphi(a)) = 0 \ \forall f \in I\} \\
&= \left\{a \in \mathbb{A}_k^n \;\middle|\; \varphi^{\#}(f)(a) = 0 \ \forall f \in I\right\}
\end{aligned}$$

Consider the set $\varphi^{\#}(I) = \{g \in \Gamma(\mathbb{A}_k^n) \mid g = f \circ \varphi\}$. This set would be the desired pre-image ideal, except it is in general not an ideal as it need not be closed under multiplication (recall the image of an ideal need not be an ideal). Thus, take $J = \langle \varphi^{\#}(I) \rangle$. Then it is clear that

$$\mathcal{Z}(J) = \varphi^{-1}(V)$$

and as V was an arbitrary closed set, φ is continuous, as we sought to show.

Note that *not all* continuous maps arise from a ring homomorphism:

Example 31.4.5: Continuous, No Associated Ring Homomorphism
Take $f : \mathbb{A}_{\mathbb{R}}^1 \to \mathbb{A}_{\mathbb{R}}^1$ given by:

$$f(a) = \begin{cases} 2 & a = 1 \\ 1 & a = 2 \\ a & \text{otherwise} \end{cases}$$

The closed sets of $\mathbb{A}_{\mathbb{R}}^1$ are the finite sets and the whole space, and f carries each of these to a set of the same size, so the pre-image of every closed set is closed and f is continuous. But f is not regular. If it were given by a polynomial p, then $p(a) = a$ for every $a \notin \{1, 2\}$, so the polynomial $p(x) - x$ would have infinitely many roots and hence be zero. That forces $p(x) = x$, which does not transpose 1 and 2.

Another such example is $k = \mathbb{C}$ with the conjugation map presented earlier.

There needs to be more information for the full and faithful correspondence between regular maps and ring homomorphisms. This extra information is a *sheaf* and a *locally ringed morphism*. We shall not define these in this book, instead the reader may refer to [5].

Example 31.4.6: Regular Maps

1. Take $V = \mathcal{Z}(y - x^2, z - x^3)$. Show that the map $f : \mathbb{A}_k^1 \to V$ given by $t \mapsto (t, t^2, t^3)$ is a regular map. Show it is not an isomorphism[a].

2. Any straight line $\mathbb{A}_k^n$ is isomorphic to any other straight-line. Similarly for any k-dimensional linear subspace.

3. The map $\mathbb{A}_k^1 \to \mathcal{Z}(y - x^2)$ given by $t \mapsto (t, t^2)$ is a regular map with inverse $(x, y) \mapsto x$, and hence is an isomorphism. Note that it was important which coordinate we project to.

4. Here is another version of the above map. Consider the global sections of $\mathbb{A}^1_k \to \mathbb{A}^1_k$ given by

$$k[y] \to k[x] \qquad y \mapsto x^2$$

(note that the k-homomorphism is the contravariant direction of the regular morphism). As a map between varieties, this gives the map $a \mapsto a^2$ (the reader should verify this!).

5. Take $f : \mathbb{A}^2_k \to \mathbb{A}^2_k$ given by $(x, y) \mapsto (x, xy)$. Show that $\mathrm{im}(f)$ is *not* an algebraic set. The image of a regular map is always a *constructible set*, a finite union of locally closed sets, which is theorem 41.3.4.

6. The reader should be careful with their geometric intuitions coming from the real numbers. Consider the map:

$$\mathcal{Z}(x^2 + y^2 - 1) \to \mathbb{A}^1_{\mathbb{C}} \qquad (x, y) \mapsto x$$

Show that this map is *surjective* (in contrast to the map certainly being *injective* when over $\mathbb{R}$). In fact, by taking different cross sessions of the resulting shape interpreted as a 2-real dimensional shape living in 4-real dimensional space, you will see that $\mathcal{Z}(x^2 + y^2 - 1)$ will have both the parabola and the hyperbolic curve as sections. The hyperbolic sections demonstrate why the map is surjective and 2-to-1.

[a] It shall later be shown to be a *birational equivalence*

Let us now take a moment to look at isomorphism of algebraic sets:

Definition 31.4.7: Isomorphism of Varieties

let X, Y be two algebraic sets. Then if there exists a regular map $f : X \to Y$ with regular inverse, then the two varieties are said to be *isomorphic*

Note that this was not called a homeomorphism between two varieties. All isomorphisms are homeomorphisms, but not all homeomorphisms are isomorphisms

Example 31.4.8: Isomorphism of Varieties

1. Consider $X = \mathcal{Z}(y - x^n) \subseteq \mathbb{A}^2_k$. Then $X \cong \mathbb{A}^1_k$ with $\varphi(x, y) = x$, which has inverse $\psi(t) = (t, t^n)$

2. Take $X = \mathcal{Z}(x^3 - y^2) \subseteq \mathbb{A}^2_k$. Then X can be parameterized by the regular map $f : \mathbb{A}^1_k \to X$ by $f(t) = (t^2, t^3)$. However, this is not an isomorphism: the inverse map is:

$$g(x, y) = \frac{y}{x}$$

however, there is no polynomial associated to g (can the reader see/prove why?), and hence it is not an isomorphism.

3. The reader may have recalled that $\mathcal{Z}(x^3 - y^2)$ has a cusp at 0, and believe this is the obstruction to the inverse. It is in fact a bit more subtle: a map can be bijective, with no singularity anywhere, and still fail to be an isomorphism. Let $\mathrm{char} k = p > 0$ and consider the Frobenius $\varphi : \mathbb{A}^1_k \to \mathbb{A}^1_k$ given by $\varphi(t) = t^p$. Over $k = \overline{k}$ this is a bijection, since $t \mapsto t^p$ is injective ($t^p = s^p$ gives $(t - s)^p = 0$) and surjective (k is algebraically closed), and it is

regular. Yet its inverse $t \mapsto t^{1/p}$ is not regular: the induced map on coordinate rings is $k[x] \to k[x]$, $x \mapsto x^p$, whose image is $k[x^p]$, and x does not lie in it. So φ is a bijective regular map that is not an isomorphism.

The hypothesis on the characteristic matters. In characteristic 0, or more generally whenever $p \nmid n$, the map $t \mapsto t^n$ is n-to-1 away from the origin over an algebraically closed field, so it is not even bijective.

On the other hand, the map given in the previous example had a rational polynomial inverse. Rational functions appear as localization of the coordinate ring, and so they offer a different class of functions and a different classification problem that turns out to be much easier, as we shall demonstrate in chapter 33.

It is in general difficult to ascertain isomorphic classes of algebraic sets or varieties. Some isomorphism classes that we shall state here but not prove are:

1. Smooth genus 0 curves are isomorphic to conic curves in $\mathbb{P}^2_k$
2. classes of elliptic curves can be classified by their j-invariant
3. Algebraic surfaces are birationally equivalent (defined in chapter 33), but have different isomorphism classes given their Kodaira dimension.

Exercise 31.4.1

1. Show that the Zariski topology on $\mathbb{A}^1_k$ is the cofinite topology, hence the coarsest topology in which points are closed. Then show that for $n \geq 2$ the Zariski topology on $\mathbb{A}^n_k$ is strictly finer than the cofinite topology, so the characterization fails in higher dimension.
2. Prove that $(I \cap J)^2 \subseteq IJ \subseteq I \cap J$, and conclude that $\mathcal{Z}(I \cap J) = \mathcal{Z}(IJ)$.
3. Let $k = \mathbb{F}_q$ be a finite field. Show that every set function $k^n \to k$ is given by a polynomial. Hint: build a characteristic function $\chi_{\overline{a}}$ with $\chi_{\overline{a}}(\overline{a}) = 1$ and 0 elsewhere, and show these span. Conclude that the Zariski topology on $\mathbb{A}^n_{\mathbb{F}_q}$ is discrete.
4. Show that the Zariski topology on $\mathbb{A}^n_k$ is Hausdorff if and only if k is finite.
5. Let $V \subseteq \mathbb{A}^n_k$ and $W \subseteq \mathbb{A}^m_k$ be algebraic sets. Show that $V \times W$ is an algebraic set in $\mathbb{A}^{n+m}_k$, and that $\mathbb{A}^n_k \times \mathbb{A}^m_k = \mathbb{A}^{n+m}_k$ as algebraic sets[14].
6. Let $f, g \in k[x, y]$ have no common *factor*. Show that $\mathcal{Z}(f, g)$ is finite. Hint: they remain coprime in $k(x)[y]$, a principal ideal domain, by a Gauss-lemma argument.
7. Show that if $V \subseteq \mathbb{A}^n_k$ and $W \subseteq \mathbb{A}^m_k$ are varieties, not merely algebraic sets, then $V \times W \subseteq \mathbb{A}^{n+m}_k$ is a variety.

[14]This easy statement becomes markedly harder for projective algebraic sets; see section 40.1.

32

The Algebra-Geometry Dictionary

Chapter 31 left $\mathcal{Z}$ and $\mathcal{I}$ almost inverse to one another: an algebraic set is recovered from its vanishing ideal, but an ideal need not be recovered from its zero set. This chapter identifies exactly which ideals are recovered. The answer, Hilbert's Nullstellensatz, is that they are the radical ideals, and it holds precisely when the ground field is algebraically closed.

From here to the end of the book, k is algebraically closed unless a result says otherwise. The hypothesis is not decorative: over $\mathbb{R}$ the ideal (x^2+1) is its own radical and has empty zero set, so the correspondence fails at once. With the Nullstellensatz in hand, the geometric side and the algebraic side can be tabulated against each other, and the rest of the book consists of extending that table.

32.1 Coordinate Rings

In this section, we explore more properties of affine sets. As affine sets correspond to ideals, we shall see that there is a correspondence between the two objects. In particular it shall will show that properties of algebraic sets will often have corresponding properties in their coordinate rings, and vice-versa. This theme shall play out often in this chapter, where a property proven in a geometric or algebraic setting shall have a corresponding result in the respective dual.

By lemma 31.4.2, the collection of regular function $\mathrm{Hom}_{\mathbf{Reg}}(\mathbb{A}^n_k, \mathbb{A}^1_k)$ can be thought of as the generalization of the dual vector-space, where polynomial functions are taken instead of linear functions. We saw that for each vector subspace $W \subseteq V$, there is a dual space $W^\vee = \mathrm{Hom}_k(W, k)$ which it is isomorphic to. It is thus natural to ask what is the appropriate dual space for an algebraic set $V \subseteq \mathbb{A}^n_k$ corresponding to the *quotient* of $\Gamma(\mathbb{A}^n_k)$[1].

[1] For the categorically inclined, since the arrows in the dual space are reversed, sub-objects correspond to quotient-objects

In order to find the dual, we must consider more the subspace topology put on $V \subseteq \mathbb{A}_k^n$. Note that all of the properties of the Zariski topology apply to A since $\mathbb{A})k^n$ is T1 (proposition 31.3.1), from which we can conclude that closed sets are preserved in the subspace topology. We will always limit ourselves to taking A to be an algebraic set, since algebraic set are the object of interest. The dual space we wish to establish is a corresponding "polynomial ring" which will generate all of the closed sets of A, just like how $k[x_1, x_2, ..., x_n]$ is needed to generate the closed sets of $\mathbb{A}^n$. Without looking at the next definition, can you see what the definition of such a ring may be?

To elucidate the coming definition, let us come back to the algebraic set V defined given by the vanishing set of $x^2 + y^2 - 1$. Then on V, the function $X^2 + Y^2 - 1$ on V will evaluate to 0, hence these two polynomials are equivalent as functions:

$$X^2 + Y^2 - 1 \equiv 0 \qquad \text{on } V$$

If g is any other polynomial corresponding to the function G, then $(X^2 + Y^2 - 1)G$ evaluates to 0 everywhere as well, showing the set of regular functions taking inputs on V is closed under multiplication by regular functions $\Gamma(\mathbb{A}_k^n)$. Furthermore, if two regular functions f, g which evaluate to zero everywhere are added, the function $f + g$ will also evaluate to zero, showing it is closed under addition. In other words, the collection of functions which evaluates to 0 forms an ideal of $\Gamma(\mathbb{A}_k^n)$. What is this ideal? If we let F be the function

$$F : k[x, y] \to \mathrm{Hom}_{\mathbf{Reg}}(V, k) \qquad f \mapsto ((a, b) \to f(a, b)) \tag{32.1}$$

where $\mathrm{Hom}_{\mathbf{Reg}}(V, k)$ is the set of regular functions which evaluate only on the points V, then we get that this map is surjective and that:

$$\frac{k[x, y]}{\ker(F)} \cong_k \mathrm{Hom}_{\mathbf{Reg}}(V, k)$$

The reader should clue in that $\ker(F)$ is by definition $\mathcal{I}(V)$ as defined in definition 31.2.8! This leads to the following definition:

Definition 32.1.1: [Affine] Coordinate Ring

Let $k[x_1, ..., x_n]$ be a polynomial ring over n variables, V an algebraic set, and $\mathcal{I}(V)$ the corresponding vanishing ideal. Then:

$$\Gamma(V) = k[V] = \frac{k[x_1, ..., x_n]}{\mathcal{I}(V)}$$

is called the *coordinate ring* of V. The corresponding regular functions shall be denoted $k[V]$ or $\Gamma(V)$.

Notice how this notation is the same as for regular functions: the coordinate ring of $V = \mathbb{A}_k^n$ is isomorphic to $k[X_1, ..., X_n]$ since $I(\mathbb{A}^n) = (0)$. In $\Gamma(V)$, we have that $\mathcal{I}(V) = \overline{(0)}$, showing the notation is consistent.

We can naturally bring the notion of regular function to algebraic sets:

Definition 32.1.2: Regular Maps on Algebraic Sets

Let $V \subseteq \mathbb{A}_k^n$, $W \subseteq \mathbb{A}_k^m$ be algebraic sets. Then a map $\varphi : V \to W$ is called a *regular map*, or *regular morphism*, if there exists polynomials $\varphi_1, \varphi_2, ..., \varphi_m \subseteq k[x_1, x_2, ..., x_m]$ such that

$$\varphi(a_1, a_2, ..., a_n) = (\varphi_1(a_1, a_2, ..., a_n), \varphi_2(a_1, a_2, ..., a_n)..., \varphi_m(a_1, a_2, ...a_n))$$

If $m = 1$, then φ is often called a *regular function*.

This definition also allows us to upgrade definition 31.1.2 and definition 31.4.1. To check that $\varphi : V \to W$ is well-defined, it suffices to check that $\varphi_1, ..., \varphi_m$ as elements of $k[V]$ satisfy the equations of W.

Proposition 32.1.3: Regular Maps Are Continuous

Let $\varphi : V \to W$ be a regular map where V, W are equipped with the Zariski topology. Then φ is continuous.

***Proof*:**
Let $S \subseteq W$ be a closed subset. It suffices to show that $\varphi^{-1}(S)$ is closed. if $\varphi^{-1}(S) = \emptyset$, it is closed by definition, so assume $\varphi^{-1}(S) \neq \emptyset$. As S is closed, there exists polynomials $f_1, ..., f_n$ such that for each $x \in S$, $g_1(x) = \cdots = g_m(x) = 0$. If $a \in \varphi^{-1}(S)$, then $\varphi(a) \in S$, so

$$g_1(\varphi(a)) = \cdots = g_m(\varphi(a)) = 0 \tag{32.2}$$

As $\varphi = (\varphi_1, ..., \varphi_n)$ is a tuple of polynomials, notice that (32.2) is a collection of polynomials in $k[V]$ where a vanishes. As this is true for all points in $\varphi^{-1}(S)$, we see that:

$$\varphi^{-1}(S) = \mathcal{Z}(g_1 \circ \varphi, ..., g_m \circ \varphi)$$

showing it is closed, as we sought to show.

As before, this is not an iff as not all continuous functions are regular functions (simply take $V = W = \mathbb{A}_{\mathbb{C}}^1$ and the conjugation map). That is because continuity in the Zariski topology is not enough to preserve the algebraic relations. In particular, any point $a \in V \subseteq \mathbb{A}_k^n$ satisfying

$$f_1(a) = \cdots f_n(a) = 0$$

must have an image that satisfies:

$$g_1(w) = \cdots = g_m(w) = 0 \qquad \text{i.e.} \qquad g_1(\varphi(a)) = \cdots g_1(\varphi(a)) = 0$$

Example 32.1.4: Coordinate Ring and Regular Mapping

1. Let $V = \mathcal{Z}(xy - 1)$, which represents the hyperbola $y = 1/x$ in $\mathbb{R}^2$. Then the coordinate ring $\mathbb{R}[V] = \mathbb{R}[x, y]/(xy - 1)$. In $\mathbb{R}[V]$, $f(x, y) = x$ and $g(x, y) = x + xy - 1$ are the same function since $f - g \in \mathcal{I}(V)$. In $\mathbb{R}[V]$, $\overline{xy} = \overline{1}$, giving us that $\mathbb{R}[V] \cong \mathbb{R}[x, 1/x]$. Thus, for any $f \in \mathbb{R}[x, y]$ and $(a, b) \in V$, $\overline{f}(a, b) = f(a, 1/a)$ for any f that maps to $\overline{f}$.

Take the regular map $\varphi : V \to \mathbb{A}^1_k$ mapping $(x, y) \mapsto x$. The image of this map is $\mathbb{A}^1_k \setminus \{0\}$, which is certainly *not* a closed set (can the reader see why). It is an open set, being the complement of the closed set $\{0\}$. However, if we take the map to be $\varphi' : V \to \mathbb{A}^2_k$ mapping $(x, y) \mapsto (x, 0)$, then the image is no longer an open set, and hence it is possible for the images of regular maps to be neither open or closed. Further exploration the properties of the image of regular maps shall be done section 39.4.

2. Take $k = \mathbb{F}_p$ and consider $\mathbb{A}^n_{\mathbb{F}_p}$. Then the map:

$$\varphi(a_1, ..., a_n) = (a_1^p, ..., a_n^p)$$

is a regular function. If $V \subseteq \mathbb{A}^n_{\mathbb{F}_p}$ is an algebraic set, notice that φ will always take V to itself, since if f is one of the polynomials that define V so that $f(a) \equiv 0 \bmod p$, then:

$$f(a_1^p, ..., a_n^p) = (f(a_1, ..., a_n))^p \equiv (0)^p \equiv 0 \bmod p$$

The regular function φ is called the *Frobenius mapping*

3. Take $t \mapsto (t^3, t^2)$. This is a regular function and maps $\mathbb{A}^1_{\mathbb{R}}$ to the curve given by $x^3 = y^2$ in $\mathbb{A}^2_{\mathbb{R}}$.

4. Take $V = \mathcal{Z}(xy - 1)$ and $W = \mathcal{Z}(v - u^2)$. We can define a map between global sections:

$$\varphi : \frac{k[u, v]}{(v - u^2)} \to \frac{k[x, y]}{(xy - 1)} \qquad u \mapsto \frac{x + y}{2}$$

and extend k-linearly (note that $v \mapsto \left(\frac{x+y}{2}\right)^2$). As the left hand side is a free k-algebra up to the quotient relation, let's insure the relation is preserved, and indeed:

$$\varphi(0) = \varphi(v - u^2) = \frac{(x + y)^2}{2} - \frac{(x + y)^2}{2} = 0$$

showing that the relation is preserved, and hence φ is well-defined. The associated map on the varieties is given by:

$$(a, 1/a) \mapsto \left(\frac{a + 1/a}{2}, \left(\frac{a + 1/a}{2}\right)^2\right)$$

Note the image is 2-to-1. If $k = \mathbb{R}$, the image is *not* surjective (ex. nothing maps to 0). However if $i \in k$ (where $i^2 = -1$), then $i + 1/i = i - i = 0$, which shows the map is surjective and 2-to-1. This demonstrates part of the importance of working over an algebraically closed field.

Theorem 32.1.5: Regular Map Algebraic Correspondence

Let $V \subseteq \mathbb{A}^n$ and $W \subseteq \mathbb{A}^m$. then there is a bijective correpspondence between the morphisms between V and W and the associated k-algebra homomorphisms, more precisely

$$\{\text{regular functions from } V \text{ to } W\} \leftrightarrow \{k\text{-algebra homomorphisms from } k[W] \text{ to } k[V]\}$$

$$\left\{\begin{matrix}\text{regular functions} \\ \varphi : V \to W\end{matrix}\right\} \underset{\psi^a}{\overset{f\circ\varphi}{\rightleftarrows}} \left\{\begin{matrix}k\text{-algebra homomorphism} \\ \varphi^\# : k[W] \to k[V]\end{matrix}\right\}$$

More generally, if k-**Aff** is the category whose objects are algebraic sets and morphism are regular functions and k-$\mathbf{Alg}_0$ is the category whose objects are coordinate rings (reduced, commutative, finite-type k-algebras) and morphisms are k-homomorphisms, then $k\text{-}\mathbf{Aff}^{\mathrm{op}} \cong k\text{-}\mathbf{Alg}_0$ (the equivalence of categories requires k algebraically closed, so that every such algebra is realized as a coordinate ring by proposition 32.5.5).

***Proof*:**
The two constructions are already in hand: lemma 31.4.2 sends a regular map φ to the k-algebra homomorphism $\varphi^\#(f) = f \circ \varphi$, and lemma 31.4.3 sends a k-algebra homomorphism back to a regular map. It remains to see they are mutually inverse.

Given $\varphi\colon V \to W$, write ψ for the regular map induced by $\varphi^\#$. Let $y_1, \ldots, y_m$ be the coordinate functions on W. By construction the ith component of ψ is $\varphi^\#(y_i) = y_i \circ \varphi = \varphi_i$, the ith component of φ; as the components agree, $\psi = \varphi$.

Conversely let $\theta\colon k[W] \to k[V]$ be a k-algebra homomorphism, φ the regular map it induces, and $\varphi^\#$ the homomorphism φ induces in turn. Both θ and $\varphi^\#$ are k-algebra maps out of $k[W]$, which is generated over k by the $\overline{y_i}$, and both send $\overline{y_i}$ to $\theta(\overline{y_i})$; a k-algebra homomorphism is determined by its values on generators, so $\varphi^\# = \theta$.

For the categorical statement, $\varphi \mapsto \varphi^\#$ is contravariant and functorial, since $(\varphi \circ \psi)^\# = \psi^\# \circ \varphi^\#$ and $\mathrm{id}^\# = \mathrm{id}$. The bijection just established makes it fully faithful, and proposition 32.5.5 makes it essentially surjective onto k-$\mathbf{Alg}_0$ when k is algebraically closed, completing the proof.

Thus, if $f : X \to Y$, then $(f^\#)^a = f$ and if $\varphi : k[Y] \to k[X]$, then $(\varphi^a)^\# = \varphi$. Similarly to the dual theory provided in section 13.3, the dual map gives information about our original map:

Proposition 32.1.6: Regular Map Properties

Let $\varphi : X \to Y$ be a regular map with $\varphi^{\#} : k[Y] \to k[X]$ the associated k-algebra homomorphism. Then:

1. The kernel of $\tilde{\varphi}$ is $\mathcal{I}(\varphi(X))$: $\ker \tilde{\varphi} = \mathcal{I}(\varphi(X))$. Furthermore, the Zariski closure of $\varphi(X)$ is $\mathcal{Z}(\ker(\tilde{\varphi}))$: $\overline{\varphi(X)} = \mathcal{Z}(\ker(\tilde{\varphi}))$.

2. As a consequence, the homomorphism $\tilde{\varphi}$ is injective if and only if $\varphi(X)$ is Zariski dense in Y, or equivalently if $\overline{f(X)} = Y$

3. $\varphi(v) = w$ if and only if $(\varphi^{\#})^{-1}(\mathcal{I}(v)) = \mathcal{I}(w)$, where $\mathcal{I}(v) \subseteq k[X]$ and $\mathcal{I}(w) \subseteq k[Y]$ are the maximal ideals of the two points

***Proof*:**

1. A function $g \in k[Y]$ satisfies $\varphi^{\#}(g) = g \circ \varphi = 0$ exactly when g vanishes at every point of $\varphi(X)$, which is to say $g \in \mathcal{I}(\varphi(X))$. Hence $\ker \varphi^{\#} = \mathcal{I}(\varphi(X))$, and applying $\mathcal{Z}$ gives $\mathcal{Z}(\ker \varphi^{\#}) = \mathcal{Z}(\mathcal{I}(\varphi(X))) = \overline{\varphi(X)}$ by corollary 31.3.3.

2. By part 1, $\varphi^{\#}$ is injective if and only if $\mathcal{I}(\varphi(X)) = 0$ in $k[Y]$, and $\mathcal{Z}(0) = Y$, so this says $\overline{\varphi(X)} = Y$: the image is dense.

3. Suppose $\varphi(v) = w$. For $g \in k[Y]$, $\varphi^{\#}(g)(v) = g(\varphi(v)) = g(w)$, so $\varphi^{\#}(g) \in \mathcal{I}(v)$ if and only if $g \in \mathcal{I}(w)$, which is exactly $(\varphi^{\#})^{-1}(\mathcal{I}(v)) = \mathcal{I}(w)$. Conversely, suppose that identity holds. Both $\mathcal{I}(v)$ and $\mathcal{I}(w)$ are maximal (theorem 32.4.3), and writing $w' = \varphi(v)$ the forward direction gives $(\varphi^{\#})^{-1}(\mathcal{I}(v)) = \mathcal{I}(w')$. Hence $\mathcal{I}(w) = \mathcal{I}(w')$, and since $\mathcal{I}$ is injective on points, $w = w' = \varphi(v)$, completing the proof.

Exercise 32.1.1

1. Show that $k[X \times Y] \cong k[X] \otimes_k k[Y]$ for algebraic sets X, Y, and deduce another proof that $\mathbb{A}_k^n \times \mathbb{A}_k^m \cong \mathbb{A}_k^{n+m}$ as objects of **Reg**.

2. Show that the product is unique up to isomorphism: if $\varphi\colon X \to X'$ and $\psi\colon Y \to Y'$ are isomorphisms, there is a unique isomorphism $X \times Y \to X' \times Y'$ commuting with the projections.

3. Let $T\colon \mathbb{A}_k^n \to \mathbb{A}_k^n$ be an affine isomorphism with $T(p) = q$. Show that $T^{\#}\colon \mathcal{O}_q(\mathbb{A}_k^n) \to \mathcal{O}_p(\mathbb{A}_k^n)$ is an isomorphism, and that for a variety V it induces $\mathcal{O}_q(V) \to \mathcal{O}_p(V^T)$. This is the precise sense in which affine space has no preferred origin.

4. Sharpen theorem 32.1.5: show that a regular map $\varphi\colon X \to Y$ is dominant if and only if $\varphi^{\#}$ is injective, and that φ is a closed immersion, meaning an isomorphism onto a closed subset, if and only if $\varphi^{\#}$ is surjective.

32.2 Radical Ideals

As we've seen in proposition 31.2.5, given some polynomial f or collection of polynomials $S \subseteq k[x_1, ..., x_n]$, the set of zero's of these polynomials does not change if we take higher powers of the polynomials (ex. $f, f^2, f^3, ...$ all produce the same zero set). Thus, studying the set $\mathcal{Z}(I^n)$ is equivalent to studying $\mathcal{Z}(I)$. This property inspires the following definition:

Definition 32.2.1: Radical Ideal

Let $I \subseteq R$ be an ideal of a commutative ring R. Then:

1. The *radical of* I in R is the collection elements for which some power of them lie in I, that is:
$$\sqrt{I} = \text{rad}(I) := \left\{a \in R \;\middle|\; a^k \in I \text{ for some } k \in \mathbb{N}\right\}$$
here $\text{rad}(I)$ stands for the radical of I. It is often written as $\sqrt{I}$.
2. The radical of the zero ideal ($\text{rad}(0) = \sqrt{(0)}$) is called the *nilradical* of R
3. An ideal I is called a *radical ideal* if $I = \text{rad}(I)$ ($I = \sqrt{I}$)

Note that the nilradical by definition contains all the nilpotent elements of R, and that $\text{rad}(\text{rad}(I)) = \text{rad}(I)$.

Proposition 32.2.2: $\text{rad}(I)$ Is An Ideal

Let $I \subseteq R$ be an ideal of a commutative ring. Then $\text{rad}(I)$ is indeed an ideal (containing I). Furthermore, $\text{rad}(I)/I$ is the nilradical of R/I.

The last statement shows us that R/I has no nilpotent elements if and only if $I = \text{rad}(I)$.

***Proof*:**
See [10].

A type of ideal in which we've seen a similar property is the prime ideal, in which $ab \in I$ if and only if $a \in I$ or $b \in I$. One important way of looking at radical is in how they correspond to prime ideals. The prime ideals in a sense "generate" the radical ideals:

Proposition 32.2.3: Radical Through Prime Ideals

Let $I \subsetneq R$ be a proper ideal of a commutative ring and $\text{rad}(I)$ the radical ideal of I. Then $\text{rad}(I)$ is the intersection of all prime ideals containing I. In particular, the nilradical is the intersection of all prime ideals of R.

***Proof*:**
This is the description of the radical as an intersection of primes; see [10].

Corollary 32.2.4: Prime Or Maximal Then Radical

Let $I \subseteq R$ be a prime ideal. Then I is a radical ideal

***Proof*:**
A prime ideal P satisfies $P = \sqrt{P}$; see [10].

Example 32.2.5: Radical Ideals

1. Let R be a UFD. Given an ideal, finds its radical.
2. The ideal $(x^3 - y^2)$ is prime in $k[x, y]$, and hence radical
3. If $l_1, l_2, ..., l_m$ is a collection of linear polynomials in $k[x_1, ..., x_n]$, then $I = (l_1, l_2, ..., l_m)$ is either $k[x_1, x_2, ..., x_n]$ or a prime ideal, and hence radical.

In the context of this chapter, since we are working over polynomials rings, we are working over Noetherian rings. Due to this, the elements of a radical ideal $\text{rad}(I)$ can be related back to elements of I quite naturally:

Proposition 32.2.6: Radical Ideals In Noetherian Rings

Let R be a noetherian ring and $I \subseteq R$ an ideal of R. Then there exists a $k \in \mathbb{N}$ such that $I^k \subseteq \text{rad}(I)$. In particular, if N is the nilradical ideal of R, then $N^k = 0$ for some $k \in \mathbb{N}$ (i.e. N is a nilpotent ideal)

***Proof*:**
See [10].

Now, returning to the fact that $\mathcal{Z}(I) = \mathcal{Z}(I^k)$, we now have that $\mathcal{Z}(I) = \mathcal{Z}(\sqrt{I})$. Remember how we were trying to characterize the ideals for which $\mathcal{Z}$ and $\mathcal{I}$ are inverses, i.e. what are the ideals $\mathcal{I}(\mathcal{Z}(I))$? We can now state one direction of this:

Proposition 32.2.7: Radical and Algebraic Sets

Let $I \subseteq R$ be an ideal where R is a polynomial ring. Then

$$\sqrt{I} \subseteq \mathcal{I}(\mathcal{Z}(I))$$

***Proof*:**
Let $f \in \sqrt{I}$, so $f^m \in I$ for some $m \geq 1$, and let $x \in \mathcal{Z}(I)$ be arbitrary. Every element of I vanishes at x, so in particular

$$f(x)^m = f^m(x) = 0$$

Since k is a field it has no nonzero nilpotents, so $f(x) = 0$. As $x \in \mathcal{Z}(I)$ was arbitrary, f vanishes on $\mathcal{Z}(I)$, which is to say $f \in \mathcal{I}(\mathcal{Z}(I))$, as we sought to show.

It would be great if we could get $\sqrt{I} = \mathcal{I}(\mathcal{Z}(I))$, however this is not true in general: consider $I = (x^2+1) \subseteq \mathbb{R}[x]$. Since $x^2 = 1$ is irreducible, it is a maximal ideal, and so $I = \sqrt{I}$. Then as we are over the reals, $\mathcal{Z}(I) = \emptyset$, so $\mathcal{I}(\mathcal{Z}(I)) = \mathbb{R}[x]$ But $I \not\supseteq \mathbb{R}[x]$, showing equality doesn't hold. The blocker here was that x^2+1 doesn't factor in $\mathbb{R}[x]$: if we were working over $\mathbb{C}$, then we would get that $(x^2+1) = ((x+i)(x-i))$, telling us that $\mathcal{Z}(I) = \{i, -i\}$, which by the basic properties of $\mathcal{I}$ we have that $\mathcal{I}(\mathcal{Z}(I)) = ((x-i)(x+i)) = (x^2+1) = I$, showing us in this case that $I = \sqrt{I}$. In fact, the underlying field k being algebraically closed is the *only* other condition we need for equality to hold; this will be theorem 32.4.4 (Hilbert's Nullstellensatz). It is tricky to show this, since it is in a sense a very simple condition: All we need to characterize the zero sets of the polynomials (over a closed field) $\mathcal{Z}(I)$ in terms of an ideal is to take the radical of I.

32.3 Irreducibility and the Decomposition Theorem

We now return to the earlier fact that $\mathcal{Z}(I) = \mathcal{Z}(f_1, ..., f_n) = \mathcal{Z}(f_1) \cap \cdots \cap \mathcal{Z}(f_n)$. This shows that we can decompose algebraic sets into smaller components. We will therefore take a moment to see if we can find "decomposition" properties of algebraic sets, and see if there is a corresponding property the coordinate ideals will poses

Definition 32.3.1: [Affine] Variety

let $V \subseteq \mathbb{A}^n$ be an algebraic set. The set V is reducible if for $V_1, V_2 \subseteq V$ where $V_1, V_2 \neq V$

$$V = V_1 \cup V_2$$

If there does not exist such sets, we say that V is *irreducible*, or more commonly an *algebraic variety*

32.3.2 Remark Note that this notion of irreducibility is in fact a topological notion, we simply choose the Zariski topology. In particular, X is irreducible if $X = X_1 \cup X_2$ for some closed subsets X_1 and X_2, then either $X = X_1$ or $X = X_2$. For those who are curious, it may be shown that irreducibility implies connected. Furthermore, the condition translates to open sets to say that if

$$\emptyset = U_1 \cap U_2$$

where U_1, U_2 open, then either $U_1 = \emptyset$ or $U_2 = \emptyset$. It follows that it is equivalent to say that every open set is *dense* in an irreducible space. This contrasts heavily with any Hausdorff space, since by definition a Hausdorff space *must* have disjoint open subsets if the space is not empty or a singleton.

The name *algebraic variety* comes from HERE. Note that this condition automatically implies that V_1 and V_2 are non-empty (do you see why?). If V is an irreducible algebraic set, then the corresponding ideal has a very nice property:

Proposition 32.3.3: Coordinate Ideal of Algebraic Variety

A set $V \subseteq \mathbb{A}^n$ is an algebraic variety if and only if $I(V)$ is a prime ideal

***Proof*:**

(⇒) Taking the contrapositive, let's assume $I(V)$ is not prime. Then there exists $a, b \in k[x_1, .., x_n]$ such that $ab \in I(V)$ but $a \notin I(V)$ and $b \notin I(V)$. But then:

$$V = (V \cap \mathcal{Z}(a)) \cup (V \cap \mathcal{Z}(b))$$

with $V \cap V(a) \subsetneq V$, $V \cap V(b) \subsetneq V$, showing that V is in fact reducible

(⇐) Doing the contrapositive again, let's assume V is reducible, so $V = V_1 \cup V_2$, where both $V_i \subsetneq V$. Then by the contra-positive correspondence of inclusion between algebraic sets and ideals, $I(V) \subsetneq I(V_i)$, $i \in \{1, 2\}$. Since the inclusion is proper, choose $a \in I(V_1)$ and $b \in I(V_b)$ such that $a, b \notin I(V)$. Then since a and b are polynomials which are zero on their respective algebraic sets and $V = V_1 \cup V_2$, $ab \in I(V)$. However, $a, b \notin I(V)$, showing that $I(V)$ is not prime

Corollary 32.3.4: Equivalent characterization

Let $V \subseteq \mathbb{A}^n$ be an algebraic set. Then V is an algebraic variety if and only if $k[V]$ is an integral domain

***Proof*:**
By definition 32.1.1, $k[V] = \Gamma(\mathbb{A}^n)/\mathcal{I}(V)$, and a quotient of a commutative ring is an integral domain exactly when the ideal quotiented out is prime. So $k[V]$ is a domain if and only if $\mathcal{I}(V)$ is prime, which by proposition 32.3.3 holds if and only if V is irreducible, as we sought to show.

This means there is a certain correspondence between being a prime and being irreducible as a variety. To deepen the conection, let's prove that every algebraic set can be decomposed into algebraic varieties:

Theorem 32.3.5: Irreducible Decomposition

Let $V \subseteq \mathbb{A}^n$ be an algebraic set. Then there exist algebraic varieties $V_1, ..., V_r$, unique up to reordering, such that

$$V = \bigcup_{i=1}^{r} V_i \qquad V_i \not\subseteq V_j \quad i \neq j$$

***Proof*:**
Existence. Since $\Gamma(\mathbb{A}^n)$ is Noetherian, every nonempty collection of ideals has a maximal element, and $\mathcal{I}$ reverses inclusion, so every nonempty collection of algebraic sets has a *minimal* element. Let

$$S = \{W \subseteq \mathbb{A}^n \text{ algebraic} \mid W \text{ is not a finite union of algebraic varieties}\}$$

and suppose for contradiction that $S \neq \emptyset$. Choose $W \in S$ minimal. Then W is not itself a variety, so $W = W_a \cup W_b$ with both $W_a, W_b \subsetneq W$. By minimality neither lies in S, so each is a finite union of varieties, say

$$W_a = V_{1,1} \cup \cdots \cup V_{1,n_1} \qquad W_b = V_{2,1} \cup \cdots \cup V_{2,n_2}$$

and then $W = \bigcup_{i,j} V_{i,j}$ is one too, contradicting $W \in S$. Hence $S = \emptyset$.

Given any finite decomposition, discard every $V_{i,j}$ contained in another; what remains still has union V and now satisfies $V_i \not\subseteq V_j$ for $i \neq j$.

Uniqueness. Suppose $V = V_1 \cup \cdots \cup V_r = W_1 \cup \cdots \cup W_m$ are two such decompositions. Fix i. Then

$$V_i = V_i \cap V = \bigcup_j (W_j \cap V_i)$$

expresses the irreducible set V_i as a finite union of closed subsets, so by irreducibility $V_i = W_j \cap V_i$ for some j, that is $V_i \subseteq W_j$. Running the same argument on W_j against the first decomposition gives $W_j \subseteq V_k$ for some k, whence $V_i \subseteq V_k$. The decomposition has no containments among distinct members, so $i = k$, and therefore $V_i = W_j$. This assigns to each i a j with $V_i = W_j$, and symmetrically to each j an i; the two assignments are mutually inverse, so $r = m$ and the two families agree up to reordering, completing the proof.

Definition 32.3.6: Irreducible Decomposition

Let V be an algebraic set. Then the *irreducible decomposition of* V is

$$V = V_1 \cup \cdots V_n \quad V_i \not\subseteq V_j i \neq j$$

for appropriate algebraic varieties V_i, $1 \leq i \leq n$

32.3.7 Remark In a more general topological setting, it can be shown that if X is a Noetherian space (i.e. satisfies A.C.C. on open sets, so D.C.C. on closed sets), X can be decomposed into irreducible sets just liked was done in theorem 32.3.5. It then follows easily that any closed subset of X has this property too.

Example 32.3.8: Affine Varieties

1. All linear polynomials produce affine varieties. Thus, the study of linear algebra can be thought as the study of varieties with multiplicity 1.
2. $\mathcal{Z}(xy-1) \subseteq \mathbb{A}^2_{\mathbb{R}}$ is an affine variety, as $\mathbb{R}[x,y]/(xy-1) \cong \mathbb{R}[x, 1/x]$. Note that $xy-1$ has two branches, however the variety is still irreducible and connected in the Zariski topology. Show that there is no variety that is only one branch of $xy-1$ (show that if two affine varieties agree on an open subset, they agree everywhere).

The decomposition of a Noetherian variety into irreducible components has an algebraic mirror: the *primary decomposition* of ideals. The minimal primes over $I(X)$ are precisely the ideals of the irreducible components of X, while embedded primes record finer, scheme-theoretic information carried by the coordinate ring but invisible to the underlying set. The full theory (primary ideals, the Lasker-Noether existence and uniqueness theorems, and the associated primes of a module) is developed in [10]; we use it freely when passing between a variety and its coordinate ring.

Exercise 32.3.1

1. Show that an algebraic set is irreducible if and only if any two nonempty open subsets of it meet, and use this to give a second proof that $\mathbb{A}_k^n$ is irreducible.
2. Show that the intersection of two irreducible algebraic sets need not be irreducible. Hint: take $\mathcal{Z}(z)$ and $\mathcal{Z}(xy-z)$ in $\mathbb{A}_k^3$.
3. Show that the closure of an irreducible subset is irreducible, and that a nonempty open subset of an irreducible set is irreducible and dense.
4. Show that a Noetherian topological space is quasicompact, and that every subspace of one is Noetherian.
5. Let X be an algebraic set with irreducible decomposition $X = X_1 \cup \cdots \cup X_r$ as in theorem 32.3.5. Show that the X_i are exactly the maximal irreducible closed subsets of X, which is why the decomposition is unique.

32.4 Hilbert's Nullstellensatz

In this section, we shall make precise the correspondence between which ideals of the coordinate ring of an affine set correspond to closed subsets of the affine set. In this exploration, we shall also produce a new tower condition on ring extensions via Noether's Normalization theorem which builds upon the theme of separating an extension to "property" and "not property" (ex. separable and inseparable, algebraic and transcendental[2]) Recall if $a_1, a_2, ..., a_n \in S$ for a k-algebra S, these elements are called *algebraically independent* if

$$k[x_1, x_2, ..., x_n] \to k[a_1, a_2, ..., a_n] \qquad x_i \mapsto a_i$$

is an isomorphism. Noether's Normalization theorem is the generalization of the idea that any field extension can be split into a transcendental extension, followed by an algebraic extension:

Theorem 32.4.1: Noether's Normalization Theorem

Let k be a field and suppose that $A = k[r_1, ..., r_n]$ is a finitely generated k-algebra. Then for some q, $0 \le q \le n$, there are algebraically independent elements $y_1, y_2, ..., y_q \in A$ such that A is integral over $k[y_1, y_2, ..., y_q]$

***Proof*:**
Induct on the number n of generators. If $n = 0$ then $A = k$ and there is nothing to prove, so let $n \ge 1$ and assume the theorem for k-algebras on fewer than n generators.

Step 1: the independent case. If $r_1, \ldots, r_n$ are algebraically independent then A is integral over itself and we may take $q = n$ and $y_i = r_i$. So assume they are algebraically dependent: there is a nonzero $f \in k[x_1, \ldots, x_n]$ with $f(r_1, \ldots, r_n) = 0$. Write d for the total degree of f, where a monomial $ax_1^{m_1} \cdots x_n^{m_n}$ has degree $m_1 + \cdots + m_n$ and $\deg f$ is the largest degree occurring.

Step 2: a change of variables making f monic in x_n. Set $\alpha_i = (1+d)^i$ for $1 \le i \le n-1$, introduce new variables $X_i = x_i - x_n^{\alpha_i}$, and define

$$g(X_1, \ldots, X_{n-1}, x_n) = f(X_1 + x_n^{\alpha_1}, \ldots, X_{n-1} + x_n^{\alpha_{n-1}}, x_n)$$

[2]Another such example will appear in the next chapter, and shall be ramified and unramified

so $g \in k[X_1, \ldots, X_{n-1}, x_n]$. A monomial $ax_1^{m_1} \cdots x_n^{m_n}$ of f contributes to g a single term in which the X_i do not appear, namely $a\, x_n^e$ with

$$e = m_n + m_1\alpha_1 + \cdots + m_{n-1}\alpha_{n-1} = m_n + m_1(1+d) + \cdots + m_{n-1}(1+d)^{n-1}$$

every other term carries at least one X_i. Since each $m_i \leq d < 1+d$, the tuple $(m_n, m_1, \ldots, m_{n-1})$ is the base-$(1+d)$ expansion of e, so distinct monomials of f give distinct e. Hence the largest such e, call it N, is attained exactly once, and

$$g = cx_n^N + \sum_{i=0}^{N-1} h_i(X_1, \ldots, X_{n-1})x_n^i$$

for some nonzero $c \in k$: after dividing by c, g is monic in x_n over $k[X_1, \ldots, X_{n-1}]$.

Step 3: descending one generator. Put $s_i = r_i - r_n^{\alpha_i}$ and $B = k[s_1, \ldots, s_{n-1}] \subseteq A$. Substituting $X_i \mapsto s_i$ and $x_n \mapsto r_n$ in g recovers $f(r_1, \ldots, r_n) = 0$, so

$$\frac{1}{c} g(s_1, \ldots, s_{n-1}, r_n) = 0$$

exhibits r_n as a root of a monic polynomial over B; thus r_n is integral over B. Each remaining $r_i = s_i + r_n^{\alpha_i}$ lies in $B[r_n]$, so $A = B[r_n]$, and A is integral over B because it is generated over B by the single integral element r_n.

Step 4: induction. B is a k-algebra generated by the $n-1$ elements $s_1, \ldots, s_{n-1}$, so by the inductive hypothesis there are algebraically independent $y_1, \ldots, y_q \in B$ with B integral over $k[y_1, \ldots, y_q]$. Integrality is transitive, so A is integral over $k[y_1, \ldots, y_q]$, and $q \leq n-1 \leq n$, as we sought to show.

Noether's Normalization can be thought of as a very geometric result: we shall see that any finitely generated k-algebra shall be an n-dimensional affine set along with some extra important information given by the integral extension, more on this in chapter 34. One very important is that if we have an extension F/k that is a finite-type k-algebra, is finitely generated if we allow for both addition and multiplication, it is finitely gendered by *just* allowing addition!!

Theorem 32.4.2: Zariski's Lemma

Let F/k be a field extension, and assume F is a finite-type k-algebra. Then F/k is a finite (and hence algebraic) extension.

***Proof*:**
Since F is a finite-type k-algebra, $F = k[r_1, r_2, ..., r_n]$. By Noether's Normalization Theorem, there exists a $q \in \mathbb{N}$ (possibly 0) enumerating variables $y_1, ..., y_q \in F$ such that F is integral over $k[y_1, y_2, ..., y_q]$, i.e. F is a finitely generated $k[y_1, ..., y_q]$-module. We can re-phrase this by saying we have a finite integral extension $F/k[y_1, y_2, ..., y_q]$. By proposition 26.1.8, since F is an integral domain, F is a field if and only if $k[y_1, y_2, ..., y_q]$ is a field. But then, $k[y_1, y_2, ..., y_q]$ is a field if and only if there are no indeterminate variables, i.e. $q = 0$. Thus, we get the finite integral extension F/k, which is equivalent to saying we get a finite field extension, as we sought to show.

The last two theorem's are true no matter what field we are working over. The following theorem requires k to be algebraically closed.

Theorem 32.4.3: Hilbert's Nullstellensatz - Weak Version

Let k be an algebraically closed field. Then M is a maximal ideal of the polynomial ring $k[x_1, ..., x_n]$ if and only if

$$M = (x_1 - a_1, x_2 - a_2, ..., x_n - a_n)$$

for any $a_1, a_2, ..., a_n \in k$. As a result, the maps:

$$\{\text{Points in } \mathbb{A}^n\} \underset{\mathcal{Z}}{\overset{\mathcal{I}}{\rightleftarrows}} \{\text{Maximal ideals in } \Gamma(\mathbb{A}^n)\}$$

are in bijective correspondence, and if $I \subseteq \Gamma(\mathbb{A}^n)$ is a proper ideal, then $\mathcal{Z}(I) \neq \emptyset$

***Proof*:**
Certainly, $(x_1 - a_1, ..., x_n - a_n)$ is a maximal ideal in $k[x_1, x_2, ..., x_n]$ since its quotient is k. Conversely, suppose M is a maximal ideal in $k[x_1, x_2, ..., x_n]$. Consider $F = k[x_1, x_2, ..., x_n]/M$. Then since $k \subseteq F$, F/k is a field extension that is also a finitely generated k-algebra (generated by $\overline{x_1}, ...\overline{x_n}$). By Zariski's lemma, F/k is a finite field extension, and hence is algebraic over k.

Now, we take advantage of the fact that k is algebraically closed, meaning there does not exist a non-trivial algebraic extension of k (lemma 19.5.2), implying that $F \cong_k k$, in particular we $\overline{x_i} \mapsto a_i$ for some $a_i \in k$. Thus, this is a k-algebra homomorphism, and $\overline{a_i} \in F$, $\overline{x_i - a_i} \mapsto a_i - a_i = 0$, which since the to fields are isomorphic, they are in bijective correspondence, implying $\overline{x_i - a_i} = \overline{0}$, i.e. $x_i - a_i \in M$. Thus, we must have $x_i - a_i \in M$ for every x_i, implying that $(x_1 - a_1, ..., x_n - a_n) \subseteq M$, which by maximality shows that $(x_1 - a_1, ..., x_n - a_n) = M$.

As a consequence, if I is any ideal of $k[x_1, x_2, ..., x_n]$, then $I \subseteq M$ for some maximal ideal M. By what we've just shown, for some $a_1, ..., a_n$, $M = (x_1 - a_1, ..., x_n - a_n)$. Thus, $(a_1, ..., a_n) \in \mathcal{Z}(I)$, showing that $\mathcal{Z}(I)$ is non-empty.

Theorem 32.4.4: Hilbert's Nullstellensatz

Let k be an algebraically closed field. Then for any ideal $I \subseteq k[x_1, x_2, ..., x_n]$,

$$\mathcal{I}(\mathcal{Z}(I)) = \sqrt{I} = \text{rad}(I)$$

Furthermore, the maps $\mathcal{I}$ and $\mathcal{Z}$ are order-reversing bijections:

$$\left\{\begin{matrix}\text{Algebraic Sets} \\ (= \text{closed sets})\end{matrix}\right\} \underset{\mathcal{Z}}{\overset{\mathcal{I}}{\rightleftarrows}} \{\text{Radical Ideals}\}$$

Under this bijection the *irreducible* algebraic sets, that is the affine varieties, correspond to the *prime* ideals, and the points to the maximal ideals.

Example 32.4.5: The Nullstellensatz in Practice

1. *Radicals are unavoidable.* In $k[x]$ take $I = (x^2)$. Then $\mathcal{Z}(I) = \{0\}$ and $\mathcal{I}(\{0\}) = (x)$, so
$$\mathcal{I}(\mathcal{Z}(I)) = (x) = \sqrt{(x^2)} \neq (x^2)$$
The theorem recovers $\sqrt{I}$ and nothing finer: passing to zero sets forgets multiplicity, which is the observation chapter 42 has to work around.

2. *A two-variable case.* Let $I = (x^2, xy) \subseteq k[x, y]$. Then $\mathcal{Z}(I) = \mathcal{Z}(x)$, the y-axis, so $\mathcal{I}(\mathcal{Z}(I)) = (x)$. Directly, $\sqrt{I} = (x)$ as well: $x^2 \in I$ gives $x \in \sqrt{I}$, and conversely every element of I is divisible by x.

3. *Algebraic closure is essential.* Over $\mathbb{R}$ take $I = (x^2 + 1) \subseteq \mathbb{R}[x]$. Then $\mathcal{Z}(I) = \emptyset$, so $\mathcal{I}(\mathcal{Z}(I)) = \mathbb{R}[x] = (1)$, while $\sqrt{I} = (x^2 + 1)$ because $(x^2 + 1)$ is maximal. The two differ, and the theorem fails. Over $\mathbb{C}$ the zero set is $\{\pm i\}$ and $\mathcal{I}(\{\pm i\}) = (x^2 + 1) = \sqrt{I}$ as promised.

4. *The bijection at work.* Under theorem 32.4.4 the maximal ideal $(x - a, y - b)$ corresponds to the point (a, b), the prime $(y - x^2)$ to the parabola, and the ideal $(y^2 - x^4) = \big((y - x^2)(y + x^2)\big)$ to the union of two parabolas. That ideal is already radical, being an intersection of the two primes $(y - x^2)$ and $(y + x^2)$, and reducibility of the zero set corresponds to the ideal failing to be prime rather than failing to be radical.

The refinement matters, and mixing the three levels up is the standard way to misread the theorem. An algebraic set is closed and corresponds to a radical ideal; it is a variety exactly when that radical ideal is prime (proposition 32.3.3); and it is a point exactly when the ideal is maximal (theorem 32.4.3). Every prime is radical, so the varieties sit inside the algebraic sets on one side just as the primes sit inside the radicals on the other, and the bijection restricts.

***Proof*:**
First, we've already shown that $\mathrm{rad}(I) \subseteq \mathcal{I}(\mathcal{Z}(I))$ (proposition 32.2.7), so it remains to show that $\mathrm{rad}(I) \supseteq \mathcal{I}(\mathcal{Z}(I))$. By Hilbert's basis theorem, we have that $I = (f_1, f_2, ..., f_n)$. Let $g \in \mathcal{I}(\mathcal{Z}(I))$. If we show that $g^N \in I$ for some $N \in \mathbb{N}_{>0}$, this would imply $g \in \mathrm{rad}(I)$. To that end, we employ the following trick known as the Rabinowitsch trick: introduce a new variable x_{n+1} and consider the ideal:
$$(f_1, ..., f_n, x_{n+1}g - 1) = I' \subseteq k[x_1, ..., x_n, x_{n+1}]$$
By definition of g, at any point where each $f_1, f_2, ..., f_n$ vanish, g also vanishes (since $g \in \mathcal{I}(\mathcal{Z}(I))$). But then, $x_{n+1}g - 1$ is nonzero! Thus, $\mathcal{Z}(I') = \emptyset$ in $\mathbb{A}^{n+1}$, and by Hilbert's weak Nullstellensatz, I' cannot be a proper ideal, and so $1 \in I'$. Since $1 \in I'$, we can write it as:
$$1 = a_1 f_1 + \cdots + a_n f_n + a_{n+1}(x_{n+1}g - 1) \qquad a_i \in k[x_1, ..., x_{n+1}]$$
For the following steps, it is insightful to expand the equation:
$$1 = \sum_i^n a_i(x_1, ..., x_{n+1}) f_1(x_1, ..., x_n) + a_{n+1}(x_1, ..., x_{n+1})(x_{n+1}g(x_1, ..., x_n) - 1)$$
Now, since x_{n+1} is a free variable, this equation still holds true if we substitute x_{n+1} for another expression (it will simply limit the equation to solution which also satisfy the substituted expression). We wil choose $x_{n+1} = 1/g(x_1, ..., x_n)$ so we get the following expression in

$k(x_1, ..., x_n)$:

$$1 = \sum_i^n a_i \left(x_1, ..., x_n, \frac{1}{g(x_1, ...x_n)} \right) f_i(x_1, ..., x_n)$$

Importantly for us, the only terms in the denominator would be some powers of $g(x_1, ..., x_n)$. Thus, for suitably large $N \in \mathbb{N}$, we may re-write the right hand side as:

$$1 = \sum_i^n \frac{b_i\,(x_1, ..., x_n)\, f_i(x_1, ..., x_n)}{g(x_1, ..., x_n)^N}$$

for suitable $b_1, ..., b_n \in k[x_1, ..., x_n]$. Multiplying both sides by the denominator gives

$$g(x_1, ..., x_n)^N = \sum_i^n b_i(x_1, ..., x_n) f_i(x_1, ..., x_n)$$

an identity between polynomials, so it holds in $k[x_1, \dots, x_n]$ and not just in the fraction field. The right hand side lies in I, so $g^N \in I$ and therefore $g \in \mathrm{rad}(I)$. As g was arbitrary, $\mathcal{I}(\mathcal{Z}(I)) \subseteq \mathrm{rad}(I)$, and with the reverse inclusion already in hand, $\mathcal{I}(\mathcal{Z}(I)) = \mathrm{rad}(I)$.

It remains to see that the two maps are mutually inverse bijections between algebraic sets and radical ideals. If I is radical then $\mathcal{I}(\mathcal{Z}(I)) = \mathrm{rad}(I) = I$, which is one composite. For the other, let A be an algebraic set, say $A = \mathcal{Z}(J)$. Then $\mathcal{Z}(\mathcal{I}(A)) = A$ by proposition 31.2.11(2). Finally $\mathcal{I}(A)$ really is radical: if $f^m \in \mathcal{I}(A)$ then $f(a)^m = 0$ for every $a \in A$, and k is a field, so $f(a) = 0$ and $f \in \mathcal{I}(A)$. Hence the maps are inverse to each other, and both reverse inclusion by proposition 31.2.5 and proposition 31.2.10.

For the refinement, $\mathcal{I}(A)$ is prime if and only if A is irreducible (proposition 32.3.3), and maximal if and only if A is a point (theorem 32.4.3), so the bijection restricts to varieties against primes and to points against maximal ideals, as we sought to show.

Corollary 32.4.6: Nullstellensatz in Algebraic Closure

Let k be any field with algebraic closure $\overline{k}$, and $I \subseteq k[x_1, ..., x_n]$ an ideal. Then:

$$\mathcal{I}_k(\mathcal{Z}_{\overline{k}}(I)) = \sqrt{I}$$

As a particular result, $I = (1)$ if and only if the polynomials in I have no common zero in $\overline{k}^n$.

***Proof*:**
Let $I^e = I\overline{k}[x_1, \dots, x_n]$ be the extension of I to the algebraic closure. Since $\overline{k}$ is algebraically closed, theorem 32.4.4 applies there and gives $\mathcal{I}_{\overline{k}}(\mathcal{Z}_{\overline{k}}(I)) = \sqrt{I^e}$. Intersecting both sides with $k[x_1, \dots, x_n]$, the left hand side becomes $\mathcal{I}_k(\mathcal{Z}_{\overline{k}}(I))$ by definition. For the right hand side, $\sqrt{I^e} \cap k[x_1, \dots, x_n] = \sqrt{I}$: the inclusion $\supseteq$ is immediate, and for $\subseteq$, if $f \in k[x_1, \dots, x_n]$ has $f^m \in I^e$ then writing f^m as an $\overline{k}$-combination of generators of I and comparing coefficients against a k-basis of $\overline{k}$ exhibits f^m as a k-combination of them, so $f^m \in I$.

For the last claim, $\mathcal{Z}_{\overline{k}}(I) = \emptyset$ gives $\mathcal{I}_k(\emptyset) = k[x_1, \dots, x_n]$, so $\sqrt{I} = (1)$ and hence $I = (1)$; the converse is immediate, as we sought to show.

32.5 The Dictionary

With these results established, let us recap all the algebraic/geometric correspondences and prove some more. There are some immediate consequence of these result. An immediate easy result is that it can finally be concluded that:

$$\mathcal{I}(A \cap B) = \sqrt{\mathcal{I}(A) + \mathcal{I}(B)}$$

Next, we now have made precise the relation between radical ideals and prime ideals, as we have fully geometrically classified radical and prime ideals. Next, we get the following dictionary between the language of geometry and the language of algebra:

Geometry	Algebra
Affine sets V	coordinate rings $k[V]$
points of V	maximal ideals of $k[V]$
closed subsets of V	radical ideals of $k[V]$
sub-varieties of V	prime ideals of $k[V]$
regular function: $\varphi : V \to W$	k-algebra homomorphism $\tilde{\varphi} : k[W] \to k[V]$

This dictionary is expanded throughout the rest of the book. As a teaser, we shall show that all PID's correspond to smooth curves (see section 50.7).

Now that we have established this definition between algebraic sets and it's algebraic counter-part, we can expand some more on the Zariski Topology:

Proposition 32.5.1: Zariski Topology is Compact

Let $\mathbb{A}^n_k$ have the Zariski topology. Then $\mathbb{A}^n_k$ is compact

***Proof*:**
Let $\mathcal{O}$ index an open cover for $\mathbb{A}^n_k$

$$\mathbb{A}^n_k = \bigcup_{\alpha\ \in \mathcal{O}} U_\alpha$$

By proposition 31.3.4, any open set is the union of sets of the form $D(f)$, so we may re-write this as:

$$\mathbb{A}^n_k = \bigcup_{\alpha \in \mathcal{O}'} D(f_\alpha)$$

appropriately re-indexing. Then taking the complement, we get that it's equivalent that:

$$\emptyset = \bigcap_{\alpha \in \mathcal{O}'} \mathcal{Z}(f_\alpha) = \mathcal{Z}\left(\sum_{\alpha \in \mathcal{O}'} (f_\alpha)\right)$$

By the Nullstellensatz, we get that it's equivalent that

$$1 \in \sum_{\alpha \in \mathcal{O}'} (f_\alpha)$$

Thus, for appropriate coefficients $g_i \in \Gamma(\mathbb{A}^n)$, not necessarily constants, it is equivalent that:

$$1 = g_1 f_1 + \cdots + g_n f_n \in (f_1, ..., f_n) = \sum_{i=1}^{n} (f_i)$$

But then

$$\emptyset = \bigcap_{i=1}^{n} \mathcal{Z}(f_i) \quad \Leftrightarrow \quad \mathbb{A}_k^n = \bigcup_{i=1}^{n} D(f_i)$$

so finitely many of the $D(f_\alpha)$ already cover $\mathbb{A}_k^n$, as we sought to show.

Recall that for any algebraic subset $V \subseteq \mathbb{A}_k^n$, the corresponding coordinate ring $k[V]$ is used to define the subspace topology. Using the Nullstellensatz, we may deduce that it cannot have nilpotent elements:

Corollary 32.5.2: Coordinate Ring Always Reduced

Let $V \subseteq \mathbb{A}_k^n$. be an algebraic subset and $k[V]$ the associated coordinate ring. Then $k[V]$ is has no nilpotent elements

***Proof*:**
Since $k[V] = k[x_1, x_2, ..., x_n]/I(V)$, and $I(V)$ is a radical ideal by the Nullstellensatz, the quotient cannot contain any nilpotent elements, as we sought to show.

Corollary 32.5.3: Nullstellensatz For Subspaces

Let $V \subseteq \mathbb{A}_k^n$ be an algebraic subset (i.e. a closed subset) with corresponding coordinate ring $k[V]$. Then we can define $\mathcal{Z}$ and $\mathcal{I}$ as:

$$\left\{ \begin{matrix} \text{affine algebraic subsets} \\ \text{of } V \end{matrix} \right\} \underset{\mathcal{Z}}{\overset{\mathcal{I}}{\rightleftarrows}} \left\{ \begin{matrix} \text{Radical Ideals} \\ \text{of } k[V] \end{matrix} \right\}$$

***Proof*:**
Write $\pi \colon \Gamma(\mathbb{A}_k^n) \to k[V]$ for the quotient by $\mathcal{I}(V)$. The correspondence theorem for rings makes $J \mapsto \pi^{-1}(J)$ a bijection between the ideals of $k[V]$ and the ideals of $\Gamma(\mathbb{A}_k^n)$ containing $\mathcal{I}(V)$, and it preserves radicals in both directions: $\pi^{-1}(\sqrt{J}) = \sqrt{\pi^{-1}(J)}$, since $f^m \in \pi^{-1}(J)$ says exactly $\pi(f)^m \in J$. So the radical ideals of $k[V]$ correspond to the radical ideals of $\Gamma(\mathbb{A}_k^n)$ containing $\mathcal{I}(V)$.

On the geometric side, the algebraic subsets of V are precisely the algebraic subsets of $\mathbb{A}_k^n$ contained in V, and $\mathcal{Z}$ reverses inclusion, so those correspond to the ideals containing $\mathcal{I}(V)$. Composing with theorem 32.4.4, which matches algebraic subsets of $\mathbb{A}_k^n$ against radical ideals of

$\Gamma(\mathbb{A}^n_k)$, gives the stated bijection, as we sought to show.

Next, sticking to the case where $k = \overline{k}$ is algebraically closed, from this section we can deduce that any coordinate ring is finitely generated, nilpotent-free (i.e. reduced) and is a k-algebra over an algebraically closed field. We can show that all such rings appear as coordinate rings for an affine set.

We first have the following:

Definition 32.5.4: Reduced Ring

Let R be a ring. Then R is said to be *reduced* if $x^n = 0$ implies $x = 0$

(finish here)

Proposition 32.5.5: Coordinate Ring Correspondence

Let R be a ring. Then if R is:

1. Finitely generated
2. Reduced
3. over an algebraically closed field

then R is isomorphic to a coordinate ring for an appropriate dimension n of $\mathbb{A}^n_{\overline{k}}$ and $\mathcal{Z}(S) \subseteq \mathbb{A}^n_{\overline{k}}$.

***Proof*:**
Write $R = \overline{k}[t_1, \ldots, t_n]$ for a choice of finitely many generators, and let

$$\pi\colon \overline{k}[x_1, \ldots, x_n] \to R \qquad x_i \mapsto t_i$$

be the resulting surjection of $\overline{k}$-algebras, with $I = \ker \pi$, so that $R \cong \overline{k}[x_1, \ldots, x_n]/I$. Since R is reduced, I is a radical ideal: if $f^m \in I$ then $\pi(f)^m = 0$ in R, so $\pi(f) = 0$ and $f \in I$.

Set $V = \mathcal{Z}(I) \subseteq \mathbb{A}^n_{\overline{k}}$. By theorem 32.4.4 the ground field being algebraically closed gives $\mathcal{I}(V) = \mathcal{I}(\mathcal{Z}(I)) = \sqrt{I} = I$. Therefore

$$\Gamma(V) = \frac{\overline{k}[x_1, \ldots, x_n]}{\mathcal{I}(V)} = \frac{\overline{k}[x_1, \ldots, x_n]}{I} \cong R$$

so R is the coordinate ring of the algebraic set V, as we sought to show.

Notice that through this result, we can define an affine set without the reference of an ambient space, similar to how we may define a manifold without the need an ambient embedding space. This idea is greatly generalized by scheme theory, which chapter 49 points the reader toward.

Exercise 32.5.1

1. Deduce the strong Nullstellensatz from the weak one by the Rabinowitsch trick: given f vanishing on $\mathcal{Z}(I)$, apply the weak form to the ideal generated by I and $1 - yf$ in one extra variable y.

2. Show that every maximal ideal of $k[x_1, \ldots, x_n]$ is $(x_1 - a_1, \ldots, x_n - a_n)$ for a unique point $(a_1, \ldots, a_n)$, and explain why this fails over $\mathbb{R}$.
3. Show that the Nullstellensatz fails over a field that is not algebraically closed, by computing $\mathcal{I}(\mathcal{Z}(x^2 + 1))$ in $\mathbb{R}[x]$ and comparing with $\sqrt{(x^2 + 1)}$.
4. Show that $\mathcal{Z}$ and $\mathcal{I}$ are mutually inverse inclusion-reversing bijections between the radical ideals of $k[x_1, \ldots, x_n]$ and the algebraic subsets of $\mathbb{A}^n_k$, and that under this correspondence prime ideals match irreducible sets.
5. Use corollary 32.4.6 to show that a system of polynomials over $\mathbb{Q}$ with no common zero in $\overline{\mathbb{Q}}$ generates the unit ideal, and contrast this with $x^2 + 1$ over $\mathbb{R}$, which has no real zero yet generates a proper ideal. What exactly does the closure give?

33

Rational Maps and Birational Classification

Isomorphism is a demanding relation between varieties, and the classification problem it poses is largely intractable. Weakening it produces a relation that can be classified: two varieties are birationally equivalent when they agree on a dense open set. Algebraically this replaces the coordinate ring by its field of fractions, and fields of transcendence degree d are far simpler objects than finitely generated reduced algebras.

The price is that the maps involved are no longer defined everywhere. This chapter pays that price carefully: it locates the domain of definition of a rational function, shows the domain is always dense, and then proves that every variety is birationally equivalent to a hypersurface.

33.1 Rational Functions and the Function Field

Let V be a variety and $k = \overline{k}$ be an algebraically closed field. In this section, we take advantage of the fact that we may localize $k[x_1, ..., x_n]/\mathcal{I}(V)$. By Noether's Normalization Theorem (theorem 32.4.1) $k[V]/k$ is a finite transcendental extension followed by an integral extension. As $k[V]$ is an integral domain we may take its field of fractions $k(V)$, and passing to fraction fields in Noether normalization exhibits $k(V)$ as a *finite* extension of the purely transcendental field $k(y_1, \dots, y_q)$. The extension is genuinely finite and not an equality: for the cuspidal cubic $V = \mathcal{Z}(y^2 - x^3)$ we have $k[V] \cong k[t^2, t^3]$, integral over $k[x]$, while $k(V) = k(t)$ is a degree-two extension of $k(x) = k(t^2)$. Were $k(V)$ always purely transcendental every variety would be rational, which item (4) of example 33.3.8 disproves. What *is* intrinsic is q, the transcendence degree, which chapter 34 identifies with $\dim V$. Geometrically $k(V)$ shall represent all *rational polynomial functions*, or all rational polynomials functions defined on V. Note that we must be a bit more careful in interpreting elements of $k(V)$ as functions: if

$f/g \in k(V)$, then if $g(p) = 0$ for some $p \in V$ then we get the undefined $f(p)/g(p) = f(p)/0$. We shall address this soon.

Rational functions supply a weaker relation than isomorphism for classifying varieties. We shall see that being *birationally equivalent* is weaker than being a (regular) isomorphism, giving coarser isomorphism classes of varieties that shall give us more information about algebraic sets. As an example, elliptic curves[1] shall require three curve invariants to be classified (the genus, the j-invariant, and the Mordell-Weil group of the curve). However, up to birational map it shall only require the genus and we can without loss of generality work with smooth elliptic curves. Though coarser, it is nonetheless a powerful tool in gathering information about varieties.

Definition 33.1.1: Rational Functions

Let V be an affine variety. Then the fraction field of $k[V]$ is called the field of *rational function on* V, and is denoted $k(V)$. An element of $k(V)$ is called a *rational function*.

The *evaluation* of a rational function at p is defined whenever it admits a representative f/g with $g(p) \neq 0$. Two rational functions are equal if they agree at every point where both are defined. The points at which *no* representative has nonvanishing denominator form the *pole set*; being a pole is a property of the function, not of a chosen representative.

If $\varphi \in k(V)$, then $\varphi = \frac{f}{g}$ where $g \notin I(V)$ (or else the denominator would be the zero function, contradicting the construction of the field of fractions). We shall say that two rational polynomial f/g and f'/g' represent the same rational function if and only if $fg' - f'g \in I(V)$. Note that if $k[V]$ is not a UFD, then there can be representatives of $k(V)$ which are not equivalent up to a unit:

Example 33.1.2: Not UFD, Different Polynomial Representatives

Let $V = \mathcal{Z}(xw - yz) \subseteq \mathbb{A}^4_k$. Show that:

$$\frac{\overline{x}}{\overline{y}} = \frac{\overline{z}}{\overline{w}} = f \in k(V)$$

if $y \neq 0$ or $w \neq 0$.

Another way of constructing $k(V)$ is the following: take the collection

$$\mathcal{O}_V = \{\varphi = f/g \in k(x_1, ..., x_n) \mid g \notin I(V)\}$$

then take the subset $\mathfrak{m}_V = \{\varphi = f/g \in \mathcal{O}_V \mid f \in I(V)\}$. This is the unique maximal ideal of the local ring $\mathcal{O}_V$, and

$$k(V) \cong \mathcal{O}_V/\mathfrak{m}_V$$

since a rational function on V is a ratio f/g of polynomials with $g \notin I(V)$, taken up to the relation that identifies two such ratios agreeing on V, and that relation is exactly congruence modulo $\mathfrak{m}_V$. This shall become the dominant way of interpreting the set of rational functions when working with quasi-projective varieties (see section 39.4)[2].

[1] for those familiar with complex geometry, genus 1 curves

[2] In Scheme theory, this shall be the rational functions at the point (closed or non-closed) associated to X

33.2 Domains of Definition and Local Rings

A key difference between regular and rational functions is that a rational function need not be defined at every point. In fact, it *has* to be the case that a rational function is not defined on a finite set of points for it to not be equivalent to a regular function (when working over an algebraically closed field). To see this, consider the following definition:

Definition 33.2.1: Regular Point

Let $\varphi \in k(X)$ be a rational function. Then a *regular point* at $x \in X$ is a point where we can represent φ as f/g, $\varphi(x) = f(x)/g(x)$, and $g(x) \neq 0$. In this case, the element $f(x)/g(x)$ in k is called the *value* of φ at x and is denoted $\varphi(x)$

For example, take $\varphi = 1/x$ on $\mathbb{A}^1_k$. Then all points except 0 are regular points. If $\varphi = f/g$ is regular at a point x, then it is regular in an open neighborhood of X, namely the neighborhood $D(g) \cap X$. Some authors put that a rational function is regular at a point if it is regular in an open neighborhood.

Proposition 33.2.2: Regular at All Points, then Regular

Let $\varphi \in k(X)$ be a rational function that is regular at all points. Then φ is a regular function

***Proof*:**
Let $\varphi \in k(X)$ be a rational function that is regular at every point. Then for each $x \in X$, we can represent φ as f_x/g_x. Take $\mathfrak{a} = (g_x)_{x\in X}$. Then as $k[X]$ is Noetherian, this is a finitely generated ideal:

$$\mathfrak{a} = (g_{x_1}, \dots, g_{x_n})$$

Note that the g_{x_i} have no common zero in X: at any $x \in X$ the chosen g_x satisfies $g_x(x) \neq 0$, and $g_x \in \mathfrak{a}$. So $\mathcal{Z}(\mathfrak{a}) = \emptyset$ in X, and the weak Nullstellensatz (theorem 32.4.3) applied in $k[X]$ forces

$$\mathfrak{a} = (1)$$

In particular, there exist $u_i \in k[X]$ such that

$$\sum_{i=1}^{n} u_i g_i = 1$$

The key point is that $\varphi = f_i/g_i$ is an identity in the field $k(X)$, so $\varphi g_i = f_i$ holds globally, not just on the neighbourhood where g_i is nonzero. Multiplying the displayed identity by φ therefore gives

$$\varphi = \varphi \cdot 1 = \sum_{i=1}^{n} u_i(\varphi g_i) = \sum_{i=1}^{n} u_i f_i$$

whose right hand side lies in $k[X]$, so φ is regular, completing the proof.

The Nullstellensatz is essential for the proof, i.e. that k is an algebraically closed field, or else this result is false!

Example 33.2.3: Rational Function That is Regular

the function
$$\frac{x}{x+100}$$
is regular on $\mathcal{Z}(x^2+y^2-1) \subseteq \mathbb{A}^2_{\mathbb{R}}$ even though it is not a polynomial: there does not exist a point (x, y) satisfying x^2+y^2-1 such that $x+100=0$ namely since all points x where $x^2+y^2-1=0$ must be within $[-1, 1]$[a]

[a] This failure is *not* present when working with the scheme $\operatorname{Spec} \mathbb{R}[x,y]/(x^2+y^2-1)$ as $x+100$ has a zero at the prime ideal $(x-100, y^2+9999)$, and so this "anomaly" is present if we don't work in the right context

Hence we always have points on which a rational function is not defined. This set of points will always be dense:

Lemma 33.2.4: Domain of Definition

Let $\varphi \in k(V)$ be a rational function. Then set of regular points is non-empty, open and dense.

***Proof*:**
Let $\varphi = f/g$ and take $U = \{p \in V \mid g(p) \neq 0\} = D(g) \cap V$. Then in V, this set is certainly open, and as g is not the zero function there is certainly at least one point $p \in V$ such that $g(p) \neq 0$.

To show U is dense, let $O \subseteq V$ be any nonempty open set; it suffices to show $U \cap O \neq \emptyset$. Suppose instead $U \cap O = \emptyset$. Then
$$V = (U \cap O)^c = U^c \cup O^c$$
writes V as a union of two closed sets, and both are *proper*: $U^c \neq V$ because $U \neq \emptyset$, and $O^c \neq V$ because $O \neq \emptyset$. That contradicts the irreducibility of V. Hence U meets every nonempty open set, which is density, completing the proof.

Definition 33.2.5: Domain of Definition

Let $\varphi \in k(V)$ a rational function. Then the set of regular point is called the *domain of definition*.

Corollary 33.2.6: Collection of Rational Points

Let $\varphi_1, ..., \varphi_n \in k(X)$ be rational functions on a variety X. Then their common domain of definition is open, non-empty, and dense.

***Proof*:**
Let U_i be the domain of definition of φ_i, which by lemma 33.2.4 is open, nonempty and dense. A finite intersection of open sets is open, so $\bigcap_i U_i$ is open.

For nonemptiness and density it is enough to show that $\bigcap_i U_i$ meets every nonempty open $O \subseteq X$, and by induction it suffices to treat $n = 2$. Given nonempty open O, density of U_1 gives $U_1 \cap O \neq \emptyset$, and $U_1 \cap O$ is open, so density of U_2 gives $U_2 \cap (U_1 \cap O) \neq \emptyset$. Hence $U_1 \cap U_2$ meets O. Taking $O = X$ gives nonemptiness, as we sought to show.

In fact, as the open sets in the Zariski topology are "huge", it suffices to define a rational mapping on any open subset:

Proposition 33.2.7: Suffice To Define On Open Set

Let $\varphi \in k(X)$ be a rational function. Then if φ is specified on some non-empty open subset $U \subseteq X$, then there is a unique extension of φ to all of X.

***Proof*:**
Take $\varphi \in k(X)$. If $\varphi = 0$, there is nothing to prove, so assume $\varphi \neq 0$. Say we specify what φ is on an open non-empty subset $U \subseteq X$, call this function $\varphi|_U$. Then let's say that φ' is a rational function where φ' on U is equal to $\varphi|_U$ (where a-priori it is unknown where there is a unique extension, and hence it may be another function). Consider $\varphi - \varphi' = \varphi''$. On U, this function is identically 0. Take the representative $\varphi'' = f/g$ on U so that $f = 0$. Then take $X_1 \subseteq X$ given by $f = 0$. This is not all of X as $\varphi \neq 0$, however we then have $X_1 \cup (X \setminus U) = X$ where both are proper subsets, contradicting irreducibility of X, completing the proof.

We are often interested in all the functions that are defined at a particular point, namely since geometrically much of the information of an object is given by collections of functions at every point (the reader familiar with differential geometry should be thinking of germs). To that end, we box the following

Definition 33.2.8: Stalk At A Point

Let V be an affine variety and $p \in V$. Then $\mathcal{O}_p(V) \subseteq k(V)$ is the collection of rational functions:

$$\mathcal{O}_p(V) = \{f/g \in k(V) \mid g(p) \neq 0\}$$

Proposition 33.2.9: Regular Functions and Stalk

$$k[V] = \bigcap_{p \in V} \mathcal{O}_p(V)$$

***Proof*:**
Certainly $k[V] \subseteq \mathcal{O}_p(V)$ for all $p \in V$ so $k[V] \subseteq \bigcap_{p \in V} \mathcal{O}_p(V)$.

For the converse, fix $\varphi \in \bigcap_{p \in V} \mathcal{O}_p(V)$ and consider its *ideal of denominators*

$$\mathfrak{d}_\varphi = \{h \in k[V] \mid h\varphi \in k[V]\}$$

This is an ideal: it is closed under addition, and if $h\varphi \in k[V]$ then $(ah)\varphi = a(h\varphi) \in k[V]$ for any $a \in k[V]$. Its zero set is exactly the pole set of φ. Indeed, if $p \notin \mathcal{Z}(\mathfrak{d}_\varphi)$ then some $h \in \mathfrak{d}_\varphi$ has $h(p) \neq 0$, and writing $\varphi = (h\varphi)/h$ exhibits a representative with nonvanishing denominator at p, so φ is regular at p; conversely a representative u/v with $v(p) \neq 0$ puts v in $\mathfrak{d}_\varphi$ with $v(p) \neq 0$, so $p \notin \mathcal{Z}(\mathfrak{d}_\varphi)$. In particular the pole set is closed.

Now φ lies in every $\mathcal{O}_p(V)$, so it is regular at every point and its pole set is empty, that is $\mathcal{Z}(\mathfrak{d}_\varphi) = \emptyset$. Lifting $\mathfrak{d}_\varphi$ to $k[x_1, \ldots, x_n]$ and applying the Nullstellensatz (theorem 32.4.3) gives

$\mathfrak{d}_\varphi = k[V]$, so $1 \in \mathfrak{d}_\varphi$ and therefore
$$\varphi = 1 \cdot \varphi \in k[V]$$
as we sought to show.

Proposition 33.2.10: Stalks Are Local Noetherian Domains

let V be a variety and $p \in V$. Then $\mathcal{O}_p(V)$ is a local Noetherian domain

***Proof*:**
Domain. $\mathcal{O}_p(V)$ is by construction a subring of the field $k(V)$, and a subring of a field has no zero divisors.

Local. Put
$$\mathfrak{m}_p = \{f/g \in \mathcal{O}_p(V) \mid f(p) = 0\}$$
This is the kernel of the evaluation map $\mathrm{ev}_p \colon \mathcal{O}_p(V) \to k$, $f/g \mapsto f(p)/g(p)$, which is well defined precisely because $g(p) \neq 0$ and is a surjective ring homomorphism; so $\mathfrak{m}_p$ is an ideal with $\mathcal{O}_p(V)/\mathfrak{m}_p \cong k$, hence maximal. Every element outside $\mathfrak{m}_p$ is a unit: if $f/g \in \mathcal{O}_p(V)$ has $f(p) \neq 0$ then g/f again has nonvanishing denominator at p, so it lies in $\mathcal{O}_p(V)$ and inverts f/g. A proper ideal contains no unit, so it is contained in $\mathfrak{m}_p$; thus $\mathfrak{m}_p$ is the unique maximal ideal and $\mathcal{O}_p(V)$ is local.

Noetherian. $\mathcal{O}_p(V)$ is the localization of $k[V]$ at the prime $\mathcal{I}(p)$, and $k[V]$ is Noetherian as a quotient of the Noetherian ring $\Gamma(\mathbb{A}^n)$. A localization of a Noetherian ring is Noetherian, since every ideal of $S^{-1}R$ is the extension of its contraction to R, and extension carries a finite generating set to a finite generating set. Hence $\mathcal{O}_p(V)$ is Noetherian, completing the proof.

In section 35.1, we shall see that when V is a curve and $\mathcal{O}_p(V)$ is a *discrete valuation ring*, then V is *smooth* at p. This requires a lot of build-up, and so we shall leave stalks alone for now.

33.3 Rational and Birational Maps

Let us now generalize regular maps:

Definition 33.3.1: Rational Map

Let X be a variety and $Y \subseteq \mathbb{A}^m_k$ a variety. A *rational mapping* $\varphi \colon X \dashrightarrow Y$ is a tuple
$$\varphi = (\varphi_1, ..., \varphi_m) \qquad \varphi_i \in k(X)$$
of rational functions such that $(\varphi_1(x), \dots, \varphi_m(x)) \in Y$ for every x in the common domain of definition of the φ_i, which by corollary 33.2.6 is a dense open subset of X. A rational mapping all of whose components happen to be regular is exactly a regular map.

Example 33.3.2: Rational Maps

1. *The circle, parameterized.* Let $C = \mathcal{Z}(x^2 + y^2 - 1) \subseteq \mathbb{A}_k^2$ and consider
$$\varphi\colon C \dashrightarrow \mathbb{A}_k^1 \qquad \varphi(x,y) = \frac{y}{1-x}$$
This is a rational map, its single component lying in $k(C)$. It is undefined where $1-x$ vanishes on C, that is at the point $(1,0)$, and defined on the dense open complement. Its inverse is the regular map
$$t \mapsto \left(\frac{t^2-1}{t^2+1}, \frac{2t}{t^2+1}\right)$$
so C and $\mathbb{A}_k^1$ are birational, though not isomorphic: the map misses $(1,0)$ in one direction and the two points with $t^2 = -1$ in the other.

2. *A rational map with no regular representative.* On $\mathbb{A}_k^2$ the assignment $(x,y) \mapsto x/y$ is a rational map $\mathbb{A}_k^2 \dashrightarrow \mathbb{A}_k^1$ whose domain of definition is $D(y)$. It cannot be extended to the origin: along the line $y = cx$ it is constantly $1/c$, so every value of k is taken arbitrarily close to the origin and no single value can be assigned there.

3. *Regular maps are the special case.* The map $(x,y) \mapsto (x^2, xy)$ has polynomial components, so its domain of definition is all of $\mathbb{A}_k^2$ and it is regular. Its image, computed in example 41.3.5, is not closed, which is a separate matter from being defined everywhere.

The image of the rational map shall be denoted $\varphi(X)$, even though not every point may be regular. Note that a rational map shall necessarily map an open subset $U \subseteq X$ into Y, and hence it is well-defined by proposition 33.2.7. By proposition 33.2.7, an equivalent definition of a rational map is that it is the collection of equivalence relations $[(\varphi, U)]$ with $U \subseteq X$ open and $\varphi : U \to Y$ where $[(\varphi, U)] \sim [(\psi, V)]$ if and only if
$$\varphi|_{U \cap V} = \psi|_{U \cap V}$$
This equivalent to definition 33.3.1, and is more prominent in the context of algebraic geometry. Its upside is that each representative is a well-defined map and hence can be seen as the appropriate quotient of some hom-sets, while its downside is that a rational mapping is now an equivalence relationship.

The following is the equivalent of being surjective:

Definition 33.3.3: Dominating Rational Map

Let $f : X \dashrightarrow Y$ be a rational map. Then f is said to be *dominating* if $f(U)$ is dense in Y.

Recall that if A, B are local rings where $A \subseteq B$ is a subring, then B dominates A if $\mathfrak{m}_A \subseteq \mathfrak{m}_B$. This vocabulary shall be made clear from the following

Proposition 33.3.4: Properties of Rational Maps

Let $F : X \dashrightarrow Y$ be a dominating rational map, $U \subseteq X$ and $V \subseteq Y$ affine open subsets where $f : U \to V$ represents F. Then:

1. If $\varphi : X \dashrightarrow Y$ and $\psi : Y \dashrightarrow Z$ are dominating rational maps, then $\psi \circ \varphi$ is dominating and $(\psi \circ \varphi)^{\#} = \varphi^{\#} \circ \psi^{\#}$.
2. $f^{\#} : k[V] \to k[U]$ is injective, hence extends to an injective map $k(V) \to k(U)$, and so gives $F^{\#} : k(Y) \hookrightarrow k(X)$, i.e. a field extension $k(X)/k(Y)$. It is an *embedding*, not an isomorphism; $F^{\#}$ is an isomorphism exactly when F is birational (proposition 33.3.6).
3. If p is in the domain of F and $F(p) = q$, then $\mathcal{O}_p(X)$ dominates $F^{\#}(\mathcal{O}_q(Y))$. Conversely, if $p \in X$ and $q \in Y$ are such that $\mathcal{O}_p(X)$ dominates $F^{\#}(\mathcal{O}_q(Y))$, then p belongs to the domain of F and $F(p) = q$.
4. Any k-homomorphism $k(Y) \to k(X)$ induces a unique dominating rational map $X \dashrightarrow Y$.

***Proof*:**

1. Let $U' \subseteq X$ be a dense open set on which φ is defined. Then $\psi\circ\varphi$ is defined on $U' \cap \varphi^{-1}(V')$ for V' a dense open set on which ψ is defined, and this is nonempty because $\varphi(U')$ is dense in Y and so meets V'. Its image contains $\psi(\varphi(U') \cap V')$, which is dense in Z since ψ is dominating and $\varphi(U') \cap V'$ is a nonempty open subset of V'. So $\psi \circ \varphi$ is dominating. Functoriality is then the computation $(\psi \circ \varphi)^{\#}(h) = h \circ \psi \circ \varphi = \varphi^{\#}(\psi^{\#}(h))$, so $(\psi \circ \varphi)^{\#} = \varphi^{\#} \circ \psi^{\#}$: pullback reverses composition.
2. Recall $f^{\#}(\varphi) = \varphi \circ f$. Suppose $f^{\#}(\varphi) = f^{\#}(\psi)$, that is $\varphi \circ f = \psi \circ f$. Then φ and ψ agree on $f(U)$, which is dense in V because F is dominating; two regular functions agreeing on a dense subset of a variety are equal (their difference vanishes on a dense set, hence on its closure). So $\varphi = \psi$ and $f^{\#}$ is injective. An injective map of integral domains extends to their fraction fields, giving $k(V) \hookrightarrow k(U)$, and since $U \subseteq X$ and $V \subseteq Y$ are nonempty open subsets of varieties we have $k(U) = k(X)$ and $k(V) = k(Y)$ by proposition 33.2.7. Hence $F^{\#}\colon k(Y) \hookrightarrow k(X)$ is a field embedding. It need not be surjective: $F^{\#}$ is onto exactly when F admits a rational inverse, which is proposition 33.3.6.
3. If $f(p) = q$, then it is a matter of pushing around the definition to see that $\mathcal{O}_p(X)$ dominates $F^{\#}(\mathcal{O}_q(Y))$. Conversely, suppose $\mathcal{O}_p(X)$ dominates $F^{\#}(\mathcal{O}_q(Y))$. Take an affine neighbourhood V of p and W of q where
$$k[W] = k[y_1, ..., y_n]$$
and $F^{\#}(y_i) = \frac{a_i}{b_i}$ for $a_i, b_i \in k[V]$ with $b_i(p) \neq 0$. Take $b = b_1 \cdots b_n$, and take the localization V_b. Then
$$F^{\#}(k[W]) \subseteq k[V_b]$$
Hence the map $F^{\#} : k[W] \to k[V_b]$ induces a unique morphism $f : V_b \to W$. Then if $g \in k[W]$ vanishes at q, $F^{\#}(g)$ vanishes at p, but then it must be that $f(p) = q$
4. If $\varphi : k(Y) \to k(X)$ wit h$\varphi(k[Y]) \subseteq k[X_b]$, for some $b \in k[X]$, φ induces a morphism $f : X_b \to Y$. Hence $f(X_b)$ is dense in Y as $f^{\#}$ is injective, giving us the desired result.

Definition 33.3.5: Birational Map

Let $\varphi : X \dashrightarrow Y$ be a rational map. Then φ is said to be *birational* if the image of φ is dense and there exists a rational map $\psi : Y \dashrightarrow X$ with dense image such that $\varphi \circ \psi = \psi \circ \varphi = \mathrm{id}$.

Proposition 33.3.6: Birational Maps and Isomorphisms

Let $\varphi\colon X \dashrightarrow Y$ be a dominating rational map of varieties. Then φ is birational if and only if $\varphi^{\#}\colon k(Y) \to k(X)$ is an isomorphism of fields over k.

***Proof*:**
Suppose φ is birational, with rational inverse ψ. Both composites are the identity, so applying proposition 33.3.4(1) gives $\varphi^{\#} \circ \psi^{\#} = (\psi \circ \varphi)^{\#} = \mathrm{id}$ and likewise $\psi^{\#} \circ \varphi^{\#} = \mathrm{id}$. Hence $\varphi^{\#}$ is an isomorphism with inverse $\psi^{\#}$.

Conversely, suppose $\varphi^{\#}$ is an isomorphism. Then $(\varphi^{\#})^{-1}\colon k(X) \to k(Y)$ is a k-homomorphism of function fields, so by proposition 33.3.4(4) it induces a dominating rational map $\psi\colon Y \dashrightarrow X$ with $\psi^{\#} = (\varphi^{\#})^{-1}$. Then $(\psi \circ \varphi)^{\#} = \varphi^{\#} \circ \psi^{\#} = \mathrm{id}$, and the uniqueness clause of proposition 33.3.4(4) forces $\psi \circ \varphi = \mathrm{id}$ as rational maps; symmetrically $\varphi \circ \psi = \mathrm{id}$. So φ is birational, as we sought to show.

The next definition uses dimension, which chapter 34 develops formally; for now read $\dim V$ as the number of independent parameters a general point of V carries.

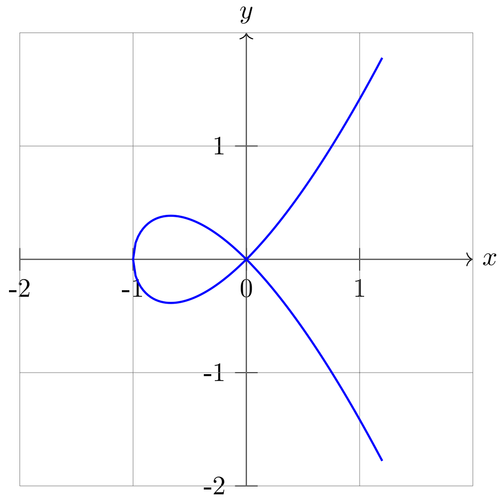
Definition 33.3.7: Rational Closed Set

Let V be a variety. Then if V is birational to $\mathbb{A}^n_k$ or $\mathbb{P}^n_k$, then V is called *affine rational* or *projective rational* respectively.

If V satisfies one of the above two conditions and has dimension $1, 2$ or codimension 1, then it is a *rational curve*, *rational surface*, or *rational hypersurface* respectively.

Example 33.3.8: Rational Mapping and Birational Maps

1. Naturally, all curves of order 1 (i.e., straight lines) are tautologically rational. More generally any linear subspace is rational.
2. Consider the curve given by
$$y^2 = x^2 + x^3$$
which looks like:



Label this curve C. Then this is a rational curve. To see this, consider that the line given by $y = tx$ passing through the origin intersects the curve at a single point (not counting the origin). Then substituting $y = tx$ into our equation, we get:

$$x^2(t^2 - x - 1) = 0$$

The root $x = 0$ corresponds to the origin, and the other roots $x = t^2 - 1$ corresponds to non-trivial solution. Hence, we get $x = t^2 - 1, y = t(t^2 - 1)$, and the rational map:

$$\varphi(t) = (t^2 - 1, t(t^2 - 1))$$

The inverse recovers the slope from the point: given $p = (x_0, y_0) \in C$ other than the origin, the line through p and the origin has slope y_0/x_0, so

$$\psi\colon C \dashrightarrow \mathbb{A}^1_k \qquad \psi(x, y) = \frac{y}{x}$$

is a rational map, regular away from $x = 0$. It inverts φ: substituting gives $\psi(\varphi(t)) = t(t^2 - 1)/(t^2 - 1) = t$ whenever $t^2 \neq 1$, and the two composites agree with the identity on dense open sets, which is equality of rational maps. So φ is birational. It is not an isomorphism: φ sends both $t = 1$ and $t = -1$ to the origin, the node of C, so it is not even injective.

If we take $k = \mathbb{Q}$, then this gives all rational solutions to the equation $y^2 = x^3 + x^2$. The case is a little more subtle if we were looking for integer solutions to the equation, which is explored in chapter 50.

3. Let us show that all irreducible curves C of order 2 are rational (note that the above example is an order 3 curve, the maximal degree being 3). Let C be given by the equation $f(x, y)$. Take any point $(x_0, y_0) \in C$, and equation of the line through (x_0, y_0) with slope t

$$y - y_0 = t(x - x_0)$$

Label this line L_t. Let us find the points of intersection of the curve C with line L_t. To do so, we substitute:

$$f(x, y_0 + t(x - x_0)) = 0$$

which is equivalent to the mapping $\mathbb{A}^1_k$ to the curve. This mapping is well-defined (notice it is still of degree 2). Then one of the roots is certainly x_0. Then let a denote the coefficient of x after dividing the equation $f(x, y_0 + t(x - x_0))$ by the coefficient of x^2. Then we can see that the remaining roots must be:

$$x + x_0 = -a \qquad \Rightarrow \qquad x = -x_0 - a$$

Then x is a rational function of t, and substituting the expression for x into L_t, we obtain an expression for y as a rational function of t.

Note that if the coordinates x_0, y_0 of $(x_0, y_0) \in C$ and the coefficients of $f(x, y)$ belong to some subfield $k_0 \subseteq k$, then so do the coefficients of the function that gives the parameterization. In this case, we can find the general form of the solution in rational numbers of a Diophantine equation of degree 2 by knowing *at least* one solution! However, it turns out that knowing there is at least one solution can sometimes be tricky[a]

4. More generally, an irreducible hypersurface $X \subseteq \mathbb{A}^n_k$ given by a polynomial of degree 2 $f(x_1, ..., x_n) = 0$ is rational. The proof generalizes the above and is left as an exercise. The geometric picture is projection the hypersurface to a hyper-plane via a projection that does not pass through some point x. The point x can be any point that is not a "vertex" on X, namely that for at least one variable:

$$\frac{\partial f}{\partial x_i}(x) \neq 0$$

5. The curve order 3 given by $x^3 + y^3 = 1$ is *not* rational. More generally, $x^n + y^n = 1$ is not rational for $n \geq 3$, provided $\mathrm{char} k \nmid n$. To see this, let C_n be the curve given by $x^n + y^n = 1$ and suppose it is rational so that there exists rational functions φ, ψ such that $x = \varphi(t), y = \psi(t)$. Then we may find polynomials p, q, r relatively prime such that

$$\varphi(t) = \frac{p(t)}{r(t)} \qquad \psi(t) = \frac{q(t)}{r(t)}$$

where the denominator is the same for simplicity (the same argument follows with more steps by taking p/r and q/s). Then plugging into our equation we get:

$$p(t)^n + q(t)^n - r(t)^n = 0$$

for all $t \in k$. Differentiating is legitimate and does not annihilate the exponent, since $\mathrm{char} k \nmid n$, and so we have:

$$p(t)^{n-1}p'(t) + q(t)^{n-1}q'(t) - r(t)^{n-1}r'(t) = 0$$

We now have two systems equations. Treating $p^{n-1}, q^{n-1}, -r^{n-1}$ as the unknowns, we can consider the above two equations as being a system of linear equations which has has the matrix:

$$\begin{pmatrix} p & q & r \\ p' & q' & r' \end{pmatrix}$$

Solving this system of linear equations, we get that $p^{n-1}, q^{n-1}, -r^{n-1}$ are proportional to

$$qr' - rq' \qquad rp' - pr' \qquad pq' - qp'$$

As p, q, r are relatively prime, we get that:

$$p^{n-1} \big| (qr' - rq') \qquad q^{n-1} \big| (rp' - pr') \qquad r^{n-1} \big| (pq' - qp')$$

Finally, if a, b, c are the degrees of p, q, r . If $a \geq b \geq c$, then

$$(n-1)a \leq b + c - 1$$

But $a \geq b \geq c$ gives $b + c - 1 \leq 2a - 1$, so $(n-1)a \leq 2a - 1$, forcing $(n-3)a \leq -1$, which is impossible for $n \geq 3$ and $a \geq 0$. The other orderings of a, b, c give the same contradiction by symmetry.

[a] see Legendre theorem in cite:HERE

33.4 Classification up to Birational Equivalence

Birational mappings are weaker than isomorphisms (of varieties). This problem can be translated algebraically to the difference between having $k[X]$ and $k[Y]$ be isomorphic and $k(X)$ and $k(Y)$ be isomorphic. The reader may recall from section 32.4 that $k[X]$ and $k[Y]$ must be finitely generated reduced k-algebras, $k(X)$ and $k(Y)$ finite field extensions. As the structure of field extensions is much simpler, we expect birational mappings to be much simpler. In fact, we may even classify all affine variety up to birational isomorphism:

Theorem 33.4.1: Classification of Varieties Up To Birational Map

Let X be a variety of dimension d over an algebraically closed field k. Then X is birationally isomorphic to a hypersurface in $\mathbb{A}_k^{d+1}$.

Thus, birationality of varieties reduces to the classification of hypersurfaces. Note that being birationally isomorphic to a hypersurface does not imply a curve or plane is rational. For example, the curve given by $x^3 + y^3 = 1$ is already a hypersurface in $\mathbb{A}^2$ as it is given by $\mathcal{Z}(x^3 + y^3 - 1)$, however it is not a rational curve. Its projective closure is the Fermat cubic $\mathcal{Z}(x^3 + y^3 - z^3)$, which is nonsingular, and corollary 44.4.3 shows no nonsingular plane cubic is birational to $\mathbb{A}_k^1$; the invariant responsible is the genus, which chapter 44 constructs.

That claim deserves more than a forward reference, because the curve looks as though it ought to yield. Every technique available at this point can be tried on it, and watching each one fail is the most economical argument for building the machinery of part IX.

Example 33.4.2: Why $x^3 + y^3 = 1$ Resists Parameterization

Assume $\operatorname{char} k \neq 3$, so that the projective closure $\mathcal{Z}(x^3 + y^3 - z^3)$ is nonsingular; the excluded case is degenerate and treated at the end. Write $C = \mathcal{Z}(x^3 + y^3 - 1) \subseteq \mathbb{A}_k^2$.

Attempt 1: the chord construction. A conic is parameterized by fixing a point on it and sweeping the lines through that point, as in example 39.4.4(3). Run the identical computation on three curves, substituting the line through the chosen point and factoring out the known root.

- The conic $x^2 + y^2 = 1$ at $(1,0)$, with $y = t(x-1)$:

$$(x-1)\left[\, x + 1 + t^2(x-1) \,\right] = 0$$

  leaving a factor *linear* in x.

- The nodal cubic $y^2 = x^2 + x^3$ at the node $(0,0)$, with $y = tx$:

$$x^2\left[\, t^2 - 1 - x \,\right] = 0$$

  again leaving a factor *linear* in x.

- The Fermat cubic $x^3 + y^3 = 1$ at $(1,0)$, with $y = t(x-1)$:

$$(x-1)\left[\, x^2 + x + 1 + t^3(x-1)^2 \,\right] = 0$$

  leaving a factor *quadratic* in x.

In the first two cases the leftover factor is linear, so x is a rational function of t and the construction succeeds: the conic gives $x = (t^2-1)/(t^2+1)$, and the nodal cubic gives $x = t^2 - 1$, recovering the parameterization of example 33.3.8.

In the third it is quadratic, and solving it introduces a square root. The reason is visible in the factorizations: the method needs the marked point to absorb all but one of the intersections. A line meets a conic twice and the marked point takes one; a line through the node of $y^2 = x^2 + x^3$ meets the cubic three times and the node, being a double point, takes two. A smooth cubic has no double point, so the marked point takes only one of three and two are left over. *The construction does not fail by accident; it fails because the curve is smooth*, which is exactly the hypothesis under which the conclusion is true.

A warning about what this does and does not show. Given *two* points of a cubic the chord still produces a third, and over $\mathbb{Q}$ that is the classical method for generating new solutions from old. That is a statement about individual points, not about parameterizing the whole curve by one variable, and the two are easy to conflate.

Attempt 2: symmetric functions. The equation is symmetric in x and y, so set $u = x + y$ and $v = xy$. Then $x^3 + y^3 = u^3 - 3uv$, and the curve becomes

$$u^3 - 3uv = 1 \qquad \text{that is} \qquad v = \frac{u^3 - 1}{3u}$$

which parameterizes the pair $\{x, y\}$ rationally by u alone. But recovering x and y individually means solving $T^2 - uT + v = 0$, whose discriminant is

$$u^2 - 4v = u^2 - \frac{4(u^3-1)}{3u} = \frac{4 - u^3}{3u}$$

So a square root of $(4 - u^3)/3u$ is needed. Setting $W = 3uw$ where w is that square root gives

$$W^2 = 3u\,(4 - u^3) = 12u - 3u^4$$

a square root of a quartic with four distinct roots, namely $u = 0$ and the three cube roots of 4. That is another curve of the same kind: section 44.4 will assign it genus 1, exactly as for C. The attempt is circular, and it is circular in a way one can see before having any theory: it trades the problem for an equal one.

Attempt 3: look for a parameterization directly. Suppose C were rational, so that $x = p(t)/r(t)$ and $y = q(t)/r(t)$ for polynomials p, q, r with no common factor, not all constant. Substituting and clearing denominators,

$$p(t)^3 + q(t)^3 = r(t)^3$$

This is Fermat's Last Theorem for polynomials in the exponent 3, and unlike its arithmetic namesake it has a short proof. The Mason–Stothers theorem states that for coprime polynomials $a + b = c$ in $k[t]$, not all constant and with $\operatorname{char} k = 0$,

$$\max(\deg a, \deg b, \deg c) \;\leq\; n_0(abc) - 1$$

where n_0 counts *distinct* roots. Apply it to $a = p^3$, $b = q^3$, $c = r^3$ and set $d = \max(\deg p, \deg q, \deg r)$. The left side is $3d$. On the right, $p^3q^3r^3$ has the same distinct roots as pqr, so $n_0 \leq \deg(pqr) \leq 3d$, giving

$$3d \;\leq\; 3d - 1$$

which is absurd. So no such p, q, r exist and C is not rational.

What the third attempt costs. It is a complete proof, but only in characteristic zero: the Mason–Stothers argument differentiates, and in characteristic $\rho > 0$ all three of p, q, r may be polynomials in t^ρ, with vanishing derivatives, and the argument collapses. And in the excluded characteristic 3 the curve is not what it appears: $x^3 + y^3 = (x + y)^3$, so $C = \mathcal{Z}((x + y)^3 - 1)$ is a union of lines and is rational after all.

So one attempt fails structurally, one is circular, and one succeeds in a single characteristic by an argument owing nothing to geometry. What is missing is an invariant of the curve that decides the question uniformly, is computable, and explains why the conic and the nodal cubic behave differently from this one. That invariant is the genus. Theorem 44.4.2 shows a plane curve is rational exactly when its genus vanishes, definition 44.4.1 computes the genus of a smooth plane curve of degree d as $\binom{d-1}{2}$, giving 1 here and 0 for a conic, and section 44.4 shows a node drops the genus by exactly 1, which is why the nodal cubic is rational and this one is not. The argument is characteristic-free.

***Proof*:**
Write $K = k(X)$, the fraction field of $k[X] = \Gamma(\mathbb{A}^n)/\mathcal{I}(X)$. It is generated over k by the images $t_1, \ldots, t_n$ of the coordinate functions, a *finite* set, so from it we may extract a maximal algebraically independent subset; after renumbering call it $t_1, \ldots, t_d$, so that $d = \operatorname{trdeg}_k K$ and every remaining t_j is algebraic over $k(t_1, \ldots, t_d)$.

Step 1: each algebraic generator can be taken separable. Fix $j > d$. Then $t_1, \ldots, t_d, t_j$ are algebraically dependent, so some irreducible $f \in k[x_1, \ldots, x_{d+1}]$ has $f(t_1, \ldots, t_d, t_j) = 0$. We claim $\partial f/\partial x_i \neq 0$ for some i. In characteristic 0 this is immediate, as f is nonconstant. In characteristic p, suppose every $\partial f/\partial x_i$ vanished. Then each variable occurs in f only in exponents divisible by p, so

$$f = \sum a_{i_1,\ldots,i_{d+1}}\, x_1^{pi_1} \cdots x_{d+1}^{pi_{d+1}}$$

Since k is algebraically closed it is perfect, so each coefficient has a p-th root: pick $b_{i_1,\ldots,i_{d+1}}$ with

$b^p_{i_1,\dots,i_{d+1}} = a_{i_1,\dots,i_{d+1}}$. Then $g = \sum b_{i_1,\dots,i_{d+1}} x_1^{i_1} \cdots x_{d+1}^{i_{d+1}}$ satisfies $f = g^p$, since the Frobenius is a ring homomorphism in characteristic p. That contradicts irreducibility of f.

Step 2: putting the separable variable last. Say $\partial f/\partial x_i \neq 0$, so x_i genuinely occurs in f and hence t_i is algebraic over the field generated by the other d elements of $\{t_1,\dots,t_d,t_j\}$. Those d elements are still algebraically independent: if they were not, the transcendence degree of $k(t_1,\dots,t_d,t_j)$ would fall below d, contradicting the independence of $t_1,\dots,t_d$. Renaming so that the independent ones are $t_1,\dots,t_d$ and the algebraic one is t_{d+1}, we have $\partial f/\partial x_{d+1} \neq 0$, which says precisely that t_{d+1} is separable over $k(t_1,\dots,t_d)$.

Step 3: collapsing to one generator. K is generated over $k(t_1,\dots,t_d)$ by the finitely many elements $t_{d+1},\dots,t_n$, each separable by Step 1 applied in turn, so $K/k(t_1,\dots,t_d)$ is a finite separable extension. The primitive element theorem then supplies a single z_{d+1} with

$$K = k(t_1,\dots,t_d)(z_{d+1})$$

Writing $z_i = t_i$ for $i \leq d$, let f be an irreducible polynomial with $f(z_1,\dots,z_{d+1}) = 0$, and set $Y = \mathcal{Z}(f) \subseteq \mathbb{A}^{d+1}_k$, a hypersurface. By construction $k(Y) \cong K = k(X)$.

Step 4: an isomorphism of function fields is a birational map. By proposition 33.3.6, a dominating rational map is birational exactly when the induced map of function fields is an isomorphism, and proposition 33.3.4(4) produces such a dominating rational map $X \dashrightarrow Y$ from the isomorphism $k(Y) \to k(X)$. Hence X and Y are birationally isomorphic, as we sought to show.

Note that $k(X)/k(z_1,...,z_d)$ is a finite separable extension, and that the primitive elements $z_1,...,z_{d+1}$ can be chosen as linear combinations of the $x_1,...,x_n$ for appropriate c_{ij}:

$$z_i = \sum_j^n c_{ij} x_j$$

Then the mapping $(x_1,...,x_n) \to (z_1,...,z_{d+1})$ is a projection from $\mathbb{A}^n_k$ that is parallel to the linear subspace given by:

$$\sum_j^n c_{ij} x_j = 0 \qquad 1 \leq i \leq d+1$$

Thus, the birational mapping theorem says that there is an appropriate hypersurface (not necessarily a hyperplane) on which we may project in a rational manner the variety! From this perspective, linear algebra results which are birationally invariant can be translated over to varieties, which we shall soon see gives us some interesting results!

One direction of the classification is easy to get wrong. A variety dominated by $\mathbb{A}^n_k$, meaning one admitting a dominant rational map from affine space, is called *unirational*; a variety birational to $\mathbb{A}^n_k$ is *rational*. Rational implies unirational trivially. In dimension one the converse also holds, and that is Lüroth's theorem. In higher dimensions it fails, which is one of the deepest facts in the subject and well outside this book.

Theorem 33.4.3: Lüroth's Theorem

Let L be a field with $k \subsetneq L \subseteq k(t)$, where t is transcendental over k. Then $L = k(u)$ for some $u \in L$.

***Proof*:**
Choose $u_0 \in L \setminus k$. Then t is algebraic over $k(u_0)$: writing $u_0 = a(t)/b(t)$ with $a, b \in k[t]$ not both constant, t is a root of $a(X) - u_0 b(X) \in k(u_0)[X]$, which is nonzero because a and b are not proportional. So t is algebraic over L as well; put $n = [k(t) : L]$ and let

$$f(X) = X^n + c_{n-1}X^{n-1} + \cdots + c_0 \ \in L[X]$$

be the minimal polynomial of t over L.

Step 1: some coefficient is not in k. If every c_j lay in k then t would be algebraic over k, contradicting transcendence. Fix j with $u = c_j \notin k$ and write $u = a(t)/b(t)$ in lowest terms, with $m = \max(\deg a, \deg b) \geq 1$.

Step 2: $[k(t) : k(u)] = m$. The polynomial $P(X) = a(X) - u\, b(X) \in k(u)[X]$ has degree m and kills t. It is irreducible over $k(u)$: regard it in $k[X][u]$, where it is linear in u and primitive, since a and b are coprime; Gauss's lemma then makes it irreducible in $k(u)[X]$. Hence it is the minimal polynomial of t over $k(u)$ and $[k(t) : k(u)] = m$.

Step 3: $n \leq m$. We have $k(u) \subseteq L \subseteq k(t)$ because $u = c_j \in L$, so $n = [k(t) : L]$ divides $[k(t) : k(u)] = m$; in particular $n \leq m$.

Step 4: $m \leq n$. Clear denominators: let $d(t) \in k[t]$ be a common denominator for $c_0, \ldots, c_{n-1}$, chosen so that $F(t, X) = d(t)f(X)$ lies in $k[t][X]$ and is primitive as a polynomial in X over $k[t]$. Its degree in X is n. Since f is the minimal polynomial of t over L and $P(X) = a(X) - ub(X)$ also kills t, we get $f \big| P$ in $L[X]$; clearing denominators and applying Gauss's lemma in $k[t][X]$,

$$F(t, X) \ \Big| \ a(X)b(t) - a(t)b(X)$$

The right side has degree at most m in t. On the other hand $u = a(t)/b(t)$ appears among the coefficients of f, so after clearing denominators F has degree at least m in t. Hence $\deg_t F = m$ and the quotient is a constant, so

$$F(t, X) = c\,\big(a(X)b(t) - a(t)b(X)\big) \qquad c \in k^*$$

Comparing degrees in X gives $n = \deg_X F = m$.

Conclusion. Steps 3 and 4 give $n = m$, so $[k(t) : L] = [k(t) : k(u)]$; since $k(u) \subseteq L$, the two fields coincide and $L = k(u)$, as we sought to show.

Corollary 33.4.4: Unirational Curves Are Rational

Let C be a curve admitting a dominant rational map $\mathbb{P}^1_k \dashrightarrow C$. Then C is rational, that is birational to $\mathbb{P}^1_k$.

***Proof*:**
A dominant rational map induces an inclusion of function fields $k(C) \hookrightarrow k(\mathbb{P}^1_k) = k(t)$ by proposition 33.3.4. Its image is a subfield of $k(t)$ containing k, and it properly contains k because C is a curve, so $k(C)$ has transcendence degree 1. Theorem 33.4.3 makes that image $k(u)$ for some u, hence $k(C) \cong k(u) \cong k(t)$, and isomorphic function fields mean birational varieties by theorem 33.4.1 and its proof.

The failure in higher dimension is worth naming, since it is easy to assume the pattern continues. There are unirational surfaces over fields of positive characteristic that are not rational, and unirational threefolds over $\mathbb{C}$ that are not rational; the first examples over $\mathbb{C}$ are due to Clemens and Griffiths, Iskovskikh and Manin, and Artin and Mumford, all in 1971–72. Deciding rationality is a live problem, and nothing in this book bears on it.

Exercise 33.4.1

1. In the proof of proposition 33.2.2, verify the identity $\varphi g_i = f_i$ in $k(X)$, and explain why it holds on all of X even though the representation $\varphi = f_i/g_i$ was chosen only near x_i.
2. Let X be irreducible. Show that any finite collection of rational functions on X has a non-empty common domain of definition
3. Show that the collection of poles of a rational function $f \in k(V)$ is an algebraic set
4. Show that for each $a \in k$, the stalk $\mathcal{O}_a(\mathbb{A}^1_k)$ at a is a DVR with uniforming parameter $x - a$.
5. Define $\mathcal{O}_\infty = \left\{ \frac{f}{g} \;\middle|\; \deg(f) \leq \deg(g) \right\}$. Show that $\mathcal{O}_\infty$ is a DVR with uniforming parameter $1/x$. (Note: This is foreshadowing that it is natural consider curves in *projective space* rather than affine space; see chapter 38).

Part VII

Local Properties

Everything in Part VI is an invariant of a whole variety. This Part changes the question: fix a point p of a variety X and ask what can be said about X near p. The answer throughout is that the local ring $\mathcal{O}_{X,p}$ carries it. The dimension of X at p is the Krull dimension of that ring, smoothness at p is the equality of that dimension with the dimension of the cotangent space $\mathfrak{m}_p/\mathfrak{m}_p^2$, the multiplicity of X at p is read from the associated graded ring, and normality is the condition that every $\mathcal{O}_{X,p}$ is integrally closed.

The passage runs both ways: A global statement is repeatedly proved by checking a condition at every point. The regular functions on a variety are the intersection of its local rings, a rational function regular everywhere is regular, and a variety is normal exactly when each of its local rings is.

Dimension theory, regular local rings, completion and the Cohen structure theorem are the commutative algebra the Part runs on, and they come from *Commutative Algebra* [10].

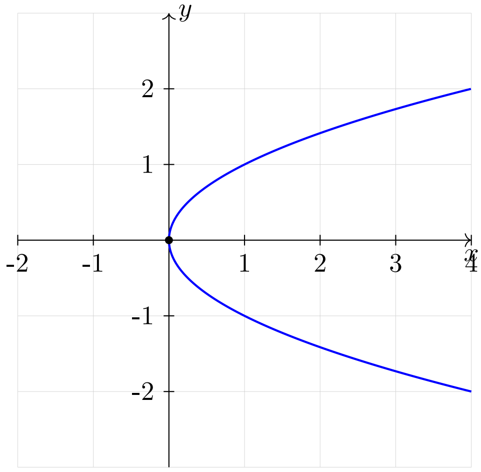
34

Dimension

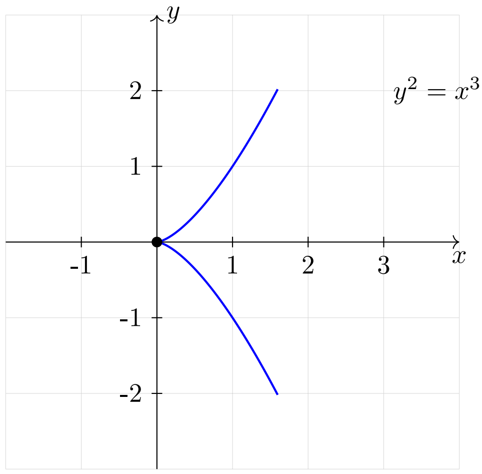
Every notion of dimension that works in analysis or topology fails on the Zariski topology, whose open sets are too large for covering arguments and whose only irreducible subsets in the Euclidean sense are points. What survives is a definition built from chains of irreducible closed subsets, and its algebraic mirror, chains of prime ideals. This chapter establishes that the two agree, and that for a variety both agree with the transcendence degree of its function field.

34.1 What Dimension Should Mean

When looking at $\mathbb{A}_k^n$, we may naturally think of this space as being n-dimensional, and in this special case this matches with our notion of a 3-dimensional vector-space. We may think that varieties also have a notion of dimension, for example the curve with coordinate ring $\frac{k[x,y]}{(y^2-x)}$ gives the impression that it should be thought of as a 1-dimensional variety:



This variety should even be thought of as being smooth, as compared to the variety with coordinate ring $\frac{k[x,y]}{(y^2-x^3)}$ which has a cusp at $(x, y) = (0, 0)$.



We may hope that the notion of dimension is as simple as it was for vector spaces, however this turns out to be a more subtle problem. For example, continuous maps do not preserve dimension (there is a continuous map from $\mathbb{R}$ onto $\mathbb{R}^2$ given by the Peano space-filling curve). Homeomorphisms between open sets of $\mathbb{R}^n$ and $\mathbb{R}^m$ do force the condition that $n = m$[1], however we still need to define the notion of dimension so that it can be preserved.

One of the most general definition of dimension can be defined in the topological level and is called the *Lebesgue covering dimension* where a space X is dimension at most n if every open cover can be refined to an open cover in which each point lies in at most $n + 1$ sets. However, this definition will fail spectacularly in the Zariski topology as most open sets are huge (in fact, most Zariski topologies will have infinite Lebesgue covering dimension).

[1]see [38]

There are a few other notion of dimension (to name some notable example: the Brouwer-Menger-Urysohn boundary dimension, the Hausdorff Dimension, Deviation of Poset). One that the Nullstellensatz has already put within reach is the transcendence degree of a field extension. For example, if we take $k[V]$ to be some coordinate ring for a variety (so that $k[V]$ is an integral domain), we may take the field of fractions $k(V)$. Then taking the transcendental degree of $k(V)/k$, we may define this to be the *transcendental dimension* of $k[V]$. This works well on smooth shapes, and theorem 40.4.4 and theorem 40.4.3 will show that every affine or projective variety admits a finite surjection onto $\mathbb{A}^n_k$ or $\mathbb{P}^m_k$ respectively, so the transcendence degree is always attained. It is also invariant under isomorphism, being defined from the function field. However this definition fails when there the coordinate ring is not an integral domain[2]. Furthermore, it does not generalize to the spectrum of an arbitrary ring: the geometric object attached to $\mathbb{Z}$ has transcendence degree 0 over no field at all, while the definition settled on below gives it dimension 1.

One limitation is that these notions of dimension are global; the entire space has some dimension. This turns out to be a misnomer, as the example of the curve $y^2 = x^3$ mentioned earlier demonstrates (it *drops* a dimension at $(x, y) = (0, 0)$). To anyone who has done differential geometry, they may already be comfortable with the fact that the dimension of a smooth manifold is given by the vector space dimension of the tangent space at a point (which for a smooth-manifold should be invariant to the choice of point, and hence a well-defined global notion). Then the reader well-versed in the equivalent definition's of tangent spaces of a manifold may recall that the tangent space can be defined as the dual of $\mathfrak{m}/\mathfrak{m}^2$ where $\mathfrak{m}$ is the maximal ideal of C^∞_a (the set of germs of C^∞ on a manifold at the point a). This definition is agnostic of the differential structure, and so can be used on varieties. It shall also capture the local subtleties; to reiterate the same example the tangent space at $(x, y) = (0, 0)$ of $y^2 = x^3$ will be dimension 2, representing the fact that it should be treated differently than the other points. Section 35.1 takes up this notion.

34.2 Krull Dimension and Transcendence Degree

What has been settled on as the most useful notion of global dimension that translates well between algebra and geometry is the following:[3]

Definition 34.2.1: Topological Krull Dimension

Let X be a topological space. Then the *Krull dimension* of X is the supremum of the length of all chains:

$$Z_0 \subsetneq Z_1 \subsetneq \cdots \subsetneq Z_n$$

of distinct irreducible closed subsets of X. The Krull dimension of a topological space will be denoted kdim

This definition is not interesting in euclidean topology, for example for $\mathbb{R}^n$ as only points are irreducible we have $\text{kdim}(\mathbb{R}^n) = 0$. If the dimension is finite, then it is equal to the length of the longest chain of irreducible closed subsets. The Krull dimension of $\mathbb{A}^1_k$ is always 1, $\text{kdim}(\mathbb{A}^1_k) = 1$, as the only irreducible closed subsets of $\mathbb{A}^1_k$ are the single point and whole space. As irreducible closed subsets

[2]It also fails as an adequate definition when localizing a ring or taking its completion, which are two important tools that we shall often use on shapes

[3]The Krull dimension turns out to be equivalent to the Brouwer-Menger-Urysohn definition of dimension, which is useful for cohomological intuitions

correspond to prime ideals, we have the equivalent notion for primes:

Definition 34.2.2: Height of Prime

Let $\mathfrak{p} \subseteq R$ be a prime ideal. Then the *height* of a prime ideal is the supremum of all integers n for which there is a chain of distinct prime ideals:

$$\mathfrak{p}_0 \subsetneq \mathfrak{p}_1 \subsetneq \cdots \subsetneq \mathfrak{p}_n = \mathfrak{p}$$

The *dimension* of R is the supremum of the heights of all prime ideals.

A field has exactly one prime, (0), of height 0, so its Krull dimension is 0. A principal ideal domain that is not a field has Krull dimension 1: its nonzero primes are maximal, so the longest chain is $(0) \subsetneq \mathfrak{p}$.

Proposition 34.2.3: Dimension of Affine Algebraic Set

Let $Y \subseteq \mathbb{A}^n_k$ be an affine algebraic set. Then $\mathrm{kdim}(Y)$ is equal to the dimension of the affine coordinate ring $k[Y]$

***Proof*:**
By proposition 32.3.3 the irreducible closed subsets of Y correspond to the prime ideals of $\Gamma(\mathbb{A}^n)$ containing $\mathcal{I}(Y)$, which by the correspondence theorem are the prime ideals of $k[Y] = \Gamma(\mathbb{A}^n)/\mathcal{I}(Y)$. The correspondence *reverses* inclusion, so a strictly ascending chain of length d of irreducible closed subsets of Y is the same thing as a strictly descending chain of length d of primes of $k[Y]$, and conversely. Taking suprema, $\mathrm{kdim}(Y) = \mathrm{kdim}(k[Y])$, as we sought to show.

When covering Noether's Normalization (theorem 32.4.1), it was observed that this can be used to define the dimension of a ring. The following definition captures this idea:

Definition 34.2.4: Transcendental Dimension

Let X be a variety. Then the *transcendence dimension* of X over k is the transcendence degree of its function field $k(X)$, and is denoted $\dim X$. (Proposition 34.2.5 shows this agrees with $\mathrm{kdim}X$, after which the two notations are used interchangeably.) The transcendence degree of a field extension, and the fact that it is well-defined, are established in *Core Algebra* [25].

Proposition 34.2.5: Transcendental and Krull Dimension

Let k be a field and R a finitely generated k-algebra which is an integral domain. Then

$$\mathrm{kdim}(R) = \mathrm{trdeg}_k \mathrm{Frac}(R)$$

Furthermore, for every prime $\mathfrak{p} \subseteq R$,

$$\mathrm{height}(\mathfrak{p}) + \mathrm{kdim}(R/\mathfrak{p}) = \mathrm{kdim}(R)$$

Neither hypothesis can be dropped. Transcendence degree is an invariant of a field extension, so R must be a domain for $\mathrm{Frac}(R)$ to exist; and finite generation cannot be dropped, as example 34.2.6

shows. The second identity is the statement that finitely generated domains over a field are catenary and equidimensional, which is what makes codimension behave additively.

***Proof*:**
This is the equality $\text{kdim}R = \text{trdeg}_k R$ for a finitely generated k-algebra, together with the dimension formula $\text{height}(\mathfrak{p}) + \text{kdim}(R/\mathfrak{p}) = \text{kdim}R$; both are proved via Noether normalization in [10].

The conditions are necessary for the equality of both equations:

Example 34.2.6: Different Krull and Transcendental Dimension
Take $R = k(x)$, the field of rational functions in one variable. As a field it has exactly one prime, so $\text{kdim}R = 0$, while $\text{trdeg}_k R = 1$. There is no contradiction with proposition 34.2.5: $k(x)$ is *not* a finitely generated k-algebra. It is finitely generated as a *field* extension, which is a strictly weaker condition, and the proposition needs the algebra version.

Geometrically the collapse is familiar to a reader who has met schemes: $k(x)$ is the local ring at the generic point of $\mathbb{A}^1_k$, a single non-closed point, and a one-point space has Krull dimension 0 however large the field attached to it.

Corollary 34.2.7: Dimension of Affine Space

The Krull dimension of affine space $\mathbb{A}^n_k$ is n.

***Proof*:**
$\Gamma(\mathbb{A}^n) \cong k[x_1, \ldots, x_n]$ is a finitely generated k-algebra and an integral domain, so proposition 34.2.5 applies and gives $\text{kdim}\mathbb{A}^n = \text{trdeg}_k k(x_1, \ldots, x_n)$. The x_i are algebraically independent over k and generate the field, so they form a transcendence basis and that degree is n, as we sought to show.

34.3 Codimension, Curves, and Surfaces

Definition 34.3.1: Codimension

Let $Y \subseteq X$ be a closed subvariety of a variety X. The *codimension* of Y in X, written $\text{codim}_X Y$, is the supremum of the lengths of chains

$$Y = Z_0 \subsetneq Z_1 \subsetneq \cdots \subsetneq Z_c \subseteq X$$

of irreducible closed subsets of X starting at Y. The ambient is part of the data and is dropped from the notation only when it is fixed by context.

Proposition 34.3.2: Codimension Is a Difference of Dimensions

Let $Y \subseteq X$ be a closed subvariety of a variety X over a field. Then

$$\operatorname{codim}_X Y = \dim X - \dim Y$$

***Proof*:**
Passing to an affine open subset changes neither side, so assume X affine with coordinate ring $k[X]$, a finitely generated domain over k. Chains of irreducible closed subsets of X starting at Y correspond, reversing inclusion, to chains of primes of $k[X]$ descending from $\mathcal{I}(Y)$, so $\operatorname{codim}_X Y = \operatorname{height}(\mathcal{I}(Y))$. Likewise $\dim Y = \operatorname{kdim}(k[X]/\mathcal{I}(Y))$ by proposition 34.2.3. The dimension formula of proposition 34.2.5 applied to $\mathfrak{p} = \mathcal{I}(Y)$ then reads $\operatorname{codim}_X Y + \dim Y = \dim X$, as we sought to show.

The identity fails without the hypotheses of proposition 34.2.5: it is exactly the statement that the ring is catenary and equidimensional, which is why codimension is *defined* by chains and only *computed* as a difference.

One consequence is used constantly and deserves a name: dimension cannot be detected locally, in the sense that shrinking to a smaller open set does not shrink the dimension.

Corollary 34.3.3: Dimension Is Determined by Any Open Subset

Let X be an irreducible variety and $U \subseteq X$ a nonempty open subset. Then $\dim U = \dim X$.

***Proof*:**
U is dense in X by irreducibility, so restriction identifies the two function fields, $k(U) = k(X)$. Both dimensions equal the transcendence degree of that common field by proposition 34.2.5, completing the proof.

Verify as a check on the definitions that planar curves have dimension 1, that finite sets have dimension 0, and that $\mathbb{A}^n_k$ and $\mathbb{P}^n_k$ have dimension n.

Definition 34.3.4: Curves and Surfaces

Let X be an algebraic variety. Then:

1. If X is 1-dimensional, it is called a *curve*
2. If X is 2-dimensional, it is called a *surface*
3. If X is 1-codimensional, it is called a *hypersurface*.

Proposition 34.3.5: Dimension Between Irreducible Varieties

Let $Y \subseteq X$ be a closed subvariety of a variety X. Then $\dim Y \leq \dim X$, with equality if and only if $Y = X$.

***Proof*:**
Every chain of irreducible closed subsets of Y is a chain of irreducible closed subsets of X, since Y is closed in X; taking suprema gives $\dim Y \leq \dim X$.

Suppose $\dim Y = \dim X = d$ and $Y \subsetneq X$. Take a chain $Z_0 \subsetneq \cdots \subsetneq Z_d = Y$ of irreducible closed subsets of Y realizing its dimension. As Y is irreducible, closed and properly contained in X, appending X extends this to

$$Z_0 \subsetneq \cdots \subsetneq Z_d = Y \subsetneq X$$

a chain of length $d + 1$ in X, so $\dim X \geq d + 1$, contradicting $\dim X = d$. Hence $Y = X$; the converse is trivial, completing the proof.

34.4 Hypersurface Sections and Codimension One

Cutting a variety by one equation ought to drop its dimension by exactly one. That is the geometric content of Krull's principal ideal theorem, and it is the fact that makes dimension usable: it lets a variety be built up, or cut down, one equation at a time. This section records it, and then proves the converse in affine space, where codimension one turns out to be the same thing as being cut out by a single polynomial.

Theorem 34.4.1: Hypersurface Sections Drop Dimension by One

Let X be a variety and $f \in k[X]$ neither zero nor a unit. Then every irreducible component of $\mathcal{Z}_X(f) = \{p \in X \mid f(p) = 0\}$ has dimension $\dim X - 1$.

***Proof*:**
The irreducible components of $\mathcal{Z}_X(f)$ correspond to the primes of $k[X]$ minimal over (f). Let $\mathfrak{p}$ be such a prime. Krull's principal ideal theorem (theorem 29.5.2) gives $\operatorname{height}(\mathfrak{p}) \leq 1$.

For the reverse bound, $k[X]$ is a domain because X is a variety (corollary 32.3.4), so (0) is prime; and $f \neq 0$ forces $\mathfrak{p} \neq (0)$. Hence $(0) \subsetneq \mathfrak{p}$ is a chain of length 1 and $\operatorname{height}(\mathfrak{p}) \geq 1$, so $\operatorname{height}(\mathfrak{p}) = 1$.

The component attached to $\mathfrak{p}$ has coordinate ring $k[X]/\mathfrak{p}$, so by proposition 34.2.5,

$$\dim \mathcal{Z}_X(\mathfrak{p}) = \operatorname{kdim}(k[X]/\mathfrak{p}) = \operatorname{kdim} k[X] - \operatorname{height}(\mathfrak{p}) = \dim X - 1$$

as we sought to show.

The theorem says nothing about $\mathcal{Z}_X(f)$ being irreducible, and it need not be: $\mathcal{Z}(xy) \subseteq \mathbb{A}^2$ is the union of the two axes, two components each of dimension 1. What the theorem guarantees is that no component is smaller than expected. The hypothesis that f be a nonunit is what keeps $\mathcal{Z}_X(f)$ nonempty in the first place, and $f \neq 0$ is what stops it from being all of X.

In affine space the converse holds, and it is a genuine dictionary entry: codimension one on the geometric side is principality on the algebraic side.

Theorem 34.4.2: Codimension One in Affine Space Means Hypersurface

Let $Y \subsetneq \mathbb{A}^n_k$ be an irreducible closed subset with $\operatorname{codim}_{\mathbb{A}^n} Y = 1$. Then there is an irreducible polynomial f, unique up to a nonzero scalar, with

$$\mathcal{I}(Y) = (f) \qquad Y = \mathcal{Z}(f)$$

Conversely $\mathcal{Z}(f)$ is irreducible of codimension 1 for every irreducible f.

***Proof*:**
$\mathcal{I}(Y)$ is prime because Y is irreducible (proposition 32.3.3), and $\operatorname{height}(\mathcal{I}(Y)) = \operatorname{codim}_{\mathbb{A}^n} Y = 1$ by proposition 34.3.2. So it suffices to show that a height-one prime $\mathfrak{p}$ in a unique factorization domain is principal, generated by an irreducible element.

The ring $\Gamma(\mathbb{A}^n) \cong k[x_1, \ldots, x_n]$ is a UFD, being a polynomial ring in finitely many variables over the field k (corollary 11.2.4). Choose $0 \neq g \in \mathfrak{p}$ and factor it into irreducibles, $g = f_1 \cdots f_r$. As $\mathfrak{p}$ is prime, some $f_i \in \mathfrak{p}$; write $f = f_i$. In a UFD an irreducible element is prime (proposition 10.1.29), so (f) is a nonzero prime ideal with

$$(0) \subsetneq (f) \subseteq \mathfrak{p}$$

If the second inclusion were strict, that chain would have length 2 and give $\operatorname{height}(\mathfrak{p}) \geq 2$. Hence $(f) = \mathfrak{p}$. Uniqueness up to a scalar is uniqueness of the generator of a principal ideal in a domain, and $Y = \mathcal{Z}(\mathcal{I}(Y)) = \mathcal{Z}(f)$ by proposition 31.2.11.

Conversely let f be irreducible. Then (f) is prime, so $\mathcal{Z}(f)$ is irreducible, and theorem 34.4.1 applied to $X = \mathbb{A}^n$ gives $\dim \mathcal{Z}(f) = n - 1$, that is codimension 1, as we sought to show.

Example 34.4.3: Hypersurfaces and Their Ideals

1. The cuspidal cubic $\mathcal{Z}(y^2 - x^3) \subseteq \mathbb{A}^2$ has $y^2 - x^3$ irreducible, so its vanishing ideal is $(y^2 - x^3)$ and it is a curve: codimension 1 in $\mathbb{A}^2$.

2. $\mathcal{Z}(xy) \subseteq \mathbb{A}^2$ is not irreducible, and correspondingly (xy) is not prime. Its two components $\mathcal{Z}(x)$ and $\mathcal{Z}(y)$ each have codimension 1, as theorem 34.4.1 predicts.

3. The twisted cubic $\mathcal{Z}(y - x^2, z - x^3) \subseteq \mathbb{A}^3$ has dimension 1, hence codimension 2, and so by theorem 34.4.2 it cannot be cut out by a single equation. This is the promise made when the twisted cubic first appeared in example 31.2.2 that it is not a hypersurface.

Exercise 34.4.1

1. Let X, Y be irreducible affine varieties. Show that $\dim(X \times Y) = \dim(X) + \dim(Y)$ directly from the transcendence-degree definition. Corollary 40.5.4 proves this a different way, by reading the projection as a family of fibres, so the two arguments are worth comparing.

2. Define the *coheight* of a prime $\mathfrak{p} \subseteq R$ to be the supremum of the lengths of chains of primes $\mathfrak{p} = \mathfrak{p}_0 \subsetneq \mathfrak{p}_1 \subsetneq \cdots \subsetneq \mathfrak{p}_d$. Show that $\operatorname{coht}(\mathfrak{p}) = \operatorname{kdim}(R/\mathfrak{p})$ for any ring, give an example where it is infinite, and show it is finite whenever R is a finitely generated k-algebra.

3. Let R be a Noetherian ring. Show that $\mathrm{kdim} R[x] = \mathrm{kdim} R + 1$.
4. Compute the dimension of $\mathcal{Z}(xy, xz) \subseteq \mathbb{A}^3_k$, identify its irreducible components, and observe that they do not all have the same dimension. Which hypothesis of proposition 34.3.2 does this violate?
5. Show that $\dim \mathcal{Z}(f) = n - 1$ for any nonconstant $f \in k[x_1, \ldots, x_n]$, and deduce that a proper closed subset of $\mathbb{A}^n_k$ has dimension at most $n - 1$.
6. Let $X \subseteq \mathbb{A}^n_k$ be an irreducible variety of dimension d. Show that X is contained in no linear subspace of dimension less than d, and give an example where it is contained in one of dimension exactly d.
7. Verify from the definitions that a plane curve has dimension 1, that a finite set has dimension 0, and that $\mathbb{A}^n_k$ has dimension n.

35

Smooth Points and Regular Local Rings

The dimension theory the previous chapter used, and which this one uses again, rests on the Hilbert-Samuel function of an $\mathfrak{m}$-primary ideal and the Dimension Theorem $\delta(R) = d(R) = \mathrm{kdim} R$ it yields, together with systems of parameters. Those are in *Commutative Algebra* [10].

This chapter takes the one consequence that carries geometric content, the tangent-space inequality $\mathrm{kdim} R \leq \dim_k(\mathfrak{m}/\mathfrak{m}^2)$, and turns it into a criterion. A point is *smooth* when the inequality is an equality, which says the tangent space has no more directions than the variety has dimensions. The Jacobian gives a computable test for this, regular local rings give an intrinsic one free of any embedding, and the two agree. The chapter closes with the case that matters most for Part IX: in dimension one, regular is the same as being a discrete valuation ring.

35.1 The Tangent Space Bound

For a Noetherian local ring $(R, \mathfrak{m})$ the Dimension Theorem ([10]) identifies the Krull dimension with the degree of the Hilbert-Samuel polynomial and with the least number of generators of an $\mathfrak{m}$-primary ideal. The one consequence we need geometrically bounds the dimension by the embedding dimension.

Corollary 35.1.1: Dimension and Tangent Space

Let $(R, \mathfrak{m})$ be a Noetherian local ring with residue field $k = R/\mathfrak{m}$. Then

$$\mathrm{kdim}(R) \leq \dim_k(\mathfrak{m}/\mathfrak{m}^2)$$

***Proof*:**
Choose $r_1, \ldots, r_s \in \mathfrak{m}$ whose images form a basis of $\mathfrak{m}/\mathfrak{m}^2$; by Nakayama they generate $\mathfrak{m}$, hence an $\mathfrak{m}$-primary ideal on s generators, so the Dimension Theorem ([10]) gives $\mathrm{kdim} R \leq s = \dim_k(\mathfrak{m}/\mathfrak{m}^2)$.

35.2 The Jacobian Criterion and Regular Local Rings

One of the properties most geometric objects carry is *smoothness*, or nonsingularity. In this section, we shall introduce an intuitive way of defining smoothness given an embedding into an affine/projective space. We shall then show a more intrinsic definition that will not require any embedding.

Definition 35.2.1: Nonsingular Variety - Embedding Variant

Let $X \subseteq \mathbb{A}^n_k$ be an affine variety of dimension r, and let $f_1, \ldots, f_t$ generate its vanishing ideal $\mathcal{I}(X) \subseteq \Gamma(\mathbb{A}^n)$. Then X is *nonsingular* at $p \in X$ if the Jacobian matrix evaluated at p,

$$J(p) = \left(\frac{\partial f_i}{\partial x_j}(p)\right)_{\substack{1 \leq i \leq t \\ 1 \leq j \leq n}}$$

has rank $J(p) = n - r$. We say X is *nonsingular* if it is nonsingular at every point.

If the reader knows differential geometry, this is saying that the tangent space must be "full rank" at every point. The matrix is called the *Jacobian matrix*. Its rank at p does not depend on the chosen generators, even though the matrix itself does: the proof of proposition 35.2.2 below identifies rank $J(p)$ with $n - \dim_k \mathfrak{m}_p/\mathfrak{m}_p^2$, an invariant of the local ring alone. This definition is geometrically well-motivated, but is not intrinsic to the variety: it is dependent on the "global" information of the embedding space and generators. We shall immediately work towards the intrinsic definition of nonsingular variety. The key is that at each point p the local ring $\mathcal{O}_{X,p}$ has a single maximal ideal $\mathfrak{m}_p$ representing where functions evaluate to 0, and $\mathfrak{m}_p/\mathfrak{m}_p^2$ "remembers" only linear terms, making it the best analogy to the cotangent space at a point. Recall that $\dim_k \mathfrak{m}_p/\mathfrak{m}_p^2 \geq \mathrm{kdim} k[X]$ by corollary 35.1.1. The inequality is strict exactly at a singular point, where the tangent space picks up extra dimensions. We thus seek conditions where the local ring at each point shall have the same dimension everywhere:

A local Noetherian ring of dimension d with maximal ideal $\mathfrak{m}$ and residue field k is *regular* when $\dim_k(\mathfrak{m}/\mathfrak{m}^2) = d$, equivalently when $\mathfrak{m}$ can be generated by d elements, equivalently when the associated graded ring $G_{\mathfrak{m}}(R)$ is the polynomial ring $k[t_1, \ldots, t_d]$ on independent degree-one indeterminates. The definition, the equivalence of the three conditions, and the consequences used below are commutative algebra and are developed in *Commutative Algebra*: see definition 29.7.2 and theorem 29.7.3.

Proposition 35.2.2: Jacobian Criterion and Regular Local Rings

Let $X \subseteq \mathbb{A}^n_k$ be a variety. Then X is nonsingular at p iff the local ring at p is regular.

***Proof*:**
Let $p = (a_1, \dots, a_n) \in \mathbb{A}^n$ and let $\mathfrak{a}_p = (x_1 - a_1, \dots, x_n - a_n)$ be the corresponding maximal ideal of $A = \Gamma(\mathbb{A}^n) \cong k[x_1, \dots, x_n]$. Define a k-linear map $\theta_p \colon A \to k^n$ by

$$\theta_p(f) = \left\langle \frac{\partial f}{\partial x_1}(p), \dots, \frac{\partial f}{\partial x_n}(p) \right\rangle$$

The images $\theta_p(x_i - a_i)$ are the standard basis vectors, so θ_p is surjective and its restriction to $\mathfrak{a}_p$ is still surjective. That restriction has kernel exactly $\mathfrak{a}_p^2$: the product rule gives $\theta_p(fg) = f(p)\theta_p(g) + g(p)\theta_p(f)$, which vanishes when $f, g \in \mathfrak{a}_p$, and conversely a Taylor expansion of $f \in \mathfrak{a}_p$ at p has linear part determined by $\theta_p(f)$. Note the restriction is essential: θ_p kills every constant, so $\ker \theta_p \neq \mathfrak{a}_p^2$ on all of A. Thus θ_p induces an isomorphism $\theta_p' \colon \mathfrak{a}_p/\mathfrak{a}_p^2 \to k^n$.

Now let $\mathfrak{b} = \mathcal{I}(X)$ be the ideal of X in A, and let $f_1, \dots, f_t$ be a set of generators of $\mathfrak{b}$. Then the rank of the Jacobian matrix $J = (\partial f_i/\partial x_j)(p)$ is just the dimension of $\theta_p(\mathfrak{b})$ as a subspace of k^n. Using the isomorphism θ_p', this is the same as the dimension of the subspace $(\mathfrak{b} + \mathfrak{a}_p^2)/\mathfrak{a}_p^2$ of $\mathfrak{a}_p/\mathfrak{a}_p^2$. On the other hand, the local ring $\mathcal{O}_p$ of p on X is obtained from A by dividing by $\mathfrak{b}$ and localizing at the maximal ideal $\mathfrak{a}_p$. Thus if $\mathfrak{m}$ is the maximal ideal of $\mathcal{O}_p$, we have

$$\mathfrak{m}/\mathfrak{m}^2 \cong \mathfrak{a}_p/(\mathfrak{b} + \mathfrak{a}_p^2)$$

Counting dimensions of vector spaces, we have $\dim \mathfrak{m}/\mathfrak{m}^2 + \operatorname{rank}(J) = n$.

But this gives what we want. If $\dim X = r$ then $\mathcal{O}_p$ has Krull dimension r, since localizing $k[X]$ at the maximal ideal of p leaves the dimension of the irreducible X unchanged by proposition 34.3.2. So $\mathcal{O}_p$ is regular if and only if $\dim_k \mathfrak{m}/\mathfrak{m}^2 = r$. But this is equivalent to $\operatorname{rank}(J) = n - r$, which says that p is a nonsingular point of X, completing the proof.

Example 35.2.3: The Jacobian Criterion on Three Curves
Each of the following is a curve in $\mathbb{A}_k^2$, and the criterion is applied at the origin, which lies on all three. Assume $\operatorname{char} k \neq 2, 3$.

1. *The parabola* $f = y - x^2$. The Jacobian is $(-2x,\ 1)$, of rank 1 everywhere. So every point is nonsingular, and $\dim_k \mathfrak{m}_p/\mathfrak{m}_p^2 = 1 = \dim C$ at each.
2. *The node* $f = y^2 - x^2 - x^3$. The Jacobian is $\left(-2x - 3x^2,\ 2y\right)$, which vanishes at $(0,0)$ and at $(-2/3, 0)$; only the first is on the curve. So the origin is the unique singular point, and there $\dim_k \mathfrak{m}/\mathfrak{m}^2 = 2 > 1$.
3. *The cusp* $f = y^2 - x^3$. The Jacobian is $\left(-3x^2,\ 2y\right)$, vanishing only at the origin, which is on the curve. Again the origin is the unique singular point with $\dim_k \mathfrak{m}/\mathfrak{m}^2 = 2$.

The criterion separates the parabola from the other two but says nothing distinguishing the node from the cusp: both are singular points of multiplicity 2. Telling them apart needs the tangent cone of definition 36.1.2, taken up in the next chapter, which sees two tangent lines at the node and one doubled at the cusp.

From this, we can extend the definition naturally to any quasiprojective variety.

Definition 35.2.4: Nonsingular Variety

Let X be a quasi-projective variety. Then X is *nonsingular* or *regular* if the local ring at every point is regular.

35.3 The Singular Locus

The set of points that are singular are usually "measure 0". We shall not go so far to prove this statement, but the following simpler result that a variety cannot *only* consist of singular points:

Proposition 35.3.1: Singular Points of Variety

Let X be a variety. Then the set of singular points of X forms a proper closed subset of X

The fact that it is closed should be unsurprising as it is essentially the pre-image of the closed set 0 for the right continuous function, the key take-away is that it must be proper.

***Proof*:**
Let $\mathrm{Sing}(X)$ denote the set of singular points of X.

Closed. Being closed is local, so covering X by affine open sets reduces us to X affine, say of dimension r in $\mathbb{A}^n$ with $\mathcal{I}(X) = (f_1, \ldots, f_t)$. Combining corollary 35.1.1 with the dimension count in the proof of proposition 35.2.2 gives rank $J(p) \leq n - r$ at every p, with equality exactly at the nonsingular points. So the singular points are those where the rank is *strictly* less than $n - r$. Thus $\mathrm{Sing}(X)$ is the algebraic set defined by the ideal generated by $I(X)$ together with all determinants of $(n-r) \times (n-r)$ sub-matrices of the matrix $\|\partial f_i/\partial x_j\|$. From these, we can see that $\mathrm{Sing}\, X$ is closed.

Proper. By theorem 33.4.1, X is birationally isomorphic to a hypersurface $Y \subseteq \mathbb{A}^{d+1}$, where $d = \dim X$; note the theorem produces an *affine* hypersurface, not a projective one. By proposition 39.5.2 a birational isomorphism identifies a dense open subset of X with one of Y, and nonsingularity is a local condition, so it is enough to exhibit a single nonsingular point on Y. Thus we may assume X is a hypersurface in $\mathbb{A}^n$ cut out by one irreducible polynomial $f(x_1, \ldots, x_n)$.

Now, $\mathrm{Sing}(X)$ is the set of points $p \in X$ such that $(\partial f/\partial x_i)(p) = 0$ for $i = 1, \ldots, n$. If $\mathrm{Sing}\, X = X$, then the functions $\partial f/\partial x_i$ are zero on X, and hence $\partial f/\partial x_i \in I(X)$ for each i. But $I(X)$ is the principal ideal generated by f, and $\deg(\partial f/\partial x_i) \leq \deg f - 1$ for each i, so we must have $\partial f/\partial x_i = 0$ for each i.

In characteristic 0 this is already impossible, because if x_i occurs in f, then $\partial f/\partial x_i \neq 0$. So we must have char $k = p > 0$, and then the fact that $\partial f/\partial x_i = 0$ implies that f is a polynomial in x_i^p. This is true for each i, so by taking pth roots of the coefficients (which is possible since k is algebraically closed), we get a polynomial $g(x_1, \ldots, x_n)$ such that $f = g^p$. But this contradicts the hypothesis that f was irreducible, and so $\mathrm{Sing}\, X \subsetneq X$, as we sought to show.

35.4 Homological Characterization: Serre's Theorem

Before stating the main theorem on regular local rings, we need a homological invariant[1]:

Definition 35.4.1: Projective and Global Dimension

Let R be a ring and M an R-module. The *projective dimension* of M, denoted $\mathrm{pd}_R(M)$, is the minimal length n of a projective resolution of M:

$$0 \to P_n \to \cdots \to P_0 \to M \to 0$$

If no finite resolution exists, $\mathrm{pd}_R(M) = \infty$.

The *global dimension* of R, denoted $\mathrm{gldim}(R)$, is the supremum of the projective dimensions of all R-modules:

$$\mathrm{gldim}(R) = \sup\{\mathrm{pd}_R(M) \big| M \in \mathrm{Mod}_R\}$$

Example 35.4.2: Infinite Global Dimension When There is A Singularity

Take $R = k[x, y]/(x^2 - y^3)$. This corresponds to the cusp singularity at the origin. Let us compute the global dimension by finding the projective dimension of the residue field $k \cong R/(x, y)$ (where $\mathfrak{m} = (\overline{x}, \overline{y})$). We begin a free resolution of k by killing $\mathfrak{m} = (\overline{x}, \overline{y})$:

$$R^2 \xrightarrow{d_1} R \to k \to 0$$

where $d_1 = \begin{pmatrix} x & y \end{pmatrix}$. The kernel of this map consists of relations between x and y. In the polynomial ring, the only relation is trivial ($yx - xy = 0$). However, in our quotient ring R, we have the relation defining the ring: $x^2 - y^3 = 0$. Thus, the kernel is generated by two elements:

1. The "Koszul" relation: $y \cdot (x) - x \cdot (y) = 0$
2. The "Curve" relation: $x \cdot (x) - y^2 \cdot (y) = 0$

So we have a map $R^2 \xrightarrow{d_2} R$ given by the matrix $\begin{pmatrix} y & x \\ -x & -y^2 \end{pmatrix}$. (Check: $\begin{pmatrix} x & y \end{pmatrix} \begin{pmatrix} y & x \\ -x & -y^2 \end{pmatrix} = \begin{pmatrix} xy - yx & x^2 - y^3 \end{pmatrix} = \begin{pmatrix} 0 & 0 \end{pmatrix}$).

Now we need the kernel of d_2. Calculating the determinant of the matrix, we find $\det(d_2) = -y^3 + x^2 = 0$. Because the determinant vanishes, the columns are linearly dependent, and the kernel is non-trivial. In fact, the resolution becomes *periodic*. The resolution continues forever with alternating matrices:

$$\cdots \xrightarrow{d_3} R^2 \xrightarrow{d_2} R^2 \xrightarrow{d_1} R \to k \to 0$$

where $d_{\text{odd}} = \begin{pmatrix} y & x \\ -x & -y^2 \end{pmatrix}$ and $d_{\text{even}} = \begin{pmatrix} y^2 & x \\ -x & -y \end{pmatrix}$. Since the resolution never terminates (it never reaches a 0 module), the projective dimension of k is infinite. Therefore, the global dimension of R is infinite.

[1] These terms will be central in section 57.2 and will be reintroduced there

For a local ring $(R, \mathfrak{m})$ with residue field k, a fundamental result of Auslander and Buchsbaum states that $\mathrm{gldim}(R) = \mathrm{pd}_R(k)$. Thus, the global dimension is determined entirely by the resolution of the residue field.

Theorem 35.4.3: Auslander-Buchsbaum-Serre Theorem

A local Noetherian ring $(R, \mathfrak{m})$ has finite global dimension if and only if it is a regular local ring.

The proof is homological rather than geometric, and belongs with the machinery that produces it: it runs through the Auslander-Buchsbaum identity $\mathrm{gldim}\, R = \mathrm{pd}_R\, k$ and the Koszul resolution of the residue field, neither of which this book develops. *Homological Algebra* [15, Auslander-Buchsbaum-Serre Theorem] gives it in full; the two books are co-requisites for exactly this reason.

***Proof Key ideas*:**
One direction is the Koszul resolution. If R is regular of dimension d, then $\mathfrak{m}$ is generated by a regular sequence $x_1, \dots, x_d$, and the Koszul complex on that sequence is a free resolution of $k = R/\mathfrak{m}$ of length d. Hence $\mathrm{pd}_R\, k \leq d < \infty$, and by the Auslander-Buchsbaum identity $\mathrm{gldim}\, R = \mathrm{pd}_R\, k$ is finite.

The converse is the substantive half. Finiteness of $\mathrm{pd}_R\, k$ forces the minimal free resolution of k to terminate, and one shows by induction on $\dim_k \mathfrak{m}/\mathfrak{m}^2$ that this can happen only when a minimal generating set of $\mathfrak{m}$ is a regular sequence, which is regularity. See [15, Auslander-Buchsbaum-Serre Theorem] for the details.

35.5 Regular Local Rings in Dimension One

Let us now take a closer look at the properties regular local rings must have, especially in low dimension. In one dimension, this shall produce a very restrictive result that will be very powerful, and in higher dimensions we shall find an algebraic condition that eventually lets the rings of integers of *Algebraic Number Theory* [7] be read as nonsingular $\mathbb{Z}$-curves. For curves this is the decisive case, and section 37.3 uses it to show that each birational class of curves contains a unique nonsingular projective model, so that a field extension K/k of transcendence degree one determines *the* curve attached to it.

One consequence is needed before the dimension-one case, and it is not visible from the numerical definition: a regular local ring has no zero-divisors. The mechanism is that regularity makes the associated graded ring a polynomial ring, hence a domain, and a domain downstairs lifts to a domain upstairs whenever $\bigcap_n \mathfrak{m}^n = 0$, which Krull's theorem supplies in any Noetherian local ring. Both steps are commutative algebra and are carried out in corollary 29.7.5.

Corollary 35.5.1: Regular Local Rings of Dimension 1

Let R be a local Noetherian ring of dimension 1. Then R is a regular local ring if and only if R is a DVR.

***Proof*:**
Suppose $(R, \mathfrak{m})$ is regular of dimension 1, so $\dim_{R/\mathfrak{m}} \mathfrak{m}/\mathfrak{m}^2 = \mathrm{kdim} R = 1$. Choose $m \in \mathfrak{m}$ whose class spans $\mathfrak{m}/\mathfrak{m}^2$. Nakayama's lemma (theorem 23.4.30) then gives $\mathfrak{m} = (m)$: the single element m generates. By the discussion above R is a domain, and a Noetherian local domain of dimension 1 with principal maximal ideal is a discrete valuation ring (theorem 27.2.2).

Conversely let R be a discrete valuation ring with uniformizer m. It is local: its unique maximal ideal is (m), the elements of positive valuation. Note it is *not* true that (m) is its only prime, as (0) is prime too; indeed the chain $(0) \subsetneq (m)$ is what makes $\mathrm{kdim} R = 1$, and no longer chain exists because every nonzero ideal is $(m)^n$. Finally $(m)/(m)^2$ is spanned by the class of m over $R/(m)$ and is nonzero, so $\dim_{R/(m)}(m)/(m)^2 = 1 = \mathrm{kdim} R$ and R is regular, as we sought to show.

This gives a nice condition for 1-dimensional regular local rings. It will sometimes be useful to identify regular local rings in different context:

Proposition 35.5.2: Equivalences of 1 Dimensional Regular Local Rings

Let $(R, \mathfrak{m})$ be a Noetherian local domain of dimension 1. Then the following are equivalent:

1. R is a regular local ring
2. R is a DVR
3. R is normal (integrally closed)
4. $\mathfrak{m}$ is principal
5. every nonzero ideal of R is a power of $\mathfrak{m}$

***Proof*:**
This is a statement about Noetherian local domains with no geometric content, and it is proved in full upstream: see theorem 27.2.2, which lists exactly these five conditions.

The equivalence of (1) and (3) is the one chapter 37 consumes. It says that for a curve, being smooth at a point and being normal at that point are the same condition, which is why normalization resolves the singularities of a curve completely while leaving higher-dimensional singularities behind.

For higher-dimensions, the situation is more subtle, and the full account belongs with divisors on a scheme; this book records only the following:

Corollary 35.5.3: Regular Local Ring, then Normal

Let R be a regular local Noetherian ring. Then R is normal.

***Proof*:**
This is corollary 29.8.7, where it is obtained from factoriality: a regular local ring is a unique factorization domain by the theorem of Auslander and Buchsbaum, and a unique factorization domain is integrally closed. In dimension one the statement is already contained in proposition 35.5.2, which is the case this book uses.

The converse does not hold in general in dimension > 1, a normal domain need not be regular, so normality alone does not detect nonsingular varieties. Chapter 37 takes up exactly this gap.

Proposition 35.5.4: Completion and Regularity

Let $(R, \mathfrak{m})$ be a Noetherian local ring and $\hat{R}$ its $\mathfrak{m}$-adic completion. Then R is regular if and only if $\hat{R}$ is regular.

***Proof*:**
This is proposition 29.7.6. The mechanism is that completion leaves the associated graded ring alone, $G_{\mathfrak{m}}(R) \cong G_{\hat{\mathfrak{m}}}(\hat{R})$, and leaves the dimension alone; the graded criterion for regularity depends on nothing else, so it holds for one exactly when it holds for the other, as we sought to show.

In this case $\hat{R}$ is again an integral domain.

Exercise 35.5.1

1. Compute the singular locus of $\mathcal{Z}(y^2 - x^3)$, of $\mathcal{Z}(y^2 - x^2 - x^3)$, and of $\mathcal{Z}(xy - z^2) \subseteq \mathbb{A}^3_k$, using proposition 35.2.2 in each case.
2. Show that the singular locus of a variety is a proper closed subset, so that the nonsingular points are dense. Where does the argument use irreducibility?
3. Let $X = \mathcal{Z}(f) \subseteq \mathbb{A}^n_k$ be a hypersurface. Show that $p \in X$ is nonsingular if and only if some $\partial f / \partial x_i$ fails to vanish at p, and give an example in characteristic $p > 0$ where every partial derivative vanishes identically on a nonconstant f.
4. Show that $\dim_k \mathfrak{m}_p / \mathfrak{m}_p^2 \geq \dim X$ at every point of a variety, with equality exactly at the nonsingular points, and identify which inequality of corollary 35.1.1 is being used.
5. Show that a regular local ring is a domain, and give an example of a Noetherian local domain that is not regular.
6. Show that the local ring of the node $\mathcal{Z}(y^2 - x^2 - x^3)$ at the origin is not a domain after completion, although it is a domain before. (This is why $\mathcal{O}_p$ alone cannot distinguish a node from a cusp, and why proposition 35.5.4 is stated for completions.)
7. Use theorem 35.4.3 to show that a regular local ring is a unique factorization domain in dimension one, and explain why the same argument does not immediately give the general case.

36

Tangent Cones and Completions

The Jacobian criterion of chapter 35 decides whether a point is smooth and stops there. It cannot tell the node $y^2 = x^2 + x^3$ from the cusp $y^2 = x^3$: both are singular at the origin, both have a one-dimensional variety with a two-dimensional cotangent space, and no invariant built from $\mathfrak{m}_p/\mathfrak{m}_p^2$ alone separates them. Yet they are visibly different curves, the node crossing itself and the cusp not.

This chapter builds the invariants that see the difference. The first keeps more of the filtration than $\mathfrak{m}_p/\mathfrak{m}_p^2$ does: the associated graded ring of $\mathcal{O}_{X,p}$, whose zero set is the cone of tangent directions and whose Hilbert polynomial is the multiplicity. The second is finer still. Passing to the completion $\hat{\mathcal{O}}_{X,p}$ makes available power series that converge in no algebraic sense, and in that larger ring the node's local equation factors while the cusp's does not; the factors are the branches of the curve through the point. The chapter closes by using regularity to show that on a smooth variety every codimension-one subvariety is locally cut out by one equation, which is the fact section 37.5 needs and could not prove.

36.1 The Tangent Cone and Multiplicity

Throughout, $X \subseteq \mathbb{A}^n_k$ is an affine variety and $p \in X$ a point, and after a translation p is the origin. Write $A = \Gamma(\mathbb{A}^n_k) = k[x_1, \ldots, x_n]$ and $\mathfrak{n} = (x_1, \ldots, x_n)$ for the maximal ideal at the origin.

Definition 36.1.1: Initial Form

For $0 \neq f \in A$ write $f = f_m + f_{m+1} + \cdots$ with each f_j homogeneous of degree j and $f_m \neq 0$. The *initial form* of f is $f_* = f_m$, and m is the *order* of f at the origin. For an ideal $I \subseteq A$, the *initial ideal* is $I_* = (f_* : 0 \neq f \in I)$, the ideal generated by all initial forms.

The initial form records the lowest-order behaviour of f at the origin, so it is the first place a curve's shape shows up: for the node $f = y^2 - x^2 - x^3$ it is $y^2 - x^2$, which factors into two distinct lines when $\operatorname{char} k \neq 2$, while for the cusp $f = y^2 - x^3$ it is y^2, a single line doubled. In characteristic 2 the node's initial form is $(y+x)^2$ and the distinction collapses, so that hypothesis is carried explicitly wherever the node is used. That the initial ideal is generated by initial forms and not merely by the initial forms of a generating set is a real distinction, and the next proposition is what makes the definition intrinsic in spite of it.

Definition 36.1.2: Tangent Cone

The *tangent cone algebra* of X at p is the associated graded ring

$$\operatorname{gr}_{\mathfrak{m}_p} \mathcal{O}_{X,p} = \bigoplus_{j \geq 0} \mathfrak{m}_p^j / \mathfrak{m}_p^{j+1}$$

a graded k-algebra. The *tangent cone* $TC_p(X) \subseteq \mathbb{A}_k^n$ is the affine algebraic set $\mathcal{Z}(I_*)$, where $I = \mathcal{I}(X)$.

Proposition 36.1.3: The Tangent Cone Algebra Is Intrinsic

With the notation above, there is an isomorphism of graded k-algebras

$$\operatorname{gr}_{\mathfrak{m}_p} \mathcal{O}_{X,p} \cong A/I_*$$

In particular the tangent cone algebra does not depend on the embedding $X \subseteq \mathbb{A}_k^n$, and $TC_p(X) = \mathcal{Z}(I_*)$ is the zero set of the tangent cone algebra.

***Proof*:**
Step 1: the local ring may be replaced by the coordinate ring. Put $R = \Gamma(X) = A/I$ with maximal ideal $\mathfrak{m} = \mathfrak{n}/I$ at the origin. Each $\mathfrak{m}^j/\mathfrak{m}^{j+1}$ is annihilated by $\mathfrak{m}$, hence is a vector space over $R/\mathfrak{m} = k$, and localizing at $\mathfrak{m}$ acts as the identity on such a module. So $\operatorname{gr}_{\mathfrak{m}_p} \mathcal{O}_{X,p} \cong \operatorname{gr}_{\mathfrak{m}} R$.

Step 2: the surjection and its kernel. The quotient $A \twoheadrightarrow R$ carries $\mathfrak{n}^j$ onto $\mathfrak{m}^j$, hence induces a surjection of graded rings $\operatorname{gr}_{\mathfrak{n}} A \twoheadrightarrow \operatorname{gr}_{\mathfrak{m}} R$, and $\operatorname{gr}_{\mathfrak{n}} A \cong A$ as graded rings since $\mathfrak{n}^j/\mathfrak{n}^{j+1}$ is the space of forms of degree j. In degree j the kernel is

$$\big((I \cap \mathfrak{n}^j) + \mathfrak{n}^{j+1}\big)/\mathfrak{n}^{j+1}$$

and it remains to identify this with the degree-j part of I_*.

Step 3: the kernel is I_*. For one inclusion, let $f \in I$ have initial form f_* of degree m and let g be homogeneous of degree $j - m$. Then $gf \in I$, and since A is a domain the initial form of gf is gf_*, of degree j; so $gf \in I \cap \mathfrak{n}^j$ and $gf \equiv gf_* \pmod{\mathfrak{n}^{j+1}}$. Hence every degree-$j$ element of I_* lies in the kernel. Conversely let $a \in I \cap \mathfrak{n}^j$. Its initial form a_* has degree at least j. If the degree exceeds j then $a \in \mathfrak{n}^{j+1}$ and a contributes 0; if it equals j then $a \equiv a_* \pmod{\mathfrak{n}^{j+1}}$ with $a_* \in I_*$. So the kernel is exactly the degree-j part of I_*, completing the proof.

The proposition is what licenses the name: the left-hand side is built from $\mathcal{O}_{X,p}$ alone and knows nothing of the embedding, so the right-hand side cannot depend on it either, even though I_* is written in the coordinates of $\mathbb{A}_k^n$.

One caution follows immediately, and it is the first place this book runs into the limitation that chapter 49 takes up. The tangent cone algebra A/I_* and the tangent cone $TC_p(X) = \mathcal{Z}(I_*)$ are not equivalent data: passing from the ideal to its zero set discards nilpotents, so the coordinate ring of $TC_p(X)$ is $A/\sqrt{I_*}$ rather than A/I_*. For the cusp $I_* = (y^2)$, so the tangent cone algebra is $k[x,y]/(y^2)$ while the tangent cone is the line $\mathcal{Z}(y)$, and the doubling is visible only in the algebra. The algebra is therefore the finer invariant, and it is the one the multiplicity is defined from.

Definition 36.1.4: Multiplicity

Let $d = \mathrm{kdim}\mathcal{O}_{X,p}$. The function $j \mapsto \ell(\mathcal{O}_{X,p}/\mathfrak{m}_p^j)$ agrees for $j \gg 0$ with a polynomial in j (proposition 29.3.2) whose degree is d (theorem 29.4.1), and whose leading coefficient may be written $e/d!$ for a positive integer e. This e is the *multiplicity* of X at p, written $m_p(X)$.

Two things in that definition need saying. First, the leading coefficient is $e/d!$ with e a positive integer because a polynomial taking integer values at all large integers is an integer combination $\sum_i c_i\binom{j}{i}$ of binomial coefficients, so its degree-d coefficient is $c_d/d!$ with $c_d \in \mathbb{Z}$; here the values $\ell(\mathcal{O}_{X,p}/\mathfrak{m}_p^j)$ are lengths, hence non-negative integers, and $c_d > 0$ since the polynomial has degree exactly d and is eventually increasing. Second, the first difference of $j \mapsto \ell(\mathcal{O}_{X,p}/\mathfrak{m}_p^j)$ is $\dim_k \mathfrak{m}_p^j/\mathfrak{m}_p^{j+1}$, the Hilbert function of the tangent cone algebra, so $m_p(X)$ is determined by that algebra, and by proposition 36.1.3 by the point alone.

Proposition 36.1.5: Multiplicity of a Hypersurface

Let $n \geq 2$ and let $X = \mathcal{Z}(f) \subseteq \mathbb{A}_k^n$ be a hypersurface with f irreducible, and let $p \in X$. Then $I_* = (f_*)$ and

$$m_p(X) = \deg f_* = \text{the order of } f \text{ at } p$$

***Proof*:**
Since f is irreducible, $\mathcal{I}(X) = (f)$ by the Nullstellensatz (theorem 32.4.4), (f) being prime hence radical. Every element of (f) is gf, and A being a domain gives $(gf)_* = g_*f_*$, so every initial form of an element of (f) is a multiple of f_*; hence $I_* = (f_*)$. Write $m = \deg f_*$.

By proposition 36.1.3 the tangent cone algebra is $A/(f_*)$, whose degree-j piece has dimension $\binom{j+n-1}{n-1} - \binom{j-m+n-1}{n-1}$ for $j \geq m$, multiplication by the form f_* being injective on a domain. This is a polynomial in j of degree $n-2$ with leading coefficient $m/(n-2)!$; the identity holds for $j \geq m$, and the finitely many earlier terms shift the sum below by a constant without affecting its degree or leading coefficient. Summing over j raises the degree by one, so $\ell(\mathcal{O}_{X,p}/\mathfrak{m}_p^j)$ has degree $n-1$ and leading coefficient $m/(n-1)!$. By the dimension theorem that degree is $\mathrm{kdim}\mathcal{O}_{X,p}$, so $d = n-1$, and definition 36.1.4 gives $m_p(X) = m$, as we sought to show.

Proposition 36.1.6: Smooth Points Have Multiplicity One

If p is a smooth point of X then $m_p(X) = 1$. For a hypersurface the converse holds: an irreducible hypersurface with $m_p(X) = 1$ is smooth at p.

***Proof*:**
If p is smooth then $\mathcal{O}_{X,p}$ is regular of dimension d (proposition 35.2.2), so its tangent cone algebra is the polynomial ring $k[t_1, \ldots, t_d]$ by theorem 29.7.3. Then $\dim_k \mathfrak{m}_p^j/\mathfrak{m}_p^{j+1} = \binom{j+d-1}{d-1}$, a polynomial of degree $d-1$ with leading coefficient $1/(d-1)!$, and summing gives leading coefficient $1/d!$; so $m_p(X) = 1$.

For the converse on a hypersurface $X = \mathcal{Z}(f)$, proposition 36.1.5 makes $m_p(X) = 1$ say that f_* is a nonzero linear form. That linear form is $\sum_i (\partial f/\partial x_i)(p)\, x_i$ up to translation, so some partial derivative of f is nonzero at p and the Jacobian has rank one there, which is smoothness by definition 35.2.1, completing the proof.

The converse in fact holds for every variety, not only for hypersurfaces, but its proof needs an unmixedness theorem of Nagata that this book does not develop; see [37]. What the general statement does require is irreducibility. On a reducible algebraic set multiplicity one can fail to force regularity: for $\mathcal{Z}(xz, yz) \subseteq \mathbb{A}^3_k$, a plane meeting a line, the local ring at the origin has $\dim_k \mathfrak{m}^j/\mathfrak{m}^{j+1} = j+2$ for $j \geq 1$, so $\ell(R/\mathfrak{m}^j) = j^2/2 + 3j/2 - 1$ and $m = 1$ at a point where the ring is not even a domain.

Example 36.1.7: Tangent Cones of the Node, the Cusp, and the Quadric Cone

1. *The node* $C = \mathcal{Z}(y^2 - x^2 - x^3)$ at the origin. The initial form is $y^2 - x^2 = (y-x)(y+x)$, so the tangent cone is the pair of lines $\mathcal{Z}(y-x) \cup \mathcal{Z}(y+x)$ and $m_0(C) = 2$. The two lines are the two directions in which the curve leaves the origin.

2. *The cusp* $C = \mathcal{Z}(y^2 - x^3)$ at the origin. The initial form is y^2, so $m_0(C) = 2$ as well, but the tangent cone algebra is $k[x,y]/(y^2)$ and the tangent cone is the single line $\mathcal{Z}(y)$, doubled. So the multiplicity does not separate the node from the cusp, while the tangent cone does: two distinct lines against one line with multiplicity two. This is the distinction example 35.2.3 promised.

3. *The quadric cone* $X = \mathcal{Z}(xy - z^2) \subseteq \mathbb{A}^3_k$ at the origin. Here f is already homogeneous, so $f_* = f$ and the tangent cone is X itself: a cone is its own tangent cone at its vertex. By proposition 36.1.5, $m_0(X) = 2$, and by proposition 36.1.6 the vertex is singular, which exercise set 35.5.1 found by computing the Jacobian.

4. *A smooth point.* For the parabola $\mathcal{Z}(y - x^2)$ at the origin the initial form is y, the tangent cone is the tangent line $\mathcal{Z}(y)$, and $m_0 = 1$. At a smooth point the tangent cone is always the tangent space: theorem 29.7.3 makes the tangent cone algebra a polynomial ring on d degree-one generators, so I_* is generated by linear forms.

For a plane curve the tangent cone is a form in two variables, hence a product of linear factors over the algebraically closed k; its factors are the *tangent lines* at the point and their exponents are the tangent multiplicities. That is the description chapter 42 works with, where the multiplicity of a plane curve at a point is defined directly as the order of its defining polynomial. For an *irreducible* f the two definitions agree, by proposition 36.1.5. They part company otherwise, and deliberately so: that chapter takes a plane curve to be a polynomial up to scalar rather than a zero set, so $\mathcal{Z}(y^2)$ carries multiplicity two there while as a variety it is the line $\mathcal{Z}(y)$, of multiplicity one.

36.2 Completion and Local Parameters

The tangent cone keeps more of the local ring than $\mathfrak{m}_p/\mathfrak{m}_p^2$ does, but it is still a linearization: it records the graded pieces $\mathfrak{m}_p^j/\mathfrak{m}_p^{j+1}$ and forgets how they are glued. The completion forgets nothing. It is the inverse limit of the quotients $\mathcal{O}_{X,p}/\mathfrak{m}_p^j$, so an element of it is a compatible system of Taylor approximations to arbitrary order, and the ring it lives in is large enough to contain power series that no polynomial fraction represents.

The construction of the $\mathfrak{m}$-adic completion, its Noetherianity, and the Cohen structure theorem are in *Commutative Algebra* [10, Completion Of A Ring] and are used here without redevelopment. Write $\hat{\mathcal{O}}_{X,p} = \varprojlim_j \mathcal{O}_{X,p}/\mathfrak{m}_p^j$ and $\hat{\mathfrak{m}}_p$ for its maximal ideal. The natural map $\mathcal{O}_{X,p} \to \hat{\mathcal{O}}_{X,p}$ is injective, since its kernel is $\bigcap_j \mathfrak{m}_p^j$, which vanishes in a Noetherian local ring by corollary 25.4.12. So completing loses no functions and adds many.

Theorem 36.2.1: The Completion at a Smooth Point

Let X be a variety over the algebraically closed field k and let $p \in X$ be a smooth point with $\dim_p X = d$. Then

$$\hat{\mathcal{O}}_{X,p} \cong k[[t_1, \ldots, t_d]]$$

as k-algebras, and the t_i may be taken to be the images of any regular system of parameters of $\mathcal{O}_{X,p}$.

***Proof*:**
Write $(R, \mathfrak{m}) = (\mathcal{O}_{X,p}, \mathfrak{m}_p)$, a Noetherian local ring of dimension d. Its residue field is k: it equals $k[X]/\mathcal{I}(p)$ for an affine neighbourhood, and Zariski's lemma (theorem 32.4.2) makes that a finite extension of the algebraically closed k, hence k itself. Since p is smooth, R is regular (proposition 35.2.2).

Step 1: the completion is complete regular local of dimension d. The ring $\hat{R}$ is Noetherian local with maximal ideal $\hat{\mathfrak{m}}$ and the same residue field k, by theorem 25.5.7, and $\mathrm{kdim}\hat{R} = \mathrm{kdim}R = d$ by corollary 25.5.9. It is regular by proposition 35.5.4.

Step 2: Cohen's theorem gives a presentation. The ring $\hat{R}$ is complete Noetherian local, contains k, and has residue field k, so theorem 25.6.1 presents it as

$$\hat{R} \cong k[[x_1, \ldots, x_n]]/I, \qquad n = \dim_k \hat{\mathfrak{m}}/\hat{\mathfrak{m}}^2$$

Regularity of $\hat{R}$ makes $n = \mathrm{kdim}\hat{R} = d$, so the presentation is $k[[x_1, \ldots, x_d]]/I$, and Cohen's theorem takes the x_i to any lift of a basis of $\hat{\mathfrak{m}}/\hat{\mathfrak{m}}^2$, which a regular system of parameters of R provides.

Step 3: the ideal vanishes. This is lemma 36.2.2 below, applied to $\hat{R}$, completing the proof.

The last step is worth isolating, since it is used again in section 36.3 and its argument is independent of the geometry.

Lemma 36.2.2: A Complete Regular Local Ring Is a Power Series Ring

Let $(T, \mathfrak{n})$ be a complete Noetherian regular local ring of dimension d containing a field k with $T/\mathfrak{n} = k$. Then $T \cong k[[x_1, \ldots, x_d]]$, the variables mapping to any regular system of parameters.

***Proof*:**
theorem 25.6.1 presents $T \cong S/I$ with $S = k[[x_1, \ldots, x_n]]$ and $n = \dim_k \mathfrak{n}/\mathfrak{n}^2$, which regularity makes d; its proof sends the variables to any lift of a basis of $\mathfrak{n}/\mathfrak{n}^2$, that is to any regular system of parameters. It remains to show $I = 0$.

First, S is a domain. A nonzero element of S has a well-defined initial form, its lowest-degree homogeneous part, and the initial form of a product is the product of the initial forms because the polynomial ring $k[x_1, \ldots, x_d]$ is a domain; so a product of nonzero elements has a nonzero initial form and is nonzero. (Completion does not preserve domains in general, so this needs the argument rather than a citation.)

Second, $\mathrm{kdim} S = d$: it is the completion of $k[x_1, \ldots, x_d]_{(x_1, \ldots, x_d)}$, whose dimension is d by corollary 34.2.7 and proposition 34.3.2, and corollary 25.5.9 applies.

Now suppose $I \neq 0$ and pick $0 \neq f \in I$; it is a nonunit, I being proper. Any chain of primes $\mathfrak{q}_0 \subsetneq \cdots \subsetneq \mathfrak{q}_e$ in S/I lifts to a chain of primes of S all containing f, hence all nonzero, so prepending the zero ideal, legitimate as S is a domain, gives a chain of length $e+1$ in S. Therefore $e + 1 \leq \mathrm{kdim} S = d$, so $\mathrm{kdim} S/I \leq d - 1$. But $S/I \cong T$ has dimension d, a contradiction. So $I = 0$, as we sought to show.

Definition 36.2.3: Local Parameters

A *system of local parameters* at a smooth point p of a d-dimensional variety is a regular system of parameters $t_1, \ldots, t_d$ of $\mathcal{O}_{X,p}$, that is, d elements of $\mathfrak{m}_p$ generating it. For a curve, $d = 1$ and a single generator t of $\mathfrak{m}_p$ is a *local parameter* or *uniformizer* at p.

Corollary 36.2.4: Power Series Expansion

Let p be a smooth point of X with local parameters $t_1, \ldots, t_d$. Every $f \in \mathcal{O}_{X,p}$ has a unique expansion as a formal power series in $t_1, \ldots, t_d$ with coefficients in k, and $f \in \mathfrak{m}_p^j$ if and only if that series has no terms of degree less than j.

***Proof*:**
The composite $\mathcal{O}_{X,p} \hookrightarrow \hat{\mathcal{O}}_{X,p} \cong k[[t_1, \ldots, t_d]]$ is injective by the remark before theorem 36.2.1, which gives existence and uniqueness of the expansion. Under the isomorphism $\hat{\mathfrak{m}}_p^j$ corresponds to the series of order at least j, so it remains to see that $\mathcal{O}_{X,p} \cap \hat{\mathfrak{m}}_p^j = \mathfrak{m}_p^j$. For a Noetherian local ring, corollary 25.1.5 identifies $\hat{\mathcal{O}}/\hat{\mathfrak{m}}_p^j$ with $\mathcal{O}_{X,p}/\mathfrak{m}_p^j$ compatibly with the map from $\mathcal{O}_{X,p}$, so that map composed with the projection is the projection $\mathcal{O}_{X,p} \to \mathcal{O}_{X,p}/\mathfrak{m}_p^j$, whose kernel is $\mathfrak{m}_p^j$, completing the proof.

For a curve the corollary is the statement that a regular function near a smooth point is a power series in the uniformizer, and the order of that series is the order of vanishing; definition 42.2.4 will take it as the definition. In higher dimension it is the algebraic substitute for a coordinate chart: a smooth point has no neighbourhood isomorphic to an open set of $\mathbb{A}^d_k$ in the Zariski topology, but it always has a formal one.

Example 36.2.5: Completions at Smooth and Singular Points

1. *Affine space.* At the origin of $\mathbb{A}^d_k$ the coordinates $x_1, \ldots, x_d$ are local parameters and $\hat{\mathcal{O}} \cong k[[x_1, \ldots, x_d]]$, which is example 25.1.2 read geometrically.
2. *A smooth plane curve.* On $C = \mathcal{Z}(y - x^2)$ the function x generates $\mathfrak{m}_0$, since $y = x^2$ in $\Gamma(C)$, so x is a uniformizer and $\hat{\mathcal{O}}_{C,0} \cong k[[x]]$. Every regular function near the origin is a power series in x alone, the variable y contributing nothing new.
3. *The parabola meets its axis.* On the same curve the function y has expansion x^2, so $\mathrm{ord}_0(y) = 2$: the axis $\mathcal{Z}(y)$ meets C doubly at the origin. That doubling is the intersection multiplicity chapter 42 will count.
4. *A singular point is not a power series ring.* At the origin of the cusp $\mathcal{Z}(y^2 - x^3)$ the ring $\hat{\mathcal{O}}$ is $k[[x, y]]/(y^2 - x^3)$, computed in proposition 36.3.1 below. It is not isomorphic to $k[[t]]$, since $\dim_k \hat{\mathfrak{m}}/\hat{\mathfrak{m}}^2 = 2$ there while $k[[t]]$ has $\dim_k (t)/(t^2) = 1$; theorem 36.2.1 of course forbids it, the origin being singular.

36.3 Formal Branches

The completion separates the node from the cusp, and it does so in the most direct way available: at the node the local equation factors, and at the cusp it does not. Neither factorization exists in $\mathcal{O}_{X,p}$, because the factors involve a square root that is a power series and not a rational function. This section makes that precise and attaches to each singular point of a curve a number, the count of branches through it.

Proposition 36.3.1: The Completion of a Plane Curve

Let $C = \mathcal{Z}(f) \subseteq \mathbb{A}^2_k$ be a plane curve with f irreducible and $p = (0, 0) \in C$. Then

$$\hat{\mathcal{O}}_{C,p} \cong k[[x, y]]/(f)$$

***Proof*:**
Write $A = k[x, y]$, $\mathfrak{n} = (x, y)$ and $R = \Gamma(C) = A/(f)$, with $\mathfrak{m}$ the image of $\mathfrak{n}$.

Step 1: the local ring may be replaced by the coordinate ring. For every j the ideal $\mathfrak{m}^j$ is $\mathfrak{m}$-primary, so $R/\mathfrak{m}^j$ is already local with maximal ideal $\mathfrak{m}/\mathfrak{m}^j$ and localizing at $\mathfrak{m}$ does nothing: $R/\mathfrak{m}^j \cong R_\mathfrak{m}/(\mathfrak{m}R_\mathfrak{m})^j$. Taking inverse limits, the $\mathfrak{m}$-adic completion of R agrees with the $\mathfrak{m}_p$-adic completion of $\mathcal{O}_{C,p} = R_\mathfrak{m}$. The same argument identifies the $\mathfrak{n}$-adic completion of A with that of $A_\mathfrak{n}$, and the former is $k[[x, y]]$ by example 25.1.2.

Step 2: complete the presentation. Completion is exact on finitely generated modules over a Noetherian ring (corollary 25.4.4), so applying it to $0 \to (f) \to A \to R \to 0$ gives $\hat{R} \cong k[[x,y]]/(f)k[[x,y]]$, and $(f)k[[x,y]]$ is the ideal generated by f there. Combining with Step 1 gives the claim, as we sought to show.

Definition 36.3.2: Formal Branch

Let p be a point of a variety X. A *formal branch* of X at p is a minimal prime of $\hat{\mathcal{O}}_{X,p}$. The number of formal branches is the number of such primes, which is finite because $\hat{\mathcal{O}}_{X,p}$ is Noetherian.

A smooth point has one branch, since $\hat{\mathcal{O}}_{X,p}$ is then a power series ring and hence a domain, whose unique minimal prime is 0. The content of the definition is at singular points, where a curve may pass through p more than once.

Proposition 36.3.3: The Node Has Two Branches, the Cusp One

At the origin,

1. if $\operatorname{char} k \neq 2$, the node $N = \mathcal{Z}(y^2 - x^2 - x^3)$ has exactly two formal branches, and $\hat{\mathcal{O}}_{N,0} \cong k[[u,v]]/(uv)$;
2. in every characteristic, the cusp $C = \mathcal{Z}(y^2 - x^3)$ has exactly one formal branch, and $\hat{\mathcal{O}}_{C,0}$ is a domain that is not a power series ring.

***Proof*:**
The node. In $k[[x]]$ the series $1+x$ has a square root when $\operatorname{char} k \neq 2$. Seek $w = 1 + \sum_{n\geq 1} a_n x^n$ with $w^2 = 1+x$; comparing coefficients gives $2a_1 = 1$ and, for $n \geq 2$,

$$2a_n = -\sum_{0<i<n} a_i a_{n-i}$$

Each equation determines a_n from its predecessors, division by 2 being the only division performed, so w exists, and it is a unit since $w(0) = 1$. Then in $k[[x,y]]$

$$y^2 - x^2 - x^3 = y^2 - x^2(1+x) = (y - xw)(y + xw)$$

Put $u = y - xw$ and $v = y + xw$. Both have order 1 with linear parts $y - x$ and $y + x$, which are linearly independent as $\operatorname{char} k \neq 2$, so their classes span $\mathfrak{n}/\mathfrak{n}^2$ for $\mathfrak{n}$ the maximal ideal of $k[[x,y]]$, and Nakayama's lemma (lemma 23.4.27) makes u, v generate $\mathfrak{n}$. Thus u, v is a regular system of parameters of the complete regular local ring $k[[x,y]]$, and lemma 36.2.2 gives $k[[x,y]] = k[[u,v]]$. By proposition 36.3.1, $\hat{\mathcal{O}}_{N,0} \cong k[[u,v]]/(uv)$, whose minimal primes are (u) and (v): these are prime because the quotients $k[[v]]$ and $k[[u]]$ are domains, they are minimal because $(uv) \subseteq \mathfrak{p}$ forces $u \in \mathfrak{p}$ or $v \in \mathfrak{p}$, and they are distinct. So there are exactly two branches.

The cusp. By proposition 36.3.1, $\hat{\mathcal{O}}_{C,0} \cong k[[x,y]]/(y^2 - x^3)$, so it suffices to show $y^2 - x^3$ is irreducible in $k[[x,y]]$; then $(y^2 - x^3)$ is prime, it is the unique minimal prime of the quotient, and the quotient is a domain. Suppose $y^2 - x^3 = gh$ with g, h nonunits. Setting $x = 0$ gives $y^2 = g(0,y)h(0,y)$ in $k[[y]]$, so the orders of $g(0,y)$ and $h(0,y)$ in y sum to 2; neither can be 0, since a factor with $g(0,y)$ a unit would be a unit in $k[[x,y]]$, so both orders are 1. The

Weierstrass preparation theorem (theorem 25.7.3) then writes each as a unit times a monic degree-one polynomial in y over $k[[x]]$, so

$$y^2 - x^3 = \varepsilon\,(y - a(x))(y - b(x)), \qquad a, b \in (x) \subseteq k[[x]]$$

with ε a unit. Here $(y-a)(y-b)$ is a Weierstrass polynomial of degree two, its lower coefficients lying in (x), and so is $y^2 - x^3$ itself; the uniqueness clause of the same theorem therefore forces $\varepsilon = 1$. Comparing coefficients gives $a + b = 0$ and $ab = -x^3$, so $b = -a$ and $a^2 = x^3$. Taking orders in $k[[x]]$ gives $2\,\mathrm{ord}(a) = 3$, impossible. So $y^2 - x^3$ is irreducible.

Finally $\hat{\mathcal{O}}_{C,0}$ is not a power series ring, since $\dim_k \hat{\mathfrak{m}}/\hat{\mathfrak{m}}^2 = 2$ while $\mathrm{kdim} = 1$, completing the proof.

Definition 36.3.4: Analytically Isomorphic Singularities

Two pointed varieties (X, p) and (Y, q) are *analytically isomorphic* if $\hat{\mathcal{O}}_{X,p} \cong \hat{\mathcal{O}}_{Y,q}$ as k-algebras.

The computation says more than that the node has two branches: it says the node is analytically isomorphic to the crossing of the two coordinate axes $\mathcal{Z}(xy)$ at the origin, since both completions are $k[[u,v]]/(uv)$, while the cusp belongs to a different class. This is a genuinely finer relation than isomorphism of neighbourhoods: the node and the pair of axes are not isomorphic as varieties near the origin, one being irreducible and the other not, and it is exactly the passage to the completion that makes them agree. The same argument applies to any plane curve whose tangent cone at p is two distinct lines, which definition 42.1.6 will call an ordinary double point.

The characteristic hypothesis in the first case is not a technicality. In characteristic two, $y^2 - x^2 - x^3 = (y+x)^2 - x^3$, so substituting $u = y + x$ turns the node's equation into $u^2 - x^3$: the curve is analytically a cusp there, with one branch rather than two. The tangent cone sees the same collapse, its initial form being the doubled line $(y+x)^2$.

The count of branches is a local invariant that a global construction also computes: the normalization of a curve separates its branches, so the points lying over p should correspond to them. Making that precise needs the normalization, which is chapter 37, and the comparison is carried out there as proposition 37.2.4.

36.4 Local Factoriality and Weil Divisors

The last local property of this chapter is the one section 37.5 depends on. A Weil divisor on a normal variety is a formal sum of codimension-one subvarieties, and the class group measures how far such a sum is from being the divisor of a function. That measurement is only meaningful once one knows when a codimension-one subvariety is cut out by a single equation near each point, and the answer is that smoothness suffices.

Corollary 36.4.1: A Smooth Variety Is Locally Factorial

Let X be a variety and $p \in X$ a smooth point. Then $\mathcal{O}_{X,p}$ is a unique factorization domain.

***Proof*:**
By proposition 35.2.2 the ring $\mathcal{O}_{X,p}$ is regular local, and a regular local ring is factorial by theorem 29.8.5, as we sought to show.

Theorem 36.4.2: Codimension One Is Locally Principal at a Smooth Point

Let X be a variety, $Y \subseteq X$ an irreducible closed subvariety of codimension one, and $p \in Y$ a point at which X is smooth. Then the ideal of Y in $\mathcal{O}_{X,p}$ is principal: there is $f \in \mathcal{O}_{X,p}$ generating it.

***Proof*:**
Step 1: reduce to the affine case. Choose an affine open $U \subseteq X$ containing p. Passing to U changes neither $\mathcal{O}_{X,p}$ nor the codimension, so $Y \cap U$ is an irreducible closed subvariety of U of codimension one, and its ideal $\mathcal{I}(Y \cap U) \subseteq k[U]$ is prime and contained in the maximal ideal at p, the point p lying on Y.

Step 2: the ideal has height one. By definition 34.3.1 the codimension of $Y \cap U$ in U is the supremum of lengths of chains of irreducible closed subsets ascending from $Y \cap U$, and the correspondence between irreducible closed subsets and prime ideals reverses inclusions (theorem 32.4.4), so such chains correspond to chains of primes descending from $\mathcal{I}(Y \cap U)$. Hence $\operatorname{ht} \mathcal{I}(Y \cap U) = \operatorname{codim}_U(Y \cap U) = 1$. Localizing at p preserves both primality and height, since the primes contained in $\mathcal{I}(Y \cap U)$ are exactly those surviving in the localization.

Step 3: apply factoriality. The ring $R = \mathcal{O}_{X,p}$ is a Noetherian local domain by proposition 33.2.10 and factorial by corollary 36.4.1, p being a smooth point. A height-one prime of a Noetherian factorial domain is principal, by lemma 29.8.1, which gives the generator f, completing the proof.

The theorem is what makes the class group of section 37.5 a global invariant on a smooth variety: every prime divisor is locally the zero set of one function, so a nonzero class records an obstruction to patching those local equations into a single global one, not a failure at any individual point. On a singular normal variety the obstruction is already local, and the quadric cone is the standard example.

Proposition 36.4.3: The Quadric Cone Is Not Locally Factorial at Its Vertex

Let $X = \mathcal{Z}(xy - z^2) \subseteq \mathbb{A}^3_k$ with vertex v the origin. Then $\mathcal{O}_{X,v}$ is not a unique factorization domain, and the ideal of the ruling $L = \mathcal{Z}(x, z) \subseteq X$ is a height-one prime of $\mathcal{O}_{X,v}$ that is not principal.

***Proof*:**
Write $R = \mathcal{O}_{X,v}$ with maximal ideal $\mathfrak{m}$, and let $\overline{x}, \overline{y}, \overline{z}$ be the images of the coordinates. Each lies in $\mathfrak{m} \setminus \mathfrak{m}^2$: the images of x, y, z form a basis of $\mathfrak{m}/\mathfrak{m}^2$, because the defining equation $xy - z^2$ lies in $(x, y, z)^2$ and so imposes no linear condition on $\mathfrak{m}/\mathfrak{m}^2$. An element of $\mathfrak{m} \setminus \mathfrak{m}^2$ is irreducible in R: a factorization into two nonunits would place it in $\mathfrak{m}^2$. So $\overline{x}, \overline{y}, \overline{z}$ are irreducible, and no two are associates, their classes in $\mathfrak{m}/\mathfrak{m}^2$ being linearly independent while an associate would differ by a unit and hence have proportional class.

The relation $\overline{x}\,\overline{y} = \overline{z}^{\,2}$ therefore exhibits two factorizations of one element into irreducibles that do not match up to associates and reordering, so R is not factorial. Here R is a Noetherian local domain by proposition 33.2.10, the cone being a variety, so lemma 29.8.1 applies and some height-one prime of R fails to be principal.

That prime may be named. The ideal $\mathfrak{p} = (\overline{x}, \overline{z})$ is the ideal of L in R, and it has height one, L having codimension one in X. Suppose $\mathfrak{p} = (f)$. Then f divides $\overline{x}$ and is a nonunit, and $\overline{x}$ is irreducible, so $(f) = (\overline{x})$. From $\overline{z} \in (\overline{x})$ write $\overline{z} = \overline{x}h$; then

$$\overline{x}\,\overline{y} = \overline{z}^{\,2} = \overline{x}^{\,2}h^2$$

and cancelling $\overline{x}$, legitimate in a domain, gives $\overline{y} = \overline{x}h^2$. As $\overline{y}$ is irreducible and $\overline{x}$ is not a unit, h^2 must be a unit, making $\overline{y}$ and $\overline{x}$ associates and contradicting what was shown above. So $\mathfrak{p}$ is not principal, as we sought to show.

The failure is concentrated at the vertex alone: away from v the cone is smooth, so theorem 36.4.2 applies at every other point of L and makes the ideal of L principal there. Note that L is nonetheless the zero set of a single function, since $\mathcal{Z}(x) \cap X = \mathcal{Z}(x, z^2) \cap X = L$ as a set; what fails is that no one function generates the ideal near v, and $\operatorname{div}(x) = 2L$ in section 37.5 is the quantitative form of that failure, the function that does vanish exactly on L vanishing to order two.

Exercise 36.4.1

1. Compute the tangent cone and the multiplicity at the origin of $C = \mathcal{Z}(y^2 - x^5) \subseteq \mathbb{A}^2_k$, and show that C is irreducible. How do the answers compare with the cusp $\mathcal{Z}(y^2 - x^3)$, and what does the comparison say about whether the tangent cone alone classifies plane singularities?
2. Let $\operatorname{char} k \neq 3$ and let $F = \mathcal{Z}(x^3 + y^3 - 3xy) \subseteq \mathbb{A}^2_k$ be the folium. Compute its tangent cone and multiplicity at the origin, and decide whether the origin is an ordinary double point.
3. Show that if $f \in k[x_1, \dots, x_n]$ is homogeneous and irreducible, then the tangent cone of $\mathcal{Z}(f)$ at the origin is $\mathcal{Z}(f)$ itself. Why is this what the word *cone* is meant to record?
4. Let p be a smooth point of a variety X with $\dim_p X = d$. Show that $TC_p(X)$ is a linear subspace of $\mathbb{A}^n_k$ of dimension d, and identify it with the tangent space.
5. The node $N = \mathcal{Z}(y^2 - x^2 - x^3)$ passes through $(-1, 0)$ as well as the origin. Show that N is smooth at $(-1, 0)$ and compute $m_{(-1,0)}(N)$ and the tangent cone there.
6. Let $\operatorname{char} k \neq 2$ and let $C = \mathcal{Z}(x^2 + y^2 - 1)$. Show that y is a local parameter at $(1, 0)$ and that $\hat{\mathcal{O}}_{C,(1,0)} \cong k[[y]]$. Write the expansion of $x - 1$ as a power series in y to order three.
7. Let C be a curve and $p \in C$. Show that if $\hat{\mathcal{O}}_{C,p} \cong k[[t]]$ then p is a smooth point of C, so the converse of theorem 36.2.1 holds in dimension one.
8. Show that $k[[x, y]]/(xy)$ has exactly two minimal primes, and conclude that the union of the two coordinate axes has two formal branches at the origin.
9. Show that analytically isomorphic pointed varieties have the same multiplicity and the same number of formal branches. Deduce that the node and the cusp are not analytically isomorphic when $\operatorname{char} k \neq 2$, and explain what happens to this deduction in characteristic two.

10. Show that every irreducible closed subvariety of $\mathbb{A}_k^n$ of codimension one is the zero set of a single irreducible polynomial, and that its ideal is principal in every local ring of $\mathbb{A}_k^n$. Which is the stronger statement, and why does theorem 36.4.2 not already give it?

11. Let p be a smooth point of X. Show that $\mathcal{O}_{X,p}$ is integrally closed in its fraction field, so that X is normal at every smooth point.

12. For the quadric cone $X = \mathcal{Z}(xy - z^2)$ and the ruling $L = \mathcal{Z}(x, z)$, show directly that $\mathcal{I}(L)$ is a prime of height one in $k[X]$, and that $\mathcal{Z}(x) \cap X = L$ as sets.

13. Show that the tangent cone algebra of the cusp at the origin is $k[x, y]/(y^2)$, and compute $\dim_k \mathfrak{m}^j/\mathfrak{m}^{j+1}$ for every j. Confirm that the multiplicity read from this Hilbert function is 2.

14. Let R be the local ring of $\mathcal{Z}(xz, yz) \subseteq \mathbb{A}_k^3$ at the origin, a plane meeting a line. Show $\dim_k \mathfrak{m}^j/\mathfrak{m}^{j+1} = j + 2$ for $j \geq 1$, deduce $m = 1$, and explain why this does not contradict proposition 36.1.6.

37

Normal Varieties and Normalization

A variety can fail to be smooth in two quite different ways, and only one of them is repairable by an isomorphism of the function field. Normality is the condition that isolates the repairable case: a variety is normal when its coordinate ring is integrally closed in its function field, and every variety admits a normalization, a finite birational map onto it from a normal one.

For curves normality and smoothness coincide, and the normalization is the smooth model. That gives this chapter's structural result: a field of transcendence degree one over k determines a unique nonsingular projective curve, so a curve and its function field are the same information.

37.1 Integrally Closed Domains and Normal Varieties

The map $\varphi\colon \mathbb{A}^1_k \to \mathcal{Z}(x^3 - y^2)$ given by $t \mapsto (t^2, t^3)$ of example 31.4.8 is a bijective birational morphism that is not an isomorphism. Its inverse is y/x, a rational function that is not regular at the origin. That failure has a purely algebraic signature. The coordinate ring is $k[t^2, t^3]$, its function field is $k(t)$, and the missing element $t = t^3/t^2$ satisfies

$$T^2 - t^2 = 0$$

a monic equation over $k[t^2, t^3]$. So t is integral over the coordinate ring without belonging to it: the ring is not integrally closed in its own fraction field. Normality is the condition that this cannot happen, and it will turn out to be exactly the condition under which the pathology above is absent.

Integral dependence, integral closure, and the term *normal domain* are developed in *Commutative Algebra* [10, Integral Elements] and are used here without redevelopment.

Definition 37.1.1: Normal Variety

Let X be an affine variety, so that $k[X]$ is an integral domain with fraction field $k(X)$. Then X is *normal* if $k[X]$ is integrally closed in $k(X)$.

More locally, X is *normal at* $p \in X$ if the stalk $\mathcal{O}_p(X)$ is integrally closed in $k(X)$.

The two notions agree, which is what makes normality a usable local condition rather than a global accident.

Proposition 37.1.2: Normality Is Local

An affine variety X is normal if and only if it is normal at every point.

***Proof*:**
By proposition 33.2.9, $k[X] = \bigcap_{p \in X} \mathcal{O}_p(X)$, and each $\mathcal{O}_p(X)$ is the localization of $k[X]$ at the maximal ideal $\mathcal{I}(p)$. Integral closure commutes with localization (corollary 26.1.10), which gives exactly the statement that a domain is normal precisely when all of its localizations at maximal ideals are, as we sought to show.

The first supply of normal varieties comes from a purely algebraic observation.

Proposition 37.1.3: Unique Factorization Implies Normal

Every unique factorization domain is integrally closed in its fraction field.

***Proof*:**
Let R be a UFD with fraction field K and let $x = a/b \in K$ be integral over R, written in lowest terms so that a and b share no irreducible factor. An integral relation

$$\left(\frac{a}{b}\right)^n + r_{n-1}\left(\frac{a}{b}\right)^{n-1} + \cdots + r_0 = 0 \qquad r_i \in R$$

becomes, after multiplying through by b^n,

$$a^n = -b\left(r_{n-1}a^{n-1} + r_{n-2}a^{n-2}b + \cdots + r_0 b^{n-1}\right)$$

So every irreducible factor of b divides a^n, and being prime in a UFD, divides a. As a and b are coprime, b has no irreducible factors at all, so b is a unit and $x \in R$, as we sought to show.

Example 37.1.4: Normal and Non-Normal Varieties

1. Affine space $\mathbb{A}^n_k$ is normal. Its coordinate ring is the polynomial ring $k[x_1, \ldots, x_n]$, a UFD (corollary 11.2.4), hence normal by proposition 37.1.3.
2. The cuspidal cubic $C = \mathcal{Z}(x^3 - y^2) \subseteq \mathbb{A}^2_k$ is not normal, by the computation opening this section: $t = y/x$ lies in $k(C)$ and is integral over $k[C]$ but not in it. The failure is concentrated at one point, the cusp: away from the origin x is invertible, so y/x is already

regular there and C is normal at every other point.

3. The nodal cubic $N = \mathcal{Z}(y^2 - x^2 - x^3)$ is not normal either. Its parameterization $t \mapsto (t^2-1, t(t^2-1))$ from example 33.3.8 identifies $k(N)$ with $k(t)$, and $t = y/x$ is again integral over $k[N]$ without lying in it. Here the failure sits at the node, where two branches of the curve cross.

4. The quadric cone $Q = \mathcal{Z}(xy - z^2) \subseteq \mathbb{A}^3_k$ *is* normal, and it is singular at the origin. Section 37.4 returns to this: in dimension one normality and smoothness coincide, but from dimension two onward normality is strictly weaker.

Items (2) and (3) share a shape. In both, the curve is the image of a map from a smooth curve that is injective away from a small set, and the element of the function field that fails to be regular is the one that undoes the map. Item (4) shows the shape is special to curves.

37.2 The Normalization of an Affine Variety

Given a variety that is not normal, one can ask for the closest normal variety above it. The construction is forced: take the integral closure of the coordinate ring, and read it back as a variety. The only thing that needs proof is that the result is again the coordinate ring of a variety, and that is exactly the finiteness theorem for integral closures.

Theorem 37.2.1: Existence of the Normalization

Let X be an affine variety and let $\widetilde{k[X]}$ be the integral closure of $k[X]$ in $k(X)$. Then there is an affine variety $\tilde{X}$, the *normalization* of X, with

$$k[\tilde{X}] = \widetilde{k[X]}$$

together with a morphism $\nu\colon \tilde{X} \to X$ induced by the inclusion $k[X] \hookrightarrow \widetilde{k[X]}$. Moreover:

1. $\tilde{X}$ is normal;
2. $k[\tilde{X}]$ is a finite $k[X]$-module;
3. ν is surjective with finite fibres, and induces an isomorphism $k(\tilde{X}) \cong k(X)$, so ν is birational;
4. ν restricts to an isomorphism over the set of points at which X is already normal.

***Proof*:**
Step 1: the ring is a coordinate ring. $k[X]$ is a domain, finitely generated as a k-algebra, so theorem 26.1.11 applies with $L = \mathrm{Frac}(k[X]) = k(X)$: the integral closure $\widetilde{k[X]}$ is a finite $k[X]$-module, and in particular is itself a finitely generated k-algebra. This gives (2). Being a subring of the field $k(X)$ it is a domain, hence reduced, so proposition 32.5.5 realizes it as the coordinate ring of an affine algebraic set $\tilde{X}$, which is a variety by corollary 32.3.4. The inclusion of k-algebras corresponds under theorem 32.1.5 to a morphism $\nu\colon \tilde{X} \to X$.

Step 2: normality. An element of $k(\tilde{X})$ integral over $\widetilde{k[X]}$ is integral over $k[X]$ by transitivity of integral dependence, hence already lies in $\widetilde{k[X]}$. So $\widetilde{k[X]}$ is integrally closed and $\tilde{X}$ is normal, giving (1).

Step 3: birational and surjective. Every element of $\widetilde{k[X]}$ lies in $k(X)$, so $k(\tilde{X}) = \mathrm{Frac}(\widetilde{k[X]}) = k(X)$ and ν is birational. Since $k[\tilde{X}]$ is integral over $k[X]$, the going-up property gives a prime of $k[\tilde{X}]$ over every prime of $k[X]$, and in particular a point of $\tilde{X}$ over every point of X; finiteness of the fibres follows because $k[\tilde{X}]$ is a finite $k[X]$-module. This is (3), and it is the first instance of what section 40.3 will call a *finite morphism*.

Step 4: isomorphism over the normal locus. Suppose X is normal at p, so $\mathcal{O}_p(X)$ is integrally closed. Localizing the inclusion $k[X] \subseteq \widetilde{k[X]}$ at $\mathcal{I}(p)$ and using that integral closure commutes with localization (corollary 26.1.10) gives $\mathcal{O}_p(X) = \widetilde{\mathcal{O}_p(X)}$, so ν is an isomorphism of local rings there and hence an isomorphism on a neighbourhood, which is (4), completing the proof.

Part (4) says the normalization only does work where the variety was not already normal, and part (3) says it never changes the function field. Together these make ν the minimal repair: it is birational, so it preserves everything visible to rational functions, and it is an isomorphism wherever no repair was needed. The universal property makes "minimal" precise.

Proposition 37.2.2: Universal Property of Normalization

Let X be an affine variety with normalization $\nu\colon \tilde{X} \to X$. If Y is a normal affine variety and $f\colon Y \to X$ is a dominating morphism, then f factors uniquely through ν:

$$f = \nu \circ g \qquad \text{for a unique morphism } g\colon Y \to \tilde{X}$$

***Proof*:**
Since f is dominating, $f^{\#}\colon k[X] \to k[Y]$ is injective (proposition 33.3.4(2)) and extends to an embedding $k(X) \hookrightarrow k(Y)$. Any $u \in \widetilde{k[X]}$ satisfies a monic equation over $k[X]$; applying $f^{\#}$ to that equation shows $f^{\#}(u)$ is integral over $k[Y]$, and $k[Y]$ is integrally closed because Y is normal, so $f^{\#}(u) \in k[Y]$. Hence $f^{\#}$ extends to $\widetilde{k[X]} = k[\tilde{X}] \to k[Y]$, and this extension is unique because $\widetilde{k[X]}$ sits inside $k(X)$, on which $f^{\#}$ is already determined. Applying theorem 32.1.5 converts this into the asserted unique factorization $f = \nu \circ g$, as we sought to show.

Example 37.2.3: Normalizing the Cuspidal and Nodal Cubics

1. For the cuspidal cubic $C = \mathcal{Z}(x^3 - y^2)$ with $k[C] = k[t^2, t^3]$, the integral closure in $k(t)$ is all of $k[t]$: it contains t by the opening computation, and $k[t]$ is normal by proposition 37.1.3, so nothing larger is needed. Hence $\tilde{C} = \mathbb{A}^1_k$ and ν is the parameterization $t \mapsto (t^2, t^3)$. It is bijective, and an isomorphism away from the cusp.

2. For the nodal cubic N, the same computation gives $\tilde{N} = \mathbb{A}^1_k$ with $\nu(t) = (t^2 - 1, t(t^2 - 1))$. Here ν is *not* injective: both $t = 1$ and $t = -1$ map to the origin. The normalization has pulled the two branches of the node apart, which is what separating a self-crossing requires.

3. Both examples end at $\mathbb{A}^1$: by corollary 37.4.2 below, normalizing a curve always produces a

smooth one.

The contrast between (1) and (2) is worth holding on to. A cusp is repaired without changing the underlying set, since ν is a bijection; a node is repaired by separating points that had been glued. Both are invisible to the function field, which is why birational geometry cannot see them and why a normal model is the right thing to ask for.

What the normalization separates is exactly what the completion sees, which is the comparison section 36.3 deferred to here.

Proposition 37.2.4: Branches and the Normalization of a Curve

Let C be an affine curve, $p \in C$ a point, and $\nu\colon \tilde{C} \to C$ the normalization of theorem 37.2.1. Then the number of formal branches of C at p, in the sense of definition 36.3.2, equals the number of points of $\tilde{C}$ lying over p.

***Proof Key ideas*:**
Both counts are computed by the same ring. The points of $\tilde{C}$ over p correspond to the maximal ideals of the integral closure $\tilde{\mathcal{O}}$ of $\mathcal{O}_{C,p}$ in its fraction field, which is finite over $\mathcal{O}_{C,p}$ by theorem 37.2.1 together with corollary 26.1.10, hence semilocal; and the minimal primes of $\hat{\mathcal{O}}_{C,p}$ correspond to the factors in the decomposition of $\hat{\tilde{\mathcal{O}}}$ into complete local rings, one for each maximal ideal of $\tilde{\mathcal{O}}$. Matching the two requires knowing that completion commutes with integral closure for a curve over a field, which is the statement that $\mathcal{O}_{C,p}$ is analytically unramified; that is a theorem about excellent rings and is not developed in this series. See [37].

The two cases this book uses are checked directly instead. Proposition 36.3.3 gives two branches at the node and one at the cusp, and the fifth and first exercises of exercise set 37.5.1 compute the normalizations: two points lie over the node, at $t = \pm 1$, and one over the cusp, at $t = 0$. So the counts agree in both cases, completing the argument in the cases used.

37.3 Curves, Function Fields, and Discrete Valuation Rings

Definition 37.3.1: Local Ring of Field

Let K/k be a field extension. Then A is a *local ring of* K if $\mathrm{Frac}(A) = K$ and $k \subseteq A$. If A is a DVR, we shall say that A is a *discrete valuation ring of* K.

The prototypical example of local fields over a ring are the stalks $\mathcal{O}_p(V)$ of $\kappa(V)$ for a variety V.

Theorem 37.3.2: Dominating DVR

Let C be a projective curve, $K = \kappa(C)$, and let L/K be a field extension. Let R be a DVR of L containing k. Assume $K \not\subseteq R$. Then there is a unique point $p \in C$ such that R dominates $\mathcal{O}_p(C)$.

***Proof*:**
First, such a point is unique; if R dominates $\mathcal{O}_p(C)$ and $\mathcal{O}_q(C)$, then choose $f \in \mathfrak{m}_p(C)$ so $1/f \in \mathcal{O}_q(C)$. Then $\operatorname{ord}(f) > 0$ and $\operatorname{ord}(1/f) \geq 0$, a contradiction.

Thus, let's show it exists. Let $C \subseteq \mathbb{P}^n_k$ be a projective curve (a closed one-dimensional subvariety. Without loss of generality, we may take C to not be strictly contained in any x_i-hyperplane so that $C \cap U_i \neq \emptyset$. Then in each $k_{\mathbb{P}}[C] = k[x_1, ..., x_n + 1]/I(C) \cong k[u_1, ..., u_{n+1}]$. Let

$$N = \max_{i,j}(x_i/x_j)$$

As the max is achieve, without loss of generality assume $\operatorname{ord}(x_j/x_{n+1}) = N$ for some j. Then for all i:

$$\operatorname{ord}\left(\frac{u_i}{u_{n+1}}\right) = \operatorname{ord}\left(\frac{(u_j/u_{n+1})}{(u_i/u_j)}\right) = N - \operatorname{ord}(u_j/u_i) \geq 0$$

If C_* is the affine curve corresponding to $C \cap U_{n+1}$, then $\Gamma(C_*)$ can be defined with

$$k\left[\frac{u_1}{u_{n+1}}, ..., \frac{u_n}{u_{n+1}}\right]$$

hence, $\Gamma(C_*) \subseteq R$.

Now, if $\mathfrak{m}$ is the maximal ideal of R, let $J = \mathfrak{m} \cap \Gamma(C_*)$. Then J is a prime ideal, so J corresponds to a closed subvariety $W \subseteq C_*$. IF $W = C_*$, $J = 0$, and every nonzero element of $\Gamma(C_*)$ is a unit of R. But then $K \subseteq R$, contradiction our assumption. So $W = \{p\}$ must be a point. Then R certainly dominates $\mathcal{O}_p(C_*)$, and hence dominates $\mathcal{O}_p(C)$, completing the proof.

Corollary 37.3.3: Rational Map of Curves

Let f be a rational map from C' to a projective curve C. Then the domain of f must include every simple point of C'. If C' is nonsingular, f is a morphism.

***Proof*:**
If F is not dominating, it is constant (as shown in corollary 40.2.4), hence its domain must be all of C. Thus, assume $F^{\#}$ embeds $K = \kappa(C)$ as subfield of $L = \kappa(C')$. Let P be a simple point of C', $R = \mathcal{O}_p(C')$. Then it is enough to show that $K \not\subseteq R$. Suppose that $K \subseteq R \subseteq L$ The L is a finite algebraic extension of K, so R is in fact a field. But DVR's are by definition *not* a field.

Corollary 37.3.4: Function Field and Morphism Correspondence

Let C be a projective curve, C' a nonsingular curve. Then there is a natural one-to-one correspondence between dominant morphisms $f : C' \to C$ and homomorphism $f^{\#} : \kappa(C) \to \kappa(C')$.

***Proof*:**
immediate

Corollary 37.3.5: Characterizing Nonsingular Projective Curves

Two nonsingular projective curves are isomorphic if and only if their function fields are isomorphic if and only if they are birationally equivalent.

***Proof*:**
immediate

Corollary 37.3.6: Function Field associated to Curve

Let C be a nonsingular projective curve over k, $K = \kappa(C)$. Then there is a natural one-to-one correspondence between the points of C and the discrete valuation rings of K, namely

$$p \in C \text{ correspond to } k \subseteq \mathcal{O}_p(C) \subseteq K$$

***Proof*:**
Certainly $\mathcal{O}_p(C)$ is a DVR in K. On the other hand if R is any such DVR, then we saw that R dominates a unique $\mathcal{O}_p(C)$, and as these are both DVR's of K, $R = \mathcal{O}_p(C)$, as we sought to show.

From this, notice that we can create the Zariski topology on the collection of DVRs: Let X be the set of all DVRS of $\kappa(C)/k$. Give the cofinite topology on X. Then the correspondence $p \mapsto \mathcal{O}_p(C)$ is a homeomorphism! We shall use this observation when looking at "smooth $\mathbb{Z}$-curves" in section 50.7 which shall how how to interpret Dedekind domains as smooth curves!

37.4 Normality and the Serre Conditions

Regularity implies normality (corollary 35.5.3), and in dimension one the implication reverses (proposition 35.5.2). The question is how far the reverse implication reaches. The answer is: exactly one codimension. A normal variety is smooth in codimension one, and no further.

Proposition 37.4.1: Normal Implies Regular in Codimension One

Let X be a normal variety and $p \in X$ a point whose closure has codimension 1 in X. Then $\mathcal{O}_p(X)$ is a discrete valuation ring, and in particular X is nonsingular at p.

***Proof*:**
The ring $\mathcal{O}_p(X)$ is a Noetherian local domain (proposition 33.2.10) of Krull dimension 1, since its dimension is the codimension of p in X by proposition 34.3.2. It is integrally closed, because X is normal and normality is local (proposition 37.1.2). A Noetherian local domain of dimension one that is integrally closed is a discrete valuation ring, and hence regular, by proposition 35.5.2. Regularity of the local ring is nonsingularity of the point by proposition 35.2.2, as we sought to show.

Corollary 37.4.2: A Normal Curve Is Smooth

A normal curve is nonsingular at every point. Consequently the normalization $\nu\colon \tilde{C} \to C$ of any curve is a smooth curve, and ν resolves every singularity of C.

***Proof*:**
Every point of a curve has codimension 1 in it, so proposition 37.4.1 applies at every point. The normalization $\tilde{C}$ is normal and is a curve, being birational to C (theorem 37.2.1(3)) and hence of the same dimension, completing the proof.

This is the structural payoff of the chapter, and it is what makes curves so much better behaved than surfaces. For a curve, "normal" and "smooth" are the same word, so a single algebraic operation, taking an integral closure, removes every singularity at once. Section 37.3 converts this into the statement that each birational class of curves contains exactly one smooth projective model.

In dimension two the two conditions come apart.

Example 37.4.3: A Normal Singular Surface

Let $\operatorname{char} k \neq 2$ and let $Q = \mathcal{Z}(xy - z^2) \subseteq \mathbb{A}^3_k$, the quadric cone. It is singular at the origin: the Jacobian of $xy - z^2$ is $(y,\ x,\ -2z)$, which vanishes at $(0,0,0)$, so by definition 35.2.1 the rank there is 0 rather than the required $3 - 2 = 1$.

Nonetheless Q is normal. Its coordinate ring is isomorphic to the subring $k[u^2, uv, v^2] \subseteq k[u,v]$ under $x \mapsto u^2$, $y \mapsto v^2$, $z \mapsto uv$, which is the ring of invariants of $k[u,v]$ under $(u,v) \mapsto (-u,-v)$. A ring of invariants of a normal domain under a finite group is normal: if w in the fraction field is integral over the invariants, it is integral over $k[u,v]$, hence lies in $k[u,v]$ by proposition 37.1.3, and it is invariant, so it lies in the invariant subring.

So Q is normal and singular, and its singular locus is a single point, of codimension 2. That is the smallest codimension proposition 37.4.1 permits.

The example is sharp, and it explains the shape of the general criterion. Normality controls codimension one and says nothing beyond it, so a full characterization needs a second condition governing the deeper codimensions. That condition is a statement about depth, and it is what Serre's criterion supplies.

A Noetherian domain is integrally closed exactly when it satisfies two conditions, one asking regularity in codimension one and one asking a lower bound on depth. Write (R_1) for "$R_{\mathfrak{p}}$ is regular for every prime $\mathfrak{p}$ of height at most one" and (S_2) for "$\operatorname{depth} R_{\mathfrak{p}} \geq \min(2, \operatorname{height} \mathfrak{p})$ for every prime $\mathfrak{p}$". The statement that normality is their conjunction is *Serre's Criterion for Normality*, theorem 57.3.2; it runs on the theory of depth and regular sequences, which is developed there and not here.

The geometric content of the two halves is worth separating. Condition (R_1) is proposition 37.4.1 read backwards: it requires smoothness exactly in codimension one. Condition (S_2) says a function defined outside a closed set of codimension two extends across it, which is why a normal variety cannot be repaired by deleting a small set and gluing differently. The quadric cone of example 37.4.3 satisfies both: it is smooth away from a codimension-two point, and it is a hypersurface, which forces (S_2).

Read geometrically, the criterion says normality is precisely "smooth in codimension one, plus no missing codimension-two information". For curves the second condition is vacuous and the first is everything, which recovers corollary 37.4.2. From dimension two onward both are needed, and normality becomes a genuinely weaker and more flexible condition than smoothness, which is why it is normality rather than smoothness that appears as a hypothesis throughout the theory of divisors in chapter 44.

37.5 Weil Divisors and the Class Group

Proposition 37.4.1 says that a normal variety is regular in codimension one, so at every codimension-one subvariety the local ring is a discrete valuation ring. That is exactly the hypothesis needed to make sense of the order of vanishing of a function along a subvariety, and with it the divisor theory of chapter 44 generalizes from curves to normal varieties of any dimension.

Definition 37.5.1: Prime Divisor and Weil Divisor

Let X be a normal variety. A *prime divisor* on X is an irreducible closed subvariety $Y \subseteq X$ of codimension one. A *Weil divisor* is a formal sum

$$D = \sum_Y n_Y Y \qquad n_Y \in \mathbb{Z}$$

over the prime divisors, with all but finitely many n_Y zero. They form a group $\mathrm{Div}(X)$.

For a curve a prime divisor is a point, and definition 44.1.1 is the case $\dim X = 1$.

Proposition 37.5.2: The Order Along a Prime Divisor

Let X be a normal variety and $Y \subseteq X$ a prime divisor. Then the local ring $\mathcal{O}_Y(X)$ of X along Y, that is the localization of $k[U]$ at the prime $\mathcal{I}(Y \cap U)$ for any affine open U meeting Y, is a discrete valuation ring with fraction field $k(X)$. Writing ord_Y for its valuation, every nonzero $f \in k(X)$ has $\mathrm{ord}_Y(f) = 0$ for all but finitely many Y.

***Proof*:**
The ring $\mathcal{O}_Y(X)$ is Noetherian and local, and it is a domain with fraction field $k(X)$, being a localization of a coordinate ring of X. Its dimension is $\mathrm{codim}_X Y = 1$ by proposition 34.3.2. It is normal, being a localization of a normal domain, as the sixth exercise of exercise set 37.5.1 shows. So proposition 35.5.2 applies through its equivalence of normality with being a discrete valuation ring.

For the finiteness, fix an affine open U and write $f = g/h$ with $g, h \in k[U]$. Then $\mathrm{ord}_Y(f) \neq 0$ forces $Y \cap U$ to lie in $\mathcal{Z}(g) \cup \mathcal{Z}(h)$, a proper closed subset of U, which has finitely many irreducible components of codimension one. Covering X by finitely many affine opens gives the claim, completing the proof.

Definition 37.5.3: Principal Divisor and the Class Group

For $f \in k(X)^*$ the *principal divisor* is $\operatorname{div}(f) = \sum_Y \operatorname{ord}_Y(f)\, Y$. The principal divisors form a subgroup $\operatorname{Prin}(X) \subseteq \operatorname{Div}(X)$, and the *class group* is

$$\operatorname{Cl}(X) = \operatorname{Div}(X)/\operatorname{Prin}(X)$$

Two divisors with the same class are *linearly equivalent.*

For a nonsingular projective curve this is $\operatorname{Pic}(C)$ of definition 44.3.1; the two names coexist because on a general variety the divisor class group and the group of line bundles are different objects, agreeing exactly when X is nonsingular. The distinction is developed in [5].

Example 37.5.4: Class Groups

1. *Affine space.* $\operatorname{Cl}(\mathbb{A}^n_k) = 0$. A prime divisor is a hypersurface $\mathcal{Z}(f)$ with f irreducible, by theorem 34.4.2, and $\operatorname{div}(f) = \mathcal{Z}(f)$; so every prime divisor is principal, and hence so is every divisor.
2. *Projective space.* $\operatorname{Cl}(\mathbb{P}^n_k) \cong \mathbb{Z}$, by degree. A prime divisor is $\mathcal{Z}(F)$ for an irreducible form F, and assigning it $\deg F$ gives a surjection $\operatorname{Div}(\mathbb{P}^n_k) \to \mathbb{Z}$. Its kernel is exactly the principal divisors: a rational function on $\mathbb{P}^n_k$ is a ratio F/G of forms of equal degree, so $\operatorname{div}(F/G) = \mathcal{Z}(F) - \mathcal{Z}(G)$ has degree zero, and conversely a degree-zero divisor $\sum n_i \mathcal{Z}(F_i)$ is $\operatorname{div}\left(\prod F_i^{n_i}\right)$.
3. *The quadric cone.* Let $X = \mathcal{Z}(xy - z^2) \subseteq \mathbb{A}^3_k$, which is normal by the fourth exercise of exercise set 37.5.1 although singular at the origin. Let $L = \mathcal{Z}(x, z) \subseteq X$ be the line through the vertex. Then L is a prime divisor, and

$$\operatorname{div}(x) = 2L$$

To see this, note $y \notin (x, z)$, so y is a unit in the local ring $\mathcal{O}_L(X)$ along L. The relation $xy = z^2$ then gives $x = z^2/y$, so the maximal ideal $(x, z)\mathcal{O}_L(X)$ is generated by z alone and $\operatorname{ord}_L(z) = 1$; hence $\operatorname{ord}_L(x) = 2\operatorname{ord}_L(z) - \operatorname{ord}_L(y) = 2$. No other prime divisor occurs, since $\mathcal{Z}(x) \cap X = L$ as a set. So $2L$ is principal but L is not, by proposition 36.4.3, and $\operatorname{Cl}(X) \cong \mathbb{Z}/2\mathbb{Z}$ generated by the class of L.

The last example is the point of the section. On a nonsingular variety every codimension-one subvariety is locally cut out by a single equation, which is theorem 36.4.2, so the class group there measures a global obstruction to patching those local equations together. On a singular normal variety the obstruction is already local, and Cl detects the singularity: the line L on the quadric cone is not the zero set of any single function near the vertex, by proposition 36.4.3, which is what $\operatorname{div}(x) = 2L$ records.

Exercise 37.5.1

1. Let $A = k[x, y]/(y^2 - x^3)$ be the coordinate ring of the cuspidal cubic and put $t = y/x$ in $\operatorname{Frac}(A)$. Show that t is integral over A but does not lie in A, so A is not normal, and show that $A[t] = k[t]$. Conclude that the normalization of the cuspidal cubic is $\mathbb{A}^1_k$.

2. Show that $\mathbb{Z}[\sqrt{-3}]$ is not integrally closed in its fraction field by exhibiting an element of $\mathbb{Q}(\sqrt{-3})$ that is integral over it but not in it. Which ring is its integral closure?

3. Let $A \subseteq B$ be an integral extension of domains with $\mathrm{Frac}(A) = \mathrm{Frac}(B)$. Show that if A is normal then $A = B$. Deduce that the normalization map of theorem 37.2.1 is an isomorphism exactly when the variety is already normal.

4. Show that the quadric cone $\mathcal{Z}(xy - z^2) \subseteq \mathbb{A}^3_k$ is normal but singular at the origin, and explain why this does not contradict corollary 37.4.2. Which of Serre's two conditions in theorem 57.3.2 is doing the work?

5. Let $C = \mathcal{Z}(y^2 - x^2 - x^3) \subseteq \mathbb{A}^2_k$ be the nodal cubic. Using the parameterization $t \mapsto (t^2 - 1,\, t(t^2 - 1))$ of example 33.3.8, show that $k[C]$ is not normal and that its normalization is $k[t]$. How many points of the normalization lie over the node, and why is that different from the cuspidal case?

6. Show that a localization of a normal domain is normal, and use this together with proposition 37.1.2 to show that normality of a variety may be checked on any affine open cover.

7. Give an example of a birational morphism of affine varieties that is bijective on points but not an isomorphism. (Chapter 47 will give a systematic supply.)

8. Let C be an irreducible curve with function field $K = k(C)$. Show that a discrete valuation ring R with $\mathrm{Frac}(R) = K$ and $k \subseteq R$ determines at most one point of any nonsingular projective model of C, and identify the valuation ring attached to the point at infinity of $\mathbb{P}^1_k$.

Part VIII

Projective Varieties

The affine theory of Part VI is incomplete in two ways that turn out to be the same way. A rational function need not be defined everywhere, and two curves of degrees m and n need not meet in mn points. Both failures are failures of points to exist, and both are repaired by the same construction: adjoin a point for every direction in which something can run off to infinity.

The resulting space is projective space, and the varieties inside it are the subject of this Part. The affine dictionary carries over almost unchanged once polynomials are replaced by homogeneous ones, but the geometry is genuinely better behaved: projective varieties are complete, so the image of a regular map out of one is closed, and the only regular functions on one are the constants. Those two facts drive everything in Part IX.

38

Projective Space and Homogeneous Ideals

This chapter builds projective space and transplants the dictionary of chapter 32 onto it. The transplant is not free: a polynomial is not a function on projective space, and the zero set of one is well defined only when the polynomial is homogeneous. Working out what replaces the coordinate ring, and which ideals the Nullstellensatz now describes, occupies the chapter.

38.1 Points at Infinity

So far, all of the algebraic sets were called *affine* algebraic sets, as they were all a subset of $\mathbb{A}_k^n$. These offer great geometrical intuition, however they are lacking in certain algebraic qualities, namely rational functions need not be defined at all points, and not all algebraic curves intersect. As an example of the former point, it is simplest to resort to an analogy from calculus. The reader may recall that the function $f : \mathbb{R} \to \mathbb{R}$ defined as $f(x) = 1/x$ is famously undefined at $x = 0$, as we cannot divide by 0.

However, we can enlarge $\mathbb{R}$ to make this function well-defined by adding a *point at infinity.* Namely, we may take $\mathbb{R}_\infty = \mathbb{R} \cup \{\infty\}$ where (for the sake of limits)[1] we extend the arithmetic of $\mathbb{R}$ to $\mathbb{R}_\infty$ by doing:

$$a + \infty = \infty + a = \infty \qquad b \cdot \infty = \infty \cdot b = \infty$$

where $a \in \mathbb{R}$ and $0 \neq b \in \mathbb{R}_\infty$. We then say that if (x_n) diverges, then

$$\lim_{n\to\infty} x_n = \infty$$

[1]The arithmetic defined here certainly does not satisfy the axioms of a ring; instead it is a structure called a *wheel*; we shall not study these in this textbook

In this case we now have that the function $1/x$ has a well-defined value of ∞ at 0, and even better under the right frame-work (provided in differential geometry), the function $f(x) = 1/x$ can be shown to be smooth on $\mathbb{R}_\infty$. In this way, we have ensured that we cannot have an undefined rational polynomials on $\mathbb{R}_\infty$! The space $\mathbb{R}_\infty$ is called the *projective line.* This "closure" on the set of definable rational polynomials makes the projective line an algebraically nicer space to work with, in a similar way that $\mathbb{C}$ is the better space to work in as all polynomials have roots.

For the latter interpretation (the one about there always being an intersection), consider a polynomial $f(z)$ defined over $\mathbb{C}$ so that there is no worry about roots not existing. Let us regard $\mathbb{C}$ as $\mathbb{R}^2$ with coordinates x, y so that $f(z) = f(x, y)$. Consider the line $L_{(a,b)}$ which has angle α given by (a, b):

$$L_{(a,b)} : t \mapsto (at, bt)$$

Then either $f = L$, in which case their intersection is an entire line, or f and $L_{(a,b)}$ intersect where these two equations agree:

$$f(at, bt) = f_n(a, b)t^n + \cdots + f_1(a, b)t + f_0(a, b)$$

where $f_n(a, b)$ represents the constant in front of the t-term. if $f_n(a, b) \neq 0$ then there are n roots, and hence n points of intersections. However, if $f_n(a, b) = 0$, then there must be a smallest m where

$$f_n(a, b) = 0,\ f_{n-1}(a, b) = 0,\ \cdots f_{m+1}(a, b) = 0,\ f_m(a, b) \neq 0$$

or else $f = L_{(a,b)}$, which we are taking to not be the case. But then we would have a number of intersections lower than the number of roots. There is a remedy to this, which is to consider the remaining $(n-m)$ points to be the intersection of L with f at a point at infinity. To see this, consider the substitution $s = 1/t$. Then:

$$f_n(a, b) + f_{n-1}(a, b)s + \cdots + f_0(a, b)s^n = 0$$

Notice that if $f_n(a, b) \neq 0$, then at $s = 0$ (conceptually can be thought of as $1/t = 0$) there is no root. But if there are $n - m$ leading terms that are zero, then this polynomial has a root of order $n - m$ at $s = 0$. Thus, we recover the intersection of L with f as an invariance! In particular, for each line $L_{(a,b)}$ we can think of there being an extra point at infinity. Notice that by combining the coordinates s, t we get a function:

$$f_n(a, b)t^n + f_{n-1}t^{n-1}s + \cdots + f_0(a, b)s^n = 0$$

which allows us to find all points of intersections at once. This motivates the use of homogeneous functions and the linear equivalence construction that shall be at the start of this section.

Why is the space called projective? This is understood from the following construction of projective space that extends to higher dimensions. Let us start in one dimension. We may define $\mathbb{R}_\infty$ as the

$$\mathbb{R}\mathrm{P}^1 := \frac{\mathbb{R}^2 \setminus \{(0, 0)\}}{\{(x, y) \sim (\lambda x, \lambda y), \forall \lambda \in \mathbb{R} \setminus \{0\}\}}$$

by taking the representative of each equivalence class to have norm 1, notice that we may associate all the points of $\mathbb{R}\mathrm{P}^1$ to S^1, and we can associate points of $\mathbb{R}$ with the equivalence class where $y \neq 0$. The equivalence class $[(\lambda x, 0)]$ can be thought of as the "point at infinity" that we've added, hence we may think of $\mathbb{R}\mathrm{P}^1$ as

$$\mathbb{R}\mathrm{P}^1 = \mathbb{R}^1 \sqcup \mathbb{R}^0 \tag{38.1}$$

The term "projective" comes from the idea that parallel lines will meet at the "point at infinity". In terms of "closure" that we talked about earlier, the reader may recall that in classical geometry, two non-parallel lines will always have a solution. In $\mathbb{R}\mathrm{P}^1$, we have added the extra property that there is one point at which parallel lines meet; they also have a solution. In terms of linear algebra, every system of linear equations has a solution.

We may extend this further, and take $\mathbb{R}^2$ and add "points at infinity". In this case, there are many possible pairs of parallel lines. We may consider that up to scaling, we have many new points that we may add. Formally, let:

$$\mathbb{R}\mathrm{P}^2 := \frac{\mathbb{R}^3 \setminus \{(0,0,0)\}}{\{(x,y,z) \sim (\lambda x, \lambda y, \lambda z), \forall \lambda \in \mathbb{R} \setminus \{0\}\}}$$

We may imagine that this insures that any rational function in two variables always has a solution. Adding multiple points at infinity insures that the function $1/y$, $1/x$, $1/xy$ and so forth are all considered distinct when evaluating ($1/y$ is zero when evaluated at one infinity but not another). In terms of parallel lines, we may think that parallel lines in any angle will have a unique solution (note that the distance between the lines does not matter, they still "meet" at the same point at infinity). The shape $\mathbb{R}\mathrm{P}^2$ can be thought of as $\mathbb{R}^2$ with a circle of points at infinity added where opposites are glued. Can the reader use this intuition to generalize (38.1) to:

$$\mathbb{R}\mathrm{P}^2 = \mathbb{R}^2 \sqcup \mathbb{R}^1 \sqcup \mathbb{R}^0$$

The reader familiar with differential geometry will know that $\mathbb{R}\mathrm{P}^2$ is a two dimensional smooth manifold, and is one of the simplest example of a shape that is 2-dimensional but cannot be embedded in 3-dimensions without self-intersection. The following image from Wikipedia may help in trying to visualize (note that this is an *immersion*, meaning it is an approximation of the shape by embedding it in a space in which there is not enough dimensions to visualize it without intersection)

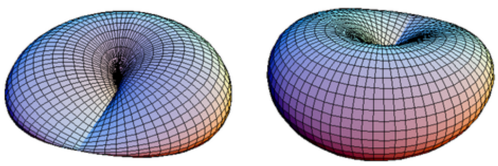


Figure 38.1: Wikipedia: Immersion of $\mathbb{R}\mathrm{P}^2$

We may continue this process and define $\mathbb{R}\mathrm{P}^n$, or more generally $\mathbb{P}^n_k$ for some field k. These shall have a larger class of functions that we may define on them (we shall eventually say that there are an "ample" amount of function[2]). Thus, we shall treat $\mathbb{P}^n_k$ as an extension of $\mathbb{A}^n_k$, and consider the equivalence classes of $\mathbb{P}^n_k$ as *points*.

An important technicality that we may encounter is that we must convert a function that is defined on $\mathbb{A}^n_k$ to a function defined on $\mathbb{P}^n_k$, namely the points of $\mathbb{P}^n_k$ are now equivalence relations and hence a function whose domain is $\mathbb{P}^n_k$ must be independent of representative. This shall form an important class of rings that are "good representatives" to extend polynomials to $\mathbb{P}^n_k$. As we are extending results from $\mathbb{A}^n_k$, this class of rings shall have a similar dictionary as that provided between affine varieties and radical ideals of coordinate rings, namely we shall show that polynomials solutions over $\mathbb{P}^n_k$ shall correspond to *graded ideals* of *graded rings*.

[2] More technically speaking, we shall say that $\mathbb{P}^n_k$ has *ample line bundles*, a notion belonging to scheme theory

38.2 Projective Space and Homogeneous Polynomials

Let us first formally define projective space

Definition 38.2.1: Projective Space

Let k be a field. Then *projective space* over k of dimension n is:

$$\mathbb{P}_k^n = \frac{k^{n+1} \setminus \{0\}}{x \sim \lambda x \text{ for all } \lambda \in k^\times}$$

The points of $\mathbb{P}_k^n$ are often denoted $[x_1 : x_2 : \cdots : x_{n+1}]$.

The first thing we must do is define the topology on this space which parallels the Zariski topology on $\mathbb{A}_k^n$. Polynomial functions need not give well-defined zero sets on projective space, for example $f(x, y) = x^2 + y$ does not map two elements in the same equivalence class to zero. We must limit to polynomials that maps the representatives of an equivalence class to the same point. These are precisely the homogeneous polynomials:

Definition 38.2.2: Homogeneous Polynomial (Algebraic Form)

Let $f \in k[x_1, \ldots, x_n]$ be a polynomial. Then f is a *homogeneous polynomial* of degree d, or an *algebraic d-form*, if every monomial occurring in f has total degree d. Equivalently, as a formal identity in $k[x_1, \ldots, x_n, \lambda]$,

$$f(\lambda x_1, \ldots, \lambda x_n) = \lambda^d f(x_1, \ldots, x_n)$$

The identity is the useful form, and it is the reason homogeneous polynomials have well defined zero sets in projective space. It characterizes homogeneity only when read formally, or when k is infinite: over $\mathbb{F}_2$ the only scalar available is $\lambda = 1$, so the identity holds vacuously for every f.

The term *algebraic form* is a bit out-dated, but is often still used for the degree two case, where such algebraic forms are called *quadratic forms*. This condition extends to a more general class of functions. The reader comfortable with some analysis will see in exercise set 38.5.1 that any (at least) d-times differential degree d homogeneous function is in fact a homogeneous polynomial[3].

Example 38.2.3: Homogeneous Polynomials

For example:

1. $f(x, y, z) = x^3 + xy^2 - 2y^3$ is homogeneous
2. $f(x, y) = x^2 + y^3$ is not homogeneous

Notice that if $f_1, \ldots, f_n$ are all homogeneous polynomial of the same degree k, then $(f_1 + \ldots + f_n)^m$ is a homogeneous polynomial of degree mk. Hence, homogeneous polynomials behave well under addition and multiplication.

[3] homogeneity does not imply differentiability or continuity, can you think of a counter-example?

If $(a_1, ..., a_n) \in k^n$ is a solution to a homogeneous polynomials, then all points on the line passing through this point and the origin is also a solution, hence homogeneous polynomials are the "natural" polynomials for the zero set of projective space, as projective space is an equivalence relation on lines. It is tempting conclude the collection of homogeneous polynomials form the set $k[\mathbb{P}^n_k]$ where the set is either equal to $\mathrm{Hom}_{\mathbf{Reg}}(\mathbb{P}^n_k, \mathbb{A}^1_k)$ or $\mathrm{Hom}_{\mathbf{Reg}}(\mathbb{P}^n_k, \mathbb{P}^1_k)$ and we get the same theory as with affine spaces. However, neither of these sets is the collection of homogeneous polynomials. If we consider $\mathrm{Hom}_{\mathbf{Reg}}(\mathbb{P}^n_k, \mathbb{P}^1_k)$, we have the problem that $[0:0] \notin \mathbb{P}^1_k$ and so we lose the fundamental property that the topology is defined by the zero-sets. The reader can further check that if $f : \mathbb{P}^n_k \to \mathbb{A}^1_k$ is homogeneous, then extending to $\mathbb{P}^1_k$ by mapping to $[f(x_1, ..., x_n) : 1]$ is not well-defined[4] (we shall see that $\mathbb{A}^1_k$ is dense in $\mathbb{P}^1_k$ and the given extension is the *only* continuous extension, and it does not work).

For the first option; even when $n = 1$ there is a problem of well-definedness. Consider $f : \mathbb{P}^1_k \to \mathbb{A}^1_k$ given by $f([x:y]) = f\left(\left[\frac{x}{y} : 1\right]\right) = \frac{x}{y}$, i.e. the closest we can do to define the identity. The reader may immediately see the problem: when $y = 0$, x/y is not defined (the codomain is $\mathbb{A}^1_k$). Even if we "ignore" this problem by simply not considering where $y = 0$, we have that f is no well-defined:

$$\frac{x}{y} = f\left(\left[\frac{x}{y} : 1\right]\right) = f([x:y]) = f\left(\left[1 : \frac{y}{x}\right]\right) = 1$$

But then f is not well-defined as it is not the case that for all $x, y \in k$, $x/y = 1$. If we instead try $f([x:y]) = xy$ then

$$xy = f([x:y]) = f\left(\left[\frac{x}{y} : 1\right]\right) = \left(\frac{x}{y}\right) \cdot 1 = \frac{x}{y}$$

but it is not always the case that $xy = \frac{x}{y}$ (ex. let $x = y = 2$), and hence it is once again not well-defined. The reader shall find that any such attempt to define a (non-constant) polynomial shall not be well-defined, namely since $f(\lambda x, \lambda y) \neq f(x, y)$.

This may unsettle the reader, as a "regular function" can't be well-defined unless the polynomial is constant. As it turns out, we can't extend (non-constant) polynomials from affine space to projective in a well-defined manner (see section 39.1) unless it is a constant, which the above exploration is a first demonstration on why this is the case.

However, we will be able to treat homogeneous polynomials as regular functions whenever we limit to one of the variables x_i to be nonzero. In particular, if $f \in k[x_1, ..., x_{n+1}]$ is homogeneous, and $x_i \neq 0$, then we can interpret f to be a regular function on $U_i = \{x \in \mathbb{P}^n_k \mid x_i \neq 0\}$:

$$f([x_1 : ... : x_i : ... : x_{n+1}]) = f\left(\left[\frac{x_1}{x_i} : ... : 1 : ... : \frac{x_{n+1}}{x_i}\right]\right) = f(x'_1, ..., x'_{n+1})$$

where we write $x'_j = \frac{x_j}{x_i}$ as $x_i \neq 0$. Notice that since the ith position can be fixed to 1 by dividing by x_i, we have chosen a representative of this equivalence relation, which is why we dropped the square brackets in the above equation. For example, for the homogeneous polynomial $x \in k[x, y]$, when $y \neq 0$, we can think of it as defining the function:

$$f(x) = x$$

and when $x \neq 0$, we can think of it as defining the function:

$$f(y) = 1$$

[4]There shall be well-defined regular maps from $\mathbb{P}^n_k$ to $\mathbb{P}^1_k$, see section 39.4, but these are not extensions of affine regular maps

This mirrors how $f([x:y]) = f([x/y:1]) = f([x':1]) = x'$ and $f([x:y]) = f([1:y/x]) = f([1:y']) = 1$. Notice that it was important to define two functions and how each behaves when $y \neq 0$ and $x \neq 0$ respectively. If we want to recover the homogeneous polynomial that defined these functions but only defined it on $x \neq 0$, we see that the $y \neq 0$ case can be λx or simply λ for any nonzero λ, showing we require the function to be defined in both situations (we shall see that having the function defined for each $x_i \neq 0$ and is compatible via "transition mappings" is enough to recover the homogeneous polynomial).

This also means that homogeneous polynomials on $\mathbb{P}^n_k$ act less like functions and more like vector-fields which "locally" are given by functions (i.e. are "local trivializations"). For the reader familiar with Differential Geometry, they may recognize that this vocabulary is used to define *vector bundles*, and that homogeneous polynomials can be seen as the sections of this vector bundle. This is in fact exactly what is going on, but it is beyond the scope of this book to get into the details of this realization (see [5] for a full exploration).

For now, we shall use homogeneous polynomial to define the closed subsets of $\mathbb{P}^n_k$. To that end, let's examine ideals generate by homogeneous elements. For each $d \in \mathbb{N}$, we can take the collection of homogeneous polynomials $R_d \subseteq k[x_1, ..., x_{n+1}]$. Then these form a k-vector space (exercise set 38.5.1(1)). Furthermore, $R_d R_e \subseteq R_{d+e}$, and so these form an $\mathbb{N}$-graded algebra over $R_0 = k$ (recall section 25.3). By proposition 16.2.7 and a bit of work, we may consider $k[x_1, ..., x_{n+1}]$ as:

$$k[x_1, ..., x_{n+1}] = R_0 \oplus R_1 \oplus \cdots \oplus R_d \oplus \cdots$$

with the $\mathbb{N}$-graded structure given by homogeneity.

Definition 38.2.4: Projective Algebraic Set

Take $R \subseteq k[x_1, x_2 ..., x_{n+1}]$ with the homogeneous $\mathbb{N}$-graded structure. Let $I \subseteq R$ be a homogeneous ideal. Then a *projective algebraic set* is:

$$\mathcal{Z}_{\mathbb{P}}(I) = \{p \in \mathbb{P}^n_k \mid f(p) = 0, \ \forall f \in I\}$$

If it is clear from context, the $\mathbb{P}$ subscript will be omitted and it shall be written as $\mathcal{Z}(I)$.

Note that we could have taken an arbitrary homogeneous set $S \subseteq R$, and then taken the homogeneous closure ${}^h\langle S \rangle = I$. By proposition 16.2.9, we get that homogeneous ideals are closed under ideal addition, multiplication, intersection, and taking the radical:

$$I + J \qquad IJ \qquad I \cap J \qquad \sqrt{I}$$

Furthermore, to check that a homogeneous ideal I is prime, it suffices to check the condition $fg \in I$ iff $f \in I$ or $g \in I$ with homogeneous polynomials (see corollary 16.2.8). The addition can naturally be extended to arbitrary sums, allowing us to define the Zariski topology on $\mathbb{P}^n_k$ when focusing on the homogeneous ideals of a $k[x_1, ..., x_{n+1}]$:

Definition 38.2.5: Zariski Topology on Projective Space

Letting the projective algebraic sets to be the closed subsets of $\mathbb{P}^n_k$, we get the [projective] Zariski topology on $\mathbb{P}^n_k$.

The vanishing ideal transplants as well, once restricted to homogeneous elements.

Definition 38.2.6: Projective Vanishing Ideal

Let $V \subseteq \mathbb{P}^n_k$. Then $\mathcal{I}_{\mathbb{P}}(V)$ is the ideal generated by

$$\{f \in R \mid f \text{ homogeneous and } f(p) = 0 \text{ for all } p \in V\}$$

It is a homogeneous ideal by construction, and it is radical by the same argument as in the affine case.

Definition 38.2.7: Projective Coordinate Ring

Let $V \subseteq \mathbb{P}^n_k$ be a projective algebraic set. Then a *projective coordinate ring* is defined as:

$$\Gamma_{\mathbb{P}}(V) = k_{\mathbb{P}}[V] = \frac{R}{(\mathcal{I}_{\mathbb{P}}(V))}$$

where we drop the $\mathbb{P}$ subscript when unambiguous.

Calling these the coordinate ring is a little misleading as the elements are not functions on V, but the similar algebraic properties between affine and projective coordinate rings shall warrant this nomenclature.

With a topology, we may define all of the topological and geometric notion such as compactness, connectedness, dimension, irreducibility, and so forth. The irreducibility condition in projective space becomes a condition on homogeneous prime ideals, from which we define

Definition 38.2.8: Projective Variety

A *projective variety* is an irreducible projective algebraic set.

As mentioned in the introduction of this section, projective space and projective algebraic sets extend affine space. Let us now make this precise. Let us first consider an algebraic subset $V = \mathcal{Z}_{\mathbb{P}}(I) \subseteq \mathbb{P}^n_k$. Then as $\mathbb{P}^n_k = \mathbb{A}^{n+1}_k / \sim$, we see that we can consider $\mathcal{Z}_{\mathbb{A}}(I) \subseteq \mathbb{A}^{n+1}_k$:

Before turning to the cone, one family of projective varieties is worth classifying completely, because it can be done with linear algebra alone and because its members supply most of the small examples in this book.

Definition 38.2.9: Quadric and Its Rank

A *quadric* in $\mathbb{P}^n_k$ is $\mathcal{Z}(Q)$ for a nonzero quadratic form $Q \in k[x_0, \ldots, x_n]$. Assume $\operatorname{char} k \neq 2$ and write $Q(x) = x^\top A x$ for the unique symmetric matrix A. The *rank* of the quadric is the rank of A.

Proposition 38.2.10: Classification of Quadrics by Rank

Let $\operatorname{char} k \neq 2$. Every quadric of rank $r+1$ in $\mathbb{P}^n_k$ is carried by a linear change of coordinates to

$$\mathcal{Z}\big(x_0^2 + x_1^2 + \cdots + x_r^2\big)$$

and two quadrics are projectively equivalent if and only if they have the same rank. The singular locus of the quadric above is the linear subspace $\mathcal{Z}(x_0, \ldots, x_r)$, of dimension $n-r-1$, so the quadric is nonsingular exactly when $r = n$, that is when A is invertible.

***Proof*:**
A symmetric bilinear form over a field of characteristic $\neq 2$ admits an orthogonal basis, by the standard diagonalization; in that basis $Q = \sum_{i=0}^{r} \lambda_i x_i^2$ with all $\lambda_i \neq 0$ and $r+1$ the rank. Since k is algebraically closed each λ_i has a square root, and rescaling $x_i \mapsto x_i/\sqrt{\lambda_i}$ brings Q to $\sum_{i \le r} x_i^2$. Rank is unchanged by an invertible linear substitution, since $A \mapsto T^\top A T$, so it is a projective invariant and two quadrics of equal rank are equivalent to the same normal form, hence to each other.

For the singular locus, $\partial Q/\partial x_i = 2x_i$ for $i \le r$ and 0 otherwise, so by proposition 35.2.2 the singular points of $\mathcal{Z}(Q)$ are those with $x_0 = \cdots = x_r = 0$, which all lie on the quadric. That is the linear subspace $\mathcal{Z}(x_0, \ldots, x_r) \cong \mathbb{P}^{n-r-1}_k$, empty exactly when $r = n$, as we sought to show.

Example 38.2.11: Quadrics of Small Rank
Take $\operatorname{char} k \neq 2$ throughout.

1. *Rank* 1. $\mathcal{Z}(x_0^2)$ is a hyperplane doubled: the same point set as $\mathcal{Z}(x_0)$, but a different curve in the sense of definition 42.1.1, which is one reason that definition keeps the polynomial rather than its zero set.
2. *Rank* 2. $\mathcal{Z}(x_0^2 + x_1^2) = \mathcal{Z}\big((x_0 + ix_1)(x_0 - ix_1)\big)$ is a pair of distinct hyperplanes meeting in a $\mathbb{P}^{n-2}$, which is its singular locus.
3. *Rank* 3 *in* $\mathbb{P}^2_k$. A nonsingular conic. It is isomorphic to $\mathbb{P}^1_k$, by the projection computed in example 39.4.4(3), and this is the genus-zero case of theorem 44.4.2.
4. *Rank* 3 *in* $\mathbb{P}^3_k$. Singular along a point, namely $\mathcal{Z}(x_0, x_1, x_2) = [0:0:0:1]$. It is the cone over a nonsingular conic with that vertex, and it is the quadric cone of example 37.4.3 in projective form: normal, and singular at one point.
5. *Rank* 4 *in* $\mathbb{P}^3_k$. A nonsingular quadric surface. Example 40.1.2 identifies it with $\mathbb{P}^1_k \times \mathbb{P}^1_k$ under the Segre embedding, and its two rulings are the two families of lines on it.

So in $\mathbb{P}^3_k$ the quadrics come in exactly four projective equivalence classes, distinguished by a single integer.

38.3 The Affine Cone and Homogenization

Definition 38.3.1: Affine Cone

Let f be a homogeneous polynomial. Then its zero set in $\mathbb{A}^{n+1}$ is called an *affine cone*.

More generally, if $V \subseteq \mathbb{P}_k^n$ is a projective algebraic subset, then $C(V) \subseteq \mathbb{A}_k^{n+1}$ is the affine cone of V.

The cone translates the projective dictionary into the affine one, and the two directions are worth recording, since section 38.4 rests on them.

Proposition 38.3.2: Cone Identities

Let $\emptyset \neq V \subseteq \mathbb{P}_k^n$ be a projective algebraic set and let I be a homogeneous ideal with $\mathcal{Z}_{\mathbb{P}}(I) \neq \emptyset$. Then

$$\mathcal{I}_{\mathbb{A}}(C(V)) = \mathcal{I}_{\mathbb{P}}(V) \qquad C(\mathcal{Z}_{\mathbb{P}}(I)) = \mathcal{Z}_{\mathbb{A}}(I)$$

***Proof*:**
For the first, a point of $C(V) \subseteq \mathbb{A}^{n+1}$ is either the origin or a nonzero vector whose class lies in V. A homogeneous f vanishes on all of $C(V)$ exactly when it vanishes at every such vector, which is exactly when it vanishes at every point of V; and f automatically vanishes at the origin, since a homogeneous polynomial of positive degree has no constant term, while a nonzero constant vanishes nowhere on the nonempty V. Since both ideals are homogeneous, agreement on homogeneous elements gives equality.

For the second, $\mathcal{Z}_{\mathbb{A}}(I)$ is stable under scaling because I is homogeneous, so it is a union of lines through the origin together with the origin itself; the set of classes of its nonzero vectors is precisely $\mathcal{Z}_{\mathbb{P}}(I)$, and taking the cone recovers $\mathcal{Z}_{\mathbb{A}}(I)$. The hypothesis $\mathcal{Z}_{\mathbb{P}}(I) \neq \emptyset$ is what rules out the degenerate case where $\mathcal{Z}_{\mathbb{A}}(I)$ is the origin alone, as we sought to show.

Let us consider a concrete example. The affine cone of the (homogeneous) polynomial $x^2 + y^2 - z^2$ is the zero-set in $\mathbb{A}_k^{2+1}$ that looks like:

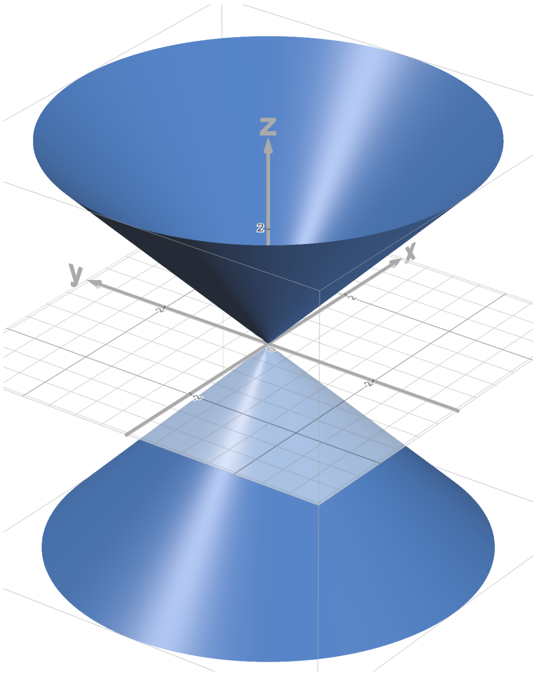


Figure 38.2: zero set of $x^2 + y^2 - z^2$

When $z \neq 0$, This equation in $\mathbb{P}^2_{\mathbb{R}}$ would resemble a circle, which we can see by intersecting the above zero-set with the plane $z = 1$, and hence the usual solution to $x^2 + y^2 = 1$ shall be in this zero set. By setting $z = 0$, we see that $x^2 + y^2 = 0$ if and only if $x = y = 0$, but $[0 : 0 : 0] \notin \mathbb{P}^2_k$, and so all the points of the zero-set $x^2 + y^2 - z^2 = 0$ are precisely the points given by lines passing through $x^2 + y^2 - 1 = 0$

Hence, we see that $x^2 + y^2 - z^2$ becomes the equation of a circle in projective space $\mathbb{P}^2_{\mathbb{R}}$. This matches how the circle has no points going "off to infinity", and so we should expect recovering the same set! However, if we instead consider $xy - z^2$, we see that when $z \neq 0$, we can take the plane where $z = 1$ and get the set $xy = 1$, which is the graph of $y = 1/x$. This has a point at infinity, and indeed if we let $z = 0$, we get the equation $xy = 0$ which is true if either x or y is zero (not that both x and y can both be zero as that would imply $[0 : 0 : 0] \in \mathbb{P}^2_{\mathbb{R}}$). Hence, we gained two additional points corresponding to the two direction the points go off to infinity (see example 38.3.4 for a visual).

The process by which we extend transform a polynomial to a homogeneous polynomial to extend the zero-set is given a name:

Definition 38.3.3: Homogenization

Let $f \in k[x_1, \dots, x_n]$ be a polynomial of degree d. Its *homogenization* is

$$f^*(x_1, \dots, x_{n+1}) = x_{n+1}^d\, f\left(\frac{x_1}{x_{n+1}}, \dots, \frac{x_n}{x_{n+1}}\right)$$

Clearing denominators shows f^* lies in $k[x_1, \dots, x_{n+1}]$ and is homogeneous of degree d. The exponent must be at least $\deg f$ for f^* to be a polynomial at all, and taking it equal to $\deg f$ is what makes f^* divisible by no power of x_{n+1}.

Conversely, a homogeneous $F \in k[x_1, \dots, x_{n+1}]$ is *dehomogenized* with respect to x_i by setting that variable to 1:

$$F_*(x_1, \dots, \widehat{x_i}, \dots, x_{n+1}) = F(x_1, \dots, x_{i-1}, 1, x_{i+1}, \dots, x_{n+1})$$

the hat marking the omitted variable. Unless said otherwise $i = n+1$. These two operations are written $(-)^*$ and $(-)_*$ throughout; the alternative f^* for f^* is not used.

Example 38.3.4: Homogenizing Polynomials

1. Take $x^2 + y^2 - 1$. Homogenizing (with degree 2) and labelling $x_3 = z$, we get $x^2 + y^2 - z^2$. If we homogenizing in degree n, we would get:

   $$z^{n-2}x^2 + z^{n-2}y^2 = z^n$$

   Homogenizing in a degree larger than $\deg f$ multiplies the result by a power of z, which adds the whole line $z = 0$ to the zero set. Those points are an artifact of the padding, so the convention is always to homogenize in degree exactly $\deg f$.

   An affine solution (x_0, y_0) with $x_0^2 + y_0^2 = 1$ becomes the projective point $[x_0 : y_0 : 1]$. For points at infinity, set $z = 0$ and solve $x^2 + y^2 = 0$. Over $\mathbb{R}$ the only solution is $x = y = 0$, which is not a point of $\mathbb{P}^2_{\mathbb{R}}$, so the real circle acquires nothing at infinity. *Over an algebraically closed field of characteristic $\neq 2$ this fails*, and that is the standing hypothesis of this book. Choosing i with $i^2 = -1$,

   $$x^2 + y^2 = (x + iy)(x - iy)$$

   so the circle meets the line at infinity in the two points

   $$[1 : i : 0] \qquad [1 : -i : 0]$$

   These are the *circular points at infinity*, and every circle in $\mathbb{P}^2_{\overline{k}}$ passes through both of them, whatever its centre or radius. The intuition that a circle is bounded and so gains nothing at infinity is an artifact of working over $\mathbb{R}$; the projective conic is a conic like any other, and chapter 43 counts its intersections accordingly.

2. Take $xy - 1$. Its shape looks like:

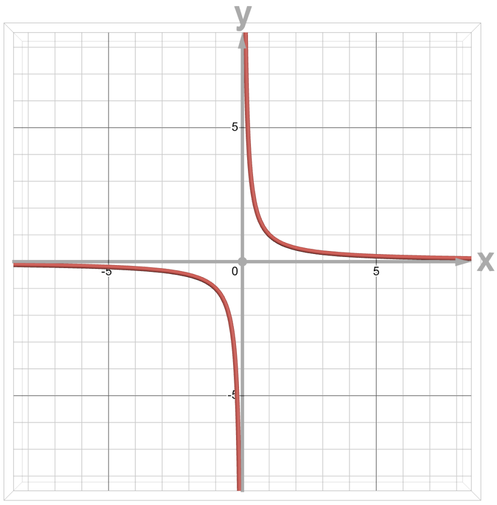


Figure 38.3: zero-set of $xy = 1$

Homogenizing, we get $xy - z^2$. A solution $x_0y_0 - 1 = 0$ in projective coordinates is of the form:

$$[x_0 : y_0 : 1]$$

setting $z = 0$, we can ask if there are any points where:

$$[x_0 : y_0 : 0]$$

is a solution. And indeed, if $x_0 = 0$ or $y_0 = 0$, we get a solution, since

$$0y_0 - 0 = 0 \qquad x_00 - 0 = 0$$

We can see by visualizing the zero set that there seems to be two extra point added at infinite, one for the y-axis and one for the x-axis (remember that opposite points are considered the same).

If we de-homogenize at x, we get $y - z^2 = 0$, equivalently $y = z^2$, and de-homogenizing at y we get $x - z^2 = 0$, equivalently $x = z^2$. In both of these cases, we see that $x = z = 0$ and $y = z = 0$ is a solution. Hence, by de-homogenizing at each variable, we can find the vanishing set of a homogeneous polynomial over projective space. This can be seen more clearly when looking at the affine cone of the homogenized polynomial:

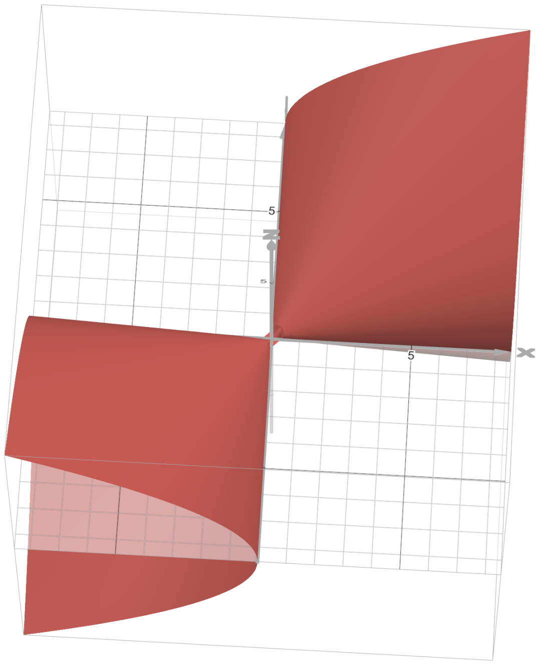


Figure 38.4: affine cone: $xy - z^2$

3. Take $x^3 + xy^2 - 2y^3 - 1$. Its shape looks like:

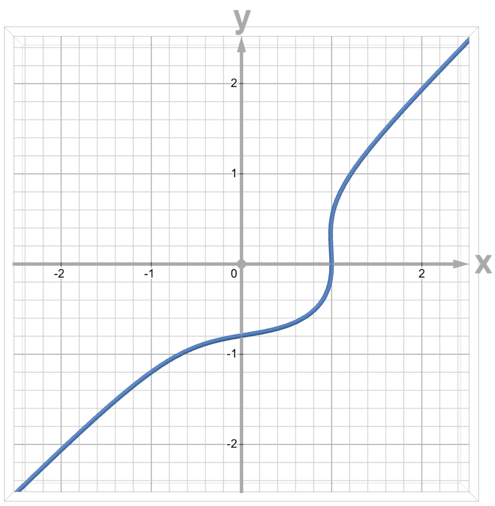


Figure 38.5: $x^3 + xy^2 - 2y^3 = 1$

When homogenizing, the affine cone gives:

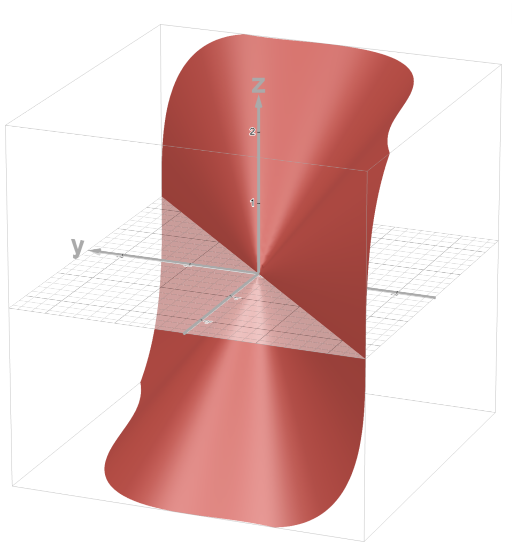


Figure 38.6: affine cone of $x^3 = xy^2 - 2y^3 = z^3$

are there any added points, and if so what are they?

4. Take $f(x) = x^2$. Its vanishing set is the single point 0. Then $f^*(x, x_\infty) = x^2$ is the degree 2 homogenization. Then this has a single zero, namely $[0 : 1]$, which correspond to the single root of x^2 in $\mathbb{A}^1_k$

5. Take $f(x, y) = x^2 - y$, whose zero set is the usual parabola. Its degree-2 homogenization in the variable z is $x^2 - zy$. Setting $z = 1$ recovers the affine points, which are

$$[x_0 : x_0^2 : 1] \qquad x_0 \in k$$

and setting $z = 0$ leaves $x^2 = 0$, so $x = 0$ and the only point at infinity is

$$[0 : 1 : 0]$$

in particular, we added a single point at infinity. This is the point where the parabola "meets". Notice that it wasn't that there was two points at infinite but just 1. The idea is that the "angle" with which the function approaches infinity tends towards $\pi/2$, and hence the edges of the graph shall meet at infinity. This shows that parabola's can be thought of as infinitely large ellipsoids[a].

Can the reader see different sorts of connections between circles, parabolas, ellipses, and hyperbolas through projective space?

[a]for those comfortable with some Riemann geometry, can the reader see why projective space can be thought of as having positive curvature?

Proposition 38.3.5: Properties of Homogenization

Let $F, G \in k[x_1, \ldots, x_{n+1}]$ be homogeneous and $f, g \in k[x_1, \ldots, x_n]$ arbitrary, with $(-)^*$ and $(-)_*$ taken at x_{n+1} as in definition 38.3.3. Then:

1. $(FG)_* = F_* G_*$ and $(fg)^* = f^* g^*$;
2. $(F+G)_* = F_* + G_*$; and writing $r = \deg f$, $s = \deg g$ and $u = \deg(f+g)$,
$$x_{n+1}^{r+s-u}\,(f+g)^* = x_{n+1}^{s} f^* + x_{n+1}^{r} g^*$$
3. if $F \neq 0$ and r is the largest power with $x_{n+1}^r \big| F$, then $x_{n+1}^r (F_*)^* = F$; in particular $(F_*)^* = F$ when $x_{n+1} \nmid F$.

***Proof*:**
Throughout write $z = x_{n+1}$ and use that dehomogenizing is the ring homomorphism $z \mapsto 1$.

1. Dehomogenization is a ring homomorphism, so $(FG)_* = F_* G_*$ is immediate. For the second, $\deg(fg) = \deg f + \deg g$ because k is a domain, and
$$(fg)^* = z^{\deg f + \deg g}(fg)\big(\tfrac{x_1}{z}, \ldots\big) = z^{\deg f} f\big(\tfrac{x_1}{z}, \ldots\big) \cdot z^{\deg g} g\big(\tfrac{x_1}{z}, \ldots\big) = f^* g^*$$
2. Additivity of $(-)_*$ is again the homomorphism property. For the second identity, all three of $f^*, g^*, (f+g)^*$ arise from the same substitution $x_i \mapsto x_i/z$ scaled by different powers of z, namely z^r, z^s and z^u. Multiplying each by the power of z that brings it to the common exponent $r+s$ gives
$$z^{\,s} f^* + z^{\,r} g^* = z^{\,r+s}\,(f+g)\big(\tfrac{x_1}{z}, \ldots\big) = z^{\,r+s-u}\,(f+g)^*$$
The correction $z^{\,r+s-u}$ is needed exactly when the top-degree terms of f and g cancel, so that $u < \max(r, s)$.
3. Write $F = z^r H$ with $z \nmid H$, so H is homogeneous of degree $\deg F - r$. Setting $z = 1$ gives $F_* = H_*$, and since $z \nmid H$ the polynomial H_* has degree exactly $\deg H$; homogenizing therefore returns $(H_*)^* = H$, whence $z^r (F_*)^* = z^r H = F$, as we sought to show.

Corollary 38.3.6: Factorization of Homogeneous Polynomials

Let $F \in k[x_1, \ldots, x_{n+1}]$ be a homogeneous polynomial. Then up to a power of x_{n+1}, factorizing F is the same as factoring F_*.

In particular, if $F \in k[x, y]$, then F factors into a product of linear factors.

***Proof*:**
The first result is immediate from proposition 38.3.5. For the second, as we are asuming k is

algebraically closed in this section we get that F_* must factor into a product of lienar terms,

$$F_* = c\prod(x - \lambda_i)$$

Then by homogenizing we get:

$$F = cy^r\prod(x - \lambda_i y)$$

as we sought to show.

From this, we are justified in decomposing projective algebraic sets by their behaviour when dehomogenizing elements of the generating ideal:

Example 38.3.7: Zero Sets of Projective Space

Example 38.3.4 contained multiple examples of zero sets. All of those examples were from a single homogeneous polynomial.

1. Every point $[a_1 : \cdots : a_{n+1}] \in \mathbb{P}^n_k$ is a projective algebraic set. A point $[x_1 : \cdots : x_{n+1}]$ equals it exactly when

   $$(x_1, \ldots, x_{n+1}) = \lambda(a_1, \ldots, a_{n+1}) \qquad \text{for some } \lambda \neq 0$$

   that is, exactly when the two vectors are proportional, which holds exactly when each 2×2 minor of the matrix

   $$\begin{pmatrix} x_1 & x_2 & \cdots & x_{n+1} \\ a_1 & a_2 & \cdots & a_{n+1} \end{pmatrix}$$

   vanish, namely that

   $$\begin{vmatrix} x_i & x_j \\ a_i & a_j \end{vmatrix} = a_j x_i - a_i x_j = 0$$

   These are $\binom{n+1}{2}$ equations, of which n already suffice (one for each variable beyond a fixed nonzero coordinate). In other words, we get a point via linear homogeneous polynomials with the appropriate coefficients. This shows how "linear" projective space is. These equations can be thought of as the intersection of all the hyper-planes passing through $[a_0 : \cdots : a_n]$. Equivalently, if you know some differential geometry, it shows that we can view $\mathbb{P}^n_k$ as the Grassmannian $G(1, n+1)$.

   The coordinate ring associated to a point would be:

   $$\frac{k[x_1, \ldots, x_{n+1}]}{\langle a_j x_i - a_i x_j \big| 1 \leq i, j \leq n+1\rangle} \cong k[x_{n+1}]$$

   That is, the coordinate ring is *not* just k at a single point! The reader may think of this as the grading condition for the coordinate rings.

2. In $\mathbb{P}^1_k$, the only closed sets are points, the empty set, and the entire space. We shall see why when we show that any closed subset of projective space can be restricted to a closed subset of affine space, in which these are the only possibilities

3. $\mathbb{P}^2_k$ has many more closed sets just like $\mathbb{A}^2_k$. Any line is given by $a_1x_1 + a_2x_2 + a_3x_3$ and any conic by $ax_1^2 + bx_1x_2 + cx_2^2 + dx_1x_3 + ex_2x_3 + fx_3^2$. A famous set of curves is the Fermat curves given by $x_1^n + x_2^n + x_3^n$. There are many more equations.

   To give one example of a coordinate ring, the coordinate ring for a line is isomorphic to $k[s, t]$ with $\deg(s) = \deg(t) = 1$.

4. In $\mathbb{P}^3_k$ the twisted cubic is cut out by three equations,

$$x_1x_3 - x_2^2, \quad x_1x_4 - x_2x_3, \quad x_2x_4 - x_3^2$$

and its homogeneous coordinate ring is

$$\frac{k[x_1, x_2, x_3, x_4]}{(x_1x_3 - x_2^2,\ x_1x_4 - x_2x_3,\ x_2x_4 - x_3^2)} \cong k[s^3, s^2t, st^2, t^3] \subseteq k[s,t]$$

the subring of $k[s,t]$ generated by the degree-three monomials. That isomorphism exhibits the parameterization $[s : t] \mapsto [s^3 : s^2t : st^2 : t^3]$, so the twisted cubic is a *rational* curve: it is the image of $\mathbb{P}^1$. What is notable about it is different: it needs three equations in $\mathbb{P}^3$ although its codimension is two, so it is not a complete intersection.

Another example in $\mathbb{P}^3_k$ are the quadratic surfaces given by $\sum_{i,j} a_{ij}x_ix_j$.

38.4 The Projective Nullstellensatz and Irrelevant Ideals

As we are considering the case where k is algebraically closed, we may naturally want the Nullstellensatz to apply between projective space and homogeneous polynomials. In particular, if $I \subseteq k[x_1, ..., x_n]$ then $Z(I) = \emptyset$ if and only if $I = (1)$. However, this is not the case:

Example 38.4.1: Nullstellensatz False in Projective Space

Let $\mathfrak{m} = (x_1, \ldots, x_{n+1})$, the ideal of all polynomials with zero constant term. Its affine zero set is the origin, and the origin is not a point of $\mathbb{P}^n_k$, so $\mathcal{Z}_{\mathbb{P}}(\mathfrak{m}) = \emptyset$ while $\mathfrak{m} \neq (1)$.

The same happens for every power: writing $\mathfrak{m}^s$ for the ideal generated by all monomials of degree s, its affine zero set is again just the origin, so $\mathcal{Z}_{\mathbb{P}}(\mathfrak{m}^s) = \emptyset$. So the projective Nullstellensatz cannot say "empty zero set implies the unit ideal"; the best it can say is "empty zero set implies the ideal contains some $\mathfrak{m}^s$", which is lemma 38.4.2.

Fortunately, the above ideals are the only ideals for which this could happen, namely because these ideals correspond to ideals that contain the origin in affine space:

Lemma 38.4.2: Characterizing Irrelevant Ideals

Let k be algebraically closed and let $I \subseteq k[x_1, \ldots, x_{n+1}]$ be a homogeneous ideal. Then $\mathcal{Z}_{\mathbb{P}}(I) = \emptyset$ if and only if $\mathfrak{m}^s \subseteq I$ for some $s > 0$, where $\mathfrak{m} = (x_1, \ldots, x_{n+1})$.

***Proof*:**

If $\mathfrak{m}^s \subseteq I$ then every x_i^s lies in I, so any point of $\mathcal{Z}_{\mathbb{P}}(I)$ has all coordinates zero, and $[0 : \cdots : 0]$ is not a point of $\mathbb{P}^n_k$. Hence $\mathcal{Z}_{\mathbb{P}}(I) = \emptyset$.

Conversely assume $\mathcal{Z}_{\mathbb{P}}(I) = \emptyset$. As $k[x_1, \ldots, x_{n+1}]$ is Noetherian, write $I = (f_1, \ldots, f_r)$ with each f_i homogeneous. Fix an index j and dehomogenize at x_j: the polynomials

$$(f_i)_* = f_i(x_1, \ldots, 1, \ldots, x_{n+1}) \qquad i = 1, \ldots, r$$

have no common root in $\mathbb{A}^n_k$, since a common root α would lift to the projective point obtained by inserting 1 in the jth slot, which would lie in $\mathcal{Z}_{\mathbb{P}}(I)$. By the weak Nullstellensatz (theorem 32.4.3) there are g_i with

$$\sum_{i=1}^{r} (f_i)_* \, g_i = 1$$

Homogenizing this identity and clearing denominators multiplies the right hand side by a power of x_j, giving $x_j^{m_j} \in I$ for some $m_j \geq 1$. Running the argument for each j produces exponents $m_1, \ldots, m_{n+1}$.

Put $m = \max_j m_j$ and $s = (n+1)(m-1)+1$. Any monomial $x_1^{a_1} \cdots x_{n+1}^{a_{n+1}}$ of total degree at least s must have some $a_j \geq m \geq m_j$, by the pigeonhole principle: were every $a_j \leq m-1$, the total degree would be at most $(n+1)(m-1) < s$. Such a monomial is therefore divisible by $x_j^{m_j} \in I$ and so lies in I. As $\mathfrak{m}^s$ is generated by the monomials of degree s, this gives $\mathfrak{m}^s \subseteq I$, as we sought to show.

Definition 38.4.3: Irrelevant Ideal

The ideal $\mathfrak{m} = (x_1, \ldots, x_{n+1}) \subseteq k[x_1, \ldots, x_{n+1}]$, consisting of the polynomials with zero constant term, is the *irrelevant ideal*. Its powers $\mathfrak{m}^s$ are the ideals generated by the monomials of degree s.

The name records the point of lemma 38.4.2: $\mathfrak{m}$ and its powers are precisely the homogeneous ideals invisible to projective space, so the projective correspondence must quotient them out. Note that $\mathfrak{m}^s$ and $(x_1^s, \ldots, x_{n+1}^s)$ are different ideals, though each contains a power of the other, which is all the lemma needs.

38.5 The Hilbert Function of a Projective Variety

An affine variety carries its coordinate ring, and dimension is read off from it as a Krull dimension. A projective variety carries a *graded* ring, and the grading holds more information than the dimension alone: counting how fast the graded pieces grow recovers the dimension, and the constant in front of that growth is a genuinely new invariant, the degree. This section extracts both, and the arithmetic genus that chapter 44 needs.

Throughout let $S = k[x_0, \ldots, x_n]$ with its grading by degree, $X \subseteq \mathbb{P}^n_k$ a projective variety, and $S(X) = S/\mathcal{I}_{\mathbb{P}}(X)$ its homogeneous coordinate ring, which inherits a grading because $\mathcal{I}_{\mathbb{P}}(X)$ is homogeneous.

Definition 38.5.1: Hilbert Function of a Projective Variety

The *Hilbert function* of X is

$$H_X \colon \mathbb{N} \to \mathbb{N} \qquad H_X(m) = \dim_k S(X)_m$$

the dimension over k of the degree-m graded piece of $S(X)$.

The definition is a count: $H_X(m)$ is the number of linearly independent forms of degree m on $\mathbb{P}^n$, modulo those vanishing on X. Equivalently it is the number of independent degree-m conditions X can impose. That it eventually behaves polynomially is Hilbert's theorem on graded modules.

Theorem 38.5.2: Hilbert Polynomial

There is a polynomial $h_X \in \mathbb{Q}[T]$, the *Hilbert polynomial* of X, and an integer m_0, such that

$$H_X(m) = h_X(m) \qquad \text{for all } m \geq m_0$$

Moreover $\deg h_X = \dim X$.

***Proof Key ideas*:**
Polynomiality is a statement about finitely generated graded modules over a polynomial ring and carries no geometric content, so it is cited rather than rebuilt: see *Commutative Algebra* [10], where it appears as the graded companion of the Hilbert-Samuel theory this book already cites in section 35.1. The argument is an induction on n, using the exact sequence obtained by multiplying by a variable.

That $\deg h_X = \dim X$ is the geometric half. Intersecting X with a general hyperplane drops the dimension by one (theorem 34.4.1) and drops the degree of the Hilbert polynomial by one, since $H_{X\cap H}(m) = H_X(m) - H_X(m-1)$ for large m and taking that difference lowers the degree of a polynomial by exactly one. Inducting until the dimension reaches zero, where the Hilbert polynomial is the constant counting points, matches the two numbers.

With polynomiality in hand, the leading coefficient is available to be named.

Definition 38.5.3: Degree and Arithmetic Genus

Let $X \subseteq \mathbb{P}^n_k$ be a projective variety of dimension d, so that h_X has degree d and its leading coefficient is c. The *degree* of X is

$$\deg X = d!\, c$$

and the *arithmetic genus* of X is

$$p_a(X) = (-1)^d\,(h_X(0) - 1)$$

The normalization by $d!$ is what makes $\deg X$ an integer and makes $\deg \mathbb{P}^d = 1$, as the first example below checks. Both invariants depend on the embedding $X \subseteq \mathbb{P}^n$ through the grading, not on X alone; a variety embedded two different ways can have two different degrees, which is why the degree of a curve is always quoted together with the projective space it sits in.

Example 38.5.4: Hilbert Polynomials

1. *Projective space.* $S(\mathbb{P}^n) = S$, and the monomials of degree m in $n+1$ variables number $\binom{m+n}{n}$, so
$$H_{\mathbb{P}^n}(m) = \binom{m+n}{n} = \frac{(m+n)(m+n-1)\cdots(m+1)}{n!}$$
This is already a polynomial in m of degree n with leading coefficient $1/n!$. So $\dim \mathbb{P}^n = n$

and $\deg \mathbb{P}^n = n! \cdot \frac{1}{n!} = 1$, and $h_{\mathbb{P}^n}(0) = \binom{n}{n} = 1$ gives $p_a(\mathbb{P}^n) = 0$.

2. *A hypersurface.* Let $X = \mathcal{Z}(F) \subseteq \mathbb{P}^n$ with F irreducible of degree e. Multiplying by F is injective on S and raises degree by e, so

$$H_X(m) = \binom{m+n}{n} - \binom{m-e+n}{n}$$

The two leading terms $m^n/n!$ cancel, leaving degree $n-1$ with leading coefficient $e/(n-1)!$. So $\dim X = n-1$ and $\deg X = e$: *the degree of a hypersurface is the degree of its equation.* This is the fact Bézout's theorem counts with, and it is the reason chapter 43 can speak of the degree of a plane curve without ambiguity.

3. *A plane curve.* Taking $n = 2$ in (2), a smooth plane curve of degree e has

$$h_X(T) = \binom{T+2}{2} - \binom{T-e+2}{2} = eT - \frac{e(e-3)}{2}$$

so $h_X(0) = -e(e-3)/2$ and

$$p_a(X) = (-1)^1\left(-\tfrac{e(e-3)}{2} - 1\right) = \frac{(e-1)(e-2)}{2}$$

This is the *genus-degree formula.* Section 44.4 takes it as the definition of the genus and puts it to work, showing that it vanishes exactly for the curves isomorphic to $\mathbb{P}^1_k$. A line and a conic have genus 0, a smooth cubic has genus 1, which is why smooth plane cubics are exactly the curves chapter 45 finds a group law on.

4. *The twisted cubic.* From example 38.3.7, $S(X) \cong k[s^3, s^2t, st^2, t^3]$, whose degree-$m$ piece is spanned by the monomials of degree $3m$ in s, t, of which there are $3m+1$. So $h_X(T) = 3T+1$: dimension 1, degree 3, and $p_a = -(1-1) = 0$. It is a rational curve of degree 3 in $\mathbb{P}^3$, matching example 38.3.7(4).

Items (1) and (4) both describe a curve of genus 0, but with different degrees, 1 and 3. That is the embedding-dependence made concrete: the twisted cubic is abstractly isomorphic to $\mathbb{P}^1$, and the two are told apart by their Hilbert polynomials only because they sit in projective space differently.

Exercise 38.5.1

1. Let $R_d \subseteq k[x_1, \ldots, x_n]$ be the space of homogeneous polynomials of degree d. Show that $\dim_k R_d = \binom{d+n-1}{n-1}$.

2. Let $T\colon \mathbb{A}^{n+1}_k \to \mathbb{A}^{n+1}_k$ be a linear isomorphism, inducing a map on $\mathbb{P}^n_k$, and let $V \subseteq \mathbb{P}^n_k$ be a subset with $V^T = T^{-1}(V)$.
   a) Show that V is a projective variety if and only if V^T is.
   b) Show that $V \cong V^T$, and that T induces an isomorphism of homogeneous coordinate rings $\Gamma_h(V) \cong \Gamma_h(V^T)$.

3. Let $V = \mathbb{P}^1_k$ with $\Gamma_h(V) = k[x, y]$ and set $t = x/y$. Show that $k(V) \cong k(t)$, and that the points of $\mathbb{P}^1_k$ correspond to the discrete valuation rings R with $\mathrm{Frac}(R) = k(V)$ and $k \subseteq R$. Which valuation ring belongs to the point at infinity?

4. Show that any two hyperplanes in $\mathbb{P}^n_k$ meet, for $n \geq 2$: if L_1, L_2 are nonzero linear forms then $\mathcal{Z}(L_1) \cap \mathcal{Z}(L_2) \neq \emptyset$. Then show the corresponding statement for two *lines* is true in $\mathbb{P}^2_k$ and false in $\mathbb{P}^3_k$, by exhibiting two disjoint lines there.

5. (Euler's relation) Let $\frac{\partial f}{\partial x_i}$ denote the formal partial derivative. Show that if $f \in k[x_0, \ldots, x_n]$ is homogeneous of degree d, then
$$df = \sum_{i=0}^{n} x_i \frac{\partial f}{\partial x_i}$$
It suffices to check both sides on a monomial, since both are k-linear.

6. Let $R = \bigoplus_{j \geq 0} R_j$ be a graded ring and write $1 = \sum_j r_j$ with $r_j \in R_j$. Show that r_0 is a multiplicative identity for R_0, and in fact that $1 = r_0$.

7. Show that $\mathcal{I}(\mathcal{Z}(I)) = \sqrt{I}$ fails for the irrelevant ideal of definition 38.4.3, and state the projective Nullstellensatz that repairs it.

8. Compute the Hilbert polynomial of the twisted cubic in $\mathbb{P}^3_k$ directly from definition 38.5.1, and check the degree and arithmetic genus against example 38.5.4(4).

9. Show that a homogeneous polynomial in two variables over an algebraically closed field factors into linear forms, and deduce corollary 38.3.6 for $n = 1$.

10. (Analysis aside, over $\mathbb{R}$.) Let $f\colon \mathbb{R}^n \to \mathbb{R}$ be d times continuously differentiable and homogeneous of degree d, meaning $f(\lambda v) = \lambda^d f(v)$ for all $\lambda > 0$. Show that f is a homogeneous polynomial of degree d. So for smooth enough functions, homogeneity in the scaling sense already forces homogeneity in the algebraic sense.

39

Quasi-Projective Varieties and Regular Maps

Projective space is covered by $n+1$ copies of affine space, and that single observation converts every local question about a projective variety into an affine one. This chapter develops the consequence: the class of varieties closed under both taking closed subsets and taking open subsets, the quasi-projective varieties, which contains the affine and the projective ones and is the setting in which the rest of the book works.

39.1 The Affine Cover of Projective Space

A common characterization of $\mathbb{P}^n_k$ is by showing that it can be constructed by gluing together $n+1$ copies of $\mathbb{A}^n_k$. Let $R = k[x_1, x_2, ..., x_{n+1}]$, and let H_i be the zero set of the (homogeneous) polynomial x_i for each i, giving us $H_1, H_2, ..., H_{n+1}$. These sets certainly cover $\mathbb{P}^n_k$ (their union is $\mathbb{P}^n_k$) and are closed as

$$H_i = \mathcal{Z}(x_i)$$

If we consider the map $\mathbb{A}^n_k \to \mathbb{P}^n_k$ mapping $(a_1, ..., a_n) \mapsto [1 : a_1 : \cdots : a_n]$, then the hyperplane H_{n+1} is called the *hyperplane at infinity*, being the hyperplane where $x_{n+1} = 0$ (it contains the points at infinity). For any hyperplane H_i, we have a map

$$H_i \to \mathbb{P}^{n-1}_k \qquad [a_1, ..., a_{i-1}, 0, a_{i+1}, ..., a_n] \to [a_1, ..., a_{i-1}, a_{i+1}, ..., a_n]$$

Once we define regular maps, the reader can verify that $H_i \cong \mathbb{P}^{n-1}_k$. Let

$$D(x_i) = U_i = \mathbb{P}^n_k - H_i$$

Then these sets are open and cover $\mathbb{P}^n_k$, as H_i is closed, and they contain all points where $x_i \neq 0$. Define a map:

$$\varphi_i : U_i \to \mathbb{A}^n_k \qquad [a_1 : \cdots : a_{n+1}] \mapsto \left(\frac{a_1}{a_i}, \ldots, \frac{a_{n+1}}{a_i}\right)$$

where we omit $\frac{a_i}{a_i}$. Notice how this is similar to de-homogenization. Verify that the map is well-defined, namely that it is independent of representative of points $U_i \subseteq \mathbb{P}^n_k$. Then:

Proposition 39.1.1: Projective Space Affine Cover

Let φ_i be defined as above. Then φ_i is a homeomorphism:

$$U_i \cong \mathbb{A}^n_k$$

***Proof*:**
First, φ_i is certainly a bijection: if

$$[x_1 : \cdots : 1_i : \cdots : x_n] = \varphi_i(x_1, ..., x_n) = \varphi(y_1, ..., y_n) = [y_1 : \cdots : 1_i : \cdots : y_n]$$

these two are equal if each $x_j = \lambda y_j$ for some $\lambda \in k$. Indeed, since $1_i = \lambda 1_i$, we must have $\lambda = 1$, which shows that $x_j = y_j$ for each coefficient, giving injectivity. For surjectivity, given any $[x_1, ..., x_i, ..., x_n]$ where $x_i \neq 0$, take:

$$[x_1 : \cdots : x_i : \cdots x_n] = \left[\frac{x_1}{x_i} : \cdots : 1_i : \cdots : \frac{x_n}{x_i}\right]$$

We now see that $(\frac{x_1}{x_i}, ..., \frac{x_n}{x_i})$ map to this point, giving surjectivity, and hence bijectivity.

For continuity, notice that $\varphi_i = \pi \circ \psi_i$ where $\psi_i : \mathbb{A}^n_k \to \mathbb{A}^{n+1}_k \setminus \{0\}$ is the natural embedding with ith-coordinate shifted by 1, and π is the projection. The map π is continuous as $\mathbb{P}^n_k$ is given the natural quotient from $\mathbb{A}^n_k$, and ψ_i is continuous as any pre-image of closed subsets of $\mathbb{A}^{n+1}_k \setminus \{0\}$ is given by the dehomogenization of the polynomials at the ith coordinate.

Furthermore, the map is open, since ψ_i is open (its image is a shifted-by-1 copy of $\mathbb{A}^n_k$, so any open set in $\mathbb{A}^n_k$ is open in $\mathcal{Z}(y_i - 1) \cap \mathbb{A}^{n+1}_k \setminus \{0\}$) and π is open since for any open subset $U \subseteq \mathbb{A}^{n+1}_k \setminus \{0\}$ the set

$$\bigcup_{\lambda \in k^*} \lambda U$$

is open, since the map $x \mapsto \lambda x$ is a homeomorphism (as is quickly verifiable). This is the pre-image of $\pi(U)$ and so since π is a quotient map we have that $\pi(U)$ is also open. Thus, the composition is open, making φ open.

Combining all the results, we get that φ_i is a homeomorphism, as we sought to show.

Given $H_i \cong \mathbb{P}^{n-1}_k$, this implies that we have a decomposition:

$$\mathbb{P}^n_k = \mathbb{A}^n_k \sqcup \mathbb{P}^{n-1}_k$$

For example, if $n = 2$, then:

$$\mathbb{P}^2_k = \mathbb{A}^2_k \sqcup \mathbb{P}^1_k$$

For this reason, we often think of $\mathbb{P}_k^2$ to be $\mathbb{A}_k^2$ along with a "line at infinity" (the reader should take the time to internalize this idea). This decomposition can be extended further, breaking down $\mathbb{P}_k^{n-1}$ into an affine and projective space. Repeating inductively, we get:

$$\mathbb{P}_k^n = \mathbb{A}_k^n \sqcup \mathbb{A}_k^{n-1} \sqcup \cdots \sqcup \mathbb{A}_k^1 \sqcup \mathbb{A}_k^0$$

Another "immediate" result from our above discussion is that projective space $\mathbb{P}_k^n$ can be covered with $n+1$ copies of $\mathbb{A}_k^n$. This shall become the dominant way of viewing projective space in modern algebraic geometry (similar to how smooth manifolds are viewed as patches of euclidean space glued together), making this result important enough to be a theorem:

Theorem 39.1.2: Projective Variety Affine Cover

Let $Y \subseteq \mathbb{P}_k^n$ be a projective algebraic set. Then the sets $Y \cap U_i$, $1 \leq i \leq n+1$, form an open cover of Y, and each one is isomorphic to an *affine* algebraic set in $\mathbb{A}_k^n$.

***Proof*:**
The U_i cover $\mathbb{P}_k^n$, since a point has some nonzero coordinate, so the $Y \cap U_i$ cover Y and are open in Y. Fix i and identify U_i with $\mathbb{A}_k^n$ by the homeomorphism φ_i of proposition 39.1.1.

Let $Y = \mathcal{Z}_{\mathbb{P}}(F_1, \ldots, F_r)$ with each F_j homogeneous. Under φ_i a point of U_i is the class of a vector whose ith coordinate is 1, and F_j vanishes there exactly when its dehomogenization $(F_j)_*$ at x_i vanishes at the corresponding point of $\mathbb{A}_k^n$. Hence

$$\varphi_i(Y \cap U_i) = \mathcal{Z}_{\mathbb{A}}\big((F_1)_*, \ldots, (F_r)_*\big)$$

an affine algebraic set. Conversely φ_i^{-1} carries that set back, so the two are isomorphic, as we sought to show.

The content is the second clause: it is not just that Y is covered by sets homeomorphic to open subsets of $\mathbb{A}^n$, but that each piece is a genuine affine algebraic set with its own coordinate ring. Every local question about a projective variety can therefore be answered affinely, which is the working method for the rest of the book.

For the reader's awareness, we box the following[1]:

[1] this shall be the transition maps to verify the gluing is well-defined

Corollary 39.1.3: Transition Between Affine Cover

For $i \neq j$ the two charts agree on the overlap: the composite

$$\varphi_{ij} = \varphi_j \circ \varphi_i^{-1} \colon \varphi_i(U_i \cap U_j) \to \varphi_j(U_i \cap U_j)$$

is an isomorphism of affine varieties, given in coordinates by:

$$\begin{aligned}(x_1, ..., \hat{x}_i, ..., x_{n+1}) &\mapsto [x_1, ..., x_{i-1}, 1, x_{i+1}, ..., x_{n+1}] \\ &\mapsto \left[\frac{x_1}{x_j}, ..., \frac{x_{i-1}}{x_j}, \frac{1}{x_j}, \frac{x_{i+1}}{x_j}, ..., \frac{x_j}{x_j}..., \frac{x_{n+1}}{x_j}\right] \\ &\mapsto \left(\frac{x_0}{x_j}, ..., \frac{x_{i-1}}{x_j}, \frac{1}{x_j}, \frac{x_{i+1}}{x_j}, ..., \hat{1}_j..., \frac{x_{n+1}}{x_j}\right)\end{aligned}$$

where the hats mark omitted variables, and $i < j$ is assumed without loss of generality.

***Proof*:**
On $U_i \cap U_j$ both x_i and x_j are nonzero, so each displayed coordinate is a ratio with nonvanishing denominator and the composite is regular. Reversing the roles of i and j gives φ_{ji}, and $\varphi_{ji} \circ \varphi_{ij} = \mathrm{id}$ because both sides come from the identity on $U_i \cap U_j$. So φ_{ij} is an isomorphism, completing the proof.

39.2 Projective Closure

Let us now formalize extending closed sets of $\mathbb{A}_k^n$ to $\mathbb{P}_k^n$, and conversely restrict closed sets of $\mathbb{P}_k^n$ to any $U_i \cong \mathbb{A}_k^n$. Let us first take a closed subset $V \subseteq \mathbb{P}_k^n$, and consider its restriction to $U_i \cong \mathbb{A}_k^n$. As $U_i = D(x_i)$ is open, $V \cap U_i = W_i$ is open in V (the subspace topology on U_i and the Zariski topology transported by φ_i agree). Let us show that W_i is closed in $U_i \cong \mathbb{A}_k^n$, that is it is the zero set of the appropriate polynomials. If $f_1, \ldots, f_r$ are the homogeneous polynomials defining V[2] so that

$$f_1 = \cdots = f_n = 0$$

with $\deg f_i = n_i$, then on U_i each f_j is converted to:

$$x_i^{n_j} f_j \left(\frac{x_1}{x_j}, ..., 1_j, ..., \frac{x_{n+1}}{x_j}\right)$$

For example $f(x, y, x_3) = xy - x_3^2$ would be converted to $xy - 1$ in U_3. Then we can see that $V = \cup W_i$, that is each V can be covered by closed subsets from each U_i.

Conversely, if $W \subseteq \mathbb{A}_k^n$ is a closed subset, we may take the closure in $\mathbb{P}_k^n$, $\overline{W}$, which is the intersection of all closed sets containing W where we interpret W as being set in one of the U_i's. Some closed W are also closed in $\mathbb{P}_k^n$. For example $x^2 + y^2 - 1$ over $\mathbb{C}[x, y]$ when embedded becomes the zero set of $x^2 + y^2 - z^2$ whose points are $[x_0 : y_0 : 1]$ with $x_0^2 + y_0^2 = 1$, together with the two circular points at

[2] Recall that we are working over Noetherian rings, and hence each ideal is finitely generated

infinity of example 38.3.4. On the other hand, if we take $xy-1$ and consider $xy-z^2$, we get that the traditional zero's become:

$$\left[a : \frac{1}{a} : 1\right]$$

and we now get a new 0 when $z=0$ giving two more points:

$$[1:0:0] \qquad [0:1:0]$$

which we can think of as adding the two points at infinity of the graph of $y=1/x$. Another example would be $y-x^2$. When extending to $\mathbb{P}^2_k$, we get $yz-x^2$. By setting $x=0$, we get that $y=0$ is a solution (which already exists in $\mathbb{A}^2_k$) as well as the new solution:

$$[0:0:1]$$

which is a new point at infinity that makes the parabola shape akin to an infinite ellipse.

Summarizing our discussion and the pertinent consequences:

Proposition 39.2.1: Closure of Affine Variety

Let $I \subseteq k[x_1,...,x_n]$ be a (not necessarily homogeneous) ideal, $I^* = \{f^* \mid f \in I\}$, $V^* = \mathcal{Z}(I^*) \subseteq \mathbb{P}^n_k$. Take the last coordinate of $\mathbb{P}^n_k$ to be the coordinate at infinity, so that $\mathbb{A}^n_k$ embeds in $\mathbb{P}^n_k$ via $\iota_{n+1}\colon \mathbb{A}^n_k \xrightarrow{\sim} U_{n+1} \subseteq \mathbb{P}^n_k$. Let $(-)_*$ be the dehomogenization operation so that if $V = \mathcal{Z}(J) \subseteq \mathbb{P}^n_k$, then $I_* = \{f_* \mid f \in J\}$ where f_* is the dehomogenization at the coordinate at infinity, and $V_* = \mathcal{Z}(I_*)$. Let H_∞ represent the hyperplane at infinity (i.e. $\mathcal{Z}(x_{n+1})$). Then:

1. $\iota_{n+1}(V) = V^* \cap U_{n+1}$
2. $(V^*)_* = V$
3. If $V \subseteq W \subseteq \mathbb{A}^n_k$, then $V^* \subseteq W^* \subseteq \mathbb{P}^n_k$
4. If $V \subseteq W \subseteq \mathbb{P}^n_k$, then $V_* \subseteq W_* \subseteq \mathbb{A}^n_k$
5. If V is irreducible in $\mathbb{A}^n_k$, then V^* is irreducible in $\mathbb{P}^n_k$
6. If $V = \cup_i V_i$ is the irreducible decomposition of V in $\mathbb{A}^n_k$, then $V^* = \cup_i V_i^*$ is the irreducible decomposition of V^* in $\mathbb{P}^n_k$
7. If $V \subseteq \mathbb{A}^n_k$, then V^* is the closure of $\iota_{n+1}(V) \subseteq \mathbb{P}^n_k$ in the Zariski topology (i.e. it is the smallest algebraic set in $\mathbb{P}^n_k$ containing $\iota_{n+1}(V)$
8. If $\emptyset \neq V \subseteq \mathbb{A}^n_k$, then no irreducible component of V^* is contained in H_∞, though components may meet it
9. If $V \subseteq \mathbb{P}^n_k$ has no irreducible component contained in H_∞, then $(V_*)^* = V$

***Proof*:**
Parts (3) and (4) are immediate from the definitions, since homogenizing and dehomogenizing preserve containment of ideals and $\mathcal{Z}$ reverses it.

(1) and (2). A point of $\mathbb{A}_k^n$ satisfies f exactly when its image in U_{n+1} satisfies f^*, because f^* restricted to $x_{n+1} = 1$ is f by proposition 38.3.5(3). Intersecting with U_{n+1} therefore recovers $\iota_{n+1}(V)$, which is (1), and dehomogenizing recovers V, which is (2).

(7). By (1), V^* is a closed set whose trace on U_{n+1} is $\iota_{n+1}(V)$, so $V^* \supseteq \overline{\iota_{n+1}(V)}$. Conversely let W be any closed set containing $\iota_{n+1}(V)$, say $W = \mathcal{Z}_{\mathbb{P}}(J)$ with J homogeneous. Each $F \in J$ vanishes on $\iota_{n+1}(V)$, so F_* vanishes on V, so $F_* \in \mathcal{I}(V)$ and hence $F = x_{n+1}^r (F_*)^* \in I^*$ for some r, again by proposition 38.3.5(3). Therefore $V^* \subseteq W$, and V^* is the smallest such closed set.

(5) and (6). If V is irreducible then $\mathcal{I}(V)$ is prime, and the homogenization of a prime ideal is prime, because a product of homogeneous elements lies in I^* exactly when the product of their dehomogenizations lies in $\mathcal{I}(V)$. So V^* is irreducible, giving (5). Applying (5) and (7) to each component of $V = \bigcup_i V_i$ gives $V^* = \bigcup_i V_i^*$ with each V_i^* irreducible, and no containments among them since (3) and (2) make $V \mapsto V^*$ inclusion-reflecting; that is (6).

(8) and (9). A component of V^* contained in H_∞ would meet U_{n+1} in the empty set, so by (1) it would contribute nothing to V, contradicting (7)'s minimality. That is (8). For (9), the hypothesis makes V the projective closure of V_*, and (7) identifies that closure with $(V_*)^*$, as we sought to show.

Definition 39.2.2: Projective Closure

Let $V \subseteq \mathbb{A}_k^n$ be an algebraic set. Then $V^* \subseteq \mathbb{P}_k^n$ as defined in proposition 39.2.1 is called the *projective closure*.

Note that the homogenization of an ideal is *not* the homogenization of its generators

Example 39.2.3: Homogenization of Ideal and Generators

Let $I = (y - x^2, z - x^3)$ and $V = \mathcal{Z}(I)$ i.e. the twisted cubic in $\mathbb{A}_k^3$. Show that

1. $zw - xy \in I(V)^*$
2. $zw - xy \notin \left((y - x^2)^*, (z - x^3)^*\right)$

In other words, the ideal $(f_1^*, ..., f_n^*)$ is too small to capture all the new relations (in particular it usually adds a line at infinity).

Next, we define $k_{\mathbb{P}}(V)$ to be the localization of $k_{\mathbb{P}}[V]$. Note that if $f/g \in k(V)$ where $\deg(f) = \deg(g) = d$, then f/g is in fact a well-defined function on V, since:

$$\frac{f(\lambda x)}{g(\lambda x)} = \frac{\lambda^d f(x)}{\lambda^d g(x)} = \frac{f(x)}{g(x)}$$

there are thus many more rational functions defined on V than regular functions. The set of well-defined functions shall be given a name:

Definition 39.2.4: Function Field

Let V be a projective variety. Then:

$$\kappa(V) = \left\{ \frac{f}{g} \;\middle|\; f, g \in k_{\mathbb{P}}[V] \text{ homogeneous of the same degree, } g \notin \mathcal{I}_{\mathbb{P}}(V) \right\}$$

taken inside the fraction field of $k_{\mathbb{P}}[V]$, and modulo the usual identification $f/g = f'/g'$ when $fg' - f'g \in \mathcal{I}_{\mathbb{P}}(V)$. Elements of $\kappa(V)$ are the *rational functions* on V.

Homogeneity of f and g is what makes the ratio well defined on V: degree is not defined on a non-homogeneous element of $k_{\mathbb{P}}[V]$, and it is equality of the two degrees that makes the scaling factors λ^d cancel.

Proposition 39.2.5: Rational Functions over Projective and Affine Variety

Let $V \subseteq \mathbb{A}^n_k$ be an affine variety. Then:

$$k(V) \cong \kappa(V^*)$$

Similarly, if $V \subseteq \mathbb{P}^n_k$ is a projective variety not contained in the hyperplane at infinity. Then:

$$\kappa(V) \cong k(V_*)$$

***Proof*:**
Define a map in the direction of the claim by homogenizing:

$$\alpha\colon k(V) \to \kappa(V^*) \qquad \alpha\left(\frac{f}{g}\right) = \frac{x_{n+1}^{\deg g - \deg f} f^*}{g^*}$$

the power of x_{n+1} inserted so that numerator and denominator have the same degree, as definition 39.2.4 requires; when $\deg f > \deg g$ the correcting factor goes on the denominator instead.

Well defined. If $f/g = f'/g'$ in $k(V)$ then $fg' - f'g \in \mathcal{I}(V)$, and homogenizing that relation, using $(uv)^* = u^*v^*$ from proposition 38.3.5(1), puts the corresponding difference in $\mathcal{I}_{\mathbb{P}}(V^*)$ up to a power of x_{n+1}, which is a unit on U_{n+1} and so does not change the class.

Inverse. Dehomogenizing gives $\beta(F/G) = F_*/G_*$, and $\beta \circ \alpha = \mathrm{id}$ because $(f^*)_* = f$ for every f by proposition 38.3.5(3). For $\alpha \circ \beta = \mathrm{id}$, part (3) of that proposition gives $(F_*)^* = F$ up to a power of x_{n+1}; that power cancels between numerator and denominator since both are corrected to a common degree. Hence α is a bijection, and it is a ring homomorphism because homogenization is multiplicative and additive up to the same powers of x_{n+1}.

The second isomorphism is the first applied to V_*, using $(V_*)^* = V$ from proposition 39.2.1(9), which is exactly the hypothesis that V is not contained in the hyperplane at infinity, as we sought to show.

On the one hand, it may seem counter-intuitive that though we are restricting to degree 0 rational polynomials, we somehow get as many rational functions as when we allowed for any degree $r - s$

rational polynomial. On the other hand, this should not be too strange for the reader: the domain of definition is a dense open subset, and we shall show that V^* is the closure of V when embedded in $\mathbb{P}^n_k$.

We can define the *stalk* at p to be

$$\mathcal{O}_p(V) = \{f/g \in \kappa(V) \mid g(p) \neq 0\}$$

The same argument as proposition 33.2.10 shows $\mathcal{O}_p(V)$ is a local Noetherian domain.

Proposition 39.2.6: Stalk Defined Locally

Let $V \subseteq \mathbb{P}^n_k$ be a projective variety not contained in the hyperplane at infinity, let V_* be its dehomogenization, and let $p \in V_*$. Then:

$$\mathcal{O}_p(V) \cong \mathcal{O}_p(V_*)$$

This shows that the properties of a stalk are inherently "local", depending only on the information around the point p.

***Proof*:**
Restrict the isomorphism α of proposition 39.2.5. A rational function on V lies in $\mathcal{O}_p(V)$ exactly when it has a representative F/G with $G(p) \neq 0$; applying α and its inverse turns such representatives into representatives f/g on V_* with $g(p) \neq 0$, since dehomogenizing at x_{n+1} does not change whether a form vanishes at a point of U_{n+1}. So α carries $\mathcal{O}_p(V)$ onto $\mathcal{O}_p(V_*)$ and is a ring isomorphism between them, as we sought to show.

Next, notice there is more than one natural set of coordinates on projective space. It is natural to consider the coordinates of $\mathbb{P}^n_k$ to be given by degree 1 homogeneous functions, namely each point $x \in \mathbb{P}^n_k$ is of the form $[x_1 : \cdots : x_{n+1}]$ where $[\lambda x_1 : \cdots : \lambda x_{n+1}] = \lambda[x_1 : \cdots : x_{n+1}]$. However, we may also just as naturally choose to represent each point in any degree of homogeneity, namely:

$$[\lambda w_1 : \cdots : \lambda w_{n+1}] = [w_1 : \cdots : w_{n+1}]$$

where we can have $w_i = (x_i)^k$.

Finally, as all homogeneous polynomials must be of the same degree, we can give a rough classification of all hypersurfaces of projective space $\mathbb{P}^n_k$ of degree d. A hypersurface is defined by a single homogeneous equation. As there are $n+1$ variables, there are $\binom{n+d}{d} = N$ possible degree d monomials. Labeling them $M_1, ..., M_N$, any degree d homogeneous polynomial is of the form:

$$\sum_i^N a_i M_i$$

These hypersurface are naturally defined up to multiplication of a constant $c \in k$. But then, the coefficients determine the hypersurface up to a constant, meaning they can be associated to a point in $\mathbb{P}^N_k$, or said differently $\mathbb{P}^N_k$ *parameterizes* the degree d hypersurfaces of $\mathbb{P}^n_k$ or is the *moduli space* for the degree d hypersurfaces of $\mathbb{P}^n_k$.

Note that some points classify the same hypersurface, for example $x^2y = 0$ and $xy^2 = 0$ in $\mathbb{P}^2_k$, both of which cut out the union of the lines $x = 0$ and $y = 0$. The difference is in the multiplicity, information that would be distinguishable when working with *closed subschemes* instead of closed subsets.

39.3 Quasi-Projective Varieties

As we saw in the above, affine and projective varieties are naturally linked to each other. Many results that are proven in classical algebraic geometry apply both to projective and affine varieties. Furthermore, the open subset of projective variety (for example $\mathbb{P}^1_k$ minus the point at infinity and zero) have natural algebraic structure but are not the vanishing set of polynomials (they are the opposite, they are the non-vanishing set of polynomials). The properties of such sets often are strongly tied to varieties, or are even varieties themselves (for example $\mathbb{P}^1_k$ without the point at infinity is the closed set $\mathbb{A}^1_k$ in $\mathbb{A}^1_k$, but is an open set in $\mathbb{P}^1_k$). The definition of quasi-projective varieties aims to unify all of these.

If $X \subseteq \mathbb{P}^n_k$ is closed and $X_i = X \cap U_i$ is open in X, they are in fact *closed* in $U_i \cong \mathbb{A}^n_k$. This can be seen by de-homogenizing: if X is given by equations $f_0, ..., f_n$ where $\deg f_i = n_i$, then U_i is given by:

$$x_i^{-n_j} f_j = f(x_1, .., x_{i-1}, 1, x_{i+1}, ..., x_{n+1}) = 0$$

If we have $X_i \subseteq U_i$ which is closed, then we may consider the closure $\overline{X_i} \subseteq \mathbb{P}^n_k$ by taking the polynomials that generate X_i and homogenizing them, namely if f is one such polynomial where $\deg f = d$, then the equations describing $\overline{X_i}$ are of the form

$$x_i^l f(x_1/x_i, ..., x_{n+1}/x_i) = 0$$

Which implies that:

$$X_i = \overline{X_i} \cap U_i$$

Hence, if $X \subseteq \mathbb{P}^n_k$, then an open subset of X will correspond to a closed subset of $\mathbb{A}^n_k$. As X is an open subset of itself, it corresponds to an open subset of a closed projective algebraic set. Thus, open subsets of closed projective sets encapsulate affine and projective varieties:

Definition 39.3.1: Quasi-projective Variety

A *quasi-projective variety* is an open subset of a closed projective variety.

Example 39.3.2: Quasi-Projective Varieties

1. *Affine varieties are quasi-projective.* $\mathbb{A}^n_k$ is the standard chart $U_0 \subseteq \mathbb{P}^n_k$, an open subset, and a closed $V \subseteq \mathbb{A}^n_k$ is $\overline{V} \cap U_0$ for its projective closure. So every affine variety is open in a projective one.

2. *Projective varieties are quasi-projective*, taking the open set to be everything.

3. *Neither affine nor projective.* $\mathbb{A}^2_k \setminus \{(0,0)\}$ is open in $\mathbb{A}^2_k$, hence quasi-projective. It is not projective, carrying the nonconstant regular function x, and it is not affine: its ring of regular functions is $k[x, y]$, whose maximal ideals include one for the missing origin. The details are the fifth exercise of exercise set 39.5.1.

4. *The complement of a hypersurface.* For a form F of degree d, the set $\mathbb{P}^n_k \setminus \mathcal{Z}(F)$ is quasi-projective and in fact affine, being isomorphic to a closed subvariety of $\mathbb{A}^N$ under the Veronese of definition 41.1.1, which turns $\mathcal{Z}(F)$ into a hyperplane section.

5. *A non-example.* Over $\mathbb{C}$, the subset $\mathbb{Z} \subseteq \mathbb{C} = \mathbb{A}^1_{\mathbb{C}}$ is not quasi-projective. A locally closed

subset of $\mathbb{A}^1_{\mathbb{C}}$ is finite or cofinite, by the sixth exercise of exercise set 41.6.1, and $\mathbb{Z}$ is neither. So it is not even constructible, and by theorem 41.3.4 it is not the image of any regular map.

A closed subset $C \subseteq U$ of a quasi-projective variety is the intersection of C with a closed subset $\mathbb{P}^n_k$. The irreducible decomposition of theorem 32.3.5 carries over verbatim, since it used only that the space is Noetherian. If $Y \subseteq X \subseteq \mathbb{P}^n_k$ with Y quasi-projective, then Y is a *subvariety* of X.

It must be noted that the collection of quasi-projective varieties is larger than the collection affine and projective varieties. For example $\mathbb{A}^2_k \setminus \{(0,0)\}$ is a quasi-projective variety, but is not affine or projective.

Let X be a quasi-projective variety and $U \subseteq X$ an open subset. Define $\Gamma(U, \mathcal{O}_X) \subseteq \kappa(X)$ to be the collection of all rational functions on X that are defined at each $p \in U$. Equivalently:

$$\Gamma(U, \mathcal{O}_X) = \bigcap_{p \in U} \mathcal{O}_p(X)$$

Note that if $U = X$, then by proposition 33.2.9 and proposition 39.2.6,

$$\Gamma(X, \mathcal{O}_X) = k[X] = \Gamma(X)$$

showing this notation is consistent with our earlier definitions.

Next, let's see that checking that a quasi-projective variety is closed is an open-local condition:

Lemma 39.3.3: Closedness of Quasi-projective Variety is Open-local

Let $Y \subseteq X$ be a quasi-projective variety. Then Y is closed if and only if $X = \cup_\alpha U_\alpha$ where U_α is open and $Y \cap U_\alpha$ is closed for every α. Namely, closedness is open-local for quasi-projective varieties

***Proof*:**
If Y is closed then $Y \cap U_\alpha$ is closed in U_α for any open cover, which is the easy direction.

Conversely let $X = \bigcup_\alpha U_\alpha$ with each U_α open and each $Y \cap U_\alpha$ closed in U_α. Write $U_\alpha = X \setminus Z_\alpha$ with Z_α closed, and $Y \cap U_\alpha = T_\alpha \cap U_\alpha$ with T_α closed in X. We claim

$$Y = \bigcap_\alpha (Z_\alpha \cup T_\alpha)$$

which exhibits Y as an intersection of closed sets and so finishes the proof.

For $\subseteq$: let $y \in Y$ and fix α. Either $y \notin U_\alpha$, in which case $y \in Z_\alpha$; or $y \in U_\alpha$, in which case $y \in Y \cap U_\alpha \subseteq T_\alpha$. Either way $y \in Z_\alpha \cup T_\alpha$.

For $\supseteq$: let y lie in every $Z_\alpha \cup T_\alpha$. The U_α cover X, so $y \in U_\beta$ for some β, and then $y \notin Z_\beta$, forcing $y \in T_\beta$. Hence $y \in T_\beta \cap U_\beta = Y \cap U_\beta \subseteq Y$, as we sought to show.

The following shows that any variety is fundamentally locally affine:

Lemma 39.3.4: Varieties Are Locally Affine

Let X be a quasi-projective variety. Then every $x \in X$ has an open neighborhood that is isomorphic to an affine variety

***Proof*:**
Take $X \subseteq \mathbb{P}^n_k$. If $x \in U_{n+1}$, then $x \in X \cap U_{n+1}$ and by definition of a quasi-projective variety:

$$X \cap U_{n+1} = Y - Y_1$$

where Y_1, Y are closed subsets of U_{n+1}. As $x \in Y$, there exists a polynomial f with coordinates in U_{n+1} where $f = 0$ on Y_1 and $f(x) \neq 0$. Let $D(f) = Y \setminus \mathcal{Z}(f)$, the subset of Y on which f does not vanish. It is open in Y and contains x, and it misses Y_1 entirely since f vanishes there, so $D(f) \subseteq Y \setminus Y_1 = X \cap U_{n+1}$ is an open neighbourhood of x inside X.

I claim this neighborhood is isomorphic to an affine variety. If $(f_1, ..., f_m)$ is the ideal associated to $Y \subseteq U_{n+1}$, then define the variety in $\mathbb{A}^{n+1}_k$ via:

$$f_1(x_1, ..., x_n) = \cdots = f_m(x_1, ..., x_n) = 0 \qquad f(x_1, ..., x_n)x_{n+1} = 1$$

i.e.

$$Z = \mathcal{Z}(f_1, ..., f_m, f \cdot x_{n+1} - 1)$$

Then the mapping

$$\varphi(x_1, ..., x_{n+1}) \to (x_1, ..., x_n)$$

gives a regular mapping from Z into $D(f)$, and

$$\psi(x_1, ..., x_n) \to (x_1, ..., x_n, f(x_1, ..., x_n)^{-1})$$

is a regular mapping of $D(f)$ into Z. The two composites are the identity: $\varphi \circ \psi$ drops the last coordinate that ψ appended, and $\psi \circ \varphi$ restores it, since a point of Z satisfies $f \cdot x_{n+1} = 1$ and so has $x_{n+1} = f^{-1}$ determined by the others. Hence $D(f) \cong Z$, an affine variety, as we sought to show.

This result also showed that principal open sets (definition 31.3.5) are naturally associated to an affine algebraic set, and that the regular function on $D(f)$ are:

$$k[D(f)] = k[X][1/f]$$

In particular f is nonzero on $D(f)$, so this is well defined. Overall, when checking if a mapping between quasi-projective is regular, it suffices to check that any affine variety in the codomain has a closed pre-image.

39.4 Regular Maps

Let us now define regular mappings on quasi-projective varieties. This definition should extend that of affine varieties where a polynomial function was given for each coordinate.

Recall that a rational map on $\mathbb{P}^n_k$ is the quotient of two homogeneous polynomials, with degree $r - s$ where r, s is the degree of the numerator and denominator respectively. A regular point x of a rational function f would be a point where f can be represented by p/q with $q(x) \neq 0$. When working over projective varieties, we shall have that most rational polynomials are not rational functions; we shall require that $r = s$, that is it must be of degree 0.

Definition 39.4.1: Regular Point [on Quasi-projective Variety]

Let X be a quasi-projective variety and $f \in \kappa(X)$ a rational function. Then f is *regular at* $x \in X$ if it admits a representative $f = p/q$ with p, q homogeneous of the same degree and $q(x) \neq 0$. Such a q is then nonzero on the whole open set $D(q)$, so being regular at x is a condition on a neighbourhood of x rather than on x alone.

If f is regular at every point of X it is called *regular* on X. The *set* of regular functions on X is written $k[X]$.

When we defined a regular function before, it was an element of the coordinate ring. We have now defined it to be locally regular at all points. When k is algebraically closed, these two are in fact the same. If X is irreducible, then this is the result of proposition 33.2.2. For the general case, say $f = p_x/q_x$ where $q_x \neq 0$ on U_x so that on U_x:

$$q_x f = p_x \tag{39.1}$$

Then this equation holds on all of X. To see this, choose a regular functions that vanishes on $X \setminus U_x$ but not at x (recall that U_x is dense in X), and multiply it by $p_x q_x$. Then we see that (39.1) must hold outside of U_x as both sides of the equality will vanish. Then like the irreducible case, we can now find points $x_1, ..., x_n$ and regular functions $h_1, ..., h_n$ such that

$$\sum_i^n q_x h_i = 1$$

Then for each x_i, multiplying by h_i and adding up we get:

$$f = \sum_i^n p_x h_i$$

and hence f is regular in the same sense given in definition 32.1.2. Recall that it was mentioned that if $X = \mathbb{P}_k^n$ that $k[\mathbb{P}_k^n]$ consists of only constants functions. We shall soon show that if $X = V \subseteq \mathbb{P}_k^n$ is any projective variety that $k[V]$ is the collection of constant functions.

Definition 39.4.2: Regular Map [On Quasi-projective Variety]

Let $f\colon X \to Y$ be a map of quasi-projective varieties with $Y \subseteq \mathbb{P}_k^m$. Then f is *regular* if for every $x \in X$ there is a standard affine chart $U_i \subseteq \mathbb{P}_k^m$ containing $f(x)$ and an open neighbourhood W of x with $f(W) \subseteq U_i$, such that each of the m coordinate functions of $f|_W$ is regular in the sense of definition 39.4.1.

The collection of regular function is denoted $k[X]$, $\mathcal{O}_X(X)$, or $\Gamma(X)$[a], and the collection of regular mapping between two quasi-projective varieties is denoted $\mathrm{Hom}_{\mathbf{Reg}}(X, Y)$.

[a]This notation shall become more prominent in modern algebraic geometry

This definition of regular function is well-defined and is independent of choice of U_i: if $f(x) = (y_0, ..., \hat{1}_i, ..., y_m) \in U_i$ (where the hat indicates omission), it is also contained in U_j when $y_j \neq 0$, and the coordinates of this point in U_i are of the form:

$$(y_0/y_j, ..., 1/y_j, ..., \hat{1}_j, ..., y_m/y_j)$$

Hence, given the mapping $f : U \to U_i$ is given by

$$(f_0, .., \hat{1}_i, .., f_m)$$

then $f : U \to U_j^m$ is given by $(f_0/f_j, ..., 1/f_j, ..., \hat{1}_j, ..., f_m/f_j)$. As $f_j(x) \neq 0$ and the set U' of points of U where $f_j = 0$ is open. Then on U', the functions $f_0/f_j, ..., 1/f_j, ..., \hat{1}_j, ..., f_m/f_j$ are regular, and so $f : U' \to U_j^m$ is regular.

If we have an affine closed set, by theorem 32.1.5 the regular mapping $f\colon X \to Y$ is determined by the pullback $f^{\#}\colon k[Y] \to k[X]$. This is not the case if X, Y are projective, for as mentioned a few times already we shall have $k[X]$ and $k[Y]$ consist of only constant functions. However, it must be noted that for a general quasi-projective variety, $k[X]$ could be quite large[3]. If X is isomorphic to a closed subset of an affine space, it shall be called an *affine variety*, and if it isomorphic to a closed subset of projective space, it shall be called *projective variety*. Not all quasi-projective varieties are affine or projective (for example $\mathbb{A}_k^2 \setminus \{(0,0)\}$).

Let us now characterize regular maps form quasi-projective varieties to projective space[4]. To understand such a map, start with a regular mapping f on X and a point $x \in X$ where $f(x) \in \mathbb{A}_k^m = U_0^m$ so that there is an open neighborhood where $f = (f_1, ..., f_m) : U \to U_i^m$. Then as it is regular we can write $f_i = p_i/q_i$ where p_i, q_i are of the same degree in homogeneous coordinates and $q_i(x) \neq 0$. Taking the l.c.m. of such fractions we get that $f_i = F_i/F_0$ where $F_0, ..., F_m$ are forms of the same degree and $F_0(x) \neq 0$; overall we can think of the point $f(x) \in U_i^m$ as the point:

$$f(x) = [F_0(x) : \cdots : F_m(x)] \in \mathbb{P}_k^m$$

in projective space. Note that there is more than one representative for such a regular function, namely if

$$F(x) = [F_0(x) : \cdots : F_m(x)] \qquad G(x) = [G_0(x) : \cdots : G_m(x)]$$

then they are the same point if and only if $F_iG_j = F_jG_i$ on X for all $0 \leq i, j \leq m$. This gives us an explicit condition for when $Y = \mathbb{P}_k^m$:

Proposition 39.4.3: Equivalent Regular Function

Let $X \subseteq \mathbb{P}_k^n$ be an irreducible quasi-projective variety and $f\colon X \to \mathbb{P}_k^m$ a map of sets. Then f is regular if and only if it can be written as

$$f = [F_0 : \cdots : F_m]$$

for forms F_i of a common degree in the homogeneous coordinates of $\mathbb{P}_k^n$, not all vanishing at any point of X. Two such presentations $[F_0 : \cdots : F_m]$ and $[G_0 : \cdots : G_m]$ define the same map exactly when

$$F_iG_j = F_jG_i \text{ on } X \text{ for all } 0 \leq i, j \leq m$$

***Proof*:**

Suppose f has such a presentation and fix $x \in X$. Some $F_j(x) \neq 0$, and on the open set $D(F_j) \cap X$ the coordinates of f in the chart U_j are the ratios F_i/F_j, each a quotient of forms of equal degree with nonvanishing denominator. By definition 39.4.1 these are regular, so f is

[3]Shafarevich mentions that Rees and Nagata constructed an example where $k[X]$ is not finitely generated

[4]This is in fact a larger exploration that culminates in the theory of line bundles on a projective scheme

regular.

Conversely let f be regular and fix x with $f(x) \in U_j$. In that chart f has coordinates $f_i = p_i/q_i$ with p_i, q_i forms of equal degree and $q_i(x) \neq 0$. Clearing denominators over $Q = q_0 \cdots q_m$ rewrites these as $f_i = F_i/F_j$ with all F_i forms of one degree and $F_j(x) \neq 0$, which is the required presentation near x. Two such local presentations agree on the overlap of their domains, which is a nonempty open subset of the irreducible X and hence dense, so they agree wherever both are defined and glue.

For the last clause, the two presentations give the same point of $\mathbb{P}_k^m$ at x exactly when the vectors $(F_0(x), \ldots, F_m(x))$ and $(G_0(x), \ldots, G_m(x))$ are proportional, which is the vanishing of every 2×2 minor $F_iG_j - F_jG_i$, as in example 38.3.7(1). Requiring it at every point of X is the stated condition, as we sought to show.

Example 39.4.4: Regular Mappings

1. *Projection*: Let $E \subseteq \mathbb{P}_k^n$ be a d-dimensional subspace given by $n-d$ linearly independent linear equations $(L_1, ..., L_{n-d})$. In homogeneous coordinates, the equations L_i are called *linear forms*. The mapping:

$$\pi(x) = (L_1(x) : \cdots : L_{n-d}(x))$$

is well-defined and is called the projection with center E. Notice this set is regular on $\mathbb{P}_k^n \setminus E$ as the L_i's do not vanish simultaneously. Then for any $X \subseteq \mathbb{P}_k^n$ where $X \cap E = \emptyset$ and X is closed, we have a regular mapping

$$\pi : X \to \mathbb{P}_k^{n-d-1}$$

Geometrically, we can interpret the mapping as follows. Take any $(n-d-1)$-dimensional subspace $H \subseteq \mathbb{P}_k^n$ with $H \cap E = \emptyset$. Through any point $x \in \mathbb{P}_k^n \setminus E$ and the space E, there passes a unique $(d+1)$-dimensional subspace E_x. Then this subspace intersects H at a unique point, namely $\pi(x)$. If X intersects E, but does not contain it, then this is a rational mapping.

We have seen the case where $d = 0$ before (think of the hyperbola to $\mathbb{A}_k^1$)

2. Take $V = \mathcal{Z}(y - x^2, z - x^3)$ (i.e. the twisted cubic). This is a rational curve with parameterization:

$$t \mapsto (t, t^2, t^3)$$

Let us now homogenize this variety. The homogenization of this map is:

$$[s : t] \mapsto \left[1 : \frac{t}{s} : \frac{t^2}{s^2} : \frac{t^3}{s^3}\right] = \left[s^3 : ts^2 : t^2s : t^3\right]$$

3. Take $V = \mathcal{Z}(x^2 + y^2 - z^2)$; the projectivization of the circle. Define the map:

$$f : \mathbb{P}_k^1 \to V \qquad [s : t] \to [s^2 - t^2 : 2st : s^2 + t^2]$$

This is an isomorphism, with inverse $[x : y : z] \mapsto [x + z : y]$. Checking the composite, $(s^2 - t^2)^2 + (2st)^2 = (s^2 + t^2)^2$, so f lands in V; and at $f([s : t])$ we have $x + z = 2s^2$

and $y = 2st$, so $[x + z : y] = [2s^2 : 2st] = [s : t]$ whenever $s \neq 0$, the case $t \neq 0$ being symmetric. So a smooth conic in $\mathbb{P}^2_k$ is isomorphic to $\mathbb{P}^1_k$, which is the genus-zero case of example 38.5.4(3).

4. *The Veronese and Segre embeddings* give further regular maps into larger projective spaces, and are the tools that turn products and degree-d hypersurfaces into linear data. Section 40.1 constructs the Segre embedding and chapter 41 the Veronese.

39.5 Rational Functions on Quasi-Projective Varieties

When working with affine varieties, we had the quotient of two regular functions. However, mentioned a few times, on $\mathbb{P}^n_k$ there are only constant regular functions, and a rational function, being the ratio of two regular functions, would still be constant.

The solution is to recall the definition of $\mathcal{O}_X$, namely:

$$\mathcal{O}_X = \{\varphi = f/g \in k(x_1, \dots, x_{n+1}) \mid f, g \text{ homogeneous of equal degree, } g \notin \mathcal{I}_{\mathbb{P}}(X)\}$$

where f and g are of the same degree. Then take the subset $\mathfrak{m}_X = \{\varphi = f/g \in \mathcal{O}_X \mid f \in I(X)\}$.

Definition 39.5.1: Rational Function On quasi-projective Variety

Let X be a quasi-projective variety. Then the set of rational functions on X is given by:

$$k(X) \cong \mathcal{O}_X/\mathfrak{m}_X$$

By a reasoning similar to proposition 33.2.7, a homogeneous function vanishes on X if and only if it vanishes on an open subset $U \subseteq X$. Hence $k(X) = k(U)$. As a consequence:

$$k(X) = k(\overline{X})$$

Due to this, when studying rational functions on quasi-projective spaces, we may reduce to studying rational functions to affine or projective varieties, which often leads most results to focus on the irreducible case. The set of regular points shall be called the *domain of definition*. If $f\colon X \to \mathbb{P}^m_k$, then it is determined by specifying $m+1$ forms $[F_0 : \cdots : F_m]$ of $n+1$ homogeneous coordinates of projective space $\mathbb{P}^n_k$ containing X (where at least one form must not vanish on X). Furthermore, if f can be given by functions $[f_0 : \cdots f_m]$ where each f_i are regular at all $x \in X$ and not all vanish at this point, then the map is regular at x, and hence determines a regular mapping of some neighborhood x into $\mathbb{P}^m_k$.

We saw before that birational maps are weaker than isomorphisms. However, rational maps are regular on an open dense set (as the reader should verify with quasi-projective varieties). This leads to the following result:

Proposition 39.5.2: Reduction To Open Subset

Let X, Y be irreducible varieties. Then X and Y are birationally isomorphic if and only if they contain isomorphic open subsets $U \subseteq X$ and $V \subseteq Y$.

***Proof*:**
If X and Y contain isomorphic open subsets then those subsets are dense, since X and Y are irreducible, and an isomorphism between dense open sets is a birational map. That is the easy direction.

Conversely let $f\colon X \dashrightarrow Y$ be birational with inverse g. Let $U_1 \subseteq X$ and $V_1 \subseteq Y$ be their domains of definition, both dense open by lemma 33.2.4. Put

$$U = U_1 \cap f^{-1}(V_1) \qquad V = V_1 \cap g^{-1}(U_1)$$

Each is open, and each is nonempty: $f(U_1)$ is dense in Y so it meets the nonempty open V_1, and symmetrically.

Shrinking once more, replace U by $U \cap g(V)$ and V by $V \cap f(U)$, which are again nonempty and open because f and g are dominating. On these sets both composites are defined everywhere and agree with the identity on a dense subset, hence equal it: two regular maps into a variety agreeing on a dense subset are equal, since their equalizer is closed. So $f|_U : U \to V$ and $g|_V : V \to U$ are mutually inverse isomorphisms, as we sought to show.

Exercise 39.5.1

1. Write down the $n+1$ standard charts of $\mathbb{P}^n_k$ explicitly and compute the transition map between two of them for $n = 1$ and $n = 2$. Verify that the transitions are regular where defined, as corollary 39.1.3 asserts.
2. Show that the projective closure of the parabola $\mathcal{Z}(y - x^2) \subseteq \mathbb{A}^2_k$ is $\mathcal{Z}(yz - x^2)$, and identify the point it acquires at infinity. Do the same for the hyperbola $\mathcal{Z}(xy - 1)$.
3. Let $V \subseteq \mathbb{A}^3_k$ be the twisted cubic, with $\mathcal{I}(V) = (y - x^2,\, z - x^3)$. Homogenize those two generators and show that their common projective zero set is strictly larger than the projective closure of V. Conclude that the projective closure is obtained by homogenizing every element of $\mathcal{I}(V)$, not just a set of generators.
4. Show that a regular function on a quasi-projective variety is continuous for the Zariski topology, and deduce that the locus where two regular maps agree is closed.
5. Show that $\mathbb{A}^2_k \setminus \{0\}$ is a quasi-projective variety and that every regular function on it extends to $\mathbb{A}^2_k$. Deduce that it is not isomorphic to any affine variety.
6. Verify the irreducible decomposition theorem for quasi-projective varieties: every quasi-projective variety is a finite union of irreducible closed subsets, none containing another, and the decomposition is unique.
7. Show that $[s : t] \mapsto [s^2 : st : t^2]$ is an isomorphism of $\mathbb{P}^1_k$ onto the conic $\mathcal{Z}(xz - y^2) \subseteq \mathbb{P}^2_k$, and compare with example 39.4.4(3).
8. Show that a rational function on an irreducible quasi-projective variety is determined by its restriction to any nonempty open subset, which is proposition 39.5.2, and give an example showing this fails for reducible varieties.

40

Completeness and Finite Maps

The property that makes projective varieties worth the trouble is completeness: the projection $X \times Y \to Y$ is a closed map whenever X is projective. It is the algebraic counterpart of compactness, and it yields at once that the image of a projective variety under a regular map is closed and that the only regular functions on a projective variety are constant.

Proving it requires knowing what a product of projective varieties is, which is not the naive thing. From there the chapter turns to finite maps, the algebraic mirror of integral extensions, and the theorem that every variety is a finite cover of a space as simple as $\mathbb{A}^m$ or $\mathbb{P}^m$. It closes by measuring a map against its fibres: the dimension of the source is the dimension of the target plus the dimension of the generic fibre, and the special fibres can only be bigger.

40.1 Products and the Segre Embedding

The direct product of algebraic sets $X \subseteq \mathbb{A}_k^n$ and $Y \subseteq \mathbb{A}_k^m$ is again algebraic, and lives in $\mathbb{A}_k^{n+m}$. If $X' \subseteq \mathbb{A}_k^p$ and $Y' \subseteq \mathbb{A}_k^q$ are algebraic sets with $X \cong X'$ and $Y \cong Y'$, then $X \times Y \cong X' \times Y' \subseteq \mathbb{A}_k^{p+q}$, showing that this definition is embedding-independent. The projective case is not so simple. There is no N with $\mathbb{P}_k^n \times \mathbb{P}_k^m \cong \mathbb{P}_k^N$; what there is instead is a *closed embedding* into $\mathbb{P}_k^N$ for $N = (n+1)(m+1)-1$, whose image is a proper subvariety cut out by explicit quadratic equations. Finding that embedding is the business of this section. Defining the product also supplies the tool that section 40.2 needs.

To start, recall that $\mathbb{A}_k^n \times \mathbb{A}_k^m \cong \mathbb{A}_k^{n+m}$. A first attempt at understanding $\mathbb{P}_k^n \times \mathbb{P}_k^m$ would be to consider $\mathbb{P}_k^{n+m}$. However, this space is dependent on the representation of the homogeneous points in $\mathbb{P}_k^n$ and $\mathbb{P}_k^m$, for example if $[2:0] = [1:0] \in \mathbb{P}_k^1$, and $[3:0] = [1:0] \in \mathbb{P}_k^1$, then:

$$[1:1:0] \neq [2:3:0] \in \mathbb{P}_k^2$$

Furthermore, if we choose $[x_1 : x_2] \in \mathbb{P}_k^1$ and $[y_1 : y_2] \in \mathbb{P}_k^1$, which point in $\mathbb{P}_k^2$ ought to represent this point? To resolve this, we shall "twist-multiply"

$$[x_1y_1 : x_1y_2 : x_2y_1 : x_2y_2] \in \mathbb{P}_k^{(1+1)(1+1)-1}$$

Notice that for $[\lambda x_1 : \lambda x_2]$ and $[\mu y_1 : \mu y_2]$ then:

$$[\lambda\mu x_1y_1 : \lambda\mu x_1y_2 : \lambda\mu x_2y_1 : \lambda\mu x_2y_2] = [x_1y_1 : x_1y_2 : x_2y_1 : x_2y_2] \in \mathbb{P}_k^3$$

and hence is independent of representation. Generalizing this, we define for any

$$x = [x_0 : \cdots : x_n], y = [y_0 : \cdots : y_m] \qquad w = [w_{ij}] \in \mathbb{P}_k^{(n+1)(m+1)-1}, \ w_{ij} = x_iy_j$$

Consider now the embedding $\varphi : \mathbb{P}_k^n \times \mathbb{P}_k^m \to \mathbb{P}_k^{(n+1)(m+1)-1}$ given by the above. The image is indeed a closed subset, for it satisfies the equations:

$$w_{ij}w_{kl} - w_{il}w_{kj} = 0 \tag{40.1}$$

which hold because both sides equal $x_ix_ky_jy_l$ once $w_{ij} = x_iy_j$ is substituted. The construction is natural in the sense that it matches the standard affine charts on both sides. Write $\mathbb{A}^n$ for $\mathbb{A}_k^n$ and let $U_i \subseteq \mathbb{P}_k^n$ be the ith standard affine chart, so $U_i \cong \mathbb{A}^n$. Then:

$$\varphi(U_i \times U_j) = W_{ij} = \mathbb{A}_{ij}^N \cap \varphi(\mathbb{P}_k^n \times \mathbb{P}_k^m)$$

where $N = (n+1)(m+1) - 1$ and $\mathbb{A}_{ij}^N$ is the set of points of $\mathbb{P}_k^N$ where $w_{ij} \neq 0$. Consider now the set $\mathbb{A}_{00}^N$. Take a point in projective space where $w_{00} \neq 0$ so that it corresponds to a point in $\mathbb{A}_{00}^N$:

$$[w_{ij}] = \varphi(x, y) \in \mathbb{P}_k^N$$

Take $z_{ij} = w_{ij}/w_{00}$, $x_i = u_i/u_0$, $y_j = v_j/v_0$ as the inhomogeneous coordinates. Then:

$$z_{i0} = x_i \qquad z_{0j} = y_j \qquad z_{ij} = x_iy_j = z_{i0}z_{0j}$$

So $\mathbb{A}_{00}^N \cong \mathbb{A}^{n+m}$ with coordinates $(x_1, \ldots, x_n, y_1, \ldots, y_m)$, the isomorphism being induced by φ. Hence φ is indeed the natural choice for the embedding of $\mathbb{P}_k^n \times \mathbb{P}_k^m$. This may feel more natural if we recall that each point of projective space $[a_0 : \cdots a_n] \in \mathbb{P}_k^n$ is closed and given by a system of linear equations

$$a_ix_j - a_jx_i = 0$$

mirroring (40.1). Notice that we may interpret the point $[w_{ij}]$ as an $(n+1) \times (m+1)$ matrix given by multiplying the points x and y as a $(n+1, 1)$ matrix and a $(1, m+1)$ matrix respectively (note the resulting rank of the matrix will be 1). From this, we may define the product of two quasi-projective varieties

Definition 40.1.1: Segre Embedding: Product of Projective Space

Let $V \subseteq \mathbb{P}_k^n, W \subseteq \mathbb{P}_k^m$ be two quasi-projective varieties. Then:

$$V \times W \subseteq \varphi(\mathbb{P}_k^n \times \mathbb{P}_k^m) \subseteq \mathbb{P}_k^{(n+1)(m+1)-1}$$

is the product of the quasi-projective varieties V, W.

Example 40.1.2: Product of Projective Lines

Let $V = W = \mathbb{P}^1_k$ and consider $\varphi(\mathbb{P}^1_k \times \mathbb{P}^1_k) \subseteq \mathbb{P}^3_k$. This set satisfies the equation:

$$w_{11}w_{00} - w_{01}w_{10} = 0$$

which is a nondegenerate quadric Q in $\mathbb{P}^3_k$. Let us consider the set

$$\varphi(\{\alpha\} \times \mathbb{P}^1_k)$$

then given $\alpha = [\alpha_0 : \alpha_1]$, these are all points in $\mathbb{P}^3_k$ such that

$$\alpha_1 w_{00} - \alpha_0 w_{10} = 0 \qquad \alpha_1 w_{01} - \alpha_0 w_{11} = 0$$

These equations give lines in $\mathbb{P}^3_k$. Similarly for $\varphi(\mathbb{P}^1_k \times \{\beta\})$. The two rulings of the quadric are visible in the following picture:

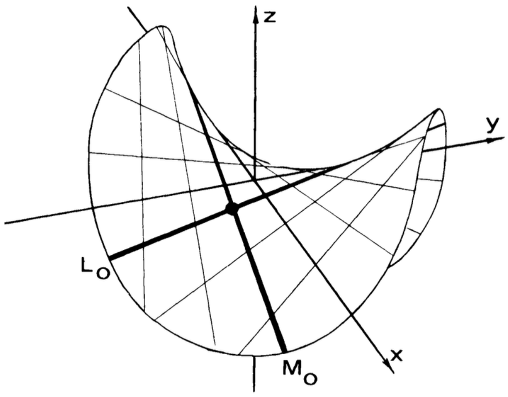


Figure 40.1: Segre Embedding of Product of Projective Lines

The closed subsets of im$\varphi \subseteq \mathbb{P}^{(n+1)(m+1)-1}_k$ correspond well with the closed subsets of $\mathbb{P}^n_k \times \mathbb{P}^m_k$. Let us start with seeing which subset of $\mathbb{P}^n_k \times \mathbb{P}^m_k$ are algebraic subsets under the Segre Embedding. Let $X \subseteq \mathbb{P}^{(n+1)(m+1)-1}_k$ be a sub-variety so that it is defined by homogeneous equations $f_i(w_{00}, ..., w_{nm})$. Then these induce the homogeneous polynomials $g_i(u_0, ..., u_n, v_0, ..., v_m)$ on $\mathbb{P}^n_k \times \mathbb{P}^m_k$ where the u_i, v_j are the same homogeneous degree. Conversely, if there are two systems of homogeneous equations taking variables of the same homogeneous degree in u_i, v_j, then there exists homogeneous polynomials taking variables in $u_i v_j$. If the degrees of the variables u_i and v_j differ, say the former are of degree r and the latter of degree s, then we can create the system of homogeneous equations:

$$v_j^{r-s} g_i(u_0, \ldots, u_n, v_0, \ldots, v_m) = 0$$

assuming $r > s$. One important consequence of this is that we can directly check for a set $Z \subseteq \mathbb{P}^n_k \times \mathbb{P}^m_k$ being closed by looking for a single polynomial in $\mathbb{P}^{(n+1)(m+1)-1}_k$ instead of looking for two polynomials in each separate projective space and verifying the set is the direct product of those two closed sets. Let us box this for future reference:

Proposition 40.1.3: Segre Embedding is Closed

Let $\varphi : \mathbb{P}^n_k \times \mathbb{P}^m_k \to \mathbb{P}^{(n+1)(m+1)-1}_k$ be the Segre embedding. Then $\mathrm{im}\varphi$ is closed. Furthermore, $X \subseteq \mathbb{P}^n_k \times \mathbb{P}^m_k$ is closed if and only if $\varphi(X) \subseteq \mathrm{im}\varphi$ is closed.

***Proof*:**
The image is closed. Substituting $w_{ij} = x_i y_j$ shows every point of $\mathrm{im}\varphi$ satisfies (40.1). Conversely let $w = [w_{ij}]$ satisfy all those relations, and pick (a, b) with $w_{ab} \neq 0$. Set $x_i = w_{ib}$ and $y_j = w_{aj}$. Then for all i, j,

$$x_i y_j = w_{ib} w_{aj} = w_{ij} w_{ab}$$

using (40.1), so $[x_i y_j] = [w_{ij}]$ after scaling by w_{ab}, and $w = \varphi([x], [y])$. Hence $\mathrm{im}\varphi = \mathcal{Z}(\{w_{ij}w_{kl} - w_{il}w_{kj}\})$ is closed. The same computation shows φ is injective, since $[x]$ and $[y]$ are recovered from w up to scalars.

Closed subsets correspond. If $\varphi(X)$ is closed in $\mathrm{im}\varphi$, then $X = \varphi^{-1}(\varphi(X))$ is closed because φ is continuous. Conversely let X be closed, cut out by bihomogeneous equations $g_i(u; v)$. By the degree-balancing displayed above we may assume each g_i has the same degree r in the u and in the v, and a bihomogeneous polynomial of bidegree (r, r) is a polynomial in the products $u_i v_j$, hence a form in the w_{ij}. Those forms cut out $\varphi(X)$ inside $\mathrm{im}\varphi$, as we sought to show.

Before leaving products, three facts about them are needed later, and none is formal. The first says the maps one expects to have are regular, the second is general topology in geometric clothing, and the third says that a product of irreducible varieties does not fall apart.

Proposition 40.1.4: Projections and Slices Are Regular

Let $X \subseteq \mathbb{P}^n_k$ and $Y \subseteq \mathbb{P}^m_k$ be quasi-projective varieties. Then both projections $\pi_1 \colon X \times Y \to X$ and $\pi_2 \colon X \times Y \to Y$ are regular. Moreover, for fixed $y \in Y$ the slice map $j_y \colon X \to X \times Y$, $x \mapsto (x, y)$, is an isomorphism onto $X \times \{y\}$, and symmetrically $i_x \colon Y \to X \times Y$ is an isomorphism onto $\{x\} \times Y$.

***Proof*:**
Work in the Segre coordinates $w_{ij} = x_i y_j$. Fix a and let $W_a \subseteq X \times Y$ be the open set where some $w_{aj} \neq 0$, which is where $x_a \neq 0$. On W_a,

$$[\, w_{a0} : \cdots : w_{am} \,] = [\, x_a y_0 : \cdots : x_a y_m \,] = [\, y_0 : \cdots : y_m \,]$$

so π_2 is given on W_a by $m + 1$ linear forms in the w_{ij}, not all zero there. The W_a cover $X \times Y$ and the formulas agree on overlaps, so π_2 is regular; the same argument with $[\, w_{0b} : \cdots : w_{nb} \,]$ handles π_1.

For the slices, fix $y = [b_0 : \cdots : b_m]$ and write $j_y(x) = [\, x_i b_j \,]$. The entries are linear forms in x, not all identically zero since some $b_j \neq 0$, so j_y is regular. Its image is $X \times \{y\}$, and $\pi_1 \circ j_y$ is the identity of X while $j_y \circ \pi_1$ is the identity of $X \times \{y\}$, so the two are mutually inverse isomorphisms. The map i_x is symmetric, as we sought to show.

Lemma 40.1.5: A Cover by Overlapping Irreducibles

Let Z be a topological space covered by open sets V_α, each irreducible, with $V_\alpha \cap V_\beta \neq \emptyset$ for every pair. Then Z is irreducible.

***Proof*:**
A space is irreducible exactly when each of its nonempty open subsets is dense. First, every V_α is dense in Z: given β, the set $V_\beta \cap V_\alpha$ is nonempty and open in V_β, hence dense in the irreducible V_β, so $V_\beta \subseteq \overline{V_\alpha}$; as the V_β cover Z this forces $\overline{V_\alpha} = Z$.

Now let $W \subseteq Z$ be nonempty and open. It meets some V_α, and $W \cap V_\alpha$ is a nonempty open subset of the irreducible V_α, hence dense in it, so $\overline{W} \supseteq V_\alpha$. Closures are closed, so $\overline{W} \supseteq \overline{V_\alpha} = Z$, as we sought to show.

Proposition 40.1.6: A Product of Irreducibles Is Irreducible

Let X and Y be irreducible quasi-projective varieties. Then $X \times Y$ is irreducible.

***Proof*:**
By proposition 40.1.4 each slice $\{x\} \times Y$ is isomorphic to Y, hence irreducible, and each slice map j_y is regular. Suppose $X \times Y = Z_1 \cup Z_2$ with both Z_i closed. For fixed x the slice $\{x\} \times Y$ is irreducible and is covered by the two relatively closed sets $Z_i \cap (\{x\} \times Y)$, so it lies wholly inside Z_1 or wholly inside Z_2. Set

$$X_i = \{x \in X \mid \{x\} \times Y \subseteq Z_i\}$$

so that $X = X_1 \cup X_2$. Each X_i is closed, being the intersection $X_i = \bigcap_{y \in Y} j_y^{-1}(Z_i)$ of closed sets. Since X is irreducible, $X = X_i$ for one of the two, and then $X \times Y = Z_i$, as we sought to show.

40.2 Completeness of Projective Varieties

The image of a regular map need not be closed: example 32.1.4 took $X = \mathcal{Z}(xy - 1)$, $Y = \mathbb{A}^1_k$ and $\pi(a, b) = a$, whose image is $\mathbb{A}^1_k \setminus \{0\}$. However, the image of a projective variety will always be projective! If the reader has done some analysis this should remind the reader of the fact that the image of compact sets are compact. Though all affine and projective algebraic sets are compact (they are closed subsets of the compact space $\mathbb{A}^n_k$ or $\mathbb{P}^n_k$), projective varieties behave more like compact sets. Scheme theory draws the distinction by calling projective varieties *proper*[1]. To see this, we require a few lemmas:

[1] In analysis a map is proper if the pre-image of every compact set is compact

Lemma 40.2.1: Graph of Regular Mapping

Let $\varphi : X \to Y$ be a regular mapping between quasi-projective varieties. Then the graph

$$\Gamma(\varphi) = \{(x, \varphi(x)) \mid x \in X\}$$

is closed in $X \times Y$

If the Zariski topology was Hausdorff, then we could have concluded using classical topological results, however some more care has to be done as the topology is not Hausdorff.

***Proof*:**
First, we may take Y to be a projective space; for an appropriate choice of m, $Y \subseteq \mathbb{P}_k^m$, so that $X \times Y \subseteq X \times \mathbb{P}_k^m$. Then by proposition 33.2.7, φ determines a mapping $\overline{\varphi} : X \to \mathbb{P}_k^m$ and $\Gamma_\varphi = \Gamma_{\overline{\varphi}} \cap (X \times Y)$. As $X \times Y$ is closed in $X \times \mathbb{P}_k^m$ by proposition 40.1.3, closed then Γ_f is closed.

Next define $(\overline{\varphi}, \mathrm{id})\colon X \times \mathbb{P}_k^m \to \mathbb{P}_k^m \times \mathbb{P}_k^m$. Then $\Gamma_{\overline{\varphi}}$ is the inverse image of the diagonal $\Delta = \Gamma_{\mathrm{id}}$ under this map. As this is a continuous map, the inverse image of a closed set is closed, and hence it suffices to show that Γ_i is closed.

Finally, the diagonal $\Delta \subseteq \mathbb{P}_k^m \times \mathbb{P}_k^m$ consists of the pairs $([u_0 : \cdots : u_m], [v_0 : \cdots : v_m])$ with $u_i v_j = u_j v_i$ for all i, j, which under the Segre embedding are exactly the points where the corresponding 2×2 minors vanish. By proposition 40.1.3 that is a closed condition, so Δ is closed, hence so is $\Gamma_{\overline{\varphi}}$ and therefore Γ_φ, completing the proof.

Proposition 40.2.2: Projection of Projective Variety

Let X be a projective variety and Y a quasi-projective variety. Then the projection map:

$$\pi_2 : X \times Y \to Y$$

is a closed map (the image of a closed set is closed)

Hence, the projection of a projective variety can always be written as a system of polynomial equations! This is not true for affine varieties: take $X = \mathcal{Z}(xy - 1)$ and $Y = \mathbb{A}_k^1$

***Proof*:**
Let us first reduce the problem: as $X \subseteq \mathbb{P}_k^n$ is closed, $X \times Y \subseteq \mathbb{P}_k^n \times Y$ is a closed subset, hence if $Z \subseteq X$ is closed, it is closed in $Z \times Y \subseteq \mathbb{P}_k^n \times Y$, and so it is enough to show it for the case of $X = \mathbb{P}_k^n$. Next, as being closed is affine local, it suffices to cover Y by affine open subsets U_i and prove the theorem on each of these. Thus, it suffices to assume that Y is an affine variety. Finally, as Y is closed in $\mathbb{A}_k^m$, $\mathbb{P}_k^n \times Y \subseteq \mathbb{P}_k^n \times \mathbb{A}_k^m$ is closed, and so we can reduce further to $Y = \mathbb{A}_k^m$.

Thus, let us characterize closed subsets $Z \subseteq \mathbb{P}_k^n \times \mathbb{A}_k^m$. By definition of the Zariski topology, and by a simple generalization of proposition 40.1.3, we see that Z is closed if and only if it is the zero set of the collection of polynomials

$$g_i([u_0 : \cdots : u_n], y_1, \ldots, y_m) \qquad 1 \leq i \leq t$$

In particular, we do not need to create two zero sets for each component and check that the their product produces Z: we can directly work with a set of polynomials. Let us write $g_i(u; y)$ to represent these polynomials with the two different inputs.

Now, take any $y_0 \in \mathbb{A}_k^m$. Then $\pi^{-1}(y_0)$ consists of all solutions (whether empty or non-empty) to the system of equations

$$g_i(u; y_0) = 0 \qquad 1 \le i \le t$$

Take the collection of y_0 such that the system of polynomials equations $g_i(u; y_0) = 0$ has a nonzero solution; call this collection $T \subseteq \mathbb{A}_k^m$. We want to show that T is closed, for then the projection π_2 is a closed map. By lemma 38.4.2, the system of polynomials equations $g_i(u; y_0) = 0$ has a nonzero solution if and only if:

$$I_s \not\subseteq (g_1(u, y_0), ..., g_t(u, y_0)) \qquad \forall s \in \mathbb{N}$$

where I_s is the irrelevant ideal of degree $s \ge 1$. Thus it suffices to show that the set T_s of points $y_0 \in \mathbb{A}_k^m$ for which $I_s \not\subseteq (g_1(u, y_0), ..., g_t(u, y_0))$ is closed, in particular if we find polynomial equations whose zero set is T_s, then as $T = \bigcap_s T_s$, T is closed which completing the proof.

To that end, let k_i be the homogeneous degree of $g_i(u; y)$ in the u_j variables. Let $(M^\alpha)_\alpha$ be the monomials of degree s in $u_0, ..., u_n$ with some total ordering (note this set is finite). Then $I_s \subseteq (g_1(u, y_0), ..., g_t(u, y_0))$ if and only if each monomial M^α can be expressed as:

$$M^\alpha = \sum_{i=1}^{t} g_i(u, y_0) F_{i,\alpha}(u) \tag{40.2}$$

As each g_i has homogeneous degree k_i, it must be that $F_{i,\alpha}$ has degree $s - k_i$ or $F_{i,\alpha} = 0$ if $k_i > s$. Then we may replace $F_{i,\alpha}$ with elements in $(N_i^\beta)_\beta$ which are monomial of degree $s - k_i$ written in some total ordering. Then (40.2) holds if and only if M^α is a linear combination of the polynomials $g_i(u, y_0)N_i^\beta$. As this is true for each M^α, this is equivalent to saying that the polynomials $g_i(u, y_0)N_i^\beta$ span the entire vector space S of homogeneous polynomials of degree s in $u_0, ..., u_n$. As $I_s \not\subseteq (g_1(u, y_0), ..., g_t(u, y_0))$, we have that the $g_i(u, y_0)N_i^\beta$ *do not* span S.

Now, take the matrix $(a_{\alpha\beta})$ that represents the coefficients of the M^α terms as a linear combination of the $g_i(u, y_0)N_i^\beta$. As not all M^α can be written as such terms, this matrix cannot be full rank. Hence all $\sigma \times \sigma$ minors (where $\sigma = \dim_k S$) are polynomials in the coefficients of the polynomials $g_i(u, y_0)$, in particular they are polynomials in the coordinates of the point y_0. So T_s is cut out by polynomial equations in y_0, hence closed.

Thus, since the arbitrary intersection of closed sets is closed $T = \bigcap_s T_s$ is closed, completing the proof.

For those interested in algebraic geometry, the above proof is in part in proving that projective varieties are *proper*.

Corollary 40.2.3: Image of Projective Variety is Closed

Let $f : X \to Y$ be a regular map where X is a projective variety. Then $\mathrm{im} f$ is closed

***Proof*:**
Let $\pi : X \times Y \to Y$ be the natural projection. Consider the graph $\Gamma_f \subseteq X \times Y$ of lemma 40.2.1. Then:

$$\mathrm{im} f = \pi(\Gamma_f)$$

and hence by proposition 40.2.2 $\mathrm{im} f$ is closed, completing the proof.

We can finally rigorously show that the collection of regular functions on projective space, and more generally projective varieties, are only constant functions:

Corollary 40.2.4: Regular Function on Projective Variety

Let φ be a regular function on an irreducible projective variety. Then $\varphi \in k$, i.e. φ is constant.

***Proof*:**
Take $\varphi : X \to \mathbb{A}^1_k$. As $\mathbb{A}^1_k$ is dense in $\mathbb{P}^1_k$, there is a unique extension to a regular map $\overline{\varphi} : X \to \mathbb{P}^1_k$. As X is a projective variety, by corollary 40.2.3 $\overline{\varphi}(X) \subseteq \mathbb{P}^1_k$ is closed. Since $\overline{\varphi}(X) \subseteq \mathbb{A}^1_k$, the point at infinity is not in the image, so $\overline{\varphi}(X)$ is a *proper* closed subset of $\mathbb{P}^1_k$. The proper closed subsets of $\mathbb{P}^1_k$ are exactly the finite sets, so $\overline{\varphi}(X)$ is finite. (It cannot be all of $\mathbb{A}^1_k$, which is not closed in $\mathbb{P}^1_k$.) Hence $\varphi(X)$ is a finite set of points. closed, their pre-image would make X the non-trivial union of closed sets, contradicting its irreducibility, and hence it must be that the image is a single point, in other words a constant, as we sought to show.

Corollary 40.2.5: Regular Map on Projective Variety

Let $f : X \to Y$ be a regular map from a projective variety to an affine variety. Then $\mathrm{im} f$ is a point

***Proof*:**
Take $f(x) = (\varphi_1(x), ..., \varphi_m(x))$ where each φ_i is a regular function. Then by corollary 40.2.4 these are all constant, but then

$$f(X) = (\alpha_1, ..., \alpha_m)$$

completing the proof.

40.2.6 Note: Completeness Is Elimination Theory The classical name for proposition 40.2.2 is *elimination theory.* Given equations in $x_1, \dots, x_n$ and $y_1, \dots, y_m$, one asks for the conditions on the y alone under which a solution in x exists; that is exactly asking for the image of a projection, and the theorem says the answer is again given by equations, provided the x range over projective space.
The classical tool is the resultant. For two one-variable polynomials

$$f = a_0 x^d + \cdots + a_d, \qquad g = b_0 x^e + \cdots + b_e$$

the resultant $\operatorname{Res}(f, g)$ is the determinant of the $(d+e) \times (d+e)$ Sylvester matrix in the coefficients, and it vanishes exactly when f and g have a common root, given a_0, b_0 not both zero. Since Res is a polynomial in the a_i and b_j, that is precisely a closed condition on the coefficients, and it eliminates x. Iterating in several variables is the classical proof of completeness, and section 19.6.3 develops the algebra.
The proof given above takes a different route, through lemma 38.4.2 and the irrelevant ideal, which is shorter and generalizes better. The resultant remains the computational tool: it is what a computer algebra system runs when asked to eliminate a variable, and example 42.4.1 could equally have been done by resultants.

40.3 Finite Maps

These maps shall be the mirror of integral extensions, and shall be one of the easiest examples of *proper morphisms* which is a somewhat complicated but fundamental type of morphism in scheme theory.

Definition 40.3.1: Finite Map [on Affine Varieties]

Let $f\colon X \to Y$ be a regular map of affine varieties with $f(X)$ dense in Y, so that the pullback $f^\#\colon k[Y] \hookrightarrow k[X]$ is injective. Then f is a *finite map* if $k[X]$ is integral over $f^\#(k[Y])$.

Example 40.3.2: Finite and Non-Finite Maps

1. *A finite map.* $\varphi\colon \mathbb{A}^1_k \to \mathbb{A}^1_k$, $\varphi(x) = x^2$. On coordinate rings it is $k[y] \to k[x]$, $y \mapsto x^2$, and $k[x]$ is generated as a $k[y]$-module by 1 and x, since $x^2 = y$. So the map is finite, and its fibres have one or two points: two over $a \neq 0$, one over 0.

2. *Finite maps are closed and surjective.* The image of φ is all of $\mathbb{A}^1_k$, as proposition 40.3.3 requires, because every a has a square root in the algebraically closed k.

3. *A non-example with finite fibres.* The projection $\mathcal{Z}(xy-1) \to \mathbb{A}^1_k$, $(x, y) \mapsto x$, has fibres of size at most one, yet it is not finite: $k[x, x^{-1}]$ is not a finitely generated $k[x]$-module, since no finite set of Laurent polynomials spans all negative powers. Its image misses the origin, so it is not even surjective. Finiteness is strictly stronger than having finite fibres.

4. *An open immersion is almost never finite.* The inclusion $D(x) \hookrightarrow \mathbb{A}^1_k$ is the previous example in disguise, by example 31.3.6(1).

The composition of finite maps is finite. If $f\colon X \to Y$ is finite then its fibres are finite. A reasonable assumption would be that a map is finite if and only if its fibers are finite, but that is not the case: an example of a non-finite map would be $\pi_1 : V(xy-1) \to \mathbb{A}^1_k$ as $k[x] \hookrightarrow k[x, 1/x]$ is not an integral extension. Instead, one way of thinking of a finite map $f : X \to Y$ is that given an element $y \in Y$, as y is moved around in Y and we look at $f^{-1}(y)$, none of the roots could tend to infinity, that is none can disappear. This intuition is captured by the following proposition:

Proposition 40.3.3: Finite Maps Are Surjective

Let $f : X \to Y$ be a finite map between affine varieties. Then f is surjective.

The tool is the lying-over property of integral extensions (theorem 26.3.2), in the following form: if B is integral over a subring A and $\mathfrak{a} \subsetneq A$ is a *proper* ideal, then $\mathfrak{a}B \neq B$. Equivalently, a proper ideal of A cannot generate the unit ideal of B. This is what stops a fibre from being empty.

***Proof*:**
Let $f : X \to Y$ be a finite map between affine varieties. Take any $y \in Y$ and let $\mathfrak{m}_y \subseteq k[Y]$ be the functions that take the value 0 at y: if $f \in \mathfrak{m}_y$, then $f(y) = 0$. Let $x_1, ..., x_n$ be coordinate functions on Y and $y = (\alpha_1, ..., \alpha_n)$ so that

$$\mathfrak{m}_y = (x_1 - \alpha_1, \ldots, x_n - \alpha_n)$$

Then the equations of the variety $f^{-1}(y)$ have the form:

$$f^{\#}(x_1) = \alpha_1, \quad \ldots, \quad f^{\#}(x_n) = \alpha_n$$

By the Nullstellensatz, $f^{-1}(y) = \emptyset$ if and only if the elements $f^{\#}(x_i) - \alpha_i$ generate the unit ideal, that is:

$$(f^*(x_1) - \alpha_1, ..., f^*(x_n) - \alpha_n) = k[X]$$

Viewing $k[Y]$ as a subring of $k[X]$, we see that the above is equivalent to:

$$(x_1 - \alpha_1, \ldots, x_n - \alpha_n)\, k[X] = k[X]$$

In other words, $f^{-1}(y) = \emptyset$ would force $\mathfrak{m}_y k[X] = k[X]$. But $\mathfrak{m}_y$ is maximal, hence a proper ideal of $k[Y]$, and $k[X]$ is integral over it, so lying over gives $\mathfrak{m}_y k[X] \neq k[X]$. The two are contradictory, so $f^{-1}(y) \neq \emptyset$. As y was arbitrary, f is surjective, completing the proof.

Corollary 40.3.4: Finite Maps Are Closed Maps

Let $f : X \to Y$ be a finite map between affine varieties. Then f is a closed map.

***Proof*:**
It suffices to check on irreducible closed subsets $Z \subseteq X$. By the above theorem the restriction of f to Z, $f|_Z : Z \to \overline{f(Z)}$, is a finite map between affine varieties, and so by surjectivity we get $f(Z) = \overline{f(Z)}$, but then $f(Z)$ is closed, completing the proof.

Let us now build-up to defining finite maps for quasi-projective varieties. For that, we start by showing that finiteness is a local property:

Lemma 40.3.5: Finiteness is Local

Let $f : X \to Y$ be a regular map of affine varieties. Then if for every point $y \in Y$, there is an affine neighborhood U of y such that $V = f^{-1}(U)$ is affine and $f : V \to U$ is finite, then f is finite

***Proof*:**
Let $k[Y] = A$ and $k[X] = B$. Take the principal open set around each point of Y, and as Y is compact take a finite subcover, label these $D(g_\alpha)$ so that $Y = \bigcup_\alpha D(g_\alpha)$. Then:

$$V_\alpha = f^{-1}(D(g_\alpha)) = D(f^{\#}(g_\alpha))$$

and

$$k[D(g_\alpha)] = A[1/g_\alpha] \qquad k[V_\alpha] = B[1/f^{\#}(g_\alpha)]$$

As f is finite on this chart, $B[1/g_\alpha]$ is a finite module over $A[1/g_\alpha]$; fix a finite generating set $w_{i,\alpha}$. (Generating set, not basis: a finite module over a ring need not be free, and freeness is never used below.) We may take $w_{i,\alpha} \in B$, since clearing the denominator of $w_{i,\alpha}/g_\alpha^{m_i}$ multiplies by a unit of $A[1/g_\alpha]$ and so preserves generation.

Now take the union of all these finite sets. We claim they generate B as an A-module. To see this, take any $b \in B$. Then for each α we can write it as :

$$b = \sum_i \frac{a_{i,\alpha}}{g_\alpha^{n_\alpha}} w_{i,\alpha}$$

As the $g_\alpha^{n_\alpha}$ generate the unit ideal of A (the sets $D(g_\alpha)$ cover Y), there exists $h_\alpha \in A$ such that

$$\sum_\alpha g_\alpha^{n_\alpha} h_\alpha = 1$$

Thus:

$$b = b \sum_\alpha g_\alpha^{n_\alpha} h_\alpha = \sum_i \sum_\alpha a_{i,\alpha} h_\alpha w_{i,\alpha}$$

which exhibits an arbitrary $b \in B$ as an A-combination of the $w_{i,\alpha}$. Hence B is a finite A-module and f is finite, as we sought to show.

Thus, we can define finite maps for quasi-projective varieties:

Definition 40.3.6: Finite Map [on Quasi-Projective Varieties]

Let $f : X \to Y$ be a regular map between quasi-projective varieties. Then f is said to be *finite* if any point $y \in Y$ has an affine local neighborhood V such that $U = f^{-1}(V)$ is affine and $f : U \to V$ is a finite map between affine varieties.

Naturally, $f^{-1}(y)$ is finite for every $y \in Y$, and it follows easily that any finite map is surjective. Note that the density condition is now a local condition. One consequence of these results that we shall not prove is that if $f : X \to Y$ between quasi-projective is a regular map (not necessarily finite) and $f(X)$ dense in Y, then $f(X)$ contains an open subset of Y. This is *not* true in analysis, the canonical example being the wrapping of the real line around the torus with an irrational period which will be a dense map with empty interior.

40.4 Noether Normalization, Geometrically

An important culminating result is that every irreducible projective variety is in some way only a "finite extension" from projective space. These results reduce many arguments about irreducible varieties to arguments about affine or projective space. These ideas return in scheme-theoretic form in *Algebraic Geometry* [5]. We first need a technical lemma:

40.4.1 Normalization and Singularities Note that this is *not* to be confused with normalization of singularities! Rather confusingly, normalization of singularities shall be the map in the *opposite direction* to the normalization map that will be defined here. The two uses of the word are unrelated, and it is not the case that a normal resolution of singularities results in a map $\mathbb{A}_k^n \to X$ or $\mathbb{P}_k^n \to X$ for a variety X (where the normalization map we shall do will be map of the form $X \twoheadrightarrow \mathbb{A}_k^n$ or $X \twoheadrightarrow \mathbb{P}_k^n$).

Lemma 40.4.2: Projection is Finite Mapping

Let X be closed in $\mathbb{P}^n$ with $X \subset \mathbb{P}^n \setminus E$, where E is a d-dimensional linear subspace. Then the projection $\pi : X \to \mathbb{P}^{n-d-1}$ with center at E determines a finite mapping $X \to \pi(X)$.

***Proof*:**
Let $y_0, \dots, y_{n-d-1}$ be homogeneous coordinates in $\mathbb{P}^{n-d-1}$ and let π be given by the formulae $y_j = L_j(x)$, $x \in X$ $(j = 0, \dots, n-d-1)$. Naturally, $U_i = \pi^{-1}(\mathbb{A}^{n-d-1}) \cap X$ is given by the condition $L_i(x) \neq 0$ and is an affine open subset of X. We want to show that $\pi : U_i \to \mathbb{A}^{n-d-1} \cap \pi(X)$ is a finite mapping. To that end, first note that every function $g \in k[U_i]$ is of the form $g = G_i(x_0, \dots, x_n)/L_i^m$, where G_i is a form of degree m. Consider the mapping $\pi_1 : X \to \mathbb{P}^{n-d} : z_j = L_j^m(x)$ $(j = 0, \dots, n-d-1)$, $z_{n-d} = G_i(x)$, where $z_0, \dots, z_{n-d}$ are homogeneous coordinates in $\mathbb{P}^{n-d}$. It is a regular mapping and its image $\pi_1(X)$ is closed in $\mathbb{P}^{n-d}$. Let $F_1 = \cdots = F_s = 0$ be its equations. Since $X \subset \mathbb{P}^n - E$, the forms $L_i (i = 0, \dots, n-d-1)$ do not vanish simultaneously on X. Hence the point $[0 : \cdots : 0 : 1]$ is not contained in $\pi_1(X)$, in other words, that the equations $z_0 = \cdots = z_{n-d-1} = F_1 = \cdots = F_s = 0$ have no solution in $\mathbb{P}^{n-d}$. Hence by lemma 38.4.2, for some $l > 0$:

$$(z_0, \dots, z_{n-d-1}, F_1, \dots, F_s) \supseteq \mathfrak{m}^l$$

in particular, $z_{n-d}^l \in (z_0, \dots, z_{n-d-1}, F_1, \dots, F_s)$. This means that

$$z_{n-d}^l = \sum_{j=0}^{n-d-1} z_j H_j + \sum_{j=1}^{s} F_j P_j,$$

where H_j and P_j are polynomials. Denote by $H^{(q)}$ the homogeneous component of H of degree q. Then:

$$\Phi(z_0, \dots, z_{n-d}) = z_{n-d}^l - \sum z_j H_j^{(l-1)} = 0 \quad \text{on } \pi_1(X) \tag{40.3}$$

The homogeneous polynomial Φ is of degree l and as a polynomial in z_{n-d} it has the leading coefficient 1:

$$\Phi = z_{n-d}^l - \sum_{j=0}^{l-1} A_{l-j}(z_0, \ldots, z_{n-d-1}) z_{n-d}^j \tag{40.4}$$

Substituting the transformation formulae for π_1 in (40.3) we find that $\Phi(L_0^m, \ldots, L_{n-d-1}^m, G_i) = 0$ on X, with Φ of the form (40.4). Dividing this relation by L_i^{ml} we obtain the required relation

$$g^l + \sum_{j=0}^{l-1} A_{l-j}(x_0, \ldots, 1, \ldots, x_{n-d-1}) g^j = 0,$$

where $x_r = y_r/y_i$ are coordinates in $\mathbb{A}^{n-d-1}$. But then we have that the mapping is finite, completing the proof.

Theorem 40.4.3: Normalization of Projective Variety

Let X be an irreducible projective variety. Then there exists a finite mapping $\varphi : X \twoheadrightarrow \mathbb{P}_k^m$ for appropriate m.

***Proof*:**
Take $X \subseteq \mathbb{P}_k^n$. If $X = \mathbb{P}_k^n$ we're done so take $X \neq \mathbb{P}_k^n$. Take $x \in \mathbb{P}_k^n \setminus X$ and take the projection from x, $\varphi_x : X \to \mathbb{P}_k^{n-1}$. This map is regular by example 39.4.4, and the image is closed, hence projective. It is furthermore finite by lemma 40.4.2. If $\varphi(X) \neq \mathbb{P}_k^{n-1}$ we repeat the argument. is closed/finite. Each projection lowers the ambient dimension by one, so after finitely many steps the image fills its ambient space: the dimension cannot fall below $\dim X$, and a proper closed subvariety of the same dimension as its ambient is impossible by proposition 34.3.5. Composing the projections gives a finite surjection

$$\varphi \colon X \twoheadrightarrow \mathbb{P}_k^m \qquad m = \dim X$$

completing the proof.

Theorem 40.4.4: Normalization of Affine Variety

Let X be an irreducible affine variety. Then there exists a finite mapping $\varphi : X \twoheadrightarrow \mathbb{A}_k^m$ for appropriate m.

Note that this is the analog of Noether's Normalization theorem for rings that are finitely generated k-algebras without zero-divisors, and in fact can be proven using it: see the fifth exercise of exercise set 40.6.1.

***Proof*:**
Take $X \subseteq \mathbb{A}_k^n$ and suppose $X \neq \mathbb{A}_k^n$, the other case being trivial. Identify $\mathbb{A}_k^n$ with the chart $U_{n+1} \subseteq \mathbb{P}_k^n$. Then X is *closed in* $\mathbb{A}_k^n$, hence locally closed in $\mathbb{P}_k^n$, and we may take its projective closure $\overline{X}$.

Since X is a proper closed subvariety of $\mathbb{A}_k^n$ we have $\dim X < n$ by proposition 34.3.5, so $\overline{X}$ cannot contain the whole hyperplane at infinity; choose $x \in H_\infty$ with $x \notin \overline{X}$. Projecting from x

gives a finite map $\overline{X} \to \mathbb{P}_k^{n-1}$ by lemma 40.4.2, and since the centre lies at infinity the projection carries X into the affine chart, giving a finite map $X \to \mathbb{A}_k^{n-1}$.

Repeating as in theorem 40.4.3 yields a finite surjection

$$\varphi\colon X \twoheadrightarrow \mathbb{A}_k^m \qquad m = \dim X$$

completing the proof.

Thus, though not all varieties are rational, all varieties are a finite cover of affine or projective space, and hence the study of which varieties are not rational will naturally lead to the question of how far away are they from being rational[2]. More on this in the exercises.

Example 40.4.5: Finite Maps and Normalization

Consider the curve $f : \mathbb{A}_k^1 \to \mathbb{A}_k^2$ given by $f(t) = (t^2, t^3)$. This is associated to the pullback on coordinate rings $f^{\#}\colon k[x, y] \to k[t]$, $x \mapsto t^2$, $y \mapsto t^3$. Taking the quotient of the domain gives $k[x, y]/(x^3 - y^2) \cong k[t^2, t^3] \hookrightarrow k[t]$, which is injective but *not surjective*: nothing maps to t. As $(x^3 - y^2)$ is prime, we take the field of fraction and extend the map, which has $t^3/t^2 = t$ we have that $\mathrm{Frac}(k[t^2, t^3]) = k(t)$ giving a field isomorphism.

This is a 1-to-1 map. Define an n-to-1[a] map $f : \mathbb{A}_k^1 \to \mathbb{A}_k^2$ given by $t \mapsto (t^{2n}, t^{3n})$. Notice the image is the same. The global section's are also isomorphic as $k[t^2, t^3] \cong k[t^{2n}, t^{3n}]$. Thus the field of fractions are isomorphic, $k(t) \cong k(t^n)$. Working with a specific n-to-1 map, the field of fraction of the global section of the image is $k(t^n)$ which has index n in the field of fractions of the domain, namely $k(t)$. This reflects the geometric nature of finite maps in algebraic properties.

Notice that $x^2 - y^3$ forms a curve with a cusp; it has a singularity at $(0, 0)$. This curve maps onto

[a] note it is important to assume that either k is algebraically closed, or at minimum that k contains all the roots of unity

40.5 Dimension of Fibres

A map between varieties should decrease dimension by the dimension of its fibres, and this section makes that precise. The statement has three parts, and only the middle one is delicate: fibres can jump *up* in dimension over special points, but never down.

Theorem 40.5.1: Dimension of Fibres

Let $f\colon X \to Y$ be a dominating regular map of irreducible varieties, with $\dim X = n$ and $\dim Y = m$. Then:

1. $m \leq n$;
2. for every $y \in f(X)$, every irreducible component of the fibre $f^{-1}(y)$ has dimension at least $n - m$;
3. there is a nonempty open $U \subseteq Y$ such that $\dim f^{-1}(y) = n - m$ for every $y \in U$.

[2] Chevalley's theorem (chapter 41) is the statement in this direction that survives without the finiteness hypothesis: the image of a regular map is constructible even when it is neither open nor closed.

***Proof*:**
(1). Since f is dominating, $f^\#$ embeds $k(Y)$ into $k(X)$ by proposition 33.3.4(2), and transcendence degree cannot decrease under a field extension, so $m = \mathrm{trdeg}_k k(Y) \leq \mathrm{trdeg}_k k(X) = n$.

(2). The question is local on Y, so replace Y by an affine neighbourhood of y and X by its preimage. The local ring $\mathcal{O}_y(Y)$ has dimension m, so it admits a system of parameters (definition 29.5.1): m functions $g_1, \dots, g_m$ generating an ideal whose zero set is $\{y\}$ near y. Then, near any point of the fibre,

$$f^{-1}(y) = \mathcal{Z}\big(f^\#(g_1), \dots, f^\#(g_m)\big)$$

is cut out of X by m equations. Applying theorem 34.4.1 once for each equation drops the dimension by at most one each time, so every component of the result has dimension at least $n - m$.

(3). Choose affine open sets so that f corresponds to an injection $k[Y] \hookrightarrow k[X]$ of finitely generated domains. Over the fraction field $K = k(Y)$, the ring $k[X] \otimes_{k[Y]} K$ is a finitely generated K-algebra of dimension $n - m$, since its fraction field is $k(X)$ and $\mathrm{trdeg}_K k(X) = n - m$. Apply Noether normalization over K: there are $t_1, \dots, t_{n-m}$ with $k[X] \otimes K$ finite over $K[t_1, \dots, t_{n-m}]$. Each of the finitely many integral equations involved has coefficients in $k(Y)$; let $g \in k[Y]$ be a common denominator. Then over the open set $U = D(g)$,

$$k[X]_g \text{ is finite over } k[Y]_g[t_1, \dots, t_{n-m}]$$

Hence for $y \in U$ the fibre $f^{-1}(y)$ is finite over $\mathbb{A}_k^{n-m}$, and a finite surjection preserves dimension by proposition 34.2.5, so $\dim f^{-1}(y) = n - m$, as we sought to show.

Part (2) is the useful half in practice, and it is worth saying what it does not claim. It gives a *lower* bound on fibre dimension, valid at every point; the matching upper bound holds only generically, which is part (3). Fibres genuinely do jump.

Example 40.5.2: Fibres That Jump
Let $X = \mathcal{Z}(xz - y) \subseteq \mathbb{A}_k^3$ and $f \colon X \to \mathbb{A}_k^1$ the projection $(x, y, z) \mapsto x$. Then X is isomorphic to $\mathbb{A}_k^2$ with coordinates (x, z), since y is determined, so $n = 2$ and $m = 1$, giving an expected fibre dimension of 1. Over $x = a \neq 0$ the fibre is the line $\{(a, az, z) \mid z \in k\}$, of dimension 1 as predicted. Over $x = 0$ it is $\{(0, 0, z) \mid z \in k\}$, again dimension 1.

Now blow the picture up: let $g \colon \mathbb{A}_k^2 \to \mathbb{A}_k^2$ be $(u, v) \mapsto (u, uv)$. Here $n = m = 2$, so the expected fibre dimension is 0. Over (a, b) with $a \neq 0$ the fibre is the single point $(a, b/a)$; but over $(0, 0)$ the fibre is the whole line $\{(0, v) \mid v \in k\}$, of dimension 1. The fibre has jumped, exactly as part (2) permits and part (3) confines to a proper closed subset. This map is the restriction to one chart of the blow-up of chapter 47, computed there as proposition 47.1.2(1).

A count that will be used in chapter 41 follows immediately by adding dimensions along a map whose fibres are all of the same dimension.

Corollary 40.5.3: Dimension of a Total Space

Let $f\colon X \to Y$ be a surjective regular map of irreducible varieties whose fibres are all irreducible of the same dimension d. Then $\dim X = \dim Y + d$.

***Proof*:**
By theorem 40.5.1(3) the generic fibre has dimension $\dim X - \dim Y$, and by hypothesis every fibre has dimension d, so $\dim X - \dim Y = d$, completing the proof.

Applied to a projection, this settles the dimension of a product, which chapter 34 could only gesture at because products of projective varieties were not yet available.

Corollary 40.5.4: Dimension of a Product

Let X and Y be irreducible quasi-projective varieties. Then $\dim(X \times Y) = \dim X + \dim Y$.

***Proof*:**
$X \times Y$ is irreducible by proposition 40.1.6, and the projection $\pi\colon X \times Y \to X$ is a surjective regular map whose fibre over x is the slice $\{x\} \times Y \cong Y$, irreducible of dimension $\dim Y$ at every point. Corollary 40.5.3 applies and gives the claim.

40.6 The Dimension of an Intersection

Theorem 34.4.1 says that cutting an affine variety by one equation costs at most one dimension. Iterating it bounds the dimension of an intersection, but only when one of the two varieties is cut out by as many equations as its codimension, and that fails in general: the twisted cubic of example 38.3.7 has codimension two in $\mathbb{P}^3_k$ and needs three equations. The repair is to intersect with the diagonal instead, and it costs nothing now that products are available.

Proposition 40.6.1: The Affine Dimension Theorem

Let $X, Y \subseteq \mathbb{A}^n_k$ be irreducible affine varieties with $\dim X = r$ and $\dim Y = s$. Then every irreducible component of $X \cap Y$ has dimension at least $r + s - n$.

***Proof*:**
Let $\Delta = \{(a, a) \mid a \in \mathbb{A}^n_k\} \subseteq \mathbb{A}^{2n}_k$ be the diagonal, which is $\mathcal{Z}(x_1 - y_1, \ldots, x_n - y_n)$, cut out by exactly n equations. The map $a \mapsto (a, a)$ is an isomorphism

$$X \cap Y \xrightarrow{\sim} (X \times Y) \cap \Delta$$

with inverse the first projection, both regular.

Now $X \times Y$ is irreducible by proposition 40.1.6 and has dimension $r + s$ by corollary 40.5.4. Cutting it by the n equations one at a time and applying theorem 34.4.1 at each step, every component of the result has dimension at least $r + s - n$, since each equation costs at most one dimension: an equation either vanishes identically on a component, leaving it untouched, or cuts it by exactly one. Transporting through the isomorphism gives the claim, as we sought to show.

Passing to affine cones turns this into the projective statement, and the projective statement is strictly stronger: it also asserts that the intersection is nonempty.

Proposition 40.6.2: Dimension of the Affine Cone

Let $\emptyset \neq V \subseteq \mathbb{P}^n_k$ be a projective variety. Then $C(V)$ is an irreducible affine variety and

$$\dim C(V) = \dim V + 1$$

***Proof*:**
Irreducibility: $\mathcal{I}_{\mathbb{A}}(C(V)) = \mathcal{I}_{\mathbb{P}}(V)$ by proposition 38.3.2, and the latter is prime because V is a variety, so $k[C(V)] = S(V)$ is a domain.

For the dimension, choose a linear form t not vanishing identically on V, possible since $V \neq \emptyset$. Every element of $\operatorname{Frac}(S(V))$ is a ratio f/g of homogeneous elements, and

$$\frac{f}{g} = \frac{f/t^{\deg f}}{g/t^{\deg g}} \cdot t^{\deg f - \deg g}$$

exhibits it as a rational function of t with coefficients in the degree-zero part, which is $k(V)$ by definition 39.2.4. So $\operatorname{Frac}(S(V)) = k(V)(t)$, and t is transcendental over $k(V)$ because it has degree 1 while $k(V)$ sits in degree 0. Hence

$$\dim C(V) = \operatorname{kdim} S(V) = \operatorname{trdeg}_k \operatorname{Frac}(S(V)) = \operatorname{trdeg}_k k(V) + 1 = \dim V + 1$$

using proposition 34.2.5 twice, as we sought to show.

Theorem 40.6.3: The Projective Dimension Theorem

Let $X, Y \subseteq \mathbb{P}^n_k$ be projective varieties with $\dim X = r$ and $\dim Y = s$. Then every irreducible component of $X \cap Y$ has dimension at least $r + s - n$. If moreover $r + s - n \geq 0$, then $X \cap Y \neq \emptyset$.

***Proof*:**
Pass to the affine cones in $\mathbb{A}^{n+1}_k$. By proposition 40.6.2 they are irreducible of dimensions $r+1$ and $s+1$, so proposition 40.6.1 says every component of $C(X) \cap C(Y)$ has dimension at least

$$(r+1) + (s+1) - (n+1) = r + s - n + 1$$

By proposition 38.3.2 the components of $C(X) \cap C(Y)$ other than the origin are the cones over the components of $X \cap Y$, whose dimensions are one less by proposition 40.6.2. That gives the bound $r + s - n$.

For nonemptiness, note that both cones contain the origin, so $C(X) \cap C(Y)$ is nonempty and the bound applies to a component through the origin: it has dimension at least $r + s - n + 1 \geq 1$. A variety of dimension at least 1 is not a single point, so that component contains some $a \neq 0$, and a determines a point of $\mathbb{P}^n_k$ lying on both X and Y, as we sought to show.

The nonemptiness clause is what separates projective space from affine space. Both cones always meet, at the origin; the content is that when the dimensions are large enough they meet in more than the origin. No such statement is available affinely: in $\mathbb{A}^2_k$ the parallel lines $\mathcal{Z}(y)$ and $\mathcal{Z}(y-1)$ have $r = s = 1$ and $n = 2$, so $r + s - n = 0$, and they are disjoint.

Example 40.6.4: Dimension Counts in Projective Space

1. *Two curves in* $\mathbb{P}^2_k$. Here $r = s = 1$ and $n = 2$, so $r + s - n = 0$ and any two projective plane curves meet. Chapter 43 refines this into a count of the intersection points; the present theorem gives only that there is one, but it needs no hypothesis beyond dimension.
2. *A curve and a surface in* $\mathbb{P}^3_k$. With $r = 1$, $s = 2$, $n = 3$ the bound is 0, so a curve and a surface in $\mathbb{P}^3_k$ always meet.
3. *Two curves in* $\mathbb{P}^3_k$. Now $r + s - n = -1$ and the theorem says nothing, correctly: two disjoint lines in $\mathbb{P}^3_k$ exist, as exercise set 38.5.1(4) exhibits.
4. *Hypersurfaces.* If $Y = \mathcal{Z}(F)$ is a hypersurface then $s = n - 1$ and the bound reads $r - 1$, with nonemptiness as soon as $\dim X \geq 1$. So a projective variety of positive dimension meets every hypersurface, which is another way of seeing why it carries no nonconstant regular function.

Exercise 40.6.1

1. Show that the image of the Segre embedding $\mathbb{P}^1_k \times \mathbb{P}^1_k \to \mathbb{P}^3_k$ is the quadric $\mathcal{Z}(w_{00}w_{11} - w_{01}w_{10})$, and that it contains two families of pairwise disjoint lines. Deduce, using exercise set 38.5.1(4), that this quadric is not isomorphic to $\mathbb{P}^2_k$.
2. Show that a variety that is both projective and affine is a single point.
3. Show that the map $\mathbb{A}^1_k \to \mathbb{A}^1_k$, $x \mapsto x^2$, is finite, and that the inclusion $\mathbb{A}^2_k \setminus \{0\} \hookrightarrow \mathbb{A}^2_k$ is not. For the second, which hypothesis of proposition 40.3.3 fails?
4. Exhibit a finite surjection $\mathcal{Z}(xy - 1) \to \mathbb{A}^1_k$, and check directly that it is finite by writing the coordinate ring as a module over the image.
5. Prove the affine normalization theorem using Noether's normalization theorem (theorem 32.4.1): every affine variety admits a finite surjective map onto some $\mathbb{A}^m_k$, with m its dimension.
6. For $f\colon \mathbb{A}^2_k \to \mathbb{A}^2_k$, $(u, v) \mapsto (u, uv)$, compute every fibre and verify all three parts of theorem 40.5.1 by hand. Which part fails if the word "component" is dropped from part (2)?
7. Using corollary 40.5.4, compute $\dim(\mathbb{P}^n_k \times \mathbb{P}^m_k)$ and check the answer against the Segre embedding for $n = m = 1$.
8. Show that the two projections $\mathbb{P}^1_k \times \mathbb{P}^1_k \to \mathbb{P}^1_k$ have all fibres isomorphic to $\mathbb{P}^1_k$, and use corollary 40.5.3 to recover the dimension of the product.

41

Projective Embeddings and Constructible Sets

Projective space carries more embeddings of itself than the linear ones, and the useful ones turn products, hypersurfaces, and linear subspaces into points of a larger projective space. This chapter collects them: the Veronese embedding, which linearizes degree, and the Plücker embedding, which realizes the Grassmannian of d-planes as a projective variety.

Then comes Chevalley's theorem. The image of a regular map is neither open nor closed in general, but it is always constructible, a finite union of locally closed sets, and that is the correct replacement for the closedness that chapter 40 supplies only in the projective case.

The chapter ends by putting the two halves together. Counting the lines that lie on a cubic surface needs the Grassmannian to parameterize the lines, projective space to parameterize the surfaces, and theorem 40.5.1 to conclude that the count is finite. It is the first argument in this book where the answer is a number nobody could have guessed.

41.1 The Veronese Embedding

A hypersurface of degree d in $\mathbb{P}^n$ is cut out by one equation, but that equation is not linear, and linear conditions are the ones that are easy to count. The Veronese embedding trades the degree for a dimension: it realizes $\mathbb{P}^n$ inside a much larger projective space in such a way that degree-d hypersurfaces downstairs become *hyperplane sections* upstairs.

Definition 41.1.1: Veronese Embedding

Fix $n, d \geq 1$ and let $M_0, \ldots, M_N$ be the monomials of degree d in $x_0, \ldots, x_n$, in some fixed order, where

$$N = \binom{n+d}{d} - 1$$

The *Veronese embedding* of degree d is

$$v_d \colon \mathbb{P}^n_k \to \mathbb{P}^N_k \qquad v_d([x_0 : \cdots : x_n]) = [M_0(x) : \cdots : M_N(x)]$$

The map is well defined: every M_i is homogeneous of the same degree d, so scaling x by λ scales every coordinate by λ^d, and the M_i never vanish simultaneously, since x_j^d is among them and some $x_j \neq 0$.

Theorem 41.1.2: The Veronese Is a Closed Embedding

v_d is injective, its image $V_{n,d} = v_d(\mathbb{P}^n_k)$ is closed in $\mathbb{P}^N_k$, and v_d is an isomorphism onto its image.

***Proof*:**
Injective. Write z_α for the coordinate of $\mathbb{P}^N$ attached to the monomial x^α. Suppose $v_d(x) = v_d(y)$. Some $x_j \neq 0$, so $z_{de_j} \neq 0$ at that point, hence $y_j \neq 0$ as well. Rescaling, take $x_j = y_j = 1$. Then the coordinate attached to $x_j^{d-1} x_i$ reads off x_i and y_i directly, so $x = y$.

Closed. The image satisfies the relations

$$z_\alpha z_\beta = z_\gamma z_\delta \qquad \text{whenever } \alpha + \beta = \gamma + \delta$$

since both sides equal $x^{\alpha+\beta}$. Conversely, suppose z satisfies them all and $z_{de_j} \neq 0$ for some j; normalize $z_{de_j} = 1$ and set $x_i = z_{(d-1)e_j + e_i}$. The relations then force $z_\alpha = x^\alpha$ for every α, by induction on how far α is from de_j. Hence the image is exactly the closed set cut out by those quadrics.

Isomorphism onto the image. The inverse is given on the open set $z_{de_j} \neq 0$ by $z \mapsto [z_{(d-1)e_j+e_0} : \cdots : z_{(d-1)e_j+e_n}]$, a tuple of forms of the same degree with no common zero there, hence regular by proposition 39.4.3. These agree on overlaps because both compute the same point, so they glue to a regular inverse, as we sought to show.

The payoff is the promised trade.

Corollary 41.1.3: Hypersurfaces Become Hyperplane Sections

Let F be a form of degree d in $x_0, \ldots, x_n$. Then there is a hyperplane $H \subseteq \mathbb{P}^N_k$ with

$$v_d\big(\mathcal{Z}(F)\big) = V_{n,d} \cap H$$

Consequently every degree-d hypersurface in $\mathbb{P}^n_k$ is the pullback of a hyperplane under v_d.

***Proof*:**
Write $F = \sum_\alpha c_\alpha x^\alpha$, the sum over monomials of degree d. Let $H = \mathcal{Z}(L)$ where $L = \sum_\alpha c_\alpha z_\alpha$, a linear form. Then $L(v_d(x)) = \sum_\alpha c_\alpha x^\alpha = F(x)$, so $v_d(x) \in H$ if and only if $F(x) = 0$, as we sought to show.

Example 41.1.4: Small Veronese Embeddings

1. $v_2 \colon \mathbb{P}^1_k \to \mathbb{P}^2_k$, $[s : t] \mapsto [s^2 : st : t^2]$. Here $N = \binom{3}{2} - 1 = 2$ and the image is the conic $\mathcal{Z}(z_0 z_2 - z_1^2)$. This is the isomorphism $\mathbb{P}^1 \cong$ smooth conic met in example 39.4.4(3), in coordinates.

2. $v_3 \colon \mathbb{P}^1_k \to \mathbb{P}^3_k$, $[s : t] \mapsto [s^3 : s^2t : st^2 : t^3]$, whose image is the twisted cubic of example 38.3.7(4). So the twisted cubic is a Veronese image, which is another way of seeing that it is rational.

3. $v_2 \colon \mathbb{P}^2_k \to \mathbb{P}^5_k$ is the *Veronese surface*, the classical example of a smooth surface in $\mathbb{P}^5$ that is not contained in any hyperplane.

The construction also gives a clean way to think about the parameter space of section 39.2: the degree-d forms on $\mathbb{P}^n$, taken up to scalar, are the points of $\mathbb{P}^N_k$, and corollary 41.1.3 says the hypersurface a point cuts out is the corresponding hyperplane section of the Veronese. The following is a first payoff.

Proposition 41.1.5: Parameterizing Reducible Hypersurfaces

Fix n and d, and let $\mathbb{P}^N_k$, $N = \binom{n+d}{d} - 1$, be the projective space parameterizing degree-d forms in $n+1$ variables up to scalar, as in section 39.2. Then the set of points corresponding to *reducible* forms is closed in $\mathbb{P}^N_k$.

***Proof*:**
A reducible form of degree d factors as a product of forms of degrees j and $d - j$ for some $1 \le j \le d - 1$. Writing Z_j for the set of points of $\mathbb{P}^N_k$ arising this way, the set of reducible forms is $\bigcup_{j=1}^{d-1} Z_j$, a finite union, so it suffices to show each Z_j is closed.

Let $N_j = \binom{n+j}{j} - 1$ and $N_{d-j} = \binom{n+d-j}{d-j} - 1$ be the corresponding parameter spaces. Multiplication of forms gives a map

$$\mu \colon \mathbb{P}^{N_j}_k \times \mathbb{P}^{N_{d-j}}_k \to \mathbb{P}^N_k$$

which is well defined, since scaling either factor scales the product, and regular, since the coefficients of a product are bilinear in the coefficients of the factors. Its image is exactly Z_j.

The source is a product of projective spaces, hence a projective variety by proposition 40.1.3, so $Z_j = \mu(\mathbb{P}^{N_j}_k \times \mathbb{P}^{N_{d-j}}_k)$ is closed by corollary 40.2.3. That is the whole point of completeness: the image of a projective variety is closed, which for an affine source would fail. Hence the reducible locus is closed, as we sought to show.

For $n = 1$ and $d = 2$ this recovers a familiar fact: a binary quadratic form $ax^2 + bxy + cy^2$ is reducible exactly when its discriminant $b^2 - 4ac$ vanishes, a single equation in the coefficients.

41.2 Grassmannians and the Plücker Embedding

The Veronese parameterizes hypersurfaces. The Grassmannian parameterizes linear subspaces, and it is the space one needs in order to ask questions like "how many lines lie on a surface". As with the Veronese, the construction realizes a set of geometric objects as the points of a projective variety.

Definition 41.2.1: Grassmannian

Let $0 < r < n$. The *Grassmannian* $G(r, n)$ is the set of r-dimensional linear subspaces of k^n. Equivalently, since an r-dimensional subspace of k^n is an $(r-1)$-dimensional linear subspace of $\mathbb{P}^{n-1}_k$, it is the set of $(r-1)$-planes in $\mathbb{P}^{n-1}_k$.

The two readings differ only by a shift, and both are used: $G(2, 4)$ is the space of 2-dimensional subspaces of k^4, which is the space of *lines* in $\mathbb{P}^3_k$. That is the case section 41.4 needs, and there the columns of k^4 are indexed by $0, 1, 2, 3$ to match the coordinates $x_0, \dots, x_3$ on $\mathbb{P}^3_k$, writing the Plücker coordinates as p_{ij} with $0 \le i < j \le 3$.

To make $G(r, n)$ a variety, represent a subspace by a basis and kill the ambiguity in that choice.

Definition 41.2.2: Plücker Embedding

Let $W \subseteq k^n$ be an r-dimensional subspace with basis $w_1, \dots, w_r$. Writing the w_i as the rows of an $r \times n$ matrix, let $p_{i_1 \cdots i_r}$ be the $r \times r$ minor on columns $i_1 < \cdots < i_r$. The *Plücker embedding* is

$$\rho\colon G(r, n) \to \mathbb{P}^{\binom{n}{r}-1}_k \qquad \rho(W) = [p_{i_1 \cdots i_r}]_{i_1 < \cdots < i_r}$$

Proposition 41.2.3: The Plücker Embedding Is Well Defined and Injective

ρ does not depend on the choice of basis, and it is injective.

***Proof*:**
Changing basis replaces the matrix A by gA for some $g \in \mathrm{GL}_r(k)$, and the Cauchy-Binet formula multiplies every maximal minor by the single scalar $\det g \neq 0$. So the tuple of minors changes by an overall scalar and its class in projective space is unchanged. Not all minors vanish, since the rows are independent and so some r columns are independent.

For injectivity, recover W from $\rho(W)$: a vector $v \in k^n$ lies in W exactly when the $(r + 1) \times n$ matrix obtained by appending v has all its maximal minors zero, and expanding those minors along the appended row writes each as a linear form in v with coefficients among the $p_{i_1 \cdots i_r}$. So the Plücker coordinates determine W as the common zero set of an explicit system of linear equations, as we sought to show.

Theorem 41.2.4: The Grassmannian Is a Projective Variety

The image of ρ is closed in $\mathbb{P}_k^{\binom{n}{r}-1}$, so $G(r,n)$ is a projective variety. Its dimension is

$$\dim G(r,n) = r(n-r)$$

***Proof*:**
Closed. The image is cut out by the *Plücker relations*, the quadratic equations expressing that a tuple of $\binom{n}{r}$ numbers arises as the maximal minors of an $r \times n$ matrix. These are quadrics in the $p_{i_1 \cdots i_r}$; example 41.2.5(2) writes them out in the first case where there is more than one. Their zero set is closed, and the argument of proposition 41.2.3 shows a point satisfying them determines an r-dimensional subspace, so the zero set is exactly imρ.

Dimension. Cover $G(r,n)$ by the open sets $U_{i_1 \cdots i_r} = \{p_{i_1 \cdots i_r} \neq 0\}$. On $U_{1 \cdots r}$, every subspace has a unique basis whose matrix is $[\, I_r \big| B\,]$ with B an arbitrary $r \times (n-r)$ matrix: row-reduce using the invertible left block. The entries of B are free coordinates, giving $U_{1 \cdots r} \cong \mathbb{A}_k^{r(n-r)}$. These charts are irreducible and pairwise meet, since any two nonempty open subsets of an irreducible space do, so $G(r,n)$ is irreducible by lemma 40.1.5, and corollary 34.3.3 reads its dimension off a single chart as $r(n-r)$, as we sought to show.

Example 41.2.5: Grassmannians

1. $G(1,n) = \mathbb{P}_k^{n-1}$: a line through the origin in k^n is a point of $\mathbb{P}^{n-1}$, and the Plücker map is the identity. Dimension $1 \cdot (n-1) = n-1$, as it should be.
2. $G(2,4)$, the lines in $\mathbb{P}_k^3$. Here $\binom{4}{2} = 6$, so ρ lands in $\mathbb{P}_k^5$ with coordinates $p_{12}, p_{13}, p_{14}, p_{23}, p_{24}, p_{34}$, and there is exactly one Plücker relation:

$$p_{12}p_{34} - p_{13}p_{24} + p_{14}p_{23} = 0$$

   So $G(2,4)$ is a quadric hypersurface in $\mathbb{P}_k^5$, of dimension $2 \cdot 2 = 4$, matching theorem 34.4.1, which gives $5 - 1 = 4$.
3. $G(r,n) \cong G(n-r,n)$, by sending a subspace to its annihilator in the dual space. So the lines in $\mathbb{P}^3$ and the planes in $\mathbb{P}^3$ form isomorphic families, both of dimension 4.

The count in item (2) is the one section 41.4 runs on: the lines in $\mathbb{P}^3$ form a 4-dimensional family, and asking how many of them lie on a given surface is a question about intersecting that family with a condition.

41.3 Constructible Sets and Chevalley's Theorem

Chapter 40 showed that the image of a *projective* variety under a regular map is closed. For a general regular map the image can be neither closed nor open: the projection of $\mathcal{Z}(xy-1)$ to the x-axis is $\mathbb{A}_k^1 \setminus \{0\}$, which is open, while the image of $(x,y) \mapsto (x,xy)$ is $(\mathbb{A}_k^1 \setminus \{0\}) \times \mathbb{A}_k^1$ together with the single point $(0,0)$, which is neither. Chevalley's theorem says the second example is as bad as it gets: images are always finite unions of pieces of that shape.

Definition 41.3.1: Locally Closed and Constructible Sets

Let X be a variety. A subset $S \subseteq X$ is *locally closed* if $S = U \cap Z$ with U open and Z closed, equivalently if S is open in its own closure. A subset is *constructible* if it is a finite union of locally closed sets.

Proposition 41.3.2: Constructible Sets Form a Boolean Algebra

The constructible subsets of X are closed under finite union, finite intersection, and complement. They are the smallest such family containing the open sets.

***Proof*:**
Closure under finite union is the definition. For intersection, $(U_1 \cap Z_1) \cap (U_2 \cap Z_2) = (U_1 \cap U_2) \cap (Z_1 \cap Z_2)$ is locally closed, and intersection distributes over the finite unions.

For complement, it suffices to complement one locally closed set, since the complement of a finite union is the finite intersection of the complements and intersections are handled. And

$$X \setminus (U \cap Z) = (X \setminus U) \cup (X \setminus Z)$$

a union of a closed set and an open set, both locally closed. Finally any family closed under these operations and containing the opens contains every $U \cap Z$, hence every constructible set, as we sought to show.

The theorem rests on the fact that a dominating map hits a dense open set. Section 40.3 stated this without proof; it follows from the generic finiteness of theorem 40.5.1(3).

Lemma 41.3.3: A Dominating Image Contains a Dense Open Set

Let $f\colon X \to Y$ be a dominating regular map of irreducible varieties. Then $f(X)$ contains a nonempty open subset of Y.

***Proof*:**
Passing to affine open subsets changes neither hypothesis nor conclusion, so assume X and Y affine. The proof of theorem 40.5.1(3) produced $g \in k[Y]$ and $t_1, \dots, t_{n-m}$ such that

$$k[X]_g \text{ is finite over } k[Y]_g[t_1, \dots, t_{n-m}]$$

Geometrically this says f restricted over $D(g)$ factors as a finite map followed by a projection $D(g) \times \mathbb{A}^{n-m}_k \to D(g)$. A finite map is surjective by proposition 40.3.3, and the projection is surjective, so the composite is surjective onto $D(g)$. Hence $f(X) \supseteq D(g)$, a nonempty open subset of Y, as we sought to show.

Theorem 41.3.4: Chevalley's Theorem

Let $f\colon X \to Y$ be a regular map of varieties. Then f carries constructible sets to constructible sets. In particular $f(X)$ is constructible.

***Proof*:**
Since a constructible set is a finite union of locally closed sets, and the image of a union is the union of the images, it suffices to treat $f(X)$ itself, with X replaced by a locally closed subset. Decomposing X into irreducible components (theorem 32.3.5) and Y into the closure of the image, we may assume X and Y irreducible and f dominating.

Argue by Noetherian induction on Y: suppose the result holds for every proper closed subvariety of Y. By lemma 41.3.3 there is a nonempty open $U \subseteq Y$ with $U \subseteq f(X)$. Then

$$f(X) = U \cup f\bigl(X \setminus f^{-1}(U)\bigr)$$

The set $X' = X \setminus f^{-1}(U)$ is closed in X, and $f(X') \subseteq Y \setminus U$, a proper closed subset of Y. Applying the inductive hypothesis to $f|_{X'}\colon X' \to Y \setminus U$, which is a regular map of varieties with strictly smaller target, shows $f(X')$ is constructible. As U is open, hence constructible, and constructible sets are closed under finite union (proposition 41.3.2), $f(X)$ is constructible.

The induction terminates because Y is a Noetherian topological space (proposition 31.3.7), so there is no infinite strictly descending chain of closed subsets, completing the proof.

Chevalley's theorem is the right general statement, and the closed-image theorem of corollary 40.2.3 is the special case where properness upgrades "constructible" to "closed". It is also the statement that survives into scheme theory unchanged, where it becomes the assertion that a finite-type morphism of Noetherian schemes has constructible image.

Example 41.3.5: The Image That Is Neither Open Nor Closed
Take $f\colon \mathbb{A}^2_k \to \mathbb{A}^2_k$, $f(x,y) = (x, xy)$. For $x = a \neq 0$ the fibre over (a,b) is the single point $(a, b/a)$, so every such (a,b) is hit; for $x = 0$ the image of the line $x = 0$ is the single point $(0,0)$. Hence

$$\operatorname{im} f = \bigl((\mathbb{A}^1_k \setminus \{0\}) \times \mathbb{A}^1_k\bigr) \;\cup\; \{(0,0)\}$$

The first piece is open, the second is closed, and the union is neither. It is constructible, being a union of two locally closed sets, exactly as theorem 41.3.4 requires. This is also the map of example 40.5.2, whose fibre over the origin jumped in dimension; the jump and the failure of openness are the same phenomenon.

41.4 Application: Lines on a Cubic Surface

The Grassmannian and the fibre-dimension theorem together answer a question that would otherwise be inaccessible: how many lines lie on a surface in $\mathbb{P}^3$? For a general cubic surface the answer is a finite number, and the argument that it is finite is a dimension count of exactly the kind this Part was built to support.

Fix the two parameter spaces. Lines in $\mathbb{P}^3_k$ form $G = G(2,4)$, of dimension 4 by theorem 41.2.4.

Cubic surfaces in $\mathbb{P}^3_k$ are the nonzero degree-3 forms in four variables up to scalar, so they form $\mathbb{P}^{19}_k$: there are $\binom{3+3}{3} = 20$ monomials of degree 3 in four variables. Note that this parameter space contains reducible and singular surfaces too, which is harmless and in fact necessary, since a parameter space has to be complete to be useful.

The strategy is to put lines and surfaces into a single variety and project it two ways. Set

$$I = \{(\ell, S) \in G \times \mathbb{P}^{19}_k \mid \ell \subseteq S\}$$

Projecting to G organizes I by the line; over a fixed line the cubics containing it form a linear system, and that side computes $\dim I$. Projecting to $\mathbb{P}^{19}$ organizes I by the surface, and its fibres are the sets of lines to be counted.

Lemma 41.4.1: The Incidence Variety

I is an irreducible projective variety of dimension 19, and $\pi_1 : I \to G$ is surjective with every fibre a linear subspace $\mathbb{P}^{15} \subseteq \mathbb{P}^{19}$.

***Proof*:**
Work chart by chart on G. By the proof of theorem 41.2.4 the open set $U \subseteq G$ where $p_{01} \neq 0$ is isomorphic to $\mathbb{A}^4_k$, with coordinates a, b, c, d naming the plane spanned by $(1, 0, a, b)$ and $(0, 1, c, d)$; the corresponding line $\ell_{abcd} \subseteq \mathbb{P}^3_k$ is the image of

$$[s : t] \longmapsto [\, s : t : as + ct : bs + dt \,]$$

Let $T = T_{abcd}$ be the linear automorphism of k^4 fixing e_2, e_3 and sending e_0, e_1 to $(1, 0, a, b)$ and $(0, 1, c, d)$. Its matrix is lower triangular with 1s on the diagonal, so $\det T = 1$, and both T and T^{-1} have entries that are polynomials in a, b, c, d. By construction T carries the line $\mathcal{Z}(x_2, x_3)$ onto ℓ_{abcd}, so for a cubic form F,

$$F \text{ vanishes on } \ell_{abcd} \iff F \circ T \text{ vanishes on } \mathcal{Z}(x_2, x_3) \iff F \circ T \text{ has no term in } x_0, x_1 \text{ alone}$$

the last step because the restriction of $F \circ T$ to $\mathcal{Z}(x_2, x_3)$ is obtained by setting $x_2 = x_3 = 0$, leaving exactly the four terms $x_0^3, x_0^2 x_1, x_0 x_1^2, x_1^3$.

Now consider

$$\Psi : U \times \mathbb{P}^{19}_k \to U \times \mathbb{P}^{19}_k \qquad \big((a, b, c, d), [F]\big) \mapsto \big((a, b, c, d), [F \circ T_{abcd}]\big)$$

The coefficients of $F \circ T$ are linear in those of F with coefficients polynomial in a, b, c, d, and $F \circ T \neq 0$ because T is invertible, so Ψ is a regular map; the same formula with T^{-1} gives its inverse, so Ψ is an isomorphism. The displayed equivalence says precisely that Ψ carries $\pi_1^{-1}(U)$ onto $U \times L$, where $L \subseteq \mathbb{P}^{19}$ is the *fixed* linear subspace of cubics whose four coefficients on $x_0^3, x_0^2 x_1, x_0 x_1^2, x_1^3$ vanish. Those four coordinates are independent, so $L \cong \mathbb{P}^{15}$, and

$$\pi_1^{-1}(U) \cong \mathbb{A}^4_k \times \mathbb{P}^{15}_k$$

Three consequences follow, all of them local statements verified on each of the six charts of G in turn, the computation being identical after permuting coordinates.

I is closed. $U \times L$ is closed in $U \times \mathbb{P}^{19}$, hence so is $\pi_1^{-1}(U)$, and being closed is a local condition on a cover. Since $G \times \mathbb{P}^{19}$ is projective by proposition 40.1.3, I is a projective variety.

π_1 is surjective with fibre $\mathbb{P}^{15}$. Under Ψ the fibre over (a, b, c, d) becomes $\{(a, b, c, d)\} \times L$, so it is a linear $\mathbb{P}^{15}$, in particular nonempty and irreducible of dimension 15.

I is irreducible of dimension 19. Each $\pi_1^{-1}(U)$ is irreducible by proposition 40.1.6, and any two of the six meet, since the charts of the irreducible G do and the fibres of π_1 are nonempty. Lemma 40.1.5 makes I irreducible, and then corollary 40.5.3 applied to π_1 gives $\dim I = 4+15 = 19$, as we sought to show.

The count now hinges on a single input from outside the dimension theory: one cubic surface whose lines can be counted by hand, and which has finitely many. The Fermat cubic is the traditional choice, but its 27 lines take work to enumerate; the following surface is singular, has only three lines, and settles the matter in half a page.

Lemma 41.4.2: A Cubic Surface With Exactly Three Lines

Let $S_0 = \mathcal{Z}(x_0x_1x_2 - x_3^3) \subseteq \mathbb{P}^3_k$. Then S_0 is an irreducible cubic surface, and the only lines contained in it are $\mathcal{Z}(x_3, x_0)$, $\mathcal{Z}(x_3, x_1)$ and $\mathcal{Z}(x_3, x_2)$.

***Proof*:**
Irreducibility. Read $x_0x_1x_2 - x_3^3$ as a polynomial in x_3 over the unique factorization domain $k[x_0, x_1, x_2]$. Its leading coefficient is -1, its middle coefficients are 0, and its constant term $x_0x_1x_2$ lies in (x_0) but not in (x_0^2). Eisenstein's criterion at the prime x_0 applies, so the polynomial is irreducible and S_0 is a variety.

The three lines lie on it. On $\mathcal{Z}(x_3, x_i)$ both $x_0x_1x_2$ and x_3^3 vanish identically.

There are no others. Let $\ell \subseteq S_0$ be a line, parameterized by $[s : t] \mapsto [g_0 : g_1 : g_2 : g_3]$, where the g_i are linear forms in s, t spanning a two-dimensional space, that being what makes the image a line rather than a point. The defining equation restricts to $g_0g_1g_2 = g_3^3$ in $k[s, t]$.

If $g_3 \equiv 0$ then $g_0g_1g_2 \equiv 0$, and $k[s, t]$ is a domain, so $g_i \equiv 0$ for some $i \leq 2$; then $\ell \subseteq \mathcal{Z}(x_3, x_i)$, and as both are lines they are equal.

If $g_3 \not\equiv 0$, change coordinates on the source $\mathbb{P}^1$ so that $g_3 = t$, which is possible because g_3 is a nonzero linear form. Then $g_0g_1g_2 = t^3$. None of g_0, g_1, g_2 is zero, and each divides t^3 in the unique factorization domain $k[s, t]$, in which t is prime; the only linear divisors of t^3 are the scalar multiples of t. So $g_i = c_it$ for every $i \leq 2$ as well, making all four g_i proportional, which contradicts the requirement that they span a two-dimensional space of linear forms. Hence this case does not occur, as we sought to show.

Theorem 41.4.3: Lines on a Cubic Surface

Every cubic surface in $\mathbb{P}^3_k$ contains a line, and there is a nonempty open subset $U \subseteq \mathbb{P}^{19}_k$ such that a cubic surface parameterized by a point of U contains only finitely many.

***Proof*:**
Write $\pi_2 \colon I \to \mathbb{P}^{19}_k$ for the second projection. By lemma 41.4.1 the source is an irreducible projective variety with $\dim I = 19 = \dim \mathbb{P}^{19}_k$, so $\pi_2(I)$ is closed by corollary 40.2.3 and irreducible, being the continuous image of an irreducible space.

π_2 *is surjective.* The fibre $\pi_2^{-1}(S_0)$ over the surface of lemma 41.4.2 is a set of three points, so it is nonempty of dimension 0; in particular $S_0 \in \pi_2(I)$. Suppose $\pi_2(I) \neq \mathbb{P}^{19}$. Then $\pi_2(I)$ is a proper closed subvariety, so $\dim \pi_2(I) \leq 18$ by proposition 34.3.5. Applying theorem 40.5.1(2) to the dominating map $\pi_2 \colon I \to \pi_2(I)$ forces every component of every fibre to have dimension at least $19 - 18 = 1$, contradicting the fibre over S_0. Hence $\pi_2(I) = \mathbb{P}^{19}$, which says exactly that every cubic surface contains a line.

The general fibre is finite. Now π_2 is a dominating map of irreducible varieties with $\dim I = \dim \mathbb{P}^{19}$, so theorem 40.5.1(3) supplies a nonempty open $U \subseteq \mathbb{P}^{19}$ with $\dim \pi_2^{-1}(S) = 19 - 19 = 0$ for every $S \in U$. Such a fibre is closed in the projective I, hence an algebraic set, and by theorem 32.3.5 it has finitely many irreducible components. Each component is closed, irreducible and of dimension 0, hence a single point: a closed irreducible set with two distinct points $p \neq q$ contains the proper closed subset $\{p\}$, giving a chain of length 1 and so dimension at least 1. The fibre is therefore finite, as we sought to show.

The dimension count says “finitely many”. The classical refinement, that a *smooth* cubic surface contains exactly 27 lines, needs more than dimension theory: one exhibits the 27 on a single smooth cubic and then shows the number is constant across the smooth locus of the family. On the Fermat cubic $\mathcal{Z}(x_0^3 + x_1^3 + x_2^3 + x_3^3)$, in characteristic zero, they are the lines

$$x_0 = \alpha x_1, \quad x_2 = \beta x_3 \qquad \alpha^3 = \beta^3 = -1$$

together with the images of these nine under the two other ways of splitting $\{x_0, x_1, x_2, x_3\}$ into pairs, giving $3 \times 9 = 27$. The count is due to Cayley and Salmon in 1849; a full treatment is in Shafarevich or in Hartshorne’s chapter on surfaces.

41.4.4 Note: Where the Numbers Come From The two numbers in the proof are worth keeping straight. The 19 is $\binom{6}{3} - 1$: cubic forms in four variables, up to scalar. The 15 is $19 - 4$, where the 4 counts the coefficients of a binary cubic, that is the number of conditions imposed by requiring a cubic to vanish on a fixed line. The coincidence $\dim I = \dim \mathbb{P}^{19}$ is exactly what makes the answer finite and nonzero. Had the incidence variety been larger, every cubic would carry infinitely many lines; had it been smaller, the general cubic would carry none, which is what happens for surfaces of degree 4 and above.

41.5 Degree, Linear Sections, and Bézout in $\mathbb{P}^n$

Definition 38.5.3 defined the degree of a projective variety as $d!$ times the leading coefficient of its Hilbert polynomial. That is a definition one can compute with and not one that says anything geometric. This section supplies the geometry: the degree counts points, and degrees multiply under intersection. Both statements need theorem 40.6.3 to know that the intersections in question are nonempty and of the right size.

Theorem 41.5.1: Degree Counts Points of a Linear Section

Let $X \subseteq \mathbb{P}^n_k$ be a projective variety of dimension r and degree d. Then for a general linear subspace $L \subseteq \mathbb{P}^n_k$ of dimension $n - r$, the intersection $X \cap L$ is a finite set of exactly d points.

***Proof*:**
The intersection is finite and nonempty. By theorem 40.6.3 every component of $X \cap L$ has dimension at least $r + (n-r) - n = 0$, and the intersection is nonempty. For a general L the dimension is exactly 0: choosing L as an intersection of r hyperplanes, each successive hyperplane may be chosen to contain no component of what remains, since a component is a proper closed subset of $\mathbb{P}^n_k$ and the hyperplanes containing a fixed subvariety form a proper linear subspace of the dual space. Each such cut drops the dimension by exactly one by theorem 34.4.1, so after r cuts the dimension is 0 and the set is finite.

The count. Write $S = S(X)$ for the homogeneous coordinate ring and let $\ell_1, \ldots, \ell_r$ be the linear forms cutting out L. Since no ℓ_i vanishes on a component of the successive intersections, each ℓ_i is a nonzerodivisor on the successive quotients, so

$$H_{X\cap L}(m) = \dim_k \left(S/(\ell_1, \ldots, \ell_r)\right)_m$$

and each quotient by a nonzerodivisor of degree 1 takes the mth graded piece to the difference of the mth and $(m-1)$st. Iterating r times, the Hilbert polynomial of $X \cap L$ is the rth finite difference of h_X. As h_X has degree r with leading coefficient $d/r!$, its rth finite difference is the constant d.

Finally $X \cap L$ is a finite set of points, and the Hilbert polynomial of a finite set of N distinct points in $\mathbb{P}^n_k$ is the constant N: the restriction map from forms of degree m to functions on the points is surjective for m large, by choosing forms separating the points. Hence $N = d$, as we sought to show.

So the degree of a plane curve is the number of points in which a general line meets it, the degree of the twisted cubic is the number of points in which a general plane meets it, which example 38.5.4(4) computed as 3, and the degree of $\mathbb{P}^r$ itself is 1, a general linear space of complementary dimension meeting it once.

The same bookkeeping gives the general form of Bézout's theorem. Stated for hypersurfaces it is a count of degrees; the plane-curve case of chapter 43 is the case $n = 2$, refined there to include local multiplicities.

Theorem 41.5.2: Bézout's Theorem in $\mathbb{P}^n$

Let $F_1, \ldots, F_r$ be forms in $k[x_0, \ldots, x_n]$ of degrees $d_1, \ldots, d_r$ with $r \le n$, and suppose $X = \mathcal{Z}(F_1, \ldots, F_r)$ has dimension exactly $n - r$, the smallest its dimension can be. Then X has degree

$$\deg X = d_1 d_2 \cdots d_r$$

counting components with multiplicity.

***Proof*:**
Induct on r. For $r = 1$, the computation of example 38.5.4(2) gives $H_X(m) = \binom{m+n}{n} - \binom{m-d_1+n}{n}$, whose leading term is $d_1 m^{n-1}/(n-1)!$, so $\deg X = d_1$: the degree of a hypersurface is the degree of its equation.

For the inductive step, let $X' = \mathcal{Z}(F_1, \ldots, F_{r-1})$, of dimension $n - r + 1$ by the hypothesis that each cut is proper, and of degree $d_1 \cdots d_{r-1}$ by induction. Cutting X' by F_r drops the dimension by exactly one, so F_r vanishes on no component of X' and is therefore a nonzerodivisor on

$S(X')$. Multiplication by F_r is then injective and raises degree by d_r, giving the exact sequence of graded pieces

$$0 \to S(X')_{m-d_r} \to S(X')_m \to S(X)_m \to 0$$

so $h_X(m) = h_{X'}(m) - h_{X'}(m - d_r)$. Write $N = n - r + 1$ and $h_{X'}(m) = \frac{e}{N!}m^N + \cdots$ with $e = \deg X'$. Since $m^N - (m - d_r)^N = N d_r m^{N-1} + \cdots$, the difference has degree $N - 1 = n - r$ with leading coefficient

$$\frac{e}{N!} \cdot N d_r \;=\; \frac{e\, d_r}{(N-1)!} \;=\; \frac{e\, d_r}{(n-r)!}$$

By definition 38.5.3 the degree of X is $(n - r)!$ times that leading coefficient, namely $e\, d_r = d_1 \cdots d_r$, as we sought to show.

Example 41.5.3: Degrees of Intersections

1. *Two quadric surfaces in* $\mathbb{P}^3_k$. Each has degree 2, and if their intersection is a curve then theorem 41.5.2 gives it degree 4. A general such curve is the elliptic quartic; it is not a plane curve, since a plane curve of degree 4 has genus 3 by definition 44.4.1 while this one has genus 1.

2. *The twisted cubic is not a complete intersection.* It has degree 3 and codimension 2 in $\mathbb{P}^3_k$. Were it $\mathcal{Z}(F_1, F_2)$ with $\deg F_i = d_i$, theorem 41.5.2 would force $d_1 d_2 = 3$, so $\{d_1, d_2\} = \{1, 3\}$ and the curve would lie in the plane $\mathcal{Z}(F_1)$. It does not: its parameterization $[s^3 : s^2t : st^2 : t^3]$ spans $\mathbb{P}^3_k$, since the four coordinate functions are linearly independent. This is the promise made in example 38.3.7.

3. *A line and a plane in* $\mathbb{P}^3_k$. Degrees 1 and 1, dimensions 1 and 2, so the intersection has dimension 0 and degree 1: exactly one point, which is the familiar statement that a line not contained in a plane meets it once.

41.6 Cones, Projections, and Secant Varieties

The embeddings of the previous sections all enlarge the ambient space. This one goes the other way: given a variety in $\mathbb{P}^N_k$, project it into a smaller projective space and ask when nothing is lost. The construction has already been used twice without being set up — to identify a conic with $\mathbb{P}^1_k$ in theorem 44.4.2, and to exhibit a finite map in section 40.3 — and the obstruction to it is a variety worth naming in its own right.

Definition 41.6.1: Cone and Join

Let $X, Y \subseteq \mathbb{P}^N_k$ be disjoint projective varieties. The *join* $J(X, Y)$ is the closure of the union of the lines $\overline{xy}$ for $x \in X$ and $y \in Y$. When $Y = \{p\}$ is a single point, $J(X, p)$ is the *cone* over X with vertex p.

Definition 41.6.2: Projection from a Point

Let $p \in \mathbb{P}^N_k$ and let $H \cong \mathbb{P}^{N-1}_k$ be a hyperplane not containing p. The *projection from* p is

$$\pi_p \colon \mathbb{P}^N_k \setminus \{p\} \to H \qquad q \mapsto \overline{pq} \cap H$$

In coordinates with $p = [0 : \cdots : 0 : 1]$ and $H = \mathcal{Z}(x_N)$, it is $[x_0 : \cdots : x_N] \mapsto [x_0 : \cdots : x_{N-1}]$, visibly regular away from p.

Proposition 41.6.3: Projection Is a Finite Morphism

Let $X \subseteq \mathbb{P}^N_k$ be a projective variety and $p \notin X$. Then π_p restricts to a morphism $X \to \mathbb{P}^{N-1}_k$ whose image is closed of the same dimension as X, and whose fibres are finite.

***Proof*:**
Since $p \notin X$, the map π_p is defined at every point of X, and it is regular there by definition 41.6.2. Its image is closed by corollary 40.2.3, X being projective.

The fibre over a point h is $X \cap \overline{ph}$, the intersection of X with a line through p. That line is not contained in X, since $p \notin X$, so the intersection is a proper closed subset of the line and hence finite. Finiteness of the fibres forces $\dim \pi_p(X) = \dim X$ by theorem 40.5.1(3), whose generic fibre dimension is then 0, as we sought to show.

Projection is injective on X exactly when no line through p meets X twice, and it is an immersion when in addition no tangent line to X passes through p. Both conditions say that p avoids a certain subvariety, and naming those subvarieties is what makes the argument work.

Definition 41.6.4: Secant and Tangent Varieties

Let $X \subseteq \mathbb{P}^N_k$ be a projective variety. The *secant variety* $\operatorname{Sec}(X)$ is the closure of the union of the lines $\overline{xy}$ over pairs of distinct points $x, y \in X$. The *tangent variety* $\operatorname{Tan}(X)$ is the closure of the union of the embedded tangent spaces $T_xX \cong \mathbb{P}^r_k$ over the nonsingular points x of X, where $r = \dim X$.

Proposition 41.6.5: Dimension of the Secant Variety

Let $X \subseteq \mathbb{P}^N_k$ be a projective variety of dimension r. Then

$$\dim \operatorname{Sec}(X) \le 2r + 1 \qquad \dim \operatorname{Tan}(X) \le 2r$$

***Proof*:**
Consider the incidence variety

$$I = \overline{\{(x, y, z) \mid x \neq y \in X,\ z \in \overline{xy}\}} \ \subseteq\ X \times X \times \mathbb{P}^N_k$$

Projecting to $X \times X$ has fibre the line $\overline{xy}$, of dimension 1, over the open set where $x \neq y$. So $\dim I \le \dim(X \times X) + 1 = 2r + 1$ by corollary 40.5.4 and theorem 40.5.1. Now $\operatorname{Sec}(X)$ is the image of I under the third projection, and the image of a projective variety is closed of dimension at most that of the source, by corollary 40.2.3 and theorem 40.5.1(1). Hence $\dim \mathrm{Sec}(X) \leq 2r+1$.

For $\mathrm{Tan}(X)$ run the same argument over X alone. The incidence $\{(x,z) \mid x \in X \text{ nonsingular},\ z \in T_xX\}$ projects to the smooth locus of X, of dimension r, with fibre the embedded tangent space T_xX, of dimension r. So it has dimension at most $2r$, and $\mathrm{Tan}(X)$, being its image, has dimension at most $2r$, as we sought to show.

The bound is what makes projection useful: as soon as the ambient space is large enough, a point can be chosen outside the secant variety, and projecting from it loses nothing.

Theorem 41.6.6: Every Projective Variety Embeds in $\mathbb{P}^{2r+1}$

Let $X \subseteq \mathbb{P}^N_k$ be a projective variety of dimension r. If $N > 2r+1$, then there is a point $p \notin \mathrm{Sec}(X)$, and π_p maps X isomorphically onto its image in $\mathbb{P}^{N-1}_k$. Iterating, X is isomorphic to a projective variety in $\mathbb{P}^{2r+1}_k$.

***Proof*:**
By proposition 41.6.5, $\dim \mathrm{Sec}(X) \leq 2r+1 < N$, so $\mathrm{Sec}(X)$ is a proper closed subset of $\mathbb{P}^N_k$ and a point p outside it exists. Note $X \subseteq \mathrm{Sec}(X)$ whenever $\dim X \geq 1$, so $p \notin X$ and proposition 41.6.3 applies.

π_p is injective on X: if $\pi_p(x) = \pi_p(y)$ with $x \neq y$ then p lies on the line $\overline{xy}$, hence in $\mathrm{Sec}(X)$, contrary to choice. It is also an immersion: a tangent line to X is a limit of secant lines, so $\mathrm{Tan}(X) \subseteq \mathrm{Sec}(X)$ and no tangent line to X passes through p either. An injective morphism of projective varieties that is an immersion is an isomorphism onto its closed image.

Repeating drops the ambient dimension by one each time and stops at $N = 2r+1$, as we sought to show.

Example 41.6.7: Projections in Practice

1. *A curve in* $\mathbb{P}^3_k$. Here $r = 1$, so $2r+1 = 3$ and theorem 41.6.6 says every projective curve embeds in $\mathbb{P}^3_k$. It does not say $\mathbb{P}^2_k$: projecting once more from a point of $\mathrm{Sec}(X)$ is unavoidable when $N = 3$, and the image acquires singularities. That is why plane curves are special and why chapter 42 must handle singular ones.

2. *The twisted cubic.* Its secant variety is all of $\mathbb{P}^3_k$, since $\dim \mathrm{Sec} \leq 3$ and the secant lines already sweep out a three-dimensional set. So no projection to $\mathbb{P}^2_k$ is an isomorphism, consistent with the twisted cubic being a rational curve of degree 3 while every plane cubic has genus 1.

3. *A conic.* Projection from a point of the conic itself, rather than from outside it, is the map of theorem 44.4.2: a line through p meets the conic once more, so the projection is a bijection to $\mathbb{P}^1_k$. This is the degenerate case $p \in X$, excluded from proposition 41.6.3, and it lowers the dimension of the image rather than preserving it.

Exercise 41.6.1

1. Write out the Veronese embedding $\mathbb{P}^1_k \to \mathbb{P}^2_k$ for $d = 2$ and identify its image. Then do $d = 3$ and show the image is the twisted cubic of example 38.3.7.
2. Compute $N = \binom{n+d}{d} - 1$ for $(n, d) = (2, 2)$ and describe the Veronese surface in $\mathbb{P}^5_k$. Show that it contains no lines, although it is covered by conics.
3. Verify corollary 41.1.3 explicitly for $n = 1$, $d = 2$: exhibit the hyperplane of $\mathbb{P}^2_k$ cutting out a prescribed pair of points of $\mathbb{P}^1_k$.
4. Take the 2-plane in k^4 spanned by $(1, 0, a, b)$ and $(0, 1, c, d)$. Compute its six Plücker coordinates and verify the relation $p_{01}p_{23} - p_{02}p_{13} + p_{03}p_{12} = 0$ directly.
5. Show that $G(1, n) \cong \mathbb{P}^{n-1}_k$ and that $G(n-1, n) \cong \mathbb{P}^{n-1}_k$, and check both against the dimension formula of theorem 41.2.4.
6. Show that a subset of $\mathbb{A}^1_k$ is constructible if and only if it is finite or cofinite. Deduce that every constructible subset of $\mathbb{A}^1_k$ is either open or closed, so the phenomenon of example 41.3.5 needs at least two dimensions.
7. Give a dominating regular map whose image is not open, and one whose image is not closed. Explain how theorem 41.3.4 accommodates both.
8. Repeat the dimension count of section 41.4 for surfaces of degree e in $\mathbb{P}^3_k$: show that the incidence variety has dimension $\binom{e+3}{3} - 1 + 3 - e$ and conclude that a general surface of degree $e \geq 4$ contains no line, while every quadric contains a one-parameter family.

Part IX

Curves

The fundamental theorem of algebra counts the roots of a polynomial in one variable, and does so exactly because $\mathbb{C}$ is algebraically closed. This Part asks for the two-variable analogue: given two plane curves, how many points do they share? Over $\mathbb{A}^2$ the question has no clean answer, since curves can miss each other or meet at infinity. In $\mathbb{P}^2$ it does, and the answer, Bézout's theorem, is that two curves of degrees m and n meet in exactly mn points.

Making that exact requires deciding what a point of intersection is worth, which is the intersection number of chapter 42. The counting then organizes itself into a group, the divisor class group of chapter 44, whose appearance on a cubic curve is the classical group law. The Part closes by removing the singularities that made the counting delicate in the first place.

42

Intersection Multiplicity

Two curves meeting at a point may cross transversally, be tangent, or meet at a singularity of one or both. Any count of intersections that is to be well behaved must weigh these cases differently. This chapter constructs the weight.

The route runs through the local ring at the point. The multiplicity of a curve at a point turns out to be readable off that ring alone, and at a nonsingular point the ring is a discrete valuation ring, so it carries an order function that already sees tangency. The intersection number of two curves is then the dimension of a single quotient of the local ring of the plane, and it is pinned down by eight properties which double as an algorithm for computing it.

42.1 Plane Curves, Multiplicity, and Tangent Lines

The fundamental theorem of algebra counts the roots of a one-variable polynomial over an algebraically closed field, and it can be read geometrically: a curve of degree n meets the line $x = 0$ in n points, provided the points are weighted correctly. This Part generalizes that reading to two curves, and the weight is what this chapter constructs.

Affine space is the wrong home for the statement. Take the two concentric circles $x^2 + y^2 - 1$ and $x^2 + y^2 - 4$ in $\mathbb{A}^2_{\mathbb{C}}$: both have degree 2, so the count should be 4, and yet they share no point at all. Passing to $\mathbb{P}^2_{\mathbb{C}}$ repairs it. The homogenizations $x^2 + y^2 - z^2$ and $x^2 + y^2 - 4z^2$ meet where $z = 0$ and $x^2 + y^2 = 0$, which is the pair of *circular points* $[1 : \pm i : 0]$ already met in example 38.3.4. Two points, not four, so the weights must be 2 each, and indeed the two circles are tangent to each other at both. The rest of this chapter builds the notion of weight that makes such counts come out, and chapter 43 proves they always do.

The development follows Fulton [54]. Throughout, k is algebraically closed.

A plane curve in the sense of this Part is not quite a variety. The point of the intersection theory is to count with multiplicity, and multiplicity is information that $\mathcal{Z}(f)$ throws away: $\mathcal{Z}(f)$ and $\mathcal{Z}(f^2)$ are the same set. So the object of study is the polynomial, taken up to scalar.

Definition 42.1.1: Plane Curve

An *affine plane curve* is an equivalence class of nonconstant polynomials in $k[x, y]$, where $f \sim g$ if and only if $f = \alpha g$ for some $\alpha \in k^*$. A *projective plane curve* is the same with $k[x, y]$ replaced by the nonconstant homogeneous polynomials of $k[x, y, z]$. The *degree* of a curve is the degree of any of its representatives.

This is a deliberate departure from definition 34.3.4, where a curve was a one-dimensional variety. The two agree on the underlying point set, since $\mathcal{Z}(f)$ for $f \in k[x, y]$ nonconstant is a hypersurface in $\mathbb{A}^2_k$ and so has dimension 1 by theorem 34.4.2; they differ in that the present notion remembers repeated factors. Homogenizing an affine plane curve and taking the closure in $\mathbb{P}^2_k$ gives a projective plane curve, as in section 38.3, and the two constructions are inverse away from the line at infinity.

Write $f = \prod_{i=1}^n f_i^{e_i}$ for the factorization into distinct irreducible factors, which is available because $k[x, y]$ and $k[x, y, z]$ are unique factorization domains. The curve f is *irreducible* if $n = 1$ and $e_1 = 1$; the component f_i is *simple* if $e_i = 1$ and *multiple* otherwise[1].

The local ring at a point is the algebraic object that stores intersection data, and it was already constructed in chapter 32.

Example 42.1.2: Curves Versus Their Zero Sets

1. Let $f = y - x^2$ and $g = (y - x^2)^2$. Then $\mathcal{Z}(f) = \mathcal{Z}(g)$ as subsets of $\mathbb{A}^2_k$, both being the parabola, yet f and g are different plane curves in the sense of definition 42.1.1: one has degree 2 and the other degree 4. Everything this Part counts distinguishes them. A general line meets $\mathcal{Z}(f)$ in 2 points and $\mathcal{Z}(g)$ in 4, once multiplicity is accounted for, which is what theorem 43.1.1 will record.

2. The multiplicity at a point behaves the same way. At the origin, $f = y - x^2$ has lowest form y, so $m_0(f) = 1$ and the origin is a nonsingular point; while $g = (y - x^2)^2$ has lowest form y^2, so $m_0(g) = 2$. The zero set is nonsingular as a set and the curve g is not.

3. *Irreducible components with multiplicity.* The curve $h = x^2 y = x \cdot x \cdot y$ has $\mathcal{Z}(h) = \mathcal{Z}(xy)$, the union of the two axes, but as a curve h has the y-axis as a *double* component and the x-axis as a simple one. The factorization $h = \prod f_i^{e_i}$ records $e = (2, 1)$, and proposition 42.1.7 will read $m_0(h) = 2 \cdot 1 + 1 \cdot 1 = 3$ off it.

Keeping the polynomial rather than its zero set is exactly what lets these distinctions be made, and it is the reason chapter 49 points at schemes, where the same information is carried intrinsically.

[1] In [5] schemes recover this information intrinsically: the scheme $\operatorname{Spec} k[x, y]/(f^2)$ is not the scheme $\operatorname{Spec} k[x, y]/(f)$, even though the two have the same points. Working with polynomials up to scalar is the classical workaround.

Definition 42.1.3: Recall: Stalk At A Point

Let V be a variety and $p \in V$ a point. The *stalk* at p is

$$\mathcal{O}_p(V) = \{f/g \in k(V) \mid g(p) \neq 0\}$$

By proposition 33.2.10 it is a Noetherian local domain, with maximal ideal $\mathfrak{m}_p(V) = \{f/g \in \mathcal{O}_p(V) \mid f(p) = 0\}$.

The intuition is that $\mathcal{O}_p(V)$ holds exactly the rational functions defined at p, which is why a denominator is allowed as long as it does not vanish there. For a curve f, write $\mathcal{O}_p(f)$ for the stalk of $\mathcal{Z}(f)$, and $\mathfrak{m}_p(f)$ for its maximal ideal.

Almost every computation below is easier at the origin, and moving a point there costs nothing.

Definition 42.1.4: Linear Coordinate Change

Let $\mathbb{A}^n_k$ be affine space with coordinate ring $k[x_1, \ldots, x_n]$. A bijective affine transformation $T \colon \mathbb{A}^n_k \to \mathbb{A}^n_k$ is called a *linear change of coordinates.*

If $T = (T_1, \ldots, T_n)$ then T acts on functions by $f(x_1, \ldots, x_n) \mapsto f(T_1, \ldots, T_n)$, and any such T decomposes into a translation followed by a bijective linear transformation. The first invariant to define this way is the multiplicity, obtained by translating the point of interest to the origin and reading off the lowest-degree term.

Definition 42.1.5: Multiplicity

Let $f \in k[x, y]$ be an affine plane curve and $p \in \mathcal{Z}(f)$. After a translation taking p to the origin, write

$$f = f_m + f_{m+1} + \cdots + f_d$$

for the decomposition into homogeneous components, where $f_m \neq 0$. The *multiplicity* of f at p, written $m_p(f)$, is m.

The definition does not depend on the translation chosen, since a translation carrying the origin to itself is the identity, and $m_p(f) \geq 1$ exactly because $f(p) = 0$. The multiplicity measures how badly the curve degenerates at p: it is 1 precisely when p is a nonsingular point of f in the sense of chapter 35, because f_1 is the differential of f at the origin and $f_1 \neq 0$ is the condition that some partial derivative survive.

The lowest form f_m carries more than its degree. Over an algebraically closed field a nonzero binary form factors completely into linear factors, so

$$f_m = \prod_{i=1}^{s} L_i^{r_i} \qquad \sum_i r_i = m$$

with the L_i pairwise non-proportional linear forms.

Definition 42.1.6: Tangent Lines of a Plane Curve

With notation as above, the lines $\mathcal{Z}(L_i)$ are the *tangent lines* to f at p, and r_i is the multiplicity of $\mathcal{Z}(L_i)$ as a tangent. A line through p that is not one of the $\mathcal{Z}(L_i)$ is assigned tangent multiplicity 0. The form f_m itself is the *tangent cone*.
The point p is an *ordinary multiple point* if f has $m = m_p(f)$ distinct tangent lines there, that is if every $r_i = 1$. An ordinary double point is a *node*.

This is the plane-curve case of definition 36.1.2: the initial form f_m generates the initial ideal of (f), so the tangent cone algebra is $k[x, y]/(f_m)$ and its zero set is the union of the tangent lines, while $m_p(f)$ agrees with the multiplicity of definition 36.1.4 by proposition 36.1.5. What the plane case adds is the factorization: a binary form splits into linear factors over the algebraically closed k, so the cone is literally a union of lines counted with multiplicity, which is false for a cone in three or more variables.

At a nonsingular point $m = 1$, so there is a single tangent line $\mathcal{Z}(f_1)$, and f_1 is the linear part of f. That recovers the familiar formula: for $p = (a, b)$ a nonsingular point of f, the tangent line is

$$\frac{\partial f}{\partial x}(p)\,(x - a) + \frac{\partial f}{\partial y}(p)\,(y - b) = 0$$

which is the definition of chapter 35 specialized to a plane curve. For a projective plane curve $F \in k[x, y, z]$ the point p is nonsingular when the three partials $\partial_x F, \partial_y F, \partial_z F$ do not all vanish at p, and the tangent line is then $\mathcal{Z}\big(\sum_i \partial_i F(p)\, x_i\big)$; passing to an affine chart recovers the two-variable statement.

Multiplicities and tangents both behave multiplicatively across the factorization of f.

Proposition 42.1.7: Multiplicity Across Components

Let $f = \prod_i f_i^{e_i}$ be an affine plane curve, factored into distinct irreducibles, and let $p \in \mathcal{Z}(f)$. Then

$$m_p(f) = \sum_i e_i\, m_p(f_i)$$

and a line L tangent to f_i with multiplicity r_i is tangent to f with multiplicity $\sum_i e_i r_i$. Consequently p is a nonsingular point of f if and only if it lies on exactly one f_i, that component is simple, and p is a nonsingular point of it.

***Proof*:**
Translate p to the origin. In the graded domain $k[x, y]$ the lowest form of a product is the product of the lowest forms, so the lowest form of f is $\prod_i (f_i)_{m_i}^{e_i}$ where $m_i = m_p(f_i)$; taking degrees gives the first claim, and reading off the linear factors gives the second.

For the last claim, $m_p(f) = 1$ forces exactly one term of $\sum_i e_i m_p(f_i)$ to be nonzero and equal to 1, so exactly one f_i passes through p, its exponent e_i is 1, and $m_p(f_i) = 1$. The converse is the same computation read backwards, as we sought to show.

42.2 Local Rings of Plane Curves

The multiplicity was defined by writing down a polynomial and looking at its lowest term, which is a coordinate-dependent recipe even if the answer is not. This section shows that $m_p(f)$ is visible inside the local ring $\mathcal{O}_p(f)$ alone, and that the nonsingular points are exactly those where that ring is as simple as a one-dimensional local ring can be.

Theorem 42.2.1: Multiplicity and Local Ring

Let f be an irreducible affine plane curve and $p \in \mathcal{Z}(f)$. Write $\mathcal{O} = \mathcal{O}_p(f)$ and $\mathfrak{m}$ for its maximal ideal. Then for all sufficiently large n,

$$m_p(f) = \dim_k \left(\frac{\mathfrak{m}^n}{\mathfrak{m}^{n+1}} \right)$$

so the multiplicity of f at p is determined by the local ring $\mathcal{O}_p(f)$.

***Proof*:**
The quotient maps $\mathcal{O}/\mathfrak{m}^{n+1} \to \mathcal{O}/\mathfrak{m}^n$ have kernel $\mathfrak{m}^n/\mathfrak{m}^{n+1}$, giving a short exact sequence of k-vector spaces

$$0 \to \mathfrak{m}^n/\mathfrak{m}^{n+1} \to \mathcal{O}/\mathfrak{m}^{n+1} \to \mathcal{O}/\mathfrak{m}^n \to 0$$

Dimension is additive along a short exact sequence of finite-dimensional vector spaces, so it suffices to prove

$$\dim_k (\mathcal{O}/\mathfrak{m}^n) = n\, m_p(f) + s$$

for a constant s independent of n and all $n \geq m_p(f)$; subtracting consecutive values then leaves $m_p(f)$.

Translate so that $p = (0,0)$ and write $I = (x, y) \subseteq k[x, y]$, so that $\mathcal{I}(\{p\}) = I$. Since $\mathcal{Z}(I^n) = \{p\}$, corollary 42.2.7 and the third exercise of exercise set 42.4.1 give

$$\frac{k[x,y]}{(I^n, f)} \cong \frac{\mathcal{O}_p(\mathbb{A}^2_k)}{(I^n, f)\mathcal{O}_p(\mathbb{A}^2_k)} \cong \frac{\mathcal{O}_p(f)}{I^n\mathcal{O}_p(f)} \cong \mathcal{O}/\mathfrak{m}^n$$

the last isomorphism because $\mathfrak{m}$ is generated by the images of x and y. So the problem is to compute $\dim_k k[x, y]/(I^n, f)$.

Let $m = m_p(f)$. Multiplication by f gives a map

$$\psi\colon \frac{k[x,y]}{I^{n-m}} \to \frac{k[x,y]}{I^n} \qquad \psi(\overline{g}) = \overline{fg}$$

which is well defined because $f \in I^m$, and injective: $k[x, y]$ is a graded domain, so the lowest form of fg is the product of those of f and g, whence $fg \in I^n$ forces $g \in I^{n-m}$. Its image is $(I^n, f)/I^n$, the kernel of the projection $\varphi\colon k[x, y]/I^n \to k[x, y]/(I^n, f)$, so

$$0 \to \frac{k[x,y]}{I^{n-m}} \xrightarrow{\psi} \frac{k[x,y]}{I^n} \xrightarrow{\varphi} \frac{k[x,y]}{(I^n, f)} \to 0$$

is exact. Now $k[x, y]/I^t$ has as a basis the monomials of degree less than t, of which there are $1 + 2 + \cdots + t = \frac{t(t+1)}{2}$, as the second exercise of exercise set 42.4.1 records. Additivity therefore gives

$$\dim_k (\mathcal{O}/\mathfrak{m}^n) = \frac{n(n+1)}{2} - \frac{(n-m)(n-m+1)}{2} = nm - \frac{m(m-1)}{2}$$

valid for all $n \geq m$. This has the required shape with $s = -\frac{m(m-1)}{2}$, a constant, as we sought to show.

Example 42.2.2: Multiplicity from the Local Ring

Take p the origin and compute $\dim_k \mathfrak{m}^n/\mathfrak{m}^{n+1}$ in each case, which theorem 42.2.1 says stabilizes at $m_p(f)$.

1. *A nonsingular point.* $f = y - x^2$. Here $\mathcal{O}_p(f) \cong k[x]_{(x)}$, since $y = x^2$ on the curve, and $\mathfrak{m} = (x)$. So $\mathfrak{m}^n/\mathfrak{m}^{n+1}$ is spanned by x^n alone and has dimension 1 for every n. That matches $m_p(f) = 1$.
2. *The node.* $f = y^2 - x^2 - x^3$, with $m_p(f) = 2$. The formula of the proof gives $\dim_k \mathcal{O}/\mathfrak{m}^n = 2n - \binom{2}{2} = 2n - 1$ for $n \geq 2$, so $\dim_k \mathfrak{m}^n/\mathfrak{m}^{n+1} = (2n+1) - (2n-1) = 2$ for $n \geq 2$, matching $m_p(f) = 2$.
3. *The cusp.* $f = y^2 - x^3$, again with $m_p(f) = 2$ since the lowest form is y^2. The same computation gives 2 again.

The node and the cusp have the same multiplicity and the same sequence of dimensions, so the local ring sees the multiplicity and not the tangent cone. What separates them is the number of tangent lines, which is definition 42.1.6, or equivalently the number of branches, which example 47.3.3 separates by blowing up.

The theorem already separates the nonsingular points from the rest, since they are the ones where the successive quotients have dimension 1. The next result says what that means ring-theoretically.

Corollary 42.2.3: Simple Points and Multiplicity

Let f be an irreducible affine plane curve and $p \in \mathcal{Z}(f)$. Then p is a nonsingular point of f if and only if $\mathcal{O}_p(f)$ is a discrete valuation ring. In that case, if L is a line through p that is not tangent to f at p, the image of its defining linear form generates $\mathfrak{m}_p(f)$.

***Proof*:**

Suppose p is nonsingular and L is a line through p that is not tangent there. Change coordinates so that $p = (0,0)$, the tangent line is $\mathcal{Z}(y)$, and $L = \mathcal{Z}(x)$. It suffices to show that $\mathfrak{m} = \mathfrak{m}_p(f)$ is generated by the image of x, since $\mathfrak{m}$ is generated by the images of x and y.

Because $\mathcal{Z}(y)$ is the tangent line, the lowest form of f is a nonzero multiple of y, and after scaling f we may take $f = y + (\text{terms of degree} \geq 2)$. Collect the terms containing y: what remains involves x alone and has degree at least 2, since the only degree-one term of f is y. Hence

$$f = yg - x^2h \qquad g = 1 + (\text{terms of positive degree}), \quad h \in k[x,y]$$

In $k[\mathcal{Z}(f)]$ this reads $yg = x^2h$, and $g(p) = 1 \neq 0$ makes g a unit in $\mathcal{O}_p(f)$, so

$$y = x^2hg^{-1} \in (x)$$

Therefore $\mathfrak{m} = (x)$ is principal. By proposition 33.2.10 the ring $\mathcal{O}_p(f)$ is a Noetherian local domain, and it has Krull dimension 1 by proposition 34.3.2, since $\mathcal{Z}(f)$ is a curve and $\{p\}$ has codimension 1 in it. So proposition 35.5.2 applies, and its implication (4) $\Rightarrow$ (2) makes $\mathcal{O}_p(f)$ a discrete valuation ring.

Conversely, if $\mathcal{O}_p(f)$ is a discrete valuation ring then $\mathfrak{m}/\mathfrak{m}^2$ is one-dimensional, generated by the image of a uniformizer, and the same holds for every $\mathfrak{m}^n/\mathfrak{m}^{n+1}$. By theorem 42.2.1 this common dimension is $m_p(f)$, so $m_p(f) = 1$ and p is nonsingular, completing the proof.

Now that $\mathcal{O}_p(f)$ is known to be a discrete valuation ring at a nonsingular point, its valuation can be named. This is the invariant that chapter 43 uses to read off intersection numbers.

Definition 42.2.4: Order Function at a Simple Point

Let f be an irreducible plane curve and p a nonsingular point of f, so that $\mathcal{O}_p(f)$ is a discrete valuation ring by corollary 42.2.3. The *order function* at p

$$\operatorname{ord}_p^f \colon \mathcal{O}_p(f) \setminus \{0\} \to \mathbb{N}$$

is the discrete valuation of $\mathcal{O}_p(f)$: writing t for a uniformizing parameter, $\operatorname{ord}_p^f(g)$ is the unique m with $g = ut^m$ for a unit $u \in \mathcal{O}_p(f)$. By convention $\operatorname{ord}_p^f(0) = \infty$. The definition extends to $k(f)$ by $\operatorname{ord}_p^f(g/h) = \operatorname{ord}_p^f(g) - \operatorname{ord}_p^f(h)$, taking values in $\mathbb{Z}$.

Geometrically $\operatorname{ord}_p^f(g)$ counts how many times g vanishes at p along f, and the computation just made says that the order function sees tangency. A line L meeting f transversally at p has $\operatorname{ord}_p^f(L) = 1$, because its linear form generates $\mathfrak{m}$. The tangent line has order at least 2: in the coordinates of the proof it is $\mathcal{Z}(y)$, and

$$\operatorname{ord}_p^f(y) = \operatorname{ord}_p^f(x^2hg^{-1}) = 2\operatorname{ord}_p^f(x) + \operatorname{ord}_p^f(hg^{-1}) \geq 2$$

since g^{-1} is a unit and $h \in \mathcal{O}_p(f)$.

The last ingredient of the chapter is a decomposition for the coordinate ring of a finite set of points. It says that such a ring sees each of its points separately, which is what will let a global intersection count be assembled from local ones.

Lemma 42.2.5: Coordinate Ring For Finite Points

Let $I \subseteq k[x_1, \ldots, x_n]$ be an ideal with $\mathcal{Z}(I) = \{p_1, \ldots, p_m\}$ finite, and write $\mathcal{O}_i = \mathcal{O}_{p_i}(\mathbb{A}_k^n)$. Then

$$\frac{k[x_1, \ldots, x_n]}{I} \cong \prod_{i=1}^{m} \frac{\mathcal{O}_i}{I\mathcal{O}_i}$$

and in particular the left side is a finite-dimensional k-vector space.

***Proof*:**
Put $R = k[x_1, \ldots, x_n]/I$. Its maximal ideals correspond by the Nullstellensatz (theorem 32.4.4) to the points of $\mathcal{Z}(I)$, so there are m of them and every prime of R is maximal, that is $\operatorname{kdim} R = 0$. As R is a finitely generated k-algebra it is Noetherian, so corollary 28.1.9 makes R Artinian, and corollary 28.1.10 makes it finite-dimensional over k.

Now theorem 28.1.6 decomposes R as a finite product of local Artinian rings, one for each maximal ideal, the factors being the localizations $R_{\mathfrak{m}_i}$. It remains only to identify $R_{\mathfrak{m}_i}$ geometrically: localizing $k[x_1, \ldots, x_n]/I$ at the maximal ideal of p_i is the same as localizing $k[x_1, \ldots, x_n]$ there and then dividing by the extended ideal, since localization is exact, and that is $\mathcal{O}_i/I\mathcal{O}_i$, as we sought to show.

This is a case where the algebra was done once upstream and should not be done again. The classical proof constructs idempotents e_i by hand and verifies the Chinese-remainder decomposition directly; the Artinian structure theorem is that construction in general form, and the only genuinely geometric step is the Nullstellensatz translation at the start.

Corollary 42.2.6: Dimension of Coordinate Ring of Finite Points

With notation as above,

$$\dim_k \frac{k[x_1, \ldots, x_n]}{I} = \sum_{i=1}^{m} \dim_k (\mathcal{O}_i/I\mathcal{O}_i)$$

***Proof*:**
Dimension is additive over a finite direct product of vector spaces, so this is lemma 42.2.5 read on dimensions, completing the proof.

Corollary 42.2.7: Coordinate Ring at a Single Point

Let $I \subseteq k[x_1, \ldots, x_n]$ satisfy $\mathcal{Z}(I) = \{p\}$. Then

$$\frac{k[x_1, \ldots, x_n]}{I} \cong \frac{\mathcal{O}_p(\mathbb{A}_k^n)}{I\,\mathcal{O}_p(\mathbb{A}_k^n)}$$

***Proof*:**
This is lemma 42.2.5 with $m = 1$, completing the proof.

One further consequence of the local rings being discrete valuation rings deserves recording here, since chapter 45 needs it: on a smooth curve, a rational map to a projective variety has no points of indeterminacy at all.

Lemma 42.2.8: Rational Maps on Smooth Curves

Let C be an irreducible curve, $V \subseteq \mathbb{P}_k^N$ a *projective* variety, and $\varphi\colon C \dashrightarrow V$ a rational map. If $p \in C$ is a nonsingular point, then φ is regular at p. If C is smooth, then φ is a regular morphism.

***Proof*:**
Since p is a nonsingular point of the curve C, the local ring $\mathcal{O}_p(C)$ is a Noetherian local domain of Krull dimension 1 which is regular, hence a discrete valuation ring by proposition 35.5.2; write ord_p for its valuation and t for a uniformizer. Represent φ near p as $[f_0 : \cdots : f_N]$ with

$f_i \in k(C)$ not all zero, and set

$$e = \min_{0 \le i \le N} \operatorname{ord}_p(f_i)$$

Then $\operatorname{ord}_p(t^{-e} f_i) \ge 0$ for every i, so each $t^{-e} f_i$ lies in $\mathcal{O}_p(C)$ and is defined at p, while $\operatorname{ord}_p(t^{-e} f_j) = 0$ for the index j achieving the minimum, so $t^{-e} f_j(p) \neq 0$ and the tuple is not identically zero at p. Since

$$[t^{-e} f_0 : \cdots : t^{-e} f_N] = [f_0 : \cdots : f_N]$$

as maps to $\mathbb{P}^N_k$, this exhibits φ as regular at p. The image lands in V because V is closed and $\varphi(C)$ is dense in it. If every point of C is nonsingular the argument applies at each, so φ is regular everywhere, as we sought to show.

Projectivity of the target is essential: the rational map $\mathbb{A}^1_k \dashrightarrow \mathbb{A}^1_k$ given by $t \mapsto 1/t$ is undefined at the origin and stays undefined, whereas the same formula into $\mathbb{P}^1_k$ extends by sending 0 to the point at infinity.

42.3 The Intersection Number

Let $f, g \in k[x, y]$ be affine plane curves and $p \in \mathbb{A}^2_k$. The goal is a number $I(p, f \cap g)$ counting how many times the two curves meet at p. Several plausible definitions present themselves, and rather than choose one and hope, the strategy is to write down the properties such a number must have and show that they pin it down uniquely. The list below is Fulton's [54, section 3.3]; for a complementary treatment leaning on complex analysis and on the local picture introduced in chapter 25, see [1].

The list is deliberately not minimal. Computations rarely need its full strength, but proving that the intersection number satisfies all of it is what makes the number usable.

Say that f and g *intersect properly* at p if they have no common component passing through p. The properties are:

1. $I(p, f \cap g)$ is a non-negative integer whenever f and g intersect properly at p, and $I(p, f \cap g) = \infty$ otherwise.
2. $I(p, f \cap g) = 0$ if and only if $p \notin \mathcal{Z}(f) \cap \mathcal{Z}(g)$. In particular $I(p, f \cap g) = 0$ whenever f or g is a nonzero constant.
3. $I(p, f \cap g)$ depends only on the components of f and of g that pass through p.
4. $I(p, f \cap g) = I(p, g \cap f)$.
5. $I(p, f \cap g)$ is invariant under a linear change of coordinates: if T is one and $q = T^{-1}(p)$, then $I(q, f^T \cap g^T) = I(p, f \cap g)$.
6. $I(p, f \cap g) \ge m_p(f)\, m_p(g)$, with equality if and only if f and g have no tangent line in common at p. In particular two curves meeting transversally at a point nonsingular on both have intersection number 1.
7. If $f = \prod_i f_i^{r_i}$ and $g = \prod_j g_j^{s_j}$, then

$$I(p, f \cap g) = \sum_{i,j} r_i s_j\, I(p, f_i \cap g_j)$$

8. $I(p, f \cap g) = I(p, f \cap (g + hf))$ for every $h \in k[x, y]$.

Property (8) is the one that does the work: it says the intersection number sees the ideal (f, g) and not the particular generators, which is exactly the licence needed to run a Euclidean algorithm on the pair.

Theorem 42.3.1: Intersection Number

There is exactly one function $I(p, f \cap g)$ of a point and a pair of affine plane curves satisfying properties (1) through (8), namely

$$I(p, f \cap g) = \dim_k \left(\frac{\mathcal{O}_p(\mathbb{A}_k^2)}{(f, g)} \right)$$

***Proof*:**
Uniqueness. Suppose $I(p, f \cap g)$ satisfies the eight properties. The properties then determine its value, by an algorithm. By (1) it is enough to treat curves intersecting properly at p, and by (5) we may translate so that $p = (0, 0)$. If p does not lie on both curves, (2) gives the value 0, so assume $0 < I(p, f \cap g) < \infty$ and argue by strong induction: suppose the value is known whenever it is smaller than n, and let $I(p, f \cap g) = n$.

Consider the one-variable restrictions $f(x, 0)$ and $g(x, 0)$, of degrees r and s respectively, with the convention that the zero polynomial has degree $-\infty$. By (4) we may assume $r \leq s$.

1. *Case* $f(x, 0) \equiv 0$. Then $y \big| f$, say $f = yh$, and (7) gives

$$I(p, f \cap g) = I(p, y \cap g) + I(p, h \cap g)$$

   Write $g(x, 0) = x^\ell(a_0 + a_1 x + \cdots)$ with $a_0 \neq 0$; here $\ell > 0$ because $g(0, 0) = 0$. Since $g(x, y) - g(x, 0)$ vanishes on the line $y = 0$, it is divisible by y, so property (8) applied to the pair (y, g) replaces g by $g(x, 0)$ without changing the value. Then (7) splits off the factor $a_0 + a_1 x + \cdots$, which does not vanish at p and so contributes nothing by (2):

$$I(p, y \cap g) = I(p, y \cap g(x, 0)) = I(p, y \cap x^\ell) = \ell\, I(p, y \cap x) = \ell$$

   the last step by (6), the two lines meeting transversally. Since $\ell > 0$, the remaining term $I(p, h \cap g)$ is strictly less than n and the induction applies.

2. *Case* $r \geq 0$. Scale so that $f(x, 0)$ and $g(x, 0)$ are monic and put $h = g - x^{s-r} f$. The leading terms cancel, so $\deg h(x, 0) = t < s$, while (8) gives

$$I(p, f \cap g) = I(p, f \cap h)$$

   Repeat on the pair (f, h), interchanging the two by (4) whenever $t < r$. Each step replaces the degree pair (r, s) by (r, t) with $t < s$, so the sum of the two degrees strictly decreases. It cannot decrease forever, so after finitely many steps one of the two restrictions is identically zero and the previous case applies.

So the eight properties leave no freedom, and at most one such function exists.

Existence. It remains to check that $\dim_k \mathcal{O}/(f, g)$, where $\mathcal{O} = \mathcal{O}_p(\mathbb{A}_k^2)$, has the eight properties. Properties (4), (5) and (8) are immediate, since the quantity depends only on the ideal $(f, g)\mathcal{O}$, which is symmetric in f and g, unchanged by $g \mapsto g + hf$, and carried to the corresponding ideal by a linear change of coordinates. Property (3) holds because a component of f not passing through p is a unit in $\mathcal{O}$ and so contributes nothing to the ideal. Property (2) holds because $\mathcal{O}/(f, g)$ is the zero ring exactly when $(f, g)\mathcal{O} = \mathcal{O}$, that is when some f or g fails to vanish at p. Translating by (5), assume from now on that $p = (0, 0)$ and every component of f and g passes through p. What is left is (1), (6) and (7).

Property (1). If f and g have no common component then $\mathcal{Z}(f, g)$ is finite, so corollary 42.2.6 makes $\dim_k \mathcal{O}/(f, g)$ finite. If instead h is a common component, then $(f, g) \subseteq (h)$ and there is a surjection

$$\frac{\mathcal{O}}{(f, g)} \twoheadrightarrow \frac{\mathcal{O}}{(h)}$$

Now $\mathcal{O}/(h) \cong \mathcal{O}_p(\mathcal{Z}(h))$ contains $k[\mathcal{Z}(h)]$, which is infinite-dimensional over k because $\mathcal{Z}(h)$ is a curve. So the left side is infinite-dimensional too.

Property (7). By induction and symmetry it is enough to prove

$$I(p, f \cap gh) = I(p, f \cap g) + I(p, f \cap h)$$

By (1) we may assume f and gh have no common component. Let $\varphi\colon \mathcal{O}/(f, gh) \to \mathcal{O}/(f, g)$ be the natural surjection and let $\psi\colon \mathcal{O}/(f, h) \to \mathcal{O}/(f, gh)$ be the k-linear map $\psi(\overline{z}) = \overline{gz}$. Since dimension is additive along a short exact sequence, it suffices to show that

$$0 \to \frac{\mathcal{O}}{(f, h)} \xrightarrow{\psi} \frac{\mathcal{O}}{(f, gh)} \xrightarrow{\varphi} \frac{\mathcal{O}}{(f, g)} \to 0$$

is exact. Surjectivity of φ is clear, and $\mathrm{im}\psi = \ker\varphi$ because $\ker\varphi = (f, g)/(f, gh)$ is generated by the images of f and g, the first being zero. For injectivity of ψ, suppose $\overline{gz} = \overline{0}$, so that $gz = uf + vgh$ with $u, v \in \mathcal{O}$. Choose $s \in k[x, y]$ with $s(p) \neq 0$ clearing all denominators, and set $A = su$, $B = sv$, $C = sz$, all in $k[x, y]$. Then $gC = Af + Bgh$, that is

$$g(C - Bh) = Af$$

Since f and g have no common factor and $k[x, y]$ is a unique factorization domain, f divides $C - Bh$, say $C - Bh = Df$. Dividing by s, which is a unit in $\mathcal{O}$,

$$z = \frac{B}{s}h + \frac{D}{s}f \in (f, h)\mathcal{O}$$

so $\overline{z} = \overline{0}$ and ψ is injective.

Property (6). Let $m = m_p(f)$, $n = m_p(g)$ and $I = (x, y) \subseteq k[x, y]$. Consider

$$\begin{tikzcd}
\frac{k[x,y]}{I^n} \times \frac{k[x,y]}{I^m} \arrow[r, "\psi"] & \frac{k[x,y]}{I^{n+m}} \arrow[r, "\varphi"] & \frac{k[x,y]}{(I^{n+m}, f, g)} \arrow[r] \arrow[d, "\alpha"] & 0 \\
& \frac{\mathcal{O}}{(f,g)} \arrow[r, "\pi"] & \frac{\mathcal{O}}{(I^{n+m}, f, g)} \arrow[r] & 0
\end{tikzcd}$$

where φ, π, α are the natural maps and $\psi(\overline{a}, \overline{b}) = \overline{af + bg}$, which is well defined because $f \in I^m$ and $g \in I^n$. The top row is exact at the middle and right, though ψ need not be injective, and α is an isomorphism by corollary 42.2.7, since $\mathcal{Z}(I^{n+m}, f, g) \subseteq \{p\}$. Exactness gives

$$\dim_k \frac{k[x, y]}{(I^{n+m}, f, g)} = \dim_k \frac{k[x, y]}{I^{n+m}} - \dim_k \ker\varphi \qquad \dim_k \ker\varphi \leq \dim_k \frac{k[x, y]}{I^n} + \dim_k \frac{k[x, y]}{I^m}$$

with equality in the second exactly when ψ is injective. Since π is surjective, chaining these and using $\dim_k k[x,y]/I^t = \frac{t(t+1)}{2}$ gives

$$\begin{aligned} I(p, f \cap g) &= \dim_k \frac{\mathcal{O}}{(f,g)} \\ &\geq \dim_k \frac{\mathcal{O}}{(I^{n+m}, f, g)} \\ &= \dim_k \frac{k[x,y]}{(I^{n+m}, f, g)} \\ &\geq \dim_k \frac{k[x,y]}{I^{n+m}} - \dim_k \frac{k[x,y]}{I^n} - \dim_k \frac{k[x,y]}{I^m} \\ &= \frac{(n+m)(n+m+1)}{2} - \frac{n(n+1)}{2} - \frac{m(m+1)}{2} = mn \end{aligned}$$

which is the inequality. Equality holds exactly when both steps are equalities, that is when π is an isomorphism, equivalently $I^{n+m} \subseteq (f,g)\mathcal{O}$, and when ψ is injective. That both hold precisely when f and g have no common tangent is lemma 42.3.2, which completes the proof.

The lemma the last step needs is where the tangent lines finally enter the argument, and it is worth isolating because both of its parts are used again in chapter 43.

Lemma 42.3.2: Technical Lemma - Intersection and Tangents

Let f, g be affine plane curves meeting properly at $p = (0,0)$, with $m = m_p(f)$, $n = m_p(g)$, and let $I = (x,y)$. Write ψ for the map of theorem 42.3.1. Then

1. If f and g have no common tangent line at p, then $I^t \subseteq (f,g)\mathcal{O}$ for every $t \geq m+n-1$.
2. ψ is injective if and only if f and g have no common tangent line at p.

***Proof*:**
Let $L_1, \ldots, L_m$ be the tangent lines of f at p, listed with multiplicity, and $M_1, \ldots, M_n$ those of g. Extend the lists by $L_i = L_m$ for $i > m$ and $M_j = M_n$ for $j > n$, and set

$$A_{ij} = L_1 L_2 \cdots L_i \, M_1 M_2 \cdots M_j$$

with $A_{00} = 1$, so A_{ij} is a form of degree $i + j$. The whole argument rests on the following claim.

Claim: if f and g have no common tangent, then $\{A_{ij} \mid i + j = t\}$ is a k-basis for the space of forms of degree t. There are $t+1$ of them and the space has dimension $t+1$, so it suffices to prove independence. Write $P_i = L_1 \cdots L_i$ and $Q_j = M_1 \cdots M_j$ and suppose $\sum_{i=0}^{t} c_i P_i Q_{t-i} = 0$ with not all c_i zero. Let a be the least and b the greatest index with $c_i \neq 0$. If $a = b$ the relation reads $c_a P_a Q_{t-a} = 0$, impossible in the domain $k[x,y]$. So $a < b$. Every P_i with $i \geq a$ is divisible by P_a, and every Q_{t-i} with $i \leq b$ is divisible by Q_{t-b}, so dividing the relation by $P_a Q_{t-b}$ leaves

$$\sum_{i=a}^{b} c_i \frac{P_i}{P_a} \cdot \frac{Q_{t-i}}{Q_{t-b}} = 0$$

Every term with $i > a$ is divisible by L_{a+1}. Hence L_{a+1} divides the $i = a$ term, which is $c_a \, Q_{t-a}/Q_{t-b}$, a nonzero scalar times a nonempty product of the M_j. But L_{a+1} is a tangent of

f and no M_j is proportional to it, contradicting unique factorization. This proves the claim, and shows exactly where the hypothesis is used.

Part (1). First, $I^t \subseteq (f,g)\mathcal{O}$ for all sufficiently large t. Write $\mathcal{Z}(f,g) = \{p, q_1, \ldots, q_s\}$, which is finite since f and g meet properly, and choose a polynomial h with $h(q_i) = 0$ for every i and $h(p) \neq 0$. Then xh and yh vanish on $\mathcal{Z}(f,g)$, so by the Nullstellensatz (theorem 32.4.4) there is N with $(xh)^N, (yh)^N \in (f,g)$. As h^N is a unit in $\mathcal{O}$, this gives $x^N, y^N \in (f,g)\mathcal{O}$, and every monomial of degree at least $2N$ is divisible by x^N or by y^N, so $I^{2N} \subseteq (f,g)\mathcal{O}$.

Now argue by descending induction on $i+j$ that $A_{ij} \in (f,g)\mathcal{O}$ whenever $i+j \geq m+n-1$, the case $i+j \geq 2N$ being covered by the previous paragraph together with the claim. Given $i+j \geq m+n-1$, one of $i \geq m$ or $j \geq n$ holds; say $i \geq m$, the other case being symmetric with g in place of f. Then $A_{ij} = A_{m0}B$ with $\deg B = i+j-m$. After scaling f so that its lowest form is exactly A_{m0}, write $f = A_{m0} + f'$ with every term of f' of degree at least $m+1$. Multiplying by B and rearranging,

$$A_{ij} = Bf - Bf'$$

Here $Bf \in (f,g)\mathcal{O}$, and every term of Bf' is a form of degree at least $i+j+1$, hence by the claim a combination of the $A_{i'j'}$ with $i'+j' \geq i+j+1$, which lie in $(f,g)\mathcal{O}$ by the inductive hypothesis. So $A_{ij} \in (f,g)\mathcal{O}$. Since the A_{ij} with $i+j=t$ span the forms of degree t, and I^t is spanned by forms of degree at least t, this gives $I^t \subseteq (f,g)\mathcal{O}$ for $t \geq m+n-1$.

Part (2). Suppose first that f and g have no common tangent, and let $\psi(\overline{A}, \overline{B}) = \overline{Af+Bg} = 0$, so that $Af + Bg \in I^{n+m}$. The domain of ψ is $\frac{k[x,y]}{I^n} \times \frac{k[x,y]}{I^m}$, so $\overline{A} \neq 0$ means $r < n$ and $\overline{B} \neq 0$ means $s < m$, where r and s are the orders of A and B, with lowest forms A_r and B_s. Assume for contradiction that $(\overline{A}, \overline{B}) \neq 0$ and consider the possibilities.

If $\overline{B} = 0$ then $B \in I^m$, so $Bg \in I^{m+n}$ and hence $Af \in I^{m+n}$. As $k[x,y]$ is a graded domain the order of Af is $r+m$, so $r+m \geq n+m$, giving $r \geq n$ and $\overline{A} = 0$, contrary to assumption. The case $\overline{A} = 0$ is symmetric.

So $\overline{A} \neq 0 \neq \overline{B}$, that is $r < n$ and $s < m$. The orders of Af and Bg are $r+m$ and $s+n$, both strictly less than $n+m$, so they must be equal, else $Af+Bg$ would have order less than $n+m$. Thus $r+m = s+n$ and the terms of that degree cancel:

$$A_r f_m = -B_s g_n$$

Now f_m and g_n have no common factor, since a common irreducible factor would be a linear form tangent to both curves. So $f_m \big| B_s$ and $g_n \big| A_r$, forcing $s \geq m$ and $r \geq n$ and contradicting $r < n$, $s < m$. Hence $(\overline{A}, \overline{B}) = 0$ and ψ is injective.

Conversely, suppose L is a common tangent and write $f_m = Lf'$, $g_n = Lg'$, so that $\deg f' = m-1$ and $\deg g' = n-1$. Then $\overline{g'} \neq 0$ in $k[x,y]/I^n$ and $\overline{f'} \neq 0$ in $k[x,y]/I^m$, and

$$\psi(\overline{g'}, \overline{-f'}) = \overline{g'f - f'g}$$

The two products $g'f$ and $f'g$ both have order $n+m-1$, and their forms of that degree are $g'f_m = g'Lf'$ and $f'g_n = f'Lg'$, which are equal. So they cancel, $g'f - f'g \in I^{n+m}$, and the image is zero on a nonzero element. Hence ψ is not injective, as we sought to show.

42.4 Computing Intersection Numbers

The proof of theorem 42.3.1 contains an algorithm, and in practice one rarely runs it to the end. Properties (7) and (8) do most of the work: (8) replaces a curve by any convenient representative of the same ideal, and (7) splits off factors. Property (6) then closes out any pair with no common tangent without further computation. The following example is worth working through slowly, because every step is one of those three.

Example 42.4.1: Intersection Number

Let

$$E = (x^2+y^2)^2 + 3x^2y - y^3 = x^4 + 2x^2y^2 + 3x^2y + y^4 - y^3$$
$$F = (x^2+y^2)^3 - 4x^2y^2 = x^6 + 3x^4y^2 + 3x^2y^4 - 4x^2y^2 + y^6$$

and $p = (0,0)$. Over the reals these are the three-leaf and four-leaf roses, $r = -\sin 3\theta$ and $r = |\sin 2\theta|$ in polar coordinates:

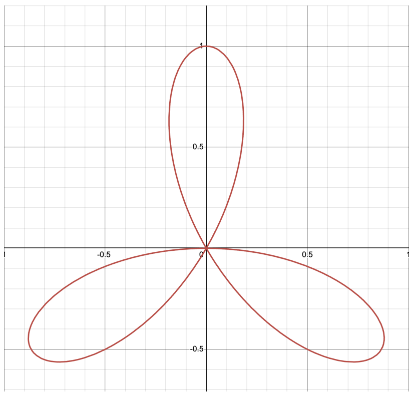


Figure 42.1: Three-Leaf Curve

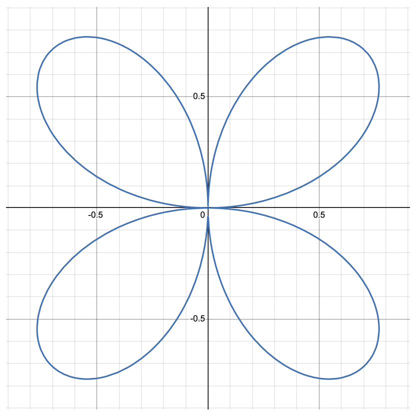


Figure 42.2: Four-Leaf Curve

Both pass through the origin, one with three branches and one with four, so the intersection number there should be substantial.

By (8) replace F with $F - (x^2+y^2)E$, which is

$$-4x^2y^2 - (x^2+y^2)(3x^2y - y^3) = y\big((x^2+y^2)(y^2-3x^2) - 4x^2y\big) = yG$$

so that by (7)

$$I(p, E\cap F) = I(p, E\cap yG) = I(p, E\cap y) + I(p, E\cap G)$$

For the second term, eliminate the quartic part of $G = -3x^4 - 2x^2y^2 - 4x^2y + y^4$ by replacing G with $G + 3E$, again legitimate by (8):

$$G + 3E = 4x^2y^2 + 5x^2y + 4y^4 - 3y^3 = y\big(5x^2 - 3y^2 + 4x^2y + 4y^3\big) = yH$$

Applying (7) once more,

$$I(p, E\cap F) = 2\,I(p, E\cap y) + I(p, E\cap H)$$

The first term is a one-variable computation: $E(x,0) = x^4$, so by (8), (7) and (6),

$$I(p, E \cap y) = I(p, x^4 \cap y) = 4\, I(p, x \cap y) = 4$$

For the second, read off the tangent cones. The lowest form of E is $3x^2y - y^3 = y(\sqrt{3}x - y)(\sqrt{3}x + y)$, so $m_p(E) = 3$ with three distinct tangents; the lowest form of H is $5x^2 - 3y^2 = (\sqrt{5}x - \sqrt{3}y)(\sqrt{5}x + \sqrt{3}y)$, so $m_p(H) = 2$ with two more. No tangent of E is proportional to one of H, so (6) applies with equality and no further computation is needed:

$$I(p, E \cap H) = m_p(E)\, m_p(H) = 3 \cdot 2 = 6$$

Altogether

$$I(p, E \cap F) = 2 \cdot 4 + 6 = 14$$

Two consequences of theorem 42.3.1 are recorded here as further properties, continuing the numbering of the list. The first identifies the intersection number with the order function of definition 42.2.4 whenever one of the curves is nonsingular at p, which is the form chapter 43 uses. The second globalizes: the intersection numbers at all points assemble into a single dimension count.

Corollary 42.4.2: Intersection Number, Order, and the Global Count

Let f, g be affine plane curves. Then

9. If p is a nonsingular point of f, then $I(p, f \cap g) = \operatorname{ord}_p^f(g)$.
10. If f and g have no common component, so that $\mathcal{Z}(f) \cap \mathcal{Z}(g)$ is finite, then

$$\sum_p I(p, f \cap g) = \dim_k \frac{k[x,y]}{(f,g)}$$

***Proof*:**
(9). Since p is a nonsingular point of f, it lies on exactly one component of f and that component is simple by proposition 42.1.7; by property (3) we may replace f by that component and assume f irreducible. Then $\mathcal{O}_p(f)$ is a discrete valuation ring by corollary 42.2.3; let t be a uniformizer and $e = \operatorname{ord}_p^f(\overline{g})$, so that $\overline{g} = ut^e$ for a unit u. Hence $(\overline{g}) = (t^e)$ and

$$\frac{\mathcal{O}_p(f)}{(\overline{g})} = \frac{\mathcal{O}_p(f)}{(t^e)}$$

has $1, t, \ldots, t^{e-1}$ as a k-basis, the residue field being k because k is algebraically closed, so its dimension is e. Finally $\mathcal{O}_p(f) \cong \mathcal{O}_p(\mathbb{A}^2_k)/(f)$ by the third exercise of exercise set 42.4.1, so

$$\frac{\mathcal{O}_p(f)}{(\overline{g})} \cong \frac{\mathcal{O}_p(\mathbb{A}^2_k)}{(f,g)}$$

whose dimension is $I(p, f \cap g)$ by theorem 42.3.1.

(10). With no common component, $\mathcal{Z}(f,g)$ is finite, so corollary 42.2.6 applied to $I = (f,g)$ gives

$$\dim_k \frac{k[x,y]}{(f,g)} = \sum_i \dim_k \frac{\mathcal{O}_{p_i}(\mathbb{A}^2_k)}{(f,g)\mathcal{O}_{p_i}(\mathbb{A}^2_k)} = \sum_i I(p_i, f \cap g)$$

and the points not in $\mathcal{Z}(f, g)$ contribute 0 by property (2), so the sum over all p is the same, completing the proof.

The second statement is the bridge to chapter 43: it converts a sum of local invariants into the dimension of a single ring, and Bézout's theorem is what happens when that ring is computed for two projective curves by a graded argument.

Exercise 42.4.1

1. Let $p = (0, \ldots, 0) \in \mathbb{A}^n_k$. Show that the maximal ideal $\mathfrak{m} \subseteq \mathcal{O}_p(\mathbb{A}^n_k)$ is generated by $x_1, \ldots, x_n$.

2. Let $I = (x, y) \subseteq k[x, y]$. Show that the monomials of degree less than n form a k-basis of $k[x, y]/I^n$, and deduce that
$$\dim_k \left(\frac{k[x, y]}{I^n} \right) = 1 + 2 + \cdots + n = \frac{n(n+1)}{2}$$

3. Let $V \subseteq \mathbb{A}^n_k$ be a variety with ideal $\mathcal{I}(V) \subseteq R = k[x_1, \ldots, x_n]$, let $p \in V$, and let $J \subseteq R$ be an ideal containing $\mathcal{I}(V)$, with image $\overline{J}$ in $k[V]$. Show that
$$\frac{\mathcal{O}_p(\mathbb{A}^n_k)}{J\, \mathcal{O}_p(\mathbb{A}^n_k)} \cong \frac{\mathcal{O}_p(V)}{\overline{J}\, \mathcal{O}_p(V)}$$
and deduce the case $J = \mathcal{I}(V)$, namely $\mathcal{O}_p(\mathbb{A}^n_k)/\mathcal{I}(V)\mathcal{O}_p(\mathbb{A}^n_k) \cong \mathcal{O}_p(V)$.

4. Compute $m_p(f)$ and the tangent lines at the origin for $f = y^2 - x^3$, $f = y^2 - x^2 - x^3$, $f = x^4 - x^2y^2 + y^4$ and $f = (x^2 + y^2)^2 + 3x^2y - y^3$. Which of the four have an ordinary singular point there?

5. Compute $I(p, f \cap g)$ at the origin for $f = y - x^2,\ g = y$; then for $f = y^2 - x^3,\ g = y$; then for $f = y^2 - x^3,\ g = x$. Check each against property (6).

6. Show that $I(p, f \cap g) = 1$ if and only if p is a nonsingular point of both curves and their tangent lines there are distinct.

7. Let f be an irreducible plane curve and p a nonsingular point of it. Show directly from definition 42.2.4 that $\operatorname{ord}^f_p(gh) = \operatorname{ord}^f_p(g) + \operatorname{ord}^f_p(h)$ and that $\operatorname{ord}^f_p(g + h) \geq \min(\operatorname{ord}^f_p(g), \operatorname{ord}^f_p(h))$, with an example where the second is strict.

43

Bézout's Theorem and Max Noether's Theorem

With the intersection number in hand the counting theorem is within reach. The proof is a dimension count in a graded ring: two curves with no common component generate an ideal whose quotient has dimension mn in every large degree, and the sum of the intersection numbers is exactly that dimension. Everything the theorem says about individual points is a consequence of that one budget being finite.

Max Noether's theorem then answers the converse question. Given that a curve passes through the points where two others meet, when is it an algebraic combination of them? The answer is local: a condition at each intersection point suffices, and checking it is cheap wherever the intersection is transversal. The chapter closes with two theorems of classical projective geometry, Pascal's and Pappus's, which fall out of a single application of that criterion to a cubic that splits into three lines.

43.1 Bézout's Theorem

Projective space was introduced so that intersections would stop disappearing, and for plane curves it delivers completely: two curves with no common component always meet, and the number of points, weighted by the intersection numbers of chapter 42, is exactly the product of the degrees.

The result is special to curves in $\mathbb{P}^2_k$, not a general feature of projective varieties. In $\mathbb{P}^3_k$ the point $\mathcal{Z}(x, y, w)$ and the plane $\mathcal{Z}(z)$ do not meet at all, and there is no weighting that would make them. What makes the plane-curve case work is that two curves in $\mathbb{P}^2_k$ have complementary dimension, so theorem 40.6.3 already guarantees they meet. Bézout's theorem sharpens that from "at least one point" to an exact count.

Theorem 43.1.1: Bézout's Theorem

Let f and g be projective plane curves of degrees m and n with no common component. Then $\mathcal{Z}(f) \cap \mathcal{Z}(g)$ is finite and

$$\sum_{p \in \mathbb{P}^2_k} I(p, f \cap g) = mn$$

***Proof*:**
Setting up. Since f and g have no common component, $\mathcal{Z}(f) \cap \mathcal{Z}(g)$ is a proper closed subset of the curve $\mathcal{Z}(f)$, hence has dimension 0 and is finite. The field k is infinite, so a linear change of coordinates moves the line at infinity off that finite set; assume from now on that no point of $\mathcal{Z}(f) \cap \mathcal{Z}(g)$ lies on $\mathcal{Z}(z)$. Write $f_* = f(x, y, 1)$ and $g_* = g(x, y, 1)$ for the dehomogenizations. Every intersection point now lies in the standard affine chart, where the intersection numbers agree with the affine ones, so property (10) of corollary 42.4.2 gives

$$\sum_p I(p, f \cap g) = \sum_p I(p, f_* \cap g_*) = \dim_k \frac{k[x, y]}{(f_*, g_*)}$$

Abbreviate

$$\Gamma_* = \frac{k[x, y]}{(f_*, g_*)} \qquad \Gamma = \frac{k[x, y, z]}{(f, g)} \qquad R = k[x, y, z]$$

Here R and Γ are graded, and R_d, Γ_d denote their degree-d pieces. The theorem follows from the two claims

$$\dim_k \Gamma_* \overset{1}{=} \dim_k \Gamma_d \overset{2}{=} mn \qquad (d \geq m + n)$$

Claim 2: $\dim_k \Gamma_d = mn$ *for* $d \geq m + n$*.* Define

$$\begin{aligned} \psi \colon R &\to R \times R & \psi(c) &= (gc, -fc) \\ \varphi \colon R \times R &\to R & \varphi(a, b) &= af + bg \\ \pi \colon R &\to \Gamma & \pi(h) &= \overline{h} \end{aligned}$$

The sequence

$$0 \to R \xrightarrow{\psi} R \times R \xrightarrow{\varphi} R \xrightarrow{\pi} \Gamma \to 0$$

is exact. Indeed ψ is injective because R is a domain; $\varphi\psi = 0$ by inspection; if $\varphi(a, b) = 0$ then $af = -bg$, and as f, g have no common factor in the unique factorization domain R we get $g \big| a$, say $a = gc$, whence $b = -fc$ and $(a, b) = \psi(c)$; finally $\operatorname{im}\varphi = (f, g) = \ker \pi$.

Everything is homogeneous, so the sequence restricts in each degree. Since $\deg f = m$ and $\deg g = n$,

$$0 \to R_{d-m-n} \xrightarrow{\psi} R_{d-m} \times R_{d-n} \xrightarrow{\varphi} R_d \xrightarrow{\pi} \Gamma_d \to 0$$

is exact, and for $d \geq m + n$ every term is a finite-dimensional space of forms, with $\dim_k R_j = \binom{j+2}{2} = \frac{(j+1)(j+2)}{2}$ for $j \geq 0$. Alternating dimensions along an exact sequence sum to zero, so

$$\dim_k \Gamma_d = \dim_k R_d - \dim_k R_{d-m} - \dim_k R_{d-n} + \dim_k R_{d-m-n}$$

Writing $P(j) = (j + 1)(j + 2)$, this is $\frac{1}{2}\big[P(d) - P(d - m) - P(d - n) + P(d - m - n)\big]$. The linear and constant parts of P cancel in that combination, leaving only the quadratic part:

$$d^2 - (d - m)^2 - (d - n)^2 + (d - m - n)^2 = 2mn$$

so $\dim_k \Gamma_d = mn$, which is Claim 2.

Multiplication by z is injective on Γ. This is the technical heart of Claim 1, and it is also used again in theorem 43.3.2. Let $\alpha\colon \Gamma \to \Gamma$ be $\alpha(\overline{h}) = \overline{zh}$ and suppose $\overline{zh} = \overline{0}$, so $zh = af + bg$ for some $a, b \in R$. For a form $u \in R$ write $u_0 = u(x, y, 0)$ for its restriction to the line at infinity. Because $\mathcal{Z}(f) \cap \mathcal{Z}(g) \cap \mathcal{Z}(z) = \emptyset$, the forms f_0 and g_0 have no common zero in $\mathbb{P}^1_k$ and so are relatively prime in $k[x, y]$. Setting $z = 0$ in $zh = af + bg$ gives

$$0 = a_0 f_0 + b_0 g_0$$

so $a_0 f_0 = -b_0 g_0$, and coprimality yields $c \in k[x, y]$ with

$$b_0 = f_0 c \qquad a_0 = -g_0 c$$

Now put $a_1 = a + cg$ and $b_1 = b - cf$. Then $a_1 f + b_1 g = af + bg + cgf - cfg = zh$, while

$$(a_1)_0 = -g_0 c + cg_0 = 0 \qquad (b_1)_0 = f_0 c - cf_0 = 0$$

so both are divisible by z, say $a_1 = za'$ and $b_1 = zb'$. Then $z(a'f + b'g) = zh$, and cancelling z in the domain R gives $h = a'f + b'g$, that is $\overline{h} = \overline{0}$. So α is injective.

Claim 1: $\dim_k \Gamma_* = \dim_k \Gamma_d$ *for* $d \geq m+n$. By the previous paragraph α restricts to an injection $\Gamma_d \hookrightarrow \Gamma_{d+1}$, and both have dimension mn by Claim 2, so it is an isomorphism. Consequently, if $\beta_1, \ldots, \beta_{mn} \in R_d$ have residues forming a basis of Γ_d, then for every $r \geq 0$ the residues of $z^r\beta_1, \ldots, z^r\beta_{mn}$ form a basis of Γ_{d+r}.

Let $b_i \in \Gamma_*$ be the residue of $(\beta_i)_* = \beta_i(x, y, 1)$. These span: given $\overline{h} \in \Gamma_*$ with $h \in k[x, y]$, let h^* be its homogenization, of some degree e, and choose r with $d + r \geq e$. Then $z^{d+r-e}h^*$ is a form of degree $d + r$, so

$$z^{d+r-e}h^* = \sum_{i=1}^{mn} \lambda_i z^r \beta_i + Af + Bg$$

for scalars λ_i and forms A, B. Dehomogenizing, that is setting $z = 1$, gives

$$h = \sum_i \lambda_i (\beta_i)_* + A_* f_* + B_* g_*$$

so $\overline{h} = \sum_i \lambda_i b_i$.

They are independent: suppose $\sum_i \lambda_i b_i = 0$, so that $\sum_i \lambda_i (\beta_i)_* = Af_* + Bg_*$ for some $A, B \in k[x, y]$. Homogenizing and clearing the powers of z that appear gives $r, s, t \geq 0$ with

$$z^r \sum_i \lambda_i \beta_i = z^s A^* f + z^t B^* g$$

so $\sum_i \lambda_i \overline{z^r \beta_i} = \overline{0}$ in Γ_{d+r}. As those residues form a basis, every $\lambda_i = 0$.

So $b_1, \ldots, b_{mn}$ is a basis of Γ_* and $\dim_k \Gamma_* = mn$, as we sought to show.

Example 43.1.2: Bézout in Practice
Every count below is over an algebraically closed field and in $\mathbb{P}^2_k$, where alone the statement holds.

1. *Two conics.* Degrees 2 and 2, so the intersection has total weight 4. For $\mathcal{Z}(x^2+y^2-z^2)$ and $\mathcal{Z}(x^2-yz)$ a direct substitution gives four distinct points, each counted once. Two general conics meet in four points.

2. *Tangency absorbs the count.* For the circle $\mathcal{Z}(x^2+y^2-z^2)$ and the line $\mathcal{Z}(y-z)$, substituting $y=z$ gives $x^2=0$, a double root: the line is tangent at $[0:1:1]$, and $I(p, C\cap L)=2$. The total is still $2\cdot 1=2$, all concentrated at one point.

3. *Where affine intuition fails.* The two parabolas $\mathcal{Z}(y-x^2)$ and $\mathcal{Z}(y-x^2-1)$ do not meet in $\mathbb{A}^2_k$ at all. Homogenizing gives $\mathcal{Z}(yz-x^2)$ and $\mathcal{Z}(yz-x^2-z^2)$, and subtracting, $z^2=0$: the two meet only where $z=0$, at $[0:1:0]$, with $I=4$. The four intersections predicted by $2\cdot 2$ are all at infinity, which is what projective space was introduced to see.

4. *A line and a cubic.* Degrees 1 and 3 give three points. For the cubic $\mathcal{Z}(y^2z-x^3+xz^2)$ and the line $\mathcal{Z}(z)$, substituting $z=0$ gives $x^3=0$: the line at infinity meets the cubic only at $[0:1:0]$, with multiplicity 3. That point is an inflection, which is why theorem 48.4.4 can put a genus-one curve in Weierstrass form with its base point there.

43.2 Consequences and Bounds on Multiplicity

Bézout's theorem is a budget: mn is all the intersection there is, and every local contribution is drawn against it. Most of its corollaries are that observation combined with the lower bound $I(p, f\cap g)\geq m_p(f)m_p(g)$ from property (6).

Corollary 43.2.1: Degree and Multiplicity

Let f, g be projective plane curves with no common component. Then

$$\sum_p m_p(f)\, m_p(g) \leq \deg(f)\deg(g)$$

***Proof*:**
Property (6) of the intersection number gives $I(p, f\cap g)\geq m_p(f)m_p(g)$ at every p, and summing over p against theorem 43.1.1 bounds the total by $\deg(f)\deg(g)$, completing the proof.

Corollary 43.2.2: Intersection and Simple Points

Let f, g be projective plane curves of degrees m and n with no common component. If $\mathcal{Z}(f)$ and $\mathcal{Z}(g)$ meet in mn distinct points, then every one of those points is a nonsingular point of both curves, and the two curves are transversal there.

***Proof*:**
The mn points each contribute at least 1 to a total of mn by theorem 43.1.1, so each contributes exactly 1. Property (6) then forces $m_p(f)m_p(g) \leq 1$, so both multiplicities are 1 and p is nonsingular on both by definition 42.1.5. Equality in (6) additionally says the curves share no tangent at p, which for two nonsingular points is transversality, completing the proof.

Corollary 43.2.3: Intersection and Common Component

If two projective plane curves of degrees m and n have more than mn points in common, they have a common component.

***Proof*:**
Were there no common component, theorem 43.1.1 would bound the number of common points by $\sum_p I(p, f \cap g) = mn$, since each contributes at least 1 by property (2), completing the proof.

The next bound goes the other way: it limits how singular a single irreducible curve can be. Both the crude and the sharp form are worth having, and the crude one is what licenses the argument for the sharp one.

Corollary 43.2.4: Bounding Multiplicity

Let f be an irreducible projective plane curve of degree n. Then f has finitely many singular points and

$$\sum_p \frac{m_p(f)\big(m_p(f)-1\big)}{2} \leq \frac{n(n-1)}{2}$$

***Proof*:**
Not every partial derivative of f vanishes identically. In characteristic zero this is because f is nonconstant. In characteristic $\rho > 0$, if all three vanished then every exponent appearing in f would be divisible by ρ, so $f \in k[x^\rho, y^\rho, z^\rho]$; as k is algebraically closed its elements are ρ-th powers, and the Frobenius is a ring homomorphism, so $f = h^\rho$ for some h, contradicting irreducibility. Let g be a nonzero partial derivative of f, a form of degree $n-1$.

At every point p, $m_p(g) \geq m_p(f) - 1$. Change coordinates so that $p = [0:0:1]$ and work in the chart $z = 1$, writing $f_* = f(x, y, 1)$ and $m = m_p(f)$. A linear change of coordinates replaces each partial derivative by a linear combination of the three partials, and multiplicity of a sum is at least the minimum of the multiplicities, so it is enough to check the bound for all three partials of the transformed form. Dehomogenization commutes with ∂_x and ∂_y, and differentiating drops the degree of each homogeneous piece of f_* by one, so $m_p(\partial_x f_*)$ and $m_p(\partial_y f_*)$ are both at least $m-1$. For the third, Euler's relation (exercise set 38.5.1(5)) $nf = x\partial_x f + y\partial_y f + z\partial_z f$ dehomogenizes to

$$(\partial_z f)_* = nf_* - x\,\partial_x f_* - y\,\partial_y f_*$$

Each term on the right has multiplicity at least m at the origin, the last two because a factor of x or y adds one to a multiplicity of at least $m-1$. So $m_p((\partial_z f)_*) \geq m$, which is more than required.

Since f is irreducible of degree n and g is a nonzero form of degree $n-1 < n$, they have no common component. Corollary 43.2.1 therefore gives

$$\sum_p m_p(f)\big(m_p(f)-1\big) \le \sum_p m_p(f)\, m_p(g) \le n(n-1)$$

which is the inequality after dividing by 2. In particular only finitely many p have $m_p(f) \ge 2$, so the singular points are finite in number, as we sought to show.

The sharp form replaces n by $n-1$ on the right, and its proof is the standard count of how many conditions singularities impose on curves of degree $n-1$.

Corollary 43.2.5: Sharp Bound on Multiplicity

Let f be an irreducible projective plane curve of degree n. Then

$$\sum_p \frac{m_p(f)\big(m_p(f)-1\big)}{2} \le \frac{(n-1)(n-2)}{2}$$

***Proof*:**
Write $m_p = m_p(f)$ and $M = \sum_p \binom{m_p}{2}$, a finite sum by corollary 43.2.4. The forms of degree $n-1$ in three variables constitute a projective space of dimension $\binom{n+1}{2} - 1 = \frac{(n-1)(n+2)}{2}$. Requiring a curve g of degree $n-1$ to satisfy $m_p(g) \ge m_p - 1$ is the vanishing of all partial derivatives of order less than $m_p - 1$ at p, which is $\binom{m_p}{2}$ linear conditions, and requiring g to pass through a further point is one more. Set

$$r = \frac{(n-1)(n+2)}{2} - M$$

This is non-negative: corollary 43.2.4 gives $M \le \frac{n(n-1)}{2}$, and $\frac{n(n-1)}{2} \le \frac{(n-1)(n+2)}{2}$. So r further conditions may be imposed, and since f has infinitely many points while only finitely many are singular, nonsingular points $q_1, \ldots, q_r \in \mathcal{Z}(f)$ may be chosen. The $M + r$ conditions cannot exhaust a projective space of dimension $M + r$, so a curve g of degree $n-1$ exists with $m_p(g) \ge m_p - 1$ at every singular point of f and $m_{q_i}(g) \ge 1$ for each i.

As before f is irreducible of degree n and g is a nonzero form of degree $n-1$, so they share no component and corollary 43.2.1 applies:

$$n(n-1) \;\ge\; \sum_p m_p(m_p-1) \;+\; r \;=\; 2M + \frac{(n-1)(n+2)}{2} - M$$

the r appearing because each q_i contributes $m_{q_i}(f)m_{q_i}(g) \ge 1$ and is not among the singular points already counted. Rearranging,

$$M \le n(n-1) - \frac{(n-1)(n+2)}{2} = (n-1)\cdot\frac{2n-n-2}{2} = \frac{(n-1)(n-2)}{2}$$

as we sought to show.

The sharp bound is attained: the curve $\mathcal{Z}(x^n + y^{n-1}z)$ is irreducible of degree n and has multiplicity $n-1$ at $[0:0:1]$, since dehomogenizing there leaves $x^n + y^{n-1}$ with lowest form y^{n-1}. That single point already contributes $\binom{n-1}{2}$, so no other singularity is available. The quantity $\frac{(n-1)(n-2)}{2}$ is the *arithmetic genus* of a plane curve of degree n, and chapter 44 will meet it again.

43.3 Max Noether's Fundamental Theorem

Bézout's theorem counts the intersection of two curves; the natural next question is what that intersection determines. The data of an intersection is a set of points together with a weight at each, and such data forms a group under addition.

Definition 43.3.1: Zero-cycle

A *zero-cycle* on $\mathbb{P}^2_k$ is a formal sum $\sum_p n_p\, p$ over points $p \in \mathbb{P}^2_k$ with $n_p \in \mathbb{Z}$ and all but finitely many $n_p = 0$. Its *degree* is $\sum_p n_p$.

Adding coefficientwise makes the zero-cycles an abelian group, and writing $\sum_p n_p p \leq \sum_p m_p p$ when $n_p \leq m_p$ for every p orders it. For projective plane curves f, g with no common component, the *intersection cycle* is

$$f \bullet g = \sum_{p \in \mathbb{P}^2_k} I(p, f \cap g)\, p$$

which is a zero-cycle of degree $\deg(f)\deg(g)$ by theorem 43.1.1. The properties of the intersection number transfer directly:

1. $f \bullet g = g \bullet f$;
2. $f \bullet (gh) = f \bullet g + f \bullet h$;
3. $f \bullet (g + af) = f \bullet g$ for every form a of degree $\deg(g) - \deg(f)$.

It is tempting to hope that every effective zero-cycle, meaning one with all $n_p \geq 0$, arises as an intersection cycle. It does not. Take three points of $\mathbb{P}^2_k$ in general position and the cycle $p_1 + p_2 + p_3$, of degree 3. Realizing it as $f \bullet g$ would need $\deg(f)\deg(g) = 3$, so one of the curves is a line, and that line would have to contain all three points, which they are not chosen to admit. The obstruction is exactly the phenomenon that chapter 44 organizes into the divisor class group: which formal sums of points are cut out by equations is a question with a non-trivial answer, and the answer is an invariant of the curve.

The question this section settles is the relative version. Suppose f, g, h are curves with $g \bullet f \leq h \bullet f$, so that h meets f at least as heavily as g does at every point. Is the difference itself an intersection cycle, that is, is there a curve b with

$$b \bullet f = h \bullet f - g \bullet f$$

Comparing degrees, such a b must have degree $\deg(h) - \deg(g)$. And it is enough to find forms a, b with

$$h = af + bg$$

for then property (3) followed by property (2) gives $h \bullet f = (bg) \bullet f = b \bullet f + g \bullet f$, which rearranges to the display. So the geometric question reduces to an algebraic one: when does h lie in the ideal generated by f and g, with the degrees working out? Max Noether's theorem answers it, and the answer is local.

Theorem 43.3.2: Max Noether's Fundamental Theorem

Let f, g, h be projective plane curves with f and g having no common component. Then there exist forms a, b of degrees $\deg(h) - \deg(f)$ and $\deg(h) - \deg(g)$ with

$$h = af + bg$$

if and only if for every $p \in \mathcal{Z}(f) \cap \mathcal{Z}(g)$ the local condition

$$h_* \in (f_*, g_*)\, \mathcal{O}_p(\mathbb{P}^2_k)$$

holds, where the subscript denotes dehomogenization in an affine chart containing p.

The local condition is called *Noether's condition* at p. That a global identity is equivalent to a finite list of local ones makes this a *local-to-global* theorem, and it is the reason the corollaries below are as easy as they are: each is a way of certifying the local condition cheaply.

***Proof*:**
One direction is immediate: if $h = af + bg$, then dehomogenizing at any p gives $h_* = a_* f_* + b_* g_*$, so h_* lies in $(f_*, g_*)\mathcal{O}_p(\mathbb{P}^2_k)$.

For the converse, change coordinates so that $\mathcal{Z}(f) \cap \mathcal{Z}(g) \cap \mathcal{Z}(z) = \emptyset$, which is possible because the intersection is finite, and dehomogenize in the chart $z = 1$. Noether's condition says the residue of h_* in $\mathcal{O}_p(\mathbb{A}^2_k)/(f_*, g_*)$ vanishes for every $p \in \mathcal{Z}(f_*, g_*)$. By lemma 42.2.5 the ring $k[x, y]/(f_*, g_*)$ is the product of those local quotients, so the residue of h_* there vanishes too:

$$h_* = \alpha f_* + \beta g_* \qquad \alpha, \beta \in k[x, y]$$

Homogenizing and clearing the powers of z introduced gives $r \geq 0$ and forms A, B with

$$z^r h = Af + Bg$$

Multiplication by z on $\Gamma = k[x, y, z]/(f, g)$ is injective, as established inside the proof of theorem 43.1.1, so cancelling z one factor at a time from $\overline{z^r h} = \overline{0}$ leaves $\overline{h} = \overline{0}$, that is $h = a' f + b' g$ for some forms a', b', not necessarily homogeneous.

Finally, extract the right graded pieces. Write $a' = \sum_i a'_i$ and $b' = \sum_i b'_i$ with a'_i, b'_i homogeneous of degree i. Comparing homogeneous components of degree $\deg h$ in $h = a' f + b' g$, and using that h is itself homogeneous, gives

$$h = a'_s f + b'_t g \qquad s = \deg(h) - \deg(f), \quad t = \deg(h) - \deg(g)$$

since those are the only pieces contributing in that degree. Taking $a = a'_s$ and $b = b'_t$ completes the proof.

Verifying Noether's condition directly is unpleasant. The following criteria cover every case that arises in practice.

Proposition 43.3.3: Noether's Criterion

Let f, g, h be projective plane curves and $p \in \mathcal{Z}(f) \cap \mathcal{Z}(g)$. Each of the following implies Noether's condition at p:

1. f and g meet transversally at p, and $p \in \mathcal{Z}(h)$;
2. p is a nonsingular point of f, and $I(p, h \cap f) \geq I(p, g \cap f)$;
3. f and g have no tangent line in common at p, and $m_p(h) \geq m_p(f) + m_p(g) - 1$.

***Proof*:**
(2). Since p is a nonsingular point of f, property (9) of corollary 42.4.2 converts both intersection numbers into orders, so the hypothesis reads $\operatorname{ord}_p^f(h_*) \geq \operatorname{ord}_p^f(g_*)$. In the discrete valuation ring $\mathcal{O}_p(f)$ that says

$$\overline{h}_* \in (\overline{g}_*) \subseteq \mathcal{O}_p(f)$$

Now $\mathcal{O}_p(f) \cong \mathcal{O}_p(\mathbb{A}^2_k)/(f_*)$ by the third exercise of exercise set 42.4.1, so $\mathcal{O}_p(f)/(\overline{g}_*) \cong \mathcal{O}_p(\mathbb{A}^2_k)/(f_*, g_*)$ and the residue of h_* there is zero, which is Noether's condition.

(3). Take $p = [0 : 0 : 1]$ and write $I = (x, y)$, $m = m_p(f)$, $n = m_p(g)$. The hypothesis $m_p(h) \geq m + n - 1$ says $h_* \in I^{m+n-1}$, and lemma 42.3.2(1) says $I^t \subseteq (f_*, g_*)\mathcal{O}_p(\mathbb{A}^2_k)$ for every $t \geq m + n - 1$ precisely because f and g have no common tangent. Hence $h_* \in (f_*, g_*)\mathcal{O}_p(\mathbb{A}^2_k)$.

(1). Transversality makes $m_p(f) = m_p(g) = 1$ with distinct tangents, so (3) applies with the requirement $m_p(h) \geq 1 + 1 - 1 = 1$, which is exactly $p \in \mathcal{Z}(h)$, completing the proof.

Corollary 43.3.4: Sufficient Condition For Max Noether's Theorem

Let f, g, h be projective plane curves with f and g having no common component, and suppose either

1. $\mathcal{Z}(f)$ and $\mathcal{Z}(g)$ meet in $\deg(f)\deg(g)$ distinct points and h passes through all of them, or
2. every point of $\mathcal{Z}(f) \cap \mathcal{Z}(g)$ is a nonsingular point of f, and $h \bullet f \geq g \bullet f$.

Then there is a curve b of degree $\deg(h) - \deg(g)$ with

$$b \bullet f = h \bullet f - g \bullet f$$

***Proof*:**
In case (1), corollary 43.2.2 says the $\deg(f)\deg(g)$ points are transversal intersections, so proposition 43.3.3(1) supplies Noether's condition at each. In case (2), the hypothesis $h \bullet f \geq g \bullet f$ is exactly $I(p, h \cap f) \geq I(p, g \cap f)$ at every p, so proposition 43.3.3(2) applies. Either way theorem 43.3.2 produces $h = af + bg$ with b of degree $\deg(h) - \deg(g)$, and the computation preceding that theorem turns the identity into the stated equation of cycles, completing the proof.

43.4 Application: Pascal's and Pappus's Theorems

Max Noether's theorem is a statement about ideals, and its geometric consequences are about incidence. Two of them are theorems of classical projective geometry, one from the fourth century and one from the seventeenth, and both come out of the same three-line computation: a cubic that splits into three lines, a conic, and the observation that Noether's condition costs nothing at a transversal intersection.

The shared setup is this. Let Q be a conic and let $p_1, \dots, p_6$ be six distinct points on $\mathcal{Z}(Q)$, thought of as the vertices of a hexagon in that cyclic order. Its six sides split into two alternating triples,

$$L_1 = \overline{p_1p_2}, \quad L_2 = \overline{p_3p_4}, \quad L_3 = \overline{p_5p_6} \qquad M_1 = \overline{p_2p_3}, \quad M_2 = \overline{p_4p_5}, \quad M_3 = \overline{p_6p_1}$$

and the three pairs of *opposite* sides are (L_1, M_2), (M_1, L_3) and (L_2, M_3). Write $C = L_1L_2L_3$ and $C' = M_1M_2M_3$, both cubics.

Lemma 43.4.1: The Hexagon Identity

With the notation above, suppose that no side of the hexagon meets $\mathcal{Z}(Q)$ outside the two vertices it joins. Then there are a constant $\lambda \in k$ and a linear form ℓ, not identically zero, with

$$M_1M_2M_3 = \lambda\, L_1L_2L_3 + \ell\, Q$$

***Proof*:**
Apply theorem 43.3.2 to $f = C$, $g = Q$ and $h = C'$. The curves C and Q have no common component, since each L_i meets $\mathcal{Z}(Q)$ in the two points p_{2i-1}, p_{2i} and so is not contained in it. By hypothesis $\mathcal{Z}(C) \cap \mathcal{Z}(Q)$ is exactly $\{p_1, \dots, p_6\}$, which is $\deg(C)\deg(Q) = 3 \cdot 2 = 6$ distinct points, so corollary 43.2.2 makes every one of them a transversal intersection. Each p_i also lies on C': the vertex p_j lies on M_1, M_2 or M_3 according to its index. So proposition 43.3.3(1) supplies Noether's condition at all six points, and theorem 43.3.2 gives

$$C' = aC + bQ$$

with $\deg a = 3 - 3 = 0$ and $\deg b = 3 - 2 = 1$; write $\lambda = a$ and $\ell = b$.

It remains to rule out $\ell = 0$, which would force $M_1M_2M_3 = \lambda L_1L_2L_3$ and hence, by unique factorization, make each M_i proportional to some L_j. That cannot happen: L_j meets $\mathcal{Z}(Q)$ in $\{p_{2j-1}, p_{2j}\}$ and M_i meets it in the corresponding pair for M_i, and the three pairs $\{p_1, p_2\}, \{p_3, p_4\}, \{p_5, p_6\}$ are disjoint from the three pairs $\{p_2, p_3\}, \{p_4, p_5\}, \{p_6, p_1\}$ as unordered pairs, the six vertices being distinct, as we sought to show.

Theorem 43.4.2: Pascal's Theorem

Let a hexagon with distinct vertices be inscribed in an irreducible conic. Then the three points in which the pairs of opposite sides meet are collinear.

***Proof*:**
Let Q be the conic and adopt the notation above. The hypothesis of lemma 43.4.1 holds: a line meets the irreducible conic $\mathcal{Z}(Q)$ in at most two points by theorem 43.1.1, since it is not a component of it, so each side meets $\mathcal{Z}(Q)$ exactly in its two vertices. Let λ and ℓ be as in that lemma.

Let r_1 be the point $L_1 \cap M_2$, the meeting of a pair of opposite sides, and define r_2 and r_3 from the pairs (M_1, L_3) and (L_2, M_3). Evaluating the identity of lemma 43.4.1 at r_1: the left side vanishes because $M_2(r_1) = 0$, and the term $\lambda L_1 L_2 L_3$ vanishes because $L_1(r_1) = 0$. Hence

$$\ell(r_1)\, Q(r_1) = 0$$

Now $Q(r_1) \neq 0$. For r_1 lies on L_1, whose only points on $\mathcal{Z}(Q)$ are p_1 and p_2, and on M_2, whose only points on $\mathcal{Z}(Q)$ are p_4 and p_5; were r_1 on $\mathcal{Z}(Q)$ it would belong to both pairs, which are disjoint. Therefore $\ell(r_1) = 0$, and the same computation gives $\ell(r_2) = \ell(r_3) = 0$. As ℓ is a nonzero linear form, $\mathcal{Z}(\ell)$ is a line containing all three points, as we sought to show.

Pappus's theorem is the case in which the conic degenerates into a pair of lines. The hexagon then has its vertices alternating between the two lines, and the argument runs unchanged.

Theorem 43.4.3: Pappus's Theorem

Let L and L' be distinct lines, and let p_1, p_2, p_3 be distinct points of $\mathcal{Z}(L)$ and q_1, q_2, q_3 distinct points of $\mathcal{Z}(L')$, none of the six equal to the point $\mathcal{Z}(L) \cap \mathcal{Z}(L')$. Put

$$r_1 = \overline{p_1 q_2} \cap \overline{q_1 p_2}, \qquad r_2 = \overline{q_2 p_3} \cap \overline{p_2 q_3}, \qquad r_3 = \overline{p_3 q_1} \cap \overline{q_3 p_1}$$

Then r_1, r_2, r_3 are collinear.

***Proof*:**
Take $Q = LL'$, a conic, and the hexagon with vertices $p_1, q_2, p_3, q_1, p_2, q_3$ in that cyclic order, all six lying on $\mathcal{Z}(Q)$ and all distinct. Its alternating side-triples are

$$\overline{p_1 q_2},\ \overline{p_3 q_1},\ \overline{p_2 q_3} \qquad \text{and} \qquad \overline{q_2 p_3},\ \overline{q_1 p_2},\ \overline{q_3 p_1}$$

and the three r_i are the intersections of the opposite pairs.

The hypothesis of lemma 43.4.1 holds. A side such as $\overline{p_1 q_2}$ joins a point of $\mathcal{Z}(L)$ to a point of $\mathcal{Z}(L')$, and it is equal to neither line, since $q_2 \notin \mathcal{Z}(L)$ and $p_1 \notin \mathcal{Z}(L')$ by the hypothesis on the six points. So it meets $\mathcal{Z}(L)$ only at p_1 and $\mathcal{Z}(L')$ only at q_2, and therefore meets $\mathcal{Z}(Q)$ exactly in its two vertices. Let λ and ℓ be as in the lemma.

Evaluating at r_1 exactly as in theorem 43.4.2 gives $\ell(r_1)Q(r_1) = 0$, so it remains to check $Q(r_1) \neq 0$, that is $r_1 \notin \mathcal{Z}(L) \cup \mathcal{Z}(L')$. Suppose $r_1 \in \mathcal{Z}(L)$. Then r_1 lies on $\overline{p_1 q_2}$, which meets $\mathcal{Z}(L)$ only at p_1, so $r_1 = p_1$; and r_1 lies on $\overline{q_1 p_2}$, which meets $\mathcal{Z}(L)$ only at p_2, so $r_1 = p_2$. These contradict $p_1 \neq p_2$. The case $r_1 \in \mathcal{Z}(L')$ is symmetric, using $q_2 \neq q_1$. Hence $\ell(r_1) = 0$, likewise for r_2 and r_3, and $\mathcal{Z}(\ell)$ is the required line, as we sought to show.

43.5 The Dual Curve and Plücker's Formulas

A nonsingular point of a plane curve has a tangent line, and a line in $\mathbb{P}^2_k$ is a point of the dual projective space. So a curve determines a second curve, and the numerical relations between the two are the oldest quantitative statements in the subject.

Definition 43.5.1: Dual Space and Dual Curve

The *dual projective space* $(\mathbb{P}^2_k)^\vee$ is the set of lines in $\mathbb{P}^2_k$, itself a $\mathbb{P}^2_k$ by sending $\mathcal{Z}(a_0x_0 + a_1x_1 + a_2x_2)$ to $[a_0 : a_1 : a_2]$. For an irreducible plane curve C, the *dual curve* $C^\vee \subseteq (\mathbb{P}^2_k)^\vee$ is the closure of the set of tangent lines to C at its nonsingular points. The *class* of C is $\deg C^\vee$.

By corollary 43.2.4 the nonsingular points of C are all but finitely many, so $C^\vee$ is the closure of the image of a map defined on a dense open set: the *Gauss map*

$$\gamma\colon C \dashrightarrow (\mathbb{P}^2_k)^\vee \qquad p \mapsto \big[\partial_0 F(p) : \partial_1 F(p) : \partial_2 F(p)\big]$$

whose coordinates are the partials of the defining form, by the tangent-line formula of definition 42.1.6. It is a rational map of curves, so $C^\vee$ is an irreducible curve by lemma 42.2.8 applied to a nonsingular model.

Proposition 43.5.2: The Class of a Nonsingular Curve

Let C be a nonsingular plane curve of degree $d \geq 2$ with $\operatorname{char} k = 0$. Then

$$\deg C^\vee = d(d-1)$$

***Proof*:**
A line of $(\mathbb{P}^2_k)^\vee$ is the set of lines of $\mathbb{P}^2_k$ through a fixed point q, so by theorem 41.5.1 the degree of $C^\vee$ is the number of points in which $C^\vee$ meets such a line for general q, counted through the Gauss map. In characteristic zero that map is birational onto its image for a nonsingular plane curve, so the count of points of $C^\vee$ equals the count of tangent lines to C through q; it is here, and only here, that the characteristic hypothesis is used.

A point $p \in C$ has its tangent line through $q = [q_0 : q_1 : q_2]$ exactly when

$$P_q(p) \;=\; q_0\,\partial_0 F(p) + q_1\,\partial_1 F(p) + q_2\,\partial_2 F(p) \;=\; 0$$

The form P_q has degree $d-1$ and is the *polar* of C with respect to q. It is not the zero form for general q: the partials $\partial_i F$ are not all zero, since $\operatorname{char} k = 0$ and F is nonconstant, and a general linear combination of forms that are not all zero is nonzero. And F does not divide P_q, because $\deg P_q = d-1 < d$ while F is irreducible. So C and $\mathcal{Z}(P_q)$ have no common component.

Theorem 43.1.1 therefore counts the intersection as $d(d-1)$ with multiplicity. For general q the intersections are transversal, so there are $d(d-1)$ distinct points and as many tangent lines through q, as we sought to show.

Example 43.5.3: Classes of Small Curves

A nonsingular conic has $d = 2$ and class 2: through a general point of the plane there pass exactly two tangent lines to a conic, and the dual of a conic is again a conic.

A nonsingular cubic has $d = 3$ and class 6. That number is the reason the dual of a smooth cubic is a sextic and not a cubic: dualizing does not preserve degree, and $(C^\vee)^\vee = C$ only because the two operations are inverse, not because the degrees match.

Singular points reduce the class, because a singular point has no well-defined tangent and the Gauss map is undefined there. The bookkeeping is Plücker's, and it is stated here rather than proved: the corrections require the genus and the behaviour of δ-invariants under duality.

43.5.4 Note: Plücker's Formulas Let C be an irreducible plane curve of degree d over a field of characteristic zero, with δ ordinary double points and κ ordinary cusps and no other singularities, and let $d^\vee$ be its class. Then

$$d^\vee = d(d-1) - 2\delta - 3\kappa$$

and dually, writing $\delta^\vee$ and $\kappa^\vee$ for the bitangents and inflections of C,

$$d = d^\vee(d^\vee - 1) - 2\delta^\vee - 3\kappa^\vee$$

The genus is common to both and is given by

$$g = \frac{(d-1)(d-2)}{2} - \delta - \kappa$$

consistently with the genus drop recorded in section 44.4, a node and a cusp each costing 1.
For a nonsingular cubic, $\delta = \kappa = 0$ gives $d^\vee = 6$, and the second formula reads $3 = 30 - 2\delta^\vee - 3\kappa^\vee$; with $\delta^\vee = 0$, since a smooth cubic has no bitangent, this gives $\kappa^\vee = 9$. Those nine are the inflection points, recovering by a completely different route the count obtained in exercise set 45.3.1 from the Hessian and theorem 45.2.2.
A proof needs the resolution of chapter 47 together with the genus, and is in [54, chapter 9] and in [29] analytically.

Exercise 43.5.1

1. Let X be an irreducible curve. Show that $k[X] = \bigcap_{p \in X} \mathcal{O}_p(X)$, the intersection being taken inside $k(X)$.

2. Show that a line and an irreducible conic are nonsingular curves, either directly from definition 42.1.5 or from one of the corollaries of section 43.2.

3. Let C and C' be cubics with $C \cdot C' = \sum_{i=1}^{9} p_i$ for nine distinct points, and let Q be a conic with $Q \cdot C = \sum_{i=1}^{6} p_i$. Assuming $p_1, \dots, p_6$ are nonsingular points of C, show that p_7, p_8, p_9 lie on a line. (Run the argument of lemma 43.4.1 with Q in place of the conic there.)

4. Show that the bound of corollary 43.2.5 fails for reducible curves by exhibiting a reducible curve of degree n violating it.

5. Verify that theorem 43.4.3 really is theorem 43.4.2 for a degenerate conic, by checking that the two proofs differ only in the verification that $r_i \notin \mathcal{Z}(Q)$.

6. Two conics meet in at most four points unless they share a component. Deduce that five points in general position determine a unique conic, and say what "general position" has to mean.

44

Divisors on Curves

The intersection cycles of section 43.3 are formal sums of points on a curve, and such sums form a group under addition. Not every formal sum is cut out by an equation, and the failure is measured by quotienting out the sums that arise from rational functions. What remains is the divisor class group, an invariant of the curve alone.

This chapter builds that group. Its degree-zero part is what chapter 45 identifies with the classical chord-tangent law on a cubic, and one book later it becomes the Jacobian. Its structure also detects rationality: a nonsingular plane curve is a copy of $\mathbb{P}^1$ exactly when distinct points on it are linearly equivalent, and that happens only in degrees 1 and 2. The invariant separating the cases is the genus, and the chapter closes by giving the genus-degree formula of section 38.5 that job.

44.1 Divisors and Their Degree

Throughout, C is an irreducible projective curve over the algebraically closed field k, and C is *nonsingular* unless said otherwise. Every local ring $\mathcal{O}_p(C)$ is then a discrete valuation ring by proposition 35.5.2, so every point carries an order function ord_p as in definition 42.2.4.

Section 37.5 already defined Weil divisors on any normal variety. A nonsingular projective curve is normal by corollary 35.5.3, and its prime divisors are its points, every point having codimension one. So the following is definition 37.5.1 specialized, restated here because the curve case is the one the rest of this Part uses.

Definition 44.1.1: Divisor on a Curve

A *divisor* on C is a formal sum

$$D = \sum_{p \in C} n_p\, p \qquad n_p \in \mathbb{Z}$$

with $n_p = 0$ for all but finitely many p. The divisors form an abelian group $\mathrm{Div}(C)$ under coefficientwise addition. The *degree* of D is $\deg D = \sum_p n_p$, and D is *effective*, written $D \geq 0$, if every $n_p \geq 0$. Write $D \geq D'$ for $D - D' \geq 0$.

A divisor is the same kind of object as a zero-cycle (definition 43.3.1), with the points confined to C. The intersection cycle of section 43.3 is the first source of divisors: if C is a nonsingular plane curve and G is a plane curve with no component in common, then $G \bullet C$ is an effective divisor on C, of degree $\deg(G)\deg(C)$ by theorem 43.1.1.

The second source is rational functions, and it is the one that matters.

Definition 44.1.2: Divisor of a Rational Function

Let $f \in k(C)^*$ be a nonzero rational function. Its *divisor* is

$$\mathrm{div}(f) = \sum_{p \in C} \mathrm{ord}_p(f)\, p$$

The divisor of zeros is $\mathrm{div}_0(f) = \sum_{\mathrm{ord}_p(f)>0} \mathrm{ord}_p(f)\, p$ and the divisor of poles is $\mathrm{div}_\infty(f) = -\sum_{\mathrm{ord}_p(f)<0} \mathrm{ord}_p(f)\, p$, so that $\mathrm{div}(f) = \mathrm{div}_0(f) - \mathrm{div}_\infty(f)$ with both parts effective.

The definition needs justification: the sum must be finite.

Proposition 44.1.3: A Rational Function Has Finitely Many Zeros and Poles

Let $f \in k(C)^*$. Then $\mathrm{ord}_p(f) = 0$ for all but finitely many $p \in C$, so $\mathrm{div}(f)$ is a divisor.

***Proof*:**
The set where $\mathrm{ord}_p(f) > 0$ is where f vanishes. Writing $f = g/h$ with g, h regular on some affine open $U \subseteq C$ and h not identically zero, the zeros of f in U lie in $\mathcal{Z}(g) \cap U$, a proper closed subset of the irreducible curve C, hence finite by proposition 34.3.5. The same argument applied to $1/f$ handles the poles, and $C \setminus U$ is finite for the same reason. So only finitely many points contribute, as we sought to show.

Example 44.1.4: Divisors on the Projective Line

Take $C = \mathbb{P}^1_k$ with coordinate x on the standard chart and ∞ the point at infinity.

1. *A function with a zero and a pole.* $\mathrm{div}(x) = 0 - \infty$, written $(0) - (\infty)$: the function x vanishes to order 1 at the origin and has a simple pole at infinity, since $1/x$ is a uniformizer there. Its degree is 0, as theorem 44.2.2 requires.

2. *Higher order.* $\operatorname{div}(x^n) = n(0) - n(\infty)$ for $n \in \mathbb{Z}$, negative n interchanging the roles. So every divisor supported on $\{0, \infty\}$ of degree 0 is principal.

3. *A general ratio.* For $f = (x-a)/(x-b)$ with $a \neq b$,

$$\operatorname{div}(f) = (a) - (b)$$

with no contribution at infinity, the numerator and denominator having equal degree. This is the computation behind proposition 44.2.4: every degree-zero divisor on the line is a combination of such differences.

4. *A divisor that is not principal.* The single point (0) has degree $1 \neq 0$, so by theorem 44.2.2 it is not $\operatorname{div}(f)$ for any f. On $\mathbb{P}^1_k$ the degree is the only obstruction, by proposition 44.2.4; on a curve of positive genus it is not, which is what theorem 44.2.5 shows.

The two sources are compatible in the case that matters, and the computation is worth doing once.

Proposition 44.1.5: Divisors from Ratios of Forms

Let $C \subseteq \mathbb{P}^2_k$ be a nonsingular plane curve and let G, H be forms of the same degree e, neither divisible by C. Then G/H defines a rational function on C and

$$\operatorname{div}\left(\frac{G}{H}\right) = G \bullet C - H \bullet C$$

***Proof*:**
The ratio G/H is a quotient of two forms of equal degree, so it is a well defined element of $k(C)$, nonzero because H does not divide G modulo C.

Fix $p \in C$. Property (9) of corollary 42.4.2 says that at a nonsingular point of C the intersection number is an order: $I(p, C \cap G) = \operatorname{ord}_p(G_*)$, where G_* is the dehomogenization of G in a chart containing p, and likewise for H. Choosing the same chart for both, $\operatorname{ord}_p(G_*/H_*) = \operatorname{ord}_p(G_*) - \operatorname{ord}_p(H_*)$ because ord_p is a valuation. The coefficient of p in $G \bullet C - H \bullet C$ is exactly that difference, and the chart cancels because G and H have the same degree, as we sought to show.

44.2 Principal Divisors and Linear Equivalence

Definition 44.2.1: Principal Divisor and Linear Equivalence

This is definition 37.5.3 for a curve. A divisor of the form $\operatorname{div}(f)$ for some $f \in k(C)^*$ is *principal*. The principal divisors form a subgroup $\operatorname{Prin}(C) \subseteq \operatorname{Div}(C)$, since $\operatorname{div}(fg) = \operatorname{div}(f) + \operatorname{div}(g)$. Two divisors D and D' are *linearly equivalent*, written $D \sim D'$, if $D - D'$ is principal.

The first structural fact is that the degree cannot see the difference between linearly equivalent divisors.

Theorem 44.2.2: Principal Divisors Have Degree Zero

Let C be a nonsingular projective plane curve and $f \in k(C)^*$. Then $\deg \operatorname{div}(f) = 0$. Consequently $D \sim D'$ implies $\deg D = \deg D'$.

***Proof*:**
Write $f = G/H$ with G, H forms of the same degree e, neither divisible by C; this is possible because $k(C)$ consists of exactly such ratios. By proposition 44.1.5, $\operatorname{div}(f) = G \bullet C - H \bullet C$. Both intersection cycles have degree $e \deg(C)$ by theorem 43.1.1, so the difference has degree 0. The consequence is immediate from additivity of the degree, completing the proof.

That Bézout's theorem is what makes this work is worth pausing on. The statement $\deg \operatorname{div}(f) = 0$ says a rational function on a projective curve has as many zeros as poles, and its analytic counterpart is the argument principle. Projectivity is essential: on $\mathbb{A}^1_k$ the function x has one zero and no poles.

The converse question is when a degree-zero divisor is principal, and the first thing to settle is which functions have no divisor at all.

Proposition 44.2.3: Functions with Trivial Divisor

Let C be a nonsingular projective curve and $f \in k(C)^*$ with $\operatorname{div}(f) = 0$. Then f is constant.

***Proof*:**
If $\operatorname{div}(f) = 0$ then $\operatorname{ord}_p(f) \geq 0$ at every p, so f lies in every local ring $\mathcal{O}_p(C)$ and is therefore a regular function on all of C. A regular function on a projective variety is constant by corollary 40.2.4, completing the proof.

On the simplest curve the converse of theorem 44.2.2 holds outright, and the computation is the model for everything later.

Proposition 44.2.4: Divisors on the Projective Line

Every divisor of degree 0 on $\mathbb{P}^1_k$ is principal. In particular any two points of $\mathbb{P}^1_k$ are linearly equivalent.

***Proof*:**
Let $D = \sum_i n_i p_i$ have degree 0, and write $p_i = [a_i : b_i]$ with $L_i = b_i x - a_i y$ the linear form vanishing exactly at p_i. Fix an index j with p_j distinct from the others. For $i \neq j$ the ratio L_i/L_j is a rational function on $\mathbb{P}^1_k$, and in the affine chart containing p_i the form L_i generates the maximal ideal at p_i while L_j is a unit there, so $\operatorname{ord}_{p_i}(L_i/L_j) = 1$; symmetrically $\operatorname{ord}_{p_j}(L_i/L_j) = -1$, and the order is 0 elsewhere. Hence $\operatorname{div}(L_i/L_j) = p_i - p_j$. Set

$$f = \prod_i \left(\frac{L_i}{L_j}\right)^{n_i}$$

a legitimate rational function since the exponents are integers. Then $\operatorname{div}(f) = \sum_i n_i(p_i - p_j) = D - (\deg D)\, p_j = D$, as we sought to show.

For curves of higher degree the answer is different, and the difference is the whole subject. The following theorem is the engine: on a plane curve of degree at least three, distinct points are never linearly equivalent. Its proof is a single application of Max Noether's theorem.

Theorem 44.2.5: Distinct Points Are Not Equivalent

Let C be a nonsingular projective plane curve of degree $d \geq 3$, and let $p, q \in C$ satisfy $p \sim q$. Then $p = q$.

***Proof*:**
Suppose $\operatorname{div}(f) = p - q$ and write $f = G/H$ with G, H forms of the same degree e, neither divisible by C. By proposition 44.1.5 this reads $G \bullet C - H \bullet C = p - q$, that is

$$G \bullet C + q = H \bullet C + p$$

an equality of effective divisors.

Choose a line L through p and write $L \bullet C = p + E$, so $E \geq 0$ has degree $d - 1 \geq 2$ by theorem 43.1.1. Then

$$(LH) \bullet C = L \bullet C + H \bullet C = p + E + H \bullet C = E + q + G \bullet C \;\geq\; G \bullet C$$

Every point of C is nonsingular, so corollary 43.3.4(2) applies to the curve C, the curve G and the curve LH, and produces a curve of degree $\deg(LH) - \deg(G) = (e + 1) - e = 1$, that is a line L', with

$$L' \bullet C = (LH) \bullet C - G \bullet C = E + q$$

So L and L' both cut E out of C, one with p left over and the other with q. Now E is effective of degree $d - 1 \geq 2$, and there are two cases. If its support contains two distinct points, then L and L' share those two points and hence coincide. Otherwise $E = (d - 1)r$ for a single point r, and since $d - 1 \geq 2$ both lines meet C at r with multiplicity at least 2, making each of them the tangent line to C at r, which is unique by definition 42.1.6; again they coincide.

With $L = L'$, comparing $L \bullet C = p + E$ against $L' \bullet C = q + E$ gives $p = q$, as we sought to show.

The hypothesis $d \geq 3$ is not an artefact. For $d = 1$ the curve is a line and proposition 44.2.4 says all its points are equivalent; for $d = 2$ the leftover divisor E has degree 1, a single point, which does not determine a line, and indeed a nonsingular conic is isomorphic to $\mathbb{P}^1_k$ by the computation in example 39.4.4. The theorem is the first place where a plane curve of degree three behaves differently from every curve seen so far.

44.3 The Picard Group

Definition 44.3.1: Picard Group

The *Picard group* of a nonsingular projective curve C, which is its class group $\mathrm{Cl}(C)$ in the sense of definition 37.5.3, is

$$\mathrm{Pic}(C) = \frac{\mathrm{Div}(C)}{\mathrm{Prin}(C)}$$

The class of a divisor D is written $[D]$. The subgroup of classes of degree zero is written $\mathrm{Pic}^0(C)$.

By theorem 44.2.2 the degree descends to a homomorphism $\deg\colon \mathrm{Pic}(C) \to \mathbb{Z}$, and it is surjective for a plane curve, since a hyperplane section has degree $d \neq 0$ and any class of degree n can be built from points. Its kernel is $\mathrm{Pic}^0(C)$, so there is a short exact sequence

$$0 \to \mathrm{Pic}^0(C) \to \mathrm{Pic}(C) \xrightarrow{\deg} \mathbb{Z} \to 0$$

All the geometry is in Pic^0.

Example 44.3.2: The Picard Group of the Line

For $C = \mathbb{P}^1_k$, proposition 44.2.4 says every degree-zero divisor is principal, so $\mathrm{Pic}^0(\mathbb{P}^1_k) = 0$ and the degree map is an isomorphism $\mathrm{Pic}(\mathbb{P}^1_k) \cong \mathbb{Z}$. The Picard group is as small as it can be, and this is exactly the statement that no formal sum of points on the line carries information beyond how many points it has.

The contrast with a curve of higher degree is total. On a nonsingular plane cubic, Pic^0 turns out to be in bijection with the points of the curve itself, and the bijection carries the chord-tangent operation of section 45.1 to addition of divisor classes. That is section 45.2, and the one general fact it needs is recorded here.

Lemma 44.3.3: Hyperplane Sections Are All Equivalent

Let C be a nonsingular plane curve and L, M lines, neither a component of C. Then $L \cdot C \sim M \cdot C$.

***Proof*:**
The ratio L/M is a quotient of forms of degree 1, so proposition 44.1.5 gives $\mathrm{div}(L/M) = L \cdot C - M \cdot C$, which is therefore principal, completing the proof.

More generally the same argument shows $G \cdot C \sim G' \cdot C$ for any two curves G, G' of equal degree meeting C properly, so the class of a plane section depends only on the degree. This is the statement that a plane curve carries a distinguished element of $\mathrm{Pic}(C)$ in each degree, and everything computed below is measured against it.

44.4 The Genus of a Plane Curve

Section 38.5 computed the arithmetic genus of a nonsingular plane curve of degree d from its Hilbert polynomial and found

$$p_a = \frac{(d-1)(d-2)}{2}$$

That computation was about the embedding. This section gives the number a job.

Definition 44.4.1: Genus of a Nonsingular Plane Curve

The *genus* of a nonsingular projective plane curve C of degree d is

$$g(C) = \frac{(d-1)(d-2)}{2}$$

which is its arithmetic genus in the sense of definition 38.5.3.

So a line and a conic have genus 0, a cubic has genus 1, a quartic genus 3, and no plane curve has genus 2: the sequence $0, 0, 1, 3, 6, 10, \ldots$ skips it. Curves of genus 2 exist, but none of them embeds in $\mathbb{P}^2_k$ as a nonsingular curve.

The genus was introduced as a number attached to an embedding, and the following theorem is the first evidence that it is measuring the curve itself.

Theorem 44.4.2: Genus Zero Means Rational

Let C be a nonsingular projective plane curve of degree d. The following are equivalent:

1. $g(C) = 0$;
2. $d \leq 2$;
3. $C \cong \mathbb{P}^1_k$;
4. some two distinct points of C are linearly equivalent.

***Proof*:**
(1) $\Leftrightarrow$ (2) is arithmetic: $(d-1)(d-2)/2 = 0$ exactly when $d = 1$ or $d = 2$.

(2) $\Rightarrow$ (3): a line is $\mathbb{P}^1_k$. For a nonsingular conic C, fix $p \in C$ and project from p onto any line not through it. A line through p meets C in one further point by theorem 43.1.1, the tangent at p being the case where that point is p itself, so the projection is a bijection $C \to \mathbb{P}^1_k$; it is regular away from p, and regular at p by lemma 42.2.8, with a regular inverse for the same reason. Example 39.4.4(3) carries the computation out explicitly for $\mathcal{Z}(x^2 + y^2 - z^2)$.

(3) $\Rightarrow$ (4): an isomorphism carries rational functions to rational functions and preserves orders, hence carries principal divisors to principal divisors and linear equivalence to linear equivalence. Any two points of $\mathbb{P}^1_k$ are linearly equivalent by proposition 44.2.4, and a curve has more than one point, so the same holds on C.

(4) $\Rightarrow$ (2): this is theorem 44.2.5 in contrapositive form, which says $d \geq 3$ forces distinct points to be inequivalent, completing the proof.

Corollary 44.4.3: Nonsingular Plane Cubics Are Not Rational

A nonsingular projective plane cubic has genus 1 and is not isomorphic to $\mathbb{P}^1_k$. In particular it is not a rational curve.

***Proof*:**
Its genus is $(3-1)(3-2)/2 = 1 \neq 0$, so theorem 44.4.2 rules out $C \cong \mathbb{P}^1_k$. A rational curve is birational to $\mathbb{P}^1_k$, and a birational map between nonsingular projective curves is an isomorphism by lemma 42.2.8 applied in both directions, completing the proof.

This settles a claim made in passing in chapter 33, that the classification of elliptic curves up to birational equivalence needs the genus: the cubics are exactly the plane curves the genus first distinguishes from $\mathbb{P}^1_k$, and theorem 45.2.2 says what replaces the vanishing Picard group when it does.

Singular curves are handled by passing to a nonsingular model, which is what chapter 37 was built for, and the genus drops in a way controlled by the singularities.

44.4.4 Note: The Genus of a Singular Plane Curve Let C be an irreducible plane curve of degree d whose singular points are all *ordinary*, meaning m_p distinct tangent lines at a point of multiplicity m_p (definition 42.1.6), and let $\tilde{C}$ be its normalization (chapter 37), a nonsingular curve birational to C. Then

$$g(\tilde{C}) = \frac{(d-1)(d-2)}{2} - \sum_p \frac{m_p(m_p-1)}{2}$$

each singular point costing exactly $\binom{m_p}{2}$. The formula is proved in theorem 47.4.1, by counting adjoint curves of degree $d-3$ rather than by any computation on a blow-up.
That the right side is non-negative is corollary 43.2.5, proved in chapter 43 by a dimension count on curves of degree $d-1$, which now reads as the statement that a genus cannot be negative. The extremal curve $\mathcal{Z}(x^d + y^{d-1}z)$ of section 43.2 attains the bound and is rational, as section 47.4 exhibits explicitly.

Exercise 44.4.1

1. Show that $\operatorname{div}(fg) = \operatorname{div}(f) + \operatorname{div}(g)$ and $\operatorname{div}(1/f) = -\operatorname{div}(f)$, so that $\operatorname{Prin}(C)$ is a subgroup of $\operatorname{Div}(C)$ and $f \mapsto \operatorname{div}(f)$ is a homomorphism $k(C)^* \to \operatorname{Div}(C)$.
2. Deduce from proposition 44.2.3 that the kernel of that homomorphism is k^*, so that $\operatorname{Prin}(C) \cong k(C)^*/k^*$.
3. Let C be a nonsingular plane curve of degree d and H a line. Show that the class $[H \cdot C]$ generates the image of deg in $\mathbb{Z}$ if and only if $d = 1$, and describe the image in general.
4. Show that on a nonsingular plane cubic C with base point O, a point p satisfies $p \oplus p \oplus p = \varphi(O, O)$ if and only if p is an inflection point. Deduce, using theorem 45.2.2, that the inflection points form a subgroup of $E_k(C, O)$ when O is itself an inflection point.

5. Show that a divisor D on a nonsingular projective curve satisfies $D \sim 0$ and $D \geq 0$ only if $D = 0$.
6. Give an example of two divisors on a nonsingular plane cubic that have the same degree but are not linearly equivalent, and prove they are not.

45

The Group Law on Cubic Curves

A line meets a nonsingular cubic in three points, and that fact alone produces a group. The construction is elementary, its verification is not: associativity is a statement about the nine intersection points of two cubics, and the proof runs through Max Noether's theorem.

The chapter then explains where the group came from. Under the divisor theory of chapter 44 the chord-tangent law is addition in $\operatorname{Pic}^0(C)$, transported to the curve by $p \mapsto [p - O]$, and the hard-won associativity becomes the associativity of a quotient group. That description also settles the one loose end of the construction: it needed a base point, and nothing distinguishes one choice from another, but the target $\operatorname{Pic}^0(C)$ mentions no base point at all. The group obtained is the object that *Elliptic Curves* [12] takes as its subject.

45.1 The Chord-Tangent Construction

Let C be a nonsingular projective plane cubic. It is irreducible: were $C = C_1C_2$ a proper factorization, theorem 43.1.1 would give $\mathcal{Z}(C_1)$ and $\mathcal{Z}(C_2)$ a common point, and proposition 42.1.7 makes any such point singular on C. So no line is a component of C, and theorem 43.1.1 says every line meets it in exactly three points counted with multiplicity. Given $p, q \in \mathcal{Z}(C)$, not necessarily distinct, let L be the line through them, taken to be the tangent line at p when $p = q$. Then

$$L \cdot C = p + q + r$$

for a unique third point r, again with multiplicity. This defines a map

$$\varphi\colon \mathcal{Z}(C) \times \mathcal{Z}(C) \to \mathcal{Z}(C) \qquad \varphi(p, q) = r$$

which is visibly symmetric, and which satisfies $\varphi(p, \varphi(p, q)) = q$, since the line realizing the left side is the same L.

Figure 45.1 shows the two cases.

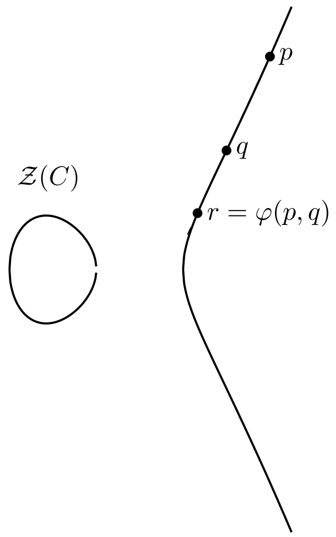


Figure 45.1: The chord through p and q meets the cubic in a third point. When $p = q$ the chord is replaced by the tangent line at p.

The map φ is a symmetric binary operation, but it is not a group law: there is no identity, since $\varphi(p, q) = q$ would force p to lie on the tangent at q for every q. The repair is to compose with φ once more against a chosen base point. Fix any $O \in \mathcal{Z}(C)$ and define

$$p \oplus q = \varphi\big(O, \varphi(p, q)\big)$$

Lemma 45.1.1: The Base Point Is an Identity

For every $p \in \mathcal{Z}(C)$, $O \oplus p = p \oplus O = p$.

***Proof*:**
Symmetry of φ gives $O \oplus p = p \oplus O$, so it is enough to compute one. Setting $q = \varphi(O, p)$, the involution property gives

$$O \oplus p = \varphi\big(O, \varphi(O, p)\big) = \varphi(O, q) = p$$

as we sought to show.

Before the group law can be proved, one point needs care. The natural guess for the inverse of p is $\varphi(p, O)$, and the natural guess for collinearity is that three collinear points sum to O. Both are false for a general base point, and the reason is that $\varphi(O, O)$, the third point on the tangent at O, need not be O itself. A point p with $\varphi(p, p) = p$ is one where the tangent line meets C triply, that is an *inflection point*, and a nonsingular cubic has only finitely many of them.

Lemma 45.1.2: Collinearity and the Sum

Let $p, q, r \in \mathcal{Z}(C)$. Then p, q, r are collinear, in the sense that $L \bullet C = p + q + r$ for some line L, if and only if

$$p \oplus q \oplus r = \varphi(O, O)$$

In particular, if the base point O is an inflection point then $\varphi(O, O) = O$ and three collinear points sum to the identity.

***Proof*:**
Suppose p, q, r are collinear, so $r = \varphi(p, q)$. Then $p \oplus q = \varphi(O, r)$, and

$$(p \oplus q) \oplus r = \varphi\Big(O, \varphi\big(\varphi(O, r), r\big)\Big)$$

The points O, r and $\varphi(O, r)$ are collinear by definition of φ, so $\varphi(\varphi(O, r), r) = O$, and the right side is $\varphi(O, O)$.

Conversely, suppose $p \oplus q \oplus r = \varphi(O, O)$, and let $r' = \varphi(p, q)$, so that p, q, r' are collinear. By the first part $p \oplus q \oplus r' = \varphi(O, O)$ as well, so r and r' have the same image under the map $x \mapsto (p \oplus q) \oplus x$. That map is injective once $\oplus$ is known to be a group law. Proposition 45.1.4 establishes this using only the involution property of φ and lemma 45.1.3, never the present converse, so there is no circularity. Hence $r = r'$ and p, q, r are collinear, as we sought to show.

The associativity of $\oplus$ is the hard point, and it rests on the following consequence of Max Noether's theorem: two cubics that agree with C in eight of their nine intersection points agree in the ninth as well.

Lemma 45.1.3: Cubic Point Counting

Let C be an irreducible cubic and C', C'' cubics with no component in common with C. Suppose

$$C' \bullet C = \sum_{i=1}^{9} p_i \qquad C'' \bullet C = \sum_{i=1}^{8} p_i + q$$

where the p_i are nonsingular points of C, not necessarily distinct. Then $q = p_9$.

***Proof*:**
Suppose $q \neq p_9$ and choose a line L through p_9 not passing through q, possible because the lines through p_9 sweep out the plane and only one of them contains q. Write $L \bullet C = p_9 + r + s$.

The quartic LC'' satisfies

$$(LC'') \bullet C = L \bullet C + C'' \bullet C = \sum_{i=1}^{9} p_i + q + r + s \ \geq\ C' \bullet C$$

Every point of $\mathcal{Z}(C') \cap \mathcal{Z}(C)$ is a nonsingular point of C by hypothesis, so corollary 43.3.4(2) applies to the triple (C, C', LC'') and produces a curve of degree $\deg(LC'') - \deg(C') = 4 - 3 = 1$, that is a line L', with

$$L' \bullet C = (LC'') \bullet C - C' \bullet C = q + r + s$$

Now L and L' coincide. If $r \neq s$, both contain the two distinct points r and s, and two distinct points determine a line. If $r = s$, then $L \bullet C \geq 2r$ and $L' \bullet C \geq 2r$, so both are tangent to C at the nonsingular point r, and the tangent line there is unique by definition 42.1.6.

But $L' = L$ forces q to lie on L, contradicting the choice of L. Hence $q = p_9$, as we sought to show.

Proposition 45.1.4: Group Law For Cubics

Let C be a nonsingular projective plane cubic and $O \in \mathcal{Z}(C)$ a base point. Then $\mathcal{Z}(C)$ with the operation $\oplus$ is an abelian group with identity O, in which the inverse of p is

$$\ominus p = \varphi\big(p, \varphi(O,O)\big)$$

Write $E_k(C,O)$ for this group.

***Proof*:**
Commutativity is the symmetry of φ, and O is an identity by lemma 45.1.1.

Inverses. Let $t = \varphi(O,O)$ and $\ominus p = \varphi(p,t)$. Only the involution property is needed: by definition of $\oplus$,

$$p \oplus \ominus p = \varphi\big(O, \varphi(p, \ominus p)\big) = \varphi\big(O, \varphi(p, \varphi(p,t))\big) = \varphi(O,t) = \varphi\big(O, \varphi(O,O)\big) = O$$

the third equality applying $\varphi(p, \varphi(p,-)) = \mathrm{id}$ and the last applying $\varphi(O, \varphi(O,-)) = \mathrm{id}$.

Associativity. Let $p, q, r \in \mathcal{Z}(C)$ and introduce six lines by

$$\begin{array}{lll} L_1 \bullet C = p + q + s' & M_1 \bullet C = O + s' + s & L_2 \bullet C = s + r + t' \\ M_2 \bullet C = q + r + u' & L_3 \bullet C = O + u' + u & M_3 \bullet C = p + u + t'' \end{array}$$

so that $s = \varphi(O,s') = p \oplus q$ and $u = \varphi(O,u') = q \oplus r$. Then

$$(p \oplus q) \oplus r = \varphi(O, \varphi(s,r)) = \varphi(O,t') \qquad p \oplus (q \oplus r) = \varphi(O, \varphi(p,u)) = \varphi(O,t'')$$

so it suffices to prove $t' = t''$. Put $C' = L_1L_2L_3$ and $C'' = M_1M_2M_3$, both cubics with no component in common with the irreducible C, since a line meets C in three points and so is not a component. Adding the three displayed cycles in each row,

$$C' \bullet C = p + q + s' + s + r + t' + O + u' + u \qquad C'' \bullet C = O + s' + s + q + r + u' + p + u + t''$$

The two agree in the eight points O, p, q, r, s, s', u, u', and every point of $\mathcal{Z}(C)$ is nonsingular, so lemma 45.1.3 gives $t' = t''$.

Thus $\oplus$ is an abelian group law, completing the proof.

The group depends on the base point only up to isomorphism, and the isomorphism is explicit. Given a second base point O', set $Q = \varphi(O,O')$ and

$$\alpha\colon E_k(C,O) \to E_k(C,O') \qquad \alpha(p) = \varphi(Q,p)$$

This sends O to O', since the line through Q and O is the line through O, O' and Q. That α is a group isomorphism is proved in section 45.3, once section 45.2 has identified $E_k(C,O)$ with a group of divisor classes and made the computation one line rather than a chain of collinearity arguments.

45.2 The Group Law via the Picard Group

The chord-tangent law was constructed by hand, and its associativity cost a nine-point argument through Max Noether. This section explains where the group comes from: it is $\operatorname{Pic}^0(C)$, transported to the curve.

Throughout, C is a nonsingular projective plane cubic with base point O, and φ, $\oplus$ are as in section 45.1. The bridge is a single linear equivalence.

Proposition 45.2.1: The Chord-Tangent Law Is Addition of Classes

For all $p, q \in C$,

$$p + q \;\sim\; (p \oplus q) + O$$

Consequently the map

$$\theta\colon E_k(C,O) \to \operatorname{Pic}^0(C) \qquad \theta(p) = [\,p - O\,]$$

is a group homomorphism.

***Proof*:**
Let $r = \varphi(p,q)$ and $s = \varphi(O,r) = p \oplus q$, and let L and M be the lines realizing them, so that

$$L \cdot C = p + q + r \qquad M \cdot C = O + r + s$$

By lemma 44.3.3 these are linearly equivalent, and subtracting the common r gives $p + q \sim O + s$, which is the display.

For the consequence, θ lands in Pic^0 because $\deg(p - O) = 0$, and

$$\theta(p) + \theta(q) = [\,p + q - 2O\,] = [\,(p \oplus q) + O - 2O\,] = [\,(p \oplus q) - O\,] = \theta(p \oplus q)$$

using the display in the middle step, as we sought to show.

Theorem 45.2.2: The Group Law Is the Picard Group

Let C be a nonsingular projective plane cubic with base point O. Then

$$\theta\colon E_k(C,O) \longrightarrow \operatorname{Pic}^0(C) \qquad \theta(p) = [\,p - O\,]$$

is an isomorphism of abelian groups.

***Proof*:**
It is a homomorphism by proposition 45.2.1.

Injective. If $\theta(p) = \theta(q)$ then $p - O \sim q - O$, so $p \sim q$, and theorem 44.2.5 gives $p = q$, the cubic having degree 3.

Surjective. Let D have degree 0 and write $D = \sum_{i=1}^{n} p_i - \sum_{j=1}^{n} q_j$, the two counts being equal because $\deg D = 0$; points may repeat, and $n = 0$ gives $D = 0 = \theta(O)$. First,

$$\sum_{i=1}^{n} p_i \;\sim\; (p_1 \oplus \cdots \oplus p_n) + (n-1)O$$

by induction on n. The case $n = 1$ is trivial. Given the statement for n, add p_{n+1} to both sides and apply proposition 45.2.1 to the pair $(p_1 \oplus \cdots \oplus p_n,\ p_{n+1})$, which replaces their sum by $(p_1 \oplus \cdots \oplus p_{n+1}) + O$; that is the statement for $n + 1$.

Writing $P = p_1 \oplus \cdots \oplus p_n$ and $Q = q_1 \oplus \cdots \oplus q_n$ and subtracting the two instances, the $(n-1)O$ terms cancel and $D \sim P - Q$. Finally

$$[P - Q] = [P - O] - [Q - O] = \theta(P) - \theta(Q) = \theta(P \ominus Q)$$

so D lies in the image, as we sought to show.

Associativity of $\oplus$, which proposition 45.1.4 obtained from lemma 45.1.3 and so ultimately from Max Noether's theorem, is now visible as the associativity of addition in a quotient group. The two proofs are not independent: theorem 44.2.5 itself runs through Max Noether. What changes is where the difficulty sits. In section 45.1 it sits in a configuration of six lines; here it sits in the statement that a rational function on a cubic cannot have a single simple pole.

45.2.3 Note: Which Facts Are Doing the Work Two theorems carry theorem 45.2.2, and they pull in opposite directions. Lemma 44.3.3 says linear equivalence is coarse enough that all line sections agree, which is what makes θ additive. Theorem 44.2.5 says it is fine enough that points stay apart, which is what makes θ injective. On a conic the second fails and Pic^0 collapses to zero; on a line there is nothing to say at all. Degree three is the smallest case where both hold, which is the precise sense in which the group law is a phenomenon of cubics.

45.3 Independence of the Base Point

The construction of $\oplus$ required choosing O, and nothing recommends one point over another. Theorem 45.2.2 settles the matter at once, because its target does not mention O.

Theorem 45.3.1: Independence of the Base Point

Let C be a nonsingular projective plane cubic and $O, O' \in C$. Then

$$E_k(C,O) \;\cong\; E_k(C,O')$$

both being isomorphic to $\operatorname{Pic}^0(C)$, which is defined without reference to a base point.

***Proof*:**
Write θ_O and $\theta_{O'}$ for the two maps of theorem 45.2.2, each an isomorphism onto $\operatorname{Pic}^0(C)$. Then $\theta_{O'}^{-1} \circ \theta_O$ is an isomorphism $E_k(C,O) \to E_k(C,O')$, completing the proof.

The explicit map promised in section 45.1 is almost that one, differing by an inversion.

Proposition 45.3.2: The Explicit Isomorphism

Let $Q = \varphi(O,O')$ and $\alpha(p) = \varphi(Q,p)$. Then $\alpha\colon E_k(C,O) \to E_k(C,O')$ is a group isomorphism, and

$$\theta_{O'}\big(\alpha(p)\big) = -\,\theta_O(p)$$

***Proof*:**
The points Q, p and $\alpha(p)$ are collinear by definition of φ, and so are O, O' and Q. By lemma 44.3.3 the two line sections are linearly equivalent:

$$Q + p + \alpha(p) \;\sim\; O + O' + Q$$

Cancelling Q and rearranging gives $\alpha(p) - O' \sim O - p$, that is

$$\theta_{O'}\big(\alpha(p)\big) = [\,\alpha(p) - O'\,] = [\,O - p\,] = -[\,p - O\,] = -\theta_O(p)$$

So $\alpha = \theta_{O'}^{-1} \circ (-\mathrm{id}) \circ \theta_O$. Inversion is an automorphism of an abelian group, and θ_O, $\theta_{O'}$ are isomorphisms, so α is one too, as we sought to show.

That the natural explicit map inverts is not a defect of the choice; no map defined by a single application of φ can avoid it, since φ itself reverses the sign. Composing twice, $p \mapsto \varphi(O', \varphi(Q,p))$ is the base-point-free isomorphism $\theta_{O'}^{-1}\theta_O$ on the nose.

45.3.3 Note: What the Group Does and Does Not Depend On The *group* $E_k(C,O)$ depends on O only up to isomorphism, and the isomorphism class is that of $\operatorname{Pic}^0(C)$. What does depend on O is the identification of points of C with group elements: which point is the identity, and hence which triples sum to zero. Choosing O to be an inflection point makes lemma 45.1.2 read in its familiar form, three collinear points summing to the identity, and this is why the literature almost always makes that choice for a curve given in Weierstrass form, where the point at infinity is an inflection.

Exercise 45.3.1

1. Show that a nonsingular *projective* plane curve must be irreducible, and give a nonsingular *affine* plane curve that is not.
2. Let C be a nonsingular plane cubic with base point O. Show that $\varphi(O, O) = O$ if and only if O is an inflection point, and that in that case the inverse of proposition 45.1.4 takes the familiar form $\ominus p = \varphi(p, O)$.
3. With O an inflection point, show that the inflection points of C are exactly the elements of order dividing 3 in $E_k(C, O)$, and bound their number using theorem 43.1.1 and the Hessian.
4. Show that a nonsingular plane cubic has no point p with $\varphi(p, p) = p$ other than its inflection points.
5. Let C be a nonsingular plane cubic and $p, q \in C$ with $p \sim q$ as divisors. Show $p = q$, and explain precisely which step of the argument fails on a nonsingular conic.
6. Compare the labour of verifying associativity of $\oplus$ directly on $y^2 = x^3 + 1$ with $O = [0 : 1 : 0]$, for three points of your choosing, against the one-line argument of theorem 45.2.2.

46

Differential Forms and the Canonical Class

Every invariant of a curve met so far has been extrinsic: the degree depends on an embedding, the multiplicity on a choice of coordinates, the divisor class group on nothing at all but says little by itself. This chapter constructs the invariant that governs the rest, the canonical class, and it does so from the one piece of machinery not yet used, differentiation.

Derivations, the module $\Omega_{B/A}$ and its universal property, and the two fundamental exact sequences are commutative algebra, developed in section 21.3; this chapter takes them as given and does the geometry. What comes out is a divisor class attached to a curve with no choices at all, whose degree is $2g-2$, and which chapter 48⋆ shows to be the error term in Riemann's inequality.

46.1 Differential Forms on a Variety

Definition 46.1.1: Differentials on a Variety

Let X be an affine variety over k. The *module of differentials* of X is $\Omega_X = \Omega_{k[X]/k}$, the module of relative differential forms of definition 21.3.2, together with the universal k-derivation $d\colon k[X] \to \Omega_X$. Its elements are the *differential forms* on X.
For $p \in X$ the *forms at* p are $\Omega_{X,p} = \Omega_X \otimes_{k[X]} \mathcal{O}_p(X)$, and for an irreducible X the *rational differential forms* are $\Omega_{k(X)/k}$, a vector space over $k(X)$.

Example 46.1.2: Modules of Differentials

1. *Affine space.* $\Omega_{\mathbb{A}^n_k}$ is free of rank n over $k[x_1, \dots, x_n]$ with basis $dx_1, \dots, dx_n$, since a k-derivation is determined by arbitrary choices of the dx_i. This is the first exercise of exercise set 46.2.1.

2. *A nonsingular plane curve.* On $C = \mathcal{Z}(y - x^2)$ the relation $dy = 2x\,dx$ holds, so Ω_C is generated by dx alone and is free of rank $1 = \dim C$, as proposition 46.1.3 predicts.

3. *The cuspidal cubic.* On $C = \mathcal{Z}(y^2 - x^3)$ the relation is $2y\,dy = 3x^2\,dx$. At the origin both coefficients vanish, so the relation imposes nothing there and $\Omega_{C,0} \otimes k$ has dimension 2, while $\dim C = 1$. By proposition 46.1.3 the origin is singular, which is the fifth exercise of exercise set 46.2.1. Away from the origin the relation is nontrivial and Ω is free of rank 1.

4. *A form with a pole.* On $\mathbb{P}^1_k$ the form dx has $\operatorname{div}(dx) = -2(\infty)$, computed in the second exercise of exercise set 46.2.1. Its degree -2 is $2g - 2$ with $g = 0$, which is theorem 46.2.4 in the smallest case.

Two facts do all the work, and both are transported from the algebra.

Proposition 46.1.3: Differentials Detect Smoothness

Let X be a variety of dimension n and $p \in X$. Then

$$\dim_k \Omega_{X,p} \otimes_{\mathcal{O}_p(X)} k \;=\; \dim_k \mathfrak{m}_p/\mathfrak{m}_p^2$$

so p is a nonsingular point if and only if that dimension is n. If X is nonsingular then $\Omega_{X,p}$ is a free $\mathcal{O}_p(X)$-module of rank n.

***Proof*:**
Apply the second fundamental exact sequence (proposition 21.3.7) to $k[X] \to \mathcal{O}_p(X) \to k$ with $I = \mathfrak{m}_p$. Since $\Omega_{k/k} = 0$, it reads

$$\mathfrak{m}_p/\mathfrak{m}_p^2 \xrightarrow{\delta} \Omega_{X,p} \otimes k \to 0$$

and δ is injective because a k-linear functional on $\mathfrak{m}_p/\mathfrak{m}_p^2$ extends to a k-derivation of $\mathcal{O}_p(X)$, by sending f to the class of $f - f(p)$; so δ has a left inverse. That gives the displayed equality, and corollary 35.1.1 identifies the right side with n exactly at the nonsingular points.

For freeness, $\Omega_{X,p}$ is a finitely generated module over the Noetherian local ring $\mathcal{O}_p(X)$ by corollary 21.3.8, whose fibre has dimension n; a finitely generated module over a regular local ring whose fibre dimension equals the generic rank is free, which is lemma 21.3.12, completing the proof.

Corollary 46.1.4: Rational Differentials on a Curve

Let C be a nonsingular projective curve. Then $\Omega_{k(C)/k}$ is a one-dimensional vector space over $k(C)$. If t is a uniformizer at a point p, then dt is a basis, and every rational differential form is $f\,dt$ for a unique $f \in k(C)$.

***Proof*:**
By proposition 46.1.3 the module $\Omega_{C,p}$ is free of rank 1 over $\mathcal{O}_p(C)$, so tensoring up to the fraction field gives $\dim_{k(C)} \Omega_{k(C)/k} = 1$.

For the basis, $\mathcal{O}_p(C)$ is a discrete valuation ring with uniformizer t and $\mathfrak{m}_p/\mathfrak{m}_p^2$ is spanned by the class of t, so dt generates $\Omega_{C,p} \otimes k$ under the isomorphism of proposition 46.1.3. Nakayama's lemma then makes dt a generator of the free rank-one module $\Omega_{C,p}$, hence a basis after tensoring with $k(C)$, completing the proof.

46.2 The Canonical Class

A nonzero rational differential form on a curve has zeros and poles, exactly as a rational function does, and they assemble into a divisor. The point of the section is that the divisor class does not depend on the form.

Definition 46.2.1: The Divisor of a Differential Form

Let C be a nonsingular projective curve and $\omega \in \Omega_{k(C)/k}$ nonzero. For $p \in C$ with uniformizer t, write $\omega = f\,dt$ with $f \in k(C)$ as in corollary 46.1.4, and set $\operatorname{ord}_p(\omega) = \operatorname{ord}_p(f)$. The *divisor* of ω is

$$\operatorname{div}(\omega) = \sum_{p \in C} \operatorname{ord}_p(\omega)\, p$$

Proposition 46.2.2: The Divisor of a Form Is Well Defined

$\operatorname{ord}_p(\omega)$ does not depend on the choice of uniformizer, and $\operatorname{ord}_p(\omega) = 0$ for all but finitely many p, so $\operatorname{div}(\omega)$ is a divisor.

***Proof*:**
Let s be another uniformizer at p, so $s = ut$ with u a unit of $\mathcal{O}_p(C)$. Then $ds = u\,dt + t\,du$, and writing $du = u'\,dt$ with $u' \in \mathcal{O}_p(C)$ gives $ds = (u + tu')\,dt$. The coefficient $u + tu'$ is a unit, being u modulo $\mathfrak{m}_p$, so $\omega = f\,dt = f(u+tu')^{-1}ds$ and the order of the coefficient is unchanged.

For finiteness, cover C by finitely many affine opens; on each, $\omega = f\,dg$ for some $f, g \in k[U]$, and $\operatorname{ord}_p(\omega) \neq 0$ only where f or dg degenerates, a proper closed subset, hence finite by proposition 34.3.5, completing the proof.

Theorem 46.2.3: The Canonical Class

Let C be a nonsingular projective curve. Any two nonzero rational differential forms on C have linearly equivalent divisors, so their common class

$$K_C = [\mathrm{div}(\omega)] \;\in\; \mathrm{Pic}\,(C)$$

is an invariant of C, the *canonical class*. A divisor in that class is a *canonical divisor*.

***Proof*:**
Let ω, ω' be nonzero. By corollary 46.1.4 the space $\Omega_{k(C)/k}$ is one-dimensional over $k(C)$, so $\omega' = f\omega$ for a unique $f \in k(C)^*$. At each p, writing $\omega = g\,dt$, gives $\omega' = fg\,dt$ and hence $\mathrm{ord}_p(\omega') = \mathrm{ord}_p(f) + \mathrm{ord}_p(\omega)$. Summing,

$$\mathrm{div}(\omega') = \mathrm{div}(f) + \mathrm{div}(\omega)$$

so the two divisors differ by a principal one and are linearly equivalent by definition 44.2.1, completing the proof.

That one line is the whole point of introducing differentials: a curve carries a distinguished divisor class, obtained without choosing an embedding, a point, or a coordinate. Its degree is the genus in disguise.

Theorem 46.2.4: The Degree of the Canonical Class

Let C be a nonsingular projective plane curve of degree d, with genus $g = \binom{d-1}{2}$ as in definition 44.4.1. Then

$$K_C = (d-3)\,H \cdot C \qquad \text{and} \qquad \deg K_C = 2g - 2$$

where H is a line.

***Proof*:**
Work in the affine chart $z = 1$ with $C = \mathcal{Z}(f)$, $f = F(x, y, 1)$, and consider the rational form

$$\omega = \frac{dx}{\partial f/\partial y}$$

On C one has $f = 0$, so $0 = df = \partial_x f\,dx + \partial_y f\,dy$ and therefore $\omega = -\,dy/\partial_x f$ as well. At a point where $\partial_y f \neq 0$ the function x is a uniformizer, since $\mathcal{O}_p(C)$ is a discrete valuation ring and dx generates $\Omega_{C,p}$ by proposition 46.1.3; there ω has neither zero nor pole. Where $\partial_y f = 0$ the second expression applies and $\partial_x f \neq 0$, again giving neither zero nor pole, since C is nonsingular and the two partials cannot both vanish. So ω has no zeros and no poles in the affine chart.

It remains to compute the behaviour on the line at infinity. Homogenizing, $x = X/Z$ and $\partial f/\partial y$ has degree $d-1$ as a form, so ω is a ratio in which the numerator contributes -2 and the denominator $-(d-1)$ in the variable Z; collecting, ω acquires a zero of order $d-3$ along each point of C at infinity. Since the points at infinity are $H \cdot C$ for the line $H = \mathcal{Z}(Z)$, this reads

$$\mathrm{div}(\omega) = (d-3)\,H \cdot C$$

Taking degrees, $\deg K_C = (d-3)\cdot d$ by theorem 43.1.1, and

$$d(d-3) = (d-1)(d-2) - 2 = 2g - 2$$

as we sought to show.

This settles what chapter 48⋆ previously had to cite Fulton for: the identification $K = (d-3)H \bullet C$ and the degree $2g-2$ are now proved, and Riemann-Roch may quote them.

Example 46.2.5: Canonical Classes of Small Curves

1. *A line*, $d = 1$: $K_C = -2H \bullet C$ of degree $-2 = 2\cdot 0 - 2$. On $\mathbb{P}^1_k$ the form dx has a double pole at infinity and no zeros, which is the whole story.
2. *A conic*, $d = 2$: $K_C = -H \bullet C$ of degree -2, consistent with a conic being isomorphic to $\mathbb{P}^1_k$ by theorem 44.4.2.
3. *A cubic*, $d = 3$: $K_C = 0$, so the canonical class is trivial and $\deg K_C = 0 = 2g-2$ with $g = 1$. A smooth cubic carries a nowhere-zero, nowhere-polar differential form, which is the invariant differential of the group law of chapter 45 and the reason theorem 48.4.4 could take $K \sim 0$.
4. *A quartic*, $d = 4$: $K_C = H \bullet C$ of degree $4 = 2\cdot 3 - 2$, so the canonical class is the hyperplane class and the canonical map is the original embedding, as exercise set 48.4.1 observes.

Exercise 46.2.1

1. Show that $\Omega_{\mathbb{A}^n_k}$ is free of rank n over $k[x_1, \dots, x_n]$ with basis $dx_1, \dots, dx_n$, directly from the universal property.
2. Compute $\operatorname{div}(dx)$ on $\mathbb{P}^1_k$ and verify $\deg K = -2$.
3. Let C be the nonsingular cubic $\mathcal{Z}(y^2z - x^3 - axz^2 - bz^3)$. Show that $\omega = dx/y$ is regular and nowhere zero on C, so that $K_C = 0$ directly.
4. Show that a nonsingular plane curve of degree $d \geq 4$ has $\deg K_C > 0$, and deduce that it admits no nonconstant map to $\mathbb{P}^1_k$ of degree 1.
5. Use proposition 46.1.3 to give a second proof that the cuspidal cubic $\mathcal{Z}(y^2 - x^3)$ is singular at the origin, by computing Ω there.
6. Show that $\operatorname{div}(\omega)$ for ω a nonzero rational form depends on ω only up to linear equivalence, and that every divisor in the class K_C arises as $\operatorname{div}(\omega)$ for some ω.

47

Blowing Up and Resolution of Curve Singularities

Chapter 37 already resolved the singularities of a curve: corollary 37.4.2 says the normalization of any curve is smooth. That proof is an existence statement obtained by taking an integral closure, and it says nothing about what the smooth model looks like or how singular the original was.

This chapter gives the geometric construction. Blowing up replaces a point by the set of directions through it, separating branches that were tangled together, and it is explicit enough to compute with: the multiplicity of a curve at a point is visible as the multiplicity with which the exceptional divisor appears in the total transform, and the tangent lines of definition 42.1.6 reappear as the points where the strict transform crosses it. The construction also survives into dimensions where normalization is not enough, which is why it, and not the integral closure, is what resolution of singularities means in general.

47.1 The Blow-Up of the Plane at a Point

A point of $\mathbb{P}^1_k$ is a direction through the origin of $\mathbb{A}^2_k$. The blow-up is the variety obtained by attaching all of those directions to the plane at once, in such a way that a curve through the origin remembers which direction it arrived from.

Definition 47.1.1: Blow-Up of the Plane at the Origin

Let $0 = (0,0) \in \mathbb{A}^2_k$ and give $\mathbb{P}^1_k$ homogeneous coordinates $[s:t]$. The *blow-up* of $\mathbb{A}^2_k$ at the origin is

$$\mathrm{Bl}_0\mathbb{A}^2_k = \big\{ \big((x,y),[s:t]\big) \in \mathbb{A}^2_k \times \mathbb{P}^1_k \;\big|\; xt = ys \big\}$$

with $\pi\colon \mathrm{Bl}_0\mathbb{A}^2_k \to \mathbb{A}^2_k$ the first projection. The *exceptional divisor* is $E = \pi^{-1}(0)$.

The defining equation says that the vector (x,y), when nonzero, points in the direction $[s:t]$. So away from the origin the second coordinate is determined by the first, and over the origin it is unconstrained.

Proposition 47.1.2: Basic Properties of the Blow-Up

Let $B = \mathrm{Bl}_0\mathbb{A}^2_k$. Then:

1. B is closed in $\mathbb{A}^2_k \times \mathbb{P}^1_k$, and is covered by two open sets each isomorphic to $\mathbb{A}^2_k$;
2. B is an irreducible quasi-projective variety of dimension 2;
3. π restricts to an isomorphism $B \setminus E \to \mathbb{A}^2_k \setminus \{0\}$;
4. $E = \{0\} \times \mathbb{P}^1_k \cong \mathbb{P}^1_k$, and π is not an isomorphism at any point of E.

***Proof*:**
(1). The equation $xt = ys$ is homogeneous of degree 1 in $[s:t]$, so its vanishing is well defined on $\mathbb{A}^2_k \times \mathbb{P}^1_k$, and being closed is a local condition on the cover by $\{s \neq 0\}$ and $\{t \neq 0\}$. On $\{s \neq 0\} \cong \mathbb{A}^2_k \times \mathbb{A}^1_k$, with $v = t/s$ and coordinates (x,y,v), the equation reads $y = xv$, whose zero set is the graph of $(x,v) \mapsto xv$ and so is closed and isomorphic to $\mathbb{A}^2_k$ with coordinates (x,v). Call this chart U_s; then

$$\pi|_{U_s}(x,v) = (x,xv)$$

Symmetrically the chart U_t where $t \neq 0$ is $\mathbb{A}^2_k$ with coordinates (u,y), where $u = s/t$ and $x = uy$, and $\pi|_{U_t}(u,y) = (uy,y)$. The two charts cover B, which is therefore closed.

(2). Each U_s, U_t is irreducible of dimension 2, being isomorphic to $\mathbb{A}^2_k$, and they meet, since the point $((1,1),[1:1])$ lies in both. Lemma 40.1.5 makes B irreducible, and corollary 34.3.3 reads its dimension off either chart as 2.

(3). For $(x,y) \neq (0,0)$ the equation $xt = ys$ determines $[s:t]$ uniquely, namely $[x:y]$, so π is bijective there with inverse $(x,y) \mapsto ((x,y),[x:y])$, which is regular because x and y do not vanish simultaneously.

(4). Setting $x = y = 0$ leaves $[s:t]$ free, so $E = \{0\} \times \mathbb{P}^1_k$. Any nonempty open neighbourhood of a point of E meets E in an infinite set, since $E \cong \mathbb{P}^1_k$ is irreducible, and π sends all of it to the origin. So π is not injective on any such neighbourhood, completing the proof.

The chart map $\pi|_{U_s}(x,v) = (x,xv)$ has been seen before. It is the map of example 40.5.2, whose fibre over the origin jumped from a point to a line, and that jump is now identified: the fibre is the exceptional divisor, and the jump is the whole point of the construction rather than a pathology.

What E records is direction, and the following makes that precise.

Proposition 47.1.3: The Exceptional Divisor Parameterizes Directions

Let $L = \mathcal{Z}(bx - ay) \subseteq \mathbb{A}^2_k$ be a line through the origin, with $[a : b] \in \mathbb{P}^1_k$. Then the closure in B of $\pi^{-1}(L \setminus \{0\})$ is a curve meeting E in the single point $(0, [a : b])$, and distinct lines meet E in distinct points.

***Proof*:**
Parameterize $L \setminus \{0\}$ by $\lambda \mapsto (\lambda a, \lambda b)$ with $\lambda \neq 0$. Its preimage under the isomorphism of proposition 47.1.2(3) is $\big((\lambda a, \lambda b), [\lambda a : \lambda b]\big) = \big((\lambda a, \lambda b), [a : b]\big)$, so the preimage lies in $\mathbb{A}^2_k \times \{[a : b]\}$, a closed set. Its closure therefore also lies there, and adding the limit $\lambda \to 0$ gives the point $(0, [a : b])$. Distinct lines through the origin have distinct $[a : b]$, completing the proof.

So two lines through the origin, which meet in the plane, become disjoint after blowing up: they are pulled apart by exactly the amount needed to record that they arrived along different directions. That is the mechanism by which a node will be resolved in section 47.3.

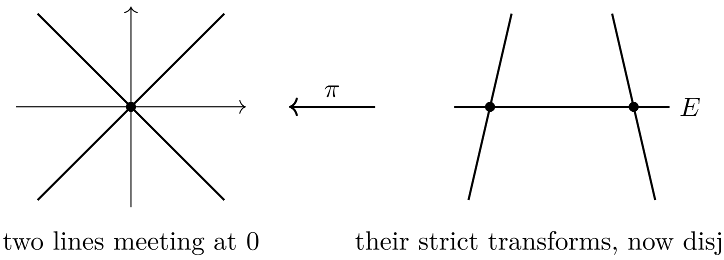


Figure 47.1: Blowing up separates the branches of a node. The exceptional divisor E records the direction along which each branch approaches the origin.

47.2 Blowing Up a Variety Along a Subvariety

Nothing in definition 47.1.1 was special to dimension two.

Definition 47.2.1: Blow-Up of Affine Space at the Origin

Give $\mathbb{P}^{n-1}_k$ homogeneous coordinates $[\ell_1 : \cdots : \ell_n]$. The blow-up of $\mathbb{A}^n_k$ at the origin is

$$\mathrm{Bl}_0\mathbb{A}^n_k = \big\{(x, [\ell]) \in \mathbb{A}^n_k \times \mathbb{P}^{n-1}_k \;\big|\; x_i \ell_j = x_j \ell_i \text{ for all } i, j\big\}$$

with π the first projection and $E = \pi^{-1}(0) \cong \mathbb{P}^{n-1}_k$.

The equations are exactly those saying the matrix with rows x and ℓ has rank at most one, which is the 2×2 minor condition of proposition 40.1.3. The proof of proposition 47.1.2 carries over verbatim with n charts in place of two: on the chart $\ell_i \neq 0$, setting $v_j = \ell_j/\ell_i$ gives $x_j = x_i v_j$ for $j \neq i$, so the chart is $\mathbb{A}^n_k$ with coordinates $(x_i, v_j)_{j \neq i}$ and

$$\pi(x_i, v) = (x_i v_1, \ldots, x_i, \ldots, x_i v_n)$$

Hence $\mathrm{Bl}_0\mathbb{A}^n_k$ is an irreducible quasi-projective variety of dimension n, isomorphic to $\mathbb{A}^n_k$ away from E, and E is a $\mathbb{P}^{n-1}_k$ parameterizing the lines through the origin.

Blowing up a subvariety rather than a point is the same construction with the point replaced by a linear subspace, at least locally.

Definition 47.2.2: Blow-Up Along a Coordinate Subspace

Let $Z = \mathcal{Z}(x_1, \ldots, x_r) \subseteq \mathbb{A}^n_k$ be a coordinate subspace, $0 < r \leq n$. The blow-up of $\mathbb{A}^n_k$ along Z is

$$\mathrm{Bl}_Z\mathbb{A}^n_k = \left\{(x, [\ell]) \in \mathbb{A}^n_k \times \mathbb{P}^{r-1}_k \;\middle|\; x_i\ell_j = x_j\ell_i,\ 1 \leq i, j \leq r\right\}$$

with π the first projection. Its exceptional locus $\pi^{-1}(Z)$ is $Z \times \mathbb{P}^{r-1}_k$.

The last $n - r$ coordinates are carried along untouched, so $\mathrm{Bl}_Z\mathbb{A}^n_k \cong \mathrm{Bl}_0\mathbb{A}^r_k \times \mathbb{A}^{n-r}_k$; the construction is the point case in the transverse directions, applied uniformly along Z. The general blow-up along an arbitrary closed subvariety is not this concrete: it is defined as the Proj of the Rees algebra $\bigoplus_{d\geq 0} I^d$ of the ideal I of the subvariety, a construction that needs the language of [5]. Everything in this chapter happens at a point of a curve, where definition 47.2.1 suffices.

What is blown up is the ambient space; what one wants is the effect on a subvariety sitting inside it.

Definition 47.2.3: Strict and Total Transform

Let $X \subseteq \mathbb{A}^n_k$ be a closed subvariety and $\pi\colon \mathrm{Bl}_0\mathbb{A}^n_k \to \mathbb{A}^n_k$ the blow-up at a point $p \in X$. The *total transform* of X is $\pi^{-1}(X)$. The *strict transform* is

$$\tilde{X} = \overline{\pi^{-1}\big(X \setminus \{p\}\big)}$$

the closure being taken in $\mathrm{Bl}_0\mathbb{A}^n_k$.

The total transform always contains the whole exceptional divisor, which carries no information about X. The strict transform discards it and keeps only what is approached from X itself. By proposition 47.1.2(3) the map $\tilde{X} \to X$ is an isomorphism away from p, so $\tilde{X}$ is birational to X; and $\tilde{X}$ is irreducible whenever X is, being the closure of a set isomorphic to the irreducible $X \setminus \{p\}$.

47.3 Resolving Plane Curve Singularities

Now specialize to a plane curve. Everything reduces to one computation, carried out in the chart $\pi(x, v) = (x, xv)$.

Theorem 47.3.1: The Blow-Up of a Plane Curve at a Point

Let $f \in k[x,y]$ be an affine plane curve with $m = m_0(f) \geq 1$, and write $f = f_m + f_{m+1} + \cdots$ for its homogeneous decomposition, so that the tangent cone is $f_m = \prod_i L_i^{r_i}$ with $\sum_i r_i = m$. Then in the chart U_s with coordinates (x,v),

$$f(x,xv) = x^m \tilde{f}(x,v) \qquad \tilde{f}(x,v) = f_m(1,v) + x\, f_{m+1}(1,v) + x^2 f_{m+2}(1,v) + \cdots$$

and $\mathcal{Z}(\tilde{f})$ is the strict transform $\tilde{C}$. Moreover:

1. the exceptional divisor occurs in the total transform with multiplicity exactly m;
2. $\tilde{C}$ meets E exactly at the points corresponding, under proposition 47.1.3, to the tangent lines $\mathcal{Z}(L_i)$, and the intersection multiplicity there is r_i; in particular $\tilde{C} \bullet E$ has degree m;
3. at the point $q_i \in E$ corresponding to L_i, the multiplicity satisfies $m_{q_i}(\tilde{C}) \leq r_i \leq m$.

***Proof*:**
Each f_j is homogeneous of degree j, so $f_j(x,xv) = x^j f_j(1,v)$, and summing over $j \geq m$ gives the displayed factorization with $\tilde{f}$ as stated. Since E is cut out by $x = 0$ in this chart, the factor x^m is the exceptional divisor with multiplicity m, which is (1). The remaining factor $\tilde{f}$ does not vanish identically on E: writing $f_m = \sum_j c_j x^{m-j} y^j$ gives $\tilde{f}(0,v) = f_m(1,v) = \sum_j c_j v^j$, which is the zero polynomial only if every $c_j = 0$, that is only if $f_m = 0$. So $\mathcal{Z}(\tilde{f})$ contains no component of E, and away from E it agrees with $\pi^{-1}(C)$; hence $\mathcal{Z}(\tilde{f})$ is the closure of $\pi^{-1}(C \setminus \{0\})$, which is $\tilde{C}$.

(2). On E the equation of $\tilde{C}$ is $f_m(1,v) = 0$. Writing $L_i = b_i x - a_i y$, so that $\mathcal{Z}(L_i)$ is the line through the origin in direction $[a_i : b_i]$, gives $f_m(1,v) = \prod_i (b_i - a_i v)^{r_i}$ up to a scalar, whose roots are $v = b_i/a_i$ when $a_i \neq 0$. Under proposition 47.1.3 the point $v = b_i/a_i$ of E is the direction $[a_i : b_i]$, that is the tangent line $\mathcal{Z}(L_i)$; the roots with $a_i = 0$ are the tangent line $\mathcal{Z}(x)$ and appear in the chart U_t. The multiplicity of the root is r_i, and $\sum_i r_i = m$.

(3). Translate so that q_i is the origin of the chart, replacing v by $v_i + w$. Then

$$\tilde{f} = f_m(1, v_i + w) + x\, f_{m+1}(1, v_i + w) + \cdots$$

The first summand has order exactly r_i in w, by (2). Every later summand is divisible by x, hence has order at least 1. So the lowest-degree part of $\tilde{f}$ has degree at most r_i, which is (3), as we sought to show.

Part (3) is the statement that blowing up never makes things worse, and for the singularities named in definition 42.1.6 it makes them disappear outright.

Corollary 47.3.2: One Blow-Up Resolves an Ordinary Singularity

Let p be an ordinary singular point of multiplicity m on a plane curve C, so that C has m distinct tangent lines at p. Then $\tilde{C}$ meets E in m distinct points, each a nonsingular point of $\tilde{C}$, and each intersection is transversal.

***Proof*:**
Ordinariness says every $r_i = 1$ and there are m of them, so theorem 47.3.1(2) gives m distinct intersection points, each with intersection multiplicity 1, which is transversality. Theorem 47.3.1(3) bounds $m_{q_i}(\tilde{C}) \leq r_i = 1$, and a multiplicity is at least 1 at a point of the curve, so each q_i is nonsingular, completing the proof.

Example 47.3.3: Resolving a Node, a Cusp, and a Higher Cusp
All three computations use theorem 47.3.1 in the chart (x, v), where $y = xv$.

1. *The node* $f = y^2 - x^2 - x^3$. Here $f_2 = y^2 - x^2 = (y - x)(y + x)$, two distinct tangents, so the point is an ordinary double point. Substituting,
$$f(x, xv) = x^2v^2 - x^2 - x^3 = x^2\big(v^2 - 1 - x\big)$$
so $\tilde{f} = v^2 - 1 - x$, whose partial derivative in x is -1, nowhere zero: the strict transform is nonsingular. It meets $E = \mathcal{Z}(x)$ at $v = \pm 1$, the two tangent directions, as corollary 47.3.2 predicts.

2. *The cusp* $f = y^2 - x^3$. Here $f_2 = y^2$, a single tangent with $r_1 = 2$, so the point is *not* ordinary. Nonetheless
$$f(x, xv) = x^2v^2 - x^3 = x^2\big(v^2 - x\big)$$
and $\tilde{f} = v^2 - x$ is nonsingular. One blow-up suffices here too, though corollary 47.3.2 does not apply; the strict transform meets E at the single point $v = 0$, with intersection multiplicity 2.

3. *A higher cusp* $f = y^2 - x^5$. Again $f_2 = y^2$, and
$$f(x, xv) = x^2v^2 - x^5 = x^2\big(v^2 - x^3\big)$$
so $\tilde{f} = v^2 - x^3$: the blow-up has replaced the singularity by an ordinary cusp, not by a nonsingular point. Blowing up once more, by (2), resolves it. Two blow-ups are needed, and no fewer.

Item (3) is the general situation: one blow-up reduces a singularity but need not remove it, and the content of the resolution theorem is that the process stops.

Theorem 47.3.4: Resolution of Plane Curve Singularities

Let C be an irreducible plane curve. Then after finitely many blow-ups at points, the strict transform is a nonsingular curve birational to C.

***Proof*:**
Each blow-up is an isomorphism away from the point blown up by proposition 47.1.2(3), so the strict transform stays birational to C throughout, and C has only finitely many singular points by corollary 43.2.4, so they may be treated one at a time. Theorem 47.3.1(3) says the multiplicity at every point above the one blown up is at most the multiplicity below, so blowing up never makes a singularity worse, and corollary 47.3.2 removes an ordinary point in a single

step.

What those facts leave open is whether the multiplicity must eventually reach 1, rather than stabilizing forever at some $m \geq 2$. The input needed is that each singular point p carries a non-negative integer δ_p, the drop in arithmetic genus recorded in section 44.4, which strictly decreases at every blow-up above p; a strictly decreasing sequence of non-negative integers is finite, so the process stops. That δ_p behaves this way is a comparison of the local rings of C and of its strict transform, the same computation section 44.4 deferred, and it is carried out in [54, chapter 7]; the cohomological account is in [5]. Granting it, the process terminates and the final strict transform is nonsingular and birational to C, as we sought to show.

The theorem is not new information: corollary 37.4.2 already produced a smooth model, and section 37.3 showed it is unique in its birational class. So the smooth curve reached by blowing up is the normalization, arrived at by a different road. What blowing up adds is that the road is explicit and that each step is measured, which is the subject of the last section.

47.4 Multiplicity, Genus, and the Effect of Blowing Up

Blowing up trades a singular point for an intersection with the exceptional divisor, and theorem 47.3.1 says the trade is exact: a point of multiplicity m becomes m points of intersection with E, counted properly. That bookkeeping is what turns resolution into a computation of the genus.

Recall from section 44.4 that a nonsingular plane curve of degree d has genus $\binom{d-1}{2}$, and that for a singular curve the genus of the smooth model drops by $\binom{m_p}{2}$ at each ordinary singular point. Blowing up is where that number comes from.

Theorem 47.4.1: The Genus Drop at an Ordinary Singularity

Let C be an irreducible plane curve of degree d whose singular points $p_1, \ldots, p_s$ are all ordinary, of multiplicities $m_1, \ldots, m_s$, and let $\tilde{C}$ be its nonsingular model. Then

$$g(\tilde{C}) \;=\; \frac{(d-1)(d-2)}{2} \;-\; \sum_{i=1}^{s} \binom{m_i}{2}$$

***Proof*:**
The genus is computed by the space of regular differential forms, whose dimension is g by theorem 48.3.2, and on a plane curve those forms are described by *adjoints*. Say a curve G of degree e is an *adjoint* of C if $m_{p_i}(G) \geq m_i - 1$ at every singular point.

Adjoints of degree $d-3$ give the regular differentials. By theorem 46.2.4 a canonical divisor on a nonsingular plane curve of degree d is cut by a curve of degree $d-3$. For a curve with ordinary singularities the same computation applies on $\tilde{C}$, with the correction that a form $G/\partial_y f$ is regular at the points of $\tilde{C}$ over p_i exactly when G vanishes to order at least $m_i - 1$ at p_i: the denominator $\partial_y f$ vanishes to order $m_i - 1$ along each of the m_i branches, and $\tilde{C}$ separates the branches by corollary 47.3.2. So the regular differential forms on $\tilde{C}$ correspond to the adjoint curves of degree $d-3$, modulo those divisible by f.

Counting them. The curves of degree $d-3$ form a space of dimension $\binom{d-1}{2}$, since $\binom{d-3+2}{2} =$

$\binom{d-1}{2}$. Requiring $m_{p_i}(G) \geq m_i - 1$ imposes $\binom{m_i}{2}$ linear conditions, being the vanishing of all partial derivatives of order less than $m_i - 1$, and for ordinary singularities in distinct positions these conditions are independent, by the same argument that established independence in corollary 43.2.5. Nothing of degree $d - 3 < d$ is divisible by f. Hence

$$g(\tilde{C}) = \binom{d-1}{2} - \sum_{i=1}^{s} \binom{m_i}{2}$$

as we sought to show.

The formula was quoted in section 44.4 and used in corollaries 47.4.3 and 47.4.4; it is now proved, and those two corollaries are unconditional.

47.4.2 Note: Two Routes to the Same Number The proof above counts conditions on adjoint curves and never mentions the blow-up. There is a second route, standard in the literature, which computes the arithmetic genus of the strict transform on the blown-up surface: the total transform is $\tilde{C} + mE$, and the correction $\frac{1}{2}m(m-1)$ falls out of the self-intersection $E \cdot E = -1$ together with $\tilde{C} \cdot E = m$ from theorem 47.3.1(2).
That route was not taken here, deliberately. It requires an intersection pairing for divisors on a surface, which this book does not build and which [5] develops as its first genuine intersection product, with the general theory in [16]. The adjoint count needs only linear systems and the canonical class, both of which are now in hand, so the hub proves what it needs without borrowing from downstream.

Two consequences can be read off without that ingredient, because they only use the non-negativity of a genus.

Corollary 47.4.3: Blowing Up Terminates for Ordinary Singularities

Let C be an irreducible plane curve of degree d all of whose singular points are ordinary. Then one blow-up at each removes them all, the total drop in arithmetic genus is $\sum_p \binom{m_p}{2}$, and the number of blow-ups is at most

$$\frac{(d-1)(d-2)}{2}$$

***Proof*:**
A blow-up at an ordinary point of multiplicity m removes that point by corollary 47.3.2, and by theorem 47.4.1 it drops the arithmetic genus by $\binom{m}{2}$, which is at least 1 because a singular point has $m \geq 2$. Summing over the singular points, finite in number by corollary 43.2.4, gives the total drop.

The arithmetic genus starts at $\binom{d-1}{2}$ by definition 44.4.1 and ends at the genus of a smooth curve, which is non-negative, so the total drop is at most $\binom{d-1}{2}$. Since each blow-up contributes at least 1 to that drop, there are at most $\binom{d-1}{2}$ of them, completing the proof.

Corollary 47.4.4: The Multiplicity Bound, Read Geometrically

For an irreducible plane curve of degree d with ordinary singularities,

$$\sum_p \binom{m_p}{2} \leq \frac{(d-1)(d-2)}{2}$$

with equality exactly when the smooth model has genus 0.

***Proof*:**
This is corollary 47.4.3 restated: the total drop is bounded by the starting genus, with equality exactly when the final genus is 0, completing the proof.

The equality case is a rationality statement, though not by way of theorem 44.4.2, which was about plane curves and the smooth model of a singular plane curve is generally not one. That a smooth projective curve of genus 0 is isomorphic to $\mathbb{P}^1_k$ is true and is one of the payoffs of chapter 48$\star$. For the extremal curve it can be seen by hand: on $\mathcal{Z}(x^d + y^{d-1}z)$ one may solve $z = -x^d/y^{d-1}$, so

$$\mathbb{P}^1_k \to \mathcal{Z}(x^d + y^{d-1}z) \qquad [x:y] \mapsto [\,xy^{d-1} : y^d : -x^d\,]$$

parameterizes it.

That inequality was already proved in chapter 43, as corollary 43.2.5, by an entirely different argument: a dimension count on the curves of degree $d-1$ passing through the singular points with prescribed multiplicity. The two proofs agree, and the agreement is informative. The Bézout argument says the bound holds because there are not enough curves of degree $d-1$ to impose more conditions; the blow-up argument says it holds because a genus cannot be negative. The extremal curve $\mathcal{Z}(x^d+y^{d-1}z)$ of section 43.2 attains the bound, and corollary 47.4.4 now explains why it is rational.

47.5 Cubic Surfaces and the Twenty-Seven Lines

Theorem 41.4.3 showed that a general cubic surface carries finitely many lines, by a dimension count that never produced a number. The number is 27, and the route to it is the blow-up: a smooth cubic surface is $\mathbb{P}^2_k$ blown up at six points, and under that description the lines can simply be listed.

Throughout, $\operatorname{char} k \neq 2$ and the six points $p_1, \ldots, p_6 \in \mathbb{P}^2_k$ are in *general position*, meaning no three collinear and not all six on a conic.

Proposition 47.5.1: The Blow-Up of Six Points Is a Cubic Surface

Let $p_1, \ldots, p_6 \in \mathbb{P}^2_k$ be in general position and let S be the blow-up of $\mathbb{P}^2_k$ at the six points. The cubics through the six points form a linear system of projective dimension 3, and the map it defines embeds S as a cubic surface in $\mathbb{P}^3_k$.

***Proof*:**
Cubic forms in three variables constitute a space of dimension $\binom{5}{3} = 10$, so the cubics of $\mathbb{P}^2_k$ form a $\mathbb{P}^9_k$. Passing through a point is one linear condition, and for six points in general position the six conditions are independent: were they dependent, some p_i would lie on every cubic through the other five, which fails because the union of a conic through five of them and a line missing the sixth, or of three lines pairing them up, gives cubics through five and not the sixth. So the system has projective dimension $9 - 6 = 3$.

Let $\varphi\colon S \to \mathbb{P}^3_k$ be the map defined by that system, which is a morphism on S because blowing up the six points resolves the base locus. Its image is a surface, and the degree of the image is the number of points in which two general members of the system meet away from the p_i: two cubics meet in 9 points by theorem 43.1.1, of which six are the base points, leaving 3. So the image is a cubic surface, and φ is birational onto it since a general pair of members separates points of $\mathbb{P}^2_k$ away from the base locus.

That φ is an embedding, rather than merely birational, is the statement that the system separates points and tangent vectors on S; it does so exactly because no three of the p_i are collinear and the six do not lie on a conic, which are the configurations making two members of the system agree to first order somewhere. See [31, chapter V.4] for the verification, as we sought to show.

With the description in hand the lines can be counted, and the count is a matter of listing three families.

Theorem 47.5.2: The Twenty-Seven Lines

Let $S \subseteq \mathbb{P}^3_k$ be a smooth cubic surface, realized as the blow-up of $\mathbb{P}^2_k$ at six points $p_1, \ldots, p_6$ in general position. Then S contains exactly 27 lines, namely

1. the six exceptional curves $E_1, \ldots, E_6$ over the six points;
2. the $\binom{6}{2} = 15$ strict transforms of the lines $\overline{p_i p_j}$;
3. the six strict transforms of the conics through five of the six points.

In total $6 + 15 + 6 = 27$.

***Proof*:**
These are lines. A curve on S is a line exactly when the linear system of proposition 47.5.1 cuts it in a single point, that is when its image under φ has degree 1. The system consists of cubics through the six points, so its degree on a curve $D \subseteq \mathbb{P}^2_k$ is $3 \deg D$ less one for each base point on D, counted with multiplicity.

For an exceptional curve E_i the system restricts to the pencil of cubics through p_i with varying tangent direction, cutting E_i once, so $\varphi(E_i)$ is a line. For the line $\overline{p_i p_j}$ we get $3 \cdot 1 - 1 - 1 = 1$. For a conic Q through five of the six, $3 \cdot 2 - 5 = 1$. So all 27 are lines.

There are no others. Let $L \subseteq S$ be a line and let $D \subseteq \mathbb{P}^2_k$ be its image, of degree $e \geq 0$, passing through p_i with multiplicity m_i. The degree computation above reads

$$3e - \sum_{i=1}^{6} m_i = 1$$

and a second constraint comes from the genus: $L \cong \mathbb{P}^1_k$ has genus 0, while by section 44.4 the normalization of a plane curve of degree e with multiplicities m_i has genus

$$\frac{(e-1)(e-2)}{2} - \sum_{i=1}^{6} \binom{m_i}{2} = 0$$

So $\sum_i m_i = 3e - 1$ and $\sum_i \binom{m_i}{2} = \binom{e-1}{2}$. The case $e = 0$ is an exceptional curve. For $e \geq 1$ these two equations already bound e. The function $m \mapsto \binom{m}{2}$ is convex, so by Jensen's inequality applied to the six values m_i, whose mean is $(3e-1)/6$,

$$\binom{e-1}{2} = \sum_{i=1}^{6} \binom{m_i}{2} \geq 6\binom{(3e-1)/6}{2} = \frac{(3e-1)(3e-7)}{12}$$

Multiplying by 12 gives $6(e-1)(e-2) \geq (3e-1)(3e-7)$, that is $3e^2 - 6e - 5 \leq 0$, so $e \leq 2$.

It remains to check $e = 1$ and $e = 2$. For $e = 1$ the genus equation reads $\sum \binom{m_i}{2} = 0$, so every $m_i \leq 1$, and $\sum m_i = 2$ picks out two of the six points: a line through p_i and p_j. For $e = 2$ again every $m_i \leq 1$, and $\sum m_i = 5$ picks out five: a conic through five of the six. Counting the choices gives 6, $\binom{6}{2} = 15$ and $\binom{6}{5} = 6$, as we sought to show.

47.5.3 Note: What the Count Is Really About Three features of the argument deserve separating.

The number 27 does not depend on the six points. Every smooth cubic surface arises as such a blow-up, so the count is intrinsic, and the configuration of the 27 lines is the same for all of them: each line meets exactly 10 others, and the whole configuration has a symmetry group of order 51840, the Weyl group of the root system E_6. That is the beginning of a story this book does not tell; see [31] and, for the representation theory, [26].

The dimension count of section 41.4 and the present listing answer different questions. The first shows finiteness for a *general* cubic without exhibiting a single line; the second exhibits all of them for a *smooth* one. Neither subsumes the other, and the first is what guarantees the second is not vacuous.

The general-position hypothesis is doing real work. If three of the six points are collinear, the strict transform of that line has $3 \cdot 1 - 3 = 0$ and is contracted rather than embedded, and the image acquires a singular point. Cubic surfaces with singularities have fewer lines: the surface $\mathcal{Z}(x_0x_1x_2 - x_3^3)$ of lemma 41.4.2 has three.

Exercise 47.5.1

1. Show that $\mathrm{Bl}_0\mathbb{A}^2_k$ is not isomorphic to $\mathbb{A}^2_k$. Suggestion: $E \cong \mathbb{P}^1_k$ is a complete curve sitting inside it, whereas a curve inside $\mathbb{A}^2_k$ carries nonconstant regular functions, which corollary 40.2.4 forbids on a projective variety.
2. Compute the strict transform of the curve $y^2 = x^4 + x^5$ under one blow-up at the origin, identify its singularities, and say how many further blow-ups are needed.
3. Show that the blow-up of $\mathbb{A}^2_k$ at the origin, restricted to the chart U_t, is the map $(u, y) \mapsto (uy, y)$, and redo example 47.3.3(2) in that chart. Explain why the answer looks different and why

the two are consistent.

4. Let C be a plane curve with an ordinary point of multiplicity m at the origin. Show that the strict transform meets E transversally in m points, and deduce that $\tilde{C} \cdot E$ has degree m whether or not the point is ordinary.

5. Show that $\mathrm{Bl}_Z \mathbb{A}^n_k \cong \mathrm{Bl}_0 \mathbb{A}^r_k \times \mathbb{A}^{n-r}_k$ for Z a coordinate subspace of codimension r, as claimed after definition 47.2.2.

6. A curve of degree 4 can have at most three double points. Prove this from corollary 47.4.4, and exhibit a quartic attaining the bound.

48⋆

Riemann-Roch for Plane Curves

⋆ *Optional. This chapter is an aside; nothing later in the book depends on it.*

Every question this book has asked about a curve reduces, sooner or later, to the same one: given a divisor D, which rational functions have poles no worse than D allows? Chapter 44 met the question in the special case $D = 0$, where proposition 44.2.3 answered it, and again in theorem 45.2.2, where the answer for $D = p - q$ decided injectivity. The Riemann-Roch theorem answers it in general, up to a correction term that vanishes once D is large.

Nothing earlier in the book depends on this chapter, and the theorem is proved in full elsewhere in the series: analytically in *Complex Analysis* [40], cohomologically in *Algebraic Geometry* [5]. What is offered here is the plane-curve account, where the statement can be made concrete and the consequences are immediate. Section 48.1 builds the vector spaces and proves what can be proved directly; sections 48.2 and 48.3 state Riemann's theorem and Riemann-Roch and say how the proof goes; section 48.4 spends the theorem, including on two debts left open in chapters 44 and 47.

Throughout, C is a nonsingular projective curve over the algebraically closed field k, and divisors are as in chapter 44.

48.1 The Space of Functions with Prescribed Poles

Definition 48.1.1: The Space $L(D)$

Let D be a divisor on C. The *Riemann-Roch space* of D is

$$L(D) = \{f \in k(C)^* \mid \operatorname{div}(f) + D \geq 0\} \cup \{0\}$$

and $\ell(D) = \dim_k L(D)$.

Writing $D = \sum_p n_p\, p$, the condition $\operatorname{div}(f) + D \geq 0$ says $\operatorname{ord}_p(f) \geq -n_p$ at every p: where $n_p > 0$ the function may have a pole of order at most n_p, and where $n_p < 0$ it is required to vanish to order at least $-n_p$. So $L(D)$ is the space of functions whose poles are bounded by D and whose zeros are forced by it.

Proposition 48.1.2: Basic Properties of $L(D)$

Let D and D' be divisors on C. Then:

1. $L(D)$ is a k-vector space;
2. $L(0) = k$, so $\ell(0) = 1$;
3. if $\deg D < 0$ then $L(D) = 0$;
4. if $D \leq D'$ then $L(D) \subseteq L(D')$ and $\ell(D') - \ell(D) \leq \deg D' - \deg D$;
5. $\ell(D) \leq \deg D + 1$ whenever $\deg D \geq 0$;
6. if $D \sim D'$ then $\ell(D) = \ell(D')$, so ℓ is a function on $\operatorname{Pic}(C)$.

***Proof*:**
(1). For nonzero $f, g \in L(D)$ and $\lambda \in k^*$, the valuation inequality $\operatorname{ord}_p(f+g) \geq \min(\operatorname{ord}_p(f), \operatorname{ord}_p(g))$ holds at every p, and $\operatorname{ord}_p(\lambda f) = \operatorname{ord}_p(f)$, so $L(D)$ is closed under addition and scaling.

(2). $L(0)$ consists of the f with $\operatorname{div}(f) \geq 0$, that is with $\operatorname{ord}_p(f) \geq 0$ at every p. Such an f lies in every local ring $\mathcal{O}_p(C)$ and so is regular on all of C, hence constant by corollary 40.2.4. Conversely every constant lies in $L(0)$.

(3). If $f \in L(D)$ is nonzero then $\operatorname{div}(f) + D \geq 0$, so $\deg \operatorname{div}(f) + \deg D \geq 0$; but $\deg \operatorname{div}(f) = 0$ by theorem 44.2.2, so $\deg D \geq 0$.

(4). The inclusion is immediate from the definition. For the inequality it suffices, by induction on $\deg D' - \deg D$, to treat $D' = D + p$ for a single point p, and to show $\ell(D+p) \leq \ell(D) + 1$. Let n be the coefficient of p in D and let t be a uniformizer at p. The map

$$L(D+p) \to k \qquad f \mapsto \left(t^{n+1} f\right)(p)$$

is well defined, since $f \in L(D+p)$ has $\operatorname{ord}_p(f) \geq -n-1$ so $t^{n+1} f$ lies in $\mathcal{O}_p(C)$, and it is k-linear. Its kernel is exactly $L(D)$, because $\left(t^{n+1} f\right)(p) = 0$ says $\operatorname{ord}_p(f) \geq -n$. So $L(D+p)/L(D)$ embeds in k and the codimension is at most 1.

(6). Suppose $D' = D + \operatorname{div}(h)$ for some $h \in k(C)^*$, and consider $f \mapsto f/h$. For $f \in L(D)$,

$$\operatorname{div}(f/h) + D' = \operatorname{div}(f) - \operatorname{div}(h) + D + \operatorname{div}(h) = \operatorname{div}(f) + D \;\geq\; 0$$

so $f/h \in L(D')$. The map is k-linear and multiplication by h inverts it, so $L(D) \cong L(D')$ and $\ell(D) = \ell(D')$.

(5). If $L(D) = 0$ then $\ell(D) = 0 \leq \deg D + 1$. Otherwise choose a nonzero $f \in L(D)$ and put $D_0 = \operatorname{div}(f) + D$, which is effective by the definition of $L(D)$ and satisfies $D_0 \sim D$, hence $\deg D_0 = \deg D$ and $\ell(D_0) = \ell(D)$ by (6). Now $0 \leq D_0$, so (4) applied to the pair $(0, D_0)$ gives

$$\ell(D_0) - \ell(0) \;\leq\; \deg D_0 - 0 = \deg D$$

and $\ell(0) = 1$ by (2), so $\ell(D) = \ell(D_0) \leq \deg D + 1$, as we sought to show.

Item (5) already shows the size of $L(D)$ is controlled by $\deg D$ from above. Riemann's theorem is the statement that it is controlled from below as well, by the same quantity shifted by an invariant of the curve.

Example 48.1.3: The Spaces $L(D)$ on the Projective Line

Take $C = \mathbb{P}^1_k$ and $D = n\,\infty$ for $n \geq 0$, where $\infty = [1:0]$. A rational function with poles only at ∞, of order at most n, is a polynomial in the affine coordinate x of degree at most n. So

$$L(n\infty) = \{a_0 + a_1 x + \cdots + a_n x^n \mid a_i \in k\} \qquad \ell(n\infty) = n + 1$$

The bound of proposition 48.1.2(5) is attained for every $n \geq 0$. Since every divisor of degree n on $\mathbb{P}^1_k$ is linearly equivalent to $n\infty$ by proposition 44.2.4, item (6) gives $\ell(D) = \deg D + 1$ for every divisor of non-negative degree on the line. The line is the curve on which $L(D)$ is as large as it can possibly be.

48.2 Riemann's Theorem

On $\mathbb{P}^1_k$ the inequality $\ell(D) \leq \deg D + 1$ is an equality. On any other curve it is not, and the deficiency is bounded.

Theorem 48.2.1: Riemann's Theorem

Let C be a nonsingular projective curve. There is a smallest integer $g \geq 0$, the *genus* of C, such that

$$\ell(D) \;\geq\; \deg D + 1 - g$$

for every divisor D on C. Moreover equality holds for all D of sufficiently large degree.

Two things are worth extracting before saying where the proof comes from. First, taking $D = 0$ gives $\ell(0) = 1 \geq 1 - g$, so $g \geq 0$ automatically; and $g = 0$ forces $\ell(D) \geq \deg D + 1$, which combined with proposition 48.1.2(5) makes it an equality, the situation of example 48.1.3. Second, the genus so defined is an invariant of C as an abstract curve, since $L(D)$ and $\deg D$ are defined from $k(C)$ and the valuations alone. That is what section 44.4 could not say about $\binom{d-1}{2}$, which was read off an embedding.

48.2.2 Note: Where the Proof Comes From For a plane curve the proof is a counting argument built on machinery this book already has. Let C be a nonsingular plane curve of degree d and H a line, so $H \cdot C$ is a divisor of degree d. One computes $\ell(mH \cdot C)$ for large m by identifying $L(mH \cdot C)$ with the space of curves of degree m modulo those containing C, and the identification is exactly the local-to-global statement of theorem 43.3.2: a form of degree m cuts out a divisor $\geq$ a given one precisely when Noether's condition holds at each point. The dimension of the space of forms of degree m in three variables is $\binom{m+2}{2}$, and subtracting the forms divisible by C, of which there are $\binom{m-d+2}{2}$, leaves

$$\binom{m+2}{2} - \binom{m-d+2}{2} = md - \frac{d(d-3)}{2} = \deg(mH \cdot C) + 1 - \frac{(d-1)(d-2)}{2}$$

which is Riemann's bound with $g = \binom{d-1}{2}$, the number definition 44.4.1 already called the genus. Extending from the divisors $mH \cdot C$ to arbitrary D is proposition 48.1.2(4) together with the fact that every divisor is dominated by some $mH \cdot C$. The details, in particular the surjectivity of the map from forms to $L(mH \cdot C)$, are in [54, chapter 8].

The computation in that box is worth pausing on, because it is the same arithmetic as example 38.5.4(3): $md - d(d-3)/2$ is the Hilbert polynomial of a plane curve of degree d. The Hilbert polynomial counts forms of degree m modulo those vanishing on C, and Riemann's theorem counts functions with poles bounded by $mH \cdot C$; Max Noether's theorem is what identifies the two counts.

Corollary 48.2.3: Genus of a Nonsingular Plane Curve

The genus of theorem 48.2.1 agrees, for a nonsingular plane curve of degree d, with the genus of definition 44.4.1:

$$g(C) = \frac{(d-1)(d-2)}{2}$$

***Proof*:**
The computation of box 48.2.2 produces the bound of theorem 48.2.1 with $g = \binom{d-1}{2}$, so the genus is at most that. It is not smaller: were the bound to hold with some $g' < g$, then taking $D = mH \cdot C$ for large m would give $\ell(D) \geq \deg D + 1 - g' > \deg D + 1 - g$, contradicting that the same computation evaluates $\ell(D)$ exactly as $\deg D + 1 - g$ for large m, completing the proof.

This is the statement section 44.4 could only assert: the number computed from the embedding is the intrinsic invariant. It is also what makes theorem 44.4.2 more than a coincidence of small degrees.

48.3 The Canonical Divisor and Riemann-Roch

Riemann's theorem leaves an inequality, and the error term has a name. It is measured by a divisor class attached to the curve itself.

Definition 48.3.1: Canonical Divisor

A *canonical divisor* on C is a divisor K with the property that

$$\ell(D) - \ell(K - D) = \deg D + 1 - g$$

for every divisor D. Its class in $\mathrm{Pic}\,(C)$ is the *canonical class.*

Stated this way the definition presumes what has to be proved, namely that such a K exists. Theorem 46.2.3 has already constructed the class: it is the divisor class of any nonzero rational differential form, and theorem 46.2.4 computes it for a plane curve. What is still unproved at this point is that the class so constructed satisfies the displayed identity, which is the content of theorem 48.3.2.

Theorem 48.3.2: Riemann-Roch

Let C be a nonsingular projective curve of genus g. Then a canonical divisor K exists, and for every divisor D,

$$\ell(D) - \ell(K - D) = \deg D + 1 - g$$

Moreover $\deg K = 2g - 2$ and $\ell(K) = g$.

The last two assertions are consequences of the first. Taking $D = 0$ gives $1 - \ell(K) = 1 - g$, so $\ell(K) = g$; taking $D = K$ gives $\ell(K) - 1 = \deg K + 1 - g$, and substituting $\ell(K) = g$ leaves $\deg K = 2g - 2$. Riemann's theorem follows too: $\ell(K - D) \geq 0$ always, so $\ell(D) \geq \deg D + 1 - g$, and once $\deg D > 2g - 2$ the divisor $K - D$ has negative degree, so $\ell(K - D) = 0$ by proposition 48.1.2(3) and the inequality becomes an equality. That is the precise form of "sufficiently large degree" in theorem 48.2.1.

Proposition 48.3.3: The Canonical Divisor of a Plane Curve

Let C be a nonsingular plane curve of degree d and H a line. Then the canonical class of C is

$$K = (d - 3)\, H \bullet C$$

***Proof*:**
This is theorem 46.2.4, proved in chapter 46 by exhibiting the form $\omega = dx/\partial_y f$ and computing its divisor: it has neither zeros nor poles in the affine chart, because C is nonsingular and its two partials cannot vanish together, and it acquires a zero of order $d - 3$ at each point of C at infinity.

The degrees are consistent with theorem 48.3.2, as they must be: $H \bullet C$ has degree d by theorem 43.1.1, so the right side has degree $d(d - 3)$, while $\deg K = 2g - 2$ with $g = \binom{d-1}{2}$ by corollary 48.2.3 gives

$$2g - 2 = (d - 1)(d - 2) - 2 = d^2 - 3d = d(d - 3)$$

completing the proof.

The degree check alone would not have sufficed: a divisor class is not determined by its degree,

$\operatorname{Pic}^0(C)$ being generally nonzero, and on a cubic theorem 45.2.2 makes it as large as the curve. It is the explicit form ω that identifies the class.

The formula explains the numerology of low-degree plane curves at a glance. For $d = 1$ and $d = 2$ the canonical class has negative degree, for $d = 3$ it is zero, and for $d \geq 4$ it is positive. Those three regimes are the trichotomy that organizes the classification of curves, and the next section is the case analysis.

48.4 Application: Curves of Low Genus

Riemann-Roch is used the same way every time: choose D so that $\ell(K - D)$ vanishes for degree reasons, read off $\ell(D)$, and interpret a function in $L(D)$ geometrically. One input is needed from outside, and it is the only one.

48.4.1 Note: The Degree of a Map to the Line Let $f \in k(C)$ be nonconstant. By lemma 42.2.8 it defines a morphism $F\colon C \to \mathbb{P}^1_k$, and the fact used below is that

$$\deg \operatorname{div}_\infty(f) = \big[\, k(C) : k(f) \,\big]$$

so that a function with a single simple pole generates the whole function field. Both sides count the fibre of F, one geometrically and one algebraically; the proof is in [54, chapter 8].

Theorem 48.4.2: Genus Zero Curves Are Rational

Let C be a nonsingular projective curve of genus 0. Then $C \cong \mathbb{P}^1_k$.

***Proof*:**
Pick any $p \in C$ and apply theorem 48.3.2 with $D = p$. Then $\deg(K - p) = -2 < 0$, so $\ell(K - p) = 0$ by proposition 48.1.2(3), and

$$\ell(p) = \deg(p) + 1 - 0 = 2$$

Since $L(0) = k$ has dimension 1 by proposition 48.1.2(2), there is a nonconstant $f \in L(p)$. Its only possible pole is a simple one at p, and it must occur, since otherwise $\operatorname{div}(f) \geq 0$ would make f constant. So $\operatorname{div}_\infty(f) = p$ has degree 1, and box 48.4.1 gives $k(C) = k(f)$, that is C is birational to $\mathbb{P}^1_k$. A birational map between nonsingular projective curves is an isomorphism by lemma 42.2.8 applied in both directions, completing the proof.

Example 48.4.3: Genus Zero, Worked
Take $C = \mathcal{Z}(x^2 + y^2 - z^2) \subseteq \mathbb{P}^2_k$, a nonsingular conic, so $d = 2$ and $g = \binom{1}{2} = 0$ by definition 44.4.1.

Follow the proof of theorem 48.4.2 with $p = [1 : 0 : 1]$. Riemann-Roch gives $\ell(p) = 2$, so there is a nonconstant f with at most a simple pole at p and no other pole. Concretely, in the affine chart $z = 1$ the function

$$f = \frac{y}{1 - x}$$

does it. Substituting the parameterization $x = (t^2 - 1)/(t^2 + 1)$, $y = 2t/(t^2 + 1)$ gives $f = t$ identically, and that parameterization sends $t = 0$ to $[-1 : 0 : 1]$ and $t = \infty$ to p. So

$$\operatorname{div}(f) = \big([-1 : 0 : 1]\big) - \big([1 : 0 : 1]\big)$$

a single simple zero and a single simple pole. Hence $\deg \operatorname{div}_\infty(f) = 1$ and box 48.4.1 gives $k(C) = k(f)$.

So the abstract argument and the explicit parameterization of example 33.3.2(1) are the same map, and the genus is what guaranteed in advance that one exists.

This settles the debt left in section 47.4: the smooth model of a plane curve attaining the bound of corollary 43.2.5 has genus 0 and is therefore rational, so the extremal curves are exactly the rational ones.

Theorem 48.4.4: Genus One Curves Are Plane Cubics

Let C be a nonsingular projective curve of genus 1 and $O \in C$. Then C is isomorphic to a nonsingular plane cubic, embedded by $L(3O)$, and O maps to an inflection point.

***Proof*:**
Here $\deg K = 2g - 2 = 0$ and $\ell(K) = g = 1$, so $L(K)$ contains a nonzero f with $\operatorname{div}(f) + K \geq 0$ of degree 0, hence $\operatorname{div}(f) + K = 0$ and $K \sim 0$. So for any D of positive degree, $\deg(K - D) < 0$ and theorem 48.3.2 reads

$$\ell(D) = \deg D$$

Applying this to O, $2O$ and $3O$ gives dimensions $1, 2, 3$. Choose $x \in L(2O) \setminus L(O)$ and $y \in L(3O) \setminus L(2O)$, so that x has a double pole at O and y a triple one.

Now count in $L(6O)$, of dimension 6. The seven functions

$$1,\ x,\ y,\ x^2,\ xy,\ x^3,\ y^2$$

all lie in $L(6O)$, their pole orders at O being $0, 2, 3, 4, 5, 6, 6$. Seven vectors in a six-dimensional space are dependent, and any dependence must involve both x^3 and y^2, since they are the only two with a pole of order exactly 6 and every other listed function has a distinct pole order. Normalizing their coefficients gives a relation

$$y^2 + a_1xy + a_3y = x^3 + a_2x^2 + a_4x + a_6$$

which is a plane cubic W in Weierstrass form.

The map $p \mapsto [x(p) : y(p) : 1]$, defined on $C \setminus \{O\}$ and regular everywhere by lemma 42.2.8, has image in W. It is birational onto W: the subfield $k(x, y) \subseteq k(C)$ satisfies $[k(C) : k(x)] = 2$ and $[k(C) : k(y)] = 3$ by box 48.4.1, since $\deg \operatorname{div}_\infty(x) = 2$ and $\deg \operatorname{div}_\infty(y) = 3$, so $[k(C) : k(x, y)]$ divides both 2 and 3 and is therefore 1.

Hence W is birational to C, and W is not rational: a rational W would make C birational to $\mathbb{P}^1_k$ and so isomorphic to it by lemma 42.2.8, giving C genus 0 rather than 1.

But a singular irreducible plane cubic *is* rational. If q is a singular point of such a cubic W', then $m_q(W') \geq 2$, so every line L through q has $I(q, W' \cap L) \geq 2$ by property (6) of the intersection number, and $\deg W' \deg L = 3$, so theorem 43.1.1 leaves exactly one further intersection point, counted with multiplicity. Sending L to that point is a birational map $\mathbb{P}^1_k \dashrightarrow W'$, the lines through q forming a $\mathbb{P}^1_k$. So W is nonsingular, and a birational map between nonsingular projective curves is an isomorphism by lemma 42.2.8 applied in both directions.

Finally, homogenizing the Weierstrass equation as $y^2z+a_1xyz+a_3yz^2 = x^3+a_2x^2z+a_4xz^2+a_6z^3$ and intersecting with the line $z=0$ leaves $x^3=0$, so that line meets W only at $[0:1:0]$ and with multiplicity 3. That point is the image of O, since x and y both blow up there with y of higher order. So O is an inflection point in the sense of definition 42.1.6, completing the proof.

That last sentence closes the loop with chapter 45. Lemma 45.1.2 took its simplest form when the base point was an inflection point, and the Weierstrass embedding produces one for free: any point of a genus-one curve can be made the inflection at infinity. So the classical picture, in which three collinear points on a cubic sum to zero, is available for every choice of base point after re-embedding.

48.4.5 Note: The Genus Trichotomy The three cases are governed by the sign of $\deg K = 2g-2$.

- $g=0$: $\deg K = -2$. The curve is $\mathbb{P}^1_k$ by theorem 48.4.2, there is exactly one such curve, and its group Pic^0 is trivial by example 44.3.2.
- $g=1$: $\deg K = 0$. The curve is a plane cubic by theorem 48.4.4, there is a one-parameter family of them, and $\operatorname{Pic}^0(C) \cong C$ by theorem 45.2.2.
- $g \geq 2$: $\deg K = 2g-2 > 0$. The canonical class itself embeds the curve for $g \geq 3$ non-hyperelliptic, the curves form a $(3g-3)$-dimensional family, and Pic^0 is an abelian variety of dimension g, the Jacobian.

Only the first two rows are proved here. The third is the beginning of the theory of moduli and of abelian varieties, and it is where this book stops.

Exercise 48.4.1

1. Show that $\ell(D)=0$ when $\deg D = 0$ and $D \not\sim 0$, and $\ell(D)=1$ when $D\sim 0$.
2. Let C have genus g and let D be effective of degree $\deg D > 2g-2$. Show $\ell(D) = \deg D + 1 - g$ exactly, and deduce $\ell(K) = g$ without using theorem 48.3.2 twice.
3. Compute $\ell(nO)$ for all $n \geq 0$ on a nonsingular plane cubic, and check the answer against example 48.1.3 to see where the two curves first differ.
4. A curve of genus 2 has $\deg K = 2$ and $\ell(K)=2$. Deduce that it admits a degree-two map to $\mathbb{P}^1_k$, so that every genus-two curve is hyperelliptic.
5. Show that a nonsingular plane quartic has genus 3 and that its canonical class is $H \bullet C$, so that the canonical map is the original embedding. Conclude that no nonsingular plane quartic is hyperelliptic.
6. Use theorem 48.4.4 to show that the group $E_k(C,O)$ of chapter 45 is defined for any nonsingular projective curve of genus 1, not just for those presented as plane cubics.

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What Next?

This book has worked over a fixed algebraically closed field, with varieties as sets of points and rings of functions on them. That framework is what makes the subject concrete, and every one of its limitations points at a successor theory.

Three Things This Book Could Not Say

Three separate difficulties have surfaced, and it is worth collecting them, because each is resolved by a different extension.

Multiplicity is not carried by the points. Chapter 42 had to define a plane curve as a polynomial up to scalar rather than as its zero set, precisely because $\mathcal{Z}(f)$ and $\mathcal{Z}(f^2)$ are the same set of points while the two curves ought to be different. Every intersection number in chapter 43 was then computed from the ideal, not from the set, and the book never had an object whose points and whose multiplicities were carried together.

The field is fixed and algebraically closed. The Nullstellensatz (theorem 32.4.4) is the foundation of the whole algebra-geometry dictionary, and it fails without algebraic closure. So nothing here applies to the geometry of equations over $\mathbb{Q}$ or over a finite field, and those are exactly the cases number theory treats.

Nilpotents are invisible. Chapter 32 matched varieties with *radical* ideals, discarding everything else. A family of varieties degenerating to a limit, say two points colliding or a smooth conic degenerating to a double line, has a limit whose natural equation is not radical, and the theory built here cannot see the difference between that limit and the reduced set underneath it.

Schemes

All three difficulties have one resolution: stop insisting that the geometric object be a set of points with a ring of functions on it, and let an arbitrary commutative ring define a space directly. The points of $\operatorname{Spec} R$ are the prime ideals of R, the structure sheaf is built from localizations, and the Nullstellensatz is no longer needed because the dictionary is definitional rather than a theorem.

That move recovers everything above as a special case. A variety over k becomes a scheme whose ring is a finitely generated reduced k-algebra, and dropping either adjective is what gives the three resolutions: dropping "reduced" makes $\operatorname{Spec} k[x,y]/(f^2)$ a genuinely different object from $\operatorname{Spec} k[x,y]/(f)$, and dropping "over an algebraically closed k" allows $\operatorname{Spec} \mathbb{Z}$ and its relatives. Chevalley's theorem (theorem 41.3.4) survives verbatim, the Picard group of section 44.3 becomes a functor, and Riemann-Roch becomes a statement about the cohomology of a sheaf.

This is *Algebraic Geometry* [5], which is the direct continuation of this book and the one to read next. Its prerequisite is exactly what has been developed here: the reason to endure the machinery of schemes is that one already knows what it is supposed to do.

Four Directions

Schemes are the trunk. Four branches lead off from the material of this book in particular.

The analytic theory. Over $\mathbb{C}$ a nonsingular projective curve is a compact Riemann surface, and the genus of chapter 48⋆ is its topological genus, the number of handles. The Riemann-Roch theorem is then a statement about holomorphic differentials, proved by analysis rather than by Max Noether's theorem, and the group law of chapter 45 becomes the quotient $\mathbb{C}/\Lambda$ by a lattice. *Complex Analysis* [40] develops this. The two theories give the same answers because the analytic and algebraic categories agree for projective varieties, which is a theorem and not merely a correspondence of vocabulary.

Intersection theory. Chapter 43 counted intersections of plane curves, and section 41.4 counted lines on a cubic surface, but both counts were run by hand. The general machinery replaces the intersection cycle of section 43.3 by the Chow ring, in which classes of subvarieties multiply and the answer to an enumerative question is a coefficient. The self-intersection self-intersection $E \bullet E = -1$, which box 47.4.2 explains this book does not need but which any surface theory must supply, is among its first computations. This is *Intersection Theory* [16].

Toric varieties. Sections 40.1 and 41.1 constructed embeddings by writing down monomials, and the varieties so obtained are governed entirely by the combinatorics of which monomials appear. Toric geometry makes that systematic: a fan of cones determines a variety, and its geometry (its singularities, its divisors, its cohomology) is read off the combinatorics. It is the best available source of examples that are complicated enough to be interesting and explicit enough to compute with. This is *Toric Geometry* [24].

Abelian varieties. Theorem 45.2.2 identified a cubic curve with its own Pic^0. For a curve of genus $g \geq 2$ that fails, and Pic^0 becomes a g-dimensional projective variety with a group law, the Jacobian. Abelian varieties are the projective group varieties, and they are where the group law of chapter 45 generalizes. This is *Abelian Varieties* [4]; the arithmetic of the genus-one case is *Elliptic Curves* [12].

Two Books to Read Alongside

Two external texts have been cited throughout and are worth naming as companions rather than as references.

Fulton's *Algebraic Curves* [54] is the source for chapters 42 and 43 and for the parts of chapter 48⋆ that were stated rather than proved. It is short, entirely classical, and available freely; a reader who wants the adjoint-curve proof of Riemann-Roch should go there first.

Shafarevich's *Basic Algebraic Geometry* [46] covers the varieties half of this book at greater length and with more examples, and its second volume treats schemes and complex manifolds side by side. Where this book has compressed, Shafarevich has room. Hartshorne [31] remains the standard reference for the scheme-theoretic account, but its first chapter is a compressed version of what has been done here, and it is easier read second.

50

Relating Algebra to Number Theory

A central object of study in mathematics is, naturally, the natural numbers $\mathbb{N}$ and, if the negatives are included, the integers $\mathbb{Z}$. So far, it is arguable to say that these objects have played a relatively minor algebraic role in this book: we know $\mathbb{N}$ (resp. $\mathbb{Z}$) is the initial abelian monoid (resp. ring) in their respective categories, that they are free objects, and that from $\mathbb{N}$ we may construct $\mathbb{Z}$, then $\mathbb{Q}$, and from there is any other field of characteristic 0. One aspect of $\mathbb{N}$ and $\mathbb{Z}$ we have barely touched on are their *primes* (recall the Fundamental Theorem of Arithmetic's, corollary 10.1.34). Exercise ref:HERE showed that there are infinitely many primes, and since $\mathbb{Z}$ is a PID we have that being irreducible is equivalent to being prime. However, we have explored little the nature of primes, and more generally have explored little of functions with integer solutions (elements of $f \in \mathbb{Z}[x_1, ..., x_n]$ are called *Diophantine equations*).

Unlike fields, it is in general much harder to ascertain whether an algebraic equation has solutions over the integers (or even integral extensions of the integers). Perhaps the most famous number theory problem demonstrating this difficulty is the following:

Show that for $n \geq 3$, the equation $x^n + y^n = z^n$ has no integer solutions

This problem is known as *Fermat's Last Theorem*. This is trivial if $n = 1$ (ex. $1 + 2 = 3$). It is more interesting to see the case of $n = 2$, that is we want to find all $(a, b, c) \in \mathbb{Z}^3$ ($\gcd(a, b, c) = 1$) such that $x^2 + y^2 = z^2$. Such a triple is called a *Pythagorean triple* and there are infinitely many. However, for $n \geq 3$, it is incredibly more difficult to find whether solutions exist or not, partly due to the fact that many rings that naturally arise in solving this case are *not* unique factorization domains, and partly because there turn out to be some "bad number" that make many of the number theory strategies fail. This particular problem has spurred a lot of the development of modern algebraic and analytic number theory, and shall be touched on later due to some interesting general theory that came from it. In this section we shall cover the mathematics that build built-up in solving Diophantine equations, including some special cases of Fermat's last theorem, and prepare the reader for Class Field theory which shall definitively answer the question of factoring integer-polynomials mod primes[1].

One final intuition that shall become more evident is that the theory of prime factorization in field extensions tie into solving irreducible polynomials mod some prime ideals. This theory extends results from the classical results of solutions or lack thereof for an irreducible polynomial over $\mathbb{Q}$ when considering it mod p for all primes $p \in \mathbb{Z}$. Recall that factorization and irreducibility of monic polynomials over $\mathbb{Q}$, more generally a finite extension $K/\mathbb{Q}$ is exactly reflected to the factorization and irreducibility of monic polynomials over $\mathbb{Z}$ (which will have a ring $\mathcal{O}_K$ that shall be the equivalent[2]) by Gauss's lemma (proposition 11.2.1). This was the historical origin of number theory, as it is much easier to work with polynomials finite fields (recall proposition 11.3.8 and example 11.3.9).

Material Overview

In this chapter, we shall start by exploring when do $\mathbb{Z}$-polynomials solutions over a field k. The space of solutions shall naturally be a subset of the field k. It shall be labeled as $\mathcal{O}_K$ and called *rings of integers*, and will have many important properties to determine. Similarly to how we extended fields to find solutions to polynomials over fields, rings of integers shall be extended (in parallel to a field extension) to find more solutions. Some key functionals such as the trace, norm, and discriminant shall become important tools to find properties of elements of $\mathcal{O}_K$ and the structure of $\mathcal{O}_K$ itself. As we classify $\mathcal{O}_K$, we shall see that we may consider $\mathcal{O}_K$ as a *lattice* when embedded in $\mathbb{R}^{r_1} \oplus \mathbb{C}^{r_2}$ for appropriate $r_1, r_1 \in \mathbb{N}$. This geometric interpretation of $\mathcal{O}_K$ shall add some important new perspectives onto the elements of $\mathcal{O}_K$[3], and will allow us to define some important inequalities.

After having done a sweep over the structure of $\mathcal{O}_K$, we shall tackle some of it's algebraic structure. Namely, we shall see that they form a new type of ring called a *Dedekind domain*. Dedekind domains are very important types of rings; for those who are keen on algebraic geometry, Dedekind domains are the algebraic counter-part to 1-dimensional smooth curves (or schemes). In fact, we have seen Dedekind domains before in the previous chapter (for example, $\mathbb{Q}[x, y]/(x - y^2)$). Hence, Dedekind domains will be studied in their own right. They shall have a very rich structure: for example, many $\mathcal{O}_K$ will *not* be unique factorization domain (a fact that led to many wrong proofs of Fermat's last theorem), however we shall be able to define an abelian group structure on the ideals of $\mathcal{O}_K$ which *will* have a unique factorization (in the case where $\mathcal{O}_K$ is a PID, we shall see the ideal group I_K is isomorphic to $\mathbb{Q}^\times$, exactly mirroring the structure of Frac($\mathbb{N}$), however in general it may be much more complicated[4]). The discussion on Dedekind domains shall finish with a discussion of the unit's in Dedekind rings, which shall be important for finding the factorization of ideals.

After we realise we ought to be focusing on the *prime ideals* of $\mathcal{O}_K$ rather than the irreducible elements of $\mathcal{O}_K$ (for they are not necessarily prime), we shall start exploring how prime may further decompose as we extend to larger rings of integers. This helps in finding solutions to $\mathbb{Z}$-polynomials for the prime ideals in larger rings will have very rigid structure that can be taken advantage of. We have already seen this before: It shall turn out that $\mathcal{O}_{\mathbb{Q}(i)} = \mathbb{Z}[i]$ (the Gaussian integers). In the

[1] In particular, the *Artin Reciprocity Law* shall give the over-arching result showing how to factorize Diophantine equations

[2] See definition 50.1.3

[3] and of the norm

[4] which is measured by the *class group*, as we shall see

Gaussian integers, we saw that 2 is not prime, for

$$2 = (1+i)(1-i) = 1^2 - i^2 = 1 + 1$$

or in terms of ideals, we shall have $(2) = (1+i)(1-i)$. Hence, 2 has further factored into new primes. In section 10.3, this was used to prove Fermat's Theorem for sums of squares. More generally, we shall see that any prime ideal $\mathfrak{p} \subseteq \mathcal{O}_K$ may factor:

$$\mathfrak{p}\mathcal{O}_L = \mathfrak{B}_1^{e_1} \cdots \mathfrak{B}_k^{e_k} \tag{50.1}$$

Though it may currently seem unrelated, the study of the factorization of primes *almost* mirrors that of the factorization of irreducible polynomials mod $\mathfrak{p}$ after extending fields. In fact, it shall turn out to be the case that *all but finitely many primes* shall factor exactly like how a polynomial factors, the "bad primes" essentially being primes for which an exponent in equation (50.1) is greater than 1, ≥ 2. This sort of exploration leads to a lot of very interesting theory that have applications beyond classical number theory, for example we shall be able to fully classify all abelian extensions of $\mathbb{Q}$.

For those who get motivation through algebraic geometry, algebraic number theory naturally ties into it; it may help to think of the main object of study of this chapter (rings of integers) as further expanding the dictionary between commutative algebra and geometry, namely they shall be important for *normal schemes* which links to the notion of a smooth curve[5]

At the end of this chapter, we shall briefly touch on some analytic number theory, for it's study is central to modern number theory and the author of this books considers that anyone that has reached this stage of the book would find it enlightening to see how the Riemann-zeta function play an important role in number theory.

50.0.1 Warning Some elementary analysis will be assumed in this section, including some properties of the real numbers.

50.1 Ring of Integers

In this chapter, whenever the notation $L/K/\mathbb{Q}$ is used for some letter's L, K, it is implied that L/K are both finite extensions of $\mathbb{Q}$, and L is an extension of K. Unless otherwise specified, the characteristic of any field is implicitly assumed to be 0. We start by defining our terms.

Definition 50.1.1: Algebraic Number Fields

Let K be a field. If $K/\mathbb{Q}$ is a finite extension of $\mathbb{Q}$, then K is called an *algebraic number field* or *number field* if unambiguous.

Example 50.1.2: Number Fields

1. Let $\zeta_n = e^{2\pi/n}$ be a root of unity. Then $\mathbb{Q}(\zeta_n)$ is a number field, called the *Cyclotomic field.*
2. Let $m \in \mathbb{Z}$ be not-a-square. Then Fields of the form $\mathbb{Q}(\sqrt{m})$ is called a *quadratic field.* Such fields always have degree 2. Notice that we can consider any $m \in \mathbb{Z}$, however we get many

[5] See a textbook in algebraic geometry for further details

equality of fields, for example $\mathbb{Q}(\sqrt{12}) = \mathbb{Q}(\sqrt{3})$. The field $\mathbb{Q}(i)$ is both a quadratic field and a Cyclotomic field since $i = \zeta_4$. As an exercise, show that $\mathbb{Q}(\sqrt{-3})$ is a Cyclotomic field.

3. The field $\mathbb{Q}(\sqrt{n}, \sqrt{m})$ is called a *biquadratic* field where n, m are both not-a-square.
4. The field $\mathbb{R}$ and $\mathbb{C}$ are *not* algebraic number fields, are they are not algebraic extensions of $\mathbb{Q}$ (they both contain transcendental elements).
5. Similarly, $k(x)$ for any field k is not an algebraic number field (it is an [algebraic] function field). However, in many ways this field is very similar to algebraic number fields, and shall be given some more attention in section 50.2.

Of key interest to this chapter the study of integer polynomials with solutions fields:

Definition 50.1.3: Ring of Integers (of Number Fields)

Let K be a field of characteristic 0 and identify $\mathbb{Z} \subseteq K$ via the (unique) map $\mathbb{Z} \hookrightarrow K$. Then the *ring of integers* of K is the integral closure of $\mathbb{Z}$ over K (definition 26.1.3):

$$\mathcal{O}_K := \overline{\mathbb{Z}}^K = \{a \in K \mid f(a) = 0,\ f(x) \in \mathbb{Z}[x]\}$$

As all fields in consideration shall be number fields $K/\mathbb{Q}$, there is already $\mathbb{Z} \subseteq K$ identified; the generality in the definition is to show that the ring of integers is defined over any field of characteristic 0. Note that there is an equivalent definition for fields of positive characteristic, however as $\operatorname{im}(\mathbb{Z}) \cong \mathbb{Z}/p\mathbb{Z}$ any integral closure of $\mathbb{Z}/p\mathbb{Z}$, $\overline{\mathbb{Z}/p\mathbb{Z}}^K$, is a field extension by proposition 26.1.8 and so there is no notion of prime ideals which essentially erases the key extra information that is studied. Instead, $\mathbb{F}_p[t]$ becomes the new $\mathbb{Z}$ and a parallel theory of finite extension $K/\mathbb{F}_p(t)$ is studied; these are called *algebraic function fields* and the corresponding $\overline{\mathbb{F}_p[t]}^K$ is the *ring of integers of function fields.*. The two theories have many parallels, but exploring these is beyond the scope of this book, see [39] for more details.

Although, once this chapter becomes it's own book, it may be a good idea to dedicate a section or chapter to this.

By corollary 26.1.6, $\mathcal{O}_K \subseteq K$ is a ring. Since we will often work over $\mathbb{Z}$, it is useful to know that instead of taking all roots of polynomials, it is equivalent to take all the roots of *irreducible polynomials*; by using Gauss's lemma, if $f = gh$, $f \in \mathbb{Z}[x]$ and $g, h \in \mathbb{Q}[x]$, then in fact $g, h \in \mathbb{Z}[x]$. Hence, the smallest degree polynomial for which α is a root will be irreducible. An important ring of integer that is worth isolating is the following:

Definition 50.1.4: Algebraic Integers [of Number Fields]

The collection of elements

$$\mathcal{O}_{\overline{\mathbb{Q}}} = \mathcal{O}_{\mathbb{C}} := \overline{\mathbb{Z}}^{\mathbb{C}}$$

are called the *algebraic integers*, and is sometimes denoted $\mathbb{A}$[a].

[a] this notation comes from the theory of *idele rings* that is an important tool in class field theory, see [39]

Just like for fields, there may be multiple "integral closures", and they are all isomorphic which motivates saying they are *the* algebraic integers.

Lemma 50.1.5: Equivalence Definition

Let $K/\mathbb{Q}$ be a finite extension and let $\mathcal{O}_\mathbb{C}$ be the algebraic integers. Then:

$$\mathcal{O}_\mathbb{C} \cap K = \mathcal{O}_K$$

***Proof*:**
exercise.

Example 50.1.6: Ring of Integers

1. $\mathcal{O}_\mathbb{Q} = \mathbb{Z}$. For this reason, the integers in algebraic number theory are sometimes called the rational integers

2. $\mathcal{O}_{\mathbb{Q}(i)} = \mathbb{Z}[i]$. To see this, let $\alpha = a + bi$, $a, b \in \mathbb{Q}$. Then for $\alpha \in \mathcal{O}_{\mathbb{Q}(i)}$, we need to show that the minimal polynomial of α:

$$\alpha^2 - 2a\alpha + (a^2 + b^2) = 0$$

has integer coefficients, that is $-2a$ and $a^2 + b^2$ are integers. First, if $c, d \in \mathbb{Z}$, $c + di \in \mathcal{O}_{\mathbb{Q}(i)}$. Since $\mathcal{O}_{\mathbb{Q}(i)}$ is a ring, if $a + bi \in \mathcal{O}_{\mathbb{Q}(i)}$, then so is $a + bi - c - di = a' + b'i$ for $c, d \in \mathbb{Z}$ where $a', b' \leq 1$. But then from $(a')^2 + (b')^2 = k \in \mathbb{Z}$, we get $a', b' \in \{-1, 0, 1\}$. Then that implies $a, b \in \mathbb{Z}$, showing that $\mathcal{O}_{\mathbb{Q}(i)} = \mathbb{Z}[i]$.

3. In general, it is difficult to find the ring of integers. The difficulty of finding the ring of integers can even show up for quadratic extensions. Consider $F = \mathbb{Q}(\sqrt{-3})$. Then we might be tempted to say that $\mathcal{O}_F = \mathbb{Z}[\sqrt{-3}]$. However, this is not the case, take:

$$\omega = \frac{-1 + \sqrt{-3}}{2} \in \mathbb{Q}(\sqrt{-3})$$

Then $\omega^2 + \omega + 1 = 0$, however $\omega \notin \mathbb{Z}[\sqrt{-3}]$.

4. In general, we will need more tools to find $\mathcal{O}_k$, but in the quadratic case, since the polynomial are relatively simple to work with, we may find all rings of integers. Let $k = \mathbb{Q}(\sqrt{m})$. Then:

$$\mathcal{O}_k = \mathbb{Z}[\sqrt{m}] \qquad m \equiv 2, 3 \bmod 4$$
$$\mathcal{O}_k = \mathbb{Z}\left[\frac{1 + \sqrt{m}}{2}\right] \qquad m \equiv 1 \bmod 4$$

To see this, let $\alpha = r + s\sqrt{m}$, $r, s \in \mathbb{Q}$. If $s = 0$, we've done, hence if $s \neq 0$ we get the monic irreducible polynomial:

$$x^2 - 2rx + r^2 - ms^2$$

Hence, α is algebraic if and only if $2r$ and $r^2 - ms^2$ are integers. Since $2r \in \mathbb{Z}$, $r \in \frac{1}{2}\mathbb{Z}$. Considering now $r^2 - ms^2 \in \mathbb{Z}$, if $r = \alpha/2$ for $\alpha \in \mathbb{Z}$, then subtracting both sides by $(\alpha^2 - 1)/4$ we get that $1/4 - ms^2 \in \mathbb{Z}$ or $m^2 = \mathbb{Z} + 1/4$. Now, this equality holds true if $s = \beta/2$ and $m \equiv 1 \bmod 4$ or $s = \beta$ and $m \equiv 2, 3 \bmod 4$. This now gives our restrictions on our coefficients.

The ring of integers $\mathcal{O}_K/\mathbb{Z}$ hold the core information of $K/\mathbb{Q}$. In particular, if $\mathbf{Fld}_0^{\text{fin}}$ is the category whose objects are finite extensions of $\mathbb{Q}$ and morphisms are field morphisms and $\mathbf{AlgInt}_0^{\text{fin}}$ is the category whose objects are the rings of integers $\mathcal{O}_K$ of finite extensions $K/\mathbb{Q}$ and whose morphisms are ring homomorphisms, these two categories are *isomorphic*. From here onto section 50.1.1 this correspondence is proven. Some of the key facts will require results that are proved later in this section (ex. that all non-trivial ring homomorphism $\mathcal{O}_K \to \mathcal{O}_L$ are injective), however knowing this correspondence is valuable enough that it is worth introducing this result early.

First, if $\varphi : F \to L$ be a field homomorphism (not necessarily a k-homomorphism), then $\varphi|_{\mathcal{O}_F} : \mathcal{O}_F \to \mathcal{O}_L$ is certainly well defined since if $\alpha \in \mathcal{O}_F$, then:

$$a_0 + a_1\alpha + \cdots + a_{n-1}\alpha^{n-1} + \alpha^n = 0 \qquad a_i \in \mathbb{Z}$$

Then $\varphi(a_i\alpha^i) = a_i\varphi(\alpha)^i$, since $a_i \in \mathbb{Z}$. Hence $\varphi(\alpha) \in \mathcal{O}_L$. From this, define $\mathcal{O}_- : \mathbf{Fld}_0^{\text{fin}} \to \mathbf{AlgInt}_0^{\text{fin}}$ which gives the commutative diagram:

$$\begin{tikzcd} F \arrow[r, hook, "\varphi"] \arrow[d, "\mathcal{O}_-"] & L \arrow[d, "\mathcal{O}_-"] \\ \mathcal{O}_F \arrow[r, "\varphi|_{\mathcal{O}_F}"] & \mathcal{O}_L \end{tikzcd}$$

Now given a map $\psi : \mathcal{O}_F \to \mathcal{O}_K$, can we find a map $\Psi : F \to K$ which is furthermore an inverse to the above functor? Yes; to find this functor, we first find how to retrieve the field from the ring of integers:

Proposition 50.1.7: Field of Fraction of Integral Closures of Extensions

Let R be an integral domain, $K = K(R)$ its field of fraction, and $S = \overline{R}^L$ where L/K is an algebraic extension. Then $L = K(S)$. Equivalently, $L = \langle S \rangle_{K(R)}$.

If $R = \mathbb{Z}$ and $K(R) = \mathbb{Q}$ so that $S = \overline{\mathbb{Z}}^L = \mathcal{O}_L$, then $L = K(\mathcal{O}_L)$.

***Proof*:**
Let R, K, S and L be defined as in the question. We shall first show that if $\alpha \in L$ is algebraic over K, then there exists a $d \in R$ such that $d\alpha$ is integral over R. Since α is algebraic over K,

$$\alpha^m + a_1\alpha^{m-1} + \cdots + a_m = 0 \qquad a_i \in K$$

Let d be the least common multiple of the denominators of a_i so that $da_i \in R$ for all i. Then multiplying the above equation by d^m, we get:

$$d^m\alpha^m + d^m a_1\alpha^{m-1} + \cdots + d^m a_m = 0 \qquad a_i \in K$$

re-writing:

$$(d\alpha)^m + da_1(d\alpha)^{m-1} + \cdots + d^m a_m = 0 \qquad a_i \in K,$$

where each $a_i d, ..., a_m d^m \in R$, hence $d\alpha$ is integral over R. Note that this immediately gives the equivalence $L = \langle S \rangle_{K(S)}$.

With that, to see that $L = K(S)$, notice that every $\alpha \in L$ can be written as $\alpha = b/d$ for $d \in R$, hence $L \subseteq K(S)$. Conversely, if $a/b \in K(S)$, where $a, b \in S$ and each satisfy an R-polynomial, then they are certainly satisfy a K-polynomial, and furthermore S is all elements of L that satisfy an R-polynomial, hence $a/b \in L$, giving equality.

Corollary 50.1.8: Basis For Number Field

Let $F/\mathbb{Q}$ be a finite extension of degree n and $(\alpha_1, \alpha_2, ..., \alpha_n)$ an $\mathbb{Q}$-basis for F. Then there exists an $m_i \in \mathbb{Z}$ such that $m_1\alpha_1, m_2\alpha_2, ..., m_n\alpha_n \in \mathcal{O}_F$

***Proof*:**
let $R = \mathbb{Z}$ in proposition 50.1.7.

Theorem 50.1.9: Isomorphic Categories: Fields and Rings of Integers

Let $K/\mathbb{Q}$ be a number field and $\mathcal{O}_K$ the ring of integers of K. Then:

1. If L/K, then $\mathcal{O}_L \cap K = \mathcal{O}_K$. In particular $\mathcal{O}_K \cap \mathbb{Q} = \mathbb{Z}$.
2. $\mathcal{O}_K \otimes_{\mathbb{Z}} \mathbb{Q} \cong K$. Namely, $\mathcal{O}_{-} \otimes_{\mathbb{Z}} \mathbb{Q} : \mathbf{AlgInt}_0^{\text{fin}} \to \mathbf{Fld}_0^{\text{fin}}$ is a functor.

Giving us a correspondence that is covariant under inclusion:

$$\left\{ \begin{matrix} \varphi : F \to K \\ \text{char}(F) = \text{char}(K) = 0 \end{matrix} \right\} \underset{(-)\otimes_{\mathbb{Z}}\mathbb{Q}}{\overset{\overline{\mathbb{Z}}^{(-)}}{\rightleftarrows}} \left\{ \begin{matrix} \varphi : \mathcal{O}_F \to \mathcal{O}_K \\ K(\mathcal{O}_F) = F, \ K(\mathcal{O}_K) = K \end{matrix} \right\}$$

which gives a bijection on Hom's:

$$\text{Hom}_{\mathbf{Fld}_0^{\text{fin}}}(L_1, L_2) \xrightarrow{\sim} \text{Hom}_{\mathbf{AlgInt}_0^{\text{fin}}}(B_1, B_2)$$

And hence there is an isomorphism of categories that preserves the lattice structure (meets and joins):

$$\mathbf{Fld}_0^{\text{fin}} \cong \mathbf{AlgInt}_0^{\text{fin}}$$

***Proof*:**
For the first two points:

1. This is essentially by definition: $\mathcal{O}_k = \overline{\mathbb{Z}}^k$ are all integral elements over $\mathbb{Z}$ that belonging k. Hence,$\mathcal{O}_k \cap \mathbb{Q}$ must be integral over k and also rational. But the only integral elements that are rational are the integers, hence we have equality. Similarly, $\mathcal{O}_L \cap k$ are all integral elements over $\mathbb{Z}$ that also belong to k, which by definition is $\mathcal{O}_k$
2. This comes from proposition 23.4.10 and its corollary.

To show the hom's are bijective, we would have to show that a non-trivial ring homomorphism $\varphi : \mathcal{O}_K \to \mathcal{O}_L$ for some two fields K, L is injective and that $L/K/\mathbb{Q}$ is a finite extension: both of these are a direct consequence of the proof of theorem 50.2.6. The lattice structure is preserved as $- \cap K$ and $- \otimes \mathbb{Q}$ preserves the lattice structures. Finally, composing these two functors certainly give the identity on $\mathbf{Fld}_0^{\text{fin}}$ and $\mathbf{AlgInt}_0^{\text{fin}}$, showing they are isomorphic (not just equivalent!) as we sought to show.

Note that finiteness is important: it shall be shown that $\mathcal{O}_K$ for $K/\mathbb{Q}$ a finite extension is a domain with factorization (or atomic domain, recall definition 10.1.9), however $\mathcal{O}_{\mathbb{C}}$ is *not* an atomic domain as shown in example 10.1.8 which is certainly desirable as we are generalizing the factorization of primes.

This shows how to relate $\mathcal{O}_K$ with K and a bit about how to move down the lattice, but we have not said much about the structure of $\mathcal{O}_K$. We know it is a ring, but is it a PID? A UFD? Every ring is a $\mathbb{Z}$-module; it is a finitely generated $\mathbb{Z}$-module? Noetherian $\mathbb{Z}$-module? Projective? The strategy in example 50.1.6 to find the structure of $\mathcal{O}_K$ gets much harder for more complicated roots of irreducible, even once we have degree 3, and especially beyond degree 4. Our first goal is to establish the structure of $\mathcal{O}_K$. To find the structure, we will require some tools for detecting elements of $\mathcal{O}_K$: we shall find numerical values for the elements of K to determine whether they are elements of $\mathcal{O}_K$, i.e. we shall find functionals. We shall see that $\mathcal{O}_K$ will be a free $\mathbb{Z}$-module of finite rank, in fact it shall be a particular $\mathbb{Z}$-module known as a *complete* lattice, as shall be demonstrated in section 50.1.4.

50.1.1 Functionals: Field Trace and Field Norms

(this link is useful: here)

Let M/k be a finite extension, and take $\alpha \in M$. We shall define two invariances on α that will detect whether α is algebraic over k or not, detect when it is an algebraic integer, and eventually help us find the structure of $\mathcal{O}_M$ as well as the factorization of elements of $\mathcal{O}_M$.

Define $L_\alpha : M \to M$ by $m \mapsto \alpha m$. This is a k-linear transformation, since:

$$L_\alpha(a + kb) = \alpha(a + kb) = \alpha a + k\alpha b = L_\alpha(a) + kL_\alpha(b)$$

Thus, we can represent this linear transformation in a matrix form (usually with basis given by α). Since the trace and determinant are basis independent calculations, it is meaningful to ponder what the trace/determinant of this transformation and what information it may provide us:

Definition 50.1.10: Trace And Norm

Let M/k be a finite extension, $L_\alpha : M \to M$ be a linear transformation defined by multiplication Then we call the trace of this linear transformation the *trace* of α and the determinant of this linear transformation the *norm* of α.

Example 50.1.11: Trace And Norm

1. Let $\mathbb{C}/\mathbb{R}$ be the extension and $\alpha = x + iy$. Pick basis $\{1, i\}$. Then multiplication is:

$$\begin{pmatrix} x & y \\ -y & x \end{pmatrix}$$

and so the trace is $2x = 2\text{Re}\alpha$ and the norm is $x^2 + y^2 = \|\alpha\|^2$. If you know some complex analysis, then we see that the norm does indeed match with our usual understanding of the norm, and the trace can be thought of as a generalization of taking the real part of the complex number (up to a factor).

2. Let $\alpha = a + b \cdot \left(\frac{1+\sqrt{-7}}{2}\right) \in \mathbb{Z}\left[\frac{1+\sqrt{-7}}{2}\right]$. Then:

$$\text{Nm}_{\mathbb{Q}(\sqrt{-7})/\mathbb{Q}}(\alpha) = \alpha^2 + \alpha\beta + 2\beta^2$$

3. Let $M = k(\alpha)$ be a finite extension, and let

$$m_{\alpha,k}(x) = x^n + a_{n-1}x^{n-1} + \cdots + a_1 x + a_0$$

be the minimal polynomial. Then this is the same as the characteristic polynomial (since plugging in L_α into the minimal polynomial produces the zero transformation by Cayley Hamilton [a]). Then the trace of α is $-a_{n-1}$ and the norm is $\pm a_0$ depending on whether $m_{\alpha,k}(x)$ is even or odd.

[a]Alternatively, choose basis $1, \alpha, ..., \alpha^{n-1}$ and compute

Naturally, if F/k, $a_1, a_2 \in F$ and $b \in k$:

$$\mathrm{tr}_{F/k}(a_1 + a_2) = \mathrm{tr}_{F/k}(a_1) + \mathrm{tr}_{F/k}(a_2) \quad \mathrm{tr}_{F/k}(ba_1) = b\mathrm{tr}_{F/k}(a_1) \quad \mathrm{tr}_{F/k}(a) = na$$

and:

$$\mathrm{Nm}_{F/k}(a_1 a_2) = \mathrm{Nm}_{F/k}(a_1)\mathrm{Nm}_{F/k}(a_2) \quad \mathrm{Nm}_{F/k}(ba_1) = b^n \mathrm{Nm}_{F/k}(a_1) \quad \mathrm{Nm}_{F/k}(b) = b^n$$

Most of the properties of the trace and norm come from linear algebra, and furthermore since every finite separable extension has a primitive element generating the entire field, the norm and trace always show up in the minimal polynomial. In section 20.2.3, we saw another way of characterizing the trace and norm, which will be emphasized here due to its importance:

Proposition 50.1.12: Trace and Norm via Galois Conjugates

Let M/k be a finite extension where $[M : k] = n$ and let $\alpha \in M$ be algebraic over k. Then:

$$\mathrm{tr}_{M/k}(\alpha) = [M : k(\alpha)] \cdot \sum_{\substack{r \\ \text{root of } m_{\alpha,k}}} r \qquad \mathrm{Nm}_{M/k}(\alpha) = \left(\prod_{\substack{r \\ \text{root of } m_{\alpha,k}}} r \right)^{[M:k(\alpha)]}$$

Proof:

Let $\sigma_i(\alpha) \in \overline{k}$ be the roots of $m_{\alpha,k}(x)$ where $\sigma_i : M \to \overline{k}$ are the embeddings of M into the algebraic closure of k. Suppose first that $M = k(\alpha)$ and is a separable extension. Pick a basis $1, \alpha, ..., \alpha^{n-1}$. Then concretely computing the trace, we get the $n-1$th coefficient of

$$\alpha^n - a_{n-1}\alpha^{n-1} - \cdots - a_1\alpha - a_0 = 0$$

which, if we factor out the above polynomial in a normal closure and collect the terms of the roots as we've done in section 20.2.3, we see that

$$\alpha_{n-1} = \sum_i \sigma_i(\alpha)$$

that is, it is the sum of the roots. Similarly, we get that $\mathrm{Nm}_{F/k}(\alpha) = a_0 = \prod_i \sigma_i(\alpha)$, giving us both results.

For the general finite extension case, we may use induction and transitivity. The inseparable case is left as an exercise.

Corollary 50.1.13: Trace and Norm Via Embeddings

Let L/k be a separable extension of degree n, an $\sigma_1, \sigma_2, ..., \sigma_n$ the embedding of L into it's Galois closure or $\overline{k}$. Then for $\alpha \in L$:

$$\mathrm{tr}_{L/k}(\alpha) = \sum_i^n \sigma_i(\alpha) \qquad \mathrm{Nm}_{L/k}(\alpha) = \prod_i^n \sigma_i(\alpha)$$

***Proof*:**
this is just a re-phrasing of the previous result. Note that there may be $\sigma_i(\alpha)$ that repeat since σ_i may be fixed in $k(\alpha)$.

This immediately shows us that for algebraic elements, $\mathrm{tr}_{M/k}(\alpha), \mathrm{Nm}_{M/k}(\alpha) \in k$, since $\pm a_{n-1}, a_0 \in k$. An easy way to remember that that the trace gives a_{n-1} and Norm gives a_0 is that it is not possible that a minimal polynomial have 0 constant term since

$$x^n + a_{n-1}x^{n-1} + \cdots + a_1 x$$

is reducible. Hence, $\mathrm{Nm}_{M/k}(\alpha) = a_1$ while $\mathrm{tr}_{M/k}(\alpha) = a_{n-1}$. To relate this back to our goal of understanding $\mathcal{O}_M$, we would like to identify when $\alpha \in M$ belongs in $\mathcal{O}_M$ using tr and Nm. This is exactly when the coefficients are integers, that is $\mathrm{tr}_{M/k}(\alpha), \mathrm{Nm}_{M/k}(\alpha) \in \mathbb{Z}$. We've in fact proved something slightly more general that is important to point out for when we are working with intermediary extensions:

Corollary 50.1.14: Trace and Norm on Rings of Integers

Let R be an integrally closed integral domain ($\overline{R}^K = R$), and let $F/K(R)$ be a finite field extension of the field of fraction of R. Then if $\alpha \in F$ is integral over R,

$$\mathrm{tr}_{F/K(R)}(\alpha) \in R \qquad \mathrm{Nm}_{F/K(R)}(\alpha) \in R$$

A particular consequence of this is $\mathrm{tr}_{L/k}(\mathcal{O}_L) \subseteq \mathcal{O}_k$

***Proof*:**
If α is integral, then certainly so are all its Galois conjugates. But then adding and multiplying all the Galois conjugates remains integral, meaning the coefficients of the corresponding polynomials remains in R, hence the trace and norm of α is in R.

In particular, if $\alpha \in \mathcal{O}_L = \overline{\mathbb{Z}}^L \subseteq \overline{k}^L$, $\mathrm{tr}_{L/k}(\alpha) \in \mathcal{O}_L \cap k = \mathcal{O}_k$ by theorem 50.1.9.

If $R = \mathbb{Z}$ and $F/\mathbb{Q}$ is a finite extension, we get can detect potential integral elements using the trace and norm! Since we may be working with intermediate fields $M/L/k$, it is useful to know how to compute the trace and norm given information about the trace and norm of the intermediate fields:

Proposition 50.1.15: Trace and Norm w.r.t. Towers

Let $M/L/K$ be finite extensions. Then:

$$\mathrm{tr}_{M/K}(\alpha) = \mathrm{tr}_{L/K}(\mathrm{tr}_{M/L}(\alpha)) \qquad \mathrm{Nm}_{M/K}(\alpha) = \mathrm{Nm}_{L/K}(\mathrm{Nm}_{M/L}(\alpha))$$

***Proof*:**
Let $\sigma_1, \sigma_2, ..., \sigma_n$ be the embeddings of L in $\mathbb{C}$ which fix K pointwise and let $\tau_1, \tau_2, ..., \tau_m$ be the embeddings in $\mathbb{C}$ fixing L pointwise. We will want to compose σ_i with the τ_j, so we will extend our embeddings as follows: Let N/M be the normal closure of M so that all σ_i, τ_j can be extended to automorphisms of N; relabel our extensions σ_i, τ_j (i.e. keep the same labeling). Then:

$$\mathrm{tr}_{L/k}(\mathrm{tr}_{M/L}(\alpha) = \sum_i^n \sigma_i \left(\sum_j^m \tau_j(\alpha) \right) = \sum_{i,j} \sigma_i \tau_j(\alpha)$$

$$\mathrm{Nm}_{L/k}(\mathrm{Nm}_{M/L}(\alpha) = \prod_i^n \sigma_i \left(\prod_j^m \tau_j(\alpha) \right) = \prod_{i,j} \sigma_i \tau_j(\alpha)$$

It only remains to show that the mappings $\sigma_i \tau_j$ when restricted to M give the embeddings of M in $\mathbb{C}$ fixing k pointwise. Certainly they fix k pointwise and there is the right number of them, so this follows by showing they are all distinct (TBD).

Note that it is not necessarily sufficient that the trace and norm is integral:

Example 50.1.16: Trace And Norm Insufficient To Detect Integral Elements
Let $L = \mathbb{Q}(\alpha)$ where $\alpha^4 + 2\alpha^3 + 1/2\alpha^2 + 1/2\alpha + 2\alpha = 0$. Note that $x^4 + 2x^3 + 1/2x^2 + 1/2x + 3$ is irreducible, hence this is a degree 4 extension. Furthermore,

$$\mathrm{tr}_{L/k}(\alpha) = 2 \qquad \mathrm{tr}_{L/K}(\alpha) = 3$$

But certainly $\alpha \notin \mathcal{O}_L$.

A notable exception to this is for quadratic extensions, since any extension of degree 2 is quadratic, giving only 2 possible coefficients:

Example 50.1.17: Rings of Integers of Quadratic Extensions
Let $K = \mathbb{Q}(\sqrt{n})$ where n is square free, and consider $\alpha = a + b\sqrt{n} \in \mathcal{O}_K$ where a, b are square free. Then:

$$\mathrm{tr}_{K/\mathbb{Q}}(\alpha) = 2a \in \mathbb{Z} \qquad \mathrm{Nm}_{K/\mathbb{Q}}(\alpha) = (a + b\sqrt{n})(a - b\sqrt{n}) = a^2 - nb^2 \in \mathbb{Z}$$

We may ask when $2a \in \mathbb{Z}$, that is when is it possible that $a = 1/2$? Let's say it was, so that

$$\left(\frac{1}{2}\right)^2 - nb^2 \in \mathbb{Z} \qquad \Leftrightarrow \qquad \frac{b^2 n}{4} \in \mathbb{Z} + \frac{1}{4}$$

hence, $4b^2 n \in \mathbb{Z}$. Since n is square-free, $n \neq m^2$ implying that we cannot have $b^2 = p^2/m^2 \cdot m^2$, so $2b \in \mathbb{Z}$. Thus, if $a = 1/2$, it must be that $b = 1/2$. Similarly, if b is an integer, a has to be an integer.

Now, is $\alpha = \frac{1}{2} + \frac{1}{2}\sqrt{n}$ an algebraic integer? For this to be the case, it would have to solve the equation:

$$\alpha^2 - \alpha + \frac{1-n}{4} = 0$$

or equivalently, $\mathrm{Nm}(\alpha) = \frac{1-n}{4}$. Hence, we need $n \equiv 1 \bmod 4$. Thus, we have

$$\mathcal{O}_{\mathbb{Q}(\sqrt{d})} = \begin{cases} \mathbb{Z}[\sqrt{d}] & d \not\equiv 1 \bmod 4 \\ \mathbb{Z}\left[\frac{1+\sqrt{n}}{2}\right] & d \equiv 1 \bmod 4 \end{cases}$$

A final result in this section is to find the group of units of $\mathcal{O}_K$. Note that in general, if $\alpha \in \mathcal{O}_K$, then $1/\alpha \notin \mathcal{O}_K$; for example $2 \in \mathcal{O}_{\mathbb{Q}}$, but certainly $1/2 \notin \mathcal{O}_{\mathbb{Q}}$. In fact, the only units in $\mathcal{O}_{\mathbb{Q}}$ are ± 1. If $\mathcal{O}_K$ contains roots of unity, then certainly their inverse is in $\mathcal{O}_K$ (for if $\alpha = \zeta_n$ $1/\zeta = \zeta_n^{n-1}$). There are in general more units in $\mathcal{O}_K$, and the group of units of a ring of integers shall be fully characterized in section 50.2.4. With the norm, we have a way of detecting units:

Proposition 50.1.18: Norm And Units

Let L/K be a finite field extension. Then

$$\alpha \in \mathcal{O}_L^* \qquad \Leftrightarrow \qquad \mathrm{Nm}_{L/K}(\alpha) \in \mathcal{O}_K^*$$

In the space case where $K = \mathbb{Q}$, then $\alpha \in \mathcal{O}_L^*$ if and only if $\mathrm{Nm}_{L/\mathbb{Q}}(\alpha) = \pm 1$.

***Proof*:**
Let $\alpha \in \mathcal{O}_L^*$ so that $\mathrm{Nm}_{L/K}(\alpha) \in \mathcal{O}_K$. If α is a unit in $\mathcal{O}_L$, then $1/\alpha \in \mathcal{O}_L$ exists. Then:

$$1 = \mathrm{Nm}_{L/K}(1) = \mathrm{Nm}_{L/K}(\alpha \cdot 1/\alpha) = \mathrm{Nm}_{L/K}(\alpha)\mathrm{Nm}_{L/K}(1/\alpha)$$

hence, $\mathrm{Nm}_{L/K}(\alpha), \mathrm{Nm}_{L/K}(1/\alpha) \in \mathcal{O}_K^\times$. Conversely, if $\mathrm{Nm}_{L/K}(\alpha) \in \mathcal{O}_K$, then there exists a β such that

$$\beta\mathrm{Nm}_{L/K}(\alpha) = 1$$

Then by definition of the norm:

$$1 = \beta\mathrm{Nm}_{L/K}(\alpha) = \beta\prod_{\sigma}\sigma(\alpha) = \alpha\left(\beta\prod_{\sigma\neq\mathrm{id}}\sigma(\alpha)\right) = \alpha y$$

where $y \in \mathcal{O}_L$ since each $\sigma(\alpha) \in \mathcal{O}_L$, which gives the result.

Proposition 50.1.19: Norm And Irreducible

Let $L/\mathbb{Q}$ be a finite extension and $\alpha \in \mathcal{O}_L$. Then if p is prime:

$$\mathrm{Nm}_{L/\mathbb{Q}}(\alpha) = p \implies \alpha \text{ is irreducible}$$

***Proof*:**
Suppose α where reducible so that $\alpha = \beta\gamma$ where β, γ or not units, hence have norm greater than 1. Then:

$$\begin{aligned}\mathrm{Nm}_{L/\mathbb{Q}}(\alpha) &= \mathrm{Nm}_{L/\mathbb{Q}}(\beta\gamma)\\ &= \mathrm{Nm}_{L/\mathbb{Q}}(\beta)\mathrm{Nm}_{L/\mathbb{Q}}(\gamma)\\ &= p\end{aligned}$$

a contradiction.

Note that the converse is not true

Example 50.1.20: Prime, But Norm Not Prime
Let $K = \mathbb{Q}(\sqrt{2})$ so that $\mathcal{O}_K = \mathbb{Z}[\sqrt{2}]$. Then $\mathrm{Nm}_{K/\mathbb{Q}}(5) = 5^2 = 25$, which is certainly not prime, however 5 is prime in $\mathbb{Z}[\sqrt{2}]$. This is at this stage a tedious calculation: assume that it factors $5 = \alpha\beta$ where $\alpha, \beta \in \mathbb{Z}[\sqrt{2}]$, $\alpha = a + bi$, $\beta = c + di$, and:

$$\mathrm{Nm}_{K/\mathbb{Q}}(\alpha) = a - 2b = 5 = c - 2d = \mathrm{Nm}_{K/\mathbb{Q}}(\beta)$$

but no such a, b, c, d exist. Later, we shall be able to deduce this almost immediately using Dedekind-Kummer's Theorem (see theorem 50.4.7).

In lemma 50.2.27, we shall give a partial converse which shall as a corollary imply:

$$\alpha \text{ prime} \implies \mathrm{Nm}_{K/\mathbb{Q}}(\alpha) = p^n$$

for appropriate n. Note how α must be prime, which is *stronger* than irreducible. However, we will often be working over rings where we have few prime elements but many irreducible elements. In section 50.2 we shall show how to "fix" this problem to make the norm more useful.

50.1.2 Functionals: Discriminant

As we saw, if $\alpha \in F/\mathbb{Q}$ is integral, then $\mathrm{tr}(\alpha), \mathrm{Nm}(\alpha) \in \mathbb{Z}$. However, the converse is not necessarily true as we saw in example 50.1.16. We will require a more powerful result, one that will verify that an element is an element of a ring of integers. For this, we shall bring in some multilinear algebra and define a new tool that shall become invaluable finding a *basis* for $\mathcal{O}_K$ (recall that it was mentioned that $\mathcal{O}_K$ shall be proven to be a finitely generated free $\mathbb{Z}$-module). Given a basis can be found, then an element can be determined to be integral using the basis. The tool that shall be used to detect a basis for $\mathcal{O}_K$ is called the *field discriminant*. The norm and the trace shall both be used in the definition of the field discriminant.

Some Multi-Linear Algebra

Recall from section 13.3.4 that multi-linear forms have matrix representations and so the determinant of multi-linear maps is well-defined:

Definition 50.1.21: [General] Discriminant

Let $\varphi : V \times V \to k$ be a symmetric k-bilinear map ($\varphi(x, y) = \varphi(y, x)$). Then given a basis $v_1, v_2, ..., v_n$, the *discriminant* of φ is:

$$\operatorname{disc}(\varphi) = \det \begin{pmatrix} \varphi(v_1, v_1) & \varphi(v_1, v_2) & \cdots & \varphi(v_1, v_n) \\ \varphi(v_2, v_1) & \varphi(v_2, v_2) & \cdots & \varphi(v_2, v_n) \\ \vdots & & \ddots & \vdots \\ \varphi(v_n, v_1) & \varphi(v_n, v_2) & \cdots & \varphi(v_n, v_n) \end{pmatrix}$$

and is denoted $\operatorname{disc}(\varphi)$.

Let $\{\beta_1, \beta_2, ..., \beta_n\}$ be a basis for V and $\{v_1, v_2, ..., v_n\}$ be some collection of elements so that

$$v_i = \sum_i a_i \beta_i$$

Then:

$$\varphi(v_k, v_l) = \sum_{i,j} \varphi(a_{ki}\beta_i, a_{lj}\beta_j) = a_{ki}\varphi(\beta_i, \beta_j)a_{kl}$$

hence, as matrices, we have the (component-wise) equality:

$$(\varphi(v_k, v_l)) = A \cdot (\varphi(e_i, e_j)) \cdot A^{\mathrm{tr}}$$

where $A = (a_{ij})$. Then if we have a change of basis, we have the determinant of the matrix is:

$$\det((\varphi(f_i, f_j)) = \det(A)^2 \cdot \det((\varphi(\beta_i, \beta_j))) \tag{50.2}$$

Since $A \in \mathrm{GL}(k)$, we see that the discriminant is defined up to the square of a unit. Hence, if $\operatorname{disc}(\varphi) \neq 0$, $\operatorname{disc}(\varphi)$ is well-defined in $k^\times/(k^\times)^2$ If $K = \mathbb{Q}$, then $(\mathbb{Q}^\times)^2 = 1$, hence the discriminant is well-defined in $\mathbb{Q}^\times$.

Invertible matrices allowed us to define an isomorphism between V and $V^\vee$ when V is finite dimensional. We shall require this same property for bilinear forms. For this, the notion of non-generate form needs to be upgraded for bilinear forms:

Definition 50.1.22: Non-degenerate Form

let φ be a bilinear form. Then φ is *non-generate* if φ has a non-zero discriminant

Proposition 50.1.23: Non-degenerate Form Equivalences

let φ be a bilinear form. Then φ is *non-generate* if any of the following equivalent conditions hold:

1. φ has a non-zero discriminant
2. the left kernel is zero: $\{v \in V \mid \varphi(v, x) = 0, \ \forall x \in V\}$
3. the right kernel is zero: $\{v \in V \mid \varphi(x, v) = 0, \ \forall x \in V\}$

***Proof*:**
exercise

If φ is non-degenerate, it always induces an isomorphism

$$V \mapsto V^\vee \qquad v \mapsto (x \mapsto \varphi(v,x))$$

Showing that non-degenerate maps φ give a "representation" of the hom-dual. The particular symmetric k-bilinear form that will be useful in finding the rank of the ring of integers is the one built from the trace, definition 19.6.30; in this chapter we call it the *trace pairing* and write it with the lower-case tr used throughout,

$$\mathrm{tr}_{F/k} : F \times F \to k \qquad (\alpha, \beta) \mapsto \mathrm{tr}_{F/k}(\alpha\beta)$$

Its nondegeneracy is a statement about *separability*, not about F alone: if F/k is separable then the trace pairing is nondegenerate (proposition 19.6.31). The hypothesis is not decorative. Over $k = \mathbb{F}_p(t)$ the extension $F = k(t^{1/p})$ has $\mathrm{tr}_{F/k} \equiv 0$, so the pairing vanishes identically. Nor can separability be dodged by arguing that $\mathrm{tr}_{F/k}(\alpha\alpha^{-1}) = \mathrm{tr}_{F/k}(1) = [F:k] \neq 0$ for $\alpha \neq 0$: that degree is computed *in* k, and it dies whenever $\mathrm{char} k \big| [F:k]$ — already for $\mathbb{F}_4/\mathbb{F}_2$, which is separable and whose trace pairing is nondegenerate all the same.

Every extension we pair below is separable, so by proposition 50.1.23 the left kernel is trivial and we have an isomorphism

$$F \xrightarrow{\cong} \mathrm{Hom}_k(F,k) = F^* \qquad x \mapsto (y \mapsto \mathrm{tr}_{F/k}(xy))$$

The submodule of F that this isomorphism carries onto a prescribed integrality condition is the *integral trace dual*, defined in definition 50.1.35 once we have the ring of integers in hand to state it for.

Field Discriminant

With a general overview of discriminants, we return to detecting algebraic integers more precisely using the discriminant. First the determinant, det, is an alternating n-form, meaning $(\det)^2$ is a symmetric n-form. Note that this is the same discriminant introduced in section 19.6.3; the equivalence shall soon be shown.

Definition 50.1.24: [Field] Discriminant

Let L/k be a finite extension of degree n, and let $\sigma_1, \sigma_2, ..., \sigma_n \in \mathrm{Emb}_k(L)$ be the k-embeddings of L in $\mathbb{C}$. Then for any n-tuple of elements $\alpha_1, \alpha_2, ..., \alpha_n \in L$, define the *discriminant* of $\alpha_1, \alpha_2, ..., \alpha_n$ to be:

$$\mathrm{disc}_{L/k}(\alpha_1, \alpha_2, ..., \alpha_n) = \det \begin{pmatrix} \sigma_1(\alpha_1) & \cdots & \sigma_1(\alpha_n) \\ \vdots & \ddots & \vdots \\ \sigma_n(\alpha_1) & \cdots & \sigma_n(\alpha_n) \end{pmatrix}^2 = \det((\sigma_i(\alpha_j))_{ij})^2$$

Notice that the square makes the discriminant independent of the ordering of the σ_i and α_j. If we chose a different set of elements $(\beta_1, \beta_2, ..., \beta_n)$ that differ from $(\alpha_1, \alpha_2, ..., \alpha_n)$ by a change of basis, that is $\vec{\alpha} = A\vec{\beta}$, then as we observed in equation 50.2, we get

$$\mathrm{disc}_{L/k}(\beta_1, \beta_2, ..., \beta_n) = \det(A)^2 \cdot \mathrm{disc}_{L/k}(\alpha_1, \alpha_2, ..., \alpha_n)$$

meaning the discriminant is not quite basis independent, meaning it is properly defined for operators in $L^\times / L^{\times 2}$ (where $L^\times$ are the units of L, since det maps invertible matrices to units, see proposition 13.4.10).

Example 50.1.25: Field Discriminant

1. Let $K = \mathbb{Q}(\sqrt{m})$ where m is square free. Then K has a basis $A = \{1, \sqrt{m}\}$ and $B = \left\{1, \frac{1+\sqrt{m}}{2}\right\}$. If $m \equiv 2, 3 \bmod 4$, then the former is a basis for $\mathcal{O}_K$ and if $m \equiv 1 \bmod 4$, then then the latter is a basis for $\mathcal{O}_K$. Computing both, we get:

$$\mathrm{disc}_{K/\mathbb{Q}}(A) = \det \begin{pmatrix} 1 & \sqrt{m} \\ 1 & -\sqrt{m} \end{pmatrix}^2 = (-\sqrt{m} - \sqrt{m})^2 = 4m$$
$$\mathrm{disc}_{K/\mathbb{Q}}(B) = \det \begin{pmatrix} 1 & \frac{1+\sqrt{m}}{2} \\ 1 & \frac{1-\sqrt{m}}{2} \end{pmatrix}^2 = \left(\frac{-2\sqrt{m}}{2}\right)^2 = m$$

Yet another basis for $\mathbb{Q}(\sqrt{m})$ is $C = \{1, \sqrt{m}/10\}$. In that case:

$$\mathrm{disc}_{K/\mathbb{Q}}(C) = \det \begin{pmatrix} 1 & \frac{\sqrt{m}}{10} \\ 1 & -\frac{\sqrt{m}}{10} \end{pmatrix} = \left(\frac{-\sqrt{m}}{5}\right)^2 = \frac{m}{25}$$

However, C is not a basis for $\mathcal{O}_K$, namely since the denominator is too big. Notice that the first two discriminant integers, while the third one is not.

2. Let $K = \mathbb{Q}(\sqrt[3]{2})$. Then $A = \{1, \sqrt[3]{2}, \sqrt[3]{3}\}$ is a basis for K, and:

$$\mathrm{disc}_{K/\mathbb{Q}}(A) = \begin{pmatrix} 1 & \sqrt[3]{2} & \sqrt[3]{4} \\ 1 & \sqrt[3]{2}\zeta_3 & \sqrt[3]{3}\zeta_3^2 \\ 1 & \sqrt[3]{2}\zeta_3^2 & \sqrt[3]{3}\zeta_3 \end{pmatrix}^2 = -108 = -2^2 \cdot 3^3$$

showing that the discriminant can be negative. We shall come back to seeing how this is useful for finding whether $\mathcal{O}_K = \mathbb{Z}[\sqrt[3]{2}]$.

The following alternative form is useful for an alternate way of computing the discriminant as well as important for deducing some theoretical properties of the elements inputted into the discriminant:

Proposition 50.1.26: Discriminant Via Trace-Pairing

$$\mathrm{disc}_{L/k}(\alpha_1, \alpha_2, ..., \alpha_n) = \det(\mathrm{tr}_{L/k}(\alpha_i \alpha_j))$$

***Proof*:**
This follows immediately from the matrix equation. If $[\sigma_i(\alpha_j)]$ represents the matrix:

$$[\sigma_j(\alpha_i)][\sigma_i(\alpha_j)] = [\sigma_1(\alpha_i\alpha_j) + \cdots + \sigma_n(\alpha_i\alpha_j)] = [\mathrm{tr}_{L/k}(\alpha_i\alpha_j)]$$

Using the fact that the determinant is multiplicative and invariant under transposes, we get our result.

In particular, the above form allows for the detection of sets of algebraic integers:

Corollary 50.1.27: Detecting Algebraic Integers using Discriminant

Let K be a number field and $\alpha_1, \alpha_2, ..., \alpha_n$ a $\mathbb{Q}$-basis. Then $\text{disc}_{K/\mathbb{Q}}(\alpha_1, ..., \alpha_n) \in \mathbb{Q}$, and if all $\alpha_1, \alpha_2, ..., \alpha_n$ are algebraic integers, then $\text{disc}(\alpha_1, \alpha_2, ..., \alpha_n) \in \mathbb{Z}$.

***Proof*:**
we know that $\text{tr}_{K/\mathbb{Q}}(\alpha_i\alpha_j) \in \mathbb{Q}$ by proposition 50.1.12 and are integers via corollary 50.1.14, hence $\text{disc}_{K/\mathbb{Q}}(\alpha_1, \alpha_2, ..., \alpha_n) \in \mathbb{Q}$, and if they are all algebraic integers then $\text{disc}_{K/\mathbb{Q}}(\alpha_1, \alpha_2, ..., \alpha_n) \in \mathbb{Z}$.

Corollary 50.1.28: Discriminant And Linear Independence

$\text{disc}_{K/\mathbb{Q}}(\alpha_1, \alpha_2, ..., \alpha_n) = 0$ if and only if $\alpha_1, \alpha_2, ..., \alpha_n$ are $\mathbb{Q}$-linearly dependent.

***Proof*:**
If the $\alpha_1, \alpha_2, ..., \alpha_n$ are $\mathbb{Q}$-linearly dependent, then two of the columns would be linearly dependent, hence $\text{disc}_{K/\mathbb{Q}}(\alpha_1, \alpha_2, ..., \alpha_n) = 0$. Conversely, assuming $\text{disc}_{K/\mathbb{Q}}(\alpha_1, \alpha_2, ..., \alpha_n) = 0$, then the rows of the matrix $[T(\alpha_i\alpha_j)]$ are linearly dependent; let's label them R_i so that for some $a_1, a_2, ..., a_n$ which are not all zero:

$$a_1 R_1 + \cdots + a_n R_n = \begin{pmatrix} 0 \\ \vdots \\ 0 \end{pmatrix}$$

For the sake of contradiction, let's say that the $\alpha_1, \alpha_2, ..., \alpha_n$ are $\mathbb{Q}$-linearly independent. Then they form a $\mathbb{Q}$-basis for K^n. Consider:

$$\alpha = a_1\alpha_n 1 + \cdots + a_n\alpha_n$$

where $\alpha \neq 0$ since some $a_i \neq 0$ and $(\alpha_1, \alpha_2, ..., \alpha_n)$ is linearly independent. Then $(\alpha\alpha_1, ..., \alpha\alpha_n)$ also forms a $\mathbb{Q}$-basis for K. By the linear dependence of the row space, we have that for each j

$$\text{tr}_{K/\mathbb{Q}}(\alpha\alpha_j) = \sum_i a_i \text{tr}_{K/\mathbb{Q}}(\alpha_i\alpha_j) = 0$$

But not for any $t \in K$, $t = \sum b_i\alpha\alpha_j$, and by our above observation that implies that $\text{tr}_{K/\mathbb{Q}}(t) = 0$, however that is a contradiction since we know that $\text{tr}_{K/\mathbb{Q}}(1) = n$. Hence, it must be that $(\alpha_1, \alpha_2, ..., \alpha_n)$ are linearly independent, completing the proof.

The corollary can be thought of as motivation for the name discriminant: If $K/\mathbb{Q}$ is a number field of dimension n, the discriminant will "discriminate" sets that are bases for $K/\mathbb{Q}$.

Since number fields are finite separable extensions, we always find a primitive element for a separable algebraic extension, the following theorem gives another important interpretation of the discriminant:

Proposition 50.1.29: Discriminant via Primitive Element

Let L/K be a finite separable field extension, with α a primitive element and $m_{\alpha,k}(x) \in K[x]$ the minimal polynomial with roots $\alpha_1, ..., \alpha_n$ in the splitting field of L. Then the *[field] discriminant* of L[a] is:

$$\Delta_K^2 = \left(\prod_{1 \leq i < j \leq n} (\alpha_i - \alpha_j) \right)^2 = \mathrm{Nm}_{L/K}(m'_{\alpha,k}(\alpha))$$

where $m'_{\alpha,k}(x)$ is the derivative of the minimal polynomial. If unambiguous the discriminant is often denoted

$$\mathrm{disc}_{L/K}(\alpha)$$

[a]In section 20.3, we called this the discriminant of $m_{\alpha,k}(x)$

***Proof*:**
Take as basis $1, \alpha, \alpha^2, ..., \alpha^{n-1}$. This comes down to noticing that $(\mathrm{tr}_{B/A}(\alpha^i \alpha^j))$ form the matrix:

$$\begin{pmatrix} 1 & \mathrm{tr}(\alpha) & \mathrm{tr}(\alpha^2) & \cdots & \mathrm{tr}(\alpha^{n-1}) \\ \mathrm{tr}(\alpha) & \mathrm{tr}(\alpha^2) & \mathrm{tr}(\alpha^3) & \cdots & \mathrm{tr}(\alpha^n) \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ \mathrm{tr}(\alpha^{n-1}) & \mathrm{tr}(\alpha^n) & \mathrm{tr}(\alpha^{n+1}) & \cdots & \mathrm{tr}(\alpha^{2n-2}) \end{pmatrix}$$

gives us:

$$\begin{pmatrix} \sum_i \alpha_i^0 & \sum_i \alpha_i^1 & \sum_i \alpha_i^2 & \cdots & \sum_i \alpha_i^{n-1} \\ \sum_i \alpha_i^1 \alpha & \sum_i \alpha_i^2 & \sum_i \alpha_i^3 & \cdots & \sum_i \alpha_i^n \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ \sum_i \alpha_i^{n-1} & \sum_i \alpha^n & \sum_i \alpha_i^2 & \cdots & \sum_i \alpha_i^{2n-2} \end{pmatrix}$$

This matrix can be split into:

$$\begin{pmatrix} 1 & 1 & \cdots \\ \alpha_1 & \alpha_2 & \cdots \\ \alpha_1^2 & \alpha_2^2 & \cdots \\ \vdots & \vdots & \ddots \end{pmatrix} \begin{pmatrix} 1 & \alpha_1 & \alpha_1^2 & \cdots \\ 1 & \alpha_2 & \alpha_2^2 & \cdots \\ \vdots & \vdots & \ddots & \end{pmatrix}$$

which are the Vandermonde matrices we saw when dealing with the symmetric group. Then by simple computation we get:

$$\det(\mathrm{tr}_{B/A}(\alpha_i \alpha_j)) = \prod_{1 \leq i < j \leq n} (x_i - x_j)$$

For the final equality, notice that:

$$\mathrm{Nm}_{L/K}(m_{\alpha,k}(\alpha)) = \prod_{r=1}^{n} \sigma_r(m'_{\alpha,k}(\alpha)) = \prod_{r=1}^{n} m'_{\alpha,k}(\alpha_r) \stackrel{!}{=} \prod_{s \neq r} (\alpha_r - \alpha_s)$$

where $\stackrel{!}{=}$ comes from differentiating $m_{\alpha,k}(x)$ and plugging in α_r, completing the proof.

If $L = K[\alpha]$, then we shall often write $\text{disc}_{L/K}(\alpha)$ instead of $\text{disc}_{L/K}(\alpha, \alpha_2, .., \alpha_n)$. Note that this theorem shows that the discriminant cannot detect a basis on a non-separable extension:

Corollary 50.1.30: Discriminant And Separability

Let F/k be a finite extension. Then $\text{disc}_{F/k}$ is a nonzero function if and only if F is separable

If you recall that the resultant is zero if a two polynomials share a root, then this shows how the discriminant is also related to the resultant.

***Proof*:**
By proposition 50.1.29, we have the discriminant is the product of the differences of roots. If all the roots are distinct, the discriminant is nonzero, and similarly if the extension is inseparable so that there are repeated roots, the discriminant is zero.

Example 50.1.31: Discriminant On Cyclotomic extension
Let ζ_n be an pth primitive root of unity and consider $K = \mathbb{Q}[\zeta_p]$. Then recall from section 19.7.1 that it's minimal polynomial is:

$$f(x) = 1 + x + x^2 + \cdots + x^{p-1}$$

Then to compute $f'(\zeta_n)$, recall that

$$x^p - 1 = (x-1)f(x) \implies px^{p-1} = f(x) + (x-1)f'(x) \implies f'(\zeta_n) = \frac{p}{\zeta_p(\zeta_p - 1)}$$

Taking the norm, we get:

$$\text{Nm}_{K/\mathbb{Q}}(f'(\zeta_p)) = \frac{\text{Nm}_{K/\mathbb{Q}}(p)}{\text{Nm}_{K/\mathbb{Q}}(\zeta_p)\text{Nm}_{K/\mathbb{Q}}(\zeta_p) - 1} = \frac{p^{p-1}}{1 \cdot p}$$

where $\text{Nm}_{K/\mathbb{Q}}(\zeta_p - 1) = p$ is left as an exercise. Hence:

$$\text{disc}_{K/\mathbb{Q}}(\zeta_p) = \text{Nm}_{K/\mathbb{Q}}(f'(\zeta_p)) = p^{p-2}$$

The precise value of $\text{disc}(\zeta_n)$ shall be covered in lemma 50.4.65, however it is relatively easy to see that for arbitrary n, $\text{disc}(\zeta_n)\big|n^{\varphi(n)}$ where φ is the Euler totient. If f is the [monic] irreducible polynomial for ζ_n over $\mathbb{Q}$ for arbitrary $n \in \mathbb{N}$, we get that

$$x^n - 1 = f(x)g(x) \qquad g(x) \in \mathbb{Z}[x]$$

Differentiating and evaluating at ζ_n, we get $n = \zeta_n f'(\zeta_n)g(\zeta_n)$. Taking the norm, we get:

$$n^{\varphi(n)} = \pm\text{disc}(\zeta_n)\text{Nm}(\zeta_n g(\zeta_n))$$

which since $\text{Nm}(\zeta_n g(\zeta_n)) \in \mathbb{Z}$, we get that $\text{disc}(\zeta_n)\big|n^{\varphi(n)}$.

Exercise 50.1.1

1. Give a second proof of proposition 19.6.31, independent of the Vandermonde argument used there: show that if F/K is finite and separable, then $\mathrm{tr}_{F/K}(a)$ cannot be 0 for all $a \in F$ (Hint: use Dedekind's lemma, corollary 20.1.40). Deduce that the bilinear form $(x, y) \mapsto \mathrm{tr}_{F/K}(xy)$ is non-degenerate.

2. (series of question for Stickelberger's theorem, that $d_K \equiv 0, 1 \bmod 4$)

50.1.3 Abelian Group Structure

We now prove that $\mathcal{O}_K$ is a free $\mathbb{Z}$-module of finite rank. From the above discussion, we can start seeing how this is useful in finding elements of $\mathcal{O}_K$: given it is finitely generated and we know the rank, we can look for generating linearly independent sets, i.e. a basis, for $\mathcal{O}_K$ (since $\mathcal{O}_K$ is a module over a PID).

We in fact can prove something slightly stronger that shall be useful later in this chapter:

Proposition 50.1.32: Generation of Ring of Integers

Let R be a normal integral domain, ($\overline{R}^{K(R)} = R$), $K = K(R)$ the field of fraction of R, F/K a separable extension of degree n, and let $S = \overline{R}^F$ where F/K. Then there exists free R-submodules M' of F such that if $M = R^n$,

$$M \subseteq S \subseteq M'$$

Hence, S is a finitely generated R-module if R is Noetherian, and it is free of rank n if R is a Principal Ideal Domain.

Hence, the integral closure over a larger field that is a finite extension shall only extend the integral closure by a relatively controlled manner; the more condition on R the better the control.

We are usually considering $\mathcal{O}_F$ for some number field $F/\mathbb{Q}$, the generality is to work over the cases that are less nice (as shall be done in algebraic geometry[6], see [12]). To see this result more concretely, it is worth stating the following immediate corollary:

Corollary 50.1.33: $\mathcal{O}_K$ is a Free $\mathbb{Z}$-Module

Let $F/\mathbb{Q}$ be a number field of degree n. Then $\mathcal{O}_F$ is a free $\mathbb{Z}$-module of degree n.

***Proof*:**
Of proposition 50.1.32
Let $K = K(R)$ and let $B = \{\beta_1, \beta_2, ..., \beta_n\}$ be a basis for F over K. We shall use this basis to squish S between two finitely generated R-modules (note that S is an R-module since S/R is an integral extension)

To define the R-module M, as we saw in the proof of proposition 50.1.7, there exists a nonzero $d \in R$ such that $d\beta_i \in S$. Clearly $dB = \{d\beta_1, d\beta_2, ..., d\beta_n\}$ is still a K-basis for F. Now, consider

$$\langle d\beta_1, d\beta_2, ..., d\beta_n\rangle_R = M$$

[6] The integral points of curves shall often have more nuanced behaviour

Since S is naturally an R-module and $d\beta_i \in S$, we have that $M \subseteq S$. Without loss of generality, we may now assume that $\beta_i \in S$ for all $1 \leq i \leq n$, that is $B \subseteq S$ and is a K-basis for F.

For the second R-module, define: the *integral trace dual*

$$^*M = M^{\mathrm{tr}\vee} = \left\{ b \in F \mid \forall \ell \in M, \mathrm{tr}_{F/K}(b\ell) \in R \right\}$$

The extension F/K is separable by hypothesis, so the trace pairing is nondegenerate (proposition 19.6.31) and there is a dual K-basis $B' = \{\beta_1', ..., \beta_n'\}$ of F given by $\mathrm{tr}(\beta_i\beta_j') = \delta_{ij}$. Both modules are free of rank n over R, so in particular $M \cong_R M^{\mathrm{tr}\vee}$. This $M^{\mathrm{tr}\vee}$ is the module called M' in the statement.

The key result we shall prove is:

$$M = R\beta_1 + R\beta_2 + \cdots + R\beta_n \subseteq S \subseteq R\beta_1' + R\beta_2' + \cdots + R\beta_n' = M^{\mathrm{tr}\vee} \tag{50.3}$$

The first inclusion has already been proved, hence we show the second inclusion. Let $b \in S$ be arbitrary. Since $S \subseteq F$ and B' is a basis for F, we may write b as

$$b = \sum_i b_i \beta_i' \qquad b_i \in K$$

If we show that each $b_i \in R$, we are done. First, Since $\beta_i, b \in S$, their product is in S since S is a ring, $b\beta_i \in S$, hence $\mathrm{tr}_{F/K}(b\beta_i) \in R$ for each i since S is integral over R. Computing, we get:

$$\mathrm{tr}(b\beta_i) = \mathrm{tr}(\sum_j b_j\beta_j'\beta_i) = \sum_j b_j \mathrm{tr}(\beta_j'\beta_i) = \sum_j b_j\delta_{ij} = b_i$$

Hence, $b_i \in R$, showing equation (50.3). If R is Noetherian, then $M^{\mathrm{tr}\vee}$ is a Noetherian R-module, and hence S would be finitely generated. If furthermore R is a Principal Ideal Domain, then any submodule of a free module (over a PID) is free, and S is free of rank less than or equal to n. Since it contains a free R-module of rank n, it has to be free rank greater than or equal to n, and so it has rank n as an R-module, completing the proof.

Note that this implies that if $\mathcal{O}_K$ is a PID and L/K a separable extension where $K = K(\mathcal{O}_K)$, then *every* nonzero finitely generated $\mathcal{O}_L$-module is a free $\mathcal{O}_K$-module of rank $[L : K]$.

Corollary 50.1.34: Maximality of Ring Of Integers for Number Fields

The ring of integers of a number field L is the largest subring that is finitely generated as a $\mathbb{Z}$-module.

***Proof*:**
By the above proposition, $\mathcal{O}_L$ is finitely generated as a $\mathbb{Z}$-module If $B \subseteq L$ is a subring that is also a finitely generated $\mathbb{Z}$-submodule, then every element of B is integral over $\mathbb{Z}$ (proposition 26.1.4(3)). Hence, $B \subseteq \mathcal{O}_L$.

With 50.1.32, we now have an "upper bound" on how far away a ring of integers is form simply being $\mathbb{Z}[\alpha]$ where $K = \mathbb{Q}(\alpha)$.

Definition 50.1.35: Integral Trace Dual

Let R be a normal integral domain, $K = K(R)$, L/K a finite separable extension, $S = \overline{R}^L$. Then the *integral trace dual* of S is

$$^*S = \left\{ b \in L \;\middle|\; \forall \ell \in S, \mathrm{tr}_{L/K}(b\ell) \in R \right\}$$

Example 50.1.36: Finding Integral Trace Dual

Let $K = \mathbb{Q}(\sqrt{5})$ considered as an extension $K/\mathbb{Q}$ so that $R = \mathbb{Z}$. Then we know that $L = \langle 1, \sqrt{5} \rangle \subseteq \mathcal{O}_K$. Let's try to bound by above the elements of $\mathcal{O}_K$. Let $b = r + s\sqrt{5} \in \mathbb{Q}(\sqrt{5})$ be an arbitrary element. Then:

$$\mathrm{tr}(b \cdot 1) = 2r \qquad r(b\sqrt{5}) = 10s$$

Hence, if $2r \in \mathbb{Z}$ and $10s \in \mathbb{Z}$, we get:

$$^*\mathcal{O}_K = \left\langle \frac{1}{2}, \frac{\sqrt{5}}{10} \right\rangle_{\mathbb{Z}}$$

Note that $^*\mathcal{O}_K$ is bigger than $\mathcal{O}_K$, for example $1/2 \notin \mathcal{O}_K$. By similar reasoning, we can check that $\frac{1}{\sqrt{5}} \in {}^*\mathcal{O}_K$, but it can be verified that it is not in $\mathcal{O}_K$, and so $\mathcal{O}_K^\vee \neq \mathcal{O}_K$. This does not contradict corollary 50.1.34 as $^*\mathcal{O}_K$ is *not* a subring (note that $1/2 \cdot 1/2 \notin {}^*\mathcal{O}_K$). Note however that they are both still rank 2.

Note that separability is a necessary hypothesis to insure the non-degeneracy of the trace-pairing, meaning this result works at least over over number fields, function fields, and perfect fields[7]. Naturally, if R is not a Principal Ideal Domain, then S need not be a free R-module. For now, we shall always let $R = \mathbb{Z}$ so that $K = \mathbb{Q}$ and $F/\mathbb{Q}$ is a number field. Note that we are only working with the addition structure of $\mathcal{O}_K$, while $\mathcal{O}_K$ is a ring. Thus, when talking about generators as a $\mathbb{Z}$-module, it must not be confounded with a set of generators for the ring. For example $(\alpha) \subseteq \mathcal{O}_K$ is a principal ideal and hence generated by a single element, but it may be generated by many elements as a $\mathbb{Z}$-module.

Definition 50.1.37: Integral Basis

Let $K/\mathbb{Q}$ be a number field. Then a $\mathbb{Z}$-basis for $\mathcal{O}_K$ is called an *integral basis* for K.

The previous proposition also gives more motivation for squaring the determinant in the definition of the discriminant. By corollary 50.1.27, if $(\alpha_1, \alpha_2, ..., \alpha_n)$ is a $\mathbb{Z}$-basis for $\mathcal{O}_F$, then $\mathrm{disc}_{F/\mathbb{Q}}(\alpha_1, \alpha_2, ..., \alpha_n) \in \mathbb{Z}$ and any change of basis A must map to a unit, hence $\det(A) \in \mathbb{Z}^\times = \{-1, 1\}$. Squaring thus guarantees that the discriminant is basis independent for rings of integers. Now that we have explicitly found that $\mathcal{O}_K$ is a finitely generated $\mathbb{Z}$-module in the case of K being a number field, the term *field discriminant*, or just *discriminant* for short, shall imply that the input is an integral basis for $\mathcal{O}_K$, unless explicitly stated otherwise:

[7] fields of positive characteristic may require some extra work

Definition 50.1.38: Discriminant For Ring of Integers

Let $K/\mathbb{Q}$ be a number field. Then if $A = \{\alpha_1, \alpha_2, ..., \alpha_n\}$ is an integral basis for $\mathcal{O}_K$, then:

$$d_K := \text{disc}(\mathcal{O}_K) := \text{disc}_{K/\mathbb{Q}}(A) = \text{disc}(\mathcal{O}_K)$$

is the *field discriminant* of K. It is sometimes also denoted $\text{disc}(K)$ if clear from context.

The discriminant can be thought of as capturing some information on how far $\mathcal{O}_K$ is from being Thus, in example 50.1.25 where $K = \mathbb{Q}(\sqrt{m})$, we have:

$$d_K = \begin{cases} 4m & m \equiv 2, 3 \text{ mod } 4 \\ m & m \equiv 1 \text{ mod } 4 \end{cases}$$

As we saw, we may have many bases for K, not all of them necessarily basis for $\mathcal{O}_K$. Given $K/\mathbb{Q}$ of dimension n, we may have many $M \subseteq B$ of rank n. Since M is of rank n, then if $\mathcal{O}_K \subseteq M$, $\mathcal{O}_K/M$ is of rank 0, and if $M \subseteq \mathcal{O}_K$, then $M/\mathcal{O}_K$ is also of rank 0 (some elementary group theory can be used to prove this, see exercise ref:HERE (fund thm of fin gen ab grp)). This naturally leads to asking how the discriminants of such $\mathbb{Z}$-modules compare:

Proposition 50.1.39: Discriminant And Comparing Basis

Let $B/\mathbb{Z}$ be a free $\mathbb{Z}$-module of rank n, and let $A \subseteq B$ be a $\mathbb{Z}$-submodule of finite index (i.e. $[B : A] < \infty$). Then if $\text{disc}_{B/\mathbb{Z}}(\gamma_1, \gamma_2, ..., \gamma_n) \neq 0$, then:

$$\text{disc}(A) = [B : A]^2 \text{disc}(B)$$

***Proof*:**
Let $\beta_1, \beta_2, ..., \beta_n$ is a basis for B. By the previous corollary, if $N \subseteq B$ is of finite index, then it must be a free module of rank n and so the discriminant is nonzero, and similarly the other way around.

Then there exists a change of basis matrix A, and we get $\text{disc}(N) = \det(A)^2 \text{disc}(B)$. It is not hard to see that $\det(A) = [B : N]$ (it will be easier to visualize after finishing section 50.1.4), which complete the proof.

Corollary 50.1.40: Subset And Same Discriminant, Then Equal

Let $F/\mathbb{Q}$ be number field, Let $A, B \subseteq F$ be free $\mathbb{Z}$-modules, $B \subseteq A$ and $\text{disc}(A) = \text{disc}(B)$. Then $B = A$. Hence, we may use discriminants to identify free $\mathbb{Z}$-modules of finite rank.

***Proof*:**
Since the discriminants match and

$$\text{disc}(B) = [B : A]^2 \text{disc}(N)$$

we have $[B : A] = 1$, meaning $B = N$.

Corollary 50.1.41: Isomorphism Check Of Number Fields

Let $\mathbb{Q}(\alpha), \mathbb{Q}(\beta)$ be two number fields. Then if the discriminant is different, the number fields are not isomorphic

***Proof*:**
If the rings of integers were the same, their index would be 1, but since their discriminant is not equal this is not the case.

Note that it is possible that there are multiple fields with the same discriminant:

Example 50.1.42: Same Discriminant, Different Fields
Take $\mathbb{Q}(\alpha), \mathbb{Q}(\beta), \mathbb{Q}(\gamma)$ where

$$a^3 - 18\alpha - 6 = 0 \qquad \beta^3 - 36\beta - 78 = 0 \qquad \gamma^3 - 58\gamma - 150 = 0$$

Then each has discriminant 22356, but these fields are not isomorphic (as shall be shown in example 50.4.20)

It should be noted that there are only finitely many fields for each discriminant value, however we shall not prove this fact[8]

The next couple of theorems give some computational tools on finding the ring of integers:

Proposition 50.1.43: Computation Of Dual For Ring Of Integers

Let K be a number field with basis $(\alpha_1, \alpha_2, ..., \alpha_n)$ of algebraic integers and let $d = \mathrm{disc}_{K/\mathbb{Q}}(\alpha_1, \alpha_2, ..., \alpha_n)$. Then every $\alpha \in \mathcal{O}_k$ can be expressed as:

$$\frac{m_1\alpha_1}{d} + \cdots + \frac{m_k\alpha_k}{d}$$

where $m_i \in \mathbb{Z}$ and each m_i^2 is divisible by d ($d \big| m_i^2$)

***Proof*:**
First, $d \neq 0$ since $\alpha_1, \alpha_2, ..., \alpha_n$ forms a basis for K and $d \in \mathbb{Z}$ since the α_i are algebraic integers (which exists by corollary 50.1.8).

Let $\alpha = x_1\alpha_1 + \cdots + x_n\alpha_n$ for some $x_i \in \mathbb{Q}$. Letting $\sigma_1, \sigma_2, ..., \sigma_n$ denote the embeddings of K into $\mathbb{C}$, we get

$$\sigma_i(\alpha) = x_1\sigma(\alpha_1) + \cdots + x_n\sigma(\alpha_n)$$

for each $1 \leq i \leq n$. We can solve for the x_j's using Cramers rule, namely

$$x_j = y_j/\delta$$

where $\delta = \det(\sigma_i(\alpha_j))$ and y_j is obtained by replacing the jth column of the previous determinant by $\sigma_i(\alpha)$. Clearly y_j and δ are algebraic integers, and furthermore $\delta^2 = \mathrm{disc}_{K/\mathbb{Q}}(\alpha_1, \alpha_2, ..., \alpha_n)$.

[8] For those interested, it is the *Hermite-Minkowski's Theorem*

Hence, letting $d = \mathrm{disc}_{K/\mathbb{Q}}(\alpha_1, \alpha_2, ..., \alpha_n)$ we have

$$dx_j = \delta y_j$$

showing that the rational number dx_j is an algebraic integer, and so $dx_j \in \mathbb{Z}$. Let $m_i = dx_j$. What's left to show is that $m_j^2/d \in \mathbb{Z}$, so it suffices to show that it is an algebraic integer. But $m_j^2/d = y_j^2$, which completes the proof.

Hence, for any $\mathcal{O}_K$, if $\alpha_1, \alpha_2, ..., \alpha_n$ is a basis for $K/\mathbb{Q}$

$$\bigoplus \alpha_i \mathbb{Z} \subseteq \mathcal{O}_K \subseteq \bigoplus \frac{\alpha_i}{d}\mathbb{Z}$$

And if $K = \mathbb{Q}(\alpha)$ so that $1, \alpha, ..., \alpha^{n-1}$ is a basis for K, then by proposition 50.1.29 $d = \left(\prod_{i \neq j}(\alpha_i - \alpha_j)\right)^2$ and:

$$\mathbb{Z}[\alpha] \subseteq \mathcal{O}_K \subseteq \mathbb{Z}\left[\frac{\alpha}{d}\right]$$

Note that in general, $\mathbb{Z}[\alpha] \neq \mathcal{O}_K$; a counter-example shall be given in example 50.4.29. In the special case where $\mathbb{Z}[\alpha] = \mathcal{O}_K$, we give such a number ring a name:

Definition 50.1.44: Monogenic Number Ring

Let $K/\mathbb{Q}$ be a number field, $\mathcal{O}_K$ be the associated number ring, and α the primitive element so that $K = \mathbb{Q}(\alpha)$. Then if

$$\mathbb{Z}[\alpha] = \mathcal{O}_K$$

Then $\mathcal{O}_K$ is called *monogenic* and is said to be a *monogenic number ring*.

Since all number fields are monogenic, that is $K = \mathbb{Q}(\alpha)$ for some appropriate α, it is reasonable to ask if $\mathcal{O}_K$ can be expressed using α. The following theorem gives this structure:

Theorem 50.1.45: Basis For Ring Of Integers

Let $\alpha \in \mathcal{O}_k$ and suppose α has degree n over $\mathbb{Q}$. Then there exists $d_1, d_2, ..., d_{n-1} \in \mathbb{Z}$, and monic polynomial $f_i \in \mathbb{Z}[x]$ of degree i for $1 \leq i \leq n-1$ such that

$$\mathcal{O}_K = \mathbb{Z} \oplus \frac{f_1(\alpha)}{d_1}\mathbb{Z} \oplus \cdots \oplus \frac{f_{n-1}(\alpha)}{d_{n-1}}\mathbb{Z}$$

is an integral basis where

$$d_1 \big| d_2 \big| \cdots \big| d_{n-1}$$

and the d_i's are unique.

***Proof*:**
Marcus p.26

Example 50.1.46: Ring Of Integers Of Cubic Extension

Let $K = \mathbb{Q}(\sqrt[3]{2})$ and $\alpha = \sqrt[3]{2}$. We want to find $\mathcal{O}_K$. First rank $\mathcal{O}_K = 3$ since $[K : \mathbb{Q}] = 3$, and $L = \mathbb{Z}[\sqrt[3]{2}] \subseteq \mathcal{O}_K$. If A is the change of basis from L to $\mathcal{O}_K$, then $\det(A) = [\mathcal{O}_K : \mathbb{Z}[\alpha]]$[a]. Thus, by doing a change of basis from $\mathbb{Z}[\alpha]$ to $\mathcal{O}_K$ by A, we get that:

$$\operatorname{disc}(\mathbb{Z}[\alpha]) = [\mathcal{O}_K : \mathbb{Z}[\alpha]]^2 \operatorname{disc}(\mathcal{O}_K)$$

Thus, computing the discriminant of the left hand side we get:

$$\operatorname{disc}(\mathbb{Z}[\alpha]) = \det \begin{pmatrix} 3 & 0 & 0 \\ 0 & 0 & 6 \\ 0 & 6 & 0 \end{pmatrix} = -3 \cdot 36 = -(2^2 \cdot 3^2) \cdot 3 = -108$$

giving us that $[\mathcal{O}_K : \mathbb{Z}[\alpha]]^2 \big| 108 \Rightarrow [\mathcal{O}_K : \mathbb{Z}[\alpha]] \big| 2 \cdot 3 = 6$. If we show the index is 1, the proof is done. We shall prove the contradiction for the case of $[\mathcal{O}_K : \mathbb{Z}[\alpha]] = 3$, the proof for $[\mathcal{O}_K : \mathbb{Z}[\alpha]] = 2$ is similar, and from that we can also conclude that $[\mathcal{O}_K : \mathbb{Z}[\alpha]] \nmid 6$.

Let's say $[\mathcal{O}_K : \mathbb{Z}[\alpha]] = 3$. Then $\mathcal{O}_K/\mathbb{Z}[\alpha] \cong \mathbb{Z}/3\mathbb{Z}$. Any element $\beta \in \mathcal{O}_K$ lands in one of the cosets of $\mathcal{O}_K/\mathbb{Z}[\alpha]$ which implies that $a, b, c \in \frac{1}{3}\mathbb{Z}$. We want to show that a, b, c must be integers (lands in the zero coset). Since the ring of integers is closed under subtraction, we may without loss of generality subtract the non-fractional part of any element β we choose so that $a, b, c \in \{-1, 0, 1\}$[b]. Hence we take the norm of $\beta = (a/3) + (b/3)\alpha + (c/3)\alpha^2$

$$N(\beta) = \det \begin{pmatrix} a/3 & 2c/3 & 2b/3 \\ b/3 & a/3 & 2c/3 \\ c/3 & b/3 & a/3 \end{pmatrix} = \frac{1}{27}(a^3 + 2b^2 + 4c^3 - 6abc)$$

Since β is algebraic, $N(\beta) \in \mathbb{Z}$ meaning $27 \big| (a^3 + 2b^2 + 4c^3 - 6abc)$. But, unless $a = b = c = 0$:

$$|a^3 + 2b^2 + 4c^3 - 6abc| \leq 15$$

showing that no such combination of a, b, c exist, meaning β is an algebraic integer if $a, b, c \in \mathbb{Z}$. A similar argument can be made in the case of $[\mathcal{O}_K : \mathbb{Z}[\alpha]] = 2$, hence we have that $\mathcal{O}_K = \mathbb{Z}[\alpha]$, completing the proof.

[a] since the index is finite, a basis for $\mathbb{Z}[\alpha]$ may be scaled to form a basis for $\mathcal{O}_K$, which translates to the determinant result

[b] Equivalently, any coset can be chosen to have such representatives

In some special cases, the ring of integers of more composite fields may be found given ring of integers of each respective field:

Theorem 50.1.47: Ring of Integers of Composites

Let $K_1/K, K_2/K$ be Galois with degree n and m and with coprime discriminants $d_{K_1} = d$ and $d_{K_2} = d'$ respectively. Let $L = K_1K_2$ and $K_1 \cap K_2 = K$. Let $\omega_1, \omega_2, ..., \omega_n$ and $\omega'_1, \omega'_2, ..., \omega'_m$ be an integral basis for K_1/K and K_2/K. Then $\omega_i\omega'_j$ is an integral basis for K_1K_2 with discriminant $(d_{K_1})^m(d_{K_2})^n = d^m(d')^n$, namely

$$\mathcal{O}_L = \mathcal{O}_{K_1}\mathcal{O}_{K_2}$$

***Proof*:**
Since $K_1 \cap K_2 = K$, $[K_1K_2 : K] = nm$, so the nm products $\omega_i\omega'_j$ form a basis of K_1K_2/K. Now, let α be an integral element of K_1K_2 over K, and write:

$$\alpha = \sum_{i,j} a_{ij}\omega_i\omega'_j \qquad a_{ij} \in K$$

We must show that $a_{ij} \in \mathcal{O}_K$. Let

$$\beta_j = \sum_i a_{ij}\omega_i$$
$$\mathrm{Gal}_{K_2}(L) = \{\sigma_1, \sigma_2, ..., \sigma_n\}$$
$$\mathrm{Gal}_{K_1}(L) = \{\tau_1, \tau_2, ..., \tau_m\}$$

Then since $[L : K] = nm$, by proposition 20.1.32$\mathrm{Gal}_K(L) = \{\sigma_k\tau_\ell \mid 1 \leq k \leq n\ 1 \leq \ell \leq m\}$. Furthermore, let:

$$T = \tau_\ell(\omega'_j) \qquad a = (\tau_1(\alpha), ..., \tau_m(\alpha))^t \qquad b = (\beta_1, ..., \beta_m)^t$$

Then by definition we have $\det(T)^2 = d'$ and $a = Tb$. If T^* is the adjoint of T, then some linear algebra manipulation gives:

$$\det(T)b = T^*a$$

Now, since T^* and a have integral entries (in L), $d'b$ has integral entries in K_1< i.e.

$$d'\beta_j = \sum_k d/a_{ij}\omega_i$$

Hence, $d'a_{ij} \in \mathcal{O}_K$. Now, changing (ω_i) and $\omega'_j)$, we get that $da_{ij} \in A$, hence since the discriminants are relatively prime, we get for appropriate x, x':

$$a_{ij}xda_{ij} + x'd'a_{ij} \in \mathcal{O}_K$$

Hence, $\omega_i\omega'_j$ form an integral basis for L/K, completing the first part of the proof.

We next find the discriminant the ring of integers of L/K. Since we have the Galois group of L over K, take the $(nm \times m)$ matrix

$$M = (\sigma_k\tau_\ell(\omega_i\omega'_j)) = (\sigma_k(\omega_i)\tau_\ell(\omega'_j))$$

Note that M can be interpreted as an $(m \times m)$-matrix with entries being $(n \times n)$-matrices where each (ℓ, j)-entry is the matrix $Q\tau_\ell\omega'_j$ where $Q = (\sigma_k\omega_i)$:

$$M = \begin{pmatrix} Q & & 0 \\ & \ddots & \\ 0 & & Q \end{pmatrix} \begin{pmatrix} E\tau_1\omega'_1 & \cdots & E\tau_m(\omega'_1) \\ \vdots & \ddots & \vdots \\ E\tau_1(\omega'_m) & \cdots & E\tau_m(\omega'_m) \end{pmatrix}$$

where E is an $(n \times n)$-unit matrix. Now, changing indices to compare the two representations of M, we get:

$$\mathrm{disc}(\mathcal{O}_L) = \det(M)^2 = \det(Q)^{2m} = \det\left((\tau_\ell(\omega'_j))\right)^{2n} = d^m(d')^n$$

as we sought to show.

Note that for the general separable case, we need to replace our assumption with

$$K_1 \cap K_1 = K \qquad K_1,\ K_2 \text{ are linearly disjoint}$$

If the discriminants are not relatively prime, then there is a partial generalization in the case where $K = \mathbb{Q}$:

Corollary 50.1.48: Ring of Integers of Composite, Partial Generalization

Let $K_1/\mathbb{Q}$ and $K_2/\mathbb{Q}$ be finite separable extensions, $d_1 = \text{disc}(\mathcal{O}_{K_1})$ and $d_2 = \text{disc}(\mathcal{O}_{K_2})$. Then if $L = K_1K_2$:

$$\mathcal{O}_L \subseteq \frac{1}{\gcd(d_1, d_2)}\mathcal{O}_{K_1}\mathcal{O}_{K_2}$$

***Proof*:**
Marcus p.25

Example: Ring of Integers of Cyclotomic Field

Let's use the above results to show that if $K = \mathbb{Q}(\zeta_n)$ then $\mathcal{O}_K = \mathbb{Z}[\zeta_n]$, that is the ring of integers of K is monogenic. Besides quadratic extensions, this is so far the only other ring of integers we have classified (in particular since they are rather difficult to classify). We first require some lemmas:

Lemma 50.1.49: Cyclotomic Field Lemmas

Let $n = p^r$ where $n \geq 3$. Then:

1. $\mathbb{Z}[\zeta_n] = \mathbb{Z}[1 - \zeta_n]$ and $\text{disc}(\zeta_n) = \text{disc}(1 - \zeta_n)$
2. $\prod_k (1 - \zeta_n^k) = p$

***Proof*:**

1. Certainly $\mathbb{Z}[\zeta_n] = \mathbb{Z}[1 - \zeta_n]$ since $\zeta_n = 1 - (1 - \zeta_n)$, implying the discriminants are equal. Similarly, by proposition 50.1.29

$$\text{disc}(\zeta_n) = \prod_{r<s}(\alpha_r - \alpha_s)^2 = \prod_{r<s}((1 - \alpha_r) - (1 - \alpha_s))^2 = \text{disc}(1 - \zeta_n)$$

2. Let

$$f(x) = \frac{x^{p^r - 1}}{x^{p^{r-1}} - 1} = 1 + x^{p^{r-1}} + x^{2p^{r-1}} + \cdots x^{(p-1)p^{r-1}}$$

Then for $p \nmid k$, ζ_n^k are the roofs of f, since they are roots of $x^{p^r} - 1$, but not of $x^{p^{r-1}} - 1$. Thus:

$$f(x) = \prod_{p \nmid k}(x - \zeta_n^k)$$

Since there is exactly $\varphi(p^r) = (p-1)p^{r-1}$ where φ is the Euler totient. Thus, setting $x = 1$ gives us the desired result.

Proposition 50.1.50: Ring Of Integers Of Prime-Power Cyclotomic Field

Let $n = p^r$ and $K = \mathbb{Q}(\zeta_n)$. Then:

$$\mathcal{O}_K = \mathbb{Z}[\zeta_n]$$

***Proof*:**
We shall show that $\mathcal{O}_K = \mathbb{Z}[1-\zeta_n]$. By theorem 50.1.45, any $\alpha \in \mathcal{O}_K$ can be written as:

$$\alpha = \frac{n_1 + n_2(1-\zeta_n) + \cdots + n_m(1-\zeta_n)^{m-1}}{\mathrm{disc}(1-\zeta_n)}$$

where $m = \varphi(p^r)$, $n_i \in \mathbb{Z}$, and $\mathrm{disc}(1-\zeta_n) = \mathrm{disc}(\zeta_n)$. As we saw in example 50.1.31, if $n = p$ is a prime, $\mathrm{disc}(\zeta_n) = p^{p-2}$. Since $n = p^r$, by example 50.1.31, d is a power of p.

Let's now assume for the sake of contradiction that $\mathcal{O}_K \neq \mathbb{Z}[1-\zeta_n]$, so that there must exist some $\alpha \in \mathcal{O}_K$ such that not all n_i are divisible by d, namely $\mathcal{O}_K$ contains an element of the form:

$$\beta = \frac{n_i(1-\zeta_n)^{i-1} + n_{i+1}(1-\zeta_n)^i + \cdots + n_m(1-\zeta_n)^{m-1}}{p}$$

where $i \leq n$, $n_i \in \mathbb{Z}$, not all n_i are divisible by p. Now, by lemma 50.1.49 since $(1-\zeta_n)\big|(1-\zeta_n)^k$ in $\mathbb{Z}[\zeta_n]$,

$$\frac{p}{(1-\zeta_n)^m} \in \mathbb{Z}[\zeta_n] \implies \frac{p}{(1-\zeta_n)^i} \in \mathbb{Z}[\zeta_n] \implies \frac{\beta p}{(1-\zeta_n)^i} \in \mathcal{O}_K$$

Subtracting the terms that are in $\mathcal{O}_K$ (by closure of multiplication) we get

$$\frac{n_i}{1-\zeta_n} \in \mathcal{O}_K \implies \mathrm{Nm}_{K/\mathbb{Q}}\left(\frac{n_i}{1-\zeta_n}\right) \in \mathbb{Z} \implies \mathrm{Nm}_{K/\mathbb{Q}}(1-\zeta_n)\big|\mathrm{Nm}_{K/\mathbb{Q}}(n_i)$$

However, this is a contradiction since $\mathrm{Nm}_{K/\mathbb{Q}}(n_i) = n_i^m$, but by lemma 50.1.49 $\mathrm{Nm}_{K/\mathbb{Q}}(1-\zeta_n) = p$, Hence

$$\mathcal{O}_K = \mathbb{Z}[\zeta_n]$$

as we sought to show.

We now look to generalize to any cyclotomic field. The key idea is that the ring of integers of cyclotomic fields will be the composite of simpler ring of integers

Proposition 50.1.51: Ring of Integers of Cyclotomic Field

Let $K = \mathbb{Q}(\zeta_n)$ for any $n \in \mathbb{N}$. Then:

$$\mathcal{O}_K \cong \mathbb{Z}[\zeta_n]$$

***Proof*:**
We shall do a form of induction. This was proven if n is a prime power, so let's assume it's not so that we may write $n = n_1 n_2$ for relatively prime $n_1, n_2 > 1$. We shall work by induction on

n_1, n_2. The base case was proposition 50.1.50, hence set:

$$\begin{aligned} \omega_1 &= \zeta_{n_1} & \omega_2 &= \zeta_{n_2} \\ K_1 &= \mathbb{Q}(\omega_1) & K_2 &= \mathbb{Q}(\omega_2) \\ \mathcal{O}_{K_1} &= \mathbb{Z}[\omega_1] & \mathcal{O}_{K_2} &= \mathbb{Z}[\omega_2] \end{aligned}$$

First, it is clear that

$$\zeta_n^{n_1} = \omega_2 \qquad \zeta_n^{n_2} = \omega_1 \implies \zeta_n = \omega_1^r \omega_2^s,\ r, s \in \mathbb{Z}$$

Thus, $L = K_1 K_2$. Next, as we saw in example 50.1.31,

$$\operatorname{disc}(\omega_1)\big|n_1 \qquad \operatorname{disc}(\omega_2)\big|n_2$$

since $\gcd(n_1, n_2) = 1$, we see the discriminants are relatively prime. Since $\varphi(n) = \varphi(n_1)\varphi(n_2)$, we by theorem 50.1.47 that

$$\mathcal{O}_K = \mathcal{O}_{K_1}\mathcal{O}_{K_2} = \mathbb{Z}[\omega_1]\mathbb{Z}[\omega_2] = \mathbb{Z}[\zeta_n]$$

as we sought to show.

Exercise 50.1.2

1. Generalize proposition 50.1.39 for modules over $\mathcal{O}_K$.
2. Give a counter-example where $\mathcal{O}_{KL} = \mathcal{O}_K\mathcal{O}_L$?

50.1.4 Lattice Structure: Geometry of Number Rings

An important word of warning must be given about free $\mathbb{Z}$-modules due to the existence of non-invertible elements. It is possible that A, B are two free $\mathbb{Z}$-module of rank 2, one contains the basis of the other, however $A \subsetneq B$! This is in contract to the case of modules over fields where if A, B are two k-vector-spaces of the same dimension and $A \subseteq B$, then $A = B$[9]. Even more behavior can happen: consider $\mathbb{Q} \otimes_{\mathbb{Z}} A$. Then this is a vector-space with a finite dimension. However, the dimension need not match the rank of A:

Example 50.1.52: Behavior of $\mathbb{Z}$-modules

1. Let $A = \langle (1,0), (0,1) \rangle_{\mathbb{Z}}$ and $B = \langle (1/2, 0), (0, 1/2) \rangle_{\mathbb{Z}}$. Then by construction they are both isomorphic as abelian groups to $\mathbb{Z}^2$, however $A \subsetneq B$.
2. Let A be a free $\mathbb{Z}$-module of rank 2. Then we may extend scalar to make A into a free $\mathbb{R}$-module; either take $\mathbb{R} \otimes_{\mathbb{Z}} A$ or consider $\langle a_1, a_1 \rangle_{\mathbb{R}}$ for some free generating set $\{a_1, a_2\}$. Then $\mathbb{R} \otimes_{\mathbb{Z}} A$ *need not be dimension 2*. For example, take
$$A = \langle (1,1), (\sqrt{2}, \sqrt{2}) \rangle_{\mathbb{Z}}$$
Then $\mathbb{R} \otimes_{\mathbb{Z}} A$ is one dimensional

[9] $b \in B$ can be appropriately shrunk to a nonzero size so that its in A (since the dimensions are the same), and then we use closure of scalar multiplication

3. By construction of $\mathcal{O}_k$, if $\langle \alpha_1, \alpha_2, ..., \alpha_n \rangle_{\mathbb{Z}} = \mathcal{O}_k$, then $\langle \alpha_1, \alpha_2, ..., \alpha_n \rangle_{\mathbb{Q}} = k$, Thus $\mathcal{O}_k$ is a little more than just a $\mathbb{Z}$-module. We shall explore this in this section.

The third example shows that the rank of the ring of integers more coincides with the dimension of the natural vector-space when extending the actions to a field. As was explored in chapter 13, [finite dimensional] vector spaces have easy visualizations using orthonormal bases. In chapter 10, this was used implicitly when it was intuited that $\mathbb{Z}[i]$ forms a nice square lattice in $\mathbb{Q}^2$. More generally, ring of integer $\mathcal{O}_K$ will form a lattice when embedded into an appropriate vector-space. To better visualize this lattice, it is easer to see it embedded into a direct sum of $\mathbb{R}$ and $\mathbb{C}$[10]. This has the extra advantage that these spaces have a well-defined notion of *measure* that gives us an area. We shall take advantage of tools from analysis to further enlighten the structure of rings of integers, however as a consequence an elementary level of topology and analysis will be assumed.

Let $n = [K : \mathbb{Q}]$ so that $\mathrm{Hom}_{\mathbb{Q}}(K, \mathbb{C}) = n$. Then every embedding of K into $\mathbb{C}$ is either embeds into $\mathbb{R}$ or does not (that is, either $K \cap (\mathbb{C} \setminus \mathbb{R}) = \emptyset$ or $\neq 0$); let's call the former *real embeddings* and the latter *complex embeddings*. For a complex embedding $\tau \in \mathrm{Hom}_{\mathbb{Q}}(K, \mathbb{C})$, we may always find another embedding $\overline{\tau}$ since for any α where $p(\alpha) = 0$ for appropriate $p(x) \in \mathbb{Q}[x]$

$$0 = \overline{\tau}(0) = \overline{\tau}(p(\alpha)) = \tau(\overline{p(\alpha)}) = \tau(p(\overline{\alpha}))$$

showing that $\overline{\tau}$ maps roots to roots. Hence the degree n may be split into $n = r_1 + 2r_2$ where r_1 represents the amount of embeddings where the image σ_i is real:

$$\alpha_1, \alpha_2, ..., \alpha_{r_1}$$

while r_2 represents the embeddings that embedded into $\mathbb{C}$ is not strictly in $\mathbb{R}$, giving us a total list of embeddings:

$$\sigma_1, \sigma_2, ..., \sigma_{r_1}, \tau_1, \tau_2, ..., \tau_{r_2}, \overline{\tau_1}, ..., \overline{\tau_{r_2}}$$

Definition 50.1.53: Archimedean Embedding

Let $K/\mathbb{Q}$ be a number field. Then its *Archimedean embedding* is defined as

$$\begin{aligned} \sigma : &K \hookrightarrow \mathbb{R}^{r_1} \oplus \mathbb{C}^{r_2} \\ &\alpha \mapsto (\sigma_1(\alpha), ..., \tau_{r_2}(\alpha)) \end{aligned}$$

The space $\mathbb{R}^{r_1} \oplus \mathbb{C}^{r_2}$ is called the *Minkowski space*, and will be labeled $K_{\mathbb{R}}$.

Example 50.1.54: Archimedean Embedding

1. Trivially, $\mathbb{Q}_{\mathbb{R}} = \mathbb{R}$, and $\mathbb{C}_{\mathbb{R}}$ is not defined since $\mathbb{C}/\mathbb{Q}$ is not a finite extension.

2. Take $K = \mathbb{Q}(i)$ so that $\mathcal{O}_K = \mathbb{Z}[i]$. Then $\mathrm{Gal}_{\mathbb{Q}}(\mathbb{Q}(i)) = \{\mathrm{id}, c\}$ where c is the conjugate. Then K is simply the embedding of K into $\mathbb{C}$

$$\sigma : K \hookrightarrow \mathbb{C}^1 \qquad k \mapsto k$$

[10] For those who may prefer $\overline{\mathbb{Q}}$ over $\mathbb{R}$ or $\mathbb{C}$, I would stipulate $\mathbb{R}$ and $\mathbb{C}$ are is a very natural object in this context: $\mathbb{R}$ is perhaps the only "natural" object for which a finite extension is an algebraic closure, this will have algebraic and geometric consequence when counting the number of embeddings of a field into the algebraically closed field $\mathbb{C}$

Notice the gaussian integers form the usual grid in $\mathbb{C}$.

3. Take $K = \mathbb{Q}(\sqrt{2}, \sqrt{3})$ Then $\mathrm{Gal}_{\mathbb{Q}}(K) \cong \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$ corresponds to the conjugating each root separately. Then

$$\begin{gathered}\sigma : K \mapsto \mathbb{R}^4 \\ (a + b\sqrt{2} + c\sqrt{3} + d\sqrt{6}) \\ \mapsto \\ (a + b\sqrt{2} + c\sqrt{3} + d\sqrt{6}, a - b\sqrt{2} + c\sqrt{3} + d\sqrt{6}, \\ a + b\sqrt{2} - c\sqrt{3} + d\sqrt{6}, a - b\sqrt{2} - c\sqrt{3} + d\sqrt{6})\end{gathered}$$

Our goal is to show that $\sigma(\mathcal{O}_K)$ behaves like the just like $\mathbb{Z}[i]$ when embedded in $\mathbb{C}$. To characterize a lattice in higher dimensions, we define it in the following way:

Definition 50.1.55: [Integral] Lattice

Let V be a finite dimensional vector space over$\mathbb{R}$. Then a *lattice* in V is a $\mathbb{Z}$-submodule satisfying any two of the following 3 properties;

1. Λ spans V as a k-vector space ($\langle\Lambda\rangle_{\mathbb{R}} = V$)
2. Λ is discrete in the topological space V (there is an open ball near the origin that intersects only the origin) and the rank of the lattice is the dimension of V[a]
3. Λ is a free $\mathbb{R}$-module on n generators

[a] some authors call lattices where $n = m$ *complete lattices*, and if $n < m$ a *lattices*

Lemma 50.1.56: Equivalence Of Conditions

Any two of these conditions implies the third

***Proof*:**
Let $L \cong_{\mathbf{Ab}} \mathbb{Z}^n$. We'll show that (2) $\Leftrightarrow$ (3). For (2) $\Rightarrow$ (1), namely

(2) L is discrete

(3) $\langle L\rangle_{\mathbb{R}} = \mathbb{R}^n$

let $\ell_1, \ell_2, ..., \ell_n$ be a $\mathbb{Z}$-basis for L. Then there exists an invertible matrix $A \in \mathrm{GL}_n(\mathbb{R})$ such that $A = (a_{ij})$ where for each e_i of the standard basis for $\mathbb{R}^n$,

$$e_i = \sum_j a_{ij}\ell_j$$

Let $v = (v_1, v_2, ..., v_n) \in \mathbb{Z}^n$. Then

$$\sum_i^n v_i\ell_i = \sum_i^n w_ie_i$$

where $w = A^{-1}v$. We have now converted our number into one in the standard basis, hence we have a notion of size we may borrow. In particular, we have that if $v = Aw$, then

$$\|w\| \geq \frac{\|v\|}{n\|A\|}$$

where we can choose our favourite matrix norm, say the max norm on entries. hence, as v grows, w get's further from 0.

Conversely, we'll do the contrapositive, so suppose $\langle \ell_1, \ell_2, ..., \ell_n \rangle_{\mathbb{R}} \neq \mathbb{R}^n$. Then for some $a_i \in \mathbb{R}$ not all zero

$$\sum_i^n a_i \ell_i = 0$$

By the Pigeon hole Principle[a] (also known as Dirichlet's box principle), there exists $N \in \mathbb{N}, N > 0$ such that $|Na_i - [Na_i]| < \epsilon$ where $[Na_i]$ is the closest integer. Thus

$$0 = \sum Na_i \ell_i = \underbrace{\sum [Na_i]\ell_i}_{\in L} + \underbrace{\sum \{Na_i\}\ell_i}_{\in B_\epsilon(0)}$$

where $\{Na_i\}$ denote the fractional part, completing the proof.

[a] see section 0.1.3

The discrete condition need only be checked at the origin due to linearity[11]. An example of a free $\mathbb{Z}$-module in $\mathbb{R}^2$ that is not discrete is the one elaborated in example 50.1.52

$$\langle (1,1), (\sqrt{2}, \sqrt{2}) \rangle \subseteq \mathbb{R}^2$$

In particular, we may get as close as we want to $(0,0)$ by adding integer-multiples these two elements.

Proposition 50.1.57: Archimedean Embedding of $\mathcal{O}_K$ is Lattice

Let $K/\mathbb{Q}$ be a number field and σ the Archimedean embedding. Then $\sigma(\mathcal{O}_k)$ is a lattice in $\mathbb{R}^{r_1} \oplus \mathbb{C}^{r_2}$, namely

$$\langle \sigma(\mathcal{O}_K) \rangle_{\mathbb{R}} = \mathbb{R}^{r_1} \oplus \mathbb{C}^{r_2}$$

***Proof*:**
We may prove this result in using any of the equivalent definitions.

1. Let's show $\sigma(\mathcal{O}_K)$ is discrete. Let $\alpha \in K$ so that

$$\mathrm{Nm}_{K/\mathbb{Q}}(\alpha) = \prod_i^{r_1} \sigma_i(\alpha) \cdot \prod_j^{r_2} \|\tau_j(\alpha)\|^2 \in \mathbb{Q}$$

Then if $\alpha \in \mathcal{O}_K$, $\mathrm{Nm}_{K/\mathbb{Q}}(\alpha) \in \mathbb{Z}$. Therefore, if $\alpha \in \mathcal{O}_K \setminus \{0\}$, we have $|\mathrm{Nm}_{K/\mathbb{Q}}(\alpha)| \geq 1$. But then $|\sigma(\alpha)| \geq 1$. Thus, taking $B \subseteq \mathbb{R}^{r_1} \oplus \mathbb{C}^{r_2}$,

$$B = \{\alpha \mid |\alpha_i| < 1,\ 1 \leq i \leq r_1 + r_2\}$$

[11] This is reminiscent to how many conditions for topological groups need only be checked at the origin, including many proofs in real and complex analysis

Then $\sigma(\mathcal{O}_K) \cap B = \{0\}$

2. Let's show that the $\mathbb{R}$-span of $\sigma(\mathcal{O}_K)$ is our entire space. Since $\mathcal{O}_K$ is $\mathbb{Z}$-finitely generated we have

$$\langle \alpha_1, \alpha_2, ..., \alpha_n \rangle_{\mathbb{Z}} = \mathcal{O}_K$$

We want to show that

$$\mathrm{span}_{\mathbb{R}}(\sigma_1(\alpha_i), ..., \sigma_{r_1}(\alpha_i), .., \tau_{r_2}(\alpha_i))_{i \in \{1,...,n\}} = \mathbb{R}^{r_1} \oplus \mathbb{C}^{r_2}$$

Then the determinant of these values would be nonzero:

$$0 \neq \det \begin{pmatrix} \sigma_1(\alpha_1) & \cdots & \sigma_{r_1}(\alpha_1) & \mathrm{Re}\tau_1(\alpha_1) & \mathrm{Im}(\tau_1(\alpha_1)) & \cdots & \mathrm{Im}\tau_{r_2}(\alpha_1) \\ \sigma_1(\alpha_2) & \cdots & \sigma_{r_1}(\alpha_2) & \mathrm{Re}\tau_2(\alpha_2) & \mathrm{Im}(\tau_2(\alpha_2)) & \cdots & \mathrm{Im}\tau_{r_2}(\alpha_2) \\ & & & \vdots & & & \\ & & & \vdots & & & \\ & & & \vdots & & & \end{pmatrix}$$

By doing some elementary operations, we get the matrix to be of the form:

$$0 \neq \det \begin{pmatrix} \sigma_1(\alpha_1) & \cdots & \sigma_{r_1}(\alpha_1) & \tau_1(\alpha_1) & \overline{\tau_1(\alpha_1)} & \cdots & \overline{\tau_{r_2}(\alpha_1)} \\ \sigma_1(\alpha_2) & \cdots & \sigma_{r_1}(\alpha_2) & \tau_2(\alpha_2) & \overline{\tau_2(\alpha_2)} & \cdots & \overline{\tau_{r_2}(\alpha_2)} \\ & & & \vdots & & & \\ & & & \vdots & & & \\ & & & \vdots & & & \end{pmatrix}$$

Now, choose a primitive element $\alpha \in K$ so that $K = Q(\alpha)$. Then we know that $\mathbb{Z}[\alpha] = \langle 1, \alpha, ..., \alpha^{n-1} \rangle_{\mathbb{Z}} \subseteq \mathcal{O}_K$. We know this is of finite index. Letting $\alpha_i = \alpha^{i-1}$, we get a Vandermonde matrix. If M is the above matrix:

$$\det(M) = \prod_{\substack{\sigma \neq \sigma' \\ \sigma, \sigma' \in \mathrm{Emb}_K(\mathbb{C})}} (\sigma(\alpha) - \sigma'(\alpha)) \neq 0$$

But then this implies that the volume of $\sigma(\mathcal{O}_K)$ is non-zero, as we sought to show.

Corollary 50.1.58: Sign Of Discriminant

Let $K/\mathbb{Q}$ be a number field, $\mathcal{O}_K$ its ring of integers, its discriminant. Then:

$$\mathrm{sign}(d_K) = (-1)^{r_2}$$

***Proof*:**
Simply keep track of the elementary operations when transitioning between both matrices in

proposition 50.1.57

$$\det\begin{pmatrix} \sigma_1(\alpha_1) & \cdots & \sigma_{r_1}(\alpha_1) & \tau_1(\alpha_1) & \overline{\tau_1(\alpha_1)} & \cdots & \overline{\tau_{r_2}(\alpha_1)} \\ \sigma_1(\alpha_2) & \cdots & \sigma_{r_1}(\alpha_2) & \tau_2(\alpha_2) & \overline{\tau_2(\alpha_2)} & \cdots & \overline{\tau_{r_2}(\alpha_2)} \\ & & & \vdots & & & \\ & & & \vdots & & & \\ & & & \vdots & & & \end{pmatrix}$$

$$= \frac{1}{(2i)^{r_2}}\det\begin{pmatrix} \sigma_1(\alpha_1) & \cdots & \sigma_{r_1}(\alpha_1) & \mathrm{Re}\tau_1(\alpha_1) & \mathrm{Im}(\tau_1(\alpha_1)) & \cdots & \mathrm{Im}\tau_{r_2}(\alpha_1) \\ \sigma_1(\alpha_2) & \cdots & \sigma_{r_1}(\alpha_2) & \mathrm{Re}\tau_2(\alpha_2) & \mathrm{Im}(\tau_2(\alpha_2)) & \cdots & \mathrm{Im}\tau_{r_2}(\alpha_2) \\ & & & \vdots & & & \\ & & & \vdots & & & \\ & & & \vdots & & & \end{pmatrix}$$

completing the proof.

Hence, any ring of integers $\mathcal{O}_K$ forms a lattice in $K_{\mathbb{R}}$, and any ideal $\mathfrak{a} \subseteq \mathcal{O}_K$ will also form a lattice in $K_{\mathbb{R}}$. Since $K_{\mathbb{R}}$ is an inner product and a lattice is discrete, it is meaningful to define the volume of a lattice:

Definition 50.1.59: Volume of Lattice

let $\mathcal{O}_K$ be a ring of integers. Then if $\Gamma = (v_1, v_2, ..., v_n)$ is an integral basis for $\mathcal{O}_K$, then the *volume* of $\mathcal{O}_K$ is defined to be

$$\mathrm{vol}(\mathcal{O}_K) := \mathrm{vol}(K_{\mathbb{R}}/\Gamma) = \mathrm{coVol}(\Gamma) = |\det(\langle v_i, v_j\rangle)|^{1/2} =$$

where $\frac{1}{2^{r_2}}$ comes from the extra complex embeddings, as seen in the proof of corollary 50.1.58.

If you know some differential geometry, we may equivalently take the canonical differential form on $\mathbb{R}^n \cong \mathbb{R}^{r_1} \oplus \mathbb{C}^{r_2}$:

$$\omega = dx_1 \wedge \cdots \wedge dx_n$$

and write:

$$\mathrm{vol}(\mathcal{O}_K) = \mathrm{coVol}(K_{\mathbb{R}}/\mathcal{O}_K) = \frac{1}{2^{r_2}}\left|\int_{K_{\mathbb{R}}/\mathcal{O}_K}\omega\right| = \frac{1}{2^{r_2}}\left|\int_P \omega\right|$$

where the $1/2^{r_2}$ comes from eliminating the extra complex dimension, and:

$$P = \left\{\sum a_i v_i \;\middle|\; 0 \leq a_i \leq 1\right\}$$

This also motivates the vocabulary of covolume. Note that any different choice of a basis will produce the same result since the change of basis matrix B will have determinant ± 1 which in the absolute value becomes 1, and hence it is well-defined. Note that every ideal $\alpha \subseteq \mathcal{O}_K$ will also produce a lattice, since it's a sublattice of $\mathcal{O}_K$, and hence we may find the volume of any sublattice. The way we shall find the volume of the lattice is by taking advantage of the discriminant and the index of the ideal:

Proposition 50.1.60: Volume of Ideal Lattice

Let $0 \neq \mathfrak{a} \subseteq \mathcal{O}_K$ be an ideal. Then $\Gamma_{\mathfrak{a}} = \sigma(\mathfrak{a})$ is a [complete] lattice in K_R and its fundamental mesh has volume

$$\operatorname{vol}(\Gamma_{\mathfrak{a}}) = \frac{1}{2^{r_2}}\sqrt{|\operatorname{disc}(K)}[\mathcal{O}_K : \mathfrak{a}]$$

As a consequence:

$$\operatorname{vol}(\mathcal{O}_K) = \frac{1}{2^{r_2}}\sqrt{|\operatorname{disc}(\mathcal{O}_K)}$$

Thus, the discriminant provides volume information for the lattice given by the ring of integers.

The theorem can equivalently be stated as:

$$\operatorname{vol}(\Gamma_{\mathfrak{a}}) = \left| \int_{\mathbb{R}^{r_1} \oplus \mathbb{R}^{r_2}/\Gamma_{\mathfrak{a}}} \omega \right|$$

***Proof*:**
Let $\alpha_1, \alpha_2, ..., \alpha_n$ be a $\mathbb{Z}$-basis of $\mathfrak{a}$ so that $\Gamma_{\mathfrak{a}} = \mathbb{Z}\sigma(\alpha_1) + \cdots + \mathbb{Z}\sigma(\alpha_n)$ is a lattice. Numbering the embeddings $\tau K \to \mathbb{C}$ as $\tau_1, \tau_2, ..., \tau_n$, define the matrix $A = (\tau_\ell \alpha_i)$ so that we get by proposition 50.1.39,

$$\operatorname{disc}(\mathfrak{a}) = \operatorname{disc}(\alpha_1, \alpha_2, ..., \alpha_n) = \det(A)^2 = [\mathcal{O}_K : \mathfrak{a}]^2 \operatorname{disc}(\mathcal{O}_K) = [\mathcal{O}_K : \mathfrak{a}]^2 d_K$$

On the other hand, we have:

$$\left(\langle \sigma(\alpha_i), \sigma(\alpha_j) \rangle\right)_{ij} = \left(\sum_{\ell=1}^{n} \tau_\ell(\alpha_i) \overline{\tau}_\ell \alpha_j \right) = A\overline{A}^t$$

Thus:

$$\operatorname{vol}(\Gamma_{\mathfrak{a}}) = \left| \det\left(\langle \sigma(\alpha_i), \sigma(\alpha_j) \rangle\right)_{ij} \right|^{1/2} = |\det(A)| = \sqrt{|d_K|}[\mathcal{O}_k : \mathfrak{a}]$$

as we sought to show.

Thus, if $K = \mathbb{Q}(\sqrt{m})$ where m is square-free:

$$\operatorname{vol}(\mathcal{O}_K) = \begin{cases} \sqrt{m} & m \equiv 1 \bmod 4 \\ 2\sqrt{m} & m \equiv 2, 3 \bmod 4 \end{cases}$$

Absolute Norm and Geometric Inequalities

Besides introducing a geometric intuition for number rings and their ideals, the key reason to introduce volumes is that they will allow us make many bounding arguments on ideals using the lattice structure of $\mathcal{O}_K$. To do this, we introduce a new "norm" on the ideals of $\mathcal{O}_K$ that shall represent the size of the ideal with respect to the lattice. This will be helpful in many algebraic arguments, but also in upgrading many clasical results about primes (such as their distribution[12], see [39]).

[12] the number of primes p where $|p| \leq X$ tends asymptotically to $\frac{X}{\log(X)}$

Definition 50.1.61: Absolute Norm

Let $0 \neq \mathfrak{a} \subseteq \mathcal{O}_K$ be a nonzero ideal of $\mathcal{O}_K$. Then the *absolute norm* of $\mathfrak{a}$ is given by

$$\mathfrak{N}_K(\mathfrak{a}) := [\mathcal{O}_K : \mathfrak{a}]$$

If unambiguous, the notation $\mathfrak{N}(\mathfrak{a})$ is also used.

The absolute norm is inspired by the fact that ideals of $\mathcal{O}_K$, when thought of as representing a sublattice of a certain sparsity as compared to the lattice of $\mathcal{O}_K$. The norm of an ideal is always finite by proposition 50.1.39. Note that the notation $\mathfrak{N}_{K/\mathbb{Q}}$ should not be used, for it shall have a different meaning than $\mathfrak{N}$[13]

Lemma 50.1.62: Norm Of Principal Ideal

Let $(\alpha) \subseteq \mathcal{O}_K$ be a principal ideal. Then:

$$\mathfrak{N}((\alpha)) = \left|N_{K/\mathbb{Q}}(\alpha)\right|$$

***Proof*:**
Let $\beta_1, \beta_2, ..., \beta_n$ be a $\mathbb{Z}$-basis for $\mathcal{O}_K$. Then $\alpha\beta_1, \alpha\beta_2, ..., \alpha\beta_n$ is a $\mathbb{Z}$-basis for $(\alpha) = \alpha\mathcal{O}_K$. Then if $A = (a_{ij})$ is the change of basis formula so

$$\alpha w_i = \sum_j a_{ij}\omega_j$$

by proposition 50.1.39 $|\det(A)| = [\mathcal{O}_K : \alpha]$. But by definition of the basis of A, namely it is left multiplication, we get the definition of the norm:

$$|\mathrm{Nm}_{K/\mathbb{Q}}(\alpha)| = \det(A)$$

giving us equality, as we sought to show.

Lemma 50.1.63: Ideal Divides Absolute Norm

Let $I \subseteq \mathcal{O}_K$ be an ideal. Then:

$$I \big| (\mathfrak{N}(I))$$

***Proof*:**
If we show $\mathfrak{N}(I) = [\mathcal{O}_K : I] \in I$, then $(\mathfrak{N}(I)) \subseteq I$, which by corollary 50.2.19 implies we're done. But this is simply some ring theory: Take $1 \in \mathcal{O}_K$. Then

$$\overline{n} = \underbrace{\overline{1 + \cdots + 1}}_{n \text{ times}} \in \mathcal{O}_K/I$$

Since $|\mathcal{O}_K/I| = n$, it must be that $\overline{n} = \overline{0}$, in particular $n \in I$. Since $n = [\mathcal{O}_K : I]$, we have the desired result.

[13] see definition 50.3.13

Lemma 50.1.64: Volume Form Reformulation

Let $I \subseteq R$ be an ideal of a number field K. Then:

$$\text{vol}(I) = \mathfrak{N}(I)\text{vol}(\mathcal{O}_K)$$

***Proof*:**
This is just reformulating proposition 50.1.60.

Theorem 50.1.65: Bounding Ideals

Let $K/\mathbb{Q}$ be a number field and $\mathcal{O}_K$ its number ring. Then there exists $\lambda \in \mathbb{R}_{>0}$ such that for every $0 \neq I \subseteq \mathcal{O}_K$, there exists a $\alpha \in I$ such that

$$|\text{Nm}_{K/\mathbb{Q}}(\alpha)| \leq \lambda\mathfrak{N}(I)$$

***Proof*:**
Let $\alpha_1, \alpha_2, ..., \alpha_n$ be an integral basis for R, and let $\sigma_1, \sigma_2, ..., \sigma_n$ denote the embeddings of K into $\mathbb{C}$. I claim that:

$$\lambda = \prod_{i=1}^{n}\left(\sum_{j=1}^{n}|\sigma_i(\alpha_j)\right)$$

Let I be any ideal, and m the unique positive integer satisfying:

$$m^n \leq \mathfrak{N}(I) < (m+10^n$$

Consider the $(m+1)^n$ elements of $\mathcal{O}_K$ where:

$$\sum_{j=1}^{n} m_j\alpha_j \qquad m_j \in \mathbb{Z} \qquad 0 \leq m_j \leq m$$

Then two of these *must* be congruent $\mod I$ since there are more than $\mathfrak{N}(I)$ of them. Taking the difference of these two elements, we get a nonzero member of I of the form:

$$\alpha = \sum_{j=1}^{n} m_j\alpha_j \qquad m_j \in \mathbb{Z} \qquad |m_j| \leq m$$

Hence:

$$\left|\text{Nm}_{K/\mathbb{Q}}(\alpha)\right| \leq \prod_{i=1}^{n}|\sigma_i(\alpha)| \leq \prod -i = 1^n\left(\sum_{j=1}^{n} m_j|\sigma_i(\alpha_j)|\right) \leq m^n\lambda \leq \mathfrak{N}(I)\lambda$$

as we sought to show.

Proposition 50.1.66: Image Of Norm Is Finite Up To Units

Let K be a field, $\mathcal{O}_K$ its ring of integers, and $a \in \mathbb{Z}_{>0}$. Then up to multiplication by units, there are only finitely many element $\alpha \in \mathcal{O}_K$ such that

$$\mathrm{Nm}_{K/\mathbb{Q}}(\alpha) = a$$

***Proof*:**
Let $a \in \mathbb{Z}_{>0}$. Recall that $\mathcal{O}_K/a\mathcal{O}_K$ is finite. I claim that, up to multiplication by units, there is at most one element α in each coset such that $|\mathfrak{N}(\alpha)| = |\mathrm{Nm}_{K/\mathbb{Q}}(\alpha)| = a$. For, if $\beta = \alpha + a\gamma$ where $\gamma \in \mathcal{O}_K$, then since $N(\beta) = a$, we can write $\beta = \alpha + N(\beta)\gamma$. Dividing both sides by β then re-arranging we get:

$$\frac{\alpha}{\beta} = 1 \pm \frac{\mathrm{Nm}_{K/\mathbb{Q}}(\beta)}{\beta}\gamma$$

since $\mathrm{Nm}_{K/\mathbb{Q}}(\beta) = \beta \prod_{\tau \neq \mathrm{id}} \tau(\beta)$, we have that $\alpha/\beta \in \mathcal{O}_K$. By diving by α, rearranging, and substituting $a = N(\alpha)$, we also get:

$$\frac{\beta}{\alpha} = 1 \pm \frac{\mathrm{Nm}_{K/\mathbb{Q}}(\alpha)}{\alpha}\gamma \in \mathcal{O}_K$$

hence, α and β are associates. Thus, up to multiplication by units, there is at most $[\mathcal{O}_K : a\mathcal{O}_K]$ elements of norm $\pm a$.

For computational purposes, we add a tighter bound on the norms of ideals. We start with the following lemma:

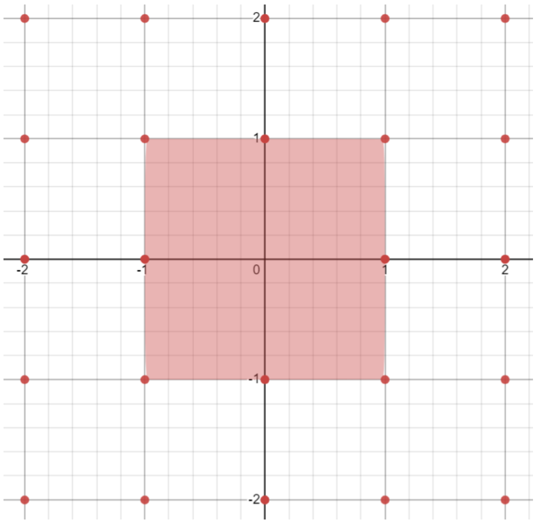
Lemma 50.1.67: Minkowski's Lattice Point Theorem

Let Γ be a discrete complete lattice in the Euclidean vector space V of dimension n and let X be a centrally symmetric convex subset of V. Then if

$$\mathrm{vol}(X) > 2^n \mathrm{vol}(\Gamma)$$

then X contains at least one nonzero lattice point $\gamma \in \Gamma$

Before proving this result, consider $\sigma(\mathbb{Z}[i])$. The smallest centrally symmetric set would



it is clear this square have volume arbitrarily close to 2^2, any large and it would intersect $\sigma(\mathbb{Z}[i])$. Similarly, if there was a long stretched area that tried to avoid the points of $\sigma(\mathbb{Z}[i])$, do to the size constraint, it will necessarily be to large to avoid intersecting the lattice:

***Proof*:**
It suffice to show that there exists two distinct lattice points $\gamma_1, \gamma_2 \in \Gamma$ such that

$$\left(\frac{1}{2}X + \gamma_1\right) \cap \left(\frac{1}{2}X + \gamma_2\right) \neq \emptyset$$

To show this, let's first say there exists such a point so that

$$\frac{1}{2}x_1 + \gamma_1 + \frac{1}{2}x_2 + \gamma_2$$

Then we may re-write to obtain:

$$\gamma = \gamma_1 - \gamma_2 = \frac{1}{2}x_1 - \frac{1}{2}x_2$$

Notice that this is in the center o the line segment joining $-x_1$ and x_2, and hence belongs in $X \cap \Gamma$.

Now, if the set $\frac{1}{2}X + \gamma$, $\gamma \in \Gamma$ were all pairwise disjoint, then the same is the case for the intersections $\Phi \cap (\frac{1}{2}X\gamma)$ for fundamental mesh Φ of Γ, hence we would have

$$\mathrm{vol}(\Phi) \geq \sum_{\gamma \in \Gamma} \mathrm{vol}\left(\Phi \cap \left(\frac{1}{2}X + \gamma\right)\right)$$

However, translation so $\Phi \cap (\frac{1}{2}X + \gamma)$ by $-\gamma$ creates the set $(\Phi - \gamma) \cap \frac{1}{2}X$ of equal volume, implying $\Phi - \gamma$ covers all of V, and hence $\frac{1}{2}X$ It follows from our assumption that

$$\mathrm{vol}(\Phi) \geq \sum)\gamma \in \Gamma \mathrm{vol}\left((\varphi - \gamma) \cap \frac{1}{2}X\right) = \mathrm{vol}(\frac{1}{2}X) = \frac{1}{2^n}X$$

contradicting the hypothesis.

Theorem 50.1.68: Minkowski's Bound

Let $[\mathfrak{a}] \in Cl(K)$ be an ideal class of degree n. Then there exists an integral ideal $\mathfrak{a}$ such that

$$\mathfrak{N}(\mathfrak{a}) \leq \frac{n!}{n^n}\left(\frac{4}{\pi}\right)^{r_2}\sqrt{|d_K|}$$

***Proof*:**
Let $\mathbb{R}^N \cong_{\mathbb{R}} \mathbb{R}^{r_1} \oplus \mathbb{R}^{r_2}$. Define the following norm on $\mathbb{R}^n$:

$$N(x) = x_1 \cdots x_r(x_{r_1+1}^2 + x_{r_1+2}^2) \cdots (x_{n-1}^2 + x_n^2)$$

(actually, type this up later, Marcus p.95)

Notice that is we re-arrange, we get:

$$\frac{n^n}{n!}\left(\frac{\pi}{4}\right)^{r_2} \leq \sqrt{|d_K}$$

Thus, we get a result that may seem obvious, but is in general rather difficult to prove:

Corollary 50.1.69: Discriminant Nonzero

Let $\mathcal{O}_K \neq \mathbb{Z}$. Then:

$$|d_K| > 1$$

***Proof*:**
immediate consequence of above since $n > 1$

50.2 Ring Structure and Dedekind Domains

So far, we have shown that rings of integers are lattices whose rank corresponds to the dimension of their corresponding field extension. Since they are rings, we may naturally ask what type of ring are they. Using example 50.1.17, we can look at the particular example of $\mathcal{O}_{\mathbb{Q}(\sqrt{-5})} = \mathbb{Z}[\sqrt{-5}]$. Is this ring a Euclidean Domain, Principal Ideal Domain, or Unique Factorization Domain? It is certainly Noetherian, being a finitely generated $\mathbb{Z}$-module (where $\mathbb{Z}$ is Noetherian). However, not all irreducible elements are prime, for example:

$$6 = 2 \cdot 3 = (1 + \sqrt{-5})(1 - \sqrt{-5})$$

showing that 2 is not prime. This may seem troubling since we may have to keep track of multiple factorizations showing there is more structure or tools that would need to be added/used to keep track of this information. Fortunately, there is a way to recover factorization. The goal of this section is to define a new type of ring for which the notion of Unique factorization is recovered, and show that all rings of integers are this type of ring. First, we may use the norm to see that a factorization into irreducible is always certainly possible in $\mathcal{O}_k$. For, if $\alpha = \beta\gamma$, then if β, γ are non-units:

$$1 < |N_{K/\mathbb{Q}}(\beta)| < |N_{K/\mathbb{Q}}(\alpha)| \qquad 1 < |N_{K/\mathbb{Q}}(\gamma)| < |N_{K/\mathbb{Q}}(\alpha)|$$

showing that it is a factorization domain. However this factorization need not be unique as we have demonstrated with $\mathcal{O}_{\mathbb{Q}(\sqrt{-5})} = \mathbb{Z}[\sqrt{-5}]$, another example being $21 = 3 \cdot 7 = (1+2\sqrt{-5})(1-2\sqrt{-5})$. We may see these are irreducible elements by using the norm. Doing the case of 3, if $3 = \alpha\beta$ for non-units α, β, then $9 = N_{K/\mathbb{Q}}(\alpha)N_{K/\mathbb{Q}}(\beta)$ would give us that $N_{K/\mathbb{Q}}(\alpha) = \pm 3$. Given $\alpha = a + b\sqrt{-5}$ we get

$$N_{K/\mathbb{Q}}(a+b\sqrt{-5}) = a^2 + 5b^2 = \pm 3$$

But this certainly has no solutions in $\mathbb{Z}$, and hence 3 is irreducible in $\mathcal{O}_K$. A similar argument can be made for $7, 1 \pm 2\sqrt{-5}$. Furthermore, the numbers 3 and 5 are not associated to $1 \pm 2\sqrt{-5}$, for we would require that

$$\frac{1 \pm 2\sqrt{-5}}{3}, \frac{1 \pm 2\sqrt{-5}}{7} \in \mathcal{O}_K$$

both of which we can check are not using the norm. Hence, we certainly have different factorizations.

The failure of unique factorization has lead Eduard Kummer to try to embed the ring of integers of K into a bigger domain of "ideal numbers" where unique factorization into "ideal prime numbers" would hold (which was inspired by the discovery of complex numbers somehow being "ideal" for roots polynomial roots). As an example, our original decomposition in $\mathbb{Z}[\sqrt{-5}]$

$$21 = 3 \cdot 7 = (1+2\sqrt{-5})(1-2\sqrt{-5})$$

would factor into ideal prime numbers $\mathfrak{p}_1, \mathfrak{p}_2, \mathfrak{p}_3, \mathfrak{p}_4$ with the relations

$$3 = \mathfrak{p}_1\mathfrak{p}_2 \qquad 7 = \mathfrak{p}_3\mathfrak{p}_4 \qquad 1+2\sqrt{-5} = \mathfrak{p}_1\mathfrak{p}_3 \qquad 1-2\sqrt{-5} = \mathfrak{p}_2\mathfrak{p}_4$$

Then we would have fixed non-uniqueness since we would get the factorization:

$$21 = (\mathfrak{p}_1\mathfrak{p}_2)(\mathfrak{p}_3\mathfrak{p}_4) = (\mathfrak{p}_1\mathfrak{p}_3)(\mathfrak{p}_2\mathfrak{p}_4)$$

Some key properties that these ideals numbers should have is that for $a, b, \lambda \in \mathcal{O}_K$, an ideal number $\mathfrak{a}$ should satisfy the following natural divisibility restrictions:

$$\mathfrak{a}\big|a \text{ and } \mathfrak{a}\big|b \Rightarrow \mathfrak{a}\big|a \pm b \qquad a\mathfrak{a}\big|a \Rightarrow \mathfrak{a}\big|\lambda a$$

An furthermore an ideal number $\mathfrak{a}$ ought to be determinable by the set of its divisors in $\mathcal{O}_K$, that is

$$\underline{\mathfrak{a}} = \left\{a \in \mathcal{O}_K \;\middle|\; \mathfrak{a}\big|a\right\}$$

But given these rules, this set is simply an ideal of $\mathcal{O}_K$ as we know the notion of ideal. It was Richard Dedekind that took the notion of ideal numbers that Kummer invented and invented the notion of *ideals* of $\mathcal{O}_K$ giving the modern definition of ideals. Then division $\mathfrak{a}\big|a$ is replaced by belonging $a \in \mathfrak{a}$, and divisibility $\mathfrak{a}\big|\mathfrak{b}$ is replaced by containment $\mathfrak{b} \subseteq \mathfrak{a}$.

Hence, we shall seek a unique factorization of *ideals* into *prime ideals* in $\mathcal{O}_K$. The 3 key properties of $\mathcal{O}_K$ that require this factorization was singled out by Dedekind and form a class of rings larger than rings of integers and are worth taking a look at on their own:

Definition 50.2.1: Dedekind Domain

Let R be an integral domain. Then if:

1. R is a Noetherian ring
2. every nonzero prime ideal of R is maximal
3. R is normal ($\overline{R}^{K(R)} = R$)

then R is called a *Dedekind Domain*

The Noetherian condition can be thought of as a natural eliminator of any infinites lurking around, while the prime implies maximal condition can be thought of as making sure that there can be a prime ideal that further contains prime ideals. In particular, we shall see that $(a) \subseteq (b)$ if and only if $b|a$, which doesn't make sense if we allow for prime ideals being contained in larger prime ideals.

50.2.2 Foreshadowing in Algebraic Geometry Notice that we have already seen a ring with these conditions before, namely if V is a smooth 1-dimensional variety, then the associated k-algebra is Noetherian, 1 [Krull] dimensional and is normal (normal implies smoothness[a]). In Algebraic Geometry, Dedekind domains can be thought of as representing 1-dimensional smooth schemes (in particular, 1-dimensional smooth $\mathbb{Z}$-curves)

[a] But only necessarily in 1 dimension; in higher dimension normalness need not imply smoothness, see [5]

Lemma 50.2.3: PID's, UFD's, and Dedekind Domain

Let R be a ring. Then:

1. If R is a PID, R is a Dedekind Domain
2. If R is a UFD, R need not be a Dedekind Domain
3. If R is a Dedekind domain and a UFD, it is a PID

The following image demonstrates the realation:

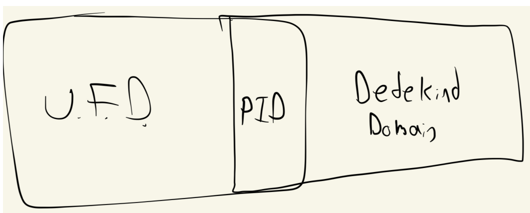


Figure 50.1: Venn diagram

***Proof*:**
R is trivially Noetherian and all prime ideals are maximal. For normalness, since R is a Unique Factorization Domain, $\overline{R}^{K(R)} = R$, hence R is a Dedekind Domain.

For the counter-example, take $\mathbb{Z}[\sqrt{-13}]$ and find two ways of decomposing 14. The last is an exercise.

For algebro-geometric purposes, the following characterizes Dedekind domains locally:

Lemma 50.2.4: Dedekind Domain As DVR

Let R be commutative ring. Then R is a Dedekind domain if and only if for each prime P, R_P is a DVR. Thus, Dedekind domains are local DVR's.

***Proof*:**
exercise.

Lemma 50.2.5: Dedekind Domains And Localization

Let R be a Dedekind domain and $S \subseteq R \setminus \{0\}$ is a multiplicative subset. Then $S^{-1}R$ is also a Dedekind Domain.

***Proof*:**
The localization of Noetherian rings is Noetherian, and by the correspondence of ideals in localizations, we have that prime implies maximal in $S^{-1}R$. Finally, $S^{-1}R$ is integrally closed since if $x \in K = K(R)$ satisfies:

$$x^n + \frac{a_1}{s_1}x^{n-1} + \cdots + \frac{a_n}{s_n} = 0 \qquad \frac{a_i}{s_i} \in S^{-1}R$$

Then multiplying with the nth power of $s_1 \cdots s_n$ gives that sx is integral over R, hence $sx \in R$, and so $x \in S^{-1}R$, as we sought to show.

Importantly for number theory, all rings of integers are Dedekind domains:

Theorem 50.2.6: Ring Of Integers Are Dedekind Domains

Let $\mathcal{O}_K$ be a ring of integers. Then it is a Dedekind domain.

***Proof*:**
By proposition 50.1.32, $\mathcal{O}_K$ is Noetherian, and by definition $\mathcal{O}_K$ is normal, namely $\mathcal{O}_K = \overline{\mathbb{Z}}^K$ meaning by corollary 26.1.6:

$$\overline{\mathcal{O}_K}^K = \overline{\overline{\mathbb{Z}}^K}^K = \overline{\mathbb{Z}}^K$$

What's left to show is that every prime ideal is maximal, which is equivalent to showing $\mathcal{O}_K/\mathfrak{p}$ is a field. In particular, since $\mathcal{O}_K/\mathfrak{p}$ is always an integral domain, if we show that it is finite then by corollary 9.1.25 we shall have that it's a field. The key intuition comes from the fact that $\mathcal{O}_K$ and its ideals form a lattice, meaning when we quotient we necessarily get a finite ring.

Furthermore, the volume of the ideal is determined by the discriminant and the index, which will determine the cardinality of the quotient ring[a]

Let $0 \neq \alpha \in \mathfrak{p}$. Then $(p) = \mathfrak{p} \cap \mathbb{Z}$ is also a prime ideal. Since $\mathfrak{p} \subseteq \mathcal{O}_K$, we know α is associated to a minimal polynomial $m_{\alpha,\mathbb{Z}}(x)$ where

$$m_{\alpha,\mathbb{Z}}(\alpha) = \alpha^n + a_{n-1}\alpha^{n-1} + \cdots + a_1\alpha + a_0 = 0 \qquad a_i \in \mathbb{Z}$$

Then $a_0 = -(\alpha^n + a_{n-1}\alpha^{n-1} + \cdots + a_1\alpha) \in \mathfrak{p} \subseteq \mathcal{O}_K$, and since $m_{\alpha,\mathbb{Q}}(x)$ is irreducible with integer coefficients, $a_0 \neq 0$ and $a_0 \in (p) = p\mathbb{Z}$. From here, we shall give two proofs: first, we may directly appeal to the fact that

$$(\mathcal{O}_K/\mathfrak{p})/(\mathbb{Z}/p\mathbb{Z})$$

is an integral extension (see example 26.1.9) where the base ring is a field, hence $\mathcal{O}_K/\mathfrak{p}$ is a field by proposition 26.1.8. Another is the following: we have a non-trivial ideal with chains of inclusions $(a_0) \subseteq \mathfrak{p} \subseteq \mathcal{O}_K$ implying

$$|\mathcal{O}_K/\mathfrak{p}| \leq |\mathcal{O}_K/(a_0)| = |\mathcal{O}_K/a_0\mathcal{O}_K|$$

Now, since we are quotienting ideals, this is a $\mathbb{Z}$-module quotient, and since we are quotienting n-rank free $\mathbb{Z}$-modules, we get that

$$|\mathcal{O}_K/(a_0)| = p^n$$

But then $\mathcal{O}_K/\mathfrak{p}$ is finite (even better, its order divides $|a_0|^n$). Thus, $\mathfrak{p}$ must be maximal, and since $\mathfrak{p}$ was arbitrary every prime ideal is maximal, completing the proof.

[a] We shall soon see that the index can be thought of as the "norm" of a general ideal; this will mean the constant term of a minimal polynomial represent the "size", sheading some light on the proof

Since in general $\mathcal{O}_K$ is not a Unique Factorization Domain, rings of integers represent a class of rings where prime implies maximal, but factorization is not unique. In this way, we see that the integral closure of $\mathbb{Z}$ with respect to K may be far from having the nice element-wise properties we are used to in PIDs, but this shall be salvaged through having an understandable ideal structure and a way to measure how far away our ring is from being a PID.

Another important example of a Dedekind domain that is not a number ring is the ring is $k[x]$ (since it's a PID). We shall later see that integral extensions of Dedekind domains are Dedekind domains, an so these types of rings will form a basis for another large class of Dedekind domains that are not number rings.

50.2.1 Fractional Ideals in Rings

Following from the idea that ideals are the generalisation of numbers, we may define the following divisibility notions usually used for the [algebraic] lattice of integers induced by divisibility. These concepts shall be defined for general commutative rings:

Definition 50.2.7: GCD, LCM, Divisibility Of Ideals

let R be a commutative ring, $\mathfrak{a}, \mathfrak{b} \subseteq R$ ideals of R. Then:

$$\gcd(\mathfrak{a}, \mathfrak{b}) = (\mathfrak{a}, \mathfrak{b}) = \mathfrak{a} + \mathfrak{b} \qquad \operatorname{lcm}(\mathfrak{a}, \mathfrak{b}) = \mathfrak{a}\mathfrak{b}$$

And we can say that $\mathfrak{a}, \mathfrak{b}$ are relatively prime when $\gcd(\mathfrak{a}, \mathfrak{b}) = (1)$. Divisibility $\mathfrak{a}\big|\mathfrak{b}$ is defined as $\mathfrak{b} \subseteq \mathfrak{a}$, and hence we have an order and the ideals similar to that given by divisibility.

As we've seen in chapter 10, gcd and lcm are usually well-behaved in special classes of rings, usually UFDs. We shall show they are well-behaved in Dedekind domains since the ideals shall have unique factorizations. Let's now prove some lemmas for this result.

One finiteness fact is needed throughout, and it is the reason the Noetherian hypothesis appears in the definition of a Dedekind domain at all: in a Noetherian domain every nonzero proper ideal *contains* a product of nonzero prime ideals, each of which contains it (corollary 23.1.6). An ideal need not factor into primes — that is the theorem we are working toward, and it needs the full strength of a Dedekind domain — but it always contains such a product, and that much is available from the Noetherian condition alone.

For the following lemmas, we bring back the notion of fractional ideal that was defined in chapter 23. These ideals were at the time a way of extending ideals into the ring of fractions, and were shown to work well with localization. Due to all the properties of Dedekind domains, the product of ideals can be made into a *group*, even an *abelian group*. Since the fractional ideal structure will have such a strong restriction, this will produce incredible results on the relations between ideals. To begin, the following lemma shows that no prime ideal acts like the identity:

Lemma 50.2.8: Fractional Ideals

Let $P \subseteq R$ be a prime ideal of the Dedekind domain R, $K = K(R)$. Define

$$(R :_K P) = P^{-1} = \{x \in K \mid xP \subseteq R\}$$

Then

$$AP^{-1} = \left\{ \sum_i a_i x_i \;\middle|\; a_i \in A,\ x_i \in P^{-1} \right\} \neq A$$

for every ideal $0 \neq A \subseteq R$

Note that by definition $R \subseteq P^{-1}$ and so P^{-1} is not in general an ideal. The simplest "fractional ideal" would be those of $\mathbb{Z}$, namely $(n)^{-1} = \frac{1}{n}\mathbb{Z}$.

***Proof*:**
Note that when we work with the inverse of elements, we are considering them as being part of $K(R)$.

We shall first show the special case of $P^{-1} \neq R$. Let $a \in P$ with $a \neq 0$. The ideal (a) is nonzero and proper, so it contains a product of nonzero primes by the fact recalled above (corollary 23.1.6); consider such a product $P_1 P_2 \cdots P_r \subseteq (a) \subseteq P$ with r minimal. Then one of the P_i, is containing in P, without loss of generality say $P_1 = P$ (since P_l is maximal). Indeed, if non of the P_i were contained in P, then for every i there would exist $a_i \in P_i \setminus P$ such that

$a_1 \cdots a_r \in P$, but P is maximal. Next, by minimality $P_2 \cdots P_r \not\subseteq (a)$, so there exists $b \in P_2 \cdots P_r$ such that $b \notin (a) = aR$, that is $a^{-1}b \notin R$. On the other hand, since $P = P_1$, $P_1 b \in P_1 P_2 \cdots P_n r$, hence we have $bP \subseteq (a) = aR$ that is $a^{-1}bP \subseteq R$, and so $a^{-1}b \in P^{-1}$. It follows that $P^{-1} \neq R$, namely $P^{-1} \supsetneq R$.

Next, let $0 \neq A \subseteq R$ be an ideal. Since R is Noetherian, let

$$A = (\alpha_1, \alpha_2, ..., \alpha_n)$$

For the sake of contradiction, say $AP^{-1} = A$. Then for every $x \in P^{-1}$

$$xa_i = \sum_j a_{ij}\alpha_j \qquad a_{ij} \in R$$

Letting M be the matrix $(x\delta_{ij} - a_{ij})$, we get $M(\alpha_1, \alpha_2, ..., \alpha_n)^t = 0$. By Cramers rule, if $d = \det(M)$, then $d\alpha_1 = \cdots = d\alpha_n = 0$, and so $d = 0$. Then x is integral over R, since its' the zero of the monic polynomial

$$f(X) = \det(X\delta_{ij} - a_{ij}) \in R[x]$$

Thus, since R is integrally closed $x \in R$. But then $P^{-1} = R$; a contradiction to your first part, as we sought to show.

Definition 50.2.9: Fractional Ideal and Quotient ideal

Let R be an integral domain and $K = K(R)$. Then a *fractional ideal* J of K is a finitely generated non-trivial R-submodule of K where there exists a $r \in R$ such that $rJ \subseteq R$ is an ideal.

If $I, J \subseteq R$ are ideals, then the *quotient ideal* of I with respect to J is:

$$(I : J) = \{x \in R \mid xJ \subseteq I\}$$

If $I = R$,then $(R : J)$ is denoted J^{-1}

Fractional ideals are multiplied in the similar way ideals are multiplied (treat them as sub-algebra multiplication). Since all ideals of R are rank 1 submodules of R, they are all fractional ideals. For those who are motivated by algebraic geometry, fractional ideals shall be shown to form line bundles over the smooth $\mathbb{Z}$-curves $\mathcal{O}_K$ [14].

Quotient ideals have a lot of interesting theory, for example to motivate their name if J is principle and $I \subseteq J$, then $I = J \cdot (I : J)$. There is no room in this chapter to explore these in detail, hence they are left as exercises 50.2.1(50.2.1 (6)).

Fractional ideals that are ideals are given a name:

[14]See section 50.7 to see how to interpret rings of integers as smooth $\mathbb{Z}$-curves, and ref:HERE for interpreting fractional ideals as line bundles

Definition 50.2.10: Integral Ideals

Let R be an integral domain and $K = K(R)$. Then a fractional ideal $I \subseteq K$ that is of rank 1 is called an *integral ideal*, and $I \subseteq R$.

Example 50.2.11: Fractional Ideal

1. Since $\mathbb{Z}$ is a Principal Ideal Domain, every ideal is of the form $a\mathbb{Z}$ for $a \in \mathbb{Z}$. Then $(a)^{-1} = a^{-1}\mathbb{Z}$. Hence, the fractional ideals are in bijective correspondence with $\mathbb{Q}^{\times}_{>0}$. Naturally, all ideals of $\mathbb{Z}$ are fractional ideals, in particular they are integral ideals.

2. More generally, if $a \in K^{\times}$, then we get fractional principal ideals $(a) = aR$, and we have a bijective correspondence between the fractional ideals and $K^{\times}/R^{\times}$.

The condition that there must exist an $r \in R$ such that $rJ \subseteq R$ is superfluous in Noetherian domains:

Corollary 50.2.12: Fractional Ideal Representative

Let R be a Noetherian Domain, $K = K(R)$, and $\mathfrak{a} \subseteq K$ a fractional ideal. Then there exists a $r \in R \setminus \{0\}$ and ideal $I \subseteq R$ such that

$$\mathfrak{a} = r^{-1}I$$

Stated differently, $\mathfrak{a} \cong_R I$

If $R = \mathcal{O}_K$, we may take $r \in \mathbb{Z}$. Furthermore, the above is an if and only if statement.

***Proof*:**
Use the fact that R is Noetherian. Let $\mathfrak{a} = (\alpha_1, \alpha_2, ..., \alpha_n)$. In particular, since $\mathfrak{a}$ is a finitely generated $\mathbb{Z}$-module:

$$\mathfrak{a} = \sum R\alpha_i \qquad\qquad \alpha_i \in K(R)$$

by definition, $\alpha_i = a_i/b_i$ for $a_i, b_i \in R$. Taking $b = b_1 b_2 \cdots b_n$, we get

$$b\mathfrak{a} \subseteq R$$

Hence, $b\mathfrak{a}$ is an ideal, say $b\mathfrak{a} = I$ Then $\mathfrak{a} \subseteq b^{-1}I$.

This immediately implies that $\mathfrak{a} \cong_R I$, for if $\mathfrak{a} = (\alpha_1, \alpha_2, ..., \alpha_n)$ so that $I = (r^{-1}\alpha_1, ..., r^{-1}\alpha_n)$, then define $\varphi(\alpha_i) = r^{-1}\alpha_i$ is an R-module isomorphism.

Notice that this generalizes the definition of ideals, if R is a Dedekind ring over itself, then the ideals of R are the finitely generated sub-modules. Then if $K = \mathrm{Frac}(R)$, the fractional ideals are the finitely generated K-modules.

Corollary 50.2.13: Inverse of Prime Ideals

Let R be a Dedekind domain and $\mathfrak{p} \subseteq R$ be a prime ideal. Then:

$$\mathfrak{p}\mathfrak{p}^{-1} = (1)$$

***Proof*:**
If $x \in \mathfrak{p}\mathfrak{p}^{-1}$, then $x = \sum p_i k_i$ for $p_i \in \mathfrak{p}$ and $k_i \in \mathfrak{p}^{-1}$. But by definition of $\mathfrak{p}^{-1}$, we have that $p_i k_i \in R$, hence x element is in R, and so $\mathfrak{p}\mathfrak{p}^{-1} \subseteq R$. Next, certainly $\mathfrak{p} \subseteq \mathfrak{p}\mathfrak{p}^{-1} \subseteq R$. Since $\mathfrak{p}$ is maximal, either $\mathfrak{p}\mathfrak{p}^{-1} = \mathfrak{p}$ or $\mathfrak{p}\mathfrak{p}^{-1} = R$. The former is a contradiction by lemma 50.2.8, and hence $\mathfrak{p}\mathfrak{p}^{-1} = (1)$, completing the proof.

Lemma 50.2.14: Invariance Number Ring Multiplication

Let $t \in K$, $\mathfrak{a} \subseteq K$ a fractional ideal. If $tI = I$, then $t \in R$

***Proof*:**
Since I is a finitely generated R module, there exists a surjective R-module homomorphism $\varphi : R^n \to I$. Next, since $tI = I$, there exists an R-module homomorphism $t : I \to I$ sending $i \mapsto ti$. Then $t \in \mathrm{Hom}_R(I, I)$. Since R^n is free, if we consider the following lift:

$$\begin{tikzcd}
R^n \arrow[r, two heads] & I \\
R^n \arrow[u, dashed, "\text{lift}"] \arrow[r, two heads, "\varphi"] \arrow[ur, two heads] & I \arrow[u, "t"']
\end{tikzcd}$$

Then $xt \in \mathrm{Hom}_R(R^n, I)$. Taking $\alpha \in \mathrm{Hom}_R(R^n, R^n)$:

$$\det(xI - \alpha)(\alpha) = 0 \Rightarrow \det(xI - \alpha)(t) = 0$$

now using that R is a Dedekind domain, namely that it's integrally closed, we get that $t \in R$, completing the proof.

50.2.2 Prime Decomposition in Dedekind Domain

With these properties, we may show the factorization of prime ideals in Dedekind domains.

Theorem 50.2.15: Unique Factorization Of Ideals In Dedekind Domains

Let R be a Dedekind Domain and $\mathfrak{a} \subseteq R$ a non-trivial proper ideal. Then there exists unique $\mathfrak{p}_1, \mathfrak{p}_2, ..., \mathfrak{p}_n$ such that

$$\mathfrak{a} = \mathfrak{p}_1\mathfrak{p}_2 \cdots \mathfrak{p}_n$$

***Proof*:**
We start with the existence of such a factorization. Let $\mathfrak{M}$ be the set of all non-trivial proper ideals which do not admit a prime ideal decomposition. If $\mathfrak{M}$ is non-empty, then since R is Noetherian there exists a maximal element, say $\mathfrak{m}$. Then this ideal is contained in some maximal ideal $\mathfrak{p}$. Since $R \subseteq \mathfrak{p}^{-1}$, we get the following chain of inclusions:

$$\mathfrak{m} \subseteq \mathfrak{m}\mathfrak{p}^{-1} \subseteq \mathfrak{p}\mathfrak{p}^{-1} \subseteq R$$

by lemma 50.2.8, we have $\mathfrak{m} \subsetneq \mathfrak{m}\mathfrak{p}^{-1}$ and $\mathfrak{p} \subsetneq \mathfrak{p}\mathfrak{p}^{-1} \subseteq R$. By maximality of $\mathfrak{p}$, $\mathfrak{p}\mathfrak{p}^{-1} = R$, and by maximality of $\mathfrak{m}$ in $\mathfrak{M}$, $\mathfrak{m}\mathfrak{p}^{-1}$ admits a prime ideal decomposition, say

$$\mathfrak{m}\mathfrak{p}^{-1} = \mathfrak{p}_1 \cdots \mathfrak{p}_r$$

But then, since $\mathfrak{p}\mathfrak{p}^{-1} = R$, $\mathfrak{m}R = \mathfrak{m}$ and so

$$\mathfrak{m} = \mathfrak{m}\mathfrak{p}^{-1}\mathfrak{p} = \mathfrak{p}_1 \cdots \mathfrak{p}_r\mathfrak{p}$$

contradicting the definition of $\mathfrak{m}$. Hence $\mathfrak{M}$ is an empty-set

For uniqueness, let

$$\mathfrak{a} = \mathfrak{p}_1\mathfrak{p}_2 \cdots \mathfrak{p}_r = \mathfrak{q}_1\mathfrak{q}_2 \cdots \mathfrak{q}_s$$

be two prime ideal factorizations of $\mathfrak{a}$. Then by the repeated application of property of prime ideals ($\mathfrak{a}\mathfrak{b} \subseteq P \Rightarrow \mathfrak{a} \subseteq P$ or $\mathfrak{b} \subseteq P$), we have that $\mathfrak{p}_1$ divides a factor of the right hand side, say $\mathfrak{q}_1$, and by maximality is thus equal to $\mathfrak{q}_1$. Then we can multiply both sides by $\mathfrak{p}_1^{-1}$ and since $\mathfrak{p}_1 \neq \mathfrak{p}_1\mathfrak{p}_1^{-1} = R$ we get

$$\mathfrak{p}_2 \cdots \mathfrak{p}_r = \mathfrak{q}_2 \cdots \mathfrak{q}_s$$

Then proceeding inductively, we will get that $r = s$ and $\mathfrak{p}_i = \mathfrak{q}_i$, up to renumbering, completing the proof.

50.2.16 Note It is in fact equivalent to define a Dedekind domain as ring in which each ideal factors uniquely, showing that this theorem characterizes Dedekind domains, see exercise ref:HERE.

From now on, we shall group the prime ideals to write

$$\mathfrak{a} = \mathfrak{p}_1^{n_1} \cdots \mathfrak{p}_r^{n_r} \qquad n_i > 0$$

If $\mathfrak{a} = (a)$ is a principal ideal, we shall use the slight abuse of notation and write

$$a = \mathfrak{p}_1^{n_1} \cdots \mathfrak{p}_r^{n_r} \qquad n_i > 0$$

Similarly, we shall often write $\mathfrak{a}\big|a$ instead of $\mathfrak{a}\big|(a)$. An important result worth emphasizing is the structure of R/I for an ideal $I \subseteq R$. First, note that if $\mathfrak{p}_i + \mathfrak{p}_j = (1)$, then $\mathfrak{p}_i^{n_i} + \mathfrak{p}_j^{n_j} = (1)$ (this was exercise ref:HERE (in CRT section)); the trick is to consider $1 = 1^{n_i+n_j+1} = (ax+by)^{n_i+n_j+1}$, use the binomial theorem, and group appropriate elements. Then by the Chinese Remainder Theorem:

$$R/I = \bigoplus_i R/P_i^{n_i}$$

By the 4th isomorphism theorem, $\overline{I} \subseteq R/P_i^{n_i}$ if and only if $P_i^{n_i} \subseteq I$, that is $I\big|P_i^{n_i}$. But by unique factorization, that means that $I = P_i^k$ for $1 \leq k \leq n_i$. Hence, every quotient has only finitely many ideals, each ideal I corresponding to a set of prime power of the decomposition of I via the isomorphism given by the CRT. This matches with the result that $R/(a)$ is a finite ring, showing there can only be finitely many ideals, and the prime factorization theorem for ideals gives us the structure of the ideals of $R/(a)$ through the correspondence given by the 4th isomorphism theorem. This also immediately shows that R/I is *Artinian*[15].

Using the fact that ideals factor and "behave" like the positive rational numbers ($-\mathfrak{p} = \mathfrak{p}$ since -1 is a unit), we can conclude some powerful results. We have shown that every ideal has a unique inverse. This immediately shows that:

$$\mathfrak{a}\mathfrak{b} = \mathfrak{a}\mathfrak{c} \quad \Longrightarrow \quad \mathfrak{b} = \mathfrak{c}$$

by the Cancellation law. Next, it is easy to see is that we may generalize the notion of finding GCD and LCM of UFD's to Dedekind domains:

Corollary 50.2.17: GCD, LCM, and Divisibility In Dedekind Domains

Let $\mathfrak{a}, \mathfrak{b} \subseteq R$ be ideals in a Dedekind domain R. Then

1. $\operatorname{lcm}(\mathfrak{a}, \mathfrak{b}) = P_1^{\max(a_1, b_1)} \cdots P_k^{\max(a_k, b_k)}$
2. $\gcd(\mathfrak{a}, \mathfrak{b}) = P_1^{\min(a_1, b_1)} \cdots P_k^{\min(a_k, b_k)}$
3. $\mathfrak{a} \big| \mathfrak{b}$ if and only if there exists a $\mathfrak{c}$ such that $\mathfrak{a}\mathfrak{c} = \mathfrak{b}$

***Proof*:**
For the first two, work with the factorization and property of ideals.

For 3, if $\mathfrak{a} \big| \mathfrak{b}$, we have that $\mathfrak{b} \subseteq \mathfrak{a}$. In a Dedekind domain, multiplying by $\mathfrak{a}^{-1}$, we get $\mathfrak{a}\mathfrak{a}^{-1} = (1)$ by extending lemma 50.2.8 with the above theorem. Hence:

$$\mathfrak{b}\mathfrak{a}^{-1} \subseteq \mathfrak{a}\mathfrak{a}^{-1} \subseteq R$$

showing that $\mathfrak{b}\mathfrak{a}^{-1} = \mathfrak{c} \subseteq R$ is an ideal, and $\mathfrak{b} = \mathfrak{c}\mathfrak{a}$, which properly generalizes the notion factorizing elements to the level of ideals.

We know that every ideal has an inverse. However, that inverse is a fractional ideal. Fortunately, there always exists an integral ideal that is "almost" the inverse, in that the product is a principal ideal[16]:

Corollary 50.2.18: Exists J such that IJ is Principal

Let R be a Dedekind Domain and $I \subseteq R$ an ideal. Then there exists an ideal $J \subseteq R$ such that $IJ = (a)$ for some element $a \in I$. In other words, there exists an ideal J such that the product IJ is principal.

[15] Equivalently, $\operatorname{kdim}(R/I) = 0$, and hence it is Artinian

[16] this almost inverse can be made into a concrete inverse when we show the principal ideals form a normal subgroup formed by the group of fractional ideals

***Proof*:**
Let $I = P_1^{n_1} \cdots P_k^{n_k}$. For each i, choose a nonzero element $r_i \in P_i^{n_i} \setminus P_i^{n_i+1}$. By the Chinese Remainder Theorem, there exists an $a \in R$ such that

$$a \equiv r_i \bmod P_i^{n_i+1} \qquad 1 \leq i \leq k$$

Thus, $a \in P_i^{n_i} \setminus P_i^{n_i+1}$ for each i, and so since $a \in P_i^{n_i}$, the power of P_i in the factorization is n_i. Now, taking the prime factorization of (a), we get:

$$(a) = P_1^{n_1} \cdots P_k^{n_k} P_{n+1}^{n_{k+1}} \cdots P_m^{n_m}$$

Letting $J = P_{n+1}^{n_{k+1}} \cdots P_m^{n_m}$ gives us

$$(a) = IJ$$

This same process can be repeated for fractional ideals; for any fractional ideal I, take $r \in R$ such that $rI = J \subseteq R$. Then proceed with the proof above, giving us an ideal L such that $(a) = JL = rIL$. Then $(r^{-1}a) = IL$, completing the proof.

In section 50.3.1, we shall find a more systematic way of finding the prime ideals that multiply to get a principal ideal using Galois groups. As seen with rings of integers, not all Dedekind domains are PIDs since they are not all UFDs. With the factorization theorem, we get that they are the next best thing:

Corollary 50.2.19: PIR, and Generation Of Ideals In Dedekind Domains

Let R be a Dedekind domain and $I \subseteq R$ an ideal. Then R/I is a Principal Ideal Ring, and I is generated by *at most* 2 elements.

***Proof*:**
For any ideal $I \subseteq R$, by the Chinese remainder theorem and prime factorization of I:

$$R/I = \bigoplus_i R/P_i^{n_i}$$

We shall show that each $R/P_i^{n_i}$ is a principal ideal ring, from which we may conclude by the isomorphism that R/I is a principal ideal ring. This shall give the desired result, for if $I \subseteq R$, then for any $J \subseteq I$ we may take R/J where $\overline{I} \subseteq R/(\alpha)$. Then since all ideals of $R/(\alpha)$ are principal, $\overline{I} = (\overline{\beta})$, that is $I = J + (\beta) = (\alpha, \beta)$. If we take $\alpha \in I$, then we would get $(\alpha) \subseteq I$ and $R/(\alpha)$, and so $I = (\alpha) + (\beta) = (\alpha, \beta)$, showing each ideal is generated by at most 2 elements.

Hence, we prove $R/P_i^{n_i}$ is a Principal ideal ring (PIR). If $\overline{J} \subseteq R/(P_i^{n_i})$, then $P_i^{n_i} \subseteq J \subseteq R$, meaning $J | P_i^{n_i}$, and so by unique factorization it must be that $J = P_i^k$ for some $1 \leq k \leq n_i$, which has image $\overline{P_i^k}$ (i.e., there are finitely many ideals).

Now, if $\overline{P_i} = \overline{P_i^2}$, then $\overline{P_i} = \overline{P_i^n} = \overline{0}$, and so $P_i = P_i^n$ implying that $R/P_i^{n_i}$ is a field, and it then automatically a PIR. If $\overline{P_i} \neq \overline{P_i^2}$, take $\alpha \in \overline{P_i} \setminus \overline{P_i^2}$. Then $\overline{\alpha} \notin \overline{P_i^2} \supseteq \overline{P_i^3} \supseteq \cdots$. Thus it must be that

$$\overline{P_i} \big| (\overline{\alpha})$$

But since the only ideals of $R/P_i^{n_i}$ are $\overline{P_i}, ..., \overline{P_i^{n_i-1}}$, it must follow that $(\overline{\alpha}) = \overline{P_i}$. From that it follows that $(\overline{\alpha}^k) = \overline{P_i^k}$, showing that $R/P_i^{n_i}$ is a PIR. But then R/I is a PIR since the isomorphism maps principal ideals to principal ideals, which then as we saw completes the proof.

As a consequence of this, if $\mathfrak{p}\big|(p)$ for some $p \in \mathbb{Z}$, then by using the construction of the proof above, we may always find an element $\alpha \in \mathfrak{p}$ such that $\mathfrak{p} = (p, \alpha)$. Hence, factorization of a prime $p \in \mathbb{Z}$ can always by recovered by adding an appropriate element to (p). More generally, if $I \subseteq J$, then $J = I + (a)$ (can you think of any conditions a needs to satisfy?). Our final result shows all pairs of ideals are, up to a multiple of $K(R)$, relatively prime. This mirrors the property that if $\gcd(a, b) = c$, we can divide a by c to get $\gcd(a/c, b) = 1$:

Corollary 50.2.20: Always Almost Relatively Prime

Let A, B be fractional ideals of the Dedekind Domain R. Then there exists elements $x, y \in K$ such that xA and yB are relative prime.

***Proof*:**
By multiplying A, B by appropriate scalars, we may assume they are integral ideal. We shall find an element c such that $cA + B = R$. First, let $a \in A$, and I be the ideal such that $AI = (a)$. Then since $BI \subseteq I$, consider:

$$\overline{I} \subseteq R/BI$$

Then since R/BI is principal, $I = BI + (b)$ for some $b \in R$. Multiplying by A, dividing by a, and letting $c = b/a$, we get

$$cA + B = R$$

as we sought to show.

With the division of ideals defined and mirroring the structure of $K^\times/R^\times$ ($\mathbb{Q}^\times_{>0}$ when working with $\mathbb{Z}$), it is tempting to say that they form a group structure, and this is indeed the case and is the focus in the remainder of this section.

One of the main metrics to measure is how many more "ideal numbers" need to be added to recover unique factorization into primes. If $\mathfrak{p}$ and $\mathfrak{q}$ are added, then if for some $t \in K$, $t\mathfrak{p} = \mathfrak{q}$, then $\mathfrak{p}$ and $\mathfrak{q}$ don't actually differ much in structure: it can be shown that $I = aJ$ ($a \in K(R)$) if and only if $I \cong_{\mathcal{O}_K} J$). Hence, we may classify "different" classes of ideals up to multiplication by a principal ideal. The more classes there are, the more a ring R fails to be a unique factorization domain. When R is a number ring, we shall show that there are only finitely many such classes.

The first result to prove is that the collection of fractional ideals of a Dedekind domain does indeed form a group:

Theorem 50.2.21: Ideal Group

Let R be a Dedekind domain, $K = K(R)$, and let J_K be the collection of fractional ideals. Then $(J_K, \cdot)$ has an abelian group structure via fractional ideal multiplication, and is called the *ideal group* of K

***Proof*:**
The operation is certainly well-defined (the product of fractional ideals is a fractional ideal), associativity and commutativity come for free, and the identity is $R = (1)$ since for any $\mathfrak{a} \in J_K$

$$\mathfrak{a}(1) = (1)\mathfrak{a} = \mathfrak{a}$$

What's left to check for is the existence of an inverse. For prime ideals, we have that $\mathfrak{p} \subsetneq \mathfrak{p}\mathfrak{p}^{-1}$, an hence by maximality we have $\mathfrak{p}\mathfrak{p}^{-1} = (1)$. For an integral ideal, if $\mathfrak{a} = \mathfrak{p}_1\mathfrak{p}_2 \cdots \mathfrak{p}_r$, then its inverse is

$$\mathfrak{b} = \mathfrak{p}_1^{-1} \cdots \mathfrak{p}_r^{-1}$$

We want to show that $\mathfrak{b} = \mathfrak{a}^{-1}$. Since $\mathfrak{b}\mathfrak{a} = (1)$, then by definition of fractional ideals $\mathfrak{b} \subseteq \mathfrak{a}^{-1}$. Conversely, to show $\mathfrak{a}^{-1} \subseteq \mathfrak{b}$, if $x\mathfrak{a} \subseteq (1)$, then $x\mathfrak{a}\mathfrak{b} \subseteq \mathfrak{b}$. Since $\mathfrak{a}\mathfrak{b} = R$, we have $aR \subseteq \mathfrak{b}$, that is $(a) \subseteq \mathfrak{b}$, and so $x \in \mathfrak{b}$. Hence, we have $\mathfrak{b} = \mathfrak{a}^{-1}$. Finally, if $\mathfrak{a}$ is an arbitrary fractional ideal and $0 \neq c \in R$ such that $c\mathfrak{a} \subseteq R$, then $(c\mathfrak{a})^{-1} = c^{-1}\mathfrak{a}^{-1}$ is the inverse of $c\mathfrak{a}$, so then $\mathfrak{a}\mathfrak{a}^{-1} = R$, completing the proof.

Corollary 50.2.22: Fractional Ideal Prime Decomposition

Every fractional ideal $\mathfrak{a}$ admits a unique representation as the product

$$\mathfrak{a} = \prod_{\mathfrak{p}} \mathfrak{p}^{n_\mathfrak{p}}$$

where $n_p \in \mathbb{Z}$ and $n_p = 0$ for almost all $\mathfrak{p}$. Thus, J_K is the free abelian group generated by the set of nonzero prime ideals $\mathfrak{p}$ of R

***Proof*:**
Notice that every fractional ideal $\mathfrak{a}$ is the quotient of two integral ideals,

$$\mathfrak{a} = \mathfrak{b}/\mathfrak{c}$$

which both have a prime decomposition. Thus, $\mathfrak{a}$ has a prime decomposition as stated in the corollary. Uniqueness follows the same argument as for integral ideals.

Definition 50.2.23: Ideal Group

Let R be a Dedekind Domain. Then J_R denotes the *ideal group* given by ideal multiplication.

We shall often write J_R interchangeably with J_K

Example 50.2.24: Ideal Group

1. The canonical example would be $J_\mathbb{Z} \cong \mathbb{Q}_+^\times$ with the bijection

   $$q\mathbb{Z} = (q) \mapsto q$$

   Note that it is important to think of the multiplicative structure of $\mathbb{Q}$, $(\mathbb{Q}, +)$ is not a free abelian group, having no two distinct elements be linearly dependent and not being cyclic.

2. More generally, let R be a Principal Ideal Domain. Then:

$$J_R \cong K^\times / R^\times$$

This isomorphism does not work in general, namely since not all ideals of R are principal.

A natural subgroup of J_K is formed by the principal ideal (a) where $a \in K^\times$. This subgroup can be thought of as representing our usual notion of factorization into elements instead of ideals As some foreshadowing, notice that the principal ideal The quotient of J_K by the subgroup of principle ideals can be thought of as measuring how far away the ideal group is from being a PID:

Definition 50.2.25: Ideal Class Group

Let J_K be the class group of K. Then

$$Cl(K) = J_K / P_K$$

is called the *ideal class group*, *class group*, or *Picard group* of K.

50.2.26 Note on Nomenclature The term Picard group shall not be used in this book, as the nomenclature is more so associated within algebraic geometry. It is pointed out here that some call the class group the Picard group as the class group is a special case of the Picard group.

These naturally fit in the following exact sequence:

$$1 \to R^\times \to K^\times \to J_K \to Cl(K) \to 1$$

where $K^\times \to J_K$ is the map $a \mapsto (a)$. Hence, we may think of the class group $Cl(K)$ as a measure of the expansion that happens when we pass from numbers to ideals, where the unit group $R^\times$ measure the contraction of the same process. Thus, we may want to understand the groups $R^\times$ and $Cl(K)$ more thoroughly. In general, these can be arbitrary abelian groups[17], but for rings of integers $\mathcal{O}_K$ of number fields K, we get that the size of the class group is always finite and that $\mathcal{O}_K^\times$ is a free $\mathbb{Z}$-module along with roots of unity contained in K, as will soon be shown.

50.2.3 Class Group of Ring of Integer

Recall that ideals $\mathfrak{a} \subseteq \mathcal{O}_K$ form a sub-lattice of $\mathcal{O}_K$ in the Archimedean embedding. The following theorem is key bound on the covolume of lattice. Using theorem 50.1.65, we may rather easily show that the class group is finite. The way we shall find the size of the class group is by showing that every element in a class group will have absolute norm less than a bound (say M), and that only finitely many ideals can have norm equal to $m < M$. Then, by considering the principal ideals that have norm less than M, we shall be able to find all prime ideals that compose these principal ideals, which shall give us an estimate on the size of the group.

[17]There are "enough" 1-dimensional smooth schemes to make arbitrary class groups on Dedekind domains

Lemma 50.2.27: Absolute Norm of Prime

Let $0 \neq \mathfrak{p} \subseteq \mathcal{O}_K$ be a prime ideal of $\mathcal{O}_K$ and let $\mathcal{O}_K$ be rank n. Then

$$\mathfrak{N}(\mathfrak{p}) = p^n$$

***Proof*:**
By theorem 26.3.2, $\mathfrak{p} \cap \mathbb{Z} = p\mathbb{Z}$. Since $\mathcal{O}_K/\mathbb{Z}$ is an integral extension, $\mathcal{O}_K/\mathfrak{p}$ is a finite field extension of $\mathbb{Z}/p\mathbb{Z}$ of degree n by example 26.1.9(3). Hence:

$$\mathfrak{N}(\mathfrak{p}) = p^n$$

Proposition 50.2.28: Absolute Norm for Ring of Integers Is Multiplicative

Let $\mathfrak{a} = \mathfrak{p}_1^{n_1} \cdots \mathfrak{p}_k^{n_s}$ be the prime Factorization of the nonzero ideal $\mathfrak{a}$. Then:

$$\mathfrak{N}(\mathfrak{a}) = \mathfrak{N}(\mathfrak{p}_1)^{n_1} \cdots \mathfrak{N}(\mathfrak{p}_k)^{n_s}$$

From this, we may conclude that

$$\mathfrak{N}(\mathfrak{a}\mathfrak{b}) = \mathfrak{N}(\mathfrak{a})\mathfrak{N}(\mathfrak{b})$$

***Proof*:**
By the Chinese remainder theorem:

$$\mathcal{O}_K/\mathfrak{a} \cong \mathcal{O}_K/\mathfrak{p}_1^{n_1} \oplus \cdots \oplus \mathcal{O}_K/\mathfrak{p}_s^{n_s}$$

Hence, we see that the index $[\mathcal{O}_K : \mathfrak{a}]$ is the product of the indexes $[\mathcal{O}_K : \mathfrak{p}_k^{n_i}]$. To find this, notice that the chain

$$\mathfrak{p}^n \subseteq \mathfrak{p}^{n-1} \subseteq \cdots \mathfrak{p} \subseteq \mathcal{O}_K$$

is a sequence of integral extensions[a], so we may use the fact that the index is multiplicative, namely:

$$[\mathcal{O}_K : \mathfrak{p}^n] = [\mathcal{O}_K : \mathfrak{p}][\mathfrak{p}^2 : \mathfrak{p}] \cdots [\mathfrak{p}^n : \mathfrak{p}^{n-1}]$$

We must now show each index has the same size. We shall take advantage of the fact that $\mathcal{O}/\mathfrak{p}$ is finite, and show that each $\mathfrak{p}^i/\mathfrak{p}^{i+1}$ is isomorphic to $\mathcal{O}/\mathfrak{p}$ (meaning they are all of the same size, hence the same index). I claim that $\mathfrak{p}^i/\mathfrak{p}^{i+1}$ is an $(\mathcal{O}_K/\mathfrak{p})$-vector space of dimension 1 with the natural left-multiplication action; this will come down to the multiplicative structure of J_K. To see this, let $a \in \mathfrak{p}^i \setminus \mathfrak{p}^{i+1}$ and take $\mathfrak{b} = (a) + \mathfrak{p}^{i+1} = \gcd((a), \mathfrak{p}^{i+1})$ so that $\overline{\mathfrak{b}}_{\mathfrak{p}^{i+1}} = \overline{(a)}_{\mathfrak{p}^{i+1}}$. Then by construction

$$\mathfrak{p}^{i+1} \subsetneq \mathfrak{b} \subseteq \mathfrak{p}^i$$

From this, we must conclude that $\mathfrak{p}^i = \mathfrak{b}$; this can be seen immediately by realising that $\mathfrak{b} = \gcd((a), \mathfrak{p}^{i+1}) = \mathfrak{p}^i$ by definition of (a). Therefore $\overline{\mathfrak{b}} = a \bmod \mathfrak{p}^{i+1}$ is a basis for the $\mathcal{O}_K/\mathfrak{p}$-vector-space $\mathfrak{p}^i/\mathfrak{p}^{i+1}$, hence $\mathfrak{p}^i/\mathfrak{p}^{i+1} \cong \mathcal{O}_K/\mathfrak{p}$. This can also be demonstrated by taking

$$\mathcal{O} \to \mathfrak{p}^i/\mathfrak{p}^{i+1} \qquad \alpha \mapsto \overline{\alpha a}$$

and seeing the map is surjective since $\mathfrak{p}^i = \gcd(\mathfrak{b}, \mathfrak{p}^{i+1})$ and has kernel $\mathfrak{p}$. Since this is true for all i, $1 \leq i \leq n$:

$$\mathfrak{N}(\mathfrak{p}^n) = [\mathcal{O}_K : \mathfrak{p}^n] = [\mathcal{O}_K : \mathfrak{p}][\mathfrak{p} : \mathfrak{p}^2] \cdots [\mathfrak{p}^{n-1} : \mathfrak{p}^n] = \mathfrak{N}(\mathfrak{p})^n$$

Thus more generally:

$$\mathfrak{N}(\mathfrak{a}\mathfrak{b}) = \mathfrak{N}(\mathfrak{a})\mathfrak{N}(\mathfrak{b})$$

as we sought to show.

[a]This is a *rng* integral extension. Show every $x \in \mathfrak{p}^i$ is a solution to a $\mathfrak{p}^{i+1}$ polynomial

You may ask what is the absolute norm of a prime $\mathfrak{p}$? If $\mathfrak{p}$ were principal, we know it is p^f for appropriate f. For the non-principal case, in corollary 50.3.16, we shall show that if $\mathfrak{p}_k$ lies over p, then $\mathfrak{N}(\mathfrak{p}_k) = p^f$ for an appropriate number f that we shall determine later.

Since the absolute norm is multiplicative and the index is always positive, and fractional ideals also form lattices, we may extend it to be defined on fractional ideals and get a group homomorphism:

$$\mathfrak{N} : J_K \to \mathbb{R}^\times_{>0} \qquad \mathfrak{a} \mapsto \mathfrak{N}(\mathfrak{a})$$

where $\mathfrak{a} = \prod_{\mathfrak{p}} \mathfrak{p}^{n_\mathfrak{p}}$ with $n_\mathfrak{p} \in \mathbb{Z}$.

Corollary 50.2.29: Volume Form Reformulation

Let $I \subseteq K$ be a fractional ideal of a number field K. Then:

$$\mathrm{vol}(I) = \mathfrak{N}(I)\mathrm{vol}(\mathcal{O}_K)$$

***Proof*:**
This is just reformulating proposition 50.1.60 or lemma 50.1.64

Perhaps include the proof in MIT p.18?

Theorem 50.2.30: Class Number

Let K be a number field and let $Cl(K) = J_K/P_k$ be the class group of K. Then $Cl(K)$ is finite.

***Proof*:**

Let $0 \neq \mathfrak{p} \subseteq \mathcal{O}_K$ be a prime ideal of $\mathcal{O}_K$. By lemma 50.2.27,$\mathfrak{N}(\mathfrak{p}) = p^n$, where $n = [\mathcal{O}_K/\mathfrak{p} : \mathbb{Z}/p\mathbb{Z}]$. Given p, there are only finitely many prime ideals $\mathfrak{p}$ such that $\mathfrak{p} \cap \mathbb{Z} = p\mathbb{Z}$ by proposition 50.1.66 (equivalently, because $p\mathcal{O}_K$ must have a number of terms in it's factorization and $\mathfrak{p}\big|p\mathcal{O}_K$). Thus, there are only finitely many prime ideal of bounded absolute norm. Since every integral ideal factors into prime ideals, we have

$$\mathfrak{N}(\mathfrak{a}) = \mathfrak{N}(\mathfrak{p}_1)^{n_1} \cdots \mathfrak{N}(\mathfrak{p}_k)^{n_k}$$

It follows that there are only a finite number ideals $\mathfrak{a}$ of $\mathcal{O}_K$ with bounded absolute norm $\mathfrak{N}(\mathfrak{a}) \leq M$.

Thus, it suffices to show that each class $[\mathfrak{a}] \in Cl(K)$ contains an integral ideal $\mathfrak{a}_1$ satisfying

$$\mathfrak{N}(\mathfrak{a}_1) \leq \lambda$$

for if there were infinitely many classes, then there would be infinitely many distinct ideals with absolute norm less than λ, contradicting our earlier discussion. By theorem 50.1.65, for any ideal $I \in [\mathfrak{a}]^{-1}$, pick $\alpha \in \mathfrak{a}$ such that

$$|\mathrm{Nm}_{K/\mathbb{Q}}(\alpha)| \leq \lambda \mathfrak{N}(I)$$

By corollary 50.2.18 there exists a J such that $(\alpha) = IJ$. By definition, $J \in [\mathfrak{a}]$. Then:

$$\mathfrak{N}(I)\mathfrak{N}(J) = \mathfrak{N}(\alpha) = |\mathrm{Nm}_{K/\mathbb{Q}}(\alpha)| \leq \lambda \mathfrak{N}(I) \implies \mathfrak{N}(J) \leq \lambda$$

but this now gives that there are only finitely many classes, and hence the class group is finite, as we sought to show.

Definition 50.2.31: Class Number

Let $Cl(K)/ = J_K/P_k$ be the class group of K. Then $h_K = [J_K : P_K]$ is called the *class number* of $Cl(K)$ or K.

A consequence of there being a finite class number is that for every ideal $I \subseteq \mathcal{O}_K$, there exists a minimal number n such that $I^n = (\alpha)$ for some $\alpha \in \mathcal{O}_K$. There is also a minimal number n' such that for *any* ideal of $\mathcal{O}_K$, the ideal to the power of n' is principal. This may encourage thoughts of converting the problem on ideals into one on principal ideals, given there is a way of going back to the original ideals. We shall see that this shall sometimes be the case.

Example 50.2.32: Class Groups

1. Naturally, $|Cl(\mathbb{Q})| = 1$ since $\mathbb{Z}$ is a PID
2. Similarly, $|Cl(\mathbb{Q}(i)| = |Cl(\mathbb{Q}(\sqrt{-3})| = 1$ since $\mathbb{Z}[i]$ and $\mathbb{Z}\left[\frac{1+\sqrt{-3}}{2}\right]$ are PID.
3. Every DVR has ideals of the form (π^n) for appropriate uniformizer π. Thus, the ideal group is isomorphic to $\mathbb{Z}$, and the class group is trivial.
4. Consider $Cl(\mathbb{Q}(\sqrt{-5})$. Since $\mathbb{Z}[\sqrt{-5}]$ is not a Unique Factorization Domain (hence Principal Ideal Domain), we will get a non-trivial class group. Let $K = \mathbb{Q}(\sqrt{-5})$. Choose an integral basis

$$\mathcal{O}_K = \langle 1, \sqrt{-5} \rangle_{\mathbb{Z}}$$

so that $D_k = -20$. Hence, the Minkowski bound gives us:

$$M_k = \sqrt{20}\frac{2}{2^2}\frac{4}{\pi} < 4$$

In fact, we have < 3, but we shall purposefully miss-guess the bound to show how to determine when two ideals belong in the same class group. Consider the ideal $(2), (3)$. Assume that $\mathfrak{p}_2 \big| (2)$. Then we know $\mathfrak{N}(\mathfrak{p}_2) \big| \mathfrak{N}((2)) = 4$. To find $\mathfrak{p}_2 \subseteq \mathcal{O}_K$ consider the map $\varphi : \mathcal{O}_K \to \mathcal{O}_K/(2)$. Since $\mathcal{O}_K/(2)$ is finite the pre-image of a prime ideal of a surjective map is prime, we may find prime ideals of $\mathcal{O}_K/(2)$. By the 4th isomorphism theorem, if $\overline{I} \subseteq \mathcal{O}_K/(2)$ if and only if $(2) \subseteq I$, that is $I\big|(2)$, and hence any prime ideal corresponding to prime ideal of $\mathcal{O}_K/(2)$ divides (2); thus we look for $\varphi^{-1}(P) = \mathfrak{p}_2$. To find such a $\mathfrak{p}_2$, think of $\mathcal{O}_K$ as $\mathcal{O}_K \cong \mathbb{Z}[x]/(x^2+5)$[a] so that we can take:

$$\mathcal{O}_K/(2) \cong \mathbb{Z}[x]/(x^2+5,2) \cong \mathbb{F}_2[x]/(x^2-1) = \mathbb{F}_2[x]/(x+1)^2)$$

Recall that the prime ideals of a ring are in corresponding with the prime ideals of the ring quotiented by the nilradical (see section 32.2). But then we immediately see that the only prime ideal in $\mathcal{O}_K/(2)$ is $(1+x)$. Hence

$$\mathfrak{p}_2 = (1+\sqrt{-5}, 2)$$

Since $\mathfrak{N}(\mathfrak{p}_2)\big|4$, the only possibility is that $\mathfrak{p}^2 = 2$, and indeed:

$$\mathfrak{p}_2^2 = (4, 2+2\sqrt{-5}, -4+2\sqrt{-5}) = (2)$$

Is $\mathfrak{p}_2$ principal? If it was, then $\mathfrak{p}_2 = (a+b\sqrt{-5})$. Then:

$$2 = \mathfrak{N}(\mathfrak{p}_2) = |\mathrm{Nm}_{K/\mathbb{Q}}(a+b\sqrt{-5})| = a^2+5b^2$$

but it's evident this is not possible, hence it is not principal! Similarly with (3), we get:

$$\mathcal{O}_K/(3) \cong \mathbb{Z}[x]/(x^2+5,3) = \mathbb{F}_3[x]/(x^2-1) = \mathbb{F}_2 \oplus \mathbb{F}_3$$

From this, we can see that $3 = \mathfrak{p}_3\mathfrak{p}_3'$ where

$$\mathfrak{p}_3 = (3, 1+\sqrt{-5}) \qquad \mathfrak{p}_3' = (3, 1-\sqrt{-5})$$

Finally, notice that $[\mathfrak{p}_2] = [\mathfrak{p}_3]$. To see this, notice their product is in fact principal:

$$\mathfrak{p}_2\mathfrak{p}_3 = (2, 1+\sqrt{-5})(3, 1+\sqrt{-5}) = (6, 1+\sqrt{-5}, (1+\sqrt{-5})^2) = (1+\sqrt{-5})$$

Hence, $\mathfrak{p}_3 = (1+\sqrt{-5})^{-1}\mathfrak{p}_2$. Evidently, $\mathfrak{p}_3, \mathfrak{p}_3'$ are in the same class. Since there are only two elements, it must be that $Cl(K) \cong \mathbb{Z}/2\mathbb{Z}$.

5. Let $K = \mathbb{Q}(\sqrt{-23})$. Choose $\mathcal{O}_K = \langle 1, \frac{1+\sqrt{-23}}{2}\rangle_{\mathbb{Z}}$ so that it's discriminant is $d_k = -23$ and the Minkowski bound is $M_K = \sqrt{23}\frac{4}{\pi}\frac{1}{2} < 4$. Hence, we again check the prime ideals (2) and (3), noting that the minimal polynomial of the generator is $x^2 - x + 6$:

$$\mathcal{O}_K/(2) \cong \mathbb{F}_2[x]/(t^2 - t)$$

Thus, if $t = \frac{1+\sqrt{-23}}{2}$:

$$(2) = (2, t-1)(2, t)$$

and similarly

$$(3) = (3, t-1)(3, t)$$

Certainly, each prime multiple of (2) and (3) are in the same equivalence class. Is $[(2,t)] = [(3, t-1)]$? Choosing good representatives and multiplying, we get

$$(3,t)(2,t) = (6,t) = (t)$$

hence, we again get that $Cl(K) = \mathbb{Z}/2\mathbb{Z}$.

6. Let $K = \mathbb{Q}(\sqrt{15})$ and $\mathcal{O}_K = \langle 1, \sqrt{15}\rangle_{\mathbb{Z}}$. Then doing the same procedure, we get:

$$(3, \sqrt{15})^2 = (3)$$

How do we know that $(3, \sqrt{15})$ is principal or not? We can see that $a^2 - 15b^2 = 3$ has no solutions using modular arithmetic's, but it will not always be feasible to use this method, so we introduce another. Let's say $(3, \sqrt{15}) = (\alpha)$. Then

$$(\alpha^2) = (3)$$

meaning they are associate, hence $\alpha^2 = 3u$ for $u \in \mathcal{O}_K^\times$. Does there exist such a unit? In section 50.2.4, we shall find this structure. For now, it can be taken for granted that $\mathcal{O}_K^\times = \pm\langle 4 + \sqrt{15}\rangle$ (notice the norm of all such numbers is 1). Then from this we see that no such unit exist, hence $(3, \sqrt{15})$ is not principal.

Notice that all of these examples are for quadratic extensions. It is much more difficult to find prime-factorization for non-quadratic extension since the ring of integers is not easy to find. In the next section, we shall see how we may get some information on the factorization of primes when $\mathcal{O}_K$ is not available

[a]Note how this relied on knowing the structure of $\mathcal{O}_K$! See comment at the end of this example

In general, we have $h_K > 1$. If $h_K = 1$ then we can think of the prime factorization of ideals being the same as the prime factorization of elements as is already well-understood. If $K = \mathbb{Q}(\sqrt{d})$ where d is square-free and negative, then the quadradic extensions in which the class number is 1 is

$$-1, -2, -3, -7, -11, -19, -43, -67, -163$$

For real quadratic extensions, there are a lot more, for example:

$$2, 3, 5, 6, 7, 11, 13, 14, 17, 19, 22, 23, 29, 31, 33, \cdots$$

it is unknown where there are infinitely many real quadratic fields of class number 1, or even if there are infinitely many algebraic number fields (of any degree) with class number 1. In general, it is in fact very difficult to determine some structure in the class group given some class of number fields, with an important notable exception cyclotomic extensions which due to the work of Kenkichi Iwasawa we know behave in a regular way.

The importance of studying cyclotomic extensions links back to Fermat's last theorem which the reader surely recalls states that the equation $x^p + y^p = z^p$ has no nonzero integer solutions for $p \geq 3$. For $p = 2$, we take the cyclotomic extension $\mathbb{Q}(\zeta_4) = \mathbb{Q}(i)$ to get

$$x^2 + y^2 = (x + iy)(x - iy)$$

and then we get the Fermat's twin prime theorem that fully solved the problem in this case. If p is prime, then considering the equation $y^p = z^p - x^p$ over $\mathbb{Q}(\zeta_p)$, we may factor this in $\mathbb{Z}[\zeta_p]$ as

$$\begin{aligned} y^p &= x^p - z^p \\ &= z^p\left(\left(\frac{x}{z}\right)^p - 1\right) \\ &= z^p \prod_{k=0}^{p-1}\left(\frac{x}{z} - \omega_k\right) \\ &= \prod_{k=0}^{p-1} z\left(\frac{x}{z} - \omega_k\right) \\ &= \prod_{k=0}^{p-1}(x - \omega_k z) \end{aligned}$$

where $\omega_k = \zeta_p^k$. Thus, if we assume an existence of a solution, we obtain two multiplicative decompositions of the same number in $\mathbb{Z}[\zeta_p]$. Using this, we can show it contradicts unique factorization *provided* that $\mathbb{Z}[\zeta_p]$ is a unique factorization domain. It was due to some mathematician erroneously assuming unique factorization (equivalently for us, that the class number is always 1) that lead to many fake proofs of Fermat's last theorem[18]. Kummer fixed this by introducing the notion of ideal primes, and translated the statement into one where we must show that $p \nmid h_p$ instead of assuming $h_p = 1$. If this is the case, such a prime is called *regular*, and *irregular* otherwise. Unfortunately, not all primes are regular. Of the first 25 primes less than 100, the primes 37, 59, and 67 are irregular. As of writing this book, it is unknown whether there are infinitely many regular primes.

The regular and irregular prime are going to play important roles in our coming theory, but they do not play much of a role in the eventual proof of Fermat's Last theorem. Due to the work of Gerhard Frey, there was a reformulation of Fermat's last theorem into a statement about elliptic curves. This lead to a new development of mathematics which is beyond the scope of this book, but the interested reader may feel free to search up the *Taniyama-Shimura-Weil Conjecture* to see how Fermat's Last Theorem got translated into a problem of elliptic curves.

We end this section with showing a certain Diophantine equation has no integer solution as it combines nicely all the machinery we have done, and some generalizations that is motivation for class field theory. Starting with the Diophantine equation:

Theorem 50.2.33: No Solution To Diophantine Equation

The equation

$$y^2 = x^3 - 5$$

has no integer solutions

***Proof*:**
Suppose the pair (x, y) is an integer solution. If x is even, then $x^3 - 5$ is odd, and we get

$$y^2 \equiv -1 \bmod 4 \equiv 3 \bmod 4$$

which is easy to check is impossible. Next, x and y must be coprime, for if they weren't then

[18] It is speculated that this is the proof Fermat was referring to as being too long for the margins of his book

their common factor must be 5 (since then $p \big| y^2 - x^3 = -5$), so

$$5^2 y_0^2 = 5^3 x_0 - 5 = 5y_0^2 = 5^2 x_0 - 1$$

But then mod 5 the equation has no solution, hence it has no solution in general.

Now, factoring in $\mathbb{Z}[\sqrt{-5}]$, we get

$$(y - \sqrt{-5})(y + \sqrt{-5}) = (x)^3$$

The ideals $(y - \sqrt{5})$ and $(y + \sqrt{5})$ are coprime; for is $\mathfrak{p}$ divides both, then $\mathfrak{p}\big|(x)^3$, so $\mathfrak{p}\big|(x)$, and since $(y - \sqrt{5}), (y + \sqrt{5}) \subseteq \mathfrak{p}$ then $(y - \sqrt{-5}) + (y + \sqrt{-5}) = 2y \in \mathfrak{p}$ so $\mathfrak{p}\big|(2y)$. Since x is odd and $\mathfrak{p}\big|(x)$, $\mathfrak{p}\big|(y)$. Then since $\mathfrak{p}\big|(y^2 - x^3) = (5)$, we must have $\mathfrak{p} = (\sqrt{-5})$. So $(\sqrt{-5})\big|(y + \sqrt{-5})$, implying $(5)\big|(x)$. But now we can again work in mod 5 and reach a contradiction.

Hence, they are co-prime. Thus, as the left hand side is the cube of (x), we must have:

$$(y - \sqrt{-5}) = \mathfrak{a}^3 \qquad (y + \sqrt{-5}) = \mathfrak{b}^3$$

Finally, since the class group is of size 2 and a third power produces a principal ideal, it must be that $\mathfrak{a}, \mathfrak{b}$ are both principal. Given that the only units are ± 1 (something we shall prove in the next section), we have that:

$$y + \sqrt{-5} = (a + b\sqrt{-5})^3 = a^3 + 3a^2 b\sqrt{-5} - 3ab^2 \cdot 5 - b^4 \cdot 5\sqrt{-5} = (a^3 - 15ab^2) + (3a^2 b - 5b^3)\sqrt{-5}$$

Manipulating and comparing the coefficients of $\sqrt{-5}$, we get that:

$$1 = 3ba^2 - 5b^3 = b(3a^2 - 5b^2)$$

implying $b = \pm 1$ since $a, b \in \mathbb{Z}$. But then:

$$3a^2 - 5 = \pm 1$$

but now this can be inspected to see that it's impossible, hence no such integer solution exists.

Note that a few things lined up well here:

- There were only two units, if there are more then the considerations are more complicated.
- The class group happened to let us deduce the ideal is principal; if the ideal was not principal there would be more to consider.
- The equation was easily factorizable in the extension (ex. $y^2 = x^5 + x + 1$ is a good deal more difficult)
- The factors are coprime.

Exercise 50.2.1

1. Let $A, B \subseteq R$ with $B \subseteq A$ and $A \neq R$. Show there exists a $\gamma \in K$ such that $\gamma B \subseteq R$, but $\gamma B \not\subseteq A$

2. Show that $IJ = (I+J)(I\cap J) = \gcd(I,J)\mathrm{lcm}(I,J)$
3. Let $\alpha \in \mathcal{O}_K \setminus \{0\}$. Show that there exists a $\beta \in \mathcal{O}_K$ st $\alpha\beta \in \mathbb{Z}\setminus\{0\}$
4. Let $K = \mathbb{Q}(\sqrt{-3})$. Show that $\mathcal{O}_K \neq \mathbb{Z}[\sqrt{-3}]$. Show that
5. Show that $3 = x^2 + 5y^2$ has no integer solutions. Note that 3 splits in $\mathbb{Z}[\sqrt{-3}]$ showing that if the ringer of integers is not UFD then it is more complicated to use factorizations.
6. exercise for quotient ideals
   a) here

50.2.4 Dirichlet's Unit Theorem

(Current intuition when it comes to solving equations: we are working with ideals, but equations use elements. We must be concious of the units at play to reduce back to elements)

As we saw in example 50.2.32, it is important to know the unit group of $\mathcal{O}_K$ to find elements in the class group. We start this section by showing that we may embed $\mathcal{O}_K^\times$ to form a lattice, and then use this to find the structure of $\mathcal{O}_K$.

The group $\mathcal{O}_K^\times$ may have some torsion elements, namely all roots of unity that lie in K; this collection is labeled as $\mu(K)$. This group will in fact be determined by the number of embeddings into Minkowski space. On a high-level, the structure of this group is within this diagram:

$$\begin{tikzcd}
K^\times \arrow[r, "\sigma"] \arrow[rr, bend left, "\sigma^\times"] \arrow[d, "N_{K/\mathbb{Q}}"] & K_\mathbb{R}^\times \arrow[r, "\log"] \arrow[d, "\mathrm{nm}"] & {[\prod_\tau \mathbb{R}]^+} \arrow[d, "\mathrm{Tr}"] \\
\mathbb{Q}^* \arrow[r, hook] & \mathbb{R}^* \arrow[r, "\log|\ |"] & \mathbb{R}
\end{tikzcd}$$

where we shall write $K_\mathbb{R}^\times = (\mathbb{R}^{r_1} \oplus \mathbb{C}^{r_2})^\times$. In particular, the top row of the above commutative diagram can be replaced by:

$$\begin{aligned}
\mathcal{O}_K^\times &= \left\{ r \in \mathcal{O}_K \;\middle|\; N_{K/\mathbb{Q}(n)=\pm 1} \right\} \\
S &= \left\{ y \in K_\mathbb{R}^\times \;\middle|\; N(y) = \pm 1 \right\} && \text{norm-one surface} \\
H &= \left\{ x \in \left[\prod_\tau \mathbb{R}\right]^+ \;\middle|\; \mathrm{tr}(x) = 0 \right\} && \text{trace-zero hyperplane}
\end{aligned}$$

Then we get the homomorphism:

$$\mathcal{O}_K^\times \xrightarrow{\sigma} S \xrightarrow{\log} H$$

where $\sigma^\times = \log \circ \sigma$.

Proposition 50.2.34: Roots Of Unity In Group Of Units

Let $\Gamma_K^\times = \sigma^\times(\mathcal{O}_K^\times) \subseteq H$. Then the sequence

$$1 \to \mu(K) \to \mathcal{O}_K^\times \xrightarrow{\sigma^\times} \Gamma_K^\times \to 0$$

is exact.

***Proof*:**
Certainly $\mu(K) \subseteq \ker(\sigma^\times)$. Conversely, say $r \in \mathcal{O}_K^\times$ where $\sigma^\times(r) = 0$. Then $|\tau(r)| = 1$ for each embedding $\gamma : K \to \mathbb{C}$, hence $\sigma(r) = (\tau(r))$ is in a bounded domain of the $\mathbb{R}$-vector space $K_\mathbb{R}$. On the other hand, $\sigma(r)$ is a point of the lattice $\sigma(\mathcal{O}_K)$ of $K_\mathbb{R}$. Thus, the kernel of $\sigma^\times$ can contain only a finite number of elements, and so since it's a finite group, it contains only the roots of unity of $K^\times$.

We now build towards the structure of $\Gamma_K^\times$. Under the right adjustments $\mathcal{O}_K^\times$ can be embedded in a similar way to $\mathcal{O}_K$. Take $\sigma(\mathcal{O}_K^\times) \subseteq \mathbb{R}^{r_1} \oplus \mathbb{C}^{r_2}$. the way a mesh is formed is by taking the logarithm, for $\log(ab) = \log(a) + \log(b)$

$$\sigma^\times : K^\times \to \mathbb{R}^{r_1} \oplus \mathbb{R}^{r_2} \qquad \sigma^\times = \log \circ \sigma$$

Proposition 50.2.35: Multiplicative Group Forms Lattice

Let H be the trace-zero hyperplane. Then $\sigma^\times(\mathcal{O}_K^\times) \subseteq W$ and it is discrete.

***Proof*:**
First, the map $(\mathbb{R}^{r_1} \oplus \mathbb{C}^{r_2})^\times \to \mathbb{R}^{r_1+r_2}$ mapping

$$(a_1, a_2, ..., a_{r_1}, b_1, b_2, ..., b_{r_2}) \mapsto (\log(a_1), ..., \log(a_{r_1}), ..., \log(b_{r_2}))$$

is a proper topological map, in particular the image of a discrete set is discrete. Since $\sigma(\mathcal{O}_K) \subseteq \mathbb{R}^{r_1} \oplus \mathbb{C}^{r_2}$ is discrete. $\sigma(\mathcal{O}_K \setminus \{0\}) \subseteq (\mathbb{R}^{r_1 \oplus r_2})^\times$ is discrete, Next, for $\epsilon > 0$, if $\sigma^\times(\alpha) \subseteq \overline{B_\epsilon}$, then $\forall \varphi \in \mathrm{Hom}_\mathbb{Q}(K, \mathbb{C})$, $e^{-\epsilon} < |\varphi(\alpha)| < e^\epsilon$. Now, since the ball is compact, by properness there are only finitely many α, but that completes the proof.

Theorem 50.2.36: Dirichlet Unit Theorem

The group of units $\mathcal{O}_K^\times$ is the direct product of the finite cyclic group $\mu(K)$ and a free abelian group of rank $r_1 + r_2 - 1$. The free-generators of $\mathcal{O}_K^\times$ are called the *fundamental units* of $\mathcal{O}_K^\times$:

$$\mathcal{O}_K^\times \cong_\mathbb{Z} \mathbb{Z}^{r_1+r_2-1} \oplus \mu(K)$$

We may also characterize $\mathcal{O}_K^\times$ by the following short exact sequence:

$$1 \to \mu(K) \to \mathcal{O}_K^\times \to \Gamma_K^\times \to 0$$

where since the right term is a free $\mathbb{Z}$-module, the sequence splits giving us $\mathcal{O}_K^\times \cong \mu(K) \oplus \Gamma_K^\times$.

***Proof*:**
(Marcus p.100 looks good)

Thus, there exist units $u_1, u_2, ..., u_t$, $t = r_1 + r_2 - 1$ such that for any other unit $u \in \mathcal{O}_K^\times$, it can be written as

$$u = \zeta u_1^{n_1} \cdots u_t^{n_t}$$

where $\zeta \in \mu(K)$ is a root of unity.

Example 50.2.37: Unit Group And Class Group

1. Let $K = \mathbb{Q}(\sqrt{2})$. Then $1 + \sqrt{2}$ generates the free part of the group of units. Similarly, for $K = \mathbb{Q}(\sqrt{3})$, $2 + \sqrt{3}$ generates the free part of the group of units.
2. Let us compute the class group and unit group of $\mathbb{Q}(\sqrt[3]{7})$ Let $K = \mathbb{Q}(\sqrt[3]{7})$ and $s = \sqrt[3]{7}$. We again look for the Minkowski bound. Doing a similar computation as before, we may workout

$$\mathcal{O}_K = \langle 1, \sqrt[3]{7}, \sqrt[3]{7}^2 \rangle$$

Hence, the discriminant is:

$$\det \begin{pmatrix} 1 & \sqrt[3]{7} & \sqrt[3]{7}^2 \\ 1 & \sqrt[3]{7}\zeta_3 & \sqrt[3]{7}^2\zeta_3^2 \\ 1 & \sqrt[3]{7}\zeta_3^2 & \sqrt[3]{7}^2\zeta_3 \end{pmatrix} = -1323$$

Thus:

$$M_K = \sqrt{1323}\frac{6}{27}\frac{4}{\pi} < 11$$

Hence, we check prime ideal $(2), (3), (5), (7)$ of $\mathbb{Z}$. Since the minimal polynomial of $\sqrt[3]{7}$ is $x^3 - 7$, we see that in $\mathbb{F}_p[x]/(x^3 - 7_p)$ we get:

(2) $(x+1)(x^2+x+1)$, hence

$$\mathfrak{p}_2 = (2, s+1) \qquad \mathfrak{p}_2' = (2, s^2+s+1)$$

(3) $(x-1)^3$, hence

$$\mathfrak{p}_3 = (3, s-1)$$

(5) $(x+2)(x^2+3x+4)$, hence

$$\mathfrak{p}_5 = (5, s+2) \qquad \mathfrak{p}_5' = (5, s^2+3s+4)$$

(7) x^3, hence

$$\mathfrak{p}_7 = (7, s) = (s)$$

Thus, just like before we have that $[\mathfrak{p}_2]$, $[\mathfrak{p}_3]$, and $[\mathfrak{p}_5]$ generate $Cl(K)$ (since $[\mathfrak{p}_7] = [1]$). Next, we see that

$$\mathfrak{p}_2\mathfrak{p}_3 = (6) \qquad \mathfrak{p}_3^2\mathfrak{p}_5 = (15)$$

Hence, $[\mathfrak{p}_3] = [\mathfrak{p}_2^{-1}]$ and $[\mathfrak{p}_3^2] = [\mathfrak{p}_5^{-1}]$. Then, with some computation we see that:

$$[\mathfrak{p}_3^3] = [\mathfrak{p}_3]^3 = [\text{id}]$$

Thus, we see that $Cl(K)$ is generated by $[\mathfrak{p}_3]$ and has order 3, hence $CL(K) \cong \mathbb{Z}/3\mathbb{Z}$. However, what if $\mathfrak{p}_3$ is in fact principal? For the sake of contradiction, let's say

$$\mathfrak{p}_3 = (\alpha) \implies \mathfrak{p}_3^3 = (3) = (\alpha^3)$$

Then $\alpha^3 = 3u$ for some unit u. To find u, we find the units of the ring of integers. To find its units, first notice that if:

$$1 = x^3 - 7 = (x - s)(x^2 + sx + s^2)$$

Then $x - s$ is a unit. But then, $x = 2$ works since $8 - 7 = 1$. Then by Dirichlet's Unit Theorem, we have that the rank of the unit group is:

$$r_1 + r_2 - 1 = 1 + 1 - 1 = 1$$

Hence, the group of units is

$$\langle 2 - \sqrt[7]{2} \rangle$$

which is the second object asked of us to find. Using this, we can finally conclude that there is no unit u that satisfies the equation $\alpha^3 = 3u$ (if $\alpha = a + bs + cs^2$, in the process of finding the coefficients a, b, c we shall be forced to have $b = c = 0$). Hence, we may properly conclude that

$$Cl(K) \cong \mathbb{Z}/3\mathbb{Z}$$

completing the proof.

In general, there is an algorithm for computing the ring of units that involved continued fractions. It is not covered in this book, but for those interested see *Borevich and Shafarevich, Number Theory* section 7.3 of chapter 2.

Proposition 50.2.38: Volume Of Unit Lattice

Let $\mathcal{O}_K$ be a ring of integers with unit group $\mathcal{O}_K^*$. Then the volume of the unit lattice $\sigma^\times(\mathcal{O}_K^\times)$ in the hyper-plane H is:

$$\text{vol}\left(\sigma^\times(\mathcal{O}_K^\times)\right) = \sqrt{r_1 + r_2}R$$

where R is the absolute value of the determinant of an arbitrary minor rank $t = r + s - 1$ of the following matrix:

$$\begin{pmatrix} \sigma_1^\times(u_1) & \cdots & \sigma_1^\times(u_t) \\ \vdots & \ddots & \ddots \\ \sigma_{t+1}^\times(u_1) & \cdots & \sigma_{t+1}^\times(u_t) \end{pmatrix}$$

***Proof*:**
computations

Definition 50.2.39: Regulator

The value of R in the above proof is called the *regulator* of $\mathcal{O}_K$.

Exercise 50.2.2

1. What number rings $\mathcal{O}_K$ have finitely many units?

50.3 Extension of Dedekind Domains and Number Fields

In section 26.3, we covered how prime ideals of a base ring correspond to prime ideals in an integral extension. Since all extensions that we are working with are integral extensions, the same theory applies. Working over Dedekind domains, we may further ask if the integral extension of a Dedekind domain is a Dedekind domain. This will be the case, and so we may further factor prime ideals in the extended ring, namely if a prime ideal $\mathfrak{p} \subseteq R$ factors in S where S/R is an integral extension. When focusing on rings of integers $\mathcal{O}_L$, the exploration of how primes factor into larger primes has a lot of consequences on the structure of $\mathcal{O}_L$ itself, and perhaps a bit surprisingly at first sight on the field extension L/K itself. We shall see in the next section that we can use the theory of factorization of prime ideals to classify all abelian extension over $\mathbb{Q}$, this is the *Kronecker-Weber Theorem* promised in example 20.1.29.

First, if R is a Dedekind domain with field of fraction K, $I \subseteq R$ an ideal, through them map $K \hookrightarrow L$ we get $I^e \subseteq \overline{R}^L$. We further generalize this for fractional ideals:

Definition 50.3.1: Extension Of Fractional Ideals

Let R be a Dedekind domain, K it's field of fractions, and L/K be a finite field extension. Then if $I \subseteq K$ is a fractional ideal and $\mathcal{O} = \overline{R}^L$,

$$I_L := I\mathcal{O} = \left\{ \sum a_i b_i \;\middle|\; a_i \in I,\ b_i \in \mathcal{O} \right\}$$

When necessary, the notation $I_{L/K}$ will be used to specify the base field over which I is taken.

It can be verified that $I_L \subseteq L$ is the minimal fractional ideal containing I, that if $M/L/K$, then $(I_L)_M = I_M$, and that $I_L = I \otimes_{\mathcal{O}_K} \mathcal{O}$. The results of lemma 26.2.3 follow quickly giving all the same results. Furthermore since $(I + J)_L = I_L + J_L$:

$$(\gcd_K(I, J))_L = \gcd_L(I_L, J_L)$$

However, by lemma 26.2.3, we so far only have

$$(\mathrm{lcm}_K(I, J))_L \subseteq \mathrm{lcm}_M(I_L, J_L)$$

Note that for $\alpha \in K$, $(\alpha)_L = (\alpha\mathcal{O}_K)_L = \alpha\mathcal{O}_L$, hence principal ideals extend to principal ideals. For reasons that shall become clearer later[19], the map

$$J_K \to J_L \qquad I \mapsto I_L$$

[19] we shall possibly increase the number of primes, hence change the "basis"

is called the *base change map*. To study the fractional ideals of L, the first immediate result we may want is that $\overline{R}^L$ remains a Dedekind domain:

Lemma 50.3.2: Extending Dedekind Domains

Let R be a Dedekind domain with field of fraction K and let L/K be a finite field extension of K. Then $\mathcal{O} = \overline{R}^L$ is a Dedekind domain.

***Proof*:**
By definition, $\mathcal{O}$ is integrally closed, and if $\mathfrak{p} \subseteq \mathcal{O}$ is prime, then $P = \mathfrak{p} \cap R$ is also prime (the contraction of a prime ideal is prime), but P is then maximal in R; hence $(\mathcal{O}/\mathfrak{p})/(R/P)$ is an integral extension over a field meaning $\mathcal{O}/\mathfrak{p}$ is a field, thus $\mathfrak{p}$ is maximal. It remains to show that $\mathcal{O}$ is Noetherian.

For now, we shall prove this for the separable case (the non-separable case has an easier proof with more machinery, and the separable case is all that is needed for number fields). Let $\alpha_1, \alpha_2, ..., \alpha_n$ be a K -basis for L. By multiplying by an appropriate constant, we may let $\alpha_1, \alpha_2, ..., \alpha_n \in \mathcal{O}$ (see proposition 50.1.7). By proposition 50.1.32,

$$\mathcal{O} \subseteq \frac{R\alpha_1}{d} + \cdots + \frac{R\alpha_n}{d}$$

Now, for any ideal $I \subseteq \mathcal{O}$, by the above it is contained in a finitely generated R-module, hence since R is Noetherian I (as a R-module) is finitely generated. But then *a fortiori*, it is a finitely generated $\mathcal{O}$-module, hence a finitely generated ideal, making $\mathcal{O}$ Noetherian.

Example 50.3.3: Dedekind Domains Via Extension
The result above shows that all rings of integers of a finite extension $L/K/\mathbb{Q}$ is a ring of integers. For an example that is not a number ring, take $R = k[x]$ for some field k. As R is a PID, it is a Dedekind domain. Then if $L/k(x)$ is an integral extension, the integral closure of R over L, $S = \overline{R}^L$, is a Dedekind domain. As $L \cong k(x)[y]/(p(y))$ for some irreducible polynomial over $k[x]$ L represents 1-dimensional curves. In section ref:HERE, we shall briefly show these curves are all 1-dimensional smooth curves.

Extending Number Fields

An immediate corollary of lemma 50.3.2 is the following:

Corollary 50.3.4: Extending Number Fields

Let L/K be a finite extension of numbers fields so that $\mathcal{O}_K = \overline{\mathbb{Z}}^K$ and $\mathcal{O} = \overline{\mathcal{O}_K}^L$. Then

$$\mathcal{O} = \mathcal{O}_L = \overline{\mathbb{Z}}^L$$

***Proof*:**
We have to show that $\mathcal{O}_L = \overline{\mathbb{Z}}^L \overset{!}{=} \overline{\mathcal{O}_k}^L = \mathcal{O}$, namely the $\overset{!}{=}$ equality. The $\subseteq$ direction is trivial, so consider the other direction: let $\alpha \in \mathcal{O}$. Then since $\mathcal{O}$ is Dedekind, it is Noetherian, it is finitely generated as a $\mathcal{O}_K$-module. Since $\mathbb{Z}[\alpha] \subseteq \mathcal{O}$:

$$\mathbb{Z}[\alpha] \subseteq \mathcal{O}_K a_1 + \cdots + \mathcal{O}_K a_n$$

Since each $\mathcal{O}_K$ is a finitely generated free $\mathbb{Z}$-module, we have that $\mathbb{Z}[\alpha]$ is a Noetherian $\mathbb{Z}$-module, hence finitely generated. But then by proposition 26.1.4 α is integral over $\mathbb{Z}$, and so $\alpha \in \overline{\mathbb{Z}}^L$, completing the equality.

Thus, when working with finite extensions of number ring L/K, there is a corresponding integral extensions are those of the ring of integers $\mathcal{O}_L/\mathcal{O}_K$ creating the diagram:

$$\begin{tikzcd} K \arrow[r, "\iota"] \arrow[d] & L \arrow[d] \\ \mathcal{O}_K \arrow[r, "\iota"] & \mathcal{O}_L \end{tikzcd}$$

As we saw, principal ideals extend to principal ideals. As it turns out, all non-principal ideals extend to be principal given a large enough extension. This result shall underpin a lot of future simplification and is good to see early on:

Lemma 50.3.5: Number Fields, Extension To Principal

Let $I \subseteq K$ be a fractional ideal where $K/\mathbb{Q}$ is a number field. Then in some finite extension L/K, I_L becomes a principal ideal.

***Proof*:**
As $K/\mathbb{Q}$ is a number fields, by theorem 50.2.30, $Cl(K)$ is finite. Thus, for every $I \in J_K$, there exists an $n \in \mathbb{N}$ such that $I^n = \alpha\mathcal{O}_K$. Let $L = K(\sqrt[n]{\alpha})$. letting $\beta = \sqrt[n]{\alpha}$, we get

$$(I_L)^n = (I_L^n) = (I^n\mathcal{O}_L) = (\alpha\mathcal{O}_L) = (\beta\mathcal{O}_L)^n$$

Thus, by unique factorization:

$$I_L = \beta\mathcal{O}_L$$

as we sought to show.

This result is quite useful; in the right cases we can pass to the field in which all our ideals become principal ideals, proof what we want for principal ideals, then try to come back down (coming back down is often called the descent argument). The following lemma gives a very common descent argument, and is important for computing the absolute norm of extended ideals: since the norm of an ideal gives a measure how space a given ideal is within its Dedekind, extending the domain shall make the given ideal sparser[20]:

[20] This means that the ideal doesn't "extend enough" to cancel the added density given by extending the ring

Lemma 50.3.6: Number Fields, Extension And Absolute Norm

Let $L/K/\mathbb{Q}$ be a finite extension of number fields and $I \subseteq K$ a fractional ideal. Then:

$$\mathfrak{N}_{L/\mathbb{Q}}(I_L) = \mathfrak{N}_{K/\mathbb{Q}}(I)^{[L:K]}$$

***Proof*:**
If $I = (\alpha) = \alpha\mathcal{O}_K$ is principal, then $\mathrm{Nm}(I) = \left|\mathrm{Nm}_{K/\mathbb{Q}}(\alpha)\right|$ and $\mathrm{Nm}(I_L) = \left|\mathrm{Nm}_{L/\mathbb{Q}}(\alpha)\right|$. Since $\alpha \in K$, by proposition 50.1.12

$$\mathrm{Nm}_{L/\mathbb{Q}}(\alpha) = \mathrm{Nm}_{K/\mathbb{Q}}(\alpha)^{[L:K]}$$

In general, $I^n = \alpha\mathcal{O}_K$, thus using the multiplicity of the absolute norm:

$$\mathfrak{N}_{L/\mathbb{Q}}(I_L)^n = \mathfrak{N}_{L/\mathbb{Q}}(I_L^n) = \mathrm{Nm}_{L/\mathbb{Q}}(\alpha\mathcal{O}_L) = \mathrm{Nm}_{L/\mathbb{Q}}(\alpha\mathcal{O}_K)^{[L:K]} = \mathfrak{N}(I^n)^{[L:K]} = \mathfrak{N}_{K/\mathbb{Q}}(I)^{n[L:K]}$$

taking the nth root, and we're done

50.3.1 Ideal Correspondence in Extension

Let L/K be a finite extension, J_K and J_L be the corresponding ideal group. Then we have a map $J_K \to J_L$ given by $I \mapsto I_L$. Do we have a converse map, and if so is it an inverse of the natural extension map? If not, to what degree is it an inverse map? Recall in proposition 26.2.2 that $I \subseteq SI \cap R$, in other words $I \subseteq I_S \cap R$. This can be made an equality for when extending number fields, and further generalized to fractional ideals. Recall that this is not true in general (a counter-example for the general case is $(2) \subsetneq \mathbb{Z} = (2)_{\mathbb{Q}} \cap \mathbb{Z}$), hence this is special for extension of Dedekind domains:

Proposition 50.3.7: Ideal Correspondence In Extensions Of Dedekind Domains

let L/K be a finite field extension with Dedekind domain $\mathcal{O}$ and o respectively. Let $I \subseteq K$ a fractional ideal. Then:

$$I_L \cap K = (I^e)^c = I$$

Hence, unlike for general integral extensions, the contraction is an inverse map to the extension.

***Proof*:**
Let $\beta \in J = I\mathcal{O} \cap K$ so that $\beta = \sum_i a_i k_i$ where $a_i \in I_L$ and $k_i \in o$ (since $\mathcal{O} \cap K = o$). Then the following equality holds:

$$\beta^n = \mathrm{Nm}_{L/K}(\beta) = \prod_\tau \tau(\beta) = \prod_\tau \left(\sum_i k_i \tau(a_i)\right)$$

where the first equality comes from $\beta \in K$, the second comes from the equivalence definition of norm (given an appropriate normal closure N/L), and the third is expanding the definition of β. On the right hand side, we have that the sum is invariant under the Galois group $\mathrm{Gal}_K(N)$ for appropriate normal closure N/L from which we took each τ. Thus, the sum is an element in o, so the product is in $\mathcal{o}$. Then we have $\beta^n \in I^n$, so $\beta \in I$, completing the proof.

Due to the importance of this new paradigm of thinking, two other proofs shall be presented. Notice that $IJ^{-1} \subseteq \mathcal{o}$ and:

$$\mathfrak{N}(IJ^{-1}) = 1$$

giving that $IJ^{-1} = \mathcal{o}$, i.e. $I = J$.

Another proof goes as follows: Since $I \subseteq J$, $I_L \subseteq J_L$. Since $I_L \subseteq J_L \subseteq I_L$ by properties of extensions, $I_L = J_L$. Thus:

$$\mathfrak{N}(J) = \mathfrak{N}(I_L)^{1/[L:K]} = \mathfrak{N}(I)$$

and since $I \subseteq J$, the norms being equal means the index between the two are the same, hence $I = J$.

Note that in the case where $\mathfrak{p}$ is prime, the result can be essentially immediately proven since by proposition 26.2.2 $\mathfrak{p} \subseteq \mathfrak{p}\mathcal{O}_L \cap K$, but since $\mathfrak{p}$ is maximal we get equality.

Corollary 50.3.8: Homomorphism between Extensions of Ideal Group

Let J_K be the ideal group of a Dedekind domain R and let L/K be a finite field extension. Then the map $J_K \to J_L$ mapping $I \mapsto I_L$ is an injective group homomorphism that preserves gcd and lcm.

On the other hand, the map $Cl(K) \to Cl(L)$ is a homomorphism that need not be injective.

For class groups of number fields:

$$\varinjlim_{\substack{K \\ [K:\mathbb{Q}]<\infty}} (Cl(K)) = 0$$

***Proof*:**
By the previous proposition, $I_L = J_K$ if $I = J$, hence the map $\varphi : \mathscr{F}(K) \to \mathscr{F}(L)$ mapping $I \mapsto I_L$ is injective and is certainly a homomorphism by a simple generalization of lemma 26.2.3 (recall that $(\mathfrak{ab})^e = \mathfrak{a}^e\mathfrak{b}^e$ for ideals).

For the map between class groups of number fields, since the extension for every fractional ideal $I \subseteq \mathbb{Q}$, there exists an extension in which it is principle, we see that the colimit can only be satisfied by the trivial group, hence $\varinjlim_{\substack{K \\ [K:\mathbb{Q}]<\infty}} (Cl(K)) = 0$, as we sought to show.

A side consequence of the colimit definition is that we may define

$$Cl(\overline{\mathbb{Q}}) := \varinjlim(Cl(K)) = 0$$

This shows that to some degree, the lack of principal ideals is tied to what algebraic extension we choose. Though the extension map is injective, it is not in general surjective:

Example 50.3.9: Lack of Surjectivity

Let $L = \mathbb{Q}(\sqrt{-5})$ and $K = \mathbb{Q}$ so $\mathcal{O}_L = \mathbb{Z}[\sqrt{-5}]$. Then the (non-principal) ideal $I = (1+\sqrt{-5}, 1-\sqrt{-5})$ are all numbers of the form $m+n\sqrt{-5}$ where m, n are even, so $I \cap \mathbb{Q} = (2)$, hence I lies over (2). However, $I \neq (2)_L$ since I is not principal (recall $(-)_L$ maps principals to principals), and no other ideal can map to I or else I would lie over some different prime, hence I is not an extension. The ideal (2) is still related to I, namely $I^2 = (2)$ (this can be used to show I is not principal).

With these results, we have better control over the extension of primes and primes lying over (resp. lying under) relate:

Proposition 50.3.10: Dedekind Domain, Lying Over And Lying Under

Let $\mathcal{O}/o$ be a finite extension of Dedekind domains given by L/K, $0 \neq \mathfrak{p} \subseteq o$ a prime, and $\mathcal{O} = \overline{o}^L$, $\mathfrak{p}$, where $\mathfrak{p}$ may factor further. Then $\mathfrak{p}$ will never be trivialized in an extension, namely

$$\mathfrak{p}\mathcal{O} \neq \mathcal{O} = (1)$$

***Proof*:**
Let $p \in \mathfrak{p} \setminus \mathfrak{p}^2$. Then

$$(p) = \mathfrak{p}\mathfrak{a} \qquad \mathfrak{p} \nmid \mathfrak{a}$$

Hence, $\mathfrak{p} + \mathfrak{a} = (1)$ $(\gcd(\mathfrak{p}, \mathfrak{a}) = (1))$, and so $a + b = 1$ for $a \in \mathfrak{p}$ and $b \in \mathfrak{a}$. Certainly $b \notin \mathfrak{p}$ so $b\mathfrak{p} \subseteq \mathfrak{p}\mathfrak{a} = (p)$. Now, if $\mathfrak{p}\mathcal{O} = \mathcal{O}$, then $b\mathcal{O} = b\mathfrak{p}\mathcal{O} \subseteq p\mathcal{O}$. hence $b\mathcal{O}_K \subseteq \mathfrak{p}$ so $b = px$ for some $x \in \mathcal{O} \cap K = \mathcal{O}_K$, implying $b \in \mathfrak{p}$ – a contradiction.

Proposition 50.3.11: Equivalence Of Prime Division In Dedekind Domain

Let $\mathfrak{p} \subseteq R$, $\mathfrak{B} \subseteq \mathcal{O}$, $\mathcal{O}/R$ an integral extension where R is a Dedekind Domain. Then the following are equivalent:

1. $\mathfrak{B} \big| \mathfrak{p}\mathcal{O}$
2. $\mathfrak{B} \supseteq \mathfrak{p}\mathcal{O}$
3. $\mathfrak{B} \supseteq \mathfrak{p}$
4. $\mathfrak{B} \cap \mathcal{O} = \mathfrak{p}$
5. $\mathfrak{B} \cap K = \mathfrak{p}$

***Proof*:**
Since we are in a Dedekind domain, $A \big| B$ if and only if $A \supseteq b$, giving us $1 \Leftrightarrow 2$. Since $\mathfrak{B}$ is an ideal of $\mathcal{O}_L$, certainly $2 \Leftrightarrow 3$. We got $3 \Leftarrow 4 \Leftrightarrow 5$ by the above injectivity argument. For $3 \implies 4$, since $\mathfrak{p}$ is maximal and $\mathfrak{B} \cap \mathcal{O}_K \supseteq \mathfrak{p}$, we have either equality or $\mathcal{O}_K$, if $\mathfrak{p} \cap \mathcal{O}_K = \mathcal{O}_K$, then $1 \in Q$, implying $Q = \mathcal{O}_L$, contradicting it's maximality.

Corollary 50.3.12: Lying Over Unique Prime

Let $\mathfrak{p}, \mathfrak{q} \subseteq \mathcal{O}$, $\mathfrak{p} \neq \mathfrak{q}$. Let:

$$\mathfrak{p}\mathcal{O}_L = \mathfrak{B}_1^{e_1} \cdots \mathfrak{B}_n^{e_n} \qquad \mathfrak{q}\mathcal{O}_L = \mathfrak{C}_1^{f_1} \cdots \mathfrak{C}_m^{f_m}$$

Then:

$$\{\mathfrak{B}_1, ..., \mathfrak{B}_n\} \cap \{\mathfrak{C}_1, ..., \mathfrak{C}_n\} = \emptyset$$

that is, extended prime ideals lie over a single prime

***Proof*:**
Let's say the intersection is non-empty, so that $\mathfrak{a}$ is a common factor. Then write:

$$\mathfrak{p}\mathcal{O}_L = \mathfrak{a}^{e_1} \cdots \mathfrak{B}_n^{e_n} \qquad \mathfrak{q}\mathcal{O}_L = \mathfrak{a}^{f_1} \cdots \mathfrak{C}_m^{f_m}$$

Then:

$$(\mathfrak{p})_L^{-1}\mathfrak{p}^{e_1-1}\mathfrak{B}_2^{-e_2} \cdots \mathfrak{B}_n^{-e_n} = \mathfrak{a} = (\mathfrak{q})_L^{-1}\mathfrak{p}^{f_1-1}\mathfrak{C}_2^{-f_2} \cdots \cdots \mathfrak{C}_m^{-f_m}$$

Thus we get:

$$\mathfrak{q}\mathcal{O}_L = (\mathfrak{p})_L\mathfrak{p}^{e_1-1}\mathfrak{B}_2^{-e_2} \cdots \mathfrak{B}_n^{-e_n}\mathfrak{p}^{f_1-1} \cdots \mathfrak{C}_m^{f_m}$$

And in particular:

$$\mathfrak{C}_1^{f_1} \cdots \mathfrak{C}_m^{f_m} = q\mathcal{O}_L \subseteq p\mathcal{O}_L = \mathfrak{B}_1^{e_1} \cdots \mathfrak{B}_n^{e_n}$$

But then, since $\mathfrak{p}\mathcal{O}_L \cap K = \mathfrak{p}$ and $\mathfrak{q}\mathcal{O}_L \cap K = \mathfrak{q}$, we have by corollary 50.3.8 that:

$$\mathfrak{q} \subseteq \mathfrak{p}$$

which since $\mathfrak{q}$ is maximal and $\mathfrak{p} \neq \mathcal{O}_K$ we have $\mathfrak{q} = \mathfrak{p}$, contradicting the fact that $\mathfrak{p} \neq \mathfrak{q}$, hence, it must be that they share no common factors.

Another proof goes as follows: if $\mathfrak{p} \subseteq \mathcal{O}_K$ is a prime and $\mathfrak{p}_L = \mathfrak{B}_1^{e_1} \cdots \mathfrak{B}_k^{e_k}$ is the factorization of $\mathfrak{p}_L$ in $\mathcal{O}_L$, then if there was a prime ideal $\mathfrak{q}$ such that $\mathfrak{q} \mapsto \mathfrak{B}_i$ so that by the above for any $1 \leq i \leq k$, then since $\mathfrak{p} \subseteq \mathfrak{B}_i \cap \mathcal{O}_k$, by maximality $\mathfrak{p} = \mathfrak{B}_i \cap \mathcal{O}_K$. Hence if $\mathfrak{q} = \mathfrak{B}_I \cap \mathcal{O}_k$, by injectivity $\mathfrak{p} = \mathfrak{q}$.

Another consequence of the above theorem is that if two prime factorizations of a prime share a factor in an extended ring of integers, they are equal.

Relative Norm

Since the ideals of Dedekind domains share factorization structure to numbers, we may hope there is an equivalent of the field norm $\mathrm{Nm}_{L/K}$ for ideals. by lemma 50.3.5, we may hope that our new relative norm map gives us some information on the power to which we may rise our ideal to get a principal ideal. We now endeavour to find a map $J_L \to J_K$ that behaves more like the field norm and generalizes the absolute norm, which can be thought of as a map:

$$\mathfrak{N}_{L/\mathbb{Q}} : J_L \to \mathbb{Q}_L \cong \mathbb{Q}_{>0}^{\times}$$

In particular, if we work of *Galois* extensions L/K, then we may hope to translate some of the fact that

$$\mathrm{Nm}_{L/K}(\alpha) = \prod_{\sigma} \sigma(\alpha)$$

to prime ideals! In particular, this allows us to bring in the Galois theory in the study of prime ideals!

We start by showing the case for Galois extensions, the general case can be inferred from the Galois case. Given a Galois extension L/K with Galois group $G = \mathrm{Gal}_K(N)$, then we can recover L from K using G by taking $L^G = K$, hence $\mathcal{O}_L^G = \mathcal{O}_K$. Certainly, if $\tau \in G$ and $\mathfrak{a} \subseteq L$, then $\tau(\mathfrak{a}) \subseteq L$ Since L/K is normal, and hence $\tau(\mathfrak{a})$ is well-defined. It may be tempting to then say that $J_L^G = J_K$ and hence defining $J_L \to J_K$ by taking

$$I \mapsto \prod_{\tau} \tau(I)$$

However, it is not the case that $J_L^G = J_K$; the ideal in example 50.3.9 is invariant under the Galois group, but certainly an ideal of $\mathbb{Z}$. We fix this by the following:

Definition 50.3.13: Relative Norm

Let L/K be a finite field extension and $I \subseteq K$ a fractional ideal in L. Then the *relative norm* or *ideal norm* of L with respect to K is:

$$\mathfrak{N}_{L/K}(I) = \left(\left\{\mathrm{Nm}_{L/K}(\alpha) \;\middle|\; \alpha \in I\right\}\right) \in J_K$$

namely, $\mathrm{Nm}_{L/K}(I)$ is the ideal generated by all norms of elements in I.

Note that this is well-defined since $\mathrm{Nm}_{L/K}(\alpha) \in K$ as every minimal polynomial $m_{\alpha,K}(x) \in K[x]$. The norm as defined above gives us a map $\mathfrak{N}_{L/K} : J_L \to J_K$. The field norm can be thought of as the ideal norm, for example take

$$\mathfrak{N}_{\mathbb{Q}(\sqrt{2})/\mathbb{Q}}((\sqrt{2})) = \left(\left\{\mathrm{Nm}_{L/K}\left(\sqrt{2}b\right) \;\middle|\; \sqrt{2}b \in \sqrt{2}\mathbb{Z}\right\}\right) = (2)$$

suggesting that $(\mathrm{Nm}_{L/K}(\alpha)) = \mathfrak{N}_{L/K}((\alpha))$, which will indeed be the case and shall be proved after the next theorem. An important reason to choose this definition is that it works well with extension: Let $J_L \xrightarrow{\mathrm{Nm}_{L/K}} J_K \xrightarrow{I_L} J_L$. What will be the image of an ideal in this chain? In the simplest case, we may take principal ideal $\alpha\mathcal{O}_L \subseteq \mathcal{O}_L$ where $\alpha \in \mathcal{O}_K$. Then it's image is $\alpha\mathcal{O}_K$, and it's extension is $\alpha\mathcal{O}_L$. Thus for the simplest type of integral principal ideal, the map is a right-inverse. For more general cases, it is not necessarily a right inverse (it cannot be, since I_L is not surjective). However, the behavior of the image is well-behaved, and in fact tied with the "almost" definition of the ideal norm presented earlier:

Theorem 50.3.14: Ideal Correspondence in Extension

Let $N/L/K$ be finite field extension where N/K is a normal extension so that $G = \mathrm{Gal}_K(N)$. Then:

$$(\mathfrak{N}_{L/K}(I))_L = \prod_{\tau \in G} (\tau I)_{L/\tau K} =: \mathrm{Nm}'_{L/K}(I)$$

In particular

$$\begin{Bmatrix} \text{Fractional ideals of} \\ \mathcal{O}_K \end{Bmatrix} \underset{\mathfrak{N}_{L/K}}{\overset{I_L}{\rightleftarrows}} \begin{Bmatrix} \text{Fractional Ideals of} \\ \mathcal{O}_L \end{Bmatrix}$$

***Proof*:**
Let $I \subseteq L$ be a fractional ideal. Without loss of generality, we may make I integral. Then, certainly an ideal is a subset of it's extension (given the natural embedding) $\sigma_i I \subseteq (\sigma_i I)_{L/\sigma_i K}$, and so by definition of $\mathrm{Nm}'_{L/K}(I)$, forall $\alpha \in I$:

$$\mathrm{Nm}_{L/K}(\alpha) = \prod_{\tau} (\tau(\alpha)) \in \mathrm{Nm}'_{L/K}(I)$$

Hence $\mathrm{Nm}_{L/K}(I) \subseteq \mathrm{Nm}'_{L/K}(I)$. Extending both sides, we get $(\mathrm{Nm}_{L/K}(I))_L = (\mathrm{Nm}'_{L/K}(I))_L = \mathrm{Nm}'_{L/K}(I)$. Hence, since we are in a Dedekind domain, there exists an [integral] ideal J such that

$$\mathrm{Nm}_{L/K}(I)_L = J\mathrm{Nm}'_{L/K}(I)$$

We want to show that $J = (1)$. We shall use lemma 50.6.1, namely that there exists an ideal $B \subseteq K$ relatively prime to $J \cap K$ such that JB is principal, say $JB = (b)$ $(b \in K)$. Then since J is fixed under $\mathrm{Gal}_K(L)$, each $(\tau B)_{L/\tau K})$ is relatively prime to J for each τ. Then since $b \in J$:

$$\begin{aligned} \mathrm{Nm}_{L/K}(I) &\supseteq \left(\prod_{\tau} \tau(b) \right) \\ &= \left(\prod_{\tau} (\tau(B))_{L/\tau K)} \right) \left(\prod_{\tau} (\tau(J))_{L/\tau K} \right) \end{aligned}$$

Thus, $\prod_{\tau} (\tau B)_{L/\tau K}$ is a multiple of J. However, $(\tau B)_{L/\tau K}$ is relatively prime to J, hence it must be that $J = (1)$, as we sought to show.

Lemma 50.3.15: Ideal Norm Properties

Let $M/L/K$ be algebraic number fields, $\mathfrak{a}, \mathfrak{b} \subseteq M$ be ideals, $I \subseteq K$ an ideal, and $a \in M$ an element. Then:

1. $\mathfrak{N}_{M/K}(a) = (\mathrm{Nm}_{M/K}(a)) = \left(\prod_{\sigma \in \mathrm{Gal}_K(M)} \sigma(a)\right)$
2. $\mathfrak{N}_{L/K}(\mathfrak{N}_{M/L}(\mathfrak{a})) = \mathfrak{N}_{M/K}(\mathfrak{a})$
3. $\mathfrak{N}_{M/k}(\mathfrak{a}\mathfrak{b}) = (\mathrm{Nm}_{M/k}(\mathfrak{a}))\mathrm{Nm}_{M/k}(\mathfrak{b})$
4. $I^{[M:K]} = \mathfrak{N}_{M/K}((I)_M)$
5. $(\mathfrak{N}_K(I)) = \mathfrak{N}_{K/\mathbb{Q}}(I)$

***Proof*:**
Note that for all of these equalities, we immediately have $\supseteq$, hence we want to show $\subseteq$ for each of these.

1. comes straight from the definition
2. exercise
3. exercise
4. Use lemma 50.3.6 and the finiteness of the class group to make an argument on principal ideals, we see that if we take $J_K \xrightarrow{I_L} J_L \xrightarrow{\mathrm{Nm}_{L/K}} J_K$, we pick up a $[L : K]$ exponent.
5. use part (1)

Corollary 50.3.16: Relative Norm Of Prime Ideal

Let L/K be a finite extension of number fields and let $\mathfrak{B}$ lie above $\mathfrak{p}$. Then if $f = [\mathcal{O}_L/\mathfrak{B} : \mathcal{O}_K/\mathfrak{p}]$, then:

$$\mathfrak{N}_{L/K}(\mathfrak{B}) = \mathfrak{p}^f$$

***Proof*:**
Using the above lemma:

$$\begin{aligned}(\mathfrak{N}_L(\mathfrak{p}_L)) &= \mathfrak{N}_{L/\mathbb{Q}}(\mathfrak{p}_L)\\ &= \mathfrak{N}_{K/\mathbb{Q}}(\mathfrak{p}^{[L:K]})\\ &= (\mathfrak{N}_K(\mathfrak{p}))^{[L:K]}\end{aligned}$$

Thus, $\mathfrak{N}_{L/K}(\mathfrak{B})$ is a power of $\mathfrak{p}$ by the 4th point in the above lemma. and by the 2nd we get:

$$\mathfrak{N}_{K/\mathbb{Q}}(\mathfrak{N}_{L/K}(\mathfrak{B})) = \mathfrak{N}_{K/\mathbb{Q}}(\mathfrak{p}^f) \implies \mathfrak{N}_{L/K}(\mathfrak{B}) = \mathfrak{p}^f$$

as we sought to show.

The relative norm can in fact be defined to be the unique group homomorphism where

$$\mathfrak{N}(\mathfrak{B}) = \mathfrak{p}^{[\mathcal{O}_L/\mathfrak{B}:\mathcal{O}_K/\mathfrak{p}]}$$

which is left as an exercise. By multiplicity, we get that in general:

$$\mathfrak{N}_{L/K}(\mathfrak{p}\mathcal{O}_L) = \prod_i \mathfrak{N}_{L/K}\mathbb{F}_{L/K}(\mathfrak{B}_i)^{e_i} = \prod_i \mathfrak{p}^{f_i e_i}$$

Corollary 50.3.17: Ideal Norm Correspondence Not Galois

Let $M/L/K$ were field extensions, M/K normal but not necessarily L/K. Then

$$(\mathrm{Nm}_{L/K}(I))_M = \prod_{\tau \in \mathrm{Gal}_K(M)/\mathrm{Gal}_L(M)} \tau(I_M)$$

***Proof*:**
exercise

With the relative norm defined, it is perhaps natural to think there is a relative discriminant:

Definition 50.3.18: Relative Discriminant

Let L/K be a finite extension of number fields. Then:

$$\mathrm{disc}'_{L/K}(\mathcal{O}_L) = \left(\left\{\mathrm{disc}_{L/K}(\alpha_1, \alpha_2, ..., \alpha_n) \;\middle|\; \alpha_1, \alpha_2, ..., \alpha_n \text{ basis for } \mathcal{O}_L \right\}\right)$$

is the *relative discriminant* of L with respect to K.

This won't come up much, but is useful in a couple of tower arguments later on.

Though we have characterized a correspondence of the ideals, it is more difficult to characterize I_L. In the following section, we characterize prime extensions, and take advantage of the fact that the factorization to give us a characterization of I_L.

50.4 Prime Factorization in Extension

Since $\mathcal{O}$ is a Dedekind domain by lemma 50.3.2, we have that $\mathfrak{p}\mathcal{O}$ decomposes uniquely into a product of prime ideals:

$$\mathfrak{p}\mathcal{O} = \mathfrak{B}_1^{e_1} \cdots \mathfrak{B}_r^{e_r}$$

If it is clear from context, we shall often write $\mathfrak{p}$ instead of $\mathfrak{p}\mathcal{O}$. Sometimes, we shall also write $\mathfrak{p}_L$ for $\mathfrak{p}\mathcal{O}_L$. We shall *almost always* find exactly how $\mathfrak{p}\mathcal{O}$ decomposes, sometimes without requiring knowledge of the ring of integers. When the extension L/K is Galois, we shall have even more control, and if L/K is not Galois, we can take $M/L/K$ to be a Galois extension, work within the Galois, and see if the results can be translated back to L/K.

Let's say $\mathfrak{B}$ lies over $\mathfrak{p}$ and that $\mathfrak{B}$ is in the prime decomposition of $\mathfrak{B}$ [21]. Then we have two important quantities associated to $\mathfrak{B}$: it's power in the prime decomposition, and the index $[\mathcal{O}/\mathfrak{B} : \mathcal{O}_K/\mathfrak{p}]$, that is the degree of the extension of the residue field. These quantities are given a name:

Definition 50.4.1: Ramification Index and Inertia Degree

Let R be a Dedekind domain, $K = \text{Frac}(R)$, L/K be a finite field extension, $\mathcal{O} = \overline{R}^L$ and $\mathfrak{p} \subseteq R$ a prime ideal. If $\mathfrak{B}_i$ is a prime in $\mathcal{O}$ lying above $\mathfrak{p}$, then it's degree is called the *ramification index* and is often denoted

$$e(\mathfrak{B}_i/\mathfrak{p}) \qquad \text{where} \qquad \mathfrak{B}^{e(\mathfrak{B}_i/\mathfrak{p})} \| \mathfrak{p}\mathcal{O}$$

or e_i for short, where $\|$ means *maximally divides.* The degree of the field extension

$$f(\mathfrak{B}_i/\mathfrak{p}) = [\mathcal{O}/\mathfrak{B}_i : R/\mathfrak{p}]$$

or f_i for short, and is called is called the *inertia degree* or *residue field degree* of $\mathfrak{B}_i$ over $\mathfrak{p}$.

For future reference, it will be important to keep track to "how much" a prime ideal splits. This will share strong similarities to the splitting of polynomials[22] (recall how "rare" an irreducible polynomial is inseparable)? This prompts the following vocabulary:

Definition 50.4.2: Split, Unramified, and Totally Ramified Primes

Let $\mathfrak{p} \subseteq R$ be a prime ideal of a Dedekind domain R, $K = K(R)$, L/K a finite extension, and $\mathcal{O} = \overline{R}^L$. Then given

$$\mathfrak{p}\mathcal{O} = \mathfrak{B}_1^{e_1} \cdots \mathfrak{B}_g^{e_g}$$

If $g = [L : K]$, then $\mathfrak{p}$ is said to *split completely* ($e_i = f_i = 1$), giving:

$$\mathfrak{p}\mathcal{O} = \mathfrak{B}_1 \cdots \mathfrak{B}_n$$

If $e_i = 1$ for all i (but some f_i are not necessarily 1), then we sy that $\mathfrak{p}$ is *unramified* and (if the residue field is separable):

$$\mathfrak{p}\mathcal{O} = \mathfrak{B}_1 \cdots \mathfrak{B}_g$$

for some $g \leq n$. If $g = 1$, then $\mathfrak{p}$ is said to be *totally ramified*, *indecomposable*, or *nonsplit* if:

$$\mathfrak{p} = \mathfrak{B}_1^{e_1}$$

and we know there is only a single prime ideal $\mathfrak{B}_i$ lying over $\mathfrak{p}$.

Most of the time, we shall work in the case of character zero extensions, hence L/K is separable. In this case, the values e_i and f_i are connected; if $\mathfrak{p}$ factors in an extension, then there is a simple equation the ramification and inertia degree must satisfy which equals the degree of the extension:

[21] we shall soon show all primes lying above $\mathfrak{p}$ will be part of the prime decomposition

[22] Though there shall be important failures when this comparison will *not* hold! This shall be elaborated soon

Proposition 50.4.3: Fundamental Identity

Let L/K be a finite separable extension, $n = [L:K]$, $\mathcal{O} = \overline{\mathcal{O}_K}^L$, and $\mathfrak{p} \subseteq \mathcal{O}_K$ a prime ideal. If

$$\mathfrak{p}\mathcal{O} = \mathfrak{B}_1^{e_1} \cdots \mathfrak{B}_g^{e_g}$$

is the prime decomposition of $\mathfrak{p}\mathcal{O}$ and f_i is the inertia degree of $\mathfrak{B}_i$ over $\mathfrak{p}$, then

$$\sum_i^g e_i f_i = n$$

***Proof*:**
Using absolute norms, this theorem becomes almost immediate, since:

$$\begin{aligned}\mathfrak{N}(\mathfrak{p})^{[L:K]} &\overset{\text{lem } 50.3.6}{=} \mathfrak{N}(\mathfrak{p}\mathcal{O}) = \mathfrak{N}\left(\prod_i \mathfrak{B}_i^{e_i}\right) \overset{\text{prop } 50.2.28}{=} \prod_i \mathfrak{N}(\mathfrak{B}_i)^{e_i} \\ &\overset{\text{cor } 50.3.16}{=} \prod_i \mathfrak{N}(\mathfrak{p})^{e_i f_i} = \mathfrak{N}(\mathfrak{p})^{\sum_i e_i f_i}\end{aligned}$$

taking log of both sides gives us:

$$n = [L:K] = \sum_I e_i f_i$$

This can also be proven directly: first by the CRT we have

$$\mathcal{O}/\mathfrak{p}\mathcal{O} \cong \bigoplus_i^r \mathcal{O}/(\mathfrak{B}_i^{e_i})$$

Next, we have that $\mathcal{O}/\mathfrak{p}\mathcal{O}$ and $\mathcal{O}/\mathfrak{B}_i^{e_i}$ are vector-spaces over $k = \mathcal{O}_K/\mathfrak{p}$. Hence, it suffices to show that

$$\dim_k(\mathcal{O}/\mathfrak{p}\mathcal{O}) = n \qquad \dim_k(\mathcal{O}/(\mathfrak{B}_i^{e_i})) = e_i f_i$$

To show the first equality, since $\mathcal{O}$ is a finitely generated $\mathcal{o}$-module, pick $x_1, x_2, ..., x_m \in \mathcal{O}$ such that $\overline{x_1}, ..., \overline{x_n}$ is a k-basis for $\mathcal{O}/\mathfrak{p}\mathcal{O}$. If $x_1, x_2, ..., x_m$ is a basis for L/K, then we're done. Let's first show they are linearly independent. For the sake of contradiction, let's say they weren't so that $x_1, x_2, ..., x_m$ are linearly dependent over K. Then they are linearly dependent over $\mathcal{O}_K$ by Gauss's lemma. Hence, there are elements $a_1, a_2, ..., a_m \in \mathcal{O}_K$ that are not all zero such that

$$a_1 x_1 + \cdots + a_m x_m = 0$$

We want to use this to show that the congruent mod $\mathfrak{p}$ is also linearly dependent. To that end, consider $\mathfrak{a} = (a_1, a_2, ..., a_m)$. Take $a \in \mathfrak{a}^{-1}$ such that $a \notin \mathfrak{a}^{-1}\mathfrak{p}$ so that $a\mathfrak{a} \not\subseteq \mathfrak{p}$. Hence, the element $aa_1, ..., aa_m$ lie in $\mathcal{O}_K$, but are not all in $\mathfrak{p}$. Hence, we have:

$$aa_1 x_1 + \cdots + aa_m x_m = 0 \text{ mod } \mathfrak{p}$$

giving linear dependence of the elements $\overline{x_1}, ..., \overline{x_m}$ over k – a contradiction.

Next, for spanning, consider the $\mathcal{O}_K$-module

$$M = \mathcal{O}_K x_1 + \cdots \mathcal{O}_K x_m$$

We shall show we can "upgrade" this to $L = Kx_1 + \cdots + Kx_m$. Consider $N = \mathcal{O}/M$. Since $\mathcal{O} = M + \mathfrak{p}\mathcal{O}$, $\mathfrak{p}N = N$. Since $\mathcal{O}$ is finitely generated (being a Dedekind domain), N is finitely generated. If $\alpha_1, \alpha_2, ..., \alpha_s$ are generators of N, then since $\mathfrak{p}N = N$:

$$\alpha_i = \sum_j a_{ij}\alpha_j \qquad a_{ij} \in \mathfrak{p}$$

Now, define the matrix $A = (a_{ij}) = I$ where I is the identity matrix of rank s. Let $B = A^*$ be the adjoint of A whose entries are the minors of rank $(s-1)$ of A. Then

$$A(\alpha_1, ..., \alpha_s)^t = 0 \qquad BA = \det(A)I = dI$$

Thus:

$$0 = BA(\alpha_1, \alpha_2, ..., \alpha_s)^t = d\alpha_1, ..., d\alpha_s)^t$$

thus, $dN = 0$, implying $d\mathcal{O} \subseteq M = \mathcal{O}_K x_1 + \mathcal{O}_K x_2 + \cdots + \mathcal{O}_K x_m$. Importantly, $d \neq 0$ since $d = \det((a_{ij}) = I) \equiv (-1)^s \mod p$ since $a_{ij} \in \mathfrak{p}$. Thus:

$$L = dL = Kx_1 + Kx_2 + \cdots + Kx_m$$

showing that it spans L/K, and hence it is a basis, establishing the first equality.

For the second, recall in proposition 50.2.28 that each $\mathfrak{B}_i^{n+1}/\mathfrak{B}_i^n \cong_{\mathcal{O}/\mathfrak{B}_i} \mathcal{O}/\mathfrak{B}_i$. Hence, since $f_i = [\mathcal{O}/\mathcal{B}_i : \mathcal{O}_k/\mathfrak{p}]$, we get

$$n = \dim_k(\mathcal{O}/\mathfrak{p}\mathcal{O}) \sum_i \dim_k(\mathcal{O}/(\mathfrak{B}_i^{e_i})) = \sum_i^r e_i f_i$$

as we sought to show.

Corollary 50.4.4: Multiplicity of Ramification And Inertia

Let $M/L/K$ be a finite extension of number fields. Then:

$$e_{M/K} = e_{M/L}e_{L/K} \qquad f_{M/K} = f_{M/L}f_{L/K}$$

***Proof*:**
immediate consequence of the arguments in the above theorem

Hence, we have a measure of how much a prime ideal splits in an extension.

Example 50.4.5: Decomposing Prime

1. It is possible that $\mathfrak{p}$ is totally ramified where $e_1 > 1$ and $f_1 = 1$: take $L = \mathbb{Q}(\sqrt{3})$ and $K = \mathbb{Q}$ with $I = (3)$. Then $(3)\mathcal{O} = (\sqrt{3})^2$

2. Similarly, it is possible that $\mathfrak{p}$ is totally ramified and $e_1 = 1$ and $f_1 > 1$: take $K = \mathbb{Q}$, $L = \mathbb{Q}(\sqrt{3})$, but $I = (5))$.

3. Let us consider the inertia degree to get some structure of the Dedekind domain. Let $L = \mathbb{Q}(\sqrt[3]{2})$ and $L/\mathbb{Q}$ be a finite extension. Since it is always the case that $\mathbb{Z}[\sqrt[3]{2}] \subseteq \mathcal{O}_L$ [a], Take $I = (\sqrt[3]{2}) \subseteq \mathcal{O}_L$, $I \cap \mathbb{Q} = (2)$, and $\mathfrak{N}((\sqrt[3]{2})) = |\mathrm{Nm}_{L/\mathbb{Q}}(\sqrt[3]{2})| = 2$, and $2 = (\sqrt[3]{2})^3$. Hence, $f_1 = 1$, meaning
$$[\mathcal{O}_L/\sqrt[3]{2}\mathcal{O}_L : \mathbb{Z}/2\mathbb{Z}] = 1$$
implying both fields only have two elements, hence
$$\mathcal{O}_L/\sqrt[3]{2}\mathcal{O}_L \cong \mathbb{Z}/2\mathbb{Z}$$
Notice how this also gave information on the ring of integers,

4. Let $\mathcal{O}_K = \mathbb{Z}[i]$ so that $2 = (1+i)(1-i)$. Then $\mathcal{O}_K/2\mathcal{O}_K$ has order 4 (look at it'snorm), and certainly $2\mathcal{O}_K \subseteq (1-i)$, hence $|\mathcal{O}_K/(1-i)| \big| 4$, which by direct computation we may see that $|\mathcal{O}_K/(1-i)| = 2$, hence $f = 1$. On the other hand, we can see that 3 is prime in $\mathcal{O}_K$, and $|\mathcal{O}_K/3\mathcal{O}_K| = 9$, giving $f_{3\mathcal{O}_K/3} = 2$.

[a] In fact, they are equal in this case, which is not needed at the moment

Since the factorization of $\mathfrak{p} \subseteq K$ in L/K is tied to the degree of L/K, we may venture into hoping that the structure of L/K may reflect something about the factorization of $\mathfrak{p}$. A natural question we may ask here is how do we find the prime factors in the extension? Finding them is split into two cases: the "good" primes and the "bad' primes. We shall see that the power of the exponentials has a lot to do with it. This is because we shall see that most primes shall decompose with ramification index 1, since it will correspond to the splitting of a polynomial in separable extension (note that all number field extensions are separable). However, as we saw it is possible that prime may have higher ramification index, however it is impossible that a polynomial splits irreducibly with multiplicity higher than 1 over a separable extension. Hence, we may think of such prime as "bad" primes. In the following we shall show how to find primes whose factorization mirrors that of a minimal polynomial, and how to detect when it is not the case.

let L/K be a finite separable extension. Then by the primitive extension theorem, there exists a $\theta \in K$
$$L = K(\theta)$$
Usually, we shall let L be Galois so that we may computationally find this element, namely it must not be fixed by any element of $\mathrm{Gal}_K(L)$ (hence, not in any intermediate extension). In fact, we may find θ so that $\theta \in \mathcal{O}_K$: if $L = K(\alpha)$, since $L = \overline{\mathcal{O}_K}^L$ we may multiply out the denominator by d to get $\alpha = \theta$; let us simply re-define θ to be in $\mathcal{O}_L$. Now we have a single element, θ with minimal polynomial $m_{K,\theta}(x)$ which characterizes L. It would be nice to say something about the nature of the decomposition of $\mathfrak{p}$ in $\mathcal{O}_L$ given this information. In the good case, we shall be able to do so, and in the bad case we shall require a bit more work. In particular, a finite number of primes ideals will be exclude; only those that are relatively prime to an ideal known as the *conductor* of the ring $\mathcal{O}_K[\theta]$ will be considered:

Definition 50.4.6: Conductor

Let $\mathcal{O}_K[\theta]$ be the ring of integers extended by θ. Then the conductor of $\mathcal{O}_K[\theta]$, is the largest ideal $\mathscr{F}_\theta \subseteq \mathcal{O}_L$ which is contained in $\mathcal{O}_K[\theta]$:

$$\mathscr{F}_\theta = \{\alpha \in \mathcal{O}_L \mid \alpha\mathcal{O}_L \subseteq \mathcal{O}_K[\theta]\}$$

In the case where $K = \mathbb{Q}$ and $\mathcal{O}_\mathbb{Q} = \mathbb{Z}$, then $\mathscr{F}_\theta = \{\alpha \in \mathcal{O}_L \mid (\alpha)_L \subseteq \mathbb{Z}[\theta]\}$. Since $\mathcal{O}_L$ is a finitely generated $\mathcal{O}_K$ module, $\mathscr{F}_\theta \neq 0$. The conductor is a measure of the "bad primes" (hence the ramification). In our case, the conductor is never 0, but it may be that $\mathscr{F}_\theta = (1)$, in which case the bad primes will be "tame" (or the field is "tamely ramified")

Theorem 50.4.7: Dedekind-Kummer Theorem

Let L/K be a separable extension, $L = K(\theta)$ where θ is a primitive element where $m_{\theta,K}(x) \in \mathcal{O}_K[x]$. Let $\mathfrak{p}$ be a prime ideal of $\mathcal{O}_K$ which is relatively prime to the conductor $\mathscr{F}$ of $\mathcal{O}_K[\theta]$ and let $m(x) = m_{\theta,K}(x)$ and consider:

$$\overline{m}(x) = \overline{m}_1(x)^{e_1}\cdots\overline{m}_r(x)^{e_r} \bmod \mathfrak{p}$$

be the factorization of the polynomial $\overline{m}(x) = m(x) \bmod \mathfrak{p}$ into irreducible over the residue field $\mathcal{O}_K/\mathfrak{p}$. Then

$$\mathfrak{B}_i = \mathfrak{p}\mathcal{O} + m_i(\theta)\mathcal{O} = (\mathfrak{p}\mathcal{O}, m_i(\theta))_L$$

are different prime ideals of $\mathcal{O}$ over $\mathfrak{p}$. The inertia degree f_i of $\mathfrak{B}_i$ is the degree of $\overline{m}_i(x)$, and

$$\mathfrak{p} = \mathfrak{B}_1^{e_1}\cdots\mathfrak{B}_r^{e_r}$$

Note that if $\mathfrak{p}$ is not relatively prime to $\mathscr{F}$, then $\mathfrak{p}$ is not relatively prime to the relative discriminant, hence $\mathfrak{p}$ will surely ramify, and hence this is a problem for primes whose ramification is somehow "more bad"; such primes are often said to have *wild ramification* and they shall be studied in section 50.4.3.

***Proof*:**
Let $\mathcal{O}' = \mathcal{O}_K[\theta]$ and $\overline{\mathcal{O}_K} = \mathcal{O}_K/\mathfrak{p}$. The key isomorphism that we can establish since $\mathfrak{p}$ is relatively prime to the conductor $\mathscr{F}_\theta$ is:

$$\mathcal{O}_L/\mathfrak{p}\mathcal{O}_L \overset{!}{\cong} \mathcal{O}_K[\theta]/\mathfrak{p}\mathcal{O}_K[\theta] \cong \overline{\mathcal{O}_K}[x]/(\overline{m}(x))$$

where the $\overset{!}{\cong}$ isomorphism is the hard one to prove; the 2nd one comes from:

$$\mathcal{O}_K[\theta]/\mathfrak{p} \cong \mathcal{O}_K[x]/(\mathfrak{p}, m(x)) \cong \overline{\mathcal{O}}_K[x]/(\overline{m}(x))$$

For the $\overset{!}{\cong}$ isomorphism, from the relative primality of $\mathfrak{p}$ and $\mathscr{F}$ we get $\mathfrak{p}\mathcal{O}_L + \mathscr{F} = \mathcal{O}_L$. Since $\mathscr{F} \subseteq \mathcal{O}' = \mathcal{O}_k[\theta]$, we get that $\mathcal{O}_L = \mathfrak{p}\mathcal{O}_L + \mathcal{O}'$. From this, we get that the homomorphism

$$\mathcal{O}' \to \mathcal{O}_L/\mathfrak{p}\mathcal{O}_L$$

is surjective (every element of $\mathcal{O}_L$ is a linear combination of an element in $\mathcal{O}'$ and $\mathfrak{p}\mathcal{O}_L$) with kernel $\mathfrak{p}\mathcal{O}_L \cap \mathcal{O}'$. What's left to show is that this kernel equals $\mathfrak{p}\mathcal{O}'$. $\mathfrak{p}\mathcal{O}' \subseteq \mathfrak{p}\mathcal{O}_L \cap \mathcal{O}'$ is immediate, so for the other direction since $\mathfrak{p} + \mathscr{F} \cap \mathcal{O}_K = \gcd(\mathfrak{p}, \mathscr{F} \cap \mathcal{O}_K) = (1)$, $1 = pq + fs$ for some $p \in \mathfrak{p}$, $f \in \mathscr{F} \cap \mathcal{O}_K$. Then certainly $1 = pq + fs \in \mathfrak{p} + \mathscr{F}$, hence $\mathcal{O}' = \mathfrak{p} + \mathscr{F}$. Then:

$$\begin{aligned}\mathfrak{p}\mathcal{O}_L \cap \mathcal{O}' &= (\mathfrak{p} + \mathscr{F})(\mathfrak{p}\mathcal{O}_L \cap \mathcal{O}') \\ &= \mathfrak{p}(\mathfrak{p}\mathcal{O}_L \cap \mathcal{O}') + \mathscr{F}(\mathfrak{p}\mathcal{O}_L \cap \mathcal{O}') \\ &\subseteq \mathfrak{p}\mathcal{O}'\end{aligned}$$

where the last inclusion comes from the fact that any element of in the 2nd equality has the form $p(p\ell + f\ell')$ where $p\ell, f\ell' \in \mathcal{O}'$ by definition (namely the intersection), and so $\mathcal{O}_L/\mathfrak{p}\mathcal{O}_L \cong \mathcal{O}'/\mathfrak{p}\mathcal{O}'$. Then by our above argument we have $\mathcal{O}'/\mathfrak{p}\mathcal{O}' \cong \overline{\mathcal{O}}_K/(\overline{m}(x))$, hence we have:

$$\mathcal{O}_L/\mathfrak{p}\mathcal{O}_L \cong \overline{\mathcal{O}_K}[x]/(\overline{m}(x))$$

Now looking at the ring on the right, since in $\bmod \mathfrak{p}$ we have $\overline{m}(x) = \prod_i^r \overline{m}_i(x)^{e_i}$, by the CRT we get:

$$\overline{\mathcal{O}}_K[x]/(\overline{m}(x)) \cong \bigoplus_i^r \overline{\mathcal{O}}_K[x]/(\overline{m}_i(x))^{e_i}$$

But now we know the structure of each direct summand, $\overline{\mathcal{O}}_k[x]/(\overline{m}_i(x))^{e_i}$ is a principal ideal ring, and by the isomorphism we may deduce the prime ideal of $R = \overline{\mathcal{O}_K}[x]/(\overline{m}(x))$ are the principal ideal

$$(\overline{m_i}) = (\overline{m}_i(x) \bmod \overline{m}(x)) \qquad 1 \leq i \leq r$$

By ring extension theory, the degree of $[R/(\overline{m}_i) : \overline{\mathcal{O}}_K]$ equals the degree of the polynomial $\overline{m}_i(x)$, and by definition of the $(\overline{m}_i)$:

$$(0) = \bigcap_i^r (\overline{m}_i)^{e_i}$$

Now, through the isomorphism $\overline{\mathcal{O}}_K[x]/(\overline{m}(x)) \cong \mathcal{O}_L/\mathfrak{p}\mathcal{O}_L$ given by $\overline{f}(x) \mapsto \overline{f}(\theta)$, we get the same result for the primes ideals of $\overline{\mathcal{O}}_L = \mathcal{O}_L/\mathfrak{p}\mathcal{O}_L$. Namely, the prime ideals $\overline{\mathfrak{B}}_i$ of $\overline{\mathcal{O}}_L$ correspond to the prime ideals

$$\overline{\mathfrak{B}}_i = (\overline{m}_i) = (m_i(\theta) \bmod \mathfrak{p}\mathcal{O}_L)$$

with degrees $[R/\overline{\mathfrak{B}}_i : \mathcal{O}_K/\mathfrak{p}]$ where $R = \overline{\mathcal{O}_K}[x]/(\overline{m(x)})$ equalling the degrees of the polynomials $\overline{m}_i(x)$. Furthermore:

$$(0) = \bigcap_i^r \overline{\mathfrak{B}}_i^{e_i}$$

Now, given the canonical projection $\mathcal{O}_L \to \mathcal{O}_L/\mathfrak{p}\mathcal{O}_L$, let $\mathfrak{B}_i = \mathfrak{p}\mathcal{O}_L + m_i(\theta)\mathcal{O}_L$, (the preimage of $\overline{\mathfrak{B}}_i$). Then since the pre-image of a prime ideal is prime, each $\mathfrak{B}_i$ is prime, and

$$\mathfrak{B}_i \cap \mathcal{O}_K = \mathfrak{p}$$

since, when written in element form, we have that

$$p\ell + m_i(\theta)\ell' \in \mathcal{O}_K$$

which is true if and only if $\ell' = 0$ and $p\ell \in \mathcal{O}_K$, showing that $\mathfrak{B}_i$, $i \leq i \leq r$, varies over the prime ideals of $\mathcal{O}_L$ above $\mathfrak{p}$, with $f_i = [\mathcal{O}_L/\mathfrak{B}_i : \mathcal{O}_K/\mathfrak{p}]$ being the degree of the polynomial $\overline{m}_i(x)$. Furthermore, $\mathfrak{B}_i^{e_i}$ is the pre-image of $(\overline{\mathfrak{B}}_i^{e_i})$ (since $e_i = \#\left\{\overline{\mathfrak{B}}^n \;\middle|\; n \in \mathbb{N}\right\}$) and

$$(0) = \cap_{i=1}^r \mathfrak{B}_i^{e_i} = \prod \mathfrak{B}_i^{e_i} \subseteq \mathfrak{p}\mathcal{O}_L$$

so that

$$\mathfrak{p}\mathcal{O}_L \mid \prod_{i=1}^{r} \mathfrak{B}_i^{e_i}$$

since $\deg(m(x)) = [L : K] = n = \sum e_i f_i$, we in fact has no other factors or powers, showing us that in fact:

$$\mathfrak{p}\mathcal{O}_L = \prod_{i=1}^{r} \mathfrak{B}_i^{e_i}$$

as we sought to show.

Corollary 50.4.8: Factorization For Number Rings (Dedekind-Kummer Theorem)

Let L/K be a finite extension of number fields, $L = K(\theta)$ for a primitive element θ where $m_{\theta,K}(x) \in \mathcal{O}_K[x]$. Let $\mathfrak{p} \subseteq \mathcal{O}_K$ a prime ideal lying over $p \in \mathbb{Z}$. If $p \nmid [\mathcal{O}_L : \mathcal{O}_K[\alpha]]$, then

$$\mathfrak{B}_i = \mathfrak{p}\mathcal{O}_L + m_i(\theta)\mathcal{O}_L = (\mathfrak{p}\mathcal{O}_L, m_i(\theta))_L$$

are different prime ideals of $\mathcal{O}$ over $\mathfrak{p}$. The inertia degree f_i of $\mathfrak{B}_i$ is the degree of $\overline{m}_i(x)$, and

$$\mathfrak{p} = \mathfrak{B}_1^{e_1} \cdots \mathfrak{B}_r^{e_r}$$

***Proof*:**
In the proof of theorem 50.4.7 we used the conductor to show that:

$$\mathcal{O}_L/\mathfrak{p}\mathcal{O}_L \cong \mathcal{O}'/\mathfrak{p}\mathcal{O}' \cong \overline{\mathcal{O}_K}[x]/(\overline{m}(x))$$

In particular, we could have used the CRT earlier and shown that:

$$\frac{\mathcal{O}_L}{\mathfrak{B}} \cong \frac{\overline{\mathcal{O}_K}[x]}{(\overline{m_i(x)})}$$

Hence, if we can show this condition, we are done. Take the map $\varphi : \mathcal{O}_K[x] \to \mathcal{O}_L/\mathfrak{B}_i$ mapping $p(x) \mapsto p(\theta)$. We claim that this map is surjective. To see this, it suffices to show that

$$\mathcal{O}_L = \mathcal{O}_K[\theta] + \mathfrak{B}_i$$

We know that $p \in \mathfrak{p} \subseteq \mathfrak{B}_i$, hence $p\mathcal{O}_L \subseteq \mathfrak{B}_i$. We now claim the slightly stronger:

$$\mathcal{O}_L = \mathcal{O}_K[\theta] + p\mathcal{O}_L$$

Now, by assumption $p \nmid |\mathcal{O}_L : \mathcal{O}_K[\theta]|$. Since the index of $\mathcal{O}_K[\theta] + p\mathcal{O}_L$ in $\mathcal{O}_L$ is the common divisor of $[\mathcal{O}_L : \mathcal{O}_K[\theta]]$ and $[\mathcal{O}_L : p\mathcal{O}_L]$, and these *must* be relatively prime since

$$[\mathcal{O}_L : p\mathcal{O}_L] = p^k$$

for some $k \in \mathbb{N}$. Thus $\mathcal{O}_K[x] \to \mathcal{O}_L/\mathfrak{B}_i$ is surjective. Now Certainly $(\mathfrak{p}, m_i) \subseteq \ker\varphi$. Furthermore, it is clear that $(\mathfrak{p}, m_i) \subseteq \mathcal{O}_K[x]$ is a maximal ideal hence $\ker\varphi$ is either $(\mathfrak{p}, m_i)$ or $\mathcal{O}_K[x]$. since the map is surjective, then either $\mathfrak{B}_i = \mathcal{O}_L = (1)_L$ (i.e. $\mathfrak{B}_i$ is *not* a prime, a contradiction), or it is indeed the desired kernel, meaning:

$$\frac{\mathcal{O}_L}{\mathfrak{B}} \cong \frac{\overline{\mathcal{O}_K}[x]}{(\overline{m_i(x)})}$$

but now the proof follows from the rest of the proof of theorem 50.4.7, as we sought to show.

Corollary 50.4.9: Monogenic Number Rings

Let L/K be a finite extension of number rings such that $\mathcal{O}_L = \mathcal{O}_K[\alpha]$ for some $\alpha \in \mathcal{O}_L$. Then all primes split as given in the above

***Proof*:**
Choose $\theta = \alpha$. Then $[\mathcal{O}_L : \mathcal{O}_K[\alpha]] = [\mathcal{O}_K[\alpha] : \mathcal{O}_K[\alpha]] = 1$. Since certainly no prime divides 1, we have that all primes factor in given their minimal polynomials and modular factorization, as we sought to show.

Note that not all number rings are of the form $\mathcal{O}_L = \mathcal{O}_K[\alpha]$, that is not all number rings are monogenic. We shall show a counter-example in example 50.4.29. This contrasts to finite extension of field where every separable finite extension of is a simple extension by theorem 19.6.20. Note how this is dependent on finding a θ and $\mathfrak{p}$ being relatively prime to $\mathscr{F}_\theta$ (or $[\mathcal{O}_L : \mathcal{O}_K[\theta]]$ in the case of number fields)[23].

As it turns out, the power of an extended prime has important geometric and algebraic consequences; it can be thought of as the measure of how much the Galois group moves a field around. We hence give it an name:

Definition 50.4.10: Ramified and Unramified Extensions

Let $\mathfrak{p} \subseteq R$ be a prime ideal, $K = K(R)$, L/K a finite extension, and $\mathcal{O} = \overline{K}^L$. Then given

$$\mathfrak{p}\mathcal{O} = \mathfrak{B}_1^{e_1} \cdots \mathcal{B}_r^{e_r}$$

Then the prime ideal $\mathfrak{B}_i$ is called *unramified* over $\mathcal{O}_k$ (or over K) if $e_i = 1$ and if the residue fields $(\mathcal{O}/\mathfrak{B}_i)/(\mathcal{O}_K/\mathfrak{p})$ is separable[a]:

$$\mathfrak{p}\mathcal{O} = \mathfrak{B}_1 \cdots \mathfrak{B}_r$$

Otherwise, $\mathfrak{B}_i$ is called *ramified*, and *totally ramified* if furthermore $f_i = 1$. The prime ideal $\mathfrak{p}$ is called *unramified* if all $\mathfrak{B}_i$ are unramified, and $\mathfrak{p}$ is ramified otherwise. The extension L/K is called *unramified* or an *unramified extension* if all prime ideals $\mathfrak{p}$ of K are unramified.

[a]which for number fields this is always the case

It is in fact rare that a prime ideal $\mathfrak{p}$ of K is ramified in L[24], however every [nontrivial] extension of $\mathbb{Q}$ is ramified (i.e. some primes ramify). Later, we shall see that there do exist [nontrivial] unramified extension L/K where $K \neq \mathbb{Q}$. To understand ramification in extension, we first show how we can factor most $\mathfrak{B}_i$. The term "ramification" comes from complex analysis, where if a holomorphic function f has a root, $f(a) = 0$, the order of the root is the ramification index[25].

[23]However, if K is a local field, as defined in [39], every extension *is* monogenic. This property, among many other similar niceties, makes developing number theory over local fields much more streamlined. The interested teacher should see [39] as a follow up to this chapter

[24]This correspond to the algebro-geometric fact that all but finitely many points of curve are *nonsingular*

[25]if $f(a) = 0$ has order n, there exists a g such that $f = g^n$ in some neighbourhood, which shows how much f "winds" around the root

Theorem 50.4.11: Counting Ramified Primes

Let L/K be a separable extension where $K = \mathrm{Frac}(R)$ or a Dedekind domain R. Then there are only finitely many prime ideals of K which are ramified in L, in particular the ideals $\mathfrak{p} \subseteq K$ that are *not* relatively prime to the discriminant of $m_{K,\theta}(x)$ and $\mathscr{F}_\theta$.

Note that it is still possible that primes factor as given in theorem 50.4.7 since a prime may be relatively prime to $\mathscr{F}_\theta$ but not the discriminant of $m_{K,\theta}(x)$.

***Proof*:**
Let $\theta \in \mathcal{O}$ be a primitive element for L, $L = K(\theta)$, and let $m(x) = m_{K,\theta}(x) \in R[x]$ be the minimal polynomial. Take the discriminant of $m(x)$

$$\mathrm{disc}(m_{\theta,K}(x)) = \mathrm{disc}_{L/K}(1, \theta, ..., \theta^{n-1}) = \prod_{i<j}(\theta_i - \theta_j)^2 \in R$$

We shall show every prime (fractional) ideal $\mathfrak{p} \subseteq K$ which is relatively prime to d and to the conductor $\mathscr{F}_\theta$ of $R[\theta]$ is unramified. Note that there are finitely many primes $\mathfrak{p} \subseteq K$ that are not relatively prime to the conductor and the discriminant since both have finitely many factors, hence this proofs the result.

By theorem 50.4.7, $\mathfrak{p}\mathcal{O}$ factors in the same way $\overline{m}_{\theta,K}(x)$ factors (mod $\mathfrak{p}$). Then the ramification indices e_i equal 1 when $\overline{m}(x)$ has no multiple roots. But this is the case since $\overline{0} \neq \overline{d} = d \bmod \mathfrak{p}$ is nonzero by assumption of being relatively prime, and the discriminant is nonzero if and only if all the roots are distinct. Thus, the residue class field extension $(\mathcal{O}/\mathfrak{B}_i)/(R/\mathfrak{p})$ are generated by $\overline{\theta} \equiv \theta \bmod \mathfrak{B}_i$, and thus are separable, showing $\mathfrak{p}$ is unramified, as we sought to show.

We can in fact be more precise than the above result: the ramified primes ideals will be given by the *relative discriminant* of $\mathcal{O}_L/\mathcal{O}_K$, which will be given by the ideal $\mathfrak{o} \subseteq \mathcal{O}_K$ which is generated by the discriminants $d(\omega_1, \omega_2, ..., \omega_n)$ for all bases $\omega_1, \omega_2, ..., \omega_n$ of L/K contained in $\mathcal{O}_L$. A special case is that of the base-field being $K = \mathbb{Q}$.

Proposition 50.4.12: Detecting Ramification In Number Fields

Let p be a prime in $\mathbb{Z}$. Suppose p is ramified. Then:

$$p \big| \mathrm{disc}(\mathcal{O}_K)$$

In proposition 50.4.16, we will show the converse, showing this is an iff.

***Proof*:**
Let $\mathfrak{B}$ lie over $\mathfrak{p}$ and $e_{\mathfrak{B}/\mathfrak{p}} > 1$. Thus

$$p\mathcal{O}_K = \mathfrak{B}I$$

where I is divisible by *all* primes of $\mathcal{O}_K$ lying over p. Let $\sigma_1, \sigma_2, ..., \sigma_n$ denote the embeddings of K in $\mathbb{C}$, and extend each σ_i to automorphisms of some extension of L/K which is normal over $\mathbb{Q}$. Let $\alpha_1, \alpha_2, ..., \alpha_n$ be some integral basis for R. What we shall do is replace one of the α_i be a suitable element that will show that $p\big|\mathrm{disc}(\mathcal{O}_K)$.

Take $\alpha \in I - p\mathcal{O}_K$. Then certainly α is in every prime ideal lying over p but *not* in $p\mathcal{O}_K$. writing

$$\alpha = m_1\alpha_1 + \cdots m_n\alpha_n$$

for appropriate $m_i \in \mathbb{Z}$, then since $\alpha \notin p\mathcal{O}_K$, we have that p does not divide each m_i. Re-arrange the terms so that $p \nmid m_1$. Then if d is the discriminant of $\mathcal{O}_K$, through direct computation, we can see that

$$\text{disc}(\alpha, \alpha_2, ..., \alpha_n) = m_1^2 d$$

Since $p \nmid m_1$, we shall show that $p \big| \text{disc}(\alpha, \alpha_2, ..., \alpha_n)$

Given $L/K/\mathbb{Q}$, since α is in every prime of $\mathcal{O}_K$ lying over p, it certainly is in every prime in $\mathcal{O}_L$ lying over p. Fixing any $\mathfrak{B}$ of S lying over p, we see that for any automorphism σ_i, $\sigma(\alpha) \in \mathfrak{B}$. To see this ,$\sigma^{-1}(\alpha)$ is a prime of $\sigma^{-1}(S) = S$ lying over p, hence must contain α. In particular, this gives $\sigma_i(\alpha) \in \mathfrak{B}$ for all i. But then $\mathfrak{B}$ contains $\text{disc}(\alpha, \alpha_2, ..., \alpha_n)$. Since the discriminant is in $\mathbb{Z}$, we get that:

$$\mathfrak{B} \cap \mathbb{Z} = p\mathbb{Z}$$

but then $p \big| \text{disc}(\mathcal{O}_K)$, as we sought to show.

Corollary 50.4.13: Ramification Over Number Fields

Let $\alpha \in R$ so that $K = \mathbb{Q}(\alpha)$ and let f be a monic polynomial over $\mathbb{Z}$ such that $f(\alpha) = 0$. Then if p is a prime such that

$$p \not\big| \text{Nm} f'(\alpha)$$

Then p is unramified in K

***Proof*:**
This follows from the above an the fact that if α is an algebraic integer and $f \in \mathbb{Z}[x]$ monic such that $f(\alpha) = 0$, then by proposition 50.1.29:

$$\text{disc}(\alpha) \big| \text{Nm}_{\mathbb{Q}(\alpha)/\mathbb{Q}}(f'(\alpha))$$

Using this result, we have an easier verification for ramification of primes of $\mathbb{Z}$ to number fields, since they must divide the discriminant, but the discriminant may only have finitely many factors. In fact, it is easy to generalize this result for extension of number fields

Corollary 50.4.14: Ramification In Number Fields

Let L/K be an extension of number fields Then only finitely many primes of $\mathcal{O}_K$ are ramified in $\mathcal{O}_L$.

***Proof*:**
Let $\mathfrak{p}$ be a prime of $\mathcal{O}_K$ which is ramified in $\mathcal{O}_L$. Then $\mathfrak{p} \cap \mathbb{Z} = p\mathbb{Z}$ is ramified in S. Hence there are only finitely many possible primes p that ramify, meaning there are only finitely many possible $\mathfrak{p}$.

The converse is also true, namely

$$p\Big|\mathrm{disc}(\mathcal{O}_K) \Leftrightarrow p \text{ ramifies in } K$$

The full proof shall be given in greater generality in 50.4.3 where it is generalized for primes $\mathfrak{p} \subseteq \mathcal{O}_K$ being ramified in $\mathcal{O}_L$ for algebraic extension L/K. However, it is instructive for the reader to prove it in the following special case that is particularly important for the study of local fields (see [39]):

Lemma 50.4.15: Partial Converse For Ramification

Let $K = \mathbb{Q}(\alpha)$ where $\alpha \in \mathcal{O}_K$ and $p \nmid [\mathcal{O}_K : \mathbb{Z}[\alpha]]$. Then if $p\Big|\mathrm{disc}(\mathcal{O}_K)$, p ramifies in K

***Proof*:**
This is a good exercise!

For the result when *only* considering primes of $\mathbb{Z}$, the result is easier:

Proposition 50.4.16: Characterizing Ramification In Number Fields

Let p be a prime in $\mathbb{Z}$. Suppose $p\Big|\mathrm{disc}(\mathcal{O}_K)$. Then p is ramified.

***Proof*:**
(difficult) exercise; or see[36, p. 79]

50.4.1 Example: Eisenstein Polynomials

We say $f \in \mathbb{Z}[x]$ is *Eisenstein* with respect to p is p divides each coefficient but p^2 does not divide the constant coefficient. Then by Eisenstein's criterion (proposition 11.3.10), such a polynomial is irreducible in $\mathbb{Q}[x]$. We shall use these to quickly determine a few useful ring of integer.

Lemma 50.4.17: Eisenstein Polynomial Modular Roots

Let $K/\mathbb{Q}$ be a number field of degree n and assume $K = \mathbb{Q}(\alpha)$ for $\alpha \in \mathcal{O}_K$ with minimal polynomial $m_{\alpha,\mathbb{Q}}$ Eisenstein at p. Then then for some $f(x) \in \mathbb{Z}[x]$ of degree at most $n-1$ with coefficients $a_0, a_1, ..., a_{n-1}$, if:

$$f(\alpha) \equiv 0 \text{ mod } p\mathcal{O}_K$$

Then:

$$a_i \equiv 0 \text{ mod } p\mathbb{Z} \qquad 1 \leq i \leq n-1$$

***Proof*:**
we shall do an argument by induction. Certainly if $a_0 \equiv 0 \text{ mod } p\mathcal{O}_K$ then $a_0 \equiv 0 \text{ mod } p\mathbb{Z}$ (since $a_0 \in \mathbb{Z}$). Thus, assume for $i < j$, $j \in \{0, 1, ..., n-1\}$ that $a_i \equiv 0 \text{ mod } p\mathbb{Z}$. Then for $i < j$:

$$a_j\alpha^j + a_{j+1}\alpha^{j+1} + \cdots + a_{n-1}\alpha^{n-1} \equiv \text{ mod } p\mathcal{O}_K$$

Multiply the above by α^{n-1-j} to make all but the first term $a_j\alpha^{n-1}$ a multiple of α^n. Since α is a root of an Eisenstein polynomial at p, we have

$$\alpha^n \equiv 0 \bmod p\mathcal{O}_K$$

Hence:

$$a_j\alpha^{n-1} \equiv p\mathcal{O}_K$$

Then $a_j\alpha^{n-1} = p\gamma$, for $\gamma \in \mathcal{O}_K$. Taking the norm, we get:

$$a_j^n \mathrm{Nm}_{K/\mathbb{Q}}(\alpha)^{n-1} = p^n \mathrm{Nm}_{K/\mathbb{Q}}(\gamma) \in \mathbb{Z}$$

the left hand side is divisible by the constant term of the minimal polynomial (since that is the value of $\mathrm{Nm}_{K/\mathbb{Q}}(\alpha)$). By Eisenstein, p divides $\mathrm{Nm}_{K/\mathbb{Q}}(\alpha)$ exactly once, so since p^n must divide the left hand side gives that $p \big| a_j^n$, thus $p \big| a_j$, thus $a_i \equiv 0 \bmod p\mathbb{Z}$ for $i < j+1$, giving us the result.

Lemma 50.4.18: Eisenstein Polynomial divisibility by p

Let $K/\mathbb{Q}$ be a number field of degree n, $K = \mathbb{Q}(\alpha)$ where $\alpha \in \mathcal{O}_K$ with minimal polynomial Eisenstein at p. Then if

$$r_0 + r_1\alpha + \cdots + r_{n-1}\alpha^{n-1} \in \mathcal{O}_K \qquad r_i \in \mathbb{Q}$$

then each r_i has no p in its denominator

***Proof*:**
For the sake of contradiction, assume some r_i has a p in its denominator. Let d be the lcm of the denominators so the r_i's so $p \big| d$ so that we may write $r_i = a_i/d$ for appropriate $a_i \in \mathbb{Z}$ where $a_i \nmid p$. Then:

$$r_0 + r_1\alpha + \cdots + r_{n-1}\alpha^{n-1} \in \mathcal{O}_K \implies \frac{a_0 + a_1\alpha + \cdots + a_{n-1}\alpha^{n-1}}{d} \in \mathcal{O}_K$$

multiplying by d,w e get:

$$a_0 + a_1\alpha + \cdots + a_{n-1}\alpha^{n-1} \in d\mathcal{O}_K \subseteq p\mathcal{O}_K$$

Then by lemma 50.4.17 we have that $a_i \in p\mathbb{Z}$ for each i, a contradiction!

Theorem 50.4.19: Eisenstein Polynomial Divisibility Test

Let $K = \mathbb{Q}(\alpha)$ where $\alpha \in \mathcal{O}_K$ a root of an Eisenstain polnomial at p of degree n. Then:

$$p \nmid [\mathcal{O}_K : \mathbb{Z}[\alpha]]$$

***Proof*:**
For the sake of contradiction, suppose $p\big|[\mathcal{O}_K : \mathbb{Z}[\alpha]]$. Then $\mathcal{O}_K/\mathbb{Z}[\alpha]$ has an element of order p, that is there is some $\gamma \in \mathcal{O}_K$ such that $\gamma \notin \mathbb{Z}[\alpha]$ but $p\gamma \in \mathbb{Z}[\alpha]$. Now, given basis $\{1, \alpha, ..., \alpha^{n-1}\}$ for $K/\mathbb{Q}$, then

$$\gamma = r_0 + r_1\alpha + \cdots + r_{n-1}\alpha^{n-1}$$

where $r_i \in \mathbb{Q}$. Since $\gamma \notin \mathbb{Z}[\alpha] <$ not all $r_i \in \mathbb{Z} >$ Since $p\gamma \in \mathbb{Z}[\alpha]$, $pr_i \in \mathbb{Z}$. Thus, r_i has p in its denominator, but that contradicts lemma 50.4.18, as we sought to show.

Example 50.4.20: Using Eisenstein Test

1. Let $K = \mathbb{Q}(\sqrt[3]{2})$. We shall quickly find $\mathcal{O}_K$. Recall that:

$$-2^2 3^3 = -108 = \operatorname{disc}(\mathbb{Z}[\sqrt[3]{2}]) = [\mathcal{O}_K : \mathbb{Z}[\sqrt[3]{2}]]^2 \operatorname{disc}(\mathcal{O}_K)$$

Since $\sqrt[3]{2}$ is a root of $x^3 - 2$, which is eisenstien at 2, we see that $2 \nmid [\mathcal{O}_K : \mathbb{Z}[\sqrt[3]{2}]]$. Furthermore, $1 + \sqrt[3]{2}$ is the root of

$$(x-1)^3 - 2 = x^3 - 3x^2 + 3x - 3$$

which is eisenstein at 3 s o

$$3 \nmid [\mathcal{O}_K : \mathbb{Z}[1 + \sqrt[3]{2}]] = [\mathcal{O}_K : \mathbb{Z}[\sqrt[3]{2}]$$

since $\mathbb{Z}[\sqrt[3]{2}] = \mathbb{Z}[1 + \sqrt[3]{2}]$. THus, the index must be 1, giving us:

$$\mathcal{O}_K = \mathbb{Z}[\sqrt[3]{2}]$$

2. Let $K = \mathbb{Q}(\sqrt[4]{2})$. Then:

$$-2^{11} = \operatorname{disc}(\mathbb{Z}[\sqrt[4]{2}]) = [\mathcal{O}_K : \mathbb{Z}[\sqrt[4]{2}]]^2 \operatorname{disc}(\mathcal{O}_K)$$

and $\sqrt[4]{2}$ is a root of $x^4 - 2$ which is eisenstein at 2, hence by the eisenstein divisibility test the idnex is 1 and

$$\mathcal{O}_K = \mathbb{Z}[\sqrt[4]{2}]$$

Let $K = \mathbb{Q}(\sqrt[5]{2})$. Then

$$2^4 5^4 = \operatorname{disc}(\mathbb{Z}[\sqrt[5]{2}]) = [\mathcal{O}_K : \mathbb{Z}[\sqrt[5]{2}]]^2 \operatorname{disc}(\mathcal{O}_K)$$

Certainly 2 doesn't divide the index. Next, the minimal polynomial of $\sqrt[5]{2} - 2$ is

$$(x+2)^5 - 2 = x^5 + 10x^4 + 40x^3 + 80x^2 + 80x + 30$$

which is Eisenstein at 5 and $\mathbb{Z}[\sqrt[5]{2} - 2] = \mathbb{Z}[\sqrt[5]{2}]$ Hence:

$$\mathcal{O}_K = \mathbb{Z}[\sqrt[5]{2}]$$

3. show that $\mathcal{O}_{\mathbb{Q}(\sqrt[3]{3})} = \mathbb{Z}[\sqrt[3]{3}]$ and $\mathcal{O}_{\mathbb{Q}(\sqrt[3]{5})} = \mathbb{Z}[\sqrt[3]{5}]$.

4. Recall the three cubic extension K_1, K_2, K_3 in example 50.1.42 with the same discriminant, but it was claimed that they where not isomorphic as fields. The three polynomials in question where:

$$f_1(x) = x^3 - 18x - 6 \qquad f_2(x) = x^3 - 36x - 78 \qquad f_3(x) = x^3 - 54x - 150$$

each having discriminant $22345 = 5 \cdot 41 \cdot 109$. It can be verified that these three polynomials have real roots (by either graphing or doing some guess-work with derivative test and evaluating), and from the above methods we can check that

$$\mathcal{O}_{K_i} = \mathbb{Z}[\alpha_i]$$

for appropriate root α_i. Since the ring of integers is Monogenic, by Dedekind-Kummers theorem the factorization of primes p mirrors the factorization of $f_i(x)$ mod p. Then it can be verified that

(a) f_1, f_2 are irreducible mod 5, but

$$f_3(x) \equiv x(x-2)(x-3) \bmod 5$$

(b) f_2, f_3 are irreducible mod 11, but

$$f_1(x) \equiv (x-3)(x-9)(x-10) \bmod 11$$

Thus, we see that these fields cannot be isomorphic, or else the prime factorization would have been equivalent (recall that ring homomorphisms map primes to primes)

It may be asked if $K = \mathbb{Q}(\sqrt[n]{2})$

$$\mathcal{O}_K = \mathbb{Z}[\sqrt[n]{2}]$$

This is true for $n \leq 1000$, but is *not always true*. For those interested, they may search for *Wieferich primes* (see Keith Conrad *The ring of integers in radical extensions*).

Eisenstein polynomials also allow us to identify when a polynomial totally ramifies:

Theorem 50.4.21: Eisenstein Polynomial And Total Ramification

Let $K = \mathbb{Q}(\alpha)$. Then α is a root of a $\mathbb{Z}$-polynomial that is Eisenstein at p iff p is totally ramified in K.

Note how there is no reference to the ring of integers

***Proof*:**
(*must* add this here! this link

Example 50.4.22: Dececting Ramification Using Eisenstein

Since $\alpha = \sqrt[3]{10}$ is a root of $x^3 - 10$, which is eisenstein at 2 and 5, then in $K = \mathbb{Q}(\alpha)$, $(2) = \mathfrak{p}^3$ and $(5) = \mathfrak{q}^3$. Note however htat

$$\mathcal{O}_K \neq \mathbb{Z}[\sqrt[3]{10}]$$

It is in fact equal to

$$\mathcal{O}_K = \mathbb{Z}[\beta] \qquad \beta = \frac{1}{3}\left(1 + \sqrt[3]{1-} + \sqrt[3]{100}\right)$$

which is a root of $x^3 - x^2 - 3x - 3$.

50.4.2 Primes in Galois Extensions

The question of prime decomposition in extension becomes more interesting when L/K is also a Galois extension since $\mathrm{Gal}_K(L)$ can act on the set of primes.

Lemma 50.4.23: Galois Group Acts On Primes

Let L/K be a finite galois extension with galois group $G = \mathrm{Gal}_K(L)$, $\mathfrak{p} \subseteq \mathcal{O}_K$ a prime ideal of $\mathcal{O}_K$, and $\mathfrak{P} = \{\mathfrak{B}_1, \mathfrak{B}_2, ..., \mathfrak{B}_n\}$ the set of primes lying above $\mathfrak{p}$. Then:

$$G \curvearrowright \mathfrak{P} \qquad \text{via} \qquad \sigma \cdot \mathfrak{B}_i = \sigma(\mathfrak{B}_i)$$

***Proof*:**
let $G = \mathrm{Gal}_K(L)$. Then for each $a \in \mathcal{O}_L$, $\tau(a) \in \mathcal{O}_L$ for all $\tau \in G$ since K is invariant under τ and all conjugates have an integral minimal polynomial (i.e. the same minimal polynomial with coefficients in $\mathbb{Z} \subseteq K$), so there is a natural action of G on $\mathcal{O}_L$; $G \curvearrowright \mathcal{O}_L$. Next, given that τ is a field isomorphism, $\tau : \mathcal{O}_L \to \mathcal{O}_L$ is a ring isomorphism, importantly it sends primes to primes. If $\mathfrak{B}$ is a prime ideal of $\mathcal{O}_L$ above $\mathfrak{p}$, then $\tau(\mathfrak{B})$ is also a prime ideal above $\mathfrak{p}$ for each prime $\mathfrak{B}$ since

$$(\tau(\mathfrak{B})) \cap \mathcal{O}_K = \tau(\mathfrak{B} \cap \mathcal{O}_K) = \tau(\mathfrak{p}) = \mathfrak{p}$$

since τ is K-invariant, hence $\mathcal{O}_K \subseteq K$ is invariant.

Definition 50.4.24: Prime Ideal Conjugate

Let $\mathcal{O}_L/\mathcal{O}_K$ be an extension of rings of integers where L/K is Galois, and $\mathfrak{B}$ lie above $\mathfrak{p}$. An ideal $\tau(\mathfrak{B})$ is called a *prime ideal conjugate* to $\mathfrak{B}$.

As we saw, most primes split similarly to a minimal polynomial. The following is further parallel between prime numbers and polynomials: just like conjugate elements, the Galois group acts transitively:

Proposition 50.4.25: Galois Group Transitive Action On Ideals

Let $G = \mathrm{Gal}_K(L)$. Then G acts transitively on the set of all ideal $\mathfrak{B}$ of $\mathcal{O}$ lying above $\mathfrak{p}$, that is the prime ideals are all conjugates of each other.

***Proof*:**
Say $\mathfrak{B}, \mathfrak{B}'$ are two primes ideals lying above $\mathfrak{p}$ where, for the sake of contradiction, $\sigma(\mathfrak{B}) \neq \mathfrak{B}'$

for any $\sigma \in G$. By the CRT, there exists an $x \in \mathcal{O}$ such that

$$x \equiv 0 \bmod \mathfrak{B}' \qquad x \equiv 1 \bmod \sigma(\mathfrak{B}) \qquad \forall \sigma \in G$$

Then

$$\mathrm{Nm}_{L/K}(x) = \prod_{\sigma \in G} \sigma(x) \overset{!}{\in} \mathfrak{B}' \cap \mathcal{O}_K = \mathfrak{p}$$

On the other hand, $x \notin \sigma(\mathfrak{B})$ for all $\sigma \in G$, hence $\sigma(x) \notin \mathfrak{B}$ for all $\sigma \in G$ since if $\sigma(x) = b \in \mathfrak{B}$, then $x = \sigma^{-1}(b) = \tau(b)$ for some $\tau \in G$. Thus,

$$\prod_{\sigma \in G} \sigma(x) \overset{!}{\notin} \mathfrak{B} \cap \mathcal{O}_K = \mathfrak{p}$$

Hence, $\prod_{\sigma \in G} \sigma(x)$ is both in and not in $\mathfrak{p}$, a contradiction.

Using the Galois group, we may encode in group-theoretic terms the number of different primes ideals a prime ideal $\mathfrak{p} \subseteq \mathcal{O}_K$ decompose to in $\mathcal{O}_L$. One of the properties of an element being acted on by a [transitive] group is it *stabalizer*:

Definition 50.4.26: Decomposition Group And Field

Let $G = \mathrm{Gal}_K(L)$. Let $\mathfrak{B} \subseteq \mathcal{O}$ be a prime ideal. Then the subgroup

$$D_{\mathfrak{B}/\mathfrak{p}} = D_{\mathfrak{B}} = \{\tau \in G \mid \tau(\mathfrak{B}) = \mathfrak{B}\} = \mathrm{stab}_G(\mathfrak{B})$$

is called the *decomposition group* of $\mathfrak{B}$ over K. The fixed field

$$L^{D_{\mathfrak{B}}} = \{x \in L \mid \tau(x) = x \ \forall \tau \in G_{\mathfrak{B}}\}$$

is called the *decomposition field* of $\mathfrak{B}$ over K, and is often labeled as $L_{\mathfrak{B}}$ or $L_{\mathfrak{B}}$ if unambiguous.

Notice that the less primes that are fixed, the larger the decomposition field is, hence there is a contravariant relation between the size of the group and the extension of the field, just like in Galois theory. Due to transitivity and the introduction of Galois theory, we immediately get many counting arguments. If $\mathfrak{B}$ is a prime ideal lying over $\mathfrak{p}$, and we let σ vary over a set of representatives from the cosets in $G/D_{\mathfrak{B}}$, then by transitivity $\sigma(\mathfrak{B})$ varies over the different primes ideals above $\mathfrak{p}$, each occurring exactly once, hence the number of prime ideals is equal to the index $[G : D_{\mathfrak{B}}]$. For reference, we shall write:

Definition 50.4.27: Prime Counter

Let L/K Then the number of prime numbers is equal to:

$$g_{L/K}(\mathfrak{p}) = [G : D_{\mathfrak{B}}]$$

for any prime $\mathfrak{B}$ over $\mathfrak{p}$. If clear from context, we shall write $g(\mathfrak{p})$.

Thus, if $D_{\mathfrak{B}} = 1$, then no $\sigma \in G$ fixes $\mathfrak{B}$ unless it's the identity and $[G : D_{\mathfrak{B}}] = |G| = [L : K]$, and so by the fundamental identity (proposition 50.4.3):

$$D_{\mathfrak{B}} = 1 \quad \iff \quad L_D = L \quad \iff \quad \mathfrak{p} \text{ totally splits}$$

On the other hand, if $D_{\mathfrak{B}} = G$, then every $\mathfrak{B}$ lying over $\mathfrak{p}$ is fixed by σ, and so there must only be one prime ideal above $\mathfrak{p}$ by transitivity giving:

$$D_{\mathfrak{B}} = G \quad \iff \quad L_D = K \quad \iff \quad \mathfrak{p} \text{ totally ramifies}$$

If $1 \subsetneq D_{\mathfrak{B}} \subsetneq G$, there is ambiguity on how $\mathfrak{p}$ splits. For example, if $[L : K] = 50 = 2 \cdot 5^2$ and $|G/D_{\mathfrak{B}}| = 5$, then either

$$\mathfrak{p}\mathcal{O}_L = \mathfrak{B}_1^2\mathfrak{B}_2^2\mathfrak{B}_3^2\mathfrak{B}_4^2\mathfrak{B}_5^2 \qquad \text{or} \qquad \mathfrak{p}\mathcal{O}_L = \mathfrak{B}_1^5\mathfrak{B}_2^5\mathfrak{B}_3^5\mathfrak{B}_4^5\mathfrak{B}_5^5$$

depending on whether the inertia degree is 5 or 2 respectively. The distinction between the two shall be given with the inertia group in proposition 50.4.37. Choosing a different prime does change the resulting index: if we were to choose a different prime $\mathfrak{B}$ lying over $\mathfrak{p}$, then $D_{\mathfrak{B}}$ is conjugate to $D_{\sigma(\mathfrak{B})}$:

$$D_{\sigma(\mathfrak{B})} = \sigma D_{\mathfrak{B}} \sigma^{-1}$$

In particular:

$$\begin{aligned} \tau \in D_{\sigma(\mathfrak{B})} &\iff \tau\sigma\mathfrak{B} = \sigma\mathfrak{B} \\ &\iff \sigma^{-1}\tau\sigma\mathfrak{B} = \mathfrak{B} \\ &\iff \sigma^{-1}\tau\sigma \in D_{\mathfrak{B}} \\ &\iff \tau \in \sigma D_{\mathfrak{B}} \sigma^{-1} \end{aligned}$$

This should make sense, for by the proceeding theorem, the inertia and ramification of each prime lying ove $\mathfrak{p}$ in a galois extension is equal. At the end of this section, we shall show how to partially recover these results when working over non-Galois extensions. A key result when working with Galois extensions is that each prime has the same behavior in it's decomposition:

Proposition 50.4.28: Galois Extension, Inertia Degree And Ramification Index Independent

Let $\mathfrak{p} \subseteq K$ be a prime ideal, L/K a Galois extension, and

$$\mathfrak{p}\mathcal{O}_L = \mathfrak{B}_1^{e_i} \cdots \mathfrak{B}_r^{e_r}$$

Then

$$f_1 = \cdots = f_r = f \qquad e_1 = \cdots = e_r = e$$

***Proof*:**
Let $\mathfrak{B} = \mathfrak{B}_1$ and $\mathfrak{B}_i = \sigma_i(\mathfrak{B})$ (which goes through all primes by transitivity). Then the isomorphism $\sigma_i : \mathcal{O}_L \to \mathcal{O}_L$ induces an isomorphism:

$$\mathcal{O}_L/\mathfrak{B} \xrightarrow{\sim} \mathcal{O}_L/\mathfrak{B}_i \qquad a \bmod \mathfrak{B} \mapsto \sigma_i(a) \bmod \sigma_i(\mathfrak{B})$$

But then:

$$f_i = [\mathcal{O}_L/\sigma_i(\mathfrak{B}) : \mathcal{O}_K/\mathfrak{p}] = [\mathcal{O}_L/\mathfrak{B} : \mathcal{O}_K/\mathfrak{p}]$$

Next, since $\sigma_i(\mathfrak{p}\mathcal{O}_L) = \mathfrak{p}\mathcal{O}_L$ (since $\mathfrak{p} \subseteq K$), we get

$$\mathfrak{B}^n \big| \mathfrak{p}\mathcal{O} \quad \iff \quad \sigma_i(\mathfrak{B}^n) \big| \sigma_i(\mathfrak{p}\mathcal{O}) \quad \iff \quad (\sigma_i(\mathfrak{B}))^n \big| \mathfrak{p}\mathcal{O}$$

showing that $e_1 = \cdots = e_r$.

Hence, the prime decomposition of $\mathfrak{p}$ can be written as :

$$\mathfrak{p} = \left(\prod_{\substack{\sigma \in S \\ S \subseteq G/D_{\mathfrak{B}}}} \sigma(\mathfrak{B}) \right)^e$$

where $S \subseteq G/D_{\mathfrak{B}}$ represents a set of representatives of $G/D_{\mathfrak{B}}$ and $|S| = g_{L/K}(\mathfrak{p})$. This information is already very powerful, the following is a use-case of such information:

Example 50.4.29: Biquadratic Extension That Is Not Monogenic

Let $K = \mathbb{Q}(\sqrt{7}, \sqrt{13})$. Without computing $\mathcal{O}_K$, it can be shown that $\mathcal{O}_K$ is *not* monogenic, that is there does not exists $\alpha \in \mathcal{O}_K$ such that $1, \alpha, \alpha^2, \alpha^3$ generates $\mathcal{O}_K$, in other words $\mathcal{O}_K$ is not isomorphic to a quotient of $\mathbb{Z}[x]$. The key is to look at the behavior of factorizations of small primes (for they can only have so many rood $\mod p$)

What we shall do is consider the factorization of (3) in $\mathcal{O}_K$, which will be our setup for a contradiction. First, notice that K has 3 intermediary fields[a]:

$$\begin{tikzcd}
& K & \\
\mathbb{Q}(\sqrt{7}) \arrow[ur, "2"] & \mathbb{Q}(\sqrt{91}) \arrow[u, "2"] & \mathbb{Q}(\sqrt{13}) \arrow[ul, "2"'] \\
& \mathbb{Q} \arrow[ul, "2"] \arrow[u, "2"] \arrow[ur, "2"'] &
\end{tikzcd}$$

Since we know the ring of integers of quadratics, by direct computation, we see that (3) splits into two primes in each quadratic extension. For example, in $\mathbb{Q}(\sqrt{7})$, we have:

$$\mathcal{O}_{\mathbb{Q}(\sqrt{7})}/3\mathcal{O}_{\mathbb{Q}(\sqrt{7})} = \frac{\mathbb{F}_3[x]}{(x-1)(x+1)}$$

Hence:

$$(3) = (3, 1+\sqrt{7})(3, 1-\sqrt{7})$$

Doing a similar computation for the other two computation, we get:

$$(3) = (3, 1+\sqrt{13})(3, 1-\sqrt{13})$$
$$(3) = (3, 1+\sqrt{91})(3, 1-\sqrt{91})$$

Thus, over K, there must be at least 2 prime ideals, and since

$$g_{K/\mathbb{Q}}(3) = [G : D_{\mathfrak{B}}] \qquad [G : D_{\mathfrak{B}}] \bigm| |\mathrm{Gal}_{\mathbb{Q}}(K)| \qquad |\mathrm{Gal}_{\mathbb{Q}}(K)| = |(\mathbb{Z}/2\mathbb{Z})^2| = 4$$

for any prime ideal $\mathfrak{B}$ over (3), we get that $g_{K/\mathbb{Q}}(3) \big| 4$. I claim that $g_{K/\mathbb{Q}}(3) = 4$. For, if $g_{K/\mathbb{Q}}(3) = 2$, then for any of the two primes lying over (3) in $\mathcal{O}_K$, say $\mathfrak{B}$:

$$\{e\} \neq D_{\mathfrak{B}} = \{\sigma \in \mathrm{Gal}_{\mathbb{Q}}(K) \mid \sigma(\mathfrak{B}) = \mathfrak{B}\} \subsetneq \mathrm{Gal}_{\mathbb{Q}}(K)$$

Since $D_{\mathfrak{B}}$ non-trivial proper subgroup acting transitively on the two primes of K lying over (3), we have $|D_{\mathfrak{B}}| = 2$, implying there is one non-trivial automorphism $\rho \in \mathrm{Gal}_{\mathbb{Q}}(K)$ fixing the prime ideals (since there are two prime ideals, if $\rho(\mathfrak{B}) = \mathfrak{B}$, then $\rho(\mathfrak{B}') = \mathfrak{B}'$, or else ρ wouldn't be bijective), and two non-trivial automorphisms σ, τ not fixing the primes. Consider now $\sigma \in \mathrm{Gal}_{\mathbb{Q}}(K)$ and take the fixed field $K^{\langle\sigma\rangle}$, which by some Galois theory we see must be one of the 3 quadratic extensions found above. Then by the 3rd fundamental theorem of Galois theory:

$$G := \mathrm{Gal}_{\mathbb{Q}}(K^{\langle\sigma\rangle}) \cong \frac{\mathrm{Gal}_{\mathbb{Q}}(K)}{\mathrm{Gal}_{K^{\langle\sigma\rangle}}(K)}$$

Importantly, since $G \cong \mathbb{Z}/2\mathbb{Z}$ and $[\sigma] = [\mathrm{id}]$, the map $\varphi : \mathrm{Gal}_{\mathbb{Q}}(K) \to \mathrm{Gal}_{\mathbb{Q}}(K^{\langle\sigma\rangle})$ maps $\rho, \tau \mapsto \rho|_{K^{\langle\sigma\rangle}}$. Since ρ is an automorphism and fixes both prime ideals in K over (3) (say $\mathfrak{B}, \mathfrak{B}'$), and all prime ideal of $K^{\langle\sigma\rangle}$ over (3) lie below $\mathfrak{B}, \mathfrak{B}'$:

$$\rho|_{K^{\langle\sigma\rangle}}(\mathfrak{B} \cap K^{\langle\sigma\rangle}) = \rho|_{K^{\langle\sigma\rangle}}(\mathfrak{B}) \cap K^{\langle\sigma\rangle} = \mathfrak{B} \cap K^{\langle\sigma\rangle}$$
$$\rho|_{K^{\langle\sigma\rangle}}(\mathfrak{B}' \cap K^{\langle\sigma\rangle}) = \rho|_{K^{\langle\sigma\rangle}}(\mathfrak{B}') \cap K^{\langle\sigma\rangle} = \mathfrak{B}' \cap K^{\langle\sigma\rangle}$$

Hence, G fixes all primes; but we saw that all intermediate fields of K have two primes, thus this contradicts the transitivity of action of the Galois group on the prime ideals. Thus, it must be that $g_{K/\mathbb{Q}}(3) = 4$.

Now, for the sake of contradiction let's say

$$\mathcal{O}_K \cong \frac{\mathbb{Z}[x]}{m_\alpha(x)}$$

for some $\alpha \in \mathcal{O}_K$ and minimal polynomial $m_\alpha(x)$ of degree 4. Then

$$\mathcal{O}_K/3\mathcal{O}_K \cong \frac{\mathbb{F}_3[x]}{(m_\alpha(x) \bmod 3)} \tag{50.4}$$

By our previous calculations, we see that there must be 4 prime ideals in the right hand side. Decomposing $m_\alpha(x)$ mod 3 into irreducible, by the CRT we get that the right hand side is isomorphic to the direct sum of $\mathbb{F}_3[x]/(m_i(x) \bmod 3)$ for each irreducible $m_i(x)$ mod 3. Since each $m_i(x)$ mod 3 is irreducible, each direct summand is a field, and hence by some ring theory we see that the number of primes on the right hand side of equation 50.4 is equal to the number of direct summands, implying $m_\alpha(x)$ splits into linear factors giving us:

$$\frac{\mathbb{F}_3[x]}{(m_\alpha(x) \bmod 3)} \cong \bigoplus_i^4 \mathbb{F}_3$$

However, this is impossible: any degree 4 polynomial over $\mathbb{F}_3$ that breaks down into linear factors has the form:

$$(x-1)^\alpha (x+1)^\beta x^\gamma$$

where one of α, β, γ is 2 and the other two are 1. But then $\mathbb{F}_3[x]/((x-1)^\alpha (x+1)^\beta x^\gamma)$ is not isomorphic to $\bigoplus_i^4 \mathbb{F}_3$ (the quotient contains nilpotent elements), showing that our assumption that $\mathcal{O}_K$ is monogenic is not possible.

[a] this can be seen by taking the fixed fields $\sqrt{7}, \sqrt{13}$, and $\sqrt{91}$, corresponding to the subgroups of $\mathrm{Gal}_{\mathbb{Q}}(K) = \mathbb{Z}_2 \oplus \mathbb{Z}_2$

Our next goal is to break down extension L/K into intermediary extensions which will "capture" the inertial degree and ramification index, similarly to how any field extension F/k may be broken down into intermediary extensions into separable, inseparable, and transcendental extensions. In our case, we shall get the tower:

$$K \xrightarrow[1]{1} L_D \xrightarrow[f]{1} L_I \xrightarrow[1]{e} L$$

where the ramification index of each field extension is on top, and the inertia degree is on the bottom.

Proposition 50.4.30: Prime Ideals in Decomposition Field

Let L/K be a finite field extension, $\mathfrak{p} \subseteq \mathcal{O}_K$ a prime ideal and $\mathfrak{B} \subseteq \mathcal{O}_L$ lying over $\mathfrak{p}$. Let $\mathfrak{B}_Z = \mathfrak{B} \cap L_D$ be the prime ideal of L_D below $\mathfrak{B}$. If $\mathfrak{B}$ has inertia degree f and ramification index e, then:

1. $\mathfrak{B}_Z$ is nonsplit in L, that is $\mathfrak{B}$ is the only prime ideal of L above $\mathfrak{B}_D$
2. $\mathfrak{B}$ over L_D has a ramification index e and inertia index f
3. The ramification index and inertia index of $\mathfrak{B}_D$ over K are both 1

Visually, we have:

$$\begin{array}{c} \mathfrak{B}^e J \\ e \Big\uparrow f \\ \mathfrak{B}_D I \\ 1 \Big\uparrow 1 \\ \mathfrak{p} \end{array}$$

where I and J are the remainder of the factorization. Note that the factorization of each prime factor of J lying above $\mathfrak{p}$ has the same e and f since L/K is Galois, but L_D/K is not necessarily Galois, and so the factorization of I may be quite different (for ex. another prime $\mathfrak{B}'_D$ in L_D lying ove $\mathfrak{B}$ may have ramification and inertia degree)

***Proof*:**

1. Notice that $\mathrm{Gal}_{L_\mathfrak{B}}(L) = D_\mathfrak{B}$: the automorphisms of L that fix $L_\mathfrak{B}$ are by definition the elements of $D_\mathfrak{B}$. Hence, the prime ideals above $\mathfrak{B}_D$ are the $\sigma(\mathfrak{B})$ for $\sigma \in \mathrm{Gal}_{L_\mathfrak{B}}(L)$, but $\sigma(\mathfrak{B}) = \mathfrak{B}$ by definition. Hence, $\mathfrak{B}_D$ is nonsplit in L.

2-3. The fundamental identity gives us:

$$n = efr$$

where $n = |\mathrm{Gal}_K(L)| = |G|$ and $r = [G : D_\mathfrak{B}]$. hence,

$$|D_\mathfrak{B}| = [L : L_\mathfrak{B}] = ef \tag{50.5}$$

Now, let e', e'' be the ramification index of $\mathfrak{B}$ over $L_\mathfrak{B}$ and $\mathfrak{B}_D$ over K respectively. Then $\mathfrak{p}$ is:

$$\mathfrak{p}\mathcal{O}_{L_\mathfrak{B}} = \mathfrak{B}_D^{e''} I$$

for some I and
$$\mathfrak{B}_D \mathcal{O}_L = \mathfrak{B}^{e'}$$
in L. Then
$$\mathfrak{p}\mathcal{O}_L = \mathfrak{B}^{e''e'} J$$
for some J. By the tower property, $e = e'e''$ and $f = f'f''$. Then the fundamental identity for the decomposition of $\mathfrak{B}_D$ in L gives us that the degree of L/L_D is:
$$[L : L_{\mathfrak{B}}] = e'f'$$
in particular, by equation (50.5), we get,
$$e'f' = ef = e'e''f'f''$$
Hence $e' = e$, $f' = f$, and $e'' = f'' = 1$, as we sought to show

This gives us:
$$K \xrightarrow[1]{1} L_{\mathfrak{B}} \xrightarrow[f]{e} L$$
Next, let us see how we can separate out an extension splitting the ramification and the inertia degree into different groups/fields. Since $\sigma(\mathcal{O}_L) = \mathcal{O}_L$ and $\sigma(\mathfrak{B}) = \mathfrak{B}$ for every $\sigma \in D_{\mathfrak{B}}$, there is an induced automorphism on the residue field $\mathcal{O}_L/\mathfrak{B}$:
$$\begin{aligned} \overline{\sigma} : \mathcal{O}_L/\mathfrak{B} &\to \mathcal{O}_L/\mathfrak{B} \\ (a \bmod \mathfrak{B}) &\mapsto (\sigma(a) \bmod \mathfrak{B}) \end{aligned}$$
Letting $\kappa(\mathfrak{B}) = \mathcal{O}_L/\mathfrak{B}$ and $\kappa(\mathfrak{p}) = \mathcal{O}_K/\mathfrak{p}$, we have that
$$f = [\kappa(\mathfrak{B}) : \kappa(\mathfrak{p})]$$
Recall that in Dedekind-Kummer's theorem, when $\mathfrak{p}$ is relatively prime to the conductor, the inertia degree of $\mathfrak{B}$ is given by the degree of $\overline{m_i}(x)$ which within the proof is derived by $\left[R/(\overline{m_i(x)}) : \overline{\mathcal{O}}_K\right]$. This shows that the above map has information about the inertia degree. The following links this quantity to $D_{\mathfrak{B}}$. To understand the result, the reader may see that $\mathrm{Gal}_K(L)$ would naturally surjects onto $\mathrm{Gal}_{\kappa(\mathfrak{p})}(\kappa(\mathfrak{B}))$ via restriction if the restriction was well-defined. The domain of this map can be restricted to the decomposition group to make it well-defined:

Proposition 50.4.31: Decomposition and Inertia Group Relation

The extension $\kappa(\mathfrak{B})/\kappa(\mathfrak{p})$ is normal and admits a surjective homomorphism:
$$D_{\mathfrak{B}} \twoheadrightarrow \mathrm{Gal}_{\kappa(\mathfrak{p})}(\kappa(\mathfrak{B}))$$
where $\mathrm{Gal}_{\kappa(\mathfrak{p})}(\kappa(\mathfrak{B})) = \mathrm{Gal}_{\mathcal{O}_K/\mathfrak{p}}(\mathcal{O}_L/\mathfrak{B}) \cong \mathrm{Gal}_{\mathbb{F}_p}(\mathbb{F}_q)$.

***Proof*:**
As we are working over finite fields, we know that $\kappa(\mathfrak{B}) = \kappa(\mathfrak{p})(\overline{\theta})$ for some θ with minimal polynomial $\overline{g}(x)$ over $\kappa(\mathfrak{p})$ ($\overline{g}(x) \in \kappa(\mathfrak{p})[x]$)). Let $\theta \in \mathcal{O}_L$ be a representative of $\overline{\theta}$, and let $f(x)$ be its minimal polynomial over $\mathcal{O}_K$. Then by definition $\overline{g}(\overline{\theta}) \equiv 0 \bmod \mathfrak{B}$. Since $f(\theta) = 0$,

$\overline{f}(\overline{\theta}) \equiv 0 \bmod \mathfrak{B}$. Hence, taking $\overline{f}(x) \in \kappa(\mathfrak{p})[x]$, we have by minimality that:

$$\overline{g}(x) \big| \overline{f}(x)$$

Next, we know that finite field extensions are normal as we saw in section 19.7.2, however here is another way of seeing this worth pointing out: since L/K is normal, $f(x)$ splits over $\mathcal{O}_L$ into linear factors. Thus, $\overline{f}(x)$ splits into linear factors over $\kappa(\mathfrak{B})$, and similarly for $\overline{g}(x)$. But then, $\kappa(\mathfrak{B})/\kappa(\mathfrak{p})$ is a normal extension.

Now, consider:

$$\overline{\sigma} \in \mathrm{Gal}_{\kappa(\mathfrak{p})}(\kappa(\mathfrak{B})) = \mathrm{Gal}_{\kappa(\mathfrak{p})}\left(\kappa(\mathfrak{p})(\overline{\theta})\right)$$

By corollary 19.2.29, these homomorphisms are determined by where they send the root θ. Then $\overline{\sigma}(\overline{\theta})$ is a root of $\overline{g}(x)$ and hence of $\overline{f}(x)$, that is there exists a root θ' of $f(x)$ such that $\theta' \equiv \overline{\sigma}(\overline{\theta}) \bmod \mathfrak{B}$. θ' is a conjugate of θ, i.e. $\theta' = \sigma(\theta)$ for some $\sigma \in D_{\mathfrak{B}}$ (as the automorphism in $D_{\mathfrak{B}}$ fix $\mathfrak{B}$ and make the map well-defined). Since $\sigma(\theta) \equiv \overline{\sigma}(\overline{\theta}) \bmod \mathfrak{B}$, the automorphism σ is mapped by the homomorphism in question to $\overline{\sigma}$, proving surjectivity, as we sought to show.

Definition 50.4.32: Inertia Group And Field

The kernel $I_{\mathfrak{B}} \subseteq D_{\mathfrak{B}}$ of the homomorphism

$$D_{\mathfrak{B}} \to \mathrm{Gal}_{\kappa(\mathfrak{p})}(\kappa(\mathfrak{B}))$$

is called the *inertia group* of $\mathfrak{B}$ over K which is a normal subgroup of $D_{\mathfrak{B}}$ (as it is the kernel of a homomorphism). Equivalently:

$$I_{\mathfrak{B}} = \{\sigma \in \mathrm{Gal}_K(L) \mid \sigma(\alpha) \equiv \alpha \bmod \mathfrak{B}\}$$

The fixed field:

$$L_I = L^{I_{\mathfrak{B}}} = \{x \in L \mid \sigma(x) = x, \ \forall \sigma \in I_{\mathfrak{B}}\}$$

is called the *inertia field* of $\mathfrak{B}$ over K.

We thus get the following tower:

$$K \subseteq L_D \subseteq L_I \subseteq L$$

and short exact sequence:

$$1 \to I_{\mathfrak{B}} \to D_{\mathfrak{B}} \to \mathrm{Gal}_{\kappa(\mathfrak{p})}(\kappa(\mathfrak{B})) \to 1$$

Corollary 50.4.33: Isomorphism Of Two Galois Groups

Let $L/K/\mathbb{Q}$ be a finite galois extension, and let $\mathfrak{B}$ be a prime lying over $\mathfrak{p} \subseteq \mathcal{O}_K$. Then:

$$\mathrm{Gal}_{L_D}(L_I) \cong \mathrm{Gal}_{\kappa(\mathfrak{p})}(\kappa(\mathfrak{B}))$$

***Proof*:**
exercise.

Corollary 50.4.34: Quotient Group of Decomposition Group

Let L/K be a finite extension. Then the group $D_{\mathfrak{B}}/I_{\mathfrak{B}}$ is cyclic of order f (hence $[D_{\mathfrak{B}} : I_{\mathfrak{B}}] = f$), and the generator $[\varphi] \in D_{\mathfrak{B}}/I_{\mathfrak{B}}$ that generates the group (and hence $\mathrm{Gal}_{\kappa(\mathfrak{p})}(\kappa(\mathfrak{B})) \cong \mathbb{Z}/m\mathbb{Z}$) is called the *Frobenius Element*. The Frobenius element $\varphi \in D_{\mathfrak{B}}$ can be represented in the simple form

$$\varphi(x) = x^{\mathfrak{N}}(\mathfrak{p})$$

Proof:
This is an immediate consequence of the surjectivity of $D_{\mathfrak{B}}$ onto the Galois group of finite field extensions, which is always cyclic.

The order of $\kappa(\mathfrak{B})/\kappa(\mathfrak{p})$ is f, giving the order of the quotient. As the galois group of a finite field is cyclic, it is generated by a Frobenius element. The Frobenius element can be explicitly described, namely

$$\varphi(x) = x^{\mathfrak{N}(\mathfrak{p})} = x^f$$

giving us the rest of the results.

Notice that this means that Frobenius elements and Cyclotomic extensions naturally appear when studying galois extensions; this motivates the study of Cyclotomic extensions in section 50.5.

Corollary 50.4.35: Equivalent Characterization Of Inertia Group

L/K be a finite Galois extension, $I_{\mathfrak{B}} \leqslant D_{\mathfrak{B}}$. Then:

$$I_{\mathfrak{B}} = \{\sigma \in \mathrm{Gal}_K(L) \mid \sigma(\mathfrak{B}) = \mathfrak{B}, \text{ and } \forall \alpha \in \mathcal{O}_L, \sigma(\alpha) - \alpha \in \mathfrak{B}\}$$

Proof:
Let $\sigma \in I_{\mathfrak{B}}$ so that $\overline{\sigma\overline{\alpha}} = \overline{\alpha}$. Then $\sigma(\alpha) = \alpha + b$ for $b \in \mathfrak{B}$. But then:

$$\sigma(\alpha) - \alpha = b \in \mathfrak{B}$$

as we sought to show.

Corollary 50.4.36: Finding Element Of Galois Group

Let K be a number field, $f(x) \in \mathcal{O}_K[x]$ an irreducible polynomial over $\mathcal{O}_K$, L/K the splitting field extension of $f(x)$. Let $\mathfrak{p} \subseteq \mathcal{O}_K$ such that

$$f(x) \bmod \mathfrak{p} \equiv g_1(x) \cdots g_m(x)$$

for distinct irreducibles with degree $\deg(g_i(x)) = d_i$. Then in $\mathrm{Gal}_K(L) \lesssim S_n$, there exists an element α with cycle type

$$(d_1, d_2, ..., d_m)$$

***Proof*:**
Since L is the splitting field, let $\alpha_1, \alpha_2, ..., \alpha_m$ be the roots of f so that

$$f(\alpha_i) = 0$$

Let $\mathfrak{B}$ in L lie above $\mathfrak{p}$. Then:

$$f(\alpha_i) \bmod \mathfrak{p} \equiv 0$$

Since each each g_i are distinct, each α_i mod $\mathfrak{p}$ are disticnt. Hence, $D_{\mathfrak{B}}$ acts on the roots of f mod $\mathfrak{p}$. Now, taking $g \in D_{\mathfrak{B}}$ that maps to the generator of $\mathrm{Gal}_{\kappa(\mathfrak{p})}(\kappa(\mathfrak{B}))$, we shall get the desired cycle type, completing the proof.

Proposition 50.4.37: Inertia Group Properties

The extension L_I/L_D is normal, and

$$\mathrm{Gal}_{L_D}(L_I) \cong \mathrm{Gal}_{\kappa(\mathfrak{p})}(\kappa(\mathfrak{B}))$$

and $\mathrm{Gal}_{L_I}(L) = I_{\mathfrak{B}}$. If the residue field extension $\kappa(\mathfrak{B})/\kappa(\mathfrak{p})$ is separable, then:

$$|I_{\mathfrak{B}}| = [L : L_I] = e \qquad [D_{\mathfrak{B}} : I_{\mathfrak{B}}] = [L_I : L_D] = f$$

and we can find the prime ideals $\mathfrak{B}_I$ of L_I bellow $\mathfrak{B}$:

1. The ramification index of $\mathfrak{B}$ over $\mathfrak{B}_I$ is e and the inertia degree is 1
2. The ramification index of $\mathfrak{B}_I$ over $\mathfrak{B}_D$ is 1, and the inertia degree if f

Hence:

$$\mathfrak{B}_Z \mathcal{O}_{L_I} = \mathfrak{B}_T I \qquad \mathfrak{B}_T \mathcal{O}_K = \mathfrak{B}^e J$$

for some appropriate ideals I, J.

***Proof*:**
The fact that:

$$|I_{\mathfrak{B}}| = [L : L_I] = e \qquad [D_{\mathfrak{B}} : I_{\mathfrak{B}}] = [L_I : L_D] = f$$

follow from $|D_{\mathfrak{B}}| = [L : L_D] = ef$ and Galois theory (namely $[L : L_I] = |\mathrm{Gal}_{L_I}(L)| = |I_{\mathfrak{B}}|$), hence we need to show $(1)-(2)$. Using the fundamental identity, it all follows from $\kappa(\mathfrak{B}_I) = \kappa(\mathfrak{B})$. Since the inertia group $I_{\mathfrak{B}}$ of $\mathfrak{B}$ over K is also the inertia group of $\mathfrak{B}$ over L_I, it follows from proposition 50.4.31 applied to L/L_I that $\mathrm{Gal}_{\kappa(\mathfrak{B}_I)}(\kappa(\mathfrak{B})) = 1$, thus

$$\kappa(\mathfrak{B}_I) = \kappa(\mathfrak{B})$$

but then the rest follows.

To sum up the results in a diagram

$$K \xrightarrow[1]{1} L_D \xrightarrow[f]{1} L_I \xrightarrow[1]{e} L$$

with a corresponding [contravariant] diagram of groups:

$$G \leftarrow D_{\mathfrak{B}} \leftarrow I_{\mathfrak{B}} \leftarrow \{e\}$$

where the ramification index of each field extension is on top, and the inertia degree is on the bottom. If the residue field extension $\kappa(\mathfrak{B})/\kappa(\mathfrak{p})$ is separable, then:

$$I_{\mathfrak{B}} = 1 \qquad \iff \qquad L_I = L \qquad \iff \qquad \mathfrak{p} \text{ is unramified in } L$$

and in this case the Galois group $\mathrm{Gal}_{\kappa(\mathfrak{p})}(\kappa(\mathfrak{B})) \cong D_{\mathfrak{B}}$, so the galois group of residue field extension can be viewed as a subgroup of $\mathrm{Gal}_K(L)$. This means that the inertia group can be thought of as the "ramification group". In fact, the generalizations of the inertia group (giving more nuanced information on the ramificaion) are called the *higher order ramification group* (which we shall cover soon). The reason the inertia group is not called the 1st ramification group is because the corresponding fixed field only effects the inertia of a prime; ultimately this is due to the contravariant relation the galois correspondence gives, hence we must make a choice on naming convention depending if fields or groups are prioritised. In this case, fields were prioritised.

From the previous proposition, we may think that L_D and L_I are the *maximal* fields with these properties with respect to $\mathfrak{B}$, and this is indeed the case, making them "natural" with respect to these properties:

Proposition 50.4.38: Intermediary Fields Are Maximal

Let L/K be a finite extension of number fields, $\mathfrak{B}$ live over $\mathfrak{p}$. Then:

1. $L_{D_{\mathfrak{B}}}$ is the smallest intermediary field such that $\mathfrak{B}$ is the only prime of $\mathcal{O}_L$ lying over $\mathfrak{B}' = \mathfrak{B} \cap \mathcal{O}_{L_{D_{\mathfrak{B}}}}$
2. $L_{D_{\mathfrak{B}}}$ is the largest intermediate such that $e(\mathfrak{B}_D/\mathfrak{p}) = f(\mathfrak{B}_I/\mathfrak{p}) = 1$
3. $L_{I_{\mathfrak{B}}}$ is the smallest intermediary field such that $\mathfrak{B}$ is totally ramified over $\mathfrak{B}_I$ (that is $e(\mathfrak{B}/\mathfrak{B}_I) = [L : L_I]$)
4. $L_{I_{\mathfrak{B}}}$ is the largest intermediary field such that $e(\mathfrak{B}_I/\mathfrak{p}) = 1$

***Proof*:**
First, L_D and L_I have these properties as we've in proposition 50.4.30 and proposition 50.4.37, hence we prove they are maximal/minimal.

Let $K' = L^H$ for some $H \leqslant \mathrm{Gal}_K(L)$ where only $\mathfrak{B} \subseteq L$ is lying over $\mathfrak{p}' \subseteq K'$. Since H acts transitively on the primes of L over K' but there is only one prime $H \subseteq D_{\mathfrak{B}}$, so $L_D \subseteq K'$, giving (1).

Next, assume $e_{\mathfrak{p}'/\mathfrak{p}} = f_{\mathfrak{p}'/\mathfrak{p}} = 1$. Consider the following diagram:

$$\begin{tikzcd}
L & \\
& L_I K' \arrow[ul, "e'"'] \\
L_I \arrow[uu, "e"] & \\
& L_D K' \arrow[uu, "f'"'] \\
L_D \arrow[uu, "f"] \arrow[ur] & \\
& K' \arrow[uu, "g'"'] \\
K \arrow[uu, "g"] \arrow[ur] &
\end{tikzcd}$$

Then by multiplicativity of towers, $e = e'$ and $f = f'$. Then by the above diagram, L_D and $L_D K'$ voth ahve the same index in L. Since one contains the other, this means they must be equal, hence

$$K' \subseteq L_D$$

giving (2). A similar arguments works for (3) and (4): if $e_{\mathfrak{p}'/\mathfrak{p}} = 1$, then $e = e'$, so $L_I = L_I K'$, so $K' \subseteq L_I$, and if $\mathfrak{B}$ is totally ramified over $\mathfrak{p}'$, then $[L : K'] = e'$, then by the above diagramwe get $K' = L_I K'$, so $L_I \subseteq K'$, as we sought to show.

Ramification

An importance consequence of this is that we may consider primes that are completely ramified/totally unramified in their galois closure. we first require the following result:

Theorem 50.4.39: Composition Field And Primes

Let L/K and M/K be finite extensions of number fields, and let $\mathfrak{p}$ be a prime of K. Then if $\mathfrak{p}$ is unramified in both L and M, then $\mathfrak{p}$ is unramified in LM. If $\mathfrak{p}$ splits completely in L and M, then $\mathfrak{p}$ splits completely in LM

***Proof*:**
Let $\mathfrak{p}$ be unramified in L and M, and take $\mathfrak{B}$ to be any prime in LM lying over $\mathfrak{p}$. We want to show $e(\mathfrak{B}/\mathfrak{p}) = 1$. Let F be the normal closure of K containing LM, and let $\mathfrak{A}$ be any prime in F lying over $\mathfrak{B}$, namely it lies over $\mathfrak{p}$. Let $I_{\mathfrak{A}/\mathfrak{p}}$ be the inertial group corresponding to the field F_I. Then we see that F_I contains both L and M since the primes $\mathfrak{A} \cap L$ and $\mathfrak{A} \cap M$ are necessarily unramified over $\mathfrak{p}$. Hence, F_I contains LM, implying that $\mathfrak{A} \cap LM$ is unramified over

$\mathfrak{p}$.

The argument for splitting follows exactly the same pattern, replacing $I_{\mathfrak{B}}$ with $D_{\mathfrak{B}}$.

Corollary 50.4.40: Primes In Normal Closure

Let L/K be a finite extension of number fields and $\mathfrak{p}$ a prime in K. Then:

1. If $\mathfrak{p}$ is unramified in L, then it is unramified in the normal closure of M of L over K
2. If $\mathfrak{p}$ splits completely in L, then it splits completely in the normal closure of M of L over K

***Proof*:**
If $\mathfrak{p}$ is unramified in L, then it is unramified in every $\sigma_i(L)$ where i runs over all the automorphism. But since $M = L\sigma_1(L)\cdots\sigma_n(L)$, we use induction on theorem 50.4.39 to get our desired result (similarly for splitting completely).

Example 50.4.41: More Decomposition Of Primes
Let $K = \mathbb{Q}(i, \sqrt{2}, \sqrt{5})$, which is abelian of degree 8. Take the prime $5 \in \mathbb{Z}$. Then it splits into tow primes in $\mathbb{Q}(i)$, stays inert in $\mathbb{Q}(\sqrt{2})$, and becomes a square in $\mathbb{Q}(\sqrt{5})$. Hence in L 5 must have ramification index at least 2, and split into at least 2 primes. Since the inertia field must be a field of degree 4 over $\mathbb{Q}$ in which 5 is unramified, the only choice is $\mathbb{Q}(i, \sqrt{2})$. Now, $(2+i)$ and $(2-i)$ remain prime in $\mathbb{Q}(\sqrt{i}, \sqrt{2})$, and become squares in L, hence 5 must have exactly 2 primes and ramification index 2.

Since normal subgroups over characteristic 0 fields are galois, it is meaningful to ask the behavior of normal subgroups of the galois subgroups:

Proposition 50.4.42: Decomposition Subgroup Is Normal

Suppose $D_{\mathfrak{B}} \trianglelefteq \mathrm{Gal}_K(L)$. Then $\mathfrak{p}\mathcal{O}_L$ splits into r distinct primes in L_D. If $I_{\mathfrak{B}} \trianglelefteq G$, then each prime remains as primes in L_I (i.e. they are inert). Furthermore, each become an eth power in L.

In general, we can only hope for:

$$\mathfrak{B}_D\mathcal{O}_{L_I} = \mathfrak{B}_I I \qquad \mathfrak{B}_I\mathcal{O}_K = \mathfrak{B}^e J$$

for some ideals, while in the normal case of D and I, we get:

$$\begin{aligned}\mathfrak{p}\mathcal{O}_{L_D} &= \mathfrak{B}_{L_1}\cdots\mathfrak{B}_{L_e}\\ \mathfrak{p}\mathcal{O}_{L_I} &= \mathfrak{B}_{L_1}\cdots\mathfrak{B}_{L_e}\\ \mathfrak{p}\mathcal{O}_L &= \mathfrak{B}_1^e\cdots\mathfrak{B}_g^e\end{aligned}$$

which also shows better why the groups $D_{\mathfrak{B}}$ and $I_{\mathfrak{B}}$ are called the “decomposition” and “inertia” groups respectively.

***Proof*:**
This is taking advantage of the fact that $L^{D_{\mathfrak{B}}}/K$ would now be a Galois extension since $D_{\mathfrak{B}} \trianglelefteq \mathrm{Gal}_K(L)$. Then we know that

$$f(\mathfrak{B}_D) = e(\mathfrak{B}_D) = 1$$

and that (since $L^{D_{\mathfrak{B}}}$ is Galois), each other prime must also have the same ramification and inertia degree. Thus, $\mathfrak{p}$ splits into exactly g distinct primes in $L^{D_{\mathfrak{B}}}$. Then by a simple counting argument, we see that each $\mathfrak{B}_D$ must lie under a unique $\mathfrak{B}_I$. It may be that some of these ramify. However, if $I_{\mathfrak{B}} \trianglelefteq \mathrm{Gal}_K(L)$, then we have that each prime must have ramification index 1, showing they are all inert. Finally, a counting argument gives that each $\mathfrak{B}_I$ becomes an eth power in L

Corollary 50.4.43: Normal Subgroup And Completely Split Prime

Let $D_{\mathfrak{B}} \trianglelefteq \mathrm{Gal}_K(L)$. Then $\mathfrak{p}$ splits completely in K' if and only if $K' \subseteq L_D$

***Proof*:**
exercise.

In the non-Galois case L/K, take $M/L/K$ where M/K is the Galois closure of L. Then M/L is Galois. Let $G = \mathrm{Gal}_K(M)$ and $H = \mathrm{Gal}_L(M)$ so that $H \leqslant G$. Take the following tower of primes:

$$\begin{tikzcd}
M \arrow[r] & \mathfrak{A} \\
L \arrow[u] \arrow[r] & \mathfrak{B} \arrow[u] \\
K \arrow[u] \arrow[r] & \mathfrak{p} \arrow[u]
\end{tikzcd}$$

Then by definition:

$$D_{\mathfrak{A}/\mathfrak{p}} \cap H = D_{\mathfrak{A}/\mathfrak{B}}$$
$$I_{\mathfrak{A}/\mathfrak{p}} \cap H = I_{\mathfrak{A}/\mathfrak{B}}$$

Proposition 50.4.44: Prime Decomposition Over Finite Extension

Let

$$\mathfrak{p}_M = \prod_{1}^{g_{M/K}} \mathfrak{A}_i^{e_i}$$

Let $\mathfrak{O}_1, \mathfrak{O}_2, ..., \mathfrak{O}_{g_{L/K}}$ be the orbits of H acting on the primes $\{\mathfrak{A}_1, \mathfrak{A}_2, ..., \mathfrak{A}_{g_{M/K}}\}$. Then:

$$\mathfrak{p}_L = \prod_{i}^{g_{L/K}} \mathfrak{B}_i^{e_{\mathfrak{B}_i/\mathfrak{p}}}$$

where each $\mathfrak{B}_i$ comes form an $\mathfrak{O}_i$. Furthermore:

$$e_{\mathfrak{B}_i/\mathfrak{p}} = \frac{|I_{\mathfrak{A}_i/\mathfrak{p}}|}{|\text{stab}_H(\mathfrak{A}_i) \cap I_{\mathfrak{A}_1/\mathfrak{p}}|} \qquad f_{\mathfrak{B}_i/\mathfrak{p}} = \frac{|D_{\mathfrak{A}_i/\mathfrak{B}_i}|}{|e_{\mathfrak{A}_i/\mathfrak{p}}|} = \frac{|\text{stab}_H(\mathfrak{A}_i)|}{|e_{\mathfrak{A}_i/\mathfrak{p}}|}$$

Proof:
exercise

It is also enlightening to have a second perspective here: for subgroup $U, V \leqslant G$, we may consider the equivalence class $\sigma \sim \sigma' \iff \sigma' = u\sigma v$ giving us the double-coset[26]:

$$U\sigma V = \{u\sigma v \mid u \in U, v \in V\}$$

This partition will give us the similar information in the non-Galois case. Let $N/L/K$ be two separable extensions where N/K is Galois with Galois group G. Let $H = \text{Gal}_L(N) \leqslant \text{Gal}_K(N)$. Let $\mathfrak{p}$ be a prime ideal of K and $P_\mathfrak{p}$ the set of all prime ideal of L above $\mathfrak{p}$. Then if $\mathfrak{B}$ is a prime ideal of N above $\mathfrak{p}$, then we get a well-defined bijection:

$$H \setminus G/D_\mathfrak{B} \to P_\mathfrak{p} \qquad H\sigma D_\mathfrak{B} \mapsto \sigma\mathfrak{B} \cap L$$

Exercise 50.4.1

1. Let L/K be a galois extension of Number fields, $G = \text{Gal}_K(L)$, $\mathfrak{B}$ lie over $\mathfrak{p}$.
   a) If $\mathfrak{B}$ is inert, then G Is cyclic
   b) If $\mathfrak{p}$ is totally ramified in each intermediary field but not totally ramified in L, then there does not exist non-trivial proper intermediary field, hence G is cyclic (of prime order)
   c) Simiarly, Suppose every intermediate field contains a unique prime lying over $\mathfrak{p}$, but L does not. Show the same as the previous
   d) Let $\mathfrak{p}$ be unramified in every proper non-trivial intermediate field, but ramified in L. Show that G has unique smallest subgroup H and that $H \trianglelefteq G$. Use this to show that $|G| = p^k$ for some k and prime p, that $|H| = p^l$ for some l, and $H \leqslant Z(G)$.
   e) Similarly, show the same for when $\mathfrak{p}$ splits compeltely in every proper non-trivial intermediate field, but not in L.

[26]warning: double cosets need not all have the same size!

f) Let $\mathfrak{p}$ be inert in every proper non-trivial intermediate field but not in L. Prove that G is a cyclic where $|G| = p^k$.

2. (the non-normal case for Frobenius element by taking normal closure, p.78 Marcus)

50.4.3 Detecting Ramification

(Part of the problem here is that there is some local field theory going on here, so I will need to separate the local field theory for [39])

(also, this strongly motivate having a separate book for number theory. This is imo the best place in my series to have this, but at the same it is too much for this textbook. So once EYNTKA gets its first version so some theorems are set in stone, it will make sense to have ll this detail)

50.4.45 AKLB setup For generality, let A be a Dedekind domain, K the field of fraction of A, L a finite separable extension of K, and B the integral closure of A in L.

Furthermore, let κ be the residue field of A and λ be the residue field for L. Assume λ/κ and B/A are separable extensions.

Definition 50.4.46: Trace Dual of Fractional Ideal

Take AKLB and let $\mathfrak{P} \subseteq B$ be a fractional ideal. Then the *trace dual* of $\mathfrak{P}$ is:

$$^*\mathfrak{P} = \{x \in L \mid \operatorname{tr}(x\mathfrak{P}) \subseteq A\}$$

By proposition 50.1.32, $^*\mathfrak{P}$ is a finitely generated B-module, and hence by definition a fractional ideal. Note that as the trace-pairing is non-degenerate:

$$^*\mathfrak{P} \xrightarrow{\cong} \operatorname{Hom}_A(\mathfrak{P}, A) \qquad x \mapsto (y \mapsto \operatorname{tr}(xy))$$

which works even when $\mathfrak{P} = (1)_B$, that is:

$$^*B \cong \operatorname{Hom}_A(B, A)$$

This particular trace dual shall be very relevant in detecting ramification:

Definition 50.4.47: Different

Assume AKLB. Then the *different* is the ideal

$$\mathfrak{C}_{B/A} = \{x \in L \mid \forall y \in A,\ \operatorname{tr}(xy) \subseteq A\}$$

is called the *Dedekind's complementary module* or *inverse different*. It's inverse:

$$\mathfrak{D}_{B/A} := \mathfrak{C}_{B/A}^{-1}$$

is called the *Different* of B/A

Since $\mathfrak{C}_{B/A} \supseteq \mathcal{O}$, $\mathfrak{D}_{B/A} \subseteq \mathcal{O}$ is an integral ideal of L. Naturally, we will often denote the different as $\mathfrak{D}_{L/K}$ and $\mathfrak{C}_{B/A}$ as $\mathfrak{C}_{L/K}$ given the rings B an A are evident from context.

Proposition 50.4.48: Tower Law For Different

Assume $AKLB$ and let M/L be a finite separable extension. Then

1. If $M/L/K$, then $\mathfrak{D}_{M/k} = \mathfrak{D}_{M/L}\mathfrak{D}_{L/K}$
2. If $S \subseteq \mathcal{o}$ is a multiplicative subset, then $\mathfrak{D}_{S^{-1}\mathcal{O}/S^{-1}\mathcal{o}} = S^{-1}\mathfrak{D}_{\mathcal{O}/\mathcal{o}}$
3. If $\mathfrak{B}/\mathfrak{p}$ is a prime ideal of $\mathcal{O}$ with respect to $\mathcal{o}$, and $\mathcal{O}_{\mathfrak{B}}/\mathcal{o}_{\mathfrak{p}}$ is the associated completion, then $\mathfrak{D}_{\mathcal{O}/\mathcal{o}}\mathcal{O}_{\mathfrak{B}} = \mathfrak{D}_{\mathcal{O}_{\mathfrak{B}}/\mathcal{o}_{\mathfrak{p}}}$

***Proof*:**

1. If we show that
$$\mathfrak{C}_{\mathcal{O}_M/\mathcal{O}_K} = \mathfrak{C}_{\mathcal{O}_M/\mathcal{O}_L}\mathfrak{C}_{\mathcal{O}_L/\mathcal{O}_K}$$
we are done. For $\supseteq$:
$$\begin{aligned}\mathrm{tr}_{M/K}(\mathfrak{C}_{\mathcal{O}_M/\mathcal{O}_L}\mathcal{O}_{L/\mathcal{O}_A}(\mathcal{O}_M)) &= \mathrm{tr}_{L/K}\mathrm{tr}_{M/L}(\mathfrak{C}_{\mathcal{O}_M/\mathcal{O}_L}\mathfrak{C}_{\mathcal{O}_L/\mathcal{O}_K}(\mathcal{O}_M)) \\ &= \mathrm{tr}_{L/K}(\mathfrak{C}_{\mathcal{O}_L/\mathcal{O}_K}\mathrm{tr}_{M/L}(\mathfrak{C}_{\mathcal{O}_M/\mathcal{O}_L)}\mathcal{O}_M) \\ &\subseteq \mathcal{O}_K\end{aligned}$$
For the $\subseteq$, since $\mathcal{O}_L\mathcal{O}_M = \mathcal{O}_M$, we get:
$$\mathrm{tr}_{M/K}(\mathfrak{C}_{\mathcal{O}_M/\mathcal{O}_K}\mathcal{O}_M) = \mathrm{tr}_{L/K}(\mathcal{O}_L\mathrm{tr}_{M/L}(\mathfrak{C}_{\mathcal{O}_M/\mathcal{O}_K}\mathcal{O}_M)) \subseteq \mathcal{O}_K$$
Hence $\mathrm{tr}_{M/L}(\mathfrak{C}_{\mathcal{O}_M/\mathcal{O}_K}\mathcal{O}_M) \subseteq \mathfrak{C}_{\mathcal{O}_L/\mathcal{O}_K}$, thus:
$$\mathrm{tr}_{M/L}(\mathfrak{C}^{-1}_{\mathcal{O}_L/\mathcal{O}_K}\mathcal{O}_{\mathcal{O}_M/C\mathcal{O}_K}\mathcal{O}_M) = \mathfrak{C}^{-1}_{\mathcal{O}_L/\mathcal{O}_K}\mathrm{tr}_{M/L}(\mathfrak{C}_{\mathcal{O}_M/\mathcal{O}_K}C) \subseteq \mathcal{O}_L$$
Thus, we get $\mathfrak{C}^{-1}_{\mathcal{O}_L/\mathcal{O}_K}\mathfrak{C}_{\mathcal{O}_M/\mathcal{O}_K} \subseteq \mathfrak{C}_{\mathcal{O}_M/\mathcal{O}}$, hence
$$\mathfrak{C}_{\mathcal{O}_M/\mathcal{O}_K} \subseteq \mathfrak{C}_{\mathcal{O}_M/\mathcal{O}_K}\mathfrak{C}_{\mathcal{O}_L/\mathcal{O}_K}$$
completing the proof.
2. follows from definition pushing
3. (Neukirch p.195)

Example 50.4.49: Different

1. Let $K = \mathbb{Q}(\sqrt{5})$ so that $\mathcal{O}_K = \mathbb{Z}\left[\frac{1+\sqrt{5}}{2}\right]$. Then:
$$\mathfrak{C}_{K/\mathbb{Q}} = \sqrt{5}^{-1}\mathcal{O}_K \implies \mathfrak{d} = \sqrt{5}\mathcal{O}_K$$

2. Let $K = \mathbb{Q}(\sqrt{3})$. So that $\mathcal{O}_K = \mathbb{Z}[\sqrt{3}]$. Then:

$$\mathfrak{C}_{K/\mathbb{Q}} = \langle \frac{1}{2}, \frac{1}{\sqrt{3}} \rangle_{\mathbb{Z}} = \frac{1}{2\sqrt{3}} \mathcal{O}_K \implies \mathfrak{d}_{K/\mathbb{Q}} = 2\sqrt{3}\mathcal{O}_K$$

Theorem 50.4.50: Norm Of Different

$$\mathfrak{N}(\mathfrak{d}_{K/\mathbb{Q}}) = |\text{disc}(\mathcal{O}_K)|$$

Proof:
good exercise!

(This next proposition gives a special case of the different and the origin of it's name)

Proposition 50.4.51: Different In Monogenic Extension

Let $\mathcal{O} = o[\alpha]$. Then the different is equal to:

$$\mathfrak{D}_{L/K} = (\delta_{L/K}(\alpha))$$

where

$$\delta_{L/K}(\alpha) = \begin{cases} f'(\alpha) & L = K(\alpha) \\ 0 & L \neq K(\alpha) \end{cases}$$

Proof:
Neukirch p.197

Theorem 50.4.52: Generation Of Different

The different $\mathfrak{D}_{L/K}$ is the ideal generated by all the different of the elements $\delta_{L/K}(\alpha)$ for all $\alpha \in \mathcal{O}$

Proof:
Neukirch p.198

Theorem 50.4.53: Detecting Ramification Using Different

A prime ideal $\mathfrak{P}$ of L is ramified over K if and only if

$$\mathfrak{B} | \mathfrak{d}_{L/K}$$

Proof:
Neukirch p. 199

(relation between different and discriminant)

Definition 50.4.54: Relative Discriminant

The *discriminant* $\mathfrak{D}_{\mathcal{O}/\mathfrak{o}}$ is the ideal of $\mathfrak{o}$ generated by the discriminants $d(\alpha_1, ..., \alpha_n)$ for all bases $\alpha_1, \alpha_2, ..., \alpha_n$ of L/K contained in $\mathcal{O}$:

$$\mathfrak{D}_{\mathcal{O}/\mathfrak{o}} = \{(d(\alpha_1, \alpha_2, ..., \alpha_n)) \mid \alpha_1, \alpha_2, ..., \alpha_n \text{ basis }, \alpha_i \in \mathcal{O}\}$$

In the case where $K = \mathbb{Q}$ and $L/\mathbb{Q}$ is a number field, then $\mathfrak{d} = (d_L)$ is the principal ideal of the usual discriminant.

Theorem 50.4.55: Relating Different And Relative Discriminant

$$\mathfrak{D}_{L/K} = \mathfrak{N}_{L/K}(\mathfrak{d}_{L/K})$$

***Proof*:**
Neukirch p.201

Corollary 50.4.56: Tower For Relative Discriminant

Let $M/L/K$. Then:

$$\mathfrak{D}_{M/K} = \mathfrak{D}_{L/K}^{[M:L]}\mathfrak{N}_{L/K}(\mathfrak{D}_{M/L})$$

***Proof*:**
Neukirch p.202

Corollary 50.4.57: Discriminant And Ramified Primes

Let $\mathfrak{p} \subseteq K$ be a prime. Then $\mathfrak{p}$ is ramified in L if and only if

$$\mathfrak{p} | \mathfrak{D}_{L/K}$$

In particular, the extension L/K is unramified if and only if

$$\mathfrak{D}_{L/K} = (1)$$

***Proof*:**
Immediate consequence

Proposition 50.4.58: Coprime Discriminants And Difference

Let K_1, K_2 be galois over $\mathbb{Q}$ with coprime discriminants and $K = K_1K_2$. Then:

$$\mathfrak{d}_{K/\mathbb{Q}} = \mathfrak{d}_{K_1/\mathbb{Q}}\mathfrak{d}_{K_2/\mathbb{Q}}$$

***Proof*:**
Using the tower

(Neukirch goes on to prove some important results which are really cool! For example, if K is an algebraic number field and S a finite set of primes of K, then there exists only finitely many extensions L/K of degree n which are unramified outside of S!)

Exercise 50.4.2

1. Show that L_D is the maximal field in which the ramification and inertial degree is 1, and similarly for L_I.
2. Let K be a number fields. Show that if disc(K) is square free, then K has no intermediary number field (this is a generlization of exercise re:HERE(galPoly))

Wild and Tame Ramification

(see this! here)

(This section is essentially here for the proof of Kronecker-Weber Theorem. May choose to move it down to that section).

We end off by exploring some more about the inertia group, namely we shall like at *wild* and *tame* inertia.

Definition 50.4.59: Wild Inertia Group

Let L/K be an extension of number fields, $I_{\mathfrak{B}} \leqslant D_{\mathfrak{B}}$ be the inertia group. Then $W_{\mathfrak{B}} \trianglelefteq I_{\mathfrak{B}}$ is defined to be:

$$W_{\mathfrak{B}} := \left\{ g \in I_{\mathfrak{B}} \;\middle|\; \forall \overline{\alpha} \in \mathcal{O}_L/\mathfrak{B}^2,\ g(\overline{\alpha}) = \overline{\alpha} \right\}$$

the wild inertia group can be thought of as a parallel to "local coordinates". In particular if we have an extension $K/\mathbb{Q}$ of number fields, then we have an extension $K/\mathbb{Q}_p$ then $W_{\mathfrak{B}}$ is the unique sylow p-subgroup of the inertia group. If you know some algebraic geometry, an intuitive reason to see why the quotient $\mathfrak{B}^2$ shows up is since the tangent space of a variety/scheme is given by $\mathfrak{m}/\mathfrak{m}^2$.

Lemma 50.4.60: Wild Inertia And Uniformizer

Let $\pi \in \mathcal{O}_L$ be the uniformizer for Q (i.e. $Q \| \pi$). Let $\sigma \in I_Q$. Then:

$$\sigma \in W_Q \qquad \Leftrightarrow \qquad \sigma(\pi) - \pi \in Q^2$$

that is, $\sigma(\pi) = \pi \bmod Q^2$

***Proof*:**
The $\Rightarrow$ direction is by definition. For $\Leftarrow$, note that we have calculated before that:

$$\left|(\mathcal{O}_L/Q^2)^{\times}\right| = \left|(\mathcal{O}_L/Q)^{\times}\right| \left|Q\mathcal{O}_L/Q^2\right| = (q-1)q$$

where $q = |\mathbb{F}_{p^r}|$ for appropriate p^r. Hence, there exists $H \leqslant (\mathcal{O}_L/Q^2)^{\times}$ such that $|H| = q - 1$

and $H \xrightarrow{\sim} \mathbb{F}_Q^\times$. Hence

$$(\mathcal{O}_L/Q^2)^\times \cong H \oplus (1+\pi\mathcal{O}_L)_{Q^2} \bmod Q^2$$

Hence $\sigma \in I_G$, then σ acts trivially on $\mathbb{F}_Q^\times$, and hence acts trivially on $H>$ Thus $\sigma \in W_Q$ if and only if σ fixes $(1+\pi\mathcal{O}_L)_{Q^2}$ if and only if $(\pi\mathcal{O}_L)_{Q^2}$ if and only if σ fixes π mod Q^2

Example 50.4.61: Example from Class for wild inertia

$K = \mathbb{Q}(\sqrt[8]{1})$, $i^4 = 1$, $\left(\frac{1+i}{\sqrt{2}}\right)^2 = i$. $[K : \mathbb{Q}] = |\mathbb{Z}/8\mathbb{Z}^\times| = 4$. and $(\mathbb{Z}/8\mathbb{Z})^\times \cong (\mathbb{Z}/2\mathbb{Z}) \oplus \mathbb{Z}/2\mathbb{Z}$. So K is biquadratic. Note that

$$K = \mathbb{Q}\left(\frac{1+i}{\sqrt{2}}\right) = \mathbb{Q}(\alpha) = \mathbb{Q}(i, \sqrt{2})$$

(diagram of the 3 quadratic extensions). Notice that 2 is the only prime that ramifies. In $\mathbb{Q}(\sqrt{d})$ where $d = -1, 2, 2$, so 2 ramifies

$$(e, f, g) = (2, 1, 1)$$

It may not be immediately clear, but in K, $2\mathcal{O}_K = \mathfrak{B}^4$, i.e. 2 totally ramifies (we have not seen cases where e increases when it's coprime). This cannot happen with unramified fields. What is $\mathfrak{B}$? Explicitly, we know that

$$\mathfrak{B} = (1-\alpha)$$

which we can directly see because $\mathfrak{B}^2 = (1 - 2\alpha + \alpha^2) = (1 + i - 2\alpha) = (1 + i)(1 - (1 - i)\alpha)$, and we see that $(1 - (1 - i)\alpha)$ is a unit.

Let $\pi = 1 - \alpha$, which is a uniformizer of $\mathfrak{B}$. Recall that we looked at $\sigma(\pi) - \pi$ for $\sigma]in\mathrm{Gal}_{\mathbb{Q}}(K)$. Computing it's valuation at $\mathfrak{B}$

$$v_{\mathfrak{B}}(\sigma(\pi) - \pi)$$

we can do this by independence of valuation on the prime. First

$$\sigma(\alpha) = \alpha^3 \implies (1-\alpha^3) - (1-\alpha) = v_{\mathfrak{B}}(\alpha^2 - 1) = 1 + v_{\mathfrak{B}}((\alpha-1)+2) = 2$$

Next, we do

$$\sigma(\alpha) = \alpha^7 = \alpha^{-1} \implies v_{\mathfrak{B}}(\alpha^7 - \alpha) = v_{\mathfrak{B}}(\alpha^{-1} - \alpha) = v_{\mathfrak{B}}(\alpha^2 - 1) = 2$$

Finally, we have

$$\sigma(\alpha) = \alpha^5 \implies v_{\mathfrak{B}}(\alpha^5 - \alpha) = v_{\mathfrak{B}}(\alpha^4 - 1) = v_{\mathfrak{B}}(\alpha - 1) + v_{\mathfrak{B}}(\alpha + 1) + v_{\mathfrak{B}}(\alpha^2 + 1) = 1 + 1 + 2 = 4$$

Hence, we have a chain:

$$\langle 1 \rangle \subseteq \langle 1, 5 \rangle \subseteq (\mathbb{Z}/\gamma\mathbb{Z})^\times = \mathrm{Gal}_{\mathbb{Q}}(K)$$

It may not jump out, but notice that 2 divides the difference (linked to a theorem called *Hasse-Arf* Theorem).

Theorem 50.4.62: Abelian Extension Wild Map

Let L/K Galois and abelian, $G = \mathrm{Gal}_K(L)$, $P \subseteq \mathcal{O}_K$ a prime ideal of $\mathcal{O}_K$, Q a prime lying over P W_Q the wild inertia group, and π a uniformizer so that $Q \| \pi$. Then:

$$\varphi : I_Q/W_Q \hookrightarrow \mathbb{F}_Q^\times \qquad [\sigma] \mapsto \frac{\sigma(\pi)}{\pi} \bmod Q$$

Assume G is abelian. Then $\mathrm{im}\varphi \subseteq \mathbb{F}_P^\times$.

***Proof*:**
Look at $\varphi(\sigma) = \varphi(\tau^{-1}\sigma\tau)$ (because abelian), so:

$$\frac{\tau^{-1}\sigma\tau(\pi)}{\pi} \bmod Q$$

and write it as:

$$\frac{\tau - 1(\sigma(\tau(\pi)))}{\tau(\pi)} \bmod Q$$

Now, it doesn't matter what uniformizer we pick. If $\tau \in D_Q$, then we can mod by Q before or after, so

$$\tau^{-1}\varphi(\sigma)$$

What kind of element can be written as τ^{-1}? We know the decomposition group surjects onto $\mathrm{Gal}_{\mathbb{F}_p}(\mathbb{F}_Q)$, we can pick τ^{-1} to be the Frobenius operator, that is

$$\varphi(\sigma) = \varphi(\sigma)^{\mathrm{Nm}(P)} \implies \varphi(\sigma) \in \mathbb{F}_p$$

Theorem 50.4.63: Wild Inertia Group Is Sylow Subgroup

Take $W_Q \leqslant \mathrm{Gal}_{\mathbb{Q}}(L)$ where Q is a prime lying over p. Then W_Q the unique p-sylow subgroup, hence $W_Q \trianglelefteq I_Q$.

***Proof*:**
check notes for now

Exercise 50.4.3

1. (Define higher-order ramification. Show that the inertia group is solvable, and when the residue fields are finite then the decomposition group is solvable).

50.4.4 Frobenius Automorphism

This is a short section to single out the Frobenius automorphisms from corollary 50.4.34 due to its importance in Class Field Theory (CFT); namely the Frobenius automorphism is central to relating the structure of abelian galois groups to the structure of the class group. The full exposition is beyond the scope of this textbook, but the reader can still get an appreciation behind why the Frobenius automorphism is so important. For a full exposition, see [39].

Assume $\mathfrak{B}$ over $\mathfrak{p}$ is unramified in $L/K/\mathbb{Q}$, so that $I_{\mathfrak{B}}$ is trivial (which by proposition 50.4.12 and proposition 50.4.16 can be shown to be the case for all put finitely many primes). Then $D_{\mathfrak{B}} \cong \mathrm{Gal}_{\kappa(\mathfrak{p})}(\kappa(\mathfrak{B})) \cong (\mathbb{Z}/n\mathbb{Z})$. By finite field theory, we know that there exists the *Frobenius generator* sending $x \mapsto x^{\mathfrak{N}(\mathfrak{B})}$ since $\mathfrak{N}(\mathfrak{B}) = f$ and the order of this group is f. Thus, for each prime $\mathfrak{B}/\mathfrak{p}$, have a corresponding automorphism in the decomposition group, and hence the Galois group:

Definition 50.4.64: Frobenius Automorphism of a Prime

Let $\sigma \in D_{\mathfrak{B}}$. Then σ is called the *Frobenius* if $\sigma(\alpha) \equiv \alpha^f \mod \mathfrak{B}$ where f is the inertia degree of $\mathfrak{B}$ over $\mathfrak{p}$. The collection of element $\sigma \in D_{\mathfrak{B}}$ with this property is denoted $\mathrm{Frob}_{\mathfrak{B}/\mathfrak{p}}$ so that is $\sigma \in \mathrm{Frob}_{\mathfrak{B}/\mathfrak{p}}$:

$$\sigma(\alpha) \equiv \alpha^f \mod \mathfrak{B}$$

Note that if $\mathfrak{B}$ were ramified, then the Frobenius automorphism is defined up to the coset $I_{\mathfrak{B}}$: $\sigma + I_{\mathfrak{B}}$. If another $\mathfrak{B}' = \sigma(\mathfrak{B})$[27] is considered, the Frobenius element differ conjugation:

$$\mathrm{Frob}_{\sigma(\mathfrak{B}/\mathfrak{p})} = \sigma \mathrm{Frob}_{\mathfrak{B}/\mathfrak{p}} \sigma^{-1}$$

If $D_{\mathfrak{B}}$ is abelian, then the Frobenius element is *only* determined by the prime $\mathfrak{p}$ as

$$\mathrm{Frob}_{\sigma(\mathfrak{B}/\mathfrak{p})} = \sigma \mathrm{Frob}_{\mathfrak{B}/\mathfrak{p}} \sigma^{-1} = \mathrm{Frob}_{\mathfrak{B}/\mathfrak{p}}$$

Thus, we may denote it as $\mathrm{Frob}_{\mathfrak{p}}$. In this case, we have:

$$\sigma(\alpha) = \alpha^{\mathfrak{N}(\mathfrak{p})} \mod \mathfrak{p}\mathcal{O}_L$$

because $\mathfrak{p}\mathcal{O}_L$ is the product of all primes $\mathfrak{B}$ over $\mathfrak{p}$. Another immediate observation is that if $\mathfrak{p}$ spits completely, then $f = 1$, and the Frobenius element for $\mathfrak{p}$ shall be the identity, and hence the Frobenius element captures behaviour in the case where a prime doesn't completely split.

The important realization is that if $\mathrm{Gal}_k(F)$ is abelian, there exist $\sigma \in \mathrm{Gal}_k(F)$ that are associated to primes $\mathfrak{p}$ in the sense that the structure of $\sigma \mod \mathfrak{B}/\mathfrak{p}$ is simply the power map (i.e. cyclic)! If each element $\sigma \in \mathrm{Gal}_k(F)$ can have such an association, then that means that primes have a strong link to the structure of an abelian galois group! Even better, the decomposition of primes is regulated by the class group. The culminating result of 20th century number theory is that any abelian galois group is isomorphic to a class group! Thus, abelian galois theory and number theory have a strong correspondence to each other.

When $\mathrm{Gal}_k(F)$ is non-abelian, the situation become more tenuous. Robert Langland is credited for finding the right framework to extend this correspondence through the use of *representation theory*. This is the origins of the famous *Langland's Conjecture*.

50.4.5 Example: Cyclotomic Fields

As we saw in example 50.1.31, the discriminant of $\mathbb{Z}[\zeta_{p^n}]$ is a prime power of p. We now generalize this result for arbitrary n, and find how p, and will find the discriminant is a power of p. Thus, p is the only prime that ramifies in K, and we shall find how p ramifies:

[27] by transitivity and definition of the decomposition group, any $\mathfrak{B}'$ is of this form

Lemma 50.4.65: Cyclotomic Fields Lemma

Let $n = p^k$ be a prime power, ζ_{p^k} a root of unity, and $\lambda = 1 - \zeta_{p^k}$. Then the principal ideal $(\lambda) \subseteq \mathcal{O}$ in the ring of integers of $\mathbb{Q}(\zeta_{p^k})$ is a prime ideal of degree 1, and:

$$p\mathcal{O} = (\lambda)^d$$

where $d = \varphi(n) = [\mathbb{Q}(\zeta_{p^k}) : \mathbb{Q}]$. Furthermore, the basis $1, \zeta_{p^k}, ..., \zeta_{p^k}^{d-1}$ of $\mathbb{Q}(\zeta_{p^k})/\mathbb{Q}$ has discriminant:

$$d_{\mathbb{Q}(\zeta_{p^k})/\mathbb{Q}}(1, \zeta_{p^k}, ..., \zeta_{p^k}^{d-1}) = \pm p^s$$

where $s = p^{k-1}(kp - k - 1)$

Note that in order for the above discriminant to be the discriminant of the ring of integers, we would have to insure the ring of integers has the above basis up to change of coordinates, which shall be the result of the next theorem.

***Proof*:**
Recall that the minimal polynomial of ζ_{p^k} over $\mathbb{Q}$ is:

$$m_{\zeta_{p^k},\mathbb{Q}}(x) = \frac{x^{l^n} - 1}{x^{l^{n-1}} - 1} = x^{l^{n-1}(l-1)} + \cdots + x^{l^{n-1}} + 1$$

Letting $x = 1$, we get

$$n = p^k = \prod_{r \in (\mathbb{Z}/n\mathbb{Z})^\times} (1 - \zeta_{p^k}^r)$$

Next, we know that

$$(1 - \zeta_{p^k}^r) = (1 - \zeta_{p^k})(1 + \zeta_{p^k} + \cdots + \zeta_{p^k}^{r-1}) = (1 - \zeta_{p^k})\epsilon_r$$

If r' is an integer such that $rr' = 1 \bmod n$, then

$$\frac{1 - \zeta_{p^k}}{1 - \zeta_{p^k}^r} = \frac{1 - (\zeta_{p^k}^r)^{r'}}{1 - \zeta_{p^k}^r} = 1 + \zeta_{p^k}^r + \cdots + (\zeta_{p^k}^r)^{r'-1}$$

is integral as well, namely ϵ_r is a unit. Thus,

$$p = \epsilon(1 - \zeta_{p^k})^{\varphi(n)}$$

with unit $\epsilon = \prod_r \epsilon_r$. Thus:

$$p\mathcal{O} = (\lambda)^{\varphi(n)}$$

Since $[\mathbb{Q}(\zeta_{p^k}) : \mathbb{Q}] = \varphi(n)$, the fundamental identity ($\sum e_i f_i$) shows that (λ) is a prime ideal of degree 1.

Now, take $\zeta_{n,1}, ..., \zeta_{n,d}$ be the conjugates of ζ_{p^k} (i.e. powers of ζ_{p^k} that are coprime with the order of ζ_{p^k}). Then the cyclotomic polynomial:

$$\Phi_{p^k}(x) = \prod_{i=1}^{d} (x - \zeta_{n,i})$$

and

$$\pm\text{disc}(1, \zeta_{p^k}, ..., \zeta_{p^k}^{d-1}) = \prod_{i\neq j}(\zeta_{n,i} - \zeta_{n,j}) = \prod_{i=1}^{d} \Phi'_{p^k}(\zeta_{n,i}) = \text{Nm}_{\mathbb{Q}(\zeta_{p^k})/\mathbb{Q}}(\Phi'_{p^k}(\zeta_{p^k}))$$

Differentiating the equation:

$$(x^{p^{k-1}} - 1)\Phi_{p^k}(x) = x^{p^k} - 1$$

and substituting ζ_{p^k} for x gives:

$$(\zeta_{p^k}^{p^{k-1}} - 1)\Phi'_{p^k}(\zeta_{p^k}) = (\eta - 1)\Phi'_{p^k}(\zeta_{p^k}) = n\zeta_{p^k}^{-1}$$

But we know $\text{Nm}_{\mathbb{Q}(\eta)/\mathbb{Q})}(\epsilon - 1) = \pm p$, thus

$$\text{Nm}_{\mathbb{Q}(\zeta_{p^k})/\mathbb{Q}}(\eta - 1) = \text{Nm}_{Q(\eta)/\mathbb{Q}}(\eta - 1)^{p^{k-1}} = \pm p^{p^{k-1}}$$

since $\zeta_{p^k}^{-1}$ has norm ± 1, we get:

$$d(1, \zeta_{p^k}, .., \zeta_{p^k}^{d-1}) = \pm\text{Nm}_{\mathbb{Q}(\zeta_{p^k})/\mathbb{Q}}\left(\Phi'_{p^k}(\zeta_{p^k})\right) = \pm p^{k^{p^{k-1}(k-1)-p^{k-1}}} = \pm p^s$$

where s is the exponent

For completeness of this section, here is a quick proof of the ring of integers of cyclotomic fields:

Proposition 50.4.66: Ring Of Integers Of Cyclotomic Fields

Let $K = \mathbb{Q}(\zeta_n)$ be a cyclotomic field. Then

$$\mathcal{O}_{K_n} \cong \mathbb{Z}[\zeta_n]$$

With discriminant:

$$\text{disc}(\mathcal{O}_{K_n}) = (-1)^{\varphi(n)/2}\frac{n^{\varphi(n)}}{\prod_{p\mid n} p^{\frac{\varphi(n)}{p-1}}}$$

***Proof*:**
Let's start with n being a prime power p^k. Since $\text{disc}(1, \zeta, ..., \zeta^{d-1}) = \pm p^s$, we have:

$$p^s\mathcal{O}_K \subseteq \mathbb{Z}[\zeta_n] \subseteq \mathcal{O}$$

Putting $\lambda = 1 - \zeta_n$, by lemma 50.4.65 we have $\mathcal{O}_K/\lambda\mathcal{O}_K \cong \mathbb{Z}/p\mathbb{Z}$, thus $\mathcal{O}_K = \mathbb{Z} + \lambda\mathcal{O}_K$, hence we *a fortiori* get

$$\lambda\mathcal{O}_K + \mathbb{Z}[\zeta] = \mathcal{O}$$

Multiplying by λ, and substituting $\lambda\mathcal{O}_K = \lambda^2\mathcal{O}_K + \lambda\mathbb{Z}[\zeta]$, we get

$$\lambda^2\mathcal{O}_K + \mathbb{Z}[\zeta] = \mathcal{O}$$

Iterating this procedure, we get:

$$\lambda^t\mathcal{O}_K + \mathbb{Z}[\zeta] = \mathcal{O} \qquad \forall t \geq 1$$

When $t = s\varphi(p^k)$, then given $p\mathcal{O}_K = \lambda^{\varphi(p^k)}\mathcal{O}_K$, we get

$$\mathcal{O} = \lambda^t\mathcal{O}_K + \mathbb{Z}[\zeta] = p^s\mathcal{O}_K + \mathbb{Z}[\zeta] = \mathbb{Z}[\zeta]$$

For the general case, let $n = p_1^{k_1}\cdots p_r^{k_r}$. Then $\zeta_i = \zeta^{n/p_i^{k_i}}$ is a primitive $p_i^{k_k}$-th root of unity, and hence:

$$Q(\zeta) = \mathbb{Q}(\zeta_1)\cdots\mathbb{Q}(\zeta_r)$$

and $\mathbb{Q}(\zeta_1)\cdots\mathbb{Q}(\zeta_{i-1})\cap\mathbb{Q}(\zeta_i) = \mathbb{Q}$. Then for each $i \in \{1, ..., r\}$, the elements $1, \zeta_i, ..., \zeta_i^{d_i-1}$, where $d_i = \varphi(p_i^{k_i})$ form an integral basis of $\mathbb{Q}(\zeta_i)/\mathbb{Q}$. Since the discriminants $d(1, \zeta_i, ..., \zeta_i^{d_i-1}) = \pm p_i^{s_i}$, are relatively prime, we get from successive uses of (Proposition 2.11 in Neukirch, I should add it to my notes!!) we get that

$$\zeta_1^{j_1}\cdots\zeta_r^{j_r}$$

where $j_i \in \{0, ..., (d_i - 1)\}$ form an integral basis of $\mathbb{Q}(\zeta)/\mathbb{Q}$. But then each of these elements is a power of ζ. Thus, every $\alpha \in \mathcal{O}$ may be written as a polynomial $\alpha = f(\zeta)$ with coefficients in $\mathbb{Z}$. Since ζ has degree $\varphi(n)$ over $\mathbb{Q}$, the degree of $f(\zeta)$ may be reduced to $\varphi(n) - 1$. But then, we may obtains:

$$\alpha = a_1 + a_1\zeta + \cdots + a_{\varphi(n)-1}\zeta^{\varphi(n)-1}$$

Thus, $1, \zeta, ..., \zeta^{\varphi(n)-1}$ is an integral basis, as we sought to show.

by theorem 50.4.53 commented this theorem out! , we know that the primes that ramify are those that divide $\mathrm{disc}(\mathcal{O}_{K_n})$, hence by the above the primes that ramify are the factors of n. The following gives exactly how they factor:

Proposition 50.4.67: Prime Decomposition In Cyclotomic Field

Enumerates all the primes $p_1, p_2, ...$, then let $n = \prod_n p_n^{n_k}$ be a prime factorization of $n \in \mathbb{Z}$. Let f_k be the smallest positive integer such that

$$p_k^{f_k} \equiv 1 \bmod (n/p_k^{n_k})$$

Then in $\mathbb{Q}(\zeta_n)$, we have:

$$p_k = (\mathfrak{p}_1\cdots\mathfrak{p}_r)^{\varphi(p_k^{n_k})}$$

where $\mathfrak{p}_1, ..., \mathfrak{p}_r$ are distinct prime ideals all of degree f_k.

Note that if $p_k \nmid n$, then $n_k = 0$, so the condition becomes

$$p_k^{f_k} \equiv 1 \bmod n$$

and $\varphi(p_k^{n_k}) = 1$, giving us:

$$p_k = \mathfrak{p}_1\cdots\mathfrak{p}_r$$

***Proof*:**
Since $\mathcal{O}_K = \mathbb{Z}[\zeta]$, we have that the conductor of $\mathbb{Z}[\zeta]$ is 1, and hence we may apply theorem 50.4.7 to any prime number p, in particular every p decompose into prime ideals in the exact same way that $\Phi_n(x)$ (the minimal polynomial of ζ_n) factors into irreducible mod p. Thus, we are reduced to showing that:

$$\Phi_n(x) \equiv (p_1(x)\cdots p_r(x))^{\varphi(p^{\nu_p})} \bmod p$$

where $p_1(x), p_2(x), ..., p_r(x)$ are distinct irreducible polynomials over $\mathbb{Z}/p\mathbb{Z}$ of degree f_p. Write n as $p^{\nu_p}m$ $(p \nmid m)$. Let ξ_i and η_j vary over the primitive roots of unity over order m and p^{ν_p} respectively. Then the product $\xi_i\eta_j$ vary over the nth root of unity, namely we have the decomposition:

$$\Phi_n(x) = \prod_{i,j}(x - \xi_i\eta_j)$$

Since $x^{p^{\nu_p}} - 1 \equiv (x-1)^{p^{\nu_p}} \bmod p$, we have that $\eta_j \equiv 1 \bmod \mathfrak{p}$ for any prime ideal where $\mathfrak{p}\big|p$. This, we get:

$$\Phi_n(x) \equiv \prod_i (x - \xi_i)^{\varphi(p^{\nu_p})} = \Phi_n(x)^{\varphi(p^{\nu_p})} \bmod \mathfrak{p}$$

Hence, we have when modding by p:

$$\Phi_n(x) \equiv \Phi_n(x)^{\varphi(p^{\nu_p})} \bmod p$$

Now, notice that f_p is the smallest positive integer such that $p^{f_p} \equiv 1 \bmod m$, hence the congruence reduces to th case where $p \nmid n$, and then $\varphi(p^{\nu_p}) = \varphi(1) = 1$.

Now, since the characteristic of $\mathcal{O}_K/\mathfrak{p}$ (i.e. p) does not divide n, the polynomial $x^n - 1$ and nx^{n-1} have no common roots in $\mathcal{O}_K/\mathfrak{p}$ by the derivative test. Thus, $x^n - 1 \bmod p$ has *no multiple roots*. Thus, we get that $\mathcal{O}_K \to \mathcal{O}_K/\mathfrak{p}$ maps the group of nth roots of unity μ_n bijectivity onto the nth roots of unity of $\mathcal{O}_K/\mathfrak{p}$. In particular ,the primitive nth roots of unity modulo $\mathfrak{p}$ ($\zeta \bmod \mathfrak{p}$) remain primitive nth roots of unity. Since the extension field of $\mathbb{F}_p = \mathbb{Z}/p\mathbb{Z}$ containing it is the field $\mathbb{F}_{p^{f_p}}$, since the multiplicative group $\mathbb{F}^*_{p^{f_p}}$ is cyclic of order $p^{f_p} - 1$. Thus, $\mathbb{F}_{p^{f_p}}$ is the field decomposition of the reduced cyclotomic polynomial:

$$\overline{\Phi}_n(x) = \Phi_n(x) \bmod p$$

Since it is a divisor of $X^n - 1 \bmod p$, this polynomial has no multiple roots, and if

$$\overline{\Phi}_n(x) = \overline{p}_1(x) \cdots \overline{p}_r(x)$$

i it's factorization into irreducible over $\mathbb{F}_p$, then every $\overline{p}_i(x)$ is the minimal polynomial of a primitive nth root of unity $\overline{\xi}\ in \mathbb{F}^*_{p^{f_p}}$. It's degree is thus f_p, as we sought to show.

Corollary 50.4.68: Special Case Of Factorization

A prime number p is ramified in $\mathbb{Q}(\zeta_n)$ iff

$$n \equiv 0 \bmod p$$

except in the case where $p = 2 = (4, n)$. A prime number $p \neq 2$ is totally split in $\mathbb{Q}(\zeta)$ iff

$$p \equiv 1 \bmod n$$

***Proof*:**
direct consequence of last proposition

Corollary 50.4.69: Subfield Of Cyclotomic Field

Let $K \subseteq \mathbb{Q}(\zeta_n) = L$ with corresponding subgroup $H \leqslant \mathrm{Gal}_K(L) \cong (\mathbb{Z}/n\mathbb{Z})^\times$ considered as a subgroup of $(\mathbb{Z}/n\mathbb{Z})^\times$. For prime p where $p \nmid n$, let f be the least positive integer such that $\overline{p^f} \in H$. Then for prime $\mathfrak{p}$ lying above p in K:

$$f = f_{\mathfrak{p}/p}$$

***Proof*:**
Notice hat $f_{\mathfrak{p}/p}$ is the order of the Frobenius automorphism.

Overall, the ring of integers of a cyclotomic field is monogenic, the only primes that ramify are those that divide n (where n is the order of the root of unity) since the divisors of the discriminant are completely dependent on n, the discriminant is a product of the powers of primes of n, and we know exactly how the each element $n \in \mathbb{Z}$ factors in cyclotomic fields!

50.4.6 Application: Quadratic Extensions and Quadratic Reciprocity

We now take a moment to solve a classical number theory problems with the developed tools. Let $p \in \mathbb{Z}$ be a prime number. In section 10.3, it was shown that p can be written as the sum of two squares if and only if $p \equiv 1 \bmod 4$. Let us now ask another question that may shed further light on the nature of primes: the square-root of primes is always an irrational number. However, when modding by $n \in \mathbb{N}$, it is possible that there is an answer. By the Chinese Remainder Theorem, it suffices to check when $n = q$ is a prime. Thus:

what primes p have the property that there exists an n such that $n^2 \equiv p \bmod q$

Example 50.4.70: Concrete Example
Let $p = 3$. Then:

mod	1^2	2^2	3^2	4^2	5^2	6^2
2	1					
3	1	1				
5	1	4	4	1		
7	1	4	2	2	4	1
11	1	4	9	5	***3***	***3***

Note the table is mirrored as $(-n)^2 = n^2$. Hence, for a reason that is yet to be explained in this book, the prime 3 is somehow tied to 11 when looking for square-roots of 3. The reader can verify that 13, 23, 37, and so forth also have a square-root of 3.

The reader should play around with different pairs (p, q) and flip them around to consider (q, p), there shall appear some symmetry that at first is not obvious.

This relation between primes and square-root in modular arithmetics was first documented to be discovered by Euler, however it took a generation before a complete proof by Gauss appeared. As it turns out, the condition translates into a condition on rings of integers of quadratic extensions!

Definition 50.4.71: Quadratic Residue

Let $n \in \mathbb{Z}$. Then if there exists an $m \in \mathbb{Z}$ such that $m^2 \equiv n \mod p$, then n is called a *quadratic residue* mod p.

First, to as whether p has a square-root $\mod q$ is is equivalent to solving $x^2 - p \equiv 0 \mod q$. Let $K = \mathbb{Q}(\sqrt{p})$. Then we we may consider $\mathcal{O}_K/q\mathcal{O}_K$; this ring is isomorphic to $(\mathbb{Z}/q\mathbb{Z})^n$ for some $n \in \mathbb{N}_{>0}$ Note that having a root in $\mathcal{O}_K$ implies having a root in $\mathcal{O}_K/p\mathcal{O}_K$ for any p (as shown in example 11.3.9, the converse does not hold even if an equation has a solution mod every prime). This shall in fact be the key property to detecting whether p has a square-root mod q. This question in a slightly more general framework where $K = \mathbb{Q}(\sqrt{m})$ where m is square-free. To start off, we have found the integral basis of the ring of integers of quadratic extension in 50.1.6[4], namely we have integral basis

$$\begin{aligned} &\{1, \sqrt{m}\} && m \not\equiv 1 \mod 4 \\ &\left\{1, \frac{1+\sqrt{m}}{2}\right\} && m \equiv 1 \mod 4 \end{aligned}$$

and $\operatorname{disc}(\mathcal{O}_K) = 4m$ in the first case and $\operatorname{disc}(\mathcal{O}_K) = m$ in the second. Now, take $(p) \subseteq \mathbb{Z}$ a prime ideal. Then we know that either p can split as:

$$p\mathcal{O}_K = \mathfrak{B} \qquad p\mathcal{O}_K = \mathfrak{B}^2 \qquad p\mathcal{O}_K = \mathfrak{B}_1\mathfrak{B}_2$$

and that only m and possibly 2 ramifies.

Theorem 50.4.72: Primes In Quadratic Extensions

Let $p \subseteq \mathbb{Z}$ be a prime ideal and $K = \mathbb{Q}(\sqrt{m})$ a quadratic extension where m is square-free. Then:

1. If $p \big| m$:
$$p\mathcal{O}_K = (p, \sqrt{m})^2$$
2. if m is odd, then 2 splits as:
$$2\mathcal{O}_K = \begin{cases} (2, 1+\sqrt{m})^2 & m \equiv 3 \mod 4 \\ \left(2, \frac{1+\sqrt{m}}{2}\right)\left(2, \frac{1-\sqrt{m}}{2}\right) & m \equiv 1 \mod 8 \\ (2) & m \equiv 5 \mod 8 \end{cases}$$

   and if p is odd and $p \nmid m$:
$$p\mathcal{O}_K = \begin{cases} (p, n+\sqrt{m})(p, n-\sqrt{m}) & m = n^2 \mod p \\ (p) & m \neq n^2 \mod p \end{cases}$$

***Proof*:**

1. Assume $p|m$. Then as it divides the discriminant, it certainly ramifies. More concretely, note this implies $m \equiv 0 \bmod p$ so $\sqrt{m} \equiv 0 \bmod p$, and $(p, \sqrt{m})$ is certainly prime by corollary 50.2.19. With this, it can be verified by hand that $(p, \sqrt{m})^2 = (p^2, p\sqrt{m}, m)$ and that (since $p|m$):
$$(p, \sqrt{m})^2 \subseteq (p)_K$$
2. let us first deal with the single prime $p = 2$ where $p \nmid m$ so that m is odd.
   (a) If $m \equiv 4 \bmod 4$, the discriminant is $4m$, hence 2 ramifies. Then by direct computation it can be verified that:
   $$(2, 1 + \sqrt{m})^2 = (2)_L$$
   giving us the splitting prime.
   (b) Assume $m \equiv 1 \pmod 8$. Then $m \equiv 1 \bmod 4$, so $\mathcal{O}_K = \mathbb{Z}\left[\frac{1+\sqrt{m}}{2}\right] = \mathbb{Z}[\omega]$. Then ω satisfies $\omega^2 - \omega + \frac{1-m}{4} = 0$. Then as $m \equiv 1 \pmod 8$, we have $\frac{1-m}{4} \equiv 0 \pmod 2$, so
   $$\omega^2 \equiv \omega \pmod 2$$
   The two prime ideals above 2 are generated by the two roots of $x^2 - x \equiv 0 \pmod 2$, which are $x \equiv 0$ and $x \equiv 1 \pmod 2$. Thus, the two prime ideals are $(2, \omega)$ and $(2, \omega - 1)$ (the reader can proof these are distinct primes)
   (c) Assuming $m \equiv 5 \bmod 8$, a similar process shows that 2 is inert.

   Now let p be any odd prime and $p \nmid m$. Note that as $p \nmid m$, $p \nmid \operatorname{disc}(\mathcal{O}_K) \in \{4m, m\}$, hence p will not ramify. Thus by corollary 50.4.8 The behavior of p in $\mathcal{O}_K$ depends on whether $f(x)$ factors over $\mathbb{F}_p$.

   (a) Assume $m \equiv n^2 \bmod p$ so that $x^2 - m$ has two roots in $\mathbb{F}_p$. Then $x^2 - m \equiv (x-n)(x+n) \pmod p$ In this case, we can show that p splits as:
   $$p\mathcal{O}_K = \mathfrak{p}_1\mathfrak{p}_2$$
   where $\mathfrak{p}_1 = (p, \sqrt{m} - n)$ and $\mathfrak{p}_2 = (p, \sqrt{m} + n)$. Indeed, $\mathcal{O}_K/\mathfrak{p}_1 \cong \mathbb{F}_p$ since $\sqrt{m} \equiv n \pmod{\mathfrak{p}_1}$ (Similarly for $\mathfrak{p}_2$). Then it is not too hard to notice they are distinct; notice that $\mathfrak{p}_1 + \mathfrak{p}_2 = (p, 2n) = \mathcal{O}_K$ (when $p \neq 2$)
   (b) Assume $m \not\equiv n^2 \bmod p$. Then $x^2 - m$ is irreducible over $\mathbb{F}_p$, so $\mathcal{O}_K/p\mathcal{O}_K \cong \mathbb{F}_{p^2}$ is a field, and so by the fundamental identity (p) must remain innert.
   This means $p\mathcal{O}_K$ is a prime ideal, so p remains inert.

Coming back to the question of the existence of quadratic reciprocity, the following notation shall be useful:

Definition 50.4.73: Legendre Symbol

Let p be an odd prime in $\mathbb{Z}$. Then for $n \in \mathbb{Z}$, $p \nmid n$, the *Legendre symbol* is defined to be:

$$\left(\frac{n}{p}\right) = \begin{cases} 1 & \exists m,\ m^2 \equiv n \bmod p \\ -1 & \text{otherwise} \end{cases}$$

The intuition behind the notation is that if n is the square mod p, it is "divisible" in mod p, motivating the $\left(\frac{n}{p}\right)$ notation.

Theorem 50.4.74: Quadratic Reciprocity

Let p, q be two distinct odd prime numbers. Then:

$$\left(\frac{q}{p}\right)\left(\frac{p}{q}\right) = (-1)^{\frac{q-1}{2}\frac{p-1}{2}}$$

In other words, if:

$$p^* = \begin{cases} 1 & p \equiv 1 \bmod 4 \\ -p & p \equiv -1 \bmod 4 \end{cases}$$

Then q is a quadratic residue modulo p if and only if p' is a quadratic residue modulo p

Quadratic reciprocity can also be interpreted as:

$$\left(\frac{2}{p}\right) = \begin{cases} 1 & p \equiv \pm 1 \bmod 8 \\ -1 & p \pm 3 \bmod 8 \end{cases}$$

and for odd $p \neq q$:

$$\left(\frac{p}{q}\right) = \begin{cases} \left(\frac{p}{q}\right) & p \text{ or} q \equiv 1 \bmod 4 \\ -\left(\frac{p}{q}\right) & p \equiv q \equiv 3 \bmod 4 \end{cases}$$

***Proof*:**
Let $q \neq p$ be odd primes. We want to show that

$$n^2 \equiv q \bmod p \qquad \text{if and only if} \qquad m^2 \equiv p' \bmod q$$

for some numbers $n, m \in \mathbb{Z}$. By theorem 50.4.72 it suffices to show hat q splits if and only if $m^2 \equiv p' \bmod q$. Let's first say that q splits, so we have $\mathfrak{B}_1, \mathfrak{B}_2$ lying above q, and $(q)_L \subsetneq \mathfrak{B}_1, \mathfrak{B}_2$. Since $p' \equiv 1 \bmod 4$, we know the number ring of L is:

$$\mathbb{Z}\left[\frac{1+\sqrt{p'}}{2}\right]$$

Hence, we have that there is an element

$$a + b\sqrt{p'} \in \mathfrak{B} - (q) \qquad a, b \in \frac{1}{2}\mathbb{Z} \tag{50.6}$$

and since q is odd, we may without loss of generality choose a, b to be in $\mathbb{Z}$ by multiplying by 2. Now, Consider the norm of the element:

$$(a+b\sqrt{p'})(a-b\sqrt{p'}) = a^2 - b^2p' \in \mathfrak{B} \cap \mathbb{Z} = (q)$$

Hence, $q \big| a^2 - b^2p'$. Notice that q cannot divide a or b by our choice in equation (50.6). Hence, get that:

$$a^2 - b^2p' \equiv 0 \bmod q$$

re-arranging (and dividing since b is invertible), we get:

$$(ab^{-1})^2 \equiv p' \bmod q$$

showing one direction of the implication.

Now, for the other direction, it suffices that q splits in L. By assumption:

$$m^2 \equiv p' \bmod q \qquad \Leftrightarrow \qquad m^2 - p' = (m-\sqrt{p'})(m+\sqrt{p'}) = cq$$

for some constant $c \in \mathbb{Z}$. I claim that $(m+\sqrt{p'}, q)$ divides q in L. Certainly

$$(q) \subsetneq (m+\sqrt{p'}, q)$$

since $m+\sqrt{p'} \notin (q)$ (since $m+\sqrt{p'}$ is not divisible by q). Furthermore, $(m+\sqrt{p'})$ is not a unit, for if it were, then if $\alpha = (m-\sqrt{p'})$ and $u = (m+\sqrt{p'})$:

$$u\alpha = (m+\sqrt{p'}, q)(m-\sqrt{p'}) = (x^2 - p', q) = (cq, q)$$

hence $q \big| u\alpha$, implying $q \big| \alpha$, but $q \not| m - \sqrt{p'}$. Thus, it must be that q splits.

But then,this implies that q is a quadratic residue modulo p if and only if p' is a quadratic residue modulo q, as we sought to show.

50.5 Application: Kronecker Weber Theorem

We now apply our finding to Field Theory and find some important consequences' on *Abelian Extension*. This shall also lead to proving one of the central theorems in field theory, The *Kronecker-Weber Theorem*, which states that all finite abelian extensions show up as sub-extensions of Cyclotomic extensions over $\mathbb{Q}$! Since Cyclotomic extensions are generated by roots of unity, every algebraic integers whose Galois group is abelian can be written as a $\mathbb{Q}$-linear combination of roots of unity! For example:

$$\sqrt{5} = e^{2\pi i/5} - e^{4\pi i/5} - e^{6\pi i/5} + e^{8\pi i/5}$$

namely, all square roots of integers may be written as $\mathbb{Q}$-linear combinations of roots of unity! We first give the setting in which we work in:

Definition 50.5.1: Universal Cyclotomic Field and Abelian Closure

1. Let $K^{\mathbf{cycl}} = \bigcup_{n\in\mathbb{N}} K(\zeta_n)$. If $K = \mathbb{Q}$, then:
$$\mathrm{Gal}(\mathbb{Q}^{\mathbf{cycl}}) = \varprojlim \mathrm{Gal}(\mathbb{Q}(\zeta_n)) = \varprojlim (\mathbb{Z}/n\mathbb{Z})^{\times} = \hat{\mathbb{Z}}^{\times}$$
is the pro-finite group given by the limit of units of cyclic groups.
2. Let K be a field. Then
$$K^{\mathbf{ab}} \subseteq \overline{K}$$
is the maximal extension for which $\mathrm{Gal}_K(K^{\mathbf{ab}})$ is abelian.

Proposition 50.5.2: Properties of Abelian Extensions

1. If $L_1, L_2 \subseteq K^{\mathrm{ab}}$ have finite abelian groups, then $\mathrm{Gal}_K(L_1L_2)$ is a finite abelian group.
2. If $G_K = \mathrm{Gal}_K(\overline{K})$ and $H = [G, G]$ (the commutator [closed] subgroup), then
$$K^{\mathrm{ab}} = \overline{K}^H$$
with Galois group G_K/H.

***Proof*:**

1. See proposition 20.1.32
2. This comes from the universal property of abelian quotients, see proposition 5.2.3

Note that in general, $K^{\mathbf{cycl}} \subsetneq K^{\mathrm{ab}}$:

Example 50.5.3: Abelian, But Not Cyclotomic

Let $k = \mathbb{Q}(i)$ and $F = \mathbb{Q}(\sqrt{1+2i})/\mathbb{Q}(i)$. Since every quadratic extension is abelian, F is an abelian extension. However, it is not a cyclotomic extension: for if it was a sub-cyclotomic extension, then since $\mathbb{Q}(i)$ is a cyclotomic extension over $\mathbb{Q}$, then $\mathbb{Q}(\sqrt{1+2i})$ would be a cyclotomic extension, but that is not possible since $\mathbb{Q}(\sqrt{1+2i})/\mathbb{Q}$ is not Galois (but all sub-extensions of a cyclotomic extension are necessarily Galois).

It is thus quite interesting that if $K = \mathbb{Q}$, we have the following:

Theorem 50.5.4: Kronecker Weber's Theorem

Let $K = \mathbb{Q}$. Then:
$$\mathbb{Q}^{\mathbf{cycl}} = \mathbb{Q}^{\mathrm{ab}}$$
In other words, every abelian extension of $\mathbb{Q}$ is contained in a cyclotomic extension.

We'll first show we can assume $\mathrm{Gal}_{\mathbb{Q}}(K)$ is a p-group for some prime p. For if $\mathrm{Gal}_{\mathbb{Q}}(K)$ is abelian, we have

$$\mathrm{Gal}_{\mathbb{Q}}(K) \cong \bigoplus_p G_p$$

where G_p is a p-group.

Proof. **Reduction #1**:

Define $K_{p'} = K^{(\bigoplus_{p \neq p'} G_p)}$. Then $K = K_2 K_3 \cdots$ is the join of all these fields. Hence, if all such fields is inside a Cyclotomic extension, so is K. This can naturally be extended to reduce to cyclic groups in the same manner. $\square$

Next, we show that p is the *only* prime which ramifies in K

Proof. **Reduction #2**:

Assume not. Assume ℓ ramifies in K as well. Let I be the inertia group (doesn't matter what prime we pick). Then the wield inertia group W is trivial (by group theory). Then by theorem 50.4.62:

$$I \hookrightarrow \mathbb{F}_\ell^\times \qquad I \cong \mathbb{Z}/\ell\mathbb{Z}, \quad e \big| \ell - 1$$

Let $L \in \mathbb{Q}(\mu_\ell)$ be the (unique) extension such that $[L : \mathbb{Q}] = e$. So ℓ totally ramifies in L. Then take:

$$\begin{tikzcd}
& M = KL & \\
L \arrow[ur] & & K \arrow[ul] \\
& \mathbb{Q} \arrow[ul] \arrow[ur] &
\end{tikzcd}$$

K and L are abelian, thus so is M. Let I_M be the inertia group of a prime over ℓ. Then

$$I_M \hookrightarrow \mathbb{Z}/e\mathbb{Z} \oplus \mathbb{Z}/\ell\mathbb{Z}$$

Hence

$$I_M \cong \mathbb{Z}/e\mathbb{Z}$$

Let $K' = M^{I_M}$. Then $K' \cap L = \mathbb{Q}$ since nothing ramifies in $K' \cap L$.

Now, $K'L/K'$ has degree at least e, in particular$\geq e$, since ℓ has to ramify (ℓ doesn't ramify in K', and ramifies in L by e). But then,

$$K'L = M = KL$$

Note that we have not introduced any new ramifications, since no other primes ramify that ramify in L, K. We can afterward inductively remove (if it ramifies, it has to have some inertia group, but it will be trivial in the intermediate extensions).

With this, let's now prove the $p = 2$ case.

Assume $K/\mathbb{Q}$ is abelian $[K : \mathbb{Q}] = 2^n$ and 2 is the only prime which ramifies. I claim that K is cyclotomic.

Without loss of generality, let's assume $\mathbb{Q}(i) \subseteq K$ (just adjoin it if need be). Then $K_0 = K \cap \mathbb{R}$ (fixed field under conjugation). This is still abelian since K is abelian, and $[K : K_0] = 2$, so $K = K_0(i)$.

Now, let $K_n/\mathbb{Q}$ be the real cyclotomic field where $\mathrm{Gal}(K_n/\mathbb{Q}) \cong \mathbb{Z}/2^n\mathbb{Z}$ (done it in comments). Consider now

$$M = KK_n$$

Then $\mathrm{Gal}_{\mathbb{Q}}(M) \subseteq \mathbb{Z}/2^n\mathbb{Z} \oplus \mathbb{Z}/2^n\mathbb{Z}$. If $K_n \neq K$, then this Galois group is *not* cyclic (too large order). Therefore, M has ≥ 3 real quadratic subfields (real, since everything is real). But there is only 2 with ramified primes – a contradiction ($\mathbb{Q}(\sqrt{2})$ is the only real quadratic extension in which 2 ramifies). ☐

Now, for the other primes, we have the following key proposition:

Proposition 50.5.5: Ramification Of Primes in $\mathbb{Z}/n\mathbb{Z}$-Extensions

Let p be an odd prime. Then there exists a unique field $K/\mathbb{Q}$ which is Galois with Galois group $\mathbb{Z}/p\mathbb{Z}$ in which p is the only ramified prime.

***Proof*:**
Let ζ be a pth primitive root of 1, let $L = \mathbb{Q}(\zeta)$. $\pi = 1 - \zeta$. So that $(\pi)^{p-1} = (p)$. Let $\gamma \in \mathcal{O}_L$ be a non pth power. Assume $\gamma \equiv 1 \bmod \pi^p$. Take the field $M = L(\sqrt[p]{\gamma})/L$. Then π is unramified. Suppose not. Then $\pi\mathcal{O}_M = Q^p$ for some prime Q where $\mathbb{F}_Q \cong \mathbb{F}_{(\pi)} \cong \mathbb{F}_p$. Let δ be such that $\delta^p = \gamma$. Let $r = \frac{\delta-1}{\pi}$. Claim that $r \in \mathcal{O}_M$. Take the conjugates of r over L, i.e. they are

$$\frac{\zeta^i\delta - 1}{\pi} \qquad i \in \mathbb{Z}/p\mathbb{Z}$$

So the minimal polynomial is the product:

$$\prod \left(x - \frac{\zeta^i\delta - 1}{\pi}\right) = \frac{1}{\pi^p}\prod_{i=1}^{p-1}((\pi x - 1) - \zeta^i\delta) = \frac{(\pi x - 1)^p - \gamma}{\pi^p}$$

So this is inside $\mathcal{O}_L[x]$, so r is inside of integers. Thus, there exists some $m \in \mathbb{Z}$ such that $r - m \in \mathbb{Q}$ (because $\mathbb{F}_Q$ is just $\mathbb{F}_p$). But then for all $\sigma \in \mathrm{Gal}_K(M)$, $\sigma(r - m) \in \mathbb{Q}$ since it's the inerta group. But

$$\frac{\zeta\gamma - 1}{\pi} - m \in \mathbb{Q}$$

and so is $\frac{\gamma-1}{\pi} \in \mathbb{Q}$. Subtracting we get

$$\frac{1-\gamma}{\pi}\gamma = \frac{\pi}{\pi}\gamma = \gamma \in \mathbb{Q}$$

But this is a contradiction of our $\gamma \equiv 1 \bmod Q$, as we sought to show.

With this, we can continue the second reduction

Proof. **Reduction #2 cont.**:

By reduction # 2, we only need to work with $K/\mathbb{Q}$ abelian of prime power order p^n, and without loss of generality $K/\mathbb{Q}$ may be assumed to be cyclic since every field is a compositum which have cyclic subgroups.

Now consider $K_n \subseteq \mathbb{Q}(\zeta_{n+1})$ be the field such that

$$[K_n : \mathbb{Q}] = p^n \qquad \mathrm{Gal}(K_n/\mathbb{Q}) \cong \mathbb{Z}/p^n\mathbb{Z}$$

Let $M + KK_n$. In general $\mathrm{Gal}(M/\mathbb{Q}) \hookrightarrow \mathbb{Z}/p^n\mathbb{Z} \oplus \mathbb{Z}/p^n\mathbb{Z}$. If $K \neq K_n$, then like for the $p = 2$ case, this Galois group is not cyclic. Thus, M contains at least $p+1$ subfields which have Galois group $\mathbb{Z}/p\mathbb{Z}$ over $\mathbb{Q}$ in which only p ramifies, a contradiction □

***Proof*:**
Of Kronecker-Weber's Theorem

We shall show that if p is an odd prime, $K/\mathbb{Q}$ is a $\mathbb{Z}/p\mathbb{Z}$ extension, and p is the only prime that ramified in K, then $K \subseteq \mathbb{Q}(\zeta_{p^2})$, which by our earlier reduction shall complete the proof.

Let $\mathrm{Gal}_{\mathbb{Q}}(M) = \langle \sigma \mid \sigma^{p(p-1)} = \mathrm{id}\rangle$, $K = M^{\langle \sigma^p \rangle}$ and $L = M^{\langle \sigma^{p-1}\rangle}$. Consider:

$$\begin{tikzcd}
 & M = KL & \\
L = \mathbb{Q}(\zeta_p) \arrow[ur] & & K \arrow[ul] \\
 & \mathbb{Q} \arrow[ul, "\mathbb{Z}/(p-1)\mathbb{Z}"] \arrow[ur, "\mathbb{Z}/p\mathbb{Z}"'] \arrow[uu, "\mathbb{Z}/p(p-1)\mathbb{Z}"] &
\end{tikzcd}$$

We hat that p is the only prime that ramifies in M. Thus, by Kummer theory, we know that $M = L(\alpha)$ where $\alpha^p = \beta \in L^\times$. Next, we know that

$$\frac{\sigma(\alpha)}{\alpha}$$

is a non-trivial rot of unity. Without loss of generality say that $\frac{\sigma^{p-1}(\alpha)}{\alpha} = \zeta_p$ (so $\langle \sigma^{p-1}\rangle = \mathrm{Gal}_L(M)$). Note that $\sigma(\zeta_p) = \zeta_p^n$, where $\overline{n} \in (\mathbb{Z}/p\mathbb{Z})^\times$ (i.e. is a primitive root).

I claim that

$$M = L\left(\frac{\sigma(\alpha)}{\alpha}\right)$$

For suppose not. Then $\frac{\sigma(a)}{\alpha} \in L$, and

$$\frac{\sigma(\alpha)}{\alpha} = \sigma^{p-1}\left(\frac{\sigma(\alpha)}{\alpha}\right) = \frac{\sigma^p(\alpha)}{\sigma^{p-1}(\alpha)} = \frac{\sigma(\zeta_p\alpha)}{\zeta_p\alpha} \implies 1 = \frac{\sigma(\zeta_p)}{\zeta_p}$$

but that is a contradiction, hence $M = L\left(\frac{\sigma(\alpha)}{\alpha}\right)$.

For the rest of this proof, we write

$$\alpha \mapsto \frac{\sigma(\alpha)}{\alpha} = \alpha' \qquad \beta \mapsto \frac{\sigma(\beta)}{\beta} = \beta'$$

Let P_k, P_L, P_M be the primes ideals above $p\mathbb{Z}$. Take the prime decomposition:

$$\beta\mathcal{O}_L = \prod Q^{n_Q} = P_L^r \cdot \prod_{Q \neq P} Q^{n_Q} \qquad n_Q \in \mathbb{Z}$$

Then:

$$\sigma(\beta)\mathcal{O}_L = P_L^r \prod_{Q \neq P_L} \sigma(Q)^{n_Q}$$
$$\frac{\sigma(\beta)\mathcal{O}_L}{\beta} = \prod_{Q \notin P_L} Q^{m_Q}$$

Hence, P_L does not occur in $\beta\mathcal{O}_L$. Take $\beta'' = \beta' N^P$ and $\alpha'' = \alpha' N$ for $N \in \mathbb{Z}$ such that $\beta'' \in \mathcal{O}_L$ Then:

$$M = L(\alpha'') \qquad (\alpha'')^P = \beta''$$

Now let $\pi = 1 - \zeta_p \in \mathcal{O}_L$ so that $P_L = (\pi)$. Then $\mathbb{F}_{P_L} \cong \mathbb{F}_P$ implies β'' mod $P_L = m$ mod P_L for $m \leq N$. Let $m_0 = m^{-1}$ mod p, so $\beta'' \to m_0\beta''$ and $\alpha'' \to m_0\alpha''$. Without loss of generality say $\beta'' = 1$ mod P_L. Then for some $c \in \mathbb{Z}/p\mathbb{Z}$

$$\beta'' = 1 + c\pi \bmod P_L^2 \qquad \zeta_p = 1 - \pi \bmod P_L^2 \qquad \zeta_p^r = 1 - r\pi \bmod P_L^2 \implies \beta'' = \zeta_p^{-c}\nu$$

where $\nu \equiv 1$ mod P_L^2. I claim that $\nu \in (L^\times)^P$, which will imply $M = \mathbb{Q}(\zeta_{p^2})$ and complete the proof. First:

$$\begin{aligned}
(\alpha'')^p &= \beta'' = \zeta_p^{-c}\nu \\
\sigma^{p-1}(\alpha'') &= \alpha''\zeta_p \\
\implies \sigma(\zeta_p) &= \zeta_p^n && n \in (\mathbb{Z}/p\mathbb{Z})^\times \\
\implies \sigma^p(\alpha'') &= \sigma(\sigma^{p-1}(\alpha'')) = \sigma(\alpha''\zeta_p) = \sigma(\alpha'') \cdot \zeta_p^n \\
\implies \sigma^{p-1}((\alpha'')^n) &= (\alpha'')^n\zeta_p^n \\
\implies \sigma^{p-1}\left(\frac{\sigma(\alpha'')}{(\alpha'')^n}\right) &= \frac{\sigma(\alpha'')}{(\alpha'')^n} \\
\implies \frac{\sigma(\alpha'')}{(\alpha'')^n} &\in M^{\langle \sigma^{p-1} \rangle} = L \\
\implies \frac{\sigma(\beta'')}{(\beta'')^n} &\in (L^\times)^P \\
\implies \frac{\sigma(\nu\zeta_p^{-c})}{\nu^n\zeta_p^{-nc}} &\in (L^\times)^p
\end{aligned}$$

Thus $\frac{\sigma(\nu)}{\nu^n} \in (L^\times)^p$.

Next, we claim that $\nu = 1$ mod P_L^p. Suppose not. Then let $p^M \| \nu - 1$ where $1 \leq m \leq p-1$, and $\nu = 1 + c\pi^m$ mod P_L^{m+1} (where $c \in (\mathbb{Z}/p\mathbb{Z})^\times$). Then by this:

$$\nu^n = 1 + nc\pi^m \bmod P_L^{m+1}$$

hence,

$$\sigma(\pi) = \sigma(1 - \zeta_p) = 1 - \zeta_p^n = (1 - \zeta_p)(1 + \zeta_p + \cdots + \zeta_p^{n-1}) = \pi n \bmod P_L^2$$

implying

$$\sigma(\pi^m) = \pi^m n^m \bmod P_L^{m+1} \implies \sigma(\nu) = 1 + cn^m\pi^m \bmod P_L^{m+1}$$

Note that $n^m \not\equiv n \bmod p$ for $n \in (\mathbb{Z}/p\mathbb{Z})^\times$. Now as a corollary:

$$\frac{\sigma(\nu)}{\nu^m} \equiv 1 + d\pi^m \bmod P_L^{m+1} \qquad d \in \mathbb{Z}/p\mathbb{Z}^\times$$

However,

$$\frac{\sigma(\nu)}{n\nu} = S^p \qquad S \in L$$

where P_L does not occur in S, and:

$$S^p \equiv 1 \bmod P_L \implies S = 1 \bmod P_L$$

Giving us $S = 1 + t$, $P_L \big| t$, so

$$S^P = 1 + pt + \cdots + t^p \equiv 1 \bmod P_L^P$$

a contradiction!

Finally, $L(\sqrt[p]{\nu})/L$ is unramified above P_L. Let

$$F = L(\sqrt[p]{\nu}),\ F \subseteq M(\zeta_{p^2}) \qquad F/Q \text{ abelian}$$

and $I \leqslant \mathrm{Gal}_{\mathbb{Q}}(\mathbb{F})$ where I is the inertia group above P, and $|I| = p-1$, so $F^I/\mathbb{Q}$ is unramified everywhere, namely $F^I = \mathbb{Q}$, so $F = L$. This implies

$$\nu \in (L^\times)^P$$

But then

$$M = \mathbb{Q}(\zeta_{p^2})$$

as we sought to show.

Let us use it in an actual example:

Example 50.5.6: Using Kronecker-Weber

Let $K/\mathbb{Q}$ be a galois extension with Galois group $\mathbb{Z}/5\mathbb{Z}$ whose discriminant is only divisible by $5, 11$, and 31. How many such number fields are there?

Proof. By Kronecker Weber, we know that K is an intermediary field of a Cyclotomic extension:

$$K \subseteq \mathbb{Q}(\zeta_n)$$

We may without loss of generailty assume n is minnimal for which this containment makes sense. Since only $5, 11, 31$, divide the discriminant, we know the discriminant of K is of the form:

$$\mathrm{disc}(\mathcal{O}_K) = 5^a 11^b 31^c$$

which also mean only $5, 11, 31$ ramify in K. Now, I claim that only $5, 11, 31$ divide n and no other number those. For the sake of contradiction, let's say $p \big| n$ but $p \notin \{5, 11, 31\}$. By minimality, we have:

$$\begin{tikzcd} & \mathbb{Q}(\zeta_n) \\ \mathbb{Q}(\zeta_{n/p}) \arrow[ur] & K \arrow[u] \\ & \mathbb{Q} \arrow[u] \arrow[ul] \end{tikzcd}$$

Let $I_{\mathfrak{B}}$ be the inertia group (for any prime $\mathfrak{B}$ lying above p). Since $p \nmid \mathrm{disc}(\mathcal{O}_K)$, p is unramified. If p is unramified in $\mathbb{Q}(\zeta_{n/p})$, then since $\mathbb{Q}(\zeta_{n/p}) \subseteq \mathbb{Q}(\zeta_n)$, we get that $\mathbb{Q}(\zeta_{n/p}) \subseteq \mathbb{Q}(\zeta_n)^{I_p} \subseteq \mathbb{Q}(\zeta_n)$

$$K \subseteq \mathbb{Q}(\zeta_n)^{I_{\mathfrak{B}}} \subseteq \mathbb{Q}(\zeta_{n/p})$$

contradicting minimality of n. Hence, we get $n = 5^a 11^b 31^c$. Thus:

$$\mathrm{Gal}_{\mathbb{Q}}(\mathbb{Q}(\zeta_n)) \cong (\mathbb{Z}/n\mathbb{Z})^\times \cong \mathbb{Z}/4\mathbb{Z} \times (\mathbb{Z}/5^{a-1}\mathbb{Z}) \times \mathbb{Z}/10\mathbb{Z} \times (\mathbb{Z}/11^{b-1}\mathbb{Z}) \times \mathbb{Z}/30\mathbb{Z} \times (\mathbb{Z}/31^{c-1}\mathbb{Z})$$

which now become a counting argument of how many subgroups of index 5 there are, counting the number of intermediary subfields, thus completing the proof. □

50.6 *Modules Over Dedekind Domains

(Useful for alg-geo, theory itself is also really nice. Also $\mathcal{O}_K$-modules appear in the next section)

Lemma 50.6.1: all Ideal co-maximal up to Module Isomorphism

Let $I, J \subseteq R$ be fractional ideals of a Dedekind domain R. Then $I \oplus J \cong_R R$

***Proof*:**
By corollary 50.2.12, we may choose in fact choose I, J to be integral ideals. We shall show that $I + j^{-1}J = R$ for some appropriate j^{-1}, which since $j^{-1}J \cong_R J$, we have $I \oplus J \cong_R R$

In particular, we shall show that for any ideal J, there exists an element $j \in J$ such that $I + jJ^{-1} = R$ (that is, every ideal is "almost" comaximal with any fractional ideal). It is equivalent to show that $IJ = (j) = J$. By corollary 50.2.19, IJ is generated by at most 2 elements, say $IJ = (a, b)$. Then since $a \in IJ \subseteq J$, we may choose a c such that $(a, c) = J$ (see proof of corollary 50.2.19 if you're not convinced we can choose such a c). Then since $b \in J$ (since $b \in IJ$)

$$IJ + (c) = (a, b) + (c) = (a, b, c) = (a, c) = J$$

hence, $I + jJ^{-1} = R$, which by corollary 50.2.12 we get $I + J \cong_R R$, completing the proof.

Proposition 50.6.2: Ideals Are Projective

Let R be a Dedekind domain and $I \subseteq R$ an ideal. Then I is a projective R-module

***Proof*:**
By lemma 50.6.1, we may choose any J such that $I + jJ^{-1} = R$, or equivalently $IJ + (j) = J$. We shall define an isomorphism $I \oplus J \to IJ \oplus R$ via

$$(x, y) \mapsto (iy - jx, x + ky)$$

This map is certainly well-defined since $iy, jx \in IJ$, so $iy - jx \in IJ$, and it is certainly R-linear. It is bijective, since it's inverse is $IJ \oplus R \to I \oplus J$

$$(x, y) \mapsto (jx + y, ix - ky)$$

Hence, we have $I \oplus J \cong IJ \oplus R$. Finally, by corollary 50.2.18, choose J so that $IJ = (a)$ is principal. Then it is a free R-module of rank 1, and so

$$I \oplus J \cong IJ \oplus R \oplus R^2$$

as we sought to show.

Proposition 50.6.3: IBN equivalent for Dedekind Domains

Let $I_1, ..., I_n$ and $J_1, ..., J_m$ be fractional ideals of a Dedekind domain R. Then

$$I_1 \oplus I_2 \oplus \cdots \oplus I_n \cong_r J_1 \oplus J_2 \oplus \cdots \oplus J_m$$

if and only if $n = m$ and

$$I_1 I_2 \cdots I_n = (a) J_1 J_2 \cdots J_n$$

that is, the product differs by a principal ideal (they belong to the same equivalence class in the class group). In particular,

$$I_1 \oplus I_2 \oplus \cdots \oplus I_n \cong_R \underbrace{R \oplus R \oplus \cdots \oplus R}_{n-1 \text{ times}} \oplus (I_1 I_2 \cdots I_n)$$

and $R^n \oplus I \cong R^n \oplus J$ if and only if $I = (a)J$ for some $a \in K(R)$.

***Proof*:**
First, let $I, J \subseteq R$ be two fractional ideals. By lemma 50.6.1, we may choose I, J to be integral ideals that are up to R-isomorphism relatively prime: $I + J = R$. Then $I \cap J = IJ$, and $\varphi : I \oplus J \to R$ given by $(x, y) \mapsto x + y$ has kernel $I \cap J = IJ$. Then:

$$0 \to IJ \to I \oplus J \to R \to 0$$

is a short exact sequence. Since R is a free R-module, it is projective, and hence the sequence splits, giving us

$$I \oplus J = IJ \oplus R$$

Theorem 50.6.4: Classifying Torsion-Free Modules Over Dedekind Domain

Let M be a torsion-free R-module over a Dedekind domain R of rank n. Then

$$M \cong R^{n-1} \oplus I$$

where we I may be replaced with any ideal J such that $I = (a)J$ (they belong in the same element in the class group). As a consequence, any torsion-free module over a Dedekind domain is projective.

***Proof*:**
If there exists a non-trivial map $\varphi : M \to R$, then since R-submodules of R are projective, we may iteratively split M as $M \cong N \oplus I$ to get a direct sum of ideals of R, which we may then use to conclude by using proposition 50.6.3.

Since M is torsion-free of rank n, we may extend scalars we get a n-dimensional $K = K(R)$ vector space $K \otimes_R M$ where $M \hookrightarrow K \otimes_R M$ naturally embeds injectively. Let $\beta_1, \beta_2, ..., \beta_n$ be a basis for $K \otimes_R M$ and $m_1, m_2, ..., m_k$ be a set of generators for M. Then

$$m_i = \sum_j (a_{ij}/b_{ij})\beta_j$$

Let $d = \prod_{ij} b_{ij}$ and $y_j = \beta_j/d$. Then

$$M = Ry_1 + \cdots + Ry_n \subseteq K\beta_1 + \cdots + K\beta_n$$

Hence, M is a R-submodule of a free R-module! Importantly, for each $m \in M$, it can be written as

$$m = \sum a_i y_i \qquad a_i \in R$$

hence, we may define $\varphi : M \to R$ by $\varphi(m) = a_n$. Then $\varphi(M) \subseteq R$ is an ideal of R, and so

$$M = \ker(M) \oplus I$$

Notice that $\text{Tor}(\ker(M)) = 0$, and so we may inductively repeat this procedure until we have products of prime ideals!

since the product of projective module is projective, any torsion-free module over a Dedekind domain is projective, as we sought to show.

50.7 *Geometric Idea: 1-Dimensional Schemes

See Neukirch, and put orders here too for their geometric interpretation. I think it is important to include (especially once this chapter becomes its own book) since (to me) this explains why ramification happens. We naturally can't prove the Riemann-Hurwitz Formula here, but this still is very interesting.

50.7.1 Localization of Dedekind Domains

For the motivation to CFT

50.7.2 Orders: Curves with Singularities

This entire section is completely inspired by Neukirch p.72

should also include how if finite L/K then $\operatorname{Spec}(\mathcal{O}_L)$ is the normalization of $\operatorname{Spec}(\mathcal{O}_K[\alpha])$ where $L = K(\alpha)$. And I would really love a discussion linking $\operatorname{Spec}(\mathcal{O}_L)$ to the "generic point" $\operatorname{Spec}(L)$ and to $y - m_{\alpha,K}(x)$ where $m_{\alpha,K}(x)$ is the minimal polynomial of L/K

All smooth 1-dimensional curves correspond to Dedekind domains. However, there are some curves with singularities. These curves are associated to sub-rings of Dedekind domains called *orders*

Definition 50.7.1: Order

Let $K/\mathbb{Q}$ be a number field of degree n. Then an order of K is a subring of $\mathcal{O}_K$ which contains an integral basis of length n. The ring $\mathcal{O}_K$ is called a *maximal order*.

By the above definition, an order is of the form:

$$\mathbb{Z}[\alpha_1, ..., \alpha_n]$$

where $K = \mathbb{Q}(\alpha_1, ..., \alpha_n)$, which motivates why $\mathcal{O}_K$ is the maximal order; it is the maximal ring with this property. This also implies that orders need not be normal, hence an order is a one-dimensional Noetherian integral domain (prime remain maximal as if $a \in \mathfrak{p} \cap \mathbb{Z}$ then $a\mathcal{O} \subseteq \mathfrak{p} \subseteq \mathcal{O}$ that is $\mathfrak{p}$ and $\mathcal{O}$ have the same rank n, and thus $\mathcal{O}/\mathfrak{p}$ is a finite integral domain, showing $\mathfrak{p}$ is maximal) . An example of an order is $\mathbb{Z} = \mathbb{Z}(\sqrt{5})$. Many order's naturally are as "rings of multipliers. For example if $\alpha_1, .., \alpha_n$ is a basis for $K/\mathbb{Q}$ and $M = \sum_i \mathbb{Z}\alpha_i$, then

$$\mathcal{O}_M = \{\alpha \in K \mid \alpha M \subseteq M\}$$

is an order.

It is important that we re-prove the Chinese remainder

Here

As orders are not Dedekind domain in general, *not* all ideals are invertible. This also means that the collection of fractional ideals does not form a group.

Proposition 50.7.2: Invertible Fracitonal Ideals

Let I be a fractional ideal of a Noetherian domain R. Then I is invertible iff its localization at every maximal ideal of R is invertible, equivalently, iff its localization at every prime ideal of R is invertible.

Proof:
MIT notes p.17

Example 50.7.3: non-invertible ideal of an order
put this example here

(I want this build up here)

Definition 50.7.4: Picard Group Of An Order

Let $J(\mathcal{O})$ be the collection of invertible ideals and $P(\mathcal{O})$ the fractional principal ideals. Then

$$\operatorname{Pic}(\mathcal{O}) = \frac{J(\mathcal{O})}{P(\mathcal{O})}$$

A lot more to be said.

The Picard group ignore s ideals that are not invertible. Another approach that gives some useful results is to define the following for A Dedekind domain:

$$\operatorname{Div}(\mathcal{O}) = \mathcal{O}_{\mathfrak{p}}\mathbb{Z}\mathfrak{p}$$

Then $\operatorname{Div}(\mathcal{O})$ is isomorphic to the ideal group. For an order, the divisor group shall be a quotient of this group. We would then need to define the divisor elements, which gives the divisor class group/chow group.

50.8 *Profinite Groups

(original title: infinite galois extensions)

I moved this from the galois theory chapter. I actually want to make this its own chapter when I extract these notes into their own book so that the many books that use profinite groups can reference this namely: the class field theory book, the p-adic rep book, and the galois cohomology book.

In this section, we shall explore the case of infinite separable normal extensions L/K. This section shall require the notion of topological groups and topological completion, which is covered in appendix C. Most of this material will be important for a course in *Class Field Theory* (see cite:HERE for a reference where this material is used).

In this section, a Galois extension is a normal separable extension, where the finiteness condition is dropped. We may want to keep the galois correspondence between the fixed fields and the subgroups, however there are simply way too many subgroups. Consider

$$\overline{\mathbb{Q}}/\mathbb{Q} \qquad \text{and} \qquad \operatorname{Gal}_{\mathbb{Q}}(\overline{\mathbb{Q}})$$

Then the reader should check that there are only countably many intermediary extension of degree 2 corresponding to the extensions of the form $\mathbb{Q}(\sqrt{\alpha})/\mathbb{Q}$ where α is either a prime p, ip, or a finite compositum of such elements. However, there are uncountably many index two subgroups. To see this, consider $\varphi_S : \operatorname{Gal}_{\mathbb{Q}}(\overline{\mathbb{Q}}) \to \mathbb{Z}/2\mathbb{Z}$ where S is a finite collection of primes where if σ fixes each $\sqrt{p}$. Then $\varphi_S(s) = 0$, and it is 1 otherwise. This is a group-homomorphism, as the composition of two maps that fix each $\sqrt{p}$ for $p \in S$ shall still fix those primes. Furthermore, There are certainly always morphisms τ such that $\varphi_S(\tau) = 1$ for each finite subset S. Thus, each φ_S is a surjective homomorphism, whose kernel is index 2. But then, there are uncountably distinct $\ker(\varphi_S)$. In fact, we don't even need S to be finite (exercise). In this case, we shall even have index two subgroups whose fixed field corresponds to $\mathbb{Q}$!

The reader may feel that there is some simple solutions that can be found, as it seems there are many galois subgroups corresponding to an intermediary. Indeed, we may still recover the galois correspondence if we appropriately limit which subgroups of $\mathrm{Gal}_{\mathbb{Q}}(\overline{\mathbb{Q}})$ we consider (and more generally which subgroups of $\mathrm{Gal}_K(L)$ we consider). This is done by taking the *Krull topology* on the galois group, which we shall naturally derive by first taking the *profinite topology*. The profinite topology shall make us limit our attention to normal subgroups of finite index (namely, such subgroups shall be open in the profinite topology), while in the Krull topology we shall limit our attention to normal subgroups of finite index which also have a [finite] galois group (namely, such subgroups shall be open in the Krull topology). Both of these topologies are achieved by showing that the infinite galois group, as a topological group, is a profinite group. The main idea is that all of the information of infinite galois extension is in fact fully captured by the finite normal extension, due to the fact galois groups represent the relationship of polynomials (where an infinite galois groups stores the information of infinitely many polynomials).

50.8.1 Profinite Groups

In this section, a brief introduction to profinite groups is given. This is not an in-depth coverage, but rather it gives enough information to describe infinite galois theory.

Definition 50.8.1: Profinite Group

Take an inverse system of finite groups with the discrete topology. Then a *profinite group* is a topological group that is the limit of the system. Given any topological group G, we can take its *profinite completion* by taking:

$$\widehat{G} = \varprojlim_N G/N$$

where N ranges over all finite index open normal subgroups ordered by containment.

Any group G can be given a topology by given it the profinite topology given by taking the cosets of finite-index normal subgroups as a basis. The reader should check that if G is a topological group and $\widehat{G}$ is its profinite completion, then the continuous homomorphism $\varphi : G \to \widehat{G}$ can be shown that G is dense in $\widehat{G}$.

Example 50.8.2: Profinite Groups

1. Let G be finite. Then all normal subgroups have finite index and there are finitely many. It follows that $G \cong \widehat{G}$.
2. Let $G = \mathbb{Z}$. Then
$$\widehat{Z} = \varprojlim_n \mathbb{Z}/n\mathbb{Z} \cong \prod_p \mathbb{Z}_p$$
where $\mathbb{Z}_p = \varprojlim_n \mathbb{Z}/p^n\mathbb{Z}$ is the p-adic completion of $\mathbb{Z}$.
3. Take $G = \mathbb{Q}$. Then $\mathbb{Q}$ has only one finite-index normal subgroup, namely itself. Hence $\widehat{\mathbb{Q}} = \{1\}$.

The above examples show that the map $\varphi : G \to \widehat{G}$ could be injective, surjective, both. The map will be injective if the intersections of all finite-index open normal subgroups of G is trivial (where we would then say G is residually finite). The completion comes equipped with a natural universal property where a continuous homomorphism $\varphi : G \to H$ with H a profinite group has a unique continuous homomorphism such that the following diagram commutes:

$$\begin{tikzcd} G \arrow[r, "\varphi"] \arrow[dr, "\varphi"'] & \widehat{G} \arrow[d, "\exists!"] \\ & H \end{tikzcd}$$

A word on strongly complete? MIT notes p.275 Remark 26.15

Theorem 50.8.3: Characterizing Profinite Groups

A topological group is profinite if and only if it is a totally disconnected compact group

***Proof*:**
exercise: Compactness comes from Tychnoff's Theorem, while totally disconnected shall come from the discreteness in each finite quotient

Corollary 50.8.4: Distinguishing Profinite Completion

Let G be a profinite group. Then G is naturally isomorphic to its profinite completion, or even stronger:

$$G \cong \varprojlim G/U$$

where U ranges over all open normal subgroups ordered by contain need.
If G is isomorphic to its profinite completion *as a group* (i.e. is strongly complete) if and only if every finite index subgroup is open

***Proof*:**
MIT notes p.276

50.8.2 Infinite Galois Theory

Let us now find which topology to put on galois groups. Let us first make sure that the notion of there being a correspondence between subfields and fixed fixed fields of subgroups of galois group is still well-defined

Lemma 50.8.5: Fixed Field Infinite Extension

Let L/K be a galois extension with galois group $G = \mathrm{Gal}_K(L)$. Then if F/K is a normal subextension of L/K, then $\mathrm{Gal}_F(L) = H \trianglelefteq G$ is a normal subgroup of G wit fixed field F. As the subgroup is normal,we have the short exact sequence:

$$1 \to \mathrm{Gal}_F(L) \to \mathrm{Gal}_K(L) \to \mathrm{Gal}_K(F) \to 1$$

which implies:

$$G/H \cong \mathrm{Gal}_K(F)$$

***Proof*:**
As F/K is normal, the map $\sigma \mapsto \sigma|_F$ is a well-defined surjective map (as every $\tau \in \mathrm{Gal}_K(F)$ can be extended to an element $\sigma \in \mathrm{Gal}_K(L)$, see exercise ref:HERE) and has kernel $\mathrm{Gal}_F(L)$ by definition. Then by definition $F \subseteq L^H$, and $L^H \subseteq F$ since if $\alpha \in L^H \setminus F$, then we can construct an element of H that sends α to a distinct root $\alpha' \neq \alpha$ of its minimal polynomials f over F, completing the proof.

Hence, every normal subextension has (at least one) corresponding normal subgroup. However, there are many normal subgroups whose fixed field F is *not* equal to $\mathrm{Gal}_F(L)$, but is contained in $\mathrm{Gal}_F(L)$ The index two example of $\mathrm{Gal}_{\mathbb{Q}}(\overline{\mathbb{Q}})$ is the prototypical example, where there are uncountably many index 2 subgroups $H \leqslant \mathrm{Gal}_{\mathbb{Q}}(\overline{\mathbb{Q}})$ whose fixed field is $\mathbb{Q}$ but naturally H is not the galois subgroup of $\overline{\mathbb{Q}}/\mathbb{Q}$, nor is it the galois group of $\overline{\mathbb{Q}}/K$ for any subextension $K/\mathbb{Q}$. Hence, the following topology limits our attention to the right groups:

Definition 50.8.6: Krull Topology On Galois Group

Let L/K be a Galois (i.e. normal separable) extension with $G = \mathrm{Gal}_K(L)$. Then the *Krull Topology* on G is the basis given by the basis consisting of cosets of the subgroups

$$H_F = \mathrm{Gal}_F(L)$$

where F ranges over finite normal extensions of K in L.

In this topology, every open normal subgroups has finite index, but not every normal subgroup of finite index is open (think $\mathrm{Gal}_{\mathbb{Q}}(\overline{\mathbb{Q}})$ again). Hence, this topology is (strictly) coarser than the porfinite topology, however it is still a profinite group:

Proposition 50.8.7: Galois Group is Profinite Group

Let L/K be a galois extension. Then under the Krull topology, the restriction maps induce a natural isomorphism of topological groups:

$$\mathrm{Gal}_K(L) \cong \varprojlim_F \mathrm{Gal}_K(F)$$

where F ranges over finite galois extension of K in L. Hence, $\mathrm{Gal}_K(L)$ is a profinite group whose open normal subgroups are those of the form $\mathrm{Gal}_F(L)$ for some finite normal extension F/K.

***Proof*:**
Let's first show the map is bijective. For injectivity, for every $\alpha \in L$ it is algebraic over K, and hence liens ins some finite normal subextensions F/K, and so every automorphism in $\mathrm{Gal}_K(L)($ is uniquely determined by all its restrictions to finite normal F/K, showing φ is injective. For surjectivity, given any $(\sigma_F) \in \varprojlim_F \mathrm{Gal}_K(F)$, we may define a corresponding $\sigma \in \mathrm{Gal}_K(L)$ that maps to it by point-wise defining:

$$\sigma(\alpha) := \sigma_F(\vec{\alpha})$$

where F is the normal closure of $K(\alpha)$. This map is well-defined by construction of the inverse limit, giving surjectivity.

For continuity, Letting $G = \mathrm{Gal}_K(L)$ and $H_F = \mathrm{Gal}_F(L)$, we can view φ as the natural continuous map:

$$\varphi : G \to \varprojlim G/H_F$$

Finally, to show the inverse is continuous, we'll show the equivalent of φ being an open map, which translates to showing that φ maps open subgroups $H \subseteq G$ (i.e. the basis) to open subsets in $\varprojlim G/H_F$. Taking $H = \mathrm{Gal}_F(L)$, we get:

$$\varphi(H) = \{(\sigma_E) \mid \sigma_{E_{E\cap F}} = \mathrm{id}_{E\cap F}\} = \pi_F^{-1}(\mathrm{id}|_F)$$

as E/K ranges over all finite normal subextension of L/K and π_F is the projection from the limit to $\mathrm{Gal}_K(F)$. Then the singleton set $\{\mathrm{id}|_F\}$ is open in the discrete group $\mathrm{Gal}_K(F)$, and so its inverse is open, completing the proof.

Theorem 50.8.8: Fundamental Theorem of Galois Theory, Infinite Extensions

Let L/K be a galois extension with galois group $G = \mathrm{Gal}_K(L)$ endowed with the Krull topology. Then the maps

$$F \mapsto \mathrm{Gal}_F(L) \qquad H \mapsto L^H$$

define inclusion reversing bijections between subextensions F/K of L/K and closed subgroups H of G. Thus, subextensions of finite degree n correspond to subgroups of finite index n, and normal subextensions F/K correspond to normal subgroups $H \trianglelefteqslant G$ such that as topological groups:

$$\mathrm{Gal}_K(F) \cong G/H$$

***Proof*:**
First off, recall that every open subgroup is clopen, as the compliment is the union of non-trivial cosets, and so closed subgroups of finite index are open. Next, the correspondence between finite galois subextensions F/K and finite index closed normal subgroups H is given by proposition 50.8.7, and furthermore we have $[F : K] = [G : H]$ as $G/H \cong \mathrm{Gal}_K(F)$ by lemma 50.8.5.

For the correspondence maps, if F/K is any finite subextension with normal closure E, then $H = \mathrm{Gal}(L/F)$ contains the normal subgroup $N = \mathrm{Gal}(L/E)$ with finite index. The subgroup N is open and therefore closed, thus H is closed since it is a finite union of cosets of N. The fixed field of H is F (by the same argument as in the proof of lemma 50.8.5), thus finite subextensions correspond to closed subgroups of finite index. Conversely, every closed subgroup H of finite index has a fixed field F of finite degree, since the intersection of its conjugates is a normal closed subgroup $N = \mathrm{Gal}(L/E)$ of finite index whose fixed field E contains F and has finite degree. The degrees and indices match because $[G : N] = [G : H][H : N]$ and $[E : K] = [F : K][E : F]$; by the previous argument for finite normal subextensions, $[E : K] = [G : N]$ and $[E : F] = [H : N]$ (for the second equality, replace L/K with L/F and G with H).]

Next, for the closed part, any subextension F/K is the union of its finite subextensions E/K. The intersection of the corresponding closed finite index subgroups $\mathrm{Gal}(L/E)$ is equal to $\mathrm{Gal}(L/F)$, which is therefore closed. Conversely, every closed subgroup H of G is an intersection of basic closed subgroups, all of which have the form $\mathrm{Gal}(L/E)$ for some finite subextension E/K, thus $H = \mathrm{Gal}(L/F)$, where F is the union of the E, completing the proof.

Finally, we see that the isomorphism $\mathrm{Gal}_K(F) \cong G/H$ of topological groups follows from lemma 50.8.5, completing the proof.

Corollary 50.8.9: Closure in Krull Topology

Let L/K be a galois extension and $H \leqslant \mathrm{Gal}_K(L)$ with fixed field F. Then the closure $\overline{H}$ in the Krull Topology is $\mathrm{Gal}_F(L)$:

$$\overline{H} = \mathrm{Gal}_F(H)$$

***Proof*:**
First, $H \subseteq \mathrm{Gal}_F(L)$ since by definition $\mathrm{Gal}_F(L)$ contains every $\sigma \in \mathrm{Gal}_K(L)$ that fixes F . Next, $\mathrm{Gal}_F(L)$ is closed and is the group characterized by $L^{\mathrm{Gal}_F(L)} = F$ by theorem 50.8.8. Thus

$$H \subseteq \overline{H} \subseteq \mathrm{Gal}_F(L)$$

Then in the bijection presented in theorem theorem 50.8.8, we have that they are equal, completing the proof.

Let us conclude with a result proven by Waterhouse in 1973 that generalizes theorem 20.2.10:

Theorem 50.8.10: Profinite Group Representable By Galois Group

Let G be a profinite group. Then G is isomorphic to the galois group of some galois extension L/K

> ***Proof*:**
> MIT notes p.279

Just like for finite groups where $K = \mathbb{Q}$ is a difficult (as of writing this book unsolved) problem called the *inverse Galois Problem*, there is little control on what base field K we can choose.

Exercise 50.8.1

1. Find $G = \mathrm{Gal}_{\mathbb{F}_p}(\overline{\mathbb{F}}_p)$ recalling that for each degree n, there is only one extension of $\mathbb{F}_p$. Show that there are too many subgroups to correspond to the fixed fields. Define the Krull topology on G.

Part X

Representation Theory

There are many domains of mathematics which have made huge amounts of progress, having very-well understood objects and morphisms, theorems, results, and properties. One of the most well-understood area's of mathematics has become linear algebra, where vector-spaces and linear transformations. One of the goals of this chapter will be do be able to study the following 3 objects where the operations becomes matrix addition and multiplication

1. (Finite) groups as a subgroup of $\mathrm{GL}(V)$, where the group operation becomes the matrix multiplication
2. Associative algebras
3. Lie Algebras

The further advantage of bringing these objects into linear algebra is that many field intersect with it: If the vector-spaces are infinite dimensional, we can make them Hilbert spaces and use tools from analysis. Vector-spaces are at the heart of differential geometry allowing us to use such theories to study groups!

For the reader who is really interested in number theory, we shall see in ref:HERE that matrix algebras have many parallels with some number theory results. The reader who goes onto learn class field theory[28] will to their delight find that these parallels continue. It was observed by Langland[29] that the study of simple k-algebras (algebras with trivial two-sided ideals) can be a higher-dimensional parallel to field extensions and the central results of class field mathematics known as the *Langland's Program*.

(The reader may want to go back to section 12.3 to review about direct sums of modules)

(Also, this entire link is really cool: from wiki. Also has some cool examples of duality!!!)

[28]See [39] for the continuation of this textbook into class field theory
[29]Or at least formalized by Langland, as these ideas were floating around during his time

51

Semisimple Rings and Modules

As mentioned in the introduction, representation theory is all about decomposing structures. In part I, we showed in chapter 3, section 3.7, that all finite groups can uniquely be represented as a series of extensions with simple factors (i.e. G_i/G_{i+1} is simple). The simplest type of composition series would be when G is a direct product of simple groups, which would be in the terminology of exact sequences a series of trivial extensions with simple factors: such a group is sometimes *semi-simple* group[1]. When it came to modules, we found that we may decompose it in different ways depending on what rings acts on it, in particular we saw in chapter 13 that if $R = k$ is a field k, then any k-module is free, and in chapter 14 we saw that if R is a Principal Ideal Domain, where we got that any R-module decompsoed into a free part R-isomorphic to R^k, and a torsion part $R/(a_1) \times \cdots \times R/(a_n)$. It is evident that the ring that acts on the abelian group can have a large impact on it's structure. In this section, we define a new type of ring, which will come to be called a *semi-simple* ring, that will force the module M to *always* be a direct product of *simple modules* (so that we are analysing the "building blocks" of modules).

To define this ring, we first shift focus and start by work with the module itself to find the "simplest" structure it may be[2], and explore in a manner which at the beginning parallels groups: we will look for *simple R-modules*, and direct product of simple modules (since they are the simplest extension of simple module).[3]. After defining such a strucutre, we will define which rings acting on moduels will *always* make the module semi-simple

[1] the term "semi-simple group" is not a completely standard terminology, as direct product of simple groups is not generally very heavily studied, see this

[2] "simple" here means no non-trivial sub-structures that can be used to build an extension, just like for simple groups or simple rings

[3] It might be tempting to say that this means we are exploring projective modules. However, we have yet to establish whether simple modules are free-modules. Are they?

51.1 Decomposing Modules

Definition 51.1.1: Decomposition Terminology

Let M be a nonzero R-module.

1. M is said to be *simple* (or *irreducible*) if the only submodules of M are 0 and M. Otherwise, M is *reducible*
2. M is said to be *indecomposable* if there do not exist submodules $M_1, M_2 \subseteq M$ such that $M = M_1 \oplus M_2$. Otherwise, M is said to be *decomposable*.
3. The module M is said to be *semisimple* (or *completely reducible*) if it is the direct summand of simple modules
4. The module M is said to be semi-indecomposable (or completely indecomposable) if it is a direct summand of indecomposable modules.
5. If M is semisimple, any direct summand of M is called a *constituent* of M (i.e. N is a constituent of M if there exists a submodulde N' such that $M = N \oplus N'$)

A moment should be taken to find the relations between these definitions:

1. Simple modules are semi-simple and indecomposable. The prototypical example of a simple module is a field k over itself.
2. Semi-Simple $\not\Rightarrow$ Simple: take k^2 over k (the 2 dimensional vector-space over k). The cannonical example of a semi-simple module would be vector-spaces, since they are all direct sums of the underlying field, which is a simple k-module.
3. Indecomposable $\not\Rightarrow$ Simple: this statement is equivalent to saying that not all short exact sequence spilt (i.e., not all modules are *projective*, see section 17.2). For a concrete example, let $\mathbb{Z}^2$ be a module over $\mathbb{Z}$ with the action:

   $$r \cdot (a, b) = (a + rb, b)$$

   Then it should be checked that it is not simple since $\mathbb{Z} \times \{0\}$ is a submodule, but it is indecomposable[4].
4. Every semi-simple module is a semi-indecomposable module, however the converse is not true: $\mathbb{Z}$ is a semi-indecomposable $\mathbb{Z}$-module (more genreally any module module over a PID is semi-indecomposable due to the Decomposition Theorem from chapter 14), however $\mathbb{Z}$ is not semi-simple a semi-simple $\mathbb{Z}$-module as $p\mathbb{Z}$ is a submodule (the general case will be immediate after lemma 51.1.7)[5]

As stated in the introduction, we want to determine a semi-simple module or semi-simple modules in terms of the ring that is acting on it. The following defintion was made for this purpose:

[4]Hint: Is it possible to write $\mathbb{Z} = n\mathbb{Z} \oplus m\mathbb{Z}$?
[5]This is a version of Krull-Schmidt's theorem for the decomposition of Noetherian or Artinian modules into the direct sum of indecomposable

Definition 51.1.2: Semi-Simple Ring

Let R be a ring. Then R is *semi-simple* if every R-module is semi-simple. Recall that R is *simple* if the only two-sided ideals are (0) and R.

Notice how a semi-simple ring was *not* defined as the direct product of simple rings. In the following proofs, we will show that the definition of semi-simple ring implies that R is the direct product of simple rings, however the converse is not true: there exists simple rings that are *not* semi-simple modules over themselves (and hence not simple R-modules). The problem is that of dimension; we will soon explore the relation between being semi-simple and Noetherian/Artinian. If we impose the condition that R be Noetherian and Artinian, then we are able to say that R is a semi-simple ring if and only if R is the direct product of simple rings.

We next establish many properties of simple and semi-simple modules, and show how closely related they are to the ring acting on them:

Lemma 51.1.3: Simple Module Characterization

Let M be an R-module. Then M is a simple R-module if and only if there exists a two-sided maximal ideal $\mathfrak{m} \subsetneq R$ such that

$$M \cong_R R/\mathfrak{m}$$

***Proof*:**

($\Rightarrow$) Let M be a simple [non-trivial] R-module so that the only sub-modules of M are $\{0\}$ and M. Consider any $0 \neq x \in M$ and the R-module homomorphism:

$$\varphi : R \to M \qquad r \mapsto r \cdot x$$

Since $\varphi(R) \subseteq M$ is a sub-module and M is a non-trivial R-module, it must be that $\varphi(R) = M$. Hence:

$$R/\ker(\varphi) \cong M$$

But then, $R/\ker(\varphi)$ cannot have any submodules, since M doesn't, and so $\ker(\varphi)$ must be maximal.

($\Leftarrow$) Let $R/\mathfrak{m} \cong M$ as R-module where $\mathfrak{m}$ is maximal. If $0 \subsetneq N \subsetneq M$, then there would exist an ideal $\subsetneq J \subsetneq R/\mathfrak{m}$, but that would make $\mathfrak{m} \subseteq J$, contradicting maximality of $\mathfrak{m}$.

If R is commutative, then $R/\mathfrak{m}$ is a field. For the non-commutative case, $R/\mathfrak{m}$ need not be a field, not even a division ring, the canonical example being $M_n(k)$ for $n \geq 2$ (see the remark ref:HERE or go over section 9.4.3). Do not confuse the isomorphism in the above statement as a ring isomorphism: the structure of M is isomorphic to the R-module structure of left-multiplication on $R/\mathfrak{m}$. Notice how this also shows that a simple R-module M is cyclic since $R \cdot \varphi(1) = \varphi(R \cdot 1) = \varphi(R/\mathfrak{m}) = M$[6]. If $R = k$ is a field, then M is isomorphic to a 1-dimensional vector-space over k. There do exist non-commutative rings R that are *not* of the form $M_n(k)$; we shall explore in ref:HERE that there is a family of quaternion algebras that are degree two extensions of a field k that is simple ring (more specifically, a simple k-algebra).

[6] In fact, any element must generate it, can you see why?

Using this result, we see that any semi-simple R-module is of the form:

$$M \cong_R \bigoplus_{i \in I} R/\mathfrak{m}_i$$

for maximal ideals $\mathfrak{m}_i \subseteq R$. In fact, we can further characterize $\mathfrak{m}_i$:

Corollary 51.1.4: Simple Module's And Annihilators

Let M be a simple left (or right) R-module. Then $M \cong R/\text{Ann}(m)$ for any nonzero m, and $\text{Ann}(m)$ is a maximal ideal.

***Proof*:**
Since M is a simple left R-module, for any $0 \neq m \in M$, $mR \subseteq M$ is a submodule (by identifying R in M by the map $r \mapsto r \cdot 1$). As it is nonzero and M is simple, $M \cong_R Rm$. Then we have a surjective R-module homomorphism

$$R \to Rm \qquad r \mapsto r \cdot m$$

Thus, $R/\mathfrak{m} \cong_R Rm$ for some ideal $\mathfrak{m}$ (note that the R-submodules of R are ideals). In particular, $\varphi(\mathfrak{m}) = \mathfrak{m}m = 0$, so $\mathfrak{m} = \text{Ann}(m)$ by definition. Thus, we have the isomorphism chian:

$$M \cong_R Rm \cong_R R/\text{Ann}(m)$$

Finally, since Rm is simple, $R/\text{Ann}(m)$ must have no other two-sided ideal,s i.e. $\text{Ann}(m)$ is a two-sided maximal ideal, completing the proof.

Since we've seen the limitations of the structure of simple modules, it is also natural to ask if there are any limitations on the structure of the hom-set given two simple modules:

Lemma 51.1.5: Schur's Lemma

Let M and N be simple R-modules, $\varphi : M \to N$ an R-module homomorphism. Then either $\varphi = 0$ or φ is an isomorphism. As a consequence, $\text{End}_R(M)$ is a division ring.

***Proof*:**
Since N is simple, either $\varphi(M) = N$ or $\varphi(M) = 0$. If $\varphi(M) = N$, it has to be that $\ker(\varphi) = 0$, since M is simple, and hence φ would be an isomorphism.

As a consequence, every element in $\text{End}_R(M)$ is either 0 or is a unit, i.e. has a multiplicative inverse, which by definition makes $\text{End}_R(M)$ a divising ring under $(+, \circ)$.

Since $M \cong_R R/\mathfrak{m}$ for some maximal ideal $\mathfrak{m} \subsetneq R$, we have that $\text{End}_R(R/\mathfrak{m})$ is always a division ring, whether R is commutative or not. In a sense, this can be thought of as "generalization" to proposition 9.4.19, where it was shown that $R/\mathfrak{m}$ is a field if R is commutative.

Notice that if $M = V$ was an 1-dimensional vector-space over k, then $\text{End}_R(V)$ is a known division ring, in particular it's ring-isomorphic to k. If V is a finite dimensional semi-simple module over k, i.e. it is n-dimensional, then $\text{End}_R(V)$ is ring-isomorphic to $M_n(k)$. We might hope for something similar for the structure of $\text{End}_R(M)$ is M is semi-simple. Take a moment to ponder this before reading the next lemma (hint: recall proposition 12.3.36)

Lemma 51.1.6: Hom's of Semisimple Modules

Let M be a semisimple R-module, so that:

$$M = S_1^{\oplus n_1} \oplus S_2^{\oplus n_2} \oplus \cdots \oplus S_k^{\oplus n_k}$$

where each S_i are simple R-modules, $S_i \not\cong S_j$. Then as a ring:

$$\text{End}_R(M) \cong_{\textbf{Ring}} \text{End}_R(S_1^{\oplus n_1}) \oplus \cdots \oplus \text{End}_R(S_k^{\oplus n_k})$$

Furthermore, for each $S_i^{\oplus n_i}$, we have that direct sums "pull out" to become matrices:

$$\text{End}_R(S_i^{\oplus n_i}) \cong_{\textbf{Ring}} M_{n_i}(\text{End}_R(S)) = M_{n_i}(D_i)$$

where $D = \text{End}_R(S_i)$ and $M_{n_i}(D_i)$ is the matrix ring over the division ring $\text{End}_R(S_i)$.

***Proof*:**
We invoke proposition 12.3.36 to get the result in terms of groups, and then use Schur's lemma and check that the ring operation is compatible:

$$\begin{aligned}
\text{End}_R(M) &= \text{Hom}_R\left(\bigoplus_i S_i^{\bigoplus n_i}, \bigoplus_j S_j^{\bigoplus n_j}\right) \\
&\cong_{\textbf{Grp}} \bigoplus_{i,j} \text{Hom}_R\left(S_i^{\bigoplus n_i}, S_j^{\bigoplus n_j}\right) \\
&\cong_{\textbf{Grp}} \bigoplus_{i,j} \text{Hom}_R\left(S_i, S_j\right)^{\bigoplus n_i n_j}
\end{aligned}$$

where when $S_i \neq S_j$, by Shur's lemma we must have:

$$\text{Hom}_R(S_i, S_j)^{n_i n_j} = 0$$

Hence we get that $\text{End}_R(M) \cong_{\textbf{Grp}} \oplus_i \text{Hom}_R(S_i^{n_i}, S_i^{n_i}) = \oplus_i \text{End}_R(S_i^{n_i})$. Next, this map is compatible with the ring structure, since the map:

$$\varphi \to (\varphi|_{S^{n_i}} : S_i^{n_i} \to S_i^{n_i} \subseteq M)$$

is clearly clearly compatible with composition, and hence

$$\text{End}_R(M) \cong_{\textbf{Ring}} \bigoplus_i \text{End}_R(S_i^{n_i})$$

For part 2, we follow the same steps as for part (1): re-indexing so that $S^n = S_1 \oplus S_2 \oplus \cdots \oplus S_n$ so that $S_i = S_j = S$, we get

$$\text{Hom}_R\left(\bigoplus_i S_i, \bigoplus_j S_j\right) \cong_{\textbf{Grp}} \bigoplus_{i,j} \text{Hom}_R(S_i, S_j) \stackrel{!}{=} \bigoplus_{i,j} D \cong_{\textbf{Grp}} M_n(D)$$

where $\overset{!}{=}$ comes from Shur's lemma with $D = \mathrm{End}_R(S)$ since $S_i = S_j = S$. The last equality comes from the fact that the matrix $M_n(D)$ when treated like a group is simply re-arranging where the addition is happening in $\bigoplus_{i,j} D$. To show that the ring structure is compatible, observe that

$$\varphi \mapsto \varphi_{ji}$$

is the desired map, where

$$\varphi \begin{pmatrix} s_1 \\ s_2 \\ \vdots \\ s_n \end{pmatrix} = \begin{pmatrix} \varphi_{11} & \varphi_{12} & \cdots & \varphi_{1n} \\ \varphi_{21} & \varphi_{22} & \cdots & \varphi_{2n} \\ \vdots & \vdots & \ddots & \cdots \\ \varphi_{n1} & \varphi_{n2} & \cdots & \varphi_{nn} \end{pmatrix} \begin{pmatrix} s_1 \\ s_2 \\ \vdots \\ s_n \end{pmatrix}$$

which is simply a formalization of what the group-isomorphism give us, showing us that composition is compatible, completing the proof. (see this for a more detailed explanation)

Hence, for any semi-simple R-module M, we get that:

$$\mathrm{End}_R(M) \cong_{\mathbf{Ring}} \prod_i M_{n_i}(D_i) \cong_{\mathbf{Ring}} \prod_i \mathrm{End}_R(S_i^{\oplus n_i})$$

and since each S_i is a simple R-module, $S \cong_R R/\mathfrak{m}_i$ for appropriate maximal ideal $\mathfrak{m}_i \subseteq R$, $\mathrm{End}_R(S_i) \cong \mathrm{End}_R(R/\mathfrak{m}_i)$. Thus:

$$\mathrm{End}_R(M) \cong_{\mathbf{Ring}} \prod_i M_{n_i}(\mathrm{End}_R(R/\mathfrak{m}_i)) = \prod_i M_{n_i}(D_i)$$

If $R = k$ is a field making $M = V$ a vector-sapce over k (say n-dimensonal), then the only simple k-submodule up to isomorphism is k, and so this result reduces to:

$$\mathrm{End}_R(V) \cong_{\mathbf{Ring}} M_n(\mathrm{End}_k(k/(0))) = M_n(k)$$

where $\mathrm{End}_k(k) \cong_{\mathbf{Ring}} k$, aligning with the results we found in secion 13.2. This shows that when generlizing from linear algebra, due to the fact that a ring is non-commutative, we may have multiple non-isomorphic maximal ideals giving us multiple products of matrix rings. With a better understanding of simple modules and their endomorphisms, we continue by characterizing semi-simple modules:

Lemma 51.1.7: Semi-Simple Module Equivalence

Let M be a semi-simple module. Then the following are equivalent:

1. M is an external direct sum of simple modules:
$$M \cong \bigoplus_{i \in I} S_i$$
2. M is is the internal direct sum of simple modules:
$$M = \bigoplus_{i \in \mathfrak{m}} S_i$$
3. If $N \subseteq M$, there exists a complimentary module, that is there exists another submodule $N' \subseteq M$ such that
$$M = N \oplus N'$$
i.e., every semi-simple R-module M is *projective* and *injective*.

***Proof*:**
(1) $\Rightarrow$ (2) is proposition 12.3.6.

For (2) $\Rightarrow$ (3), Let $N \subseteq M$. Then using Zorn's lemma, we take a maximal submodule $N' \subseteq M$ ordered by inclusion with the property that $N \cap N' = 0$ (upper-bounds naturally exist in this set, and 0 is in this set, and hence Zorn's lemma does apply). If $M = N + N'$, we're done, since we can use the isomorphism from proposition 12.3.6. For the sake of contradiction, let's say $N + N' \subsetneq M$. Then since $M = \sum_i S_i$, there must exist some $i \in I$ and $s_i \in S_i$ such that $s_i \notin N + N'$, since $0 \in S_j$ for all $j \in I$. Hence $S_i \subsetneq N + N'$, so $S_i \cap (N + N') \subsetneq S_i$, however S_i is simple, and hence:
$$S_i \cap (N + N') = 0$$
But then:
$$(S_i + N') \cap N \subseteq N' \cap N = 0$$
Thus, by maximality of N', $S_i + N' = N'$, so $S_i \subseteq N' \subseteq N + N'$, contradicting the assumption that $s_i \notin N + N'$, as we sought to show.

For (3) $\Rightarrow$ (1), Let $\mathcal{S}$ be the collection of simple submodule of M where their direct sum maps injectively into M. Since $0 \in \mathcal{S}$ and taking the sum's of all the modules is an upper bound, Let N be the maximal submodule with this property. By proposition 12.3.6,
$$N = \sum_{i \in I} S_i \cong \bigoplus_{i \in I} S_i$$
If the map is surjective, we are done. For the sake of contradiction, suppose it is not. Then by part (3), there exists an N' such that $M = N + N'$. If $N' = 0$, we found a contradiction. So, let's say $N' \neq 0$. Then it is enough to show that N' contains a simple module, say S, since then we can make the collection $\{(S_i)_{i \in I}, S\}$ which will map injectively into M, contradicting maximality.

(here is what the professor posted Suppose every submodule N of M has a complement (i.e. a submodule N' of M such that $M = N \oplus N'$ (internal direct sum)). Then every nonzero submodule N' of M contains a simple submodule.

Proof: first we show that every submodule K of M has the same property (every submodule has a complement): suppose L is a submodule of K. by assumption, $M = L \oplus L'$ for some submodule L' of M. then note that $K = L \oplus (L' \cap K)$.

Now, suppose N' is nonzero submodule of M. Take a nonzero $x \in N'$, so Rx is a nonzero submodule and by 1st isomorphism theorem we have $Rx \cong_R R/I$ for some proper left ideal I. By the previous paragraph, every submodule of Rx has a complement, i.e. every submodule of R/I has a complement. Take any maximal left ideal $\mathfrak{m}$ containing I. Then M/I is a submodule of $R/\mathfrak{m}$ with complement L' in R/I, i.e. $R/I = \mathfrak{m}/I \oplus L'$, so $L' \cong (R/I)/(\mathfrak{m}/I \cong R/\mathfrak{m}$ is simple, i.e. L' is a simple submodule of N' (via the isomorphism $Rx \cong R/I$))

This theorems shows that every vector-space over k is a semisimple k-module, however by remark internalSumVsSumRmk, $\mathbb{Z}$ is *not* a semi-simple $\mathbb{Z}$-module, since $2\mathbb{Z}$ is a submodule with no compliment (showing that not all free modules are semi-simple).We can characterize semi-simple rings by saying that R is semi-simple if and only if every R-module is projective. This also shows us that $\mathbb{Z}$ is not a semi-simple ring; though $\mathbb{Z}$ is projective over itself, any finite ablıean group is a non-projective $\mathbb{Z}$-module. Thus, a module of a semi-simple ring is stronger then being a projective module. Hence being semi-simple is stronger (in a sense much stronger) than being projective, but may not be free: even though many example of semi-simple modules are free-modules, we would require the underlying ring to be semi-simple over itself. In fact, it is enough that the underlying ring be a semi-simple module over itself, giving us an easier condition for defining a semi-simple ring! We first intorduce an important result we will use:

Corollary 51.1.8: Extension Closure Of Semisimple Modules

Let M an R-module. Then submodule and quotient modules of M are semisimple if and only if M is semi-simple. Furthermore, if $(M_i)_{i\in I}$ is a collection of semi-simple modules, then $\bigoplus_{i\in I} M_i$ is a semi-simple module.

***Proof*:**
The direct sum of semi-simple modules is semi-simple by definition, and so we check the other two statements.

Let M be a semi-simple R-module so that $M = \bigoplus_{i\in I} S_i$ is by lemma 51.1.7. First, consider a surjection $M \twoheadrightarrow N$.

$$\varphi(M) = \varphi\left(\bigoplus_{i\in I} S_i\right) = \bigoplus_{i\in I} \varphi(S_i)$$

then by Schur's lemma, either $\varphi(S_i) = 0$ or $\varphi(S_i) \cong S_i$, and hence N is semisimple.

If $N \subseteq M$, then by lemma 51.1.7, we have that $M = N \oplus N'$, and so by proposition 12.3.6

$$M/N' \cong N$$

which we just showed makes N semi-simple.

Using these two facts, it is not difficult to show that if $N \subseteq M$ and $M/N \cong N'$ are semi-simple, then so is M, where if $x \in M$, $x = \sum_i n_i + \sum_j n'_j$.

These properties of semi-simple modules allow us to characterize semi-simple rings much more easily:

Corollary 51.1.9: Semi-Simple Rings Condition

Let R be a ring. Then R is a semi-simple ring if and only if R is a semi-simple R-module, that is we only need to check that R is a semi-simple R-module for one R-module structure (namely R itself)

***Proof*:**
By definition, R if R is a semi-simple ring then any R-module is a semi-simple R-module, including R over itself.

Now R is given some R-action becomes a semi-simple R-module, then by corollary 51.1.8 so is $\oplus_{i\in I} R$. We will use this to show that any R-module M is semi-simple, which will make R semi-simple by definition.

Let M is any other R-module, by by the universal property of the direct sum and free-modules:

$$M = \bigoplus_i Rm_i$$

for appropriate m_i (for ex. we can pick all nonzero elemetnts of M). Then we get a surjective map:

$$\bigoplus_{i\in I} R \to M \qquad (r_i)_{i\in I} \mapsto \sum r_i m_i$$

hence, M is the quotient of a semi-simple module, and hence semi-simple. Thus, R is semi-simple, as we sought to show.

Due to this, we get a nice description of semi-simple rings which is even stronger than semi-simple modules:

Lemma 51.1.10: Decomposing Semi-Simple Rings

Let R be a semi-simple ring. Then:

$$R = I_1 \oplus \cdots \oplus I_n$$

for a finite collection of I_k, where I_k are simple left-submodules of R, i.e. *minimal* left ideals[a] of R

[a]These are *left ideals* as we are working with *left* submodules

***Proof*:**
Since R is a semi-simple ring, it is a semi-simple R-module and so by definition:

$$R = \bigoplus_{i\in\mathcal{C}} I_i$$

for simple submodules I_i (i.e. minimal left ideals I_i). To make this a finite sum, since $1 \in R$, we get for an appropraite finite subset $A \subseteq \mathcal{C}$

$$1 \in I_{\alpha_1} + \cdots + I_{\alpha_n}$$

But then, since these are all ideals,

$$R = R \cdot 1 \subseteq I_{\alpha_1} \oplus \cdots \oplus I_{\alpha_n} \subseteq R$$

and hence $R = I_{\alpha_1} \oplus \cdots \oplus I_{\alpha_n}$, completing the proof.

51.1.11 Remark As a consequence of this, any semi-simple ring R is a left-Noetherian/Artinian ring. Interestingly, we will soon show that this also implies that R is a *right*-Noetherian/Artinian ring (see rightNoethArtRemark)

Furthermore, this shows how it is necessary for the condition that R be left-Noetherian/Artinian so that the notion of being semi-simple coincide between the two definitions.

Since we characterized simple R-modules for any ring R to be R-isomorphic to $R/\mathfrak{m}$, and we've given a structure of R if R is a semisimple ring, we can make lemma 51.1.3 more precise:

Corollary 51.1.12: Structure Of Simple Modules Of Semi-Simple Ring

Let R be a semi-simple ring so that $R = I_1 \oplus I_2 \oplus \cdots \oplus I_n$ for minimal left ideal I_k (equivalently, simple left R-modules). Then if M is a simple left R-module,

$$M \cong_{R\text{-}\mathbf{Mod}} I_i$$

for some $1 \leq i \leq n$.

***Proof*:**
Let M be a simple nonzero left R-module, and consider the R-module homomorphism:

$$\varphi : R \to M \qquad r \mapsto r \cdot x$$

where $x \neq 0$. Since M is simple, it is nonzero, so φ is surjective. Since $R = \bigoplus_i^n I_i$, for some i's

$$\varphi|_{I_i} \neq 0$$

and since $\varphi|_{I_i} = \pi_i \circ \varphi$ we have the R-module homomorphism:

$$\varphi|_{I_i} I_i \to M$$

and since both modules are simple and this is a nonzero homomorphism, by Schur's lemma this map must be an isomorphism, as we sought to show.

Note that there can be multiple I_i's that are isomrophic to M, which we will show in a following theorem.

This shows that any simple R-module over a semi-simple ring R is in fact isomorphic to a simple

R-module over R (i.e. R-isomorphic to a minimal ideal of R). Thus, studying simple R-modules over semi-simple rings comes down to studying the ring R itself, and that there are only *finitely many* simple R-modules. Furthermore, we have that $R/(\mathfrak{m}_1 \oplus \cdots \mathfrak{m}_{n-1}) \cong_R \mathfrak{m}_n$. Here is a good example of where the ring structure of the left hand side cannot exist in the right hand side. For example, if R is commutative, then $R/\mathfrak{m}$ is a field that is R-module isomorphic to a minimal ideal, but ideals cannot be fields (over the same identity[7]). Thus, in the case where R is semi-simple, for a maximal ideal $\mathfrak{m}$, $R/\mathfrak{m}$ is R-isomorphic to a minimal ideal of R.

Exercise 51.1.1

1. Let S be a simple module. Show that any nonzero submodule of the injective enveloppe $E(S)$ is indecomposable, but not simple.
2. Show that if M is a simple R-module, then it necessarily has to be cyclic (the converse, however, is not true).
3. Show that $A = \mathbb{Q}[x,y]/(xy - yx - 1)$ is a simple ring, but no a semi-simple ring (this is a special example of an algebra called a *Weyl* Algebra)
4. Let R be a ring, not necessarily commutative, and let $[R, R] = \{rs - sr \mid r, s \in R\}$. Show that the trace map induces an isomorphism

$$\frac{M_n(R)}{[M_n(R), M_n(R)]} \cong \frac{R}{[R, R]}$$

From this perspective, matrix rings are the natural “nonabelian” extension of a ring!

51.2 Artin-Wederburn

We just showed how studying simple and semi-simple R-modules comes down to studying the ring R, especially if the ring R is semi-simple. Thus, it is fruitful to classify semi-simple rings, which leads us to the main classification result for semi-simple rings, namely we build on lemma 51.1.10 and show that we can find more precisely how to represent each minimal ideal $\mathfrak{m}_i$, and furthermore show that this classification is exhaustive, and that any semi-simple ring uniquely breaks down into the result we'll show. We will then finish by showing that if we upgrade our ring to k-algebras (that is, we can put a vector-space structure on our ring), we get some stronger results due to the fact we can count dimensions.

First, we try to relate what we've shown for the hom-sets of simple modules with the strucure of simple modules. You may perhaps recall that we commented in exercise ref:HERE (in 9.1.1) and example 9.1.15 that R is note necessarily isomorphic to R^{op}. There is in fact a very nice way of linking R^{op} and R, which will be demonstrated after formally define R^{op} for reference

[7] $k \times \{0\} \subseteq k \times k$ is an ideal and a field with identity $(1, 0)$

Definition 51.2.1: Opposite Ring

Let R be a ring. Then we define a new ring labelled R^{op} that has the same underlying set and addition operation, whith the multiplication operation $*$ being:

$$x * y = yx$$

where the multiplication yx happens in R

If R is commutative, then $R = R^{\text{op}}$, and so we only get non-trivial results when R is not commutative. An example of $R \ncong R^{\text{op}}$ would be a free-algebra over two variables (say $k\langle x, y\rangle$).

Proposition 51.2.2: Properties Of Opposite Ring

Let R be a ring and R^{op} its oppposite ring. Then:

1. $(R^{\text{op}})^{\text{op}} = R$
2. $(R \times S)^{\text{op}} = R^{\text{op}} \times S^{\text{op}}$
3. $M_n(R)^{\text{op}} = M_n(R^{\text{op}})$ where the isomorphism is $A \mapsto A^t$ (essentially since $(AB)^t = B^tA^t$)

***Proof*:**
exercise

With this, we can make a link between R and R^{op}:

Lemma 51.2.3: Yonneda's Lemma for opposite Rings

Let R be a ring and consider the ring $\text{End}_R(R)$ to be the set of all R-module homomorphisms from R (as an R-module) to itself. Then $\varphi : R^{\text{op}} \to \text{End}_R(R)$, $r \mapsto (x \mapsto xr)$ is a ring isomorphism:

$$R^{\text{op}} \cong_{\textbf{Ring}} \text{End}_R(R)$$

***Proof*:**
Let φ_r be the given map. Let's first check that $\varphi_r \in \text{End}_R(R)$. Then we see that:

$$\begin{aligned}\varphi_r(\lambda x + \lambda y) &= (\lambda x + \lambda y)r \\ &= \lambda(xr) + \lambda(yr) \\ &= \lambda\varphi_r(x) + \lambda\varphi_r(y)\end{aligned}$$

showing that $\varphi_r \in \text{End}_R(R)$. Next, we check that φ is ineed a ring-homomorphism. By the properties of modules, it is easy to see that:

1. $\varphi_1 = \text{id}$, and hence φ sends the identity to the identity.
2. $\varphi_{r+r'} = \varphi_r + \varphi_{r'}$
3. $\varphi_{rr'} = \varphi_{r'} \circ \varphi_r$ (don't forget we are working with R^{op})

Finally, showing bijectivity, for injectivity we see that if $\varphi_r = 0$, then $0 = \varphi_r(1) = r$, and hence $\varphi_r = \varphi_0$. For surjectivity, let $\varphi \in \mathrm{End}_R(R)$. Let's say $\varphi(1) = r$. Then $\varphi(x) = x\varphi(1) = xr$, so $\varphi = \varphi_r$, completing the proof.

In the special case where R is commutative, then $R \cong R^{\mathrm{op}}$, showing that in this case $R \cong \mathrm{End}_R(R)$. Using this, we will use lemma 51.2.3 along with lemma 51.1.6 to upgrade lemma 51.1.10:

Theorem 51.2.4: Artin-Wedderburn, one Direction

Let R be a semisimple ring. Then:

$$R \cong_{\mathbf{Ring}} M_{n_1}(D_1) \times \cdots \times M_{n_k}(D_k)$$

where each D_i is a division ring and $M_{n_i}(D_i)$ is the matrix ring over D_i

The key realization about this theorem is that while the multiplication on the left hand side can "send you" all over the place, the multiplication on the right hand side only sends you back into your particular component. To see this more clearly, let's say R was a k-algebra for a field k so that it has a basis. If R was a semi-simple ring, then that means that when multiplying elements of R (so figuring out where the basis elements go while multiplying), we can isolate the problem down to the components on the right hand side, and figure out where they go through matrix multiplication!

***Proof*:**
First, by lemma 51.1.10, for we get simple R-submodule of R, i.e. minimal ideals of R,

$$R = I_1 \oplus \cdots \oplus I_n$$

Now, we can always re-order the sums (i.e. $I_1 \oplus I_2 \cong I_2 \oplus I_1$, since $R \cong I_1 \oplus \cdots \oplus I_n$ by proposition 12.3.6). Hence, we will group all the I_i's that belong to the same isomrophism class (as R-modules), letting us re-write R as:

$$R \cong_{\mathbf{Ring}} I_1^{n_1} \oplus \cdots \oplus I_k^{n_k}$$

where $I_i \ncong I_j$ for $i \neq j$. Hence by lemma 51.2.3 and lemma 51.1.6:

$$\begin{aligned} R^{\mathrm{op}} &\cong_{\mathbf{Ring}} \mathrm{End}_R(R) \\ &\cong_{\mathbf{Ring}} M_{n_1}(D_1) \times \cdots \times M_{n_k}(D_k) \end{aligned}$$

where $D_i = \mathrm{End}_R(I_i)$ is a divison ring. Finally, Taking the opposite ring on both sides, we get:

$$\begin{aligned} R &\cong_{\mathbf{Ring}} (R^{\mathrm{op}})^{\mathrm{op}} \\ &\cong_{\mathbf{Ring}} M_{n_1}(D_1)^{\mathrm{op}} \times \cdots \times M_{n_k}(D_k)^{\mathrm{op}} \\ &\cong_{\mathbf{Ring}} M_{n_1}(D_1^{\mathrm{op}}) \times \cdots \times M_{n_k}(D_k^{\mathrm{op}}) \end{aligned}$$

and it is clear the D_i^{op} is still a division ring, hence completing the proof.

Thus, if we have a simple-ring then the ring breaks down into the product of matrix rings over division rings. In the special case where $R = k$ is a field, then k has only one minimal ideal, namely itself, and $\mathrm{End}_k(k) \cong M_1(k) \cong k$, and hence we reduce to $k \cong k$. If $R = k^n$ is a product of the same field, then the minimal ideals of R are $R = k \oplus \cdots \oplus k$, so that we get back $k^n \cong_{\mathbf{Ring}} k^n$.

We will now show that the converse of this theorem is true, that the product of matrix rings over division rings is semi simple. We start by showing that matrix ring are simple modules[8].

Proposition 51.2.5: Matrix Ring over Division Rings are semi-simple

Let $R = M_n(D)$ be a matrix ring over a divison ring. Let $C = D^{\oplus n}$ be the left $M_n(D)$-module where the action is left matrix multiplication (imagine C as a column). Then C is a simple $M_n(D)$-module, and

$$M_n(D) \cong_{M_n(D)\text{-}\mathbf{Mod}} C^{\oplus n} = (D^{\oplus n})^{\oplus n}$$

showing that $M_n(D)$ is semisimple ring and $M_n(D) \cong_{\mathbf{Ring}} C^{\oplus n}$

***Proof*:**
Let $R = M_n(D)$ and $C = D^{\oplus n}$ (C for *column*). For notational simplicity, let $E_{ij} \in R$ be the elementary matrix which is 1 in position ij and zero everywehre else. Let $e_i \in C$ be the column where it is 1 at position i and zero everywehre else:

$$E_{ij} = \begin{pmatrix} 0 & & & & 0 \\ & \ddots & & \iddots & \\ & & 1 & & \\ & \iddots & & \ddots & \\ 0 & & & & 0 \end{pmatrix} \qquad e_i = \begin{pmatrix} 0 \\ \vdots \\ 1 \\ \vdots \\ 0 \end{pmatrix}$$

Then $E_{ij}e_k = \delta_{jk}e_i$ where δ_{jk} is the Kronecker Delta:

$$\delta_{jk} = \begin{cases} 1 & j = k \\ 0 & j \neq k \end{cases}$$

Now, we will start by showing that C is a simple $M_n(D)$-module. Since $C = D^{\oplus n}$,

$$C = \bigoplus_i De_i$$

for some basis $e_1, e_2, ..., e_n$. To show that C is simlpe, it is enough to show that for any $0 \neq x \in C$, $C = Rx$ since then any nonzero submodule would in fact be C. Given our above notation, we have that:

$$x = \sum_k d_k e_k$$

for appropriate $d_k \in D$, where at least one of these is nonzero, let's say $d_j \neq 0$. Then consider $E_{ij}x \in Rx$:

$$E_{ij}x = \sum_k \delta_{jk} d_k e_k = d_j e_i$$

since $d_j \neq 0$. Since D is a division ring, we get that $d_j^{-1}e_i \in Rx$, for all $1 \leq i \leq n$. But then, we get:

$$C = \sum De_i \subseteq Rx \subseteq C$$

[8]This will be an example of R being both a simple R-module and a simple ring

showing that $C = Rx$ for any $0 \neq x \in C$, showing that C is indeed simple.

Now, we can re-write this ring as the internal direct sum of matrices I_i which are zero outisde the ith column:

$$R = M_n(D) = I_1 \oplus \cdots \oplus I_n$$

$$I_k = \begin{pmatrix} 0 & \cdots & a_{1,k} & \cdots & 0 \\ 0 & & a_{2,k} & & 0 \\ \vdots & & \vdots & & \vdots \\ 0 & & a_{2,n-1} & & 0 \\ 0 & \cdots & a_{n,k} & \cdots & 0 \end{pmatrix}$$

This decomposition works since each I_i is stable under *left* multiplication by elements of $M_n(D)$ (since left multiplication is a *column* transformation, right multiplication would be *row* transformation). Finally, $C \cong_{R\textbf{-mod}} I_i$ by simply moving the column to the right place, and so each I_i is simple. Hence, we have decomposed $M_n(D)$ as a dierct product of simple left $M_n(D)$-modules, and hence $M_n(D)$ is a semisimple $M_n(D)$-module, which by corollary 51.1.9 makes $M_n(D)$ a semi-simple ring. Finally, since $C^{\oplus n}$ represents the n rows of n-length columns, we see that the action of $M_n(D)$ on $C^{\oplus n}$ is identical to that of matrix multiplying elements of $C^{\oplus n}$, which gives us that $M_n(D) \cong_{\textbf{Ring}} C^{\oplus n}$, as we sought to show.

51.2.6 Remark In fact, $M_n(D)$ is a simple ring, meaning the only *two-sided* ideals of R are 0 and R; the above proof can be modified to show this. This gives us a nice class of non-commutative simple rings (since all simple commutative rings are fields).

Notice that it is quite sepcial that the ring opration and the module operation coincided, this because of the special way in which we decompoed the $M_n(D)$-module $M_n(D)$ and by the fact that it is a module over itself. In the previous proposition, we showed that $D^{\oplus n}$ is a simple $M_n(D)$-module. In fact, all simple modules of $M_n(D)$ are isomorphic to $D^{\oplus n}$:

Corollary 51.2.7: Structure of Simple $M_n(D)$-Module

Let $R = M_n(D)$ be a matrix ring over a division ring. Then any simple $M_n(D)$-module is isomorphic to $D^{\oplus n}$.

***Proof*:**
Let M be a simple $M_n(D)$-module. By proposition 51.2.5

$$M_n(D) \cong_{\textbf{Ring}} D^{\oplus n} \oplus \cdots_{\textbf{Ring}} \oplus D^{\oplus n}$$

And by corollary 51.1.12,

$$M \cong D^{\oplus n}$$

Corollary 51.2.8: Endomorphism Of Simple $M_n(D)$-Module

Let S be a simple $M_n(D)$-module. Then

$$\mathrm{End}_{M_n(D)}(S) \cong_{\mathbf{Ring}} D^{\mathrm{op}}$$

***Proof*:**
Since S is simple, $S \cong_R D^{\oplus n}$ for some n. Let $\varphi_d : S \to S$, $\varphi_d(x) = x \cdot d$ where $x \cdot d$ is x times dI. Then:

$$\varphi_d(Ax) = (Ax) \cdot d = A(x \cdot d) = A\varphi_d(x)$$

and

$$\varphi_d(x+y) = (x+y) \cdot d = (x+y)dI = xdI + ydI = x \cdot d + y \cdot d = \varphi_d(x) + \varphi_d(y)$$

hence $\varphi_d \in \mathrm{End}_R(S)$. Conversely, if $\varphi \in \mathrm{End}_R(S)$, Since $S \cong D^{\oplus n}$, and

$$D^{\oplus n} \cong \begin{pmatrix} D^{\oplus n} & 0^{\oplus n} & \cdots & 0^{\oplus n} \end{pmatrix}$$

we will treat S as a $M_n(D)$ submodule of $M_n(D)$. Now, since φ is a group homomorphism, we know that:

$$\varphi \begin{pmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix} = \begin{pmatrix} \varphi_{11} & \cdots & \varphi_{n1} \\ \vdots & \ddots & \vdots \\ \varphi_{n1} & \cdots & \varphi_{nn} \end{pmatrix} \begin{pmatrix} 1 \\ \vdots \\ 0 \end{pmatrix} = \begin{pmatrix} \varphi_{11}(1) + \cdots + \varphi_{n1}(1) \\ \vdots \\ 0 \end{pmatrix}$$

hence let $d = \sum_i^n \varphi_{i1}(1)$, and let the right-action on $\mathrm{End}_{M_n(D)}(S)$ be right multiplication by the identity matrix dI. Then: Then for all $x \in S$, we have:

$$\begin{aligned}
\varphi \begin{pmatrix} x_1 & \cdots & 0 \\ \vdots & \ddots & \vdots \\ x_n & \cdots & 0 \end{pmatrix} &= \varphi \left(\begin{pmatrix} x_1 & \cdots & 0 \\ \vdots & \ddots & \vdots \\ x_n & \cdots & 0 \end{pmatrix} \begin{pmatrix} 1 & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & 0 \end{pmatrix} \right) \\
&\overset{!}{=} \begin{pmatrix} x_1 & \cdots & 0 \\ \vdots & \ddots & \vdots \\ x_n & \cdots & 0 \end{pmatrix} \varphi \begin{pmatrix} 1 & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & 0 \end{pmatrix} \\
&= \begin{pmatrix} x_1 & \cdots & 0 \\ \vdots & \ddots & \vdots \\ x_n & \cdots & 0 \end{pmatrix} \cdot d
\end{aligned}$$

where $\overset{!}{=}$ comes from the fact that we can treat that matrix as an element of $M_n(D)$, and since φ is an $M_n(D)$-module homomorphism, we can pull it out. But then, the action is completely dependent on on d, an so we have a bijection between right multiplication by $d \in D$ an $M_n(D)$-endomrophisms. It is an easy taks to check that:

$$\varphi_{dd'} = \varphi_{d'} \circ \varphi_d$$

showing that

$$D^{\mathrm{op}} \cong_{\mathbf{Ring}} \mathrm{End}_{M_n(D)}(S)$$

as we sought to show.

The next thing we will need is to show that if R, S are semisimple rings, then so is $R \times S$. Since R and S are semi-simple R-modules and S-modules, if we can show that $R \times S$ is a semi-simple $(R \times S)$-module, we are done. For this to be the case, we must be able to decompose $R \times S$ into simple $(R \times S)$-submodules. Forunately, many universal propeties play in our favour to be able to easily identify simple $(R \times S)$-modules of $R \times S$ given we know the simple R-modules and S-modules.

To see the key idea of the following proof, we define some terms

Definition 51.2.9: Idempotent And Orthogonal Elements

Let R be a ring. Then $e \in R$ is said to be *idempotent* if

$$e^2 = e$$

Furthermore, if e_1, e_2 are idempotent elements in R, if

$$e_1 e_2 = e_2 e_1 = 0$$

then they are said to be *orthogonal.* If e is idempotent and cannot be written as the sum of two (commuting) idempotent elements, then e is said to be a *primitive* idempotent element.

Example 51.2.10: Idempotent Elements

Naturally, for any ring R, $1 \in R$ is an idempotent element.

For a more interesting example the product ring $R \times S$, and take $e = (1, 0) \in R \times S$, $f = (0, 1) \in R \times S$. Then:

1. $e^2 = e$ and $f^2 = f$ (i.e. they are both *idempotent*)
2. $ef = fe = 0$ (implying also that $ef \in Z(R \times S)$)
3. $e + f = 1$ (i.e. e and f are *not* primitive)

What happens if the number of terms in the product increases?

Next, if M is an R-module an N is an S-module, we can M and N to be $(R \times S)$-module by ignoring what happens to S (resp. R), i.e. think of the action being defined by the following function

$$R \times S \twoheadrightarrow R \to \mathrm{End}_{\mathbf{ab}}(M)$$

and similarly for a S-module N. Hence, we can define $M \oplus N$ as an $(R \times S)$-module. Conversely, if M is an $(R \times S)$-module, then using what we've shown in example 51.2.10, given $e = (1, 0)$ and $f = (0, 1)$:

$$M = eM + fM$$

Notice that eM is killed by Rf, so it's a $A \cong A \times B/(\{0\} \times B)$-module, and similarly for fM. In other words, idempoent elements characterize the splitting of modules! This gives us the following:

Proposition 51.2.11: Direct Product Decomposition

If A-**Mod** and B-**Mod** are the categoies of A-modules and B-modules, and A-**Mod** $\times$ B-**Mod** is the category where every object and morphism is a product of objects or morphisms. Then there is an equivalence:

$$A\textbf{-Mod} \times B\textbf{-Mod} \underset{\psi}{\overset{\varphi}{\rightleftarrows}} (A \times B)\textbf{-Mod}$$

sending:

$$(M, N) \mapsto M \oplus N$$
$$(eM, fM) \leftarrow\!\shortmid M$$

where we further get: $\varphi(\psi(M \oplus N)) \cong M \oplus N$ and $\psi(\varphi(M)) \cong M$

***Proof*:**
easy to check

Proposition 51.2.12: Classifying Simple $(A \times B)$-Modules

Let P be a simple $(A \times B)$-module. Then

$$P \cong_{A\times B\text{-Mod}} M \oplus (0) \qquad \text{or} \qquad P \cong_{A\times B\text{-Mod}} (0) \oplus N$$

where M is a simple A-module and N is a simple B-module.

***Proof*:**
By proposition 51.2.11, $P \cong_{A\times B\text{-Mod}} M \oplus N$ where M is an A-module and N is a B-module. BUt then:

$$P \cong_{A\times B\text{-Mod}} (M \oplus (0)) \oplus ((0) \oplus N)$$

since P is simple, one of these has to be zero, and the other one has to be simple.

Corollary 51.2.13: Product Of Semi-Simple Rings

Let R and S be semi-simple rings. Then $R \times S$ is a semi-simple ring. Furtheromre, if $R_1, R_2, ..., R_n$ are the simple R-module up to isomorphism and $S_1, S_2, ..., S_m$ are the simple S-modules, then the simple $R \times S$-modules are of the form:

$$R_i \oplus (0) \qquad (0) \oplus (S_j)$$

***Proof*:**
By proposition 51.2.11, since R is a semi-simple R-module and S is a semi-simple S-module, $R \times S$ is a semi-simple $(R \times S)$-module, and so $R \times S$ is a semi-simple ring. Then by proposition 51.2.12,

if $P \subseteq R \times S$ is a simple $(R \times S)$-module, then they it must be of the form:

$$R_i \oplus (0) \qquad (0) \oplus (S_j)$$

Corollary 51.2.14: Structure Of $\prod_i M_{N_i}(D_i)$: Artin Wedderburn part II

Let $R = M_{n_1}(D_1) \times \cdots \times M_{n_k}(D_k)$. Then

1. R is semi-simple
2. There exist exactly k simple R-modules up to isomorphism, each of the form:

$$0 \oplus \cdots \oplus D_i^{\oplus n_i} \oplus \cdots \oplus 0$$

***Proof*:**
The ring R is semi-simpleby corollary 51.2.13 using induction. Furthermore, each $M_{n_i}(D_i)$ has $D_i^{\oplus n_i}$ is it's unique simple $M_{n_i}(D_i)$-module, since there are k matrix rings in the productt, there must be exactly k simple R-modules, as we sought to show.

51.2.15 Remark We can as before get that

$$R^{\text{op}} \cong M_{n_1}(D_1^{\text{op}}) \times \cdots \times M_{n_k}(D_k^{\text{op}})$$

and hence R^{op} is left Noetherian/Artinian, which we know now is true if and only if R is right Noetherian/Artinian (Since a left R^{op})-module is a right R-module). By leftNoethArtRemark, we have that any semisimple ring is left-Noetherian/Artinian, and hence any semisimple ring is both left and right Noetherian/Artinian, even though R is not necesarily commutative!

Thus, we have the following statemnt, were the last thing we need to prove is uniquness

Theorem 51.2.16: Artin-Wedderburn

Let R be a ring. Then R is a semi-simple ring if and only if

$$R \cong_{\mathbf{Ring}} M_{n_1}(D_1) \times \cdots \times M_{n_k}(D_k)$$

where each D_k is a division ring and $M_{n_i}(D_i)$ is the matrix ring over D_i, and each D_i n_i are unique up to isomorphism

***Proof*:**
All that is left to prove is the uniqueness. Let:

$$M_{n_1}(D_1) \oplus \cdots \oplus M_{n_s}(D_s) \cong_{\mathbf{Ring}} R \cong_{\mathbf{Ring}} M_{n'_1}(D'_1) \oplus \cdots \oplus M_{n_s}(D'_s)$$

By corollary 51.2.14, R has exactly s and t simple module, and so $s = t$. Hence, for any $D_i^{\oplus n_i}$, there must exist a $(D'_i)^{\oplus n'_i}$ such that

$$D_i^{\oplus n_i} \cong_R (D'_i)^{\oplus n'_i}$$

Since $D_i^{\oplus n_i} \cong_R (D_i')^{\oplus n_i'}$, $\mathrm{End}_{M_{n_i}(D_i)}(D_i^{\oplus n_i}) \cong_{\mathbf{Ring}} \mathrm{End}_{M_{n_i'}(D_i')}((D_i')^{\oplus n_i'})$. By corollary 51.2.8,

$$D_i^{\mathrm{op}} \cong_{\mathbf{Ring}} \mathrm{End}_{M_{n_i}(D_i)}(D_i^{\oplus n_i}) \cong_{\mathbf{Ring}} \mathrm{End}_{M_{n_i'}(D_i')}((D_i')^{\oplus n_i'}) \cong_{\mathbf{Ring}} (D_i')^{\mathrm{op}}$$

which, taking op on both sides, we see that:

$$D_i \cong_{\mathbf{Ring}} D_i'$$

Finally, we need to show that $D^{\oplus n} \cong_R D^{\oplus m}$ implies $n = m$. Treating $D^{\oplus n}$ and $D^{\oplus m}$ as D-modules, then we get that each have a basis of dimension n and m. Importantly, the results of the Replacement Theorem (theorem 13.1.12) applies to divions rings (notice that commutativity was not necessary), and so corollary 13.1.14 and theorem 13.1.8 give us that $n = m$, completing the proof.

This theorem also shows us that if R is a [finitely-generated] semi-simple ring, then R is indeed the direct product of simple rings (since $M_{n_i}(D_i)$ is a simple ring). Notice too that this theorem just gave use the answer to the classification of all *finite simple rings*: namely they would have ot be of the form $M_n(\mathbb{F}_q)$[9] for some finite field of order q (which is a power of some prime p). Contrast this to how hard it is to find all finite simple groups! Furthermore, the finite dimensional assumption was key: the *Weyl Algebra*:

$$W = k\langle x, y\rangle/(xy - yx - 1)$$

is a simple algebra that is not the matrix ring of a division algebra.

Example 51.2.17: Artin Wedderburn

$\mathrm{End}_{M_2(k)}(M_2(k)) = \mathrm{End}_{M_2(k)}((k^2)^{\oplus 2}) = M_2(\mathrm{End}_{M_2(k)}(k^2)) = M_2(k)$

Corollary 51.2.18: Artin-Wedderburn For k-Algebras

Let R be a semi-simple k-algebra for some field k. Then

$$R \cong_{k\text{-}\mathbf{Alg}} M_{n_1}(D_1) \times \cdots \times M_{n_l}(D_l)$$

where each D_i is a divsion k-algebra, finite dimensional as a k-vector space:

$$\dim_k R = \sum n_i^2 \dim_k(D_i)$$

and has exactly l simple R-modules as we've classified before.

***Proof*:**

By the Artin-Wedderburn theorem, we have that

$$R \cong_{\mathbf{Ring}} M_{n_1}(D_1) \times \cdots \times M_{n_l}(D_l)$$

Let's now further assume R is a k-algebra for some field k, so that there is a map $k \hookrightarrow Z(R)$.

[9]Why is the division ring a field? The answer lies in proposition 9.1.20

Then by the above isomorphism:

$$\begin{aligned} k \mapsto Z(R) &\cong Z(M_{n_1}(D_1) \times \cdots \times M_{n_l}(D_l)) \\ &\cong Z(M_{n_1}(D_1)) \times \cdots \times Z(M_{n_l}(D_l)) \\ &\cong Z(D_1) \times \cdots \times Z(D_l) \end{aligned}$$

where the last one comes form the fact that $Z(M_{n_i}(D_i))$ are the diagonal elements, and $z \mapsto zI$. This shows that D_i each have a natural k-algebra strucure induced by R, and so the above isomrophsim was already a k-algebra isomorphism:

$$R \cong_{k\text{-}\mathbf{Alg}} M_{n_1}(D_1) \times \cdots \times M_{n_l}(D_l)$$

Finally, treating $M_{n_i}(D_i)$ as k-vector spaces we have:

$$M_{n_i}(D_i) \cong_{k\text{-}\mathbf{Vect}} D^{\oplus n_i^2}$$

which gives the desired result, completing the proof.

Corollary 51.2.19: Special Case: k Algebraically Closed

Let $k = \overline{k}$ be an algebraically clsoed field and D a k-algebra that is finite-dimensional as a k-vector-space. Then $D = k$

***Proof*:**
Let's say $k \subsetneq D$. Then pick any $\delta \in D \setminus k$. Then

$$k \subseteq k[\delta] \subseteq D$$

the key thing to observe is that $k[\delta]$ is still commutative!! This is essentially the same reasoning for why any cyclic group is commutative! Since D is a finite-dimensional vector-sapce over k, some power of δ^n must be a k-linear combination of every smaller term. Since D is a division ring, every δ has an inverse, and so every element of $k[\delta]$ is in fact a unit. Hence $k[\delta]/k$ is an algebraic field extension. However, k is an *algebraically cloased field*, and hence $k[\delta] = k$. Since δ was arbitary, we must have that $D = k$, as we sought to show.

Corollary 51.2.20: Artin-Wedderburn For $\overline{k}$-Algebras

Let R be a semi-simple k-algebra for some algebraically closed field $k = \overline{k}$. Then

$$R \cong_{k\text{-}\mathbf{Alg}} M_{n_1}(k) \times \cdots \times M_{n_l}(k)$$

showing that R is a finite dimensional vector-space over R with dimension:

$$\dim_k(R) = \sum_{i}^{l} n_i^2$$

and has exactly l simple R-modules as we've classified before.

As a consequence of this, if R is a semi-simple $\mathbb{C}$-algebra, the simple R-modules are all R-isomorphic to $\mathbb{C}^{n_i}$ for appropriate power n_i. As a final word of warning, it is natural to wonder if all simple rings are of the form $M_n(D_i)$

Exercise 51.2.1

1. Let k be a field, k^n be a direct product of fields a k^n-module over itslef. Let $0 \oplus \cdots \oplus k \oplus \cdots \oplus 0 = k_i \subseteq k^n$. Then $\text{End}_{k^n}(k_i) \cong_{\mathbf{Ring}} k$, $\text{End}_{k^n}(k^n) \cong_{\mathbf{Ring}} k^n$, and $\text{End}_{k^n}(k^n) \cong_{\mathbf{Ring}} M_n(\text{End}_{k^n}(k_i)) \cong_{\mathbf{Ring}} M_n(k)$. But that would imply $k^n \cong_{\mathbf{Ring}} M_n(k)$ – what's the contradiction?

2. Let k be an algebraically closed field, and let R be a finite-dimensional k-algebra. Let M be a simple R-module. Prove that if r is an element of the centre $Z(R)$ of R, there exists $c \in k$ such that $rm = cm$ for all $m \in M$

51.3 Central Simple Algebras

Definition 51.3.1: Algebra Representation

Let A be a finite dimensional (associative) algebra over k. Then a *representation* of A is a k-homomorphism:

$$\rho : A \to \text{End}_k(V)$$

for some vector space V.

Recall that one definition of an R-module is given by a ring homomorphism $\rho : R \to \text{End}_{\mathbf{Ab}}(M)$ for an abelian group M. If we restrict to k-homomorphisms and k-endomorphisms, we can define the A-module $\rho : A \to \text{End}_k(V)$, hence we can think of representation of and algebra A as a left A-modules. This makes sense, as representation theory can be thought of the study of associative algebras (the study of group representation is the study of group algebras, the study of the representation of lie algebras is universal enveloping algebras, the study of the representation of quiver algebras is the study of path algebras, and so forth). Because of this, we shall see that many of the results for the representation of associative algebras outlined bellow to be familiar to anyone who has taken a basic course in representation theory.

Next, we define the object that shall represent the generalization of number rings:

Definition 51.3.2: Central Simple Algebra

Let A be an algebra over a field k. Then A is called *simple* if it is a simple ring. It is called a *central simple algebra* (CSA) if it is simple, is finite dimensional over k, and its center is k.

The idea behind this definition is that it generalizes field extensions F/k, for we have a distinguished field k, and we are interested in the "algebraic case", hence the extension ought to be finite. Contrasting to fields, a CSA need not be commutative and need be a division ring

Example 51.3.3: Simple and Central Simple Algebras

1. Naturally, $M_n(k)$ is a central simple algebra of dimension n^2
2. It is possible to be simple and have center k but not be finite: the Weyl algebra $k[x, \delta_x]$ is infinite dimensional over k.
3. $\mathbb{C}$ is a central simple algebra over $\mathbb{C}$, but not over $\mathbb{R}$ (as it's center contains more than $\mathbb{R}$
4. Let $A = \mathbb{H}$ be the quaternions (or Hamiltonians). Then this is a CSA over $\mathbb{R}$ with dimension 4. We shall soon define a quaternion algebra, and see that all 4-dimensional central simples algebra's over a field k is a quaternion algebra, and it shall be limited to 2-by-2 matrices or division algebras.

Let us now generalize some of basic representation theory results. Let E be an irreducible representation of A (take for example a minimal left ideal of A).

Lemma 51.3.4: Schur's Lemma - updated

Let E be an irreducible representation of a k-algebra A, and let $K \subseteq \text{End}_k(E)$ be the endomorphisms that commute with the operations of A: it is the centralizer of A in $\text{End}_k(E)$, $K = C_{\text{End}_k(E)}(A)$. Then K is a division ring.

Note that this makes K an A-algebra

***Proof*:**
Take $u \in K$. We want to show that $u^{-1} \in K$. Consider $u^{-1}(0) = V \subseteq E$. For $x \in V$, we have:

$$0 = a \cdot u(x) = u(a \cdot x)$$

hence, V is stable under A, and as $u \neq 0$ we must have $V = 0$ as u is invertible in $\text{End}_k(E)$. It follows that $u^{-1} \in K$.

Notice from this that we may say that the representation E must be faithful (the map must be injective). Indeed, the kernel of this representation would be an ideal of A but

Lemma 51.3.5: Minimal Centralizer of Representations

Let E be a semi-simple an faithful representation of A, $K = C_{\mathrm{End}_k(E)}(A)$, $B = C_{\mathrm{End}_k(E)}(K)$ (i.e. the centralizer of A and K in $\mathrm{End}_k(E)$). Then

$$A = B$$

***Proof*:**
Let $x \in E$ and $b \in B$. We shall show that there exists an $a \in A$ such that $a(x) = b(x)$.

Let $Ax = V \subseteq E$ be a stable subspace. Then V admits a (stable) compliment: $E = V \oplus V^{\perp}$. Let $\pi : E \to V$ be the natural projection map. Then certainly π is A-linear and commutes with K, namely $\pi \in K$. Then:

$$\pi(b(x)) = b(\pi(x)) = b(x)$$

namely, $b(x) \in V = aX$, implying $a(x) = b(x)$, as we sought to show.

Lemma 51.3.6: Generator lemma

Let E be a semi-simple an faithful representation of A, $K = C_{\mathrm{End}_k(E)}(A)$, $B = C_{\mathrm{End}_k(E)}(K)$ (i.e. the centralizer of A and K in $\mathrm{End}_k(E)$). Let $x_1, ..., x_n \in E$ and $b \in B$. Then there exits values $a_i \in A$ such that:

$$ax_1 = bx_1, \; ax_2 = bx_2 \qquad \cdots \qquad ax_n = bx_n$$

***Proof*:**
Let F be the direct sum space of n copies of representation E. Then F is formed of elements $(y_1, ..., y_n)$, $y_i \in E$, on which A acts by

$$a \cdot (y_1, ..., y_n) = (ay_1, ..., ay_n)$$

The representation F is still semi-simple. The centralizer of A in $\mathrm{End}_k(F)$ is formed of matrices (u_{ij}), $u_{ij} \in K$, and it follows that, if we let B operate on F by the formula:

$$b \cdot (y_1, ..., y_n) = (by_1, ..., by_n)$$

the elements of $\mathrm{End}_k(F)$ thus defined will still be in the centralizer of the centralizer of A (in the theory of semi-groups, this is sometimes called the bicommutant). By lemma 51.3.5 applying to the point $x = (x_1, ..., x_n) \in F$, and to $b \in B$ giving us the desired result.

Serre write K_n for $M_n(K)$ for a division ring K, but we shall stick to $M_n(D)$ where our division ring is D instead of K. If D contains k in its center, we have:

$$M_n(D) = M_n(k) \otimes D$$

as can be easily seen.

We present one more important result that has two useful corollaries. Let A be a k-algebra. Then A acting on itself by left multiplication gives something called the *regular left representation*. If

$A = M_n(D)$, it has basis e_{ij}, $1 \leq i, j \leq n$. The elements e_{i,j_0} for a fixed j_0 gives a subspace of $M_n(D)$, say $N_{j_0} \subseteq M_n(D)$. It is easy to see that N_{j_0} is a minimal left ideal (namely, an irreducible representation). Then we see that for any other choice of j_0, say j_1, we get an isomorphic irreducible representation N_{j_1}. Then this gives the following classical result:

Theorem 51.3.7: Representation of Simple Algebra

Let A be a simple k-algebra. Then all representation of A is isomorphic to the direct sum of irreducible representation

***Proof*:**
use corollary 51.2.20.

Corollary 51.3.8: Representation of Simple Algebra

Let A be a simple k-algebra. Then:

1. All representations of A are semi-simple (completely reducible in Serre's words)
2. Two representation of A which have the same dimension are isomorphic

***Proof*:**
immediate from the above

Let us now turn towards the tensor product of simple algebras.

Lemma 51.3.9: Centralizer of Tensor Product of Algebras

Let A, A' be two k-algebras, B, B' be sub k-algebras of A, A' (respectively) with C, C' their centralizers on A, A' (respectively). Then the centralizer of $B \otimes B'$ in $A \otimes A'$ is $C \otimes C'$.

***Proof*:**
Let us first find the centralizer of $B \otimes 1$. For this, let a_i' be a basis of A'. Any $x \in A \otimes A'$ can be written uniquely in the form:

$$x = \sum_i a_i \otimes a_i' \qquad \text{for appropriate } a_i \in A$$

If x commutes with $b \otimes 1$, then $a_i b = b a_i$ that is $a_i \in C$ for all i. We thus see that the centralizer of $B \otimes 1$ is $C \otimes A'$. Similarly, that of $1 \otimes B'$ is $A \otimes C'$. It follows that the centralizer of $B \otimes B'$ is equal to the intersection of the centralizers of $B \otimes 1$ and $1 \otimes B'$, namely

$$(C \otimes A') \cap (A \otimes C') = C \otimes C'$$

as we sought to show.

Thus, the centre of the tensor product of two algebras is the tensor product of the centre of the two algebras. Applying this to the special case of $A = M_n(k) \otimes D$ for some division ring D, then:

$$Z(A) = k \otimes Z(D)$$

Corollary 51.3.10: Centre of simple algebra is a Field

Let A be a simple k-algebra, finitely generated over k. Then $Z(A)$ is a field.

***Proof*:**
Consider $Z(A)$. As A is a finitely generated k-algebra, $Z(A)$ is a finitely generated k-algebra. For any $0 \neq z \in Z(A)$, consider zA. As z is central,we have that zA is two-sided ideal. As A is simple, it must be that $zA = A$, that is z is there exists a $w \in A$ such that $za = 1$. Therefore, $Z(A)$ is a division ring. It is by definition commutative, and hence it is a field. Since z was an arbitrary nonzero element, $Z(A)$ is a division ring. It is by definition commutative, and hence it is a field.

We shall next require to see the interaction between the tensor product of central simple algebras. We require the following lemma:

Theorem 51.3.11: Ideal Generation Tensor Product

Let A be a k-algebra and K' a left division ring with center k. Then every ideal $N \subseteq A \otimes K'$ is generated (as a left module over K') by its intersection with $A \otimes 1$.

Note that neither A nor K' are assumed finite over k

***Proof*:**
We will first note that we can make K' operate on the left on $A \otimes K'$ by:

$$k'(a \otimes k'') = a \otimes k'k''$$

If N is an ideal of the algebra $A \otimes K'$, it is in particular a K'-module subspace of $A \otimes K'$.

Let a_i be a basis of A over k; the elements $a_i \otimes 1$ equally form a basis of the K'-module $A \otimes K'$. Let x be a primitive element of N with respect to this basis (Bourbaki, Alg.II, paragraph 5):

$$x = \sum_i a_i \otimes k'_i$$

Consider the element $x \cdot m$ ($m \in K'$); we have $x \cdot m \in N$ (N is an ideal), and on the other hand:

$$x \cdot m = \sum_i a_i \otimes k'_i m$$

It then follows from the properties of primitive elements that we have:

$$k'_i m = n k'_i \text{ for all } i \quad (n \in K')$$

Since one of the k'_i is equal to 1, we have $m = n$, and the preceding equality simply means that k'_i commutes with all $m \in K'$, i.e., that $k'_i \in k$. We can then write: $x = a \otimes 1$, by setting: $a = \sum_i k'_i a_i$. We have thus shown that every primitive element x is in $A \otimes 1$; the theorem follows immediately since every vector space is generated by its primitive elements.

Theorem 51.3.12: Tensor Product CSA

Let A, A' be simple k-algebras finitely generated over k, where one of them is a CSA. Then $A \otimes A'$ is a simple k-algebra

***Proof*:**
Suppose that $A' = M_n(k) \otimes K'$ is central, in other words that the center of the left field K' is reduced to k. It will suffice to show that $A \otimes K'$ is a simple algebra. Indeed, if this is demonstrated, $A \otimes K' = M_m(k) \otimes K''$, and $A \otimes A' = M_m(k) \otimes M_n(k) \otimes K'' = M_{mn}(k) \otimes K'' = M_{nm}(K'')$. The algebra $A \otimes A'$ will thus be isomorphic to a matrix algebra for which simplicity is immediately verified. Then the general case of $A \otimes K'$ being simple follow from theorem 51.3.11, completing the proof.

Corollary 51.3.13: Tensor Product of Two CSA's

The tensor product of two CSA is a CSA

***Proof*:**
In the previous proof, it was enough for one of them to be a CSA.

Corollary 51.3.14: Opposite Algebra Tensoring

Let A be a CSA over k and A^{op} the opposite algebar. Then:

$$A \otimes A^{\mathrm{op}} \cong M_{n^2}(k)$$

***Proof*:**
Let us denote by E the algebra A considered as a vector space over k. We can make A act naturally on E by the left or right by multiplication to create a left or right A-module. This amounts to identifying A and A^{op} as sub-algebras of the algebra $\mathrm{End}_k(E)$ of k-endomorphisms of E. Since the algebras A and A^{op} commute in $\mathrm{End}_k(E)$, we can define a canonical homomorphism of $A \otimes A^{\mathrm{op}}$ into $\mathrm{End}_k(E)$ which extends the isomorphisms: $A \to \mathrm{End}_k(E)$ and $A^{\mathrm{op}} \to \mathrm{End}_k(E)$. The algebra $A \otimes A^{\mathrm{op}}$ being simple, this homomorphism is an isomorphism; moreover, as;

$$[A \otimes A^{\mathrm{op}} : k] = [A : k]^2 = n^2 = [\mathrm{End}_k(E) : k]$$

this isomorphism maps $A \otimes A^{\mathrm{op}}$ onto $\mathrm{End}_k(E) = M_{n^2}(k)$, as we sought to show.

51.3.1 Brauer Group

Let A be a CSA over k. As A is simple, we get by Artin-Wedderburn that it is isomorphic to $M_n(D) \cong M_n(k) \otimes D$. For example, if $n = 1$ and $k = \mathbb{R}$ we get (by a classical theorem from algebraic topology) that $A \cong \mathbb{R}$ or $A \cong \mathbb{H}$ (the quaternions). If $n > 1$, we get $M_n(\mathbb{R})$ and $M_n(\mathbb{H})$ (which are no longer division algebras). We see that the division ring is one of the "characterizing" feature of the CSA (the other being its matrix ring), especially that up to Morita-equivalence we get the same category. We thus have the following natural equivalence we may define:

Definition 51.3.15: Brauer Equivalence

Let k-**CSA** be a collection of central simple algebras over k. Then define the equivalence relation $A \sim A'$ where the two are equivalent if and only if their associated division rings are isomorphic.

Proposition 51.3.16: Brauer Group

The Brauer equivalence on a collection of CSA's forms a commutative group under the tensor product. This group will be denoted $\mathrm{Br}(k)$.

In Serre, this group is denoted G_k.

***Proof*:**
By corollary 51.3.13, it is closed under the tensor product, the equivalence class containing k is certainly the identity, and corollary 51.3.14 shows that every equivalence class has an inverse. Associativity and commutativity is a classical result for tensor products.

Example 51.3.17: Brauer Groups

1. Let $k = \overline{k}$. Then $\mathrm{Br}(\overline{k}) = 0$ by corollary 51.2.20.
2. If $k = \mathbb{R}$, then we shall show in the next section that $\mathrm{Br}(\mathbb{R}) = \mathbb{Z}/2\mathbb{Z}$, namely we shall have $\mathbb{R}$ and $\mathbb{H}$, the quaternions.
3. Let k be a finite field. Then $\mathrm{Br}(k) = 0$, as all finite division rings are fields.
4. If $k = \mathbb{Q}_p$, then $\mathrm{Br}(k) \cong \mathbb{Q}/\mathbb{Z}$ (Serre references Algebren - Chap. VII - 2nd paragarph).
5. If $k = \mathbb{Q}$, $\mathrm{Br}(\mathbb{Q})$ will be a subgroup of
$$\mathbb{Z}/2\mathbb{Z} \oplus (\mathbb{Q}/\mathbb{Z})^{\oplus \mathbb{N}}$$
which can be thought of as a result similar to that in number theory where we study the "global field" $\mathbb{Q}$ by looking at all its localizations (note that $\mathbb{Z}/2\mathbb{Z}$ comes from $\mathrm{Br}(\mathbb{R})$!) Serre references Deuring for the proof (see Sere p.8)

Theorem 51.3.18: Field Extensions and CSA

Let A be a CSA over k, L/k a field extension (either finite or infinite). Then $A \otimes L$ is simple

***Proof*:**
If A is a division ring, then all ideals of $A \otimes L$ will be given with in part with the intersection of the ideals of L, which is simply (0), making $A \otimes L$ simple.

If A is any CSA, write $A \cong M_n(k) \otimes D$. Then:

$$A \otimes L \cong M_n(k) \otimes (D \otimes L) \stackrel{!}{=} M_n(k) \otimes M_m(D')$$

where for $\stackrel{!}{=}$, D' is the division ring naturally given by $D \otimes_k L$! But this gives the result.

Corollary 51.3.19: Index of CSA

Let A be a simple, central and finite algebra over k. Then $[A : k]$ is a perfect square.

***Proof*:**
Let L be an algebraically closed extension of k. The algebra $A \otimes L$ can be endowed with an algebra structure over L, and we have:

$$[A \otimes L : L] = [A : k]$$

But $A \otimes L$ is a simple algebra, central over L, and finite over L. It then follows from example 51.3.17[1] that it is a matrix algebra over L, and $[A \otimes L : L]$ is indeed a perfect square.

Looking at the field extension of a k-algebra, it is natural to consider that $-\otimes_k L$ gives a homomorphism of Brauer groups $\varphi_{k,L} : \mathrm{Br}(k) \to \mathrm{Br}(L)$, namely due to the following result from linear algebra:

$$(A \otimes_k L) \otimes_L (A' \otimes_k L) \cong (A \otimes_k A') \otimes_k L$$

The kernel $\ker \varphi_{k,L}$ consists of all k-algebra's that become trivial (matrix algebras) when extending to L. Serre calls these algebras the algebras that *admit L as a field of decomposition.* (or admit a decomposition by L). More recently, the term *splitting field* is adopted due to the parallels to number theory. Naturally, if $L = \overline{k}$, then every algebra admits a decomposition as $\mathrm{Br}(\overline{k}) = 0$. We only require a finite extension:

Theorem 51.3.20: Splitting Field is Finite

The Brauer group $\mathrm{Br}(k)$ is the union of $\ker \varphi_{k,L}$ going over each L/k where L is a finite field extension. In particular, there exists a finite extension L/k such that for a CSA A over k, $[A \otimes_k L] = [1]$ in $\mathrm{Br}(L)$.

***Proof*:**
By example 51.3.17[1] we have that for a CSA A over $\overline{k}$, $A \otimes_k \overline{k} \cong M_n(D)$. Then the right hand side is generated by finitely many elements of the left hand side. We need only take the elements that map to the generators of the right hand side and to choose the appropriate field elements to get find L/k, as we sought to show.

51.4 Build up to Quaternion Algebras

Theorem 51.4.1: Skolem-Noether Theorem

Let A, B be CSA's over k, and let $f, g : A \to B$ be k-algebra isomorphism. Then there exists a unit $b \in B$ such that:

$$g(a) = bf(a)b^{-1}$$

In particular, every automorphism is an inner automorphism

***Proof*:**
Let's first prove the special case of $B = M_n(k)$. Then the isomorphisms $f, g : A \to M_n(k)$ define actions on k^n, or said differently these define two A-modules (with two different actions): V_f, V_g. Then using f, g we have an A-module isomorphism $g \circ f^{-1} : V_f \to V_g$. But any isomorphism between vector-spaces is representable as (unit) matrix in $M_n(k)$, and so we we get our value by linear algebra.

For the general case, we know that $B \otimes_k B^{\text{op}} \cong M_n(k)$ and $A \otimes_k B^{\text{op}}$ is simple as B^{op} is still simple (use theorem 51.3.12). As $B \otimes_k^B$ op is a (finite dimensional) matrix algebra, we may apply our special case to:

$$f \otimes 1,\ g \otimes 1 : A \otimes_k B^{\text{op}} \to B \otimes_k B^{\text{op}} \cong M_n(k)$$

which shows that:

$$(f \otimes 1)(a \otimes c) = b(g \otimes 1(a \otimes c)b^{-1}$$

which is true for all $a \in A$ and $c \in B^{\text{op}}$. Thus, it is true for $a = 1$, giving us:

$$1 \otimes c = b(1 \otimes c)b^{-1}$$

for all $c \in B^{\text{op}}$. Thus,

$$b \in Z_{B \otimes_k B^{\text{op}}}(k \otimes B^{\text{op}}) = B \otimes k$$

Hence, as constants move through the tensor we get that $b = b \otimes_k 1$. And so, by setting $z = 1$, we find:

$$f(a) = b' g(a) b'^{-1}$$

as we sought to show.

Theorem 51.4.2: Centralizer of Subalgebras

Let A be a CSA over k, $B \subseteq A$ a simple subalgebra, $C = C_A(B)$ the centralizer of B in A. Then C is simple, $B = C_A(C)$, and:

$$[B : k][C : k] = [A : k]$$

***Proof*:**
Let E be B with only the structure of a k-vector space. Then B acts on E via left multiplication; this action isomorphically embeds B into $\text{End}_k(E)$. Then the centralizer of B in $\text{End}_k(E)$ is nothing more than the algebra of right translations by E, that is B^{op}.

Given this, we can consider the simple algebra $A \otimes_k \mathrm{End}_k(E)$. We can now embed B into this tensored algebra in two ways: via $B \otimes 1$ or $1 \otimes B$. Their centralizers are $C \otimes \mathrm{End}_k(E)$ and $A \otimes B^{\mathrm{op}}$ respectively.

By theorem 51.4.1, there exists an inner automorphism of $A \otimes_k \mathrm{End}_k(E)$ mapping $B \otimes 1$ to $1 \otimes B$; it is then a simple algebra exercise to see that this inner automorphism induces a map on the centralizers. As a consequence, as $A \otimes B^{\mathrm{op}}$ is simple, so is $C \otimes \mathrm{End}_k(E)$, which must make C simple.

Furthermore:

$$[C \otimes \mathrm{End}_k(E) : k] = [C : k][B : k]^2 \qquad \text{and} \qquad [A \otimes B^{\mathrm{op}} : k][A : k][B : k]$$

From which we get $[A : k] = [B : k][C : k]$. What's left to show is the centralizer of C in A is B. Say that B' is this centralizer. Then $B \subseteq B'$. Then as $[A : k] = [B : k][C : k]$, we get:

$$[B' : k] = [B : k]$$

But then as they contain we must get $B = B'$, completing the proof.

52

The Jacobson Radical and Non-Semisimple Algebras

The Artin-Wedderburn theorem (theorem 51.2.16) is a complete answer to a narrow question: it classifies the semisimple rings and, in the process, shows that a semisimple k-algebra is nothing more than a finite product of matrix rings over division algebras. When R is semisimple, every module splits into simples, every short exact sequence splits, and the entire representation theory of R collapses onto the finite list of its simple modules. This is the best possible situation, and it is also the exception. The algebras whose representation theory occupies most of this book are precisely the ones that fail to be semisimple: the group algebra $k[G]$ when $\mathrm{char}(k)$ divides $|G|$ (where Maschke's theorem breaks down), the truncated polynomial algebra $k[x]/(x^n)$, universal enveloping algebras of Lie algebras, and the path algebra of a quiver with an oriented cycle. For all of these, some module fails to be a direct sum of simples, some sequence refuses to split, and Wedderburn has nothing to say.

This chapter builds the tools that take over where Wedderburn stops. The first of these, and the subject of the present section, is the *Jacobson radical* $\mathrm{Jac}(A)$: the two-sided ideal that measures, in a single invariant, exactly how far A is from being semisimple. It is the obstruction that vanishes precisely when A is semisimple, and dividing it out always produces a semisimple ring $A/\mathrm{Jac}(A)$ whose simple modules are the same as those of A. In the sections that follow, the radical becomes the engine behind the theory of indecomposable modules and the Krull-Schmidt theorem (which replaces "decomposition into simples" with "decomposition into indecomposables"), and behind the construction of projective covers, which repair the failure of projectivity that semisimplicity had guaranteed for free. The radical is where the non-semisimple story begins.

52.1 The Jacobson Radical

Throughout chapter 51 the simple modules did all the work: a semisimple ring was assembled out of its minimal left ideals, and every module was a direct sum of simples. When a ring is *not* semisimple, the simple modules are still present, but they no longer see everything. Some elements of the ring act as zero on every simple module without being zero themselves, and these elements form an ideal that records the discrepancy. To isolate them, we begin one level down, with modules, and ask which part of a module is invisible to all of its simple quotients.

The maximal submodules of a module M are exactly the kernels of the surjections $M \twoheadrightarrow S$ onto simple modules: a submodule $N \subsetneq M$ is maximal precisely when M/N is simple, and every simple quotient of M is of the form M/N for such an N (the classification of simples as quotients by maximal ideals is lemma 51.1.3). An element lying in every maximal submodule is one that every simple quotient sends to zero. Collecting these gives the radical of a module.

Definition 52.1.1: Radical of a Module

Let M be a left A-module. The *radical* of M, written $\operatorname{rad}(M)$, is the intersection of all maximal submodules of M:

$$\operatorname{rad}(M) = \bigcap_{\substack{N \subsetneq M \\ N \text{ maximal}}} N$$

If M has no maximal submodules, the intersection is empty and $\operatorname{rad}(M) := M$ by convention.

Since each maximal submodule is the kernel of a map onto a simple module, an equivalent description is

$$\operatorname{rad}(M) = \bigcap_{\substack{\varphi\colon M \to S \\ S \text{ simple}}} \ker(\varphi)$$

so that $\operatorname{rad}(M)$ is the set of elements killed by *every* homomorphism from M into a simple module. The quotient $M/\operatorname{rad}(M)$ is the largest quotient of M that embeds into a product of simple modules; it is the *semisimple quotient* of M, the part of M that its simple quotients can detect. Everything inside $\operatorname{rad}(M)$ is invisible to them.

Example 52.1.2: The Radical of a Cyclic $\mathbb{Z}$-Module

Take $A = \mathbb{Z}$ and $M = \mathbb{Z}/12\mathbb{Z}$. The maximal submodules of M correspond to the maximal ideals of $\mathbb{Z}$ containing $12\mathbb{Z}$, that is to the primes dividing 12, namely 2 and 3. The corresponding maximal submodules are $2\mathbb{Z}/12\mathbb{Z}$ and $3\mathbb{Z}/12\mathbb{Z}$, with simple quotients $\mathbb{Z}/2\mathbb{Z}$ and $\mathbb{Z}/3\mathbb{Z}$. Their intersection is

$$\operatorname{rad}(\mathbb{Z}/12\mathbb{Z}) = \frac{2\mathbb{Z}}{12\mathbb{Z}} \cap \frac{3\mathbb{Z}}{12\mathbb{Z}} = \frac{6\mathbb{Z}}{12\mathbb{Z}}$$

so $\operatorname{rad}(M) = 6\mathbb{Z}/12\mathbb{Z} \cong \mathbb{Z}/2\mathbb{Z}$, and the semisimple quotient is $M/\operatorname{rad}(M) \cong \mathbb{Z}/6\mathbb{Z} \cong \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/3\mathbb{Z}$. The radical has stripped $\mathbb{Z}/12\mathbb{Z}$ down to its squarefree quotient: the extra factors of 2 and 3 that no simple quotient can see are exactly what lies in $6\mathbb{Z}/12\mathbb{Z}$.

Applying this construction to the ring itself, viewed as a module over itself, produces the central object of the chapter. The radical of A as a left A-module is the intersection of its maximal *left* ideals.

Definition 52.1.3: Jacobson Radical

Let A be a ring. The *Jacobson radical* of A, written $\mathrm{Jac}(A)$, is the radical of A regarded as a left module over itself, that is the intersection of all maximal left ideals of A:

$$\mathrm{Jac}(A) = \mathrm{rad}({}_A A) = \bigcap_{\substack{\mathfrak{m} \subseteq A \\ \mathfrak{m} \text{ maximal left ideal}}} \mathfrak{m}$$

The definition is one-sided: it uses left ideals, and a priori $\mathrm{Jac}(A)$ need only be a left ideal. One of the first surprises is that the resulting set is symmetric, a two-sided ideal that could equally have been defined with maximal *right* ideals. This symmetry, and several other faces of the same object, are the content of the next theorem. Before stating it, recall that an element $u \in A$ is *left-invertible* if there exists $v \in A$ with $vu = 1$.

Theorem 52.1.4: Characterizations of the Jacobson Radical

Let A be a ring and let $a \in A$. The following subsets of A coincide, and their common value is $\mathrm{Jac}(A)$:

1. the intersection of all maximal left ideals of A;
2. the intersection of the annihilators $\mathrm{Ann}_A(S)$ of all simple left A-modules S;
3. the set $\{a \in A \big| 1 - xa \text{ is left-invertible for all } x \in A\}$.

Consequently, $\mathrm{Jac}(A)$ is a two-sided ideal of A, and it is the largest two-sided ideal of A that annihilates every simple left A-module.

The three descriptions look at the same ideal from three angles. The first is the definition. The second recasts it representation-theoretically: $\mathrm{Jac}(A)$ is exactly the set of elements that act as zero on every simple module, which is why it is invisible to the simple representation theory. The third is an intrinsic ring-theoretic criterion that never mentions modules at all, and it is the one that makes the two-sidedness fall out for free, since it is symmetric in a way the definition is not.

***Proof*:**
We prove (1) = (2), then (2) = (3), and finally read off the two-sidedness and maximality statements.

Step 1: (1) = (2). Let J_1 denote the intersection of all maximal left ideals and J_2 the intersection of the annihilators of all simple left modules. If S is a simple left A-module, then for any $0 \neq s \in S$ we have $S = As$ (as As is a nonzero submodule of the simple module S), and by corollary 51.1.4 the map $a \mapsto as$ gives $S \cong A/\mathrm{Ann}_A(s)$ with $\mathrm{Ann}_A(s)$ a maximal left ideal. The annihilator of the whole module is

$$\mathrm{Ann}_A(S) = \bigcap_{0 \neq s \in S} \mathrm{Ann}_A(s)$$

an intersection of maximal left ideals, so $J_1 \subseteq \mathrm{Ann}_A(S)$ for every simple S, giving $J_1 \subseteq J_2$. Conversely, if $\mathfrak{m}$ is any maximal left ideal, then $S = A/\mathfrak{m}$ is a simple left module and $\mathrm{Ann}_A(S) \subseteq \mathfrak{m}$ (since $\mathrm{Ann}_A(S)$ kills the coset $1 + \mathfrak{m}$, so $\mathrm{Ann}_A(S) \cdot 1 \subseteq \mathfrak{m}$). Hence $J_2 \subseteq \mathfrak{m}$ for every maximal left ideal $\mathfrak{m}$, giving $J_2 \subseteq J_1$. Thus $J_1 = J_2$.

Step 2: (2) $\subseteq$ (3). Let $a \in J_2$, so a annihilates every simple left module, and fix $x \in A$. We must show $1 - xa$ is left-invertible. If not, then $A(1 - xa)$ is a proper left ideal, hence contained in some maximal left ideal $\mathfrak{m}$ (using that $1 \notin A(1 - xa)$ and Zorn's lemma). The quotient $S = A/\mathfrak{m}$ is simple, so a annihilates it: $a \cdot (1 + \mathfrak{m}) = 0$, meaning $a \in \mathfrak{m}$. But $xa \in \mathfrak{m}$ as well, since $\mathfrak{m}$ is a left ideal, and $1 - xa \in \mathfrak{m}$ since $1 - xa \in A(1 - xa) \subseteq \mathfrak{m}$. Adding, $1 = (1 - xa) + xa \in \mathfrak{m}$, contradicting that $\mathfrak{m}$ is proper. Hence $1 - xa$ is left-invertible for every x, so $a \in J_3$, the set in (3).

Step 3: (3) $\subseteq$ (2). Let $a \in J_3$, and suppose toward a contradiction that a does *not* annihilate some simple left module S. Then $as \neq 0$ for some $s \in S$, and since As is a nonzero submodule of the simple module S we have $As = S$. In particular $s \in As$, so $s = xas$ for some $x \in A$, giving $(1 - xa)s = 0$. By hypothesis $1 - xa$ is left-invertible, say $v(1 - xa) = 1$; applying v gives $s = v(1 - xa)s = 0$, contradicting the choice of s. Hence a annihilates every simple module, so $a \in J_2$.

Combining the three steps, $J_1 = J_2 = J_3 = \mathrm{Jac}(A)$.

Step 4: two-sidedness and maximality. As the intersection of the left ideals $\mathrm{Ann}_A(S)$, description (2) shows $\mathrm{Jac}(A)$ is a left ideal. But each $\mathrm{Ann}_A(S)$ is in fact a two-sided ideal: if $a \in \mathrm{Ann}_A(S)$ and $b \in A$, then for all $s \in S$ we have $(ab)s = a(bs) = 0$, so $ab \in \mathrm{Ann}_A(S)$. An intersection of two-sided ideals is two-sided, so $\mathrm{Jac}(A)$ is a two-sided ideal. Finally, if I is any two-sided ideal annihilating every simple left module, then $I \subseteq \mathrm{Ann}_A(S)$ for every simple S, hence $I \subseteq \bigcap_S \mathrm{Ann}_A(S) = \mathrm{Jac}(A)$; thus $\mathrm{Jac}(A)$ is the largest such ideal, completing the proof.

The left-invertibility criterion (3) deserves a second look, because it is what makes the radical robust. It says a lies in the radical exactly when $1 - xa$ can never be turned into a non-unit by any left multiplier x. A standard manipulation upgrades "left-invertible" to "two-sided invertible" for these particular elements: if $a \in \mathrm{Jac}(A)$ and $v(1 - xa) = 1$, then $v = 1 + vxa = 1 - (-vx)a$, and since $a \in \mathrm{Jac}(A)$ the element v is itself left-invertible by the same criterion; a left-invertible element with a left-invertible left inverse is a unit. So for $a \in \mathrm{Jac}(A)$ the element $1 - xa$ is a two-sided unit for every x, and by the right-handed version of the whole theorem $1 - ax$ is a unit as well. This is the working form of the radical: $\mathrm{Jac}(A)$ is the set of elements a such that $1 - xay$ is a unit for all $x, y \in A$, and it is exactly this that will drive Nakayama's lemma below.

Description (2) also settles the two-sided nature of the radical without any of the asymmetry of the original definition, and it makes precise the sense in which the radical is invisible to representation theory. If A is semisimple, every module is a sum of simples, so an element killing all simples kills everything: $\mathrm{Jac}(A) = 0$. The content of the non-semisimple theory is that $\mathrm{Jac}(A)$ can be nonzero, and it is this ideal that the simple modules fail to detect.

Before turning to that theory, the radical of a ring is anchored by two computations. The first is the archetype of a local non-semisimple algebra.

Example 52.1.5: The Radical of a Truncated Polynomial Algebra

Let k be a field and let $A = k[x]/(x^n)$ with $n \geq 1$, writing $\bar{x}$ for the image of x. This is a commutative local ring: an element $f(\bar{x}) = c_0 + c_1\bar{x} + \cdots + c_{n-1}\bar{x}^{n-1}$ is a unit if and only if $c_0 \neq 0$, since when $c_0 = 0$ the element is nilpotent ($\bar{x}^n = 0$ forces $f(\bar{x})^n = 0$), and when $c_0 \neq 0$ one inverts using the geometric series, which terminates because $\bar{x}^n = 0$:

$$(c_0 + \bar{x}g)^{-1} = c_0^{-1}\big(1 + c_0^{-1}\bar{x}g\big)^{-1} = c_0^{-1}\sum_{j=0}^{n-1}(-c_0^{-1}\bar{x}g)^j$$

The non-units are therefore exactly the elements with $c_0 = 0$, which is the ideal $(\bar{x})$. In a commutative ring the maximal ideals are the maximal left ideals, and here there is a single maximal ideal, namely $(\bar{x})$, because $A/(\bar{x}) \cong k$ is a field while any element outside $(\bar{x})$ is a unit. Hence

$$\mathrm{Jac}(A) = (\bar{x}) = \{c_1\bar{x} + c_2\bar{x}^2 + \cdots + c_{n-1}\bar{x}^{n-1} \big| c_i \in k\}$$

This ideal is nilpotent: $(\bar{x})^n = (\bar{x}^n) = 0$, so already $\mathrm{Jac}(A)^n = 0$. The quotient $A/\mathrm{Jac}(A) = A/(\bar{x}) \cong k$ is a field, hence semisimple, which is the general phenomenon established in theorem 52.1.9. The single simple A-module is $k = A/(\bar{x})$, and $\bar{x}$ acts on it as zero, confirming that $\mathrm{Jac}(A) = (\bar{x})$ annihilates every simple module.

The second computation is the model of a non-commutative, non-local radical, where the one-sided origin of the definition is easiest to see in matrices.

Example 52.1.6: The Radical of Upper-Triangular Matrices

Let k be a field and let $A = T_n(k)$ be the ring of $n \times n$ upper-triangular matrices over k. Write $N \subseteq A$ for the strictly upper-triangular matrices (zero on and below the diagonal). The claim is $\mathrm{Jac}(A) = N$.

N annihilates every simple module. For each $1 \leq i \leq n$ there is a one-dimensional A-module $S_i = k$ on which an upper-triangular matrix $a = (a_{pq})$ acts by its ith diagonal entry, $a \cdot v = a_{ii}v$; this is a ring homomorphism $A \to k$ because the (i, i) entry of a product of upper-triangular matrices is the product of the (i, i) entries. The S_i are pairwise non-isomorphic simple A-modules. Any strictly upper-triangular matrix has all diagonal entries zero, so it acts as 0 on each S_i; thus $N \subseteq \mathrm{Ann}_A(S_i)$ for all i, and $N \subseteq \mathrm{Jac}(A)$ by theorem 52.1.4.

Nothing outside N lies in the radical. If $a \in A \setminus N$, some diagonal entry a_{ii} is nonzero, so a acts invertibly on S_i, and a does not annihilate the simple module S_i. By part (2) of theorem 52.1.4, $a \notin \mathrm{Jac}(A)$. Combining, $\mathrm{Jac}(A) = N$.

As a check on the intrinsic criterion, N is nilpotent with $N^n = 0$ (each multiplication by a strictly upper-triangular matrix pushes nonzero entries one further diagonal above the main diagonal), so for $a \in N$ the element $1 - xa$ has the form $1-$ (nilpotent) and is a unit via the terminating geometric series $\sum_{j\geq 0}(xa)^j$. The quotient is $A/N \cong k^n$, the diagonal matrices, which is semisimple, and there are exactly n simple modules $S_1, \ldots, S_n$, matching the n factors of k^n.

The two examples share a feature: in each, $\mathrm{Jac}(A)$ turned out to be nilpotent, and the geometric series that inverts $1 - xa$ terminated precisely because a power of the radical vanished. This is not special to small examples. For the finite-dimensional and, more generally, left Artinian algebras that concern us, the radical is always nilpotent. The route to that fact runs through Nakayama's lemma, which converts the invertibility criterion into a statement about modules being forced to zero.

The lemma answers a question that recurs throughout module theory: if multiplying a module by the radical does not shrink it, must the module already be zero? The commutative version, over a local ring or with a general ideal contained in the radical, is proved in the companion volume 'NateCommAlg' [10]and will not be re-derived here; the module form over an arbitrary ring is the form needed here, and its proof uses only the invertibility criterion from theorem 52.1.4.

Lemma 52.1.7: Nakayama's Lemma

Let A be a ring and M a finitely generated left A-module.

1. If $\mathrm{Jac}(A)\,M = M$, then $M = 0$.
2. If $N \subseteq M$ is a submodule with $M = N + \mathrm{Jac}(A)\,M$, then $N = M$.

***Proof*:**
Write $J = \mathrm{Jac}(A)$.

1. Suppose $M \neq 0$ and $JM = M$. Since M is finitely generated, choose a generating set $m_1, \ldots, m_r$ with r minimal, so no fewer than r elements generate M; as $M \neq 0$ we have $r \geq 1$. From $M = JM$ we may write the last generator as

$$m_r = \sum_{i=1}^{r} a_i m_i \qquad \text{with } a_i \in J$$

   Isolating m_r gives $(1 - a_r)m_r = \sum_{i=1}^{r-1} a_i m_i$. Because $a_r \in J = \mathrm{Jac}(A)$, the element $1 - a_r$ is a unit (the working form of theorem 52.1.4: $1 - a_r$ is left-invertible, and for radical elements this upgrades to a two-sided inverse). Multiplying on the left by $(1 - a_r)^{-1}$ expresses m_r as an A-combination of $m_1, \ldots, m_{r-1}$, so these $r-1$ elements already generate M, contradicting the minimality of r. Hence $M = 0$.

2. Apply part (1) to the finitely generated module M/N. Its finite generation is inherited from M (the images of generators of M generate the quotient). The hypothesis $M = N + JM$ gives, in the quotient, $M/N = J\,(M/N)$, since the image of JM is $J(M/N)$ and the image of N is 0. By part (1), $M/N = 0$, that is $N = M$, completing the proof.

The finite-generation hypothesis is not optional. Over the local ring $A = \mathbb{Z}_{(p)}$ (integers localized at p, whose radical is $p\mathbb{Z}_{(p)}$) the module $M = \mathbb{Q}$ satisfies $pM = M$ yet $M \neq 0$; it fails to be finitely generated, and the minimal-generator argument has nothing to grip. Part (2) is the form used in practice: it says that generators of $M/\mathrm{Jac}(A)M$ lift to generators of M. Once $\mathrm{Jac}(A)M$ is known (below) to be a "small" submodule, this becomes the statement that a minimal generating set of M is detected by the semisimple quotient $M/\mathrm{Jac}(A)M$, which is the seed of the theory of projective covers in a later section.

We now specialize to the setting where the radical is best behaved. A ring A is *left Artinian* if it satisfies the descending chain condition on left ideals; every finite-dimensional algebra over a field is left Artinian, since a descending chain of left ideals is a descending chain of subspaces and dimension cannot drop forever. In this setting the radical is not only an intersection of maximal ideals but a nilpotent ideal, and it is the largest one.

Theorem 52.1.8: The Radical Is Nilpotent for Artinian Rings

Let A be a left Artinian ring. Then $\mathrm{Jac}(A)$ is nilpotent: there exists $m \geq 1$ with $\mathrm{Jac}(A)^m = 0$. Moreover, $\mathrm{Jac}(A)$ contains every nilpotent left ideal and every nilpotent two-sided ideal of A, so it is the largest nilpotent one-sided or two-sided ideal.

***Proof*:**
Write $J = \mathrm{Jac}(A)$.

Step 1: J is nilpotent. The descending chain of two-sided ideals

$$J \supseteq J^2 \supseteq J^3 \supseteq \cdots$$

is a descending chain of left ideals, so by the descending chain condition it stabilizes: there is m with $I := J^m = J^{m+1} = J\,I$. Suppose for contradiction that $I \neq 0$. Consider the set Σ of left ideals $L \subseteq A$ with $IL \neq 0$; it is nonempty because $I \cdot I = J^{2m} = I \neq 0$ shows $I \in \Sigma$ (using $J^{2m} = (J^m)^2 = I \cdot I$ and $J^{2m} = J^m = I$ since the chain stabilized). By the descending chain condition, Σ has a minimal element L_0. Choose $x \in L_0$ with $Ix \neq 0$; then $Ix \subseteq L_0$ is a left ideal with $I(Ix) = I^2x = Ix \neq 0$ (as $I^2 = I$), so $Ix \in \Sigma$, and minimality forces $Ix = L_0$. In particular $x \in L_0 = Ix$, so $x = ax$ for some $a \in I \subseteq J$. Then $(1-a)x = 0$, and since $a \in J$ the element $1 - a$ is a unit, giving $x = 0$ and hence $Ix = 0$, a contradiction. Therefore $I = J^m = 0$.

Step 2: J contains every nilpotent one-sided or two-sided ideal. Let I be a nilpotent left ideal, say $I^t = 0$. For any $a \in I$ and any $x \in A$, the element xa lies in I (a left ideal absorbs left multiplication), so $(xa)^t \in I^t = 0$; thus xa is nilpotent. A nilpotent element y with $y^t = 0$ makes $1 - y$ a unit, with inverse the terminating sum $1 + y + y^2 + \cdots + y^{t-1}$. Hence $1 - xa$ is a unit, in particular left-invertible, for every $x \in A$, so $a \in \mathrm{Jac}(A)$ by part (3) of theorem 52.1.4. This holds for every $a \in I$, so $I \subseteq J$. A nilpotent two-sided ideal is in particular a nilpotent left ideal, so the same conclusion applies, completing the proof.

For a left Artinian ring, then, "the Jacobson radical" and "the largest nilpotent ideal" name the same object; the two definitions of the radical current in the literature (intersection of maximal left ideals versus largest nilpotent ideal) agree exactly when A is Artinian. The nilpotence is what powers the terminating geometric series seen in the truncated-polynomial and upper-triangular examples, and it is the property that fails for general rings. It also gives a concrete way to detect non-semisimplicity: a single nonzero nilpotent element of a two-sided ideal already forces the radical to be nonzero. This is exactly the obstruction that appears for modular group algebras.

Having pinned down $\mathrm{Jac}(A)$ as a nilpotent ideal, the pieces are in place for the structural payoff. Dividing A by its radical removes precisely the elements that the simple modules could not see, and what remains is semisimple.

Theorem 52.1.9: Semisimplicity of the Quotient by the Radical

Let A be a left Artinian ring. Then $A/\mathrm{Jac}(A)$ is a semisimple ring, and hence Artin-Wedderburn (theorem 51.2.16) applies to it. Moreover, the simple left A-modules are exactly the simple left $A/\mathrm{Jac}(A)$-modules: every simple A-module is annihilated by $\mathrm{Jac}(A)$ and so factors through $A/\mathrm{Jac}(A)$, and every simple $A/\mathrm{Jac}(A)$-module is a simple A-module via the quotient map $A \twoheadrightarrow A/\mathrm{Jac}(A)$.

***Proof*:**
Write $J = \mathrm{Jac}(A)$ and $\bar{A} = A/J$.

Step 1: $\bar{A}$ is semisimple. By corollary 51.1.9 it suffices to show that $\bar{A}$ is semisimple as a left module over itself, that is that $\mathrm{rad}(\bar{A}) = 0$. The maximal left ideals of $\bar{A}$ are exactly the images $\mathfrak{m}/J$ of the maximal left ideals $\mathfrak{m}$ of A (which all contain $J = \mathrm{Jac}(A)$), by the correspondence theorem. Their intersection is

$$\mathrm{Jac}(\bar{A}) = \bigcap_{\mathfrak{m} \supseteq J} \mathfrak{m}/J = \Big(\bigcap_{\mathfrak{m}} \mathfrak{m}\Big)\Big/ J = J/J = 0$$

So $\mathrm{Jac}(\bar{A}) = 0$. It remains to convert this into semisimplicity, which is where the Artinian hypothesis is spent: $\bar{A}$ is Artinian (a quotient of the Artinian ring A), and for an Artinian ring $\mathrm{Jac}(\bar{A}) = 0$ forces $\bar{A}$ to be semisimple. Concretely, since $\bar{A}$ is Artinian we may pick finitely many maximal left ideals $\mathfrak{m}_1/J, \dots, \mathfrak{m}_r/J$ whose intersection is already 0 (take a minimal finite subintersection, which exists by the descending chain condition and equals the full intersection $\mathrm{Jac}(\bar{A}) = 0$). The map

$$\bar{A} \longrightarrow \bigoplus_{i=1}^{r} \bar{A}/(\mathfrak{m}_i/J)$$

is injective, since its kernel is $\bigcap_i(\mathfrak{m}_i/J) = 0$. Each summand $\bar{A}/(\mathfrak{m}_i/J)$ is a simple $\bar{A}$-module, so $\bar{A}$ embeds as a submodule of a semisimple module and is therefore semisimple by corollary 51.1.8. Thus $\bar{A}$ is a semisimple $\bar{A}$-module, and by corollary 51.1.9 a semisimple ring. Artin-Wedderburn now applies to $\bar{A}$.

Step 2: the simple modules coincide. Let S be a simple left A-module. By part (2) of theorem 52.1.4, $J = \mathrm{Jac}(A)$ annihilates every simple A-module, so $JS = 0$. Hence the A-action on S factors through the quotient $\bar{A} = A/J$: setting $(\bar{a})s := as$ is well-defined because J acts as zero. The A-submodules and $\bar{A}$-submodules of S are the same subsets (a subgroup of S is A-stable if and only if it is $\bar{A}$-stable, since the actions agree), so S is simple as an $\bar{A}$-module precisely because it is simple as an A-module.

Conversely, let T be a simple left $\bar{A}$-module. Pulling back the action along the surjection $\pi\colon A \twoheadrightarrow \bar{A}$, that is defining $a \cdot t := \pi(a)\, t$, makes T an A-module. Its A-submodules are exactly its $\bar{A}$-submodules (again the two actions have the same invariant subgroups, because π is surjective), so T is simple as an A-module. These two passages are mutually inverse, so the simple A-modules and the simple $\bar{A}$-modules are identified, as we sought to show.

This is the structural role of the radical: it is the exact obstruction one divides out to recover the Wedderburn world. The simple modules of A are unchanged by the passage to $A/\mathrm{Jac}(A)$, so the finite list of simples that a semisimple ring enjoys is available for *any* Artinian ring, read off from its semisimple quotient. What the radical takes away is not the simples but the way they fit together: over A the simples can be glued into indecomposable modules that do not split, and the gluing data lives in $\mathrm{Jac}(A)$. The theorem says that for an Artinian A there are finitely many simple A-modules, one for each simple factor of the Wedderburn decomposition of $A/\mathrm{Jac}(A)$, and this is the starting point for the character theory and the block theory developed later.

The special case $\mathrm{Jac}(A) = 0$ returns to chapter 51.

Corollary 52.1.10: Semisimplicity via the Radical

Let A be a left Artinian ring. Then A is semisimple if and only if $\mathrm{Jac}(A) = 0$.

***Proof*:**
If A is semisimple, then by corollary 51.1.9 it is a semisimple left module over itself, so its radical (the intersection of its maximal left ideals) is 0; that is, $\mathrm{Jac}(A) = \mathrm{rad}({}_A A) = 0$. Indeed, a semisimple module is a direct sum of simples, and its radical, the intersection of maximal submodules, is 0: writing $A = \bigoplus_i I_i$ as a sum of minimal left ideals (lemma 51.1.10), each $\bigoplus_{j \neq i} I_j$ is a maximal submodule, and their intersection is 0.

Conversely, if $\mathrm{Jac}(A) = 0$, then $A = A/\mathrm{Jac}(A)$, and by theorem 52.1.9 the quotient $A/\mathrm{Jac}(A)$ is semisimple; hence A itself is semisimple, completing the proof.

The corollary makes the radical the precise measure of the failure of semisimplicity: A is semisimple exactly when the measurement returns zero, and Artin-Wedderburn (theorem 51.2.16) describes what A looks like in that case, namely a product $M_{n_1}(D_1) \times \cdots \times M_{n_k}(D_k)$. For a general Artinian A, the same product describes $A/\mathrm{Jac}(A)$ rather than A, and the difference between A and this product is stored in the nilpotent ideal $\mathrm{Jac}(A)$. The truncated-polynomial algebra $k[x]/(x^n)$ with $n \geq 2$ is the simplest case where the measurement is nonzero: its radical $(\bar{x})$ is a nonzero nilpotent ideal, so it is not semisimple, even though its semisimple quotient is the field k.

The most consequential source of non-semisimple algebras in representation theory is the group algebra $k[G]$ when the characteristic of k divides the order of the group. Maschke's theorem guarantees semisimplicity precisely when $\mathrm{char}(k) \nmid |G|$; the radical detects, and measures, the failure in the opposite case.

Example 52.1.11: The Radical of a Modular Group Algebra
Let $G = \mathbb{Z}/p\mathbb{Z}$ be cyclic of order p and let k be a field. Writing g for a generator, the group algebra is

$$k[G] \cong k[t]/(t^p - 1)$$

via $g \mapsto t$, since g has order p. The structure of $\mathrm{Jac}(k[G])$ depends entirely on whether $\mathrm{char}(k)$ divides $|G| = p$.

The modular case $\mathrm{char}(k) = p$. In characteristic p the Frobenius identity gives $t^p - 1 = (t-1)^p$, so

$$k[G] \cong k[t]/\big((t-1)^p\big) \cong k[s]/(s^p) \qquad (s := t - 1)$$

the last isomorphism being the change of variable $s = t - 1$. This is precisely the truncated polynomial algebra of example 52.1.5 with $n = p$, so its radical is

$$\mathrm{Jac}(k[G]) = (s) = (g - 1)$$

the ideal generated by $g - 1$, which is the augmentation ideal (the kernel of the augmentation map $k[G] \to k$ sending $g \mapsto 1$). It is nilpotent, $(g-1)^p = g^p - 1 = 0$, and nonzero, so $k[G]$ is not semisimple: Maschke's theorem fails exactly because $p = \mathrm{char}(k)$ divides $|G|$. The unique simple $k[G]$-module is the trivial module k, on which g acts as the identity, so $g - 1$ acts as zero, confirming $\mathrm{Jac}(k[G]) = (g-1)$ annihilates the only simple module.

The coprime case $\mathrm{char}(k) \nmid p$. If the characteristic does not divide p, then the polynomial $t^p - 1$ is separable (its derivative pt^{p-1} is nonzero and shares no root with it), so it has no repeated factors. Over a field containing a primitive pth root of unity ζ it splits as $t^p - 1 = \prod_{i=0}^{p-1}(t - \zeta^i)$, and the Chinese remainder theorem gives

$$k[G] \cong \prod_{i=0}^{p-1} k[t]/(t - \zeta^i) \cong k^p$$

a product of p copies of k, which is semisimple with radical $\mathrm{Jac}(k[G]) = 0$. (Over a field without the roots of unity, $t^p - 1$ still factors into distinct irreducible separable factors and $k[G]$ is a product of finite separable field extensions of k, again semisimple with zero radical.) This is Maschke's theorem in the coprime case, seen through the radical: separability of $t^p - 1$ is exactly the condition $\mathrm{char}(k) \nmid |G|$, and it forces $\mathrm{Jac}(k[G]) = 0$.

The contrast in the two cases is the whole point of the chapter in miniature. When $\mathrm{char}(k) \nmid |G|$ the group algebra is semisimple, its radical is zero, and the representation theory of G over k is governed by Wedderburn and character theory. When $\mathrm{char}(k)\big||G|$ the radical becomes a nonzero nilpotent ideal, the group algebra acquires indecomposable modules that are not simple, and one enters modular representation theory, where the radical, the indecomposables, and the projective covers built in the following sections are the working tools.

52.2 Indecomposable Modules and Krull-Schmidt

The previous chapter rested on a single organizing principle: over a semisimple ring, every module splits into simple pieces, and Artin-Wedderburn turns that splitting into a matrix description of the ring itself. The trouble is that semisimplicity is a strong hypothesis. The ring $k[x]/(x^2)$ is a two-dimensional k-algebra, and the module $k[x]/(x^2)$ over itself has a proper nonzero submodule $(x)/(x^2)$, yet that submodule has no complement: there is no way to write $k[x]/(x^2) = (x)/(x^2) \oplus N$. So this module does not break into simples at all. Once the semisimple world is left behind, the simple modules stop being the building blocks, and the decomposition theory of lemma 51.1.7 collapses.

What survives is a weaker notion of "building block". A module that cannot be split into two nonzero summands is *indecomposable* (definition 51.1.1), and every module of finite length is at least a finite direct sum of indecomposables, for the trivial reason that one keeps splitting until nothing splits further. The content of this section is that this crude decomposition is in fact canonical: the list of indecomposable summands, taken up to isomorphism, is uniquely determined by the module. This is the Krull-Schmidt theorem, and it is the correct replacement for the semisimple decomposition when simples are no longer available. The engine driving uniqueness is a structural fact about the endomorphism ring of an indecomposable module of finite length, that it is a *local ring*, which is itself a consequence of Fitting's lemma. So the section runs Fitting's lemma to the local-ring theorem to Krull-Schmidt, each result feeding the next.

Throughout, A is an algebra over a field k (the arguments need only that modules have finite length, so the results apply verbatim to modules of finite length over any ring). Modules are left modules. The starting point is worth isolating, since it is the gap between this chapter and the last. A simple module is indecomposable, because a simple module has no proper nonzero submodule at all, let alone a complemented one. The converse fails, and it fails already in dimension two: the $k[x]$-module $k[x]/(x^2)$ has the unique proper nonzero submodule $(x)/(x^2)$, so it is not simple, but that submodule is not a summand (any complement would have to be one-dimensional and x-stable, and $(x)/(x^2)$ is the only such subspace), so the module is indecomposable. An indecomposable module of finite length can therefore sit strictly between "simple" and "decomposable", and it is exactly this middle ground that Krull-Schmidt organizes.

The first tool is a splitting result for a single endomorphism. Given $\varphi \in \mathrm{End}_A(M)$, the iterated kernels $\ker(\varphi) \subseteq \ker(\varphi^2) \subseteq \cdots$ increase and the iterated images $\mathrm{im}(\varphi) \supseteq \mathrm{im}(\varphi^2) \supseteq \cdots$ decrease. When M has finite length, both chains must stabilize, and Fitting's lemma says that at a common stabilization point the module splits into the eventual kernel and the eventual image.

Lemma 52.2.1: Fitting's Lemma

Let M be an A-module of finite length and let $\varphi \in \mathrm{End}_A(M)$. Then there exists a positive integer n such that

$$M = \ker(\varphi^n) \oplus \mathrm{im}(\varphi^n)$$

Consequently, if M is indecomposable, then every $\varphi \in \mathrm{End}_A(M)$ is either nilpotent or an automorphism.

***Proof*:**
Write $\ell = \ell(M)$ for the length of M, which is finite by hypothesis.

Step 1: The kernel and image chains stabilize. The submodules $\ker(\varphi^i)$ form an increasing chain and the submodules $\mathrm{im}(\varphi^i)$ form a decreasing chain:

$$\ker(\varphi) \subseteq \ker(\varphi^2) \subseteq \cdots \qquad \mathrm{im}(\varphi) \supseteq \mathrm{im}(\varphi^2) \supseteq \cdots$$

A module of finite length is both Noetherian and Artinian, so an increasing chain of submodules cannot strictly ascend more than ℓ times and a decreasing chain cannot strictly descend more than ℓ times. Setting $n = \ell$ makes both chains stationary from index n onward: for all $m \geq n$,

$$\ker(\varphi^m) = \ker(\varphi^n) \qquad \mathrm{im}(\varphi^m) = \mathrm{im}(\varphi^n)$$

Indeed, once the kernel chain repeats, it is stationary forever after: if $\ker(\varphi^i) = \ker(\varphi^{i+1})$ and $x \in \ker(\varphi^{i+2})$, then $\varphi(x) \in \ker(\varphi^{i+1}) = \ker(\varphi^i)$, so $x \in \ker(\varphi^{i+1}) = \ker(\varphi^i)$, giving $\ker(\varphi^{i+2}) = \ker(\varphi^{i+1})$; iterating shows $\ker(\varphi^j) = \ker(\varphi^i)$ for all $j \geq i$. A chain that strictly increased past step ℓ would produce more than ℓ strict inclusions, violating finite length, so the first repeat occurs by step ℓ. The same bound applies to the image chain, and $n = \ell$ dominates both stabilization thresholds.

Step 2: The two submodules intersect trivially. Fix n as above and take $y \in \ker(\varphi^n) \cap \mathrm{im}(\varphi^n)$. Write $y = \varphi^n(x)$ for some $x \in M$. Then $\varphi^n(y) = 0$ gives $\varphi^{2n}(x) = 0$, so $x \in \ker(\varphi^{2n}) = \ker(\varphi^n)$ by the stabilization of the kernel chain. Hence $y = \varphi^n(x) = 0$, and $\ker(\varphi^n) \cap \mathrm{im}(\varphi^n) = 0$.

Step 3: The two submodules span M. Take any $z \in M$. Since $\mathrm{im}(\varphi^n) = \mathrm{im}(\varphi^{2n})$, the element $\varphi^n(z) \in \mathrm{im}(\varphi^n)$ also lies in $\mathrm{im}(\varphi^{2n})$, so $\varphi^n(z) = \varphi^{2n}(w)$ for some $w \in M$. Then

$$z = \varphi^n(w) + \big(z - \varphi^n(w)\big)$$

where $\varphi^n(w) \in \mathrm{im}(\varphi^n)$, and the second term lies in $\ker(\varphi^n)$ because $\varphi^n(z - \varphi^n(w)) = \varphi^n(z) - \varphi^{2n}(w) = 0$. Thus $M = \ker(\varphi^n) + \mathrm{im}(\varphi^n)$, and together with Step 2 this gives the internal direct sum $M = \ker(\varphi^n) \oplus \mathrm{im}(\varphi^n)$.

Step 4: The dichotomy for indecomposable M. Suppose now that M is indecomposable. The decomposition $M = \ker(\varphi^n) \oplus \mathrm{im}(\varphi^n)$ forces one of the two summands to be M and the other to be 0. If $\ker(\varphi^n) = M$, then $\varphi^n = 0$, so φ is nilpotent. If instead $\ker(\varphi^n) = 0$ and $\mathrm{im}(\varphi^n) = M$, then φ is injective, since $\ker(\varphi) \subseteq \ker(\varphi^n) = 0$, and φ is surjective, since $\mathrm{im}(\varphi) \supseteq \mathrm{im}(\varphi^n) = M$. An injective and surjective endomorphism is an automorphism. In either case φ is nilpotent or an automorphism, completing the proof.

The decomposition in Fitting's lemma is the "eventual kernel" and "eventual image" of φ, and it works for any single endomorphism of any finite-length module, indecomposable or not. The sharp consequence appears only in the indecomposable case: an indecomposable module of finite length admits no interesting endomorphisms in the sense that the only two behaviors available to a $\varphi \in \mathrm{End}_A(M)$ are "eventually kills everything" (nilpotent) and "moves everything bijectively" (automorphism). There is no room in between, precisely because any intermediate behavior would produce a nontrivial Fitting decomposition and split the module. This is the mechanism that will make the endomorphism ring so rigid.

To see the dichotomy in a case where it can be checked by hand, take the endomorphism given by multiplication by x on $M = k[x]/(x^3)$. Multiplication by x has kernel $(x^2)/(x^3)$ and image $(x)/(x^3)$, which are not complementary. But the lemma is about a *power*: φ^3 is multiplication by $x^3 = 0$, so $\ker(\varphi^3) = M$ and $\mathrm{im}(\varphi^3) = 0$, giving the trivial decomposition $M = M \oplus 0$. This is the nilpotent branch of the dichotomy, and indeed multiplication by x is nilpotent. Multiplication by $1 + x$, on the other hand, is invertible (with inverse $1 - x + x^2$), landing in the automorphism branch. Every element of $\mathrm{End}_{k[x]}(M)$ is one or the other.

The endomorphism ring $\mathrm{End}_A(M)$ of an indecomposable finite-length module is close to being a division ring but not quite: Schur's lemma (lemma 51.1.5) gave a division ring only for *simple* M, and indecomposable is weaker. The right substitute is that $\mathrm{End}_A(M)$ is a *local ring*. A ring is local when its non-units form a two-sided ideal, equivalently when it has a unique maximal left ideal; the intuition is that "being a non-unit" is closed under addition, so failures of invertibility do not conspire to produce an invertible element. Fitting's lemma is exactly what closes the non-units under addition here, because a non-unit of $\mathrm{End}_A(M)$ is precisely a nilpotent endomorphism, and controlling sums of nilpotents is where the argument lives.

Theorem 52.2.2: Endomorphism Rings of Indecomposables are Local

Let M be a nonzero indecomposable A-module of finite length. Then $\mathrm{End}_A(M)$ is a local ring: the set of non-units of $\mathrm{End}_A(M)$ is a two-sided ideal, and it consists exactly of the nilpotent endomorphisms of M.

***Proof*:**
Write $E = \mathrm{End}_A(M)$ and let $\mathfrak{m} \subseteq E$ be the set of endomorphisms that are not units, that is, not automorphisms of M.

Step 1: The non-units are the nilpotents. By lemma 52.2.1, every $\varphi \in E$ is either an automorphism or nilpotent, and these are mutually exclusive (a nilpotent $\varphi \neq 0$ kills some nonzero element, and $\varphi = 0$ is not invertible since $M \neq 0$). Hence $\mathfrak{m}$ is exactly the set of nilpotent endomorphisms.

Step 2: $\mathfrak{m}$ is closed under left and right multiplication by E. Let $\varphi \in \mathfrak{m}$, so φ is not an automorphism, and let $\psi \in E$ be arbitrary. If $\psi\varphi$ were an automorphism, then φ would be injective (as $\psi\varphi$ is) and ψ surjective; an injective endomorphism of a finite-length module is bijective (its image has the same length as M, hence equals M), so φ would be an automorphism, a contradiction. Thus $\psi\varphi \in \mathfrak{m}$. Symmetrically, if $\varphi\psi$ were an automorphism, then φ would be surjective, hence bijective by the length argument, again contradicting $\varphi \in \mathfrak{m}$. So $\varphi\psi \in \mathfrak{m}$, and $\mathfrak{m}$ absorbs multiplication on both sides.

Step 3: $\mathfrak{m}$ is closed under addition. This is the step that uses indecomposability. Let $\varphi, \psi \in \mathfrak{m}$ and suppose, for contradiction, that $\varphi + \psi$ is an automorphism. Set

$$\alpha = (\varphi + \psi)^{-1}\varphi \qquad \beta = (\varphi + \psi)^{-1}\psi$$

so that $\alpha + \beta = (\varphi + \psi)^{-1}(\varphi + \psi) = \mathrm{id}_M$. By Step 2, $\mathfrak{m}$ absorbs left multiplication, so $\alpha, \beta \in \mathfrak{m}$; in particular α is nilpotent, say $\alpha^r = 0$. From $\beta = \mathrm{id}_M - \alpha$ the standard geometric-series identity gives an inverse for β:

$$(\mathrm{id}_M - \alpha)\big(\mathrm{id}_M + \alpha + \alpha^2 + \cdots + \alpha^{r-1}\big) = \mathrm{id}_M - \alpha^r = \mathrm{id}_M$$

so $\beta = \mathrm{id}_M - \alpha$ is an automorphism of M. But $\beta \in \mathfrak{m}$ means β is not an automorphism, a contradiction. Hence $\varphi + \psi \in \mathfrak{m}$.

Step 4: Conclusion. Steps 2 and 3 show $\mathfrak{m}$ is a two-sided ideal of E, and Step 1 identifies it as the set of non-units. A ring whose non-units form a two-sided ideal is local: any element outside $\mathfrak{m}$ is a unit, so $\mathfrak{m}$ is the unique maximal left ideal (a proper left ideal contains no units, hence lies in $\mathfrak{m}$). Therefore $E = \mathrm{End}_A(M)$ is local, as we sought to show.

The theorem sharpens Schur's lemma in exactly the direction needed. For a simple module, lemma 51.1.5 gave a division ring, where every nonzero element is a unit. For an indecomposable module of finite length, one gets the next best thing: a local ring, where the non-units are corralled into a single ideal and every element outside it is a unit. The residue $\mathrm{End}_A(M)/\mathfrak{m}$ is a division ring, and one can think of the local ring as a division ring with a nilpotent "fuzz" $\mathfrak{m}$ attached. The content used below is the geometric series in Step 3, which turns "id_M minus a nilpotent" into a unit; this is what will let one element of a decomposition be swapped for another in the exchange argument below.

The running example makes the local structure explicit.

Example 52.2.3: The Endomorphism Ring of a Truncated Polynomial Module

Fix $n \geq 1$ and let $M = k[x]/(x^n)$, viewed as a module over $A = k[x]$ (equivalently over $k[x]/(x^n)$, which acts through the same operators). This module is indecomposable of length n: its submodules are exactly the ideals $(x^j)/(x^n)$ for $0 \leq j \leq n$, forming a single chain

$$0 = (x^n)/(x^n) \subseteq (x^{n-1})/(x^n) \subseteq \cdots \subseteq (x)/(x^n) \subseteq M$$

and a chain of submodules has no way to split off a complementary summand (a decomposition $M = M_1 \oplus M_2$ with both $M_i \neq 0$ would give two submodules neither containing the other).

Every A-endomorphism of M is multiplication by some polynomial: since M is generated by the image of 1, a homomorphism φ is determined by $\varphi(1) = \overline{f(x)}$, and then $\varphi(\overline{g}) = \overline{gf}$. So

$$\mathrm{End}_A(M) \cong k[x]/(x^n)$$

as rings, the isomorphism sending φ to $\varphi(1)$. This is a local ring with maximal ideal $\mathfrak{m} = (x)/(x^n)$: an element $\overline{f} = \overline{a_0 + a_1 x + \cdots + a_{n-1}x^{n-1}}$ is a unit if and only if $a_0 \neq 0$ (with inverse computed by the geometric series when $a_0 = 1$), and it is nilpotent if and only if $a_0 = 0$. The non-units are precisely the nilpotents, matching theorem 52.2.2, and $\mathrm{End}_A(M)/\mathfrak{m} \cong k$ is the residue division ring (here a field, since A is commutative).

The local endomorphism ring is now used to prove the main theorem. Existence of a decomposition into indecomposables is easy, by induction on length; the content is uniqueness. The classical route to uniqueness is an *exchange argument*: given two decompositions of the same module, one shows that each indecomposable summand of the first can be swapped into the second, one at a time, without changing the ambient module, and the local endomorphism rings are exactly what make each swap possible. The theorem is due to Krull and Schmidt for finite groups and finite-length modules, with Azumaya's extension to any module that is a direct sum of submodules with local endomorphism rings.

Theorem 52.2.4: Krull-Schmidt(-Azumaya)

Let M be a nonzero A-module of finite length. Then:

1. M is a finite direct sum of indecomposable submodules,
$$M = M_1 \oplus M_2 \oplus \cdots \oplus M_r$$
with each M_i indecomposable.
2. This decomposition is unique up to isomorphism and reordering: if
$$M = M_1 \oplus \cdots \oplus M_r = N_1 \oplus \cdots \oplus N_s$$
are two decompositions into indecomposable submodules, then $r = s$ and there is a permutation σ of $\{1, \ldots, r\}$ with $M_i \cong N_{\sigma(i)}$ for every i.

***Proof*:**

1. *Existence.* Argue by induction on the length $\ell(M)$. If M is indecomposable, then $M = M$ is already such a decomposition, with one summand. Otherwise $M = M_1 \oplus M_2$ with both M_i nonzero, and then $\ell(M_1), \ell(M_2) < \ell(M)$ because length is additive over direct sums and both summands are nonzero (hence of positive length). By the inductive hypothesis each M_i is a finite direct sum of indecomposables, and concatenating the two lists expresses M as a finite direct sum of indecomposables. The base case $\ell(M) = 1$ is a simple, hence indecomposable, module. This proves existence.
2. *Uniqueness.* Suppose
$$M = M_1 \oplus \cdots \oplus M_r = N_1 \oplus \cdots \oplus N_s$$
with all M_i and N_j indecomposable of finite length, so each $\mathrm{End}_A(M_i)$ and $\mathrm{End}_A(N_j)$ is local by theorem 52.2.2. Induct on r. The strategy is to exchange M_1 for one of the N_j, peel it off both sides, and apply the inductive hypothesis to what remains.

   Let $\iota_i \colon M_i \hookrightarrow M$ and $\pi_i \colon M \twoheadrightarrow M_i$ be the inclusion and projection for the first decomposition, and $\iota'_j \colon N_j \hookrightarrow M$, $\pi'_j \colon M \twoheadrightarrow N_j$ those for the second. These satisfy
$$\sum_{i=1}^{r} \iota_i \pi_i = \mathrm{id}_M \qquad \sum_{j=1}^{s} \iota'_j \pi'_j = \mathrm{id}_M$$
   Restricting the second identity to M_1 and projecting back to M_1 gives, in $\mathrm{End}_A(M_1)$,
$$\mathrm{id}_{M_1} = \pi_1 \Big(\sum_{j=1}^{s} \iota'_j \pi'_j \Big) \iota_1 = \sum_{j=1}^{s} (\pi_1 \iota'_j)(\pi'_j \iota_1)$$
   Now $\mathrm{End}_A(M_1)$ is local, so its non-units form an ideal and cannot sum to the unit id_{M_1}; at least one summand $(\pi_1 \iota'_j)(\pi'_j \iota_1)$ must be a unit of $\mathrm{End}_A(M_1)$. After reindexing the N_j

we may assume this happens for $j = 1$: the composite

$$\theta = (\pi_1 \iota_1')(\pi_1' \iota_1) \colon M_1 \longrightarrow M_1$$

is an automorphism of M_1, where $\pi_1' \iota_1 \colon M_1 \to N_1$ and $\pi_1 \iota_1' \colon N_1 \to M_1$.

Exchange step. Consider the maps

$$f = \pi_1' \iota_1 \colon M_1 \to N_1 \qquad g = \pi_1 \iota_1' \colon N_1 \to M_1$$

so $gf = \theta$ is an automorphism of M_1. Then f is a split monomorphism, with left inverse $\theta^{-1} g$, since $(\theta^{-1} g) f = \theta^{-1}(gf) = \mathrm{id}_{M_1}$. The endomorphism $e = f\theta^{-1} g \colon N_1 \to N_1$ is then idempotent:

$$e^2 = f\theta^{-1}(gf)\theta^{-1} g = f\theta^{-1}\theta\theta^{-1} g = f\theta^{-1} g = e$$

and its image is $\mathrm{im}(f)$ (from $ef = f\theta^{-1} gf = f$), so $N_1 = \mathrm{im}(e) \oplus \ker(e) = \mathrm{im}(f) \oplus \ker(e)$ is the splitting associated to the idempotent e. But N_1 is indecomposable and $\mathrm{im}(f) \neq 0$ (since f is injective and $M_1 \neq 0$), which forces $\mathrm{im}(f) = N_1$, making f surjective as well. Hence $f \colon M_1 \to N_1$ is an isomorphism, and in particular

$$M_1 \cong N_1$$

Peeling off the summand. It remains to remove M_1 and N_1 from the two decompositions and match the rest. Set $M' = M_2 \oplus \cdots \oplus M_r$, so $M = M_1 \oplus M'$ and M' is the kernel of the projection $\pi_1 \colon M \to M_1$. The claim is

$$M = N_1 \oplus M'$$

viewing $N_1 = \iota_1'(N_1) \subseteq M$ as the submodule it names in the second decomposition. The key observation is that the restriction of π_1 to N_1 is an isomorphism onto M_1: on N_1 this restriction is $\pi_1 \iota_1' = g$, and $g = \theta f^{-1}$ is a composite of the automorphism θ with the isomorphism f^{-1}, hence itself an isomorphism $N_1 \to M_1$. Granting that $\pi_1|_{N_1} \colon N_1 \to M_1$ is an isomorphism, the claim follows:

- *Trivial intersection.* If $y \in N_1 \cap M'$, then $\pi_1(y) = 0$ because $M' = \ker(\pi_1)$, and $\pi_1|_{N_1}$ is injective, so $y = 0$. Thus $N_1 \cap M' = 0$.
- *Spanning.* Given $z \in M$, set $a = \pi_1(z) \in M_1$. Since $\pi_1|_{N_1}$ is onto M_1, choose $y \in N_1$ with $\pi_1(y) = a$. Then $\pi_1(z - y) = a - a = 0$, so $z - y \in \ker(\pi_1) = M'$, and $z = y + (z - y) \in N_1 + M'$.

Hence $M = N_1 \oplus M'$, establishing the claim.

Induction. Quotienting $M = N_1 \oplus M'$ by the summand N_1 gives $M/N_1 \cong M'$, and quotienting the original $M = N_1 \oplus N'$ by the same submodule N_1 gives $M/N_1 \cong N'$. Therefore

$$M' \cong N'$$

Now M' carries the decomposition $M_2 \oplus \cdots \oplus M_r$ into $r - 1$ indecomposables, and the isomorphism $M' \cong N'$ transports the decomposition $N_2 \oplus \cdots \oplus N_s$ onto M' as a second decomposition into indecomposables. Applying the inductive hypothesis to M', whose first decomposition has $r - 1$ summands, yields $r - 1 = s - 1$, so $r = s$, together with a bijection matching $M_2, \ldots, M_r$ against $N_2, \ldots, N_s$ up to isomorphism. Combined with $M_1 \cong N_1$ from the exchange step, this produces the permutation σ with $M_i \cong N_{\sigma(i)}$ for all i. The base case $r = 1$ has $M = M_1$ indecomposable, forcing $s = 1$ and $N_1 = M$, completing the induction and the proof.

The theorem says that the finite-length representation theory of *any* algebra, semisimple or not, has canonical building blocks: the indecomposable modules. Over a semisimple algebra these are the simple modules and Krull-Schmidt recovers the decomposition of lemma 51.1.7, but the statement here needs no semisimplicity. What it does need is finite length, and the two places that hypothesis enters are worth marking. Finite length gave the Fitting decomposition (via the ascending and descending chain conditions), which produced the local endomorphism rings; and finite length is what let “injective endomorphism” upgrade to “automorphism” at several points. Drop finiteness and uniqueness can fail. A concrete failure: over the ring $R = \mathbb{R}[x, y, z]/(x^2 + y^2 + z^2 - 1)$ of real polynomial functions on the 2-sphere, the module R^3 splits both as $R \oplus P$, where P is the (indecomposable, non-free) module of tangent vector fields, and as $R \oplus R \oplus R$; the two decompositions have non-isomorphic summands, so uniqueness is lost once the summands are no longer of finite length. Krull-Schmidt is thus not a formality but a theorem with content tied to its hypotheses.

The exchange argument also explains *why* the answer is unique, not just *that* it is. The reason each M_i has a unique isomorphism partner is that when the identity of M_1 is written as a sum of composites through the N_j, locality of $\mathrm{End}_A(M_1)$ forbids the sum from being “spread out” among non-units; one composite must already be invertible, and that single composite pins down the partner. Locality is doing the work that a division ring did in the simple case, and it is the exact amount of structure required.

Reading off the multiplicities gives the invariant form of the theorem that is used in practice.

Corollary 52.2.5: Uniqueness of Multiplicities

Let M be a nonzero A-module of finite length. For each isomorphism class of indecomposable module I, the number of summands isomorphic to I appearing in a decomposition of M into indecomposables is independent of the decomposition. That is, writing

$$M \cong I_1^{\oplus m_1} \oplus I_2^{\oplus m_2} \oplus \cdots \oplus I_t^{\oplus m_t}$$

with the I_j pairwise non-isomorphic indecomposables, the classes $I_1, \ldots, I_t$ and the multiplicities $m_1, \ldots, m_t$ are uniquely determined by M.

***Proof*:**
By theorem 52.2.4, any two decompositions of M into indecomposables have the same number of summands and are matched by a permutation σ respecting isomorphism type. Grouping the summands of a decomposition by isomorphism class, the multiplicity of a class I is the number of indices i with $M_i \cong I$. Since σ carries each M_i to an $N_{\sigma(i)} \cong M_i$ and is a bijection, it restricts to a bijection between the summands isomorphic to I in the first decomposition and those isomorphic to I in the second. Hence the two decompositions assign I the same multiplicity, and this holds for every isomorphism class I. The set of classes that occur and their multiplicities are therefore invariants of M, completing the proof.

The multiplicities m_j are the numbers one computes when decomposing a representation, and the corollary is what guarantees the computation has a well-defined answer regardless of the method used to obtain it. In the language of the Grothendieck-style count, M determines a formal $\mathbb{N}$-linear combination $\sum_j m_j[I_j]$ of indecomposable classes, and two modules are isomorphic if and only if these combinations agree. This is the precise sense in which the indecomposables are a basis for the finite-length modules of A.

The full classification is transparent for $A = k[x]/(x^n)$, where the indecomposables can be listed and the uniqueness of multiplicities checked against an explicit computation.

Example 52.2.6: Modules over the Truncated Polynomial Algebra $k[x]/(x^n)$

Fix $n \geq 1$ and let $A = k[x]/(x^n)$. A finite-dimensional A-module is the same datum as a finite-dimensional k-vector space V together with a k-linear operator T, the action of the class of x, satisfying $T^n = 0$, so that T is nilpotent of nilpotency index at most n. Being finite-dimensional over k, every such module has finite length, so Krull-Schmidt applies.

The indecomposables. Since A is a quotient of the principal ideal domain $k[x]$, the structure theorem for finitely generated modules over $k[x]$ describes every finite-length A-module as a direct sum of cyclic modules $k[x]/(x^m)$. Only the exponents $1 \leq m \leq n$ occur, because x^n annihilates the module. Each

$$J_m := k[x]/(x^m) \qquad 1 \leq m \leq n$$

is indecomposable: as in example 52.2.3, its submodules form the single chain $0 \subseteq (x^{m-1}) \subseteq \cdots \subseteq (x) \subseteq J_m$, which admits no splitting into two nonzero summands. So the indecomposable A-modules are exactly

$$J_1, J_2, \ldots, J_n$$

up to isomorphism, with $\dim_k J_m = m$; these are the Jordan blocks for the nilpotent operator, J_m being a single Jordan block of size m with eigenvalue 0. There are precisely n of them.

A decomposition and its uniqueness. Take $n = 3$ and the module $M = A \oplus J_1$, that is, $V = k^4$ with the operator

$$T = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

a single Jordan block of size 3 in the top-left together with a 1×1 zero block. As an A-module,

$$M \cong J_3 \oplus J_1$$

with multiplicities $m_3 = 1$, $m_1 = 1$, and $m_2 = 0$. corollary 52.2.5 asserts these multiplicities are forced by M, and they can be read off invariantly from the ranks of powers of T, which do not depend on any chosen decomposition. The number of Jordan blocks of size at least j equals $\operatorname{rank}(T^{j-1}) - \operatorname{rank}(T^j)$, so the multiplicity of J_m is

$$m_m = \operatorname{rank}(T^{m-1}) - 2\operatorname{rank}(T^m) + \operatorname{rank}(T^{m+1})$$

Here $\operatorname{rank}(T^0) = \operatorname{rank}(\mathrm{id}) = 4$, $\operatorname{rank}(T) = 2$, $\operatorname{rank}(T^2) = 1$, and $\operatorname{rank}(T^3) = 0$. Substituting,

$$m_1 = 4 - 2 \cdot 2 + 1 = 1, \qquad m_2 = 2 - 2 \cdot 1 + 0 = 0, \qquad m_3 = 1 - 2 \cdot 0 + 0 = 1$$

recovering $M \cong J_3 \oplus J_1$. Since the ranks $\operatorname{rank}(T^j)$ are intrinsic to the operator T (hence to the module M), the multiplicities are intrinsic too, exactly as Krull-Schmidt predicts. Any presentation of M as a direct sum of J_m's, arrived at by any change of basis, must yield one copy of J_3, one copy of J_1, and no copy of J_2.

52.3 Idempotents, Projective Covers, and the Regular Representation

The Artin–Wedderburn theorem told us how to read off the simple modules of a semisimple algebra: they are exactly the summands of the regular module A, because a semisimple A splits as a direct sum of minimal left ideals and each minimal left ideal is simple. Once A is not semisimple this picture breaks. The regular module A still decomposes into indecomposable summands (the algebra is finite-dimensional, so a dimension count forces any module to be a finite direct sum of indecomposables), but those summands are no longer simple. They are the *indecomposable projective modules*, and the simple modules sit one level down, as their tops. This section builds the dictionary that replaces the semisimple one: the summands of A are cut out by a decomposition $1 = e_1 + \cdots + e_n$ of the identity into primitive orthogonal idempotents, each summand Ae_i is an indecomposable projective covering a single simple module, and the whole configuration is governed, through the quotient $A/\mathrm{Jac}(A)$, by the semisimple theory of chapter 51.

Throughout this section A is a finite-dimensional algebra over a field k (so in particular A is left Artinian and $\mathrm{Jac}(A)$ is nilpotent). Modules are left modules. The tool that transports information from the semisimple quotient $A/\mathrm{Jac}(A)$ back up to A is the lifting of idempotents, so that is where we begin.

The projection $\pi\colon A \to A/\mathrm{Jac}(A)$ collapses the radical. A decomposition of $A/\mathrm{Jac}(A)$ into a sum of simple summands is a statement about idempotents in $A/\mathrm{Jac}(A)$, and to pull such a decomposition back to A we need to produce idempotents in A that reduce to the given ones. The obstruction is that a preimage of an idempotent need not be idempotent. The nilpotence of $\mathrm{Jac}(A)$ removes the obstruction: a Newton-style correction converts any element that is idempotent modulo $\mathrm{Jac}(A)$ into a genuine idempotent lying over it.

Theorem 52.3.1: Lifting Idempotents Modulo the Radical

Let A be a finite-dimensional k-algebra and write $J = \mathrm{Jac}(A)$. Let $\pi\colon A \to A/J$ be the quotient map.

1. If $\bar{e} \in A/J$ satisfies $\bar{e}^2 = \bar{e}$, then there exists an idempotent $e \in A$ with $\pi(e) = \bar{e}$.
2. If $\bar{e}_1, \ldots, \bar{e}_m \in A/J$ are pairwise orthogonal idempotents with $\bar{e}_1 + \cdots + \bar{e}_m = \bar{1}$, then there exist pairwise orthogonal idempotents $e_1, \ldots, e_m \in A$ with $\pi(e_i) = \bar{e}_i$ and $e_1 + \cdots + e_m = 1$.

***Proof*:**
Since A is left Artinian, $J = \mathrm{Jac}(A)$ is nilpotent by theorem 52.1.8; fix N with $J^N = 0$.

Step 1: Lifting a single idempotent. Choose any $a \in A$ with $\pi(a) = \bar{e}$. Then $a^2 - a \in J$, since $\pi(a^2 - a) = \bar{e}^2 - \bar{e} = 0$. The idea is to correct a inside the coset $a + J$ until the error $a^2 - a$ is pushed into higher and higher powers of J, where it eventually vanishes. Set

$$e = 3a^2 - 2a^3$$

which lies in $a + J$, since $e - a = (3a^2 - 2a^3) - a = -(2a - 1)(a^2 - a) \in J$, so $\pi(e) = \bar{e}$. A direct computation gives

$$e^2 - e = (3a^2 - 2a^3)^2 - (3a^2 - 2a^3) = (a^2 - a)^2\,(4a^2 - 4a - 3)$$

Writing $u = a^2 - a \in J$, this says $e^2 - e \in (u^2) \subseteq J^2$. Thus the operation $a \mapsto 3a^2 - 2a^3$ sends an element whose defect $a^2 - a$ lies in J^r to an element whose defect lies in J^{2r}, while staying in the same coset modulo J. Starting from $a_0 = a$ with defect in J^1 and iterating,

$$a_{t+1} = 3a_t^2 - 2a_t^3$$

produces $a_t \in a + J$ whose defect $a_t^2 - a_t$ lies in J^{2^t}. Once $2^t \geq N$ we have $a_t^2 - a_t \in J^N = 0$, so $e := a_t$ is a genuine idempotent with $\pi(e) = \bar{e}$. This proves (1).

Step 2: Lifting an orthogonal family. We argue by induction on m. For $m = 1$ the statement is $e_1 = 1$, which is immediate. Suppose the result holds for families of size $m-1$, and let $\bar{e}_1, \ldots, \bar{e}_m$ be pairwise orthogonal idempotents summing to $\bar{1}$. Apply the inductive hypothesis to the shorter orthogonal family $\bar{e}_1, \ldots, \bar{e}_{m-2}, \bar{e}_{m-1} + \bar{e}_m$ (its members are pairwise orthogonal idempotents summing to $\bar{1}$, since $\bar{e}_{m-1}$ and $\bar{e}_m$ are orthogonal). This yields pairwise orthogonal idempotents $e_1, \ldots, e_{m-2}, f \in A$ lifting them, with $e_1 + \cdots + e_{m-2} + f = 1$ and $\pi(f) = \bar{e}_{m-1} + \bar{e}_m$.

It remains to split f. Consider the algebra fAf, which has identity element f. Its radical is $\mathrm{Jac}(fAf) = fJf$, and the reduction $fAf \to fAf/fJf$ identifies with the corner $\bar{f}(A/J)\bar{f}$ of the semisimple algebra A/J, where $\bar{f} = \bar{e}_{m-1} + \bar{e}_m$. In that corner, $\bar{e}_{m-1}$ and $\bar{e}_m$ are orthogonal idempotents summing to the identity $\bar{f}$. Applying part (1) inside the finite-dimensional algebra fAf to the idempotent $\bar{e}_{m-1} \in fAf/fJf$ produces an idempotent $e_{m-1} \in fAf$ with $\pi(e_{m-1}) = \bar{e}_{m-1}$. Set $e_m = f - e_{m-1}$. Then e_m is idempotent, since

$$e_m^2 = (f - e_{m-1})^2 = f - 2e_{m-1} + e_{m-1} = f - e_{m-1} = e_m$$

using $fe_{m-1} = e_{m-1}f = e_{m-1}$ (because $e_{m-1} \in fAf$) and $e_{m-1}^2 = e_{m-1}$; moreover $\pi(e_m) = \bar{f} - \bar{e}_{m-1} = \bar{e}_m$, and $e_{m-1}e_m = e_{m-1}f - e_{m-1}^2 = e_{m-1} - e_{m-1} = 0$, with $e_m e_{m-1} = 0$ symmetrically. Finally $e_{m-1}, e_m \in fAf$ are orthogonal to each of $e_1, \ldots, e_{m-2}$, since f is, and $e_1 + \cdots + e_{m-2} + e_{m-1} + e_m = e_1 + \cdots + e_{m-2} + f = 1$. This completes the induction and the proof.

The polynomial $3a^2 - 2a^3$ is the second Newton iterate for the equation $x^2 - x = 0$; the quadratic doubling of the exponent is exactly Newton's quadratic convergence, made to terminate by nilpotence rather than to converge by a metric. The content of the theorem is that the arithmetic of idempotents in A is completely controlled by the arithmetic of idempotents in the semisimple quotient $A/\mathrm{Jac}(A)$: any way of splitting the identity in $A/\mathrm{Jac}(A)$ lifts to a way of splitting it in A, so the summand structure of the regular module is governed from below by Artin–Wedderburn. The next result turns this into a decomposition of A itself.

Before stating it, recall the mechanism from example 51.2.10: an idempotent $e \in A$ produces a direct sum decomposition $A = Ae \oplus A(1-e)$ of the regular module, because every $a \in A$ writes uniquely as $a = ae + a(1-e)$ with $ae \in Ae$ and $a(1-e) \in A(1-e)$. More generally an orthogonal decomposition $1 = e_1 + \cdots + e_n$ gives $A = Ae_1 \oplus \cdots \oplus Ae_n$. The summand Ae_i is indecomposable precisely when e_i cannot be split further, which is the meaning of *primitive*. Pushing the splitting as far as it will go decomposes the regular module into indecomposables.

Theorem 52.3.2: Primitive Idempotent Decomposition of the Regular Module

Let A be a finite-dimensional k-algebra. Then the identity admits a decomposition

$$1 = e_1 + e_2 + \cdots + e_n$$

into pairwise orthogonal primitive idempotents, and correspondingly the regular module decomposes as

$$A = Ae_1 \oplus Ae_2 \oplus \cdots \oplus Ae_n$$

where each Ae_i is an indecomposable projective left A-module. For an idempotent $e \in A$, the summand Ae is indecomposable if and only if e is primitive, if and only if the corner algebra eAe has no idempotents other than 0 and e.

***Proof*:**
Step 1: Idempotent decompositions correspond to direct-sum decompositions. Let $e \in A$ be an idempotent. Direct summands of the module Ae correspond to idempotents of its endomorphism ring: if $Ae = P \oplus Q$, the projection onto P along Q is an idempotent $\varepsilon \in \mathrm{End}_A(Ae)$, and conversely an idempotent $\varepsilon \in \mathrm{End}_A(Ae)$ splits $Ae = \varepsilon(Ae) \oplus (1-\varepsilon)(Ae)$. By the same computation as in lemma 51.2.3, the map $eAe \to \mathrm{End}_A(Ae)$ sending eae to right multiplication $x \mapsto x(eae)$ is a ring isomorphism

$$\mathrm{End}_A(Ae) \cong (eAe)^{\mathrm{op}}$$

The idempotents of $(eAe)^{\mathrm{op}}$ are the idempotents of eAe. So Ae decomposes nontrivially if and only if eAe has an idempotent other than 0 and $e = 1_{eAe}$. Writing such an idempotent as g and its complement $e - g$ (which is idempotent and orthogonal to g, by the computation in Step 2 of theorem 52.3.1), the decomposition $Ae = Ag \oplus A(e-g)$ corresponds to the refinement $e = g + (e-g)$ of e into orthogonal idempotents. Thus e is primitive (cannot be written as a sum of two nonzero orthogonal idempotents) if and only if eAe has no idempotent besides $0, e$, if and only if Ae is indecomposable. This proves the final three-way equivalence.

Step 2: The decomposition terminates. Start from 1. If 1 is primitive, take $n = 1$. Otherwise $1 = f + f'$ with f, f' nonzero orthogonal idempotents, giving $A = Af \oplus Af'$ with both summands nonzero. Since A is finite-dimensional and the summands Af, Af' are proper (each has strictly smaller k-dimension than A, as the other is nonzero), repeating the splitting on any summand that is still decomposable strictly decreases dimension at each step and so must stop. When it stops, every summand Ae_i is indecomposable, so by Step 1 each e_i is primitive. The resulting $e_1, \ldots, e_n$ are pairwise orthogonal (an idempotent split off from a corner Ae is orthogonal to every idempotent orthogonal to e) and sum to 1.

Step 3: Projectivity. Each Ae_i is a direct summand of the free module A, hence projective. A direct summand of a free module is projective because the splitting $A = Ae_i \oplus A(1-e_i)$ exhibits Ae_i as a retract of A, and retracts of free modules satisfy the lifting property defining projectivity; the general statement is proved in the companion volume `NateCoreAlgebra` [25] on projective modules. This gives the decomposition of A into indecomposable projectives, completing the proof.

The decomposition $A = \bigoplus_i Ae_i$ is the non-semisimple replacement for the Wedderburn splitting of a semisimple algebra into minimal left ideals. In the semisimple case each Ae_i was already simple, and the e_i were the matrix units picking out columns of the matrix blocks (proposition 51.2.5). In general each Ae_i is only indecomposable, and it will turn out to have a whole composition series rather than length one. The indecomposable summands that appear are the objects the rest of the section studies, so they deserve a name.

Definition 52.3.3: Indecomposable Projective Module

Let A be a finite-dimensional k-algebra and $1 = e_1 + \cdots + e_n$ a decomposition of the identity into primitive orthogonal idempotents. The indecomposable projective left A-modules are the summands

$$P_i = Ae_i$$

Its *radical* is $\mathrm{rad}(P_i) = \mathrm{Jac}(A)\, P_i$, and its *top* (or *head*) is the quotient

$$\mathrm{top}(P_i) = P_i/\mathrm{rad}(P_i) = Ae_i/\mathrm{Jac}(A)Ae_i$$

The module P_i is called the *projective cover* of its top; the top is a simple module.

The claim that the top is simple is the hinge of the whole correspondence, so we verify it at once. Reducing modulo $\mathrm{Jac}(A)$ turns the primitive idempotent e_i into a primitive idempotent $\bar{e}_i \in A/\mathrm{Jac}(A)$, and over the semisimple algebra $A/\mathrm{Jac}(A)$ a primitive idempotent cuts out a simple module.

Example 52.3.4: The Top of Ae_i

Let $A = k[x]/(x^3)$, a local algebra with $\mathrm{Jac}(A) = (x)/(x^3)$. Here 1 is the only nonzero idempotent (a commutative local ring has no idempotents besides 0 and 1), so the identity decomposition is trivial, $n = 1$ and $e_1 = 1$, giving a single indecomposable projective $P_1 = A$. Its radical is $\mathrm{rad}(P_1) = \mathrm{Jac}(A)\, A = (x)/(x^3)$, and its top is

$$P_1/\mathrm{rad}(P_1) = \big(k[x]/(x^3)\big)\big/\big((x)/(x^3)\big) \cong k[x]/(x) = k$$

a one-dimensional, hence simple, module. So the single projective $P_1 = A$ (of dimension 3) covers the single simple module k (of dimension 1); the two extra dimensions live in the radical $\mathrm{rad}(P_1)$, whose composition factors are again copies of k. This is the smallest example in which a projective indecomposable is strictly larger than the simple it covers.

The passage $P_i \mapsto \mathrm{top}(P_i)$ from indecomposable projectives to simple modules, and its reverse, is the organizing principle of non-semisimple representation theory. Simple modules are visible in the semisimple quotient $A/\mathrm{Jac}(A)$, where Artin–Wedderburn counts them; primitive idempotents in A are the summands of the regular module; and lifting idempotents connects the two. The following theorem assembles these into a single correspondence.

Theorem 52.3.5: Simples, Indecomposable Projectives, and Primitive Idempotents

Let A be a finite-dimensional k-algebra, and let $1 = e_1 + \cdots + e_n$ be a decomposition of the identity into primitive orthogonal idempotents. There are mutually inverse bijections among the following three finite sets:

1. isomorphism classes of simple left A-modules;
2. isomorphism classes of indecomposable projective left A-modules;
3. conjugacy classes of primitive idempotents of A, where e and e' are conjugate if $e' = ueu^{-1}$ for a unit $u \in A^\times$.

The bijection from (2) to (1) sends $P_i \mapsto S_i := P_i/\mathrm{rad}(P_i)$; the bijection from (3) to (2) sends the class of e_i to $P_i = Ae_i$. If A is split (for instance $k = \bar{k}$), the number of classes equals the number of blocks in the Artin–Wedderburn decomposition of $A/\mathrm{Jac}(A)$, and

$$A/\mathrm{Jac}(A) \cong M_{d_1}(k) \times \cdots \times M_{d_r}(k), \qquad d_i = \dim_k S_i$$

where $S_1, \ldots, S_r$ are the simple modules and P_i appears $d_i = \dim_k S_i$ times in $A = \bigoplus_i P_i^{\oplus m_i}$.

***Proof*:**
Write $J = \mathrm{Jac}(A)$ and $\bar{A} = A/J$, which is semisimple by theorem 52.1.9.

Step 1: Simple modules are the same for A and $\bar{A}$. A simple A-module S is annihilated by J: the submodule $JS \subseteq S$ is either 0 or S, and $JS = S$ is impossible by Nakayama's lemma (lemma 52.1.7, applied to the cyclic module S: $JS = S$ would force $S = 0$). Hence $JS = 0$ and S is an $\bar{A}$-module; conversely every $\bar{A}$-module is an A-module through $\pi\colon A \to \bar{A}$. Since a submodule for A is the same as a submodule for $\bar{A}$ on a module killed by J, simplicity is preserved in both directions. Thus the simple A-modules are exactly the simple $\bar{A}$-modules, and by corollary 51.1.12 there are finitely many, one for each Wedderburn block of $\bar{A}$.

Step 2: $\mathrm{top}(P_i)$ is simple, and $P_i \mapsto \mathrm{top}(P_i)$ lands in (1). Reducing the decomposition $1 = e_1 + \cdots + e_n$ modulo J gives $\bar{1} = \bar{e}_1 + \cdots + \bar{e}_n$, a decomposition into orthogonal idempotents of $\bar{A}$. Applying the right-exact functor $\bar{A} \otimes_A -$ (equivalently reducing mod J) to $P_i = Ae_i$ gives

$$P_i/JP_i = Ae_i/JAe_i \cong \bar{A}\bar{e}_i$$

as $\bar{A}$-modules. Now $\bar{e}_i$ is primitive in $\bar{A}$: an orthogonal splitting $\bar{e}_i = \bar{g} + \bar{h}$ in $\bar{A}$ would lift, by theorem 52.3.1 applied inside the corner e_iAe_i, to an orthogonal splitting of e_i in A, contradicting primitivity of e_i. Over the semisimple algebra $\bar{A}$ a primitive idempotent generates a simple module, since $\bar{A}\bar{e}_i$ is an indecomposable summand of the semisimple module $\bar{A}$ and indecomposable semisimple means simple. Hence $\mathrm{top}(P_i) = P_i/\mathrm{rad}(P_i) = \bar{A}\bar{e}_i =: S_i$ is simple.

Step 3: The map (2) $\to$ (1) is a bijection. Distinct indecomposable projectives have distinct tops: if $Ae_i \cong Ae_j$ as A-modules then reducing mod J gives $\bar{A}\bar{e}_i \cong \bar{A}\bar{e}_j$, so $S_i \cong S_j$; conversely if $S_i \cong S_j$ then the two primitive idempotents $\bar{e}_i, \bar{e}_j$ generate isomorphic simple summands of $\bar{A}$, hence are conjugate in $\bar{A}$ (two primitive idempotents of a semisimple algebra generating isomorphic simples lie in the same Wedderburn block and differ by an inner automorphism), and a conjugating unit lifts to a unit of A because $\pi\colon A^\times \to \bar{A}^\times$ is surjective (an element of A mapping to a unit of $\bar{A}$ is itself a unit, since J is nilpotent: if $\bar{a}$ is invertible then a has a left and right inverse modulo the nilpotent ideal J, and $1+J \subseteq A^\times$). Conjugate idempotents give isomorphic projectives $Ae_i \cong Ae_j$ via right multiplication by the conjugating unit. So $P_i \cong P_j$ if and only if $S_i \cong S_j$, giving a bijection between (2) and (1). Every simple module arises as some S_i: a simple $\bar{A}$-module is a summand of $\bar{A} = \bigoplus_i \bar{A}\bar{e}_i$, hence isomorphic to some $\bar{A}\bar{e}_i = S_i$.

Step 4: Conjugacy classes of primitive idempotents. By the last paragraph, $Ae_i \cong Ae_j$ as modules if and only if e_i and e_j are conjugate in A: one direction lifts a conjugation, the other reduces an isomorphism to a conjugation via $\mathrm{End}_A(A) \cong A^{\mathrm{op}}$ (lemma 51.2.3), under which a module isomorphism $Ae_i \to Ae_j$ is right multiplication by an element u with $ue_iu^{-1} = e_j$. Thus the class of e_i under conjugacy corresponds to the isomorphism class of P_i, giving the bijection between (3) and (2).

Step 5: The split case. If A is split, Artin–Wedderburn over the semisimple $\bar{A}$ (corollary 51.2.20) gives $\bar{A} \cong \prod_{i=1}^r M_{d_i}(k)$ with r the number of simple modules and $d_i = \dim_k S_i$, since the simple module of the block $M_{d_i}(k)$ is k^{d_i}. The multiplicity of P_i in $A = \bigoplus_j Ae_j$ equals the number of j with e_j conjugate to e_i, which reducing mod J equals the multiplicity of the column module k^{d_i} inside the block $M_{d_i}(k)$, namely d_i; so $m_i = d_i$. This gives the stated dimension bookkeeping, as we sought to show.

The correspondence is the reason the count of simple modules is finite and computable: it equals the number of Wedderburn blocks of $A/\mathrm{Jac}(A)$, which Artin–Wedderburn hands us. In the semisimple case the three sets coincide with the columns of the matrix blocks and the map $P_i \mapsto S_i$ is the identity; the passage to a general finite-dimensional algebra keeps the same indexing set but separates the projective P_i from the simple S_i it covers. The word "cover" is precise, and the next definition makes it so. A cover should be the most economical projective surjecting onto a module, with no wasted projective sitting over the radical.

Definition 52.3.6: Superfluous Submodule and Projective Cover

Let A be a ring and M an A-module. A submodule $L \subseteq M$ is *superfluous* (or *small*) if for every submodule $N \subseteq M$ the equality $L + N = M$ forces $N = M$. A surjection $f\colon P \to M$ is *essential* if $\ker f$ is superfluous in P.
A *projective cover* of M is a projective module P together with an essential surjection $f\colon P \to M$; that is, f is surjective, P is projective, and $\ker f$ is superfluous in P.

The condition that $\ker f$ be superfluous is what pins the cover down. Any module has a surjection from a projective (a free module surjects onto it), but without the superfluous condition one could pad the source with a free summand mapping to zero and never have a canonical choice. Superfluousness forbids exactly this padding: no proper submodule of P maps onto M, so P is as small a projective as can surject onto M. Over a finite-dimensional algebra the radical supplies a ready source of superfluous submodules, and this makes covers exist and be unique.

Theorem 52.3.7: Existence of Projective Covers

Let A be a finite-dimensional k-algebra and M a finitely generated (equivalently, finite-length) left A-module. Then M has a projective cover $f\colon P \to M$, unique up to isomorphism over M. For a simple module $S = S_i$, the projective cover is the indecomposable projective $P_i = Ae_i$ with its quotient map $P_i \twoheadrightarrow P_i/\mathrm{rad}(P_i) = S_i$.

***Proof*:**
Write $J = \mathrm{Jac}(A)$.

Step 1: JP is superfluous in a finitely generated P. Let P be finitely generated and suppose $JP + N = P$ for a submodule $N \subseteq P$. Then $J \cdot (P/N) = (JP + N)/N = P/N$, so the finitely generated module P/N satisfies $J \cdot (P/N) = P/N$. Nakayama's lemma (the module form, lemma 52.1.7; concretely, using the nilpotence of J: if $J^N = 0$ and $JQ = Q$ then $Q = JQ = J^2Q = \cdots = J^N Q = 0$) forces $P/N = 0$, i.e. $N = P$. Hence JP is superfluous in P.

Step 2: Construction of the cover. The top $\bar{M} = M/JM$ is a module over the semisimple algebra $\bar{A} = A/J$, hence semisimple: $\bar{M} \cong \bigoplus_i S_i^{\oplus c_i}$ for multiplicities $c_i \geq 0$. Set

$$P = \bigoplus_i P_i^{\oplus c_i}$$

with each $P_i = Ae_i$ the indecomposable projective covering S_i. Then $P/JP = \bigoplus_i (P_i/JP_i)^{\oplus c_i} = \bigoplus_i S_i^{\oplus c_i} = \bar{M}$, so the projection $P \to P/JP = \bar{M}$ agrees with $M \to \bar{M}$ on tops. Because P is projective, the surjection $M \twoheadrightarrow \bar{M}$ lifts through $P \to \bar{M}$ to a homomorphism $f\colon P \to M$ with $M \to \bar{M}$ equal to $P \xrightarrow{f} M \to \bar{M}$. The image $f(P)$ satisfies $f(P) + JM = M$ (their images in $\bar{M}$ agree and equal all of $\bar{M}$), and JM is superfluous in M by Step 1 applied to M; hence $f(P) = M$ and f is surjective.

Step 3: f is essential. Let $K = \ker f$. Suppose $K + N = P$ for a submodule N. Then $f(N) = f(P) = M$, so $N + JP \supseteq N + \ker(P \to \bar{M})$ surjects onto $\bar{M} = P/JP$, giving $N + JP = P$; since JP is superfluous (Step 1), $N = P$. Thus K is superfluous and $f\colon P \to M$ is a projective cover.

Step 4: Uniqueness. Let $f\colon P \to M$ and $f'\colon P' \to M$ be two projective covers. Projectivity of P' lifts f' through f to $g\colon P' \to P$ with $fg = f'$. Then $f(g(P')) = f'(P') = M$, so $g(P') + \ker f = P$; since $\ker f$ is superfluous, $g(P') = P$ and g is surjective. As P' is projective, the surjection g splits, $P' = \ker g \oplus P''$ with $P'' \cong P$; but $\ker g \subseteq \ker f' = \ker(fg)$ is superfluous in P' and a superfluous direct summand is zero (a summand L with complement C satisfies $L + C = P'$, forcing $C = P'$ hence $L = 0$). So g is an isomorphism over M.

Step 5: The simple case. For $M = S_i$ the top $\bar{M} = S_i$ has $c_i = 1$ and all other multiplicities 0, so the construction returns $P = P_i$ with $f\colon P_i \twoheadrightarrow P_i/\mathrm{rad}(P_i) = S_i$, the stated cover, completing the proof.

The theorem certifies the terminology in definition 52.3.3: $P_i \to S_i$ really is *the* projective cover, and every finite-length module is covered by an explicit direct sum of the P_i read off from its top. Uniqueness makes the assignment $M \mapsto P(M)$ well defined and turns the correspondence of theorem 52.3.5 into a computational tool: to find the projective cover of any module, compute its top $M/\mathrm{Jac}(A)M$, decompose that semisimple module into simples, and sum the corresponding P_i. What the projective P_i contributes beyond the simple top S_i is recorded in its composition series, and tabulating those multiplicities across all i produces the central numerical invariant of the algebra.

Definition 52.3.8: Cartan Matrix of an Algebra

Let A be a finite-dimensional k-algebra with simple modules $S_1, \ldots, S_r$ and indecomposable projectives $P_1, \ldots, P_r$, indexed so that $P_i/\mathrm{rad}(P_i) = S_i$. The *Cartan matrix* of A is the $r \times r$ matrix $C = (C_{ij})$ of nonnegative integers

$$C_{ij} = [P_i : S_j]$$

the multiplicity of the simple module S_j among the composition factors of a composition series of P_i.

The multiplicities $[P_i : S_j]$ are well defined by the Jordan–Hölder theorem, which guarantees the composition factors of a finite-length module are independent of the chosen composition series. The Cartan matrix measures how far A is from semisimple in a single array: when A is semisimple every $P_i = S_i$ has length one, so $C_{ij} = \delta_{ij}$ is the identity matrix, and any off-diagonal or large diagonal entry records simples fused together by the radical. The diagonal entry $C_{ii} = [P_i : S_i]$ is at least 1 (the top contributes one copy of S_i), and $C_{ii} > 1$ signals a self-extension living in the radical. The next example computes the whole apparatus of this section for the smallest non-semisimple algebra with two non-isomorphic simple modules, so that the Cartan matrix is not diagonal.

Before the example, one further point for completeness: the entire theory has a dual half. Everything above was built from the regular module A as a source of projectives; dualizing over k turns projectives into injectives and covers into envelopes.

52.3.9 Remark: The Dual Theory of Injective Envelopes Dual to a projective cover is an *injective envelope*: an injective module E with an essential injection $M \hookrightarrow E$, meaning every nonzero submodule of E meets the image of M nontrivially. Over a finite-dimensional algebra A these exist and are unique, and the duality $D = \mathrm{Hom}_k(-, k)$ carries left A-modules to right A-modules (equivalently left A^{op}-modules), exchanging projective covers of A-modules with injective envelopes of A^{op}-modules. The indecomposable injective A-modules are thus $D(e_i A)$, and each has a simple *socle* (its largest semisimple submodule) rather than a simple top; the socle plays for injectives the role the top plays for projectives. The general theory of injective and projective modules over a ring, including existence of injective envelopes via injective hulls, is developed in the companion volume `NateCoreAlgebra` [25]; the present section uses only the projective side.

Example 52.3.10: Projective Covers over the Upper-Triangular Algebra $T_2(k)$

Let

$$A = T_2(k) = \left\{ \begin{pmatrix} a & b \\ 0 & c \end{pmatrix} : a, b, c \in k \right\}$$

the algebra of 2×2 upper-triangular matrices over a field k, a three-dimensional k-algebra. We work out its idempotents, indecomposable projectives, simple modules, projective covers, and Cartan matrix.

Idempotents. The matrix units

$$e_1 = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \qquad e_2 = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}$$

are orthogonal idempotents with $e_1+e_2=1$. Each is primitive: the corner algebra $e_1Ae_1 = k\,e_1 \cong k$ has no idempotents but $0, e_1$, and likewise $e_2Ae_2 \cong k$, so by theorem 52.3.2 both Ae_1 and Ae_2 are indecomposable. Thus $1 = e_1 + e_2$ is a primitive orthogonal decomposition and there are two indecomposable projectives.

The radical. The Jacobson radical is the strictly upper-triangular part,

$$\mathrm{Jac}(A) = \begin{pmatrix} 0 & k \\ 0 & 0 \end{pmatrix}$$

a one-dimensional nilpotent ideal with $\mathrm{Jac}(A)^2 = 0$, and the quotient is

$$A/\mathrm{Jac}(A) \cong \begin{pmatrix} k & 0 \\ 0 & k \end{pmatrix} \cong k \times k$$

a product of two copies of k, confirming via theorem 52.3.5 that there are exactly two simple modules, both one-dimensional.

The indecomposable projectives. Left multiplication by A on e_i gives, with $E_{12} = \left(\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}\right)$,

$$P_1 = Ae_1 = \begin{pmatrix} k & 0 \\ 0 & 0 \end{pmatrix}, \qquad P_2 = Ae_2 = \begin{pmatrix} 0 & k \\ 0 & k \end{pmatrix}$$

so $\dim_k P_1 = 1$ and $\dim_k P_2 = 2$, and indeed $\dim_k A = 1 + 2 = 3 = \dim_k P_1 + \dim_k P_2$, consistent with $A = P_1 \oplus P_2$.

The simple modules and covers. The tops are

$$S_1 = P_1/\mathrm{rad}(P_1) = P_1/\mathrm{Jac}(A)P_1 = P_1 \cong k, \qquad S_2 = P_2/\mathrm{rad}(P_2) = P_2/\mathrm{Jac}(A)P_2 \cong k$$

Here $\mathrm{Jac}(A)P_1 = 0$, so $P_1 = S_1$ is already simple and is its own projective cover, an essential surjection $P_1 \xrightarrow{\cong} S_1$. For P_2, the radical $\mathrm{Jac}(A)P_2 = \left(\begin{smallmatrix} 0 & k \\ 0 & 0 \end{smallmatrix}\right)$ is one-dimensional, and

$$\mathrm{rad}(P_2) = \begin{pmatrix} 0 & k \\ 0 & 0 \end{pmatrix} \cong S_1, \qquad P_2/\mathrm{rad}(P_2) \cong S_2$$

So P_2 is a length-two module with a nonsplit exact sequence

$$0 \to S_1 \to P_2 \to S_2 \to 0$$

and the quotient map $P_2 \twoheadrightarrow S_2$ is the projective cover of S_2, with kernel $\mathrm{rad}(P_2) = S_1$ superfluous in P_2 (it lies inside $\mathrm{Jac}(A)P_2$). The two simple modules S_1 and S_2 are not isomorphic, since e_1 and e_2 act as 1 on one and 0 on the other; the algebra is not semisimple, as $\mathrm{Jac}(A) \neq 0$.

The Cartan matrix. Reading off composition factors: P_1 has the single factor S_1, while P_2 has factors S_2 (its top) and S_1 (its radical). Hence

$$[P_1 : S_1] = 1, \quad [P_1 : S_2] = 0, \quad [P_2 : S_1] = 1, \quad [P_2 : S_2] = 1$$

and the Cartan matrix is the upper-triangular integer matrix

$$C = \begin{pmatrix} C_{11} & C_{12} \\ C_{21} & C_{22} \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}$$

Its determinant is 1 and its off-diagonal entry $C_{21} = 1$ records the single copy of S_1 that the radical of P_2 contributes: the exact numerical trace of the failure of $T_2(k)$ to be semisimple, where the Cartan matrix would have been the identity.

53

Representation of Finite Groups

All modules will be left R-modules for a not necessarily commutative ring R. Furthermore, all actions will be non-trivial actions unless specified otherwise.

53.1 Representations of Finite Groups

Similarly to how we studied groups and by how they acted on themselves through group-actions (ex. how a group acted on itself by conjugation gave us Sylow's Theorem), we will now try to study groups using *linear algebra*. Many really difficult results in classical group theory will become much easier when studied through this new "representation" of groups (and hence why it's called *represenation theory*). Furthermore, vector-spaces afford many more structures than a module structures (ex. a complex vector-space is the starting point for the Nullstellensatz, Complex Geometry, Analysis, and much more!) Hence, one large advantage of Representation theory will be the intersection of many mathematical diciplines which contribute towards our study (an they all feedback on each other)

In this part of the group, G will always be a finite group[1], k will always be a field (usually $\mathbb{C}$, but can be any field of any character), $\mathrm{Aut}_{k\text{-}\mathbf{Vect}}(V) = \mathrm{GL}(V)$ represents all automorphisms from the k-vector space V to itself, which when $\dim(V) < \infty$ is equivalent to the group of all nonsingular linear transformations from V to itself $\mathrm{GL}_n(k)$ where $n = \dim(V)$.

Definition 53.1.1: Representation

Let G be a finite group and V a vector space over k. Then a *linear representation* is any homomorphism $\varphi : G \to \mathrm{GL}(V)$. The *degree* of the representation is the dimension of V. A *matrix representation* of G is any homomorphism $\varphi : G \to \mathrm{GL}_n(k)$

[1] Not sure about where to address infinite groups

Usually, when we say representation we will mean *linear representation.* If a vector-space V has a representation of G, we will often call it a G-representation. To be as rigorous as possible we would say we have a representation of G on V over k, if the context requires it. Furthermore, most of the time we will have map from G to $\mathrm{GL}(V)$ to be *injective*; non-injective representations will be more common when representing infinite groups that are "trivial" on some subgroup (see [20]). Just like for group actions, A representation is *faithful* if it is injective.

As usual, we can curry or uncurry an action[2]: we can take $\varphi : G \to \mathrm{GL}(V)$ and make it $G \times V \to V$ where $g : V \to V$ with $v \mapsto g \cdot v$ is linear over k for all g. It is left for the curious to see that this is indeed the proper uncurrying of this action.

Using the definition of $\varphi : G \to \mathrm{GL}(V)$, it is easy to define a representation homomorphism mirroring the definition of group-actions and modules (which are ring actions on abelian groups):

Definition 53.1.2: Representation Homomorphism

Let V, W be two vectors spaces over k with a linear representation by G. Then $\varphi : V \to W$ a *representation homomorphism* if φ k-linear and for any $g \in G$:

$$\varphi(g \cdot v) = g \cdot \varphi(v)$$

such a homomorphism will be called G-linear.

From this perspective, it might be hard to see how our previous discussion on semi-simple rings might give us any insight. We thus introduce a perspective better suited to the results of chapter 51: every representation of G on V over k is in bijective correspondence with all $k[G]$-modules over V, where $k[G]$ represents the *group ring* as seen in section 9.2. This second perspective will also make it easier to define a couple of examples, and so we will delay giving examples till the correspondence has been established. As a reminder, a group ring $k[G]$ over a ring k (which in our case will always be a field) is the set of all formal sums:

$$\sum_{g_i \in G} a_i g_i$$

where addition and multiplication are defined in the natural way[3]:

$$\sum_{i}^{n} a_i g_i + \sum b_i g_i = \sum_{i} (a_i + b_i) g_i$$

$$\left(\sum_{i}^{n} a_i g_i \right) \left(\sum b_i g_i \right) = \sum_{k}^{n} \left(\sum_{\substack{i,j \\ g_i g_j = g_k}} a_i b_j \right) g_k$$

Another common notation for groups rings is kG, where the brackets are dropped. A couple of important acts about group rings should be recalled (or the excises ref:HERE can be completed)

1. The group G naturally injects into $k[G]$ by identifying g_i with $1 g_i$ which is the formal sum $\sum_j \delta_{ij} g_i$ (where δ_{ij} is the Kronecker delta).

[2] See remark 4.1.3

[3] A word on this really being a natural way?

2. The field k naturally injects in $k[G]$ by corresponding an element $a \in k$ with $ag_1 = ae$ where $g_1 = e$ is the identity. If it is clear from context, we will sometimes simply write $a \in k[G]$ if it is understood $a = ae = ag_1$.

3. There is a natural k action on $k[G]$ defined by

$$b \cdot \left(\sum_i a_i g_i \right) = \sum_i (ba_i) g_i$$

due to this, $k[G]$ is a vector space over k with the elements of G as a basis, and hence is finite dimensional if $|G| < \infty$.

4. By definition, The elements of k identified as kg_1 commute with all the elements of $k[G]$, hence it is in the center of $k[G]$: $k \subseteq Z(k[G])$. Thus, $k[G]$ is in fact a k-algebra.

5. A group ring the adjoint given by the forgetful functor from **Ring** to **Grp**, hence is free with respect to groups, justifying the name “formal sum” for it’s elements.

Using these, we show how representations of G on V over k correspond to $k[G]$-modules: let $\varphi : G \to \mathrm{GL}(V)$ be a representation, with $G = \{g_1, g_2, ..., g_n\}$. Thus, each $\varphi(g_i)$ is linear transformation (in particular an automorphism). Define the $k[G]$ action on V by defining:

$$\left(\sum_i^n a_i g_i \right) \cdot v = \left(\sum_i^n a_i \varphi(g_i) \right)(v) = \sum_i^n a_i \varphi(g_i)(v)$$

Then this defines a $k[G]$-module (as should be verified if it doesn’t seem like it). A particular reminder for those who are checking that $\varphi(g_i g_j) = \varphi(g_i) \circ \varphi(g_j)$, since φ is a group homomorphism onto $\mathrm{GL}(V)$ under composition. Notice how this action in fact extends the action of k on $k[G]$ since $a \cdot v = av$ (similarly to how the $F[x]$ action extended the action of F on V in section 14.3).

Conversely, let’s say V is an $k[G]$-module. Seeking to make V a vector space over k and finding a group homomorphism $\varphi : G \to \mathrm{GL}(V)$, the first thing is that V is already a K vector-space by limiting to the action of $k \subseteq k[G]$. Furthermore, for each $g \in G$, we can obtain a map $\varphi : V \to V$ by defining:

$$\varphi(g)(v) = g \cdot v$$

Then it is easy to check that $\varphi(g)$ is a linear transformation: for $\alpha, \beta \in k$, $v, w \in V$

$$\begin{aligned} \varphi(g)(\alpha v + \beta w) &= g \cdot (\alpha v + \beta w) \\ &= \alpha(g \cdot v) + \beta(g \cdot w) \\ &= \alpha\varphi(g)(v) + \beta\varphi(g)(w) \end{aligned}$$

and by the associativity axiom of modules, we get

$$\varphi(g_i g_j)(v) = (\varphi(g_i) \circ \varphi(g_j))(v)$$

Thus, $\varphi : G \to \mathrm{GL}(V)$ is indeed a group homomorphism, and we have:

$$\left\{ V \text{ an } k[G]\text{-module} \right\} \rightleftarrows \left\{ \begin{array}{c} V \text{ a vector space over } k \\ \text{and} \\ \varphi : G \to \mathrm{GL}(V) \text{ a representation} \end{array} \right\}$$

Note that this also means that this also shows that linear representations also have kernel objects, since modules have kernels. Hence, the kernel of a G-representation homomorphism between V and W is G-invariant.

Definition 53.1.3: Affording Representation

Let V be a $k[G]$-module. Then given the previous discussion, we say that V *affords* the representation φ of G.

Given this new perspective, we can also characterize sub-actions. If we have a $k[G]$-module V, then a $k[G]$-submodule $W \subseteq V$ would have to be a k-submodule, and hence a subspace, but it also has to be closed under the action of g on W, i.e. $g \cdot W \subseteq W$. Hence, any $k[G]$-submodule is a G-stable subspace of V. Conversely, if $W \subseteq V$ is G-stable, then it is closed under the action of G and is a subspace, and hence a $k[G]$-submodule. Hence

$$\left\{ W \subseteq V \text{ a } k[G]\text{-submodule} \right\} \rightleftarrows \left\{ W \subseteq V \text{ a } G\text{-stable subspace} \right\}$$

Next, we should know when two representations are equivalent. Recall that when studying a linear transformation T on V to itself, a basis for V can be found that gives a "canonical form" to the matrix representation of T. Changing basis for V kept V the same, but changed the matrix representation of T up to similarity (i.e. changed the isomorphism from $\mathrm{GL}(V)$ to $\mathrm{GL}_n(k)$). The following is the analogous term:

Definition 53.1.4: Equivalent Representation

Let φ and ψ be two representations of G over k. Then we say these two represetnations are *equivalent* (or *similar*) if the $k[G]$-module affording them are isomorphic as modules. If two representations are not equivalent, they are called *inequivalent* (or *not similar*).

When the field is known[a], if V and W are equivalent (i.e. isomorphic as $k[G]$-modules), then we write $V \cong_G W$.

[a]In most cases, the field will be $\mathbb{C}$

Note that two different representations can mean two different vector-spaces over k. If $\varphi : G \to \mathrm{GL}(V)$ and $\psi : G \to \mathrm{GL}(W)$ are equivalent representations, it would make sense if there is a relation between V and W. If $T : V \to W$ is the $k[G]$-module isomorphism between them, It is a k-module isomorphism, and hence $V \cong W$ as vector spaces, implying $\dim(V) = \dim(W)$. Furthermore, by definition of T, $T(g \cdot v) = g \cdot T(v)$. By definition of the action, we have that $T(\varphi(g)(v)) = \psi(g)(T(v))$, i.e.

$$T \circ \varphi(g) = \psi(g) \circ T \quad \Leftrightarrow \quad T \circ \varphi(g) \circ T^{-1} = \psi(g) \qquad \forall g \in G$$

That is, T is the *simultaneous change of basis* for all $\varphi(g)$, $g \in G$. In terms of matrices, there must exist a fixed invertible matrix P such that for all $g \in G$,

$$P\varphi(g)P^{-1} = \psi(g)$$

The function T (or matrix P) are said to *intertwine* φ and ψ (it gives the rules on how to relate φ and ψ).

Example 53.1.5: Representations

1. Let G be any finite group, V any 1 dimensional vector-space over k. Define the $k[G]$-module structure on V by
$$g \cdot v = v$$
for all $g \in G$, $v \in V$. This called the *trivial representation* of G. This module affords the representation $\varphi : G \to \mathrm{GL}(V)$ by $\varphi(g) = I$ (the corresponding matrix representation from G to $\mathrm{GL}_1(k)$ sends $g \mapsto I$).

2. Let G be any finite group and k any field. Let $V = k[G]$ and consider it as a $k[G]$-module over itself. Then $\dim_k(V) = |G| = n$, with the basis being the elements of G. Then we see that each $g \in G$ permute the basis elements:
$$g \cdot g_i = gg_i = g_j \in G$$
Thus, the matrix representation for the action by the element g is an $n \times n$ matrix with 1 in the ij component if $g \cdot g_j = g_i$ (hence n 1's) and zero everywhere else. This representation of g is called the *regular representation* of G. Note how the underlying field didn't matter: the coefficients in the matrices can only ever by 1 or 0, and we would simply say "Let φ be the regular representation of G" without specifying which field the group-ring is.

3. Let $G = S_n$ ($n \in \mathbb{N}_{>0}$) and k be any field. Then let V be a vector-space over k of dimension n, $(e_1, e_2, ..., e_n)$ a basis for V, and define the action:
$$\sigma \cdot e_i = e_{\sigma(i)}$$
that is, σ permute the basis of V. Through this, we see that we afford a faithful representation $\varphi : S_n \hookrightarrow \mathrm{GL}(V)$. Furthermore, since any finite group is a subgroup of S_n for an appropriate n, we see that a group of order n we can define $\varphi : G \to S_n \to \mathrm{GL}(V)$, giving another perspective on how to embed any finite group into a matrix group. Notice too how we made a $k[G]$-module given the action of a $k[S_n]$-module. This is called the *permutation representation* of G.

   Note that this means that *any* finite groups acts on a vector space, and by using this construction we see that any finite group G of size n can be embedded into $\mathrm{GL}_n(k)$ showing how much structure the matrix groups embody.

4. Generlizing the last result, if $\psi : H \to \mathrm{GL}(V)$ is any representation and $\varphi : G \to H$ is a group homomorphism, then $\psi \circ \varphi$ is a representation of G on V.

5. Recall that in section 20.1.3 we defined a *group character* to be a homomorphism from G to $L^\times$. This is in fact a 1-dimensional representation of G, since $\mathrm{GL}_1(L) = L^\times$. For a concrete example, let $G = C_n$ be the cyclic group of order n and $\zeta_n \in \mathbb{C}^\times$ the nth root of unity. Then the map:
$$\chi : G \to \mathbb{C}^\times \qquad a^i \mapsto \zeta_n^i$$
is a faithful representation of G.

6. Let $G = D_8$. We already have the regular represenation of dimension 8 and premutation represetntation of dimension 8, but can we do better? In fact we can. In order to see this, let's first write out D_8 in terms of generators and relations:
$$D_8 = \langle r, s \mid r^4 = s^2 = 1,\ rs = sr^{-1} \rangle$$

Let $k = \mathbb{R}$ and consider $\mathrm{GL}_2(\mathbb{R})$. If we can find matrices that satisfy these relations, than we would have a faithful representation of degree 2. This can be laborious, but with some work we can find:

$$r \mapsto \begin{pmatrix} \cos(2\pi/n) & -\sin(2\pi/n) \\ \sin(2\pi/n) & \cos(2\pi/n) \end{pmatrix} \qquad s \mapsto \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

Let $G = Q_8$. We will show how different fiels can produce different represetnations. First, recall the following two generator and relation set for Q_8:

$$Q_8 = \langle i, j \mid i^4 = j^4 = 1,\ i^2 = j^2,\ i^{-1}ji = j^{-1} \rangle$$

$$Q_8 = \langle i, j, k \mid i^2 = j^2 = k^2 = ijk = 1 \rangle$$

If $k = \mathbb{C}$, and we work over $\mathrm{GL}_2(\mathbb{C})$ we can use the first representation an find:

$$i \mapsto \begin{pmatrix} \sqrt{-1} & 0 \\ 0 & -\sqrt{-1} \end{pmatrix} \qquad j \mapsto \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$$

can you find one if $k = \mathbb{R}$ and $\mathrm{GL}_4(\mathbb{R})$? (Hint: think of the Hamiltonian's $\mathbb{H}$)

7. Let $H \trianglelefteq G$ which is further an abelian p-group for some prime p. Then $V = H$ is a vector-space over $\mathbb{F}_p$ with the action of $a \cdot v = v^a$. Then define the action of G on H by conjugation: $g \cdot h = ghg^{-1}$. This action is well-defined since $gh^a g^{-1} = (ghg^{-1})^a$. The kernel of this action is $C_G(H)$. Such an action is called a *module representation*.

8. Let V be the $k[S_n]$-module which affords the permutation representation. We will find 2 S_n-invariant subspaces: Let $N \subseteq V$ be the subspace defined by:

$$N = \{\alpha_1 e_1 + \cdots + \alpha_n e_n \mid \alpha_1 = \cdots = \alpha_n\}$$

Then this a subspace that is also S_n-invariant. As practice, show that if $n \geq 3$, then N is the unique 1-dimension subspace which is S_n-stable. This submodule is called the *trace* submodule of V.

Another natural S_n-stable subspace is the following: take $I \subseteq V$ such that

$$I = \{\alpha_1 e_1 + \cdots + \alpha_n e_n \mid \alpha_1 + \cdots + \alpha_n = 0\}$$

notice that $I = \ker(\varphi)$ where $\varphi : V \to \mathbb{C}$, $\varphi(\alpha_1 e_1 \alpha_2 e_2 \cdots \alpha_n e_n) = \sum_i \alpha_i$, and so I is $n-1$ dimensional. Since φ is called the *augmentation map* (was in section 7.3 of D&F), I is called the *augmentation module*.

9. If V is a regular representation of G so that $V = k[G]$, then V has the same two natural G-invariant subspaces:

$$\begin{aligned} N &= \{\alpha_1 e_1 + \cdots + \alpha_n e_n \mid \alpha_1 = \cdots = \alpha_n\} \\ I &= \{\alpha_1 e_1 + \cdots + \alpha_n e_n \mid \alpha_1 + \cdots + \alpha_n = 0\} \end{aligned}$$

these are both in fact *two-sided ideals*, even if G is not abelian. Since they appear naturally, they are usually given a name: N is called the *trace ideal* and I is called the *augment ideal*.

Corollary 53.1.6: Schur's Lemma For Representations

Let V, W be irreducible G-representations (over $\mathbb{C}$) with represetnations ρ_V and ρ_W. Then:

1. If $V \not\cong_G W$, then there does not exist a non-trivial G-linear map between V and W
2. Let's say $V = W$, are finite dimensional vecor-spaces with $\rho_V = \rho_W$, then the only non-trivial G-linear maps from V to W are scalars: $\lambda I : V \to W$, $\lambda I(v) = \lambda v$.

***Proof*:**

1. Since V and W are irreducible G-representations, they are simple $k[G]$-modules, and hence we apply the original Schur's lemma.
2. Let $V = W$ of the same dimension with matching actions $\rho_V = \rho_W$, and let $f : V \to W$. Since f is a linear transformation over $\mathbb{C}$, there always exists an eigenvalue, say λ so that $f(x) = \lambda x$. Consider:
$$\varphi = f - \lambda I$$
so that $\varphi(x) = 0$. Since the difference of two G-linear maps is G-linear, then $\ker(\varphi)$ is a G-invariant subspace. But V is irreducible, and $\ker(\varphi)$ is non-empty, and hence must be all of V, and so $\varphi(V) = 0$, telling us that $0 = f(x) - \lambda I(x)$, i.e.
$$f(x) = \lambda x$$
completing the proof.

Recall that if V is a simple R-module over a semi-simple ring R, then $\text{End}_R(V) \cong_{\mathbf{Ring}} D^{\text{op}}$ is a division ring, and that any finite-dimensional division $\mathbb{C}$-algebra is in fact equal to $\mathbb{C}$. This lemma now gives an explicit isomorphism between these two!

Since we are working over a vector-space, it is often convenient when possible to bring in an inner product that give further relations between our objects. The key the following inner product is that it is invariant on G-actions:

Definition 53.1.7: G-Invariant Inner Product

Let V be $k[G]$-module. Then a *G-invariant inner product* is an inner product $\langle \cdot, \cdot \rangle$ along with the additional property of:
$$\langle g \cdot v, g \cdot w \rangle = \langle v, w \rangle$$

Instead of giving examples of this inner product, we will soon show if V is a semi-simple $k[G]$-module, we can transform any inner product into a G-invariant inner product (hint: can you see how to use the "averaging" idea in Machke's Theorem to do so? The answer is given in corollary 53.1.12, page 1475).

Now, the study of linear representations in themseves can difficult; we are essentially studying all possible $\mathbb{C}[G]$-module. We would rather be able to decompose them into smaller pieces, and show that any represetnation of G is some how build form these simpler pieces (similarly to our exploration of semi-simple modules). The rest of this chapter is dedicated to seeing if we can bring

Artin-Wedderburn's theorem and all of it's consequences to understanding linear representations of finite groups:

Definition 53.1.8: Decomposing Representations

A representation is called *simple, reducible, indecomposable, decomposable*, or *semisimple* if the $k[G]$-module affording it has the corresponding property

Recall that indecomposable in general does not imply irreducible. The example under definition 51.1.1 could is in fact a representation of $\mathbb{Z}$ defined by:

$$n \mapsto \begin{pmatrix} 1 & n \\ 0 & 1 \end{pmatrix}$$

which is reducible (since it is in block upper-triangular form) but not indecomposable (since it's not in block-diagonal form). Notice that any 1-dimensional representation is automatically an irreducible, since any $k[G]$-submodule of k would be a k-subspace of k, but k is already irreducible as vector-space over k.

To get an idea of how $k[G]$-submodules interact with the definitions above, let V be an n-dimensional representation of G affording the represetnation ρ, and $W \subseteq V$ an m-dimensional $k[G]$-submodule (i.e. m-dimensional G-invariant subspace) with $0 \neq m \neq n$. Then choosing a basis $\{w_1, w_2, ..., w_m\}$ for W, we can extend it to a basis $\{w_1, w_2, ..., w_m, v_{m+1}, ..., v_n\}$. Since W is G-invariant, for any $w \in W$, $\rho(g)(w) \in W$. In matrix represetntation, $\varphi(g)(w^i) = \rho^i_j w^j$, and the coefficeints of $v^{m+1}, ..., v_n$ are zero. Hence we get the matrix representation of $\rho(g)$ as:

$$\begin{pmatrix} \alpha(g) & \beta(g) \\ 0 & \gamma(g) \end{pmatrix}$$

Notice too that:

$$\begin{aligned}
\begin{pmatrix} \alpha(gh) & \beta(gh) \\ 0 & \gamma(gh) \end{pmatrix} &= \rho(gh) \\
&= \rho(g)\rho(h) \\
&= \begin{pmatrix} \alpha(g) & \beta(g) \\ 0 & \gamma(g) \end{pmatrix} \begin{pmatrix} \alpha(h) & \beta(h) \\ 0 & \gamma(h) \end{pmatrix} \\
&= \begin{pmatrix} \alpha(g)\alpha(h) & \alpha(g)\beta(h) + \beta(g)\gamma(h) \\ 0 & \gamma(g)\gamma(h) \end{pmatrix}
\end{aligned}$$

which shows the the converse is also true: if we find a basis so that the matrices are block upper triangular, then the subspace spanned by the first m are G-invariant. As an example of this, let V be the permutation representation of S_3. Then the set of all vectors that are invariant to the action of $G = S_3$ is $V^G = \text{span}(e_1 + e_2 + e_3)$. Let $W = \text{span}(e_1 + e_2 + e_3)$. Since $G \curvearrowright W$ is of the form $g \cdot w = w$, we see that W is in fact the *trivial representation*. Since V is semi-simple, we have that $V = V^G \oplus V^{\perp}$. Making some new basis for V, say $(e_1 + e_2 + e_3, e_2, e_3)$, we see that $S_3 \curvearrowright V$ will have matrix representations that are of the above form:

$$\begin{pmatrix} 1 & * & * \\ 0 & * & * \\ 0 & * & * \end{pmatrix}$$

If V were irreducible, then no such basis would exist. If $V = U \oplus W$, then we would get a block diagonal form. It turns out that all representations of groups over either a character zero ring or fields where $|G| \nmid \text{char}(k)$ are block-diagonal, i.e. are semisimple!

Theorem 53.1.9: Maschkle's Theorem

Let G be a finite group and let k be a field whose characteristic does not divide $|G|$ or $\text{char}(k) = 0$. Then if V is a $k[G]$-module, it is a semi-simple $k[G]$-module.

The condition that $|G| \nmid \text{ch}(k)$ is equivalent to $|G| \in k^{\times}$. Note how this theorem is "surprising" as a finite group G certainly cannot be reduced to a direct sum of irreducible (or simple) groups, but as extensions of simple groups, but furthermore (as of writing this book) we do not know how to construct all finite group via extensions of simple groups. Hence, having a very computable decomposition into irreducible representations provides a very potent tool to analyze the structure of finite groups.

***Proof*:**
We will use lemma 51.1.7[3] to show that for any $U \subseteq V$, there exists a compliment W such that $V = U \oplus W$

We will find a projection onto U: $\pi : V \to U$. Thus, we need to show it has the following properties:

1. $\pi(u) = u$ for all $u \in U$
2. $\pi(\pi(v)) = \pi(v)$ for all $v \in V$ [a]

As a reminder on why this is sufficient[b], let π be such that the function defined on the $k[G]$-module that satisfies these properties. Let $W = \ker(\pi)$. Then $W \subseteq V$ is a submodule, and W is the comliment of U. First, let $v \in U \cap W$. Then since $v \in U$, $\pi(v) = v$ and since $v \in W$, $\pi(v) = 0$, hence $v = 0$, meaning $U \cap W = 0$. Next, let $v \in V$ be an arbitrary element, and write $v = \pi(v) + (v - \pi(v))$. By definition, $\pi(v) \in U$, and by the 2nd property:

$$\pi(v - \pi(v)) = \pi(v) - \pi(\pi(v)) = \pi(v) - \pi(v) = 0$$

and hence $v - \pi(v) \in W$. Hence $v \in U + W$, and so $V = U + W$, hence $V \cong U \oplus W$. Hence, it suffices to find an $k[G]$-module projection π.

Let $U \subseteq V$ be a $k[G]$-submodule. Then it is already a k-subspace, and so it has a vector-space direct sum compliment $W_0 \subseteq V$ (take a basis β for U, extend it to a basis $\mathcal{B}$ for V, then take $\mathcal{B} - \beta$) Thus, $V = U \oplus W_0$ as a vector-space. However, W_0 need not be G-stable (i.e. a $k[G]$-submodule), for example $G = C_2$ on $\mathbb{R}^2$ with the refection across $y = x$ which has the G-subspace $y = x$ and compliment spanned by $(1, 0)$ which is certainly not G-invariant. We will show how to find such the appropriate $k[G]$-module given W_0. First, let $\pi_0 : V \to U$ be the vector-space projection map:

$$\pi_0(u + w_0) = u$$

Here is the key to finding a proper compliment: we will "average" π_0 over G as follows:

$$g\pi_0 g^{-1} : V \to U \qquad g\pi_0 g^{-1}(v) = g \cdot \pi_0(g^{-1} \cdot v)$$

Since π_0 maps V to U, and U is a $k[G]$-submodule, U is stable under this G-action, hence $g\pi_0 g^{-1}$ is well-defined. Note that both g and g^{-1} act as k-linear transformations, so $g\pi_0 g^{-1]}$

is a k-linear transformation. Furthermore, if $u \in U$, then so must $g^{-1}u$, so by definition of π_0, $\pi_0(g^{-1}u) = g^{-1}u$. Thus, for all $g \in G$:

$$g\pi_0 g^{-1}(u) = u$$

hence, $g\pi_0 g^{-1}$ is also a vector-space projection of V onto U!

Let $n = |G|$, and viewing n as an element of k $(n = 1 + \cdots + 1)$, since $\mathrm{ch}(k) \nmid |G|$ $n \neq 0$, i.e. $n \in k^\times$. Thus, n has an inverse in k, meaning we can define:

$$\pi = \frac{1}{n} \sum_{g \in G} g\pi_i g^{-1}$$

We can immediately deduce that π has the following properties:

1. By definition, π is still k-linear from V to U.
2. Restricting π to U would restrict each summand in the definition of π, and so as we've pointed out earlier each summand would be the identity map. Hence, we get $1/n$ times n copes of u, meaning $\pi|_U = \mathrm{id}$
3. π is well-defined, hence $\pi(v) \in U$, so $\pi(v)^2 = \pi(v)$

It remains to show that π is in fact a $k[G]$-module homomorphism (i.e. is $k[G]$ linear). It suffices to show that for any $h \in G$, $\pi(hv) = h\pi(v)$. This is simply a computation:

$$\begin{aligned}
\pi(hv) &= \frac{1}{n} \sum_{g \in G} g\pi_0(g^{-1}hv) \\
&= \frac{1}{n} \sum_{g \in G} h(h^{-1}g)\pi_0(g^{-1}hv) \\
&= \frac{1}{n} \sum_{\substack{k=h^{-1}g \\ g \in G}} h(k\pi_0(k^{-1}v)) \\
&\overset{!}{=} h\frac{1}{n} \sum_{\substack{k=h^{-1}g \\ g \in G}} (k\pi_0(k^{-1}v)) \\
&= h\pi(v)
\end{aligned}$$

where $\overset{!}{=}$ comes form the fact that as g goes over all elements of G, then so will $k = h^{-1}g$ (i.e. groups acting on themselves are always transitive actions). Hence, π is indeed a $k[G]$-homomorphism, and so it is a projection map, completing the proof.

[a] In fact, this is implied by 1

[b] see ref:HERE for the original development of this

Thus, if we take any representation V over k of G, and consider the $k[G]$-module, then we know through $k[G]$ and G being finite and k a field (so V is a vector-space) that there are only *finitely many* simple $k[G]$-modules (i.e. finitely many irreducible representations of G), meaning we only have a finite number of "building blocks" from which we can build) any representation for G! Even better, the way we put together irreducible representation is through direct sums the simplest of ways of putting together algebraic objects! Then through the Artin-Wedderburn decomposition of V, we can find all *irreducible* representations that make up the representation V of G, giving us a simple finite list of irreducible representations per finite group! Note that if $p = \text{ch}(k) > 0$ and $p \big| |G|$, then $k[G]$ is *not* semisimple, or if G is infinite, it need not be semi-simple

Example 53.1.10: Non-Simple Group-Ring

For the infinite case, we've presented $\mathbb{Z} \curvearrowright \mathbb{Z}^2$ above. For the finite case where $p \big| |G|$,

1. let $G = C_n$ be the cyclic group of order n. Show that $k[C_p] \cong k[a]/(a^n - 1)$
2. Show that if char$k \nmid |G|$, then $k[C_n]$ is isomorphic product of fields. Since $M_1(k) = k$, by the uniqueness of Artin Wedderburn, this is the Artin-Wedderburn decomposition. Conclude that since $k[C_n]$ is abelian, any Artin-Wedderburn decomposition *has* to be a product of fields.
3. Show that if $\text{char}(k) = p$, then $k[C_p]$ has nilpotent elements, and hence cannot be a product of fields.

If $k = \mathbb{C}$, then Maschkle's Theorem always applies, and we can further use Artin Wedderburn over closed k-algebras (corollary 51.2.20) to deduce a relatively simple structure! Hence, from now on, we will always have $k = \mathbb{C}$, meaning all of our representations of finite groups will be over the complex vector-field of finite dimension.

Corollary 53.1.11: Matrix Representation Consequences

Let $\rho : G \to \text{GL}(V)$ be a representation over a field k where $\text{char}(k) \nmid |G|$. Then there exists a basis for V such that for all $g \in G$, the matrix $\rho(g)$ is of block-diagonal form:

$$\begin{pmatrix} \rho_1(g) & & & \\ & \rho_2(g) & & \\ & & \ddots & \\ & & & \rho_m \end{pmatrix}$$

where ρ_i is an irreducible representation of G, $1 \leq i \leq m$.

***Proof*:**
Let V be the $k[G]$-module representing ρ. By Maschkle's Theorem, we may write $V = U_1 \oplus \cdots \oplus U_m$. where U_i is simple $k[G]$-module. For each U_i, choose a basis β_i. Let $\beta = \bigcup_i^m \beta_i$. then each matrix representation of $\rho(g)$ is of the form in the corollary, with $\rho(g)|_{U_i} = \rho_i(g)$, completing the proof.

Corollary 53.1.12: Defining G-Invariant Inner Product

Let V be a $k[G]$-module and $\langle \cdot, \cdot \rangle$ be an inner product. Then this inner product induces a G-invariant inner product

***Proof*:**
If $\langle \cdot, \cdot \rangle$ is an inner product, then:

$$\langle v, w \rangle_G = \frac{1}{|G|} \sum_{g \in G} \langle gv, gw \rangle$$

is a well-defined inner product (as can be checked), and for any $h \in G$

$$\begin{aligned}\langle hv, hw \rangle_G &= \frac{1}{|G|} \sum_{g \in G} \langle ghv, ghw \rangle \\ &= \frac{1}{|G|} \sum_{k = gh \in G} \langle kv, kw \rangle \\ &= \langle v, w \rangle_G\end{aligned}$$

In other words, there is always an inner product on our space in which the group actions keeps all elements of the same length and the same angle between them, showing how we can interpret a group as somehow rotating and reflecting an object in a vector-space. Now that we know any represetnation of G is composed in some way by finitely many irreducible representations, it would be useful to be able to quickly deduce the Artin-Wedderburn decomposition of $\mathbb{C}[G]$:

Proposition 53.1.13: Properties Of Representations

Let $k = \mathbb{C}$ and G be finite. Let $S_1, S_2, ..., S_r$ be the irreducible G-representations (up to isomorphism). Then:

1. the number of irreducible G-representations is equal to the number of conjugacy classes in G
2. For every representation V of G,
$$V \cong S_1^{\oplus \ell_1} \oplus \cdots \oplus S_r^{\ell_r}$$
where $\ell_i > 0$ and is unique
3. $\mathbb{C}[G] \cong \bigoplus_{i=1}^n \left(S_i^{\oplus \dim_k S_i} \right)$ as regular representation
4. $|G| = \sum (\dim_k S_i)^2$

***Proof*:**
part (3) and (4) come from corollary 51.2.20.

For part (1), we first use Artin-Wedderburn to conclude that $Z(\mathbb{C}[G]) = \prod_i^r \mathbb{C}$, so $\dim_{\mathbb{C}}(Z(\mathbb{C}[G])) = r$, so the dimension of the center is equal to the number of irreducible G-representations.

Now, let $\mathcal{K}_1, ..., \mathcal{K}_s$ represent the conjugacy classes (recall they partition G). Define:

$$\sigma_i = \sum_{g \in \mathcal{K}_i} g$$

Note that any pair of elements σ_i and σ_j have no common terms since the conjugacy classes partition G. Furthermore, by definition, for any $g \in G$, $g\sigma_i g^{-1} = \sigma_i$, and so $\sigma_i \in Z(\mathbb{C}[G])$. If the σ_i's form a basis for $Z(\mathbb{C}[G])$, then we would have $s = t$. As a $\mathbb{C}$-vector-space, they are already linearly independent since the group elements from a basis, so it suffices to show they span. If $\sum_{g \in G} \lambda_g a_g \in Z(\mathbb{C}[G])$. Then by definition for all $h \in G$,

$$\begin{aligned}
\Leftrightarrow\ h\left(\sum_{g\in G}\lambda_g g\right) &= \left(\sum_{g\in G}\lambda_g g\right)h \\
\Leftrightarrow \left(\sum_{g\in G}\lambda_g g\right) &= h^{-1}\left(\sum_{g\in G}\lambda_g g\right)h \\
\Leftrightarrow \sum_{g\in G}\lambda_g g &= \left(\sum_{g\in G}\lambda_g (h^{-1}gh)\right) \\
\overset{!}{\Leftrightarrow} \sum_{g\in G}\lambda_g g &= \sum_{g\in G}\lambda_{hgh^{-1}} g \\
\Leftrightarrow \lambda_{hgh^{-1}} &= \lambda_g \qquad \forall g \in G
\end{aligned}$$

since h arbitrary, we see that the coefficients of elements in the same conjugacy class remain with an element that's in that conjugacy class. Equivalently, the function $G \to \mathbb{C}$, $g \mapsto \lambda_g$ is constant on conjugacy classes. Thus:

$$Z(\mathbb{C}[G]) = \operatorname{span}\left\{\sum_{g\in G} f([g])g\right\}$$

where f is a function from the conjugacy classes of G to $\mathbb{C}$. Hence $s = t$.

for (2), Since $\mathbb{C}[G]$ is a semi-simple ring, V is a semi-simple $\mathbb{C}[G]$-module, giving us existence of the decomposition by AW. For uniqueness, first note that:

$$\operatorname{End}_{\mathbb{C}[G]}(S_i) = \mathbb{C}$$

since $\mathbb{C}$ is the only [finite-dimensional?] division $\mathbb{C}$-algebra. Then, we get by Schur's lemma:

$$\operatorname{Hom}_{\mathbb{C}[G]}(S_i, V) \overset{\text{Schur}}{\cong} \operatorname{Hom}_{\mathbb{C}[G]}(S_i, S_i)^{\oplus l_i} \cong \mathbb{C}^{l_i}$$

Corollary 53.1.14: Abelian Groups And Degree 1 Decomposition

1. If A is an abelian group, then every irreducible complex representation of A is 1-dimensional (i.e. a homomorphism from A to $\mathbb{C}^\times$), and A has $|A|$ inequivalent irreducible complex representations.
2. The number of inequivalent (irreducible) 1-dimensional complex representations of any finite group G is equal to $|G/G'|$

***Proof*:**

1. Since A is abelian, so is $\mathbb{C}[A]$. By the Artin-Wedderburn decomposition, it must be a product of matrices, which since $\mathbb{C}[A]$ abelian is so must each matrix ring, which is only true if and only if $n_i = 1$. Then, by corollary 51.2.20, $\mathbb{C}[A]$ is a product of fields. It follows that any irreducible representation (i.e. simple $\mathbb{C}[A]$-module) is 1-dimensional. Conversely, if A were not abelian, $\mathbb{C}[A]$ cannot be a product of fields, hence one of the dimensions of the matrix rings *must* be greater than 1

2. Let G be an arbitrary group. We want to show that there is a bijective correspondence between the 1-dimension representations of G and the 1-dimensional representations of G/G'. $\rho : G/G' \to \mathbb{C}^\times$ is a 1-dimensional representations of G/G', then we have the natural map $G \to G/G'$, which shows that each 1-dimensional representations of G/G' is also a 1-dimensional representations of G. Convexly, if f $\rho : G \to \mathbb{C}^\times$ is a 1-dimensional representations of G, then by the universal property, this factors through:

$$\begin{tikzcd} G/G' \arrow[r] & \mathbb{C}^\times \\ G \arrow[u] \arrow[ur] & \end{tikzcd}$$

showing that any 1-dimensional representation of G is associated to a 1-dimensional representations of G/G', showing there can't be more 1-dimensional representations of G than there are of G/G', completing the proof.

A more general technique can be adopted from this proof: If $H \trianglelefteq G$ and V is an irreducible representation of G/H, then V is an irreducible representation of G.

Example 53.1.15: Decomposition Of Modules

1. DnF p. 847, and a bit about $\dim(V) = 1$ example right under definition

Finally, for those who read chapter 50, the following results are a glimpse at how representation theory is a "generalization" of number theory. First, if $H \leqslant G$ is a subgroup and (ρ, V) is an irreducible representation, $\rho|_H$ breaks up into the direct sum of representations, not necessarily of the same size or multiplicity. This is similar to how for a prime $\mathfrak{p} \subseteq F$ it may split to multiple primes with varying inertia degree and ramification numbers in a finite separable extension L/F. If L/F is furthermore normal (and hence Galois) then all the inertia degree's and ramification numbers match. In a similar vein:

Theorem 53.1.16: Clifford's Theorem

G be a group, $N \trianglelefteq G$ a normal subgroup, K a field, $\pi : G \to \mathrm{GL}_n(K)$ be an irreducible representation. Then the restriction $\pi|_N$ breaks up into a direct sum of irreducible representations of N of equal dimensions (i.e. it is semisimple and each representation is of the same dimension). These irreducible representations of N lie in one orbit for the action of G by conjugation on the equivalence classes of irreducible representations of N (i.e. they are isotypic components with the same multiplicity).

***Proof*:**
Let W be an irreducible subrepresentation of $V|_H$. Since H is normal in G, for $g \in G$, H acts irreducibly on gW by $(h, gw) \mapsto hgh^{-1}hw$. The subspace $\sum_{g \in G} gW$ is a nonzero subrepresentation of V. Since V is irreducible, it is equal to V. Since a representation which is a sum of irreducible sub-representations is semisimple, $V|_H$ is semisimple. A similar argument follows for the 2nd assertion, completing the proof.

Exercise 53.1.1

1. If χ_1 and χ_2 are both 1 dimensional representations, then $\chi_1 \cong \chi_2 \Leftrightarrow \chi_1 = \chi_2$. Use the property of equivalent representations to see this.
2. Let V be a non-trivial irreducible G-representation (i.e. not $\chi(g) = 1$ for all $g \in G$). Show that there are $\dim_{\mathbb{C}}(V^G) = 0$
3. By Caley's Theorem, all finite groups are linear. Do there exists infinite non-linear groups? (Hard: Do there exists finitely generated non-linear groups?)
4. Show that if ρ is irreducible and $N \trianglelefteq G$ is normal, if N fixes a point v, N acts trivially on all of V.

53.2 Character Theory

We know now we can decompose a G-representation V into the direct sum of r different types of irreducible G-modules. How do we know proceed to find these? These can be quite unwiedly: if we have a 10 or 100-dimensional represenation of a group G, then we have a matrix with 100 and $10'000$ entries! We'd then have to multiply these and figure out what relation's they have between each other, quite a hassle! Furthermore, how do we do this in such way that is independent of equivalent representations? A convinient basis might be nice, but to find a nice basis for every $\rho(g) \in \rho(G)$, we'd have to find a basis that can *simultaneously* diagonalize every element in $\rho(G)$! If there was some invariant we can find on a representation that we can use to keep track of the decomposition information between representations that is invariant of representation, that would be useful to this endevaour. The following does exactly this (Move the discussion of trace here to build-up to it? Give some of D&F examples to demonstrate it does get quite ugly sometimes?)

Definition 53.2.1: Class Function and Character

Let V be a representation of G over an arbitrary field F. Then

1. A function $f : G \to F$ that is constant on conjugacy classes is called a *class function* of G

2. If φ is a representation of G afforded by the $F[G]$-module V, then a function:

$$\chi_V : G \to \mathbb{C} \qquad g \mapsto \operatorname{tr}(g : V \to V) = \operatorname{tr}(\varphi(g))$$

   is called the *character* of φ, where $\operatorname{tr}(\varphi(g))$ is the trace of φ with respect to some basis for V. A character is called *irreducible* if φ is irreducible (and reducible oherwise). The *degree* of a character is the degree of any representation affording it.

For ease of reference, recall these properties of the trace:

1. $\operatorname{tr}(A+B) = \operatorname{tr}(A) + \operatorname{tr}(B)$ an $\operatorname{tr}(cA) = c\operatorname{tr}(A)$
2. $\operatorname{tr}(A) = \operatorname{tr}(A^t)$
3. $\operatorname{tr}(AB) = \operatorname{tr}(BA)$

Note that in general, a character is *not* a homomorphism from G into the additive or multiplicative structure of k. Since $\operatorname{tr}(AB) = \operatorname{tr}(BA)$, $\operatorname{tr}(ABA^{-1}) = \operatorname{tr}(B)$, we see that the trace (and hence the character) is independent of choice of basis, and so equivalent representations have the same character, meaning the character is indeed an invariant. Furthermore, for any representation φ of G,

$$\varphi(g^{-1}xg) = \varphi(g)^{-1}\varphi(x)\varphi(g)$$

hence, we can take the trace and use the 3rd property above to show that characters an an example of *class functions*. Furthermore, since the trace is linear, we can extend the domain of a character to be the trace over any element in $\mathbb{C}[G]$. If $F = \mathbb{C}$, then the collection of class function's form a complex vector-space, since the sum of two class functions is a class function, and multiplying by a scalar $z \in \mathbb{C}$ does not change that a class function is invariant.

Example 53.2.2: Character

1. Given the trivial representation of G (i.e. any 1-dim vector space where $g \mapsto I$), then if φ is the representating affording the trivial representation, it has $\chi_\varphi(g) = 1$ for all g. This is called the *trivial character*.

2. If we have a degree 1-representation, then the character and the representation are the same. Thus for abelian group, complex representations and their characters are the same. For example, take the representation $\varphi : C_p \to \mathbb{C}^\times$, $\varphi(a^i) = \zeta_p^i$. Then:

$$\chi(a^i) = \zeta_p^i$$

3. Let $\Pi : G \to S_n$ be a permutation representation of G and φ the linear of G of degree n on V where:

$$\varphi(g)(e_i) = e_{\Pi(g)(i)}$$

with respect to the given basis, $\varphi(g)$ has 1 in the ii-entry if $\Phi(g)$ fixes i, and zero otherwise. Thus, we can say that the character χ is:

$$\chi(g) = \#\{\text{number of fixed points of } g \text{ on } \{1, 2, ..., n\}\}$$

If in particular the permuation representation of G into S_n is by left multiplication, then the character is called the *pemutation character*.

4. Consider any regular representation of G, φ. Then the character is:

$$\chi(g) = \begin{cases} 0 & g \neq 1 \\ |G| & g = 1 \end{cases}$$

This character is called the *regular character* of G (note how this character is not a group homomorphism into k or $k^\times$)

5. Take the representation $\varphi : D_n \to \mathrm{GL}_2(\mathbb{R})$ from example 53.1.5. Then:

$$\chi(r) = 2\cos(2\pi/n) \qquad \chi(s) = 0 \qquad \chi(1) = 2$$

where the last one is since $1 \in G$ must map to the identity matrix $I \in \mathrm{GL}_2(\mathbb{R})$.

From now on, all our representation will be complex.

Lemma 53.2.3: Properties Of Character

Let $\rho_V : G \to \mathrm{GL}(V)$ be a complex representation. Then:

1. $\chi_V(1) = \dim_{\mathbb{C}}(V)$
2. $\rho_V(g)$ is diagonalizable for all $g \in G$ and $\chi(g)$ is integral over the integers: $\chi(g) \in \overline{\mathbb{Z}}^{\mathbb{C}}$
3. If $g \in G$ has order n, then $\chi_V(g)$ is a sum of $\dim(V)$ number of nth roots of 1
4. $|\chi_V(g)| \leq \chi_V(1)$ for all $g \in G$
5. $\chi_V(g^{-1}) = \overline{\chi_V(g)}$

***Proof*:**

1. We showed this in the above examples.
2. The criterion we need from linear algebra is that $A \in M_n(\mathbb{C})$ is diagonlizable if and only if the minimal polynomial has no repeated roots, which can be implied if $M^k = I$ for some k, which it certainly is in our case since every element is finite.
3. If $g^n = 1$, then $\rho_V(g)^n = 1$, so by (2)

$$\rho_V(g) \sim \begin{pmatrix} z_1 & & \\ & \ddots & \\ & & z_k \end{pmatrix}$$

where $z_i^n = 1$ for all $1 \leq i \leq k$. Furthermore, this shows that these trace values in the matrix representation are always integral over the integers, being the sum of integral elements.

4.
$$|\chi_V(g)| = \left|\sum z_i\right| \leq \sum |z_i| = \dim(\chi_V(1))$$

5. Since $\rho_V(g^{-1}) = \rho_V(g)^{-1}$ and

$$\rho_V(g)^{-1} \sim \begin{pmatrix} z_1^{-1} & & \\ & \ddots & \\ & & z_k^{-1} \end{pmatrix}$$

$\chi_V(g^{-1}) = \sum_i z_i^{-1}$, but $\|z_i\| = 1$, so $z_i^{-1} = \overline{z_i}$, so

$$\chi_V(g^{-1}) = \sum_i z_i^{-1} = \sum_i \overline{z_i} = \overline{\chi_V(g)}$$

Now, if we have some G-representation V, it has some character χ. In the following, we will show how we can always break-down a character to be the sums of characterse on the irreducible G-representations:

Proposition 53.2.4: Linearly Independent Characters

Let G be a finite group and V a semi-simple $k[G]$-module, and φ the representation that V affords. Then:

1. The character of V is the sum of the characters of the componens appearing in the direct sum decomposition.
2. The irreducible characters of G are linearly independent set in the set of class functions. Furthermore, the irreducible characters of G form a basis of the vector-space of class functions.

***Proof*:**
For the 1st point, By the Artin-Wedderburn decoposition, we have:

$$V \cong (S_1)^{n_1} \oplus \cdots \oplus (S_k)^{n_k}$$

where S_i are simple $k[G]$-modules (irreducible G-representations). If we focus on the case where $M = S_1 \oplus S_2$, then we can choose a basis for S_1 and S_2 which will give us:

$$\varphi(g) = \begin{pmatrix} \varphi_1(g) & 0 \\ 0 & \varphi_2(g) \end{pmatrix}$$

where φ_1, φ_2 are the representations afforded by M_1 and M_2 respectively. Then if ψ_1 and ψ_2 are the characters of φ_1 and φ_2 an χ is the character of φ then we get:

$$\chi = \psi_1 + \psi_2$$

using induction get's us the result.

For the 2nd point, first notice that the class functions defined by being 1 on a conjugacy class and 0 otherise forms a basis for class functions of dimension r. It thus suffices to sho that the r irreducible characters are independent. To that end, define:

$$\widetilde{\chi_V} : \mathbb{C}[G] \to G \qquad x \mapsto \mathrm{tr}(x : V \to V)$$

Note that this map is $\mathbb{C}$-linear since $\mathbb{C}$ commutes with $\mathbb{C}[G]$, and $\widetilde{\chi_V}|_G = \chi_V$. Now, say $S_1, ..., S_r$ are simple $\mathbb{C}[G]$-modules and:

$$\sum_i^r \lambda_i \chi_{S_i} = 0 \qquad \lambda_i \in \mathbb{C}$$

Then this is equivalent to:

$$\sum_i^r \lambda_i \widetilde{\chi_{S_i}}(g) = 0 \qquad \forall g \in G$$

and since the maps are $\mathbb{C}$-linear, by linearity:

$$\sum_i^r \lambda_i \widetilde{\chi_{S_i}}(x) = 0 \qquad \forall x \in \mathbb{C}[G]$$

which by plugging into the definition we get:

$$\sum_i^r \lambda_i \mathrm{tr}(x : S_i \to S_i) = 0 \qquad \forall x \in \mathbb{C}[G]$$

Next, by Artin Wederburn, we have the decomposition:

$$C[G] \cong M_{n_1}(\mathbb{C}) \times \cdots \times M_{n_r}(\mathbb{C})$$

where each $M_{n_i}(\mathbb{C}) \cong S_i^{\oplus n_i}$, with $S_i = \mathbb{C}^{\oplus n_i}$. Using this, take the element:

$$x_i = (0, ..., I, ..., 0)$$

where I is the identity matrix[a]. Then $x_i : S_j \to S_j$ is 0 if $i \neq j$, and the identity marix if $i = j$. Thus, the trace is:

$$\mathrm{tr}(x_j : S_j \to S_j) = \begin{cases} 0 & i \neq j \\ n_i & i = j \end{cases}$$

and since n_j is at least 1, we must have that

$$\lambda_1 = \cdots = \lambda_r = 0$$

where we took advantage of the fact we are in characeristic 0

[a] If the field had finite characteristic, it is better to take the elementary matrix E_{11}

53.2.5 Remark We can prove this directly without using Artin-Wedderburn, which we will

Corollary 53.2.6: Detecting Isomorphic Representations Using Characters

Let V_1, V_2 be representations that are semi-simple modules.

1. Every character of G is uniquely of the form

$$\sum_{i}^{r} \lambda_i \chi_i \qquad \lambda_i \geq 0$$

where χ_i are irreducible characters.

2. the representations V_1 and V_2 are isomorphic if and only if $\chi_{V_1} = \chi_{V_2}$

***Proof*:**

1. This comes straight from the fact that irreducible characters form a basis, and that any G-representation V will be some direct sum of a finite list of known simple G-modules

2. If $V_1 \cong_G V_2$, there is a matrix P that simultaneously diagonalizes so that if A and B are basis matrices, $A = PBP^{-1}$. But the trace is independent of conjugation, and so the characters are the same. For the othe way around, treating a representation as a $k[G]$-module, then by Machke's theorem, we *know* the list simple $k[G]$-modules and the actions of $k[G]$ on them. Since any $k[G]$-module can be broken down into a direct summand of simlpe $k[G]$-modules, and the two chracters are the same, the list of $k[G]$-modules and their multiplicity are the same.

Thus, we have translated the problem of finding the irreducible representations of a group G to finding the irreducible characters of G! Since a character of a representation is determined by the irreducible characters, we give a way to systematize the tracking of irreducible characters:

Definition 53.2.7: Character Table

Let $\varphi : G \to \mathrm{GL}(V)$ be a representaiton of G an χ_V it's character. Then we call the *character table* a table with the rows being irreducible characters and columns being an element in the conjugacy class. Common practice is to make the first column the identity conjugacy class and the first row the trivial representation

Example 53.2.8: Character Table

Consider $G = S_3$ Then we can start creating it's character table:

	e	$(1\ 2)$	$(1\ 2\ 3)$
1	1	1	1
χ_{sgn}	1	-1	1
$\vdots$	$\vdots$	$\vdots$	$\vdots$

Since G has 3 conjugacy classes, exactly 3 irreducible characters of G. How would we go about finding it? The comning propositions work towards this.

To be able to find the missing character, and more generally find new characters for a character table, we may look to construct new character given old one, and hope that we break down those characters into a linear combination of irreducible characters. In a sense, we are finding a way to create represetnations without doing it "by hand". We will do this in two way: the first is by taking advatange of group-actions, since are relatively easy to define and their representations are matrices with 1's and 0's. Since all groups embedd into a symmetric group, this can be done systematically. We then see how to work directly with known representation to create new represenations.

Definition 53.2.9: Precaution Representation

Let G act on a finite set X. Define $\mathbb{C}X$ to be the set of formal sums:

$$\mathbb{C}X = \left\{ \sum_{x \in X} \lambda_x x \;\middle|\; \lambda_x \in \mathbb{C} \right\}$$

So that we get a $\mathbb{C}$-algebra with $\dim_{\mathbb{C}}(\mathbb{C}X) = |X|$. Then we can make this a G-represetnation by defining:

$$g \cdot \left(\sum \lambda_x x \right) = \sum \lambda_x (g \cdot x)$$

The 3 usual premutation representations is $G \curvearrowright G$ by left multiplication (which gives the regular representation) or conjugation, and $G \curvearrowright \{*\}$ trivially (which gives the trivial representation).

Lemma 53.2.10: Character Of Permutation Representation

Let G act on X, $G \curvearrowright X$. Then

$$\chi_{\mathbb{C}X}(g) = \# \{ x \in X \mid gx = x \}$$

***Proof*:**
FIx $g \in G$. Then the set of x's is a basis for the representation. To make the computation simpler, we will find a convinient order for them. In particular, by choosing any $x_1 \in X$, we can find the orbit of the element:

$$\begin{tikzcd} x_1 \arrow[r, "g"] & x_2 \arrow[r] & \cdots \arrow[r] & x_k \arrow[lll, bend left] & x_{k+1} \arrow[r, "g"] & \cdots \end{tikzcd}$$

so the basis representation will look like:

$$\begin{pmatrix} 0 & & & \\ 1 & 0 & & \\ 0 & 1 & \ddots & \\ & & 1 & 0 \end{pmatrix}$$

so we get that the trace is 0 if the order of the orbit is greater than 1, hence:

$$\chi_{\mathbb{C}X}(g) = \#\{g\text{-orbits of size } 1\}$$

An immediate consequence of this is that we have another way of computing the character of the regular representation

$$\chi_{\mathbb{C}G}(g) = \begin{cases} |G| & g = e \\ 0 & g \neq e \end{cases}$$

which makes sense, since the $\dim_{\mathbb{C}}(\mathbb{C}G) = |G|$.

Next, we turn towards a more systematic way of constructing characters by creating them out of pervious ones, that is, given some representations V and W of a group G, we can make new representations by creating new representations from V and G:

Definition 53.2.11: Constructing Representations from smaller ones

Let V, W be G-representations. Then

1. we can make a $V \oplus W$ representation by:
$$g \cdot (v, w) = (g \cdot v, g \cdot w) \; [(\rho_V(g)v, \rho_W(g)w)]$$
2. we can make $V \otimes_{\mathbb{C}} W$ a representations by:
$$g \cdot (v \otimes w) \mapsto (gv) \otimes (gw)$$
3. we can make $\mathrm{Hom}_{\mathbb{C}}(V, W)$ a representations by:
$$g \cdot f : V \to W \quad v \mapsto g \cdot f(g^{-1} \cdot v)$$
4. we can make $V^* = \mathrm{Hom}_{\mathbb{C}}(V, \mathbb{C})$ a representaiton by:
$$g \cdot f : V \to \mathbb{C}, \; v \mapsto f(g^{-1} \cdot v)$$

Note that the last construction is the same as the 3rd if we take $\mathbb{C}$ as the trivial representation. Before we show what the character's of these new representations are, we first find a relatively simpler way of representing the hom representation of V and W:

Lemma 53.2.12: Hom-Tensor Representation Dual

Let V, W be G representations. Then:

$$\mathrm{Hom}_{\mathbb{C}}(V, W) \cong_G V^* \otimes_{\mathbb{C}} W$$

as G-representations.

Note that this is not true in general, even for vector-spaces. It is important that the spaces are finite-dimensional. In this way, this is a stronger result then the hom-tensor adjoint presented in theorem 15.3.9 for the case of finite-dimensional vector-spaces.

Proof:
By the universal property:

$$\theta : V^* \otimes_{\mathbb{C}} W \to \mathrm{Hom}_{\mathbb{C}}(V, W) \qquad \lambda \otimes w \mapsto (v \mapsto \lambda(v)w)$$

you should check that this is well-define (note that it is bilinear).

Now, pick basis $e_1, e_2, ..., e_n$ for V, $f_1, f_2, ..., f_m$ for W, and the natural dual $e_1^*, ..., e_n^*$ for V^*. Then we get a basis for $V^* \otimes W$:

$$(e_i^* \otimes f_j) \qquad 1 \leq i \leq n,\ 1 \leq j \leq m$$

Now, notice that:

$$\theta(e_i^* \otimes f_j) \qquad v \mapsto e_i^*(v) f_j \sim \begin{pmatrix} 0 & \cdots & 0 & \cdots & 0 \\ \vdots & & \vdots & & 0 \\ 0 & & 1 & & 0 \\ \vdots & & \vdots & & \vdots \\ 0 & \cdots & 0 & \cdots & 0 \end{pmatrix} = E_{ij}$$

which is a matrix with 1 in the ij position and zero evywehre else. This is clearly a basis, hence θ sends a basis to a basis, and hence θ is a vector-space isomorphism. We next check that θ is G-linear:

$$\begin{aligned} \theta(g \cdot (\lambda \otimes w)) &= \theta(g\lambda \otimes gw) \\ &\equiv v \mapsto (g\lambda)(v) gw \\ &= v \mapsto (g\lambda)(g^{-1}v) w \\ &= g(\lambda(g^{-1}v)w) \\ &\equiv g(\theta(\lambda \otimes w) g^{-1} v) \\ &= g \cdot \theta(\lambda \otimes w) \end{aligned}$$

Lemma 53.2.13: Creating New Characters

1. $\chi_{V \oplus W} = \chi_V + \chi_W$
2. $\chi_{V \otimes W} = \chi_V \cdot \chi_W$
3. $\chi_{\mathrm{Hom}_{\mathbb{C}}(V,W)} = \overline{\chi_V} \chi_W$
4. $\chi_{V^*} = \overline{\chi_V}$

Proof:

1. obvious

2. It is enough to show that if $f : V \to V$ and $g : W \to W$ are linear functions, then:

$$\mathrm{tr}(f \otimes h) = \mathrm{tr}(f)\mathrm{tr}(h)$$

Now if we pick basis (e_i) for V and (f_j) for W, then $f(e_i) = \sum a_{ij}e_j$ and $h(f_k) = \sum b_{lk}f_l$. Then:

$$(f \otimes h)(e_i \otimes f_j) = \left(\sum_j a_{ij}e_j\right) \otimes \left(\sum_l b_{kl}f_l\right) = \sum_{j,l} a_{ij}b_{lk}(e_j \otimes f_l)$$

Notice the cofficient of $e_i \otimes f_j$ is $\sum_{i,k} a_{ii}b_{kk} = (\sum_i a_{ii})(\sum_k b_{kk}) = \mathrm{tr}(f)\mathrm{tr}(h)$

3. Recall that for $f \in V^*$,

$$(gf)(v) = f(g^{-1} \cdot v)$$

that is

$$\rho_{V^*}(g)(f) = f \circ \rho_V(g^{-1})$$

that is, it is the transpose:

$$\rho_{V^*}(g) = (\rho_V(g^{-1}))^t$$

So, since the transpose matrix has the same trace, we have

$$\chi_{V^*}(g) = \chi_V(g^{-1}) = \overline{\chi_V(g)}$$

we then apply lemma 53.2.12 and part (3)to get:

$$\chi_{\mathrm{Hom}_{\mathbb{C}}(V,W)} = \overline{\chi_V} \cdot \chi_W$$

4. $\chi_{V^*} = \chi_{\mathrm{Hom}(V,\mathbb{C})} = \overline{\chi_V} \cdot 1 = \overline{\chi_V}$.

With this, we can create new characters! However, we have yet to establish how we may find the irreducible components of these chracters. For example, if we take $V \otimes V$, we may find that $\chi_{V \otimes V}$ is a linear combination of some characters already present in our table, but then we have a left-over part. How would we determine if the left-over is irreducible? We will show this by defining an *inner product* on the vector-space of class equations, and show that the irreducible form an *orthogonal basis*!

Definition 53.2.14: Inner Product On Class functions

Let $\chi_1, \chi_2 : G \to \mathbb{C}$ be class functions. Then define:

$$\langle \chi_1, \chi_2 \rangle = \frac{1}{|G|} \sum_{g \in G} \chi_1(g)\overline{\chi_2(g)} \in \mathbb{C}$$

Then this is a [Hermitian] *inner product.*

It should be verified that this is indeeed an inner product. Since class functions are by definition constant on conjugacy classes, we only need to run through representatives of conjgacy classe times the order of each conjugacy class:

$$\langle \chi_1, \chi_2 \rangle = \frac{1}{|G|} \sum_{\substack{\mathcal{K} \in \mathrm{conj}(G) \\ g \in \mathcal{K}}} |\mathcal{K}| \chi_1(g) \overline{\chi_2(g)} \in \mathbb{C}$$

where we pick a single represetnative $g \in \mathcal{K}$ and sum over the number of conjugacy classes, i.e. $|\mathrm{conj}(G)|$. Looking a the inner product like this is desirable to make computation easier (ex. if taking the inner product of a character with itself, we would simply take the squrae of each element in a row multiplied by the size of it's conjugcay class). Notice how this inner product mirrors the G-invariant inner product we induced in corollary 53.1.12 if we let $\langle, \cdot, \cdot \rangle$ be the usual conjugate dot product on $\mathbb{C}^n$. (TBD)

Lemma 53.2.15: Inner Product With Trivial Character

Let V be a G-representation. Then if $\mathbb{1}$ is the trivial representation,

$$\langle \chi_V, \chi_{\mathbb{1}} \rangle = \dim_{\mathbb{C}}(V^G)$$

where $V^G = \{v \in V \mid g \cdot v = v \; \forall g \in G\}$ is the subspace fixed by G.

***Proof*:**
Define the linear map $\pi : V \to V$ where

$$v \mapsto \frac{1}{|G|} \sum_{g \in G} g \cdot v$$

then:

$$\pi|_{V^G} = \mathrm{id}$$

and

$$\mathrm{im}(\pi) \subseteq V^G$$

which implies $\pi^2 = \pi$, i.e. a projection. By linear linear algebra:

$$V = \ker(\pi) \oplus \mathrm{im}(\pi)$$

where on the kernel, $\pi = 0$, and on the image $\pi = \mathrm{id}$. Thus,

$$\frac{1}{|G|} \sum_g \chi_V(g) = \frac{1}{|G|} \sum_g \mathrm{tr}(g : V \to V) = \mathrm{tr}(\pi) = \dim_{\mathbb{C}}(V^G)$$

but the left hand side is $\langle \chi_V, \chi_{\mathbb{1}} \rangle$, completing the proof.

Corollary 53.2.16: Row Orthogonality

If V and W are any representations of G, then

$$\langle \chi_V, \chi_W \rangle = \dim_{\mathbb{C}}(\hom_{\mathbb{C}[G]}(W, V))$$

In particualr, if V an W are irreducible, then

$$\langle \chi_V, \chi_W \rangle = \begin{cases} 1 & V \cong W \\ 0 & \text{otherwise} \end{cases}$$

that is, irreducible character's are orthogonal to each other.

***Proof*:**
We reduce to the previous lemma:

$$\begin{aligned}
\langle \chi_V, \chi_W \rangle &= \frac{1}{|G|} \sum_{g \in G} \chi_V(g) \overline{\chi_W(g)} \\
&= \frac{1}{|G|} \sum_{g \in G} \chi_{\mathrm{Hom}_{\mathbb{C}}(W,V)}(g) \\
&= \langle \chi_{\mathrm{Hom}_{\mathbb{C}}(W,V)}, 1 \rangle \\
&= \dim_{\mathbb{C}}(\mathrm{Hom}_{\mathbb{C}}(W, V)^G)
\end{aligned}$$

Note that $f \in \mathrm{Hom}_{\mathbb{C}}(W, V)$ is G-invariant, so $g \cdot f = gfg^{-1} = f$ for all $g \in G$ (recall the definition of $g \cdot f$ on $\mathrm{Hom}_{\mathbb{C}}(W, V)$). Thus this is true iff

$$gf = fg$$

Thus

$$gf(w) = f(gw) \qquad \forall g \forall w \in W$$

so f is $\mathbb{C}[G]$-linear so f is in the $\mathrm{Hom}_{\mathbb{C}[G]}(W, V)$.

There is also a more concrete proof worth remarking: since characters are linear combination's of irreducible characters, the direct product of irreducible modules translates to the sum of an irreducible character with the multiplicity of the irreducible module afording the representation being the multiplicity of the ireducibile characters. Hence:

$$\begin{aligned}
\langle \chi_V, \chi_W \rangle &= \langle \sum_i m_i \chi^{(i)}, \sum_j m_j \chi^{(j)} \rangle \\
&= \sum_{i,j} m_i m_j \langle \chi^{(i)}, \chi^{(j)} \rangle
\end{aligned}$$

where we have now reduced it to $\langle \chi^{(i)}, \chi^{(j)} \rangle$. Expanding the inner product and using the hom-set definition, we can apply Schur's Lemma and get:

$$\langle \chi^{(i)}, \chi^{(j)} \rangle = \delta_{ij}$$

Thus, our formula becomes:

$$\begin{aligned}\langle \chi_V, \chi_W \rangle &= \langle \sum_i m_i \chi^{(i)}, \sum_j m_j \chi^{(j)} \rangle \\ &= \sum_{i,j} m_i m_j \langle \chi^{(i)}, \chi^{(j)} \rangle \\ &= \sum_{i,j} m_i m_j \delta_{ij}\end{aligned}$$

To find $\langle \chi^{(i)}, \chi^{(i)} \rangle$ for an irreducible character, seeing it through multiplicites, then we know that an irreducible character corresponds to an irreducible representation, which necessarily has to then have multiplicity one, so if in particular:

$$\langle \chi^{(i)}, \chi^{(i)} \rangle = m_i^2 = 1 \cdot 1 = 1$$

Thus, the irreducible character produce and orthogonal basis of class functions!

Corollary 53.2.17: Consequence of Orthogonality

Let $V = \bigoplus_i^r S_i^{l_i}$ where each S_i irreducible and $S_i \ncong S_j$. Then

1. the multipliity of l_i of S_i is: $l_i = \langle \chi_V, \chi_{S_i} \rangle$
2. $\langle \chi_V, \chi_V \rangle = \sum_i l_i^2$
3. $\hom_G(W, V) \cong \hom_G(V, W)^*$

***Proof*:**

1. Use the Artin-Wedderburn decomposition and the fact that direct sums of modules becomes sums of characters
2. this is just taking advantage of the summing formula
3. This is taking advantage of the fact that $\langle \chi, \psi \rangle = \overline{\langle \psi, \chi \rangle}$.

The second line is useful enough to be given it's own name:

Definition 53.2.18: Character Norm

Let V be a G-representation and let $\langle \cdot, \cdot \rangle$ be the inner product on the class functions of G. Then for any class function $\psi = \sum_i \lambda_i \chi_i$,

$$\|\psi\| = \left(\sum_i \lambda_i^2 \right)^{1/2}$$

We will use all this informaiton to deduce new characters:

Example 53.2.19: Orthogonality Relation

Let's go back to our table for S_3:

	e	(1 2)	(1 2 3)
1	1	1	1
χ_{sgn}	1	-1	1
$\vdots$	$\vdots$	$\vdots$	$\vdots$

Since S_3 has 3 conjugacy classes, it can at most have one more irreducible character. What can this character be? We may start by looking at the regular represetnation of S_3 and take it's character χ_V. Then:

$$\langle \chi_V, 1\rangle = 1$$
$$\langle \chi_V, \chi_{\text{sng}}\rangle = 0$$

showing the regular character χ_V has one copy of the trivial representation (which makes sense, since $\chi_V(e) = 1$). Since

$$\chi_V(e) = 3 \qquad \chi_V((1\ 2)) = 1 \qquad \chi_V((1\ 2\ 3)) = 0$$

then we can take a "compliment" character by subtracting the trivial character, which defines a character which gives:

$$\chi^{\perp}(e) = 2 \qquad \chi^{\perp}(1\ 2) = 0 \qquad \chi^{\perp}(1\ 2\ 3) = -1$$

So, if there were to be another irreducible character, it would be of this form. But does there exist a degree 2 irreducible representation with this character? In fact, there has to: we saw that the number of irreducible representations is equal to the number of conjugacy classes, meaning the number of characters must be equal to the number of irreducible representations.

More generally, there can be a few approaches to finding it. One is to tensor two characters we know and see if it is irreducible, or one of it's linear components is irreducible. Another is to find the compliment of a represetnation of an existing character (since $k[G]$-modules are semi-simple). This compliment method is in general less desirable, and usually harder, but is managable if working with permutation representations.To demonstrate, let V be the degree three permutation representation of S_3[a] with basis $\{e_1, e_2, e_3\}$. Then if $\mathbb{1}$ is the trivial action, $V^{\mathbb{1}} = \text{span}(e_1+e_2+e_3)$, Thus, by Machke's theorem, $V^{\mathbb{1}}$ has a compliment. To see it, we simply have to extend $e_1+e_2+e_3$ to an orthogonal basis for V, for example, choose $(e_1+e_2+e_3, e_2-e_1, e_3-e_1)$. Then we can calculate what the action ρ does on this basis. For example:

$$\begin{aligned}\rho(1\ 2)(e_1+e_2+e_3) &= e_1+e_2+e_3\\ \rho(1\ 2)(e_2-e_1) &= e_1-e_2 = -(e_2-e_1)\\ \rho(1\ 2)(e_3-e_1) &= e_3-e_2 = -(e_2-e_1)+(e_3-e_1)\end{aligned}$$

If we write out ρ for an element in each conjugacy class, we will find that it's trace will be

$$\chi^{\perp}(e) = 2 \qquad \chi^{\perp}((1\ 2)) = 0 \quad \chi^{\perp}((1\ 2\ 3)) = -1$$

showing we have found the desired irreducible representation! Finally, by Artin-Wedderburn, we have that $\mathbb{C}S_3 \cong \mathbb{1} \oplus \text{sgn} \oplus V^{\perp}$

Now that we know the character table for S_3, we can construct any other character of representations of S_3, and hence any other representation of S_3. For example, we know that $V^\perp \otimes V^\perp$ has as character $\chi_{V^\perp} \cdot \chi_{V^\perp}$. Using this, we can compute that:

$$\langle V^\perp \otimes V^\perp, \mathbb{1} \rangle = 1$$
$$\langle V^\perp \otimes V^\perp, \mathrm{sgn} \rangle = 1$$
$$\langle V^\perp \otimes V^\perp, V^\perp \rangle = 1$$

showing that

$$\chi_{V^\perp \otimes V^\perp} = \chi_{\mathbb{1}} + \chi_{\mathrm{sgn}} + \chi_{V^\perp}$$

[a]Which in this case, is just the regular representation

Recall from linear algebra that if the rows are orthogonal in a square matrix, then so would be the column (and vice-versa). We bring to light this relation for the character table. As in linear algebra, it comes down to proper normalizing of a different basis: recall that the class equations of the form $\delta_{\mathcal{K}} : G \to \mathbb{C}^\times$ where:

$$\delta_{\mathcal{K}}(g) = \begin{cases} 1 & g \in \mathcal{K} \\ 0 & \text{otherwise} \end{cases}$$

this naturally forms a basis for the class equations too. The following theorem will also show how to go between this basis and the irreducible character basis:

Theorem 53.2.20: Column Orthogonality

If $\chi_1, \chi_2, ..., \chi_r$ are the irreducible characters of G and $g_1, g_2, ..., g_r$ represent the conjugacy classes of G, then

$$\sum_i \chi_i(g_s)\overline{\chi_i(g_t)} = \delta_{st}\frac{|G|}{|\mathrm{conj}(g_s)|}$$

which by Orbt-stabalizer,

$$\sum_i \chi_i(g_s)\overline{\chi_i(g_t)} = \delta_{st}|C_G(g_s)|$$

If $g_s, g_t \in \mathrm{conj}(g)$, then we get that

$$\sum_i \chi_i(g_s)\overline{\chi_i(g_t)} = \sum_i |\chi_i(g_s)|^2 = |C_G(g_s)|$$

in particular, we must have:

$$\sum_i |\chi_i(e)|^2 = |G|$$

***Proof*:**
Define $f_s : G \to \mathbb{C}$,

$$g \mapsto \begin{cases} 1 & g \in \mathrm{conj}(g_s) \\ 0 & \text{otherwise} \end{cases}$$

which is natural basis for the class functions. since it's a class function, we know we can write it as the sum of orhtogonal ireducibile characters:

$$f_s = \sum_i^r \lambda_{is}\chi_i \qquad \lambda_{is} \in \mathbb{C}$$

since these are orthornormal, we know that:

$$\lambda_{is} = \langle f_s, \chi_i \rangle$$

so computing:

$$\begin{aligned}\lambda_{is} &= \langle f_s, \chi_i \rangle \\ &= \frac{1}{|G|}\sum_{g\in G} f_s(g)\overline{\chi_i(g)} \\ &= \frac{1}{|G|}\sum_{g\in \mathrm{conj}(g_s)} \overline{\chi_i(g)} \\ &= \frac{|\mathrm{conj}(g_s)|}{|G|}\overline{\chi_i(g_s)}\end{aligned}$$

so

$$f_s = \frac{\mathrm{conj}(g_s)|}{|G|}\sum_i^r \overline{\chi_i(g_s)}\chi_i$$

we know evaluate at g_t.

Since both the rows and columns of a character table are all orthogonal to each other, the character table is in fact invertible!

Example 53.2.21: Using Column Orthogonality

Let $G = S_4$. We want to find it's character table. We first pick what conjugacy classes to index the columns by, say:

conj. class	1	(1 2)	(1 2 3)	(1 2)(3 4)	(1 2 3 4)
size	1	6	8	3	6

Next, we can immediately deduce that $\chi_{\mathbb{1}}$ and χ_{sgn} are irreducible characters, giving us the current begining of a character table:

	1	6	8	3	6
	1	(1 2)	(1 2 3)	(1 2)(3 4)	(1 2 3 4)
$\chi_{\mathbb{1}}$	1	1	1	1	1
χ_{sgn}	1	-1	1	1	-1

To find more characters, we can first take the pemutation character of $\mathbb{C}X$. The character of each conjugacy class is the number of fixed points, giving us:

	1	6	8	3	6
	1	(1 2)	(1 2 3)	(1 2)(3 4)	(1 2 3 4)
$\chi_{\mathbb{C}X}$	4	2	1	0	0

We can see that the trivial character is part of linear decomposition of $\chi_{\mathbb{C}X}$, since:

$$\langle \chi_{\mathbb{C}X}, \chi_{\mathbb{1}} \rangle = (1(4 \cdot 1) + 6(2 \cdot 1) + 8(1 \cdot 1))/24 = 24/2 = 1$$

and so $\chi_{\mathbb{C}X} - \chi_{\mathbb{1}}$ is a 3 dimensional character, let's label this χ_3. Then $\|\chi_3\|^2 = 1$, showing that it's irreducible, hence we expand our table:

	1 1	6 (1 2)	8 (1 2 3)	3 (1 2)(3 4)	6 (1 2 3 4)
$\chi_{\mathbb{1}}$	1	1	1	1	1
χ_{sgn}	1	-1	1	1	-1
$\chi_{\mathbb{C}X} - \chi_{\mathbb{1}} = \chi_3$	3	2	0	-1	-1

To find the next character, we can multiply χ_{sgn} and χ_3 (note we can do χ_3 with itself, but it will be 9 dimensional, which we'll show in the next proposition cannot be an irreducible representation). Let$\chi_4 = \chi_{\text{sgn}} \cdot \chi_3$. Then $\|\chi_4\|^2 = 1$, meaning it's irreducible, and so we add another row to our table:

	1 1	6 (1 2)	8 (1 2 3)	3 (1 2)(3 4)	6 (1 2 3 4)
$\chi_{\mathbb{1}}$	1	1	1	1	1
χ_{sgn}	1	-1	1	1	-1
χ_3	3	2	0	-1	-1
$\chi_{\text{sng}} \cdot \chi_3 = \chi_4$	3	-1	0	-1	1

Finally, since $|G| = \sum_i \chi_i(e)^2$, we can deduce that $\chi_5(e) = 2$. We can't do this same trick exactly for the other columns since we will get the value up to a complex number z with $\|z\| = 1$. To find the rest of the character values, we can either use column orthogonality For example, to find $\chi_5(1\ 2)$, noticet that the inner product of the first and second column is:

$$1 \cdot 1 + (-1) \cdot 1 + 3 \cdot 1 + 3 \cdot (-1) = 0$$

and so $\chi_5(2)$ must be 0. For the next column, we'd get :

$$1 \cdot 1 + 1 \cdot 1 + 3 \cdot 0 + 3 \cdot 0 = 2$$

and so, since $2 \cdot -1 = -2$, we get that $\chi_5(1\ 2) = -1$. Continuing this way, we get that the rest of the table will be:

	1 1	6 (1 2)	8 (1 2 3)	3 (1 2)(3 4)	6 (1 2 3 4)
$\chi_{\mathbb{1}}$	1	1	1	1	1
χ_{sgn}	1	-1	1	1	-1
χ_3	3	2	0	-1	-1
χ_4	3	-1	0	-1	1
χ_5	2	0	-1	2	0

We can verify this is indeed irreducible by taking $\|\chi_5\|^2 = 1$.

Another method would be to use the regular representation, where we get:

$$\chi_{\mathbb{C}[G]} = \sum_i \chi_i(1)\chi_i$$

These can be verified by taking advantage of the fact that the "column norm" should give you $C_G(g)$ (i.e. the number on top of the table).

As a fun aside, it can be shown that the represetned by χ_4 is the action of the rotation of a cube that is, if $C \subseteq \mathbb{R}^3$ are the vertex points of the cube, then χ_3 is the character of $S_4 \curvearrowright \mathrm{GL}_3(\mathbb{R})$ of all rotations C.

Notice how the dimension of each irreducible representation divided the order of the group. This is in fact true in general. It is not a trivial fact, the only "order" information we've shown so far is that the sums of the squares of the dimensions of the ireducibile representation sum to $|G|$. It is also relatively eays to show that if $W \subseteq V$ is an irreducible $k[G]$-module then the dimension of W is $< |G|$. The following is thus a much stronger result:

Proposition 53.2.22: Dimension Theorem of Irreducible Representations

Let V be an irreducible G-representation. Then:

$$\dim_{\mathbb{C}}(V) \,\big|\, |G|$$

that is, the dimension of V divides the order of G.

***Proof*:**
Let $g_1, g_2, ..., g_r$ be representatives of the conjugacy class. Recall the following trick from proposition 53.1.13:

$$e_i = \left(\sum_{g \in \mathrm{conj}(g)} g \right) \in Z(\mathbb{C}[G])$$

Note that e_i in fact lies in $e_i \in Z(\mathbb{Z}[G])$, since $ge_ig^{-1} = e_i$ (by definition), and so extending linearly all $xe_ix^{-1} = e_i$. Now we have a commutative ring that is a finitely generated $\mathbb{Z}$-module (one way to see this is as we've done in proposition 53.1.13, another way is to notice that $\mathbb{Z}[G]$ is a finitely generated $\mathbb{Z}$-module and $\mathbb{Z}$ is Noetherian, so the center is finitely generated)

Thus, by proposition 26.1.4, $\mathbb{Z} \subseteq Z(\mathbb{Z}[G])$ is an *integral extension*. Next, we have a ring homomorphism:

$$\varphi : Z(\mathbb{C}[G]) \to \mathrm{End}_{\mathbb{C}[G]}(V) \overset{!}{\cong} \mathbb{C} \qquad z \mapsto (z : V \to V)$$

where $\overset{!}{=}$ comes from corollary 51.2.19 and Schur's lemma, and $z(v) = zv$. Hence, each element of $\varphi(Z(\mathbb{C}[G]))$ is $\mathbb{C}$-linear, and must be invertible, and so φ is in fact a map from $Z(\mathbb{C}[G]) \to \mathrm{Aut}_{\mathbb{C}}(V) = \mathrm{GL}(V)$, which is simply a restriction of the origional G-action ρ to $Z(\mathbb{C}[G])$. Hence $\rho(e_i) = \lambda_i I$ for appropriate $\lambda_i = \varphi(e_i)$, meaning $\chi(e_i) = \mathrm{tr}(e_i : V \to V) = n\varphi(e_i)$ where $n = \dim(V)$ since by what we've just shown, $\varphi(e_i)$ acts as scalar multiplication on the representation V.

Now, since $e_i \in Z(\mathbb{C}[G])$ is integral over $\mathbb{Z}$, $\varphi(e_i) \in \mathbb{C}$ is integral over $\mathbb{Z}$, that is $\varphi(e_i) \in \overline{\mathbb{Z}}^{\mathbb{C}} \subseteq \mathbb{C}$ (i.e. $\varphi(e_i)$ is algebraic). Next, we can express $\chi(e_i)$ in a different way:

$$\sum_{g \in \mathrm{conj}(g_i)} \chi_V(g) = |\mathrm{conj}(g_i)| \cdot \chi_V(g_i) = \chi(e_i)$$

so:

$$\frac{\text{conj}(g_i)}{n}\chi_V(g_i) = \varphi(e_i) \in \overline{\mathbb{Z}}^{\mathbb{C}}$$

which is telling us that if we divide it by n, we still get an algebraic integer! Taking a nice linear combination and multiplying by $\overline{\chi_V(g_i)}$ we get:

$$\sum_i^r \frac{\text{conj}(g_i)}{n}\chi_V(g_i)\overline{\chi_V(g_i)}$$

Then we've just shown that $\frac{\text{conj}(g_i)}{n}\chi_V(g_i)$ is an algebraic integer, and $\overline{\chi_V(g_i)}$ is an algebraic integer (being the sum of roots of unity). Now, notice this is the form of the inner product, giving us:

$$\frac{|G|}{n}\langle\chi_V,\chi_V\rangle = \frac{|G|}{n} \in \overline{\mathbb{Z}}^{\mathbb{C}}$$

as a V irreducible. Thus, we have that $\frac{|G|}{n}$ is some algebraic integer. We further have that it is certainly a rational number ($|G|, n \in \mathbb{N}$), but then we have:

$$\frac{|G|}{n} \in \overline{\mathbb{Z}}^{\mathbb{C}} \cap \mathbb{Q} \overset{!}{=} \mathbb{Z}$$

where $\overset{!}{=}$ comes from the fact that if R is a Unique Factorization Domain, $\alpha \in \text{Frac}(R)$, α is integral over R if and only if $\alpha \in R$, completing the proof.

This will make the computation of characters of abelian groups paticularly simple! (see rep theory of finite groups: An introductory Appraoch p. 47!)

Exercise 53.2.1

1. Let G act on the n-dimensional vector-space V by basis permutation. Show that $\dim(V^G)$ equals the number of orbits of G.

53.2.1 More Character Tables

In this section, we go over a few more exmaples of character tables in order to see some techniques used to find them. For any abelian, group, the character table is straight forward. As a really simple example, consider $\mathbb{Z}/3\mathbb{Z}$:

	1	1	1
	1	2	3
χ_1	1	1	1
χ_2	1	ζ_3	ζ_3^2
χ_3	1	ζ_3^2	ζ_3

If we take $\mathbb{Z}/3\mathbb{Z} \times \mathbb{Z}/3\mathbb{Z}$, we will get a 9 by 9 table. The table bellow fills in a few of the characters (note that these were found by taking $\chi_i(g_i)\chi_j(g_j)$)

	1	1	1	1	1	1	1	1	1
$\mathbb{Z}/3\mathbb{Z} \times \mathbb{Z}/3\mathbb{Z}$	$(1,1)$	$(2,1)$	$(3,1)$	$(1,2)$	$(2,2)$	$(3,2)$	$(1,3)$	$(2,3)$	$(3,3)$
$\chi_{(1,1)}$	1	1	1	1	1	1	1	1	1
$\chi_{(2,1)}$	1	ζ_3	ζ_3^2	1	ζ_3	ζ_3^2	1	ζ_3	ζ_3^2
$\chi_{(2,2)}$	1	ζ_3	ζ_3^2	ζ_3	ζ_3^2	1	ζ_3^2	1	ζ_3
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$

Note that the root ζ_3 might change if we take the direct product of cyclic groups of differnet orders, for example the character table of $\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/3\mathbb{Z}$ has roots of unity of order 3 and 6 (as you can verify).

More generally, if G is abelian, we break G down into a product of cyclic groups and construct the table like above. Thus, we focus on non-abelian groups. We have already found the character table for S_3:

S_3	e	$(1\ 2)$	$(1\ 2\ 3)$
$\chi_{\mathbb{1}}$	1	1	1
χ_{sgn}	1	-1	1
χ_3	2	0	-1

and the character table for S_4:

	1	6	8	3	6
S_4	1	$(1\ 2)$	$(1\ 2\ 3)$	$(1\ 2)(3\ 4)$	$(1\ 2\ 3\ 4)$
$\chi_{\mathbb{1}}$	1	1	1	1	1
χ_{sgn}	1	-1	1	1	-1
χ_3	3	2	0	-1	-1
χ_4	3	-1	0	-1	1
χ_5	2	0	-1	2	0

Next, let $G = D_4$. Then: $D' = \{e, r^2\}$, so G has 4 degree 1-irreducible representations. We also found in example 53.1.5 that D_4 has a degree 2 irreducible representation. These are all of the representations, sinc $1+1+1+1+2^2 = 8$. To find the 1-dimensional irreducible representations, notice that $D_4/D_4' \cong \mathbb{Z}_2 \times \mathbb{Z}_2$, meaning we can find the irreducible representations of $\mathbb{Z}/2\mathbb{Z}$, which are:

$\mathbb{Z}/2\mathbb{Z}$	0	1
χ_0	1	1
χ_1	1	-1

and take the tensor product to create:

$\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$	$(0,0)$	$(1,0)$	$(0,1)$	$(1,1)$
$\chi_{(0,0)}$	1	1	1	1
$\chi_{(1,0)}$	1	-1	1	-1
$\chi_{(0,1)}$	1	-1	-1	1
$\chi_{(1,1)}$	1	1	-1	-1

and using this, we can complete the table for D_4:

D_4	e	r	r^2	σ	σr
χ_1	1	1	1	1	1
χ_2	1	1	1	-1	-1
χ_3	1	-1	1	1	-1
χ_4	1	1	1	-1	1
χ_5	2	0	-2	0	0

More generally, for D_n, the character table will conssite of characters of degree 1 and 2.

Let's now let $G = Q_8$. Taking the presentation:

$$Q_8 = \langle i, j | i^4 = 1,\ i^2 = j^2,\ i^{-1}ji = j^{-1} \rangle$$

with $k = ij$ and $i^2 = -1$, we get the conjugacy classes can be represented by $1, -1, i, j, k$, with sizes 1, 1, 2, 2, 2. Since $Q_8/Q_8' \cong \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$, there are 4 degree 1 characters. The final irreducible character thus *must* have degree 2; let's label if χ_5. We have in fact explicitely found this character before in example 53.1.5, but we can find it without actually manually finding a representation. Let φ be an irreducible representation of degree 2. Since $-1 \in Z(Q_8)$, by Schur's lemma $\varphi(-1)$ is a 2×2 scalar matrix of order 2, hence it is $\pm I$, and so $\chi_5(-1) = \pm 2$. Next, if $\chi_5(i) = a$, $\chi_5(j) = b$, and $\chi_5(k) = c$, then since χ_5 is orthogonal, we have that:

$$1 = \langle \chi_5, \chi_5 \rangle = \frac{1}{8}(2^2 + (\pm 2)^2 + 2a\overline{a} + 2b\overline{b} + 2c\overline{c})$$

and since $a\overline{a}, b\overline{b}, c\overline{c} \in \mathbb{R}^+$, they must all be zero. Furthermore, we must have that:

$$0 = \langle \chi_1, \chi_5 \rangle = \frac{1}{8}(2(\pm 2) + 0 + 0 + 0 + 0)$$

implying that $\chi_5(-1) = -2$. Thus, we have that the character table is:

Q_8	1	i	-1	j	k
χ_1	1	1	1	1	1
χ_2	1	1	1	-1	-1
χ_3	1	-1	1	1	-1
χ_4	1	1	1	-1	1
χ_5	2	0	-2	0	0

Notice that this character table is identical to the character table of D_8. Hence, identical character tables does not imply isomorphic groups! Finally, we could have found the last row of the above table by applying column orthogonality to find the missing values, once we knew $\chi_5(1) = 2$.

Next, let $G = A_4$. Since we found the character table for S_4, we can use it it to find the character table of A_4. We first note that A_4 has 4 conjugacy classes (represented by 1, (1 2)(3 4), (1 2 3) and (1 3 2)), and $[A_4 : A_4'] = 3$, meaning A_4 has 3 chacacters of degree 1, and the only other character must be of degree 3. Next, using the orthogonality relation, notice that χ_3 in the table for S_4 when restricted to A_4 is still irreducible (this is certainly not generally the case, for ex. the degree 2 representaiton of S_3 will become reducible since any proper subgroup of S_3 is abelian). Overall, we get:

A_4	1	(1 2)(3 4)	(1 2 3)	(1 3 2)
χ_1	1	1	1	1
χ_2	1	1	ζ_3	ζ_3^3
χ_3	1	1	ζ_3^2	ζ_3
χ_5	3	-1	0	0

Finally, we will do the character table for S_5:

S_5	1	(1 2)	(1 2 3)	(1 2 3 4)	(1 2 3 4 5)	(1 2)(3 4)	(1 2)(3 4 5)
χ_1	1	1	1	1	1	1	1
χ_2	1	-1	1	-1	1	1	-1
χ_3	4	2	1	0	-1	0	-1
χ_4	4	-2	1	0	-1	0	1
χ_5	5	-1	-1	1	0	1	-1
χ_6	5	1	-1	-1	0	1	1
χ_7	6	0	0	0	1	-2	0

TBD

53.2.2 Burnside's Theorem

We now move towards Burnside's Theorem, which shows that any group of order $p^a q^b$ for primes p, q and numbers $a, b \in \mathbb{N}$ is a solvable group! This proof is relatively simple in the context of representation theory, but it took over 70 years before a proof was found using just group theory! This is a good example of the power perspective: simply switching to the context of representation theory and seeing groups as embedded into $GL(V)$ let us prove a claim that from another perspective is much much harder!

Lemma 53.2.23: Burnside build-up Lemma

Suppose that V is an irreducible representation is of dimensin n (i.e. $\chi_1(V) = n$). If $\gcd(n, |\text{conj}(g)|) = 1$, then:

$$|\chi_V(g)| = 0 \text{ or } n$$

***Proof*:**
By the above proof, we have:

$$\frac{\text{conj}(g)}{n}\chi_V(g) \in \overline{\mathbb{Z}}^{\mathbb{C}}$$

where $\chi_V(g) \in \overline{\mathbb{Z}}^{\mathbb{C}}$ as well, so by Bézout $(1 = am + b|\text{conj}(g)|)$

$$\frac{1}{n}\chi_V(g) \in \overline{\mathbb{Z}}^{\mathbb{C}}$$

Let $\alpha = \frac{1}{n}\chi_V(g)$. Furthermore, notice that:

$$|\alpha| = \left|\frac{\chi_V(g)}{\chi_V(1)}\right| \leq 1$$

Now, let N be the Galois closure of $\mathbb{Q}(\alpha)/\mathbb{Q}$ so that $G = \text{Gal}(N/\mathbb{Q})$ is the glaois group of N. Let

$$\beta = \prod_{\sigma \in G} \sigma(\alpha)$$

Then certainly β is in the fixed field of the galois group, $\beta \in N^G = \mathbb{Q}$. Notice that each $\sigma(\alpha)$ satisfies the same monic polynomials as that of α, and so $\sigma(\alpha) \in \overline{\mathbb{Z}}^{\mathbb{C}}$, and so $\beta \in \overline{\mathbb{Z}}^{\mathbb{C}}$, but then $\beta \in \overline{\mathbb{Z}}^{\mathbb{C}} \cap \mathbb{Q} = \mathbb{Z}$.

Now, recall that:

$$\alpha = \frac{\chi_V(g)}{n} = \frac{\zeta_1 + \zeta_2 + ... + \zeta_n}{n}$$

which are roots of unity, which means that:

$$\sigma(\alpha) = \frac{\sigma(\zeta_1) + \cdots + \sigma(\zeta_n)}{n}$$

so $|\sigma(\zeta_1)| \leq 1$, so $|\beta| \leq 1$. Since β is an integer, we either get that $|\beta| \in \{0, 1\}$, If $|\beta| = 0$, then $\alpha = 0$, and if $|\beta| = 1$, then $|\alpha| = 1$, and so plugging back into our original equation we get that:

$$\chi_V \in \{0, n\}$$

as we sought to show.

We can also treat this lemma as another check that we have given the correct character table (verify it for the character table of S_4 we found in example 53.2.21).

Theorem 53.2.24: Burnside's Theorem

Let $p \neq q$ be prime numbers, $a, b \in \mathbb{N}$. If $|G| = p^a q^b$, then G is solvable.

Notice how this immedaitely implies that the smallest possible order of a non-abelian simple group is 30, since $30 = 2 \cdot 3 \cdot 5$, which matches with the fact that the smallest non-abelian group is A_5 which has order 60! Recall also that if $|G| = p^a$, then we would need that $a = 1$ (namely, any p-group has a non-trivial center)

***Proof*:**
Take the composition series:

$$1 = G_0 \trianglelefteq G_1 \trianglelefteq \cdots \trianglelefteq G_k = G$$

where G_i/G_{i-1}. Then we'll show that if G is simple, then G is *abelian*. For the sake of contradiction, suppos that G is non-abelian and simple with order $p^a q^b$ (where we may as well assume $a, b \in \mathbb{N}_{>0}$). Let $\chi_1, \chi_2, ..., \chi_r$ be the irreducible representation for the group G, and $\chi_1 = \chi_{\mathbb{1}}$. Let $Z_i = \{g \in G \mid |\chi_i(g)| = \chi_i(1),\ \forall i\}$. The condition in side doesn't happen so often, we can equivanelty say that $\rho_i(g)$ is a scalar by using the triangle inequality. Thus,

$$Z_i \trianglelefteq G$$

If $Z_i = G$, then G acts by scalars on V_i. But then every subspace of V_i is a subrepresentation, so the dimension of all the irreducible is 1. And so each character is of the form:

$$\chi_i : G \to \mathbb{C}^\times$$

Since G is simple, each each i is injective or trivial. If each i is injective, then G is abelian, a contradiction, and if it is trivial we get $i = 1$ (the trivial representation). Thus, it must be that:

$$Z_i = 1 \qquad \forall i > 1$$

Now, take the sylow q-subgroup Q: (notice we are using that $a, b > 0$)

$$1 \subsetneq Q \subsetneq G$$

So Q is a p-group, and so has a non-trivial center: $Z(Q) \neq 1$. Take $h \in Z(Q) \setminus \{e\}$. Hence $Q \leqslant C_G(h)$. , so $[G : C_G(h)] \big| [G : Q] = p^a$, where $[G : C_G(h)] = |\text{conj}(h)$. Now recall we've shown that:

$$\chi_1 \big| |G| = p^a q^b$$

Now, if $p \nmid \chi_i(1)$, then by the previous line we have that $\chi_1(1) \big| q^b$. So, $\gcd(\chi_i(1), |\text{conj}(h)) = 1$, and so by lemma 53.2.23

$$\chi_i(h) = 0 \qquad \text{or} \qquad \chi_i(1)$$

But if $|\chi_i(h)| = \chi_i(1)$, $h \in Z_i$, and so $i = 1$

So, if $p \nmid \chi_i(1)$ and $i > 0$, then $\chi_i(h) = 0$. Using this, we can get our contradiction: as h is not the trivial element, we get:

$$\begin{aligned} 0 &= \sum_i \overline{\chi_i(h)} \chi_i(1) \\ &= \sum_i \chi_i(h) \chi_i(1) \\ &\stackrel{!}{=} 1 + \sum_{\substack{i>1 \\ p \big| \chi_i(1)}} \underbrace{\chi_i(1)}_{\in p\mathbb{Z}} \underbrace{\chi_i(h)}_{\in \overline{\mathbb{Z}}^{\mathbb{C}}} \\ &= 1 + p\alpha && \text{some } \alpha \in \overline{\mathbb{Z}}^{\mathbb{C}} \end{aligned}$$

where the one in the $\stackrel{!}{=}$ equality came from when $i = 1$. Thus,

$$a = -1/p \in \overline{\mathbb{Z}}^{\mathbb{C}} \cap \mathbb{Q} = \mathbb{Z}$$

a contradiction, completing the proof.

(huge list of character tables: commented)

53.3 Induced Representation

This is another way of constructing representations. We saw that if $H \leqslant G$ and V is a G representation, then we can make V an H-representation through the inclusion map $H \hookrightarrow G \to \text{GL}(V)$. We may ask if we may go the other way: given an H representation W, can we find a G-representation? If φ is the representations afforded by H, can we find a Φ such that $\Phi|_H = \varphi$? It turns out we cannot: take $A_3 \subseteq S_3$ and consider a faithful representation of degree 1 (recall it's a cyclic group). Then notice that any 1 dimensional representations of S_3 will have to have A_3 in it's kernel, meaning we can't extend the action of A_3 to an action of S_3. A little more generally, if we let G be a simple group, $H \leqslant G$, and φ a representation of H where $\ker(\varphi)$ is a proper nontrivial subgroup of H, then if we somehow extend the representation φ of H to Φ of G, then $\ker(\Phi)$ would have a proper kernel too, contradicting that G is simple. So instead, we will not be extending representation from H to G, but *inducing* a representation in a natural way. This method of inducing a representation will work even in the two above cases, meaning an induced representation will not always be an extension.

Let $H \leqslant G$, and consider some H-representation V, which can be represented as a $k[H]$-module V.

Recall that we can "extend scalars" and define a new $k[G]$-module:

$$k[G] \otimes_{k[H]} V$$

This is (the outer tensor product for representatives, which I haven't talked about yet, but probably should have it here to motivate the idea)

Definition 53.3.1: Induced Representation

Let $H \leqslant G$ of a finite group G and let V be a $k[H]$-module affording the representation φ. Then the $k[G]$-module $k[G] \otimes_{k[H]} V$ is called the *induced module* of V, the representation is called the *induced representation*, and if χ is the character of φ, then the character of the induced representation is called the *induced character*, and is written as $\mathrm{Ind}_H^G(\psi)$.

Another intuition behind defining the induced module is the following: If W is an H-representation, then let our new module be:

$$\mathrm{Ind}_H^G(W) = \{f \in \mathrm{Hom}_{\mathbf{Set}}(G, W) \mid f(hg) = h \cdot f(g),\ \forall g \in G, h \in H\}$$

Then $\mathrm{Ind}_H^G(W)$ is clearly a complex vector space under pointwise addition and scalar multiplication, and the G-action on this space would be:

$$\gamma \cdot f : G \to W \qquad g \mapsto f(g\gamma)$$

It should be verified that this is indeed G-linear. In this case, we get that:

$$\mathrm{Ind}_H^G(W) = \mathrm{Hom}_{\mathbb{C}[H]}(\mathbb{C}[G], W)$$

where we may use a variant of lemma 53.2.12 to show that this is the same thing as we've just defined. It is also easy to see the dimension of $\mathrm{Ind}_H^G(W)$, namely:

$$\dim_{\mathbb{C}}(\mathrm{Ind}_H^G(W)) = [G : H] \dim_{\mathbb{C}}(W)$$

with the $\mathbb{C}$-vector space isomorphism being:

$$\mathrm{Ind}_H^G(W) \to W^{\oplus[G:H]} \qquad f \mapsto f(g_1), ..., f(g_{[G:H]})$$

where we treat $G = \coprod_i Hg_i$ for the cosets Hg_i. By letting $g_1 = e$ being the identity, there is a natural canonical map:

$$\mathrm{Ind}_H^G(W) \to W \qquad f \mapsto f(e)$$

This motivates the following adjoint of the restriction functor:

Theorem 53.3.2: Frobenius Reciprocity

Let $H \leqslant G$. Then if V is a G-representation and W and H-representation, then:

$$\mathrm{Hom}_G(V, \mathrm{Ind}_H^G W) \cong_{\mathbb{C}} \mathrm{Hom}_H(V|_H, W)$$

where $V|_H$ is the restriction of the action to H

***Proof*:**
We will explicitly construct inverse bijections Φ from left to right and Θ from right to left. Let $\alpha : \mathrm{Ind}_H^G(W) \to W$, $f \mapsto f(1)$. Then α is H-linear. Then let

$$\Phi(f) = \alpha \circ f$$

Then $\Phi(f)$ is H-linear (check $\mathbb{C}$-linearity of Φ). For the other way, let $\psi : V|_H \to W$ be H-linear,

$$\Theta(\psi) : V \to \mathrm{Ind}_H^G(W) \qquad v \mapsto (g \mapsto \psi(gv))$$

Then this is well-defined. What's left is to check they are inverses of each other

53.3.3 Left Vs Right Adjoint This proof shows that induction is right-adjoint to restriction. It can be shown to also be *left adjoint*. A pair of functors (F, G) where the functor F is both left and right adjoint to a functor G (and hence G is also right/left adjoint to F) is sometimes called a *Frobenius pairing*. The functor Ind is usually only right-adjoint to the restriction restriction functor when G is infinite (The *compact induction* functor is usually left-adjoint to the restriction functor, see [20]).

Corollary 53.3.4: Frobenius Reciprocity On Characters

Let $H \leqslant G$, χ be the character of a G representation and ψ a character of an H-representation. Then:

$$\langle \chi|_H, \psi \rangle_H = \langle \chi, \mathrm{Ind}_H^G(\psi) \rangle_G$$

***Proof*:**
https://unapologetic.wordpress.com/2010/12/03/dual-frobenius-reciprocity/

Example 53.3.5: Induced Representations

1. $\mathrm{Ind}_G^G(W) = W$
2. $\mathrm{Ind}_{\mathbb{1}}^G(\mathbb{C}) = \mathbb{C}[G]$
3. $\mathrm{Ind}_H^G(\mathbb{1}) = \mathbb{C}[G/H]$

Another way of thinking of this result is that if V is an irreducible representation of G and W is an irreducible representation of W, then the number of times W occurs in $\mathrm{Ind}_H^G(W)$ is equal to the number of times W occurs in $V|_H$.

Proposition 53.3.6: Properties Of Induced Representation

Let $K \leqslant H \leqslant G$, W be an K-representation, $\operatorname{Ind}_K^H(W)$ the induced H representation, $\operatorname{Ind}_H^G(\operatorname{Ind}_K^H(W))$ the induced G-representation. Then:

1. If χ, χ' are two characters of W,
$$\operatorname{Ind}_K^H(\chi + \chi') = \operatorname{Ind}_K^H(\chi) + \operatorname{Ind}_K^H(\chi')$$
2. Induction is transitive:
$$\operatorname{Ind}_H^G(\operatorname{Ind}_K^H(W)) = \operatorname{Ind}_K^G(W)$$

***Proof*:**

Theorem 53.3.7: Computing Induced Representation

Let $H \leqslant G$ of a finite group G, and $g_1, g_2, ..., g_m$ be representatives for distinct left cosets of H in G. Let V be an H-representation of degree n. Then the induced G-representation $W = k[G] \otimes_{k[H]} V$ is an nm degree representation, and if Φ repersented the induced representation, then there exists a basis for W such that W affords the matrix representation defined for each $g \in G$ by:
$$\Phi(g) = \begin{pmatrix} \varphi(g_1^{-1}gg_1) & \cdots & \varphi(g_1^{-1}gg_m) \\ \vdots & \vdots & \cdots \\ \varphi(g_m^{-1}gg_1) & \cdots & \varphi(g_m^{-1}gg_m) \end{pmatrix}$$
where each $\varphi(g_i^{-1}gg_j)$ is an $n \times n$ block, with $\varphi(g_i^{-1}gg_j) = 0$ if $g_i^{-1}gg_j \notin H$.

***Proof*:**
First, $k[G]$ is a free right $k[H]$-module with decomposition:
$$k[G] = g_1k[H] \oplus \cdots \oplus g_mk[H]$$
Since tensor products commute with direct sums, we get:
$$W = k[G] \otimes_{k[H]} V \cong (g_1 \otimes V) \oplus \cdots \oplus (g_m \otimes V)$$
Since k is in the center of $k[G]$, the above isomorphism is also a k-vector space isomorphism. Thus, if $v_1, v_2, ..., v_n$ is a basis for V, then $\{g_i \otimes v_j \mid 1 \leq i \leq m,\ 1 \leq j \leq m\}$ is a basis for W, and hence $\dim_k(W) = nm$.

To show $\Phi(g)$ is of the given form, order the basis into m of size n as:
$$g_1 \otimes v_1, g_1 \otimes v_2, ..., g_1 \otimes v_n, g_2 \otimes v_1, ..., g_m \otimes v_n$$
Using this order, we can compute the matrix $\varphi(g)$ for each g acting on W given this basis. Fixing j and g, let $g_j = g_ih$ for some index i and $h \in H$. Then for every k:
$$g(g_j \otimes v_k) = (gg_j \otimes v_k) = g_i \otimes v_k = \sum_{t=1}^{n} a_{tk}(h)(g_i \otimes v_t)$$

where a_{tk} is the t, k coefficient of the matrix h acting on V with respect to the basis $v_1, v_2, ..., v_n$. Notice that the action of g on W maps the jth block of n basis vectors of W to the ith block of basis vectors, and then has the matrix $\varphi(h)$ as that block. Since $h = g_i^{-1} g g_j$, this fully describes $\Phi(g)$, as we sought to show.

Corollary 53.3.8: Computing The Induced Character

Let $H \leqslant G$ where G is a finite group, V be an H-representation with character χ, and $\mathrm{Ind}_H^G(\chi)$ the induced character. Then:

$$\mathrm{Ind}_H^G(\chi) = \sum_{i=1}^{m} \chi(g_i^{-1} g g_i)$$

where $\chi(g_i^{-1} g g_i) := 0$ if $g_i^{-1} g g_i \notin H$. Hence, $\mathrm{Ind}_H^G(\chi)(g) = 0$ if g is not conjugate of some element in H in G, in particular if $H \trianglelefteq G$ then $\mathrm{Ind}_H^G(\chi)$ is zero on all elements $G - H$.

Proposition 53.3.9: Induced Index Two Induced Representation

Let $H \leqslant G$ where $[G : H]$ has index two[a] (ex. $A_n \leqslant S_n$ or $C_n \leqslant D_{2n}$). Then if V is a finite-dimensional irreducible representation of G, either $V|_H$ is irreducible or $V|_H = W_1 \oplus W_2$ where $W_1 \not\cong W_2$, are both ireducibile, and have the same dimension. Moreover, the second option implies that $V \cong V \otimes \chi$ where $\chi : G \twoheadrightarrow G/H \cong \{\pm 1\}$ (i.e. a 1-dimensional representation)

[a] Recall that this makes H a normal subgroup

In the case where $V|_H$ is not irreducible, we can imagine it splitting into a diagram like so:

$$\begin{tikzcd}
& W_1 \\
V \arrow[ur, dash] \arrow[dr, dash] & \\
& W_2
\end{tikzcd}$$

while in the case where $V|_H$ is irreducible, what we actually have (and didn't have time to prove), is that we have *two* irreducible representation's of G (V and V') that both get restricted to W:

$$\begin{tikzcd}
V \arrow[dr, dash] & \\
& W \\
V' \arrow[ur, dash] &
\end{tikzcd}$$

and then the induced representation of W is the direct sum of V and V'.

***Proof*:**
Let $W \subseteq V|_H$ be an irreducible H-representation. Then Since $H \trianglelefteq G$, gW is is an H-stable space: gW is some subspace of V (it satisfies the subspace axioms since the action is linear), and for any $h \in H$, $h \cdot gW = hgW = g'hW$ that is $gW = hg'h^{-1}W$, and since $hg'h^{-1} \in H$, we have that $h \cdot gW = gW$. Now, gW is irreducible: since W is irreducible and $\varphi_g(W) = gW$ is a G-linear isomorphism (with inverse$\varphi_{g^{-1}}(gW) = W$), gW is irreducible. Finally, the structure of gW only depends on that of the coset gH

Thus, fix $\gamma \in G \setminus H$ so that

$$\sum_{g \in G} gW = W + \gamma W \subseteq V$$

where the $\subseteq$ comes from the fact that this is a sum of two subspaces. Now, notice that the left hand side is a G-stable space, but V is irreducible meaning it does not have a G-stable subspace, and therefore, we actually have the equality:

$$W + \gamma W = V$$

meaning the $H + gH = G$ is parallelled in the action! We know have two possibilities: either $W = \gamma W$, and so $V|_H = W$, and we have $\dim V = \dim W$ (which is case (a)).

on the otherhand, if $W \neq \gamma W$, then $W \cap \gamma W = (0)$ (since they are both irrreducible), and so $V = W \oplus \gamma W$ (i.e. direct sum as H-representations), and so V is the direct sum of two irreducible H-representations.

Now, by Frobeinus reciprocity, we get:

$$\langle W, V|_H \rangle_H = \langle \mathrm{Ind}_H^G(W), V \rangle_G$$

Since W occurs in $V|_H$ by definition, we get that $1 \leq \langle W, V|_H \rangle_H$, while since V is irreducible, meaning by a dimension argument we get $\langle \mathrm{Ind}_H^G(W), V \rangle_G \leq 1$, and hence:

$$1 = \langle W, V|_H \rangle_H = \langle \mathrm{Ind}_H^G(W), V \rangle_G = 1$$

Thus, from the first one, we get that $W \ncong \gamma W$ (or else the left hand side would be equal to 2), and we must have $V = \mathrm{Ind}_H^G(W)$ (or else the right hand side would be equal to something greater than 1)

If part (b) holds, then

$$V \oplus \chi|_H \cong V|_H$$

and

$$V \cong \mathrm{Ind}_H^G(W) \cong V \oplus \chi$$

Conversely, if $V \cong V \oplus \chi$, then $\chi_V(g) = \chi_V(g)\chi(g)$ for all $g \in G$. Thus, by simple arithmetics we see that $\chi(g) = 0$ for all $g \notin H$. And so:

$$\|\chi_{V|_H}\|_H^2 = 2\|\chi_V\|_G^2$$

and so it is a direct sum of irreducibles, as we sought to show.

53.4 *Application to Harmonic Analysis

In this section, we take a brief moment to explore the representation of locally compact abelian groups, such as $\mathbb{R}$. This section will assume some understanding of topological groups, and is mainly to motivate for the reader the connection between the representation theory of this chapter with *Fourier analysis* and more broadly *Harmonic analysis*.

Every group G is always a topological group with the discrete topology (see appendix C.4). The group G is finite iff it is a compact discrete group. Many groups of interest are either compact (like the torus or compact lie groups), or locally compact (like the real line). The representation theory of these groups rely on the correct framing of the representation in the finite case. For the case of infinite dimension *non-abelian* groups, the build-up is a good deal too complicated for this expository section[4], however many of the major ideas for the abelian are simple enough to exposit, and even better naturally motivate the Fourier transform and abstract harmonic analysis through the lens of representation theory.

Let G be a finite abelian group so that all complex irreducible representations are 1-dimensional, and hence characters and necessarily a homomorphism. As G is finite, say $n = |G|$, then $\chi(g^n) = \chi(e) = 1$ for all $g \in G$, and so $\chi^n = \text{id}$, showing that $\chi : G \to \mathbb{C}^*$ restricts to a map $\chi : S^1$. This shows that all characters are *unitary representations*. All characters for a finite group are necessarily unitary representations, while for non-finite groups there can be non-unitary representations[5]. Such groups which have a (general) representation theory that doesn't generalize as well as infinite unitary representations (many representation are indecomposable but not irreducible, for the details see the companion Lie groups notes). Thus, for this sections we stick to maps to S^1 with the natural multiplication structure as the subgroup of $\mathbb{C}^*$: label this collection $\hat{G} = \text{Hom}_{\mathbf{Ab}}(G, S^1)$. As the codomain is a group, $\hom_{\mathbf{Ab}}(G, S^1)$ has a group structure given by point-wise multiplication.

Definition 53.4.1: Pontryagin Group

Let G be locally compact abelian group. Then the *Pontryagin dual* of G is the group:

$$\text{Hom}_{\text{Top}}(G, S^1)$$

with the operation being point-wise multiplication.

This is just another way of representing the characters of the group. Let us stick to the special case where G is a finite abelian group and re-frame the previous theory using the Pontryagin group:

[4] see [26] for the EYNTKA series on geometric representation theory

[5] $\text{SL}_2(\mathbb{R})$ acting on $\mathbb{R}^2$ need not be unitary

Proposition 53.4.2: Pontryagin Dual

Let G be a finite abelian group, $\hat{G} = \mathrm{Hom}_{\mathbf{Ab}}(G, S^1)$ the group of characters under point-wise multiplication. Then there is an isomorphism:

$$G \cong \widehat{G}$$

Furthermore, there is a natural isomorphism:

$$G \cong \hat{\hat{G}}$$

If G is infinite, it is rarely the case that $G \cong \widehat{G}$ (think $\widehat{L^p} \cong L^q$). Even when G is compact this G need not be isomorphic to $\widehat{G}$. What is most important is that $\widehat{G}$ contains the necessary information to recover G, as the double-dual recovers G. We can think of $\widehat{(-)}$ as an operation that goes back and forth from a group to a unitary representation, and as we are working with unitary representations this collection also forms a group[6]. We can thus imagine going back and forth between these two:

$$\text{Group} \overset{\widehat{(-)}}{\rightleftarrows} \text{Unitary Rep.}$$

***Proof*:**
As G is finite abelian,

$$G \cong \mathbb{Z}/n_1\mathbb{Z} \times \mathbb{Z}/n_2\mathbb{Z}/ \times \cdots \times \mathbb{Z}/n_k\mathbb{Z}$$

where $n_1 \big| n_2 \big| \cdots \big| n_k$. Choose generators, $a_1, ..., a_k$. Define the indicator characters on the generators:

$$\chi_i(a_j) = \begin{cases} e^{\frac{2\pi i}{n_i}} & i = j \\ 1 & i \neq j \end{cases}$$

These certainly generate $\hat{G}$. Define the map on the generators:

$$\Phi(a_i) = \chi_i$$

This is an isomorphism.

For the double-dual, take the natural map:

$$\Phi(g_i) = \mathrm{ev}_{g_i}$$

where as usual $\mathrm{ev}_{g_i}(\chi) = \chi(g_i)$. This is easy to check that is an isomorphism, completing the proof.

Now, when studying spaces we study all (continuous) functions on a space. In our case, our space is finite abelian groups. Let G have the discrete topology so that the set of continuous functions from G to $\mathbb{C}$ is all set functions. We want a $\mathbb{C}$-algebra isomorphism between $\mathbb{C}[G]$ and $\mathrm{Hom}_{\mathbf{Top}}(G, \mathbb{C})$.

[6] the collection of representations also form a ring called the *representation ring* and has fascinating ties to K-theory and intersection theory, but it is again too much for this section and the connections are discussed in [5, Part II] of the EYTNKA series

These two sets are clearly bijective via the natural mapping:

$$\sum_i a_i g_i \mapsto (g_i \mapsto a_i)$$

And this map preserves addition if addition is defined point-wise in the hom-set:

$$\begin{aligned}
&\Phi\left(\sum_i a_i g_i + \sum_i b_i g_i\right)\\
&= \Phi\left(\sum_i (a_i + b_i) g_i\right)\\
&\mapsto (g_i \mapsto a_i + b_i)\\
&= (g_i \mapsto a_i) + (g_i \mapsto b_i)\\
&= \Phi\left(\sum_i a_i g_i\right) + \Phi\left(\sum_i b_i g_i\right)
\end{aligned}$$

But it does *not* preserve multiplication if the operation is pointwise multiplication:

$$\begin{aligned}
&\Phi\left(\sum_{g\in G} a_i g \sum_{g\in G} b_i g\right)\\
&= \Phi\left(\sum_{g\in G} \sum_{g_k g_\ell = g} (a_k b_\ell) g\right)\\
&\mapsto (g_i \mapsto \sum_{g_k g_\ell = g_i} (a_k b_\ell))\\
&\neq (g_i \mapsto a_i b_i)\\
&\neq (g_i \mapsto a_i) \cdot (g_i \mapsto b_i)\\
&= \Phi\left(\sum_i a_i g_i\right) \cdot \Phi\left(\sum_i b_i g_i\right)
\end{aligned}$$

This is because the points g_i act on each other. There is a natural multiplication function defined on $\mathrm{Hom}_{\mathbf{Top}}(G, \mathbb{C})$ called *convolution*:

Definition 53.4.3: Convolution

Let $p, q \in \mathrm{Hom}_{\mathbf{Top}}(G, \mathbb{C})$. Then their *convolution* is define pointwise as:

$$(p * q)(g) = \sum_{g_i g_j = g} p(g_i) q(g_j) = \sum_{h \in G} p(h) q(h^{-1} g)$$

Lemma 53.4.4: Isomorphism With Convolution

Let G be a finite abelian group. Then:

$$(\mathbb{C}[G], +, \cdot) \cong_{k\text{-}\mathbf{Alg}} (\mathrm{Hom}_{\mathbf{Top}}(G, \mathbb{C}), +, *)$$

where the left hand side is the usual group ring and the right hand side has convolution as multiplication

***Proof*:**
refer to the above calculations.

Now, $(\mathrm{Hom}_{\mathbf{Top}}(G, \mathbb{C}), +, *)$ is a $|G|$-dimensional $\mathbb{C}$ vector-space with the natural basis given by the indicator function:

$$\delta_h(g) = \begin{cases} 1 & g = h \\ 0 & g \neq h \end{cases}$$

Namely for any $f : G \to \mathbb{C}$ there is a unique decomposition given the functionals defined by each $g \in G$:

$$f = \sum_{h \in G} f(h)\delta_h \qquad \text{so that} \qquad f(g) = \sum_{h \in G} f(h)\delta_h(g) = f(g) \cdot 1 = f(g)$$

The key feature of this basis is that it acts like evaluation when taking the inner product. Define:

$$\langle f, g \rangle = \sum_{h \in G} f(h)\overline{g(h)}$$

Then $\{\delta_g\}_{g \in G}$ is an orthogonal basis and:

$$\langle f, \delta_{g_i} \rangle = f(g_i) \cdot \overline{\delta_{g_i}(g_i)} = f(g_i) \cdot 1 = f(g_i) = a_i$$

This is a special case of the Riesz representation theorem: every linear functional here is inner-product representable using these δ_{g_i}. This inner product can be thought as expressly chosen so that the collection δ_{g_i} forms an orthonormal basis, or conversely given this standard inner product, the indicator function will always be an orthonormal basis. By Pontryagin duality, $G \cong \hat{G}$, but this group isomorphism does not *a-priori* preserve the orthogonality of the basis given by G. The key insight of Fourier analysis is that it does!

Proposition 53.4.5: Character Form Orthogonal Basis: Fourier Basis

The characters $\widehat{G}$ form an orthogonal basis for $\mathrm{Hom}_{\mathbf{Top}}(G, \mathbb{C})$. Normalizing by $1/\sqrt{|G|}$ forms an orthonormal basis.

***Proof*:**
If $i = j$:

$$\begin{aligned}\langle \chi_i, \chi_i \rangle &= \sum_{g \in G} \chi_i(g)\overline{\chi_i(g)} \\ &= \sum_{g \in G} |\chi_i(g)|^2 \\ &= \sum_{g \in G} 1^2 \\ &= |G|\end{aligned}$$

if $i \neq j$, Then as $\overline{\chi_j} = \chi_j^{-1}$ (as we are mapping to the unit circle), we get that $\chi' = \chi_i \chi_j^{-1}$ is a non-trivial character, so there is some h such that $\chi'(h) \neq 1$. Let

$$\langle \chi_i, \chi_j \rangle = S = \sum_{g \in G} \chi'(g)$$

Then:

$$\chi'(h)S = \chi'(h) \sum_{g \in G} \chi'(g) = \sum_{g \in G} \chi'(h)\chi'(g) = \sum_{g \in G} \chi'(h + g)$$

As $h + G = G$, this is just a re-arranging of terms, hence:

$$\chi'(h)S = \sum_{g \in G} \chi'(g) = S \qquad \text{so} \qquad S(\chi'(h) - 1) = 0$$

Now, by assumption $\chi'(h) \neq 1$, so $\chi'(h) - 1 \neq 0$, so as $\mathbb{C}$ is an integral domain it must be that $S = 0$, implying:

$$\langle \chi_i, \chi_j \rangle = 0$$

as we sought to show.

This inner product can be thought of as extracting the "phase" information.

Example 53.4.6: Phase-Space Basis
Let $G = \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z} = \{(0,0), (1,0), (0,1), (1,1)\}$. A character $\chi : G \to \mathbb{C}^*$ is determined by its values on the generators $g_1 = (1,0)$ and $g_2 = (0,1)$. Since both generators have order 2, we must have $\chi(g_1)^2 = \chi(g_1^2) = \chi(0,0) = 1$ and likewise for g_2. Thus, the character's values on the generators must be either 1 or -1. This gives $2 \times 2 = 4$ possible characters, which we can index by pairs $(k_1, k_2) \in \{0,1\}^2$.

The four characters χ_{k_1,k_2} are defined by their values on the generators:

$$\chi_{k_1,k_2}(1,0) = (-1)^{k_1} \qquad \text{and} \qquad \chi_{k_1,k_2}(0,1) = (-1)^{k_2}$$

From this, the value for any element $(a,b) \in G$ is $\chi_{k_1,k_2}(a,b) = (-1)^{k_1 a + k_2 b}$.

This gives the following character table, which explicitly lists the functions that form our new "basis":

$\hat{G}$	**(0,0)**	**(1,0)**	**(0,1)**	**(1,1)**
χ_{00}	1	1	1	1
χ_{10}	1	-1	1	-1
χ_{01}	1	1	-1	-1
χ_{11}	1	-1	-1	1

We can verify the orthogonality relation from proposition 53.4.5 directly. Let us compute $\langle \chi_{10}, \chi_{11} \rangle$:

$$\begin{aligned}\langle \chi_{10}, \chi_{11} \rangle &= \sum_{g \in G} \chi_{10}(g)\overline{\chi_{11}(g)} \\ &= \chi_{10}(0,0)\chi_{11}(0,0) + \chi_{10}(1,0)\chi_{11}(1,0) + \chi_{10}(0,1)\chi_{11}(0,1) + \chi_{10}(1,1)\chi_{11}(1,1) \\ &= (1)(1) + (-1)(-1) + (1)(-1) + (-1)(1) \\ &= 1 + 1 - 1 - 1 = 0\end{aligned}$$

Using the above example, let us change perspectives on how to interpret this change of basis. The function $f : G \to \mathbb{C}$ takes the "points" $g \in G$ and maps them (continuously) to the complex numbers, and those points are evaluated by the Riesz representation of f against the indicator functions that form an orthonormal basis. But we have just found another orthonormal basis for $\mathrm{Hom}_{\mathbf{Top}}(G, \mathbb{C})$ against which we can take the inner product. We may thus think of the characters $\chi \in \hat{G}$ as points in their own right: points carrying "phase" information rather than the "location" information that the indicator functions carry. This motivates the following definition:

Definition 53.4.7: Fourier Transform (Gelfand Transform)

Let G be a finite abelian group with the discrete topology. Then the *Fourier transform* is:

$$\widehat{(-)} : \mathrm{Hom}_{\mathbf{Top}}(G, \mathbb{C}) \mapsto \mathrm{Hom}_{\mathbf{Top}}(\hat{G}, \mathbb{C}) \qquad f \mapsto (\chi \mapsto \langle f, \chi \rangle)$$

where the image of f is often written as:

$$\hat{f}(\chi) = \langle f, \chi \rangle = \sum_{g \in G} f(g)\overline{\chi(g)}$$

The name *Gelfand transform* comes from the fact that the Fourier transform is traditionally about real periodic functions, or Legesgue measurable functions. It wasn't until later where mathematicians like Folland and Gelfand generalized it to the context of locally compact abelian groups, where the above is the special case where G is finite.

Now, certainly as vector spaces $\mathrm{Hom}_{\mathbf{Top}}(G, \mathbb{C}) \cong_{\mathbb{C}} \mathrm{Hom}_{\mathbf{Top}}(\hat{G}, \mathbb{C})$ mapping:

$$(g \mapsto \langle f, \delta_g \rangle) \mapsto (\chi \mapsto \langle f, \chi \rangle)$$

But does this map preserve the inner product, and more so can the convolution product of $\mathrm{Hom}_{\mathbf{Top}}(G, \mathbb{C})$ be preserved given the appropriate choice of product on $\mathrm{Hom}_{\mathbf{Top}}(\hat{G}, \mathbb{C})$? The answer is yes, and can be thought of as the corner-stone result of harmonic analysis on discrete abelian groups:

Theorem 53.4.8: Fourier Reciprocity

The map given in definition 53.4.7 is a unitary $\mathbb{C}$-algebra isomorphism,

$$(\mathrm{Hom}_{\mathbf{Top}}(G,\mathbb{C}),+,*) \cong (\mathrm{Hom}_{\mathbf{Top}}(\hat{G},\mathbb{C}),+,\cdot)$$

with $\cdot$ in $(\mathrm{Hom}_{\mathbf{Top}}(\hat{G},\mathbb{C}),+,\cdot)$ being point-wise multiplication.

Recall that unitary means $\langle \hat{f},\hat{g}\rangle = \langle f,g\rangle$. In a conventional real analysis course:

- the $\mathbb{C}$-algebra homomorphism is called the *Convolution Theorem*,
- the unitary property is called *Plancherel's Theorem*,
- invertibility (i.e. bijectivity) is called the *Fourier Inversion Theorem*.

***Proof*:**

1. *1. Linearity:* Let $f,h \in \mathrm{Hom}_{\mathbf{Top}}(G,\mathbb{C})$ and $a,b\in\mathbb{C}$.

$$\begin{aligned}
\widehat{af+bh}(\chi) &= \langle af+bh,\chi\rangle && \text{by definition}\\
&= \sum_{g\in G}(af+bh)(g)\overline{\chi(g)}\\
&= a\sum_{g\in G} f(g)\overline{\chi(g)} + b\sum_{g\in G} h(g)\overline{\chi(g)} && \text{by linearity of sums}\\
&= a\langle f,\chi\rangle + b\langle h,\chi\rangle\\
&= a\hat{f}(\chi)+b\hat{h}(\chi)
\end{aligned}$$

Since this holds for all $\chi\in\hat{G}$, the transform is a linear map.

2. $\mathbb{C}$-algebra Homomorphism (The Convolution Theorem): We must show that the Fourier transform converts convolution into pointwise multiplication.

$$\begin{aligned}
\widehat{f*h}(\chi) &= \langle f*h,\chi\rangle\\
&= \sum_{k\in G}(f*h)(k)\overline{\chi(k)} && \text{by definition of inner product}\\
&= \sum_{k\in G}\left(\sum_{g\in G} f(g)h(g^{-1}k)\right)\overline{\chi(k)} && \text{by definition of convolution}\\
&= \sum_{g\in G} f(g)\sum_{k\in G} h(g^{-1}k)\overline{\chi(k)} && \text{swapping order of summation}
\end{aligned}$$

Let $\ell = g^{-1}k$, so $k=g\ell$. As k runs over G, so does ℓ. The inner sum becomes:

$$\sum_{\ell\in G} h(\ell)\overline{\chi(g\ell)} = \sum_{\ell\in G} h(\ell)\overline{\chi(g)\chi(\ell)} = \overline{\chi(g)}\sum_{\ell\in G} h(\ell)\overline{\chi(\ell)} = \overline{\chi(g)}\hat{h}(\chi)$$

Substituting this back into the main expression:

$$\begin{aligned}\widehat{f*h}(\chi) &= \sum_{g\in G} f(g)\left(\overline{\chi(g)}\hat{h}(\chi)\right)\\ &= \hat{h}(\chi)\sum_{g\in G} f(g)\overline{\chi(g)}\\ &= \hat{h}(\chi)\hat{f}(\chi)\end{aligned}$$

Thus, $\widehat{f*h} = \hat{f}\cdot\hat{h}$, proving the map is an algebra homomorphism.

3. Isomorphism (The Fourier Inversion Theorem): We show the map is invertible by constructing its inverse. Let $\mathcal{F}(f) = \hat{f}$. We claim the inverse is given by $\mathcal{F}^{-1}(\hat{f})(g) = \frac{1}{|G|}\sum_{\chi\in\hat{G}}\hat{f}(\chi)\chi(g)$.

$$\begin{aligned}\frac{1}{|G|}\sum_{\chi\in\hat{G}}\hat{f}(\chi)\chi(g) &= \frac{1}{|G|}\sum_{\chi\in\hat{G}}\left(\sum_{h\in G} f(h)\overline{\chi(h)}\right)\chi(g) && \text{by definition of } \hat{f}\\ &= \frac{1}{|G|}\sum_{h\in G} f(h)\sum_{\chi\in\hat{G}}\chi(g)\overline{\chi(h)} && \text{swapping order of summation}\\ &= \frac{1}{|G|}\sum_{h\in G} f(h)\sum_{\chi\in\hat{G}}\chi(gh^{-1})\end{aligned}$$

The inner sum is an example of the *Second Orthogonality Relation* and is a standard exercise the reader should do. In particular, the sum of all characters evaluated at a group element (giving all powers of a certain root of unity) is $|G|$ if $x = e$ and 0 otherwise. Thus $\sum_{\chi\in\hat{G}}\chi(gh^{-1}) = |G|\delta_{g,h}$.

$$\frac{1}{|G|}\sum_{h\in G} f(h)(|G|\delta_{g,h}) = \sum_{h\in G} f(h)\delta_{g,h} = f(g)$$

Since we can recover f uniquely from $\hat{f}$, the map is an isomorphism.

4. Unitary Property (Plancherel's Theorem): Note that here is where it is important for the inner product to be normalized. Let us calculate the inner product for the given definition.

$$\begin{aligned}
\langle \hat{f}, \hat{h} \rangle_{\hat{G}} &= \sum_{\chi \in \hat{G}} \hat{f}(\chi)\overline{\hat{h}(\chi)} \\
&= \sum_{\chi \in \hat{G}} \left(\sum_{g \in G} f(g)\overline{\chi(g)} \right) \overline{\left(\sum_{k \in G} h(k)\overline{\chi(k)} \right)} \\
&= \sum_{\chi \in \hat{G}} \left(\sum_{g \in G} f(g)\overline{\chi(g)} \right) \left(\sum_{k \in G} \overline{h(k)}\chi(k) \right) \\
&= \sum_{g \in G} \sum_{k \in G} f(g)\overline{h(k)} \sum_{\chi \in \hat{G}} \chi(k)\overline{\chi(g)} && \text{re-arranging terms} \\
&= \sum_{g \in G} \sum_{k \in G} f(g)\overline{h(k)}(|G|\delta_{g,k}) && \text{by Second Orthogonality} \\
&= |G| \sum_{g \in G} f(g)\overline{h(g)} \\
&= |G|\langle f, h \rangle_G
\end{aligned}$$

This proves *Plancherel's Theorem.* To make the map strictly unitary, one must normalize the transform. If we define the unitary transform $\mathcal{U}(f) = \frac{1}{\sqrt{|G|}}\hat{f}$, then it follows immediately that $\langle \mathcal{U}(f), \mathcal{U}(h) \rangle = \langle f, h \rangle$, completing the proof.

Example 53.4.9: Fourier Transform

Take $G = \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$ form example 53.4.6. Let us now compute the Fourier transform of a concrete function. Let $f = \delta_{(1,0)}$, so that $f(1,0) = 1$ and is zero elsewhere. This is a basis vector in the standard "position" basis. We find its coordinates in the character basis by computing $\hat{f}(\chi) = \langle f, \chi \rangle$ for each character.

$$\begin{aligned}
\hat{f}(\chi) = \langle \delta_{(1,0)}, \chi \rangle &= \sum_{g \in G} \delta_{(1,0)}(g)\overline{\chi(g)} \\
&= 1 \cdot \overline{\chi(1,0)} + 0 + 0 + 0 = \chi(1,0)
\end{aligned}$$

Using the character table, we can compute this value for each χ:

- $\hat{f}(\chi_{00}) = \chi_{00}(1,0) = 1$
- $\hat{f}(\chi_{10}) = \chi_{10}(1,0) = -1$
- $\hat{f}(\chi_{01}) = \chi_{01}(1,0) = 1$
- $\hat{f}(\chi_{11}) = \chi_{11}(1,0) = -1$

Thus, the original function $f = \delta_{(1,0)}$, which is a spike at one point in the "position" basis, has been transformed into a new function $\hat{f}$ on $\hat{G}$ with values $(1, -1, 1, -1)$.

Recall the inversion formula from the proof of theorem 53.4.8: $f(g) = \frac{1}{|G|}\sum_{\chi\in\hat{G}}\hat{f}(\chi)\chi(g)$ (here $|G| = 4$). Let's check the value at $g = (1, 0)$:

$$\begin{aligned}
f(1,0) &= \frac{1}{4}\left(\hat{f}(\chi_{00})\chi_{00}(1,0) + \hat{f}(\chi_{10})\chi_{10}(1,0) + \hat{f}(\chi_{01})\chi_{01}(1,0) + \hat{f}(\chi_{11})\chi_{11}(1,0)\right)\\
&= \frac{1}{4}\left((1)(1) + (-1)(-1) + (1)(1) + (-1)(-1)\right)\\
&= \frac{1}{4}(1+1+1+1) = 1
\end{aligned}$$

Now let's check the value at $g = (0, 1)$, where we expect 0:

$$\begin{aligned}
f(0,1) &= \frac{1}{4}\left(\hat{f}(\chi_{00})\chi_{00}(0,1) + \hat{f}(\chi_{10})\chi_{10}(0,1) + \hat{f}(\chi_{01})\chi_{01}(0,1) + \hat{f}(\chi_{11})\chi_{11}(0,1)\right)\\
&= \frac{1}{4}\left((1)(1) + (-1)(1) + (1)(-1) + (-1)(-1)\right)\\
&= \frac{1}{4}(1-1-1+1) = 0
\end{aligned}$$

Let us now connect this to Representation Theory. So far, we have established that for a finite abelian group G, the Fourier transform is a unitary $\mathbb{C}$-algebra isomorphism between the algebra of functions on G with convolution, $(\mathrm{Hom}_{\mathbf{Top}}(G,\mathbb{C}), *)$, and the algebra of functions on the dual group $\hat{G}$ with pointwise multiplication. The representation theory perspective shows that Fourier analysis is the simplest case of a Artin-Wedderburn decomposition. The key is to view the $\mathbb{C}$-group algebra $\mathbb{C}[G]$ with the natural G-action given by left multiplication (convolution), namely $\rho : G \to \mathrm{GL}(\mathbb{C}[G])$ given by:

$$\rho(g) : f \mapsto g * f$$

From this viewpoint, the study of the $\mathbb{C}$-algebra $(\mathbb{C}[G], +, *)$ and the $\mathbb{C}[G]$-module $\mathbb{C}[G]$ is the same thing that is the study of the regular representation of G. Now, this representation can be broken down into the direct sum of 1-dimensional irreducible representations. Namely, for a finite abelian group, the set of irreducible representations is exactly the dual group: $\mathrm{Irr}(G) = \widehat{G}$ and so we get:

$$\mathbb{C}[G] \cong \bigoplus_{\chi\in\widehat{G}} V_\chi$$

where each V_χ is a 1-dimensional subspace corresponding to the character χ.

This decomposition is made concrete by the Fourier transform. The Fourier Inversion Formula, $f = \frac{1}{|G|}\sum_{\chi\in\hat{G}}\hat{f}(\chi)\chi$, is the literal expression of this decomposition. It writes the function f as a sum of components, where each term $\hat{f}(\chi)\chi$ is the projection of f onto the 1-dimensional irreducible subspace V_χ. Then the fourier coefficient $\hat{f}(\chi)$ is the coordinate of f in the irreducible component corresponding to χ.

Note that we can define projection functions $p_\chi = \frac{1}{|G|}\sum_{g\in G}\overline{\chi(g)}g$ that we can construct in $\mathbb{C}[G]$ that project onto the irreducible subspace V_χ. Convolving any function f with p_χ annihilates all other "frequency components", leaving the part of f that lies in V_χ.

Hence, the Fourier basis *the* unique basis that diagonalizes the natural action of the group on its own algebra of functions.

If G is a *non-abelian* finite group, the concept of a dual group $\hat{G}$ and a simple transform onto it ceases to be useful (it is now a set with no natural group structure). However, the representation theory perspective remains: the generalized Fourier transform (given by the *Peter-Weyl Theorem*) decomposes a function f on the group into a sum of *matrix coefficients* corresponding to these higher-dimensional matrix representations.

53.4.1 Generalization to Locally Compact Abelian Groups

What happens when we wish to analyze functions on abelian groups like $(\mathbb{R}, +)$, $(\mathbb{Z}, +)$, or the circle group $(S^1, \cdot)$? These are examples of *Locally Compact Abelian (LCA)* groups. Moving from the finite to the infinite setting requires us to replace our finite sums with integrals, but to integrate on an arbitrary group, we first need a notion of "volume" or "measure". The shape of the answer is already visible in the finite case: read the delta functions δ_h as *measures*, and every f becomes a sum of weighted point masses. What we want, then, is a measure on the group that the group operation cannot see. The essential new ingredient is the *Haar measure*. For any LCA group G, there exists a measure μ that gives a consistent way to integrate functions on the group.

Definition 53.4.10: Haar Measure

A *Haar measure* on an LCA group G is a measure μ on the Borel σ-algebra of G that is left-translation-invariant. That is, for any measurable set $E \subseteq G$ and any element $g \in G$:

$$\mu(gE) = \mu(E)$$

This measure is unique up to a positive scaling constant. For abelian groups, it is also right-translation-invariant.

See [22] for details on how to construct such a measure on any LCA that is unique up to scaling. This measure is the direct generalization of taking the order of the group.

Example 53.4.11: Haar Measure

1. For a finite group with the discrete topology, the Haar measure is the counting measure, and $\mu(G) = |G|$.
2. For $(\mathbb{R}, +)$, the Haar measure is the standard Lebesgue measure dx.
3. For $(S^1, \cdot)$, the Haar measure can be taken as $d\theta$.
4. For $(\mathbb{Z}, +)$, the Haar measure is again the counting measure.

Note that in the infinite case we can no longer represent points by indicator functions; the *Dirac delta distributions* take on that role instead. See [42] for the theory of distributions.

With the Haar measure, we can redefine our main objects of study by replacing every sum over G with an integral with respect to $d\mu$.

1. *The Group Algebra $L^1(G)$:* The space of all functions becomes too large. The natural replacement for the convolution algebra $\mathbb{C}[G]$ is the space of absolutely integrable functions, $L^1(G, d\mu)$. For

two functions $f, h \in L^1(G)$, their convolution is defined by the integral:

$$(f * h)(g) = \int_G f(x)h(x^{-1}g)d\mu(x)$$

This space $(L^1(G), +, *)$ forms a Banach algebra.

2. *The Dual Group $\widehat{G}$:* It was proven that the correct codomain should be S^1. The dual group is the set of all *continuous* group homomorphisms into S^1.

$$\widehat{G} = \mathrm{Hom}_{\mathbf{Top}}(G, S^1)$$

The dual group $\widehat{G}$ is itself an LCA group. It is crucial to note that, unlike the finite case, it is rarely true that $G \cong \widehat{G}$. For example:

$$\widehat{\mathbb{R}} \cong \mathbb{R}, \qquad \widehat{\mathbb{Z}} \cong S^1, \qquad \widehat{S^1} \cong \mathbb{Z}$$

This highlights why it is essential to treat $\hat{G}$ as a distinct space. The theory that governs this relationship is *Pontryagin Duality*, which states that the double-dual is canonically isomorphic to the original group: $\widehat{\widehat{G}} \cong G$.

3. *The Fourier Transform:* The definition of the Fourier transform is the direct analogue of the finite case, replacing the sum with an integral. For a function $f \in L^1(G) \cap L^2(G)$, its Fourier transform is a function $\hat{f}$ on the dual group $\hat{G}$:

$$\hat{f}(\chi) = \int_G f(g)\overline{\chi(g)}d\mu(g)$$

Similarly, the inner product that defines the Hilbert space structure on $L^2(G)$ is:

$$\langle f, h \rangle = \int_G f(g)\overline{h(g)}d\mu(g)$$

With these definitions, the entire suite of theorems proven in the finite case carries over:

Theorem 53.4.12: Fourier Reciprocity on LCA Groups

Let G be an LCA group with Haar measure μ. Then there exists a unique Haar measure ν on the dual group $\widehat{G}$ such that the Fourier transform, appropriately normalized, is a unitary $\mathbb{C}$-algebra isomorphism. The main results are:

1. *The Fourier Inversion Theorem:* For a sufficiently well-behaved function f (e.g., $f \in L^1 \cap L^2$ such that $\hat{f} \in L^1$), the original function can be recovered by an integral over the dual group:
$$f(g) = \int_{\widehat{G}} \hat{f}(\chi)\chi(g)d\nu(\chi)$$
2. *Plancherel's Theorem:* The Fourier transform extends uniquely from $L^1 \cap L^2$ to a unitary isomorphism of Hilbert spaces, $\mathcal{F} : L^2(G, \mu) \to L^2(\widehat{G}, \nu)$. This means it preserves the inner product:
$$\langle \hat{f}, \hat{h} \rangle_{L^2(\widehat{G})} = \langle f, h \rangle_{L^2(G)}$$
3. *The Convolution Theorem:* For $f, h \in L^1(G)$, the Fourier transform maps convolution to pointwise multiplication:
$$\widehat{f * h}(\chi) = \hat{f}(\chi) \cdot \hat{h}(\chi)$$

***Proof*:**
This is the Pontryagin duality theorem; a full proof via the Gelfand transform belongs to the harmonic analysis of locally compact abelian groups and is not reproduced in this starred section.

The correspondence between the finite and general LCA settings can be summarized as follows:

Finite Abelian Case	**LCA Case**
Group G	LCA Group G
Order $\lvert G\rvert$	Haar Measure μ
Group Algebra $\mathbb{C}[G]$	Banach Algebra $L^1(G, \mu)$
Hilbert Space $\mathrm{Hom}(G, \mathbb{C})$	Hilbert Space $L^2(G, \mu)$
Summation $\sum_{g \in G}$	Integration $\int_G d\mu(g)$
Dual Group $\widehat{G} = \mathrm{Hom}(G, \mathbb{C}^\times)$	Dual Group $\widehat{G} = \mathrm{Hom}_{\mathbf{Top}}(G, S^1)$
Inner Product $\sum f(g)\overline{h(g)}$	Inner Product $\int f(g)\overline{h(g)}d\mu(g)$
Fourier Transform $\sum f(g)\overline{\chi(g)}$	Fourier Transform $\int f(g)\overline{\chi(g)}d\mu(g)$

Further interesting facts can be found here: https://mathoverflow.net/questions/37021/is-fourier-analysis-a-special-case-of-representation-theory-or-an-analogue

53.4.2 The Algebraic-Geometric Meaning of the Dual Group

In algebraic geometry, a fundamental principle is that the geometry of a space can be recovered from its algebra of functions. For an affine space k^n, the corresponding algebra is the polynomial ring $k[x_1, \dots, x_n]$. The points of the space correspond to the maximal ideals of this algebra. Specifically, for any point $p = (a_1, \dots, a_n) \in k^n$, the evaluation map:
$$\mathrm{ev}_p : k[x_1, \dots, x_n] \to k \qquad f \mapsto f(p)$$

is an algebra homomorphism whose kernel, $m_p = (x_1 - a_1, \ldots, x_n - a_n)$, is a maximal ideal. The quotient $k[x_1, \ldots, x_n]/m_p \cong k$ recovers the value of the function at that point.

We have established a similar setup for a finite abelian group G, with the convolution algebra $(\mathrm{Hom}_{\mathbf{Top}}(G, \mathbb{C}), *, +)$ taking the role of the function algebra. A natural question arises: can we recover the "points" of our space from the maximal ideals of this algebra? Let's first attempt a direct analogy. We might assume the points are the elements of the group G itself. For a point $g_i \in G$, we can define the point-evaluation map, which is a linear map:

$$\mathrm{ev}_{g_i} : \mathrm{Hom}_{\mathbf{Top}}(G, \mathbb{C}) \to \mathbb{C} \qquad f \mapsto f(g_i)$$

The kernel of this map is the vector subspace of functions that are zero at g_i. However, for this subspace to be an ideal, it must be closed under convolution. Let $f \in \ker(\mathrm{ev}_{g_i})$ (so $f(g_i) = 0$) and let h be any function in our algebra. Consider their convolution:

$$(h * f)(g_i) = \sum_{k \in G} h(k) f(k^{-1} g_i)$$

Just because $f(g_i) = 0$ does not imply that the sum on the right-hand side is zero. The value of f at other points, $f(k^{-1} g_i)$, can be non-zero. Therefore, the kernel of the point-evaluation map is *not* an ideal with respect to convolution. This implies the points of G do not correspond to the maximal ideals of its convolution algebra.

The key insight is that the correct "geometric points" for the convolution algebra are not the elements of G, but the characters of the dual group $\widehat{G}$.

Proposition 53.4.13: Characters and Maximal Ideals

Let G be a finite abelian group. The maximal ideals of the convolution algebra $(\mathrm{Hom}_{\mathbf{Top}}(G, \mathbb{C}), +, *)$ are in bijective correspondence with the characters $\chi \in \widehat{G}$.

***Proof*:**

For each character $\chi \in \widehat{G}$, define a Fourier evaluation map:

$$\widehat{\mathrm{ev}}_\chi : \mathrm{Hom}_{\mathbf{Top}}(G, \mathbb{C}) \to \mathbb{C} \qquad f \mapsto \hat{f}(\chi)$$

This map takes a function and returns its Fourier coefficient at the frequency χ. By 53.4.8 $\widehat{\mathrm{ev}}_\chi$ is a $\mathbb{C}$-algebra homomorphism. Since $\widehat{\mathrm{ev}}_\chi$ is a surjective algebra homomorphism onto a field ($\mathbb{C}$), its kernel, which we denote by I_χ, is a maximal ideal:

$$I_\chi = \ker(\widehat{\mathrm{ev}}_\chi) = \{f \in \mathrm{Hom}_{\mathbf{Top}}(G, \mathbb{C}) \big| \hat{f}(\chi) = 0\}$$

By the First Isomorphism Theorem for rings, we have:

$$\mathrm{Hom}_{\mathbf{Top}}(G, \mathbb{C})/I_\chi \cong \mathbb{C}$$

This shows that every character $\chi \in \widehat{G}$ defines a maximal ideal. Conversely, After applying the Fourier transform we get the usual multiplication and ring structure, and then an approach similar to the Gelfand-Kolmogorov theorem gives the desired result.

For the infinite case, the maximal ideals of the Banach algebra $L^1(G, \mu)$ arise in this same way from homomorphisms into $\mathbb{C}$ (see the *Gelfand-Mazur Theorem*).

Thus, with the convolution algebra $\mathbb{C}[G]$ as our primary object, the natural geometric space that emerges is not the group G itself, but its Pontryagin dual, $\widehat{G}$. The Fourier transform is the mechanism that allows us to evaluate the functions of our algebra at the points of this "true" underlying space. This perspective, generalized by Gelfand, is the foundation of non-commutative geometry.

Concepts	**Classical Algebraic Geometry**
Algebra	Polynomial Ring $(k[x_1, \ldots, x_n], \cdot)$
Product	Polynomial Multiplication
Geometric Space	Affine Space k^n
A "Point" p	A tuple $(a_1, \ldots, a_n) \in k^n$
Evaluation Map	$\mathrm{ev}_p : f \mapsto f(p)$
Maximal Ideal for p	$\ker(\mathrm{ev}_p) = (x_i - a_i)$

Concepts	**Harmonic Analysis on G**
Algebra	Group Algebra $(\mathbb{C}[G], *)$
Product	Convolution
Geometric Space	The Dual Group $\widehat{G}$
A "Point" p	A character $\chi \in \widehat{G}$
Evaluation Map	Fourier "Evaluation" $\widehat{\mathrm{ev}}_\chi : f \mapsto \hat{f}(\chi)$
Maximal Ideal for p	$\ker(\widehat{\mathrm{ev}}_\chi) = I_\chi$

54

Quiver Representations and Path Algebras

(Currently, this chapter really focuses on Quiver theory. I will put my exploration of Lie Algebra Representations here)

(This motivation is if you know some differential geometry) Quiver's have their origin in the classification of Lie groups. A Lie group is a group $(G, \cdot)$ G is a *smooth manifold* and $\cdot : G \times G \to G$ is a *smooth map*. The classification of these types of groups (in particular, the continuous ones, since discrete lie groups is just the classification of all groups) is a problem completed by Killing. Out of his classification, there came out 5 exotic Lie groups lying outside of the other classifications. Why did these Lie groups come up? The answer of this question might come from a change in perspective: Any Lie group has a Lie Algebra – the tangent space at the identity of the group which also has an extra operation called the *lie braked* which allows to capture in some of the smooth properties of G into T_eG. These Lie brackets allow us to define a linear map $\text{ad}_x : T_eG \to \mathbb{C}$, $\text{ad}_x(y) = [x, y]$. The image of these operations are called the *ideals* of a lie algebra (not unlike rings, since an ideal $I \subseteq L$ is the set of all elements where for all $x \in L$ and $a \in I$, $[x, a] \in I$). Using this fact, we can ask for the classification of *simple lie algebras*. As an example of one:

$$\text{sl}(3, \mathbb{C}) = \left\{ \begin{matrix} a & b & c \\ d & e & f \\ g & h & i \end{matrix} \middle| \, a + e + i = 0 \right\}$$

which is closed under Lie braked $[A, B] = AB - BA$ (but not closed under multiplication). Now, since these are linear maps, they can be diagonalized. If we take $h_1, ..., h_k \in \text{sl}(3, \mathbb{C})$ to be diagonal elements (elements that are zero everywhere except possibly the diagonal), then $\text{ad}_{h_1}, ..., \text{ad}_{h_k}$ are all simultaneously diagonalizable. This leads us to be able to define *root systems*, which allows for *root decomposition* and *root systems*. This leads to two important theorems for classification of *complex lie algebras*:

(Cartan's Thorems)

1. Up to isomorphism, there is just one complex semisimple Lie algebra for each root system
2. With 5 exceptions, every finite dimensional semisimple Lie algebra over $\mathbb{C}$ is isomorphic to one of the classical Lie Algebras:

$$\mathrm{sl}(n, \mathbb{C}) \quad \mathrm{so}(n, \mathbb{C}) \quad \mathrm{sp}(2n, \mathbb{C})$$

and 5 excpetional Lie algebras, labeled: e_6, e_7, e_8, f_4, g_2

54.1 Quiver's and their Decomposition

The following is just an accumulation of common terms in graph theory that we will need. The one notable exception is that we will rename oriented graphs to *quivers*

Definition 54.1.1: Quiver

A *quiver* Q is an oriented graph (V, E, s, t) where V represents the vertices (the vertex set), E is the set of edges (the "arrows") and $s : E \to V$ with $t : E \to V$ represents the orientation (s for source, t for terget), that is for any $e \in E$, if we have:

$$\overset{i}{\bullet} \xrightarrow{e} \overset{j}{\bullet}$$

then $s(e) = i$ and $t(e) = j$.

Usually, we will abuse notation say $Q = (V, E)$ is a quiver and ommit s and t. Most of the time, we will draw quivers instead of explicitely define them through a 4-tuple or 2-tuple

Example 54.1.2: Quivers

1. If $V = \{1\}$, $E = \{e\}$, and $s(e) = t(e) = 1$, then we can draw this quiver as

$$\begin{tikzcd} 1 \arrow[loop above] \end{tikzcd}$$

2. If $V = \{1, 2\}$, $E = \{e\}$, $s(e) = 1$, and $t(e) = 2$, then we can draw this quiver as

$$1 \xrightarrow{e} 2$$

3. If $V = \{1, 2\}, E = \{e, f\}$, with $s(e) = 1$, $t(e) = 2$ and $s(f) = 2$, $t(f) = 1$, then we can draw this quiver as

$$\begin{tikzcd} 1 \arrow[r, bend left, "e"] & 2 \arrow[l, bend left, "f"] \end{tikzcd}$$

If you are still uncomfortable, with quivers, try to come up with some as a challenge.

Definition 54.1.3: Path and Cycle

Let Q be a quiver.

1. A path γ is a sequence of edges
$$\gamma = e_k e_{k-1} \cdots e_1$$
where $t(e_i) = s(e_{i+1})$. for $1 \leq i \leq k-1$.
2. If γ is a path where $s(e_1) = a$ and $t(e_k) = b$, then we say that γ is a path from a to b, and we write $s(\gamma) = a$, $t(\gamma) = b$.
3. The number of number of edges in a path is called the *length* of he path.
4. If γ is a path and $s(\gamma) = t(\gamma)$, we call γ a *cycle*.
5. A cycle of length 1 is called a *loop*

Definition 54.1.4: Sink And Source

Let Q be a quiver

1. Let $e \in E$ and $i \in V$. We say that e *contains* i if $s(e) = i$ or $t(e) = i$. We say that e is *incoming at* i if $t(e) = i$ and that e is *outgoing at* i if $s(e) = i$
2. We say a vertex $i \in V$ is a *sink* if all edges containing i are incoming. Intuitvely, this can be visualized as:
$$\begin{tikzcd} {} \arrow[dr] & {} \arrow[d] & {} \arrow[dl] \\ {} \arrow[r] & i & {} \arrow[l] \\ {} \arrow[ur] & {} \arrow[u] & {} \arrow[ul] \end{tikzcd}$$
3. We say a vertex $i \in V$ is a *source* if all edges containing i are outgoing. Intuitvely, this can be visualized as:
$$\begin{tikzcd} {} & {} & {} \\ {} & i \arrow[ul] \arrow[u] \arrow[ur] \arrow[l] \arrow[r] \arrow[dl] \arrow[d] \arrow[dr] & {} \\ {} & {} & {} \end{tikzcd}$$
4. Let s be a sink or a source. Then define sQ to be a new quiver where all the arrows at s are reversed.

54.1.1 Quiver Representation and Varieties

Definition 54.1.5: Quiver Representation

Let Q be a quiver. Then a *quiver representation* V *of* Q consists of

1. a vector space $V(i)$ over a fixed field $\mathbb{F}$ for each vertex of Q
2. a linear transofmration $T_e : V(i) \to V(j)$ for each $e \in E$

Let $\text{Rep}_{\mathbb{F}}(Q)$, or $\text{Rep}(Q)$ be the collection of all quiver representations of Q.

Example 54.1.6: Quiver Representations

1. Let

$$1 \xrightarrow{e} 2$$

be a quiver. Then any linear transformation $T : V \to W$ is a representation of this quiver

2. Let

$$\begin{tikzcd} 1 \arrow[r, "e"] & 2 \arrow[loop above, "f"] \end{tikzcd}$$

be a quiver. Then we can take any linear transformation $T : V \to W$ and an identity map $\text{id} : W \to W$. Note that f need not be represented by the identity: any linear transformation from W to itself is valid.

3. Take

$$1 \xrightarrow{e} 2 \xrightarrow{f} 3$$

Then one class of possible representations of this quiver consits of vector-spaces $V(1)$, $V(2)$, $V(3)$ where the sequence is exact at $V(2)$. Another representation can have T_e and T_f be scalar multiplications. Yet another can be the set of vector spaces and linear transformations T_e and T_f where $T_e, T_f \neq \text{id}$ but $T_f \circ T_e = \text{id}$. There are many more possibilities

Since each vertex has a representation, we can capture a notion of dimension of a quiver representation

Definition 54.1.7: Quiver Dimension

Let Q be a quiver and V a representation of Q. Then the *dimension* or *quiver dimension* of V is the tuple that is denoted:

$$\dim_Q V = (\dim V(i))_{i \in I}$$

where each i is a vertex. If there are finitely many vertices, then we can biject them to natural numbers and have:

$$\dim_Q V = (\dim V(1), \dim V(2), ..., \dim V(n))$$

Definition 54.1.8: Quiver Subrepresentation

Let Q be a quiver and V be a quiver representation. Then a *subrepresentation* W of V consists of a subspace $W(i) \subseteq V(i)$ for each $i \in V$ and a linear transformations restricted to the new domain, $T_e|_{W(i)} = S_e$ for each $e \in E$, such that each S_e, $S_e(W_i) \subseteq W_j$

the last condition is so that each T_e is still well-defined. As an exercise, see if you can come up with quivers and subrepresentations (you can keep it simple and consider a quiver $\bullet \to \bullet$).

Definition 54.1.9: Quiver Morphisms

Let Q be a quiver and let V and V be representations of Q. Then we say that $\varphi : V \to W$ is a *quiver morphism* if for every edge e with representation $T_e : V(i) \to V(j)$ in V and representation $S_e : W(i) \to W(j)$ in W, there exists linear transformations $\varphi(i)$ and $\varphi(j)$ such that the following diagram commutes:

$$\begin{tikzcd} V(i) \arrow[r, "T_e"] \arrow[d, "\varphi(i)"] & V(j) \arrow[d, "\varphi(j)"] \\ W(i) \arrow[r, "S_e"] & W(j) \end{tikzcd}$$

collection of all quiver morphisms between V and W is denoted $\text{Hom}_{\mathbf{Q}}(V, W)$ or $\text{Hom}(V, W)$ if it is unambiguous in context.

Proposition 54.1.10: Structure Of $\text{Hom}(V, W)$

Let $V, W \in \text{Rep}(Q)$. Then $\text{Hom}(V, W)$ forms a vector-space structure over the base field $\mathbb{F}$.

***Proof*:**
left as an exercise

Example 54.1.11: Quiver Morphism

1. Let $V = \overset{\mathbb{F}^1}{\bullet} \xrightarrow{1} \overset{\mathbb{F}^1}{\bullet}$ and $W = \overset{\mathbb{F}^1}{\bullet} \xrightarrow{0} \overset{\{0\}}{\bullet}$ be two quiver representations. Then we can define a morphism:

$$\begin{tikzcd} V \arrow[d, "\varphi"] & \mathbb{F} \arrow[r, "1"] \arrow[d, "1"] & \mathbb{F} \arrow[d, "0"] \\ W & \mathbb{F} \arrow[r, "0"] & \{0\} \end{tikzcd}$$

Does there exist a nonzero morphism $\varphi : W \to V$?

2. Let $V = \overset{\mathbb{R}}{\bullet} \xrightarrow{1} \overset{\mathbb{R}}{\bullet} \xrightarrow{1} \overset{\mathbb{R}}{\bullet}$ and $W = \overset{\mathbb{R}^2}{\bullet} \xrightarrow{\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}} \overset{\mathbb{R}^2}{\bullet} \xrightarrow{\begin{pmatrix} 1 & 1 \end{pmatrix}} \overset{\mathbb{R}}{\bullet}$. Then: (commented to compile) is a quiver morphism. Does there exist a nonzero morphism $\varphi : W \to V$?

What we have technically defined is a *quiver representation morphism.* A quiver morphisms would usually be between two direct graphs where if $Q = (V, E, s, t)$ and $R = (V', E', s', t')$ are two quivers, then a quiver morphism would consit two functions $m_v : V \to V'$ and $m_e : E \to E'$ such that $m_v \circ s = s' \circ m_e$ and $m_v \circ t = t' \circ m_e$. This is ostensibly the same concept with different objects. All mosrphisms between Q and R would be represented as $\mathrm{Hom}_{\mathbf{Quiv}}(Q, R)$. Most of the time, we will care about the representation of quivers, and hence leaving the name of "quiver morphism" to representations. In either case, the collection of all quivers along with the just-defined quiver morphism defines a category **Quiv**, and the collection of all representations of a quiver $\mathrm{Rep}(Q)$ along with the previously-defined quiver morphism defines a category **Quiv(Q)** (you can check these as exercises)

The category **Quiv(Q)** has a a 0, namely the representation where all vertices are assigned the zero vector-sapces and the linear transformations are the identity. Since 0 exists, we can define kernels. Images also exist in this category:

Definition 54.1.12: Kernels And Images

Let $\varphi : V \to W$ be a quiver morphism. Then $\ker(\varphi)$ is the subrepresetnation of V consisting of $\ker(\varphi(i)) \subseteq V(i)$, and $\mathrm{im}(\varphi)$ is the subrepresentation of W consisting of $\mathrm{im}(V(i)) \subseteq W_i$

Note that $\ker(\varphi)$ and $\mathrm{im}(\varphi)$ are indeed subrepresentations: the fact that the vertices exist is a direct consequence of linear algebra. It remains to show that the diagram still commutes.

Definition 54.1.13: Quiver Isomorphisms

Let $V, W \in \mathrm{Rep}(Q)$. Then V is isomorphic to W if there exists a quiver morphism $\varphi : V \to W$ and $\psi : W \to V$. We denote two isomorphic quivers as $V \cong W$

Example 54.1.14: Quiver Isomorphism

In Jamail's notes. Some pretty cool ones. In parituclar, the classification of quiver representations of the form $\bullet \xrightarrow{e} \bullet$ with each vector-space being finite dimensional.

Next, we define the dual of sub-representations. Since for every subspace of a vector space $W \subseteq V$ we can define V/W, the notion of a quotient quiver is defined similarly

Definition 54.1.15: Quotient Quiver

Let $V \in \mathrm{Rep}(Q)$ and $W \subseteq V$ a subrepresentation. Then V/W is defined as a new quiver where every vertex of V is replaced by the quotient of $V(i)/W(i)$.

Proposition 54.1.16: 1st Isomorphism for Quivers

Let $V, W \in \mathrm{Rep}(Q)$ and $\varphi : V \to W$ be a quiver morphism. Then:

$$\mathrm{im}(\varphi) = V/\ker(\varphi)$$

***Proof*:**
left as an exercise

Definition 54.1.17: Product Quiver

Let $V, W \in \text{Rep}(Q)$. Define $V \oplus W \in \text{Rep}(Q)$ as

1. $(V \oplus W)(i) = V(i) \oplus W(i)$
2. $T_e : (V \oplus W)(i) \to (V \oplus W)(j)$ is given by $T_e = T_{e,v} \oplus T_{e,w}$ (recall that $T \oplus S(a,b) = (T(a), S(b))$)

Definition 54.1.18: Quiver Variety

Let $Q = (I, E, s, t)$ be a finite quiver. and $\underline{d} \in \mathbb{N}^n$. Then a *quiver variety* is:

$$\text{Rep}(\underline{d}, Q) := \prod_{e \in E} M_{d_j, d_i}(\mathbb{F})$$

where $i = s(e)$ and $j = t(e)$. Every point in $\text{Rep}(\underline{d}, Q)$ is a representation of Q with dimension vector $\underline{d}$.

A quiver variety let's us reduce a quiver where each vertex can have a different dimension (i.e. any possible total dimension) to only thinking about quivers with the same total dimension

Proposition 54.1.19: Action On Quiver Variety

Let $Q = (I, E, s, t)$ be a finite quiver with $I = \{1, ..., n\}$ and $E = \{e_1, ..., e_n\}$. Let $\underline{d} = (d_1, ..., d_n) \in \mathbb{N}^n$, and define the group $G(\underline{d}) = \prod_i GL_{d_i}(\mathbb{F})$. Then define the $G(\underline{d})$ action on $\text{Rep}(Q, \underline{d})$ as follows:

$$g = (\varphi_1, ..., \varphi_n)$$
$$(g \cdot V)_e = (\varphi_j A_e \varphi_i^{-1})_e$$

where $s(e) = i$ and $t(e) = j$.

***Proof*:**
If $g = (I)_e$ for each edge, then

$$(g \cdot V)_e = (I_j A_e I_i)_e = (A_e)_e$$

showing that $I \cdot V = V$. Associativity comes for free from assocaitivty of composition of linear maps.

I think the group in the theorme is called a *Weyl Group*.

Theorem 54.1.20: Isomorphism Equivalence

Let $\text{Rep}(Q, \underline{d})$ be a quiver variety and $G(\underline{d})$ be the group as defined in proposition 54.1.19. Then two representations $V, W \in \text{Rep}(Q, \underline{d})$ are isomorphic if and only if there exists a $g \in G(\underline{d})$ such that $g \cdot V$. That is, two quivers are isomorphic if they lie in the same orbit:

$$\{G(\underline{d})\text{-orbit of } \text{Rep}(Q, \underline{d})\} \rightleftarrows \{\text{iso classes of rep with dim } \underline{d}\}$$

***Proof*:**

As we've seen in example 54.1.14, two quiver represenations (of the same dimension) of $\bullet \xrightarrow{e} \bullet$ are equivalent if T_e and S_e are equivalent, i.e., their range is of the same dimension, i.e. $\varphi_j A_e \varphi_i^{-1} = B_e$ for two invertible matrices φ_i and φ_j. However, this is exactly the condition for the orbits of this group-actions.

54.1.2 Decomposing Quivers

Similarly to all other algebraic structures we've encountered, we have a notion of simple objects:

Definition 54.1.21: Simple Representation

Let Q be a quiver. Then $V \in \text{Rep}(Q)$ is called a *simple* or *irreducible* representation if the only subrepresentations of V are $\{0\}$ and V

The notion of simple objects is essentially trivial for vectorspaces since $V \cong \mathbb{F}^n$. In other words, every vector-space is decomposable into simple vector-spaces. The analogous notion of decomposability exists for quivers, however, they can be more complicated for quivers

Definition 54.1.22: Indecomposable Quiver

Let Q be a quiver. Then $V \in \text{Rep}(V)$ said to be *decomposable* if there exists non-trival subrepresentation $V_1, V_2 \subseteq V$ such that:

$$V \cong V_1 \oplus V_2$$

and is said to be *indecomposable* otherwise

Example 54.1.23: Simple Quivers and decomposability

Let $Q = \bullet \xrightarrow{e} \bullet$. Then:

$$0 \xrightarrow{\ 0\ } \mathbb{F} \qquad \mathbb{F} \xrightarrow{\ 0\ } 0$$

are both simple representations. Note that the left quivers is a subrepresentations of $V = \mathbb{F} \xrightarrow{\text{id}} \mathbb{F}$, and along with $\mathbb{F} \xrightarrow{\text{id}} \mathbb{F}$ and $0 \xrightarrow{0} 0$, $\mathbb{F} \xrightarrow{\text{id}} \mathbb{F}$ has 3 sub-quivers. Thus, V is not a simple quiver. However, V is *indecomposable*. To see this, notice that if V were decomposable, then we get:

$$\dim_Q(V_1 \oplus V_2) = \dim_Q(V_1) + \dim_Q(V_2)$$

and for the representation to be nontrival, we would have that one representation would have dimension $(1, 0)$, while the other has $(0, 1)$. However,V has no representation of dimension $(1, 0)$.

This means that the extension of quivers will be more elaborate than split exact sequences.

Theorem 54.1.24: Quiver Representation Decomposition

Let Q be a quiver and V be a representaiton where $\sum_i \dim V(i) < \infty$. Then there exists a finite set of indecomposable subrepesentations I_1, ..., I_n such that

$$V = I_1 \oplus I_2 \oplus \cdots \oplus I_n$$

***Proof*:**
Define the *total dimension* $\mathrm{Tdim}(V) = \sum_i \dim V(i)$. If $\mathrm{Tdim}(V) = 1$, then V must be only contain one vector-space $\mathbb{F}$ with all other vecor-spcaes being the trivial vector spaces. Assume for a moment that V has no oriented cycles. Then all maps from edges that are being the trivial map. Clearly, V is indecomposable, and so it is it's own decomposition.

On the other hand, if V has an oriented cycle, then without loss of generality we can consider

$$T = T_n \circ T_{n-1} \circ \cdots \circ T_1$$

so that we have $T : \mathbb{F} \to \mathbb{F}$. Since $\mathbb{F}$ has no non-trival subspace (being an indecomposable vector-space over $\mathbb{F}$), V is still indecomposable, and so is it's own decomposition.

Using strong induction, assume that all vector-spaces of total dimension $\mathrm{Tdim}(V) = n$ can be decomposed into a direct sum of indecomposable representations, and consider a representation of total dimension $n + 1$. If V is indecomposable, then we're done, so let's assume V is decomposableso that there exists two non-trival subrepresentations V_1, V_2 such that

$$V = V_1 \oplus V_2$$

Then since V_1, V_2 are both non-trivial, their total dimension must be strictly greater than and strictly less than that of V and nonzero. But then, by the inductive hypothesis, V_1 and V_2 can de decomposed into a finite direct sum of indecomposable modules. Hence:

$$V = V_1 \oplus V_2 = (I_{1,1} \oplus \cdots \oplus I_{1,n_1}) \oplus (I_{2,1} \oplus \cdots \oplus I_{2,n_2})$$

completing the proof.

Example 54.1.25: Decomposing Representations

1. Let $Q = \bullet \xrightarrow{e} \bullet$ be a quiver. Find all simple and indecomposable representations of Q. Using this knowledge, decompose any representation $V \xrightarrow{T_e} W$.

2. Let $Q = \bullet \xrightarrow{e} \bullet \xrightarrow{f} \bullet$. Find all simple and indecomposable representation of Q. Using this knoweldge, decompose any representation $V \xrightarrow{T_e} W \xrightarrow{T_f} L$.

3. Let $Q = \begin{tikzcd} \bullet \arrow[loop above] \end{tikzcd}$ be a quiver. Given $\mathbb{F}$ is algebraically closed, find all simple and indecomposable representations of Q (this is heavily reliant on Jordan canonical forms from chapter 14). Using this knowledge, decompose any representation $T : V \to V$.

54.1.3 Root Systems

Recall that if $(V, \langle \cdot, \cdot \rangle)$ ia an inner product space and $v \in V$, then

$$\frac{\langle w, v\rangle}{\langle v, v\rangle} v$$

is the projection of w onto v. Furthermore, define:

$$s_v(w) = w - 2\frac{\langle w, v\rangle}{\langle v, v\rangle} v$$

which reflects w across the hyperplane orthogonal to v.

Definition 54.1.26: Root System

Let V be an inner product space over $\mathbb{R}$. Then a *(reduced) root system* $R \subseteq V$ is a finite collection of non-zero vectors such that

1. $\operatorname{span}(R) = V$
2. For all $\alpha, \beta \in R$, $\eta_{\alpha,\beta} = \frac{2\langle \alpha, \beta\rangle}{\langle \beta, \beta\rangle} \in \mathbb{Z}$. In terms of angles, recall that $\theta = \cos^{-1}\left(\frac{\langle \alpha, \beta\rangle}{\|\alpha\|\|\beta\|}\right)$, so $n_{\alpha\beta} = \|\alpha\| \cos(\theta)$
3. For each $\alpha \in R$, the corresponding reflection $s_\alpha : V \to V$ has the property that for all $\beta \in R$, $s_\alpha(\beta) \in R$, i.e., R is symmetric under all α reflections.
4. If both $\alpha, c\alpha \in R$ then $c = \pm 1$ (this is true for reduced systems only)

Example 54.1.27: Root Systems

Let $V = \{(x, y, z) \in V \mid x + y + z = 0\}$, where V inherits the usual inner product from $\mathbb{R}^3$. Let

$$R = \{\pm(e_1 - e_2), \pm(e_2 - e_3), \pm(e_1, e_3)\} = \{\alpha, \beta, \gamma\}$$

Then R is a reduced root system. This set spans V because HERE.

Furthermore:

$$\eta_{\pm\alpha,\pm\beta} = \frac{2\langle \pm(e_1 - e_2), \pm(e_2 - e_3)\rangle}{\pm\langle (e_2 - e_3), \pm(e_2 - e_3)\rangle}$$

(finish rest later)

Include these

Notice that any root system can be simplified:

Definition 54.1.28: Polarization And Simple

Let R be a root sytem. Then if there exists a vector v such that $\langle v, \alpha \rangle \neq 0$ for all $v \in R$ then $R_+ = \{\alpha \in R \mid \langle v, \alpha \rangle > 0\}$ along with $R_+ = \{\alpha \in R \mid \langle v, \alpha \rangle < 0\}$ is called the *polarization of* R, and $R = R_+ \sqcup R_-$. The elements of R_+ are called *positive roots*. A positive root is called *simple* if it cannot be expressed as the sume of other positive roots.

Example 54.1.29: Simple Roots

homework, exercise 8 in Jimails notes

Theorem 54.1.30: Simple Root Decomposition

Let R, V be a (reduced) root system with polarization $R = R_+ \sqcup R_-$. Let Π denote the set of simple roots for this polarization. Then:

1. Every $\alpha \in R_+$ is a sum of simple roots
2. If $\alpha, \beta \in R_+$, then $\langle \alpha, \beta \rangle \leq 0$
3. The set Π of simple roots froms a basis for V

***Proof*:**

exercise 9

(words about root systems being isomorphic, which I'm guessing is a map from one vec sp from another s.t. all the properties of root systems are preserved)

Definition 54.1.31: Cartan Matrix

Let R, V be a (reduced) root system with polarization $R = R_+ \sqcup R_-$, let Π denote the set of simple roots for this polarization, and let $r = |\Pi|$. Then define *Cartan matrix* A to be the $r \times r$ matrix defined by:

$$A_{ij} = n_{\alpha_j \alpha_i} = \frac{2\langle \alpha_i, \alpha_j}{\alpha_i, \alpha_i}$$

Example 54.1.32: Cartan Matrix

exercise 10

(word to motivate the following visualization)

Definition 54.1.33: Dynkin Diagrams

Let R, V be a (reduced) root system with polarization $R = R_+ \sqcup R_-$, let Π denote the set of simple roots for this polarization, and identify Π with the set $\{1, ..., k\}$. Then we can construct a graph following these steps:

1. The vertices are given by simple roots Π (or equivalently, by indices $i \in \{1, ..., k\}$)
2. We join a vertex i and j by n_{ij} edges, which depends on the angle θ where

$$\theta = \frac{\pi}{2} \Rightarrow n_{ij} = 0 \qquad \theta = \frac{2\pi}{3} \Rightarrow n_{ij} = 1 \qquad \theta = \frac{3\pi}{4} \Rightarrow n_{ij} = 2 \qquad \theta = \frac{5\pi}{6} \Rightarrow n_{ij} = 3$$

3. Roots joined by a single edge have the same length. Roots joined by muliple edges have an arrowhead ">" on the edge pointing towards the root with sorther length (imagine this as an inequality between the length of the roots)

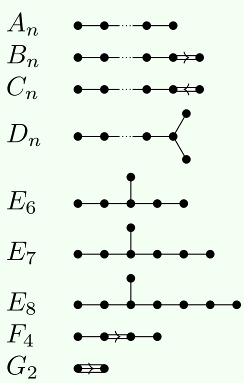
Example 54.1.34: Dynkin Diagrams

1. finding dynkin diagram for many root systems

We know proceed to classify root systems. As with other definition, if R is a root system and $R = R_1 \sqcup R_2$ where R_1, R_2 are proper subsets of R that are root systems, we say that R is reducible, and irreducible otherwise.

Theorem 54.1.35: Classification of Root Systems

Let R be a (reduced) irreducible root system. Then it's dynkin Diagram is one of the following:



***Proof*:**
here

(representing root systems via it's symmetries)

Definition 54.1.36: Weyl Group

Let $R \subseteq V$ be a root system. Let $W \subseteq \mathrm{GL}(V)$ be the group generated by the reflections s_α, $\alpha \in R$. Then W is called the *Weyl group* of R.

Since it is a group generated by reflections, every element acts linearly on V. Furthermore, it perserves lengths and angles, and so it is an element of $O(V)$ (definition 1.2.32)

Example 54.1.37: Weyl Group
p. 11 Jimail

Definition 54.1.38: Simple Reflection

Let $R \subseteq V$ be a root system with Weyl group W, and $R = R_+ \sqcup R_-$ is a polarizatoin with simple roots Π.
Then for each $\alpha_i \in \Pi$, the corresponding reflection $s_i = s_{\alpha_i} \in W$ is called a *simple reflection.*

Theorem 54.1.39: Weyl Groups Representing Root Systems

Let $R \subseteq V$ be a root system with Weyl group W, and $R = R_+ \sqcup R_-$ a polarization with simple roots Π. Then:

1. The simple roots s_i (for $i \in \Pi$) generate W
2. $W(\Pi) = R$. Hence, the root system can be recovered from the simple roots; we apply all Weyl group elements to Π to produce the full list of possible roots

***Proof*:**
exercise?

(reduced expressions and longest elements)

Definition 54.1.40: Length Of Weyl Group Elements

Let W be the Weyl group of a root system R, and Π the simple roots. Then the *length* of $w \in W$ is the shortest length of word for w as a product of simple reflections. We denote the length of w by $l(w)$.

In other words, if $w = s_{i_\ell} \cdots s_{i_1}$ and no shorts sequence exists, then $l(w) = \ell$.

Theorem 54.1.41: Generating Roots Using Weyl Group

Let W be the Weyl group of a root system R with polarization $R = R_+ \sqcup R_-$ and Π the corresponding simple roots.

1. There exists a unique element $w_0 \in W$ such that $\ell(w_0) = |R_+|$
2. The element w_0 has the property that $w_0(R_+) = R_-$
3. If $w = s_{i_k} \cdots s_{i_1}$ is an expression for w_0 of length $l(w_0)$, then
$$\begin{aligned} \gamma_1 &= \alpha_{i_1} \\ \gamma_2 &= s_{i_1}(\alpha_{i_2}) \\ &\vdots \\ \gamma_k &= s_{i_k} \cdots s_{i_1}(\alpha_{i_k}) \end{aligned}$$
is an enumeration of the positive roots
4. The elements γ_j defined above satisfy the propety that the roots γ_j, $s_{i_1}(\gamma_j)$, ..., $s_{i_1}(\gamma_j) \in R_+$ are positive, but the roots $s_{i_j} s_{i_{j-1}} \cdots s_{i_1}(\gamma_j) \in R_-$ are negative.

***Proof*:**
Jimail didn't prove – we don't have enough tools yet

Now that we have a better undersatnding of Weyl groups, we seek to link some of their structure to structure of simple roots. As we saw, the relations between simple roots can be captured via Dynkin Diagrams. We will seek for elements of the Weyl group which are "compatible" with the Dynkin diagrams. (for some reaosn), let Γ be a dynkin diagram of type A, D, or E (i.e. a Dynkin diagram with no multiple edges). Let R denote the corresponding root system. For any orientation Ω of Γ, Let (γ, Ω) represent the correspondign quiver. Recall that if i is a sink or source in Ω, we have an operator $s_i^{\pm}$ that reverses the arrows at i. Denote thie operation by

$$(\Gamma, \Omega) \mapsto (\Gamma, s_i^{\pm}\Omega)$$

where we would put a $+$ if it is a sink-to-source operation, and a $-$ if it is a source-to-sink operation

Definition 54.1.42: Adapted Word

Let Γ be a Dynkin diagram of type A, D, or E (i.e. a diagram with no multiple edges, or "simply laced"), W the corresponding Weyl group, and Ω an orientation of Γ. Let $w \in W$. Then the reduced expression $w = s_{i_k} \cdots s_{i_1}$ for w is *adapted for* Ω if

1. The verex i_1 is a sink for (Γ, Ω)
2. The verex i_2 is a sink for $(\Gamma, s^+_{i_1}\Omega)$

   $\vdots$

3. The verex i_k is a sink for $(\Gamma, s^+_{i_{k-1}} \cdots s^+_{i_1}\Omega)$

i.e. at each step, we can apply the operator $s^+_{i_j}$ to the quiver $(\Gamma, s^+_{i_{j-1}} \cdots s^+_{i_1}\Omega)$

Example 54.1.43: Adapted Dynkin Diagrams
p. 15 Jaimal

Theorem 54.1.44: Lusztig's Theorem

Let R be an A, D, or E root system with dynkin diagram Γ and Weyl group W. For any orientation Ω of Γ, there exists an expession for the longest element $w_0 \in W$ which is adapted for Ω

***Proof*:**
exercise 19

54.1.4 Gabriel's Theorem

In this section, we come back to Quiver's. Recall in example 54.1.25 it was asked to show how find the indecomposable quivers with underlying graphs of type A_1, A_2, and a cyclic quiver. The result was that A_2 and A_3 (with the given orientation) reduced to finitely many indecomposable representations. However, if there is a cycle, the quiver did not afford such a representation. A quiver which has finite decomposition is naturally much easier to study (TBD). The question then becomes:

> Which quivers Q can be decomposed into finitely many indecomposable representations?
> Can we classify all such quivers?

The answer of this question is *Gabriel's Theorem*; the goal of this section is to prove this result. Gabriel's Theorem is the key result linking the representation of quivers to Lie algebras, a pivotal tool in studying manifolds with group structures.

(a,b) To make the connection more precise, some terms need to be defined:

Definition 54.1.45: Collection Of Indecomposables

Let Q be a quiver, and let $\text{Rep}(Q)$ be it's representations. Then the set $\text{Ind}(Q) = \{[X] \mid X \in \text{Rep}(Q) \text{ is indecomposable }\}$ is the set of isomrophism classes of indecomposable quivers of $\text{Rep}(Q)$.
We say that a quiver is of *finite type* if $\text{Ind}(Q)$ is finite.

Before continuing with proving he result, it is useful to at least see how such a connection might even be hypothesized:

Example 54.1.46: Motivating Connection

Consider the quiver $\overset{1}{\bullet} \longrightarrow \overset{2}{\bullet} \longrightarrow \overset{3}{\bullet}$. The underlying graph is A_3, so let α_1, α_2, and α_3 denote the simple roots for the root system A_3. Using these, we can deduce the positive roots are:

$$\{\alpha_1, \alpha_2, \alpha_3, \alpha_1 + \alpha_2, \alpha_2 + \alpha_3, \alpha_1 + \alpha_2 + \alpha_3\}$$

We furthermore saw in example 54.1.25 that the indecomposable representations are:

$$\mathbb{F} \xrightarrow{0} 0 \xrightarrow{0} 0 \qquad \mathbb{F} \xrightarrow{1} \mathbb{F} \xrightarrow{0} 0$$

$$0 \xrightarrow{0} \mathbb{F} \xrightarrow{0} 0 \qquad 0 \xrightarrow{0} \mathbb{F} \xrightarrow{1} \mathbb{F}$$

$$0 \xrightarrow{0} 0 \xrightarrow{0} \mathbb{F} \qquad \mathbb{F} \xrightarrow{1} \mathbb{F} \xrightarrow{1} \mathbb{F}$$

Here is the key insight: notice we can correspond the *dimension vector* of these repesenations to the positive roots!

Theorem 54.1.47: Gabriel's Theorem

Let Q be a quiver. then Q is of finite type if and only if the underlying graph of Q is a Dynkin Diagram of type A, D, or E. In this case, the dimension vector:

$$\dim_Q : \text{Ind}(Q) \to R_+ \qquad [I] \mapsto \dim_Q(I)$$

bijects the indecomposable representations (up to isomorphism) and positive roots in the corresponding root system.

The proof will be relegated to page pageref:HERE

The theorem hints to us that indecomposable quivers somehow relate to the simple reflections of a Weyl group. However, a Dynkin Diagram is inherently unoriented, while Quivers are by definition an oriented graph. The first thing we need to do to fix is find the equivalent of a "reflection" for quivers.

The idea for the following definition is as follows: Take the quiver and the reflection

$$\overset{1}{\bullet} \xrightarrow{e} \overset{2}{\bullet} \qquad \overset{1}{\bullet} \xleftarrow{f} \overset{2}{\bullet}$$

Now, given a representation assigning $V(1)$, $V(2)$, and T_e, how should we assign values to the reflection? One way of doing it is by taking $T_e : V(1) \to V(2)$, and mapping it to $S_f : \ker(T_e) \to V(1)$ where $S_f(x) = 0$. Visually, we see that:

$$\left(\overset{V(1)}{\bullet} \xrightarrow{T_e} \overset{V(2)}{\bullet} \right) \mapsto \left(\overset{V(1)}{\bullet} \xleftarrow{S_f} \overset{\ker(T_e)}{\bullet} \right)$$

this map considers what happens at a sink. Another approach maps by considering the source:

$$\left(\overset{V(1)}{\bullet} \xrightarrow{T_e} \overset{V(2)}{\bullet} \right) \mapsto \left(\overset{\operatorname{coker}(T_e)}{\bullet} \xleftarrow{S_f} \overset{V(2)}{\bullet} \right)$$

hwere $S_f(x) = \overline{T_e(x)}$. Both of these asignements are in fact functorial, If we have a quiver morphism $\varphi : V \to W$ represented by:

$$\begin{tikzcd} V(1) \arrow[r, "T_e"] \arrow[d, "\varphi(1)"] & V(2) \arrow[d, "\varphi(2)"] \\ W(1) \arrow[r, "S_e"] & W(2) \end{tikzcd}$$

Then the appropriate morphim φ maps to is represented in these diagrams:

$$\begin{tikzcd} V(1) \arrow[d, "\varphi(1)"] & \ker(T_e) \arrow[l, "T_f|_{\ker(T_e)}"'] \arrow[d, "\varphi(1)|_{\ker(T_e)}"] \\ W(1) & \ker(S_e) \arrow[l, "S_f|_{\ker(S_e)}"'] \end{tikzcd} \qquad \begin{tikzcd} \operatorname{coker}(T_e) \arrow[d, "\overline{\varphi(2)}"] & V(2) \arrow[l, "\overline{T_f}"'] \arrow[d, "\varphi(2)"] \\ \operatorname{coker}(S_e) & W(2) \arrow[l, "\overline{S_f}"'] \end{tikzcd}$$

which both commute (where it was shown that $\varphi|_{\ker(T_e)}(\ker(T_e)) \subseteq \ker(S_e)$ when we showed the cateogry of quivers contains kernels). This will essentially represent how a reflection functor between two quivers will be defined – locally.

The different between these two functors is where there is a sink or a source. If i is a sink, then for any given quiver representatin $V \in \operatorname{Rep}(Q)$, let

$$\Psi_i : \bigoplus_{e:i \to j} V(j) \to V(i)$$

represent the collection of all linear transformation, where for each summand we have $T_e : V(i) \to V(j)$. If i is a source, then let:

$$\Phi_i : V(i) \to \bigoplus_{e:i \to j} V(j)$$

where each summand is mapped to T_e.

Definition 54.1.48: Reflection Functor

Let Q be a quiver. Define a functor from $\text{Rep}(Q)$ to $\text{Rep}(s_i^{\pm}Q)$ as follows:

1. For any $V \in \text{Rep}(Q)$, define $S(V)_i^+ \in \text{Rep}(s_i^+(Q))$ for a sink i, where at each vertex k,

$$S_i^+(V)(k) = \begin{cases} V(k) & i \neq k \\ \ker(\Psi_i) & i = k \end{cases}$$

and for each linear transformation T_e:

$$S_i^+(T_e) = \begin{cases} T_e & s(k) \neq i \\ \pi_k \circ \iota_i & \text{if } f : i \to k \text{ is in } s_i^+(Q) \end{cases}$$

where $\iota_i : \ker(\Psi) \to \bigoplus_{e:j\to i} V(j)$ is the inclusion map and $\pi_k : \bigoplus_{e:j\to i} V(j) \to V(k)$ is the projection map onto the kth component.

2. For any $V \in \text{Rep}(Q)$, define $S(V)_i^- \in \text{Rep}(s_i^n(Q))$ for a source i, where at each vertex k,

$$S_i^+(V)(k) = \begin{cases} V(k) & i \neq k \\ \text{coker}(\Phi_i) & i = k \end{cases}$$

and for each linear transformation T_e:

$$S_i^+(T_e) = \begin{cases} T_e & s(k) \neq i \\ q_k \circ \iota_i & \text{if } f : i \to k \text{ is in } s_i^+(Q) \end{cases}$$

where $\iota_i : V(k) \to \bigoplus_{e:j\to i} V(j)$ is the inclusion map and $q : \bigoplus_{e:j\to i} V(j) \to \text{coker}(\Phi_i)$ is the quotient map

Then these two functors, $S_i^{\pm} : \text{Rep}(Q) \to \text{Rep}(s_i^{\pm}Q)$ are called the *Bernstein-Gelfand-Ponomarev reflection functor*, or *reflection functor* for short.

Example 54.1.49: Reflection Functor

p. 70 Jaimal. The key idea is that these functors give different quiver representations, even if it on A_2 with the only orientation and doing S_1^+ then S_2^-.

A couple of properties of the functors needs to be established. For notational simplicity, let $\text{Rep}^{i,-}(Q)$ denote the full subcategory consisting of representations of Q for which the map

$$\Psi_i : \bigoplus_{e:j\to i} V(j) \to V(i)$$

is surjective, while $\text{Rep}^{i,+}(Q)$ denotes the full subcategory consisting of representations of Q for which the map

$$\Phi_i : V(i) \to \bigoplus_{e:i\to j} V(j)$$

is injective.

Proposition 54.1.50: Properties Of Reflection Functor

Let Q be a quiver.

1. If i is a sink, then for any $V \in \mathrm{Rep}(Q)$, $S_i^+ \in \mathrm{Rep}^{i,+}(s_i^+(Q))$ (similarly for source)
2. If i a sink, then for any $V \in \mathrm{Rep}(Q)$, $S_j^- S_i^+ \in \mathrm{Rep}^{i,-}(Q)$ (similarly for source)
3. The restriction of the reflection functors are inverse equivalences (adjoints?)
$$S_i^+ : \mathrm{Rep}^{i,-}(Q) \to \mathrm{Rep}^{i,+}(s_i^+ Q)$$
$$S_i^- : \mathrm{Rep}^{i,+}(Q) \to \mathrm{Rep}^{i,-}(s_i^- Q)$$
4. If V is simple at vertex i, then $S_i^{\pm}(V) = 0$.
5. If V is a non-zero indecomposable representation of Q that is not simple at i, then $S_i^{\pm}(V)$ is a nonzero indecomposable representation of $s_i^{\pm}(Q)$, and $V \cong S_j^{\mp} S_i^{\pm}(V)$.
6. $S_i^{\pm}(V)$ is a nonzero indecomposable representation of $s_i^{\pm}(V)$ if and only if $s_i(\dim_Q(V)) \geq 0$
7. If $V \in \mathrm{Rep}^{i,\pm}(Q)$ is a nonzero indecomposable representation, then $\dim_Q(S_i^{\pm}(V)) = s_i \dim_Q(V)$

***Proof*:**
exercise

Finally, we define a new object with purpose HERE (Apparently, these definitions can be given if we use some homological algebra. Perhaps revamp this after you do the homological algebra section, or keep it at this level with the mention of the version of greater generality for those who want to skip Hom Alg)

Definition 54.1.51: Root Lattice

Let Q be a quiver with vertex set I. Then if the underlying graph of Q is a dynkin diagram, $\mathbb{Z}^I$ is called the *root lattice*. For $(v_i)_{i\in I} \in \mathbb{Z}^I$, we say that $v \geq 0$ if $v_i \geq 0$ for all $i \in I$. Furthermore, we let $\mathbb{Z})+^I := \left\{v \in \mathbb{Z}^I \;\middle|\; v \geq 0\right\}$

(The following theorem seems to make more sense with some homological algebra, where the domain of the function should actually be the Grothendieck group of a $\mathrm{Rep}(Q)$) (Are objects in the following theorem representations?)

Theorem 54.1.52: Root Lattice Map

here exists a map $\dim_Q : \{\text{iso classes of Rep's}\} \to \mathbb{Z}_+^I$

***Proof*:**
not proved

Definition 54.1.53: Euler And Tits Form

Let $Q = (I, E)$ be a quiver. Define a bilinar form on $\mathbb{Z}^I$ as follows: for any $v, w \in \mathbb{Z}^I$:

$$\langle v, w\rangle_Q = \sum_i v_i w_i - \sum_e v_{s(e)} w_{t(e)}$$

This is called the *Euler form* of Q. The *symmetrized Euler form* of Q is given by

$$(v, w)_Q = \langle v, w\rangle_Q + \langle w, v\rangle_Q$$

Finally, the *tits form* of Q is the quadratic form $q_Q(v) = \langle v, v\rangle = \frac{1}{2}(v, v)$

Example 54.1.54: Euler And Tits Forms

Let $Q = \overset{1}{\bullet} \xrightarrow{e} \overset{2}{\bullet} \xrightarrow{f} \overset{3}{\bullet}$. Then the Euler's form is:

$$\langle v, w\rangle_Q = \sum_{i=1}^{3} v_i w_i - v_1 w_2 - v_2 w_3$$

$$q_Q(v) = \sum_{i=1}^{3} v_i v_i - v_1 v_2 - v_2 v_3$$

The tits form is also positive definite, since if it weren't, we would be able to pick 3 positive integer's satisfying:

$$v_2(v_1 + v_3) > v_1^2 + v_2^2 + v_3^2$$

However, this is no possible (we can show this with elementary calculus and optimization, or take a couple of space cases and induction).

Now let $Q' = \overset{1}{\bullet} \xleftarrow{e} \overset{2}{\bullet} \xrightarrow{f} \overset{3}{\bullet}$. Then:

$$\langle v, w\rangle_Q = \sum_{i}^{3} v_i w_i - v_2 w_1 - v_2 w_3$$

and:

$$q_Q(v) = \sum_{i}^{3} v_i v_i - v_2 v_1 - v_2 v_3$$

then we see that the tits form of Q and Q' agree.

Theorem 54.1.55: Positive Definite Forms And Dynkin Diagrams

Let Γ be a connected graph. The Tits form q_Γ is positive definite if and only if Γ is a Dynkin diagram of type A, D, or E

Proof:
ommited by Jimail

(exercise showing not true for not A, D, E)

Finally, we will state one more theorem we need for the proof

Theorem 54.1.56: Coxeter Element Lemma

Let R be a root system, $\Pi = \{\alpha_1, \alpha_2, ..., \alpha_n\}$ a set of simple roots and W its Weyl group. Let $s_i = s_{\alpha_i}$ denote the simple reflections corresponding to the simple root α_i. let $C = s_{i_n} \cdots s_{i_1}$ where $i_1, i_2, ..., i_n$ is some ordering of the simple roots (such an element is called a *Coxeter element*).

Then for any $v \in \mathbb{Z}^I$, there exists a $m \geq 0$ such that $C^m(v) \ngeq 0$

Proof:
ommited

Proof:
(of Gabriels theorem) here

54.1.5 Path Algebra of Quivers

(there are some really important insights that feel like they should actually be the priority of this section!! Essentially, we build-up more on how to think about algebras in terms of quotients of free modules, build-up how "path algebras" naturally come into play in representing quivers, and finish off with the Morita equivalence between R-Mod and R/I-Mod where $I = \text{Ann}_R(M)$).

Let $Q = (V, E)$ be a quiver. For vertices i, j of Q, Define the set $\text{Path}(i, j) = \text{span}\{\text{paths } \gamma \text{ from } i \text{ to } j\}$ (i.e. consider the path as a basis to generate a vector-space over k, where k is usually understood), which we will call the *path space*. the length of a particular path is the number of edges between each point. The length of the path from i to i is zero. If γ is a path, and if we let i, j represent a path from it to itself, then if γ is a path from i to j, we will say that γi and $j\gamma$ are the same path.

Lemma 54.1.57: Properties Of Path

here

***Proof*:**

1. We want to show that P_i has the structure of a representation $\text{Rep}(Q)$, which consists of a collection of vector-spaces and linear transformation's between them. First, each $P_i(j)$ is a vector-space, as pointed out earlier. Next, we may define a "linear" map $T_e : P_i(j) \to P_i(k)$ to be $\gamma \mapsto e\gamma$ (i.e. append another edge to the path). Notice that since we are treating paths as basis elements to some freely generated vector-space $P_i(j)$, then we get by construciton that $T_e(c \cdot \gamma) = ec \cdot \gamma = ce\gamma = cT_e(\gamma)$. Next, taking $\gamma + \epsilon$ to be a formal sum of two paths, we can naturally extend the action of concatinating to the formal sum:

$$e(\gamma + \epsilon) = e\gamma + e\epsilon$$

   which will mirror the linear maps within a represetnation $\text{Rep}(Q)$. Furthermore, if e, f are neigbhouring edges, then $T_{ef} = T_e T_f$. This now mirrors how $\text{Rep}(Q)$ is both a collection of vector-spaces and a collection of maps between them.

2. We are now going to treat P_i as a representation, since we considered them to be related to them in the above example. We want to prove that the collection of vector-spces P_i along with the linear transformations T_e are indecomposable, that is we cannot find subrepresentations. This comes from the construction of our map, in particular that we have that for any collection of paths that form a basis for $P_i(j)$, we have that they are eached mapped to by some paths through the appropriate T_e for an appropriate edge e. To make it more intuitive to see, consider for example P_i of the bellow quiver with a "cycle":

$$\begin{tikzcd} & j \arrow[dr, "b"] & \\ i \arrow[ur, "a"] \arrow[rr, "c"'] & & k \end{tikzcd}$$

   Then $\dim(V(i)) = 1$, $\dim(V(j))) = 1$, and $\dim(V(k)) = 2$, with the path i mapping to ci in $V(k)$ and mapping to bi in $V(j)$, while the path bi of $V(j)$ maps to ibc, in $V(k)$. Then notice that $V(k)$ is "surjected" upon by both $V(i)$ and $V(j)$ combined, and all the maps are injective. If the above quiver were to split, then either the dimension of $V(i)$ or $V(j)$ would have to get smaller (since $V(k)$ would have to diminish a dimension): if $\dim(V(j)) = 0$, then that would break the surejctivity of T_a, while if $\dim(V(i)) = 0$, then the map $T_c \neq T_c|_A \oplus T_c|_B$ for the two subquivers A and B.

   If the quiver has a proper cycle (i.e. $P_i(i)$ has more than $\{i\}$ as a basis), then since every map must be injective (mapping onto the appropriate basis) we have that each vector-space is 1-dimensional and an isomorphism, which we've shown in chapter 1 with a dimension argument that this representation is indecomposable.

   If The quiver does not have a cycle:

$$\begin{tikzcd} j & & k \\ & i \arrow[ul, "a"] \arrow[ur, "b"'] & \end{tikzcd}$$

   Then if for the sake of contradiction P_i were decomposable, then if $\dim(V(i)) = 0$ in one of the decomposibition, so would $V(j)$ and $V(k)$, making it a trivial decomposition, while if $\dim(V(i)) = 1$, both $\dim(V(j))$ and $\dim(V(k))$ would have to be 1 or else the map wouldn't be well-defined, and hence this quiver is indecomposable.

More generally, any quiver Q can be broken down into one of these two cases: If the quiver has a "cycle", then since every map is injective and must surject onto the vertex where two edges meet, we can repeat the above argument to show that the two scenarios which we may have while trying to decompose a quiver lead to a contradiction, and if the quiver has no cycles, we repeat the argument we've done for the above quiver.

3. If P_i were not a sink, then if e is an edge where that makes i not a sink, say $s(e) = i$ and $t(e) = j$, then we can take a subquiver where the vector-space at $P_i(i) = 0$, and we have a zero-map from $P_i(i)$ to $P_i(j)$, giving us a subquiver.

   Conversely, if P_i were a sink, the only path from i would be to i, and so all other vertices would have a 0 dimensional vector-space, while $P_i(i)$ would be 1-dimensional, which makes P_i into a simple representation

4. For notational simplicity, if $\gamma = k_n k_{n-1} \cdots k_1$ is some path, let

$$f_\gamma(v) = f_{k_n}(f_{n-1}(\cdots f_1(v)))$$

   so that $f_{ab}(v) = f_a(f_b(v))$.

   For the proof, pick any $v \in V(i)$, and consider the following commutative diagram:

$$\begin{tikzcd} P_i(a) \arrow[r, "T_e"] \arrow[d, "\varphi_a"] & P_i(b) \arrow[d, "\varphi_b"] \\ V(a) \arrow[r, "S_e"] & V(b) \end{tikzcd}$$

   Then define $\varphi^v : P_i \to V$, $\varphi^v(\gamma) = S_\gamma(v)$. Then

$$S_e(\varphi_a^v(\gamma)) = S_e S_\gamma(v) = S_{e\gamma}(v) = \varphi_b^v(e\gamma) = \varphi_b^v(T_e(\gamma))$$

   proving commutativity of the diagram. If $v \neq w$, then since $\varphi_i^v(i) = v \neq w = \varphi_i^w(i)$, $\varphi^v \neq \varphi^w$. Next, if $\varphi \in \mathrm{Hom}_Q(P_i, V)$, then if $(\gamma_1, \gamma_2, ..., \gamma_n)$ is a basis for $V(i)$, consider

$$(\varphi_i(\gamma_1), ..., \varphi_i(\gamma_n)) = (v_1, ..., v_n)$$

   and define $v = \sum_i v_i$. Then extending linearly we get that $\varphi_i\left(\sum_i c_i\gamma_i\right) = \sum_i c_i v_i$, and then extending φ_i to any other φ_j through a chain of commutative diagrams, we get that all of φ is determined by where v maps to in any path in V. Thus, $\varphi = \varphi^v$, and so $\mathrm{Hom}(P_i, V)$ and $V(i)$ are in bijective correspondence. Finally, By linearly of each map in φ, and each map assigned to an edge in P_i and $V(i)$, it we see that:

$$\varphi^{v+cw} = \varphi^v + c\varphi^w \in \mathrm{Hom}(P_i, V)$$

   giving us that the map is in fact a linear map, completing the proof.

Definition 54.1.58: Path Algebra

Let $Q = (V, E)$ be a quiver. Then the *path algebra* $P(Q)$ is defined as follows:

1. As a vector-space $P(Q) = \text{span}\{\text{paths in } Q\}$, where each vertex $i \in V$ we include an element e_i representing the constant path $i \to i$
2. multiplication is defined via concatenation when legal, that is:

$$p \cdot q = \begin{cases} pq & \text{if } t(q) = s(p) \\ 0 & \text{otherwise} \end{cases}$$

54.2 Bound Quiver Algebras

The path algebra of definition 54.1.58 takes a combinatorial object, a quiver, and turns it into an associative algebra whose modules are exactly the quiver representations of definition 54.1.5. This is already a bridge between two worlds, but it runs in only one direction: every quiver produces an algebra. The converse question is the one that completes the picture. Given an abstract finite-dimensional algebra A over a field k, is there a quiver Q whose path algebra is A, or close enough to A that the two have the same representation theory? A path algebra is rarely finite-dimensional (any quiver with a cycle has paths of every length), so one cannot hope for equality on the nose. The fix is to impose relations, cutting the path algebra down by an ideal, and the content of this section is that this always suffices: over an algebraically closed field, every finite-dimensional algebra is, up to the coarsening supplied by Morita equivalence, a path algebra modulo relations. This is the structure theorem of Gabriel, and it makes the theory of quivers the universal combinatorial model for finite-dimensional representation theory.

The route to this statement passes through three reductions. First we cut a path algebra by an *admissible* ideal, which is what makes it finite-dimensional while keeping its arrows intact. Then we isolate the class of algebras for which the reconstruction is cleanest, the *basic* algebras, and show via Morita theory that restricting to them loses nothing. Finally we read a quiver directly off an abstract algebra A (its *Gabriel quiver*) using the Jacobson radical of chapter 52, and prove that A is the path algebra of that quiver modulo an admissible ideal. Throughout, A denotes a finite-dimensional algebra over k, modules are left modules, and $\text{Jac}(A)$ is the Jacobson radical. We write kQ for the path algebra $P(Q)$ of definition 54.1.58, following the standard notation, and recall its multiplication convention: for paths p and q the product $p \cdot q$ is the concatenation "q then p", nonzero exactly when $t(q) = s(p)$. In particular a path γ from i to j satisfies $e_j \gamma e_i = \gamma$, so it lives in $e_j(kQ)e_i$; this ordering of the idempotents matters when counting arrows.

The first task is to identify, inside kQ, the piece built from arrows of positive length. The constant paths e_i span a semisimple part, a product of copies of k, and everything else is generated by the arrows. Collecting the arrow-generated part into an ideal gives the object that measures "length at least one", and its powers measure length.

Definition 54.2.1: Arrow Ideal and Admissible Ideal

Let Q be a finite quiver and kQ its path algebra over k.

1. The *arrow ideal* $R_Q \subseteq kQ$ is the two-sided ideal generated by the arrows (the paths of length 1).

2. A two-sided ideal $I \subseteq kQ$ is *admissible* if there exists an integer $m \geq 2$ with

$$R_Q^m \subseteq I \subseteq R_Q^2$$

Two facts make this definition run, both immediate from the way paths concatenate. First, R_Q^ℓ is spanned by the paths of length at least ℓ: a product of ℓ arrows is a path of length ℓ (when the concatenation is legal) and zero otherwise, and every path of length ℓ factors as such a product. Consequently $kQ = kQ_0 \oplus R_Q$ as vector spaces, where $kQ_0 = \bigoplus_i ke_i$ is the span of the constant paths, and more generally the paths of a fixed length span a complement to $R_Q^{\ell+1}$ inside R_Q^ℓ. Second, the sandwiching $R_Q^m \subseteq I \subseteq R_Q^2$ has a concrete meaning at each end. The lower bound $R_Q^m \subseteq I$ forces I to contain all sufficiently long paths, which is exactly what truncates kQ/I to finite dimension. The upper bound $I \subseteq R_Q^2$ forbids relations of length 0 or 1: no relation may involve a constant path or set a single arrow equal to a combination of other arrows. This is the condition that keeps the arrows of Q visible in the quotient, so that Q can be recovered from the algebra kQ/I rather than being partly erased by the relations.

Definition 54.2.2: Bound Quiver Algebra

Let Q be a finite quiver and $I \subseteq kQ$ an admissible ideal. The quotient

$$kQ/I$$

is called the *bound quiver algebra* (or the path algebra of Q *bound by* the relations I). The pair (Q, I) is called a *bound quiver*.

Because $R_Q^m \subseteq I$, the quotient kQ/I is spanned by the images of the paths of length less than m, of which there are finitely many when Q is finite; so a bound quiver algebra is always finite-dimensional. The constant paths e_i descend to a complete set of orthogonal idempotents summing to 1, and R_Q/I is a nilpotent ideal with $(R_Q/I)^m = 0$. In fact R_Q/I is precisely the Jacobson radical of kQ/I: the quotient by it is $kQ_0 = \prod_i k$, a product of copies of k, which is semisimple, and a nilpotent ideal is always contained in the radical, so the two coincide. This last observation is the seed of the reconstruction theorem: the arrows of Q will be recovered from $\mathrm{Jac}(kQ/I)/\mathrm{Jac}(kQ/I)^2$.

Example 54.2.3: Loops, Truncation, and the A_2 Quiver

We compute three bound quiver algebras from the ground up.

1. *The dual numbers as a bound one-loop quiver.* Let Q be the quiver with one vertex 1 and one loop $\alpha \colon 1 \to 1$:

$$\begin{tikzcd} 1 \arrow[loop above, "\alpha"] \end{tikzcd}$$

The paths in Q are $e_1, \alpha, \alpha^2, \alpha^3, \ldots$, so as a vector space $kQ = k[\alpha]$ is the polynomial ring in one variable (identifying e_1 with 1 and concatenation of the loop with multiplication). The arrow ideal is $R_Q = (\alpha)$, and $R_Q^m = (\alpha^m)$. Take $I = (\alpha^2) = R_Q^2$, which is admissible with $m = 2$ (indeed $R_Q^2 \subseteq I \subseteq R_Q^2$). Then

$$kQ/I = k[\alpha]/(\alpha^2)$$

is the algebra of *dual numbers*, two-dimensional over k with basis $\{e_1, \alpha\}$ and the single relation $\alpha^2 = 0$. Its radical is $R_Q/I = (\alpha)$, one-dimensional, with square zero, and kQ/I modulo its radical is k: there is exactly one simple module, and it is one-dimensional. This is the smallest example of a bound quiver algebra that is not itself a path algebra, the relation $\alpha^2 = 0$ being unavoidable if one wants a finite-dimensional quotient of $k[\alpha]$.

2. *Truncating a longer loop.* Keeping the same one-loop quiver but taking instead $I = (\alpha^3) = R_Q^3$ gives $kQ/I = k[\alpha]/(\alpha^3)$, three-dimensional with basis $\{e_1, \alpha, \alpha^2\}$. Here $m = 3$; the admissibility bound $I \subseteq R_Q^2$ holds because $\alpha^3 \in R_Q^3 \subseteq R_Q^2$, and the relation still involves no constant path and no bare arrow. More generally $k[\alpha]/(\alpha^n)$ is the one-loop quiver bound by (α^n) for each $n \geq 2$, and these exhaust the local algebras arising from a single loop.

3. *The A_2 quiver, hereditary case.* Let Q be the quiver

$$1 \xrightarrow{\alpha} 2$$

with two vertices and a single arrow $\alpha \colon 1 \to 2$. The only paths are e_1, e_2, and α, so kQ is already three-dimensional and $R_Q^2 = 0$ (there is no legal concatenation of two arrows). The only admissible ideal is therefore $I = 0$: admissibility requires $I \subseteq R_Q^2 = 0$, so nothing may be quotiented, and $R_Q^2 = 0 = I$ supplies the lower bound with $m = 2$. Thus $kQ/I = kQ$, a path algebra with no relations. An algebra whose only admissible bound is the zero ideal, equivalently one whose radical squares to zero and imposes no relations, is called *hereditary*; the A_2 path algebra is the prototype.

Concretely kQ is isomorphic to the lower-triangular matrix algebra

$$\begin{pmatrix} k & 0 \\ k & k \end{pmatrix}$$

via $e_1 \mapsto E_{11}$, $e_2 \mapsto E_{22}$, and $\alpha \mapsto E_{21}$, where E_{ij} is the elementary matrix with a 1 in position (i, j). Lower- rather than upper-triangular is forced by the multiplication convention: the arrow $\alpha \colon 1 \to 2$ satisfies $e_2 \alpha e_1 = \alpha$, so its image must land in position $(2, 1)$, the strictly lower-triangular slot. The check is direct on the multiplication table: $e_2 \cdot \alpha \cdot e_1 = \alpha$ matches $E_{22}E_{21}E_{11} = E_{21}$, and every other product of basis elements is forced to zero on both sides. (The upper-triangular algebra is the opposite algebra, the A_2 path algebra with the arrow reversed to $2 \to 1$; we recover it from the radical in example 54.2.9.) We return to this identification once we have located the indecomposable projectives.

The dual-numbers example already shows the shape of the reconstruction: a local algebra with radical square zero came out as a single loop bound by α^2, and the loop is exactly a basis vector of $\mathrm{Jac}/\mathrm{Jac}^2$. Before making this into a theorem, one must decide which algebras to reconstruct. Not every finite-dimensional algebra is a bound quiver algebra on the nose. The matrix algebra $M_2(k)$, for instance, is simple, so its only simple module is two-dimensional, whereas a bound quiver algebra always has all its simples one-dimensional over k when k is algebraically closed (they are the vertex simples S_i). The class for which the reconstruction is exact is singled out by demanding that the semisimple quotient $A/\mathrm{Jac}(A)$ be as small as possible: a product of division rings, with no matrix blocks of size larger than one.

Definition 54.2.4: Basic Algebra

A finite-dimensional k-algebra A is *basic* if the quotient $A/\mathrm{Jac}(A)$ is a product of division rings,

$$A/\mathrm{Jac}(A) \cong D_1 \times \cdots \times D_n$$

with each D_i a division k-algebra. Equivalently, in the Artin–Wedderburn decomposition of $A/\mathrm{Jac}(A)$ (corollary 51.2.18) every matrix block $M_{n_i}(D_i)$ has $n_i = 1$.

When k is algebraically closed the finite-dimensional division k-algebras collapse to k itself (corollary 51.2.20), so a basic algebra over such a k has

$$A/\mathrm{Jac}(A) \cong k \times \cdots \times k$$

a product of n copies of k. Two consequences are worth stating plainly. First, the simple A-modules are the n simple modules of this product, each one-dimensional over k; a basic algebra over an algebraically closed field has *all simple modules one-dimensional*. Second, the number n of simple modules equals the number of factors, and it becomes the number of vertices of the reconstructed quiver. The condition "every matrix block has size one" is precisely the condition that the regular representation of $A/\mathrm{Jac}(A)$ contains each simple exactly once, so that A is, in the language of idempotents, a sum of pairwise non-isomorphic indecomposable projectives.

Example 54.2.5: Basic and Non-Basic Algebras

The bound quiver algebras of example 54.2.3 are all basic: for $k[\alpha]/(\alpha^2)$ the radical quotient is k, and for the A_2 path algebra it is $k \times k$, both products of copies of k. The upper-triangular algebra of 2×2 matrices is basic for the same reason. By contrast $M_2(k)$ is *not* basic: it is already semisimple, so $\mathrm{Jac}(M_2(k)) = 0$ and $M_2(k)/\mathrm{Jac}(M_2(k)) = M_2(k)$ is a single 2×2 block. The group algebra $k[S_3]$ over $k = \mathbb{C}$ is not basic either, since by Maschke's theorem it is semisimple and Artin–Wedderburn gives $\mathbb{C} \times \mathbb{C} \times M_2(\mathbb{C})$, a product with a 2×2 block coming from the two-dimensional irreducible representation of S_3. These non-basic examples are not pathological; they are the reason the theorem is stated up to Morita equivalence rather than isomorphism.

The point of Morita equivalence is that it does not see the difference between an algebra and its basic model. Two algebras are Morita equivalent when their module categories are equivalent, and Morita's theorem identifies exactly which algebras share a module category with a given one: they are the endomorphism algebras of progenerators. That theory, including generators, progenerators, and the classification of Morita equivalences by bimodules, is developed in *Core Algebra* [25, Morita theory]; what is used here is the criterion for a corner ring, that Ae is a progenerator exactly when the idempotent e is full, in which case A and eAe are Morita equivalent. The block structure of the radical quotient is this book's own and is settled below.

Proposition 54.2.6: Reduction to the Basic Case

Let A be a finite-dimensional k-algebra. Then there is a basic finite-dimensional k-algebra A^b, the *basic algebra of* A, that is Morita equivalent to A: the categories of left modules A**-Mod** and A^b**-Mod** are equivalent. The algebra A^b is unique up to isomorphism, and A and A^b have the same number of simple modules and the same radical quotient block *data* in the sense that $A^b/\mathrm{Jac}(A^b) \cong D_1 \times \cdots \times D_n$ where $M_{n_i}(D_i)$ are the Artin–Wedderburn blocks of $A/\mathrm{Jac}(A)$.

***Proof*:**
Step 1: the idempotent. Write $1 = e_1 + \cdots + e_r$ as a sum of primitive orthogonal idempotents, following theorem 52.3.2, so that $A = \bigoplus_j Ae_j$ expresses the regular module as a direct sum of indecomposable projectives $P_j = Ae_j$. Two of these are isomorphic exactly when the corresponding simples $P_j/\mathrm{rad}(P_j)$ and $P_{j'}/\mathrm{rad}(P_{j'})$ agree, so grouping the e_j by the isomorphism class of P_j and choosing one representative from each class produces an idempotent

$$e = e_{i_1} + \cdots + e_{i_n}$$

with n the number of simple modules. It is idempotent because the e_{i_j} are orthogonal. Set $A^b := eAe$.

Step 2: e is full. Each e_{i_j} satisfies $e_{i_j}e = e_{i_j}$, again by orthogonality, so $Ae_{i_j}A = Ae_{i_j}eA \subseteq AeA$. For an arbitrary l the projective Ae_l is isomorphic to Ae_{i_j} for some j by the choice of representatives, and isomorphic projectives generate the same two-sided ideal, so $Ae_lA = Ae_{i_j}A \subseteq AeA$. Summing,

$$1 = \sum_{l=1}^{r} e_l \in \sum_{l=1}^{r} Ae_lA \subseteq AeA$$

so $AeA = A$ and e is full.

Step 3: the equivalence. By theorem 18.3.6, a full idempotent makes Ae a progenerator with $\mathrm{End}_A(Ae)^{\mathrm{op}} \cong eAe$, so A and $A^b = eAe$ are Morita equivalent. The equivalence is $\mathrm{Hom}_A(Ae, -)$, which sends an A-module M to eM.

Step 4: A^b is basic, with the stated radical quotient. The equivalence carries the indecomposable projectives of A to those of A^b and preserves isomorphism, and $Ae = \bigoplus_j Ae_{i_j}$ retains exactly one indecomposable projective from each isomorphism class. So the regular representation of A^b contains each simple exactly once, which is what it means for A^b to be basic, and its radical quotient has all blocks of size one. The division rings are unchanged: the equivalence induces a bijection on simple modules and a ring isomorphism on their endomorphism rings, so the blocks are over the same $D_i = \mathrm{End}_A(S_i)^{\mathrm{op}}$ furnished by Schur's lemma (lemma 51.1.5). Uniqueness of A^b up to isomorphism follows because any two choices of representatives give isomorphic modules Ae, hence isomorphic endomorphism rings, completing the proof.

The reduction lets us prove the structure theorem only for basic algebras and inherit it for all finite-dimensional algebras through the equivalence of module categories: whatever we say about the representations of $A^b = kQ_A/I$ transfers verbatim to the representations of A. From now on A is basic, and to make the reconstruction concrete assume k is algebraically closed, so that all simple modules are one-dimensional and the radical quotient is a product of copies of k. This is the setting of Gabriel's theorem, and it is where the assumption $k = \overline{k}$ earns its place: over a non-closed field the division algebras D_i reappear and the vertices of the quiver would have to be weighted by them.

The quiver attached to an abstract algebra can now be defined. Its vertices are the simple modules; its arrows count how simples fail to split off one another, measured either homologically by Ext^1 or internally by the radical filtration. That these two counts agree is the bridge between the abstract and combinatorial descriptions.

Definition 54.2.7: Ext-Quiver (Gabriel Quiver)

Let A be a basic finite-dimensional algebra over an algebraically closed field k, with simple modules $S_1, \dots, S_n$ up to isomorphism and a complete set of primitive orthogonal idempotents $e_1, \dots, e_n$ with $S_i = Ae_i/\mathrm{rad}(Ae_i)$. The *Ext-quiver* (or *Gabriel quiver*) Q_A is the quiver with

1. vertex set $\{1, \dots, n\}$, one vertex for each simple module S_i, and
2. exactly
$$\dim_k \mathrm{Ext}^1_A(S_i, S_j) = \dim_k\, e_j\big(\mathrm{Jac}(A)/\mathrm{Jac}(A)^2\big)e_i$$
arrows from i to j.

The order of the idempotents in the formula is fixed by the multiplication convention of definition 54.1.58 and is worth pausing on, since it is easy to get backwards. In the path algebra a would-be arrow from i to j satisfies $e_j \alpha e_i = \alpha$, so it lives in $e_j(kQ)e_i$; matching this, the arrows from i to j in Q_A are counted by $e_j(\mathrm{Jac}/\mathrm{Jac}^2)e_i$, with e_j on the left and e_i on the right. A source using the opposite (left-to-right) convention for path multiplication would write $e_i(\mathrm{Jac}/\mathrm{Jac}^2)e_j$ instead; the two conventions transpose the quiver, so one must commit to one. Here the commitment is to the book's convention throughout. The equality of the two counts, homological and idempotent-theoretic, is the following lemma, proved here because it is what makes the definition well-posed: the arrow count must not depend on the choice of description.

Lemma 54.2.8: Arrows Count Radical Extensions

Let A be a finite-dimensional k-algebra with primitive orthogonal idempotents $e_1, \dots, e_n$ and simple modules $S_i = Ae_i/\mathrm{rad}(Ae_i)$, and assume k is algebraically closed. Then for all i, j,
$$\mathrm{Ext}^1_A(S_i, S_j) \cong e_j\big(\mathrm{Jac}(A)/\mathrm{Jac}(A)^2\big)e_i$$
as k-vector spaces. In particular the two are equidimensional.

***Proof*:**
Write $J = \mathrm{Jac}(A)$ and $P_i = Ae_i$, the indecomposable projective with top $S_i = P_i/\mathrm{rad}(P_i)$; recall from definition 52.3.3 that $\mathrm{rad}(P_i) = Je_i$, so that $S_i = Ae_i/Je_i$.

Step 1: A projective presentation of S_i. The surjection $P_i \twoheadrightarrow S_i$ has kernel $\mathrm{rad}(P_i) = Je_i$, giving the short exact sequence
$$0 \to Je_i \to P_i \to S_i \to 0$$
Applying $\mathrm{Hom}_A(-, S_j)$ produces the exact sequence
$$0 \to \mathrm{Hom}_A(S_i, S_j) \to \mathrm{Hom}_A(P_i, S_j) \to \mathrm{Hom}_A(Je_i, S_j) \to \mathrm{Ext}^1_A(S_i, S_j) \to 0$$

where the last term is $\mathrm{Ext}^1_A(S_i, S_j)$ and not $\mathrm{Ext}^1(P_i, S_j)$ because P_i is projective, so $\mathrm{Ext}^1_A(P_i, S_j) = 0$ and the connecting map surjects onto $\mathrm{Ext}^1(S_i, S_j)$. The definition and basic properties of Ext^1 as the obstruction to lifting homomorphisms along a projective presentation are developed in the companion volume on core algebra, `NateCoreAlgebra` [25]; we use only that a projective resolution computes it and that projectives have no higher self-extensions.

Step 2: Evaluating the outer terms. For any simple S_j and any module M, a homomorphism $M \to S_j$ factors through $M/\mathrm{rad}(M)$ because $\mathrm{rad}(M)$ maps into $\mathrm{rad}(S_j) = 0$. Applying this to P_i gives $\mathrm{Hom}_A(P_i, S_j) = \mathrm{Hom}_A(S_i, S_j)$, so the first two terms of the sequence in Step 1 cancel and the map between them is an isomorphism. The sequence therefore collapses to

$$0 \to \mathrm{Hom}_A(Je_i, S_j)/\big(\text{image of } \mathrm{Hom}_A(P_i, S_j)\big) \to \mathrm{Ext}^1_A(S_i, S_j) \to 0$$

but the image of $\mathrm{Hom}_A(P_i, S_j)$ in $\mathrm{Hom}_A(Je_i, S_j)$ is zero: a homomorphism $P_i \to S_j$ vanishes on $\mathrm{rad}(P_i) = Je_i$ by the factoring just noted. Hence the connecting map is an isomorphism

$$\mathrm{Ext}^1_A(S_i, S_j) \cong \mathrm{Hom}_A(Je_i, S_j)$$

Step 3: The Hom into a simple is a radical layer. Again by the factoring through the top, any homomorphism $Je_i \to S_j$ kills $\mathrm{rad}(Je_i) = J^2 e_i$, so

$$\mathrm{Hom}_A(Je_i, S_j) = \mathrm{Hom}_A\big(Je_i/J^2e_i,\ S_j\big)$$

The module Je_i/J^2e_i is semisimple, being a module over the semisimple algebra A/J, and S_j is simple, so by Schur's lemma (lemma 51.1.5) and $\mathrm{End}_A(S_j) = k$ (from $k = \overline{k}$) the space $\mathrm{Hom}_A(Je_i/J^2e_i, S_j)$ has dimension equal to the multiplicity of S_j in Je_i/J^2e_i. That multiplicity is $\dim_k e_j(Je_i/J^2e_i)$, since for a semisimple module N over $A/J = \prod_\ell k$ the multiplicity of S_j in N is $\dim_k e_jN$: the idempotent e_j projects N onto its S_j-isotypic part, which over an algebraically closed field is a sum of copies of S_j each one-dimensional. Therefore

$$\dim_k \mathrm{Ext}^1_A(S_i, S_j) = \dim_k e_j(Je_i/J^2e_i) = \dim_k e_j(J/J^2)e_i$$

the last equality because $e_j(J/J^2)e_i = e_j(Je_i)/e_j(J^2e_i)$ and right-multiplication by e_i identifies Je_i/J^2e_i with the e_i-column of J/J^2. Assembling the three isomorphisms gives $\mathrm{Ext}^1_A(S_i, S_j) \cong e_j(J/J^2)e_i$, as we sought to show.

The lemma says the arrows of Q_A have two readings. Homologically, an arrow $i \to j$ records a one-dimensional worth of non-split extension of S_i by S_j: a module of length two with S_j as a submodule and S_i on top that does not break as $S_i \oplus S_j$. Internally, the same arrow is a basis vector of the (j, i) block of the first radical layer $\mathrm{Jac}/\mathrm{Jac}^2$. The first reading is the reason the quiver is called the Ext-quiver and explains why it is an invariant of the module category, hence of the Morita equivalence class; the second reading is the one to compute with, because $\mathrm{Jac}/\mathrm{Jac}^2$ is a finite matrix of idempotent-sandwiched subspaces. The running examples compute it.

Example 54.2.9: Gabriel Quivers of the Running Algebras

Reading off Q_A for two algebras confirms it recovers the starting quiver.

1. *The dual numbers.* Let $A = k[\alpha]/(\alpha^2)$. This is local: its only maximal ideal is (α), so $\mathrm{Jac}(A) = (\alpha)$ and there is a single simple module $S_1 = A/(\alpha) = k$, giving one vertex. The first radical layer is $\mathrm{Jac}/\mathrm{Jac}^2 = (\alpha)/(\alpha^2)$, one-dimensional with basis the image of α, and

with $e_1 = 1$ the block $e_1(\mathrm{Jac}/\mathrm{Jac}^2)e_1$ is all of it. So Q_A has one vertex and one arrow, a loop:

$$\begin{tikzcd} 1 \arrow[loop above, "\alpha"] \end{tikzcd}$$

exactly the quiver we bound in example 54.2.3. The relation to be imposed will be $\alpha^2 = 0$, recovering A.

2. *The upper-triangular algebra.* Let $A = \begin{pmatrix} k & k \\ 0 & k \end{pmatrix}$, the 2×2 upper-triangular matrices. Its radical is the strictly upper-triangular part, $\mathrm{Jac}(A) = kE_{12}$, one-dimensional with square zero. The idempotents $e_1 = E_{11}$ and $e_2 = E_{22}$ are primitive orthogonal with $e_1 + e_2 = 1$, and $A/\mathrm{Jac}(A) \cong k \times k$, so there are two vertices with simples S_1, S_2. Since $\mathrm{Jac}^2 = 0$ we have $\mathrm{Jac}/\mathrm{Jac}^2 = kE_{12}$, and we locate E_{12} by its idempotent sandwich:

$$e_1 E_{12} e_2 = E_{11} E_{12} E_{22} = E_{12}, \qquad e_2 E_{12} e_1 = E_{22} E_{12} E_{11} = 0$$

so the one basis vector of $\mathrm{Jac}/\mathrm{Jac}^2$ sits in the block $e_1(\mathrm{Jac}/\mathrm{Jac}^2)e_2$. By definition 54.2.7 the block $e_j(\cdots)e_i$ counts arrows $i \to j$, so with $j = 1$, $i = 2$ this is one arrow $2 \to 1$:

$$2 \xrightarrow{\alpha} 1$$

Under the identification $\alpha \mapsto E_{12}$ this is the A_2 quiver of example 54.2.3, up to relabelling the vertices, and $\mathrm{Jac}^2 = 0$ forces the admissible ideal to be $I = 0$. The algebra is hereditary, and Q_A carries no relations: the upper-triangular algebra *is* its own path algebra kQ_A.

The two examples describe the same three-dimensional hereditary algebra, once as a bound quiver algebra and once reconstructed from its radical, and the indecomposable projectives match under the identification. Take the bound-quiver presentation of example 54.2.3, the A_2 path algebra with arrow $\alpha\colon 1 \to 2$ realized as the lower-triangular matrices via $e_1 \mapsto E_{11}$, $e_2 \mapsto E_{22}$, $\alpha \mapsto E_{21}$. The indecomposable projective $P_i = kQ \cdot e_i$ is spanned by the paths starting at i: P_1 has basis $\{e_1, \alpha\}$, the trivial path at 1 together with the arrow out of 1, and is two-dimensional with top S_1 and socle S_2, while P_2 has basis $\{e_2\}$ and is the simple S_2 itself. On the matrix side these are the column modules AE_{11} and AE_{22}: the first column of the lower-triangular algebra is two-dimensional, spanned by E_{11} and E_{21} (the entries in rows 1 and 2 of column 1), and the second column is one-dimensional, spanned by E_{22}. So the indecomposable projectives read off from the primitive idempotents of theorem 52.3.2 are exactly the projective quiver representations P_i of the path algebra. The Gabriel reconstruction of example 54.2.9 applied instead the upper-triangular representative, whose radical kE_{12} sits in the opposite corner and so produces the arrow $2 \to 1$; the two matrix algebras are opposite to one another (transpose exchanges upper and lower triangular), which is exactly the transposition of the quiver that the two multiplication conventions differ by. Either way the representation theory of A_2 is the same, and the two constructions coincide.

Everything is now in place for the structure theorem. It asserts that the passage from a basic algebra to its Gabriel quiver and back to a bound quiver algebra is a round trip: A is isomorphic to kQ_A modulo an admissible ideal. The proof builds a surjection $kQ_A \twoheadrightarrow A$ by sending vertices to the primitive idempotents and arrows to a lift of the radical layer, then shows the kernel is admissible using that $\mathrm{Jac}(A)$ is nilpotent.

Theorem 54.2.10: Structure of Basic Algebras (Gabriel)

Let A be a basic finite-dimensional algebra over an algebraically closed field k, with Gabriel quiver Q_A (definition 54.2.7). Then there is an admissible ideal $I \subseteq kQ_A$ and an isomorphism of k-algebras

$$A \cong kQ_A/I$$

Consequently, by the reduction to the basic case (proposition 54.2.6), every finite-dimensional k-algebra over an algebraically closed field is Morita equivalent to a bound quiver algebra kQ/I.

The quiver Q_A is determined by A, but the ideal I is not: different admissible ideals can present isomorphic algebras, so the theorem produces *some* admissible presentation rather than a canonical one. This is the one point where the correspondence between algebras and bound quivers fails to be a clean bijection, and it is the reason bound quiver algebras are studied through their quiver plus a choice of relations rather than through the quiver alone.

***Proof*:**
Write $J = \mathrm{Jac}(A)$, and recall J is nilpotent, say $J^m = 0$ for some $m \geq 2$, because A is finite-dimensional (a nilpotent radical is the Artinian case established in theorem 52.1.8). Let $e_1, \dots, e_n$ be the complete set of primitive orthogonal idempotents underlying Q_A, with $1 = e_1 + \cdots + e_n$ and $S_i = Ae_i/Je_i$.

Step 1: Send vertices to idempotents and arrows to a radical lift. For each vertex i of Q_A set the constant path $\varepsilon_i \in kQ_A$ to map to $e_i \in A$. For the arrows, fix for each pair (i, j) a k-basis of the block $e_j(J/J^2)e_i$; by definition 54.2.7 and lemma 54.2.8 the arrows of Q_A from i to j are in bijection with this basis. For each arrow $\alpha\colon i \to j$ choose a lift $x_\alpha \in e_jJe_i \subseteq A$ of the corresponding basis vector of $e_j(J/J^2)e_i$; such a lift exists because $e_j(J/J^2)e_i$ is a quotient of e_jJe_i. Note $x_\alpha \in e_jJe_i$ means $e_jx_\alpha e_i = x_\alpha$, which is the algebra image of the path relation $e_j\alpha e_i = \alpha$, so the assignment respects the idempotent bookkeeping.

Step 2: Extend to an algebra homomorphism. Because kQ_A is free on paths, an algebra homomorphism out of it is determined by the images of the vertices and arrows, subject only to the relations $\varepsilon_i\varepsilon_{i'} = \delta_{ii'}\varepsilon_i$, $\sum_i \varepsilon_i = 1$, and the source-target matching for arrows. The chosen images e_i and x_α satisfy all of these: the e_i are orthogonal idempotents summing to 1, and $x_\alpha \in e_jAe_i$ matches the source i and target j of α. Extend multiplicatively, sending a path $\alpha_r \cdots \alpha_1$ to the product $x_{\alpha_r} \cdots x_{\alpha_1} \in A$ (in the algebra's own multiplication, so a path from i to j lands in e_jAe_i). This defines a k-algebra homomorphism

$$\varphi\colon kQ_A \to A$$

Step 3: φ is surjective. The image contains all e_i and, since each x_α together with the x_α-products spans, it contains a lift of a basis of J/J^2. We show $\mathrm{im}\varphi = A$ by showing it contains all of $A = A/J^0 \supseteq J \supseteq J^2 \supseteq \cdots \supseteq J^m = 0$ layer by layer. The images $\varphi(\varepsilon_i) = e_i$ generate $A/J = \prod_i ke_i$ over k, so $\mathrm{im}\varphi + J = A$. Next, the products of length one, $x_\alpha = \varphi(\alpha)$, form lifts of a basis of J/J^2, so $\mathrm{im}\varphi + J^2 \supseteq J$, hence $\mathrm{im}\varphi + J^2 = A$. Inductively suppose $\mathrm{im}\varphi + J^\ell = A$; then J^ℓ is spanned modulo $J^{\ell+1}$ by products of ℓ elements of J, each of which may be taken from the spanning set $\{x_\alpha\}$ of J modulo J^2, so products $\varphi(\alpha_\ell \cdots \alpha_1)$ of ℓ arrows span J^ℓ modulo $J^{\ell+1}$. Thus $\mathrm{im}\varphi + J^{\ell+1} = A$. Since $J^m = 0$, after m steps $\mathrm{im}\varphi = \mathrm{im}\varphi + J^m = A$, so φ is surjective.

Step 4: The kernel is admissible. Let $I = \ker\varphi$, so $A \cong kQ_A/I$. It remains to check $R_Q^{m'} \subseteq I \subseteq$

R_Q^2 for some m', where R_Q is the arrow ideal of kQ_A.

For the lower bound, $\varphi(R_Q^m) \subseteq J^m = 0$: a path of length at least m is a product of at least m arrows, and each arrow maps into J, so a length-$\geq m$ path maps into $J^m = 0$. Hence $R_Q^m \subseteq \ker \varphi = I$, giving the lower bound with $m' = m$.

For the upper bound $I \subseteq R_Q^2$, suppose $u \in I$ and write $u = u_0 + u_1 + u_2$ where $u_0 \in kQ_{A,0}$ is the constant-path part, u_1 is the arrow part (length exactly one), and $u_2 \in R_Q^2$ collects the length-≥ 2 paths. We must show $u_0 = 0$ and $u_1 = 0$. Applying φ and reducing modulo J: $\varphi(u_1) \in J$ and $\varphi(u_2) \in J^2 \subseteq J$, so $0 = \varphi(u) \equiv \varphi(u_0) \pmod{J}$. But φ restricted to constant paths is the isomorphism $kQ_{A,0} = \prod_i k\varepsilon_i \xrightarrow{\sim} \prod_i ke_i = A/J$ (the e_i are linearly independent modulo J because they are orthogonal idempotents with distinct simple tops), so $\varphi(u_0) \equiv 0 \pmod{J}$ forces $u_0 = 0$. Now reduce modulo J^2: with $u_0 = 0$ we have $0 = \varphi(u) = \varphi(u_1) + \varphi(u_2)$, and $\varphi(u_2) \in J^2$, so $\varphi(u_1) \equiv 0 \pmod{J^2}$. But $\varphi(u_1)$ is the image in J/J^2 of a linear combination of arrows, and the arrows were chosen to map to a *basis* of J/J^2; a linear combination of basis vectors that is zero in J/J^2 has all coefficients zero, so $u_1 = 0$. Therefore $u = u_2 \in R_Q^2$, proving $I \subseteq R_Q^2$.

Combining the two bounds, $R_Q^m \subseteq I \subseteq R_Q^2$, so I is admissible and $A \cong kQ_A/I$ is a bound quiver algebra. The final clause on general algebras follows by composing with the Morita reduction of proposition 54.2.6: an arbitrary finite-dimensional A is Morita equivalent to its basic algebra A^b, and $A^b \cong kQ_{A^b}/I$ by what was just proved, completing the proof.

Three features of the argument deserve a closing word. The surjectivity in Step 3 is a radical-filtration induction: because $\mathrm{Jac}(A)$ is nilpotent, generating $A/\mathrm{Jac}(A)$ by the idempotents and the first radical layer $\mathrm{Jac}/\mathrm{Jac}^2$ by the arrows suffices to generate all of A, since higher layers are spanned by products. The admissibility of the kernel in Step 4 is the exact counterpart of the two bounds in definition 54.2.1: the lower bound $R_Q^m \subseteq I$ is nilpotence of the radical, and the upper bound $I \subseteq R_Q^2$ is the choice of arrows as a basis of $\mathrm{Jac}/\mathrm{Jac}^2$, which is precisely why an admissible ideal is forbidden to contain constant paths or bare arrows. And the theorem is the structure theorem of Gabriel, to be kept distinct from Gabriel's other theorem in this chapter (theorem 54.1.47), which classifies the quivers of finite representation type as the simply-laced Dynkin diagrams A, D, E. The two results are complementary: the present one says every finite-dimensional algebra is a bound quiver algebra, and the Dynkin classification says which of the unbound path algebras have only finitely many indecomposable modules.

Part XI

Homological Algebra

Throughout this book, the reader has encountered many functors, some notable examples being $\operatorname{Hom}_R(-, N)$, $\operatorname{Hom}_R(M, -)$ and $M \otimes_R -$ among many others. Each of these functors give some information about the objects and the morphisms passing through the functor. As we saw for the Hom and $M \otimes_R -$ functors, certain appropriate objects preserve exactness, while most preserve it only partially. Perhaps a crowning achievement of mid 20th century mathematics is to be able to quantify this lack of exactness, and rather spectacularly use the information given by this quantification to extract a (sometimes copious) amount of information about the objects of study. The great effectiveness with which homological algebra can produce interesting information may seem surprising at first, for example, using homological algebra the notion of *dimension* as it is understood in geometry is readily translatable to algebra, or that we may measure how far away a module is from being a vector-space (details are in K-theory)[1] How this is the case is perhaps best motivated by going through learning the subject to see for oneself, but it may be helpful for some readers to have some of my (the authors) intuition presented to stand on some ground before proceeding.

If we have two objects M_1 and M_2, we may wish to consider a way of constructing a new object M given M_1 and M_2. Ideally, we may form a short exact sequence like so:

$$0 \to M_1 \to M \to M_2 \to 0$$

More generally, we may have a collection of objects $\{M_i\}_{i \in I}$ which we use to construct an object via a long exact sequence. However, it is often not the case that such an exact sequence is too strong, that is, it is not the case that $\operatorname{im}(\varphi_i) = \ker(\varphi_{i+1})$. Even if such a construction exists, we often then want to "transform" this information via a functor which shall then lose us exactness (one can think of a functor as asking some sort of "question" whose output is the answer). In this case, it has been eerily common that the image of φ_i is contained in the kernel of φ_{i+1}:

$$\operatorname{im}(\varphi_i) \subseteq \ker(\varphi_{i+1}) \tag{54.1}$$

Conceptually, this can be thought as the object "being obstructed" from fully being an extension. This new type of sequence is called a *cochain complex.* The reader with a good memory may recall definition 12.2.11 where it was given that another name for a long exact sequence is an exact cochain complex. In this way, the condition given by equation 54.1 is a generalization of the exact condition. The theoretical heart of homological algebra can be thought of how to extract "quantitatively" how a cochain complex *fails* to be an exact cochain complex. For example, in chapter 17⋆ it was have eluded that through the use of homology theory, the Ext functor gives us all the ways two modules M_1, M_2 may be extended. In this case, the failure of $\operatorname{Hom}(M, -)$ or $\operatorname{Hom}(-, N)$ preserving injectivity or surjectivity (equivalently, the "obstruction" to these functors being exact) gave rise to the Ext functor and the study of its homology.

The term "algebraic object" has been is a bit ambiguous. What type of algebraic objects can be used to in the theory of homological algebra? As it turns out, not all algebraic objects satisfy the right conditions for the theory of homological algebra (for example, **Grp** and **Fld** will be shown to be inappropriate for a theory of homology). However, there are non-algebraic objects for which a homological theory can be induced (famously, topological spaces and smooth manifolds admit a theory of homological algebra). This spurred the need to generalize the theory of homological algebra to the categorical level, where homological theory is well-defined on *abelian categories.* We thus shall start homological algebra with an overview of abelian categories. For those who are already familiar with some homological algebra, they may have studied homological algebra within the category of

[1] This is overviewed in chapter 57

R-**Mod**, which is a concrete abelian category. Doing so will in fact not loose any generality due to Freyd-Mitchell Theorem (theorem 55.1.15) which states that any [small] abelian category A admits a fully faithful exact functor to a category R-**Mod** for a suitable ring R. Working with the concrete category R-**Mod** brings it benefits as it may be easier to conceptualize the manipulation of elements, and so the reader may ask for the benefit in sticking to working with general abelian categories. As the author, I believe the benefits comes in being comfortable with more advanced category theory. Future algebraists are going to encounter more complicated problems involving category theory (higher order categories, topoi's, and stacks, are all examples the reader may see in the not too distant future). Hence, introducing a categorical perspective will familiarize the reader for more complicated categorical notions in their not too distant future[2].

The reader may wonder why a cochain complex is not called a chain complex. This is because we can work in the dual category, where all arrows are flipped. In the dual category, we shall get chain complexes. Homological algebra in the dual category is theoretically the same, however we shall see that cohomology is often computationally easier to work in, permitting more useful algebraic structures we can use to study. It is for this reason that most mathematicians work with cohomology rather than homology, even though every theorem we shall show for homology translates directly to cohomology[3]. In this book, we shall start with homology, and switch to cohomology when starting to work with more computational parts of the theory to get the reader accustomed to both modes of thinking.

Goal

Chapter 55 shall introduce the theory of abelian categories which shall be the types of categories in which homological algebra is well-defined.

The main result Chapter 56 of the book is trying to convey is what information from R-modules give rise to certain properties in its respective chain complex(es), and conversely what information from it's chain complex(es) gives us information about modules. A (co)chain complex need not be exact, and (co)homology shall be the algebraic object quantifying this failure of exactness. We shall see that given an object M within an abelian category, there are in general an infinite number of ways we may associate a [co]chain complex to it. Thus, there will have to be a way to either associate a canonical [co]chain complex (in a natural functorial manner), or we assign [co]chain complexes to an object up to appropriate notion of isomorphism. The latter shall be the approach in this book, where we shall see that we will want to consider [co]chain complexes up to something called *quasi-isomorphism* of [co]chain complexes. We shall then see what cohomological information we may deduce given different choices of functors and how they transform [co]chain complexes.

After having developed the ground work for extracting cohomological information, in Chapter 57 we shall put this knowledge to good use and apply cohomology theory to G-invariant functors and see what this this can tell us about groups. This shall also finally address what was foreshadowed in box 20.4.5 where it was said the units of a Galois field form a trivial 1st homology group. We shall define the notion of homological dimension which shall line up with the results presented in the section on commutative algebra. Finally, we present an interesting technical result to check flatness of a module over a local ring that builds on the same intuitions the Nakayama lemma (lemma 52.1.7)

[2]assuming, naturally, that the reader holds great interest in algebra and shall keep pursuing branches within the field

[3]there is even an equivalent algebraic structure for the additional algebraic structure that is induced in cohomology, however it is less useful

presented.

Finally, this Part concludes with the introduction to spectral sequences in chapter 58, which are the quintessential tool for many concrete computations in homological algebra and any serious work in many disciplines of algebra and geometry. The main idea is that most object of study can be broken down into a filtration of simpler objects whose cohomology is more computed. The spectral sequence will put piece together this cohomological information.

Connection to Singular Homology/Cohomology

For readers familiar with algebraic topology and the theory of (co)homology, the connection between the above abstraction and the homology theory of topological spaces comes from noticing that singular (co)homology on a manifold can be interpreted as taking the projective resolution $C_n(X) = \mathrm{Hom}_{\mathbf{Top}}(\Delta^n, X)$. In particular, given M, we can apply $C_*(-)$ to it and get a projective (even free) resolution from which we extract homological information. This should also motivate the use of $\mathrm{Hom}_R(M, -)$ and $\mathrm{Hom}_R(-, N)$ as the "right" functors to be considering when generalizing classical algebraic topology to the algebraic setting. As the tensor product it adjoint (theorem 15.3.9) it is another natural candidate.

54.2.11 Foreshadowing: Yoneda's Lemma The Hom is very useful as it is "represented" by an element (either in the domain or codomain). Even more is true: it can be used to represent other functors. In theorem 60.2.1 this is studied *representable functors* which are functors where $F(-) \Rightarrow \mathrm{Hom}_C(A, -)$ for natural transformation $\Rightarrow$, category C, object $A \in C$, and $F : C \to \mathbf{Set}$. The Yoneda lemma will give a way of representing objects using Hom.

Historical Origins

While modern homological algebra is phrased in the language of categories, its machinery evolved from two distinct 19th-century sources which converged in the mid-20th century.

The algebraic lineage begins around 1890 with David Hilbert's Invariant Theory. To prove the Basis Theorem, Hilbert studied the relations ("syzygies") between generators of an ideal. He constructed a chain of free modules to resolve these relations:

$$0 \to F_n \to \cdots \to F_1 \to F_0 \to M \to 0$$

Hilbert proved the *Syzygy Theorem*, showing that for polynomial rings, this chain of relations eventually terminates in an exact sequence of free modules. However, this theory encountered a significant obstruction when applied to rings with singularities (such as coordinate rings of singular curves). In these non-regular settings, Hilbert's resolutions do not terminate but continue infinitely. This failure shifted the mathematical focus from simply computing invariants to essentially classifying the severity of a singularity by the infinite length of its resolution.

Parallel to this, in the 1930s, number theorists like Emmy Noether, Richard Brauer, and Helmut Hasse were attempting to classify division algebras (generalizations of the quaternions) over number fields. They classified these algebras using "Crossed Products," which relied on complex systems of functions called "Factor Sets" ($c : G \times G \to K^\times$). While they successfully defined the *Brauer Group* of a field to organize these algebras, the calculations were notoriously opaque. The "Factor Sets" satisfied a messy algebraic identity that seemed arbitrary, acting as a computational blocker that obscured the underlying structure of the fields they were studying.

Simultaneously, from 1895 onwards, the topological roots of the subject were forming as Henri Poincaré classified manifolds. He decomposed spaces into simplices and represented their boundaries using incidence matrices. He defined Betti numbers as the ranks of these matrices. However, Poincaré's numerical approach had a blind spot: rank obliterates "torsion." A twisted shape like the projective plane appeared numerically identical to a trivial one (both have rank 0 in dimension 1).

In 1925, Emmy Noether revolutionized this view. She realized that the matrices topologists were diagonalizing were actually presentations of abelian groups. By shifting focus from the numerical rank to the *Homology Group* itself ($\ker \partial / \operatorname{im} \partial$), she captured the torsion information, effectively "algebraizing" the topology. She showed that the matrix was not just a calculator for a number, but a presentation of a structure.

The convergence of these fields occurred in the 1940s when algebraists attempted to classify *Group Extensions*. Saunders Mac Lane and Samuel Eilenberg attempted to classify these extensions and rediscovered the same "Factor Sets" used by Brauer and Hasse. They discovered that the algebraic identity these functions satisfied was *identical* to the boundary formula for a 2-simplex in topology; essentially they realized they were calculating $H^2(G, N)$ (see [13] for this computation in the EYNTKA series). This was the turning point: it proved that "cohomology" was not just a property of geometric shapes, but a fundamental algebraic structure that arose whenever one tried to "extend" one object by another, whether in topology, group theory, or number theory.

The grand unification arrived in 1956 when Cartan and Eilenberg realized that the topological concept of "boundaries" and the algebraic concept of "extensions" were instances of the same phenomenon: the *failure of exactness*. They observed that while Hilbert's resolution is exact, applying a functor (like the tensor product or Hom) usually destroys that exactness. They realized that the "homology" groups were simply the mathematical terms required to repair this broken exactness. This birth of the *Derived Functor* (Tor and Ext) transformed the field from the study of static objects to the study of how functors distort structure. To build this machinery, they faced a technical blocker: Hilbert's reliance on "Free Modules." While free modules worked for Homology (Tor), they failed for the dual theory of Cohomology (Ext), as the dual of a free module is rarely free. To achieve the necessary symmetry, they replaced the specific condition of "Freeness" with the categorical properties of *Projectivity* and *Injectivity*. This allowed them to define derived functors for any ring, regardless of whether a basis existed.

With this machinery established, the period from 1955 to 1980 saw these newly developed tools being used to solving the original obstructions in geometry and number theory. In geometry, Jean-Pierre Serre proved that a local ring is regular (non-singular) if and only if it has finite global homological dimension (theorem 57.2.8). This transformed the "problem" of infinite resolutions into a precise detection tool: the non-terminating resolution was effectively the algebraic signature of a geometric singularity. Alexander Grothendieck later introduced *Local Cohomology* ($H^i_{\mathfrak{m}}$) to quantify the depth and severity of these singularities, eventually leading to Intersection Homology which restored Poincaré Duality to singular spaces.

In number theory, the mysterious "Factor Sets" of the 1930s were identified as elements of the second Galois Cohomology group, $H^2(\mathrm{Gal}(L/K), L^\times)$. This realization allowed Emil Artin and John Tate to rewrite Class Field Theory entirely in the language of cohomology. The obscure identities of the past were revealed to be standard cohomological relations, and the *Tate Cohomology* groups ($\hat{H}^i$) were developed specifically to handle the arithmetic of finite groups, solving the classification problems that had stalled the previous generation (see [39] for the EYNTKA series exposition of modern class field theory).

55

Abelian Categories

This chapter focuses on the category in which we may define "cohomological theories" as not all categories are well-suited for this endeavor

55.1 Definitions and Properties

We have extensively worked with categories which have a multitudes of different properties. Some important terms to recall are the following:

- *zero-object*: an object 0 in a category C $0 \in \mathrm{Obj}(C)$ that is both the initial and final object. Such an object is also called a terminal object. Similarly, a zero morphism is a morphism $f : X \to Y$ such that for any other morphisms $g, h : X \to Y$, $fg = fh$ and $hf = gf$.
- *Products* and *Coproducts*: Given two objects $a, b \in \mathrm{Obj}(C)$, if there exists an object $c \in \mathrm{Obj}(C)$ that is the limit (resp. colimit) of the diagram:

$$\overset{A}{\bullet} \qquad \overset{B}{\bullet}$$

  then c is the product (resp. coproduct) of a and b, and is usually written as $a \times b$ (resp. $a \amalg b$)
- *kernels* and *cokernels*: Given a morphism $f : a \to b$, the kernel (resp. cokernel) of f is an object k and a morphism $\iota : k \to a$ such that k is the limit (resp. colimit) of the diagarm:

$$\overset{a}{\bullet} \xrightarrow{f} \overset{b}{\bullet}$$

  with the added restriction that the map $\varphi : k \to y$ must be the zero morphism (namely, by commutativity, $f\iota = 0_{kb}$ must be the zero morphism). In terms of diagrams, we get that that for any $\iota' : K \to A$ such that $f\pi\iota' = 0$, that there exists a morphism φ st the following diagram commutes:

$$\begin{tikzcd}
K' \arrow[r, "\iota'"'] \arrow[rr, bend left, "0"] \arrow[dr, dashed, "\varphi"'] & A \arrow[r, "\varphi"'] & B \\
& K \arrow[u, "\iota"'] &
\end{tikzcd}$$

and simmilarly for cokernels:

$$\begin{tikzcd}
& K' \arrow[d, "\pi'"] \arrow[dr, dashed, "\psi"] & \\
A \arrow[r, "\varphi"] \arrow[rr, bend right, "0"'] & B \arrow[r, "\pi"] & K
\end{tikzcd}$$

Kernels and cokernels do not necessarily exist in a category, or not all morphisms have kernels (ex. the category of finitely generated R-modules for any ring R need has morphisms without kernels). Furthermore, monomorphism need not be kernels (not all subgroups are normal) and epimorphisms need not be cokernels (recall $\mathbb{Z} \to \mathbb{Q}$ is an epimorphism in **Ring**)

Not all categories we have worked with have these properties. For example, **Fld** doesn't have products (what would be the characteristic of the product of $\mathbb{F}_p$ and $\mathbb{F}_q$ for $p \neq q$?), some categories don't have zero morphisms (set being an example), some have monomorphisms/epimorphisms that do not correspond to kernels (as showed above). We first limit our attention to categories which have these nice properties:

Definition 55.1.1: Additive Category

Let $\mathbf{A}$ be a category. Then $\mathbf{A}$ is said to be *additive* if it has a zero-object, has both finite products and coproducts, and each $\text{Hom}_A(X, Y)$ can be given an abelian group structure such that the composition of maps:

$$\text{Hom}_A(X, Y) \times \text{Hom}_A(Y, Z) \to \text{Hom}_A(X, Z)$$

is bilinear. A functor between additive categories is *additive* if it is a group homomorphism on hom-sets.

A functor between two additive categories is called an *additive functor* if the maps between the hom-sets is a homomorphism.

The term "additive" comes froms the fact that we may interpret the additive category as a monoidal category (a category which we may add a functor $\otimes : C \times C \to C$ that is given the property of a monoid operation, for an additive category this would be the direct product), where the hom-sets are "enriched" to also be abelian groups. Another way of looking at additive categories is in which the algebra of matrices is well-defined.

Example 55.1.2: Additive Categories

1. The category of R-modules is an example of an additive category.

2. **Grp** is not additive, for $\mathrm{Hom}_{\mathbf{Grp}}(A, B)$ doesn't have an abelian group strucuture satisfying the above condition.

3. The category of matrices over a ring is an additive category.

4. The category of finitely generated R-modules is an additive category

A key realisation is that in the absence of kernels and cokernels as objects monomorphisms and epimorphisms take their place:

Lemma 55.1.3: Additive Categories: Monomorphisms and Kernels I

Let A be an additive category. Then kernels are monomorphisms and cokernels are epimorphisms

***Proof*:**
Let $k\colon K \to A$ be a kernel of $\varphi\colon A \to B$, and let $f, g\colon X \to K$ satisfy $k \circ f = k \circ g$. Because the hom-sets of an additive category are abelian groups and composition is bilinear, this says $k \circ (f - g) = 0$.

Now apply the universal property of the kernel to the zero morphism $0\colon X \to A$. It satisfies $\varphi \circ 0 = 0$, so there is a *unique* morphism $u\colon X \to K$ with $k \circ u = 0$. Both $f - g$ and the zero morphism $X \to K$ have this property, so uniqueness forces $f - g = 0$, that is $f = g$. Hence k is a monomorphism.

The statement for cokernels is the same argument in the opposite category: a cokernel of φ in A is a kernel of φ in A^{op}, an additive category, and a monomorphism there is an epimorphism in A, completing the proof.

Note that kernels and cokernels need not exist in an additive category, hence the converse of each accession need not be the case (in a vacuous way). If a morphism φ has a kernel then we may identify exactly what this kernel is:

Lemma 55.1.4: Additive Categories: Monomorphisms and Kernels II

Let A be an additive category and $\varphi : X \to Y$ a morphism. Then:

1. If φ has a kernel, then φ is a monomorphism if and only if $0 \to X$ is its kernel

2. If φ has a cokernel, then φ is an epimorphism if and only if $Y \to 0$ is its cokernel

***Proof*:**

1. Suppose first that the zero morphism $0 \to X$ is a kernel of φ, and let $f, g\colon Z \to X$ satisfy $\varphi \circ f = \varphi \circ g$. Then $\varphi \circ (f - g) = 0$, so $f - g$ factors through the kernel: there is $u\colon Z \to 0$ with $f - g = 0 \circ u = 0$. Hence $f = g$ and φ is a monomorphism.

   Conversely suppose φ is a monomorphism and let $k\colon K \to X$ be a kernel of φ. From

$\varphi \circ k = 0 = \varphi \circ 0$ and φ monic it follows that $k = 0$. But k is itself a monomorphism by lemma 55.1.3, and $k \circ \mathrm{id}_K = 0 = k \circ 0$, so $\mathrm{id}_K = 0$. An object whose identity is the zero morphism is a zero object, so $K \cong 0$ and the kernel is $0 \to X$.

2. This is the statement of (1) in the opposite category, where kernels and monomorphisms become cokernels and epimorphisms.

This completes the proof.

As mentioned, it is not guaranteed that kernels and cokernels exist. We shall shortly axiomatically add them to our category by adding them into the definition of a new category that supersedes additive categories, but first we show a few more important results about abelian categories. Namely, in an additive category, products and coproducts are equivalent:

Lemma 55.1.5: Additive Categories: Products and Coproducts

Let A be an additive category and $X, Y \in \mathrm{Obj}(A)$. Then if $X \times Y$ exists, so does $X \amalg Y$, and if $X \amalg Y$ exists, so does $X \times Y$. Furthermore,

$$X \times Y \cong X \amalg Y$$

***Proof*:**
Suppose $P = X \times Y$ exists, with projections $p_X : P \to X$ and $p_Y : P \to Y$. Let $i_X : X \to P$ be the unique morphism with $p_X i_X = \mathrm{id}_X$ and $p_Y i_X = 0$, and let $i_Y : Y \to P$ be defined symmetrically; both exist by the universal property of the product.

Step 1: The key identity $i_X p_X + i_Y p_Y = \mathrm{id}_P$. Two morphisms out of P into P agree as soon as their composites with p_X and with p_Y agree, again by the universal property of the product. Compute

$$p_X(i_X p_X + i_Y p_Y) = (p_X i_X)p_X + (p_X i_Y)p_Y = \mathrm{id}_X \circ p_X + 0 \circ p_Y = p_X$$

using bilinearity of composition, and symmetrically $p_Y(i_X p_X + i_Y p_Y) = p_Y$. Since id_P has the same two composites, the identity holds.

Step 2: (P, i_X, i_Y) is a coproduct. Let $f : X \to Z$ and $g : Y \to Z$ be given and set $h = f p_X + g p_Y : P \to Z$. Then

$$h i_X = f(p_X i_X) + g(p_Y i_X) = f, \qquad h i_Y = f(p_X i_Y) + g(p_Y i_Y) = g$$

so h is a morphism with the required restrictions. For uniqueness, suppose h' also satisfies $h' i_X = f$ and $h' i_Y = g$. Then by Step 1,

$$h' = h' \mathrm{id}_P = h'(i_X p_X + i_Y p_Y) = (h' i_X)p_X + (h' i_Y)p_Y = f p_X + g p_Y = h$$

Hence P with i_X, i_Y satisfies the universal property of $X \amalg Y$, and in particular $X \times Y \cong X \amalg Y$.

If instead $X \amalg Y$ is assumed to exist, the same argument in the opposite category produces the product, completing the proof.

Hence, additive categories behave very much like abelian groups, however as we pointed out kernels and cokernels need not exist. We thus define a new category that forces these conditions:

Definition 55.1.6: Abelian Category

[Abelian Category] Let A be an additive category. Then A is an *abelian* category if

1. kernels and cokernels exist in A;
2. every monomorphism is the kernel of some morphism, and every epimorphism is the cokernel of some morphism.

The second condition is not a consequence of the first, and it is the one that does the work. An additive category satisfying only (1) is called *pre-abelian*, and there the canonical map from the coimage to the image of a morphism (theorem 55.1.9) need not be an isomorphism; the category of torsion-free abelian groups is pre-abelian but not abelian, since $\mathbb{Z} \xrightarrow{2} \mathbb{Z}$ is there both a monomorphism and an epimorphism without being an isomorphism. Condition (2) rules this out, and is what makes lemma 55.1.8 true.

Note that lemma 55.1.4 is a weaker statement, not this one: it characterizes a monomorphism by having *zero* kernel, which says nothing about whether the monomorphism is itself a kernel.

Since each kernel is associated to a monomorphism, we may intuitively think of the association:

$$\text{kernel} \;\Leftrightarrow\; \text{submodule}$$

and similarly:

$$\text{cokernel} \;\Leftrightarrow\; \text{quotient module}$$

In many ways, an abelian category should be thought of as the minimum number of conditions necessary to define exact sequence[1]. To see this, we require the following:

Lemma 55.1.7: Abelian Category: Kernels And Cokernels

Let A be an abelian category. Then every kernel is the kernel of its cokernel, and every cokernel is the cokernel of its kernel.

***Proof*:**
Let $k\colon K \to A$ be a kernel, say of $\varphi\colon A \to B$, and let $c\colon A \to C$ be a cokernel of k. The claim is that k is a kernel of c.

First, $c \circ k = 0$ holds by the definition of a cokernel. Now let $t\colon T \to A$ satisfy $c \circ t = 0$; a unique factorization of t through k is needed. Since $\varphi \circ k = 0$ and c is the cokernel of k, the morphism φ factors as $\varphi = \bar{\varphi} \circ c$ for some $\bar{\varphi}\colon C \to B$. Therefore

$$\varphi \circ t = \bar{\varphi} \circ c \circ t = 0$$

and since k is the kernel of φ, there is a unique $u\colon T \to K$ with $k \circ u = t$. Uniqueness is automatic because k is a monomorphism by lemma 55.1.3. So k satisfies the universal property of $\ker c$.

[1] Those familiar with some algebraic topology may find it insightful that an abelian category is the the category with ninimum number of condition so that the snake lemma (theorem 56.1.15) holds

The dual statement, that every cokernel is the cokernel of its kernel, is this argument in the opposite category, completing the proof.

Lemma 55.1.8: Abelian Categories: Isomorphisms

[Abelian Categories, Isomorphisms] Let $\mathcal{A}$ be an abelian category and $\varphi : A \to B$ a morphism in $\mathcal{A}$. Then if φ is a monomorphism and an epimorphism, it is an isomorphism

Note that this does not hold in additive categories, nor even in pre-abelian ones: condition (2) of definition 55.1.6 is exactly what the proof below consumes.

***Proof*:**
Since φ is an epimorphism, condition (2) of definition 55.1.6 makes it the cokernel of some morphism $\psi\colon C \to A$. By the definition of a cokernel $\varphi \circ \psi = 0$, so ψ factors through $\ker \varphi$ by the universal property of the kernel.

Now φ is also a monomorphism, so $\ker \varphi = 0$ by lemma 55.1.4. The factorization of the previous paragraph then runs through the zero object, so $\psi = 0$ outright and φ is a cokernel of the zero morphism $0 \to A$. But the identity $\mathrm{id}_A\colon A \to A$ is also a cokernel of $0 \to A$: any $f\colon A \to X$ with $f \circ 0 = 0$, which is every f, factors uniquely through id_A. Cokernels are unique up to unique isomorphism, so there is an isomorphism $A \to B$ commuting with φ and id_A, which forces φ to be that isomorphism, as we sought to show.

Theorem 55.1.9: Abelian Category: Canonical Decomposition

[Abelian Category, Canonical Decomposition] Let $\mathcal{A}$ be an abelian category and $\varphi : A \to B$ a morphism. Write $C = \operatorname{coker}(\ker \varphi)$ and $K = \ker(\operatorname{coker}\varphi)$. Then φ decomposes in the following cannonical way:

$$\begin{tikzcd} A \arrow[r, two heads] \arrow[rrr, bend left, "\varphi"] & C \arrow[r, "\tilde{\varphi}"'] & K \arrow[r, tail] & B \end{tikzcd}$$

where the first morphism is an epimorphism, the last a monomorphism, and the middle morphism $\tilde{\varphi}$ is an *isomorphism*.

***Proof*:**
Write $k\colon N \to A$ for $\ker \varphi$ and $p\colon A \to C$ for its cokernel, and $q\colon B \to Q$ for $\operatorname{coker}\varphi$ and $\iota\colon K \to B$ for its kernel. By lemma 55.1.3 the morphism p is an epimorphism and ι is a monomorphism.

Step 1: φ factors through ι. Since q is the cokernel of φ, $q \circ \varphi = 0$, and since ι is the kernel of q there is a unique $\varphi'\colon A \to K$ with $\varphi = \iota \circ \varphi'$.

Step 2: φ' factors through p. From $\iota \circ \varphi' \circ k = \varphi \circ k = 0$ and ι monic it follows that $\varphi' \circ k = 0$. As p is the cokernel of k, there is a unique $\tilde{\varphi}\colon C \to K$ with $\varphi' = \tilde{\varphi} \circ p$. Combining, $\varphi = \iota \circ \tilde{\varphi} \circ p$, which is the displayed factorization.

Step 3: ι is the smallest subobject through which φ factors. Let $m\colon V \to B$ be any monomorphism with $\varphi = m \circ \psi$ for some ψ. By condition (2) of definition 55.1.6 the monomorphism m is a kernel, so by lemma 55.1.7 it is the kernel of its own cokernel; write $d\colon B \to D$ for coker$m$, so that $m = \ker d$. Then

$$d \circ \varphi = d \circ m \circ \psi = 0$$

so d factors through the cokernel of φ, say $d = \bar{d} \circ q$. Since $\iota = \ker q$ gives $q \circ \iota = 0$, it follows that $d \circ \iota = \bar{d} \circ q \circ \iota = 0$, and therefore ι factors through $\ker d = m$.

Step 4: φ' is an epimorphism. Let $c\colon K \to W$ be the cokernel of φ' and $n\colon V \to K$ its kernel. From $c \circ \varphi' = 0$ the morphism φ' factors as $\varphi' = n \circ \psi$, whence $\varphi = \iota \circ n \circ \psi$. The composite $\iota \circ n$ is a monomorphism, being a composite of two (each is a kernel, so lemma 55.1.3 applies), so Step 3 gives a t with $\iota = \iota \circ n \circ t$. As ι is monic this yields $n \circ t = \mathrm{id}_K$, so n is a split epimorphism as well as a monomorphism, hence an isomorphism by lemma 55.1.8. Then $c \circ n = 0$ with n invertible forces $c = 0$; but c is a cokernel, hence an epimorphism by lemma 55.1.3, and $\mathrm{id}_W \circ c = 0 = 0 \circ c$ then gives $\mathrm{id}_W = 0$, so $W \cong 0$. A morphism with zero cokernel is an epimorphism, so φ' is one.

Step 5: $\tilde{\varphi}$ is an isomorphism. By Step 4 the composite $\tilde{\varphi} \circ p = \varphi'$ is an epimorphism, and if $u \circ \tilde{\varphi} = 0$ then $u \circ \tilde{\varphi} \circ p = 0$ and hence $u = 0$; so $\tilde{\varphi}$ is an epimorphism. Dually, running Steps 1 to 4 in the opposite category shows that the induced morphism $C \to B$, which is $\iota \circ \tilde{\varphi}$, is a monomorphism; and a composite $g \circ h$ can be a monomorphism only if h is, so $\tilde{\varphi}$ is a monomorphism. Being both, it is an isomorphism by lemma 55.1.8, completing the proof.

From the above, we may think of ker(coker) : $K \to B$ as representing the "image" of φ:

Lemma 55.1.10: Abelian Category: Images

Let $\varphi : A \to B$ be a morphism in an abelian category, and let $\iota : K \to B$ be the kernel of the cokernel of φ. Then:

1. ι is a monomorphism
2. φ factors through ι
3. ι is initial with these properties

***Proof*:**
Part (1) is lemma 55.1.3, since ι is a kernel. Part (2) is Step 1 of the proof of theorem 55.1.9: writing $q = \mathrm{coker}\varphi$, the identity $q \circ \varphi = 0$ makes φ factor uniquely through $\iota = \ker q$.

Part (3) is the minimality established in Step 3 of that proof: if $m\colon V \to B$ is any monomorphism through which φ factors, then ι factors through m. Concretely, m is a kernel by condition (2) of definition 55.1.6, hence the kernel of its own cokernel d by lemma 55.1.7; then $d \circ \varphi = 0$ makes d factor through q, so $d \circ \iota = 0$ and ι factors through $\ker d = m$, as we sought to show.

Definition 55.1.11: Abelian Category: Image

Let $\varphi : A \to B$ be a morphism in an abelian category. Then the *iamge* of φ is $\ker(\operatorname{coker}(\varphi))$ and is denoted $\operatorname{im}(\varphi)$. Similarly, its co-image is $\operatorname{coker}(\ker(\varphi))$ and is dentoed $\operatorname{coim}(\varphi)$.

With our intuition from working with R-modules, given a morphism $\varphi : A \to B$ with image $\iota : K \to B$, we may imagine there is an induced epimorphism $\overline{\varphi} : A \to K$. This is indeed the case:

Proposition 55.1.12: Abelian Category: sub object and quotient objects

Let $\varphi : A \to B$ be a morphism in an abelian category with image $\operatorname{im}\varphi\colon K \to B$ and coimage $\operatorname{coim}\varphi\colon A \to C$. Then the induced morphism $A \to K$ is an epimorphism and the induced morphism $C \to B$ is a monomorphism.

***Proof*:**
The induced morphism $A \to K$ is the morphism φ' of theorem 55.1.9, shown to be an epimorphism in Step 4 of that proof. The induced morphism $C \to B$ is the composite $\iota \circ \tilde{\varphi}$, a composite of a monomorphism with an isomorphism, hence a monomorphism, completing the proof.

From the cannonical decomposition, we may say that the image and coimage are "isomorphic" (which is not strictly true since these are morphisms, however by the Freyd-Mitchell Theorem this is a safe way of looking at it).

Theorem 55.1.13: Abelian Category: Exactness

Consider a sequence of objects in an abelian cateogy

$$\cdots \to A \xrightarrow{\varphi} B \xrightarrow{\psi} C \to \cdots$$

Then this sequence is *exact* if:

1. $\psi \circ \varphi = 0$
2. $\operatorname{coker}\varphi \circ \ker\psi = 0$

Example 55.1.14: Categorical Exactness

1. Aluffi p.577

Freyd-Mitchell Theorem

As mentioned, R-mod is an abelian category, and as we shall develop a bit later it the categories R-mod "represent" all abelian categories.

Theorem 55.1.15: Freyd-Mitchell Theorem

Let A be a small abelian category. Then there is a fully faithful exact functor $A \to R$-Mod for a suitable ring R. Such a functor reflects exactness: a sequence in A is exact if and only if its image in R-Mod is.

***Proof Key ideas*:**
The proof constructs R as the endomorphism ring of a suitable injective cogenerator in the category of left exact functors $A^{\mathrm{op}} \to \mathbf{Ab}$, and occupies a chapter's worth of category theory that this book does not otherwise need; it is given in full in *An Introduction to Homological Algebra* [53], §1.6. The idea is that A embeds into its category of left exact presheaves by Yoneda, that this larger category is a Grothendieck category with enough injectives, and that a single injective cogenerator there converts the whole category into modules over its endomorphism ring.

The clause about reflecting exactness is immediate from the rest and is what the theorem is used for: a fully faithful exact functor reflects kernels and cokernels, so a diagram chase valid in R-Mod transfers back to A. This is what licenses the element-based arguments in lemma 56.1.17 and throughout chapter 58.

55.2 Localization of Categories

The constructions of this book repeatedly produce morphisms that ought to be invertible but are not. The model for what to do about it is already familiar from commutative algebra: a ring R and a multiplicative set S of elements one wishes to invert are replaced by $S^{-1}R$, together with a map $R \to S^{-1}R$ that is universal among ring maps sending S into the units. Nothing in that description mentions elements of a ring rather than morphisms of a category, and the same universal property can be imposed one level up.

The payoff comes in chapter 56, where the class to be inverted is the class of quasi-isomorphisms and the resulting category is the derived category. The construction is recorded here, before it is needed, because it is a statement about categories and not about complexes.

Definition 55.2.1: Localization of a Category

Let $\mathcal{C}$ be a category and W a class of morphisms of $\mathcal{C}$. A *localization* of $\mathcal{C}$ at W is a category $\mathcal{C}[W^{-1}]$ together with a functor $Q\colon \mathcal{C} \to \mathcal{C}[W^{-1}]$ such that

1. $Q(w)$ is an isomorphism for every $w \in W$;
2. for every category $\mathcal{D}$ and every functor $F\colon \mathcal{C} \to \mathcal{D}$ carrying each element of W to an isomorphism, there is a functor $\overline{F}\colon \mathcal{C}[W^{-1}] \to \mathcal{D}$ with $\overline{F} \circ Q = F$, and $\overline{F}$ is unique with this property.

As with every universal property, the definition determines the answer up to canonical isomorphism as soon as an answer exists: two localizations receive unique comparison functors from one another whose composites must be the identity, by the uniqueness clause applied to Q itself. Existence is the substantive point, and it is settled by a construction that simply writes down the morphisms that formal inversion forces to exist.

Theorem 55.2.2: Existence of the Localization

Let $\mathcal{C}$ be a category and W a class of morphisms. Then a localization $Q\colon \mathcal{C} \to \mathcal{C}[W^{-1}]$ exists.

***Proof*:**

Step 1: The construction. Let $\mathcal{C}[W^{-1}]$ have the same objects as $\mathcal{C}$. For objects X and Y, consider finite *zigzags* from X to Y: alternating strings of morphisms of $\mathcal{C}$

$$X = X_0 \;\vdash\!\!-\; X_1 \;\vdash\!\!-\; \cdots \;\vdash\!\!-\; X_n = Y$$

in which each link $X_i \vdash\!\!- X_{i+1}$ is either a morphism $X_i \to X_{i+1}$ of $\mathcal{C}$, called a forward link, or a morphism $X_{i+1} \to X_i$ that belongs to W, called a backward link. Enlarging W to contain all identities and to be closed under composition changes neither the class of functors inverting it nor, therefore, the localization, so assume this without loss of generality; it is what makes the second move below well formed. Impose on the collection of zigzags the smallest equivalence relation generated by: replacing two consecutive forward links by their composite; replacing two consecutive backward links by their composite; deleting a forward or backward link that is an identity; and deleting a pair of adjacent links w forward followed by w backward, or w backward followed by w forward, for $w \in W$. Let $\mathrm{Hom}_{\mathcal{C}[W^{-1}]}(X, Y)$ be the collection of equivalence classes, with composition given by concatenation, which is well defined because the generating relations are local to the links they touch. Concatenation is associative and the empty zigzag is an identity, so this is a category. Let Q send an object to itself and a morphism to the one-link forward zigzag.

Step 2: Q inverts W. For $w \in W$ the backward one-link zigzag on w is a morphism in the opposite direction, and concatenating it with the forward one-link zigzag on w in either order produces a pair of adjacent links which the last generating relation deletes, leaving the empty zigzag. So $Q(w)$ is an isomorphism.

Step 3: The universal property. Let $F\colon \mathcal{C} \to \mathcal{D}$ carry W into isomorphisms. Define $\overline{F}$ to be F on objects, and on a zigzag to be the composite in $\mathcal{D}$ obtained by applying F to each forward link and $F(-)^{-1}$ to each backward link. Each generating relation is respected: composites go to composites by functoriality, identities to identities, and an adjacent pair w, w of opposite orientation goes to $F(w)F(w)^{-1}$ or $F(w)^{-1}F(w)$, both identities. So $\overline{F}$ is well defined on equivalence classes, and it is a functor because concatenation goes to composition. By construction $\overline{F} \circ Q = F$.

For uniqueness, suppose $G \circ Q = F$ for a functor G. Every morphism of $\mathcal{C}[W^{-1}]$ is a concatenation of one-link zigzags, each of which is $Q(f)$ or $Q(w)^{-1}$; a functor is determined by its values on these, and $G(Q(f)) = F(f)$ forces $G(Q(w)^{-1}) = G(Q(w))^{-1} = F(w)^{-1}$. So G agrees with $\overline{F}$ on every morphism, completing the proof.

55.2.3 Warning: Localization and Set Theory The construction above builds each $\mathrm{Hom}_{\mathcal{C}[W^{-1}]}(X,Y)$ out of *all* zigzags, and for a category $\mathcal{C}$ that is not small there is no guarantee that the result is a set rather than a proper class. The property at risk is *local smallness*: the localization of a locally small category need not be locally small.

Two remedies are standard. The first is to assume $\mathcal{C}$ small, which is of no use here since $\mathrm{Ch}(\mathbf{A})$ never is. The second is to exhibit a calculus of fractions as in theorem 55.2.5, which replaces arbitrary zigzags by roofs; note that this alone does not bound anything, since the roofs over a fixed object are indexed by a proper class. What is needed in addition is that the W-morphisms into each object admit a *set* that is cofinal among them, and a multiplicative system with that property is what makes the localization locally small.

Neither remedy applies to $\mathrm{Ch}(\mathbf{A})$ localized at quasi-isomorphisms, because the quasi-isomorphisms there are *not* a multiplicative system. Both conditions (2) and (3) of definition 55.2.4 fail: taking $C^\bullet = (\mathbb{Z} \xrightarrow{\mathrm{id}} \mathbb{Z})$ in degrees $-1, 0$, which is acyclic, and $h\colon \mathbb{Z}[0] \to C^\bullet$ the chain map that is the identity in degree 0, every chain map $t\colon X^\bullet \to \mathbb{Z}[0]$ has only a degree-zero component, so $h \circ t = 0$ forces $t = 0$, which is not a quasi-isomorphism. The standard route is instead to pass to the homotopy category first: quasi-isomorphisms *are* a multiplicative system in $K(\mathbf{A})$, and $D(\mathbf{A}) = K(\mathbf{A})[\mathrm{qis}^{-1}]$, which is how section 56.5 presents it.

The zigzag description settles existence and is useless for computation: a morphism is an equivalence class of arbitrarily long strings, and deciding when two strings agree is not something one does by hand. Under a mild hypothesis on W every zigzag collapses to a single roof, and then morphisms become genuinely manipulable.

Definition 55.2.4: Multiplicative System

A class W of morphisms in a category $\mathcal{C}$ is a *multiplicative system* if

1. W contains every identity and is closed under composition;
2. (Ore condition) for every $f\colon Z \to B$ in $\mathcal{C}$ and every $t\colon V \to B$ in W there are $t'\colon Z' \to Z$ in W and $f'\colon Z' \to V$ in $\mathcal{C}$ with $f \circ t' = t \circ f'$;
3. for every pair $f, g\colon X \to Y$ and every $s\colon Y \to Y'$ in W with $s \circ f = s \circ g$, there is $t\colon X' \to X$ in W with $f \circ t = g \circ t$.

Condition (2) is what lets two roofs be composed, and condition (3) is what makes the resulting notion of equality between roofs transitive. Both are visible in the ring case: for a commutative ring every S is a multiplicative system, and the conditions are the familiar manipulations that put two fractions over a common denominator.

Theorem 55.2.5: Calculus of Fractions

Let W be a multiplicative system in $\mathcal{C}$. Then:

1. every morphism $A \to B$ in $\mathcal{C}[W^{-1}]$ is represented by a *roof*
$$\begin{tikzcd} & Z \arrow[dl, "s"'] \arrow[dr, "f"] & \\ A & & B \end{tikzcd}$$
with $s \in W$, standing for $Q(f) \circ Q(s)^{-1}$;
2. if two roofs (s, f) and (s', f') are *commonly dominated*, meaning there are $u \colon Z'' \to Z$ and $u' \colon Z'' \to Z'$ with $su = s'u' \in W$ and $fu = f'u'$, then they represent the same morphism;
3. conversely, two roofs representing the same morphism are commonly dominated, so that roofs modulo common domination compute $\mathrm{Hom}_{\mathcal{C}[W^{-1}]}(A, B)$.

***Proof*:**

Step 1: Roofs compose. Given roofs $A \xleftarrow{s} Z \xrightarrow{f} B$ and $B \xleftarrow{t} V \xrightarrow{g} C$, apply the Ore condition to $f \colon Z \to B$ and $t \in W$ to obtain $t' \colon Z' \to Z$ in W and $f' \colon Z' \to V$ with $ft' = tf'$. Then $A \xleftarrow{st'} Z' \xrightarrow{gf'} C$ is a roof, since $st' \in W$ by closure under composition, and in $\mathcal{C}[W^{-1}]$ it represents
$$Q(g)Q(f')Q(t')^{-1}Q(s)^{-1} = Q(g)Q(t)^{-1}Q(f)Q(s)^{-1}$$
the identity following by rearranging $Q(f)Q(t') = Q(t)Q(f')$, both $Q(t)$ and $Q(t')$ being invertible. This is the composite of the two given morphisms.

Step 2: Part (2). Suppose (s, f) and (s', f') are commonly dominated by u, u'. Since $su = s'u'$ lies in W and $Q(s)$ is invertible, $Q(u)$ is invertible too, and then
$$Q(f)Q(s)^{-1} = Q(fu)Q(su)^{-1} = Q(f'u')Q(s'u')^{-1} = Q(f')Q(s')^{-1}$$
so the two roofs represent the same morphism.

Step 3: Common domination is an equivalence relation. Reflexivity and symmetry are immediate. For transitivity, suppose Z_1 dominates (s, f) and (s', f') through u_1, v_1 with $\sigma_1 := su_1 = s'v_1 \in W$, and Z_2 dominates (s', f') and (s'', f'') through u_2, v_2 with $\sigma_2 := s'u_2 = s''v_2 \in W$. Apply the Ore condition to $\sigma_1 \colon Z_1 \to A$ and $\sigma_2 \in W$ to get $t' \colon Z_3 \to Z_1$ in W and $w \colon Z_3 \to Z_2$ with $\sigma_1 t' = \sigma_2 w$. Then
$$s'(v_1 t') = \sigma_1 t' = \sigma_2 w = s'(u_2 w)$$
so condition (3) of definition 55.2.4, applied to the parallel pair $v_1 t', u_2 w \colon Z_3 \rightrightarrows Z'$ and to $s' \in W$, yields $t \colon Z_4 \to Z_3$ in W with $v_1 t' t = u_2 w t$. The roof with middle leg $\sigma_1 t' t \in W$ and legs $u_1 t' t$ and $v_2 w t$ then dominates both (s, f) and (s'', f''): the middle leg lies in W by closure, and the two outer compatibilities follow because $fu_1 = f'v_1$ and $f''v_2 = f'u_2$ both factor through f'.

Step 4: Part (1). Every morphism of $\mathcal{C}[W^{-1}]$ is a zigzag, and it suffices to show every zigzag is equivalent to one of the form backward-then-forward. Assign to a zigzag the number of pairs $i < j$ with link i forward and link j backward. Composing two adjacent links of the same orientation strictly decreases the length and does not increase this number; and given an adjacent pair $Z \xrightarrow{f} B \xleftarrow{t} V$ with $t \in W$, the Ore condition supplies $Z \xleftarrow{t'} Z' \xrightarrow{f'} V$ with $t' \in W$ and $ft' = tf'$, which represents the same morphism because $Q(t)^{-1}Q(f) = Q(f')Q(t')^{-1}$, and which strictly decreases the count. The count cannot decrease forever, so the process terminates at a zigzag with no forward link preceding a backward one; composing along each run leaves a single roof.

Step 5: Part (3). What remains is the converse of Step 2. The standard route builds the category whose morphisms are roofs modulo common domination, with composition as in Step 1, and checks that this is well defined on classes, associative and unital, and that it satisfies the universal property of definition 55.2.1; comparing with $\mathcal{C}[W^{-1}]$ through that universal property then forces the two to agree, which is exactly the converse. Those verifications are bookkeeping of the same kind as Steps 1 and 3, carrying no further idea, and they are carried out in full in *An Introduction to Homological Algebra* [53]. This completes the proof.

The construction now applies verbatim to the situation this book treats, and it is recorded as a corollary so that later chapters may cite existence rather than assume it.

Corollary 55.2.6: Existence of the Derived Category

Let $\mathbf{A}$ be an abelian category and let qis denote the class of quasi-isomorphisms in $\mathrm{Ch}(\mathbf{A})$. Then the localization

$$D(\mathbf{A}) := \mathrm{Ch}(\mathbf{A})[\mathrm{qis}^{-1}]$$

exists, and the canonical functor $Q\colon \mathrm{Ch}(\mathbf{A}) \to D(\mathbf{A})$ is universal among functors carrying quasi-isomorphisms to isomorphisms.

***Proof*:**
Apply theorem 55.2.2 with $\mathcal{C} = \mathrm{Ch}(\mathbf{A})$ and $W = \mathrm{qis}$; no hypothesis on W is needed for existence. The universal property is clause (2) of definition 55.2.1.

The construction says nothing about local smallness, and by box 55.2.3 neither remedy applies here, so $D(\mathbf{A})$ is produced only as a possibly non-locally-small category. The route that repairs this, and the one section 56.5 follows, is to localize $K(\mathbf{A})$ rather than $\mathrm{Ch}(\mathbf{A})$, where the quasi-isomorphisms do form a multiplicative system and theorem 55.2.5 applies, as we sought to show.

56

Complexes and (co)Homology

Let $\mathbf{A}$ be a fixed abelian category, which by theorem 55.1.15 may without loss of generality it can be considered ${}_R\mathrm{Mod}$. In this chapter, we shall focus on studying (co)chains of such modules, that is sequence of modules that respect equation (54.1):

$$\mathrm{im}(d^n) \subseteq \ker(d^{n+1})$$

The collection of all chains shall be denoted $\mathrm{Ch}(\mathbf{A})$. We shall see that $\mathrm{Ch}(\mathbf{A})$ is an abelian category, the notion of extensions (and snakes lemma) is well defined, which will be explored. In the right categories, it is relatively easy to construct a concrete complex given the category has enough objects, for example $M \in \mathrm{Obj}(R\text{-}\mathbf{Mod})$ always has a complex known as a *free resolution* (see definition 12.3.27). There will be many different ways of creating complexes given an object, giving us different types of resolutions. We shall see that when a category has enough projective objects (resp. Injective objects), then "up to homotopy equivalence", projective resolutions shall be *initial* (resp. final) in the category of complexes $\mathrm{Ch}(A)$. Thus, we shall take the time to develop the homology theory in categories with enough projective or injective objects. We shall see that under appropriate circumstances, taking (co)homology of a projective resolution of an object can give us some important information about said object.

Homology and cohomology are duals of each other, which means on the theoretical level they are the same. As cohomology is the more computationally useful tool, we shall be focusing on cohomology on this chapter, with the idea that the reader shall be comfortable being able to switch indices when necessary. Taking the cohomology of a complex shall be shown to be an additive functorial process, meaning for a given morphism of complexes $\alpha^\bullet : A^\bullet \to B^\bullet$, we shall get a morphism each cohomology $H^i(\alpha) : H^i(A^\bullet) \to H^i(B^\bullet)$ which will be shown can be to form a new complex $H^\bullet(\alpha) : H^\bullet(A^\bullet) \to H^\bullet(B^\bullet)$.

The goal of this chapter is to show how to extract information given a complexes (co)homology. Given this, it is important to know what information a morphism of complexes preserves in cohomology, that is given a morphism of complexes, $\alpha^\bullet : A^\bullet \to B^\bullet$, when is $H^\bullet(\alpha) : H^\bullet(A^\bullet) \to H^\bullet(B^\bullet)$ an isomorphism? This shall lead us to the notion of *quasi-isomorphisms*, and their "representation" through mapping cones. As it turns out, quasi-isomorphisms do not behave nicely in general. What we shall then seek out is a stricter concept known as a *homotopy equivalence* that will still be a quasi-isomorphism, but will be invertible in the appropriate category.

56.1 Basic Definition and Properties

Starting with defining the generalization of an exact sequence:

Definition 56.1.1: Chain Complex

A *chain complex* $(C_\bullet, d_\bullet)$ is a sequence of objects and morphisms:

$$\cdots \xleftarrow{d_i} M_i \xleftarrow{d_{i+1}} M_{i+1} \xleftarrow{d_{i+2}} \cdots$$

with the indices running through $\mathbb{Z}$, such that

$$d_i \circ d_{i+1} = 0$$

and each d_i is called a *boundary map*. Similarly, a *cochain complex* $(C^\bullet, d^\bullet)$ is a sequence of objects

$$\cdots \xrightarrow{d^i} M^i \xrightarrow{d_{i-1}} M^{i-1} \xrightarrow{d^{i+2}} \cdots$$

with the indices running through $\mathbb{Z}$, such that

$$d^i \circ d^{i-1} = 0$$

where $d^\bullet$ are is called the *differential*[a].

[a]Due to it's origin's in differential geometry and algebraic topology

Chains and cochains can be interchanged by taking

$$M_i = M^{-i}$$

Working with one or the other makes essentially no difference for homological algebra. In certain cases, some structures present themselves more in the cohomological case (ex. a natural ring structure called the *cohomology ring* shall become important to us), and so we opt to work with the cohomology, even though at this stage this can be considered to be an arbitrary choice. The condition that $d^i \circ d^{i-1} = 0$ is the same as $\mathrm{im} d^{i-1} \subseteq \ker d^i$. If the complex is exact, then $\mathrm{im} d^{i-1} = \ker d^i$ for all i. We also often write $C^\bullet$ instead of $(C^\bullet, d^\bullet)$ if the mappings $d^\bullet$ are clear from context or implied. In some textbooks, even the bullet is dropped and M represents the complex. Sometime, for homology, the notation $\partial_\bullet$ shall be used instead of $d_\bullet$ due to it's connection to being the boundary map for topological spaces when creating complexes out of topological spaces[1].The condition that $d^i \circ d^{i-1} = 0$ (resp. $d_i \circ d_{i+1} = 0$) is often abbreviated to $d \circ d = 0$.

Due to historical origins in algebraic topology, the kernel and image of the differential

[1]Similarly, $d^\bullet$ stands for the *differential* due to it's link to the derivative, which would be explored in a class in algebraic topology, or see EYNTKA Algebraic Topology

Definition 56.1.2: Cycles and Boundaries

Let $C_\bullet$ (resp. $C^\bullet$) be a chain complex (resp. cochain complex). Then:

1. $\ker(d_n)$ is called the n-*cycles* of $C_\bullet$ and $\mathrm{im}(d_{n+1})$ is called the n-boundary of $C_\bullet$. The n-cycle is represented as $Z_n = Z_n(C_\bullet)$ and the n-*boundary* is represented as $B_n = B_n(C_\bullet)$
2. $\ker(d^n)$ is called the n-*cocycles* of $C^\bullet$ and $\mathrm{im}(d^{n-1})$ is called the n-*coboundary* of $C^\bullet$. The n-cocycle is represented as $Z^n = Z_n(C_\bullet)$ and the n-coboundary is represented as $B_n = B_n(C_\bullet)$

The name cycles and boundaries has it's origin in algebraic topology:

Example 56.1.3: Cycles and Boundaries

(Put that image from p.67 from alg top notes)

The collection of all complexes forms a Category:

Definition 56.1.4: Category Of Complexes

Let $\mathbf{A}$ be an abelian category. Then $\mathrm{Ch}^\bullet(\mathbf{A})$ is a new category called the *category of complexes of A* where

1. $\mathrm{Obj}(\mathrm{Ch}^\bullet(\mathbf{A})) = \{\text{cochain complexes in } \mathbf{A}\}$
2. For two complexes $C^\bullet$ and $N^\bullet$, a morphism $\alpha^\bullet \in \mathrm{Hom}_{\mathrm{Ch}(\mathbf{A})}(C^\bullet, N^\bullet)$ consists of morphisms st the following diagram commute:

$$\begin{array}{ccccccccc}
\cdots & \longrightarrow & M^{i-1} & \xrightarrow{d^{i-1}_{C^\bullet}} & M^i & \xrightarrow{d^i_{C^\bullet}} & M^{i+1} & \longrightarrow & \cdots \\
 & & \downarrow{\scriptstyle \alpha^{i-1}} & & \downarrow{\scriptstyle \alpha^{i-1}} & & \downarrow{\scriptstyle \alpha^{i-1}} & & \\
\cdots & \longrightarrow & N^{i-1} & \xrightarrow{d^{i-1}_{N^\bullet}} & N^i & \xrightarrow{d^i_{N^\bullet}} & N^{i+1} & \longrightarrow & \cdots
\end{array}$$

Similarly for $\mathrm{Ch}_\bullet(\mathbf{A})$ with chain complexes.

If unambiguous, the bullet is dropped giving $\mathrm{Ch}(\mathbf{A})$. It should be checked that $\mathrm{Ch}(A)$ is indeed a category. There are many variations of $\mathrm{Ch}(A)$ which will be important:

1. $C^+(A)$ is the full subcategory of $\mathrm{Ch}(A)$ of complexes which are bounded from bellow, so that $L^i = 0$ for $i \ll 0$
2. $C^-(A)$ is the full subcategory of $\mathrm{Ch}(A)$ of complexes which are bounded from above, so that $L^i = 0$ for $i \gg 0$
3. $C^{\geq 0}(A)$ (resp. $C^{\leq 0}$) for which $L^i = 0$ for $i \leq 0$ (resp. $i \geq 0$)
4. P, I are full subcategories of A consisting of projective and injective objects in A respectively, and hence we have $\mathrm{Ch}(P)$, $\mathrm{Ch}(I)$, and all its variants

There are also a few simple functors we may immediately define:

1. Take $\iota_r : \mathbf{A} \to \mathrm{Ch}(\mathbf{A})$ that sends an object A form $\mathbf{A}$ to a complex that is zero every and A in the rth degree. We shall commonly identify A with the complex given by ι_0.
2. Take $s_r : \mathrm{Ch}(\mathbf{A}) \to \mathrm{Ch}(\mathbf{A})$ which shifts over the complex, namely
$$C^\bullet \mapsto M[r]^\bullet$$
where $M[r]^i = M^{i+r}$ and $d^i_{M[r]^\bullet} = (-1)^r d^{i+r}_{C^\bullet}$ (the $(-1)^r$ shall be explained shortly).
3. the truncation functor shall truncate the complex in the natural way.

More importantly, $\mathrm{Ch}(A)$ is an abelian category, allowing the notion of exactness between complexes to be well-defined:

Lemma 56.1.5: Complex Category Is Abelian

Let A be an abelian category and $\mathrm{Ch}(A)$ the category of complexes of A. Then $\mathrm{Ch}(A)$ is an abelian category

***Proof*:**
Left as an exercise, check that:

1. morphisms between two complexes form an abelian group, namely since $\alpha^i + \beta^i$ will be well-defined
2. Finite products and coporducts exist in $\mathrm{Ch}(A)$ as a consequence of them existing in A
3. kernels and cokernls will come from the commutativity of the diagram defining morphisms of complexes

Since it is an abelian category, we shall be able to consider sequences of complexes, and in particular exact sequences of complexes:
$$\cdots \to L^\bullet \to C^\bullet \to N^\bullet \to \cdots$$
which give a commutative diagram like so:

$$\begin{tikzcd}
 & 0 \arrow[d] & 0 \arrow[d] & 0 \arrow[d] & 0 \arrow[d] & \\
\cdots \arrow[r] & A_{n+2} \arrow[r] \arrow[d, "\iota"] & A_{n+1} \arrow[r] \arrow[d, "\iota"] & A_n \arrow[r] \arrow[d, "\iota"] & A_{n-1} \arrow[r] \arrow[d, "\iota"] & \cdots \\
\cdots \arrow[r] & B_{n+2} \arrow[r] \arrow[d, "\pi"] & B_{n+1} \arrow[r] \arrow[d, "\pi"] & B_n \arrow[r] \arrow[d, "\pi"] & B_{n-1} \arrow[r] \arrow[d, "\pi"] & \cdots \\
\cdots \arrow[r] & C_{n+2} \arrow[r] \arrow[d] & C_{n+1} \arrow[r] \arrow[d] & C_n \arrow[r] \arrow[d] & C_{n-1} \arrow[r] \arrow[d] & \cdots \\
 & 0 & 0 & 0 & 0 &
\end{tikzcd} \tag{56.1}$$

Example 56.1.6: Complexes

1. Given any object M, we may always create the trivial complex
$$\cdots 0 \to 0 \to M \to 0 \to 0 \to \cdots$$
Verify that this is indeed a complex.

2. As mentioned, In R-**Mod**, the free resolution is an example of a complex.

3. Any short exact sequence can be thought of an (exact) complex if we add 0's that extend arbitrarily to the left and right.

4. Let $C^n = \mathbb{Z}/8\mathbb{Z}$ for $n \geq 0$ and $C^n = 0$ for $n < 0$. Let each d^n be defined as:
$$\overline{x} \mapsto \overline{4x}$$
Show that this is a cochain complex.

5. Let $\{V_i, W_i\}_{i\in\mathbb{Z}}$ be a collection of vector spaces. Define:
$$A_n = V_n \oplus W_n \oplus V_{n-1}$$
Combining the projection and inclusion maps $A_n \to V_{n-1} \to A_{n-1}$, show that $A_\bullet$ forms a chain complex.

6. Let $C^\bullet$ be a cochain complex, choose an object $A \in \mathbf{A}$, and apply $\mathrm{Hom}_A(-, A)$ to each object. Show that this is a cochain complex.

Since $\mathrm{Ch}(\mathbf{A})$ is an abelian category, kernels and cokernels exist and match with monomorphisms/epimorphisms, hence giving us the following:

Definition 56.1.7: Sub-complexes And Quotient Complexes

Let $C^\bullet$ be a (co)chain complex. Then a (co)chain complex $N^\bullet$ is called a *subcomplex* if each N_n is a sub-object of C_n and the differential on B_n is the restriction of the differential in C_n (categorically, the inclusion map $\iota : N^\bullet \to C^\bullet$ is a morphism of complexes).

Furthermore, C^n/B^n is a *quotient-complex*. Furthermore, by the canonical decomposition of morphisms in abelian categories, for any complex-morphism $\alpha^\bullet$, $\{\ker(\alpha^\bullet)\}$ and $\{\mathrm{coker}(\alpha^\bullet)\}$ is a sub-complex and quotient complex respectively.

56.1.1 Homology and Cohomology

Chain complexes need not be exact. The cohomology of a complex measures how far away a complex is from being exact, that is how far away a complex is form representing a sequence of extensions

Definition 56.1.8: Homology and Cohomology Of A Complex

Let $(C^\bullet, d^\bullet)$ be a complex. Then the ith homology and cohomology of the complex at M^i is, respectively:

$$H_i(C^\bullet) = \frac{\ker d_i}{\mathrm{im} d_{i+1}} \qquad H^i(C^\bullet) := \frac{\ker d^i}{\mathrm{im} d^{i-1}}$$

Though homology and cohomology are similar, it is not necessarily the case that $H^i(C^\bullet) = H_{n-i}(C^\bullet)$ for some appropriate n around which the chain can "pivot". We shall see such examples soon, and later also show under what conditions we may compare homology and cohomology[2]

Example 56.1.9: Cohomology Of Complex

Let's compute the cohomology of sequences in example 56.1.6.

1. The homology and cohomology at the 0th entry is:

$$\frac{\ker(d_0)}{\mathrm{im}(d_1)} = \frac{\ker(d^1)}{\mathrm{im}(d^0)} = M$$

   which shows that at this position, the chain is "off" by M in being exact.

2. As we have defined exact free resolutions, they are exact, meaning $\mathrm{im}(d^i) = \ker(d^{i-1})$, hence their homology and cohomology is always 0
3. Any exact sequence has trivial homology and cohomology
4. This sequence is not exact, and yet it's homoogy/cohomology is trivial (as the reader should verify). Such a sequence is called *acyclic*.
5. The reader should calculate this cohomology as an exercise
6. This cohomology shall be calculated later in this chapter.

Lemma 56.1.10: Cohomology Functor

For every $i \in \mathbb{Z}$, there exists a functor:

$$H^i : \mathrm{Ch}(A) \to A \qquad C^\bullet \mapsto H^i(C^\bullet)$$

which is additive and covariant.

This shows that taking cohomology is indeed a way of analyzing resolutions, for it preserves enough of the structures of modules and complexes. Notice it is not necessarily an exact functor, hence it will not in general preserve extensions.

[2]The universal coefficient theorem and Poincaré duality being two famous examples the reader may have heard of if they took algebraic topology

***Proof*:**
Let $\alpha^\bullet : C^\bullet \to N^\bullet$ be a morphism of complexes and take $H^i(C^\bullet)$ and $H^i(N^\bullet)$. Let $x \in H^i(H^\bullet)$ so that $x = k + \text{im}(d_M^{i-1})$ for d^{i-1}. Since $d^i(x) = 0$, by commutativity it must be that $\alpha^i(x) \in \ker(d_N^i)$. Furthermore, by commutativity we see that $\overline{\alpha^i}$ is zero on the image of $\text{im}(d_M^{i-1})$, hence $H^i(\alpha^\bullet)$ is well-defined.

This functor is certainly covariant, and is additive since $\text{Hom}_A(H^i(C^\bullet), H^i(N^\bullet))$ is an abelian group.

Notice that each H^i takes d^i to the zero map. Hence, we may consider the following functor:

$$H^\bullet : \text{Ch}(\mathbf{A}) \to \text{Ch}(\mathbf{A})$$

by taking $H^i(C^\bullet)$ for each i and connecting them by the zero-morphism. This is the cohomology of this complex then equals that of $C^\bullet$, namely:

$$H^\bullet(H^\bullet(C^\bullet)) = H^\bullet(C^\bullet)$$

Lemma 56.1.11: Cohomology And Products

Let $\{A_\alpha^\bullet\}$ be a collection of complexes in $\mathbf{A}$, and suppose the coproduct (respectively the product) indexed by α exists and is an exact functor. Then

$$H^n\Big(\bigoplus_\alpha A_\alpha^\bullet\Big) = \bigoplus_\alpha H^n(A_\alpha^\bullet) \qquad H^n\Big(\prod_\alpha A_\alpha^\bullet\Big) = \prod_\alpha H^n(A_\alpha^\bullet)$$

The exactness hypothesis is automatic for *finite* index sets, where the two agree by lemma 55.1.5, and it holds for arbitrary index sets in R-Mod.

***Proof*:**
Treat the coproduct; the product is the same argument in the opposite category. The complex $\bigoplus_\alpha A_\alpha^\bullet$ has n-th term $\bigoplus_\alpha A_\alpha^n$ and differential $\bigoplus_\alpha d_\alpha^n$.

An exact functor preserves kernels and images, so

$$\ker\Big(\bigoplus_\alpha d_\alpha^n\Big) = \bigoplus_\alpha \ker d_\alpha^n, \qquad \text{im}\Big(\bigoplus_\alpha d_\alpha^{n-1}\Big) = \bigoplus_\alpha \text{im} d_\alpha^{n-1}$$

and it preserves the quotient of the second by the first, being exact. Hence

$$H^n\Big(\bigoplus_\alpha A_\alpha^\bullet\Big) = \frac{\bigoplus_\alpha \ker d_\alpha^n}{\bigoplus_\alpha \text{im} d_\alpha^{n-1}} = \bigoplus_\alpha \frac{\ker d_\alpha^n}{\text{im} d_\alpha^{n-1}} = \bigoplus_\alpha H^n(A_\alpha^\bullet)$$

For a finite index set the coproduct is a finite biproduct, which every additive functor preserves, so no hypothesis is needed; in R-Mod both $\bigoplus$ and $\prod$ are exact and kernels and images are computed componentwise, giving the statement for arbitrary index sets. This completes the proof.

Now, given a morphism of complexes $\alpha^\bullet : C^\bullet \to N^\bullet$, we may ask what (co)homological information such a morphism preserves. In other words, given a morphism $\alpha^\bullet$, when is $H^\bullet(\alpha^\bullet)$ an isomorphism? We give such morphisms a name:

Definition 56.1.12: Quasi-Isomorphism

Let $\alpha^\bullet : C^\bullet \to N^\bullet$ be a morphism of complexes. Then if $H^\bullet(\alpha^\bullet) : H^\bullet(C^\bullet) \to H^\bullet(N^\bullet)$ is an isomorphism, $\alpha^\bullet$ is called a *quasi-isomorphism*

One may wonder if the fact that the (co)homology is isomorphism is strong enough of a condition for the chains to be isomorphic. This turns out to be rather far from the case:

Example 56.1.13: Quasi-Isomorphism

1. The following counter example shall come up with many different minor variations, and so the reader should keep this in mind. Take the following morphism of complexes:

$$\begin{tikzcd}
\cdots \arrow[r] & 0 \arrow[r] \arrow[d] & \mathbb{Z} \arrow[r, "\cdot 2"] \arrow[d] & \mathbb{Z} \arrow[r] \arrow[d, "\pi"] & 0 \arrow[r] \arrow[d] & \cdots \\
\cdots \arrow[r] & 0 \arrow[r] & 0 \arrow[r] & \mathbb{Z}/2\mathbb{Z} \arrow[r] & 0 \arrow[r] & \cdots
\end{tikzcd}$$

   This is a quasi-isomorphism, as the reader should check, however it does not have an inverse. More generally, it is not even an equivalence relation, failing to be symmetric (show there is no morphism the other way around that is a quasi-isomorphism).

2. The natural maps $H^\bullet(C^\bullet) \to C^\bullet$ and $C^\bullet \to H^\bullet(C^\bullet)$ are quasi-isomorphisms, given the property of $H^\bullet(H^\bullet(C^\bullet)) = H^\bullet(C^\bullet)$ discussed earlier.

3. Let $\iota(A) \in \mathrm{Obj}(\mathrm{Ch}(\mathbf{A}))$ be the complex with A at the 0th position and zero's everywhere else. Say that $C^\bullet$ is a resolution such that $C^\bullet \to \iota(A)$ is a quasi-isomorphism. Show that this implies $C^\bullet$ is exact everywhere except possibly at $H^0(M^0)$

The lack of symmetry means that quasi-isomorphisms do not form an equivalence relation, but a *weak equivalence relation* (a relation that is reflective and transitive, but not symmetric). This is rather a pity, since if it did form an equivalence relation, then the equivalence class of morphisms up to weak isomorphism would hold all the necessary (co)homological information. Alas, this was not mean to be. Thus, in the pursuit of finding (co)homological invariants, it productive to ask how (co)homological information can be detected between (co)chain complexes. One idea is to map $\mathrm{Ch}(\mathbf{A})$ into another category $D(\mathbf{A})$ in which quasi-isomorphisms become isomorphisms. If quasi-isomorphisms formed equivalence relationships, we could have formed a congruence category which would satisfy this condition, however as just mentioned this cannot be the case.

Such a functor has already been encountered: the (co)homology functor is already one such example. It is more fruitful to as if this can be done in a *univerasl* way, that is if there is a category $D(\mathbf{A})$ where for any category D for which there exists an additive functor $\mathscr{F} : \mathrm{Ch}(\mathbf{A}) \to D$ where every quasi-isomorphism in $\mathrm{Ch}(\mathbf{A})$ becomes an isomorphism in D, there exists a unique functor such that the following diagram commutes:

$$\begin{tikzcd} \mathrm{Ch}(\mathbf{A}) \arrow[r, "\mathscr{F}"] \arrow[d] & D \\ D(\mathbf{A}) \arrow[ur, "\exists!"'] & \end{tikzcd} \tag{56.2}$$

This category does indeed exist, by corollary 55.2.6, and is called the *derived category* (hence the $D(-)$ notation). This category shall further be explored in the coming sections. In some nice categories $\mathbf{A}$, quasi-isomorphisms will line up with the nicer notion of *homotopy-equivalence*, in which case the derived category will be eminently understandable. We shall explore these all in greater detail section 56.2.

Though quasi-isomorphisms may not be ideal in computing (co)homological invariants, they are still important as the defining concept for (co)homological information. For example, the following shows the relationship between quasi-isomorphism and exactness:

Lemma 56.1.14: Quasi-isomorphisms And Exactness

Let $C^\bullet$ be a cochain complex. Then the following are equivalent:

1. $C^\bullet$ is exact
2. $C^\bullet$ is acyclic, that is $H^\bullet(C^\bullet) = 0$
3. The morphism of complexes $0 \to C^\bullet$ is a quasi-isomorphism

***Proof*:**
(1) $\Leftrightarrow$ (2): exactness of $C^\bullet$ at degree n means $\ker d^n = \mathrm{im} d^{n-1}$, and $H^n(C^\bullet) = \ker d^n / \mathrm{im} d^{n-1}$ by definition 56.1.8. A quotient object is zero exactly when the two are equal, so $C^\bullet$ is exact at every degree if and only if $H^n(C^\bullet) = 0$ for every n.

(2) $\Leftrightarrow$ (3): the zero complex has $H^n(0) = 0$ for every n, and the unique morphism $0 \to C^\bullet$ induces the unique morphism $0 \to H^n(C^\bullet)$ in each degree. That morphism is an isomorphism precisely when $H^n(C^\bullet) = 0$. So $0 \to C^\bullet$ is a quasi-isomorphism, meaning it induces isomorphisms in every degree, if and only if $H^\bullet(C^\bullet) = 0$, completing the proof.

56.1.2 Snake Lemma

We often have multiple ways of constructions a resolution for an object M. We shall often extend resolutions using smaller resolutions. We then get cohomology of this new extended resolution by simply applying the cohomology functor. However, is is this functor exact? More generally, let's say we have a short exact sequence of complexes, then after applying cohomology do we get a short exact sequence of homology complexes? The answer is no: if

$$0 \to L^\bullet \to C^\bullet \to N^\bullet \to 0$$

is a short exact sequence, then we get the (not necessarily exact) sequence:

$$H^i(L^\bullet) \to H^i(C^\bullet) \to H^i(N^\bullet)$$

In terms of morphisms, if we have:

$$\alpha^\bullet : C^\bullet \to N^\bullet$$

Then the morphism after applying the cohomology functor $H^\bullet(\alpha^\bullet) : H^\bullet(C^\bullet) \to H^\bullet(N^\bullet)$ need not be exact, i.e. it doesn't preserve kernels and cokernels. Can we "fix" this in some way by recovering some exact sequence from this and thus allowing our analysis of building more complicated resolutions for M to be independent of cohomology. The answer is yes, Namely, if we take the cohomology of complexes, we may "snake" like in the following diagram:

$$\begin{tikzcd}
 & 0 \arrow[d, dotted] & 0 \arrow[d, dotted] & 0 \arrow[d, dotted] & 0 \arrow[d, dotted] & \\
\cdots \arrow[r, dotted] & H_{n+2}(A) \arrow[r, dotted] \arrow[d, "\iota"] & H_{n+1}(A) \arrow[r, dotted] \arrow[d, "\iota"] & H_n(A) \arrow[r, dotted] \arrow[d, "\iota"] & H_{n-1}(A) \arrow[r, dotted] \arrow[d, "\iota"] & \cdots \\
\cdots \arrow[r, dotted] & H_{n+2}(B) \arrow[r, dotted] \arrow[d, "\pi"] & H_{n+1}(B) \arrow[r, dotted] \arrow[d, "\pi"] & H_n(B) \arrow[r, dotted] \arrow[d, "\pi"] & H_{n-1}(B) \arrow[r, dotted] \arrow[d, "\pi"] & \cdots \\
\cdots \arrow[r, dotted] & H_{n+2}(C) \arrow[r, dotted] \arrow[d, dotted] \arrow[uur] & H_{n+1}(C) \arrow[r, dotted] \arrow[d, dotted] \arrow[uur] & H_n(C) \arrow[r, dotted] \arrow[d, dotted] \arrow[uur] & H_{n-1}(C) \arrow[r, dotted] \arrow[d, dotted] & \cdots \\
 & 0 & 0 & 0 & 0 &
\end{tikzcd}$$

Theorem 56.1.15: Snake Lemma

Given a short exact sequence of complexes $0 \to A \to B \to C \to 0$, we may form a long exact sequence of modules using cohomology:

$$\cdots \to H^n(A) \xrightarrow{i^*} H^n(B) \xrightarrow{\pi^*} H^n(C) \xrightarrow{\partial} H^{n+1}(A) \xrightarrow{i^*} H^{n+1}(B) \xrightarrow{\pi^*} H^{n+1}(C) \to \cdots$$

Equivalently, in homological indexing, where the differential lowers degree and H_n is written for the homology in degree n, the same statement reads

$$\cdots \to H_n(A) \xrightarrow{i_*} H_n(B) \xrightarrow{\pi_*} H_n(C) \xrightarrow{\partial} H_{n-1}(A) \xrightarrow{i_*} H_{n-1}(B) \to \cdots$$

and it is this form that the proof below constructs. The two are the same statement under $H_n = H^{-n}$, and both are used in this book.

The following proof can be found (in the Homological variant) in EYNTKA Algebraic Topology

***Proof*:**
We first define the map $\partial : H_n(C) \to H_{n-1}(A)$. We should keep in mind the following diagram:

$$\begin{tikzcd} & A_{n-1} \arrow[d, "\iota"] \\ B_n \arrow[r, "\partial"] \arrow[d, "\pi"] & B_{n-1} \\ C_n & \end{tikzcd}$$

Let $c \in C_n$ be a cycle so that $c \in \ker \partial_n$. Since π is surjective, $c = \pi(b)$ for some $b \in B_n$. Then by definition $\partial : B_n \to B_{n-1}$ so $\partial(b) \in B_{n-1}$. Then by commutativity $j(\partial(b)) = \partial(j(b)) = \partial(c) = 0$ (since c is a cycle), thus $\partial b \in \ker(\pi)$. Since $\ker j = \mathrm{im} i$, $\partial b = i(a)$ for some $a \in A_{n-1}$. Since $i(\partial a) = \partial i(a) = \partial\partial b = 0$ since i is injective, $\partial a = 0$. Thus, define $\partial : H_n(C) \to H_{n-1}(A)$ by $[c] \mapsto [a]$. This is well defined since:

1. The element a is uniquely determined by ∂b since i is injective
2. for differnet choice b' for b, we get $j(b') = j(b)$, os $b' - b \in \ker(j)$ which is equal to $\mathrm{im}(i)$. Thus, $b' - b = i(a')$ for some a', ehcen $b' = b + i(a')$. But then this means we just chanage a to another representative (a homologous element) $a + \partial a'$, namely since
$$i(a + \partial a') = i(a) + i(\partial a') = \partial b + \partial i(a') = \partial(b = i(a'))$$
3. A difference choice of c within its homology class would have hte form $c = \partial c'$. To see this, since $c' = j(b')$ for some b',
$$c + \partial c' = c\partial j(b') = c + j(\partial b') = j(b + \partial b')$$
thus b is replaced by $b = \partial b'$, which gives ∂b and thus a is unchanged.

Finally, $\partial : H_n(C) \to H_{n-1}(A)$ is a homomorphism, for if $\partial[c_1] = [a_1]$ and $\partial[c_2] = [a_2]$ via elements b_1, b_2, then
$$j(b_1 + b_2) = j(b_1) + j(b_2) = c_1 = c_2$$
and
$$i(a_1 + a_2) = i(a_1) + i(a_2) = \partial b_1 + \partial b_2 = \partial(b_1 + b_2)$$
hence
$$\partial([c_1] + [c_2]) = [a_1] + [a_2]$$
giving us the map ∂ is indeed a homomorphism. What's left to show is the following inclusions:

1. $\mathrm{im} i_* \subseteq \ker j_*$. Since $ji = 0$, $j_* i_* = 0$
2. $\mathrm{im} j_* \subseteq \ker \partial$. By definition of ∂, $\partial b = 0$ so $\partial j_* = 0$.
3. $\ker j_* \subseteq \mathrm{im} i_*$. Let $b \in B_n$ with boundary represent a homology class in $\ker j_*$, so $j(b) = \partial c'$ for some $c' \in C_{n+1}$. Since j is surjectiv,e $c' = j(b')$ for some $b' \in B_{n+1}$. Then since $\partial j(b') = \partial c' = j(b)$,
$$j(b - \partial b') = j(b) - j(\partial b') = j(b) - \partial j(b') = 0$$

Thus, $b - \partial b' = i(a)$ for some $a \in A - n$. Since $i(\partial a) = \partial i(a) = \partial(b - \partial b') = \partial b = 0$ and i is injective, a is a cycle. Thus:

$$\iota_*[a] = [b - \partial b'] = [b]$$

showing that i_* maps onto $\ker j_*$

4. $\ker \partial \subseteq \mathrm{im} j_*$. By definition of ∂, if c represents a homology class in $\ker \partial$, then $a = \partial a'$ for some $a' \in A_n$. The element $b - i(a')$ is a cycle since $\partial(b - i(a')) = \partial b - \partial i(a') = \partial b - i(\partial a') = \partial b - i(a) = 0$. Since

$$j(b - i(a')) = j(b) - ji(a') = j(b) = c$$

we get that $j_*[b - i(a')] = [c]$

5. $\ker i_* \subseteq \mathrm{im}\partial$. Given a cycle $a \in A_{n-1}$ such that $i(a) = \partial b$ for seom $b \in B_n <$ then $j(b)$ is a cycle since $\partial j(b) = j(\partial b) = ji(a) = 0$, hence $\partial[j(b)] = [a]$

Doing this shall completing the proof.

Example 56.1.16: Uses Of Snake Lemma

1. *The six-term kernel-cokernel sequence.* The classical statement that carries the name "snake" is a special case of theorem 56.1.15. Suppose given a commutative diagram with exact rows

$$\begin{tikzcd}
0 \arrow[r] & A^0 \arrow[r] \arrow[d, "f"] & B^0 \arrow[r] \arrow[d, "g"] & C^0 \arrow[r] \arrow[d, "h"] & 0 \\
0 \arrow[r] & A^1 \arrow[r] & B^1 \arrow[r] & C^1 \arrow[r] & 0
\end{tikzcd}$$

Read each column as a complex concentrated in degrees 0 and 1, so that $A^\bullet = (A^0 \xrightarrow{f} A^1)$ and likewise for $B^\bullet$ and $C^\bullet$. Exactness of the two rows says precisely that $0 \to A^\bullet \to B^\bullet \to C^\bullet \to 0$ is a short exact sequence of complexes, and for a two-term complex $H^0 = \ker$ and $H^1 = \mathrm{coker}$. Since H^n vanishes outside degrees 0 and 1, the long exact sequence of theorem 56.1.15 has only six nonzero terms:

$$0 \to \ker f \to \ker g \to \ker h \xrightarrow{\partial} \mathrm{coker} f \to \mathrm{coker} g \to \mathrm{coker} h \to 0$$

The connecting morphism ∂ is the "snake" that crosses from the kernel row to the cokernel row.

2. *A computation.* Apply the six-term sequence to multiplication by a fixed integer $n \geq 1$ on the short exact sequence $0 \to \mathbb{Z} \to \mathbb{Q} \to \mathbb{Q}/\mathbb{Z} \to 0$. Multiplication by n is injective on $\mathbb{Z}$ and bijective on $\mathbb{Q}$, and $\mathbb{Q}/\mathbb{Z}$ is divisible, so

$$\ker f = 0, \quad \ker g = 0, \quad \ker h = \tfrac{1}{n}\mathbb{Z}/\mathbb{Z} \cong \mathbb{Z}/n\mathbb{Z}$$

$$\mathrm{coker} f = \mathbb{Z}/n\mathbb{Z}, \quad \mathrm{coker} g = 0, \quad \mathrm{coker} h = 0$$

The six-term sequence collapses to $0 \to \mathbb{Z}/n\mathbb{Z} \xrightarrow{\partial} \mathbb{Z}/n\mathbb{Z} \to 0$, forcing ∂ to be an isomorphism. The explicit description confirms it: an element of $\ker h$ is the class of some $q \in \mathbb{Q}$ with $nq \in \mathbb{Z}$, and ∂ sends it to the class of nq in $\mathbb{Z}/n\mathbb{Z}$; the generator $1/n$ goes to the generator 1.

3. *Where the machine is used later.* Every long exact sequence in this book is an instance of theorem 56.1.15. The long exact sequence of derived functors (theorem 56.3.10) applies it to a short exact sequence of resolutions, the exact triangle of a mapping cone (proposition 56.1.19) applies it to the cone sequence, and the exact couple that generates a spectral sequence (theorem 58.2.2) is nothing but all the long exact sequences of a filtration assembled at once. The topological applications, the Mayer-Vietoris sequence and the long exact sequence of a pair, are developed where they are used, in *Algebraic Topology* [38].

The most-used consequence of the snake lemma is a criterion for recognizing an isomorphism in the middle of a diagram from isomorphisms on either side. It is stated here because it is needed whenever two exact sequences are compared, and in particular in the comparison theorem for spectral sequences (theorem 58.2.9).

Lemma 56.1.17: Five Lemma

Consider a commutative diagram in an abelian category with exact rows

$$\begin{tikzcd} A_1 \arrow[r] \arrow[d, "f_1"] & A_2 \arrow[r] \arrow[d, "f_2"] & A_3 \arrow[r] \arrow[d, "f_3"] & A_4 \arrow[r] \arrow[d, "f_4"] & A_5 \arrow[d, "f_5"] \\ B_1 \arrow[r] & B_2 \arrow[r] & B_3 \arrow[r] & B_4 \arrow[r] & B_5 \end{tikzcd}$$

If f_2 and f_4 are isomorphisms, f_1 is an epimorphism, and f_5 is a monomorphism, then f_3 is an isomorphism.
More precisely, f_3 is a monomorphism as soon as f_2 and f_4 are monomorphisms and f_1 is an epimorphism, and f_3 is an epimorphism as soon as f_2 and f_4 are epimorphisms and f_5 is a monomorphism.

***Proof*:**
By theorem 55.1.15 the diagram may be chased with elements. Label the horizontal maps $\alpha_i\colon A_i \to A_{i+1}$ and $\beta_i\colon B_i \to B_{i+1}$.

Step 1: f_3 is a monomorphism. Assume f_2, f_4 monomorphisms and f_1 an epimorphism, and let $a \in A_3$ with $f_3(a) = 0$. Then $f_4(\alpha_3(a)) = \beta_3(f_3(a)) = 0$, and f_4 is a monomorphism, so $\alpha_3(a) = 0$. By exactness of the top row at A_3 there is $b \in A_2$ with $\alpha_2(b) = a$. Now $\beta_2(f_2(b)) = f_3(\alpha_2(b)) = f_3(a) = 0$, so by exactness of the bottom row at B_2 there is $c \in B_1$ with $\beta_1(c) = f_2(b)$. Since f_1 is an epimorphism, $c = f_1(e)$ for some $e \in A_1$. Then

$$f_2(\alpha_1(e)) = \beta_1(f_1(e)) = \beta_1(c) = f_2(b)$$

and f_2 is a monomorphism, so $\alpha_1(e) = b$. Therefore $a = \alpha_2(b) = \alpha_2(\alpha_1(e)) = 0$, the last equality by exactness of the top row at A_2.

Step 2: f_3 is an epimorphism. Assume f_2, f_4 epimorphisms and f_5 a monomorphism, and let $b \in B_3$. Since f_4 is an epimorphism, $\beta_3(b) = f_4(a)$ for some $a \in A_4$. Then

$$f_5(\alpha_4(a)) = \beta_4(f_4(a)) = \beta_4(\beta_3(b)) = 0$$

by exactness of the bottom row at B_4, and f_5 is a monomorphism, so $\alpha_4(a) = 0$. By exactness of the top row at A_4 there is $c \in A_3$ with $\alpha_3(c) = a$. The element $b - f_3(c)$ satisfies

$$\beta_3(b - f_3(c)) = \beta_3(b) - f_4(\alpha_3(c)) = f_4(a) - f_4(a) = 0$$

so by exactness of the bottom row at B_3 there is $e \in B_2$ with $\beta_2(e) = b - f_3(c)$. Since f_2 is an epimorphism, $e = f_2(g)$ for some $g \in A_2$, and then

$$f_3\big(\alpha_2(g) + c\big) = \beta_2(f_2(g)) + f_3(c) = \big(b - f_3(c)\big) + f_3(c) = b$$

so b lies in the image of f_3.

Under the hypotheses of the first statement both steps apply, so f_3 is both a monomorphism and an epimorphism, hence an isomorphism by lemma 55.1.8, completing the proof.

There is another way of looking at chain complexes which shall be important in elucidating triangle categories in section 56.5.2.

Mapping Cones

Using a particular total complex, we can detect when a quasi-isomorphism is an isomorphism. This result requires the snake lemma to prove, hence the need to put this section after the snake lemma.

Definition 56.1.18: Mapping Cone

Let $\alpha^\bullet : L^\bullet \to M^\bullet$ be a morphism of complexes. Then the *mapping cone* is the complex denoted by $MC(\alpha^\bullet)$ defined by:

$$MC(\alpha)^i := L[1]^i \oplus M^i = L^{i+1} \oplus M^i$$

where the morphism $d^i_{MC(\alpha)} : MC(\alpha)^i \to MC(\alpha)^{i+1}$ are given by:

$$d^i_{MC(\alpha)}(l, m) := (-d_L^{i+1}(l), \alpha^{i+1}(l) + d^i_M(m))$$

The term mapping cone comes from the topological notion of a mapping cone of a function. Visually, the mapping cone can be seen as the following commutative diagram:

$$\begin{tikzcd}
\cdots \arrow[r] \arrow[ddr] & L^{i+1} \arrow[r, "-d_L^{i+1}"] \arrow[ddr, "\alpha^{i+1}"] & L^{i+2} \arrow[r] \arrow[ddr] & \cdots \\
& \oplus & \oplus & \\
\cdots \arrow[r] & M^i \arrow[r, "d_M^i"'] & M^{i+1} \arrow[r] & \cdots
\end{tikzcd}$$

Given $MC(\alpha)^\bullet$, we may define two natural chain maps $C^\bullet \to MC(\alpha)^\bullet$ and $MC(\alpha)^\bullet \to L[1]^\bullet$ induced by the natural exact morphisms $M^i \to L^{i+1} \oplus M^i \to L^{i+1}$ which implies that:

$$0 \to C^\bullet \to MC(\alpha)^\bullet \to L[1]^\bullet \to 0$$

is exact.

Proposition 56.1.19: Mapping Cone And Exact Triangles

There is an exact triangle

$$\begin{tikzcd} & H^\bullet(C^\bullet) \arrow[dr] & \\ H^\bullet(L[1]^\bullet) \arrow[ur, "+1\ \delta"] & & H^\bullet(MC(\alpha)^\bullet) \arrow[ll] \end{tikzcd}$$

where δ is the morphism induced by cohomology

***Proof*:**
The short exact sequence of complexes displayed above,

$$0 \to C^\bullet \to MC(\alpha)^\bullet \to L[1]^\bullet \to 0$$

is exact, so theorem 56.1.15 produces a long exact sequence in cohomology

$$\cdots \to H^n(C^\bullet) \to H^n(MC(\alpha)^\bullet) \to H^n(L[1]^\bullet) \xrightarrow{\delta} H^{n+1}(C^\bullet) \to \cdots$$

Exactness at each of the three kinds of term is precisely exactness of the displayed triangle at each of its three corners, the degree shift $H^n(L[1]^\bullet) = H^{n+1}(L^\bullet)$ being what the label $+1$ records.

The connecting morphism is the one induced by α, with no intervening sign; the projection $MC(\alpha)^\bullet \to L[1]^\bullet$ is a chain map precisely because the shift carries the differential $-d_L$, which is what cancels against the sign in the cone. To see this, unwind definition 56.1.18 on representatives: a class in $H^n(L[1]^\bullet)$ is represented by some $l \in L^{n+1}$ with $d_L l = 0$, and its preimage in $MC(\alpha)^n = L^{n+1} \oplus C^n$ may be taken to be $(l, 0)$. Applying the differential of the cone,

$$d_{MC(\alpha)}(l, 0) = \big(-d_L l,\ \alpha^{n+1}(l)\big) = \big(0,\ \alpha^{n+1}(l)\big)$$

which is the image of $\alpha^{n+1}(l) \in C^{n+1}$ under the inclusion. Hence $\delta[l] = [\alpha^{n+1}(l)]$, that is $\delta = H^{n+1}(\alpha)$, completing the proof.

56.1.20 Remark Not all categories have this triangle property, categories which will have this will be called *triangulated categories*

Proposition 56.1.21: Quasi-isomorphism And Isomorphism

Let $\alpha^\bullet : C^\bullet \to N^\bullet$. Then $H^\bullet(\alpha^\bullet)$ is an isomorphism if and only if $MC(\alpha)^\bullet$ is an exact complex.

***Proof*:**
By proposition 56.1.19, applied with $\alpha\colon C^\bullet \to N^\bullet$, the cone sits in a long exact sequence

$$\cdots \to H^n\big(MC(\alpha)^\bullet\big) \to H^n\big(C[1]^\bullet\big) \xrightarrow{H^{n+1}(\alpha)} H^{n+1}(N^\bullet) \to H^{n+1}\big(MC(\alpha)^\bullet\big) \to \cdots$$

the connecting morphism having been identified there as the map induced by α, and $H^n(C[1]^\bullet) = H^{n+1}(C^\bullet)$.

Suppose $MC(\alpha)^\bullet$ is exact, so that $H^\bullet(MC(\alpha)^\bullet) = 0$ by lemma 56.1.14. Deleting the vanishing terms leaves

$$0 \to H^{n+1}(C^\bullet) \xrightarrow{H^{n+1}(\alpha)} H^{n+1}(N^\bullet) \to 0$$

exact for every n, so each $H^{n+1}(\alpha)$ is an isomorphism.

Conversely suppose every $H^n(\alpha)$ is an isomorphism. Fix n and read the sequence around $H^n(MC(\alpha)^\bullet)$:

$$H^{n-1}(C[1]^\bullet) \xrightarrow{H^n(\alpha)} H^n(N^\bullet) \to H^n\big(MC(\alpha)^\bullet\big) \to H^n(C[1]^\bullet) \xrightarrow{H^{n+1}(\alpha)} H^{n+1}(N^\bullet)$$

The left-hand map is surjective and the right-hand map is injective, both being isomorphisms. Exactness at $H^n(N^\bullet)$ then makes the morphism $H^n(N^\bullet) \to H^n(MC(\alpha)^\bullet)$ zero, and exactness at $H^n(C[1]^\bullet)$ makes the morphism $H^n(MC(\alpha)^\bullet) \to H^n(C[1]^\bullet)$ zero. Exactness at $H^n(MC(\alpha)^\bullet)$ therefore gives

$$H^n\big(MC(\alpha)^\bullet\big) = \ker\big(\text{zero morphism}\big) = \operatorname{im}\big(\text{zero morphism}\big) = 0$$

As n was arbitrary the cone is acyclic, hence exact by lemma 56.1.14, completing the proof.

56.1.3 Chain Homotopies and Homotopic Categories

The notion of homotopies originated in topology as a weaker alternative to continuous functions[3]. The topological notion of homotopy settled on the level of continuous functions, where $f, g : X \to Y$ are homotopic (denoted $f \simeq g$) if there exists a continuous function $F : X \times [0,1] \to Y$ where $F(\cdot, t) = f_t$ is continuous, $f_0 = f$ and $f_1 = g$. If we represent X, Y as one dimensional lines, then we may visualize a homotopy as:

[3] every homeomorphic is a homotopy equivalence, however not every homotopy equivalence is a homeomorphism, hence homotopy can be seen as weaker than continuity

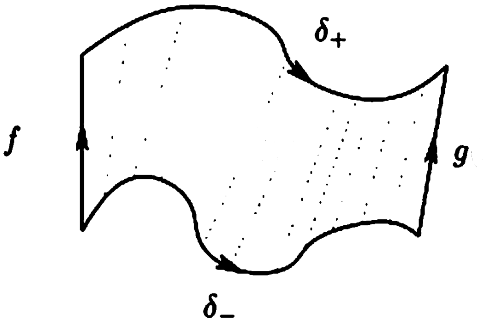


Figure 56.1: homotopy

In the development of homotopy theory, it was found that the (co)homology and homotopy of topological spaces (or their representations as chain complexes) is homology-invariant, hence homotopies are examples of quasi-isomorphisms. The proofs of these results turn out to be independent of the underlying topological structures and is purely algebraic, and hence the notion of homotopy has found a generalization as a purely algebraic notion between two (co)chain complexes.

Due to its origin in topology, the intuitions behind the following definitions will proceed more naturally with a more geometric build-up. Since homotopy may be interpreted purely algebraically, this geometrica build-up will be followed by a more algebraic interpretation. First, the reader may take for granted that given a topological space X, we may induce a chain $C_\bullet$, which is a topological object[4]. Given two continuous maps $f, g : X \to Y$, we may push forward the chain like so:

update image

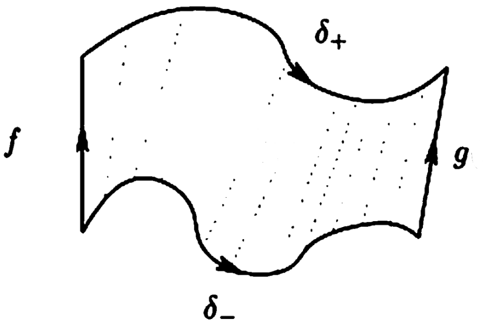


Figure 56.2: homotopy on chain

The space $h(C_\bullet)$ is a topological space, hence we may ask for it's border, $\partial h(C_\bullet)$. By inspecting the image, we see that:

$$\partial h(C_\bullet) = g(C_\bullet) - \delta_+ - f(C_\bullet) + \delta_- = g(C_\bullet) - f(C_\bullet) - h(\partial C_\bullet)$$

re-ordering, we get:

$$g(C_\bullet) - f(C_\bullet) = \partial h(C_\bullet) + h(\partial C_\bullet)$$

[4]The construction of singular chains is in *Algebraic Topology* [38]; [32] is the standard outside reference.

Abstracting notation, we get:

$$g - f = \partial h + h\partial \tag{56.3}$$

Now, recall that in a chain complex, the image of the function ∂ is called the boundary, and that by definition of homology the boundary becomes trivial. In a topological setting, this vocabulary reflects the concrete of the boundary map ∂: it maps a chain to it's boundary. What this means for us is that when taking homology on both sides, we get:

$$H^k(g|_{C_\bullet} - f|_{C_\bullet}) = \overline{0}$$

meaning the two maps are equivalent up to homology! In other words, this "geometric intuition" shows that a homotopy is an example of a quasi-isomorphism. Even better, homotopies shall form an equivalence relation, hence they are an example of a quasi-isomorphism that shall capture some (co)homological information (and in the case where all quasi-isomorphisms are homotopies, they could fully capture the (co)homological information!)

Though this is the origins of chain homotopies being (co)homology-invariant, since they may be interpreted in pure algebraic terms, it is worthwhile taking a moment to have an algebraic build-up as well. Algebraically, homotopies can be seen as forming equivalence classes of (co)chain complexes up to *obstruction from being split-exact*. Recall that a split exact sequence is a short exact sequence that is also split at each map, that is there exists a section morphism $s^n : C^n \to C^{n-1}$ such that $d^{n-1} \circ s^n \circ d^{n-1} = d$, or if we drop the indices and collect all the section maps:

$$d = dsd$$

In other words, it is possible to "inverse" the map d via the map s in such a way that it "cancels" the effect of d (for we may re-apply the map d after sd and get the same result). The simplest source of split-exact (co)chains are exact (co)chains of vector-spaces, for any vector-space maps split[5]. Given a cochain, we may write:

$$C^n = Z^n \oplus B' \qquad (B^n)' \cong C^n/Z^n = d(C^n) = B^{n+1}$$
$$Z^n = B^n \oplus (H^n)' \qquad (H^n)' = Z^n/B^n = H^n(C)$$

Hence, $C^n \cong Z^n \oplus B^{n+1}$. This allows for the definition of the section map via the composition:

$$C^n \hookrightarrow Z^n \hookrightarrow B^n \cong d(C^n) \cong B^{n+1} \subseteq C^{n+1}$$

We may thus define the maps $d \circ s$ and $s \circ d$ where $d \circ s(C^n) = B^n$ and $s \circ d(C^n)(B^n)' \cong B^{n+1}$. Adding these maps together, we get a map $ds + sd : C^n \to C^n$ whose kernel is the homology:

$$\ker(ds + sd)_n = H^n(C^\bullet)$$

The kernel (and cokernel) of the chain map $ds + sd$ is naturally $H^\bullet(C^\bullet)$. Working over more general R-modules (and more generally in an arbitrary abelian category), we loose the natural section maps, for example $0 \to \mathbb{Z} \twoheadrightarrow \mathbb{Z}/2\mathbb{Z} \to 0$ does note have a section map for each map. Instead, we may define arbitrary maps $s^n : C^n \to D^{n-1}$ where we may put them in the following (*not* necessarily commutative) diagram:

$$\begin{tikzcd}
\cdots \arrow[r] & C^{n-1} \arrow[r, "d"] \arrow[dl] & C^n \arrow[r, "d"] \arrow[dl, "s^n"] & C^{n+1} \arrow[r] \arrow[dl, "s^{n+1}"] & \cdots \arrow[dl] \\
\cdots \arrow[r] & D^{n-1} \arrow[r, "d"] & D^n \arrow[r, "d"] & D^{n+1} \arrow[r] & \cdots
\end{tikzcd}$$

[5] More generally, any maps of projective modules split

Given any such collection of morphisms, we may define $f^n = s^{n+1} \circ d^n + d^{n-1} \circ s^n$ (or, if dropping indices and abbreviate, $f = sd + ds$). Notice that even though the maps $s^\bullet$ do not commute, $f^\bullet$ *will* create a commutative diagram, in particular it is a chain map. To see this, notice that (after dropping indices):

$$df = d(ds + sd) = dsd = (ds + sd)d = fd$$

showing it is indeed a chain map:

$$\begin{tikzcd}
\cdots \arrow[r] & C^{n-1} \arrow[r, "d"] \arrow[d, "f^{n-1}"] \arrow[dl, dotted] & C^n \arrow[r, "d"] \arrow[d, "f^n"] \arrow[dl, dotted, "s^n"] & C^{n+1} \arrow[r] \arrow[d, "f^{n+1}"] \arrow[dl, dotted, "s^{n+1}"] & \cdots \arrow[dl, dotted] \\
\cdots \arrow[r] & D^{n-1} \arrow[r, "d"] & D^n \arrow[r, "d"] & D^{n+1} \arrow[r] & \cdots
\end{tikzcd}$$

Thus, given any collection of morphisms $s^\bullet$, we may naturally define a chain map using the differentials. Conversely, is every morphism of complexes representable as $ds + sd$. This is not in general the case; maps being representable like so are given a name:

Definition 56.1.22: Null-homotopic Maps

Let $f^\bullet : C^\bullet \to D^\bullet$ be a morphism of complexes. Then f is called *null-homotopic* if there exists a collection $\{s^i\}_{i\in\mathbb{Z}}$ of morphisms $s^i : C^i \to D^{i-1}$ such that

$$f^\bullet = ds + sd$$

The maps $s^\bullet$ are called the *chain contraction* of $f^\bullet$

To show that not all maps are null-homotopic, consider the following lemma:

Lemma 56.1.23: Null-homotopic And Split Exact Complexes

Let $C^\bullet$ be a complex. Then $C^\bullet$ is split exact if and only if the identity map is null-homotopic.

***Proof*:**
Throughout, split exact means exact together with a family $s^n : C^n \to C^{n-1}$ satisfying $dsd = d$, and null-homotopic means $\mathrm{id} = ds + sd$ for some such family.

Step 1: Null-homotopic implies split exact. Suppose $\mathrm{id} = dt + td$. For exactness, let $c \in \ker d^n$; then

$$c = dt(c) + td(c) = dt(c)$$

so $c \in \mathrm{im} d^{n-1}$, and the reverse inclusion $\mathrm{im} d^{n-1} \subseteq \ker d^n$ holds because $d \circ d = 0$. For the splitting, compute

$$dtd = d(td) = d(\mathrm{id} - dt) = d - d^2 t = d$$

so t is a splitting family.

Step 2: Split exact implies null-homotopic. Suppose $C^\bullet$ is exact and $dsd = d$. Set

$$\varphi = ds + sd$$

Then $d\varphi = dsd = d$ and $\varphi d = dsd = d$, both using $d \circ d = 0$; in particular φ commutes with d.

Put $\psi = \mathrm{id} - \varphi$. From the two identities just proved, $d\psi = d - d = 0$ and $\psi d = d - d = 0$. The first says $\mathrm{im}\psi \subseteq \ker d$, and exactness gives $\ker d = \mathrm{im} d$, so $\mathrm{im}\psi \subseteq \mathrm{im} d$; the second says ψ vanishes on $\mathrm{im} d$. Composing, $\psi \circ \psi = 0$.

Therefore $\varphi = \mathrm{id} - \psi$ is invertible, with inverse $\mathrm{id} + \psi$, and φ^{-1} commutes with d because φ does. Set $t = \varphi^{-1} s$. Then

$$dt + td = d\varphi^{-1}s + \varphi^{-1}sd = \varphi^{-1}(ds + sd) = \varphi^{-1}\varphi = \mathrm{id}$$

so the identity is null-homotopic with contraction t, completing the proof.

The name "null-homotopic" comes from how the (co)homology of such maps are trivial:

Proposition 56.1.24: Null-homotopic And Cohomology

Let $f^\bullet : C^\bullet \to D^\bullet$ be null-homotopic. Then $H^\bullet(f)$ is the zero map

***Proof*:**
Since f is null-homotopic, let $f = ds + sd$. Take a n-cycle representative x of an equivalence class of $H^n(C^\bullet)$. Then:

$$f(x) = d(s(x))$$

Hence, $f(x)$ is an n-boundary in D^n, hence $\overline{f(x)} = \overline{0}$ in $H^n(D^\bullet)$, as we sought to show. (this is an elementary result, but a similar proof is given in Aluffi p.612 if needed)

Thus, maps that are null-homotopic induce zero (co)homology maps![6] Thus, we may think of maps that are null-homotopic as being the "zero" maps. This leads us to the following definition:

Definition 56.1.25: Chain-homotopy

Let $f^\bullet, g^\bullet : C^\bullet \to D^\bullet$ be two morphisms of complexes. Then we say that these two maps are *chain homotopic* or just *homotopic* if there exists maps $h^i : C^i \to D^{i-1}$ such that

$$g^i - f^i = d^{i-1}_{D^\bullet} \circ h^i + h^{i+1} \circ d^i_{C^\bullet}$$

or, dropping indices:

$$g^\bullet - f^\bullet = dh + hd$$

If $f^\bullet, g^\bullet$ are homotopic, we write $f^\bullet \sim g^\bullet$

The s notation has been replaced with h notation to emphasize now the "homotopy" aspect of it rather than the "section" property that was evident in the case of vector spaces. If we were working with chains, then we would have

$$g_\bullet - f_\bullet = \partial h + h\partial$$

which, the reader may recognize as equation (56.3). If we were working with chains, then $h_i : C_i \to D_{i+1}$ would shift up a degree, which would be interpreted in the geometrical setting as moving *up* a dimension, in the same way that homotopies are a "dimension more" than the original spaces X, Y.

[6]TBD How much can we say about the converse, if a map induces the zero morphism, then is it null-homotopic?

Furthermore, the reader should verify the "up to homotopy" *is* an equivalence relation, as the reader should verify!

Importantly, two chain-homotopic maps induce the same (co)homology:

Proposition 56.1.26: Chain-Homotopy and Cohomology

Let $f^\bullet, g^\bullet : C^\bullet \to D^\bullet$ be chain homotopic. Then

$$H^\bullet(f^\bullet) = H^\bullet(g^\bullet)$$

***Proof*:**
Since $f^\bullet$ and $g^\bullet$ are chain-homotopic, by proposition 56.1.24 for each element in $H^k(C^\bullet)$ (since it is an additive functor):

$$H^k(f^\bullet)(\overline{x}) - H^k(g^\bullet)(\overline{x}) = 0 \qquad \Rightarrow \qquad H^k(f^\bullet)(\overline{x}) = H^k(g^\bullet)(\overline{x})$$

as we sought to show.

The above result shows promise in using chain-homotopies to hold (co)homological information. The following definition is the important connection idea:

Definition 56.1.27: Homotopy Equivalent Chains

Let $f^\bullet : C^\bullet \to D^\bullet$ be a morphism of complexes. Then $f^\bullet$ is a *homotopy equivalence* if there exists a morphisms of complexes $g^\bullet : D^\bullet \to C^\bullet$ such that

$$f^\bullet \circ g^\bullet \sim 1_{D^\bullet} \qquad \text{and} \qquad g^\bullet \circ f^\bullet \sim 1_{C^\bullet}$$

If $f^\bullet : C^\bullet \to D^\bullet$ is a homotopy equivalence, then $C^\bullet$ and $D^\bullet$ are said to be *homotopy equivalent*

Proposition 56.1.28: Homotopy Equivalence Ans Cohomology

Let $f^\bullet : C^\bullet \to D^\bullet$ be a homotopy equivalence. Then $f^\bullet$ is a quasi-isomorphism, that is $H^\bullet(f^\bullet)$ is an isomorphism

***Proof*:**
Since $f^\bullet$ is a homotopy equivalence, we have $g^\bullet$ that is it's homotopic inverse. Then the maps $f^\bullet \circ g^\bullet$ and $g^\bullet \circ f^\bullet$ induce inverse morphisms of cohomology by proposition 56.1.26, and hence give an isomorphism, as we sought to show.

As demonstrated in example 56.1.13 not all quasi-isomorphisms are invertible even up to homotopy equivalence, hence quasi-isomorphisms are more general. As mentioned earlier, in nice enough categories these two concepts shall coincide.

Now that homotopy equivalences is established, can be asked if given an (additive) functor $\mathscr{F} : \mathbf{A} \to \mathbf{B}$, if there are two homotopic complexes in $\mathrm{Ch}(\mathbf{A})$, will the (co)homology be isomorphic in $\mathbf{B}$? In other words, when trying to define the derived category that was mentioned earlier, can this category be defined at least up to homotopy? The answer is yes! The proof of this result essentially follows from the following lemma:

Lemma 56.1.29: Homotopy Equivalence And Additive Functors

Let $\mathscr{F} : \mathbf{A} \to \mathbf{B}$ be an additive functor between two abelian categories, $C(\mathscr{F})$ the induced functor on chains. Then if $f^\bullet \sim g^\bullet$ in $\mathrm{Ch}(\mathbf{A})$, $C(\mathscr{F})(f^\bullet) \sim \mathrm{Ch}(\mathbf{B})(g^\bullet)$ in $\mathrm{Ch}(\mathbf{B})$. Furthermore, if $C^\bullet$ and $D^\bullet$ are homotopy equivalence, $C(\mathscr{F})(C^\bullet)$ and $C(\mathscr{F})(D^\bullet)$ are homotopy equivalent

***Proof*:**
Given the first assertion, the second is an immediate consequence, hence we prove the first. Let $f^\bullet \sim g^\bullet$ so that :

$$f^i - g^i = d_{D^\bullet}^{i-1} \circ h^i + h^{i+1} \circ d_{C^\bullet}^i$$

Since $\mathscr{F}$ is an additive functor:

$$\mathscr{F}(f^i) - \mathscr{F}(g^i) = \mathscr{F}\left(d_{D^\bullet}^{i-1}\right) \circ \mathscr{F}(h^i) + \mathscr{F}\left(h^{i+1}\right) \circ \mathscr{F}\left(d_{C^\bullet}^i\right)$$

hence, the morphisms $\mathscr{F}(h^i)$ gives a homotopy between $\mathscr{F}(f^\bullet)$ and $\mathscr{F}(g^\bullet)$, as we sought to show.

Note this does *not* hold for quasi-isomorphisms (exercise 4.15 in Aluffi), needing the stronger condition of exact functors.

Theorem 56.1.30: Derived Category And Homotopy Equivalence

Let $\mathscr{F} : \mathbf{A} \to \mathbf{B}$ be an additive functor between two abelian categories. Then if $C^\bullet$ and $D^\bullet$ are homotopy-equivalent, then:

$$H^\bullet(\mathscr{F}(C^\bullet)) \cong H^\bullet(\mathscr{F}(D^\bullet))$$

that is, their cohomology is isomorphic.

***Proof*:**
By lemma 56.1.29, an additive functor carries homotopy-equivalent complexes to homotopy-equivalent complexes, so $\mathscr{F}(C^\bullet)$ and $\mathscr{F}(D^\bullet)$ are homotopy equivalent in $\mathrm{Ch}(\mathbf{B})$. By proposition 56.1.28 a homotopy equivalence is a quasi-isomorphism, so the induced map on cohomology is an isomorphism, giving $H^\bullet(\mathscr{F}(C^\bullet)) \cong H^\bullet(\mathscr{F}(D^\bullet))$, as we sought to show.

Hence, up to homotopy, the choice of cochain shall give the same cohomology information!

Homotopic Categories

With chain homotopies being a nicer sub-class of quasi-isomorphisms, their exploration is worth further pursuing. They are in fact almost the solution to the universal problem given in equation (56.2). Let us take for granted that a category $D(A)$ exists that satisfies the universal property. What necessary conditions can be deduced about $D(A)$?

Lemma 56.1.31: Derived Category And Additive Functors

Let $\mathbf{D}$ be an additive category, $\mathscr{F} : \mathrm{Ch}(\mathbf{A}) \to \mathbf{D}$ be an additive functor that takes quasi-isomorphisms to isomorphisms.

1. Let $C^\bullet$ be an exact complex in $\mathrm{Ch}(\mathbf{A})$. Then $\mathscr{F}(C^\bullet) = 0$, that is $\mathscr{F}(C^\bullet)$ is the zero object in $\mathbf{D}$.
2. Let $\alpha^\bullet : C^\bullet \to E^\bullet$ be a morphism of complexes that factors through an exact complex:

$$\begin{tikzcd} C^\bullet \arrow[rr, "\alpha^\bullet"] \arrow[dr] & & E^\bullet \\ & D^\bullet \arrow[ur] & \end{tikzcd}$$

   where $D^\bullet$ is exact. Then $\mathscr{F}(\alpha^\bullet)$ is the zero morphism.

Proof:

1. Since $C^\bullet$ is exact, the zero morphism $0^\bullet : C^\bullet \to C^\bullet$ is a quasi-isomorphism, and so $\mathscr{F}(0^\bullet)$ maps to an isomorphism (i.e. an invertible morphism in $\mathbf{D}$). Thus:

$$\begin{tikzcd} \mathscr{F}(C^\bullet) \arrow[r, "\mathscr{F}(0^\bullet)"'] \arrow[rr, bend left, "\mathrm{id}_{\mathscr{F}(C^\bullet)}"] & \mathscr{F}(C^\bullet) \arrow[r, "\mathscr{F}(0^\bullet)^{-1}"'] & \mathscr{F}(C^\bullet) \end{tikzcd}$$

   Since $\mathscr{F}$ is additive functor, in particular a group homomorphism on hom-sets, $\mathscr{F}(0) = 0$. Thus:

$$\mathrm{id}_{\mathscr{F}(C^\bullet)} = 0$$

   It follows that $\mathscr{F}(C^\bullet)$ must be the zero object: an object whose identity morphism is zero admits only the zero endomorphism, so it is isomorphic to 0

2. Applying $\mathscr{F}$ to the diagram in the question, we get the diagram:

$$\begin{tikzcd} \mathscr{F}(C^\bullet) \arrow[rr] \arrow[dr] & & \mathscr{F}(E^\bullet) \\ & \mathscr{F}(D^\bullet) \arrow[ur] & \end{tikzcd}$$

   Since $\mathscr{F}(D^\bullet) = 0$ by the above result, we see that $\mathscr{F}(\alpha^\bullet)$ must be the zero morphisms, completing the proof.

Lemma 56.1.32: Derived Categories And Homotopies

Let $\mathscr{F} : \mathrm{Ch}(\mathbf{A}) \to \mathbf{D}$ be an additive functor that maps quasi-isomorphisms to isomorphisms. Let $f^\bullet, g^\bullet : C^\bullet \to D^\bullet$ be homotopic morphisms in $\mathrm{Ch}(\mathbf{A})$. Then

$$\mathscr{F}(f^\bullet) = \mathscr{F}(g^\bullet)$$

Note that this was shown explicitly when $\mathscr{F} = H^i(-)$.

***Proof*:**
Since $\mathscr{F}$ is additive, it suffices to show that if $f^\bullet \sim 0$, then $\mathscr{F}(f^\bullet) = 0$. By lemma 56.1.31, it suffices to show $f^\bullet$ factors through an exact complex. The exact complex that shall be used is the mapping cone $MC(\mathrm{id}_{C^\bullet})$ for the identity map $\mathrm{id}_{C^\bullet} : C^\bullet \to C^\bullet$, namely

$$MC(\mathrm{id}_{C^\bullet})^i = C^{i+1} \oplus C^i$$

$$d^i_{MC(\mathrm{id}_{C^\bullet})}(a,b) = (-d^{i+1}_{C^\bullet}(a), a + d^i_{C^\bullet}(b))$$

By proposition 56.1.21, since the identity map $\mathrm{id}_{C^\bullet}$ is (trivially) a quasi-isomorphism, $MC(\mathrm{id}_{C^\bullet})$ is an exact complex.

Now, the goal is to find morphism of complexes where:

$$C^\bullet \to MC(\mathrm{id}_{C^\bullet}) \to D^\bullet$$

the morphism of complexes $C^\bullet \to MC(\mathrm{id}_{C^\bullet})$ is readily available: for each i:

$$b \mapsto (0, b)$$

On the other hand, it is not immediately clear there is a morphism of complexes $MC(\mathrm{id}_{C^\bullet}) \to D^\bullet$. To find such a morphism, we take advantage of the homotopy $f^\bullet \sim 0$, that is there exists maps:

$$h^i : C^i \to D^{i-1}$$

such that $f^\bullet = fh + hd$. With these, define the maps:

$$\pi^i : MC(\mathrm{id}_{C^\bullet})i \to D^i \qquad (a,b) \mapsto h^{i+1}(a) + f^i(b)$$

If this is indeed a morphism of complexes, then the map $f : C^\bullet \to D^\bullet$ factored through an exact complex and hence must be trivial. What must be verified is that:

$$d^{i-1}_{D^\bullet} \circ \pi^{i+1} = \pi^i \circ d^{i-1}_{MC(\mathrm{id}_{C^\bullet})} \tag{56.4}$$

The left hand side of equation (56.4) gives:

$$(a,b) \mapsto h^i(a) + f^{i-1}(b) \mapsto d^{i-1}_{D^\bullet}(h^i(a) + f^{i-1}(b))$$

while the right hand side gives:

$$(a,b) \mapsto (-d^i_{C^\bullet}(a), a + d^{i-1}_{L^\bullet}(b)) \mapsto -h^{i+1}(d^i_{C^\bullet}(a) + f^i(a + d^{i-1}_{C^\bullet}(b))$$

These two sides must be equal. Simplifying the equality, we get:

$$(d^{i-1}_{D^\bullet} \circ h^i + h^{i+1} \circ d^i_{C^\bullet} - f^i)(a) = (f^i \circ d^{i-1}_{C^\bullet} - d^{i-1}_{D^\bullet} \circ f^{-1})(a)$$

for all $a \in C^{i+1}, b \in C^i$. But notice that both sides are zero: the right hand side b $f^\bullet$ is a morphism of complexes, while the left hand side because h is a homotopy and $f^\bullet \sim 0$. But then this gives us our desired result, completing the proof.

Thus, behave well for the desired universal property presented in equation 56.2. We formalize the category in question:

Definition 56.1.33: Homotopy Category

Let $\mathbf{A}$ be an abelian category. Then the *homotopy category* of $\mathbf{A}$, denoted $K(\mathbf{A})$ is the category whose objects ar ethe (co)chain complexes in $\mathbf{A}$ (i.e. $\mathrm{Ch}(\mathbf{A})$), and whose morphisms are:

$$\mathrm{Hom}_{K(\mathbf{A})}(C^\bullet, D^\bullet) = \mathrm{Hom}_{\mathrm{Ch}(\mathbf{A})}(C^\bullet, D^\bullet)/\sim$$

where $\sim$ is the homotopy relation, that is, the morphisms are morphisms of complexes up to homotopy.

These are indeed morphisms, namely $\sim$ is transitive, it respects composition (exercise). Naturally, the equivalent bounded variations $K^-(\mathbf{A})$ and $K^+(\mathbf{A})$ are obtained from $\mathrm{Ch}^+(\mathbf{A})$ and $\mathrm{Ch}^-(\mathbf{A})$.

Lemma 56.1.34: Homotopy Category Is Additive

Let $\mathbf{A}$ be an abelian category. Then $K(\mathbf{A})$ is an *additive category*

***Proof*:**
Recall from definition 56.1.33 that $K(\mathbf{A})$ has the same objects as $\mathrm{Ch}(\mathbf{A})$ and $\mathrm{Hom}_{K(\mathbf{A})}(C^\bullet, D^\bullet) = \mathrm{Hom}_{\mathrm{Ch}(\mathbf{A})}(C^\bullet, D^\bullet)/\sim$, the quotient by chain homotopy.

Step 1: The hom-sets are abelian groups. The null-homotopic morphisms form a subgroup of $\mathrm{Hom}_{\mathrm{Ch}(\mathbf{A})}(C^\bullet, D^\bullet)$: if $f = ds + sd$ and $f' = ds' + s'd$ then $f - f' = d(s - s') + (s - s')d$ is null-homotopic. Being a quotient of an abelian group by a subgroup, each hom-set of $K(\mathbf{A})$ is an abelian group, and $f \sim g$ exactly when $f - g$ is null-homotopic.

Step 2: Composition is well defined and bilinear. Suppose $f \sim f'$, say $f - f' = ds + sd$, and let g be a chain map composable on the left. Since g commutes with the differentials,

$$gf - gf' = g(ds + sd) = d(gs) + (gs)d$$

so $gf \sim gf'$; composing on the right with a chain map h gives $fh - f'h = d(sh) + (sh)d$ likewise. Hence composition descends to the quotient, and it is bilinear because it is bilinear in $\mathrm{Ch}(\mathbf{A})$ and the quotient maps are group homomorphisms.

Step 3: Zero object and biproducts. The zero complex is a zero object in $\mathrm{Ch}(\mathbf{A})$ and remains one after the quotient, since its hom-sets are already zero. For two complexes $C^\bullet$ and $D^\bullet$, the degreewise direct sum $C^\bullet \oplus D^\bullet$ carries the four structure morphisms i_C, i_D, p_C, p_D, which are chain maps and satisfy the biproduct identities $p_C i_C = \mathrm{id}$, $p_D i_D = \mathrm{id}$, $p_C i_D = 0$, $p_D i_C = 0$ and $i_C p_C + i_D p_D = \mathrm{id}$ in $\mathrm{Ch}(\mathbf{A})$. Identities between morphisms are preserved by the quotient functor, so they continue to hold in $K(\mathbf{A})$, and an object with such data is a biproduct there as well.

Together these are the axioms of an additive category, completing the proof.

Note that in general homotopic categories are *not* abelian. Indeed, homotopic maps $f^\bullet \sim g^\bullet$ need not have the same kernel/cokernel, giving no natural definition of these concepts. The homotopic category is a type of category that is stronger than additive but weaker than abelian, known as a *triangulated category* (recall box 56.1.20).

In $K(\mathbf{A})$, homotopy equivalences become isomorphisms. This comes straight from the definition, for if $f^\bullet \circ g^\bullet \sim \text{id}$ in $\text{Ch}(\mathbf{A})$, then $f^\bullet \circ g^\bullet = \text{id}$ in $K(\mathbf{A})$, which can be thought of as formally inverting homotopy equivalences. Using this, we have the following almost universal property, or factorization

It looks like a universal property because it is one, though not the one for quasi-isomorphisms. Comparing with definition 55.2.1, the homotopy category is precisely the localization

$$K(\mathbf{A}) \cong \text{Ch}(\mathbf{A})\big[\{\text{homotopy equivalences}\}^{-1}\big]$$

The proposition below is stated with the hypothesis that $\mathscr{F}$ inverts *quasi-isomorphisms*, which is strictly stronger than it needs to be, and that mismatch is what makes it read as an "almost" universal property. Its proof uses only that $\mathscr{F}$ inverts homotopy equivalences. Inverting all quasi-isomorphisms is the stronger requirement that produces $D(\mathbf{A})$ instead (corollary 55.2.6), and the gap between the two is exactly example 56.1.13: a quasi-isomorphism need not be a homotopy equivalence.

Proposition 56.1.35: Homotopy Category Factorization

Let $\mathscr{F} : \text{Ch}(\mathbf{A}) \to D$ be an additive functor that maps quasi-isomorphisms to isomorphisms. Then $\mathscr{F}$ factors uniquely through $K(\mathbf{A})$:

$$\begin{tikzcd} \text{Ch}(\mathbf{A}) \arrow[r, "\mathscr{F}"] \arrow[d] & D \\ K(\mathbf{A}) \arrow[ur, "\exists!"'] & \end{tikzcd}$$

***Proof*:**
Since every homotopy equivalence is a quasi-isomorphism (proposition 56.1.28), the hypothesis in particular sends homotopy equivalences to isomorphisms, and that is all the proof uses. Because $\mathscr{F}$ is defined on $\text{Ch}(\mathbf{A})$ and $K(\mathbf{A})$ has the same objects, the only thing to establish is that $f \sim g$ implies $\mathscr{F}(f) = \mathscr{F}(g)$; the factorization and its uniqueness are then forced, since the quotient functor $\text{Ch}(\mathbf{A}) \to K(\mathbf{A})$ is the identity on objects and surjective on morphisms.

Step 1: The cylinder of a complex. For a complex $C^\bullet$ define

$$\text{Cyl}(C)^n = C^n \oplus C^{n+1} \oplus C^n, \qquad d(a,b,c) = (da - b,\ -db,\ dc + b)$$

This squares to zero, since $d^2(a,b,c) = (d(da-b)+db,\ d(db),\ d(dc+b)-db) = (0,0,0)$. Define chain maps

$$i_0(a) = (a,0,0), \qquad i_1(c) = (0,0,c), \qquad p(a,b,c) = a + c$$

Each is compatible with the differentials: $d\,i_0(a) = (da,0,0) = i_0(da)$, similarly for i_1, and $p(d(a,b,c)) = (da-b)+(dc+b) = d(a+c)$. Moreover $pi_0 = \text{id}_C = pi_1$.

Step 2: p is a homotopy equivalence. The composite $i_0p - \text{id}$ sends (a,b,c) to $(c,-b,-c)$, and setting $s(a,b,c) = (0,-c,0)$ gives

$$ds(a,b,c) = (c,\ dc,\ -c), \qquad sd(a,b,c) = (0,\ -(dc+b),\ 0)$$

whose sum is $(c,-b,-c)$. So $i_0 p \sim \mathrm{id}_{\mathrm{Cyl}(C)}$ while $pi_0 = \mathrm{id}_C$, making i_0 and p mutually inverse homotopy equivalences. The symmetric computation applies to i_1, with the contracting homotopy $s'(a,b,c) = (0,a,0)$ in place of s, since $i_1 p - \mathrm{id}$ sends (a,b,c) to $(-a,-b,a)$.

Step 3: Homotopies are morphisms out of the cylinder. Let $f,g\colon C^\bullet \to D^\bullet$ with $g - f = ds + sd$. Define $H\colon \mathrm{Cyl}(C) \to D^\bullet$ by $H(a,b,c) = f(a) + s(b) + g(c)$. It is a chain map: expanding both sides, $H(d(a,b,c)) - dH(a,b,c) = (g - f - ds - sd)(b) = 0$. By construction $Hi_0 = f$ and $Hi_1 = g$.

Step 4: Conclusion. Apply $\mathscr{F}$. By Step 2 the morphisms i_0 and i_1 are homotopy equivalences, hence quasi-isomorphisms, so $\mathscr{F}(i_0)$ and $\mathscr{F}(i_1)$ are isomorphisms by hypothesis. From $pi_0 = \mathrm{id} = pi_1$ and functoriality, $\mathscr{F}(p)\mathscr{F}(i_0) = \mathrm{id} = \mathscr{F}(p)\mathscr{F}(i_1)$, and since $\mathscr{F}(i_0)$ is invertible so is $\mathscr{F}(p)$; cancelling it gives $\mathscr{F}(i_0) = \mathscr{F}(p)^{-1} = \mathscr{F}(i_1)$. Therefore

$$\mathscr{F}(f) = \mathscr{F}(Hi_0) = \mathscr{F}(H)\mathscr{F}(i_0) = \mathscr{F}(H)\mathscr{F}(i_1) = \mathscr{F}(Hi_1) = \mathscr{F}(g)$$

so $\mathscr{F}$ identifies homotopic morphisms and factors uniquely through $K(\mathbf{A})$, completing the proof.

Hence, in the case where $K(\mathbf{A})$ is *not* the universal object satisfying equation (56.2), there must be a unique functor from $K(\mathbf{A})$ to $D(\mathbf{A})$.

56.2 Projective and Injective Resolutions

It has been alluded to a few times that for special categories, homotopy equivalence and quasi-isomorphism coincide. This shall now be covered in this section. In particular, this shall linked to the idea of an abelian category $\mathbf{A}$ being bounded from above or bellow, and having *enough projective objects* or *enough injective objects*. These categories shall furthermore be shown to be universal in some appropriate way, making their choice not arbitrary. Recall that in $\mathbf{A} = R$-**Mod**, an R-module M Is called projective if $\mathrm{Hom}_R(M,-)$ is exact, and injective if $\mathrm{Hom}_R(-,M)$ is exact. Recall that projective/injective modules nicely lend themselves to a splitting condition on short exact sequences for which they are one of the modules forming them (recall theorem 17.2.4 and theorem 17.3.4). This will be important, due to the link of chain homotopies with split exact sequences.

Definition 56.2.1: Projective And Injective Objects

Let $\mathbf{A}$ be an abelian category. Then

1. An object $P \in \mathbf{A}$ is said to be *projective* if the functor $\mathrm{Hom}_A(P,-)$ is exact.
2. An object $Q \in \mathbf{A}$ is said to be *injective* if the functor $\mathrm{Hom}_A(-,Q)$ is exact

theorem 17.2.4 and theorem 17.3.4 hold for abelian categories more generally (in particular the splitting remarks), as the reader may venture to check. Since the axioms for an abelian category are self-dual, so that $\mathbf{A}^{\mathrm{op}}$ is abelian whenever $\mathbf{A}$ is, the theory on projective objects is dual to that of injective object, and the functor $(-)^{\text{op}}$ maps projective/injectives to injectives/projectives respectively. Thus, many proofs shall simply proof the result for projectives, the result for injectives coming automatically. Note that as we've seen in chapter 17⋆ in a fixed [abelian] category $\mathbf{A}$, projective and injective may have very different features.

In what follows, let $\mathbf{P}$ denote the full subcategory of $\mathbf{A}$ determined by the projective objects, and let $\mathbf{I}$ denote the full subcategory of $\mathbf{A}$ determined by the injective objects. Let $K^-(\mathbf{P})$ denote the full subcategory of $K(\mathbf{A})$ consisting of bounded-above complexes of projective objects of $\mathbf{A}$, and $K^+(\mathbf{I})$ be the full subcategory of $K(\mathbf{A})$ consisting of bounded-below complexes of injective objects of $\mathbf{A}$. Note that these are generally not abelian categories, and furthermore projective and injective objects need not exist:

Example 56.2.2: Category With And Without Projective/Injective Resolutions

1. The category of finite abelian groups has no projective or injective, hence no projective/injective resolution.
2. The category of finitely generated abelian groups has projectives, but no injectives, namely since $\mathbb{Z}^n$ for every n are objects this category. Thus, it is possible to create a free, and hence projective, resolution.
3. the dual of the above category has enough injectives but not enough projectives.
4. The category of R-modules has both projective and injective resolutions (projective resolutions comes from free resolutions, while injective resolutions comes from repeated application of theorem 17.3.19)

It is thus helpful to add the following condition to an abelian category

Definition 56.2.3: Enough Projective And Enough Injectives

Let $\mathbf{A}$ be an abelian category. Then:

1. we say that $\mathbf{A}$ as *enough projectives* if for every object A of $\mathbf{A}$, there exists a projective object P of $\mathbf{A}$ and an epimorphism $P \twoheadrightarrow A$.
2. we say that $\mathbf{A}$ as *enough injectives* if for every object A of $\mathbf{A}$, there exists a injective object Q of $\mathbf{A}$ and a monomorphism $A \hookrightarrow Q$.

With the important definitions established, the main objective of this section may start going under way. The goal is to show that if we have a quasi-isomorphism between two bounded-above complexes of projectives $f^\bullet : P_0^\bullet \to P_1^\bullet$ (resp. quasi-isomorphism between two bounded-bellow complexes of injectives $f^\bullet : Q_0^\bullet \to Q_1^\bullet$), then $f^\bullet$ is in fact a homotopy-equivalence, from which it can be concluded that in $K^-(\mathbf{P})$ and $K^+(\mathbf{I})$ (homotopy-classes of) quasi-isomorphisms are isomorphisms, which shall give us the desired derived category $D^-(\mathbf{A})$ and $D^+(\mathbf{A})$ in the case where A has enough projectives (resp. enough injectives).

Let us now revisit this splitting condition. Recall that if $C^\bullet$ in $\mathrm{Ch}(\mathbf{A})$ is exact, then the identity map $\mathrm{id}_{C^\bullet}$ and the trivial map induce the same morphisms in (co)homology. However, there do exist exact complexes that are *not* homotopic to 0. As discussed earlier, this directly links to the splitting condition, namely the identity is homotopic to 0 if the sequence is split exact. Being projective is exactly the condition that causes splitting behavior. With this in mind, let us prove the first lemma:

Lemma 56.2.4: Zero-morphism In Cohomology And Homotopy

Let $\mathbf{A}$ be an abelian category, $P^\bullet \in \mathrm{Ch}^-(\mathbf{P})$ (i.e. $P^i = 0$ for $i > 0$), and let $C^\bullet$ be a complex in $\mathrm{Ch}(\mathbf{A})$ such that $H^i(C^\bullet) = 0$ for $i < 0$ (i.e. has trivial cohomology in negative coefficients). Let $f^\bullet : P^\bullet \to C^\bullet$ be a morphism of complexes that induces the zero-morphism in cohomology: $H^\bullet(f^\bullet) = 0$. Then

$$f^\bullet \sim 0$$

i.e., $f^\bullet$ is null-homotopic.

The injective version of this statement is comes from dualizing the appropriate terms.

***Proof*:**
To show $f^\bullet$ is null-homotopic, we must show there exists maps $h^i : P^i \to C^{i-1}$ such that

$$f^i = d_{C^\bullet}^{i-1} \circ h^i + h^{i+1} \circ d_{P^\bullet}^i \tag{56.5}$$

as a diagram:

$$\begin{tikzcd}
\cdots \arrow[r] & P^{-2} \arrow[r] \arrow[d, "f^{-2}"] \arrow[dl, "h^{-2}"'] & P^{-1} \arrow[r] \arrow[d, "f^{-1}"] \arrow[dl, "h^{-1}"'] & P^0 \arrow[r] \arrow[d, "f^0"'] \arrow[dl, "h^0"'] & 0 \arrow[r] \arrow[d] \arrow[dl, "h^1=0"'] & \cdots \\
\cdots \arrow[r] & C^{-2} \arrow[r] & C^{-1} \arrow[r] & C^0 \arrow[r] & C^1 \arrow[r] & \cdots
\end{tikzcd}$$

Naturally, $h^i = 0$ for $i > 0$. For the rest, we shall use induction. The base case is $i = 0$. Since the cohomology induced by $f^\bullet$ is zero, f^0 factors through the image of $d_{C^\bullet}^{-1}$:

$$\begin{tikzcd}
 & P^0 \arrow[d, "f^0"] & \\
C^{-1} \arrow[r, "d_{C^\bullet}^{-1}"] & \mathrm{im} d_{C^\bullet}^{-1} \arrow[r] & 0
\end{tikzcd}$$

Now, since P^0 is projective, there exists a morphism $h^0 : P^0 \to C^{-1}$ such that $f^0 = d_{C^\bullet}^{-1} \circ h^0$, as needed.

Next, assume h^i and h^{i+1} satisfies equation (56.5). The goal is to show h^{i-1} satisfies it too. By equation (56.5) and the fact that $P^\bullet$ is a complex:

$$d_{C^\bullet}^{i-2} \circ h^i \circ d_{P^\bullet}^{i-1} = (f^i - h^{i+1} \circ d_{P^\bullet}^{i-1}) = f^i \circ d_{P^\bullet}^{i-1} - h^{i+1} \circ 0 = f^i \circ d_{P^\bullet}^{i-1}$$

which can be seen by staring at the following diagram:

$$\begin{tikzcd}
\cdots \arrow[r, dotted] & P^{i-1} \arrow[r, "d_{P^\bullet}^{i-1}"] \arrow[d, dotted] & P^i \arrow[r, "d_{P^\bullet}^i"] \arrow[d, "f^i"'] \arrow[dl, "h^i"'] & P^{i+1} \arrow[r, dotted] \arrow[d, dotted] \arrow[dl, "h^{i+1}"'] & {} \\
\cdots \arrow[r, dotted] & C^{i-1} \arrow[r, "d_{C^\bullet}^{i-1}"'] & C^i \arrow[r, dotted] & C^{i+1} \arrow[r, dotted] & {}
\end{tikzcd}$$

Thus, since f^b is a morphism of cochain complexes:

$$d_{C^\bullet}^{i-1} \circ (f^{i-1} - h^i \circ d_{P^\bullet}^{i-1}) = d_{C^\bullet}^{i-1} \circ f^{i-1} - f^i \circ d_{P^\bullet}^{i-1} = 0$$

Importantly, $(f^{i-1} - h^i \circ d_{P^\bullet}^{i-1})$ factors through $\ker d_{C^\bullet}^{i-1}$. Since $C^\bullet$ is exact at C^{i-1} for $i \leq 0$, $\ker d_{C^\bullet} = \mathrm{im}_{C^\bullet}^{i-2}$. Thus, we get:

$$\begin{tikzcd} & P^{i-1} \arrow[d, "f^{i-1} - h^i \circ d_{P^\bullet}^{i-1}"] & \\ C^{i-2} \arrow[r, "d_{C^\bullet}^{i-2}"] & \mathrm{im}(d_{C^\bullet}^{i-2}) \arrow[r] & 0 \end{tikzcd}$$

Now, since P^{i-1} is projective, there exists a lift $h^{i-1} : P^{i-1} \to C^{i-2}$ such that

$$f^{i-1} - h^i \circ d_{P^\bullet}^{i-1} = d_{C^\bullet}^{i-2} \circ h^{i-1}$$

which, shuffling around, gives us

$$f^{i-1} = d_{C^\bullet}^{i-2} \circ h^{i-1} + h^i \circ d_{P^\bullet}^{i-1}$$

as we sought to show.

Corollary 56.2.5: Exactness Cohomology And Homotopy

Let $\mathbf{A}$ be an abelian category, $P^\bullet \in \mathrm{Ch}^-(\mathbf{A})$, and $C^\bullet \in \mathrm{Ch}(\mathbf{S})$ an exact complex. Then every morphism $P^\bullet \to L^\bullet$ is homotopic to 0

***Proof*:**
Since every morphism to an exact complex induces zero-morphism in cohomology, this is a direct application of lemma 56.2.4.

Note the importance of the bounding from above/bellow (depending on whether working with projectives or injective) in the inductive argument, without it there wouldn't be an appropriate base case. Using the above results, the first quasi-isomorphism to homotopy equivalence can be established:

Corollary 56.2.6: Projectives With Quasi-isomorphism And Zero Morphisms

Let $P^\bullet$ (resp. $Q^\bullet$) be a bounded-above exact complex of projectives (resp. a bounded bellow exact complex of injectives). Then $P^\bullet$ (resp. $Q^b b$) is homotopy-equivalent to the zero-complex.

***Proof*:**
Treat the projective case; the injective one is the same argument in the opposite category. A complex is homotopy equivalent to the zero complex exactly when its identity morphism is null-homotopic, since a homotopy inverse to the zero complex forces id ~ 0 and conversely. By lemma 56.1.23 it therefore suffices to show that $P^\bullet$ is split exact.

Write $Z^n = \ker d^n$ and $B^n = \mathrm{im} d^{n-1}$; exactness gives $Z^n = B^n$, so for every n there is a short exact sequence

$$0 \to B^n \to P^n \to B^{n+1} \to 0$$

Since $P^\bullet$ is bounded above there is an N with $P^n = 0$ for $n > N$, whence $B^n = 0$ for $n > N + 1$.

Argue by descending induction that each B^n is projective. For $n > N + 1$ it is zero. Suppose B^{n+1} is projective. Then the displayed short exact sequence splits, because a surjection onto a projective object admits a section, so $P^n \cong B^n \oplus B^{n+1}$ and B^n is a direct summand of the projective object P^n, hence projective itself.

Every one of the short exact sequences therefore splits. Choosing a section $\sigma^{n+1} \colon B^{n+1} \to P^n$ of d^n for each n and letting $s^{n+1} \colon P^{n+1} \to P^n$ be σ^{n+1} composed with the projection $P^{n+1} \twoheadrightarrow B^{n+1}$ gives $dsd = d$, since $d^n \sigma^{n+1}$ is the identity on B^{n+1} and d factors through B^{n+1}. So $P^\bullet$ is split exact, and lemma 56.1.23 makes $\mathrm{id}_{P^\bullet}$ null-homotopic, completing the proof.

Another important result that will be used in the next section is that quasi-isomorphisms are "non-zero-divisors" up to homotopy wrt morphisms from commplexes of projectives:

Lemma 56.2.7: Quasi-isomorphisms Of Projectives And Zero-divisor

Let $\mathbf{A}$ be an abelian category, $f^\bullet : C^\bullet \to D^\bullet$ be a quasi-isomorphism. Let $P^\bullet$ be a bounded-above complex of projectives, and let $\alpha^\bullet : P^\bullet \to C^\bullet$ be a morphism of cochain complexes such that the composition:

$$P^\bullet \xrightarrow{\alpha^\bullet} C^\bullet \xrightarrow{f^\bullet} D^\bullet$$

is homotopic to the zero-morphism. Then $\alpha^\bullet \sim 0$

***Proof*:**
Let $h^i : P^i \to D^{i-1}$ define the homotopy between $f^\bullet \circ \alpha^\bullet$ and 0, namely:

$$-f^i \circ \alpha^i = d^{i-1}_{C^\bullet} \circ h^i + h^{i+1} \circ d^i_{P^\bullet}$$

Take the mapping cone $MC(f)^\bullet$ of $f^\bullet$ and define the morphisms:

$$\beta^i = (\alpha' h^i) : P^i \to MC(f)^{i-1}$$

which may be visualized in the following diagram:

$$\begin{tikzcd}
\cdots \arrow[r] & P^{i-1} \arrow[r, "d^{i-1}_{P^\bullet}"] \arrow[d, dotted, "\beta^{i-1}"] & P^i \arrow[r, "d^i_{P^\bullet}"] \arrow[d, "\beta^i"'] & P^{i+1} \arrow[r] \arrow[d, "\beta^{i+1}"] & {} \\
\cdots \arrow[r] & C^{i-1} \oplus D^{i-2} \arrow[r, "d^{i-2}_{MC(f)^\bullet}"'] & C^i \oplus D^{i-1} \arrow[r, "d^{i-1}_{MC(f)^\bullet}"'] & C^{i+1} \oplus D^i \arrow[r] & {}
\end{tikzcd}$$

These give a morphism of cochain complexes $P^\bullet \to MC(f)[-1]^\bullet$. Indeed:

$$\beta^{i+1} \circ d^i_{P^\bullet} = (\alpha^{i+1} \circ d^i_{P^\bullet}, h^{i+1} \circ d^i_{P^\bullet})$$
$$-d^{i-1}_{MC(f)} \circ \beta^i = (d^i_{C^\bullet} \circ \alpha^i, -f^i \circ \alpha^i - d^{i-1}_{M^\bullet} \circ h^i)$$

Then by definition of h^i and by the morphism property of $\alpha^\bullet$, these are equal. Now, by hypothesis, $f^\bullet$ is a quasi-isomorphism. Then by proposition 56.1.21 $MC(f)^\bullet$ is an exact complex. Then by corollary 56.2.6 $\beta^\bullet$ is homotopic to 0. Since $\alpha^\bullet$ is the composition:

$$P^\bullet \xrightarrow{\beta^\bullet} MC(f)[-1]^\bullet = L^\bullet \oplus M[-1]^\bullet \to L^\bullet$$

it follows that $\beta^\bullet \sim 0$ implies $\alpha^\bullet \sim 0$, as we sought to show.

Though $\alpha^\bullet$ must be homotopic to 0, it need not be the case that $\alpha^\bullet = 0$. Take for example:

$$\begin{tikzcd}
\cdots \arrow[r] & 0 \arrow[r] \arrow[d] & \mathbb{Z} \arrow[r] \arrow[d, "\mathrm{id}"] & \mathbb{Z} \arrow[r] \arrow[d, "\cdot 2"] & 0 \arrow[r] \arrow[d] & \cdots \\
\cdots \arrow[r] & 0 \arrow[r] \arrow[d] & \mathbb{Z} \arrow[r, "\cdot 2"] \arrow[d] & \mathbb{Z} \arrow[r] \arrow[d, "\pi"] & 0 \arrow[r] \arrow[d] & \cdots \\
\cdots \arrow[r] & 0 \arrow[r] & 0 \arrow[r] & \mathbb{Z}/2\mathbb{Z} \arrow[r] & 0 \arrow[r] & \cdots
\end{tikzcd}$$

In this case, $f^\bullet$ is a quasi-isomorphism and $f^\bullet \circ \alpha^\bullet = 0$, and so $\alpha^\bullet \sim 0$, even though it is not equal to 0. With this, one more result is required before the desired goal:

Proposition 56.2.8: projectives, and one-sided inverses

Let $\mathbf{A}$ be an abelian category, $L^\bullet \in \mathrm{Ch}(\mathbf{A})$, $P^\bullet \in \mathrm{Ch}^-(\mathbf{P})$, and $f^\bullet : C^\bullet \to P^\bullet$ a quasi-isomorphism. Then there exists a morphism of complexes $g^\bullet : P^\bullet \to C^\bullet$ such that:

$$f^\bullet \circ g^\bullet \sim \mathrm{id}_{P^\bullet}$$

For injective, $Q^\bullet \in \mathrm{Ch}^+(\mathbf{I})$, $f^\bullet : Q^\bullet \to C^\bullet$ is a quasi-isomorphism, for which then there exists a morphism $g^\bullet : C^\bullet \to Q^\bullet$ such that $g^\bullet \circ f^\bullet \sim \mathrm{id}_{Q^\bullet}$. Notice that this is a one-sided inverse, hence is a slightly more general result than what is being built up to.

***Proof*:**
Since $f^\bullet : C^\bullet \to P^\bullet$ is a quasi-isomorphism, $MC(f)^\bullet = L[1]^\bullet \oplus P^\bullet$ is an exact complex. Define $\rho^\bullet = (0, \mathrm{id}_{P^\bullet})$ to be the morphism:

$$\rho^\bullet : P^\bullet \to MC(f)^\bullet$$

By corollary 56.2.5, since $MC(f)^\bullet$ is exact, $\rho^\bullet$ is homotopic to the zero morphism. Thus, there exists morphisms:

$$\widehat{h}^i : P^i \to MC(f)^{i-1} = L^i \oplus P^{i-1}$$

such that

$$\rho^i = d^{i-1}_{MC(f)^\bullet} \circ \widehat{h}^i + \widehat{h}^{i+1} \circ d^i_{P^\bullet}$$

Visually, we may see this as:

$$\begin{tikzcd}
\cdots \arrow[r] & P^{i-1} \arrow[r, "d^{i-1}_{P^\bullet}"] \arrow[dl, "\widehat{h}^{i-1}"'] \arrow[d, "\rho^{i-1}"] & P^i \arrow[r, "d^i_{P^\bullet}"] \arrow[dl, "\widehat{h}^i"'] \arrow[d, "\rho^i"] & P^{i+1} \arrow[r] \arrow[dl, "\widehat{h}^{i+1}"'] \arrow[d, "\rho^{i+1}"] & \cdots \\
\cdots \arrow[r] & C^i \oplus P^{i-1} \arrow[r, "d^{i-2}_{MC(f)^\bullet}"'] & C^{i+1} \oplus P^i \arrow[r, "d^{i-1}_{MC(f)^\bullet}"'] & C^{i+2} \oplus P^{i+1} \arrow[r] & \cdots
\end{tikzcd}$$

Now, write out $\widehat{h}^i$ out in components so that $\widehat{h}^i = (g^i, h^i)$ where $g^i : P^i \to C^i$ and $h^i : P^i \to P^{i-1}$. We'll show that the collection $\{g^i\}_{i\in\mathbb{Z}}$ is a morphism of cochain complexes (i.e. they form commutative squares $g^{i+1} \circ d^i_{P^\bullet} = d^i_{C^\bullet} \circ g^i$) such that $f^\bullet \circ g^\bullet \sim \mathrm{id}_{P^\bullet}$. Indeed, since

$$\begin{aligned}(0, \mathrm{id}_{P^i}) &= \rho^i \\ &= d^{i-1}_{MC(f)^\bullet} \circ \widehat{h}^i + \widehat{h}^{i+1} \circ d^i_{P^\bullet} \\ &= (d_{MC}(f)^\bullet)^{i-1} \circ (g^i, h^i) + (g^{i+1}, h^{i+1}) \circ d^i_{P^\bullet} \\ &= (-d^i_{C^\bullet} \circ g^i, f^i \circ g^i + d^{i-1}_{P^\bullet} \circ h^i) + (\beta^{i+1} \circ d^i_{P^\bullet}, h^{i+1} \circ d^i_{P^\bullet})\end{aligned}$$

It follows that:

$$\begin{aligned}g^{i+1} \circ d^i_{P^\bullet} - d^i_{C^\bullet} \circ g^i = 0 \\ \mathrm{id}_{P^i} - f^i \circ g^i = d^{i-1}_{P^\bullet} \circ h^i + h^{i+1} \circ d^i_{P^\bullet}\end{aligned}$$

which is exactly what we need, hence the morphism of complexes $g^\bullet : P^\bullet - > C^\bullet$ is the needed homotopy right-inverse of $f^\bullet$, as we sought to show.

With this, we can finally prove the desired result:

Theorem 56.2.9: Projective, Quasi-isomorphism Becomes Homotopy

Let $\mathbf{A}$ be an abelian category, and let $\alpha^\bullet : P_0^\bullet - > P_1^\bullet$ be a quasi-isomorphism between bounded-above complexes of projectives (resp. quasi-isomorphism between two bounded-bellow complexes of injectives $f^\bullet : Q_0^\bullet \to Q_1^\bullet$). Then $\alpha^b b$ is a homotopy-equivalence.

***Proof*:**
The argument for projectives shall be given, the injective argument following by appropriate dualizing. By proposition 56.2.8, since $P_1^\bullet$ is in $C^-(\mathbf{P})$ and $\alpha^\bullet : P_0^\bullet \to P_1^\bullet$ is a quasi-isomorphism, there exists a morphism of complexes $\beta^\bullet : P_1^\bullet \to p_0^\bullet$ such that

$$\alpha^\bullet \circ \beta^\bullet \sim \mathrm{id}^\bullet_{P_1}$$

Then by proposition 56.1.26, $H^\bullet(\alpha^\bullet) \circ H^\bullet(\beta^\bullet) = \mathrm{id}$. Since $H^\bullet(\alpha^\bullet)$ is invertible, so is $H^\bullet(\beta^\bullet)$, and so $\beta^\bullet$ must be a quasi-isomorphism. Since $P_0^\bullet$ is also in $\mathrm{Ch}^-(\mathbf{P})$, we have that there exists a morphism $(\alpha^\bullet)' : P_0^\bullet \to P_1^\bullet$ such that $\beta^\bullet \circ (\alpha^\bullet)' \sim \mathrm{id}_{P_0^\bullet}$. Since:

$$(\alpha^\bullet)' \sim (\alpha^\bullet \circ \beta^\bullet) \circ (\alpha^\bullet)' = \alpha^\bullet(\beta^\bullet \circ (\alpha^\bullet)') \sim \alpha^\bullet$$

it follows that $\beta^\bullet \sim \alpha^\bullet \sim_{P_0^\bullet}$ (exercise 4.6 in Aluffi). Thus, $\alpha^\bullet$ is indeed a homotopy equivalence, as we sought to show.

Corollary 56.2.10: Projective In Homotopy Category

In $K^-(\mathbf{P})$ and $K^+(\mathbf{I})$, (homotopy classes of) quasi-isomorphissm are isomorphisms.

***Proof*:**
follows immediately form theorem 56.2.9.

56.2.1 Resolutions

With the important equivalence between quasi-isomorphisms and homotopy-equivalence established in the case of complexes of projective (resp. projective) objects, we turn back our attention to the question of finding (co)homological invariants, focusing on categories with enough projective s(resp. Injectives). In particular, given an object A in an abelian category $\mathbf{A}$, it wouldb e desirable to associate to it a (co)chain of projective (resp. injective) objects in some universal way (up to homotopy) from which information can be extracted.

Let us first refine the definition of projective/injective resolutions of an object. Recall that there i a copy of $\mathbf{A}$ in $\mathrm{Ch}(\mathbf{A})$ given by mapping $A \mapsto \iota(A)$.

Definition 56.2.11: Projective And Injective Resolution

Let $A \in \mathrm{Obj}(\mathbf{A})$ be an object in an abelian category A. Then:

1. A *resolution* of A is a quasi-isomorphism $C^\bullet \to \iota(A)$ for some complex $C^\bullet \in \mathrm{Obj}(\mathrm{Ch}(\mathbf{A}))$.
2. A *projective resolution* of A is a quasi-isomorphism $P^\bullet \to \iota(A)$ where $P^\bullet$ is a complex in $\mathrm{Ch}^{\leq 0}(\mathbf{P})$
3. An *injective resolution* of A is a quasi-isomorphism $\iota(A) \to Q^\bullet$ where $Q^\bullet$ is a complex in $\mathrm{Ch}^{\geq 0}(\mathbf{I})$

Often, $P^\bullet$ and $Q^\bullet$ shall be referred to as the projective resolution and injective resolution respectively, omitting the quasi-isomorphism.

56.2.12 Remark There is a point of confusion that may arise. It may seem that the resolutions *themselves* are projective/injective objects in the abelian category $\mathrm{Ch}(\mathbf{A})$. However, this is not the case: a projective object of $\mathrm{Ch}(\mathbf{A})$ must be a complex of projectives that is moreover split exact in the relevant range, which a resolution is not.

Recall further that any resolution $C^\bullet$ will be exact everywhere except possibly at C^0 (exercise, or double-check example 56.1.13). Hence, the non-trivial cohomological information lies in $H^0(C^\bullet)$. Let us now focus on resolutions for a particular object A: consider the category of homotopy classes of quasi-isomorphisms with *target* $\iota(A)$. Then we shall show that, if they exist, projective resolution are initial in this category.

$$\begin{tikzcd}
P^\bullet \arrow[d, "\exists!"] \arrow[dr] & \\
C^\bullet \arrow[r, "\text{q-iso}"'] & \iota(A)
\end{tikzcd}$$

Analogously, injective resolutions are final in homotopic category of quasi-isomorphisms with *source* $\iota(A)$. Given this universal property, this means that each projective resolution maps uniquely (in $K(\mathbf{A})$) to *every* resolution of A. The following lemma proves this:

Lemma 56.2.13: Projective Resolution As Initial Objects

Let $A \in \mathbf{A}$, let $C^\bullet$ be a resolution of A (so that $C^i = 0$ for $i > 0$, $H^i(C^\bullet) = 0$ for $i < 0$, and $H^0(C^\bullet) = A$), and let $P^\bullet \in \mathrm{Ch}^{\leq 0}(\mathbf{P})$ be a bounded-above complex of projectives. Then any morphism

$$\varphi : H^0(P^\bullet) \to H^0(C^\bullet) \; [= A]$$

is induced by a morphism of complexes $\varphi^\bullet : P^\bullet \to C^\bullet$, and any two such $\varphi^\bullet$ are homotopic.

***Proof*:**
Since both complexes vanish in positive degrees, $H^0(P^\bullet) = \mathrm{coker} d_P^{-1}$ and $H^0(C^\bullet) = \mathrm{coker} d_C^{-1}$; write $\pi_P \colon P^0 \twoheadrightarrow H^0(P^\bullet)$ and $\pi_C \colon C^0 \twoheadrightarrow A$ for the two quotient maps.

Step 1: Degree 0. The composite $\varphi \circ \pi_P \colon P^0 \to A$ is a morphism out of a projective object, and π_C is an epimorphism, so by definition 56.2.1 it lifts: there is $\varphi^0 \colon P^0 \to C^0$ with $\pi_C \circ \varphi^0 = \varphi \circ \pi_P$.

Step 2: Degree -1. Compute

$$\pi_C \circ \varphi^0 \circ d_P^{-1} = \varphi \circ \pi_P \circ d_P^{-1} = 0$$

since $\pi_P \circ d_P^{-1} = 0$. Hence $\varphi^0 \circ d_P^{-1}$ factors through $\ker \pi_C = \mathrm{im} d_C^{-1}$. As P^{-1} is projective and d_C^{-1} is an epimorphism onto its image, the map $P^{-1} \to \mathrm{im} d_C^{-1}$ lifts to $\varphi^{-1} \colon P^{-1} \to C^{-1}$ with $d_C^{-1} \circ \varphi^{-1} = \varphi^0 \circ d_P^{-1}$.

Step 3: The induction. Suppose φ^{-i} has been constructed for $i < n$ with every square commuting. Then

$$d_C^{-n+1} \circ \left(\varphi^{-n+1} \circ d_P^{-n}\right) = \varphi^{-n+2} \circ d_P^{-n+1} \circ d_P^{-n} = 0$$

so $\varphi^{-n+1} \circ d_P^{-n}$ lands in $\ker d_C^{-n+1}$, which for $n \geq 2$ equals $\mathrm{im} d_C^{-n}$ because $C^\bullet$ is exact in degree $-n+1 < 0$; the case $n = 1$ is Step 2, where $\ker \pi_C$ plays that role. Projectivity of P^{-n} again lifts it along the epimorphism $C^{-n} \twoheadrightarrow \mathrm{im} d_C^{-n}$, producing φ^{-n} with $d_C^{-n} \circ \varphi^{-n} = \varphi^{-n+1} \circ d_P^{-n}$. This constructs $\varphi^\bullet$ in every degree, and Step 1 arranges that it induces φ on H^0.

Step 4: Uniqueness up to homotopy. Let $\varphi^\bullet$ and $\psi^\bullet$ both induce φ, and set $f^\bullet = \varphi^\bullet - \psi^\bullet$. On H^0 it induces $\varphi - \varphi = 0$, and in every negative degree $H^i(C^\bullet) = 0$, so $H^i(f^\bullet) = 0$ there as well; in positive degrees both complexes vanish. Hence $H^\bullet(f^\bullet) = 0$. The hypotheses of lemma 56.2.4 are met, $P^\bullet$ being a bounded-above complex of projectives and $C^\bullet$ having vanishing cohomology in negative degrees, so $f^\bullet \sim 0$, that is $\varphi^\bullet \sim \psi^\bullet$, completing the proof.

Since these are initial (resp. final) objects in the category, we have a unique isomorphism beetween any two solutions to the universal problem. In this case, this may be stated as followed:

Proposition 56.2.14: Unique Projection Resolution

Let $\mathbf{A}$ be an abelian category. Then any two projective (resp. injective) resolutions of an object A in $\mathbf{A}$ are homotopy equivalent

***Proof*:**
Let $P^\bullet$ and $Q^\bullet$ be two projective resolutions of A, so that each is a bounded-above complex of projectives with $H^0 = A$. Apply lemma 56.2.13 twice to the morphism id_A: once with source $P^\bullet$ and target $Q^\bullet$, giving $f^\bullet\colon P^\bullet \to Q^\bullet$, and once the other way, giving $g^\bullet\colon Q^\bullet \to P^\bullet$. Each is unique up to homotopy.

The composite $g^\bullet \circ f^\bullet\colon P^\bullet \to P^\bullet$ induces id_A on H^0, and so does $\mathrm{id}_{P^\bullet}$. Both are therefore lifts of the same morphism id_A, and the uniqueness clause of lemma 56.2.13 gives $g^\bullet \circ f^\bullet \sim \mathrm{id}_{P^\bullet}$. Symmetrically $f^\bullet \circ g^\bullet \sim \mathrm{id}_{Q^\bullet}$. Hence $f^\bullet$ is a homotopy equivalence with homotopy inverse $g^\bullet$, as we sought to show. The injective case is dual.

A final important observation is that this lift is functorial!

Proposition 56.2.15: Projective Resolution Functoriality

Let $\mathbf{A}$ be an abelian category, and let A_0, A_1 be objects of $\mathbf{A}$. Let $P_0^\bullet$ and $P_1^\bullet$ be projective resolutions for A_0 and A_1 respectively. Then every morphism $\varphi : A_0 \to A_1$ in $\mathbf{A}$ induces a morphism $\alpha^\bullet : P_0^\bullet \to P_1^\bullet$ that is unique up to homotopy

***Proof*:**
Notice that φ can be interpreted as a morphism $\varphi : H^0(P_0^\bullet) \to H^0(P_1^\bullet)$. The complex $P_0^\bullet$ consists of projectives, and $P1^\bullet$ is a resolution of A_1. Thus, by lemma 56.2.13, there exists a list α^b that is unique up to homotopy, as we sought to show.

Corollary 56.2.16: Full Subcategory Of Homotopic Category

Let $\mathbf{A}$ be an abelian category with enough projectives (resp. enough injectives). Then $\mathbf{A}$ is equivalent to a full subcategory of $K^-(\mathbf{P})$ (resp. $K^+(\mathbf{I})$).

***Proof*:**
Treat the projective case. Choose for each object A of $\mathbf{A}$ a projective resolution $P_A^\bullet$, possible because $\mathbf{A}$ has enough projectives, and send $A \mapsto P_A^\bullet$. By proposition 56.2.15 every $\varphi\colon A_0 \to A_1$ lifts to a morphism of resolutions, well defined up to homotopy, so this is a functor $R\colon \mathbf{A} \to K^-(\mathbf{P})$: composites and identities are preserved because a lift of a composite and the composite of lifts both lift the same morphism and so agree up to homotopy, which is equality in $K^-(\mathbf{P})$.

R is faithful. If $R(\varphi) = 0$ then a lift $\varphi^\bullet$ of φ is null-homotopic, so it induces the zero morphism on cohomology by proposition 56.1.24. Since $H^0(\varphi^\bullet) = \varphi$ under the identifications $H^0(P_{A_i}^\bullet) = A_i$, this gives $\varphi = 0$.

R is full. Let $f^\bullet\colon P_{A_0}^\bullet \to P_{A_1}^\bullet$ be any morphism in $K^-(\mathbf{P})$, represented by a chain map. It induces $\varphi := H^0(f^\bullet)\colon A_0 \to A_1$, and $f^\bullet$ is then a lift of φ. By the uniqueness clause of lemma 56.2.13 any two lifts of φ are homotopic, so $R(\varphi) = [f^\bullet] = f^\bullet$ in $K^-(\mathbf{P})$.

A fully faithful functor is an equivalence onto its essential image, which is a full subcategory of $K^-(\mathbf{P})$. Hence $\mathbf{A}$ is equivalent to a full subcategory of $K^-(\mathbf{P})$, and dually to one of $K^+(\mathbf{I})$ when $\mathbf{A}$ has enough injectives, completing the proof.

56.2.2 Derived Category with Enough Projectives

Let us finally confirm that the solution to the universal problem in equation 56.2. Recall that the derived category is a category $D(\mathbf{A})$ where for any category D for which there exists an additive functor $\mathscr{F} : \mathrm{Ch}(\mathbf{A}) \to D$ where every quasi-isomorphism in $\mathrm{Ch}(\mathbf{A})$ becomes an isomorphism in D, there exists a unique functor such that the following diagram commutes:

$$\begin{tikzcd} \mathrm{Ch}(\mathbf{A}) \arrow[r, "\mathscr{F}"] \arrow[d] & D \\ D(\mathbf{A}) \arrow[ur, "\exists!"'] & \end{tikzcd}$$

In the case where $\mathbf{A}$ has enough projectives (resp. enough injectives) and is bounded from above (resp. from bellow), then $K^-(\mathbf{P})$ (resp. $K^+(\mathbf{I})$) is the solution to the above universal problem. In particular, there is an equivalence of categories between $K^-(\mathbf{P})$ and $D^-(\mathbf{A})$. The formal definition of the derived category shall be given in section 56.5.1, hence this equivalence shall not be proven, however the universal property is readily provable, and once the derived category is formally defined it will be the equivalence essentially come to proper manipulation of universal properties.

To start, a general procedure for finding projective/injective resolutions for any (co)chain must be explored. The discussion so far has been for projective/injective resolution, that is quasi-isomorphisms $P^\bullet \to \iota(A)$ or $\iota(A) \to Q^\bullet$. Is it possible to replace $\iota(A)$ with any bounded complex? The answer is yes, though rather technical. The importance of this will be the existence of a functor p that maps $\mathrm{Ch}^-(\mathbf{P})$ into $K^-(\mathbf{P})$, which shall be important for solving the universal problem above, as well as finding functors between derived categories. Such functors shall also be key in solving the "extension problem" that was mentioned all the way back in chapter 17$\star$.

Theorem 56.2.17: Resolution And Complexes

Let $\mathbf{A}$ be an abelian category with enough projectives and $C^\bullet$ a complex in $C^-(\mathbf{A})$. Then there exists a founded-above complex of projective $P^\bullet$ and a quasi-isomorphism $P^\bullet \to C^\bullet$, where $P^\bullet$ is uniquely defined up to homotopy equivalence. Furthermore, every morphism $\alpha^\bullet$ of complexes in $C^-(\mathbf{A})$ lifts to a morphism of the corresponding projective resolutions in $K^-(\mathbf{P})$, which is uniquely defined up to homotopy.

***Proof Key ideas*:**
The uniqueness and lifting halves are already available; only the existence half needs a construction, and that construction is a descending induction whose bookkeeping is long without being conceptually informative, so it is given in outline here and in full in *An Introduction to Homological Algebra* [53].

Existence. Let $C^\bullet$ be bounded above, say $C^n = 0$ for $n > N$. Build P^n by descending induction on n, starting with $P^n = 0$ for $n > N$. At each stage one has already constructed the terms above degree n together with the maps to $C^\bullet$, and one chooses a projective object surjecting onto the pullback of C^n against the cycles constructed one degree up. That single choice simultaneously arranges surjectivity on cohomology in degree n and injectivity in degree $n+1$, which is what makes the resulting map a quasi-isomorphism; enough projectives is exactly what supplies the surjection at each stage.

Uniqueness up to homotopy equivalence. Two such projective complexes over $C^\bullet$ are homotopy equivalent by the argument of proposition 56.2.14, with its object-level input lemma 56.2.13 replaced by its complex-level counterpart proposition 56.2.8: each maps to the other over $C^\bullet$, and both composites lift the identity, hence are homotopic to it.

Lifting of morphisms. Given $\alpha^\bullet\colon C^\bullet \to D^\bullet$ and projective complexes $P^\bullet \to C^\bullet$, $Q^\bullet \to D^\bullet$, the composite $P^\bullet \to C^\bullet \to D^\bullet$ is a morphism from a bounded-above complex of projectives into a complex quasi-isomorphic to $Q^\bullet$. Existence of a factorization through $Q^\bullet$ comes from corollary 56.2.5, applied to the cone of $Q^\bullet \to D^\bullet$, and its uniqueness up to homotopy from the cancellation property lemma 56.2.7. Together they give a well-defined morphism in $K^-(\mathbf{P})$.

Thus, given an abelian category $\mathbf{A}$ with enough projective, there is a functor

$$\mathscr{P} : C^-(\mathbf{A}) \to K^-(\mathbf{P})$$

associating each (bounded-above) complex $C^\bullet$ to a projective resolution $\mathscr{P}(C^\bullet)$ in a functorial manner (so that morphisms $f^\bullet$, $g^\bullet$ are lifted and $\mathscr{P}(g^\bullet \circ f^\bullet) = \mathscr{P}(g^\bullet) \circ \mathscr{P}(f^\bullet)$ since both sides are classes of homotopy lifts of $g^\bullet \circ f^\bullet$). The uniqueness of $\mathscr{P}$ is up to natural transformation, that is if another functor $\mathscr{P}'$ is chosen that assigns a different choice of projective resolution, then by theorem 56.2.17 there is a homotopy equivalence between $\mathscr{P}(C^\bullet)$ and $\mathscr{P}'(C^\bullet)$, hence an isomorphism $\varphi_{C^\bullet} : \mathscr{P}(C^\bullet) \to \mathscr{P}'(C^\bullet)$ in $K^-(\mathbf{P})$. Since the identity on $C^\bullet$ lifts to a unique map up to homotopy, the isomorphism $\varphi_{C^\bullet}$ is unique. Furthermore, given a morphism $f^\bullet : C^\bullet \to D^\bullet$, the diagram:

$$\begin{tikzcd}
\mathscr{P}(C^\bullet) \arrow[r, "\mathscr{P}(f^\bullet)"] \arrow[d, "\varphi_{C^\bullet}"', "\sim"] & \mathscr{P}(D^\bullet) \arrow[d, "\varphi_{D^\bullet}", "\sim"'] \\
\mathscr{P}'(C^\bullet) \arrow[r, "\mathscr{P}'(f^\bullet)"'] & \mathscr{P}'(D^\bullet)
\end{tikzcd}$$

commutes (again by the uniqueness of lifts up to homotopy, can you see why)? Thus, we have the unique natural isomorphism for any two choices of functors giving a projective resolution.

Theorem 56.2.18: Universal Property Of Derived Category With Enough Projectives

Let $\mathbf{A}$ be an abelian category with enough projectives, and $\mathscr{P}$ a functor sending bounded-above cochains to projective resolutions. Then:

1. If $f^\bullet$ is a quasi-isomorphism in $C^-(\mathbf{A})$, then $\mathscr{P}(f^\bullet)$ is an isomorphism in $K^-(\mathbf{P})$
2. If $\mathscr{F} : C^-(\mathbf{A}) \to D$ is an additive functor that takes quasi-isomorphisms to isomorphisms, then there exists a unique functor $\widetilde{\mathscr{F}} : K^-(\mathbf{P}) \to D$ unique up to natural isomorphism such that the following diagram commutes up to natural isomorphism:

$$\begin{tikzcd} C^-(\mathbf{A}) \arrow[r, "\mathscr{F}"] \arrow[d, "\mathscr{P}"'] & D \\ K^-(\mathbf{P}) \arrow[ur, "\widetilde{\mathscr{F}}"'] & \end{tikzcd}$$

where commuting up to natural transformation means there exists a natural transformation $\nu : \widetilde{\mathscr{F}} \circ \mathscr{P} \rightsquigarrow \mathscr{F}$ such that:

$$\nu_{C^\bullet} : \widetilde{\mathscr{F}} \circ \mathscr{P}(C^\bullet) \xrightarrow{\sim} \mathscr{F}(C^\bullet)$$

is an isomorphism for every complex $C^\bullet$ in $C^-(\mathbf{A})$.

The fact that the commutative diagram commutes up to natural isomorphism reflects that the homotopic category is categorically equivalent to the derived category. If $\mathbf{A}$ has enough injective, then we may instead define the functor $\mathscr{I} : C^+(\mathbf{A}) \to K^+(\mathbf{I})$ that shall satisfy the dual universal property.

***Proof*:**

1. Let $f^\bullet : C^\bullet \to D^\bullet$ be any morphism in $C^-(\mathbf{A})$. By theorem 56.2.17, $f^\bullet$ lifts to a morphism of resolutions which commutes up to homotopy in the following diagram:

$$\begin{tikzcd} \mathscr{P}(C^\bullet) \arrow[r, "\mathscr{P}(f^\bullet)"] \arrow[d, "\mu_{C^\bullet}"'] & \mathscr{P}(C^\bullet) \arrow[d, "\mu_{D^\bullet}"] \\ C^\bullet \arrow[r, "f^\bullet"] & D^\bullet \end{tikzcd}$$

where $\mu_{C^\bullet}, \mu_{D^\bullet}$ are the quasi-isomorphisms of the projective resolutions. Since the diagram commute up to homootpy, it commutes after taking cohomology, thus since $f^\bullet$ is a quasi-isomorphism, so is $\mathscr{P}(f^\bullet)$. Then by corollary 56.2.10, $\mathscr{P}(f^\bullet)$ is an isomorphism.

2. Define $\widetilde{\mathscr{F}} : K^-(\mathbf{P}) \to D$ via:

$$\widetilde{\mathscr{F}}(P^\bullet) := \mathscr{F}(P^\bullet) \qquad \widetilde{\mathscr{F}}(f^\bullet) := \mathscr{F}(f^\bullet)$$

where $\widetilde{\mathscr{F}}(f^\bullet)$ can be any representative of $f^\bullet$. This map is well-defined, since homotopic morphisms have the same image in D (lemma 56.1.32). Naturally, $\widetilde{\mathscr{F}}$ is a functor. We need to check the commutativity up to morphism of the diagram and the uniqueness of $\widetilde{\mathscr{F}}$ up to natural isomorphism.

First, let $C^\bullet \in C^-(\mathbf{A})$. Then (by construction of $\mathscr{P}$, there exists a quasi-isomorphism

$$\mu_{C^\bullet} : \mathscr{P}(C^\bullet) \to C^\bullet$$

Since $\mathscr{F}$ maps quasi-isomorphisms to isomorphism, the induced morphism $\nu_{C^\bullet} := \mathscr{F}(\mu_{C^\bullet})$:

$$\nu_{C^\bullet} : \widetilde{\mathscr{F}}(\mathscr{P}(C^\bullet)) = \mathscr{F}(\mathscr{P}(C^\bullet)) \to \mathscr{F}(C^\bullet)$$

is an isomorphism. What's left to check is that $\nu_{C^\bullet}$ defines a natural transformation. Let $f^\bullet : C^\bullet \to D^\bullet$ in $C^-(\mathbf{A})$. We want to show the following diagram commutes:

$$\begin{tikzcd}
\widetilde{\mathscr{F}}(\mathscr{P}(C^\bullet)) \arrow[r, "\widetilde{\mathscr{F}}(\mathscr{P}(f^\bullet))"] \arrow[d, "\nu_{C^\bullet}"'] & \widetilde{\mathscr{F}}(\mathscr{P}(D^\bullet)) \arrow[d, "\nu_{D^\bullet}"] \\
\mathscr{F}(C^\bullet) \arrow[r, "\mathscr{F}(f^\bullet)"] & \mathscr{F}(D^\bullet)
\end{tikzcd}$$

This diagram is obtained by applying $\mathscr{F}$ to the diagram discussed above the theorem:

$$\begin{tikzcd}
\mathscr{P}(C^\bullet) \arrow[r, "\mathscr{P}(f^\bullet)"] \arrow[d, "\varphi_{C^\bullet}"', "\sim"] & \mathscr{P}(D^\bullet) \arrow[d, "\varphi_{D^\bullet}", "\sim"'] \\
\mathscr{P}'(C^\bullet) \arrow[r, "\mathscr{P}'(f^\bullet)"'] & \mathscr{P}'(D^\bullet)
\end{tikzcd}$$

This diagram commutes up to homotopy, hence the other diagram commutes directly (lemma 56.1.32).

For uniqueness up to isomorphism, say $\overline{\mathscr{F}} : K^-(\mathbf{P}) \to D$ is another functor satisfying the properties. Then there exists a natural isomorphism:

$$\overline{\nu} : \overline{\mathscr{F}} \circ \mathscr{P} \rightsquigarrow \mathscr{F}$$

Now, for any $P^\bullet \in K^-(\mathbf{P})$, take the composition:

$$\overline{ff}(P^\bullet) \overset{\overline{ff}(\mu_{P^\bullet})^{-1}}{\underset{\sim}{\to}} \overline{ff}(\mathscr{P}((P^\bullet)) \overset{\overline{\nu}_{P^\bullet}}{\underset{\sim}{\to}} = \widetilde{ff}(P^\bullet)$$

If φ is this composition, then it is easy to verify that $\varphi : \overline{\mathscr{F}} \to \widetilde{\mathscr{F}}$ is a natural isomorphism, completing the proof.

56.3 Derived Functors: Ext and Tor

Let $\mathscr{F} : \mathbf{A} \to \mathbf{B}$ be an additive functor. Often, we study information about $\mathbf{A}$ by passing it through $\mathbf{B}$ (think of $- \otimes N$ or $\mathrm{Hom}_R(N, -)$). As alluded to in section 17.6, such information may be the number of extensions two objects may have, the amount of torsion an object may have. In section 17.6, it was claimed that this comes from "fixing" the lack of exactness of a left- or right-exact functor using homological algebra. We are finally in a position to make this precise. In this section, we shall show how to take a (not-necessarily left/right exact) additive functor between two abelian categories with $\mathbf{A}$ having enough projectives (resp. injectives) $\mathscr{F} : \mathbf{A} \to \mathbf{B}$ and get a *derived functor*:

$$D^-(\mathscr{F}) : D^-(\mathbf{A}) \to D^-(\mathbf{B}) \qquad (\text{resp. } D^-(\mathscr{F}) : D^-(\mathbf{A}) \to D^-(\mathbf{B}))$$

This functor shall be used to develop a long-exact sequence in $\mathbf{B}$, given a short exact sequence of objects in $\mathbf{A}$.

56.3.1 Derived Functor

Let $\mathscr{F} : \mathbf{A} \to \mathbf{B}$ be an additive functor between two abelian categories. Then $\mathscr{F}$ extends to a functor:

$$\mathrm{Ch}(\mathscr{F}) : \mathrm{Ch}(\mathbf{A}) \to \mathrm{Ch}(\mathbf{B})$$

namely, if $C^\bullet$ is a complex:

$$\mathscr{F}(d^{i+1}_{C^\bullet}) \circ \mathscr{F}(d^i_{C^\bullet}) = \mathscr{F}(d^{i+1}_{C^\bullet} \circ d^i_{C^\bullet}) = \mathscr{F}(0) = 0$$

hence, $\mathrm{Ch}(\mathscr{F})(C^\bullet)$ is indeed a complex. Then since homotopic morphisms are sent to homotopic morphisms (lemma 56.1.29), $\mathrm{Ch}(\mathscr{F})$ induces a morphism $K(\mathscr{F})$:

$$K(\mathscr{F}) : K(\mathbf{A}) \to K(\mathbf{B})$$

From the results we have proved, we have the following commutative diagram:

$$\begin{tikzcd} \mathbf{A} \arrow[r, "\mathscr{F}"] \arrow[d, hook, "\iota"'] & \mathbf{B} \arrow[d, hook, "\iota"] \\ K(\mathbf{A}) \arrow[r, "K(\mathscr{F})"] & K(\mathbf{B}) \end{tikzcd}$$

Furthermore, if the complexes are bounded, then certainly $\mathrm{Ch}(\mathscr{F})$ sends bounded complexes to bounded complexes, and hence the same holds for $K(\mathscr{F})$. Now, if $\mathbf{A}$ has enough projectives so that we may consider $K^-(\mathbf{P})$, can we restrict $K(\mathbf{A})$ to $K(P(\mathbf{A}))$, and map to $K(P(\mathbf{B}))$ (where $P(\mathbf{A})$ and $P(\mathbf{B})$ are the projectives of $\mathbf{A}$ and $\mathbf{B}$ respectively)? This is not the case in general; for example we saw that $- \otimes N$ in R-**Mod** does not preserve projectives. If $\mathbf{B}$ has enough projectives, then we may choose a projective resolution functor $\mathscr{P}_B : \mathrm{Ch}^-(B) \to K^-(\mathbf{P}^{-1})$ mapping each complex to a projective resolution in a functorial manner. Since $\mathscr{P}_B$ sends homotopic morphisms to homotopic morphisms (lemma 56.2.7), $\mathscr{P}_B$ descends to a functor $K(\mathscr{P}_B)$. Combining these facts, we may define the following functor:

Definition 56.3.1: Left-Derived (right-Derived) Functor

Let $\mathscr{F} : \mathbf{A} \to \mathbf{B}$ be an additive functor. Then the *left-derived functor* of $\mathscr{F}$ is the composition:

$$\begin{tikzcd} K^-(P(\mathbf{A})) \arrow[r, "\iota_A"'] \arrow[rrr, bend left, "\mathbf{L}\mathscr{F}"] & K^-(\mathbf{A}) \arrow[r, "K(\mathscr{F})"'] & K^-(\mathbf{B}) \arrow[r, "K(\mathscr{P}_B))"'] & K^-(P(\mathbf{B})) \end{tikzcd}$$

where ι_A is the inclusion of $K^-(P(\mathbf{A}))$ as a full subcategory of $K^-(\mathbf{A})$. The resulting functor is denoted $\mathbf{L}\mathscr{F}$.

It may seem that the natural notation would be $D^-(\mathscr{F})$, however the $\mathbf{L}\mathscr{F}$ notation was established before the abstraction given above, and has stuck. Similarly, if $\mathbf{A}$, $\mathbf{B}$ has enough injectives, we may define the right-derived functor $\mathbf{R}\mathscr{F}$.

It is natural to want to characterize $\mathbf{L}\mathscr{F}$ and $\mathbf{R}\mathscr{F}$ in terms of some universal property. Naturally, this universal property would be defined up to natural isomorphism, for $\mathscr{P}_A$ and $\mathscr{P}_B$ are chosen up to natural isomorphism. This turns to be a bit subtle. For example, there is no natural functor which shall always make the following diagram commute:

$$\begin{tikzcd}
\hat{\mathbf{A}} \arrow[r, dashed, "??"] \arrow[d, hook] & \hat{\mathbf{B}} \arrow[d, hook] \\
K^-(P(\mathbf{A})) \arrow[r, "\mathbf{L}\mathscr{F}"] & K^-(P(\mathbf{B}))
\end{tikzcd}$$

The reason is, it is not necessarily the case that a complex whose cohomology is concentrated at 0 would map via $\mathbf{L}\mathscr{F}$ to a complex with the same property. In fact, the reader may expect this, for the image of a functor $\mathscr{F}$ may present some non-trivial cohomological information, (think of $\mathscr{F} := \mathrm{Hom}_R(N, -)$ where N is non-projective). If the functor $\mathscr{F}$ is *exact*, then then there is essentially no "change in cohomology". This is captured in the following proposition:

Proposition 56.3.2: Derived Functors And Exactness

Let $\mathscr{F}$ be an additive *exact* functor. Then $\hat{\mathbf{A}}$ is in fact sent to $\hat{\mathbf{B}}$ via $\mathbf{L}\mathscr{F}$

***Proof*:**
Let $P^\bullet \in \hat{\mathbf{A}}$ so that $H^i(P^\bullet) = 0$ for all $i \neq 0$. Then $\mathbf{L}\mathscr{F}(P^\bullet)$ is given by choosing a projective resolution $P^\bullet_{\mathscr{F}(P^\bullet)}$ of $\mathscr{F}(P^\bullet)$. Since quasi-isomorphissm preserve cohomology and (key!) exact functors *commute* with cohomology, an exact functor carrying kernels to kernels and images to images and hence subquotients to subquotients:

$$H^i(\mathbf{L}\mathscr{F}(P^\bullet)) = H^i(P^\bullet_{\mathscr{F}(P^\bullet)}) \cong H^i(\mathscr{F}(P^\bullet)) \cong \mathscr{F}(H^i(P^\bullet)) = \mathscr{F}(0) \overset{i\neq 0}{=} 0$$

Thus, $\mathbf{L}\mathscr{F}(P^\bullet)$ does indeed map to an object of $\hat{\mathbf{B}}$.

Even more is true if $\mathscr{F}$ is exact: $H^i(\mathbf{L}\mathscr{F}(P^\bullet)) \cong \mathscr{F}(H^0(P^\bullet))$, saying that the restriction of $\mathbf{L}\mathscr{F}$ agrees with $\mathscr{F}$ (modulo equivalence of categories). Hence, in general we shall be interested in non-exact functors, usually functors that are only left or right exact.

Returning to finding an appropriate universal property for derived functors, another natural candidate would be given by taking a "step back" from the above diagram and consider instead if the following diagram commutes (up to natural isomorphism)

$$\begin{tikzcd}
K^-(\mathbf{A}) \arrow[r, "K(\mathscr{F})"] \arrow[d, "\mathscr{P}_A"] & K^-(\mathbf{B}) \arrow[d, "\mathscr{P}_B"] \\
K^-(P(\mathbf{A})) \arrow[r, "\mathbf{L}\mathscr{F}"'] & K^-(P(\mathbf{B}))
\end{tikzcd}$$

in particular, that there may be a natural isomorphism:

$$\mathbf{L}\mathscr{F} \circ \mathscr{P}_A \overset{\sim}{\rightsquigarrow} \mathscr{P}_B \circ K(\mathscr{F})$$

Unfortunately, this isn't in general the case

Example 56.3.3: Lacking Natural Isomorphism
(exercise 7.3 in Aluffi, should I put solution here?)

The following is the best we can hope for:

Proposition 56.3.4: Universal Property of (left-)Derived Functor

Let $\mathscr{F} : \mathbf{A} \to \mathbf{B}$ be an additive functor between abelian categories. Then $\mathbf{L}\mathscr{F}$ satisfies the following universal property:

1. There is a natural transformation:

$$\mathbf{L}\mathscr{F} \circ \mathscr{P}_A \overset{\sim}{\rightsquigarrow} \mathscr{P}_B \circ K(\mathscr{F})$$

2. for every functor $\mathscr{G} : K^-(P(\mathbf{A})) \to K^-(P(\mathbf{B}))$ and every natural tarnsformation $\gamma : \mathscr{G} \circ \mathscr{P}_A \rightsquigarrow \mathscr{P}_B \circ K(\mathscr{F})$, there is a unique (up to natural isomorphism) natural transformation $\mathscr{G} \rightsquigarrow \mathbf{L}\mathscr{F}$ inducign a factorization:

$$\gamma : \mathscr{G} \circ \mathscr{P}_A \rightsquigarrow \mathbf{L}\mathscr{F} \circ \mathscr{P}_A \rightsquigarrow \mathscr{P}_B \circ K(\mathscr{F})$$

A similar result holds for right-derived functors.

***Proof Key ideas*:**
The argument is the usual one for a Kan-extension-style universal property, and it is stated here rather than carried out because the ambient apparatus is not yet in place: the definition of $D(\mathbf{A})$ used below rests on the localization machinery that section 55.2 defers. The full treatment is in *An Introduction to Homological Algebra* [53].

For (1), the natural transformation is the comparison map induced by functoriality of projective resolutions: $\mathscr{P}_A$ picks a resolution, $K(\mathscr{F})$ applies $\mathscr{F}$ termwise, and proposition 56.2.15 makes the resulting square commute up to homotopy, which is commutativity in K^-.

For (2), given $\mathscr{G}$ and γ, the factorization is forced on objects because a projective resolution is initial among complexes of projectives mapping to the object (lemma 56.2.13), so any transformation out of $\mathscr{G} \circ \mathscr{P}_A$ is determined by its value on resolutions; uniqueness up to natural isomorphism is then the uniqueness up to homotopy of those comparison maps (proposition 56.2.14).

(a word on composition)

56.3.2 Long Exact Sequences of Derived Functors

With the derived functor established and being the best candidate for holding (co)homology information, we proceed with extracting this informaiton: we may take $\mathbf{L}_i\mathscr{F} := H^{-i} \circ \mathbf{L}\mathscr{F}$. The reason for taking H^{-i} instead of i is since the indexing works out better. This produced a functor

$\mathbf{L}_i\mathscr{F} : D^-(\mathbf{A}) \to B$. Notice that B doesn't need to have enough projectives for:

$$\mathbf{L}_i\mathscr{F}(P^\bullet) = H^{-i}(\mathscr{P}_B(\mathrm{Ch}(\mathscr{F})(P^\bullet))) \cong H^{-i}(\mathrm{Ch}(\mathscr{F})(P^\bullet))$$

where the last isomorphism comes from $\mathscr{P}_B(\mathrm{Ch}(\mathscr{F})(P^\bullet)))$ being quasi-isomorphic to $\mathrm{Ch}(\mathscr{F})(P^\bullet)$. Though this abstract approach is useful, the following more concrete version of $\mathbf{L}_i\mathscr{F}$ shall prove more useful for computations.

Definition 56.3.5: ith Left-Derived Functor

Let $\mathscr{F} : \mathbf{A} \to \mathbf{B}$ be an additive functor between two abelian categories where $\mathbf{A}$ has enough projectives. Then the *ith left-derived functor* $\mathbf{L}_i\mathscr{F} : A \to B$ is given by:

$$\mathbf{L}_i\mathscr{F} := H^{-1} \circ \mathbf{L}\mathscr{F} \circ \mathscr{P}_A \circ \iota_A$$

The complex in $\mathrm{Ch}(\mathbf{B})$ with $\mathbf{L}_i\mathscr{F}(M)$ in the $-i$th degree with vanishing differentials is denoted $\mathbf{L}_\bullet\mathscr{F}(M)$.

To be precise, if $M \in \mathbf{A}$, then $\mathbf{L}_i(M)$ is found by

1. finding any projective resolution $P_M^\bullet$ of M
2. Applying $\mathrm{Ch}(\mathscr{F})$ to $P_M^\bullet$, $\mathrm{Ch}(\mathscr{F})(P_M^\bullet)$
3. take the $(-i)$th cohomology of this complex, $H^{-i}(\mathrm{Ch}(\mathscr{F})(P_M^\bullet))$

With all the build-up done so far, the resulting cohomology is independent of choice, for:

1. Any two projective resolutions are homotopy-equivalent
2. the image of homotopy-equivalent complexes via additive functors have isomorphic cohomology

Naturally, the *ith right-derived functor* $\mathbf{R}^i\mathscr{F}$ of an additive functor $\mathscr{F}$ and the assocaited complex $\mathbf{R}^\bullet\mathscr{F}$ is defined simplify (given $\mathbf{A}$ has enough injectives), namely

$$\mathbf{R}^i\mathscr{F} := H^i \circ \mathbf{R}\mathscr{F} \circ \mathscr{Q}_A \circ \iota_A$$

A note should be said about *contravriant functors*: since they can always be viewed as *covariant* functor $\mathbf{A}^{\mathrm{op}} \to B$, $\mathbf{A}$ would instead have to have the appropriate injectives or projectives swapped. The right-derived functor of an additive *contravariant functor* would need $\mathbf{A}$ to have enough *projectives*, since $\mathbf{A}^{\mathrm{op}}$ would have to have enough injectives. The $\mathbf{L}_i$ and $\mathbf{R}_i$ notation can be thought of as "looking" at the cohomology provided by $\mathscr{F}$. As mentioned earlier, if $\mathscr{F}$ is exact, then the derived functors are *trivial* in terms of cohomology, hence the derived functors are only interesting if $\mathscr{F}$ is not exact, at which point these functors give you interesting information.

Example 56.3.6: ith Derived Functors

It is finally time to bring back section 17.6

1. Let $\mathbf{A} = R\text{-}\mathbf{Mod}$ for a commutative ring R, pick an R-module N, and define

$$- \otimes_R N$$

The left-derived functor of $-\otimes_R N$ is labeled

$$-\overset{\mathbf{L}}{\otimes}_R N : D^-(R\text{-Mod}) \to D^-(R\text{-Mod})$$

The ith derived left-functor of $-\otimes_R N$ is denoted:

$$\mathrm{Tor}_i^R(-\otimes_R N) = H^{-i}((-)_\bullet \otimes N)$$

where the right hand side of the inequality comes from the result elaborated in section 17.6. It was mentioned that a flat resolution may be used instead of a projective resolution, however this was does not currently fit within the theory. This shall be addressed in the next section.

2. Similarly, $\mathrm{Hom}_R(-, N)$ and $\mathrm{Hom}_R(N, -)$ admits a right derived functors

$$\mathrm{Ext}_R^i(M, N) = H^i(\mathrm{Hom}_R(M, N^\bullet))$$

For $\mathrm{Hom}_R(-, N)$, don't forget that one must take the opposite category and a projective resolution.

The most important part of left-/right-derived functors is their use in fixing exactness after applying a not necessarily exact functor $\mathscr{F}$. On a high level, since the derived category there is not an abelian category (hence not necessarily containing kernels and cokernels, nor is it really feasible to investigate such categories), the notion of *triangulated categories* is useful for categories where given a "short exact sequence", we can create a long exact sequence (that is, a category in which snakes lemma can be applied). We shall not investigate the idea of triangulated categories, but the following is a concrete results needed for any rigorous exploration of triangulated categories.

Assuming the whole that that our category $\mathbf{A}$ has enough projectives, take a short exact sequence:

$$0 \to L \to M \to N \to 0$$

Since there is enough projectives, can the projective resolutions be arranged to create a short exact sequence? The answer is yes, and given the following apt name:

Lemma 56.3.7: Horseshoe Lemma

Let

$$0 \to L \to M \to N \to 0$$

be an exact sequence in an abelian category $\mathbf{A}$ with enough projectives, and let $P_L^\bullet, P_N^\bullet$ be projective resolutions of L, N respectively. Then there exists an exact sequence:

$$0 \to P_L^\bullet \to P_M^\bullet \to P_N^\bullet \to 0$$

where $P_M^\bullet$ is a projective resolution of M, inducing the original sequence in cohomology (i.e. applying H^0). Moreover the sequence is *degreewise split*: the construction produces $P_M^i = P_L^i \oplus P_N^i$ in each degree, with the two displayed maps the inclusion and the projection.

The dual statement holds in an abelian category with enough injectives: given injective resolutions $L \to I_L^\bullet$ and $N \to I_N^\bullet$, there is a degreewise split exact sequence $0 \to I_L^\bullet \to I_M^\bullet \to I_N^\bullet \to 0$ with $I_M^\bullet$ an injective resolution of M. It is this form that theorem 58.4.2 uses, and it follows by applying the projective statement in the opposite category.

***Proof*:**
Visually, we want to find objects and maps to fill in this commutative diagram:

$$\begin{tikzcd}
 & \vdots \arrow[d] & \vdots \arrow[d, dotted] & \vdots \arrow[d] & \\
0 \arrow[r] & P_L^{-2} \arrow[r, dotted] \arrow[d] & P_M^{-2} \arrow[r, dotted] \arrow[d, dotted] & P_N^{-2} \arrow[r] \arrow[d] & 0 \\
0 \arrow[r] & P_L^{-1} \arrow[r, dotted] \arrow[d] & P_M^{-1} \arrow[r, dotted] \arrow[d, dotted] & P_N^{-1} \arrow[r] \arrow[d] & 0 \\
0 \arrow[r] & L \arrow[r] & M \arrow[r] & N \arrow[r] & 0
\end{tikzcd}$$

where the dotted lines and the P_M^i are the goal (the name of the lemma comes from this horseshoe shaped diagram). I claim that $P_M^i = P_L^i \oplus P_N^i$ (which the reader should recall or reprove is projective) with the natural inclusion and projection maps shall work. The argument is done via induction. Since P_N^0 is projective and $M \to N$ is an epimorphism, the morphism $P_N^0 \to N$ lifts to a isomorphism m $\beta : P_N^0 \to M$. On the other hand, P_L^0 maps to M via $\alpha : P_L^0 \to L \to M$. Thus, there is a morphism:

$$\pi_M := \alpha \oplus \beta : P_L^0 \oplus P_N^0 \to M$$

diving us the diagram

$$\begin{tikzcd}
 & \vdots \arrow[d] & \vdots \arrow[d, dotted] & \vdots \arrow[d] & \\
0 \arrow[r] & P_L^{-2} \arrow[r, dotted] \arrow[d] & P_M^{-2} \arrow[r, dotted] \arrow[d, dotted] & P_N^{-2} \arrow[r] \arrow[d] & 0 \\
0 \arrow[r] & P_L^{-1} \arrow[r, dotted] \arrow[d] & P_M^{-1} \arrow[r, dotted] \arrow[d, dotted] & P_N^{-1} \arrow[r] \arrow[d] & 0 \\
0 \arrow[r] & P_L^0 \arrow[r] \arrow[d] & P_L^0 \oplus P_N^0 \arrow[r] \arrow[d] & P_N^0 \arrow[r] \arrow[d] & 0 \\
0 \arrow[r] & L \arrow[r] & M \arrow[r] & N \arrow[r] & 0
\end{tikzcd}$$

The snake lemma and π_L being an epimorphism gives that the kernels of the vertical maps form an exact sequence, and hence we have an epimorphism from P_L^{-1} and P_N^{-1} to the respective kernels since the column of the original diagram are exact:

$$\begin{tikzcd}
 & P_L^{-1} \arrow[d, "\pi_L^{-1}"] & & P_N^{-1} \arrow[d, "\pi_N^{-1}"] & \\
0 \arrow[r] & \ker \pi_L \arrow[r] & \ker \pi_M \arrow[r] & \ker \pi_N \arrow[r] & 0
\end{tikzcd}$$

Now, repeat as before, defining $P_M^{-1} = P_L^{-1} \oplus P_N^{-1}$ and define $\pi_M^{-1} : P_L^{-1} \oplus P_N^{-1} \to \ker \pi_M \to P_L^0 \oplus P_N^0$ giving the diagram:

$$\begin{tikzcd}
0 \arrow[r] & P_L^{-1} \arrow[r] \arrow[d, "\pi_L^{-1}"] & P_L^{-1} \oplus P_N^{-1} \arrow[r] \arrow[d, "\pi_M^{-1}"] & P_N^{-1} \arrow[r] \arrow[d, "\pi_N^{-1}"] & 0 \\
0 \arrow[r] & P_L^0 \arrow[r] \arrow[d, "\pi_L"] & P_L^0 \oplus P_N^0 \arrow[r] \arrow[d, "\pi_M"] & P_N^0 \arrow[r] \arrow[d, "\pi_N"] & 0 \\
0 \arrow[r] & L \arrow[r] & M \arrow[r] & N \arrow[r] & 0
\end{tikzcd}$$

By then by the snake lemma, $P_L^{-1} \oplus P_N^{-1} \to \ker \pi_M$ is an epimorphism, and hence we get exactness at $P_L^0 \oplus P_N^0$. Continuing inductively constructs the rest of the diagram.

It is tempting to say from this result that the mapping $\mathbf{A} \to K(\mathbf{A})$ is exact, however $K(\mathbf{A})$ is not abelian and hence this is simply undefined. The above is should instead be interpreted as follows: $P_M^\bullet$ is the mapping cone of a morphism: $\rho^\bullet : P_N[-1]^\bullet \to P_L^\bullet$, giving the distinguished triangle:

$$\begin{tikzcd}
& P_L^\bullet \arrow[dr] & \\
P_N^\bullet \arrow[ur, "+1"] & & MC(\rho)^\bullet [= P_M^\bullet] \arrow[ll]
\end{tikzcd}$$

This is as close as we can get to preserving exactness.

Corollary 56.3.8: Resolution Of Projective Resolutions

Let $M^\bullet$ be a complex bounded from above in a category $\mathbf{A}$ with enough projectives. Then there exists a complex $P^b_{M^\bullet}b$:

$$\cdots \to P^\bullet_{M^{-3}} \to P^\bullet_{M^{-2}} \to P^\bullet_{M^{-1}} \to\to P^\bullet_{M^0} \to 0$$

where each $P^\bullet_{M^i}$ is a projection resolution of M^i and inducing $M^\bullet$ in cohomology. If $M^\bullet$ is exact, then $P^\bullet_{M^\bullet}$ can be chosen to be exact.

***Proof*:**
Since $M^\bullet$ is a complex, it can be broken down into two short exact sequences:

$$0 \to K^i \to M^i \to I^{i+1} \to 0 \qquad 0 \to I^i \to K^i \to H^i \to 0$$

where K^i is the kernel of $M^i \to M^{i+1}$, I^{i+1} is its image, and H^i the cohomology at M^i. Chose arbitrary projective resolutions for each I^i and H^i for all i. Then by the horseshoe lemma, we get a projective resolution for K^i that fits within the short exact sequence:

$$0 \to P^\bullet_{I^i} \to P^\bullet_{K^i} \to P^\bullet_{H^i} \to 0$$

Applying the horseshoe lemma gain, we get a short exact sequence:

$$0 \to P^\bullet_{K^i} \to P^\bullet_{M^i} \to P^\bullet_{I^{i+1}} \to 0$$

Using these, the $P^\bullet_{M^i}$ can be used to construct a cochain complex, the differentials obtains by composition. If $M^\bullet$ is exact, then $I^i = K^i$, and so the construction gives $P^\bullet_{I^i} = P^\bullet_{K^i}$, so that $P^\bullet_{M^\bullet}$ is also exact.

Lemma 56.3.9: Projective Resolution and Functors

Let $\mathbf{A}$ be an abelian category and let

$$0 \to L^\bullet \to M^\bullet \to P^\bullet \to 0$$

be an exact sequence of complexes in $\mathbf{A}$, where P^i is projective for all i. Let $\mathscr{F} : \mathbf{A} \to \mathbf{B}$ be an additive functor between abelian categories. Then the sequence:

$$0 \to \mathscr{F}(L^b) \to \mathscr{F}(M^\bullet) \to \mathscr{F}(P^\bullet) \to 0$$

is exact

Note that there is nothing about $\mathscr{F}$ being an exact functor! This can be thought of as the generalization of the properties of projective modules which preserve exactness!!

***Proof*:**
Since P^i is projective, the sequence

$$0 \to L^i \to M^i \to P^i \to 0$$

splits. Thus

$$0 \to \mathscr{F}(L^i) \to \mathscr{F}(M^i) \to \mathscr{F}(P^i) \to 0$$

is split exact for all i (as the reader should check). Since exactness of a sequence of complexes is given by exactness at each degree, this gives the desired result.

Theorem 56.3.10: Long Exact Sequence From Derived Functor

Let $\mathscr{F} : \mathbf{A} \to \mathbf{B}$ be an additive functor of abelian categories, and assume $\mathbf{A}$ has enough projective. Then every short exact sequence in $\mathbf{A}$

$$0 \to L \to M \to N \to 0$$

[covariantly] functorially induces a long exact sequence in $\mathbf{B}$:

$$\begin{tikzcd}
0 \arrow[r] & L_2\mathscr{F}(L) \arrow[r] & L_2\mathscr{F}(M) \arrow[r] & L_2\mathscr{F}(N) \arrow[dll, "\delta_2"] & \\
& L_1\mathscr{F}(L) \arrow[r] & L_1\mathscr{F}(M) \arrow[r] & L_1\mathscr{F}(N) \arrow[dll, "\delta_1"] & \\
& L_0\mathscr{F}(L) \arrow[r] & L_0\mathscr{F}(M) \arrow[r] & L_0\mathscr{F}(N) \arrow[r] & 0
\end{tikzcd}$$

***Proof*:**
The functorial assignment will be left as an exercise. Given the short exact sequence, we get short exact sequence of projective resolutions, which by lemma 56.3.9 creates a short exact sequence:

$$0 \to \mathscr{F}(P_L^\bullet) \to \mathscr{F}(P_M^\bullet) \to \mathscr{F}(P_N^\bullet) \to 0$$

Since the cohomology of these complexes is precisely the left-derifevd functors of $\mathscr{F}$, the long exact cohomology sequence is given by Snakes lemma, completing the proof.

The change from short exact sequence to long exact sequence can be unified under the notion of triangles, namely this theorem can be seen as the triangle:

$$\begin{tikzcd} & \iota(L) \arrow[dr] & \\ \iota(N) \arrow[ur, "+1"] & & \iota(M) \arrow[ll] \end{tikzcd}$$

inducing the triangle

$$\begin{tikzcd} & \mathbf{L}_\bullet(L) \arrow[dr] & \\ \mathbf{L}_\bullet(N) \arrow[ur, "-1", "\delta"'] & & \mathbf{L}_\bullet(M) \arrow[ll] \end{tikzcd}$$

where the vertices of this triangle are complexes in $C^{\leq 0}(\mathbf{B})$. Naturally, the dual situation happens for right derived functors for categories that have enough injectives, iin which case the above triangle induces a triangle:

$$\begin{tikzcd} & \mathbf{R}^\bullet(R) \arrow[dr] & \\ \mathbf{R}^\bullet(N) \arrow[ur, "-1", "\delta"'] & & \mathbf{R}^\bullet(M) \arrow[ll] \end{tikzcd}$$

where the vertices of this triangle are complexes in $C^{\geq 0}(\mathbf{B})$.

The dual statement is what the right-derived functors satisfy, and it is recorded separately because Ext in its second variable is computed that way.

Corollary 56.3.11: Long Exact Sequence From a Right-Derived Functor

Let $\mathscr{F} : \mathbf{A} \to \mathbf{B}$ be an additive functor of abelian categories with $\mathbf{A}$ having enough injectives. Then every short exact sequence $0 \to L \to M \to N \to 0$ in $\mathbf{A}$ induces a long exact sequence

$$0 \to R^0\mathscr{F}(L) \to R^0\mathscr{F}(M) \to R^0\mathscr{F}(N) \xrightarrow{\delta^0} R^1\mathscr{F}(L) \to R^1\mathscr{F}(M) \to R^1\mathscr{F}(N) \xrightarrow{\delta^1} R^2\mathscr{F}(L) \to \cdots$$

functorial in the short exact sequence.

***Proof*:**
Apply theorem 56.3.10 in the opposite categories. An injective object of $\mathbf{A}$ is a projective object of $\mathbf{A}^{op}$, so $\mathbf{A}$ having enough injectives is $\mathbf{A}^{op}$ having enough projectives; the short exact sequence reverses to $0 \to N \to M \to L \to 0$ in $\mathbf{A}^{op}$; and the left-derived functors of $\mathscr{F}^{op}$ are

the right-derived functors of $\mathscr{F}$ with the arrows reversed, by definition 56.3.1. Transporting the conclusion back gives the displayed sequence, completing the proof.

These consideration were for general additive functors. In most cases we saw, there is usually a left-exact or right-exact functor (indeed, the nature of the left-derived and right-derived series almost force us to choose "a side"). If a functor is given to be either left- or right-exact, then it can be directly recovered from their derived functors, as we shall now show.

Proposition 56.3.12: Right-Exactness And Derived Functors

L $\mathbf{A}$ be an abelian category with enough projective and $\mathscr{F} : \mathbf{A} \to \mathbf{B}$ be a right-exact additive functor. Then

1. $\mathbf{L}_i\mathscr{F} = 0$ for $i < 0$
2. $L_0\mathscr{F}$ is naturally isomorphic to $\mathscr{F}$

The dual result holds for the category with enough injectives. Do note that the *right*-exact functor is associated to the *left* derived functors, and vice-versa for left-exact functors which would have the associated *right* derived functors. This is because the derived functors are "fixing" the exactness on the other side.

***Proof*:**
Since projective resolution $P^\bullet$ of an object M in $\mathbf{A}$ is in $C^{\leq 0}(\mathbf{A})$, $C(\mathscr{F})(P^bb)$ is 0 in positive degree, and hence so is its cohomology. Since $H_i = H^{-i}$, the first claim is immediate.

For the second, an exact sequence:

$$P^{-1} \to P^0 \to M \to 0$$

will map to a exact sequence by $\mathscr{F}$ (due to right-exactness)

$$\mathscr{F}(P^{-1}) - > \mathscr{F}(P^0) - > \mathscr{F}(M) - > 0$$

This induces an isomorphism: $\nu_L : \mathbf{L}_0\mathscr{F}(M) = H^0(C(\mathscr{F})(P^\bullet)) \xrightarrow{\sim} \mathscr{F}(M)$. If $\varphi : M_0 - > M_1$ is a morphism, then there exists a morphism of projective resolutions $\alpha^\bullet : P_0^\bullet - > P_1^b b$:

$$\begin{tikzcd}
\cdots \arrow[r] & P_0^{-1} \arrow[r] \arrow[d, "\alpha^{-1}"] & P_1^0 \arrow[r] \arrow[d, "\alpha^0"] & M_0 \arrow[r] \arrow[d, "\varphi"] & 0 \\
\cdots \arrow[r] & P_1^{-1} \arrow[r] & P_1^0 \arrow[r] & M_1 \arrow[r] & 0
\end{tikzcd}$$

induced by φ where $H^0(\alpha^b b)$ agrees with φ. Applying $\mathscr{F}$ and truncating to $\alpha^\bullet$, we get the commutative diagram:

$$\begin{tikzcd}
\mathscr{F}(P_0^{-1}) \arrow[r] \arrow[d, "\mathscr{F}(\alpha^{-1})"] & \mathscr{F}(P_1^0) \arrow[r] \arrow[d, "\mathscr{F}(\alpha^0)"] & \mathscr{F}(M_0) \arrow[r] \arrow[d, "\mathscr{F}(\varphi)"] & 0 \\
\mathscr{F}(P_1^{-1}) \arrow[r] & \mathscr{F}(P_1^0) \arrow[r] & \mathscr{F}(M_1) \arrow[r] & 0
\end{tikzcd}$$

with exact rows since $\mathscr{F}$ is right-exact. But then the following diagram commutes:

$$\begin{tikzcd}
\mathbf{L}_0\mathscr{F}(M_0) \arrow[r, "\mathbf{L}_0\mathscr{F}(\varphi)"] \arrow[d, "\nu_{M_0}"] & \mathbf{L}_0\mathscr{F}(M_1) \arrow[d, "\nu_{M_1}"] \\
\mathscr{F}(M_0) \arrow[r, "\mathscr{F}(\varphi)"'] & \mathscr{F}(M_1)
\end{tikzcd}$$

completing the proof.

This finally gives the desired recovery of exactness promised way back. Given a short exact sequence

$$0 \to L \to M \to N \to 0$$

then applying a right-exact functor

$$\mathscr{F}(L) \to \mathscr{F}(M)\mathscr{F}(N) \to 0$$

will give an exact sequence, which can be stretched out to form a long exact sequence using the derived functors:

$$\begin{tikzcd}
0 \arrow[r] & L_2\mathscr{F}(L) \arrow[r] & L_2\mathscr{F}(M) \arrow[r] & L_2\mathscr{F}(N) \arrow[dll, "\delta_2"] & \\
& L_1\mathscr{F}(L) \arrow[r] & L_1\mathscr{F}(M) \arrow[r] & L_1\mathscr{F}(N) \arrow[dll, "\delta_1"] & \\
& \mathscr{F}(L) \arrow[r] & \mathscr{F}(M) \arrow[r] & \mathscr{F}(N) \arrow[r] & 0
\end{tikzcd}$$

In this way, $\mathbf{L}_1\mathscr{F}$ measures the extent to which $\mathscr{F}$ fails to be left-exact, and each progressive $\mathbf{L}_i\mathscr{F}$ gives further measurement, further information, about this failure. When in a geometrical setting, this failure is often thought of as some form of obstruction (famously, in algebraic topology, this obstruction is interpreted as "'holes", in K-theory).

Similarly, $\mathbf{R}^{-}\mathscr{F}$ is naturally isomorphic to $\mathscr{F}$ if $\mathscr{F}$ is left-exact and the right-derived functors fit into the diagram:

$$\begin{tikzcd}
0 \arrow[r] & \mathscr{F}(L) \arrow[r] & \mathscr{F}(M) \arrow[r] & \mathscr{F}(N) \arrow[dll, "\delta_1"] & \\
& R^1\mathscr{F}(R) \arrow[r] & R^1\mathscr{F}(M) \arrow[r] & R^1\mathscr{F}(N) \arrow[dll, "\delta_0"] & \\
& R^2\mathscr{F}(R) \arrow[r] & R^2\mathscr{F}(M) \arrow[r] & R^2\mathscr{F}(N) \arrow[r] & \cdots
\end{tikzcd}$$

Thus, we have successfully captured the cohomological information that a functor may provide!

Acyclic Functors

There still remain some questions to be answered, one that was mentioned was that instead of using a projective resolution for Tor, a flat resolution may be used instead and give the same answer.

Definition 56.3.13: Acyclic Functor

Let $\mathscr{F}$ be an additive functor. An object M of $\mathbf{A}$ is *$\mathscr{F}$-acyclic* (with respect to left-derived functors) if $\mathbf{L}_i\mathscr{F}(M) = 0$ for all $i \neq 0$.

Similarly, an object M is $\mathscr{F}$-acyclic with respect to right-derived functors if $\mathbf{R}^i\mathscr{F}(M) = 0$ for $i \neq 0$. Naturally, projectives are acyclic for every right-exact functor, and injective acyclic for every left-exact functor, however such a functor may admit other $\mathscr{F}$-acyclic objects.

Example 56.3.14: Functor-acyclic Objects

Let R be a commutative ring. Recall that an R-module M is *flat* if $- \otimes_R M$ is exact (equivalently, by symmetry of $\otimes$ that $M \otimes_R -$ is exact). Then flat modules are *acyclic* with respect to tensor products, namely if N is flat:

$$\mathrm{Tor}^i_R(M, N) = 0 \qquad \forall i \neq 0,\ \forall M$$

and symmetrically if M is flat. Thus, flat modules are $(- \otimes_R N)$-acyclic for all modules N.

What shall be built up to now is that $\mathscr{F}$-acyclic resolutions can be used to compute $\mathbf{L}_i\mathscr{F}$.

Theorem 56.3.15: Derived Functors And Acylcic Functors

Let $\mathscr{F} : \mathbf{A} \to \mathbf{B}$ be a right-exact functor of abelian groups where $\mathbf{A}$ has enough projectives. Let $A^\bullet$ be a resolution of M where each A^i is $\mathscr{F}$-acyclic. Then for all i:

$$\mathbf{L}_i\mathscr{F}(M) \cong H^{-i}(\mathrm{Ch}(\mathscr{F})(A^\bullet))$$

Note that if each A^i is projective, the statement follows by definition.

***Proof*:**
The 0th and -1th position will be computed directly, and each successive position comes from induction. Take the exact sequence

$$A^{-1} \to A^0 \to M \to 0$$

which creates another exact sequence in $\mathbf{B}$ after applying the right-exact functor $\mathscr{F}$:

$$\mathscr{F}(A^{-1}) \to \mathscr{F}(A^0) \to \mathscr{F}(M) \to 0$$

Then:

$$H^0(\mathrm{Ch}(\mathscr{F})(A^\bullet)) \cong \mathscr{F}(M) \cong \mathbf{L}_0\mathscr{F}(M)$$

Now, if K is the kernel of $A^0 \to M$, then we have a short exact sequence:

$$0 \to K \to A^0 \to M \to 0$$

Thus applying the left-derived functors we form the long exact sequence:

$$\mathbf{L}_1\mathscr{F}(A^0) \to \mathbf{L}_1\mathscr{F}(M) \to \mathscr{F}(K) \to \mathscr{F}(A^0) \to \mathscr{F}(M) \to 0$$

Since A^0 is acyclic, $L_1\mathscr{F}(A^0) = 0$, hence by exactness

$$\mathbf{L}_1\mathscr{F}(M) \cong \ker(\mathscr{F}(K) \to \mathscr{F}(A^0)) \cong H^{-1}(\mathrm{Ch}(\mathscr{F})(A^\bullet))$$

where the last isomorphism is some manipulation of definition. For the rest, we do induction. Assume now that for all object of $\mathbf{A}$ and all indices $< i$. Truncating and shifting $\mathbf{A}^\bullet$ so that A^{-1} is in the 0 degree:

$$\cdots \to A^{-3} \to A^{-2} \to A^{-1} \to 0 \to \cdots$$

we get an $\mathscr{F}$-acyclic resolution of K, and so by the induction hypothesis $\mathbf{L}_{i-1}\mathscr{F}(K)$ can be computed by applying $\mathrm{Ch}(\mathscr{F})$ to the complex and taking cohomology, namely:

$$\mathbf{L}_{i-1}\mathscr{F}(K) \cong H^{-i}(A^\bullet)$$

On the other hand, the other terms in the long exact sequence obtained using the left derived functors gives:

$$\cdots \to \mathbf{L}_i\mathscr{F}(A^0) \to L_i\mathscr{F}(M) \to \mathbf{L}_{i-1}\mathscr{F}(K) \to \mathbf{L}_{i-1}\mathscr{F}(A^0) \to \cdots$$

since A^0 is $\mathscr{F}$-acyclic and $i > 1$, we get:

$$\mathbf{L}_i\mathscr{F}(M) \cong \mathbf{L}_{i-1}\mathscr{F}(K)$$

giving the desired result.

Thus, for the Tor functor, we may use a flat resolution to compute it the left-derived functors. Since $\mathscr{F}$-acyclic objects suffice in the computation, one can imagine that there is not need for $\mathbf{A}$ to have enough projectives to compute it. This is indeed the case, given that $\mathbf{A}$ has “enough $\mathscr{F}$-acyclic's”. (this is an exercise in Aluffi, 8.14, with bifunctors).

56.4 Double Complexes

This final section shall be the basis for being able to compute many complexes. It was mentioned in box 56.2.12 that $\mathrm{Ext}^i_R(M, N)$ may be computed by using either a projective resolution of M or an injective resolution on N, both give the same answer. More generally, we may compute the resolution of either M or N and get the same value. This section will now prove this result. This will make many computations much simpler, for usually one object in is readily understandable while the other is being analyzed.

To motivate this, switching from projective to $\mathscr{F}$-acyclic resolutions can be viewed in the following important alternative way. Consider the following diagram where the top row and right column are cochain:

$$\begin{tikzcd}
\cdots \arrow[r] & \mathscr{F}(A^{-2}) \arrow[r] & \mathscr{F}(A^{-1}) \arrow[r] & \mathscr{F}(A^{-0}) \arrow[r] & \mathscr{F}(M) \\
\cdots \arrow[r, dotted] & {} \arrow[u, dotted] \arrow[r, dotted] & {} \arrow[u, dotted] \arrow[r, dotted] & {} \arrow[u, dotted] \arrow[r, dotted] & \mathscr{F}(P_M^0) \arrow[u] \\
\cdots \arrow[r, dotted] & {} \arrow[u, dotted] \arrow[r, dotted] & {} \arrow[u, dotted] \arrow[r, dotted] & {} \arrow[u, dotted] \arrow[r, dotted] & \mathscr{F}(P_M^{-1}) \arrow[u] \\
\cdots \arrow[r, dotted] & {} \arrow[u, dotted] \arrow[r, dotted] & {} \arrow[u, dotted] \arrow[r, dotted] & {} \arrow[u, dotted] \arrow[r, dotted] & \mathscr{F}(P_M^{-2}) \arrow[u] \\
& \vdots \arrow[u, dotted] & \vdots \arrow[u, dotted] & \vdots \arrow[u, dotted] & \vdots \arrow[u]
\end{tikzcd}$$

in some way, there was a reason to go from the top chain to the right chain, with the intermediate values "interpolating" the result. The natural candidate for this would be a complex of complex. Needing to bound it in one direction, say we are looking at $\mathrm{Ch}^{\leq 0}(\mathrm{Ch}^{\leq 0}(\mathbf{A}))$. Such an object would be of the form:

$$\cdots \to M^{-3,\bullet} \xrightarrow{d_h^{-3,\bullet}} M^{-2,\bullet} \xrightarrow{d_h^{-2,\bullet}} M^{-1,\bullet} \xrightarrow{d_h^{-1,\bullet}} M^{0,\bullet} \to 0 \to \cdots$$

The h in the subscript is to represent the fact we are going *horizontally*, and the respective morphism of each $d_h^{-i,\bullet}$ will be going vertically and are denoted $d_v^{i,j}$, or simply d_v to simply show the function's direction:

$$\begin{tikzcd}
\cdots \arrow[r] & M^{-3,0} \arrow[r] & M^{-2,0} \arrow[r] & M^{-1,0} \arrow[r] & M^{0,0} \\
\cdots \arrow[r] & M^{-3,-1} \arrow[u] \arrow[r] & M^{-2,-1} \arrow[u] \arrow[r] & M^{-1,-1} \arrow[u] \arrow[r] & M^{0,-1} \arrow[u] \\
\cdots \arrow[r] & M^{-3,-2} \arrow[u] \arrow[r] & M^{-2,-2} \arrow[u] \arrow[r] & M^{-1,-2} \arrow[u] \arrow[r] & M^{0,-2} \arrow[u] \\
& \vdots \arrow[u] & \vdots \arrow[u] & \vdots \arrow[u] & \vdots \arrow[u]
\end{tikzcd}$$

Given a complex of complex, it would be nice to reduce this to being a complex, a more familiar setting. One strategy the reader may come up with is to take:

$$\bigoplus_{i+j=n} M^{i,j}$$

with the natural morphisms between them. However these morphisms *almost* complexes: instead of commutativity for each square, they are *anti-commutative*. To fix this, create a new type of object that will satisfy this condition:

Definition 56.4.1: Double Complex

Let $A = \{A_{p,q}\}$ be a family of objects from $\mathbf{A}$. Then A together with maps

$$d^h : A_{p,q} \to A_{p-1,q} \qquad \delta^v : A_{p,q} \to A_{p,q-1}$$

satisfying $d^h \circ d^h = 0$, $\delta^v \circ \delta^v = 0$ and $d^h \circ \delta^v + \delta^v \circ d^h = 0$ is called a *double complex* or *bicomplex*.
Both differentials *lower* an index; this homological convention is the one used throughout, and in particular in chapter 58. A cochain double complex, whose differentials raise indices, is the same thing read through $A_{p,q} = A^{-p,-q}$.

We may visualize a double complex like so, with d^h running to the right and δ^v running downwards, each lowering its index:

$$\begin{tikzcd}
 & \vdots \arrow[d] & \vdots \arrow[d] & \vdots \arrow[d] & \\
\cdots \arrow[r] & A_{p+1,q+1} \arrow[r] \arrow[d] & A_{p,q+1} \arrow[r] \arrow[d] & A_{p-1,q+1} \arrow[r] \arrow[d] & \cdots \\
\cdots \arrow[r] & A_{p+1,q} \arrow[r] \arrow[d] & A_{p,q} \arrow[r] \arrow[d] & A_{p-1,q} \arrow[r] \arrow[d] & \cdots \\
\cdots \arrow[r] & A_{p+1,q-1} \arrow[r] \arrow[d] & A_{p,q-1} \arrow[r] \arrow[d] & A_{p-1,q-1} \arrow[r] \arrow[d] & \cdots \\
 & \vdots & \vdots & \vdots &
\end{tikzcd} \tag{56.6}$$

Note this is not a commutative diagram but an *anti-commutative* one. That is a convention, not a restriction, and it is worth pinning down once because both forms occur in practice. A grid whose squares *commute*, which is what a complex of complexes literally gives, is converted into a double complex in the sense above by replacing the vertical differential with

$$\delta^v_{p,q} := (-1)^p \, \tilde{\delta}^v_{p,q}$$

where $\tilde{\delta}^v$ is the commuting one. The sign makes the squares anticommute, which is exactly what makes $d^h + \delta^v$ square to zero on the total complex. Every later construction, here and in chapter 58, assumes the anticommuting form; when a source of double complexes produces commuting squares, this twist is understood to have been applied.

As mentioned, we can form a total complex from this:

Proposition 56.4.2: Total Complex

Let $A = \{A^{p,q}\}$ be a double complex. Then define the total complex $\mathrm{Tot}^{\Pi}(A)$ and $\mathrm{Tot}^{\oplus}(A)$ by:

$$\mathrm{Tot}^{\Pi}(A)_n = \prod_{p+q=n} A_{p,q} \qquad \mathrm{Tot}^{\oplus}(A)_n = \bigoplus_{p+q=n} A_{p,q}$$

with $d^{\text{tot}} = d^h + \delta^v$, which squares to zero precisely because the two differentials anticommute. The two agree whenever only finitely many $A_{p,q}$ are nonzero on each diagonal $p + q = n$, which is the case for every double complex considered here.

***Proof*:**
Computing:

$$d_{tot} \circ d_{tot} = (d_h + \delta_v) \circ (d_h + \delta_v) = d_h \circ d_h + d_h \circ \delta_v + \delta_v \circ d_h + \delta_v \circ \delta_v = 0$$

Note that the total complex does not exist in all categories, namely it may not exist in categories which are not closed under infinite products and coproducts (ex. in the category of finite abelian groups). A category which is closed for infinite products is called *complete* and *cocomplete* if it is closed for infinite sums. We can visualize a total complex by taking the diagram in equation (56.6) and moving it like so:

$$\begin{tikzcd}
 & & & M^{0,0} \\
 & & M^{-1,0} \arrow[ur, "d_h"] & M^{0,-1} \arrow[u, "\delta_v"'] \\
 & M^{-2,0} \arrow[ur, "d_h"] & M^{-1,-1} \arrow[u, "\delta_v"] \arrow[ur, "d_h"] & M^{0,-2} \arrow[u, "\delta_v"'] \\
M^{-3,0} \arrow[ur, "d_h"] & M^{-2,-1} \arrow[u] \arrow[ur, "d_h"] & M^{-1,-2} \arrow[u, "\delta_v"] \arrow[ur, "d_h"] & M^{0,-2} \arrow[u, "\delta_v"'] \\
\vdots \arrow[u, "\delta_v"] \arrow[ur, "d_h"'] & \vdots \arrow[u, "\delta_v"] \arrow[ur, "d_h"'] & \vdots \arrow[u, "\delta_v"] \arrow[ur, "d_h"'] & \vdots \arrow[u, "\delta_v"']
\end{tikzcd}$$

It is then easy to visualize how we get a total complex form this.

Example 56.4.3: Total Complexes

1. Consider a double complex:

$$\cdots \to 0 \to L^\bullet \xrightarrow{\alpha^\bullet} M^\bullet \to 0 \to \cdots$$

which we may represent as:

$$\begin{tikzcd}
 & \vdots & \vdots & \vdots & \vdots & \\
\cdots \arrow[r] & 0 \arrow[r] \arrow[u] & L^{i+1} \arrow[r, "\alpha^{i+1}"] \arrow[u, "d_L^{i+1}"] & M^{i+1} \arrow[r] \arrow[u, "d_M^{i+1}"'] & 0 \arrow[r] \arrow[u] & \cdots \\
\cdots \arrow[r] & 0 \arrow[r] \arrow[u] & L^{i} \arrow[r, "\alpha^{i}"] \arrow[u, "d_L^{i}"] & M^{i} \arrow[r] \arrow[u, "d_M^{i}"'] & 0 \arrow[r] \arrow[u] & \cdots \\
\cdots \arrow[r] & 0 \arrow[r] \arrow[u] & L^{i-1} \arrow[r, "\alpha^{i-1}"] \arrow[u, "d_L^{i-1}"] & M^{i-1} \arrow[r] \arrow[u, "d_M^{i-1}"'] & 0 \arrow[r] \arrow[u] & \cdots \\
 & \vdots \arrow[u] & \vdots \arrow[u] & \vdots \arrow[u] & \vdots \arrow[u] &
\end{tikzcd}$$

we shall get the following:

$$\begin{tikzcd}
 & \vdots & \vdots & \vdots & \vdots & & & \vdots & & \vdots \\
\cdots \arrow[ur] & 0 \arrow[u] \arrow[ur] & L^{i+2} \arrow[u, "d_L^{i+1}"] \arrow[ur, "\alpha^{i+2}"] & M^{i+1} \arrow[u, "d_M^{i+1}"'] \arrow[ur] & 0 \arrow[u] & \cdots & & L^{i+2} \arrow[u] \arrow[urr] & \oplus & M^{i+1} \arrow[u] \\
\cdots \arrow[ur] & 0 \arrow[u] \arrow[ur] & L^{i+1} \arrow[u, "d_L^{i}"] \arrow[ur, "\alpha^{i+1}"] & M^{i} \arrow[u, "d_M^{i}"'] \arrow[ur] & 0 \arrow[u] \arrow[ur] & \cdots & \Rightarrow & L^{i+1} \arrow[u] \arrow[urr] & \oplus & M^{i} \arrow[u] \\
\cdots \arrow[ur] & 0 \arrow[u] \arrow[ur] & L^{i} \arrow[u, "d_L^{i-1}"] \arrow[ur, "\alpha^{i}"] & M^{i-1} \arrow[u, "d_M^{i-1}"'] \arrow[ur] & 0 \arrow[u] \arrow[ur] & \cdots & & L^{i} \arrow[u] \arrow[urr] & \oplus & M^{i-1} \arrow[u] \\
 & \vdots \arrow[u] \arrow[ur] & \vdots \arrow[u] \arrow[ur] & \vdots \arrow[u] \arrow[ur] & \vdots \arrow[u] \arrow[ur] & & & \vdots \arrow[u] \arrow[urr] & & \vdots \arrow[u]
\end{tikzcd}$$

which is exactly the definition of a mapping cone!

2. The operators $\mathrm{Hom}_A(-,-)$ and $- \otimes -$ both give rise to double complexes. Focusing on the $\mathrm{Hom}_A(-,-)$ example, consider a chain in $L^\bullet \in \mathrm{Ch}^{\leq 0}(\mathbf{A})$ and $M^\bullet \in \mathrm{Ch}^{\geq 0}(\mathbf{A})$. These can be used to form a double complex, or isomorphically a complex in $\mathrm{Ch}^{\leq 0}(\mathrm{Ch}^{\leq 0}(\mathbf{A}))$:

$$\cdots \to \mathrm{Hom}_A(L^\bullet, M^0) \to \mathrm{Hom}_A(L^\bullet, M^1) \to \mathrm{Hom}_A(L^\bullet, M^2) \to \cdots$$

which creates the following upper-quadrant grid:

$$\begin{tikzcd}
\vdots & \vdots & \vdots & \\
\mathrm{Hom}_A(L^{-2}, M^0) \arrow[u] \arrow[r] & \mathrm{Hom}_A(L^{-2}, M^1) \arrow[u] \arrow[r] & \mathrm{Hom}_A(L^{-2}, M^2) \arrow[u] \arrow[r] & \cdots \\
\mathrm{Hom}_A(L^{-1}, M^0) \arrow[u] \arrow[r] & \mathrm{Hom}_A(L^{-1}, M^1) \arrow[u] \arrow[r] & \mathrm{Hom}_A(L^{-1}, M^2) \arrow[u] \arrow[r] & \cdots \\
\mathrm{Hom}_A(L^{0}, M^0) \arrow[u] \arrow[r] & \mathrm{Hom}_A(L^{0}, M^1) \arrow[u] \arrow[r] & \mathrm{Hom}_A(L^{0}, M^2) \arrow[u] \arrow[r] & \cdots
\end{tikzcd}$$

Flipping the above diagram across the main diagonal, we would get the double complex:

$$\cdots \to \mathrm{Hom}_A(L^0, M^\bullet) \to \mathrm{Hom}_A(L^{-1}, M^\bullet) \to \mathrm{Hom}_A(L^{-2}, M^\bullet) \to \cdots$$

These two double complexes are isomorphic (as the reader should check). Due to this, by slight abuse of notation the total complex of both can be denoted $TC(\mathrm{Hom}_A(L, M))^\bullet$, where the degree i term is:

$$\mathrm{Hom}_A(L^{-i}, M^0) \oplus \mathrm{Hom}_A(L^{-i+1}, M^1) \oplus \cdots \oplus 0\mathrm{Hom}_A(L^{-1}, M^{i-1}) \oplus 0\mathrm{Hom}_A(L^0, M^i)$$

The reader should verify that:

(a) $\ker d^i_{tot}$ is equal to morphisms of cochains $L^\bullet \to M[i]^\bullet$

(b) The image $\mathrm{im} d^{i-1}_{tit}$ consists of morphisms that are homotopy equivalent to 0

(c) Thus, the cohomology of $TC(\mathrm{Hom}_A(M, N))^\bullet$ "paramaterizes" the morphisms in the homotopic category

To ground this, consider $TC(\mathrm{Hom}_A(M, N))^0 = \mathrm{Hom}_A(L^0, M^0)$. Then $d_{tot}(\alpha) = 0$ means that both $\alpha \circ d^{-1}_{L^\bullet} \in \mathrm{Hom}_A(L^{-1}, M^0)$ and $d^0_{M^\bullet} \in \mathrm{Hom}_A(L^0, M^1)$ both vanish. The reader should draw out this diagram, and notice that this will make α have the condition to be a morphism of cochain complexes $L^\bullet \to M^\bullet$!

$$H^0(TC(\mathrm{Hom}_A(M, N))^\bullet) = \mathrm{Hom}_{\mathrm{Ch}(\mathbf{A})}(L^\bullet, M^\bullet) \overset{!}{=} \mathrm{Hom}_{K(\mathbf{A})}(L^\bullet, M^\bullet)$$

where the $\overset{!}{=}$ equality comes from the fact that there is no nontrivial homotopies in this case! For $H^i(TC(\mathrm{Hom}_A(M, N))^\bullet)$, with some care for signs, it can be identified with $\mathrm{Hom}_{K(\mathbf{A})}(L^\bullet, M[i]^\bullet)$

The cohomology of double complexes requires some book-keeping through the use of *spectral sequences*, which shall be explored in greater depth in chapter 58. For the current goal, it is enough to consider when a double-complex is exact:

Proposition 56.4.4: Exactness Of Complex Of Complex

Let $\mathbf{A}$ be an abelian category and let

$$\cdots \to M^{-3,\bullet} \to M^{-2,\bullet} \to M^{-1,\bullet} \to M^{0,\bullet} \to 0$$

be a complex in $\mathrm{Ch}^{\leq 0}(\mathrm{Ch}^{\leq 0}(\mathbf{A}))$, regarded as a double complex $M^{\bullet,\bullet}$ with $M^{p,q} = 0$ unless $p \leq 0$ and $q \leq 0$, and let $T^\bullet$ denote its total complex. Then $T^\bullet$ is exact if *either* of the following holds:

1. the displayed complex is exact in $\mathrm{Ch}^{\leq 0}(\mathbf{A})$, equivalently every row $M^{\bullet,q}$ is exact;
2. every column $M^{p,\bullet}$ is exact.

Neither hypothesis implies the other, and either one alone suffices; this is the form in which the result is used below and in chapter 58. The same statement holds in $\mathrm{Ch}^{\geq 0}(\mathrm{Ch}^{\geq 0}(\mathbf{A}))$, by reversing the direction of every index. Note further that only one of the two directions need be bounded: case (2) uses only $p \leq 0$ and case (1) only $q \leq 0$, the bound being what makes the following staircase terminate; with only one bound the diagonals are infinite, so there the total complex must be taken to be $\mathrm{Tot}^{\oplus}$, since the staircase needs a least index in the support of a given element. The result is known as the *acyclic assembly lemma*; see *An Introduction to Homological Algebra* [53] for the variants in the other quadrants and for the distinction between Tot^{Π} and $\mathrm{Tot}^{\oplus}$, which does not arise here because each diagonal of $M^{\bullet,\bullet}$ carries only finitely many nonzero terms.

***Proof*:**
Step 1: Reduction to case (2). Transposing the double complex, that is exchanging the two indices and the two differentials, converts rows into columns and leaves the total complex unchanged up to the sign of the differential, which does not affect exactness. So case (1) follows from case (2) applied to the transpose, and only the latter needs proof.

Step 2: The staircase. Assume every column is exact, write d_h and d_v for the two differentials and $d = d_h + d_v$ for the differential of $T^\bullet$. Let $x \in T^n$ satisfy $dx = 0$; the claim is that $x = dy$ for some $y \in T^{n-1}$. Write $x = \sum_{p+q=n} x^{p,q}$, a finite sum.

Let p_1 be least with $x^{p_1,q_1} \neq 0$, where $q_1 = n - p_1$; if there is no such p_1 then $x = 0$ and there is nothing to prove. Consider the component of the equation $dx = 0$ in bidegree $(p_1, q_1 + 1)$. Only two terms can contribute to it, namely $d_v x^{p_1,q_1}$ and $d_h x^{p_1-1,q_1+1}$, and the second vanishes because $p_1 - 1 < p_1$ and p_1 was least. Hence

$$d_v x^{p_1,q_1} = 0$$

so x^{p_1,q_1} is a cocycle in the column $M^{p_1,\bullet}$. That column is exact, so there is $y^{p_1,q_1-1} \in M^{p_1,q_1-1}$ with $d_v y^{p_1,q_1-1} = x^{p_1,q_1}$.

Step 3: Termination. Replace x by

$$x' = x - d\big(y^{p_1,q_1-1}\big) = x - d_v y^{p_1,q_1-1} - d_h y^{p_1,q_1-1}$$

This is again a cocycle, since it differs from x by a coboundary. Its component in bidegree (p_1, q_1) is $x^{p_1,q_1} - d_v y^{p_1,q_1-1} = 0$, and $d_h y^{p_1,q_1-1}$ lives in bidegree $(p_1 + 1, q_1 - 1)$, so no component with $p < p_1$ is disturbed and all of those were already zero. The least index with a nonzero component has therefore increased by at least one.

Because $M^{p,q} = 0$ for $p > 0$, this least index cannot exceed 0, so after at most $|p_1| + 1$ repetitions every component vanishes. Summing the finitely many y's produced along the way gives $y \in T^{n-1}$ with $dy = x$, so $T^\bullet$ is exact at degree n. As n was arbitrary, this completes the proof.

Example 56.4.5: Exactness Of Double Complex

Let $P^\bullet \in C^{\leq 0}(\mathbf{A})$ be a complex and $L^\bullet \in C^{\geq 0}(\mathbf{A})$ an exact complex. Since each P^i is projective, $\mathrm{Hom}_{\mathbf{A}}(P^i, L^\bullet)$ for each i. Hence, the complex:

$$0 \to \mathrm{Hom}_{\mathbf{A}}(P^0, L^\bullet) \to \mathrm{Hom}_{\mathbf{A}}(P^{-1}, L^\bullet) \to \mathrm{Hom}_{\mathbf{A}}(P^{-2}, L^\bullet) \to \cdots$$

is exact. Using example 56.4.3, we see that:

$$\mathrm{Hom}_{K(\mathbf{A})}(P^\bullet, L[i]^\bullet) = 0 \qquad \forall i \in \mathbb{Z}$$

This implies that every cochain morphism from a bounded-above projective complex to a bounded-bellow exact complex is homotopic to 0. This just proved corollary 56.2.5 (with an extra boundedness condition)! In some textbooks, due to the speed at which some results can be proved using double-complexes, they are introduced earlier. However, the more "down-to-earth" introduction has is favored in this chapter since it demonstrated key homological concepts.

Next, we look at resolutions of complexes.. As we saw in definition 56.2.11, a (projective) resolution is a quasi-isomorphism $M^\bullet \to \iota(A)$ for some complex $M^\bullet \in \mathrm{Obj}(\mathrm{Ch}(\mathbf{A}))$, and more generally a projective resolution for a general complex is a quasi-isomorphism $P^\bullet \to C^\bullet$ for some projective complex $P^\bullet$. Generalizing the definition of resolution of the first kind, $\iota(A)$, and say that resolution of a complex $N^\bullet \in \mathrm{Ch}^\bullet(A) = A'$ is a complex of complex $M^{\bullet,\bullet} \in \mathrm{Ch}^{\leq 0}(A')$ and a quasi-isomorphism

$$\begin{tikzcd} \cdots \arrow[r] & M^{-2,\bullet} \arrow[r, "d_h"] \arrow[d] & M^{-2,\bullet} \arrow[r, "d_h"] \arrow[d] & M^{-2,\bullet} \arrow[r] \arrow[d] & 0 \arrow[r] \arrow[d] & \cdots \\ \cdots \arrow[r] & 0 \arrow[r] & 0 \arrow[r] & N^\bullet \arrow[r] & 0 \arrow[r] & \cdots \end{tikzcd} \tag{56.7}$$

where the diagram commutes. We shall show that this definition of a resolution induces a resolution as given in theorem 56.2.17. To see this, notice that in equation (56.7), $\alpha^\bullet \circ d_h = 0$, hence we can collapse the diagram like so:

$$\cdots \to M^{-2,\bullet} \xrightarrow{-d_h} M^{-1,\bullet} \xrightarrow{d_h} M^{0,\bullet} \xrightarrow{\alpha^\bullet} N^\bullet \to 0 \cdots$$

Call this complex $M_N^{\bullet,\bullet}$. Two important diagrams can be derived here. The first is by taking the total complex of $M^{\bullet,\bullet}$ and taking a morphism from this to $N^\bullet$:

$$\begin{tikzcd}
 & & & & 0 \arrow[r] & 0 \\
 & & & & M^{0,0} \arrow[u] \arrow[r] & N^0 \arrow[u] \\
 & & M^{-1,0} \arrow[urr] & \oplus & M^{0,-1} \arrow[u] \arrow[r] & N^{-1} \arrow[u] \\
M^{-2,0} \arrow[urr] & \oplus & M^{-1,-1} \arrow[u] \arrow[urr] & \oplus & M^{0,-2} \arrow[u] \arrow[r] & N^{-2} \arrow[u] \\
{} \arrow[u] \arrow[urr] & & {} \arrow[u] \arrow[urr] & & {} \arrow[u] & {} \arrow[u]
\end{tikzcd}$$

This gives a morphism:

$$TC(\alpha^\bullet) : TC(M^{\bullet,\bullet}) \to N^\bullet$$

where the morphism is defined since $TC(-)$ is a functorial. Another diagram we can create is by considering the total complex of $M_N^{\bullet,\bullet}$:

$$\begin{tikzcd}
 & & & & & & N^0 \\
 & & & & M^{0,0} \arrow[urr] & \oplus & N^{-1} \arrow[u] \\
 & & M^{-1,0} \arrow[urr] & \oplus & M^{0,-1} \arrow[u] \arrow[urr] & \oplus & N^{-2} \arrow[u] \\
M^{-2,0} \arrow[urr] & \oplus & M^{-1,-1} \arrow[u] \arrow[urr] & \oplus & M^{0,-2} \arrow[u] \arrow[urr] & \oplus & N^{-3} \arrow[u] \\
{} \arrow[u] \arrow[urr] & & {} \arrow[u] \arrow[urr] & & {} \arrow[u] \arrow[urr] & & {} \arrow[u]
\end{tikzcd}$$

These two are related in the following way:

Proposition 56.4.6: Total Complex Of Resolution

Let $M^{\bullet,\bullet}$ be a resolution of $N^\bullet$ and let $M_N^{\bullet,\bullet}$ be defined as above. Then

$$TC(M_N)^\bullet \cong MC\big(TC(\alpha)\big)[-1]^\bullet$$

canonically, equivalently $TC(M_N)[1]^\bullet \cong MC(TC(\alpha))^\bullet$.

***Proof*:**
Both sides are computed degreewise, so it suffices to match the terms and the differentials.

Step 1: The terms. The complex $M_N^{\bullet,\bullet}$ is $M^{\bullet,\bullet}$ with $N^\bullet$ appended in the column $p = 1$, to the right of $M^{0,\bullet}$ and joined to it by the augmentation $\alpha^\bullet$. A term N^j therefore sits in total degree

$1 + j$, so its contribution to total degree n is N^{n-1} and

$$TC(M_N)^n = \Big(\bigoplus_{i+j=n} M^{i,j}\Big) \oplus N^{n-1} = TC(M)^n \oplus N^{n-1}$$

which is what the diagram above the proposition displays, $M^{0,0}$ being grouped with N^{-1}. On the other side $TC(\alpha)\colon TC(M)^\bullet \to N^\bullet$ is the totalized augmentation, and since definition 56.1.18 puts the *source* of a morphism in the shifted slot,

$$MC\big(TC(\alpha)\big)^n = TC(M)^{n+1} \oplus N^n$$

Comparing the two, $TC(M_N)^n = MC(TC(\alpha))^{n-1}$, which is the shift asserted in the statement.

Step 2: The differentials. On $TC(M_N)^\bullet$ the differential is the total differential of the augmented double complex: on the M part it is $d_h + d_v$, the component carrying $M^{0,\bullet}$ into the appended column is $\alpha^\bullet$, and on the N part it is $-d_N$, the sign being the $(-1)^p$ twist at $p = 1$ that makes the augmented grid anticommute. So $d_{TC(M_N)}(m, n) = (d_{TC(M)}m,\ \alpha(m) - d_N n)$. By definition 56.1.18 the cone differential is $d_{MC}(l, n) = (-d_{TC(M)}l,\ \alpha(l) + d_N n)$, so under the identification of Step 1 the two differ by an overall sign.

The shift convention $d_{[r]} = (-1)^r d$ gives $d_{MC(TC(\alpha))[-1]}(m, n) = -\big(-d_{TC(M)}m,\ \alpha(m) + d_N n\big) = (d_{TC(M)}m,\ -\alpha(m) - d_N n)$, so the two differentials agree in the $TC(M)$ component and differ exactly in the sign of the α-component. They are therefore not equal on the nose, but the map

$$\varphi(m, n) = (m, -n)$$

is an isomorphism of complexes intertwining them: $\varphi\big(d_{TC(M_N)}(m, n)\big) = (d_{TC(M)}m,\ -\alpha(m) + d_N n) = d_{MC(TC(\alpha))[-1]}\big(\varphi(m, n)\big)$. Hence $TC(M_N)^\bullet \cong MC(TC(\alpha))[-1]^\bullet$, canonically, completing the proof.

Using this, we can see how the resolution in $\mathrm{Ch}^{\leq 0}(A')$ leads to a resolution as defined in theorem 56.2.17:

Theorem 56.4.7: Total Complexes And Resolution Of Complexes

Let $\mathbf{A}$ be an abelian category, $\mathbf{A}' = \mathrm{Ch}^{\leq 0}(\mathbf{A})$, and $N^\bullet \in A'$. Let $M^{\bullet,\bullet} \in \mathrm{Ch}^{\leq 0}(\mathbf{A}')$ with total complex $T^\bullet$.

1. Assume that $M^{\bullet,\bullet}$ is a resolution of $N^\bullet$ in $\mathrm{Ch}^{\leq 0}(\mathbf{A}')$. Then $T^\bullet$ is quasi-isomorphic to $N^\bullet$ in $\mathbf{A}'$.

2. Assuming the cohomology of $M^{i,\bullet}$ is concentrated at 0. Then $T^\bullet$ is quasi-isomorphic to the complex of cohomology objects induced by the complex $M^{\bullet,\bullet}$:

$$\cdots \to H^0(M^{-3,\bullet}) \to H^0(M^{-2,\bullet}) \to H^0(M^{-1,\bullet}) \to H^0(M^{0,\bullet}) \to 0 \cdots$$

Notice that this theorem is a direct extensions of proposition 56.4.4. This is also a special case of results we shall see in chapter 58

***Proof*:**
Notice that (2) follows from (1) by flipping the corresponding double-complex across the diagonal, hence we prove the first. For the first, if $M_N^{\bullet,\bullet}$ is quasi-isomorphic, then $TC(M_N)^\bullet$ is exact by proposition 56.4.4, which by proposition 56.4.6 is the mapping cone of the morphism $T^\bullet \to N^\bullet$, which by proposition 56.1.21 shows that $\alpha^\bullet : T^\bullet \to N^\bullet$ is a quasi-isomorphism, as we sought to show.

Using this theorem, a simple alternative to resolving complexes (theorem 56.2.17). By the horseshoe lemma, every bounded-above complex $N^\bullet$ in an abelian category with enough projectives may be viewed as a complex induced by H^0 from a complex:

$$\cdots \to P^{-3,\bullet} \to P^{-2,\bullet} \to P^{-1,\bullet} \to P^{0,\bullet} \to 0$$

where $P^{i,\bullet}$ are projective resolutions of N^i. Then by theorem 56.4.7(2) the total complex of $P^{\bullet,\bullet}$ is a projective resolution of $N^\bullet$. This then can give theorem 56.2.17. It is still worth presenting theorem 56.2.17 to firmly grasp the fundamentals of homological algebra.

56.4.1 Applications to Computations of Ext and Tor

Let us now show that we may compute $\mathrm{Ext}(M,N)$ or $\mathrm{Tor}(M,N)$ using a resolution of either term. Going back to the $\mathscr{F}$-acyclic resolution of an object M:

$$\cdots \to A^{-2} \to A^{-1} \to A^0 \to M \to 0 \to \cdots$$

We may create a complex of projective resolution by corollary 56.3.8:

$$\cdots \to P^{-2,\bullet} \to P^{-1,\bullet} \to P^{0,\bullet} \to P_M^\bullet \to 0 \to \cdots$$

where $P^{i,\bullet}$ is a projective resolution of A^{-i}. Since the resolution by projective resolution's is exact, the rows of the double-complex corresponding to $P^{\bullet,\bullet}$ are projective resolutions of P_M^i. Now, applying the functor $\mathscr{F}$, we get the diagram:

$$\begin{tikzcd}
\cdots \arrow[r] & \mathscr{F}(A^{-2}) \arrow[r] & \mathscr{F}(A^{-1}) \arrow[r] & \mathscr{F}(A^{0}) \arrow[r] & \mathscr{F}(M) \\
\cdots \arrow[r] & \mathscr{F}(P_M^{-2,0}) \arrow[r] \arrow[u] & \mathscr{F}(P_M^{-1,0}) \arrow[r] \arrow[u] & \mathscr{F}(P_M^{0,0}) \arrow[r] \arrow[u] & \mathscr{F}(P_M^{0}) \arrow[u] \\
\cdots \arrow[r] & \mathscr{F}(P_M^{-2,-1}) \arrow[r] \arrow[u] & \mathscr{F}(P_M^{-1,-1}) \arrow[r] \arrow[u] & \mathscr{F}(P_M^{0,-1}) \arrow[r] \arrow[u] & \mathscr{F}(P_M^{-1}) \arrow[u] \\
\cdots \arrow[r] & \mathscr{F}(P_M^{-2,-2}) \arrow[r] \arrow[u] & \mathscr{F}(P_M^{-1,-2}) \arrow[r] \arrow[u] & \mathscr{F}(P_M^{0,-2}) \arrow[r] \arrow[u] & \mathscr{F}(P_M^{-2}) \arrow[u] \\
& \vdots \arrow[u] & \vdots \arrow[u] & \vdots \arrow[u] & \vdots \arrow[u]
\end{tikzcd}$$

Since each row and column of this diagram (excluding the top row and right column) is exact, by theorem 56.4.7 the total complex of:

$$\cdots \mathscr{F}(P^{-2,\bullet}) \to \mathscr{F}(P^{-1,\bullet}) \to \mathscr{F}(P^{0,\bullet}) \to 0 \to \cdots$$

is quasi-isomorphic to both $\text{Ch}(\mathscr{F})(A^\bullet)$ and $\text{Ch}(\mathscr{F})(P^\bullet)$, that is:

$$TC(\mathscr{F}(P))^\bullet \sim \text{Ch}(\mathscr{F})(A^\bullet) \sim \text{Ch}(\mathscr{F})(P^\bullet)$$

Thus, by definition of quasi-isomorphism

$$\mathbf{L}_i\mathscr{F}(M) \cong H^i(\text{Ch}(\mathscr{F})(A^\bullet)) \cong H^i(\text{Ch}(\mathscr{F})(P^\bullet))$$

Applying these results to Ext and Tor will finally give us the following results:

Theorem 56.4.8: Computation Of Tor and Ext

Let M, N be R-modules for a commutative ring R, and let $P_M^\bullet$ and $P_N^\bullet$ be projections resolutions of M and N respectively, and $Q_N^\bullet$ an injective resolution of N. Then:

$$\text{Tor}_i^R(M, N) \cong H^i(P_M^\bullet \otimes_R N) \cong H^i(M \otimes_R P_N^\bullet)$$

and

$$\text{Ext}_R^i(M, N) \cong H^i(\text{Hom}_R(P_M^\bullet, N)) \cong H^i(\text{Hom}_R(M, Q_N^\bullet))$$

***Proof*:**
We shall show the result for Tor, the result following very similarly for Ext. Take the complex:

$$\cdots \to P_M^{-2} \otimes_R P_N^\bullet \to P_M^{-1} \otimes_R P_N^\bullet \to P_M^0 \otimes_R P_N^\bullet \to 0 \cdots$$

Since each P_N^j is projective, the cohomology of the complex is concentrated at 0 and equals $M \otimes_R P_N^\bullet$. By theorem 56.4.7 we have that :

$$TC(P_M^\bullet \otimes_R P_N^\bullet)^\bullet \qquad \text{is quasi-isomorphic to} \qquad M \otimes_R P_N^\bullet$$

Similarly, since each P_M^i is projective, the complex $P_M^i \otimes_R P_N^\bullet$ is a resolution of $P_M^i \otimes N$, i.e. the cohomology of each $P_M^i \otimes_R P_N^\bullet$ is concentrated at degree 0 and equals $P_M^i \otimes_R N$. By theorem 56.4.7 we have that :

$$TC(P_M^\bullet \otimes_R P_N^\bullet)^\bullet \qquad \text{is quasi-isomorphic to} \qquad P_M^\bullet \otimes_R N$$

but then, the cohomologies are identical, as we sought to show.

56.5 *Further Topic in Homological Algebra

look at commented line here:

In this bonus section, we shall introduce formally give the definition of a derived category and introduce the notion of triangulated categories.

56.5.1 Derived Categories, Formally

As defined in this chapter, derived categories required enough projectives/injectives. However, it was mentioned that derived categories may be defined without this restriction, allowing for a category that satisfies the universal property for abelian categories more generally. The theoretical build-up needed for this was carried out in section 55.2: the general derived category is the localization $D(\mathbf{A}) = \mathrm{Ch}(\mathbf{A})[\mathrm{qis}^{-1}]$, which exists by theorem 55.2.2 and is recorded as corollary 55.2.6. What follows revisits that construction concretely, in the roof-diagram form that theorem 55.2.5 justifies, so that the reader sees the mechanism rather than only the universal property. This section is heavily inspired by *Aluffi Chapter 0* which presented these idea's very concisely.

We shall have the objects still be $\mathrm{C}(\mathbf{A})$, however we need the quasi-isomorphisms to be invertible. Recall lemma 56.2.7 where it was shown that quasi-isomorphisms of projective resolutions have the property of being acting like non-zero divisors. We shall take full advantage of this idea, and essentially "formally invert" quasi-isomorphisms in the same vein as the construction of localization's for rings and modules. In fact, from an abstract perspective, the study of homological algebra can be thought of as the study of the localization of the category of (co)chain complexes (localized at the class of quasi-isomorphisms). When there is enough projectives, the localization of $\mathrm{C}(\mathbf{A})$ becomes $\mathrm{K}(\mathbf{a})$.

For the more general case, consider the following *roof diagram*:

$$\begin{tikzcd} & Z \arrow[dl, "\alpha"'] \arrow[dr, "\beta"] & \\ A & & B \end{tikzcd}$$

Then we may formally add a morphism $\beta \circ \alpha^{-1} : A \to B$:

$$\begin{tikzcd} & Z \arrow[dl, "\alpha"'] \arrow[dr, "\beta"] & \\ A \arrow[rr, dashed] & & B \end{tikzcd}$$

This can be thought of as the fraction. There are many technicalities being swept under the rug, like how do we insure these are well-defined morphisms (ex. how would we compose roofs?), while more fundamental problems like the amount of morphisms shouldn't exceed a *set*-amount (note that $\mathbf{A}$ and $\mathbf{B}$ are classes). Furthermore, we may ask if this is possible on any possible category, or should we restrict to abelian categories? Can any morphism be invertible, or do we need to restrict to a special subset (similar to a multiplicative set)? Giving this technical difficulties, we may imagine that from this the diagram

$$\begin{tikzcd} & Z \arrow[dl, "\alpha"'] \arrow[dr, "\mathrm{id}_Z"] & \\ A \arrow[rr, dashed] & & Z \end{tikzcd}$$

would formally invert a morphism. For $\mathrm{D}(\mathbf{A})$, we would localize at the class of quasi-isomorphism of $\mathrm{K}(\mathbf{A})$. The category $\mathrm{D}(\mathbf{A})$ shall be the homotopic category of $\mathrm{C}(\mathbf{A})$ where a morphism $L^\bullet \to M^\bullet$ can be represented as a roof:

$$\begin{tikzcd} & N^\bullet \arrow[dl, "\text{quasi-iso}"'] \arrow[dr] & \\ L^\bullet & & M^\bullet \end{tikzcd}$$

(A word on how many abelian categories may induce the same derived categories. However, there is a special "triangulatedd structure", or t-structure which allows you to recover A)

56.5.2 Triangulated Categories

Homotopic categories are certainly almost never not abelian, for the kernels and cokernels of homotopic maps need not match (hence not being well-defined in the equivalence class). However, a derived functors $\mathrm{D}(\mathscr{F})$ always has a long exact sequence associated to it, which shows some weaker property these categories possess. These considerations are explored in this chapter. Take a short exact sequence of complexes:

$$0 \to L^\bullet \to M^\bullet \to N^\bullet \to 0$$

Then snakes lemma says that after taking homology, there is an interesting connection between $N^\bullet$ and $L^\bullet$. When working with the complex, there is no interesting connection between $N^\bullet$ and $L^\bullet$. We may connect these two complexes via the zero morphism: $N^\bullet \xrightarrow{0} L^\bullet$. From this, the short exact sequence of complexes may be interpreted as these three exact sequences:

$$N^\bullet \xrightarrow{0} L^\bullet \to M^\bullet$$
$$L^\bullet \to M^\bullet \to N^\bullet$$
$$M^\bullet \to N^\bullet \xrightarrow{0} L^\bullet$$

These three exact sequences are essentially can be combined if we fold any of them into the following triangle:

Definition 56.5.1: Exact Triangle

Let $L^\bullet, M^\bullet, N^\bullet \in \mathrm{Ch}(\mathbf{A})$ be three complexes. Then if there exists morphisms st the following diagram is exact (at each vertex):

$$\begin{tikzcd} & L^\bullet \arrow[dr] & \\ N^\bullet \arrow[ur] & & M^\bullet \arrow[ll] \end{tikzcd}$$

Then the diagram is called an *exact triangle* of $\mathrm{Ch}(\mathbf{A})$. This diagram is also sometimes denoted

$$L^\bullet \to M^\bullet \to N^\bullet \to$$

if it is easier to write it out flat.

Hence, the particular exact triangle given by folding any of the above exact sequences is:

$$\begin{tikzcd} & L^\bullet \arrow[dr] & \\ N^\bullet \arrow[ur, "+1"] & & M^\bullet \arrow[ll] \end{tikzcd}$$

where the $+1$ morphism represents the zero map. The $+1$ notation is to hint that this map will be a sort of shift from $N^\bullet$ to $L^\bullet[1]$. This can be stretched out into a single long sequence:

$$\cdots \to N^{i-1} \xrightarrow{0} L^i \to M^i \to N^i \xrightarrow{0} L^{i+1} \to \cdots$$

More explicitly, the triangle can be "unwrapped" to create the following nice diagram:

here, v good visual intuition

Applying the cohomology functor to the above long exact sequence and replacing the zero maps with the connecting morphism, we get:

$$\cdots \to H^{i-1}(N^\bullet) \xrightarrow{\delta^{i-1}} H^i(L^\bullet) \xrightarrow{0} H^i(M^\bullet) \xrightarrow{0} H^i(N^\bullet) \xrightarrow{\delta^i} H^{i+1}(L^\bullet) \to \cdots$$

Note that the maps δ^i did not come from the cohomology functor. In a sense, this is because $\mathrm{Ch}(\mathbf{A})$ is the wrong category to apply the cohomology functor, rather it is better to apply it in $\mathrm{D}(\mathbf{A})$. To make this connection, the notion of *distinguished triangles* would have to be developed in some level of detail. I do wish to do so since I'm interested in some of the connections they have, but for now I shall follow Aluffi's example and simply state the ingredients necessary on a shallow enough level to explain how the morphism δ would arise without needing any special considerations.

Let $\mathbf{A}$ be any Abelian category. We shall define the notion of a *distinguished triangle*. We first define a functor which is called the *translation functor*. Though it is called a translation functor, it shall be characterized by the axioms of a distinguished triangle. For the categories $\mathrm{K}(\mathbf{A})$ and $\mathrm{D}(\mathbf{A})$ this is the shift functor $L^\bullet \xrightarrow{+1} L^\bullet[1]$. Distinguished triangle are triangles that must satisfy 4 axioms:

1. The triangle:

$$\begin{tikzcd} & 0 \arrow[dr] & \\ A \arrow[ur, "+1"] & & A \arrow[ll, "\mathrm{id}_A"] \end{tikzcd}$$

   has to be part of the collection of distinguished triangles

2. every morphism $\alpha : A \to B$ must be the side of a distinguished triangle.

3. Triangles that are isomorphic (as triangle diagrams) to distinguished triangles must also be distinguished

4. Distinguished triangles must be able to rotate, namely if:

$$\begin{tikzcd} & A \arrow[dr, "\alpha"] & \\ C \arrow[ur, "\gamma^{+1}"] & & B \arrow[ll, "\beta"] \end{tikzcd}$$

   is a distinguished triangle if and only if

$$\begin{tikzcd} & A \arrow[dr, "\beta"] & \\ A[1] \arrow[ur, "-\alpha^{+1}"] & & B \arrow[ll, "\gamma"] \end{tikzcd}$$

   is a distinguished triangle (the rotation is best pictured as turning the triangle one third of a turn, which is what the sign on α^{+1} records)

5. Finally, distinguished triangles must satisfy an axiom called the *octahedral axiom* which will not be necessary to cover for the remarks this section endeavors to explicate.

A category that contains a collection of distinguished triangles is called a *triangulated category*. The category $\mathrm{K}(\mathbf{A})$ is an example of such a category where the translation functor is the shift functor and all distinguished triangles are isomorphic to:

$$\begin{tikzcd}
& L^\bullet \arrow[dr, "\alpha^\bullet"] & \\
MC(\alpha^\bullet) \arrow[ur, "+1"] & & M^\bullet \arrow[ll]
\end{tikzcd}$$

Note how this diagram also fulfills the 3rd axiom: any map $\alpha^\bullet : L^\bullet \to M^\bullet$ is the edge of the above distinguished triangle where the other morphisms are the natural ones. Note also how the axioms prevent $\mathrm{Ch}(\mathbf{A})$ from being a triangulated category. Since

$$\begin{tikzcd}
& 0 \arrow[dr] & \\
A \arrow[ur, "+1"] & & A \arrow[ll, "\mathrm{id}_A"]
\end{tikzcd}$$

needs to be a distinguished triangle, then since $L^\bullet$ is the mapping cone of the zero-morphism $0 \to L^\bullet$ the rotation would axiom would require

$$\begin{tikzcd}
& L^\bullet \arrow[dr, "\mathrm{id}_{L^\bullet}"] & \\
0 \arrow[ur, "+1"] & & L^\bullet \arrow[ll]
\end{tikzcd}$$

The mapping cone of the identity is not 0, and hence it cannot be a distinguished category. However, the mapping cone of the identity is *homotopically-equivalent* to 0, and hence the above triangle is indeed distinguished in $\mathrm{K}(\mathbf{A})$. Furthermore, a triangulated category, the cohomology functor will map distinguished triangles to exact triangles (in the case of $\mathrm{K}(\mathbf{A})$, this is because all distinguished triangles are isomorphic to the triangles with mapping cones), and hence give long exact sequences. Two notable examples of cohomological functors that map quasi-isomorphisms to isomorphisms for any triangulated category $\mathbf{A}$ are $\mathrm{Hom}(A, -)$ and $\mathrm{Hom}(=, A)$ for any object $A \in \mathbf{A}$.

The derived category $\mathrm{D}(\mathbf{A})$ is naturally a triangulated category, and the natural functor $\mathrm{K}(\mathbf{A}) \to \mathrm{D}(\mathbf{A})$ maps distinguished triangles to distinguished triangles and maps the translation functor to the translation functor. Derived categories are less restrictive than homotopic categories in the following important way: if

$$0 \to L^\bullet \xrightarrow{\alpha^\bullet} M^\bullet \xrightarrow{\beta^\bullet} N^\bullet \to 0$$

is a short exact sequence, then in general there is no distinguished triangle

$$\begin{tikzcd}
& L^\bullet \arrow[dr, "\alpha^\bullet"] & \\
N^\bullet \arrow[ur, "\gamma^\bullet\ +1"] & & M^\bullet \arrow[ll, "\beta^\bullet"]
\end{tikzcd}$$

In $\mathrm{K}(\mathbf{A})$ for any choice $\gamma^\bullet$ (even more is the case, $N^\bullet$ need not be homotopically equivalent to $MC(\alpha^\bullet)$). However, such a triangle *does* always exist in $\mathrm{D}(\mathbf{A})$), namely because there does exist a quasi-isomorphism $MC(\alpha)^\bullet \to N^\bullet$. Then applying the cohomology functor to this triangle shall give

$$\begin{tikzcd} & H^\bullet(L^\bullet) \arrow[dr] & \\ H^\bullet((N^\bullet) \arrow[ur, "+1"] & & H^\bullet((M^\bullet) \arrow[ll] \end{tikzcd}$$

where $+1$ is the connection homomorphism δ up to isomorphism. Furthermore, the 0 morphisms that connect the above in when applying the $H^\bullet(-)$ turned out to be particular choices in the more general case.

Finally, it should be noted that any functor $\mathscr{F} : \mathbf{A} \to \mathbf{B}$ between Abelian categories shall induce a functor $\mathrm{K}(\mathscr{F}) : \mathrm{K}(\mathbf{A}) \to \mathrm{K}(\mathbf{B})$, and this functor is in fact a functor of triangulated categories. Hence, the bounded equivalents, as well as $\mathrm{L}(\mathscr{F}), \mathrm{R}(\mathscr{F}) : \mathrm{D}^\pm(\mathbf{A}) \to \mathrm{D}^\pm(\mathbf{B})$ are functor of triangulated categories. From this perspective, the results proven about derived functors is a consequence of this fact.

56.5.3 Stable Module Theory

I really want to talk about $f, g : M \to N$ being "homotopic" if $h = f - g$ can factor through a projective module $h : M \to P \to N$; and how this is equivalent to the homotopy equivalence we talked about earlier. This brings back the intuition from the geometric setting were we didn't need a resolution of modules to have two modules be homotopy equivalent! This also naturally leads to naturally define stably equivalent, namely M, N are stably equivalent if there exists a P such that:

$$M \oplus P \cong N \oplus P$$

This is further developed in [43], but it is definitely worth mentioning it here since it is an excellent prelude!

57

Applications of (co)Homology

With cohomology theory more securely under our belt, it is time to use it! All the work in this chapter can be learnt without the need for any algebraic topology background, as it is the goal of the book to minimize the need for external material. However, the proceeding ideas are abstractions of ideas in algebraic topology, and ignoring the source of inspiration for these ideas is itself a disservice to the pedagogy of the material. Therefore, the section will be written in such a way that a background in homology/cohomology will be advantageous. Namely, it is best if the reader is familiar with simplicial and singular homology, how cohomology is derived from $\mathrm{Hom}(C_\bullet(X), G) = C^\bullet(X; G)$ and the relation of homology to cohomology (ex. the universal coefficient theorem for cohomology). The reader may choose to consult [38] or [32] for the details.

This chapter does *not* present an in-depth coverage of any of the material, in particular these subjects are a large source of modern development in mathematics (as of the early 21st century). The material presented is to demonstrate to the reader the power of cohomological theories so that the reader has an easier time when reading the more dense books containing this information. For a book in the EYNTKA series covering group and galois cohomology, see [13], and for K-theory see [43].

In this chapter we:

1. Shall use the above theory to find cohomological invariants of groups
2. Shall show some ways of interpreting Ext and Tor
3. Shall show that the notion of dimension from the geometric setting can be brought over to the algebraic!
4. Some K-theory will certainly interest the reader

We shall not do sheaf cohomology, as (naturally) it required sheaf theory that was not covered, and it is an exploration for EYNTKA Algebraic Geometry [5].

57.1 Group Cohomology

We start with no link to algebraic topology, and shall include the relation to singular cohomology before the examples.

Given a (not necessarily abelian) group G, what would it mean to extract cohomological information from it? A (co)chain would have to naturally be formed, and a functor must be found to derive. Groups themselves don't form an abelian category, but it was shown that representations, that is $k[G]$-modules, do form them. This is perhaps a first place in which we may think we can study groups, however this category is too nice (think of (co)homology with $\mathbb{Z}$ vs. $\mathbb{R}/\mathbb{C}$ coefficients). What is more useful is it's generalization:

Definition 57.1.1: G-Modules

Let G be an arbitrary group, and M an abelian group. Then M is a G-*module* if there exists a [left] group-action on M. Equivalently, if there exists a $\mathbb{Z}[G]$-module structure on M.

The reader is free to parse through the technicalities of the group action. All $k[G]$-modules presented in chapter 53 are $\mathbb{Z}[G]$-modules with the action restricted. An example of a G-module that is not a representation is the action of the group $\text{Gal}_K(L)$ on the module L/K (a galois extension of L over K). All finite groups G have a non-trivial G-module given by the action of G on the group-ring $k[G]$ (note the addition is abelian). From a geometric perspective, modules represent sheaves (or in special cases vector bundles) over some schemes,

As we have seen many times at this point, the object acts on an abelian group can give lots of information on the object acting on it and the object being acted on. Given a G-module M, we can define the following:

Definition 57.1.2: G-Invariant subgroup

Let M be a G-module. Then the G-*invariant* set of M is the set:

$$M^G := \{m \in M \mid g \cdot m = m, \ \forall g \in G\}$$

which is the largest trivial G-submodule of M.

It is easy to see that $(-)^G$ is a functor. The following will help in computing cohomology

Corollary 57.1.3: Hom Representation of G-invariant

$$M^G \cong \text{Hom}_{\mathbb{Z}}(\mathbb{Z}, M^G) \cong \text{Hom}_G(\mathbb{Z}, M)$$

***Proof*:**
Give $\mathbb{Z}$ the trivial G-action, which is the action the statement requires.

For any abelian group N, evaluation at 1 is an isomorphism $\text{Hom}_{\mathbb{Z}}(\mathbb{Z}, N) \to N$, with inverse sending x to the map $m \mapsto mx$. Taking $N = M^G$ gives the first isomorphism.

For the second, let $\varphi\colon \mathbb{Z} \to M$ be $\mathbb{Z}[G]$-linear and put $m = \varphi(1)$. Equivariance gives $\varphi(g \cdot 1) = g\varphi(1)$, and $g \cdot 1 = 1$ because the action on $\mathbb{Z}$ is trivial, so $m = gm$ for every $g \in G$, that is $m \in M^G$. Conversely, any $m \in M^G$ defines a $\mathbb{Z}[G]$-linear map $\mathbb{Z} \to M$, $k \mapsto km$, since $g(km) = k(gm) = km$. These assignments are mutually inverse and additive, so $\mathrm{Hom}_G(\mathbb{Z}, M) \cong M^G$, completing the proof.

Similarly, we can define:

Definition 57.1.4: G-Coinvariant subgroup

Let M be a G-module. Then the G-*coinvariant* of M is the set:

$$M_G := \frac{M}{\langle (gm - m) \mid g \in G, m \in M \rangle}$$

and is the maximal invariant quotient module with the trivial action.

Just like $(-)^G$, $(-)_G$ is also a functor. The reader should further verify that

Proposition 57.1.5: Tensor Representation of Coinvariant Module

$$M_G \cong_G \mathbb{Z} \otimes_{\mathbb{Z}[G]} M$$

Proposition 57.1.6: Invariants Is Left-exact

The functor $(-)^G$ is left-exact, the functor $(-)_G$ is right exact.

***Proof*:**
First, prove that $(-)^G$ is right-adjoint to the trivial-action functor.

$$\mathrm{Hom}(T(M), N) \cong \mathrm{Hom}(M, N^G)$$

And similarly for $(-)_G$; it is left-adjoint to the trivial-action functor. As $T(-)$ is an exact functor, $(-)^G$ and $(-)_G$ are left-exact and right-exact respectively, completing the proof.

Both of these functors are not exact:

Example 57.1.7: Lack Of Exactness's
Let $G = C_2$ be the cyclic group of order 2 and:

$$0 \to I_G \to \mathbb{Z}[G] \xrightarrow{\epsilon} \mathbb{Z} \to 0$$

where $\epsilon(1 \cdot a + \sigma \cdot b) = a + b$ and I_G is the kernel. This sequence is exact. Take $(-)^G$:

$$0 \to (I_G)^G \to (\mathbb{Z}[G])^G \xrightarrow{\epsilon^G} (\mathbb{Z})^G \to 0$$

i.e.:

$$0 \to 0 \to \Delta \xrightarrow{\epsilon^G} \mathbb{Z} \to 0$$

where $\Delta = \{(a, a) \mid a \in \mathbb{Z}\}$. Then ϵ^G is no longer surjective as $\epsilon^G(a, a) = 2a$. The above example can also be used to show $(-)_G$ is not left-exact. More generally, if $0 \to M_2^G \to M_2 \to M_3 \to 0$ is exact, taking $(-)^G$ doesn't change $M - 2^G$, but now we get an isomorphism between the 1st and second terms, so it cannot be surjective.

More generally, let G act trivially on a coset $[m]$ of a quotient M/L. Can this action always be lifted so that G acts trivially on a representative m?

Thus, it is fruitful to take the ith right-derived functor of $(-)^G$ (similarly the left-derived functors of $(-)_G$):

$$\mathbf{R}^i(-)^G := H^i(G, -)$$

which would give us the long exact sequence of abelian groups:

$$0 \to L^G \to M^G \to N^G \to H^1(G, L) \to H^1(G, M) \to \cdots$$

Since $M^G = \mathrm{Hom}_G(\mathbb{Z}, M)$ by taking an injective resolution of M, or a projective resolution of $\mathbb{Z}$, we can write:

$$H^i(G, M) = \mathrm{Ext}^i_{\mathbb{Z}[G]}(\mathbb{Z}, M)$$

Note that $H^0(G, M) = M^G$, hence the higher order cohomology groups represent some failure for points to be fixed. Let us compute some examples to show the power of the machinery developed in chapter 56 before concretely demonstrating how the cohomology groups represent the obstruction of points being fixed.

Example 57.1.8: Group Cohomologies – Homological Algebra Computations

1. The trivial group has trivial cohomology
2. (Integers) Let $G = \mathbb{Z}$. Then $\mathbb{Z}[G] \cong \mathbb{Z}[x, x^{-1}]$. Then since $\mathbb{Z}[x, x^{-1}]/(1 - x) \cong_G \mathbb{Z}$ with the trivial action, define the following free (hence projective) resolution:

$$\cdots \to 0 \to \mathbb{Z}[x, x^{-1}] \xrightarrow{\cdot(1-x)} \mathbb{Z}[x, x^{-1}] \to 0 \to \cdots$$

   Applying the (contravariant) $\mathrm{Hom}_{\mathbb{Z}[x,x^{-1}]}(-, M)$, we can compute the cohomology of

$$\cdots \to 0 \to M \xrightarrow{\cdot(1-x)} M \to 0 \to \cdots$$

   Then we can conclude by direct computation:

$$H^0(\mathbb{Z}, M) \cong \ker(M \xrightarrow{\cdot(1-x)} M) = M^{\mathbb{Z}}$$

   and:

$$H^1(\mathbb{Z}, M) = \frac{M}{(1 - x)M}$$

   and furthermore $H^i(\mathbb{Z}, M) = 0$ for $i \notin \{0, 1\}$
3. Let $G = C_n$ be a finite cyclic group of over n with generator σ. Let us form a free resolution of $\mathbb{Z}[G]$ by taking advantage of the fact that $\sigma - 1$ and $N = \sum_{i=0}^{n-1} \sigma^i$ are zero divisors:

$$N \cdot (\sigma - 1) = (1 + \sigma + \cdots + \sigma^{n-1})(\sigma - 1) = \sigma^n - 1 = 1 - 1 = 0$$

This gives a resolution:

$$\cdots \xrightarrow{\sigma-1} \mathbb{Z}[G] \xrightarrow{N} \mathbb{Z}[G] \xrightarrow{\sigma-1} \mathbb{Z}[G] \xrightarrow{\epsilon}\!\!\to \mathbb{Z} \to 0$$

where

$$\epsilon\left(\sum_g a_g g\right) = \sum_g a_g$$

is the usual augmentation map. Then:

$$H^k(C_n, M) = \begin{cases} \frac{M^G}{NM} & k \text{ even, } k \geq 2 \\ \frac{{}_NM}{(\sigma-1)M} & k \text{ odd, } k \geq 1 \end{cases} \qquad \text{where} \qquad {}_NM = \{m \in M \mid N \cdot m = 0\}$$

For example, if $M = \mathbb{Z}$, then:

$$H^k(C_n, \mathbb{Z}) = \begin{cases} \mathbb{Z}/m\mathbb{Z} & k \text{ even, } k \geq 2 \\ 0 & \text{otherwise} \end{cases}$$

4. Let $G = *_{s\in S}\mathbb{Z}$ be the free product of S copies of $\mathbb{Z}$. This has a simple free resolution, namely take:

$$\epsilon : \mathbb{Z}[G] \to \mathbb{Z}$$

where the kernel is:

$$I_S = \sum_{s\in S} \mathbb{Z}[G] \cdot (s-1)$$

It should be verified that I_S is a free G-module, hence we have a free-resoluion:

$$0 \to I_S \to \mathbb{Z}[G] \xrightarrow{\epsilon}\!\!\to \mathbb{Z} \to 0$$

and cohomology:

$$H^k(*_{s\in S}\mathbb{Z}, \mathbb{Z}^G) = \begin{cases} \mathbb{Z} & k = 0 \\ \bigoplus_{s\in S} \mathbb{Z} & k = 1 \\ 0 & k \geq 2 \end{cases}$$

5. (If you know some algebraic geometry) *Galois Cohomology*: Take the category of varieties W over k such that when extending scalars to $\overline{k}$ we have $V/\overline{k} \cong W/\overline{k}$. Let's label it $\mathbf{Var}_{k\to\overline{k}}$, and say that the objects are:

$$\mathrm{Obj}(\mathbf{Var}_{k\to\overline{k}}) = \left\{W/k \;\middle|\; V/\overline{k} \cong W/\overline{k}\right\}$$

and the morphism are regular morphisms. Then I claim that:

$$\mathbf{Var}_{k\to\overline{k}} \cong H^1(\mathrm{Gal}_k(\overline{k}), \mathrm{Aut}(V/\overline{k}))$$

To see this, first note that an element $\sigma \in \mathrm{Gal}_k(\overline{k})$ acts on a $\overline{k}$-variety $W/\overline{k} = W_k \times_{\mathrm{Spec}\,(k)} \mathrm{Spec}\,(\overline{k})$ via:

$$\mathrm{id}_{W_k} \times \mathrm{Spec}\,(\sigma^{-1}) : W/\overline{k} \to W/\overline{k}$$

Or, for those less comfortable with fibered produced from algebraic geometry, they can think of this map as the "equivalent":

$$\mathrm{id}_{W_k} \otimes_k \sigma^{-1} : k[W] \otimes_k \overline{k}$$

With the action defined, let us look at the correspondence. Take $W_k \in \mathbf{Var}_{k\to\overline{k}}$ so that $\varphi : W_{\overline{k}} \xrightarrow{\sim} V_{\overline{k}}$ is an isomorphism. Then notice that this isomorphism induces a map $\varphi \circ \sigma^{-1} \circ \varphi^{-1}$, which is an automorphism on $W_{\overline{k}}$. The collection of all these maps as σ varies can uniquely define W_k (k is the fixed subfield over all σ). This is true up to a choice of $\mathrm{Aut}_k(V)$, and so we quotient out by it. Then we can see this creates the correspondence between these two categories, and shows how far away the category $\mathbf{Var}_{k\to\overline{k}}$ is from being invariant to the group-action of the galois group.

6. (if you know elliptic curves) An excellent example of where galois cohomology is immediately useful is in the theory of elliptic curves. On a high level, it is relatively easy to find the solution of an elliptic curve over $\mathbb{C}$, but it becomes much more difficult to solve it over finite extensions of $\mathbb{Q}$, or $\mathbb{Q}$ itself. If the reader has read (or is reading) [12], then the have encountered the short exact sequence of $\mathrm{Gal}_K(\overline{K})$-modules:

$$0 \to E[m] \to E(\overline{K}) \xrightarrow{[m]} E(\overline{K}) \to 0$$

where $E[m]$ are the m-torsion points of an elliptic curve ($m \geq 2$), K is a number field, $E(\overline{K})$ are the points of the elliptic curve over $\overline{K}$. Let $G = \mathrm{Gal}_K(\overline{K})$. Then when considering solutions over K, the short exact sequence sequence becomes a long exact sequences:

$$\begin{aligned} 0 \to E(K)[m] \to E(K) \xrightarrow{[m]} E(K) \xrightarrow{\delta} H^1(G, E[m]) \\ \to H^1(G, E(\overline{K})) \xrightarrow{[m]} H^1(G, E(\overline{K}) \to \cdots \end{aligned}$$

This is the foundation on which the torsion-free points of E/K will be computed!

7. If the reader knows about the classifying space BG, then:

$$H^n(BG, \mathbb{Z}) \cong H^n(G, \mathbb{Z})$$

where he left hand side is singular cohomology and the right hand side is group cohomology.

What obstruction information does this capture? Since there is a failure in surjectivity $\varphi : M \twoheadrightarrow N$ when applying $(-)^G$, this means that a fixed element $n \in N^G$ does *not* necessarily come form a fixed element $m \in M^G$. Thus, even though $\varphi(m) = n$ exists, it need not be the case that there exists an $m' \in M^G$ such that $\varphi^G(m') = n$. To measure this failure, take the short exact sequence

$$0 \to L \xrightarrow{\iota} M \xrightarrow{\varphi} N \to 0$$

and let $n \in N^G$. As $N^G \subseteq N$, there exists an $m \in M$ such that $\varphi(m) = n$. Consider any $g \in G$ and take $g \cdot m - m$. Then applying the G-homomorphism:

$$\varphi(g \cdot m - m) = g \cdot \varphi(m) - \varphi(m) = g \cdot n - n \stackrel{!}{=} n - n = 0$$

where $\stackrel{!}{=}$ comes from $n \in N^G$. By exactness $\ker \varphi = \mathrm{im}\iota$, so there must exist an element $\lambda_m(g) \in L$ such that

$$\iota(\lambda_m(g)) = g \cdot m - m$$

Now, $n \in N^G$ can be lifted if and only if $g \cdot m - m = 0$ for all $g \in G$, since then $m \in M^G$. In terms of elements of L:

$$\forall g \in G,\ \iota(\lambda_m(g)) = 0$$

by injectivity, it must be that $\lambda_m(g) = 0$ for all $g \in G$. Thus, the function $\lambda_m : G \to L$ is non-trivial if the lift does not exist. Let us explore the properties of λ_m. First, if $g_1, g_2 \in G$ then:

$$\begin{aligned}
\iota(\lambda_m(g_1g_2)) &= g_1g_2 \cdot m - m \\
&= g_1g_2 \cdot m - g_1 \cdot m + g_1 \cdot m - m \\
&= g_1 \cdot (g_2 \cdot m - m) + (g_1 \cdot m - m) \\
&= g_1\iota(\lambda_m(g_2)) + \iota(\lambda_m(g_1)) \\
&= \iota(g_1\lambda_m(g_2)) + \iota(\lambda_m(g_1)) \\
&= \iota(g_1\lambda_m(g_2) + \lambda_m(g_1))
\end{aligned}$$

Thus, as f is injective (more specifically a monomorphism) we get:

$$\lambda_m(g_1g_2) = \lambda_m(g_1) + g_1\lambda_m(g_2)$$

A function with this property is given a name:

Definition 57.1.9: 1-Cocycle

A function $f : G \to M$ is a called a *1-cocycle*, *1-derivation*, or *crossed homomorphism* if:

$$f(gh) = f(g) + gf(h)$$

The set of all derivations is denoted $Z^1(G, M)$. If it is clear from context, the "1-" is dropped.

Now, the above λ_m was dependent on a choice of m. But $M \hookrightarrow N$ is surjective, and hence there may be another m' such that $\varphi(m') = n$. As φ is a G-homomorphism

$$\varphi(m - m') = c - c = 0$$

and so $m - m' \in \ker\varphi = \operatorname{im}\iota$. Thus, there is an $\ell \in L$ such that

$$m - m' = \iota(\ell) \qquad \text{or} \qquad m' = m + \iota(\ell)$$

Thus, let us relate $\lambda_{m'}$ to λ_m:

$$\begin{aligned}
\iota(\lambda_{m'}(g)) &= g \cdot m' - m' \\
&= g \cdot (m + \iota(\ell)) - m - \iota(\ell) \\
&= g \cdot m + g \cdot \iota(\ell)) - m - \iota(\ell) \\
&= (g \cdot m - m) + g \cdot \iota(\ell) - \iota(\ell) \\
&= \iota(\lambda_m(g)) + \iota(g \cdot \ell) - \iota(\ell) \\
&= \iota(\lambda_m(g) + g \cdot \ell - \iota(\ell))
\end{aligned}$$

Thus, as ι is a monomorphism:

$$\lambda_{m'}(g) = \lambda_m(g) + g \cdot \ell - \ell$$

Thus, the term $g \cdot \ell - \ell$ can be seen as the "trivial" part of the obstruction, as it does not change the difficulty in finding an obstruction. This is given a name:

Definition 57.1.10: 1-Coboundary

A function $f_m : G \to M$ is a called a *1-coboundary* or *principal derivation* if:

$$f_m(g) = g \cdot m - m$$

The set of all boundary maps is denoted $B^1(G, M)$.If it is clear from context, the "1-" is dropped.

Note that a coboundary is a cocycle:

$$f_m(gh) = gh \cdot m - m = gh \cdot m - h \cdot m + h \cdot m - m = f_m(g) + gf_m(h)$$

Hence $B^1(G, M) \subseteq Z^1(G, M)$. It shall be shown in a moment that:

$$H^1(G, M) \cong \frac{Z^1(G, M)}{B^1(G, M)}$$

First, let us show there is a map $L^G \to H^1(G, M)$. Indeed there is a well-defined G-module homomorphism:

$$\ell \mapsto [g \mapsto g \cdot m - m]$$

where m is in the fiber of ℓ.

Let us next motivate the 2nd order cohomology group. Let us assume the cocycle-representation and the derived functor representation are isomorphic for the next part. Let us start with the long exact sequence we know is given:

$$0 \to L^G \xrightarrow{\alpha} M^G \xrightarrow{\beta} N^G \xrightarrow{\delta^0} H^1(G, L) \xrightarrow{\alpha_*} H^1(G, M) \xrightarrow{\beta_*} H^1(G, N) \xrightarrow{\delta^1} H^2(G, L) \cdots$$

Consider $[\nu] \in H^1(G, N)$; let us call it a 1-obstruction. Then can it be lifted some $[\mu] \in H^1(G, M)$, namely does there exist $[\mu]$ such that $\beta_*([\mu]) = [\nu]$. To find out, first take some representative cocycle $\nu : G \to N$ (notice we are now working with a map, not an element). As $M \hookrightarrow N$ is surjective, we have a diagram:

$$\begin{tikzcd} & M \arrow[d, "\beta"] \\ G \arrow[ur, dashed, "\mu?"] \arrow[r, "\nu"'] & N \end{tikzcd}$$

We can certainly lift each point *locally*, namely given a point $\nu(g) = n \in N$ there is a point (or points) m such that $\beta(m) = n$[1]. But this lifting can fail *globally*, namely such a lift is not guaranteed to be a 1-cocycle. This mirrors the general pattern in cohomology where having local data of a function does not imply the existence of a global function that restricts to all the local data. Let us measure this failure of lifting. If the lifted μ is a cocycle, it would satisfy:

$$\mu(g_1g_2) - \mu(g_1) - g_1\mu(g_2) = 0$$

Thus, define the function:

$$\psi : G \times G \to M, \qquad \psi(g_1, g_2) = \mu(g_1g_2) - \mu(g_1) - g_1\mu(g_2)$$

[1]Technically, the axiom of choice is invoked here for making a potentially uncountable number of choices

Let us show that for any $(g_1, g_2) \in G \times G$, $\psi(g_1, g_2) \in \ker \beta$. Indeed:

$$\begin{aligned}
\beta(\psi(g_1, g_2)) &= \beta(\mu(g_1 g_2) - \mu(g_1) - g_1\mu(g_2) \\
&= \beta(\mu(g_1 g_2)) - \beta(\mu(g_1)) - \beta(g_1\mu(g_2)) \\
&\overset{!}{=} \nu(g_1 g_2)) - \nu(g_1) - g_1\nu(g_2)) \\
&= 0 && \nu \text{ is a cocycle}
\end{aligned}$$

where $\overset{!}{=}$ comes from μ being defined as a lift of ν and hence $\beta \circ \mu = \nu$. Thus, by exactness $\ker \beta = \mathrm{im}\alpha$ and so for each pair $(g_1, g_2) \in G \times G$, there exists a unique element in L that maps to $\psi(g_1, g_2)$, namely there exists a function $\Omega : G \times G \to L$ such that

$$\alpha(\Omega(g_1, g_2)) = \psi(g_1, g_2)$$

This Ω would now be our new 2-cocycle and are denoted $Z^2(G, M)$. Doing a similar but longer computations such as the one for the 1-cycle, we can find that:

$$g_1\Omega(g_2, g_3) - \Omega(g_1 g_2, g_3) + \Omega(g_1, g_2 g_3) - \Omega(g_1, g_2) = 0$$

Similarly, an explicit computation would fine that 2-boundaries (given a 1-chain ψ) are of the form

$$\Omega(g_1, g_2) = g_1 \cdot \psi(g_2) - \psi(g_1 g_2) + \psi(g_1)$$

These are label $B^2(G, M)$ and:

$$H^2(G, M) \cong \frac{Z^2(G, M)}{B^2(G, M)}$$

The same process continues for higher order cohomology. For the reader who knows some algebraic topology, the following informal discussion grounds the higher order group-cohomology as the singular (or simplicial) cohomology of BG, the classifying space of G with the property that $\pi_1(BG) = G$ and $\pi_n(BG) = \{0\}$ for all $n > 1$ (see [38]). In particular:

$$H^n_{\mathrm{Sing}}(BG, M) \cong H^n(G, M)$$

Note that if M has a non-trivial G-module action, it corresponds to cohomology with local coefficients, but the geometric picture remains the same (the details are left for [13]). Using the nerve or bar construction on BG and considering the usual covering $EG \to BG$ we get the following correspondence:

Geometric Object in EG	Correspondence in BG	Algebraic equivalent
0-Simplices (Vertices)	A single 0-simplex, base point $*$.	elements of G
1-Simplices (Edges)	Directed edge for each element $g \in G$. An arrow from $*$ to $*$ labeled by g.	Functions on G
2-Simplices (Triangles)	Oriented triangle for each pair $(g_1, g_2) \in G \times G$. Edges are the 1-simplices for g_1, g_2, and the product $g_1 g_2$.	Functions on $G \times G$
3-Simplices (Tetrahedral)	Oriented tetrahedron for each triple $(g_1, g_2, g_3) \in G \times G \times G$. Faces are the triangles for (g_2, g_3), $(g_1, g_2 g_3)$, etc.	Functions on $G \times G \times G$

n-Simplices	n-tuple $(g_1, \ldots, g_n) \in G^n$.	Functions on G^n

Where "functions" here means set-function (to take into account the "failed" lifting), and the transition maps d_i captures the G-module homomorphism information. Geometrically, the G-module homomorphisms correspond to G-equivariant maps $f : G^n \to M$.

Using this intuition, we can construct a standard way of defining a projective resolution of $\mathbb{Z}$ by G-modules. First, let $\mathbb{Z}[G^n]$ be a G-module with the diagonal action given by multiplying on the left, or more specifically:

$$g \cdot (g_1, ..., g_n) = (gg_1, ..., gg_n)$$

Note that this diagonal map corresponds to the natural action of left multiplication by g on EG. These modules are free G-modules; clearly $\mathbb{Z}[G]$ is a $\mathbb{Z}$-module with basis given by the elements G. Then for $n \geq 0$:

$$\mathbb{Z}[G^{n+1}] \cong \bigoplus_{(g_1,...,g_n)\in G^n} \mathbb{Z}[G] \cdot (1, g_1, ..., g_n)$$

which shows that these are free. From this, we can consider the following projective (even free) resolution:

Definition 57.1.11: Standard Resolution

Let G be a group. Then the *standard resolution* of $\mathbb{Z}$ by G-modules is the exact sequence of G-module homomorphism:

$$\cdots \to \mathbb{Z}[G^{n+1}] \xrightarrow{d_n} \mathbb{Z}[G^n] \xrightarrow{d_{n-1}} \cdots \xrightarrow{d_1} \mathbb{Z}[G] \xrightarrow{\epsilon} \mathbb{Z} \to 0$$

where $\mathbb{Z}[G^{n+1}]$ carries the diagonal G-action and, on the basis G^{n+1},

$$d_n(g_0, ..., g_n) = \sum_{i=0}^{n} (-1)^i (g_0, ..., \hat{g}_i, ..., g_n)$$

the term $\hat{g}_i$ being omitted from the tuple, while $\epsilon(g) = 1$ for every $g \in G$. The map ϵ is also called the *augmentation map*.

Note that each $\mathbb{Z}[G^{n+1}]$ is a free $\mathbb{Z}[G]$-module, the diagonal action permuting a basis freely. We claimed this is an exact sequence, however this ought to be verified.

Lemma 57.1.12: Standard Resolution is Exact

The standard resolution of $\mathbb{Z}$ by G-modules is exact

This shows that the nomenclature of cocycles and coboundaries above is used correctly.

***Proof*:**
Exactness is a statement about the underlying abelian groups, so it is enough to produce a contracting homotopy by $\mathbb{Z}$-linear maps; these need not be G-equivariant, and indeed the ones below are not.

Step 1: The maps. Write $e \in G$ for the identity element and define

$$s_{-1}\colon \mathbb{Z} \to \mathbb{Z}[G], \quad 1 \mapsto (e), \qquad s_n\colon \mathbb{Z}[G^{n+1}] \to \mathbb{Z}[G^{n+2}], \quad (g_0, \dots, g_n) \mapsto (e, g_0, \dots, g_n)$$

extended $\mathbb{Z}$-linearly.

Step 2: The contracting identity in positive degrees. Fix $n \geq 1$ and compute $d_{n+1}s_n(g_0, \dots, g_n) = d_{n+1}(e, g_0, \dots, g_n)$. Omitting the entry in position 0 gives $(g_0, \dots, g_n)$ with sign $+1$; omitting the entry in position $j \geq 1$, which is g_{j-1}, gives $(e, g_0, \dots, \hat{g}_{j-1}, \dots, g_n)$ with sign $(-1)^j$. Reindexing by $i = j - 1$ and using $(-1)^{i+1} = -(-1)^i$,

$$d_{n+1}s_n(g_0, \dots, g_n) = (g_0, \dots, g_n) - \sum_{i=0}^{n} (-1)^i (e, g_0, \dots, \hat{g}_i, \dots, g_n)$$

The subtracted sum is exactly $s_{n-1}d_n(g_0, \dots, g_n)$, since s_{n-1} prepends e to each term of $d_n(g_0, \dots, g_n) = \sum_i (-1)^i (g_0, \dots, \hat{g}_i, \dots, g_n)$. Hence

$$d_{n+1}s_n + s_{n-1}d_n = \mathrm{id}$$

Step 3: The bottom of the complex. At the augmentation, $\epsilon s_{-1}(1) = \epsilon(e) = 1$, so $\epsilon s_{-1} = \mathrm{id}_{\mathbb{Z}}$. In degree zero,

$$d_1 s_0(g_0) = d_1(e, g_0) = (g_0) - (e), \qquad s_{-1}\epsilon(g_0) = s_{-1}(1) = (e)$$

so $d_1 s_0 + s_{-1}\epsilon = \mathrm{id}$ as well.

Step 4: Conclusion. The augmented complex therefore has its identity morphism null-homotopic as a complex of abelian groups. By Step 1 of the proof of lemma 56.1.23, a complex whose identity is null-homotopic is exact. Hence the standard resolution is exact, completing the proof.

Hence, after taking the contravariant functor $\mathrm{Hom}_{\mathbb{Z}[G]}(-, M)$, we get the cochain complex:

$$0 \to \mathrm{Hom}_{\mathbb{Z}[G]}(\mathbb{Z}[G], M) \to \cdots \to \mathrm{Hom}_{\mathbb{Z}[G^n]}(-, M) \xrightarrow{d_n^*} \mathrm{Hom}_{\mathbb{Z}[G^{n+1}]}(-, M) \to \cdots$$

where d_n^* maps $\varphi \mapsto \varphi \circ d_n$. Similarly, we can take the covariant functor

$$\cdots \to \mathbb{Z}[G^3] \otimes_{\mathbb{Z}[G]} A \xrightarrow{d_{2*}} \mathbb{Z}[G^2] \otimes_{\mathbb{Z}[G]} A \xrightarrow{d_{1*}} \mathbb{Z}[G] \otimes_{\mathbb{Z}[G]} A \to 0$$

where d_{n*} is given by $(g_0, ..., g_n) \otimes a \mapsto d_n(g_0, ..., g_n) \otimes a$, the map ϵ is the augmentation map given by $\sum a_g g \mapsto \sum a_g$, and we view the modules as *right* $\mathbb{Z}[G]$-modules with the action

$$(g_1, ..., g_n) \cdot g = (g_1 g, ..., g_n g)$$

Let us now better understand these hom-sets and relate to the above discussion. Let

Definition 57.1.13: coChain

Let

$$C^n(G,M) = (\mathrm{Hom}_{\mathbf{Set}}(G^n, M), +)$$

with group operation defined on the maps given by point-wise addition. This group is called the *n-cochain group*.

From this, we can define a boundary map:

$$\begin{aligned} d^n : C^n(G,M) &\to C^{n+1}(G,M) \\ d^n(f)(g_0,...,g_n) &= g_0 f(g_1,...,g_n) - f(g_0g_1, g_2, ..., g_n) \\ &\quad + f(g_0, g_1g_2, ..., g_n) - \cdots + (-1)^n f(g_0, ..., g_{n-2}, g_{n-1}g_n) + (-1)^{n+1} f(g_0, ..., g_{n-1}) \end{aligned}$$

Then $d^{n+1} \circ d^n = 0$, giving us a cochain complex. We can then have:

Definition 57.1.14: Cohomology Group – using Cocycles and Coboundaries

$$H^n(G,M) = \frac{\ker d^n}{\mathrm{im} d^{n-1}} = \frac{Z^n(G,M)}{B^n(G,M)}$$

We sum up many of the results that we've shown before using the new notation:

1. $C^0(G,M) = M$, and $H^0(G,M) \cong M^G$
2. $d^0 : C^0(G,M) \to C^1(G,M)$ is given by $d^0(a)(g) = ga - a$
3. $H^0(G,M) \cong \ker d^0 = M^G$
4. $\mathrm{im} d^0 = \{f : G \to M \mid \exists a \in M \text{ such that } f(g) = ga - a,\ \forall g \in G\}$ (principal crossed homomorphism)
5. $d^1 : C^1(G,M) \to C^2(G,M)$ given by $d^1(f)(g,h) = gf(h) - f(gh) + f(g)$
6. $\ker d^1 = \{f : G \to M \mid f(gh) = f(g) + gf(h)\}$ (crossed homomorphisms)
7. If $0 \to A \to B \to C \to 0$ is exact, we get an exact sequence of cochains for every n:

$$0 \to C^n(G,A) \to C^n(G,B) \to C^n(G,C) \to 0$$

We shall show that the collection of complexes given by the above is isomorphic to those given by the standard resolution

Proposition 57.1.15: Characterizing Hom of Resolution

Let M be a G-module. Then for every $n \geq 0$ there is an isomorphism:

$$\Phi^n : \mathrm{Hom}_{\mathbb{Z}[G]}(\mathbb{Z}[G^{n+1}], M) \xrightarrow{\cong} C^n(G, M)$$

where the map $\varphi : \mathbb{Z}[G^{n+1}] \to M$ goes to $f : G^n \to M$ given by:

$$f(g_1, ..., g_n) = \varphi(1, g_1, g_1g_2, g_1g_2g_3, ..., g_1g_2g_3 \cdots g_n)$$

The isomorphism Φ^n commutes with boundary maps, that is

$$\Phi^{n+1} \circ d^*_{n+1} = d^n \circ \Phi^n$$

Hence the two cochain complexes agree: the one built from the standard resolution and the one built from cocycles and coboundaries are isomorphic, so the two descriptions of $H^n(G, M)$ coincide.

***Proof*:**
Step 1: Φ^n is a bijection. The group G acts diagonally on the basis G^{n+1} of $\mathbb{Z}[G^{n+1}]$, and this action is free, since $g \cdot (h_0, \ldots, h_n) = (h_0, \ldots, h_n)$ forces $gh_0 = h_0$ and hence $g = e$. Each orbit contains exactly one tuple whose first entry is e, namely $h_0^{-1} \cdot (h_0, \ldots, h_n)$. A $\mathbb{Z}[G]$-linear map out of $\mathbb{Z}[G^{n+1}]$ is therefore determined freely and without constraint by its values on the tuples $(e, k_1, \ldots, k_n)$, so restriction to those tuples is a bijection onto $\mathrm{Hom}_{\mathbf{Set}}(G^n, M) = C^n(G, M)$.

The substitution in the statement is the reparametrization $k_i = g_1g_2 \cdots g_i$ of that indexing. It is a bijection $G^n \to G^n$, with inverse $g_1 = k_1$ and $g_i = k_{i-1}^{-1}k_i$ for $i \geq 2$. Composing the two bijections gives Φ^n, and it is additive because both steps are, so Φ^n is an isomorphism of abelian groups.

Step 2: Compatibility with the differentials. Let $\varphi \in \mathrm{Hom}_{\mathbb{Z}[G]}(\mathbb{Z}[G^{n+1}], M)$ and set $f = \Phi^n(\varphi)$. Write $h_i = g_1 \cdots g_i$, so that $f(g_1, \ldots, g_n) = \varphi(e, h_1, \ldots, h_n)$. Applying Φ^{n+1} to $d^*_{n+1}\varphi = \varphi \circ d_{n+1}$ and evaluating at $(g_0, \ldots, g_n)$ gives $\varphi\big(d_{n+1}(e, k_1, \ldots, k_{n+1})\big)$ where $k_i = g_0g_1 \cdots g_{i-1}$.

Expand d_{n+1} by definition 57.1.11. The term omitting position 0 is $(k_1, \ldots, k_{n+1})$; pulling out the common left factor $k_1 = g_0$ by equivariance of φ turns it into $g_0 \cdot \varphi(e, g_0^{-1}k_2, \ldots) = g_0 f(g_1, \ldots, g_n)$. The term omitting position j for $1 \leq j \leq n$ merges g_{j-1} and g_j, contributing $(-1)^j f(g_0, \ldots, g_{j-1}g_j, \ldots, g_n)$, and the term omitting the last position drops g_n, contributing $(-1)^{n+1} f(g_0, \ldots, g_{n-1})$. Summing,

$$\Phi^{n+1}(d^*_{n+1}\varphi)(g_0, \ldots, g_n) = g_0 f(g_1, \ldots, g_n) + \sum_{j=1}^{n} (-1)^j f(g_0, \ldots, g_{j-1}g_j, \ldots, g_n) + (-1)^{n+1} f(g_0, \ldots, g_{n-1})$$

which is precisely $d^n(f)(g_0, \ldots, g_n)$ as defined above. Hence $\Phi^{n+1} \circ d^*_{n+1} = d^n \circ \Phi^n$.

The maps $\Phi^\bullet$ are therefore isomorphisms commuting with the differentials, that is an isomorphism of cochain complexes, completing the proof.

Due to the relevance of the standard representation, we have the following definition:

Definition 57.1.16: Induced and Coinduced G-module

Let G be a group and A an abelian group. Then:

$$\mathrm{CoInd}^G(A) = \mathrm{Hom}_{\mathbb{Z}}(\mathbb{Z}[G], A)$$

with the G-action defined as $(g\varphi)(z) = \varphi(zg)$ is called the *coinduced G-module* associated to A. Similarly

$$\mathrm{Ind}^G(A) = \mathbb{Z}[G] \otimes_{\mathbb{Z}} A$$

with G-action defined by $g(z \otimes a) = (gz) \otimes a$ is the is the induced G-module associated to A.

A result that is common in many Cohomologies theories is some version of the following:

Lemma 57.1.17: Induced and Coinduced Equivalence

Let G be a finite group and A an abelian group. Then

$$\mathrm{Ind}^G(A) \cong \mathrm{CoInd}^G(A)$$

As a consequence, if $B = \mathrm{Ind}^G(A)$ or $B = \mathrm{CoInd}^G(A)$, then

$$H_0(G, B) \cong H^0(G, B) \cong A$$

and they are both zero for $n \geq 1$.

***Proof*:**
Define the map: $\alpha : \mathrm{Coind}^G(A) \to \mathrm{Ind}(G)(A)$ given by:

$$\varphi \mapsto \sum_{g \in G} g^{-1} \otimes \varphi(g)$$

$$(g^{-1} \mapsto a) \leftarrow\!\shortmid g \otimes a$$

with $(g^{-1} \mapsto a)(h) = 0$ for $h \in G \setminus \{g^{-1}\}$. These are certainly inverses, and easily verifiable to be G-module homomorphisms (this is a similar technique used in Maschkle's Theorem (see theorem 53.1.9).

For the last part, write $B = \mathrm{CoInd}^G(A) = \mathrm{Hom}_{\mathbb{Z}}(\mathbb{Z}[G], A)$, which the first part identifies with $\mathrm{Ind}^G(A)$. Let $P_\bullet \to \mathbb{Z}$ be a resolution by free $\mathbb{Z}[G]$-modules; each P_i is then also free as an abelian group. The adjunction

$$\mathrm{Hom}_{\mathbb{Z}[G]}\big(M, \mathrm{Hom}_{\mathbb{Z}}(\mathbb{Z}[G], A)\big) \cong \mathrm{Hom}_{\mathbb{Z}}(M, A)$$

of theorem 15.3.9, natural in M, turns the complex computing $H^\bullet(G, B)$ into $\mathrm{Hom}_{\mathbb{Z}}(P_\bullet, A)$. Its cohomology is $\mathrm{Ext}^\bullet_{\mathbb{Z}}(\mathbb{Z}, A)$, which is A in degree 0 and vanishes in positive degrees because $\mathbb{Z}$ is projective over $\mathbb{Z}$. Dually $P_\bullet \otimes_{\mathbb{Z}[G]} (\mathbb{Z}[G] \otimes_{\mathbb{Z}} A) \cong P_\bullet \otimes_{\mathbb{Z}} A$ computes $\mathrm{Tor}^{\mathbb{Z}}_\bullet(\mathbb{Z}, A)$, which is A in degree 0 and zero above. So $H^0(G, B) \cong H_0(G, B) \cong A$ and both vanish for $n \geq 1$, as we sought to show.

Proposition 57.1.18: Torsion of H^1

if G has order n and is finite, then every element of $H^1(G, M)$ has order divisible by n.

***Proof*:**
Let $\delta \in Z^1$ so that $\delta(gh) = \delta(g) + g\delta(h)$ for all $g \in G$. Sum both sides over $h \in G$ keeping g fixed. Then as multiplying by g is translation:

$$\sum_{h\in G} \delta(gh) = \sum_{h\in G} \delta(h) = S$$

As $\delta(g)$ is a constant, we can re-arrange and get:

$$(1-g)S = |G|\delta(g)$$

Now, re-write it suggestively as:

$$g(-S) - (-S) = |G|\delta(g)$$

But now the left hand side is a coboundary, and hence the equation is 0 in $H^1(G, M)$ showing that the order of the group zero's any cocycle, completing the proof.

Proposition 57.1.19: M finitely generated, then H1 finite

Let M be a finitely generated G-module and G finite. Then $H^1(G, M)$ is finite.

***Proof*:**
Let $\langle m_1, ..., m_r\rangle = M$. Then C^1 is finitely generated as a $\mathbb{Z}$-module (note that G had to be finite). Then $Z^1(G, M)$ (as $\mathbb{Z}$-submodule) is finitely generated. But now by proposition 57.1.19, every element has order divisible by $n = |G|$. Thus, take the lcm of the order of the generators, giving a bound of the order of H^1, completing the proof.

Finally, let us prove the foreshadowing in box 20.4.5. Let L/K be a cyclic extension so that $G = \mathrm{Gal}_K(L) \cong \mathbb{Z}/m\mathbb{Z}$. The G acts on $L^\times$ in the natural way, and hence we find it's group cohomology. First:

Lemma 57.1.20: Cocycles using Norm and Trace

Let L/K be a cyclic extension $G = \mathrm{Gal}_K(L) \cong \mathbb{Z}/m\mathbb{Z}$ with generator σ, and $Z^1(G, L^\times)$ the 1-cocycles. Let $U_{L/K} = \left\{ l \in L \;\middle|\; \mathrm{Nm}_{L/K}(l) = 1 \right\}$ (sometimes this is denoted L^1). Then as groups:

$$\begin{gathered} Z^1(G, L^\times) \cong U_{L/K} \\ c \mapsto c(\sigma) \end{gathered}$$

Similarly:

$$Z^1(G, L) \cong T_{L/K}$$

where $T_{L/K}$ is the traceless elements[a]

[a] There is less common notation, sometimes it is denoted L^0

***Proof*:**
Let us first show the map is well-defined. Note first that a cocycle is determined by where it maps σ, namely for an arbitrary $c \in Z^1(G, L^\times)$:

$$\begin{aligned} c(\sigma^2) &= c(\sigma) \cdot \sigma(c(\sigma)) \\ c(\sigma^3) &= c(\sigma) \cdot \sigma(c(\sigma^2)) \\ &\vdots \end{aligned}$$

Next, note that $c(\sigma^m) = c(e) = 1$ since:

$$c(e) = c(e \cdot e) = c(e) \cdot e(\sigma(e)) = c(e) \cdot c(e)$$

as these map to $L^\times$, all elements are invertible, hence:

$$c(e) = 1 \in L^\times$$

Then:

$$1 = c(\sigma^n) = c(\sigma) \cdot \sigma(c(\sigma)) \cdot \sigma^2(c(\sigma)) \cdots \sigma^{n-1}(c(\sigma))$$

Letting $\beta = c(\sigma)$:

$$1 = c(\sigma^n) = \beta \cdot \sigma(\beta) \cdot \sigma^2(\beta) \cdots \sigma^{n-1}(\beta) = \prod_{g \in G} g(\beta) = \mathrm{Nm}_{L/K}(\beta)$$

showing the map is well-defined. As it is determined by where σ maps to, if $c_1, c_2 \in Z^1(G, L^\times)$ and $c_1(\sigma) = c_2(\sigma)$ then they agree on each σ^k and hence $c_1 = c_2$, showing injectivity. Surjectivity can be shown by defining $c(\sigma) = \beta$ for $\beta \in U_{L/K}$ and showing it satisfies the cocycle condition, which will be left as an exercise.

A similar argument can be done using the trace.

Theorem 57.1.21: Hilbert's 90th Theorem – Cohomology Upgrade

Using the above lemma's notation:

$$H^1(G, L^\times) \cong \{1\}$$

***Proof*:**
By Hilbert's 90th theorem (theorem 20.4.4), every norm 1 element is of the form:

$$\frac{\alpha}{\sigma^k(\alpha)}$$

But this is a coboundary element, hence all cocycles are coboundaries, completing the proof.

Note this theorem generalizes to arbitrary finite Galois extensions; see [13]. This result should now make intuitive sense: the elements of K are *exactly* the $\mathrm{Gal}_K(L)$-invariant element. Note that if there were more complicated groups instead of $L^\times$, there can be more interesting cohomology groups, see [13] for more details.

Exercise 57.1.1

1. Let A, B be any G-modules. Show that $H^n(G, A \oplus B) \cong H^n(G, A) \oplus H^n(G, B)$
2. Show that $H^0(G, \mathrm{CoInd}^G(A)) \cong A$ and $H^n(G, \mathrm{CoInd}^G(A)) \cong 0$ for all $n \geq 1$ (hint: consider $\zeta : \mathrm{Hom}_{\mathbb{Z}[G]}(\mathbb{Z}[G^n], \mathrm{CoInd}^G(A)) \to \mathrm{Hom}_{\mathbb{Z}}(\mathbb{Z}[G^n], A)$ given by $\varphi \mapsto (z \mapsto \varphi(z)(1))$. What is the inverse of this map?)
3. Show that $H_0(G, \mathrm{Ind}^G(A)) \cong A$ and $H_n(G, \mathrm{Ind}^G(A)) \cong 0$ for all $n \geq 1$.

57.2 Homological Dimension

We now turn to measuring the "complexity" of modules and rings using the length of their resolutions. Before defining these measures rigorously, it is worth stepping back to 1890 to understand why mathematicians began counting the length of resolutions in the first place.

Historical Motivation: Invariant Theory

In the late 19th century, algebra was dominated by *Invariant Theory*. Imagine a group G acting on a polynomial ring $S = \mathbb{C}[x_1, \ldots, x_n]$ (for example, by linear substitutions of variables). Mathematicians were looking for *invariant ring* S^G:

$$S^G = \{f \in S \,|\, g \cdot f = f \text{ for all } g \in G\}$$

The central question was if S^G finitely generated? Can we find a finite set of basic invariant polynomials $f_1, \ldots, f_m$ such that every other invariant is just a polynomial combination of these?

Hilbert showed we can map a polynomial ring onto the invariants:

$$\varphi : \mathbb{C}[y_1, \ldots, y_m] \twoheadrightarrow S^G \qquad y_i \mapsto f_i$$

and then the kernel $I = \ker(\varphi)$ represents the *relations* between the invariants. Hence, the study of invariants reduced to the study of the kernel. To find these invariants, 19th-century computationalists (like Paul Gordan) asked tried computing them and found the following problem:

1. **First Syzygies:** The ideal I describes the relations. By the Basis Theorem, I is finitely generated.
2. **Second Syzygies:** Are these relations independent? Or are there "relations between the relations"? If we have a relation $Ar_1 + Br_2 = 0$, that is a second syzygy.
3. **Higher Syzygies:** We can keep going. We can form a module of second syzygies and ask for its generators, then ask for the relations between *those* generators.

Mathematicians feared this chain of relations might continue forever, becoming infinitely complex. Hilbert's *Syzygy Theorem* proved that if you are working with a polynomial ring in n variables, this process *must stop*. Specifically, after at most n steps, the module of relations becomes free—there are no more "hidden" relations.

This was part of David Hilbert's seminal paper laying the foundation of commutative geometry. In particular, he proved three theorems: The *Basis Theorem*, the *Nullstellensatz*, and the *Syzygy Theorem*.

57.2.1 Etymology The word *Syzygy* comes from the Greek *syzygia*, meaning "yoked together". In astronomy, it refers to the alignment of three celestial bodies (like the Sun, Earth, and Moon). In algebra, it refers to a relation that "yokes" generators together to zero.

57.2.1 Projective and Global Dimension

We can now formalize Hilbert's discovery using the language of resolutions.

Definition 57.2.2: Projective Dimension

Let M be an R-module. The *projective dimension* of M, denoted $\mathrm{pd}_R(M)$, is the minimal length n of a projective resolution of M:

$$0 \to P_n \to \cdots \to P_1 \to P_0 \to M \to 0$$

If no finite resolution exists, we say $\mathrm{pd}_R(M) = \infty$.

Definition 57.2.3: Global Dimension

The *global dimension* of a ring R, denoted $\mathrm{gldim}(R)$, is the supremum of the projective dimensions of all R-modules:

$$\mathrm{gldim}(R) = \sup\{\mathrm{pd}_R(M) \big| M \in R\textbf{-Mod}\}$$

Theorem 57.2.4: Global Dimension of Polynomial Rings

Let R be a commutative ring. Then

$$\operatorname{gldim}(R[x]) = \operatorname{gldim}(R) + 1$$

***Proof*:**
We will prove this by establishing both the upper bound $\operatorname{gldim}(R[x]) \leq \operatorname{gldim}(R) + 1$ and the lower bound $\operatorname{gldim}(R[x]) \geq \operatorname{gldim}(R) + 1$. If $\operatorname{gldim}(R) = \infty$, the equality trivially holds, so we may assume $\operatorname{gldim}(R) = n < \infty$.

Step 1: The Characteristic Exact Sequence Let M be any $R[x]$-module. We can view M as an R-module by restricting the scalars. We can then construct the extended $R[x]$-module $R[x] \otimes_R M$.

Define a sequence of $R[x]$-modules:

$$0 \to R[x] \otimes_R M \xrightarrow{\alpha} R[x] \otimes_R M \xrightarrow{\beta} M \to 0$$

where the maps are defined by:

- $\alpha(f(x) \otimes m) = xf(x) \otimes m - f(x) \otimes xm$
- $\beta(f(x) \otimes m) = f(x)m$

We must prove this sequence is exact.

- *Surjectivity of β:* For any $m \in M$, $\beta(1 \otimes m) = m$, so β is surjective.
- *Exactness at the middle:* Clearly $\beta(\alpha(1 \otimes m)) = \beta(x \otimes m - 1 \otimes xm) = xm - xm = 0$, so $\operatorname{im}(\alpha) \subseteq \ker(\beta)$. Conversely, any element in $R[x] \otimes_R M$ can be written as $\sum_{i=0}^{d} x^i \otimes m_i$. If it lies in $\ker(\beta)$, then $\sum_{i=0}^{d} x^i m_i = 0$. We can rewrite our original element as:

  $$\sum_{i=0}^{d} x^i \otimes m_i = \sum_{i=0}^{d} (x^i \otimes m_i - 1 \otimes x^i m_i)$$

  Notice that $x^i \otimes m - 1 \otimes x^i m = \alpha\left(\sum_{j=0}^{i-1} x^{i-1-j} \otimes x^j m\right)$. Thus, the element is in the image of α.
- *Injectivity of α:* Suppose $\sum_{i=0}^{d} x^i \otimes m_i \in \ker(\alpha)$. Then:

  $$\sum_{i=0}^{d} x^{i+1} \otimes m_i - \sum_{i=0}^{d} x^i \otimes xm_i = 0$$

  Because $R[x]$ is free over R, we can compare the coefficients of $x^k \otimes -$ degree by degree. For degree 0, we have $-1 \otimes xm_0 = 0 \implies xm_0 = 0$. For degree $k > 0$, we have $m_{k-1} - xm_k = 0 \implies m_{k-1} = xm_k$. For degree $d+1$, we have $m_d = 0$. Working backward from $m_d = 0$, we get $m_{d-1} = xm_d = 0$, and by induction, $m_i = 0$ for all i. Hence, α is strictly injective.

Step 2: The Upper Bound
Let M be an $R[x]$-module. View M as an R-module, and take a projective resolution $P_\bullet \to M \to 0$ over R of length at most n (since $\text{gldim}(R) = n$).

Because $R[x]$ is free (and thus flat) over R, tensoring the resolution with $R[x]$ preserves exactness. Therefore, $R[x] \otimes_R P_\bullet$ is a projective resolution of $R[x] \otimes_R M$ over $R[x]$. This proves that:

$$\text{pd}_{R[x]}(R[x] \otimes_R M) \leq \text{pd}_R(M) \leq n$$

Now, apply the functor $\text{Ext}^*_{R[x]}(-, N)$ for an arbitrary $R[x]$-module N to our characteristic exact sequence. We obtain the long exact sequence:

$$\cdots \to \text{Ext}^i_{R[x]}(R[x] \otimes_R M, N) \to \text{Ext}^{i+1}_{R[x]}(M, N) \to \text{Ext}^{i+1}_{R[x]}(R[x] \otimes_R M, N) \to \cdots$$

Since $\text{pd}_{R[x]}(R[x] \otimes_R M) \leq n$, the Ext groups vanish for all indices $\geq n+1$. Therefore, setting $i = n+1$, we get:

$$0 \to \text{Ext}^{n+2}_{R[x]}(M, N) \to 0$$

Since $\text{Ext}^{n+2}_{R[x]}(M, N) = 0$ for all modules N, we conclude $\text{pd}_{R[x]}(M) \leq n+1$. Because this holds for all $R[x]$-modules, $\text{gldim}(R[x]) \leq n+1$.

Step 3: The Lower Bound
Since $\text{gldim}(R) = n$, there exists an R-module M and an R-module K such that $\text{Ext}^n_R(M, K) \neq 0$. We can elevate both M and K to the status of $R[x]$-modules by declaring that x acts trivially on them (i.e., $xM = 0$ and $xK = 0$).

Apply the characteristic exact sequence to M (viewed as this specific $R[x]$-module) and take the long exact sequence for $\text{Hom}_{R[x]}(-, K)$. By the Tensor-Hom Adjunction, there is a natural isomorphism:

$$\text{Hom}_{R[x]}(R[x] \otimes_R P_\bullet, K) \cong \text{Hom}_R(P_\bullet, K)$$

Because $R[x] \otimes_R P_\bullet$ resolves $R[x] \otimes_R M$ over $R[x]$, taking the cohomology of both sides gives an isomorphism of the derived functors:

$$\text{Ext}^i_{R[x]}(R[x] \otimes_R M, K) \cong \text{Ext}^i_R(M, K) \quad \text{for all } i \geq 0$$

Let us examine the connecting map $\alpha^* : \text{Ext}^i_{R[x]}(R[x] \otimes_R M, K) \to \text{Ext}^i_{R[x]}(R[x] \otimes_R M, K)$ induced by α. On the level of the 0-th Hom group, for any $R[x]$-linear map $f : R[x] \otimes_R M \to K$:

$$(\alpha^* f)(1 \otimes m) = f(\alpha(1 \otimes m)) = f(x \otimes m - 1 \otimes xm)$$

Since x acts trivially on M, $1 \otimes xm = 0$. By $R[x]$-linearity, $f(x \otimes m) = x \cdot f(1 \otimes m)$. Because x also acts trivially on the codomain K, we get $x \cdot f(1 \otimes m) = 0$. Thus, the map α^* is identically the zero map on the entire cochain complex, and consequently, it is the zero map on the Ext groups.

Because the maps α^* are zero, our long exact sequence splits into short exact sequences for each i:

$$0 \to \text{Ext}^i_R(M, K) \to \text{Ext}^{i+1}_{R[x]}(M, K) \to \text{Ext}^{i+1}_R(M, K) \to 0$$

Setting $i = n$, we get:

$$0 \to \text{Ext}^n_R(M, K) \to \text{Ext}^{n+1}_{R[x]}(M, K) \to \text{Ext}^{n+1}_R(M, K) \to 0$$

Since $\text{Ext}^n_R(M, K) \neq 0$, it directly follows that $\text{Ext}^{n+1}_{R[x]}(M, K) \neq 0$. This proves that $\text{pd}_{R[x]}(M) \geq n+1$.

Combining the upper and lower bounds gives $\text{gldim}(R[x]) = \text{gldim}(R) + 1$, completing the proof.

Hilbert's theorem is effectively the calculation of the global dimension of polynomial rings.

Theorem 57.2.5: Hilbert's Syzygy Theorem

Let k be a field and $R = k[x_1, \dots, x_n]$ be the polynomial ring in n variables. Then every finitely generated R-module has finite projective dimension, and:

$$\operatorname{gldim}(R) = n$$

***Proof*:**
While Hilbert's original proof was highly computational, the modern approach is much cleaner. We must show two things: that the global dimension is at least n, and that it is at most n.

First, we show $\operatorname{gldim}(R) \geq n$ by finding a module with projective dimension exactly n. This module is the residue field $k \cong R/(x_1, \dots, x_n)$. We can explicitly construct a free resolution for k called the *Koszul Complex*. Let V be a k-vector space of dimension n with basis $e_1, \dots, e_n$. The resolution is given by the exterior algebra $\bigwedge^\bullet V \otimes_k R$:

$$0 \to \bigwedge^n V \otimes R \xrightarrow{d_n} \dots \xrightarrow{d_2} \bigwedge^1 V \otimes R \xrightarrow{d_1} R \xrightarrow{\epsilon} k \to 0$$

where the differential is given by $d_1(e_i \otimes 1) = x_i$. Because the variables $x_1, \dots, x_n$ form a regular sequence, this complex is exact. It has length exactly n, so $\operatorname{pd}_R(k) = n$.

Second, to show that *no* module has a resolution longer than n, we proceed by induction on n. For $n = 0$, $R = k$ is a field; every module is free, so the dimension is 0. For $n = 1$, $R = k[x]$ is a PID. Submodules of free modules are free, so any resolution stops at step 1. For the inductive step, we write $R = S[x_n]$ where $S = k[x_1, \dots, x_{n-1}]$.

By theorem 57.2.4 ($\operatorname{gldim}(S[x]) = \operatorname{gldim}(S) + 1$) and our inductive hypothesis, $\operatorname{gldim}(S) = n - 1$, which forces $\operatorname{gldim}(R) \leq n$, completing the proof.

This theorem has implications beyond just counting relations:

1. *Degree of the Hilbert Polynomial:* Hilbert originally used this theorem to prove that the Hilbert Function $H_M(t) = \dim_k(M_t)$ of a graded module eventually becomes a polynomial. The finite resolution allows us to calculate $H_M(t)$ as an alternating sum of the Hilbert functions of free modules, which are essentially binomial coefficients $\binom{t+k}{k}$.

2. *Structure of Projective Modules:* A celebrated problem posed by Serre (related to this theorem) asked if every projective module over $k[x_1, \dots, x_n]$ is free. This is the algebraic analogue of "vector bundles over affine space are trivial". This was solved affirmatively in 1976 (the *Quillen-Suslin Theorem*). This implies that over polynomial rings, "projective resolution" and "free resolution" are synonymous.

57.2.2 Dimension of Local Rings

Computing the global dimension for general rings seems daunting, as one must check *every* module. However, we can simplify this task using derived functors.

Proposition 57.2.6: Detecting Projective Dimension with Tor

Let R be a ring and M an R-module. The following are equivalent:

1. $\mathrm{pd}_R(M) \leq n$
2. $\mathrm{Tor}_i^R(M, N) = 0$ for all R-modules N and all $i > n$.
3. $\mathrm{Tor}_{n+1}^R(M, N) = 0$ for all R-modules N.

***Proof*:**
(1) $\Rightarrow$ (2): If $\mathrm{pd}_R(M) \leq n$, there is a resolution $P_\bullet$ of length n. Computing Tor using this resolution involves terms P_i which are zero for $i > n$, hence the homology vanishes. (2) $\Rightarrow$ (3): Trivial. (3) $\Rightarrow$ (1): Let $\cdots \to P_1 \to P_0 \to M \to 0$ be a projective resolution. Let $K_n = \ker(P_{n-1} \to P_{n-2})$. A standard dimension shifting argument (using the long exact sequence of Tor) shows that for any N:

$$\mathrm{Tor}_1^R(K_n, N) \cong \mathrm{Tor}_{n+1}^R(M, N)$$

By assumption (3), $\mathrm{Tor}_1^R(K_n, N) = 0$ for all N. This implies K_n is flat (and if R is Noetherian and M finitely generated, K_n is projective). Thus the resolution can stop at K_n, giving a length n projective resolution $0 \to K_n \to P_{n-1} \cdots \to M \to 0$.

It is more common that we consider rings that already have certain properties and verify how these conditions affect the global dimension. For example, given a local ring, the work is significantly reduced. In the local case, we can verify condition (3) by checking a *single* module: the residue field.

Lemma 57.2.7: Projective Dimension over Local Rings

Let $(R, \mathfrak{m}, k)$ be a local Noetherian ring and M a finitely generated R-module. Then:

$$\mathrm{pd}_R(M) \leq n \iff \mathrm{Tor}_{n+1}^R(M, k) = 0$$

Consequently, $\mathrm{pd}_R(M)$ is the smallest n such that $\mathrm{Tor}_i^R(M, k) = 0$ for all $i > n$.

***Proof*:**
The forward direction is immediate from proposition 57.2.6. For the converse, let $\cdots \to P_1 \to P_0 \to M \to 0$ be a *minimal* free resolution (meaning the matrices of the maps have entries in $\mathfrak{m}$). Then the differentials in the complex $P_\bullet \otimes_R k$ are all zero maps. Thus:

$$\mathrm{Tor}_i^R(M, k) \cong H_i(P_\bullet \otimes k) \cong P_i \otimes k \cong k^{\beta_i}$$

where β_i is the rank of the free module P_i. If $\mathrm{Tor}_{n+1}^R(M, k) = 0$, then $P_{n+1} \otimes k = 0$. By Nakayama's Lemma, this implies $P_{n+1} = 0$, so the resolution stops at n.

This leads to the fundamental equality attributed to Auslander, Buchsbaum, and Serre, which connects the homological algebra back to the geometry of the ring.

Theorem 57.2.8: Auslander-Buchsbaum-Serre Theorem

Let $(R, \mathfrak{m}, k)$ be a local Noetherian ring. Then the global dimension is determined by the residue field:

$$\operatorname{gldim}(R) = \operatorname{pd}_R(k)$$

Furthermore, $\operatorname{gldim}(R)$ is finite if and only if R is a *regular local ring*, that is, if and only if $\mathfrak{m}$ can be generated by $\dim R$ elements ([10, Regular Local Ring]).

This theorem is the bridge between algebra and geometry. Regularity is a condition on the number of generators of $\mathfrak{m}$, equivalently on the associated graded ring ([10, Characterizations of a Regular Local Ring]), while global dimension is a homological condition on resolutions. Geometrically the regular local rings are the local rings at the nonsingular points of a variety, which is the reading [27] develops.

***Proof*:**
By definition, $\operatorname{gldim}(R) \geq \operatorname{pd}_R(k)$. We must show the reverse. Let $n = \operatorname{pd}_R(k)$. Let M be any finitely generated R-module. We check its projective dimension using lemma 57.2.7.

$$\operatorname{Tor}_{n+1}^R(M, k)$$

Since Tor is symmetric, $\operatorname{Tor}_{n+1}^R(M, k) \cong \operatorname{Tor}_{n+1}^R(k, M)$. Since $\operatorname{pd}_R(k) = n$, by proposition 57.2.6, $\operatorname{Tor}_{n+1}^R(k, -)$ is the zero functor. Thus $\operatorname{Tor}_{n+1}^R(M, k) = 0$. By lemma 57.2.7, this implies $\operatorname{pd}_R(M) \leq n$. Since this holds for all finitely generated M, $\operatorname{gldim}(R) = n$ (a standard limit argument extends this to non-finitely generated modules).

57.2.3 Beyond Regularity: The Hierarchy of Singularities

The Auslander-Buchsbaum-Serre theorem (theorem 57.2.8) presents a stark dichotomy: a local ring either has finite global dimension (and is geometrically smooth), or its global dimension is completely infinite.

For decades, algebraists sought to understand the "infinite" case. If a ring is singular, just *how* singular is it? How chaotic do the infinite free resolutions become? This inquiry led to a classification of local Noetherian rings, a hierarchy that measures the severity of singularities based on homological behavior.

To properly measure the complexity of rings that fail to be regular, we need to formalize the notion of a regular sequence:

Definition 57.2.9: Regular Sequence

Let R be a ring and M an R-module. A sequence of elements $x_1, \ldots, x_n \in R$ is called an *M-regular sequence* if:

1. $(x_1, \ldots, x_n)M \neq M$
2. For $i = 1, \ldots, n$, the element x_i is not a zero-divisor on the quotient module $M/(x_1, \ldots, x_{i-1})M$.

Intuitively, a regular sequence is a sequence of elements that "cut down" the dimension of the module as cleanly and efficiently as possible, without creating any hidden nilpotent. This allows us to define a new homological invariant.

Definition 57.2.10: Depth

Let $(R, \mathfrak{m})$ be a local Noetherian ring and M a finitely generated R-module. The *depth* of M, denoted $\operatorname{depth}(M)$, is the maximum length of an M-regular sequence contained in the maximal ideal $\mathfrak{m}$.

Depth is always bounded above by the Krull dimension ($\operatorname{depth}(R) \leq \operatorname{kdim}(R)$). The gap between a ring's depth and its dimension is one of the primary ways we measure how "bad" a singularity is.

As defined, depth is a maximum over regular sequences, which makes it awkward to compute and awkward to use: knowing that *some* regular sequence has length n says nothing about whether a *given* regular element extends to one. Rees's theorem removes both difficulties at once by identifying depth with a vanishing degree for Ext, from which it follows that every maximal regular sequence has the same length.

Theorem 57.2.11: Depth via Ext (Rees)

Let $(R, \mathfrak{m}, k)$ be a Noetherian local ring and $M \neq 0$ a finitely generated R-module. If $x_1, \ldots, x_n \in \mathfrak{m}$ is an M-regular sequence, then

$$\operatorname{Ext}^i_R(k, M) = 0 \text{ for } i < n, \qquad \operatorname{Ext}^n_R(k, M) \cong \operatorname{Hom}_R\big(k,\ M/(x_1, \ldots, x_n)M\big)$$

Consequently every maximal M-regular sequence in $\mathfrak{m}$ has length $\min\big\{i \bigm| \operatorname{Ext}^i_R(k, M) \neq 0\big\}$, and in particular they all have the same length, namely $\operatorname{depth}(M)$.

***Proof*:**
Step 1: $\mathfrak{m}$ acts as zero on $\operatorname{Ext}^i_R(k, M)$. For $x \in \mathfrak{m}$, multiplication by x on $\operatorname{Ext}^i_R(k, M)$ is induced by multiplication by x on the first argument, which is the zero map $k \to k$; functoriality makes the induced map zero.

Step 2: the displayed formulas, by induction on n. For $n = 0$ there is nothing to prove, $\operatorname{Ext}^0_R(k, M) \cong \operatorname{Hom}_R(k, M)$ by proposition 56.3.12. Let $n \geq 1$ and put $M' = M/x_1M$, which is nonzero by Nakayama's lemma. As x_1 is a non-zero divisor on M, there is a short exact sequence

$$0 \to M \xrightarrow{x_1} M \to M' \to 0$$

and corollary 56.3.11, applied to $\operatorname{Hom}_R(k, -)$, gives a long exact sequence in which every map induced by x_1 vanishes by Step 1. It therefore breaks into short exact sequences

$$0 \to \operatorname{Ext}^i_R(k, M) \to \operatorname{Ext}^i_R(k, M') \to \operatorname{Ext}^{i+1}_R(k, M) \to 0$$

for every i. Now $x_2, \ldots, x_n$ is an M'-regular sequence of length $n-1$, so by induction $\operatorname{Ext}^i_R(k, M') = 0$ for $i < n-1$ and $\operatorname{Ext}^{n-1}_R(k, M') \cong \operatorname{Hom}_R(k, M/(x_1, \ldots, x_n)M)$. Reading the displayed sequence for $i < n-1$ forces $\operatorname{Ext}^i_R(k, M) = 0$ and $\operatorname{Ext}^{i+1}_R(k, M) = 0$, so $\operatorname{Ext}^j_R(k, M) = 0$ for all $j < n$. Reading it at $i = n-1$, the left-hand term vanishes, so $\operatorname{Ext}^n_R(k, M) \cong \operatorname{Ext}^{n-1}_R(k, M')$, which is the claimed Hom.

Step 3: maximality. A regular sequence $x_1, \dots, x_n$ is maximal exactly when no element of $\mathfrak{m}$ is a non-zero divisor on $N = M/(x_1, \dots, x_n)M$, that is when $\mathfrak{m}$ consists of zero-divisors on N. The zero-divisors on N are the union of its associated primes ([10, The Zero-Divisors Are the Union of the Associated Primes]), of which there are finitely many ([10, Finiteness of Associated Primes]), so prime avoidance ([10, Prime Avoidance]) makes this equivalent to $\mathfrak{m} \in \mathrm{Ass}(N)$, that is to $\mathrm{Hom}_R(k, N) \neq 0$. By Step 2 that says $\mathrm{Ext}^n_R(k, M) \neq 0$, while $\mathrm{Ext}^i_R(k, M) = 0$ for $i < n$. So n is the least degree in which $\mathrm{Ext}^\bullet_R(k, M)$ is nonzero, a quantity independent of the sequence chosen, completing the proof.

Two consequences are immediate and are used throughout what follows: a given M-regular sequence in $\mathfrak{m}$ extends to one of length $\mathrm{depth}(M)$, since any maximal extension of it has that length; and $\mathrm{depth}(M) = 0$ exactly when $\mathfrak{m} \in \mathrm{Ass}(M)$.

Using this, we can make the Auslander-Buchsbaum-Serre theorem more refined by looking at modules over R:

Theorem 57.2.12: Auslander-Buchsbaum Formula

Let $(R, \mathfrak{m}, k)$ be a local Noetherian ring, and let M be a non-zero finitely generated R-module with finite projective dimension. Then:

$$\mathrm{pd}_R(M) + \mathrm{depth}(M) = \mathrm{depth}(R)$$

***Proof Key ideas*:**
The proof is an induction on $\mathrm{depth}(M)$ whose steps are standard commutative algebra rather than homological algebra; it is given in full in *An Introduction to Homological Algebra* [53].

The base case is $\mathrm{depth}(M) = 0$, where $\mathfrak{m}$ is an associated prime of M and one shows $\mathrm{pd}_R(M) = \mathrm{depth}(R)$ directly from a minimal free resolution: the length of that resolution is detected by $\mathrm{Tor}^R_i(M, k)$ against the residue field (lemma 57.2.7), and the depth of R bounds where those groups can survive.

The inductive step reduces the depth of M by one. If $\mathrm{depth}(M) > 0$ and $\mathrm{depth}(R) > 0$, prime avoidance applied to $\mathrm{Ass}(R) \cup \mathrm{Ass}(M)$ produces an $x \in \mathfrak{m}$ regular on both M and R; passing to M/xM over $R/(x)$ drops both $\mathrm{depth}(M)$ and $\mathrm{depth}(R)$ by one while leaving pd unchanged, so the formula for the quotient gives the formula for M. The remaining case is $\mathrm{depth}(R) = 0$, handled separately: there finite projective dimension forces M to be free, so both sides of the formula vanish.

Using depth and other structural properties, local rings can be classified into a strict chain of implications:

$$\text{Regular} \implies \text{Complete Intersection} \implies \text{Gorenstein} \implies \text{Cohen-Macaulay}$$

The chain stops at Cohen-Macaulay: it does *not* continue to normal or reduced. The hypersurface $k[[x, y]]/(xy)$ is a complete intersection, hence Gorenstein and Cohen-Macaulay, and it is reduced, but it is not normal, being not even a domain. By Serre's criterion (theorem 57.3.2) normality is the conjunction of R_1 and S_2; Cohen-Macaulay supplies S_2 and says nothing about R_1, which is where the implication fails. The correct companion chain runs

$$\text{Regular} \implies \text{Normal} \implies \text{Reduced}$$

While these definitions are strictly for *local* rings, we say a general commutative ring has one of these properties if its localization at every prime ideal has it.

Complete Intersections: These are the "nicest" singularities. Geometrically, they are spaces cut out by exactly the correct number of equations (e.g., a 1D curve in $\mathbb{C}^3$ defined by exactly two equations). Algebraically, R is the quotient of a regular local ring by a regular sequence. While modules over these rings can have infinite projective dimension, their free resolutions are highly predictable, eventually becoming periodic.

Gorenstein Rings: These rings represent spaces with "symmetric" singularities. They are exactly the spaces where a generalized form of Poincaré Duality still holds. Algebraically, a local ring is Gorenstein if it has finite injective dimension over itself. The free resolutions over Gorenstein rings exhibit a mirror symmetry when the functor $\mathrm{Hom}_R(-, R)$ is applied.

Cohen-Macaulay Rings: This is perhaps the most important class of singular rings in algebraic geometry, and the one the subsection below treats properly. A local ring is Cohen-Macaulay if its depth equals its Krull dimension (definition 57.2.13). Geometrically, these spaces are "equidimensional" and lack hidden, embedded fuzzy components. Here, the Auslander-Buchsbaum formula perfectly aligns with the geometric dimension.

Normal Rings: A ring is normal if it is integrally closed in its total ring of fractions. Geometrically, this means the space is "smooth in codimension 1". For example, a 2D normal surface can have an isolated singular point (like the tip of a cone), but it cannot have a singular crease or folded line running through it.

Reduced Rings: A ring is reduced if it contains no non-zero nilpotents ($x^n = 0 \implies x = 0$). Geometrically, this simply means the space has no "thickness" or "fuzz"; it is a classical geometric space uniquely determined by the values of functions at its points.

57.2.4 Cohen-Macaulay Rings

Of the classes just listed, Cohen-Macaulay is the one downstream geometry actually consumes, and it deserves more than a description. Two facts are wanted: that a system of parameters in such a ring is automatically a regular sequence, so that the naive count of equations is the honest one; and that the ring carries no embedded associated primes, so that a subscheme cut out this way has no hidden components. Both follow from a single inequality relating depth to the dimensions of the associated primes. Associated primes, prime avoidance, and the identification of the zero-divisors with the union of the associated primes are those of [10], a co-requisite of this book.

Definition 57.2.13: Cohen-Macaulay Ring

A Noetherian local ring $(R, \mathfrak{m})$ is *Cohen-Macaulay* if $\mathrm{depth}(R) = \mathrm{kdim}(R)$. A Noetherian ring is Cohen-Macaulay if its localization at every prime is. More generally a finitely generated module $M \neq 0$ over a Noetherian local ring is *Cohen-Macaulay* if $\mathrm{depth}(M) = \mathrm{kdim}(M)$, the dimension of $R/\mathrm{Ann}(M)$.

Lemma 57.2.14: Depth Drops by One Modulo a Non-Zero Divisor

Let $(R, \mathfrak{m})$ be Noetherian local, $M \neq 0$ finitely generated, and $x \in \mathfrak{m}$ a non-zero divisor on M. Then $\operatorname{depth}(M/xM) = \operatorname{depth}(M) - 1$.

***Proof*:**
Prepending x to any M/xM-regular sequence in $\mathfrak{m}$ yields an M-regular sequence, so $\operatorname{depth}(M) \geq \operatorname{depth}(M/xM) + 1$ directly from definition 57.2.10.

For the reverse, extend x to a maximal M-regular sequence $x, y_1, \ldots, y_m$ in $\mathfrak{m}$; this is possible, and the result has length $\operatorname{depth}(M)$, by theorem 57.2.11. Then $y_1, \ldots, y_m$ is an M/xM-regular sequence, so $\operatorname{depth}(M/xM) \geq m = \operatorname{depth}(M) - 1$. Combining the two, both are equalities, as we sought to show.

Proposition 57.2.15: Depth Is Bounded by the Dimension of Every Associated Prime

Let $(R, \mathfrak{m})$ be Noetherian local, $M \neq 0$ finitely generated, and $\mathfrak{p} \in \operatorname{Ass}(M)$. Then

$$\operatorname{depth}(M) \ \leq \ \operatorname{kdim}(R/\mathfrak{p})$$

***Proof*:**
Induct on $n = \operatorname{depth}(M)$. For $n = 0$ there is nothing to prove. Let $n > 0$ and let $x \in \mathfrak{m}$ be a non-zero divisor on M, which exists because the depth is positive. Since $\mathfrak{p}$ consists of zero-divisors on M, [10, The Zero-Divisors Are the Union of the Associated Primes] gives $x \notin \mathfrak{p}$.

Write $\mathfrak{p} = \operatorname{Ann}(m)$ with $m \neq 0$. We first arrange $m \notin xM$. If $m = xm_1$ then $\operatorname{Ann}(m_1) = \mathfrak{p}$: an r killing m_1 kills m, and conversely $rm = 0$ gives $rxm_1 = 0$, whence $rm_1 = 0$ as x is a non-zero divisor. Iterating produces $m = x^k m_k$ with $\operatorname{Ann}(m_k) = \mathfrak{p}$ for each k, and the submodules $Rm \subseteq Rm_1 \subseteq Rm_2 \subseteq \cdots$ ascend. As M is Noetherian the chain stabilizes, say $Rm_k = Rm_{k+1}$, so $m_{k+1} = rm_k = rx\,m_{k+1}$ for some r; then $(1 - rx)m_{k+1} = 0$, and $1 - rx$ is a unit because $x \in \mathfrak{m}$, forcing $m_{k+1} = 0$ and hence $m = 0$. That contradiction means the iteration halts, so we may take $m \notin xM$.

Let $N = Rm$ and let $\overline{N}$ be its image in M/xM, nonzero by the previous paragraph. Every element of $\mathfrak{p}$ kills N, hence $\overline{N}$; and x kills $\overline{N}$ too, since $xN \subseteq xM$. So $\operatorname{Ann}(\overline{N}) \supseteq \mathfrak{p} + (x)$. Choose $\mathfrak{q} \in \operatorname{Ass}(\overline{N}) \subseteq \operatorname{Ass}(M/xM)$, nonempty by [10, Zero-Divisors and Annihilators]. Then $\mathfrak{q} \supseteq \mathfrak{p} + (x)$, and $\mathfrak{q} \neq \mathfrak{p}$ because $x \notin \mathfrak{p}$, so $\mathfrak{q} \supsetneq \mathfrak{p}$ and $\operatorname{kdim}(R/\mathfrak{q}) \leq \operatorname{kdim}(R/\mathfrak{p}) - 1$. By lemma 57.2.14 and the inductive hypothesis applied to M/xM and $\mathfrak{q}$,

$$n - 1 \ = \ \operatorname{depth}(M/xM) \ \leq \ \operatorname{kdim}(R/\mathfrak{q}) \ \leq \ \operatorname{kdim}(R/\mathfrak{p}) - 1$$

which is the assertion.

Corollary 57.2.16: Depth Is at Most the Dimension

For $M \neq 0$ finitely generated over a Noetherian local ring, $\operatorname{depth}(M) \leq \operatorname{kdim}(M)$. In particular $\operatorname{depth}(R) \leq \operatorname{kdim}(R)$, so Cohen-Macaulay is the extremal case.

***Proof*:**
$\mathrm{Ass}(M)$ is nonempty ([10, Zero-Divisors and Annihilators]); pick $\mathfrak{p}$ in it. Then $\mathfrak{p} \supseteq \mathrm{Ann}(M)$, so $\mathrm{kdim}(R/\mathfrak{p}) \leq \mathrm{kdim}(R/\mathrm{Ann}(M)) = \mathrm{kdim}(M)$, and proposition 57.2.15 bounds the depth by the left-hand side.

This is the inequality asserted informally above, and it is what makes the definition of Cohen-Macaulay a demand rather than a coincidence.

Corollary 57.2.17: A Cohen-Macaulay Ring Has No Embedded Associated Primes

Let $(R, \mathfrak{m})$ be a Cohen-Macaulay local ring of dimension d. Then every $\mathfrak{p} \in \mathrm{Ass}(R)$ satisfies $\mathrm{kdim}(R/\mathfrak{p}) = d$, and every such $\mathfrak{p}$ is a minimal prime of R. In particular R has no embedded associated primes, and R is equidimensional.

***Proof*:**
For $\mathfrak{p} \in \mathrm{Ass}(R)$, proposition 57.2.15 gives $d = \mathrm{depth}(R) \leq \mathrm{kdim}(R/\mathfrak{p}) \leq d$, so $\mathrm{kdim}(R/\mathfrak{p}) = d$. If $\mathfrak{p}$ were not minimal it would strictly contain a prime $\mathfrak{p}_0$, and then $\mathrm{kdim}(R/\mathfrak{p}) < \mathrm{kdim}(R/\mathfrak{p}_0) \leq d$, a contradiction. An associated prime that is minimal is by definition not embedded.

Theorem 57.2.18: A System of Parameters in a Cohen-Macaulay Local Ring Is a Regular Sequence

Let $(R, \mathfrak{m})$ be a Cohen-Macaulay local ring of dimension d and let $x_1, \ldots, x_d \in \mathfrak{m}$ be a system of parameters, that is d elements generating an $\mathfrak{m}$-primary ideal. Then $x_1, \ldots, x_d$ is a regular sequence, and $R/(x_1, \ldots, x_d)$ is Cohen-Macaulay of dimension zero.

***Proof*:**
Induct on d, the case $d = 0$ being the empty sequence. Let $d > 0$. Since $x_1, \ldots, x_d$ generate an $\mathfrak{m}$-primary ideal, the images of $x_2, \ldots, x_d$ generate an $\mathfrak{m}$-primary ideal of $R/(x_1)$, so $\mathrm{kdim} R/(x_1) \leq d - 1$ by the dimension theorem of [10]; and $\mathrm{kdim} R/(x_1) \geq d - 1$ by Krull's height theorem there. So $\mathrm{kdim} R/(x_1) = d - 1$.

Now x_1 is a non-zero divisor. Suppose $x_1 \in \mathfrak{p}$ for some $\mathfrak{p} \in \mathrm{Ass}(R)$. By corollary 57.2.17, $\mathrm{kdim}(R/\mathfrak{p}) = d$; but $\mathfrak{p} \supseteq (x_1)$ gives a surjection $R/(x_1) \twoheadrightarrow R/\mathfrak{p}$ and hence $\mathrm{kdim} R/(x_1) \geq d$, contradicting the previous paragraph. So x_1 lies outside every associated prime and is a non-zero divisor by [10, The Zero-Divisors Are the Union of the Associated Primes].

By lemma 57.2.14, $\mathrm{depth}\, R/(x_1) = d - 1 = \mathrm{kdim} R/(x_1)$, so $R/(x_1)$ is Cohen-Macaulay of dimension $d - 1$, and the images of $x_2, \ldots, x_d$ are a system of parameters for it. The inductive hypothesis makes them a regular sequence there, which is exactly the statement that $x_1, \ldots, x_d$ is a regular sequence on R and that the final quotient is Cohen-Macaulay of dimension zero.

The form geometry reaches for is slightly more general: the elements need not be a full system of parameters, only as many as the height of the ideal they generate.

Corollary 57.2.19: Height-Many Elements in a Cohen-Macaulay Ring Form a Regular Sequence

Let R be a Cohen-Macaulay local ring and let $x_1, \dots, x_h \in R$ generate an ideal $I = (x_1, \dots, x_h)$ of height h. Then $x_1, \dots, x_h$ is a regular sequence.

***Proof*:**
Let $\mathfrak{p}$ be a prime minimal over I with $\operatorname{ht}\mathfrak{p} = h$, which exists because the height of I is the minimum of $\operatorname{ht}\mathfrak{q}$ over primes $\mathfrak{q}$ minimal over I. Localizing, $R_\mathfrak{p}$ is Cohen-Macaulay of dimension h (localization of a Cohen-Macaulay local ring at a prime is Cohen-Macaulay, and $\dim R_\mathfrak{p} = \operatorname{ht}\mathfrak{p} = h$), and the images of $x_1, \dots, x_h$ generate an ideal primary to the maximal ideal of $R_\mathfrak{p}$, since $\mathfrak{p}$ is minimal over I. They are therefore a system of parameters of $R_\mathfrak{p}$, hence a regular sequence there by theorem 57.2.18.

To descend, induct on i: suppose $x_1, \dots, x_{i-1}$ is a regular sequence on R and that x_i is a zero divisor on $M = R/(x_1, \dots, x_{i-1})$. Then x_i lies in an associated prime $\mathfrak{q}$ of M ([10]), and M is Cohen-Macaulay of dimension $\dim R - i + 1$ by theorem 57.2.18 applied step by step, so $\mathfrak{q}$ has no embedded components by corollary 57.2.17 and $\dim R/\mathfrak{q} = \dim R - i + 1$. But $\mathfrak{q} \supseteq (x_1, \dots, x_i)$ forces $\operatorname{ht}\mathfrak{q} \geq$ the height of that ideal, and comparing with the localized statement above bounds $\dim R/\mathfrak{q}$ strictly below $\dim R - i + 1$, a contradiction. So each x_i is a non-zero divisor on the preceding quotient, as we sought to show.

57.3 Serre's Criterion for Normality

Depth and regularity are conditions on a local ring; imposing them at every prime, with a strength that grows with the height, produces the two families of conditions Serre introduced. Their point is the theorem below: normality, which is a statement about integral closure and has no homological content on its face, is exactly the conjunction of a regularity condition in codimension one and a depth condition everywhere.

Definition 57.3.1: Serre's Conditions

Let R be a Noetherian ring and $k \geq 0$ an integer. Then R satisfies

- (R_k) if $R_\mathfrak{p}$ is a regular local ring for every prime $\mathfrak{p}$ with $\operatorname{ht}\mathfrak{p} \leq k$;
- (S_k) if $\operatorname{depth}(R_\mathfrak{p}) \geq \min\big(k,\ \operatorname{ht}\mathfrak{p}\big)$ for every prime $\mathfrak{p}$.

Both conditions weaken as attention moves to primes of large height: (R_k) says nothing about primes of height above k, and (S_k) asks only for depth k once the height reaches k. A regular ring satisfies (R_k) for every k, and a Cohen-Macaulay ring satisfies (S_k) for every k, since there $\operatorname{depth} R_\mathfrak{p} = \operatorname{kdim} R_\mathfrak{p} = \operatorname{ht}\mathfrak{p}$.

Theorem 57.3.2: Serre's Criterion for Normality

Let R be a Noetherian domain. Then R is integrally closed in its fraction field if and only if it satisfies (R_1) and (S_2).

***Proof*:**
Necessity. Suppose R is normal. For (R_1), a prime of height zero is (0) and $R_{(0)}$ is a field, which is regular; and for $\operatorname{ht}\mathfrak{p} = 1$ the ring $R_\mathfrak{p}$ is a discrete valuation ring by [10, A Normal Domain Is the Intersection of Its Height-One Localizations], hence regular.

For (S_2) let $\mathfrak{p}$ be a prime. If $\operatorname{ht}\mathfrak{p} = 0$ there is nothing to prove. If $\operatorname{ht}\mathfrak{p} = 1$ then $R_\mathfrak{p}$ is a discrete valuation ring, whose maximal ideal is generated by a non-zero divisor, so its depth is at least $1 = \min(2, 1)$. Suppose $\operatorname{ht}\mathfrak{p} \geq 2$, so that $\min(2, \operatorname{ht}\mathfrak{p}) = 2$. The ring $R_\mathfrak{p}$ is a Noetherian normal domain, so any nonzero $b \in \mathfrak{p}R_\mathfrak{p}$ is a non-zero divisor and $\operatorname{depth}(R_\mathfrak{p}) \geq 1$. By [10, Associated Primes of a Principal Ideal Have Height One], every associated prime of $bR_\mathfrak{p}$ has height one, so $\mathfrak{p}R_\mathfrak{p}$, of height at least two, is not one of them; hence $\operatorname{depth}(R_\mathfrak{p}/bR_\mathfrak{p}) \neq 0$ by the second consequence of theorem 57.2.11. Then lemma 57.2.14 gives $\operatorname{depth}(R_\mathfrak{p}) = \operatorname{depth}(R_\mathfrak{p}/bR_\mathfrak{p}) + 1 \geq 2$.

Sufficiency. Suppose R satisfies (R_1) and (S_2), and write K for its fraction field. Every prime of height one has $R_\mathfrak{p}$ regular local of dimension one by (R_1), hence a discrete valuation ring, hence integrally closed. An intersection of integrally closed subrings of K each with fraction field K is integrally closed, so it suffices to prove

$$R = \bigcap_{\operatorname{ht}\mathfrak{p}=1} R_\mathfrak{p}$$

One inclusion is immediate. For the other, let x lie in every such $R_\mathfrak{p}$, write $x = a/b$ with $b \neq 0$, and suppose $a \notin (b)$. Then $\operatorname{Ann}(\overline{a}) = (b : a)$ is a proper ideal consisting of zero-divisors on $R/(b)$, so as in [10, The Zero-Divisors Are the Union of the Associated Primes] together with prime avoidance it lies in a single $\mathfrak{q} \in \operatorname{Ass}(R/(b))$.

The height of $\mathfrak{q}$ is one. It is at least one because R is a domain and $b \neq 0$. If it were at least two, then (S_2) would give $\operatorname{depth}(R_\mathfrak{q}) \geq 2$, and b is a non-zero divisor in $\mathfrak{q}R_\mathfrak{q}$, so lemma 57.2.14 would give $\operatorname{depth}(R_\mathfrak{q}/bR_\mathfrak{q}) \geq 1$, whence $\mathfrak{q}R_\mathfrak{q} \notin \operatorname{Ass}(R_\mathfrak{q}/bR_\mathfrak{q})$ by theorem 57.2.11. But associated primes localize, so $\mathfrak{q} \in \operatorname{Ass}(R/(b))$ makes $\mathfrak{q}R_\mathfrak{q}$ associated to $R_\mathfrak{q}/bR_\mathfrak{q}$, a contradiction.

So $\operatorname{ht}\mathfrak{q} = 1$ and $x \in R_\mathfrak{q}$ by hypothesis; that produces $s \notin \mathfrak{q}$ with $sa \in bR$, so $s \in (b : a) \subseteq \mathfrak{q}$, a contradiction. Hence $a \in (b)$ and $x \in R$, completing the proof.

Read through the geometry, (R_1) says a normal variety is smooth away from a closed set of codimension at least two, and (S_2) says a function defined outside such a set extends across it. That reading, and the divisor theory it supports, is [27].

57.4 The Local Criterion for Flatness

One of the most powerful applications of Nakayama's Lemma and homological algebra is the Local Criterion for Flatness. Checking whether a module is flat directly from the definition (that tensoring preserves exactness) can be extremely difficult. The local criterion reduces this to checking exactness against a single module (the residue field), or checking that regular sequences map to regular sequences.

Theorem 57.4.1: Local Criterion for Flatness

Let $\varphi : (R, \mathfrak{m}) \to (S, \mathfrak{n})$ be a local homomorphism of Noetherian local rings, and let M be a finitely generated S-module. Then the following are equivalent:

1. M is flat over R.
2. $\mathrm{Tor}_1^R(M, R/\mathfrak{m}) = 0$.
3. The natural multiplication map $\mathfrak{m} \otimes_R M \to \mathfrak{m}M$ is an isomorphism.

***Proof*:**

The implication (1) $\Rightarrow$ (2) is trivial since M is flat implies $\mathrm{Tor}_i^R(M, -) = 0$ for all $i > 0$. The equivalence (2) $\iff$ (3) follows from applying $- \otimes_R M$ to the short exact sequence $0 \to \mathfrak{m} \to R \to R/\mathfrak{m} \to 0$, which gives the long exact sequence in Tor:

$$0 = \mathrm{Tor}_1^R(R, M) \to \mathrm{Tor}_1^R(R/\mathfrak{m}, M) \to \mathfrak{m} \otimes_R M \to R \otimes_R M \to R/\mathfrak{m} \otimes_R M \to 0$$

Since $R \otimes_R M \cong M$, the kernel of $M \to M/\mathfrak{m}M$ is precisely $\mathfrak{m}M$. Thus $\mathrm{Tor}_1^R(R/\mathfrak{m}, M) = 0$ if and only if $\mathfrak{m} \otimes_R M \to \mathfrak{m}M$ is injective (and it is always surjective).

For (2) $\Rightarrow$ (1), we must show that $\mathrm{Tor}_1^R(M, N) = 0$ for all finitely generated R-modules N. We proceed by induction on the length of N. By Nakayama's Lemma (lemma 23.4.27), if $N \neq 0$, we can find a non-zero quotient $N \to R/\mathfrak{m}$. Let K be the kernel, so we have $0 \to K \to N \to R/\mathfrak{m} \to 0$. The long exact sequence in Tor gives:

$$\mathrm{Tor}_1^R(K, M) \to \mathrm{Tor}_1^R(N, M) \to \mathrm{Tor}_1^R(R/\mathfrak{m}, M) = 0$$

Thus, $\mathrm{Tor}_1^R(K, M)$ surjects onto $\mathrm{Tor}_1^R(N, M)$. By iterating this process (using the Artin-Rees lemma and the $\mathfrak{m}$-adic filtration on N; theorem 25.4.1), one can show that $\mathrm{Tor}_1^R(N, M) \subseteq \mathfrak{m}\mathrm{Tor}_1^R(N, M)$. Since M is a finitely generated S-module and S is Noetherian, $\mathrm{Tor}_1^R(N, M)$ is a finitely generated S-module. By Nakayama's Lemma (applied over S, since $\mathfrak{m}S \subseteq \mathfrak{n}$), we conclude $\mathrm{Tor}_1^R(N, M) = 0$.

In algebraic geometry, we often deal with regular local rings and regular sequences. The Local Criterion gives us an incredibly useful corollary for checking flatness in terms of regular sequences. This is the algebraic engine behind an infamous result known as the "Miracle Flatness" theorem, an important "shortcut" for proving flatness in [5].

Corollary 57.4.2: Regular Sequence Criterion for Flatness

Let $\varphi : (R, \mathfrak{m}) \to (S, \mathfrak{n})$ be a local homomorphism of Noetherian local rings, and M a finitely generated S-module. Let $x \in \mathfrak{m}$ be a non-zero divisor on R. Then M is flat over R if and only if both of the following hold if and only if:

1. x is a non-zero divisor on M, and
2. M/xM is flat over $R/(x)$.

By induction, if $x_1, \dots, x_d \in \mathfrak{m}$ is a regular sequence in R, then M is flat over R if and only if $x_1, \dots, x_d$ forms a regular sequence on M and $M/(x_1, \dots, x_d)M$ is flat over $R/(x_1, \dots, x_d)$.

***Proof*:**
($\Rightarrow$): If M is flat over R, applying $- \otimes_R M$ to the exact sequence $0 \to R \xrightarrow{\cdot x} R \to R/(x) \to 0$ yields the exact sequence $0 \to M \xrightarrow{\cdot x} M \to M/xM \to 0$. Thus x is a non-zero divisor on M. Furthermore, base change preserves flatness, so $M/xM \cong M \otimes_R R/(x)$ is flat over $R/(x)$.

($\Leftarrow$): Assume x is regular on M and M/xM is flat over $R/(x)$. We want to show M is flat. Since M/xM is flat over $R/(x)$, $\mathrm{Tor}_1^{R/(x)}(R/\mathfrak{m}, M/xM) = 0$. Combined with the fact that x is regular on M, this gives $\mathrm{Tor}_1^R(R/\mathfrak{m}, M) = 0$: taking a free resolution $F_\bullet \to M$ over R, regularity of x on both R and M makes $F_\bullet \otimes_R R/(x)$ a free resolution of M/xM over $R/(x)$, so $\mathrm{Tor}_i^{R/(x)}(N, M/xM) \cong \mathrm{Tor}_i^R(N, M)$ for every $R/(x)$-module N, and $N = R/\mathfrak{m}$ is one. By the Local Criterion for Flatness (theorem 57.4.1), M is flat over R.

The variant that geometry actually asks for reverses the roles: rather than a regular sequence downstairs, one is handed an element that is a nonzerodivisor *on the fibre*, and wants to conclude that it was a nonzerodivisor upstairs all along. This is how flatness of a family cut out by equations gets verified one equation at a time.

Corollary 57.4.3: Fibrewise Nonzerodivisor Criterion for Flatness

Let $\varphi : (R, \mathfrak{m}) \to (S, \mathfrak{n})$ be a local homomorphism of Noetherian local rings with $\kappa = R/\mathfrak{m}$, let M be a finitely generated S-module that is flat over R, and let $u \in \mathfrak{n}$. If u is a non-zero divisor on $M/\mathfrak{m}M = M \otimes_R \kappa$, then u is a non-zero divisor on M, and M/uM is flat over R.

***Proof*:**
Write $K = \ker(u : M \to M)$ and $M' = M/K$, so that multiplication by u factors as a surjection $M \twoheadrightarrow M'$ followed by an injection $M' \hookrightarrow M$, and $N := M/uM$ is the cokernel of the latter. Both sequences

$$0 \to K \to M \to M' \to 0, \qquad 0 \to M' \xrightarrow{u} M \to N \to 0$$

are exact by construction.

Step 1: N is flat. Apply $- \otimes_R \kappa$ to the second sequence. Flatness of M gives $\mathrm{Tor}_1^R(\kappa, M) = 0$, so the long exact sequence begins

$$0 \to \mathrm{Tor}_1^R(\kappa, N) \to M' \otimes_R \kappa \to M \otimes_R \kappa \to N \otimes_R \kappa \to 0$$

Now right-exactness applied to the first sequence makes $M \otimes_R \kappa \to M' \otimes_R \kappa$ surjective, and composing it with $M' \otimes_R \kappa \to M \otimes_R \kappa$ recovers multiplication by u on $M \otimes_R \kappa$, which is injective by hypothesis. A surjection whose composite with a further map is injective is itself injective, so $M \otimes_R \kappa \to M' \otimes_R \kappa$ is an isomorphism, and therefore $M' \otimes_R \kappa \to M \otimes_R \kappa$ is injective as well. Hence $\mathrm{Tor}_1^R(\kappa, N) = 0$, and N is flat over R by theorem 57.4.1.

Step 2: $K = 0$. Since M and N are both flat over R, the second sequence forces M' to be flat too: for any R-module T the long exact sequence reads $0 = \mathrm{Tor}_2^R(T, N) \to \mathrm{Tor}_1^R(T, M') \to \mathrm{Tor}_1^R(T, M) = 0$. In particular $\mathrm{Tor}_1^R(\kappa, M') = 0$, so applying $- \otimes_R \kappa$ to the first sequence leaves $K \otimes_R \kappa \to M \otimes_R \kappa$ injective. But Step 1 showed $M \otimes_R \kappa \to M' \otimes_R \kappa$ is an isomorphism, so exactness at $M \otimes_R \kappa$ makes the image of $K \otimes_R \kappa$ zero. An injective map with zero image is zero, so $K/\mathfrak{m}K = 0$. As K is a submodule of the finitely generated module M over the Noetherian ring S, it is finitely generated, and $\mathfrak{m}S \subseteq \mathfrak{n}$, so Nakayama gives $K = 0$.

A third consequence is the one deformation arguments in geometry lean on: a resolution of a flat object stays a resolution after restriction to a fibre. Without it a resolution built on a total space says nothing about the special fibres it was built to compare.

Proposition 57.4.4: Restriction of a Resolution to a Fibre

Let R be a ring, A an R-algebra, and M an A-module that is flat over R. Let

$$0 \to G_n \to G_{n-1} \to \cdots \to G_0 \to M \to 0$$

be an exact sequence of A-modules with every G_i flat over R. Then for every R-algebra $R \to k$, applying $- \otimes_R k$ leaves the sequence exact:

$$0 \to G_n \otimes_R k \to \cdots \to G_0 \otimes_R k \to M \otimes_R k \to 0$$

In particular, if $X \to S$ is a morphism of schemes, $\mathscr{F}$ a quasi-coherent sheaf on X flat over S, and $G_\bullet \to \mathscr{F}$ a resolution by quasi-coherent sheaves flat over S, then the restriction of $G_\bullet \to \mathscr{F}$ to any fibre X_s is a resolution of $\mathscr{F}|_{X_s}$.

***Proof*:**
Break the sequence into short exact pieces $0 \to Z_{i+1} \to G_i \to Z_i \to 0$ with $Z_0 = M$ and $Z_{n+1} = 0$. We show by induction on i that every Z_i is flat over R, the base case $Z_0 = M$ being the hypothesis. Given that Z_i is R-flat, the sequence $0 \to Z_{i+1} \to G_i \to Z_i \to 0$ has flat right-hand and middle terms, so the long exact sequence of $\mathrm{Tor}^R(-, N)$ gives $0 = \mathrm{Tor}_2^R(Z_i, N) \to \mathrm{Tor}_1^R(Z_{i+1}, N) \to \mathrm{Tor}_1^R(G_i, N) = 0$ for every N, whence Z_{i+1} is flat. The induction therefore gives flatness of every syzygy.

Now tensor each short exact piece with k over R. Flatness of Z_i makes $\mathrm{Tor}_1^R(Z_i, k) = 0$, so $0 \to Z_{i+1} \otimes_R k \to G_i \otimes_R k \to Z_i \otimes_R k \to 0$ is exact. Splicing the pieces back together reassembles the long sequence, still exact. The sheaf statement is this one applied on affine opens, flatness and exactness both being local, as we sought to show.

Exercise 57.4.1

1. Let $R = k[x, y]$. Find a free resolution for the module $M = R/(x^2, xy, y^2)$. What is the length

of this resolution?

2. Show that over $R = \mathbb{Z}$, the module $M = \mathbb{Q}$ has projective dimension 1, but does not have a finite *free* resolution if we require finitely generated free modules (which is impossible since $\mathbb{Q}$ isn't finitely generated).

3. Why does the Syzygy theorem fail for $R = k[x_1, x_2, \dots]$ (infinite variables)? Find a module with infinite projective dimension.

57.5 The Limit of a Tower and Its First Derived Functor

The functors derived so far in this chapter have been Hom and $\otimes$, and the objects they were applied to were modules over a ring. The functor of this section is neither, and its source category is not a category of modules: it is the limit, applied to a sequence of abelian groups indexed by $\mathbb{N}$ with maps running backwards. The limit of such a sequence is left exact and not exact, so it has derived functors like any other left-exact functor. What is special is how few of them there are. Exactly one is nonzero, it admits a description with no resolutions in it at all, and it vanishes under a hypothesis satisfied in most of the situations where the limit is taken.

That last point is why the section exists. A cohomology theory evaluated on an infinite complex, the cohomology of an infinite-dimensional classifying space, the completion of a ring: each is computed as a limit over a sequence of finite approximations, and in each the answer is the naive limit *provided* a single obstruction group vanishes. Naming that group is what allows the provision to be checked rather than assumed.[2]

Definition 57.5.1: Tower of Abelian Groups

A *tower* of abelian groups is a sequence $A_\bullet = (A_n)_{n\geq 0}$ of abelian groups together with homomorphisms

$$A_0 \xleftarrow{f_1} A_1 \xleftarrow{f_2} A_2 \xleftarrow{f_3} \cdots$$

called the *transition maps*. For $m \geq n$ write $f_{n,m} \colon A_m \to A_n$ for the composite $f_{n+1} \circ \cdots \circ f_m$, with $f_{n,n} = \mathrm{id}_{A_n}$.
A *morphism of towers* $u \colon A_\bullet \to B_\bullet$ is a family of homomorphisms $u_n \colon A_n \to B_n$ with $u_n \circ f_{n+1} = g_{n+1} \circ u_{n+1}$ for every n, where $g_\bullet$ denotes the transition maps of $B_\bullet$. Towers and their morphisms form an abelian category, in which kernels, cokernels, images and exactness are all computed degreewise.

That the category is abelian with everything computed degreewise is what makes the section possible: a sequence of towers is exact exactly when each of its levels is exact, so the hypotheses of every result below can be checked one group at a time. The limit itself is the group of coherent families, and the observation that turns it into something derivable is that this group is the kernel of a single explicit map.

[2]The older literature writes this group $\varprojlim^1$ and calls it the first derived functor of the inverse limit. This series says *limit* and names the shape of the diagram separately, so the notation here is $\varprojlim^1$ and the phrase is "the limit of a tower".

Definition 57.5.2: The Limit and $\varprojlim^1$ of a Tower

Let $A_\bullet$ be a tower of abelian groups. Define

$$d^A\colon \prod_{n\geq 0} A_n \to \prod_{n\geq 0} A_n, \qquad d^A\big((a_n)_n\big) = \big(a_n - f_{n+1}(a_{n+1})\big)_n$$

The *limit* of the tower is $\varprojlim A_\bullet := \ker d^A$, and the *first derived functor of the limit* is

$$\varprojlim{}^1 A_\bullet := \operatorname{coker} d^A$$

Both are functors from towers to abelian groups: a morphism $u\colon A_\bullet \to B_\bullet$ induces $\prod u_n$ on products, and $d^B \circ \prod u_n = \prod u_n \circ d^A$ because $u_n f_{n+1} = g_{n+1} u_{n+1}$, so $\prod u_n$ carries $\ker d^A$ into $\ker d^B$ and descends to cokernels.

The kernel is the usual object under a different description. An element $(a_n) \in \prod A_n$ lies in $\ker d^A$ precisely when $a_n = f_{n+1}(a_{n+1})$ for every n, which is the condition that the family be compatible with the transition maps; so $\ker d^A$ is the group of coherent sequences, with the projections $\varprojlim A_\bullet \to A_n$ obtained by restricting the product projections. The cokernel measures how far d^A is from being surjective, and the content of the next theorem is that this single group is the entire obstruction to the limit being exact.

Theorem 57.5.3: The Six-Term Sequence of the Limit

Let $0 \to A_\bullet \to B_\bullet \to C_\bullet \to 0$ be a short exact sequence of towers of abelian groups. Then there is an exact sequence

$$0 \to \varprojlim A_\bullet \to \varprojlim B_\bullet \to \varprojlim C_\bullet \xrightarrow{\partial} \varprojlim{}^1 A_\bullet \to \varprojlim{}^1 B_\bullet \to \varprojlim{}^1 C_\bullet \to 0$$

natural in the short exact sequence.

The sequence says two things worth separating. Reading the left half, the limit is left exact but its failure to preserve surjections is entirely accounted for by the connecting map ∂ into $\varprojlim^1 A_\bullet$: if that group vanishes, $\varprojlim B_\bullet \to \varprojlim C_\bullet$ is onto. Reading the right half, the sequence *stops*. There is no seven-term continuation, and proposition 57.5.4 explains why: the higher derived functors are zero, so no further obstruction can appear.

***Proof*:**
Write $P(A) = \prod_{n\geq 0} A_n$ and likewise for B and C, and let $u\colon A_\bullet \to B_\bullet$, $v\colon B_\bullet \to C_\bullet$ be the given morphisms.

Step 1: The two rows are short exact. By definition 57.5.1 exactness of a sequence of towers is checked degreewise, so $0 \to A_n \to B_n \to C_n \to 0$ is exact for every n. A product of short exact sequences of abelian groups is short exact, since kernels, images and surjectivity of a family of maps into a product are all tested coordinatewise (this is the exactness of arbitrary products in R-Mod recorded in lemma 56.1.11). Hence

$$0 \to P(A) \xrightarrow{\prod u_n} P(B) \xrightarrow{\prod v_n} P(C) \to 0$$

is exact, and the same sequence serves as both rows of the diagram below.

Step 2: The squares commute. Take the vertical maps to be d^A, d^B, d^C:

$$\begin{tikzcd}
0 \arrow[r] & P(A) \arrow[r, "\prod u_n"] \arrow[d, "d^A"] & P(B) \arrow[r, "\prod v_n"] \arrow[d, "d^B"] & P(C) \arrow[r] \arrow[d, "d^C"] & 0 \\
0 \arrow[r] & P(A) \arrow[r, "\prod u_n"] & P(B) \arrow[r, "\prod v_n"] & P(C) \arrow[r] & 0
\end{tikzcd}$$

Commutativity of the left square is the computation already made in definition 57.5.2: for $(a_n) \in P(A)$,

$$d^B\big((u_n(a_n))_n\big) = \big(u_n(a_n) - g_{n+1}(u_{n+1}(a_{n+1}))\big)_n = \big(u_n(a_n) - u_n(f_{n+1}(a_{n+1}))\big)_n$$

using $g_{n+1}u_{n+1} = u_n f_{n+1}$, and the right-hand side is $\prod u_n$ applied to $d^A((a_n))$. The right square is the same computation for v.

Step 3: Apply the snake lemma and identify the terms. Read each of the three vertical maps as a complex concentrated in degrees 0 and 1, so that the diagram of Step 2 is a short exact sequence of two-term complexes. This is exactly the reading carried out in example 56.1.16(1): for a two-term complex $H^0 = \ker$ and $H^1 = \operatorname{coker}$, and $H^n = 0$ otherwise, so the long exact sequence of theorem 56.1.15 has only six nonzero terms,

$$0 \to \ker d^A \to \ker d^B \to \ker d^C \xrightarrow{\partial} \operatorname{coker} d^A \to \operatorname{coker} d^B \to \operatorname{coker} d^C \to 0$$

The connecting map is the one built in the proof of theorem 56.1.15, by the recipe: given $c \in \ker d^C$, choose $b \in P(B)$ with $\prod v_n(b) = c$, note that $d^B(b)$ has zero image in $P(C)$ and so equals $\prod u_n(a)$ for a unique $a \in P(A)$, and set $\partial(c) := [a] \in \operatorname{coker} d^A$. By definition 57.5.2 the three kernels are $\varprojlim A_\bullet$, $\varprojlim B_\bullet$, $\varprojlim C_\bullet$ and the three cokernels are $\varprojlim^1 A_\bullet$, $\varprojlim^1 B_\bullet$, $\varprojlim^1 C_\bullet$, which is the displayed sequence.

Step 4: Naturality. Let (s, t, r) be a morphism from $0 \to A_\bullet \to B_\bullet \to C_\bullet \to 0$ to a second short exact sequence $0 \to \tilde{A}_\bullet \to \tilde{B}_\bullet \to \tilde{C}_\bullet \to 0$, so $s\colon A_\bullet \to \tilde{A}_\bullet$ and so on, with both squares commuting. Each of s, t, r induces a map on products commuting with the relevant d, by the computation of Step 2, so the four maps other than ∂ are natural for the reason that they are restrictions and quotients of $\prod s_n$, $\prod t_n$, $\prod r_n$.

For ∂, take $c \in \varprojlim C_\bullet$ and run the recipe of Step 3, producing b and a with $\partial(c) = [a]$. Applying s gives $\big[\prod s_n(a)\big]$. Running the recipe instead on $\prod r_n(c)$, one *may* take $\prod t_n(b)$ as the required lift, since $\prod \tilde{v}_n(\prod t_n(b)) = \prod r_n(\prod v_n(b)) = \prod r_n(c)$; and then

$$d^{\tilde{B}}\Big(\prod t_n(b)\Big) = \prod t_n(d^B(b)) = \prod t_n\Big(\prod u_n(a)\Big) = \prod \tilde{u}_n\Big(\prod s_n(a)\Big)$$

so the unique preimage is $\prod s_n(a)$ and $\tilde{\partial}(\prod r_n(c)) = \big[\prod s_n(a)\big]$ as well. The two agree, and the value does not depend on the lift chosen, that being the well-definedness already established for $\tilde{\partial}$ in the target sequence. Hence the square through ∂ commutes, completing the proof.

The name given to $\operatorname{coker} d^A$ in definition 57.5.2 has so far been unearned: nothing above identifies it with a derived functor in the sense of definition 56.3.5 and the right-derived construction recorded alongside it. The identification is genuine, and it is what licenses both the notation and the assertion made after theorem 57.5.3 that the sequence terminates.

Proposition 57.5.4: $\varprojlim^1$ Is the First Derived Functor of the Limit

The category of towers of abelian groups has enough injectives, and the functor $\varprojlim$ is left exact. Writing $R^n \varprojlim$ for its right derived functors,

$$R^0 \varprojlim \cong \varprojlim, \qquad R^1 \varprojlim \cong \varprojlim{}^1, \qquad R^n \varprojlim = 0 \text{ for } n \geq 2$$

***Proof Key ideas*:**
Two ingredients are used here and not proved: that the category of towers has enough injectives, and that the right derived functors of $\varprojlim$ are computed by the two-term complex

$$P(A) \xrightarrow{d^A} P(A)$$

placed in degrees 0 and 1. Both are established in [53, §3.5]. The deferral is deliberate rather than incidental: constructing the injectives is long, and nothing in this book or in the books citing this section uses more than the two-term description, which definition 57.5.2 and theorem 57.5.3 supply on their own.

Granting the two ingredients, the rest is a reading-off. Left exactness of $\varprojlim$ is the left half of theorem 57.5.3, proved above without any of this. The displayed complex has $H^0 = \ker d^A = \varprojlim A_\bullet$ and $H^1 = \operatorname{coker} d^A = \varprojlim^1 A_\bullet$ by definition 57.5.2, and it is concentrated in degrees 0 and 1, so $H^n = 0$ for $n \geq 2$. These three groups are the values of $R^0 \varprojlim$, $R^1 \varprojlim$ and $R^n \varprojlim$ for $n \geq 2$, which is the statement.

With the sequence in hand the practical question is when the obstruction vanishes. The condition is not that the transition maps be surjective, which would be too strong to be useful, but that the images they produce inside any fixed level eventually stop shrinking.

Definition 57.5.5: Mittag-Leffler Condition

A tower $A_\bullet$ satisfies the *Mittag-Leffler condition* if for every n the decreasing chain of subgroups

$$A_n = \operatorname{im} f_{n,n} \supseteq \operatorname{im} f_{n,n+1} \supseteq \operatorname{im} f_{n,n+2} \supseteq \cdots$$

is eventually constant: there is an $m_0 \geq n$ with $\operatorname{im} f_{n,m} = \operatorname{im} f_{n,m_0}$ for all $m \geq m_0$. The common value $A'_n := \bigcap_{m \geq n} \operatorname{im} f_{n,m}$ is the *stable image* in level n.

The condition is deliberately weak. It permits the transition maps to be far from surjective, and it permits the stable image to be a proper subgroup of A_n; it forbids only an infinite strictly decreasing chain of images inside a single level. A tower whose transition maps are all surjective satisfies it with $A'_n = A_n$, and a tower whose transition maps are eventually zero in each level satisfies it with $A'_n = 0$.

Theorem 57.5.6: Mittag-Leffler Vanishing

If a tower $A_\bullet$ of abelian groups satisfies the Mittag-Leffler condition, then $\varprojlim^1 A_\bullet = 0$.

***Proof*:**
The claim is that d^A is surjective. Two special cases are treated first, and the general case reduces to them.

Step 1: Surjective transition maps. Suppose every f_{n+1} is surjective, and let $(b_n) \in P(A)$ be given. Set $a_0 := 0$, and having chosen a_n, use surjectivity of f_{n+1} to choose $a_{n+1} \in A_{n+1}$ with $f_{n+1}(a_{n+1}) = a_n - b_n$. Then

$$a_n - f_{n+1}(a_{n+1}) = a_n - (a_n - b_n) = b_n$$

for every n, so $d^A((a_n)) = (b_n)$ and d^A is surjective.

Step 2: Transition maps eventually zero in each level. Suppose instead that for each n there is an $m_1 \geq n$ with $f_{n,m} = 0$ for all $m \geq m_1$. Given $(b_n) \in P(A)$, the sum

$$a_n := \sum_{k \geq n} f_{n,k}(b_k)$$

has only finitely many nonzero terms and so defines an element of A_n. Since $f_{n+1} \circ f_{n+1,k} = f_{n,k}$ for $k \geq n+1$,

$$f_{n+1}(a_{n+1}) = \sum_{k \geq n+1} f_{n,k}(b_k) = a_n - f_{n,n}(b_n) = a_n - b_n$$

so again $a_n - f_{n+1}(a_{n+1}) = b_n$ and d^A is surjective.

Step 3: The general Mittag-Leffler case. Let A'_n be the stable image of definition 57.5.5, and let $m_0(n)$ be an index with $\mathrm{im} f_{n,m} = A'_n$ for all $m \geq m_0(n)$. The subgroups A'_n form a subtower: for $m \geq n+1$ one has $f_{n+1}(\mathrm{im} f_{n+1,m}) = \mathrm{im} f_{n,m}$, so $f_{n+1}(A'_{n+1}) \subseteq \bigcap_{m \geq n+1} \mathrm{im} f_{n,m} = A'_n$.

That subtower has surjective transition maps. Let $x \in A'_n$ and choose $m \geq \max\{m_0(n), m_0(n+1)\}$. Since $x \in \mathrm{im} f_{n,m}$ there is $z \in A_m$ with $f_{n,m}(z) = x$; put $y := f_{n+1,m}(z)$. Then $y \in \mathrm{im} f_{n+1,m} = A'_{n+1}$ because $m \geq m_0(n+1)$, and $f_{n+1}(y) = f_{n+1}(f_{n+1,m}(z)) = f_{n,m}(z) = x$. So $f_{n+1} \colon A'_{n+1} \to A'_n$ is onto, and Step 1 gives $\varprojlim^1 A'_\bullet = 0$.

The quotient tower $(A_n/A'_n)_n$ has transition maps that are eventually zero in each level: for $m \geq m_0(n)$ the image of $A_m/A'_m \to A_n/A'_n$ is $(\mathrm{im} f_{n,m} + A'_n)/A'_n = A'_n/A'_n = 0$. So Step 2 gives $\varprojlim^1 (A_\bullet/A'_\bullet) = 0$.

Now apply theorem 57.5.3 to the short exact sequence of towers $0 \to A'_\bullet \to A_\bullet \to A_\bullet/A'_\bullet \to 0$, which is exact degreewise by construction. Its last three terms read

$$\varprojlim{}^1 A'_\bullet \to \varprojlim{}^1 A_\bullet \to \varprojlim{}^1 (A_\bullet/A'_\bullet) \to 0$$

with the two outer groups zero, so $\varprojlim^1 A_\bullet = 0$, completing the proof.

Corollary 57.5.7: Surjective Towers Have Vanishing $\varprojlim^1$

If every transition map of a tower $A_\bullet$ is surjective, then $\varprojlim^1 A_\bullet = 0$. Consequently, for a short exact sequence of towers $0 \to A_\bullet \to B_\bullet \to C_\bullet \to 0$ in which $A_\bullet$ has surjective transition maps, the sequence of limits

$$0 \to \varprojlim A_\bullet \to \varprojlim B_\bullet \to \varprojlim C_\bullet \to 0$$

is exact.

***Proof*:**
A surjective tower satisfies the Mittag-Leffler condition with $A'_n = A_n$, so theorem 57.5.6 applies; equivalently this is Step 1 of that proof on its own. Substituting $\varprojlim^1 A_\bullet = 0$ into theorem 57.5.3 makes the connecting map ∂ land in the zero group, so the first four terms of the six-term sequence give the displayed short exact sequence, as we sought to show.

The corollary is the form in which the result is usually met, and it is the statement proved directly, without the surrounding theory, in proposition 25.1.3, where it is what makes completion along a surjective system an exact operation. The theory developed here names the group that appears once the surjectivity hypothesis is dropped: the cokernels left unlabelled in that argument are the $\varprojlim^1$ terms.

Example 57.5.8: The Tower of Residue Rings $\mathbb{Z}/p^n\mathbb{Z}$
Fix a prime p and let $T_\bullet$ be the tower $T_n = \mathbb{Z}/p^n\mathbb{Z}$ with transition maps the natural reductions $\mathbb{Z}/p^{n+1}\mathbb{Z} \twoheadrightarrow \mathbb{Z}/p^n\mathbb{Z}$ (so $T_0 = 0$). A coherent family is a sequence of residues agreeing under reduction, which is precisely a p-adic integer, so

$$\varprojlim T_\bullet = \mathbb{Z}_p$$

Every transition map is surjective, so corollary 57.5.7 gives $\varprojlim^1 T_\bullet = 0$. This is the typical situation: the obstruction is absent and the limit behaves as expected.

The next example shows that the obstruction is not always absent, and that it can be large. It is worth working through, because a reader who meets only towers with surjective transition maps will reasonably conclude that $\varprojlim^1$ is a formality.

Example 57.5.9: A Tower With Nonvanishing $\varprojlim^1$
Keep p fixed and let $S_\bullet$ be the tower of subgroups $S_n = p^n\mathbb{Z} \subseteq \mathbb{Z}$, with transition maps the inclusions $p^{n+1}\mathbb{Z} \hookrightarrow p^n\mathbb{Z}$. Rescaling by $p^n\mathbb{Z} \cong \mathbb{Z}$ identifies $S_\bullet$ with the tower

$$\mathbb{Z} \xleftarrow{p} \mathbb{Z} \xleftarrow{p} \mathbb{Z} \xleftarrow{p} \cdots$$

Mittag-Leffler fails: inside $S_n = p^n\mathbb{Z}$ the image $\mathrm{im} f_{n,m}$ is $p^m\mathbb{Z}$, and these subgroups shrink strictly as m grows, so the chain in definition 57.5.5 is never constant. The limit vanishes, since $\varprojlim S_\bullet \subseteq \bigcap_{n\geq 0} p^n\mathbb{Z} = 0$.

To compute $\varprojlim^1 S_\bullet$, let $\underline{\mathbb{Z}}$ denote the constant tower with $\underline{\mathbb{Z}}_n = \mathbb{Z}$ and all transition maps the identity, and let $T_\bullet$ be the tower of example 57.5.8. The quotient maps $\mathbb{Z} \to \mathbb{Z}/p^n\mathbb{Z}$ assemble into

a short exact sequence of towers
$$0 \to S_\bullet \to \underline{\mathbb{Z}} \to T_\bullet \to 0$$
exact in each level because $p^n\mathbb{Z}$ is the kernel of $\mathbb{Z} \to \mathbb{Z}/p^n\mathbb{Z}$, and compatible with the transition maps by construction. The identity transition maps of $\underline{\mathbb{Z}}$ are surjective, so $\varprojlim \underline{\mathbb{Z}} = \mathbb{Z}$ and $\varprojlim^1 \underline{\mathbb{Z}} = 0$ by corollary 57.5.7; and $\varprojlim T_\bullet = \mathbb{Z}_p$ with $\varprojlim^1 T_\bullet = 0$ by example 57.5.8. Feeding all of this into theorem 57.5.3 leaves
$$0 \to 0 \to \mathbb{Z} \to \mathbb{Z}_p \xrightarrow{\partial} \varprojlim{}^1 S_\bullet \to 0 \to 0 \to 0$$
where the map $\mathbb{Z} \to \mathbb{Z}_p$ is the canonical inclusion. Exactness identifies ∂ as a surjection whose kernel is the image of $\mathbb{Z}$, so
$$\varprojlim{}^1 S_\bullet \cong \mathbb{Z}_p/\mathbb{Z}$$
This is nonzero, and not by a small margin: $\mathbb{Z}_p$ is uncountable while $\mathbb{Z}$ is countable, so the quotient is uncountable. The failure of $\varprojlim$ to be exact on this sequence is exactly the failure of the inclusion $\mathbb{Z} \hookrightarrow \mathbb{Z}_p$ to be onto.

Exercise 57.5.1

1. Let A be an abelian group and $\underline{A}$ the constant tower with every level A and every transition map the identity. Compute $\varprojlim \underline{A}$ and $\varprojlim^1 \underline{A}$ directly from definition 57.5.2, without invoking corollary 57.5.7, by exhibiting a solution of $d^{\underline{A}}((a_n)) = (b_n)$ for an arbitrary (b_n).

2. Show that a tower in which every A_n is a finite group satisfies the Mittag-Leffler condition, and conclude that $\varprojlim^1 A_\bullet = 0$. Where does the argument fail for a tower of finitely generated abelian groups?

3. Give a morphism of towers $u\colon B_\bullet \to C_\bullet$ that is surjective in every level but for which $\varprojlim B_\bullet \to \varprojlim C_\bullet$ is not surjective. Name the group in theorem 57.5.3 that receives the resulting obstruction, and say which tower it is computed from.

4. Let $A_\bullet$ be a tower and let $A_\bullet^{(2)}$ be the tower $A_0 \leftarrow A_2 \leftarrow A_4 \leftarrow \cdots$ obtained by keeping the even levels and composing transition maps in pairs. Show that $\varprojlim A_\bullet \cong \varprojlim A_\bullet^{(2)}$ and that $A_\bullet$ satisfies the Mittag-Leffler condition if and only if $A_\bullet^{(2)}$ does.

5. Let $A_\bullet$ be the tower of subgroups $A_n = (n!)\mathbb{Z} \subseteq \mathbb{Z}$ with the inclusions as transition maps, so that rescaling identifies it with $\mathbb{Z} \xleftarrow{1} \mathbb{Z} \xleftarrow{2} \mathbb{Z} \xleftarrow{3} \cdots$, the n-th transition map being multiplication by n. Show that Mittag-Leffler fails and that $\varprojlim A_\bullet = 0$, then run the argument of example 57.5.9 with $\mathbb{Z}/(n!)\mathbb{Z}$ in place of $\mathbb{Z}/p^n\mathbb{Z}$ to compute $\varprojlim^1 A_\bullet$. The limit of the quotient tower is the profinite completion $\widehat{\mathbb{Z}}$ rather than $\mathbb{Z}_p$; say which property of the sequence $(n!)$ is responsible, and give $\varprojlim^1 A_\bullet$ itself.

6. In the situation of example 57.5.9, make the connecting map $\partial\colon \mathbb{Z}_p \to \varprojlim^1 S_\bullet$ explicit by running the construction of ∂ from theorem 57.5.3: given a p-adic integer presented as a coherent family of residues, produce a representative in $\prod_n S_n$ of the class it maps to, and check that the class does not depend on the lifts chosen.

58

Spectral Sequences

The derived functors of chapter 56 measure the failure of a single functor to be exact, and the long exact sequence of theorem 56.3.10 turns that measurement into something computable. Both tools carry the same limitation: each relates the cohomology of one object to the cohomology of two others, and no more. An object assembled from many pieces produces not one long exact sequence but a whole family of them at once, one for each stage of the assembly, and no finite amount of splicing turns that family into an answer. A module carrying a filtration $M = M_0 \supseteq M_1 \supseteq \cdots$ (section 25.2) is the basic case; a functor that factors through a third abelian category is another, and the double complexes of section 56.4 are a third.

A spectral sequence is the bookkeeping that assembles such a family. It produces a sequence of *pages* $E_1, E_2, E_3, \ldots$, each the cohomology of the one before it, beginning with a crude first approximation to the answer and correcting it repeatedly until the corrections run out. This chapter builds the machine and then supplies the three standard sources of one. Section 58.1 isolates the algebraic mechanism, Massey's exact couple, from which every page and every differential is generated by a single operation. Section 58.2 constructs the spectral sequence of a filtered complex and proves the convergence and comparison theorems that make it usable. Section 58.3 treats the two spectral sequences of a double complex, completing the computations of section 56.4.1 that were deferred to this chapter. Section 58.4 constructs the spectral sequence of a composite of derived functors, the form in which the machine is most often applied. Section 58.5 closes with hypercohomology, the derived functors of a functor applied to a complex rather than an object.

58.1 Exact Couples and Derived Couples

The obstacle a spectral sequence exists to overcome is visible already in the simplest case. A short exact sequence of complexes $0 \to A_\bullet \to C_\bullet \to C_\bullet/A_\bullet \to 0$ yields a long exact sequence relating the three cohomologies, but that sequence does not determine $H_\bullet(C)$ from $H_\bullet(A)$ and $H_\bullet(C/A)$: it determines only the two ends of a short exact sequence

$$0 \to \operatorname{coker}\partial \to H_n(C) \to \ker \partial \to 0$$

and the extension itself is extra information the sequence does not carry. With one layer the ambiguity is a single extension problem. With a filtration of many layers there is one such problem per layer, and there are also correction terms of a different kind: a cohomology class of one layer may fail to extend over the next layer but succeed over the one after, and nothing in any individual long exact sequence detects that.

The device that organizes all of these at once packages every long exact sequence of the filtration into a single self-regenerating triangle. Its two vertices carry different roles: one is the answer being approximated, the other the auxiliary object that remembers how the layers are glued.

Definition 58.1.1: Exact Couple

Let $\mathcal{A}$ be an abelian category. An *exact couple* in $\mathcal{A}$ is a pair of objects D and E together with three morphisms

$$\begin{tikzcd} D \arrow[rr, "i"] & & D \arrow[dl, "j"] \\ & E \arrow[ul, "k"] & \end{tikzcd}$$

such that the triangle is exact at each of its three corners:

$$\operatorname{im} i = \ker j, \qquad \operatorname{im} j = \ker k, \qquad \operatorname{im} k = \ker i$$

The composite $d := j \circ k \colon E \to E$ is the *differential* of the couple, and the couple is written (D, E, i, j, k).

The data should be read as a compression of a long exact sequence, or rather of all of them simultaneously. In the filtered case treated in section 58.2, the object D collects the cohomologies of the stages of the filtration, the object E collects the cohomologies of the successive layers, the map i is induced by the inclusion of one stage in the next, the map j is the passage to the layer, and k is the connecting homomorphism. Exactness at the three corners is precisely the exactness of every one of those long exact sequences. Of the two objects only E is meant to be looked at, since it is the approximation to the answer; D is the auxiliary object, and it is the auxiliary object that makes the approximation improvable.

Two of the three exactness conditions are spent constructing the next stage, and one is spent immediately, on the claim that d deserves its name.

Lemma 58.1.2: The Differential of an Exact Couple Squares to Zero

Let (D, E, i, j, k) be an exact couple. Then $d \circ d = 0$, so (E, d) is a differential object and its cohomology

$$H(E, d) = \frac{\ker d}{\operatorname{im} d}$$

is defined.

***Proof*:**
Expand the composite and regroup:

$$d \circ d = (j \circ k) \circ (j \circ k) = j \circ (k \circ j) \circ k$$

Exactness at the corner E gives $\mathrm{im} j = \ker k$, so every morphism factoring through j is annihilated by k, that is $k \circ j = 0$. Hence $d \circ d = j \circ 0 \circ k = 0$, as we sought to show.

Only exactness at E was used. The remaining two conditions are exactly what is needed to promote $H(E, d)$ from a bare cohomology object to the E-term of a new exact couple, and it is that promotion which makes the construction repeat. The new auxiliary object is the image of i, and the new maps are induced by the old ones.

58.1.3 Convention: Elements in an Abelian Category The constructions below are phrased with elements, in the style of a diagram chase. This is legitimate in any abelian category: by the Freyd-Mitchell theorem (theorem 55.1.15), a small abelian category admits a fully faithful exact embedding into R-Mod for some ring R, and a diagram whose exactness is verified in R-Mod is exact in $\mathcal{A}$ because the embedding reflects exactness. Every statement in this section may therefore be proved for modules and read off for a general $\mathcal{A}$. The standard sources of exact couples, filtered complexes and double complexes of modules or of sheaves, are in any case module-like.

Definition 58.1.4: Derived Couple

Let (D, E, i, j, k) be an exact couple with differential $d = j \circ k$. Its *derived couple* (D', E', i', j', k') consists of the objects

$$D' := \mathrm{im} i, \qquad E' := H(E, d) = \frac{\ker d}{\mathrm{im} d}$$

together with the morphisms

1. $i' := i|_{D'} : D' \to D'$, the restriction of i to its image;
2. $j' : D' \to E'$ defined by $j'(i(x)) := [\,j(x)\,]$ for $x \in D$;
3. $k' : E' \to D'$ defined by $k'([e]) := k(e)$ for $e \in \ker d$.

The name is older than, and unrelated to, the derived categories of section 56.5; it originates in Massey's work in algebraic topology. Three of the four items above require an argument before the definition is even meaningful: that i carries D' into itself, that j' is independent of the choice of preimage, and that k' both lands in D' and is independent of the representative cycle. Those checks and the exactness of the result are one theorem.

Theorem 58.1.5: The Derived Couple is an Exact Couple

Let (D, E, i, j, k) be an exact couple. Then the morphisms i', j' and k' of definition 58.1.4 are well defined, and the triangle

$$\begin{tikzcd} D' \arrow[rr, "i'"] & & D' \arrow[dl, "j'"] \\ & E' \arrow[ul, "k'"] & \end{tikzcd}$$

is an exact couple.

***Proof*:**
Step 1: The three maps are well defined. For i', note that $i(D') = i(i(D)) \subseteq i(D) = D'$, so restricting i to D' does indeed produce an endomorphism of D'.

For j', two things must be checked. First, $j(x)$ is a cycle for every $x \in D$: applying the differential gives $d(j(x)) = j(k(j(x))) = 0$, since $k \circ j = 0$ by exactness at E. So the class $[j(x)] \in E'$ makes sense. Second, the class does not depend on the choice of x. Suppose $i(x) = i(x')$. Then $x - x' \in \ker i = \mathrm{im} k$ by exactness at the target corner, say $x - x' = k(e)$ for some $e \in E$. Applying j gives

$$j(x) - j(x') = j(k(e)) = d(e) \in \mathrm{im} d$$

so $[j(x)] = [j(x')]$ in $E' = \ker d / \mathrm{im} d$.

For k', again two checks. First, k carries cycles into D': if $d(e) = 0$ then $j(k(e)) = 0$, so $k(e) \in \ker j = \mathrm{im} i = D'$ by exactness at the source corner. Second, the value is independent of the representative. If $e - e' = d(f) = j(k(f))$ for some $f \in E$, then

$$k(e) - k(e') = k(j(k(f))) = (k \circ j)(k(f)) = 0$$

again because $k \circ j = 0$. So k' is well defined.

Step 2: Exactness at the corner receiving i', that is $\mathrm{im} i' = \ker j'$. For the inclusion $\mathrm{im} i' \subseteq \ker j'$, take $y = i(x) \in D'$ and compute

$$j'(i'(y)) = j'(i(i(x))) = [\, j(i(x))\,] = [0] = 0$$

the last equality because $\mathrm{im} i = \ker j$. For the reverse inclusion, let $y = i(x) \in D'$ satisfy $j'(y) = 0$, that is $[j(x)] = 0$, so $j(x) = d(e) = j(k(e))$ for some $e \in E$. Then $j(x - k(e)) = 0$, so $x - k(e) \in \ker j = \mathrm{im} i$, say $x - k(e) = i(z)$. Applying i and using $i \circ k = 0$ (exactness at the corner where k arrives) gives

$$y = i(x) = i(k(e)) + i(i(z)) = i(i(z)) = i'(i(z))$$

and $i(z) \in D'$, so $y \in \mathrm{im} i'$.

Step 3: Exactness at E', that is $\mathrm{im} j' = \ker k'$. For $\mathrm{im} j' \subseteq \ker k'$, take $y = i(x) \in D'$ and compute

$$k'(j'(y)) = k'([\, j(x)\,]) = k(j(x)) = 0$$

using $k \circ j = 0$. For the reverse inclusion, let $[e] \in E'$ satisfy $k'([e]) = k(e) = 0$. Then $e \in \ker k = \mathrm{im} j$, say $e = j(x)$ for some $x \in D$. The element $i(x)$ lies in D' and $j'(i(x)) = [j(x)] = [e]$, so $[e] \in \mathrm{im} j'$.

Step 4: Exactness at the corner receiving k', that is $\mathrm{im} k' = \ker i'$. For $\mathrm{im} k' \subseteq \ker i'$, take $[e] \in E'$; then $i'(k'([e])) = i(k(e)) = 0$ since $\mathrm{im} k = \ker i$. For the reverse inclusion, let $y = i(x) \in D'$ satisfy $i'(y) = 0$, that is $i(i(x)) = 0$. Then $i(x) \in \ker i = \mathrm{im} k$, say $i(x) = k(e)$ for some $e \in E$. This e is a cycle:

$$d(e) = j(k(e)) = j(i(x)) = 0$$

because $\mathrm{im} i = \ker j$. Hence $[e] \in E'$ and $k'([e]) = k(e) = i(x) = y$, so $y \in \mathrm{im} k'$.

All six inclusions hold, so the derived triangle is exact at each corner, completing the proof.

The theorem is the basis of the entire chapter. Because the derived couple is again an exact couple, the construction applies to it in turn, and iterating produces an infinite sequence of couples whose E-terms form a sequence of objects each of which is the cohomology of its predecessor. That sequence is the object the chapter is named for.

Definition 58.1.6: Spectral Sequence

Let $\mathcal{A}$ be an abelian category. A *spectral sequence* in $\mathcal{A}$ is a family $\{(E_r, d_r)\}_{r \geq r_0}$ of objects E_r of $\mathcal{A}$, each equipped with an endomorphism $d_r \colon E_r \to E_r$ satisfying $d_r \circ d_r = 0$, together with isomorphisms

$$E_{r+1} \cong H(E_r, d_r) = \frac{\ker d_r}{\mathrm{im} d_r}$$

for every $r \geq r_0$. The object E_r is the *r-th page*, d_r is the *r-th differential*, and r_0 is the *initial page.*

The spectral sequence is *bigraded* if each E_r carries a decomposition $E_r = \bigoplus_{p,q} E_r^{p,q}$ and each d_r is homogeneous of a fixed bidegree. Two conventions are in use. In the *homological* convention the pages are written $E^r_{p,q}$ and

$$d_r \colon E^r_{p,q} \to E^r_{p-r,\, q+r-1}$$

while in the *cohomological* convention they are written $E_r^{p,q}$ and

$$d_r \colon E_r^{p,q} \to E_r^{p+r,\, q-r+1}$$

Both differentials change the total degree $p+q$ by exactly one, downward in the homological convention and upward in the cohomological one. The two are exchanged by $E_r^{p,q} = E^r_{-p,-q}$.

The bidegrees are not a separate convention imposed on the definition; they are forced by the exact couple that generates the sequence, as the next result records. Every spectral sequence appearing in this book, and every one cited from it elsewhere in the series, arises from an exact couple in the manner described.

Theorem 58.1.7: The Spectral Sequence of an Exact Couple

Let $(D, E, i, j, k) = (D^1, E^1, i^{(1)}, j^{(1)}, k^{(1)})$ be an exact couple, and for $r \geq 1$ let $(D^{r+1}, E^{r+1}, i^{(r+1)}, j^{(r+1)}, k^{(r+1)})$ be the derived couple of the r-th one. Then $\{(E^r, d_r)\}_{r\geq 1}$, with $d_r = j^{(r)} \circ k^{(r)}$, is a spectral sequence, and $D^{r+1} = i^r(D)$ for every r.
Suppose in addition that D and E are bigraded and that i, j, k are homogeneous of bidegrees

$$\deg i = (1, -1), \qquad \deg j = (0, 0), \qquad \deg k = (-1, 0)$$

with the pages of the derived couples indexed by $D^{r+1}_{p,q} = i\big(D^r_{p-1,q+1}\big)$. Then

$$\deg i^{(r)} = (1, -1), \qquad \deg j^{(r)} = \big(-(r-1),\, r-1\big), \qquad \deg k^{(r)} = (-1, 0)$$

and consequently d_r is homogeneous of bidegree $(-r,\, r-1)$, so the sequence is bigraded in the homological convention of definition 58.1.6.

***Proof*:**
Step 1: The pages form a spectral sequence. By theorem 58.1.5 each derived couple is again an exact couple, so the construction may be iterated and every d_r satisfies $d_r \circ d_r = 0$ by lemma 58.1.2. By definition 58.1.4 the E-term of the derived couple is $E^{r+1} = H(E^r, d_r)$, which is the required isomorphism. The identity $D^{r+1} = i^r(D)$ follows by induction: it holds for $r = 0$ by convention, and if $D^r = i^{r-1}(D)$ then $D^{r+1} = i^{(r)}(D^r) = i(i^{r-1}(D)) = i^r(D)$, since $i^{(r)}$ is the restriction of i.

Step 2: The bidegree of $i^{(r)}$ and $k^{(r)}$. Each $i^{(r)}$ is a restriction of i to a subobject, so it is homogeneous of the same bidegree $(1, -1)$ for every r. Similarly $E^{r+1}_{p,q}$ is a subquotient of $E^r_{p,q}$ in the same bidegree, and $k^{(r+1)}$ is induced by $k^{(r)}$ on representatives, so $\deg k^{(r)} = (-1, 0)$ for every r by induction.

Step 3: The bidegree of $j^{(r)}$ shifts by $(-1, 1)$ at each stage. Fix r and suppose $\deg j^{(r)} = (a, b)$. By definition 58.1.4 the map $j^{(r+1)}$ is defined on an element of D^{r+1} written as $i(x)$ with $x \in D^r$, by $j^{(r+1)}(i(x)) = [\,j^{(r)}(x)\,]$. With the stated indexing, an element of $D^{r+1}_{p,q}$ is of the form $i(x)$ for $x \in D^r_{p-1,q+1}$, and $j^{(r)}(x)$ then lies in $E^r_{p-1+a,\,q+1+b}$. Hence $j^{(r+1)}$ carries $D^{r+1}_{p,q}$ into $E^{r+1}_{p+a-1,\,q+b+1}$, that is $\deg j^{(r+1)} = (a-1, b+1)$. Since $\deg j^{(1)} = (0, 0)$, induction gives $\deg j^{(r)} = (-(r-1), r-1)$.

Step 4: The bidegree of d_r. Combining Steps 2 and 3,

$$\deg d_r = \deg j^{(r)} + \deg k^{(r)} = \big(-(r-1),\, r-1\big) + (-1, 0) = (-r,\, r-1)$$

so $d_r \colon E^r_{p,q} \to E^r_{p-r,\,q+r-1}$, which is the homological convention of definition 58.1.6, completing the proof.

The bidegree computation is worth pausing over, because it explains what the higher pages are for. The differential d_1 moves one step in the filtration direction, d_2 moves two steps, and d_r moves r steps. Each derivation of the couple replaces D by $i(D)$, which is to say it discards the classes that survive only one stage further and keeps those that survive r stages; the reach of the differential grows in step with that. A class killed by d_r and by no earlier differential encodes a gluing that spans exactly r layers of the filtration, information no single long exact sequence of the family can see. This is the precise sense in which the spectral sequence contains more than the collection of long exact sequences it was built from.

The simplest instance is a filtration with only two stages, where there is a single differential and the machine reduces to something already familiar.

Proposition 58.1.8: The Two-Step Filtration Recovers the Long Exact Sequence

Let $0 \to A_\bullet \xrightarrow{\iota} C_\bullet \xrightarrow{\pi} C_\bullet/A_\bullet \to 0$ be a short exact sequence of complexes in $\mathcal{A}$, and let $D = H_\bullet(A) \oplus H_\bullet(C)$ and $E = H_\bullet(A) \oplus H_\bullet(C/A)$ be the terms of the exact couple attached to the two-step filtration $0 \subseteq A_\bullet \subseteq C_\bullet$. Then:

1. the first page is $E^1 = H_\bullet(A) \oplus H_\bullet(C/A)$, with d_1 the connecting homomorphism $\partial\colon H_n(C/A) \to H_{n-1}(A)$ of theorem 56.1.15;
2. $E^2 = E^\infty$, that is $d_r = 0$ for all $r \geq 2$;
3. E^2 is the associated graded of $H_\bullet(C)$ for the induced two-step filtration, namely

$$E^2 \cong \operatorname{coker}\partial \oplus \ker\partial$$

with $\operatorname{coker}\partial$ the image of $H_n(A) \to H_n(C)$ and $\ker\partial$ its quotient.

***Proof*:**
Step 1: Identification of the first page. The filtration has stages $F_0 = A_\bullet$ and $F_1 = C_\bullet$, with $F_{-1} = 0$ and $F_p = C_\bullet$ for $p \geq 1$. The layers are $F_0/F_{-1} = A_\bullet$ and $F_1/F_0 = C_\bullet/A_\bullet$, and all other layers vanish. Since E^1 is by construction the sum of the cohomologies of the layers, $E^1 = H_\bullet(A) \oplus H_\bullet(C/A)$. The map k is the connecting homomorphism of the long exact sequence attached to the layer and j is the projection onto the layer, so on the summand $H_n(C/A)$ the composite $d_1 = j \circ k$ is the connecting homomorphism ∂ followed by the identification of $H_{n-1}(A)$ with the layer at filtration level zero, which is the identity. Hence $d_1 = \partial$ on that summand, and $d_1 = 0$ on $H_\bullet(A)$ because k vanishes there (its target is the cohomology of a layer at filtration level -1, which is zero).

Step 2: Degeneration after the second page. By theorem 58.1.7 the differential d_r has bidegree $(-r, r-1)$, so it decreases the filtration index by r. Only the filtration indices 0 and 1 carry nonzero terms, so for $r \geq 2$ the source and target of d_r cannot both be nonzero: a nonzero source at index 1 would have target at index $1 - r \leq -1$, and a source at index 0 would have target at index $-r < 0$. Hence $d_r = 0$ for $r \geq 2$ and the pages stabilize at E^2.

Step 3: Identification of the limit. By Step 1 the second page is the cohomology of $d_1 = \partial$, which is $\ker\partial$ in filtration index 1 and $\operatorname{coker}\partial$ in filtration index 0. The long exact sequence of theorem 56.1.15 reads

$$\cdots \to H_n(A) \xrightarrow{\iota_*} H_n(C) \xrightarrow{\pi_*} H_n(C/A) \xrightarrow{\partial} H_{n-1}(A) \to \cdots$$

Exactness at $H_n(C)$ gives $\mathrm{im}\iota_* = \ker \pi_*$, and exactness at $H_n(A)$ identifies $\mathrm{im}\iota_* \cong H_n(A)/\mathrm{im}\partial = \mathrm{coker}\partial$ in degree n; exactness at $H_n(C/A)$ identifies the quotient $H_n(C)/\mathrm{im}\iota_* \cong \mathrm{im}\pi_* = \ker \partial$. These two are precisely the graded pieces of $H_n(C)$ for the filtration induced by $A_\bullet \subseteq C_\bullet$, and they agree with the two summands of E^2 computed above, as we sought to show.

The proposition is the sanity check for the whole apparatus: in the one case where the answer is already known, the spectral sequence returns exactly the long exact sequence and nothing more. It also isolates what the machine does *not* do. The second page records $\ker \partial$ and $\mathrm{coker}\partial$ as separate graded pieces, but the extension

$$0 \to \mathrm{coker}\partial \to H_n(C) \to \ker \partial \to 0$$

is not part of the output. A spectral sequence delivers the associated graded of the answer, never the answer itself; recovering the latter requires solving the extension problems left behind. This is a permanent feature, not a defect of the two-step case, and it is stated in general in theorem 58.2.6.

A worked instance makes the bookkeeping concrete.

Example 58.1.9: The Spectral Sequence of a Filtered Complex of Abelian Groups

Let $C_\bullet$ be the complex of abelian groups concentrated in degrees 1 and 0,

$$C_1 = \mathbb{Z} \xrightarrow{\cdot 2} C_0 = \mathbb{Z}$$

and let $A_\bullet \subseteq C_\bullet$ be the subcomplex with $A_1 = 0$ and $A_0 = 2\mathbb{Z}$. This is a subcomplex because $\partial(A_1) = 0 \subseteq A_0$. Filter by $F_0 = A_\bullet$ and $F_1 = C_\bullet$.

The cohomology of $C_\bullet$ itself is immediate: $H_1(C) = \ker(\cdot 2) = 0$ and $H_0(C) = \mathbb{Z}/2\mathbb{Z}$. The spectral sequence recomputes this from the two layers. The layer at level 0 is $A_\bullet$, with $H_0(A) = 2\mathbb{Z} \cong \mathbb{Z}$ and $H_1(A) = 0$. The layer at level 1 is $C_\bullet/A_\bullet$, which is $\mathbb{Z}$ in degree 1 and $\mathbb{Z}/2\mathbb{Z}$ in degree 0, with induced differential $1 \mapsto 2 \bmod 2 = 0$; so $H_1(C/A) = \mathbb{Z}$ and $H_0(C/A) = \mathbb{Z}/2\mathbb{Z}$. Indexing by $E^1_{p,q} = H_{p+q}(F_p/F_{p-1})$, the nonzero entries of the first page are

$$E^1_{0,0} = H_0(A) \cong \mathbb{Z}, \qquad E^1_{1,0} = H_1(C/A) \cong \mathbb{Z}, \qquad E^1_{1,-1} = H_0(C/A) \cong \mathbb{Z}/2\mathbb{Z}$$

The only possibly nonzero differential on this page is $d_1 \colon E^1_{1,0} \to E^1_{0,0}$, the connecting homomorphism $H_1(C/A) \to H_0(A)$. Compute it on the generator: the class of $1 \in (C/A)_1$ lifts to $1 \in C_1$, whose boundary is $2 \in C_0$, and 2 is the generator of $A_0 = 2\mathbb{Z}$. So d_1 sends a generator to a generator and is an isomorphism $\mathbb{Z} \to \mathbb{Z}$.

The second page therefore has $E^2_{1,0} = E^2_{0,0} = 0$, while $E^2_{1,-1} = \mathbb{Z}/2\mathbb{Z}$ survives untouched (the differential out of the spot $(1,-1)$ lands in $E^1_{0,-1} = H_{-1}(A) = 0$, and the one into it comes from $E^1_{2,-1} = 0$). By proposition 58.1.8 the pages stabilize here. Reading off total degrees: the diagonal $p+q = 1$ is entirely zero, giving $H_1(C) = 0$, and the diagonal $p+q = 0$ contains the single group $E^\infty_{1,-1} = \mathbb{Z}/2\mathbb{Z}$, giving $H_0(C) = \mathbb{Z}/2\mathbb{Z}$. Both agree with the direct computation, and because each diagonal contains at most one nonzero group there is no extension problem to solve.

Not every exact couple comes from a filtration. A single short exact sequence of complexes, read the other way round, produces one whose iteration does not terminate and whose limit is a statement about torsion.

Example 58.1.10: The Bockstein Exact Couple

Let p be a prime and $C_\bullet$ a complex of free abelian groups. Multiplication by p is injective on each C_n, so there is a short exact sequence of complexes

$$0 \to C_\bullet \xrightarrow{\cdot p} C_\bullet \to C_\bullet/pC_\bullet \to 0$$

Its long exact sequence in cohomology is exactly an exact couple, with

$$D = H_\bullet(C), \qquad E = H_\bullet(C/pC), \qquad i = (\cdot p), \qquad j = \text{reduction}, \qquad k = \partial$$

The differential $d_1 = j \circ k$ is the *Bockstein homomorphism*. By theorem 58.1.7 the r-th auxiliary object is $D^{r+1} = i^r(D) = p^r H_\bullet(C)$, so passing to the derived couple repeatedly divides out by higher and higher powers of p. When $H_\bullet(C)$ is finitely generated, an element of p-power torsion is annihilated by p^r for some r and therefore disappears from D^{r+1}, while a free generator survives every stage; the limit page is $\big(H_\bullet(C)/\text{torsion}\big) \otimes \mathbb{F}_p$.

The smallest case exhibits the mechanism. Take $C_1 = \mathbb{Z} \xrightarrow{\cdot p} C_0 = \mathbb{Z}$, so that $H_1(C) = 0$ and $H_0(C) = \mathbb{Z}/p\mathbb{Z}$ is pure torsion. Then $C_\bullet/pC_\bullet$ has zero differential, so $E^1 = H_\bullet(C/pC)$ is $\mathbb{F}_p$ in degrees 1 and 0. The connecting map k sends the degree-one generator to the class of 1 in $H_0(C) = \mathbb{Z}/p\mathbb{Z}$ (lift $\bar{1}$ to $1 \in C_1$, take the boundary p, and divide by p), and j sends that class to a generator of $H_0(C/pC) = \mathbb{F}_p$. So d_1 is an isomorphism $E_1^1 \to E_0^1$, giving $E^2 = 0 = E^\infty$. This matches the general description: $H_\bullet(C)$ is entirely torsion, so $(H_\bullet(C)/\text{torsion}) \otimes \mathbb{F}_p = 0$.

Exercise 58.1.1

1. Let (D, E, i, j, k) be an exact couple in which i is an isomorphism. Show that $E = 0$. Deduce that every page of the associated spectral sequence vanishes.
2. Let (D, E, i, j, k) be an exact couple. Show that $d = j \circ k$ vanishes if and only if $\ker i \subseteq \operatorname{im} i$. Then exhibit an exact couple with $d = 0$ but $k \neq 0$, so that a vanishing differential does not force the connecting map to vanish. (Take $D = \mathbb{Z}/4\mathbb{Z}$ and i multiplication by 2.)
3. Verify directly from definition 58.1.4 that the derived couple of the derived couple has $D'' = i^2(D)$, and that i'' is the restriction of i to $i^2(D)$, without appealing to theorem 58.1.7.
4. Let $C_\bullet$ be the complex $C_1 = \mathbb{Z} \xrightarrow{\cdot 6} C_0 = \mathbb{Z}$ filtered by $F_0 = A_\bullet$ with $A_1 = 0$, $A_0 = 3\mathbb{Z}$, and $F_1 = C_\bullet$. Compute the first page, the differential d_1, and the limit page, and check the answer against a direct computation of $H_\bullet(C)$.
5. In the Bockstein couple of example 58.1.10, take $C_\bullet$ to be $C_1 = \mathbb{Z} \xrightarrow{\cdot p^2} C_0 = \mathbb{Z}$. Show that $d_1 = 0$ but $d_2 \neq 0$, and that $E^3 = E^\infty = 0$. Explain how this illustrates the statement that d_r detects gluing spanning r stages.
6. Let $\mathcal{A}$ be an abelian category and (D, E, i, j, k) an exact couple with $E = 0$. Show that i is an isomorphism, so that the converse of exercise 58.1.1 (1) holds.

58.2 The Spectral Sequence of a Filtered Complex

The exact couples of section 58.1 were treated as abstract data. Their principal source is a complex equipped with a filtration, and this section carries out that construction, describes the pages explicitly enough to compute with, and proves the two theorems that make the output usable: a convergence theorem saying what the limit page computes, and a comparison theorem saying when two spectral sequences agree. These are the results cited from this chapter across the series, so both are proved in the generality in which they are used, for increasing filtrations of chain complexes and, in the restatement at the end of the section, for decreasing filtrations of cochain complexes.

Definition 58.2.1: Filtered Complex

Let $\mathcal{A}$ be an abelian category. A *filtered complex* is a chain complex $C_\bullet$ in $\mathcal{A}$ together with a family of subcomplexes $\{F_pC_\bullet\}_{p\in\mathbb{Z}}$ satisfying $F_pC_\bullet \subseteq F_{p+1}C_\bullet$ for every p. The filtration is called

1. *exhaustive* if $\bigcup_p F_pC_n = C_n$ for every n;
2. *bounded below* if for each n there is an $s(n)$ with $F_{s(n)}C_n = 0$;
3. *bounded* if it is bounded below and for each n there is a $t(n)$ with $F_{t(n)}C_n = C_n$.

The *associated graded* of the filtration is the complex $\mathrm{gr}_pC_\bullet = F_pC_\bullet/F_{p-1}C_\bullet$, and the filtration induced on homology is

$$F_pH_n(C) = \mathrm{im}\big(H_n(F_pC) \to H_n(C)\big)$$

Each inclusion $F_{p-1}C_\bullet \hookrightarrow F_pC_\bullet$ is a subcomplex inclusion in the sense of definition 56.1.7, so $\mathrm{gr}_pC_\bullet$ is again a complex.

Boundedness is the hypothesis that does the work later, and it is worth seeing why before it is used. In each fixed degree n a bounded filtration passes from 0 to all of C_n in finitely many steps, so any bookkeeping indexed by the filtration terminates in that degree after finitely many stages. Without some such condition the pages of the spectral sequence need not stabilize and the limit need not compute anything.

Every filtered complex produces an exact couple in one step: the long exact sequence of each pair $(F_pC_\bullet, F_{p-1}C_\bullet)$ supplies the three maps, and the filtration index supplies the bigrading.

Theorem 58.2.2: The Exact Couple of a Filtered Complex

Let $(C_\bullet, F_\bullet)$ be a filtered complex. Set

$$D^1_{p,q} = H_{p+q}(F_pC), \qquad E^1_{p,q} = H_{p+q}(\mathrm{gr}_pC)$$

and let i be induced by the inclusion $F_{p-1}C \hookrightarrow F_pC$, let j be induced by the projection $F_pC \to \mathrm{gr}_pC$, and let k be the connecting homomorphism of theorem 56.1.15 applied to

$$0 \to F_{p-1}C_\bullet \to F_pC_\bullet \to \mathrm{gr}_pC_\bullet \to 0$$

Then (D^1, E^1, i, j, k) is an exact couple with

$$\deg i = (1, -1), \qquad \deg j = (0, 0), \qquad \deg k = (-1, 0)$$

Consequently theorem 58.1.7 yields a spectral sequence with first page $E^1_{p,q} = H_{p+q}(\mathrm{gr}_pC)$ and differentials $d_r \colon E^r_{p,q} \to E^r_{p-r,\,q+r-1}$.
It is convenient to record one page earlier. Set

$$E^0_{p,q} := \mathrm{gr}_pC_{p+q}$$

with d_0 the differential induced by ∂ on the associated graded; then $E^1 = H(E^0, d_0)$, which is the identification just made, and the sequence is indexed from $r = 0$.

***Proof*:**
Step 1: Exactness. The displayed short exact sequence of complexes gives, by theorem 56.1.15, the long exact sequence

$$\cdots \to H_n(F_{p-1}C) \xrightarrow{i} H_n(F_pC) \xrightarrow{j} H_n(\mathrm{gr}_pC) \xrightarrow{k} H_{n-1}(F_{p-1}C) \to \cdots$$

Assembling these over all p and all n, the three maps i, j, k are defined on all of D^1 and E^1, and exactness of the triangle at each corner is exactness of the long exact sequences at the three types of term. This is exactly definition 58.1.1.

Step 2: Bidegrees. Write $n = p + q$. The map i carries $H_n(F_{p-1}C) = D^1_{p-1,q+1}$ to $H_n(F_pC) = D^1_{p,q}$, since $(p-1) + (q+1) = n$; hence $\deg i = (1, -1)$. The map j carries $H_n(F_pC) = D^1_{p,q}$ to $H_n(\mathrm{gr}_pC) = E^1_{p,q}$, so $\deg j = (0, 0)$. The connecting map k carries $H_n(\mathrm{gr}_pC) = E^1_{p,q}$ to $H_{n-1}(F_{p-1}C) = D^1_{p-1,q}$, since $(p-1) + q = n - 1$; hence $\deg k = (-1, 0)$. These are the hypotheses of theorem 58.1.7, which then gives the spectral sequence and the stated bidegree of d_r, completing the proof.

The construction is quick, but the pages it produces are described only implicitly, as iterated subquotients of the first one. Computing with them, and proving anything about the limit, requires a description of E^r in terms of the original complex. The right notion is that of a chain which is a cycle only up to a given depth in the filtration.

Definition 58.2.3: Approximate Cycles

Let $(C_\bullet, F_\bullet)$ be a filtered complex and write $n = p + q$. For $r \geq 0$ set

$$Z^r_{p,q} = \{\, x \in F_pC_n \;:\; \partial x \in F_{p-r}C_{n-1} \,\}$$

the object of chains at filtration level p whose boundary drops r levels. By convention $Z^{-1}_{p,q} = F_pC_n$.

The objects $Z^r_{p,q}$ decrease in r, and their intersection over all r is, when the filtration is bounded below, the cycles of $C_\bullet$ lying in F_pC_n. An element of $Z^r_{p,q}$ is a class that survives to the r-th page: it is a cycle in the associated graded to accuracy r, and its failure to be a cycle of $C_\bullet$ is what d_r records.

Proposition 58.2.4: Explicit Description of the Pages

Let $(C_\bullet, F_\bullet)$ be a filtered complex and let $\{(E^r, d_r)\}$ be the spectral sequence of theorem 58.2.2. For every $r \geq 0$ there are isomorphisms

$$E^r_{p,q} \cong \frac{Z^r_{p,q}}{Z^{r-1}_{p-1,q+1} + \partial Z^{r-1}_{p+r-1,\,q-r+2}}$$

under which the differential d_r is induced by the boundary map ∂ of $C_\bullet$.

***Proof*:**
Write $\mathcal{E}^r_{p,q}$ for the right-hand side and $n = p + q$. The proof is an induction on r; the inductive statement is that $E^r \cong \mathcal{E}^r$ compatibly with the differentials, the differential on $\mathcal{E}^r$ being the one induced by ∂.

Step 1: The map induced by ∂ is a differential on $\mathcal{E}^r$. If $x \in Z^r_{p,q}$ then $\partial x \in F_{p-r}C_{n-1}$ and $\partial(\partial x) = 0 \in F_{p-2r}C_{n-2}$, so $\partial x \in Z^r_{p-r,\,q+r-1}$. Hence ∂ carries $Z^r_{p,q}$ into $Z^r_{p-r,q+r-1}$, and it carries the denominator at (p, q) into the denominator at $(p-r, q+r-1)$: the summand $Z^{r-1}_{p-1,q+1}$ is sent to $\partial Z^{r-1}_{p-1,q+1}$, which is literally the second summand of the target denominator, and the summand $\partial Z^{r-1}_{p+r-1,q-r+2}$ is sent to zero because $\partial\partial = 0$. So ∂ induces a map $\mathcal{E}^r_{p,q} \to \mathcal{E}^r_{p-r,q+r-1}$ squaring to zero.

Step 2: The base case $r = 0$. With the convention $Z^{-1}_{p,q} = F_pC_n$, the numerator is $Z^0_{p,q} = F_pC_n$ and the denominator is $F_{p-1}C_n + \partial F_{p-1}C_{n+1} = F_{p-1}C_n$, since $\partial F_{p-1}C_{n+1} \subseteq F_{p-1}C_n$. So $\mathcal{E}^0_{p,q} = F_pC_n/F_{p-1}C_n = \mathrm{gr}_pC_n$, which is $E^0_{p,q}$, and the induced differential is the one ∂ induces on the associated graded. Taking homology recovers $E^1_{p,q} = H_n(\mathrm{gr}_pC)$, matching theorem 58.2.2; the differential $d_1 = j \circ k$ is by construction the connecting homomorphism, which on representatives is ∂.

Step 3: The inductive step. Assume $E^r \cong \mathcal{E}^r$ compatibly with differentials. Then $E^{r+1} = H(E^r, d_r) \cong H(\mathcal{E}^r, \partial)$, so it suffices to identify the latter with $\mathcal{E}^{r+1}$. Abbreviate the denominator $D = Z^{r-1}_{p-1,q+1} + \partial Z^{r-1}_{p+r-1,q-r+2}$.

First the kernel. Let $x \in Z^r_{p,q}$ represent a class killed by d_r, so that ∂x lies in the denominator at $(p-r, q+r-1)$, namely $\partial x = z + \partial w$ with $z \in Z^{r-1}_{p-r-1,q+r}$ and $w \in Z^{r-1}_{p-1,q+1}$. Then $x - w$ still lies in F_pC_n, because $w \in F_{p-1}C_n$, and

$$\partial(x-w) = \partial x - \partial w = z \in F_{p-r-1}C_{n-1}$$

so $x - w \in Z^{r+1}_{p,q}$. Since $w \in D$, the classes of x and $x - w$ agree in $\mathcal{E}^r_{p,q}$. Hence $\ker d_r = (Z^{r+1}_{p,q} + D)/D$.

Next the image. A class in $\mathcal{E}^r_{p+r,q-r+1}$ is represented by some $y \in Z^r_{p+r,q-r+1}$, and d_r sends it to the class of ∂y. Hence $\mathrm{im} d_r = (\partial Z^r_{p+r,q-r+1} + D)/D$, and therefore

$$H(\mathcal{E}^r, \partial)_{p,q} = \frac{Z^{r+1}_{p,q} + D}{\partial Z^r_{p+r,q-r+1} + D} \cong \frac{Z^{r+1}_{p,q}}{Z^{r+1}_{p,q} \cap \left(\partial Z^r_{p+r,q-r+1} + D\right)}$$

the isomorphism being the second isomorphism theorem.

It remains to show that the intersection appearing there equals $Z^r_{p-1,q+1} + \partial Z^r_{p+r,q-r+1}$, which is exactly the denominator of $\mathcal{E}^{r+1}_{p,q}$. For the inclusion $\supseteq$, note $Z^r_{p-1,q+1} \subseteq Z^{r+1}_{p,q}$ (an element lies in $F_{p-1} \subseteq F_p$ and its boundary lies in F_{p-1-r}) and $Z^r_{p-1,q+1} \subseteq Z^{r-1}_{p-1,q+1} \subseteq D$; while $\partial Z^r_{p+r,q-r+1} \subseteq Z^{r+1}_{p,q}$ because such a boundary lies in F_pC_n and is itself a cycle. For the inclusion $\subseteq$, let $x \in Z^{r+1}_{p,q}$ be written as $x = \partial y + z + \partial w$ with $y \in Z^r_{p+r,q-r+1}$, $z \in Z^{r-1}_{p-1,q+1}$ and $w \in Z^{r-1}_{p+r-1,q-r+2}$. Then $z = x - \partial y - \partial w$ lies in $F_{p-1}C_n$ and $\partial z = \partial x \in F_{p-r-1}C_{n-1}$, so $z \in Z^r_{p-1,q+1}$. Moreover $w \in Z^{r-1}_{p+r-1,q-r+2}$ satisfies $w \in F_{p+r-1} \subseteq F_{p+r}$ and $\partial w \in F_{(p+r-1)-(r-1)} = F_p$, so $w \in Z^r_{p+r,q-r+1}$ and hence $\partial w \in \partial Z^r_{p+r,q-r+1}$. Therefore $x = z + \partial(y+w) \in Z^r_{p-1,q+1} + \partial Z^r_{p+r,q-r+1}$, as required.

Substituting gives $H(\mathcal{E}^r, \partial)_{p,q} \cong \mathcal{E}^{r+1}_{p,q}$, which closes the induction and completes the proof.

The formula repays reading slowly, because it makes the interpretive claim of section 58.1 precise. The numerator at page r consists of the chains whose boundary drops r filtration levels, and the denominator discards those already accounted for one level down together with the boundaries of chains that drop $r-1$ levels. Passing from page r to page $r+1$ tightens the requirement on the boundary by one level. A class that survives to page r and dies there is one that could be corrected to a cycle modulo F_{p-r} but not modulo F_{p-r-1}, which is the precise sense in which d_r detects gluing across r layers.

With the pages described, the limit can be identified. This is the theorem the rest of the series cites from this section.

Definition 58.2.5: Convergence of a Spectral Sequence

A bigraded spectral sequence $\{(E^r, d_r)\}$ *degenerates* at the spot (p, q) if the differentials into and out of that spot vanish for all sufficiently large r; the common value of $E^r_{p,q}$ for large r is then written $E^\infty_{p,q}$. The spectral sequence *converges* to a graded object $H_\bullet$ equipped with a filtration $F_\bullet H_\bullet$, written

$$E^1_{p,q} \implies H_{p+q}$$

if it degenerates at every spot, the filtration $F_\bullet H_n$ is bounded for each n (that is $F_p H_n = 0$ for p small and $F_p H_n = H_n$ for p large), and $E^\infty_{p,q} \cong F_p H_{p+q}/F_{p-1} H_{p+q}$ for all p, q. The boundedness clause is what lets a vanishing family of graded pieces be read back as a vanishing abutment; without it the graded pieces constrain only the successive quotients.

Theorem 58.2.6: Convergence for a Bounded Filtration

Let $(C_\bullet, F_\bullet)$ be a filtered complex whose filtration is bounded in the sense of definition 58.2.1. Then the spectral sequence of theorem 58.2.2 converges to $H_\bullet(C)$: it degenerates at every spot, and

$$E^\infty_{p,q} \cong \frac{F_p H_{p+q}(C)}{F_{p-1} H_{p+q}(C)}$$

for the filtration $F_p H_n(C) = \operatorname{im}(H_n(F_p C) \to H_n(C))$ of definition 58.2.1. Moreover that filtration of $H_n(C)$ is itself bounded, so $H_n(C)$ has a finite filtration whose graded pieces are the terms of the limit page on the diagonal $p + q = n$.

***Proof*:**
Fix n and let $s(m), t(m)$ be the bounds of definition 58.2.1 in degree m.

Step 1: Degeneration. Fix a spot (p, q) with $p + q = n$. The differential out of it lands in $E^r_{p-r,q+r-1}$, a subquotient of $E^1_{p-r,q+r-1} = H_{n-1}(\mathrm{gr}_{p-r} C)$. Once r is large enough that $p-r \le s(n-1)$, the complex $\mathrm{gr}_{p-r} C$ vanishes in degree $n-1$, so the target is zero. The differential into (p, q) comes from $E^r_{p+r,q-r+1}$, a subquotient of $H_{n+1}(\mathrm{gr}_{p+r} C)$; once $p + r - 1 \ge t(n + 1)$ one has $F_{p+r-1} C_{n+1} = F_{p+r} C_{n+1} = C_{n+1}$, so $\mathrm{gr}_{p+r} C$ vanishes in degree $n + 1$ and the source is zero. Both hold for all large r, so $E^{r+1}_{p,q} = E^r_{p,q}$ eventually and $E^\infty_{p,q}$ is defined.

Step 2: The large-r value of the explicit description. Take r large enough that $F_{p-r} C_{n-1} = 0$ and $F_{p+r-1} C_{n+1} = C_{n+1}$, which is possible by boundedness. Write $Z_n = \ker(\partial \colon C_n \to C_{n-1})$ and $B_n = \operatorname{im}(\partial \colon C_{n+1} \to C_n)$. Then:

1. $Z^r_{p,q} = \{x \in F_p C_n : \partial x \in F_{p-r} C_{n-1} = 0\} = Z_n \cap F_p C_n$;
2. $Z^{r-1}_{p-1,q+1} = \{x \in F_{p-1} C_n : \partial x \in F_{p-r} C_{n-1} = 0\} = Z_n \cap F_{p-1} C_n$;
3. $\partial Z^{r-1}_{p+r-1,q-r+2} = \partial\{y \in F_{p+r-1} C_{n+1} : \partial y \in F_p C_n\} = \partial(C_{n+1}) \cap F_p C_n = B_n \cap F_p C_n$, using $F_{p+r-1} C_{n+1} = C_{n+1}$.

Substituting into proposition 58.2.4,

$$E^{\infty}_{p,q} \cong \frac{Z_n \cap F_pC_n}{(Z_n \cap F_{p-1}C_n) + (B_n \cap F_pC_n)}$$

Step 3: The same object is the graded piece of homology. By definition $F_pH_n(C)$ is the image of $H_n(F_pC)$ in $H_n(C)$, that is

$$F_pH_n(C) = \frac{(Z_n \cap F_pC_n) + B_n}{B_n} \cong \frac{Z_n \cap F_pC_n}{B_n \cap F_pC_n}$$

the isomorphism being the second isomorphism theorem. Taking the quotient by $F_{p-1}H_n(C)$, which under this identification is the image of $Z_n \cap F_{p-1}C_n$, gives

$$\frac{F_pH_n(C)}{F_{p-1}H_n(C)} \cong \frac{Z_n \cap F_pC_n}{(Z_n \cap F_{p-1}C_n) + (B_n \cap F_pC_n)}$$

which is the object computed in Step 2. Hence $E^{\infty}_{p,q} \cong \mathrm{gr}_pH_n(C)$.

Both sides were computed as subquotients of F_pC_n by the same formula, and the isomorphism between them is induced by the identity on representatives. A morphism of filtered complexes carries representatives to representatives and preserves every object appearing in that formula, so the identification is natural in $(C_\bullet, F_\bullet)$; this is what theorem 58.2.9 uses when it compares the two filtrations of homology.

Step 4: The induced filtration is bounded. If $p \leq s(n)$ then $F_pC_n = 0$, so $H_n(F_pC) = 0$ and $F_pH_n(C) = 0$. If $p \geq t(n)$ then $F_pC_n = C_n$; taking p also at least $t(n+1)$ makes $F_pC_{n+1} = C_{n+1}$ as well, so every cycle and every boundary of degree n already lies at filtration level p and $F_pH_n(C) = H_n(C)$. Hence the filtration of $H_n(C)$ passes from 0 to everything in finitely many steps, completing the proof.

Two features of the statement deserve emphasis, both of them permanent. The limit page computes the *associated graded* of $H_n(C)$, not $H_n(C)$ itself; recovering the latter means solving the extension problems between successive graded pieces, exactly as in the two-step case of proposition 58.1.8. And convergence is a theorem about the filtration, not about the complex: boundedness is what forces the pages to stabilize, and without it the limit page can fail to compute anything.

The applications elsewhere in the series are stated for cochain complexes with decreasing filtrations, so the translation is recorded explicitly rather than left to the reader.

Corollary 58.2.7: Convergence for a Decreasing Filtration of a Cochain Complex

Let $(C^\bullet, d)$ be a cochain complex with a decreasing filtration $C^\bullet = F^0C^\bullet \supseteq F^1C^\bullet \supseteq \cdots$ compatible with d, that is $d(F^pC^n) \subseteq F^pC^{n+1}$, and suppose the filtration is exhaustive and bounded, so that for each n there is a P with $F^PC^n = 0$. Then there is a spectral sequence with

$$E_0^{p,q} = \frac{F^pC^{p+q}}{F^{p+1}C^{p+q}}, \qquad d_r \colon E_r^{p,q} \to E_r^{p+r,\, q-r+1}$$

each page the cohomology of the one before it, converging to $H^{p+q}(C)$:

$$E_1^{p,q} \implies H^{p+q}(C), \qquad E_\infty^{p,q} \cong \mathrm{gr}_F^p H^{p+q}(C) = \frac{F^pH^{p+q}(C)}{F^{p+1}H^{p+q}(C)}$$

where $F^pH^n(C) = \mathrm{im}\big(H^n(F^pC^\bullet) \to H^n(C^\bullet)\big)$.

***Proof*:**
Set $C_n = C^{-n}$ and $F_pC_\bullet = F^{-p}C^\bullet$. The differential d raises cochain degree by one and therefore lowers chain degree by one, so $(C_\bullet, F_\bullet)$ is a filtered chain complex in the sense of definition 58.2.1: the filtration is increasing because $F^\bullet$ is decreasing, it is exhaustive because $F^\bullet$ is, and it is bounded because for each n there is a P with $F^PC^n = 0$, giving $F_{-P}C_{-n} = 0$, while $F^0C^n = C^n$ gives $F_0C_{-n} = C_{-n}$.

Theorem 58.2.6 applies and gives a convergent spectral sequence with $E^0_{p,q} = \mathrm{gr}_pC_{p+q}$ and d_r of bidegree $(-r, r-1)$. Re-indexing by $E_r^{p,q} = E^r_{-p,-q}$, as in definition 58.1.6, turns the bidegree of d_r into $(r, -r+1)$ and turns the graded piece $\mathrm{gr}_{-p}H_{-p-q}(C_\bullet)$ into $\mathrm{gr}_F^pH^{p+q}(C^\bullet)$, which is the statement, as we sought to show.

The same re-indexing carries every other result of this section across, and it is worth saying so once rather than hedging at each use, since section 58.3 onward works exclusively with cochain complexes.

Corollary 58.2.8: The Cochain Form of the Section

Let $\mathcal{T}$ denote the translation $C_n = C^{-n}$, $F_p = F^{-p}$, $E_r^{p,q} = E^r_{-p,-q}$ of corollary 58.2.7. Under $\mathcal{T}$:

1. Theorem 58.2.9 holds for morphisms of cochain complexes with bounded decreasing filtrations: a morphism inducing an isomorphism on E_{r_0} for some r_0 induces one on E_∞ and on $H^\bullet$.

2. Definition 58.2.3, proposition 58.2.4 and definition 58.2.5 hold with the *bigrading* negated, the page index r unchanged. Explicitly $Z_r^{p,q} = \{x \in F^pC^{p+q} : dx \in F^{p+r}C^{p+q+1}\}$ and

$$E_r^{p,q} \cong \frac{Z_r^{p,q}}{Z_{r-1}^{p+1,q-1} + d\, Z_{r-1}^{p-r+1,\, q+r-2}}$$

***Proof*:**
$\mathcal{T}$ is an isomorphism between the category of cochain complexes with bounded decreasing filtrations and the category of chain complexes with bounded increasing filtrations, carrying morphisms to morphisms and the spectral sequence of one to the spectral sequence of the other, by the computation in the proof of corollary 58.2.7. A statement proved for one is therefore proved for the other; no hypothesis in theorem 58.2.9 or proposition 58.2.4 refers to the direction of the grading beyond boundedness, which $\mathcal{T}$ preserves, completing the proof.

The remaining tool is the one that lets a spectral sequence be recognized rather than computed: if two filtered complexes are compared by a map, and the comparison is an isomorphism on any single page, then it is an isomorphism on the abutment. This is what makes a spectral sequence an identification technique and not only a calculation.

Theorem 58.2.9: Comparison Theorem

Let $f\colon (C_\bullet, F_\bullet) \to (C'_\bullet, F'_\bullet)$ be a morphism of filtered complexes, that is a chain map with $f(F_pC_\bullet) \subseteq F'_pC'_\bullet$, and suppose both filtrations are bounded. Then f induces a morphism of spectral sequences $f^r\colon E^r_{p,q} \to E'^r_{p,q}$ commuting with the differentials, and if f^{r_0} is an isomorphism for some $r_0 \geq 0$, then:

1. f^r is an isomorphism for every $r \geq r_0$, and hence so is f^∞;
2. $f_*\colon H_n(C) \to H_n(C')$ is an isomorphism for every n.

***Proof*:**
Step 1: Functoriality. On the zeroth page f induces f^0 on gr_pC_{p+q} directly, and the identification $E^1 = H(E^0, d_0)$ of theorem 58.2.2 is natural in the filtered complex, so the case $r_0 = 0$ reduces to $r_0 = 1$. For the remaining pages: because f preserves the filtration, it carries the short exact sequences of theorem 58.2.2 for C to those for C', and the connecting homomorphism of theorem 56.1.15 is natural. So f induces a morphism of exact couples, and the construction of the derived couple in definition 58.1.4 is built from i, j, k alone and is therefore natural in the couple. Hence f induces f^r on every page, commuting with d_r.

Step 2: Isomorphism on all later pages. Suppose f^r is an isomorphism. Since f^r commutes with the differentials, it carries $\ker d_r$ isomorphically onto $\ker d'_r$ and $\mathrm{im} d_r$ isomorphically onto $\mathrm{im} d'_r$, so the induced map on the quotients is an isomorphism. That induced map is f^{r+1}, so the claim follows by induction from r_0.

Step 3: Isomorphism on the limit page. Both filtrations are bounded, so by Step 1 of the proof of theorem 58.2.6 each spot degenerates: there is an R depending on (p, q) with $E^r_{p,q} = E^\infty_{p,q}$ and $E'^r_{p,q} = E'^\infty_{p,q}$ for $r \geq R$. Taking $r \geq \max(R, r_0)$ and applying Step 2 gives $f^\infty_{p,q}$ an isomorphism.

Step 4: Isomorphism on homology. Fix n. By theorem 58.2.6 the filtrations $F_\bullet H_n(C)$ and $F'_\bullet H_n(C')$ are finite, and f_* respects them. Proceed by induction up the filtration. The bottom stage is zero on both sides. If f_* is an isomorphism $F_{p-1}H_n(C) \to F'_{p-1}H_n(C')$, then in the commutative diagram of short exact sequences

$$\begin{tikzcd}
0 \arrow[r] & F_{p-1}H_n(C) \arrow[r] \arrow[d] & F_pH_n(C) \arrow[r] \arrow[d] & E^{\infty}_{p,n-p} \arrow[r] \arrow[d] & 0 \\
0 \arrow[r] & F'_{p-1}H_n(C') \arrow[r] & F'_pH_n(C') \arrow[r] & E'^{\infty}_{p,n-p} \arrow[r] & 0
\end{tikzcd}$$

the outer vertical maps are isomorphisms, the left one by the inductive hypothesis and the right one by Step 3. Each row is already a five-term exact sequence, so lemma 56.1.17 applies directly, with the outer maps $0 \to 0$ trivially epi and mono, and forces the middle map to be an isomorphism. Since the filtration is finite and exhausts $H_n(C)$, finitely many steps give that $f_*\colon H_n(C) \to H_n(C')$ is an isomorphism, completing the proof.

A first application of the machinery, and the one that recurs in section 58.5, is the crudest filtration a complex carries: truncate it.

Example 58.2.10: The Stupid Filtration of a Cochain Complex

Let $(C^\bullet, d)$ be a cochain complex concentrated in degrees $0 \le n \le N$, and define the *stupid filtration* (the *filtration bête*) by

$$F^pC^n = \begin{cases} C^n & n \ge p \\ 0 & n < p \end{cases}$$

This is a decreasing filtration compatible with d, since d raises degree, and it is exhaustive and bounded because $F^0C^\bullet = C^\bullet$ and $F^{N+1}C^\bullet = 0$. The graded pieces are $\mathrm{gr}^p_F C^n = C^n$ if $n = p$ and 0 otherwise, so the zeroth page has a single nonzero entry in each column,

$$E_0^{p,q} = \begin{cases} C^p & q = 0 \\ 0 & q \neq 0 \end{cases}$$

The differential d_0 has bidegree $(0,1)$ and so runs between entries with q differing by one; every such pair has a zero member, so $d_0 = 0$ and $E_1 = E_0$. The differential d_1 has bidegree $(1,0)$ and is the map induced by d, namely $d\colon C^p \to C^{p+1}$. Hence

$$E_2^{p,q} = \begin{cases} H^p(C^\bullet) & q = 0 \\ 0 & q \neq 0 \end{cases}$$

and every later differential has target or source zero, so $E_2 = E_\infty$. By corollary 58.2.7 the sequence converges to $H^{p+q}(C^\bullet)$, and indeed the single surviving entry on the diagonal $p+q = n$ is $H^n(C^\bullet)$, with no extension problem. The stupid filtration therefore returns the cohomology of the complex and nothing more, which is the correct answer and a useful check on the indexing conventions. It becomes informative only when the entries C^p are themselves replaced by objects with cohomology of their own, which is what section 58.5 does.

Exercise 58.2.1

1. Let $(C_\bullet, F_\bullet)$ be a filtered complex whose filtration has exactly three nonzero stages, $0 = F_{-1} \subseteq F_0 \subseteq F_1 \subseteq F_2 = C_\bullet$. Show that $d_r = 0$ for $r \ge 3$, so that $E^3 = E^\infty$. Generalize to a filtration

with N stages.

2. Show that the filtration induced on $H_n(C)$ in definition 58.2.1 is increasing, and that a morphism of filtered complexes $f\colon (C_\bullet, F_\bullet) \to (C'_\bullet, F'_\bullet)$ satisfies $f_*\big(F_pH_n(C)\big) \subseteq F'_pH_n(C')$. (This is the compatibility used in Step 4 of the proof of theorem 58.2.9.)

3. A spectral sequence *degenerates at* E^r if $d_s = 0$ for all $s \geq r$. Show that if the spectral sequence of a bounded filtered complex degenerates at E^1, then for every n the object $H_n(C)$ has a finite filtration with graded pieces $H_n(\mathrm{gr}_pC)$. Show by example that this does not force $H_n(C) \cong \bigoplus_p H_n(\mathrm{gr}_pC)$.

4. Verify proposition 58.2.4 directly at $r = 1$ by computing both sides for the filtered complex of example 58.1.9, and confirm that the differential induced by ∂ is the map computed there.

5. Let $f\colon (C_\bullet, F_\bullet) \to (C'_\bullet, F'_\bullet)$ be a morphism of bounded filtered complexes which is an isomorphism on E^2 but not on E^1. Show that f_* is still an isomorphism on homology, and construct such an f with $E^1 \neq E'^1$.

58.3 The Two Spectral Sequences of a Double Complex

A double complex carries two filtrations of its total complex, one by columns and one by rows, and section 58.2 converts each into a spectral sequence. Both converge to the homology of the same total complex, and that coincidence is the source of most of the identifications homological algebra makes: whenever one filtration degenerates for an easy reason and the other computes something of interest, the two are forced to agree. Section 56.4 promised this machinery twice, once for the cohomology of double complexes and once for the balancing of Tor and Ext, and this section delivers both.

Throughout, $A = \{A_{p,q}\}$ is a double complex in the sense of definition 56.4.1, with horizontal differential $d^h\colon A_{p,q} \to A_{p-1,q}$ and vertical differential $\delta^v\colon A_{p,q} \to A_{p,q-1}$ satisfying $d^hd^h = 0$, $\delta^v\delta^v = 0$ and $d^h\delta^v + \delta^vd^h = 0$. The total complex is $\mathrm{Tot}(A)_n = \bigoplus_{p+q=n} A_{p,q}$ with differential $d = d^h + \delta^v$, as in proposition 56.4.2; anticommutativity is exactly what makes $d \circ d = 0$. Write

$$H^h_p(A)_\bullet, \qquad H^v_q(A)_\bullet$$

for the homology of the rows and of the columns respectively, each of which is again a complex under the differential the other direction induces.

Definition 58.3.1: The Column and Row Filtrations

Let A be a double complex. The *column filtration* and the *row filtration* of $\mathrm{Tot}(A)$ are

$${}^{\mathrm{I}}F_p\mathrm{Tot}(A)_n = \bigoplus_{\substack{p'+q'=n\\ p'\leq p}} A_{p',q'}, \qquad {}^{\mathrm{II}}F_p\mathrm{Tot}(A)_n = \bigoplus_{\substack{p'+q'=n\\ q'\leq p}} A_{p',q'}$$

The double complex is *first-quadrant* if $A_{p,q} = 0$ whenever $p < 0$ or $q < 0$.

Both are filtrations by subcomplexes, and for the reason that each differential moves in only one of the two directions: d^h lowers p' and leaves q' alone, while δ^v lowers q' and leaves p' alone. So d carries ${}^{\mathrm{I}}F_p$ into itself, and likewise for ${}^{\mathrm{II}}F_p$. The two differ only in which coordinate is being watched, which is why the same total complex acquires two different, and generally inequivalent, approximations to its homology.

Theorem 58.3.2: The Two Spectral Sequences of a Double Complex

Let A be a first-quadrant double complex. Then the column and row filtrations of $\mathrm{Tot}(A)$ give two spectral sequences, both converging to $H_\bullet(\mathrm{Tot}(A))$, with second pages

$$^{\mathrm{I}}E^2_{p,q} = H^h_p\big(H^v_q(A)\big) \implies H_{p+q}(\mathrm{Tot}(A))$$

$$^{\mathrm{II}}E^2_{p,q} = H^v_p\big(H^h_q(A)\big) \implies H_{p+q}(\mathrm{Tot}(A))$$

***Proof*:**
Step 1: Boundedness. Fix n. Since A is first-quadrant, $A_{p',q'} = 0$ unless $0 \le p' \le n$ on the diagonal $p' + q' = n$. Hence $^{\mathrm{I}}F_{-1}\mathrm{Tot}(A)_n = 0$ and $^{\mathrm{I}}F_n\mathrm{Tot}(A)_n = \mathrm{Tot}(A)_n$, so the column filtration is bounded in the sense of definition 58.2.1; the same argument applies to the row filtration. Theorem 58.2.6 therefore applies to each and gives convergence to $H_\bullet(\mathrm{Tot}(A))$ once the pages are identified.

Step 2: The zeroth and first pages of the column filtration. By definition 58.2.1 the graded piece is

$$\mathrm{gr}_p\mathrm{Tot}(A)_n = \frac{^{\mathrm{I}}F_p\mathrm{Tot}(A)_n}{^{\mathrm{I}}F_{p-1}\mathrm{Tot}(A)_n} = A_{p,\,n-p}$$

so $^{\mathrm{I}}E^0_{p,q} = A_{p,q}$. The induced differential is the part of $d = d^h + \delta^v$ that preserves the filtration index p. The summand d^h lowers p, so it vanishes on the graded piece; the summand δ^v preserves p. Hence $d^0 = \delta^v$ and

$$^{\mathrm{I}}E^1_{p,q} = H_q\big(A_{p,\bullet},\, \delta^v\big) = H^v_q(A)_p$$

the homology of the p-th column.

Step 3: The second page of the column filtration. By theorem 58.2.2 the differential d^1 has bidegree $(-1, 0)$, so it runs $^{\mathrm{I}}E^1_{p,q} \to {}^{\mathrm{I}}E^1_{p-1,q}$; by proposition 58.2.4 it is induced by the differential of $\mathrm{Tot}(A)$ on representatives. A representative of a class in $^{\mathrm{I}}E^1_{p,q}$ is an element of $A_{p,q}$ killed by δ^v, and applying $d = d^h + \delta^v$ to it gives d^h of it, an element of $A_{p-1,q}$. So d^1 is the map induced by d^h on vertical homology, and

$$^{\mathrm{I}}E^2_{p,q} = H_p\big(H^v_q(A)_\bullet,\, d^h\big) = H^h_p\big(H^v_q(A)\big)$$

Step 4: The row filtration. The argument is identical with the two coordinates exchanged. The graded piece of the row filtration at index p in total degree n is $A_{n-p,\,p}$, so $^{\mathrm{II}}E^0_{p,q} = A_{q,p}$, and the differential preserving the index p is now d^h. Hence $^{\mathrm{II}}E^1_{p,q} = H_q(A_{\bullet,p}, d^h) = H^h_q(A)_p$, and the next differential is induced by δ^v, giving $^{\mathrm{II}}E^2_{p,q} = H^v_p(H^h_q(A))$. Both converge to the homology of the same total complex by Step 1, completing the proof.

The two sequences are not related by any map; they are two different routes to one answer, and the content of the theorem is that the answer is the same. The standard way to exploit this is to arrange for one of them to collapse. A double complex whose columns are exact except in one spot, for instance, has a first page concentrated in a single row, and then every differential from the second page onward has a zero source or a zero target.

Lemma 58.3.3: Collapse Along a Single Row

Let $\{(E^r, d_r)\}$ be a spectral sequence with $E^2_{p,q} = 0$ for all $q \neq 0$. Then $E^2 = E^\infty$, and if the sequence converges to $H_\bullet$ then $H_n \cong E^2_{n,0}$ for every n.

***Proof*:**
The differential d_r has bidegree $(-r, r-1)$ by theorem 58.1.7. For $r \geq 2$ the differential out of the spot $(p, 0)$ lands in $(p-r,\, r-1)$, which has second index $r-1 \geq 1$ and so vanishes; the differential into $(p, 0)$ comes from $(p+r,\, 1-r)$, which has negative second index and so vanishes. Every other spot is already zero. Hence $d_r = 0$ for all $r \geq 2$ and $E^2 = E^\infty$.

For the abutment, the diagonal $p + q = n$ of the limit page contains the single nonzero group $E^\infty_{n,0}$. By definition 58.2.5 the filtration of H_n is bounded and has these as graded pieces; a bounded filtration whose graded pieces vanish outside one step has exactly one nonzero step, so $H_n \cong E^\infty_{n,0} = E^2_{n,0}$, as we sought to show.

Both results are used in section 58.4 and section 58.5 for cochain double complexes, whose differentials raise indices. As with corollary 58.2.8, the translation is recorded once.

Corollary 58.3.4: The Cochain Form of the Double-Complex Sequences

Let $A^{\bullet,\bullet}$ be a cochain double complex concentrated in non-negative bidegrees, with differentials raising the two indices. Then the column and row filtrations of $\mathrm{Tot}(A)$ give two spectral sequences converging to $H^{p+q}(\mathrm{Tot}(A))$, with second pages

$$^{\mathrm{I}}E_2^{p,q} = H^p_h\big(H^q_v(A)\big), \qquad ^{\mathrm{II}}E_2^{p,q} = H^p_v\big(H^q_h(A)\big)$$

and lemma 58.3.3 holds in the form: if $E_2^{p,q} = 0$ for all $q \neq 0$ then $E_2 = E_\infty$ and $H^n \cong E_2^{n,0}$; symmetrically if $E_2^{p,q} = 0$ for all $p \neq 0$ then $H^n \cong E_2^{0,n}$.

***Proof*:**
Apply the translation $\mathcal{T}$ of corollary 58.2.8 in both indices, setting $A_{p,q} = A^{-p,-q}$. This turns a cochain double complex concentrated in non-negative bidegrees into a chain double complex concentrated in non-positive ones, which is bounded on diagonals in the sense of exercise 58.3.1 (5); theorem 58.3.2 and lemma 58.3.3 apply to it, and negating every index returns the displayed statements. The column form of the collapse is exercise 58.3.1 (4), translated the same way, completing the proof.

The lemma turns theorem 58.3.2 into an identification technique, and the standard application is the one section 56.4 deferred: that Tor may be computed by resolving either variable.

Corollary 58.3.5: Balancing Tor

Let R be a ring, M a right R-module and N a left R-module, and let $P_\bullet \to M$ and $Q_\bullet \to N$ be projective resolutions. Then

$$H_n\big(P_\bullet \otimes_R N\big) \cong H_n\big(\mathrm{Tot}(P_\bullet \otimes_R Q_\bullet)\big) \cong H_n\big(M \otimes_R Q_\bullet\big)$$

for every n. In particular $\mathrm{Tor}_n^R(M, N)$ may be computed by resolving either variable, recovering the commutative case of theorem 56.4.8 as the degenerate case of theorem 58.3.2.

***Proof*:**
Let $A_{p,q} = P_p \otimes_R Q_q$, with horizontal differential induced by that of $P_\bullet$ and vertical differential induced by that of $Q_\bullet$, the latter twisted by the sign $(-1)^p$ so that the two anticommute as definition 56.4.1 requires. Both resolutions are supported in non-negative degrees, so A is first-quadrant and theorem 58.3.2 applies.

Step 1: The column filtration. Each P_p is projective, hence a direct summand of a free module, hence flat, so the functor $P_p \otimes_R -$ is exact. Therefore it commutes with homology and

$${}^{\mathrm{I}}E^1_{p,q} = H_q\big(P_p \otimes_R Q_\bullet\big) \cong P_p \otimes_R H_q(Q_\bullet)$$

Since $Q_\bullet \to N$ is a resolution, $H_q(Q_\bullet)$ is N for $q = 0$ and zero otherwise. So the first page is concentrated in the row $q = 0$, where it is $P_p \otimes_R N$, and the second page is

$${}^{\mathrm{I}}E^2_{p,0} = H_p\big(P_\bullet \otimes_R N\big)$$

with ${}^{\mathrm{I}}E^2_{p,q} = 0$ for $q \neq 0$. By lemma 58.3.3, $H_n(\mathrm{Tot}(A)) \cong H_n(P_\bullet \otimes_R N)$.

Step 2: The row filtration. Symmetrically, each Q_q is flat, so ${}^{\mathrm{II}}E^1_{p,q} = H_q(P_\bullet \otimes_R Q_p) \cong H_q(P_\bullet) \otimes_R Q_p$, which is $M \otimes_R Q_p$ for $q = 0$ and zero otherwise. Hence ${}^{\mathrm{II}}E^2_{p,0} = H_p(M \otimes_R Q_\bullet)$ with the other rows zero, and lemma 58.3.3 gives $H_n(\mathrm{Tot}(A)) \cong H_n(M \otimes_R Q_\bullet)$.

Both computations describe the homology of the same total complex, so the two are isomorphic, completing the proof.

The argument is worth contrasting with the one given in section 56.4. There the comparison was made by hand, through an explicit quasi-isomorphism between the total complex and each of its edges. Here it is a two-line consequence of a general theorem, and the same two lines apply verbatim to Ext, to group cohomology, and to any other derived functor of a bifunctor. That is the characteristic economy of the method: the work moves once into theorem 58.2.6 and is then reused without repetition.

Exercise 58.3.1

1. Let A be a first-quadrant double complex all of whose columns are exact. Show that $\mathrm{Tot}(A)$ is acyclic. Deduce proposition 56.4.4 for first-quadrant double complexes.

2. State and prove the analogue of corollary 58.3.5 for Ext, using a projective resolution of M and an injective resolution of N. Identify which functor plays the role of $P_p \otimes_R -$ in Step 1, and which exactness property replaces flatness.

3. Let A be the first-quadrant double complex with $A_{0,0} = A_{1,0} = \mathbb{Z}$, $A_{0,1} = A_{1,1} = \mathbb{Z}$, horizontal differentials the identity, and vertical differentials the identity, with the sign twist making it anticommute. Compute both spectral sequences and $H_\bullet(\mathrm{Tot}(A))$, and check that the two agree.

4. Show that a spectral sequence with $E^2_{p,q} = 0$ for all $p \neq 0$ also collapses, and that $H_n \cong E^2_{0,n}$. (No quadrant hypothesis is needed.) (This is the column analogue of lemma 58.3.3.)

5. The first-quadrant hypothesis in theorem 58.3.2 is used in exactly one place. Identify it, and show that the theorem still holds for any double complex that is *bounded on diagonals*, meaning that for each n only finitely many $A_{p,q}$ with $p + q = n$ are nonzero. Then show that a double complex with $A_{p,q} \neq 0$ for every $p, q \in \mathbb{Z}$ fails this condition, so that theorem 58.2.6 does not apply to either filtration.

58.4 The Grothendieck Spectral Sequence

Derived functors were built in section 56.3 one functor at a time. In practice a functor of interest is often a composite: global sections of a sheaf factor through pushforward along a map, invariants under a group factor through invariants under a normal subgroup, and global Hom factors through sheaf Hom. In each case the derived functors of the two factors are understood and the derived functors of the composite are what is wanted. The relation between them is not an equality, and it is not captured by any exact sequence; it is a spectral sequence, and it is the form in which the machinery of this chapter is most often used elsewhere in the series.

The construction needs one tool beyond section 58.3: a way to resolve a whole complex, rather than a single object, by injectives, in a manner compatible with the complex's own cohomology. That is the Cartan-Eilenberg resolution, and it is built by applying the horseshoe lemma twice.

Definition 58.4.1: Cartan-Eilenberg Resolution

Let $\mathcal{A}$ be an abelian category with enough injectives and $C^\bullet$ a cochain complex in $\mathcal{A}$ bounded below. A *Cartan-Eilenberg resolution* of $C^\bullet$ is a double complex $I^{\bullet,\bullet}$ of injective objects, bounded below, together with an augmentation $C^p \to I^{p,0}$ for each p, such that for every p the induced augmentations make each of

$$I^{p,\bullet}, \qquad B^p(I^{\bullet,\bullet}), \qquad Z^p(I^{\bullet,\bullet}), \qquad H^p(I^{\bullet,\bullet})$$

an injective resolution of C^p, of $B^p(C^\bullet)$, of $Z^p(C^\bullet)$ and of $H^p(C^\bullet)$ respectively, where B^p, Z^p and H^p are taken in the first (horizontal) direction.

The three extra conditions are what make the resolution usable: without them a levelwise injective resolution of each C^p carries no information about the cohomology of $C^\bullet$, and the whole point is that the resolution should compute that cohomology as well as the individual terms.

Theorem 58.4.2: Existence of Cartan-Eilenberg Resolutions

Every bounded-below cochain complex in an abelian category with enough injectives admits a Cartan-Eilenberg resolution.

***Proof*:**
Step 1: Choices. For each p, use that $\mathcal{A}$ has enough injectives (definition 56.2.3) to choose injective resolutions

$$B^p(C^\bullet) \to \mathcal{B}^{p,\bullet}, \qquad H^p(C^\bullet) \to \mathcal{H}^{p,\bullet}$$

Since $C^\bullet$ is bounded below, so are all of these.

Step 2: Resolving the cycles. The definition of cohomology gives a short exact sequence

$$0 \to B^p(C^\bullet) \to Z^p(C^\bullet) \to H^p(C^\bullet) \to 0$$

Its two ends are already resolved by Step 1, so the horseshoe lemma (lemma 56.3.7), in its injective form, produces an injective resolution $Z^p(C^\bullet) \to \mathcal{Z}^{p,\bullet}$ fitting into a degreewise split short exact sequence of complexes $0 \to \mathcal{B}^{p,\bullet} \to \mathcal{Z}^{p,\bullet} \to \mathcal{H}^{p,\bullet} \to 0$.

Step 3: Resolving the terms. The complex $C^\bullet$ also gives, for each p, a short exact sequence

$$0 \to Z^p(C^\bullet) \to C^p \to B^{p+1}(C^\bullet) \to 0$$

whose outer terms are resolved by $\mathcal{Z}^{p,\bullet}$ from Step 2 and $\mathcal{B}^{p+1,\bullet}$ from Step 1. Applying lemma 56.3.7 again yields an injective resolution $C^p \to I^{p,\bullet}$ with $I^{p,q} \cong \mathcal{Z}^{p,q} \oplus \mathcal{B}^{p+1,q}$ in each degree, compatibly with the two inclusions.

Step 4: The horizontal differential. Define $I^{p,q} \to I^{p+1,q}$ as the composite

$$I^{p,q} \twoheadrightarrow \mathcal{B}^{p+1,q} \hookrightarrow \mathcal{Z}^{p+1,q} \hookrightarrow I^{p+1,q}$$

using the splitting of Step 3 for the first map and the inclusion of Step 2 for the second. The composite of two consecutive such maps is zero, because the image of the first lands in $\mathcal{Z}^{p+1,q}$, which the projection to $\mathcal{B}^{p+2,q}$ annihilates. So $I^{\bullet,\bullet}$ is a double complex of injectives, after the usual sign twist making the two differentials anticommute as definition 56.4.1 requires.

Step 5: The conditions hold. By construction $I^{p,\bullet}$ is an injective resolution of C^p, which is the first condition. By Step 4 the horizontal cycles are $Z^p(I^{\bullet,q}) = \mathcal{Z}^{p,q}$ and the horizontal boundaries are $B^p(I^{\bullet,q}) = \mathcal{B}^{p,q}$, which are injective resolutions of $Z^p(C^\bullet)$ and $B^p(C^\bullet)$ by Steps 2 and 1. The quotient $\mathcal{Z}^{p,q}/\mathcal{B}^{p,q} \cong \mathcal{H}^{p,q}$ is an injective resolution of $H^p(C^\bullet)$ by Step 1 and the splitting of Step 2. The augmentations are compatible with the horizontal differential by construction, since d^h was defined through the factorization $C^p \twoheadrightarrow B^{p+1} \hookrightarrow Z^{p+1} \hookrightarrow C^{p+1}$ that the differential of $C^\bullet$ itself factors through. All four conditions of definition 58.4.1 hold, completing the proof.

Existence is only half of what the resolution is used for. The spectral sequences built from it in section 58.4 and section 58.5 are asserted to be natural, and proposition 58.5.2 compares the resolutions of two different complexes; both need to know that a morphism of complexes lifts to the resolutions. The object-level comparison theorem (lemma 56.2.13 and its dual) does not supply this, since a Cartan-Eilenberg resolution is constrained not only degreewise but also on cycles, boundaries and cohomology. The lift has to be built through the same two horseshoes that produced the resolution, and the first step is to record that the horseshoe construction is itself functorial.

Lemma 58.4.3: Functoriality of the Horseshoe Construction

Consider a morphism of short exact sequences in an abelian category with enough injectives,

$$\begin{tikzcd} 0 \arrow[r] & L \arrow[r] \arrow[d, "\lambda"] & M \arrow[r] \arrow[d, "\mu"] & N \arrow[r] \arrow[d, "\nu"] & 0 \\ 0 \arrow[r] & L' \arrow[r] & M' \arrow[r] & N' \arrow[r] & 0 \end{tikzcd}$$

and injective resolutions $I_L^\bullet, I_N^\bullet, I_{L'}^\bullet, I_{N'}^\bullet$ of the four outer objects, and injective resolutions $I_M^\bullet$, $I_{M'}^\bullet$ of M and M' sitting in *degreewise split* short exact sequences $0 \to I_L^\bullet \to I_M^\bullet \to I_N^\bullet \to 0$ and $0 \to I_{L'}^\bullet \to I_{M'}^\bullet \to I_{N'}^\bullet \to 0$ lifting the two given rows. (This is what lemma 56.3.7 produces, but the argument uses only the splitting, which is automatic whenever the outer terms are injective.) Given lifts $\lambda^\bullet\colon I_L^\bullet \to I_{L'}^\bullet$ and $\nu^\bullet\colon I_N^\bullet \to I_{N'}^\bullet$ of λ and ν, there is a lift $\mu^\bullet\colon I_M^\bullet \to I_{M'}^\bullet$ of μ compatible with both, that is making the induced diagram of resolutions commute.

***Proof*:**
By lemma 56.3.7 the middle resolutions are $I_M^q = I_L^q \oplus I_N^q$ and $I_{M'}^q = I_{L'}^q \oplus I_{N'}^q$ in each degree, with the displayed maps the inclusion and the projection. Compatibility with those two maps forces $\mu^\bullet$ to have the shape

$$\mu^q = \begin{pmatrix} \lambda^q & \sigma^q \\ 0 & \nu^q \end{pmatrix}$$

so the only freedom is the family $\sigma^q\colon I_N^q \to I_{L'}^q$, and the content of the lemma is that it can be chosen so that $\mu^\bullet$ is a chain map lifting μ.

Write the differential of $I_M^\bullet$ as $\begin{pmatrix} d_L & \tau \\ 0 & d_N \end{pmatrix}$ and that of $I_{M'}^\bullet$ as $\begin{pmatrix} d_{L'} & \tau' \\ 0 & d_{N'} \end{pmatrix}$, where τ and τ' are the off-diagonal terms the horseshoe construction produces. The chain-map condition $\mu^{q+1}d = d\mu^q$ reads, in the only entry that is not automatic,

$$\lambda^{q+1}\tau^q + \sigma^{q+1}d_N^q = d_{L'}^q\sigma^q + \tau'^q\nu^q$$

so, writing $\theta^q := d_{L'}^q\sigma^q + \tau'^q\nu^q - \lambda^{q+1}\tau^q$, the family σ must satisfy $\sigma^{q+1} \circ d_N^q = \theta^q$ for every q.

The base case. Write the augmentation of $I_M^\bullet$ through the splitting as $\varepsilon_M = (\varphi,\ \varepsilon_N\pi)^\mathsf{T}$, where $\pi\colon M \to N$ is the given epimorphism and $\varphi\colon M \to I_L^0$ restricts to ε_L on L; similarly for M'. The morphism σ^0 is not given by the hypothesis, it has to be built. Set $\theta^{-1} := \varphi'\mu - \lambda^0\varphi\colon M \to I_{L'}^0$. On the subobject L this is $\varepsilon_{L'}\lambda - \varepsilon_{L'}\lambda = 0$, so θ^{-1} kills $L = \ker\pi$ and factors as $\bar\theta \circ \pi$ for a unique $\bar\theta\colon N \to I_{L'}^0$. Since $\varepsilon_N\colon N \to I_N^0$ is a monomorphism and $I_{L'}^0$ is injective, $\bar\theta$ extends along it to $\sigma^0\colon I_N^0 \to I_{L'}^0$ with $\sigma^0\varepsilon_N\pi = \theta^{-1}$, which is the degree-zero instance of the required identity and is the only place the compatibility of μ with λ and ν is used.

The inductive step. Suppose $\sigma^{\leq q}$ has been chosen. To extend θ^q along d_N^q it suffices that θ^q vanish on $\ker d_N^q = \mathrm{im} d_N^{q-1}$, and this is a direct computation. Squaring the differential of $I_M^\bullet$ gives $d_L\tau + \tau d_N = 0$, and likewise $d_{L'}\tau' + \tau' d_{N'} = 0$; using these, the inductive hypothesis $\sigma^q d_N^{q-1} = \theta^{q-1}$, the chain-map identities for λ and ν, and $d \circ d = 0$,

$$\theta^q \circ d_N^{q-1} = \left(\lambda^{q+1} d_L^q - d_{L'}^q \lambda^q\right) \circ \tau^{q-1} = 0$$

the last equality because $\lambda^\bullet$ is a chain map. For $q = 0$ the same conclusion holds with $\ker d_N^0 = \mathrm{im}\varepsilon_N$ in place of $\mathrm{im} d_N^{-1}$: applying θ^0 to $\varepsilon_N \pi$ and using $d_L^0 \varphi + \tau^0 \varepsilon_N \pi = 0$, which is the first component of $d_M^0 \varepsilon_M = 0$, together with the base case, gives $\theta^0 \varepsilon_N \pi = (\lambda^1 d_L^0 - d_{L'}^0 \lambda^0)\varphi = 0$, and π is an epimorphism. Hence in every degree θ^q factors through $\mathrm{im} d_N^q$, and since $\mathrm{im} d_N^q \hookrightarrow I_N^{q+1}$ is a monomorphism and $I_{L'}^{q+1}$ is injective, definition 56.2.1 extends the factored morphism to $\sigma^{q+1} \colon I_N^{q+1} \to I_{L'}^{q+1}$ with $\sigma^{q+1} d_N^q = \theta^q$, as required.

The resulting $\mu^\bullet$ is a chain map by construction, is compatible with the inclusion and projection by its shape, and lifts μ because it does so in degree -1, that is on the augmentations, completing the proof.

Theorem 58.4.4: Comparison Theorem for Cartan-Eilenberg Resolutions

Let $f^\bullet \colon C^\bullet \to \widetilde{C}^\bullet$ be a morphism of bounded-below cochain complexes in an abelian category with enough injectives, and let $J^{\bullet,\bullet}$ and $\widetilde{J}^{\bullet,\bullet}$ be Cartan-Eilenberg resolutions of $C^\bullet$ and $\widetilde{C}^\bullet$. Then $f^\bullet$ lifts to a morphism of double complexes

$$\tilde{f}^{\bullet,\bullet} \colon J^{\bullet,\bullet} \to \widetilde{J}^{\bullet,\bullet}$$

compatible with the augmentations, and any two such lifts are homotopic. Consequently the induced maps on Z^p, B^p and H^p in the horizontal direction are the resolutions of $Z^p(f)$, $B^p(f)$ and $H^p(f)$.

***Proof*:**
Step 1: Lifts on the building blocks. The morphism $f^\bullet$ induces $B^p(f) \colon B^p(C^\bullet) \to B^p(\widetilde{C}^\bullet)$ and $H^p(f) \colon H^p(C^\bullet) \to H^p(\widetilde{C}^\bullet)$ for every p. By definition 58.4.1 the objects $\mathcal{B}^{p,\bullet}$ and $\mathcal{H}^{p,\bullet}$ occurring in J are injective resolutions of $B^p(C^\bullet)$ and $H^p(C^\bullet)$, and likewise for $\widetilde{J}$. The dual of lemma 56.2.13 therefore lifts $B^p(f)$ and $H^p(f)$ to morphisms of those resolutions, each unique up to homotopy.

Step 2: Lifts on cycles. Apply lemma 58.4.3 to the morphism of short exact sequences

$$\left(0 \to B^p(C) \to Z^p(C) \to H^p(C) \to 0\right) \longrightarrow \left(0 \to B^p(\widetilde{C}) \to Z^p(\widetilde{C}) \to H^p(\widetilde{C}) \to 0\right)$$

induced by $f^\bullet$, with the lifts of Step 1 on the two ends. This produces a lift $\mathcal{Z}^{p,\bullet} \to \widetilde{\mathcal{Z}}^{p,\bullet}$ of $Z^p(f)$.

Step 3: Lifts on the terms. Apply lemma 58.4.3 again, now to

$$\big(0 \to Z^p(C) \to C^p \to B^{p+1}(C) \to 0\big) \longrightarrow \big(0 \to Z^p(\widetilde{C}) \to \widetilde{C}^p \to B^{p+1}(\widetilde{C}) \to 0\big)$$

with the lift of Step 2 on the left end and that of Step 1 on the right. This produces $\tilde{f}^{p,\bullet}\colon J^{p,\bullet} \to \widetilde{J}^{p,\bullet}$ lifting f^p.

Step 4: Compatibility with the horizontal differential. For any Cartan-Eilenberg resolution the horizontal differential factors as $J^{p,q} \twoheadrightarrow \mathcal{B}^{p+1,q} \hookrightarrow \mathcal{Z}^{p+1,q} \hookrightarrow J^{p+1,q}$: it factors through its image $\mathcal{B}^{p+1,q}$ by definition, and that image lies in $\mathcal{Z}^{p+1,q}$ because $d^h \circ d^h = 0$. Each of the three maps is respected by the lifts of Steps 1 to 3, the projection and the two inclusions being the structure maps of the splittings. Hence $\tilde{f}^{\bullet,\bullet}$ commutes with d^h, and it commutes with d^v because each $\tilde{f}^{p,\bullet}$ is a morphism of resolutions. So it is a morphism of double complexes.

Step 5: Uniqueness up to homotopy. Let $\tilde{f}$ and $\tilde{g}$ both lift $f^\bullet$. In each column their difference lifts the zero morphism $C^p \to \widetilde{C}^p$, so by the dual of lemma 56.2.4 it is null-homotopic: there are $s^{p,\bullet}$ with $d^v s + s d^v = \tilde{f} - \tilde{g}$ in each column.

Producing a homotopy of *double* complexes, that is one which additionally satisfies $sd^h = d^h s$, requires choosing those column homotopies compatibly. This is a third induction of the same shape as lemma 58.4.3: choose null-homotopies on $\mathcal{B}^{p,\bullet}$ and $\mathcal{H}^{p,\bullet}$ by the object-level uniqueness of Step 1, extend over the first horseshoe to $\mathcal{Z}^{p,\bullet}$ by an obstruction computation of the same shape as the one for θ^q, and extend again over the second to $J^{p,\bullet}$. The bookkeeping is routine and carries no idea not already in lemma 58.4.3, so it is not repeated; it is carried out in *An Introduction to Homological Algebra* [53].

For the final clause, let $\tilde{f}$ be any lift. Its restriction to horizontal cocycles, coboundaries and cohomology is determined by its compatibility with the augmentations, since those subquotients are computed from d^h and $\tilde{f}$ commutes with d^h by Step 4; so on H^p it is a lift of $H^p(f)$, whether or not it is the one built in Steps 1 to 3. This completes the proof.

With the tool in hand, the theorem itself is a two-filtration argument of exactly the kind section 58.3 was written for. The hypothesis to watch is the one relating the two functors: the first must carry injectives to objects on which the second's higher derived functors vanish.

Definition 58.4.5: Acyclic for a Left-Exact Functor

Let $G\colon \mathcal{B} \to \mathcal{C}$ be a left-exact additive functor between abelian categories. An object B of $\mathcal{B}$ is *G-acyclic* if $R^iG(B) = 0$ for all $i \geq 1$. This is the right-derived counterpart of definition 56.3.13. Injective objects are G-acyclic for every left-exact G, since an injective I has $0 \to I \to I \to 0$ as an injective resolution; and a resolution by G-acyclic objects computes $R^\bullet G$. The latter is the dual of theorem 56.3.15, which is stated there for right-exact functors and left-derived functors; the statement needed here is its image under passage to the opposite categories, where projectives become injectives and $\mathbf{L}_i$ becomes R^i.

Theorem 58.4.6: The Grothendieck Spectral Sequence

Let $\mathcal{A}$, $\mathcal{B}$, $\mathcal{C}$ be abelian categories with $\mathcal{A}$ and $\mathcal{B}$ having enough injectives, and let

$$F\colon \mathcal{A} \to \mathcal{B}, \qquad G\colon \mathcal{B} \to \mathcal{C}$$

be left-exact additive functors. Suppose F carries injective objects of $\mathcal{A}$ to G-acyclic objects of $\mathcal{B}$. Then for every object A of $\mathcal{A}$ there is a first-quadrant cohomological spectral sequence, natural in A,

$$E_2^{p,q} = R^pG\big(R^qF(A)\big) \implies R^{p+q}(G \circ F)(A)$$

***Proof*:**
Step 1: Setup. Choose an injective resolution $A \to I^\bullet$ in $\mathcal{A}$. Then $R^qF(A) = H^q(F(I^\bullet))$ by definition of the right-derived functors (definition 56.3.1 in its dual form). The complex $F(I^\bullet)$ is bounded below, so by theorem 58.4.2 it admits a Cartan-Eilenberg resolution $J^{\bullet,\bullet}$. Apply G to obtain a first-quadrant double complex $G(J^{\bullet,\bullet})$ in $\mathcal{C}$, and consider the two spectral sequences of corollary 58.3.4 converging to $H^\bullet(\mathrm{Tot}(G(J^{\bullet,\bullet})))$.

Step 2: The filtration that collapses. Take first the filtration whose first page is obtained by computing cohomology in the resolution direction. For each fixed p, the complex $J^{p,\bullet}$ is an injective resolution of $F(I^p)$, so

$$H^q\big(G(J^{p,\bullet})\big) = R^qG\big(F(I^p)\big)$$

By hypothesis I^p is injective and F carries injectives to G-acyclics, so $F(I^p)$ is G-acyclic and this vanishes for $q \geq 1$, leaving $G(F(I^p))$ when $q = 0$. Hence this first page is concentrated in the single row $q = 0$, where it is the complex $G(F(I^\bullet))$, and its second page is

$$H^p\big(GF(I^\bullet)\big) = R^p(G \circ F)(A)$$

the last equality because $I^\bullet$ is an injective resolution of A. By corollary 58.3.4,

$$H^n\big(\mathrm{Tot}(G(J^{\bullet,\bullet}))\big) \cong R^n(G \circ F)(A)$$

Step 3: The filtration that computes. Take now the other filtration, whose first page is the cohomology in the p direction. Because $J^{\bullet,\bullet}$ is a Cartan-Eilenberg resolution, the horizontal cocycles, coboundaries and cohomology are injective resolutions of those of $F(I^\bullet)$. The short exact sequences relating them therefore have injective outer terms in each degree, hence are degreewise split, exactly as spelled out in Step 2 of theorem 58.5.3. A functor preserves split exactness, so applying G commutes with taking horizontal cohomology:

$$H^p\big(G(J^{\bullet,q})\big) \cong G\big(H^p(J^{\bullet,q})\big)$$

By definition 58.4.1 the complex $H^a(J^{\bullet,\bullet})$ is an injective resolution of $H^a(F(I^\bullet)) = R^aF(A)$, where a denotes the degree in $F(I^\bullet)$ and the resolution degree is the remaining direction. This is the row filtration of corollary 58.3.4, whose index swap makes the filtration index, here the resolution degree, the *first* spectral index. Writing p for that resolution degree and q for a, taking cohomology in the resolution direction therefore gives

$$E_2^{p,q} = R^pG\big(R^qF(A)\big)$$

Step 4: Conclusion. Both spectral sequences converge to the cohomology of the same total complex by corollary 58.3.4. Step 2 identifies that cohomology as $R^n(G \circ F)(A)$ and Step 3 identifies the second page of the other as $R^pG(R^qF(A))$, giving

$$E_2^{p,q} = R^pG\big(R^qF(A)\big) \implies R^{p+q}(G \circ F)(A)$$

For naturality in A, let $a \colon A \to A'$ be a morphism. The dual of lemma 56.2.13 lifts it to a morphism of injective resolutions $I^\bullet \to I'^\bullet$, unique up to homotopy; applying F gives a morphism $F(I^\bullet) \to F(I'^\bullet)$, and theorem 58.4.4 lifts that to a morphism of the chosen Cartan-Eilenberg resolutions, again unique up to homotopy. Applying G and totalizing yields a morphism of filtered complexes, hence a morphism of spectral sequences by Step 1 of theorem 58.2.9. Homotopic lifts induce the same map on E_r for every $r \geq 2$. This needs an argument, because the homotopy has bidegree $(0, -1)$ and the filtration in play is by resolution degree, so it does *not* preserve the filtration: it lowers it by one. The standard filtration-shift estimate covers the gap. For a class represented by $x \in Z_r^p$, the two totalized lifts differ by $Ds(x) + s(Dx)$, where s is the totalized homotopy and D the total differential; here $s(x) \in F^{p-1}$ so $Ds(x) \in dF^{p-1}$, which dies in E_r for $r \geq 2$, and $Dx \in F^{p+r}$ so $s(Dx) \in F^{p+r-1}$, which also dies in E_r for $r \geq 2$. Hence the morphism of spectral sequences from page two onward depends only on a, which is all the naturality statement asserts. This completes the proof.

The shape of the conclusion deserves comment, because it is easy to misread. The spectral sequence does not say that $R^n(GF)$ is built from the groups $R^pG(R^qF)$ by any simple rule; it says there is a filtration of $R^n(GF)(A)$ whose graded pieces are subquotients of those groups, obtained after running the differentials. Only in the corners is the information immediate, and the standard exact sequence extracted from those corners is the following, which is how the theorem is most often applied in low degrees.

Nothing in that extraction uses where the spectral sequence came from. It uses only that the sequence is first-quadrant and converges, so it is worth recording at that generality: the same five terms appear for the spectral sequence of a double complex, or of any filtered complex meeting the hypotheses of corollary 58.2.7, and consumers outside this book need it in that form.

Both maps at the ends of that sequence are instances of a general construction worth naming, because downstream books identify them with concrete maps and need a statement to point at. In a first-quadrant spectral sequence the corner spots $(p, 0)$ and $(0, q)$ each meet the abutment in only one way.

Proposition 58.4.7: Edge Morphisms

Let $E_2^{p,q} \Rightarrow H^{p+q}$ be a first-quadrant cohomological spectral sequence converging in the sense of definition 58.2.5. Then for every $p, q \geq 0$ there are canonical morphisms

$$E_2^{p,0} \longrightarrow H^p, \qquad H^q \longrightarrow E_2^{0,q}$$

called the *edge morphisms*, given respectively by the composites

$$E_2^{p,0} \twoheadrightarrow E_\infty^{p,0} = F^p H^p \hookrightarrow H^p, \qquad H^q \twoheadrightarrow H^q/F^1 H^q = E_\infty^{0,q} \hookrightarrow E_2^{0,q}$$

They are natural in the spectral sequence.
For the Grothendieck spectral sequence of theorem 58.4.6 the two edge morphisms are the natural maps

$$R^p G\big(F(A)\big) \longrightarrow R^p(G \circ F)(A), \qquad R^q(G \circ F)(A) \longrightarrow G\big(R^q F(A)\big)$$

***Proof*:**
Step 1: The two composites are defined. Fix p. Every differential out of $(p,0)$ lands in $(p+r, 1-r)$, which has negative second index for $r \geq 2$, and every differential into it comes from $(p-r, r-1)$, which has negative first index once $r > p$. So $E_\infty^{p,0}$ is a quotient of $E_2^{p,0}$, the successive pages being obtained by killing images. The filtration of H^p is finite with $F^{p+1}H^p = 0$, because $E_\infty^{a,b} = 0$ for $a > p$ on the diagonal $a + b = p$ by first-quadrantness, so $E_\infty^{p,0} = F^p H^p / F^{p+1} H^p = F^p H^p$, a subobject of H^p. Composing gives the first morphism.

Dually, every differential into $(0,q)$ comes from $(-r, q+r-1)$, which has negative first index, so $E_\infty^{0,q}$ is a subobject of $E_2^{0,q}$, obtained by taking successive kernels. And $E_\infty^{0,q} = F^0 H^q / F^1 H^q = H^q / F^1 H^q$, since $F^0 H^q = H^q$ by exhaustiveness. Composing gives the second morphism.

Step 2: Naturality. A morphism of spectral sequences compatible with the abutments respects the pages, hence the successive kernels and images defining E_∞, and respects the filtration of the abutment. Each edge morphism is built only from those data, so the two squares commute.

Step 3: The Grothendieck case. Substituting $E_2^{p,q} = R^p G(R^q F(A))$ and $H^n = R^n(G \circ F)(A)$ gives $E_2^{p,0} = R^p G(R^0 F(A)) = R^p G(F(A))$, since $R^0 F = F$ by left exactness of F, and $E_2^{0,q} = R^0 G(R^q F(A)) = G(R^q F(A))$ likewise. So the two edge morphisms take the displayed shape. That they are the *natural* maps, that is the ones induced by the comparison of an injective resolution of A with a Cartan-Eilenberg resolution of F applied to it, is Step 2 together with the naturality of theorem 58.4.6, completing the proof.

Proposition 58.4.8: The Five-Term Exact Sequence of a First-Quadrant Spectral Sequence

Let $E_2^{p,q} \Rightarrow H^{p+q}$ be a first-quadrant cohomological spectral sequence, so that $E_r^{p,q} = 0$ whenever $p < 0$ or $q < 0$, converging to a filtered graded object $H^\bullet$ in the sense of definition 58.2.5. Then there is an exact sequence

$$0 \to E_2^{1,0} \to H^1 \to E_2^{0,1} \xrightarrow{d_2} E_2^{2,0} \to H^2$$

natural in the spectral sequence: a morphism of first-quadrant spectral sequences compatible with the abutment induces a morphism of these five-term sequences.

***Proof*:**
The spectral sequence is first-quadrant, so the filtration of H^n has graded pieces $E_\infty^{p,q}$ with $p+q=n$ and $0 \le p \le n$.

Step 1: Degree one. The diagonal $p+q=1$ contains the two spots $(1,0)$ and $(0,1)$. On the second page, the differentials touching $(1,0)$ run to $(3,-1)$ and from $(-1,1)$, both outside the first quadrant, so $E_\infty^{1,0} = E_2^{1,0}$. The differentials touching $(0,1)$ are $d_2\colon E_2^{0,1} \to E_2^{2,0}$ and one arriving from $(-2,2)$, which is zero; so $E_\infty^{0,1} = \ker\big(d_2\colon E_2^{0,1} \to E_2^{2,0}\big)$. The filtration of H^1 therefore gives a short exact sequence

$$0 \to E_\infty^{1,0} \to H^1 \to E_\infty^{0,1} \to 0$$

that is $0 \to E_2^{1,0} \to H^1 \to \ker d_2 \to 0$. The first map here is the edge morphism $E_2^{1,0} \to H^1$ of proposition 58.4.7 and the second is the edge morphism $H^1 \to E_2^{0,1}$ followed by nothing, both by construction; this is what lets consumers identify the two ends of the five-term sequence with concrete maps.

Step 2: Splicing. Substituting $E_\infty^{0,1} = \ker d_2$ turns the short exact sequence into exactness of

$$0 \to E_2^{1,0} \to H^1 \to E_2^{0,1} \xrightarrow{d_2} E_2^{2,0}$$

the last map being d_2 and exactness at $E_2^{0,1}$ being the statement that the image of H^1 is exactly $\ker d_2$.

Step 3: The final term. It remains to show exactness at $E_2^{2,0}$, that is that the kernel of $E_2^{2,0} \to H^2$ is the image of d_2. The filtration of H^2 has $E_\infty^{2,0}$ as its bottom graded piece, and the map $E_2^{2,0} \to H^2$ is the composite $E_2^{2,0} \twoheadrightarrow E_\infty^{2,0} \hookrightarrow H^2$. The second map is a monomorphism, so the kernel of the composite is the kernel of the first, which is the image of the only differential arriving at $(2,0)$, namely $d_2\colon E_2^{0,1} \to E_2^{2,0}$. Hence the sequence is exact there.

Step 4: Naturality. Let $\{E_r\} \to \{E'_r\}$ be a morphism of first-quadrant spectral sequences compatible with the abutments. Every object in the five-term sequence was constructed from data the morphism respects: the terms $E_2^{1,0}$, $E_2^{0,1}$, $E_2^{2,0}$ receive the induced map on the second page; the map d_2 is respected because a morphism of spectral sequences commutes with the differentials; $E_\infty^{p,q}$ receives the induced map because it is a subquotient of $E_2^{p,q}$ cut out by those same differentials; and H^1, H^2 receive the given map on abutments, which by hypothesis is compatible with the filtrations and hence with the identifications $E_\infty^{p,q} \cong F^p H^{p+q}/F^{p+1}H^{p+q}$. Each of the four maps in the sequence was defined from exactly these, so all four squares commute, completing the proof.

Specializing to theorem 58.4.6 recovers the form in which the sequence is usually quoted, and it is the one example 58.4.10 and the group-cohomology literature use.

Corollary 58.4.9: The Five-Term Exact Sequence of a Composite

Under the hypotheses of theorem 58.4.6, there is an exact sequence

$$0 \to R^1G\big(F(A)\big) \to R^1(GF)(A) \to G\big(R^1F(A)\big) \to R^2G\big(F(A)\big) \to R^2(GF)(A)$$

natural in A.

***Proof*:**
The Grothendieck spectral sequence is first-quadrant and converges to $R^\bullet(GF)(A)$, so proposition 58.4.8 applies. Substituting its second page, $E_2^{p,q} = R^pG(R^qF(A))$, gives $E_2^{1,0} = R^1G(F(A))$ since $R^0F = F$, then $E_2^{0,1} = G(R^1F(A))$ since $R^0G = G$, and $E_2^{2,0} = R^2G(F(A))$; and H^1, H^2 are $R^1(GF)(A)$, $R^2(GF)(A)$. Naturality in A follows from naturality of the spectral sequence in A, as we sought to show.

The machine can now be put to work. The instance closest to hand uses the group cohomology of section 57.1: invariants under a group factor through invariants under a normal subgroup.

Example 58.4.10: The Lyndon-Hochschild-Serre Spectral Sequence
Let G be a group, $N \trianglelefteq G$ a normal subgroup, and M a G-module. Taking N-invariants produces a G/N-module, because N acts trivially on M^N and the residual action of G factors through G/N, and taking G/N-invariants of that returns M^G. So the functor $(-)^G$ factors as

$$\mathrm{Mod}_G \xrightarrow{(-)^N} \mathrm{Mod}_{G/N} \xrightarrow{(-)^{G/N}} \mathbf{Ab}$$

Both factors are left exact: for $(-)^{G/N}$ this is proposition 57.1.6, and for $(-)^N : \mathrm{Mod}_G \to \mathrm{Mod}_{G/N}$ the same argument applies, exactness being tested on the underlying abelian groups. That $R^p(-)^{G/N} = H^p(G/N, -)$ is section 57.1 by definition.

The identification $R^q(-)^N = H^q(N, -)$ needs one step more than it looks. The left-hand side is computed from an injective resolution in Mod_G, while the right-hand side is computed from one in Mod_N, so the two agree only because *restriction to a subgroup preserves injectives.* That holds here because $\mathbb{Z}[G]$ is free as a $\mathbb{Z}[N]$-module, on any set of coset representatives, so the relative induction functor $\mathrm{Ind}_N^G = \mathbb{Z}[G] \otimes_{\mathbb{Z}[N]} -$, which is the evident generalization of the absolute one in definition 57.1.16, is exact and is left adjoint to restriction; a right adjoint to an exact functor preserves injectives, by the adjunction $\mathrm{Hom}_N(-, \mathrm{Res}I) \cong \mathrm{Hom}_G(\mathrm{Ind}(-), I)$, which is then a composite of two exact functors. Hence restricting an injective G-module gives an injective N-module, and an injective resolution in Mod_G restricts to one in Mod_N, computing $H^q(N, -)$.

The hypothesis of theorem 58.4.6 is that $(-)^N$ carries injective G-modules to $(-)^{G/N}$-acyclic G/N-modules, and in fact it carries them to injectives. Let I be an injective G-module, let

$0 \to A \to B$ be a monomorphism of G/N-modules, and let $\varphi\colon A \to I^N$ be G/N-equivariant. Regard A and B as G-modules through $G \twoheadrightarrow G/N$; then $A \to I^N \hookrightarrow I$ is G-equivariant, so injectivity of I extends it to $\psi\colon B \to I$. For $b \in B$ and $n \in N$ one has $n \cdot \psi(b) = \psi(n \cdot b) = \psi(b)$, since N acts trivially on B; so ψ lands in I^N and is the required extension. Hence I^N is injective, and injectives are acyclic for any left-exact functor by definition 58.4.5.

Theorem 58.4.6 therefore gives a first-quadrant spectral sequence

$$E_2^{p,q} = H^p\big(G/N,\, H^q(N,M)\big) \implies H^{p+q}(G,M)$$

the *Lyndon-Hochschild-Serre spectral sequence*, and corollary 58.4.9 specializes to the inflation-restriction sequence

$$0 \to H^1(G/N, M^N) \to H^1(G,M) \to H^1(N,M)^{G/N} \to H^2(G/N, M^N) \to H^2(G,M)$$

using $H^0(G/N, H^1(N,M)) = H^1(N,M)^{G/N}$. The same argument runs verbatim for a profinite group acting on discrete modules, once the category of discrete modules is known to have enough injectives; that is the setting in which the sequence is used in Galois cohomology.

Exercise 58.4.1

1. Show that if $R^qF(A) = 0$ for all $q \geq 1$, that is if A is F-acyclic, then the Grothendieck spectral sequence collapses and $R^n(GF)(A) \cong R^nG(F(A))$ for every n.
2. Show that if G is exact, then $R^n(GF)(A) \cong G(R^nF(A))$ for every n. Identify which page of the spectral sequence collapses and why.
3. In the situation of example 58.4.10 take $N = G$, so that G/N is trivial. Verify directly that the spectral sequence degenerates and returns $H^n(G,M) = H^n(G,M)$, and explain which hypothesis makes the E_2 page concentrated in one column.
4. Prove the dual statement of theorem 58.4.6 for right-exact functors and left-derived functors: if $F\colon \mathcal{A} \to \mathcal{B}$ and $G\colon \mathcal{B} \to \mathcal{C}$ are right exact, $\mathcal{A}$ and $\mathcal{B}$ have enough projectives, and F carries projectives to G-acyclics, then $E^2_{p,q} = L_pG(L_qF(A)) \implies L_{p+q}(G \circ F)(A)$.
5. Show that the hypothesis that F carries injectives to G-acyclics cannot simply be dropped, by explaining exactly which step of the proof of theorem 58.4.6 fails without it and what the first filtration computes in its absence.

58.5 Hypercohomology

The derived functors of section 56.3 take an object as input. Many constructions instead present a whole complex and ask for the derived functor of that: a complex of sheaves rather than a single sheaf, a complex of modules rather than a module. Replacing the complex by its cohomology objects loses information, as the last exercise of section 58.4 showed, and the correct construction resolves the complex as a complex. The tool is again the Cartan-Eilenberg resolution, and the output carries two spectral sequences relating it to the two things one might naively have computed instead.

Definition 58.5.1: Hypercohomology

Let $F\colon \mathcal{A} \to \mathcal{B}$ be a left-exact additive functor between abelian categories with $\mathcal{A}$ having enough injectives, and let $C^\bullet$ be a bounded-below cochain complex in $\mathcal{A}$. Choose a Cartan-Eilenberg resolution $J^{\bullet,\bullet}$ of $C^\bullet$ (theorem 58.4.2), where the first index is the degree in $C^\bullet$ and the second the resolution degree. The *hypercohomology* of $C^\bullet$ with respect to F is

$$\mathbb{R}^n F(C^\bullet) := H^n\Big(\mathrm{Tot}\big(F(J^{\bullet,\bullet})\big)\Big)$$

The definition is only worth making if it does not depend on the resolution, and the comparison theorem of section 58.2 is exactly the instrument for showing that it does not.

Proposition 58.5.2: Hypercohomology is Well Defined

The objects $\mathbb{R}^n F(C^\bullet)$ of definition 58.5.1 are independent of the choice of Cartan-Eilenberg resolution, up to canonical isomorphism, and a quasi-isomorphism $C^\bullet \to \widetilde{C}^\bullet$ of bounded-below complexes induces an isomorphism $\mathbb{R}^n F(C^\bullet) \cong \mathbb{R}^n F(\widetilde{C}^\bullet)$ for every n.

***Proof*:**
Let $J^{\bullet,\bullet}$ and $\widetilde{J}^{\bullet,\bullet}$ be Cartan-Eilenberg resolutions of $C^\bullet$ and $\widetilde{C}^\bullet$ respectively, and let $f\colon C^\bullet \to \widetilde{C}^\bullet$ be a quasi-isomorphism; taking f to be the identity handles the first claim.

Step 1: A comparison map exists. By theorem 58.4.4, f lifts to a morphism of double complexes $J^{\bullet,\bullet} \to \widetilde{J}^{\bullet,\bullet}$, uniquely up to homotopy, and the lift induces $H^q(f)$ on the horizontal cohomology resolutions. Applying F and totalizing gives a map of filtered complexes $\mathrm{Tot}(F(J)) \to \mathrm{Tot}(F(\widetilde{J}))$ compatible with the filtration by resolution degree.

Step 2: The comparison is an isomorphism on a page. By theorem 58.5.3 below, whose proof uses only definition 58.4.1 and not the present proposition, the filtration by resolution degree has second page $R^pF(H^q(C^\bullet))$. By the final clause of theorem 58.4.4 the lift of Step 1 restricts on horizontal cohomology to a resolution of $H^q(f)$, so the map it induces on that second page is exactly $R^pF(H^q(f))$. Since f is a quasi-isomorphism, $H^q(f)$ is an isomorphism for every q, and hence so is $R^pF(H^q(f))$ by functoriality of the derived functors. So the induced map is an isomorphism on the second page.

Step 3: Conclusion. Both complexes are bounded below and both Cartan-Eilenberg resolutions are first-quadrant, so both filtrations are bounded and theorem 58.2.9 applies with $r_0 = 2$, in the cochain form of corollary 58.2.8. It gives that the map is an isomorphism on the abutment, that is $\mathbb{R}^n F(C^\bullet) \cong \mathbb{R}^n F(\widetilde{C}^\bullet)$ for every n. Independence of the resolution is the case $f = \mathrm{id}$, and canonicity follows because the lift of Step 1 is unique up to homotopy and homotopic maps agree on cohomology, completing the proof.

The two filtrations of the defining double complex give the two spectral sequences that make hypercohomology computable. One approximates it by the derived functors of the cohomology of the complex, the other by the cohomology of the complex of derived functors; neither is generally the answer, and the difference between them is the reason hypercohomology is not redundant.

Theorem 58.5.3: The Two Hypercohomology Spectral Sequences

With the hypotheses of definition 58.5.1, there are two spectral sequences with bounded filtrations, first-quadrant when $C^\bullet$ is concentrated in non-negative degrees, converging to $\mathbb{R}^{p+q}F(C^\bullet)$:

$$^{\mathrm{I}}E_2^{p,q} = R^pF\big(H^q(C^\bullet)\big) \implies \mathbb{R}^{p+q}F(C^\bullet)$$

$$^{\mathrm{II}}E_1^{p,q} = R^qF\big(C^p\big) \implies \mathbb{R}^{p+q}F(C^\bullet)$$

the first filtered by resolution degree, the second by the degree in $C^\bullet$.

***Proof*:**
Both are instances of corollary 58.3.4 applied to the double complex $F(J^{\bullet,\bullet})$, which is a cochain double complex, and the work is in identifying the pages.

Step 1: Filtration by the degree in $C^\bullet$. The graded piece at level p is the column $F(J^{p,\bullet})$, and its cohomology in the resolution direction is

$$H^q\big(F(J^{p,\bullet})\big) = R^qF(C^p)$$

because $J^{p,\bullet}$ is an injective resolution of C^p by definition 58.4.1. This is the first page of that filtration, so $^{\mathrm{II}}E_1^{p,q} = R^qF(C^p)$, with d_1 induced by the differential of $C^\bullet$.

Step 2: Filtration by resolution degree. The graded piece at level q is the row $F(J^{\bullet,q})$, whose cohomology is taken in the direction of the degree in $C^\bullet$. By definition 58.4.1 the short exact sequences

$$0 \to Z^p(J^{\bullet,q}) \to J^{p,q} \to B^{p+1}(J^{\bullet,q}) \to 0$$

$$0 \to B^p(J^{\bullet,q}) \to Z^p(J^{\bullet,q}) \to H^p(J^{\bullet,q}) \to 0$$

have injective outer terms and are therefore split. An additive functor preserves split exact sequences, so F commutes with forming Z^p, B^p and H^p in this direction:

$$H^p\big(F(J^{\bullet,q})\big) \cong F\big(H^p(J^{\bullet,q})\big)$$

By definition 58.4.1 the complex $H^p(J^{\bullet,\bullet})$ is an injective resolution of $H^p(C^\bullet)$ as the resolution degree varies. Taking cohomology in that remaining direction therefore gives $R^\bullet F(H^p(C^\bullet))$, and indexing this sequence with the resolution degree as the filtration index yields $^{\mathrm{I}}E_2^{p,q} = R^pF(H^q(C^\bullet))$.

Step 3: Convergence. The complex $C^\bullet$ is bounded below and the resolution is bounded below, so $F(J^{\bullet,\bullet})$ is concentrated in non-negative bidegrees after a shift, and corollary 58.3.4 gives that both converge to the cohomology of $\mathrm{Tot}(F(J^{\bullet,\bullet}))$, which is $\mathbb{R}^{p+q}F(C^\bullet)$ by definition 58.5.1, completing the proof.

Two special cases recover the ordinary theory and explain the name. Both follow by collapsing one of the two sequences, and together they say that hypercohomology interpolates between the two computations one might have attempted directly.

Corollary 58.5.4: Hypercohomology Recovers the Ordinary Derived Functors

With the hypotheses of definition 58.5.1:

1. If $C^\bullet$ is concentrated in degree 0, with $C^0 = A$, then $\mathbb{R}^n F(C^\bullet) \cong R^n F(A)$ for every n.
2. If every C^p is F-acyclic, then $\mathbb{R}^n F(C^\bullet) \cong H^n\big(F(C^\bullet)\big)$ for every n.

***Proof*:**

1. If $C^\bullet$ is concentrated in degree 0 then $H^q(C^\bullet) = 0$ for $q \neq 0$ and $H^0(C^\bullet) = A$. So ${}^{\mathrm{I}}E_2^{p,q} = R^p F(H^q(C^\bullet))$ vanishes unless $q = 0$, where it is $R^p F(A)$. The second page is concentrated in the row $q = 0$, so by corollary 58.3.4 the sequence collapses and the diagonal $p + q = n$ carries the single group $R^n F(A)$. Hence $\mathbb{R}^n F(C^\bullet) \cong R^n F(A)$.
2. If every C^p is F-acyclic then $R^q F(C^p) = 0$ for $q \geq 1$ by definition 58.4.5, so ${}^{\mathrm{II}}E_1^{p,q} = R^q F(C^p)$ vanishes unless $q = 0$, where it is $F(C^p)$. The first page is concentrated in the row $q = 0$, and its differential d_1 is induced by the differential of $C^\bullet$, so the second page is $H^p(F(C^\bullet))$ in that row and zero elsewhere. Collapsing by corollary 58.3.4 gives $\mathbb{R}^n F(C^\bullet) \cong H^n(F(C^\bullet))$.

This completes the proof.

The second case is what makes hypercohomology a practical tool: a complex of acyclic objects may be substituted for the original complex, and the functor is then applied termwise. It is also the precise sense in which the definition generalizes the ordinary theory, since a resolution of a single object is a complex of acyclics quasi-isomorphic to that object, and proposition 58.5.2 says the answer depends only on the quasi-isomorphism class.

Example 58.5.5: Hypercohomology of a Complex with Zero Differential

Let $C^\bullet$ be a bounded-below complex with zero differential. Then $H^q(C^\bullet) = C^q$, and the first spectral sequence reads

$$ {}^{\mathrm{I}}E_2^{p,q} = R^p F(C^q) \implies \mathbb{R}^{p+q} F(C^\bullet) $$

The second has ${}^{\mathrm{II}}E_1^{p,q} = R^q F(C^p)$ with d_1 induced by the zero differential, hence zero, so ${}^{\mathrm{II}}E_2 = {}^{\mathrm{II}}E_1$. A Cartan-Eilenberg resolution may be taken to be the direct sum of injective resolutions of the individual C^p, since with zero differential the conditions of definition 58.4.1 on Z^p, B^p and H^p reduce to the condition on C^p itself. The resulting double complex splits as a direct sum, so every differential from the second page onward vanishes and there are no extension problems:

$$ \mathbb{R}^n F(C^\bullet) \cong \bigoplus_{p+q=n} R^p F(C^q) $$

The example is the sanity check for the definition: when the complex carries no gluing, hypercohomology is the direct sum of the ordinary derived functors of its terms, and the content of the general case is exactly the failure of that splitting.

Exercise 58.5.1

1. Show that if F is exact then $\mathbb{R}^n F(C^\bullet) \cong F(H^n(C^\bullet))$ for every n, using whichever of the two spectral sequences of theorem 58.5.3 collapses.
2. Let $C^\bullet$ be a bounded-below complex with $H^q(C^\bullet) = 0$ for all q. Show that $\mathbb{R}^n F(C^\bullet) = 0$ for every n, and explain why this is the expected answer given proposition 58.5.2.
3. Write down the five-term exact sequence attached to the first hypercohomology spectral sequence of theorem 58.5.3, by the argument of corollary 58.4.9.
4. Show that theorem 58.4.6 may be restated as the assertion that $\mathbb{R}^n G(F(I^\bullet)) \cong R^n(GF)(A)$ when F carries injectives to G-acyclics, where $I^\bullet$ is an injective resolution of A. Which part of corollary 58.5.4 is being used?
5. Let $C^\bullet$ be the two-term complex $A \xrightarrow{f} B$ in degrees 0 and 1. Show that $\mathbb{R}^0 F(C^\bullet) \cong F(\ker f)$, by both spectral sequences of theorem 58.5.3.

58.6 Filtered Complexes and the Filtered Derived Category

Section 58.2 extracted a spectral sequence from a filtered complex and stopped there. That is enough when one filtration is present. It is not enough when two are, and two is the case that arises: a complex carrying both a weight filtration and a Hodge filtration is the object mixed Hodge theory is built from, and the question there is whether the second filtration passes to the pages of the spectral sequence of the first without ambiguity. Answering it needs the filtered complexes themselves organised into a category, with the right notion of equivalence.

Definition 58.6.1: Filtered Object and Strict Morphism

Let $\mathcal{A}$ be an abelian category. A *filtered object* is a pair (A, F) with F a decreasing chain of subobjects $\cdots \supseteq F^p A \supseteq F^{p+1} A \supseteq \cdots$. A morphism $f\colon (A, F) \to (B, F)$ is a morphism $f\colon A \to B$ with $f(F^p A) \subseteq F^p B$ for all p; it is *strict* if

$$f(A) \cap F^p B = f(F^p A)$$

for all p, that is, if every element of the image that lies in $F^p B$ is the image of something already in $F^p A$.

Strictness is the condition that makes filtered algebra behave. Without it the category of filtered objects is not abelian: a morphism can be a monomorphism and an epimorphism without being an isomorphism, because the filtration on the image can be finer than the filtration inherited from the target.

Proposition 58.6.2: Strictness and Exactness

A morphism $f\colon (A, F) \to (B, F)$ of filtered objects is strict if and only if the induced sequence

$$0 \to F^p(\ker f) \to F^pA \to F^pB$$

has gr_F exact, equivalently if and only if gr_F applied to $0 \to \ker f \to A \to B$ stays exact. Strict morphisms are closed under composition only when the intermediate filtrations agree; strictness is not a two-out-of-three property.

***Proof*:**
If f is strict and $b = f(a) \in F^pB$, then $b = f(a')$ with $a' \in F^pA$, so $\mathrm{gr}^p_F f$ is surjective onto the image, and its kernel is $\mathrm{gr}^p_F(\ker f)$ because $a - a' \in \ker f$ lies in F^pA when both do. Conversely if gr_F preserves exactness, an element of $f(A) \cap F^pB$ lifts step by step through the graded pieces from a large p downwards, the filtration being exhaustive and separated in each degree.

For the last claim, $g \circ f$ can fail to be strict when f and g are: the image of f may meet F^pB in more than $f(F^pA)$ once g has collapsed part of it. The standard example is a two-step filtration on a three-dimensional space, and the failure is exactly what theorem 58.6.5 is designed to rule out.

Definition 58.6.3: Filtered Quasi-Isomorphism

A morphism $f\colon (K^\bullet, F) \to (L^\bullet, F)$ of filtered complexes is a *filtered quasi-isomorphism* if $\mathrm{gr}^p_F f$ is a quasi-isomorphism of complexes for every p.

This is strictly stronger than being a quasi-isomorphism: a map can induce an isomorphism on cohomology while distorting the filtration, and such maps must not become invertible, or the spectral sequence would not be an invariant of the object.

Theorem 58.6.4: The Filtered Derived Category

Let $\mathcal{A}$ be an abelian category with enough injectives. The localization of the homotopy category of bounded-below filtered complexes over $\mathcal{A}$ at the filtered quasi-isomorphisms exists; call it the *filtered derived category* $D^+F(\mathcal{A})$. The forgetful functor $D^+F(\mathcal{A}) \to D^+(\mathcal{A})$ is well defined, and for each r the assignment $(K, F) \mapsto E_r(K, F)$ of theorem 58.2.2 factors through $D^+F(\mathcal{A})$: a filtered quasi-isomorphism induces an isomorphism of spectral sequences from the E_1-page onwards.

***Proof*:**
Existence of the localization is corollary 55.2.6 applied to the filtered setting: filtered quasi-isomorphisms form a multiplicative system, the verification being the same calculus-of-fractions argument, with the octahedral completion carried out one graded piece at a time. The only point needing care is that a filtered mapping cone is filtered by the cones of the graded pieces, which holds because the cone construction is degreewise a direct sum and gr_F is additive.

For the second claim, the E_1-page of the spectral sequence of (K, F) is $E_1^{p,q} = H^{p+q}(\mathrm{gr}^p_F K)$ by theorem 58.2.2, and a filtered quasi-isomorphism induces isomorphisms on exactly these groups by definition 58.6.3. Since every later page is computed from E_1 by taking subquotients along differentials that are natural in the filtered complex, the isomorphism propagates. It does not extend to E_0, which sees $\mathrm{gr}_F K$ itself rather than its cohomology.

The lemma below is the reason this section exists. It is the tool that lets a complex carrying two filtrations be handled at all, and it is what mixed Hodge theory consumes: there W is a weight filtration, F is a Hodge filtration, and the conclusion is that F passes unambiguously to the pages of the spectral sequence of W.

Theorem 58.6.5: The Lemma on Two Filtrations

Let $K^\bullet$ be a bounded-below complex in an abelian category $\mathcal{A}$, carrying an increasing filtration W by subcomplexes that is biregular (in each degree, finite) and a decreasing filtration F by subcomplexes that is biregular as well. Give the subcomplex $F^pK^\bullet$ the filtration $W_n(F^pK) = W_nK \cap F^pK$ and the quotient $K^\bullet/F^pK^\bullet$ the filtration $W_n(K/F^pK) = \mathrm{im}(W_nK)$, and write ${}_WE_r(-)$ for the spectral sequences of theorem 58.2.2. The three filtrations F induces on ${}_WE_r(K)$ are

$$F^p_d = \mathrm{im}\big({}_WE_r(F^pK) \to {}_WE_r(K)\big), \qquad F^p_{d^*} = \ker\big({}_WE_r(K) \to {}_WE_r(K/F^pK)\big)$$

on ${}_WE_r(K)$ for r finite, and on ${}_WE_\infty(K)$ by the same two formulas with $r = \infty$, the maps being the stabilized ones; and the *recursive* filtration F_{rec}: on ${}_WE_0$ it is the filtration F induces on $\mathrm{gr}^W_n K^\bullet$, and on ${}_WE_{r+1} = H({}_WE_r, d_r)$ it is the filtration a subquotient inherits from F_{rec} on ${}_WE_r$. Then:

1. $F^p_d \subseteq F^p_{\mathrm{rec}} \subseteq F^p_{d^*}$ on ${}_WE_r$ for every $r \geq 0$ and on ${}_WE_\infty$, and the three coincide on ${}_WE_0$ and on ${}_WE_1$, with no hypothesis beyond the above;

2. let $r_0 \geq 0$ and suppose that d_s is strictly compatible with F_{rec} on ${}_WE_s$ for every $s \leq r_0$. Then for every p the sequence

$$0 \to {}_WE_r(F^pK) \to {}_WE_r(K) \to {}_WE_r(K/F^pK) \to 0$$

is exact for every $r \leq r_0 + 1$, and the three filtrations coincide on ${}_WE_r$ for every $r \leq r_0 + 2$;

3. if d_s is strictly compatible with F_{rec} for every $s \geq 0$, then the three filtrations coincide on every page and on ${}_WE_\infty$, the differential of $K^\bullet$ is strictly compatible with F in the sense of definition 58.6.1, and the spectral sequence of $(K^\bullet, F)$ degenerates at E_1.

The hypothesis of (2) at $s = 0$ says exactly that the differential of $\mathrm{gr}^W_n K^\bullet$ is strictly compatible with the filtration induced by F, for every n; on its own it therefore gives coincidence on ${}_WE_1$ and ${}_WE_2$, and example 58.6.6 shows that it gives nothing more.

***Proof*:**
Conventions. By theorem 55.1.15 the arguments below may be run with elements. Index the spectral sequence of W by the weight n and the cohomological degree m: setting $C_m = K^{-m}$ and $F_pC_\bullet = W_pK^{-\bullet}$ presents $(K^\bullet, W)$ as a filtered chain complex in the sense of definition 58.2.1,

and theorem 58.2.2 produces

$$ {}_W E_0^{n,m} = \mathrm{gr}_n^W K^m, \qquad {}_W E_1^{n,m} = H^m(\mathrm{gr}_n^W K^\bullet), \qquad d_r \colon {}_W E_r^{n,m} \to {}_W E_r^{n-r,\, m+1} $$

Biregularity of W and boundedness below of $K^\bullet$ make that filtration bounded in each degree, so theorem 58.2.6 applies: the pages stabilize at every spot and ${}_W E_\infty^{n,m} \cong \mathrm{gr}_n^W H^m(K^\bullet)$ for the filtration $W_n H^m(K) = \mathrm{im}\big(H^m(W_n K) \to H^m(K)\big)$, itself finite in each degree. The same holds for $F^p K^\bullet$ and for $K^\bullet/F^p K^\bullet$: a biregular filtration restricts and corestricts to biregular filtrations, and both complexes are bounded below.

Fix p for the whole argument except where stated, and abbreviate

$$ A_r = {}_W E_r(F^p K^\bullet), \qquad B_r = {}_W E_r(K^\bullet), \qquad C_r = {}_W E_r(K^\bullet/F^p K^\bullet) $$

writing $\alpha_r \colon A_r \to B_r$ and $\beta_r \colon B_r \to C_r$ for the maps induced by the inclusion and the projection. The construction of theorem 58.2.2 is functorial in the filtered complex, so α_r and β_r commute with d_r, and α_{r+1}, β_{r+1} are the maps they induce on d_r-cohomology. By definition $F_d^p B_r = \mathrm{im}\alpha_r$ and $F_{d^*}^p B_r = \ker \beta_r$.

Step 1: the sequence at $r = 0$, and the three filtrations there. For every n and m the sequence

$$ 0 \to W_n K^m \cap F^p K^m \to W_n K^m \to W_n (K/F^p K)^m \to 0 $$

is exact, by the definition of the two induced filtrations. These sequences are compatible with the inclusions $W_{n-1} \subseteq W_n$, and the vertical maps from the sequence at $n-1$ to the sequence at n are monomorphisms. Applying theorem 56.1.15 to that map of short exact sequences, all three kernels vanish and the six-term sequence collapses to

$$ 0 \to \mathrm{gr}_n^W (F^p K)^m \to \mathrm{gr}_n^W K^m \to \mathrm{gr}_n^W (K/F^p K)^m \to 0 $$

which is therefore exact. This is $0 \to A_0 \to B_0 \to C_0 \to 0$ at the spot (n,m), and the maps commute with d_0, so it is a short exact sequence of complexes in the cohomological degree m, one for each n.

For the filtrations, $F_d^p B_0 = \mathrm{im}\alpha_0$ is $\big(W_n K^m \cap F^p K^m + W_{n-1} K^m\big)/W_{n-1} K^m$, which is the filtration F induces on $\mathrm{gr}_n^W K^m$, that is $F_{\mathrm{rec}}^p B_0$ by definition; and $F_{d^*}^p B_0 = \ker \beta_0$ is the same subobject by the exactness just proved. So all three agree on ${}_W E_0$, and in particular the phrase “d_0 is strict” is unambiguous and says exactly that the differential of each $\mathrm{gr}_n^W K^\bullet$ is strict for the induced F.

Step 2: the two inclusions, and equality on ${}_W E_1$. We prove $F_d^p \subseteq F_{\mathrm{rec}}^p \subseteq F_{d^*}^p$ on B_r by induction on r, the case $r = 0$ being the equality of Step 1.

For the first inclusion let $\xi \in F_d^p B_{r+1}$, say $\xi = \alpha_{r+1}(\eta)$ with $\eta \in A_{r+1} = H(A_r, d_r)$. Write $\eta = [a]$ with $a \in A_r$, $d_r a = 0$. Then $\alpha_r(a)$ is killed by d_r and lies in $\mathrm{im}\alpha_r = F_d^p B_r$, which by induction lies in $F_{\mathrm{rec}}^p B_r$; and $\xi = [\alpha_r(a)]$. Thus ξ is the class of an element of $\ker d_r \cap F_{\mathrm{rec}}^p B_r$, which is precisely what $F_{\mathrm{rec}}^p B_{r+1}$ consists of.

For the second let $\xi \in F_{\mathrm{rec}}^p B_{r+1}$, say $\xi = [b]$ with $b \in \ker d_r \cap F_{\mathrm{rec}}^p B_r$. By induction $b \in F_{d^*}^p B_r =$

$\ker \beta_r$, so $\beta_{r+1}(\xi) = [\beta_r(b)] = 0$ and $\xi \in \ker \beta_{r+1} = F^p_{d^*} B_{r+1}$. Since each spot of each page equals its limit value for large r, and the three filtrations are defined at $r = \infty$ as those common values, the inclusions persist on ${}_W E_\infty$.

Equality on ${}_W E_1$ needs the short exact sequence of Step 1. For fixed n the objects $B_0^{n,\bullet}$ with d_0 form a complex with $H^m = B_1^{n,m}$, and likewise for A and C; theorem 56.1.15 gives the long exact sequence

$$\cdots \to A_1^{n,m} \xrightarrow{\alpha_1} B_1^{n,m} \xrightarrow{\beta_1} C_1^{n,m} \to A_1^{n,m+1} \to \cdots$$

Exactness at $B_1^{n,m}$ reads $\mathrm{im}\alpha_1 = \ker \beta_1$, that is $F^p_d = F^p_{d^*}$ on ${}_W E_1$, and F_{rec} is squeezed between them. This proves (1).

Step 3: two equivalences. (Throughout, theorem 56.1.15 is applied to the E_r-pages with differential d_r, which for each fixed residue class of the weight index form ordinary complexes, so the lemma applies as stated.) For $r \geq 0$ write

$$(\star_r)\colon \qquad 0 \to A_r \to B_r \to C_r \to 0 \quad \text{is exact}$$

a statement about complexes, since α_r and β_r commute with d_r. Step 1 is $(\star_0)$.

(a) $(\star_r)$ implies $F^p_d = F^p_{d^*}$ on ${}_W E_{r+1}$. By theorem 56.1.15 the sequence

$$\cdots \to A_{r+1} \xrightarrow{\alpha_{r+1}} B_{r+1} \xrightarrow{\beta_{r+1}} C_{r+1} \xrightarrow{\delta_r} A_{r+1} \to \cdots$$

is exact; exactness at B_{r+1} is $\mathrm{im}\alpha_{r+1} = \ker \beta_{r+1}$. Combined with Step 2 this forces all three filtrations to agree on ${}_W E_{r+1}$.

(b) Assume $(\star_r)$. Then $(\star_{r+1})$ holds if and only if d_r is strictly compatible with F^p_d on ${}_W E_r$, that is, if and only if $d_r(B_r) \cap F^p_d B_r = d_r(F^p_d B_r)$ at every spot. By the long exact sequence of (a), α_{r+1} is a monomorphism exactly when the δ_r landing in A_{r+1} vanishes and β_{r+1} is an epimorphism exactly when the δ_r leaving C_{r+1} vanishes; so $(\star_{r+1})$ is equivalent to $\delta_r = 0$. By $(\star_r)$ the map α_r is a monomorphism, so δ_r is described as follows: for $c \in C_r$ with $d_r c = 0$, choose $b \in B_r$ with $\beta_r(b) = c$; then $\beta_r(d_r b) = d_r c = 0$, so $d_r b \in \ker \beta_r = \mathrm{im}\alpha_r$ and there is a unique $a \in A_r$ with $\alpha_r(a) = d_r b$; it satisfies $d_r a = 0$, and $\delta_r[c] = [a]$.

Suppose d_r is strict for F^p_d, and let c, b, a be as above. Then $d_r b \in \mathrm{im}\alpha_r = F^p_d B_r$, so strictness gives $d_r b = d_r b'$ with $b' \in F^p_d B_r$, say $b' = \alpha_r(a')$. Then $\alpha_r(a) = d_r \alpha_r(a') = \alpha_r(d_r a')$, so $a = d_r a'$ and $\delta_r[c] = 0$. Conversely suppose $\delta_r = 0$ and let $b \in B_r$ with $d_r b \in F^p_d B_r = \ker \beta_r$. Then $d_r \beta_r(b) = \beta_r(d_r b) = 0$, so $c := \beta_r(b)$ is a d_r-cocycle and $\delta_r[c] = [a]$ with $\alpha_r(a) = d_r b$. Since $\delta_r = 0$ we get $a = d_r a'$ for some $a' \in A_r$, whence $d_r b = \alpha_r(d_r a') = d_r\big(\alpha_r(a')\big)$ with $\alpha_r(a') \in F^p_d B_r$. Thus $d_r(B_r) \cap F^p_d B_r \subseteq d_r(F^p_d B_r)$, and the reverse inclusion is automatic.

Step 4: the induction, giving (2). Assume d_s is strictly compatible with F_{rec} on ${}_W E_s$ for every $s \leq r_0$. We show by induction on $s \leq r_0 + 1$, simultaneously for every p, that

$$(\mathrm{I}_s)\colon (\star_s) \text{ holds}, \qquad (\mathrm{II}_s)\colon F_d = F_{\mathrm{rec}} = F_{d^*} \text{ on } {}_W E_s$$

Step 1 supplies (I_0) and (II_0). Let $s \le r_0$ and assume both at stage s. By (II_s) the hypothesis at stage s says that d_s is strict for F_d^p, so Step 3(b) upgrades $(\star_s)$ to $(\star_{s+1})$, which is (I_{s+1}); and (I_s) with Step 3(a) gives (II_{s+1}). This delivers $(\star_r)$ for every $r \le r_0+1$ and coincidence on ${}_WE_r$ for every $r \le r_0+1$. Applying Step 3(a) once more to $(\star_{r_0+1})$ gives coincidence on ${}_WE_{r_0+2}$ as well. Each stage consumes the strictness produced by the previous one and produces the exactness that makes the next stage's strictness hypothesis unambiguous, which is why the two conclusions cannot be separated.

Step 5: part (3), first the limit page. Assume now that d_s is strict for every $s \ge 0$. Then $(\star_r)$ holds for every r and every p, and the three filtrations agree on every page. Fix p and a spot (n,m). By convergence there is an R with $A_r^{n,m} = A_\infty^{n,m}$, $B_r^{n,m} = B_\infty^{n,m}$ and $C_r^{n,m} = C_\infty^{n,m}$ for $r \ge R$, compatibly with α and β, so

$$0 \to {}_WE_\infty^{n,m}(F^pK) \to {}_WE_\infty^{n,m}(K) \to {}_WE_\infty^{n,m}(K/F^pK) \to 0$$

is exact and the three filtrations agree on ${}_WE_\infty$.

Step 6: from the limit page to the complex. By theorem 58.2.6 and the naturality recorded in its proof, $\alpha_\infty^{n,m}$ is gr_n^W of the map $H^m(F^pK) \to H^m(K)$ induced by the inclusion, for the filtrations $W_nH^m(-) = \mathrm{im}\big(H^m(W_n-) \to H^m(-)\big)$, both finite in each degree. Apply theorem 56.1.15 to the map of short exact sequences

$$\begin{array}{ccccccccc}
0 & \to & W_{n-1}H^m(F^pK) & \to & W_nH^m(F^pK) & \to & \mathrm{gr}_n^W H^m(F^pK) & \to & 0 \\
 & & \downarrow & & \downarrow & & \downarrow & & \\
0 & \to & W_{n-1}H^m(K) & \to & W_nH^m(K) & \to & \mathrm{gr}_n^W H^m(K) & \to & 0
\end{array}$$

The sequence $0 \to \ker(W_{n-1}) \to \ker(W_n) \to \ker(\mathrm{gr}_n^W)$ is exact, the third kernel vanishes by Step 5, and $W_nH^m(F^pK) = 0$ for n small; so induction upwards on n gives $\ker\big(W_nH^m(F^pK) \to W_nH^m(K)\big) = 0$ for every n, and since the filtration is exhaustive in each degree, $H^m(F^pK) \to H^m(K)$ is a monomorphism for every p and m.

Strictness of d follows at once. Let $z \in d(K^{m-1}) \cap F^pK^m$, say $z = dx$ with $x \in K^{m-1}$. Then z is a cocycle of $F^pK^\bullet$ whose class maps to $[z] = 0$ in $H^m(K)$, so by injectivity $z = dy$ for some $y \in F^pK^{m-1}$. Hence $d(K^{m-1}) \cap F^pK^m = d(F^pK^{m-1})$, which is strict compatibility with F in the sense of definition 58.6.1.

Step 7: degeneration of the F-sequence. The filtration F is decreasing and biregular and $K^\bullet$ is bounded below, so after a shift of the index corollary 58.2.7 applies, and by corollary 58.2.8(2)

$$E_r^{p,q}(K,F) \cong \frac{Z_r^{p,q}}{Z_{r-1}^{p+1,q-1} + d\,Z_{r-1}^{p-r+1,\,q+r-2}}, \qquad Z_r^{p,q} = \{\, x \in F^pK^{p+q} : dx \in F^{p+r}K^{p+q+1} \,\}$$

Write $Z_\infty^{p,q} = \{x \in F^pK^{p+q} : dx = 0\}$. For $r \ge 1$ we claim $Z_r^{p,q} = Z_\infty^{p,q} + Z_{r-1}^{p+1,q-1}$. Given $x \in Z_r^{p,q}$, strictness at level $p+r$ gives $dx \in d(K^{p+q}) \cap F^{p+r}K^{p+q+1} = d(F^{p+r}K^{p+q})$, say $dx = dy$ with $y \in F^{p+r}K^{p+q}$. Then $x - y \in Z_\infty^{p,q}$, while $y \in F^{p+1}K^{p+q}$ has $dy \in F^{p+r}K^{p+q+1} = F^{(p+1)+(r-1)}K^{p+q+1}$, so $y \in Z_{r-1}^{p+1,q-1}$, which is a summand of the denominator. Hence every class of $E_r^{p,q}(K,F)$ is represented by a cocycle of $K^\bullet$, on which d vanishes, so $d_r = 0$ for every $r \ge 1$ and the spectral sequence of $(K^\bullet, F)$ degenerates at E_1. This proves (3), completing the proof.

The proof never manipulated the objects Z_r of definition 58.2.3 on the W-side, and it is worth saying what those would show. By proposition 58.2.4, with $Z_r^{n,m} = \{x \in W_n K^m : dx \in W_{n-r}K^{m+1}\}$ and $D_r^{n,m} = Z_{r-1}^{n-1,m} + d\,Z_{r-1}^{n+r-1,m-1}$, one has ${}_WE_r^{n,m} = Z_r^{n,m}/D_r^{n,m}$, and F_d^p is the image of $F^pK^m \cap Z_r^{n,m}$, while $F_{d^*}^p$ is the image of $Z_r^{n,m} \cap \big(F^pK^m + \widehat{D}_r^{n,m}\big)$, where $\widehat{D}$ is built from the objects $\{y \in W_aK^b : dy \in W_{a-s}K^{b+1} + F^pK^{b+1}\}$ in place of $Z_s^{a,b}$. The gap between the two is exactly the failure of the boundary condition to be detectable modulo F^p, and closing it by hand is the several-page computation the argument above replaces with a long exact sequence.

Example 58.6.6: Sharpness of the Hypothesis

Take $\mathcal{A}$ the category of $\mathbb{Q}$-vector spaces and let $K^\bullet$ be $\mathbb{Q}e$ in degree 0 and $\mathbb{Q}f$ in degree 1, with $de = f$. Filter it by

$$W_{-1}K = 0, \quad W_0K^0 = 0,\ W_0K^1 = \mathbb{Q}f, \quad W_1K = K$$

$$F^0K = K, \quad F^1K^0 = 0,\ F^1K^1 = \mathbb{Q}f, \quad F^2K = 0$$

Both are biregular filtrations by subcomplexes. The graded pieces are $\mathrm{gr}_0^W K = \mathbb{Q}f$ in degree 1 and $\mathrm{gr}_1^W K = \mathbb{Q}e$ in degree 0, each with zero differential, so the differential of every $\mathrm{gr}_n^W K^\bullet$ is trivially strict for F: the hypothesis of theorem 58.6.5(2) holds with $r_0 = 0$ and no more.

It holds with no more. The differential $d_1\colon {}_WE_1^{1,0} = \mathbb{Q}e \to {}_WE_1^{0,1} = \mathbb{Q}f$ is the connecting map of $0 \to W_0K \to K \to \mathrm{gr}_1^W K \to 0$, hence $e \mapsto f$, an isomorphism; its source has $F^1 = 0$ and its target has F^1 equal to everything, so d_1 is not strict. Likewise $H^\bullet(K) = 0$ while $\mathrm{gr}_F^0 K = \mathbb{Q}e$ in degree 0 and $\mathrm{gr}_F^1 K = \mathbb{Q}f$ in degree 1, so the spectral sequence of (K, F) has $E_1^{0,0} = \mathbb{Q}e$ and $E_1^{1,0} = \mathbb{Q}f$ with $E_\infty = 0$ and does not degenerate at E_1; equivalently $d(K^0) \cap F^1K^1 = \mathbb{Q}f$ while $d(F^1K^0) = 0$, so d is not strict for F. Conclusion (2) of theorem 58.6.5 with $r_0 = 0$ is therefore exactly right: here ${}_WE_2 = 0$, so the three filtrations do agree on ${}_WE_1$ and ${}_WE_2$, and nothing beyond that is true.

The example is not pathological, and a consumer should read it as a warning about the shape of the input. The strictness of d_0 is a statement about the graded pieces of W alone and can never see how those pieces are glued; the strictness of d_1 is a statement about the gluing, and it has to be supplied. In mixed Hodge theory it is supplied by weights: the terms of ${}_WE_1$ are cohomologies of smooth projective varieties, d_1 is a morphism of pure Hodge structures of a single weight, and such a morphism is strict, while d_r for $r \geq 2$ lowers the weight and vanishes. With all d_r strict, part (3) applies and the Hodge filtration degenerates at E_1.

The last of the three exercises below is used often enough elsewhere to be worth stating as a result in its own right.

Lemma 58.6.7: Degeneration at E_1 and Strictness

Let $(K^\bullet, F)$ be a complex with F a biregular decreasing filtration by subcomplexes. The spectral sequence of $(K^\bullet, F)$ degenerates at E_1 if and only if the differential of $K^\bullet$ is strictly compatible with F.

***Proof*:**
This is exercise 58.6.1 (3), proved there in both directions. No finiteness or field hypothesis is needed: biregularity alone makes the correction argument in the reverse direction terminate.

Exercise 58.6.1

1. Exhibit filtered vector spaces and strict morphisms f and g with $g \circ f$ not strict, confirming the last claim of proposition 58.6.2.
2. Give a morphism of filtered complexes that is a quasi-isomorphism but not a filtered quasi-isomorphism.
3. Show that the spectral sequence of a filtered complex (K, F) degenerates at E_1 if and only if the differential of K is strictly compatible with F.

Part XII

Category Theory

This final part gathers the categorical language that the rest of the book has been using informally and develops it in its own right. The change of viewpoint is the whole point: an object is understood not through its internal makeup but through its relationships to every other object of its kind. A group is a set with a multiplication, but what a group *is*, categorically, is recorded in the homomorphisms into and out of it. It is often said that category theory is ten percent about objects and ninety percent about the maps between them; the Yoneda lemma, the central result of this part, makes that slogan a theorem.

Much of what we have proved separately for groups, rings, and modules is one statement in disguise. The isomorphism theorems recur almost verbatim; a group action is a group of a special shape; a free module is a direct sum of copies of its ring; the Galois correspondence matches intermediate fields with subgroups. Category theory names the pattern behind these parallels. Because the language reaches well beyond algebra, the examples are drawn also from topology, order theory, and set theory; a reader who knows those subjects will find the categorical vocabulary already at work in them, and a reader who does not can take the algebraic instances as the primary thread.

59

Categories, Functors, and Natural Transformations

Category theory begins from a change of viewpoint: an object is understood not through its internal constitution but through its relationships to every other object of its kind. A group is a set carrying a multiplication; but what a group *is*, categorically, is recorded in the homomorphisms into and out of it, not in the elements it happens to contain. This chapter installs the three notions that make the viewpoint precise and usable. A *category* packages a collection of objects together with the morphisms between them and a law for composing those morphisms. A *functor* is the matching notion of map between categories, carrying objects to objects and morphisms to morphisms while respecting composition and identities. A *natural transformation*, in turn, is a map between functors; with it the word "canonical", used informally for a construction that rests on no arbitrary choice, acquires an exact meaning.

The reward for the abstraction is uniformity. The isomorphism theorems for groups, rings, and modules; the sense in which a group action is a group of a special shape, or a free module a direct sum of copies of its ring; the Galois correspondence between intermediate fields and subgroups: each is an instance of one categorical pattern rather than a coincidence restated in three notations. The examples in this chapter are drawn deliberately from across mathematics and not from algebra alone, so a reader who knows some topology or order theory will meet the language already at work in familiar disguises. The thread running through the chapter, which the Yoneda lemma will later promote to a theorem, is that an object is pinned down completely by the morphisms it receives from every other object, or dually by those it sends to them.

59.1 Categories, Morphisms, and Size

A category is a bookkeeping device. It records a collection of mathematical objects together with the transformations allowed between them, and it insists on almost nothing else: that transformations compose, that the composition is associative, and that each object carries a transformation doing nothing. From this austere data an entire discipline grows, because the axioms are met not only by sets and functions but by groups and homomorphisms, spaces and continuous maps, and by structures in which the "transformations" are not maps of any underlying set at all. The definition below is the one on which every later chapter rests, so it is worth reading with the care its brevity conceals.

Definition 59.1.1: Category

A *category* C consists of

- a collection of *objects*; and
- a collection of *morphisms* (also called *arrows*), each equipped with a specified *domain* object and a specified *codomain* object. That f is a morphism with domain X and codomain Y is written $f\colon X \to Y$;

subject to the following data. Each object X carries an *identity morphism* $1_X\colon X \to X$. For each pair of morphisms $f\colon X \to Y$ and $g\colon Y \to Z$, in which the codomain of f is the domain of g, there is a specified *composite morphism* $gf\colon X \to Z$. These are required to satisfy two axioms.

1. *Unitality.* For every $f\colon X \to Y$, the composites $1_Y f$ and $f 1_X$ are both equal to f.
2. *Associativity.* For every composable triple $f\colon W \to X$, $g\colon X \to Y$, $h\colon Y \to Z$, the composites $h(gf)$ and $(hg)f$ are equal, so that the triple composite may be written hgf without ambiguity.

Two points in this definition carry most of its weight. The first is the choice of the word *collection* rather than *set*, deliberate and returned to at the end of this section: a category may have so many objects or morphisms that they do not form a set, and the definition is engineered to tolerate this. The second is that a morphism $f\colon X \to Y$ *need not be a function.* Nothing in the axioms mentions elements, images, or preimages; a morphism is a primitive arrow from one object to another, and its only obligation is to compose. Several of the examples below realize morphisms as functions with extra structure, but others, matrices, order relations, homotopy classes, are morphisms that are not functions on any underlying set, and the theory treats them on exactly the same footing.

A word on the order of composition, fixed once and used throughout. The composite gf means "g after f": one applies $f\colon X \to Y$ first and $g\colon Y \to Z$ second, and the notation is read right to left, matching the classical convention $(g \circ f)(x) = g(f(x))$ for functions. The equivalent notation $g \circ f$ is used interchangeably when the circle aids legibility. The requirement that the codomain of f agree with the domain of g before gf is defined is precisely what makes the arrows chain: composition is a partial operation, defined exactly when the arrows line up head to tail.

The identity morphisms deserve a remark, because they let one dispense with objects entirely. Since 1_X is the unique morphism $e\colon X \to X$ with $ef = f$ and $ge = g$ for all composable f, g, distinct objects have distinct identities, and objects stand in bijection with their identity morphisms. Some treatments exploit this and define a category using morphisms alone, recovering each object as its identity arrow. The object/morphism formulation is retained here as the more transparent one, but the observation underlies the slogan that category theory is about the arrows: even the objects are secretly morphisms.

The definition matters only through its instances, and the supply is enormous. The first family of examples is the algebraic one, where objects are sets equipped with operations and morphisms are the maps respecting those operations. These are the categories a reader of [25] has been working inside all along, now named as such.

Example 59.1.2: Algebraic Categories

Each of the following is a category; in each case the identity morphism on an object is the identity function on its underlying set, and composition is composition of functions, so the two axioms of definition 59.1.1 are inherited from the associativity and unitality of function composition.

1. **Set** has sets as objects and functions (each with a specified domain and codomain) as morphisms. This is the ambient category of classical mathematics, and many of the categories below are built by equipping its objects with structure and cutting down its morphisms to those preserving that structure.

2. **Grp** has groups as objects and group homomorphisms as morphisms. That a composite of homomorphisms is a homomorphism, and that the identity function on a group is a homomorphism, are the two facts needed; both are immediate from the definition of a homomorphism (see [25]). The word *morphism* itself descends etymologically from *homomorphism*.

3. **Ring** has (unital) rings as objects and ring homomorphisms as morphisms.

4. **Field** has fields as objects and ring homomorphisms between them as morphisms. A field is a commutative unital division ring, so a field homomorphism is nothing but a ring homomorphism whose domain and codomain happen to be fields; changing which objects are admitted, while keeping the same notion of morphism, produces a different category.

5. $\mathbf{Mod}_R$, for a fixed ring R, has left R-modules as objects and R-module homomorphisms as morphisms. Two cases are frequent enough to earn their own names. When $R = k$ is a field, an R-module is a k-vector space and the category is written $\mathbf{Vect}_k$. When $R = \mathbb{Z}$, a $\mathbb{Z}$-module is precisely an abelian group, and the category is written **Ab**.

In every algebraic category the pattern is the same: an object is a set carrying structure, a morphism is a function preserving that structure, and the category axioms reduce to the observation that structure-preservation is closed under composition and holds for identity functions. This is worth verifying once, as above for **Grp**, and thereafter taken for granted. A category assembled in this way, one whose objects have underlying sets and whose morphisms are the structure-preserving functions between them, is called *concrete*; the examples of example 59.1.2 are all concrete. The reach of the definition, however, extends well past algebra and well past concreteness.

The next family shows that concreteness has nothing to do with algebra: these categories carry topological, measure-theoretic, differential, or order-theoretic structure, yet each is still a set with structure and a structure-preserving function.

Example 59.1.3: Non-Algebraic Concrete Categories

Each of the following is a concrete category; as before, identities are identity functions and composition is function composition, so the axioms of definition 59.1.1 hold automatically once one checks that the chosen class of functions is closed under composition and contains the identities.

1. **Top** has topological spaces as objects and continuous functions as morphisms. A composite of continuous functions is continuous and every identity function is continuous, which is all the axioms require.

2. $\mathbf{Top}_*$ has pointed topological spaces, that is, pairs (X, x_0) of a space with a chosen basepoint, as objects, and basepoint-preserving continuous functions as morphisms. The analogous construction on **Set** gives $\mathbf{Set}_*$, the category of pointed sets.

3. **Meas** has measurable spaces as objects and measurable functions as morphisms.

4. **Man** has smooth manifolds as objects and smooth maps as morphisms.

5. **Poset** has partially ordered sets as objects and order-preserving (monotone) functions as morphisms: a function $f\colon (P, \leq) \to (Q, \leq)$ with $x \leq y$ implying $f(x) \leq f(y)$.

Every category met so far is concrete, and the uniformity can mislead. It would be easy to conclude that a morphism is always a function of some kind, decorated with a preservation property, and that a category is always a collection of structured sets. Both conclusions are false, and the examples that break them are not exotic. In the categories below the morphisms are matrices, or comparisons in an order, or homotopy classes of maps, none of which is a function from the domain object to the codomain object, indeed in some of them the objects are not sets at all.

Example 59.1.4: Categories Whose Morphisms Are Not Functions

1. Fix a unital ring R. The category $\mathbf{Mat}_R$ has the natural numbers $0, 1, 2, \ldots$ as objects, and a morphism $n \to m$ is an $m \times n$ matrix with entries in R. Composition is matrix multiplication: given $A\colon n \to m$ (an $m \times n$ matrix) and $B\colon m \to k$ (a $k \times m$ matrix), the composite $BA\colon n \to k$ is the $k \times n$ product matrix, formed along the arrow chain

$$n \xrightarrow{A} m \xrightarrow{B} k$$

   The identity morphism on n is the $n \times n$ identity matrix. Associativity of composition is associativity of matrix multiplication, and unitality is the fact that the identity matrix acts trivially; both are standard, so $\mathbf{Mat}_R$ is a category. A morphism here is not a function from n to m: the objects n and m are natural numbers, not sets, and it would make no sense to feed an "element of n" to the matrix A. The arrows are matrices, full stop.

2. Any partially ordered set $(P, \leq)$ is itself a category. The objects are the elements of P, and between elements x and y there is a *unique* morphism $x \to y$ when $x \leq y$ and no morphism otherwise. Reflexivity, $x \leq x$, supplies the identity morphism on each object; transitivity, $x \leq y \leq z$ implies $x \leq z$, supplies the composite of $x \to y$ and $y \to z$. Because there is at most one morphism between any two objects, the associativity and unitality axioms hold trivially: any two morphisms with matching domain and codomain are equal. This is a category whose morphisms are comparisons, not functions, and it should be contrasted with **Poset**, whose *objects* are posets and whose morphisms are monotone functions; here a single poset *is* a category.

3. The homotopy category **hTop** has topological spaces as objects, but a morphism $X \to Y$ is a *homotopy class* of continuous maps, two maps being identified when one deforms continuously into the other. Composition is well defined because composing homotopic maps yields homotopic composites. A morphism is now an equivalence class of maps rather than a single map, so **hTop** is not concrete in the naive way; it is the natural home of much of algebraic topology, where a space matters only up to homotopy.

4. There is a category **Cat** whose objects are (small, in the sense defined below) categories and whose morphisms are the functors between them, the structure-preserving maps of categories taken up later in this chapter. That the objects are categories rather than sets is exactly the kind of self-reference that must be handled with care, and the size restriction to *small* categories, deferred to the size discussion that closes this section, is what keeps **Cat** from collapsing into paradox.

These last examples make precise a slogan already stated: the objects of a category are almost beside the point, and it is the morphisms that carry the content. A category is named for its objects only by convention and convenience, because in most cases the intended morphisms are clear once the objects are fixed, one says "the category of groups" and the reader supplies "and group homomorphisms". The convention is harmless but occasionally misleading, for the same collection of objects can support wildly different categories depending on the morphisms chosen. The natural numbers are the objects both of the discrete category with only identity arrows and of $\mathbf{Mat}_R$ with all matrices as arrows, and these are not the same category; the underlying objects settle nothing. What a category theorist studies is how objects relate, and relation is the business of arrows.

The word *collection* in definition 59.1.1 has been flagged twice as deliberate, and it is time to explain it. The difficulty is that the objects of **Set** are all sets, and there are too many sets to form a set. Attempting to gather them into one leads directly to contradiction, a warning that naive set formation is unsafe.

59.1.5 Russell's Paradox Suppose there were a set S of all sets. Consider

$$R = \{x \in S \big| x \notin x\}$$

the set of all sets that are not members of themselves. Since S contains every set, R is a set, so $R \in S$, and one may ask whether $R \in R$. If $R \in R$, then R satisfies the defining condition $R \notin R$, a contradiction. If $R \notin R$, then R satisfies the condition for membership in R, so $R \in R$, again a contradiction. Either horn is absurd, so no set S of all sets can exist.

The paradox is not a defect of category theory but of unrestricted set formation, and the standard remedy, adopted throughout the series, is to distinguish two sizes of aggregate. A *set* is an aggregate small enough to be handled by the usual axioms without contradiction; a *proper class*, or informally a *collection*, may be larger, large enough to hold all sets, or all groups, or all topological spaces, and it is precisely not required to be a set. The objects of **Set** form a proper class, not a set, and definition 59.1.1 was written with "collection" so that **Set** counts as a category despite this. The same care applies to **Cat**: were its objects allowed to include all categories, **Cat** would contain itself and the paradox would return, which is why its objects are restricted to *small* categories.

This forces a distinction between categories that are set-sized and those that have only set-sized pieces. The strictest condition asks that all of a category's arrows form a set.

Definition 59.1.6: Small Category

A category is *small* if its collection of morphisms is a set (rather than a proper class).

Since each object has its own identity morphism, and distinct objects have distinct identities, a small category also has only a set's worth of objects: the objects inject into the morphisms via $X \mapsto 1_X$. Smallness is therefore a single condition controlling the whole category at once. A poset viewed as a category (the second item of example 59.1.4) is small, as is any finite category and any group regarded as a one-object category. But **Set**, **Grp**, and their relatives are not small: already their objects form a proper class, so a fortiori their morphisms do.

Smallness is often too much to ask, and a weaker condition suffices for nearly everything the theory needs. It requires only that the arrows between each *fixed* pair of objects form a set, while allowing the objects, and hence all the arrows together, to be a proper class.

Definition 59.1.7: Locally Small Category

A category C is *locally small* if for every ordered pair of objects (X, Y) the morphisms from X to Y form a set. This set is the *hom-set* from X to Y, written

$$\mathrm{Hom}(X, Y) \qquad \text{or} \qquad \mathrm{Hom}_C(X, Y) \qquad \text{or} \qquad C(X, Y)$$

when the ambient category C must be named.

The name *hom-set* is a fossil of the algebraic origin of the notion, where $\mathrm{Hom}(X, Y)$ was literally a set of homomorphisms; in a general category its elements are morphisms, which need not be homomorphisms or even functions, but the name has stuck. The notation is convenient enough that $\mathrm{Hom}(X, Y)$ and $C(X, Y)$ are written even for categories that are not locally small, with the understanding that the aggregate in question is then a proper class rather than a set. Local smallness is the standing assumption behind almost every construction in this book, and for good reason: it is exactly the hypothesis that lets one apply set-level tools, functions, bijections, cardinalities, to the arrows between two objects, and the Yoneda lemma of the next chapter will turn each hom-set into a probe that reads off an object from its arrows.

The distinction between small and locally small is exactly the distinction between **Cat** and its examples. **Set** is locally small but not small: its objects form a proper class, so it is not small, yet for any two sets X and Y the functions from X to Y form a genuine set (a subset of the power set $\mathcal{P}(X \times Y)$), so it is locally small. The same holds for **Grp**, **Ring**, **Top**, and every other category of example 59.1.2 and example 59.1.3: each has a proper class of objects but only a set of morphisms between any fixed pair. These are the categories one works in daily, and they are locally small without exception; smallness, by contrast, is a special property enjoyed only by categories one can in principle write out.

Example 59.1.8: The Hom-Set of a Group

Hom-set notation names constructions the reader already knows. Let G and H be groups, regarded as objects of **Grp**. Then $\mathrm{Hom}_{\mathbf{Grp}}(G, H)$ is the set of all group homomorphisms $G \to H$, exactly the object studied under that name in [25]. A concrete instance illuminates a familiar idea: take $H = S_A$, the symmetric group of a set A. A group homomorphism $G \to S_A$ is by definition an action of G on A by permutations, so the hom-set

$$\mathrm{Hom}_{\mathbf{Grp}}(G, S_A)$$

is precisely the set of G-actions on A (see [25] for the action-homomorphism correspondence). A single piece of categorical notation thus repackages a definition from group theory: an action is a morphism into a symmetric group, and the collection of actions is a hom-set. The same phenomenon recurs throughout the book, and recognizing prior constructions as hom-sets is the first step toward the representability viewpoint developed in later chapters, where an object is recovered from the hom-sets into or out of it.

59.2 Isomorphisms, Monomorphisms, and Epimorphisms

The previous section built the vocabulary of a category: objects, morphisms, composition, identities. What that vocabulary conspicuously lacks is a notion of two objects being "the same". In the category **Set** two sets ought to count as interchangeable when they have the same cardinality; in **Grp** two groups when they are isomorphic; in **Top** two spaces when they are homeomorphic. Each of these is a bespoke definition phrased in terms of the internal anatomy of the objects. The thread of this book is that such internal descriptions are avoidable: sameness, injectivity, and surjectivity can all be read off the morphisms alone. This section makes good on that promise for the first three, defining an isomorphism, a monomorphism, and an epimorphism purely by how a morphism composes with others, and then harvesting the definitions for a categorical account of groups and of the invertible part of any category.

Two objects should count as the same when there is a morphism between them that can be undone. In **Set** a bijection $f\colon X \to Y$ has a two-sided inverse function; in **Grp** an isomorphism has an inverse homomorphism. The common feature, visible without ever mentioning elements, is the existence of a morphism running the other way that composes to the identity on both sides.

Definition 59.2.1: Isomorphism

Let C be a category. A morphism $f\colon X \to Y$ in C is an *isomorphism* if there exists a morphism $g\colon Y \to X$ with

$$gf = 1_X \qquad \text{and} \qquad fg = 1_Y$$

Such a g is called the *inverse* of f; two objects X and Y are *isomorphic*, written $X \cong Y$, when there exists an isomorphism between them.
An *endomorphism* of X (a morphism $X \to X$) that is an isomorphism is called an *automorphism* of X. The automorphisms of X form a set (in a locally small category) denoted $\mathrm{Aut}(X)$.

The inverse deserves the definite article: if g and g' are both inverses of f, then

$$g = g\,1_Y = g(fg') = (gf)g' = 1_X\,g' = g'$$

so an inverse, when it exists, is unique. This little computation is the categorical form of the familiar uniqueness of inverses in a group, and it uses nothing but associativity and the identity laws. The notation $g = f^{-1}$ is therefore unambiguous. The relation $X \cong Y$ is an equivalence relation: reflexivity comes from 1_X, symmetry from the fact that f^{-1} is again an isomorphism with inverse f, and transitivity from the observation that a composite of isomorphisms is an isomorphism, which is worked out in lemma 59.2.7 below.

The point of the definition is that it is category-relative. The morphism f carries no elements and no structure of its own; whether it is an isomorphism is a question about the category it lives in, and the answer can change dramatically as the category changes even when the underlying function stays fixed. The following computation of isomorphisms across the standing cast makes this concrete, and already exhibits the phenomenon that will recur throughout the book: a bijective structure-preserving map need not be an isomorphism.

Example 59.2.2: Isomorphisms in the Standing Cast

The isomorphisms of a category are computed by unwinding what a two-sided inverse must be.

1. In **Set** the isomorphisms are exactly the bijections. A function $f: X \to Y$ with a two-sided inverse function is injective (if $f(x) = f(x')$ then $x = f^{-1}f(x) = f^{-1}f(x') = x'$) and surjective (every $y = f(f^{-1}(y))$ is a value), and conversely a bijection has the inverse function as its two-sided inverse. Consequently the only invariant a set carries up to isomorphism is its cardinality: the particular names of the elements are invisible to the category.

2. In **Grp**, **Ring**, and $\mathbf{Mod}_R$ the isomorphisms are exactly the bijective homomorphisms. If a homomorphism f is bijective, its set-theoretic inverse f^{-1} is automatically a homomorphism: for $y, y' \in Y$, writing $x = f^{-1}(y)$ and $x' = f^{-1}(y')$, the identity $f(xx') = f(x)f(x') = yy'$ gives $f^{-1}(yy') = xx' = f^{-1}(y)f^{-1}(y')$, and similarly for the additive and scalar operations, see [25]. So f^{-1} is a morphism in the same category and f is an isomorphism. Here a bijective morphism *is* an isomorphism.

3. In **Top** the isomorphisms are the *homeomorphisms*: continuous bijections whose inverse is again continuous. This is strictly stronger than being a continuous bijection. The map from the half-open interval $[0, 1)$ to the circle S^1 sending t to $(\cos 2\pi t, \sin 2\pi t)$ is a continuous bijection, but its inverse is discontinuous at the point $(1, 0)$, where the circle is torn open. There is no isomorphism $[0, 1) \cong S^1$ in **Top**, even though the two underlying sets are isomorphic in **Set**. This is the first place where a bijective "structure-preserving" map fails to be an isomorphism, and it is the reason the categorical definition, which requires an inverse morphism rather than a bijective morphism, is the correct one.

4. In a poset $(P, \leq)$ regarded as a category, the only isomorphisms are the identities. A morphism $x \to y$ exists precisely when $x \leq y$, so an isomorphism $x \to y$ forces both $x \leq y$ and $y \leq x$, and antisymmetry gives $x = y$. A poset therefore has no nontrivial isomorphisms at all.

The contrast between the second and third items is the lesson to carry forward. In an algebraic category a bijective morphism inverts to a morphism for free, because the algebraic operations are equations that transport across the bijection; in **Top** continuity is not an equation but a genuine extra constraint on the inverse, and it can fail. Defining "sameness" as the existence of an inverse morphism, rather than as bijectivity of the morphism, is what makes the notion track the right equivalence in every category at once.

The poset computation, where every isomorphism is trivial, is one extreme. The opposite extreme, where *every* morphism is an isomorphism, is important enough to name, because it turns out to be the categorical home of group theory.

Definition 59.2.3: Groupoid

A *groupoid* is a category in which every morphism is an isomorphism.

A groupoid is the categorical distillation of "a system of reversible processes". Any category has, sitting inside it, a largest groupoid, obtained by throwing away every non-invertible morphism; that construction is lemma 59.2.7 below. But the single-object groupoids already recover an entire algebraic theory, and identifying which one is the first genuine payoff of the categorical language: a group is nothing but a groupoid with one object.

Example 59.2.4: A Group as a One-Object Groupoid

Let $\mathcal{G}$ be a groupoid with a single object, call it $*$. All the data of $\mathcal{G}$ lives in the single hom-set

$$G := \mathrm{Hom}_{\mathcal{G}}(*, *) = \mathrm{Aut}(*)$$

where the second equality holds because $\mathcal{G}$ is a groupoid, so every endomorphism of $*$ is an isomorphism, hence an automorphism. Composition of morphisms gives a binary operation

$$G \times G \to G \qquad (h, g) \mapsto hg$$

and the claim is that $(G, \cdot)$ is a group. Each group axiom is exactly one of the category axioms specialized to the one object.

Associativity. For any three morphisms $f, g, h \in G$, composition is defined (all domains and codomains equal $*$) and the associativity axiom of a category gives $h(gf) = (hg)f$. This is the associativity of the group operation.

Identity. The object $*$ carries an identity morphism $1_* \in G$, and the identity axiom of a category gives $1_* g = g = g\,1_*$ for every $g \in G$. So 1_* is a two-sided identity element, and it is the unique such element by the general uniqueness of identities in a category, which is the same argument as the uniqueness of a group identity, see [25].

Inverses. Because $\mathcal{G}$ is a groupoid, every $g \in G$ is an isomorphism, so there is a morphism $g^{-1} \in G$ with $gg^{-1} = 1_*$ and $g^{-1}g = 1_*$. Since there is only one object, 1_* is the only identity available, so this g^{-1} is a two-sided inverse of g in the group sense.

The three category axioms have produced exactly the three group axioms, and no data was left over: a one-object groupoid *is* a group, and conversely a group G gives a one-object groupoid $\mathbf{B}G$ with $\mathrm{Hom}(*, *) = G$ and composition the group multiplication (every element is invertible, so it is a groupoid). This is not an analogy but an identification, and it is the first sign of the book's thread at work. The group axioms are not an arbitrary list to be memorized; they are what the category axioms say when there is a single object and every arrow is reversible.

The one-object-groupoid picture reframes group theory as the study of the automorphisms of a single point, and this vantage point pays dividends later: a group action becomes a functor out of $\mathbf{B}G$, and the homomorphisms $G \to H$ become the functors $\mathbf{B}G \to \mathbf{B}H$. For now the identification stands as evidence that categorical definitions, far from being empty generalities, can reconstruct familiar structures exactly.

Groupoids and one-object groupoids are examples of a general move: carving a smaller category out of a larger one by restricting the objects and morphisms. When the restriction is compatible with the categorical structure, the result is again a category.

Definition 59.2.5: Subcategory

A *subcategory* D of a category C consists of a subcollection of the objects of C and a subcollection of the morphisms of C, such that

1. the domain and codomain of every morphism in D are objects of D;
2. the identity morphism 1_X of every object X of D lies in D; and
3. the composite gf of any composable pair of morphisms f, g of D lies in D.

A subcategory D is *full* if for every pair of objects X, Y of D one has $\mathrm{Hom}_D(X, Y) = \mathrm{Hom}_C(X, Y)$, that is, D omits objects but no morphisms between the objects it keeps.

The three conditions are exactly what is needed for D, equipped with the composition and identities inherited from C, to satisfy the category axioms in its own right: they are the closure requirements guaranteeing that composition never leads outside D and that every object has its identity. Associativity and the unit laws are inherited from C for free, since D's morphisms are C's morphisms and are composed the same way. A subcategory is thus the categorical analogue of a subgroup or a subspace: a sub-collection closed under the ambient operations.

Example 59.2.6: Rings with and without Identity

The categories of rings sit in a chain of subcategories

$$\mathbf{CRing} \subseteq \mathbf{Ring} \subseteq \mathbf{Rng}$$

where **Rng** has as objects the (not necessarily unital) rings with morphisms the ring homomorphisms not required to preserve a unit, **Ring** has as objects the unital rings with morphisms the unit-preserving homomorphisms, and **CRing** restricts **Ring** to the commutative unital rings. Each inclusion satisfies the three conditions: a homomorphism between unital rings has unital rings as domain and codomain, the identity homomorphism of a unital ring is unit-preserving, and a composite of unit-preserving homomorphisms is unit-preserving, so **Ring** is a subcategory of **Rng**; the same checks with commutativity added give $\mathbf{CRing} \subseteq \mathbf{Ring}$. The inclusion $\mathbf{CRing} \subseteq \mathbf{Ring}$ is *full*, since a ring homomorphism between commutative rings is automatically a morphism of **Ring** with no further condition. The inclusion $\mathbf{Ring} \subseteq \mathbf{Rng}$ is *not* full: between two unital rings there are non-unital homomorphisms (for instance the zero map $\mathbb{Z} \to \mathbb{Z}$, which sends 1 to 0) that live in **Rng** but not in **Ring**.

The distinction between full and non-full subcategories, illustrated cleanly by the rings example, matters throughout: a full subcategory is determined by its objects alone, whereas a non-full one records a genuine choice of which morphisms to keep. The one construction that packages the isomorphisms of a category always produces a subcategory, and identifying it settles the transitivity of $\cong$ left open after definition 59.2.1.

Every category, no matter how few of its morphisms are invertible, has a canonical groupoid inside it: keep all the objects but only the isomorphisms. That this collection really is a subcategory, and a groupoid, is the content of the next lemma. The proof is a matter of checking that identities are isomorphisms and that isomorphisms are closed under composition, but the second point carries the useful formula for the inverse of a composite.

Lemma 59.2.7: The Maximal Groupoid

Every category C contains a *maximal groupoid* $C^{\cong}$: the subcategory whose objects are all the objects of C and whose morphisms are exactly the isomorphisms of C. It is a groupoid, and it is maximal in the sense that any subcategory of C that is a groupoid is contained in $C^{\cong}$.

***Proof*:**
Write $C^{\cong}$ for the collection consisting of all objects of C together with those morphisms of C that are isomorphisms. The verification proceeds in three steps: that $C^{\cong}$ is a subcategory, that it is a groupoid, and that it is maximal.

Step 1: $C^{\cong}$ is a subcategory. The three closure conditions of definition 59.2.5 must hold. Every isomorphism of C has its domain and codomain among the objects of C, which are all present in $C^{\cong}$, so condition (1) holds. For condition (2), each identity 1_X is an isomorphism, being its own two-sided inverse ($1_X 1_X = 1_X$), so every identity of C lies in $C^{\cong}$. For condition (3), let $f\colon X \to Y$ and $g\colon Y \to Z$ be composable isomorphisms, with inverses f^{-1} and g^{-1}. Then $f^{-1}g^{-1}$ is a two-sided inverse of gf:

$$(f^{-1}g^{-1})(gf) = f^{-1}(g^{-1}g)f = f^{-1}1_Y f = f^{-1}f = 1_X$$

and symmetrically $(gf)(f^{-1}g^{-1}) = g(ff^{-1})g^{-1} = g1_Y g^{-1} = gg^{-1} = 1_Z$. Hence gf is an isomorphism, with inverse the reversed composite $(gf)^{-1} = f^{-1}g^{-1}$, and condition (3) holds. So $C^{\cong}$ is a subcategory of C.

Step 2: $C^{\cong}$ is a groupoid. Every morphism of $C^{\cong}$ is by construction an isomorphism of C; its inverse is again an isomorphism of C, hence again a morphism of $C^{\cong}$. So every morphism of $C^{\cong}$ is invertible *within* $C^{\cong}$, which is precisely the condition in definition 59.2.3 for a groupoid.

Step 3: maximality. Let D be any subcategory of C that is a groupoid. Every morphism f of D is invertible in D, so it has an inverse g in D satisfying $gf = 1$ and $fg = 1$; since g is in particular a morphism of C, this exhibits f as an isomorphism of C. Thus every morphism of D is a morphism of $C^{\cong}$, and D is contained in $C^{\cong}$, completing the proof.

The inverse formula extracted in Step 1, $(gf)^{-1} = f^{-1}g^{-1}$, is the categorical version of the "socks and shoes" rule familiar from group theory: to undo two operations performed in order, undo them in the reverse order. It also closes the loose end from earlier: transitivity of the isomorphism relation is exactly the statement that a composite of isomorphisms is an isomorphism, which Step 1 establishes. The maximal groupoid is a functor of C in a sense to be made precise later, and it is the object one names when speaking of "the isomorphisms of C" as a category in their own right; the one-object case recovers example 59.2.4, since the maximal groupoid of a one-object category with all endomorphisms invertible is that group.

Isomorphisms capture sameness. The finer information carried by a morphism, whether it identifies distinct inputs or misses part of its target, is captured by two weaker conditions, each phrased entirely through composition. The clue is the elementary characterization of injections and surjections of sets by cancellation. A function f is injective exactly when it can be cancelled on the left ($fh = fk$ forces $h = k$), and surjective exactly when it can be cancelled on the right ($hf = kf$ forces $h = k$). These cancellation properties make no reference to elements and so define morphisms in any category.

Definition 59.2.8: Monomorphism and Epimorphism

Let $f\colon X \to Y$ be a morphism in a category C.

1. f is a *monomorphism* (or is *monic*) if it is left-cancellable: for every pair of morphisms $h, k\colon W \to X$,
$$fh = fk \quad\Longrightarrow\quad h = k$$
A monomorphism is often written $f\colon X \rightarrowtail Y$.

2. f is an *epimorphism* (or is *epic*) if it is right-cancellable: for every pair of morphisms $h, k\colon Y \to Z$,
$$hf = kf \quad\Longrightarrow\quad h = k$$
An epimorphism is often written $f\colon X \twoheadrightarrow Y$.

The two definitions are word-for-word the same with the composition order reversed, which is the first hint of a pervasive symmetry: a monomorphism in C is the same thing as an epimorphism in the category C^{op} with all arrows reversed, and vice versa. Reversing the arrows turns left-cancellation into right-cancellation, so every statement about monomorphisms has a mirror statement about epimorphisms obtained by passing to the opposite category. This is the duality principle, the organizing tool of the next section; here it already halves the work, since anything proved for monomorphisms holds for epimorphisms by reversing arrows.

Every isomorphism is both monic and epic: if f has an inverse g, then $fh = fk$ gives $h = gfh = gfk = k$, and $hf = kf$ gives $h = hfg = kfg = k$. The converse fails in general, and the failure is exactly what makes mono and epi worth naming separately from iso. In the concrete algebraic categories the two conditions usually coincide with injectivity and surjectivity, but the coincidence is not automatic, and epimorphisms are where it breaks first.

Example 59.2.9: Epic Need Not Mean Surjective

In many concrete categories a monomorphism is the same as an injective map and an epimorphism the same as a surjective map. In **Set** this holds on both counts: a function is monic exactly when it is injective and epic exactly when it is surjective, since one can probe it with maps out of and into a one-point set. In **Grp**, **Ab**, and $\mathbf{Mod}_R$ the monomorphisms are again the injective homomorphisms, and in these categories the epimorphisms are also exactly the surjective homomorphisms. The agreement can fail, however, and the standard counterexample is the inclusion of the integers into the rationals.

Consider **Ring** (unital rings, unit-preserving homomorphisms) and the inclusion $\iota\colon \mathbb{Z} \to \mathbb{Q}$. This map is not surjective, since $\frac{1}{2}$ is not in its image. It is nonetheless an epimorphism. To see this, let R be any unital ring and let $h, k\colon \mathbb{Q} \to R$ be ring homomorphisms with $h\iota = k\iota$, that is, h and k agree on every integer. For a rational number $\frac{a}{b}$ with $b \neq 0$, the element $h(b) = k(b)$ is invertible in R (it is the image of the unit $b \in \mathbb{Q}$, and a unit-preserving homomorphism sends units to units), and
$$h\left(\tfrac{a}{b}\right) = h(a)\,h(b)^{-1} = k(a)\,k(b)^{-1} = k\left(\tfrac{a}{b}\right)$$
because $h(a) = k(a)$ and $h(b) = k(b)$ force $h(b)^{-1} = k(b)^{-1}$. Thus $h = k$ on all of $\mathbb{Q}$, so ι is right-cancellable and hence epic. The homomorphism out of $\mathbb{Q}$ is pinned down by its values on the integers because the rationals are generated from the integers by inverting nonzero elements, and inverses are forced once the elements they invert are known. A ring epimorphism can therefore fail

to be surjective; indeed there is no nonzero ring homomorphism $\mathbb{Q} \to \mathbb{Z}$ at all, so ι has no hope of being "undone" set-theoretically.

The lesson is that mono and epi are categorical conditions: they test a morphism against all the other morphisms of the category rather than inspecting its effect on elements, and this test can diverge from injectivity or surjectivity precisely when the category's morphisms are constrained in a way that lets a non-surjective map still be cancellable. In **Ring** the constraint is that a homomorphism is determined by less data than its full graph, because it must respect inverses; the inclusion $\mathbb{Z} \to \mathbb{Q}$ exploits exactly that. Keeping the categorical definitions rather than the element-wise ones is what allows a single notion of "injective-like" and "surjective-like" morphism to make sense in every category, including those such as **Poset** or the arrow-only categories where elements are not available to inspect at all.

59.3 Duality and Opposite Categories

The philosophy of category theory is that an object is known by its morphisms, and the morphisms carry a direction: an arrow $f\colon X \to Y$ is not the same datum as an arrow $Y \to X$. This directedness is why morphisms are drawn as arrows rather than undirected lines. It also raises a question that turns out to organize a large part of the subject. What happens if every arrow is reversed? If reversal always produced the same theory, the direction would be inessential and $X - Y$ would be adequate notation. It does not: in most categories reversal yields a different category, and comparing a category with its reverse is one of the most productive moves available. The reversal is packaged into a single construction, the opposite category, and the payoff is a metatheorem, the duality principle, that halves the labor of category theory by turning every proved statement into a second proved statement for free.

The construction reverses each arrow while leaving the objects untouched. Composition must be reversed with it, since the composite of $X \to Y \to Z$ should become an arrow $Z \to X$ built from the reversed pieces.

Definition 59.3.1: Opposite Category

Let C be a category (definition 59.1.1). The *opposite category* (or *dual category*) C^{op} is the category with

- the same objects as C;
- for each morphism $f\colon X \to Y$ of C, a morphism $f^{\mathrm{op}}\colon Y \to X$ of C^{op}, with domain and codomain interchanged; every morphism of C^{op} arises this way, so that

  $$\mathrm{Hom}_{C^{\mathrm{op}}}(X, Y) = \mathrm{Hom}_C(Y, X)$$

  as sets;

subject to the requirements

1. for each object X, the identity in C^{op} is $1_X^{\mathrm{op}} = 1_X$;
2. for morphisms $f\colon X \to Y$ and $g\colon Y \to Z$ of C, whose composite $gf\colon X \to Z$ is defined, the reversed morphisms $g^{\mathrm{op}}\colon Z \to Y$ and $f^{\mathrm{op}}\colon Y \to X$ are composable in C^{op}, with composite
   $$f^{\mathrm{op}} g^{\mathrm{op}} = (gf)^{\mathrm{op}}$$

The composition rule is forced by the direction of the arrows and is worth reading slowly. In C the composite gf runs $X \to Y \to Z$. Reversing each arrow turns f into $f^{\mathrm{op}}\colon Y \to X$ and g into $g^{\mathrm{op}}\colon Z \to Y$, and the only way to compose these two is $Z \to Y \to X$, that is $f^{\mathrm{op}} g^{\mathrm{op}}$. The reversed composite therefore takes the label $(gf)^{\mathrm{op}}$, the reverse of the original composite. The rule $(gf)^{\mathrm{op}} = f^{\mathrm{op}} g^{\mathrm{op}}$ says exactly that reversal turns the order of composition around, the same order-reversal seen in the transpose of matrices $(AB)^\top = B^\top A^\top$ or the inverse of a product $(gh)^{-1} = h^{-1} g^{-1}$ in a group. The following square, the two ways of reading the composable pair $X \xrightarrow{f} Y \xrightarrow{g} Z$ and its reverse, records the construction.

$$\begin{tikzcd}
X \arrow[r, "f"] \arrow[rr, bend left, "gf"] & Y \arrow[r, "g"] & Z \\
X & Y \arrow[l, "f^{\mathrm{op}}"'] & Z \arrow[l, "g^{\mathrm{op}}"'] \arrow[ll, bend right, "(gf)^{\mathrm{op}} = f^{\mathrm{op}} g^{\mathrm{op}}"']
\end{tikzcd}$$

The top row lives in C and the bottom row in C^{op}; the two rows carry the same objects and the same underlying arrows, only the directions and the composition order differ.

The construction is an involution: reversing twice restores the original category. Interchanging domain and codomain twice returns each arrow to its starting orientation, $1_X^{\mathrm{opop}} = 1_X$, and the composition rule applied twice gives $(f^{\mathrm{op}} g^{\mathrm{op}})^{\mathrm{op}} = g^{\mathrm{opop}} f^{\mathrm{opop}} = gf$, so $(C^{\mathrm{op}})^{\mathrm{op}} = C$ on the nose. This is what makes duality a tool rather than a mere relabeling: a category and its opposite stand on equal footing, and any construction available in one is available in the other by symmetry. There is no distinguished “forward” direction, only a chosen one.

The definition is uniform across all categories, but its content varies sharply with the category. The next example runs it through three of the standard cast to show that the opposite can be isomorphic to the original, isomorphic in a less obvious way, or a category of an entirely different kind.

Example 59.3.2: Opposites of Posets, Groups, and Sets

1. *Posets.* A partially ordered set $(P, \leq)$ is a category with one object per element and a unique arrow $x \to y$ exactly when $x \leq y$ (definition 59.1.1). Reversing the arrows reverses the order: P^{op} has a unique arrow $x \to y$ exactly when $y \leq x$, so P^{op} is the poset $(P, \geq)$ with the order turned upside down. For the chain $(\mathbb{Z}, \leq)$ the covering arrows run

$$\cdots \longrightarrow -1 \longrightarrow 0 \longrightarrow 1 \longrightarrow 2 \longrightarrow \cdots$$

   and in $(\mathbb{Z}, \leq)^{\mathrm{op}} = (\mathbb{Z}, \geq)$ they run the other way,

$$\cdots \longleftarrow -1 \longleftarrow 0 \longleftarrow 1 \longleftarrow 2 \longleftarrow \cdots$$

   The opposite of a poset is again a poset, but usually a different one: a least element becomes a greatest element, an infimum becomes a supremum. This is the categorical root of the fact that every order-theoretic notion comes in a matched pair.

2. *Groups.* A group G is a one-object groupoid with $\mathrm{Hom}(*, *) = G$, composition given by the group multiplication. Its opposite G^{op} is again a one-object groupoid, hence again a group: the underlying set is the same, but the product is $g^{\mathrm{op}}h^{\mathrm{op}} = (hg)^{\mathrm{op}}$, so multiplication in G^{op} is the multiplication of G with the factors in reversed order. This G^{op} is the *opposite group*. It is isomorphic to G, but the isomorphism is not the identity map on the underlying set unless G is abelian; the natural isomorphism is inversion,

$$\varphi\colon G \to G^{\mathrm{op}}, \qquad \varphi(g) = g^{-1}$$

   which is a homomorphism precisely because inversion reverses products,

$$\varphi(gh) = (gh)^{-1} = h^{-1}g^{-1}$$

   the right-hand side being the product $\varphi(g)\varphi(h)$ computed in G^{op}. Passing to the opposite group is the same operation as passing between left and right actions. A left action of G on a set X is a rule with $(gh) \cdot x = g \cdot (h \cdot x)$; a right action satisfies $x \cdot (gh) = (x \cdot g) \cdot h$. Setting $g \cdot x := x \cdot g^{-1}$, or equivalently reading a right action of G as a left action of G^{op}, converts one into the other, and the order-reversal in G^{op} is exactly what makes the two associativity laws match. Dropping the requirement of inverses leaves a one-object category rather than a one-object groupoid, that is, a monoid; the opposite monoid M^{op} is defined by the same reversed product, and here M^{op} need not be isomorphic to M.

3. *Sets.* The opposite of **Set** is not **Set**. A morphism $X \to Y$ in $\mathbf{Set}^{\mathrm{op}}$ is, by definition, a function $Y \to X$, so composition in $\mathbf{Set}^{\mathrm{op}}$ is composition of functions read backwards. One is tempted to picture a morphism $f\colon X \to Y$ of $\mathbf{Set}^{\mathrm{op}}$ as "f run in reverse", a map $Y \to X$, but a function need not be invertible, so there is no such reverse function in general. What the reversed arrows track is not functions but relations. A function $f\colon A \to B$ is a relation that is single-valued (each $a \in A$ has exactly one image) and total (every $a \in A$ has an image); its reversed graph $\{(f(a), a) : a \in A\} \subseteq B \times A$ is a relation that is injective-on-the-nose (distinct a give distinct pairs) and surjective (every a appears), but is single-valued and total only when f is a bijection. So the arrows of $\mathbf{Set}^{\mathrm{op}}$ are reversed graphs of functions, which

> are relations of a constrained type, not functions. The precise statement is that $\mathbf{Set}^{\mathrm{op}}$ is equivalent to the category of complete atomic Boolean algebras (the powersets PX with their function-induced maps), not to **Set** itself. The takeaway needs no such machinery: reversing arrows in **Set** leaves the world of sets and functions entirely, so "the opposite" is a strictly new category, not a repackaging of the old one.

The three cases exhibit the full range. For a poset the opposite is another object of the same kind, obtained by an evident flip. For a group the opposite is again of the same kind and even isomorphic to the original, though by a map (inversion) that is invisible if one looks only at underlying sets. For **Set** the opposite is a category of a different nature altogether. In every case the objects are unchanged and only the direction of the arrows moves, yet the categories that result can be as close as isomorphic or as far apart as sets and Boolean algebras. This variability is what gives the next principle its force: a single argument, reinterpreted in the opposite category, can say something new precisely because the opposite category is different.

Everything so far concerns one category at a time. The reason opposite categories are indispensable is a metatheorem about the whole theory. Because the axioms of a category are self-dual, reversing every arrow in a valid argument produces another valid argument, so any statement proved for all categories automatically holds for all opposite categories, which is to say for all categories again with the arrows reversed. Stating this cleanly halves the work of the subject.

59.3.3 The Duality Principle Let S be a statement about categories, expressed in the language of objects, morphisms, domains, codomains, composites, and identities. Let S^{op} be the *dual statement*, obtained from S by reversing the direction of every morphism and the order of every composite (and leaving the objects, identities, and logical structure unchanged). Then S holds in a category C if and only if S^{op} holds in C^{op}. In particular, if S holds in *every* category, then so does S^{op}: each theorem of category theory yields a second theorem, its *dual*, for free.

The justification is immediate from the involution $(C^{\mathrm{op}})^{\mathrm{op}} = C$. A proof of S valid in every category is valid in particular in C^{op}; unwinding the definition of C^{op} (each arrow reversed, each composite reordered) turns the assertion "S holds in C^{op}" into the assertion "S^{op} holds in C". Since C was arbitrary, S^{op} holds in every category. There is nothing to check beyond the bookkeeping of reversal, and that bookkeeping is exactly the passage to the opposite category. The practical consequence is a discipline: a construction and its dual, conventionally distinguished by the prefix "co-", are literally the same construction performed in the opposite category. A colimit is a limit in C^{op}, a coproduct is a product in C^{op}, an initial object is a terminal object in C^{op}. To prove a fact about colimits, prove the corresponding fact about limits and read it in C^{op}; no separate argument is ever needed for the dual.

The cleanest illustration available at this stage is the pairing of monomorphisms with epimorphisms. Recall the definitions (definition 59.2.8): a morphism $f\colon X \to Y$ is a *monomorphism* if $fh = fk$ implies $h = k$ for all h, k into X, and an *epimorphism* if $hf = kf$ implies $h = k$ for all h, k out of Y. Written side by side, the epimorphism condition is the monomorphism condition with every composite reordered, which is to say read in the opposite category.

Proposition 59.3.4: Monomorphisms are Epimorphisms in the Opposite

Let C be a category and $f\colon X \to Y$ a morphism of C, with corresponding morphism $f^{\mathrm{op}}\colon Y \to X$ of C^{op}. Then f is a monomorphism in C if and only if f^{op} is an epimorphism in C^{op}. Dually, f is an epimorphism in C if and only if f^{op} is a monomorphism in C^{op}.

***Proof*:**
Fix $f\colon X \to Y$ in C. By definition 59.2.8, f is a monomorphism in C precisely when, for every object W and every pair of morphisms $h, k\colon W \to X$ of C,

$$fh = fk \implies h = k$$

Passing to C^{op}, the morphisms $h, k\colon W \to X$ of C become morphisms $h^{\mathrm{op}}, k^{\mathrm{op}}\colon X \to W$ of C^{op}, and $f^{\mathrm{op}}\colon Y \to X$ of C^{op} is composable before them. Applying the composition rule of definition 59.3.1 to the hypothesis and conclusion,

$$fh = fk \iff (fh)^{\mathrm{op}} = (fk)^{\mathrm{op}} \iff h^{\mathrm{op}} f^{\mathrm{op}} = k^{\mathrm{op}} f^{\mathrm{op}}$$

and $h = k \iff h^{\mathrm{op}} = k^{\mathrm{op}}$, since reversal is a bijection on morphisms. Substituting, the defining implication of a monomorphism for f in C reads

$$h^{\mathrm{op}} f^{\mathrm{op}} = k^{\mathrm{op}} f^{\mathrm{op}} \implies h^{\mathrm{op}} = k^{\mathrm{op}}$$

for all $h^{\mathrm{op}}, k^{\mathrm{op}}\colon X \to W$ of C^{op}. As W ranges over all objects and $h^{\mathrm{op}}, k^{\mathrm{op}}$ over all such pairs, this is exactly the condition that f^{op} be an epimorphism in C^{op} (definition 59.2.8). Hence f is a monomorphism in C if and only if f^{op} is an epimorphism in C^{op}. The second equivalence follows by applying the first to C^{op} in place of C and using $(C^{\mathrm{op}})^{\mathrm{op}} = C$, so that f^{op} is a monomorphism in C^{op} if and only if f is an epimorphism in C, as we sought to show.

The proposition is duality in miniature. Monomorphism and epimorphism are not two unrelated conditions that happen to resemble each other; they are a single condition read in the two directions, one the dual of the other. Anything proved about monomorphisms in every category, by the duality principle (box 59.3.3), holds verbatim for epimorphisms once every arrow is reversed, and conversely. This is why the two notions appear throughout the subject as an inseparable pair, and why the prefix "co-" recurs so insistently: it is the marker of a statement obtained by passing to the opposite category.

Duality is not confined to category theory as an internal bookkeeping device. Across mathematics one meets theorems asserting that two apparently unrelated classes of objects are opposite to each other, in the precise sense that one category is equivalent to the opposite of another. Each such theorem lets questions about one class be translated into questions about the other, often turning a hard geometric or analytic problem into a tractable algebraic one, and each is evidence that the reversal of arrows is a pervasive phenomenon. Three named examples indicate the range. *Gelfand duality* identifies the category of compact Hausdorff spaces with the opposite of the category of commutative unital C^*-algebras, sending a space X to its algebra $C(X)$ of continuous complex functions, so that the topology of X is recovered entirely from an algebra of functions on it. *Stone duality* identifies the category of Boolean algebras with the opposite of the category of compact Hausdorff totally disconnected spaces (Stone spaces), the algebraic and the topological faces of the same data. *Pontryagin duality* sends a locally compact abelian group G to its dual group $\hat{G} = \mathrm{Hom}(G, S^1)$ of continuous characters and exhibits the category of locally compact abelian groups as equivalent to its own opposite, with the double dual $\hat{\hat{G}}$ canonically G again; it is the structural home of the Fourier transform. The proofs of these three belong to functional analysis, general topology, and harmonic analysis respectively and lie well beyond this chapter; they are cited here only as evidence that the opposite category, introduced above as a formal reversal of arrows, names a structure that recurs at the center of several mathematical disciplines.

59.4 Functors

Categories were built to record how objects of one kind relate to one another through their morphisms. The next question is the one that always follows the introduction of a new mathematical structure: what are the structure-preserving maps between two such structures? A map from one category to another must carry objects to objects and morphisms to morphisms, and it must respect the two pieces of data a category carries beyond its raw collections, composition and identities. Such a map is a *functor*, and it is the vehicle by which one part of mathematics is transported into another: the fundamental group turns spaces into groups, the spectrum turns rings into spaces, and the derivative turns smooth maps into linear ones. Each of these is a functor, and in each case the price of admission is exactly that composition be respected.

Definition 59.4.1: Functor

A *functor* $F\colon C \to D$ between categories C and D consists of

- an object Fc of D for each object c of C, and
- a morphism $Ff\colon Fc \to Fc'$ of D for each morphism $f\colon c \to c'$ of C, so that the domain and codomain of Ff are, respectively, F applied to the domain and codomain of f,

subject to the two axioms

1. (functoriality) $F(gf) = Fg\,Ff$ for every composable pair $f\colon c \to c'$, $g\colon c' \to c''$ of morphisms of C, and
2. (identities) $F(1_c) = 1_{Fc}$ for every object c of C.

A functor is thus a map of the underlying data of a category that preserves everything a category is made of: the assignment of domains and codomains, composition, and identities. The functoriality axiom is the substantive one. It says that F cannot see a composite gf as anything other than the composite $Fg\,Ff$ of the images, so a functor can never manufacture a relation between images that was not already present between the sources, nor destroy one that was. This is what makes a functor a faithful conduit of structure rather than an arbitrary reassignment: a commutative triangle in C is sent to a commutative triangle in D, so any equation between composites that holds in the source survives into the target. When $C = D$ the functor is an *endofunctor* of C.

The examples below are the standing cast the rest of the book returns to. Several are functors the reader has already met under other names; the point is that a single word now covers the powerset operation, the act of forgetting algebraic structure, the fundamental group, the derivative, and the notion of a group action all at once.

Example 59.4.2: Functors

1. *The covariant powerset endofunctor.* There is an endofunctor $P\colon \mathbf{Set} \to \mathbf{Set}$ sending a set A to its powerset PA, the set of all subsets of A, and sending a function $f\colon A \to B$ to the *direct image* function
$$Pf\colon PA \to PB \qquad A' \mapsto f(A')$$
where $f(A') = \{f(a) \big| a \in A'\}$. This is a functor: for $A' \subseteq A$ one has $(Pg\,Pf)(A') = g(f(A')) = (gf)(A') = P(gf)(A')$ because the image of a set under gf is the image under g of its image under f, and $P(1_A)(A') = 1_A(A') = A' = 1_{PA}(A')$, so P carries identities to identities.

2. *Forgetful functors.* There is a functor $U\colon \mathbf{Grp} \to \mathbf{Set}$ carrying each group to its underlying set and each group homomorphism to its underlying function. Functoriality and preservation of identities are immediate, since composition and identities of homomorphisms are computed as composition and identities of the underlying functions. A functor that discards part of the structure of its source is a *forgetful functor*. The same pattern gives $\mathbf{Mod}_R \to \mathbf{Ab}$ (forget the R-action, remember the underlying abelian group) and $\mathbf{Ring} \to \mathbf{Ab}$ (forget the multiplication, remember the additive group). These need not be inclusions: a single abelian group underlies many different rings, so $\mathbf{Ring} \to \mathbf{Ab}$ is far from injective on objects.

3. *The free-group functor.* In the opposite direction there is the *free functor* $F\colon \mathbf{Set} \to \mathbf{Grp}$ sending a set X to the free group FX on X and a function $f\colon X \to Y$ to the unique homomorphism $Ff\colon FX \to FY$ extending $x \mapsto f(x)$ on generators, see [25]. Uniqueness of the extending homomorphism forces $F(gf)$ and $Fg\,Ff$ to agree, since both extend $x \mapsto g(f(x))$, and $F(1_X)$ and 1_{FX} both extend $x \mapsto x$, so F is a functor.

4. *The fundamental group.* The fundamental group is the functor $\pi_1\colon \mathbf{Top}_* \to \mathbf{Grp}$ from pointed spaces and basepoint-preserving continuous maps to groups: it sends a pointed space (X, x_0) to its fundamental group $\pi_1(X, x_0)$ of homotopy classes of loops at x_0, and a pointed map $f\colon (X, x_0) \to (Y, y_0)$ to the induced homomorphism $f_*\colon \pi_1(X, x_0) \to \pi_1(Y, y_0)$, $[\gamma] \mapsto [f \circ \gamma]$. That $(g \circ f)_* = g_*\,f_*$ and $(1_X)_* = 1_{\pi_1(X,x_0)}$ is exactly the functoriality of π_1, developed in [38].

5. *The derivative and the chain rule.* Let $\mathbf{Euclid}_*$ be the category whose objects are pointed finite-dimensional Euclidean spaces $(\mathbb{R}^n, a)$ and whose morphisms $(\mathbb{R}^n, a) \to (\mathbb{R}^m, b)$ are differentiable maps f with $f(a) = b$. The derivative is a functor
$$D\colon \mathbf{Euclid}_* \to \mathbf{Mat}_{\mathbb{R}} \qquad D(\mathbb{R}^n, a) = n, \qquad Df = \text{the Jacobian of } f \text{ at } a$$
sending a pointed Euclidean space to its dimension (an object of $\mathbf{Mat}_{\mathbb{R}}$) and a pointed differentiable map to its Jacobian matrix at the marked point. For composable $f\colon (\mathbb{R}^n, a) \to (\mathbb{R}^m, b)$ and $g\colon (\mathbb{R}^m, b) \to (\mathbb{R}^k, c)$, functoriality reads
$$D(gf) = Dg\,Df$$
the Jacobian of the composite equals the product of the Jacobian of g at b with the Jacobian of f at a. This equation is the chain rule; that D preserves identities is the statement that the derivative of an identity map is the identity matrix. The chain rule is the functoriality of the derivative.

6. *A group as a one-object category.* A group G is a category $\mathbf{B}G$ with a single object $*$, one morphism $* \to *$ for each element of G, composition given by the group operation, and 1_* the identity element. A functor $X\colon \mathbf{B}G \to C$ is then precisely an *action of G on an object of C*: it chooses an object $X*$ of C and, for each $g \in G$, an endomorphism $g_*\colon X* \to X*$, subject to
$$(hg)_* = h_*\, g_* \qquad \text{and} \qquad e_* = 1_{X*}$$
which are the functoriality and identity axioms written out. These are exactly the axioms of an action. Specializing the target recovers the familiar notions: a functor $\mathbf{B}G \to \mathbf{Set}$ is a G-set, a functor $\mathbf{B}G \to \mathbf{Vect}_k$ is a k-linear representation of G, and a functor $\mathbf{B}G \to \mathbf{Top}$ is a G-space. The three classical settings in which a group acts are three choices of target for one functor.

The last two examples repay a second look. The chain rule, which is usually presented as a formula to be memorized, is nothing more than the assertion that one particular assignment respects composition; once the derivative is recognized as a functor, the chain rule is not a theorem to be proved separately but the definition of what it means for D to be one. The one-object viewpoint is the same economy applied to group actions: rather than three definitions of a G-set, a G-representation, and a G-space, there is a single definition, a functor out of $\mathbf{B}G$, whose three instances differ only in where the arrows land. This is the recurring dividend of the categorical language, that constructions which look unrelated turn out to be one construction viewed through different targets.

Duality (definition 59.3.1) supplies a second kind of functor. Reversing the arrows of the source category before applying a functor produces a map that turns composition around, and this arrow-reversing behavior is so common that it is given its own name.

Definition 59.4.3: Contravariant Functor

A *contravariant functor* F from C to D is a functor $F\colon C^{\mathrm{op}} \to D$. Unwinding the definition of the opposite category (definition 59.3.1), a contravariant functor assigns

- an object Fc of D to each object c of C, and
- a morphism $Ff\colon Fc' \to Fc$ of D to each morphism $f\colon c \to c'$ of C, reversing domain and codomain,

subject to

1. $F(gf) = Ff\,Fg$ for every composable pair $f\colon c \to c'$, $g\colon c' \to c''$ of C, and
2. $F(1_c) = 1_{Fc}$ for each object c of C.

For emphasis, a functor in the sense of definition 59.4.1 is called *covariant*.

The single point of difference is the reversal in the first axiom: a covariant functor sends gf to $Fg\,Ff$, whereas a contravariant functor sends it to $Ff\,Fg$, varying *contrary* to the direction of composition. This is not a new species of object but a bookkeeping convenience: a contravariant functor $C \to D$ is literally a covariant functor $C^{\mathrm{op}} \to D$, so every theorem about functors applies to contravariant ones after the source is replaced by its opposite. The examples are as ubiquitous as the covariant ones, and each is built by the same move, sending a map to a map that goes the other way.

Example 59.4.4: Contravariant Functors

1. *The linear dual.* The dual-space construction is a contravariant functor $(-)^*\colon \mathbf{Vect}_k^{\mathrm{op}} \to \mathbf{Vect}_k$ carrying a vector space V to its dual $V^* = \mathrm{Hom}_k(V,k)$ and a linear map $f\colon V \to W$ to the transpose $f^*\colon W^* \to V^*$ given by precomposition,

$$f^*(\varphi) = \varphi \circ f \qquad (\varphi\colon W \to k)$$

The reversal is forced by precomposition: since f^* eats a functional on W and returns one on V, it points from W^* to V^*. For composable $f\colon U \to V$ and $g\colon V \to W$ one computes $(gf)^*(\varphi) = \varphi \circ g \circ f = f^*(g^*(\varphi))$, so $(gf)^* = f^*g^*$, and $(1_V)^*(\varphi) = \varphi = 1_{V^*}(\varphi)$; the arrows reverse exactly as the definition requires.

2. *The contravariant powerset.* There is a contravariant functor $P\colon \mathbf{Set}^{\mathrm{op}} \to \mathbf{Set}$ sending a set A to PA and a function $f\colon A \to B$ to the *preimage* function

$$f^{-1}\colon PB \to PA \qquad B' \mapsto f^{-1}(B') = \{a \in A \big| f(a) \in B'\}$$

In contrast to the covariant powerset of example 59.4.2, which uses direct image and preserves the direction of f, the preimage points the other way. Functoriality holds because $(gf)^{-1}(C') = f^{-1}(g^{-1}(C'))$ for $C' \subseteq C$, which is the identity $(gf)^{-1} = f^{-1}g^{-1}$, and $(1_A)^{-1} = 1_{PA}$.

3. *The prime spectrum.* The spectrum is a contravariant functor $\mathrm{Spec}: \mathbf{CRing}^{\mathrm{op}} \to \mathbf{Top}$ sending a commutative ring R to its prime spectrum $\mathrm{Spec}\, R$ and a ring homomorphism $\varphi\colon R \to S$ to the map

$$\varphi^*\colon \mathrm{Spec}\, S \to \mathrm{Spec}\, R \qquad \mathfrak{p} \mapsto \varphi^{-1}(\mathfrak{p})$$

taking a prime $\mathfrak{p}$ of S to its preimage $\varphi^{-1}(\mathfrak{p})$, which is a prime of R because the preimage of a prime ideal under a ring homomorphism is prime. The contravariance is again a matter of preimages: φ runs from R to S, but φ^* runs from $\mathrm{Spec}\, S$ to $\mathrm{Spec}\, R$. Functoriality follows from $(\psi\varphi)^{-1}(\mathfrak{p}) = \varphi^{-1}(\psi^{-1}(\mathfrak{p}))$.

4. *The opens of a space.* For a topological space X, let $\mathcal{O}(X)$ be the set of open subsets of X, partially ordered by inclusion and so a poset. This is a contravariant functor $\mathcal{O}\colon \mathbf{Top}^{\mathrm{op}} \to \mathbf{Poset}$: a continuous map $f\colon X \to Y$ is sent to the preimage map

$$f^{-1}\colon \mathcal{O}(Y) \to \mathcal{O}(X) \qquad U \mapsto f^{-1}(U)$$

which lands in $\mathcal{O}(X)$ precisely because f is continuous, and which preserves inclusions since $U \subseteq V$ gives $f^{-1}(U) \subseteq f^{-1}(V)$. Functoriality is once more the composite-of-preimages identity $(gf)^{-1}(U) = f^{-1}(g^{-1}(U))$.

The four contravariant examples are variations on a single theme: each sends a map to its associated preimage or precomposition, and preimage inverts the direction of the arrow. The dual space, the contravariant powerset, the spectrum, and the opens of a space are the same movement performed in four categories. Set against the covariant powerset, which uses direct image and so preserves direction, they display in the sharpest possible terms the difference between forward transport by image and backward transport by preimage.

One property holds for every functor, covariant or contravariant, and it is the first indication that a functor is a genuine conduit of categorical structure rather than an arbitrary map: a functor cannot tell isomorphic objects apart, because it carries the very equations that witness an isomorphism.

Lemma 59.4.5: Functors Preserve Isomorphisms

Let $F\colon C \to D$ be a functor. If $f\colon x \to y$ is an isomorphism in C, then $Ff\colon Fx \to Fy$ is an isomorphism in D, with inverse $F(f^{-1})$. In particular, $x \cong y$ in C implies $Fx \cong Fy$ in D.

***Proof*:**
Let $f\colon x \to y$ be an isomorphism in C with inverse $g = f^{-1}\colon y \to x$, so that $gf = 1_x$ and $fg = 1_y$ (definition 59.2.1). Applying F and using the functoriality and identity axioms of definition 59.4.1,

$$Fg\,Ff = F(gf) = F(1_x) = 1_{Fx}$$

so $Fg\colon Fy \to Fx$ is a left inverse of Ff. The same computation with the roles of f and g interchanged gives

$$Ff\,Fg = F(fg) = F(1_y) = 1_{Fy}$$

so Fg is also a right inverse of Ff. Thus Ff is an isomorphism with two-sided inverse $Fg = F(f^{-1})$, and $Fx \cong Fy$, as we sought to show.

The proof used only the two functor axioms, and it applies verbatim to a contravariant functor: there the images of f and g compose in the opposite order, but the two products $Ff\,Fg$ and $Fg\,Ff$ still equal 1_{Fx} and 1_{Fy} in some order, so Ff remains invertible. Every functor therefore respects the categorical notion of sameness, which is why a functorial invariant, one whose value is a functor, can be used to distinguish objects: if $Fx \ncong Fy$ then $x \ncong y$. This is the mechanism behind the fundamental group as a tool in topology, since π_1, being a functor, sends homeomorphic pointed spaces to isomorphic groups, so non-isomorphic fundamental groups certify that two spaces are not homeomorphic.

Isomorphisms are the only class of maps a functor is guaranteed to preserve. Functors need not preserve monomorphisms or epimorphisms (definition 59.2.8). The spectrum makes the failure concrete: the inclusion $\mathbb{Z} \hookrightarrow \mathbb{Q}$ is an epimorphism in **CRing** (example 59.2.9), yet Spec reverses it, and its image $\operatorname{Spec}\mathbb{Q} \to \operatorname{Spec}\mathbb{Z}$ is a monomorphism rather than an epimorphism, so the epi property is not carried across. What a functor does preserve, by the identical one-line argument that proves lemma 59.4.5, is any mono or epi that *splits*: if f has a one-sided inverse s with $sf = 1$, then applying F gives $Fs\,Ff = 1$, so Ff retains that one-sided inverse and remains a split monomorphism, and dually for a split epimorphism. The equations a splitting provides survive because a functor preserves exactly the equations between composites, and nothing more.

59.5 Natural Transformations

Functors compare categories; natural transformations compare functors. This third layer is what makes the categorical vocabulary powerful enough to say precisely what mathematicians have always meant by a construction being *canonical*, or *basis-free*, or *independent of choices*. The guiding example is one every reader has met in linear algebra. A finite-dimensional vector space V is isomorphic to its dual V^*, but only after a basis is chosen; change the basis and the isomorphism changes. Yet V is also isomorphic to its double dual V^{**}, and this time no choice is needed at all: a single formula, valid for every V and compatible with every linear map, produces the isomorphism. The distinction between these two isomorphisms, one requiring a choice and one not, is not a vague matter of taste. It is captured exactly by whether a family of maps assembles into a natural transformation. The section develops the double-dual example first, extracts the general definition, packages functors and their transformations into a category of their own, and closes with the notion of sameness that naturality makes available for categories themselves: equivalence.

The dual space is the starting point. For a field k and a vector space V, the linear dual is $V^* = \mathrm{Hom}_k(V, k)$, the space of linear functionals on V; this is the object developed in Core Algebra, and its basic properties are cited rather than reproved here (see [25]). When V is finite-dimensional, V^* has the same dimension as V, so the two are isomorphic. An isomorphism is built by choosing a basis $e_1, \ldots, e_n$ of V and sending e_i to the dual basis vector e_i^*, the functional determined by $e_i^*(e_j) = \delta_{ij}$. This map is a linear isomorphism, but it depends on the basis: a different basis of V yields a different isomorphism $V \to V^*$, and there is no way to single one out. The double dual behaves differently, and pinning down that difference is the object of the example.

Example 59.5.1: The Double Dual of a Vector Space

Let k be a field and V a vector space over k. Its *double dual* is $V^{**} = \mathrm{Hom}_k(V^*, k) = \mathrm{Hom}_k(\mathrm{Hom}_k(V, k), k)$. For each vector $v \in V$ define the *evaluation map* $\mathrm{ev}_V(v)\colon V^* \to k$ by

$$\mathrm{ev}_V(v)(\varphi) = \varphi(v) \qquad (\varphi \in V^*)$$

so $\mathrm{ev}_V(v)$ is the functional on V^* that reads off the value of a given functional at v. This assembles into a single map $\mathrm{ev}_V\colon V \to V^{**}$, $v \mapsto \mathrm{ev}_V(v)$, defined without any reference to a basis.

Step 1: $\mathrm{ev}_V(v)$ is linear for each v. For $\varphi, \psi \in V^*$ and $c \in k$,

$$\mathrm{ev}_V(v)(\varphi + \psi) = (\varphi + \psi)(v) = \varphi(v) + \psi(v) = \mathrm{ev}_V(v)(\varphi) + \mathrm{ev}_V(v)(\psi)$$

and $\mathrm{ev}_V(v)(c\varphi) = (c\varphi)(v) = c\,\varphi(v) = c\,\mathrm{ev}_V(v)(\varphi)$, using the vector-space structure on V^*. So $\mathrm{ev}_V(v) \in V^{**}$.

Step 2: ev_V is linear in v. For $v, w \in V$ and $c \in k$, the identity of functionals $\mathrm{ev}_V(v + w) = \mathrm{ev}_V(v) + \mathrm{ev}_V(w)$ is checked by evaluating both sides at an arbitrary $\varphi \in V^*$:

$$\mathrm{ev}_V(v + w)(\varphi) = \varphi(v + w) = \varphi(v) + \varphi(w) = \mathrm{ev}_V(v)(\varphi) + \mathrm{ev}_V(w)(\varphi)$$

and likewise $\mathrm{ev}_V(cv)(\varphi) = \varphi(cv) = c\,\varphi(v) = c\,\mathrm{ev}_V(v)(\varphi)$, so $\mathrm{ev}_V(cv) = c\,\mathrm{ev}_V(v)$. The linearity of ev_V is inherited from the linearity of the functionals it evaluates against.

Step 3: ev_V is injective. Suppose $v \neq 0$. Extend v to a basis of V and let $\varphi \in V^*$ be the coordinate functional with $\varphi(v) = 1$ (see [25] for the existence of such a functional). Then $\mathrm{ev}_V(v)(\varphi) = \varphi(v) = 1 \neq 0$, so $\mathrm{ev}_V(v) \neq 0$. Thus $\ker \mathrm{ev}_V = 0$.

Step 4: in finite dimension ev_V is an isomorphism. If $\dim V = n < \infty$ then $\dim V^* = n$ and $\dim V^{**} = n$, so ev_V is an injective linear map between spaces of equal finite dimension and is therefore an isomorphism. Hence $V \cong V^{**}$ with no choice of basis.

Step 5: the evaluation maps are compatible with every linear map. Given a linear map $f\colon V \to W$, the dual map $f^*\colon W^* \to V^*$ is $f^*(\psi) = \psi \circ f$, and the double dual map $f^{**}\colon V^{**} \to W^{**}$ is

$f^{**} = (f^*)^*$, so $f^{**}(\Phi) = \Phi \circ f^*$ for $\Phi \in V^{**}$. The square

$$\begin{tikzcd} V \arrow[r, "\mathrm{ev}_V"] \arrow[d, "f"'] & V^{**} \arrow[d, "f^{**}"] \\ W \arrow[r, "\mathrm{ev}_W"'] & W^{**} \end{tikzcd}$$

commutes. Indeed, for $v \in V$ and any $\psi \in W^*$,

$$f^{**}(\mathrm{ev}_V(v))(\psi) = \mathrm{ev}_V(v)(f^*\psi) = (f^*\psi)(v) = \psi(f(v)) = \mathrm{ev}_W(f(v))(\psi)$$

so $f^{**} \circ \mathrm{ev}_V = \mathrm{ev}_W \circ f$ as maps $V \to W^{**}$.

The five steps isolate exactly what is meant by canonicity. The map ev_V is defined by one formula that mentions only the data present in every vector space, so it exists for all V simultaneously (Steps 1 and 2), it is an isomorphism whenever V is finite-dimensional (Steps 3 and 4), and Step 5 shows that the whole family $\{\mathrm{ev}_V\}_V$ is compatible with the maps between spaces: transporting a vector along f and then evaluating gives the same answer as evaluating and then transporting along f^{**}. The isomorphism $V \cong V^*$ has no analogue of Step 5, precisely because it is built from a basis rather than from a formula uniform in V. The commuting square of Step 5 is the whole content of naturality, and abstracting it gives the central definition.

Definition 59.5.2: Natural Transformation

Let C and D be categories and let $F, G\colon C \to D$ be functors. A *natural transformation* $\alpha\colon F \Rightarrow G$ assigns to each object c of C a morphism

$$\alpha_c\colon Fc \to Gc$$

in D, called the *component* of α at c, such that for every morphism $f\colon c \to c'$ in C the *naturality square*

$$\begin{tikzcd} Fc \arrow[r, "\alpha_c"] \arrow[d, "Ff"'] & Gc \arrow[d, "Gf"] \\ Fc' \arrow[r, "\alpha_{c'}"'] & Gc' \end{tikzcd}$$

commutes, that is,

$$Gf \circ \alpha_c = \alpha_{c'} \circ Ff$$

A natural transformation $\alpha\colon F \Rightarrow G$ is a *natural isomorphism* if every component α_c is an isomorphism in D. When a natural isomorphism $F \Rightarrow G$ exists, the functors are *naturally isomorphic*, written $F \cong G$.

A natural transformation is a family of morphisms, one per object of the source category, wired together by the naturality condition so that the family respects every morphism of the source. The double arrow $\Rightarrow$ marks the new altitude: an ordinary morphism $\to$ connects objects, a functor $\to$ connects categories, and a natural transformation $\Rightarrow$ connects functors. It is often drawn as a 2-cell filling the region between two parallel functors,

$$\begin{tikzcd} C \arrow[r, bend left, "F"{name=U}] \arrow[r, bend right, "G"'{name=L}] & D \arrow[from=U, to=L, Rightarrow, "\alpha"] \end{tikzcd}$$

which records the shape of the data: two functors $C \to D$ and a transformation between them. The naturality square is the substantive clause. Without it a family $\{\alpha_c\}$ is just an unstructured collection of morphisms; with it, α interacts coherently with all of C, and this coherence is what earns the word "natural". A useful reading of the square is that α commutes with change of object: whether one applies α before or after transporting along f, the result is the same.

The double-dual example is now an instance of the definition, and naming the two functors involved makes the fit exact.

Example 59.5.3: Evaluation as a Natural Isomorphism

Take $C = D = \mathbf{Vect}_k$. The identity functor $\mathrm{Id}\colon \mathbf{Vect}_k \to \mathbf{Vect}_k$ sends each space to itself and each linear map to itself. The *double dual functor* $(-)^{**}\colon \mathbf{Vect}_k \to \mathbf{Vect}_k$ sends V to V^{**} and $f\colon V \to W$ to $f^{**}\colon V^{**} \to W^{**}$; it is a covariant functor because dualizing twice restores the original variance (each application of $(-)^*$ is contravariant, and the composite of two contravariant functors is covariant, cf. definition 59.4.3).

The evaluation maps of example 59.5.1 are the components of a natural transformation

$$\mathrm{ev}\colon \mathrm{Id} \Rightarrow (-)^{**}$$

with component $\mathrm{ev}_V\colon V \to V^{**}$ at each space V. The naturality square at a linear map $f\colon V \to W$ is

$$\begin{array}{ccc} V & \xrightarrow{\mathrm{ev}_V} & V^{**} \\ {\scriptstyle f}\big\downarrow & & \big\downarrow{\scriptstyle f^{**}} \\ W & \xrightarrow[\mathrm{ev}_W]{} & W^{**} \end{array}$$

which is exactly the commuting square proved in Step 5 of example 59.5.1. Restricting to the full subcategory of finite-dimensional spaces, every component ev_V is an isomorphism (Step 4), so ev is a natural isomorphism there and $\mathrm{Id} \cong (-)^{**}$. This is the precise sense in which a finite-dimensional space is canonically its own double dual: not only that each V is isomorphic to V^{**}, but that the isomorphisms are the components of a single natural transformation.

The single dual admits no such transformation, and the reason is worth isolating carefully, since it is a matter of variance. Each application of $(-)^*$ is contravariant, so the single dual is a functor $(-)^*\colon \mathbf{Vect}_k^{\mathrm{op}} \to \mathbf{Vect}_k$, not an endofunctor of $\mathbf{Vect}_k$; the symbol $\mathrm{Id} \Rightarrow (-)^*$ does not even typecheck, because a natural transformation compares two functors of the same variance and the naturality square would need its right edge $f^*\colon W^* \to V^*$ to point the same way as $f\colon V \to W$, which it does not. The comparison is made on the groupoid $\mathbf{Vect}_k^{\mathrm{fd},\cong}$ of finite-dimensional spaces and their isomorphisms, where every morphism is invertible and variance is immaterial; there both Id and $(-)^*$ are genuine endofunctors, so a natural isomorphism $\mathrm{Id} \cong (-)^*$ is a meaningful thing to ask for. None exists: the family of basis-dependent isomorphisms $V \to V^*$ does *not* assemble into a natural transformation, because the naturality square fails for the isomorphisms of V that rescale the chosen basis. Naturality is exactly the property the basis-free construction has and the basis-dependent one lacks.

The contrast in the example is worth stating as a principle. A construction is canonical precisely when it is natural, that is, when its instances at the various objects are the components of a natural transformation. Any construction that requires an arbitrary choice, a basis, a splitting, a lift, fails to be natural, because different choices produce components that violate the naturality square. The vocabulary of natural transformations turns the informal "no choices were made" into a checkable condition on a commuting diagram.

Once natural transformations are in hand, functors between two fixed categories become the objects of a category in their own right, with natural transformations as the morphisms. This packaging is not a formality: it is the setting in which diagrams, presheaves, and much of the rest of the subject live, and it is where the double-dual isomorphism becomes an isomorphism in the literal categorical sense. To build it, one needs to know how natural transformations compose. Given $\alpha\colon F \Rightarrow G$ and $\beta\colon G \Rightarrow H$ between functors $F, G, H\colon C \to D$, the composite is defined componentwise by $(\beta\alpha)_c = \beta_c \circ \alpha_c$, and the identity transformation $1_F\colon F \Rightarrow F$ has components $(1_F)_c = 1_{Fc}$.

Definition 59.5.4: Functor Category

Let C and D be categories. The *functor category* $[C, D]$ (also written D^C) has

- as objects, the functors $F\colon C \to D$;
- as morphisms $F \to G$, the natural transformations $\alpha\colon F \Rightarrow G$.

The identity morphism on a functor F is the natural transformation 1_F with components $(1_F)_c = 1_{Fc}$, and the composite of $\alpha\colon F \Rightarrow G$ with $\beta\colon G \Rightarrow H$ is the natural transformation $\beta\alpha\colon F \Rightarrow H$ with components $(\beta\alpha)_c = \beta_c \circ \alpha_c$.

The definition asserts that $[C, D]$ is a category, and this must be verified: the proposed composite and identities have to be natural transformations, and they have to satisfy the category axioms. The verification is short, and doing it in full shows how the naturality squares of the factors combine.

***Proof*:**
Two things require checking: that $\beta\alpha$ and 1_F are natural transformations, and that composition is associative and unital.

Step 1: the componentwise composite is natural. Let $\alpha\colon F \Rightarrow G$ and $\beta\colon G \Rightarrow H$ be natural, and let $f\colon c \to c'$ be a morphism in C. Naturality of α gives $Gf \circ \alpha_c = \alpha_{c'} \circ Ff$, and naturality of β gives $Hf \circ \beta_c = \beta_{c'} \circ Gf$. Then

$$Hf \circ (\beta\alpha)_c = Hf \circ \beta_c \circ \alpha_c = \beta_{c'} \circ Gf \circ \alpha_c = \beta_{c'} \circ \alpha_{c'} \circ Ff = (\beta\alpha)_{c'} \circ Ff$$

so the naturality square for $\beta\alpha$ commutes. Diagrammatically, the two naturality squares stack vertically and the outer rectangle is what commutes:

$$\begin{tikzcd} Fc \arrow[r, "\alpha_c"] \arrow[d, "Ff"'] & Gc \arrow[r, "\beta_c"] \arrow[d, "Gf"] & Hc \arrow[d, "Hf"] \\ Fc' \arrow[r, "\alpha_{c'}"'] & Gc' \arrow[r, "\beta_{c'}"'] & Hc' \end{tikzcd}$$

Step 2: the identity is natural. For 1_F with components 1_{Fc} and any $f\colon c \to c'$, both routes around the square equal Ff, since $Ff \circ 1_{Fc} = Ff = 1_{Fc'} \circ Ff$. So 1_F is a natural transformation.

Step 3: the category axioms. Composition and identities in $[C, D]$ are defined componentwise, and in each component they are ordinary composition and identities in D. Associativity and the unit laws therefore hold component by component, inherited from the corresponding axioms in D. For a third composable transformation $\gamma\colon H \Rightarrow K$ and each object c,

$$((\gamma\beta)\alpha)_c = \gamma_c\beta_c\alpha_c = (\gamma(\beta\alpha))_c, \qquad (1_G\,\alpha)_c = \alpha_c = (\alpha\,1_F)_c$$

A pair of natural transformations agreeing in every component are equal, so these componentwise identities give the required equalities of natural transformations. Thus $[C, D]$ satisfies the category axioms, completing the proof.

The functor category is the natural home for the double-dual isomorphism. In $[\mathbf{Vect}_k, \mathbf{Vect}_k]$ the transformation $\mathrm{ev}\colon \mathrm{Id} \Rightarrow (-)^{**}$ of example 59.5.3 is a morphism, and because every component is an isomorphism (in finite dimensions) it is an isomorphism in the functor category: the componentwise inverse ev^{-1}, with components $(\mathrm{ev}_V)^{-1}$, is again natural and is a two-sided inverse to ev. So the statement "V is canonically isomorphic to V^{**}" becomes the statement "Id and $(-)^{**}$ are isomorphic objects of the functor category". The composition just defined, taking place along the objects of C, is the *vertical* composition of natural transformations; there is a second, *horizontal* composition that composes transformations between functors that themselves compose, developed when it is needed. Functor categories are already familiar in disguise: when C is a small category and $D = \mathbf{Set}$, an object of $[C, \mathbf{Set}]$ is a diagram of sets shaped like C, and morphisms of diagrams are exactly natural transformations.

Natural isomorphism is also the ingredient needed to fix the notion of sameness for categories themselves. The obvious candidate, isomorphism of categories, asks too much. Two categories C and D would be isomorphic if there were functors $F\colon C \to D$ and $G\colon D \to C$ with $GF = \mathrm{Id}_C$ and $FG = \mathrm{Id}_D$ on the nose, meaning $GFc = c$ for every object c and $GFf = f$ for every morphism. In practice this equality-on-objects requirement is almost never met by the functors one cares about, because the objects of C and D are typically built from different raw material even when the categories carry the same mathematical content. What one wants is that GFc be canonically isomorphic to c, not equal to it, uniformly in c; and "canonically isomorphic, uniformly in c" is precisely a natural isomorphism $GF \cong \mathrm{Id}_C$. Weakening equality to natural isomorphism gives the right notion.

Definition 59.5.5: Equivalence of Categories

An *equivalence of categories* between C and D consists of functors $F\colon C \to D$ and $G\colon D \to C$ together with natural isomorphisms

$$\eta\colon \mathrm{Id}_C \Rightarrow GF \qquad \text{and} \qquad \varepsilon\colon FG \Rightarrow \mathrm{Id}_D$$

(each component an isomorphism). When such data exist, F is an *equivalence*, G is a *quasi-inverse* of F, and the categories C and D are *equivalent*, written $C \simeq D$.

An equivalence relaxes the two rigid equations $GF = \mathrm{Id}_C$ and $FG = \mathrm{Id}_D$ of an isomorphism to the two natural isomorphisms $GF \cong \mathrm{Id}_C$ and $FG \cong \mathrm{Id}_D$. The functor F need not hit every object of D on the nose, nor establish a bijection between objects; it need only do so up to isomorphism and in a way compatible with all morphisms. This is the correct notion of sameness for categories: it identifies categories that carry the same mathematical information even when their objects are packaged differently, and essentially every "two descriptions of the same theory" result in mathematics is an equivalence rather than an isomorphism of categories.

There is a second characterization of equivalences that is usually easier to check than exhibiting a quasi-inverse, phrased entirely in terms of the single functor F. Recall from definition 59.4.1 that F acts on hom-sets by $f \mapsto Ff$; call F *full* if each of these maps $\mathrm{Hom}_C(c, c') \to \mathrm{Hom}_D(Fc, Fc')$ is surjective, *faithful* if each is injective, and *fully faithful* if each is a bijection. Call F *essentially surjective* if every object d of D is isomorphic to Fc for some object c of C. The two conditions together say that F is a bijection on morphisms between any two objects and hits every object up to isomorphism.

Proposition 59.5.6: Characterization of Equivalences

A functor $F: C \to D$ is an equivalence of categories if and only if F is full, faithful, and essentially surjective.

The forward direction is elementary and can be given now; the converse requires building a quasi-inverse object by object, and its construction is deferred to the chapter on the Yoneda lemma, where the tools for choosing the components coherently are in place.

***Proof*:**
Only the forward implication is proved here; for the converse, the construction of a quasi-inverse from full faithfulness and essential surjectivity is carried out once the necessary machinery is available, and is taken up later in the book.

Suppose F is an equivalence, with quasi-inverse G and natural isomorphisms $\eta: \mathrm{Id}_C \Rightarrow GF$ and $\varepsilon: FG \Rightarrow \mathrm{Id}_D$.

Step 1: F is essentially surjective. For any object d of D, the component $\varepsilon_d: FGd \to d$ is an isomorphism, so $d \cong Fc$ for $c = Gd$. Every object of D is thus isomorphic to one in the image of F.

Step 2: F is faithful. Let $f, g: c \to c'$ satisfy $Ff = Fg$. Applying G gives $GFf = GFg$. Naturality of η at the morphism f gives $GFf \circ \eta_c = \eta_{c'} \circ f$, and likewise for g, so

$$\eta_{c'} \circ f = GFf \circ \eta_c = GFg \circ \eta_c = \eta_{c'} \circ g$$

Since $\eta_{c'}$ is an isomorphism it is monic, and cancelling it gives $f = g$. Thus F is faithful. The identical argument applied to G (with the natural isomorphism ε in place of η) shows G is faithful as well.

Step 3: F is full. Let $h: Fc \to Fc'$ be any morphism in D. Set $f = \eta_{c'}^{-1} \circ Gh \circ \eta_c : c \to c'$, a morphism of C built from $Gh: GFc \to GFc'$ and the isomorphisms η_c, $\eta_{c'}$. Naturality of η at f reads $GFf \circ \eta_c = \eta_{c'} \circ f$, and substituting the definition of f gives $\eta_{c'} \circ f = Gh \circ \eta_c$, whence $GFf = Gh$. Because G is faithful (Step 2), $GFf = Gh$ forces $Ff = h$. So every $h: Fc \to Fc'$ is Ff for some f, and F is full, as we sought to show.

The two descriptions of an equivalence serve different purposes. The definition in terms of a quasi-inverse and two natural isomorphisms is the one to use when the inverse construction is at hand and one wants to transport structure back and forth. The characterization by full faithfulness and essential surjectivity is the one to use when only F is given and building an explicit G would be laborious: three conditions on a single functor replace the search for a partner. A concrete instance shows the second criterion at work.

Example 59.5.7: Matrices as an Equivalence

Let k be a field, let $\mathbf{Vect}_k^{\mathrm{fd}}$ be the category of finite-dimensional k-vector spaces and linear maps, and let $\mathbf{Mat}_k$ be the category whose objects are the natural numbers $0, 1, 2, \ldots$ and whose morphisms $m \to n$ are the $n \times m$ matrices over k, with composition given by matrix multiplication and identities the identity matrices (this is the matrix category $\mathbf{Mat}_k$ of example 59.1.4, specialized to $R = k$). Define $F\colon \mathbf{Mat}_k \to \mathbf{Vect}_k^{\mathrm{fd}}$ by $F(n) = k^n$ on objects and by sending an $n \times m$ matrix A to the linear map $k^m \to k^n$, $x \mapsto Ax$.

The claim is that F is an equivalence, verified through proposition 59.5.6.

- *Fully faithful.* A linear map $k^m \to k^n$ is represented, with respect to the standard bases, by a unique $n \times m$ matrix, and every such matrix arises this way. So $\mathbf{Mat}_k(m, n) \to \mathrm{Hom}_k(k^m, k^n)$, $A \mapsto (x \mapsto Ax)$, is a bijection: F is full and faithful.
- *Essentially surjective.* Every finite-dimensional space V has a basis of some size n, and choosing one gives an isomorphism $V \cong k^n = F(n)$ (see [25]). So every object of $\mathbf{Vect}_k^{\mathrm{fd}}$ is isomorphic to one in the image of F.

Hence F is an equivalence, and $\mathbf{Mat}_k \simeq \mathbf{Vect}_k^{\mathrm{fd}}$.

The equivalence in the example is not an isomorphism of categories, and the reason is instructive. The objects of $\mathbf{Mat}_k$ are natural numbers, while the objects of $\mathbf{Vect}_k^{\mathrm{fd}}$ are vector spaces; there are a proper class of two-dimensional spaces but only one object 2 in $\mathbf{Mat}_k$, so no functor can be a bijection on objects and $GF = \mathrm{Id}$ cannot hold on the nose. Yet the two categories carry identical information: linear algebra on finite-dimensional spaces is the theory of matrices up to change of basis, and the equivalence is the exact expression of that identity. This is the recurring pattern. Equivalence, not isomorphism, is the notion of sameness for categories, because it records that two categories have the same objects and morphisms up to canonical isomorphism, which is all that the categorical viewpoint, where an object is known only through its morphisms, can or should ask.

60

The Yoneda Lemma and Representability

Chapter 1 built the categorical language and returned to one thread: an object is determined by its morphisms. This chapter makes it a theorem. The *representable functor* $\mathrm{Hom}(c, -)$ records how an object c maps to every other; the *Yoneda lemma* computes its natural transformations into any set-valued functor, and the *Yoneda embedding* recovers c from the presheaf it represents. Along the way we settle questions left open in Chapter 1, among them the converse of the equivalence characterization (proposition 59.5.6).

60.1 Representable Functors

The first step toward the Yoneda lemma is to package the morphisms out of a fixed object as a functor. Fixing an object c of a locally small category and letting the second slot of the hom-set vary produces a set-valued functor; the functors that arise this way, and the objects that produce them, are the subject of this section. A functor isomorphic to one of these is *representable*, and the isomorphism is pinned down by a single element, the universal element.

We work throughout in a locally small category C (definition 59.1.7), so that each collection $\mathrm{Hom}_C(x, y)$ is a set.

Definition 60.1.1: Representable Functor

Let C be locally small and c an object of C. The *covariant hom-functor*

$$\mathrm{Hom}(c,-)\colon C \to \mathbf{Set}$$

sends an object x to the set $\mathrm{Hom}(c,x)$ and a morphism $f\colon x \to y$ to *post-composition*

$$f_*\colon \mathrm{Hom}(c,x) \to \mathrm{Hom}(c,y), \qquad f_*(h) = fh$$

The *contravariant hom-functor*

$$\mathrm{Hom}(-,c)\colon C^{\mathrm{op}} \to \mathbf{Set}$$

sends x to $\mathrm{Hom}(x,c)$ and $f\colon x \to y$ to *pre-composition*

$$f^*\colon \mathrm{Hom}(y,c) \to \mathrm{Hom}(x,c), \qquad f^*(h) = hf$$

A functor $F\colon C \to \mathbf{Set}$ is *representable* if there is an object c and a natural isomorphism $F \cong \mathrm{Hom}(c,-)$; the object c is a *representing object* for F. Dually, a functor $F\colon C^{\mathrm{op}} \to \mathbf{Set}$ is *representable* if $F \cong \mathrm{Hom}(-,c)$ for some c.

Functoriality of $\mathrm{Hom}(c,-)$ is the two composition laws of C: for $f\colon x \to y$ and $g\colon y \to z$ one has $(gf)_*(h) = gfh = g_*(f_*(h))$, so $(gf)_* = g_*f_*$, and $(1_x)_*(h) = h$, so $(1_x)_* = 1_{\mathrm{Hom}(c,x)}$. The same identities read backwards give $(gf)^* = f^*g^*$, the order-reversal that makes $\mathrm{Hom}(-,c)$ contravariant. A representing object, if one exists, is a solution to the problem posed by F: it is an object whose maps out of it (or into it) reproduce the sets Fx compatibly with all the morphisms of C. Chapter 1 already showed how much a hom-set encodes, since a morphism is an isomorphism exactly when post-composition with it is a bijection of hom-sets for every test object (example 59.1.8 recorded the first instances); here the hom-set is promoted from a single set to a functor.

Example 60.1.2: The Underlying Set of a Group

The forgetful functor $U\colon \mathbf{Grp} \to \mathbf{Set}$, sending a group to its underlying set and a homomorphism to its underlying function, is representable, and the integers $\mathbb{Z}$ represent it. A homomorphism $\mathbb{Z} \to G$ is determined by the image of the generator 1, and any element of G is a legitimate image, so

$$\mathrm{Hom}_{\mathbf{Grp}}(\mathbb{Z},G) \cong UG, \qquad \varphi \mapsto \varphi(1)$$

This bijection is natural in G: for a homomorphism $\psi\colon G \to H$ the square comparing $\varphi \mapsto \varphi(1)$ on either side commutes, since $(\psi\varphi)(1) = \psi(\varphi(1))$. So $U \cong \mathrm{Hom}_{\mathbf{Grp}}(\mathbb{Z},-)$. That $\mathbb{Z}$ is the free group on one generator ([25], free groups) is exactly the statement that it represents the underlying-set functor.

A representing object is a compressed description of a whole functor, and the examples of most interest name a construction the reader already knows. Before collecting more of them it is worth isolating the contravariant case, where the objects of the represented functor are the presheaves.

Definition 60.1.3: Presheaf

A *presheaf* on a category C is a functor $F\colon C^{\mathrm{op}} \to \mathbf{Set}$. Presheaves on C, together with natural transformations between them, form the *presheaf category*

$$\mathbf{PSh}(C) = [C^{\mathrm{op}}, \mathbf{Set}]$$

the functor category (definition 59.5.4) of contravariant set-valued functors on C.

For each object c the contravariant hom-functor $\mathrm{Hom}(-, c)$ is a presheaf, the *presheaf represented by* c; a presheaf isomorphic to some $\mathrm{Hom}(-, c)$ is representable in the sense of definition 60.1.1. The two variances are best seen together. Holding neither slot of the hom-set fixed gives a functor of both arguments at once.

Definition 60.1.4: The Two-Sided Hom-Bifunctor

Let C be locally small. The *hom-bifunctor*

$$\mathrm{Hom}(-,-)\colon C^{\mathrm{op}} \times C \to \mathbf{Set}$$

sends an object (x, y) to $\mathrm{Hom}(x, y)$ and a morphism $(f, g)\colon (x, y) \to (x', y')$, i.e. a pair $f\colon x' \to x$ in C and $g\colon y \to y'$ in C, to

$$\mathrm{Hom}(x, y) \to \mathrm{Hom}(x', y'), \qquad h \mapsto ghf$$

Functoriality is one line: given a second pair $(f', g')\colon (x', y') \to (x'', y'')$, composing the induced maps sends h to $g'(ghf)f' = (g'g)h(ff')$, which is the map induced by the composite $(f', g')(f, g) = (ff', g'g)$ in $C^{\mathrm{op}} \times C$; and the identity pair $(1_x, 1_y)$ induces $h \mapsto h$. Fixing the first slot at c recovers $\mathrm{Hom}(c, -)$ and fixing the second recovers $\mathrm{Hom}(-, c)$, so the bifunctor is the common source of both hom-functors.

The definition of representability quantifies over all objects and all elements at once, through the natural isomorphism. It has a pointwise reformulation carried by a single piece of data. An isomorphism $F \cong \mathrm{Hom}(c, -)$ is pinned down by where the identity 1_c goes, and the element of Fc it corresponds to satisfies a universal property that says exactly "every element of every Fd factors through it uniquely."

Definition 60.1.5: Universal Element

Let $F\colon C \to \mathbf{Set}$. A *universal element* of F is a pair (c, x) with c an object of C and $x \in Fc$ such that for every object d and every $y \in Fd$ there is a unique morphism $f\colon c \to d$ with $(Ff)(x) = y$.

The pair (c, x) is universal in that every element of the functor is obtained from x by transport along a unique morphism. The following makes precise that a universal element is the same data as a representation.

Proposition 60.1.6: Representability by a Universal Element

A functor $F\colon C \to \mathbf{Set}$ is representable if and only if it has a universal element. Explicitly, a natural isomorphism $\alpha\colon \mathrm{Hom}(c,-) \Rightarrow F$ corresponds to the universal element (c,x) with $x = \alpha_c(1_c)$, and conversely every universal element arises this way from a unique natural isomorphism.

***Proof*:**
Step 1: A representation gives a universal element. Suppose $\alpha\colon \mathrm{Hom}(c,-) \Rightarrow F$ is a natural isomorphism, and set $x = \alpha_c(1_c) \in Fc$. Fix an object d and an element $y \in Fd$. Naturality of α at a morphism $f\colon c \to d$ is the commuting square

$$\begin{tikzcd} \mathrm{Hom}(c,c) \arrow[r, "\alpha_c"] \arrow[d, "f_*"'] & Fc \arrow[d, "Ff"] \\ \mathrm{Hom}(c,d) \arrow[r, "\alpha_d"'] & Fd \end{tikzcd}$$

Chasing 1_c around it gives $\alpha_d(f) = \alpha_d(f_*(1_c)) = (Ff)(\alpha_c(1_c)) = (Ff)(x)$. Thus for every $f \in \mathrm{Hom}(c,d)$,

$$\alpha_d(f) = (Ff)(x) \tag{60.1}$$

Since α_d is a bijection, for the given $y \in Fd$ there is exactly one $f\colon c \to d$ with $\alpha_d(f) = y$, that is, exactly one f with $(Ff)(x) = y$. So (c,x) is a universal element.

Step 2: A universal element gives a representation. Conversely, let (c,x) be a universal element of F. Define, for each object d, a map

$$\alpha_d\colon \mathrm{Hom}(c,d) \to Fd, \qquad \alpha_d(f) = (Ff)(x)$$

The universal property says precisely that for each $y \in Fd$ there is a unique $f\colon c \to d$ with $(Ff)(x) = y$, which is the statement that α_d is a bijection. The maps α_d are natural: for $g\colon d \to d'$ and $f\colon c \to d$,

$$\alpha_{d'}(g_*(f)) = \alpha_{d'}(gf) = (F(gf))(x) = (Fg)((Ff)(x)) = (Fg)(\alpha_d(f))$$

so the naturality square for g commutes. Hence $\alpha\colon \mathrm{Hom}(c,-) \Rightarrow F$ is a natural isomorphism, and F is representable.

Step 3: The two constructions are inverse. A representation α produces $x = \alpha_c(1_c)$, and (60.1) shows that the representation rebuilt from x in Step 2 has components $f \mapsto (Ff)(x) = \alpha_d(f)$, recovering α. In the other direction, the representation built from (c,x) in Step 2 sends 1_c to $(F1_c)(x) = x$, recovering the element. So the correspondence is a bijection between representations and universal elements, completing the proof.

The proposition converts a question about a natural isomorphism, which involves a component at every object, into a question about one element and its universal property. Verifying representability then reduces to exhibiting a single x and checking that every y factors through it uniquely, as the following examples show.

Example 60.1.7: Representable Functors

1. The identity functor on **Set** is representable, with representing object a one-point set $1 = \{*\}$. A function $1 \to X$ is a choice of a point of X, so $\mathrm{Hom}_{\mathbf{Set}}(1, X) \cong X$ by $\varphi \mapsto \varphi(*)$, naturally in X. The universal element is the pair $(1, *)$: every element y of every set X is $(Ff)(*) = f(*)$ for the unique $f\colon 1 \to X$ with $f(*) = y$.

2. The contravariant powerset $P\colon \mathbf{Set}^{\mathrm{op}} \to \mathbf{Set}$, sending a set X to its powerset PX and a function $f\colon X \to Y$ to the preimage map $f^{-1}\colon PY \to PX$, is a representable presheaf. Let $2 = \{0, 1\}$. A subset $A \subseteq X$ is the same data as its indicator function $\chi_A\colon X \to 2$, with $\chi_A(a) = 1$ exactly when $a \in A$, and this correspondence is a bijection

$$PX \cong \mathrm{Hom}_{\mathbf{Set}}(X, 2), \qquad A \mapsto \chi_A$$

It is natural in X: for $f\colon X \to Y$ and $B \subseteq Y$, the indicator of $f^{-1}(B)$ is $\chi_B \circ f$, which is exactly $f^*(\chi_B)$. So $P \cong \mathrm{Hom}_{\mathbf{Set}}(-, 2)$, and 2 is the representing object. The universal element is the pair $(2, U)$ with $U = \{1\} \subseteq 2$: every subset $A \subseteq X$ is the preimage $\chi_A^{-1}(U)$ under a unique map $X \to 2$.

60.2 The Yoneda Lemma

The representable functor $\mathrm{Hom}(c, -)$ of definition 60.1.1 records how c maps to every object of C. This section computes the natural transformations out of it. The answer, the Yoneda lemma, is that a natural transformation from $\mathrm{Hom}(c, -)$ to a set-valued functor F is the same thing as a single element of Fc: the whole transformation is pinned down by where it sends the identity 1_c. The embedding of the next section and everything that follows rest on this computation.

The mechanism is visible before any proof. A natural transformation $\alpha\colon \mathrm{Hom}(c, -) \Rightarrow F$ has, at each object d, a component $\alpha_d\colon \mathrm{Hom}(c, d) \to Fd$. The set $\mathrm{Hom}(c, c)$ holds a distinguished element, the identity 1_c, and α_c sends it to some $\alpha_c(1_c) \in Fc$. Naturality forces every other value of every component to be determined by this one element: for $f\colon c \to d$, naturality applied to f and evaluated at 1_c expresses $\alpha_d(f)$ through $\alpha_c(1_c)$. Reading off $\alpha_c(1_c)$ is thus a map from natural transformations to Fc, and the lemma is that it is a bijection.

Theorem 60.2.1: The Yoneda Lemma

Let C be a locally small category, c an object of C, and $F\colon C \to \mathbf{Set}$ a functor. Evaluation at the identity,

$$\Phi\colon\ \mathrm{Nat}(\mathrm{Hom}(c, -), F) \to Fc, \qquad \Phi(\alpha) = \alpha_c(1_c)$$

is a bijection, natural in both c and F.

The proof produces an explicit inverse and checks that the two maps compose to the identity in each order. The inverse takes an element $x \in Fc$ and builds a natural transformation from it; the functoriality of F is what makes that transformation natural.

***Proof*:**
Step 1: the inverse map. Fix $x \in Fc$. Define a family $\Psi(x)$ by giving, at each object d, the component

$$\Psi(x)_d\colon \mathrm{Hom}(c,d) \to Fd, \qquad \Psi(x)_d(f) = (Ff)(x)$$

for $f\colon c \to d$, where $Ff\colon Fc \to Fd$ is the action of F on f. This is well-defined: Ff is a function $Fc \to Fd$ and $x \in Fc$, so $(Ff)(x) \in Fd$.

The claim is that $\Psi(x)$ is a natural transformation $\mathrm{Hom}(c,-) \Rightarrow F$. Let $g\colon d \to d'$ be a morphism of C. The functor $\mathrm{Hom}(c,-)$ sends g to post-composition $g_*\colon \mathrm{Hom}(c,d) \to \mathrm{Hom}(c,d')$, $g_*(f) = gf$. The naturality square to check is

$$\begin{tikzcd}
\mathrm{Hom}(c,d) \arrow[r, "\Psi(x)_d"] \arrow[d, "g_*"'] & Fd \arrow[d, "Fg"] \\
\mathrm{Hom}(c,d') \arrow[r, "\Psi(x)_{d'}"'] & Fd'
\end{tikzcd}$$

Chase $f\colon c \to d$ around both routes. Down then across gives $\Psi(x)_{d'}(g_*(f)) = \Psi(x)_{d'}(gf) = (F(gf))(x)$; across then down gives $(Fg)(\Psi(x)_d(f)) = (Fg)((Ff)(x))$. These agree because F is a functor, $F(gf) = Fg \circ Ff$, so

$$(F(gf))(x) = (Fg \circ Ff)(x) = (Fg)((Ff)(x))$$

The square commutes for every g, so $\Psi(x)$ is natural, and $\Psi\colon Fc \to \mathrm{Nat}(\mathrm{Hom}(c,-),F)$, $x \mapsto \Psi(x)$, is a well-defined candidate inverse to Φ.

Step 2: Φ and Ψ are mutually inverse. There are two composites to compute.

First, $\Phi\Psi = 1_{Fc}$. For $x \in Fc$,

$$\Phi(\Psi(x)) = \Psi(x)_c(1_c) = (F1_c)(x) = 1_{Fc}(x) = x$$

using that F preserves identities, $F1_c = 1_{Fc}$. This direction is immediate.

Second, $\Psi\Phi$ is the identity on $\mathrm{Nat}(\mathrm{Hom}(c,-),F)$. Let $\alpha\colon \mathrm{Hom}(c,-) \Rightarrow F$ be given and put $x = \Phi(\alpha) = \alpha_c(1_c)$. Two natural transformations agree exactly when all components agree, so it suffices to show $\Psi(x)_d = \alpha_d$ for every d. Fix d and $f\colon c \to d$. Naturality of α at f is the commuting square

$$\begin{tikzcd}
\mathrm{Hom}(c,c) \arrow[r, "\alpha_c"] \arrow[d, "f_*"'] & Fc \arrow[d, "Ff"] \\
\mathrm{Hom}(c,d) \arrow[r, "\alpha_d"'] & Fd
\end{tikzcd}$$

Evaluate it at $1_c \in \mathrm{Hom}(c,c)$. Down then across gives $\alpha_d(f_*(1_c)) = \alpha_d(f1_c) = \alpha_d(f)$; across then down gives $(Ff)(\alpha_c(1_c)) = (Ff)(x)$. Equating the routes,

$$\alpha_d(f) = (Ff)(x) = \Psi(x)_d(f)$$

Since f and d were arbitrary, $\alpha_d = \Psi(x)_d$ for all d, so $\alpha = \Psi(\Phi(\alpha))$. Both composites are identities, so Φ is a bijection with inverse Ψ.

Step 3: naturality of the bijection. The source and target of Φ each depend on the pair (c, F), and the claim is that Φ commutes with change of F and change of c. Consider F first, with c

fixed. A natural transformation $\theta\colon F \Rightarrow G$ acts on the target by its component $\theta_c\colon Fc \to Gc$ and on the source by post-composition $\alpha \mapsto \theta\alpha$, carrying $\mathrm{Hom}(c,-) \Rightarrow F$ to $\mathrm{Hom}(c,-) \Rightarrow G$. Naturality in F is the square

$$\begin{tikzcd} \mathrm{Nat}(\mathrm{Hom}(c,-),F) \arrow[r, "\Phi_F"] \arrow[d, "\theta\circ(-)"'] & Fc \arrow[d, "\theta_c"] \\ \mathrm{Nat}(\mathrm{Hom}(c,-),G) \arrow[r, "\Phi_G"'] & Gc \end{tikzcd}$$

For $\alpha\colon \mathrm{Hom}(c,-) \Rightarrow F$, down then across gives $\Phi_G(\theta\alpha) = (\theta\alpha)_c(1_c) = \theta_c(\alpha_c(1_c))$, since composition of natural transformations is componentwise (definition 59.5.4); across then down gives $\theta_c(\Phi_F(\alpha)) = \theta_c(\alpha_c(1_c))$. The two agree, so Φ is natural in F.

Now fix F and vary c. A morphism $h\colon c \to c'$ acts on the target by $Fh\colon Fc \to Fc'$. On the source it acts by precomposition with the representable's own functoriality: h induces $\mathrm{Hom}(h,-)\colon \mathrm{Hom}(c',-) \Rightarrow \mathrm{Hom}(c,-)$, whose component at d is $\mathrm{Hom}(h,d)\colon \mathrm{Hom}(c',d) \to \mathrm{Hom}(c,d)$, $k \mapsto kh$, and a transformation $\alpha\colon \mathrm{Hom}(c,-) \Rightarrow F$ is carried to $\alpha\circ\mathrm{Hom}(h,-)\colon \mathrm{Hom}(c',-) \Rightarrow F$. Naturality in c is the square

$$\begin{tikzcd} \mathrm{Nat}(\mathrm{Hom}(c,-),F) \arrow[r, "\Phi_c"] \arrow[d, "(-)\circ\mathrm{Hom}(h,-)"'] & Fc \arrow[d, "Fh"] \\ \mathrm{Nat}(\mathrm{Hom}(c',-),F) \arrow[r, "\Phi_{c'}"'] & Fc' \end{tikzcd}$$

Take $\alpha\colon \mathrm{Hom}(c,-) \Rightarrow F$. Across then down gives $Fh(\Phi_c(\alpha)) = (Fh)(\alpha_c(1_c))$. Down then across gives

$$\Phi_{c'}(\alpha \circ \mathrm{Hom}(h,-)) = \big(\alpha \circ \mathrm{Hom}(h,-)\big)_{c'}(1_{c'}) = \alpha_{c'}\big(\mathrm{Hom}(h,c')(1_{c'})\big) = \alpha_{c'}(1_{c'}\, h) = \alpha_{c'}(h)$$

To match the two, apply naturality of α (a transformation $\mathrm{Hom}(c,-) \Rightarrow F$) at the morphism $h\colon c \to c'$: its square is

$$\begin{tikzcd} \mathrm{Hom}(c,c) \arrow[r, "\alpha_c"] \arrow[d, "h_*"'] & Fc \arrow[d, "Fh"] \\ \mathrm{Hom}(c,c') \arrow[r, "\alpha_{c'}"'] & Fc' \end{tikzcd}$$

Evaluated at $1_c \in \mathrm{Hom}(c,c)$, down then across gives $\alpha_{c'}(h_*(1_c)) = \alpha_{c'}(h1_c) = \alpha_{c'}(h)$; across then down gives $(Fh)(\alpha_c(1_c))$. Hence $\alpha_{c'}(h) = (Fh)(\alpha_c(1_c))$, which is exactly the equality of the two routes around the outer square. So Φ is natural in c, completing the proof.

Out of a lone element $x \in Fc$ the map Ψ manufactures an entire natural family $\{(Ff)(x)\}_f$, one function per object, coherent across every morphism. Nothing beyond the functoriality of F went into the construction, which is why the correspondence is as tight as it is: the data of a natural transformation, a priori a large amount of information spread over all objects, collapses to one point of a single set.

A first payoff sets $F = \mathrm{Hom}(c',-)$ for a second object c'. Then $Fc = \mathrm{Hom}(c',c)$, and the lemma reads $\mathrm{Nat}(\mathrm{Hom}(c,-),\mathrm{Hom}(c',-)) \cong \mathrm{Hom}(c',c)$: natural transformations between two representables are exactly the morphisms of C in the reverse direction. This is the computation on which the next section builds the Yoneda embedding.

Example 60.2.2: The Underlying Set as a Representable
Let $F\colon \mathbf{Grp} \to \mathbf{Set}$ be the forgetful functor, sending a group to its underlying set. It is represented by the free group $\mathbb{Z}$ on one generator: a homomorphism $\mathbb{Z} \to G$ is determined by, and freely choosable as, the image of the generator 1, so evaluation at 1 gives a bijection $\mathrm{Hom}_{\mathbf{Grp}}(\mathbb{Z}, G) \cong FG$ natural in G (definition 60.1.1). The Yoneda lemma names the element of $F\mathbb{Z} = \mathbb{Z}$ that encodes this: it is

$$\Phi(\alpha) = \alpha_{\mathbb{Z}}(1_{\mathbb{Z}}) \in F\mathbb{Z} = \mathbb{Z}$$

the image, under the natural isomorphism $\alpha\colon \mathrm{Hom}(\mathbb{Z}, -) \Rightarrow F$, of the identity homomorphism $1_{\mathbb{Z}}$. Tracing the definitions, $\alpha_{\mathbb{Z}}(1_{\mathbb{Z}})$ is the generator $1 \in \mathbb{Z}$, and the whole isomorphism α is recovered from it by $\alpha_G(\varphi) = (F\varphi)(1) = \varphi(1)$, the image of the generator under $\varphi\colon \mathbb{Z} \to G$. The single element $1 \in \mathbb{Z}$ carries the entire natural family.

The presheaf version is the same statement read in C^{op}. Recall the duality principle (box 59.3.3): a statement provable for every locally small category holds equally for every opposite. Applying the Yoneda lemma in C^{op} turns the covariant representable $\mathrm{Hom}_{C^{\mathrm{op}}}(c, -)$ into the contravariant $\mathrm{Hom}_C(-, c)$, and a covariant functor $C^{\mathrm{op}} \to \mathbf{Set}$ is a presheaf on C (definition 60.1.3). No new argument is needed.

Corollary 60.2.3: The Yoneda Lemma, Presheaf Form

Let C be a locally small category, c an object of C, and $F\colon C^{\mathrm{op}} \to \mathbf{Set}$ a presheaf. Evaluation at the identity,

$$\mathrm{Nat}(\mathrm{Hom}(-, c), F) \to Fc, \qquad \alpha \mapsto \alpha_c(1_c)$$

is a bijection, natural in c and F.

***Proof*:**
The Yoneda lemma holds in every locally small category, and by the duality principle it applies in particular to C^{op}, which is again locally small (box 59.3.3). Applied there to the object c and the functor F, the covariant representable of C^{op} at c is $\mathrm{Hom}_{C^{\mathrm{op}}}(c, -) = \mathrm{Hom}_C(-, c)$ read in the free variable, and a functor $F\colon C^{\mathrm{op}} \to \mathbf{Set}$ is precisely a covariant functor on C^{op}. So theorem 60.2.1 yields the bijection $\mathrm{Nat}(\mathrm{Hom}_C(-, c), F) \cong Fc$, with the same evaluation-at-1_c map and the same naturality in c and F, as required.

The two forms are one theorem seen from two sides, and both will be used: the covariant form for functors out of C, the presheaf form for the embedding $c \mapsto \mathrm{Hom}(-, c)$ into presheaves.

The lemma also closes the loop with the universal element of section 60.1. An element $x \in Fc$ is called universal when every $y \in Fd$ arises as $(Ff)(x)$ for a unique $f\colon c \to d$ (definition 60.1.5), and F is representable exactly when such an x exists. Under the Yoneda bijection x corresponds to the transformation $\Psi(x)\colon \mathrm{Hom}(c, -) \Rightarrow F$ with $\Psi(x)_d(f) = (Ff)(x)$, and x is universal precisely when every $\Psi(x)_d$ is a bijection, that is, when $\Psi(x)$ is a natural isomorphism. So the representable-iff-universal-element statement is the Yoneda lemma specialized to a representable F: the universal element is the point of Fc that Yoneda hands back, and for $F = \mathrm{Hom}(c, -)$ itself it is the identity 1_c, the element from which the whole representable is generated.

60.3 The Yoneda Embedding

The Yoneda lemma of the previous section computes the natural transformations out of a representable functor. Read one way, it is a calculation; read another, it says that the assignment $c \mapsto \mathrm{Hom}(-, c)$ loses no information. This section makes that reading precise. Packaging the contravariant representables into a single functor $\mathbf{y}\colon C \to \mathbf{PSh}(C)$, the presheaf form of the lemma becomes the statement that $\mathbf{y}$ is fully faithful, and full faithfulness in turn says that an object is recovered, up to isomorphism, from the presheaf it represents.

Recall from definition 60.1.3 that $\mathbf{PSh}(C) = [C^{\mathrm{op}}, \mathbf{Set}]$ is the category of presheaves on C, whose objects are the contravariant set-valued functors and whose morphisms are natural transformations. Each object c of C gives such a presheaf, the represented $\mathrm{Hom}(-, c)$ of definition 60.1.1; each morphism of C gives a natural transformation between two of them. Assembling these into a functor is the content of the first definition.

Definition 60.3.1: The Yoneda Embedding

Let C be a locally small category (definition 59.1.7). The *Yoneda embedding* is the functor

$$\mathbf{y}\colon C \to \mathbf{PSh}(C)$$

defined as follows. On an object c, it is the represented presheaf

$$\mathbf{y}(c) = \mathrm{Hom}(-, c) \in \mathbf{PSh}(C)$$

On a morphism $f\colon c \to d$, it is the natural transformation

$$\mathbf{y}(f) = \mathrm{Hom}(-, f)\colon \mathrm{Hom}(-, c) \Rightarrow \mathrm{Hom}(-, d)$$

whose component at an object x is post-composition with f,

$$\mathrm{Hom}(-, f)_x\colon \mathrm{Hom}(x, c) \to \mathrm{Hom}(x, d), \qquad g \mapsto fg$$

That $\mathbf{y}$ is a functor is exactly the statement that its two clauses respect composition and identities, and both follow from the category axioms of C. The verification is short and worth doing once, because it exhibits how the unit and associativity laws of composition become the functoriality of $\mathbf{y}$.

***Proof*:**
Two checks are needed: that each $\mathbf{y}(f)$ is a natural transformation, and that $\mathbf{y}$ preserves composites and identities.

Step 1: $\mathbf{y}(f)$ is natural. Fix $f\colon c \to d$ and a morphism $h\colon x' \to x$ in C. The presheaf $\mathrm{Hom}(-, c)$ sends h to pre-composition $\mathrm{Hom}(h, c)\colon \mathrm{Hom}(x, c) \to \mathrm{Hom}(x', c)$, $g \mapsto gh$, and similarly for $\mathrm{Hom}(-, d)$. Naturality of $\mathbf{y}(f) = \mathrm{Hom}(-, f)$ at h is the commuting of

$$\begin{tikzcd}
\mathrm{Hom}(x, c) \arrow[r, "g \mapsto fg"] \arrow[d, "g \mapsto gh"'] & \mathrm{Hom}(x, d) \arrow[d, "g \mapsto gh"] \\
\mathrm{Hom}(x', c) \arrow[r, "g \mapsto fg"'] & \mathrm{Hom}(x', d)
\end{tikzcd}$$

Both routes send $g: x \to c$ to $f(gh)$: down-then-across gives $f(gh)$, across-then-down gives $(fg)h$, and these agree by associativity of composition in C. So $\mathbf{y}(f)$ is natural.

Step 2: $\mathbf{y}$ *preserves composition.* Let $f: c \to d$ and $f': d \to e$. Both $\mathbf{y}(f'f)$ and $\mathbf{y}(f')\mathbf{y}(f)$ are natural transformations $\mathrm{Hom}(-, c) \Rightarrow \mathrm{Hom}(-, e)$, so it suffices to compare their components at each object x. On $g: x \to c$,

$$\mathbf{y}(f'f)_x(g) = (f'f)g, \qquad \big(\mathbf{y}(f')\mathbf{y}(f)\big)_x(g) = f'(fg)$$

and $(f'f)g = f'(fg)$ by associativity. Hence $\mathbf{y}(f'f) = \mathbf{y}(f')\mathbf{y}(f)$.

Step 3: $\mathbf{y}$ *preserves identities.* The component of $\mathbf{y}(1_c)$ at x sends $g: x \to c$ to $1_c g = g$, so it is the identity function on $\mathrm{Hom}(x, c)$. As this holds for every x, $\mathbf{y}(1_c) = 1_{\mathrm{Hom}(-,c)}$, the identity natural transformation on $\mathbf{y}(c)$.

Naturality of each $\mathbf{y}(f)$, preservation of composites, and preservation of identities together make $\mathbf{y}$ a functor, completing the proof.

The proof isolates a small dictionary: associativity of composition in C becomes both the naturality of each $\mathbf{y}(f)$ and the functoriality of $\mathbf{y}$, and the left unit law becomes the preservation of identities. The embedding carries an object to the record of all maps into it and a morphism to post-composition with that morphism, and this record transforms functorially because composition does. A concrete category makes the abstract assignment tangible.

Example 60.3.2: The Yoneda Embedding of a Poset

Let P be a poset, regarded as a category with a single morphism $x \to y$ whenever $x \le y$ and none otherwise (definition 59.1.1). For objects x, y the hom-set $\mathrm{Hom}(x, y)$ is a one-element set when $x \le y$ and empty otherwise, so a presheaf $\mathrm{Hom}(-, c) = \mathbf{y}(c)$ on P is the indicator of the down-set of c: for each object x,

$$\mathrm{Hom}(x, c) = \begin{cases} \{*\} & x \le c \\ \varnothing & x \not\le c \end{cases}$$

Thus $\mathbf{y}(c)$ records precisely which elements lie below c, that is, the principal down-set $\downarrow c = \{x \in P \big| x \le c\}$. A morphism $c \le d$ in P gives the natural transformation $\mathbf{y}(c \le d): \mathrm{Hom}(-, c) \Rightarrow \mathrm{Hom}(-, d)$, which at each x is the unique function from $\mathrm{Hom}(x, c)$ to $\mathrm{Hom}(x, d)$; such a function exists as a morphism of presheaves exactly because $x \le c \le d$ forces $x \le d$, so the down-set of c is contained in that of d. The embedding $\mathbf{y}: P \to \mathbf{PSh}(P)$ is therefore the assignment $c \mapsto \downarrow c$ sending each element to its principal down-set, ordered by inclusion. Since $c \le d$ if and only if $\downarrow c \subseteq \downarrow d$, distinct elements have distinct down-sets and the order is faithfully reproduced: $\mathbf{y}$ is the poset embedding of P into its lattice of down-sets. This is the order-theoretic case of the full faithfulness proved next.

The example already displays the pattern of the general theorem. The embedding of a poset is injective on objects and reflects the order, so the whole of P can be read off from its image in $\mathbf{PSh}(P)$. For a general category the statement is not that $\mathbf{y}$ is injective on objects, which is too rigid, but that it is a bijection on morphisms between any two objects, full and faithful in the sense recalled from proposition 59.5.6. This is where the Yoneda lemma does the work: its presheaf form, applied to a represented presheaf, is precisely the assertion that $\mathbf{y}$ is fully faithful.

Theorem 60.3.3: The Yoneda Embedding is Fully Faithful

Let C be a locally small category. For all objects c, d of C the function

$$\mathrm{Hom}_C(c,d) \to \mathrm{Hom}_{\mathbf{PSh}(C)}\big(\mathbf{y}(c), \mathbf{y}(d)\big) = \mathrm{Nat}\big(\mathrm{Hom}(-,c), \mathrm{Hom}(-,d)\big), \qquad f \mapsto \mathbf{y}(f)$$

is a bijection. Equivalently, the Yoneda embedding $\mathbf{y}\colon C \to \mathbf{PSh}(C)$ is full and faithful.

***Proof*:**
Apply the presheaf form of the Yoneda lemma, corollary 60.2.3, to the presheaf $F = \mathrm{Hom}(-,d)$ and the object c. It supplies a bijection

$$\mathrm{Nat}\big(\mathrm{Hom}(-,c), \mathrm{Hom}(-,d)\big) \xrightarrow{\cong} \big(\mathrm{Hom}(-,d)\big)(c) = \mathrm{Hom}_C(c,d) \tag{60.2}$$

natural in c and d. The bijection of (60.2) sends a natural transformation $\alpha\colon \mathrm{Hom}(-,c) \Rightarrow \mathrm{Hom}(-,d)$ to the element $\alpha_c(1_c) \in \mathrm{Hom}_C(c,d)$, the image of the identity 1_c under the component of α at c. The claimed map $f \mapsto \mathbf{y}(f)$ runs the other way, so it suffices to check that the two are mutually inverse. In fact $f \mapsto \mathbf{y}(f)$ is exactly the inverse produced by corollary 60.2.3, and identifying it is the whole content of the theorem.

Step 1: the Yoneda bijection sends $\mathbf{y}(f)$ to f. Take $\alpha = \mathbf{y}(f) = \mathrm{Hom}(-,f)$ for a morphism $f\colon c \to d$. Its component at c is post-composition with f, so it sends the identity to

$$\big(\mathbf{y}(f)\big)_c(1_c) = f \circ 1_c = f$$

Thus the composite $\mathrm{Hom}_C(c,d) \to \mathrm{Nat}(\mathrm{Hom}(-,c), \mathrm{Hom}(-,d)) \to \mathrm{Hom}_C(c,d)$, first $f \mapsto \mathbf{y}(f)$ and then the bijection of (60.2), is the identity on $\mathrm{Hom}_C(c,d)$.

Step 2: the other composite is the identity. The map of (60.2) is already a bijection by corollary 60.2.3, and Step 1 shows $f \mapsto \mathbf{y}(f)$ is a right inverse to it. A right inverse to a bijection is its two-sided inverse, so $f \mapsto \mathbf{y}(f)$ is itself a bijection, inverse to the Yoneda bijection. Concretely, starting from a natural transformation α, its image $f = \alpha_c(1_c)$ satisfies $\mathbf{y}(f) = \alpha$: for any object x and any $g\colon x \to c$, naturality of α at g applied to 1_c gives

$$\alpha_x(g) = \alpha_x\big(\mathrm{Hom}(g,c)(1_c)\big) = \mathrm{Hom}(g,d)\big(\alpha_c(1_c)\big) = \alpha_c(1_c) \circ g = fg = \big(\mathbf{y}(f)\big)_x(g)$$

so $\alpha = \mathbf{y}(f)$, recovering the second half of the inverse relation.

The two composites being identities, $f \mapsto \mathbf{y}(f)$ is a bijection $\mathrm{Hom}_C(c,d) \xrightarrow{\cong} \mathrm{Nat}(\mathrm{Hom}(-,c), \mathrm{Hom}(-,d))$ for every pair c, d. This is exactly the statement that $\mathbf{y}$ is full (the map is surjective, so every natural transformation between representables is $\mathbf{y}(f)$ for some f) and faithful (the map is injective, so $\mathbf{y}(f)$ determines f), as we sought to show.

Full faithfulness is the sharpest possible statement that $\mathbf{y}$ loses no information at the level of morphisms. Every natural transformation between the presheaves $\mathrm{Hom}(-,c)$ and $\mathrm{Hom}(-,d)$ is post-composition with a unique morphism $c \to d$: there are no natural transformations between representables beyond the ones the category already supplies. The functor $\mathbf{y}$ therefore identifies the hom-set $\mathrm{Hom}_C(c,d)$ with the hom-set $\mathrm{Hom}_{\mathbf{PSh}(C)}(\mathbf{y}(c), \mathbf{y}(d))$ without collapsing or inventing morphisms, so C sits inside $\mathbf{PSh}(C)$ as a full subcategory (up to the identification of objects with the presheaves they represent). The step from this to objects is one general fact about fully faithful functors: they detect isomorphisms as well as reflect the morphisms that witness them.

Before stating the corollary, it is worth pinning the one property of fully faithful functors it rests on. A functor always preserves isomorphisms (lemma 59.4.5): if f is invertible, so is Ff. A fully faithful functor also *reflects* them, and this is the new content. Suppose $Ff\colon Fc \to Fd$ is an isomorphism with inverse $k\colon Fd \to Fc$. Because F is full, $k = Fg$ for some $g\colon d \to c$; because F is faithful, the equations $Fg \circ Ff = 1_{Fc} = F(1_c)$ and $Ff \circ Fg = 1_{Fd} = F(1_d)$, which say $F(gf) = F(1_c)$ and $F(fg) = F(1_d)$, force $gf = 1_c$ and $fg = 1_d$. So f is an isomorphism (definition 59.2.1) with inverse g. A morphism whose image under a fully faithful functor is invertible is therefore already invertible.

Corollary 60.3.4: Objects are Determined by their Points

Let C be a locally small category. Two objects c, d of C are isomorphic in C if and only if the represented presheaves $\mathbf{y}(c) = \mathrm{Hom}(-, c)$ and $\mathbf{y}(d) = \mathrm{Hom}(-, d)$ are isomorphic in $\mathbf{PSh}(C)$. An object of C is determined, up to isomorphism, by the presheaf it represents, its *functor of points*.

***Proof*:**
The Yoneda embedding $\mathbf{y}$ is full and faithful by theorem 60.3.3. If $c \cong d$ in C, then $\mathbf{y}$ preserves the isomorphism (lemma 59.4.5), giving $\mathbf{y}(c) \cong \mathbf{y}(d)$ in $\mathbf{PSh}(C)$. Conversely, suppose $\varphi\colon \mathbf{y}(c) \to \mathbf{y}(d)$ is an isomorphism of presheaves. By fullness $\varphi = \mathbf{y}(f)$ for some $f\colon c \to d$, and $\mathbf{y}(f)$ is an isomorphism, so by the reflection of isomorphisms established above f is itself an isomorphism. Hence $c \cong d$ in C, as we sought to show.

This is the precise form of the chapter's thread: an object is known, up to isomorphism, by how everything maps into it. The functor of points $\mathrm{Hom}(-, c)$ carries the full isomorphism type of c, so two objects with naturally isomorphic points are the same object seen twice.

60.4 Applications

The embedding of section 60.3 converts three recurring questions into questions about hom-functors, where they become easy. This section collects the payoffs. First, a representing object is pinned down not just up to isomorphism but up to a *unique* isomorphism, so "the" representing object is a legitimate phrase. Second, the deferred converse from Chapter 1 falls out: a full, faithful, essentially surjective functor is an equivalence (proposition 59.5.6). Third, a concrete instance of corollary 60.3.4: an object of a category is recovered, up to isomorphism, from its functor of points.

A representing object for a set-valued functor F is any object c together with a natural isomorphism $F \cong \mathrm{Hom}(c, -)$. Two such objects are isomorphic by the general principle that represented functors determine their representing object (corollary 60.3.4), but more is true: the isomorphism between them is forced, with no freedom of choice. This is what licenses treating the representing object as canonical data attached to F.

Corollary 60.4.1: Uniqueness of Representing Objects

Let C be a locally small category and $F\colon C \to \mathbf{Set}$ a functor. If $\varphi\colon F \Rightarrow \mathrm{Hom}(c,-)$ and $\varphi'\colon F \Rightarrow \mathrm{Hom}(c',-)$ are natural isomorphisms, then there is a unique isomorphism $u\colon c' \to c$ in C for which the induced natural isomorphism $\mathrm{Hom}(u,-)\colon \mathrm{Hom}(c,-) \Rightarrow \mathrm{Hom}(c',-)$ equals $\varphi'\varphi^{-1}$. In particular the representing object of F, when it exists, is unique up to a unique isomorphism.

***Proof*:**
Write ψ for the composite $\varphi'\varphi^{-1}\colon \mathrm{Hom}(c,-) \Rightarrow \mathrm{Hom}(c',-)$, a natural isomorphism, being a composite of natural isomorphisms.

The covariant hom-functors are the objects in the image of the covariant embedding $c \mapsto \mathrm{Hom}(c,-)$, which sends C^{op} fully faithfully into $[C,\mathbf{Set}]$. This is the covariant analog of theorem 60.3.3 and is read off the Yoneda lemma directly: applying theorem 60.2.1 to the functor $F = \mathrm{Hom}(c',-)$ gives a bijection

$$\mathrm{Nat}\big(\mathrm{Hom}(c,-),\, \mathrm{Hom}(c',-)\big) \;\cong\; \mathrm{Hom}(c',-)(c) \;=\; \mathrm{Hom}(c',c)$$

natural in both variables. Tracing the Yoneda bijection, a morphism $v\colon c' \to c$ corresponds to the natural transformation $\mathrm{Hom}(v,-)$ whose component at x is $g \mapsto gv$; and every natural transformation $\mathrm{Hom}(c,-) \Rightarrow \mathrm{Hom}(c',-)$ arises this way from a unique such v. Applying this to ψ yields a unique $u\colon c' \to c$ with $\mathrm{Hom}(u,-) = \psi = \varphi'\varphi^{-1}$.

It remains to check u is an isomorphism. Applying the same bijection to $\psi^{-1}\colon \mathrm{Hom}(c',-) \Rightarrow \mathrm{Hom}(c,-)$ produces a unique $w\colon c \to c'$ with $\mathrm{Hom}(w,-) = \psi^{-1}$. Since $c \mapsto \mathrm{Hom}(c,-)$ is a functor $C^{\mathrm{op}} \to [C,\mathbf{Set}]$, precomposition is sent to composition of natural transformations, so $\mathrm{Hom}(uw,-) = \mathrm{Hom}(w,-)\mathrm{Hom}(u,-) = \psi^{-1}\psi = \mathrm{id}$ and likewise $\mathrm{Hom}(wu,-) = \mathrm{id}$. Faithfulness of the embedding (again theorem 60.2.1, whose bijection is injective) forces $uw = \mathrm{id}_c$ and $wu = \mathrm{id}_{c'}$, so u is an isomorphism with inverse w. Uniqueness of u is the uniqueness already supplied by the Yoneda bijection, completing the proof.

The statement isolates the mechanism behind the familiar slogan that universal objects are unique "up to canonical isomorphism". The isomorphism u is not merely one of possibly several: it is singled out by the requirement that it intertwine the two representations φ, φ', and full faithfulness of the embedding says there is exactly one morphism doing so. Every construction recognized by a universal property in the chapters to come (products, kernels, tensor products) inherits this uniqueness for free, which is why one speaks of *the* product rather than a product.

The second application discharges a debt from Chapter 1. There the notion of an equivalence of categories was introduced (definition 59.5.5), and proposition 59.5.6 showed that an equivalence is necessarily full, faithful, and essentially surjective. The converse was deferred, because constructing the quasi-inverse requires choosing, for each object of the target, an object of the source mapping to it, and then transporting morphisms along those choices coherently. The bookkeeping is exactly the naturality that the Yoneda circle of ideas trains one to handle, so this is the natural place to settle it.

Proposition 60.4.2: Full, Faithful, Essentially Surjective Implies Equivalence

Let $F\colon C \to D$ be a functor that is full, faithful, and essentially surjective. Then F is an equivalence of categories: there exist a functor $G\colon D \to C$ and natural isomorphisms $\varepsilon\colon FG \Rightarrow \mathrm{id}_D$ and $\eta\colon \mathrm{id}_C \Rightarrow GF$.

***Proof*:**
Step 1: Define G on objects. For each object d of D, essential surjectivity supplies an object of C mapping isomorphically to d. Choose one such object, call it Gd, together with an isomorphism $\varepsilon_d\colon F(Gd) \to d$ in D. This is one simultaneous selection over all objects of D, the only appeal to choice in the argument; it is harmless, and when D is small it is a choice from a set.

Step 2: Define G on morphisms. Given $g\colon d \to d'$ in D, consider the morphism

$$\varepsilon_{d'}^{-1}\, g\, \varepsilon_d\colon F(Gd) \to F(Gd')$$

Because F is full and faithful, the map $\mathrm{Hom}_C(Gd, Gd') \to \mathrm{Hom}_D(F(Gd), F(Gd'))$ induced by F is a bijection. Hence there is a unique morphism $Gg\colon Gd \to Gd'$ in C with

$$F(Gg) \;=\; \varepsilon_{d'}^{-1}\, g\, \varepsilon_d \tag{60.3}$$

Existence gives G a value; uniqueness is what makes the value well-defined and, as we now see, functorial.

Step 3: G is a functor. For the identity id_d, the morphism $\varepsilon_d^{-1}\,\mathrm{id}_d\,\varepsilon_d = \mathrm{id}_{F(Gd)} = F(\mathrm{id}_{Gd})$, so by the uniqueness in (60.3) the unique preimage is id_{Gd}, giving $G(\mathrm{id}_d) = \mathrm{id}_{Gd}$. For composability $d \xrightarrow{g} d' \xrightarrow{g'} d''$, faithfulness lets us compare $G(g'g)$ and $(Gg')(Gg)$ by comparing their images under F. Applying F and using (60.3) termwise,

$$F\big((Gg')(Gg)\big) = F(Gg')\,F(Gg) = \big(\varepsilon_{d''}^{-1}\, g'\, \varepsilon_{d'}\big)\big(\varepsilon_{d'}^{-1}\, g\, \varepsilon_d\big) = \varepsilon_{d''}^{-1}\, g'\, g\, \varepsilon_d = F\big(G(g'g)\big)$$

where the middle equality cancels $\varepsilon_{d'}\varepsilon_{d'}^{-1} = \mathrm{id}$. Faithfulness then gives $(Gg')(Gg) = G(g'g)$, so G is a functor.

Step 4: $\varepsilon\colon FG \Rightarrow \mathrm{id}_D$ is a natural isomorphism. Each component $\varepsilon_d\colon F(Gd) \to d$ is an isomorphism by construction, so it suffices to check naturality. For $g\colon d \to d'$, rewriting (60.3) as $g\,\varepsilon_d = \varepsilon_{d'}\,F(Gg)$ is exactly the statement that the square

$$\begin{tikzcd} F(Gd) \arrow[r, "\varepsilon_d"] \arrow[d, "F(Gg)"'] & d \arrow[d, "g"] \\ F(Gd') \arrow[r, "\varepsilon_{d'}"'] & d' \end{tikzcd}$$

commutes. Thus ε is natural, and being componentwise an isomorphism it is a natural isomorphism.

Step 5: Construct $\eta\colon \mathrm{id}_C \Rightarrow GF$. For an object c of C, the object $GF(c)$ comes with the chosen isomorphism $\varepsilon_{Fc}\colon F(GFc) \to Fc$. Since F is full and faithful, the isomorphism $\varepsilon_{Fc}^{-1}\colon Fc \to F(GFc)$ has a unique preimage $\eta_c\colon c \to GF(c)$ with

$$F(\eta_c) \;=\; \varepsilon_{Fc}^{-1} \tag{60.4}$$

Each η_c is an isomorphism: its inverse is the preimage of ε_{Fc} under the same bijection, and faithfulness turns the identities $F(\eta_c)\,\varepsilon_{Fc} = \mathrm{id}$ and vice versa into identities in C, as in the argument of corollary 60.4.1. For naturality, let $f\colon c \to c'$. Applying F to both candidate composites and using (60.4) together with (60.3) for the morphism Ff,

$$F\big(GF(f)\big)\,F(\eta_c) = \big(\varepsilon_{Fc'}^{-1}\,(Ff)\,\varepsilon_{Fc}\big)\,\varepsilon_{Fc}^{-1} = \varepsilon_{Fc'}^{-1}\,(Ff) = F(\eta_{c'})\,F(f)$$

so $F\big(GF(f)\,\eta_c\big) = F\big(\eta_{c'}\,f\big)$, and faithfulness gives $GF(f)\,\eta_c = \eta_{c'}\,f$. Hence η is a natural isomorphism $\mathrm{id}_C \Rightarrow GF$.

With G a functor and ε, η natural isomorphisms, F and G are quasi-inverse, so F is an equivalence of categories, as we sought to show.

Together with proposition 59.5.6 this completes the characterization: a functor is an equivalence if and only if it is full, faithful, and essentially surjective. The "if" direction just proved is the useful one in practice, since it lets one certify an equivalence without ever exhibiting the inverse functor. One checks three local conditions, two of them (fullness, faithfulness) about hom-sets and one (essential surjectivity) about the image on objects, and the quasi-inverse is then guaranteed to exist. The proof also shows the quasi-inverse is far from unique as bare data, since it depends on the choices in Step 1, yet any two such choices yield equivalent functors, matching the uniqueness pattern of corollary 60.4.1.

The last application makes corollary 60.3.4 tangible. That corollary says an object is determined up to isomorphism by the functor it represents, its functor of points; two objects with naturally isomorphic points are isomorphic. On a category of algebraic objects this recovers the object from the ways of mapping it to arbitrary test objects, which is the content of the "functor of points" viewpoint that pervades algebraic geometry.

Example 60.4.3: Recovering a Ring from Its Points

Work in $C = \mathbf{Ring}$ (commutative rings with unit). Each ring R represents the covariant functor of points $\mathrm{Hom}(R,-)\colon \mathbf{Ring} \to \mathbf{Set}$, sending a ring A to the set $\mathrm{Hom}(R,A)$ of ring homomorphisms out of R. Suppose R and S are rings and

$$\Phi\colon \mathrm{Hom}(R,-) \Rightarrow \mathrm{Hom}(S,-)$$

is a natural isomorphism of these functors. Then $R \cong S$ as rings, and the isomorphism is canonical.

The point is the covariant form of corollary 60.3.4: represented functors determine their representing object, and here the representation is covariant, so the roles are dual to the presheaf statement. Concretely, apply corollary 60.4.1 with $F = \mathrm{Hom}(R,-)$. The two natural isomorphisms $\mathrm{id}\colon F \Rightarrow \mathrm{Hom}(R,-)$ and $\Phi\colon F \Rightarrow \mathrm{Hom}(S,-)$ represent the same F, so there is a unique isomorphism $u\colon S \to R$ with $\mathrm{Hom}(u,-) = \Phi$.

The isomorphism u can be named by hand, which shows the abstraction costs nothing. Chase the identity through the components. At $A = R$ the map $\Phi_R\colon \mathrm{Hom}(R,R) \to \mathrm{Hom}(S,R)$ carries id_R to a homomorphism $u\colon S \to R$, that is, set $u = \Phi_R(\mathrm{id}_R)$. For any A and any $f\colon R \to A$, naturality of Φ along f gives the square

$$\begin{tikzcd}
\mathrm{Hom}(R,R) \arrow[r, "\Phi_R"] \arrow[d, "f\circ(-)"'] & \mathrm{Hom}(S,R) \arrow[d, "f\circ(-)"] \\
\mathrm{Hom}(R,A) \arrow[r, "\Phi_A"'] & \mathrm{Hom}(S,A)
\end{tikzcd}$$

Evaluating both routes at id_R gives $\Phi_A(f) = fu$, so Φ_A is precomposition with u, that is, $\Phi = \mathrm{Hom}(u, -)$. Running the same chase on Φ^{-1} produces $v = (\Phi^{-1})_S(\mathrm{id}_S)\colon R \to S$ with $\Phi^{-1} = \mathrm{Hom}(v, -)$. Then $\mathrm{Hom}(uv, -) = \mathrm{Hom}(v, -)\mathrm{Hom}(u, -) = \Phi^{-1}\Phi = \mathrm{id}$ forces $uv = \mathrm{id}_R$, and symmetrically $vu = \mathrm{id}_S$, so $u\colon S \to R$ is an isomorphism of rings with inverse v. The functor of points has recovered the ring.

The computation is the whole Yoneda lemma in miniature: the natural transformation Φ is completely determined by its value on a single element, the identity id_R, and that value is the morphism representing it. The same argument works verbatim in any locally small category, with **Ring** replaced by **Grp**, **Top**, or the category of schemes, and it is the reason the functor of points is a faithful record of the object rather than a lossy one.

61

Universal Properties, Limits, and Colimits

Chapters 1 and 2 built the categorical language and its central theorem; this chapter turns to construction. Products, quotients, kernels, unions, gluings: most objects mathematics builds are pinned down not by their elements but by a mapping property. This chapter isolates that pattern as a *universal property*, organizes the constructions it governs into *limits* and *colimits*, and computes them in the categories at hand.

61.1 Universal Properties and Universal Arrows

An object is often pinned down not by naming its elements but by a rule describing how it maps to, or receives maps from, every other object. The empty set is the set with a unique map to every set; the free group on a set X is the group into which every function from X extends uniquely. This section isolates the two building blocks of such descriptions. The first is the pair of extremal objects, *initial* and *terminal*, which sit at the ends of a category. The second, the *universal arrow*, relativizes them along a functor and turns out to be the general form of the universal element from definition 60.1.5.

Definition 61.1.1: Initial and Terminal Objects

Let C be a category. An object 0 of C is *initial* if for every object X there is exactly one morphism $0 \to X$. An object 1 of C is *terminal* if for every object X there is exactly one morphism $X \to 1$. An object that is both initial and terminal is a *zero object*.

The two notions are dual: a terminal object of C is an initial object of C^{op}, so every statement about one specializes to a statement about the other by box 59.3.3. Neither need exist, and when one does it carries no further data beyond the bare mapping property; that austerity is what makes it useful.

Example 61.1.2: Initial and Terminal Objects in the Standing Cast
The following run through the standing cast.

1. In **Set** the empty set $\varnothing$ is initial: the empty function is the unique map $\varnothing \to X$ for every X. Every singleton $\{*\}$ is terminal, since a map $X \to \{*\}$ has no choice in where to send its elements. There is no zero object: $\varnothing$ is not terminal (no map $\{*\} \to \varnothing$ exists).

2. In **Grp** and in $\mathbf{Mod}_R$ the trivial object (the group $\{e\}$, the zero module 0) is both initial and terminal, hence a zero object. Every homomorphism to or from it is forced, since a homomorphism preserves the identity; this is the algebraic fact recorded in [25] (trivial group, zero module).

3. In **Ring** (unital rings, $1 \neq 0$ not required) the ring $\mathbb{Z}$ is initial: for any ring R the unique homomorphism $\mathbb{Z} \to R$ sends n to $n \cdot 1_R$, forced because a ring map preserves 1. The zero ring, in which $1 = 0$, is terminal. Here the two extremes are distinct, so **Ring** has no zero object.

4. A poset $(P, \leq)$, viewed as a category with a unique arrow $x \to y$ when $x \leq y$, has an initial object exactly when P has a least element and a terminal object exactly when it has a greatest element.

The **Ring** example already exhibits the pattern the rest of the chapter formalizes: the map $\mathbb{Z} \to R$ is not just *some* distinguished morphism, it is the *unique* one, and that uniqueness is the whole content. A universal property is always a uniqueness statement of this kind.

The defining clause of an initial object, "exactly one morphism", has an immediate consequence: an initial object is determined by the property, not only up to isomorphism but up to a *unique* isomorphism. This is the prototype every universal-property uniqueness argument imitates, so it is worth seeing once in full.

Proposition 61.1.3: Initial and Terminal Objects are Unique up to Unique Isomorphism

If 0 and $0'$ are both initial objects of a category C, then there is a unique isomorphism $0 \to 0'$. Dually, any two terminal objects are related by a unique isomorphism.

***Proof*:**
Since 0 is initial, there is a unique morphism $u\colon 0 \to 0'$; since $0'$ is initial, there is a unique morphism $v\colon 0' \to 0$. The composite $vu\colon 0 \to 0$ is a morphism from 0 to itself, and 0 is initial, so there is only one such morphism. The identity id_0 is one, hence $vu = \mathrm{id}_0$. The same argument in C with the roles of 0 and $0'$ exchanged gives $uv = \mathrm{id}_{0'}$. Thus u is an isomorphism with inverse v, and it is the only morphism $0 \to 0'$ by initiality of 0, so it is the unique isomorphism. The terminal statement is the initial statement read in C^{op}, completing the proof.

The argument uses nothing about 0 except that maps out of it are forced. Every "unique up to unique isomorphism" claim for a universal construction, for products, kernels, free objects, limits, runs the same way: two solutions each map uniquely to the other, and the round trips are identities because they compete with id for a slot the property allows only one morphism to fill. The reader may reduce any such uniqueness proof to proposition 61.1.3 by exhibiting the construction as an initial or terminal object of a suitable category, which is precisely the reduction the next two definitions make available.

Initial and terminal objects describe an object by its maps to or from *everything*. Most constructions of interest carry data along a functor: the free group on X is built from the set X through the forgetful functor $U\colon \mathbf{Grp} \to \mathbf{Set}$. The universal arrow relativizes the extremal notion to this setting.

Definition 61.1.4: Universal Arrow

Let $U\colon C \to D$ be a functor and d an object of D. A *universal arrow from d to U* is a pair (c, η) consisting of an object c of C and a morphism $\eta\colon d \to Uc$ in D with the following property: for every object x of C and every morphism $f\colon d \to Ux$ in D, there is a unique morphism $g\colon c \to x$ in C such that $f = (Ug) \circ \eta$. The defining equation is the commuting triangle

$$\begin{tikzcd} d \arrow[r, "\eta"] \arrow[dr, "f"'] & Uc \arrow[d, "Ug"] \\ & Ux \end{tikzcd}$$

Dually, a *universal arrow from U to d* is a pair (c, ε) with $\varepsilon\colon Uc \to d$ such that every $f\colon Ux \to d$ factors as $f = \varepsilon \circ (Ug)$ for a unique $g\colon x \to c$.

The pair (c, η) is universal in that η is the most efficient map from d into the image of U: any other map f from d into U is recovered from η by post-composing with a C-morphism, and in exactly one way. Setting $U = \mathrm{id}_D$ recovers the extremal case, an initial object of the coslice under d; the functor U is what makes the notion carry structure across categories.

This connects directly to the representability material of the previous chapter. A universal element of a functor $F\colon C \to \mathbf{Set}$ (definition 60.1.5) is an element $u \in Fc$ such that every $x \in Fa$ equals $Fg(u)$ for a unique $g\colon c \to a$; that is the special case of definition 61.1.4 in which $D = \mathbf{Set}$, the object d is a singleton $\{*\}$, and a map $\{*\} \to Fc$ is named by the element it picks out. Both say the same thing: the datum η (or u) generates every map into U (or every element of F) freely and uniquely. Equivalently, a universal arrow from d to U makes the functor $D(d, U-)$ representable (definition 60.1.1), represented by c with the element η corresponding under Yoneda; the connection to preservation of limits is taken up later in the chapter.

Example 61.1.5: The Universal Arrow of the Free Group

Take $U\colon \mathbf{Grp} \to \mathbf{Set}$ the forgetful functor and X a set. The free group FX carries the inclusion of generators $\eta\colon X \to UFX$. Given any group H and any function $f\colon X \to UH$, the universal property of the free group ([25]) supplies a unique homomorphism $g\colon FX \to H$ extending f on generators, so $f = (Ug) \circ \eta$. Thus (FX, η) is a universal arrow from X to U: the free group is exactly the solution to this universal problem, with η marking the generators.

The example is the template for "free" constructions in general, and the next definition packages the search for a universal arrow into a single object of a single category, so that finding one becomes finding an initial object.

Definition 61.1.6: Comma Category

Let $F\colon A \to C$ and $G\colon B \to C$ be functors into a common category C. The *comma category* $F \downarrow G$ has as objects the triples (a, b, h) with a an object of A, b an object of B, and $h\colon Fa \to Gb$ a morphism of C. A morphism $(a, b, h) \to (a', b', h')$ is a pair $(p\colon a \to a',\ q\colon b \to b')$ making the square

$$\begin{tikzcd} Fa \arrow[r, "h"] \arrow[d, "Fp"'] & Gb \arrow[d, "Gq"] \\ Fa' \arrow[r, "h'"'] & Gb' \end{tikzcd}$$

commute; composition and identities are computed componentwise in A and B.

Two special cases recur. When $B = C$ and $G = \mathrm{id}_C$, an object is a map $Fa \to c$ with c fixed, giving the *slice category* C/c of objects over c; dually $F = \mathrm{id}_C$ and a fixed c give the *coslice* c/C of objects under c. The unifying point is a single reformulation of definition 61.1.4.

Proposition 61.1.7: Universal Arrows are Initial Objects of a Comma Category

Let $U\colon C \to D$ be a functor and d an object of D, viewed as a functor $d\colon \mathbf{1} \to D$ from the one-object one-morphism category. A universal arrow from d to U is the same thing as an initial object of the comma category $d \downarrow U$.

***Proof*:**
An object of $d \downarrow U$ is a pair $(c, \eta\colon d \to Uc)$ with c an object of C, and a morphism $(c, \eta) \to (x, f)$ is a C-morphism $g\colon c \to x$ with $(Ug) \circ \eta = f$. To say (c, η) is initial is to say that for every (x, f) there is exactly one such g, which is verbatim the defining property of a universal arrow from d to U in definition 61.1.4, as we sought to show.

proposition 61.1.7 discharges the uniqueness of universal arrows into proposition 61.1.3: a universal arrow, being an initial object, is determined up to a unique isomorphism, so the free group, the initial ring, and every other solution of a universal problem is unique in the strongest sense available. The comma category is the “neighbouring category” whose initial object is the construction one is after, and the sections that follow build limits and colimits by finding exactly such extremal objects.

Example 61.1.8: Universal Arrows Across the Standing Cast

Each construction below is a universal arrow, read off from its factorization property.

1. *The integers as initial ring.* Let $U\colon \mathbf{Ring} \to \mathbf{Set}$ be the forgetful functor and $\{*\}$ a one-point set. A universal arrow from $\{*\}$ to U is a ring c together with a chosen element $\eta(*) \in Uc$ such that every pointed ring (R, r), with $r = f(*) \in UR$, receives a unique ring map $g\colon c \to R$ sending $\eta(*)$ to r. Take $c = \mathbb{Z}[x]$ with $\eta(*) = x$: a ring map $\mathbb{Z}[x] \to R$ is determined by, and freely specifies, the image of x ([25], universal property of the polynomial ring), so $(\mathbb{Z}[x], x)$ is the universal arrow. Forgetting the marked point instead, and taking U with no chosen element, the empty universal problem is solved by $\mathbb{Z}$ itself: $\mathbb{Z}$ is the initial object of $\mathbf{Ring}$ (example 61.1.2), the universal arrow from the empty set to U.
2. *Abelianization.* Let $U\colon \mathbf{Ab} \to \mathbf{Grp}$ be the inclusion of abelian groups into groups and let G

be a group. The abelianization $G^{\mathrm{ab}} = G/[G,G]$ carries the quotient map $\eta\colon G \to U(G^{\mathrm{ab}})$. Every homomorphism $f\colon G \to UA$ into an abelian group A kills commutators, so it factors as $f = (Ug)\circ\eta$ through a unique homomorphism $g\colon G^{\mathrm{ab}} \to A$ ([25]). Thus (G^{ab}, η) is a universal arrow from G to U, the best abelian approximation of G from the left.

The two items sit on opposite sides of a forgetful (or inclusion) functor. The polynomial ring is universal *from* a set *to* U, a free construction; abelianization is likewise a universal arrow from an object to an inclusion, a reflection into a subcategory. Both are computed the same way, by recognizing that a map out of the constructed object is forced by its effect on the universal datum η. The organizing principle behind such matched pairs of universal arrows, one for each object, is the subject of the chapter on adjunctions.

61.2 Limits

An initial object receives a unique map from nothing, a terminal object receives one from everything; both are universal against the empty diagram. A limit asks for more: a universal way to map compatibly into a whole configuration of objects and morphisms at once. Products, intersections, kernels, and fibered products are all this one requirement read against different shapes. This section isolates the shape as a small category J, packages a configuration as a functor out of it, and defines the limit as the terminal object among the compatible maps into it. The atomic cases, products and equalizers, are then shown to generate every limit.

We first fix what "a configuration" means. A configuration of objects and morphisms in $\mathbf{C}$ is exactly a functor into $\mathbf{C}$ whose domain records the shape; the domain is small so that the configuration has only a set's worth of pieces.

Definition 61.2.1: Diagram, Cone, and Limit

Let $\mathbf{C}$ be a category and J a small category, the *index category*. A *diagram* of shape J in $\mathbf{C}$ is a functor $D\colon J \to \mathbf{C}$ (definition 59.4.1); the object $D(j)$ is written Dj.
A *cone* over D with *apex* X is an object X of $\mathbf{C}$ together with a family of morphisms $\lambda_j\colon X \to Dj$, one for each object j of J, such that for every morphism $u\colon j \to j'$ in J the triangle

$$\begin{tikzcd} & X \arrow[dl, "\lambda_j"'] \arrow[dr, "\lambda_{j'}"] & \\ Dj \arrow[rr, "Du"'] & & Dj' \end{tikzcd}$$

commutes, that is $Du \circ \lambda_j = \lambda_{j'}$. A morphism of cones $(X, \{\lambda_j\}) \to (Y, \{\mu_j\})$ is a morphism $f\colon X \to Y$ with $\mu_j \circ f = \lambda_j$ for every j. Cones over D and their morphisms form a category $\mathrm{Cone}(D)$.
A *limit* of D is a terminal object of $\mathrm{Cone}(D)$: a cone $(L, \{\pi_j\})$, the *limit cone*, such that every cone $(X, \{\lambda_j\})$ factors through it by a unique morphism $\langle\lambda_j\rangle\colon X \to L$ with $\pi_j \circ \langle\lambda_j\rangle = \lambda_j$ for all j. The apex L is written $\lim D$ or $\lim_J D$, and the π_j are its *projections*.

The commuting condition is the whole content of "compatible": a cone is not just a map into each Dj but a family that respects every morphism the shape imposes among the Dj. The limit is the cone through which all others factor, so a map into L is precisely a compatible family of maps into the diagram. This is the universal-arrow pattern of definition 61.1.4 for the diagonal functor $\Delta\colon \mathbf{C} \to [J, \mathbf{C}]$ sending X to the constant diagram at X: a cone with apex X is a natural transformation $\Delta X \Rightarrow D$ (definition 59.5.2), and the limit is the universal such X.

Being terminal in $\mathrm{Cone}(D)$ settles uniqueness at once.

Proposition 61.2.2: Uniqueness of Limits

If $(L, \{\pi_j\})$ and $(L', \{\pi'_j\})$ are both limits of $D\colon J \to \mathbf{C}$, there is a unique isomorphism $L \to L'$ commuting with the projections.

***Proof*:**
Both are terminal objects of $\mathrm{Cone}(D)$, and terminal objects are unique up to a unique isomorphism by proposition 61.1.3. The isomorphism is a morphism of cones, so it commutes with the projections by definition of $\mathrm{Cone}(D)$, completing the proof.

The limit is thus determined by D alone, and one speaks of "the" limit. The rest of the section reads specific shapes J off this single definition, then shows two of them suffice to build every other.

The simplest shapes carry no morphisms beyond identities, so the cone condition is vacuous and a cone is just a chosen map into each object.

Definition 61.2.3: Product

Let J be a *discrete* category (only identity morphisms) with objects indexed by a set I, and let D pick out a family $\{X_i\}_{i\in I}$. The limit of D, when it exists, is the *product* $\prod_{i\in I} X_i$, with projections $\pi_i\colon \prod_{i\in I} X_i \to X_i$. Its universal property: for every object X and every family $f_i\colon X \to X_i$ there is a unique $\langle f_i\rangle\colon X \to \prod_{i\in I} X_i$ with $\pi_i \circ \langle f_i\rangle = f_i$ for all i. When $I = \{1, 2\}$ the product is written $X_1 \times X_2$; when $I = \varnothing$ the limit over the empty diagram is a terminal object 1, the *empty product*.

A product is a single object out of which one can name any tuple of maps to the factors and nothing more; the projections read the coordinates back. There are no compatibility constraints because a discrete shape imposes none, so a map into $\prod_i X_i$ is exactly an unconstrained family of maps into the factors. The empty case is worth pausing on: with no factors to hit, the universal property says every object admits a unique map to the limit, which is the definition of a terminal object (definition 61.1.1).

Next introduce the shape with two parallel arrows between two objects. A cone must equalize the two composites, so it selects the part of the source on which they agree.

Definition 61.2.4: Equalizer

Let J be the category $\{\bullet \rightrightarrows \bullet\}$ with two objects and two parallel non-identity morphisms, and let D send it to a parallel pair $f, g\colon A \to B$ in $\mathbf{C}$. A cone over D is an object E with maps to A and to B making both triangles commute; the map to B is forced to be the common composite, so a cone is a morphism $e\colon E \to A$ with $fe = ge$. The limit is the *equalizer* of f and g: a morphism $e\colon E \to A$ with $fe = ge$ such that every $h\colon X \to A$ with $fh = gh$ factors as $h = e \circ \bar{h}$ for a unique $\bar{h}\colon X \to E$.

The equalizer is the categorical "locus where two maps agree": e is the universal way to map into A so that f and g become equal after the map. That universality forces e to be injective on maps.

Proposition 61.2.5: Equalizers are Monic

The equalizer $e\colon E \to A$ of a parallel pair $f, g\colon A \to B$ is a monomorphism.

***Proof*:**
Let $p, q\colon X \to E$ satisfy $ep = eq$, and set $h = ep$. Then $fh = fep = gep = gh$, so $h\colon X \to A$ satisfies $fh = gh$ and factors uniquely through e. Both p and q are factorizations of h through e: indeed $e \circ p = h$ and $e \circ q = eq = ep = h$. Uniqueness of the factorization gives $p = q$, so e is monic, as required.

That equalizers are monic recovers the intuition that "where f and g agree" is a subobject of the source A, and it is the ingredient that later lets equalizers cut a limit out of a product.

The third shape is the cospan: two arrows into a common target. Its limit combines a product with an agreement condition.

Definition 61.2.6: Pullback

Let J be the *cospan* category $\{\bullet \to \bullet \leftarrow \bullet\}$, and let D send it to a cospan $f\colon X \to Z$, $g\colon Y \to Z$ in $\mathbf{C}$. The limit is the *pullback* $X \times_Z Y$, with projections $p\colon X \times_Z Y \to X$ and $q\colon X \times_Z Y \to Y$ satisfying $fp = gq$, universal among such squares:

$$\begin{tikzcd} X \times_Z Y \arrow[r, "q"] \arrow[d, "p"'] & Y \arrow[d, "g"] \\ X \arrow[r, "f"'] & Z \end{tikzcd}$$

For every object W with maps $a\colon W \to X$, $b\colon W \to Y$ such that $fa = gb$, there is a unique $\langle a, b\rangle\colon W \to X \times_Z Y$ with $p\langle a, b\rangle = a$ and $q\langle a, b\rangle = b$.

A pullback is a product "constrained to agree over Z": it is the subobject of $X \times Y$ on which the two ways to reach Z coincide. When $Z = 1$ is terminal the agreement condition is vacuous and the pullback collapses to the product $X \times Y$, so the pullback refines the product. This one shape specializes to intersections of subobjects, preimages, fibers of a map, and fibered products of schemes, which is why it recurs throughout the series.

Each of these shapes should be seen at work in the category the reader knows best.

Example 61.2.7: Limits in Set
In **Set** each of the three limits above has a concrete description, and the universal property is verified by hand.

1. *Products.* The product $\prod_{i \in I} X_i$ is the cartesian product, the set of tuples $(x_i)_{i \in I}$ with $x_i \in X_i$, and π_i is the i-th coordinate. Given $f_i\colon X \to X_i$, the map $\langle f_i\rangle(x) = (f_i(x))_{i \in I}$ satisfies $\pi_i\langle f_i\rangle = f_i$, and any g with $\pi_i g = f_i$ has i-th coordinate f_i, so $g = \langle f_i\rangle$. The empty product is a one-element set $\{*\}$, terminal in **Set**. In **Grp**, **Ab**, and $\mathbf{Mod}_R$ the

same underlying cartesian product carries the pointwise (coordinatewise) operations, and the projections are homomorphisms.

2. *Equalizers.* The equalizer of $f, g\colon A \to B$ is the subset

$$E = \{\, a \in A \,|\, f(a) = g(a) \,\}$$

with e the inclusion. If $h\colon X \to A$ satisfies $fh = gh$ then $f(h(x)) = g(h(x))$ for all x, so $h(x) \in E$; corestricting h to E gives the unique $\bar{h}$ with $e\bar{h} = h$. The inclusion is injective, illustrating proposition 61.2.5.

3. *Pullbacks.* The pullback of $f\colon X \to Z$ and $g\colon Y \to Z$ is the subset of $X \times Y$

$$X \times_Z Y = \{\, (x, y) \in X \times Y \,|\, f(x) = g(y) \,\}$$

with p, q the coordinate maps. Given $a\colon W \to X$ and $b\colon W \to Y$ with $fa = gb$, the map $w \mapsto (a(w), b(w))$ lands in $X \times_Z Y$ and is the unique factorization. Taking $g\colon \{z\} \hookrightarrow Z$ the inclusion of a point recovers the fiber $f^{-1}(z)$, and taking $X, Y \subseteq Z$ with f, g the inclusions recovers the intersection $X \cap Y$.

The three examples share a pattern: the limit is a subset of a product cut out by the equations the shape requires. The next theorem promotes that pattern to a construction, showing that products and equalizers alone build every limit.

Products handle the objects of a diagram, equalizers handle the constraints its morphisms impose. Stack every Dj into one product, stack every constraint into a second, and the two maps that express "constraint satisfied versus not" have an equalizer that is exactly the limit. This reduces the existence of all limits to two special cases.

Theorem 61.2.8: Limits from Products and Equalizers

Let $\mathbf{C}$ have all small products and all equalizers. Then $\mathbf{C}$ has all small limits. Explicitly, for a diagram $D\colon J \to \mathbf{C}$ with J small, form the products

$$P = \prod_{j \in J} Dj, \qquad Q = \prod_{u\colon j \to j' \text{ in } J} D(\operatorname{cod} u)$$

indexed by the objects and by the morphisms of J respectively. Let $s, t\colon P \to Q$ be the morphisms determined on the u-component $(u\colon j \to j')$ by

$$\rho_u \circ s = D(u) \circ \mathrm{pr}_j, \qquad \rho_u \circ t = \mathrm{pr}_{j'}$$

where $\mathrm{pr}_j\colon P \to Dj$ and $\rho_u\colon Q \to D(\operatorname{cod} u) = Dj'$ are the product projections. Then the equalizer $e\colon L \to P$ of s and t is a limit of D, with projections $\pi_j = \mathrm{pr}_j \circ e$.

***Proof*:**
The proof runs in three stages: define the two maps whose disagreement measures failure of the cone condition, check that the equalizer carries a cone, and check that this cone is terminal.

Step 1: The maps s and t. By the universal property of the product Q (definition 61.2.3), a morphism into Q is determined by its composite with each projection ρ_u. So the two displayed families of composites define s and t uniquely. Reading them off: for a point of P, thought of as an arbitrary family $(x_j)_j$ with $x_j : T \to Dj$ for a test object T, the u-component of s is $D(u) \circ x_j$ (push the source coordinate along $D(u)$), while the u-component of t is $x_{j'}$ (read off the target coordinate). A family (x_j) satisfies $s = t$ on the u-component precisely when $D(u) \circ x_j = x_{j'}$, which is the cone condition for the morphism u. Thus s and t agree on (x_j) if and only if (x_j) is a cone over D.

Step 2: The equalizer is a cone. Let $e\colon L \to P$ equalize s and t, so $se = te$. Set $\pi_j = \mathrm{pr}_j \circ e\colon L \to Dj$. For any $u\colon j \to j'$, compose $se = te$ with ρ_u:

$$\rho_u \circ s \circ e = \rho_u \circ t \circ e$$

The left side is $D(u) \circ \mathrm{pr}_j \circ e = D(u) \circ \pi_j$ and the right side is $\mathrm{pr}_{j'} \circ e = \pi_{j'}$. Hence $D(u) \circ \pi_j = \pi_{j'}$ for every u, so $(L, \{\pi_j\})$ is a cone over D.

Step 3: The cone is terminal. Let $(X, \{\lambda_j\})$ be any cone over D. By the universal property of P the family $\{\lambda_j\}$ determines a unique morphism $\ell\colon X \to P$ with $\mathrm{pr}_j \circ \ell = \lambda_j$ for all j. The cone condition $D(u) \circ \lambda_j = \lambda_{j'}$ says, on each u-component,

$$\rho_u \circ s \circ \ell = D(u) \circ \mathrm{pr}_j \circ \ell = D(u) \circ \lambda_j = \lambda_{j'} = \mathrm{pr}_{j'} \circ \ell = \rho_u \circ t \circ \ell$$

Since these hold for every u, and morphisms into Q are determined by their ρ_u-components, $s\ell = t\ell$. So ℓ equalizes s and t and factors uniquely as $\ell = e \circ \bar{\ell}$ for a unique $\bar{\ell}\colon X \to L$. Then

$$\pi_j \circ \bar{\ell} = \mathrm{pr}_j \circ e \circ \bar{\ell} = \mathrm{pr}_j \circ \ell = \lambda_j$$

so $\bar{\ell}$ is a morphism of cones $(X, \{\lambda_j\}) \to (L, \{\pi_j\})$. For uniqueness, suppose $\varphi\colon X \to L$ is any cone morphism, so $\pi_j \circ \varphi = \lambda_j$ for all j. Then $\mathrm{pr}_j \circ (e \circ \varphi) = \pi_j \circ \varphi = \lambda_j = \mathrm{pr}_j \circ \ell$ for every j, and the uniqueness clause of P's universal property forces $e \circ \varphi = \ell$. Since e is monic (proposition 61.2.5), $e \circ \varphi = e \circ \bar{\ell}$ gives $\varphi = \bar{\ell}$. Thus $\bar{\ell}$ is the unique cone morphism $X \to L$, so $(L, \{\pi_j\})$ is terminal in $\mathrm{Cone}(D)$ and is a limit of D, as we sought to show.

The construction is the abstract form of the **Set** computation in example 61.2.7: P is the product of the objects, s and t encode the two sides of each morphism-constraint, and the equalizer $L \subseteq P$ is the sub-configuration on which every constraint holds. What was “a subset of a product cut out by equations” in **Set** is now “an equalizer of two maps out of a product” in any category with those two limits, and the same argument gives the universal property. A category with all products and equalizers is therefore complete for all small diagrams, and to check completeness one need only supply products and equalizers.

A limit is the universal way to map compatibly into an entire diagram at once, so that a single morphism into the apex is the same data as a family of morphisms into the diagram respecting its shape. Products and equalizers are the atomic cases, the first handling objects without constraint and the second handling a constraint without extra objects, and theorem 61.2.8 shows these two generate all the rest.

61.3 Colimits

Every construction of the previous section has a mirror image obtained by reversing all arrows. Where a limit sits at the tip of an incoming cone, its dual sits at the tip of an outgoing one; where a limit is a terminal cone, its dual is an initial cocone. This section develops that mirror image. The payoff is not only a second family of constructions but a method: because a colimit in C is exactly a limit in C^{op}, every theorem about limits transports across the duality principle (box 59.3.3) with no new proof, and the constructions themselves become recognizable as the coproducts, quotients, and gluings the reader already meets throughout algebra and topology.

The starting point is the dual of a cone. A cone over a diagram receives a compatible family of maps from a single apex; a cocone under a diagram emits a compatible family of maps into a single nadir.

Definition 61.3.1: Cocone and Colimit

Let $D\colon J \to C$ be a diagram. A *cocone* under D with *nadir* X is an object X of C together with a family of morphisms

$$\iota_j\colon Dj \to X, \qquad j \in J$$

one for each object of J, such that for every morphism $u\colon j \to k$ of J the triangle

$$\iota_k \circ Du = \iota_j$$

commutes. A *morphism of cocones* $(X, \{\iota_j\}) \to (Y, \{\kappa_j\})$ is a morphism $f\colon X \to Y$ of C with $f\iota_j = \kappa_j$ for every j.

A *colimit* of D is an *initial* cocone: a cocone $(L, \{\iota_j\})$ such that for every cocone $(X, \{\kappa_j\})$ under D there is a unique morphism $f\colon L \to X$ with $f\iota_j = \kappa_j$ for all j. The nadir L is written $\operatorname{colim}_J D$, and the maps ι_j are the *structure maps* (or *coprojections*) of the colimit.

The definition is definition 61.2.1 read backwards. A cocone under D is precisely a cone over the diagram $D^{\mathrm{op}}\colon J^{\mathrm{op}} \to C^{\mathrm{op}}$ obtained by reversing every arrow, and an initial cocone is a terminal cone in C^{op}. This is the organizing fact of the section.

Proposition 61.3.2: Colimits are Limits in the Opposite Category

Let $D\colon J \to C$ be a diagram and write $D^{\mathrm{op}}\colon J^{\mathrm{op}} \to C^{\mathrm{op}}$ for the same assignment on objects, regarded as a diagram in C^{op}. An object L with maps $\iota_j\colon Dj \to L$ is a colimit of D in C if and only if L with the same maps, now read as $\iota_j\colon L \to Dj$ in C^{op}, is a limit of D^{op} in C^{op}. Thus

$$\operatorname*{colim}_J D \text{ in } C \quad = \quad \lim_{J^{\mathrm{op}}} D^{\mathrm{op}} \text{ in } C^{\mathrm{op}}$$

***Proof*:**
Reversing arrows sends the compatibility condition $\iota_k \circ Du = \iota_j$ for a cocone under D to the condition $D^{\mathrm{op}}u \circ \iota_k = \iota_j$ for a cone over D^{op} in C^{op}, since composition in C^{op} is composition in C in the opposite order and each Du becomes $D^{\mathrm{op}}u$ pointing the other way. So cocones under D in C are the same data as cones over D^{op} in C^{op}, and a morphism of cocones $f\colon X \to Y$ in C is a morphism of cones $f\colon Y \to X$ in C^{op}. An initial object among cocones is therefore a terminal object among cones, so L is a colimit of D exactly when it is a limit of D^{op} in C^{op}, completing the proof.

This proposition licenses a mechanical translation. Every notion of the previous section has a dual with the prefix “co”: the terminal cone becomes the initial cocone, and each specific limit shape becomes a colimit shape by reversing the diagram J. The three shapes worth naming are the duals of the product, the equalizer, and the pullback. Their definitions are stated directly, but each one could equally be obtained by applying proposition 61.3.2 to the corresponding definition of section 61.2.

The first is the dual of the product (definition 61.2.3): a discrete diagram $\{X_i\}_{i\in I}$ with no nontrivial morphisms, whose colimit assembles the objects side by side rather than cutting out their common part.

Definition 61.3.3: Coproduct

Let $\{X_i\}_{i\in I}$ be a family of objects of C, viewed as a diagram on the discrete index category I. A *coproduct* of the family is a colimit: an object $\coprod_{i\in I} X_i$ together with *injections*

$$\iota_i\colon X_i \to \coprod_{i\in I} X_i$$

such that for every object Y and every family of morphisms $f_i\colon X_i \to Y$ there is a unique morphism $f\colon \coprod_{i\in I} X_i \to Y$ with $f\iota_i = f_i$ for all i. A binary coproduct is written $X_1 \sqcup X_2$.

A coproduct is the object into which every X_i maps and out of which maps are prescribed by naming one map from each summand independently. Where a product is determined by projections and its universal property governs *maps in*, a coproduct is determined by injections and governs *maps out*. The empty coproduct, indexed by $I = \varnothing$, is an object with a unique map to every Y and no data to specify: it is an initial object 0 (definition 61.1.1).

Example 61.3.4: Coproducts in the Standing Cast

The coproduct instantiates as a familiar assembly in each concrete category.

1. In **Set** the coproduct is the *disjoint union* $\coprod_i X_i$, the set of pairs (i, x) with $x \in X_i$, and $\iota_i(x) = (i, x)$. A family of functions $f_i\colon X_i \to Y$ assembles into the single function $(i, x) \mapsto f_i(x)$, which is the only function restricting to each f_i on the i-th copy. For two sets this is $X_1 \sqcup X_2$.

2. In **Grp** the coproduct is the *free product* $G_1 * G_2$: its elements are reduced words alternating between nonidentity elements of G_1 and of G_2, with ι_1, ι_2 the inclusions of the one-letter words. A pair of homomorphisms $f_i\colon G_i \to H$ extends to the unique homomorphism sending a word to the product of the images of its letters, each letter mapped by the f_i of the factor it comes from, and this assignment is forced ([25], free products). The free group on a set S is the coproduct of $|S|$ copies of $\mathbb{Z}$.

3. In **Ab** and in $\mathbf{Mod}_R$ the coproduct of a family is the *direct sum* $\bigoplus_i M_i$, the submodule of the product $\prod_i M_i$ consisting of tuples with finitely many nonzero entries, with ι_i the insertion into the i-th coordinate. A family $f_i\colon M_i \to N$ assembles into $(m_i)_i \mapsto \sum_i f_i(m_i)$, a finite sum because only finitely many m_i are nonzero ([25], direct sums). For finitely many summands the direct sum and the direct product coincide, so the biproduct $M_1 \oplus M_2$ is at once a product and a coproduct; this coincidence is a feature of additive categories, not a general phenomenon.

The contrast with products is sharpest in **Grp**: the product $G_1 \times G_2$ forces the two factors to commute, while the coproduct $G_1 * G_2$ imposes no relations at all between them.

The direct-sum example already shows the divergence between limits and colimits concretely: in **Set** the product is a Cartesian product and the coproduct a disjoint union, unrelated constructions, and in **Grp** the gap between $G_1 \times G_2$ and $G_1 * G_2$ is the gap between imposing all cross-relations and imposing none. The next shape is the dual of the equalizer (definition 61.2.4). Where an equalizer selects the sub-object on which two maps agree, its dual forms the quotient on which two maps are forced to agree.

Definition 61.3.5: Coequalizer

Let $f, g\colon A \to B$ be a parallel pair of morphisms in C. A *coequalizer* of f and g is a morphism $q\colon B \to Q$ with $qf = qg$ that is universal with this property: for every morphism $h\colon B \to Y$ satisfying $hf = hg$ there is a unique morphism $\bar{h}\colon Q \to Y$ with $\bar{h}q = h$.

The coequalizer is the colimit of the two-object diagram $A \rightrightarrows B$ whose only nontrivial morphisms are f and g: the condition $qf = qg$ is the cocone compatibility for that diagram, and universality is initiality among such cocones. The map q is the largest quotient of B on which f and g become equal, the categorical quotient. It is always epic, the dual of the fact that an equalizer is always monic.

Lemma 61.3.6: A Coequalizer is Epic

The coequalizer map $q\colon B \to Q$ of any parallel pair $f, g\colon A \to B$ is an epimorphism.

***Proof*:**
Suppose $s, t\colon Q \to Y$ satisfy $sq = tq$. Set $h = sq = tq\colon B \to Y$. Then $hf = sqf = sqg = hg$ using $qf = qg$, so h satisfies the coequalizing condition and factors uniquely through q. Both s and t are such factorizations, since $sq = h$ and $tq = h$, so uniqueness forces $s = t$. Thus q is epic, as we sought to show.

Example 61.3.7: The Coequalizer in Sets is a Quotient
For functions $f, g\colon A \to B$ in **Set**, let $\sim$ be the smallest equivalence relation on B containing every pair $(f(a), g(a))$ for $a \in A$. The coequalizer is the quotient set $Q = B/{\sim}$ with $q\colon B \to Q$ the quotient map. A function $h\colon B \to Y$ satisfies $hf = hg$ exactly when $h(f(a)) = h(g(a))$ for all a, that is, when h is constant on each generating pair, hence on each $\sim$-class; such an h descends to a unique $\bar{h}\colon Q \to Y$ with $\bar{h}q = h$. The generated equivalence relation is not just the set of pairs $(f(a), g(a))$: it is their reflexive, symmetric, transitive closure, since a quotient map respecting the relations must respect everything they entail. Taking g constant recovers, as a special case, the quotient of B collapsing a subset to a point.

The equivalence closure in example 61.3.7 is the concrete face of a general fact: a coequalizer imposes the relations $f = g$ together with every consequence forced by the category's own composition, exactly as a coproduct in **Grp** imposes no relations while a quotient group imposes stated ones. The last named shape is the dual of the pullback (definition 61.2.6), which glues two objects along a shared source rather than intersecting them over a shared target.

Definition 61.3.8: Pushout

Let $f\colon Z \to X$ and $g\colon Z \to Y$ be a pair of morphisms with common *source* Z. A *pushout* of f and g is a colimit of this diagram: an object $X \sqcup_Z Y$ with morphisms $i\colon X \to X \sqcup_Z Y$ and $j\colon Y \to X \sqcup_Z Y$ satisfying $if = jg$, universal among such cocones. For every object W with maps $p\colon X \to W$ and $q\colon Y \to W$ satisfying $pf = qg$ there is a unique $h\colon X \sqcup_Z Y \to W$ with $hi = p$ and $hj = q$.

A pushout glues X and Y along the images of Z: the map f marks a copy of Z inside X, the map g marks a copy inside Y, and $X \sqcup_Z Y$ is the result of identifying the two marked copies. In **Set** it is the quotient of $X \sqcup Y$ by the relation $f(z) \sim g(z)$, which is why the pushout is the formal model of every "glue two pieces along a common part" construction. When $Z = 0$ is initial the constraint is vacuous and the pushout is the coproduct $X \sqcup Y$, exactly as a pullback over a terminal object is a product.

Example 61.3.9: Pushouts as Gluing

Two instances make the gluing concrete.

1. In **Set**, if $Z \hookrightarrow X$ and $Z \hookrightarrow Y$ are inclusions of a common subset, the pushout $X \sqcup_Z Y$ is the union of X and Y overlapping exactly along Z: form the disjoint union $X \sqcup Y$ and identify each copy of a point of Z. Explicitly it is $(X \sqcup Y)/\sim$ with $\iota_1 f(z) \sim \iota_2 g(z)$, the coequalizer of the two maps $Z \rightrightarrows X \sqcup Y$ given by $\iota_1 f$ and $\iota_2 g$.
2. In **Top** the same colimit glues spaces. Taking X and Y to be two closed disks D^2 and $Z = S^1$ their common boundary circle, mapping into each by the boundary inclusion, the pushout $D^2 \sqcup_{S^1} D^2$ is the 2-sphere S^2: two disks glued along their boundary. This is the mechanism behind attaching cells and building spaces from simpler pieces, the staple of the algebraic topology of CW complexes.

The two constructions from which every colimit is built are now in hand: coproducts assemble a family of objects, and coequalizers impose relations. The dual of theorem 61.2.8 states that these two suffice, and its proof is that theorem read in the opposite category. The motivation is the same as in the limit case: to check that a category has all small colimits, it is enough to check two constructions rather than one for every diagram shape.

Theorem 61.3.10: Colimits from Coproducts and Coequalizers

Let C be a category with all small coproducts and all coequalizers. Then C has all small colimits. Concretely, the colimit of a diagram $D\colon J \to C$ with J small is the coequalizer of a canonical parallel pair of maps between two coproducts,

$$\coprod_{(u\colon j \to k) \in \mathrm{mor}\, J} Dj \rightrightarrows \coprod_{j \in \mathrm{ob}\, J} Dj \longrightarrow \operatorname*{colim}_J D$$

where the first coproduct is indexed by the morphisms of J and the second by its objects.

***Proof*:**
By proposition 61.3.2, a colimit of $D\colon J \to C$ is a limit of $D^{\mathrm{op}}\colon J^{\mathrm{op}} \to C^{\mathrm{op}}$. The hypotheses transport across the duality principle (box 59.3.3): a coproduct in C is a product in C^{op} and a coequalizer in C is an equalizer in C^{op}, so C having all small coproducts and coequalizers is C^{op} having all small products and equalizers. Theorem 61.2.8 then applies in C^{op} and produces the limit of D^{op}, hence the colimit of D. This proves existence; it remains only to record what the construction of that theorem becomes when read back in C.

The construction of theorem 61.2.8 builds a limit as an equalizer of two maps between the product $\prod_j Dj$ over the objects and the product $\prod_u D(\operatorname{cod} u)$ over the morphisms. Dualizing every arrow turns each product into a coproduct and the equalizer into a coequalizer, giving the coproduct $\coprod_j Dj$ over the objects, the coproduct $\coprod_u D(\operatorname{dom} u)$ over the morphisms, and a coequalizer of two maps between them. Concretely, for a morphism $u\colon j \to k$ of J the u-th summand Dj of the source coproduct is sent by the two maps to the object coproduct in two ways: one injection includes Dj as the j-th summand directly, and the other sends it through $Du\colon Dj \to Dk$ into the k-th summand. Writing $\iota_j\colon Dj \to \coprod_j Dj$ for the coproduct injections, the two maps $s, t\colon \coprod_u Dj \to \coprod_j Dj$ are determined on the u-th summand by

$$s \circ \iota_u = \iota_j, \qquad t \circ \iota_u = \iota_k \circ Du$$

Their coequalizer $q\colon \coprod_j Dj \to Q$ forces $q\iota_j$ and $q\iota_k \circ Du$ to become equal for every u, since $qs = qt$ reads $q\iota_j = q\iota_k \circ Du$ on the u-th summand. This is exactly the cocone compatibility for the maps $q\iota_j$. So the maps $q\iota_j\colon Dj \to Q$ form a cocone under D, and the coequalizer's universal property makes it the initial one: a cocone $(Y, \{\kappa_j\})$ is a family with $\kappa_k \circ Du = \kappa_j$, which is a map $\langle \kappa_j \rangle\colon \coprod_j Dj \to Y$ coequalizing s and t, hence factoring uniquely through q. Thus $Q = \operatorname{colim}_J D$, completing the proof.

The theorem and its proof are the mirror of the limit case in the strict sense that not one line of new mathematics was needed: proposition 61.3.2 moves the problem to C^{op}, the duality principle converts the hypotheses, and theorem 61.2.8 does the work. The dual construction is worth carrying explicitly all the same, because the coequalizer of two maps between coproducts is how a colimit is computed: assemble every object of the diagram into one coproduct, then force the relations that the morphisms of J impose. The equalizer-of-products picture cut a sub-object out of an assembly; the coequalizer-of-coproducts picture builds an assembly and then collapses it.

This is the interpretive content of the whole duality. Limits and colimits are two readings of one definition, exchanged by reversing arrows, and the exchange is not a formal trick but a difference in operation. A limit intersects and cuts out: the product is the joint object seen from all its projections, the equalizer is the locus where two maps agree, the pullback is the fibered intersection over a shared target. A colimit glues and quotients: the coproduct sets objects side by side, the coequalizer imposes relations, the pushout glues along a shared source. The product cuts $\prod X_i$ down by an equalizer to enforce compatibility; the coproduct is quotiented by a coequalizer to enforce it. Each theorem, each construction, and each computation of section 61.2 therefore has a twin here, obtained for free by the duality principle, and the two families are the same mathematics viewed from opposite sides.

61.4 Computing (Co)limits

The previous sections built limits and colimits abstractly and reduced their existence to two special cases: products and equalizers on one side, coproducts and coequalizers on the other (theorem 61.2.8 and its dual). This section makes the constructions concrete: a name for the categories in which every diagram has a limit; the working principle that in the algebraic categories a limit is computed on underlying sets while a colimit usually is not; the pointwise computation in functor and presheaf categories; and the functors that carry limits to limits, the representable ones foremost. That last result closes the loop with Chapter 2: an object sees limits through its hom-functor exactly because of Yoneda.

We fix conventions once. A category has *all small limits* if every diagram $D\colon J \to C$ indexed by a small category J has a limit, and likewise for colimits.

Definition 61.4.1: Complete and Cocomplete Categories

A category C is *complete* if it has all small limits, that is, if every diagram $D\colon J \to C$ with J small has a limit $\lim D$. It is *cocomplete* if it has all small colimits, that is, if every such diagram has a colimit $\operatorname{colim}_J D$.

Completeness is a strong hypothesis stated compactly, but it need never be checked diagram by diagram. Theorem 61.2.8 constructed the limit of an arbitrary diagram from a pair of products and one equalizer, so a category that has all small products and all equalizers already has all small limits. This is the only test one runs in practice.

Proposition 61.4.2: Completeness from Products and Equalizers

A category C is complete if and only if it has all small products and all equalizers. Dually, C is cocomplete if and only if it has all small coproducts and all coequalizers.

***Proof*:**
If C is complete it has, in particular, all products (limits of discrete diagrams) and all equalizers (limits over the parallel-pair shape), so the forward direction is immediate. Conversely, suppose C has all small products and all equalizers. Given any diagram $D\colon J \to C$ with J small, the construction of theorem 61.2.8 builds $\lim D$ as the equalizer of two maps between the product $\prod_j Dj$ over the objects of J and the product $\prod_u D(\operatorname{cod} u)$ over the morphisms of J; both products are small because J is small. Hence $\lim D$ exists, and C is complete. The colimit statement is the dual, obtained by applying the same argument in C^{op} (box 59.3.3): a coproduct in C is a product in C^{op}, a coequalizer in C is an equalizer in C^{op}, and completeness of C^{op} is cocompleteness of C, completing the proof.

The categories of the standing cast are all complete and cocomplete: **Set**, **Grp**, **Ab**, **Ring**, $\mathbf{Mod}_R$, and **Top** each have all small limits and colimits. For **Set** this is direct, and completeness of the rest follows from the next result once the limits are located inside **Set**. Cocompleteness holds too, but its proof is not dual in content: by proposition 61.4.2 it suffices to have all small coproducts and all coequalizers, and in the algebraic categories these are the free constructions and quotients of [25] (free products in **Grp**, direct sums in **Ab** and $\mathbf{Mod}_R$, and quotients by the relations generated for coequalizers), while **Top** carries the quotient and coproduct topologies. The second half of this section does not reprove these but illustrates the point that colimits are built by their own universal properties rather than read off underlying sets.

The pattern that makes limits computable in the algebraic categories is that the forgetful functor to **Set** does more than preserve them, it *creates* them: the underlying set of a limit is the limit of the underlying sets, and the algebraic structure on it is forced. A product of groups is the product set with coordinatewise multiplication; the equalizer of two homomorphisms is the subset where they agree, which is a subgroup because the operations are computed coordinatewise. The precise statement follows; the notion of creation is made formal later in this section (definition 61.4.6), but the content here is elementary and self-contained.

Proposition 61.4.3: Limits in the Algebraic Categories

Let C be any of **Grp**, **Ab**, **Ring**, or $\mathbf{Mod}_R$, and let $U\colon C \to \mathbf{Set}$ be the forgetful functor. Then C is complete, and U *creates* all small limits: for a diagram $D\colon J \to C$, every limit cone of UD in **Set** lifts to a unique cone in C lying over it, and that cone is a limit. In particular the underlying set of $\lim D$ is $\lim(UD)$, computed in **Set**, with the algebraic operations defined coordinatewise. Concretely:

1. *Products.* The product $\prod_i X_i$ has underlying set the Cartesian product $\prod_i UX_i$, with the operations applied coordinatewise; the projections $\pi_j\colon \prod_i X_i \to X_j$ are the coordinate projections.

2. *Equalizers.* The equalizer of two morphisms $f, g\colon X \to Y$ is the subobject $E = \{x \in UX \,|\, f(x) = g(x)\}$ with the operations inherited from X; the equalizing morphism $E \to X$ is the inclusion.

The same holds for **Top**, with the limit set carrying the initial topology for the cone maps.

***Proof*:**
By proposition 61.4.2 it suffices to treat products and equalizers and to assemble the general limit from these by theorem 61.2.8, since creation is inherited stage by stage: a functor that creates products and equalizers creates every limit built from them. For each of the two shapes the argument shows more than preservation. Starting from a limit cone in **Set** (the Cartesian product with its projections, or the agreement subset with its inclusion), the algebraic operations that turn it into a cone in C are forced coordinatewise, so the lift is unique; the lifted cone is then verified to be a limit in C; and U returns it to the given cone in **Set**. That is creation (definition 61.4.6), and preservation and completeness of C follow.

Step 1: Products. Take a small family $(X_i)_{i\in I}$ in C. Let P be the Cartesian product $\prod_i UX_i$ of underlying sets, an element of which is a family $(x_i)_i$ with $x_i \in UX_i$. Define the algebraic operations coordinatewise: for **Grp**, $(x_i)(y_i) = (x_i y_i)$ with identity (e_i) and inverse $(x_i)^{-1} = (x_i^{-1})$; for a ring or module the addition and multiplication (or scalar action) are applied in each coordinate. The group, ring, or module axioms hold in P because they hold in each X_i and are tested coordinatewise. The coordinate projections $\pi_j\colon P \to X_j$, $(x_i) \mapsto x_j$, are morphisms of C because the operations on P are coordinatewise. Given any Z in C with morphisms $h_i\colon Z \to X_i$, the set map $h\colon UZ \to P$, $z \mapsto (h_i(z))_i$, is the unique function with $\pi_j h = h_j$ for all j (this is the universal property of the product in **Set**), and it is a morphism of C because each h_i is and the operations on P are coordinatewise. So P with the π_j is the product in C, and

$UP = \prod_i UX_i$ is the product in **Set**; thus U preserves it. The lift is unique: any operations on the set $\prod_i UX_i$ making the coordinate projections into morphisms of C must agree with each X_j in that coordinate, hence be the coordinatewise ones, so this is the only structure lying over the **Set** product. That is creation for the product shape.

Step 2: Equalizers. Take $f, g\colon X \to Y$ in C and set $E = \{x \in UX \,|\, f(x) = g(x)\} \subseteq UX$, which is the equalizer of Uf and Ug in **Set**. The subset E is closed under the operations of X: if $f(x) = g(x)$ and $f(x') = g(x')$ then, since f and g are morphisms, $f(xx') = f(x)f(x') = g(x)g(x') = g(xx')$, and likewise f and g agree on the identity, on inverses, and on scalar multiples. Hence E is a subgroup, subring, or submodule of X, and with these inherited operations the inclusion $e\colon E \to X$ is a morphism of C with $fe = ge$. If $h\colon Z \to X$ satisfies $fh = gh$ then h lands in E at the level of sets, so it factors uniquely as $h = eh'$ through the set map $h'\colon UZ \to E$; and h' is a morphism of C because its composite with the inclusion is and E carries the inherited operations. So E is the equalizer in C, and UE is the equalizer in **Set**. Again the lift is unique: the subset E inherits its operations from X by restriction, the only structure making the inclusion a morphism, so the cone over the **Set** equalizer is forced. That is creation for the equalizer shape.

Step 3: Topological spaces. For **Top** the same two constructions work on underlying sets, and the topology is forced: the product carries the product topology (the coarsest making every projection continuous) and the equalizer carries the subspace topology. In each case this is the coarsest topology making the cone maps continuous, so a map out of a test space is continuous exactly when its composites with the cone maps are, which is the universal property. The initial topology is the unique one making the cone maps continuous while giving the universal property, so here too the lift over the **Set** limit is unique: $U\colon \mathbf{Top} \to \mathbf{Set}$ creates these limits as well. By proposition 61.4.2 every category listed is complete with limits computed on underlying sets, completing the proof.

The proposition is what lets one compute in these categories at all: a limit is found by taking the limit of the underlying sets and then observing that the algebraic structure has nowhere else to go. The equalizer clause is the categorical form of a familiar move, that the locus where two homomorphisms agree is a subgroup. The kernel of a group or module homomorphism $f\colon G \to H$ is the special case where g is the trivial homomorphism (written $g = 0$ in the additive case), so such kernels are equalizers, and the general limit packages products with these agreement loci. In **Ring** the zero map is not a morphism, so a ring-map kernel is an ideal rather than an equalizer; the equalizer description here stays in the group and module setting.

The contrast the reader should carry forward is that colimits are not computed this way. The forgetful functor does not create colimits: the coproduct in **Grp** is the free product, whose underlying set is far larger than the disjoint union of the underlying sets.

Example 61.4.4: The Coproduct in Grp Is Not the Disjoint Union

In **Set** the coproduct is the disjoint union $X \sqcup Y$. In **Grp** the coproduct of G and H is the *free product* $G * H$, whose elements are the reduced words alternating between nontrivial elements of G and of H ([25], free products). Its universal property is the one required of a coproduct: a pair of homomorphisms $G \to K$ and $H \to K$ extends uniquely to $G * H \to K$. Take $G = H = \mathbb{Z}/2$, generated by a and b respectively. The disjoint union of the underlying sets has four elements, while $\mathbb{Z}/2 * \mathbb{Z}/2$ is infinite: it contains the distinct reduced words $a, b, ab, ba, aba, bab, \ldots$ and is isomorphic to the infinite dihedral group. So $U(G * H) \neq UG \sqcup UH$, and the forgetful functor does not preserve this coproduct. In **Ab** the coproduct of G and H is instead the direct sum $G \oplus H$, which for two groups agrees with the product, and its underlying set is $UG \times UH$, not $UG \sqcup UH$: even the abelian coproduct is not the set-level coproduct.

The lesson is that "compute on underlying sets" is a statement about limits, and that colimits in the algebraic categories must be built by their own universal property (free constructions, quotients, amalgamation) rather than read off the forgetful functor. Coequalizers, for instance, are quotients rather than subobjects.

The functor and presheaf categories behave better: there both limits and colimits are computed pointwise, one object of the indexing category at a time. This is what makes presheaf theory computable, and the reason presheaf categories inherit completeness and cocompleteness from **Set**.

Proposition 61.4.5: Pointwise Limits in Functor Categories

Let J be a small category and C a category, and let $[J, C]$ be the functor category (definition 59.5.4). Suppose C is complete. Then $[J, C]$ is complete, and limits are computed *pointwise*: for a diagram $F\colon I \to [J, C]$ indexed by a small category I, the limit $\lim_I F$ is the functor

$$\left(\lim_I F\right)(j) = \lim_I \big(F(-)(j)\big)$$

sending each object j of J to the limit in C of the diagram $i \mapsto F(i)(j)$. Dually, if C is cocomplete then $[J, C]$ is cocomplete, with colimits computed pointwise. In particular the presheaf category $\mathbf{PSh}(C) = [C^{\mathrm{op}}, \mathbf{Set}]$ (definition 60.3.1) is complete and cocomplete, and both are computed pointwise from **Set**.

***Proof*:**
Write $G_j = \lim_I \big(F(-)(j)\big)$ for the object of C obtained by taking, for a fixed j, the limit of the diagram $i \mapsto F(i)(j)$; it exists because C is complete. Let $\lambda_i^j\colon G_j \to F(i)(j)$ be the limit cone.

Step 1: The pointwise limits assemble into a functor. For a morphism $u\colon j \to j'$ in J and each i, the maps $F(i)(u)\colon F(i)(j) \to F(i)(j')$ are compatible with the diagram in I, since each $F(i)$ is a functor and the arrows of the diagram over I are natural transformations, so the family $\big(F(i)(u) \circ \lambda_i^j\big)_i$ is a cone over $i \mapsto F(i)(j')$. By the universal property of $G_{j'}$ there is a unique map $G(u)\colon G_j \to G_{j'}$ with $\lambda_i^{j'} \circ G(u) = F(i)(u) \circ \lambda_i^j$ for all i. Uniqueness makes G functorial: $G(1_j) = 1_{G_j}$ and $G(u'u) = G(u')G(u)$, because both sides satisfy the equations defining the induced map. So $G\colon J \to C$ is a functor, an object of $[J, C]$, and the maps $\lambda_i^\bullet\colon G \Rightarrow F(i)$ are natural transformations, a cone over F in $[J, C]$.

Step 2: This cone is a limit. Let H be a functor in $[J, C]$ with a cone $\mu_i\colon H \Rightarrow F(i)$. For each fixed j the components $(\mu_i)_j\colon H(j) \to F(i)(j)$ form a cone over $i \mapsto F(i)(j)$ in C, so there is a unique map $\varphi_j\colon H(j) \to G_j$ with $\lambda_i^j \circ \varphi_j = (\mu_i)_j$ for all i. The components φ_j are natural in j: for $u\colon j \to j'$, both $G(u)\varphi_j$ and $\varphi_{j'}H(u)$ are maps $H(j) \to G_{j'}$ that, after composing with each $\lambda_i^{j'}$, give $(\mu_i)_{j'}H(u) = F(i)(u)(\mu_i)_j$, so they agree by the uniqueness in the universal property of $G_{j'}$. Hence $\varphi\colon H \Rightarrow G$ is a natural transformation with $\lambda_i^\bullet \circ \varphi = \mu$, and it is the unique such because each φ_j was forced. So G with the cone $\lambda_i^\bullet$ is the limit of F in $[J, C]$. The colimit statement is the dual (box 59.3.3), and applying both to $C = \mathbf{Set}$, which is complete and cocomplete, gives the claim for $\mathbf{PSh}(C)$, completing the proof.

The proposition says that a (co)cone in $[J, C]$ is exactly an objectwise-compatible family of (co)cones in C: naturality is what glues the pointwise data into a morphism of functors, and the universal property is checked one object at a time. For presheaves this means every limit and colimit is computed in **Set** and then reassembled, so all the set-level formulas transfer verbatim. A product of presheaves is the objectwise product of sets, a colimit of presheaves is the objectwise colimit, and representable presheaves, while not closed under these operations, sit inside a category where every limit and colimit exists.

Having located limits in specific categories, the next results treat how limits behave under functors. Three conditions arrange the possibilities: a functor may carry limit cones to limit cones, may detect a limit from its image, or may lift limits from the target to the source. These are preservation, reflection, and creation.

Definition 61.4.6: Preserving, Reflecting, and Creating Limits

Let $G\colon C \to D$ be a functor and $F\colon J \to C$ a diagram.

1. G *preserves* the limit of F if, whenever $(\lim F, \lambda)$ is a limit cone in C, its image $(G(\lim F), G\lambda)$ is a limit cone over GF in D.
2. G *reflects* limits if a cone (L, λ) over F in C is a limit cone whenever its image $(GL, G\lambda)$ is a limit cone over GF in D.
3. G *creates* the limit of F if, given any limit cone (M, μ) over GF in D, there is a cone (L, λ) over F in C with $GL = M$ and $G\lambda = \mu$, this cone is a limit of F, and it is the unique cone in C (up to the induced isomorphism) lying over (M, μ).

The dual notions, for colimits and cocones, are named identically.

Creation is the strongest of the three and is what proposition 61.4.3 really established: the forgetful functor $U\colon \mathbf{Grp} \to \mathbf{Set}$ creates limits, because a limit cone downstairs in **Set** lifts to a unique cone of groups, and that cone is a limit. Creation implies preservation and reflection together. Preservation alone is the condition that appears most often, and the hom-functor is the instance developed next. A morphism into a limit is the same data as a compatible family of morphisms into the diagram, which is precisely what the hom-set of the limit computes.

Proposition 61.4.7: Representable Functors Preserve Limits

Let C be a locally small category and c an object. The covariant hom-functor $\mathrm{Hom}(c, -)\colon C \to \mathbf{Set}$ (definition 60.1.1) preserves all limits that exist in C: for any diagram $D\colon J \to C$ with a limit,

$$\mathrm{Hom}\big(c, \lim_J D\big) \;\cong\; \lim_J \mathrm{Hom}(c, D-)$$

naturally in c, where the right-hand limit is taken in **Set**. Dually, $\mathrm{Hom}(-, c)\colon C^{\mathrm{op}} \to \mathbf{Set}$ sends colimits in C to limits in **Set**.

***Proof*:**
Let $(\lim_J D, \lambda)$ be the limit cone in C, with legs $\lambda_j\colon \lim_J D \to Dj$. Apply $\mathrm{Hom}(c,-)$: the images $(\lambda_j)_*\colon \mathrm{Hom}(c, \lim_J D) \to \mathrm{Hom}(c, Dj)$ form a cone over $\mathrm{Hom}(c, D-)$ in **Set**, since post-composition is functorial and the λ_j commute with the diagram. It remains to show this cone is a limit in **Set**, that is, that $\mathrm{Hom}(c, \lim_J D) \to \lim_J \mathrm{Hom}(c, Dj)$ is a bijection.

Step 1: The target is the set of compatible families. A limit in **Set** is the set of compatible families over the diagram (the equalizer inside a product built in theorem 61.2.8), so an element of $\lim_J \mathrm{Hom}(c, Dj)$ is a family $(g_j)_j$ with $g_j \in \mathrm{Hom}(c, Dj)$ such that for every morphism $u\colon j \to j'$ in J one has $Du \circ g_j = g_{j'}$. That is exactly a cone over D with apex c: a family of maps $g_j\colon c \to Dj$ compatible with the diagram.

Step 2: Cones with apex c correspond to maps into the limit. The universal property of $\lim_J D$ says precisely that cones over D with apex c are in bijection with morphisms $c \to \lim_J D$: to each cone (g_j) there is a unique $g\colon c \to \lim_J D$ with $\lambda_j g = g_j$, and every $g\colon c \to \lim_J D$ yields the cone $(\lambda_j g)_j$. Under the identification of Step 1, this bijection is exactly the map $g \mapsto ((\lambda_j)_* g)_j = (\lambda_j g)_j$, so $(\lambda_j)_*\colon \mathrm{Hom}(c, \lim_J D) \to \lim_J \mathrm{Hom}(c, Dj)$ is a bijection. Hence the cone is a limit, and $\mathrm{Hom}(c,-)$ preserves the limit of D. Naturality in c is functoriality of the two-sided hom-bifunctor (definition 60.1.4): a morphism $c' \to c$ induces compatible pre-composition maps on both sides.

Step 3: The dual. Applying the statement in C^{op} (box 59.3.3) gives the second claim. A colimit in C is a limit in C^{op}, and $\mathrm{Hom}_C(-, c) = \mathrm{Hom}_{C^{\mathrm{op}}}(c, -)$; the preservation of C^{op}-limits by $\mathrm{Hom}_{C^{\mathrm{op}}}(c,-)$ reads in C as $\mathrm{Hom}_C(\mathrm{colim}_J D, c) \cong \lim_J \mathrm{Hom}_C(D-, c)$, so $\mathrm{Hom}(-,c)$ carries colimits to limits, completing the proof.

This is the categorical version of a fact used constantly: to map into a limit is to map compatibly into each stage, and to map out of a colimit is to map compatibly out of each stage. It is why a homomorphism into a product is a tuple of homomorphisms, and why a homomorphism out of a quotient or a pushout is a compatible pair. The result also reads from the Yoneda side. By the Yoneda lemma (theorem 60.2.1) an object is determined by the functor $\mathrm{Hom}(-,c)$ it represents, and that functor turns colimits into limits, while $\mathrm{Hom}(c,-)$ preserves limits. So an object's behaviour under limits is legible through its hom-functors, the precise sense in which Chapter 2's lens on objects extends to the constructions of this chapter.

The results combine into a single computation. In **Set** a pullback is the fibered product, and the hom-functor preservation says a map into it is a compatible pair of maps into its factors. This one construction computes intersections and preimages at once.

Example 61.4.8: Pullbacks in Set as Fibered Products
Let $f\colon X \to Z$ and $g\colon Y \to Z$ be functions. Their pullback (definition 61.2.6) in **Set** is the fibered product

$$X \times_Z Y = \{(x,y) \in X \times Y \,\big|\, f(x) = g(y)\}$$

with the two projections to X and Y. This is a limit, computed on the underlying sets exactly as proposition 61.4.3 prescribes: it is the equalizer of $f \circ \pi_X$ and $g \circ \pi_Y$ inside the product $X \times Y$, so it is the subset where the two composites agree.

Two special cases recover familiar operations.

1. *Intersection.* Let $A, B \subseteq Z$ be subsets, with inclusions $f\colon A \hookrightarrow Z$ and $g\colon B \hookrightarrow Z$. Then $f(a) = g(b)$ forces $a = b$ as elements of Z, so

$$A \times_Z B = \{(a, b) \in A \times B \big| a = b\} \cong A \cap B$$

   The pullback of two subobjects along their inclusions is their intersection.

2. *Preimage.* Let $g\colon Y \to Z$ be arbitrary and take $A \subseteq Z$ with inclusion $f\colon A \hookrightarrow Z$. Then $A \times_Z Y = \{(a, y) \big| a = g(y)\}$, and the projection to Y identifies it with

$$\{y \in Y \big| g(y) \in A\} = g^{-1}(A)$$

   the preimage. Pulling back a subobject along a map is taking its preimage.

The preservation result reads off the maps into $X \times_Z Y$: for a set W,

$$\mathrm{Hom}(W, X \times_Z Y) \cong \{(p, q) \big| p\colon W \to X,\ q\colon W \to Y,\ fp = gq\}$$

which is the equalizer of f_* and g_* inside $\mathrm{Hom}(W, X) \times \mathrm{Hom}(W, Y)$, precisely the pullback in **Set** of the induced hom-sets. So a map into the fibered product is a matching pair of maps into the factors, the concrete face of proposition 61.4.7.

62

Adjunctions

Chapters 1 through 3 built the vocabulary; adjunctions turn it into an organizing principle. An *adjunction* makes precise the sense in which one functor is the best approximate inverse of another: free constructions are adjoint to forgetful ones, colimits and limits to the diagonal, tensoring to hom. This chapter gives the definition in its three equivalent forms, works the examples, and proves the theorem every adjunction obeys, that right adjoints preserve limits, before asking when an adjoint exists at all.

62.1 Adjoint Functors

The previous chapter recognized free constructions, abelianization, and the initial ring as universal arrows, one for each object of a category (definition 61.1.4). When a functor supplies such an arrow at *every* object in a coherent way, the two functors involved stand in the relationship this section defines. We give the definition first through a natural isomorphism of hom-sets, then repackage it through a unit and a counit, and prove that these two presentations and the object-by-object universal-arrow condition are the same data. The free-monoid construction runs alongside as the guiding example.

Definition 62.1.1: Adjunction

Let $F\colon C \to D$ and $G\colon D \to C$ be functors. An *adjunction* from F to G is a natural isomorphism

$$\Phi\colon \mathrm{Hom}_D(F-,-) \cong \mathrm{Hom}_C(-,G-)$$

of functors $C^{\mathrm{op}} \times D \to \mathbf{Set}$; componentwise, for each object c of C and d of D a bijection

$$\Phi_{c,d}\colon \mathrm{Hom}_D(Fc,d) \xrightarrow{\cong} \mathrm{Hom}_C(c,Gd)$$

natural in c and in d. When such an adjunction exists we say F is *left adjoint* to G and G is *right adjoint* to F, written $F \dashv G$. For a morphism $f\colon Fc \to d$, the corresponding morphism $\Phi_{c,d}(f)\colon c \to Gd$ is its *transpose* (or *adjunct*), and likewise $\Phi^{-1}_{c,d}(g)$ is the transpose of $g\colon c \to Gd$.

Naturality is the content of the definition. It says that transposition commutes with composition on both sides: for $u\colon c' \to c$ in C and $v\colon d \to d'$ in D, the transpose of $v \circ f \circ Fu$ is $Gv \circ \Phi_{c,d}(f) \circ u$. Concretely, the naturality square in the C-variable,

$$\begin{tikzcd}
\mathrm{Hom}_D(Fc,d) \arrow[r, "\Phi_{c,d}"] \arrow[d, "(Fu)^*"'] & \mathrm{Hom}_C(c,Gd) \arrow[d, "u^*"] \\
\mathrm{Hom}_D(Fc',d) \arrow[r, "\Phi_{c',d}"'] & \mathrm{Hom}_C(c',Gd)
\end{tikzcd}$$

commutes, where $(Fu)^*$ is precomposition with Fu and u^* is precomposition with u; the analogous square in the D-variable uses postcomposition with v and with Gv. The intuition is compact: a map out of the free object Fc is the same datum as a map into the underlying object Gd. The bijection lets one pass between the world where c lives and the world where d lives without loss, and naturality guarantees the passage respects every morphism in sight.

A first structural consequence, stated here and proved in section 62.3, is that either functor determines the other up to isomorphism: if $F \dashv G$ and $F \dashv G'$ then $G \cong G'$, and dually for left adjoints. The reason is that Gd represents the functor $c \mapsto \mathrm{Hom}_D(Fc,d)$, and a representing object is unique up to unique isomorphism (corollary 60.4.1). An adjoint is thus not extra structure to be chosen but a property that F may or may not have, pinned down when it holds.

Example 62.1.2: The Free Monoid and Underlying Set

Let $U\colon \mathbf{Mon} \to \mathbf{Set}$ send a monoid to its underlying set and a homomorphism to its underlying function, and let $F\colon \mathbf{Set} \to \mathbf{Mon}$ send a set X to the free monoid FX of finite words in the alphabet X under concatenation, with the empty word as identity ([25]). A monoid homomorphism $FX \to M$ is determined by its values on the length-one words, which are the elements of X, and any assignment of those values extends uniquely to a homomorphism because a word is a product of letters. Restriction to length-one words and free extension are mutually inverse, so

$$\mathrm{Hom}_{\mathbf{Mon}}(FX,M) \cong \mathrm{Hom}_{\mathbf{Set}}(X,UM)$$

and the bijection is natural in X and M: precomposing a homomorphism with Ff for $f\colon Y \to X$ corresponds to precomposing the associated function with f, and postcomposing with a homomorphism $M \to N$ corresponds to postcomposing with its underlying function. Thus $F \dashv U$.

The two hom-sets in example 62.1.2 are compared across a change of category, and the bijection is exactly the statement that FX is free on X. Every "free object" theorem in algebra is an instance of this pattern, and the adjunction packages all of them at once. The transpose of the identity of FX under this bijection is the map $X \to UFX$ sending each element to its length-one word, the insertion of generators; the transpose of the identity of UM is a homomorphism $FUM \to M$ evaluating a word of elements of M to their product. These two distinguished maps are the general phenomenon isolated next.

Passing to hom-sets and back is often less convenient than working with two natural transformations, one in each category, from which the whole bijection can be reconstructed. Taking transposes of identity morphisms produces them.

Definition 62.1.3: Unit and Counit

Let $F \dashv G$ with natural isomorphism Φ as in definition 62.1.1. The *unit* of the adjunction is the natural transformation

$$\eta\colon \mathrm{id}_C \Rightarrow GF$$

whose component at c is $\eta_c = \Phi_{c,Fc}(\mathrm{id}_{Fc})\colon c \to GFc$, the transpose of id_{Fc}. The *counit* is the natural transformation

$$\varepsilon\colon FG \Rightarrow \mathrm{id}_D$$

whose component at d is $\varepsilon_d = \Phi^{-1}_{Gd,d}(\mathrm{id}_{Gd})\colon FGd \to d$, the transpose of id_{Gd}.

That η and ε are natural transformations follows from the naturality of Φ: naturality in c, applied to the identity, moves η_c along any morphism $c \to c'$ as a component of GF requires, and the counit argument is dual. The unit $\eta_c\colon c \to GFc$ is a universal arrow from c to G in the sense of definition 61.1.4, as the equivalence theorem below makes precise, and the counit $\varepsilon_d\colon FGd \to d$ is a universal arrow from F to d. In example 62.1.2 the unit is the insertion of generators $X \to UFX$ and the counit $FUM \to M$ multiplies out a word of monoid elements.

The unit and counit recover the full hom-set bijection. The transpose of a morphism is obtained by whiskering with the appropriate one and then applying the functor, as the next proposition records; the formulas are the computational core of everything that follows.

Proposition 62.1.4: Transposition via the Unit and Counit

Let $F \dashv G$ with unit η and counit ε as in definition 62.1.3. For every $f\colon Fc \to d$ in D and every $g\colon c \to Gd$ in C,

$$\Phi_{c,d}(f) = Gf \circ \eta_c\colon c \to Gd \tag{62.1}$$

and

$$\Phi^{-1}_{c,d}(g) = \varepsilon_d \circ Fg\colon Fc \to d \tag{62.2}$$

These two assignments are mutually inverse.

***Proof*:**
Step 1: The formula for Φ. Fix $f\colon Fc \to d$. Naturality of Φ in the D-variable, applied to $f\colon Fc \to d$ regarded as a morphism advancing the second argument, gives a commuting square

$$\begin{tikzcd}
\mathrm{Hom}_D(Fc, Fc) \arrow[r, "\Phi_{c,Fc}"] \arrow[d, "f_*"'] & \mathrm{Hom}_C(c, GFc) \arrow[d, "(Gf)_*"] \\
\mathrm{Hom}_D(Fc, d) \arrow[r, "\Phi_{c,d}"'] & \mathrm{Hom}_C(c, Gd)
\end{tikzcd}$$

where f_* is postcomposition with f and $(Gf)_*$ is postcomposition with Gf. Chasing id_{Fc} around the square: down then right gives $\Phi_{c,d}(f \circ \mathrm{id}_{Fc}) = \Phi_{c,d}(f)$, while right then down gives $Gf \circ \Phi_{c,Fc}(\mathrm{id}_{Fc}) = Gf \circ \eta_c$ by the definition of the unit. The two are equal, which is (62.1).

Step 2: The formula for Φ^{-1}. The dual computation uses naturality of Φ in the C-variable. Fix $g\colon c \to Gd$ and apply naturality to g advancing the first argument:

$$\begin{tikzcd}
\mathrm{Hom}_D(FGd, d) \arrow[r, "\Phi_{Gd,d}"] \arrow[d, "(Fg)^*"'] & \mathrm{Hom}_C(Gd, Gd) \arrow[d, "g^*"] \\
\mathrm{Hom}_D(Fc, d) \arrow[r, "\Phi_{c,d}"'] & \mathrm{Hom}_C(c, Gd)
\end{tikzcd}$$

where $(Fg)^*$ and g^* are precomposition with Fg and g. Chase $\varepsilon_d \in \mathrm{Hom}_D(FGd, d)$ around the square. Right then down gives $g^*\big(\Phi_{Gd,d}(\varepsilon_d)\big) = g^*(\mathrm{id}_{Gd}) = g$, since $\Phi_{Gd,d}(\varepsilon_d) = \mathrm{id}_{Gd}$ by definition 62.1.3. Down then right gives $\Phi_{c,d}\big(\varepsilon_d \circ Fg\big)$. Equating the two, $\Phi_{c,d}(\varepsilon_d \circ Fg) = g$, and applying $\Phi_{c,d}^{-1}$ yields $\Phi_{c,d}^{-1}(g) = \varepsilon_d \circ Fg$, which is (62.2).

Step 3: Mutual inverseness. Because $\Phi_{c,d}$ is a bijection with inverse $\Phi_{c,d}^{-1}$, the two assignments (62.1) and (62.2) are mutually inverse by construction. This can also be checked directly from the triangle identities established in theorem 62.1.5, and the two verifications agree, completing the proof.

Formulas (62.1) and (62.2) are how one computes with an adjunction in practice: to transpose a map out of Fc, apply G and precompose with the unit; to transpose back, apply F and postcompose with the counit. Since the two are inverse, feeding one into the other must return the original morphism, and unwinding what that requires of η and ε alone gives a characterization of adjunctions with no mention of hom-sets. That characterization is the equivalence theorem.

The theorem collects three descriptions of the same situation. The first is the hom-set isomorphism of definition 62.1.1. The second replaces it by the unit and counit subject to two equations, the *triangle identities*, which require only that transposing a map and transposing it back be the identity. The third is the object-by-object universal-arrow condition, tying the chapter back to definition 61.1.4. Their equivalence is what lets one build, recognize, and use adjunctions from whichever description is at hand.

Theorem 62.1.5: Equivalent Formulations of an Adjunction

Let $F\colon C \to D$ and $G\colon D \to C$ be functors. The following data are equivalent, each determining the others.

1. A natural isomorphism $\Phi\colon \mathrm{Hom}_D(F-,-) \cong \mathrm{Hom}_C(-,G-)$, that is, an adjunction $F \dashv G$.
2. Natural transformations $\eta\colon \mathrm{id}_C \Rightarrow GF$ and $\varepsilon\colon FG \Rightarrow \mathrm{id}_D$ satisfying the *triangle identities*
$$(G\varepsilon)(\eta G) = \mathrm{id}_G, \qquad (\varepsilon F)(F\eta) = \mathrm{id}_F \tag{62.3}$$
as natural transformations $G \Rightarrow G$ and $F \Rightarrow F$.
3. A natural transformation $\eta\colon \mathrm{id}_C \Rightarrow GF$ whose component $\eta_c\colon c \to GFc$ is a universal arrow from c to G for every object c of C (definition 61.1.4).

***Proof*:**
We prove (1) $\Rightarrow$ (2) $\Rightarrow$ (3) $\Rightarrow$ (1).

Step 1: From the hom-set isomorphism to the triangle identities. Assume (1), and define η and ε as in definition 62.1.3. They are natural transformations, as noted after that definition. The triangle identities are the componentwise statements

$$G\varepsilon_d \circ \eta_{Gd} = \mathrm{id}_{Gd}, \qquad \varepsilon_{Fc} \circ F\eta_c = \mathrm{id}_{Fc} \tag{62.4}$$

for all c and d. For the first, apply (62.1) of proposition 62.1.4 to the morphism $\varepsilon_d\colon FGd \to d$ with source object Gd: its transpose is $\Phi_{Gd,d}(\varepsilon_d) = G\varepsilon_d \circ \eta_{Gd}$. But $\varepsilon_d = \Phi^{-1}_{Gd,d}(\mathrm{id}_{Gd})$ by definition 62.1.3, so $\Phi_{Gd,d}(\varepsilon_d) = \mathrm{id}_{Gd}$, giving $G\varepsilon_d \circ \eta_{Gd} = \mathrm{id}_{Gd}$. For the second, apply (62.2) to $\eta_c\colon c \to GFc$ with target object Fc: its transpose is $\Phi^{-1}_{c,Fc}(\eta_c) = \varepsilon_{Fc} \circ F\eta_c$, and $\eta_c = \Phi_{c,Fc}(\mathrm{id}_{Fc})$, so this transpose is id_{Fc}. Thus (62.4) holds, which is (62.5).

Step 2: From the triangle identities to universal arrows. Assume (2). Fix an object c; we show $\eta_c\colon c \to GFc$ is a universal arrow from c to G. Given any object d of D and any morphism $g\colon c \to Gd$ in C, we must produce a unique $f\colon Fc \to d$ with $Gf \circ \eta_c = g$.

Existence. Set $f = \varepsilon_d \circ Fg\colon Fc \to d$. Then

$$Gf \circ \eta_c = G\varepsilon_d \circ GFg \circ \eta_c$$

Naturality of η applied to $g\colon c \to Gd$ gives $GFg \circ \eta_c = \eta_{Gd} \circ g$, so

$$Gf \circ \eta_c = G\varepsilon_d \circ \eta_{Gd} \circ g = (G\varepsilon_d \circ \eta_{Gd}) \circ g = \mathrm{id}_{Gd} \circ g = g$$

using the first triangle identity in (62.4). This f solves the factorization problem.

Uniqueness. Suppose $f'\colon Fc \to d$ also satisfies $Gf' \circ \eta_c = g$. Then

$$\varepsilon_d \circ F(Gf' \circ \eta_c) = \varepsilon_d \circ Fg$$

The left side equals $\varepsilon_d \circ FGf' \circ F\eta_c$. Naturality of ε applied to $f'\colon Fc \to d$ gives $\varepsilon_d \circ FGf' = f' \circ \varepsilon_{Fc}$, so the left side is $f' \circ \varepsilon_{Fc} \circ F\eta_c = f' \circ \mathrm{id}_{Fc} = f'$ by the second triangle identity. The right side is our chosen f. Hence $f' = f$, and the universal arrow is unique. Since c was arbitrary, η_c is universal for every c, which is (3).

Step 3: From universal arrows to the hom-set isomorphism. Assume (3): $\eta\colon \mathrm{id}_C \Rightarrow GF$ is a natural transformation whose every component $\eta_c\colon c \to GFc$ is a universal arrow from c to G. Fix c and d and define

$$\Phi_{c,d}\colon \mathrm{Hom}_D(Fc, d) \to \mathrm{Hom}_C(c, Gd), \qquad \Phi_{c,d}(f) = Gf \circ \eta_c$$

The universal property of η_c says exactly that for each $g\colon c \to Gd$ there is a unique $f\colon Fc \to d$ with $Gf \circ \eta_c = g$, that is, $\Phi_{c,d}$ is a bijection. It remains to check naturality in c and d. For $v\colon d \to d'$, both $\Phi_{c,d'}(v \circ f)$ and $Gv \circ \Phi_{c,d}(f)$ equal $Gv \circ Gf \circ \eta_c$, since G is a functor; this is naturality in d. For $u\colon c' \to c$, naturality of η gives $GFu \circ \eta_{c'} = \eta_c \circ u$, whence

$$\Phi_{c',d}(f \circ Fu) = G(f \circ Fu) \circ \eta_{c'} = Gf \circ GFu \circ \eta_{c'} = Gf \circ \eta_c \circ u = \Phi_{c,d}(f) \circ u$$

which is naturality in c. Thus Φ is a natural isomorphism and $F \dashv G$, returning to (1) and closing the cycle, completing the proof.

The theorem is the working definition of an adjunction: one verifies whichever of the three is easiest and gains the other two. Presentation (1) is the cleanest to state and the one that makes symmetry between F and G visible. Presentation (2) is the most convenient in proofs, since the triangle identities are two equations to check rather than a family of bijections, and it survives into contexts where hom-sets are not available. Presentation (3) is the bridge from the last chapter: an adjunction is precisely a functorial choice of universal arrow, one at each object, so the free monoid, abelianization, and the initial ring of section 61.1 were adjunctions seen one object at a time. There is a symmetric fourth form, dual to (3), asking that each counit component $\varepsilon_d\colon FGd \to d$ be a universal arrow from F to d; it follows by applying the theorem in the opposite categories, since $F \dashv G$ in C, D is $G^{\mathrm{op}} \dashv F^{\mathrm{op}}$ in $D^{\mathrm{op}}, C^{\mathrm{op}}$ (box 59.3.3).

The name "triangle identity" comes from the diagrams the equations in (62.4) assert commute. The first says the outer path in

$$\begin{tikzcd} Gd \arrow[r, "\eta_{Gd}"] \arrow[dr, "\mathrm{id}_{Gd}"'] & GFGd \arrow[d, "G\varepsilon_d"] \\ & Gd \end{tikzcd}$$

is the identity, and the second says the same of

$$\begin{tikzcd} Fc \arrow[r, "F\eta_c"] \arrow[dr, "\mathrm{id}_{Fc}"'] & FGFc \arrow[d, "\varepsilon_{Fc}"] \\ & Fc \end{tikzcd}$$

Read aloud, they say that inserting generators into Gd and then evaluating, or building the free object on Fc and then collapsing it, returns where one started. In example 62.1.2 the first identity says: take a set of generators inside a monoid M (the map η_{UM}), form the free monoid on them, multiply each word out (ε_M), and the underlying-set effect is the identity on UM. The equations look slight, but by theorem 62.1.5 they carry the entire adjunction.

62.2 Examples of Adjunctions

The definition and its three forms are in place; the work of this section is to show how much of the mathematics already met is an adjunction in disguise. Four families cover most of the recurring instances: free constructions adjoint to forgetful functors, tensoring adjoint to hom, colimits and limits adjoint to the diagonal, and Galois connections between posets. Each is presented as a hom-set bijection $\mathrm{Hom}_D(Fc, d) \cong \mathrm{Hom}_C(c, Gd)$ (definition 62.1.1), natural in both variables, with the naturality left to the one-line check it always reduces to. The through-line stated at the definition returns here in its standard form: left adjoints are free constructions.

The first family is the one the definition was built to capture. A forgetful functor discards structure; its left adjoint builds the freest structure on the bare data it was handed. The universal property of a free construction is exactly a hom-set bijection, so each free-forgetful pair is an adjunction with no extra argument to make.

Example 62.2.1: Free and Forgetful Adjunctions
Across the standing cast, each free construction is left adjoint to the forgetful functor it inverts.

1. *Free group $\dashv$ forgetful.* Let $U\colon \mathbf{Grp} \to \mathbf{Set}$ be the forgetful functor (example 59.4.2) and $F\colon \mathbf{Set} \to \mathbf{Grp}$ the free-group functor. The universal property of the free group FX says that a set function $X \to UG$ extends uniquely to a homomorphism $FX \to G$, and every homomorphism $FX \to G$ restricts to such a function on generators ([25]). These two operations are mutually inverse, giving a bijection
$$\mathrm{Hom}_{\mathbf{Grp}}(FX, G) \cong \mathrm{Hom}_{\mathbf{Set}}(X, UG)$$
so $F \dashv U$. Naturality in X and in G is the statement that restriction to generators commutes with precomposing a function on the left and postcomposing a homomorphism on the right, which holds because both sides are "evaluate on generators, then extend."

2. *Free R-module $\dashv$ forgetful.* For a ring R, let $U\colon \mathbf{Mod}_R \to \mathbf{Set}$ forget the module structure and F send a set X to the free module $R^{(X)}$ on X. The universal property of the free module ([25]) gives, for every module M,
$$\mathrm{Hom}_{\mathbf{Mod}_R}(R^{(X)}, M) \cong \mathrm{Hom}_{\mathbf{Set}}(X, UM)$$
a function $X \to UM$ extending uniquely to an R-linear map on the basis X, so $F \dashv U$. The same check applies with $\mathbf{Vect}_k$ in place of $\mathbf{Mod}_R$: every vector space is free, so the left adjoint here sends a set to the space with that basis.

3. *Abelianization $\dashv$ inclusion.* Let $\iota\colon \mathbf{Ab} \to \mathbf{Grp}$ be the inclusion of abelian groups among all groups and $\mathrm{ab}\colon \mathbf{Grp} \to \mathbf{Ab}$ send G to its abelianization $G^{\mathrm{ab}} = G/[G, G]$. A homomorphism from G to an abelian group A kills every commutator, hence factors uniquely through the quotient $G \to G^{\mathrm{ab}}$ ([25]), which is the bijection
$$\mathrm{Hom}_{\mathbf{Ab}}(G^{\mathrm{ab}}, A) \cong \mathrm{Hom}_{\mathbf{Grp}}(G, \iota A)$$
so $\mathrm{ab} \dashv \iota$. Here the "structure" the right adjoint remembers is commutativity, and the left adjoint imposes it as freely as possible, by quotienting out only what commutativity forces.

4. *Discrete $\dashv$ forgetful $\dashv$ indiscrete.* Let $U\colon \mathbf{Top} \to \mathbf{Set}$ forget the topology. It has adjoints on both sides. The *discrete* functor $\mathrm{disc}\colon \mathbf{Set} \to \mathbf{Top}$ equips a set with its discrete topology, in which every subset is open; the *indiscrete* functor indisc equips it with the indiscrete topology, whose only opens are $\varnothing$ and the whole set. Every function out of a discrete space is continuous and every function into an indiscrete space is continuous, so

$$\mathrm{Hom}_{\mathbf{Top}}(\mathrm{disc}\, X, Y) \cong \mathrm{Hom}_{\mathbf{Set}}(X, UY), \qquad \mathrm{Hom}_{\mathbf{Set}}(UY, X) \cong \mathrm{Hom}_{\mathbf{Top}}(Y, \mathrm{indisc}\, X)$$

The first bijection makes $\mathrm{disc} \dashv U$; the second makes $U \dashv \mathrm{indisc}$. Thus U is at once a right adjoint and a left adjoint, so it preserves both limits and colimits, a fact the next section will read off directly.

The first three items instantiate one pattern: the forgetful functor is a right adjoint and its left adjoint is the free construction, which is why free objects are recovered by the recipe "the value of the left adjoint on a set." The last item is different. A forgetful functor need not sit on only one side of an adjunction; when the structure being forgotten can be added in a freest and a cofreest way, it acquires adjoints on both. The point to carry forward is that a left adjoint is a free construction and a right adjoint is a forgetful or cofree one, with the hom-set bijection the precise content of "free."

The second family trades one bifunctor for two others. Fixing one input of a product and asking to map out of it is the same as mapping into a function object; this is currying, and in its module form it is the tensor-hom adjunction that pervades homological algebra. The prototype is in **Set**, where the function object is a plain set of functions.

Example 62.2.2: The Tensor-Hom Adjunction
Fix a set B. Currying converts a function of two variables into a function valued in functions: a map $A \times B \to C$ is the same as a map $A \to C^B$ sending a to the function $b \mapsto f(a,b)$, where $C^B = \mathrm{Hom}_{\mathbf{Set}}(B, C)$ is the set of functions $B \to C$. This assignment is a bijection, natural in A and C,

$$\mathrm{Hom}_{\mathbf{Set}}(A \times B, C) \cong \mathrm{Hom}_{\mathbf{Set}}(A, C^B)$$

so the functor $(-) \times B$ is left adjoint to $(-)^B = \mathrm{Hom}_{\mathbf{Set}}(B, -)$. This is the *cartesian closure* of **Set**: each product functor has the corresponding function-object functor as a right adjoint.

The module version replaces the product by the tensor product. Let R be a commutative ring and fix an R-module B. The defining property of the tensor product ([25]) is that R-bilinear maps $A \times B \to C$ correspond to R-linear maps $A \otimes_R B \to C$; currying the second variable identifies a bilinear map with an R-linear map $A \to \mathrm{Hom}_R(B, C)$, giving the bijection

$$\mathrm{Hom}_R(A \otimes_R B, C) \cong \mathrm{Hom}_R(A, \mathrm{Hom}_R(B, C))$$

natural in A and C. Thus $(-) \otimes_R B$ is left adjoint to $\mathrm{Hom}_R(B, -)$.

The tensor-hom adjunction is why tensoring and hom behave as complementary halves of the same structure: the left adjoint $(-) \otimes_R B$ and the right adjoint $\mathrm{Hom}_R(B, -)$ are tied by a natural bijection, and the preservation theorem of the next section will explain from this alone why tensoring is right exact and hom left exact. The **Set** prototype is the same phenomenon with $\times$ for $\otimes$ and the bare function set for the internal hom; a category carrying such an adjunction for each fixed object is called closed, a notion the monoidal chapter returns to.

The third family exhibits limits and colimits themselves as adjoints. A single diagram-valued functor, the diagonal, has the limit on one side and the colimit on the other, so the two constructions of the previous chapter are the two adjoints of one functor.

Recall the diagonal functor $\Delta\colon \mathbf{C} \to [J, \mathbf{C}]$ sending an object X to the constant diagram at X, the diagram taking every object of J to X and every morphism to 1_X (section 61.2). A natural transformation $\Delta X \Rightarrow D$ is exactly a cone over D with apex X, and a natural transformation $D \Rightarrow \Delta X$ is exactly a cocone under D with nadir X. This identification turns the universal properties of the limit and the colimit into hom-set bijections.

Example 62.2.3: Limits and Colimits as Adjoints to the Diagonal
Let J be a small category and suppose $\mathbf{C}$ has all limits and all colimits of shape J, so that $\lim\colon [J, \mathbf{C}] \to \mathbf{C}$ and $\operatorname{colim}\colon [J, \mathbf{C}] \to \mathbf{C}$ are defined. Then

$$\operatorname{colim} \;\dashv\; \Delta \;\dashv\; \lim$$

the colimit is left adjoint to the diagonal and the limit is right adjoint to it.

The two adjunctions are dual, and each is immediate once the naturality bijection is unwound. Only the $\Delta \dashv \lim$ side is proved; the $\operatorname{colim} \dashv \Delta$ side is the same argument in $\mathbf{C}^{\mathrm{op}}$ by the duality principle (box 59.3.3), since a colimit in $\mathbf{C}$ is a limit in $\mathbf{C}^{\mathrm{op}}$ (proposition 61.3.2) and Δ is self-opposite.

***Proof*:**
Fix an object X of $\mathbf{C}$ and a diagram D in $[J, \mathbf{C}]$. A morphism in the functor category $\mathrm{Hom}_{[J,\mathbf{C}]}(\Delta X, D)$ is a natural transformation $\alpha\colon \Delta X \Rightarrow D$ (definition 59.5.4): a family of components $\alpha_j\colon X \to Dj$, one for each object j of J, such that for every morphism $u\colon j \to j'$ of J the naturality square commutes. Since ΔX sends u to 1_X, that square reads $Du \circ \alpha_j = \alpha_{j'} \circ 1_X = \alpha_{j'}$, which is exactly the cone condition (definition 61.2.1). A natural transformation $\Delta X \Rightarrow D$ is therefore the same data as a cone over D with apex X.

By the universal property of the limit, a cone over D with apex X corresponds to a unique morphism $X \to \lim D$ commuting with the projections, and every morphism $X \to \lim D$ arises this way by composing with the limit cone. This correspondence is a bijection

$$\mathrm{Hom}_{\mathbf{C}}(X, \lim D) \cong \mathrm{Hom}_{[J,\mathbf{C}]}(\Delta X, D)$$

It is natural in X: precomposing a morphism $X' \to X$ transports a cone with apex X to one with apex X' by the same precomposition on each component, and the induced map to $\lim D$ transforms accordingly. It is natural in D: a morphism of diagrams $D \Rightarrow D'$ postcomposes each cone component and the induced maps to the limits commute by the universal property. The bijection, natural in both variables, is the defining data of an adjunction (definition 62.1.1), so $\Delta \dashv \lim$. The dual identity $\operatorname{colim} \dashv \Delta$ follows by applying this to $\mathbf{C}^{\mathrm{op}}$, completing the proof.

This recovers the limits and colimits of the previous chapter as the two adjoints of a single functor, and it explains a fact used repeatedly downstream: the limit functor is a right adjoint, so it preserves limits. The general preservation theorem is deferred to the next section, but the shape of the consequence is already visible, that a limit of limits may be computed in either order because the outer limit is a right adjoint. Dually the colimit functor is a left adjoint and preserves colimits. The diagonal, sitting between its two adjoints, is the categorical device that makes “take the limit” and “take the colimit” into functors with formal properties rather than case-by-case constructions.

The last family lives one dimension down, where the categories are posets and a functor is a monotone map. An adjunction between two posets is a pair of monotone maps tied by the single inequality that a hom-set, being either empty or a point, can express.

A poset $(P, \leq)$ is a category with one morphism $p \to q$ when $p \leq q$ and none otherwise (example 59.4.2 treats such thin categories). A functor between posets is a monotone map, and a hom-set $P(p, q)$ has at most one element, present exactly when $p \leq q$. The hom-set bijection defining an adjunction therefore collapses to a biconditional between inequalities.

Definition 62.2.4: Galois Connection

Let P and Q be posets, viewed as categories. A *(monotone) Galois connection* between P and Q is an adjunction $f \dashv g$ between them: a pair of monotone maps $f : P \to Q$ and $g : Q \to P$ such that for all $p \in P$ and $q \in Q$,

$$f(p) \leq q \quad \Longleftrightarrow \quad p \leq g(q)$$

The map f is the *left adjoint* (or lower adjoint) and g the *right adjoint* (or upper adjoint).

The biconditional is the poset case of the natural bijection $\mathrm{Hom}_Q(fp, q) \cong \mathrm{Hom}_P(p, gq)$: each hom-set is either empty or a single point, so the bijection between them is precisely the equivalence of the two inequalities, and naturality is automatic because there is nothing to compare beyond presence or absence of a morphism. A Galois connection is an adjunction with the analytic content stripped away, leaving the order-theoretic skeleton visible.

A single monotone connection already produces a closure operator, and the classical Galois correspondence is the archetype once one factor is passed to its opposite.

Example 62.2.5: The Galois Correspondence

Let L/k be a finite Galois extension with Galois group $G = \mathrm{Gal}(L/k)$. Write $\mathcal{S}$ for the poset of subgroups of G ordered by inclusion and $\mathcal{F}$ for the poset of intermediate fields $k \subseteq E \subseteq L$ ordered by inclusion. The two maps

$$\mathrm{Fix} : \mathcal{S} \to \mathcal{F}, \quad H \mapsto L^H, \qquad \mathrm{Gal} : \mathcal{F} \to \mathcal{S}, \quad E \mapsto \mathrm{Gal}(L/E)$$

each reverse inclusions: a larger subgroup fixes a smaller field, and a larger field is fixed by a smaller subgroup ([25]). They satisfy

$$H \subseteq \mathrm{Gal}(L/E) \quad \Longleftrightarrow \quad E \subseteq L^H$$

both sides asserting that every element of H fixes every element of E. Because each map reverses order, the pair is an *antitone* Galois connection, which is a monotone one after replacing $\mathcal{F}$ by its opposite $\mathcal{F}^{\mathrm{op}}$: then $\mathrm{Fix} : \mathcal{S} \to \mathcal{F}^{\mathrm{op}}$ and $\mathrm{Gal} : \mathcal{F}^{\mathrm{op}} \to \mathcal{S}$ are monotone and the biconditional above is definition D.1.11. The fundamental theorem of Galois theory ([25]) states that here both composites are the identity, so the connection is an order-reversing bijection between subgroups and intermediate fields.

The example carries two lessons about Galois connections in general. First, an antitone connection is an adjunction only after an opposite category is introduced; the raw pair of order-reversing maps becomes a genuine $f \dashv g$ once one side is flipped, which is the categorical account of why order-reversing correspondences are as natural as order-preserving ones. Second, even when the composites are not the identity, a Galois connection $f \dashv g$ always yields a *closure operator* $gf\colon P \to P$: it is monotone, satisfies $p \le gf(p)$, and is idempotent, $gfgf = gf$. The Galois correspondence is the degenerate case where the closure is trivial because gf is the identity; in a general connection the closure operator gf records the "saturation" a subset acquires, exactly as the double annihilator $(-)^{\perp\perp}$ closes a subspace W to $W^{\perp\perp}$. Every adjunction produces such a monad on the domain of its left adjoint, of which the closure operator is the poset instance, a thread the chapter on monads develops.

62.3 Units, Counits, and Preservation

The three formulations of an adjunction are on the table (theorem 62.1.5), and the examples of the previous section show how often one is in play. This section pays off the unit/counit formulation. The unit and counit are tied together by two equations, the triangle identities, and those equations are exactly the data that lets one reconstruct the hom-set bijection. Once the identities are in hand, two structural facts follow with almost no work: an adjoint, when it exists, is determined up to a unique natural isomorphism, and a right adjoint carries every limit that exists to a limit. The second of these, right adjoints preserve limits, explains at one stroke why so many limits in algebra and topology are computed on underlying sets.

Throughout, $F \dashv G$ is an adjunction with $F\colon \mathbf{C} \to \mathbf{D}$ and $G\colon \mathbf{D} \to \mathbf{C}$, unit $\eta\colon \mathrm{id}_{\mathbf{C}} \Rightarrow GF$ and counit $\varepsilon\colon FG \Rightarrow \mathrm{id}_{\mathbf{D}}$, in the notation of definition 62.1.3.

The bijection $\mathrm{Hom}_{\mathbf{D}}(Fc, d) \cong \mathrm{Hom}_{\mathbf{C}}(c, Gd)$ moves a morphism across the adjunction; call the image of $g\colon Fc \to d$ its *transpose* $g^\flat\colon c \to Gd$, and the image of $f\colon c \to Gd$ its transpose $f^\sharp\colon Fc \to d$. The two operations are mutually inverse, so transposing twice returns the original morphism. In terms of unit and counit, the transpose of $g\colon Fc \to d$ is $Gg \circ \eta_c$ and the transpose of $f\colon c \to Gd$ is $\varepsilon_d \circ Ff$; this is how theorem 62.1.5 recovered the bijection from the two natural transformations. The content of the next theorem is that the round trip really is the identity, and that this fact is captured by two equations relating η and ε.

Theorem 62.3.1: Triangle Identities

Let $F \dashv G$ with unit $\eta\colon \mathrm{id}_{\mathbf{C}} \Rightarrow GF$ and counit $\varepsilon\colon FG \Rightarrow \mathrm{id}_{\mathbf{D}}$. Then the two whiskered composites

$$G \overset{\eta G}{\Longrightarrow} GFG \overset{G\varepsilon}{\Longrightarrow} G, \qquad F \overset{F\eta}{\Longrightarrow} FGF \overset{\varepsilon F}{\Longrightarrow} F$$

are the respective identity natural transformations:

$$(G\varepsilon)(\eta G) = \mathrm{id}_G, \qquad (\varepsilon F)(F\eta) = \mathrm{id}_F \tag{62.5}$$

Conversely, given functors F, G and natural transformations $\eta\colon \mathrm{id}_{\mathbf{C}} \Rightarrow GF$, $\varepsilon\colon FG \Rightarrow \mathrm{id}_{\mathbf{D}}$ satisfying (62.5), the assignment $g \mapsto Gg \circ \eta_c$ is a bijection $\mathrm{Hom}_{\mathbf{D}}(Fc, d) \cong \mathrm{Hom}_{\mathbf{C}}(c, Gd)$ natural in c and d, so η and ε present an adjunction $F \dashv G$.

The two equations (62.5) are named for the shape of the diagrams that express them. Reading the whiskered composites ηG, $G\varepsilon$ and $F\eta$, εF as edges, each identity says that a triangle commutes:

$$\begin{tikzcd} G \arrow[r, Rightarrow, "\eta G"] \arrow[dr, Rightarrow, "\mathrm{id}_G"'] & GFG \arrow[d, Rightarrow, "G\varepsilon"] \\ & G \end{tikzcd} \qquad \begin{tikzcd} F \arrow[r, Rightarrow, "F\eta"] \arrow[dr, Rightarrow, "\mathrm{id}_F"'] & FGF \arrow[d, Rightarrow, "\varepsilon F"] \\ & F \end{tikzcd}$$

Componentwise at objects $d \in \mathbf{D}$ and $c \in \mathbf{C}$, these read $G\varepsilon_d \circ \eta_{Gd} = \mathrm{id}_{Gd}$ and $\varepsilon_{Fc} \circ F\eta_c = \mathrm{id}_{Fc}$.

***Proof*:**
The forward direction is the statement that transposing twice returns the original morphism, read off at the two extreme objects.

Step 1: The identities are the two double-transpose calculations. Fix $c \in \mathbf{C}$. The morphism $\eta_c \colon c \to GFc$ is the transpose of id_{Fc}, by the description of the transpose in theorem 62.1.5. Transposing back sends $f \colon c \to GFc$ to $\varepsilon_{Fc} \circ Ff$; applied to $f = \eta_c$ this gives $\varepsilon_{Fc} \circ F\eta_c$. Since transposing twice is the identity, $\varepsilon_{Fc} \circ F\eta_c = \mathrm{id}_{Fc}$. As c was arbitrary, $(\varepsilon F)(F\eta) = \mathrm{id}_F$. Dually, fix $d \in \mathbf{D}$: the morphism $\varepsilon_d \colon FGd \to d$ is the transpose of id_{Gd}, and transposing forward sends $g \colon FGd \to d$ to $Gg \circ \eta_{Gd}$; applied to $g = \varepsilon_d$ this gives $G\varepsilon_d \circ \eta_{Gd}$, equal to id_{Gd} because transposing twice is the identity. As d was arbitrary, $(G\varepsilon)(\eta G) = \mathrm{id}_G$.

Step 2: The converse. Assume now only that η, ε are natural transformations satisfying (62.5). Define

$$\Phi_{c,d} \colon \mathrm{Hom}_{\mathbf{D}}(Fc, d) \to \mathrm{Hom}_{\mathbf{C}}(c, Gd), \qquad g \mapsto Gg \circ \eta_c$$

and

$$\Psi_{c,d} \colon \mathrm{Hom}_{\mathbf{C}}(c, Gd) \to \mathrm{Hom}_{\mathbf{D}}(Fc, d), \qquad f \mapsto \varepsilon_d \circ Ff$$

Both are natural in c and d. For Φ, naturality in d is functoriality of G: for $k \colon d \to d'$, $\Phi_{c,d'}(kg) = G(kg)\,\eta_c = Gk\,(Gg\,\eta_c) = Gk \circ \Phi_{c,d}(g)$; naturality in c is naturality of η: for $h \colon c' \to c$, $\Phi_{c',d}(g\,Fh) = G(g\,Fh)\,\eta_{c'} = Gg\,(GFh\,\eta_{c'}) = Gg\,(\eta_c\,h) = \Phi_{c,d}(g) \circ h$, the middle equality by the naturality square of η at h. The naturality of Ψ is dual, using functoriality of F and naturality of ε. It remains to check that Φ and Ψ are mutually inverse, and here the triangle identities enter.

Take $g \colon Fc \to d$. Then

$$\Psi_{c,d}(\Phi_{c,d}(g)) = \varepsilon_d \circ F(Gg \circ \eta_c) = \varepsilon_d \circ FGg \circ F\eta_c$$

Naturality of ε applied to $g \colon Fc \to d$ gives $\varepsilon_d \circ FGg = g \circ \varepsilon_{Fc}$, so the right side equals $g \circ \varepsilon_{Fc} \circ F\eta_c$. By the second identity in (62.5), $\varepsilon_{Fc} \circ F\eta_c = \mathrm{id}_{Fc}$, hence $\Psi_{c,d}(\Phi_{c,d}(g)) = g$.

Take $f \colon c \to Gd$. Then

$$\Phi_{c,d}(\Psi_{c,d}(f)) = G(\varepsilon_d \circ Ff) \circ \eta_c = G\varepsilon_d \circ GFf \circ \eta_c$$

Naturality of η applied to $f \colon c \to Gd$ gives $GFf \circ \eta_c = \eta_{Gd} \circ f$, so the right side equals $G\varepsilon_d \circ \eta_{Gd} \circ f$. By the first identity in (62.5), $G\varepsilon_d \circ \eta_{Gd} = \mathrm{id}_{Gd}$, hence $\Phi_{c,d}(\Psi_{c,d}(f)) = f$.

Thus Φ is a natural bijection with inverse Ψ, which is the hom-set formulation of $F \dashv G$ (definition 62.1.1), completing the proof.

The identities encode a single fact about the two transpose operations: they are inverse. Everything about how F and G interact through the adjunction, including the naturality of the correspondence, is compressed into these two equations. This makes the unit/counit formulation the computational one, since checking an adjunction reduces to producing η, ε and verifying two triangle equations, with no bijection to construct by hand.

Example 62.3.2: The Free Group Triangle Identities
Let $F\colon \mathbf{Set} \to \mathbf{Grp}$ be the free-group functor and $U\colon \mathbf{Grp} \to \mathbf{Set}$ the forgetful functor, $F \dashv U$ (example 62.1.2). The unit $\eta_S\colon S \to UFS$ includes a set S as the generators of the free group FS. The counit $\varepsilon_G\colon FUG \to G$ is the homomorphism from the free group on the underlying set of G that sends each generator g to the element $g \in G$; it evaluates a formal word in the elements of G as an actual product in G.

The first identity $(U\varepsilon)(\eta U) = \mathrm{id}_U$ reads, at a group G, as follows. Start with an element $g \in UG$. The unit η_{UG} views g as a generator of the free group FUG, that is, as the length-one word g. Then $U\varepsilon_G$ evaluates that word in G, returning the element g. The composite is the identity on UG, as the identity requires.

The second identity $(\varepsilon F)(F\eta) = \mathrm{id}_F$ reads, at a set S, as follows. A reduced word $w = s_1^{\pm 1} \cdots s_n^{\pm 1}$ in FS is sent by $F\eta_S$ to the same formal word, now regarded as living in $FUFS$ (each generator s_i first viewed as a generator of FS, then as an element of UFS, then as a generator of $FUFS$). Applying ε_{FS} evaluates that word back in FS, recovering w. The composite is the identity on FS. The two identities are the two ways "generator" and "evaluate" cancel.

An adjunction determines its right adjoint completely: if a functor has a left adjoint at all, the resulting right adjoint is fixed up to a unique natural isomorphism. This is what licenses the article "the" in "the right adjoint". The proof is a direct application of the Yoneda embedding (definition 60.3.1): a functor value Gd is pinned down by the functor it represents, $\mathrm{Hom}_{\mathbf{C}}(-, Gd)$, and the adjunction identifies that representable with one built from F alone.

Proposition 62.3.3: Uniqueness of Adjoints

Let $F\colon \mathbf{C} \to \mathbf{D}$. If $F \dashv G$ and $F \dashv G'$, then there is a unique natural isomorphism $\theta\colon G \Rightarrow G'$ compatible with the adjunctions, in the sense that it identifies the two transpose bijections. Dually, if $G \dashv F$ and $G' \dashv F$, the two left adjoints G, G' are isomorphic by a unique such natural isomorphism.

***Proof*:**
Fix $d \in \mathbf{D}$. The two adjunctions give, for every $c \in \mathbf{C}$, bijections

$$\mathrm{Hom}_{\mathbf{C}}(c, Gd) \cong \mathrm{Hom}_{\mathbf{D}}(Fc, d) \cong \mathrm{Hom}_{\mathbf{C}}(c, G'd)$$

natural in c, the first from $F \dashv G$ and the second from $F \dashv G'$. Their composite is a natural isomorphism

$$\mathrm{Hom}_{\mathbf{C}}(-, Gd) \cong \mathrm{Hom}_{\mathbf{C}}(-, G'd)$$

of functors $\mathbf{C}^{\mathrm{op}} \to \mathbf{Set}$. The Yoneda embedding $\mathbf{y}\colon \mathbf{C} \to \mathbf{PSh}(\mathbf{C})$ is fully faithful (definition 60.3.1), so this isomorphism of representables is $\mathbf{y}(\theta_d)$ for a unique isomorphism $\theta_d\colon Gd \to G'd$ in $\mathbf{C}$.

It remains to see that the θ_d assemble into a natural transformation $\theta\colon G \Rightarrow G'$. Let $k\colon d \to d'$ in $\mathbf{D}$. Both composite bijections are natural in d as well as in c, since the transpose correspondences are natural in both variables (definition 62.1.1); consequently the square of representables

$$\begin{tikzcd} \mathrm{Hom}_{\mathbf{C}}(-, Gd) \arrow[r, "\cong"] \arrow[d, "\mathrm{Hom}(-,Gk)"'] & \mathrm{Hom}_{\mathbf{C}}(-, G'd) \arrow[d, "\mathrm{Hom}(-,G'k)"] \\ \mathrm{Hom}_{\mathbf{C}}(-, Gd') \arrow[r, "\cong"] & \mathrm{Hom}_{\mathbf{C}}(-, G'd') \end{tikzcd}$$

commutes. Applying full faithfulness of $\mathbf{y}$ to this commuting square of representables yields $\theta_{d'} \circ Gk = G'k \circ \theta_d$, which is naturality of θ. Uniqueness of each θ_d, hence of θ, is the injectivity of $\mathbf{y}$ on morphisms. The compatibility with the transpose bijections holds by construction, since θ was read off from them.

For the dual statement, if $G \dashv F$ and $G' \dashv F$, then for every c the bijections $\mathrm{Hom}_{\mathbf{D}}(Gc, -) \cong \mathrm{Hom}_{\mathbf{C}}(c, F-) \cong \mathrm{Hom}_{\mathbf{D}}(G'c, -)$ are natural, so the covariant representables $\mathrm{Hom}_{\mathbf{D}}(Gc, -)$ and $\mathrm{Hom}_{\mathbf{D}}(G'c, -)$ are isomorphic; the covariant Yoneda embedding (definition 60.3.1, applied in $\mathbf{D}^{\mathrm{op}}$) delivers a unique natural isomorphism $Gc \cong G'c$ by the same argument, as we sought to show.

Two constructions each solving the same universal problem cannot differ in any observable way, and the isomorphism between them is forced rather than chosen. This is the categorical form of a phenomenon seen repeatedly in chapter 62: the free group on a set, the tensor product of two modules, the coproduct of a family of objects are each unique up to unique isomorphism, and here the reason is uniform, since all three are adjoint (or universal) constructions and Yoneda pins down what an adjoint represents. From now on we speak of *the* left adjoint and *the* right adjoint of a functor, when one exists.

The structural payoff of the section is that right adjoints preserve limits. The mechanism is the same Yoneda argument, now run against a diagram: a right adjoint commutes with a limit because its defining bijection turns the limit-detecting property of Gd into the limit-detecting property inherited through F from representables, which already preserve limits (proposition 61.4.7). The abbreviation RAPL, "right adjoints preserve limits", is standard.

Recall from definition 61.4.6 what preservation means: a functor $G\colon \mathbf{D} \to \mathbf{C}$ *preserves* the limit of $D\colon \mathbf{J} \to \mathbf{D}$ if, whenever $\lim D$ exists with limit cone $(\lambda_j\colon \lim D \to Dj)_j$, the image cone $(G\lambda_j\colon G(\lim D) \to G(Dj))_j$ is a limit cone for $G \circ D$, so that $G(\lim D) \cong \lim(G \circ D)$ canonically.

Theorem 62.3.4: Right Adjoints Preserve Limits

Let $F \dashv G$ with $G\colon \mathbf{D} \to \mathbf{C}$, and let $D\colon \mathbf{J} \to \mathbf{D}$ be a diagram whose limit $\lim D$ exists in $\mathbf{D}$. Then G preserves this limit: the canonical comparison morphism $G(\lim D) \to \lim(G \circ D)$ is an isomorphism, so

$$G(\lim D) \cong \lim_{j \in \mathbf{J}} G(Dj)$$

Dually, a left adjoint preserves all colimits that exist in its domain.

***Proof*:**
Write $L = \lim D$ in $\mathbf{D}$. The claim is that GL, with the cone $(G\lambda_j)_j$, is a limit of $G \circ D$ in $\mathbf{C}$. By the universal property of a limit (definition 61.2.1) this is equivalent to the statement that

for every $c \in \mathbf{C}$ the map

$$\mathrm{Hom}_{\mathbf{C}}(c, GL) \to \lim_j \mathrm{Hom}_{\mathbf{C}}(c, G(Dj)), \qquad f \mapsto (G\lambda_j \circ f)_j$$

is a bijection, where the limit on the right is the limit in **Set** of the diagram $j \mapsto \mathrm{Hom}_{\mathbf{C}}(c, G(Dj))$, that is, the set of cones over $G \circ D$ with apex c. Fix c and compute this hom-set through a chain of bijections, each natural in c.

Step 1: Cross the adjunction. The adjunction $F \dashv G$ gives

$$\mathrm{Hom}_{\mathbf{C}}(c, GL) \cong \mathrm{Hom}_{\mathbf{D}}(Fc, L)$$

natural in c (definition 62.1.1).

Step 2: Use that representables preserve limits. Since $L = \lim D$ in $\mathbf{D}$, the representable $\mathrm{Hom}_{\mathbf{D}}(Fc, -)$ preserves it (proposition 61.4.7):

$$\mathrm{Hom}_{\mathbf{D}}(Fc, L) \cong \lim_j \mathrm{Hom}_{\mathbf{D}}(Fc, Dj)$$

This is the step where the hypothesis that $\lim D$ exists is used: the limit-preservation of G is pulled back from the limit-preservation of a representable.

Step 3: Cross the adjunction back, inside the limit. Applying $F \dashv G$ objectwise in j, and using that these bijections are natural in j so they assemble into an isomorphism of the diagrams whose limits are being taken,

$$\lim_j \mathrm{Hom}_{\mathbf{D}}(Fc, Dj) \cong \lim_j \mathrm{Hom}_{\mathbf{C}}(c, G(Dj))$$

Step 4: Reassemble. Composing the three bijections gives an isomorphism

$$\mathrm{Hom}_{\mathbf{C}}(c, GL) \cong \lim_j \mathrm{Hom}_{\mathbf{C}}(c, G(Dj))$$

natural in c. Tracing an element through the chain shows the composite is $f \mapsto (G\lambda_j \circ f)_j$, the comparison map above. By the Yoneda embedding (definition 60.3.1) a natural isomorphism of the representables $\mathrm{Hom}_{\mathbf{C}}(-, GL)$ and $\mathrm{Hom}_{\mathbf{C}}(-, \lim(G \circ D))$ comes from a unique isomorphism $GL \cong \lim(G \circ D)$, and by naturality this isomorphism is the comparison morphism. Hence G preserves the limit.

The dual statement follows from the duality principle (box 59.3.3): a left adjoint $F\colon \mathbf{C} \to \mathbf{D}$ is $F \dashv G$, so $F^{\mathrm{op}}\colon \mathbf{C}^{\mathrm{op}} \to \mathbf{D}^{\mathrm{op}}$ is a right adjoint (its left adjoint is G^{op}, as one checks by dualizing the hom-set bijection), and a colimit in $\mathbf{C}$ is a limit in $\mathbf{C}^{\mathrm{op}}$. The result just proved, applied to F^{op}, says F^{op} preserves limits in $\mathbf{C}^{\mathrm{op}}$, which is exactly the statement that F preserves colimits in $\mathbf{C}$, completing the proof.

RAPL is a source of both computations and obstructions. On the computational side it says a right adjoint may be evaluated on a limit by evaluating it factorwise: G commutes with products, pullbacks, equalizers, and all other limits. On the obstruction side it gives a fast way to prove a functor is *not* a right adjoint, by exhibiting one limit it fails to preserve. The tensor product $M \otimes_R -$ does not preserve limits (it fails to preserve infinite products in general), so it cannot be a right adjoint; and indeed it is a left adjoint, adjoint to $\mathrm{Hom}_R(M, -)$, and preserves colimits instead (example 62.2.2). A functor that preserves neither all limits nor all colimits can be neither a left nor a right adjoint, which is one reason adjointness is a strong condition.

The corollary the reader has already met is that a forgetful functor which happens to be a right adjoint computes limits on underlying sets.

Corollary 62.3.5: Limits on Underlying Sets

Let $U\colon \mathbf{D} \to \mathbf{Set}$ be a forgetful functor that has a left adjoint (a free functor $F \dashv U$). Then U preserves all limits that exist in $\mathbf{D}$: the underlying set of a limit in $\mathbf{D}$ is the corresponding limit of underlying sets, with its structure determined by that of the diagram.

***Proof*:**
Immediate from theorem 62.3.4 applied to $F \dashv U$, since U is the right adjoint, as required.

This recovers, and now explains, the pattern computed directly when limits were introduced (definition 61.2.1): the underlying set of a product of groups is the product of the underlying sets with coordinatewise multiplication, the underlying set of an equalizer is the equalizer of underlying sets, and the limit of a diagram of modules is computed on underlying sets. Each forgetful functor in question, from **Grp**, **Ab**, **Ring**, $\mathbf{Mod}_R$, or **Top** to **Set**, is a right adjoint (example 62.2.1), so RAPL forces its compatibility with limits, in the sense of preservation from definition 61.4.6.

Example 62.3.6: Products and Kernels of Abelian Groups
The forgetful functor $U\colon \mathbf{Ab} \to \mathbf{Set}$ has the free-abelian-group functor as left adjoint, so U preserves limits. Two instances make this concrete.

A product $\prod_i A_i$ in **Ab** has underlying set $\prod_i UA_i$, the cartesian product of the underlying sets, because U preserves the product. The group structure is coordinatewise, which is precisely the structure the limit universal property forces once the underlying set is fixed as the product.

An equalizer of two homomorphisms $f, g\colon A \to B$ is the subgroup $\{a \in A \big| f(a) = g(a)\}$; its underlying set is $\{x \in UA \big| Uf(x) = Ug(x)\}$, the equalizer of Uf, Ug in **Set**, again because U preserves the equalizer. Taking $g = 0$ specializes this to the kernel, whose underlying set is $\{a \in UA \big| Uf(a) = 0\}$. Both computations are consequences of RAPL, not separate verifications.

62.4 Adjoint Functor Theorems

The RAPL theorem (theorem 62.3.4) makes limit-preservation a necessary condition for a functor $G\colon \mathbf{D} \to \mathbf{C}$ to be a right adjoint. The adjoint functor theorems address the converse: when a limit-preserving G is guaranteed to have a left adjoint. Preservation of limits alone is not enough, because a proper class of candidate arrows can obstruct the construction. The extra ingredient is a smallness hypothesis. The General Adjoint Functor Theorem asks for a set of arrows that jointly dominates every arrow out of a fixed object, the solution-set condition; the Special Adjoint Functor Theorem replaces that condition with a global structural hypothesis on $\mathbf{C}$. We prove the general theorem in full and record the special one with its key ideas.

The whole chapter turns on one reformulation, established in section 62.1: a left adjoint to G is the same data as a universal arrow $\eta_c\colon c \to Gd_c$ from each object c of $\mathbf{C}$ to G. A universal arrow from c to G (definition 61.1.4) is exactly an initial object of the comma category (c/G), whose objects are pairs (d, f) with $f: c \to Gd$ and whose morphisms $(d, f) \to (d', f')$ are arrows $h: d \to d'$ in $\mathbf{D}$ with $Gh \circ f = f'$. Constructing a left adjoint therefore reduces to producing an initial object in each (c/G). The solution-set condition is precisely the smallness input that lets a completeness argument produce one.

Definition 62.4.1: Solution-Set Condition

Let $G: \mathbf{D} \to \mathbf{C}$ be a functor and c an object of $\mathbf{C}$. Then G satisfies the *solution-set condition* at c if there is a *set* of morphisms

$$\{\, f_i: c \to Gd_i \,\}_{i \in I}$$

such that every morphism $f: c \to Gd$ factors as $f = Gh \circ f_i$ for some index i and some morphism $h: d_i \to d$ in $\mathbf{D}$. The functor G *satisfies the solution-set condition* if it does so at every object c of $\mathbf{C}$.

In the language of the comma category, the condition says the objects (d_i, f_i) form a set that is *weakly initial* in (c/G): every object receives at least one morphism from a member of the set. The word "weakly" marks that the arrow witnessing the factorization is neither required to be unique nor to come from a single chosen source. The force of the definition is entirely its smallness. In a locally small $\mathbf{D}$ the objects (d, f) of (c/G) already form a proper class, and there is no reason a class-sized family should admit a set-sized dominating subfamily; the solution-set condition asserts that one exists. That is the gap the theorem fills between the necessary condition (RAPL) and sufficiency.

Example 62.4.2: The Solution Set for Free Groups

Let $G: \mathbf{Grp} \to \mathbf{Set}$ be the forgetful functor and c a set. A single arrow already works: take the free group Fc on c with its insertion of generators $\eta_c: c \to G(Fc)$, and let $\{f_i\}$ be the one-element family $\{\eta_c\}$. Every function $f: c \to G(H)$ into the underlying set of a group H extends uniquely to a homomorphism $\bar{f}: Fc \to H$ with $G(\bar{f}) \circ \eta_c = f$, so $f = G(\bar{f}) \circ \eta_c$ exhibits the required factorization. Here the solution set has one element because a genuine universal arrow already exists; the theorem below runs the implication in the other direction, manufacturing such a universal arrow from any solution set, however large.

The example makes the shape of the argument visible. When a left adjoint is present, the singleton solution set is the universal arrow itself. The content of the General Adjoint Functor Theorem is that a solution set of *any* size, together with completeness of $\mathbf{D}$ and preservation of limits by G, suffices to assemble a universal arrow that was not given in advance. The assembly is a completeness argument inside the comma category, and its engine is a lemma about initial objects, isolated first.

The lemma is where a set-sized weakly initial family is compressed to a single genuine initial object. In a complete category one can form the product of the family; the product receives an arrow to every object, so it is again weakly initial, and now it is a single object. Passing from weakly initial to initial then requires killing the ambiguity in those arrows, which is what an equalizer of endomorphisms does.

Lemma 62.4.3: Initial Object from a Weakly Initial Set

Let $\mathbf{A}$ be a complete, locally small category. Suppose $\mathbf{A}$ has a *weakly initial set*: a set $\{a_i\}_{i\in I}$ of objects such that every object of $\mathbf{A}$ admits at least one morphism from some a_i. Then $\mathbf{A}$ has an initial object.

***Proof*:**
Step 1: Form a weakly initial object. Since I is a set and $\mathbf{A}$ is complete, the product $P = \prod_{i\in I} a_i$ exists, with projections $\pi_i \colon P \to a_i$. Given any object a, weak initiality supplies some i and a morphism $g\colon a_i \to a$; then $g \circ \pi_i \colon P \to a$ is a morphism from P to a. Thus P is a weakly initial *object*: every object receives at least one morphism from P.

Step 2: Equalize the endomorphisms of P. Local smallness makes $\mathbf{A}(P, P)$ a set, so the family of all endomorphisms of P is a set of parallel arrows $P \rightrightarrows P$. Completeness gives their equalizer

$$e\colon E \to P$$

the joint equalizer of $\{\, u\colon P \to P \,\}$, characterized by $u \circ e = e$ for every endomorphism u of P, and universal among arrows into P with this property. We claim E is initial.

Step 3: Existence of a morphism from E. For any object a, weak initiality of P gives a morphism $P \to a$, and precomposing with e yields a morphism $E \to a$. So E admits at least one morphism to every object.

Step 4: Uniqueness. Let $s, t\colon E \to a$ be two morphisms to a common object a; we must show $s = t$. Form the equalizer $k\colon K \to E$ of s and t, so $s \circ k = t \circ k$ and k is universal with this property. Since P is weakly initial there is a morphism $g\colon P \to K$, and composing produces an endomorphism

$$v = e \circ k \circ g\colon P \to P$$

Now e is the joint equalizer of all endomorphisms of P, and v is one such endomorphism, so $v \circ e = e$, that is $e \circ k \circ g \circ e = e$. The morphism e is a monomorphism, being an equalizer, so it is left-cancellable: from $e \circ (k \circ g \circ e) = e = e \circ \mathrm{id}_P$ we cancel e to get

$$k \circ (g \circ e) = \mathrm{id}_E$$

Thus k is a split epimorphism, with section $g \circ e$. But k is also a monomorphism, being an equalizer, and a split epimorphism that is monic is an isomorphism: composing $k \circ (g \circ e) = \mathrm{id}_E$ with k on the right gives $k \circ (g \circ e) \circ k = k = k \circ \mathrm{id}_K$, and cancelling the monic k on the left yields $(g \circ e) \circ k = \mathrm{id}_K$, so $g \circ e$ is a two-sided inverse of k. Since $s \circ k = t \circ k$ and k is an isomorphism, right-cancelling k gives $s = t$.

Combining Steps 3 and 4, every object of $\mathbf{A}$ admits exactly one morphism from E, so E is initial, completing the proof.

The lemma is the whole mechanism in miniature. The product collapses a set of objects into one weakly initial object; the equalizer of its endomorphisms then removes the freedom in the arrows out of it, leaving a single arrow to each target. The two hypotheses do disjoint work: completeness supplies the product and the equalizers, local smallness guarantees that the endomorphisms of P form a set so that their equalizer exists. Drop either and the argument fails. With the lemma in hand the theorem is a matter of transporting it into the comma category (c/G), which is where the smallness of the solution set becomes the weakly initial set the lemma requires.

The remaining input is that (c/G) inherits completeness from $\mathbf{D}$. This is exactly where the hypothesis that G preserves limits is spent: limits in (c/G) are computed in $\mathbf{D}$, and the cone over c that makes the limit an object of the comma category exists precisely because G carries the $\mathbf{D}$-limit to the corresponding $\mathbf{C}$-limit.

Theorem 62.4.4: General Adjoint Functor Theorem

Let $\mathbf{D}$ be a locally small, complete category and $G\colon \mathbf{D} \to \mathbf{C}$ a functor that preserves all small limits. Then G has a left adjoint if and only if G satisfies the solution-set condition (definition 62.4.1).

***Proof*:**
Necessity. Suppose G has a left adjoint $F \dashv G$, with unit η. For each object c of $\mathbf{C}$, the component $\eta_c\colon c \to G(Fc)$ is a universal arrow from c to G (theorem 62.1.5): every $f\colon c \to Gd$ factors as $f = Gh \circ \eta_c$ for a unique $h\colon Fc \to d$. The singleton family $\{\eta_c\}$ is then a solution set at c, so G satisfies the solution-set condition.

Sufficiency. Assume G preserves small limits and satisfies the solution-set condition. By theorem 62.1.5 it suffices to produce, for each object c of $\mathbf{C}$, a universal arrow from c to G, equivalently an initial object of the comma category (c/G) (definition 61.1.4). Fix c; we produce that initial object by applying lemma 62.4.3 to $\mathbf{A} = (c/G)$, which requires that (c/G) be locally small and complete and possess a weakly initial set.

Step 1: (c/G) is locally small. A morphism $(d, f) \to (d', f')$ in (c/G) is a morphism $h\colon d \to d'$ in $\mathbf{D}$ subject to the equation $Gh \circ f = f'$. Thus the hom-set of (c/G) is a subset of $\mathbf{D}(d, d')$, which is a set because $\mathbf{D}$ is locally small. So (c/G) is locally small.

Step 2: (c/G) is complete, using preservation of limits. Let $D\colon \mathbf{J} \to (c/G)$ be a small diagram, writing $D(j) = (d_j, f_j)$ with structure arrows $f_j\colon c \to Gd_j$; a morphism $j \to j'$ in $\mathbf{J}$ gives $\alpha_{jj'}\colon d_j \to d_{j'}$ with $G\alpha_{jj'} \circ f_j = f_{j'}$. The projection to $\mathbf{D}$ sends D to the diagram $j \mapsto d_j$, which has a limit $L = \lim_{\mathbf{J}} d_j$ in $\mathbf{D}$ by completeness, with limit cone $\lambda_j\colon L \to d_j$.

Because G preserves this limit, $G(L)$ with the cone $G\lambda_j\colon G(L) \to Gd_j$ is a limit of $j \mapsto Gd_j$ in $\mathbf{C}$. The arrows $f_j\colon c \to Gd_j$ form a cone over $j \mapsto Gd_j$ with apex c: for a morphism $j \to j'$ one has $G\alpha_{jj'} \circ f_j = f_{j'}$, which is exactly the compatibility condition for a cone. By the universal property of the limit $G(L)$, this cone factors through a unique morphism

$$f_L\colon c \to G(L)$$

satisfying $G\lambda_j \circ f_L = f_j$ for every j. The pair (L, f_L) is then an object of (c/G), and each $\lambda_j\colon L \to d_j$ satisfies $G\lambda_j \circ f_L = f_j$, so λ_j is a morphism $(L, f_L) \to (d_j, f_j)$ in (c/G); these assemble into a cone over D.

This cone is a limit cone in (c/G). Given any cone over D with apex (d, f), its legs are morphisms $\mu_j\colon (d, f) \to (d_j, f_j)$, hence morphisms $\mu_j\colon d \to d_j$ in $\mathbf{D}$ forming a cone over $j \mapsto d_j$; the limit L in $\mathbf{D}$ yields a unique $u\colon d \to L$ with $\lambda_j \circ u = \mu_j$. It remains to check u is a morphism $(d, f) \to (L, f_L)$, that is $Gu \circ f = f_L$. Both sides are morphisms $c \to G(L)$ into the limit $G(L)$; they agree after composing with each projection $G\lambda_j$, since $G\lambda_j \circ Gu \circ f = G(\lambda_j u) \circ f = G\mu_j \circ f = f_j$ and $G\lambda_j \circ f_L = f_j$. By the uniqueness clause of the limit $G(L)$, $Gu \circ f = f_L$. So u is the unique factorization in (c/G), and (L, f_L) is a limit of D. Hence (c/G) has all small limits: it is complete.

Step 3: the solution set is a weakly initial set for (c/G)*.* By hypothesis G satisfies the solution-set condition at c: there is a set $\{f_i : c \to Gd_i\}_{i \in I}$ such that every $f : c \to Gd$ factors as $f = Gh \circ f_i$ for some i and some $h : d_i \to d$. Read in (c/G), the objects $\{(d_i, f_i)\}_{i \in I}$ form a set, and the factorization $f = Gh \circ f_i$ says exactly that $h : (d_i, f_i) \to (d, f)$ is a morphism of (c/G). So every object (d, f) receives a morphism from some (d_i, f_i): the set $\{(d_i, f_i)\}$ is weakly initial.

Step 4: apply the lemma. By Steps 1-3 the comma category (c/G) is locally small, complete, and has a weakly initial set. Lemma 62.4.3 produces an initial object (d_c, η_c) of (c/G). An initial object of (c/G) is a universal arrow $\eta_c : c \to Gd_c$ from c to G (definition 61.1.4). This holds for every c, so by theorem 62.1.5 the assignment $c \mapsto d_c$ extends to a left adjoint $F \dashv G$, completing the proof.

The theorem isolates the two obstructions to being a right adjoint and names both. The soft obstruction is failure to preserve limits, ruled out by RAPL and assumed away here. The hard obstruction is size: a limit-preserving functor between well-behaved categories can still fail to have a left adjoint if the comma categories are too large to admit an initial object, and the solution-set condition is exactly the hypothesis that excludes this. When one has a candidate construction in hand, the theorem is often applied by exhibiting an explicit small solution set (frequently a single arrow, as in example 62.4.2) rather than by verifying an abstract smallness principle.

A recurring difficulty in practice is that verifying the solution-set condition can be as much work as constructing the adjoint by hand. The Special Adjoint Functor Theorem removes this burden under stronger global hypotheses on $\mathbf{C}$: a small cogenerating set does the work of every solution set at once. The cost is machinery, well-poweredness and cogenerators phrased through subobject lattices, that this book does not develop, so the theorem is stated with its key ideas rather than proved in full.

Theorem 62.4.5: Special Adjoint Functor Theorem

Let $\mathbf{D}$ be a locally small, complete, well-powered category with a small cogenerating set, and let $G : \mathbf{D} \to \mathbf{C}$ be a functor between locally small categories that preserves all small limits. Then G has a left adjoint.

***Proof Key ideas*:**
The full argument requires two notions this book does not develop: a category is *well-powered* when every object has only a set of subobjects (its subobjects form a set rather than a proper class), and a set $\{s_k\}$ of objects is *cogenerating* when any two distinct parallel morphisms $u \neq v : d \rightrightarrows d'$ are separated by some morphism $d' \to s_k$ that distinguishes them. These are hypotheses on the lattice of subobjects, whose systematic development belongs to a treatment of well-powered categories and their factorization systems; see Mac Lane, *Categories for the Working Mathematician*, Chapter V.8.

Granting them, the shape of the proof matches the general theorem, with the cogenerating set replacing the solution-set condition. Fix an object c of $\mathbf{C}$ and form, inside $\mathbf{D}$, the product $\prod_k s_k^{\mathbf{C}(c, Gs_k)}$ indexed by the cogenerating set and, for each factor, by the set of morphisms $c \to G(s_k)$. The candidate universal object is the smallest subobject of this product through which the canonical arrow from c factors; well-poweredness guarantees a smallest such subobject exists, since the subobjects form a set closed under the intersections that completeness provides, and the cogenerating property forces the resulting arrow to be universal. Preservation of limits by G is used, as before, to place the construction in the comma category (c/G). The upshot is that a small cogenerating set functions as a uniform solution set: it supplies, in one stroke, the smallness input that the general theorem requires separately at each object, completing the sketch.

The two theorems are the standard existence criteria for left adjoints, and their hypotheses trade off against each other. The general theorem requires little of the target categories but requires a solution set verified case by case; the special theorem front-loads strong structural hypotheses on $\mathbf{D}$, a small cogenerating set and well-poweredness, and then asks nothing further of the individual objects. Many categories of everyday interest, **Set**, **Grp**, **Ab**, **Top**, and $\mathbf{Mod}_R$, are complete and well-powered with a small cogenerating set, so the special theorem applies to limit-preserving functors out of them without any per-object verification.

A final application of the general theorem recurs across the series: producing a reflection onto a subcategory.

Example 62.4.6: A Reflective Subcategory via GAFT

Let $\mathbf{D}$ be a locally small, complete category and $\mathbf{C} \subseteq \mathbf{D}$ a full subcategory that is closed under small limits, meaning the inclusion $G\colon \mathbf{C} \hookrightarrow \mathbf{D}$ preserves and reflects small limits and every $\mathbf{D}$-limit of a diagram in $\mathbf{C}$ again lies in $\mathbf{C}$. Suppose in addition that $\mathbf{C}$ is complete and that G satisfies the solution-set condition: for each object d of $\mathbf{D}$ there is a set of morphisms $\{d \to c_i\}$ with c_i in $\mathbf{C}$ through which every morphism $d \to c$ into an object of $\mathbf{C}$ factors. Then theorem 62.4.4, applied to $G\colon \mathbf{C} \to \mathbf{D}$, produces a left adjoint $L \dashv G$: a *reflection* of $\mathbf{D}$ onto $\mathbf{C}$.

Concretely, take $\mathbf{D} = \mathbf{Grp}$ and $\mathbf{C} = \mathbf{Ab}$ the full subcategory of abelian groups, with $G\colon \mathbf{Ab} \hookrightarrow \mathbf{Grp}$ the inclusion. A limit of abelian groups computed in **Grp** is again abelian, so G preserves limits and **Ab** is closed under them; **Ab** is complete. For the solution-set condition at a group H, a single arrow suffices: the quotient map $q_H\colon H \to H^{\mathrm{ab}} = H/[H,H]$ onto the abelianization. Any homomorphism $H \to A$ into an abelian group A kills the commutator subgroup $[H,H]$, so it factors uniquely as $\bar{\varphi} \circ q_H$ with $\bar{\varphi}\colon H^{\mathrm{ab}} \to A$. The general theorem then returns the left adjoint $H \mapsto H^{\mathrm{ab}}$, recovering abelianization as the reflection of **Grp** onto **Ab**. As with example 62.4.2, the singleton solution set is present because the universal arrow already exists; the value of the theorem is that a solution set of any size would have sufficed.

63

Monads and their Algebras

A monad is the algebraic structure an adjunction leaves on one side: from $F \dashv G$ the composite GF inherits a unit and a multiplication, and this structure alone suffices to rebuild a category of algebras. This chapter defines *monads*, constructs the two canonical adjunctions that realize any monad, the Eilenberg-Moore and Kleisli categories, and asks the converse, *monadicity*: when is a forgetful functor the forgetful functor of a monad? Beck's theorem answers it, and explains why the algebraic categories over **Set** are exactly the monadic ones.

63.1 Monads from Adjunctions

Every adjunction $F \dashv G$ leaves a trace on the category where its two functors meet: the composite GF is an endofunctor, the unit is a natural transformation $\mathrm{id} \Rightarrow GF$, and the counit supplies a second natural transformation $GFGF \Rightarrow GF$. This section isolates that trace as a structure in its own right, the *monad*, defines it by the two laws it must obey, and shows that every adjunction produces one. The free-monoid, powerset, and error monads run alongside as the guiding examples.

Definition 63.1.1: Monad

A *monad* on a category C is an endofunctor $T\colon C \to C$ together with two natural transformations, the *unit*

$$\eta\colon \mathrm{id}_C \Rightarrow T$$

and the *multiplication*

$$\mu\colon T^2 \Rightarrow T$$

where T^2 denotes the composite TT, subject to two laws. The *associativity law* is the equality of natural transformations $T^3 \Rightarrow T$

$$\mu \circ T\mu = \mu \circ \mu T \tag{63.1}$$

and the *unit laws* are the equalities of natural transformations $T \Rightarrow T$

$$\mu \circ T\eta = \mathrm{id}_T = \mu \circ \eta T \tag{63.2}$$

Here $T\mu$ and μT are the whiskered composites: $T\mu$ has component $T(\mu_c)\colon T^3c \to T^2c$ at c, and μT has component $\mu_{Tc}\colon T^3c \to T^2c$; similarly $T\eta$ and ηT have components $T(\eta_c)$ and η_{Tc}. The two laws are the commutativity of the associativity square

$$\begin{tikzcd}
T^3 \arrow[r, Rightarrow, "T\mu"] \arrow[d, Rightarrow, "\mu T"'] & T^2 \arrow[d, Rightarrow, "\mu"] \\
T^2 \arrow[r, Rightarrow, "\mu"'] & T
\end{tikzcd}$$

and the two unit triangles

$$\begin{tikzcd}
T \arrow[r, Rightarrow, "\eta T"] \arrow[dr, Rightarrow, "\mathrm{id}_T"'] & T^2 \arrow[d, Rightarrow, "\mu"] & T \arrow[l, Rightarrow, "T\eta"'] \arrow[dl, Rightarrow, "\mathrm{id}_T"] \\
 & T &
\end{tikzcd}$$

The pattern is that of a monoid: μ plays the role of a multiplication and η the role of a unit element, with (63.1) the associativity of the multiplication and (63.2) its two-sided unit law. The slogan is that a monad is a monoid in the category of endofunctors of C under composition, with the functor category $[C, C]$ (definition 59.5.4) supplying the ambient category and functor composition supplying the tensor product. The definition is the monoid axioms transported into that setting, with vertical composition of natural transformations for equality of maps and horizontal composition (whiskering) for the two ways of associating T^3.

The endofunctor alone carries no content; the unit and multiplication are what make a monad an algebraic object. Before the examples, the source of every such structure is an adjunction.

The chapter opening framed a monad as the structure an adjunction leaves on one side. That statement is now a proposition: the composite GF of an adjunction, equipped with the adjunction unit and the multiplication built from the counit, satisfies definition 63.1.1. The verification is a diagram chase in which every square commutes by naturality of the counit or by a triangle identity.

Proposition 63.1.2: The Monad of an Adjunction

Let $F\colon C \to D$ and $G\colon D \to C$ be functors with $F \dashv G$, unit $\eta\colon \mathrm{id}_C \Rightarrow GF$ and counit $\varepsilon\colon FG \Rightarrow \mathrm{id}_D$ (definition 62.1.1, definition 62.1.3). Then

$$T = GF\colon C \to C, \qquad \eta\colon \mathrm{id}_C \Rightarrow T, \qquad \mu = G\varepsilon F\colon T^2 \Rightarrow T$$

is a monad on C, where μ is the whiskered natural transformation with component $\mu_c = G\varepsilon_{Fc}\colon GFGFc \to GFc$ at c.

***Proof*:**
The unit of the monad is the adjunction unit, already a natural transformation $\mathrm{id}_C \Rightarrow GF = T$ (definition 62.1.3). The multiplication $\mu = G\varepsilon F$ is the whiskering of the natural transformation $\varepsilon\colon FG \Rightarrow \mathrm{id}_D$ by G on the left and F on the right, hence a natural transformation $GFGF \Rightarrow GF$, that is, $T^2 \Rightarrow T$. It remains to check the associativity law and the two unit laws.

Step 1: Associativity. Both $\mu \circ T\mu$ and $\mu \circ \mu T$ are natural transformations $T^3 = GFGFGF \Rightarrow GF = T$; the claim is that their components agree at every c. Writing out the components,

$$(\mu \circ T\mu)_c = G\varepsilon_{Fc} \circ GF(G\varepsilon_{Fc}) = G\varepsilon_{Fc} \circ GFG\varepsilon_{Fc}$$

and

$$(\mu \circ \mu T)_c = G\varepsilon_{Fc} \circ G\varepsilon_{FGFc}$$

Applying the functor G to the naturality square of ε reduces both to a common expression. Naturality of $\varepsilon\colon FG \Rightarrow \mathrm{id}_D$ at the morphism $\varepsilon_{Fc}\colon FGFc \to Fc$ in D gives the commuting square

$$\begin{tikzcd} FGFGFc \arrow[r, "\varepsilon_{FGFc}"] \arrow[d, "FG\varepsilon_{Fc}"'] & FGFc \arrow[d, "\varepsilon_{Fc}"] \\ FGFc \arrow[r, "\varepsilon_{Fc}"'] & Fc \end{tikzcd}$$

that is, $\varepsilon_{Fc} \circ FG\varepsilon_{Fc} = \varepsilon_{Fc} \circ \varepsilon_{FGFc}$ in D. Applying G to this equation yields $G\varepsilon_{Fc} \circ GFG\varepsilon_{Fc} = G\varepsilon_{Fc} \circ G\varepsilon_{FGFc}$, which is precisely $(\mu \circ T\mu)_c = (\mu \circ \mu T)_c$. As c was arbitrary, (63.1) holds. Associativity of the monad is thus the naturality of the counit, pushed through G.

Step 2: The unit laws. The component of $\mu \circ T\eta$ at c is

$$(\mu \circ T\eta)_c = G\varepsilon_{Fc} \circ GF(\eta_c) = G\varepsilon_{Fc} \circ GF\eta_c = G(\varepsilon_{Fc} \circ F\eta_c)$$

using that G is a functor. By the second triangle identity (theorem 62.3.1), $\varepsilon_{Fc} \circ F\eta_c = \mathrm{id}_{Fc}$, so the component is $G(\mathrm{id}_{Fc}) = \mathrm{id}_{GFc} = (\mathrm{id}_T)_c$. Hence $\mu \circ T\eta = \mathrm{id}_T$. For the other unit law, the component of $\mu \circ \eta T$ at c is

$$(\mu \circ \eta T)_c = G\varepsilon_{Fc} \circ \eta_{GFc}$$

the composite $GFc \to GFGFc \to GFc$. This is exactly the first triangle identity (theorem 62.3.1) evaluated at the object Fc of D, namely $G\varepsilon_d \circ \eta_{Gd} = \mathrm{id}_{Gd}$ with $d = Fc$, giving $G\varepsilon_{Fc} \circ \eta_{GFc} = \mathrm{id}_{GFc}$. Hence $\mu \circ \eta T = \mathrm{id}_T$, and both equalities in (63.2) hold, completing the proof.

The proof shows the two monad laws are repackaged adjunction data: associativity is the naturality of the counit, and each unit law is one of the two triangle identities. This is the origin of every monad met in practice. A monad is not usually written down axiomatically and then studied; it arises as GF for a free-forgetful adjunction, and its multiplication is the counit describing how a free object built on a free object collapses. The examples make this concrete, and each recovers a familiar operation.

Example 63.1.3: The Free-Monoid, Powerset, and Error Monads

Three monads on **Set**, each the monad of a free-forgetful adjunction (proposition 63.1.2) or checked directly against definition 63.1.1.

1. *The free-monoid monad.* Let TX be the set of finite words $(x_1, \ldots, x_n)$, $n \geq 0$, in the alphabet X, including the empty word; for $f\colon X \to Y$, Tf applies f letterwise. This is GF for the free-monoid adjunction $F \dashv G$ with $G\colon \mathbf{Mon} \to \mathbf{Set}$ the underlying-set functor and F the free-monoid functor ([25]). The unit $\eta_X\colon X \to TX$ sends x to the length-one word (x). The multiplication $\mu_X\colon T^2X \to TX$ takes a word of words and concatenates: a length-one word of words $\big((x_1, \ldots, x_k)\big)$ flattens to $(x_1, \ldots, x_k)$, and in general
$$\mu_X\big((w_1, \ldots, w_m)\big) = w_1 \cdot w_2 \cdots w_m$$
the concatenation of the constituent words $w_i \in TX$. Associativity (63.1) is the associativity of concatenation: flattening a triple-nested word by first concatenating the inner groupings or first concatenating the outer groupings gives the same word. The unit laws (63.2) say that turning each letter into a length-one word and then flattening, or wrapping a whole word into a length-one word of words and then flattening, both return the original word.

2. *The powerset monad.* Let $T = P$ be the covariant powerset functor: PX is the set of subsets of X, and $Pf(S) = f(S)$ is the image of $S \subseteq X$ under $f\colon X \to Y$. The unit $\eta_X\colon X \to PX$ is the singleton map $x \mapsto \{x\}$. The multiplication $\mu_X\colon P^2X \to PX$ is the union: for a set $\mathcal{S}$ of subsets of X,
$$\mu_X(\mathcal{S}) = \bigcup_{S \in \mathcal{S}} S$$
Associativity is the equality of the two ways to flatten a set of sets of sets by union, which both give the union of all the innermost members. The unit laws read $\bigcup_{x \in S}\{x\} = S$ (union of the singletons of the elements of S) and $\bigcup\{S\} = S$ (union of the one-element family $\{S\}$). This is the monad of the adjunction between **Set** and the category of complete sup-lattices, whose algebras are taken up in the next section.

3. *The error monad.* Fix a one-point set $\{*\}$ and let $TX = X \sqcup \{*\}$, the disjoint union adjoining a single point $*$ read as an error value; for $f\colon X \to Y$, Tf acts as f on X and fixes $*$. The unit $\eta_X\colon X \to X \sqcup \{*\}$ is the inclusion of X. The multiplication $\mu_X\colon T^2X \to TX$, where $T^2X = X \sqcup \{*_1\} \sqcup \{*_2\}$ carries two error points ($*_1$ from the inner T, $*_2$ from the outer), acts as the identity on X and sends both $*_1$ and $*_2$ to the single $*$, merging the two error points. Associativity holds because on the X-part every map in sight is the identity and on the error part every composite lands on $*$; the unit laws hold because η adjoins an error point that μ immediately absorbs. This is the monad GF for the free-pointed-set adjunction, F adjoining a basepoint and G forgetting it.

The three examples share a shape: TX is a syntax of "X-terms" (words, subsets, values-or-error), the unit embeds each element as a trivial term, and the multiplication evaluates a term of terms into a single term. A monad is thus a theory of substitution, with μ the operation that carries out one round of it, and the two laws are exactly what make repeated substitution unambiguous. This reading anticipates the theory of algebras for a monad, developed in the next section, where an algebra supplies the missing evaluation of a term into an actual element.

Reversing every arrow in definition 63.1.1 produces the dual notion, obtained by applying the definition in the opposite category. A comonad packages the trace an adjunction casts on the *other* side, the composite FG with counit and comultiplication, and is the structure whose coalgebras encode "co-substitution" in the same way a monad's algebras encode substitution.

Definition 63.1.4: Comonad

A *comonad* on a category C is a monad on C^{op}: an endofunctor $T\colon C \to C$ with a *counit* $\varepsilon\colon T \Rightarrow \mathrm{id}_C$ and a *comultiplication* $\delta\colon T \Rightarrow T^2$ satisfying the coassociativity law $T\delta \circ \delta = \delta T \circ \delta$ and the counit laws $\varepsilon T \circ \delta = \mathrm{id}_T = T\varepsilon \circ \delta$, the diagrams of definition 63.1.1 with every arrow reversed. Dually to proposition 63.1.2, an adjunction $F \dashv G$ induces a comonad FG on D with counit ε and comultiplication $F\eta G$ (box 59.3.3).

Everything proved for monads holds for comonads by the duality principle, with the roles of unit and multiplication interchanged and arrows reversed. The comonad and its coalgebras are the opposite-category translation of the monad theory developed here and in the sections that follow, and are not pursued further.

63.2 Algebras for a Monad

Proposition 63.1.2 produced a monad from every adjunction. This section runs the construction the other way. Given a monad (T, η, μ) on $\mathbf{C}$ (definition 63.1.1), an *algebra* for it is an object equipped with an action of T compatible with η and μ; the algebras assemble into a category, the Eilenberg-Moore category $\mathbf{C}^T$, and the free-algebra construction supplies an adjunction whose induced monad is the original T. A monad thus records an algebraic theory, and $\mathbf{C}^T$ is its category of models.

The definition is the categorical distillation of what it means to be an algebra for an operation. If T is the free-monoid endofunctor on $\mathbf{Set}$, then TA is the set of finite words in A; an action $a\colon TA \to A$ that respects the unit and multiplication is precisely a rule multiplying words, that is, a monoid structure. The two laws below are exactly associativity and unitality of that multiplication, transcribed into the language of T.

Definition 63.2.1: Algebra for a Monad

Let (T, η, μ) be a monad on a category $\mathbf{C}$. A *T-algebra* is a pair (A, a) consisting of an object A of $\mathbf{C}$ and a morphism $a\colon TA \to A$, called the *structure map*, making the two diagrams

$$\begin{tikzcd} A \arrow[r, "\eta_A"] \arrow[dr, "\mathrm{id}_A"'] & TA \arrow[d, "a"] \\ & A \end{tikzcd} \qquad \begin{tikzcd} T^2A \arrow[r, "\mu_A"] \arrow[d, "Ta"'] & TA \arrow[d, "a"] \\ TA \arrow[r, "a"'] & A \end{tikzcd}$$

commute, that is, $a \circ \eta_A = \mathrm{id}_A$ (the *unit law*) and $a \circ \mu_A = a \circ Ta$ (the *associativity law*).

A *morphism of T-algebras* $(A, a) \to (B, b)$ is a morphism $f\colon A \to B$ in $\mathbf{C}$ commuting with the structure maps, that is, making

$$\begin{tikzcd} TA \arrow[r, "Tf"] \arrow[d, "a"'] & TB \arrow[d, "b"] \\ A \arrow[r, "f"'] & B \end{tikzcd}$$

commute: $f \circ a = b \circ Tf$.

The unit law forces the structure map to restore an element inserted by η_A without altering it: the free elements act as they must. The associativity law compares the two ways of collapsing a doubly-applied T down to A (first collapse the outer layer with μ_A, then act by a; or first act by a inside the outer T, then act again) and requires they agree. For the free-monoid monad these are the associativity and unit axioms of a monoid, as the next passage makes precise; the definition abstracts that pattern away from any one theory.

Example 63.2.2: The Trivial Algebra and the Free Algebra

Two algebras exist for every monad. First, (TA, μ_A) is a T-algebra for each object A: the unit law $\mu_A \circ \eta_{TA} = \mathrm{id}_{TA}$ is one of the monad unit laws, and the associativity law $\mu_A \circ \mu_{TA} = \mu_A \circ T\mu_A$ is the monad associativity law (definition 63.1.1). This is the *free T-algebra* on A. Second, for any algebra (A, a) the associativity law read as a commuting square makes the structure map $a\colon (TA, \mu_A) \to (A, a)$ itself a morphism of T-algebras.

The algebras and their morphisms form a category, and the underlying-object assignment forgets back to $\mathbf{C}$.

Definition 63.2.3: Eilenberg-Moore Category

Let (T, η, μ) be a monad on $\mathbf{C}$. The *Eilenberg-Moore category* $\mathbf{C}^T$ has T-algebras as objects and morphisms of T-algebras as arrows, with composition and identities computed in $\mathbf{C}$. The *forgetful functor*

$$U^T : \mathbf{C}^T \to \mathbf{C}, \qquad (A, a) \mapsto A, \qquad (f : (A, a) \to (B, b)) \mapsto (f : A \to B)$$

sends an algebra to its underlying object and an algebra morphism to its underlying morphism.

Composition is well-defined because a composite of algebra morphisms is again one: stacking the two commuting squares of definition 63.2.1 for $f : (A, a) \to (B, b)$ and $g : (B, b) \to (C, c)$ gives the square for gf, using $T(gf) = Tg \circ Tf$. The identity of $\mathbf{C}$ on A is an algebra endomorphism of (A, a). So $\mathbf{C}^T$ is a genuine category and U^T a functor; the content lies not in these formalities but in what U^T admits as a partner. The free-algebra assignment $A \mapsto (TA, \mu_A)$ of example 63.2.2 is the object part of its left adjoint.

The motivation for the adjunction is the recovery program. Proposition 63.1.2 showed an adjunction breeds a monad; the reverse question is whether every monad comes from an adjunction, and if so from which one. The Eilenberg-Moore construction answers yes and produces a canonical such adjunction, built entirely out of T itself.

Theorem 63.2.4: The Eilenberg-Moore Adjunction

Let (T, η, μ) be a monad on $\mathbf{C}$. The assignment

$$F^T : \mathbf{C} \to \mathbf{C}^T, \qquad c \mapsto (Tc, \mu_c), \qquad (g : c \to c') \mapsto (Tg : (Tc, \mu_c) \to (Tc', \mu_{c'}))$$

is a functor, the *free T-algebra functor*, and it is left adjoint to the forgetful functor U^T. The adjunction $F^T \dashv U^T$ induces (via proposition 63.1.2) the original monad (T, η, μ).

***Proof*:**

Step 1: F^T is a functor. On objects $F^T c = (Tc, \mu_c)$ is a T-algebra by example 63.2.2. On a morphism $g : c \to c'$, the arrow $Tg : Tc \to Tc'$ is an algebra morphism $(Tc, \mu_c) \to (Tc', \mu_{c'})$: the required square $Tg \circ \mu_c = \mu_{c'} \circ T(Tg)$ is naturality of $\mu : T^2 \Rightarrow T$ applied to g. Functoriality of F^T then follows from functoriality of T, since F^T acts as T on morphisms.

Step 2: The transposition bijection. Fix an object c of $\mathbf{C}$ and a T-algebra (A, a). Define

$$\Phi : \mathrm{Hom}_{\mathbf{C}^T}\big((Tc, \mu_c), (A, a)\big) \longrightarrow \mathrm{Hom}_{\mathbf{C}}(c, A), \qquad \Phi(f) = f \circ \eta_c$$

which precomposes an algebra map out of the free algebra with the unit component $\eta_c : c \to Tc$; this is a morphism of $\mathbf{C}$ because $U^T f = f$ and η_c live there. In the other direction define

$$\Psi : \mathrm{Hom}_{\mathbf{C}}(c, A) \longrightarrow \mathrm{Hom}_{\mathbf{C}^T}\big((Tc, \mu_c), (A, a)\big), \qquad \Psi(g) = a \circ Tg$$

Two claims must be checked: that $\Psi(g)$ is an algebra morphism, and that Φ and Ψ are mutually inverse.

Step 3: $\Psi(g)$ is an algebra morphism. Write $h = a \circ Tg : Tc \to A$. The algebra-morphism square for $h\colon (Tc,\mu_c) \to (A,a)$ requires $h \circ \mu_c = a \circ Th$. Compute the right-hand side:

$$a \circ Th = a \circ T(a \circ Tg) = a \circ Ta \circ T^2 g$$

Apply the associativity law $a \circ Ta = a \circ \mu_A$ of (A,a):

$$a \circ Ta \circ T^2 g = a \circ \mu_A \circ T^2 g$$

Naturality of $\mu\colon T^2 \Rightarrow T$ applied to g gives $\mu_A \circ T^2 g = Tg \circ \mu_c$, so

$$a \circ \mu_A \circ T^2 g = a \circ Tg \circ \mu_c = h \circ \mu_c$$

which is the identity required. Hence $\Psi(g)$ is a morphism $(Tc,\mu_c) \to (A,a)$ of T-algebras.

Step 4: Φ and Ψ are inverse. For $g\colon c \to A$,

$$\Phi(\Psi(g)) = (a \circ Tg) \circ \eta_c = a \circ Tg \circ \eta_c = a \circ \eta_A \circ g = g$$

where the third equality is naturality of $\eta\colon \mathrm{id}_{\mathbf{C}} \Rightarrow T$ applied to g (namely $Tg \circ \eta_c = \eta_A \circ g$) and the fourth is the unit law $a \circ \eta_A = \mathrm{id}_A$ of (A,a). Conversely, for an algebra morphism $f\colon (Tc,\mu_c) \to (A,a)$,

$$\Psi(\Phi(f)) = a \circ T(f \circ \eta_c) = a \circ Tf \circ T\eta_c$$

Because f is an algebra morphism, $a \circ Tf = f \circ \mu_c$, so

$$a \circ Tf \circ T\eta_c = f \circ \mu_c \circ T\eta_c = f \circ \mathrm{id}_{Tc} = f$$

the middle step using the monad unit law $\mu_c \circ T\eta_c = \mathrm{id}_{Tc}$ (definition 63.1.1). Thus Φ and Ψ are mutually inverse bijections.

Step 5: Naturality. The bijection must be natural in c and in (A,a), so that it is an adjunction (definition 62.1.1). For naturality in the algebra variable, let $\varphi\colon (A,a) \to (B,b)$ be an algebra morphism. Postcomposition by φ acts on the left-hand hom-set (in $\mathbf{C}^T$) and by $U^T\varphi = \varphi$ on the right (in $\mathbf{C}$); for $f\colon (Tc,\mu_c) \to (A,a)$,

$$\Phi(\varphi \circ f) = \varphi \circ f \circ \eta_c = \varphi \circ \Phi(f)$$

which is the naturality square for the target variable. For naturality in c, let $k\colon c' \to c$ be a morphism of $\mathbf{C}$. Precomposition acts by k on the right-hand hom-set and by $F^T k = Tk$ on the left; for $f\colon (Tc,\mu_c) \to (A,a)$,

$$\Phi(f \circ Tk) = f \circ Tk \circ \eta_{c'} = f \circ \eta_c \circ k = \Phi(f) \circ k$$

using naturality of η applied to k (namely $Tk \circ \eta_{c'} = \eta_c \circ k$). Both squares commute, so Φ is a natural isomorphism and $F^T \dashv U^T$.

Step 6: The induced monad is T. By proposition 63.1.2, the monad induced by $F^T \dashv U^T$ has endofunctor $U^T F^T$, unit the adjunction unit, and multiplication $U^T \varepsilon^T F^T$, where ε^T is the counit of $F^T \dashv U^T$. On objects, $U^T F^T c = U^T(Tc,\mu_c) = Tc$, and on morphisms $U^T F^T g = Tg$, so $U^T F^T = T$ as endofunctors. The adjunction unit is the transpose of the identity on $F^T c = (Tc,\mu_c)$, which by Step 2 is $\Phi(\mathrm{id}) = \mathrm{id}_{Tc} \circ \eta_c = \eta_c$; the induced unit is η. The counit component at (A,a) is the transpose of id_A under $F^T \dashv U^T$, that is $\varepsilon^T_{(A,a)} = \Psi(\mathrm{id}_A) = a \circ T\mathrm{id}_A = a$, the structure map itself. Hence the induced multiplication at c is $U^T \varepsilon^T_{F^T c} = \varepsilon^T_{(Tc,\mu_c)} = \mu_c$. Endofunctor, unit, and multiplication all agree with (T,η,μ), completing the proof.

The counit computation in Step 6 isolates a useful fact: the counit of $F^T \dashv U^T$ at an algebra (A, a) is its own structure map a, viewed as the algebra morphism $(TA, \mu_A) \to (A, a)$. This recovers the second observation of example 63.2.2 as the counit of the adjunction, and it says that every algebra is a canonical quotient of a free one, namely the free algebra on its underlying object, acted on by a. The theorem thus does two things at once: it realizes an arbitrary monad by an adjunction, closing the loop opened in section 63.1, and it does so through a category $\mathbf{C}^T$ assembled purely from T, with no auxiliary data.

The examples make the slogan "a monad is an algebraic theory, and $\mathbf{C}^T$ is its category of models" concrete. Each monad from example 63.1.3 has an Eilenberg-Moore category that is, up to equivalence, a familiar category of structured sets.

Example 63.2.5: Eilenberg-Moore Categories of Standard Monads

1. *The free-monoid monad.* Let T be the free-monoid monad on **Set**, so TA is the set of finite words $(a_1, \ldots, a_n)$ in A, with $\eta_A(a) = (a)$ the length-one word and μ_A concatenating a word of words. A T-algebra (A, a) has a structure map $a \colon TA \to A$ assigning an element to each word. Write $a_1 \cdots a_n$ for $a(a_1, \ldots, a_n)$. The unit law $a \circ \eta_A = \mathrm{id}_A$ says $a(a) = a$: single letters are unchanged. The associativity law $a \circ \mu_A = a \circ Ta$ says that evaluating a word of words either by first concatenating (via μ_A) or by first evaluating each inner word (via Ta) gives the same answer; taking the empty outer word gives a distinguished element $e = a()$ and taking a two-word split gives $(a_1 \cdots a_i)(a_{i+1} \cdots a_n) = a_1 \cdots a_n$. These are exactly the associativity and unit axioms of a monoid with unit e and product $xy = a(x, y)$. A morphism of T-algebras is a map preserving the product and unit, that is, a monoid homomorphism ([25]). Hence $\mathbf{Set}^T$ is equivalent to **Mon**, the category of monoids.

2. *The powerset monad.* Let $T = P$ be the covariant powerset monad on **Set**, with $\eta_A(a) = \{a\}$ and μ_A taking a union of a set of subsets. A P-algebra is a map $a \colon PA \to A$ sending each subset $S \subseteq A$ to an element $\bigvee S := a(S)$. The unit law $a(\{x\}) = x$ pins the singletons. Setting $x \leq y$ iff $\bigvee\{x, y\} = y$ defines a partial order on A, and the associativity law $a \circ \mu_A = a \circ Pa$ forces $a(S)$ to be the least upper bound of S for every subset S, including the empty set (whose supremum is the bottom element $a(\varnothing)$). An algebra structure on A is thus a rule assigning a supremum to *every* subset, which is exactly a complete sup-lattice; a P-algebra morphism is a map preserving arbitrary suprema. So $\mathbf{Set}^P$ is the category of complete sup-lattices and sup-preserving maps.

3. *The maybe monad.* For the maybe monad $MA = A \sqcup \{*\}$ (example 63.1.3), an M-algebra $a \colon A \sqcup \{*\} \to A$ is determined by its value $a(*)$ together with the unit law fixing A; the associativity law imposes nothing further, so an M-algebra is a set with a chosen basepoint $a(*)$, and $\mathbf{Set}^M$ is the category of pointed sets.

In each case the structure map is the operation of the theory: word-multiplication for monoids, supremum for sup-lattices, basepoint selection for pointed sets. This is the sense in which a monad on **Set** encodes an algebraic theory whose operations are "T-many arguments in, one out", and $\mathbf{C}^T$ is its category of models with the theory-preserving maps. The converse question, which categories over **Set** arise this way, is the subject of the monadicity theorem later in the chapter.

One interpretive point closes the section. Theorem 63.2.4 realizes a given monad by an adjunction, but it is not the only such realization: $\mathbf{C}^T$ is the *terminal* one. Among all adjunctions $F \dashv U$ inducing the given monad, the Eilenberg-Moore adjunction receives a unique comparison functor from every other, making it the largest category of T-algebras compatible with the monad. The next section constructs the initial realization, the Kleisli category, together with the comparison functor between them; the two together bound every adjunction that induces T.

63.3 The Kleisli Category

Where the Eilenberg-Moore category $\mathbf{C}^T$ (section 63.2) realizes T from *every* T-algebra, this section builds the opposite extreme from the smallest possible supply, the free ones. The construction packages the free algebras into a category $\mathbf{C}_T$ using only η and μ, exhibits it as a second adjunction that induces T, and locates it against $\mathbf{C}^T$: $\mathbf{C}_T$ is initial and $\mathbf{C}^T$ terminal among all adjunctions inducing T.

The idea is to read a morphism $c \to Tc'$ not as a map landing in Tc' but as a map from c to c' that carries a T-shaped effect: a function returning a formal word, a subset, or a possibly-undefined value rather than a single element. Composing two such effectful maps requires flattening the doubled effect, and μ is exactly the flattening.

Definition 63.3.1: Kleisli Category

Let (T, η, μ) be a monad on a category $\mathbf{C}$ (definition 63.1.1). The *Kleisli category* $\mathbf{C}_T$ has the same objects as $\mathbf{C}$. A morphism $c \to c'$ in $\mathbf{C}_T$ is a $\mathbf{C}$-morphism $f : c \to Tc'$, called a *Kleisli arrow*. The identity on c is the unit component

$$\eta_c : c \to Tc$$

and the composite of $f : c \to Tc'$ with $g : c' \to Tc''$ (in that order, g after f) is the Kleisli arrow

$$g \odot f \;=\; \mu_{c''} \circ Tg \circ f \;:\; c \to Tc'' \tag{63.3}$$

where $\odot$ denotes composition in $\mathbf{C}_T$ to distinguish it from composition $\circ$ in $\mathbf{C}$.

The formula (63.3) reads left to right through the effect: apply f to reach Tc', apply Tg to push each intermediate value through g and land in TTc'', then apply $\mu_{c''}$ to collapse the doubled effect down to Tc''. That $\mathbf{C}_T$ is a category is not automatic; it is exactly the monad laws in disguise, and the verification shows why associativity and unitality of μ and η are the right axioms to impose.

Proposition 63.3.2: The Kleisli Category is a Category

With the composition and identities of definition 63.3.1, $\mathbf{C}_T$ is a category.

***Proof*:**
Write $\odot$ for Kleisli composition and $\circ$ for composition in $\mathbf{C}$; naturality of η and μ and the monad laws (definition 63.1.1) are used throughout.

Step 1: Left unit. Let $f : c \to Tc'$ be a Kleisli arrow, so its target identity is $\eta_{c'} : c' \to Tc'$. Then

$$\eta_{c'} \odot f = \mu_{c'} \circ T\eta_{c'} \circ f = (\mu_{c'} \circ T\eta_{c'}) \circ f = \mathrm{id}_{Tc'} \circ f = f$$

using the right unit law $\mu \circ T\eta = \mathrm{id}_T$ of the monad.

Step 2: Right unit. With $\eta_c \colon c \to Tc$ the identity on the source,

$$f \odot \eta_c = \mu_{c'} \circ Tf \circ \eta_c$$

Naturality of $\eta \colon \mathrm{id}_{\mathbf{C}} \Rightarrow T$ applied to $f \colon c \to Tc'$ gives $Tf \circ \eta_c = \eta_{Tc'} \circ f$, so

$$f \odot \eta_c = \mu_{c'} \circ \eta_{Tc'} \circ f = (\mu_{c'} \circ \eta_{Tc'}) \circ f = \mathrm{id}_{Tc'} \circ f = f$$

now using the left unit law $\mu \circ \eta T = \mathrm{id}_T$.

Step 3: Associativity. Let $f \colon c \to Tc'$, $g \colon c' \to Tc''$, and $h \colon c'' \to Tc'''$. Expanding the two bracketings,

$$h \odot (g \odot f) = \mu_{c'''} \circ Th \circ (\mu_{c''} \circ Tg \circ f) \tag{63.4}$$

$$(h \odot g) \odot f = \mu_{c'''} \circ T(\mu_{c'''} \circ Th \circ g) \circ f \tag{63.5}$$

In (63.5), functoriality of T splits $T(h \odot g) = T\mu_{c'''} \circ TTh \circ Tg$, so the right side becomes $\mu_{c'''} \circ T\mu_{c'''} \circ TTh \circ Tg \circ f$. In (63.4), naturality of $\mu \colon TT \Rightarrow T$ applied to $h \colon c'' \to Tc'''$ gives $Th \circ \mu_{c''} = \mu_{Tc'''} \circ TTh$, so the left side becomes $\mu_{c'''} \circ \mu_{Tc'''} \circ TTh \circ Tg \circ f$. The two agree once $\mu_{c'''} \circ T\mu_{c'''} = \mu_{c'''} \circ \mu_{Tc'''}$, which is precisely the associativity law $\mu \circ T\mu = \mu \circ \mu T$ of the monad, evaluated at c'''. Hence (63.4) and (63.5) coincide, and Kleisli composition is associative, completing the proof.

The proof is the monad laws read backward: the two unit laws become the two identity axioms, and the single associativity square becomes associativity of composition. A monad is thus exactly the data needed to make Kleisli arrows compose, which is one sense in which a monad is an "algebraic theory" presented by its effects. A concrete instance makes the effects visible.

Example 63.3.3: The Kleisli Category of the Powerset Monad

Let $P \colon \mathbf{Set} \to \mathbf{Set}$ be the covariant powerset monad (example 63.1.3), with unit $\eta_X \colon X \to PX$ sending x to the singleton $\{x\}$ and multiplication $\mu_X \colon PPX \to PX$ sending a set of subsets to its union. A Kleisli arrow $f \colon X \to PY$ is a function assigning to each $x \in X$ a subset $f(x) \subseteq Y$, that is, a *relation* between X and Y. The Kleisli identity η_X is the diagonal relation $x \mapsto \{x\}$. Given $f \colon X \to PY$ and $g \colon Y \to PZ$, the Kleisli composite is

$$(g \odot f)(x) = \mu_Z\big((Pg)(f(x))\big) = \bigcup_{y \in f(x)} g(y)$$

which is exactly the composite relation: x is related to z if and only if some y has $x \ f \ y$ and $y \ g \ z$. Thus $\mathbf{Set}_P$ is the category $\mathbf{Rel}$ of sets and relations, with relational composition read off the monad structure. The maybe monad $X \mapsto X + 1$ (add a disjoint basepoint) gives instead the category of partial functions, and the free-monoid monad gives functions returning formal words.

Every Kleisli category comes with a canonical adjunction, and it induces the monad one started from. The left adjoint reinterprets an ordinary map as a pure Kleisli arrow, one with trivial effect, and the right adjoint sends an object to the free algebra Tc.

Theorem 63.3.4: The Kleisli Adjunction

Let (T, η, μ) be a monad on $\mathbf{C}$ with Kleisli category $\mathbf{C}_T$. Define functors

$$F_T\colon \mathbf{C} \to \mathbf{C}_T, \qquad U_T\colon \mathbf{C}_T \to \mathbf{C}$$

as follows. The functor F_T is the identity on objects, and sends $f\colon c \to c'$ in $\mathbf{C}$ to the Kleisli arrow $\eta_{c'} \circ f\colon c \to Tc'$. The functor U_T sends an object c to Tc, and a Kleisli arrow $f\colon c \to Tc'$ to the $\mathbf{C}$-morphism $\mu_{c'} \circ Tf\colon Tc \to Tc'$. Then $F_T \dashv U_T$, and the monad induced by this adjunction (proposition 63.1.2) is (T, η, μ).

***Proof*:**
Step 1: F_T and U_T are functors. For F_T: the identity on c in $\mathbf{C}$ goes to $\eta_c \circ \mathrm{id}_c = \eta_c$, the Kleisli identity. For composability, given $f\colon c \to c'$ and $g\colon c' \to c''$ in $\mathbf{C}$, the Kleisli composite of their images is

$$(\eta_{c''} \circ g) \odot (\eta_{c'} \circ f) = \mu_{c''} \circ T(\eta_{c''} \circ g) \circ \eta_{c'} \circ f$$

Naturality of $\eta\colon \mathrm{id}_{\mathbf{C}} \Rightarrow T$ at the morphism $\eta_{c''} \circ g\colon c' \to Tc''$ gives $T(\eta_{c''} \circ g) \circ \eta_{c'} = \eta_{Tc''} \circ \eta_{c''} \circ g$, whence

$$(\eta_{c''} \circ g) \odot (\eta_{c'} \circ f) = \mu_{c''} \circ \eta_{Tc''} \circ \eta_{c''} \circ g \circ f = \eta_{c''} \circ (g \circ f)$$

using $\mu_{c''} \circ \eta_{Tc''} = \mathrm{id}_{Tc''}$ (left unit law). This is $F_T(g \circ f)$, so F_T is a functor. For U_T: the identity η_c goes to $\mu_c \circ T\eta_c = \mathrm{id}_{Tc}$ (right unit law); and U_T preserves composition because, for $f\colon c \to Tc'$ and $g\colon c' \to Tc''$,

$$U_T(g \odot f) = \mu_{c''} \circ T(\mu_{c''} \circ Tg \circ f) = \mu_{c''} \circ T\mu_{c''} \circ TTg \circ Tf$$

while $U_T(g) \circ U_T(f) = \mu_{c''} \circ Tg \circ \mu_{c'} \circ Tf$. Naturality of μ at g gives $Tg \circ \mu_{c'} = \mu_{Tc''} \circ TTg$, and associativity $\mu_{c''} \circ \mu_{Tc''} = \mu_{c''} \circ T\mu_{c''}$ turns the second expression into the first. Thus U_T is a functor.

Step 2: The hom-set bijection. By construction, for objects c of $\mathbf{C}$ and c' of $\mathbf{C}_T$,

$$\mathrm{Hom}_{\mathbf{C}_T}(F_T c, c') = \mathrm{Hom}_{\mathbf{C}_T}(c, c') = \mathrm{Hom}_{\mathbf{C}}(c, Tc') = \mathrm{Hom}_{\mathbf{C}}(c, U_T c') \tag{63.6}$$

the middle equality being the definition of a Kleisli arrow and the last the definition of U_T on objects. Take $\Phi_{c,c'}$ to be this identification.

Step 3: Naturality of the bijection. Naturality in c requires that for $u\colon c_0 \to c$ in $\mathbf{C}$ and a Kleisli arrow $k\colon c \to c'$, precomposing in $\mathbf{C}_T$ with $F_T u$ matches precomposing in $\mathbf{C}$ with u under (63.6). The left side is $k \odot (\eta_c \circ u) = \mu_{c'} \circ Tk \circ \eta_c \circ u$; naturality of η at k gives $Tk \circ \eta_c = \eta_{Tc'} \circ k$, so this is $\mu_{c'} \circ \eta_{Tc'} \circ k \circ u = k \circ u$, the right side. Naturality in c' requires that for a Kleisli arrow $v\colon c' \to c''$, postcomposing in $\mathbf{C}_T$ with v matches postcomposing in $\mathbf{C}$ with $U_T v = \mu_{c''} \circ Tv$. The left side is $v \odot k = \mu_{c''} \circ Tv \circ k = (U_T v) \circ k$, the right side. Thus Φ is natural and $F_T \dashv U_T$.

Step 4: The induced monad is T. The composite $U_T F_T$ sends an object c to $U_T c = Tc$ and a morphism $f\colon c \to c'$ to $U_T(\eta_{c'} \circ f) = \mu_{c'} \circ T(\eta_{c'} \circ f) = \mu_{c'} \circ T\eta_{c'} \circ Tf = Tf$, using the right unit law. Hence $U_T F_T = T$ as an endofunctor. The unit of the adjunction is the transpose of the identity Kleisli arrow on $F_T c$: under (63.6) the identity $\mathrm{id}_{F_T c} = \eta_c \in \mathrm{Hom}_{\mathbf{C}_T}(c, c)$ corresponds to $\eta_c \in \mathrm{Hom}_{\mathbf{C}}(c, Tc)$, so the induced unit is η. The induced multiplication is $U_T \varepsilon F_T$ (proposition 63.1.2), where the counit $\varepsilon_{c'}\colon F_T U_T c' \to c'$ is the transpose of $\mathrm{id}_{U_T c'} = \mathrm{id}_{Tc'}$; under (63.6) at object Tc' this is the Kleisli arrow $\mathrm{id}_{Tc'}\colon Tc' \to Tc'$. Then $U_T \varepsilon_{c'} = \mu_{c'} \circ T\mathrm{id}_{Tc'} = \mu_{c'}$, so the induced multiplication at c' is $\mu_{c'}$. The induced monad is (T, η, μ), as we sought to show.

Together with theorem 63.2.4, the Kleisli adjunction proves the converse promised in section 63.1: every monad arises from an adjunction, and now from at least two. The left adjoint F_T names each object by its free algebra and each map by its pure lift; the counit at c is the identity Kleisli arrow on Tc, which under U_T is μ_c, so the counit *is* the multiplication seen through the underlying category. Two adjunctions inducing the same T raises the question of how they relate, and the answer places $\mathbf{C}_T$ inside $\mathbf{C}^T$.

The bridge is the comparison functor. Every free T-algebra (Tc, μ_c) lives in $\mathbf{C}^T$, and a Kleisli arrow induces a canonical map between free algebras; assembling these identifies $\mathbf{C}_T$ with the free-algebra part of $\mathbf{C}^T$.

Theorem 63.3.5: Kleisli as the Free Algebras

Let (T, η, μ) be a monad on $\mathbf{C}$. The assignment

$$K\colon \mathbf{C}_T \to \mathbf{C}^T$$

sending an object c to the free T-algebra (Tc, μ_c) and a Kleisli arrow $f\colon c \to Tc'$ to the $\mathbf{C}$-morphism $\mu_{c'} \circ Tf\colon Tc \to Tc'$ is a fully faithful functor. Its image is the full subcategory of $\mathbf{C}^T$ on the free algebras, so K identifies $\mathbf{C}_T$ with that subcategory. Moreover, among all adjunctions inducing T, the Kleisli adjunction is initial and the Eilenberg-Moore adjunction (theorem 63.2.4) is terminal.

***Proof*:**

Step 1: K lands in $\mathbf{C}^T$ and is a functor. That (Tc, μ_c) is a T-algebra (definition 63.2.1) is the content of the free-algebra construction: the unit law $\mu_c \circ \eta_{Tc} = \mathrm{id}_{Tc}$ and the associativity law $\mu_c \circ \mu_{Tc} = \mu_c \circ T\mu_c$ are two of the monad laws. For a Kleisli arrow $f\colon c \to Tc'$, the morphism $Kf = \mu_{c'} \circ Tf$ is an algebra map $(Tc, \mu_c) \to (Tc', \mu_{c'})$: the required square $Kf \circ \mu_c = \mu_{c'} \circ T(Kf)$ reads $\mu_{c'} \circ Tf \circ \mu_c = \mu_{c'} \circ T(\mu_{c'} \circ Tf)$, and both sides equal $\mu_{c'} \circ \mu_{Tc'} \circ TTf$ by naturality of μ at f (left) and functoriality of T together with associativity of μ (right). On objects and morphisms K agrees with U_T followed by the free-algebra data, so it preserves identities and composition exactly as U_T does (theorem 63.3.4, Step 1). Thus K is a functor.

Step 2: K is faithful. A Kleisli arrow $f\colon c \to Tc'$ can be recovered from $Kf = \mu_{c'} \circ Tf$ by precomposing with the unit:

$$Kf \circ \eta_c = \mu_{c'} \circ Tf \circ \eta_c = \mu_{c'} \circ \eta_{Tc'} \circ f = f$$

using naturality of η at f and the left unit law $\mu \circ \eta T = \mathrm{id}$. Hence $Kf = Kg$ forces $f = Kf \circ \eta_c = Kg \circ \eta_c = g$, so K is injective on each hom-set.

Step 3: K is full. Let $h\colon (Tc, \mu_c) \to (Tc', \mu_{c'})$ be any algebra map, so $h \circ \mu_c = \mu_{c'} \circ Th$. Set $f = h \circ \eta_c\colon c \to Tc'$, a Kleisli arrow. Then

$$Kf = \mu_{c'} \circ Tf = \mu_{c'} \circ T(h \circ \eta_c) = \mu_{c'} \circ Th \circ T\eta_c$$

The algebra-map condition rewrites $\mu_{c'} \circ Th = h \circ \mu_c$, so

$$Kf = h \circ \mu_c \circ T\eta_c = h \circ \mathrm{id}_{Tc} = h$$

using the right unit law $\mu \circ T\eta = \mathrm{id}$. Thus every algebra map between free algebras is Kf for some Kleisli arrow f, and combined with Step 2, K is fully faithful.

Step 4: The image and the identification. By Steps 2 and 3, K is injective on objects up to the free-algebra assignment and a bijection on each hom-set $\mathrm{Hom}_{\mathbf{C}_T}(c, c') \to \mathrm{Hom}_{\mathbf{C}^T}\big((Tc, \mu_c), (Tc', \mu_{c'})\big)$. Its objects are exactly the free algebras (Tc, μ_c), and fullness says every $\mathbf{C}^T$-morphism between two free algebras lies in the image. Hence K is an isomorphism of $\mathbf{C}_T$ onto the full subcategory of $\mathbf{C}^T$ spanned by the free algebras.

Step 5: Initial and terminal placement. Fix the monad T and let $\mathbf{Adj}(T)$ be the category whose objects are adjunctions inducing T: a category $\mathbf{D}$ with $F \dashv G$, $F\colon \mathbf{C} \to \mathbf{D}$ and $G\colon \mathbf{D} \to \mathbf{C}$, such that $GF = T$ and the induced unit and multiplication (proposition 63.1.2) are η and μ. A morphism from $(\mathbf{D}, F, G)$ to $(\mathbf{D}', F', G')$ is a functor $R\colon \mathbf{D} \to \mathbf{D}'$ with $RF = F'$ and $G'R = G$.

Step 5a: The Kleisli adjunction is initial. Fix a target $(\mathbf{D}', F', G')$ with counit ε'; write $\Phi\colon \mathbf{D}'(F'c, F'c') \to \mathbf{C}(c, G'F'c')$ for the adjunction bijection of $F' \dashv G'$, which sends g to $G'g \circ \eta_c$ (definition 62.1.1), using that the induced unit of $F' \dashv G'$ is the monad unit η. A morphism $R\colon \mathbf{C}_T \to \mathbf{D}'$ must satisfy $RF_T = F'$ and $G'R = U_T$. Since F_T is the identity on objects, $Rc = RF_Tc = F'c$ on objects. For a Kleisli arrow $f\colon c \to Tc'$, note $Tc' = G'F'c'$, so f lies in $\mathbf{C}(c, G'F'c')$; the constraint $G'(Rf) = U_Tf = \mu_{c'} \circ Tf$ determines the transpose of Rf, because

$$\Phi(Rf) = G'(Rf) \circ \eta_c = \mu_{c'} \circ Tf \circ \eta_c = \mu_{c'} \circ \eta_{Tc'} \circ f = f$$

using naturality of η at f and the left unit law $\mu \circ \eta T = \mathrm{id}$. As Φ is a bijection, Rf is forced to be $\Phi^{-1}(f) = \varepsilon'_{F'c'} \circ F'f\colon F'c \to F'c'$. Both the object and the morphism assignment are thus determined, so at most one R exists.

This assignment is a morphism of adjunctions. It is a functor: the Kleisli identity η_c goes to $\varepsilon'_{F'c} \circ F'\eta_c = \mathrm{id}_{F'c}$ by the triangle identity $\varepsilon'F' \circ F'\eta = \mathrm{id}_{F'}$, and for Kleisli arrows $f\colon c \to Tc'$, $g\colon c' \to Tc''$ the two morphisms $R(g \odot f)$ and $Rg \circ Rf$ have the same transpose under Φ: the computation above gives $\Phi(R(g \odot f)) = g \odot f$, while

$$\Phi(Rg \circ Rf) = G'(Rg) \circ G'(Rf) \circ \eta_c = U_Tg \circ (U_Tf \circ \eta_c) = U_Tg \circ f = \mu_{c''} \circ Tg \circ f = g \odot f$$

where $U_Tf \circ \eta_c = \mu_{c'} \circ Tf \circ \eta_c = f$ as before, so $R(g \odot f) = Rg \circ Rf$. The constraint $G'R = U_T$ holds by construction: $G'(Rf) = G'\varepsilon'_{F'c'} \circ G'F'f = \mu_{c'} \circ Tf = U_Tf$, using that the induced multiplication is $\mu = G'\varepsilon'F'$, so $\mu_{c'} = G'\varepsilon'_{F'c'}$. Finally $RF_T = F'$: on a $\mathbf{C}$-morphism $f\colon c \to c'$, the image $F_Tf = \eta_{c'} \circ f$ has $R(F_Tf) = \varepsilon'_{F'c'} \circ F'\eta_{c'} \circ F'f = F'f$ by the same triangle identity. Hence R exists and is unique, and $\mathbf{C}_T$ is initial.

Step 5b: The Eilenberg-Moore adjunction is terminal. Fix a source $(\mathbf{D}, F, G)$ with counit ε. Define $K^{\mathbf{D}}\colon \mathbf{D} \to \mathbf{C}^T$ on an object d by $K^{\mathbf{D}}d = (Gd, G\varepsilon_d)$ and on a morphism $p\colon d \to d'$ by $K^{\mathbf{D}}p = Gp$. That $(Gd, G\varepsilon_d)$ is a T-algebra (definition 63.2.1) uses the two adjunction facts. Its unit law $G\varepsilon_d \circ \eta_{Gd} = \mathrm{id}_{Gd}$ is the triangle identity $G\varepsilon \circ \eta G = \mathrm{id}_G$ at d. Its associativity law reads $G\varepsilon_d \circ \mu_{Gd} = G\varepsilon_d \circ T(G\varepsilon_d)$; since $\mu_{Gd} = G\varepsilon_{FGd}$ and $T(G\varepsilon_d) = GFG\varepsilon_d$, both sides are G applied to $\varepsilon_d \circ \varepsilon_{FGd}$ and $\varepsilon_d \circ FG\varepsilon_d$ respectively, equal by naturality of $\varepsilon\colon FG \Rightarrow \mathrm{id}$ at ε_d. For $p\colon d \to d'$, the morphism Gp is an algebra map $(Gd, G\varepsilon_d) \to (Gd', G\varepsilon_{d'})$: the condition $Gp \circ G\varepsilon_d = G\varepsilon_{d'} \circ T(Gp)$ is G applied to $p \circ \varepsilon_d = \varepsilon_{d'} \circ FGp$, again naturality of ε at p. Being G on morphisms, $K^{\mathbf{D}}$ preserves identities and composition, so it is a functor.

It is a morphism of adjunctions. On objects $U^TK^{\mathbf{D}}d = Gd$ and on morphisms $U^TK^{\mathbf{D}}p = Gp$, so $U^TK^{\mathbf{D}} = G$. For F: on an object c, $K^{\mathbf{D}}(Fc) = (GFc, G\varepsilon_{Fc}) = (Tc, \mu_c) = F^Tc$, using $\mu_c = G\varepsilon_{Fc}$; on a morphism f, $K^{\mathbf{D}}(Ff) = GFf = Tf = F^Tf$, so $K^{\mathbf{D}}F = F^T$.

For uniqueness, let $S\colon \mathbf{D} \to \mathbf{C}^T$ be any morphism of adjunctions, so $SF = F^T$ and $U^TS = G$. The second constraint forces the underlying object of Sd to be $U^T(Sd) = Gd$, so $Sd = (Gd, x_d)$

for some structure map $x_d\colon TGd \to Gd$; only x_d remains to be pinned. Apply S to the counit $\varepsilon_d\colon FGd \to d$: since $SF = F^T$, the source is $SFGd = F^T(Gd) = (TGd, \mu_{Gd})$, so $S\varepsilon_d$ is an algebra map $(TGd, \mu_{Gd}) \to (Gd, x_d)$ whose underlying $\mathbf{C}$-morphism is $U^T(S\varepsilon_d) = G\varepsilon_d$. Under the free-forgetful bijection of theorem 63.2.4, an algebra map out of the free algebra (TGd, μ_{Gd}) equals $x_d \circ T(-)$ applied to its restriction along η_{Gd}; that restriction is $G\varepsilon_d \circ \eta_{Gd} = \mathrm{id}_{Gd}$ by the triangle identity, so

$$G\varepsilon_d = x_d \circ T(\mathrm{id}_{Gd}) = x_d$$

Hence $x_d = G\varepsilon_d$ and $Sd = (Gd, G\varepsilon_d) = K^{\mathbf{D}}d$ on objects. On a morphism p, $U^T(Sp) = Gp = U^T(K^{\mathbf{D}}p)$, and U^T is faithful (a $\mathbf{C}^T$-morphism is determined by its underlying $\mathbf{C}$-morphism, definition 63.2.3), so $Sp = K^{\mathbf{D}}p$. Thus $S = K^{\mathbf{D}}$, and $\mathbf{C}^T$ is terminal.

Composing the two extremes, every $(\mathbf{D}, F, G)$ inducing T receives the unique morphism $R\colon \mathbf{C}_T \to \mathbf{D}$ from $\mathbf{C}_T$ and admits the unique morphism $K^{\mathbf{D}}\colon \mathbf{D} \to \mathbf{C}^T$ to $\mathbf{C}^T$, so the comparison factors as $\mathbf{C}_T \to \mathbf{D} \to \mathbf{C}^T$. Taking $\mathbf{D} = \mathbf{C}^T$ shows the composite $K^{\mathbf{C}^T} \circ R\colon \mathbf{C}_T \to \mathbf{C}^T$ is the unique morphism $\mathbf{C}_T \to \mathbf{C}^T$, which is the K of Steps 1 to 4. This establishes the initial and terminal placement, completing the proof.

The theorem exhibits $\mathbf{C}_T$ and $\mathbf{C}^T$ as the two ends of a spectrum. Every construction of T from an adjunction factors as $\mathbf{C}_T \to \mathbf{D} \to \mathbf{C}^T$: the Kleisli category adjoins nothing beyond the free algebras, the Eilenberg-Moore category adjoins every algebra, and any other realization sits between. When T is the free-monoid monad, $\mathbf{C}_T$ is the category whose objects are sets and whose maps $X \to Y$ are functions $X \to (\text{words in } Y)$, while $\mathbf{C}^T = \mathbf{Mon}$; the inclusion $\mathbf{C}_T \hookrightarrow \mathbf{Mon}$ is the free monoids sitting inside all monoids. Fullness on the free part is what makes $\mathbf{C}_T$ an object of study rather than a bookkeeping device: a Kleisli arrow and the algebra map it induces carry the same information, so one may compute with whichever is convenient. When the comparison functor $\mathbf{D} \to \mathbf{C}^T$ is not just unique but an equivalence, the category $\mathbf{D}$ is $\mathbf{C}^T$ up to equivalence, so the adjunction recovers the full algebraic content of T; whether a given forgetful functor has this property is the question of *monadicity*.

63.4 Monadicity

The previous sections travel one direction: every monad T on $\mathbf{C}$ produces a category of algebras $\mathbf{C}^T$, and the forgetful functor $U^T\colon \mathbf{C}^T \to \mathbf{C}$ carries the monad back (definition 63.2.3, theorem 63.2.4). This section runs the converse. Given a functor $G\colon \mathbf{D} \to \mathbf{C}$ with a left adjoint, G remembers a monad $T = GF$ on $\mathbf{C}$; the question is whether $\mathbf{D}$ is nothing more than the algebras for that monad. When it is, G is *monadic*: the objects of $\mathbf{D}$ are algebraic structures on their images, and G forgets exactly the operations that a T-algebra structure records. Beck's theorem gives the exact test.

Fix an adjunction $F \dashv G$ with $F\colon \mathbf{C} \to \mathbf{D}$, $G\colon \mathbf{D} \to \mathbf{C}$, unit η, and counit ε. By proposition 63.1.2 this induces the monad $T = GF$ on $\mathbf{C}$, with unit $\eta\colon \mathrm{id}_C \Rightarrow T$ and multiplication $\mu = G\varepsilon F$. Every object d of $\mathbf{D}$ then carries a canonical T-algebra structure on its image Gd: the counit component $\varepsilon_d\colon FGd \to d$ has image $G\varepsilon_d\colon TGd = GFGd \to Gd$, and the triangle identities and naturality of the counit make $(Gd, G\varepsilon_d)$ a T-algebra. This assignment is functorial, and it is the object of study.

Definition 63.4.1: Monadic Functor

Let $G\colon \mathbf{D} \to \mathbf{C}$ have a left adjoint $F \dashv G$, and let $T = GF$ be the induced monad on $\mathbf{C}$. The *comparison functor* is

$$K\colon \mathbf{D} \to \mathbf{C}^T, \qquad K(d) = (Gd,\, G\varepsilon_d), \qquad K(h\colon d \to d') = Gh$$

into the Eilenberg-Moore category of T. The functor G is *monadic* if it has a left adjoint and the comparison functor K is an equivalence of categories.

The comparison functor always exists and always commutes with the forgetful functors: $U^T K = G$, since $U^T(Gd, G\varepsilon_d) = Gd$. So K factors G through the algebras, presenting $\mathbf{D}$ as sitting over $\mathbf{C}^T$. Monadicity is the requirement that this factorization lose nothing, that K be an equivalence. When it is, $\mathbf{D}$ is, up to equivalence, precisely $\mathbf{C}^T$: every object of $\mathbf{D}$ is a T-algebra, every T-algebra comes from an object of $\mathbf{D}$, and morphisms match. The functor G is then a bona fide forgetful functor, and T is the complete record of the structure it forgets.

Before stating Beck's test, it helps to see why a naive expectation fails. One might hope that any G with a left adjoint is monadic. It is not: the free-forgetful adjunction between **Set** and **Top** has $GF = \mathrm{id}_{\mathbf{Set}}$ (the free topology on a set is discrete, whose underlying set is the set again), so T is the identity monad, $\mathbf{Set}^T \cong \mathbf{Set}$, and $K\colon \mathbf{Top} \to \mathbf{Set}$ is the forgetful functor, which is not an equivalence. A topology is not encoded by any operations on the underlying set. Beck's theorem isolates the exact structure whose presence certifies monadicity.

The certifying structure is a condition on how G interacts with certain coequalizers (definition 61.3.5). A parallel pair $f, g\colon d \rightrightarrows d'$ in $\mathbf{D}$ is a *G-split pair* if its image $Gf, Gg\colon Gd \rightrightarrows Gd'$ admits a split coequalizer in $\mathbf{C}$: a coequalizer $q\colon Gd' \to c$ together with sections $s\colon c \to Gd'$ and $t\colon Gd' \to Gd$ satisfying $qs = \mathrm{id}_c$, $Gf\,t = \mathrm{id}_{Gd'}$, and $Gg\,t = sq$. A split coequalizer is preserved by every functor, because the equations defining it are equational: applying any functor to q, s, t produces the split coequalizer of the image pair. This is the property that makes the condition usable, and it is why the coequalizers Beck asks for are exactly the G-split ones.

Theorem 63.4.2: Beck's Monadicity Theorem

A functor $G\colon \mathbf{D} \to \mathbf{C}$ is monadic if and only if

1. G has a left adjoint, and
2. G creates coequalizers of G-split pairs (definition 61.4.6): for every G-split pair $f, g\colon d \rightrightarrows d'$ in $\mathbf{D}$, the coequalizer cocone of Gf, Gg in $\mathbf{C}$ has a unique lift along G to a cocone on f, g in $\mathbf{D}$, and that lifted cocone is a coequalizer.

***Proof Key ideas*:**
The full proof is a long diagram bookkeeping (the *precise* and *crude* monadicity theorems in the literature); it is deferred to Mac Lane's *Categories for the Working Mathematician* (VI.7) and Riehl's *Category Theory in Context* (5.5), which is why only the ideas are recorded here (BOOK.md §6 sanctions this deferral). Three points carry the argument.

The canonical presentation of a T-algebra. Every T-algebra (A, a) is a coequalizer of free algebras.

The free algebras (T^2A, μ_{TA}) and (TA, μ_A) sit in a parallel pair

$$(T^2A,\, \mu_{TA}) \rightrightarrows (TA,\, \mu_A)$$

with legs μ_A and Ta, and (A, a) is its coequalizer in $\mathbf{C}^T$. Applying U^T gives the pair $\mu_A, Ta\colon T^2A \rightrightarrows TA$ in $\mathbf{C}$, which is split by η_{TA} and η_A: the algebra axioms $a \circ \eta_A = \mathrm{id}_A$ and $a \circ \mu_A = a \circ Ta$ supply the section and coequalizing equations, the remaining splitting data coming from the monad unit laws and the naturality of η. So every T-algebra is a U^T-split pair's coequalizer, and it is its own free presentation.

Constructing the inverse to K. The comparison functor K has a left adjoint $L\colon \mathbf{C}^T \to \mathbf{D}$ built by forming these coequalizers in $\mathbf{D}$. Given (A, a), transport its free presentation $\mu_A, Ta\colon T^2A \rightrightarrows TA$ through F to the parallel pair $FTA \rightrightarrows FA$ in $\mathbf{D}$ (the legs are $F\mu_A$ and the composite FTa followed by ε_{FA}, the transported free presentation); this pair is G-split, its splitting descending from the split coequalizer of μ_A, Ta established above. Condition (2) creates the coequalizer of this pair in $\mathbf{D}$, and $L(A, a)$ is defined to be that coequalizer. Creation makes L well-defined and functorial.

Why the G-split condition forces an equivalence. The unit and counit of $L \dashv K$ are the comparison maps between an object and the coequalizer of its canonical presentation. On the algebra side, $KL(A, a) \cong (A, a)$ because K preserves the created coequalizer (split coequalizers are absolute, preserved by every functor) and (A, a) is the coequalizer of its own presentation. On the $\mathbf{D}$ side, $LK(d) \cong d$ because d is the coequalizer of the G-split pair $\varepsilon_{FGd}, FG\varepsilon_d\colon FGFGd \rightrightarrows FGd$ (the presentation of d), and creation delivers exactly that object. Thus K and L are mutually inverse up to isomorphism, so K is an equivalence and G is monadic. Conversely, if G is monadic then G inherits condition (2) from U^T, which creates these coequalizers by the free-presentation computation above, completing the proof.

The theorem trades the global requirement "K is an equivalence" for a local, checkable one: create the coequalizers of G-split pairs. The mechanism is the canonical presentation. Every algebra is the coequalizer of free algebras built from it, and G knows the free objects because it has a left adjoint; the only thing G needs to supply on its own is these coequalizers, lifted uniquely from $\mathbf{C}$. Where G can build them, $\mathbf{D}$ reassembles from the free part exactly as the algebras do, and the two categories coincide. This is why the condition is about coequalizers and not general colimits: the presentation of an algebra is a single reflexive coequalizer, so that one shape suffices.

The payoff is that the standard forgetful functors of algebra pass the test, and this is the precise content of the slogan that groups, rings, and modules are algebraic structures.

Example 63.4.3: Groups, Rings, and Modules are Monadic over Set

The forgetful functors

$$U\colon \mathbf{Grp} \to \mathbf{Set}, \qquad U\colon \mathbf{Ring} \to \mathbf{Set}, \qquad U\colon \mathbf{Mod}_R \to \mathbf{Set}$$

are each monadic. Take $U\colon \mathbf{Grp} \to \mathbf{Set}$; the other two are identical in form.

The left adjoint is the free-group functor F, so condition (1) holds by the free-forgetful adjunction ([25]). The induced monad $T = UF$ sends a set X to the underlying set of the free group on X, and its algebras are groups: a T-algebra structure on a set is precisely a choice of multiplication, inverse, and identity satisfying the group axioms, recovered from the operations that the free-group monad encodes ([25] for the free group and the group axioms). So the comparison functor $K\colon \mathbf{Grp} \to \mathbf{Set}^T$ is already the identification of groups with T-algebras.

For condition (2), let $f, g\colon G \rightrightarrows H$ be group homomorphisms whose underlying maps Uf, Ug admit a split coequalizer of sets. The coequalizer of Uf, Ug in **Set** is the quotient of UH by the equivalence relation generated by $Uf(x) \sim Ug(x)$. The split hypothesis is what guarantees this set-level quotient is compatible with the operations, so a group structure descends onto it: under that hypothesis the set-level quotient coincides with UH modulo the normal subgroup generated by $\{f(x)g(x)^{-1}\}$, which is the coequalizer of f, g in **Grp**. So the quotient carries a unique group structure making q a homomorphism, and with that structure it is the coequalizer in **Grp**. The lift exists, is unique, and is a coequalizer, so U creates the coequalizer. Both conditions hold, and Beck's theorem gives monadicity. The rings and modules cases run the same way, with the relevant quotient (by an ideal, by a submodule) descending along the split coequalizer. This is what "algebraic over **Set**" means: the category is the algebras for a monad on **Set** that its forgetful functor remembers.

By contrast, $U\colon \mathbf{Top} \to \mathbf{Set}$ is not monadic. As noted above, the free topology on a set is the discrete one, whose underlying set is the original set, so the induced monad is the identity and $\mathbf{Set}^T \cong \mathbf{Set}$; the comparison functor is U itself, sending a space to its point-set and collapsing all topologies on a fixed set to one object. No monad on **Set** has topological spaces as its algebras, because a topology is not a system of finitary operations on the underlying set. Beck's theorem locates the failure precisely in condition (2): U does *not* create coequalizers of U-split pairs. Given a U-split pair with set-coequalizer $q\colon UY \to Q$, the lift is not unique. The final topology on Q (the quotient topology) does make q a coequalizer in **Top**, but every topology on Q coarser than the final one also makes q continuous, so the coequalizer cocone admits many lifts along U, only one of which is a coequalizer. Creation requires a *unique* lift, and uniqueness fails. This is the same multiplicity-of-topologies phenomenon that also breaks conservativity: U does not reflect isomorphisms, because a continuous bijection from a finer to a coarser topology has underlying map a set-isomorphism yet is not a homeomorphism, and a monadic functor is conservative. Both failures trace to one fact: many topologies sit over a single set, and U cannot tell them apart. The dividing line Beck's theorem draws is exactly the line between structure recorded by operations and structure, like a topology, that is not.

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Kan Extensions

Given a functor $F\colon \mathbf{C} \to \mathbf{E}$ and a functor $K\colon \mathbf{C} \to \mathbf{D}$, a *Kan extension* extends F along K universally, in a left and a right variant. This is the last construction the book needs: limits, colimits, adjunctions, and the Yoneda lemma are all Kan extensions. This chapter defines the two Kan extensions as adjoints to restriction, computes them by a pointwise (co)limit formula, derives that slogan, and closes with the calculus of ends and coends and the density theorem, which writes every presheaf as a colimit of the representables the Yoneda embedding produced.

64.1 Left and Right Kan Extensions

A functor $F\colon \mathbf{C} \to \mathbf{E}$ records data indexed by the objects and morphisms of $\mathbf{C}$. Given a second functor $K\colon \mathbf{C} \to \mathbf{D}$, one may ask to transport that data along K: to produce a functor on $\mathbf{D}$ that agrees with F on the image of K, and does so in the best possible way. When K is an inclusion this is a genuine extension of F from a subcategory to the whole of $\mathbf{D}$; in general K need not be injective on objects, so the word "extension" is optimistic, but the universal formulation survives intact. Two such best approximations exist, one approximating F from the left and one from the right, and each is characterized by a universal property. This section defines them and shows that the two universal properties are precisely the two halves of a chain of adjunctions with restriction along K.

Throughout, fix $K\colon \mathbf{C} \to \mathbf{D}$. Restriction along K is the functor

$$K^*\colon [\mathbf{D}, \mathbf{E}] \to [\mathbf{C}, \mathbf{E}], \qquad H \mapsto HK$$

sending a functor $H\colon \mathbf{D} \to \mathbf{E}$ to the composite $HK\colon \mathbf{C} \to \mathbf{E}$ and a natural transformation $\theta\colon H \Rightarrow H'$ to its whiskering θK (the natural transformation with component θ_{Kc} at c). Left and right Kan extension will turn out to be the two adjoints of K^*.

Definition 64.1.1: Left and Right Kan Extension

Let $K\colon \mathbf{C} \to \mathbf{D}$ and $F\colon \mathbf{C} \to \mathbf{E}$ be functors. A *left Kan extension* of F along K is a functor

$$\operatorname{Lan}_K F\colon \mathbf{D} \to \mathbf{E}$$

together with a natural transformation $\eta\colon F \Rightarrow (\operatorname{Lan}_K F)K$, the *unit*, that is universal from F to K^*: for every functor $G\colon \mathbf{D} \to \mathbf{E}$ and every natural transformation $\alpha\colon F \Rightarrow GK$, there is a unique natural transformation $\beta\colon \operatorname{Lan}_K F \Rightarrow G$ with

$$(\beta K) \circ \eta = \alpha$$

where βK is the whiskering of β by K.

Dually, a *right Kan extension* of F along K is a functor $\operatorname{Ran}_K F\colon \mathbf{D} \to \mathbf{E}$ with a natural transformation $\varepsilon\colon (\operatorname{Ran}_K F)K \Rightarrow F$, the *counit*, universal to F from K^*: for every $G\colon \mathbf{D} \to \mathbf{E}$ and every $\alpha\colon GK \Rightarrow F$, there is a unique $\beta\colon G \Rightarrow \operatorname{Ran}_K F$ with $\varepsilon \circ (\beta K) = \alpha$.

The two triangles record the same shape with the 2-cell pointing opposite ways. For the left Kan extension the unit fills the extension triangle

$$\begin{tikzcd}
\mathbf{C} \arrow[r, "F"] \arrow[d, "K"'] \arrow[dr, phantom, "\Searrow\eta"] & \mathbf{E} \\
\mathbf{D} \arrow[ur, "\operatorname{Lan}_K F"']
\end{tikzcd}
\qquad
\begin{tikzcd}
\mathbf{C} \arrow[r, "F"] \arrow[d, "K"'] \arrow[dr, phantom, "\varepsilon\Nwarrow"] & \mathbf{E} \\
\mathbf{D} \arrow[ur, "\operatorname{Ran}_K F"']
\end{tikzcd}$$

and the universal property says every competing pair (G, α) factors through $(\operatorname{Lan}_K F, \eta)$ by a unique 2-cell β. This is the defining shape of a universal arrow from F into the functor K^*, and universal objects are unique up to unique isomorphism. Applied here: if $(\operatorname{Lan}_K F, \eta)$ and (L', η') both left-Kan-extend F along K, the universal property of each produces a comparison, and the uniqueness clause forces the two comparisons to be mutually inverse. So the left Kan extension, when it exists, is determined up to a unique natural isomorphism compatible with the units; likewise the right Kan extension.

Existence, on the other hand, is not automatic: for a fixed K and F there may be no functor on $\mathbf{D}$ carrying a universal unit, and whether one exists depends on $\mathbf{E}$ having enough (co)limits. That question is settled in section 64.2, which constructs the extensions explicitly as (co)limits and pins down exactly when they exist. The present section treats the extensions where they do exist and extracts the structural consequence: the universal property of definition 64.1.1 is an adjunction.

Before the adjunction, one worked example, to see that "extension" is sometimes literal. It also isolates the mechanism, a coproduct that copies Fc once for each arrow $Kc \to d$, that section 64.2 will generalize into the pointwise formula.

Example 64.1.2: Extension Along a Coproduct Inclusion

Let $\mathbf{C}$ be the discrete category on one object $*$, so a functor $F\colon \mathbf{C} \to \mathbf{E}$ is the choice of a single object $e = F*$ of $\mathbf{E}$. Let $\mathbf{D}$ be the discrete category on a two-element set $\{0, 1\}$, and let $K\colon \mathbf{C} \to \mathbf{D}$ pick out the object 0. Assume $\mathbf{E}$ has an initial object and binary coproducts.

A functor $G\colon \mathbf{D} \to \mathbf{E}$ is a pair of objects $(G0, G1)$, and GK is the single object $G0$; a natural transformation $F \Rightarrow GK$ is a single morphism $e \to G0$. The left Kan extension is

$$(\mathrm{Lan}_K F)(0) = e, \qquad (\mathrm{Lan}_K F)(1) = 0_{\mathbf{E}}$$

where $0_{\mathbf{E}}$ is the initial object, with unit $\eta\colon e \to e$ the identity. Given $\alpha\colon e \to G0$, the unique $\beta\colon \mathrm{Lan}_K F \Rightarrow G$ satisfying $(\beta K) \circ \eta = \alpha$ has component $\beta_0 = \alpha$ at 0, forced by the equation, and component $\beta_1\colon 0_{\mathbf{E}} \to G1$ the unique map out of the initial object. So the extension copies F where K lands and fills the rest with the initial object, the "empty" value that imposes no constraint. The right Kan extension is the mirror image: $(\mathrm{Ran}_K F)(0) = e$ and $(\mathrm{Ran}_K F)(1) = 1_{\mathbf{E}}$, the terminal object.

The mechanism sharpens when K is not injective. Take $\mathbf{C}$ discrete on $\{a, b\}$ and $\mathbf{D} = \mathbf{1}$ the terminal category, so K sends both objects to the single object $*$. A functor F is a pair (Fa, Fb) of objects of $\mathbf{E}$; a natural transformation $F \Rightarrow GK$ is a pair of maps $Fa \to G*$, $Fb \to G*$ into the one object $G*$. Such a pair is exactly a map out of the coproduct $Fa \sqcup Fb$, so

$$(\mathrm{Lan}_K F)(*) = Fa \sqcup Fb$$

with unit the two coprojections. This is the colimit of F, computed by hand. That a left Kan extension along the functor to $\mathbf{1}$ is a colimit is the first of the identifications developed in section 64.3.

The example shows the universal property doing local work; the following theorem shows it is global. Read across all F at once, the correspondence $\alpha \leftrightarrow \beta$ of definition 64.1.1 is exactly the defining bijection of an adjunction between Lan_K and K^*.

Theorem 64.1.3: Kan Extensions as Adjoints to Restriction

Fix $K\colon \mathbf{C} \to \mathbf{D}$ and a category $\mathbf{E}$. Suppose the left Kan extension $\mathrm{Lan}_K F$ exists for every $F\colon \mathbf{C} \to \mathbf{E}$. Then $F \mapsto \mathrm{Lan}_K F$ is the object part of a functor $\mathrm{Lan}_K\colon [\mathbf{C}, \mathbf{E}] \to [\mathbf{D}, \mathbf{E}]$ that is left adjoint to restriction,

$$\mathrm{Lan}_K \dashv K^*$$

Dually, if $\mathrm{Ran}_K F$ exists for every F, then Ran_K is right adjoint to K^*, so that

$$\mathrm{Lan}_K \dashv K^* \dashv \mathrm{Ran}_K$$

The unit η of definition 64.1.1 is the unit of the first adjunction, and the counit ε is the counit of the second.

***Proof*:**
The two claims are dual (box 59.3.3): a right Kan extension of F along K is a left Kan extension of F^{op} along K^{op} into $\mathbf{E}^{\mathrm{op}}$, and Ran_K $\dashv$-ness for $\mathbf{E}$ is $\mathrm{Lan}_{K^{\mathrm{op}}}$ $\dashv$-ness for $\mathbf{E}^{\mathrm{op}}$. We prove the left case; the right case follows by replacing every category by its opposite and reversing every 2-cell.

Step 1: the hom-set bijection. Fix $F\colon \mathbf{C} \to \mathbf{E}$ and let $(\mathrm{Lan}_K F, \eta_F)$ be its left Kan extension. For any $G\colon \mathbf{D} \to \mathbf{E}$, the universal property of definition 64.1.1 states that the assignment

$$[\mathbf{D}, \mathbf{E}](\mathrm{Lan}_K F, G) \longrightarrow [\mathbf{C}, \mathbf{E}](F, K^*G), \qquad \beta \longmapsto (\beta K) \circ \eta_F \tag{64.1}$$

is a bijection: existence of a unique β for each α on the right is exactly the statement that this map is bijective, with inverse sending α to the induced β. Here $[\mathbf{D}, \mathbf{E}]$ and $[\mathbf{C}, \mathbf{E}]$ are the functor categories of definition 59.5.4 and the elements are natural transformations in the sense of definition 59.5.2.

Step 2: Lan_K *is a functor.* Given $\varphi\colon F \Rightarrow F'$ in $[\mathbf{C}, \mathbf{E}]$, define $\operatorname{Lan}_K \varphi\colon \operatorname{Lan}_K F \Rightarrow \operatorname{Lan}_K F'$ to be the unique natural transformation β obtained by applying the universal property of $(\operatorname{Lan}_K F, \eta_F)$ to the competitor

$$F \xRightarrow{\varphi} F' \xRightarrow{\eta_{F'}} (\operatorname{Lan}_K F')K$$

that is, $\operatorname{Lan}_K \varphi$ is the unique 2-cell with $((\operatorname{Lan}_K \varphi)\, K) \circ \eta_F = \eta_{F'} \circ \varphi$. Uniqueness in the universal property gives functoriality directly: for $\varphi = \mathrm{id}_F$ the transformation $\mathrm{id}_{\operatorname{Lan}_K F}$ satisfies the defining equation, so $\operatorname{Lan}_K \mathrm{id}_F = \mathrm{id}$; and for $F \xRightarrow{\varphi} F' \xRightarrow{\psi} F''$ both $\operatorname{Lan}_K(\psi \circ \varphi)$ and $(\operatorname{Lan}_K \psi) \circ (\operatorname{Lan}_K \varphi)$ solve the universal problem for $\eta_{F''} \circ \psi \circ \varphi$, hence coincide.

Step 3: naturality of the bijection. An adjunction is a bijection (64.1) natural in both variables (definition 62.1.1). Naturality in G: for $\gamma\colon G \Rightarrow G'$ in $[\mathbf{D}, \mathbf{E}]$ and $\beta\colon \operatorname{Lan}_K F \Rightarrow G$, post-composing then restricting gives $((\gamma \circ \beta)K) \circ \eta_F = (\gamma K) \circ (\beta K) \circ \eta_F$, since whiskering by K preserves vertical composition; this is exactly $K^*\gamma$ applied to the image of β, so the square commutes. Naturality in F: for $\varphi\colon F' \Rightarrow F$ and $\beta\colon \operatorname{Lan}_K F \Rightarrow G$, one must check

$$\big((\beta \circ \operatorname{Lan}_K \varphi)\, K\big) \circ \eta_{F'} = \big((\beta K) \circ \eta_F\big) \circ \varphi$$

The left side is $(\beta K) \circ \big((\operatorname{Lan}_K \varphi)\, K\big) \circ \eta_{F'}$, and by the defining equation of $\operatorname{Lan}_K \varphi$ from Step 2 the tail $\big((\operatorname{Lan}_K \varphi)\, K\big) \circ \eta_{F'}$ equals $\eta_F \circ \varphi$; substituting gives $(\beta K) \circ \eta_F \circ \varphi$, the right side. So (64.1) is natural in both variables, and $\operatorname{Lan}_K \dashv K^*$.

Step 4: the unit. Under the correspondence of definition 62.1.3, the unit of $\operatorname{Lan}_K \dashv K^*$ at F is the image of $\mathrm{id}_{\operatorname{Lan}_K F}$ under (64.1), namely $(\mathrm{id}\, K) \circ \eta_F = \eta_F$. So the unit of the adjunction is the unit η of definition 64.1.1, as claimed. Dualizing, the counit of $K^* \dashv \operatorname{Ran}_K$ is the counit ε, completing the proof.

The theorem reorganizes the definition rather than adding to it: the phrase "universal among functors restricting to something receiving F" is the phrase "left adjoint to restriction", once one reads the universal property as a natural hom-set bijection. This is the productive form of the definition. Adjoints are unique when they exist, which re-derives the uniqueness noted after definition 64.1.1; a right adjoint preserves limits (theorem 62.3.4) and, dually, a left adjoint preserves colimits, so Lan_K and Ran_K inherit exactness properties for free; and composites of Kan extensions along composable functors become composites of adjoints. The hypothesis, that the extension exist for *every* F, is what upgrades the object-level universal property into a functor and hence an adjunction. When only some F admit an extension there is no global adjoint, only the pointwise universal arrows of definition 64.1.1.

64.2 The Pointwise Formula

The adjoint characterization of theorem 64.1.3 shows that Kan extensions exist, but it does not say what they are: knowing that Lan_K is left adjoint to restriction gives no recipe for the value $(\operatorname{Lan}_K F)(d)$ at a particular object d of $\mathbf{D}$. This section supplies the recipe. When the target category

E has enough colimits, the left Kan extension is computed one object at a time as a colimit, and dually the right Kan extension as a limit. The indexing shape of that colimit is a comma category, so the formula reduces the abstract extension to a concrete diagram assembled out of the values of F.

Fix functors $K\colon \mathbf{C} \to \mathbf{D}$ and $F\colon \mathbf{C} \to \mathbf{E}$. For an object d of $\mathbf{D}$, the relevant indexing category is the comma category $(K \downarrow d)$ of definition 61.1.6: its objects are pairs (c, φ) with c an object of $\mathbf{C}$ and $\varphi\colon Kc \to d$ a morphism of $\mathbf{D}$, and a morphism $(c, \varphi) \to (c', \varphi')$ is a morphism $f\colon c \to c'$ of $\mathbf{C}$ with $\varphi' \circ Kf = \varphi$. Projecting to the first coordinate gives a functor

$$\Pi_d\colon (K \downarrow d) \to \mathbf{C}, \qquad (c, \varphi) \mapsto c, \quad f \mapsto f$$

and composing with F gives the diagram $F \circ \Pi_d\colon (K \downarrow d) \to \mathbf{E}$ whose colimit will be the value at d. The dual indexing category $(d \downarrow K)$ has objects $(c, \psi\colon d \to Kc)$ with the projection $\Pi^d\colon (d \downarrow K) \to \mathbf{C}$ defined the same way.

The colimit of $F \circ \Pi_d$ carries more structure than its value at a single d: as d varies it assembles into a functor, and that functor is the Kan extension. The next theorem proves this, and in doing so exhibits the universal 2-cell of definition 64.1.1 directly from the colimit injections.

Theorem 64.2.1: The Pointwise Formula for Kan Extensions

Let $K\colon \mathbf{C} \to \mathbf{D}$ and $F\colon \mathbf{C} \to \mathbf{E}$ be functors, and suppose that for every object d of $\mathbf{D}$ the colimit of $F \circ \Pi_d\colon (K \downarrow d) \to \mathbf{E}$ exists. Then the left Kan extension $\operatorname{Lan}_K F$ exists and is computed pointwise by

$$(\operatorname{Lan}_K F)(d) \cong \operatorname{colim}_{(K\downarrow d)} F \circ \Pi_d$$

Dually, if for every d the limit of $F \circ \Pi^d\colon (d \downarrow K) \to \mathbf{E}$ exists, then the right Kan extension $\operatorname{Ran}_K F$ exists and is computed pointwise by

$$(\operatorname{Ran}_K F)(d) \cong \lim_{(d\downarrow K)} F \circ \Pi^d$$

***Proof*:**
The right Kan extension case follows from the left by duality (box 59.3.3), so the proof concentrates on $\operatorname{Lan}_K F$. The argument has four steps: define the object function on $\mathbf{D}$, extend it to a functor $L\colon \mathbf{D} \to \mathbf{E}$, build a natural transformation $\eta\colon F \Rightarrow L \circ K$, and verify that (L, η) satisfies the universal property of the left Kan extension.

Step 1: The value at each object. For each object d of $\mathbf{D}$, write

$$Ld = \operatorname{colim}_{(K\downarrow d)} F \circ \Pi_d$$

which exists by hypothesis. The colimit comes with injection morphisms

$$\iota^d_{(c,\varphi)}\colon Fc \to Ld$$

one for each object (c, φ) of $(K \downarrow d)$, and these form a cocone: for a morphism $f\colon (c, \varphi) \to (c', \varphi')$ in $(K \downarrow d)$ one has $\iota^d_{(c',\varphi')} \circ Ff = \iota^d_{(c,\varphi)}$. The colimit is universal among such cocones: a family of morphisms $\lambda_{(c,\varphi)}\colon Fc \to X$ compatible with all the Ff factors uniquely through a single morphism $Ld \to X$.

Step 2: Functoriality in d. A morphism $g\colon d \to d'$ in $\mathbf{D}$ induces a functor

$$g_*\colon (K \downarrow d) \to (K \downarrow d'), \qquad (c, \varphi) \mapsto (c, g \circ \varphi)$$

which acts as the identity on the underlying morphisms of $\mathbf{C}$, so that $\Pi_{d'} \circ g_* = \Pi_d$ and hence $F \circ \Pi_{d'} \circ g_* = F \circ \Pi_d$. The injections for d' therefore restrict along g_* to a cocone on $F \circ \Pi_d$ with vertex Ld', whose leg at (c, φ) is $\iota^{d'}_{(c,g\circ\varphi)} \colon Fc \to Ld'$. By the universal property of Ld this cocone factors through a unique morphism

$$Lg \colon Ld \to Ld'$$

characterized by $Lg \circ \iota^{d}_{(c,\varphi)} = \iota^{d'}_{(c,g\circ\varphi)}$ for every (c, φ). Functoriality is now a uniqueness argument: both $L(\mathrm{id}_d)$ and id_{Ld} satisfy the defining equation for $g = \mathrm{id}_d$, so they agree, and for composable $g \colon d \to d'$, $h \colon d' \to d''$ both $L(h \circ g)$ and $Lh \circ Lg$ send $\iota^{d}_{(c,\varphi)}$ to $\iota^{d''}_{(c,(h\circ g)\circ\varphi)}$, so they agree by the uniqueness clause of the universal property. Thus $L \colon \mathbf{D} \to \mathbf{E}$ is a functor.

Step 3: The universal 2*-cell.* An object c of $\mathbf{C}$ gives a canonical object (c, id_{Kc}) of $(K \downarrow Kc)$, and the injection there defines a morphism

$$\eta_c = \iota^{Kc}_{(c,\mathrm{id}_{Kc})} \colon Fc \longrightarrow L(Kc)$$

To see that $\eta \colon F \Rightarrow L \circ K$ is natural, let $f \colon c \to c'$ be a morphism of $\mathbf{C}$. The map f is a morphism $(c, Kf) \to (c', \mathrm{id}_{Kc'})$ in $(K \downarrow Kc')$, since $\mathrm{id}_{Kc'} \circ Kf = Kf$, so the cocone condition for $L(Kc')$ gives

$$\iota^{Kc'}_{(c',\mathrm{id}_{Kc'})} \circ Ff = \iota^{Kc'}_{(c,Kf)}$$

On the other hand $Kf \colon Kc \to Kc'$ acts by Step 2 with $(Kf)_*(c, \mathrm{id}_{Kc}) = (c, Kf)$, so

$$L(Kf) \circ \eta_c = L(Kf) \circ \iota^{Kc}_{(c,\mathrm{id}_{Kc})} = \iota^{Kc'}_{(c,Kf)}$$

Comparing the two displays, $L(Kf) \circ \eta_c = \eta_{c'} \circ Ff$, which is exactly the naturality square for η.

Step 4: The universal property. It remains to show that (L, η) is initial among pairs (G, γ) consisting of a functor $G \colon \mathbf{D} \to \mathbf{E}$ and a natural transformation $\gamma \colon F \Rightarrow G \circ K$; this is the universal property of definition 64.1.1. Given such a pair, a natural transformation $\alpha \colon L \Rightarrow G$ satisfies $(\alpha K) \circ \eta = \gamma$ if and only if, for every c,

$$\alpha_{Kc} \circ \eta_c = \gamma_c$$

The claim is that there is exactly one such α. Fix an object d of $\mathbf{D}$. For each object (c, φ) of $(K \downarrow d)$ the morphism $\varphi \colon Kc \to d$ gives, by naturality of the candidate α against φ and the required equation at c, the composite

$$Gd \xleftarrow{G\varphi} G(Kc) \xleftarrow{\gamma_c} Fc$$

and the family $\{G\varphi \circ \gamma_c\}_{(c,\varphi)}$ is a cocone on $F \circ \Pi_d$ with vertex Gd: for a morphism $f \colon (c, \varphi) \to (c', \varphi')$ one has $\varphi' \circ Kf = \varphi$, so naturality of γ gives

$$(G\varphi' \circ \gamma_{c'}) \circ Ff = G\varphi' \circ G(Kf) \circ \gamma_c = G(\varphi' \circ Kf) \circ \gamma_c = G\varphi \circ \gamma_c$$

which is the cocone compatibility condition. By the universal property of the colimit Ld there is a unique morphism

$$\alpha_d \colon Ld \to Gd, \qquad \alpha_d \circ \iota^{d}_{(c,\varphi)} = G\varphi \circ \gamma_c$$

Taking $d = Kc$ and $(c, \varphi) = (c, \mathrm{id}_{Kc})$ recovers $\alpha_{Kc} \circ \eta_c = G(\mathrm{id}_{Kc}) \circ \gamma_c = \gamma_c$, so any α with components so defined satisfies the required equation; and conversely, for any such α, naturality against φ and the equation at c force

$$\alpha_d \circ \iota^{d}_{(c,\varphi)} = \alpha_d \circ L\varphi \circ \iota^{Kc}_{(c,\mathrm{id}_{Kc})} = G\varphi \circ \alpha_{Kc} \circ \iota^{Kc}_{(c,\mathrm{id}_{Kc})} = G\varphi \circ \gamma_c$$

which is exactly the defining equation, so α_d is the only possibility. Naturality of α against a morphism $g\colon d \to d'$ is checked on injections: using the defining equations for $\alpha_{d'}$, Lg, and α_d,

$$\alpha_{d'} \circ Lg \circ \iota^d_{(c,\varphi)} = \alpha_{d'} \circ \iota^{d'}_{(c,g\circ\varphi)} = G(g \circ \varphi) \circ \gamma_c = Gg \circ G\varphi \circ \gamma_c = Gg \circ \alpha_d \circ \iota^d_{(c,\varphi)}$$

Since the $\iota^d_{(c,\varphi)}$ are jointly epic (a colimit cocone is), $\alpha_{d'} \circ Lg = Gg \circ \alpha_d$, so α is natural. This exhibits (L, η) as initial in the category of pairs (G, γ), hence as the left Kan extension $\mathrm{Lan}_K F$ with universal 2-cell η, completing the proof.

The comma-category colimit deserves a plain reading. An object (c, φ) of $(K \downarrow d)$ is a way of mapping the image Kc into d; the value $(\mathrm{Lan}_K F)(d)$ glues together all the values Fc over all such approaches to d, identifying Fc with Fc' whenever a morphism of $\mathbf{C}$ carries one approach to another. The extension therefore builds the value at d out of "everything $\mathbf{C}$ knows about d through K", and does so in the most efficient way, which is what a colimit is. When K is the inclusion of a full subcategory and $\varphi = \mathrm{id}$ is available (that is, when $d = Kc$ lies in the image of K), the object (c, id_{Kc}) is terminal in $(K \downarrow d)$, the colimit collapses to Fc, and the extension extends F; the general formula measures how far the Kan extension must interpolate when no such terminal approach exists.

The hypothesis of theorem 64.2.1 is a supply of colimits indexed by the categories $(K \downarrow d)$. When $\mathbf{C}$ is small these indexing categories are small, so a cocomplete target guarantees every required colimit at once.

Corollary 64.2.2: Existence of Kan Extensions

Let $K\colon \mathbf{C} \to \mathbf{D}$ be a functor with $\mathbf{C}$ small. If $\mathbf{E}$ is cocomplete, then $\mathrm{Lan}_K F$ exists for every functor $F\colon \mathbf{C} \to \mathbf{E}$; if $\mathbf{E}$ is complete, then $\mathrm{Ran}_K F$ exists for every F. In either case the extension along K is defined for all functors simultaneously, so restriction $K^*\colon [\mathbf{D}, \mathbf{E}] \to [\mathbf{C}, \mathbf{E}]$ acquires the adjoints $\mathrm{Lan}_K \dashv K^* \dashv \mathrm{Ran}_K$ of theorem 64.1.3.

***Proof*:**
For each object d of $\mathbf{D}$ the comma category $(K \downarrow d)$ is small: its objects are pairs (c, φ) with c ranging over the small object collection of $\mathbf{C}$ and φ over the set $\mathbf{D}(Kc, d)$ (a set since $\mathbf{D}$ is locally small, as throughout), and its morphisms form a set for the same reason. A cocomplete category has a colimit for every functor out of a small category (definition 61.4.1), so $F \circ \Pi_d$ has a colimit for every d, and theorem 64.2.1 produces $\mathrm{Lan}_K F$. Since this holds for every F, restriction has a left adjoint on all of $[\mathbf{C}, \mathbf{E}]$, which is the content of the left half of theorem 64.1.3. The complete case is dual (box 59.3.3): $(d \downarrow K)$ is small by the same count, a complete $\mathbf{E}$ supplies the limits, and restriction gains its right adjoint Ran_K, completing the proof.

Most target categories met so far are cocomplete and complete: **Set**, **Grp**, **Ab**, **Ring**, $\mathbf{Mod}_R$, and **Top** all have all small limits and colimits, so along any functor out of a small category the Kan extensions into these categories always exist. The pointwise formula is thus not an existence caveat in practice but a computation: it turns the question "what is $\mathrm{Lan}_K F$" into the question "what is this colimit", which the concrete structure of $\mathbf{E}$ answers. The following example runs that computation to the end.

Example 64.2.3: Left Kan Extension along an Inclusion of Ordinals

Let $\mathbf{C}$ be the two-object discrete category $\{0, 1\}$ (no morphisms beyond identities), let $\mathbf{D}$ be the poset $\{0 < 1 < 2\}$ regarded as a category, and let $K\colon \mathbf{C} \to \mathbf{D}$ send $0 \mapsto 0$ and $1 \mapsto 1$. Take $\mathbf{E} = \mathbf{Set}$ and let $F\colon \mathbf{C} \to \mathbf{Set}$ pick out two sets, $F0 = A$ and $F1 = B$. Since $\mathbf{C}$ is small and $\mathbf{Set}$ is cocomplete, corollary 64.2.2 guarantees $\operatorname{Lan}_K F$ exists, and theorem 64.2.1 computes it object by object.

Compute each comma category $(K \downarrow d)$. Its objects are pairs $(c, \varphi\colon Kc \to d)$, and in a poset there is at most one morphism between any two objects, so φ exists precisely when $Kc \leq d$.

- For $d = 0$: the only c with $Kc \leq 0$ is $c = 0$ (since $K0 = 0 \leq 0$ but $K1 = 1 \not\leq 0$), and φ is the unique arrow $0 \leq 0$. So $(K \downarrow 0)$ has a single object $(0, \varphi)$, and the colimit of $F \circ \Pi_0$ is just $F0 = A$.
- For $d = 1$: both $K0 = 0 \leq 1$ and $K1 = 1 \leq 1$, giving two objects $(0, \varphi_0)$ and $(1, \varphi_1)$. There are no morphisms between them ($\mathbf{C}$ is discrete), so $(K \downarrow 1)$ is the discrete two-object category and the colimit is the coproduct $F0 \coprod F1 = A \coprod B$, the disjoint union.
- For $d = 2$: again both $K0 = 0 \leq 2$ and $K1 = 1 \leq 2$, so $(K \downarrow 2)$ is discrete on two objects and the colimit is $A \coprod B$.

The value function is therefore $(\operatorname{Lan}_K F)(0) = A$ and $(\operatorname{Lan}_K F)(1) = (\operatorname{Lan}_K F)(2) = A \coprod B$. The functor structure is read off Step 2 of the proof of theorem 64.2.1: the morphism $0 \leq 1$ of $\mathbf{D}$ induces the coprojection $A \hookrightarrow A \coprod B$ into the first summand, while the morphism $h\colon 1 \leq 2$ induces the identity $A \coprod B \to A \coprod B$, since the functor $h_*\colon (K \downarrow 1) \to (K \downarrow 2)$ sends $(c, \varphi) \mapsto (c, h \circ \varphi)$, matching the two objects of each comma category by their $\mathbf{C}$-component, so the induced map on colimits carries each coprojection to itself. The unit $\eta\colon F \Rightarrow (\operatorname{Lan}_K F) \circ K$ is read off the colimit injections at the identity approaches $(0, \mathrm{id})$ and $(1, \mathrm{id})$: $\eta_0\colon A \to (\operatorname{Lan}_K F)(0) = A$ is the identity, an isomorphism, while $\eta_1\colon B \to (\operatorname{Lan}_K F)(1) = A \coprod B$ is the coprojection into the second summand, which is monic but not an isomorphism unless $A = \emptyset$; there $\eta_1\colon B \to \emptyset \coprod B$ is itself an isomorphism. So η is not invertible. This is consistent with the general principle that runs in one direction: when K is fully faithful the pointwise unit η is a natural isomorphism, so F is recovered on the nose. Here K is not full, since $\mathbf{C}(0, 1) = \emptyset$ while $\mathbf{D}(K0, K1) = \{0 \leq 1\}$ is nonempty, and the direct computation shows η fails to be invertible. The converse fails in general: fullness is sufficient but not necessary for η to be an isomorphism, as the case $A = \emptyset$ of this same example shows, where η is a natural isomorphism while K remains non-full. The extension therefore fails to restrict back to F on the nose.

The picture matches the interpretive reading above. At objects in the image of K (namely 0 and 1) the extension records the data F prescribes, assembled by coproduct where K maps two objects below the same d; at the new object 2, which lies above both 0 and 1 but is not in the image of K, the extension is forced by the colimit to be the coproduct $A \coprod B$, the freest set receiving compatible maps from A and B. A left Kan extension fills in unseen values as freely as the diagram permits, which is the precise sense in which it is the left, rather than the right, adjoint to restriction.

64.3 All Concepts are Kan Extensions

The machinery of the previous two sections was built for one payoff: to show that the central constructions of this book are Kan extensions in disguise. Limits and colimits, adjoints, and the Yoneda lemma each reappear here as an instance of Lan or Ran, computed by the universal property of definition 64.1.1 or the pointwise formula of theorem 64.2.1. The identifications are short, so the section is a sequence of corollaries rather than a development of new theory.

The first identification is the cleanest. A functor $F\colon \mathbf{J} \to \mathbf{E}$ out of a small category has a colimit exactly when a certain Kan extension exists, and the two objects agree. The trick is to extend F not along an inclusion but along the collapse $\mathbf{J} \to \mathbf{1}$ to the terminal category, the category with one object $*$ and only its identity arrow.

Proposition 64.3.1: Limits and Colimits as Kan Extensions

Let $\mathbf{J}$ be a small category and $!\colon \mathbf{J} \to \mathbf{1}$ the unique functor to the terminal category. For $F\colon \mathbf{J} \to \mathbf{E}$:

1. the colimit of F exists if and only if $\mathrm{Lan}_! F$ exists, and then $(\mathrm{Lan}_! F)(*) \cong \operatorname{colim} F$;
2. the limit of F exists if and only if $\mathrm{Ran}_! F$ exists, and then $(\mathrm{Ran}_! F)(*) \cong \lim F$.

***Proof*:**
We prove (1); part (2) follows by the duality principle (box 59.3.3), reading the statement in $\mathbf{E}^{\mathrm{op}}$.

A functor $\mathbf{1} \to \mathbf{E}$ is the same datum as an object of $\mathbf{E}$: it picks out the image $G(*)$ of the unique object, and its action on the sole arrow is forced to be id. Write $e = G(*)$. Precomposition with $!$ sends such a functor to the constant functor $\Delta e\colon \mathbf{J} \to \mathbf{E}$ at e, since every object of $\mathbf{J}$ maps to $*$ and thence to e, and every arrow to id_e. A natural transformation $F \Rightarrow (\Delta e)$ assigns to each object j of $\mathbf{J}$ a map $\lambda_j\colon Fj \to e$, naturally in j: for every arrow $u\colon j \to j'$ the square

$$\begin{tikzcd} Fj \arrow[r, "Fu"] \arrow[dr, "\lambda_j"'] & Fj' \arrow[d, "\lambda_{j'}"] \\ & e \end{tikzcd}$$

commutes, that is, $\lambda_{j'} \circ Fu = \lambda_j$. This is precisely the data of a cocone under F with summit e (definition 61.3.1). So a natural transformation $F \Rightarrow (\Delta e) = G \circ !$ is the same thing as a cocone under F with summit e.

Now unwind definition 64.1.1. The left Kan extension $\mathrm{Lan}_! F$ is a functor $\mathbf{1} \to \mathbf{E}$, hence an object $L = (\mathrm{Lan}_! F)(*)$, equipped with a universal natural transformation $\eta\colon F \Rightarrow (\mathrm{Lan}_! F) \circ ! = \Delta L$. By the previous paragraph η is a cocone under F with summit L. Universality of η says: every natural transformation $F \Rightarrow G \circ !$, equivalently every cocone (λ_j) under F with some summit e, factors uniquely as η followed by a natural transformation $(\mathrm{Lan}_! F) \Rightarrow G$; the latter is a single map $L \to e$ in $\mathbf{E}$. Uniqueness of the factoring map is exactly the statement that (L, η) is initial among cocones under F, which is the universal property of the colimit (definition 61.3.1, definition 61.1.1). Hence $L \cong \operatorname{colim} F$, and the objects exist together, completing the proof.

This is the pointwise formula (theorem 64.2.1) at the single object $*$. The comma category $(! \downarrow *)$ has an object of $\mathbf{J}$ for each way of mapping $!\,j = *$ to $*$ in $\mathbf{1}$, of which there is exactly one, so $(! \downarrow *) \cong \mathbf{J}$ and the projection to $\mathbf{J}$ is the identity; the colimit formula $(\mathrm{Lan}_! F)(*) \cong \mathrm{colim}_{(!\downarrow *)} F\pi$ collapses to $\mathrm{colim}_{\mathbf{J}} F$. The proposition says a (co)limit is the universal way to compress a diagram to a single object, and Kan extension along $\mathbf{J} \to \mathbf{1}$ is the universal compression. Every general Kan extension is then, by the pointwise formula, a (co)limit computed one object of the target at a time.

The second identification places adjunctions inside the same frame. An adjoint is determined by a universal property, and a universal property is what a Kan extension records; the content of the next result is that a right adjoint is literally the Kan extension of the identity along its left adjoint.

Proposition 64.3.2: Adjoints as Kan Extensions

Let $F\colon \mathbf{C} \to \mathbf{D}$ be left adjoint to $G\colon \mathbf{D} \to \mathbf{C}$, with unit $\eta\colon \mathrm{id}_{\mathbf{C}} \Rightarrow G \circ F$ and counit $\varepsilon\colon F \circ G \Rightarrow \mathrm{id}_{\mathbf{D}}$. Then G is the left Kan extension of $\mathrm{id}_{\mathbf{C}}$ along F,

$$G \cong \mathrm{Lan}_F\, \mathrm{id}_{\mathbf{C}}$$

with the unit $\eta\colon \mathrm{id}_{\mathbf{C}} \Rightarrow G \circ F$ serving as the universal cell. Dually, F is the right Kan extension of $\mathrm{id}_{\mathbf{D}}$ along G, $F \cong \mathrm{Ran}_G\, \mathrm{id}_{\mathbf{D}}$, with the counit as its universal cell.

***Proof*:**
We prove the first statement; the second is its dual under box 59.3.3, read in the opposite categories, exchanging the left and right adjoints. By definition 64.1.1, exhibiting G as $\mathrm{Lan}_F\, \mathrm{id}_{\mathbf{C}}$ means producing a natural transformation $u\colon \mathrm{id}_{\mathbf{C}} \Rightarrow G \circ F$ such that for every functor $H\colon \mathbf{D} \to \mathbf{C}$ the map

$$\Phi_H\colon [\mathbf{D}, \mathbf{C}](G, H) \longrightarrow [\mathbf{C}, \mathbf{C}](\mathrm{id}_{\mathbf{C}}, H \circ F), \qquad \theta \longmapsto (\theta F) \circ u$$

is a bijection natural in H, where $(\theta F) \circ u$ has components $\theta_{Fc} \circ u_c\colon c \to GFc \to HFc$. We take $u = \eta$, the unit, and produce an inverse to Φ_H from the adjunction $F \dashv G$ (definition 62.1.1).

Step 1: the inverse. Given $\beta\colon \mathrm{id}_{\mathbf{C}} \Rightarrow H \circ F$ with components $\beta_c\colon c \to HFc$, define

$$\Psi_H(\beta)_d \;=\; H\varepsilon_d \circ \beta_{Gd}\colon\; Gd \xrightarrow{\beta_{Gd}} HFGd \xrightarrow{H\varepsilon_d} Hd$$

for each object d of $\mathbf{D}$, using the counit $\varepsilon_d\colon FGd \to d$. Naturality of β and of ε makes $\Psi_H(\beta)\colon G \Rightarrow H$ a natural transformation: for $g\colon d \to d'$ the square for $\Psi_H(\beta)$ commutes because β is natural at Gg and ε is natural at g. This defines a map Ψ_H in the reverse direction.

Step 2: Ψ_H inverts Φ_H. Take $\theta\colon G \Rightarrow H$ and compute $\Psi_H(\Phi_H(\theta))$. Its component at d is

$$H\varepsilon_d \circ (\Phi_H\theta)_{Gd} \;=\; H\varepsilon_d \circ \theta_{FGd} \circ \eta_{Gd}$$

Naturality of θ at $\varepsilon_d\colon FGd \to d$ gives $H\varepsilon_d \circ \theta_{FGd} = \theta_d \circ G\varepsilon_d$, so the component equals $\theta_d \circ G\varepsilon_d \circ \eta_{Gd}$. The triangle identity $G\varepsilon_d \circ \eta_{Gd} = \mathrm{id}_{Gd}$ (definition 62.1.3) collapses this to θ_d. Hence $\Psi_H \circ \Phi_H = \mathrm{id}$.

Conversely take $\beta\colon \mathrm{id}_{\mathbf{C}} \Rightarrow HF$ and compute $\Phi_H(\Psi_H(\beta))$. Its component at c is

$$(\Psi_H\beta)_{Fc} \circ \eta_c \;=\; H\varepsilon_{Fc} \circ \beta_{GFc} \circ \eta_c$$

Naturality of β at $\eta_c\colon c \to GFc$ gives $\beta_{GFc} \circ \eta_c = HF\eta_c \circ \beta_c$, so the component equals $H\varepsilon_{Fc} \circ HF\eta_c \circ \beta_c = H(\varepsilon_{Fc} \circ F\eta_c) \circ \beta_c$. The other triangle identity $\varepsilon_{Fc} \circ F\eta_c = \mathrm{id}_{Fc}$ (definition 62.1.3) collapses this to β_c. Hence $\Phi_H \circ \Psi_H = \mathrm{id}$.

So Φ_H is a bijection, and it is natural in H because it is given by postcomposition with the fixed cell η. By definition 64.1.1 this exhibits (G, η) as the left Kan extension $\mathrm{Lan}_F \mathrm{id}_{\mathbf{C}}$, completing the proof.

An adjoint is a Kan extension. The unit $\eta\colon \mathrm{id}_{\mathbf{C}} \Rightarrow GF$, which one meets first as the insertion of generators of the free construction, is exactly the universal cell exhibiting the right adjoint G as $\mathrm{Lan}_F \mathrm{id}_{\mathbf{C}}$; the right Kan extension $\mathrm{Ran}_G \mathrm{id}_{\mathbf{D}} \cong F$ recovers the left adjoint dually, with the counit as its universal cell. The converse holds only under an extra hypothesis: if $\mathrm{Lan}_F \mathrm{id}_{\mathbf{C}}$ exists and is *preserved* by F, it is right adjoint to F; without the preservation clause a Kan extension of the identity need not arise from an adjunction.

The last identification returns to the Yoneda lemma. The Yoneda embedding $\mathbf{y}\colon \mathbf{C} \to \mathbf{PSh}(\mathbf{C})$ (definition 60.3.1) is fully faithful (theorem 60.2.1), so restricting a functor along $\mathbf{y}$ loses nothing that the functor does on representables. The precise statement is that a functor out of the presheaf category which respects colimits is recovered as the left Kan extension of its restriction to representables.

Proposition 64.3.3: Yoneda via Kan Extension

Let $\mathbf{C}$ be small and $L\colon \mathbf{PSh}(\mathbf{C}) \to \mathbf{E}$ a colimit-preserving functor into a cocomplete category. Then L is the left Kan extension along $\mathbf{y}$ of its restriction to representables:

$$L \cong \mathrm{Lan}_{\mathbf{y}}(L \circ \mathbf{y})$$

In particular L is determined up to canonical isomorphism by the composite $L \circ \mathbf{y}\colon \mathbf{C} \to \mathbf{E}$.

***Proof*:**
Every presheaf is canonically a colimit of representables: for $P \in \mathbf{PSh}(\mathbf{C})$ there is a natural isomorphism $P \cong \mathrm{colim}_{(\mathbf{y}\downarrow P)}\, \mathbf{y}\pi$, the colimit of the representables mapping into P. This is the density theorem, proved in the next section; grant it here. Because L preserves colimits,

$$L(P) \cong L\big(\mathrm{colim}_{(\mathbf{y}\downarrow P)}\, \mathbf{y}\pi\big) \cong \mathrm{colim}_{(\mathbf{y}\downarrow P)} (L \circ \mathbf{y})\pi$$

The right-hand side is the pointwise colimit formula (theorem 64.2.1) for the left Kan extension of $L\circ\mathbf{y}$ along $\mathbf{y}$, evaluated at P. Hence $L(P) \cong (\mathrm{Lan}_{\mathbf{y}}(L\circ\mathbf{y}))(P)$, naturally in P, so $L \cong \mathrm{Lan}_{\mathbf{y}}(L\circ\mathbf{y})$. The composite $L \circ \mathbf{y}$ therefore determines L, as we sought to show.

The presheaf category is the free cocompletion of $\mathbf{C}$: a colimit-preserving functor out of $\mathbf{PSh}(\mathbf{C})$ is no more and no less than an arbitrary functor out of $\mathbf{C}$, extended by the universal formula. This is the sense in which the Yoneda lemma is a Kan extension. The density theorem that supplies the canonical presentation $P \cong \mathrm{colim}\, \mathbf{y}\pi$ is the subject of the next section, where the co-Yoneda identity $P \cong \int^c Pc \cdot \mathbf{y}(c)$ makes the same statement in the language of coends.

64.4 Ends, Coends, and Density

The pointwise formula (theorem 64.2.1) writes a Kan extension as a colimit over a comma category. That comma-category colimit has a more compact description once one has a piece of notation for integrating a functor of two variables that is contravariant in one and covariant in the other. This section develops that notation, the calculus of ends and coends, and applies it to presheaves. The results are the *density* theorem, which recovers every presheaf as a colimit of representables, and a one-line coend formula for the left Kan extension.

A functor $T\colon \mathbf{C}^{\mathrm{op}} \times \mathbf{C} \to \mathbf{E}$ pairs each object c with itself along the diagonal, giving objects $T(c,c)$, but a morphism $f\colon c \to c'$ acts on T in two clashing directions: covariantly in the second slot, $T(c,f)\colon T(c,c) \to T(c,c')$, and contravariantly in the first, $T(f,c')\colon T(c',c') \to T(c,c')$. A single object of $\mathbf{E}$ that sees all the $T(c,c)$ at once must reconcile these two actions. The device that does so is a wedge.

Definition 64.4.1: End and Coend

Let $T\colon \mathbf{C}^{\mathrm{op}} \times \mathbf{C} \to \mathbf{E}$ be a functor, with $\mathbf{C}$ small. A *wedge* from an object $e \in \mathbf{E}$ to T is a family of morphisms $w_c\colon e \to T(c,c)$, one for each object c, that is *dinatural*: for every morphism $f\colon c \to c'$ the square

$$\begin{tikzcd} e \arrow[r, "w_{c'}"] \arrow[d, "w_c"'] & T(c',c') \arrow[d, "T(f,c')"] \\ T(c,c) \arrow[r, "T(c,f)"'] & T(c,c') \end{tikzcd}$$

commutes, that is $T(c,f) \circ w_c = T(f,c') \circ w_{c'}$. The *end* of T, written $\int_c T(c,c)$, is a universal wedge: an object $\int_c T(c,c)$ with a wedge $\pi_c\colon \int_c T(c,c) \to T(c,c)$ such that every wedge $w_c\colon e \to T(c,c)$ factors as $w_c = \pi_c \circ u$ through a unique $u\colon e \to \int_c T(c,c)$.

Dually, a *cowedge* from T to an object e is a dinatural family $\iota_c\colon T(c,c) \to e$, meaning $\iota_c \circ T(f,c) = \iota_{c'} \circ T(c',f)$ for every $f\colon c \to c'$. The *coend* of T, written $\int^c T(c,c)$, is a universal cowedge: an object $\int^c T(c,c)$ with a cowedge $\iota_c\colon T(c,c) \to \int^c T(c,c)$ through which every cowedge factors uniquely.

The universal property is the abstract characterization; the computational handle is a (co)equalizer. A wedge $w_c\colon e \to T(c,c)$ is exactly a map $e \to \prod_c T(c,c)$ whose two composites into $\prod_{f\colon c\to c'} T(c,c')$ agree, one composite applying $T(c,f)$ in the codomain and the other $T(f,c')$. The end is therefore the equalizer

$$\int_c T(c,c) \longrightarrow \prod_c T(c,c) \rightrightarrows \prod_{f\colon c\to c'} T(c,c') \tag{64.2}$$

of these two maps (definition 61.2.4), so an end exists whenever $\mathbf{E}$ has the products indexed by the objects and the morphisms of $\mathbf{C}$ together with equalizers. Dualizing every arrow, the coend is the coequalizer

$$\coprod_{f\colon c\to c'} T(c',c) \rightrightarrows \coprod_c T(c,c) \longrightarrow \int^c T(c,c) \tag{64.3}$$

of the two maps sending the f-summand $T(c',c)$ into $T(c,c)$ by $T(f,c)$ and into $T(c',c')$ by $T(c',f)$ (definition 61.3.5). A coend is thus $\coprod_c T(c,c)$ with the two ways of moving along a morphism identified. This is the description used in every computation below.

Two pieces of notation package the copies of an object indexed by a set. For a set S and an object $e \in \mathbf{E}$, the *copower* $S \cdot e$ is the S-indexed coproduct $\coprod_S e$, and the *power* e^S is the S-indexed product $\prod_S e$. When $\mathbf{E} = \mathbf{Set}$ the copower $S \cdot e = S \times e$ and the power e^S is the set of functions $S \to e$.

These appear the moment a coend involves a hom-set: a coend $\int^c \mathbf{C}(c', c) \cdot Fc$ has each summand $\mathbf{C}(c', c) \cdot Fc$ a copower of Fc by a hom-set.

The end over a Hom-valued functor is the object that justifies the notation: it is the set of natural transformations.

Example 64.4.2: Natural Transformations as an End

Let $F, G \colon \mathbf{C} \to \mathbf{Set}$ be functors on a small category $\mathbf{C}$. The assignment $(c', c) \mapsto \mathbf{Set}(Fc', Gc)$ is a functor $\mathbf{C}^{\mathrm{op}} \times \mathbf{C} \to \mathbf{Set}$, contravariant in c' (precompose with Ff) and covariant in c (postcompose with Gf). A wedge from a one-point set to this functor is a choice, for each c, of an element $\alpha_c \in \mathbf{Set}(Fc, Gc)$, that is a map $\alpha_c \colon Fc \to Gc$; dinaturality at $f \colon c \to c'$ reads

$$Gf \circ \alpha_c = \alpha_{c'} \circ Ff$$

which is exactly the naturality square of α. By (64.2) the end is the equalizer of

$$\prod_c \mathbf{Set}(Fc, Gc) \rightrightarrows \prod_{f \colon c \to c'} \mathbf{Set}(Fc, Gc')$$

whose elements are the families (α_c) satisfying every naturality square, so

$$\int_c \mathbf{Set}(Fc, Gc) \;\cong\; \mathrm{Nat}(F, G)$$

The set of natural transformations is an end. This is the identity that makes the Yoneda lemma, and the density theorem below, coend computations rather than ad hoc arguments.

The dual computation is a coend that identifies elements rather than selecting them. For a functor $M \colon \mathbf{C}^{\mathrm{op}} \to \mathbf{Set}$ and a functor $N \colon \mathbf{C} \to \mathbf{Set}$ on a small category $\mathbf{C}$, the coend $\int^c Mc \times Nc$ is the set of pairs $(x \in Mc,\, y \in Nc)$ modulo the relation identifying $(x \cdot f, y)$ with $(x, f \cdot y)$ for $f \colon c \to c'$ and $x \in Mc'$, by (64.3) with $T(c', c) = Mc' \times Nc$. This is the tensor product of the two functors over $\mathbf{C}$: the same coequalizer shape, taken over a one-object category enriched in abelian groups (a ring R) with $\otimes$ in place of $\times$, produces the module tensor product $M \otimes_R N$ by identifying (xf, y) with (x, fy). The coend is the categorical tensor product, which is why the copower notation borrows the multiplicative sign.

That tensor reading underlies the density theorem. Every presheaf decomposes as a coend of representables, and the coend is a colimit, so every presheaf is a colimit of representables: a presheaf is assembled from the representables $\mathbf{y}(c) = \mathbf{C}(-, c)$ by gluing along $\mathbf{C}$'s own morphisms.

Theorem 64.4.3: Density (Co-Yoneda)

Let $\mathbf{C}$ be small and $F \colon \mathbf{C}^{\mathrm{op}} \to \mathbf{Set}$ a presheaf (definition 60.1.3). Then there is a canonical isomorphism

$$F \;\cong\; \int^c Fc \cdot \mathbf{y}(c) \tag{64.4}$$

in $\mathbf{PSh}(\mathbf{C})$, where $\mathbf{y}(c) = \mathbf{C}(-, c)$ is the representable presheaf (definition 60.1.1) under the Yoneda embedding $\mathbf{y}$ (definition 60.3.1) and $Fc \cdot \mathbf{y}(c)$ is the copower. Consequently every presheaf is a colimit of representables, and the Yoneda embedding $\mathbf{y} \colon \mathbf{C} \to \mathbf{PSh}(\mathbf{C})$ is dense.

***Proof*:**
The integrand $(c', c) \mapsto Fc' \cdot \mathbf{y}(c) = Fc' \times \mathbf{C}(-, c)$ is a functor $\mathbf{C}^{\mathrm{op}} \times \mathbf{C} \to \mathbf{PSh}(\mathbf{C})$, contravariant in the first slot through F and covariant in the second through $\mathbf{y}$; its coend is computed in $\mathbf{PSh}(\mathbf{C})$, where colimits are formed objectwise (definition 61.3.1). It suffices to produce, for each presheaf G, a bijection

$$\mathbf{PSh}(\mathbf{C})\Big(\int^{c} Fc \cdot \mathbf{y}(c),\, G\Big) \;\cong\; \mathbf{PSh}(\mathbf{C})(F, G)$$

natural in G: two objects of $\mathbf{PSh}(\mathbf{C})$ whose covariant hom-functors are naturally isomorphic are themselves isomorphic, this being the Yoneda lemma (theorem 60.2.1) applied in the opposite category $\mathbf{PSh}(\mathbf{C})^{\mathrm{op}}$, where the covariant hom-functors become the representables.

Step 1: Unwind the left-hand side to a wedge. By the universal property of the coend, a map $\int^c Fc \cdot \mathbf{y}(c) \to G$ is a cowedge, a dinatural family of maps $Fc \cdot \mathbf{y}(c) \to G$. A map out of a copower $Fc \cdot \mathbf{y}(c) = \coprod_{x \in Fc} \mathbf{y}(c)$ is a choice, for each $x \in Fc$, of a map $\mathbf{y}(c) \to G$ in $\mathbf{PSh}(\mathbf{C})$. Thus

$$\mathbf{PSh}(\mathbf{C})\big(Fc \cdot \mathbf{y}(c),\, G\big) \;\cong\; \prod_{x \in Fc} \mathbf{PSh}(\mathbf{C})\big(\mathbf{y}(c), G\big) \;\cong\; \mathbf{Set}\big(Fc,\; \mathbf{PSh}(\mathbf{C})(\mathbf{y}(c), G)\big)$$

Step 2: Apply Yoneda. The Yoneda lemma (theorem 60.2.1) gives a bijection $\mathbf{PSh}(\mathbf{C})(\mathbf{y}(c), G) \cong Gc$, natural in c. Substituting,

$$\mathbf{PSh}(\mathbf{C})\big(Fc \cdot \mathbf{y}(c),\, G\big) \;\cong\; \mathbf{Set}(Fc, Gc)$$

and a cowedge into G becomes a wedge from the one-point set to the functor $(c', c) \mapsto \mathbf{Set}(Fc', Gc)$, the dinaturality conditions matching under the Yoneda bijection because that bijection is natural in c.

Step 3: Recognize the end. By example 64.4.2 the wedges from a point to $(c', c) \mapsto \mathbf{Set}(Fc', Gc)$ are exactly the natural transformations $F \Rightarrow G$:

$$\int_c \mathbf{Set}(Fc, Gc) \;\cong\; \mathrm{Nat}(F, G) = \mathbf{PSh}(\mathbf{C})(F, G)$$

Composing the three bijections gives $\mathbf{PSh}(\mathbf{C})\big(\int^c Fc \cdot \mathbf{y}(c), G\big) \cong \mathbf{PSh}(\mathbf{C})(F, G)$, natural in G. By the Yoneda lemma applied in $\mathbf{PSh}(\mathbf{C})$, the two presheaves are isomorphic, and the isomorphism is the canonical one sending $x \in Fc$ to the pair (x, id_c) in the copower, as we sought to show.

For the final clause, the coend (64.4) is the coequalizer (64.3), hence a colimit, of a diagram of representables. So F is a colimit of representables, and since F was arbitrary the representables generate $\mathbf{PSh}(\mathbf{C})$ under colimits, which is the statement that $\mathbf{y}$ is dense, completing the proof.

The density theorem is the promised converse to the Yoneda embedding. Yoneda embeds $\mathbf{C}$ fully faithfully into $\mathbf{PSh}(\mathbf{C})$; density says the image is not lost inside a larger category but generates it, every presheaf being reachable from the representables by colimits. Together they identify $\mathbf{PSh}(\mathbf{C})$ as the *free cocompletion* of $\mathbf{C}$: freely adjoining colimits to $\mathbf{C}$ produces exactly the presheaf category, with $\mathbf{y}$ the universal inclusion. The isomorphism (64.4) makes this concrete: a presheaf F is rebuilt from the sets Fc of its elements and the representables that receive them.

The concrete content is a formula for the elements of a presheaf and how they patch, seen most on a poset.

Example 64.4.4: Density over a Poset

Let $\mathbf{C}$ be a poset, viewed as a category with a single arrow $c \to c'$ when $c \le c'$, and let $F\colon \mathbf{C}^{\mathrm{op}} \to \mathbf{Set}$ be a presheaf (an order-reversing assignment of sets with restriction maps $Fc' \to Fc$ for $c \le c'$). The representable $\mathbf{y}(c) = \mathbf{C}(-,c)$ sends b to the one-point set if $b \le c$ and to the empty set otherwise, so $\mathbf{y}(c)$ is the presheaf represented by the down-set of c. The coend (64.4) identifies, for each $x \in Fc$ and each $b \le c$, the "germ" of x at b with the actual restriction $x|_b \in Fb$: the copower $Fc \cdot \mathbf{y}(c)$ contributes one copy of the down-set of c for every section $x \in Fc$, and the coequalizer glues the copy attached to x to the copy attached to $x|_b$ along b. The result reassembles F from its sections and their restrictions, the poset analogue of recovering a sheaf from its stalks, that is from its local data together with the compatibilities that let the pieces be glued.

The same coend calculus rewrites the pointwise formula. The colimit over the comma category $(K \downarrow d)$ in theorem 64.2.1 is a coend once the comma category is unwound, and the coend form is the one that composes with the density theorem and generalizes to the enriched setting.

Proposition 64.4.5: Coend Formula for Kan Extensions

Let $K\colon \mathbf{C} \to \mathbf{D}$ with $\mathbf{C}$ small, $F\colon \mathbf{C} \to \mathbf{E}$, and suppose the relevant (co)limits exist in $\mathbf{E}$. Then the left Kan extension is the coend

$$(\mathrm{Lan}_K F)(d) \;\cong\; \int^{c} \mathbf{D}(Kc,d)\cdot Fc \tag{64.5}$$

and the right Kan extension is the end

$$(\mathrm{Ran}_K F)(d) \;\cong\; \int_{c} Fc^{\mathbf{D}(d,Kc)} \tag{64.6}$$

where $\mathbf{D}(Kc,d)\cdot Fc$ is the copower and $Fc^{\mathbf{D}(d,Kc)}$ the power.

***Proof*:**

Fix $d \in \mathbf{D}$. By theorem 64.2.1 the value $(\mathrm{Lan}_K F)(d)$ is the colimit of the composite $(K \downarrow d) \xrightarrow{\pi} \mathbf{C} \xrightarrow{F} \mathbf{E}$, where the comma category $(K \downarrow d)$ has objects the pairs (c,g) with $g\colon Kc \to d$ and morphisms $(c,g) \to (c',g')$ the maps $h\colon c \to c'$ with $g' \circ Kh = g$. The claim is that this colimit is computed by the coequalizer (64.3) for the integrand $T(c',c) = \mathbf{D}(Kc',d)\cdot Fc$.

Step 1: Match the coproducts. The coproduct $\coprod_c T(c,c) = \coprod_c \mathbf{D}(Kc,d)\cdot Fc$ is $\coprod_c \coprod_{g\colon Kc\to d} Fc$, one copy of Fc for each pair (c,g), that is one copy of Fc for each object of $(K \downarrow d)$. This is the coproduct over the objects of $(K \downarrow d)$ of $F\pi$, the vertex coproduct in the coequalizer presentation of the colimit $\mathrm{colim}_{(K\downarrow d)} F\pi$ (definition 61.3.1).

Step 2: Match the relations. The parallel pair in (64.3) identifies, for each $h\colon c \to c'$ in $\mathbf{C}$, the summand $T(c',c) = \mathbf{D}(Kc',d)\cdot Fc$ under the two maps $T(h,c)$ and $T(c',h)$. Concretely a summand indexed by $h\colon c \to c'$ and $g'\colon Kc' \to d$ is sent, on one side, to the copy of Fc at $(c, g' \circ Kh)$ acting by the identity on Fc, and on the other to the copy of Fc' at (c',g') acting by $Fh\colon Fc \to Fc'$. This is exactly the relation imposed in the colimit of $F\pi$: a morphism

$h\colon (c, g' \circ Kh) \to (c', g')$ of $(K \downarrow d)$ identifies the image of Fc in the vertex with the image of Fc' precomposed by Fh. The two coequalizers therefore present the same quotient, giving (64.5).

Step 2 (dual): the right extension. Applying the duality principle (box 59.3.3) to the argument just given, $(\mathrm{Ran}_K F)(d)$ is the limit over $(d \downarrow K)$ of $F\pi$, presented by the equalizer (64.2) for the integrand $(c', c) \mapsto Fc^{\mathbf{D}(d,Kc')}$. The vertex product $\prod_c Fc^{\mathbf{D}(d,Kc)}$ is one copy of Fc for each map $d \to Kc$, that is one copy for each object of $(d \downarrow K)$, and the equalizer imposes the morphisms of $(d \downarrow K)$ exactly as in the left case with all arrows reversed. This gives (64.6), completing the proof.

Read against density, the coend formula (64.5) says the left Kan extension along K is what one gets by resolving F into representables and then relabeling their indices along K: the coend integrates Fc against the hom-set $\mathbf{D}(Kc, d)$, the copower weighting each Fc by the ways Kc maps to d. Taking $K = \mathbf{y}$ the Yoneda embedding and $F = \mathbf{y}$ itself, (64.5) computes $(\mathrm{Lan}_{\mathbf{y}}\, \mathbf{y})(G) \cong \int^c \mathbf{PSh}(\mathbf{C})(\mathbf{y}(c), G) \cdot \mathbf{y}(c) \cong \int^c Gc \cdot \mathbf{y}(c)$ at a presheaf G, using $\mathbf{PSh}(\mathbf{C})(\mathbf{y}(c), G) \cong Gc$; this is the density isomorphism (64.4) for G. Density is the case of the coend formula where the extension is along Yoneda. The end and coend are thus not a separate topic appended to Kan extensions but the notation in which the pointwise formula, the Yoneda lemma, and density are one computation.

65

Monoidal Categories

A *monoidal category* carries a tensor product $\otimes$, a binary operation on objects that is associative and unital up to coherent isomorphism. It is the structure behind every tensor product in mathematics, and behind a category acting on itself. This chapter defines monoidal categories and the coherence theorem that tames their associativity isomorphisms, adds the symmetry and closedness most examples carry, and identifies the monoids that live inside a monoidal category, whose instance among endofunctors is the monad of chapter 63.

65.1 Monoidal Categories and Coherence

A tensor product multiplies objects. On R-modules it is $\otimes_R$; on sets it is the cartesian product; on the endofunctors of a category it is composition. Each is associative and unital, but only up to a natural isomorphism, and those isomorphisms are themselves data that must fit together. This section isolates that data as the notion of a monoidal category, states the two axioms (pentagon and triangle) that control it, and proves the coherence theorem, which shows the axioms suffice to make every reassociation unambiguous. We write $\otimes$ for the tensor product and I for the unit object throughout.

The associativity isomorphism $(X \otimes Y) \otimes Z \cong X \otimes (Y \otimes Z)$ is called the *associator*, and the isomorphisms $I \otimes X \cong X$ and $X \otimes I \cong X$ the *left* and *right unitors*. Requiring these to be natural in their arguments is the first constraint; the pentagon and triangle axioms are the second, and they are what distinguishes a coherent choice of associativity from an arbitrary one.

Definition 65.1.1: Monoidal Category

A *monoidal category* is a category C equipped with

- a functor $\otimes\colon C \times C \to C$, the *tensor product*;
- an object I of C, the *unit object*;
- a natural isomorphism α, the *associator*, with components

$$\alpha_{X,Y,Z}\colon (X \otimes Y) \otimes Z \to X \otimes (Y \otimes Z)$$

- natural isomorphisms λ and ρ, the *left* and *right unitors*, with components

$$\lambda_X\colon I \otimes X \to X, \qquad \rho_X\colon X \otimes I \to X$$

subject to two axioms. The *pentagon axiom* requires that the diagram

$$\begin{tikzcd}
 & (W \otimes X) \otimes (Y \otimes Z) \arrow[dr, "\alpha_{W,X,Y\otimes Z}"] & \\
\big((W \otimes X) \otimes Y\big) \otimes Z \arrow[ur, "\alpha_{W\otimes X,Y,Z}"] \arrow[d, "\alpha_{W,X,Y}\otimes \mathrm{id}_Z"'] & & W \otimes \big(X \otimes (Y \otimes Z)\big) \\
\big(W \otimes (X \otimes Y)\big) \otimes Z \arrow[rr, "\alpha_{W,X\otimes Y,Z}"'] & & W \otimes \big((X \otimes Y) \otimes Z\big) \arrow[u, "\mathrm{id}_W \otimes \alpha_{X,Y,Z}"']
\end{tikzcd}$$

commutes for all objects W, X, Y, Z, and the *triangle axiom* requires that the diagram

$$\begin{tikzcd}
(X \otimes I) \otimes Y \arrow[rr, "\alpha_{X,I,Y}"] \arrow[dr, "\rho_X \otimes \mathrm{id}_Y"'] & & X \otimes (I \otimes Y) \arrow[dl, "\mathrm{id}_X \otimes \lambda_Y"] \\
 & X \otimes Y &
\end{tikzcd}$$

commutes for all objects X, Y.

The pentagon governs a fourfold tensor product. There are five ways to fully parenthesize $W \otimes X \otimes Y \otimes Z$, and the associator supplies an isomorphism between adjacent parenthesizations; the pentagon says the two paths around its perimeter, from $\big((W \otimes X) \otimes Y\big) \otimes Z$ to $W \otimes \big(X \otimes (Y \otimes Z)\big)$, agree. The triangle relates the associator to the unitors, forcing the two ways of cancelling a unit factor wedged between X and Y to coincide. These two small diagrams are the entire input; the coherence theorem below shows they already force every larger diagram of associators and unitors to commute.

When the associator and unitors are not just coherent but are identity morphisms, the tensor product is strictly associative and strictly unital, and the bookkeeping vanishes.

Definition 65.1.2: Strict Monoidal Category

A monoidal category is *strict* if the associator α and the unitors λ, ρ are all identity morphisms, so that $(X \otimes Y) \otimes Z = X \otimes (Y \otimes Z)$, $I \otimes X = X$, and $X \otimes I = X$ on the nose.

In a strict monoidal category the pentagon and triangle axioms hold trivially, being equalities of identity morphisms, and one may write iterated tensor products $X_1 \otimes \cdots \otimes X_n$ without parentheses. Most monoidal categories arising from a tensor construction are not strict, since $(X \otimes Y) \otimes Z$ and $X \otimes (Y \otimes Z)$ are built by different constructions, isomorphic but not equal. The endofunctor example below is a case where strictness does hold.

The unit object is determined by the structure. It is not extra data to be pinned down by a further choice, but is fixed up to a unique compatible isomorphism, in the same way a terminal object is (definition 61.1.1).

Proposition 65.1.3: Uniqueness of the Unit Object

Let $(C, \otimes, I, \alpha, \lambda, \rho)$ be a monoidal category, and suppose I' is another object admitting isomorphisms $\lambda'_X : I' \otimes X \to X$ and $\rho'_X : X \otimes I' \to X$ natural in X and satisfying the triangle axiom with α. Then there is a unique isomorphism $u : I \to I'$ satisfying $u \circ \lambda_I = \lambda'_{I'} \circ (u \otimes u)$; it is the canonical comparison of the two unit structures.

The compatibility condition packages the two units into a single square: it says u intertwines the multiplication $\lambda_I : I \otimes I \to I$ that the unit induces on itself with the corresponding multiplication $\lambda'_{I'} : I' \otimes I' \to I'$ on I'. Three standard facts about the unitors drive the proof. Two are recorded once and for all in Mac Lane, *Categories for the Working Mathematician*, Chapter VII: the left and right unitors agree at the unit, $\lambda_I = \rho_I$ (and likewise $\lambda'_{I'} = \rho'_{I'}$), so the induced multiplication is unambiguous; and the two unit structures are compared by the left-unit coherence identity $\lambda'_X \circ (u \otimes \mathrm{id}_X) = \lambda_X$, a consequence of the pentagon and triangle axioms. The third is elementary: tensoring by an object whose unitors are isomorphisms is an equivalence, hence *faithful*, which lets one cancel a factor $(-) \otimes \mathrm{id}_I$ from an equation.

***Proof Key ideas*:**
Write $v = \lambda_I = \rho_I : I \otimes I \to I$ and $v' = \lambda'_{I'} = \rho'_{I'} : I' \otimes I' \to I'$ for the two induced multiplications, using $\lambda_I = \rho_I$ (and its primed analogue) cited above.

Existence. Set

$$u = \rho_{I'} \circ (\lambda'_I)^{-1} : I \to I' \otimes I \to I',$$

an isomorphism as a composite of isomorphisms. The compatibility square $u \circ v = v' \circ (u \otimes u)$ reduces to a single unitor identity. Indeed, write $u \otimes u = (u \otimes \mathrm{id}_{I'}) \circ (\mathrm{id}_I \otimes u)$; naturality of λ at $u : I \to I'$ gives $u \circ \lambda_I = \lambda_{I'} \circ (\mathrm{id}_I \otimes u)$, so $u \circ v = u \circ \lambda_I$ and $v' \circ (u \otimes u) = \lambda'_{I'} \circ (u \otimes \mathrm{id}_{I'}) \circ (\mathrm{id}_I \otimes u)$ agree if and only if, after cancelling the isomorphism $\mathrm{id}_I \otimes u$,

$$\lambda'_{I'} \circ (u \otimes \mathrm{id}_{I'}) = \lambda_{I'}$$

This is the instance at $X = I'$ of the left-unit coherence identity $\lambda'_X \circ (u \otimes \mathrm{id}_X) = \lambda_X$, which relates the two unit structures through α and holds in any monoidal category by the pentagon and triangle axioms; it is a companion to the coherence theorem (theorem 65.1.5), deferred with it to Mac Lane, Chapter VII, which is why the existence half is given in outline. Granting it, the square commutes and u is a compatible comparison.

> *Uniqueness.* Suppose $f\colon I \to I'$ is another isomorphism with $f \circ v = v' \circ (f \otimes f)$. Both u and f are then isomorphisms $I \to I'$ of the two multiplications v, v', so $g = u^{-1} \circ f\colon I \to I$ is an isomorphism satisfying $g \circ v = v \circ (g \otimes g)$, an automorphism of the multiplication v. Naturality of λ at g gives $g \circ \lambda_I = \lambda_I \circ (\mathrm{id}_I \otimes g)$, that is $g \circ v = v \circ (\mathrm{id}_I \otimes g)$; comparing with $g \circ v = v \circ (g \otimes g)$ and cancelling the isomorphism v yields $g \otimes g = \mathrm{id}_I \otimes g$. By functoriality of $\otimes$, $g \otimes g = (g \otimes \mathrm{id}_I) \circ (\mathrm{id}_I \otimes g)$, so the equation reads $(g \otimes \mathrm{id}_I) \circ (\mathrm{id}_I \otimes g) = \mathrm{id}_I \otimes g$; precomposing with the inverse of the isomorphism $\mathrm{id}_I \otimes g$ gives $g \otimes \mathrm{id}_I = \mathrm{id}_{I \otimes I} = \mathrm{id}_I \otimes \mathrm{id}_I$, whence $g = \mathrm{id}_I$ by faithfulness of $(-) \otimes I$. Thus $f = u$, and the unit object is unique up to a unique compatible isomorphism, as we sought to show.

The proposition is the reason one speaks of *the* unit object: any two choices are canonically identified, so the ambiguity carries no content. This is the monoidal analogue of the uniqueness of a terminal object (definition 61.1.1), and it is what allows I to be treated as a fixed piece of the structure rather than a variable one.

The standing cast supplies the primary examples. Each is a category already in use, now viewed as carrying a tensor product.

Example 65.1.4: Monoidal Structures on the Standing Cast

Four monoidal categories, each read against the data and axioms of definition 65.1.1.

1. *Sets under cartesian product.* The category **Set** is monoidal with $\otimes = \times$ the cartesian product (definition 61.2.3) and $I = 1$ any singleton, terminal in **Set** (definition 61.1.1). The associator $\alpha_{X,Y,Z}\colon (X \times Y) \times Z \to X \times (Y \times Z)$ sends $((x, y), z)$ to $(x, (y, z))$, and the unitors $\lambda_X\colon 1 \times X \to X$, $\rho_X\colon X \times 1 \to X$ delete the singleton coordinate. These are bijections, natural in their arguments, and the pentagon and triangle are the evident equalities of tuple regroupings. This is not strict: $(X \times Y) \times Z$ is a set of pairs whose first entry is a pair, not equal to $X \times (Y \times Z)$.

2. *Vector spaces under tensor product.* The category $\mathbf{Vect}_k$ of vector spaces over a field k is monoidal with $\otimes = \otimes_k$ the tensor product over k ([25]) and $I = k$ the ground field as a one-dimensional space. The associator is the canonical isomorphism $(U \otimes V) \otimes W \to U \otimes (V \otimes W)$ sending $(u \otimes v) \otimes w$ to $u \otimes (v \otimes w)$, and the unitors $k \otimes V \to V$, $V \otimes k \to V$ send $c \otimes v$ and $v \otimes c$ to cv. The pentagon and triangle hold because both sides of each act identically on the generating simple tensors. Again not strict.

3. *Modules under tensor product.* The same construction gives a monoidal structure on $\mathbf{Mod}_R$ for a commutative ring R, with $\otimes = \otimes_R$ and $I = R$ ([25]); the associator and unitors are the standard isomorphisms of module tensor products, and $\mathbf{Vect}_k$ is the special case $R = k$.

4. *Endofunctors under composition.* For any category C, the functor category $[C, C]$ (definition 59.5.4) of endofunctors is monoidal with $\otimes = \circ$ functor composition and $I = \mathrm{id}_C$ the identity functor. Composition of functors is associative on the nose, $(F \circ G) \circ H = F \circ (G \circ H)$, and the identity functor is a strict two-sided unit, $\mathrm{id}_C \circ F = F = F \circ \mathrm{id}_C$, so the associator and unitors are identity natural transformations. Thus $([C, C], \circ, \mathrm{id}_C)$ is a *strict* monoidal category (definition 65.1.2).

The endofunctor example is the one to keep in mind. It is where strictness appears without contrivance, composition being an operation that is equal, not just isomorphic, when reassociated, and it is the setting in which the monad (T, η, μ) met earlier will reappear as a monoid object. The strictification argument in the coherence proof is modelled on exactly this category, replacing each object by its translation endofunctor. The first three examples, by contrast, are the reason coherence is needed at all: their associators and unitors are nontrivial isomorphisms, and without the pentagon and triangle there would be no guarantee that a long chain of reassociations returns an unambiguous answer.

More generally, any category with finite products is monoidal under $(\times, 1)$, with $\times$ the binary product (definition 61.2.3) and 1 a terminal object (definition 61.1.1) as unit; the first example is the case $C = \mathbf{Set}$. This is the *cartesian monoidal* structure, and it is available on **Grp**, **Top**, **Cat**, and every other member of the standing cast with finite products.

The pentagon and triangle are two small diagrams. It is not obvious that they suffice: a diagram built from a dozen associators and unitors offers many paths between its corners, and nothing so far forces those paths to agree. Mac Lane's coherence theorem is the guarantee that they do. It has two equivalent formulations, one internal (every formal diagram commutes) and one structural (every monoidal category is equivalent to a strict one), and the second is the route to proving the first.

Theorem 65.1.5: Mac Lane's Coherence Theorem

Let C be a monoidal category. Then:

1. *Coherence.* Every diagram in C whose edges are built by tensoring identities, associators $\alpha^{\pm 1}$, and unitors $\lambda^{\pm 1}$, $\rho^{\pm 1}$ commutes. Any two morphisms between the same pair of iterated tensor products built from these constituents are equal.
2. *Strictification.* There is a strict monoidal category C^{st} and a monoidal equivalence $C \simeq C^{\mathrm{st}}$.

The full proof of part (1) is a combinatorial argument about a word problem: one shows that any two formal composites of associators and unitors with the same source and target are provably equal from the pentagon and triangle, and the complete verification is long. It is given in Mac Lane, *Categories for the Working Mathematician*, Chapter VII. What follows establishes part (2) in full and shows how it yields part (1), which is the conceptual content.

***Proof Key ideas*:**
The strictification construction is given in full; the word-problem proof of part (1) from part (2) is sketched, with the complete combinatorial argument cited to Mac Lane, Chapter VII, as noted above.

Step 1: The strict category of lists. Let C^{st} be the category whose objects are finite lists $\underline{X} = (X_1, \dots, X_n)$ of objects of C (including the empty list, of length 0), and whose morphisms

$$C^{\mathrm{st}}(\underline{X}, \underline{Y}) = \mathrm{Hom}_C\big(\overline{X}, \overline{Y}\big)$$

are the C-morphisms between the tensorings $\overline{X} = ((X_1 \otimes X_2) \otimes \cdots) \otimes X_n$ associated to the left in a fixed order (the empty list tensoring to I), with composition and identities taken in C. Concatenation of lists, $\underline{X} \otimes \underline{Y} = (X_1, \dots, X_n, Y_1, \dots, Y_m)$, is an associative operation on objects with strict unit the empty list, and on morphisms it is defined by transporting the tensor of the underlying C-morphisms across the associators that compare $\overline{\underline{X} \otimes \underline{Y}}$ with $\overline{X} \otimes \overline{Y}$; the pentagon axiom makes this transport well-defined and functorial. Thus C^{st} is a *strict* monoidal category (definition 65.1.2): concatenation is associative and unital on the nose. This is the list model of the endofunctor picture, where a length-one list (X) plays the role of the translation $X \otimes -$ and concatenation the role of composition, but built so that the bookkeeping isomorphisms live inside the definition of the tensor rather than obstructing strictness.

Step 2: L is a monoidal equivalence. Define $L\colon C \to C^{\mathrm{st}}$ on objects by $L(X) = (X)$, the length-one list, and on morphisms by $L(f) = f$, viewing $f\colon X \to Y$ as a morphism $\overline{(X)} = X \to Y = \overline{(Y)}$ in C. Because $C^{\mathrm{st}}((X),(Y)) = \mathrm{Hom}_C(X,Y)$ by definition, L is *fully faithful* with no further argument, its action on each hom-set being the identity map of $\mathrm{Hom}_C(X,Y)$. It is essentially surjective: every list $\underline{X}$ is isomorphic in C^{st} to the length-one list $(\overline{\underline{X}})$, the isomorphism being the identity of $\overline{\underline{X}}$ regarded as a morphism $\underline{X} \to (\overline{\underline{X}})$. A functor that is fully faithful and essentially surjective is an equivalence (proposition 59.5.6), so L is an equivalence of categories. Finally L is *strong monoidal*: the structure isomorphism $L(X) \otimes L(Y) = (X,Y) \to (X \otimes Y) = L(X \otimes Y)$ is the identity of $X \otimes Y$ regarded as a morphism $\overline{(X,Y)} = X \otimes Y \to X \otimes Y$, and the unit comparison $I_{C^{\mathrm{st}}} = () \to (I)$ is id_I, since both $\overline{()}$ and $\overline{(I)}$ equal I; the pentagon and triangle axioms of C are exactly the coherence identities these comparisons must satisfy. Hence L is a monoidal equivalence $C \simeq C^{\mathrm{st}}$ with strict target, proving part (2).

Step 3: Coherence from strictification. Given a formal diagram in C built from associators and unitors, apply the equivalence $L\colon C \simeq C^{\mathrm{st}}$. Since L is *strong* (not strict) monoidal, it does not send an associator to an identity; rather, the strong-monoidal coherence axioms and the strictness of the target force $L(\alpha_{X,Y,Z})$ to equal a fixed composite of the structure isomorphisms $\varphi_{A,B}\colon L(A)\otimes L(B) \cong L(A\otimes B)$ of Step 2 (each of which, in the list model, is an identity morphism of C regarded across a change of tensoring), and likewise $L(\lambda_X)$, $L(\rho_X)$ to fixed composites of the $\varphi_{A,B}$ and the unit comparison. Because concatenation in C^{st} is strictly associative and unital, these image composites depend only on the source and target parenthesizations, not on the path taken: any two formal composites of associators and unitors sharing a source and target map under L to the *same* morphism of C^{st}. As L is faithful, they are already equal in C, so the formal diagram commutes. The content that Step 3 defers is exactly the path-independence just used: that the structure isomorphisms assemble into a well-defined map depending only on the endpoints is the word-problem statement, that the free monoidal category on the objects involved has at most one morphism between any two parenthesizations. This is the combinatorial argument deferred to Mac Lane, Chapter VII; granting it, every formal diagram of associators and unitors commutes, completing the proof of part (1).

Coherence licenses a controlled sloppiness. Because every formal diagram of associators and unitors commutes, one may drop parentheses from iterated tensor products and suppress the unit isomorphisms entirely, writing $X \otimes Y \otimes Z$ and $I \otimes X = X$ as though the category were strict. No ambiguity results: whichever way a suppressed reassociation is reinstated, part (1) guarantees the same morphism. In practice one works in a monoidal category as if it were strict, invoking coherence to justify the omitted brackets, and the strictification of part (2) is what makes this rigorous rather than convenient. Every monoidal category met later in this chapter is handled this way.

65.2 Braided and Symmetric Monoidal Categories

A monoidal structure (definition 65.1.1) relates $X \otimes Y$ and $Y \otimes X$ by nothing at all; in $(\mathbf{Vect}_k, \otimes, k)$ they are visibly isomorphic by the swap $x \otimes y \mapsto y \otimes x$, coherently with the tensor. This section isolates that extra datum. A *braiding* is a natural isomorphism $\beta_{X,Y}\colon X \otimes Y \to Y \otimes X$ compatible with the associator, and a braiding is *symmetric* when swapping twice returns to the start. The compatibility is governed not by one axiom but by two hexagons, one for each way of moving a factor past a tensor product.

The compatibility a braiding must satisfy is forced by coherence. Given three objects, $X \otimes (Y \otimes Z)$ can be rewritten as $(Y \otimes Z) \otimes X$ in two ways: apply the braiding to the whole factor $Y \otimes Z$ at once, or slide X past Y and then past Z one factor at a time, reassociating as needed. A braiding is required to make these two routes agree. Each route is a hexagon of associators and braidings, and there are two of them because X may be moved to the right (past $Y \otimes Z$) or an object may be moved to the left (past $X \otimes Y$); the second hexagon is the analogue of the first for the reverse braiding.

Definition 65.2.1: Braided Monoidal Category

A *braided monoidal category* is a monoidal category $(\mathbf{C}, \otimes, I, \alpha, \lambda, \rho)$ (definition 65.1.1) equipped with a natural isomorphism

$$\beta_{X,Y} \colon X \otimes Y \to Y \otimes X$$

called the *braiding*, natural in X and Y (definition 59.5.2), such that the two hexagon diagrams commute for all objects X, Y, Z. The first hexagon

$$\begin{tikzcd}
 & (X \otimes Y) \otimes Z \arrow[r, "\beta_{X,Y} \otimes \mathrm{id}_Z"] & (Y \otimes X) \otimes Z \arrow[dr, "\alpha"] & \\
X \otimes (Y \otimes Z) \arrow[ur, "\alpha^{-1}"] \arrow[dr, "\beta_{X,Y \otimes Z}"'] & & & Y \otimes (X \otimes Z) \arrow[dl, "\mathrm{id}_Y \otimes \beta_{X,Z}"] \\
 & (Y \otimes Z) \otimes X \arrow[r, "\alpha"'] & Y \otimes (Z \otimes X) &
\end{tikzcd}$$

expresses the compatibility of β with α when the object X is braided past the tensor product $Y \otimes Z$. The second hexagon

$$\begin{tikzcd}
 & X \otimes (Y \otimes Z) \arrow[r, "\mathrm{id}_X \otimes \beta_{Y,Z}"] & X \otimes (Z \otimes Y) \arrow[dr, "\alpha^{-1}"] & \\
(X \otimes Y) \otimes Z \arrow[ur, "\alpha"] \arrow[dr, "\beta_{X \otimes Y,Z}"'] & & & (X \otimes Z) \otimes Y \arrow[dl, "\beta_{X,Z} \otimes \mathrm{id}_Y"] \\
 & Z \otimes (X \otimes Y) \arrow[r, "\alpha^{-1}"'] & (Z \otimes X) \otimes Y &
\end{tikzcd}$$

expresses the same compatibility when Z is braided past $X \otimes Y$.

Naturality of β means that for morphisms $f \colon X \to X'$ and $g \colon Y \to Y'$, the square

$$\begin{tikzcd}
X \otimes Y \arrow[r, "\beta_{X,Y}"] \arrow[d, "f \otimes g"'] & Y \otimes X \arrow[d, "g \otimes f"] \\
X' \otimes Y' \arrow[r, "\beta_{X',Y'}"'] & Y' \otimes X'
\end{tikzcd}$$

commutes: braiding and then acting by $g \otimes f$ agrees with acting by $f \otimes g$ and then braiding. The two hexagons are the coherence conditions for the braiding against the associator; together with Mac Lane's coherence theorem (theorem 65.1.5) for the underlying monoidal structure, they guarantee that every diagram built from associators, unitors, and braidings that one might expect to commute does commute, provided its two sides braid the factors into the same permutation by isotopic routes. The proviso is the point: unlike the associator and unitors, the braiding is not forced to be its own inverse, so the order in which factors are braided can matter.

In a strict monoidal category (definition 65.1.2) the associators are identities and each hexagon collapses to a triangle. The first becomes

$$\beta_{X,Y\otimes Z} = (\mathrm{id}_Y \otimes \beta_{X,Z}) \circ (\beta_{X,Y} \otimes \mathrm{id}_Z)$$

and the second becomes

$$\beta_{X\otimes Y,Z} = (\beta_{X,Z} \otimes \mathrm{id}_Y) \circ (\mathrm{id}_X \otimes \beta_{Y,Z})$$

so that a braiding past a tensor product is determined by braiding past each factor in turn. These are the relations one draws as strands crossing: $\beta_{X,Y}$ is the strand carrying X passing over the strand carrying Y, and the hexagons say that a strand passing over two others equals its passing over them one at a time.

The extra condition that makes a braiding into a symmetry is the one the swap map satisfies: undoing the crossing recovers the identity.

Definition 65.2.2: Symmetric Monoidal Category

A braided monoidal category (definition 65.2.1) is *symmetric* if its braiding satisfies

$$\beta_{Y,X} \circ \beta_{X,Y} = \mathrm{id}_{X\otimes Y}$$

for all objects X and Y. A braiding with this property is called a *symmetry*.

For a symmetry, $\beta_{Y,X} = \beta_{X,Y}^{-1}$, so the two hexagons of definition 65.2.1 become equivalent: each is obtained from the other by inverting the braidings, and symmetry identifies a braiding with its inverse. A symmetric monoidal category therefore carries a single hexagon axiom, and the strand carrying X over the strand carrying Y is indistinguishable from X under Y. The distinction between over and under crossings is what a general braiding retains and a symmetry discards.

The condition $\beta_{Y,X}\beta_{X,Y} = \mathrm{id}$ is the categorical form of a familiar identity: it says the swap is an involution. Symmetry is the monoidal counterpart of commutativity. Just as a commutative operation lets a product $a_1a_2\cdots a_n$ be formed without regard to the order of its factors, a symmetric braiding lets a tensor $X_1 \otimes \cdots \otimes X_n$ be reindexed by any permutation, with any two reindexings related by a canonical isomorphism independent of how the permutation is factored into transpositions. This is coherence for the symmetric group: the symmetric group S_n acts on the n-fold tensor, and the action is well-defined precisely because $\beta_{Y,X}\beta_{X,Y} = \mathrm{id}$ makes a transposition square to the identity. In a braided category that is not symmetric the braid group B_n acts in place of S_n, and B_n is infinite, so the order of braiding is remembered.

Example 65.2.3: The Standing Cast as Symmetric Monoidal Categories

The commutative examples of example 65.1.4, $(\mathbf{Vect}_k, \otimes, k)$ and $(\mathbf{Set}, \times, 1)$, each carry a symmetry; the third example there, the endofunctor category $([\mathbf{C}, \mathbf{C}], \circ, \mathrm{id})$, is deliberately excluded, since functor composition is non-commutative ($F \circ G$ is not naturally isomorphic to $G \circ F$ in general) and so admits no braiding, a point that returns when a monad is identified as a monoid object in that category. The vector-space case extends to modules over any commutative ring.

1. In $(\mathbf{Vect}_k, \otimes, k)$ the braiding is the swap

$$\beta_{V,W} \colon V \otimes W \to W \otimes V, \qquad v \otimes w \mapsto w \otimes v$$

extended k-bilinearly. This is a well-defined linear isomorphism because $(v, w) \mapsto w \otimes v$ is k-bilinear, and its inverse is $\beta_{W,V}$. Applying it twice returns $v \otimes w \mapsto w \otimes v \mapsto v \otimes w$, so $\beta_{W,V}\beta_{V,W} = \mathrm{id}_{V \otimes W}$ and the structure is symmetric. The hexagon axioms hold because both routes send $u \otimes v \otimes w$ (with the associators identifying the two bracketings) to the same reordered pure tensor.

2. For a commutative ring R, the category $(\mathbf{Mod}_R, \otimes_R, R)$ is symmetric with the same swap $m \otimes n \mapsto n \otimes m$. Well-definedness uses commutativity of R: for $r \in R$ the assignment must respect $rm \otimes n = m \otimes rn$, and the target $n \otimes rm = rn \otimes m$ agrees only because r is central, which it is. The tensor product itself is the tensor product of R-modules cited from [25]; here it is repackaged as a symmetric monoidal structure. The case $R = k$ a field recovers the previous item.

3. In $(\mathbf{Set}, \times, 1)$ the braiding is the swap of ordered pairs

$$\beta_{A,B} \colon A \times B \to B \times A, \qquad (a, b) \mapsto (b, a)$$

which is its own inverse: applying it twice returns $(a, b) \mapsto (b, a) \mapsto (a, b)$, so it is a symmetry. Every category with finite products carries this symmetry on its cartesian monoidal structure $(\mathbf{C}, \times, 1)$ (the structure of definition 61.2.3): the braiding $\beta_{X,Y} \colon X \times Y \to Y \times X$ is the unique map whose projections to Y and X are the projections of $X \times Y$ in the other order, and its square is the identity because both maps have the same projections. A cartesian monoidal category is thus symmetric with no further choice.

Braidings that are not symmetric exist. The category of finite-dimensional representations of a quantum group $U_q(\mathfrak{g})$ carries a braiding, built from the R-matrix solving the Yang-Baxter equation, for which $\beta_{Y,X}\beta_{X,Y} \neq \mathrm{id}$: braiding past twice is a nontrivial automorphism, and the braid group B_n acts faithfully on tensor powers. This is the source of the quantum knot invariants, the Jones polynomial among them, where an over-crossing and an under-crossing of strands must be assigned different operators precisely because β and β^{-1} differ. These braided-not-symmetric categories are the natural home of that theory and are developed in the literature on quantum groups; the present book stays with the symmetric case, where the standing cast lives.

The three examples share a mechanism. In each, the braiding is a swap of coordinates, its square is the identity, and the hexagons hold because reassociating and swapping commute on pure tensors or pairs. The symmetry is not an added structure to be checked case by case so much as a reflection of the fact that the underlying product, of vector spaces, of modules, of sets, is commutative up to canonical isomorphism. Symmetric monoidal categories are the ambient setting in which the algebra of tensor products behaves like the algebra of a commutative operation.

65.3 Closed Monoidal Categories

The tensor product $\otimes$ combines two objects into one. A closed structure supplies the inverse move: an object $[X, Y]$ inside the category whose maps out of any A are exactly the maps $A \otimes X \to Y$. This section makes that require precise as an adjunction (definition 62.1.1), identifies the two canonical maps it carries, and reads off the closure of the two examples the adjunction chapter already met in another guise.

The condition to impose is that tensoring with a fixed object have a right adjoint. Naming that right adjoint $[X, -]$ turns the informal "object of maps $X \to Y$" into a functor internal to the category.

Definition 65.3.1: Closed Monoidal Category

A monoidal category $(\mathbf{C}, \otimes, I)$ (definition 65.1.1) is *closed* if for every object X the functor

$$- \otimes X : \mathbf{C} \to \mathbf{C}$$

has a right adjoint (definition 62.1.1), written $[X, -] : \mathbf{C} \to \mathbf{C}$ and called the *internal hom*. Explicitly, for each X there is a bijection

$$\mathbf{C}(A \otimes X, Y) \;\cong\; \mathbf{C}(A, [X, Y]) \tag{65.1}$$

natural in A and in Y. The object $[X, Y]$ is the *internal hom* from X to Y.

The word "closed" names the fact that the hom-set $\mathbf{C}(X, Y)$, an object of **Set** external to $\mathbf{C}$, has been promoted to an object $[X, Y]$ of $\mathbf{C}$ itself: the category is closed under forming its own hom-objects. (65.1) is the defining feature. It says a map from A into the internal hom $[X, Y]$ is the same datum as a map from $A \otimes X$ to Y, so $[X, Y]$ represents the functor $A \mapsto \mathbf{C}(A \otimes X, Y)$. The external hom is recovered by taking $A = I$: since $I \otimes X \cong X$ through the left unitor, (65.1) gives $\mathbf{C}(X, Y) \cong \mathbf{C}(I, [X, Y])$, so the ordinary maps $X \to Y$ are the "points" $I \to [X, Y]$ of the internal hom.

The definition asks $- \otimes X$ to have a right adjoint, tensoring on the right. In a symmetric monoidal category (definition 65.2.2) the braiding gives a natural isomorphism $A \otimes X \cong X \otimes A$, so a right adjoint to $- \otimes X$ is at once a right adjoint to $X \otimes -$: left and right closedness coincide, and one speaks simply of a closed symmetric monoidal category.[1]

The adjunction packaged by (65.1) carries a unit and a counit (definition 62.1.3). Named and unwound in the present notation, they are the two structural maps a closed category always provides: an evaluation and a currying map.

[1]Without symmetry the two conditions may differ, and a category can be right closed without being left closed. Every example below is symmetric, so the distinction does not arise.

Theorem 65.3.2: The Tensor-Hom Adjunction

Let $(\mathbf{C}, \otimes, I)$ be a closed monoidal category (definition 65.3.1), and fix an object X. The adjunction $- \otimes X \dashv [X, -]$ has:

1. counit the *evaluation* morphism, natural in Y,
$$\mathrm{ev}_{X,Y} \colon [X, Y] \otimes X \to Y$$
the transpose of $\mathrm{id}_{[X,Y]}$;
2. unit the *coevaluation* (or *currying*) morphism, natural in A,
$$\mathrm{coev}_{A,X} \colon A \to [X, A \otimes X]$$
the transpose of $\mathrm{id}_{A \otimes X}$.

The bijection (65.1) sends $f \colon A \otimes X \to Y$ to $[X, f] \circ \mathrm{coev}_{A,X} \colon A \to [X, Y]$, and is natural in A and in Y.

***Proof*:**
By definition 65.3.1 the functors $- \otimes X$ and $[X, -]$ are adjoint, so the general theory of section 62.1 applies with $F = - \otimes X$ and $G = [X, -]$; the content of the theorem is to name the unit and counit in this setting and confirm the transpose formula.

Step 1: The counit is evaluation. By definition 62.1.3 the counit of $- \otimes X \dashv [X, -]$ has component at Y the transpose of $\mathrm{id}_{[X,Y]}$ under the inverse of (65.1). Setting $A = [X, Y]$, that bijection reads $\mathbf{C}([X, Y] \otimes X, Y) \cong \mathbf{C}([X, Y], [X, Y])$, and the transpose of $\mathrm{id}_{[X,Y]}$ is a morphism
$$\mathrm{ev}_{X,Y} \colon [X, Y] \otimes X \to Y$$
which is the counit, hence natural in Y. It evaluates the internal hom: for $\mathbf{C} = \mathbf{Set}$ with $[X, Y] = \mathbf{Set}(X, Y)$, ev is the map $(g, x) \mapsto g(x)$.

Step 2: The unit is coevaluation. Dually, the unit has component at A the transpose of $\mathrm{id}_{A \otimes X}$ under (65.1) with $Y = A \otimes X$: the bijection $\mathbf{C}(A \otimes X, A \otimes X) \cong \mathbf{C}(A, [X, A \otimes X])$ sends $\mathrm{id}_{A \otimes X}$ to a morphism
$$\mathrm{coev}_{A,X} \colon A \to [X, A \otimes X]$$
the unit, natural in A. In $\mathbf{Set}$ it sends a to the function $x \mapsto (a, x)$, the map that curries the second coordinate.

Step 3: The transpose formula. Proposition 62.1.4 computes the transpose of a morphism through the unit and counit: for $F \dashv G$ with unit η, the transpose of $f \colon FA \to Y$ is $Gf \circ \eta_A$. With $F = - \otimes X$, $G = [X, -]$, $\eta_A = \mathrm{coev}_{A,X}$, and $Gf = [X, f]$, the transpose of $f \colon A \otimes X \to Y$ is
$$A \xrightarrow{\mathrm{coev}_{A,X}} [X, A \otimes X] \xrightarrow{[X,f]} [X, Y] \tag{65.2}$$
which is the asserted formula: curry the identity, then apply the internal hom to f.

Step 4: Naturality. Naturality of (65.1) in A and Y is naturality of the adjunction isomorphism $\Phi \colon \mathbf{C}(F-, -) \cong \mathbf{C}(-, G-)$ of definition 62.1.1, which holds by the definition of an adjunction. Concretely, for $u \colon A' \to A$ the square relating $\Phi_{A,Y}$ and $\Phi_{A',Y}$ commutes because (65.2) is built from the natural transformation coev and the functor $[X,-]$; the Y-variable is dual, using ev. Thus the bijection is natural in both variables, completing the proof.

The evaluation and coevaluation maps are the internal forms of two operations on ordinary functions. Evaluation $[X,Y]\otimes X \to Y$ applies a map to an argument; coevaluation $A \to [X, A\otimes X]$ holds an element of A and waits for an argument in X, producing the pair. The triangle identities of theorem 62.1.5 record that currying and then evaluating returns the original map, which is the content of (65.2) run forward and back. This is the tensor-hom adjunction of example 62.2.2, now read as structure carried by a single category rather than a bijection between two external hom-sets: the module Hom-tensor identity $\mathrm{Hom}_R(A\otimes_R X, Y) \cong \mathrm{Hom}_R(A, \mathrm{Hom}_R(X,Y))$ and the currying isomorphism $\mathbf{Set}(A\times X, Y) \cong \mathbf{Set}(A, Y^X)$ are exactly (65.1) for $\mathbf{Mod}_R$ and for $\mathbf{Set}$. What the adjunction chapter presented as two adjoint functors, the closed structure presents as the category's ability to internalize its own hom.

When the monoidal structure is the cartesian one built from finite products (definition 61.2.3), closedness has a classical name and the internal hom is the exponential object.

Definition 65.3.3: Cartesian Closed Category

A category $\mathbf{C}$ with finite products is *cartesian closed* if its cartesian monoidal structure $(\mathbf{C}, \times, 1)$ is closed (definition 65.3.1): for every object X the functor $-\times X$ has a right adjoint $(-)^X\colon \mathbf{C}\to\mathbf{C}$, the *exponential*. The object $Y^X = [X,Y]$ is the *exponential object*, and the adjunction reads

$$\mathbf{C}(A\times X, Y) \;\cong\; \mathbf{C}(A, Y^X) \tag{65.3}$$

natural in A and Y.

Cartesian closedness is the special case where the tensor is the product and the unit is the terminal object 1. The bijection (65.3) is currying in its familiar form: a map of two variables $A\times X\to Y$ is the same as a map $A\to Y^X$ into the object of maps $X\to Y$. The evaluation counit $Y^X\times X\to Y$ applies a function to an argument, and taking $A=1$ recovers the global elements of Y^X as the maps $X\to Y$, so the exponential is a genuine internal object of functions.

Example 65.3.4: Set and Vect as Closed Categories

1. $\mathbf{Set}$ *is cartesian closed.* The cartesian monoidal structure is $(\mathbf{Set}, \times, 1)$ with 1 a one-point set. For sets X and Y the exponential is the set of all functions,

$$Y^X = \mathbf{Set}(X,Y)$$

and currying is the bijection

$$\mathbf{Set}(A\times X, Y) \;\cong\; \mathbf{Set}(A, Y^X)$$

sending $f\colon A\times X\to Y$ to $a\mapsto (x\mapsto f(a,x))$. This is a bijection because a function of two arguments is determined by, and determines, its curried form; it is natural in A and Y because currying commutes with precomposition on A and postcomposition on Y. Thus $-\times X \dashv (-)^X$, and the evaluation counit $Y^X\times X\to Y$ is $(g,x)\mapsto g(x)$.

2. $\mathbf{Vect}_k$ *is closed monoidal, not cartesian.* The relevant monoidal structure on vector spaces over k is $(\mathbf{Vect}_k, \otimes_k, k)$, the tensor product with unit the ground field k, not the cartesian

product. It is closed with internal hom the space of linear maps,

$$[X, Y] = \mathrm{Hom}_k(X, Y)$$

itself a k-vector space. The bijection

$$\mathrm{Hom}_k(A \otimes_k X, Y) \;\cong\; \mathrm{Hom}_k(A, \mathrm{Hom}_k(X, Y))$$

is the tensor-hom adjunction over k (example 62.2.2): a bilinear map $A \times X \to Y$ corresponds to a linear map $A \otimes_k X \to Y$ and, curried, to a linear map $A \to \mathrm{Hom}_k(X, Y)$. The evaluation counit $\mathrm{Hom}_k(X, Y) \otimes_k X \to Y$ is $g \otimes x \mapsto g(x)$. The cartesian product $X \times Y$ (the direct sum, since finite products and coproducts coincide here) does *not* make $\mathbf{Vect}_k$ cartesian closed: the exponential would need $\mathbf{Vect}_k(A \times X, Y) \cong \mathbf{Vect}_k(A, Y^X)$, but the left side is not additive in A while the representable right side is. Since $A \times X = A \oplus X$, the left side is $\mathbf{Vect}_k(A \oplus X, Y) \cong \mathbf{Vect}_k(A, Y) \times \mathbf{Vect}_k(X, Y)$, which carries the constant summand $\mathbf{Vect}_k(X, Y)$: at $A = 0$ it returns $\mathbf{Vect}_k(X, Y)$, not a one-point set, so with $X = Y = k$ it fails to send the zero space to a terminal set. A representable functor $\mathbf{Vect}_k(-, Y^X)$ does send 0 to a point, so the two cannot be naturally isomorphic and no cartesian exponential exists. Closedness here is a feature of $\otimes_k$, not of $\times$.

The two examples separate the cartesian case from the general one. In **Set** the tensor is the product, so the internal hom is the exponential and the category is cartesian closed. In $\mathbf{Vect}_k$ the closure lives on the tensor product, and the cartesian product carries no matching exponential; the category is closed monoidal but not cartesian closed. The pattern recurs: the tensor-hom adjunction of example 62.2.2 for $\mathbf{Mod}_R$ makes each module category closed monoidal under $\otimes_R$, while a topos such as **Set** or a presheaf category is cartesian closed under $\times$.

Cartesian closed categories are the categorical semantics of the typed lambda calculus: types are objects, terms are morphisms, the product $\times$ interprets pairing, the exponential Y^X interprets the function type $X \to Y$, and the evaluation and currying maps interpret function application and lambda abstraction, with the adjunction (65.3) the statement that a function of two arguments and its curried form are interchangeable.[2]

65.4 Monoids, Comonoids, and Duality

The definition of a monoid transcribes verbatim into any monoidal category once the product is a functor $\otimes$ rather than a set-theoretic multiplication, and the reward is that the same three axioms name at once monoids, rings, algebras, and, in the endofunctor category of example 65.1.4, monads. This section makes that transcription, dualizes it to comonoids, and closes with a taste of the finiteness condition, dualizability, that lets an object have a linear dual.

Fix a monoidal category $(\mathbf{C}, \otimes, I)$ with associator α and unitors λ, ρ (definition 65.1.1). A monoid in **Set** is a set M together with a multiplication $M \times M \to M$ and a chosen unit element, which is the same as a map $1 \to M$ from the terminal set. Replacing $\times$ by $\otimes$ and the terminal set by the monoidal unit I gives the general definition.

[2]This correspondence, that cartesian closed categories model simply typed lambda calculus, is due to Lambek; a standard reference is Lambek and Scott, *Introduction to Higher Order Categorical Logic*.

Definition 65.4.1: Monoid Object

A *monoid object* in a monoidal category $(\mathbf{C}, \otimes, I)$ is an object M of $\mathbf{C}$ together with two morphisms

$$\mu\colon M \otimes M \to M \qquad \text{and} \qquad \eta\colon I \to M$$

the *multiplication* and the *unit*, such that the associativity diagram

$$\begin{tikzcd}
(M \otimes M) \otimes M \arrow[r, "\alpha_{M,M,M}"] \arrow[d, "\mu \otimes \mathrm{id}_M"'] & M \otimes (M \otimes M) \arrow[r, "\mathrm{id}_M \otimes \mu"] & M \otimes M \arrow[d, "\mu"] \\
M \otimes M \arrow[rr, "\mu"'] & & M
\end{tikzcd}$$

and the two unit diagrams

$$\begin{tikzcd}
I \otimes M \arrow[r, "\eta \otimes \mathrm{id}_M"] \arrow[dr, "\lambda_M"'] & M \otimes M \arrow[d, "\mu"] & M \otimes I \arrow[l, "\mathrm{id}_M \otimes \eta"'] \arrow[dl, "\rho_M"] \\
& M &
\end{tikzcd}$$

commute.

The associativity square says the two ways of multiplying a triple $M \otimes M \otimes M$ agree once the parenthesization is corrected by the associator, and the two triangles say that inserting the unit on either side and multiplying is the same as forgetting the unit factor via λ or ρ. When $\mathbf{C}$ is strict the associator and unitors are identities and the diagrams reduce to the familiar equations $\mu(\mu \otimes \mathrm{id}) = \mu(\mathrm{id} \otimes \mu)$, $\mu(\eta \otimes \mathrm{id}) = \mathrm{id} = \mu(\mathrm{id} \otimes \eta)$; theorem 65.1.5 guarantees no generality is lost by picturing the strict case. A morphism of monoid objects $(M, \mu, \eta) \to (M', \mu', \eta')$ is a map $f\colon M \to M'$ with $f\mu = \mu'(f \otimes f)$ and $f\eta = \eta'$; monoid objects and their morphisms form a category $\mathrm{Mon}(\mathbf{C})$.

Reversing every arrow gives the dual notion. Rather than a multiplication that fuses two copies of M into one, a comonoid carries a comultiplication that splits one copy into two, and in place of a unit picking out an element it has a counit that measures an element against the ground object I.

Definition 65.4.2: Comonoid Object

A *comonoid object* in $(\mathbf{C}, \otimes, I)$ is a monoid object in the opposite monoidal category $(\mathbf{C}^{\mathrm{op}}, \otimes, I)$: an object C with a *comultiplication* $\delta\colon C \to C \otimes C$ and a *counit* $\varepsilon\colon C \to I$ making the coassociativity diagram

$$\begin{tikzcd} C \arrow[r, "\delta"] \arrow[d, "\delta"'] & C \otimes C \arrow[d, "\mathrm{id}_C \otimes \delta"] \\ C \otimes C \arrow[r, "\delta \otimes \mathrm{id}_C"'] & C \otimes C \otimes C \end{tikzcd}$$

and the two counit diagrams

$$\begin{tikzcd} & C \arrow[dl, "\lambda_C^{-1}"'] \arrow[d, "\delta"] \arrow[dr, "\rho_C^{-1}"] & \\ I \otimes C & C \otimes C \arrow[l, "\varepsilon \otimes \mathrm{id}_C"] \arrow[r, "\mathrm{id}_C \otimes \varepsilon"'] & C \otimes I \end{tikzcd}$$

commute (associators suppressed in the strict picture).

By the duality principle (box 59.3.3), every statement about monoid objects has a mirror statement about comonoid objects, obtained by reversing the arrows in $\mathbf{C}$; there is nothing to prove separately. Comonoids are the less familiar of the pair only because the everyday examples, sets and modules, carry no interesting comultiplication under $\times$ or $\otimes$: the diagonal $C \to C \otimes C$ is forced. They become substantial in categories where $\otimes$ is genuinely non-cartesian, where a comonoid is the categorical form of a coalgebra, the structure dual to an algebra that organizes, for instance, the representation theory of a group through its group algebra.

The point of the definition is that a single set of axioms, read in different monoidal categories, returns structures that look unrelated on their face. The following example collects the payoff.

Example 65.4.3: Monoids, Rings, Algebras, and Monads

A monoid object is a monoid, a ring, an algebra, or a monad according to where it is read.

1. *In* $(\mathbf{Set}, \times, 1)$. A monoid object is exactly a monoid in the ordinary sense. The map $\mu\colon M \times M \to M$ is a binary operation, $\eta\colon 1 \to M$ picks out an element $e = \eta(*)$, the associativity square is $\mu(\mu(a, b), c) = \mu(a, \mu(b, c))$, and the unit triangles say $ea = a = ae$. The cartesian structure $(\mathbf{C}, \times, 1)$ of definition 61.2.3 is what makes this the definition of a monoid; the same reading in **Top** or **Grp** gives topological monoids and monoids internal to **Grp**.

2. *In* $(\mathbf{Ab}, \otimes_{\mathbb{Z}}, \mathbb{Z})$. A monoid object is exactly a ring. An abelian group R with a $\mathbb{Z}$-bilinear multiplication $\mu\colon R \otimes_{\mathbb{Z}} R \to R$ and a unit $\eta\colon \mathbb{Z} \to R$, $\eta(1) = 1_R$, is precisely a ring: bilinearity is built into the tensor product, associativity and unitality are the two diagrams. Replacing $\mathbb{Z}$ by a commutative ring k and **Ab** by $(\mathbf{Mod}_k, \otimes_k, k)$ gives a k-algebra (the multiplication and unit of k-modules cited to [25]). Thus rings and algebras are monoid objects, one monoidal category up from monoids.

3. *In* $([\mathbf{C}, \mathbf{C}], \circ, \mathrm{id})$. A monoid object in the strict monoidal category of endofunctors of $\mathbf{C}$, with $\otimes$ the composition of functors and unit object the identity functor (example 65.1.4, definition 59.5.4), is exactly a monad on $\mathbf{C}$. An object of $[\mathbf{C}, \mathbf{C}]$ is an endofunctor T; the

multiplication $\mu\colon T \circ T \to T$ and unit $\eta\colon \mathrm{id} \to T$ are natural transformations; and the associativity and unit diagrams of definition 65.4.1 are, symbol for symbol, the associativity and unit laws of a monad (T, η, μ) from definition 63.1.1. This is the identity

a monad is a monoid in the category of endofunctors

stated as the theorem it is: the two definitions are not analogous, they are the same definition read in two monoidal categories.

The third reading is the reason the notion of monoid object is worth isolating. Chapter 5 built monads by hand, checking the associativity and unit squares of natural transformations directly; definition 65.4.1 now shows those squares were never special to endofunctors. Any construction valid for monoids in an arbitrary monoidal category applies at once to monads: the category of monoid objects, the free monoid on an object, morphisms of monoids. A monad morphism, for instance, is nothing but a morphism of monoid objects in $[\mathbf{C}, \mathbf{C}]$.

A monoid in $(\mathbf{Set}, \times, 1)$ becomes a group exactly when every element has an inverse, and the inverse is itself a morphism $i\colon G \to G$ subject to a diagram. This upgrades the monoid-object definition in a cartesian monoidal category to a group object, the notion that lets "group" be interpreted inside geometry.

Example 65.4.4: Group Objects

Let $(\mathbf{C}, \times, 1)$ be a category with finite products (definition 61.2.3), regarded as a cartesian monoidal category. A *group object* is a monoid object (G, μ, η) together with an *inverse* morphism $i\colon G \to G$ such that

$$\begin{tikzcd}
G \arrow[r, "(\mathrm{id}_G, i)"] \arrow[d, "e"'] & G \times G \arrow[d, "\mu"] & G \arrow[l, "(i, \mathrm{id}_G)"'] \arrow[d, "e"] \\
1 \arrow[r, "\eta"'] & G & 1 \arrow[l, "\eta"]
\end{tikzcd}$$

commutes, where $e\colon G \to 1$ is the unique map to the terminal object and (id_G, i), (i, id_G) are the maps into the product with the indicated components. The two squares say $\mu(g, i(g)) = \eta e(g) = \mu(i(g), g)$: multiplying an element by its inverse on either side returns the unit. In $(\mathbf{Set}, \times, 1)$ a group object is a group. In $(\mathbf{Top}, \times, 1)$ it is a topological group, a group whose multiplication and inversion are continuous; in the category **Man** of smooth manifolds with the product, it is a Lie group, a group whose operations are smooth, the central object of the differential-geometric theory in [50].

Group objects show that "group" is not a set-theoretic notion but a diagrammatic one: the same diagrams, interpreted in **Top** or **Man**, define topological and Lie groups without a word changing. Comultiplication reverses the picture once more, and a group object with a compatible comonoid structure in a symmetric monoidal category (definition 65.2.2) is a Hopf algebra, the structure underlying quantum groups; the symmetry is what lets the compatibility between μ and δ be stated.

The remaining topic is a finiteness property of the object itself, orthogonal to the algebraic structure above. In $\mathbf{Vect}_k$ a finite-dimensional space V has a dual V^*, and the pairing $V^* \otimes V \to k$ together with the "identity element" $k \to V \otimes V^*$ obey two triangle identities that reconstruct V from V^* and conversely. Abstracting those two identities gives dualizability.

Definition 65.4.5: Dualizable Object

An object X of a monoidal category $(\mathbf{C}, \otimes, I)$ is *(left) dualizable* if there is an object $X^\vee$, the *dual*, together with morphisms

$$\mathrm{ev}\colon X^\vee \otimes X \to I \qquad \text{and} \qquad \mathrm{coev}\colon I \to X \otimes X^\vee$$

the *evaluation* and *coevaluation*, such that the two *zigzag identities*

$$\begin{tikzcd} X \arrow[r, "\mathrm{coev}\otimes\mathrm{id}_X"] \arrow[dr, "\mathrm{id}_X"'] & X \otimes X^\vee \otimes X \arrow[d, "\mathrm{id}_X\otimes\mathrm{ev}"] \\ & X \end{tikzcd} \qquad \begin{tikzcd} X^\vee \arrow[r, "\mathrm{id}\otimes\mathrm{coev}"] \arrow[dr, "\mathrm{id}"'] & X^\vee \otimes X \otimes X^\vee \arrow[d, "\mathrm{ev}\otimes\mathrm{id}"] \\ & X^\vee \end{tikzcd}$$

commute (associators and unitors suppressed in the strict picture). A monoidal category in which every object is dualizable is called *rigid*.

The two triangles are the categorical form of the linear-algebra identity that expresses id_V through a basis and its dual basis. Reading them in $\mathbf{Vect}_k$ makes the abstraction concrete, and pins down exactly which spaces qualify.

Example 65.4.6: Dualizable Vector Spaces

In $(\mathbf{Vect}_k, \otimes_k, k)$ a vector space V is dualizable if and only if it is finite-dimensional, with dual $V^\vee = V^* = \mathrm{Hom}_k(V, k)$. The evaluation is the pairing $\mathrm{ev}(\varphi \otimes v) = \varphi(v)$, and the coevaluation is

$$\mathrm{coev}\colon k \to V \otimes V^*, \qquad 1 \mapsto \sum_i e_i \otimes e_i^*$$

for a basis (e_i) with dual basis (e_i^*); the element $\sum_i e_i \otimes e_i^*$ is independent of the basis, being the image of id_V under the isomorphism $\mathrm{End}_k(V) \cong V \otimes V^*$ that holds precisely when V is finite-dimensional. The first zigzag sends $v \mapsto \sum_i e_i \otimes e_i^* \otimes v \mapsto \sum_i e_i\, e_i^*(v) = v$, so it is id_V; the second is the symmetric computation. When V is infinite-dimensional no coevaluation exists: a map $k \to V \otimes V^*$ has image an element of $V \otimes V^*$, necessarily a finite sum of simple tensors, and no such finite sum can satisfy the first zigzag for every v. Thus dualizability is exactly the finiteness that makes $V \cong V^{**}$ canonically.

Dualizable objects are where the linear-algebraic notion of trace survives into a symmetric monoidal category (definition 65.2.2): for an endomorphism $f\colon X \to X$ of a dualizable object one forms the composite $I \xrightarrow{\mathrm{coev}} X \otimes X^\vee \xrightarrow{f\otimes\mathrm{id}} X \otimes X^\vee \xrightarrow{\beta_{X,X^\vee}} X^\vee \otimes X \xrightarrow{\mathrm{ev}} I$, an endomorphism of I that specializes to the ordinary trace in $\mathbf{Vect}_k$. The symmetry β is what supplies the swap $X \otimes X^\vee \to X^\vee \otimes X$; in a bare monoidal category no such canonical map exists, so the trace needs at least a braiding. Rigid symmetric monoidal categories, in which every object is dualizable, are the setting for Tannakian duality, which reconstructs a group from its rigid tensor category of representations; both developments belong to the next stage of the theory and are not pursued here.

66

Enriched and 2-Categories

Two facts about the previous chapters were left unremarked. The hom-*sets* of a category are objects of **Set**, but nothing forced that choice; and **Cat** carried natural transformations, a second layer of morphism an ordinary category lacks. This chapter follows both threads. *Enriched category theory* replaces hom-sets by objects of a monoidal category, recovering additive, topological, and simplicial categories as the base varies. *Two-category theory* takes the morphisms between morphisms seriously, with **Cat** as the guiding example, and recovers adjunctions and monads as statements internal to a 2-category.

The two threads are developed to different depths. The enriched theory is built in full through section 66.3: tensors and cotensors, the enriched functor category, the enriched Yoneda lemma with its proof, and weighted limits, which subsume ordinary limits and both of the first two. The 2-categorical theory is developed only as far as the enriched theory already carries it, since a strict 2-category is the **Cat**-enriched case of definition 66.1.1 and the structures whose coherence is genuine, rather than provable, are the subject of the sequel. The chapter, and the book, close at the edge of the subject, where the strict equalities weaken to coherent isomorphism, pointing beyond it to higher category theory.

66.1 Enriched Categories

The hom-set of a category is a set, and nothing more. Yet the standing cast carries hom-sets with extra structure that the plain definition discards: the morphisms between two abelian groups form an abelian group under pointwise addition, the maps between two topological spaces form a topological space, and between two categories the functors form a category (with natural transformations as its own morphisms). *Enriched category theory* takes this extra structure seriously. It replaces the hom-*set* $\mathbf{C}(x, y)$ by a hom-*object* living in a fixed monoidal category $\mathbf{V}$ (definition 65.1.1), whose tensor product supplies the composition and whose unit supplies the identities. The one requirement is that $\mathbf{V}$ be monoidal: composition pairs two hom-objects, and $\otimes$ is exactly the operation for pairing objects of $\mathbf{V}$.

Definition 66.1.1: Enriched Category

Let $(\mathbf{V}, \otimes, I)$ be a monoidal category. A *category enriched over* $\mathbf{V}$, or a $\mathbf{V}$-*category*, $\mathbf{C}$ consists of

- a collection of *objects* $\mathrm{Ob}(\mathbf{C})$;
- for each ordered pair (x, y) of objects, a *hom-object* $\mathbf{C}(x, y)$ of $\mathbf{V}$;
- for each ordered triple (x, y, z), a *composition* morphism of $\mathbf{V}$
$$\circ_{xyz} \colon \mathbf{C}(y, z) \otimes \mathbf{C}(x, y) \to \mathbf{C}(x, z)$$
- for each object x, an *identity* morphism of $\mathbf{V}$
$$j_x \colon I \to \mathbf{C}(x, x)$$

subject to the associativity and unit laws, which are the following diagrams in $\mathbf{V}$. Associativity is the commuting square (writing α for the associator of $\mathbf{V}$)

$$\begin{tikzcd}
\big(\mathbf{C}(z,w) \otimes \mathbf{C}(y,z)\big) \otimes \mathbf{C}(x,y) \arrow[r, "\alpha"] \arrow[d, "\circ\otimes\mathrm{id}"'] & \mathbf{C}(z,w) \otimes \big(\mathbf{C}(y,z) \otimes \mathbf{C}(x,y)\big) \arrow[d, "\mathrm{id}\otimes\circ"] \\
\mathbf{C}(y,w) \otimes \mathbf{C}(x,y) \arrow[d, "\circ"'] & \mathbf{C}(z,w) \otimes \mathbf{C}(x,z) \arrow[d, "\circ"] \\
\mathbf{C}(x,w) \arrow[r, equal] & \mathbf{C}(x,w)
\end{tikzcd}$$

and the two unit laws are the commuting triangles (writing λ, ρ for the left and right unitors of $\mathbf{V}$)

$$\begin{tikzcd}
I \otimes \mathbf{C}(x,y) \arrow[r, "j_y\otimes\mathrm{id}"] \arrow[dr, "\lambda"'] & \mathbf{C}(y,y) \otimes \mathbf{C}(x,y) \arrow[d, "\circ"] & \mathbf{C}(x,y) \otimes I \arrow[l, "\mathrm{id}\otimes j_x"'] \arrow[dl, "\rho"] \\
& \mathbf{C}(x,y) &
\end{tikzcd}$$

The two diagrams are the ordinary associativity and unit laws with every equation of *elements* replaced by a commuting diagram of *morphisms in* $\mathbf{V}$, and every reassociation or unit-insertion routed through the coherence isomorphisms α, λ, ρ that a monoidal category supplies for exactly this purpose. Nowhere does the definition speak of an element of $\mathbf{C}(x, y)$: $\mathbf{C}(x, y)$ is an object of $\mathbf{V}$, and $\mathbf{V}$ need not be a category of structured sets. Composition is not a function applied to a pair of morphisms but a single arrow out of a tensor product, and the identity at x is not a chosen element but an arrow $I \to \mathbf{C}(x, x)$ selecting it in whatever sense $\mathbf{V}$ allows. This is the shift the definition forces: from *elements and functions* to *objects and morphisms of* $\mathbf{V}$.

An ordinary category is the special case $\mathbf{V} = (\mathbf{Set}, \times, 1)$. Then $\mathbf{C}(x, y)$ is a hom-set, the composition arrow $\mathbf{C}(y, z) \times \mathbf{C}(x, y) \to \mathbf{C}(x, z)$ is the usual composition function $(g, f) \mapsto gf$, and the identity arrow $j_x \colon 1 \to \mathbf{C}(x,x)$ from the one-point set picks out id_x. The associativity and unit diagrams reduce to $h(gf) = (hg)f$ and $f\,\mathrm{id}_x = f = \mathrm{id}_y f$. Enrichment loses nothing and is a genuine generalization: it remembers structure on the hom-objects that the passage to underlying sets throws away.

That passage is itself a construction. Given a $\mathbf{V}$-category, one recovers an ordinary category by asking $\mathbf{V}$ for the elements of each hom-object, where "element" means a morphism out of the monoidal unit.

Definition 66.1.2: Underlying Ordinary Category

Let $\mathbf{C}$ be a $\mathbf{V}$-category. Its *underlying ordinary category* $\mathbf{C}_0$ has the same objects as $\mathbf{C}$, and hom-sets

$$\mathrm{Hom}_{\mathbf{C}_0}(x,y) = \mathbf{V}\big(I, \mathbf{C}(x,y)\big)$$

the set of $\mathbf{V}$-morphisms from the unit I to the hom-object $\mathbf{C}(x,y)$. The identity of x in $\mathbf{C}_0$ is the enriched identity $j_x \colon I \to \mathbf{C}(x,x)$. The composite in $\mathbf{C}_0$ of $f \colon I \to \mathbf{C}(x,y)$ and $g \colon I \to \mathbf{C}(y,z)$ is

$$I \xrightarrow{\lambda^{-1}} I \otimes I \xrightarrow{g \otimes f} \mathbf{C}(y,z) \otimes \mathbf{C}(x,y) \xrightarrow{\circ} \mathbf{C}(x,z)$$

where $\lambda^{-1} \colon I \to I \otimes I$ is the inverse left unitor.

The associativity and unit laws of $\mathbf{C}_0$ are not assumed; they are *consequences* of the enriched laws of definition 66.1.1. Composing three elements in $\mathbf{C}_0$ produces two bracketings of the same triple tensor, and the enriched associativity square, together with the coherence (theorem 65.1.5) that makes all instances of α, λ, ρ agree, identifies them; the enriched unit triangles give the identity laws in $\mathbf{C}_0$. The underlying category is what the $\mathbf{V}$-category induces on $\mathbf{Set}$ by probing each hom-object with the unit. When $\mathbf{V} = \mathbf{Set}$ it changes nothing, since $\mathbf{Set}(1, S) \cong S$: an ordinary category is its own underlying category. The interesting cases are those where $\mathbf{C}_0$ forgets real structure, and the enriched data is strictly richer than the ordinary category it sits over.

Example 66.1.3: Enrichment over the Standing Cast

Each choice of monoidal base $\mathbf{V}$ names a familiar species of category.

1. **Set**-*enriched categories are ordinary categories.* With $\mathbf{V} = (\mathbf{Set}, \times, 1)$ the data of definition 66.1.1 is precisely objects, hom-sets, a composition function, and chosen identities obeying associativity and unit laws. This is the definition of a locally small category, and the underlying category $\mathbf{C}_0$ is $\mathbf{C}$ itself.

2. **Ab**-*enriched categories are preadditive categories.* Take $\mathbf{V} = (\mathbf{Ab}, \otimes_{\mathbb{Z}}, \mathbb{Z})$, the tensor product of abelian groups with unit $\mathbb{Z}$. Each hom-object $\mathbf{C}(x,y)$ is then an abelian group, and composition is a morphism of abelian groups

$$\mathbf{C}(y,z) \otimes_{\mathbb{Z}} \mathbf{C}(x,y) \to \mathbf{C}(x,z)$$

which is exactly a $\mathbb{Z}$-bilinear map $\mathbf{C}(y,z) \times \mathbf{C}(x,y) \to \mathbf{C}(x,z)$, by the universal property of $\otimes_{\mathbb{Z}}$. Composition is therefore biadditive: $h(g+g') = hg + hg'$ and $(h+h')g = hg + h'g$. A category with abelian-group hom-sets and biadditive composition is a *preadditive category*, and the underlying category recovers the hom-set as $\mathbf{Ab}(\mathbb{Z}, \mathbf{C}(x,y)) \cong \mathbf{C}(x,y)$. This is the enrichment that homological algebra rests on: an additive category is a preadditive category with a zero object and biproducts, and **Ab**-enrichment is the first axiom (see [25] for the module-theoretic origin of $\otimes_{\mathbb{Z}}$).

3. **Top**-*enriched categories topologize their hom-sets.* With $\mathbf{V} = (\mathbf{Top}, \times, *)$ each $\mathbf{C}(x, y)$ is a topological space and composition $\mathbf{C}(y, z) \times \mathbf{C}(x, y) \to \mathbf{C}(x, z)$ is continuous. The identity $* \to \mathbf{C}(x, x)$ from the one-point space picks out id_x. The underlying category has hom-sets $\mathbf{Top}(*, \mathbf{C}(x, y)) \cong \mathbf{C}(x, y)$ as sets, discarding the topology; the enriched structure remembers, for instance, when two maps are homotopic through a continuous path of morphisms.

4. **Cat**-*enriched categories are strict* 2-*categories.* Take $\mathbf{V} = (\mathbf{Cat}, \times, \mathbf{1})$, categories under product with the terminal category $\mathbf{1}$ as unit. Each hom-object $\mathbf{C}(x, y)$ is now a *category*: its objects are the morphisms $x \to y$, and its own morphisms are a second layer of structure between them. Composition is a functor $\mathbf{C}(y, z) \times \mathbf{C}(x, y) \to \mathbf{C}(x, z)$. This is exactly a strict 2-category, the subject of section 66.4: the prototype is **Cat** itself, whose hom-categories are the functor categories $[\mathbf{C}, \mathbf{D}]$ (definition 59.5.4) with natural transformations as the second layer.

5. *Lawvere metric spaces.* Let $\mathbf{V} = ([0, \infty], +, 0)$ be the extended nonnegative reals, viewed as a category by the order $\geq$ (an arrow $a \to b$ exactly when $a \geq b$), monoidal under addition with unit 0. A category enriched over this $\mathbf{V}$ has objects, and for each pair a number $\mathbf{C}(x, y) \in [0, \infty]$ to be read as a *distance*. The composition arrow $\mathbf{C}(y, z) + \mathbf{C}(x, y) \to \mathbf{C}(x, z)$ is, in this order category, precisely the inequality

$$\mathbf{C}(x, z) \leq \mathbf{C}(y, z) + \mathbf{C}(x, y)$$

the triangle inequality, and the identity $0 \to \mathbf{C}(x, x)$ is the inequality $\mathbf{C}(x, x) \leq 0$, forcing $\mathbf{C}(x, x) = 0$. Such a structure is a *Lawvere metric space*: a metric space that drops symmetry and the separation axiom (distinct points may sit at distance 0) and permits ∞. Composition of morphisms and the triangle inequality are the same axiom seen through two monoidal bases.

The five bases run from the trivial to the exotic and read off five theories at a stroke. **Set** recovers the ordinary notion; **Ab** recovers the preadditive categories of homological algebra; **Cat** recovers strict 2-categories; and the order category $[0, \infty]$ shows that enrichment need not be over structured sets at all, turning the triangle inequality into an instance of composition. The last example is the sharpest test of the definition, since $[0, \infty]$ has no underlying sets in the naive sense: what matters is only that it is monoidal, and that alone is enough to carry categorical composition. The lesson is that "category" is not tied to **Set**; it is tied to whatever monoidal $\mathbf{V}$ one composes in.

66.2 Enriched Functors and Naturality

A $\mathbf{V}$-category (definition 66.1.1) carries its structure in $\mathbf{V}$: a functor between two such categories must act on hom-objects by morphisms of $\mathbf{V}$, and a natural transformation must have components that live in $\mathbf{V}$ rather than in Hom-sets. This section installs the two notions, shows how a monoidal functor transports the whole theory from one base to another, and records the enriched form of the Yoneda lemma.

An ordinary functor sends each hom-set $\mathrm{Hom}(x, y)$ to $\mathrm{Hom}(Fx, Fy)$ by post- and pre-composition, and the two functoriality axioms say this map respects composition and identities. The enriched version replaces those set functions with **V**-morphisms and rewrites the two axioms as commuting diagrams in **V**.

Definition 66.2.1: Enriched Functor

Let **V** be a monoidal category and let **C**, **D** be **V**-categories. A **V**-*functor* (or *enriched functor*) $F\colon \mathbf{C} \to \mathbf{D}$ consists of

- a map $x \mapsto Fx$ on objects, and
- for each pair of objects x, y of **C**, a **V**-morphism

$$F_{x,y}\colon \mathbf{C}(x,y) \to \mathbf{D}(Fx, Fy)$$

such that the following two diagrams in **V** commute. Writing $\circ$ for the composition morphisms of **C** and **D** and $j_x\colon I \to \mathbf{C}(x,x)$ for the identity morphisms, compatibility with composition is the commutativity of

$$\begin{tikzcd}
\mathbf{C}(y,z) \otimes \mathbf{C}(x,y) \arrow[r, "F_{y,z} \otimes F_{x,y}"] \arrow[d, "\circ"'] & \mathbf{D}(Fy,Fz) \otimes \mathbf{D}(Fx,Fy) \arrow[d, "\circ"] \\
\mathbf{C}(x,z) \arrow[r, "F_{x,z}"'] & \mathbf{D}(Fx,Fz)
\end{tikzcd}$$

and compatibility with identities is the commutativity of

$$\begin{tikzcd}
I \arrow[r, "j_x"] \arrow[dr, "j_{Fx}"'] & \mathbf{C}(x,x) \arrow[d, "F_{x,x}"] \\
& \mathbf{D}(Fx,Fx)
\end{tikzcd}$$

Reading the squares in $\mathbf{V} = \mathbf{Set}$ recovers the ordinary axioms: the first says $F(g \circ f) = Fg \circ Ff$ and the second $F(\mathrm{id}_x) = \mathrm{id}_{Fx}$. The content is that these hold at the level of **V**, so the functor is compatible with whatever extra structure the hom-objects carry.

Example 66.2.2: Additive Functors as Ab-Functors

Let $\mathbf{V} = \mathbf{Ab}$, so that **C** and **D** are preadditive categories (example 66.1.3): each Hom-set is an abelian group and composition is bilinear. An **Ab**-functor $F\colon \mathbf{C} \to \mathbf{D}$ is an ordinary functor whose action on morphisms

$$F_{x,y}\colon \mathbf{C}(x,y) \to \mathbf{D}(Fx, Fy)$$

is a group homomorphism for every pair x, y, that is, $F(f+g) = Ff + Fg$. These are exactly the *additive functors*. The composition square is the statement that F is bilinear-compatible, and the identity triangle that $F(\mathrm{id}_x) = \mathrm{id}_{Fx}$; both are automatic for a functor once additivity on hom-groups is imposed. For $\mathbf{V} = \mathbf{Cat}$, an enriched functor is a strict 2-functor, sending objects, 1-cells, and 2-cells compatibly (this is developed in section 66.4).

An enriched functor moves objects and hom-objects; a natural transformation must compare two such functors at each object. In the unenriched setting a component is a morphism $\alpha_x\colon Fx \to Gx$, and morphisms out of x are elements of a hom-set. In **V** there are no elements; a "point" of the hom-object

$\mathbf{D}(Fx, Gx)$ is a $\mathbf{V}$-morphism $I \to \mathbf{D}(Fx, Gx)$ out of the unit, matching the underlying-category convention $\mathrm{Hom}(x, y) = \mathbf{V}(I, \mathbf{C}(x, y))$ from definition 66.1.2. A component is therefore such a morphism, and naturality becomes a single diagram in $\mathbf{V}$.

Definition 66.2.3: Enriched Natural Transformation

Let $F, G \colon \mathbf{C} \to \mathbf{D}$ be $\mathbf{V}$-functors. A $\mathbf{V}$*-natural transformation* (or *enriched natural transformation*) $\alpha \colon F \Rightarrow G$ assigns to each object x of $\mathbf{C}$ a $\mathbf{V}$-morphism

$$\alpha_x \colon I \to \mathbf{D}(Fx, Gx)$$

(the *component* at x) such that for every pair x, y the following diagram in $\mathbf{V}$ commutes:

$$\begin{tikzcd}
\mathbf{C}(x,y) \arrow[r, "\rho^{-1}"] \arrow[d, "\lambda^{-1}"'] & \mathbf{C}(x,y) \otimes I \arrow[r, "G_{x,y} \otimes \alpha_x"] & \mathbf{D}(Gx,Gy) \otimes \mathbf{D}(Fx,Gx) \arrow[d, "\circ"] \\
I \otimes \mathbf{C}(x,y) \arrow[r, "\alpha_y \otimes F_{x,y}"'] & \mathbf{D}(Fy,Gy) \otimes \mathbf{D}(Fx,Fy) \arrow[r, "\circ"'] & \mathbf{D}(Fx,Gy)
\end{tikzcd}$$

Here λ and ρ are the unitors of $\mathbf{V}$ (definition 65.1.1), inserting the unit I so that a component may be composed against a hom-object.

The two legs of the diagram are the enriched versions of the two paths around the ordinary naturality square. The top-right leg applies G to a morphism $f \colon x \to y$ and pre-composes with α_x, producing the path $Gf \cdot \alpha_x$; the bottom-left leg applies F and post-composes with α_y, producing $\alpha_y \cdot Ff$. Equality of the legs is precisely $\alpha_y \circ Ff = Gf \circ \alpha_x$ written internally to $\mathbf{V}$. In $\mathbf{V} = \mathbf{Set}$ each component $\alpha_x \colon I = \{*\} \to \mathbf{D}(Fx, Gx)$ names an element $\alpha_x \in \mathrm{Hom}(Fx, Gx)$, and the diagram collapses to the familiar commuting square.

Example 66.2.4: Additive Natural Transformations

For $\mathbf{V} = \mathbf{Ab}$, a component $\alpha_x \colon \mathbb{Z} \to \mathbf{D}(Fx, Gx)$ is a group homomorphism out of $\mathbb{Z}$, hence pinned down by the single element $\alpha_x(1) \in \mathrm{Hom}(Fx, Gx)$: an ordinary component morphism. The enriched naturality diagram then says nothing more than ordinary naturality, because both composition maps of $\mathbf{Ab}$ are $\mathbb{Z}$-bilinear and evaluating at $1 \otimes 1$ returns the set-level square. An $\mathbf{Ab}$-natural transformation between additive functors is thus an ordinary natural transformation, with no further condition to check.

The definitions of $\mathbf{V}$-category, $\mathbf{V}$-functor, and $\mathbf{V}$-natural transformation depend on $\mathbf{V}$ only through its monoidal structure. A functor between monoidal categories that preserves that structure should therefore transport enriched categories along it. The correct hypothesis is weaker than strict preservation of $\otimes$: it is enough that $\mathbf{V}$-tensors map to $\mathbf{W}$-tensors up to a coherent comparison morphism, which is exactly a lax monoidal functor. That comparison is what carries composition across the base.

Proposition 66.2.5: Change of Enriching Base

Let $(\Phi, \varphi, \varphi_0)\colon \mathbf{V} \to \mathbf{W}$ be a lax monoidal functor: Φ is a functor equipped with a natural comparison $\varphi_{a,b}\colon \Phi a \otimes \Phi b \to \Phi(a \otimes b)$ and a unit comparison $\varphi_0\colon I_{\mathbf{W}} \to \Phi(I_{\mathbf{V}})$, satisfying the associativity and unit coherence laws. Then Φ induces a functor

$$\Phi_*\colon \mathbf{V}\text{-}\mathbf{Cat} \to \mathbf{W}\text{-}\mathbf{Cat}$$

called *change of base*, sending a $\mathbf{V}$-category $\mathbf{C}$ to the $\mathbf{W}$-category $\Phi_*\mathbf{C}$ with the same objects and hom-objects $(\Phi_*\mathbf{C})(x, y) = \Phi\big(\mathbf{C}(x, y)\big)$.

***Proof*:**
Fix a lax monoidal functor $(\Phi, \varphi, \varphi_0)$. The construction proceeds in two stages: first the object $\Phi_*\mathbf{C}$, then the action of Φ_* on enriched functors, checking the axioms at each stage.

Step 1: The category $\Phi_*\mathbf{C}$. Take the objects of $\mathbf{C}$ and the hom-objects $\Phi(\mathbf{C}(x, y))$ in $\mathbf{W}$. The composition morphism of $\Phi_*\mathbf{C}$ is the composite

$$\Phi(\mathbf{C}(y, z)) \otimes \Phi(\mathbf{C}(x, y)) \xrightarrow{\varphi} \Phi\big(\mathbf{C}(y, z) \otimes \mathbf{C}(x, y)\big) \xrightarrow{\Phi(\circ)} \Phi(\mathbf{C}(x, z))$$

which first uses the comparison φ to gather the two hom-objects inside Φ and then applies Φ to the composition of $\mathbf{C}$. The identity morphism is

$$I_{\mathbf{W}} \xrightarrow{\varphi_0} \Phi(I_{\mathbf{V}}) \xrightarrow{\Phi(j_x)} \Phi(\mathbf{C}(x, x))$$

using the unit comparison followed by Φ applied to the identity of $\mathbf{C}$. Associativity of composition in $\Phi_*\mathbf{C}$ follows from associativity in $\mathbf{C}$ (apply Φ to its associativity square) glued to the associativity coherence of the lax structure φ; the two unit laws follow likewise from the unit laws in $\mathbf{C}$ glued to the unit coherence of φ_0. Each verification is a diagram in $\mathbf{W}$ built from a Φ-image of a $\mathbf{C}$-axiom and one lax-coherence cell sharing a common edge: the coherence cell rebrackets the φ-comparisons to expose Φ applied to the tensor of the hom-objects, and the Φ-image of the $\mathbf{C}$-axiom equates the two composites there, so the two cells paste along that edge into the required square. Thus $\Phi_*\mathbf{C}$ is a $\mathbf{W}$-category.

Step 2: The action on functors and naturality. Given a $\mathbf{V}$-functor $F\colon \mathbf{C} \to \mathbf{D}$, define Φ_*F to agree with F on objects and to act on hom-objects by

$$(\Phi_*F)_{x,y} = \Phi(F_{x,y})\colon \Phi(\mathbf{C}(x, y)) \to \Phi(\mathbf{D}(Fx, Fy))$$

Its composition square is Φ applied to the composition square of F, with the naturality of φ used to slide $\Phi(F_{y,z}) \otimes \Phi(F_{x,y})$ past the two copies of the comparison; the identity triangle is Φ applied to that of F, precomposed with φ_0. Both commute, so Φ_*F is a $\mathbf{W}$-functor. Functoriality of Φ_*, preservation of enriched composition and identities of functors, is immediate, since Φ_* acts as Φ on hom-objects and Φ is a functor. This gives the assignment on $\mathbf{V}\text{-}\mathbf{Cat}$, completing the proof.

Change of base is the mechanism by which enriched category theory specializes to ordinary category theory, and by which one kind of enrichment passes to another. The main instance takes $\mathbf{W} = \mathbf{Set}$.

Example 66.2.6: The Underlying Category as Change of Base
Let $\mathbf{V}$ be monoidal. The representable functor

$$\mathbf{V}(I,-)\colon \mathbf{V} \to \mathbf{Set}, \qquad a \mapsto \mathbf{V}(I,a)$$

is lax monoidal: its comparison sends a pair $(u\colon I \to a,\ v\colon I \to b)$ to the composite $I \xrightarrow{\lambda^{-1}} I \otimes I \xrightarrow{u \otimes v} a \otimes b$, and its unit comparison $\{*\} \to \mathbf{V}(I,I)$ picks out id_I. Change of base along it,

$$\big(\mathbf{V}(I,-)\big)_*\colon \mathbf{V}\text{-}\mathbf{Cat} \to \mathbf{Set}\text{-}\mathbf{Cat} = \mathbf{Cat}$$

sends a $\mathbf{V}$-category $\mathbf{C}$ to a category with hom-sets $\mathbf{V}(I, \mathbf{C}(x,y))$, exactly the underlying category $\mathbf{C}_0$ of definition 66.1.2. The construction of $\mathbf{C}_0$ is thus not an ad hoc device but the $\mathbf{W} = \mathbf{Set}$ case of this proposition. Concretely, for $\mathbf{V} = \mathbf{Ab}$ the functor $\mathbf{Ab}(\mathbb{Z},-)$ is the forgetful functor $\mathbf{Ab} \to \mathbf{Set}$, and change of base along it strips a preadditive category down to its underlying ordinary category, forgetting the abelian-group structure on the hom-sets while keeping the same objects and morphisms.

A $\mathbf{V}$-natural transformation, as definition 66.2.3 defines it, is a family of morphisms, and families form a set. Every other hom in this chapter has been an object of $\mathbf{V}$, and the collection of $\mathbf{V}$-natural transformations ought to be one too. Supplying that object is the first task of section 66.3, and it is what the enriched Yoneda lemma needs before it can even be stated.

66.3 Enriched Functor Categories and Weighted Limits

Two constructions stand between the definitions of the previous section and a working enriched theory. The first is the *end*, a limit in $\mathbf{V}$ that cuts the product of the hom-objects $\mathbf{D}(Fc, Gc)$ down to the part respecting naturality; it is what promotes the $\mathbf{V}$-natural transformations from a set to an object of $\mathbf{V}$, and with it the $\mathbf{V}$-functors $\mathbf{C} \to \mathbf{D}$ become a $\mathbf{V}$-category $[\mathbf{C}, \mathbf{D}]$. The second is the *tensor* and *cotensor*, by which an object of $\mathbf{V}$ acts on an object of a $\mathbf{V}$-category. Together they make the enriched Yoneda lemma provable, and they yield one definition, the weighted limit, that has ordinary limits, tensors and cotensors as three instances.

Throughout this section $\mathbf{V}$ is symmetric monoidal closed (definition 65.2.2, definition 65.3.1) with all small limits and colimits, and $\mathbf{C}$ is small. Closedness supplies the internal hom $[a,b]$; symmetry is what allows $\mathbf{C}^{\mathrm{op}}$ to be formed as a $\mathbf{V}$-category and what orients the second of the two morphisms in definition 66.3.3; completeness supplies the products and equalizers that the end is built from, and cocompleteness the coproducts and coequalizers its dual needs. The bracket $[a,b]$ for the internal hom and $[\mathbf{C}, \mathbf{D}]$ for the functor category are distinguished by their arguments, objects in the first case and categories in the second.

Ordinary category theory copies an object along a set: the copower $S \cdot e$ of section 64.4 is S-many copies of e, characterized by $\mathrm{Hom}(S \cdot e, d) \cong \mathbf{Set}(S, \mathrm{Hom}(e,d))$. Enriched over $\mathbf{V}$ the indexing set becomes an object of $\mathbf{V}$ and the hom-sets become hom-objects, and the same two universal properties read in $\mathbf{V}$ define the two constructions below.

Definition 66.3.1: Tensor and Cotensor

Let $\mathbf{C}$ be a $\mathbf{V}$-category, c an object of $\mathbf{C}$, and v an object of $\mathbf{V}$. The *tensor* of c by v is an object $v \odot c$ of $\mathbf{C}$ together with an isomorphism

$$\mathbf{C}(v \odot c,\ d)\ \cong\ \big[v,\ \mathbf{C}(c,d)\big]$$

of objects of $\mathbf{V}$, natural in d. The *cotensor* of c by v is an object c^v of $\mathbf{C}$ together with an isomorphism

$$\mathbf{C}(d,\ c^v)\ \cong\ \big[v,\ \mathbf{C}(d,c)\big]$$

natural in d. The category $\mathbf{C}$ is *tensored* if $v \odot c$ exists for every v and c, and *cotensored* if c^v always exists.

Each is defined by a representing property, so each is unique up to a unique isomorphism compatible with the displayed natural isomorphism, by the argument of theorem 60.2.1 applied to the underlying category. The tensor is v-indexed copies of c and the cotensor is v-indexed families in c, with "v-indexed" interpreted by $\mathbf{V}$ rather than by a set. Note which variable each is natural in: the tensor represents a functor of the target and the cotensor a functor of the source, which is why the tensor is a colimit-like construction and the cotensor a limit-like one.

Example 66.3.2: Tensors and Cotensors over the Standing Bases

1. $\mathbf{V}$ *over itself.* The base $\mathbf{V}$ is a $\mathbf{V}$-category with hom-objects $[a,b]$, and there $v \odot a = v \otimes a$ and $a^v = [v,a]$. The first is the internal form of (65.1): the required isomorphism $[v \otimes a, b] \cong [v,[a,b]]$ holds because, for every u, the adjunction gives $\mathbf{V}(u,[v\otimes a,b]) \cong \mathbf{V}(u \otimes v \otimes a, b) \cong \mathbf{V}(u,[v,[a,b]])$ naturally in u, and theorem 60.2.1 turns a natural isomorphism of representables into one of the representing objects. The second follows from the first by symmetry of $\otimes$.

2. $\mathbf{V} = \mathbf{Set}$. For an ordinary category $\mathbf{C}$ the defining isomorphism reads $\mathrm{Hom}(S \odot c, d) \cong \mathbf{Set}(S, \mathrm{Hom}(c,d))$, and an S-indexed family of maps $c \to d$ is a single map out of the coproduct. So $S \odot c = \coprod_S c$ and, dually, $c^S = \prod_S c$: the tensor and cotensor are the copower and power of section 64.4, and a category is tensored and cotensored over $\mathbf{Set}$ exactly when it has all small coproducts and products.

3. $\mathbf{V} = \mathbf{Ab}$. For $\mathbf{C} = \mathbf{Ab}$ itself, $A \odot B = A \otimes_{\mathbb{Z}} B$ and $B^A = \mathrm{Hom}_{\mathbb{Z}}(A,B)$, by (1), since $\mathbf{Ab}$ is closed with internal hom $\mathrm{Hom}_{\mathbb{Z}}$. For $\mathbf{C}$ the R-modules, the tensor of a module M by an abelian group A is $A \otimes_{\mathbb{Z}} M$ with R acting through M, because a homomorphism $A \to \mathrm{Hom}_R(M,N)$ is the same as an R-linear map $A \otimes_{\mathbb{Z}} M \to N$.

The three bases show the pattern: over $\mathbf{Set}$ the tensor is a coproduct of copies, over $\mathbf{Ab}$ it is a tensor product, and over $\mathbf{V}$ itself it is $\otimes$. In every case the operation is the one that lets an object of the base act on an object of the enriched category, and that action is what definition 66.3.7 will weight a diagram by.

The second construction is the one the previous section stopped short of. A $\mathbf{V}$-natural transformation $F \Rightarrow G$ is a family of morphisms $I \to \mathbf{D}(Fc, Gc)$ satisfying the naturality condition of definition 66.2.3. Replacing "family of morphisms out of I" by "object of $\mathbf{V}$ mapping to each $\mathbf{D}(Fc,Gc)$", and imposing naturality as an equation between two morphisms into a product, turns that collection into a limit in

V.

Definition 66.3.3: The Object of Enriched Natural Transformations

Let $F, G\colon \mathbf{C} \to \mathbf{D}$ be $\mathbf{V}$-functors. The *object of* $\mathbf{V}$*-natural transformations* from F to G, also called the *end* of $c \mapsto \mathbf{D}(Fc, Gc)$ and written

$$[\mathbf{C}, \mathbf{D}](F, G) \;=\; \int_c \mathbf{D}(Fc, Gc)$$

is the equalizer in $\mathbf{V}$ (definition 61.2.4) of the two morphisms

$$\prod_c \mathbf{D}(Fc, Gc) \;\rightrightarrows\; \prod_{c,c'} \big[\, \mathbf{C}(c, c'),\; \mathbf{D}(Fc, Gc')\,\big]$$

whose components at a pair (c, c') are the transposes under (65.1) of

$$\textstyle\prod_b \mathbf{D}(Fb, Gb) \otimes \mathbf{C}(c, c') \xrightarrow{\pi_{c'} \otimes F_{c,c'}} \mathbf{D}(Fc', Gc') \otimes \mathbf{D}(Fc, Fc') \xrightarrow{\circ} \mathbf{D}(Fc, Gc')$$

and

$$\textstyle\prod_b \mathbf{D}(Fb, Gb) \otimes \mathbf{C}(c, c') \xrightarrow{\pi_c \otimes G_{c,c'}} \mathbf{D}(Fc, Gc) \otimes \mathbf{D}(Gc, Gc') \xrightarrow{\sigma} \mathbf{D}(Gc, Gc') \otimes \mathbf{D}(Fc, Gc) \xrightarrow{\circ} \mathbf{D}(Fc, Gc')$$

where π_c is a product projection and σ the symmetry of $\mathbf{V}$. The products are indexed by the objects and the ordered pairs of objects of the small category $\mathbf{C}$, and both they and the equalizer exist because $\mathbf{V}$ is complete.

The two morphisms are the two routes from a component at one object to a value at another: the first carries the component at c' backwards along F, the second carries the component at c forwards along G, and a family is natural exactly when the two agree. Writing α_c for a component, the first is $\alpha_{c'} \circ F(f)$ and the second $G(f) \circ \alpha_c$ in the notation of an ordinary category; the symmetry σ appears in the second because $\circ$ takes its arguments in the order (later, earlier) while the projection produces them in the order (earlier, later).

Probing the equalizer with the unit recovers the unenriched notion. The functor $\mathbf{V}(I, -)$ is representable, hence preserves limits, so $\mathbf{V}\big(I, [\mathbf{C}, \mathbf{D}](F, G)\big)$ is the equalizer of the corresponding pair of functions, which is precisely the set of families $\alpha_c\colon I \to \mathbf{D}(Fc, Gc)$ satisfying definition 66.2.3. The new object therefore refines the old set rather than replacing it.

Proposition 66.3.4: The Enriched Functor Category

Let $\mathbf{C}$ be a small $\mathbf{V}$-category and $\mathbf{D}$ a $\mathbf{V}$-category. The $\mathbf{V}$-functors $\mathbf{C} \to \mathbf{D}$ are the objects of a $\mathbf{V}$-category $[\mathbf{C}, \mathbf{D}]$ whose hom-objects are those of definition 66.3.3. Its underlying ordinary category (definition 66.1.2) is the category of $\mathbf{V}$-functors and $\mathbf{V}$-natural transformations, and for $\mathbf{V} = \mathbf{Set}$ it is the functor category of definition 59.5.4.

***Proof*:**
Write $E(F,G) = [\mathbf{C},\mathbf{D}](F,G)$ and let $\pi_c\colon E(F,G) \to \mathbf{D}(Fc,Gc)$ denote the composite of the equalizer inclusion with the projection at c.

Step 1: Composition. The componentwise composites

$$E(G,H) \otimes E(F,G) \xrightarrow{\pi_c \otimes \pi_c} \mathbf{D}(Gc,Hc) \otimes \mathbf{D}(Fc,Gc) \xrightarrow{\circ} \mathbf{D}(Fc,Hc)$$

are the components of a morphism into $\prod_c \mathbf{D}(Fc,Hc)$. It factors through the equalizer $E(F,H)$, and this is the one point needing argument. Transposing, the claim is that the two routes from $E(G,H) \otimes E(F,G) \otimes \mathbf{C}(c,c')$ to $\mathbf{D}(Fc,Hc')$ agree. Take the route carrying the composite component at c' backwards along F. Since $E(G,H)$ and $E(F,G)$ are equalizers, their own defining equations apply: the β-factor may be moved from "at c', transported along G" to "at c, transported along G" and then the α-factor from "at c' along F" to "at c along H". Concretely, insert $\mathbf{C}(c,c')$ between the two factors using symmetry, apply the equalizer equation of $E(F,G)$ to replace $\alpha_{c'} \circ F$ by $G \circ \alpha_c$, then apply the equalizer equation of $E(G,H)$ to replace $\beta_{c'} \circ G$ by $H \circ \beta_c$. The two applications are along different tensor factors, so they commute, and associativity of $\circ$ in $\mathbf{D}$ closes the square onto the second route. Hence the composite lands in $E(F,H)$.

Step 2: Identities. The morphisms $j_{Fc}\colon I \to \mathbf{D}(Fc,Fc)$ assemble into $I \to \prod_c \mathbf{D}(Fc,Fc)$, and this factors through $E(F,F)$: the two routes out of $I \otimes \mathbf{C}(c,c')$ are $\mathrm{id}_{Fc'} \circ F(f)$ and $F(f) \circ \mathrm{id}_{Fc}$, equal by the unit laws of $\mathbf{D}$. This gives $j_F\colon I \to E(F,F)$.

Step 3: The axioms. Associativity and the unit laws for the composition of Step 1 are equations between morphisms into $E(F,K)$, and the projections π_c out of an equalizer of a product are jointly monic, so it suffices to check them after each π_c. There they read as associativity and the unit laws of $\mathbf{D}$ at the object Fc, which hold by definition 66.1.1. So $[\mathbf{C},\mathbf{D}]$ is a $\mathbf{V}$-category.

Step 4: The underlying category and the **Set** *case.* By definition 66.1.2 the underlying category has hom-sets $\mathbf{V}(I, E(F,G))$, identified above with the $\mathbf{V}$-natural transformations $F \Rightarrow G$, and Steps 1 and 2 restrict to their usual composition and identities. For $\mathbf{V} = \mathbf{Set}$ the internal hom is the hom-set, the product is the cartesian product and the equalizer the subset on which the two functions agree, so $E(F,G)$ is the set of natural transformations and $[\mathbf{C},\mathbf{D}]$ is definition 59.5.4.

The construction is the ordinary functor category with two substitutions: the product of hom-sets becomes a product in $\mathbf{V}$, and the naturality condition, stated in the ordinary case as an equation holding at each morphism, becomes the equalizer that imposes it. Everything else, composition and identities, is inherited componentwise from $\mathbf{D}$, which is why Step 3 reduces to axioms already available.

One base makes both constructions concrete at once.

Example 66.3.5: Modules over a Ring as an Enriched Functor Category
Let R be a ring and let $\mathbf{R}$ be the $\mathbf{Ab}$-category with a single object $*$ and hom-object $\mathbf{R}(*,*) = R$, its composition the multiplication of R and its identity $j_*\colon \mathbb{Z} \to R$ the map sending 1 to 1_R. This is definition 66.1.1 over $\mathbf{V} = (\mathbf{Ab}, \otimes_{\mathbb{Z}}, \mathbb{Z})$, and the associativity and unit diagrams are the ring axioms.

An $\mathbf{Ab}$-functor $F\colon \mathbf{R} \to \mathbf{Ab}$ is an abelian group $M = F(*)$ together with a homomorphism $R \to \mathrm{Hom}_{\mathbb{Z}}(M,M)$ respecting multiplication and the unit, which is exactly a left R-module.

For two such, $F \leftrightarrow M$ and $G \leftrightarrow N$, definition 66.3.3 has a single-factor product $\mathrm{Hom}_{\mathbb{Z}}(M,N)$ and equalizes the two maps $\mathrm{Hom}_{\mathbb{Z}}(M,N) \to \mathrm{Hom}_{\mathbb{Z}}\big(R, \mathrm{Hom}_{\mathbb{Z}}(M,N)\big)$ sending f to $r \mapsto f \circ (r \cdot -)$ and to $r \mapsto (r \cdot -) \circ f$. Their equalizer is the subgroup of f with $f(rm) = rf(m)$, so

$$[\mathbf{R}, \mathbf{Ab}](F,G) \;=\; \mathrm{Hom}_R(M,N)$$

and proposition 66.3.4 says the left R-modules form an **Ab**-category whose hom-objects are the R-linear maps. The **Ab**-enrichment of R-modules that homological algebra takes as given is thus an instance of the enriched functor category.

With $[\mathbf{C}, \mathbf{V}]$ available, the enriched Yoneda lemma can be stated and proved. The ordinary statement (theorem 60.2.1) identified the natural transformations out of a representable with the elements of Fx; the enriched one identifies an object of $\mathbf{V}$ with an object of $\mathbf{V}$, and the proof is the same evaluation-at-the-identity argument written in $\mathbf{V}$.

Theorem 66.3.6: Enriched Yoneda Lemma

Let $\mathbf{C}$ be a small $\mathbf{V}$-category and $F\colon \mathbf{C} \to \mathbf{V}$ a $\mathbf{V}$-functor into the base. For each object x of $\mathbf{C}$ there is an isomorphism

$$[\mathbf{C}, \mathbf{V}]\big(\mathbf{C}(x,-),\, F\big) \;\cong\; Fx$$

of objects of $\mathbf{V}$, natural in x and in F. For $\mathbf{V} = \mathbf{Set}$ it is the bijection $\mathrm{Nat}(\mathrm{Hom}(x,-),F) \cong Fx$ of theorem 60.2.1.

***Proof*:**
Write $E = [\mathbf{C}, \mathbf{V}]\big(\mathbf{C}(x,-), F\big) = \int_y [\mathbf{C}(x,y), Fy]$, with $\pi_y\colon E \to [\mathbf{C}(x,y), Fy]$ the map of definition 66.3.3. Two morphisms are constructed and shown to be mutually inverse.

Step 1: The map $\psi\colon E \to Fx$. Take the component at $y = x$ and evaluate at the identity. The identity $j_x\colon I \to \mathbf{C}(x,x)$ induces $[\,j_x, Fx\,]\colon [\mathbf{C}(x,x), Fx] \to [I, Fx]$, and $[I, Fx] \cong Fx$ since I is the unit. Set $\psi = \big([\,j_x, Fx\,] \circ \pi_x\big)$ followed by that isomorphism.

Step 2: The map $\varphi\colon Fx \to E$. The $\mathbf{V}$-functor F supplies $F_{x,y}\colon \mathbf{C}(x,y) \to [Fx, Fy]$, whose transpose under (65.1) is an action $\mathbf{C}(x,y) \otimes Fx \to Fy$. Transposing that in the other variable gives $\varphi_y\colon Fx \to [\mathbf{C}(x,y), Fy]$ for each y. These form a wedge: the equalizer condition of definition 66.3.3 at (y,z) is, after transposition, the assertion that acting by a morphism of $\mathbf{C}(y,z)$ after acting by one of $\mathbf{C}(x,y)$ agrees with acting by their composite, which is the composition axiom of the $\mathbf{V}$-functor F. So the φ_y factor through E, defining φ.

Step 3: $\psi\varphi = \mathrm{id}_{Fx}$. Composing, $\psi\varphi$ is the transpose of $\mathbf{C}(x,x) \otimes Fx \to Fx$ precomposed with $j_x \otimes \mathrm{id}$, that is, the action of the identity. The unit axiom of the $\mathbf{V}$-functor F says $F_{x,x} \circ j_x = j_{Fx}$, and j_{Fx} is the transpose of the left unitor, so the action of the identity is λ and $\psi\varphi = \mathrm{id}$.

Step 4: $\varphi\psi = \mathrm{id}_E$. The projections π_z are jointly monic, so it suffices that $\pi_z\varphi\psi = \pi_z$ for every z. Transpose to morphisms $E \otimes \mathbf{C}(x,z) \to Fz$. Now take the equalizer equation of definition 66.3.3 at the pair (x,z) and transpose it to an equation between morphisms $E \otimes \mathbf{C}(x,z) \otimes \mathbf{C}(x,x) \to Fz$: one side applies π_x and then acts by $\mathbf{C}(x,z)$, the other composes $\mathbf{C}(x,z) \otimes \mathbf{C}(x,x) \to \mathbf{C}(x,z)$ and then applies π_z. Precompose both with $\mathrm{id}_E \otimes \mathrm{id} \otimes j_x$ and use the unit law of $\mathbf{C}$. The left side becomes the transpose of $\pi_z\varphi\psi$, since ψ is exactly π_x followed by evaluation at j_x and φ is the action; the right side becomes the transpose of π_z. Hence $\pi_z\varphi\psi = \pi_z$ for all z, and $\varphi\psi = \mathrm{id}_E$.

Step 5: Naturality. Naturality in F holds because ψ was built from π_x and j_x, neither of which involves F, so a $\mathbf{V}$-natural transformation $F \Rightarrow F'$ commutes with it by construction. Naturality in x follows from Step 4 read for the two objects together with the composition axiom of $\mathbf{C}$, in the same way that Step 2 used the composition axiom of F.

Nothing in the argument is new relative to theorem 60.2.1: Step 1 is "evaluate at id_x", Step 2 is "transport along the action", and Steps 3 and 4 are the two round trips. What the enrichment changes is that each step is a morphism of $\mathbf{V}$ rather than a function on elements, and the naturality condition that was a family of equations is now the equalizer through which the maps must factor. Setting $\mathbf{V} = \mathbf{Set}$ collapses every internal object to a set and returns the original proof verbatim.

The last definition of the chapter's enriched material unifies the two constructions above. An ordinary limit is a universal cone, and a cone over F with vertex d is a natural transformation from the constant functor at a point into $\mathrm{Hom}(d, F-)$. Nothing requires the constant functor: any $\mathbf{V}$-functor W may take its place, and the resulting notion is strictly more general because W may assign different objects of $\mathbf{V}$ to different objects of $\mathbf{C}$.

Definition 66.3.7: Weighted Limit and Weighted Colimit

Let $\mathbf{C}$ be a small $\mathbf{V}$-category, $\mathbf{D}$ a $\mathbf{V}$-category and $F\colon \mathbf{C} \to \mathbf{D}$ a $\mathbf{V}$-functor. For a *weight* $W\colon \mathbf{C} \to \mathbf{V}$, the *limit of F weighted by W* is an object $\lim^W F$ of $\mathbf{D}$ with an isomorphism

$$\mathbf{D}\big(d,\ \lim^W F\big) \ \cong\ [\mathbf{C}, \mathbf{V}]\big(W,\ \mathbf{D}(d, F-)\big)$$

of objects of $\mathbf{V}$, natural in d. For a weight $W\colon \mathbf{C}^{\mathrm{op}} \to \mathbf{V}$, the *colimit of F weighted by W*, written $W \star F$, is an object with

$$\mathbf{D}\big(W \star F,\ d\big) \ \cong\ [\mathbf{C}^{\mathrm{op}}, \mathbf{V}]\big(W,\ \mathbf{D}(F-, d)\big)$$

natural in d. Here $\mathbf{C}^{\mathrm{op}}$ is the $\mathbf{V}$-category with the same objects, hom-objects $\mathbf{C}^{\mathrm{op}}(c, c') = \mathbf{C}(c', c)$, and composition that of $\mathbf{C}$ precomposed with the symmetry σ.

Both are representing objects, so both are unique up to canonical isomorphism when they exist. The weight is the diagram's shape data: where an ordinary cone offers one morphism $d \to Fc$ for each c, a W-weighted cone offers an object Wc worth of them, and the naturality built into $[\mathbf{C}, \mathbf{V}]$ requires those to be compatible along the morphisms of $\mathbf{C}$. The definition of $\mathbf{C}^{\mathrm{op}}$ is where symmetry of $\mathbf{V}$ is needed, since reversing composition means feeding $\circ$ its two arguments in the opposite order.

Example 66.3.8: Three Weighted Limits

1. *Ordinary limits are the constant weight.* Take $\mathbf{V} = \mathbf{Set}$ and W the constant functor at the one-point set. Then $[\mathbf{C}, \mathbf{Set}]\big(W, \mathrm{Hom}(d, F-)\big)$ is the set of natural transformations from the constant functor into $\mathrm{Hom}(d, F-)$, which is the set of cones over F with vertex d. The defining isomorphism becomes $\mathrm{Hom}(d, \lim^W F) \cong \{\text{cones with vertex } d\}$, the universal property of the ordinary limit. Definition 61.2.1 is thus the special case in which every object of $\mathbf{C}$ carries the same one-point weight.

2. *Tensors and cotensors are weighted (co)limits.* Let $\mathbf{I}$ be the unit $\mathbf{V}$-category, one object $*$ with $\mathbf{I}(*, *) = I$. A $\mathbf{V}$-functor $F\colon \mathbf{I} \to \mathbf{D}$ is an object c of $\mathbf{D}$, and a weight $W\colon \mathbf{I} \to \mathbf{V}$ is an object v of $\mathbf{V}$. Unwinding definition 66.3.7, the products in definition 66.3.3 have one factor and the equalizer is trivial, so the defining isomorphism reads $\mathbf{D}(d, \lim^W F) \cong [v, \mathbf{D}(d, c)]$,

which is definition 66.3.1: $\lim^W F = c^v$ and dually $W \star F = v \odot c$.

3. *Tensor product of modules.* In the setting of example 66.3.5, a weight $W \colon \mathbf{R}^{\mathrm{op}} \to \mathbf{Ab}$ is a right R-module and F a left R-module M. For an abelian group A, the group $\mathrm{Hom}_{\mathbb{Z}}(M, A)$ is a right R-module by $(f \cdot r)(m) = f(rm)$, and the defining isomorphism reads

$$\mathrm{Hom}_{\mathbb{Z}}(W \star F,\ A) \ \cong\ \mathrm{Hom}_{R^{\mathrm{op}}}\big(W,\ \mathrm{Hom}_{\mathbb{Z}}(M, A)\big)$$

the right-hand side being the right R-module maps, by example 66.3.5 applied to $\mathbf{R}^{\mathrm{op}}$. That is the universal property of $W \otimes_R M$, so $W \star F = W \otimes_R M$: the tensor product over a ring is a weighted colimit.

Three constructions that arose separately, the limit of a diagram, the power of an object and the tensor product of two modules, are one construction read over three bases with three weights. The generality is used again once homotopy enters: a homotopy limit is a weighted limit whose weight is built from simplicial sets, and the two-sided bar construction computing it is a weighted colimit. Both are developed in *Higher Category Theory* [14], which takes definition 66.3.7 as given.

66.4 Strict 2-Categories

The enriching base of the preceding sections has been an arbitrary monoidal category $\mathbf{V}$. Take $\mathbf{V} = \mathbf{Cat}$, and a $\mathbf{V}$-category acquires an extra layer of morphisms: each hom-object is now itself a category, so between two arrows there sit arrows-of-arrows. This is the setting in which the natural transformation lives on the same footing as the functor, and in which the adjunction of definition 62.1.1 and the monad of definition 63.1.1 become structures one can name on any object at all. This section makes the extra layer explicit, gives it its own composition laws, and identifies it with the $\mathbf{Cat}$-enriched case of definition 66.1.1.

Definition 66.4.1: Strict 2-Category

A *strict 2-category* $\mathbf{K}$ consists of

- a class of 0-*cells* (or *objects*) $A, B, C, \ldots$;
- for each ordered pair A, B of 0-cells, a class of 1-*cells* $f\colon A \to B$;
- for each parallel pair $f, g\colon A \to B$ of 1-cells, a class of 2-*cells* $\alpha\colon f \Rightarrow g$,

together with two ways of composing 2-cells. *Vertical composition* takes 2-cells sharing a 1-cell boundary,

$$\alpha\colon f \Rightarrow g, \qquad \beta\colon g \Rightarrow h,$$

to a 2-cell $\beta \cdot \alpha\colon f \Rightarrow h$; *horizontal composition* takes 2-cells sharing a 0-cell boundary,

$$\alpha\colon f \Rightarrow g\ (A \to B), \qquad \alpha'\colon f' \Rightarrow g'\ (B \to C),$$

to a 2-cell $\alpha' \circ \alpha\colon f'f \Rightarrow g'g\ (A \to C)$. Both compositions are strictly associative and unital: vertical composition has, for each 1-cell f, an identity 2-cell $1_f\colon f \Rightarrow f$; horizontal composition has, for each 0-cell A, an identity 2-cell 1_{id_A} on the identity 1-cell id_A. The two compositions are compatible through the *interchange law*: whenever the composites are defined,

$$(\beta' \cdot \alpha') \circ (\beta \cdot \alpha) = (\beta' \circ \beta) \cdot (\alpha' \circ \alpha) \tag{66.1}$$

The two compositions are pictured most cleanly by drawing 1-cells as arrows and 2-cells as double arrows between them. Vertical composition stacks two 2-cells inside a single hom-class $\mathbf{K}(A, B)$:

$$\begin{tikzcd}
A \arrow[r, bend left=70, "f"] \arrow[r, "g" description] \arrow[r, bend right=70, "h"'] \arrow[r, phantom, shift left=3ex, "\Downarrow\alpha"] \arrow[r, phantom, shift right=3ex, "\Downarrow\beta"] & B
\end{tikzcd}$$

Horizontal composition pastes two 2-cells along a shared 0-cell B:

$$\begin{tikzcd}
A \arrow[r, bend left=70, "f"] \arrow[r, bend right=70, "g"'] \arrow[r, phantom, "\Downarrow\alpha"] & B \arrow[r, bend left=70, "f'"] \arrow[r, bend right=70, "g'"'] \arrow[r, phantom, "\Downarrow\alpha'"] & C
\end{tikzcd}$$

The interchange law is the assertion that the four-fold pasting diagram

$$\begin{tikzcd}
A \arrow[r, bend left=70, "f"] \arrow[r, "g" description] \arrow[r, bend right=70, "h"'] \arrow[r, phantom, shift left=3ex, "\Downarrow\alpha"] \arrow[r, phantom, shift right=3ex, "\Downarrow\beta"] & B \arrow[r, bend left=70, "f'"] \arrow[r, "g'" description] \arrow[r, bend right=70, "h'"'] \arrow[r, phantom, shift left=3ex, "\Downarrow\alpha'"] \arrow[r, phantom, shift right=3ex, "\Downarrow\beta'"] & C
\end{tikzcd}$$

evaluates to a single 2-cell $f'f \Rightarrow h'h$, no matter whether one first composes vertically within each column and then horizontally across, or first composes horizontally within each row and then vertically down. Equation (66.1) says the two readings agree. This is the one axiom that couples the two operations; associativity and unitality govern each in isolation.

The definition looks like a bookkeeping exercise until one reads it through the lens of section 66.1. Assemble, for each pair A, B, the 1-cells $A \to B$ as objects and the 2-cells between them as morphisms; vertical composition makes this data a category $\mathbf{K}(A, B)$, its identities the 1_f. Horizontal composition of 1-cells and of 2-cells then assembles into functors $\mathbf{K}(B, C) \times \mathbf{K}(A, B) \to \mathbf{K}(A, C)$, and the interchange law is exactly the statement that horizontal composition respects the categorical structure of each factor, that is, that it is a functor on the product and not only a function on objects. This is the hinge of the section.

Proposition 66.4.2: 2-Categories are Cat-enriched Categories

A strict 2-category is exactly a category enriched over $(\mathbf{Cat}, \times, \mathbf{1})$ in the sense of definition 66.1.1: the hom-object $\mathbf{K}(A, B)$ is the category whose objects are the 1-cells $A \to B$ and whose morphisms are the 2-cells between them.

***Proof*:**
Fix the correspondence between the two packages of data and check that the axioms match term by term.

From a 2-category to enrichment. Given a strict 2-category $\mathbf{K}$, for each pair A, B let $\mathbf{K}(A, B)$ be the category with the 1-cells $A \to B$ as objects, the 2-cells as morphisms, and vertical composition as composition; associativity and unitality of vertical composition are precisely the category axioms, with 1_f the identity on f. Horizontal composition assigns to a pair of objects (f', f) the object $f'f$ and to a pair of 2-cells (α', α) the 2-cell $\alpha' \circ \alpha$. That this assignment is a functor

$$c_{A,B,C} \colon \mathbf{K}(B, C) \times \mathbf{K}(A, B) \to \mathbf{K}(A, C)$$

is the content of the interchange law (66.1): it sends the identity morphism $(1_{f'}, 1_f)$ to $1_{f'f}$, which is unitality of horizontal composition, and it respects composition of morphisms in the product category, where composition is componentwise vertical, so functoriality is the equation $(\beta' \cdot \alpha') \circ (\beta \cdot \alpha) = (\beta' \circ \beta) \cdot (\alpha' \circ \alpha)$ of (66.1). The identity 1-cell id_A together with 1_{id_A} is a functor $\mathbf{1} \to \mathbf{K}(A, A)$ from the terminal category, the unit map of the enrichment. Associativity and unitality of horizontal composition of 1-cells become the associativity and unit diagrams required of an enriched category.

From enrichment to a 2-category. Conversely, a **Cat**-enriched category supplies for each pair a category $\mathbf{K}(A, B)$: its objects are declared the 1-cells, its morphisms the 2-cells, its composition the vertical composition. The composition functors $c_{A,B,C}$ act on objects to give horizontal composition of 1-cells and on morphisms to give horizontal composition of 2-cells; functoriality of $c_{A,B,C}$ is the interchange law, and the enriched associativity and unit axioms are the associativity and unitality of horizontal composition. The two passages are mutually inverse, matching each datum and each axiom, so the two notions coincide, as we sought to show.

The proposition is the reason the tensor product $\times$ on **Cat** mattered. An ordinary category is enriched over $(\mathbf{Set}, \times)$; promoting the enriching base one level, from sets to categories, adds exactly one dimension of morphism and produces the strict 2-category. The interchange law is not an extra decoration but the naturality already built into enrichment, read in the base **Cat**. Every general fact about **V**-categories from section 66.1 specializes: a 2-category has an underlying ordinary category $\mathbf{K}_0$ (definition 66.1.2), obtained by discarding the 2-cells and keeping only the 0-cells and 1-cells, since $\mathbf{Cat}(\mathbf{1}, \mathbf{K}(A, B))$ is the set of objects of $\mathbf{K}(A, B)$, namely the 1-cells $A \to B$.

The definition of a 2-category was abstracted from one example, in which all three layers of cell are already familiar and the two compositions of 2-cells have standing names.

Example 66.4.3: Cat as a 2-Category
Let **Cat** denote the 2-category whose

- 0-cells are (small) categories $\mathbf{C}, \mathbf{D}, \ldots$;
- 1-cells $F \colon \mathbf{C} \to \mathbf{D}$ are functors (definition 59.4.1);
- 2-cells $\alpha \colon F \Rightarrow G$ are natural transformations (definition 59.5.2) between parallel functors.

The hom-object $\mathbf{Cat}(\mathbf{C}, \mathbf{D})$ is the functor category $[\mathbf{C}, \mathbf{D}]$ (definition 59.5.4). Vertical composition is the composition of natural transformations: for $\alpha \colon F \Rightarrow G$ and $\beta \colon G \Rightarrow H$, the transformation $\beta \cdot \alpha$ has component $(\beta \cdot \alpha)_c = \beta_c \circ \alpha_c$ at each object c. Horizontal composition is the Godement product: for $\alpha \colon F \Rightarrow G$ between functors $\mathbf{C} \to \mathbf{D}$ and $\alpha' \colon F' \Rightarrow G'$ between functors $\mathbf{D} \to \mathbf{E}$, the component of $\alpha' \circ \alpha$ at $c \in \mathbf{C}$ is the common value of the two paths around the naturality square of α' at α_c:

$$(\alpha' \circ \alpha)_c = G'(\alpha_c) \circ \alpha'_{Fc} = \alpha'_{Gc} \circ F'(\alpha_c) \tag{66.2}$$

The interchange law is the middle-four exchange for natural transformations.

The two expressions in (66.2) are equal precisely because $\alpha' \colon F' \Rightarrow G'$ is natural: applied to the arrow $\alpha_c \colon Fc \to Gc$ of **D**, naturality of α' is the commuting square

$$\begin{tikzcd}
F'Fc \arrow[r, "\alpha'_{Fc}"] \arrow[d, "F'(\alpha_c)"'] & G'Fc \arrow[d, "G'(\alpha_c)"] \\
F'Gc \arrow[r, "\alpha'_{Gc}"'] & G'Gc
\end{tikzcd}$$

whose two directed paths from $F'Fc$ to $G'Gc$ are the two formulas for $(\alpha' \circ \alpha)_c$. That **Cat** satisfies the axioms of definition 66.4.1 is the content of proposition 66.4.2 together with the fact that $[\mathbf{C}, \mathbf{D}]$ is a category (definition 59.5.4) and that these composites are natural. The interchange law $(\beta' \cdot \alpha') \circ (\beta \cdot \alpha) = (\beta' \circ \beta) \cdot (\alpha' \circ \alpha)$ is verified componentwise from (66.2) by a direct diagram chase; it is the classical middle-four exchange, so named because the four inner terms of the two sides pair off.

A frequent special case of horizontal composition fixes one of the two 2-cells to be an identity. This is common enough, and geometrically clean enough, to deserve its own name.

Definition 66.4.4: Whiskering

Let $\alpha\colon f \Rightarrow g$ be a 2-cell between 1-cells $f, g\colon A \to B$ in a 2-category $\mathbf{K}$. Given a 1-cell $h\colon B \to C$, the *left whiskering* of α by h is the horizontal composite of α with the identity 2-cell on h,

$$h * \alpha \coloneqq 1_h \circ \alpha\colon hf \Rightarrow hg,$$

a 2-cell between 1-cells $A \to C$. Given a 1-cell $k\colon Z \to A$, the *right whiskering* of α by k is

$$\alpha * k \coloneqq \alpha \circ 1_k\colon fk \Rightarrow gk,$$

a 2-cell between 1-cells $Z \to B$.

Pictorially, whiskering appends a 1-cell to one side of the 2-cell α without introducing a second nondegenerate 2-cell:

$$\begin{tikzcd}
A \arrow[r, bend left=50, "f"] \arrow[r, bend right=50, "g"'] \arrow[r, phantom, "\Downarrow\alpha"] & B \arrow[r, "h"] & C
\end{tikzcd}
\qquad\qquad
\begin{tikzcd}
Z \arrow[r, "k"] & A \arrow[r, bend left=50, "f"] \arrow[r, bend right=50, "g"'] \arrow[r, phantom, "\Downarrow\alpha"] & B
\end{tikzcd}$$

In **Cat**, where $\alpha\colon F \Rightarrow G$ is a natural transformation and H a functor, left whiskering $H * \alpha$ has component $(H * \alpha)_c = H(\alpha_c)$ at each object c, obtained by applying H to each component of α; right whiskering $\alpha * K$ has component $(\alpha * K)_z = \alpha_{Kz}$, obtained by reindexing α along K. Every horizontal composite factors through whiskering: for $\alpha\colon F \Rightarrow G$ and $\alpha'\colon F' \Rightarrow G'$ as in (66.2),

$$\alpha' \circ \alpha = (G' * \alpha) \cdot (\alpha' * F) = (\alpha' * G) \cdot (F' * \alpha),$$

the two factorizations being the two paths of the naturality square, so the two whiskerings together with vertical composition recover the full Godement product. Whiskering is therefore the elementary operation from which horizontal composition is built, and it is the notation in which the constructions below are cleanest.

With the two compositions and whiskering in hand, one can transcribe the central notions of the core theory into any 2-category, referring only to 0-, 1-, and 2-cells and never to elements. The payoff is that an adjunction (definition 62.1.1) and a monad (definition 63.1.1) become structures an object can carry in any $\mathbf{K}$, of which the definitions of the earlier chapters are the **Cat** case.

Proposition 66.4.5: Adjunctions and Monads in a 2-Category

Let $\mathbf{K}$ be a 2-category.

1. An *adjunction* in $\mathbf{K}$ consists of 1-cells $f\colon A \to B$ and $g\colon B \to A$ and 2-cells $\eta\colon \mathrm{id}_A \Rightarrow gf$ and $\varepsilon\colon fg \Rightarrow \mathrm{id}_B$ satisfying the triangle identities
$$(\varepsilon * f) \cdot (f * \eta) = 1_f, \qquad (g * \varepsilon) \cdot (\eta * g) = 1_g \tag{66.3}$$
2. A *monad* on an object A consists of a 1-cell $t\colon A \to A$ and 2-cells $\eta\colon \mathrm{id}_A \Rightarrow t$ and $\mu\colon tt \Rightarrow t$ satisfying the associativity and unit laws
$$\mu \cdot (\mu * t) = \mu \cdot (t * \mu), \qquad \mu \cdot (\eta * t) = 1_t = \mu \cdot (t * \eta) \tag{66.4}$$
3. Every adjunction in $\mathbf{K}$ induces a monad on A, with underlying 1-cell $t = gf$, unit the adjunction unit η, and multiplication $\mu = g * \varepsilon * f\colon gfgf \Rightarrow gf$.

In $\mathbf{K} = \mathbf{Cat}$, an adjunction is an adjunction of functors in the sense of definition 62.1.1 with unit and counit as in definition 62.1.3, a monad is a monad in the sense of definition 63.1.1, and the induced monad of part (3) is the monad an adjunction of functors induces on its source.

***Proof*:**

Parts (1) and (2) are definitions internal to $\mathbf{K}$; the content is part (3), that the data they package assemble into a monad, and that in $\mathbf{Cat}$ everything reduces to the earlier chapters.

Step 1: The induced multiplication and unit. From an adjunction $(f, g, \eta, \varepsilon)$ set $t = gf$. Left whiskering by g and right whiskering by f turn the counit $\varepsilon\colon fg \Rightarrow \mathrm{id}_B$ into a 2-cell
$$\mu = g * \varepsilon * f\colon gfgf \Rightarrow gf,$$
that is, $\mu\colon tt \Rightarrow t$, where $g*\varepsilon*f$ abbreviates the whiskering $g*(\varepsilon*f) = (g*\varepsilon)*f$ (associativity of horizontal composition makes the two readings equal). The unit of the monad is the adjunction unit $\eta\colon \mathrm{id}_A \Rightarrow t$.

Step 2: The unit laws. The two unit laws in (66.4) are the two triangle identities (66.3), whiskered into place. Whisker the first triangle identity $(\varepsilon * f) \cdot (f * \eta) = 1_f$ on the left by g: using functoriality of whiskering with respect to vertical composition,
$$(g * \varepsilon * f) \cdot (gf * \eta) = 1_{gf},$$
and since $t*\eta = gf*\eta$ and $\mu = g*\varepsilon*f$, this reads $\mu\cdot(t*\eta) = 1_t$. Whiskering the second triangle identity $(g * \varepsilon) \cdot (\eta * g) = 1_g$ on the right by f gives, by the same computation, $\mu \cdot (\eta * t) = 1_t$. Together these are the unit laws.

Step 3: Associativity. Both sides of $\mu\cdot(\mu*t) = \mu\cdot(t*\mu)$ are 2-cells $ttt \Rightarrow t$, that is, $gfgfgf \Rightarrow gf$. Expand using $\mu = g * \varepsilon * f$ and $t = gf$: the left side, $\mu \cdot (\mu * gf)$, whiskers the outer copy of ε over the inner one on the left, while the right side, $\mu \cdot (gf * \mu)$, whiskers on the right; the two inner whiskerings apply ε at the two different copies of the pair fg in $fgfg$. Both reduce to a single whiskered double application of the counit,
$$g * (\varepsilon \circ \varepsilon) * f\colon gfgfgf \Rightarrow gf,$$

where $\varepsilon \circ \varepsilon \colon fgfg \Rightarrow \mathrm{id}_B$ is the horizontal composite of ε with itself. That this horizontal composite is unambiguous, that whiskering the outer ε over the inner one on the left equals whiskering it on the right, is precisely the interchange law (66.1) applied to the two copies of ε: $(\varepsilon \circ 1_{\mathrm{id}_B}) \cdot (1_{fg} \circ \varepsilon) = (\varepsilon \circ \varepsilon) = (1_{\mathrm{id}_B} \circ \varepsilon) \cdot (\varepsilon \circ 1_{fg})$. Hence the two sides agree and (t, η, μ) satisfies (66.4), a monad on A.

Step 4: The **Cat** *case.* Take $\mathbf{K} = \mathbf{Cat}$. A 1-cell $f \colon A \to B$ is a functor, a 2-cell is a natural transformation, and the triangle identities (66.3), read componentwise via the whiskering formulas $(H * \alpha)_c = H(\alpha_c)$ and $(\alpha * K)_z = \alpha_{Kz}$ of definition 66.4.4, are exactly the triangle identities of definition 62.1.3: at an object a the first reads $\varepsilon_{fa} \circ f(\eta_a) = \mathrm{id}_{fa}$. So an adjunction in **Cat** is an adjunction of functors (definition 62.1.1). Likewise $t = gf$ is the composite endofunctor, and the multiplication $\mu = g * \varepsilon * f$ has component $\mu_a = g(\varepsilon_{fa})$ at $a \in A$, which is the multiplication of the monad an adjunction of functors induces (definition 63.1.1). The construction of part (3) used only 2-cells, whiskering, and the interchange law, which is why the derivation for **Cat** was, without saying so, purely 2-categorical. This completes the proof.

The proposition is the abstract residue of the monad theory of the core chapters. The construction of a monad from an adjunction (definition 63.1.1) never touched an element of an object; it composed the counit with functors on the left and right and invoked naturality, and those are exactly whiskering and the interchange law. Reread through proposition 66.4.5, that construction proved a fact about **Cat** that holds verbatim in any 2-category: the endomorphism gf of the source of any adjunction carries a canonical monad structure. The same transcription runs the other way for the Eilenberg-Moore and Kleisli constructions (section 63.2, section 63.3), which recover an adjunction from a monad by building new objects, and it is the systematic version of this dictionary, one layer of categorical structure at a time, that the sequel takes up.

66.5 Toward Weak and Higher Categories

A strict 2-category asks the composition of 1-cells to be strictly associative and strictly unital, exactly as in an ordinary category. In the archetype **Cat** this holds on the nose. But many of the 2-dimensional structures that arise in practice do not: composing 1-cells there produces objects that are associative and unital only *up to* a specified invertible 2-cell, and one cannot wish those 2-cells away without discarding the mathematics that forced them. The passage from the strict laws of definition 66.4.1 to laws-up-to-coherent-isomorphism is the last structural step this book takes, and it is the door to the sequel.

The relaxation is governed by exactly the coherence discipline of chapter 65. There the associator and unitors of a monoidal category were required to satisfy the pentagon and triangle axioms, and Mac Lane's coherence theorem (theorem 65.1.5) guaranteed that every formal diagram then commutes. A bicategory imports that discipline one dimension up: the associativity and unit *equalities* of 1-cell composition become invertible 2-cells subject to a pentagon and a triangle, and the same kind of coherence theorem results.

Definition 66.5.1: Bicategory

A *bicategory* $\mathbf{B}$ consists of

- a collection of *0-cells* (objects) $A, B, C, \ldots$;
- for each ordered pair (A, B) a category $\mathbf{B}(A, B)$, whose objects are the *1-cells* $f\colon A \to B$ and whose morphisms are the *2-cells* $\alpha\colon f \Rightarrow g$ (composition within $\mathbf{B}(A, B)$ is *vertical* composition of 2-cells);
- for each triple (A, B, C) a *horizontal composition* functor

$$\circ\colon \mathbf{B}(B, C) \times \mathbf{B}(A, B) \to \mathbf{B}(A, C)$$

  and for each A an identity 1-cell $\mathrm{id}_A \in \mathbf{B}(A, A)$;
- for all composable 1-cells f, g, h a natural *associator* 2-cell

$$a_{h,g,f}\colon (h \circ g) \circ f \;\Rightarrow\; h \circ (g \circ f)$$

  and for each $f\colon A \to B$ natural *left* and *right unitor* 2-cells

$$\lambda_f\colon \mathrm{id}_B \circ f \Rightarrow f, \qquad \rho_f\colon f \circ \mathrm{id}_A \Rightarrow f$$

  each of which is invertible,

subject to two coherence axioms: the *pentagon*, requiring that the two ways of reassociating a fourfold composite $((k \circ h) \circ g) \circ f$ to $k \circ (h \circ (g \circ f))$ agree, that is

$$(\mathrm{id}_k \circ a_{h,g,f}) \cdot a_{k,h\circ g,f} \cdot (a_{k,h,g} \circ \mathrm{id}_f) \;=\; a_{k,h,g\circ f} \cdot a_{k\circ h,g,f}$$

(where $\cdot$ is vertical composition and $\circ$ here denotes whiskering, definition 66.4.4); and the *triangle*, requiring for composable $g\colon B \to C$, $f\colon A \to B$ that

$$(\mathrm{id}_g \circ \lambda_f) \cdot a_{g,\mathrm{id}_B,f} \;=\; \rho_g \circ \mathrm{id}_f$$

A *strict 2-category* (definition 66.4.1) is the special case in which a, λ, and ρ are all identity 2-cells.

The two axioms are the pentagon and triangle of theorem 65.1.5, transplanted from the 1-cells of a monoidal category to the 1-cells of $\mathbf{B}$. This is not an analogy imposed after the fact: a bicategory with a single 0-cell is precisely a monoidal category, its one hom-category $\mathbf{B}(A, A)$ being the underlying category, horizontal composition the tensor, id_A the unit, and a, λ, ρ the monoidal associator and unitors. A bicategory is thus a monoidal category with many objects, and the coherence data is the same data seen one dimension higher.

The force of the pentagon and triangle is again a coherence theorem, which frees the working mathematician from tracking associators and unitors explicitly.

Theorem 66.5.2: Coherence for Bicategories

Every bicategory is biequivalent to a strict 2-category. Consequently, in any bicategory every diagram of 2-cells built from associators, unitors, their inverses, whiskerings, and identities commutes.

***Proof Key ideas*:**
The statement, and the notion of biequivalence it uses, belong to the weak 2-dimensional theory developed in the sequel; the full argument is beyond the light treatment here. The mechanism mirrors the strictification of a monoidal category behind theorem 65.1.5: one embeds $\mathbf{B}$ into a strict 2-category of $\mathbf{B}$-representations by a bicategorical Yoneda construction, sending each 0-cell to a strict object whose 1-cells compose on the nose, and checks the embedding is a biequivalence. A complete proof is given in the sequel ([14]) and in Mac Lane's and Bénabou's original treatments, to which the details are deferred.

Coherence has a practical consequence worth stating plainly: a bicategory may be manipulated as though it were strict. Parenthesization of a composite of 1-cells can be suppressed, and any two reassociations of the same string are identified by a canonical invertible 2-cell. The weakness is real, but coherence renders it invisible in computation, precisely as in the monoidal case.

The examples that make bicategories necessary are the ones where the associator fails to be an identity. Two are canonical.

Example 66.5.3: The Bicategory of Bimodules

Bimodules. Let **Bimod** have as 0-cells the (unital, associative) rings. A 1-cell $R \to S$ is an (S, R)-bimodule ${}_SM_R$, that is, an abelian group carrying commuting left S- and right R-actions ([25]). A 2-cell ${}_SM_R \Rightarrow {}_SN_R$ is a bimodule homomorphism. Vertical composition is composition of bimodule maps, so $\mathbf{Bimod}(R, S)$ is the category of (S, R)-bimodules.

Horizontal composition is the tensor product over the middle ring: the composite of $R \xrightarrow{M} S$ and $S \xrightarrow{N} T$ is the (T, R)-bimodule

$$N \circ M \;=\; N \otimes_S M$$

The identity 1-cell on R is the bimodule ${}_RR_R$, since $R \otimes_R M \cong M$ and $M \otimes_R R \cong M$ naturally.

Here associativity is not an equality. For bimodules ${}_SM_R$, ${}_TN_S$, ${}_UP_T$ the two triple tensor products

$$(P \otimes_T N) \otimes_S M \qquad \text{and} \qquad P \otimes_T (N \otimes_S M)$$

are distinct abelian groups: their elements are cosets in different quotients of $P \otimes N \otimes M$. They are canonically isomorphic by $(p \otimes n) \otimes m \mapsto p \otimes (n \otimes m)$, and this isomorphism is the associator $a_{P,N,M}$; it is invertible but not the identity. The unitors are the canonical maps $R \otimes_R M \cong M$ and $M \otimes_R R \cong M$. That these satisfy the pentagon and triangle is the statement that the standard tensor-product isomorphisms cohere, which they do because each is built from the universal property of $\otimes$. Thus **Bimod** is a bicategory that is not a strict 2-category, and no reasonable change of definition makes it one: the tensor product of bimodules is associative up to canonical isomorphism and no more.

The second canonical example needs only a category with pullbacks, and it shows the same phenomenon geometrically rather than algebraically.

Example 66.5.4: The Bicategory of Spans

Let $\mathbf{C}$ be a category with pullbacks. A *span* from A to B is a diagram $A \leftarrow S \rightarrow B$, an object S with a leg to each of A and B; this is a 1-cell $A \rightarrow B$. A 2-cell between spans $A \leftarrow S \rightarrow B$ and $A \leftarrow S' \rightarrow B$ is a morphism $S \rightarrow S'$ commuting with both legs.

Horizontal composition composes spans by pullback. Given $A \leftarrow S \rightarrow B$ and $B \leftarrow T \rightarrow C$, form the pullback $S \times_B T$ of the two legs over B (definition 61.2.6):

$$\begin{tikzcd} & & S \times_B T \arrow[dl] \arrow[dr] & & \\ & S \arrow[dl] \arrow[dr] & & T \arrow[dl] \arrow[dr] & \\ A & & B & & C \end{tikzcd}$$

The composite span $A \leftarrow S \times_B T \rightarrow C$ takes the outer legs. The identity 1-cell on A is the span $A \xleftarrow{\mathrm{id}} A \xrightarrow{\mathrm{id}} A$.

Associativity again holds only up to isomorphism. Composing three spans in the two possible orders forms two iterated pullbacks, $(S \times_B T) \times_C U$ and $S \times_B (T \times_C U)$; these are canonically isomorphic because both compute the limit of the same diagram, but the isomorphism is supplied by the universal property of the pullback and is not an identity. It is the associator, and the pentagon it satisfies is exactly the coherence of iterated pullback limits. So $\mathbf{Span}(\mathbf{C})$ is a bicategory, strict only in the degenerate cases where pullbacks can be chosen strictly functorially.

In both examples the weakness is not a defect of the definition to be engineered away but a feature of the mathematics: tensor product and pullback are operations defined by universal properties, and a universal property determines its output only up to canonical isomorphism. Any 2-dimensional structure organized by such operations is a bicategory and not, on the nose, a strict 2-category. The equalities of definition 66.4.1 were an idealization; the isomorphisms of definition 66.5.1 are what the objects satisfy.

This is where strictness gives way to coherence, and where the present book ends. Everything developed here has been strict 1-category theory. Composition of morphisms is strictly associative and strictly unital; a diagram commutes or it does not; two objects are the same when they are equal or, at the outermost, when they are isomorphic. Sameness has meant equality or isomorphism, and nothing weaker. The structured categories of the last two chapters have pushed against that boundary, introducing associators and unitors as data, but coherence has kept them tame: by theorem 65.1.5 and theorem 66.5.2 the strict theory recovers the weak one up to equivalence, and the strict picture remains adequate.

The subject changes character when equality weakens past this point. When the associativity of composition is asked to hold only up to a coherent *homotopy*, and that homotopy up to a further homotopy, and so on through every dimension, one is no longer doing strict 1-category theory but *higher category theory*. There the objects of study are model categories, quasi-categories, and ∞-categories, in which morphisms compose only up to coherent higher cells and the coherence data is infinite rather than a single pentagon. The theorems of this book do not disappear beyond it; they reappear in homotopy-coherent form. Limits and colimits become homotopy limits and colimits, taken up to coherent equivalence rather than on the nose. Adjunctions, monads, and the Yoneda lemma each have ∞-categorical statements that specialize back to the strict ones proved here. The duality principle of box 59.3.3 survives verbatim, opposite ∞-categories and all.

The sequel is *Higher Category Theory* ([14]), which takes up simplicial sets and nerves, Quillen's model categories, the quasi-category model of ∞-categories after Joyal, Lurie, and Riehl, homotopy (co)limits, and the stable and derived structures that connect back to homological algebra. It is the natural continuation for a reader who now recognizes a universal property on sight and wants to see what becomes of these constructions when equality is replaced throughout by coherent equivalence. The strict theory built here is the ground from which that generalization is read.

Part XIII

Appendix

In the appendix, certain section will take liberties on how much the reader know's. In particular, the reader is expected to be comfortable with (some analysis or topology in certain appendicies, like category theory and geometric group theory).

A

Axiom of Choice

Look into this explanation of truth in mathematics, looks good!:

In mathematics, we are constantly making choices. We can make algorithmic choices (for example, if you're given the number n, choose the number $n+1$), or we make arbitrary choices (choose some natrual number n). The fact that we can make an arbitrary choice is uncontroversial in the axioms mathematicians used, that is ZF (short for Zermelo-Frankel axioms). So too is making a finite number of choices. These two "abilities" were also uncontroversial in other foundational formulations of mathematics, not just ZF. It might seem very natural then to be able to make an infinite, or more generally an arbitrary (including uncountably infinite or other cardinalities) amount of choices. However, making an arbitrary amount of choices turns out to have some very strange consequences. The German mathematician Ernst Zermelo axiomatically used the fact that you can make an arbitrary number of choices (which we will soon define what this means mathematically) to prove a very controversial conjecture called the *well-ordering Theorem* (which we shall cover later in this appendix), sparking a debate on whether mathematics should allow for arbitrary choice. One famous "counter-example" developed to "prove" that we shouldn't allow arbitray choice is a proof by Banach and Tarski that you can take a sphere, partition it into 6 subsets (which uses the axiom of choice), rotate them in certain ways, and recombine them to form 2 identical spheres! Importantly, we did *not* add any more points, yet we have somehow "doubled" the surface area of the points we've chosen [1]! This is called the *Banach-Tarski Paradox*.

Though this result is strange, as time went on, more and more mathematicians found that making an arbitrary number of choices produces some "natural" results. For example, the fact that every vector-space has a basis comes down to the ability of making an arbitrary number of choices. Another example (HERE: from that stackexchagne, I think it's the one saying "if $|A| = |B|$, then we need the AC to say there is an injective function from A to B"). In 1963, a mathematician named Paul Cohen proved that the making an arbitrary number of choices is independent of the standard system

[1]For those who have taken measure theory, the chosen subsets turn out to be *unmeasurable* sets, and so the concept of surface area does not apply to these sets which allows us to produce this strange result

of axioms ZF, meaning if we assume the we can do an arbitrary number of choices, it will not cause any more contradiction and ZF would already cause [2], and if we assume we can't make an arbitrary number of choices, it will *also* not cause any more contradiction than ZF would already cause. Through this result, most mathematicians now accept the axiom of choice as part of the standard model we use, which is represented as ZFC (ZF + the axiom of **C**hoice).

In Algebra, the axiom of choice most often appears in a special form called *Zorn's Lemma*. The aim of this chapter is to be able show that Zorn's lemma is equivalent to the axiom of choice.

A.1 Build-up

Definition A.1.1: Cartesian Product and Choice Function

Let $\{A_i\}_{i\in I}$ be any family of sets. Then the *Cartesian product* of this family, written $\prod_{i\in I} A_i$, is the set of all function $x : I \to \bigcup_{i\in I} A_i$, where $x(i) \in A_i$ for all $i \in I$ The function x is called a *choice function.*

Though we've defined what the cartesian product is, it does not mean that the function x exist! Naturally, if $I = \emptyset$ or if $A_i = \emptyset$ for some $i \in I$ (since then $\prod_{i\in I} A_i = \emptyset$), then this function exists. If $I \neq \emptyset$ or $A_i \neq \emptyset$, then it turns out the existence of this function is independent of the ZF axioms! Thus, the axiom of choice was formulated:

Axiom A.1.2: Axiom of Choice

The cartesian product of any non-empty family of non-empty sets is a non-empty set. In particular, if $\{A_i\}_{i\in I}$ is a family of sets such that $I \neq \emptyset$ and $A_i \neq \emptyset$ for each $i \in I$, then there exists at least one choice function for $\{A_i\}_{i\in I}$

Since the cartesian product exists (axiomatically), we will make some shorthand notation for it similar to that defined in chapter 5. If $\{A_i\}_{i\in I}$ is a family of sets, let $A^I = \prod_{i\in I} A_i$ where $A_i = A$ for each $i \in I$. If $|I|$ is finite, and so $|I| = n$ for some $n \in \mathbb{N}$, then we write A^n instead. If $|I|$ is countable, we sometimes use the A^∞ or $A^{\mathbb{N}}$ notation. the elements of A^n can be represented as n-tuples, that is

$$A^n = \{(a_1, a_2, ..., a_n \mid a_i \in A\}$$

and the elements of $A^{\mathbb{N}}$ can be represents as sequences

$$A^{\mathbb{N}} = \{(a_1, a_2, ..., a_n, ...) \mid a_i \in A\}$$

Example A.1.3: Cartesian Products

1. An example we've already encountered is $\mathbb{R}^n$ and $\mathbb{R}^{\mathbb{N}}$. This caretsian product is called *Euclidan Space*
2. Let $A = \{0, 1\}$. Then $A^{\mathbb{N}}$ is the set of all sequences with either 0 or 1 in each component. If we change notation a bit and let $A = \mathbb{Z}/2\mathbb{Z}$, and define addition point-wise, then this would

[2]The careful wording here is because it has *not* been proven that the axioms of ZF don't contradict each other – this remains an open problem in mathematics

be a group

(words here)

Definition A.1.4: Partially Ordered Sets

1. Let P be a non-mepty set. A *partial order* on P is a relation $\leq$ ($\leq \subseteq P \times P$) on P such that
   (a) **(reflective)** $x \leq x$ for all $x \in A$
   (b) **(antisymmetric)** if $x \leq y$ and $y \leq x$, then $x = y$
   (c) **(transitive)** if $x \leq y$ and $y \leq z$ then $x \leq z$

   Where we usually say P is a partiallly ordered set under the orering $\leq$ or that A is partially ordered by $\leq$.

   If $\leq$ also satisfies

   (d) **(trichotomy)** If $x, y \in P$, then $x \leq y$ or $y \leq x$,

   then $\leq$ is called a *total* (or linear, simple, complete) ordering on P. If further

   (e) **(Well-Ordering)**For all $\emptyset \neq A \subseteq P$, there exists an element $a \in A$ such that $a \leq x$ for all $x \in A$

   then we say that $\leq$ is a *well-ordering* of P, or that P is well-ordered by $\leq$
2. Let $(P, \leq)$ be a partially ordered set and let $A \subseteq P$. An element $u \in P$ is called an *upper bound* for A if for all $x \in A$, $x \leq u$. Similarly, $l \in P$ is called a *lower bound* if for all $x \in A$, $l \leq x$.
3. Let $(P, \leq)$ be a partially oredered set and let $A \subseteq P$. An element $m \in P$ is called a *maximal element* of P if whenever $m \leq x$ with $x \in P$, then $m = x$. Similarly, an element $m \in P$ is called a *minimal element* if whenever $x \leq m$ for $x \in P$, then $x = m$.
4. Let $(P, \leq)$ be a partially ordered set. Then a subset $C \subseteq P$ is called a *chain* in P if C is totally ordered under the order relation of $\leq$, that is, if $x, y \in C$, then either $x \leq y$ or $y \leq x$.

Note that it's called a *partial* order because becaue not all elements *necessarily* need to to have a relation between them: if $x, y \in P$ and $\leq$ is a parital order on P, then it need not be the case that $x \leq y$ or $y \leq x$. If this were the case, we would call $\leq$ a *total order*.

In the following examples we'll show the difference between being maximal and being an upper bound. Importantly, it is possible that *not all* $x \in P$ have the relation $x \leq m$ (i.e. m is not necessarily the "*largest*" or "*maximum*" element)

Example A.1.5: Partially Ordered Sets

1. Let S be an arbitrary set, and $P(S)$ be it's powerset. Then $(P(S), \subseteq)$ forms a partially ordered set. This set is *only* a partial order (it is not a total order) if $|S| > 1$: for two distinct

elements $x, y \in S$ $\{x\} \not\subseteq \{y\}$ and $\{y\} \not\subseteq \{x\}$. A chain in $(P(S), \subseteq)$ would look like:

$$S_1 \subseteq S_2 \subseteq S_3 \subseteq S_4 \subseteq \cdots$$

If $T \subseteq S$, then then T has an upperbound U:

$$U = \bigcup_{x \in T} \{x\}$$

Furthermore, S is also the maximal element of $P(S)$.

2. Let $S = \{a, b, b', c'\}$ be the set, and $\leq$ be the relation visualized by the graph:

$$\begin{tikzcd}
 & & c' \\
b & & b' \arrow[u] \\
 & a \arrow[ul] \arrow[ur] &
\end{tikzcd}$$

   Then both b and c' are maximal elements. S has no upper bound, but it has a as a lower bound.

3. Let S be the collection of *proper subsets* of $\mathbb{N}$ ordered by $\subseteq$. Notice that the chain

$$\{1\} \subseteq \{1, 2\} \, \{1, 2, 3\} \subseteq \cdots$$

   does not have an upper bound. However, S *does* have maximal elements. $(\mathbb{N} - \{n\}) \in S$ for any $n \in \mathbb{N}$ is a maximal element of S

4. Let $S = \mathbb{R}$ with the usual order $\leq$. Notice that $\leq$ is a *total order* in this case. some sets of $\mathbb{R}$ have upper bound (ex. $[0, 1]$ or $\{0, 1, 2, 3\}$), while other's don't ($\mathbb{Z}, (0, \infty), (0, 1)$). $\mathbb{R}$ does not have maximal elements in this order.

When we say we give a partial order to an arbitrary set S, we will assume that this is the partial order on the power set given by $\subseteq$.

Next, we will build-up some facts to bootstrap the result

Definition A.1.6: Finite Character

Let $\mathcal{F}$ be a family of sets. Then $\mathcal{F}$ is said to be a *family of finite character* if for any set S, $S \in \mathcal{F}$ if and only if each finite subset of S is in $\mathcal{F}$.

Lemma A.1.7: Finite Character and Chains

Let $\mathcal{F}$ be a family of finite character and let $\mathcal{B}$ be a chain in $\mathcal{F}$. Then $\cup\mathcal{B} \in \mathcal{F}$

***Proof*:**
Since $\mathcal{F}$ is of finite character, if we show that every finite subset of $\mathcal{B}$ is an element of $\mathcal{F}$. Let $\{x_1, x_2, ..., x_n\} = F \subseteq \cup\mathcal{B}$. Then there must exists sts $\mathcal{B}_1, ...\mathcal{B}_n \in \mathcal{B}$ such that $x_i \in \mathcal{B}_i$. Since $\mathcal{B}$ is a chain, there exists a $1 \leq j \leq n$ such that $B_i \subseteq B_j$, for all $i \in \{1, 2, ...n\}$. But then, $F \subseteq B_j \in \mathcal{F}$. Since $\mathcal{F}$ is of finite character, and $\mathcal{B}_j \in \mathcal{F}$, $F \in \mathcal{F}$. Since F was arbitrary, then this holds true for any finite subset of $\cup\mathcal{B}$, and so $\cup\mathcal{B} \in \mathcal{F}$, as we sought to show.

A.2 Axiom of Choice Equivalences

Note that terms like "lemma" or "theorem" for the following names are due to historical precedents rather than actually representing a lemma or theorem, since all of the following will be equivalent to the axiom of choice.

Theorem A.2.1: Turkey's Lemma

Every nomepty family of finite character has a maximal member

Theorem A.2.2: Hausdorff Maximality Principle

Every nonempty partially ordered set contains a maximal chain, i.e., every totally ordered subset is contained in a maximally totaly ordered subset

Maximally totally ordered here means that if it is enlraged in any way, it does not remained totally ordered

Theorem A.2.3: Zorn's Lemma

Every nonemtpy partially ordered set in which each chain has an upper bound has a maximal element

Theorem A.2.4: Well-Ordering Theorem

Every set can be well-orderd, i.e., if S is a set, then there exists some well-ordering $\leq$ on S

A well-ordering on a set S is a total order (i.e. a chain) on S with the additional property that every non-empty subset of S has a lower-bound. For example, given the usual orderings for the following examples, $\mathbb{N}$ is well-ordered (every subset has a lower-bound), while $\mathbb{Z}$ is not (namely, $\mathbb{Z}$ itself doens't have a lower-bound). A trickier example would be $\mathbb{R}$. The set (a, b) for $a, b \in \mathbb{R}$, $a \neq b$, does *not* have a minimal element, and so $\mathbb{R}$ is not well-ordered with the usual ordering.

Theorem A.2.5: Axiom of Choice Equivalences

The following are equivalent

1. The axiom of choice
2. Turkey's Lemma
3. The Hausdroff maximality principle
4. Zorn's Lemma
5. Well-ordering Theorem

Credit to the proof to

***Proof*:**
Step (1) $\Rightarrow$ (2) will be the longest step requiring many techincal details. the other steps will come much more naturally. The proof will also be long since we will not be using heavy machinery from a course in set-theory (for ex. we will not use transfinite induction [directly?? are we secretly using it?])

(1) $\Rightarrow$ (2) For the sake of contradiction, let's say Turkey's lemma was false – there exists a non-empty family of sets of finite characetr that has no maximal element. let's say $\mathcal{F}$ was such a set. That means for every $A \in \mathcal{F}$, there exists a B such that $A \subsetneq B$. Let $\mathcal{O}_F = \{E \in \mathcal{F} \mid F \subsetneq E\}$. Then $\{\mathcal{O}_F \mid F \in \mathcal{F}\}$ is a set of sets indexed by $\mathcal{F}$. Thus, by the axiom of choice, there exists a choice function f defined on $\mathcal{F}$ such that $f(F) \in \mathcal{O}_F$ (i.e. $F \subsetneq f(F)$ for all F).

Using f, we'll define a new concept: a subfamily $\mathscr{F} \subseteq \mathcal{F}$ is f-inductive if:

(a) $\emptyset \in \mathscr{F}$

(b) if $A \in \mathscr{F}$ then $f(A) \in \mathscr{F}$

(c) if $\mathcal{C} \subseteq \mathscr{F}$ is a chain, then $\cup\mathcal{C} \in \mathscr{F}$

Notice that $\mathcal{F}$ is itself an f-inductive set since: $\mathcal{F}$ is non-empty (by assumption), so some $A \in \mathcal{F}$ exists, and $\emptyset \subseteq A$ is a finite set, and so $\emptyset \in \mathcal{F}$ since $\mathcal{F}$ has finite character, fulfilling the first condition. The second conditions holds true since the image of f is in $\mathcal{F}$, and the third condition holds by lemma A.1.7. Notice too that if the family $\mathscr{F}_i \subseteq \mathcal{F}$ is f-inductive (for some indexing set I), then so too is $\bigcap_i \mathscr{F}_i$. Thus, define the smallest f-inductive set of $\mathcal{F}$:

$$\mathscr{F}_0 = \{A \in \mathcal{F} \mid A \in \mathscr{F} \text{ for every } f\text{-inductive subset of } \mathcal{F}\}$$

The fact that $\mathscr{F}_0$ is the smallest f-inductive set will allow us to force certain properties on to it. Our goal is to show that $\mathscr{F}_0$ *must* be a chain, which we will use to arrive at a contradiction.

To show that $\mathscr{F}_0$ is a chain, we will have to go through some technical steps. First, let $\mathcal{H} = \{A \in \mathscr{F}_0 \mid \text{if } B \in \mathscr{F}_0 \text{ and } B \subsetneq A \text{ then } f(B) \subseteq A\}$. The set $\mathcal{H}$ is the set of all elements $A \in \mathscr{F}_0$ such that B element is an element of $\mathscr{F}_0$ and is properly contained in A, f of that elements is contained in or equal to A. However, some oddities occurs with this set too: note that $\mathcal{H}$ is non-empty since vacusously, $\emptyset \in \mathcal{H}$ (since (False $\Rightarrow$ True) and (False $\Rightarrow$ False) are both *true* statements!!) This oddity is necessary, since we are guaranteed that $\mathcal{H}$ is non-empty. Now, we'll show that if $A \in \mathcal{H}$ and $C \in \mathscr{F}_0$, then one of two relations *must* hold between A and C: either $C \subseteq A$ or $f(A) \subseteq C$. Notice we are forcing a subset relation between all elements of $\mathscr{F}_0$ and $\mathcal{H}$ – wel'l use this to create a total order later. First, let's prove the last assertion:

Proof. Define $\mathcal{G}_A = \{C \in \mathscr{F}_0 \mid C \subseteq A \text{ or } f(A) \subseteq C\}$. By definition, $\mathcal{G}_A \subseteq \mathscr{F}_0$. If we show that $\mathcal{G}_A$ is f-inductive, then $\mathscr{F}_0 \subseteq \mathcal{G}_A$, and since $\mathscr{F}_0$ is the smallest f-inductive subset, it must be that $\mathcal{G}_A = \mathscr{F}_0$ which will tell us that any element $C \in \mathscr{F}_0$ must be either contained in A or contain $f(A)$. For notational convinience, let's label the condition $C \subseteq A$ as [1] and condition $F(A) \subseteq C$ as [2]

(a) Since $\emptyset \in \mathscr{F}_0$, $\emptyset \subseteq A$, and so by [1] $\emptyset \in \mathcal{G}_A$

(b) Let $C \in \mathcal{G}_A$. We must show that $f(C) \in \mathcal{G}_A$. Since $C \in \mathcal{G}_A$, then we have 3 possibilites (we're splitting [1] into 2 seperate cases:

  i. $C \subsetneq A$: Since $C \subsetneq A$ and $A \in \mathcal{H}$, then $f(C) \subseteq A$, so by [1] $f(C) \in \mathcal{G}_A$

  ii. $C = A$: Since $C = A$ then $f(C) = f(A)$. Which is equivalent to $f(C) \subseteq f(A)$ and $f(A) \subseteq f(C)$. For our purposes, we care that $f(A) \subseteq f(C)$. By condition [2], $f(C) \in \mathcal{G}_A$

  iii. $f(A) \subseteq C$, then since $C \subsetneq f(C)$ by definition of f, $f(A) \subseteq f(C)$, and so by condition [2], $f(C) \in \mathcal{G}_A$

  in all cases, $f(C) \in \mathcal{G}_A$

(c) Let $\mathcal{C} \subseteq \mathcal{G}_A$ be a chain in $\mathcal{G}_A$. We must show that $\cup\mathcal{C} \in \mathcal{G}_A$. Then, either for all $C \in \mathcal{C}$, $C \subseteq A$, in which case $\cup\mathcal{C} \subseteq A$, and so by [1] $\cup\mathcal{C} \in \mathcal{G}_A$, or there is some $C \in \mathcal{C}$ $f(A) \subseteq C \subseteq \cup\mathcal{C}$, in which case by [2] $\cup\mathcal{C} \in \mathcal{G}_A$. In both cases, the chain is in $\mathcal{G}_A$

Thus, $\mathcal{G}_A$ is indeed f-inductive for any A, and so $\mathscr{F}_0 = \mathcal{G}_A$

□

Using the previous result, we will force one more important result: $\mathcal{H} = \mathscr{F}_0$.

Proof. As before, it is already given that $\mathcal{H} \subseteq \mathscr{F}_0$. If we show $\mathcal{H}$ is f-inductive, we'll have that $\mathcal{H} = \mathscr{F}_0$:

(a) As mentioned before, $\emptyset$ has no proper subset, so $(F \Rightarrow T)$ is vacuously true, meaning $\emptyset \in \mathcal{H}$

(b) Let $A \in \mathcal{H}$. We must show $f(A) \in \mathcal{H}$, that is if $B \in \mathscr{F}_0$ and $B \subsetneq f(A)$ then $f(B) \subseteq f(A)$. Since $B \in \mathscr{F}_0 = \mathcal{G}_A$, then either $B \subseteq A$ or $f(A) \subseteq B$. Since $B \subsetneq f(A)$, it cannot be the latter, thus we must have that $B \subseteq A$. From here, we have two possibilities:

  i. $B \subsetneq A$: If so, then since $A \in \mathcal{H}$, by definition $f(B) \subseteq A \subseteq f(A)$, where the last $\subseteq$ comes from the definition of f. Thus, $f(B) \subseteq f(A)$

  ii. $B = A$: Then $f(A) = f(B)$, in particular $f(B) \subseteq f(A)$

  in either case, we get $f(B) \subseteq f(A)$

(c) Let $\mathcal{C} \subseteq \mathcal{H}$ be a chain. To show $\cup\mathcal{C} \in \mathcal{H}$, it must be that for any $B \in \mathscr{F}_0$ where $B \subsetneq \cup\mathcal{C}$, then $f(B) \subseteq \cup\mathcal{C}$. We will do a sneaky trick for this proof using the law of excluded middle [a]: For each $C \in \mathcal{C}$, either "$C \subseteq A$ for some A", or "$A \subsetneq C$ for all A" (which is it's direct negation). If the latter was true then

$$B \subsetneq \cup\mathcal{C} \subseteq \cup\{f(A) \mid A \in \mathcal{C}\} \subseteq B$$

implying $B \subsetneq B$ – a contradiction. Thus, it must be that for some $A \in \mathcal{C}$, $B \subseteq A$. There are again two cases

i. $B \subsetneq A$: Since $A \in \mathcal{H}$, then $f(B) \subseteq A \subseteq \cup\mathcal{C}$, so $f(B) \subseteq \cup\mathcal{C}$

ii. $B = A$: Then we have $B \in \mathcal{H}$, so $\mathcal{G}_B$ is defined and $\mathscr{F}_0 = \mathcal{G}_B$. Since $\mathcal{G}_B$ is f-inductive, $\cup\mathcal{C} \in \mathcal{G}_B = \mathscr{F}_0$. Thus, either $\cup\mathcal{C} \subseteq B$ or $f(B) \subseteq \cup\mathcal{C}$, and since $\cup\mathcal{C} \subseteq B$ is not possible since $B \subsetneq \cup\mathcal{C}$, it must be that $f(B) \subseteq \cup\mathcal{C}$.

In either case, $f(B) \subseteq \cup\mathcal{C}$, showing that $\cup\mathcal{C} \in \mathcal{H}$, showing that $\mathcal{H}$ is f-inducitve

Thus, $\mathscr{F}_0 = \mathcal{H}$ as we sought to show □

Finally, we get to combine all these facts. If $A \in \mathscr{F}_0 = \mathcal{H}$ and $B \in \mathscr{F}_0 = \mathcal{G}_A$, then either $B \subseteq A$ or $A \subseteq f(A) \subseteq B$. We couldn't do this earlier since we didn't know that $\mathscr{F}_0 = \mathcal{H}$ and $\mathscr{F}_0 = \mathcal{G}_F$ for any $F \in \mathscr{F}_0$. Thus, we have that $A \subseteq B$ or $B \subseteq A$, showing that $\mathscr{F}_0$ is a chain. We can now define $M = \cup\mathscr{F}_0$. Since $\mathscr{F}_0$ is f-inductive, then by the 3rd condition $M \in \mathscr{F}_0$. By the 2nd condition:

$$\cup\mathscr{F}_0 = M \subsetneq f(M) \in \mathscr{F}_0$$

But $f(M) \subseteq \cup\mathscr{F}_0 = M$, so $M \subsetneq f(M) \subseteq M$ implying $M \subsetneq M$ – a contradiction. The source of the contradiction was assuming an f existed which let's as choose an ever bigger value, and so no such function can exist, meaning for some A, there does not exist a B such that $A \subsetneq B$, but then A is maximal. And so, we have that (1) ⇒ (2)

(2) ⇒ (3) Let $(P, \leq)$ be a non-empty partialy orderd set. We want to show that $(P, \leq)$ contains a maximal chain (i.e. a chain C that cannot be enlarged further without loosing its total order). Let $\mathcal{F}$ be the set of all chains in P, orderd by inclusion (so if $P = A \leq B$ is a chain and $Q = A \leq B \leq C$, then $P \leq Q$). This set is non-empty, since $\emptyset \in \mathcal{F}$ and $\{x\} \in \mathcal{F}$ for all $x \in P$. Furthermore, Since any finite subchain of any infinite chain in $\mathcal{F}$ would be in $\mathcal{F}$, $\mathcal{F}$ has finite character. Thus, by Turkey's Lemma, $\mathcal{F}$ has a maximal element, meaning P has a maximal chain.

(3) ⇒ (4) Let $(P, \leq)$ be a non-empty partially ordered set where each chain has an upper bound. Then by the Hausdroff Maximality Principle, there exists a maximal chain $M \subseteq P$. Since M is a chain, it has an upper-bound – let's say m. Then m is must be a maximal element of P: If there exists some $x \in P$ such that $m \leq x$ and $m \neq x$, then $M \cup \{x\}$ is a chain that is properly included in M, but M is maximal – a contradiction. Thus, m is maximal in P

(4) ⇒ (5) Let S be any set, and let $\mathcal{L}$ be the family of all well-ordered set $(W, \leq)$ such that $W \subseteq S$. This set is non-empty, since for each $x \in S$, $\{\{x\}, \{x, x\}\} \in \mathcal{L}$. In the upcoming proof, we will give an ordering to $\mathcal{L}$ and use Zorns Lemma to show that the maximal element will be $(S, \leq)$ with appropriate $\leq$

Define $(W_1, \leq_1) \preceq (W_2, \leq_2)$ if either

(a) $W_1 = W_2$ and $\leq_1 = \leq_2$ or

(b) there exists a $a \in W_2$ such that $W_1 = \{x \in W_2 \mid x \leq_2 a,\ x \neq a\}$, and $\leq_2$ agrees with $\leq_1$ on W_1 ($\leq_1 \subseteq \leq_2$ [b]). In this case, W_2 is called a continuition of $(W_1, \leq_1)$

It should be relatively clear that $(\mathcal{L}, \preceq)$ is a partial ordering on $\mathcal{L}$. Note that it is not a total ordering: If W_i and W_j do not have the same minimal element, then it $W_i \not\preceq W_j$ and $W_j \not\preceq W_i$. This means that any chain of elements will end up with the same minimal element. Now, let $\ell = \{(W_i, \leq_i)\}_{i \in I}$ be any non-empty chain in $\mathcal{L}$ (relative to $\preceq$). We want to show it must have an upper-bound. Set $W = \bigcup_{i \in I} W_i$ and $\leq = \bigcup_{i \in I} \leq_i$ (see definition 0.1.12). Then $(W, \leq)$ is a total order (see exercise ref:HERE). We'll show it is a well-ordering. Let A be a non-empty subset of W. Since it's non-empty, there exists some $i \in I$ such that $A \cap W_i \neq \emptyset$. Since $(W_i, \leq_i)$ is well-ordered, there is an element $a \in A \cap W_i$ such that $a \leq_i x$ for all $x \in A \cap W_i$. Suppose now that there is an element $b \in A$ such that $b \leq a$. Then by definition of $\preceq$, it must be that $b \in W_i$, so $b \leq_i a$, so $b = a$. Thus, A has a smallest element a in $(W, \leq)$, meaning $(W, \leq) \in \mathcal{L}$. Furthermore, it should be clear that $(W, \leq)$ is an upper-bound for ℓ

Thus, by Zorn's Lemma, $\mathcal{L}$ has a maximal element $(W_0, \leq_0)$. If $W_0 = S$, we're done – $\leq_0$ is our well-ordering of S. If $W_0 \neq S$, let $z \in S \cap W_0$. Define $(W_0 \cup \{z\}, \leq)$, where

$$\leq = \leq_0 \cup \{(x, z \mid x \in W_0 \cup \{z\}\}$$

that is, place z after everything in W_0. Then we can see that $(W_0 \cup \{z\}, \leq) \in \mathcal{L}$ and $(W_0, \leq_0) \preceq (W_0 \cup \{z\}, \leq)$ and $(W_0 \cup \{z\}, \leq) \not\preceq (W_0, \leq_0)$. However, this contradicts the maximality of $(W_0, \leq_0)$, and so it must be that $W_0 = S$, showing that S does indeed have a well-ordering.

(5) $\Rightarrow$ (1) Let $\{A_i\}_{i \in I}$ be any non-empty family of non-empty sets, and let $S = \cup_{i \in I} A_i$. By the well-ordering principle, there exists an $\leq$ that well-orders S. Now, for each $i \in I$, let $f(i)$ be the smallest element of A_i (relative to $\leq$). Then f is a chocie function on the family $\{A_i\}_{i \in I}$

[a] For every proposition P, either P is true or $\neg P$ is true. This is implied by ZF

[b] If these seems strange, review the technical definition of a relation in section 0.1.4

This textbook contains many cases here zorn's lemma must be used, for example:

1. All proper ideals in a ring [with identity] is contained in a maximal ideal
2. All vector-spaces have a basis

(The proof of Schröder-Bernstein Theorem requires Zorn's lemma, so I can show it here, it's in Follands Real Analysis preliminary chapter)

(maybe also show Shcröder-Bernstein's lemma in the finite case after introducing cardinality?

A.3 Zorn's Lemma and Transfinite Induction

For the purposes of this textbook, the most important consequence of the Axiom of Choice is Zorn's lemma. I think a section on Zorn's lemma is important to be able to reference it and explain how it is a sort of "super induction".

(Include transfinite induction, used a bit when showing IBN for vector spaces)

(there is a surprising amount of progress made in fields because of a single illustrative observation: In representation theorem,the fact that any $N \subseteq M$ has a compliment implies M is the direct sum of simple modules gives rise to a ton of rep theory. The fact that any R-module over a vector-space is free gives rise to liner algebra. The fact k-homomorphism from from k to F can always be extended to a k-homomorphism to algebraically closed field over k $\Phi : F \to L$ leads to results about normal extensions that are abused like crazy. I'm sure there is something more with Baer's criterion. I just realized that what I actually just rambled about is how Zorn's lemma opened up many new avenues of thinking...)

B

Sub-Group of a Free Group is Free

In this book, the fact that a subgroup of a free-group is free doesn't come into play. It is interesting tangentially as an example of a property which is closed under taking subgroups, contrasting to how being finitely generated is is not a property closed under taking subgroups, or that a sub-module of a free module is not always free, but we don't use this fact anywhere else. However, some might be curious as to how we would prove this fact. The goal of this section is to do just that using only group-theoretical knowledge.

B.1 Using Cosets

There are in fact many proofs of this fact, including a famous one by Serre topological properties of trees (connected graphs with no cycles). Such a proof would be covered in geometric group theory.

Lemma B.1.1: Finite Index Subgroup Of Finitely Generated Subgroup

Let G be a finitely generated subgroup, and let $H \leqslant G$ be a subgroup of finite index (i.e. $|G : H| = s$ for some $s \in \mathbb{N}$). Then H is finitely generated

***Proof*:**
Let $G = \langle x_1, x_2, ..., x_n \rangle$ be generated by the elements $X = \{x_1, x_2, ..., x_n\}$, $H \leqslant G$ so that $|G : H| = s$, and let $A = \{a_1 = 1, ..., a_s\}$ be a set of representative sfor the right cosets of H (also known as a transversal for H in G).

Let $x \in H$. Since $H \leqslant G$, then $x \in G$. Since G is finitely generated we have that

$$x = y_1 y_2 \cdots y_n$$

where y_i is either a generator or an inverse of a generator. Since $y_1, y_2, ..., y_n$ are not necessarily in H, this is not a finite representation of x. Instead, we'll show that from these we can create a representation with elements in H. Take the representative a_{i_1} which represents the coset Hy_1, so $a_{i_1} \in Hy_1$, meaning $a_{i_1} = h_1 y_1$ for some $h_1 \in H$. Then

$$x = y_1 a_{i_1}^{-1} a_{i_1} y_2 y_3 \cdots y_n$$

Next, let a_{i_2} be the represetnative for $Ha_{i_1} y_2$ so that $a_{i_2} = h_2 a_{i_1} y_2$. In particular, $h_2^{-1} = a_{i_1} y_2 a_{i_2}^{-1}$ so that this conjugation is part of H. Now, re-write

$$x = y_1 a_{i_1}^{-1} a_{i_1} y_2 a_{i_2}^{-1} a_{i_2} y_3 \cdots y_n$$

Similarly for a_{i_3}, picking a representative from $Ha_{i_2} y_3$. Going on in this fashion, we'll end up with an equation

$$x = y_1 a_{i_1}^{-1} a_{i_1} y_2 a_{i_2}^{-1} a_{i_2} y_3 a_{i_3}^{-1} a_{i_3} \cdots y_n a_{i_n}^{-1} a_{i_n} \tag{B.1}$$

Thus, every element $x \in H$ can be in written out as in equation B.1. Now, as we noticed earlier: $a_j y_k a_l^{-1} \in H$. If $a_{i_n} \in H$. Thus, every triple $a_j y_k a_l^{-1}$ in equation B.1 is in H. The only element that might not be in H is a_{i_n} (if a_{i_n} is a representative for a coset zH for $z \neq 1$). If $a_{i_n} \in H$, we'd have a finite representation, so suppose $a_{i_n} \notin H$. Then we can "squish" everything down to

$$x = ha_{i_n}$$

where $h = y_1 a_{i_1}^{-1} a_{i_1} y_2 a_{i_2}^{-1} a_{i_2} y_3 a_{i_3}^{-1} a_{i_3} \cdots y_n a_{i_n}^{-1} \in H$. But then, this implies that $x \in Ha_{i_n}$, and since cosets partition groups (ref:HERE (preliminary chapter on equivalence relation)), we have $x \notin H$ – a contradiction to our original assumption. Therefore, $a_{i_n} \in H$, and so x is finitely represented by the set of triplet products $a_j y_k a_l^{-1}$, wherre $a_j, a_l \in A$ an $y_k \in X \cup X^{-1}$. Since x was arbitrary, this is true for any element of H, and so H is finitely generated.

This theorem is interesting in it's own right as a partial result to when a subgroup of a finitely generated subgroup is finitely generated [1]. It also has application in geometric group theory where we are interested in finding what is the growth type of the function $\mathfrak{g}(G)(n)$ (ex. polynomial growth, exponential growth) which represents the number of words which require n elements from the generators to represent it. In this case, what we found was if $m(H)$ represents the minimum number of generators for the group H, then $m(H) \leq 2m(G)|G : H|$. For more, see cite:HERE

The part that is more intriguing for us is the way we have constructed the elements of H. Though we've limited to $|H : S|$ being finite, we can still represent the elements of $H \leqslant G$ with the same trick! Using this fact, we will show that the elements of a subgroup of a free-group will have to be relation less:

Theorem B.1.2: Nielsen-Schreier's Theorem

Let G be a free group and $H \leqslant G$. Then H is a free group

[1] recall that $F_2' \leqslant F_2$, the commutator subgroup of the free group on 2 elements, is not finitely generated (see ref:HERE)

***Proof*:**
Let F be any free group freely generated [a] by the set of elements X. Note that X need not be finite, or even countable. By the well-ordering principle (ref:HERE), fix some ordering of the generators and their inverses. Then, order the elements of F, first by lengh of their words, and for words with the same length, order them lexicographically (i.e. the same way you order a dictionary). This is now a well-ordering of F.

Let $H \leqslant F$. Choose as representatives for the cosets of H the element which is first in the ordering. Letting A be the set of these representatives, then we saw in B.1.1 H is generated by elements of the form $a_j x_k a_l^{-1}$, $a_j, a_l \in A$, $x_k \in X$. Let's label the element $a_j x_k a_l^{-1}$ as $y_{j,k}$, and let Y represent the set of all these elements. We will show that Y is the set of free generators of H by showing: (1) the generators cannot be reduced, (2) the words the generators form cannot be reducted

1. The elements $a_j x_k a_l^{-1}$ are in reduced form.

   Suppose not. Then some terms in the word must cancel, which would only be possible if x_k is cancelled out somehow. Let's say it was cancelled out on the left. Then $a_j = u x_k^{-1}$ and u $u \in Ha_j$. But then, $u = a_j x_k$, meaning that u is the representative of $Ha_j x_k$, that is $u = a_l$. Thus, we get

   $$a_j x_k a_l^{-1} = u x_k^{-1} x_k a_l^{-1} = a_l x_k^{-1} x_k a_l^{-1} = 1$$

   However, no such relation exists between these elements in F – a contradiction. The same argument applies if x_k is cancelled from the right, or take the inverse of $a_j x_k a_l^{-1}$

2. If x is any reduced word by non-identity elements in Y, then when writing x as a word from X, the elements x_k (or their inerses) in the middle of $y_{j,k}$ are not cancelled

   Let's say some x_k is cancelled by some y on the right, say in the product $y_{j,k} y_{r,s}$. Then the a_l^{-1} occuring in $y_{j,k}$ must be cancelled first, then x_k would be cancelled. One way for this to happen is for a_r to start with $a l x_k^{-1}$. Then like before, x_k^{-1} is also a representative of its coset. BUt looking at $y_{j,k}^{-1} = a l x_k^{-1} a_j^{-1}$, we see that a_j represents the coset of $a_l x_k^{-1}$. That means that $a_j = a_l x_k^{-1}$, and thus $y_{j,k} = 1$.

   If x_k is cancelled by x_s in $y_{r,s}^{-1}$ to its right, then $y_{r,s}^{-1}$ msut start with a_l and $x_s = x_k$, but then $y_{r,s} = y_{j,k}$, and so the word x is not reducced in Y, contradiction our assumption.

   Finally, it might be that x_k is cancelled a letter in the a_s^{-1} that ends $y_{r,s}$. Then inverting x, we get a wrod in whic $y_{r,s}^{-1} y_{j,k}^{-1}$ occurs, and the x_s^{-1} in $y_{r,s}$ is cancelled in the same manner as in the the first case for x_k, and just like before this implies that $y_{r,s} = 1$.

   Similarly if x_k is cancelled from the left, showing that x cannot be reduced using elements from Y

From this, we see that Y freely generates H, and so H is free, as we sought to show.

[a] as in the elements of X can be treated as letters

B.2 Using Tree's

A perhaps more intuitive proof of this theorem comes from some interesting results on groups acting on trees. HERE (Clara Löh's proof is really concise and I like it!)

C

Basic Topology and Topological Groups

This appendix is by no means supposed to be a subsitute for a proper introduction of Topology. Rather, it should be treated as either a revision or a test of knowledge to see if the prerequisite Topology knowledge for chapter ref:HERE has been aquired. To that end, this will be a speed-run of definitions and basic concepts with some exercises sprinkled in to give the reader the opportunity to check their knowledge. Good books to get a proper grasp of topology would be Munkre's Topology or ________.

C.1 Definition and Properties

Definition C.1.1: Topology

Let X be a set and $\mathcal{T}_X \subseteq P(X)$ be a subset of the powerset of X such that

1. $\emptyset$ and X are in $\mathcal{T}_X$
2. For any arbitrary collection of elements in $\mathcal{T}_X$, their union is in $\mathcal{T}_X$
3. for any finite collection of elements in $\mathcal{T}_X$, their intersection is in $\mathcal{T}_X$

Then $(X, \mathcal{T}_X)$ is call a *topological space* and $\mathcal{T}_X$ is called a *topology on X*. Elements of $\mathcal{T}_X$ are called *open sets*.

Usually we omit the $\mathcal{T}_X$ and simply call X the topology, thus we say that $Y \subseteq X$ is open if and only if $Y \in \mathcal{T}_X$. If $x \in X$, then a *neighbhorhood* of X will be an *open set* $U \in \mathcal{T}$ such that $x \in U$. In some book, a neighbhorhood is simply some set containing x and need not be open; sometimes to emphasize that the neighbhorhood is open, it is called an *open neighbhorhood*. The *neighbhorhood system* of an element $x \in X$ is the collection of all open neighbhorhoods containing x. Topologies can also be defined in terms of closed sets:

Proposition C.1.2

Let X be a set and $\mathcal{T}_X^C \subseteq P(X)$ be subset of the powerset such that

1. $\emptyset$ and X are in $\mathcal{T}_X^C$
2. For any finite collection of elements in $\mathcal{T}_X^C$, their union is in $\mathcal{T}_X^C$
3. for any arbitrary collection of elements in $\mathcal{T}_X^C$, their intersection is in $\mathcal{T}_X^C$

Then $\mathcal{T}_X^C$ forms a topology on X identical to the topology $\mathcal{T}_X$ if every set is replaced with its copmliment.

***Proof*:**
This should be done as an exercise

Many times, we define a topology through the use of a topological basis (not to be confused with a vector-space basis)

Definition C.1.3: Topological Basis

Let X be a set. Then a subset of the powerset $\mathcal{B} \subseteq P(X)$ is called a *topological basis* (or just *basis* if unambiguous) where:

1. for every $x \in X$, there is some $U \in \mathbb{C}\mathrm{P}$ such that $x \in U$
2. For every $U, V \in \mathcal{B}$ and every $x \in U \cap V$, there exists a $B \in \mathcal{B}$ such that $x \in B \subseteq U \cap V$

If only the first condition is met, then $\mathcal{B}$ is called a *topological subbasis* (or *subbasis* for short). The topology $\mathcal{T}_{\mathcal{B}}$ on X is the topology where $U \in \mathcal{T}_{\mathcal{B}}$ if and only if U is the union of some subfamily of $\mathcal{B}$. If $\mathcal{B}$ is a subbasis, then U is some union and intersection of a subfamily of $\mathcal{B}$.

Two topologies that can be naturally defined on any set X are the *trivial topology* of $\mathcal{T}(X) = \{\emptyset, X\}$ and the *discrete topology* of $\mathcal{T}(X) = P(X)$. The Euclidean topology is a common topology defined on $\mathbb{R}^n$ for $n \geq 1$. Another common topology is defined on real or complex vector-spaces using a function called a *norm* (very similar to the euclidean norm used in when defining euclidean domains): a norm is a function $\|\cdot\| : V \to \mathbb{R}$ where

1. $\|x + y\| \leq \|x\| + \|y\|$
2. $\|c \cdot x\| = |c| \cdot \|x\|$
3. $\|x\| = 0$ if and only if $x = 0$

Then then the collection of all $B_r(x) = \{y \in X \mid \|x - y\| \leq r\}$ forms a topological basis for a topology. Another common way of defining a topological space is by using a metric: if (X, d) is a metric space, then the collection of all $B_r(x) = \{y \in X \mid d(x, y) < r\}$ forms a basis for X.

We relate two topologies by taking *continuous functions* between them:

Definition C.1.4: Continuous Function

Let X, Y be topological spaces and $f : X \to Y$ be a function. Then f is called a continuous function is the pre-iamge of every open set of Y is in X:

$$f^{-1}(U) \in \mathcal{T}(X)$$

Common examples of continuous functions are the polynomials functions from $\mathbb{R}^n$ to $\mathbb{R}^m$ given the euclidean topology in both cases. If a function is also bijective, then we call such a function a bijective continuous function. Unlike homomorphisms (be it groups, rings, etc.) a bijective continuous function does *not* imply the inverse function is continuous, or equivalently, that the image of open sets are open (can you think of an easy example from $\mathbb{R}^2$ to $\mathbb{R}$?), and so we give a special name to a bicontinuous bijective function:

Definition C.1.5: Homeomorphim

Let $(X, \mathcal{T}_X)$ and $(Y, \mathcal{T}_Y)$ be two topological spaces. Then a function $f : X \to Y$ is called a *homeomorphism* if f is bijective and

$$f^{-1}(\mathcal{T}_Y) = \mathcal{T}_X$$

examples of homeomorphisms are any bijective function if the codomain has the indiscrete topology or the domain has the discrete topology, the identity function or $f(x) = x^3$ on $\mathbb{R}$ in the discrete, indiscrete, and euclidean topology. A functions that is bijective continuous but are not homeomorphisms is the identity function from X to X where the domain has the discrete topology and the codomain has the trivial topology. Note that for any topological space X, the identity function is continuous, the composition of continuous functions is continuous, and associativty of continuous functions follows from the associativity of functions. Hence, the collection of topological spaces and continuous maps from a category

Definition C.1.6: Category Of Top

The collection of all topologies on any set and the collection of continuous maps between topologies forms a category, labelled as **Top**, with objects being topologies and morphisms begin continuous maps. Homeomorphisms are the isomorphisms of **Top**.

If the codomain of a continuous function is an algebra, (usually $\mathbb{R}$ or $\mathbb{C}$), we can perform arithmetics on continuous functions:

Proposition C.1.7: Arithmetics of Continuous Functions

Let $f : X \to A$ and $g : X \to A$ be two continuous functions with A being an R-algebra (usually $\mathbb{R}$ or $\mathbb{C}$). Then $f + g$, fg and cf for $c \in A$ are continuous functions.

This fact is used in a couple of examples in the book, and so proving this proposition is an important exercise. If the fact that the codomain is an R-algebra makes the proof seem complicated, simply let $A = \mathbb{R}$ or $\mathbb{C}$ with the euclidean topology (it is all that is really necessary to prove).

Given an arbitrary set $U \subseteq X$, we may consider the set of all closed subsets containing U. Since the arbitrary intersection of closed sets is closed, there is a notion of smallest closed subsets *containing* U. Similarly, we may consider all open subsets *contained* in U, and take their union, giving the dual

Definition C.1.8: Closure And Interior

Let $U \subseteq X$ be an arbitrary set. Then

1. The *closure* of U, denoted $\overline{U}$ is the intersection of all closed sets containing U
2. The *interior* of U, denoted $\text{int}(U)$, is the union of all open sets contained in U.

Given a topological space, injective and surjective maps give rise to natural spaces, and the product of sets has a natural topology:

Definition C.1.9: Subspace, Quotient Space, and Product Space

Let X be a topological space. Then

1. the subset $Y \subseteq X$ is a *subsapce* of X if it is endowed with the topology given by the sets $\{U \cap Y \mid U \subseteq X \text{ open}\}$
2. a set of equivalence space $Y = X/\sim$ is a *quotient topology* if it is endowed with the topology where $U \subseteq Y$ is open if and only if $\pi^{-1}(U) \subseteq X$ is open (where $\pi : X \to Y$ is the natural projection)
3. Given a collection of topological spaces $\{X_i\}_{i \in I}$ for an indexing set I, the product topology is the topology endowed on products of the sets $Y = \prod_{i \in I} X_i$ where $\pi_i^{-1}(U)$ forms a sub-basis for Y.

The reader may verify these satisfy the necessary universal properties: the universal property of subspaces, the universal property of quotient spaces (which is the dual of the former universal property), and the universal property of products.

Recall that a topological space has varying notion of separation defined on it. The most prominent of which is the Hausdorff axiom (also known as T_1):

Definition C.1.10: Hausdorff space (T_2)

Let X be a topological space, and let $x, y \in X$ be arbitrary different points. Then X is called *Hausdorff* if there always exists open sets U, V such that $x \in U$, $y \in V$, and $U \cap V = \emptyset$.

There are two weaker conditions that do show up in this book:

1. Féchet (T_1) means that for each $x, y \in X$, there exists open neigbhorhoods U, V such that $x \in U$, $y \in V$, but $y \notin V$ and $x \notin U$. Note that U and V need not be disjoint. Key feature of T_1 is that all singletons are closed.
2. Even weaker is Kolmogorov (T_0) (Zariski topologies are T_0 but not T_1)

Proposition C.1.11: Hausdorff And Diagonals

Let X be a topological space. Then X is Hausdorff if the diagonal $\Delta \subseteq X \times X$ with the product topology is closed.

***Proof*:**
exercise

(define it, show that if $A \subseteq X$ is closed, then A is compact)

A final tool that is very common in topology is that of the *sequences*[1]

Definition C.1.12: Sequence

Let X be any set. Then a *sequence*, is a function $f : \mathbb{N} \to X$. Usually, the element $f(n)$ is labeled as x_n for some dummy variable x, and the function f is represented as $(x_n)_{n=1}^{\infty}$ or simply (x_n) for short.

Sequences become interesting once they are in a topological space, and there is a notion of them "approaching" a point. In a metric space, we can define this notion using epsilon-balls. However, there is a more general notion of convergence for topological spaces that are non-metrizable

Definition C.1.13: Convergent Sequence

Let X be a topological space, (x_n) a sequence in X, and $x \in X$ an element of X. Then (x_n) is said to *converge to x* if for every neighborhoods U of x, there exists a natural number $S(U) \in \mathbb{N}$ such that

$$x_n \in U \qquad \forall n \geq S(U)$$

Proposition C.1.14: Convergent Sequence In Hausdroff Spaces

Let X be a topological space and (x_n) a convergent sequence. Then (x_n) converges to *at most* 1 point. The resulting point is called the *limit point*.

***Proof*:**
exercise

For most cases that are considered in algebra and analysis, every point $x \in X$ has a "countable neighborhood basis", which essentially translates to every open set is "locally" a countable union of open sets. This property is called *first-countable*. It is not important for this book, but it is important to be able to state the following result:

[1]More generally, ultra-filters or nets may be more appropriate for the topological setting, however they are not necessary for this review of topology

Proposition C.1.15: Closed Sets In Hausdroff Spaces w.r.t. Sequences

Let $U \subseteq X$ be any subset of a first-countable Hausdorff topological space X. Then if $\lim U$ represents the collection of all limit points of all possible convergent sequences $(x_n) \subseteq X$, then:

$$\overline{U} = \lim U$$

***Proof*:**
exercise

Most spaces covered in a topological class, and all topological spaces covered in this book, have this property and hence closedness can be characterized with sequences. If $\overline{U} = X$ (i.e. the closure gives the entire space), then we give such a set a name

Definition C.1.16: Dense Set

Let X be a topological space. Then if $U \subseteq X$ is a subset of X such that $\overline{U} = X$, then U is called a *dense set.*

These are very important in real and functional anslysis, and they show up in the discussion of completion in section 25.

If X is a metric space, we can define the notion of a cauchy sequence:

Definition C.1.17: Cauchy Sequence

Let X be a metric space with metric d. Then (x_n) is a *cauchy sequence* if for every $\epsilon > 0$, there exists an $N \in \mathbb{N}$ such that for all $n, m > N$,

$$d(x_n, x_m) < \epsilon$$

Given any metric space, we can "complete" it by adding all of the cauchy sequence that do not converge in the following way: for any two cauchy sequences, they are called *equivalent* if $x_n - y_n \to 0$ in G. Then the collection of all equivalent cauchy sequence is called the *completion of the metric space.* It should be checked that this is equivalence to the following definition:

Definition C.1.18: Complete Metric Space

Let X be a metric space. Then X is compelte if all cauchy sequences are convergent sequences

Example C.1.19: Complete Spaces
$\mathbb{R}$ is a complete space, while $\mathbb{Q}$ is not

The notion of completeness given in section 25 is in fact a generalization of the above notion, as we'll show in section C.4. If is often the case that when working with complete spaces, we want to have.

For comparison to other definitions of dimension presented, the following can be considered one of the "most broad" definitions:

Definition C.1.20: Topological Dimension

Let X be a topological space. The *order* of an open cover $\{U_i\}$ of X is the smallest number n for which each point $x \in X$ belongs to at most n open sets in the open cover. A *refinement* of an open cover $\{U_i\}$ is an open cover $\{V_j\}$ such that each $V_j \subseteq U_i$ for some i. The *Topological dimension* or *Lebesgue Covering Dimension* is the minimum number n such that every finite open cover has a refinement of order $n+1$.

C.2 Construction Topologies

The subspace, product, and quotient topology are all special examples of the following two constructions:

Definition C.2.1: Weak [initial] Topology

Let $F = \{f : X \to Y_\alpha\}_{\alpha \in A}$ be a collection of maps from a set A to topological spaces Y_α indexed by $\alpha \in A$. Then the coarsest topology on X which makes each f continuous is called the *weak topology.* In particular, a basis for the weak topology generated by F consists of $f_\alpha(U)$ for all open sets $U \subseteq Y_\alpha$ for every $\alpha \in A$.

Note that it is not interesting to ask what is the finnest topology on X that makes all the maps continuous; that would simply be the discrete topology on X. Note too that since the weak topology $\mathcal{T}$ is the coarest topology on X that makes all the maps continuous, if $\mathcal{T}'$ is another topology on X that makes each map in F continuous, then $\mathcal{T}' \subseteq \mathcal{T}$. Note too that we showed in the definition how to actually create the weak topology, however for the skeptical reader it should be verified that the weak topology does indeed exist

Example C.2.2: Weak Topology

1. Let $\{X_\alpha\}_{\alpha \in A}$ be some collection of topologies and $X = \prod_{\alpha \in A} X_\alpha$. Let $\Pi = \{\pi_\alpha : X \to X_\alpha\}$ be the collection of functions. Then the weak topology on X generated by F is *precisely* the product topology of X This is be

2. Let X be any topological space and $A \subseteq X$ any subset. Let $\iota : A \to X$ be the inclusion map. Then the weak topology on A generated by ι is the *subspace* topology.

One of the great advantages of of defining weak topologies is that we can make the collection of f_α and spaces X_α anything we want. This flexibility will the definition's of many interesting topologies on funcitonal spaces (ex. collection of smooth functions on compact spaces, collection of bounded continuous functions, or simply collection of bounded functions, and so forth). All such spaces are of key importance in funtional analysis. Weak topologies are also a special case of a much more general construction, which we'll explore at the end of this chapter.

Finally, there is also the dual of weak topologies where we take maps that map *into* X.

Definition C.2.3: Final Topologies

Let Y be a set, $\{X_\alpha\}_{\alpha\in A}$ be a collection of topological spaces, and $F = \{f_\alpha : X_\alpha \to Y\}$ be a collection of maps into Y. Then the the finnest topology on Y that makes each f_α continuous is called the *final topology*. In particular, $U \subseteq X$ is open if and only if $f_\alpha(U)^{-1}$ is open for each $\alpha \in A$

Again, it should be verified that this is indeed a topology. While the weak (or initial) topology seeks for a domain with the coarsest topology, final topologies seek a codomain with the finnest topology. This is because the coarsest topology on the co-domain will always create continuous maps. Unlike the initial topology, the final topology is not given the name "strong" to contrast the name "weak" for initial topologies. There is in fact a topology known as the strong topolgoy on a space X: If $\|\cdot\|$ is a norm on X[2], the strong topology on X is the topology generated by all the balls $B_r(x)$ for all $x \in X$ and $r > 0$. The contrast between strong and weak topologies will make more sense when studying functional analysis, where there we would much rather a space have a strong topology, but it will sometimes be unfeasible (ex. the space of all smooth functions, or infinite dimensional vector spaces won't have a good norm defined on them), and so we will "weaken" the topology that will create topology that will not be induced by something as strong as a norm (usually, it the weak topology will be induced by a countable collection of seminorms, see Folland chapter 5 for more information).

Example C.2.4: Final Topologies

Let X be an arbitrary topological space an $A \subseteq X$ a subspace. Let $\pi : X \to A$ be the projection map. Then the final topology on A is the *quotient topology*.

Both the initial and final topologies can be generalized even further. If you have read XII, both the weak and strong topologies are simple examples of *limits* and *colimits* in **Top**. These two also go by the name of *projective limit* and *injective limit*, or *colimit* and *limit*.

Exercise C.2.1

1. There are some cool exercises Hossain gave that I'd like to include for those who have covered the section on commutative algebra and would like to know some more topological properties of Spec (R).
2. Let X be a topological space. Show that X is Hausdorff if and only if the diagonal $\triangle = \{(x, y) \in X \times X \mid x = y\}$ is closed.

C.3 Compactness and Connectedness

Definition C.3.1: Compact Space

Let X be a topological space. If $\{U_i\}_{i\in I}$ is a collection of open sets such that $\bigcup_{i\in I} U_I = X$, then $\{U_i\}_{i\in I}$ is called an *open cover*. An *open subcover*, or *subcover* for short, is a subst of an open cover. If any open cover of X adminits a finite subcover, then X is *compact*.

[2]See definition ref:HERE

Proposition C.3.2: Closed Subset Of Compact Spaces

Let $U \subseteq X$ be a closed subset of a compact space. Then U is compact is a subspace.

Proof:
exercise

Definition C.3.3: Connected Space

Let X be a topological space. Then if there exists no non-trivial proper clopen subsets, X is connected. Equivalently, X is connected if there does not exist two non-empty open sets $U_1, U_2 \subseteq X$ such that $X = U_1 \cup U_2$.

C.4 Topological Groups

In this book, topological groups show up when talking about the completion of rings in section 25 and when talking about the representation of non-finite groups in chapter 53. The topological considerations in both of those cases were purposefully scarce to not have any topological requirements for learning those materials. However, having a topological perspective brings out a deeper perspective, and so they are further explored in this section.

Definition C.4.1: Topological Group

Let G be a topological space. Then if there exists a binary operation $+ : G \times G \to G$ satisfying the axioms of a group that is also *continuous* (where $G \times G$ has the product topology), and a $(\_)^{-1} : G \to G$ mapping $g \mapsto g^{-1}$ that is *continuous*, then G is a *topological group*

Though we have defined topological groups, the same definition can be done for any other algebraic object we've worked with, notably *topological rings*, *topological R-modules*, and *topological R-algebras*. Topological groups (resp. rings, modules, algebras) is a special example of a *group object* in the category of **Top** (see ref:HERE, currently at the end of Cat theory part).

Example C.4.2: Topological Group

1. For any group G, if G is equipped with the discrete topology it is a topological group, and all homomorphisms are continuous between any other topological group. In this way, the theory of topological groups can be thought as subsuming group theory. Any group with the indiscrete topology (also known as trivial topology) is also a topological group.

2. If $H \leqslant G$ is also a topological space, then H is a topological group with the subspace topology.

3. The real numbers $\mathbb{R}$ and $\mathbb{R}^\times$ under multiplication is a topological group (can you prove continuity of $+$, $\cdot$, and $(\_)^{-1}$?). More generally, $\mathbb{R}^n$ is a topological group, or any *topological vector-space* (a vector-space where the action is continuous as well) is a topological group

4. The circle S^1, and the n-torus $(S^1)^n$ are topological groups. Show that $\mathbb{R}/\mathbb{Z}$ is both

homeomorphic and isomorphic to S^1. If we take S^1 as a subset of $\mathbb{C}$, then we $S^1 = U(1) = \{x \in \mathbb{C} \mid |x| = 1\}$.

5. The general linear group over the reals $\mathrm{GL}_n(\mathbb{R})$ is a topological group. Similarly, the special linear group,$\mathrm{SL}_n(\mathbb{R}) = \{A \in \mathrm{GL}_n \mid \det(A) = 1\}$, is a topological group.

6. Take $O(n) \subseteq \mathrm{GL}(n)$, where $O(n) = \{A \in \mathrm{GL}_n(\mathbb{R}) \mid AA^t = I\}$. This is a topological group known as the *orthogonal group*, since it consists of all orthonormal basis of $\mathbb{R}^n$. Similarly, we can define $U(n) = \left\{A \in \mathrm{GL}_n(\mathbb{R}) \;\middle|\; A\overline{A}^t = I\right\}$ is a topological group known as the *unitary group*. Similarly to $\mathrm{SL}_n(\mathbb{R})$, we have:

$$\mathrm{SO}(n) = \mathrm{SL}_n(\mathbb{R}) \cap O(n) \qquad \mathrm{SU}(n) = \mathrm{SL}_n(\mathbb{R}) \cap U(n)$$

   Note that all these groups are *compact*.

7. (If you know some Differential Geometry) Any Lie group is naturally a topological group. An example of a topological group that is not a Lie group is $\mathbb{Q}$ with the subspace topology induced by $\mathbb{R}$.

8. If you have covered the completion chapter of *Commutative Algebra* [10], then the p-adic integers, $\mathbb{Z}_p$ has the induced inverse-limit topology (in this case, the p-adic topology). This group is compact, homeomorphic to the Cantor set, but is totally disconnected (hence cannot be a Lie Group)

9. Similarly to $\mathbb{Z}_p$, giving $\mathbb{Q}$ a different p-adic metric, we can take a different completion to form $\mathbb{Q}_p$, which is also a Lie Group.

Proposition C.4.3: Topological Criterion

Let G be a group and a topological space. Then G is a topological group if and only if the map $\cdot : H \times H \to H$, $(x, y) \mapsto xy^{-1}$ is continuous

***Proof*:**
exercise

Definition C.4.4: Topological Homomorphism/Isomorphism

Let $f : G \to H$ be a function between topological groups. Then if f is a continuous homomorphism (resp. a homeomorphic isomorphism) then f is called a *topological homomorphism* (resp. *topological isomorphism*)

Example C.4.5: Topological Isomorphism

Naturally, the identity map is a topological isomorphism. If the topology is discrete, every homomorphism/isomorphism is a topological homomorphism/isomorphism.

It is possible that a homomorphism/continuous map is not a continuous map/homomorphism. As an example of a continuous, even homeomorphic map, that is not a homomorphism is the following: let $g \in G$ and $\varphi_g : G \to G$ be the map sending $x \mapsto gx$. Show that φ_g is a homeomorphism. This example is particularly interesting since it shows that every $g \in G$ induces a homeomorphism! A space in which the maps φ_g are homeomorphisms is called a *homogeneous space*. As a consequence of this, if U is a neighborhood of 1, then gU is a neighborhood of g, meaning that the topology of G is determined by the neighborhoods of 1!

As an exercise, it is worth seeing how if $H \leqslant G$, then G/H has the quotient topology, is a homogeneous space, and if $H \trianglelefteq G$, then G/H is a topological group. Furthermore, There exists continuous homomorphisms $f : G \to H$ where $G/\ker(f) \ncong f(H)$ (i.e. there exists continuous bijective homomorphisms that are not isomorphisms in the category of topological groups). However, the 3rd isomorphism theorem is essentially the same, as can be done as an exercise.

Some authors make being Hausdorff (or at least T_1) part of the definition of the group. This limitation is not imposed in this definition, since in algebraic geometry, there will be non-Hausdorff spaces (ex. the a-adic topology on certain topological rings are non-Hausdorff). For Topological groups, there is a way to always find a quotient topology that is Hausdorff:

Proposition C.4.6

Let G be a topological group, $e \in g$ the identity, and $H = \bigcap_{\substack{U \subseteq G \text{ open} \\ e \in U}} U$ Then:

1. H is a subgroup
2. H is the closure of e, i.e. $\overline{e} = H$
3. G/H is Hausdorff
4. G is Hausdorff if and only if $H = \{e\}$

***Proof*:**
exercise

Topological groups are explored in a great many areas; we will list a few places topological groups show up without going into any detail. Perhaps the most studied topological groups are *locally compact topological groups* (topological groups where for any point $x \in G$, there exists a compact neighborhood). These are very useful in analysis: one can always define a measure on them (called the *Haar measure*), and the representation of topological groups leads to the *Fourier transform*, which is the decomposition into irreducible representations of the continuous group action of $\mathbb{R}$ on the Hilbert space $L^2(\mathbb{R})$. (Something with homotopy theory. Don't enough yet to understand). Finally, topological groups show up in *Hilbert's fifth problem*. First, it was proven that if $f : G \to H$ is a continuous homomorphism between lie groups, then f is in fact *smooth*(!). Thus, if G is a topological group, if there can exist a smooth structure on it, making it a lie group, then that structure must be unique. Hilbert's fifth problem asks if any topological group is also a topological manifold (i.e. locally homeomorphic to $\mathbb{R}^n$), then is it a lie group (i.e. the operation is smooth, and the group is locally diffeomorphic to $\mathbb{R}^n$). The answer was shown in 1955 to in fact be yes(!) by Andrew Gleason, Deane Montgomery, and Leo Zippin. Even better, G will always be *real analytic*.

C.5 Completion in Topology

The last subject of topological groups we will explore here is the *completion* of topological groups, since it generalizes some of the results of *Commutative Algebra* [10] (completion). This definition of Cauchy generalizes that of the definition of a Cauchy sequence in a metric space by realizing that we only need to have a notion of "continuous subtraction" to define a Cauchy sequence:

Definition C.5.1: Cauchy Sequence In Topological Group

Let G be a topological group and $(x_n) \subseteq G$ be a sequence. Then (x_n) is a *Cauchy sequence* if for every neighborhood U around 0, there exists a natural number $s(U) \in \mathbb{N}$ such that

$$x_n - x_m \in U \qquad \forall n, m > S(U)$$

Just like for metric spaces, the collection of all equivalent Cauchy sequence is the completion of the group G, which we'll denote as $\hat{G}$ (notice that if x_n, y_n are Cauchy sequence, so is $(x_n - y_n)$, and so $\hat{G}$ depends only on the equivalence classes). For any $g \in G$, we can define the constant sequence (g), which is an element of $\hat{G}$. We thus identify G in $\hat{G}$ by the map $\iota : G \to \hat{G}$ via $g \mapsto [(g)]$. Note that in general that ι is *not* in general injective. In fact, we have that:

$$\ker \iota = \bigcap_{\substack{0 \in U \\ U \text{ open}}} U$$

and so ι is injective if and only if G is Hausdorff. Completion is functorial: Since continuous homomorphisms preserve Cauchy sequences, $f : G \to H$ maps to $\hat{f} : \hat{G} \to \hat{H}$, and $\widehat{g \circ f} = \hat{g} \circ \hat{f}$.

This construction is quite general: we can allow for $G = \mathbb{Q}$ with the euclidean topology, then $\hat{G} = \mathbb{R}$. We can also allow for $G = \mathbb{Q}$, and define a different metric (known as the p-metric), which will give us $\hat{G} = \mathbb{Q}_p$. We will focus on a particular case that useful for commutative algebra and show how it relates to completion.

Define a topology on a commutative group G by taking a descending sequence of subgroups[3]:

$$G = G_0 \supseteq G_1 \supseteq \cdots \supseteq G_n \supseteq \cdots$$

This is called a *filtration* of the group G. Let the collection of (G_n) define the *local neighborhood* of 0, and $x + G_n$ be an open neighborhood of x. Then this generates a topology, which we'll call the *subgroup topology*: $U \subseteq G$ is a neighborhood of 0 if and only if U contains some G_n. An example of such a topology would be:

$$\mathbb{Z} \supseteq p\mathbb{Z} \supseteq p^2\mathbb{Z} \supseteq \cdots \supseteq p^n\mathbb{Z} \supseteq \cdots$$

The subgroup topology is interesting in that every subgroup G_n is open and closed (that is, clopen and hence connected). First, for any $g \in G_n$, $g + G_n \subseteq G_n$, and so G_n is open. Thus, for any $h \notin G_n$, $h + G_n$ is open (since translation is a homeomorphism), and so:

$$H = \bigcup_{h \notin G_n} h + G_n$$

is open and disjoint from G_n. In fact, $H^c = G_n$, showing that G_n is *closed.*

[3] If G were not commutative, we would take a descending sequence of normal subgroups

Now, since G with the subgroup topology is a topological group, we can define $\hat{G}$ in the exact same way. Since the open sets are derived from a descending chain of subgroups G, we see can observe that cauchy sequences will be induce a special type of sequence called a *coherent sequence* which shall be equivalent to Cauchy sequence in the subgroup topology. Let (x_n) be a Cauchy sequence in G with the subgroup topology generated by (G_n). Then in G/G_n, the image of(x_n) must be eventually constant; say it is ζ_n. When considering G_{n+1}, we may get different constant, say ζ_{n+1}. When considering the relation between these two, it is natural to see that the natural projection map ζ_{n+1} to ζ_n, that is the map:

$$\pi_{n+1} : G/G_{n+1} \to G_n$$

Now, for all n, we will get some ζ_n. The collection of these ζ_n's is called a *coherent sequence*. We chose the maps to be projection maps, however we may imagine a similar such sequence with any such functions, as long as they map to the right elements. Thus, being a little more general, a coherence sequence is a collection (ζ_n) such that for a collection of functions f_n, we have:

$$f_{n+1}(\zeta_{n+1}) = \zeta_n \qquad \forall n \in \mathbb{N}$$

It is left as an exercise that equivalent Cauchy sequences define the same coherent sequence, and that a coherent sequence defines a Cauchy sequence, and hence these two concepts capture the same idea. For this reason, the collection of all coherent sequences is also the completion of G, and is also denoted $\hat{G}$.

This now brings us to the same definition as given in section 25 in the case of Abelian groups: an *inverse system* is a collection of subgroups $\{G_n\}$ and functions f_n such that

$$f_{n+1} : G_{n+1} \to G_n$$

Then the group of all coherent sequences, $\hat{G}$, is called the *limit*, and is denoted $\varprojlim G_n$, where the homomorphism f_n are usually implicitly understood. With this notation, we have that:

$$\hat{G} \cong \varprojlim G/G_n$$

The most common subgroup topology and inverse system is when $G = R$ and $G_n = I^n$ is an ideal. In this case, the subgroup topology of R is called the *I-adic topology*, or simply the *I-topology*. Recall that we have defined a norm on a vector-space to define a topology. We can similarly define a norm on any topological group defined via the I-adic topology which will generate the I-adic topology: Define

$$\|x\| = C^{-n} \qquad \text{if} \qquad x \in I^n,\ x \notin I^{n+1}$$

for some $C \in \mathbb{N}$. If $x \in I^n$ for all $n \in \mathbb{N}$, then we will say that $\|x\| = 0$. Then through this norm, we can define a metric:

$$d(x, y) = \|x - y\|$$

and so we can define a completion, $\hat{G}$ as defined in definition C.1.18, which the reader should verify is the same as $\varprojlim G/G_n$. This new metric we have defined is also more powerful than the usual euclidean metric, namely:

$$\|x + y\| \leq \max(\|x\|, \|y\|) \qquad d(x, y) \leq \max(d(x, z), d(z, y))$$

This metric condition is *much* stronger and is called an *ultra-meric*. The reader is invited to prove the following incredible properties of topologies with ultra-metrics:

1. Given a collection of any three points $x, y, z \in X$ one of the following three equalities must hold:
$$d(x,y) = d(y,z) \qquad d(x,z) = d(y,z) \qquad d(x,y) = d(z,x)$$
Hence, every triple *must* form an isosceles triangle.

2. If $B_r(x)$ is an open ball then every point inside the ball is it's center: if $d(x,y) < r$:
$$B_r(x) = B_r(y)$$
Every intersection must contain each other:
$$B_r(x) \cap B_s(y) \Rightarrow B_r(x) \subseteq B_s(y) \qquad \text{or} \qquad B_y(s) \subseteq B_r(x)$$
If $r > 0$, then $B_r(x)$ is clopen.

What is notable is that completion is the beginning of real analysis (and hence also PDEs), and given the strong algebraic structure introduced we also have variants such completions coming up in complex analysis. In [39], we shall use different completion of $\mathbb{Z}$ with p-adic metrics to construct fields in which we are "keeping track' of a single prime that is splitting or ramifying.

Theorem C.5.2: Universal Property Of Completion

Here

***Proof*:**
here

C.5.1 Classifying Complete Archimidean Fields

(Should I show that $\mathbb{R}$ and $\mathbb{C}$ are the only complete archimidean fields upto isomorphism?

Exercise C.5.1

1. (Relate this to DVR topologies?)

D

Lattice Theory and Universal Algebras

(here: Two important classifying objects but not really central to any current big field of study. Lattice theory going to be presented as build-up to Stone Duality, while Universal algebra presented as a start to Model Theory)

D.1 Lattice Theory

We first generalize the notion of gcd and lcm for partial orders:

Definition D.1.1: GCD and LCM for Partially ordered sets

Let $(S, \leq)$ be a partially ordered set. Then

1. a *lowerbound* of a subset $T \subseteq S$ is an element $l \in S$ such that $l \leq t$ for all $s \in T$. A *greatest lower bound* (gcd) is a lowerbound g such that for any other lower-bound l, $l \leq g$. In lattice theory, the greatest lower bound is also called the *meet* or *infimum*
2. an *upperbound* of a subset $T \subseteq S$ is an element $u \in S$ such that $t \leq u$ for all $s \in T$. A *least upper bound* (gcd) is a upper-bound ℓ such that for any other upper-bounds u, $\ell \leq u$. In lattice theory, the greatest lower bound is also called the *join* or *supremum*

The reader should check that if a meet/join exists, then it is unique. If the meet/join of a subset exists for a subset $T \subseteq S$, then it is denoted:

$$\bigwedge_{t \in T} t \qquad \text{and} \qquad \bigvee_{t \in T} t$$

Definition D.1.2: Lattice

Let $(S, \leq)$ be a partially ordered set. Then

1. If for any two $a, b \in S$, $a \wedge b$ exists, then S is called a *lower semi-lattice*
2. If for any two $a, b \in S$, $a \vee b$ exists, then S is called a *upper semi-lattice*
3. If S is both a lower and upper semi-lattice, S is called a *lattice*.

Proposition D.1.3: Properties Of Lattice Binary Operation

Let S be a semi-lattice. Then $\vee$ and $\wedge$ as binary operations are

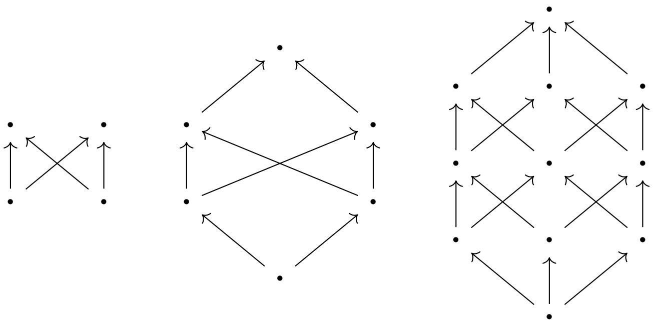
idempotent, commutative, associative, and order preserving.

Furthermore, by using induction, every finite subset has a meet and join

***Proof*:**
Exercise. Note that it is enough to prove it for $\wedge$ as $(S, \leq)^{\text{op}} = (S, \leq^{op})$ turns any meet into a join.

Example D.1.4: Lattices

1. $\mathbb{N}, \mathbb{Z}, \mathbb{Q}, \mathbb{R}$ and real fields (see section 21.1) with the usual $\leq$ all form lattices; what is $a \wedge b$ and $a \vee b$ for any a, b in these sets?
2. Let S be a set with powerset $P(S)$. Then with $\cup$ and $\cap$, $P(S)$ is a lattice
3. Let S be the collection of all (subgroups/ideals/submodules) of (group/ring/module). Then S forms a lattice with $A \cap B \in S$ and $\langle A \cup B \rangle \in S$. In the ideal/submodule case, $\langle A \cup B \rangle = A + B$.
4. Any partially ordered set can be represented by a graph where an edge represents where $x \leq y$ if $x \to y$. Then the following are examples of non-lattices:



5. Let X be a topological space. Then the set of open sets forms a lattice, for the union of two open/closed sets is an open/closed set.

Unlike most algebraic structures encountered with two binary operations, lattices are *not* distributive in general (can the reader find a counter-example?). They have the following alternative:

Proposition D.1.5: Absorption Law

Let L be a lattice. Then:

$$x \wedge (x \vee y) = x \qquad x \vee (x \wedge y) = x$$

***Proof*:**
exercise

Using the above result, we can formulate lattices purely algebraically:

Proposition D.1.6: Algebraic Definition Of A Lattice

Let L be a set with binary operations $\vee$ and $\wedge$ that are associative and commutative. Then if these satisfy the absorption laws, then L forms a lattice

***Proof*:**
exercise

Naturally, a lattice homomorphism if a order-homomorphism such that $\varphi(x \vee y) = \varphi(x) \vee \varphi(y)$ and $\varphi(x \wedge v) = \varphi(x) \wedge \varphi(y)$. Note that not all order-homomorphisms are lattice homomorphisms (as the reader should come up with a counter-example). However an order-isomorphism is a lattice-isomorphism (see exercise ref:HERE).

Lattices as they are can be considered as two commutative *semigroups* with a compatible structure (see section 7); the absorption being the unique feature of lattices. If furthermore the lattice is bounded, then it shall have a monoid structure.

Definition D.1.7: Bounded Lattice

Let L be a lattice. Then L is said to be *bounded* if there exists an minimal element (denoted 0 or $\bot$) and a maximal element (denoted 1 or $\top$) such that for all $x \in L$:

$$\bot \leq x \leq \top \qquad \text{equiv.} \qquad 0 \leq x \leq 1$$

As an algebraic structures, we can think of a bounded lattice as a set $(L, \vee, \wedge, 0, 1)$ such that $(L, \vee, \wedge)$ is a lattice, 0 is the identity element for the join operation $\vee$, and 1 is the identity element for the meet operation $\wedge$:

$$a \vee 0 = a \qquad a \wedge 1 = a$$

If we require the stronger condition that every subset $S \subseteq L$ of a lattice has a supremum and infimum, we get the following stronger notion:

Definition D.1.8: Complete Lattice

Let L be a lattice. Then if every subset $S \subseteq L$ has a gcd and lcm, then L is said to be a *complete lattice*

This makes L a bounded lattice, for $L \subseteq L$ must have a gcd and lcm. Every finite set can be made into a complete lattice, however $\mathbb{N}, \mathbb{Q}, \mathbb{R}$ are incomplete. Every closed set of $\mathbb{R}$ is complete. More notably, $P(X)$ for any set X, as well as the collection of (subgroups/ideals/submodules) of (groups/rings/modules) forms a complete lattice.

Given any lattice, it can be embedded in a complete lattice. Rather interestingly, this shall link to Galois connections and the closure map:

Definition D.1.9: Closure Map

Let $(S, \leq)$ be a partially ordered set. Then the *closure map* is an order-homomorphism $\Gamma S \to S$ that is idempotent, $(\Gamma \circ \Gamma) = \Gamma$, and expanding, $((\Gamma(x) \geq x)$, for all $x \in X)$. An element $x \in S$ is *closed relative to* Γ if $\Gamma(x) = x$.

The closure map is so named since the closure map $A \mapsto \overline{A}$ is the quintessential closure map. As we shall see when discussing pointless topology, lattices and topologies shall be intimately related, elucidating better this example.

Lemma D.1.10: Closed Element Infimum and Supremum

Let L be a complete lattice and Γ a closure map. Then the set of closed elements relative to Γ is closed under infimums and supremums.

***Proof*:**
exercise

Next, we define the Galois connection more generally:

Definition D.1.11: Galois Connection

Let X, Y be two partially ordered sets. Then a *galois connection* between X, Y is a pair of order-reversing homomorphism, say $\alpha : X \to Y$ and $\beta : Y \to X$ such that for all $x \in X$ and $y \in Y$:

$$(\beta \circ \alpha)(x) \geq x \qquad (\alpha \circ \beta)(y) \geq y$$

Naturally, the Galois correspondence is a Galois connection, and the generalization to closed groups shows the correspondence in the infinite case.

Lemma D.1.12: Galois Connection And Lattices

Let X, Y be partially ordered sets and (α, β) a Galois connection. Then

1. α, β, induce order-reversing mutually inverse bijections between $\mathrm{im}(\alpha)$ and $\mathrm{im}(\beta)$
2. $\alpha \circ \beta$ and $\beta \circ \alpha$ are closure maps
3. If X, Y are complete lattices, $\mathrm{im}(\alpha)$ and $\mathrm{im}(\beta)$ are complete lattices

***Proof*:**
exercise

Theorem D.1.13: Lattice Embedding Theorem

Let $(L, \leq)$ be a partially ordered set. Then L can be embedded in a complete lattice $\hat{L}$ in such a way that all infimums and supremums of L are preserved.

***Proof*:**
Grillet (p.544, rather standard argument)

Examples of Lattices

Lattices are rather general objects. As it turns out, when working the set of sub-objects of an R-module with $(+, \cap)$, we get some more interesting behavior:

Definition D.1.14: Modular Lattice

Let L be a lattice. Then L is called *modular* if whenever $x \leq z$,

$$x \vee (y \wedge z) = (x \vee y) \wedge z$$

Note that for any lattice, if $x \leq z$ then x and $y \wedge z$ are lower bounds of $\{x \vee y, z\}$ and hence:

$$x \vee (y \wedge z) \leq (x \vee y) \wedge z$$

Thus, a lattice is modular if and only if $x \leq z$ implies $x \vee (y \wedge z) \geq (x \vee y) \wedge z$. Note that the opposite of a modular lattice is modular, hence the theory is self-dual. As mentioned, the name modular lattice comes straight form the lattice of submodules:

Example D.1.15: Modular Lattices
Letting $(\vee, \wedge) = (\cap, +)$, and taking the lattice of submodules of an R-module. Then if $A \subseteq C$, then taking $x \in (A + B) \cap C$ we have $x = a + b \in C$ for some appropriate $a \in A \subseteq C$. Then $b = x - a \in C$, and $x = a + b \in A + (B \cap C)$

Detecting when a lattice fails to be modular is similar to how to detect a complete bipartite graph:

Proposition D.1.16: Detecting Modular Lattice

Let L be a lattice. Then L is modular if and only if it does not contain a sub-lattice isomorphic to N_5:

$$\begin{tikzcd}
 & 1 & \\
 & & c \arrow[ul] \\
a \arrow[uur] & & \\
 & & b \arrow[uu] \\
 & 0 \arrow[uul] \arrow[ur] &
\end{tikzcd}$$

***Proof*:**
Certainly this lattice is not modular (as the reader should check). For the converse, use the fact the lattice is not modular to find elements that do not satisfy the modularity property and construct N_5

Modular lattices help generalizes results such as theorem 3.7.10(2) (i.e. that all maximal normal series have unique maximal length).

Proposition D.1.17: Modular Lattices And Finite Maximal Chains

Let L be a modular lattice. Then any two finite maximal chains have the same length

***Proof*:**
exercise (or p.547 of Grillet)

A little more restrictive than modular lattices are *distributive* lattices:

Definition D.1.18: Distributive Lattice

Let $(L, \vee, \wedge)$ be a lattice. Then L is a *distributive lattice* if one of the two equivalent conditions hold for all $x, y, z \in L$:

1. $x \vee (y \wedge z) = (x \vee y) \wedge (y \wedge z)$
2. $x \wedge (y \vee z) = (x \vee y) \wedge (y \vee z)$

Proposition D.1.19: Detecting Distributive Lattices

Let L be a lattice. Then L is a distributive lattice if and only if L does not have a sublattice isomorphic to N_5 or M_5, where M_5 is the lattice:

$$\begin{tikzcd} & 1 & \\ a \arrow[ur] & b \arrow[u] & c \arrow[ul] \\ & 0 \arrow[ul] \arrow[u] \arrow[ur] & \end{tikzcd}$$

***Proof*:**
exercise

The power-set $P(X)$ is the canonical distributive lattice with $(\cup, \cap)$. In fact a lattice is distributive if and only if it is isomorphic to a sublattice of $P(X)$ for some appropriate X. To see this, we require the following build-up

Definition D.1.20: Lattice Irreducible Elements

Let L be a lattice. Then an element $i \in L$ is called *sup-irreducible* (or just *irreducible*) if i is not a minimal element and:

$$x \vee y = i \quad \Rightarrow \quad x = i \text{ or } y = i$$

The collection of irreducible elements is denoted $\mathrm{Irr}(L)$.

In $P(X)$, singletons are irreducible.

Proposition D.1.21: Lattice And Irreducible

Let L be a lattice satisfying the descending chain condition. Then every elemetn $x \in L$ is the join of the set of irreducible elements of L

***Proof*:**
exercise (see Grillet p.551)

Definition D.1.22: Order Ideal

Let S be a partially ordered set. Then a [left] order ideal $I \subseteq S$ is a subset such that if $y \in I$ and:

$$x \leq y \in I \quad \Rightarrow \quad x \in I$$

Order ideals naturally create a distributive lattice given any partially ordered set:

Proposition D.1.23: Order Ideals And Partially Ordered Sets

Let S be a partially ordered set. Then the order ideals, ordered by inclusion, produce a distributive lattice, denoted $\text{Ideals}(S)$

***Proof*:**
exercise.

Theorem D.1.24: Finite Distributive Lattices And Ideals

Let L be a finite lattice. Then L is a distributive lattice if and only if

$$L \cong \text{Ideals}(\text{Irr}(L))$$

***Proof*:**
exercise

hello world!

This result can be generalized to all ideals, see Grillet for further explanation.

A final important lattice naturally arises in mathematical logic, known as a *boolean lattice*. In a Boolean lattice, the meet, join, and "complement" correspond to conjunction, disjunction, and negation in mathematical logic:

Definition D.1.25: Boolean Lattice

Let L be a bounded lattice (hence containing a 0 and 1). Then a *complement* of an element $a \in L$ is an element a' such that $a \wedge a' = 0$ and $a \vee a' = 1$. Then a lattice is a *Boolean lattice* if every element has as compliment

The reader should verify that compliments are unique, and that the operation of taking the compliment, $(-)'$ is a lattice-homomorphism.

Example D.1.26: Boolean Lattice

For an easy example of a Boolean lattice take the power-set $P(X)$ with compliment $Y' = X \setminus Y$. For a more interesting example, let R be a Boolean ring (i.e. a ring such that for $x \in R$, $x^2 = x$). Then define a partial order on R via:

$$x \leq y \qquad \text{if and only if} \qquad xy = x$$

Then we get a boolean lattice via

$$x \wedge y = xy \qquad x \vee y = x + y + xy \qquad x' = 1 - x$$

Interestingly, the all Boolean lattices can be converted into Boolean rings, for a Boolean lattice $(L, \vee, \wedge, (-)', 0, 1)$ define:

$$x = y = (x' \wedge y) \vee (x \wedge y') \qquad xy = x \wedge y$$

denote this ring $R(L)$. Then

$$\mathrm{L}(\mathrm{R}(L)) = L \qquad \mathrm{R}(\mathrm{L}(R)) = R$$

Since power-sets are naturally boolean lattices, they offer themselves to the following classification theorem:

Theorem D.1.27: Finite Boolean Lattice Classification

Let L be a finite lattice. Then L is a boolean lattice if and only if there exists a set X such that

$$L \cong P(X)$$

***Proof*:**
build off of the same result for distributive lattices.

To generalize to general boolean lattices, we require the following lemmas:

Lemma D.1.28: Stone Space Basis

Let L be a Boolean lattice and let X be the set of all prime ideals. Then the set

$$V(a) = \{P \in X \mid a \notin P\}$$

is a basis for the topology on X. Furthermore, $V(0) = \emptyset$, $V(1) = X$, and $V(a') = V(a)'$

This topology shall be given a name soon

***Proof*:**
exercise

Theorem D.1.29: Boolean Lattice Classification

Let L be a lattice. Then L is a Boolean lattice if and only if there exists a set X such that L is isomorphic to a subset of $P(X)$.

***Proof*:**
build off of the same result for distributive lattices.

As mentioned, the above topology has a special name:

Definition D.1.30: Stone Space

Let L be a Boolean lattice and X the collection of prime ideals of L. Then with the above basis, X is called the *stone space*.

Proposition D.1.31: Stone Space Properties

Let X be a stone space. Then X is compact, Hausdorff, and totally disconnected

***Proof*:**
For compactness, it is recommended the reader proves it by showing every ultra-filer converges.

Given a space with these properties, note that every finite union/intersection of closed/open sets is closed/open, and similarly for compliment, hence it can be shown that it forms a Boolean lattice $\mathrm{L}(X)$, and so a stone space can be defined with the above properties.

D.2 Pointless Topology

Using some lattice theory, we may find the closest result to an "algebraic dual" to topological spaces (mirroring the Stone duality and the duality presented in section 32.5). The main idea is that we can focus

(The overall goal is to get the adjoint pair (Ω, pt) between **Top** and Loc, and that it restricts to *Stone Duality*)

Heyting Algebras

(they deserve a mention due to their importance in logic)

D.3 Universal Algebras

(generalizes many theorem that have the same pattern, including isomorphism theorems, free constructions, and that there will exist sub-algebras, products, and quotient algebras).

E

Model Theory

1. I want to build up the basics of Model theory for it's interesting use in Algebraic Geometry. In particualr, I want to write here the build up and proof of the Ax-Grothendeick theorem.
2. I would like to have the Lefschetz Principle and Löwenheim-Skolem theorem (the Löwenheim-Skolem theorem is doubly interesting since it can be used to formally show there is no "algebraic"' construction of $\mathbb{R}$ from $\mathbb{Q}$)
3. perhaps as a mini bonus or as exercises, I can include some other infinite algebraic contructions that can be achieved using Model Theory (ex infinite torsion group).

E.1 Löwenheim-Skolem Theorem

Part XIV

Glossary, Index, Bibliography

Glossary of Algebraic Structures

In this chapter, we list many of the algebraic structures discussed throughout the book. These are categorized by subject and are roughly grouped by the order in which they appear with priority given to grouping properties together and examples together. To find where these objects appear in the textbook, see the appendix (which is sorted in alphabetical order).

Fundamental Structures

- Set; Relation, Equivalence Relation, Partial Order, Total Order
- Function (Injective, Surjective, Bijective); Cardinality
- Number Systems ($\mathbb{N}$, $\mathbb{Z}$, $\mathbb{Q}$, $\mathbb{R}$, $\mathbb{C}$); Modular Arithmetic ($\mathbb{Z}/n\mathbb{Z}$)
- Graph, Lattice; Matrix

Groups

Basic Group Types and Properties

- Group; Abelian Group; Subgroup; Normal Subgroup; Characteristic Subgroup
- Quotient Group; Homomorphism, Isomorphism, Endomorphism, Automorphism (including Inner Automorphism, Outer Automorphism)
- Kernel, Image; Center, Centralizer, Normalizer
- Order of a Group, Order of an Element; Generators and Relations, Presentations of Groups

Examples of Groups

- Cyclic Group (C_n, $\mathbb{Z}/n\mathbb{Z}$); Symmetric Group (S_n); Alternating Group (A_n)
- Dihedral Group (D_n); Quaternion Group (Q_8)

- Matrix Groups (e.g., $GL_n(F)$, $SL_n(F)$, $O_n(F)$, $SO_n(F)$)
- Free Group; Free Abelian Group

Group Actions

- Group Action; Orbit, Stabilizer; Class Equation; Cayley Graph

Constructions and Decompositions

- Direct Product of Groups; Semi-Direct Product of Groups; Internal Direct Product
- Extensions of Groups; Fundamental Theorem of Finitely Generated Abelian Groups

Specialized Group Types

- Simple Group; Solvable Group; Nilpotent Group; Polycyclic Group; Perfect Group

Generalizations of Groups

- Monoid; Groupoid

Rings

$$\text{Rngs} \supset \text{Rings (Unital)} \supset \text{Commutative Rings} \supset \text{Integral Domains}$$
$$\supset \text{Factorization Domains} \supset \text{Unique Factorization Domains (UFDs)}$$
$$\supset \text{Principal Ideal Domains (PIDs)} \supset \text{Euclidean Domains}$$
$$\supset \text{Fields} \supset \text{Algebraically Closed Fields}$$

Basic Types

- Division Rings (Skew Fields); Noetherian Rings; Artinian Rings
- Dedekind Domains; Valuation Rings, Discrete Valuation Rings (DVRs)

Examples of Rings

- Polynomial Rings ($R[x]$, $R[x_1, \ldots, x_n]$); Formal Power Series ($R[[x]]$)
- Matrix Rings ($M_n(R)$); Group Rings ($R[G]$)
- Ring of Fractions / Localization ($D^{-1}R$, R_S, R_f, R_P); Completion of a Ring ($\hat{R}_I$)

Ring Concepts

- Ring Homomorphisms, Isomorphisms
- Ideals (including Principal Ideal; Prime Ideal; Maximal Ideal; Radical Ideal, Nilradical; Comaximal Ideals; Homogeneous Ideals; Fractional Ideals)
- Quotient Rings; Units, Zero Divisors, Nilpotent Elements
- Characteristic of a Ring; Integral Extensions, Integrally Closed Domains (Normal Domains)

Modules

Basic Module Types and Properties

- R-Module (Left, Right, Bimodule); Submodule; Quotient Module
- Module Homomorphism (Linear Transformation), Isomorphism
- Simple Module (Irreducible Module); Semisimple Module; Indecomposable Module
- Free Module, Rank of a Free Module; Projective Module; Injective Module; Flat Module
- Finitely Generated Module; Noetherian Module, Artinian Module
- Cyclic Module; Torsion Elements, Torsion Module, Torsion-Free Module; Annihilator of a Module

Constructions

- Direct Sum / Direct Product of Modules; Tensor Product of Modules ($M \otimes_R N$)
- Exact Sequences (Short, Long, Split)

Special Modules

- Vector Spaces (over a field k) (including Subspace, Quotient Space; Basis, Dimension; Dual Space (V^*), Dual Basis, Double Dual (V^{**}); Linear Functionals, Bilinear/-Multilinear Forms; Inner Product Spaces, Normed Spaces)
- R-Algebras (as R-modules)

Fields

- Field Extension (K/F) (including Degree of an Extension ($[K : F]$); Finite Extension; Algebraic Element, Transcendental Element; Algebraic Extension, Transcendental Extension; Simple Extension ($F(\alpha)$), Primitive Element; Normal Extension; Separable Extension, Separable Degree ($[K : F]_s$); Galois Extension)
- Minimal Polynomial ($m_{\alpha,F}(x)$); Splitting Field; Algebraic Closure ($\bar{F}$)
- Galois Group ($\mathrm{Gal}(K/F)$, $\mathrm{Aut}_F(K)$) (including Fixed Field; Galois Correspondence)
- Roots of Unity, Cyclotomic Polynomials ($\Phi_n(x)$), Cyclotomic Fields ($\mathbb{Q}(\zeta_n)$)
- Finite Fields / Galois Fields ($\mathbb{F}_q$, $\mathbb{F}_{p^n}$); Ordered Fields, Real Fields
- Absolute Values on Fields, p-adic Valuation; Trace ($\mathrm{Tr}_{K/F}(\alpha)$), Norm ($N_{K/F}(\alpha)$), Discriminant

Algebras

- Associative Algebra (R-Algebra, k-Algebra); Commutative Algebra; Division Algebra
- Lie Algebra; Path Algebra (of a Quiver)
- Tensor Algebra ($T(M)$); Symmetric Algebra ($S(M)$); Exterior Algebra ($\Lambda(M)$)
- Graded Algebra; Central Simple Algebra (CSA); Brauer Group ($\mathrm{Br}(k)$)

Structures from Commutative Algebra & Algebraic Geometry

- Noetherian Ring, Artinian Ring; Dedekind Domain; Valuation Ring, Discrete Valuation Ring (DVR)
- Localization (R_P, R_f, $S^{-1}R$); Completion ($\hat{R}_I$)
- Integral Extension, Integrally Closed Domain (Normal Domain)
- Algebraic Set (Affine, Projective); Variety (Affine, Projective, Quasi-Projective)
- Coordinate Ring ($k[V]$, $k^P[V]$); Zariski Topology
- Regular Function/Map, Rational Function/Map; Dimension (Krull, Transcendental)
- Hilbert Function, Hilbert Polynomial; Stalk ($\mathcal{O}_{X,p}$)
- Spectrum of a Ring ($\mathrm{Spec}(R)$), Affine Scheme

Number Theoretic Structures

- Algebraic Number Field, Ring of Integers ($\mathcal{O}_K$); Algebraic Number, Algebraic Integer
- Discriminant (of a field, of a basis, Δ_K)
- Norm (of an element $N_{K/F}(\alpha)$, of an ideal $N(\mathfrak{a})$); Trace (of an element $\mathrm{Tr}_{K/F}(\alpha)$)
- Ideal Class Group ($\mathrm{Cl}(K)$), Class Number (h_K)
- Roots of Unity, Cyclotomic Polynomials ($\Phi_n(x)$), Cyclotomic Fields ($\mathbb{Q}(\zeta_n)$)
- Fractional Ideals; Ramification Index (e), Inertia Degree (f) (including Decomposition Group (D_P), Inertia Group (I_P); Frobenius Element/Automorphism (Frob_P))

Representation Theory Structures

- Representation (Linear, Matrix) (including Irreducible Representation, Simple Module; Semisimple Representation, Semisimple Module; Indecomposable Representation; Trivial Representation; Regular Representation; Permutation Representation; Induced Representation)
- G-Module ($k[G]$-module); Character, Character Table; Class Function
- Semisimple Ring (Artin-Wedderburn Theorem); Quiver, Quiver Representation (including Path Algebra)

Homological Algebra Structures

- Chain Complex, Cochain Complex (including Cycles, Boundaries (Cocycles, Coboundaries))
- Homology ($H_n(C_\bullet)$), Cohomology ($H^n(C^\bullet)$)
- Exact Sequence (Long Exact Sequence in

(Co)homology) (including Snake Lemma)

- Resolution (Projective, Injective, Free, Flat)
- Derived Functor (L_iF, R^iF) (including Ext Functors ($\mathrm{Ext}^i_R(M,N)$); Tor Functors ($\mathrm{Tor}^R_i(M,N)$))
- Homotopy (Chain Homotopy, Homotopy Equivalence); Mapping Cone
- Derived Category ($D(A)$), Homotopy Category ($K(A)$), Triangulated Category; Spectral Sequence

Category Theory Concepts

- Category (Small, Locally Small); Object
- Morphism (Homomorphism, Isomorphism, Monomorphism, Epimorphism, Endomorphism, Automorphism)
- Functor (Covariant, Contravariant) (including Additive Functor, Exact Functor; Faithful Functor, Full Functor, Essentially Surjective Functor)
- Natural Transformation, Natural Isomorphism; Equivalence of Categories
- Duality, Opposite Category (C^{op}); Commutative Diagram

Universal Properties

- Universal Property (general concept); Initial Object, Final Object (Zero Object)
- Product, Coproduct (Direct Product, Direct Sum, Free Product)
- Kernel, Cokernel (in Abelian Categories); Equalizer, Coequalizer
- Limit, Colimit (Inverse/Projective Limit, Direct/Injective Limit)
- Free Objects (Free Group, Free Module, Free R-algebra, Polynomial Ring as free commutative R-algebra)
- Adjoint Functors (e.g., Free-Forgetful Adjoints, Hom-Tensor Adjoint)

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What Next?

Congratulations on reaching the end of the book[1]! Having parsed through the material of the book, the reader must be on the edge of their seat wondering what comes next! From my current understanding, the ultimate goal is to understand better the properties of numbers or the properties of geometric objects that arise out of polynomials (or more generally out of commutative or non-commutative rings). This results in classification results, tools to better understand these results, new ways of solving equations, and so forth. The number of branching points is vast; this book shall address what is currently some common general starting points, given the reader is in the 21st century.

The reader should be aware that at this stage, it is important to gain some experience in many other fields of mathematics such as algebraic topology [38], differential geometry [41], Lie Groups,real analysis [45](particularly Fourier analysis), PDEs [18] (for at least the idea of elliptic operators and harmonic PDEs), and complex analysis [40]. All these fields interconnect to various extents in the study of number theory, representation theory, and algebraic geometry.

It must be pointed out that the topics covered here are *not* a comprehensive list of areas of research, simply large ones. Other areas that are not covered in detail here that have active research include, but are not limited to, the model theory of algebraically and differentially closed fields, probabilistic linear optimization, mathematics of AI (which can be thought of as a continuation of linear algebra and functional analysis), higher order category theory, P vs. NP as an Invariance Theory problem, and so forth.

Algebraic Geometry

A lot of focus is within the domain of *algebraic geometry*. The reader has the background material to start reading the next EYTNKA textbook *EYNTKA Algebraic Geometry* ([5]), or read classical texts such as *Hartshorne's Algebraic geometry* ([31]) or *Vakil's The Rising Sea* ([52]). The starting point of modern algebraic geometry is the *affine scheme* (developed in *Algebraic Geometry* [5]) which generalizes varieties by defining the geometric counter-part to number rings, rings with nilpotents or

[1] or welcome curious reader to how the book ends!

zero-divisors, rings that are not finitely generated, and overall any commutative ring. These shall be the "model spaces" that are glued together to create *schemes*, where "glued together" is geometrically understood as forming equivalence classes between the points that are being glued and is algebraically understood as adding compatibility relations between functions (essentially, think of the example of projective space where $f \in \text{Hom}(U_1, R), g \in \text{Hom}(U_2, R)$ such that $f(z) = g(1/z)$ on $U_1 \cap U_2$). A lot of the build-up in modern algebraic geometry is translating the concepts of differential geometry over $\mathbb{R}$ or $\mathbb{C}$ (ex. smoothness, dimensions, irreducibility, compactness, Hausdorffness, Poincaré duality, monodromy, Riemann-Roch) into the non-analytic setting of schemes. This translation abstracts many of the notions that will require a good deal of build-up, but the reward is a unified language to deal with number theory and varieties which have many of the geometric intuitions one would expect, as well as novel geometric intuitions unique to schemes with not analytic counter-part. A lot of similarities shall be shared between differential geometry and scheme theory, but important differences shall also emerge (ex. points containing other points, existence of distinct points that are arbitrarily closed to each other, functions that are not determined by their value at every point).

Scheme are the starting point for the study of many different modern fields, but it can be studied in it of itself. For example to find classification of different types of schemes; this research often involves the use of *moduli spaces* and the theory of *stacks*.

Another very important problem in Algebraic Geometry is the *Hodge Conjecture* which roughly says says that homological information about compact complex manifolds X (which from an algebraic view point, are objects that very closed resemble smooth projective varieties and often are categorically equivalent due to GAGA [30]) can be given by nice subvarieties of X. In particular, without defining any of our terms, it asks which cohomology classes of $H^{k,k}(X)$ can be represented by closed subvarieties Z where X is given the natural Kähler manifold structure. Solving the conjecture or finding a counter-example would be rewarded the reader \$1 million by the Clay Institute. The reader interested in the Hodge Conjecture wanting to continue the EYTNKA series can read *K-theory* for an introduction to vector/fiber-bundles and their algebraic generalizations [43], *Algebraic Topology* for the cohomology background [38], *Algebraic Geometry* for the algebraic background [5], *Differential Topology* for the Kähler manifold background [41], and (eventually cite hodge theory notes) for putting this all together.

Elliptic Curves

The advantage of scheme theory is in full effect when studying curves (Noetherian integral separated 1-dimensional schemes of finite type over k). The study of curves splits to the study of curves of genus g by the Riemann-Roch Theorem (see [40] or [5]). Curves of genus 1 are called *elliptic curves* which by GAGA results ([30]) is equivalent to studying non-singular degree 3 polynomial equations. When in char $\neq 2, 3$ we can algebraically say that the study of elliptic curves is the study of equation $y^2 = x^3 + ax + b$. Finding all rational solutions to such polynomials is still an open problem; the *Birch and Swinnerton-Dyer Conjecture* would, if proven, explain how many solutions there exists, and the person(s) who succeeds to prove it would win \$1 million from the Clay Institute of Mathematics[2]. Due to scheme theory there are natural ways of understanding "bad reductions" (think of ramified primes), of defining a covering space using galois theory (like in homotopy theory), and keeping track of when curves intersect non-transversally (with multiplicity higher than 1).

The reader interested can start on the EYTNKA textbook *Elliptic Curves* ([12]) or look at *Silvermen's Arithmetic of Elliptic Curves* ([48]).

[2]It is one of the Millennium problems set at the beginning of the 21st century

Number Theory and Arithmetic Geometry

Scheme theory over number fields or number rings is called *Arithmetic Geometry.* A huge amount of progress in modern theory was made by using tools in arithmetic geometry, for example the proof of Fermat Last Theorem (that there exists no natural number $n \geq 3$ such that $x^n + y^n = y^n$ has an integer solution) was proved through the lens of Arithmetic Geometry and analytic number theory. Much of modern number theory builds directly on the algebraic foundations presented in this book.

A major area of research is the *Langlands program* which aims to generalize Artin's reciprocity (see *Class Field Theory*, [39]) which can be thought of the 1-dimensional case of Langland's reciprocity. The 2-dimensional case strongly involved elliptic curves and the related modular forms and automorphic forms. An introduction to the required background covering these topics includes [12], [5], and [47].

With the advent of complex analysis, many problems in number theory can be studied using certain meromorphic functions. Of central interest is the *Riemann-Zeta function*

$$\zeta(s) = \sum_{n=1}^{\infty} \frac{1}{n^s} = \prod_{p \text{ prime}} \frac{1}{1 - p^{-s}}$$

which can be analytically extended to $\mathbb{C} \setminus \{1\}$ and captures an incredible amount of information about the distribution of primes (and by the proven *Weil Conjecture* any abelian variety over $\mathbb{F}_p$ has an associated Zeta function capturing its arithmetic properties). The *Riemann Hypothesis*, one of the Millennium problems, states that all the [non-trivial][3] roots of $\zeta(s)$ lie on the line $L = \{z \in \mathbb{C} \mid \text{im}(z) = 1/2\}$. By some application of Fourier analysis, this would drastically improve our understand of how many primes $p \leq n$ there exists for any $n \in \mathbb{N}$, and would resolve many open problems. Proving that all [nontrivial] zero's are on L or that there is at least one [nontrivial] zero that is not on L would give the person(s) who proved it $1 million by the Clay Institute.

Geometric Group Theory

part I mainly focused on finite groups. Infinite groups are much harder to grasp, however there are many infinite groups that naturally show up in geometric settings and have a metric that can be imposed on them. Studying these groups falls under *Geometric group theory* which views infinite groups themselves as geometric objects. By equipping a group with a metric, one can study its large-scale geometry to deduce its algebraic properties. This approach has led to the resolution of major conjectures in topology (most famously, Thurston's Geometrization Conjecture implying the *Poincaré Conjecture*, another Millennium problem).

Derived Algebraic Geometry and Higher Algebra

Schemes are incredibly power, but are not adept with "singular" spaces or when taking "non-regular" quotients; categorically they do not behave well under limits and colimits. *Derived geometry* is a redevelopment of commutative algebra inspired abstract topology. It replaces rings with derived objects that retain homological information, allowing for a more robust theory of intersections and moduli spaces that works in greater generality. This field often draws on *higher category theory* to create a unified language for geometry.

[3] There are roots that are given by the Gamma function which are considered to be related to the prime at infinity, and is not important for the usual, finite, prime

Motivic and Chromatic Homotopy Theory

Algebraic topology uses algebraic invariants, such as homology groups, to classify and distinguish topological spaces. The algebraic tools themselves have become the focus of intense study. *Chromatic homotopy theory* seeks to organize how shapes can be mapped into one another (homotopy) by filtering it through algebraic data coming from the theory of formal groups. *Motivic homotopy theory* builds a direct bridge, or "motive," between algebraic geometry and topology. The ultimate goal is a structural understanding of shape itself, with the calculation of the *stable homotopy groups of spheres* remaining one of the field's major goals.

Anabelian Geometry

Anabelian geometry is asking if we can recover an arithmetic variety X given information about the (étale) fundamental group of X. While Motivic Homotopy Theory seeks to build a dictionary of algebraic invariants to describe a space (using its entire tower of homotopy groups π_n of the sphere spectrum), anabelian geometry focuses on the first étale homotopy group, $\pi_1^{\text{ét}}$, and posits that for a special class of "hyperbolic" varieties, $\pi_1^{\text{ét}}$ contains all the necessary information to recover the variety.

The central algebraic invariant in this field is the *étale fundamental group*, denoted $\pi_1^{\text{ét}}(X)$. Its connection to the homotopy $\pi^{\mathbf{Top}}$ group comes from how the universal cover $E \to X$ and the intermediate covers has a galois correspondence to the subgroups of $\pi^{\mathbf{Top}}$. An example of a simple étale fundamental group is those for a field k, where $\pi^{\text{ét}}(k) = \text{Gal}_k(k_s)$ where k_s is the maximal separable extension of k. More generally, for a variety X defined over a number field like $\mathbb{Q}$, its étale fundamental group is a profinite group that captures not only the topological shape of X (as seen over the complex numbers) but also the entirety of its arithmetic structure.

The "anabelian" hypothesis is the conjecture that this reconstruction is only possible when the group is sufficiently complex and far from being abelian.

- **Abelian Case (Fails):** For an object like an elliptic curve, whose fundamental group is abelian, this principle fails. Many different elliptic curves can have the same fundamental group, so the group does not uniquely determine the geometry.
- **Anabelian Case (Conjectured to Succeed):** For a "hyperbolic" variety, such as a curve of genus two or more, the étale fundamental group is a non-abelian group. The anabelian hypothesis posits that there is enough information to uniquely determines the original variety.

This makes anabelian geometry a subfield of arithmetic geometry in its goals, but leaning more towards the philosophy of geometric group theory – it seeks to understand geometry by studying the algebraic properties of an associated group. An unsolved problem (as of writing this book) is the *Section Conjecture* which attempts to use the structure of $\pi_1^{\text{ét}}(X)$ to describe the set of rational points on X, offering a new (non-abelian) approach to solving Diophantine equations.

Thanks for reading!